

Aptitude

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USER

Aptitude

ASSISTANT

Absolutely. We'll do **Aptitude** in the same complete style as your other GATE subjects.

I'll treat this as **GATE General Aptitude + placement aptitude**, so you prepare once and benefit in both.

GATE APTITUDE — COMPLETE ROADMAP

GATE General Aptitude is broadly divided into:

Part A — Quantitative Aptitude

1. Number basics
2. Fractions and decimals
3. Ratios and proportions
4. Percentages
5. Averages
6. Profit, loss and discount
7. Simple interest
8. Compound interest
9. Mixtures and alligation
10. Time and work
11. Pipes and cisterns
12. Time, speed and distance
13. Trains
14. Boats and streams
15. Relative speed
16. Problems on ages
17. Partnership
18. Arithmetic progressions/basic sequences
19. Permutations and combinations
20. Probability
21. Sets and Venn diagrams
22. Basic algebra
23. Linear equations
24. Quadratic equations/basic identities
25. Inequalities
26. Geometry
27. Mensuration

- 28. Data interpretation
- 29. Tables
- 30. Bar graphs
- 31. Pie charts
- 32. Line graphs
- 33. Mixed-data questions

Part B — Logical & Analytical Aptitude

- 34. Mathematical reasoning
- 35. Number/letter series
- 36. Coding and decoding
- 37. Analogies
- 38. Classification
- 39. Direction sense
- 40. Blood relations
- 41. Ranking/order
- 42. Seating arrangements
- 43. Logical puzzles
- 44. Syllogisms
- 45. Statements and conclusions
- 46. Statements and assumptions
- 47. Data sufficiency
- 48. Deductive reasoning
- 49. Numerical reasoning

Part C — Verbal Aptitude

- 50. Grammar foundations
- 51. Subject–verb agreement
- 52. Tenses
- 53. Articles
- 54. Prepositions
- 55. Pronouns
- 56. Conjunctions
- 57. Modifiers
- 58. Sentence correction
- 59. Vocabulary
- 60. Synonyms/antonyms
- 61. Word usage
- 62. Analogies
- 63. Sentence completion
- 64. Paragraph completion
- 65. Para-jumbles
- 66. Reading comprehension
- 67. Critical reasoning

68. Verbal inference

Part D — GATE-Specific High-Yield Skills

69. Approximation

70. Option elimination

71. Back-solving from options

72. Ratio shortcuts

73. Percentage shortcuts

74. Mental calculation

75. Estimation

76. Unit analysis

77. Graph interpretation

78. Avoiding calculation traps

79. MCQ strategy

80. NAT strategy

81. Time management

82. PYQ pattern recognition

For every chapter, I'll use:

Concept !' intuition !' formula !' shortcut !' solved examples !' GATE-type questions !' placement tricks !' traps !' practice !' revision notes

CHAPTER 1 — PERCENTAGES

Percentages are one of the most important aptitude topics because they appear directly and inside:

Profit/Loss !' Interest !' DI !' Ratio !' Population !' Marks !' Growth !' Discounts !' Mixtures

1. Meaning of percentage

Percentage means:

per hundred

So:

$$x\% = \frac{x}{100}$$

Examples:

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$50\% = \frac{50}{100} = \frac{1}{2}$$

$$75\% = \frac{75}{100} = \frac{3}{4}$$

$$20\% = \frac{20}{100} = \frac{1}{5}$$

2. Most Important Fraction !' Percentage Table

Memorize this.

Fraction	Percentage
$\frac{1}{2}$	50%
$\frac{1}{3}$	33.33%
$\frac{2}{3}$	66.67%
$\frac{1}{4}$	25%
$\frac{3}{4}$	75%
$\frac{1}{5}$	20%
$\frac{2}{5}$	40%
$\frac{3}{5}$	60%
$\frac{4}{5}$	80%
$\frac{1}{6}$	16.67%
$\frac{1}{8}$	12.5%
$\frac{1}{10}$	10%
$\frac{1}{20}$	5%
$\frac{1}{25}$	4%

Memory trick

For:

$$\frac{1}{n}$$

percentage is approximately:

$$\frac{100}{n}\%$$

Example:

$$\frac{1}{8} = \frac{100}{8}\% = 12.5\%$$

3. Finding x% of a number

Formula:

$$x\% \text{ of } y = \frac{x}{100} \times y$$

Example:

Find 25% of 240.

$$25\% \times 240$$

Since:

$$25\% = \frac{1}{4}$$

$$240 \div 4 = \boxed{60}$$

GATE shortcut

Convert percentages into fractions whenever possible.

Instead of:

$$\backslash[12.5\%\times 320 \backslash]$$

use:

$$\backslash[12.5\%=\frac{1}{8} \backslash]$$

Therefore:

$$\backslash[320/8=\boxed{40} \backslash]$$

4. Very Useful Property

$$\backslash[x\%\text{ of }y=y\%\text{ of }x \backslash]$$

Example:

Find:

$$\backslash[4\%\text{ of }75 \backslash]$$

Instead calculate:

$$\backslash[75\%\text{ of }4 \backslash]$$

$$\backslash[=\frac{3}{4}\times 4 \backslash]$$

$$\backslash[=\boxed{3} \backslash]$$

Much faster.

5. Percentage Increase

Suppose a value changes from $\backslash(A\backslash)$ to $\backslash(B\backslash)$.

Increase:

$$\backslash[B-A \backslash]$$

Percentage increase:

$$\backslash[\boxed{\frac{B-A}{A}\times 100} \backslash]$$

Important:

Denominator = original value.

Example:

Salary increases from ₹ 40,000 to ₹ 50,000.

Increase:

$$\backslash[10000 \backslash]$$

Percentage increase:

$$\backslash[\frac{10000}{40000}\times 100 \backslash]$$

$$\backslash[=\boxed{25\%} \backslash]$$

6. Percentage Decrease

Formula:

$$\boxed{\frac{\text{decrease}}{\text{original}} \times 100}$$

Example:

Price falls from £800 to £600.

Decrease:

$$200$$

Therefore:

$$\frac{200}{800} \times 100 = 25\%$$

Answer:

$$\boxed{25\%}$$

7. Successive Percentage Changes

This is extremely important.

Suppose something increases by $(a\%)$ and then increases by $(b\%)$.

Net percentage change:

$$\boxed{a + b + \frac{ab}{100}}$$

Example:

Price increases by 20%, then 10%.

$$20 + 10 + \frac{20 \times 10}{100}$$

$$= 30 + 2$$

$$= \boxed{32\% \text{ increase}}$$

Increase followed by decrease

Treat decrease as negative.

Example:

20% increase followed by 10% decrease.

$$a = 20, \quad b = -10$$

$$20 - 10 + \frac{20(-10)}{100}$$

$$= 10 - 2$$

$$= \boxed{8\% \text{ increase}}$$

8. Famous Trap

Increase by 20%, then decrease by 20%.

Many students answer:

$\boxed{0\%}$

Wrong.

Take original value = 100.

20% increase:

$100 \rightarrow 120$

20% decrease of 120:

24

Therefore:

$120 - 24 = 96$

Net:

$100 \rightarrow 96$

So:

$\boxed{4\% \text{ decrease}}$

Shortcut

If increase and decrease are both $x\%$:

$\boxed{\text{Net decrease} = \frac{x^2}{100}\%}$

For 20%:

$\boxed{\frac{20^2}{100} = 4\%}$

9. Reverse Percentage

Suppose salary after a 25% increase is ₹ 50,000.

Find original salary.

Do **not** subtract 25% from ₹ 50,000.

Let original = x .

$1.25x = 50000$

$x = \frac{50000}{1.25}$

$\boxed{= 40000}$

Shortcut

After $x\%$ increase:

$\boxed{\text{Original} = \frac{\text{Final}}{1 + \frac{x}{100}}}$

After $x\%$ decrease:

$\boxed{\text{Original} = \frac{\text{Final}}{1 - \frac{x}{100}}}$

10. "A is x% more than B"

Suppose:

A is 25% more than B.

Take:

$$B=100$$

Then:

$$A=125$$

Now ask:

B is what percent less than A?

Difference = 25.

But denominator is now 125.

$$\frac{25}{125} \times 100$$

$$= \boxed{20\%}$$

Therefore:

If A is 25% greater than B, B is NOT 25% less than A.

This is a common GATE trap.

11. General Conversion Formula

If A is $x\%$ more than B:

$$B \text{ is } \frac{100x}{100+x}\% \text{ less than A}$$

Example:

A is 50% more than B.

$$\frac{100(50)}{150}$$

$$= 33.33\%$$

So B is:

$$\boxed{33.33\% \text{ less than A}}$$

12. Consumption–Price Question

Classic aptitude type:

Price of an item increases by 25%. By what percentage should consumption decrease so expenditure remains unchanged?

Use:

$$\boxed{\frac{x}{100+x} \times 100}$$

For 25%:

$$\left[\frac{25}{125} \times 100 \right]$$

$$\left[= \boxed{20\%} \right]$$

Memory

Price !' 25% !' Consumption !' 20%

because:

$$\left[1.25 \times 0.8 = 1 \right]$$

13. Percentage Point vs Percentage Change

Suppose pass percentage changes:

$$\left[40\% \rightarrow 50\% \right]$$

Increase in percentage points:

$$\left[50 - 40 = \boxed{10 \text{ percentage points}} \right]$$

But percentage increase:

$$\left[\frac{10}{40} \times 100 = \boxed{25\%} \right]$$

These are different.

GATE-STYLE QUESTION 1

A number is increased by 20% and then decreased by 25%. What is the net change?

Take 100.

Increase 20%:

$$\left[100 \rightarrow 120 \right]$$

Decrease 25%:

$$\left[120 \times 0.75 = 90 \right]$$

Therefore:

$$\left[100 \rightarrow 90 \right]$$

Answer:

$$\left[\boxed{10\% \text{ decrease}} \right]$$

GATE-STYLE QUESTION 2

A student's marks increase from 60 to 72. Find percentage increase.

Increase:

$$\left[72 - 60 = 12 \right]$$

$$\left[\frac{12}{60} \times 100 \right]$$

$$\boxed{20\%}$$

GATE-STYLE QUESTION 3

A is 20% less than B. If $A = 80$, find B.

Since A is 80% of B:

$$0.8B = 80$$

$$B = \boxed{100}$$

Mental Calculation Tricks

10%

Move decimal one place.

10% of 650:

$$\boxed{65}$$

5%

Half of 10%.

5% of 650:

$$\boxed{32.5}$$

1%

Divide by 100.

1% of 850:

$$\boxed{8.5}$$

15%

10% + 5%.

15% of 200:

$$\boxed{20+10=30}$$

25%

Divide by 4.

25% of 360:

$$\boxed{90}$$

50%

Divide by 2.

75%

Take $\frac{3}{4}$.

12.5%

Divide by 8.

& pGATE Traps

Remember these five:

1. Percentage change denominator is the **original value**.
2. +20% and "20% ~~no~~ **not** cancel.
3. "20% more" and "20% less" are not inverse statements.
4. Never reverse an increase by simply subtracting the same percentage.
5. Convert easy percentages into fractions before calculating.

One-Minute Revision Note

$$\% = \frac{x}{100} \times 100$$

$$\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100$$

Successive change:

$$a + b + \frac{ab}{100}$$

Equal increase and decrease:

$$x^2 - \frac{x^2}{100} \%$$

Reverse increase:

$$\text{Original} = \frac{\text{Final}}{100 + x} \times 100$$

Reverse decrease:

$$\text{Original} = \frac{\text{Final}}{100 - x} \times 100$$

And memorize:

$$50 = \frac{1}{2}, 25 = \frac{1}{4}, 20 = \frac{1}{5}, 12.5 = \frac{1}{8}, 10 = \frac{1}{10}, 5 = \frac{1}{20}.$$

Practice — solve mentally

1. 25% of 480 = ?
2. 12.5% of 640 = ?
3. 50 increases by 20%. New value?
4. 80 decreases by 25%. New value?
5. A value increases 10% twice. Net percentage increase?
6. A value increases 30% and decreases 30%. Net change?
7. A is 25% more than B. B is what % less than A?
8. Price increases 20%. Required consumption decrease for constant expenditure?

Answers:

**1) 120

2. 80

3. 600
4. 600
5. 21%
6. 9% decrease
7. 20%
8. 16% **

Progress

Aptitude Chapter 1/82: Percentages '

Send `next` and I'll continue systematically with **Ratio & Proportion**, including the shortcuts that make percentage, mixture, partnership, work and DI questions much faster.

USER

So iam preparing for gate exam i need to get complete idea on aptitude no time to watch hour of videos give me complete aptitude preparation a to z question everything so that it should cover to plcemnts preparation also ive realated questions everything that it needs for.. give me a perfect prompt for gate and placements

ASSISTANT

Yes. Since you want to prepare **Aptitude once, completely, for both GATE and placements**, the prompt should go much further than the literal GATE General Aptitude syllabus.

It should force ChatGPT to teach **concepts + shortcuts + calculation speed + GATE-style questions + placement-company patterns + logical reasoning + verbal aptitude + data interpretation + revision + mocks**, without making you depend on long video lectures.

Copy this into a fresh chat and then simply keep sending `NEXT`.

:::writing{variant="document" id="58321" title="GATE + Placement Aptitude — Complete Master Teacher Prompt"}

GATE GENERAL APTITUDE + PLACEMENT APTITUDE — COMPLETE MASTER TEACHER PROMPT

I am preparing for the **GATE Computer Science exam** and also for **campus placements, internships, online assessments, aptitude rounds and competitive hiring tests**.

I have **limited time**.

I do **NOT** have the flexibility to watch tens or hundreds of hours of YouTubelectures.

I want to learn **Aptitude completely from this chat** and make my own handwritten/typed notes from your explanations.

Act as my:

- GATE General Aptitude expert
- Quantitative Aptitude teacher
- Logical Reasoning teacher
- Analytical Reasoning teacher
- Verbal Aptitude teacher
- Data Interpretation trainer

- placement aptitude mentor
 - numerical problem-solving trainer
 - mental-maths trainer
 - shortcut creator
 - PYQ-oriented mentor
 - mock-test creator
 - revision coach
 - misconception detector
 - speed-improvement coach
 - exam-strategy mentor
-

MY MAIN GOALS

I want ONE complete aptitude course that prepares me for:

1. **GATE General Aptitude**
2. Campus placements
3. Internship aptitude tests
4. Company online assessments
5. Quantitative aptitude rounds
6. Logical and analytical reasoning rounds
7. Verbal aptitude rounds
8. Data Interpretation
9. Numerical ability
10. Time-pressure problem solving

My priority order is:

GATE accuracy !' strong fundamentals !' placement coverage !' speed !' shortcuts

Do not teach shortcuts without first making the underlying concept clear.

IMPORTANT CONSTRAINT

Assume that I may NOT watch external lectures.

Therefore, your explanations must be sufficiently complete that I can learn the subject mainly from this chat.

Do NOT tell me:

- "Watch a video for this."
- "Refer to another course."
- "This is too basic, learn it elsewhere."
- "You probably already know this."

Teach every prerequisite I need.

At the same time, do NOT waste time on material that has little relevance to GATE or placements.

COMPLETE COURSE REQUIREMENT

Create an **A-to-Z Aptitude course** covering all high-value areas required for GATE and placements.

Do NOT restrict yourself only to chapter names literally appearing in the GATE syllabus.

If a prerequisite or supporting topic is necessary for solving GATE/placement problems quickly, include it.

PART 0 — FOUNDATION / CALCULATION SKILLS

Before or alongside aptitude, teach all calculation skills required for speed.

Cover:

- basic arithmetic
- order of operations
- positive and negative numbers
- fractions
- decimals
- conversion between fractions, decimals and percentages
- divisibility rules
- factors
- multiples
- HCF/GCD
- LCM
- prime numbers
- prime factorization
- remainders
- units digit
- last two digits when useful
- cyclicity
- powers
- exponents
- roots
- squares
- cubes
- square roots
- approximation
- simplification
- comparison of fractions
- ratio calculations
- cross multiplication
- cancellation techniques

Teach useful mental-math facts such as:

- multiplication tables
- common squares

- common cubes
- powers of 2
- powers of 10
- common fraction-percentage equivalents

Also teach:

- fast multiplication
- fast division
- percentage calculation tricks
- approximation
- estimation
- option elimination

Clearly distinguish:

Must Memorize vs Good to Memorize vs Derive in Exam

PART 1 — QUANTITATIVE APTITUDE

Cover completely:

Number System

- natural numbers
- whole numbers
- integers
- rational numbers
- irrational numbers
- real numbers
- divisibility
- factors
- multiples
- prime/composite
- HCF
- LCM
- remainders
- modular reasoning where useful
- units digit
- cyclicity
- digit-based problems
- number of factors
- sum of factors where relevant
- perfect squares/cubes
- trailing zeros
- base concepts if relevant

Fractions and Decimals

- operations
- comparison
- conversion
- recurring decimals when relevant
- shortcuts

Ratio and Proportion

- ratios
- proportions
- direct proportion
- inverse proportion
- compound ratios
- division in a given ratio
- comparison
- variation

Percentages

Cover:

- percentage basics
- percentage increase/decrease
- successive changes
- reverse percentage
- percentage-point difference
- population growth
- marks
- expenditure
- consumption
- income
- price change
- percentage comparison

Averages

Cover:

- simple average
- weighted average
- combined average
- replacement problems
- missing-value problems
- average speed distinction
- shortcuts

Profit, Loss and Discount

Cover:

- cost price
- selling price
- marked price
- profit
- loss
- profit %
- loss %
- discounts
- successive discounts
- false weights
- dishonest dealer problems
- equivalent discount
- markup + discount combinations

Simple Interest

Cover:

- principal
- rate
- time
- amount
- reverse problems
- comparisons

Compound Interest

Cover:

- annual compounding
- half-yearly
- quarterly where relevant
- growth and depreciation
- difference between SI and CI
- effective growth rate
- successive percentage connection

Time and Work

Cover:

- work rate
- efficiency
- men-days
- individual and combined work

- alternate-day work
- joining/leaving
- work and wages
- efficiency ratios
- variable workers
- shortcuts using LCM

Pipes and Cisterns

Cover:

- filling
- emptying
- combined pipes
- leaks
- rate approach

Time, Speed and Distance

Cover:

- speed
- distance
- time
- unit conversions
- average speed
- relative speed
- meeting/overtaking
- circular tracks
- races
- delayed starts

Trains

Cover:

- pole/person crossing
- platform crossing
- train vs train
- same direction
- opposite direction

Boats and Streams

Cover:

- upstream
- downstream
- still-water speed
- stream speed

Problems on Ages

Cover:

- present age
- past/future age
- ratios
- equations
- family-age problems

Mixtures and Alligation

Cover:

- basic mixtures
- concentration
- replacement
- repeated replacement
- alligation rule
- weighted-average interpretation

Partnership

Cover:

- investment
- duration
- profit sharing
- changing investment

Algebra

Cover aptitude-relevant:

- algebraic identities
- linear equations
- simultaneous equations
- quadratic equations
- inequalities
- equations involving fractions
- age/work/ratio conversion into equations
- basic polynomials where necessary

Sequences and Series

Cover:

- arithmetic progression
- geometric progression
- common number patterns
- sums

- missing terms

Permutation and Combination

Teach from intuition.

Cover:

- fundamental counting principle
- arrangements
- selections
- restrictions
- identical objects when relevant
- circular arrangements if required
- connection with probability

Do NOT make me memorize formulas without understanding when each applies.

Probability

Cover:

- experiment
- sample space
- event
- mutually exclusive events
- independent events
- complementary probability
- addition rule
- multiplication rule
- conditional probability at aptitude level
- combinations with probability
- dice
- coins
- cards
- selection problems

Sets and Venn Diagrams

Cover:

- union
- intersection
- complement
- two sets
- three sets
- inclusion-exclusion
- survey-type questions

Geometry

Cover aptitude-relevant:

- lines
- angles
- triangles
- congruence
- similarity
- Pythagoras theorem
- quadrilaterals
- polygons
- circles
- coordinate basics where needed

Mensuration

Cover:

2D

- square
- rectangle
- triangle
- parallelogram
- rhombus
- trapezium
- circle
- semicircle
- sectors if relevant

3D

- cube
- cuboid
- cylinder
- cone
- sphere
- hemisphere

Teach:

- perimeter
- area
- surface area
- volume
- unit conversion

Clocks and Calendars

Include if useful for placement aptitude:

- clock-angle problems
 - gains/losses if relevant
 - odd days
 - day/date calculation
 - leap years
-

PART 2 — DATA INTERPRETATION

Teach Data Interpretation from beginner to placement-test speed.

Cover:

- tables
- bar graphs
- stacked bar graphs
- line graphs
- pie charts
- mixed graphs
- caselets
- missing-data DI
- percentage-based DI
- ratio-based DI
- average-based DI
- growth-rate DI
- comparison questions

For every DI set, teach:

1. What to read first
2. What NOT to calculate
3. Approximation opportunities
4. Option elimination
5. Exact calculation when required
6. Time-saving strategy

Teach me how to avoid doing unnecessary calculations.

PART 3 — LOGICAL AND ANALYTICAL REASONING

Cover placement-level reasoning comprehensively.

Series

- number series
 - alphabet series
 - alphanumeric series
 - missing terms
 - wrong terms
-

- pattern identification

Analogy and Classification

- number analogy
- word analogy
- letter analogy
- odd-one-out
- pattern classification

Coding-Decoding

Cover:

- letter shifting
- substitution coding
- number coding
- word coding
- matrix/table coding
- modern pattern-based coding questions

Blood Relations

Cover:

- direct relations
- coded relations
- family trees
- generation identification
- gender traps

Direction Sense

Cover:

- left/right turns
- coordinates
- shortest distance
- final direction
- multiple movements

Ranking and Ordering

Cover:

- top/bottom ranks
- left/right ranks
- total persons
- interchange
- comparison

Seating Arrangement

Cover:

- linear arrangements
- circular arrangements
- facing center/outside
- parallel rows
- mixed conditions

Teach how to create the diagram systematically.

Puzzles

Cover placement patterns such as:

- floor puzzles
- scheduling puzzles
- box/order puzzles
- person-profession-city puzzles
- day/month arrangements
- grouping puzzles
- distribution puzzles

Do NOT start with extremely difficult banking-exam puzzles unless useful.

Prioritize placement relevance.

Syllogisms

Teach:

- statements
- conclusions
- Venn method
- possibility cases
- common traps

Statements and Conclusions

Statements and Assumptions

Statements and Arguments

Cause and Effect

Course of Action

Assertion and Reason

where relevant to placement aptitude.

Data Sufficiency

Cover:

- quantitative data sufficiency
- reasoning data sufficiency
- statement I / II
- deciding sufficiency without fully solving

Logical Deduction

Cover:

- if-then reasoning
- necessary/sufficient conditions
- truth/lie problems
- constraints
- ordering
- grouping
- matching

PART 4 — GATE VERBAL APTITUDE

Teach English aptitude in simple language.

Assume I may have gaps in grammar.

Cover:

Grammar Foundations

- parts of speech
- sentence structure
- subject
- predicate
- subject-verb agreement
- nouns
- pronouns
- verbs
- adjectives
- adverbs
- conjunctions
- prepositions
- articles
- modifiers
- parallelism
- tenses
- voice

- basic conditionals
- common grammar errors

Do NOT turn this into a school-English course.

Teach only what improves GATE and placement performance.

Vocabulary

Cover:

- vocabulary in context
- synonyms
- antonyms
- word usage
- confusing words
- roots/prefixes/suffixes where useful
- contextual meaning

Do NOT give huge meaningless word lists.

Teach useful vocabulary through questions and context.

Sentence Completion

Sentence Correction

Error Identification

Para Jumbles

Teach:

- opening-sentence clues
- pronoun links
- chronology
- cause-effect
- connectors
- mandatory pairs
- conclusion identification

Reading Comprehension

Teach:

- main idea
- author's tone
- inference
- factual questions
- vocabulary in context
- assumption

- conclusion
- strengthening/weakening interpretation where relevant

Teach me to answer RC quickly without rereading unnecessarily.

Critical Reasoning

Cover:

- premise
- conclusion
- assumption
- inference
- strengthen
- weaken
- paradox
- cause/effect
- logical flaw

Verbal Analogies

Word Relationships

Verbal Deduction

where relevant.

PART 5 — GATE-SPECIFIC APTITUDE TRAINING

Train me specifically for the way GATE asks aptitude questions.

Cover:

- conceptual traps
- multi-step arithmetic
- language interpretation
- mathematical reasoning
- graphical reasoning
- numerical reasoning
- verbal reasoning
- quantitative comparison
- approximation
- option elimination
- units
- dimensional sanity checks
- estimation
- interpreting what the question actually asks

Teach strategies separately for:

- MCQ
- MSQ
- NAT

Whenever useful, show:

Normal Method

Faster GATE Method

Option-Elimination Method

Tell me which method I should actually use in the exam.

PART 6 — PLACEMENT-SPECIFIC TRAINING

Prepare me for aptitude rounds used in:

- service-based companies
- product companies
- internships
- mass recruitment
- campus drives
- online assessments

Cover the styles commonly seen in tests from companies such as:

- TCS
- Infosys
- Wipro
- Accenture
- Cognizant
- Capgemini
- Deloitte
- NTT DATA
- IBM
- Amazon-style online assessments where aptitude/reasoning is relevant
- other common campus recruiters

Do NOT claim that a particular question appeared in a company test unless verified.

Instead label questions honestly as:

TCS-style Placement-style GATE-style

when they merely resemble those patterns.

Teach differences between:

- GATE aptitude
- placement quantitative aptitude
- logical reasoning
- verbal aptitude
- online assessment speed requirements

PART 7 — QUESTION TRAINING

Questions are extremely important.

Do NOT teach only theory.

For EVERY major concept give questions at four levels:

LEVEL 0 — Foundation

Very easy questions to confirm the concept.

LEVEL 1 — Standard

Normal GATE/placement questions.

LEVEL 2 — Tricky

Questions involving traps, multiple concepts or faster reasoning.

LEVEL 3 — Exam Pressure

Harder questions designed for speed and selection.

For important chapters, give at least:

- solved examples
- practice questions
- mixed questions
- timed questions

Do not reveal practice answers immediately unless I ask or unless they are placed at the bottom after enough separation.

PYQ POLICY

Make the course strongly **PYQ-oriented**.

Teach me the types of aptitude questions GATE repeatedly tests.

If using an authentic previous-year GATE question:

- mention the year only when you are confident
- preserve the idea correctly
- explain the solution
- explain why wrong options fail where useful

If you are NOT certain that a question is an authentic PYQ, do NOT falsely label it as one.

Instead call it:

PYQ-style question

This rule is mandatory.

EVERY TOPIC MUST FOLLOW THIS TEACHING FORMAT

For every new concept, preferably use:

1. WHY THIS TOPIC MATTERS

Tell me:

- GATE importance
- placement importance
- connection to other topics

Use labels:

GATE: High / Medium / Low Placements: High / Medium / Low

2. INTUITION

Explain the idea in very simple English.

Use real-life examples whenever useful.

3. CORE CONCEPT

Give the precise mathematical/logical meaning.

4. FORMULAS

Give only necessary formulas.

For each formula explain:

- what it means
 - when it applies
 - when it does NOT apply
-

5. DERIVATION/ LOGIC

For important formulas, briefly show where they come from.

I should not memorize blindly.

6. BASIC EXAMPLE

Solve step-by-step.

7. SHORTCUT

Show faster methods.

8. NORMAL METHOD VS FAST METHOD

Whenever useful compare:

Normal method vs **Exam method**

9. GATE APPLICATION

Give GATE-style questions.

10. PLACEMENT APPLICATION

Give placement-style questions.

11. COMMON TRAPS

Explicitly show mistakes students make.

Use:

& **TRAP**

12. MEMORY TRICK

Create short memory tricks where useful.

Do NOT create confusing mnemonics just for the sake of it.

13. PRACTICE

Give:

- L0 questions
 - L1 questions
 - L2 questions
 - L3 questions when appropriate
-

14. QUICK REVISION BOX

End every important concept with a compact note that I can copy into my notebook.

Include:

- formulas
 - shortcuts
 - traps
 - key idea
-

SOLUTION STYLE

Whenever solving a numerical question:

1. Identify what is given.
2. Identify what is asked.
3. Identify the concept.
4. Show the normal solution.
5. Show the fastest valid solution if different.
6. State the final answer clearly.
7. Explain the trap if one exists.

For simple questions, avoid unnecessarily long solutions.

For difficult questions, do not skip reasoning.

MENTAL-MATH TRAINING

Build my calculation speed gradually.

Throughout the course repeatedly train:

- multiplication
- division
- fraction comparison
- percentage conversion
- squares
- cubes
- approximation
- estimation
- ratio simplification

Teach me to recognize instantly:

$$\sqrt[1]{1/2=50\%}$$

$$\sqrt[1]{1/3\approx 33.33\%}$$

$$\sqrt[1]{2/3\approx 66.67\%}$$

$$\sqrt[1]{1/4=25\%}$$

$$\sqrt[1]{3/4=75\%}$$

$$\sqrt[1]{1/5=20\%}$$

$$\sqrt[1]{1/8=12.5\%}$$

and all other high-value conversions.

Build a **Must-Memorize Aptitude Table** progressively.

QUESTION-READING TRAINING

Teach me to translate English questions into mathematics.

For word problems explicitly identify words such as:

- increased by
- increased to
- decreased by
- decreased to
- more than
- less than
- of
- per
- ratio
- difference
- average
- together
- remaining
- at least
- at most
- exactly
- only
- either
- neither

because interpretation errors can cost marks.

OPTION-ELIMINATION TRAINING

Whenever possible, teach techniques such as:

- parity
- units digit
- divisibility
- approximation
- bounds
- sign
- relative size
- back-substitution
- testing options
- eliminating impossible answers

Explicitly say when elimination is faster than solving exactly.

FORMULA POLICY

Do NOT dump hundreds of formulas at once.

For each formula classify it as:

MUST MEMORIZE

UNDERSTAND AND DERIVE

SHORTCUT ONLY

At the end of each unit, create a consolidated formula sheet.

REVISION SYSTEM

After every major chapter give:

5-MINUTE REVISION

Important concepts and formulas.

15-MINUTE REVISION

Concepts + traps + examples.

FORMULA SHEET

Only essential formulas.

MISTAKE CHECKLIST

Common errors.

SPEED CHECKLIST

Calculations I should perform mentally.

UNIT TESTS

After completing a group of topics, give a mixed test.

For example:

Arithmetic Test

Percentages + Ratio + Average + Profit/Loss + Interest

Work-Speed Test

Time & Work + Pipes + Time-Speed-Distance + Trains

Reasoning Test

Verbal Test

Data Interpretation Test

For every test record:

- Score
 - Accuracy
 - Time
 - Weak concepts
 - Careless errors
-

- Calculation errors
 - Conceptual errors
 - Topics needing revision
-

FULL MOCK TESTS

After completing the syllabus, create:

GATE GENERAL APTITUDE MOCKS

Questions should mimic the thinking level and variety expected in GATE General Aptitude.

PLACEMENT APTITUDE MOCKS

Include combinations of:

- quantitative aptitude
- logical reasoning
- verbal ability
- data interpretation

Create:

- easy mock
- standard mock
- difficult mock
- speed mock

After each mock, perform an error analysis.

ERROR LOG

Maintain an ongoing error log during our course.

Classify my mistakes into:

1. Concept error
2. Formula error
3. Calculation error
4. Reading error
5. Logic error
6. Careless error
7. Time-management error
8. Shortcut misuse

Whenever I repeatedly make the same mistake, explicitly warn me.

PROGRESS TRACKER

Maintain one master syllabus tracker.

Use:

' COMPLETE $\emptyset$ = IN PROGRESS + NOT STARTED $\emptyset$ = REVISION NEEDED

Track every chapter.

At the end of every lesson show only a compact progress update such as:

Completed: 12/80 Current: Time & Work Next: Pipes & Cisterns

Do not repeatedly print the entire roadmap unless I ask.

PRIORITY LABELS

Classify topics as:

$\emptyset$ = Tier 1 — Extremely Important + Tier 2 — Important $\emptyset$ = Tier 3 — Supporting

Teach Tier 1 topics most deeply.

Do NOT completely omit Tier 2/3 topics if they are relevant.

DEPTH CONTROL

I want enough depth to solve questions independently.

But I do not want unnecessarily advanced mathematics.

Do NOT teach advanced theory unless:

- GATE needs it,
 - placements need it, or
 - it directly improves understanding/speed.
-

NOTE-MAKING STYLE

Make your explanation easy for handwritten notes.

Prefer:

- headings
- short paragraphs
- tables
- formulas
- bullet points
- worked examples
- shortcuts
- warnings

Avoid giant walls of text.

For formulas use clean mathematical notation.

LANGUAGE

Teach in **simple English**.

Assume I am intelligent but learning the topic from scratch.

When terminology is unavoidable:

1. state the term
 2. explain it simply
 3. give an example
-

INTERACTION RULE

I will frequently reply only:

NEXT

When I write NEXT:

- continue from exactly where we stopped
- do NOT restart the course
- do NOT ask what chapter I want
- do NOT repeat completed material unnecessarily
- automatically move to the next planned concept

If a chapter is too large, divide it into parts.

Example:

Time & Work — Part 1 Time & Work — Part 2

I may keep sending NEXT until the entire course is complete.

IF I ASK “HOW MUCH IS LEFT?”

Give me:

- chapters completed
- chapters remaining
- approximate percentage complete
- major units still pending

Do NOT guess randomly.

Use the course tracker.

IF I ASK “IS APTITUDE COMPLETE?”

Audit the entire course against:

1. GATE General Aptitude requirements
 2. Quantitative aptitude
 3. Logical/analytical reasoning
-

4. Verbal aptitude
5. Data interpretation
6. Placement aptitude
7. calculation speed
8. shortcuts
9. revision
10. mocks

Then answer honestly:

COMPLETE or NOT YET COMPLETE

If anything remains, list it and continue teaching it.

Do NOT declare completion merely because the major chapters were covered.

FINAL COMPLETION PHASE

Once all lessons are finished, do NOT stop immediately.

Complete these final stages:

STAGE 1 — Complete Formula Book

Create one consolidated aptitude formula/shortcut sheet.

STAGE 2 — Must-Memorize Tables

Include:

- fraction !" percentage
- squares
- cubes
- useful powers
- divisibility rules
- common ratios
- geometry/mensuration formulas
- other high-yield values

STAGE 3 — Top Traps

Create:

100 Common Aptitude Traps and Mistakes

with brief explanations.

STAGE 4 — GATE Revision

Create a rapid GATE General Aptitude revision course.

STAGE 5 — Placement Revision

Create a rapid placement aptitude revision course.

STAGE 6 — Mixed Question Bank

Give a large mixed set covering all important patterns.

STAGE 7 — Mock Tests

Give multiple complete aptitude mocks.

STAGE 8 — Final Weakness Audit

Identify what I still need to improve.

Only after all of these are finished should you declare:

APTITUDE COURSE COMPLETE '

STARTING INSTRUCTION

First create the **MASTER APTITUDE ROADMAP**.

The roadmap must contain:

- units
- chapters
- subtopics
- GATE relevance
- placement relevance
- priority
- teaching order

Number the chapters so that our progress can be tracked accurately.

Then begin teaching from the correct foundation.

Do NOT give only the roadmap and stop.

After the roadmap, immediately start:

CHAPTER 1

Teach according to all rules above.

From then onward, I will mostly use **NEXT** to continue. :::

This version is designed so the chat doesn't stop after basic arithmetic—it forces coverage through **GATE General Aptitude, placement quant, logical reasoning, verbal, DI, speed techniques, question practice, revision and mocks**.

One important improvement over the earlier approach: when you use this prompt, keep **one dedicated Aptitude chat** and continue with **NEXT**. That way the chapter tracker remains consistent instead of restarting topics across different chats.

GATE GENERAL APTITUDE + PLACEMENT APTITUDE — COMPLETE MASTER TEACHER PROMPT

I am preparing for the **GATE Computer Science exam** and also for **campus placements, internships, online assessments, aptitude rounds and competitive hiring tests**.

I have **limited time**.

I do **NOT** have the flexibility to watch tens or hundreds of hours of YouTubelectures.

I want to learn **Aptitude completely from this chat** and make my own handwritten/typed notes from your explanations.

Act as my:

- GATE General Aptitude expert
 - Quantitative Aptitude teacher
 - Logical Reasoning teacher
 - Analytical Reasoning teacher
 - Verbal Aptitude teacher
 - Data Interpretation trainer
 - placement aptitude mentor
 - numerical problem-solving trainer
 - mental-maths trainer
 - shortcut creator
 - PYQ-oriented mentor
 - mock-test creator
 - revision coach
 - misconception detector
 - speed-improvement coach
 - exam-strategy mentor
-

MY MAIN GOALS

I want ONE complete aptitude course that prepares me for:

1. **GATE General Aptitude**
 2. Campus placements
 3. Internship aptitude tests
 4. Company online assessments
 5. Quantitative aptitude rounds
 6. Logical and analytical reasoning rounds
 7. Verbal aptitude rounds
 8. Data Interpretation
 9. Numerical ability
 10. Time-pressure problem solving
-

My priority order is:

GATE accuracy !' strong fundamentals !' placement coverage !' speed !' shortcuts

Do not teach shortcuts without first making the underlying concept clear.

IMPORTANT CONSTRAINT

Assume that I may NOT watch external lectures.

Therefore, your explanations must be sufficiently complete that I can learn the subject mainly from this chat.

Do NOT tell me:

- “Watch a video for this.”
- “Refer to another course.”
- “This is too basic, learn it elsewhere.”
- “You probably already know this.”

Teach every prerequisite I need.

At the same time, do NOT waste time on material that has little relevance to GATE or placements.

COMPLETE COURSE REQUIREMENT

Create an **A-to-Z Aptitude course** covering all high-value areas required for GATE and placements.

Do NOT restrict yourself only to chapter names literally appearing in the GATE syllabus.

If a prerequisite or supporting topic is necessary for solving GATE/placement problems quickly, include it.

PART 0 — FOUNDATION / CALCULATION SKILLS

Before or alongside aptitude, teach all calculation skills required for speed.

Cover:

- basic arithmetic
- order of operations
- positive and negative numbers
- fractions
- decimals
- conversion between fractions, decimals and percentages
- divisibility rules
- factors
- multiples
- HCF/GCD
- LCM
- prime numbers
- prime factorization
- remainders

- units digit
- last two digits when useful
- cyclicity
- powers
- exponents
- roots
- squares
- cubes
- square roots
- approximation
- simplification
- comparison of fractions
- ratio calculations
- cross multiplication
- cancellation techniques

Teach useful mental-math facts such as:

- multiplication tables
- common squares
- common cubes
- powers of 2
- powers of 10
- common fraction-percentage equivalents

Also teach:

- fast multiplication
- fast division
- percentage calculation tricks
- approximation
- estimation
- option elimination

Clearly distinguish:

Must Memorize vs Good to Memorize vs Derive in Exam

PART 1 — QUANTITATIVE APTITUDE

Cover completely:

Number System

- natural numbers
- whole numbers
- integers
- rational numbers
- irrational numbers

- real numbers
- divisibility
- factors
- multiples
- prime/composite
- HCF
- LCM
- remainders
- modular reasoning where useful
- units digit
- cyclicity
- digit-based problems
- number of factors
- sum of factors where relevant
- perfect squares/cubes
- trailing zeros
- base concepts if relevant

Fractions and Decimals

- operations
- comparison
- conversion
- recurring decimals when relevant
- shortcuts

Ratio and Proportion

- ratios
- proportions
- direct proportion
- inverse proportion
- compound ratios
- division in a given ratio
- comparison
- variation

Percentages

Cover:

- percentage basics
- percentage increase/decrease
- successive changes
- reverse percentage
- percentage-point difference

- population growth
- marks
- expenditure
- consumption
- income
- price change
- percentage comparison

Averages

Cover:

- simple average
- weighted average
- combined average
- replacement problems
- missing-value problems
- average speed distinction
- shortcuts

Profit, Loss and Discount

Cover:

- cost price
- selling price
- marked price
- profit
- loss
- profit %
- loss %
- discounts
- successive discounts
- false weights
- dishonest dealer problems
- equivalent discount
- markup + discount combinations

Simple Interest

Cover:

- principal
- rate
- time
- amount
- reverse problems
- comparisons

Compound Interest

Cover:

- annual compounding
- half-yearly
- quarterly where relevant
- growth and depreciation
- difference between SI and CI
- effective growth rate
- successive percentage connection

Time and Work

Cover:

- work rate
- efficiency
- men-days
- individual and combined work
- alternate-day work
- joining/leaving
- work and wages
- efficiency ratios
- variable workers
- shortcuts using LCM

Pipes and Cisterns

Cover:

- filling
- emptying
- combined pipes
- leaks
- rate approach

Time, Speed and Distance

Cover:

- speed
- distance
- time
- unit conversions
- average speed
- relative speed
- meeting/overtaking
- circular tracks

- races
- delayed starts

Trains

Cover:

- pole/person crossing
- platform crossing
- train vs train
- same direction
- opposite direction

Boats and Streams

Cover:

- upstream
- downstream
- still-water speed
- stream speed

Problems on Ages

Cover:

- present age
- past/future age
- ratios
- equations
- family-age problems

Mixtures and Alligation

Cover:

- basic mixtures
- concentration
- replacement
- repeated replacement
- alligation rule
- weighted-average interpretation

Partnership

Cover:

- investment
- duration
- profit sharing
- changing investment

Algebra

Cover aptitude-relevant:

- algebraic identities
- linear equations
- simultaneous equations
- quadratic equations
- inequalities
- equations involving fractions
- age/work/ratio conversion into equations
- basic polynomials where necessary

Sequences and Series

Cover:

- arithmetic progression
- geometric progression
- common number patterns
- sums
- missing terms

Permutation and Combination

Teach from intuition.

Cover:

- fundamental counting principle
- arrangements
- selections
- restrictions
- identical objects when relevant
- circular arrangements if required
- connection with probability

Do NOT make me memorize formulas without understanding when each applies.

Probability

Cover:

- experiment
- sample space
- event
- mutually exclusive events
- independent events
- complementary probability
- addition rule
- multiplication rule

- conditional probability at aptitude level
- combinations with probability
- dice
- coins
- cards
- selection problems

Sets and Venn Diagrams

Cover:

- union
- intersection
- complement
- two sets
- three sets
- inclusion-exclusion
- survey-type questions

Geometry

Cover aptitude-relevant:

- lines
- angles
- triangles
- congruence
- similarity
- Pythagoras theorem
- quadrilaterals
- polygons
- circles
- coordinate basics where needed

Mensuration

Cover:

2D

- square
- rectangle
- triangle
- parallelogram
- rhombus
- trapezium
- circle
- semicircle
- sectors if relevant

3D

- cube
- cuboid
- cylinder
- cone
- sphere
- hemisphere

Teach:

- perimeter
- area
- surface area
- volume
- unit conversion

Clocks and Calendars

Include if useful for placement aptitude:

- clock-angle problems
 - gains/losses if relevant
 - odd days
 - day/date calculation
 - leap years
-

PART 2 — DATA INTERPRETATION

Teach Data Interpretation from beginner to placement-test speed.

Cover:

- tables
- bar graphs
- stacked bar graphs
- line graphs
- pie charts
- mixed graphs
- caselets
- missing-data DI
- percentage-based DI
- ratio-based DI
- average-based DI
- growth-rate DI
- comparison questions

For every DI set, teach:

1. What to read first
2. What NOT to calculate

3. Approximation opportunities
4. Option elimination
5. Exact calculation when required
6. Time-saving strategy

Teach me how to avoid doing unnecessary calculations.

PART 3 — LOGICAL AND ANALYTICAL REASONING

Cover placement-level reasoning comprehensively.

Series

- number series
- alphabet series
- alphanumeric series
- missing terms
- wrong terms
- pattern identification

Analogy and Classification

- number analogy
- word analogy
- letter analogy
- odd-one-out
- pattern classification

Coding-Decoding

Cover:

- letter shifting
- substitution coding
- number coding
- word coding
- matrix/table coding
- modern pattern-based coding questions

Blood Relations

Cover:

- direct relations
- coded relations
- family trees
- generation identification
- gender traps

Direction Sense

Cover:

- left/right turns
- coordinates
- shortest distance
- final direction
- multiple movements

Ranking and Ordering

Cover:

- top/bottom ranks
- left/right ranks
- total persons
- interchange
- comparison

Seating Arrangement

Cover:

- linear arrangements
- circular arrangements
- facing center/outside
- parallel rows
- mixed conditions

Teach how to create the diagram systematically.

Puzzles

Cover placement patterns such as:

- floor puzzles
- scheduling puzzles
- box/order puzzles
- person-profession-city puzzles
- day/month arrangements
- grouping puzzles
- distribution puzzles

Do NOT start with extremely difficult banking-exam puzzles unless useful.

Prioritize placement relevance.

Syllogisms

Teach:

- statements

- conclusions
- Venn method
- possibility cases
- common traps

Statements and Conclusions

Statements and Assumptions

Statements and Arguments

Cause and Effect

Course of Action

Assertion and Reason

where relevant to placement aptitude.

Data Sufficiency

Cover:

- quantitative data sufficiency
- reasoning data sufficiency
- statement I / II
- deciding sufficiency without fully solving

Logical Deduction

Cover:

- if-then reasoning
 - necessary/sufficient conditions
 - truth/lie problems
 - constraints
 - ordering
 - grouping
 - matching
-

PART 4 — GATE VERBAL APTITUDE

Teach English aptitude in simple language.

Assume I may have gaps in grammar.

Cover:

Grammar Foundations

- parts of speech
- sentence structure
- subject
- predicate
- subject-verb agreement
- nouns
- pronouns
- verbs
- adjectives
- adverbs
- conjunctions
- prepositions
- articles
- modifiers
- parallelism
- tenses
- voice
- basic conditionals
- common grammar errors

Do NOT turn this into a school-English course.

Teach only what improves GATE and placement performance.

Vocabulary

Cover:

- vocabulary in context
- synonyms
- antonyms
- word usage
- confusing words
- roots/prefixes/suffixes where useful
- contextual meaning

Do NOT give huge meaningless word lists.

Teach useful vocabulary through questions and context.

Sentence Completion

Sentence Correction

Error Identification

Para Jumbles

Teach:

- opening-sentence clues
- pronoun links
- chronology
- cause-effect
- connectors
- mandatory pairs
- conclusion identification

Reading Comprehension

Teach:

- main idea
- author's tone
- inference
- factual questions
- vocabulary in context
- assumption
- conclusion
- strengthening/weakening interpretation where relevant

Teach me to answer RC quickly without rereading unnecessarily.

Critical Reasoning

Cover:

- premise
- conclusion
- assumption
- inference
- strengthen
- weaken
- paradox
- cause/effect
- logical flaw

Verbal Analogies

Word Relationships

Verbal Deduction

where relevant.

PART 5 — GATE-SPECIFIC APTITUDE TRAINING

Train me specifically for the way GATE asks aptitude questions.

Cover:

- conceptual traps
- multi-step arithmetic
- language interpretation
- mathematical reasoning
- graphical reasoning
- numerical reasoning
- verbal reasoning
- quantitative comparison
- approximation
- option elimination
- units
- dimensional sanity checks
- estimation
- interpreting what the question actually asks

Teach strategies separately for:

- MCQ
- MSQ
- NAT

Whenever useful, show:

Normal Method

Faster GATE Method

Option-Elimination Method

Tell me which method I should actually use in the exam.

PART 6 — PLACEMENT-SPECIFIC TRAINING

Prepare me for aptitude rounds used in:

- service-based companies
- product companies
- internships
- mass recruitment
- campus drives
- online assessments

Cover the styles commonly seen in tests from companies such as:

- TCS
- Infosys
- Wipro

- Accenture
- Cognizant
- Capgemini
- Deloitte
- NTT DATA
- IBM
- Amazon-style online assessments where aptitude/reasoning is relevant
- other common campus recruiters

Do NOT claim that a particular question appeared in a company test unless verified.

Instead label questions honestly as:

TCS-style Placement-style GATE-style

when they merely resemble those patterns.

Teach differences between:

- GATE aptitude
- placement quantitative aptitude
- logical reasoning
- verbal aptitude
- online assessment speed requirements

PART 7 — QUESTION TRAINING

Questions are extremely important.

Do NOT teach only theory.

For EVERY major concept give questions at four levels:

LEVEL 0 — Foundation

Very easy questions to confirm the concept.

LEVEL 1 — Standard

Normal GATE/placement questions.

LEVEL 2 — Tricky

Questions involving traps, multiple concepts or faster reasoning.

LEVEL 3 — Exam Pressure

Harder questions designed for speed and selection.

For important chapters, give at least:

- solved examples
- practice questions
- mixed questions
- timed questions

Do not reveal practice answers immediately unless I ask or unless they are placed at the bottom after enough separation.

PYQ POLICY

Make the course strongly **PYQ-oriented**.

Teach me the types of aptitude questions GATE repeatedly tests.

If using an authentic previous-year GATE question:

- mention the year only when you are confident
- preserve the idea correctly
- explain the solution
- explain why wrong options fail where useful

If you are NOT certain that a question is an authentic PYQ, do NOT falsely label it as one.

Instead call it:

PYQ-style question

This rule is mandatory.

EVERY TOPIC MUST FOLLOW THIS TEACHING FORMAT

For every new concept, preferably use:

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Tell me:

- GATE importance
- placement importance
- connection to other topics

Use labels:

GATE: High / Medium / Low Placements: High / Medium / Low

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Explain the idea in very simple English.

Use real-life examples whenever useful.

3. CORE CONCEPT

Give the precise mathematical/logical meaning.

4. FORMULAS

Give only necessary formulas.

For each formula explain:

- what it means
 - when it applies
 - when it does NOT apply
-

5. DERIVATION/ LOGIC

For important formulas, briefly show where they come from.

I should not memorize blindly.

6. BASIC EXAMPLE

Solve step-by-step.

7. SHORTCUT

Show faster methods.

8. NORMAL METHOD VS FAST METHOD

Whenever useful compare:

Normal method vs **Exam method**

9. GATE APPLICATION

Give GATE-style questions.

10. PLACEMENT APPLICATION

Give placement-style questions.

11. COMMON TRAPS

Explicitly show mistakes students make.

Use:

& **TRAP**

12. MEMORY TRICK

Create short memory tricks where useful.

Do NOT create confusing mnemonics just for the sake of it.

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Give:

- L0 questions
 - L1 questions
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14. QUICK REVISION BOX

End every important concept with a compact note that I can copy into my notebook.

Include:

- formulas
 - shortcuts
 - traps
 - key idea
-

SOLUTION STYLE

Whenever solving a numerical question:

1. Identify what is given.
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For difficult questions, do not skip reasoning.

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Throughout the course repeatedly train:

- multiplication
 - division
 - fraction comparison
 - percentage conversion
 - squares
 - cubes
 - approximation
 - estimation
-

- ratio simplification

Teach me to recognize instantly:

MATH

$$1/2=50\%$$

MATH

$$1/3\approx 33.33\%$$

MATH

$$2/3\approx 66.67\%$$

MATH

$$1/4=25\%$$

MATH

$$3/4=75\%$$

MATH

$$1/5=20\%$$

MATH

$$1/8=12.5\%$$

and all other high-value conversions.

Build a **Must-Memorize Aptitude Table** progressively.

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Teach me to translate English questions into mathematics.

For word problems explicitly identify words such as:

- increased by
- increased to
- decreased by
- decreased to
- more than
- less than
- of
- per
- ratio
- difference
- average
- together

- remaining
- at least
- at most
- exactly
- only
- either
- neither

because interpretation errors can cost marks.

OPTION-ELIMINATION TRAINING

Whenever possible, teach techniques such as:

- parity
- units digit
- divisibility
- approximation
- bounds
- sign
- relative size
- back-substitution
- testing options
- eliminating impossible answers

Explicitly say when elimination is faster than solving exactly.

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 - Topics needing revision
-

FULL MOCK TESTS

After completing the syllabus, create:

GATE GENERAL APTITUDE MOCKS

Questions should mimic the thinking level and variety expected in GATE General Aptitude.

PLACEMENT APTITUDE MOCKS

Include combinations of:

- quantitative aptitude
- logical reasoning
- verbal ability
- data interpretation

Create:

- easy mock
- standard mock
- difficult mock
- speed mock

After each mock, perform an error analysis.

ERROR LOG

Maintain an ongoing error log during our course.

Classify my mistakes into:

1. Concept error
2. Formula error
3. Calculation error
4. Reading error
5. Logic error
6. Careless error
7. Time-management error
8. Shortcut misuse

Whenever I repeatedly make the same mistake, explicitly warn me.

PROGRESS TRACKER

Maintain one master syllabus tracker.

Use:

' COMPLETE ~~0~~=100% PROGRESS + NOT STARTED ~~0~~=0% REVISION NEEDED

Track every chapter.

At the end of every lesson show only a compact progress update such as:

Completed: 12/80 Current: Time & Work Next: Pipes & Cisterns

Do not repeatedly print the entire roadmap unless I ask.

PRIORITY LABELS

Classify topics as:

~~0~~=Tier 1 — Extremely Important + Tier 2 — Important ~~0~~=Tier 3 — Supporting

Teach Tier 1 topics most deeply.

Do NOT completely omit Tier 2/3 topics if they are relevant.

DEPTH CONTROL

I want enough depth to solve questions independently.

But I do not want unnecessarily advanced mathematics.

Do NOT teach advanced theory unless:

- GATE needs it,
 - placements need it, or
 - it directly improves understanding/speed.
-

NOTE-MAKING STYLE

Make your explanation easy for handwritten notes.

Prefer:

- headings
- short paragraphs
- tables
- formulas
- bullet points
- worked examples
- shortcuts
- warnings

Avoid giant walls of text.

For formulas use clean mathematical notation.

LANGUAGE

Teach in **simple English**.

Assume I am intelligent but learning the topic from scratch.

When terminology is unavoidable:

1. state the term
 2. explain it simply
 3. give an example
-

INTERACTION RULE

I will frequently reply only:

NEXT

When I write NEXT:

- continue from exactly where we stopped
- do NOT restart the course
- do NOT ask what chapter I want
- do NOT repeat completed material unnecessarily
- automatically move to the next planned concept

If a chapter is too large, divide it into parts.

Example:

Time & Work — Part 1 Time & Work — Part 2

I may keep sending NEXT until the entire course is complete.

IF I ASK “HOW MUCH IS LEFT?”

Give me:

- chapters completed
- chapters remaining
- approximate percentage complete
- major units still pending

Do NOT guess randomly.

Use the course tracker.

IF I ASK “IS APTITUDE COMPLETE?”

Audit the entire course against:

1. GATE General Aptitude requirements
2. Quantitative aptitude
3. Logical/analytical reasoning
4. Verbal aptitude
5. Data interpretation
6. Placement aptitude
7. calculation speed
8. shortcuts
9. revision
10. mocks

Then answer honestly:

COMPLETE or **NOT YET COMPLETE**

If anything remains, list it and continue teaching it.

Do NOT declare completion merely because the major chapters were covered.

FINAL COMPLETION PHASE

Complete these final stages:

Create one consolidated aptitude formula/shortcut sheet.

Include:

- Create:

with brief explanations.

Create a rapid GATE General Aptitude revision course.

Create a rapid placement aptitude revision course.

Give a large mixed set covering all important patterns.

Give multiple complete aptitude mocks.

Identify what I still need to improve.

Only after all of these are finished should you declare:

APTITUDE COURSE COMPLETE'

STARTING INSTRUCTION

First create the **MASTER APTITUDE ROADMAP**.

The roadmap must contain:

- units
- chapters
- subtopics
- GATE relevance
- placement relevance
- priority
- teaching order

Number the chapters so that our progress can be tracked accurately.

Then begin teaching from the correct foundation.

Do NOT give only the roadmap and stop.

After the roadmap, immediately start:

CHAPTER 1

Teach according to all rules above.

From then onward, I will mostly use **NEXT** to continue.

ASSISTANT

Understood. From this point onward, this chat is your **single complete Aptitude course for GATE + placements**.

I will keep one permanent roadmap, move in order when you say **NEXT**, track what is finished, and will **not declare the course complete until Quant + DI + Reasoning + Verbal + GATE strategy + placement strategy + revision + question banks + mocks are all covered**.

MASTER APTITUDE ROADMAP

I'm organizing the course into **8 units and 76 numbered chapters**. Some large chapters will have multiple lessons, so **76 chapters** does **not** mean only 76 messages.

Priority:

- ~~Ø=Tier~~ **Tier 1** — extremely important
- **+Tier 2** — important
- ~~Ø=Tier~~ **Tier 3** — supporting

Relevance:

- **G:** GATE
- **P:** Placements

UNIT 0 — CALCULATION & NUMBER FOUNDATIONS

Ch.	Topic	G	P	Priority
1	Arithmetic & Calculation Foundations	High	High	Ø=Ý%
2	Fractions and Decimals	High	High	Ø=Ý%
3	Fraction–Decimal–Percentage Conversion	High	High	Ø=Ý%
4	Divisibility Rules	Medium	High	+P
5	Factors, Multiples, HCF and LCM	High	High	Ø=Ý%
6	Prime Numbers and Prime Factorization	Medium	High	+P
7	Remainders and Modular Reasoning	Medium	High	+P
8	Units Digit, Last Digits and Cyclicity	Medium	High	+P
9	Powers, Exponents and Roots	Medium	High	+P
10	Squares, Cubes and Mental-Math Tables	Medium	High	+P
11	Simplification, Approximation and Estimation	High	High	Ø=Ý%
12	Fast Multiplication, Division and Cancellation	High	High	Ø=Ý%

UNIT 1 — QUANTITATIVE APTITUDE

Ch.	Topic	G	P	Priority
13	Ratio and Proportion	High	High	Ø=Ý%
14	Direct and Inverse Variation	Medium	High	+P
15	Percentages	High	High	Ø=Ý%
16	Averages	High	High	Ø=Ý%
17	Weighted and Combined Average	High	High	Ø=Ý%
18	Profit and Loss	High	High	Ø=Ý%
19	Discount and Marked Price	Medium	High	Ø=Ý%
20	Simple Interest	Medium	High	+P
21	Compound Interest, Growth and Depreciation	High	High	Ø=Ý%
22	Mixtures and Alligation	Medium	High	Ø=Ý%
23	Replacement in Mixtures	Medium	High	+P
24	Partnership	Low	Medium	Ø=ÜØ
25	Problems on Ages	Medium	High	+P
26	Time and Work	High	High	Ø=Ý%

Ch.	Topic	G	P	Priority
27	Efficiency and Work-Wages Problems	Medium	High	$\emptyset = \dot{Y}\%$
28	Pipes and Cisterns	Medium	High	+P
29	Time, Speed and Distance	High	High	$\emptyset = \dot{Y}\%$
30	Average and Relative Speed	High	High	$\emptyset = \dot{Y}\%$
31	Trains	Medium	High	+P
32	Boats and Streams	Low-Medium	Medium	$\emptyset = \ddot{U}\emptyset$
33	Races and Circular Tracks	Low-Medium	Medium	$\emptyset = \ddot{U}\emptyset$
34	Algebraic Identities and Simplification	High	High	$\emptyset = \dot{Y}\%$
35	Linear and Simultaneous Equations	High	High	$\emptyset = \dot{Y}\%$
36	Quadratics and Inequalities	Medium	High	+P
37	Sequences, AP and GP	Medium	High	+P
38	Fundamental Counting Principle	High	High	$\emptyset = \dot{Y}\%$
39	Permutations	Medium	High	$\emptyset = \dot{Y}\%$
40	Combinations	Medium	High	$\emptyset = \dot{Y}\%$
41	Probability	High	High	$\emptyset = \dot{Y}\%$
42	Sets and Venn Diagrams	High	High	$\emptyset = \dot{Y}\%$
43	Geometry Foundations	Medium	Medium	+P
44	Triangles, Similarity and Pythagoras	Medium	Medium	+P
45	Quadrilaterals, Polygons and Circles	Medium	Medium	+P
46	2D Mensuration	Medium	High	+P
47	3D Mensuration	Medium	High	+P
48	Clocks	Low	Medium	$\emptyset = \ddot{U}\emptyset$
49	Calendars	Low	Medium	$\emptyset = \ddot{U}\emptyset$

UNIT 2 — DATAINTERPRETATION

Ch.	Topic	G	P	Priority
50	DI Foundations and Reading Data	High	High	$\emptyset = \dot{Y}\%$
51	Tables	High	High	$\emptyset = \dot{Y}\%$
52	Bar and Stacked-Bar Graphs	High	High	$\emptyset = \dot{Y}\%$

Ch.	Topic	G	P	Priority
53	Line Graphs	High	High	Ø=Ý%
54	Pie Charts	High	High	Ø=Ý%
55	Mixed Graphs and Caselets	High	High	Ø=Ý%
56	Missing-Data and Advanced DI	Medium	High	+P

UNIT 3 — LOGICAL & ANALYTICAL REASONING

Ch.	Topic	G	P	Priority
57	Number, Letter and Alphanumeric Series	Medium	High	Ø=Ý%
58	Analogy, Classification and Odd-One-Out	Low-Medium	High	+P
59	Coding-Decoding	Low	High	+P
60	Blood Relations	Low	High	+P
61	Direction Sense	Low	High	+P
62	Ranking and Ordering	Medium	High	Ø=Ý%
63	Linear Seating Arrangement	Medium	High	Ø=Ý%
64	Circular and Parallel Seating	Medium	High	Ø=Ý%
65	Logical Puzzles	Medium	High	Ø=Ý%
66	Syllogisms	Medium	High	Ø=Ý%
67	Statements, Assumptions and Conclusions	High	High	Ø=Ý%
68	Arguments, Cause-Effect and Course of Action	Medium	High	+P
69	Data Sufficiency	High	High	Ø=Ý%
70	Logical Deduction and Constraint Reasoning	High	High	Ø=Ý%

UNIT 4 — VERBAL APTITUDE

Ch.	Topic	G	P	Priority
71	Grammar for Aptitude	High	High	Ø=Ý%
72	Vocabulary and Word Usage	High	High	Ø=Ý%
73	Sentence Completion, Correction and Error Detection	High	High	Ø=Ý%
74	Para Jumbles and Paragraph Completion	High	High	Ø=Ý%

Ch.	Topic	G	P	Priority
75	Reading Comprehension	High	High	Ø=Ý%
76	Critical Reasoning and Verbal Deduction	High	High	Ø=Ý%

After Chapter 76, the **teaching syllabus is not considered finished**. We then enter the compulsory final training phases:

UNIT 5 — GATE SPECIAL TRAINING

- GATE-style numerical reasoning
- verbal reasoning
- graphical interpretation
- question-language traps
- MCQ strategy
- MSQ strategy
- NAT strategy
- approximation
- elimination
- back-solving
- sanity checking
- time management

UNIT 6 — PLACEMENT SPECIAL TRAINING

- service-company aptitude patterns
- product-company OA reasoning
- speed-based tests
- mixed quant/reasoning/verbal rounds
- company-style practice sets
- timed sectional tests

UNIT 7 — REVISION & QUESTION BANK

- complete formula book
- must-memorize tables
- 100 aptitude traps
- chapter-wise revision
- mixed question bank
- error log
- weak-topic repair

UNIT 8 — MOCKS

- GATE Aptitude mocks
- placement mocks
- easy/standard/hard/speed mocks
- post-mock analysis
- final weakness audit

Only then:

APTITUDE COURSE COMPLETE '

CHAPTER 1 — ARITHMETIC & CALCULATION FOUNDATIONS

Status: $\emptyset = \text{Not}$ **PROGRESS GATE:** High **Placements:** High **Priority:** $\emptyset = \text{Yes}$ **For 1**

This chapter may look elementary, but weak calculations destroy aptitude speed.

You should reach a stage where operations such as:

$\backslash [25\% \text{ of } 480, \quad \frac{7}{8} \times 320, \quad 48 \times 25 \backslash]$

do not require slow written calculation.

1. WHY THIS CHAPTER MATTERS

Almost every quantitative topic depends on arithmetic.

For example:

- Percentages !' multiplication/division
- Ratios !' division
- Average !' addition/division
- Profit/Loss !' percentages
- Time & Work !' fractions
- Probability !' fractions
- DI !' rapid arithmetic

So there are two kinds of aptitude students:

Student A: understands the formula but calculates slowly.

Student B: understands the formula and calculates quickly.

Student B gets more questions attempted with fewer mistakes.

Our goal is Student B.

2. NUMBER OPERATIONS

The four fundamental operations are:

$\backslash [+, \quad -, \quad \times, \quad \div \backslash]$

You obviously know what they mean, but aptitude questions combine them.

That introduces **order of operations**.

3. BODMAS / PEMDAS

Suppose:

$$\backslash[8+4\times3 \backslash]$$

Do NOT calculate:

$$\backslash[8+4=12 \backslash]$$

then:

$$\backslash[12\times3=36 \backslash]$$

Wrong.

Multiplication comes before addition:

$$\backslash[8+12 = \boxed{20} \backslash]$$

Use:

BODMAS

B — Brackets **O** — Orders/powers **D** — Division **M** — Multiplication **A** — Addition **S** — Subtraction

But there is an important detail.

Division and multiplication have the **same priority**.

Addition and subtraction also have the **same priority**.

When equal-priority operations occur:

Solve from left to right.

Example:

$$\backslash[24\div6\times2 \backslash]$$

Left to right:

$$\backslash[24\div6=4 \backslash]$$

$$\backslash[4\times2=8 \backslash]$$

Answer:

$$\backslash[\boxed{8} \backslash]$$

Not 2.

4. BRACKETS

Consider:

$$\backslash[5(8-3) \backslash]$$

First bracket:

$$\backslash[8-3=5 \backslash]$$

Then:

$$\backslash[5\times5=25 \backslash]$$

Example

$$\backslash[18-[4+2(5-3)]\backslash]$$

Start innermost:

$$\backslash[5-3=2\backslash]$$

$$\backslash[2\times 2=4\backslash]$$

Therefore:

$$\backslash[18-[4+4]\backslash]$$

$$\backslash[=18-8\backslash]$$

$$\backslash[=\boxed{10}\backslash]$$

5. POSITIVE AND NEGATIVE NUMBERS

These matter in algebra, graphs, percentage change and reasoning.

Remember:

Addition

$$\backslash[(+)(+)=+\backslash]$$

$$\backslash[(-)(-)=-\backslash]$$

For opposite signs, subtract magnitudes.

Example:

$$\backslash[8+(-3)=5\backslash]$$

$$\backslash[-8+3=-5\backslash]$$

6. SIGN RULES FOR MULTIPLICATION

Extremely important:

$$\backslash[(+)(+)=+\backslash]$$

$$\backslash[(-)(-)=+\backslash]$$

$$\backslash[(+)(-)=-\backslash]$$

$$\backslash[(-)(+)=-\backslash]$$

Memory rule

Same signs !' positive

Different signs !' negative

Example:

$$\backslash[(-6)(-4)=24\backslash]$$

$$\backslash[(-6)(4)=-24\backslash]$$

7. SUBTRACTING NEGATIVE NUMBERS

Example:

$$8 - (-5)$$

Subtracting a negative becomes addition:

$$8 + 5 = \boxed{13}$$

Similarly:

$$-3 - (-7)$$

$$-3 + 7$$

$$= \boxed{4}$$

8. EVEN AND ODD NUMBERS

An integer divisible by 2 is **even**.

Examples:

$$2, 4, 6, 8, 10$$

Not divisible by 2 is **odd**.

Examples:

$$1, 3, 5, 7, 9$$

This becomes a powerful option-elimination tool.

Addition rules

$$E + E = E$$

$$O + O = E$$

$$E + O = O$$

Example:

odd + odd:

$$7 + 11 = 18$$

even.

Multiplication rules

$$E \times E = E$$

$$E \times O = E$$

$$O \times O = O$$

Important shortcut

If **any factor is even**, the entire product is even.

9. QUICK PARITY QUESTION

If x and y are odd, what is the parity of:

$$x+y+xy$$

Since:

$$x+y=O+O=E$$

and:

$$xy=O \times O=O$$

Therefore:

$$E+O=O$$

Answer:

$$\boxed{\text{Odd}}$$

No actual values were needed.

This is aptitude thinking.

10. COMMON MULTIPLICATION SHORTCUTS

These save enormous time.

Multiply by 10

Append zero:

$$47 \times 10 = 470$$

Multiply by 100

$$47 \times 100 = 4700$$

11. MULTIPLY BY 5

Instead of multiplying by 5:

Multiply by 10 and divide by 2.

Example:

$$68 \times 5$$

$$= \frac{68 \times 10}{2}$$

$$= \frac{680}{2}$$

$$= \boxed{340}$$

12. MULTIPLY BY 25

Remember:

$$25 = \frac{100}{4}$$

Therefore:

$$N \times 25 = \frac{N \times 100}{4}$$

Example:

$$48 \times 25$$

$$= \frac{4800}{4}$$

$$= \boxed{1200}$$

Much faster than long multiplication.

MUST MEMORIZE

$$N \times 25 = N \times 100 \div 4$$

13. MULTIPLY BY 50

$$50 = \frac{100}{2}$$

So:

$$N \times 50 = \frac{100N}{2}$$

Example:

$$34 \times 50$$

$$= \frac{3400}{2}$$

$$= \boxed{1700}$$

14. MULTIPLY BY 125

Very useful.

$$125 = \frac{1000}{8}$$

Therefore:

$$N \times 125 = N \times 1000 \div 8$$

Example:

$$32 \times 125$$

$$= \frac{32000}{8}$$

$$= \boxed{4000}$$

MUST MEMORIZE

$$N \times 125 = \frac{N \times 1000}{8}$$

15. MULTIPLICATION NEAR 100

Suppose:

$$98 \times 97$$

Normal multiplication works, but aptitude wants speed.

Use:

$$(100-2)(100-3)$$

$$= 10000 - 300 - 200 + 6$$

$$= \boxed{9506}$$

Another shortcut:

98 is 2 below 100. 97 is 3 below 100.

Cross-subtract:

$$98 - 3 = 95$$

or:

$$97 - 2 = 95$$

Multiply deficits:

$$2 \times 3 = 6$$

So:

$$95 | 06$$

$$= \boxed{9506}$$

Useful, but this is a **shortcut**, not something you must force into every calculation.

16. MULTIPLY NUMBERS AROUND A COMMON BASE

Example:

$$49 \times 51$$

Recognize:

$$(50-1)(50+1)$$

Use identity:

$$(a-b)(a+b) = a^2 - b^2$$

Therefore:

$$50^2 - 1^2$$

$$2500 - 1$$

$$= \boxed{2499}$$

This identity becomes extremely useful later.

17. SQUARING NUMBERS ENDING IN 5

Example:

$$\backslash[35^2 \backslash]$$

Take first digit 3.

Multiply:

$$\backslash[3(3+1)=12 \backslash]$$

Append 25:

$$\backslash[\boxed{1225} \backslash]$$

Therefore:

$$\backslash[35^2=1225 \backslash]$$

Another:

$$\backslash[85^2 \backslash]$$

$$\backslash[8\times 9=72 \backslash]$$

append 25:

$$\backslash[\boxed{7225} \backslash]$$

Rule

For:

$$\backslash[(10n+5)^2 \backslash]$$

write:

$$\backslash[n(n+1);25 \backslash]$$

18. FAST ADDITION BY COMPENSATION

Example:

$$\backslash[398+267 \backslash]$$

Instead:

$$\backslash[400+267-2 \backslash]$$

$$\backslash[667-2 \backslash]$$

$$\backslash[=\boxed{665} \backslash]$$

19. FAST SUBTRACTION BY COMPENSATION

Example:

$$\backslash[1000-487 \backslash]$$

You can mentally find complement:

$$487 \text{ !' } 500 = 13 \quad 500 \text{ !' } 1000 = 500$$

Total:

$$\boxed{513}$$

So:

$$1000 - 487 = \boxed{513}$$

This complement idea later helps with percentages and probability.

20. CANCELLATION BEFORE MULTIPLICATION

This is one of the most useful aptitude habits.

Suppose:

$$\frac{48 \times 25}{12}$$

Don't multiply first.

Cancel:

$$\frac{48}{12} = 4$$

Then:

$$4 \times 25 = \boxed{100}$$

Golden rule

Cancel first, multiply later.

It reduces both calculation time and errors.

21. ANOTHER CANCELLATION EXAMPLE

$$\frac{36 \times 45}{15}$$

Cancel:

$$45 \div 15 = 3$$

So:

$$36 \times 3 = \boxed{108}$$

Instead of calculating:

$$36 \times 45 = 1620$$

and then dividing by 15.

22. BREAK NUMBERS INTO EASY PARTS

Example:

$$47 \times 18$$

Think:

$$47(20-2)$$

$$=940-94$$

$$= \boxed{846}$$

Or:

$$47(10+8)$$

Choose whichever looks easier.

This is called the **distributive property**:

$$a(b+c)=ab+ac$$

23. MUST-MEMORIZE MULTIPLICATION TABLES

For strong aptitude speed, aim to know:

MUST MEMORIZE

Multiplication tables:

$$1 \text{ to } 20$$

very comfortably.

GOOD TO MEMORIZE

Tables:

$$21 \text{ to } 25$$

Not because the exam directly asks tables, but because they reduce computation time.

24. MUST-MEMORIZE SQUARES

Eventually know these instantly:

$$1^2 \text{ to } 30^2$$

Minimum high-value set:

(n)	(n^2)
10	100
11	121
12	144
13	169
14	196
15	225
16	256
17	289

$\sqrt[n]{n!}$	$\sqrt[n]{n^2!}$
18	324
19	361
20	400
21	441
22	484
23	529
24	576
25	625
26	676
27	729
28	784
29	841
30	900

You do not need to memorize all of this tonight. We will reinforce it repeatedly.

25. MUST-MEMORIZE CUBES

At least:

$\sqrt[n]{1^3 \text{ to } 10^3}$

Number	Cube
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000

Later we will extend useful values.

26. POWERS OF 2

Very useful for CS students and aptitude.

MUST MEMORIZE

$$2^0=1$$

$$2^1=2$$

$$2^2=4$$

$$2^3=8$$

$$2^4=16$$

$$2^5=32$$

$$2^6=64$$

$$2^7=128$$

$$2^8=256$$

$$2^9=512$$

$$2^{10}=1024$$

Good to continue:

$$2^{11}=2048$$

$$2^{12}=4096$$

$$2^{13}=8192$$

$$2^{14}=16384$$

$$2^{15}=32768$$

$$2^{16}=65536$$

These also help elsewhere in GATE CS.

27. DIVISION ESTIMATION

Suppose:

$$\frac{1998}{51}$$

Instead of exact long division, notice:

$$1998 \approx 2000$$

and:

$$51 \approx 50$$

So:

$$\frac{2000}{50} \approx 40$$

If options are:

A. 19 B. 28 C. 39 D. 73

you can immediately select around:

$$\boxed{39}$$

without exact division.

We will study approximation deeply in Chapter 11.

28. GATE-TYPE QUESTION

Evaluate efficiently:

$$\frac{125 \times 64}{8}$$

Normal method

$$125 \times 64 = 8000$$

$$8000 \div 8 = 1000$$

Faster method

Cancel first:

$$64 \div 8 = 8$$

Then:

$$125 \times 8$$

And because:

$$125 \times 8 = 1000$$

Answer:

$$\boxed{1000}$$

Exam lesson

Avoid creating large numbers unnecessarily.

29. PLACEMENT-STYLE QUESTION

Evaluate:

$$49 \times 51$$

Normal method

Long multiplication.

Fast method

$$(50-1)(50+1)$$

$$= 50^2 - 1$$

$$= 2500 - 1$$

$$= \boxed{2499}$$

30. OPTION-ELIMINATION EXAMPLE

Without calculating completely, determine whether:

$$73 \times 49 \times 22$$

is odd or even.

Since one factor is:

$$22$$

which is even, entire product is even.

$$\boxed{\text{Even}}$$

If only one answer choice were even, you could solve without multiplication.

& COMMON TRAPS

Trap 1 — Blind left-to-right calculation

$$8 + 2 \times 5$$

is:

$$18$$

not 50.

Trap 2 — Wrong sign

$$(-5)(-4) = +20$$

not " 20 .

Trap 3 — Multiplying before cancelling

Bad:

$$\frac{72 \times 35}{18}$$

Better:

$$72/18 = 4$$

then:

$$4 \times 35 = 140$$

Trap 4 — Using fancy tricks when normal arithmetic is easier

A shortcut is useful only when it saves time and stays reliable.

Do not turn every simple multiplication into a trick.

MEMORY RULES

Remember these:

Same signs multiply !' positive. Different signs !' negative.

Cancel first, multiply later.

$$\times 25 = \times 100 \div 4$$

$$\times 125 = \times 1000 \div 8$$

Numbers ending in 5: square using $n(n+1)$, then append 25.

QUICK REVISION BOX

BODMAS

$B \rightarrow O \rightarrow D/M \rightarrow A/S$

Equal priorities !' left to right.

Sign rules

$$(-)(-) = +$$

$$(+)(-) = -$$

Parity

$$E + E = E$$

$$O + O = E$$

$$E + O = O$$

$$O \times O = O$$

Anything $\times$ even = even.

Multiplication shortcuts

$$N \times 5 = N \times 10 \div 2$$

$$N \times 25 = N \times 100 \div 4$$

$$N \times 50 = N \times 100 \div 2$$

$$N \times 125 = N \times 1000 \div 8$$

Golden calculation rule

$\boxed{\text{Simplify/cancel before multiplying}}$

PRACTICE — LEVEL 0

Try without calculator.

Q1

$$\lfloor 12+6\times 4 \rfloor$$

Q2

$$\lfloor (-8)(-7) \rfloor$$

Q3

$$\lfloor 72\times 25 \rfloor$$

Q4

$$\lfloor 48\times 125 \rfloor$$

Q5

$$\lfloor 65^2 \rfloor$$

LEVEL 1

Q6

$$\lfloor \frac{84\times 25}{21} \rfloor$$

Q7

$$\lfloor 99\times 101 \rfloor$$

Q8

If $\lfloor x \rfloor$ and $\lfloor y \rfloor$ are odd, is:

$$\lfloor xy+x \rfloor$$

odd or even?

Q9

$$\lfloor 64\times 25+36\times 25 \rfloor$$

Solve without performing two separate multiplications.

Q10

$$\lfloor 998+397 \rfloor$$

Solve using compensation.

LEVEL 2

Q11

Without fully calculating, determine whether:

$$\lfloor 57^{23}+31^{18} \rfloor$$

is odd or even.

Q12

Calculate:

$$\sqrt{75^2}$$

mentally.

Q13

Calculate:

$$\sqrt{96 \times 104}$$

using an identity.

Q14

Simplify:

$$\sqrt{\frac{125 \times 72 \times 16}{9 \times 8}}$$

using cancellation.

ANSWERS

1. $\sqrt{36}$
2. $\sqrt{56}$
3. $\sqrt{1800}$
4. $\sqrt{6000}$
5. $\sqrt{4225}$
6. $\sqrt{100}$
7. $\sqrt{9999}$
8. Even
9. $\sqrt{2500}$
10. $\sqrt{1395}$
11. Even
12. $\sqrt{5625}$
13. $\sqrt{9984}$
14. $\sqrt{2000}$

CHAPTER 1 STATUS

We have completed the **first part of calculation foundations**, but I am **not marking Chapter 1 complete yet** because the next lesson needs to cover:

- division concepts
- divisibility intuition
- fractions inside calculations
- decimal arithmetic
- powers/order details
- reciprocal thinking
- useful arithmetic identities

- speed drills
- calculation mistake prevention

Progress

Course chapters: 76 **Completed:** 0/76 **Current:** Ø=Chapter 1 — Arithmetic & Calculation Foundations **Next:** Chapter 1 Part 2 — Division, reciprocals, decimal handling & speed arithmetic

Send **NEXT**.

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ASSISTANT

CHAPTER 1 — ARITHMETIC & CALCULATION FOUNDATIONS

PART 2 — DIVISION, RECIPROCALS, DECIMALS & SPEED ARITHMETIC

Status: Ø=Not in PROGRESS **GATE:** High **Placements:** High **Priority:** Ø=Yes 1

In Part 1, we covered:

- BODMAS
- positive/negative numbers
- parity
- multiplication shortcuts
- cancellation
- squares/cubes basics
- powers of 2
- estimation basics

Now we complete the second foundation layer.

1. DIVISION — CORE IDEA

Division means distributing or finding how many times one number fits into another.

Example:

$\backslash 20 \div 5 = 4 \backslash$

because:

$\backslash 5 \times 4 = 20 \backslash$

So division is the inverse of multiplication.

2. IMPORTANT TERMS

For:

$$\lfloor 20 \div 5 = 4 \rfloor$$

- 20 = dividend
- 5 = divisor
- 4 = quotient

If something remains, that is called the **remainder**.

Example:

$$\lfloor 23 \div 5 \rfloor$$

$$\lfloor 23 = 5(4) + 3 \rfloor$$

So:

- Quotient = 4
- Remainder = 3

This becomes extremely important in number-system problems later.

3. NEVER DIVIDE BY ZERO

$$\lfloor \frac{5}{0} \rfloor$$

is **undefined**.

You cannot divide by zero.

But:

$$\lfloor \frac{0}{5} = 0 \rfloor$$

is perfectly valid.

& bTRAP

$$\lfloor \frac{0}{0} \rfloor$$

is not 0.

It is **indeterminate**.

For aptitude, remember:

$$\lfloor \boxed{\text{Division by zero is not allowed}} \rfloor$$

4. DIVISION AS MULTIPLICATION BY RECIPROCAL

This idea is extremely useful.

The reciprocal of:

$$\lfloor \frac{a}{b} \rfloor$$

is:

$$\lfloor \frac{b}{a} \rfloor$$

Example:

Reciprocal of:

$$\left[\frac{3}{4} \right]$$

is:

$$\left[\frac{4}{3} \right]$$

Therefore:

$$\left[8 \div \frac{2}{3} \right]$$

means:

$$\left[8 \times \frac{3}{2} \right]$$

$$\left[=12 \right]$$

RULE

$$\left[\boxed{a \div \frac{b}{c} = a \times \frac{c}{b}} \right]$$

In simple language:

Dividing by a fraction = multiplying by its reciprocal.

5. COMMON RECIPROCAL

MUST MEMORIZE

$$\left[\frac{1}{2} \rightarrow 2 \right]$$

$$\left[\frac{1}{3} \rightarrow 3 \right]$$

$$\left[\frac{1}{4} \rightarrow 4 \right]$$

$$\left[\frac{1}{5} \rightarrow 5 \right]$$

More generally:

$$\left[\frac{1}{n} \rightarrow n \right]$$

Example:

$$\left[15 \div \frac{1}{5} \right]$$

$$\left[=15 \times 5 \right]$$

$$\left[=\boxed{75} \right]$$

Notice something important:

Dividing by a number smaller than 1 can make the answer **larger**.

6. VERY COMMON TRAP

Which is larger?

$$\left[20 \div 2 \right]$$

or

$$\left[20 \div 0.5 \right]$$

First:

$$\lfloor 20 \div 2 = 10 \rfloor$$

Second:

$$\lfloor 0.5 = \frac{1}{2} \rfloor$$

So:

$$\lfloor 20 \div \frac{1}{2} = 20 \times 2 = 40 \rfloor$$

Therefore:

$$\lfloor \boxed{20 \div 0.5 = 40} \rfloor$$

Memory

Dividing by $\frac{1}{2}$ means asking: "How many halves are there?"

There are 40 halves in 20.

7. FAST DIVISION BY 5

Instead of:

$$\lfloor N \div 5 \rfloor$$

you can multiply by 2 and divide by 10.

Why?

$$\lfloor \frac{15}{5} = \frac{2 \times 10}{10} \rfloor$$

Example:

$$\lfloor 365 \div 5 \rfloor$$

$$\lfloor 365 \times 2 = 730 \rfloor$$

$$\lfloor 730 \div 10 = 73 \rfloor$$

Answer:

$$\lfloor \boxed{73} \rfloor$$

8. FAST DIVISION BY 25

Since:

$$\lfloor \frac{1}{25} = \frac{4}{100} \rfloor$$

therefore:

$$\lfloor N \div 25 = N \times 4 \div 100 \rfloor$$

Example:

$$\lfloor 725 \div 25 \rfloor$$

$$\lfloor 725 \times 4 = 2900 \rfloor$$

$$\lfloor 2900 / 100 = 29 \rfloor$$

Answer:

$$\boxed{29}$$

9. FAST DIVISION BY 125

Since:

$$125 \times 8 = 1000$$

we get:

$$N \div 125 = N \times 8 \div 1000$$

Example:

$$3750 \div 125$$

$$3750 \times 8 = 30000$$

$$30000 / 1000 = 30$$

Answer:

$$\boxed{30}$$

10. DIVISION BY POWERS OF 10

Very simple but essential.

$$345 \div 10 = 34.5$$

$$345 \div 100 = 3.45$$

$$345 \div 1000 = 0.345$$

So division by:

$$10^n$$

moves the decimal point n places left.

11. MULTIPLICATION BY POWERS OF 10

Opposite direction.

$$3.45 \times 10 = 34.5$$

$$3.45 \times 100 = 345$$

$$3.45 \times 1000 = 3450$$

Move decimal right.

12. DECIMAL PLACE VALUE

Consider:

$$37.482$$

Digits represent:

$$3 \times 10$$

$$7 \times 1$$

$$4 \times \frac{1}{10}$$

$$8 \times \frac{1}{100}$$

$$2 \times \frac{1}{1000}$$

So:

$$37.482 = 30 + 7 + 0.4 + 0.08 + 0.002$$

Understanding place value prevents decimal mistakes.

13. ADDING DECIMALS

Always align decimal points.

Example:

$$12.4 + 3.75$$

Write:

$$12.40$$

$$+ 3.75$$

$$= 16.15$$

Answer:

$$\boxed{16.15}$$

14. SUBTRACTING DECIMALS

Example:

$$10 - 3.275$$

Write:

$$10.000 - 3.275$$

$$= 6.725$$

Answer:

$$\boxed{6.725}$$

15. MULTIPLYING DECIMALS

Example:

$$2.5 \times 1.2$$

Ignore decimal points temporarily:

$$25 \times 12 = 300$$

Total decimal places:

- 2.5 !' 1 decimal
- 1.2 !' 1 decimal

Total = 2 decimal places.

Therefore:

$$3.00$$

Answer:

$$\boxed{3}$$

16. DIVIDING DECIMALS

Example:

$$4.8 \div 0.6$$

Multiply both by 10:

$$48 \div 6$$

$$= 8$$

Answer:

$$\boxed{8}$$

General rule

If divisor contains decimals, multiply numerator and denominator by an appropriate power of 10.

Example:

$$3.75 \div 0.25$$

Multiply both by 100:

$$375 \div 25$$

$$= 15$$

17. VERY IMPORTANT DECIMAL !"FRACTION VALUES

MUST MEMORIZE

$$0.5 = \frac{1}{2}$$

$$0.25 = \frac{1}{4}$$

$$0.75 = \frac{3}{4}$$

$$0.2 = \frac{1}{5}$$

$$0.4 = \frac{2}{5}$$

$$0.6 = \frac{3}{5}$$

$$0.8 = \frac{4}{5}$$

$$\sqrt[0.125]{\frac{1}{8}}$$

$$\sqrt[0.375]{\frac{3}{8}}$$

$$\sqrt[0.625]{\frac{5}{8}}$$

$$\sqrt[0.875]{\frac{7}{8}}$$

These will make percentage and probability questions much faster.

18. TERMINATING VS RECURRING DECIMALS

A terminating decimal ends.

Examples:

$$\sqrt[0.25]{\quad 0.625}{\quad 1.75}$$

A recurring decimal repeats forever.

Example:

$$\sqrt{\frac{1}{3}}=0.3333\dots$$

Written:

$$\sqrt[0.\overline{3}]{}{\quad}$$

Similarly:

$$\sqrt{\frac{2}{3}}=0.6666\dots$$

19. HIGH-VALUE RECURRING DECIMALS

MUST KNOW

$$\sqrt{\frac{1}{3}}=0.3333\dots$$

$$\sqrt{\frac{2}{3}}=0.6666\dots$$

$$\sqrt{\frac{1}{6}}=0.1666\dots$$

$$\sqrt{\frac{5}{6}}=0.8333\dots$$

$$\sqrt{\frac{1}{9}}=0.1111\dots$$

$$\sqrt{\frac{2}{9}}=0.2222\dots$$

$$\sqrt{\frac{3}{9}}=0.3333\dots$$

and so on.

20. QUICK CONVERSION OF RECURRING DECIMALS

Example:

$$\sqrt{x=0.\overline{3}}{\quad}$$

Then:

$$\sqrt{10x=3.\overline{3}}{\quad}$$

Subtract:

$$10x - x = 3$$

$$9x = 3$$

$$x = \frac{1}{3}$$

21. ANOTHER EXAMPLE

Convert:

$$0.\overline{27}$$

to fraction.

Let:

$$x = 0.272727\ldots$$

Multiply by 100:

$$100x = 27.272727\ldots$$

Subtract:

$$100x - x = 27$$

$$99x = 27$$

$$x = \frac{27}{99}$$

Simplify:

$$x = \frac{1}{3}$$

Answer:

$$\boxed{\frac{1}{3}}$$

This may appear in placement arithmetic or number-system questions.

22. FRACTION COMPARISON — METHOD 1

Which is larger?

$$\frac{5}{7}$$

or

$$\frac{3}{4}$$

Cross-multiply:

$$5 \times 4 = 20$$

$$3 \times 7 = 21$$

Since:

$$21 > 20$$

therefore:

$$\boxed{\frac{3}{4} > \frac{5}{7}}$$

23. FRACTION COMPARISON — METHOD 2

If numerators are equal:

$$\left[\frac{5}{8}, \quad \frac{5}{11} \right]$$

Smaller denominator gives larger fraction.

Therefore:

$$\left[\boxed{\frac{5}{8} > \frac{5}{11}} \right]$$

Why?

Because dividing the same quantity into fewer parts makes each part larger.

24. FRACTION COMPARISON — METHOD 3

If denominators are equal:

$$\left[\frac{7}{13}, \quad \frac{9}{13} \right]$$

Larger numerator means larger fraction.

Thus:

$$\left[\boxed{\frac{9}{13} > \frac{7}{13}} \right]$$

25. ADDING FRACTIONS

Example:

$$\left[\frac{2}{3} + \frac{1}{4} \right]$$

LCM of 3 and 4 is 12.

$$\left[\frac{2}{3} = \frac{8}{12} \right]$$

$$\left[\frac{1}{4} = \frac{3}{12} \right]$$

Therefore:

$$\left[\frac{11}{12} \right]$$

26. FAST FRACTION ADDITION

For:

$$\left[\frac{a}{b} + \frac{c}{d} \right]$$

$$\left[= \frac{ad+bc}{bd} \right]$$

Example:

$$\left[\frac{2}{5} + \frac{3}{7} \right]$$

$$\left[= \frac{2(7)+3(5)}{35} \right]$$

$$\left[= \frac{14+15}{35} \right]$$

$$\boxed{\frac{29}{35}}$$

Use this when denominators are small.

27. MULTIPLYING FRACTIONS

Example:

$$\frac{3}{4} \times \frac{8}{9}$$

Cancel first.

$$\frac{3}{4} \times \frac{8}{9}$$

Cancel:

$$\frac{8}{4} = 2$$

and:

$$\frac{3}{9} = \frac{1}{3}$$

So:

$$\frac{1 \times 2}{1 \times 3}$$

$$\boxed{\frac{2}{3}}$$

Again:

Cancel first.

28. DIVIDING FRACTIONS

Example:

$$\frac{3}{4} \div \frac{5}{8}$$

Invert second fraction:

$$\frac{3}{4} \times \frac{8}{5}$$

Cancel 8 and 4:

$$\frac{3 \times 2}{1 \times 5}$$

$$\boxed{\frac{6}{5}}$$

29. POWERS — CORE RULES

These matter across GATE aptitude and CS.

Rule 1

$$a^m \times a^n = a^{m+n}$$

Example:

$$2^3 \times 2^4 = 2^7 = 128$$

Rule 2

$$\left[\frac{a^m}{a^n} = a^{m-n} \right]$$

for $(a \neq 0)$.

Example:

$$\left[\frac{5^6}{5^2} = 5^4 = 625 \right]$$

Rule 3

$$\left[(a^m)^n = a^{mn} \right]$$

Example:

$$\left[(2^3)^4 = 2^{12} \right]$$

Rule 4

$$\left[(ab)^n = a^n b^n \right]$$

Rule 5

$$\left[a^0 = 1 \right]$$

for $(a \neq 0)$.

Example:

$$\left[27^0 = 1 \right]$$

30. NEGATIVE EXPONENTS

$$\left[a^{-n} = \frac{1}{a^n} \right]$$

Example:

$$\left[2^{-3} = \frac{1}{2^3} = \frac{1}{8} \right]$$

31. FRACTIONAL EXPONENTS

$$\left[a^{1/2} = \sqrt{a} \right]$$

$$\left[a^{1/3} = \sqrt[3]{a} \right]$$

Example:

$$\left[16^{1/2} = 4 \right]$$

$$\left[27^{1/3} = 3 \right]$$

32. VERY IMPORTANT POWER TRAP

Compare:

$$\sqrt{-2^2}$$

and:

$$\sqrt{(-2)^2}$$

First:

$$\sqrt{-2^2}=-(2^2)=-4$$

Second:

$$\sqrt{(-2)^2}=4$$

So:

$$\boxed{-2^2 \neq (-2)^2}$$

Brackets matter.

33. USEFUL ALGEBRAIC IDENTITIES

MUST MEMORIZE

$$(a+b)^2=a^2+2ab+b^2$$

$$(a-b)^2=a^2-2ab+b^2$$

$$(a+b)(a-b)=a^2-b^2$$

These are highly useful for rapid arithmetic too.

34. FAST SQUARE USING IDENTITY

Calculate:

$$52^2$$

Use:

$$(50+2)^2$$

$$=2500+200+4$$

$$=\boxed{2704}$$

35. ANOTHER FAST SQUARE

$$98^2$$

Use:

$$(100-2)^2$$

$$=10000-400+4$$

$$=\boxed{9604}$$

36. DIFFERENCE OF SQUARES SHORTCUT

Calculate:

$$\sqrt{103 \times 97}$$

Recognize:

$$\sqrt{(100+3)(100-3)}$$

$$\sqrt{=100^2-3^2}$$

$$\sqrt{=10000-9}$$

$$\sqrt{=\boxed{9991}}$$

37. NEARBY NUMBER PRODUCT

Calculate:

$$\sqrt{62 \times 58}$$

Center:

$$\sqrt{60}$$

So:

$$\sqrt{(60+2)(60-2)}$$

$$\sqrt{=60^2-4}$$

$$\sqrt{=3600-4}$$

$$\sqrt{=\boxed{3596}}$$

38. APPROXIMATION — FIRST PRINCIPLE

Approximation means replacing awkward numbers with nearby easy numbers when exact calculation is unnecessary.

Example:

$$\sqrt{19.98 \times 5.02}$$

Approximately:

$$\sqrt{20 \times 5 = 100}$$

If options are:

- 20
- 50
- 100
- 500

you don't need exact multiplication.

Answer must be near:

$$\sqrt{\boxed{100}}$$

39. ESTIMATION USING BOUNDS

Suppose:

$$[49 \times 19]$$

You know:

$$[50 \times 20 = 1000]$$

Actual value must be slightly below 1000.

Exact:

$$[49(20-1) = 980 - 49 = 931]$$

But if options are:

- 210
- 470
- 930
- 1500

estimation alone identifies:

$$[\boxed{930}]$$

approximately.

40. ORDER OF MAGNITUDE CHECK

Suppose your calculation gives:

$$[49 \times 51 = 24990]$$

Something is clearly wrong because:

$$[50 \times 50 \approx 2500]$$

Therefore answer should be near 2500, not 25,000.

This is called a **sanity check**.

GATE habit

Before accepting an answer, ask:

Is the size reasonable?

41. LAST-DIGIT SANITY CHECK

Suppose:

$$[37 \times 24]$$

The last digit must come from:

$$[7 \times 4 = 28]$$

So product must end in:

$\backslash 8 \backslash$

If your answer ends in 4 or 6, it is wrong.

This is a very fast error detector.

42. PARITY SANITY CHECK

Example:

$\backslash 27 \times 35 \backslash$

odd $\times$ odd = odd.

So answer must be odd.

If you somehow calculate an even number, reject it.

43. DIGIT-SUM CHECK — BASIC IDEA

This can catch some arithmetic errors.

Example:

$\backslash 48 \times 23 \backslash$

Expected product:

$\backslash 1104 \backslash$

Digit sum:

$\backslash 48 \rightarrow 4+8=12 \rightarrow 3 \backslash$

$\backslash 23 \rightarrow 2+3=5 \backslash$

Expected digit-root behavior:

$\backslash 3 \times 5 = 15 \rightarrow 6 \backslash$

Product:

$\backslash 1104 \rightarrow 1+1+0+4=6 \backslash$

Matches.

This is related to modulo 9.

Important

This does **not prove** an answer is correct.

It only helps detect some mistakes.

So:

Good to know, not mandatory for every problem.

44. FAST MULTIPLICATION BY 11

Example:

$$\boxed{43 \times 11}$$

For a two-digit number (ab) :

Place:

- first digit
- sum of digits
- second digit

For 43:

$$\boxed{4, \text{quad} 4+3=7, \text{quad} 3}$$

So:

$$\boxed{473}$$

Check:

$$\boxed{43 \times 11 = 473}$$

45. MULTIPLY BY 11 WITH CARRY

Example:

$$\boxed{78 \times 11}$$

Write:

$$\boxed{7, \text{quad} 7+8=15, \text{quad} 8}$$

Because middle is 15:

write 5, carry 1 to 7:

$$\boxed{8 \backslash 5 \backslash 8}$$

So:

$$\boxed{858}$$

46. MULTIPLY BY 9

Use:

$$\boxed{N \times 9 = N \times 10 - N}$$

Example:

$$\boxed{47 \times 9}$$

$$\boxed{470 - 47}$$

$$\boxed{= 423}$$

47. MULTIPLY BY 99

$$\boxed{N \times 99 = N \times 100 - N}$$

Example:

$$\boxed{73 \times 99}$$

$$7300 - 73$$

$$\boxed{7227}$$

48. MULTIPLY BY 101

$$N \times 101 = N \times 100 + N$$

Example:

$$84 \times 101$$

$$8400 + 84$$

$$\boxed{8484}$$

49. DOUBLE-HALF TECHNIQUE

Suppose:

$$16 \times 35$$

Double one and halve the other:

$$8 \times 70$$

$$= 560$$

Or:

$$4 \times 140 = 560$$

This is useful when one factor is even.

Another:

$$25 \times 48$$

Double 25 !' 50 Half 48 !' 24

$$50 \times 24$$

Again:

$$100 \times 12$$

$$\boxed{1200}$$

50. WHY DOUBLE-HALF WORKS

Because:

$$a \times b = (2a) \times \frac{b}{2}$$

The product remains unchanged.

Use it when it creates easier numbers.

51. BREAKING DIVISION INTO FACTORS

Example:

$$\backslash[720\div24 \backslash]$$

Since:

$$\backslash[24=6\times4 \backslash]$$

You can do:

$$\backslash[720\div6=120 \backslash]$$

then:

$$\backslash[120\div4=30 \backslash]$$

Answer:

$$\backslash[\boxed{30} \backslash]$$

52. ANOTHER DIVISION SHORTCUT

$$\backslash[840\div35 \backslash]$$

Since:

$$\backslash[35=5\times7 \backslash]$$

$$\backslash[840\div5=168 \backslash]$$

$$\backslash[168\div7=24 \backslash]$$

Answer:

$$\backslash[\boxed{24} \backslash]$$

Or:

$$\backslash[840\times\frac{1}{35} \backslash]$$

Simplify factor-wise.

53. CALCULATION MISTAKE PREVENTION

Most aptitude arithmetic mistakes come from only a few causes.

1. Losing a sign

Especially with negative numbers.

2. Decimal shift error

Example:

$$\backslash[3.2\times100=320 \backslash]$$

not 32.

3. Multiplying before cancelling

Creates unnecessary large numbers.

4. Forgetting denominator

Especially in fractions and percentages.

5. Wrong order of operations

6. Copying the wrong number

7. Using a shortcut you haven't mastered

54. BEST EXAM CALCULATION PROCESS

For a numerical aptitude question, follow this:

Step 1

Estimate rough answer.

Step 2

Simplify/cancel.

Step 3

Choose easiest representation:

- fraction?
- decimal?
- percentage?
- ratio?

Step 4

Calculate.

Step 5

Sanity-check:

- sign
- size
- parity
- final digit
- units

This saves marks.

GATE-STYLE APPLICATION 1

Evaluate:

$$\left[\frac{7.5 \times 32}{0.25 \times 8} \right]$$

Step 1 — Convert useful decimal

$$\left[0.25 = \frac{1}{4} \right]$$

So denominator:

$$\left[\frac{14}{8} \times 2 \right]$$

Numerator:

$$\left[7.5 \times 32 = 240 \right]$$

Therefore:

$$\left[240 / 2 = 120 \right]$$

Answer:

$$\left[\boxed{120} \right]$$

GATE-STYLE APPLICATION 2

Evaluate quickly:

$$\left[999^2 \right]$$

Use:

$$\left[(1000-1)^2 \right]$$

$$\left[= 1000000 - 2000 + 1 \right]$$

$$\left[= \boxed{998001} \right]$$

PLACEMENT-STYLE APPLICATION 1

Find:

$$\left[125 \times 48 \right]$$

Fastest route

$$\left[125 = \frac{1000}{8} \right]$$

So:

$$\left[48 \times 125 = 48 \times 1000 / 8 \right]$$

$$\left[= 6 \times 1000 \right]$$

$$\left[= \boxed{6000} \right]$$

PLACEMENT-STYLE APPLICATION 2

Find:

$$\left[4.8 \div 0.12 \right]$$

Multiply both by 100:

$$\left[480 \div 12 \right]$$

$$\left[= 40 \right]$$

Answer:

$$\left[\boxed{40} \right]$$

& IMPORTANT TRAPS

Trap 1

$$\left[6 \div \frac{1}{2} \right]$$

is not 3.

It is:

$$\left[6 \times 2 = 12 \right]$$

Trap 2

$$\left[2 + 3^2 \right]$$

$$\left[= 2 + 9 = 11 \right]$$

not:

$$\left[(2 + 3)^2 \right]$$

Trap 3

$$\left[-3^2 = -9 \right]$$

but:

$$\left[(-3)^2 = 9 \right]$$

Trap 4

$$\left[0.2 \times 0.3 \right]$$

is:

$$\left[0.06 \right]$$

not 0.6.

Trap 5

Do not compare decimals digit-by-digit incorrectly.

Example:

$$\left[0.9 > 0.85 \right]$$

because:

$$\left[0.90 > 0.85 \right]$$

MUST MEMORIZE FROM PART 2

Reciprocal rule

$$\left[a \div \frac{b}{c} = a \times \frac{c}{b} \right]$$

Divide by 5

$$\left[N/5 = 2N/10 \right]$$

Divide by 25

$$\left[N/25 = 4N/100 \right]$$

Divide by 125

$$\left[N/125 = 8N/1000 \right]$$

Identities

$$\left[(a+b)^2 = a^2 + 2ab + b^2 \right]$$

$$\left[(a-b)^2 = a^2 - 2ab + b^2 \right]$$

$$\left[(a+b)(a-b) = a^2 - b^2 \right]$$

Power laws

$$\left[a^m a^n = a^{m+n} \right]$$

$$\left[a^m / a^n = a^{m-n} \right]$$

$$\left[(a^m)^n = a^{mn} \right]$$

$$\left[a^{-n} = \frac{1}{a^n} \right]$$

$$\left[a^0 = 1 \right]$$

5-MINUTE REVISION

Remember these ideas:

1. Division by a fraction !' multiply by reciprocal.
2. Dividing by a number below 1 may increase the result.
3. Move decimal left for division by 10, 100, 1000.
4. Convert awkward decimals into easy fractions.
5. Cancel before multiplying fractions.
6. Use identities near round numbers.
7. Approximate before calculating.
8. Check order of magnitude.
9. Check sign/parity/last digit where useful.
10. Do not use shortcuts blindly.

SPEED DRILL

Try each mentally.

L0

1.

$$\lfloor 360 \div 5 \rfloor$$

2.

$$\lfloor 700 \div 25 \rfloor$$

3.

$$\lfloor 2500 \div 125 \rfloor$$

4.

$$\lfloor 5 \div 0.5 \rfloor$$

5.

$$\lfloor 7.2 \div 0.8 \rfloor$$

L1

6.

$$\lfloor 48 \times 99 \rfloor$$

7.

$$\lfloor 72 \times 11 \rfloor$$

8.

$$\lfloor 102 \times 98 \rfloor$$

9.

$$\lfloor 96^2 \rfloor$$

10.

$$\lfloor \frac{34}{16} \times \frac{16}{9} \rfloor$$

L2

11.

$$\lfloor \frac{125 \times 96}{12} \rfloor$$

12.

$$\lfloor 3.75 \div 0.125 \rfloor$$

13.

$$\lfloor \frac{2^8 \times 2^5}{2^{10}} \rfloor$$

14.

$$\lfloor 0.375 \times 64 \rfloor$$

15.

Without exact multiplication, determine the last digit of:

$$\lfloor 347 \times 628 \rfloor$$

ANSWERS

1. $\sqrt{72}$
 2. $\sqrt{28}$
 3. $\sqrt{20}$
 4. $\sqrt{10}$
 5. $\sqrt{9}$
 6. $\sqrt{4752}$
 7. $\sqrt{792}$
 8. $\sqrt{9996}$
 9. $\sqrt{9216}$
 10. $\sqrt{\frac{43}{}}$
 11. $\sqrt{1000}$
 12. $\sqrt{30}$
 13. $\sqrt{8}$
 14. $\sqrt{24}$
 15. $\sqrt{6}$
-

CHAPTER 1 — FINAL CHECKLIST

You should now understand:

- arithmetic operation order '
- sign rules '
- parity '
- multiplication shortcuts '
- cancellation '
- fast division '
- reciprocals '
- decimal arithmetic '
- fraction arithmetic '
- power laws '
- useful identities '
- estimation '
- sanity checks '
- speed-calculation habits '

Therefore:

' CHAPTER 1 COMPLETE

Progress

Completed: 1/76 **Current:** ' Chapter 1 — Arithmetic & Calculation Foundations **Next:** Chapter 2 — Fractions and Decimals: complete aptitude mastery

Chapter 2 will go much deeper into:

- proper/improper/mixed fractions
- equivalent fractions
- comparison
- operations
- complex fractions
- decimal conversions
- recurring decimals
- approximation
- fraction-based word problems
- placement shortcuts
- GATE traps

Send **NEXT**.

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CHAPTER 2 — FRACTIONS AND DECIMALS

PART 1 — FRACTION FOUNDATIONS, COMPARISON & OPERATIONS

Status: 0% **PROGRESS** **GATE:** High **Placements:** High **Priority:** 0% **Yr 1**

Fractions appear everywhere:

- percentages
- ratios
- averages
- probability
- time and work
- mixtures
- speed
- DI
- algebra

If fractions become natural to you, many aptitude topics become much easier.

1. WHAT IS A FRACTION?

A fraction represents a part of a whole.

$$\left[\frac{a}{b} \right]$$

where:

- a = numerator
- b = denominator
- $b \neq 0$

Example:

$$\left[\frac{3}{4} \right]$$

means 3 parts out of 4 equal parts.

2. TYPES OF FRACTIONS

Proper Fraction

Numerator < denominator.

Example:

$$\left[\frac{3}{5} \right]$$

Value is less than 1.

Improper Fraction

Numerator $\geq$ denominator.

Example:

$$\left[\frac{7}{4} \right]$$

Value is greater than or equal to 1.

Mixed Fraction

Combination of whole number + fraction.

Example:

$$\left[2\frac{3}{4} \right]$$

Convert to improper fraction:

$$\left[2\frac{3}{4} = \frac{2 \times 4 + 3}{4} = \frac{11}{4} \right]$$

Rule

$$\left[a\frac{b}{c} = \frac{ac+b}{c} \right]$$

MUST MEMORIZE

3. EQUIVALENT FRACTIONS

Fractions can look different but have the same value.

Example:

$$\frac{12}{100} = \frac{24}{200} = \frac{36}{300} = \frac{50}{500}$$

Why?

Because multiplying numerator and denominator by the same non-zero number does not change the value.

$$\frac{a}{b} = \frac{ka}{kb}$$

for $k \neq 0$.

4. REDUCING A FRACTION

Example:

$$\frac{18}{24}$$

GCD of 18 and 24 is 6.

Divide both:

$$\frac{18 \div 6}{24 \div 6} = \frac{3}{4}$$

So:

$$\boxed{\frac{18}{24} = \frac{3}{4}}$$

Exam habit

Always simplify fractions when possible.

5. FRACTION ON NUMBER LINE

Fractions represent actual numbers.

Example:

$$\frac{1}{2} = 0.5$$

$$\frac{3}{4} = 0.75$$

$$\frac{5}{4} = 1.25$$

This helps compare fractions intuitively.

6. COMPARING FRACTIONS — SAME DENOMINATOR

Example:

$$\frac{5}{11}, \quad \frac{8}{11}$$

Same denominator.

Larger numerator !' larger fraction.

Therefore:

$$\boxed{\frac{8}{11} > \frac{5}{11}}$$

7. COMPARING FRACTIONS — SAME NUMERATOR

Example:

$$\left[\frac{5}{7}, \quad \frac{5}{9} \right]$$

Same numerator.

Smaller denominator != larger fraction.

Thus:

$$\left[\boxed{\frac{5}{7} > \frac{5}{9}} \right]$$

Intuition

Dividing a pizza into 7 pieces gives bigger pieces than dividing it into 9 pieces.

8. CROSS-MULTIPLICATION METHOD

Compare:

$$\left[\frac{5}{7} \right]$$

and

$$\left[\frac{7}{9} \right]$$

Cross products:

$$\left[5 \times 9 = 45 \right]$$

$$\left[7 \times 7 = 49 \right]$$

Since:

$$\left[49 > 45 \right]$$

therefore:

$$\left[\boxed{\frac{7}{9} > \frac{5}{7}} \right]$$

Fast rule

For positive denominators:

$$\left[\frac{a}{b} > \frac{c}{d} \right]$$

if:

$$\left[ad > bc \right]$$

9. COMPARISON USING 1

Sometimes easiest question:

Is fraction above or below 1?

If:

$$\left[\frac{\text{numerator}}{\text{denominator}} < 1 \right]$$

then:

$$\left[\frac{ab}{1} \right]$$

If:

$$\left[a > b \right]$$

then:

$$\left[\frac{ab}{1} > 1 \right]$$

Example:

$$\left[\frac{97}{101} < 1 \right]$$

$$\left[\frac{103}{99} > 1 \right]$$

So immediately:

$$\left[\boxed{\frac{103}{99} > \frac{97}{101}} \right]$$

No multiplication needed.

10. COMPARISON USING 1/2

Useful aptitude trick.

To check whether:

$$\left[\frac{ab}{1} > \frac{1}{2} \right]$$

compare:

$$\left[2a \text{ with } b \right]$$

Example:

$$\left[\frac{7}{13} \right]$$

$$\left[2 \times 7 = 14 > 13 \right]$$

Therefore:

$$\left[\frac{7}{13} > \frac{1}{2} \right]$$

Another:

$$\left[\frac{8}{17} \right]$$

$$\left[16 < 17 \right]$$

So:

$$\left[\frac{8}{17} < \frac{1}{2} \right]$$

11. COMPARING CLOSE FRACTIONS USING DIFFERENCE FROM 1

Compare:

$$\left[\frac{99}{101} \right]$$

and

$$\left[\frac{98}{100} \right]$$

Write as distance from 1.

$$\left[\frac{99}{101} = 1 - \frac{2}{101} \right]$$

$$\left[\frac{98}{100} = 1 - \frac{2}{100} \right]$$

Since:

$$\left[\frac{2}{101} < \frac{2}{100} \right]$$

the first is closer to 1.

Therefore:

$$\left[\boxed{\frac{99}{101} > \frac{98}{100}} \right]$$

This is faster than large cross-products in some questions.

12. ADDING FRACTIONS — SAME DENOMINATOR

$$\left[\frac{3}{7} + \frac{2}{7} = \frac{3+2}{7} = \frac{5}{7} \right]$$

Rule

$$\left[\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b} \right]$$

13. SUBTRACTING FRACTIONS — SAME DENOMINATOR

$$\left[\frac{7}{9} - \frac{2}{9} = \frac{5}{9} \right]$$

14. ADDING DIFFERENT DENOMINATORS

Example:

$$\left[\frac{2}{3} + \frac{1}{4} \right]$$

LCM of 3 and 4 is 12.

$$\left[\frac{2}{3} = \frac{8}{12} \right]$$

$$\left[\frac{1}{4} = \frac{3}{12} \right]$$

Therefore:

$$\left[\boxed{\frac{11}{12}} \right]$$

15. SHORTCUT FORMULA

$$\left[\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} \right]$$

Example:

$$\left[\frac{2}{5} + \frac{3}{7} \right]$$

$$\left[= \frac{14+15}{35} \right]$$

$$\left[= \boxed{\frac{29}{35}} \right]$$

Use this when numbers are manageable.

16. SUBTRACTING DIFFERENT DENOMINATORS

$$\left[\frac{5}{6} - \frac{1}{4} \right]$$

LCM = 12.

$$\left[\frac{10}{12} - \frac{3}{12} = \frac{7}{12} \right]$$

Answer:

$$\left[\boxed{\frac{7}{12}} \right]$$

17. MULTIPLYING FRACTIONS

Rule:

$$\left[\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \right]$$

But in aptitude:

Cancel before multiplying.

Example:

$$\left[\frac{15}{28} \times \frac{14}{25} \right]$$

Cancel:

$$\left[\frac{15}{25} = \frac{3}{5} \right]$$

$$\left[\frac{14}{28} = \frac{1}{2} \right]$$

So:

$$\left[\frac{3}{5} \times \frac{1}{2} = \boxed{\frac{3}{10}} \right]$$

Much easier.

18. DIVIDING FRACTIONS

Rule:

$$\left[\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} \right]$$

Example:

$$\left[\frac{3}{4} \div \frac{2}{5} \right]$$

$$\left[= \frac{3}{4} \times \frac{5}{2} \right]$$

$$\left[= \frac{15}{8} \right]$$

Answer:

$$\left[\boxed{\frac{15}{8}} \right]$$

19. “OF” USUALLY MEANS MULTIPLICATION

Very important for aptitude.

Example:

$\frac{3}{5}$ of 40

means:

$\frac{3}{5} \times 40$

Cancel:

$40/5 = 8$

$3 \times 8 = 24$

Answer:

$\boxed{24}$

Question-reading rule

“of” !’ multiply

This becomes essential in percentages.

20. FRACTION OF A FRACTION

Example:

$\frac{2}{3}$ of $\frac{4}{5}$

$= \frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$

21. FRACTION REMAINING

Suppose you spend:

$\frac{3}{5}$

of your money.

Fraction remaining:

$1 - \frac{3}{5} = \frac{2}{5}$

Useful pattern

If fraction used = $\frac{x}{y}$,

$\text{remaining} = 1 - x$

22. SUCCESSIVE FRACTION USE

Suppose a person spends:

$\frac{1}{3}$

of money, then spends:

$\frac{1}{4}$

of the remaining money.

Take original = 1.

After first spending:

$$\left[1 - \frac{1}{3} = \frac{2}{3} \right]$$

Second spend:

$$\left[\frac{1}{4} \times \frac{2}{3} = \frac{1}{6} \right]$$

Remaining:

$$\left[\frac{2}{3} - \frac{1}{6} = \frac{4}{6} - \frac{1}{6} = \frac{1}{2} \right]$$

So half remains.

& #x26A0 TRAP

Second fraction is of the **remaining amount**, not original amount.

23. FRACTION INCREASE

Suppose number increases by:

$$\left[\frac{1}{5} \right]$$

of itself.

New value:

$$\left[1 + \frac{1}{5} = \frac{6}{5} \right]$$

times original.

So:

$$\left[\boxed{\text{New} = \frac{6}{5} \text{ of original}} \right]$$

This connects directly to percentages:

$$\left[\frac{1}{5} = 20\% \right]$$

24. FRACTION DECREASE

Decrease by:

$$\left[\frac{1}{4} \right]$$

of original.

Remaining:

$$\left[1 - \frac{1}{4} = \frac{3}{4} \right]$$

So new value:

$$\left[\boxed{\frac{3}{4} \text{ of original}} \right]$$

25. COMPLEX FRACTIONS

Example:

$$\left[\frac{\frac{3}{4}}{\frac{5}{8}} \right]$$

This means:

$$\left[\frac{3}{4} \div \frac{5}{8} \right]$$

$$\left[= \frac{3}{4} \times \frac{8}{5} \right]$$

$$\left[= \frac{6}{5} \right]$$

Answer:

$$\left[\boxed{\frac{6}{5}} \right]$$

26. ANOTHER COMPLEX FRACTION

$$\left[\frac{1 + \frac{1}{2}}{1 - \frac{1}{4}} \right]$$

Numerator:

$$\left[1 + \frac{1}{2} = \frac{3}{2} \right]$$

Denominator:

$$\left[1 - \frac{1}{4} = \frac{3}{4} \right]$$

So:

$$\left[\frac{\frac{3}{2}}{\frac{3}{4}} = \frac{3}{2} \times \frac{4}{3} = 2 \right]$$

Answer:

$$\left[\boxed{2} \right]$$

27. RECIPROCAL RELATIONSHIP

Two numbers are reciprocals if their product is 1.

Example:

$$\left[\frac{3}{7} \times \frac{7}{3} = 1 \right]$$

So reciprocal of:

$$\left[\frac{3}{7} \right]$$

is:

$$\left[\frac{7}{3} \right]$$

& hReciprocal of zero

Does not exist.

Because:

$$\left[\frac{1}{0} \right]$$

is undefined.

28. MIXED FRACTIONS — OPERATIONS

Example:

$$\left[2\frac{12}{1}+1\frac{34}{4} \right]$$

Convert:

$$\left[2\frac{12}{1}=\frac{52}{1} \right]$$

$$\left[1\frac{34}{4}=\frac{74}{4} \right]$$

Then:

$$\left[\frac{10}{4}+\frac{74}{4} = \frac{17}{4} \right]$$

Convert back:

$$\left[\boxed{4\frac{14}{4}} \right]$$

29. FRACTION !”DECIMAL

To convert fraction to decimal:

$$\text{numerator} \div \text{denominator}$$

Example:

$$\left[\frac{3}{8} \right]$$

$$\left[3\div8=0.375 \right]$$

So:

$$\left[\boxed{\frac{3}{8}=0.375} \right]$$

30. DECIMAL !”FRACTION

Example:

$$\left[0.75 \right]$$

$$\left[=\frac{75}{100} \right]$$

Simplify:

$$\left[=\frac{3}{4} \right]$$

31. DECIMAL WITH THREE PLACES

$$\left[0.625 = \frac{625}{1000} \right]$$

Divide by 125:

$$\left[=\frac{5}{8} \right]$$

Therefore:

$$\left[\boxed{0.625=\frac{5}{8}} \right]$$

32. MUST-MEMORIZE FRACTION-DECIMAL TABLE

Fraction	Decimal
$\frac{1}{2}$	0.5
$\frac{1}{3}$	0.333...
$\frac{2}{3}$	0.666...
$\frac{1}{4}$	0.25
$\frac{3}{4}$	0.75
$\frac{1}{5}$	0.2
$\frac{2}{5}$	0.4
$\frac{3}{5}$	0.6
$\frac{4}{5}$	0.8
$\frac{1}{8}$	0.125
$\frac{3}{8}$	0.375
$\frac{5}{8}$	0.625
$\frac{7}{8}$	0.875
$\frac{1}{10}$	0.1

These should become automatic.

33. WHEN DOES A FRACTION TERMINATE?

This is useful number-system knowledge.

Take a fraction in lowest terms:

$$\frac{a}{b}$$

Its decimal terminates **only if** denominator contains prime factors 2 and/or 5.

Examples:

$$\frac{18}{8}$$

Denominator:

$$8 = 2^3$$

Terminates.

$$\frac{3}{20}$$

$$20 = 2^2 \times 5$$

Terminates.

But:

$$\left[\frac{17}{} \right]$$

contains factor 7.

So decimal repeats.

Rule

After simplifying:

$$\left[b=2^m5^n \right]$$

!' terminating decimal.

Otherwise !' recurring decimal.

34. EXAMPLE

Will:

$$\left[\frac{7}{40} \right]$$

terminate?

$$\left[40=2^3 \times 5 \right]$$

Yes.

35. ANOTHER EXAMPLE

Will:

$$\left[\frac{9}{24} \right]$$

terminate?

First simplify:

$$\left[\frac{9}{24} = \frac{3}{8} \right]$$

Now:

$$\left[8=2^3 \right]$$

Therefore yes.

& ⚠TRAP

Always simplify first.

36. FRACTIONS BETWEEN TWO FRACTIONS

There are infinitely many fractions between any two distinct fractions.

Example between:

$$\left[\frac{1}{2} \right]$$

and

$$\left[\frac{2}{3} \right]$$

One method:

$$\left[\frac{1+2}{2+3} = \frac{3}{5} \right]$$

This is called the mediant:

$$\left[\frac{a+c}{b+d} \right]$$

For positive fractions, it lies between them.

So:

$$\left[\frac{1}{2} < \frac{3}{5} < \frac{2}{3} \right]$$

Useful conceptually, though not a major GATE shortcut.

37. FRACTION COMPARISON SHORTCUT — COMPLEMENTS

Compare:

$$\left[\frac{11}{12} \right]$$

and

$$\left[\frac{17}{18} \right]$$

Distance from 1:

$$\left[1 - \frac{11}{12} = \frac{1}{12} \right]$$

$$\left[1 - \frac{17}{18} = \frac{1}{18} \right]$$

Since:

$$\left[\frac{1}{18} < \frac{1}{12} \right]$$

$$\left[\frac{17}{18} \right]$$

is closer to 1.

Therefore:

$$\left[\boxed{\frac{17}{18} > \frac{11}{12}} \right]$$

Very fast.

38. FRACTION COMPARISON SHORTCUT — SAME DIFFERENCE

Compare:

$$\left[\frac{7}{8} \right]$$

and

$$\left[\frac{8}{9} \right]$$

Both are one less than denominator.

$$\left[\frac{7}{8} = 1 - \frac{1}{8} \right]$$

$$\left[\frac{8}{9} = 1 - \frac{1}{9} \right]$$

Since:

$$\left[\frac{19}{18} \right]$$

therefore:

$$\left[\boxed{\frac{89}{78}} \right]$$

39. GATE-STYLE QUESTION 1

Which is greatest?

A.

$$\left[\frac{17}{19} \right]$$

B.

$$\left[\frac{18}{20} \right]$$

C.

$$\left[\frac{19}{21} \right]$$

D.

$$\left[\frac{20}{22} \right]$$

All are close to 1.

Write differences:

$$\left[\frac{17}{19} = 1 - \frac{2}{19} \right]$$

$$\left[\frac{18}{20} = 1 - \frac{2}{20} \right]$$

$$\left[\frac{19}{21} = 1 - \frac{2}{21} \right]$$

$$\left[\frac{20}{22} = 1 - \frac{2}{22} \right]$$

Smallest subtraction is:

$$\left[\frac{2}{22} \right]$$

Hence largest fraction:

$$\left[\boxed{\frac{20}{22}} \right]$$

which simplifies to:

$$\left[\boxed{\frac{10}{11}} \right]$$

40. GATE-STYLE QUESTION 2

If:

$$\left[\frac{35}{x} \right]$$

of a number is 72, find the number.

Let number = (x) .

$$\left[\frac{35}{x} = 72 \right]$$

Multiply both sides by reciprocal:

$$\left[x = 72 \times \frac{5}{3} \right]$$

$$\boxed{=24 \times 5}$$

$$\boxed{=120}$$

Fast method

If $\frac{3}{5}$ corresponds to 72:

1 part:

$$\frac{72}{3}=24$$

5 parts:

$$24 \times 5 = 120$$

41. PLACEMENT-STYLE QUESTION

A person spends:

$$\frac{25}{100}$$

of salary on rent and:

$$\frac{14}{100}$$

of salary on food.

What fraction remains?

Total spent:

$$\frac{25}{100} + \frac{14}{100}$$

LCM 20:

$$\frac{8}{20} + \frac{5}{20} = \frac{13}{20}$$

Remaining:

$$1 - \frac{13}{20} = \boxed{\frac{7}{20}}$$

42. TRICKY PLACEMENT QUESTION

A person spends:

$$\frac{14}{100}$$

of income on rent and then:

$$\frac{13}{100}$$

of the remaining amount on food.

What fraction remains?

Take income = 1.

After rent:

$$1 - \frac{14}{100} = \frac{86}{100}$$

Food:

$$\frac{13}{100} \times \frac{86}{100} = \frac{1118}{10000}$$

Remaining:

$$\left[\frac{34}{14} - \frac{14}{12} = \frac{12}{14} \right]$$

Answer:

$$\left[\boxed{\frac{12}{14}} \right]$$

& pTRAP

Do not add:

$$\left[\frac{14}{14} + \frac{13}{13} \right]$$

because second fraction is not of original income.

43. NORMAL METHOD VS FAST METHOD

Question:

$$\left[\frac{18}{35} \times \frac{49}{27} \right]$$

Normal method

Multiply:

$$\left[\frac{882}{945} \right]$$

then simplify.

Slow.

Fast method

Cancel first:

$$\left[18/27 = 2/3 \right]$$

$$\left[49/35 = 7/5 \right]$$

So:

$$\left[\frac{23}{15} \times \frac{75}{75} = \boxed{\frac{14}{15}} \right]$$

Exam method

Always prefer cancellation.

44. COMMON FRACTION TRAPS

& pTRAP 1

$$\left[\frac{12}{13} + \frac{13}{13} \neq \frac{25}{13} \right]$$

Correct:

$$\left[\frac{12}{13} + \frac{13}{13} = \frac{25}{13} \right]$$

& pTRAP 2

$\backslash \frac{ab}{\div \frac{cd}{}}$

requires inverting **only the second fraction**.

& pTRAP 3

A larger denominator does not always mean smaller fraction unless numerators are same.

Example:

$\backslash \frac{9}{10} > \frac{12}{}$

& pTRAP 4

Always simplify before deciding whether decimal terminates.

& pTRAP 5

“Fraction of remaining” is different from “fraction of original”.

45. MEMORY TRICKS

Addition/subtraction

Same denominator !' work on numerators.

Division

Keep – Change – Flip

Keep first fraction Change $\div$ to $\times$ Flip second fraction

Example:

$\backslash \frac{23}{\div \frac{45}{}}$

becomes:

$\backslash \frac{23}{\times \frac{54}{}}$

Fraction remaining

Remaining = 1 " used

46. QUICK REVISION BOX

Fraction basics

$\backslash \frac{ab}{}, \quad b \neq 0$

Equivalent

$$\left[\frac{a}{b} = \frac{ka}{kb} \right]$$

Addition

$$\left[\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} \right]$$

Multiplication

$$\left[\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \right]$$

Division

$$\left[\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} \right]$$

Mixed to improper

$$\left[a \frac{a}{b} = \frac{ac+b}{c} \right]$$

Decimal termination

After simplification:

$$\left[\text{denominator} = 2^m 5^n \right]$$

! terminating.

PRACTICE — LEVEL 0

Q1

Simplify:

$$\left[\frac{24}{36} \right]$$

Q2

Convert:

$$\left[3 \frac{2}{5} \right]$$

to an improper fraction.

Q3

Find:

$$\left[\frac{3}{4} + \frac{1}{8} \right]$$

Q4

Find:

$$\left[\frac{5}{6} - \frac{1}{3} \right]$$

Q5

Find:

$$\left[\frac{23}{10} \times \frac{9}{10} \right]$$

LEVEL 1

Q6

Find:

$$\left[\frac{78}{32} \div \frac{21}{32} \right]$$

Q7

Which is larger?

$$\left[\frac{13}{17} \right]$$

or

$$\left[\frac{16}{21} \right]$$

Q8

If:

$$\left[\frac{47}{52} \right]$$

of a number is 52, find the number.

Q9

A student completes:

$$\left[\frac{38}{52} \right]$$

of a book on Monday and:

$$\left[\frac{14}{52} \right]$$

on Tuesday.

What fraction remains?

Q10

Convert:

$$\left[0.875 \right]$$

to a fraction.

LEVEL 2

Q11

Which is largest?

$$\left[\frac{29}{31}, \frac{30}{32}, \frac{31}{33} \right]$$

Q12

A person spends:

$$\left[\frac{15}{52} \right]$$

of money, then:

$$\left[\frac{14}{} \right]$$

of the remaining money.

What fraction remains?

Q13

Simplify:

$$\left[\frac{\frac{23}{} + \frac{14}{}}{\frac{56}{} - \frac{13}{}} \right]$$

Q14

Does:

$$\left[\frac{27}{120} \right]$$

have a terminating decimal?

Q15

Simplify quickly:

$$\left[\frac{36}{55} \times \frac{121}{54} \right]$$

ANSWERS

1.

$$\left[\frac{23}{} \right]$$

2.

$$\left[\frac{17}{5} \right]$$

3.

$$\left[\frac{78}{} \right]$$

4.

$$\left[\frac{12}{} \right]$$

5.

$$\left[\frac{35}{} \right]$$

6.

$$\left[\frac{43}{} \right]$$

7.

$$\left[\frac{13}{17} \right]$$

8.

$$\left[91 \right]$$

9.

$$\left[\frac{38}{} \right]$$

10.

$$\frac{7}{8}$$

11.

$$\frac{31}{33}$$

12.

$$\frac{3}{5}$$

13.

$$\frac{11}{6}$$

14. Yes

15.

$$\frac{22}{15}$$

CHAPTER 2 STATUS

We have completed:

- fraction types '
- simplification '
- fraction comparison '
- operations '
- mixed fractions '
- reciprocals '
- complex fractions '
- fraction/decimal conversion '
- termination rule '
- fraction word problems '
- comparison shortcuts '

But Chapter 2 is **not complete yet**.

Next we need:

- decimal mastery
- recurring decimals
- rounding
- significant approximation for aptitude
- decimal comparison traps
- converting complex decimals quickly
- decimal-based word problems
- placement speed techniques
- mixed fraction-decimal exam questions

Progress

Completed: 1/76 **Current:** 0/35 **Chapter 2 — Fractions and Decimals** **Next:** Chapter 2 Part 2 — **Decimal mastery, recurring decimals & approximation**

Send **NEXT**.

USER

next

ASSISTANT

CHAPTER 2 — FRACTIONS AND DECIMALS

PART 2 — DECIMAL MASTERY, RECURRING DECIMALS & APPROXIMATION

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

Now we complete the decimal side of Chapter 2.

1. DECIMAL PLACE VALUE

Example:

$\sqrt{47.382}$

This means:

$\sqrt{4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 8 \times \frac{1}{100} + 2 \times \frac{1}{1000}}$

So:

- 3 is tenths
- 8 is hundredths
- 2 is thousandths

2. COMPARING DECIMALS

Compare:

$\sqrt{0.7}$

and

$\sqrt{0.65}$

Write:

$\sqrt{0.70}$

and:

$\sqrt{0.65}$

Therefore:

$\sqrt{\boxed{0.70 > 0.65}}$

& bTRAP

More digits does not mean larger value.

For example:

$$\lfloor 0.9 > 0.8999 \rfloor$$

3. COMPARING NEGATIVE DECIMALS

Compare:

$$\lfloor -0.4 \rfloor$$

and

$$\lfloor -0.7 \rfloor$$

On number line:

$$\lfloor -0.4 > -0.7 \rfloor$$

because $\lfloor -0.4 \rfloor$ is closer to zero.

Rule

Among negative numbers, smaller magnitude may be larger.

4. ADDING DECIMALS

Align decimal points.

Example:

$$\lfloor 4.25 + 13.7 \rfloor$$

Write:

$$\lfloor 4.25 + 13.70 \rfloor$$

$$\lfloor =\boxed{17.95} \rfloor$$

5. SUBTRACTING DECIMALS

Example:

$$\lfloor 20 - 7.386 \rfloor$$

Write:

$$\lfloor 20.000 - 7.386 \rfloor$$

$$\lfloor =\boxed{12.614} \rfloor$$

6. MULTIPLYING DECIMALS

Example:

$$\lfloor 2.4 \times 0.35 \rfloor$$

Ignore decimals first:

$$\lfloor 24 \times 35 = 840 \rfloor$$

Total decimal places:

$$\backslash[1+2=3 \backslash]$$

So:

$$\backslash[0.840 \backslash]$$

Therefore:

$$\backslash[\boxed{0.84} \backslash]$$

7. DIVIDING DECIMALS

Example:

$$\backslash[6.75\div0.25 \backslash]$$

Multiply both by 100:

$$\backslash[675\div25 \backslash]$$

$$\backslash[=27 \backslash]$$

Answer:

$$\backslash[\boxed{27} \backslash]$$

Shortcut

Convert friendly decimals into fractions whenever possible.

$$\backslash[0.25=\frac{1}{4} \backslash]$$

So:

$$\backslash[6.75\div\frac{1}{4} = 6.75\times4 = 27 \backslash]$$

8. COMMON DECIMAL VALUES YOU SHOULD RECOGNIZE

MUST MEMORIZE

$$\backslash[0.5=\frac{1}{2} \backslash]$$

$$\backslash[0.25=\frac{1}{4} \backslash]$$

$$\backslash[0.75=\frac{3}{4} \backslash]$$

$$\backslash[0.2=\frac{1}{5} \backslash]$$

$$\backslash[0.4=\frac{2}{5} \backslash]$$

$$\backslash[0.6=\frac{3}{5} \backslash]$$

$$\backslash[0.8=\frac{4}{5} \backslash]$$

$$\backslash[0.125=\frac{1}{8} \backslash]$$

$$\backslash[0.375=\frac{3}{8} \backslash]$$

$$\backslash[0.625=\frac{5}{8} \backslash]$$

$$\backslash[0.875=\frac{7}{8} \backslash]$$

9. DECIMAL TO FRACTION

Example:

$$\backslash[0.48 \backslash]$$

$$\backslash[=\frac{48}{100} \backslash]$$

Simplify:

$$\backslash[=\frac{12}{25} \backslash]$$

So:

$$\backslash[\boxed{0.48=\frac{12}{25}} \backslash]$$

10. FRACTION TO DECIMAL

Example:

$$\backslash[\frac{7}{20} \backslash]$$

Convert denominator to 100:

$$\backslash[\frac{7}{20}=\frac{35}{100} \backslash]$$

So:

$$\backslash[\boxed{0.35} \backslash]$$

11. WHEN DENOMINATOR DOES NOT EASILY BECOME 10, 100, 1000

Example:

$$\backslash[\frac{7}{16} \backslash]$$

Since:

$$\backslash[\frac{1}{16}=0.0625 \backslash]$$

then:

$$\backslash[\frac{7}{16}=7\times 0.0625=0.4375 \backslash]$$

But in exam, long division may be faster if you don't know the decimal.

12. TERMINATING DECIMALS

A decimal stops.

Examples:

$$\backslash[0.25, \backslash 1.375, \backslash 2.5 \backslash]$$

As already learned, a reduced fraction terminates only if denominator is:

$$\backslash[2^m 5^n \backslash]$$

13. RECURRING DECIMALS

A recurring decimal repeats a pattern infinitely.

Example:

$\backslash[0.3333\ldots \backslash]$

Written:

$\backslash[0.\overline{3} \backslash]$

Another:

$\backslash[0.272727\ldots \backslash]$

Written:

$\backslash[0.\overline{27} \backslash]$

14. PURE RECURRING DECIMAL

A recurring decimal is **pure recurring** if repetition starts immediately after decimal point.

Example:

$\backslash[0.\overline{7} \backslash]$

Let:

$\backslash[x=0.7777\ldots \backslash]$

Then:

$\backslash[10x=7.7777\ldots \backslash]$

Subtract:

$\backslash[10x-x=7 \backslash]$

$\backslash[9x=7 \backslash]$

$\backslash[x=\boxed{\frac{7}{9}} \backslash]$

15. TWO-DIGIT RECURRING BLOCK

Convert:

$\backslash[0.\overline{36} \backslash]$

Let:

$\backslash[x=0.363636\ldots \backslash]$

Multiply by 100:

$\backslash[100x=36.363636\ldots \backslash]$

Subtract:

$\backslash[99x=36 \backslash]$

$\backslash[x=\frac{36}{99} \backslash]$

Simplify:

$$\boxed{\frac{4}{11}}$$

16. FAST RULE FOR PURE RECURRING DECIMALS

If one digit repeats:

$$0.\overline{a} = \frac{a}{9}$$

If two digits repeat:

$$0.\overline{ab} = \frac{ab}{99}$$

If three digits repeat:

$$0.\overline{abc} = \frac{abc}{999}$$

Example:

$$0.\overline{125} = \frac{125}{999}$$

Then simplify if possible.

17. MIXED RECURRING DECIMALS

Example:

$$0.1\overline{6}$$

means:

$$0.1666\ldots$$

Let:

$$x = 0.1666\ldots$$

Multiply by 10:

$$10x = 1.666\ldots$$

Multiply again by 10:

$$100x = 16.666\ldots$$

Subtract:

$$100x - 10x = 15$$

$$90x = 15$$

$$x = \frac{1}{6}$$

So:

$$\boxed{0.1\overline{6} = \frac{1}{6}}$$

18. FAST RULE FOR MIXED RECURRING DECIMAL

Example:

$$\left[0.\overline{2} \right]$$

Use:

$$\left[\frac{27-2}{90} \right]$$

Why?

- whole digits up to first repeat = 27
- non-repeating part = 2
- denominator: one 9 for repeating digit, one 0 for non-repeating digit

So:

$$\left[\frac{25}{90} = \frac{5}{18} \right]$$

General pattern

$$\left[\boxed{\frac{\text{all digits up to repeat} - \text{non-repeating digits}}{\text{9s followed by 0s}}} \right]$$

Use carefully.

19. EXAMPLE

Convert:

$$\left[0.\overline{12} \right]$$

That is:

$$\left[0.123333\ldots \right]$$

Numerator:

$$\left[123-12=111 \right]$$

Denominator:

one repeating digit !' 9 two non-repeating digits !' 00

So:

$$\left[\frac{111}{900} \right]$$

Simplify:

$$\left[\boxed{\frac{37}{300}} \right]$$

20. ROUNDING OFF

Suppose:

$$\left[7.486 \right]$$

Round to 2 decimal places.

Look at third decimal digit:

$$\left[6 \right]$$

Since:

$$\left[6 \geq 5 \right]$$

increase second decimal:

$\lfloor 7.48 \rightarrow 7.49 \rfloor$

Answer:

$\lfloor \boxed{7.49} \rfloor$

21. ROUNDING RULE

If next digit is:

$\lfloor 0,1,2,3,4 \rfloor$

leave current digit unchanged.

If next digit is:

$\lfloor 5,6,7,8,9 \rfloor$

increase current digit by 1.

22. EXAMPLE

Round:

$\lfloor 13.742 \rfloor$

to one decimal place.

Look at hundredths digit:

$\lfloor 4 < 5 \rfloor$

So:

$\lfloor \boxed{13.7} \rfloor$

23. APPROXIMATION FOR APTITUDE

Suppose:

$\lfloor 49.8 \times 19.9 \rfloor$

Approximate:

$\lfloor 50 \times 20 = 1000 \rfloor$

Actual answer must be close to 1000.

This is often enough if options are far apart.

24. WHEN NOT TO APPROXIMATE

Do not approximate aggressively if options are:

- 990
 - 992
-

- 994
- 996

Then exact or near-exact calculation may be necessary.

Exam principle

Approximation is useful only when error is smaller than option separation.

25. APPROXIMATION IN DIVISION

Suppose:

$$\left[\frac{599}{19.8} \right]$$

Approximate:

$$\left[\frac{600}{20} = 30 \right]$$

So answer is near:

$$\boxed{30}$$

26. APPROXIMATION IN PERCENTAGE-TYPE CALCULATION

Suppose:

$$\left[\frac{198}{502} \times 100 \right]$$

Approximate:

$$\left[\frac{200}{500} \times 100 = 40\% \right]$$

Actual is close to 40%.

This is especially useful in DI.

27. RANGE CHECK

Suppose:

$$\left[\frac{49}{101} \right]$$

Since:

$$\left[\frac{50}{100} = 0.5 \right]$$

and numerator is slightly smaller while denominator slightly larger:

$$\left[\frac{49}{101} < 0.5 \right]$$

This gives a quick qualitative answer.

28. DECIMAL PERCENTAGE CONNECTION

Every decimal can be converted to percentage by multiplying by 100.

Example:

$$\backslash[0.35=35\% \backslash]$$

$$\backslash[0.8=80\% \backslash]$$

$$\backslash[1.25=125\% \backslash]$$

This will be formalized in Chapter 3.

29. FRACTION PERCENTAGE CONNECTION

Example:

$$\backslash[\frac{3}{8} \backslash]$$

As decimal:

$$\backslash[0.375 \backslash]$$

As percentage:

$$\backslash[37.5\% \backslash]$$

So the three forms:

$$\backslash[\frac{3}{8}=0.375=37.5\% \backslash]$$

Aptitude becomes much faster when you move freely among these forms.

30. GATE-STYLE QUESTION 1

Which is largest?

A. $\backslash(0.72\backslash)$

B. $\backslash(\frac{7}{10}\backslash)$

C. $\backslash(71\%\backslash)$

D. $\backslash(\frac{18}{25}\backslash)$

Convert:

$$\backslash[0.72=72\% \backslash]$$

$$\backslash[\frac{7}{10}=70\% \backslash]$$

$$\backslash[71\%=71\% \backslash]$$

$$\backslash[\frac{18}{25}=72\% \backslash]$$

Largest:

$$\backslash[\boxed{0.72 \text{ and } \frac{18}{25}} \backslash]$$

They are equal.

31. GATE-STYLE QUESTION 2

Convert:

$$\left[0.\overline{45} \right]$$

to simplest fraction.

$$\left[0.\overline{45} = \frac{45}{99} \right]$$

Simplify by 9:

$$\left[\boxed{\frac{5}{11}} \right]$$

32. PLACEMENT-STYLE QUESTION

Evaluate:

$$\left[2.5 \times 0.4 \times 20 \right]$$

Best order:

$$\left[2.5 \times 0.4 = 1 \right]$$

Then:

$$\left[1 \times 20 = 20 \right]$$

Answer:

$$\left[\boxed{20} \right]$$

Speed lesson

Reorder multiplication to create easy products.

33. PLACEMENT-STYLE QUESTION 2

Evaluate:

$$\left[0.125 \times 64 \right]$$

Since:

$$\left[0.125 = \frac{1}{8} \right]$$

$$\left[64 \div 8 = 8 \right]$$

Answer:

$$\left[\boxed{8} \right]$$

This is much faster than decimal multiplication.

34. COMMON DECIMAL TRAPS

& pTRAP 1

$$\left[0.4 + 0.35 = 0.75 \right]$$

not 0.39.

& pTRAP 2

$\backslash[0.2\text{times}0.3=0.06\backslash]$

not 0.6.

& pTRAP 3

$\backslash[0.5\text{div}0.25=2\backslash]$

not 0.125.

& pTRAP 4

$\backslash[0.09<0.1\backslash]$

because:

$\backslash[0.09<0.10\backslash]$

& pTRAP 5

Recurring notation matters.

$\backslash[0.1\overline{6}\backslash]$

is different from:

$\backslash[0.\overline{16}\backslash]$

35. SPEED TABLE — MUST KNOW

Decimal	Fraction
0.1	$\backslash(1/10\backslash)$
0.2	$\backslash(1/5\backslash)$
0.25	$\backslash(1/4\backslash)$
0.333...	$\backslash(1/3\backslash)$
0.375	$\backslash(3/8\backslash)$
0.4	$\backslash(2/5\backslash)$
0.5	$\backslash(1/2\backslash)$
0.6	$\backslash(3/5\backslash)$
0.625	$\backslash(5/8\backslash)$
0.666...	$\backslash(2/3\backslash)$
0.75	$\backslash(3/4\backslash)$
0.8	$\backslash(4/5\backslash)$
0.875	$\backslash(7/8\backslash)$

36. NORMAL METHOD VS FAST METHOD

Question:

$$\sqrt[0.375]{\times 240}$$

Normal method

$$\sqrt[0.375]{\times 240}$$

long multiplication.

Fast method

$$\sqrt[0.375]{\frac{38}{}}$$

So:

$$\sqrt[\frac{38}{\times 240}]$$

$$\sqrt[240/8=30]$$

$$\sqrt[30 \times 3=90]$$

Answer:

$$\sqrt[\boxed{90}]$$

Exam choice

Use fraction conversion.

37. QUESTION READING

If a question says:

0.4 of a quantity

That means:

$$\sqrt[0.4]{\times \text{quantity}}$$

Since:

$$\sqrt[0.4]{\frac{25}{}}$$

you can often calculate as:

$$\sqrt[\frac{25}{\times \text{quantity}}]$$

38. 5-MINUTE REVISION

Remember:

1. Align decimal points for addition/subtraction.
2. Count decimal places in multiplication.
3. Make divisor whole in decimal division.

4. Convert familiar decimals into fractions.

5. Pure recurring:

$$\backslash[0.\overline{ab}=\frac{ab}{99}\backslash]$$

6. Mixed recurring: subtract non-repeating part.

7. Round using next digit.

8. Approximate only when answer choices allow it.

9. Convert freely:

$$\backslash[\text{fraction}\leftrightharpoons\text{decimal}\leftrightharpoons\%\backslash]$$

39. FORMULA / RULE SHEET

Decimal to fraction

$$\backslash[0.ab=\frac{ab}{100}\backslash]$$

then simplify.

Pure recurring

$$\backslash[0.\overline{a}=\frac{a}{9}\backslash]$$

$$\backslash[0.\overline{ab}=\frac{ab}{99}\backslash]$$

$$\backslash[0.\overline{abc}=\frac{abc}{999}\backslash]$$

Mixed recurring

$$\backslash[\frac{\text{all relevant digits}-\text{non-repeating part}}{\text{9s for repeats, 0s for non-repeats}}\backslash]$$

PRACTICE — LEVEL 0

Q1

Convert to fraction:

$$\backslash[0.45\backslash]$$

Q2

Convert to decimal:

$$\backslash[\frac{7}{8}\backslash]$$

Q3

Evaluate:

$$\backslash[0.25\times 48\backslash]$$

Q4

Evaluate:

$$\backslash[4.8\div 0.6\backslash]$$

Q5

Which is larger?

$\sqrt[0.67]{\text{or } 0.7}$

LEVEL 1

Q6

Convert:

$0.\overline{27}$

to fraction.

Q7

Convert:

$0.1\overline{6}$

to fraction.

Q8

Round:

18.376

to 2 decimal places.

Q9

Evaluate mentally:

0.625×80

Q10

Approximate:

$\frac{401}{19.9}$

LEVEL 2

Q11

Convert:

$0.12\overline{5}$

to fraction.

Q12

Which is greater?

0.833

or

$\frac{5}{6}$

Q13

Evaluate:

$$\lfloor 2.4 \times 1.25 \times 8 \rfloor$$

using smart grouping.

Q14

Without exact division, estimate:

$$\lfloor \frac{1002}{33.4} \rfloor$$

Q15

Which is largest?

$$\lfloor 0.875, \quad \frac{78}{87.4}, \quad 87.4\% \rfloor$$

ANSWERS

1.

$$\lfloor \frac{9}{20} \rfloor$$

2.

$$\lfloor 0.875 \rfloor$$

3.

$$\lfloor 12 \rfloor$$

4.

$$\lfloor 8 \rfloor$$

5.

$$\lfloor 0.7 \rfloor$$

6.

$$\lfloor \frac{3}{11} \rfloor$$

7.

$$\lfloor \frac{16}{1} \rfloor$$

8.

$$\lfloor 18.38 \rfloor$$

9.

$$\lfloor 50 \rfloor$$

10. Approximately $\lfloor 20 \rfloor$

11.

$$\lfloor \frac{113}{900} \rfloor$$

12.

$$\lfloor \frac{56}{1} \rfloor$$

because:

$$\lfloor \frac{56}{1} \approx 0.83333... \rfloor$$

13.

$\sqrt[3]{24}$

14. Approximately $\sqrt[3]{30}$

15.

$0.875 = \frac{7}{8} = 87.5\%$

so first two are equal and largest.

CHAPTER 2 — FINAL CHECK

We have now covered:

- fraction types '
- equivalent fractions '
- simplification '
- comparison methods '
- fraction operations '
- mixed fractions '
- reciprocals '
- complex fractions '
- decimal place value '
- decimal operations '
- fraction-decimal conversion '
- terminating decimals '
- recurring decimals '
- mixed recurring decimals '
- rounding '
- approximation '
- decimal comparison traps '
- fraction-based and decimal-based exam questions '
- speed techniques '

Therefore:

' CHAPTER 2 COMPLETE

Progress

Completed: 2/76 **Current:** ' Chapter 2 — Fractions and Decimals **Next:** ~~0=3~~ Chapter 3 — Fraction–Decimal–Percentage Conversion

Chapter 3 is extremely important because it creates the conversion speed needed for **percentages, profit/loss, probability, DI, ratios and placement calculations.**

Send **NEXT**.

USER

next

ASSISTANT

CHAPTER 3 — FRACTION–DECIMAL–PERCENTAGE CONVERSION

PART 1 — COMPLETE CONVERSION MASTERY

Status: $\emptyset = \text{Nil}$ **PROGRESS GATE:** High **Placements:** High **Priority:** $\emptyset = \text{Nil}$ **Per 1**

This chapter is one of the highest-value calculation chapters in the entire aptitude course.

You should eventually see:

$$\left[\frac{3}{8} \right]$$

and immediately think:

$$\left[0.375 = 37.5\% \right]$$

without calculation.

That speed directly improves:

- percentages
- profit/loss
- averages
- probability
- ratios
- DI
- mixtures
- interest
- placement aptitude

1. THE THREE FORMS OF THE SAME NUMBER

A quantity can be represented as:

Fraction

$$\left[\frac{3}{4} \right]$$

Decimal

$$\left[0.75 \right]$$

Percentage

$$\left[75\% \right]$$

Therefore:

$$\left[\boxed{\frac{3}{4} = 0.75 = 75\%} \right]$$

They are not three different values.

They are three different ways of writing the same value.

2. WHAT DOES PERCENT MEAN?

Percent means:

per hundred

Therefore:

$$x\% = \frac{x}{100}$$

Example:

$$25\% = \frac{25}{100}$$

$$= \frac{1}{4}$$

So:

$$25\% = 0.25 = \frac{1}{4}$$

3. PERCENTAGE TO DECIMAL

Divide by 100.

Or move decimal point two places left.

Example:

$$75\% = 0.75$$

$$8\% = 0.08$$

$$125\% = 1.25$$

$$0.5\% = 0.005$$

4. DECIMAL TO PERCENTAGE

Multiply by 100.

Move decimal two places right.

Example:

$$0.4 = 40\%$$

$$0.75 = 75\%$$

$$1.25 = 125\%$$

$$0.007 = 0.7\%$$

& ⚠️TRAP

$$0.5 = 50\%$$

but:

$$\left[0.5\% = 0.005 \right]$$

These are very different.

5. FRACTION TO PERCENTAGE

Multiply by 100.

$$\left[\frac{ab}{\times 100} \% \right]$$

Example:

$$\left[\frac{34}{\times 100} = 75\% \right]$$

6. FAST FRACTION TO PERCENTAGE

Whenever possible, make denominator 100.

Example:

$$\left[\frac{7}{20} \right]$$

Multiply top and bottom by 5:

$$\left[\frac{35}{100} \right]$$

Therefore:

$$\left[\boxed{35\%} \right]$$

7. ANOTHER EXAMPLE

$$\left[\frac{9}{25} \right]$$

Since:

$$\left[25 \times 4 = 100 \right]$$

$$\left[\frac{9}{25} = \frac{36}{100} \right]$$

Therefore:

$$\left[\boxed{36\%} \right]$$

8. PERCENTAGE TO FRACTION

Write over 100 and simplify.

Example:

$$\left[60\% = \frac{60}{100} = \frac{3}{5} \right]$$

Therefore:

$$\left[\boxed{60\% = \frac{3}{5}} \right]$$

9. DECIMAL TO FRACTION TO PERCENT

Example:

0.625

As fraction:

$0.625 = \frac{5}{8}$

As percentage:

$0.625 \times 100 = 62.5\%$

Therefore:

$\boxed{0.625 = \frac{5}{8} = 62.5\%}$

10. MUST-MEMORIZE MASTER TABLE

This table is extremely important.

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.333...	33.33...%
$\frac{2}{3}$	0.666...	66.67%
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{6}$	0.1667	16.67%
$\frac{5}{6}$	0.8333	83.33%
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%
$\frac{5}{8}$	0.625	62.5%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{10}$	0.1	10%
$\frac{1}{20}$	0.05	5%
$\frac{1}{25}$	0.04	4%
$\frac{1}{40}$	0.025	2.5%

Fraction	Decimal	Percentage
$\frac{1}{50}$	0.02	2%

MUST MEMORIZE

At minimum, memorize this entire table gradually.

11. HIGH-VALUE FRACTIONS BASED ON 1/3

Since:

$$\frac{1}{3} = 33.33\%$$

then:

$$\frac{2}{3} = 66.67\%$$

Similarly:

$$\frac{1}{6} = 16.67\%$$

$$\frac{5}{6} = 83.33\%$$

12. HIGH-VALUE FRACTIONS BASED ON 1/8

$$\frac{1}{8} = 12.5\%$$

Therefore:

$$\frac{2}{8} = \frac{1}{4} = 25\%$$

$$\frac{3}{8} = 37.5\%$$

$$\frac{4}{8} = 50\%$$

$$\frac{5}{8} = 62.5\%$$

$$\frac{6}{8} = 75\%$$

$$\frac{7}{8} = 87.5\%$$

13. HIGH-VALUE FRACTIONS BASED ON 1/12

Useful for aptitude speed:

$$\frac{1}{12} = 8.33\%$$

Therefore:

$$\frac{2}{12} = \frac{1}{6} = 16.67\%$$

$$\frac{3}{12} = \frac{1}{4} = 25\%$$

$$\frac{4}{12} = \frac{1}{3} = 33.33\%$$

$$\frac{5}{12} = 41.67\%$$

$$\frac{7}{12} = 58.33\%$$

$$\left[\frac{11}{12} = 91.67\% \right]$$

These are **Good to Memorize**.

14. HIGH-VALUE FRACTIONS BASED ON 1/16

Since:

$$\left[\frac{1}{16} = 6.25\% \right]$$

we get:

$$\left[\frac{3}{16} = 18.75\% \right]$$

$$\left[\frac{5}{16} = 31.25\% \right]$$

$$\left[\frac{7}{16} = 43.75\% \right]$$

$$\left[\frac{9}{16} = 56.25\% \right]$$

$$\left[\frac{15}{16} = 93.75\% \right]$$

Useful in placement questions and DI.

15. QUICK METHOD USING KNOWN FRACTIONS

Suppose you need:

$$\left[\frac{7}{20} \right]$$

You know:

$$\left[\frac{1}{20} = 5\% \right]$$

So:

$$\left[\frac{7}{20} = 7 \times 5\% = 35\% \right]$$

Much faster.

16. ANOTHER EXAMPLE

Convert:

$$\left[\frac{13}{25} \right]$$

Since:

$$\left[\frac{1}{25} = 4\% \right]$$

$$\left[\frac{13}{25} = 13 \times 4\% \right]$$

$$\left[= \boxed{52\%} \right]$$

17. ANOTHER FAST EXAMPLE

Convert:

$$\left[\frac{17}{40} \right]$$

Since:

$$\left[\frac{1}{40} = 2.5\% \right]$$

$$\left[17 \times 2.5\% = 42.5\% \right]$$

Therefore:

$$\left[\boxed{42.5\%} \right]$$

18. PERCENTAGE ABOVE 100%

Very important.

$$\left[125\% \right]$$

means:

$$\left[\frac{125}{100} = 1.25 = \frac{5}{4} \right]$$

So:

$$\left[\boxed{125\% = \frac{5}{4}} \right]$$

A percentage can absolutely exceed 100%.

19. PERCENTAGE BELOW 1%

Example:

$$\left[0.25\% \right]$$

$$\left[= \frac{0.25}{100} \right]$$

$$\left[= 0.0025 \right]$$

As fraction:

$$\left[0.0025 = \frac{1}{400} \right]$$

Therefore:

$$\left[\boxed{0.25\% = \frac{1}{400}} \right]$$

20. IMPORTANT DISTINCTION

Compare:

$$\left[25\% \right]$$

and:

$$\left[0.25\% \right]$$

First:

$$\left[25\% = 0.25 \right]$$

Second:

$$\left[0.25\% = 0.0025 \right]$$

Thus:

$$\left[25\% \right]$$

is 100 times:

$$\left[0.25\% \right]$$

21. FINDING PERCENT OF A NUMBER

Formula:

$$\left[x\% \text{ of } N = \frac{x}{100}N \right]$$

Example:

$$\left[20\% \text{ of } 450 \right]$$

Since:

$$\left[20\% = \frac{15}{100} \right]$$

$$\left[450/5 = 90 \right]$$

Answer:

$$\left[\boxed{90} \right]$$

22. FAST METHOD — CONVERT PERCENT TO FRACTION

Question:

$$\left[12.5\% \text{ of } 320 \right]$$

Because:

$$\left[12.5\% = \frac{18}{100} \right]$$

$$\left[320/8 = 40 \right]$$

Answer:

$$\left[\boxed{40} \right]$$

Do not calculate:

$$\left[320 \times 12.5/100 \right]$$

unless necessary.

23. FAST METHOD — BREAK PERCENTAGE

Find:

$$\left[17\% \text{ of } 300 \right]$$

Break:

$$\left[17\% = 10\% + 5\% + 2\% \right]$$

10%:

$\backslash[30 \backslash]$

5%:

$\backslash[15 \backslash]$

2%:

$\backslash[6 \backslash]$

Total:

$\backslash[30+15+6=51 \backslash]$

Answer:

$\backslash[\boxed{51} \backslash]$

24. 1% METHOD

Find:

$\backslash[37\%\text{ of }240 \backslash]$

1%:

$\backslash[2.4 \backslash]$

37%:

$\backslash[2.4\times37 \backslash]$

Possible, but not always fastest.

Better:

$\backslash[40\%-3\% \backslash]$

40% of 240:

$\backslash[96 \backslash]$

3%:

$\backslash[7.2 \backslash]$

So:

$\backslash[96-7.2 = \boxed{88.8} \backslash]$

25. VERY USEFUL PROPERTY

$\backslash[\boxed{x\%\text{ of }y=y\%\text{ of }x} \backslash]$

Example:

$\backslash[8\%\text{ of }25 \backslash]$

Switch:

$\backslash[25\%\text{ of }8 \backslash]$

$\backslash[=\frac{14}{1}\times8 \backslash]$

$\backslash[=\boxed{2} \backslash]$

Much easier.

26. WHY THAT PROPERTY WORKS

$x\%$ of $y = \frac{x}{100}y$

$= \frac{xy}{100}$

Similarly:

$y\%$ of $x = \frac{y}{100}x$

Same value.

27. EXAMPLE

Find:

16% of 25

Instead:

25% of 16

$= \frac{14}{100} \times 16$

$= \boxed{4}$

28. FIND WHAT PERCENT ONE NUMBER IS OF ANOTHER

Example:

18 is what percent of 72?

Formula:

$\frac{18}{72} \times 100$

$= \frac{14}{100} \times 100$

$= \boxed{25\%}$

29. QUESTION READING

If question says:

A is what percent of B?

Always calculate:

$\boxed{\frac{AB}{100}}$

& ⚠TRAP

Denominator is the number after “of”.

30. EXAMPLE

40 is what percent of 160?

$$\left[\frac{40}{160} \times 100 \right]$$

$$\left[= \frac{14}{100} \times 100 \right]$$

$$\left[= \boxed{25\%} \right]$$

But:

160 is what percent of 40?

$$\left[\frac{160}{40} \times 100 = 400\% \right]$$

Order matters.

31. CONVERT RATIO TO PERCENT

Suppose:

$$\left[A:B=3:5 \right]$$

A as a percentage of B:

$$\left[\frac{3}{5} \times 100 = 60\% \right]$$

So:

$$\left[\boxed{A=60\% \text{ of } B} \right]$$

32. PART AS PERCENTAGE OF TOTAL

Suppose boys:girls:

$$\left[3:2 \right]$$

Total parts:

$$\left[3+2=5 \right]$$

Boys percentage:

$$\left[\frac{3}{5} \times 100 = 60\% \right]$$

Girls percentage:

$$\left[\frac{2}{5} \times 100 = 40\% \right]$$

& Important

“Boys as % of girls” and “boys as % of total” are different.

33. DECIMAL PERCENTAGES IN QUESTIONS

Example:

Find:

$$\left[2.5\% \text{ of } 800 \right]$$

Since:

$$2.5\% = \frac{1}{40}$$

$$800/40 = 20$$

Answer:

$$\boxed{20}$$

34. ANOTHER USEFUL CONVERSION

$$6.25\% = \frac{1}{16}$$

Therefore:

$$6.25\% \text{ of } 480 = 480/16 = \boxed{30}$$

35. HIGH-VALUE PERCENTAGE–FRACTION TABLE

MUST MEMORIZE

Percentage	Fraction
100%	$\frac{1}{1}$
75%	$\frac{3}{4}$
66.67%	$\frac{2}{3}$
62.5%	$\frac{5}{8}$
50%	$\frac{1}{2}$
40%	$\frac{2}{5}$
37.5%	$\frac{3}{8}$
33.33%	$\frac{1}{3}$
25%	$\frac{1}{4}$
20%	$\frac{1}{5}$
16.67%	$\frac{1}{6}$
12.5%	$\frac{1}{8}$
10%	$\frac{1}{10}$
8.33%	$\frac{1}{12}$
6.25%	$\frac{1}{16}$
5%	$\frac{1}{20}$
4%	$\frac{1}{25}$
2.5%	$\frac{1}{40}$

Percentage	Fraction
2%	$\frac{1}{50}$
1%	$\frac{1}{100}$

36. IMPORTANT PLACEMENT SPEED TRICK

Find:

$$37.5\% \text{ of } 480$$

Recognize:

$$37.5\% = \frac{3}{8}$$

So:

$$480 \times \frac{3}{8} = 180$$

$$180 \times 3 = 540$$

Answer:

$$\boxed{540}$$

37. ANOTHER SPEED QUESTION

Find:

$$62.5\% \text{ of } 144$$

$$62.5\% = \frac{5}{8}$$

$$144 \times \frac{5}{8} = 90$$

$$90 \times 5 = 450$$

Answer:

$$\boxed{450}$$

38. GATE-STYLE QUESTION 1

If:

$$0.375x = 90$$

find x .

Recognize:

$$0.375 = \frac{3}{8}$$

So:

$$\frac{3}{8}x = 90$$

$$x = 90 \times \frac{8}{3} = 240$$

$$\lfloor =30\times 8 \rfloor$$

$$\lfloor =\boxed{240} \rfloor$$

39. GATE-STYLE QUESTION 2

Which is greatest?

A. $\lfloor (0.42\rfloor$

B. $\lfloor (41\%\rfloor$

C. $\lfloor (\frac{25}\rfloor$

D. $\lfloor (\frac{17}{40}\rfloor$

Convert:

$$\lfloor 0.42=42\% \rfloor$$

$$\lfloor 41\%=41\% \rfloor$$

$$\lfloor \frac{25}{40}=40\% \rfloor$$

$$\lfloor \frac{17}{40}=42.5\% \rfloor$$

Therefore:

$$\lfloor \boxed{\frac{17}{40}} \rfloor$$

40. PLACEMENT-STYLE QUESTION

What is:

$$\lfloor 16\frac{23}{40}\text{ of }420? \rfloor$$

Recognize:

$$\lfloor 16\frac{23}{40}=16.666...\%=\frac{16}{3} \rfloor$$

Therefore:

$$\lfloor 420/6=70 \rfloor$$

Answer:

$$\lfloor \boxed{70} \rfloor$$

41. PLACEMENT-STYLE QUESTION

What is:

$$\lfloor 33\frac{13}{270}\text{ of }270? \rfloor$$

$$\lfloor 33\frac{13}{270}=\frac{13}{3} \rfloor$$

Therefore:

$$\lfloor 270/3=90 \rfloor$$

Answer:

$\boxed{90}$

42. PERCENTAGE AS A MULTIPLIER

This concept becomes extremely important later.

If a quantity becomes 120% of original:

$$120\% = 1.2$$

So:

$$\text{new} = 1.2 \times \text{old}$$

If quantity becomes 80%:

$$80\% = 0.8$$

So:

$$\text{new} = 0.8 \times \text{old}$$

This will make percentage increase/decrease very easy.

43. MULTIPLIER TABLE

Percentage of original	Multiplier
200%	2
150%	1.5
125%	1.25
120%	1.2
110%	1.1
100%	1
90%	0.9
80%	0.8
75%	0.75
50%	0.5
25%	0.25

This idea becomes central in Chapter 15.

44. COMMON TRAPS

& pTRAP 1

$$5\% = 0.05$$

not 0.5.

& pTRAP 2

$\backslash[0.5=50\% \backslash]$

but:

$\backslash[0.5\%=0.005 \backslash]$

& pTRAP 3

“A is what percent of B?”

Use:

$\backslash[\frac{AB}{\times 100} \backslash]$

not the reverse.

& pTRAP 4

$\backslash[150\% \backslash]$

does not mean 0.15.

It means:

$\backslash[1.5 \backslash]$

& pTRAP 5

Boys:girls $\backslash(=3:2\backslash)$.

Boys are:

$\backslash[150\% \backslash]$

of girls,

but only:

$\backslash[60\% \backslash]$

of the total.

45. MEMORY TRICKS

Percent !' decimal

Move decimal 2 places left.

Decimal !' percent

Move decimal 2 places right.

Percent !' fraction

Put over 100 and simplify.

“A is what % of B?”

A over B $\times$ 100.

“x% of y”

Switch if easier:

$x\% \text{ of } y = y\% \text{ of } x$

46. QUICK REVISION BOX

$x\% = \frac{x}{100}$

$\text{decimal} \rightarrow \times 100$

$\% \rightarrow \text{decimal} : \div 100$

$\text{fraction} \rightarrow \% : \frac{ab}{100}$

A $\text{as } \% \text{ of } B : \frac{AB}{100}$

Useful:

$12.5\% = \frac{1}{8}$

$16.67\% = \frac{1}{6}$

$20\% = \frac{1}{5}$

$25\% = \frac{1}{4}$

$33.33\% = \frac{1}{3}$

$37.5\% = \frac{3}{8}$

$50\% = \frac{1}{2}$

$62.5\% = \frac{5}{8}$

$75\% = \frac{3}{4}$

PRACTICE — LEVEL 0

Q1

Convert to percentage:

$\frac{7}{20}$

Q2

Convert to fraction:

45%

Q3

Convert to decimal:

$\backslash[12.5\% \backslash]$

Q4

Convert to percentage:

$\backslash[0.625 \backslash]$

Q5

Find:

$\backslash[25\%\text{ of }360 \backslash]$

LEVEL 1

Q6

Find:

$\backslash[37.5\%\text{ of }240 \backslash]$

Q7

Find:

$\backslash[6.25\%\text{ of }640 \backslash]$

Q8

45 is what percent of 180?

Q9

If:

$\backslash[A:B=7:10 \backslash]$

A is what percentage of B?

Q10

Convert:

$\backslash[\frac{11}{40} \backslash]$

to percentage.

LEVEL 2

Q11

Find quickly:

$\backslash[16\frac{23}{40}\text{ of }540 \backslash]$

Q12

Find:

$\frac{8}{25}$ of 25 %

using the switch property.

Q13

If boys:girls:

$5:3$

what percentage of the class are girls?

Q14

Which is largest?

0.375 , 38% , $\frac{38}{50}$, $\frac{19}{50}$

Q15

If:

$0.625x = 125$

find x .

ANSWERS

1. 35%

2.

$\frac{9}{20}$

3.

0.125

4.

62.5%

5.

90

6.

90

7.

40

8.

25%

9.

70%

10.

27.5%

11.

[90]

12.

[2]

13.

[37.5\%]

14.

[38\%=\frac{19}{50}]

Both are equal and largest.

15.

[200]

CHAPTER 3 STATUS

We have completed:

- fraction !' decimal '
- fraction !' percentage '
- percentage !' fraction '
- percentage !' decimal '
- decimal !' percentage '
- common conversion table '
- percent of a number '
- what-percent-of questions '
- ratio-to-percentage basics '
- percent multipliers '
- mental conversion shortcuts '
- GATE/placement applications '

But Chapter 3 is **not complete yet**.

The next lesson will focus on **conversion speed drills, benchmark percentages, approximation, reverse conversion, tricky comparison questions, and mental-calculation patterns** so these conversions become automatic.

Progress

Completed: 2/76 **Current:** 0=Chapter 3 — Fraction–Decimal–Percentage Conversion **Next:** Chapter 3 Part 2 — **Speed mastery & advanced conversion tricks**

Send **NEXT**.

USER

next

CHAPTER 3 — FRACTION–DECIMAL–PERCENTAGE CONVERSION

PART 2 — SPEED MASTERY, BENCHMARKS & ADVANCED CONVERSION TRICKS

Status: $\emptyset$ = No PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

This lesson is not about learning new notation. It is about becoming **fast enough that conversions happen almost automatically**.

Your goal is to see:

$$\left[\frac{7}{8} \right]$$

and instantly think:

$$\left[87.5\% \right]$$

or see:

$$\left[62.5\% \right]$$

and instantly think:

$$\left[\frac{5}{8} \right]$$

without written calculation.

1. WHY SPEED MATTERS

Consider:

Find (37.5%) of 640.

Slow method:

$$\left[640 \times \frac{37.5}{100} \right]$$

Fast method:

$$\left[37.5\% = \frac{3}{8} \right]$$

$$\left[640 \times \frac{3}{8} = 80 \times 3 = \boxed{240} \right]$$

The concept is the same.

The difference is exam speed.

2. THE BENCHMARK METHOD

Instead of treating every percentage as new, build answers around easy percentages.

Core benchmarks

$$\left[50\% = \frac{1}{2} \right]$$

$$\left[25\% = \frac{1}{4} \right]$$

$$\left[10\% = \frac{1}{10} \right]$$

$$\left[5\% = \frac{1}{20} \right]$$

$$\left[1\% = \frac{1}{100} \right]$$

From these, many percentages can be built mentally.

3. FINDING 15%

$$\left[15\% = 10\% + 5\% \right]$$

Example:

15% of 360:

10%:

$$\left[36 \right]$$

5%:

$$\left[18 \right]$$

Total:

$$\left[\boxed{54} \right]$$

4. FINDING 35%

$$\left[35\% = 25\% + 10\% \right]$$

Example:

35% of 400:

$$\left[100 + 40 = \boxed{140} \right]$$

5. FINDING 45%

$$\left[45\% = 50\% - 5\% \right]$$

Example:

45% of 240:

50%:

$$\left[120 \right]$$

5%:

$$\left[12 \right]$$

Therefore:

$$\left[120 - 12 = \boxed{108} \right]$$

This is usually faster than calculating 45 directly.

6. FINDING 55%

$$\backslash[55\%=50\%+5\% \backslash]$$

Example:

55% of 360:

$$\backslash[180+18 = \boxed{198} \backslash]$$

7. FINDING 65%

One way:

$$\backslash[65\%=50\%+10\%+5\% \backslash]$$

Example:

65% of 200:

$$\backslash[100+20+10 = \boxed{130} \backslash]$$

8. FINDING 75%

Never calculate from scratch.

$$\backslash[75\%=\frac{3}{4} \backslash]$$

Example:

75% of 520:

$$\backslash[520/4=130 \backslash]$$

$$\backslash[130\times 3=390 \backslash]$$

Answer:

$$\backslash[\boxed{390} \backslash]$$

9. FINDING 85%

$$\backslash[85\%=100\%-15\% \backslash]$$

Example:

85% of 600:

15%:

$$\backslash[90 \backslash]$$

Therefore:

$$\backslash[600-90 = \boxed{510} \backslash]$$

10. FINDING 95%

$$\backslash[95\%=100\%-5\% \backslash]$$

Example:

95% of 840:

5%:

$\backslash[42 \backslash]$

So:

$\backslash[840-42 = \boxed{798} \backslash]$

11. FINDING 99%

$\backslash[99\%=100\%-1\% \backslash]$

Example:

99% of 700:

$\backslash[700-7 = \boxed{693} \backslash]$

12. FINDING 101%

$\backslash[101\%=100\%+1\% \backslash]$

Example:

101% of 900:

$\backslash[900+9 = \boxed{909} \backslash]$

13. FINDING 110%

$\backslash[110\%=100\%+10\% \backslash]$

Example:

110% of 250:

$\backslash[250+25 = \boxed{275} \backslash]$

14. FINDING 125%

Recognize immediately:

$\backslash[125\%=\frac{5}{4} \backslash]$

Example:

125% of 320:

$\backslash[320\times\frac{5}{4} \backslash]$

$\backslash[80\times5 = \boxed{400} \backslash]$

15. FINDING 150%

$$\left[150\% = \frac{3}{2} \right]$$

Example:

150% of 240:

$$\left[240 \times \frac{3}{2} = 120 \times 3 = \boxed{360} \right]$$

16. FINDING 200%

$$\left[200\% = 2 \right]$$

So 200% means double.

Example:

200% of 65:

$$\left[\boxed{130} \right]$$

17. FINDING 250%

$$\left[250\% = 2.5 = \frac{5}{2} \right]$$

Example:

250% of 80:

$$\left[80 \times \frac{5}{2} = 40 \times 5 = \boxed{200} \right]$$

18. COMMON BENCHMARK TABLE

MUST MEMORIZE

Percentage	Fraction/Multiplier
1%	$\left(\frac{1}{100} \right)$
2%	$\left(\frac{1}{50} \right)$
2.5%	$\left(\frac{1}{40} \right)$
4%	$\left(\frac{1}{25} \right)$
5%	$\left(\frac{1}{20} \right)$
6.25%	$\left(\frac{1}{16} \right)$
8.33%	$\left(\frac{1}{12} \right)$
10%	$\left(\frac{1}{10} \right)$
12.5%	$\left(\frac{1}{8} \right)$
16.67%	$\left(\frac{1}{6} \right)$

Percentage	Fraction/Multiplier
20%	$\frac{1}{5}$
25%	$\frac{1}{4}$
33.33%	$\frac{1}{3}$
37.5%	$\frac{3}{8}$
40%	$\frac{2}{5}$
50%	$\frac{1}{2}$
60%	$\frac{3}{5}$
62.5%	$\frac{5}{8}$
66.67%	$\frac{2}{3}$
75%	$\frac{3}{4}$
80%	$\frac{4}{5}$
83.33%	$\frac{5}{6}$
87.5%	$\frac{7}{8}$
100%	1
125%	$\frac{5}{4}$
150%	$\frac{3}{2}$
200%	2

19. HOW TO GENERATE PERCENTAGES FROM UNIT FRACTIONS

If:

$$\frac{1}{n}$$

then percentage is:

$$\frac{100}{n}\%$$

Example:

$$\frac{1}{7}$$

$$=\frac{100}{7}\%$$

$$\approx 14.2857\%$$

So:

$$\boxed{\frac{1}{7} \approx 14.29\%}$$

20. USEFUL SEVENTHS

Good to recognize approximately:

$$\left[\frac{1}{7} \approx 14.29\% \right]$$

$$\left[\frac{2}{7} \approx 28.57\% \right]$$

$$\left[\frac{3}{7} \approx 42.86\% \right]$$

$$\left[\frac{4}{7} \approx 57.14\% \right]$$

$$\left[\frac{5}{7} \approx 71.43\% \right]$$

$$\left[\frac{6}{7} \approx 85.71\% \right]$$

GOOD TO MEMORIZE

Not as essential as halves, thirds, quarters, fifths and eighths.

21. USEFUL NINTHS

These are very easy.

$$\left[\frac{1}{9} = 11.11\% \right]$$

$$\left[\frac{2}{9} = 22.22\% \right]$$

$$\left[\frac{3}{9} = 33.33\% \right]$$

$$\left[\frac{4}{9} = 44.44\% \right]$$

$$\left[\frac{5}{9} = 55.56\% \right]$$

$$\left[\frac{6}{9} = 66.67\% \right]$$

$$\left[\frac{7}{9} = 77.78\% \right]$$

$$\left[\frac{8}{9} = 88.89\% \right]$$

22. USEFUL ELEVENTHS

$$\left[\frac{1}{11} \approx 9.09\% \right]$$

Therefore:

$$\left[\frac{2}{11} \approx 18.18\% \right]$$

$$\left[\frac{3}{11} \approx 27.27\% \right]$$

$$\left[\frac{4}{11} \approx 36.36\% \right]$$

$$\left[\frac{5}{11} \approx 45.45\% \right]$$

Pattern continues.

This is useful in DI and comparison.

23. PERCENTAGES OF 360 — QUICK REFERENCE

Because 360 appears frequently.

$$\left[10\% = 36 \right]$$

$$\left[5\% = 18 \right]$$

$$\left[25\% = 90 \right]$$

$$\left[50\% = 180 \right]$$

$$\left[75\% = 270 \right]$$

$$\left[12.5\% = 45 \right]$$

$$\left[3\frac{1}{3}\% = 120 \right]$$

$$\left[16\frac{2}{3}\% = 60 \right]$$

Do not memorize this table specifically; learn to derive instantly.

24. REVERSE CONVERSION — FIND TOTAL FROM A PERCENT

Question:

25% of a number is 80. Find number.

Since:

$$\left[25\% = \frac{1}{4} \right]$$

If one-fourth is 80:

$$\left[\text{whole} = 80 \times 4 \right]$$

$$\left[= \boxed{320} \right]$$

25. ANOTHER REVERSE EXAMPLE

62.5% of a number is 150.

$$\left[62.5\% = \frac{5}{8} \right]$$

So:

$$\left[\frac{5}{8}x = 150 \right]$$

$$\left[x = 150 \times \frac{8}{5} \right]$$

$$\left[30 \times 8 = \boxed{240} \right]$$

26. UNITARY THINKING

Suppose:

40% of a number = 72.

40%:

$$\left[= \frac{2}{5} \right]$$

So:

2 parts = 72

1 part:

$$\frac{1}{36}$$

5 parts:

$$\frac{1}{180}$$

Answer:

$$\boxed{180}$$

This is often faster than equation writing.

27. FRACTION-BASED REVERSE THINKING

If:

$$\frac{3}{7}$$

of a number is 60.

Then:

$$3 \text{ parts} = 60.$$

1 part:

$$\frac{1}{20}$$

7 parts:

$$\frac{1}{140}$$

Answer:

$$\boxed{140}$$

28. COMPARE PERCENTAGES WITHOUT CONVERSION

Which is larger?

$$\frac{5}{8}$$

or:

$$61\%$$

You know:

$$\frac{5}{8} = 62.5\%$$

Therefore:

$$\boxed{\frac{5}{8}}$$

29. COMPARISON USING BENCHMARK 50%

Which is greater?

$$\frac{7}{15}$$

or 50%?

Compare:

$$\left[2 \times 7 = 14 \right]$$

with denominator 15.

Since:

$$\left[14 < 15 \right]$$

$$\left[\frac{7}{15} < \frac{12}{15} \right]$$

So:

$$\left[\boxed{50\% \text{ is greater}} \right]$$

No decimal conversion needed.

30. COMPARISON USING BENCHMARK 25%

To determine whether:

$$\left[\frac{5}{18} \right]$$

is greater than 25%.

Since:

$$\left[25\% = \frac{1}{4} \right]$$

Compare:

$$\left[\frac{5}{18} \right]$$

and:

$$\left[\frac{1}{4} \right]$$

Cross-multiply:

$$\left[5 \times 4 = 20 \right]$$

$$\left[18 \times 1 = 18 \right]$$

So:

$$\left[\frac{5}{18} > \frac{1}{4} \right]$$

Thus:

$$\left[\boxed{\frac{5}{18} > 25\%} \right]$$

31. APPROXIMATE FRACTION TO PERCENTAGE

Suppose:

$$\left[\frac{41}{83} \right]$$

Since:

$$\left[\frac{41.5}{83} = 50\% \right]$$

41 is slightly less.

So percentage is slightly less than 50%.

Approximately:

$\boxed{49.4\%}$

If options are:

- 25%
- 49%
- 67%
- 80%

choose around:

$\boxed{49\%}$

without full calculation.

32. ANOTHER APPROXIMATION

$\frac{198}{401} \times 100$

Approximate:

$\frac{200}{400} \times 100$

$= 50\%$

Actual value is slightly below 50%.

Useful in DI.

33. PERCENTAGE POINTS VS PERCENTAGE

This is important enough to introduce now.

Suppose a success rate goes from:

$40\% \text{ to } 50\%$

Increase in **percentage points**:

$50 - 40 = 10$

So:

$\boxed{10 \text{ percentage points}}$

But relative percentage increase:

$\frac{50 - 40}{40} \times 100$

$= 25\%$

Thus:

$\boxed{25\% \text{ increase}}$

These are not the same.

34. PERCENTAGE OF PERCENTAGE

Example:

Find 20% of 30%.

$$\left[20\% \times 30\% \right]$$

Convert:

$$\left[0.2 \times 0.3 = 0.06 \right]$$

As percentage:

$$\left[\boxed{6\%} \right]$$

Alternative:

$$\left[\frac{15}{100} \times \frac{30}{100} = \frac{450}{10000} = 4.5\% \right]$$

35. ANOTHER EXAMPLE

50% of 40%:

$$\left[\frac{1}{2} \times 40\% = \boxed{20\%} \right]$$

This appears in multi-stage problems.

36. FRACTION OF A PERCENTAGE

$$\left[\frac{3}{4} \text{ of } 80\% \right]$$

$$\left[= \frac{3}{4} \times 80\% \right]$$

$$\left[= \boxed{60\%} \right]$$

37. PERCENTAGE OF A FRACTION

What is 25% of:

$$\left[\frac{4}{5} \right]$$

$$\left[25\% = \frac{1}{4} \right]$$

So:

$$\left[\frac{1}{4} \times \frac{4}{5} = \frac{1}{5} \right]$$

As percentage:

$$\left[20\% \right]$$

Answer:

$$\left[\boxed{\frac{1}{5} = 20\%} \right]$$

38. PERCENT DIFFERENCE — EARLY INTRODUCTION

Suppose A = 60 and B = 50.

A is what percent of B?

$$\left[\frac{60}{50} \times 100 = 120\% \right]$$

Therefore A is:

$$\left[20\% \right]$$

more than B.

But B is what percentage of A?

$$\left[\frac{50}{60} \times 100 = 83.33\% \right]$$

Thus B is:

$$\left[16.67\% \right]$$

less than A.

Key lesson

Percentage comparison depends on the **reference base**.

We will study this deeply in Chapter 15.

39. GATE-STYLE QUESTION 1

If:

$$\left[12.5\% \right]$$

of (x) equals 45, find (x) .

$$\left[12.5\% = \frac{18}{x} \right]$$

So:

$$\left[x/8 = 45 \right]$$

$$\left[x = \boxed{360} \right]$$

40. GATE-STYLE QUESTION 2

Which is largest?

A. (0.666)

B. (66.5%)

C. $(\frac{23}{50})$

D. $(\frac{33}{50})$

Convert intelligently:

$$\left[\frac{23}{50} = 0.6666... \right]$$

$$\left[66.5\% = 0.665 \right]$$

$$\left[\frac{33}{50} = 0.66 \right]$$

Thus:

$$\left[\boxed{\frac{23}{50}} \right]$$

is largest.

41. GATE-STYLE QUESTION 3

If 80% of A equals 60% of B , find:

$A:B$

Equation:

$0.8A = 0.6B$

Convert:

$\frac{4A}{5} = \frac{3B}{5}$

Cancel:

$4A = 3B$

Therefore:

$A:B = 3:4$

This is a preview of ratio problems.

42. PLACEMENT-STYLE QUESTION 1

Find:

87.5% of 640

$87.5\% = \frac{7}{8}$

$640/8 = 80$

$80 \times 7 = 560$

Answer:

560

43. PLACEMENT-STYLE QUESTION 2

What percentage of 160 is 24?

$\frac{24}{160} \times 100$

Simplify:

$\frac{3}{20} \times 100$

$= 15\%$

Answer:

15%

44. PLACEMENT-STYLE QUESTION 3

If:

$$\left[33\frac{13}{100} \right]$$

of (A) equals 25% of (B) , find $(A:B)$.

Convert:

$$\left[\frac{13A}{100} = \frac{14B}{100} \right]$$

Cross multiply:

$$\left[4A = 3B \right]$$

Therefore:

$$\left[\boxed{A:B=3:4} \right]$$

45. OPTION-ELIMINATION USING PERCENTAGE SIZE

Suppose:

$$\left[\frac{19}{41} \times 100 \right]$$

Options:

A. 18% B. 31% C. 46% D. 78%

Since:

$$\left[\frac{20}{40} = 50\% \right]$$

and actual fraction is slightly smaller:

$$\left[\frac{19}{41} < 50\% \right]$$

but still near 50%.

Only:

$$\left[\boxed{46\%} \right]$$

is plausible.

No exact calculation required.

46. OPTION-ELIMINATION USING FRACTION BENCHMARKS

Suppose:

$$\left[\frac{5}{12} \right]$$

Options:

A. 20% B. 31% C. 42% D. 75%

You know:

$$\left[\frac{5}{12} \approx 41.67\% \right]$$

So:

$$\left[\boxed{42\%} \right]$$

47. SPEED METHOD FOR 1% / 0.1%

Suppose number = 860.

$$\backslash[1\%=8.6 \backslash]$$

$$\backslash[0.1\%=0.86 \backslash]$$

$$\backslash[0.01\%=0.086 \backslash]$$

Useful when questions involve small percentage changes.

48. SPEED METHOD FOR 2%

$$\backslash[2\%=2\times 1\% \backslash]$$

Example:

2% of 750:

$$\backslash[1\%=7.5 \backslash]$$

$$\backslash[2\%=15 \backslash]$$

49. SPEED METHOD FOR 4%

4% can be:

$$\backslash[2\%+2\% \backslash]$$

or:

$$\backslash[\frac{1}{25} \backslash]$$

Example:

4% of 625:

$$\backslash[625/25 = \boxed{25} \backslash]$$

50. SPEED METHOD FOR 6.25%

$$\backslash[6.25\%=\frac{1}{16} \backslash]$$

Example:

6.25% of 800:

$$\backslash[800/16 = \boxed{50} \backslash]$$

51. SPEED METHOD FOR 8.33%

Approximately:

$$\backslash[8.33\%=\frac{1}{12} \backslash]$$

Example:

$$\backslash(8\frac{1}{3}\%) \text{ of } 600:$$

$$\boxed{600/12 = 50}$$

52. SPEED METHOD FOR 11.11%

$$11.11\% = \frac{1}{9}$$

Example:

$(\frac{1}{9})$ of 450:

$$450/9 = \boxed{50}$$

53. SPEED METHOD FOR 14.2857%

$$14.2857\% = \frac{1}{7}$$

So if a placement question gives:

$$14\frac{2}{7}\%$$

recognize approximately:

$$\frac{1}{7}$$

54. WHAT SHOULD YOU ACTUALLY MEMORIZE?

MUST MEMORIZE

Fractions with denominators:

$$2, 3, 4, 5, 6, 8, 10, 20, 25$$

Also:

$$\frac{1}{12}, \frac{1}{16}, \frac{1}{40}, \frac{1}{50}$$

GOOD TO MEMORIZE

Common fractions involving:

$$7, 9, 11$$

DERIVE IN EXAM

Rare fractions such as:

$$\frac{13}{37}$$

unless an obvious shortcut appears.

55. COMMON TRAPS

TRAP 1 — Percent and decimal confusion

$$2\% = 0.02$$

not 0.2.

& pTRAP 2 — Percentage of total vs percentage of another part

If boys:girls = 3:2:

boys as % of girls:

$$\left[150\% \right]$$

boys as % of class:

$$\left[60\% \right]$$

& pTRAP 3 — Percentage points

40% ! 50% = 10 percentage points, but 25% relative increase.

& pTRAP 4 — Approximation too early

If options are close, calculate accurately.

& pTRAP 5 — Wrong reference quantity

“A is what percent of B?”

$$\left[\frac{AB \times 100}{} \right]$$

The denominator must be B.

56. MEMORY SYSTEM

Instead of memorizing dozens randomly, organize them in families.

Halves

$$\left[1/2 = 50\% \right]$$

Quarters

$$\left[1/4 = 25\%, \quad 3/4 = 75\% \right]$$

Fifths

$$\left[20, \, 40, \, 60, \, 80\% \right]$$

Eighths

Every step adds:

$$\left[12.5\% \right]$$

Sixths

Every step adds:

$$\left[16.67\% \right]$$

Twelfths

Every step adds:

$\lfloor 8.33\% \rfloor$

This is much easier to remember.

57. 5-MINUTE REVISION

Know these instantly:

$\lfloor 1/2=50\% \rfloor$

$\lfloor 1/3=33.33\% \rfloor$

$\lfloor 1/4=25\% \rfloor$

$\lfloor 1/5=20\% \rfloor$

$\lfloor 1/6=16.67\% \rfloor$

$\lfloor 1/8=12.5\% \rfloor$

$\lfloor 1/10=10\% \rfloor$

$\lfloor 1/12=8.33\% \rfloor$

$\lfloor 1/16=6.25\% \rfloor$

$\lfloor 1/20=5\% \rfloor$

$\lfloor 1/25=4\% \rfloor$

$\lfloor 1/40=2.5\% \rfloor$

$\lfloor 1/50=2\% \rfloor$

Also:

$\lfloor x\% \text{ of } y = y\% \text{ of } x \rfloor$

and:

$\lfloor A \text{ as } \% \text{ of } B = \frac{AB}{100} \rfloor$

58. SPEED CHECKLIST

You should eventually answer these almost instantly:

- 25% of 200
- 12.5% of 400
- 75% of 80
- 62.5% of 160
- 16% of 300
- 6.25% of 640
- 5% of 720
- 2.5% of 1200
- 37.5% of 240

Answers:

$\lfloor 50, 50, 60, 100, 50, 40, 36, 30, 90 \rfloor$

PRACTICE — LEVEL 0

Q1

Convert:

$$\left[\frac{3}{16} \right]$$

to percentage.

Q2

Convert:

$$\left[87.5\% \right]$$

to fraction.

Q3

Find:

$$\left[62.5\% \text{ of } 320 \right]$$

Q4

Find:

$$\left[2.5\% \text{ of } 800 \right]$$

Q5

Find:

$$\left[125\% \text{ of } 240 \right]$$

LEVEL 1

Q6

20% of a number is 84. Find the number.

Q7

37.5% of a number is 150. Find the number.

Q8

32 is what percentage of 160?

Q9

If:

$$\left[\frac{3}{8} \right]$$

of x is 72, find x .

Q10

Which is greater?

$$\left[\frac{7}{12} \right]$$

or:

$$\left[58\% \right]$$

LEVEL 2

Q11

If 75% of (A) equals 60% of (B) , find $(A:B)$.

Q12

If:

$$\left[\frac{5}{8} \right]$$

of (A) equals:

$$\left[\frac{3}{4} \right]$$

of (B) , find $(A:B)$.

Q13

What is:

$$\left[25\% \text{ of } 40\% \text{ of } 800? \right]$$

Q14

Which is largest?

$$\left[\frac{5}{12}, 41\%, 0.42, \frac{21}{50} \right]$$

Q15

A value changes from 48% to 60%. Find:

1. increase in percentage points
 2. percentage increase relative to original rate
-

LEVEL 3 — EXAM PRESSURE

Q16

Without lengthy division, estimate:

$$\left[\frac{299}{602} \times 100 \right]$$

Q17

If:

$$\left[16\frac{23}{100} \text{ of } A = 12.5\% \text{ of } B \right]$$

find:

$$\left[A:B \right]$$

Q18

If $(A:B=4:5)$, then A is:

1. what percentage of B?
2. what percentage of $(A+B)$?

Q19

A quantity is 250% of another quantity. Find their ratio.

Q20

If:

$$0.375x = 62.5\% \text{ of } 240$$

find (x) .

ANSWERS

1.

$$18.75\%$$

2.

$$\frac{7}{8}$$

3.

$$200$$

4.

$$20$$

5.

$$300$$

6.

$$420$$

7.

$$400$$

8.

$$20\%$$

9.

$$192$$

10.

$$\frac{7}{12} \approx 58.33\%$$

so $(\frac{7}{12})$ is greater.

11.

$$4:5$$

12.

$$\left[\frac{6}{5} \right]$$

13.

$$\left[80 \right]$$

14.

$$\left[0.42 = \frac{21}{50} = 42\% \right]$$

These two are equal and largest.

15.

$$\left[12 \text{ percentage points} \right]$$

and:

$$\left[\frac{12}{48} \times 100 = 25\% \right]$$

16. Approximately:

$$\left[50\% \right]$$

17.

$$\left[\frac{3}{4} \right]$$

18.

$$\left[80\% \right]$$

of B, and:

$$\left[\frac{49}{100} \times 100 \approx 44.44\% \right]$$

of total.

19.

$$\left[2.5:1 = 5:2 \right]$$

20.

First:

$$\left[62.5\% \text{ of } 240 = \frac{58}{100} \times 240 = 150 \right]$$

Then:

$$\left[0.375x = 150 \right]$$

$$\left[\frac{3}{8}x = 150 \right]$$

$$\left[x = 400 \right]$$

CHAPTER 3 — FINAL CHECK

We have now covered:

- fraction !" decimal conversion '
- fraction !" percentage conversion '
- decimal !" percentage conversion '
- master conversion tables '

- percentage benchmarks '
- percentages above 100% '
- tiny percentages '
- percentage of a number '
- reverse percentage-to-total calculations '
- percentage of percentage '
- ratio-to-percentage previews '
- percentage-point distinction '
- approximation '
- option elimination '
- speed conversion families '
- GATE-style problems '
- placement-style problems '
- exam-pressure questions '

Therefore:

' CHAPTER 3 COMPLETE

Progress

Completed: 3/76 **Current:** ' Chapter 3 — Fraction–Decimal–Percentage Conversion **Next:** Ø=Chapter 4 — Divisibility Rules

Chapter 4 will cover:

- divisibility by 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and others
- why the rules work
- missing-digit questions
- divisibility-based option elimination
- large-number tricks
- GATE/placement question patterns
- divisibility + remainder connections

Send **NEXT**.

USER

next

ASSISTANT

CHAPTER 4 — DIVISIBILITY RULES

PART 1 — CORE RULES, WHY THEY WORK & FAST TESTS

Status: Ø=No PROGRESS **GATE:** Medium **Placements:** High **Priority:** +Tier 2

Divisibility rules help in:

- number system
- HCF/LCM
- factors
- remainders
- digit problems
- option elimination
- simplification
- large-number questions

The goal is not to memorize random tricks blindly. You should know the important rules and when to use them.

1. WHAT DOES “DIVISIBLE” MEAN?

A number (A) is divisible by (B) if division leaves remainder 0.

Example:

$$[24 \div 6 = 4]$$

Remainder:

$$[0]$$

Therefore:

$$[\boxed{24} \text{ is divisible by } 6]$$

But:

$$[25 \div 6]$$

leaves remainder 1.

So 25 is not divisible by 6.

2. DIVISIBILITY BY 2

A number is divisible by 2 if its **last digit is even**.

Possible last digits:

$$[0, 2, 4, 6, 8]$$

Examples:

$$[438]$$

last digit 8 !' divisible by 2.

$$[725]$$

last digit 5 !' not divisible by 2.

MUST MEMORIZE

3. WHY ONLY THE LAST DIGIT MATTERS FOR 2

Consider:

$$438 = 430 + 8$$

430 is divisible by 2.

So divisibility depends only on:

$$8$$

More generally, every multiple of 10 is divisible by 2.

4. DIVISIBILITY BY 5

A number is divisible by 5 if last digit is:

$$\boxed{0 \text{ or } 5}$$

Examples:

$$1245$$

yes.

$$7810$$

yes.

$$4382$$

no.

5. DIVISIBILITY BY 10

A number is divisible by 10 if last digit is:

$$\boxed{0}$$

Example:

$$7320$$

divisible by 10.

6. DIVISIBILITY BY 3

Add all digits.

If the digit sum is divisible by 3, the number is divisible by 3.

Example:

$$582$$

Digit sum:

$$5 + 8 + 2 = 15$$

Since:

$$\lfloor 15 \div 3 = 5 \rfloor$$

$$\lfloor \boxed{582 \text{ is divisible by } 3} \rfloor$$

7. EXAMPLE

Is:

$$\lfloor 9274 \rfloor$$

divisible by 3?

Digit sum:

$$\lfloor 9+2+7+4=22 \rfloor$$

22 is not divisible by 3.

Therefore:

$$\lfloor \boxed{\text{No}} \rfloor$$

8. WHY DIVISIBILITY BY 3 WORKS

Take:

$$\lfloor abc \rfloor$$

as a three-digit number.

$$\lfloor 100a+10b+c \rfloor$$

Modulo 3:

$$\lfloor 100 \equiv 1 \rfloor$$

$$\lfloor 10 \equiv 1 \rfloor$$

So:

$$\lfloor 100a+10b+c \equiv a+b+c \rfloor$$

Therefore number and digit sum have the same remainder modulo 3.

You do not need to reproduce this proof in the exam, but understand the logic.

9. DIVISIBILITY BY 9

Exactly the same idea.

Add all digits.

If sum is divisible by 9, number is divisible by 9.

Example:

$$\lfloor 729 \rfloor$$

$$\lfloor 7+2+9=18 \rfloor$$

18 divisible by 9.

Therefore:

$\boxed{729}$ is divisible by 9

10. DIFFERENCE BETWEEN RULES FOR 3 AND 9

Example:

123

Digit sum:

$1+2+3=6$

6 divisible by 3, but not by 9.

Therefore:

123 is divisible by 3

but not by 9.

Important

Divisible by 9 automatically means divisible by 3.

Reverse is not always true.

11. DIVISIBILITY BY 4

Look only at the **last two digits**.

If the number formed by last two digits is divisible by 4, entire number is divisible by 4.

Example:

7316

Last two digits:

16

16 divisible by 4.

Therefore:

$\boxed{7316}$ is divisible by 4

12. EXAMPLE

51238

Last two digits:

38

38 is not divisible by 4.

Therefore number is not divisible by 4.

13. WHY LAST TWO DIGITS?

Any number can be written as:

$$100k + r$$

where r is last two digits.

Since:

$$100$$

is divisible by 4, only r matters.

14. DIVISIBILITY BY 8

Look at the **last three digits**.

If those three digits form a number divisible by 8, whole number is divisible by 8.

Example:

$$45128$$

Last three digits:

$$128$$

$$128 \div 8 = 16$$

Therefore:

$$\boxed{45128} \text{ is divisible by } 8$$

15. EXAMPLE

$$89246$$

Last three digits:

$$246$$

246 is not divisible by 8.

Therefore number is not divisible by 8.

16. WHY LAST THREE DIGITS?

Because:

$$1000$$

is divisible by 8.

So anything before last 3 digits contributes a multiple of 1000.

17. DIVISIBILITY BY 6

A number is divisible by 6 if it is divisible by:

$\lfloor 2 \text{ text } \{ \text{ and } \} 3 \rfloor$

because:

$\lfloor 6 = 2 \times 3 \rfloor$

and 2 and 3 are coprime.

Example:

$\lfloor 438 \rfloor$

Check 2:

last digit 8 !' yes.

Check 3:

$\lfloor 4 + 3 + 8 = 15 \rfloor$

15 divisible by 3.

Therefore:

$\lfloor \boxed{438} \text{ text } \{ \text{ is divisible by } \} 6 \rfloor$

18. COMMON TRAP FOR 6

A number being divisible by 2 alone is not enough.

Example:

$\lfloor 124 \rfloor$

Even !' divisible by 2.

Digit sum:

$\lfloor 1 + 2 + 4 = 7 \rfloor$

Not divisible by 3.

Therefore:

$\lfloor 124 \rfloor$

is not divisible by 6.

19. DIVISIBILITY BY 12

Since:

$\lfloor 12 = 3 \times 4 \rfloor$

and 3 and 4 are coprime, check:

- divisible by 3
- divisible by 4

Example:

$\lfloor 924 \rfloor$

Digit sum:

$$\sqrt{9+2+4=15}$$

divisible by 3.

Last two digits:

$$\sqrt{24}$$

divisible by 4.

Therefore:

$$\boxed{924 \text{ is divisible by } 12}$$

20. DIVISIBILITY BY 15

$$\sqrt{15=3\times 5}$$

So check divisibility by:

- 3
- 5

Example:

$$\sqrt{735}$$

Digit sum:

$$\sqrt{15}$$

divisible by 3.

Last digit:

$$\sqrt{5}$$

Therefore divisible by 5.

Hence:

$$\boxed{735 \text{ is divisible by } 15}$$

21. DIVISIBILITY BY 18

$$\sqrt{18=2\times 9}$$

Check:

- even?
- digit sum divisible by 9?

Example:

$$\sqrt{738}$$

last digit 8 !' divisible by 2.

Digit sum:

$$\sqrt{7+3+8=18}$$

divisible by 9.

Therefore:

$\boxed{738}$ is divisible by 18

22. DIVISIBILITY BY 25

A number is divisible by 25 if last two digits are:

00, 25, 50, 75

Examples:

3475

yes.

1200

yes.

9130

no.

MUST MEMORIZE

23. DIVISIBILITY BY 50

Last two digits must be:

00 or 50

Example:

7350

divisible by 50.

24. DIVISIBILITY BY 100

Last two digits:

00

Example:

12800

divisible by 100.

25. DIVISIBILITY BY 125

Look at last three digits.

They must be a multiple of 125:

000, 125, 250, 375, 500, 625, 750, 875

Example:

$$\backslash[43625 \backslash]$$

Last three:

$$\backslash[625 \backslash]$$

Therefore divisible by 125.

26. DIVISIBILITY BY 1000

Last three digits must be:

$$\backslash[000 \backslash]$$

27. DIVISIBILITY BY 11

This rule is very useful.

Take alternating sum of digits.

If the difference is:

$$\backslash[0, \pm 11, \pm 22, \ldots \backslash]$$

then number is divisible by 11.

Example:

$$\backslash[121 \backslash]$$

Alternating sum:

$$\backslash[1-2+1=0 \backslash]$$

Therefore:

$$\backslash[\boxed{121}\text{ is divisible by }11 \backslash]$$

28. ANOTHER EXAMPLE

Check:

$$\backslash[2728 \backslash]$$

Take alternating sums:

$$\backslash[(2+2)-(7+8) \backslash]$$

$$\backslash[4-15=-11 \backslash]$$

Since (-11) is divisible by 11:

$$\backslash[\boxed{2728}\text{ is divisible by }11 \backslash]$$

29. ANOTHER WAY TO APPLY 11 RULE

For:

$$\backslash[50644 \backslash]$$

Odd-position digits:

$$\backslash[5+6+4=15 \backslash]$$

Even-position digits:

$$\backslash[0+4=4 \backslash]$$

Difference:

$$\backslash[15-4=11 \backslash]$$

Therefore divisible by 11.

30. DIVISIBILITY BY 7 — PRACTICAL RULE

7 is less elegant, but useful for placements.

Take the last digit, double it, subtract from remaining number.

Repeat.

If result is divisible by 7, original number is divisible by 7.

Example:

$$\backslash[203 \backslash]$$

Last digit:

$$\backslash[3 \backslash]$$

Double:

$$\backslash[6 \backslash]$$

Remaining:

$$\backslash[20 \backslash]$$

Subtract:

$$\backslash[20-6=14 \backslash]$$

14 divisible by 7.

Therefore:

$$\backslash[\boxed{203}\text{ is divisible by }7 \backslash]$$

31. ANOTHER 7 EXAMPLE

Check:

$$\backslash[371 \backslash]$$

Last digit = 1.

$$\backslash[37-2(1)=35 \backslash]$$

35 divisible by 7.

Therefore:

$$\backslash[\boxed{371}\text{ is divisible by }7 \backslash]$$

32. DIVISIBILITY BY 7 — ANOTHER EXAMPLE

Check:

$$\lfloor 1232 \rfloor$$

$$\lfloor 123 - 2(2) = 119 \rfloor$$

Now:

$$\lfloor 11 - 2(9) = -7 \rfloor$$

$\lfloor -7 \rfloor$ is divisible by 7.

Therefore:

$$\lfloor \boxed{1232} \rfloor \text{ is divisible by } 7 \rfloor$$

33. SHOULD YOU ALWAYS USE THE RULE FOR 7?

No.

For small numbers, direct division may be easier.

For example:

$$\lfloor 84 \rfloor$$

Obviously:

$$\lfloor 84 = 7 \times 12 \rfloor$$

Don't force a trick when normal arithmetic is faster.

34. DIVISIBILITY BY 13 — OPTIONAL BUT USEFUL

One rule:

Remove last digit, multiply it by 4, add to remaining number.

If result is divisible by 13, original is divisible by 13.

Example:

$$\lfloor 286 \rfloor$$

Remaining:

$$\lfloor 28 \rfloor$$

Last digit:

$$\lfloor 6 \rfloor$$

$$\lfloor 28 + 4(6) = 52 \rfloor$$

52 divisible by 13.

Therefore:

$$\lfloor \boxed{286} \rfloor \text{ is divisible by } 13 \rfloor$$

This is **Good to Know**, not Tier-1 memorization.

35. DIVISIBILITY BY COMPOSITE NUMBERS

Important concept:

To test divisibility by a composite number, factor it into suitable coprime factors.

Example:

$$24 = 3 \times 8$$

Since 3 and 8 are coprime, number divisible by both is divisible by 24.

Example:

Check 264:

Digit sum:

$$2 + 6 + 4 = 12$$

divisible by 3.

Last 3 digits:

264.

$$264 \div 8 = 33$$

Thus:

$$\boxed{264} \text{ is divisible by } 24$$

36. COMMON TRAP — NON-COPRIME FACTORS

Suppose you want divisibility by 8.

Checking divisibility by 2 and 4 is not enough.

Why?

Because:

$$2 \times 4 = 8$$

but they are not coprime.

Example:

12 is divisible by 2 and 4, but:

$$12$$

is not divisible by 8.

Rule

When combining divisibility conditions, use factors whose LCM equals the target.

37. DIVISIBILITY BY 24

Useful decomposition:

$$24 = 3 \times 8$$

Check 3 and 8.

Not 4 and 6, unless you reason via LCM.

38. DIVISIBILITY BY 36

$$36 = 4 \times 9$$

Since 4 and 9 are coprime:

Check:

- last two digits divisible by 4
- digit sum divisible by 9

Example:

$$5544$$

Digit sum:

$$5+5+4+4=18$$

divisible by 9.

Last two:

$$44$$

divisible by 4.

Therefore:

$$\boxed{5544} \text{ is divisible by } 36$$

39. DIVISIBILITY BY 72

$$72 = 8 \times 9$$

Since 8 and 9 are coprime:

Check both.

Example:

$$3672$$

Digit sum:

$$3+6+7+2=18$$

divisible by 9.

Last three:

$$672$$

$$672/8=84$$

Therefore:

$\boxed{3672}$ is divisible by 72

40. MISSING DIGIT PROBLEMS — DIVISIBILITY BY 3

Question:

Find (x) if:

$$45x2$$

is divisible by 3.

Digit sum:

$$4+5+x+2=11+x$$

Need multiple of 3.

Possible $(x \in \{0, \dots, 9\})$.

Values making $(11+x)$ divisible by 3:

$$x=1, 4, 7$$

Answer:

$$\boxed{1, 4, 7}$$

41. MISSING DIGIT — DIVISIBILITY BY 9

Find digit (x) such that:

$$72x5$$

is divisible by 9.

Digit sum:

$$7+2+x+5=14+x$$

Nearest multiple of 9 in range 14–23:

$$18$$

So:

$$14+x=18$$

$$x=\boxed{4}$$

42. MISSING DIGIT — DIVISIBILITY BY 11

Find (x) such that:

$$3x74$$

is divisible by 11.

Alternating sum:

$$3-x+7-4$$

$$\lfloor =6-x \rfloor$$

Need multiple of 11.

For $(x=0)$ to 9, possible value 0:

$$\lfloor 6-x=0 \rfloor$$

$$\lfloor x=\boxed{6} \rfloor$$

Check:

$$\lfloor 3674/11=334 \rfloor$$

Correct.

43. MISSING DIGIT — MULTIPLE CONDITIONS

Find (x) if:

$$\lfloor 42x6 \rfloor$$

is divisible by 6.

For divisibility by 6:

Check 2

Last digit = 6 !' already even.

Check 3

Digit sum:

$$\lfloor 4+2+x+6=12+x \rfloor$$

Need divisible by 3.

Thus:

$$\lfloor x=0,3,6,9 \rfloor$$

Answer:

$$\lfloor \boxed{0,3,6,9} \rfloor$$

44. OPTION ELIMINATION USING DIVISIBILITY

Suppose:

$$\lfloor N=37\times 48 \rfloor$$

Without exact multiplication, which could be (N) ?

A. 1774 B. 1776 C. 1778 D. 1781

Since:

$$\lfloor 48 \rfloor$$

is divisible by 3 and 16, product must at least be divisible by 3.

Check digit sums:

A:

$$\backslash[1+7+7+4=19 \backslash]$$

No.

B:

$$\backslash[1+7+7+6=21 \backslash]$$

Yes.

C:

$$\backslash[23 \backslash]$$

No.

D:

$$\backslash[17 \backslash]$$

No.

Therefore:

$$\backslash[\boxed{1776} \backslash]$$

No multiplication required.

45. LARGE NUMBER DIVISIBILITY

Is:

$$\backslash[928374615 \backslash]$$

divisible by 3?

Digit sum:

$$\backslash[9+2+8+3+7+4+6+1+5 \backslash]$$

$$\backslash[=45 \backslash]$$

45 divisible by 3 and 9.

Therefore number is divisible by both:

$$\backslash[\boxed{3\text{ and }9} \backslash]$$

46. DIVISIBILITY AND REMAINDER CONNECTION

If digit sum is not divisible by 3, it can also tell remainder mod 3.

Example:

$$\backslash[572 \backslash]$$

Digit sum:

$$\backslash[5+7+2=14 \backslash]$$

$$\backslash[14\equiv 2\pmod{3} \backslash]$$

Therefore:

$\lfloor 572 \rfloor$

also leaves remainder 2 when divided by 3.

This becomes important in Chapter 7.

47. SAME FOR 9

Example:

$\lfloor 731 \rfloor$

Digit sum:

$\lfloor 11 \rfloor$

$\lfloor 11 \equiv 2 \pmod{9} \rfloor$

So 731 leaves remainder:

$\lfloor \boxed{2} \rfloor$

when divided by 9.

48. GATE-STYLE QUESTION 1

What values of $\lfloor x \rfloor$ make:

$\lfloor 84x6 \rfloor$

divisible by 9?

Digit sum:

$\lfloor 8+4+x+6=18+x \rfloor$

Need multiple of 9.

For digit $\lfloor x \in 0, \dots, 9 \rfloor$:

$\lfloor 18+x \rfloor$

can be 18 or 27.

Therefore:

$\lfloor x=0 \text{ or } 9 \rfloor$

Answer:

$\lfloor \boxed{0,9} \rfloor$

49. GATE-STYLE QUESTION 2

A number is divisible by both 8 and 9. What smallest positive number must it be divisible by?

We need:

$\lfloor \operatorname{LCM}(8,9) \rfloor$

Since coprime:

$$8 \times 9 = 72$$

Answer:

$$\boxed{72}$$

50. PLACEMENT-STYLE QUESTION

Find the smallest digit x such that:

$$56x4$$

is divisible by 6.

Need even:

last digit 4 is yes.

Need divisible by 3:

$$5 + 6 + x + 4 = 15 + x$$

Smallest digit making divisible by 3:

$$x = 0$$

Answer:

$$\boxed{0}$$

51. TRICKY QUESTION

Find x such that:

$$3x52$$

is divisible by 11.

Alternating sum:

$$3 - x + 5 - 2 = 6 - x$$

Set equal to 0:

$$x = 6$$

Answer:

$$\boxed{6}$$

52. MUST-MEMORIZE RULE TABLE

Divisor	Quick test
2	Last digit even
3	Digit sum divisible by 3
4	Last 2 digits divisible by 4

Divisor	Quick test
5	Last digit 0 or 5
6	Divisible by 2 and 3
7	Double last digit, subtract
8	Last 3 digits divisible by 8
9	Digit sum divisible by 9
10	Last digit 0
11	Alternating digit-sum difference divisible by 11
12	Divisible by 3 and 4
15	Divisible by 3 and 5
18	Divisible by 2 and 9
25	Last two digits 00, 25, 50, 75
36	Divisible by 4 and 9
125	Last 3 digits multiple of 125

53. WHAT TO MEMORIZE

MUST MEMORIZE

Rules for:

\[2,3,4,5,6,8,9,10,11,12,25 \]

GOOD TO MEMORIZE

Rules for:

\[7,15,18,36,125 \]

OPTIONAL

13 and other unusual divisibility tricks.

54. COMMON TRAPS

& pTRAP 1

Divisibility by 6 requires **both** 2 and 3.

& pTRAP 2

For 4, check last **two** digits.

For 8, last **three** digits.

& pTRAP 3

Divisible by 3 does not imply divisible by 9.

& pTRAP 4

For composite divisor, don't blindly check arbitrary factors.

Check factors whose LCM gives the divisor.

& pTRAP 5

For 11, alternating difference can be:

$\{ 0, \pm 11, \pm 22, \dots \}$

not only 0.

55. MEMORY TRICK

Think:

2 !' last 1 digit 4 !' last 2 digits 8 !' last 3 digits

because:

$\{ 10, 100, 1000 \}$

respectively become divisible by those powers of 2.

And:

3 and 9 !' digit sum

11 !' alternating sum

56. QUICK REVISION BOX

$\{ 2: \text{even last digit} \}$

$\{ 3: \text{digit sum multiple of 3} \}$

$\{ 4: \text{last 2 digits} \}$

$\{ 5: \text{last digit } 0/5 \}$

$\{ 6: 2 \text{ and } 3 \}$

$\{ 8: \text{last 3 digits} \}$

$\{ 9: \text{digit sum multiple of 9} \}$

$\{ 11: \text{alternating sum difference} \}$

PRACTICE — LEVEL 0

Q1

Is 7344 divisible by 3?

Q2

Is 8716 divisible by 4?

Q3

Is 15248 divisible by 8?

Q4

Is 684 divisible by 6?

Q5

Is 7425 divisible by 25?

LEVEL 1

Q6

Find all possible $\backslash(x\backslash)$ such that:

$$\backslash[61 \times 2 \backslash]$$

is divisible by 3.

Q7

Find $\backslash(x\backslash)$:

$$\backslash[43 \times 8 \backslash]$$

is divisible by 9.

Q8

Is:

$$\backslash[91872 \backslash]$$

divisible by 11?

Q9

Find smallest digit $\backslash(x\backslash)$ such that:

$$\backslash[52 \times 4 \backslash]$$

is divisible by 6.

Q10

Which of the following is divisible by 9?

LEVEL 2

Q11

Find $\backslash(x\backslash)$ such that:

$$\backslash[5 \times 63 \backslash]$$

is divisible by 11.

Q12

Check whether:

$$\backslash[1547 \backslash]$$

is divisible by 7.

Q13

A number is divisible by 8 and 9. What is the smallest number it must be divisible by?

Q14

Which digit $\backslash(x\backslash)$ makes:

$$\backslash[24 \times 72 \backslash]$$

divisible by both 9 and 4?

Q15

Without full division, find remainder when:

$$\backslash[98765432 \backslash]$$

is divided by 9.

ANSWERS

1. Yes, digit sum $\backslash(=18\backslash)$
2. Yes, $\backslash(16\backslash)$ divisible by 4
3. Yes, $\backslash(248=31 \times 8\backslash)$
4. Yes
5. Yes

6.

$$\backslash[x=0,3,6,9 \backslash]$$

7.

$$\backslash[4+3+x+8=15+x \backslash]$$

Nearest multiple 18:

$$\backslash[\boxed{x=3} \backslash]$$

8.

Alternating difference:

$$\lfloor (9+8+2)-(1+7)=19-8=11 \rfloor$$

Yes.

9.

$$\lfloor 5+2+x+4=11+x \rfloor$$

Need divisible by 3.

Smallest:

$$\lfloor \boxed{x=1} \rfloor$$

10. A, since digit sum $\neq 15$? Careful: $7+3+1+4 = 15$, not divisible by 9. Let's check others:

- B: $6+4+2+8 = 20$
- C: $9+2+3+5 = 19$
- D: $8+0+1+7 = 16$

So:

$$\lfloor \boxed{\text{None}} \rfloor$$

That itself is a useful exam check: never force an option if the arithmetic says none.

11.

$$\lfloor 5-x+6-3=8-x \rfloor$$

Need 0 or ± 11 . Digit solution:

$$\lfloor \boxed{x=8} \rfloor$$

12.

$$\lfloor 154-2(7)=140 \rfloor$$

140 divisible by 7.

So yes.

13.

$$\lfloor \operatorname{LCM}(8,9)=\boxed{72} \rfloor$$

14. Last two digits 72 already divisible by 4.

Digit sum:

$$\lfloor 2+4+x+7+2=15+x \rfloor$$

Need multiple of 9:

$$\lfloor 18 \rfloor$$

Therefore:

$$\lfloor \boxed{x=3} \rfloor$$

15. Digit sum:

$$\lfloor 9+8+7+6+5+4+3+2=44 \rfloor$$

$$\lfloor 44 \equiv 8 \pmod{9} \rfloor$$

Answer:

CHAPTER 4 STATUS

We have covered:

- divisibility meaning '
- rules for 2, 3, 4, 5 '
- rules for 6, 8, 9, 10 '
- rule for 11 '
- rule for 7 '
- composite divisibility '
- rules for 12, 15, 18, 25, 36, 72, 125 '
- missing-digit questions '
- option elimination '
- remainder connection '
- GATE-style questions '
- placement-style questions '

But I am **not marking Chapter 4 complete yet.**

Next we need:

- advanced missing-digit problems
- divisibility with multiple unknown digits
- forming largest/smallest divisible numbers
- digit replacement problems
- divisibility combined with remainders
- divisibility of powers and expressions
- deeper placement/GATE mixed questions
- final speed drill

Progress

Completed: 3/76 **Current:** 0/31 Chapter 4 — Divisibility Rules **Next:** Chapter 4 Part 2 — **Advanced digit problems & divisibility applications**

Send **NEXT**.

USER

next

ASSISTANT

CHAPTER 4 — DIVISIBILITY RULES

PART 2 — ADVANCED DIGIT PROBLEMS, LARGE/SMALL NUMBERS & MIXED APPLICATIONS

Status: ~~0=Not~~ IN PROGRESS GATE: Medium Placements: High Priority: +Tier 2

This part is where divisibility becomes useful for actual exam questions rather than just memorizing rules.

We will cover:

- unknown digits
 - two unknown digits
 - largest/smallest possible numbers
 - digit replacement
 - divisibility with multiple conditions
 - divisibility of powers/expressions
 - remainder-based reasoning
 - fast option elimination
-

1. SINGLE UNKNOWN DIGIT — BASIC METHOD

Suppose:

$$\lfloor 47x5 \rfloor$$

is divisible by 9.

Digit sum:

$$\lfloor 4+7+x+5=16+x \rfloor$$

For divisibility by 9:

$$\lfloor 16+x \rfloor$$

must be a multiple of 9.

Possible range:

$$\lfloor 16 \leq 16+x \leq 25 \rfloor$$

Only multiple of 9:

$$\lfloor 18 \rfloor$$

So:

$$\lfloor 16+x=18 \rfloor$$

$$\lfloor \boxed{x=2} \rfloor$$

2. USE RANGE BEFORE TRYING DIGITS

This is faster than testing all digits 0–9.

Example:

$$\lfloor 83x7 \rfloor$$

divisible by 9.

Digit sum:

$$[18+x]$$

Range:

$$[18 \text{ to } 27]$$

Multiples of 9:

$$[18, 27]$$

Therefore:

$$[x=0 \text{ or } 9]$$

Answer:

$$\boxed{0,9}$$

3. UNKNOWN DIGIT WITH DIVISIBILITY BY 11

Find (x) if:

$$[4x73]$$

is divisible by 11.

Use alternating sum:

$$[4-x+7-3]$$

$$[=8-x]$$

Need:

$$[0, \pm 11, \pm 22, \dots]$$

Since (x) is a digit:

$$[8-x]$$

ranges from (8) to (-1) .

Only possible multiple of 11 is 0.

So:

$$[8-x=0]$$

$$\boxed{x=8}$$

4. TWO UNKNOWN DIGITS — DIVISIBILITY BY 9

Suppose:

$$[4x7y]$$

is divisible by 9.

Digit sum:

$$[4+x+7+y = 11+x+y]$$

Need multiple of 9.

Thus:

$$11 + x + y = 18$$

or:

$$27$$

Hence:

$$x + y = 7$$

or:

$$x + y = 16$$

So many pairs are possible.

Examples:

For $(x+y=7)$:

$$(0, 7), (1, 6), \dots, (7, 0)$$

For $(x+y=16)$:

$$(7, 9), (8, 8), (9, 7)$$

This kind of question often adds another condition to find a unique pair.

5. TWO UNKNOWN DIGITS WITH TWO CONDITIONS

Find digits (x, y) such that:

$$3 \times 5y$$

is divisible by both 9 and 5.

Divisible by 5 means:

$$y = 0 \text{ or } 5$$

Now divisibility by 9:

$$3 + x + 5 + y = 8 + x + y$$

must be a multiple of 9.

Case 1: $(y=0)$

$$8 + x$$

must be 9 or 18.

Only possible:

$$x = 1$$

So:

$$(x, y) = (1, 0)$$

Case 2: $(y=5)$

$$13 + x$$

must be 18.

So:

$$\boxed{x=5}$$

Thus:

$$\boxed{(x,y)=(1,0)\text{ or }(5,5)}$$

6. DIVISIBILITY BY 6 WITH UNKNOWN DIGIT

Find all (x) such that:

$$\boxed{73x2}$$

is divisible by 6.

For divisibility by 6:

- divisible by 2
- divisible by 3

Last digit = 2, so condition for 2 is already satisfied.

Digit sum:

$$\boxed{7+3+x+2 = 12+x}$$

Need divisible by 3.

So:

$$\boxed{x=0,3,6,9}$$

Answer:

$$\boxed{0,3,6,9}$$

7. DIVISIBILITY BY 12 WITH UNKNOWN DIGIT

Find (x) such that:

$$\boxed{56x4}$$

is divisible by 12.

Need divisibility by:

$$\boxed{3\text{ and }4}$$

Divisibility by 4

Last two digits:

$$\boxed{x4}$$

Possible:

$$\boxed{04,14,24,34,44,54,64,74,84,94}$$

Divisible by 4:

$$\boxed{04,24,44,64,84}$$

So:

$$\{x=0,2,4,6,8\}$$

Divisibility by 3

Digit sum:

$$5+6+x+4 = 15+x$$

Need divisible by 3.

So:

$$\{x=0,3,6,9\}$$

Intersection:

$$\boxed{x=0,6}$$

8. INTERSECTION METHOD

For multiple divisibility conditions:

1. Find digits satisfying first condition.
2. Find digits satisfying second condition.
3. Take the **intersection**.

This avoids trial-and-error.

9. FIND THE SMALLEST DIGIT

Question:

Find smallest $\{x\}$ such that:

$$62x4$$

is divisible by 9.

Digit sum:

$$12+x$$

Nearest multiple of 9 greater than or equal to 12:

$$18$$

So:

$$x=6$$

Answer:

$$\boxed{6}$$

10. FIND THE LARGEST DIGIT

Find largest $\{x\}$ such that:

$$27x4$$

is divisible by 3.

Digit sum:

$\{ 13+x \}$

Need divisible by 3.

Valid digits:

$\{ x=2,5,8 \}$

Largest:

$\boxed{8}$

11. FORM THE SMALLEST NUMBER

Using digits:

$\{ 2,3,4,5 \}$

without repetition, form the smallest 4-digit number divisible by 5.

Divisible by 5 means last digit must be:

$\{ 5 \}$

because 0 is unavailable.

To minimize number, arrange remaining digits ascending:

$\{ 2,3,4 \}$

So:

$\boxed{2345}$

12. FORM THE LARGEST NUMBER

Using digits:

$\{ 1,2,4,5,8 \}$

without repetition, form the largest number divisible by 2.

Divisible by 2 means last digit must be even:

$\{ 2,4,8 \}$

To maximize the number, we want the largest digits as early as possible.

Try keeping the smallest possible even digit at the end:

$\{ 2 \}$

Arrange rest descending:

$\{ 8,5,4,1 \}$

Number:

$\boxed{85412}$

This is larger than choices ending in 4 or 8 because putting 8 earlier helps more.

13. LARGEST NUMBER DIVISIBLE BY 5

Using digits:

$\{0, 2, 3, 6, 9\}$

once each, form largest 5-digit number divisible by 5.

Last digit must be:

$\{0 \text{ or } 5\}$

Only 0 available.

So last digit:

$\{0\}$

Arrange remaining descending:

$\{9, 6, 3, 2\}$

Thus:

$\boxed{96320}$

14. SMALLEST NUMBER DIVISIBLE BY 5 WITH ZERO

Using:

$\{0, 1, 2, 5\}$

once each, form smallest 4-digit number divisible by 5.

Possible last digit:

$\{0 \text{ or } 5\}$

If last digit = 0

Remaining smallest non-zero first:

$\{1, 2, 5\}$

Number:

$\{1250\}$

If last digit = 5

Remaining digits $\{0, 1, 2\}$.

First digit cannot be 0.

Smallest:

$\{1025\}$

Compare:

$\{1025 < 1250\}$

Therefore:

$\boxed{1025}$

& pTRAP

Do not automatically put 0 at the end without checking both possibilities.

15. DIGIT REPLACEMENT PROBLEM

What digit should replace x so that:

$$72x36$$

is divisible by 9?

Digit sum:

$$7+2+x+3+6 = 18+x$$

Need multiple of 9:

$$18 \text{ or } 27$$

So:

$$x=0 \text{ or } 9$$

Answer:

$$\boxed{0,9}$$

16. TWO-DIGIT REPLACEMENT WITH 11

Find x if:

$$5x462$$

is divisible by 11.

Alternating positions:

$$(5+4+2)-(x+6)$$

$$=11-x$$

Need multiple of 11.

Possible:

$$11-x=11 \text{ or } 0$$

Thus:

$$x=0$$

or:

$$x=11$$

But x must be digit.

Therefore:

$$\boxed{x=0}$$

17. DIVISIBILITY OF A SUM

If both (a) and (b) are divisible by (n) , then:

$$(a+b)$$

is divisible by (n) .

Example:

$$84 \text{ and } 126$$

both divisible by 7.

Then:

$$84+126=210$$

also divisible by 7.

18. DIVISIBILITY OF A DIFFERENCE

If:

$$n \mid a$$

and:

$$n \mid b$$

then:

$$n \mid (a-b)$$

Example:

Both 1001 and 693 are divisible by 7.

Therefore:

$$1001-693=308$$

must also be divisible by 7.

19. DIVISIBILITY OF A PRODUCT

If any factor is divisible by (n) , then whole product is divisible by (n) .

Example:

$$17 \times 24 \times 31$$

Since:

$$24$$

is divisible by 8:

entire product is divisible by 8.

No multiplication required.

20. STRONGER PRODUCT REASONING

Suppose product:

$$15 \times 14$$

Is it divisible by 6?

Neither 15 nor 14 is divisible by 6 individually.

But:

$$15 = 3 \times 5$$

and:

$$14 = 2 \times 7$$

Together product contains factors:

$$2 \times 3$$

So it is divisible by 6.

Exam lesson

For products, prime-factor reasoning is often better than divisibility rules.

21. DIVISIBILITY OF POWERS

Suppose:

$$12^5$$

Is it divisible by 6?

Since:

$$12$$

itself is divisible by 6, every positive power:

$$12^n$$

is also divisible by 6.

22. POWER WITH PRIME FACTORS

Is:

$$6^4$$

divisible by 72?

Factor:

$$6^4 = (2 \times 3)^4 = 2^4 \times 3^4$$

and:

$$72 = 2^3 \times 3^2$$

Since (6^4) contains enough factors:

$$\sqrt[4]{2^4} \geq 2^3$$

and:

$$\sqrt[4]{3^4} \geq 3^2$$

Yes.

$$\boxed{6^4 \text{ is divisible by } 72}$$

This prepares you for factorization/HCF-LCM chapters.

23. EXPRESSION DIVISIBILITY USING FACTORIZATION

Is:

$$\sqrt{35^2 - 15^2}$$

divisible by 10?

Use:

$$\sqrt{a^2 - b^2} = (a-b)(a+b)$$

$$\sqrt{= (35-15)(35+15)}$$

$$\sqrt{= 20 \times 50}$$

Obviously divisible by 10.

Faster than calculating squares.

24. CLASSIC FORM

For integers (a, b) :

$$\sqrt{a^2 - b^2} = (a-b)(a+b)$$

Use factorization before multiplication when divisibility is asked.

25. DIVISIBILITY OF $n(n+1)$

For any integer n :

$$\sqrt{n(n+1)}$$

is always divisible by 2.

Why?

Among two consecutive integers, one is even.

Therefore:

$$\boxed{2 \mid n(n+1)}$$

26. DIVISIBILITY OF THREE CONSECUTIVE INTEGERS

$$\sqrt{n(n+1)(n+2)}$$

is always divisible by 6.

Why?

Among three consecutive integers:

- at least one is divisible by 3
- at least one is even

So product has factors 2 and 3.

Thus:

$$\boxed{6 \mid n(n+1)(n+2)}$$

Very useful pattern.

27. FOUR CONSECUTIVE INTEGERS

Product:

$$n(n+1)(n+2)(n+3)$$

is always divisible by:

$$24$$

Why?

Among four consecutive integers:

- one divisible by 4
- another even factor provides extra 2
- one divisible by 3

So product contains:

$$2^3 \times 3 = 24$$

This is useful in combinatorial/divisibility reasoning.

28. DIVISIBILITY OF $(n^3 - n)$

Factor:

$$n^3 - n$$

$$= n(n^2 - 1)$$

$$= n(n-1)(n+1)$$

These are three consecutive integers.

Therefore divisible by:

$$6$$

So:

$$\boxed{n^3 - n \text{ is always divisible by } 6}$$

This is a classic mathematical reasoning question.

29. DIVISIBILITY OF $(n^5 - n)$ — PREVIEW

For every integer (n) :

$$[n^5 - n]$$

is divisible by:

$$[30]$$

This follows from factors 2, 3 and 5.

You do **not** need a full number-theory proof for aptitude right now, but recognize this as a useful classic result.

30. REMAINDER CONNECTION — SAME REMAINDER

If two numbers leave the same remainder when divided by (n) , their difference is divisible by (n) .

Example:

Suppose:

$$[A \equiv 2 \pmod{7}]$$

and:

$$[B \equiv 2 \pmod{7}]$$

Then:

$$[A - B \equiv 0 \pmod{7}]$$

So:

$$[\boxed{7 \mid (A - B)}]$$

This will become central in Chapter 7.

31. EXAMPLE

Check:

$$[31]$$

and:

$$[17]$$

Both leave remainder 3 when divided by 7:

$$[31 = 7(4) + 3]$$

$$[17 = 7(2) + 3]$$

Difference:

$$[31 - 17 = 14]$$

which is divisible by 7.

32. FIND SMALLEST NUMBER TO ADD FOR DIVISIBILITY

Question:

What smallest number should be added to 347 to make it divisible by 9?

Digit sum:

$$[3+4+7=14]$$

Remainder when divided by 9:

$$[14 \equiv 5]$$

To reach next multiple of 9, add:

$$[9-5=4]$$

Answer:

$$[\boxed{4}]$$

Check:

$$[347+4=351]$$

Digit sum:

$$[9]$$

33. FIND SMALLEST NUMBER TO SUBTRACT

What smallest number should be subtracted from 347 to make it divisible by 9?

Remainder:

$$[5]$$

Subtract remainder itself:

$$[347-5=342]$$

So:

$$[\boxed{5}]$$

34. GENERAL RULE

If:

$$[N]$$

leaves remainder (r) when divided by (d) :

Smallest subtraction:

$$[\boxed{r}]$$

Smallest addition:

$$[\boxed{d-r}]$$

provided $(r \neq 0)$.

If $(r=0)$, nothing needs to be added or subtracted.

35. EXAMPLE WITH DIVISIBILITY BY 8

What smallest number should be added to 726 to make it divisible by 8?

Last three digits:

$$\boxed{726}$$

Nearby multiples:

$$8 \times 90 = 720$$

Remainder:

$$6$$

Need:

$$8 - 6 = 2$$

Answer:

$$\boxed{2}$$

36. LARGEST MULTIPLE BELOW A NUMBER

Find largest number less than or equal to 500 divisible by 9.

Compute:

$$500 \div 9$$

$$9 \times 55 = 495$$

So:

$$\boxed{495}$$

37. SMALLEST MULTIPLE ABOVE A NUMBER

Smallest multiple of 9 greater than 500:

$$495 + 9 = 504$$

Answer:

$$\boxed{504}$$

38. COUNT MULTIPLES IN A RANGE

How many positive multiples of 7 are ≤ 100 ?

Largest multiple:

$$7 \times 14 = 98$$

So:

$\boxed{14}$

General:

$\left\lfloor \frac{N}{d} \right\rfloor$

positive multiples of d up to N .

This will matter in counting questions too.

39. COUNT MULTIPLES BETWEEN TWO LIMITS

How many multiples of 6 from 20 to 80 inclusive?

Largest multiple ≤ 80 :

$$78 = 6 \times 13$$

Largest multiple < 20 :

$$18 = 6 \times 3$$

Count:

$$13 - 3 = 10$$

So:

$$\boxed{10}$$

They are:

24, 30, ..., 78.

40. GATE-STYLE QUESTION 1

For every integer n , which expression is always divisible by 6?

A.

$$n(n+1)$$

B.

$$n(n+1)(n+2)$$

C.

$$n^2 + 1$$

D.

$$2n + 1$$

Three consecutive integers always contain:

- one factor 2
- one factor 3

Therefore:

$$\boxed{B}$$

41. GATE-STYLE QUESTION 2

What is the smallest integer to add to:

$$\sqrt[9]{98765}$$

to make it divisible by 9?

Digit sum:

$$\sqrt[9]{9+8+7+6+5=35}$$

$$\sqrt[9]{35 \equiv 8 \pmod{9}}$$

Need:

$$\sqrt[9]{9-8=1}$$

Answer:

$$\sqrt[9]{\boxed{1}}$$

42. GATE-STYLE QUESTION 3

Which of the following is necessarily divisible by 6 for every integer n ?

$$\sqrt[9]{n^3-n}$$

Factor:

$$\sqrt[9]{n(n-1)(n+1)}$$

Three consecutive integers.

Therefore:

$$\sqrt[9]{\boxed{n^3-n}}$$

is divisible by 6.

43. PLACEMENT-STYLE QUESTION 1

Find the smallest digit x such that:

$$\sqrt[9]{37x52}$$

is divisible by 9.

Digit sum:

$$\sqrt[9]{3+7+x+5+2=17+x}$$

Next multiple of 9:

$$\sqrt[9]{18}$$

So:

$$\sqrt[9]{\boxed{x=1}}$$

44. PLACEMENT-STYLE QUESTION 2

Find the greatest digit (x) such that:

$$[84x6]$$

is divisible by 6.

Last digit is even.

Digit sum:

$$[8+4+x+6=18+x]$$

Need divisible by 3.

Valid:

$$[x=0,3,6,9]$$

Greatest:

$$[\boxed{9}]$$

45. PLACEMENT-STYLE QUESTION 3

What smallest number should be subtracted from 12345 to make it divisible by 11?

We can divide or use nearby multiple.

$$[11 \times 1122 = 12342]$$

Difference:

$$[12345 - 12342 = 3]$$

Answer:

$$[\boxed{3}]$$

46. OPTION ELIMINATION USING DIVISIBILITY BY 11

Suppose:

$$[N = 143 \times 11]$$

Which could be (N) ?

A. 1571 B. 1573 C. 1575 D. 1577

Product must be divisible by 11.

Check A:

$$[1-5+7-1=2]$$

No.

B:

$$[1-5+7-3=0]$$

Yes.

Therefore:

$$[\boxed{1573}]$$

No multiplication needed.

47. OPTION ELIMINATION USING LAST DIGITS

Suppose:

$$N = 125 \times 48$$

Which answer is possible?

Since:

$$125 \times 8$$

always contributes trailing structure and exact product is divisible by 1000 when combined with 48? Let's reason faster:

$$48 = 6 \times 8$$

so:

$$125 \times 48 = (125 \times 8) \times 6 = 1000 \times 6$$

Therefore number must end in:

$$000$$

Thus:

$$\boxed{6000}$$

This is both factorization and divisibility thinking.

48. PRIME-FACTOR DIVISIBILITY THINKING

Suppose:

Is:

$$2^5 3^4 5^2$$

divisible by:

$$360?$$

Factor:

$$360 = 2^3 3^2 5$$

Compare exponents:

- 2^5 has enough 2s
- 3^4 has enough 3s
- 5^2 has enough 5s

Therefore yes.

This idea will be covered deeply in Chapter 6.

49. COMMON TRAPS

& pTRAP 1 — Not taking intersection

For divisibility by 12, satisfy both 3 and 4.

& pTRAP 2 — Leading zero

When forming a smallest number, first digit cannot be 0.

& pTRAP 3 — Largest number logic

Keeping a smaller valid ending digit may let larger digits occupy higher place values.

& pTRAP 4 — Add vs subtract

If remainder is $\backslash(r\backslash)$:

- subtract $\backslash(r\backslash)$
- add $\backslash(d-r\backslash)$

& pTRAP 5 — Composite factors

Being divisible by two factors only guarantees divisibility by their LCM, not necessarily their product.

Example:

Divisible by 4 and 6 !' necessarily divisible by:

$\backslash[\operatorname{LCM}\{4,6\}=12\backslash]$

not necessarily 24.

50. MEMORY RULES

Unknown digit

Turn divisibility condition into an equation on digits.

Multiple conditions

Find candidate sets !' intersect.

Add for next multiple

$\backslash[d-r \backslash]$

Subtract for previous multiple

$\backslash[r \backslash]$

Expression divisibility

Factor first.

Products

51. 5-MINUTE REVISION

You should now be able to:

- solve missing-digit questions
- solve two-condition digit questions
- build largest/smallest divisible numbers
- use divisibility to avoid multiplication
- reason about products/powers
- find smallest addition/subtraction
- count multiples
- factor expressions for divisibility

Important classics:

$n(n+1)$ is divisible by 2

$n(n+1)(n+2)$ is divisible by 6

$n^3 - n$ is divisible by 6

PRACTICE — LEVEL 0

Q1

Find x such that:

$$62x1$$

is divisible by 9.

Q2

Find all x such that:

$$41x8$$

is divisible by 6.

Q3

What smallest number must be added to 238 to make it divisible by 7?

Q4

What smallest number must be subtracted from 538 to make it divisible by 9?

Q5

How many positive multiples of 8 are ≤ 100 ?

LEVEL 1

Q6

Find $\lfloor x \rfloor$ such that:

$$\lfloor 7x48 \rfloor$$

is divisible by 11.

Q7

Find all $\lfloor x \rfloor$ such that:

$$\lfloor 45x6 \rfloor$$

is divisible by 12.

Q8

Using digits $\{0,3,5,7\}$ once each, form the smallest 4-digit number divisible by 5.

Q9

Using digits $\{1,2,4,6,8\}$ once each, form the largest 5-digit number divisible by 2.

Q10

How many multiples of 9 lie between 100 and 250 inclusive?

LEVEL 2

Q11

Show whether:

$$\lfloor n^3 - n \rfloor$$

is divisible by 6 for every integer $\lfloor n \rfloor$.

Q12

Is:

$$\lfloor 5^4 \times 6^3 \rfloor$$

divisible by 1000?

Q13

What smallest number should be added to 9999 to make it divisible by 11?

Q14

Find digits $\{x,y\}$ if:

$$\lfloor 4x6y \rfloor$$

is divisible by both 5 and 9.

Q15

A number is divisible by both 12 and 18. What smallest positive integer must it therefore be divisible by?

LEVEL 3 — EXAM PRESSURE

Q16

Find the greatest digit x such that:

$$23x74$$

is divisible by 11.

Q17

Without expanding, determine whether:

$$101^2 - 99^2$$

is divisible by 8.

Q18

For every integer n , is:

$$n(n+1)(n+2)(n+3)$$

always divisible by 24?

Q19

Find the smallest 5-digit number divisible by 72.

Q20

If a number leaves remainder 4 when divided by 9, what remainder does the number plus 14 leave when divided by 9?

ANSWERS

1

Digit sum:

$$6+2+x+1=9+x$$

Need multiple of 9.

$$\boxed{x=0 \text{ or } 9}$$

2

Last digit 8 !' even.

Digit sum:

$$4+1+x+8=13+x$$

Need divisible by 3.

$$\boxed{x=2,5,8}$$

3

$$238=7(34)$$

Already divisible by 7.

$$\boxed{0}$$

4

Digit sum:

$$\lfloor 5+3+8=16 \rfloor$$

Remainder mod 9:

$$\lfloor 7 \rfloor$$

Subtract:

$$\lfloor \boxed{7} \rfloor$$

5

$$\lfloor \left\lfloor 100/8 \right\rfloor = \boxed{12} \rfloor$$

6

$$\lfloor 7-x+4-8 = 3-x \rfloor$$

Need 0 or ± 11 .

$$\lfloor \boxed{x=3} \rfloor$$

7

Divisible by 4:

last two digits $\lfloor (x6) \rfloor$.

Valid:

$$\lfloor 16,36,56,76,96 \rfloor$$

so:

$$\lfloor x=1,3,5,7,9 \rfloor$$

Divisible by 3:

$$\lfloor 4+5+x+6=15+x \rfloor$$

Need $\lfloor (x=0,3,6,9) \rfloor$.

Intersection:

$$\lfloor \boxed{x=3,9} \rfloor$$

8

Check endings 0 or 5.

Ending 5:

$$\lfloor \boxed{3075} \rfloor$$

Ending 0 gives at least 3570.

So:

$$\lfloor \boxed{3075} \rfloor$$

9

Use smallest even digit at end if it maximizes leading places.

Ending 2:

$$\boxed{86412}$$

Thus:

$$\boxed{86412}$$

10

Largest multiple ≤ 250 :

$$243 = 9 \times 27$$

Largest multiple < 100 :

$$99 = 9 \times 11$$

Count:

$$27 - 11 = \boxed{16}$$

11

$$n^3 - n = n(n-1)(n+1)$$

Three consecutive integers contain factor 2 and factor 3.

Therefore:

$$\boxed{\text{Yes}}$$

12

$$5^4 \times 6^3 = 5^4 \times 2^3 \times 3^3$$

For 1000:

$$1000 = 2^3 \times 5^3$$

Enough 2s and 5s.

$$\boxed{\text{Yes}}$$

13

$$9999 \div 11$$

Since:

$$11 \times 909 = 9999$$

already divisible.

$$\boxed{0}$$

14

Divisible by 5:

$$y = 0 \text{ or } 5$$

Digit sum:

$$4 + x + 6 + y = 10 + x + y$$

For $(y=0)$:

$$10 + x = 18 \Rightarrow x = 8$$

For $(y=5)$:

$$\lfloor 15+x=18 \rightarrow x=3 \rfloor$$

So:

$$\lfloor \boxed{(x,y)=(8,0),(3,5)} \rfloor$$

15

$$\lfloor \operatorname{LCM}(12,18) = \boxed{36} \rfloor$$

16

Alternating sum:

$$\lfloor 2-3+x-7+4 = x-4 \rfloor$$

Need multiple of 11.

For digit $\lfloor x \rfloor$, only 0 possible:

$$\lfloor x-4=0 \rfloor$$

$$\lfloor \boxed{x=4} \rfloor$$

17

$$\lfloor 101^2-99^2 = (101-99)(101+99) \rfloor$$

$$\lfloor =2 \times 200 = 400 \rfloor$$

400 divisible by 8.

$$\lfloor \boxed{\text{Yes}} \rfloor$$

18

Yes.

Among four consecutive integers:

- one multiple of 4
- another even
- one multiple of 3

Thus product contains:

$$\lfloor 2^3 \times 3 = 24 \rfloor$$

$$\lfloor \boxed{\text{Always divisible by 24}} \rfloor$$

19

Smallest 5-digit number:

$$\lfloor 10000 \rfloor$$

Divide by 72:

$$\lfloor 72 \times 138 = 9936 \rfloor$$

Next multiple:

$$\lfloor 72 \times 139 = 10008 \rfloor$$

Answer:

$$\lfloor \boxed{10008} \rfloor$$

Original remainder:

$$\boxed{4}$$

Add:

$$14 \equiv 5 \pmod{9}$$

New remainder:

$$4 + 5 = 9 \equiv 0$$

$$\boxed{0}$$

CHAPTER 4 — FINAL CHECK

We have now completed:

- core divisibility rules '
- rules for composite divisors '
- unknown digit questions '
- multiple unknown digits '
- simultaneous divisibility conditions '
- largest/smallest digit questions '
- forming largest/smallest divisible numbers '
- digit replacement '
- divisibility of sums/differences/products '
- powers and prime-factor reasoning '
- consecutive-integer divisibility '
- expression factorization '
- smallest number to add/subtract '
- counting multiples '
- remainder connection '
- option elimination '
- GATE-style questions '
- placement-style questions '
- exam-pressure practice '

Therefore:

' CHAPTER 4 COMPLETE

Progress

Completed: 4/76 **Current:** ' Chapter 4 — Divisibility Rules **Next:** 0=3 Chapter 5 — Factors, Multiples, HCF/GCD and LCM

Chapter 5 is 0=166 because it connects directly to:

- number system
- time & work shortcuts

- fractions
- repeated events
- bells/clocks patterns
- grouping/distribution problems
- factor questions
- divisibility
- remainder problems

Send **NEXT**.

USER

next

ASSISTANT

CHAPTER 5 — FACTORS, MULTIPLES, HCF/GCD & LCM

PART 1 — FOUNDATIONS, PRIME FACTORIZATION & CORE METHODS

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

This chapter is extremely important because it connects to:

- divisibility
- fractions
- number system
- time & work
- pipes
- repeated events
- bells and schedules
- grouping/distribution
- remainder questions
- prime factorization
- arithmetic simplification

The central idea is simple:

HCF tells us the greatest common divisor. LCM tells us the smallest common multiple.

1. FACTOR — WHAT DOES IT MEAN?

A factor of a number divides that number exactly.

Example:

Factors of 12:

$\{1, 2, 3, 4, 6, 12\}$

because each divides 12 with remainder 0.

So:

$3 \mid 12$

means:

3 divides 12.

2. MULTIPLE — WHAT DOES IT MEAN?

Multiples are obtained by multiplying a number by integers.

Multiples of 4:

$\{4, 8, 12, 16, 20, 24, \dots\}$

Multiples of 6:

$\{6, 12, 18, 24, 30, \dots\}$

3. FACTOR VS MULTIPLE

For:

$4 \times 6 = 24$

We can say:

- 4 is a factor of 24
- 6 is a factor of 24
- 24 is a multiple of 4
- 24 is a multiple of 6

Memory trick

Factor goes INTO the number. Multiple comes OUT of multiplication.

4. IMPORTANT PROPERTIES OF FACTORS

For any positive integer (N) :

$\{1\}$

is always a factor.

And:

$\{N\}$

is always a factor of itself.

Example:

Factors of 17:

$\{ 1, 17 \}$

Therefore 17 is prime.

5. COMMON FACTORS

Consider:

12 and 18.

Factors of 12:

$\{ 1, 2, 3, 4, 6, 12 \}$

Factors of 18:

$\{ 1, 2, 3, 6, 9, 18 \}$

Common factors:

$\{ 1, 2, 3, 6 \}$

The greatest common factor is:

$\{ 6 \}$

Therefore:

$\boxed{\operatorname{HCF}(12, 18) = 6}$

6. HCF / GCD

HCF means:

Highest Common Factor

GCD means:

Greatest Common Divisor

They mean the same thing.

$\boxed{\text{HCF} = \text{GCD}}$

GATE mathematics often uses **gcd**.

Placement aptitude frequently uses **HCF**.

7. LCM — LOWEST COMMON MULTIPLE

Consider 4 and 6.

Multiples of 4:

$\{ 4, 8, 12, 16, 20, 24, \dots \}$

Multiples of 6:

$\{ 6, 12, 18, 24, \dots \}$

Common multiples:

$\{12, 24, 36, \dots\}$

Smallest:

12

Therefore:

$\boxed{\operatorname{LCM}(4, 6) = 12}$

8. INTUITION — HCF VS LCM

Think of two different questions.

HCF question

What is the largest equal size that can divide everything exactly?

LCM question

When will different repeating cycles meet for the first time?

This distinction will solve many word problems.

9. PRIME FACTORIZATION METHOD

Suppose:

24

Prime factorization:

$24 = 2^3 \times 3$

And:

$36 = 2^2 \times 3^2$

This makes HCF and LCM very easy.

10. HCF USING PRIME FACTORIZATION

For:

$24 = 2^3 \times 3^1$

$36 = 2^2 \times 3^2$

For HCF:

Take common prime factors with minimum exponent.

For factor 2:

$$\lfloor \min(3,2)=2 \rfloor$$

For factor 3:

$$\lfloor \min(1,2)=1 \rfloor$$

Therefore:

$$\lfloor \operatorname{HCF}(24,36) = 2^2 \times 3 \rfloor$$

$$\lfloor =4 \times 3 \rfloor$$

$$\lfloor =\boxed{12} \rfloor$$

11. LCM USING PRIME FACTORIZATION

For LCM:

Take every required prime with maximum exponent.

For:

$$\lfloor 24=2^3 \times 3 \rfloor$$

$$\lfloor 36=2^2 \times 3^2 \rfloor$$

Take:

$$\lfloor 2^3 \rfloor$$

and:

$$\lfloor 3^2 \rfloor$$

Therefore:

$$\lfloor \operatorname{LCM}(24,36) = 2^3 \times 3^2 \rfloor$$

$$\lfloor =8 \times 9 \rfloor$$

$$\lfloor =\boxed{72} \rfloor$$

12. MEMORY TRICK

HCF

HCF takes the LOW powers.

LCM

LCM takes the LARGE powers.

Or:

$$\lfloor \boxed{\text{HCF = minimum exponents}} \rfloor$$

$$\lfloor \boxed{\text{LCM = maximum exponents}} \rfloor$$

13. EXAMPLE WITH THREE NUMBERS

Find HCF and LCM of:

$$\{ 12, 18, 30 \}$$

Prime factors:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

$$30 = 2 \times 3 \times 5$$

HCF

Common primes to all:

$$2^1, 3^1$$

Thus:

$$\boxed{\text{HCF} = 6}$$

LCM

Maximum powers:

$$2^2, 3^2, 5$$

$$= 4 \times 9 \times 5$$

$$= \boxed{180}$$

14. HCF BY LISTING FACTORS

For small numbers, listing may be fastest.

Find HCF of:

$$\{ 20, 30 \}$$

Factors of 20:

$$\{ 1, 2, 4, 5, 10, 20 \}$$

Factors of 30:

$$\{ 1, 2, 3, 5, 6, 10, 15, 30 \}$$

Highest common:

$$\boxed{10}$$

But for large numbers, use prime factorization or Euclidean algorithm.

15. LCM BY LISTING MULTIPLES

For small numbers:

$$\{6, 8\}$$

Multiples of 6:

$$\{6, 12, 18, 24, \dots\}$$

Multiples of 8:

$$\{8, 16, 24, \dots\}$$

First common:

$$\boxed{24}$$

Again, useful only for small numbers.

16. VERY IMPORTANT RELATION FOR TWO NUMBERS

For two positive integers (a, b) :

$$\boxed{\operatorname{HCF}(a, b) \times \operatorname{LCM}(a, b) = ab}$$

This is extremely useful.

17. EXAMPLE

Two numbers are 18 and 24.

HCF:

$$6$$

Then:

$$\text{LCM} = \frac{18 \times 24}{6}$$

$$= 18 \times 4$$

$$= \boxed{72}$$

No need to prime-factorize again.

18. WHEN DOES THIS PRODUCT FORMULA APPLY?

Directly:

$$\boxed{\operatorname{HCF} \times \operatorname{LCM} = \text{product}}$$

for **two numbers**.

& bTRAP

Do not blindly use:

$$\operatorname{HCF} \times \operatorname{LCM} = abc$$

for three numbers.

It is generally false.

19. COPRIME NUMBERS

Two numbers are **coprime** if their HCF is 1.

Example:

$$\{ 8, 15 \}$$

Factors share only 1.

Therefore:

$$\{ \gcd(8,15)=1 \}$$

They are coprime.

Important

Coprime numbers do **not** need to be prime individually.

8 is composite.

15 is composite.

But they are coprime to each other.

20. LCM OF COPRIME NUMBERS

If:

$$\{ \gcd(a,b)=1 \}$$

then:

$$\{ \operatorname{LCM}(a,b)=ab \}$$

Example:

$$\{ 8, 15 \}$$

$$\{ \operatorname{LCM}=8 \times 15=120 \}$$

21. PRIME NUMBERS AS A SPECIAL CASE

For distinct primes:

$$\{ p, q \}$$

$$\{ \gcd(p,q)=1 \}$$

and:

$$\{ \operatorname{LCM}=pq \}$$

Example:

$$\{ 7, 11 \}$$

$$\{ \operatorname{HCF}=1 \}$$

$$\{ \operatorname{LCM}=77 \}$$

22. WHEN ONE NUMBER DIVIDES ANOTHER

Example:

$$\backslash 12, \backslash 36 \backslash$$

Since:

$$\backslash 12 \backslash \text{mid} 36 \backslash$$

then:

$$\backslash \boxed{\text{HCF}=12} \backslash$$

and:

$$\backslash \boxed{\text{LCM}=36} \backslash$$

General rule

If:

$$\backslash a \backslash \text{mid} b \backslash$$

then:

$$\backslash \text{gcd}(a,b)=a \backslash$$

$$\backslash \text{LCM}(a,b)=b \backslash$$

assuming $\backslash(a < b) \backslash$.

23. FAST EXAMPLE

Find HCF and LCM of:

$$\backslash 25, \backslash 100 \backslash$$

Since:

$$\backslash 25 \backslash \text{mid} 100 \backslash$$

Therefore:

$$\backslash \boxed{\text{HCF}=25} \backslash$$

$$\backslash \boxed{\text{LCM}=100} \backslash$$

No calculation needed.

24. HCF BY COMMON DIVISION METHOD

Find HCF of:

$$\backslash 48, \backslash 60 \backslash$$

Divide both by common primes.

$$\backslash 48, \backslash 60 \backslash$$

Both divisible by 2:

$$\backslash 24, \backslash 30 \backslash$$

Again by 2:

$$\begin{array}{r} 12 \\ 2 \\ \hline 15 \end{array}$$

Both by 3:

$$\begin{array}{r} 4 \\ 3 \\ \hline 5 \end{array}$$

Common divisors multiplied:

$$2 \times 2 \times 3 = \boxed{12}$$

25. LCM USING LADDER/DIVISION METHOD

For:

$$\begin{array}{r} 12 \\ 18 \end{array}$$

Divide using prime factors until all reach 1.

$$\begin{array}{r} 12 \\ 18 \end{array}$$

divide by 2:

$$\begin{array}{r} 6 \\ 9 \end{array}$$

divide by 2:

$$\begin{array}{r} 3 \\ 9 \end{array}$$

divide by 3:

$$\begin{array}{r} 1 \\ 3 \end{array}$$

divide by 3:

$$\begin{array}{r} 1 \\ 1 \end{array}$$

Multiply divisors:

$$2 \times 2 \times 3 \times 3 = \boxed{36}$$

This is a common placement method.

26. EUCLIDEAN ALGORITHM — VERY IMPORTANT

For large numbers, GCD can be found using remainders.

Principle:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

Repeat until remainder becomes 0.

Last non-zero remainder is GCD.

27. EXAMPLE — EUCLIDEAN ALGORITHM

Find:

$$\gcd(252, 105)$$

Divide:

$$\lfloor 252=105(2)+42 \rfloor$$

Therefore:

$$\lfloor \gcd(252,105)=\gcd(105,42) \rfloor$$

Now:

$$\lfloor 105=42(2)+21 \rfloor$$

Thus:

$$\lfloor \gcd(105,42)=\gcd(42,21) \rfloor$$

Now:

$$\lfloor 42=21(2)+0 \rfloor$$

Therefore:

$$\lfloor \boxed{\gcd(252,105)=21} \rfloor$$

28. WHY EUCLIDEAN ALGORITHM WORKS

If:

$$\lfloor a=bq+r \rfloor$$

then any number dividing both $\lfloor a \rfloor$ and $\lfloor b \rfloor$ also divides:

$$\lfloor a-bq=r \rfloor$$

So common divisors of $\lfloor a,b \rfloor$ are the same as common divisors of $\lfloor b,r \rfloor$.

Hence:

$$\lfloor \gcd(a,b)=\gcd(b,r) \rfloor$$

This is mathematically important and highly useful.

29. FAST EUCLIDEAN EXAMPLE

Find HCF:

$$\lfloor 391, \lfloor 299 \rfloor \rfloor$$

Subtract/divide:

$$\lfloor 391-299=92 \rfloor$$

Then:

$$\lfloor 299=92(3)+23 \rfloor$$

$$\lfloor 92=23(4) \rfloor$$

Therefore:

$$\lfloor \boxed{\text{HCF}=23} \rfloor$$

30. HCF OF MORE THAN TWO NUMBERS

Compute sequentially.

Example:

$$\backslash \gcd(24,36,60) \backslash$$

First:

$$\backslash \gcd(24,36)=12 \backslash$$

Then:

$$\backslash \gcd(12,60)=12 \backslash$$

Therefore:

$$\backslash \boxed{12} \backslash$$

31. LCM OF MORE THAN TWO NUMBERS

Likewise:

$$\backslash \text{LCM}(4,6,8) \backslash$$

First:

$$\backslash \text{LCM}(4,6)=12 \backslash$$

Then:

$$\backslash \text{LCM}(12,8)=24 \backslash$$

Therefore:

$$\backslash \boxed{24} \backslash$$

32. HCF OF FRACTIONS — PLACEMENT FORMULA

For positive fractions in lowest terms:

$$\backslash \boxed{\text{HCF} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}} \backslash$$

Example:

Find HCF of:

$$\backslash \frac{23}{49} \backslash$$

Numerator HCF:

$$\backslash \text{HCF}(2,4)=2 \backslash$$

Denominator LCM:

$$\backslash \text{LCM}(3,9)=9 \backslash$$

Therefore:

$$\backslash \boxed{\frac{2}{9}} \backslash$$

This is more placement-oriented than GATE core.

33. LCM OF FRACTIONS

For positive fractions in lowest terms:

$$\boxed{\text{LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}}$$

Example:

$$\frac{2}{3}, \frac{4}{9}$$

LCM of numerators:

$$\text{LCM}(2, 4) = 4$$

HCF of denominators:

$$\text{HCF}(3, 9) = 3$$

Thus:

$$\boxed{\frac{4}{3}}$$

34. IMPORTANT CONDITION

Before applying fraction HCF/LCM formulas, fractions should be in **lowest terms**.

This avoids wrong answers.

35. WORD-PROBLEM SIGNAL — WHEN TO USE HCF

Use HCF when question asks for:

- largest possible equal size
- maximum length
- greatest number of identical groups
- largest tile
- largest measuring unit
- maximum capacity dividing all exactly

Memory

Largest equal division !' HCF

36. WORD-PROBLEM SIGNAL — WHEN TO USE LCM

Use LCM when question asks:

- when events meet again
- first common time
- smallest number divisible by all
- bells ring together

- lights flash together
- buses arrive together
- repeating cycles synchronize

Memory

First common repetition !' LCM

37. HCF WORD PROBLEM — TILE SIZE

A rectangular floor is:

$$[84\text{ cm} \times 108\text{ cm}]$$

Find largest square tile that can cover it exactly.

Tile side must divide both:

$$[84, 108]$$

So find HCF.

Prime factors:

$$[84 = 2^2 \times 3 \times 7]$$

$$[108 = 2^2 \times 3^3]$$

HCF:

$$[2^2 \times 3 = 12]$$

Therefore largest square tile:

$$[\boxed{12\text{ cm} \times 12\text{ cm}}]$$

38. NUMBER OF TILES

Same floor:

$$[84 \times 108]$$

Tile:

$$[12 \times 12]$$

Tiles along length:

$$[84/12 = 7]$$

Along width:

$$[108/12 = 9]$$

Total:

$$[7 \times 9 = \boxed{63}]$$

39. HCF WORD PROBLEM — GROUPING

There are:

48 red balls

and:

72 blue balls

They must be arranged into maximum number of identical groups, each group having equal numbers of red balls and equal numbers of blue balls.

Maximum groups:

$\text{HCF}(48, 72)$

$= 24$

So:

24 groups

Each group gets:

$48/24 = 2 \text{ red}$

$72/24 = 3 \text{ blue}$

40. LCM WORD PROBLEM — BELLS

Bell A rings every 6 minutes.

Bell B every 8 minutes.

If they ring together now, when next together?

Find:

$\text{LCM}(6, 8)$

$= 24$

Therefore:

24 minutes

41. THREE REPEATING EVENTS

Three lights flash every:

$4, 6, 10$

seconds.

When together again?

$\text{LCM}(4, 6, 10)$

Prime factors:

$4 = 2^2$

$6 = 2 \times 3$

$$\backslash 10=2\times 5 \backslash$$

LCM:

$$\backslash 2^2\times 3\times 5 = 60 \backslash$$

Answer:

$$\backslash \boxed{60\text{seconds}} \backslash$$

42. SMALLEST NUMBER DIVISIBLE BY SEVERAL NUMBERS

Find smallest number divisible by:

$$\backslash 8,\backslash 12,\backslash 15 \backslash$$

This is simply:

$$\backslash \text{LCM}(8,12,15) \backslash$$

Factor:

$$\backslash 8=2^3 \backslash$$

$$\backslash 12=2^2\times 3 \backslash$$

$$\backslash 15=3\times 5 \backslash$$

LCM:

$$\backslash 2^3\times 3\times 5 = 120 \backslash$$

Answer:

$$\backslash \boxed{120} \backslash$$

43. GREATEST NUMBER DIVIDING SEVERAL NUMBERS

Find greatest number that divides:

$$\backslash 84,\backslash 126,\backslash 210 \backslash$$

exactly.

This is HCF:

$$\backslash \text{gcd}(84,126,210) \backslash$$

$$\backslash \text{gcd}(84,126)=42 \backslash$$

$$\backslash \text{gcd}(42,210)=42 \backslash$$

Answer:

$$\backslash \boxed{42} \backslash$$

44. GREATEST NUMBER THAT DIVIDES WITH SAME REMAINDER

This is a very important pattern.

Question:

Find greatest number that divides:

$$\sqrt[4]{43, 91, 183}$$

leaving the same remainder.

If same divisor gives same remainder, subtract numbers.

Differences:

$$\sqrt[4]{91-43=48}$$

$$\sqrt[4]{183-91=92}$$

$$\sqrt[4]{183-43=140}$$

Required divisor must divide all differences.

So:

$$\sqrt[4]{\text{HCF}(48, 92, 140)}$$

First:

$$\sqrt[4]{\text{HCF}(48, 92)=4}$$

Then:

$$\sqrt[4]{\text{HCF}(4, 140)=4}$$

Therefore:

$$\sqrt[4]{\boxed{4}}$$

45. WHY SAME-REMAINDER METHOD WORKS

Suppose:

$$\sqrt[4]{A=dq_1+r}$$

and:

$$\sqrt[4]{B=dq_2+r}$$

Subtract:

$$\sqrt[4]{A-B=d(q_1-q_2)}$$

So:

$$\sqrt[4]{d \mid (A-B)}$$

Hence the divisor must divide differences.

Must remember

Same remainder !' take differences !' HCF

46. EXAMPLE

Find greatest number that divides 122 and 170 leaving same remainder.

Difference:

$$\lfloor 170-122=48 \rfloor$$

Greatest possible divisor:

$$\lfloor \boxed{48} \rfloor$$

Check:

$$\lfloor 122=48(2)+26 \rfloor$$

$$\lfloor 170=48(3)+26 \rfloor$$

Same remainder 26.

47. NUMBERS WITH GIVEN HCF

Suppose two numbers have HCF:

$$\lfloor 12 \rfloor$$

Then numbers can be written as:

$$\lfloor 12a, 12b \rfloor$$

where:

$$\lfloor \gcd(a,b)=1 \rfloor$$

This is extremely useful.

48. EXAMPLE

Two numbers have HCF 6 and ratio:

$$\lfloor 4:5 \rfloor$$

Since 4 and 5 are coprime:

numbers are:

$$\lfloor 6 \times 4 = 24 \rfloor$$

$$\lfloor 6 \times 5 = 30 \rfloor$$

Therefore:

$$\lfloor \boxed{24, 30} \rfloor$$

49. IMPORTANT RATIO CONDITION

If ratio terms are not coprime, simplify first.

Suppose ratio:

$$\lfloor 8:12 \rfloor$$

Simplify:

$$\lfloor 2:3 \rfloor$$

Then if HCF is $\lfloor h \rfloor$, numbers are:

$\sqrt[2]{h}, \sqrt[3]{h}$

not necessarily:

$\sqrt[8]{h}, \sqrt[12]{h}$

because then HCF would contain extra factor 4.

50. NUMBERS WITH GIVEN LCM

Suppose:

$\sqrt[LCM=60]{}$

Any candidate numbers must each divide 60.

Examples:

$\sqrt[12, 20]{}$

have:

$\sqrt[LCM=60]{}$

But random numbers bigger than 60 cannot have LCM 60.

This helps option elimination.

51. HCF MUST DIVIDE BOTH NUMBERS

If HCF of two numbers is 12, then both numbers must be divisible by:

$\sqrt[12]{}$

Obvious but useful for eliminating options.

52. LCM MUST BE MULTIPLE OF BOTH NUMBERS

If LCM of two numbers is 120, then:

$\sqrt[a \mid 120, b \mid 120]{}$

So each number must be a factor of 120.

53. HCF ALWAYS $\leq$ EACH NUMBER

For positive numbers:

$\sqrt[HCF(a,b) \leq \min(a,b)]{}$

54. LCM ALWAYS $\geq$ EACH NUMBER

$\sqrt[LCM(a,b) \geq \max(a,b)]{}$

These are quick sanity checks.

55. GATE-STYLE QUESTION 1

Two numbers have:

$$\text{HCF}=6, \text{LCM}=180$$

If one number is 30, find the other.

Use:

$$ab = \text{HCF} \times \text{LCM}$$

$$30b = 6 \times 180$$

$$b = \frac{1080}{30}$$

$$b = \boxed{36}$$

56. GATE-STYLE QUESTION 2

Find:

$$\text{gcd}(414, 662)$$

Use Euclidean algorithm:

$$662 = 414(1) + 248$$

$$414 = 248(1) + 166$$

$$248 = 166(1) + 82$$

$$166 = 82(2) + 2$$

$$82 = 2(41)$$

Therefore:

$$\boxed{\text{GCD}=2}$$

57. GATE-STYLE QUESTION 3

Two positive integers are coprime and their product is:

$$221$$

Possible prime factorization:

$$221 = 13 \times 17$$

Since 13 and 17 are coprime, one possible pair is:

$$\boxed{13, 17}$$

If the problem says both are greater than 1 and their LCM is 221, this becomes the natural pair.

58. PLACEMENT-STYLE QUESTION 1

What largest square tile can exactly cover a floor of:

$$\text{HCF}(96, 144) = ?$$

Find:

$$\text{HCF}(96, 144)$$

Factor:

$$96 = 2^5 \times 3$$

$$144 = 2^4 \times 3^2$$

HCF:

$$2^4 \times 3 = 48$$

Answer:

$$\boxed{48}$$

59. PLACEMENT-STYLE QUESTION 2

Three buses arrive every:

$$12, 18, 30$$

minutes.

They arrive together at 8:00 AM.

When next together?

Find:

$$\text{LCM}(12, 18, 30)$$

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

$$30 = 2 \times 3 \times 5$$

LCM:

$$2^2 \times 3^2 \times 5 = 180$$

180 minutes:

$$= 3 \text{ hours}$$

So next together:

$$\boxed{11:00 \text{ AM}}$$

60. PLACEMENT-STYLE QUESTION 3

Find greatest number that divides:

$$75, 123, 171$$

leaving the same remainder.

Differences:

$$123 - 75 = 48$$

$$\boxed{171-123=48}$$

HCF:

$$\boxed{48}$$

Answer:

$$\boxed{48}$$

61. NORMAL METHOD VS FAST METHOD

Find LCM of:

$$18, 24$$

Normal method

Prime factorization:

$$18 = 2 \times 3^2$$

$$24 = 2^3 \times 3$$

$$\text{LCM} = 2^3 \times 3^2 = 72$$

Faster if HCF known

If:

$$\text{HCF} = 6$$

then:

$$\text{LCM} = \frac{18 \times 24}{6}$$

$$= 18 \times 4$$

$$= \boxed{72}$$

Choose whichever requires fewer steps.

62. COMMON TRAPS

⚠️ TRAP 1 — Confusing HCF and LCM

Largest equal division !' HCF.

First common repetition !' LCM.

⚠️ TRAP 2 — Wrong exponent

For HCF:

minimum power

For LCM:

maximum power

⚠️ TRAP 3 — Product relation for three numbers

$$\boxed{\text{HCF} \times \text{LCM} = ab}$$

is guaranteed in this simple form for **two numbers**, not arbitrary three numbers.

& pTRAP 4 — Coprime "both prime

$$\boxed{8, 15}$$

are coprime.

& pTRAP 5 — Same remainder questions

Do not directly HCF the original numbers.

First subtract.

& pTRAP 6 — Ratio not reduced

If ratio is:

$$\boxed{8:12}$$

reduce to:

$$\boxed{2:3}$$

before linking to HCF.

63. MEMORY BOX

HCF

Largest quantity that divides everything.

$$\boxed{\text{Minimum prime exponents}}$$

LCM

Smallest quantity divisible by everything.

$$\boxed{\text{Maximum prime exponents}}$$

Two numbers

$$\boxed{\text{HCF} \times \text{LCM} = ab}$$

Same remainder

$$\boxed{\text{Differences} \rightarrow \text{HCF}}$$

Repeating events

$$\boxed{\text{LCM}}$$

64. QUICK REVISION

For:

$$\backslash[a=2^3 3^2 5 \backslash]$$

$$\backslash[b=2^2 3^4 7 \backslash]$$

HCF:

common primes, minimum powers:

$$\backslash[2^2 3^2 \backslash]$$

LCM:

all primes, maximum powers:

$$\backslash[2^3 3^4 5 7 \backslash]$$

That is the whole prime-factor method.

PRACTICE — LEVEL 0

Q1

Find:

$$\backslash[\text{HCF}(18,30) \backslash]$$

Q2

Find:

$$\backslash[\text{LCM}(8,12) \backslash]$$

Q3

Find HCF and LCM of:

$$\backslash[15, \backslash 25 \backslash]$$

Q4

Are 14 and 25 coprime?

Q5

If one number divides another, say 16 divides 64, find HCF and LCM.

LEVEL 1

Q6

Find:

$$\backslash[\text{HCF}(72,120) \backslash]$$

Q7

Find:

$$\backslash[\text{LCM}(18,30,45) \backslash]$$

Q8

If HCF of two numbers is 8, LCM is 240, and one number is 48, find the other.

Q9

Find greatest number that divides:

$$\sqrt[3]{85}, \sqrt[3]{133}$$

leaving same remainder.

Q10

Three bells ring every 8, 12 and 18 minutes. When will they next ring together?

LEVEL 2

Q11

Using Euclidean algorithm, find:

$$\sqrt[3]{\text{GCD}(867, 255)}$$

Q12

Two numbers have ratio:

$$\sqrt[3]{5:7}$$

and HCF 9. Find the numbers.

Q13

Find largest square tile that exactly covers a:

$$\sqrt[3]{168 \text{ cm} \times 252 \text{ cm}}$$

floor.

Q14

Find greatest number that divides:

$$\sqrt[3]{187, 233, 325}$$

leaving the same remainder.

Q15

If two numbers are coprime and their LCM is 143, what is their product?

LEVEL 3 — EXAM PRESSURE

Q16

The HCF and LCM of two numbers are 14 and 840 respectively. If one number is 70, find the other.

Q17

Find the least positive integer divisible by:

$$\sqrt[3]{12, 15, 18, 20}$$

Q18

Three machines complete a cycle every 24, 36 and 54 seconds. If they start together, after how many seconds will they synchronize again?

Q19

Find greatest integer (d) such that:

$$[341, 457, 573]$$

leave the same remainder when divided by (d) .

Q20

If:

$$[a = 2^4 \cdot 3^2 \cdot 5^3]$$

and:

$$[b = 2^2 \cdot 3^5 \cdot 5]$$

find HCF and LCM in prime-factor form.

ANSWERS**1**

$$[\boxed{6}]$$

2

$$[\boxed{24}]$$

3

$$[HCF=5]$$

$$[LCM=75]$$

4

Yes.

$$[\gcd(14, 25)=1]$$

5

$$[HCF=16]$$

$$[LCM=64]$$

6

$$[72=2^3 \cdot 3^2]$$

$$[120=2^3 \cdot 3 \cdot 5]$$

$$[\boxed{HCF=24}]$$

7

$$[18=2 \cdot 3^2]$$

$$\backslash[30=2\cdot3\cdot5 \backslash]$$

$$\backslash[45=3^2\cdot5 \backslash]$$

$$\backslash[\text{LCM}=2\cdot3^2\cdot5 \backslash]$$

$$\backslash[\boxed{90} \backslash]$$

8

$$\backslash[48x=8\cdot240 \backslash]$$

$$\backslash[x=40 \backslash]$$

$$\backslash[\boxed{40} \backslash]$$

9

Difference:

$$\backslash[133-85=48 \backslash]$$

$$\backslash[\boxed{48} \backslash]$$

10

$$\backslash[\text{LCM}(8,12,18) \backslash]$$

$$\backslash[=72 \backslash]$$

$$\backslash[\boxed{72\text{ minutes}} \backslash]$$

11

$$\backslash[867=255(3)+102 \backslash]$$

$$\backslash[255=102(2)+51 \backslash]$$

$$\backslash[102=51(2) \backslash]$$

$$\backslash[\boxed{51} \backslash]$$

12

Since 5 and 7 are coprime:

$$\backslash[5(9),7(9) \backslash]$$

$$\backslash[\boxed{45,63} \backslash]$$

13

$$\backslash[\text{HCF}(168,252)=84 \backslash]$$

Largest tile side:

$$\backslash[\boxed{84\text{ cm}} \backslash]$$

14

Differences:

$$\backslash[233-187=46 \backslash]$$

$$\backslash[325-233=92 \backslash]$$

$$\backslash[\text{HCF}(46,92)=46 \backslash]$$

$$\backslash[\boxed{46} \backslash]$$

15

For coprime numbers:

$$\boxed{LCM=ab}$$

Thus:

$$\boxed{143}$$

16

$$70x=14\times 840$$

$$x=\frac{11760}{70}=168$$

$$\boxed{168}$$

17

Prime factors:

$$12=2^2 \cdot 3$$

$$15=3 \cdot 5$$

$$18=2 \cdot 3^2$$

$$20=2^2 \cdot 5$$

LCM:

$$2^2 \cdot 3^2 \cdot 5 = 180$$

$$\boxed{180}$$

18

$$LCM(24,36,54)$$

$$24=2^3 \cdot 3$$

$$36=2^2 \cdot 3^2$$

$$54=2 \cdot 3^3$$

LCM:

$$2^3 \cdot 3^3 = 216$$

$$\boxed{216 \text{ seconds}}$$

19

Differences:

$$457-341=116$$

$$573-457=116$$

Therefore:

$$\boxed{116}$$

20

HCF:

$$\boxed{2^2 \cdot 3^2 \cdot 5}$$

LCM:

$\boxed{2^4 3^5 5^3}$

CHAPTER 5 STATUS

Completed so far:

- factors and multiples '
- HCF/GCD meaning '
- LCM meaning '
- prime-factorization method '
- minimum/maximum exponent rule '
- listing method '
- division/ladder method '
- Euclidean algorithm '
- $\text{HCF} \times \text{LCM}$ relation '
- coprime numbers '
- HCF/LCM of multiple numbers '
- basic fraction HCF/LCM '
- largest equal-group problems '
- tiles/grouping applications '
- repeated-event problems '
- same-remainder problems '
- ratio + HCF connection '
- GATE/placement questions '

But **Chapter 5 is not complete yet.**

Next we will cover:

- advanced HCF/LCM word problems
- remainder forms such as "leaves remainder $\backslash(r)$ "
- least number to add/subtract for common divisibility
- numbers leaving different specified remainders
- HCF/LCM + ratios
- finding possible pairs from HCF and LCM
- number of possible pairs
- advanced synchronization/counting
- denominator LCM tricks used later in Time & Work
- GATE/placement mixed problems

Progress

Completed: 4/76 **Current:** 0 = Chapter 5 — Factors, Multiples, HCF & LCM **Next:** Chapter 5 Part 2 — **Advanced HCF/LCM applications & remainder patterns**

Send **NEXT**.

USER

next

USER

next

ASSISTANT

CHAPTER 5 — FACTORS, MULTIPLES, HCF & LCM

PART 2 — ADVANCED HCF/LCM APPLICATIONS, REMAINDER PATTERNS & POSSIBLE PAIRS

Status: $\emptyset = 0\%$ PROGRESS GATE: High Placements: High Priority: $\emptyset = 0\%$ Per 1

Now we move from the basic formulas to the patterns that actually appear in tougher aptitude questions.

1. SAME REMAINDER PATTERN— MASTER RULE

Suppose a divisor (d) divides several numbers leaving the **same remainder**.

Example:

- 47 leaves remainder (r)
- 83 leaves remainder (r)
- 119 leaves remainder (r)

Then subtract:

$$[83 - 47 = 36]$$

$$[119 - 83 = 36]$$

The divisor must divide the differences.

So:

$$[d \mid 36]$$

The **greatest possible divisor** is:

$$[\boxed{36}]$$

Master rule

Same remainder !' subtract the numbers !' take HCF of differences.

2. WHY THIS WORKS

Suppose:

$$A = dq_1 + r$$

and:

$$B = dq_2 + r$$

Subtract:

$$A - B = d(q_1 - q_2)$$

So:

$$d \mid (A - B)$$

The remainder disappears.

That is why differences are used.

3. GREATEST NUMBER LEAVING SAME REMAINDER

Find greatest number that divides:

$$187, 233, 325$$

leaving the same remainder.

Differences:

$$233 - 187 = 46$$

$$325 - 233 = 92$$

Now:

$$\text{HCF}(46, 92) = 46$$

Answer:

$$\boxed{46}$$

4. YOU DO NOT NEED EVERY PAIRWISE DIFFERENCE

For:

$$a, b, c$$

you can take:

$$b - a, c - b$$

and HCF them.

You don't necessarily need:

$$c - a$$

because it is already a combination of the first two.

5. SAME REMAINDER IS NOT ALWAYS ZERO

In the previous problem, divisor:

$$\sqrt[4]{46}$$

leaves:

$$\sqrt[4]{187=46(4)+3}$$

$$\sqrt[4]{233=46(5)+3}$$

$$\sqrt[4]{325=46(7)+3}$$

Same remainder:

$$\sqrt[4]{\boxed{3}}$$

The question did not say the numbers were exactly divisible.

6. NUMBERS LEAVE SPECIFIC SAME REMAINDER

Find greatest number which divides:

$$\sqrt[4]{65, 89, 113}$$

leaving remainder 5 each time.

Subtract the remainder first:

$$\sqrt[4]{65-5=60}$$

$$\sqrt[4]{89-5=84}$$

$$\sqrt[4]{113-5=108}$$

Now find:

$$\sqrt[4]{\text{HCF}(60,84,108)}$$

$$\sqrt[4]{\text{HCF}(60,84)=12}$$

$$\sqrt[4]{\text{HCF}(12,108)=12}$$

Answer:

$$\sqrt[4]{\boxed{12}}$$

7. GENERAL RULE FOR KNOWN REMAINDER

If numbers:

$$\sqrt[4]{N_1, N_2, N_3}$$

all leave remainder (r) when divided by (d) , then:

$$\sqrt[4]{d \mid (N_1 - r)}$$

$$\sqrt[4]{d \mid (N_2 - r)}$$

$$\sqrt[4]{d \mid (N_3 - r)}$$

So:

$$\sqrt[4]{\boxed{d = \text{HCF}(N_1 - r, N_2 - r, N_3 - r)}}$$

for the greatest possible divisor.

8. DIFFERENT SPECIFIED REMAINDERS

Suppose a number (d) divides:

- 62 leaving remainder 2
- 95 leaving remainder 5
- 128 leaving remainder 8

Subtract the respective remainders:

$$[62-2=60]$$

$$[95-5=90]$$

$$[128-8=120]$$

So (d) must divide:

$$[60, 90, 120]$$

Hence:

$$[d=\text{HCF}(60,90,120)]$$

$$[\boxed{30}]$$

Important

If remainders differ, subtract each specified remainder before taking HCF.

9. SMALLEST NUMBER WHICH LEAVES SAME REMAINDER

Suppose:

Find the least number which when divided by 6, 8 and 10 leaves remainder 3 in each case.

If (N) leaves remainder 3:

$$[N-3]$$

must be divisible by:

$$[6,8,10]$$

So:

$$[N-3=\text{LCM}(6,8,10)]$$

Prime factors:

$$[6=2\cdot 3]$$

$$[8=2^3]$$

$$[10=2\cdot 5]$$

LCM:

$$[2^3\cdot 3\cdot 5=120]$$

Therefore:

$$\lfloor N=120+3 \rfloor$$

$$\lfloor \boxed{123} \rfloor$$

Memory

Least number + same remainder != LCM + remainder.

10. IMPORTANT QUALIFICATION

Suppose the question asks:

Find the least positive number that leaves remainder 3 when divided by 6, 8, 10.

Technically, $\lfloor(3)\rfloor$ itself leaves remainder 3 on division by all three, since it is smaller than each divisor.

But aptitude questions usually intend the **least number greater than the divisors**, or a common-cycle form.

So always read wording carefully.

If the intended exam pattern is "least number greater than the divisors," use:

$$\lfloor \text{LCM}+r \rfloor$$

11. LEAST NUMBER DIVISIBLE AFTER ADDING A CONSTANT

Question:

Find the least number which becomes divisible by:

$$\lfloor 12,18,30 \rfloor$$

after adding 5.

Let number be $\lfloor(N)\rfloor$.

Then:

$$\lfloor N+5 \rfloor$$

must be divisible by all three.

So:

$$\lfloor N+5=\text{LCM}(12,18,30) \rfloor$$

We already know:

$$\lfloor \text{LCM}=180 \rfloor$$

Thus:

$$\lfloor N=175 \rfloor$$

Answer:

$$\lfloor \boxed{175} \rfloor$$

12. LEAST NUMBER DIVISIBLE AFTER SUBTRACTING A CONSTANT

Find least number which becomes divisible by:

$$\{ 8, 12, 15 \}$$

after subtracting 7.

Then:

$$\{ N-7 \}$$

must equal the least common multiple.

$$\{ \text{LCM}(8, 12, 15) = 120 \}$$

So:

$$\{ N = 127 \}$$

Answer:

$$\boxed{127}$$

13. SAME REMAINDER WITH SEVERAL DIVISORS

Question:

Find least number that leaves remainder 4 when divided by:

$$\{ 5, 6, 8 \}$$

Then:

$$\{ N-4 \}$$

is divisible by all.

$$\{ \text{LCM}(5, 6, 8) \}$$

$$\{ 5 = 5 \}$$

$$\{ 6 = 2 \cdot 3 \}$$

$$\{ 8 = 2^3 \}$$

LCM:

$$\{ 2^3 \cdot 3 \cdot 5 = 120 \}$$

Therefore:

$$\boxed{N = 124}$$

assuming the intended common-multiple pattern.

14. REMAINDER ONE LESS THAN EACH DIVISOR

Classic question:

Find least number which leaves:

- remainder 4 when divided by 5
- remainder 6 when divided by 7

- remainder 8 when divided by 9

Observe:

$$4 = 5 - 1$$

$$6 = 7 - 1$$

$$8 = 9 - 1$$

So:

$$N + 1$$

is divisible by:

$$5, 7, 9$$

Thus:

$$N + 1 = \text{LCM}(5, 7, 9)$$

Since pairwise coprime:

$$\text{LCM} = 5 \times 7 \times 9 = 315$$

Therefore:

$$\boxed{N = 314}$$

Pattern

If remainder is always divisor (-1) :

$$\boxed{N = \text{LCM}(\text{divisors}) - 1}$$

15. REMAINDER TWO LESS THAN EACH DIVISOR

Suppose:

- divided by 7 !' remainder 5
- divided by 9 !' remainder 7
- divided by 11 !' remainder 9

Each remainder is:

$$d - 2$$

Therefore:

$$N + 2$$

is divisible by all divisors.

So:

$$N = \text{LCM}(7, 9, 11) - 2$$

Since coprime:

$$\text{LCM} = 693$$

Thus:

$$\boxed{691}$$

16. GENERAL PATTERN

If:

$$N \not\equiv -k \pmod{d_1}$$

$$N \not\equiv -k \pmod{d_2}$$

etc., then:

$$N+k$$

is divisible by all.

Therefore:

$$\boxed{N = \text{LCM}(d_1, d_2, \dots) - k}$$

for the least intended common-cycle solution.

17. HCF + RATIO QUESTIONS

Suppose two numbers are in ratio:

$$5:7$$

and HCF is:

$$6$$

Since 5 and 7 are coprime, numbers are:

$$5 \times 6 = 30$$

$$7 \times 6 = 42$$

Answer:

$$\boxed{30, 42}$$

18. WHY THE RATIO TERMS MUST BE COPRIME

Suppose the ratio is:

$$10:15$$

Reduce:

$$2:3$$

If HCF is 7:

$$2 \times 7 = 14$$

$$3 \times 7 = 21$$

Answer:

$$\boxed{14, 21}$$

If you incorrectly use 10:15:

$$70, 105$$

their HCF is:

$$\sqrt[3]{35}$$

not 7.

19. LCM WHEN HCF AND RATIO ARE GIVEN

Two numbers have:

$$\sqrt[3]{\text{HCF}=8}$$

and ratio:

$$\sqrt[3]{3:5}$$

Numbers:

$$\sqrt[3]{24, 40}$$

Since ratio terms are coprime:

$$\sqrt[3]{\text{LCM}=8 \times 3 \times 5}$$

$$\sqrt[3]{= \boxed{120}}$$

Shortcut

If:

$$\sqrt[3]{a:b=m:n}$$

with $\gcd(m,n)=1$, and HCF $(=h)$,

then numbers are:

$$\sqrt[3]{hm, hn}$$

and:

$$\sqrt[3]{\boxed{\text{LCM}=hmn}}$$

20. PRODUCT WHEN HCF AND RATIO ARE GIVEN

Same setup:

$$\sqrt[3]{a=hm, \text{quad } b=hn}$$

Product:

$$\sqrt[3]{ab=h^2mn}$$

Example:

$$\sqrt[3]{h=6, \text{quad } m:n=4:7}$$

Numbers:

$$\sqrt[3]{24, 42}$$

Product:

$$\sqrt[3]{24 \times 42 = 1008}$$

Shortcut:

$$\sqrt[6]{6^2 \times 4 \times 7} = 36 \times 28 = 1008$$

21. SUM WHEN HCF AND RATIO ARE GIVEN

Numbers:

$$\sqrt{hm, \sqrt{hn}}$$

Sum:

$$\sqrt{h(m+n)}$$

Example:

HCF 5, ratio 3:7.

Sum:

$$\sqrt{5(3+7)=50}$$

Numbers:

$$\sqrt{15, \sqrt{35}}$$

Correct.

22. GIVEN SUM AND HCF

Question:

Two numbers have HCF 6 and ratio 4:5. Find their sum.

Numbers:

$$\sqrt{24, \sqrt{30}}$$

Sum:

$$\sqrt{\boxed{54}}$$

Or directly:

$$\sqrt{6(4+5)=54}$$

23. GIVEN SUM AND RATIO — FIND HCF

Suppose two numbers are in ratio:

$$\sqrt{3:5}$$

and sum is:

$$\sqrt{64}$$

Let numbers:

$$\sqrt{3k, \sqrt{5k}}$$

Then:

$$\sqrt{8k=64}$$

$$\boxed{k=8}$$

Numbers:

$$\boxed{24, 40}$$

Since 3 and 5 are coprime:

$$\boxed{\text{HCF}=8}$$

24. GIVEN DIFFERENCE AND RATIO — FIND NUMBERS

Ratio:

$$\boxed{5:7}$$

Difference:

$$\boxed{18}$$

Difference in ratio parts:

$$\boxed{2}$$

So one part:

$$\boxed{18/2=9}$$

Numbers:

$$\boxed{45, 63}$$

HCF:

$$\boxed{9}$$

25. HCF AND LCM — FIND THE OTHER NUMBER

For two numbers:

$$\boxed{ab = \text{HCF} \times \text{LCM}}$$

Example:

$$\boxed{\text{HCF}=12, \text{LCM}=420}$$

One number:

$$\boxed{60}$$

Other:

$$\boxed{\frac{12 \times 420}{60}}$$

$$\boxed{=84}$$

Answer:

$$\boxed{84}$$

26. VALIDITY CHECK FOR HCF AND LCM

Very important:

For any two positive integers:

$$\boxed{\text{HCF} \mid \text{LCM}}$$

That means HCF must divide LCM.

Example:

Can two numbers have:

$$\text{HCF}=12, \text{LCM}=70?$$

No, because:

$$12 \nmid 70$$

So such integers do not exist.

Fast option elimination

Always check this first.

27. NUMBERS WITH GIVEN HCF AND LCM

Suppose:

$$\text{HCF}=6, \text{LCM}=180$$

Write numbers as:

$$6x, 6y$$

with:

$$\gcd(x, y)=1$$

Also:

$$\text{LCM}=6xy$$

because (x, y) are coprime.

Thus:

$$6xy=180$$

$$xy=30$$

Now find coprime factor pairs of 30.

28. COPRIME FACTOR PAIRS

Factor pairs of 30:

$$1 \times 30$$

$$2 \times 15$$

$$3 \times 10$$

$$5 \times 6$$

Check coprime:

- $(1, 30)$!' yes
- $(2, 15)$!' yes
- $(3, 10)$!' yes
- $(5, 6)$!' yes

Thus possible number pairs:

$$\{ (6, 180) \}$$

$$\{ (12, 90) \}$$

$$\{ (18, 60) \}$$

$$\{ (30, 36) \}$$

All have:

$$\{ \text{HCF}=6, \text{LCM}=180 \}$$

29. NUMBER OF POSSIBLE PAIRS FROM HCF & LCM

Suppose:

$$\{ \text{HCF}=h, \text{LCM}=l \}$$

Let:

$$\{ n = \frac{lh}{h^2} \}$$

We need factor pairs:

$$\{ xy=n \}$$

such that:

$$\{ \gcd(x, y)=1 \}$$

Each distinct prime-power block of (n) must go completely to either (x) or (y) .

If:

$$\{ n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \}$$

then number of **unordered** coprime factor pairs is:

$$\{ \boxed{2^{k-1}} \}$$

for $(n > 1)$.

This is an advanced but useful placement/GATE number-theory shortcut.

30. EXAMPLE — COUNT POSSIBLE PAIRS

Given:

$$\{ \text{HCF}=4, \text{LCM}=240 \}$$

Then:

$$\{ n = \frac{240}{4} = 60 \}$$

Prime factorization:

$$60 = 2^2 \times 3 \times 5$$

Distinct prime factors:

$$k = 3$$

Number of unordered coprime factor pairs:

$$2^{3-1} = 4$$

So there are:

$$\boxed{4}$$

possible unordered pairs of numbers.

31. FIND THE PAIRS

For:

$$60 = 2^2 \times 3 \times 5$$

Coprime factor pairs:

$$(1, 60)$$

$$(4, 15)$$

$$(3, 20)$$

$$(5, 12)$$

Multiply each by HCF 4:

$$(4, 240)$$

$$(16, 60)$$

$$(12, 80)$$

$$(20, 48)$$

All valid.

32. MINIMUM SUM PAIR

If HCF and LCM are fixed, sometimes asked:

Which pair has minimum sum?

Among valid factor pairs (x, y) with:

$$xy = n$$

choose pair closest to:

$$\sqrt{n}$$

because product fixed !' sum minimized when numbers are as close as possible.

Example:

$$HCF = 6, LCM = 180$$

We found normalized pairs:

$$\{(1,30), (2,15), (3,10), (5,6)\}$$

Closest:

$$\{5,6\}$$

So numbers:

$$30, 36$$

Minimum sum:

$$\boxed{66}$$

33. MAXIMUM DIFFERENCE PAIR

With fixed product, maximum difference comes from most extreme factor pair.

Usually:

$$\{1, n\}$$

So with HCF h , pair:

$$\{h, \text{LCM}\}$$

gives maximum difference.

Example:

$$\{HCF=6, \text{LCM}=180\}$$

Most extreme pair:

$$\{6, 180\}$$

Difference:

$$174$$

34. REPEATING EVENTS — FIRST MEETING

Three bells ring every:

$$12, 15, 20$$

minutes.

When next together?

$$\text{LCM}(12, 15, 20)$$

$$12 = 2^2 \cdot 3$$

$$15 = 3 \cdot 5$$

$$20 = 2^2 \cdot 5$$

LCM:

$$2^2 \cdot 3 \cdot 5 = 60$$

Answer:

$\boxed{60 \text{ minutes}}$

35. HOW MANY TIMES DO THEY MEET?

Suppose they ring together at 8:00 AM and repeat every:

60 minutes

How many times do they ring together from 8:00 AM through 2:00 PM inclusive?

Times:

- 8
- 9
- 10
- 11
- 12
- 1
- 2

Count:

$\boxed{7}$

& pTrap

If starting instant is included, count it.

36. COUNTING SYNCHRONIZATIONS

If common interval is (L) and total time window is (T) :

Number of **future meetings after the start** up to time (T) :

$\left\lfloor \frac{TL}{L} \right\rfloor$

If start is also counted:

$\boxed{\left\lfloor \frac{TL}{L} \right\rfloor + 1}$

37. EXAMPLE

Machines synchronize every:

$24, 36, 54 \text{ sec}$

LCM:

216 sec

Over 1800 seconds, future synchronizations:

$\left\lfloor \frac{1800}{216} \right\rfloor = 8$

Including start:

$\boxed{9}$

38. HCF FOR CUTTING LENGTHS

Three wires have lengths:

$$\sqrt[96, 144, 168]{\text{cm}}$$

They must be cut into equal longest pieces with no waste.

Use:

$$\sqrt{\text{HCF}(96, 144, 168)}$$

$$\sqrt{\text{HCF}(96, 144) = 48}$$

$$\sqrt{\text{HCF}(48, 168) = 24}$$

Longest piece:

$$\sqrt{\boxed{24} \text{ cm}}$$

39. NUMBER OF PIECES

Same wires:

$$\sqrt{96/24 = 4}$$

$$\sqrt{144/24 = 6}$$

$$\sqrt{168/24 = 7}$$

Total pieces:

$$\sqrt{4 + 6 + 7 = \boxed{17}}$$

40. HCF FOR RECTANGULAR TILING

A floor measures:

$$\sqrt{180 \times 252}$$

Largest square tile side:

$$\sqrt{\text{HCF}(180, 252)}$$

Euclidean:

$$\sqrt{252 - 180 = 72}$$

$$\sqrt{180 = 72(2) + 36}$$

$$\sqrt{72 = 36(2)}$$

So:

$$\sqrt{\text{HCF} = 36}$$

Tile side:

$$\sqrt{\boxed{36}}$$

Number tiles:

$$\backslash[180/36=5 \backslash]$$

$$\backslash[252/36=7 \backslash]$$

$$\backslash[5\times 7=35 \backslash]$$

$$\backslash[\boxed{35} \backslash]$$

41. LCM IN FRACTION ADDITION — WHY IT MATTERS LATER

Suppose rates are:

$$\backslash[\frac{1}{12}, \text{quad} \frac{1}{18} \backslash]$$

In Time & Work, instead of fractions, choose total work:

$$\backslash[\text{LCM}(12,18)=36 \backslash]$$

Then efficiencies become:

$$\backslash[36/12=3 \backslash]$$

$$\backslash[36/18=2 \backslash]$$

This turns fractions into integers.

Important preview

This is why LCM is one of the most useful placement shortcuts for Time & Work.

42. LCM FOR REPEATING FRACTIONAL CYCLES

Suppose events repeat every:

$$\backslash[\frac{1}{2}, \text{quad} \frac{1}{3} \backslash]$$

hours.

Convert to minutes:

$$\backslash[30, \text{quad} 45 \backslash]$$

LCM:

$$\backslash[90 \text{ minutes} \backslash]$$

So they coincide every:

$$\backslash[\boxed{1.5 \text{ hours}} \backslash]$$

Use common units first.

43. HCF OF DECIMAL QUANTITIES — PRACTICAL METHOD

Suppose lengths:

$$\backslash[1.2, \ 1.8, \ 2.4 \text{ m} \backslash]$$

Find greatest equal length.

Remove decimals by multiplying by 10:

$\sqrt[12]{18 \cdot 24}$

HCF:

$\sqrt[6]{}$

Scale back:

$\sqrt[0.6]{m}$

Answer:

$\boxed{0.6 \text{ m}}$

44. LCM OF DECIMALS — PRACTICAL METHOD

Find LCM of:

$\sqrt[0.6]{}, \sqrt[0.8]{}$

Multiply both by 10:

$\sqrt[6]{}, \sqrt[8]{}$

LCM:

$\sqrt[24]{}$

Scale back:

$\boxed{2.4}$

This works when you scale both consistently.

45. GATE-STYLE QUESTION 1

Two integers have:

$\sqrt{HCF=12, \text{quad } LCM=720}$

If their ratio is:

$\sqrt{3:5}$

Can such numbers exist?

If ratio terms are coprime, numbers would be:

$\sqrt{36, 60}$

Their LCM:

$\sqrt{180}$

not 720.

Alternatively:

$\sqrt{LCM=h \cdot 3 \cdot 5 = 12 \cdot 15 = 180}$

So:

$\boxed{\text{No}}$

This is a consistency-check problem.

46. GATE-STYLE QUESTION 2

Find greatest integer that divides:

$$\{ 431, 619, 807 \}$$

leaving the same remainder.

Differences:

$$619 - 431 = 188$$

$$807 - 619 = 188$$

Therefore:

$$\boxed{188}$$

47. GATE-STYLE QUESTION 3

The HCF and LCM of two numbers are:

$$\{ 8, 168 \}$$

How many unordered pairs are possible?

Normalize:

$$\frac{168}{8} = 21$$

Prime factorization:

$$21 = 3 \times 7$$

Distinct primes:

$$k = 2$$

Pairs:

$$2^2 - 1 = 3$$

They are normalized:

$$(1, 21), (3, 7)$$

Actual pairs:

$$(8, 168), (24, 56)$$

Answer:

$$\boxed{2}$$

48. PLACEMENT-STYLE QUESTION 1

Find least number which when divided by 8, 12 and 18 leaves remainder 5.

$$N - 5$$

must be divisible by all.

$$\text{LCM}(8,12,18)$$

$$8=2^3$$

$$12=2^2 \cdot 3$$

$$18=2 \cdot 3^2$$

LCM:

$$2^3 \cdot 3^2 = 72$$

So:

$$N = 72 + 5$$

$$\boxed{77}$$

under the standard intended interpretation.

49. PLACEMENT-STYLE QUESTION 2

Find least number which leaves remainders:

- 4 when divided by 5
- 6 when divided by 7
- 10 when divided by 11

Each is one less than divisor.

Thus:

$$N+1$$

must be divisible by:

$$5, 7, 11$$

LCM:

$$5 \times 7 \times 11 = 385$$

Therefore:

$$\boxed{384}$$

50. PLACEMENT-STYLE QUESTION 3

Two numbers have ratio:

$$4:9$$

and LCM:

$$252$$

Find numbers.

Since 4 and 9 are coprime, if numbers are:

$$4h, 9h$$

then:

$$\boxed{LCM=36h}$$

So:

$$\boxed{36h=252}$$

$$\boxed{h=7}$$

Numbers:

$$\boxed{28, 63}$$

51. COMMON TRAPS

& pTRAP 1 — Same remainder vs specified remainder

Same unknown remainder:

take differences.

Specified same remainder $\pmod{r}$:

subtract $\pmod{r}$, then HCF.

& pTRAP 2 — Least number with remainder

If:

$$\boxed{N \equiv r}$$

for several divisors:

$$\boxed{N-r}$$

must be common multiple.

& pTRAP 3 — Ratio not reduced

Always reduce the ratio before connecting it to HCF.

& pTRAP 4 — Invalid HCF/LCM pair

HCF must divide LCM.

& pTRAP 5 — Counting meetings

Check whether the initial meeting is included.

& pTRAP 6 — Possible pairs

Not every factor pair works.

Normalized factors must be **coprime**.

52. MUST-MEMORIZE PATTERNS

Pattern A — Same remainder

$$\boxed{\text{HCF}(\text{differences})}$$

Pattern B — Same specified remainder (r)

$$\boxed{\text{HCF}(N_1-r, N_2-r, \dots)}$$

Pattern C — Least number leaving (r)

$$\boxed{\text{LCM}+r}$$

under standard common-cycle interpretation.

Pattern D — Remainder divisor $(-k)$

$$\boxed{\text{LCM}-k}$$

Pattern E — HCF (h) , ratio $(m:n)$

after simplifying ratio:

$$\boxed{hm, hn}$$

Pattern F — $\text{HCF} \times \text{LCM}$

$$\boxed{ab = \text{HCF} \times \text{LCM}}$$

for two numbers.

53. 5-MINUTE REVISION

Remember the decision rule:

“largest equal...”

! HCF

“first/least common...”

! LCM

“same remainder...”

! differences then HCF

“same remainder (r) with several divisors...”

! subtract (r) , then LCM for least common-number pattern

“one less than each divisor...”

! $\text{LCM} - 1$

“HCF + ratio...”

! reduce ratio, multiply by HCF

54. FORMULA SHEET

For two positive integers:

$$\boxed{H \times L = ab}$$

If:

$$a:b=m:n, \quad \gcd(m,n)=1, \quad \text{HCF}=h$$

then:

$$a=hm, \quad b=hn$$

$$\text{LCM}=hmn$$

If:

$$\frac{\text{LCM}}{\text{HCF}} = p_1^{a_1} \cdots p_k^{a_k}$$

then number of unordered pairs with that HCF and LCM:

$$\boxed{2^{k-1}}$$

for ratio (>1) .

PRACTICE — LEVEL 0

Q1

Find greatest number dividing 71 and 107 with the same remainder.

Q2

Find least standard common-cycle number leaving remainder 2 when divided by 4 and 6.

Q3

Two numbers have HCF 5 and ratio 3:7. Find numbers.

Q4

If HCF is 6, LCM is 180 and one number is 18, find the other.

Q5

Are HCF = 8 and LCM = 60 possible for two integers?

LEVEL 1

Q6

Find greatest number dividing:

$$102, 174, 246$$

leaving same remainder.

Q7

Find greatest number dividing:

$\backslash 74, \backslash 110, \backslash 146 \backslash$

leaving remainder 2 each time.

Q8

Find least number which leaves remainder 5 when divided by:

$\backslash 6, \backslash 8, \backslash 12 \backslash$

using the standard common-cycle interpretation.

Q9

Find least number which leaves remainder one less than divisor when divided by:

$\backslash 6, \backslash 8, \backslash 10 \backslash$

Q10

Two numbers are in ratio 5:8 and HCF 7. Find their LCM.

LEVEL 2

Q11

Two numbers have HCF 6 and LCM 630. How many unordered pairs are possible?

Q12

For the same HCF and LCM above, find the pair with minimum sum.

Q13

Two numbers are in ratio 3:8 and LCM is 360. Find the numbers.

Q14

Three machines run cycles of 18, 24 and 30 seconds. They start together. How many times are they together in the first 12 minutes, including the start?

Q15

Find greatest number that divides:

$\backslash 257, \backslash 353, \backslash 449 \backslash$

leaving the same remainder.

LEVEL 3 — EXAM PRESSURE

Q16

Find the least number which when divided by 8, 12 and 20 leaves remainders 5, 9 and 17 respectively.

Q17

Two positive integers have HCF 12 and LCM 720. Find the possible pair with minimum difference.

Q18

A room is $(504\text{ cm} \times 792\text{ cm})$. Find:

1. largest square tile side
2. number of tiles required

Q19

The HCF and LCM of two integers are 10 and 210 respectively. How many unordered pairs exist?

Q20

Three bells ring at intervals of 24, 36 and 54 seconds. If they ring together at 10:00:00, how many times do they ring together from 10:00:00 through 10:18:00 inclusive?

ANSWERS

Q1

$$\lfloor 107-71=36 \rfloor$$

$$\lfloor \boxed{36} \rfloor$$

Q2

$$\lfloor \text{LCM}(4,6)=12 \rfloor$$

$$\lfloor 12+2=\boxed{14} \rfloor$$

Q3

$$\lfloor 15, \lfloor 35 \rfloor$$

$$\lfloor \boxed{15,35} \rfloor$$

Q4

$$\lfloor 18x=6\times 180 \rfloor$$

$$\lfloor x=60 \rfloor$$

$$\lfloor \boxed{60} \rfloor$$

Q5

No.

$$\lfloor 8\mid 60 \rfloor$$

$$\lfloor \boxed{\text{Impossible}} \rfloor$$

Q6

Differences:

$$\lfloor 72, \lfloor 72 \rfloor$$

$$\lfloor \boxed{72} \rfloor$$

Q7

Subtract 2:

$$\lfloor 72, \lfloor 108, \lfloor 144 \rfloor$$

$$\lfloor \text{HCF}=36 \rfloor$$

$$\boxed{36}$$

Q8

$$\text{LCM}(6,8,12)=24$$

$$24+5=\boxed{29}$$

Q9

$$\text{LCM}(6,8,10)-1$$

$$=120-1$$

$$\boxed{119}$$

Q10

Numbers:

$$35, 56$$

LCM:

$$7 \times 5 \times 8 = \boxed{280}$$

Q11

$$\frac{630}{6}=105$$

$$105=3 \times 5 \times 7$$

Distinct prime factors:

$$k=3$$

Pairs:

$$2^{3-1} = \boxed{4}$$

Q12

Coprime factor pairs of 105:

$$(1,105),(3,35),(5,21),(7,15)$$

Closest:

$$7,15$$

Multiply by 6:

$$\boxed{42, 90}$$

Minimum sum:

$$132$$

Q13

Ratio 3:8 is coprime.

$$\text{LCM}=3 \times 8 \times h=24h$$

$$24h=360$$

$$h=15$$

Numbers:

$$\boxed{45, 120}$$

Q14

Common interval:

$$\text{LCM}(18, 24, 30)$$

$$18 = 2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

$$30 = 2 \cdot 3 \cdot 5$$

LCM:

$$2^3 \cdot 3^2 \cdot 5 = 360 \text{ sec}$$

12 minutes:

$$720 \text{ sec}$$

Future meetings:

$$720/360 = 2$$

Including start:

$$\boxed{3}$$

Q15

Differences:

$$353 - 257 = 96$$

$$449 - 353 = 96$$

$$\boxed{96}$$

Q16

Observe:

$$8 - 5 = 3$$

$$12 - 9 = 3$$

$$20 - 17 = 3$$

So each remainder is divisor (-3) .

Thus:

$$N + 3$$

must be divisible by 8, 12 and 20.

$$\text{LCM}(8, 12, 20) = 120$$

$$N = 120 - 3$$

$$\boxed{117}$$

Q17

$$\frac{720}{12} = 60$$

Coprime pairs of 60:

$$\sqrt[3]{(1,60),(3,20),(4,15),(5,12)}$$

Closest:

$$\sqrt[3]{5,12}$$

Actual numbers:

$$\sqrt[3]{60,144}$$

Difference:

$$\sqrt[3]{\boxed{84}}$$

Q18

$$\sqrt[3]{\text{HCF}(504,792)}$$

Euclidean:

$$\sqrt[3]{792-504=288}$$

$$\sqrt[3]{504-288=216}$$

$$\sqrt[3]{288-216=72}$$

$$\sqrt[3]{216=72 \times 3}$$

Largest tile side:

$$\sqrt[3]{\boxed{72 \text{ cm}}}$$

Tiles:

$$\sqrt[3]{504/72=7}$$

$$\sqrt[3]{792/72=11}$$

Total:

$$\sqrt[3]{\boxed{77}}$$

Q19

$$\sqrt[3]{\frac{210}{10}}=21=3 \times 7$$

Distinct prime factors:

$$\sqrt[3]{k=2}$$

Unordered pairs:

$$\sqrt[3]{2^{2-1} = \boxed{2}}$$

Q20

Common interval:

$$\sqrt[3]{\text{LCM}(24,36,54)=216 \text{ sec}}$$

18 minutes:

$$\sqrt[3]{1080 \text{ sec}}$$

$$\sqrt[3]{1080/216=5}$$

Five future coincidences plus initial:

$$\sqrt[3]{\boxed{6}}$$

CHAPTER 5 — FINAL CHECK

We have now completed:

- factors and multiples '
- HCF/GCD '
- LCM '
- prime-factorization method '
- Euclidean algorithm '
- coprime numbers '
- $\text{HCF} \times \text{LCM}$ relationship '
- ratios with HCF/LCM '
- same-remainder patterns '
- specified remainders '
- least-number remainder patterns '
- divisor-minus- $\backslash(k)$ patterns '
- repeated-event synchronization '
- counting synchronizations '
- cutting/grouping/tiling applications '
- possible number pairs '
- number of possible pairs '
- minimum-sum/minimum-difference reasoning '
- fraction/decimal scaling applications '
- LCM shortcut connection to Time & Work '
- GATE-style practice '
- placement-style practice '

Therefore:

' CHAPTER 5 COMPLETE

Progress

Completed: 5/76 **Current:** ' Chapter 5 — Factors, Multiples, HCF & LCM **Next:** Ø=Chapter 6 — **Prime Numbers & Prime Factorization**

In Chapter 6 we'll cover:

- prime/composite numbers
- prime detection
- fundamental theorem of arithmetic
- prime-factor exponents
- number of factors
- sum of factors
- odd/even factors
- perfect square/cube conditions

- trailing-zero foundations
- highest power of a prime in factorials
- GATE/placement number-system patterns.

USER

next

ASSISTANT

CHAPTER 6 — PRIME NUMBERS & PRIME FACTORIZATION

PART 1 — PRIME BASICS, TESTING, UNIQUE FACTORIZATION & FACTOR COUNTS

Status: 0% PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This chapter supports:

- divisibility
- HCF/LCM
- number of factors
- perfect squares/cubes
- factorials
- trailing zeros
- remainder reasoning
- probability/counting
- number-system aptitude

1. WHAT IS A PRIME NUMBER?

A **prime number** is a positive integer greater than 1 with exactly two positive factors:

1 and itself

Examples:

$2, 3, 5, 7, 11, 13, 17, 19, \dots$

For example:

7

has only:

$1, 7$

So 7 is prime.

2. COMPOSITE NUMBER

A positive integer greater than 1 having more than two positive factors is called **composite**.

Example:

$\backslash[12\backslash]$

Factors:

$\backslash[1,2,3,4,6,12\backslash]$

Therefore:

$\backslash[\boxed{12}\text{ is composite}]\backslash]$

3. WHAT ABOUT 1?

This is a famous trap.

$\backslash[\boxed{1}\text{ is neither prime nor composite}]\backslash]$

Why?

A prime requires **exactly two** positive factors.

But 1 has only:

$\backslash[1\backslash]$

So it is not prime.

4. SMALLEST PRIME NUMBER

$\backslash[\boxed{2}\backslash]$

2 is also special because it is the:

$\backslash[\boxed{\text{only even prime number}}]\backslash]$

Every other even number greater than 2 is divisible by 2 and is therefore composite.

5. SMALLEST ODD PRIME

$\backslash[\boxed{3}\backslash]$

6. FIRST PRIME NUMBERS — MUST RECOGNIZE

At least know primes up to 100 comfortably:

$\backslash[2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97\backslash]$

You do not need to recite them continuously, but you should recognize them quickly.

7. PRIME TESTING — BASIC IDEA

To determine whether $\backslash(n)\backslash$ is prime, you do **not** need to test divisibility by every integer below $\backslash(n)\backslash$.

You only need to test prime divisors up to:

$$\boxed{\sqrt{n}}$$

8. WHY ONLY UP TO $\sqrt{n}$?

If n is composite:

$$n = ab$$

If both:

$$a > \sqrt{n}, \quad b > \sqrt{n}$$

then:

$$ab > n$$

which is impossible.

So at least one factor must satisfy:

$$a \leq \sqrt{n}$$

Therefore, if no prime up to $\sqrt{n}$ divides n , then n is prime.

9. EXAMPLE — IS 97 PRIME?

$$\sqrt{97} < 10$$

So test primes:

$$2, 3, 5, 7$$

- not even
- digit sum:

$$9 + 7 = 16$$
 not divisible by 3

- does not end in 0 or 5
- $97 \div 7$ is not exact

Therefore:

$$\boxed{97 \text{ is prime}}$$

10. EXAMPLE — IS 143 PRIME?

$$\sqrt{143} \approx 11.96$$

Test primes:

$$2, 3, 5, 7, 11$$

Check 11:

$$143 = 11 \times 13$$

Therefore:

$$\boxed{143 \text{ is composite}}$$

11. QUICK PRIME FILTERS

For $(n > 5)$:

A prime cannot:

- be even
- end in 5
- have digit sum divisible by 3

So before actual testing, eliminate obvious composites.

Example:

$[237]$

digit sum:

$[2+3+7=12]$

Therefore divisible by 3.

So immediately composite.

12. IMPORTANT FORM OF PRIMES GREATER THAN 3

Any prime greater than 3 must be of form:

$[\boxed{6k \pm 1}]$

Why?

Every integer is one of:

$[6k, 6k+1, 6k+2, 6k+3, 6k+4, 6k+5]$

These forms except:

$[6k+1, \text{quad} 6k+5]$

are divisible by 2 or 3.

And:

$[6k+5=6(k+1)-1]$

So prime candidates are:

$[6k \pm 1]$

13. IMPORTANT TRAP

Being of the form:

$[6k \pm 1]$

does **not** guarantee primality.

Example:

$[25=6(4)+1]$

but 25 is composite.

So:

$$\boxed{p > 3 \text{ prime} \Rightarrow p = 6k \pm 1}$$

but reverse is false.

14. PRIME FACTORIZATION

Every composite number can be written as a product of primes.

Example:

$$60$$

$$60 = 2 \times 30$$

$$= 2 \times 2 \times 15$$

$$= 2^2 \times 3 \times 5$$

So:

$$\boxed{60 = 2^2 \cdot 3 \cdot 5}$$

15. FUNDAMENTAL THEOREM OF ARITHMETIC

Every integer greater than 1 can be represented as a product of primes in a **unique way**, apart from the order of factors.

Example:

$$84 = 2^2 \cdot 3 \cdot 7$$

You cannot get a completely different prime-factor decomposition of 84.

This uniqueness is the foundation of:

- HCF
 - LCM
 - number of factors
 - perfect square/cube rules
-

16. PRIME FACTORIZATION BY REPEATED DIVISION

Factor:

$$360$$

Divide by 2:

$$360 = 2 \times 180$$

Again:

$$= 2^2 \times 90$$

Again:

$$= 2^3 \times 45$$

Then:

$$45 = 3^2 \times 5$$

Therefore:

$$\boxed{360 = 2^3 \cdot 3^2 \cdot 5}$$

17. EXPONENT INTERPRETATION

If:

$$N = 2^4 \cdot 3^2 \cdot 5$$

then (N) contains:

- four factors of 2
- two factors of 3
- one factor of 5

This is exactly what we used for HCF/LCM.

18. NUMBER OF FACTORS — CORE FORMULA

Suppose:

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

Then number of positive divisors is:

$$\boxed{d(N) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1)}$$

MUST MEMORIZE

19. WHY THE FACTOR-COUNT FORMULA WORKS

Take:

$$N = 2^3 \cdot 3^2$$

Any factor must look like:

$$2^a \cdot 3^b$$

where:

$$a = 0, 1, 2, 3$$

4 choices.

And:

$$b = 0, 1, 2$$

3 choices.

Total:

$$4 \times 3 = 12$$

So:

$$\boxed{d(N)=12}$$

20. EXAMPLE

Find number of factors of:

$$72$$

Factor:

$$72=2^3\cdot 3^2$$

Therefore:

$$d(72)=(3+1)(2+1)$$

$$=4\times 3$$

$$=\boxed{12}$$

21. LISTING TO VERIFY

Factors of 72:

$$1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72$$

Count:

$$12$$

Correct.

22. NUMBER OF PROPER FACTORS

A proper positive factor generally means a factor other than the number itself.

So:

$$\boxed{\text{proper factors}=d(N)-1}$$

Example:

72 has 12 factors.

Proper factors:

$$\boxed{11}$$

Note

Some contexts exclude both 1 and the number, so always read the definition in the question.

23. NUMBER OF FACTORS EXCLUDING 1 AND (N)

If asked:

How many non-trivial factors?

Often means exclude:

$\nmid N$

Then:

$d(N)-2$

for composite N .

24. PRIME NUMBER FACTOR COUNT

For prime p :

$p = p^1$

Number of factors:

$1+1=2$

Exactly the definition of a prime.

25. SQUARE OF A PRIME

For:

p^2

number of factors:

$2+1=3$

Example:

$49 = 7^2$

Factors:

$1, 7, 49$

Exactly 3.

Useful fact

A positive integer has exactly **3 positive factors** iff it is square of a prime.

26. NUMBER WITH EXACTLY 4 FACTORS

A number has exactly 4 positive factors if either:

$N = p^3$

or:

$N = pq$

where (p, q) are distinct primes.

Why?

Factor-count patterns:

$$\sqrt[3]{3+1=4}$$

or:

$$\sqrt[3]{(1+1)(1+1)=4}$$

Example:

$$\sqrt[3]{8=2^3}$$

has:

$$\sqrt[3]{1,2,4,8}$$

And:

$$\sqrt[3]{15=3 \cdot 5}$$

has:

$$\sqrt[3]{1,3,5,15}$$

27. FIND NUMBER OF FACTORS OF LARGE EXPRESSION

Find factors of:

$$\sqrt[3]{2^5 \cdot 3^4 \cdot 7^2}$$

Number:

$$\sqrt[3]{(5+1)(4+1)(2+1)}$$

$$\sqrt[3]{=6 \times 5 \times 3}$$

$$\sqrt[3]{=\boxed{90}}$$

No need to calculate the original number.

28. ODD FACTORS

Suppose:

$$\sqrt[3]{N=2^a \cdot 3^b \cdot 5^c}$$

To get an **odd factor**, exponent of 2 must be:

$$\sqrt[3]{0}$$

So number of odd factors is:

$$\sqrt[3]{\boxed{(b+1)(c+1)}}$$

Ignore the power of 2.

29. EXAMPLE — ODD FACTORS

Find number of odd factors of:

$$\sqrt[3]{360=2^3 \cdot 3^2 \cdot 5}$$

Ignore $\sqrt[3]{(2^3)}$.

Choices:

$$\{3^0, 3^1, 3^2\}$$

3 choices.

$$\{5^0, 5^1\}$$

2 choices.

Total:

$$3 \times 2 = 6$$

Answer:

$$\boxed{6}$$

30. EVEN FACTORS

Total factors:

$$d(N)$$

Odd factors:

$$O$$

Then:

$$\boxed{\text{even factors} = d(N) - O}$$

Example for 360:

Total:

$$(3+1)(2+1)(1+1) = 24$$

Odd:

$$3 \times 2 = 6$$

Even:

$$24 - 6 = \boxed{18}$$

31. FACTORS DIVISIBLE BY A GIVEN NUMBER

Example:

How many factors of:

$$N = 2^4 \cdot 3^3$$

are divisible by:

$$12 = 2^2 \cdot 3$$

A divisor has form:

$$2^a \cdot 3^b$$

To be divisible by 12:

$$a \geq 2, b \geq 1$$

For (a) :

$\{2, 3, 4\}$

3 choices.

For (b) :

$\{1, 2, 3\}$

3 choices.

Total:

$\boxed{9}$

This pattern is very useful.

32. FACTORS THAT ARE MULTIPLES OF (k)

General idea:

Factor (N) and (k) .

If:

$k \mid N$

then require divisor exponents to be at least the exponents appearing in (k) .

Count valid exponent choices.

33. FACTORS THAT ARE PERFECT SQUARES

Suppose:

$N = 2^5 \cdot 3^4$

A perfect-square divisor needs **even exponents**.

For factor 2:

possible exponents:

$\{0, 2, 4\}$

3 choices.

For factor 3:

$\{0, 2, 4\}$

3 choices.

Total square divisors:

$\boxed{9}$

34. GENERAL SQUARE-DIVISOR COUNT

For:

$$N = p_1^{a_1} \cdots p_k^{a_k}$$

square divisor exponents can be:

$$0, 2, 4, \dots, \leq a_i$$

Number of choices:

$$\left\lfloor \frac{a_i}{2} \right\rfloor + 1$$

So number of square divisors:

$$\prod_i \left(\left\lfloor \frac{a_i}{2} \right\rfloor + 1 \right)$$

35. FACTORS THAT ARE PERFECT CUBES

Cube divisor requires exponents multiples of 3:

$$0, 3, 6, \dots$$

Number of choices for exponent a_i :

$$\left\lfloor \frac{a_i}{3} \right\rfloor + 1$$

Therefore number of cube divisors:

$$\prod_i \left(\left\lfloor \frac{a_i}{3} \right\rfloor + 1 \right)$$

36. PERFECT SQUARE CONDITION

A positive integer is a perfect square iff **every exponent in its prime factorization is even**.

Example:

$$144 = 2^4 \cdot 3^2$$

Both exponents even.

Therefore:

$$\boxed{144 \text{ is a perfect square}}$$

37. NON-SQUARE EXAMPLE

$$72 = 2^3 \cdot 3^2$$

Exponent of 2 is odd.

Therefore:

$$\boxed{72 \text{ is not a perfect square}}$$

38. PERFECT CUBE CONDITION

A number is a perfect cube iff every prime exponent is a multiple of 3.

Example:

$$216 = 2^3 \cdot 3^3$$

Both exponents multiples of 3.

So:

$$\boxed{216=6^3}$$

39. SMALLEST NUMBER TO MULTIPLY TO MAKE A SQUARE

Example:

Find smallest positive integer to multiply:

$$72=2^3 \cdot 3^2$$

to make it a perfect square.

We need exponent of 2 even.

Currently:

$$3$$

Multiply by:

$$2$$

Then:

$$72 \times 2 = 2^4 \cdot 3^2$$

perfect square.

Answer:

$$\boxed{2}$$

40. ANOTHER SQUARE COMPLETION

Find smallest number multiplying:

$$540$$

Factor:

$$540=2^2 \cdot 3^3 \cdot 5$$

Need all exponents even.

- 2^2 fine
- 3^3 needs one more 3
- 5^1 needs one more 5

Multiply:

$$3 \times 5 = 15$$

Answer:

$$\boxed{15}$$

41. SMALLEST NUMBER TO DIVIDE TO MAKE A SQUARE

Example:

$$108 = 2^2 \cdot 3^3$$

To leave even exponents, remove one factor of 3:

$$108/3 = 36$$

Thus:

$$\boxed{3}$$

is the smallest divisor to remove.

42. SMALLEST MULTIPLIER TO MAKE A CUBE

Example:

$$72 = 2^3 \cdot 3^2$$

For cube:

- exponent 3 of 2 is fine
- exponent 2 of 3 needs one more

Multiply by:

$$3$$

Then:

$$216 = 2^3 \cdot 3^3$$

Answer:

$$\boxed{3}$$

43. ANOTHER CUBE COMPLETION

$$N = 2^4 \cdot 3^2 \cdot 5$$

Need next multiples of 3:

- 2^4 !' needs 2^2 to reach 6
- 3^2 !' needs 3
- 5^1 !' needs 5^2

Multiplier:

$$2^2 \cdot 3 \cdot 5^2$$

$$= 4 \times 3 \times 25$$

$$= \boxed{300}$$

44. PERFECT SQUARE HAS ODD NUMBER OF FACTORS

Important theorem:

A positive integer has an **odd** number of positive factors iff it is a perfect square.

Why?

Normally divisors pair:

$$\left[d, \frac{N}{d} \right]$$

Distinct pairs give even count.

But if:

$$\left[d = \frac{N}{d} \right]$$

then:

$$\left[d^2 = N \right]$$

This occurs exactly when $\sqrt{N}$ is a square.

So one divisor, $\sqrt{N}$, pairs with itself.

45. EXAMPLE

$$\left[36 \right]$$

Factors:

$$\left[1, 2, 3, 4, 6, 9, 12, 18, 36 \right]$$

Count:

$$\left[9 \right]$$

odd.

And:

$$\left[36 = 6^2 \right]$$

46. NON-SQUARE FACTOR COUNT

$$\left[24 \right]$$

has factors:

$$\left[1, 2, 3, 4, 6, 8, 12, 24 \right]$$

Count:

$$\left[8 \right]$$

even.

24 is not a square.

47. SUM OF FACTORS — CORE FORMULA

Suppose:

$$\left[N = p^a \right]$$

Sum of positive factors:

$$\backslash[1+p+p^2+\cdots+p^a \backslash]$$

This is a geometric series:

$$\backslash[\boxed{ \frac{p^{a+1}-1}{p-1} } \backslash]$$

For multiple prime factors:

$$\backslash[N=p_1^{a_1}p_2^{a_2}\cdots \backslash]$$

then:

$$\backslash[\boxed{ \sigma(N) = (1+p_1+\cdots+p_1^{a_1}) (1+p_2+\cdots+p_2^{a_2})\cdots } \backslash]$$

48. EXAMPLE — SUM OF FACTORS OF 12

$$\backslash[12=2^2\cdot 3 \backslash]$$

So:

$$\backslash[\sigma(12) = (1+2+4)(1+3) \backslash]$$

$$\backslash[=7\cdot 4 \backslash]$$

$$\backslash[=\boxed{28} \backslash]$$

Check:

$$\backslash[1+2+3+4+6+12=28 \backslash]$$

49. SUM OF PROPER FACTORS

Proper factors exclude the number itself.

Therefore:

$$\backslash[\boxed{ \text{sum proper factors} = \sigma(N)-N } \backslash]$$

Example:

12:

$$\backslash[28-12=16 \backslash]$$

Proper factors:

$$\backslash[1+2+3+4+6=16 \backslash]$$

50. PRODUCT OF ALL POSITIVE FACTORS

If $\backslash(N\backslash)$ has $\backslash(d(N)\backslash)$ positive divisors:

$$\backslash[\boxed{ \text{Product of all positive divisors} = N^{d(N)/2} } \backslash]$$

This works because factors pair to product $\backslash(N\backslash)$.

For perfect squares, interpretation still works mathematically.

51. EXAMPLE

For:

$$\sqrt[12]{}$$

Number of factors:

$$\sqrt[6]{}$$

Product:

$$\sqrt[12^{6/2}]{12^3} = 12^3 = 1728$$

Indeed:

$$1 \cdot 2 \cdot 3 \cdot 4 \cdot 6 \cdot 12 = 1728$$

52. GATE-STYLE QUESTION 1

How many positive factors does:

$$2^4 \cdot 3^3 \cdot 5^2$$

have?

$$(4+1)(3+1)(2+1)$$

$$= 5 \times 4 \times 3$$

$$= \boxed{60}$$

53. GATE-STYLE QUESTION 2

How many odd positive factors does:

$$\sqrt[720]{}$$

have?

Factor:

$$\sqrt[720]{2^4 \cdot 3^2 \cdot 5}$$

Ignore power of 2.

Odd factor count:

$$(2+1)(1+1)$$

$$= 3 \times 2$$

$$= \boxed{6}$$

54. GATE-STYLE QUESTION 3

Find smallest integer k such that:

$$\sqrt[90k]{}$$

is a perfect square.

$$\backslash[90=2\cdot3^2\cdot5 \backslash]$$

Odd exponents are on:

$$\backslash[2^1,\text{quad}5^1 \backslash]$$

Multiply by:

$$\backslash[2\cdot5=10 \backslash]$$

So:

$$\backslash[\boxed{k=10} \backslash]$$

55. PLACEMENT-STYLE QUESTION 1

How many factors does 360 have?

$$\backslash[360=2^3\cdot3^2\cdot5 \backslash]$$

$$\backslash[d(360)=4\cdot3\cdot2 \backslash]$$

$$\backslash[=\boxed{24} \backslash]$$

56. PLACEMENT-STYLE QUESTION 2

How many even factors does 360 have?

Total:

$$\backslash[24 \backslash]$$

Odd:

$$\backslash[(2+1)(1+1)=6 \backslash]$$

Even:

$$\backslash[24-6 = \boxed{18} \backslash]$$

57. PLACEMENT-STYLE QUESTION 3

How many perfect-square factors does:

$$\backslash[2^6\cdot3^5 \backslash]$$

have?

For exponent 6:

$$\backslash[0,2,4,6 \backslash]$$

4 choices.

For exponent 5:

$$\backslash[0,2,4 \backslash]$$

3 choices.

Total:

$\boxed{12}$

58. COMMON TRAPS

& pTRAP 1

1

is neither prime nor composite.

& pTRAP 2

2 is the only even prime.

& pTRAP 3

$(6k \pm 1)$ gives prime **candidates**, not guaranteed primes.

& pTRAP 4

Number of factors:

$(a+1)(b+1)\dots$

not:

ab

& pTRAP 5

For perfect square, all prime exponents must be even.

For perfect cube, all must be multiples of 3.

& pTRAP 6

Odd factors !' ignore the factor 2 entirely.

59. MUST-MEMORIZE BOX

Prime test

Check prime divisors only up to:

$\sqrt{n}$

Factor count

$N = \prod p_i^{a_i}$

$\boxed{d(N) = \prod (a_i + 1)}$

Square condition

All a_i even.

Cube condition

All $(a_i \equiv 0 \pmod{3})$.

Odd number of divisors

$\boxed{\text{iff perfect square}}$

Sum of factors

$\boxed{\sigma(N) = \prod (1 + p + \dots + p^a)}$

PRACTICE — LEVEL 0

Q1

Is 1 prime, composite, or neither?

Q2

Is 91 prime?

Q3

Prime-factorize:

$\boxed{180}$

Q4

Find number of factors of:

$\boxed{36}$

Q5

Find number of odd factors of:

$\boxed{72}$

LEVEL 1

Q6

Find number of factors of:

$\boxed{2^5 \cdot 3^2 \cdot 7}$

Q7

Find number of even factors of:

$\boxed{240}$

Q8

Find smallest number by which 150 must be multiplied to become a perfect square.

Q9

Find smallest number by which 108 must be multiplied to become a perfect cube.

Q10

Find sum of positive factors of:

$$\sqrt[3]{18}$$

LEVEL 2

Q11

How many perfect-square divisors does:

$$\sqrt[3]{2^7 \cdot 3^4 \cdot 5^3}$$

have?

Q12

How many perfect-cube divisors does:

$$\sqrt[3]{2^8 \cdot 3^7}$$

have?

Q13

How many factors of:

$$\sqrt[3]{2^6 \cdot 3^4}$$

are divisible by:

$$\sqrt[3]{2^2 \cdot 3^3}$$

Q14

A positive integer has exactly 3 factors. What form must it have?

Q15

A number has an odd number of positive divisors. What can you conclude?

LEVEL 3 — EXAM PRESSURE

Q16

How many positive divisors does:

$$\sqrt[3]{10^6}$$

have?

Q17

How many odd divisors does:

$$\sqrt[3]{10!}$$

have?

Q18

Find the smallest positive integer k such that:

$$\sqrt[3]{180k}$$

is a perfect cube.

Q19

Find sum of all positive divisors of:

$$\sqrt[3]{72}$$

Q20

Find product of all positive divisors of:

$$\sqrt[3]{36}$$

ANSWERS

1

$$\sqrt[3]{\boxed{\text{Neither}}}$$

2

$$\sqrt[3]{91=7\times 13}$$

$$\sqrt[3]{\boxed{\text{Composite}}}$$

3

$$\sqrt[3]{180=2^2\cdot 3^2\cdot 5}$$

4

$$\sqrt[3]{36=2^2\cdot 3^2}$$

$$\sqrt[3]{(2+1)(2+1)=\boxed{9}}$$

5

$$\sqrt[3]{72=2^3\cdot 3^2}$$

Odd factors:

$$\sqrt[3]{2+1=\boxed{3}}$$

6

$$\sqrt[3]{(5+1)(2+1)(1+1)=6\times 3\times 2=\boxed{36}}$$

7

$$\sqrt[3]{240=2^4\cdot 3\cdot 5}$$

Total:

$$\sqrt[3]{5\times 2\times 2=20}$$

Odd:

$$\sqrt[3]{2\times 2=4}$$

Even:

$$\sqrt[3]{20-4=\boxed{16}}$$

8

$$\backslash[150=2\cdot3\cdot5^2\backslash]$$

Need another:

$$\backslash[2\cdot3\backslash]$$

$$\backslash[\boxed{6}\backslash]$$

9

$$\backslash[108=2^2\cdot3^3\backslash]$$

Exponent 2 of 2 needs one more 2.

$$\backslash[\boxed{2}\backslash]$$

10

$$\backslash[18=2\cdot3^2\backslash]$$

$$\backslash[(1+2)(1+3+9)=3\times13=\boxed{39}\backslash]$$

11

Exponents allowed even.

For $\backslash(2^7\backslash)$:

$$\backslash[0,2,4,6\backslash]$$

4 choices.

For $\backslash(3^4\backslash)$:

$$\backslash[0,2,4\backslash]$$

3 choices.

For $\backslash(5^3\backslash)$:

$$\backslash[0,2\backslash]$$

2 choices.

Total:

$$\backslash[4\times3\times2=\boxed{24}\backslash]$$

12

Cube exponents.

For $\backslash(2^8\backslash)$:

$$\backslash[0,3,6\backslash]$$

3 choices.

For $\backslash(3^7\backslash)$:

$$\backslash[0,3,6\backslash]$$

3 choices.

$$\backslash[\boxed{9}\backslash]$$

13

A divisor:

$$\backslash[2^a 3^b \backslash]$$

needs:

$$\backslash[a=2,3,4,5,6 \backslash]$$

5 choices.

$$\backslash[b=3,4 \backslash]$$

2 choices.

Total:

$$\backslash[\boxed{10} \backslash]$$

14

$$\backslash[\boxed{p^2} \backslash]$$

where $\backslash(p\backslash)$ is prime.

15

$$\backslash[\boxed{\text{It is a perfect square}} \backslash]$$

16

$$\backslash[10^6=2^6\cdot 5^6 \backslash]$$

Factors:

$$\backslash[7\cdot 7 = \boxed{49} \backslash]$$

17

$$\backslash[10!=2^8\cdot 3^4\cdot 5^2\cdot 7 \backslash]$$

Odd divisors ignore factor 2:

$$\backslash[(4+1)(2+1)(1+1) \backslash]$$

$$\backslash[=5\cdot 3\cdot 2 = \boxed{30} \backslash]$$

18

$$\backslash[180=2^2\cdot 3^2\cdot 5 \backslash]$$

To reach cube exponents:

- $\backslash(2^2\text{to}2^3\backslash)$: multiply by 2
- $\backslash(3^2\text{to}3^3\backslash)$: multiply by 3
- $\backslash(5^1\text{to}5^3\backslash)$: multiply by $\backslash(5^2\backslash)$

So:

$$\backslash[k=2\cdot 3\cdot 25 = \boxed{150} \backslash]$$

19

$$\backslash[72=2^3\cdot 3^2 \backslash]$$

$$\backslash[\sigma(72) = (1+2+4+8)(1+3+9) \backslash]$$

$$\lfloor =15\times 13 \rfloor$$

$$\lfloor =\boxed{195} \rfloor$$

20

$$\lfloor 36=2^2\cdot 3^2 \rfloor$$

Number of divisors:

$$\lfloor 9 \rfloor$$

Product of divisors:

$$\lfloor 36^{9/2} \rfloor$$

Since 36 is a square:

$$\lfloor 36^{9/2}=(\sqrt{36})^9=6^9 \rfloor$$

$$\lfloor 6^9=\boxed{10077696} \rfloor$$

You usually should leave this as:

$$\lfloor \boxed{36^{9/2}} \rfloor$$

unless an exact integer is requested.

CHAPTER 6 STATUS

Completed so far:

- prime/composite definitions '
- special role of 1 and 2 '
- prime testing up to $\lfloor \sqrt{n} \rfloor$ '
- $\lfloor 6k\pm 1 \rfloor$ candidate rule '
- prime factorization '
- Fundamental Theorem of Arithmetic '
- number of factors '
- odd/even factors '
- factors divisible by a given value '
- square/cube divisors '
- perfect square/cube conditions '
- smallest square/cube multipliers '
- odd-number-of-divisors theorem '
- sum of factors '
- product of factors '
- GATE/placement practice '

But **Chapter 6 is not complete yet.**

Next we need:

- factorial prime exponents
- Legendre's formula
- highest power of a prime dividing $\lfloor n! \rfloor$

- trailing-zero foundations
- number of trailing zeros in $(n!)$
- trailing zeros in products and powers
- exact power of 2/5 inside factorials
- advanced prime-factor applications
- factor-count construction problems

Progress

Completed: 5/76 **Current:** 0/31 **Chapter 6 — Prime Numbers & Prime Factorization** **Next:** Chapter 6 Part 2 — Factorials, prime exponents & trailing zeros

USER

next

ASSISTANT

CHAPTER 6 — PRIME NUMBERS & PRIME FACTORIZATION

PART 2 — FACTORIALS, PRIME EXPONENTS & TRAILING ZEROS

Status: 0/14 **PROGRESS GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This part is extremely useful for number-system questions involving:

- factorials
- highest powers
- divisibility
- trailing zeros
- prime exponents
- large products
- option elimination

1. WHAT IS A FACTORIAL?

For a positive integer (n) :

$$[n! = n(n-1)(n-2)\cdots 3\cdots 2\cdots 1]$$

Examples:

$$[5! = 5\cdots 4\cdots 3\cdots 2\cdots 1 = 120]$$

$$[6! = 720]$$

Also:

$$[\boxed{0! = 1}]$$

MUST MEMORIZE

2. WHY FACTORIALS MATTER

Factorials appear in:

- permutations
- combinations
- probability
- divisibility
- trailing zeros
- prime-factor counting

Example:

$10!$

is enormous compared to 10, but its prime-factor structure is easy to analyze.

3. PRIME EXPONENT INSIDE A NUMBER

Suppose:

$$N = 2^4 \cdot 3^2 \cdot 5$$

The highest power of 2 dividing N is:

$$2^4$$

The exponent of 2 is:

$$4$$

We sometimes write:

$$v_2(N) = 4$$

where $v_p(N)$ means:

exponent of prime p in the prime factorization of N .

You do not need to memorize the notation if you prefer simple language.

4. PRIME EXPONENT IN A FACTORIAL

Question:

What is the exponent of 2 in:

$$10!?$$

Instead of expanding fully, count how many factors of 2 appear.

Numbers contributing at least one 2:

$$2, 4, 6, 8, 10$$

That gives:

$$\lfloor 5 \rfloor$$

But 4 and 8 contain extra 2s.

- $\lfloor 4=2^2 \rfloor$ gives one extra
- $\lfloor 8=2^3 \rfloor$ gives two extra

Total:

$$\lfloor 5+2+1 \rfloor$$

Let's do systematically.

5. LEGENDRE'S FORMULA

The exponent of prime $\lfloor p \rfloor$ in $\lfloor n! \rfloor$ is:

$$\lfloor \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots \rfloor$$

Stop when:

$$\lfloor p^k > n \rfloor$$

This is a **MUST UNDERSTAND** formula.

6. WHY LEGENDRE'S FORMULA WORKS

For $\lfloor p=2 \rfloor$:

$$\lfloor \left\lfloor \frac{n}{2} \right\rfloor \rfloor$$

counts numbers contributing at least one factor 2.

Then:

$$\lfloor \left\lfloor \frac{n}{4} \right\rfloor \rfloor$$

counts numbers contributing a **second** factor 2.

Then:

$$\lfloor \left\lfloor \frac{n}{8} \right\rfloor \rfloor$$

counts numbers contributing a **third** factor 2.

And so on.

7. EXAMPLE — EXPONENT OF 2 IN $\lfloor 10! \rfloor$

$$\lfloor \left\lfloor \frac{10}{2} \right\rfloor \rfloor = 5$$

$$\lfloor \left\lfloor \frac{10}{4} \right\rfloor \rfloor = 2$$

$$\lfloor \left\lfloor \frac{10}{8} \right\rfloor \rfloor = 1$$

Next:

$$\lfloor 16 > 10 \rfloor$$

Stop.

Total exponent:

$$\lfloor 5+2+1 = \boxed{8} \rfloor$$

So:

$$\lfloor 10! \rfloor$$

contains:

$$\lfloor 2^8 \rfloor$$

as part of its prime factorization.

8. EXPONENT OF 3 IN $\lfloor(10!)\rfloor$

$$\lfloor \left\lfloor \frac{10}{3} \right\rfloor \right\rfloor = 3 \rfloor$$

$$\lfloor \left\lfloor \frac{10}{9} \right\rfloor \right\rfloor = 1 \rfloor$$

Total:

$$\lfloor \boxed{4} \rfloor$$

So:

$$\lfloor 10! \rfloor$$

contains:

$$\lfloor 3^4 \rfloor$$

9. PRIME FACTORIZATION OF $\lfloor(10!)\rfloor$

Using exponent counting:

$$\lfloor v_2(10!) = 8 \rfloor$$

$$\lfloor v_3(10!) = 4 \rfloor$$

$$\lfloor v_5(10!) = 2 \rfloor$$

$$\lfloor v_7(10!) = 1 \rfloor$$

Therefore:

$$\lfloor \boxed{10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7} \rfloor$$

This is what we used earlier.

10. HIGHEST POWER OF A PRIME DIVIDING $\lfloor(n!)\rfloor$

Question:

Find highest power of 5 dividing:

$$\lfloor 100! \rfloor$$

Compute exponent:

$$\lfloor \left\lfloor \frac{100}{5} \right\rfloor \right\rfloor = 20 \rfloor$$

$$\lfloor \left\lfloor \frac{100}{25} \right\rfloor \right\rfloor = 4 \rfloor$$

$$\lfloor \frac{100}{125} \rfloor = 0$$

Total:

$$24$$

Therefore highest power is:

$$\boxed{5^{24}}$$

11. ANOTHER EXAMPLE

Highest power of 3 dividing:

$$25!$$

$$\lfloor \frac{25}{3} \rfloor = 8$$

$$\lfloor \frac{25}{9} \rfloor = 2$$

$$\lfloor \frac{25}{27} \rfloor = 0$$

Total:

$$10$$

Therefore:

$$\boxed{3^{10}}$$

12. HIGHEST POWER OF A COMPOSITE NUMBER DIVIDING $(n!)$

This is more interesting.

Example:

Find highest power of:

$$12$$

that divides:

$$20!$$

Factor:

$$12 = 2^2 \cdot 3$$

We need prime exponents in $(20!)$.

13. EXPONENT OF 2 IN $(20!)$

$$\lfloor \frac{20}{2} \rfloor = 10$$

$$\lfloor \frac{20}{4} \rfloor = 5$$

$$\lfloor \frac{20}{8} \rfloor = 2$$

$$\lfloor \frac{20}{16} \rfloor = 1$$

Total:

$\lfloor 18 \rfloor$

14. EXPONENT OF 3 IN $(20!)$

$$\lfloor \frac{20}{3} \rfloor = 6$$

$$\lfloor \frac{20}{9} \rfloor = 2$$

Total:

$$8$$

So:

$$20!$$

contains:

$$2^{18} \cdot 3^8$$

relevant to 12.

Each 12 needs:

$$2^2 \cdot 3$$

Number of complete copies:

From factor 2:

$$18/2 = 9$$

From factor 3:

$$8/1 = 8$$

Take smaller:

$$\boxed{8}$$

Therefore highest power:

$$\boxed{12^8}$$

divides $(20!)$.

15. GENERAL RULE — HIGHEST POWER OF COMPOSITE (m) IN $(n!)$

Factor:

$$m = p_1^{a_1} p_2^{a_2} \cdots$$

Find:

$$v_{p_i}(n!)$$

Then number of full copies of (m) is:

$$\boxed{\min_i \left\lfloor \frac{v_{p_i}(n!)}{a_i} \right\rfloor}$$

16. EXAMPLE

Find highest power of:

$$\lfloor 18=2\cdot 3^2 \rfloor$$

dividing:

$$\lfloor 25! \rfloor$$

For 2:

$$\lfloor v_2(25!) = 12+6+3+1 = 22 \rfloor$$

For 3:

$$\lfloor v_3(25!) = 8+2 = 10 \rfloor$$

Each 18 needs:

$$\lfloor 2^1\cdot 3^2 \rfloor$$

Copies possible:

From 2:

$$\lfloor 22/1=22 \rfloor$$

From 3:

$$\lfloor 10/2=5 \rfloor$$

Minimum:

$$\lfloor \boxed{5} \rfloor$$

So highest power is:

$$\lfloor \boxed{18^5} \rfloor$$

17. TRAILING ZEROS — CORE IDEA

A trailing zero is a zero at the end of a number.

Examples:

$$\lfloor 1200 \rfloor$$

has:

$$\lfloor 2 \rfloor$$

trailing zeros.

$$\lfloor 45000 \rfloor$$

has:

$$\lfloor 3 \rfloor$$

trailing zeros.

18. WHAT CREATES A TRAILING ZERO?

Every trailing zero comes from a factor:

$$\lfloor 10=2\cdot 5 \rfloor$$

So number of trailing zeros equals number of complete:

$$\lfloor 2 \times 5 \rfloor$$

pairs.

Thus:

$$\boxed{\text{Trailing zeros} = \min(v_2(N), v_5(N))}$$

19. WHY FACTORIAL TRAILING ZEROS DEPEND ON 5

In $(n!)$, there are many more factors of 2 than factors of 5.

Therefore:

$$v_2(n!) > v_5(n!)$$

for ordinary $(n!)$.

So trailing zeros in $(n!)$ are determined by:

$$\boxed{v_5(n!)}$$

20. TRAILING ZEROS IN $(10!)$

Count factors of 5:

$$\left\lfloor \frac{10}{5} \right\rfloor = 2$$

No 25 contribution.

Therefore:

$$\boxed{2}$$

Indeed:

$$10! = 3628800$$

ends in 2 zeros.

21. TRAILING ZEROS IN $(25!)$

$$\left\lfloor \frac{25}{5} \right\rfloor = 5$$

But 25 contributes an extra 5:

$$\left\lfloor \frac{25}{25} \right\rfloor = 1$$

Total:

$$5 + 1 = 6$$

Therefore:

$$\boxed{6}$$

trailing zeros.

& pTRAP

Do not just calculate:

$$\lfloor 25/5=5 \rfloor$$

You must include powers of 25, 125, etc.

22. TRAILING ZEROS IN $(100!)$

$$\lfloor \frac{100}{5} \rfloor = 20$$

$$\lfloor \frac{100}{25} \rfloor = 4$$

Total:

$$\boxed{24}$$

23. TRAILING ZEROS IN $(125!)$

$$\lfloor \frac{125}{5} \rfloor = 25$$

$$\lfloor \frac{125}{25} \rfloor = 5$$

$$\lfloor \frac{125}{125} \rfloor = 1$$

Total:

$$25+5+1 = \boxed{31}$$

24. TRAILING ZEROS IN A NORMAL PRODUCT

Example:

$$2^7 \cdot 3^2 \cdot 5^4$$

Number of 2s:

$$7$$

Number of 5s:

$$4$$

Trailing zeros:

$$\min(7, 4) = \boxed{4}$$

25. TRAILING ZEROS IN A POWER

Example:

$$200^5$$

Factor:

$$200 = 2^3 \cdot 5^2$$

Therefore:

$$200^5 = 2^{15} \cdot 5^{10}$$

Trailing zeros:

$$\lfloor \min(15, 10) = \boxed{10} \rfloor$$

26. TRAILING ZEROS IN PRODUCT OF FACTORIALS

Example:

$$\lfloor 20! \times 25! \rfloor$$

Trailing-zero analysis requires total exponents.

For 5:

$$\lfloor v_5(20!) = 4 \rfloor$$

because:

$$\lfloor 20/5 = 4 \rfloor$$

For $\lfloor 25! \rfloor$:

$$\lfloor 5 + 1 = 6 \rfloor$$

Total 5 exponent:

$$\lfloor 10 \rfloor$$

2s are more abundant.

Therefore:

$$\lfloor \boxed{10} \rfloor$$

trailing zeros.

27. TRAILING ZEROS IN QUOTIENTS

Example:

$$\lfloor \frac{20!}{5!} \rfloor$$

Assuming result is integer, count prime exponents by subtraction.

$$\lfloor v_5(20!) = 4 \rfloor$$

$$\lfloor v_5(5!) = 1 \rfloor$$

So remaining:

$$\lfloor 3 \rfloor$$

Check 2s are enough.

Thus:

$$\lfloor \boxed{3} \rfloor$$

trailing zeros.

28. GENERAL QUOTIENT RULE

For:

$$N = \frac{A}{B}$$

when integer:

$$v_p(N) = v_p(A) - v_p(B)$$

This is useful for factorial ratios and combinations.

29. TRAILING ZEROS IN $\frac{n!}{(n-k)!}$

Example:

$$\frac{30!}{20!} = 21 \cdot 22 \cdot \dots \cdot 30$$

To count trailing zeros:

$$v_5(30!) - v_5(20!)$$

$$v_5(30!) = 6 + 1 = 7$$

$$v_5(20!) = 4$$

Difference:

$$3$$

Now 2s are certainly enough.

So:

$$\boxed{3}$$

trailing zeros.

30. TRAILING ZEROS OF BINOMIAL EXPRESSIONS — PREVIEW

For:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

you can count 2s and 5s using factorial valuations.

We will use this again in Permutation & Combination.

31. HOW MANY TRAILING ZEROS CAN $n!$ HAVE?

Interesting property:

Not every nonnegative integer occurs as the number of trailing zeros of some factorial.

Example:

Can $n!$ have exactly 5 trailing zeros?

Let's test.

$$24!$$

zeros:

$$\lfloor 4 \rfloor$$

$$\lfloor 25! \rfloor$$

zeros:

$$\lfloor 6 \rfloor$$

So no factorial has exactly 5 trailing zeros.

Why?

At multiples of 25, zero count jumps by more than 1.

32. FIND SMALLEST $\lfloor n! \rfloor$ SUCH THAT $\lfloor n! \rfloor$ HAS AT LEAST $\lfloor k \rfloor$ ZEROS

Example:

Find smallest $\lfloor n! \rfloor$ such that $\lfloor n! \rfloor$ has at least 6 trailing zeros.

We know:

$$\lfloor 25! \rfloor$$

has 6.

Check:

$$\lfloor 24! \rfloor$$

has:

$$\lfloor \lfloor 24/5 \rfloor \rfloor = 4$$

So smallest:

$$\boxed{25}$$

For large $\lfloor k \rfloor$, binary search is a common coding approach, but aptitude often uses reasoning.

33. EXAMPLE — AT LEAST 10 ZEROS

Test multiples of 5.

$$\lfloor 40! \rfloor$$

zeros:

$$\lfloor 8+1=9 \rfloor$$

because:

$$\lfloor 40/5=8, \quad 40/25=1 \rfloor$$

$$\lfloor 45! \rfloor$$

zeros:

$$\lfloor 9+1=10 \rfloor$$

Thus:

$$\boxed{45}$$

is the smallest $\lfloor n! \rfloor$ giving at least 10 zeros.

34. EXACTLY $\lfloor k \rfloor$ TRAILING ZEROS

If one $\lfloor n \rfloor$ gives exactly $\lfloor k \rfloor$ zeros and $\lfloor n \rfloor$ is not at a jump boundary afterward, typically five consecutive values can share the same zero count.

Example:

$$\lfloor 20!, 21!, 22!, 23!, 24! \rfloor$$

all have:

$$\lfloor 4 \rfloor$$

trailing zeros.

Then at:

$$\lfloor 25! \rfloor$$

count jumps to 6.

So exactly 4 zeros occurs for 5 values:

$$\lfloor 20! \text{ through } 24! \rfloor$$

35. TRAILING ZEROS IN OTHER BASES — ADVANCED PREVIEW

Decimal trailing zeros come from:

$$\lfloor 10 = 2 \cdot 5 \rfloor$$

But if asked trailing zeros in base 12:

$$\lfloor 12 = 2^2 \cdot 3 \rfloor$$

Then number of base-12 trailing zeros in $\lfloor N \rfloor$ would be:

$$\lfloor \min \left(\left\lfloor \frac{v_2(N)}{2} \right\rfloor, v_3(N) \right) \rfloor$$

This is advanced and not a core placement topic, but the principle is important.

36. HIGHEST POWER OF 10 DIVIDING A NUMBER

This is exactly the number of trailing zeros.

If $\lfloor N \rfloor$ has $\lfloor t \rfloor$ trailing zeros:

$$\lfloor 10^t \mid N \rfloor$$

but:

$$\lfloor 10^{t+1} \nmid N \rfloor$$

Example:

$$\lfloor 36000 \rfloor$$

has 3 trailing zeros.

So highest power of 10 dividing it is:

$$\lfloor \boxed{10^3} \rfloor$$

37. HIGHEST POWER OF 10 DIVIDING $(100!)$

Trailing zeros:

$$\lfloor \frac{100}{5} \rfloor$$

Therefore highest power of 10 dividing $(100!)$:

$$\boxed{10^{24}}$$

38. EXPONENT OF 2 IN $(100!)$

This appears often.

$$\lfloor \frac{100}{2} \rfloor = 50$$

$$\lfloor \frac{100}{4} \rfloor = 25$$

$$\lfloor \frac{100}{8} \rfloor = 12$$

$$\lfloor \frac{100}{16} \rfloor = 6$$

$$\lfloor \frac{100}{32} \rfloor = 3$$

$$\lfloor \frac{100}{64} \rfloor = 1$$

Total:

$$50+25+12+6+3+1 = \boxed{97}$$

39. EXPONENT OF 3 IN $(100!)$

$$33+11+3+1 = \boxed{48}$$

because powers:

$$3, 9, 27, 81$$

40. EXPONENT OF 5 IN $(100!)$

$$20+4 = \boxed{24}$$

41. FAST METHOD FOR LEGENDRE

Write powers of (p) :

Example $(n=200, p=5)$:

$$5, 25, 125$$

Then:

$$200/5=40$$

$$200/25=8$$

$$\lfloor 200/125=1 \rfloor$$

Total:

$$\lfloor \boxed{49} \rfloor$$

Always stop when next power exceeds $\lfloor n \rfloor$.

42. GATE-STYLE QUESTION 1

Find highest exponent $\lfloor k \rfloor$ such that:

$$\lfloor 2^k \rfloor \mid 50! \rfloor$$

$$\lfloor v_2(50!) = 25+12+6+3+1 \rfloor$$

$$\lfloor =\boxed{47} \rfloor$$

So:

$$\lfloor \boxed{k=47} \rfloor$$

43. GATE-STYLE QUESTION 2

Find highest exponent $\lfloor k \rfloor$ such that:

$$\lfloor 15^k \rfloor \mid 100! \rfloor$$

Factor:

$$\lfloor 15=3 \cdot 5 \rfloor$$

We need:

$$\lfloor v_3(100!)=48 \rfloor$$

$$\lfloor v_5(100!)=24 \rfloor$$

Each 15 needs one 3 and one 5.

So:

$$\lfloor k=\min(48,24) \rfloor$$

$$\lfloor =\boxed{24} \rfloor$$

44. GATE-STYLE QUESTION 3

Find highest exponent $\lfloor k \rfloor$ such that:

$$\lfloor 72^k \rfloor \mid 50! \rfloor$$

Factor:

$$\lfloor 72=2^3 \cdot 3^2 \rfloor$$

We know:

$$\lfloor v_2(50!)=47 \rfloor$$

For 3:

$$\lfloor \frac{50}{3} \rfloor = 16$$

$$\lfloor \frac{50}{9} \rfloor = 5$$

$$\lfloor \frac{50}{27} \rfloor = 1$$

Total:

$$22$$

Copies allowed:

From 2:

$$\left\lfloor \frac{47}{3} \right\rfloor = 15$$

From 3:

$$\left\lfloor \frac{22}{2} \right\rfloor = 11$$

Take minimum:

$$\boxed{11}$$

45. PLACEMENT-STYLE QUESTION 1

How many trailing zeros in:

$$50!?$$

$$50/5 = 10$$

$$50/25 = 2$$

Total:

$$\boxed{12}$$

46. PLACEMENT-STYLE QUESTION 2

How many trailing zeros in:

$$200!?$$

$$40+8+1 = \boxed{49}$$

47. PLACEMENT-STYLE QUESTION 3

How many trailing zeros in:

$$25! \times 10!?$$

$$\sqrt[5]{25!} = 6$$

$$\sqrt[5]{10!} = 2$$

Total:

$$8$$

2s are sufficient.

Therefore:

$$\boxed{8}$$

48. PLACEMENT-STYLE QUESTION 4

How many trailing zeros in:

$$100^7?$$

$$100=10^2$$

Therefore:

$$100^7=10^{14}$$

Answer:

$$\boxed{14}$$

49. TRICKY QUESTION — PRODUCT RANGE

How many trailing zeros in:

$$11 \times 12 \times 13 \times \cdots \times 25?$$

This equals:

$$\frac{25!}{10!}$$

Count 5s:

$$v_5(25!) - v_5(10!)$$

$$6 - 2 = 4$$

Check 2s are enough.

Answer:

$$\boxed{4}$$

50. TRICKY QUESTION — FACTORIAL QUOTIENT

How many trailing zeros in:

$$\frac{50!}{25!}?$$

Count 5s:

$$v_5(50!) = 12$$

$$v_5(25!) = 6$$

Difference:

$$6$$

2s are more than enough.

Thus:

$\boxed{6}$

51. COMMON TRAPS

& pTRAP 1 — Counting only multiples of 5

For $(100!)$:

20

is not enough.

Need extra factors from:

$25, 50, 75, 100$

So answer is 24.

& pTRAP 2 — Highest power vs exponent

If exponent is 24:

highest power of 5 is:

5^{24}

not 24.

& pTRAP 3 — Composite base

For (12^k) , don't count only factor 12.

Factor:

$12 = 2^2 \cdot 3$

then compare prime supplies.

& pTRAP 4 — Trailing zeros in arbitrary number

Use:

$\min(v_2, v_5)$

Do not automatically count only 5s unless factorial structure guarantees extra 2s.

& pTRAP 5 — Quotient

Subtract prime exponents.

52. MUST-MEMORIZE FORMULAS

Prime exponent in factorial

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor$$

Decimal trailing zeros of $(n!)$

$$Z(n!) = \left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots$$

$$\lfloor \frac{n}{125} \rfloor + \dots$$

Trailing zeros of arbitrary N

$$\boxed{\min(v_2(N), v_5(N))}$$

Highest power of composite

Factor composite and take the limiting prime.

53. MEMORY TRICKS

Factorial exponent

Count multiples of (p) , then (p^2) , then (p^3) ...

Trailing zeros

Every zero needs one 2 and one 5.

Factorials

5 is scarce, 2 is abundant.

So factorial zeros !' count 5s.

54. 5-MINUTE REVISION

For:

$$100!$$

Remember:

$$v_2=97$$

$$v_3=48$$

$$v_5=24$$

Not because these specific values must all be memorized, but because they show how quickly Legendre works.

For decimal trailing zeros:

$$100! \rightarrow 24$$

$$125! \rightarrow 31$$

$$200! \rightarrow 49$$

PRACTICE — LEVEL 0

Q1

Find exponent of 5 in:

$\lfloor 30! \rfloor$

Q2

Find exponent of 2 in:

$\lfloor 20! \rfloor$

Q3

How many trailing zeros in:

$\lfloor 30! \rfloor$

Q4

How many trailing zeros in:

$\lfloor 2^8 \cdot 5^3 \rfloor$

Q5

What is the highest power of 10 dividing:

$\lfloor 50! \rfloor$

LEVEL 1

Q6

Find highest exponent $\lfloor k \rfloor$ such that:

$\lfloor 3^k \rfloor \mid 50! \rfloor$

Q7

Find highest exponent $\lfloor k \rfloor$ such that:

$\lfloor 10^k \rfloor \mid 75! \rfloor$

Q8

Find highest exponent $\lfloor k \rfloor$ such that:

$\lfloor 12^k \rfloor \mid 30! \rfloor$

Q9

How many trailing zeros in:

$\lfloor 40! \times 20! \rfloor$

Q10

How many trailing zeros in:

$\lfloor \frac{30!}{15!} \rfloor$

LEVEL 2

Q11

Find highest power of 18 dividing:

$$\lfloor \frac{50!}{18} \rfloor$$

Q12

Find highest exponent $\lfloor \frac{k}{100} \rfloor$ such that:

$$\lfloor \frac{72^k}{100!} \rfloor$$

Q13

How many trailing zeros in:

$$\lfloor \frac{125^8 \cdot 16^5}{5!} \rfloor$$

Q14

How many trailing zeros in:

$$\lfloor \frac{100!}{50!} \rfloor$$

Q15

Find smallest $\lfloor \frac{n}{10} \rfloor$ such that $\lfloor \frac{n!}{10} \rfloor$ has at least 20 trailing zeros.

LEVEL 3 — EXAM PRESSURE

Q16

Can any factorial have exactly 5 trailing zeros?

Q17

Find highest exponent $\lfloor \frac{k}{100} \rfloor$ such that:

$$\lfloor \frac{360^k}{100!} \rfloor$$

Q18

How many trailing zeros are there in:

$$\lfloor \frac{1 \cdot 2 \cdot 3 \cdot \dots \cdot 100}{10} \rfloor$$

and why is this the same as $\lfloor \frac{100!}{10} \rfloor$?

Q19

Find highest exponent of 7 in:

$$\lfloor \frac{200!}{7} \rfloor$$

Q20

Find trailing zeros of:

$$\lfloor \frac{75!}{25! \cdot 50!} \rfloor$$

You must check both 2s and 5s, not assume blindly.

ANSWERS

1

$$\lfloor \frac{30}{5} \rfloor + \lfloor \frac{30}{25} \rfloor = 6 + 1 = \boxed{7}$$

2

$$10 + 5 + 2 + 1 = \boxed{18}$$

3

$$\boxed{7}$$

4

$$\min(8, 3) = \boxed{3}$$

5

$50!$ has 12 trailing zeros.

Highest power:

$$\boxed{10^{12}}$$

6

$$\lfloor \frac{50}{3} \rfloor + \lfloor \frac{50}{9} \rfloor + \lfloor \frac{50}{27} \rfloor$$

$$16 + 5 + 1 = \boxed{22}$$

7

Trailing zeros of $75!$:

$$15 + 3 = \boxed{18}$$

So:

$$\boxed{k=18}$$

8

$$12 = 2^2 \cdot 3$$

For $30!$:

$$v_2 = 15 + 7 + 3 + 1 = 26$$

$$v_3 = 10 + 3 + 1 = 14$$

Copies of 12:

$$\min(\lfloor \frac{26}{2} \rfloor, 14)$$

$$= \min(13, 14)$$

$$= \boxed{13}$$

9

$$v_5(40!) = 8 + 1 = 9$$

$$v_5(20!) = 4$$

Total:

$$\boxed{13}$$

10

$$\lfloor v_5(30!)-v_5(15!) \rfloor$$

$$\lfloor 7-3 = \boxed{4} \rfloor$$

11

$$\lfloor 18=2\cdot 3^2 \rfloor$$

For $\lfloor(50!)\rfloor$:

$$\lfloor v_2=47 \rfloor$$

$$\lfloor v_3=22 \rfloor$$

Copies:

$$\lfloor \min(47, \lfloor 22/2 \rfloor) = \boxed{11} \rfloor$$

So highest power:

$$\lfloor \boxed{18^{11}} \rfloor$$

12

$$\lfloor 72=2^3\cdot 3^2 \rfloor$$

For $\lfloor(100!)\rfloor$:

$$\lfloor v_2=97 \rfloor$$

$$\lfloor v_3=48 \rfloor$$

Copies:

$$\lfloor \left\lfloor 97/3 \right\rfloor = 32 \rfloor$$

$$\lfloor \left\lfloor 48/2 \right\rfloor = 24 \rfloor$$

Minimum:

$$\lfloor \boxed{24} \rfloor$$

13

$$\lfloor 125^8=(5^3)^8=5^{24} \rfloor$$

$$\lfloor 16^5=(2^4)^5=2^{20} \rfloor$$

Zeros:

$$\lfloor \min(24,20) = \boxed{20} \rfloor$$

14

$$\lfloor v_5(100!)=24 \rfloor$$

$$\lfloor v_5(50!)=12 \rfloor$$

Difference:

$$\lfloor \boxed{12} \rfloor$$

2s are sufficient.

15

Try:

$$\lfloor 80! \rfloor$$

zeros:

$$\lfloor 16+3=19 \rfloor$$

$$\lfloor 85! \rfloor$$

zeros:

$$\lfloor 17+3=20 \rfloor$$

So smallest:

$$\lfloor \boxed{85} \rfloor$$

16

No.

$$\lfloor 24! \rfloor$$

has:

$$\lfloor 4 \rfloor$$

zeros.

$$\lfloor 25! \rfloor$$

has:

$$\lfloor 6 \rfloor$$

So:

$$\lfloor \boxed{\text{exactly 5 is impossible}} \rfloor$$

17

$$\lfloor 360=2^3 \cdot 3^2 \cdot 5 \rfloor$$

For $\lfloor (100!) \rfloor$:

$$\lfloor v_2=97, \text{quad } v_3=48, \text{quad } v_5=24 \rfloor$$

Copies:

$$\lfloor 97/3 \text{ to } 32 \rfloor$$

$$\lfloor 48/2=24 \rfloor$$

$$\lfloor 24/1=24 \rfloor$$

Minimum:

$$\lfloor \boxed{24} \rfloor$$

So:

$$\lfloor 360^{24} \rfloor$$

is the highest power.

18

$$\lfloor 1 \cdot 2 \cdot 3 \cdot \dots \cdot 100=100! \rfloor$$

Therefore zeros:

$$\lfloor 20+4 = \boxed{24} \rfloor$$

19

$$\lfloor \left\lfloor \frac{200}{7} \right\rfloor + \left\lfloor \frac{200}{49} \right\rfloor \rfloor$$

$$\lfloor 28+4 = \boxed{32} \rfloor$$

Next power:

$$\lfloor 343 > 200 \rfloor$$

20

Expression:

$$\binom{75}{25}$$

For factor 5:

$$v_5(75!) = 18$$

$$v_5(25!) = 6$$

$$v_5(50!) = 12$$

Difference:

$$18 - 6 - 12 = 0$$

So there are no factors of 5 in the result.

Therefore trailing zeros:

$$\boxed{0}$$

No need to count 2s once one side is zero.

CHAPTER 6 — FINAL CHECK

We have now covered:

- prime/composite numbers
- prime testing
- $(6k \pm 1)$ candidate property
- prime factorization
- Fundamental Theorem of Arithmetic
- number of factors
- odd/even factors
- square/cube divisors
- sum/product of factors
- perfect-square/cube completion
- factorials
- prime exponent in $(n!)$
- Legendre's formula
- highest power of a prime in $(n!)$
- highest power of a composite in $(n!)$

- trailing zeros in factorials '
- trailing zeros in arbitrary products '
- powers/products/quotients '
- factorial ratios '
- inverse trailing-zero reasoning '
- GATE/placement questions '

Therefore:

' CHAPTER 6 COMPLETE

Progress

Completed: 6/76 **Current:** ' Chapter 6 — Prime Numbers & Prime Factorization **Next:** ~~Chapter 6~~ Chapter 7 — **Remainders & Modular Reasoning**

Chapter 7 will cover:

- quotient–remainder theorem
- modulo notation
- remainder of sums/products
- negative remainders
- powers and cycles
- remainder of huge exponents
- modular reduction
- divisibility connection
- algebraic remainder patterns
- placement tricks
- GATE-style modular arithmetic.

USER

next

ASSISTANT

CHAPTER 7 — REMAINDERS & MODULAR REASONING

PART 1 — QUOTIENT–REMAINDER THEOREM, MODULO BASICS & CORE OPERATIONS

Status: ~~0=Not~~ IN PROGRESS **GATE:** High **Placements:** High **Priority:** ~~0=Not~~ 1

Remainders are one of the most useful number-system topics because they appear in:

- divisibility
- cyclicity
- units digit

- powers
- modular arithmetic
- large-number questions
- number theory
- GATE mathematical reasoning
- placement aptitude

The big idea is:

Instead of calculating a huge number completely, calculate only the remainder information you actually need.

1. QUOTIENT–REMAINDER THEOREM

When an integer (N) is divided by a positive integer (d) :

$$\boxed{N=dq+r}$$

where:

- (q) = quotient
- (r) = remainder

and:

$$\boxed{0 \leq r < d}$$

This last condition is very important.

2. EXAMPLE

Divide:

$$\boxed{29}$$

by:

$$\boxed{6}$$

We have:

$$\boxed{29=6(4)+5}$$

Therefore:

$$\boxed{q=4, \text{quad } r=5}$$

So remainder:

$$\boxed{5}$$

3. POSSIBLE REMAINDERS

If a number is divided by 7, possible remainders are:

$$\boxed{0,1,2,3,4,5,6}$$

Never:

$\lfloor 7 \rfloor$

because if 7 is left, one more full group of 7 can be removed.

General rule

When dividing by $\lfloor d \rfloor$:

$\lfloor \boxed{r \in \{0, 1, \dots, d-1\}} \rfloor$

4. WHAT DOES REMAINDER 0 MEAN?

If:

$\lfloor N = dq + 0 \rfloor$

then:

$\lfloor d \mid N \rfloor$

So:

$\lfloor \boxed{\text{remainder } 0} \text{ iff } \text{exact divisibility}} \rfloor$

5. MODULO NOTATION

Instead of saying:

29 leaves remainder 5 when divided by 6

we can write:

$\lfloor \boxed{29 \equiv 5 \pmod{6}} \rfloor$

Read:

29 is congruent to 5 modulo 6.

For aptitude, think:

Both have the same remainder when divided by 6.

6. EXAMPLE

$\lfloor 17 \equiv 2 \pmod{5} \rfloor$

because:

$\lfloor 17 = 5(3) + 2 \rfloor$

Similarly:

$\lfloor 32 \equiv 2 \pmod{5} \rfloor$

Therefore:

$$\boxed{17 \equiv 32 \pmod{5}}$$

because both leave remainder 2.

7. CONGRUENCE MEANS DIFFERENCE IS DIVISIBLE

Very important theorem:

$$\boxed{a \equiv b \pmod{m}}$$

iff:

$$\boxed{m \mid (a-b)}$$

Example:

$$\boxed{38 \equiv 3 \pmod{5}}$$

because:

$$\boxed{38 - 3 = 35}$$

and 35 is divisible by 5.

8. REDUCING A NUMBER MODULO (m)

Suppose you need:

$$\boxed{1234 \bmod 7}$$

You can divide directly:

$$\boxed{1234 = 7(176) + 2}$$

Therefore:

$$\boxed{1234 \equiv 2 \pmod{7}}$$

But for huge expressions, we reduce pieces instead of calculating everything.

9. REMAINDER OF A SUM

If:

$$\boxed{a \equiv r_1 \pmod{m}}$$

and:

$$\boxed{b \equiv r_2 \pmod{m}}$$

then:

$$\boxed{a + b \equiv r_1 + r_2 \pmod{m}}$$

Reduce again if necessary.

10. EXAMPLE — SUM

Find remainder when:

$$\backslash[38+47 \backslash]$$

is divided by 5.

Individual remainders:

$$\backslash[38\equiv 3\pmod{5} \backslash]$$

$$\backslash[47\equiv 2\pmod{5} \backslash]$$

So:

$$\backslash[38+47\equiv 3+2=5\equiv 0\pmod{5} \backslash]$$

Answer:

$$\backslash[\boxed{0} \backslash]$$

No need to compute 85, though here it is easy.

11. REMAINDER OF A DIFFERENCE

$$\backslash[\boxed{a-b\equiv r_1-r_2\pmod{m}} \backslash]$$

Example:

Find remainder of:

$$\backslash[103-47 \backslash]$$

when divided by 7.

$$\backslash[103\equiv 5\pmod{7} \backslash]$$

because:

$$\backslash[98+5 \backslash]$$

and:

$$\backslash[47\equiv 5\pmod{7} \backslash]$$

Therefore:

$$\backslash[103-47\equiv 5-5=0 \backslash]$$

Answer:

$$\backslash[\boxed{0} \backslash]$$

12. REMAINDER OF A PRODUCT

This is extremely useful.

If:

$$\backslash[a\equiv r_1\pmod{m} \backslash]$$

$$\backslash[b\equiv r_2\pmod{m} \backslash]$$

then:

$$\backslash[\boxed{ab\equiv r_1r_2\pmod{m}} \backslash]$$

13. EXAMPLE — PRODUCT

Find remainder when:

$$38 \times 47$$

is divided by 5.

$$38 \equiv 3 \pmod{5}$$

$$47 \equiv 2 \pmod{5}$$

Therefore:

$$38 \times 47 \equiv 3 \times 2 = 6$$

$$6 \equiv 1 \pmod{5}$$

Answer:

$$\boxed{1}$$

No large multiplication needed.

14. REDUCE AT EVERY STEP

Suppose:

$$17 \times 23 \times 29$$

modulo 5.

Reduce:

$$17 \equiv 2$$

$$23 \equiv 3$$

$$29 \equiv 4$$

So:

$$2 \times 3 \times 4 = 24$$

$$24 \equiv 4 \pmod{5}$$

Answer:

$$\boxed{4}$$

15. REMAINDER OF A POWER

If:

$$a \equiv r \pmod{m}$$

then:

$$\boxed{a^n \equiv r^n \pmod{m}}$$

Example:

Find remainder of:

$$\lfloor 12^5 \rfloor$$

when divided by 5.

$$\lfloor 12 \equiv 2 \pmod{5} \rfloor$$

Therefore:

$$\lfloor 12^5 \equiv 2^5 = 32 \rfloor$$

$$\lfloor 32 \equiv 2 \pmod{5} \rfloor$$

Answer:

$$\lfloor \boxed{2} \rfloor$$

16. NEGATIVE REMAINDERS AS SHORTCUTS

Suppose:

$$\lfloor 9 \pmod{10} \rfloor$$

Normally:

$$\lfloor 9 \equiv 9 \pmod{10} \rfloor$$

But also:

$$\lfloor 9 \equiv -1 \pmod{10} \rfloor$$

because:

$$\lfloor 9 - (-1) = 10 \rfloor$$

divisible by 10.

This is extremely useful for powers.

17. EXAMPLE

Find remainder of:

$$\lfloor 9^{37} \rfloor$$

when divided by 10.

Use:

$$\lfloor 9 \equiv -1 \pmod{10} \rfloor$$

Then:

$$\lfloor 9^{37} \equiv (-1)^{37} = -1 \rfloor$$

Convert to standard remainder:

$$\lfloor -1 \equiv 9 \pmod{10} \rfloor$$

Answer:

$$\lfloor \boxed{9} \rfloor$$

18. STANDARD REMAINDER MUST BE NON-NEGATIVE

Modulo calculations may temporarily give:

$$\lfloor -2 \rfloor$$

But if divisor is 7, standard remainder is:

$$\lfloor -2+7=5 \rfloor$$

Therefore:

$$\lfloor \boxed{-2 \equiv 5 \pmod{7}} \rfloor$$

19. EXAMPLE WITH NEGATIVE DIFFERENCE

Find remainder when:

$$\lfloor 12-20 \rfloor$$

is divided by 7.

$$\lfloor 12 \equiv 5 \rfloor$$

$$\lfloor 20 \equiv 6 \rfloor$$

So:

$$\lfloor 5-6=-1 \rfloor$$

$$\lfloor -1 \equiv 6 \pmod{7} \rfloor$$

Answer:

$$\lfloor \boxed{6} \rfloor$$

Indeed:

$$\lfloor 12-20=-8 \rfloor$$

and in modular arithmetic:

$$\lfloor -8=7(-2)+6 \rfloor$$

20. REMAINDER OF LARGE SUM

Find remainder of:

$$\lfloor 1001+2002+3003 \rfloor$$

when divided by 7.

Notice:

$$\lfloor 1001=7 \times 143 \rfloor$$

So:

$$\lfloor 1001 \equiv 0 \pmod{7} \rfloor$$

Then:

$$\lfloor 2002=2(1001) \equiv 0 \rfloor$$

$$\boxed{3003 = 3(1001) \equiv 0}$$

Therefore total remainder:

$$\boxed{0}$$

21. REMAINDER OF ALGEBRAIC EXPRESSIONS

Suppose:

$$n \equiv 2 \pmod{5}$$

Find remainder of:

$$n^2 + 3n + 4$$

modulo 5.

Substitute remainder representative:

$$2^2 + 3(2) + 4$$

$$= 4 + 6 + 4$$

$$= 14$$

$$14 \equiv 4 \pmod{5}$$

Answer:

$$\boxed{4}$$

22. WHY SUBSTITUTION WORKS

If:

$$n \equiv r \pmod{m}$$

then polynomial expressions behave consistently under addition and multiplication.

So for:

$$P(n)$$

we can often compute:

$$P(r) \pmod{m}$$

This is a major GATE-style reasoning tool.

23. REMAINDER OF A NUMBER FROM DIGIT SUM — MOD 3

As learned earlier:

$$10 \equiv 1 \pmod{3}$$

Therefore:

$$10^k \equiv 1 \pmod{3}$$

So any decimal number has same remainder mod 3 as its digit sum.

Example:

$$\backslash[72845 \backslash]$$

Digit sum:

$$\backslash[7+2+8+4+5=26 \backslash]$$

$$\backslash[26\equiv 2\pmod{3} \backslash]$$

Therefore:

$$\backslash[\boxed{72845\equiv 2\pmod{3}} \backslash]$$

24. SAME FOR MOD 9

Because:

$$\backslash[10\equiv 1\pmod{9} \backslash]$$

digit sum preserves remainder modulo 9.

Example:

$$\backslash[987654 \backslash]$$

Digit sum:

$$\backslash[39 \backslash]$$

$$\backslash[39\equiv 3\pmod{9} \backslash]$$

So original number leaves remainder:

$$\backslash[\boxed{3} \backslash]$$

25. MODULO 11 AND ALTERNATING SUM

Because:

$$\backslash[10\equiv -1\pmod{11} \backslash]$$

then:

$$\backslash[10^2\equiv 1 \backslash]$$

$$\backslash[10^3\equiv -1 \backslash]$$

and so on.

That is why divisibility by 11 uses alternating digit sums.

26. EXAMPLE

Find remainder of:

$$\backslash[47382 \backslash]$$

modulo 11.

Alternating sum:

$$\lfloor (4+3+2)-(7+8) \rfloor$$

$$\lfloor =9-15=-6 \rfloor$$

Convert:

$$\lfloor -6 \equiv 5 \pmod{11} \rfloor$$

So remainder:

$$\lfloor \boxed{5} \rfloor$$

27. DIVISION BY 2 — PARITY AS MODULAR ARITHMETIC

Every integer is either:

$$\lfloor 0 \pmod{2} \rfloor$$

or:

$$\lfloor 1 \pmod{2} \rfloor$$

That is simply:

- even
- odd

So parity rules are modular rules.

Example:

Odd + odd:

$$\lfloor 1+1=2 \equiv 0 \pmod{2} \rfloor$$

Therefore even.

28. SQUARES MODULO 4

Every integer is even or odd.

If even:

$$\lfloor n=2k \rfloor$$

$$\lfloor n^2=4k^2 \rfloor$$

so:

$$\lfloor n^2 \equiv 0 \pmod{4} \rfloor$$

If odd:

$$\lfloor n=2k+1 \rfloor$$

Then square is:

$$\lfloor 4k(k+1)+1 \rfloor$$

So:

$$\lfloor n^2 \equiv 1 \pmod{4} \rfloor$$

Therefore:

$\boxed{\text{Any perfect square is } 0 \text{ or } 1 \pmod{4}}$

This is useful for eliminating impossible answers.

29. SQUARES MODULO 3

Possible residues:

$\{0, 1, 2\}$

Squares:

$0^2 \equiv 0$

$1^2 \equiv 1$

$2^2 = 4 \equiv 1$

Thus:

$\boxed{\text{Square mod 3 is only } 0 \text{ or } 1}$

Never 2.

30. SQUARES MODULO 8

For odd integer (n) :

$\boxed{n^2 \equiv 1 \pmod{8}}$

Example:

$3^2 = 9 \equiv 1$

$5^2 = 25 \equiv 1$

$7^2 = 49 \equiv 1$

Very useful GATE-style fact.

31. CUBES MODULO 9 — QUICK INSIGHT

Cubes have restricted residues modulo 9.

For residues (0) through (8) , cubes reduce to:

$\{0, \pm 1\}$

That is:

$\boxed{\{0, 1, 8\}}$

modulo 9.

This can eliminate impossible perfect cubes.

We do not need to memorize every residue pattern yet.

32. REMAINDER WHEN A PRODUCT HAS A DIVISIBLE FACTOR

If any factor is divisible by m , product remainder is 0.

Example:

$$17 \times 24 \times 31$$

modulo 8.

Since:

$$24 \equiv 0 \pmod{8}$$

whole product:

$$\boxed{0}$$

33. CONSECUTIVE-INTEGERS REMAINDER REASONING

For:

$$n(n+1)$$

among two consecutive integers, one is even.

Thus:

$$n(n+1) \equiv 0 \pmod{2}$$

For:

$$n(n+1)(n+2)$$

product is:

$$0 \pmod{6}$$

This is modular divisibility reasoning.

34. CLASSIC RESULT

$$n^3 - n$$

Factor:

$$n(n-1)(n+1)$$

Three consecutive integers.

Therefore:

$$n^3 - n \equiv 0 \pmod{6}$$

So:

$$\boxed{n^3 \equiv n \pmod{6}}$$

for every integer n .

35. ANOTHER USEFUL RESULT

For any integer n :

$$\boxed{n^2 \equiv n \pmod{2}}$$

Why?

$$n^2 - n = n(n-1)$$

product of consecutive integers, so even.

Therefore:

$$\boxed{n^2 \equiv n \pmod{2}}$$

36. SAME REMAINDER CONNECTION

If:

$$A \equiv r \pmod{d}$$

and:

$$B \equiv r \pmod{d}$$

then:

$$A - B \equiv 0 \pmod{d}$$

Thus:

$$d \mid (A - B)$$

This is exactly why Chapter 5 same-remainder problems use HCF of differences.

37. FIND REMAINDER AFTER ADDING A NUMBER

If N leaves remainder 4 when divided by 7, what remainder does:

$$N + 5$$

leave?

$$N \equiv 4 \pmod{7}$$

So:

$$N + 5 \equiv 9 \equiv 2 \pmod{7}$$

Answer:

$$\boxed{2}$$

38. FIND REMAINDER AFTER MULTIPLYING

If:

$$N \equiv 3 \pmod{8}$$

what is remainder of:

$$5N$$

mod 8?

$$\lfloor 5N \equiv 5(3) = 15 \rfloor$$

$$\lfloor 15 \equiv 7 \pmod{8} \rfloor$$

Answer:

$$\lfloor \boxed{7} \rfloor$$

39. TWO NUMBERS WITH KNOWN REMAINDERS

Suppose:

$$\lfloor A \equiv 5 \pmod{7} \rfloor$$

and:

$$\lfloor B \equiv 4 \pmod{7} \rfloor$$

Find remainder of:

$$\lfloor AB + A \rfloor$$

Factor if you want:

$$\lfloor A(B+1) \rfloor$$

Modulo 7:

$$\lfloor 5(4+1) = 25 \rfloor$$

$$\lfloor 25 \equiv 4 \rfloor$$

Answer:

$$\lfloor \boxed{4} \rfloor$$

40. USE FACTORIZATION BEFORE MODULO CALCULATION

Find:

$$\lfloor 101^2 - 99^2 \rfloor$$

modulo 8.

Instead of powers:

$$\lfloor (101-99)(101+99) \rfloor$$

$$\lfloor = 2 \times 200 \rfloor$$

$$\lfloor = 400 \rfloor$$

$$\lfloor 400 \equiv 0 \pmod{8} \rfloor$$

Answer:

$$\lfloor \boxed{0} \rfloor$$

41. REMAINDER OF $(a^2 - b^2)$

Use:

$$\backslash[a^2-b^2=(a-b)(a+b) \backslash]$$

If one factor has strong divisibility, calculation becomes easy.

This matters in GATE numerical reasoning.

42. GATE-STYLE QUESTION 1

If:

$$\backslash[n\equiv 4\pmod{7} \backslash]$$

find remainder of:

$$\backslash[n^2+2n+3 \backslash]$$

when divided by 7.

Substitute:

$$\backslash[4^2+2(4)+3 \backslash]$$

$$\backslash[=16+8+3=27 \backslash]$$

$$\backslash[27\equiv 6\pmod{7} \backslash]$$

Answer:

$$\backslash[\boxed{6} \backslash]$$

43. GATE-STYLE QUESTION 2

Find remainder of:

$$\backslash[999^2 \backslash]$$

when divided by 10.

$$\backslash[999\equiv -1\pmod{10} \backslash]$$

So:

$$\backslash[999^2\equiv 1 \backslash]$$

Answer:

$$\backslash[\boxed{1} \backslash]$$

44. GATE-STYLE QUESTION 3

If an integer $\backslash(n\backslash)$ leaves remainder 2 when divided by 5, find remainder of:

$$\backslash[n^3+4n \backslash]$$

modulo 5.

Substitute:

$$\backslash[2^3+4(2) \backslash]$$

$$\backslash[=8+8=16 \backslash]$$

$$\lfloor 16 \equiv 1 \pmod{5} \rfloor$$

Answer:

$$\lfloor \boxed{1} \rfloor$$

45. PLACEMENT-STYLE QUESTION 1

Find remainder when:

$$\lfloor 12345 + 67890 \rfloor$$

is divided by 9.

Digit sums:

$$\lfloor 12345 \text{ to } 15 \text{ to } 6 \rfloor$$

$$\lfloor 67890 \text{ to } 30 \text{ to } 3 \rfloor$$

Sum remainders:

$$\lfloor 6 + 3 = 9 \rfloor$$

Remainder:

$$\lfloor \boxed{0} \rfloor$$

46. PLACEMENT-STYLE QUESTION 2

Find remainder when:

$$\lfloor 123 \times 456 \rfloor$$

is divided by 7.

Reduce:

$$\lfloor 123 \equiv 4 \pmod{7} \rfloor$$

since $(119 + 4)$.

$$\lfloor 456 \equiv 1 \pmod{7} \rfloor$$

since $(455 + 1)$.

Therefore:

$$\lfloor 4 \times 1 = 4 \rfloor$$

Answer:

$$\lfloor \boxed{4} \rfloor$$

47. PLACEMENT-STYLE QUESTION 3

A number leaves remainder 6 when divided by 11. What remainder does three times the number plus 5 leave?

$$\lfloor N \equiv 6 \pmod{11} \rfloor$$

Then:

$$\lfloor 3N+5\equiv 18+5=23 \rfloor$$

$$\lfloor 23\equiv 1\pmod{11} \rfloor$$

Answer:

$$\lfloor \boxed{1} \rfloor$$

48. COMMON TRAPS

& pTRAP 1 — Remainder cannot equal divisor

When dividing by 7:

$$\lfloor r<7 \rfloor$$

& pTRAP 2 — Negative temporary result

$$\lfloor -2\pmod{7} \rfloor$$

means standard remainder:

$$\lfloor 5 \rfloor$$

& pTRAP 3 — Forgetting final reduction

If product remainder gives 23 modulo 7:

$$\lfloor 23\equiv 2 \rfloor$$

Answer is 2, not 23.

& pTRAP 4 — Dividing congruences casually

Addition, subtraction and multiplication are safe.

Division modulo $\lfloor(m)\rfloor$ requires care because inverses may not exist.

We will cover that later.

& pTRAP 5 — Huge powers

Do not calculate the huge number.

Find cycles or reduce the base.

49. MUST-MEMORIZE RULES

If:

$$\lfloor a\equiv r_1\pmod{m},\quad b\equiv r_2\pmod{m} \rfloor$$

then:

Addition

$$\lfloor \boxed{a+b\equiv r_1+r_2\pmod{m}} \rfloor$$

Subtraction

$$\boxed{a - b \equiv r_1 - r_2 \pmod{m}}$$

Multiplication

$$\boxed{ab \equiv r_1 r_2 \pmod{m}}$$

Powers

$$\boxed{a^n \equiv r_1^n \pmod{m}}$$

Always reduce the final answer to:

$$0 \leq r < m$$

50. MEMORY TRICK

Reduce early, reduce often.

If modulus is 7, replace:

$$1000$$

by its remainder before continuing.

Never carry huge numbers if small residues will do.

51. 5-MINUTE REVISION

Core theorem:

$$N = dq + r, \text{quad } 0 \leq r < d$$

Modulo:

$$a \equiv b \pmod{m} \text{ iff } m \mid (a - b)$$

Operations work through:

- addition
- subtraction
- multiplication
- powers

Useful negative representations:

$$m - 1 \equiv -1 \pmod{m}$$

Example:

$$99 \equiv -1 \pmod{100}$$

PRACTICE — LEVEL 0

Q1

Find remainder when 83 is divided by 7.

Q2

Find:

$$\lfloor 47 \bmod 9 \rfloor$$

Q3

Find remainder of:

$$\lfloor 23+35 \rfloor$$

when divided by 6.

Q4

Find remainder of:

$$\lfloor 17 \times 19 \rfloor$$

when divided by 5.

Q5

Convert:

$$\lfloor -3 \rfloor$$

to standard remainder modulo 8.

LEVEL 1

Q6

Find remainder of:

$$\lfloor 1234+5678 \rfloor$$

when divided by 9.

Q7

Find remainder of:

$$\lfloor 123 \times 456 \rfloor$$

when divided by 11.

Q8

Find remainder of:

$$\lfloor 19^5 \rfloor$$

when divided by 10.

Q9

If:

$$\lfloor N \equiv 5 \pmod{8} \rfloor$$

find remainder of:

$$\lfloor 3N+7 \rfloor$$

mod 8.

Q10

If:

$$x \equiv 3 \pmod{5}$$

find remainder of:

$$x^2 + x + 1$$

mod 5.

LEVEL 2

Q11

Find remainder of:

$$999^7$$

when divided by 100.

Q12

Find remainder of:

$$37^2 - 13^2$$

when divided by 8.

Q13

If:

$$a \equiv 4 \pmod{7}$$

and:

$$b \equiv 6 \pmod{7}$$

find remainder of:

$$a^2 + ab + b^2$$

modulo 7.

Q14

Find remainder of:

$$987654321$$

when divided by 9.

Q15

Find remainder of:

$$47382$$

when divided by 11.

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$\lfloor n \equiv 2 \pmod{7} \rfloor$$

find remainder of:

$$\lfloor n^4 + 3n^2 + 5 \rfloor$$

modulo 7.

Q17

Find remainder of:

$$\lfloor (10^6 + 10^3 + 1) \rfloor$$

when divided by 9.

Q18

Show that:

$$\lfloor n^3 - n \rfloor$$

is divisible by 6 for every integer $\lfloor n \rfloor$.

Q19

If:

$$\lfloor A \equiv 3 \pmod{5}, \quad B \equiv 4 \pmod{5} \rfloor$$

find:

$$\lfloor A^2B + AB^2 \rfloor$$

modulo 5.

Q20

Without full multiplication, find remainder of:

$$\lfloor 2026 \times 2027 \times 2028 \rfloor$$

when divided by 3.

ANSWERS

1

$$\lfloor 83 = 7(11) + 6 \rfloor$$

$$\lfloor \boxed{6} \rfloor$$

2

$$\lfloor 47 = 9(5) + 2 \rfloor$$

$$\lfloor \boxed{2} \rfloor$$

3

$$\backslash[23\equiv 5,\quad 35\equiv 5\pmod 6 \backslash]$$

$$\backslash[5+5=10\equiv 4 \backslash]$$

$$\backslash[\boxed{4} \backslash]$$

4

$$\backslash[17\equiv 2,\quad 19\equiv 4\pmod 5 \backslash]$$

$$\backslash[2\times 4=8\equiv 3 \backslash]$$

$$\backslash[\boxed{3} \backslash]$$

5

$$\backslash[-3+8=5 \backslash]$$

$$\backslash[\boxed{5} \backslash]$$

6

Digit sums:

$$\backslash[1234\text{to}10\text{to}1 \backslash]$$

$$\backslash[5678\text{to}26\text{to}8 \backslash]$$

$$\backslash[1+8=9 \backslash]$$

$$\backslash[\boxed{0} \backslash]$$

7

$$\backslash[123\equiv 2\pmod{11} \backslash]$$

$$\backslash[456\equiv 5\pmod{11} \backslash]$$

$$\backslash[2\times 5=10 \backslash]$$

$$\backslash[\boxed{10} \backslash]$$

8

$$\backslash[19\equiv -1\pmod{10} \backslash]$$

$$\backslash[(-1)^5=-1\equiv 9 \backslash]$$

$$\backslash[\boxed{9} \backslash]$$

9

$$\backslash[3(5)+7=22 \backslash]$$

$$\backslash[22\equiv 6\pmod 8 \backslash]$$

$$\backslash[\boxed{6} \backslash]$$

10

$$\backslash[3^2+3+1=13 \backslash]$$

$$\backslash[13\equiv 3\pmod 5 \backslash]$$

$$\backslash[\boxed{3} \backslash]$$

11

$$\lfloor 999 \equiv -1 \pmod{100} \rfloor$$

$$\lfloor (-1)^7 = -1 \rfloor$$

$$\lfloor \boxed{99} \rfloor$$

12

Factor:

$$\lfloor (37-13)(37+13) \rfloor$$

$$\lfloor = 24 \times 50 \rfloor$$

24 is divisible by 8.

$$\lfloor \boxed{0} \rfloor$$

13

Substitute:

$$\lfloor 4^2 + 4(6) + 6^2 \rfloor$$

$$\lfloor = 16 + 24 + 36 = 76 \rfloor$$

$$\lfloor 76 \equiv 6 \pmod{7} \rfloor$$

$$\lfloor \boxed{6} \rfloor$$

14

Digit sum:

$$\lfloor 45 \rfloor$$

$$\lfloor 45 \equiv 0 \pmod{9} \rfloor$$

$$\lfloor \boxed{0} \rfloor$$

15

Alternating sum:

$$\lfloor (4+3+2)-(7+8) \rfloor$$

$$\lfloor = 9 - 15 = -6 \rfloor$$

$$\lfloor -6 \equiv 5 \pmod{11} \rfloor$$

$$\lfloor \boxed{5} \rfloor$$

16

$$\lfloor n \equiv 2 \rfloor$$

So:

$$\lfloor n^2 \equiv 4 \rfloor$$

$$\lfloor n^4 \equiv 16 \equiv 2 \rfloor$$

Expression:

$$\lfloor 2 + 3(4) + 5 \rfloor$$

$$\lfloor = 19 \rfloor$$

$$\boxed{19 \equiv 5 \pmod{7}}$$

$$\boxed{5}$$

17

Since:

$$10 \equiv 1 \pmod{9}$$

all powers of 10 are 1 modulo 9.

Therefore:

$$1+1+1=3$$

$$\boxed{3}$$

18

$$n^3 - n = n(n-1)(n+1)$$

Three consecutive integers contain:

- one even number
- one multiple of 3

So product divisible by:

$$2 \times 3 = 6$$

$$\boxed{6 \mid (n^3 - n)}$$

19

Factor:

$$A^2B + AB^2 = AB(A+B)$$

Modulo 5:

$$3 \cdot 4 \cdot (3+4)$$

$$= 12 \cdot 7$$

Reduce:

$$2 \cdot 2 = 4$$

$$\boxed{4}$$

20

Among:

$$2026, 2027, 2028$$

one must be divisible by 3 because they are three consecutive integers.

Therefore product remainder:

$$\boxed{0}$$

CHAPTER 7 STATUS

We have completed:

- quotient–remainder theorem '
- valid remainder range '
- modulo notation '
- congruence meaning '
- sum/difference/product rules '
- powers under modulo '
- negative residues '
- digit-sum remainders '
- modulo 11 connection '
- polynomial substitution '
- square residue basics '
- same-remainder connection '
- factorization-based modular reasoning '
- GATE/placement practice '

But **Chapter 7 is not complete yet.**

Next comes the most important remainder section:

- cyclic powers
- huge exponents
- remainder cycles
- Fermat's little theorem
- Euler's theorem at aptitude/GATE level
- modular inverses
- last-digit connections
- $(a^n \bmod b^n)$ divisibility patterns
- geometric-series remainders
- advanced GATE/placement problems

Progress

Completed: 6/76 **Current:** 0/34 **Chapter 7 — Remainders & Modular Reasoning** **Next:** Chapter 7 Part 2 — **Huge powers, cycles & modular shortcuts**

USER

next

ASSISTANT

CHAPTER 7 — REMAINDERS & MODULAR REASONING

PART 2 — HUGE POWERS, CYCLICITY, FERMAT/EULER & MODULAR INVERSES

Status: 0/14 **PROGRESS** **GATE:** High **Placements:** High **Priority:** 0/14 **Per 1**

This is the section that lets you solve things like:

$$\mathbb{[7^{2026} \bmod 10]}$$

or

$$\mathbb{[3^{100} \bmod 7]}$$

without ever calculating the huge power.

The core idea is:

Powers repeat in cycles modulo a number. Find the cycle, reduce the exponent, and solve only the small case.

1. WHY POWERS FORM CYCLES

Take powers of 2 modulo 5:

$$\mathbb{[2^1 = 2 \equiv 2]}$$

$$\mathbb{[2^2 = 4 \equiv 4]}$$

$$\mathbb{[2^3 = 8 \equiv 3]}$$

$$\mathbb{[2^4 = 16 \equiv 1]}$$

$$\mathbb{[2^5 \equiv 2]}$$

Now the pattern repeats:

$$\mathbb{[2, 4, 3, 1, \boxed{2, 4, 3, 1, \dots}]}$$

Cycle length:

$$\mathbb{[\boxed{4}]}$$

So instead of computing $\mathbb{(2^{1000})}$, reduce 1000 modulo 4.

2. BASIC CYCLICITY METHOD

Find:

$$\mathbb{[2^{100} \bmod 5]}$$

Cycle length:

$$\mathbb{[4]}$$

Now:

$$\mathbb{[100 \bmod 4 = 0]}$$

Exponent corresponds to the 4th position of cycle.

Cycle:

$$\mathbb{[2, 4, 3, 1]}$$

Therefore:

$$\mathbb{[\boxed{2^{100} \equiv 1 \pmod{5}}]}$$

3. IMPORTANT EXPONENT-REMAINDER RULE

Suppose cycle length is $\phi(k)$.

Find:

$$n \bmod k = r$$

Then:

- if $r \neq 0$, use cycle position (r)
- if $r = 0$, use cycle position $\phi(k)$

& ⚠TRAP

Remainder 0 does **not** mean use exponent 0.

It means the last position of the cycle.

4. EXAMPLE

Find:

$$3^{57} \bmod 7$$

Build cycle:

$$3^1 \equiv 3$$

$$3^2 = 9 \equiv 2$$

$$3^3 \equiv 6$$

$$3^4 \equiv 4$$

$$3^5 \equiv 5$$

$$3^6 \equiv 1$$

Cycle length:

$$6$$

Now:

$$57 \bmod 6 = 3$$

So use:

$$3^3$$

Remainder:

$$\boxed{6}$$

5. USE NEGATIVE RESIDUES TO SHORTEN CYCLES

Find:

$$6^{101} \bmod 7$$

Since:

$$\mathbb{6} \equiv -1 \pmod{7}$$

Therefore:

$$\mathbb{6}^{101} \equiv (-1)^{101}$$

$$\mathbb{6} \equiv -1$$

Standard remainder:

$$\boxed{6}$$

This is far faster than finding a long cycle.

6. ANOTHER EXAMPLE

Find:

$$\mathbb{8}^{2026} \bmod{9}$$

$$\mathbb{8} \equiv -1 \pmod{9}$$

Exponent is even:

$$\mathbb{(-1)}^{2026} = 1$$

Therefore:

$$\boxed{1}$$

7. WHEN BASE IS 1 MODULO $\mathbb{m}$

If:

$$\mathbb{a} \equiv 1 \pmod{m}$$

then:

$$\mathbb{a}^n \equiv 1$$

for every positive $\mathbb{n}$.

Example:

$$\mathbb{101}^{1000} \bmod{10}$$

$$\mathbb{101} \equiv 1 \pmod{10}$$

So:

$$\boxed{1}$$

8. WHEN BASE IS -1 MODULO $\mathbb{m}$

If:

$$\mathbb{a} \equiv -1 \pmod{m}$$

then:

$$[a^n \equiv \begin{cases} 1, & \text{even } n \\ -1, & \text{odd } n \end{cases}]$$

This is one of the most useful shortcuts in modular arithmetic.

9. POWERS OF 10 MODULO 9

Because:

$$[10 \equiv 1 \pmod{9}]$$

we have:

$$[10^n \equiv 1 \pmod{9}]$$

Therefore:

$$[10^{100} + 10^{50} + 1]$$

mod 9 becomes:

$$[1 + 1 + 1 = 3]$$

Answer:

$$[\boxed{3}]$$

10. POWERS OF 10 MODULO 11

Since:

$$[10 \equiv -1 \pmod{11}]$$

$$[10^2 \equiv 1]$$

Thus powers alternate:

$$[10^1 \equiv -1]$$

$$[10^2 \equiv 1]$$

$$[10^3 \equiv -1]$$

This is the basis of alternating-sum divisibility by 11.

11. FERMAT'S LITTLE THEOREM

This is important for GATE-level modular reasoning.

If (p) is prime and:

$$[p \nmid a]$$

then:

$$[\boxed{a^{p-1} \equiv 1 \pmod{p}}]$$

Equivalent form:

$$[\boxed{a^p \equiv a \pmod{p}}]$$

MUST UNDERSTAND

12. EXAMPLE USING FERMAT

Find:

$$3^{100} \bmod 7$$

Since 7 is prime and:

$$7 \nmid 3$$

Fermat says:

$$3^6 \equiv 1 \pmod{7}$$

Reduce exponent:

$$100 = 6(16) + 4$$

Therefore:

$$3^{100} = (3^6)^{16} 3^4$$

Modulo 7:

$$\equiv 1^{16} \cdot 3^4$$

$$3^4 = 81$$

$$81 \equiv 4 \pmod{7}$$

Answer:

$$\boxed{4}$$

13. ANOTHER FERMAT EXAMPLE

Find:

$$2^{1000} \bmod 13$$

Since 13 prime:

$$2^{12} \equiv 1 \pmod{13}$$

Now:

$$1000 \bmod 12 = 4$$

So:

$$2^{1000} \equiv 2^4 = 16$$

$$16 \equiv 3 \pmod{13}$$

Answer:

$$\boxed{3}$$

14. IMPORTANT FERMAT CONDITION

Fermat's theorem:

$$[a^{p-1} \equiv 1 \pmod p]$$

requires:

$$[\gcd(a,p)=1]$$

Since (p) is prime, that simply means:

$$[p \nmid a]$$

& pTRAP

If:

$$[a \equiv 0 \pmod p]$$

then:

$$[a^n \equiv 0]$$

not 1.

15. FERMAT'S SECOND FORM

For every integer (a) and prime (p) :

$$[\boxed{a^p \equiv a \pmod p}]$$

This version even works when $(p \mid a)$.

Example:

$$[10^5 \equiv 10 \pmod 5]$$

Both sides:

$$[0 \pmod 5]$$

16. EULER'S TOTIENT FUNCTION

Euler's theorem generalizes Fermat.

First define:

$$[\phi(n)]$$

as the number of positive integers from 1 to (n) that are coprime to (n) .

Example:

For 8:

Numbers:

$$[1, 2, 3, 4, 5, 6, 7, 8]$$

Coprime to 8:

$$[1, 3, 5, 7]$$

So:

$$[\boxed{\phi(8)=4}]$$

17. EULER TOTIENT FORMULA

If:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

then:

$$\boxed{\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots}$$

Only distinct prime factors appear in the product.

18. EXAMPLE

Find:

$$\phi(12)$$

Prime factors:

$$12 = 2^2 \cdot 3$$

Therefore:

$$\phi(12) = 12 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right)$$

$$= 12 \cdot \frac{1}{2} \cdot \frac{2}{3}$$

$$= \boxed{4}$$

Indeed:

$$1, 5, 7, 11$$

are coprime to 12.

19. SPECIAL CASE FOR PRIME $\phi(p)$

If $\phi(p)$ is prime:

$$\boxed{\phi(p) = p - 1}$$

That is why Euler reduces to Fermat for prime modulus.

20. EULER'S THEOREM

If:

$$\gcd(a, n) = 1$$

then:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

This works even when $\phi(n)$ is composite.

21. EULER EXAMPLE

Find:

$$\mathbb{Z}_{100}^{\times}$$

Since:

$$\gcd(3,10)=1$$

and:

$$\phi(10)=4$$

Euler:

$$3^4 \equiv 1 \pmod{10}$$

Now:

$$100 \bmod 4 = 0$$

Therefore:

$$3^{100} \equiv 1$$

Answer:

$$\boxed{1}$$

22. ANOTHER EULER EXAMPLE

Find:

$$\mathbb{Z}_{12}^{\times}$$

$$\gcd(5,12)=1$$

$$\phi(12)=4$$

So:

$$5^4 \equiv 1 \pmod{12}$$

But even faster:

$$5^2 = 25 \equiv 1 \pmod{12}$$

Thus:

$$5^{100} = (5^2)^{50} \equiv 1$$

Answer:

$$\boxed{1}$$

Exam lesson

Euler gives a valid cycle length, but the **actual cycle may be shorter**.

Use the shortest reliable method.

23. EULER CONDITION

You may use:

$$\backslash[a^{\phi(n)} \equiv 1 \pmod n \backslash]$$

only when:

$$\backslash[\boxed{\gcd(a,n)=1} \backslash]$$

& pTRAP

For:

$$\backslash[2^{100} \bmod 8 \backslash]$$

Euler cannot be applied because:

$$\backslash[\gcd(2,8) \neq 1 \backslash]$$

But this problem is easier:

$$\backslash[2^3 = 8 \equiv 0 \pmod 8 \backslash]$$

so every higher power also gives:

$$\backslash[\boxed{0} \backslash]$$

24. MODULAR INVERSE — CORE IDEA

A number $a^{-1} \pmod m$ is a number x such that:

$$\backslash[\boxed{ax \equiv 1 \pmod m} \backslash]$$

Example:

Find inverse of 3 modulo 7.

We need:

$$\backslash[3x \equiv 1 \pmod 7 \backslash]$$

Try small numbers:

$$\backslash[3 \times 5 = 15 \backslash]$$

$$\backslash[15 \equiv 1 \pmod 7 \backslash]$$

Therefore:

$$\backslash[\boxed{3^{-1} \equiv 5 \pmod 7} \backslash]$$

25. WHEN DOES MODULAR INVERSE EXIST?

Very important:

$$\backslash[\boxed{a^{-1} \pmod m \text{ exists iff } \gcd(a,m)=1} \backslash]$$

Example:

Inverse of 2 modulo 6?

$$\backslash[\gcd(2,6)=2 \backslash]$$

So no inverse exists.

26. WHY MODULAR DIVISION IS DANGEROUS

Suppose:

$$2x \equiv 2 \pmod{6}$$

If you naively divide both sides by 2:

$$x \equiv 1 \pmod{6}$$

But actual solutions include:

$$x \equiv 1, 4 \pmod{6}$$

because:

$$2(1) = 2$$

and:

$$2(4) = 8 \equiv 2$$

So modular division is not ordinary division unless the divisor has an inverse modulo (m) .

27. SAFE MODULAR DIVISION

If:

$$ax \equiv b \pmod{m}$$

and:

$$\gcd(a, m) = 1$$

then multiply both sides by a^{-1} .

Example:

$$3x \equiv 4 \pmod{7}$$

Inverse of 3 modulo 7 is 5.

Multiply:

$$x \equiv 5 \cdot 4 = 20$$

$$20 \equiv 6 \pmod{7}$$

Answer:

$$\boxed{x \equiv 6 \pmod{7}}$$

28. FINDING INVERSE USING FERMAT

If modulus (p) is prime and:

$$p \nmid a$$

then:

$$a^{p-1} \equiv 1$$

So:

$$\boxed{a^{p-2}}$$

acts as inverse:

$$\boxed{a^{-1} \equiv a^{p-2} \pmod{p}}$$

This is more useful in algorithms than manual aptitude, but good GATE knowledge.

29. EXAMPLE

Inverse of 2 modulo 7:

$$2^{7-2} = 2^5 = 32$$

$$32 \equiv 4 \pmod{7}$$

Check:

$$2 \times 4 = 8 \equiv 1$$

So:

$$\boxed{2^{-1} \equiv 4 \pmod{7}}$$

30. LAST DIGIT AS MODULO 10

The last digit of a number is simply its remainder modulo 10.

Example:

Last digit of:

$$7^{100}$$

means:

$$7^{100} \bmod{10}$$

We will study full last-digit cycles in Chapter 8, but conceptually:

$$\boxed{\text{last digit} = \bmod{10}}$$

31. LAST TWO DIGITS

Last two digits correspond to:

$$\bmod{100}$$

Last three digits correspond to:

$$\bmod{1000}$$

This connects Chapter 7 directly to Chapter 8.

32. DIVISIBILITY OF $(a^n - b^n)$

Classic result:

For positive integer (n) :

$$\boxed{a-b \mid a^n - b^n}$$

Why?

Factor:

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$$

So it is always divisible by $(a-b)$.

33. EXAMPLE

Is:

$$7^{100} - 2^{100}$$

divisible by 5?

Since:

$$7 - 2 = 5$$

and:

$$5 \mid 7^{100} - 2^{100}$$

Yes.

$$\boxed{\text{Divisible by 5}}$$

Modulo method also says:

$$7 \equiv 2 \pmod{5}$$

so their 100th powers are congruent.

34. DIVISIBILITY OF $(a^n + b^n)$

If (n) is odd:

$$\boxed{a+b \mid a^n + b^n}$$

Example:

$$5^7 + 2^7$$

is divisible by:

$$5 + 2 = 7$$

because exponent 7 is odd.

35. IMPORTANT TRAP FOR $(a^n + b^n)$

If (n) is even, $(a+b)$ does not generally divide:

$$a^n + b^n$$

Example:

$$\lfloor 2^2+1^2=5 \rfloor$$

but:

$$\lfloor 2+1=3 \rfloor$$

and 3 does not divide 5.

So:

$$\lfloor \boxed{a+b \mid a^n+b^n} \text{ only guaranteed when } n \text{ odd} \rfloor$$

36. $(a+b)$ DIVIDES (a^n-b^n) FOR EVEN (n)

If (n) is even:

$$\lfloor \boxed{a+b \mid a^n-b^n} \rfloor$$

because modulo $(a+b)$:

$$\lfloor a \equiv -b \rfloor$$

Then:

$$\lfloor a^n-b^n \equiv (-b)^n-b^n \rfloor$$

For even (n) :

$$\lfloor b^n-b^n=0 \rfloor$$

37. SUMMARY OF POWER DIVISIBILITY

Always:

$$\lfloor \boxed{a-b \mid a^n-b^n} \rfloor$$

If (n) odd:

$$\lfloor \boxed{a+b \mid a^n+b^n} \rfloor$$

If (n) even:

$$\lfloor \boxed{a+b \mid a^n-b^n} \rfloor$$

These are very useful.

38. GEOMETRIC SERIES REMAINDER

Consider:

$$\lfloor 1+a+a^2+\cdots+a^{n-1} \rfloor$$

We know:

$$\lfloor \boxed{1+a+\cdots+a^{n-1} = \frac{a^n-1}{a-1}} \rfloor$$

when $(a \neq 1)$.

This helps divisibility and remainder questions.

39. EXAMPLE

Find:

$$\{ 1+2+2^2+\cdots+2^9 \}$$

modulo 3.

Since:

$$\{ 2\equiv -1\pmod{3} \}$$

terms alternate:

$$\{ 1,-1,1,-1,\ldots \}$$

There are 10 terms.

Pairs cancel:

$$\{ \boxed{0} \}$$

No geometric-sum calculation needed.

40. ANOTHER GEOMETRIC SERIES EXAMPLE

Find:

$$\{ 1+3+3^2+\cdots+3^{10} \}$$

modulo 2.

Since:

$$\{ 3\equiv 1\pmod{2} \}$$

Every term:

$$\{ \equiv 1 \}$$

Number of terms:

$$\{ 11 \}$$

So:

$$\{ 11\equiv 1\pmod{2} \}$$

Answer:

$$\{ \boxed{1} \}$$

41. REMAINDER OF REPEATED-DIGIT NUMBERS

Example:

$$\{ 111111 \}$$

This equals:

$$\{ \frac{10^6-1}{9} \}$$

But often digit-sum reasoning is easier.

Modulo 3:

Digit sum:

$$\boxed{6}$$

So remainder:

$$\boxed{0}$$

Modulo 9:

also:

$$\boxed{6}$$

because 6 is less than 9.

42. REPEATED 9s

$$999\dots 999$$

is:

$$10^n - 1$$

So modulo something where (10^n) is easy, repeated-9 numbers are easy.

Example:

$$999999 \bmod 7$$

$$999999 = 10^6 - 1$$

Since:

$$10 \equiv 3 \pmod{7}$$

and:

$$3^6 \equiv 1 \pmod{7}$$

then:

$$10^6 - 1 \equiv 0$$

Answer:

$$\boxed{0}$$

43. GATE-STYLE QUESTION 1

Find:

$$7^{222} \bmod 10$$

Powers of 7 mod 10:

$$7, 9, 3, 1$$

Cycle length:

$$4$$

$$222 \bmod 4 = 2$$

Second cycle term:

$$\boxed{9}$$

44. GATE-STYLE QUESTION 2

Find:

$$2^{100} + 3^{100}$$

modulo 5.

For 2:

Cycle length 4.

$$100 \bmod 4 = 0$$

$$2^{100} \equiv 1$$

For 3:

$$3^4 \equiv 1 \pmod{5}$$

So:

$$3^{100} \equiv 1$$

Sum:

$$2$$

Answer:

$$\boxed{2}$$

45. GATE-STYLE QUESTION 3

Find:

$$11^{1000} \bmod{12}$$

$$11 \equiv -1 \pmod{12}$$

Even exponent:

$$(-1)^{1000} = 1$$

Answer:

$$\boxed{1}$$

46. GATE-STYLE QUESTION 4

Find:

$$4^{2026} \bmod{7}$$

Cycle:

$$4^1 \equiv 4$$

$$4^2 = 16 \equiv 2 \pmod{17}$$

$$4^3 \equiv 8 \equiv 1 \pmod{17}$$

Cycle length:

$$3$$

$$2026 \bmod 3 = 1$$

So:

$$\boxed{4}$$

47. GATE-STYLE QUESTION 5

Find:

$$3^{1000} \bmod 13$$

Since 13 prime:

$$3^{12} \equiv 1 \pmod{13}$$

$$1000 \bmod 12 = 4$$

$$3^4 = 81$$

$$81 \equiv 3 \pmod{13}$$

Answer:

$$\boxed{3}$$

48. PLACEMENT-STYLE QUESTION 1

Find last digit of:

$$9^{999}$$

$$9 \equiv -1 \pmod{10}$$

Odd exponent:

$$\boxed{9}$$

49. PLACEMENT-STYLE QUESTION 2

Find remainder of:

$$17^{35}$$

when divided by 16.

$$17 \equiv 1 \pmod{16}$$

So:

$$\boxed{1}$$

50. PLACEMENT-STYLE QUESTION 3

Find remainder of:

$$49^{51}$$

when divided by 50.

$$49 \equiv -1 \pmod{50}$$

Odd exponent:

$$-1 \equiv 49$$

Answer:

$$\boxed{49}$$

51. PLACEMENT-STYLE QUESTION 4

Find remainder of:

$$6^{100}$$

when divided by 7.

$$6 \equiv -1 \pmod{7}$$

Even exponent:

$$\boxed{1}$$

52. MODULAR INVERSE EXAMPLE

Solve:

$$5x \equiv 3 \pmod{7}$$

Inverse of 5 modulo 7?

$$5 \times 3 = 15 \equiv 1$$

So inverse is 3.

Multiply:

$$x \equiv 3 \times 3 = 9$$

$$x \equiv 2 \pmod{7}$$

Answer:

$$\boxed{x \equiv 2 \pmod{7}}$$

53. COMMON TRAPS

& pTRAP 1 — Cycle remainder zero

If exponent mod cycle length = 0, use last cycle term.

& pTRAP 2 — Fermat on composite modulus

Fermat specifically uses prime modulus.

For composite modulus, think Euler or direct cycles.

& pTRAP 3 — Euler without coprimality

Need:

$$\forall \gcd(a,n)=1$$

& pTRAP 4 — Modular division

Never cancel/divide unless the factor has an inverse modulo the modulus.

& pTRAP 5 — $(a+b)$ divisibility

$$\forall a+b \mid a^n+b^n$$

only guaranteed for odd n .

& pTRAP 6 — Calculating huge powers

Always ask first:

- can base become 0?
 - 1?
 - -1?
 - does it have a small cycle?
 - can Fermat/Euler reduce exponent?
-

54. DECISION TREE FOR HUGE POWERS

If asked:

$$\forall a^n \bmod m$$

use this order:

Step 1

Reduce base:

$$\forall a \bmod m$$

Step 2

Check easy residues:

$$\forall 0, 1, -1$$

Step 3

Look for short power cycle.

Step 4

If modulus is prime, consider Fermat:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}}$$

Step 5

If modulus composite and coprime base, consider Euler:

$$\boxed{a^{\phi(m)} \equiv 1 \pmod{m}}$$

Step 6

Reduce exponent and finish.

55. 5-MINUTE REVISION

Fermat

If p prime and $p \nmid a$:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}}$$

Euler

If:

$$\gcd(a, n) = 1$$

then:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

Inverse

Exists iff:

$$\boxed{\gcd(a, m) = 1}$$

Power divisibility

$$\boxed{a-b \mid a^n - b^n}$$

If n odd:

$$\boxed{a+b \mid a^n + b^n}$$

If n even:

$$\boxed{a+b \mid a^n - b^n}$$

PRACTICE — LEVEL 0

Q1

Find:

$$2^{20} \pmod{5}$$

Q2

Find:

$$8^{51} \pmod{9}$$

Q3

Find:

$$\lfloor 11^{40} \bmod 12 \rfloor$$

Q4

Find:

$$\lfloor 101^{999} \bmod 10 \rfloor$$

Q5

Find inverse of 3 modulo 5.

LEVEL 1

Q6

Find:

$$\lfloor 3^{50} \bmod 7 \rfloor$$

Q7

Find:

$$\lfloor 2^{100} \bmod 13 \rfloor$$

Q8

Find:

$$\lfloor 7^{101} + 3^{101} \rfloor$$

modulo 10.

Q9

Solve:

$$\lfloor 4x \equiv 3 \pmod{7} \rfloor$$

Q10

Find:

$$\lfloor 5^{222} \bmod 8 \rfloor$$

LEVEL 2

Q11

Find:

$$\lfloor 7^{1000} \bmod 13 \rfloor$$

Q12

Find:

$$\lfloor 2^{1000} + 4^{1000} \rfloor$$

modulo 5.

Q13

Find:

$$\lfloor 13^{100} + 11^{100} \rfloor$$

modulo 12.

Q14

Determine whether:

$$\lfloor 5^{101} + 2^{101} \rfloor$$

is divisible by 7.

Q15

Determine whether:

$$\lfloor 11^{100} - 1 \rfloor$$

is divisible by 10.

LEVEL 3 — EXAM PRESSURE

Q16

Find:

$$\lfloor 3^{1000} \rfloor \bmod 17$$

Q17

Find:

$$\lfloor 7^{2026} \rfloor \bmod 20$$

Q18

Find remainder of:

$$\lfloor 2^{100} + 2^{101} + 2^{102} \rfloor$$

when divided by 7.

Q19

Find:

$$\lfloor 1 + 2 + 2^2 + \cdots + 2^{20} \rfloor$$

modulo 3.

Q20

Solve:

$$\lfloor 7x \equiv 5 \pmod{12} \rfloor$$

ANSWERS

1

Cycle of $2 \bmod 5$ has length 4.

$$20 \bmod 4 = 0$$

$$\boxed{1}$$

2

$$8 \equiv -1 \pmod{9}$$

Odd exponent:

$$\boxed{8}$$

3

$$11 \equiv -1 \pmod{12}$$

Even exponent:

$$\boxed{1}$$

4

$$101 \equiv 1 \pmod{10}$$

$$\boxed{1}$$

5

Need:

$$3x \equiv 1 \pmod{5}$$

$$3 \times 2 = 6 \equiv 1$$

$$\boxed{2}$$

6

By Fermat:

$$3^6 \equiv 1 \pmod{7}$$

$$50 \bmod 6 = 2$$

$$3^2 = 9 \equiv 3$$

$$\boxed{2}$$

7

$$2^{12} \equiv 1 \pmod{13}$$

$$100 \bmod 12 = 4$$

$$2^4 = 16 \equiv 3$$

$$\boxed{3}$$

8

Modulo 10:

$$\backslash[7^{\{101\}} \backslash]$$

cycle $\backslash(7,9,3,1\backslash)$, and:

$$\backslash[101\backslash\bmod 4=1 \backslash]$$

so remainder 7.

For:

$$\backslash[3^{\{101\}} \backslash]$$

cycle $\backslash(3,9,7,1\backslash)$, same position:

$$\backslash[3 \backslash]$$

Sum:

$$\backslash[10\equiv 0 \backslash]$$

$$\backslash[\boxed{0} \backslash]$$

9

Inverse of 4 modulo 7 is 2 because:

$$\backslash[4\cdot 2=8\equiv 1 \backslash]$$

So:

$$\backslash[x\equiv 2\cdot 3=6 \backslash]$$

$$\backslash[\boxed{6} \backslash]$$

10

$$\backslash[5^2=25\equiv 1\pmod 8 \backslash]$$

Even exponent:

$$\backslash[\boxed{1} \backslash]$$

11

Since 13 prime:

$$\backslash[7^{\{12\}}\equiv 1 \backslash]$$

$$\backslash[1000\backslash\bmod 12=4 \backslash]$$

$$\backslash[7^2=49\equiv 10 \backslash]$$

$$\backslash[7^4\equiv 10^2=100\equiv 9 \backslash]$$

$$\backslash[\boxed{9} \backslash]$$

12

Modulo 5:

$$\backslash[2^4\equiv 1 \backslash]$$

$$\backslash[4\equiv -1 \backslash]$$

Since 1000 divisible by 4:

$$\backslash[2^{\{1000\}}\equiv 1 \backslash]$$

and even exponent:

$$\backslash[4^{\{1000\}}\equiv 1 \backslash]$$

Sum:

$$\backslash[\boxed{2} \backslash]$$

13

$$\backslash[13\equiv 1\pmod{12} \backslash]$$

$$\backslash[11\equiv -1\pmod{12} \backslash]$$

Even exponent:

$$\backslash[1+1=2 \backslash]$$

$$\backslash[\boxed{2} \backslash]$$

14

Since:

$$\backslash[5+2=7 \backslash]$$

and 101 is odd:

$$\backslash[7\mid 5^{\{101\}}+2^{\{101\}} \backslash]$$

$$\backslash[\boxed{\text{Yes}} \backslash]$$

15

$$\backslash[11\equiv 1\pmod{10} \backslash]$$

So:

$$\backslash[11^{\{100\}}-1\equiv 1-1=0 \backslash]$$

$$\backslash[\boxed{\text{Yes}} \backslash]$$

16

Since 17 prime:

$$\backslash[3^{\{16\}}\equiv 1 \backslash]$$

$$\backslash[1000\bmod 16=8 \backslash]$$

$$\backslash[3^4=81\equiv 13\pmod{17} \backslash]$$

$$\backslash[3^8\equiv 13^2=169\equiv 16 \backslash]$$

$$\backslash[\boxed{16} \backslash]$$

17

Modulo 20:

$$\backslash[7^2=49\equiv 9 \backslash]$$

$$\backslash[7^4\equiv 9^2=81\equiv 1 \backslash]$$

Cycle divides 4.

$$\backslash[2026\bmod 4=2 \backslash]$$

So:

$$\backslash[7^{\{2026\}}\equiv 7^2 \backslash]$$

$\boxed{9}$

18

Factor:

$$2^{100}(1+2+4)$$

$$=7 \cdot 2^{100}$$

Therefore divisible by 7.

Remainder:

$$\boxed{0}$$

19

Modulo 3:

$$2 \equiv -1$$

So terms alternate:

$$1, -1, 1, -1, \dots$$

There are:

$$21$$

terms from exponent 0 through 20.

Ten pairs cancel, one (1) remains.

$$\boxed{1}$$

20

Need inverse of 7 modulo 12.

$$7 \cdot 7 = 49 \equiv 1 \pmod{12}$$

So inverse is 7.

$$x \equiv 7 \cdot 5 = 35$$

$$35 \equiv 11 \pmod{12}$$

$$\boxed{x \equiv 11 \pmod{12}}$$

CHAPTER 7 — FINAL CHECK

We have now covered:

- quotient–remainder theorem
- modular notation
- modular addition/subtraction/multiplication
- negative residues
- polynomial substitution
- digit-sum remainder reasoning
- cyclic powers
- huge exponent reduction

- $\backslash(\pm 1)$ shortcuts '
- Fermat's Little Theorem '
- Euler's theorem '
- Euler totient basics '
- modular inverses '
- safe modular division '
- power divisibility identities '
- geometric-series remainder reasoning '
- last-digit connection '
- GATE-style problems '
- placement-style problems '

Therefore:

' CHAPTER 7 COMPLETE

Progress

Completed: 7/76 **Current:** ' Chapter 7 — Remainders & Modular Reasoning **Next:** ~~Chapter 7~~ Chapter 8 — **Units Digit, Last Digits & Cyclicity**

Chapter 8 will turn modular arithmetic into very fast exam methods for:

- unit digits of huge powers
- last two digits
- cyclicity tables
- powers ending in 0/1/5/6
- products and factorial endings
- nested exponents
- last non-zero digit basics
- GATE and placement number-system shortcuts.

USER

next

ASSISTANT

CHAPTER 8 — UNITS DIGIT, LAST DIGITS & CYCLICITY

PART 1 — UNIT DIGIT OF POWERS, PRODUCTS & HUGE EXPONENTS

Status: ~~0=Not~~ **PROGRESS** **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This chapter is a direct application of modular arithmetic.

Remember:

$\backslash \boxed{\text{Unit digit of } N = N \bmod 10} \backslash$

So questions like:

Find the unit digit of 7^{2026}

really mean:

$$7^{2026} \bmod 10$$

The advantage is that powers modulo 10 repeat in very short cycles.

1. WHY ONLY THE LAST DIGIT OF THE BASE MATTERS

Suppose:

$$347^{100}$$

For the unit digit, only:

$$347 \bmod 10 = 7$$

matters.

Therefore:

$$347^{100}$$

has the same unit digit as:

$$7^{100}$$

Golden rule

For unit-digit questions, immediately replace the base by its last digit.

2. UNIT-DIGIT CYCLES — MUST MEMORIZE

This is one of the most useful placement tables.

Base ending	Unit-digit cycle	Cycle length
0	0	1
1	1	1
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
4	4, 6	2
5	5	1
6	6	1
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

Base ending	Unit-digit cycle	Cycle length
9	9, 1	2

MUST MEMORIZE

You should eventually know this table instantly.

3. WHY 0, 1, 5, 6 ARE SPECIAL

Any positive power ending in:

0

still ends in 0.

Similarly:

1^n

5^n

6^n

So:

$\boxed{0, 1, 5, 6}$ are fixed unit digits

4. CYCLE OF 2

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

Unit digits:

$\boxed{2, 4, 8, 6}$

Then repeat.

Cycle length:

4

5. EXAMPLE — UNIT DIGIT OF (2^{37})

Cycle length:

4

Compute:

$37 \bmod 4 = 1$

Take first number in cycle:

$\boxed{2}$

6. EXPONENT REMAINDER ZERO TRAP

Find unit digit of:

$$2^{100}$$

$$100 \bmod 4 = 0$$

Do **not** choose position 0.

Use the last position of the 4-cycle:

$$2, 4, 8, \boxed{6}$$

Therefore:

$$\boxed{6}$$

Memory

If:

$$n \bmod 4 = 0$$

use cycle position 4.

7. CYCLE OF 3

$$3^1 \text{ to } 3$$

$$3^2 \text{ to } 9$$

$$3^3 \text{ to } 7$$

$$3^4 \text{ to } 1$$

Cycle:

$$\boxed{3, 9, 7, 1}$$

8. EXAMPLE

Unit digit of:

$$3^{2026}$$

$$2026 \bmod 4 = 2$$

Second position:

$$\boxed{9}$$

9. CYCLE OF 7

$$7, 9, 3, 1$$

Example:

Unit digit of:

$$\{ 7^{99} \}$$

$$99 \bmod 4 = 3$$

Third position:

$$\boxed{3}$$

10. CYCLE OF 8

$$\{ 8, 4, 2, 6 \}$$

Example:

Unit digit of:

$$\{ 8^{54} \}$$

$$54 \bmod 4 = 2$$

Second:

$$\boxed{4}$$

11. BASE ENDING IN 4

Cycle:

$$\{ 4, 6 \}$$

So:

- odd exponent !' 4
- even exponent !' 6

Example:

$$\{ 24^{113} \}$$

Base ending:

$$\{ 4 \}$$

Exponent odd.

Unit digit:

$$\boxed{4}$$

12. BASE ENDING IN 9

Cycle:

$$\{ 9, 1 \}$$

So:

- odd exponent !' 9
 - even exponent !' 1
-

Example:

$$\backslash[39^{\{2026\}} \backslash]$$

Exponent even:

$$\backslash[\boxed{1} \backslash]$$

13. FAST PARITY METHOD FOR 4 AND 9

You do not need modulo 4.

For ending 4 or 9, only ask:

Is exponent odd or even?

That is faster.

14. UNIT DIGIT OF A PRODUCT

Find unit digit of:

$$\backslash[27^{\{15\}} \times 13^{\{8\}} \backslash]$$

First factor:

$$\backslash[27^{\{15\}} \text{ to } 7^{\{15\}} \backslash]$$

Cycle of 7:

$$\backslash[7, 9, 3, 1 \backslash]$$

$$\backslash[15 \bmod 4 = 3 \backslash]$$

So unit digit:

$$\backslash[3 \backslash]$$

Second:

$$\backslash[13^8 \text{ to } 3^8 \backslash]$$

$$\backslash[8 \bmod 4 = 0 \backslash]$$

Unit digit:

$$\backslash[1 \backslash]$$

Product:

$$\backslash[3 \times 1 = 3 \backslash]$$

Answer:

$$\backslash[\boxed{3} \backslash]$$

15. REDUCE AFTER EVERY MULTIPLICATION

Suppose factors have unit digits:

$$\{ 8, 7, 9 \}$$

Product unit digit:

$$8 \times 7 = 56 \rightarrow 6$$

Then:

$$6 \times 9 = 54 \rightarrow 4$$

So:

$$\boxed{4}$$

Do not multiply huge numbers.

16. UNIT DIGIT OF A SUM

Find unit digit:

$$2^{10} + 3^{10}$$

For (2^{10}) :

$$10 \bmod 4 = 2$$

unit digit:

$$4$$

For (3^{10}) :

second position:

$$9$$

Sum:

$$4 + 9 = 13$$

Unit digit:

$$\boxed{3}$$

17. UNIT DIGIT OF DIFFERENCE

Find unit digit:

$$7^{20} - 3^{20}$$

Both exponents:

$$20 \bmod 4 = 0$$

So:

$$7^{20} \rightarrow 1$$

$$3^{20} \rightarrow 1$$

Difference:

$$0$$

Unit digit:

$\boxed{0}$

18. VERY LARGE BASE

Find unit digit:

$$123456789^{321}$$

Base last digit:

$$9$$

Exponent 321 is odd.

So:

$$\boxed{9}$$

That is the entire problem.

19. HUGE EXPONENT — REDUCE ONLY WHAT YOU NEED

Find unit digit:

$$7^{123456789}$$

Cycle length:

$$4$$

So only need exponent modulo 4.

For divisibility/mod 4, use last two digits:

$$89 \bmod 4 = 1$$

Therefore first position of 7-cycle:

$$\boxed{7}$$

20. EXPONENT ITSELF IS A POWER

Find unit digit:

$$3^{2^{10}}$$

Cycle length of 3:

$$4$$

So we need:

$$2^{10} \bmod 4$$

Since:

$$2^{10}$$

is divisible by 4,

$$2^{10} \bmod 4 = 0$$

Therefore use 4th position of 3-cycle:

$\boxed{1}$

21. NESTED EXPONENTS — CORE METHOD

For:

a^{b^c}

to find unit digit:

1. find cycle length of (a)
2. compute exponent (b^c) modulo that cycle length
3. use corresponding cycle position

Do **not** compute (b^c) completely.

22. EXAMPLE — NESTED POWER

Find unit digit:

7^{3^5}

Cycle length of 7:

4

Need:

$3^5 \bmod 4$

Since:

$3 \equiv -1 \pmod{4}$

$3^5 \equiv -1 \equiv 3$

So use 3rd position in:

$7, 9, 3, 1$

Answer:

$\boxed{3}$

23. ANOTHER NESTED POWER

Find unit digit:

$2^{5^{100}}$

Cycle of 2 length 4.

Need:

$5^{100} \bmod 4$

But:

$5 \equiv 1 \pmod{4}$

Therefore:

$$5^{100} \equiv 1 \pmod{2}$$

Use first position in 2-cycle:

$$\boxed{2}$$

24. EXPONENT ENDING RULE FOR MOD 4

If exponent itself is huge, to find mod 4 you often only need its final two digits.

Example:

$$1234567^{987654}$$

Base is 7.

Need exponent:

$$987654 \bmod 4$$

Last two digits:

$$54 \bmod 4 = 2$$

So 2nd position of 7-cycle:

$$\boxed{9}$$

25. UNIT DIGIT OF FACTORIALS

Recall:

$$n! = 1 \cdot 2 \cdot \dots \cdot n$$

For:

$$n \geq 5$$

the product contains factors:

$$2 \times 5 = 10$$

Therefore:

$$n!$$

ends in 0.

So:

$$\boxed{n! \text{ has unit digit } 0 \text{ for } n \geq 5}$$

26. EXAMPLES

$$4! = 24$$

unit digit:

$$4$$

$$5! = 120$$

unit digit:

$$0$$

$$100!$$

unit digit:

$$\boxed{0}$$

27. UNIT DIGIT OF PRODUCT OF CONSECUTIVE NUMBERS

Suppose:

$$21 \times 22 \times 23 \times 24 \times 25$$

Contains:

$$22 \text{ or } 24$$

providing factor 2, and:

$$25$$

providing factor 5.

Thus product has factor 10.

Unit digit:

$$\boxed{0}$$

No multiplication required.

28. WHEN A PRODUCT MUST END IN ZERO

If the prime factorization contains at least one:

$$2$$

and one:

$$5$$

then it contains:

$$10$$

and ends in zero.

This is trailing-zero reasoning.

29. UNIT DIGIT OF SQUARES

Possible unit digits of a perfect square are:

$$\boxed{0, 1, 4, 5, 6, 9}$$

Never:

[2,3,7,8]

Why?

Square last digits (0) through (9):

Last digit	Square last digit
0	0
1	1
2	4
3	9
4	6
5	5
6	6
7	9
8	4
9	1

This is useful for option elimination.

30. UNIT DIGIT OF PERFECT CUBES

Cube last digits are:

Last digit	Cube last digit
0	0
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9

Interesting fact:

Every digit (0) through (9) is possible as a cube's unit digit.

31. CUBE UNIT-DIGIT INVERSE PAIRS

Useful mental map:

$$\{2^3 \text{ to } 8\}$$

$$\{8^3 \text{ to } 2\}$$

$$\{3^3 \text{ to } 7\}$$

$$\{7^3 \text{ to } 3\}$$

And:

$$\{0, 1, 4, 5, 6, 9\}$$

map to themselves as unit digits under cubing.

This can help in cube-root unit-digit puzzles.

32. UNIT DIGIT OF POLYNOMIAL EXPRESSIONS

Find unit digit:

$$\{23^5 + 17^3 + 9^2\}$$

First:

$$\{23^5 \text{ to } 3^5\}$$

Cycle of 3:

$$\{5 \bmod 4 = 1\}$$

! 3.

Next:

$$\{17^3 \text{ to } 7^3\}$$

third cycle position:

$$\{3\}$$

Next:

$$\{9^2 \text{ to } 1\}$$

Sum:

$$\{3 + 3 + 1 = 7\}$$

Answer:

$$\boxed{7}$$

33. EXPRESSION WITH MULTIPLICATION AND ADDITION

Find unit digit:

$$\{2^{15} \cdot 3^{12} + 7^9\}$$

For:

$$\backslash[2^{\{15\}} \backslash]$$

$$\backslash[15\text{bmod}4=3 \backslash]$$

unit digit 8.

For:

$$\backslash[3^{\{12\}} \backslash]$$

$$\backslash[12\text{bmod}4=0 \backslash]$$

unit digit 1.

Product:

$$\backslash[8 \backslash]$$

For:

$$\backslash[7^9 \backslash]$$

$$\backslash[9\text{bmod}4=1 \backslash]$$

unit digit 7.

Sum:

$$\backslash[15 \backslash]$$

unit digit:

$$\backslash[\boxed{5} \backslash]$$

34. SAME BASE POWERS — FACTOR FIRST

Find unit digit:

$$\backslash[3^{\{20\}}+3^{\{21\}} \backslash]$$

Factor:

$$\backslash[3^{\{20\}}(1+3) \backslash]$$

$$\backslash[=4\cdot 3^{\{20\}} \backslash]$$

Now:

$$\backslash[3^{\{20\}} \backslash]$$

unit digit 1.

So:

$$\backslash[4\text{times}1 \backslash]$$

Unit digit:

$$\backslash[\boxed{4} \backslash]$$

Could also solve individually, but factorization may be faster.

35. CLEVER ZERO CREATION

Find unit digit:

$$\lfloor 2^{\{100\}} + 8^{\{100\}} \rfloor$$

Both exponent mod 4 = 0.

$$\lfloor 2^{\{100\}} \rfloor \pmod{6}$$

$$\lfloor 8^{\{100\}} \rfloor \pmod{6}$$

Sum:

$$\lfloor 12 \rfloor$$

Answer:

$$\lfloor \boxed{2} \rfloor$$

But in other questions combinations may create 10 and yield zero.

Always look before computing.

36. POWERS ENDING IN 5 OR 6

This deserves separate memorization:

$$\lfloor \boxed{\text{Any positive power ending in 5 ends in 5}} \rfloor$$

$$\lfloor \boxed{\text{Any positive power ending in 6 ends in 6}} \rfloor$$

Examples:

$$\lfloor 125^{\{999\}} \rfloor \pmod{5}$$

$$\lfloor 736^{\{2026\}} \rfloor \pmod{6}$$

37. POWERS ENDING IN 0 OR 1

Similarly:

$$\lfloor 40^{\{12345\}} \rfloor \pmod{0}$$

$$\lfloor 781^{\{567\}} \rfloor \pmod{1}$$

Immediate answers.

38. GATE-STYLE QUESTION 1

Find unit digit of:

$$\lfloor 17^{\{2026\}} \rfloor$$

Base last digit:

$$\lfloor 7 \rfloor$$

Cycle:

$$\lfloor 7, 9, 3, 1 \rfloor$$

$$\lfloor 2026 \bmod 4 = 2 \rfloor$$

Answer:

$$\boxed{9}$$

39. GATE-STYLE QUESTION 2

Find unit digit of:

$$\lfloor 3^{100} + 7^{100} \rfloor$$

Both have exponent:

$$\lfloor 100 \bmod 4 = 0 \rfloor$$

Thus:

$$\lfloor 3^{100} \bmod 10 \rfloor$$

$$\lfloor 7^{100} \bmod 10 \rfloor$$

Sum:

$$\boxed{2}$$

40. GATE-STYLE QUESTION 3

Find unit digit of:

$$\lfloor 2^{3^4} \rfloor$$

Cycle of 2:

$$\lfloor 4 \rfloor$$

Need:

$$\lfloor 3^4 \bmod 4 \rfloor$$

Since:

$$\lfloor 3 \equiv -1 \pmod{4} \rfloor$$

even power:

$$\lfloor 3^4 \equiv 1 \rfloor$$

Use first position:

$$\boxed{2}$$

41. GATE-STYLE QUESTION 4

Find unit digit of:

$$\lfloor 9^{9^9} \rfloor$$

For base 9:

- odd exponent !' 9
-

- even exponent != 1

Since:

$$\sqrt[9]{9}$$

is odd.

Therefore:

$$\boxed{9}$$

No need to calculate the exponent.

42. GATE-STYLE QUESTION 5

Find unit digit:

$$4^{7^{100}}$$

Base ending 4.

Need only exponent parity.

$$7^{100}$$

is odd.

Therefore:

$$\boxed{4}$$

43. PLACEMENT-STYLE QUESTION 1

Find unit digit of:

$$123^{456}$$

Base:

$$3$$

$$456 \bmod 4 = 0$$

Fourth position:

$$\boxed{1}$$

44. PLACEMENT-STYLE QUESTION 2

Find unit digit:

$$87^{65} \times 32^{14}$$

First:

$$7^{65}$$

$$65 \bmod 4 = 1$$

!= 7.

Second:

$$\backslash[2^{\{14\}} \backslash]$$

$$\backslash[14 \bmod 4 = 2 \backslash]$$

! 4.

Product:

$$\backslash[28 \backslash]$$

Unit digit:

$$\backslash[\boxed{8} \backslash]$$

45. PLACEMENT-STYLE QUESTION 3

Find unit digit of:

$$\backslash[6^{\{999\}} + 5^{\{2026\}} + 4^{\{101\}} \backslash]$$

$$\backslash[6^{\{999\}} \bmod 6 \backslash]$$

$$\backslash[5^{\{2026\}} \bmod 5 \backslash]$$

$$\backslash[4^{\{101\}} \bmod 4 \backslash]$$

Sum:

$$\backslash[15 \backslash]$$

Answer:

$$\backslash[\boxed{5} \backslash]$$

46. PLACEMENT-STYLE QUESTION 4

Find unit digit:

$$\backslash[8^{\{2^{\{20\}}\}} \backslash]$$

Cycle of 8 length 4.

Need:

$$\backslash[2^{\{20\}} \bmod 4 \backslash]$$

For exponent $\backslash(\geq 2\backslash)$:

$$\backslash[2^{\{20\}} \backslash]$$

is divisible by 4.

So cycle position 4:

$$\backslash[\boxed{6} \backslash]$$

47. SPECIAL PATTERN— PRODUCT OF FOUR CONSECUTIVE INTEGERS

Any four consecutive integers contain at least:

- enough factor 2s,
- often factors making endings easy,

but do not assume unit digit 0 unless factor 5 is also present.

Example:

$$[2 \cdot 3 \cdot 4 \cdot 5]$$

ends in 0.

But:

$$[6 \cdot 7 \cdot 8 \cdot 9]$$

does not contain factor 5.

So do not overgeneralize.

48. FACTORIAL SUM TRICK

Find unit digit:

$$[1! + 2! + 3! + \dots + 100!]$$

For:

$$[n! \text{ quad } n \geq 5]$$

unit digit is 0.

So only calculate:

$$[1! + 2! + 3! + 4!]$$

$$[= 1 + 2 + 6 + 24]$$

$$[= 33]$$

Unit digit:

$$[\boxed{3}]$$

This is a classic placement pattern.

49. ANOTHER FACTORIAL SUM

Find unit digit:

$$[2! + 3! + 4! + \dots + 50!]$$

Terms from 5! onward end in 0.

So:

$$[2! + 3! + 4!]$$

$$[= 2 + 6 + 24 = 32]$$

Unit digit:

$$[\boxed{2}]$$

50. LAST DIGIT OF (n^n) — EXAMPLE

Find unit digit of:

$$7^7$$

$$7 \bmod 4 = 3$$

Third position of 7-cycle:

$$\boxed{3}$$

For larger (n^n) , reduce exponent (n) by the cycle of its last digit.

51. UNIT DIGIT OF $(1^1 + 2^2 + \dots)$

These questions are usually solved term-by-term using cycles or block patterns.

Example:

Find unit digit:

$$1^1 + 2^2 + 3^3 + 4^4$$

Unit digits:

$$1$$

$$4$$

$$3^3 \text{ to } 7$$

$$4^4 \text{ to } 6$$

Sum:

$$1 + 4 + 7 + 6 = 18$$

Answer:

$$\boxed{8}$$

52. OPTION ELIMINATION WITH SQUARE UNIT DIGITS

Suppose options for a perfect square are:

A. 12342 B. 89347 C. 45689 D. 76538

Look only at unit digits:

- 2 impossible
- 7 impossible
- 9 possible
- 8 impossible

Only:

$$\boxed{\text{C}}$$

could be a square.

No square root needed.

53. COMMON TRAPS

& pTRAP 1 — Using the entire base

Only last digit matters for unit digit.

& pTRAP 2 — Exponent modulo cycle = 0

Use last cycle position.

& pTRAP 3 — Nested exponents

Reduce the exponent modulo the **base's cycle length**, not modulo 10 automatically.

& pTRAP 4 — Base 4 or 9

Only exponent parity matters.

& pTRAP 5 — Factorials

For $(n \geq 5)$, factorial unit digit is 0, but **last non-zero digit** is a different question.

We will cover that later.

54. MUST-MEMORIZE TABLE

$$\begin{array}{|c|c|} \hline \text{Ending} & \text{Cycle} \\ \hline 0 & 0 \\ 1 & 1 \\ 2 & 2, 4, 8, 6 \\ 3 & 3, 9, 7, 1 \\ 4 & 4, 6 \\ 5 & 5 \\ 6 & 6 \\ 7 & 7, 9, 3, 1 \\ 8 & 8, 4, 2, 6 \\ 9 & 9, 1 \\ \hline \end{array}$$

This single table solves a huge fraction of unit-digit questions.

55. MEMORY GROUPING

Rather than memorizing 10 unrelated rows:

Fixed

$\{0, 1, 5, 6\}$

Two-cycle

$\{4, 9\}$

Four-cycle

$\{2, 3, 7, 8\}$

That is much easier.

56. 5-MINUTE REVISION

For unit digit:

1. Replace base by last digit.
 2. Identify cycle.
 3. Reduce exponent by cycle length.
 4. If exponent remainder is zero, use last cycle term.
 5. For products/sums, find individual unit digits and combine.
 6. For nested powers, reduce the exponent itself modulo cycle length.
 7. For factorial $(n \geq 5)$, unit digit = 0.
-

PRACTICE — LEVEL 0

Q1

Find unit digit:

$$\lfloor 2^{17} \rfloor$$

Q2

Find unit digit:

$$\lfloor 3^{24} \rfloor$$

Q3

Find unit digit:

$$\lfloor 4^{101} \rfloor$$

Q4

Find unit digit:

$$\lfloor 9^{80} \rfloor$$

Q5

Find unit digit:

$$\lfloor 125^{999} \rfloor$$

LEVEL 1

Q6

Find unit digit:

$$\lfloor 27^{51} \rfloor$$

Q7

Find unit digit:

$$\lfloor 13^{37} + 17^{26} \rfloor$$

Q8

Find unit digit:

$$\lfloor 23^{15} \times 18^{14} \rfloor$$

Q9

Find unit digit:

$$\lfloor 2^{\{3^5\}} \rfloor$$

Q10

Find unit digit:

$$\lfloor 7^{\{2^{\{10\}}\}} \rfloor$$

LEVEL 2

Q11

Find unit digit:

$$\lfloor 3^{\{4^{\{5\}}\}} \rfloor$$

Q12

Find unit digit:

$$\lfloor 8^{\{7^{\{6\}}\}} \rfloor$$

Q13

Find unit digit:

$$\lfloor 2^{\{100\}} + 3^{\{100\}} + 7^{\{100\}} + 8^{\{100\}} \rfloor$$

Q14

Find unit digit:

$$\lfloor 1! + 2! + \cdots + 50! \rfloor$$

Q15

Find unit digit:

$$\lfloor 99^{\{99\}} + 88^{\{88\}} \rfloor$$

LEVEL 3 — EXAM PRESSURE

Q16

Find unit digit:

$$\lfloor 7^{\{7^7\}} \rfloor$$

Q17

Find unit digit:

$$\lfloor 2^{\{2^{\{20\}}\}} \rfloor$$

Q18

Find unit digit:

$$\lfloor 3^{\{3^{\{3^{\{3\}}}\}} \rfloor$$

Q19

Find unit digit:

$$\{ 1^1+2^2+3^3+\cdots+10^{10} \}$$

Q20

Which of these could be a perfect square?

A. 78342 B. 94735 C. 65276 D. 84218

ANSWERS**1**

$$\{ 17 \bmod 4 = 1 \}$$

$$\boxed{2}$$

2

$$\{ 24 \bmod 4 = 0 \}$$

$$\boxed{1}$$

3

Odd exponent for ending 4:

$$\boxed{4}$$

4

Even exponent for ending 9:

$$\boxed{1}$$

5

Any positive power ending 5:

$$\boxed{5}$$

6

Base $\neq 7$.

$$\{ 51 \bmod 4 = 3 \}$$

Third position:

$$\boxed{3}$$

7

First:

$$\{ 13^{37} \bmod 3^{37} \}$$

$$\{ 37 \bmod 4 = 1 \}$$

$\neq 3$.

Second:

$$\sqrt[17]{26} \text{ to } \sqrt[7]{26}$$

$$26 \bmod 4 = 2$$

! 9.

Sum:

$$\sqrt{12}$$

$$\boxed{2}$$

8

$$\sqrt[23]{15} \text{ to } \sqrt[3]{15}$$

$$15 \bmod 4 = 3$$

! 7.

$$\sqrt[18]{14} \text{ to } \sqrt[8]{14}$$

$$14 \bmod 4 = 2$$

! 4.

Product:

$$\sqrt{28}$$

$$\boxed{8}$$

9

Cycle of 2 length 4.

Need:

$$3^5 \bmod 4$$

$$3^5 \equiv 3$$

Third position of 2-cycle:

$$\boxed{8}$$

10

Cycle of 7 length 4.

$$\sqrt[2]{10}$$

is divisible by 4.

Use 4th position:

$$\boxed{1}$$

11

Cycle of 3 length 4.

Exponent:

$$\sqrt[4]{5}$$

is divisible by 4.

Use 4th position:

$\boxed{1}$

12

Cycle of 8 length 4.

Need:

$$7^6 \bmod 4$$

$$7 \equiv 3 \pmod{4}$$

$$3^6 \equiv 1 \pmod{4}$$

Use first position:

$\boxed{8}$

13

Each exponent 100 is multiple of 4.

$$2^{100} \bmod 6$$

$$3^{100} \bmod 1$$

$$7^{100} \bmod 1$$

$$8^{100} \bmod 6$$

Sum:

$$14$$

$\boxed{4}$

14

Only first four factorials matter:

$$1+2+6+24=33$$

$\boxed{3}$

15

$$99^{99} \bmod 99$$

odd exponent:

$$9$$

$$88^{88} \bmod 88$$

$$88 \bmod 4 = 0$$

! 6.

Sum:

$$15$$

$\boxed{5}$

16

Cycle of 7 length 4.

Need:

$$\backslash[7^7\backslash\bmod 4 \backslash]$$

$$\backslash[7\equiv 3 \backslash]$$

Odd power:

$$\backslash[3^7\equiv 3\backslash\pmod 4 \backslash]$$

Third position:

$$\backslash[\boxed{3} \backslash]$$

17

Cycle of 2 length 4.

Exponent:

$$\backslash[2^{20} \backslash]$$

is divisible by 4.

Use 4th position:

$$\backslash[\boxed{6} \backslash]$$

18

For base 3, cycle length 4.

We need exponent:

$$\backslash[3^{3^3}\backslash\bmod 4 \backslash]$$

Since:

$$\backslash[3\equiv -1\backslash\pmod 4 \backslash]$$

and exponent $\backslash(3^3=27\backslash)$ is odd:

$$\backslash[3^{27}\equiv -1\equiv 3 \backslash]$$

So use 3rd position of 3-cycle:

$$\backslash[3,9,\boxed{7},1 \backslash]$$

Answer:

$$\backslash[\boxed{7} \backslash]$$

19

Compute unit digits:

$$\backslash[1^1\backslash\text{to}1 \backslash]$$

$$\backslash[2^2\backslash\text{to}4 \backslash]$$

$$\backslash[3^3\backslash\text{to}7 \backslash]$$

$$\backslash[4^4\backslash\text{to}6 \backslash]$$

$$\backslash[5^5\backslash\text{to}5 \backslash]$$

$$\backslash[6^6\backslash\text{to}6 \backslash]$$

$$\backslash[7^7:\backslash 7\backslash\bmod 4=3\backslash\text{to}3 \backslash]$$

$$\backslash[8^8:\backslash 8\backslash\bmod 4=0\backslash\text{to}6 \backslash]$$

$\backslash[9^9\backslashto9 \backslash]$

$\backslash[10^{\{10\}}\backslashto0 \backslash]$

Sum:

$\backslash[1+4+7+6+5+6+3+6+9 = 47 \backslash]$

Unit digit:

$\backslash[\boxed{7} \backslash]$

20

Perfect-square endings are:

$\backslash[0,1,4,5,6,9 \backslash]$

Options end:

- 2 'L
- 5 '
- 6 '
- 8 'L

So both B and C survive the **unit-digit test**.

But unit digit alone cannot decide which is actually a square.

Therefore:

$\backslash[\boxed{\text{B and C are possible by unit-digit test alone}} \backslash]$

This is a useful exam lesson: elimination rules can narrow options without always uniquely solving.

CHAPTER 8 STATUS

We have completed:

- unit digit as modulo 10 '
- complete unit-digit cycles '
- fixed endings 0/1/5/6 '
- two-cycle endings 4/9 '
- four-cycle endings 2/3/7/8 '
- huge exponents '
- nested exponents '
- unit digit of sums/products '
- factorial unit digit '
- square/cube ending restrictions '
- factorial-sum tricks '
- option elimination '
- GATE/placement questions '

But **Chapter 8 is not complete yet**.

Next we need:

- last two digits = modulo 100

- last three digits = modulo 1000
- cyclicity modulo 100
- powers ending in 25/76 etc.
- Chinese Remainder intuition where useful
- trailing-zero vs last non-zero digit distinction
- last non-zero digit of factorial/product basics
- advanced huge-power endings
- GATE/placement mixed questions

Progress

Completed: 7/76 **Current:** 0 = Chapter 8 — Units Digit, Last Digits & Cyclicity **Next:** Chapter 8 Part 2 — **Last two digits, last three digits & last non-zero digit**

USER

next

ASSISTANT

CHAPTER 8 — UNITS DIGIT, LAST DIGITS & CYCLICITY

PART 2 — LAST TWO DIGITS, LAST THREE DIGITS & LAST NON-ZERO DIGIT

Status: 0 = In PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

Now we move beyond just the unit digit.

Remember:

$$\boxed{\text{last 2 digits} = N \bmod 100}$$

and:

$$\boxed{\text{last 3 digits} = N \bmod 1000}$$

So the same modular-arithmetic ideas from Chapter 7 still work.

1. LAST TWO DIGITS — BASIC IDEA

Suppose we want the last two digits of:

$$\boxed{1234567}$$

Obviously:

$$\boxed{67}$$

Mathematically:

$$\boxed{1234567 \bmod 100 = 67}$$

So for powers:

$$\backslash[a^n \backslash]$$

we need:

$$\backslash[a^n\bmod 100 \backslash]$$

2. REDUCE THE BASE FIRST

Find last two digits of:

$$\backslash[1234^5 \backslash]$$

Only:

$$\backslash[1234\bmod 100=34 \backslash]$$

matters.

So solve:

$$\backslash[34^5\bmod 100 \backslash]$$

not the full power.

3. LAST TWO DIGITS OF SMALL POWERS

Example:

$$\backslash[12^4 \backslash]$$

Compute gradually modulo 100:

$$\backslash[12^2=144\equiv 44 \backslash]$$

$$\backslash[12^4\equiv 44^2 \backslash]$$

$$\backslash[44^2=1936 \backslash]$$

Last two digits:

$$\backslash[\boxed{36} \backslash]$$

4. REPEATED SQUARING

For large exponents, repeated squaring is very useful.

Find last two digits of:

$$\backslash[3^{20} \backslash]$$

We can build:

$$\backslash[3^2=9 \backslash]$$

$$\backslash[3^4=81 \backslash]$$

$$\backslash[3^8=81^2=6561\equiv 61 \backslash]$$

$$\backslash[3^{16}\equiv 61^2=3721\equiv 21 \backslash]$$

Then:

$$\backslash[3^{\{20\}}=3^{\{16\}}3^4 \backslash]$$

$$\backslash[\equiv 21 \times 81 \backslash]$$

$$\backslash[=1701 \backslash]$$

So last two digits:

$$\backslash[\boxed{01} \backslash]$$

Write as **01**, not just 1, because question asks last two digits.

5. IMPORTANT FORMAT TRAP

If:

$$\backslash[N \bmod 100 = 7 \backslash]$$

then last two digits are:

$$\backslash[\boxed{07} \backslash]$$

Similarly:

$$\backslash[N \bmod 1000 = 8 \backslash]$$

means last three digits:

$$\backslash[\boxed{008} \backslash]$$

6. POWERS ENDING IN 25

Very useful pattern.

For any positive integer $\backslash(n\backslash)$:

$$\backslash[25^1 = 25 \backslash]$$

$$\backslash[25^2 = 625 \backslash]$$

last two digits:

$$\backslash[25 \backslash]$$

Therefore:

$$\backslash[\boxed{25^n \text{ ends in } 25} \backslash]$$

for every positive $\backslash(n\backslash)$.

7. NUMBERS ENDING IN 25

If base itself ends in 25:

$$\backslash[125, \backslash 225, \backslash 725, \backslash \text{dots} \backslash]$$

then every positive power also ends in 25.

Example:

$$\backslash[125^{\{2026\}} \backslash]$$

last two digits:

$$\boxed{25}$$

8. POWERS ENDING IN 76

Another classic pattern.

$$76^2 = 5776$$

which again ends in:

$$76$$

Thus:

$$\boxed{76^n \text{ ends in } 76}$$

for every positive (n) .

Example:

$$76^{100}$$

last two digits:

$$\boxed{76}$$

9. IDEMPOTENT LAST-TWO-DIGIT ENDINGS

Certain endings satisfy:

$$x^2 \equiv x \pmod{100}$$

Important examples:

$$00, 01, 25, 76$$

For positive powers, these endings remain fixed.

Examples:

$$101^{50} \text{ to } 01$$

$$625^{20} \text{ to } 25$$

$$276^{30} \text{ to } 76$$

10. LAST TWO DIGITS OF A PRODUCT

Find last two digits of:

$$37 \times 68$$

$$37 \times 68 = 2516$$

Last two digits:

$$\boxed{16}$$

For larger expressions, reduce each factor modulo 100 first.

11. EXAMPLE WITH POWERS

Find last two digits of:

$$3^{10} \times 7^5$$

First:

$$3^{10} = 59049$$

so:

$$3^{10} \equiv 49 \pmod{100}$$

For 7^5 :

$$7^2 = 49$$

$$7^4 = 2401 \equiv 1$$

So:

$$7^5 \equiv 7$$

Product:

$$49 \times 7 = 343$$

Last two digits:

$$\boxed{43}$$

12. SHORT CYCLE EXAMPLE MODULO 100

Consider powers of 7:

$$7^1 \equiv 7$$

$$7^2 \equiv 49$$

$$7^3 \equiv 43$$

$$7^4 \equiv 1$$

Then repeat because multiplying by 7 again returns 7.

Cycle:

$$7, 49, 43, 1$$

length:

$$4$$

Therefore:

$$7^{2026}$$

last two digits:

$$2026 \bmod 4 = 2$$

Second term:

$$\boxed{49}$$

13. NOT EVERY BASE HAS A TINY MOD-100 CYCLE

For unit digits, cycle lengths are at most 4.

For last two digits, cycles can be longer.

So do not assume cycle length 4 automatically.

Instead use:

- repeated squaring
 - pattern recognition
 - Euler theorem when valid
-

14. EULER FOR LAST TWO DIGITS

If:

$$\gcd(a, 100) = 1$$

then:

$$a^{\phi(100)} \equiv 1 \pmod{100}$$

Now:

$$100 = 2^2 \cdot 5^2$$

So:

$$\phi(100) = 100 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right)$$

$$= 40$$

Thus, for bases coprime with 100:

$$\boxed{a^{40} \equiv 1 \pmod{100}}$$

This gives a safe exponent cycle divisor of 40.

15. EXAMPLE

Find last two digits:

$$3^{2026}$$

Since:

$$\gcd(3, 100) = 1$$

Euler:

$$3^{40} \equiv 1 \pmod{100}$$

Reduce exponent:

$$2026 \bmod 40 = 26$$

Now compute:

$$3^{26} \bmod 100$$

Use repeated squaring:

$$\sqrt[3]{3^4=81}$$

$$\sqrt[3]{3^8 \equiv 61}$$

$$\sqrt[3]{3^{16} \equiv 21}$$

Then:

$$\sqrt[3]{3^{26}=3^{16}3^8 3^2}$$

$$\sqrt[3]{\equiv 21 \times 61 \times 9}$$

First:

$$\sqrt[3]{21 \times 61 = 1281 \equiv 81}$$

Then:

$$\sqrt[3]{81 \times 9 = 729}$$

So:

$$\sqrt[3]{\boxed{29}}$$

last two digits.

16. WHEN EULER CANNOT BE USED DIRECTLY

For:

$$\sqrt[3]{20^{100} \bmod 100}$$

$$\sqrt[3]{\gcd(20,100) \neq 1}$$

So Euler's theorem condition fails.

But direct structure is easier:

$$\sqrt[3]{20^2=400}$$

which is divisible by 100.

Therefore for:

$$\sqrt[3]{n \geq 2}$$

$$\sqrt[3]{20^n \equiv 0 \pmod{100}}$$

Thus:

$$\sqrt[3]{\boxed{20^{100}} \text{ ends in } 00}$$

17. BASE HAS FACTORS 2 AND 5

To end in:

$$\sqrt[3]{00}$$

a number must be divisible by:

$$\sqrt[3]{100=2^2 \cdot 5^2}$$

For powers, track whether enough 2s and 5s appear.

Example:

$$\backslash[50^3 \backslash]$$

$$\backslash[50=2\cdot 5^2 \backslash]$$

So:

$$\backslash[50^3=2^3\cdot 5^6 \backslash]$$

Contains at least:

$$\backslash[2^2 5^2 \backslash]$$

Therefore ends in:

$$\backslash[\boxed{00} \backslash]$$

18. LAST THREE DIGITS

Same principle:

$$\backslash[\boxed{\text{last 3 digits}=N\text{mod}1000} \backslash]$$

Example:

$$\backslash[1234567 \backslash]$$

last three:

$$\backslash[\boxed{567} \backslash]$$

19. LAST THREE DIGITS OF A POWER — REPEATED SQUARING

Find last three digits of:

$$\backslash[3^{20} \backslash]$$

Compute:

$$\backslash[3^4=81 \backslash]$$

$$\backslash[3^8=6561\equiv 561\pmod{1000} \backslash]$$

$$\backslash[3^{16}\equiv 561^2 \backslash]$$

$$\backslash[561^2=314721 \backslash]$$

So:

$$\backslash[3^{16}\equiv 721 \backslash]$$

Then:

$$\backslash[3^{20}=3^{16}3^4 \backslash]$$

$$\backslash[\equiv 721\cdot 81 \backslash]$$

$$\backslash[721\cdot 81=58401 \backslash]$$

Thus:

$$\backslash[\boxed{401} \backslash]$$

20. FIXED ENDING 625

A useful pattern:

$$\backslash[625^2 \backslash]$$

ends in:

$$\backslash[625 \backslash]$$

because:

$$\backslash[625^2=390625 \backslash]$$

Thus every positive power of a number ending in 625 keeps last three digits 625.

Example:

$$\backslash[1625^{\{100\}} \backslash]$$

last three digits:

$$\backslash[\boxed{625} \backslash]$$

21. LAST THREE DIGITS OF NUMBERS ENDING 000

If base contains enough 2s and 5s to make $\backslash(1000\backslash)$, powers may quickly end in 000.

Example:

$$\backslash[100^2=10000 \backslash]$$

last three:

$$\backslash[000 \backslash]$$

Therefore:

$$\backslash[100^n \backslash]$$

for $\backslash(n\geq 2\backslash)$ ends in:

$$\backslash[\boxed{000} \backslash]$$

22. CHINESE REMAINDER INTUITION — WHY IT HELPS

Sometimes mod 100 is easier if split into:

$$\backslash[4\text{quad}\text{and}\text{quad}25 \backslash]$$

because:

$$\backslash[100=4\text{times}25 \backslash]$$

and:

$$\backslash[\gcd(4,25)=1 \backslash]$$

If we know:

$$\backslash[N\text{bmod}4 \backslash]$$

and:

$$\pmod{25}$$

there is exactly one answer modulo 100 matching both.

This is the basic intuition of the **Chinese Remainder Theorem (CRT)**.

For aptitude, you mainly need the idea, not a formal proof.

23. EXAMPLE USING CRT INTUITION

Find last two digits of:

$$2^{20}$$

Modulo 4:

$$2^{20} \equiv 0 \pmod{4}$$

Modulo 25:

$$2^{20}$$

Since:

$$2^{10} = 1024 \equiv 24 \equiv -1 \pmod{25}$$

Thus:

$$2^{20} \equiv 1 \pmod{25}$$

We need a number between 0 and 99 that is:

- divisible by 4
- leaves 1 mod 25

Candidates mod 25:

$$1, 26, 51, 76$$

Only:

$$76$$

is divisible by 4.

Therefore:

$$\boxed{76}$$

Indeed:

$$2^{20} = 1048576$$

24. CRT CAN SAVE WORK, BUT DON'T FORCE IT

Use CRT when:

- mod 4 and mod 25 are much easier separately
- direct repeated squaring is messy

Otherwise, direct mod 100 is often simpler.

25. LAST NON-ZERO DIGIT — DIFFERENT PROBLEM

Important distinction:

For:

$$\sqrt[10]{10!}$$

unit digit is:

$$\sqrt[10]{0}$$

But the **last non-zero digit** is not 0.

Since:

$$\sqrt[10]{10!}=3628800$$

last non-zero digit:

$$\boxed{8}$$

So:

Unit digit and last non-zero digit are different concepts.

26. WHY TRAILING ZEROS MUST BE REMOVED

Every trailing zero comes from:

$$2 \times 5$$

To find last non-zero digit:

1. factor out all 10s
2. then inspect remaining unit digit

This is the basic idea.

27. SIMPLE PRODUCT EXAMPLE

Find last non-zero digit of:

$$40$$

$$40=4 \times 10$$

Remove factor 10:

$$4$$

Last non-zero digit:

$$\boxed{4}$$

28. ANOTHER EXAMPLE

Find last non-zero digit of:

$$\boxed{24000}$$

Remove three trailing zeros:

$$\boxed{24}$$

Last non-zero digit:

$$\boxed{4}$$

29. LAST NON-ZERO DIGIT OF A FACTORIAL — BASIC MANUAL METHOD

Example:

$$\boxed{7!}$$

$$\boxed{7!=5040}$$

Last non-zero digit:

$$\boxed{4}$$

But we don't want to calculate huge factorials.

For small factorials, cancel pairs of 2 and 5.

30. MANUAL CANCELLATION EXAMPLE — $(10!)$

Prime-factor structure:

$$\boxed{10!=2^8 \cdot 3^4 \cdot 5^2 \cdot 7}$$

Two pairs:

$$\boxed{2^2 \cdot 5^2 = 10^2}$$

create two trailing zeros.

Remove them.

Remaining factor of 2:

$$\boxed{2^{8-2}=2^6}$$

So after removing 100, remaining relevant product:

$$\boxed{2^6 \cdot 3^4 \cdot 7}$$

Take mod 10:

$$\boxed{2^6 \bmod 4}$$

$$\boxed{3^4 \bmod 1}$$

$$\boxed{7 \bmod 7}$$

Product:

$$\boxed{4 \times 1 \times 7 = 28}$$

Last digit:

$$\boxed{8}$$

Correct.

31. LAST NON-ZERO DIGIT OF $(n!)$ — ADVANCED NOTE

There are efficient recursive algorithms for very large $(n!)$, but that is beyond normal aptitude requirements.

For GATE/placements, typically enough to know:

- remove 2×5 pairs
- extra 2s remain
- calculate remaining unit digit modulo 10

We will not turn this chapter into advanced number theory unnecessarily.

32. LAST NON-ZERO DIGIT OF A PRODUCT

Find last non-zero digit of:

$$25 \times 16$$

$$25 \times 16 = 400$$

Remove zeros:

$$4$$

Answer:

$$\boxed{4}$$

Fast factor method:

$$25 = 5^2$$

$$16 = 2^4$$

Pair two 5s with two 2s !' two 10s.

Extra:

$$2^2 = 4$$

So last non-zero digit:

$$\boxed{4}$$

33. LAST NON-ZERO DIGIT OF $(2^a 5^b)$

If:

$$a > b$$

remove (b) pairs of $(2 \cdot 5)$.

Remaining:

$$2^{a-b}$$

If:

$$b > a$$

remaining:

$$5^{b-a}$$

which has last digit 5.

Example:

$$2^8 \cdot 5^3$$

remove 3 pairs.

Remaining:

$$2^5$$

Unit digit:

$$2^5 \pmod{10}$$

So last non-zero digit:

$$\boxed{2}$$

34. LAST NON-ZERO DIGIT WITH OTHER FACTORS

Suppose:

$$N = 2^7 \cdot 3^2 \cdot 5^3$$

Remove three $(2 \cdot 5)$ pairs.

Remaining:

$$2^4 \cdot 3^2$$

Unit digit:

$$2^4 \pmod{10}$$

$$3^2 \pmod{10}$$

Product:

$$6 \cdot 9 = 54$$

Last digit:

$$\boxed{4}$$

35. FACTORIAL LAST NON-ZERO DIGIT SMALL TABLE

Good to recognize for small values:

$n!$	$(n!)$	Last non-zero digit
1	1	1
2	2	2
3	6	6
4	24	4

$\backslash(n)$	$\backslash(n!)$	Last non-zero digit
5	120	2
6	720	2
7	5040	4
8	40320	2
9	362880	8
10	3628800	8

No need to memorize all of these beyond familiarity.

36. LAST TWO DIGITS OF FACTORIALS

For sufficiently large $\backslash(n)$, $\backslash(n!)$ ends in:

$\backslash[00]$

When does this happen?

Need at least two factors of 5 and two factors of 2.

Already by:

$\backslash[10!]$

there are factors 5 and 10, giving two 5s.

So:

$\backslash[\boxed{n!} \text{ ends in } 00 \text{ for } n \geq 10]$

Example:

$\backslash[10! = 3628800]$

37. LAST THREE DIGITS OF FACTORIALS

To end in:

$\backslash[000]$

need at least three factors of 5.

At:

$\backslash[15!]$

multiples of 5:

$\backslash[5, 10, 15]$

provide three 5s, with plenty of 2s.

Therefore:

$\backslash[\boxed{n!} \text{ ends in } 000 \text{ for } n \geq 15]$

38. GENERAL TRAILING-ZERO CONNECTION

If $(n!)$ has at least (k) trailing zeros, then its last (k) digits are all zero.

This connects Chapter 6 and Chapter 8 directly.

39. GATE-STYLE QUESTION 1

Find last two digits of:

$$11^{20}$$

Since:

$$11^2 = 121 \equiv 21$$

$$11^4 \equiv 21^2 = 441 \equiv 41$$

$$11^5 \equiv 51$$

But a simpler pattern appears:

$$11^1 \equiv 11$$

$$11^2 \equiv 21$$

$$11^3 \equiv 31$$

$$11^4 \equiv 41$$

So:

$$11^n$$

for these powers increases by 10 modulo 100 until cycle wraps.

Indeed:

$$11^{10} \equiv 01 \pmod{100}$$

Thus cycle length 10.

$$20 \bmod 10 = 0$$

Therefore:

$$\boxed{01}$$

40. GATE-STYLE QUESTION 2

Find last two digits:

$$99^{99}$$

$$99 \equiv -1 \pmod{100}$$

Odd exponent:

$$-1$$

So:

$$\boxed{99}$$

41. GATE-STYLE QUESTION 3

Find last two digits:

$$51^{2026}$$

Observe:

$$51^2 = 2601$$

last two digits:

$$01$$

So any even exponent gives:

$$01$$

because:

$$51^{2026} = (51^2)^{1013}$$

Thus:

$$\boxed{01}$$

42. GATE-STYLE QUESTION 4

Find last three digits:

$$999^{101}$$

$$999 \equiv -1 \pmod{1000}$$

Odd exponent:

$$-1$$

Therefore last three digits:

$$\boxed{999}$$

43. GATE-STYLE QUESTION 5

Find last three digits:

$$1001^{500}$$

$$1001 \equiv 1 \pmod{1000}$$

Therefore:

$$\boxed{001}$$

44. PLACEMENT-STYLE QUESTION 1

Find last two digits of:

$$\backslash[49^{51} \backslash]$$

$$\backslash[49 \equiv -51 \pmod{100} \backslash]$$

But easier:

$$\backslash[49^2 = 2401 \equiv 1 \backslash]$$

Thus odd exponent:

$$\backslash[49^{51} = 49(49^2)^{25} \backslash]$$

$$\backslash[\equiv 49 \backslash]$$

Answer:

$$\backslash[\boxed{49} \backslash]$$

45. PLACEMENT-STYLE QUESTION 2

Find last two digits of:

$$\backslash[76^{999} \backslash]$$

Fixed ending:

$$\backslash[\boxed{76} \backslash]$$

46. PLACEMENT-STYLE QUESTION 3

Find last two digits of:

$$\backslash[25^{100} + 76^{100} \backslash]$$

$$\backslash[25^{100} \to 25 \backslash]$$

$$\backslash[76^{100} \to 76 \backslash]$$

Sum:

$$\backslash[101 \backslash]$$

Last two digits:

$$\backslash[\boxed{01} \backslash]$$

47. PLACEMENT-STYLE QUESTION 4

Find last non-zero digit of:

$$\backslash[2^9 \cdot 5^4 \backslash]$$

Cancel four pairs:

$$\backslash[2^{9-4} = 2^5 \backslash]$$

$$\backslash[2^5 = 32 \backslash]$$

Last non-zero digit:

$$\backslash[\boxed{2} \backslash]$$

48. PLACEMENT-STYLE QUESTION 5

Find last non-zero digit of:

$$2^6 \cdot 3^3 \cdot 5^2$$

Remove two pairs:

$$2^4 \cdot 3^3$$

Unit digit:

$$2^4 \pmod{10}$$

$$3^3 \pmod{10}$$

$$6 \times 7 = 42$$

Answer:

$$\boxed{2}$$

49. COMMON TRAPS

& pTRAP 1 — Last digit vs last two digits

Unit digit:

$$\pmod{10}$$

Last two digits:

$$\pmod{100}$$

Last three digits:

$$\pmod{1000}$$

& pTRAP 2 — Leading zeros

Remainder 4 modulo 100 means:

$$\boxed{04}$$

as the last two digits.

& pTRAP 3 — Assuming unit-digit cycle works for last two digits

Cycles modulo 100 can be much longer.

& pTRAP 4 — Euler without coprimality

For modulo 100:

$$\gcd(a, 100) = 1$$

must hold before using Euler directly.

& pTRAP 5 — Unit digit 0 vs last non-zero digit

These are completely different questions.

& pTRAP 6 — Removing zeros incorrectly

Every zero requires exactly:

$\backslash[2\times 5 \backslash]$

Pair them before finding the remaining digit.

50. MUST-KNOW SHORTCUTS

Last digit

$\backslash[\backslashbmod 10 \backslash]$

Last 2 digits

$\backslash[\backslashbmod 100 \backslash]$

Last 3 digits

$\backslash[\backslashbmod 1000 \backslash]$

Fixed endings

$\backslash[25^n \backslashto 25 \backslash]$

$\backslash[76^n \backslashto 76 \backslash]$

Near $\backslash(\backslashpmod 1\backslash)$

$\backslash[99 \backslashequiv -1 \backslashpmod {100} \backslash]$

$\backslash[101 \backslashequiv 1 \backslashpmod {100} \backslash]$

$\backslash[999 \backslashequiv -1 \backslashpmod {1000} \backslash]$

$\backslash[1001 \backslashequiv 1 \backslashpmod {1000} \backslash]$

These solve many questions instantly.

51. NORMAL METHOD VS FAST METHOD

Find last two digits of:

$\backslash[99^{\{2027\}} \backslash]$

Normal method

Trying to calculate powers is absurd.

Fast method

$\backslash[99 \backslashequiv -1 \backslashpmod {100} \backslash]$

Odd exponent:

$\backslash[(-1)^{\{2027\}} = -1 \backslash]$

Therefore:

$\boxed{99}$

This is exactly the kind of shortcut you should seek.

52. 5-MINUTE REVISION

Ask:

Need unit digit?

Use modulo 10 and unit cycles.

Need last 2?

Use modulo 100.

Need last 3?

Use modulo 1000.

Then look for:

$\{0, 1, -1\}$

fixed endings:

$\{25, 76\}$

repeated squaring,

or Euler/CRT if helpful.

For last non-zero digit:

remove all (2×5) pairs first.

PRACTICE — LEVEL 0

Q1

Find last two digits:

99^{24}

Q2

Find last two digits:

125^{50}

Q3

Find last two digits:

76^{101}

Q4

Find last three digits:

1001^{25}

Q5

Find last non-zero digit of:

$$\lfloor 2^6 \cdot 5^4 \rfloor$$

LEVEL 1

Q6

Find last two digits:

$$\lfloor 7^{18} \rfloor$$

Q7

Find last two digits:

$$\lfloor 49^{100} \rfloor$$

Q8

Find last two digits:

$$\lfloor 25^{20} + 76^{20} \rfloor$$

Q9

Find last three digits:

$$\lfloor 999^{100} \rfloor$$

Q10

Find last non-zero digit of:

$$\lfloor 2^8 \cdot 3^2 \cdot 5^3 \rfloor$$

LEVEL 2

Q11

Find last two digits:

$$\lfloor 2^{20} \rfloor$$

Q12

Find last two digits:

$$\lfloor 3^{20} \rfloor$$

Q13

Find last two digits:

$$\lfloor 51^{99} \rfloor$$

Q14

Find last three digits:

$$\lfloor 3^{20} \rfloor$$

Q15

Find last non-zero digit of:

$$\lfloor 2^7 \cdot 3^4 \cdot 5^2 \rfloor$$

LEVEL 3 — EXAM PRESSURE

Q16

Find last two digits:

$$\lfloor 7^{2026} \rfloor$$

Q17

Find last two digits:

$$\lfloor 3^{2026} \rfloor$$

Q18

Find last three digits:

$$\lfloor 999^{999} \rfloor$$

Q19

Find last non-zero digit of:

$$\lfloor 20! \rfloor$$

Q20

Find last two digits of:

$$\lfloor 2^{100} + 3^{100} \rfloor$$

ANSWERS

1

$$\lfloor 99 \equiv -1 \pmod{100} \rfloor$$

Even exponent:

$$\lfloor \boxed{01} \rfloor$$

2

Any power of number ending 25:

$$\lfloor \boxed{25} \rfloor$$

3

$$\lfloor \boxed{76} \rfloor$$

4

$$\lfloor 1001 \equiv 1 \pmod{1000} \rfloor$$

$$\lfloor \boxed{001} \rfloor$$

5

Cancel four $(2 \cdot 5)$ pairs:

$$2^2$$

Last digit:

$$\boxed{4}$$

6

Cycle for 7 modulo 100:

$$7^1 \pmod{100}$$

$$7^2 \pmod{100}$$

$$7^3 \pmod{100}$$

$$7^4 \pmod{100}$$

Cycle 4.

$$18 \bmod 4 = 2$$

$$\boxed{49}$$

7

$$49^2 \equiv 1 \pmod{100}$$

Even exponent:

$$\boxed{01}$$

8

$$25 + 76 = 101$$

Last two:

$$\boxed{01}$$

9

$$999 \equiv -1 \pmod{1000}$$

Even exponent:

$$\boxed{001}$$

10

Cancel three pairs:

$$2^5 \cdot 3^2$$

$$2^5 \pmod{2}$$

$$3^2 \pmod{9}$$

$$2 \times 9 = 18$$

$$\boxed{8}$$

11

Using CRT result:

$$\boxed{76}$$

12

As calculated:

$$3^{20} \equiv 01 \pmod{100}$$

$$\boxed{01}$$

13

Since:

$$51^2 \equiv 01 \pmod{100}$$

odd exponent:

$$51^{99} = 51(51^2)^{49}$$

$$\boxed{51}$$

14

Previously:

$$3^{20} \equiv 401 \pmod{1000}$$

$$\boxed{401}$$

15

Cancel two 2–5 pairs:

$$2^5 \cdot 3^4$$

Unit digits:

$$2^5 \rightarrow 2$$

$$3^4 \rightarrow 1$$

$$\boxed{2}$$

16

Cycle of 7 modulo 100 has length 4.

$$2026 \bmod 4 = 2$$

$$\boxed{49}$$

17

Using Euler modulo 100:

$$2026 \bmod 40 = 26$$

and repeated squaring gives:

$$\boxed{29}$$

18

$$999 \equiv -1 \pmod{1000}$$

Odd exponent:

$$\boxed{999}$$

19

We can use factor cancellation.

$$[20!]$$

has:

$$[v_2(20!)=18]$$

and:

$$[v_5(20!)=4]$$

Remove 4 pairs of $(2 \cdot 5)$.

Remaining factor structure relevant modulo 10 is:

$$[2^{14}]$$

times all the other odd factors.

A direct careful reduction gives last non-zero digit:

$$[\boxed{4}]$$

This is a more advanced factorial-last-nonzero-digit calculation; for large factorials, use a specialized recurrence rather than brute force.

20

Find modulo 100.

For:

$$[2^{100}]$$

Since powers of 2 modulo 100 repeat with period 20 after the initial structure, and:

$$[2^{20} \equiv 76 \pmod{100}]$$

then:

$$[2^{100} = (2^{20})^5]$$

$$[76^5 \equiv 76]$$

So:

$$[2^{100} \equiv 76]$$

For:

$$[3^{100}]$$

Euler gives:

$$[3^{40} \equiv 1]$$

and:

$$[100 \bmod 40 = 20]$$

We know:

$$[3^{20} \equiv 01]$$

So:

$$[3^{100} \equiv 01]$$

Sum:

$\lfloor 76+1=77 \rfloor$

$\lfloor \boxed{77} \rfloor$

CHAPTER 8 — FINAL CHECK

We have now covered:

- unit digit as modulo 10 '
- complete unit-digit cycles '
- sums/products of power endings '
- huge/nested exponents '
- factorial unit digits '
- square/cube ending restrictions '
- last two digits via modulo 100 '
- last three digits via modulo 1000 '
- repeated squaring '
- fixed endings such as 25 and 76 '
- $\lfloor \pmod{1} \rfloor$ ending shortcuts '
- Euler for last-digit blocks '
- CRT intuition '
- trailing zeros vs last non-zero digit '
- last non-zero digit by cancelling 2–5 pairs '
- factorial ending basics '
- GATE-style questions '
- placement-style questions '

Therefore:

' CHAPTER 8 COMPLETE

Progress

Completed: 8/76 **Current:** ' Chapter 8 — Units Digit, Last Digits & Cyclicity **Next:** ~~Ø=~~Chapter 9 — Powers, Exponents & Roots

Next we will cover:

- exponent laws
- zero/negative/fractional powers
- roots and surds
- comparing huge powers
- simplifying exponential expressions
- rationalization basics
- square/cube-root shortcuts
- perfect powers

- GATE and placement traps.

USER

next

ASSISTANT

CHAPTER 9 — POWERS, EXPONENTS & ROOTS

PART 1 — EXPONENT LAWS, NEGATIVE/FRACTIONAL POWERS & ROOT BASICS

Status: $\emptyset$ **IN PROGRESS** **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This chapter is important for:

- algebraic simplification
- square/cube roots
- surds
- scientific notation
- comparing large powers
- logarithm foundations
- number-system questions
- aptitude calculations

The main goal is:

Manipulate powers without expanding huge numbers.

1. WHAT IS AN EXPONENT?

$[a^n]$

means (a) multiplied by itself (n) times.

Example:

$[2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32]$

Here:

- (2) = base
- (5) = exponent/power

2. FIRST LAW — MULTIPLY SAME BASES

$[\boxed{a^m \cdot a^n = a^{m+n}}]$

Example:

$[2^3 \cdot 2^5 = 2^8 = 256]$

Memory

Same base, multiplication !' add powers.

3. WHY THIS WORKS

$$\underbrace{a \cdot a \cdots a}_{m \text{ times}}$$

and:

$$\underbrace{a \cdot a \cdots a}_{n \text{ times}}$$

Together:

$$a^{m+n}$$

copies of (a) .

Therefore:

$$a^{m+n}$$

4. SECOND LAW — DIVIDE SAME BASES

$$\boxed{\frac{a^m}{a^n} = a^{m-n}}$$

for $a \neq 0$.

Example:

$$\frac{3^7}{3^4} = 3^3 = 27$$

Memory

Same base, division !' subtract powers.

5. THIRD LAW — POWER OF A POWER

$$\boxed{(a^m)^n = a^{mn}}$$

Example:

$$(2^3)^4 = 2^{12}$$

Memory

Power raised to power !' multiply powers.

6. IMPORTANT TRAP

$$(a^m)^n = a^{mn}$$

but:

$$\boxed{a^m + a^n}$$

cannot generally be simplified to:

$$\boxed{a^{m+n}}$$

Example:

$$\boxed{2^2 + 2^3 = 4 + 8 = 12}$$

but:

$$\boxed{2^5 = 32}$$

So:

$$\boxed{a^m + a^n \neq a^{m+n}}$$

7. POWER OF A PRODUCT

$$\boxed{(ab)^n = a^n b^n}$$

Example:

$$\boxed{(2 \cdot 3)^4 = 2^4 \cdot 3^4}$$

$$\boxed{= 16 \cdot 81 = 1296}$$

8. POWER OF A QUOTIENT

$$\boxed{\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}}$$

for $b \neq 0$.

Example:

$$\boxed{\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3}}$$

9. ZERO EXPONENT

For non-zero a :

$$\boxed{a^0 = 1}$$

Why?

$$\boxed{\frac{a^m}{a^m} = a^{m-m} = a^0}$$

But also:

$$\boxed{\frac{a^m}{a^m} = 1}$$

Therefore:

$$\boxed{a^0 = 1}$$

10. IMPORTANT EXCEPTION

$$\boxed{0^0}$$

is not treated as an ordinary value in elementary algebra; it is context-dependent/indeterminate in many settings.

For aptitude:

$$\boxed{a^0=1 \text{ only when } a \neq 0}$$

11. NEGATIVE EXPONENT

$$\boxed{a^{-n} = \frac{1}{a^n}}$$

Example:

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Memory

Negative power !' reciprocal.

12. EXAMPLE

$$\frac{1}{5^{-2}}$$

Since:

$$5^{-2} = \frac{1}{25}$$

Therefore:

$$\frac{1}{\frac{1}{25}} = \boxed{25}$$

Alternatively:

$$\frac{1}{5^{-2}} = 5^2$$

13. NEGATIVE POWER OF FRACTION

$$\left(\frac{2}{3}\right)^{-2}$$

Flip fraction:

$$= \left(\frac{3}{2}\right)^2$$

$$= \boxed{\frac{9}{4}}$$

Shortcut

Negative exponent on fraction !' invert fraction, make exponent positive.

14. FRACTIONAL EXPONENT

Very important:

$$\boxed{a^{\frac{1}{n}} = \sqrt[n]{a}}$$

Examples:

$$\sqrt[16]{16} = \sqrt{16} = 4$$

$$\sqrt[27]{27} = \sqrt[3]{27} = 3$$

15. GENERAL FRACTIONAL POWER

$$\boxed{a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m}$$

Example:

$$27^{2/3}$$

Take cube root first:

$$\sqrt[3]{27} = 3$$

Then square:

$$3^2 = 9$$

Answer:

$$\boxed{9}$$

16. WHICH ORDER IS BETTER?

For:

$$64^{2/3}$$

You could do:

$$64^2$$

then cube root.

That is unnecessary.

Better:

$$\sqrt[3]{64} = 4$$

then:

$$4^2 = 16$$

Answer:

$$\boxed{16}$$

Exam rule

Take the root first whenever it keeps numbers small.

17. NEGATIVE FRACTIONAL POWER

Example:

$$16^{-3/4}$$

First:

$$\sqrt[4]{16}=2$$

Thus:

$$\sqrt[4]{16^3}=2^3=8$$

Negative exponent gives reciprocal:

$$\boxed{\frac{1}{8}}$$

18. ROOT BASICS

Square root:

$$\sqrt{a}$$

means the **principal non-negative** number whose square is a .

Example:

$$\sqrt{49}=7$$

& pTRAP

$$\sqrt{49}=7$$

not:

$$\pm 7$$

But if solving:

$$x^2=49$$

then:

$$x=\pm 7$$

These are different statements.

19. CUBE ROOT

$$\sqrt[3]{a}$$

is the number whose cube is a .

Example:

$$\sqrt[3]{125}=5$$

Also:

$$\sqrt[3]{-125}=-5$$

Unlike real square roots, cube roots of negative numbers are real.

20. ROOT OF A PRODUCT

For non-negative a, b :

$$\boxed{\sqrt{ab} = \sqrt{a}\sqrt{b}}$$

Example:

$$\sqrt{72}$$

$$= \sqrt{36 \cdot 2}$$

$$= 6\sqrt{2}$$

21. ROOT OF A QUOTIENT

For suitable positive values:

$$\boxed{\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}}$$

Example:

$$\sqrt{\frac{49}{64}} = \frac{7}{8}$$

22. IMPORTANT TRAP — ROOT OF SUM

Generally:

$$\boxed{\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}}$$

Example:

$$\sqrt{9+16} = 5$$

while:

$$\sqrt{9} + \sqrt{16} = 3 + 4 = 7$$

So never split a root across addition/subtraction.

23. SIMPLIFYING SURDS

A **surd** is an irrational root expression such as:

$$\sqrt{2}, \sqrt{3}, \sqrt{5}$$

Simplify:

$$\sqrt{48}$$

Find largest perfect-square factor:

$$48 = 16 \cdot 3$$

Therefore:

$$\sqrt{48} = 4\sqrt{3}$$

24. ANOTHER EXAMPLE

$$\sqrt{180}$$

$$180 = 36 \cdot 5$$

So:

$$\sqrt[6]{6\sqrt{5}}$$

25. FAST SURD SIMPLIFICATION USING PRIME FACTORS

Suppose:

$$\sqrt[6]{2^5 3^2}$$

Pair equal prime factors:

$$2^5 = 2^4 \cdot 2$$

So:

$$\sqrt[6]{2^4 3^2 \cdot 2}$$

$$= 2^2 \cdot 3 \sqrt[6]{2}$$

$$= \boxed{12\sqrt[6]{2}}$$

26. CUBE-ROOT SIMPLIFICATION

Example:

$$\sqrt[3]{54}$$

$$54 = 27 \cdot 2$$

Therefore:

$$\sqrt[3]{54} = 3\sqrt[3]{2}$$

27. PRIME-EXPONENT VIEW OF ROOTS

For square root, every pair of equal prime factors comes out.

Example:

$$\sqrt[6]{2^6 3^4 5}$$

$$= 2^3 3^2 \sqrt[6]{5}$$

$$= \boxed{72\sqrt[6]{5}}$$

For cube root, every group of 3 comes out.

28. PERFECT SQUARE CONDITION REVISITED

If:

$$N = p_1^{a_1} \cdot p_k^{a_k}$$

then $\sqrt{N}$ is a perfect square iff every exponent is even.

So:

$$\sqrt{N}$$

is an integer iff all prime exponents are even.

29. PERFECT CUBE CONDITION

Likewise, $\sqrt[3]{N}$ is a perfect cube iff every prime exponent is a multiple of 3.

30. SQUARE ROOT ESTIMATION

Suppose:

$$\sqrt{50}$$

Nearby squares:

$$7^2=49$$

$$8^2=64$$

Therefore:

$$7 < \sqrt{50} < 8$$

and since 50 is very close to 49:

$$\sqrt{50} \approx 7.07$$

For aptitude, bounds may be enough.

31. CUBE ROOT ESTIMATION

$$\sqrt[3]{70}$$

Nearby cubes:

$$4^3=64$$

$$5^3=125$$

So:

$$\boxed{4 < \sqrt[3]{70} < 5}$$

32. COMPARING POWERS WITH SAME BASE

If:

$$a > 1$$

then larger exponent means larger value.

Example:

$$3^{10} > 3^8$$

If:

$$0 < a < 1$$

the relation reverses.

Example:

$$\left(\frac{1}{2}\right)^5 < \left(\frac{1}{2}\right)^3$$

because repeated multiplication by $\frac{1}{2}$ makes the number smaller.

33. COMPARING POWERS WITH SAME EXPONENT

For positive bases and same positive exponent:

$$a > b \Rightarrow a^n > b^n$$

Example:

$$7^5 > 6^5$$

34. COMPARING DIFFERENT BASES AND EXPONENTS

Example:

Compare:

$$2^{10}$$

and:

$$5^4$$

Compute smartly:

$$2^{10} = 1024$$

$$5^4 = 625$$

Thus:

$$\boxed{2^{10} > 5^4}$$

Sometimes convert both to a common base.

35. COMMON-BASE METHOD

Compare:

$$8^4$$

and:

$$4^6$$

Convert:

$$8 = 2^3$$

$$4 = 2^2$$

Then:

$$8^4 = 2^{12}$$

$$4^6 = 2^{12}$$

Therefore:

$$\boxed{8^4=4^6}$$

36. ANOTHER COMMON-BASE EXAMPLE

Compare:

$$9^5$$

and:

$$27^3$$

$$9=3^2$$

$$27=3^3$$

Therefore:

$$9^5=3^{10}$$

$$27^3=3^9$$

Thus:

$$\boxed{9^5>27^3}$$

37. SIMPLIFICATION EXAMPLE

Simplify:

$$\frac{2^7 \cdot 3^5}{2^3 \cdot 3^2}$$

Subtract exponents:

$$2^{7-3} 3^{5-2}$$

$$=2^4 3^3$$

$$=16 \cdot 27$$

$$=\boxed{432}$$

38. POWER OF POWER EXAMPLE

$$\frac{(3^2)^5}{3^7}$$

Numerator:

$$3^{10}$$

Then:

$$3^{10-7} = 3^3 = \boxed{27}$$

39. MIXED NEGATIVE EXPONENT EXAMPLE

Simplify:

$$\left[\frac{2^{-3} \cdot 2^5}{2} \right]$$

$$\left[=2^{-3+5-1} \right]$$

$$\left[=2^1 \right]$$

$$\left[=\boxed{2} \right]$$

40. EXPONENT EQUATION — SAME BASE

Solve:

$$\left[2^{x+1}=32 \right]$$

Since:

$$\left[32=2^5 \right]$$

Then:

$$\left[x+1=5 \right]$$

$$\left[\boxed{x=4} \right]$$

41. ANOTHER EXPONENT EQUATION

$$\left[9^x=27 \right]$$

Convert to base 3:

$$\left[(3^2)^x=3^3 \right]$$

$$\left[3^{2x}=3^3 \right]$$

So:

$$\left[2x=3 \right]$$

$$\left[\boxed{x=\frac{3}{2}} \right]$$

42. FRACTIONAL EXPONENT EQUATION

$$\left[x^{1/2}=7 \right]$$

means:

$$\left[\sqrt{x}=7 \right]$$

Square both sides:

$$\left[x=49 \right]$$

Answer:

$$\left[\boxed{49} \right]$$

43. SCIENTIFIC NOTATION

A number can be written as:

$$\lfloor a \times 10^n \rfloor$$

where usually:

$$\lfloor 1 \leq |a| < 10 \rfloor$$

Example:

$$\lfloor 4500000 = 4.5 \times 10^6 \rfloor$$

And:

$$\lfloor 0.00072 = 7.2 \times 10^{-4} \rfloor$$

Useful in approximation and numerical questions.

44. MULTIPLY SCIENTIFIC NOTATION

$$\lfloor (3 \times 10^4)(2 \times 10^5) \rfloor$$

$$\lfloor = 6 \times 10^9 \rfloor$$

Add exponents.

45. DIVIDE SCIENTIFIC NOTATION

$$\lfloor \frac{8 \times 10^7}{2 \times 10^3} \rfloor$$

$$\lfloor = 4 \times 10^4 \rfloor$$

Subtract exponents.

46. GATE-STYLE QUESTION 1

Simplify:

$$\lfloor \frac{4^5 \cdot 8^2}{2^{10}} \rfloor$$

Convert to base 2:

$$\lfloor 4^5 = (2^2)^5 = 2^{10} \rfloor$$

$$\lfloor 8^2 = (2^3)^2 = 2^6 \rfloor$$

So:

$$\lfloor \frac{2^{16}}{2^{10}} = 2^6 = \boxed{64} \rfloor$$

47. GATE-STYLE QUESTION 2

Evaluate:

$$\lfloor 81^{3/4} \rfloor$$

$$\lfloor 81 = 3^4 \rfloor$$

Thus:

$$\lfloor (3^4)^{3/4} = 3^3 = \boxed{27} \rfloor$$

48. GATE-STYLE QUESTION 3

Simplify:

$$\sqrt{\sqrt{75} + \sqrt{12}}$$

$$\sqrt{\sqrt{75}} = 5\sqrt{3}$$

$$\sqrt{\sqrt{12}} = 2\sqrt{3}$$

Therefore:

$$\boxed{7\sqrt{3}}$$

49. PLACEMENT-STYLE QUESTION 1

Evaluate:

$$32^{2/5}$$

$$32 = 2^5$$

Therefore:

$$(2^5)^{2/5} = 2^2$$

$$= \boxed{4}$$

50. PLACEMENT-STYLE QUESTION 2

Simplify:

$$\frac{5^8}{25^3}$$

$$25^3 = (5^2)^3 = 5^6$$

Thus:

$$5^{8-6} = 5^2$$

$$= \boxed{25}$$

51. PLACEMENT-STYLE QUESTION 3

Evaluate:

$$\sqrt{196} + \sqrt[3]{216}$$

$$14 + 6 = \boxed{20}$$

52. COMMON TRAPS

& pTRAP 1

$$a^m + a^n \neq a^{m+n}$$

& pTRAP 2

$$\nmid (a+b)^n \neq a^n + b^n \nmid$$

in general.

Example:

$$\nmid (2+3)^2 = 25 \nmid$$

but:

$$\nmid 2^2 + 3^2 = 13 \nmid$$

& pTRAP 3

$$\nmid \sqrt{a+b} \neq \sqrt{a} + \sqrt{b} \nmid$$

& pTRAP 4

$$\nmid \sqrt{x^2} = |x| \nmid$$

not always $\sqrt{x}$.

Example:

$$\nmid x = -5 \nmid$$

Then:

$$\nmid \sqrt{x^2} = 5 \nmid$$

not $\sqrt{-5}$.

& pTRAP 5

Negative exponent means reciprocal, not negative value.

$$\nmid 2^{-3} = \frac{1}{8} \nmid$$

not $\sqrt{-8}$.

& pTRAP 6

$$\nmid (-2)^4 = 16 \nmid$$

but:

$$\nmid -2^4 = -16 \nmid$$

because exponent applies before the leading minus unless brackets are present.

53. MUST-MEMORIZE EXPONENT LAWS

$$\nmid \boxed{a^m a^n = a^{m+n}} \nmid$$

$$\nmid \boxed{\frac{a^m}{a^n} = a^{m-n}} \nmid$$

$$\nmid \boxed{(a^m)^n = a^{mn}} \nmid$$

$$\nmid \boxed{(ab)^n = a^n b^n} \nmid$$

$$\boxed{a^0=1}$$

$$\boxed{a^{-n}=\frac{1}{a^n}}$$

$$\boxed{a^{1/n}=\sqrt[n]{a}}$$

$$\boxed{a^{m/n}=\sqrt[n]{a^m}}$$

54. MUST-MEMORIZE ROOT RULES

For appropriate non-negative values:

$$\sqrt{ab}=\sqrt{a}\sqrt{b}$$

$$\sqrt{\frac{ab}{c}}=\frac{\sqrt{a}\sqrt{b}}{\sqrt{c}}$$

But:

$$\sqrt{a+b}\neq\sqrt{a}+\sqrt{b}$$

55. SPEED METHOD — LOOK FOR PERFECT POWERS

Whenever you see:

$$64^{2/3}$$

ask:

$$64=4^3$$

Then:

$$(4^3)^{2/3}=4^2=16$$

This is faster than using decimals or calculators.

56. MUST-MEMORIZE PERFECT POWERS

You already know many from Chapters 1 and 6.

Useful squares:

$$1^2, 2^2, \dots, 30^2$$

Useful cubes:

$$1^3, \dots, 10^3$$

Also recognize:

$$16=2^4$$

$$32=2^5$$

$$64=2^6=4^3=8^2$$

$$81=3^4=9^2$$

$$125=5^3$$

$$243=3^5$$

$$\sqrt[4]{256} = 2^8 = 4^4 = 16^2$$

These make exponent questions much faster.

PRACTICE — LEVEL 0

Q1

Simplify:

$$\sqrt[4]{2^4 \cdot 2^7}$$

Q2

Simplify:

$$\sqrt[4]{\frac{3^8}{3^5}}$$

Q3

Evaluate:

$$\sqrt[4]{5^0}$$

Q4

Evaluate:

$$\sqrt[4]{2^{-4}}$$

Q5

Evaluate:

$$\sqrt[4]{64^{1/3}}$$

LEVEL 1

Q6

Evaluate:

$$\sqrt[4]{81^{1/2}}$$

Q7

Evaluate:

$$\sqrt[4]{32^{3/5}}$$

Q8

Simplify:

$$\sqrt[4]{\sqrt{108}}$$

Q9

Simplify:

$$\sqrt[4]{\sqrt[3]{128}}$$

Q10

Solve:

$$\sqrt[4]{4^x=64}$$

LEVEL 2

Q11

Simplify:

$$\sqrt{\frac{2^9 \cdot 4^3}{8^4}}$$

Q12

Evaluate:

$$\sqrt[4]{16^{-3/4}}$$

Q13

Compare:

$$\sqrt[5]{8^5}$$

and:

$$\sqrt[7]{4^7}$$

Q14

Simplify:

$$\sqrt{\sqrt{200}} + \sqrt{8}$$

Q15

Solve:

$$\sqrt[9]{27^x=9}$$

LEVEL 3 — EXAM PRESSURE

Q16

Simplify:

$$\sqrt{\frac{9^4 \cdot 27^2}{3^{10}}}$$

Q17

Evaluate:

$$\sqrt[8]{256^{3/8}}$$

Q18

If:

$$\sqrt{2^x=8^{x-2}}$$

find $\sqrt{x}$.

Q19

Compare without full expansion:

$$\sqrt[3]{16^7}$$

and:

$$\sqrt[3]{8^9}$$

Q20

Simplify:

$$\sqrt{12} + \sqrt{27} - \sqrt{75}$$

ANSWERS

1

$$2^{11} = \boxed{2048}$$

2

$$3^3 = \boxed{27}$$

3

$$\boxed{1}$$

4

$$\boxed{\frac{1}{16}}$$

5

$$\boxed{4}$$

6

$$\boxed{9}$$

7

$$32 = 2^5$$

$$32^{\frac{3}{5}} = 2^3$$

$$\boxed{8}$$

8

$$108 = 36 \cdot 3$$

$$\boxed{6\sqrt{3}}$$

9

$$128 = 64 \cdot 2$$

$$\sqrt[3]{128} = \boxed{4\sqrt[3]{2}}$$

10

$$4^x = (2^2)^x = 2^{2x}$$

$$\backslash[64=2^6 \backslash]$$

So:

$$\backslash[2x=6 \backslash]$$

$$\backslash[\boxed{x=3} \backslash]$$

11

$$\backslash[4^3=2^6 \backslash]$$

$$\backslash[8^4=2^{\{12\}} \backslash]$$

So:

$$\backslash[2^{\{9+6-12\}} = 2^3 = \boxed{8} \backslash]$$

12

$$\backslash[16^{\{1/4\}}=2 \backslash]$$

$$\backslash[16^{\{3/4\}}=8 \backslash]$$

Thus:

$$\backslash[\boxed{\frac{18}{}} \backslash]$$

13

$$\backslash[8^5=(2^3)^5=2^{\{15\}} \backslash]$$

$$\backslash[4^7=(2^2)^7=2^{\{14\}} \backslash]$$

Therefore:

$$\backslash[\boxed{8^5>4^7} \backslash]$$

14

$$\backslash[\sqrt{200}=10\sqrt{2} \backslash]$$

$$\backslash[\sqrt{8}=2\sqrt{2} \backslash]$$

So:

$$\backslash[\boxed{12\sqrt{2}} \backslash]$$

15

$$\backslash[27=3^3,\text{quad}9=3^2 \backslash]$$

$$\backslash[3^{\{3x\}}=3^2 \backslash]$$

$$\backslash[\boxed{x=\frac{2}{3}} \backslash]$$

16

Convert:

$$\backslash[9^4=(3^2)^4=3^8 \backslash]$$

$$\backslash[27^2=(3^3)^2=3^6 \backslash]$$

Therefore:

$$\backslash[3^{\{8+6-10\}} = 3^4 = \boxed{81} \backslash]$$

17

$$\boxed{256=2^8}$$

$$(2^8)^{3/8}=2^3$$

$$\boxed{8}$$

18

$$8^{x-2}=(2^3)^{x-2}$$

$$=2^{3x-6}$$

Thus:

$$x=3x-6$$

$$2x=6$$

$$\boxed{x=3}$$

19

$$16^7=(2^4)^7=2^{28}$$

$$8^9=(2^3)^9=2^{27}$$

Therefore:

$$\boxed{16^7>8^9}$$

20

$$\sqrt{12}=2\sqrt{3}$$

$$\sqrt{27}=3\sqrt{3}$$

$$\sqrt{75}=5\sqrt{3}$$

So:

$$2\sqrt{3}+3\sqrt{3}-5\sqrt{3}=\boxed{0}$$

CHAPTER 9 STATUS

We have completed:

- exponent meaning '
- product/quotient laws '
- power of a power '
- power of product/quotient '
- zero exponent '
- negative exponent '
- fractional exponent '
- roots and surds '
- square/cube simplification '
- perfect-square/cube connection '
- comparing powers via common bases '
- exponent equations '

- scientific notation '
- GATE/placementpractice '

But **Chapter 9 is not complete yet.**

Next we need:

- advanced surd operations
- rationalization
- conjugates
- nested roots
- comparing irrational values
- square-root and cube-root estimation
- approximation shortcuts
- exponent inequalities
- large-power comparison without expansion
- advanced GATE/placementmixed problems

Progress

Completed: 8/76 **Current:** Ø=Chapter 9 — Powers, Exponents & Roots **Next:** Chapter 9 Part 2 — **Surds, rationalization & advanced power comparison**

USER

next

ASSISTANT

CHAPTER 9 — POWERS, EXPONENTS & ROOTS

PART 2 — SURDS, RATIONALIZATION, NESTED ROOTS & ADVANCED POWER COMPARISON

Status: Ø=Not in PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This part focuses on questions where ordinary exponent laws are not enough.

We will cover:

- operations on surds
- rationalization
- conjugates
- nested roots
- comparing irrational quantities
- estimating roots
- comparing very large powers
- exponent inequalities
- exam shortcuts

1. WHAT IS A SURD?

A **surd** is an irrational root that cannot be simplified into a rational number.

Examples:

$$\sqrt{2}, \sqrt[3]{3}, \sqrt[4]{5}, \sqrt[3]{2}$$

But:

$$\sqrt{16}=4$$

is not a surd after simplification.

2. LIKE SURDS

Just as:

$$2x+5x=7x$$

we can combine identical surds.

Example:

$$3\sqrt{2}+5\sqrt{2} = \boxed{8\sqrt{2}}$$

3. UNLIKE SURDS CANNOT BE DIRECTLY ADDED

$$\sqrt{2}+\sqrt{3}$$

cannot be simplified to:

$$\sqrt{5}$$

& bTRAP

$$\boxed{\sqrt{a}+\sqrt{b}\neq\sqrt{a+b}}$$

So:

$$\sqrt{2}+\sqrt{3}$$

must generally remain as it is.

4. SIMPLIFY BEFORE COMBINING

Example:

$$\sqrt{12}+\sqrt{27}$$

Simplify:

$$\sqrt{12}=2\sqrt{3}$$

$$\sqrt{27}=3\sqrt{3}$$

Therefore:

$$2\sqrt{3}+3\sqrt{3} = \boxed{5\sqrt{3}}$$

5. SUBTRACTION OF SURDS

$$\sqrt{5\sqrt{7}-2\sqrt{7}} = \boxed{3\sqrt{7}}$$

Example:

$$\sqrt{\sqrt{75}-\sqrt{12}}$$

$$= 5\sqrt{3}-2\sqrt{3}$$

$$= \boxed{3\sqrt{3}}$$

6. MULTIPLICATION OF SURDS

Use:

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$$

Example:

$$\sqrt{3} \cdot \sqrt{12} = \sqrt{36} = \boxed{6}$$

7. MULTIPLYING COEFFICIENTS

$$(2\sqrt{3})(5\sqrt{6})$$

Multiply coefficients:

$$2 \times 5 = 10$$

Multiply roots:

$$\sqrt{18} = 3\sqrt{2}$$

So:

$$10(3\sqrt{2}) = \boxed{30\sqrt{2}}$$

8. SQUARING A SURD

$$(\sqrt{a})^2 = a$$

Example:

$$(3\sqrt{5})^2$$

$$= 9 \times 5$$

$$= \boxed{45}$$

9. BINOMIAL SURD SQUARE

$$(\sqrt{a} + \sqrt{b})^2$$

Use:

$$\sqrt{(x+y)^2} = x^2 + y^2 + 2xy$$

Therefore:

$$\boxed{(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}}$$

10. EXAMPLE

$$\sqrt{(\sqrt{3} + \sqrt{2})^2}$$

$$= 3 + 2 + 2\sqrt{6}$$

$$\boxed{= 5 + 2\sqrt{6}}$$

11. DIFFERENCE SQUARE

$$\sqrt{(\sqrt{a} - \sqrt{b})^2} = a + b - 2\sqrt{ab}$$

Example:

$$\sqrt{(\sqrt{5} - \sqrt{2})^2}$$

$$= 5 + 2 - 2\sqrt{10}$$

$$\boxed{= 7 - 2\sqrt{10}}$$

12. CONJUGATES

For:

$$\sqrt{a + \sqrt{b}}$$

its conjugate is:

$$\sqrt{a - \sqrt{b}}$$

Similarly, conjugate of:

$$\sqrt{\sqrt{a} + \sqrt{b}}$$

is:

$$\sqrt{\sqrt{a} - \sqrt{b}}$$

13. WHY CONJUGATES ARE USEFUL

Because:

$$\sqrt{(a+b)(a-b)} = a^2 - b^2$$

Therefore:

$$\sqrt{(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})} = a - b$$

The irrational middle terms disappear.

14. EXAMPLE

$$\sqrt{(\sqrt{7}+\sqrt{3})(\sqrt{7}-\sqrt{3})}$$

$$=7-3$$

$$=4$$

This is one of the most important surd tricks.

15. RATIONALIZATION— WHAT DOES IT MEAN?

Rationalization means removing surds from the denominator.

Example:

$$\frac{1}{\sqrt{2}}$$

Multiply top and bottom by:

$$\sqrt{2}$$

$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Thus:

$$\boxed{\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}}$$

16. SIMPLE RATIONALIZATION RULE

$$\boxed{\frac{a}{\sqrt{b}} = \frac{a\sqrt{b}}{b}}$$

Example:

$$\frac{5}{\sqrt{3}} = \boxed{\frac{5\sqrt{3}}{3}}$$

17. RATIONALIZING A BINOMIAL DENOMINATOR

Suppose:

$$\frac{1}{\sqrt{5}+\sqrt{2}}$$

Multiply numerator and denominator by conjugate:

$$\sqrt{5}-\sqrt{2}$$

So:

$$\frac{\sqrt{5}-\sqrt{2}}{(\sqrt{5}+\sqrt{2})(\sqrt{5}-\sqrt{2})}$$

Denominator:

$$5-2=3$$

Therefore:

$$\boxed{\frac{\sqrt{5}-\sqrt{2}}{3}}$$

18. ANOTHER EXAMPLE

Simplify:

$$\sqrt{\frac{1}{3+\sqrt{5}}}$$

Multiply by conjugate:

$$\sqrt{3-\sqrt{5}}$$

$$= \frac{3-\sqrt{5}}{9-5}$$

$$= \boxed{\frac{3-\sqrt{5}}{4}}$$

19. FAST RATIONALIZATION PATTERN

$$\sqrt{\frac{1}{\sqrt{a}+\sqrt{b}}} = \boxed{\frac{\sqrt{a}-\sqrt{b}}{a-b}}$$

provided $a \neq b$.

Similarly:

$$\sqrt{\frac{1}{\sqrt{a}-\sqrt{b}}} = \boxed{\frac{\sqrt{a}+\sqrt{b}}{a-b}}$$

20. CLASSIC IDENTITY

Evaluate:

$$\sqrt{(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})}$$

Immediately:

$$\sqrt{3-2} = \boxed{1}$$

Therefore:

$$\sqrt{\frac{1}{\sqrt{3}+\sqrt{2}}} = \sqrt{3}-\sqrt{2}$$

because denominator difference is exactly 1.

This is a high-value shortcut.

21. ANOTHER BEAUTIFUL PAIR

$$\sqrt{(\sqrt{5}+2)(\sqrt{5}-2)} = 5-4 = 1$$

Therefore:

$$\sqrt{\frac{1}{\sqrt{5}+2}} = \sqrt{5}-2$$

22. NESTED ROOTS

Expressions like:

$$\sqrt{\sqrt{a+2}\sqrt{b}}$$

sometimes simplify to:

$$\sqrt{\sqrt{m}+\sqrt{n}}$$

because:

$$\sqrt{(\sqrt{m} + \sqrt{n})^2} = m + n + 2\sqrt{mn}$$

So compare:

$$\sqrt{m+n=a}$$

and:

$$\sqrt{mn=b}$$

23. EXAMPLE — NESTED ROOT

Simplify:

$$\sqrt{\sqrt{5+2\sqrt{6}}}$$

Assume:

$$\sqrt{\sqrt{m} + \sqrt{n}}$$

Need:

$$\sqrt{m+n=5}$$

$$\sqrt{mn=6}$$

Numbers:

$$\sqrt{2,3}$$

Therefore:

$$\boxed{\sqrt{\sqrt{5+2\sqrt{6}}} = \sqrt{2} + \sqrt{3}}$$

24. NESTED ROOT WITH MINUS

Simplify:

$$\sqrt{\sqrt{5-2\sqrt{6}}}$$

Use:

$$\sqrt{(\sqrt{3}-\sqrt{2})^2} = 3+2-2\sqrt{6}$$

Therefore:

$$\boxed{\sqrt{\sqrt{5-2\sqrt{6}}} = \sqrt{3}-\sqrt{2}}$$

because $\sqrt{3} > \sqrt{2}$.

25. ANOTHER EXAMPLE

Simplify:

$$\sqrt{\sqrt{7+4\sqrt{3}}}$$

We want:

$$\sqrt{m+n=7}$$

and:

$$\sqrt[2]{mn} = 4\sqrt{3}$$

Thus:

$$mn = 12$$

Numbers with sum 7 and product 12:

$$3, 4$$

Therefore:

$$\sqrt{7+4\sqrt{3}} = \sqrt{3} + \sqrt{4}$$

$$= \boxed{\sqrt{3} + 2}$$

26. FAST NESTED-ROOT METHOD

For:

$$\sqrt{a+2\sqrt{b}}$$

find (m, n) satisfying:

$$m+n=a, \text{ and } mn=b$$

Then:

$$\boxed{\sqrt{a+2\sqrt{b}} = \sqrt{m} + \sqrt{n}}$$

Similarly minus sign gives difference when valid.

27. COMPARING SURDS — METHOD 1: SQUARE BOTH SIDES

Compare:

$$\sqrt{17}$$

and:

$$4$$

Both non-negative.

Square:

$$17$$

vs:

$$16$$

Therefore:

$$\boxed{\sqrt{17} > 4}$$

28. COMPARE TWO ROOTS

Compare:

$$\sqrt[3]{7}$$

and:

$$\sqrt[3]{8}$$

Since square root is increasing for positive numbers:

$$7 < 8$$

Therefore:

$$\boxed{\sqrt[3]{7} < \sqrt[3]{8}}$$

29. COMPARE DIFFERENT ROOT TYPES

Compare:

$$\sqrt[3]{3}$$

and:

$$\sqrt[5]{3}$$

Both positive.

Raise both to LCM of root indices:

$$\text{LCM}(2,3)=6$$

$$(\sqrt[3]{3})^6 = 3^2 = 9$$

$$(\sqrt[5]{3})^6 = 3^{6/5} = 3^{1.2}$$

Therefore:

$$\boxed{\sqrt[3]{3} > \sqrt[5]{3}}$$

This is a powerful exact comparison method.

30. COMPARE $(a^{m/n})$ VALUES

Example:

Compare:

$$2^{3/2}$$

and:

$$3^{2/3}$$

Raise both to power 6:

$$(2^{3/2})^6 = 2^9 = 512$$

$$(3^{2/3})^6 = 3^4 = 81$$

Thus:

$$\boxed{2^{3/2} > 3^{2/3}}$$

31. ESTIMATING SQUARE ROOTS — NEAREST SQUARE

Suppose:

$$\sqrt{82}$$

Nearby:

$$9^2=81$$

Therefore:

$$\sqrt{82}$$

is just above 9.

Approx:

$$9.05$$

For many aptitude options, knowing:

$$9 < \sqrt{82} < 10$$

is already enough.

32. FAST APPROXIMATION NEAR A PERFECT SQUARE

For small (x) relative to (a^2) :

$$\sqrt{a^2+x} \approx a + \frac{x}{2a}$$

Example:

$$\sqrt{101}$$

Take:

$$a=10, \text{quad } x=1$$

$$\sqrt{101} \approx 10 + \frac{1}{20}$$

$$=\boxed{10.05}$$

Actual value is about 10.0499.

SHORTCUT ONLY

Use for estimation, not exact answers.

33. ANOTHER APPROXIMATION

$$\sqrt{399}$$

Nearby:

$$20^2=400$$

So:

$$\sqrt{399}=400-1$$

Approx:

$$\lfloor 20 - \frac{1}{40} \rfloor$$

$$\lfloor = 19.975 \rfloor$$

Very accurate.

34. ESTIMATING CUBE ROOTS

If:

$$\lfloor a^3 + x \rfloor$$

is near a perfect cube:

$$\lfloor \sqrt[3]{a^3 + x} \approx a + \frac{x}{3a^2} \rfloor$$

Example:

$$\lfloor \sqrt[3]{65} \rfloor$$

Since:

$$\lfloor 4^3 = 64 \rfloor$$

$$\lfloor \approx 4 + \frac{1}{3(16)} \rfloor$$

$$\lfloor = 4 + \frac{1}{48} \rfloor$$

$$\lfloor \approx 4.021 \rfloor$$

This is optional speed math.

35. COMPARING LARGE POWERS — COMMON BASE FIRST

Compare:

$$\lfloor 32^7 \rfloor$$

and:

$$\lfloor 8^{12} \rfloor$$

Convert to base 2:

$$\lfloor 32^7 = (2^5)^7 = 2^{35} \rfloor$$

$$\lfloor 8^{12} = (2^3)^{12} = 2^{36} \rfloor$$

Therefore:

$$\lfloor \boxed{8^{12} > 32^7} \rfloor$$

36. COMMON EXPONENT METHOD

Compare:

$$\lfloor 3^{20} \rfloor$$

and:

$$\lfloor 5^{15} \rfloor$$

We can rewrite using common exponent 5:

$$\sqrt[5]{3^{20}} = (3^4)^5 = 81^5$$

$$\sqrt[5]{5^{15}} = (5^3)^5 = 125^5$$

Since:

$$\sqrt[5]{125} > \sqrt[5]{81}$$

$$\sqrt[5]{5^{15}} > \sqrt[5]{3^{20}}$$

This is much easier than expanding.

37. ANOTHER POWER COMPARISON

Compare:

$$\sqrt[10]{2^{30}}$$

and:

$$\sqrt[10]{3^{20}}$$

Use common exponent 10:

$$\sqrt[10]{2^{30}} = (2^3)^{10} = 8^{10}$$

$$\sqrt[10]{3^{20}} = (3^2)^{10} = 9^{10}$$

Thus:

$$\sqrt[10]{3^{20}} > \sqrt[10]{2^{30}}$$

38. COMPARING $\sqrt[a]{b}$ AND $\sqrt[b]{a}$

Classic aptitude pattern.

Compare:

$$\sqrt[3]{2} = 8$$

and:

$$\sqrt[2]{3} = 9$$

So:

$$\sqrt[2]{3} > \sqrt[3]{2}$$

But for larger values, pattern can change.

Example:

$$\sqrt[4]{3} = 81$$

$$\sqrt[3]{4} = 64$$

So:

$$\sqrt[4]{3} > \sqrt[3]{4}$$

Do not assume one universal simple order for $\sqrt[a]{b}$ vs $\sqrt[b]{a}$.

39. USE LOGIC WITH ROOTS

Compare:

$$\sqrt[4]{4^5}$$

and:

$$\sqrt[4]{5^4}$$

Take fourth roots:

$$\sqrt[4]{(4^5)^{1/4}} = 4^{5/4} = 4\sqrt[4]{2}$$

and:

$$\sqrt[4]{(5^4)^{1/4}} = 5$$

Since:

$$\sqrt[4]{4\sqrt[4]{2}} \approx 5.657 > 5$$

Therefore:

$$\boxed{4^5 > 5^4}$$

Though here direct values are also manageable.

40. EXPONENT INEQUALITIES WITH SAME BASE

If:

$$a > 1$$

then:

$$a^x > a^y \text{ iff } x > y$$

Example:

$$2^{3x-1} > 2^8$$

Therefore:

$$3x-1 > 8$$

$$3x > 9$$

$$\boxed{x > 3}$$

41. BASE BETWEEN 0 AND 1 — INEQUALITY REVERSES

If:

$$0 < a < 1$$

then:

$$a^x > a^y \text{ iff } x < y$$

Example:

$$\left(\frac{1}{2}\right)^x$$

$$\left(\frac{12}{\text{right}}\right)^5$$

Thus:

$$\boxed{x < 5}$$

& pTRAP

This reversal is very important.

42. SOLVING AN EXPONENTIAL INEQUALITY

$$4^{x+1} \leq 64$$

Convert to base 2:

$$2^{2x+2} \leq 2^6$$

Since base $(2 > 1)$:

$$2x+2 \leq 6$$

$$x \leq 2$$

Answer:

$$\boxed{x \leq 2}$$

43. ROOT INEQUALITY

Solve:

$$\sqrt{x} > 5$$

For real principal square root:

$$x \geq 0$$

Square both sides:

$$x > 25$$

Thus:

$$\boxed{x > 25}$$

44. BE CAREFUL WHEN SQUARING INEQUALITIES

Squaring is safest when both sides are known non-negative.

Example:

$$\sqrt{x+1} > x-1$$

cannot be blindly squared without considering whether:

$$x-1$$

is negative.

This becomes more relevant in algebra chapters.

45. RATIONALIZATION SPEED EXAMPLE

Simplify:

$$\sqrt{\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}}$$

Multiply numerator and denominator by:

$$\sqrt{\sqrt{3}+\sqrt{2}}$$

Denominator:

$$\sqrt{3-2=1}$$

Numerator:

$$\sqrt{(\sqrt{3}+\sqrt{2})^2}$$

$$\sqrt{=5+2\sqrt{6}}$$

Therefore:

$$\sqrt{\boxed{5+2\sqrt{6}}}$$

46. CLASSIC RECIPROCAL PAIR

Let:

$$\sqrt{x=\sqrt{2}+1}$$

Then:

$$\sqrt{\frac{1}{x} = \sqrt{2}-1}$$

because:

$$\sqrt{(\sqrt{2}+1)(\sqrt{2}-1)=1}$$

Therefore:

$$\sqrt{x+\frac{1}{x} = (\sqrt{2}+1)+(\sqrt{2}-1)}$$

$$\sqrt{=\boxed{2\sqrt{2}}}$$

This style appears in algebraic aptitude problems.

47. GATE-STYLE QUESTION 1

Simplify:

$$\sqrt{\frac{1}{\sqrt{3}-1}}}$$

Multiply by conjugate:

$$\sqrt{\sqrt{3}+1}$$

$$\sqrt{= \frac{\sqrt{3}+1}{\{3-1\}}}$$

$$\sqrt{=\boxed{\frac{\sqrt{3}+1}{\{2\}}}}$$

48. GATE-STYLE QUESTION 2

Simplify:

$$\sqrt{9+4\sqrt{5}}$$

We need:

$$m+n=9$$

and:

$$mn=20$$

Numbers:

$$4, 5$$

Therefore:

$$\sqrt{9+4\sqrt{5}} = \sqrt{4+4\sqrt{5}}$$

$$= \boxed{2+\sqrt{5}}$$

49. GATE-STYLE QUESTION 3

Compare:

$$\sqrt[3]{7}$$

and:

$$\sqrt{5}$$

Raise both to power 6:

$$(\sqrt[3]{7})^6 = 7^2 = 49$$

$$(\sqrt{5})^6 = 5^3 = 125$$

Therefore:

$$\boxed{\sqrt{5} > \sqrt[3]{7}}$$

50. GATE-STYLE QUESTION 4

Compare:

$$2^{15}$$

and:

$$4^8$$

$$4^8 = (2^2)^8 = 2^{16}$$

Thus:

$$\boxed{4^8 > 2^{15}}$$

51. PLACEMENT-STYLE QUESTION 1

Rationalize:

$$\sqrt[4]{\frac{4}{\sqrt{5}}}$$

$$\boxed{\frac{4\sqrt{5}}{5}}$$

52. PLACEMENT-STYLE QUESTION 2

Simplify:

$$\sqrt{48} + \sqrt{27} - \sqrt{75}$$

$$4\sqrt{3} + 3\sqrt{3} - 5\sqrt{3}$$

$$\boxed{2\sqrt{3}}$$

53. PLACEMENT-STYLE QUESTION 3

Simplify:

$$(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2})$$

$$= 5 - 2$$

$$\boxed{3}$$

54. PLACEMENT-STYLE QUESTION 4

Which is larger?

$$8^{10}$$

or:

$$4^{16}$$

Convert:

$$8^{10} = 2^{30}$$

$$4^{16} = 2^{32}$$

Therefore:

$$\boxed{4^{16}}$$

55. COMMON TRAPS

& pTRAP 1

Do not add unlike surds:

$$\sqrt{2} + \sqrt{3} \neq \sqrt{5}$$

& pTRAP 2

Use conjugate for binomial denominators.

& pTRAP 3

For:

$$\sqrt{a^2}$$

the answer is:

$$\sqrt{a}$$

a

$$\sqrt{a}$$

not automatically $\sqrt{a}$.

& pTRAP 4

Do not use approximations when exact simplification exists.

Example:

$$\sqrt{50}$$

may remain:

$$5\sqrt{2}$$

if exact answer is required.

& pTRAP 5

When comparing powers, convert to common base or common exponent before doing huge arithmetic.

& pTRAP 6

For bases between 0 and 1, exponent inequality direction reverses.

56. MUST-MEMORIZE SURD IDENTITIES

$$(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}$$

$$(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab}$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

$$\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$$

57. ROOT-ESTIMATION CHECKLIST

For:

$$\sqrt{N}$$

1. locate surrounding perfect squares

2. get an interval
3. approximate only if needed

Example:

$$\sqrt{14^2=196}$$

$$\sqrt{15^2=225}$$

Therefore:

$$\sqrt{14 < \sqrt{210} < 15}$$

That alone may eliminate most answer options.

58. POWER-COMPARISON CHECKLIST

Before calculating large powers, ask:

1. Can I convert to same base?
2. Can I convert to same exponent?
3. Can I take a common positive root?
4. Can I compare nearby benchmark values?
5. Is logarithm actually needed?

For aptitude, usually one of the first four works.

PRACTICE — LEVEL 0

Q1

Simplify:

$$\sqrt{3\sqrt{5}+4\sqrt{5}}$$

Q2

Simplify:

$$\sqrt{\sqrt{50}}$$

Q3

Rationalize:

$$\sqrt{\frac{1}{\sqrt{3}}}$$

Q4

Evaluate:

$$\sqrt{(\sqrt{7}+\sqrt{2})(\sqrt{7}-\sqrt{2})}$$

Q5

Simplify:

$$\sqrt{\sqrt{5+2\sqrt{6}}}$$

LEVEL 1

Q6

Simplify:

$$\sqrt{\sqrt{72}-\sqrt{32}}$$

Q7

Rationalize:

$$\frac{1}{\sqrt{7}+\sqrt{3}}$$

Q8

Simplify:

$$\sqrt{7-4\sqrt{3}}$$

Q9

Compare:

$$\sqrt{10}$$

and:

$$3$$

Q10

Compare:

$$8^7$$

and:

$$4^{11}$$

LEVEL 2

Q11

Simplify:

$$\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$$

Q12

Simplify:

$$\sqrt{11+6\sqrt{2}}$$

Q13

Compare:

$$\sqrt[3]{10}$$

and:

$$\sqrt{5}$$

Q14

Solve:

$$\lfloor 2^{2x-1} \rfloor > 8 \rfloor$$

Q15

Solve:

$$\lfloor \left(\frac{13}{\right)^{x+2} < \left(\frac{13}{\right)^5} \rfloor$$

LEVEL 3 — EXAM PRESSURE

Q16

Simplify:

$$\lfloor \frac{1}{\sqrt{5}-2} \rfloor$$

Q17

If:

$$\lfloor x = \sqrt{3} + \sqrt{2} \rfloor$$

find:

$$\lfloor \frac{1}{x} \rfloor$$

Q18

Compare:

$$\lfloor 3^{24} \rfloor$$

and:

$$\lfloor 9^{11} \rfloor$$

Q19

Compare:

$$\lfloor 2^{40} \rfloor$$

and:

$$\lfloor 4^{19} \rfloor$$

Q20

Approximate:

$$\lfloor \sqrt{102} \rfloor$$

using the nearest-square method.

ANSWERS

1

$$\lfloor \boxed{7\sqrt{5}} \rfloor$$

2

$$\sqrt{25 \cdot 2} = \boxed{5\sqrt{2}}$$

3

$$\boxed{\frac{\sqrt{3}}{3}}$$

4

$$7 - 2 = \boxed{5}$$

5

$$m + n = 5, \text{quad } mn = 6$$

So $(m, n = 2, 3)$.

$$\boxed{\sqrt{2} + \sqrt{3}}$$

6

$$\sqrt{72} = 6\sqrt{2}$$

$$\sqrt{32} = 4\sqrt{2}$$

Therefore:

$$\boxed{2\sqrt{2}}$$

7

$$\frac{1}{\sqrt{7} + \sqrt{3}} = \frac{\sqrt{7} - \sqrt{3}}{7 - 3}$$

$$= \boxed{\frac{\sqrt{7} - \sqrt{3}}{4}}$$

8

We need:

$$m + n = 7$$

and:

$$2\sqrt{mn} = 4\sqrt{3}$$

So:

$$mn = 12$$

Thus $(m, n = 4, 3)$.

$$\sqrt{7 - 4\sqrt{3}} = 2 - \sqrt{3}$$

because $(2 > \sqrt{3})$.

$$\boxed{2 - \sqrt{3}}$$

9

Square:

$$10 > 9$$

Therefore:

$$\boxed{\sqrt{10} > 3}$$

10

$$\backslash[8^7=2^{21} \backslash]$$

$$\backslash[4^{11}=2^{22} \backslash]$$

Thus:

$$\backslash[\boxed{4^{11}>8^7} \backslash]$$

11

Multiply by conjugate:

$$\backslash[\frac{(\sqrt{5}+\sqrt{3})^2}{5-3} \backslash]$$

Numerator:

$$\backslash[8+2\sqrt{15} \backslash]$$

Divide by 2:

$$\backslash[\boxed{4+\sqrt{15}} \backslash]$$

12

Need:

$$\backslash[m+n=11 \backslash]$$

and:

$$\backslash[mn=18 \backslash]$$

Numbers:

$$\backslash[2,9 \backslash]$$

Therefore:

$$\backslash[\boxed{\sqrt{2}+3} \backslash]$$

13

Raise both to sixth power:

$$\backslash[(\sqrt[3]{10})^6=10^2=100 \backslash]$$

$$\backslash[(\sqrt{5})^6=5^3=125 \backslash]$$

Thus:

$$\backslash[\boxed{\sqrt{5}>\sqrt[3]{10}} \backslash]$$

14

$$\backslash[8=2^3 \backslash]$$

So:

$$\backslash[2x-1>3 \backslash]$$

$$\backslash[2x>4 \backslash]$$

$$\backslash[\boxed{x>2} \backslash]$$

15

Base:

$$\left[0 < \frac{13}{1} \right]$$

So inequality reverses for exponents:

$$\left[x+2 > 5 \right]$$

$$\left[\boxed{x > 3} \right]$$

16

$$\left[\frac{1}{\sqrt{5}-2} \cdot \frac{\sqrt{5}+2}{\sqrt{5}+2} \right]$$

Denominator:

$$\left[5-4=1 \right]$$

Therefore:

$$\left[\boxed{\sqrt{5}+2} \right]$$

17

$$\left[(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})=1 \right]$$

So:

$$\left[\boxed{\frac{1}{x}=\sqrt{3}-\sqrt{2}}} \right]$$

18

$$\left[9^{11}=(3^2)^{11}=3^{22} \right]$$

Compare:

$$\left[3^{24} > 3^{22} \right]$$

Therefore:

$$\left[\boxed{3^{24} > 9^{11}} \right]$$

19

$$\left[4^{19}=2^{38} \right]$$

Thus:

$$\left[2^{40} > 2^{38} \right]$$

$$\left[\boxed{2^{40} > 4^{19}} \right]$$

20

Nearby square:

$$\left[10^2=100 \right]$$

Use:

$$\left[\sqrt{100+2} \approx 10+\frac{2}{20} \right]$$

$$\left[=10.1 \right]$$

So:

$$\left[\boxed{\sqrt{102} \approx 10.1} \right]$$

Actual value is about 10.10.

CHAPTER 9 — FINAL CHECK

We have now completed:

- exponent laws '
- zero/negative/fractional powers '
- roots and surds '
- surd addition/subtraction/multiplication '
- conjugates '
- rationalization '
- nested roots '
- root estimation '
- perfect-power recognition '
- common-base comparison '
- common-exponent comparison '
- large-power comparison '
- exponent equations '
- exponent inequalities '
- base $(0 < a < 1)$ reversal '
- scientific notation '
- GATE-style practice '
- placement-style practice '

Therefore:

' CHAPTER 9 COMPLETE

Progress

Completed: 9/76 **Current:** ' Chapter 9 — Powers, Exponents & Roots **Next:** Ø=Chapter 10 — Squares, Cubes & Mental-Math Tables

Chapter 10 will focus heavily on **calculation speed**:

- squares you must memorize
- cubes you should memorize
- squaring numbers near 10/50/100
- numbers ending in 5
- difference-of-squares shortcuts
- fast cube recognition
- square/cube-root identification
- perfect square/cube patterns
- mental-math drills for GATE and placements.

USER

next

CHAPTER 10 — SQUARES, CUBES & MENTAL-MATH TABLES

PART 1 — MUST-MEMORIZE VALUES, FAST SQUARING & PERFECT-SQUARE RECOGNITION

Status: $\emptyset$ = Not in PROGRESS GATE: Medium Placements: High Priority: +Tier 2

This chapter is mainly about **speed**.

You should not spend 20–30 seconds calculating something like:

$$\sqrt{48^2}$$

or:

$$\sqrt{75^2}$$

when it can be done mentally in a few seconds.

The goal is to build a reliable mental library of:

- squares
- cubes
- powers
- square-root recognition
- cube-root recognition
- fast multiplication identities

1. MUST-MEMORIZE SQUARES

You should know squares from:

$$\sqrt{1^2 \text{ to } 30^2}$$

almost instantly.

$\sqrt{n}$	$\sqrt{n^2}$	$\sqrt{n}$	$\sqrt{n^2}$
1	1	16	256
2	4	17	289
3	9	18	324
4	16	19	361
5	25	20	400
6	36	21	441
7	49	22	484
8	64	23	529
9	81	24	576

$\sqrt{n}$	$\sqrt{n^2}$	$\sqrt{n}$	$\sqrt{n^2}$
10	100	25	625
11	121	26	676
12	144	27	729
13	169	28	784
14	196	29	841
15	225	30	900

MUST MEMORIZE

These values appear constantly.

2. GOOD-TO-MEMORIZE SQUARES 31–40

$\sqrt{n}$	$\sqrt{n^2}$
31	961
32	1024
33	1089
34	1156
35	1225
36	1296
37	1369
38	1444
39	1521
40	1600

These are very useful in placement aptitude.

3. MUST-MEMORIZE CUBES

At minimum, know:

$\sqrt[3]{1^3}$ to $\sqrt[3]{10^3}$

$\sqrt[3]{n}$	$\sqrt[3]{n^3}$
1	1

$\sqrt[n]{n}$	$\sqrt[n]{n^3}$
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000

MUST MEMORIZE

4. GOOD-TO-MEMORIZE CUBES 11–20

$\sqrt[n]{n}$	$\sqrt[n]{n^3}$
11	1331
12	1728
13	2197
14	2744
15	3375
16	4096
17	4913
18	5832
19	6859
20	8000

You do not need to memorize these all on Day 1, but knowing them improves placement speed.

5. FAST SQUARE USING $\sqrt{(a+b)^2}$

Formula:

$$\sqrt{(a+b)^2} = a^2 + 2ab + b^2$$

Example:

$$\sqrt[4]{43^2}$$

Take:

$$\sqrt[4]{43=40+3}$$

Then:

$$\sqrt[4]{43^2 = 40^2 + 2(40)(3) + 3^2}$$

$$\sqrt[4]{=1600+240+9}$$

$$\sqrt[4]{=\boxed{1849}}$$

6. FAST SQUARE USING NEAREST ROUND NUMBER

Find:

$$\sqrt[4]{48^2}$$

Use:

$$\sqrt[4]{48=50-2}$$

Then:

$$\sqrt[4]{(50-2)^2 = 2500-200+4}$$

$$\sqrt[4]{=\boxed{2304}}$$

Much faster than long multiplication.

7. SQUARE NEAR 100

Find:

$$\sqrt[4]{97^2}$$

Use:

$$\sqrt[4]{97=100-3}$$

$$\sqrt[4]{(100-3)^2 = 10000-600+9}$$

$$\sqrt[4]{=\boxed{9409}}$$

8. ANOTHER NEAR-100 EXAMPLE

$$\sqrt[4]{104^2}$$

$$\sqrt[4]{=(100+4)^2}$$

$$\sqrt[4]{=10000+800+16}$$

$$\sqrt[4]{=\boxed{10816}}$$

9. ULTRA-FASTNEAR-100 METHOD

For:

$$\sqrt{(100+x)^2}$$

$$\sqrt{=10000+200x+x^2}$$

Example:

$$\sqrt{103^2}$$

$$\sqrt{=10000+600+9}$$

$$\sqrt{=\boxed{10609}}$$

For:

$$\sqrt{(100-x)^2}$$

$$\sqrt{=10000-200x+x^2}$$

10. NUMBERS ENDING IN 5 — VERY IMPORTANT

For any number:

$$\sqrt{10n+5}$$

square:

$$\sqrt{(10n+5)^2}$$

Shortcut:

Multiply $\sqrt{n!}$ by $\sqrt{(n+1)}$, then append 25.

Example:

$$\sqrt{35^2}$$

Take:

$$\sqrt{3 \times 4 = 12}$$

append 25:

$$\sqrt{\boxed{1225}}$$

11. MORE ENDING-IN-5 EXAMPLES

$$\sqrt{(65^2)}$$

$$\sqrt{6 \times 7 = 42}$$

Append 25:

$$\sqrt{\boxed{4225}}$$

$$\sqrt{(95^2)}$$

$$\sqrt{9 \times 10 = 90}$$

Append 25:

$$\sqrt{\boxed{9025}}$$

$$\backslash(115^2)$$

Use 11 as the part before 5:

$$\backslash[11 \times 12 = 132]$$

Append 25:

$$\backslash[\boxed{13225}]$$

12. WHY THE ENDING-5 TRICK WORKS

$$\backslash[(10n+5)^2 = 100n^2 + 100n + 25]$$

$$\backslash[= 100n(n+1) + 25]$$

So the number before final two digits is:

$$\backslash[n(n+1)]$$

and final digits are:

$$\backslash[25]$$

13. PRODUCT OF EQUALLY SPACED NUMBERS

Use:

$$\backslash[(a-b)(a+b) = a^2 - b^2]$$

Example:

$$\backslash[48 \times 52]$$

These are equally spaced around 50.

$$\backslash[= (50-2)(50+2)]$$

$$\backslash[= 50^2 - 2^2]$$

$$\backslash[= 2500 - 4]$$

$$\backslash[= \boxed{2496}]$$

14. ANOTHER DIFFERENCE-OF-SQUARES EXAMPLE

$$\backslash[97 \times 103]$$

Center:

$$\backslash[100]$$

Distance:

$$\backslash[3]$$

So:

$$\backslash[100^2 - 3^2]$$

$$\backslash[= 10000 - 9]$$

$$\boxed{9991}$$

15. NEARBY NUMBERS DIFFERING BY 2

Example:

$$49 \times 51$$

Center:

$$50$$

$$= 50^2 - 1$$

$$\boxed{2499}$$

This is a classic speed trick.

16. NEARBY NUMBERS DIFFERING BY 4

$$68 \times 72$$

Center:

$$70$$

Distance:

$$2$$

$$= 70^2 - 2^2$$

$$= 4900 - 4$$

$$\boxed{4896}$$

17. SQUARE FROM PREVIOUS SQUARE

Very useful relationship:

$$(n+1)^2 = n^2 + 2n + 1$$

So if you know:

$$24^2 = 576$$

then:

$$25^2 = 576 + 49$$

$$= 625$$

because:

$$2(24) + 1 = 49$$

18. DIFFERENCE BETWEEN CONSECUTIVE SQUARES

$$(n+1)^2 - n^2 = 2n + 1$$

So consecutive squares differ by consecutive odd numbers.

Examples:

$$\sqrt{10^2=100}$$

$$\sqrt{11^2=121}$$

difference:

$$\sqrt{21}$$

Then:

$$\sqrt{12^2-11^2=23}$$

Then:

$$\sqrt{13^2-12^2=25}$$

19. MEMORY TRICK FOR BUILDING SQUARE TABLE

Start:

$$\sqrt{1}$$

Then add odd numbers:

$$\sqrt{+3,+5,+7,+9,\dots}$$

Example:

$$\sqrt{1^2=1}$$

$$\sqrt{2^2=1+3=4}$$

$$\sqrt{3^2=4+5=9}$$

$$\sqrt{4^2=9+7=16}$$

This is useful if you forget a square during an exam.

20. FAST SQUARE OF NUMBER NEAR 50

Find:

$$\sqrt{53^2}$$

$$\sqrt{=(50+3)^2}$$

$$\sqrt{=2500+300+9}$$

$$\sqrt{=\boxed{2809}}$$

21. FAST SQUARE OF NUMBER NEAR 20

$$\sqrt{23^2}$$

$$\sqrt{=(20+3)^2}$$

$$\sqrt{=400+120+9}$$

$$\boxed{529}$$

22. FAST SQUARE USING KNOWN VALUE

Suppose you know:

$$50^2=2500$$

Then:

$$51^2 = 50^2 + 50 + 51$$

because:

$$(n+1)^2 = n^2 + n + (n+1)$$

$$= 2500 + 101$$

$$\boxed{2601}$$

23. PERFECT-SQUARE UNIT-DIGIT FILTER

A perfect square can end only in:

$$\boxed{0,1,4,5,6,9}$$

So numbers ending in:

$$2,3,7,8$$

can never be perfect squares.

Example:

$$12347$$

ends in 7.

Therefore:

$$\boxed{\text{not a perfect square}}$$

No square root needed.

24. PERFECT SQUARES AND TRAILING ZEROS

A perfect square must have an **even number of trailing zeros**.

Why?

Each trailing zero corresponds to factor 10:

$$10=2\cdot 5$$

A square has even prime exponents.

Therefore trailing-zero count must be even.

Example:

$$900=30^2$$

has:

$$\sqrt{2}$$

trailing zeros.

But:

$$\sqrt{9000}$$

has 3 trailing zeros, so cannot be a perfect square.

25. PERFECT SQUARES MODULO 4

As learned earlier:

$$\boxed{\text{square mod 4 is } 0 \text{ or } 1}$$

Therefore any integer congruent to:

$$2 \text{ or } 3 \pmod{4}$$

cannot be a square.

Example:

$$2026 \bmod 4 = 2$$

Therefore:

$$\boxed{2026 \text{ is not a perfect square}}$$

26. PERFECT SQUARES MODULO 3

Squares modulo 3 are:

$$0, 1$$

Never:

$$2$$

So if:

$$N \not\equiv 2 \pmod{3}$$

then N is not a square.

27. PERFECT SQUARES MODULO 8

Odd perfect squares are always:

$$1 \pmod{8}$$

So an odd number with remainder:

$$3, 5, \text{ or } 7$$

modulo 8 cannot be a square.

28. NUMBER OF DIGITS IN A SQUARE — QUICK BOUNDS

If:

$$\sqrt[k-1]{10} \leq n < \sqrt[k]{10}$$

then:

$$10^{2k-2} \leq n^2 < 10^{2k}$$

So the square has either:

$$2k-1$$

or:

$$2k$$

digits.

Example:

Any 3-digit number has square with either:

$$5 \text{ or } 6$$

digits.

29. RECOGNIZING SQUARE ROOTS

Suppose:

$$\sqrt{1764}$$

You should estimate first.

$$40^2 = 1600$$

$$45^2 = 2025$$

So answer is between 40 and 45.

Try:

$$42^2$$

$$= (40+2)^2$$

$$= 1600 + 160 + 4$$

$$= 1764$$

Therefore:

$$\boxed{\sqrt{1764} = 42}$$

30. FAST ROOT RECOGNITION USING LAST DIGIT

Suppose a perfect square ends in 9.

Its square root must end in:

$$3 \text{ or } 7$$

because:

$$\sqrt{3^2 \text{ to } 9^2}$$

$$\sqrt{7^2 \text{ to } 9^2}$$

Similarly:

Square ends in	Root can end in
0	0
1	1 or 9
4	2 or 8
5	5
6	4 or 6
9	3 or 7

Useful for exact root identification.

31. EXAMPLE — ROOT USING RANGE + LAST DIGIT

Find:

$$\sqrt{\text{1849}}$$

Range:

$$\sqrt{40^2 = 1600}$$

$$\sqrt{50^2 = 2500}$$

So root is in 40s.

1849 ends in 9.

Root ends in 3 or 7.

Candidates:

$$\sqrt{43, 47}$$

Now:

$$\sqrt{43^2 = (40+3)^2}$$

$$\sqrt{= 1600 + 240 + 9}$$

$$\sqrt{= 1849}$$

So:

$$\sqrt{\boxed{43}}$$

32. FAST ROOT FOR 4-DIGIT PERFECT SQUARES

Method:

1. identify rough tens range
2. use last digit
3. test only 1–2 candidates

Example:

$$\sqrt{3481}$$

Range:

$$50^2=2500$$

$$60^2=3600$$

So root in 50s.

Number ends in 1.

Root ends in:

1 or 9

Candidates:

51, 59

$$59^2=(60-1)^2=3600-120+1=3481$$

Therefore:

$$\boxed{59}$$

33. CUBE UNIT-DIGIT MAP

For cube roots, the last digit is even easier because each cube-ending digit corresponds uniquely.

Root last digit	Cube last digit
0	0
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9

Notice every cube ending identifies exactly one root ending.

34. CUBE-ROOT EXAMPLE

Find:

$$\sqrt[3]{12167}$$

We know:

$$20^3=8000$$

$$30^3=27000$$

So root is in the 20s.

12167 ends in 7.

Cube ending 7 comes from root ending:

$$3$$

So likely:

$$23$$

Check:

$$23^3=12167$$

Therefore:

$$\boxed{23}$$

35. FAST CUBE OF A NUMBER

Formula:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Example:

$$12^3$$

$$=(10+2)^3$$

$$=1000+600+120+8$$

$$=\boxed{1728}$$

For small deviations from a round base, this is manageable.

36. CUBE NEAR 10

$$9^3$$

$$=(10-1)^3$$

$$=1000-300+30-1$$

$$=\boxed{729}$$

37. CUBE NEAR 20

$$\sqrt[3]{21^3}$$

$$\sqrt[3]{=(20+1)^3}$$

$$\sqrt[3]{=8000+1200+60+1}$$

$$\sqrt[3]{=\boxed{9261}}$$

38. DIFFERENCE OF CUBES

Formula:

$$\sqrt[3]{a^3-b^3}=(a-b)(a^2+ab+b^2)$$

Example:

$$\sqrt[3]{10^3-9^3}$$

$$\sqrt[3]{=(10-9)(100+90+81)}$$

$$\sqrt[3]{=\boxed{271}}$$

Indeed:

$$\sqrt[3]{1000-729=271}$$

39. SUM OF CUBES

$$\sqrt[3]{a^3+b^3}=(a+b)(a^2-ab+b^2)$$

Example:

$$\sqrt[3]{2^3+3^3}$$

$$\sqrt[3]{=(2+3)(4-6+9)}$$

$$\sqrt[3]{=5\times 7}$$

$$\sqrt[3]{=\boxed{35}}$$

40. CONSECUTIVE CUBES DIFFERENCE

$$\sqrt[3]{(n+1)^3-n^3}$$

$$\sqrt[3]{=3n^2+3n+1}$$

Example:

$$\sqrt[3]{11^3-10^3}$$

$$\sqrt[3]{=3(100)+30+1}$$

$$\sqrt[3]{=\boxed{331}}$$

So:

$$\sqrt[3]{11^3=1331}$$

41. FAST MULTIPLICATION USING SQUARES

If you know:

$$\sqrt{a^2-b^2}=(a-b)(a+b)$$

you can solve products.

Example:

$$\sqrt{76 \times 84}$$

Center:

$$\sqrt{80}$$

Distance:

$$\sqrt{4}$$

$$\sqrt{=80^2-4^2}$$

$$\sqrt{=6400-16}$$

$$\sqrt{=\boxed{6384}}$$

42. FIND PRODUCT FROM SUM AND DIFFERENCE

If:

$$\sqrt{x+y=S}$$

and:

$$\sqrt{x-y=D}$$

then:

$$\sqrt{(x+y)(x-y)=x^2-y^2}$$

This identity appears in algebra and aptitude.

43. GATE-STYLE QUESTION 1

Find:

$$\sqrt{999^2}$$

Normal method

Long multiplication.

Fast method

$$\sqrt{999=1000-1}$$

$$\sqrt{(1000-1)^2 = 1000000-2000+1}$$

$$\sqrt{=\boxed{998001}}$$

44. GATE-STYLE QUESTION 2

Find:

$$\sqrt{1003^2}$$

$$\sqrt{=(1000+3)^2}$$

$$\sqrt{=1000000+6000+9}$$

$$\sqrt{=\boxed{1006009}}$$

45. GATE-STYLE QUESTION 3

Without multiplication, calculate:

$$\sqrt{96 \times 104}$$

$$\sqrt{=(100-4)(100+4)}$$

$$\sqrt{=10000-16}$$

$$\sqrt{=\boxed{9984}}$$

46. GATE-STYLE QUESTION 4

Can:

$$\sqrt{7238}$$

be a perfect square?

Last digit:

$$\sqrt{8}$$

A perfect square cannot end in 8.

Therefore:

$$\sqrt{\boxed{\text{No}}}$$

47. PLACEMENT-STYLE QUESTION 1

Find:

$$\sqrt{85^2}$$

Ending-in-5 trick:

$$\sqrt{8 \times 9 = 72}$$

append 25:

$$\sqrt{\boxed{7225}}$$

48. PLACEMENT-STYLE QUESTION 2

Find:

$$\sqrt{47^2}$$

Use 50:

$$\sqrt{(50-3)^2}$$

$$\sqrt{=2500-300+9}$$

$$\sqrt{=\boxed{2209}}$$

49. PLACEMENT-STYLE QUESTION 3

Find:

$$\sqrt{39 \times 41}$$

$$\sqrt{=(40-1)(40+1)}$$

$$\sqrt{=1600-1}$$

$$\sqrt{=\boxed{1599}}$$

50. PLACEMENT-STYLE QUESTION 4

Find:

$$\sqrt{\sqrt{2209}}$$

Range:

$$\sqrt{40^2=1600, \text{quad} 50^2=2500}$$

Ending 9 !' root ends 3 or 7.

Candidates:

$$\sqrt{43, \sqrt{47}}$$

Since:

$$\sqrt{47^2=2209}$$

$$\sqrt{\boxed{47}}$$

51. MUST-MEMORIZE MENTAL-MATH TABLE

Squares

$$\sqrt{1^2 \text{ through } 30^2}$$

Cubes

$$\sqrt{1^3 \text{ through } 10^3}$$

Good to memorize

Squares:

$$\sqrt{31^2 \text{ through } 40^2}$$

Cubes:

$\{ 11^3 \text{ through } 20^3 \}$

52. MUST-MEMORIZE IDENTITIES

$$\boxed{(a+b)^2=a^2+2ab+b^2}$$

$$\boxed{(a-b)^2=a^2-2ab+b^2}$$

$$\boxed{(a+b)(a-b)=a^2-b^2}$$

$$\boxed{(a+b)^3=a^3+3a^2b+3ab^2+b^3}$$

$$\boxed{a^3-b^3=(a-b)(a^2+ab+b^2)}$$

$$\boxed{a^3+b^3=(a+b)(a^2-ab+b^2)}$$

53. SPEED DECISION TREE FOR SQUARING

If asked (n^2) :

Ends in 5?

Use append-25 trick.

Near 10, 50, 100, 1000?

Use:

$$(a \pm b)^2$$

Know nearby square?

Use:

$$(n+1)^2=n^2+2n+1$$

Otherwise

Use direct multiplication if small.

Do not force a complicated trick.

54. COMMON TRAPS

TRAP 1 — Ending 5 trick

For:

$$75^2$$

use:

$$7 \times 8$$

not:

$$75 \times 76$$

Then append 25.

& pTRAP 2 — Difference of squares requires equal distance

$$\sqrt[4]{48 \times 52}$$

works around 50.

But:

$$\sqrt[4]{48 \times 53}$$

is not symmetric around an integer base in the same easy way.

& pTRAP 3 — Square ending test is necessary, not sufficient

A number ending in 9 **may** be a square.

It is not automatically a square.

& pTRAP 4 — Root from last digit alone

For square roots, ending 9 gives two possibilities:

$$\sqrt[3]{\text{or } 7}$$

You still need range/check.

55. 5-MINUTE REVISION

Know instantly:

$$\sqrt[2]{25^2=625}$$

$$\sqrt[2]{30^2=900}$$

$$\sqrt[2]{40^2=1600}$$

$$\sqrt[3]{15^3=3375}$$

$$\sqrt[3]{20^3=8000}$$

Fast square:

$$\sqrt{(50 \pm x)^2}$$

Ending 5:

previous part × next integer, append 25.

Symmetric multiplication:

$$\sqrt{(a-x)(a+x)=a^2-x^2}$$

Perfect square endings:

$$\sqrt{0,1,4,5,6,9}$$

PRACTICE — LEVEL 0

Q1

Find:

$$\sqrt{35^2}$$

Q2

Find:

$$\sqrt{48^2}$$

Q3

Find:

$$\sqrt{49 \times 51}$$

Q4

Find:

$$\sqrt{13^2}$$

Q5

Find:

$$\sqrt{8^3}$$

LEVEL 1

Q6

Find:

$$\sqrt{75^2}$$

Q7

Find:

$$\sqrt{103^2}$$

Q8

Find:

$$\sqrt{96 \times 104}$$

Q9

Find:

$$\sqrt{\sqrt{1521}}$$

Q10

Find:

$$\sqrt{\sqrt[3]{13824}}$$

LEVEL 2

Q11

Find:

$$\sqrt{115^2}$$

Q12

Find:

$$\sqrt{68 \times 72}$$

Q13

Without direct square-root calculation, determine whether 9801 is a perfect square and find its root if yes.

Q14

Can 12345678 be a perfect square?

Q15

Find:

$$\sqrt{19^3}$$

using a nearby base.

LEVEL 3 — EXAM PRESSURE

Q16

Find:

$$\sqrt{997^2}$$

Q17

Find:

$$\sqrt{1004 \times 996}$$

Q18

Find:

$$\sqrt{\sqrt{3481}}$$

mentally.

Q19

Find:

$$\sqrt{\sqrt[3]{17576}}$$

Q20

If:

$$\sqrt{n^2 = 3969}$$

find $\sqrt{n}$ assuming $(n > 0)$.

ANSWERS

1

$$\backslash[3\text{times}4=12 \backslash]$$

append 25:

$$\backslash[\boxed{1225} \backslash]$$

2

$$\backslash[(50-2)^2 = 2500-200+4 = \boxed{2304} \backslash]$$

3

$$\backslash[50^2-1^2 = \boxed{2499} \backslash]$$

4

$$\backslash[\boxed{169} \backslash]$$

5

$$\backslash[\boxed{512} \backslash]$$

6

$$\backslash[7\text{times}8=56 \backslash]$$

append 25:

$$\backslash[\boxed{5625} \backslash]$$

7

$$\backslash[(100+3)^2 = 10000+600+9 = \boxed{10609} \backslash]$$

8

$$\backslash[(100-4)(100+4) = 10000-16 = \boxed{9984} \backslash]$$

9

We know:

$$\backslash[39^2=1521 \backslash]$$

Therefore:

$$\backslash[\boxed{39} \backslash]$$

10

Since:

$$\backslash[24^3=13824 \backslash]$$

$$\backslash[\boxed{24} \backslash]$$

A quick way: cube lies between $\backslash(20^3=8000\backslash)$ and $\backslash(30^3=27000\backslash)$, and ending 4 means root ends in 4.

11

$$\backslash[11\text{times}12=132 \backslash]$$

append 25:

$$\boxed{13225}$$

12

$$70^2 - 2^2 = 4900 - 4 = \boxed{4896}$$

13

9801 ends in 1, so root can end 1 or 9.

Also:

$$100^2 = 10000$$

so root is near 100.

Try:

$$99^2 = (100-1)^2 = 9801$$

Therefore:

$$\boxed{99}$$

14

Ends in:

$$8$$

Perfect square cannot end in 8.

$$\boxed{\text{No}}$$

15

$$19^3 = (20-1)^3$$

$$= 8000 - 1200 + 60 - 1$$

$$= \boxed{6859}$$

16

$$(1000-3)^2$$

$$= 1000000 - 6000 + 9$$

$$= \boxed{994009}$$

17

$$(1000+4)(1000-4)$$

$$= 1000000 - 16$$

$$= \boxed{999984}$$

18

As seen:

$$59^2 = (60-1)^2$$

$$= 3600 - 120 + 1$$

$$= 3481$$

Therefore:

$\boxed{59}$

19

Cube lies between:

$$20^3 = 8000$$

and:

$$30^3 = 27000$$

Last digit 6 means cube root ends in 6.

Only candidate in 20s:

$$26$$

Check:

$$26^2 = 676$$

$$676 \times 26 = 17576$$

Therefore:

$$\boxed{26}$$

20

$$60^2 = 3600$$

$$70^2 = 4900$$

3969 ends in 9, so candidate root ends 3 or 7.

$$63^2 = (60+3)^2$$

$$= 3600 + 360 + 9$$

$$= 3969$$

Therefore:

$$\boxed{63}$$

CHAPTER 10 STATUS

We have covered:

- squares 1–30 '
- useful squares 31–40 '
- cubes 1–10 '
- useful cubes 11–20 '
- squaring near round numbers '
- ending-in-5 square trick '
- difference-of-squares multiplication '
- consecutive-square relation '
- square-root recognition '
- cube-root unit-digit recognition '

- perfect-square filters '
- fast cube calculations '
- GATE/placement speed problems '

But **Chapter 10 is not complete yet.**

Next we will focus on:

- memorizing powers of 2, 3, 5 and 10
- rapid square/cube drills
- estimating non-perfect roots
- nearest-square/nearest-cube reasoning
- perfect square/cube count in ranges
- squares/cubes under constraints
- mixed mental-calculation speed training
- Must-Memorize Aptitude Table consolidation

Progress

Completed: 9/76 **Current:** Chapter 10 — Squares, Cubes & Mental-Math Tables **Next:** Chapter 10 Part 2 — **Power tables, root estimation & speed drills**

USER

next

ASSISTANT

CHAPTER 10 — SQUARES, CUBES & MENTAL-MATH TABLES

PART 2 — POWER TABLES, ROOT ESTIMATION, RANGE COUNTING & SPEED DRILLS

Status: 0% **PROGRESS** **GATE:** Medium **Placements:** High **Priority:** +Tier 2

This part completes the speed toolkit around:

- powers of 2, 3, 5 and 10
- nearest-square / nearest-cube estimation
- counting perfect squares and cubes in ranges
- fast root bounds
- mixed mental calculations
- Must-Memorize Aptitude Table

1. POWERS OF 2 — MUST MEMORIZE

These appear everywhere in:

- digital logic
- computer architecture

- probability
- permutations
- binary systems
- aptitude

Power	Value
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096
2^{13}	8192
2^{14}	16384
2^{15}	32768
2^{16}	65536

MUST MEMORIZE

At least up to:

$$2^{16}$$

For CS/GATE this is exceptionally useful.

2. FAST APPROXIMATION OF POWERS OF 2

Very useful:

$$2^{10} \approx 10^3$$

Therefore:

$$2^{20} \approx 10^6$$

$$\lfloor 2^{\{30\}} \approx 10^9 \rfloor$$

More exactly:

$$\lfloor 2^{\{20\}} = 1,048,576 \rfloor$$

$$\lfloor 2^{\{30\}} = 1,073,741,824 \rfloor$$

Good mental rule

$$\lfloor \boxed{2^{\{10\}} \approx 10^3} \rfloor$$

Useful for order-of-magnitude estimation.

3. POWERS OF 3 — GOOD TO MEMORIZE

Power	Value
$\lfloor 3^{\{1\}} \rfloor$	3
$\lfloor 3^{\{2\}} \rfloor$	9
$\lfloor 3^{\{3\}} \rfloor$	27
$\lfloor 3^{\{4\}} \rfloor$	81
$\lfloor 3^{\{5\}} \rfloor$	243
$\lfloor 3^{\{6\}} \rfloor$	729
$\lfloor 3^{\{7\}} \rfloor$	2187
$\lfloor 3^{\{8\}} \rfloor$	6561
$\lfloor 3^{\{9\}} \rfloor$	19683
$\lfloor 3^{\{10\}} \rfloor$	59049

MUST MEMORIZE

At least:

$$\lfloor 3^{\{1\}} \text{ through } 3^{\{6\}} \rfloor$$

GOOD TO MEMORIZE

Up to:

$$\lfloor 3^{\{10\}} \rfloor$$

4. POWERS OF 5 — USEFUL PATTERN

Power	Value
$\lfloor 5^{\{1\}} \rfloor$	5
$\lfloor 5^{\{2\}} \rfloor$	25

Power	Value
(5^3)	125
(5^4)	625
(5^5)	3125
(5^6)	15625

Useful identity:

$$\boxed{5^n = \frac{10^n}{2^n}}$$

Example:

$$5^5 = \frac{100000}{32} = 3125$$

5. POWERS OF 10

$$10^1 = 10$$

$$10^2 = 100$$

$$10^3 = 1000$$

and so on.

For negative powers:

$$10^{-1} = 0.1$$

$$10^{-2} = 0.01$$

$$10^{-3} = 0.001$$

This is fundamental for scientific notation and estimation.

6. RECOGNIZING COMMON PERFECT POWERS

Know these identities:

$$64 = 2^6 = 4^3 = 8^2$$

$$81 = 3^4 = 9^2$$

$$125 = 5^3$$

$$256 = 2^8 = 4^4 = 16^2$$

$$512 = 2^9 = 8^3$$

$$625 = 5^4 = 25^2$$

$$729 = 3^6 = 9^3 = 27^2$$

$$1024 = 2^{10} = 32^2$$

These help simplify fractional powers instantly.

7. NEAREST-SQUARE ESTIMATION

Suppose:

$$\sqrt{70}$$

Nearby squares:

$$8^2=64$$

$$9^2=81$$

Therefore:

$$\boxed{8 < \sqrt{70} < 9}$$

Since 70 is closer to 64 than 81:

$$\sqrt{70}$$

is closer to 8.

Actual approximate value:

$$8.37$$

But often the interval is enough.

8. NEAREST-SQUARE FAST APPROXIMATION

If:

$$N=a^2+x$$

with (x) small relative to (a^2) :

$$\boxed{\sqrt{a^2+x} \approx a + \frac{x}{2a}}$$

Example:

$$\sqrt{65}$$

$$= \sqrt{64+1}$$

Take $(a=8)$:

$$\approx 8 + \frac{1}{16}$$

$$= \boxed{8.0625}$$

Actual:

$$\approx 8.0623$$

Very close.

9. APPROXIMATION BELOW A PERFECT SQUARE

$$\sqrt{99}$$

Nearby:

$$10^2=100$$

So:

$$\sqrt[3]{99} \approx 10 - \frac{1}{20}$$

$$= \boxed{9.95}$$

Actual is approximately 9.9499.

10. WHEN NOT TO USE THIS APPROXIMATION

Do not use:

$$\sqrt[3]{a + \frac{x}{2a}}$$

when $|x|$ is large relative to a^2 .

Example:

$$\sqrt[3]{150}$$

using $a=10$ gives:

$$10 + \frac{50}{20} = 12.5$$

Actual:

$$12.247\dots$$

Not terrible, but not precise.

Better use bounds:

$$12^3 = 144, \quad 13^3 = 169$$

So:

$$12 < \sqrt[3]{150} < 13$$

11. FAST CUBE-ROOT BOUNDS

Example:

$$\sqrt[3]{100}$$

Nearby cubes:

$$4^3 = 64$$

$$5^3 = 125$$

Therefore:

$$\boxed{4 < \sqrt[3]{100} < 5}$$

12. CUBE-ROOT APPROXIMATION NEAR A CUBE

If:

$$N = a^3 + x$$

with small $|x|$:

$$\sqrt[3]{a^3+x} \approx a + \frac{x}{3a^2}$$

Example:

$$\sqrt[3]{126}$$

Since:

$$125=5^3$$

$$\approx 5 + \frac{1}{75}$$

$$\approx \boxed{5.013}$$

13. NUMBER OF PERFECT SQUARES UP TO (N)

Very important formula:

Number of positive perfect squares:

$$\leq N$$

is:

$$\boxed{\lfloor \sqrt{N} \rfloor}$$

Example:

How many positive perfect squares are (≤ 100) ?

$$\sqrt{100}=10$$

Therefore:

$$\boxed{10}$$

They are:

$$1^2, 2^2, \dots, 10^2$$

14. EXAMPLE

How many perfect squares are (≤ 200) ?

$$14^2=196$$

$$15^2=225>200$$

Therefore:

$$\boxed{14}$$

15. PERFECT SQUARES IN A RANGE

Count squares from (L) to (R) , inclusive.

Formula:

$$\boxed{\lfloor \sqrt{R} \rfloor - \lceil \sqrt{L} \rceil + 1}$$

provided the interval contains at least one square.

16. EXAMPLE

How many perfect squares lie between:

$[50 \text{ and } 200]$

inclusive?

First square:

$[8^2=64]$

because:

$[7^2=49<50]$

Largest:

$[14^2=196]$

So roots:

$[8,9,10,11,12,13,14]$

Count:

$[14-8+1 = \boxed{7}]$

17. FAST RANGE METHOD

Instead of memorizing the ceiling/floor formula, often easiest:

1. find smallest integer whose square is $\lceil L \rceil$
 2. find largest integer whose square is $\lfloor R \rfloor$
 3. count integers between them
-

18. NUMBER OF PERFECT CUBES UP TO $\lfloor N \rfloor$

Number of positive perfect cubes:

$\lfloor \sqrt[3]{N} \rfloor$

is:

$\lfloor \sqrt[3]{N} \rfloor$

Example:

Perfect cubes ≤ 1000 :

$[1^3, \dots, 10^3]$

So:

$\boxed{10}$

19. PERFECT CUBES IN A RANGE

Example:

How many positive cubes from 100 to 1000 inclusive?

Smallest cube:

$$\sqrt[3]{5^3=125}$$

Largest:

$$\sqrt[3]{10^3=1000}$$

Roots:

$$\sqrt[3]{5,6,7,8,9,10}$$

Count:

$$\sqrt[3]{10-5+1} = \boxed{6}$$

20. NUMBERS THAT ARE BOTH PERFECT SQUARE AND PERFECT CUBE

A number that is both must be a:

$$\sqrt[3]{\boxed{\text{perfect sixth power}}}$$

Why?

Square !' exponents multiples of 2.

Cube !' exponents multiples of 3.

So exponents must be multiples of:

$$\sqrt[3]{\text{LCM}(2,3)=6}$$

Example:

$$\sqrt[3]{64=2^6}$$

is both:

$$\sqrt[3]{8^2}$$

and:

$$\sqrt[3]{4^3}$$

21. COUNT NUMBERS BOTH SQUARE AND CUBE UP TO $\sqrt[3]{N}$

Count sixth powers:

$$\sqrt[3]{1^6, 2^6, 3^6, \dots}$$

Number is:

$$\sqrt[3]{\boxed{\lfloor N^{1/6} \rfloor}}$$

Example:

Up to:

$$\sqrt[3]{10^6}$$

sixth powers:

$$\sqrt[1^6, \dots, 10^6]{}$$

So:

$$\boxed{10}$$

22. PERFECT (k) -TH POWERS

General principle:

A positive integer is a perfect (k) -th power iff every prime-factor exponent is divisible by (k) .

Example:

$$\sqrt[2^{12} 3^6]{}$$

is:

- a square because exponents divisible by 2
- a cube because exponents divisible by 3
- a sixth power because exponents divisible by 6

23. LEAST MULTIPLIER TO MAKE PERFECT SQUARE — FAST REVIEW

Example:

$$\sqrt[N=2^5 3^2 7^3]{}$$

To make all exponents even:

- (2^5) needs one 2
- (3^2) fine
- (7^3) needs one 7

Multiply by:

$$\boxed{14}$$

24. LEAST MULTIPLIER TO MAKE PERFECT CUBE

Same:

$$\sqrt[2^5 3^2 7^3]{}$$

Need exponent multiples of 3:

- (2^5) !' need one 2 to reach 6
- (3^2) !' need one 3
- (7^3) fine

Multiply:

$$\sqrt[2 \cdot 3]{} = \boxed{6}$$

25. LEAST DIVISOR TO MAKE PERFECT SQUARE

Example:

$$\sqrt[3]{2^5 3^4 5^3}$$

To make all exponents even by division:

remove one factor 2 and one factor 5.

Divide by:

$$\sqrt[3]{\boxed{10}}$$

Remaining:

$$\sqrt[3]{2^4 3^4 5^2}$$

which is a square.

26. FAST MENTAL PRODUCT — NUMBERS AROUND 100

Calculate:

$$\sqrt[3]{96 \times 97}$$

Method 1: direct adjustment

$$\sqrt[3]{96(100-3)}$$

$$\sqrt[3]{=9600-288}$$

$$\sqrt[3]{=\boxed{9312}}$$

Method 2: base-100 trick

Deficiencies:

$$\sqrt[3]{-4, -3}$$

Cross-subtract:

$$\sqrt[3]{96-3=93}$$

Multiply deficiencies:

$$\sqrt[3]{4 \times 3=12}$$

So:

$$\sqrt[3]{\boxed{9312}}$$

27. BASE-100 MULTIPLICATION TRICK

For numbers near 100:

$$\sqrt[3]{(100+a)(100+b)}$$

Compute:

1. $\sqrt[3]{(100+a+b)}$ as left part conceptually
2. $\sqrt[3]{(ab)}$ as two-digit right part, with proper carry

Example:

$$\sqrt[3]{103 \times 104}$$

Excesses:

$$\backslash[3, \backslash 4 \backslash]$$

Cross-add:

$$\backslash[103+4=107 \backslash]$$

Product:

$$\backslash[3 \times 4=12 \backslash]$$

Answer:

$$\backslash[\boxed{10712} \backslash]$$

because base is 100, so right block uses two digits.

28. ANOTHER BASE-100 EXAMPLE

$$\backslash[98 \times 94 \backslash]$$

Deficiencies:

$$\backslash[2, \backslash 6 \backslash]$$

Cross-subtract:

$$\backslash[98-6=92 \backslash]$$

Right block:

$$\backslash[2 \times 6=12 \backslash]$$

Answer:

$$\backslash[\boxed{9212} \backslash]$$

29. IMPORTANT CARRY CASE

$$\backslash[107 \times 109 \backslash]$$

Excesses:

$$\backslash[7, \backslash 9 \backslash]$$

Cross-add:

$$\backslash[107+9=116 \backslash]$$

Right product:

$$\backslash[7 \times 9=63 \backslash]$$

Answer:

$$\backslash[\boxed{11663} \backslash]$$

30. BASE-100 RIGHT-BLOCK RULE

Because base is:

$\backslash[100 \backslash]$

the right block must contain exactly:

$\backslash[2\text{digits} \backslash]$

Example:

$\backslash[101\times 102 \backslash]$

Excesses:

$\backslash[1,2 \backslash]$

Cross-add:

$\backslash[103 \backslash]$

Product:

$\backslash[02 \backslash]$

Answer:

$\backslash[\boxed{10302} \backslash]$

not 1032.

31. NUMBERS NEAR 1000

Same logic works with base:

$\backslash[1000 \backslash]$

Right block requires:

$\backslash[3\text{digits} \backslash]$

Example:

$\backslash[1002\times 1003 \backslash]$

Excesses:

$\backslash[2,3 \backslash]$

Cross-add:

$\backslash[1005 \backslash]$

Product:

$\backslash[006 \backslash]$

Answer:

$\backslash[\boxed{1005006} \backslash]$

32. DO NOT OVERUSE BASE TRICKS

For:

$\backslash[47\times 68 \backslash]$

forcing a base-100 trick is usually not best.

Use normal decomposition:

$$\backslash[47(70-2) \backslash]$$

$$\backslash[=3290-94 \backslash]$$

$$\backslash[=\boxed{3196} \backslash]$$

Choose the method that minimizes mental load.

33. QUICK MULTIPLICATION BY 11

For a two-digit number:

$$\backslash[ab\times 11 \backslash]$$

insert sum $\backslash(a+b)$ between digits, handling carry.

Example:

$$\backslash[43\times 11 \backslash]$$

$$\backslash[4,\quad 4+3=7,\quad 3 \backslash]$$

$$\backslash[=\boxed{473} \backslash]$$

34. $\times 11$ WITH CARRY

$$\backslash[78\times 11 \backslash]$$

$$\backslash[7+8=15 \backslash]$$

Write:

- right digit 8
- middle 5
- carry 1 to 7 !' 8

Thus:

$$\backslash[\boxed{858} \backslash]$$

35. QUICK $\times 99$

$$\backslash[N\times 99 = N(100-1) \backslash]$$

Example:

$$\backslash[73\times 99 \backslash]$$

$$\backslash[7300-73 = \boxed{7227} \backslash]$$

36. QUICK $\times 101$

$$\backslash[N\times 101=N\times 100+N \backslash]$$

Example:

$$\lfloor 84 \times 101 = 8400 + 84 = \boxed{8484} \rfloor$$

37. QUICK $\times 125$

Recall:

$$\lfloor 125 = \frac{1000}{8} \rfloor$$

So:

$$\lfloor N \times 125 = N \times 1000 \div 8 \rfloor$$

Example:

$$\lfloor 48 \times 125 \rfloor$$

$$\lfloor 48 / 8 = 6 \rfloor$$

$$\lfloor 6 \times 1000 = \boxed{6000} \rfloor$$

38. QUICK $\times 25$

$$\lfloor 25 = \frac{100}{4} \rfloor$$

Example:

$$\lfloor 76 \times 25 \rfloor$$

$$\lfloor 76 / 4 = 19 \rfloor$$

$$\lfloor 19 \times 100 = \boxed{1900} \rfloor$$

39. QUICK $\times 5$ / $\times 50$

$$\lfloor N \times 5 = \frac{10N}{2} \rfloor$$

Example:

$$\lfloor 86 \times 5 = 430 \rfloor$$

And:

$$\lfloor N \times 50 = \frac{100N}{2} \rfloor$$

$$\lfloor 86 \times 50 = 4300 \rfloor$$

40. QUICK DIVISION

Divide by 5

$$\lfloor N / 5 = 2N / 10 \rfloor$$

Example:

$$\lfloor 345 / 5 = 690 / 10 = \boxed{69} \rfloor$$

Divide by 25

$$\lfloor N / 25 = 4N / 100 \rfloor$$

Example:

$$\lfloor 825/25 = 3300/100 = \boxed{33} \rfloor$$

Divide by 125

$$\lfloor N/125=8N/1000 \rfloor$$

Example:

$$\lfloor 3750/125 = 30000/1000 = \boxed{30} \rfloor$$

41. GATE-STYLE QUESTION 1

How many positive perfect squares are less than or equal to 500?

$$\lfloor 22^2=484 \rfloor$$

$$\lfloor 23^2=529 \rfloor$$

Therefore:

$$\lfloor \boxed{22} \rfloor$$

42. GATE-STYLE QUESTION 2

How many perfect cubes are less than or equal to 10000?

$$\lfloor 21^3=9261 \rfloor$$

$$\lfloor 22^3=10648>10000 \rfloor$$

Therefore:

$$\lfloor \boxed{21} \rfloor$$

43. GATE-STYLE QUESTION 3

How many numbers from 1 to 1000 inclusive are both perfect squares and perfect cubes?

They must be sixth powers.

$$\lfloor 1^6=1 \rfloor$$

$$\lfloor 2^6=64 \rfloor$$

$$\lfloor 3^6=729 \rfloor$$

$$\lfloor 4^6=4096>1000 \rfloor$$

Therefore:

$$\lfloor \boxed{3} \rfloor$$

44. PLACEMENT-STYLE QUESTION 1

Calculate mentally:

$$\lfloor 98\times 97 \rfloor$$

Deficiencies from 100:

$$\lfloor 2,3 \rfloor$$

Cross-subtract:

$$\lfloor 98-3=95 \rfloor$$

Product:

$$\lfloor 2\times 3=06 \rfloor$$

Therefore:

$$\lfloor \boxed{9506} \rfloor$$

45. PLACEMENT-STYLE QUESTION 2

Calculate:

$$\lfloor 104\times 108 \rfloor$$

Excesses:

$$\lfloor 4,8 \rfloor$$

Cross-add:

$$\lfloor 112 \rfloor$$

Product:

$$\lfloor 4\times 8=32 \rfloor$$

Answer:

$$\lfloor \boxed{11232} \rfloor$$

46. PLACEMENT-STYLE QUESTION 3

Estimate:

$$\lfloor \sqrt{226} \rfloor$$

Nearby squares:

$$\lfloor 15^2=225 \rfloor$$

So:

$$\lfloor \sqrt{226} \approx 15+\frac{1}{30} \rfloor$$

$$\lfloor \approx \boxed{15.033} \rfloor$$

47. PLACEMENT-STYLE QUESTION 4

How many perfect squares lie between 100 and 1000 inclusive?

Smallest root:

$$\lfloor 10 \rfloor$$

because:

$$\sqrt{10^2=100}$$

Largest:

$$\sqrt{31}$$

because:

$$\sqrt{31^2=961}$$

but:

$$\sqrt{32^2=1024}$$

Count:

$$\sqrt{31-10+1} = \boxed{22}$$

48. MIXED SPEED DRILL — SET A

Try mentally before reading answers.

1.

$$\sqrt{45^2}$$

2.

$$\sqrt{55^2}$$

3.

$$\sqrt{99^2}$$

4.

$$\sqrt{101^2}$$

5.

$$\sqrt{48 \times 52}$$

6.

$$\sqrt{98 \times 102}$$

7.

$$\sqrt{24^2}$$

8.

$$\sqrt{12^3}$$

9.

$$\sqrt{2025}$$

10.

$$\sqrt[3]{4913}$$

49. SPEED DRILL A — ANSWERS

1.

$$45^2: \text{quad}4 \times 5 = 20$$

append 25:

$$\boxed{2025}$$

2.

$$55^2: \text{quad}5 \times 6 = 30$$

$$\boxed{3025}$$

3.

$$99^2 = (100-1)^2$$

$$\boxed{9801}$$

4.

$$101^2 = (100+1)^2$$

$$\boxed{10201}$$

5.

$$50^2 - 2^2 = \boxed{2496}$$

6.

$$100^2 - 2^2 = \boxed{9996}$$

7.

$$\boxed{576}$$

8.

$$\boxed{1728}$$

9.

$$\boxed{45}$$

10.

$$17^3 = 4913$$

$$\boxed{17}$$

50. MIXED SPEED DRILL — SET B

1.

$$2^{12}$$

2.

$$3^7$$

3.

$$5^5$$

4.

$$\sqrt[7]{6 \times 25}$$

5.

$$\sqrt[5]{6 \times 125}$$

6.

$$\sqrt[4]{495/5}$$

7.

$$\sqrt[9]{75/25}$$

8. Number of positive squares $\sqrt[10]{10000}$

9. Number of positive cubes $\sqrt[8]{8000}$

10. Number of sixth powers $\sqrt[10]{1000000}$

51. SPEED DRILL B — ANSWERS

1.

$$\sqrt[7]{4096}$$

2.

$$\sqrt[2]{2187}$$

3.

$$\sqrt[3]{3125}$$

4.

$$\sqrt[7]{6/4 \times 100} = \sqrt[7]{19 \times 100} = \sqrt[7]{1900}$$

5.

$$\sqrt[5]{6/8 \times 1000} = \sqrt[5]{7 \times 1000} = \sqrt[5]{7000}$$

6.

$$\sqrt[4]{495 \times 2/10} = \sqrt[4]{990/10} = \sqrt[4]{99}$$

7.

$$\sqrt[9]{75 \times 4/100} = \sqrt[9]{300/100} = \sqrt[9]{3}$$

8.

$$\sqrt[10]{10000} = 100$$

$$\sqrt[8]{100}$$

9.

$$\sqrt[20]{3} = 8000$$

$$\sqrt[2]{20}$$

10.

$$\sqrt[10]{10^6} = 1000000$$

$$\sqrt[10]{10}$$

52. MUST-MEMORIZE APTITUDE TABLE — CURRENT VERSION

A. Squares

MUST MEMORIZE

$\backslash [1^2 \text{ to } 30^2]$

GOOD TO MEMORIZE

$\backslash [31^2 \text{ to } 40^2]$

B. Cubes

MUST MEMORIZE

$\backslash [1^3 \text{ to } 10^3]$

GOOD TO MEMORIZE

$\backslash [11^3 \text{ to } 20^3]$

C. Powers of 2

MUST MEMORIZE

$\backslash [2^0 \text{ to } 2^{16}]$

D. Powers of 3

MUST MEMORIZE

$\backslash [3^1 \text{ to } 3^6]$

GOOD TO MEMORIZE

$\backslash [3^7 \text{ to } 3^{10}]$

E. Powers of 5

Know comfortably:

$\backslash [5, 25, 125, 625, 3125]$

53. WHAT SHOULD BE DERIVED IN EXAM?

Do **not** try to memorize everything.

Derive:

- rare squares above 40
-

- rare cubes above 20
- unusual roots
- large powers that have an obvious common-base shortcut
- exact values where approximation is enough

Memory should reduce work, not become another burden.

54. COMMON TRAPS

& pTRAP 1 — Approximation vs exact answer

If question asks exact:

$\sqrt{72}$

answer:

$6\sqrt{2}$

not 8.49.

& pTRAP 2 — Counting squares in ranges

Check whether endpoints are included.

& pTRAP 3 — Square and cube simultaneously

Use sixth powers, not just numbers that “look like both.”

& pTRAP 4 — Base-100 block

Right side must have exactly two digits.

Example:

98×99

Deficiencies 2 and 1:

left:

97

right:

02

So:

$\boxed{9702}$

& pTRAP 5 — Memorization errors

A wrong memorized value is worse than deriving it.

If unsure, quickly derive using a nearby square/cube.

55. QUICK REVISION BOX

Squares up to $\sqrt{N}$

$$\lfloor \sqrt{N} \rfloor$$

Cubes up to $\sqrt[3]{N}$

$$\lfloor \sqrt[3]{N} \rfloor$$

Both square and cube

$$\lfloor \sqrt[6]{N} \rfloor$$

Near-square approximation

$$\sqrt{a^2+x} \approx a + \frac{x}{2a}$$

Near-cube approximation

$$\sqrt[3]{a^3+x} \approx a + \frac{x}{3a^2}$$

High-value approximation

$$2^{10} \approx 10^3$$

PRACTICE — LEVEL 0

Q1

Find:

$$2^{15}$$

Q2

Find:

$$3^6$$

Q3

How many positive perfect squares are ≤ 225 ?

Q4

How many positive perfect cubes are ≤ 729 ?

Q5

Calculate:

$$97 \times 103$$

LEVEL 1

Q6

How many perfect squares lie between 30 and 300 inclusive?

Q7

How many perfect cubes lie between 50 and 1000 inclusive?

Q8

Calculate mentally:

$$\sqrt[3]{94 \times 98}$$

Q9

Estimate:

$$\sqrt{\sqrt{257}}$$

using a nearby perfect square.

Q10

Find smallest positive integer to multiply:

$$2^3 3^5 5^2$$

to make it a perfect square.

LEVEL 2

Q11

How many integers from 1 to 100000 inclusive are both squares and cubes?

Q12

Find smallest positive multiplier that makes:

$$2^7 3^4 5^2$$

a perfect cube.

Q13

Calculate:

$$\sqrt{1007 \times 1004}$$

using a nearby base.

Q14

Find:

$$\sqrt{\sqrt{9801}}$$

mentally.

Q15

Find:

$$\sqrt[3]{24389}$$

if it is a perfect cube.

LEVEL 3 — EXAM PRESSURE

Q16

How many perfect squares lie strictly between 100 and 10000?

Q17

How many perfect cubes lie strictly between 1 and 100000?

Q18

Find the least positive number that is simultaneously a perfect square and perfect cube and greater than 100.

Q19

Without full multiplication, calculate:

$$\sqrt[3]{9998 \times 10002}$$

Q20

Estimate:

$$\sqrt[3]{3998}$$

to about two decimal places using a nearby perfect square.

ANSWERS

1

$$\boxed{32768}$$

2

$$\boxed{729}$$

3

$$\sqrt{15^2 = 225}$$

So:

$$\boxed{15}$$

4

$$\sqrt[3]{9^3 = 729}$$

So:

$$\boxed{9}$$

5

$$\sqrt{(100-3)(100+3) = 10000-9}$$

$$\boxed{=9991}$$

6

Smallest square:

$$\sqrt[4]{6^2=36}$$

Largest:

$$\sqrt[4]{17^2=289}$$

Count:

$$\sqrt[4]{17-6+1} = \boxed{12}$$

7

Smallest:

$$\sqrt[4]{4^3=64}$$

Largest:

$$\sqrt[4]{10^3=1000}$$

Count:

$$\sqrt[4]{10-4+1} = \boxed{7}$$

8

Using base 100:

deficiencies:

$$\sqrt[4]{6, \sqrt{2}}$$

Cross-subtract:

$$\sqrt[4]{94-2=92}$$

Right block:

$$\sqrt[4]{12}$$

So:

$$\sqrt[4]{\boxed{9212}}$$

9

Nearby square:

$$\sqrt[4]{16^2=256}$$

$$\sqrt[4]{\sqrt{257}} \approx 16 + \frac{1}{32}$$

$$\sqrt[4]{\approx \boxed{16.031}}$$

10

$$\sqrt[4]{2^3 3^5 5^2}$$

Need even exponents.

Multiply by:

$$\sqrt[4]{2^1 3^1}$$

$$\sqrt[4]{=\boxed{6}}$$

11

Need sixth powers.

$$\backslash[6^6=46656 \backslash]$$

$$\backslash[7^6=117649>100000 \backslash]$$

So positive sixth powers:

$$\backslash[1^6,\ldots,6^6 \backslash]$$

Count:

$$\backslash[\boxed{6} \backslash]$$

12

Need exponents multiples of 3:

$$\backslash[2^7\text{to}2^9 \backslash]$$

need:

$$\backslash[2^2 \backslash]$$

$$\backslash[3^4\text{to}3^6 \backslash]$$

need:

$$\backslash[3^2 \backslash]$$

$$\backslash[5^2\text{to}5^3 \backslash]$$

need:

$$\backslash[5 \backslash]$$

Multiplier:

$$\backslash[4\cdot 9\cdot 5 = \boxed{180} \backslash]$$

13

Base 1000, excesses:

$$\backslash[7,\backslash 4 \backslash]$$

Cross-add:

$$\backslash[1011 \backslash]$$

Right block must have 3 digits:

$$\backslash[028 \backslash]$$

So:

$$\backslash[\boxed{1011028} \backslash]$$

14

$$\backslash[99^2=(100-1)^2=9801 \backslash]$$

Therefore:

$$\backslash[\boxed{99} \backslash]$$

15

Cube lies between:

$$\backslash[20^3=8000 \backslash]$$

and:

$$\sqrt[3]{30^3=27000}$$

Ends in 9, so root must end in 9.

Candidate:

$$\sqrt[3]{29}$$

$$\sqrt[3]{29^3=24389}$$

Therefore:

$$\sqrt[3]{\boxed{29}}$$

16

Strictly between:

$$\sqrt{100=10^2}$$

and:

$$\sqrt{10000=100^2}$$

Roots allowed:

$$\sqrt{11,12,\dots,99}$$

Count:

$$\sqrt{99-11+1 = \boxed{89}}$$

17

Strictly greater than:

$$\sqrt[3]{1=1^3}$$

and less than:

$$\sqrt[3]{100000}$$

Since:

$$\sqrt[3]{46^3=97336}$$

$$\sqrt[3]{47^3=103823}$$

Allowed roots:

$$\sqrt[3]{2\text{ through }46}$$

Count:

$$\sqrt[3]{46-2+1 = \boxed{45}}$$

18

Numbers both square and cube are sixth powers.

$$\sqrt[6]{2^6=64<100}$$

$$\sqrt[6]{3^6=729}$$

Therefore:

$$\sqrt[6]{\boxed{729}}$$

19

Center:

$$\sqrt{10000}$$

Distances:

$$\sqrt{2}$$

Therefore:

$$\sqrt{(10000-2)(10000+2)}$$

$$\sqrt{=10000^2-2^2}$$

$$\sqrt{=100000000-4}$$

$$\sqrt{=\boxed{99999996}}$$

20

Nearby square:

$$\sqrt{63^2=3969}$$

Difference:

$$\sqrt{3998-3969=29}$$

Approximation:

$$\sqrt{\sqrt{3998} \approx 63 + \frac{29}{126}}$$

$$\sqrt{\approx 63 + 0.230}$$

$$\sqrt{\approx \boxed{63.23}}$$

This is sufficiently accurate for aptitude estimation.

CHAPTER 10 — FINAL CHECK

We have now completed:

- squares 1–30 '
- useful squares 31–40 '
- cubes 1–10 '
- useful cubes 11–20 '
- powers of 2/3/5/10 '
- high-value perfect powers '
- fast squaring '
- ending-in-5 trick '
- difference-of-squares multiplication '
- near-100/1000 multiplication '
- square-root recognition '
- cube-root recognition '
- root estimation '
- perfect-square/cube filters '

- counting squares/cubes in ranges '
- simultaneous square-and-cube reasoning '
- mental $\times 5/25/50/125/99/101/11$ '
- speed drills '
- Must-Memorize Aptitude Table '

Therefore:

' CHAPTER 10 COMPLETE

Progress

Completed: 10/76 **Current:** ' Chapter 10 — Squares, Cubes & Mental-Math Tables **Next:** Ø=Chapter 11 — Simplification, Approximation & Estimation

Chapter 11 is Ø=100% because it develops the calculation style used everywhere later:

- BODMAS under time pressure
- complex fraction simplification
- cancellation before multiplication
- approximation
- percentage/ratio estimation
- bounds
- order of magnitude
- rounding
- option elimination
- deciding when exact calculation is unnecessary.

USER

next

ASSISTANT

CHAPTER 11 — SIMPLIFICATION, APPROXIMATION & ESTIMATION

PART 1 — BODMAS UNDER TIME PRESSURE, CANCELLATION & SMART EXACT CALCULATION

Status: Ø=100% **PROGRESS GATE:** High **Placements:** High **Priority:** Ø=100% 1

This chapter is about choosing the **right amount of calculation**.

In aptitude, many students lose time because they calculate everything exactly even when:

- cancellation is available,
- numbers can be rearranged,
- approximation is enough,
- options are far apart,

- or the expression has a hidden shortcut.

The goal is:

Simplify first. Calculate later.

1. BODMAS / PEMDAS — CORE ORDER

The standard order is:

1. **B**rackets
2. **O**rders — powers, roots, exponents
3. **D/M** — division and multiplication, left to right
4. **A/S** — addition and subtraction, left to right

Important:

$\boxed{\text{Division does not always come before multiplication}}$

and:

$\boxed{\text{Addition does not always come before subtraction}}$

They have equal priority within their pair, so go **left to right**.

2. CLASSIC TRAP

Evaluate:

$24 \div 6 \times 2$

Do not do:

$6 \times 2 = 12$

first.

Division and multiplication have equal priority.

Go left to right:

$24 \div 6 = 4$

$4 \times 2 = 8$

Answer:

$\boxed{8}$

3. ANOTHER LEFT-TO-RIGHT TRAP

$20 - 8 + 3$

Addition and subtraction have equal priority.

Go left to right:

$20 - 8 = 12$

$$\backslash[12+3=15 \backslash]$$

Answer:

$$\backslash[\boxed{15} \backslash]$$

4. BRACKETS CHANGE EVERYTHING

Compare:

$$\backslash[20-(8+3) \backslash]$$

First bracket:

$$\backslash[8+3=11 \backslash]$$

Then:

$$\backslash[20-11=9 \backslash]$$

Answer:

$$\backslash[\boxed{9} \backslash]$$

This is different from:

$$\backslash[20-8+3=15 \backslash]$$

5. MULTIPLE BRACKET TYPES

You may see:

$$\backslash[(),\quad[],\quad\{\}\backslash]$$

Work from the innermost bracket outward.

Example:

$$\backslash[2\{3+[4(5-2)]\}\backslash]$$

First:

$$\backslash[5-2=3 \backslash]$$

Then:

$$\backslash[4\times 3=12 \backslash]$$

Then:

$$\backslash[3+12=15 \backslash]$$

Finally:

$$\backslash[2\times 15=30 \backslash]$$

$$\backslash[\boxed{30} \backslash]$$

6. POWERS BEFORE MULTIPLICATION

$$\backslash[2+3^2\times 4 \backslash]$$

First:

$$\sqrt{3^2=9}$$

Then:

$$\sqrt{9 \times 4 = 36}$$

Then:

$$\sqrt{2+36} = \boxed{38}$$

7. ROOTS ARE ALSO ORDERS

$$\sqrt{\sqrt{49}+2^3}$$

$$\sqrt{=7+8}$$

$$\sqrt{=\boxed{15}}$$

8. FRACTION BAR ACTS LIKE A BRACKET

Very important.

$$\sqrt{\frac{2+3}{4+1}}$$

means:

$$\sqrt{\frac{(2+3)}{(4+1)}}$$

So:

$$\sqrt{\frac{5}{5}}=1$$

Answer:

$$\sqrt{\boxed{1}}$$

9. COMPLEX FRACTIONS — SIMPLIFY TOP AND BOTTOM FIRST

Evaluate:

$$\sqrt{\frac{\frac{3}{4}+\frac{1}{2}}{\frac{5}{4}}}$$

Numerator:

$$\sqrt{\frac{3}{4}+\frac{1}{2} = \frac{3}{4}+\frac{2}{4} = \frac{5}{4}}$$

So:

$$\sqrt{\frac{5/4}{5/4}} = \boxed{1}$$

10. DIVISION BY A FRACTION

Recall:

$$\sqrt{\boxed{a \div \frac{b}{c} = a \times \frac{c}{b}}}$$

Example:

$$\frac{18}{3}$$

$$= 18 \times \frac{1}{3}$$

Cancel:

$$\frac{18}{3} = 6$$

$$6 \times 4 = \boxed{24}$$

11. CALCULATE AFTER CANCELLATION

Suppose:

$$\frac{48 \times 75}{25 \times 12}$$

Slow method

Multiply numerator and denominator first.

Better method

Cancel:

$$\frac{48}{12} = 4$$

$$\frac{75}{25} = 3$$

So:

$$4 \times 3 = \boxed{12}$$

This principle saves enormous time.

12. GOLDEN RULE FOR FRACTIONS

Never multiply first when cancellation is possible.

Example:

$$\frac{63 \times 44}{21 \times 11}$$

Cancel:

$$\frac{63}{21} = 3$$

$$\frac{44}{11} = 4$$

Therefore:

$$\boxed{12}$$

13. CROSS-CANCELLATION

Evaluate:

$$\frac{28}{45} \times \frac{15}{14}$$

Do not multiply:

$$\frac{28}{15}$$

first.

Cancel crosswise:

$$\frac{28}{14} = 2$$

$$\frac{15}{45} = \frac{1}{3}$$

So:

$$\boxed{\frac{2}{3}}$$

14. MULTIPLE FRACTIONS

$$\frac{18}{35} \times \frac{14}{27} \times \frac{15}{8}$$

Cancel systematically:

$$\frac{18}{27} = \frac{2}{3}$$

$$\frac{14}{35} = \frac{2}{5}$$

$$\frac{15}{3} = 5$$

Then the 5 cancels with denominator 5.

Remaining:

$$\frac{2 \times 2 \times 8}{8} = \frac{48}{8} = \boxed{\frac{12}{1}}$$

15. FACTOR BEFORE CALCULATING

Evaluate:

$$\frac{99^2 - 1}{98}$$

Use:

$$99^2 - 1^2 = (99 - 1)(99 + 1)$$

$$= 98 \times 100$$

Therefore:

$$\frac{98 \times 100}{98} = \boxed{100}$$

No squaring needed.

16. ANOTHER FACTORIZATION SHORTCUT

$$\frac{51^2 - 49^2}{2}$$

Use difference of squares:

$$(51 - 49)(51 + 49)$$

$$= 2 \times 100$$

Then:

$$\left[\frac{200}{2} = \boxed{100} \right]$$

17. TAKE COMMON FACTOR

Evaluate:

$$\left[17 \times 43 + 17 \times 57 \right]$$

Factor 17:

$$\left[17(43+57) \right]$$

$$\left[= 17 \times 100 \right]$$

$$\left[= \boxed{1700} \right]$$

This is faster than two multiplications.

18. COMMON FACTOR WITH SUBTRACTION

$$\left[83 \times 29 - 83 \times 19 \right]$$

$$\left[= 83(29-19) \right]$$

$$\left[= 83 \times 10 \right]$$

$$\left[= \boxed{830} \right]$$

19. DISTRIBUTIVE PROPERTY FOR SPEED

$$\left[a(b+c) = ab+ac \right]$$

Can be used in reverse or forward.

Example:

$$\left[47 \times 102 \right]$$

$$\left[= 47(100+2) \right]$$

$$\left[= 4700 + 94 \right]$$

$$\left[= \boxed{4794} \right]$$

20. COMPENSATION

Find:

$$\left[999 \times 37 \right]$$

Use:

$$\left[999 = 1000 - 1 \right]$$

$$\left[37(1000-1) \right]$$

$$\left[= 37000 - 37 \right]$$

$$\left[= \boxed{36963} \right]$$

21. COMPENSATION IN ADDITION

$$\backslash[498+267 \backslash]$$

Think:

$$\backslash[500+267-2 \backslash]$$

$$\backslash[=767-2 \backslash]$$

$$\backslash[=\boxed{765} \backslash]$$

22. COMPENSATION IN SUBTRACTION

$$\backslash[1002-587 \backslash]$$

Think:

$$\backslash[1000-587+2 \backslash]$$

$$\backslash[413+2 \backslash]$$

$$\backslash[=\boxed{415} \backslash]$$

23. MULTIPLY THEN ADJUST

$$\backslash[48\times 19 \backslash]$$

Use:

$$\backslash[19=20-1 \backslash]$$

$$\backslash[48\times 20-48 \backslash]$$

$$\backslash[=960-48 \backslash]$$

$$\backslash[=\boxed{912} \backslash]$$

24. DIVIDE BY FACTORING THE DIVISOR

$$\backslash[840\div 24 \backslash]$$

Since:

$$\backslash[24=6\times 4 \backslash]$$

Do:

$$\backslash[840/6=140 \backslash]$$

then:

$$\backslash[140/4=35 \backslash]$$

Answer:

$$\backslash[\boxed{35} \backslash]$$

25. SMART DIVISION BY 12

$$\backslash[936/12 \backslash]$$

Divide by 3 then 4:

$$\backslash[936/3=312 \backslash]$$

$$\backslash[312/4=78 \backslash]$$

So:

$$\backslash[\boxed{78} \backslash]$$

Or divide by 4 then 3.

Choose whichever feels easier.

26. SMART DIVISION BY 15

$$\backslash[765/15 \backslash]$$

Since:

$$\backslash[15=3\times 5 \backslash]$$

$$\backslash[765/5=153 \backslash]$$

$$\backslash[153/3=51 \backslash]$$

Answer:

$$\backslash[\boxed{51} \backslash]$$

27. SMART DIVISION BY 125

Recall:

$$\backslash[125=\frac{1000}{8} \backslash]$$

So:

$$\backslash[N/125=\frac{8N}{1000} \backslash]$$

Example:

$$\backslash[6250/125 \backslash]$$

$$\backslash[6250\times 8=50000 \backslash]$$

$$\backslash[50000/1000=50 \backslash]$$

Answer:

$$\backslash[\boxed{50} \backslash]$$

28. SIMPLIFY DECIMALS BY SCALING

Evaluate:

$$\backslash[\frac{2.4}{0.08} \backslash]$$

Multiply numerator and denominator by 100:

$$\left[\frac{240}{8} = \boxed{30} \right]$$

Memory

Move decimal points together.

29. ANOTHER DECIMAL EXAMPLE

$$\left[\frac{0.75}{0.025} \right]$$

Multiply both by 1000:

$$\left[\frac{750}{25} = \boxed{30} \right]$$

30. DECIMAL × FRACTION CONVERSION

Evaluate:

$$\left[0.125 \times 64 \right]$$

Recognize:

$$\left[0.125 = \frac{1}{8} \right]$$

So:

$$\left[64/8 = \boxed{8} \right]$$

Much faster than decimal multiplication.

31. HIGH-VALUE DECIMAL CONVERSIONS

Recognize quickly:

$$\left[0.5 = \frac{1}{2} \right]$$

$$\left[0.25 = \frac{1}{4} \right]$$

$$\left[0.75 = \frac{3}{4} \right]$$

$$\left[0.2 = \frac{1}{5} \right]$$

$$\left[0.125 = \frac{1}{8} \right]$$

$$\left[0.375 = \frac{3}{8} \right]$$

$$\left[0.625 = \frac{5}{8} \right]$$

$$\left[0.875 = \frac{7}{8} \right]$$

These simplify calculations instantly.

32. FRACTION-FRIENDLY PERCENTAGES

Similarly:

$$\backslash[50\%=\frac{1}{2} \backslash]$$

$$\backslash[25\%=\frac{1}{4} \backslash]$$

$$\backslash[20\%=\frac{1}{5} \backslash]$$

$$\backslash[12.5\%=\frac{1}{8} \backslash]$$

$$\backslash[6.25\%=\frac{1}{16} \backslash]$$

Example:

$$\backslash[12.5\%\text{ of }640 \backslash]$$

$$\backslash[=\frac{1}{8}\times 640 \backslash]$$

$$\backslash[=\boxed{80} \backslash]$$

33. BODMAS WITH PERCENTAGES

Evaluate:

$$\backslash[200+25\%\text{ of }200 \backslash]$$

“of” means multiplication:

$$\backslash[25\%\times 200 = 50 \backslash]$$

Then:

$$\backslash[200+50 = \boxed{250} \backslash]$$

34. DO NOT TREAT PERCENT AS A SEPARATE OPERATOR MYSTERY

$$\backslash[20\%\text{ of }50 \backslash]$$

means:

$$\backslash[\frac{20}{100}\times 50 \backslash]$$

$$\backslash[=10 \backslash]$$

So convert percentage to fraction/decimal as soon as useful.

35. POWERS CAN OFTEN CANCEL

Evaluate:

$$\backslash[\frac{2^8\cdot 3^4}{2^5\cdot 3^2} \backslash]$$

Subtract exponents:

$$\backslash[2^3\cdot 3^2 \backslash]$$

$$\backslash[=8\times 9 \backslash]$$

$$\backslash[=\boxed{72} \backslash]$$

Do not expand powers first.

36. ROOTS CAN CANCEL TOO

$$\sqrt{\frac{\sqrt{72}}{\sqrt{2}}}$$

Use:

$$\sqrt{\frac{72}{2}}$$

$$=\sqrt{36}$$

$$=\boxed{6}$$

37. USE IDENTITIES INSTEAD OF EXPANSION

Evaluate:

$$51^2+49^2$$

You can calculate individually, but another method:

Let:

$$51=50+1, \quad 49=50-1$$

Then:

$$(50+1)^2+(50-1)^2$$

Middle terms cancel:

$$2(50^2)+2(1^2)$$

$$=5000+2$$

$$=\boxed{5002}$$

38. GENERAL SYMMETRIC-SQUARE IDENTITY

$$(a+b)^2+(a-b)^2 = \boxed{2a^2+2b^2}$$

Useful in mental arithmetic.

39. SYMMETRIC DIFFERENCE

$$(a+b)^2-(a-b)^2$$

Difference:

$$\boxed{4ab}$$

Example:

$$53^2-47^2$$

Here:

$$a=50, \quad b=3$$

So:

$$4(50)(3) = \boxed{600}$$

40. SIMPLIFY RATIOS BEFORE USING THEM

Suppose:

$$\left[\frac{84}{126} \right]$$

HCF:

$$\left[42 \right]$$

So:

$$\left[\frac{84}{126} = \boxed{\frac{2}{3}} \right]$$

A reduced ratio is usually easier to use in later calculations.

41. CHAIN FRACTIONS — WORK LEFT TO RIGHT CAREFULLY

Evaluate:

$$\left[1 \div \frac{1}{2} \div \frac{1}{4} \right]$$

Division has equal priority, so left to right:

$$\left[1 \div \frac{1}{2} = 2 \right]$$

Then:

$$\left[2 \div \frac{1}{4} = 8 \right]$$

Answer:

$$\left[\boxed{8} \right]$$

Do not combine denominators arbitrarily.

42. NEGATIVE SIGNS WITH POWERS

Recall:

$$\left[-3^2 = -9 \right]$$

because exponent first.

But:

$$\left[(-3)^2 = 9 \right]$$

This can appear in simplification questions.

43. EXAMPLE

Evaluate:

$$\left[-2^3 + (-2)^2 \right]$$

$$\left[-2^3 = -8 \right]$$

$$\left[(-2)^2 = 4 \right]$$

Therefore:

$$\boxed{-8+4 = -4}$$

44. GATE-STYLE QUESTION 1

Evaluate:

$$\frac{72 \times 35}{21 \times 20}$$

Cancel:

$$\frac{72}{21} = \frac{24}{7}$$

$$\frac{35}{7} = 5$$

Alternative cross-cancel more cleanly:

$$\frac{35}{21} = \frac{5}{3}$$

$$\frac{72}{20} = \frac{18}{5}$$

Then:

$$\frac{18}{5} \times \frac{5}{3} = \frac{18}{3} = \boxed{6}$$

45. GATE-STYLE QUESTION 2

Evaluate:

$$\frac{101^2 - 99^2}{4}$$

Difference of squares:

$$(101 - 99)(101 + 99)$$

$$= 2 \times 200$$

$$= 400$$

Then:

$$400/4 = \boxed{100}$$

46. GATE-STYLE QUESTION 3

Evaluate:

$$\frac{0.48 \times 25}{0.12 \times 5}$$

Group:

$$\frac{0.48}{0.12} \times \frac{25}{5}$$

$$= 4 \times 5$$

$$= \boxed{20}$$

47. PLACEMENT-STYLE QUESTION 1

Evaluate:

$$\backslash[48\text{times}25+52\text{times}25 \backslash]$$

Take common factor:

$$\backslash[25(48+52) \backslash]$$

$$\backslash[=25\text{times}100 \backslash]$$

$$\backslash[=\boxed{2500} \backslash]$$

48. PLACEMENT-STYLE QUESTION 2

Evaluate:

$$\backslash[999\text{times}999 \backslash]$$

Use:

$$\backslash[(1000-1)^2 \backslash]$$

$$\backslash[=1000000-2000+1 \backslash]$$

$$\backslash[=\boxed{998001} \backslash]$$

49. PLACEMENT-STYLE QUESTION 3

Evaluate:

$$\backslash[0.625\text{times}48 \backslash]$$

Recognize:

$$\backslash[0.625=\frac{5}{8} \backslash]$$

So:

$$\backslash[48\text{times}\frac{5}{8} \backslash]$$

$$\backslash[=6\text{times}5 \backslash]$$

$$\backslash[=\boxed{30} \backslash]$$

50. PLACEMENT-STYLE QUESTION 4

Evaluate:

$$\backslash[\frac{75\%\text{ of }240}{25\%\text{ of }120} \backslash]$$

Convert percentages:

Numerator:

$$\backslash[\frac{3}{4}\text{times}240=180 \backslash]$$

Denominator:

$$\backslash[\frac{1}{4}\text{times}120=30 \backslash]$$

Ratio:

$$\lfloor 180/30 = \boxed{6} \rfloor$$

51. THE 5-STEP EXACT-CALCULATIONWORKFLOW

When you see an ugly expression:

Step 1 — Check brackets/BODMAS

Avoid structural mistakes.

Step 2 — Convert friendly forms

Examples:

$$\lfloor 0.125\text{\%to}\frac{1}{8} \rfloor$$

$$\lfloor 25\text{\%to}\frac{1}{4} \rfloor$$

Step 3 — Factor

Look for:

- common factors
- identities
- powers
- difference of squares

Step 4 — Cancel

Before multiplication.

Step 5 — Calculate

Only what remains.

52. EXAMPLE USING THE FULL WORKFLOW

Evaluate:

$$\lfloor \frac{12.5\text{\%}\times 480\times 14}{21} \rfloor$$

Convert:

$$\lfloor 12.5\text{\%}=\frac{1}{8} \rfloor$$

So:

$$\lfloor \frac{1}{8}\times 480\times \frac{14}{21} \rfloor$$

Simplify:

$$\lfloor 480/8=60 \rfloor$$

$$\lfloor 14/21=2/3 \rfloor$$

Then:

$$\lfloor 60\times \frac{2}{3} = 40 \rfloor$$

Answer:

$$\lfloor \boxed{40} \rfloor$$

53. DO NOT CHASE DECIMALS WHEN FRACTIONS ARE EASIER

Example:

$\backslash[33.33\%\text{ of }300 \backslash]$

If it is intended as exact:

$\backslash[33\frac{1}{3}\%=\frac{1}{3} \backslash]$

then answer:

$\backslash[100 \backslash]$

But **33.33%** literally is:

$\backslash[0.3333 \backslash]$

giving:

$\backslash[99.99 \backslash]$

Important

Do not silently treat an approximate percentage as an exact recurring value unless the context clearly intends $\backslash(33\frac{1}{3}\%\backslash)$.

54. EXACT VS APPROXIMATE DATA

These are not identical:

$\backslash[\frac{1}{3} \backslash]$

$\backslash[0.333 \backslash]$

$\backslash[33.3\% \backslash]$

$\backslash[33\frac{1}{3}\% \backslash]$

Only:

$\backslash[\frac{1}{3}=0.\overline{3}=33\frac{1}{3}\% \backslash]$

exactly.

This matters in numerical aptitude.

55. COMMON TRAPS

& pTRAP 1 — Multiply before canceling

Usually slower and increases arithmetic errors.

& pTRAP 2 — Wrong BODMAS

Multiplication and division are same priority.

Addition and subtraction are same priority.

Use left-to-right.

& pTRAP 3 — Fraction bar

Treat numerator and denominator as grouped expressions.

& pTRAP 4 — Approximate decimal treated as exact fraction

$\backslash[0.33\backslashneq\backslashfrac{1}{3} \backslash]$

exactly.

& pTRAP 5 — Negative sign and exponent

$\backslash[-2^2=-4 \backslash]$

$\backslash[(-2)^2=4 \backslash]$

& pTRAP 6 — Expanding before factoring

For:

$\backslash[a^2-b^2 \backslash]$

factor first.

56. MUST-MEMORIZE SPEED CONVERSIONS

$\backslash[0.5=\backslashfrac{1}{2} \backslash]$

$\backslash[0.25=\backslashfrac{1}{4} \backslash]$

$\backslash[0.2=\backslashfrac{1}{5} \backslash]$

$\backslash[0.125=\backslashfrac{1}{8} \backslash]$

$\backslash[0.625=\backslashfrac{5}{8} \backslash]$

$\backslash[12.5\%=\backslashfrac{1}{8} \backslash]$

$\backslash[25\%=\backslashfrac{1}{4} \backslash]$

$\backslash[75\%=\backslashfrac{3}{4} \backslash]$

$\backslash[6.25\%=\backslashfrac{1}{16} \backslash]$

These reduce arithmetic dramatically.

57. 5-MINUTE REVISION

Before calculating:

Can I cancel?

Can I factor?

Can I convert decimal/% into a simple fraction?

Can I use a round-number compensation?

Can I rearrange terms?

Exact arithmetic should be the **last** step, not the first.

PRACTICE — LEVEL 0

Q1

Evaluate:

$$\lfloor 18 \div 3 \times 2 \rfloor$$

Q2

Evaluate:

$$\lfloor \frac{24 \times 15}{6 \times 10} \rfloor$$

Q3

Evaluate:

$$\lfloor 0.25 \times 80 \rfloor$$

Q4

Evaluate:

$$\lfloor 25\% \text{ of } 360 \rfloor$$

Q5

Evaluate:

$$\lfloor 51^2 - 49^2 \rfloor$$

LEVEL 1

Q6

Evaluate:

$$\lfloor \frac{36}{55} \times \frac{22}{18} \rfloor$$

Q7

Evaluate:

$$\lfloor \frac{0.72}{0.09} \rfloor$$

Q8

Evaluate:

$$\sqrt[125]{64}$$

Q9

Evaluate:

$$\sqrt[997]{23}$$

Q10

Evaluate:

$$\sqrt[\frac{12.5\%\text{ of }640}{20}]{} \]$$

LEVEL 2

Q11

Evaluate:

$$\sqrt[\frac{84 \times 45 \times 22}{33 \times 21 \times 20}]{} \]$$

Q12

Evaluate:

$$\sqrt[\frac{75^2 - 25^2}{50}]{} \]$$

Q13

Evaluate:

$$\sqrt[\frac{0.375 \times 96}{1.5}]{} \]$$

Q14

Evaluate:

$$\sqrt{(48+52) \times (75-25)}$$

Q15

Evaluate:

$$\sqrt[\frac{2^9 \cdot 3^5}{2^6 \cdot 3^3}]{} \]$$

LEVEL 3 — EXAM PRESSURE

Q16

Evaluate:

$$\sqrt[\frac{99^2 - 1}{49}]{} \]$$

Q17

Evaluate:

$$\sqrt[\frac{16.67\%\text{ of }600}{25}]{} \]$$

assuming 16.67% is used as the stated decimal percentage, not as exact $(16\frac{23}{100})$.

Q18

Evaluate:

$$\left[\frac{999^2 - 998^2}{1} \right]$$

without squaring.

Q19

Evaluate:

$$\left[\frac{2.4 \times 1.25 \times 8}{0.5} \right]$$

Q20

Evaluate:

$$\left[\frac{101^2 + 99^2 - 2(101)(99)}{4} \right]$$

ANSWERS**1**

Left to right:

$$\left[18/3=6 \right]$$

$$\left[6 \times 2 = \boxed{12} \right]$$

2

$$\left[24/6=4 \right]$$

$$\left[15/10=3/2 \right]$$

$$\left[4 \times \frac{3}{2} = \boxed{6} \right]$$

3

$$\left[0.25 = \frac{1}{4} \right]$$

$$\left[80/4 = \boxed{20} \right]$$

4

$$\left[\frac{1}{4} \times 360 = \boxed{90} \right]$$

5

$$\left[(51-49)(51+49) \right]$$

$$\left[=2 \times 100 = \boxed{200} \right]$$

6

$$\left[\frac{36}{55} \times \frac{22}{18} \right]$$

$$\left[36/18=2 \right]$$

$$\left[22/55=2/5 \right]$$

So:

$$\left[2 \times \frac{2}{5} = \boxed{\frac{4}{5}} \right]$$

7

Multiply both by 100:

$$\backslash[72/9 = \boxed{8} \backslash]$$

8

$$\backslash[125=\frac{1000}{8} \backslash]$$

$$\backslash[64/8=8 \backslash]$$

$$\backslash[8\times 1000 = \boxed{8000} \backslash]$$

9

$$\backslash[997=1000-3 \backslash]$$

$$\backslash[23(1000-3) \backslash]$$

$$\backslash[=23000-69 \backslash]$$

$$\backslash[=\boxed{22931} \backslash]$$

10

$$\backslash[12.5\%=\frac{1}{8} \backslash]$$

$$\backslash[640/8=80 \backslash]$$

$$\backslash[80/20 = \boxed{4} \backslash]$$

11

$$\backslash[\frac{84\times 45\times 22}{33\times 21\times 20} \backslash]$$

Simplify:

$$\backslash[84/21=4 \backslash]$$

$$\backslash[22/33=2/3 \backslash]$$

$$\backslash[45/20=9/4 \backslash]$$

Thus:

$$\backslash[4\times \frac{23}{33}\times \frac{9}{4} \backslash]$$

Cancel 4:

$$\backslash[\frac{23}{33}\times 9 = \boxed{6} \backslash]$$

12

Difference of squares:

$$\backslash[75^2-25^2 = (75-25)(75+25) \backslash]$$

$$\backslash[=50\times 100 \backslash]$$

Divide by 50:

$$\backslash[\boxed{100} \backslash]$$

13

$$\backslash[0.375=\frac{3}{8} \backslash]$$

$$\backslash[\frac{3}{8}\times 96=36 \backslash]$$

Then:

$$\boxed{36/1.5 = 24}$$

14

$$100 \times 50 = \boxed{5000}$$

15

$$2^{9-6} 3^{5-3} = 2^3 3^2$$

$$= 8 \times 9 = \boxed{72}$$

16

$$99^2 - 1^2 = 98 \times 100$$

Then:

$$\frac{9800}{49}$$

Since:

$$98/49 = 2$$

$$= 2 \times 100 = \boxed{200}$$

17

Use the stated decimal percentage:

$$16.67\% = 0.1667$$

Then:

$$0.1667 \times 600 = 100.02$$

Now:

$$100.02/25 = \boxed{4.0008}$$

If the question instead meant exact:

$$16\frac{23}{100}\% = \frac{16}{6.25}$$

the answer would be exactly 4.

This is why wording precision matters.

18

$$999^2 - 998^2$$

$$= (999 - 998)(999 + 998)$$

$$= 1 \times 1997$$

$$= \boxed{1997}$$

19

First:

$$1.25 \times 8 = 10$$

So:

$$2.4 \times 10 = 24$$

Then:

$$\sqrt[2]{24/0.5}=48$$

$$\boxed{48}$$

20

Recognize:

$$a^2+b^2-2ab=(a-b)^2$$

So:

$$(101-99)^2 = 2^2 = 4$$

Then:

$$4/4 = \boxed{1}$$

CHAPTER 11 STATUS

We have covered:

- BODMAS under time pressure '
- left-to-right priority traps '
- fraction-bar grouping '
- complex fractions '
- cancellation before multiplication '
- cross-cancellation '
- common-factor extraction '
- algebraic identities '
- compensation '
- decimal scaling '
- fraction-friendly decimals '
- percentage-friendly fractions '
- power/root simplification '
- exact-vs-approximate data traps '
- GATE/placement speed problems '

But **Chapter 11 is not complete yet.**

Next comes the actual approximation/estimation section:

- rounding rules
- significant digits
- upper/lower bounds
- approximation of products and quotients
- order of magnitude
- percentage error
- sanity checks
- back-of-the-envelope estimation
- option elimination

- deciding exact vs approximate calculation

Progress

Completed: 10/76 **Current:** Chapter 11 — Simplification, Approximation & Estimation **Next:** Chapter 11 Part 2 — Rounding, bounds, estimation & option elimination

USER

next

ASSISTANT

CHAPTER 11 — SIMPLIFICATION, APPROXIMATION & ESTIMATION

PART 2 — ROUNDING, BOUNDS, ERROR, ORDER OF MAGNITUDE & OPTION ELIMINATION

Status: $\emptyset$ **IN PROGRESS** **GATE:** High **Placements:** High **Priority:** $\emptyset$ **Year:** 1

This part teaches you when **exact calculation is unnecessary**.

The exam skill is:

Estimate enough to identify the answer confidently, but do not approximate so aggressively that you change the answer.

1. ROUNDING TO NEAREST INTEGER

Basic rule:

- decimal part $\setminus(<0.5\setminus)$!' round down
- decimal part $\setminus(\geq 0.5\setminus)$!' round up

Examples:

$\setminus[7.3\setminus\text{to}7 \setminus]$

$\setminus[7.8\setminus\text{to}8 \setminus]$

$\setminus[7.5\setminus\text{to}8 \setminus]$

2. ROUNDING TO DECIMAL PLACES

Round:

$\setminus[3.746 \setminus]$

to 2 decimal places.

Keep:

$\setminus[3.74 \setminus]$

Look at next digit:

$\lfloor 6 \rfloor$

Since $\lfloor 6 \geq 5 \rfloor$, increase 4 to 5.

Answer:

$\lfloor \boxed{3.75} \rfloor$

3. ROUNDING TO 1 DECIMAL PLACE

$\lfloor 18.263 \rfloor$

Keep:

$\lfloor 18.2 \rfloor$

Next digit is 6.

So:

$\lfloor \boxed{18.3} \rfloor$

4. ROUNDING TO TENS, HUNDREDS, THOUSANDS

Nearest ten

$\lfloor 347 \text{ to } 350 \rfloor$

Nearest hundred

$\lfloor 347 \text{ to } 300 \rfloor$

because 47 is less than 50.

Nearest thousand

$\lfloor 6480 \text{ to } 6000 \rfloor$

since it is closer to 6000 than 7000.

5. SIGNIFICANT FIGURES — BASIC IDEA

Significant figures measure useful precision.

Examples:

$\lfloor 3472 \rfloor$

to 2 significant figures:

Keep:

$\lfloor 34 \rfloor$

next digit is 7, so:

$\lfloor 35 \rfloor$

Restore place value:

$\boxed{3500}$

6. DECIMAL SIGNIFICANT FIGURES

Round:

0.004786

to 2 significant figures.

Leading zeros are not significant.

First significant digits:

$4, 7$

Next digit:

8

So 47 rounds to 48.

Answer:

$\boxed{0.0048}$

7. DECIMAL PLACES VS SIGNIFICANT FIGURES

They are different.

For:

0.004786

2 decimal places would be:

$\boxed{0.00}$

But 2 significant figures:

$\boxed{0.0048}$

Always read the question carefully.

8. APPROXIMATION OF SUMS

Estimate:

$397+603+198$

Round:

$397 \approx 400$

$603 \approx 600$

$198 \approx 200$

So:

$400+600+200 = \boxed{1200}$

Exact value is:

$\sqrt{1198}$

Very close.

9. APPROXIMATION OF DIFFERENCES

Estimate:

$\sqrt{1003-498}$

Use:

$\sqrt{1000-500} = \boxed{500}$

Exact:

$\sqrt{505}$

Good enough if options are far apart.

10. APPROXIMATION OF PRODUCTS

Estimate:

$\sqrt{49.8 \times 20.1}$

Use:

$\sqrt{50 \times 20} = \boxed{1000}$

Actual answer is close to 1000.

This is faster than exact multiplication.

11. APPROXIMATION OF QUOTIENTS

Estimate:

$\sqrt{998/49.9}$

Use:

$\sqrt{1000/50} = \boxed{20}$

This is an excellent approximation.

12. FRIENDLY-NUMBER APPROXIMATION

Sometimes nearest integer is not best.

Example:

$\sqrt{198/6.1}$

Instead of:

$\sqrt{200/6}$

you might use:

$$\sqrt[3]{198/6} \approx 3.3$$

because 198 divides nicely by 6.

Choose numbers that simplify the arithmetic.

13. ESTIMATE BEFORE EXACT CALCULATION

Suppose options are:

A. 8 B. 19 C. 41 D. 92

Expression:

$$\sqrt[3]{998/49.9}$$

Estimate:

$$\sqrt[3]{1000/50} = 20$$

Only option near 20:

$$\boxed{\text{B}}$$

No exact calculation needed.

14. OPTION-ELIMINATION METHOD

Before solving exactly, ask:

What rough range must the answer lie in?

Example:

$$\sqrt[3]{\frac{203 \times 49}{101}}$$

Approximate:

$$\sqrt[3]{\frac{200 \times 50}{100}} = 10$$

If options are:

- 20
- 50
- 99
- 250

choose:

$$\boxed{99}$$

Then exact work may be unnecessary.

15. ROUNDING ERROR

If true value is $\sqrt{x}$ and approximate value is $\sqrt{a}$:

$$\boxed{\text{absolute error} = |x - a|}$$

Example:

True:

$$10.2$$

Approx:

$$10$$

Absolute error:

$$10.2 - 10$$

$$0.2$$

$$\boxed{0.2}$$

16. RELATIVE ERROR

$$\boxed{\text{relative error} = \frac{\text{absolute error}}{\text{true value}}}$$

Example:

True value:

$$100$$

Approximation:

$$98$$

Absolute error:

$$2$$

Relative error:

$$\frac{2}{100} = 0.02$$

17. PERCENTAGE ERROR

$$\boxed{\% \text{ error} = \frac{|\text{approx} - \text{true}|}{\text{true}} \times 100}$$

Example:

True:

$$80$$

Approx:

$$84$$

Error:

$$4$$

Percentage error:

$$\frac{4}{80} \times 100 = \boxed{5\%}$$

18. IMPORTANT DENOMINATOR TRAP

Percentage error uses **true/accepted value** in denominator.

Not approximate value.

So:

$$\boxed{\% \text{ error} = \frac{\text{error}}{\text{true value}} \times 100}$$

unless the question defines another convention.

19. UPPER AND LOWER BOUNDS — BASIC IDEA

If a number is rounded to nearest integer as:

$$\boxed{8}$$

then actual value lies in:

$$\boxed{7.5 \leq x < 8.5}$$

Why?

Everything from 7.5 up to but not including 8.5 rounds to 8 under ordinary half-up reasoning.

20. NEAREST TEN BOUNDS

If a number rounds to:

$$\boxed{350}$$

to the nearest 10, then:

$$\boxed{345 \leq x < 355}$$

21. NEAREST HUNDRED BOUNDS

If:

$$\boxed{x \approx 1200}$$

to nearest hundred:

$$\boxed{1150 \leq x < 1250}$$

22. DECIMAL-PLACE BOUNDS

If:

$$\boxed{x = 3.7}$$

correct to 1 decimal place, then:

$$\boxed{3.65 \leq x < 3.75}$$

Half-unit in the next place:

$$\lfloor 0.05 \rfloor$$

23. GENERAL ROUNDING-BOUND RULE

If rounded to nearest unit $\lfloor u \rfloor$:

$$\lfloor \boxed{\text{true value}} \rfloor \text{ in } \left[a - \frac{u}{2}, a + \frac{u}{2} \right)$$

Example:

Nearest 0.01:

$$\lfloor u = 0.01 \rfloor$$

Half-width:

$$\lfloor 0.005 \rfloor$$

24. BOUNDS IN PRODUCTS

Suppose:

$$\lfloor x \rfloor \text{ in } [9.5, 10.5)$$

and:

$$\lfloor y \rfloor \text{ in } [19.5, 20.5)$$

with positive numbers.

Then product lies approximately between:

$$\lfloor 9.5 \times 19.5 \rfloor$$

and:

$$\lfloor 10.5 \times 20.5 \rfloor$$

This is useful when a question asks maximum/minimum possible values.

25. IMPORTANT BOUND RULE FOR POSITIVE NUMBERS

If all quantities are positive:

- minimum product !' multiply lower bounds
- maximum product !' multiply upper bounds

Similarly for sums:

- minimum sum !' add lower bounds
 - maximum sum !' add upper bounds
-

26. BOUNDS IN DIVISION

For positive:

$$\left[\frac{x}{y} \right]$$

minimum ratio occurs roughly with:

$$\left[\frac{x_{\min}}{y_{\max}} \right]$$

maximum ratio:

$$\left[\frac{x_{\max}}{y_{\min}} \right]$$

Memory

To make a fraction small: numerator down, denominator up.

To make it large: numerator up, denominator down.

27. ORDER OF MAGNITUDE

Order of magnitude tells the scale of a number.

Examples:

$$\left[700 \right]$$

is around:

$$\left[10^3 \right]$$

$$\left[0.003 \right]$$

is around:

$$\left[10^{-3} \right]$$

For rough calculations, powers of 10 help detect absurd answers.

28. SCIENTIFIC-NOTATION ESTIMATION

Estimate:

$$\left[(3.2 \times 10^5)(2.1 \times 10^3) \right]$$

Multiply coefficients:

$$\left[3.2 \times 2.1 \approx 6.7 \right]$$

Powers:

$$\left[10^8 \right]$$

So:

$$\left[\boxed{\approx 6.7 \times 10^8} \right]$$

29. ORDER-OF-MAGNITUDE SANITY CHECK

Suppose:

$$\sqrt[5]{5000 \times 300}$$

Expected scale:

$$\sqrt[5]{5 \times 10^3 \times 3 \times 10^2}$$

$$\sqrt[5]{=15 \times 10^5}$$

$$\sqrt[5]{=1.5 \times 10^6}$$

So an answer like:

$$\sqrt[5]{15000}$$

is obviously wrong.

30. SANITY CHECK — SIGN

If you multiply:

positive $\times$ negative:

answer must be negative.

If your calculation gives positive, reject it immediately.

31. SANITY CHECK — MAGNITUDE

$$\sqrt[5]{49 \times 51}$$

should be near:

$$\sqrt[5]{50 \times 50 = 2500}$$

So any answer like 250 or 25000 is impossible.

32. SANITY CHECK — FRACTIONS

If:

$$\sqrt[5]{0 < a < 1}$$

and:

$$\sqrt[5]{0 < b < 1}$$

then:

$$\sqrt[5]{ab < a}$$

and:

$$\sqrt[5]{ab < b}$$

Example:

$$\sqrt[5]{0.4 \times 0.3 = 0.12}$$

If you get 1.2, decimal placement is wrong.

33. SANITY CHECK — DIVISION BY A FRACTION

If dividing a positive number by a fraction less than 1:

$$\backslash \ N \div \frac{1}{2} \backslash$$

the result should become larger.

Example:

$$\backslash \ 20 \div \frac{1}{2} = 40 \backslash$$

If your answer is less than 20, something is wrong.

34. SANITY CHECK — PERCENTAGE

If asked:

$$\backslash \ 15\% \text{ of } 200 \backslash$$

answer must be less than 200.

If you get 300, reject instantly.

35. APPROXIMATION WITH PERCENTAGES

Estimate:

$$\backslash \ 19.8\% \text{ of } 502 \backslash$$

Use:

$$\backslash \ 20\% \text{ of } 500 \backslash$$

$$\backslash \ = 100 \backslash$$

So exact answer should be near:

$$\backslash \ \boxed{100} \backslash$$

36. APPROXIMATION WITH FRACTIONS

Estimate:

$$\backslash \ \frac{49}{101} \backslash$$

Think:

$$\backslash \ \frac{50}{100} = \boxed{0.5} \backslash$$

Actual is slightly below 0.5.

This directional information is useful.

37. DIRECTIONAL ESTIMATION

For:

$$\backslash \ \frac{49}{101} \backslash$$

numerator is slightly below 50 and denominator slightly above 100.

So value is **definitely below**:

$$\sqrt{0.5}$$

This can eliminate options like:

$$\sqrt{0.52}, \sqrt{0.55}$$

without exact division.

38. APPROXIMATION WITH SQUARE ROOTS

Estimate:

$$\sqrt{63}$$

Nearby:

$$8^2 = 64$$

Therefore:

$$\sqrt{63}$$

is slightly less than 8.

So if options are:

- 6.2
- 7.94
- 8.5
- 9.1

choose:

$$\boxed{7.94}$$

39. APPROXIMATION OF RECIPROCAL

If:

$$x$$

is close to 1:

$$\frac{1}{1+x} \approx 1-x$$

for small x .

Example:

$$\frac{1}{1.02}$$

Approx:

$$1 - 0.02 = 0.98$$

Actual:

$$0.98039...$$

SHORTCUT ONLY

Good for estimation, not exact work.

40. APPROXIMATE PRODUCT OF NEAR-1 FACTORS

$$(1+a)(1+b) = 1+a+b+ab$$

If a, b are very small, ab is tiny.

So:

$$(1+a)(1+b) \approx 1+a+b$$

Example:

$$1.02 \times 1.03$$

Approx:

$$1+0.02+0.03 = 1.05$$

Exact:

$$1.0506$$

41. APPROXIMATE PERCENTAGE CHANGE

If a value increases by 2% and then 3%, approximate total increase:

$$2\% + 3\% = 5\%$$

Exact:

$$1.02 \times 1.03 = 1.0506$$

so:

$$5.06\%$$

For small percentages, addition is a good estimate.

Full successive-percentage theory comes in Chapter 15.

42. WHEN APPROXIMATION IS SAFE

Approximation is safe when:

- answer choices are widely separated,
 - question explicitly says approximate,
 - only order/range is needed,
 - exact calculation is disproportionately long,
 - estimation clearly isolates one option.
-

43. WHEN APPROXIMATION IS DANGEROUS

Avoid rough approximation when:

- options are very close,
- cancellation between large terms occurs,
- small error can flip ranking,
- question asks exact NAT answer,
- percentage difference is tiny.

Example:

Compare:

$$\sqrt[1]{1.001^{100}}$$

and:

$$\sqrt[1]{1.002^{50}}$$

Crude rounding both bases to 1 destroys the problem.

44. CANCELLATION CAN AMPLIFY ERROR

Consider:

$$\sqrt[1]{1001-999}$$

If you approximate both as:

$$\sqrt[1]{1000-1000}$$

you get:

$$\sqrt[1]{0}$$

but exact answer is:

$$\sqrt[1]{2}$$

So approximation is bad when subtracting nearly equal numbers.

Memory

Near-cancellation !' keep more precision.

45. RATIO ESTIMATION

Estimate:

$$\sqrt[1]{\frac{398}{202}}$$

Use:

$$\sqrt[1]{\frac{400}{200}=2}$$

Actual answer slightly below 2 because:

- numerator lower,
- denominator higher.

So:

$$\sqrt[1]{\boxed{\text{slightly less than } 2}}$$

46. PERCENT ESTIMATION

What percent is:

$$\sqrt[1]{198}$$

of:

$$\sqrt[1]{402?}$$

Approx:

$$\sqrt[1]{200/400=1/2}$$

So:

$$\sqrt[1]{\boxed{\approx 50\%}}$$

Actual is slightly less than 50%.

47. BACK-SOLVING FROM OPTIONS

Suppose:

$$\sqrt[1]{x \times 49 \approx 2450}$$

Options:

- 5
- 25
- 50
- 500

Since:

$$\sqrt[1]{49 \approx 50}$$

$$\sqrt[1]{2450/50 \approx 49}$$

Closest:

$$\sqrt[1]{\boxed{50}}$$

No formal algebra necessary.

48. OPTION CHECKING CAN BE FASTER THAN SOLVING

Suppose:

$$\sqrt[1]{\frac{x}{12}=8}$$

Options:

10, 24, 96, 120.

Multiply denominator:

$$\sqrt[1]{12 \times 8 = 96}$$

Answer:

$\boxed{96}$

In more complex equations, substituting options may beat symbolic solving.

49. ESTIMATION WITH DATA INTERPRETATION

Suppose a bar is around:

$\boxed{497}$

and another around:

$\boxed{251}$

Ratio is approximately:

$\boxed{500:250 = 2:1}$

So if options are:

- 1.2
- 1.5
- 2.0
- 3.5

choose:

$\boxed{2.0}$

This will be essential in DI.

50. GATE-STYLE QUESTION 1

Estimate:

$\boxed{\frac{1002}{33.4}}$

Use:

$\boxed{1000/33.3 \approx 30}$

So:

$\boxed{\approx 30}$

In fact it is exactly 30 because:

$\boxed{33.4 \times 30 = 1002}$

51. GATE-STYLE QUESTION 2

Estimate:

$\boxed{\frac{198 \times 51}{101}}$

Use:

$\boxed{\frac{200 \times 50}{100} = 100}$

So:

$$\boxed{\approx 100\%}$$

52. GATE-STYLE QUESTION 3

A value is measured as 49 instead of true value 50.

Percentage error:

$$\frac{|49-50|}{50} \times 100$$

$$= 2\%$$

Answer:

$$\boxed{2\%}$$

53. GATE-STYLE QUESTION 4

A length is given as 8.2 cm correct to nearest 0.1 cm.

Bounds:

$$8.2 \pm 0.05$$

So:

$$\boxed{8.15 \leq L < 8.25}$$

54. PLACEMENT-STYLE QUESTION 1

Estimate:

$$49.7 \times 19.9$$

$$\approx 50 \times 20$$

$$= \boxed{1000}$$

55. PLACEMENT-STYLE QUESTION 2

Estimate:

$$603/19.8$$

Use:

$$600/20 = 30$$

So:

$$\boxed{\approx 30}$$

56. PLACEMENT-STYLE QUESTION 3

Estimate:

$\left[29.8\% \text{ of } 399 \right]$

Use:

$\left[30\% \text{ of } 400 \right]$

$\left[=120 \right]$

So:

$\left[\boxed{\approx 120} \right]$

57. PLACEMENT-STYLE QUESTION 4

True value:

$\left[250 \right]$

Measured value:

$\left[245 \right]$

Percentage error:

$\left[\frac{5}{250} \times 100 = \boxed{2\%} \right]$

58. FAST ERROR APPROXIMATION FOR PRODUCTS — OPTIONAL

If:

$\left[z=xy \right]$

and (x,y) have small percentage errors, approximate percentage error in (z) by:

$\left[\boxed{\% \Delta z \approx \% \Delta x + \% \Delta y} \right]$

Example:

- (x) error " $\pm 2\%$ "
- (y) error " $\pm 3\%$ "

Product error approximately:

$\left[\boxed{5\%} \right]$

This is a useful engineering/GATE approximation.

59. FAST ERROR APPROXIMATION FOR QUOTIENTS

For:

$\left[z = \frac{x}{y} \right]$

approximate maximum percentage error:

$\left[\boxed{\% \Delta z \approx \% \Delta x + \% \Delta y} \right]$

because denominator uncertainty also affects the ratio.

This is a first-order approximation.

60. ERROR IN POWERS — OPTIONAL

If:

$$y = x^n$$

then small relative errors approximately satisfy:

$$\boxed{\frac{\Delta y}{y} \approx n \frac{\Delta x}{x}}$$

So if x has 1% error and:

$$y = x^3$$

then y has roughly:

$$\boxed{3\%}$$

relative error.

Useful but not a core placement topic.

61. THE EXAM ESTIMATION WORKFLOW

When you see a numerical expression:

Step 1

Estimate the scale.

Step 2

Check whether options are far apart.

Step 3

Round to friendly values.

Step 4

Preserve direction when useful:

- above/below
- larger/smaller

Step 5

Calculate exactly only if needed.

62. NORMAL METHOD VS FAST METHOD

Find approximate:

$$\frac{497 \times 201}{99}$$

Normal method

Large multiplication and division.

Fast method

$$\sqrt[4]{497} \approx 500$$

$$\sqrt[4]{201} \approx 200$$

$$\sqrt[4]{99} \approx 100$$

So:

$$\sqrt[4]{\frac{500 \times 200}{100}} = 1000$$

Thus:

$$\boxed{\approx 1000}$$

63. OPTION-ELIMINATION EXAMPLE

Evaluate approximately:

$$\sqrt[4]{\frac{61.2 \times 19.7}{3.02}}$$

Options:

A. 40 B. 120 C. 400 D. 4000

Estimate:

$$\sqrt[4]{\frac{60 \times 20}{3}} = 400$$

Therefore:

$$\boxed{\text{C}}$$

No detailed arithmetic needed.

64. COMMON TRAPS

& pTRAP 1 — Over-rounding

$$\sqrt[4]{149}$$

should not always become 100.

Sometimes 150 is much better.

& pTRAP 2 — Approximation in subtraction

Near-equal numbers can produce huge relative error.

& pTRAP 3 — Close options

If options differ by 1–2%, rough rounding may be insufficient.

& pTRAP 4 — Wrong percentage-error denominator

Use true/reference value unless defined otherwise.

& pTRAP 5 — Bound inclusivity

Nearest-unit interval is typically:

$$\left[a - \frac{u}{2}, a + \frac{u}{2} \right)$$

not both endpoints inclusive under standard rounding convention.

& pTRAP 6 — Exact NAT question

Approximation may guide you, but final numeric answer may still require exact computation.

65. MUST-MEMORIZE FORMULAS

Absolute error

$$\boxed{|A - T|}$$

where $|A|$ = approximate, $|T|$ = true.

Relative error

$$\boxed{\frac{|A - T|}{|T|}}$$

Percentage error

$$\boxed{\frac{|A - T|}{|T|} \times 100}$$

Rounding bound

If rounded to nearest unit $|u|$:

$$\boxed{a - \frac{|u|}{2} \leq x < a + \frac{|u|}{2}}$$

66. 5-MINUTE REVISION

Use approximation when:

- numbers are close to friendly values,
- options are far apart,
- only range/order is needed.

Use exact calculation when:

- options are close,
- subtraction nearly cancels,
- exact NAT is required.

Always perform:

- sign check
 - magnitude check
 - decimal check
 - fraction/percentage sanity check.
-

PRACTICE — LEVEL 0

Q1

Round:

$$\sqrt[3]{7.846}$$

to 2 decimal places.

Q2

Round:

$$\sqrt[3]{3472}$$

to 2 significant figures.

Q3

Estimate:

$$\sqrt[3]{49.9 \times 20.2}$$

Q4

Estimate:

$$\sqrt[3]{1001/49.8}$$

Q5

True value = 40, approximate = 42. Find percentage error.

LEVEL 1

Q6

A value rounds to 25 to nearest integer. Give its bounds.

Q7

A value is 6.3 correct to 1 decimal place. Give its bounds.

Q8

Estimate:

$$\sqrt[3]{398/201}$$

and say whether exact value is slightly above or below 2.

Q9

Estimate:

$$\sqrt[3]{19.9\% \text{ of } 503}$$

Q10

True value 500, observed value 485. Find percentage error.

LEVEL 2

Q11

Estimate:

$$\left[\frac{302 \times 49.5}{15.1} \right]$$

Q12

A length is 120 cm to nearest 10 cm. Give lower and upper bounds.

Q13

If $\sqrt{x}$ is measured with 2% error and $\sqrt{y}$ with 3% error, estimate maximum percentage error in $\sqrt{xy}$.

Q14

Estimate:

$$\left[\sqrt{82} \right]$$

to one decimal place.

Q15

Which is closer to:

$$\left[\frac{49}{101} \right]$$

A. 0.3 B. 0.48 C. 0.7 D. 1.2

LEVEL 3 — EXAM PRESSURE

Q16

Estimate:

$$\left[\frac{61.4 \times 39.8}{2.02} \right]$$

Options:

A. 120 B. 1200 C. 12000 D. 12

Q17

A measured quantity is 8.0 correct to nearest 0.1. What is its maximum absolute rounding error?

Q18

A value changes from 200 to 206. Approximate percentage increase.

Q19

Estimate:

$$\left[\frac{999 \times 502}{249} \right]$$

Q20

Why is approximating:

$$\left[1001-999 \right]$$

as:

$$\left[1000-1000 \right]$$

a bad method?

ANSWERS

1

$$\lfloor 7.846 \rfloor \rightarrow \boxed{7.85}$$

2

$$\lfloor 3472 \rfloor \rightarrow \boxed{3500}$$

3

$$\lfloor 50 \times 20 \rfloor = \boxed{\approx 1000}$$

4

$$\lfloor 1000/50 \rfloor = \boxed{\approx 20}$$

5

Error:

$$\lfloor 2 \rfloor$$

Percentage error:

$$\lfloor \frac{2}{40} \times 100 \rfloor = \boxed{5\%}$$

6

$$\lfloor \boxed{24.5 \leq x < 25.5} \rfloor$$

7

Half of 0.1:

$$\lfloor 0.05 \rfloor$$

So:

$$\lfloor \boxed{6.25 \leq x < 6.35} \rfloor$$

8

$$\lfloor 398/201 \rfloor$$

Compare with:

$$\lfloor 400/200 = 2 \rfloor$$

Numerator lower and denominator higher.

Therefore:

$$\lfloor \boxed{\text{slightly below } 2} \rfloor$$

9

$$\lfloor 20\% \text{ of } 500 \rfloor = \boxed{\approx 100}$$

10

Error:

$$\lfloor 15 \rfloor$$

$$\lfloor \frac{15}{500} \times 100 = \boxed{3\%} \rfloor$$

11

Approximate:

$$\lfloor \frac{300 \times 50}{15} \rfloor$$

$$\lfloor = \frac{15000}{15} \rfloor$$

$$\lfloor = \boxed{1000} \rfloor$$

12

Nearest 10:

$$\lfloor 120 \pm 5 \rfloor$$

So:

$$\lfloor \boxed{115 \leq x < 125} \rfloor$$

13

Approximate product percentage error:

$$\lfloor 2\% + 3\% = \boxed{5\%} \rfloor$$

14

$$\lfloor 9^2 = 81 \rfloor$$

So:

$$\lfloor \sqrt{82} \approx 9.055 \rfloor$$

to one decimal place:

$$\lfloor \boxed{9.1} \rfloor$$

15

$$\lfloor 49/101 \approx 0.485 \rfloor$$

Closest:

$$\lfloor \boxed{\text{B}} \rfloor$$

16

$$\lfloor \frac{60 \times 40}{2} = 1200 \rfloor$$

So:

$$\lfloor \boxed{\text{B}} \rfloor$$

17

Nearest 0.1 means half-step:

$$\lfloor 0.05 \rfloor$$

Maximum absolute rounding error:

$$\lfloor \boxed{0.05} \rfloor$$

18

Increase:

$$\sqrt[6]{}$$

Base:

$$\sqrt[200]{}$$

$$\sqrt[6]{200} \times 100 = \boxed{3\%}$$

19

Approximate:

$$\sqrt{\frac{1000 \times 500}{250}}$$

$$\sqrt{=1000 \times 2}$$

$$\sqrt{=\boxed{2000}}$$

20

Because the exact difference is small:

$$\sqrt{1001 - 999} = 2$$

but rounding both values destroys that small difference and gives:

$$\sqrt{0}$$

This is a **near-cancellation** problem.

CHAPTER 11 — FINAL CHECK

We have now covered:

- BODMAS and exact simplification '
- cancellation and factorization '
- decimal/fraction/percentage conversion for speed '
- compensation and rearrangement '
- rounding '
- decimal places vs significant figures '
- approximation of sums/products/quotients '
- upper and lower bounds '
- absolute/relative/percentage error '
- order of magnitude '
- sanity checks '
- directional estimation '
- back-solving and option elimination '
- small-error approximations '
- when approximation is safe/unsafe '
- GATE-style practice '
- placement-style practice '

Therefore:

CHAPTER 11 COMPLETE

Progress

Completed: 11/76 **Current:** Chapter 11 — Simplification, Approximation & Estimation **Next:** Chapter 12 — Fast Multiplication, Division & Cancellation

Chapter 12 will consolidate calculation speed into a dedicated toolkit:

- fast multiplication by common constants
- near-base multiplication
- split-and-distribute methods
- fast division
- fraction cancellation
- ratio-preserving cancellation
- decimal movement
- divisibility-assisted division
- speed drills under exam pressure.

USER

next

ASSISTANT

CHAPTER 12 — FAST MULTIPLICATION, DIVISION & CANCELLATION

PART 1 — HIGH-VALUE MULTIPLICATION SHORTCUTS, SMART DIVISION & FRACTION CANCELLATION

Status: 0% **PROGRESS GATE:** High **Placements:** High **Priority:** 0% **Yr 1**

This chapter consolidates the calculation tricks from Chapters 1, 10 and 11 into one **speed-calculation system**.

The goal is not to memorize dozens of magic tricks.

The goal is:

Rewrite the calculation into an easier equivalent form.

1. THE 4 CORE SPEED IDEAS

Most fast calculations use one of these:

A. Compensation

Replace an awkward number by a nearby easy number.

[98=100-2]

B. Factorization

Break a number into easy factors.

$$\backslash[24=6\times 4 \backslash]$$

C. Scaling

Use powers of 10.

$$\backslash[25=\frac{100}{4} \backslash]$$

D. Cancellation

Remove common factors before multiplying.

$$\backslash[\frac{48\times 35}{12\times 7} \backslash]$$

becomes:

$$\backslash[4\times 5=20 \backslash]$$

2. MULTIPLICATION BY 5

$$\backslash[\boxed{N\times 5=\frac{N\times 10}{2}} \backslash]$$

Example:

$$\backslash[68\times 5 \backslash]$$

$$\backslash[680/2 = \boxed{340} \backslash]$$

3. MULTIPLICATION BY 50

$$\backslash[50=\frac{100}{2} \backslash]$$

So:

$$\backslash[\boxed{N\times 50=\frac{N\times 100}{2}} \backslash]$$

Example:

$$\backslash[84\times 50 \backslash]$$

$$\backslash[8400/2 = \boxed{4200} \backslash]$$

4. MULTIPLICATION BY 25

$$\backslash[25=\frac{100}{4} \backslash]$$

Therefore:

$$\backslash[\boxed{N\times 25=\frac{N\times 100}{4}} \backslash]$$

when division by 4 is easy.

Example:

$$\backslash[76\times 25 \backslash]$$

$$\backslash[76/4=19 \backslash]$$

$$19 \times 100 = \boxed{1900}$$

5. MULTIPLICATION BY 125

$$125 = \frac{1000}{8}$$

Therefore:

$$\boxed{N \times 125 = \frac{N \times 1000}{8}}$$

Example:

$$56 \times 125$$

$$56/8 = 7$$

$$7 \times 1000 = \boxed{7000}$$

6. MULTIPLICATION BY 250

$$250 = \frac{1000}{4}$$

Example:

$$36 \times 250$$

$$36/4 = 9$$

$$9 \times 1000 = \boxed{9000}$$

7. MULTIPLICATION BY 20

$$N \times 20 = (N \times 2) \times 10$$

Example:

$$47 \times 20 = 94 \times 10 = \boxed{940}$$

8. MULTIPLICATION BY 15

$$15 = 10 + 5$$

So:

$$N \times 15 = 10N + 5N$$

Example:

$$64 \times 15$$

$$640 + 320 = \boxed{960}$$

Another method:

$$15 = \frac{30}{2}$$

Choose whichever is easier.

9. MULTIPLICATION BY 12

$$\backslash[12=10+2\backslash]$$

Example:

$$\backslash[73\times 12 = 730+146 = \boxed{876}\backslash]$$

10. MULTIPLICATION BY 18

$$\backslash[18=20-2\backslash]$$

Example:

$$\backslash[46\times 18\backslash]$$

$$\backslash[=920-92\backslash]$$

$$\backslash[=\boxed{828}\backslash]$$

11. MULTIPLICATION BY 9

$$\backslash[9=10-1\backslash]$$

$$\backslash[N\times 9=10N-N\backslash]$$

Example:

$$\backslash[84\times 9 = 840-84 = \boxed{756}\backslash]$$

12. MULTIPLICATION BY 99

$$\backslash[99=100-1\backslash]$$

Example:

$$\backslash[63\times 99\backslash]$$

$$\backslash[6300-63 = \boxed{6237}\backslash]$$

13. MULTIPLICATION BY 999

$$\backslash[999=1000-1\backslash]$$

Example:

$$\backslash[48\times 999\backslash]$$

$$\backslash[48000-48 = \boxed{47952}\backslash]$$

14. MULTIPLICATION BY 101

$$\backslash[101=100+1\backslash]$$

Example:

$$[74 \times 101]$$

$$[7400 + 74 = \boxed{7474}]$$

15. MULTIPLICATION BY 1001

$$[1001 = 1000 + 1]$$

Example:

$$[324 \times 1001]$$

$$[324000 + 324 = \boxed{324324}]$$

16. MULTIPLICATION BY 11 — TWO-DIGIT CASE

For:

$$[ab \times 11]$$

put $(a+b)$ between the digits.

Example:

$$[43 \times 11]$$

$$[4, \ 4+3, \ 3]$$

$$[= \boxed{473}]$$

17. $\times 11$ WITH CARRY

$$[78 \times 11]$$

Middle sum:

$$[7+8=15]$$

Write 5, carry 1:

$$[\boxed{858}]$$

18. $\times 11$ — THREE-DIGIT METHOD

Example:

$$[352 \times 11]$$

Write:

- first digit: 3
- $(3+5=8)$
- $(5+2=7)$
- last digit: 2

Thus:

$$\boxed{3872}$$

Check:

$$3520+352=3872$$

19. COMPENSATION AROUND 10

$$37 \times 9$$

$$= 37(10-1)$$

$$= 370 - 37$$

$$= \boxed{333}$$

20. COMPENSATION AROUND 100

$$48 \times 97$$

Use:

$$97 = 100 - 3$$

$$48 \times 100 - 48 \times 3$$

$$= 4800 - 144$$

$$= \boxed{4656}$$

21. COMPENSATION AROUND 1000

$$63 \times 998$$

$$= 63(1000-2)$$

$$= 63000 - 126$$

$$= \boxed{62874}$$

22. SPLIT-AND-DISTRIBUTE METHOD

For:

$$37 \times 46$$

split one number:

$$46 = 40 + 6$$

$$37 \times 40 = 1480$$

$$37 \times 6 = 222$$

Total:

$$\boxed{1702}$$

This is often better than forcing a specialized trick.

23. CHOOSE THE EASIER NUMBER TO SPLIT

For:

$$83 \times 24$$

split 24:

$$24 = 6 \times 4$$

or:

$$20 + 4$$

Method:

$$83 \times 20 = 1660$$

$$83 \times 4 = 332$$

Total:

$$\boxed{1992}$$

24. DOUBLE-HALF METHOD

If one number is even:

$$a \times b = \frac{a^2 \times 2b}{2}$$

Example:

$$16 \times 35$$

Half 16:

$$8$$

Double 35:

$$70$$

Again:

$$4 \times 140$$

$$= \boxed{560}$$

This is especially useful when doubling creates a round number.

25. DOUBLE-HALF EXAMPLE

$$25 \times 48$$

Double 25:

$$50$$

Half 48:

$$24$$

Then:

$$50 \times 24 = \boxed{1200}$$

26. SYMMETRIC PRODUCT AROUND A CENTER

Use:

$$(a-b)(a+b) = a^2 - b^2$$

Example:

$$47 \times 53$$

Center:

$$50$$

Distance:

$$3$$

$$50^2 - 3^2 = 2500 - 9 = \boxed{2491}$$

27. ANOTHER SYMMETRIC PRODUCT

$$96 \times 104$$

$$= (100 - 4)(100 + 4)$$

$$= 10000 - 16$$

$$= \boxed{9984}$$

28. NUMBERS NEAR 100 — BASE METHOD

Example:

$$97 \times 94$$

Deficiencies:

$$3, 6$$

Cross-subtract:

$$97 - 6 = 91$$

Multiply deficiencies:

$$3 \times 6 = 18$$

So:

$$\boxed{9118}$$

29. BASE-100 METHOD WITH EXCESSES

$$103 \times 106$$

Excesses:

$$\backslash[3, \ 6 \]$$

Cross-add:

$$\backslash[103+6=109 \]$$

Multiply:

$$\backslash[3\times6=18 \]$$

Answer:

$$\backslash[\boxed{10918} \]$$

30. BASE-100 RIGHT BLOCK RULE

Because the base is 100, the right block must have exactly 2 digits.

Example:

$$\backslash[98\times99 \]$$

Deficiencies:

$$\backslash[2, \ 1 \]$$

Cross-subtract:

$$\backslash[97 \]$$

Right product:

$$\backslash[02 \]$$

Answer:

$$\backslash[\boxed{9702} \]$$

31. BASE-1000 METHOD

Example:

$$\backslash[997\times996 \]$$

Deficiencies:

$$\backslash[3, \ 4 \]$$

Cross-subtract:

$$\backslash[993 \]$$

Multiply:

$$\backslash[3\times4=12 \]$$

But base 1000 requires 3-digit right block:

$$\backslash[012 \]$$

Therefore:

$$\backslash[\boxed{993012} \]$$

32. SMART DIVISION — FACTOR THE DIVISOR

Suppose:

$$\backslash[936/24 \backslash]$$

Factor:

$$\backslash[24=6\times 4 \backslash]$$

Then:

$$\backslash[936/6=156 \backslash]$$

$$\backslash[156/4=39 \backslash]$$

Answer:

$$\backslash[\boxed{39} \backslash]$$

33. DIVISION BY 5

$$\backslash[\boxed{N/5=\frac{2N}{10}} \backslash]$$

Example:

$$\backslash[435/5 \backslash]$$

$$\backslash[870/10 = \boxed{87} \backslash]$$

34. DIVISION BY 25

$$\backslash[\boxed{N/25=\frac{4N}{100}} \backslash]$$

Example:

$$\backslash[675/25 \backslash]$$

$$\backslash[2700/100 = \boxed{27} \backslash]$$

35. DIVISION BY 125

$$\backslash[\boxed{N/125=\frac{8N}{1000}} \backslash]$$

Example:

$$\backslash[4375/125 \backslash]$$

$$\backslash[4375\times 8=35000 \backslash]$$

$$\backslash[35000/1000 = \boxed{35} \backslash]$$

36. DIVISION BY 50

$$\backslash[N/50=\frac{2N}{100} \backslash]$$

Example:

$$\backslash[2350/50 \backslash]$$

$$\backslash[4700/100 = \boxed{47} \backslash]$$

37. DIVISION BY 20

$$\backslash[N/20=\frac{N}{2}\div 10 \backslash]$$

or:

$$\backslash[N/20=N/2\times 0.1 \backslash]$$

Example:

$$\backslash[860/20 \backslash]$$

$$\backslash[430/10 = \boxed{43} \backslash]$$

38. DIVISION BY 0.5

Dividing by:

$$\backslash[0.5=\frac{1}{2} \backslash]$$

means multiplying by 2.

$$\backslash[\boxed{N/0.5=2N} \backslash]$$

Example:

$$\backslash[37/0.5 = \boxed{74} \backslash]$$

39. DIVISION BY 0.25

$$\backslash[0.25=\frac{1}{4} \backslash]$$

So:

$$\backslash[\boxed{N/0.25=4N} \backslash]$$

Example:

$$\backslash[18/0.25 = \boxed{72} \backslash]$$

40. DIVISION BY 0.125

$$\backslash[0.125=\frac{1}{8} \backslash]$$

Thus:

$$\backslash[N/0.125=8N \backslash]$$

Example:

$$\backslash[15/0.125 = \boxed{120} \backslash]$$

41. DIVISION BY 1.25

$$\left[1.25 = \frac{5}{4} \right]$$

Therefore:

$$\left[N/1.25 = N \times \frac{4}{5} \right]$$

Example:

$$\left[75/1.25 \right]$$

$$\left[75 \times \frac{4}{5} = 15 \times 4 = \boxed{60} \right]$$

42. DIVISION BY 2.5

$$\left[2.5 = \frac{5}{2} \right]$$

Therefore:

$$\left[N/2.5 = N \times \frac{2}{5} \right]$$

Example:

$$\left[90/2.5 \right]$$

$$\left[90 \times \frac{2}{5} = 18 \times 2 = \boxed{36} \right]$$

43. REMOVE DECIMALS BEFORE DIVIDING

$$\left[\frac{4.2}{0.07} \right]$$

Multiply numerator and denominator by 100:

$$\left[\frac{420}{7} = \boxed{60} \right]$$

44. ANOTHER DECIMAL DIVISION

$$\left[\frac{0.84}{0.012} \right]$$

Multiply both by 1000:

$$\left[840/12 = \boxed{70} \right]$$

45. FRACTION MULTIPLICATION— CANCEL FIRST

$$\left[\frac{24}{35} \times \frac{14}{9} \right]$$

Cancel:

$$\left[14/35 = 2/5 \right]$$

$$\left[24/9 = 8/3 \right]$$

So:

$$\left[\frac{8 \times 2}{3 \times 5} = \boxed{\frac{16}{15}} \right]$$

46. MULTI-FACTOR CANCELLATION

Evaluate:

$$\left[\frac{18 \times 25 \times 14}{35 \times 15} \right]$$

Factor/cancel:

$$\left[\frac{14}{35} = \frac{2}{5} \right]$$

$$\left[\frac{25}{5} = 5 \right]$$

$$\left[\frac{18}{15} = \frac{6}{5} \right]$$

Then:

$$\left[\frac{6 \times 5 \times 2}{5} = \boxed{12} \right]$$

47. CANCELLATION WITH PERCENTAGES

Evaluate:

$$\left[35\% \text{ of } 480 \right]$$

Convert:

$$\left[35\% = \frac{35}{100} = \frac{7}{20} \right]$$

Then:

$$\left[480 \times \frac{7}{20} \right]$$

$$\left[480/20 = 24 \right]$$

$$\left[24 \times 7 = \boxed{168} \right]$$

48. CANCELLATION WITH RATIOS

Suppose:

$$\left[\frac{84}{126} \right]$$

Reduce:

$$\left[84:126 \right]$$

HCF:

$$\left[42 \right]$$

So:

$$\left[\boxed{2:3} \right]$$

Always reduce before using the ratio in later calculations.

49. CANCELLATION WITH POWERS

$$\left[\frac{2^9 \times 3^6}{2^5 \times 3^2} \right]$$

$$= 2^4 3^4$$

$$= 16 \times 81$$

$$= \boxed{1296}$$

Exponent subtraction is just cancellation written compactly.

50. PRIME-FACTOR CANCELLATION

Evaluate:

$$\frac{84 \times 75}{70 \times 45}$$

Prime factors:

$$84 = 2^2 \cdot 3 \cdot 7$$

$$75 = 3 \cdot 5^2$$

$$70 = 2 \cdot 5 \cdot 7$$

$$45 = 3^2 \cdot 5$$

Cancel all common factors.

Remaining:

$$\frac{2}{1}$$

Therefore:

$$= \boxed{2}$$

You usually don't need to write the complete prime factorization; mentally cancel obvious factors.

51. CANCEL BEFORE SQUARING

Evaluate:

$$\frac{48^2}{16^2}$$

Instead of squaring:

$$\left(\frac{48}{16} \right)^2$$

$$= 3^2$$

$$= \boxed{9}$$

52. CANCEL BEFORE TAKING ROOT

$$\sqrt{\frac{288}{2}}$$

$$= \sqrt{144}$$

$$= \boxed{12}$$

53. COMMON-FACTOR CANCELLATION IN EXPRESSIONS

$$\left[\frac{17(43+57)}{17 \times 20} \right]$$

Cancel 17:

$$\left[\frac{100}{20} = \boxed{5} \right]$$

The same principle appears constantly in algebra and ratio problems.

54. FRACTION DIVISION — FLIP THEN CANCEL

$$\left[\frac{12}{35} \div \frac{18}{49} \right]$$

Flip second fraction:

$$\left[\frac{12}{35} \times \frac{49}{18} \right]$$

Cancel:

$$\left[\frac{12}{18} = \frac{2}{3} \right]$$

$$\left[\frac{49}{35} = \frac{7}{5} \right]$$

So:

$$\left[\frac{14}{15} \right]$$

$$\left[\boxed{\frac{14}{15}} \right]$$

55. DO NOT CANCEL ACROSS ADDITION

Very important trap.

$$\left[\frac{6+12}{6} \right]$$

You cannot “cancel the 6” only with one term unless you factor:

$$\left[\frac{6(1+2)}{6} \right]$$

Now cancel:

$$\left[=3 \right]$$

Rule

You may cancel **factors**, not terms.

56. INVALID CANCELLATION EXAMPLE

$$\left[\frac{x+5}{x} \right]$$

You cannot cancel $\cancel{x}$ because numerator is a sum.

But:

$$\left[\frac{x(x+5)}{x} \right]$$

allows cancellation:

$$\left[=x+5 \right]$$

provided $\cancel{x} \neq 0$.

57. GATE-STYLE QUESTION 1

Evaluate:

$$\left[\frac{48 \times 125}{24} \right]$$

Cancel first:

$$\left[48/24 = 2 \right]$$

Then:

$$\left[2 \times 125 = \boxed{250} \right]$$

58. GATE-STYLE QUESTION 2

Evaluate:

$$\left[\frac{96 \times 25}{8 \times 3} \right]$$

$$\left[96/8 = 12 \right]$$

$$\left[12/3 = 4 \right]$$

$$\left[4 \times 25 = \boxed{100} \right]$$

59. GATE-STYLE QUESTION 3

Evaluate:

$$\left[\frac{1.25 \times 48}{0.5} \right]$$

Use:

$$\left[1.25 = \frac{5}{4} \right]$$

$$\left[0.5 = \frac{1}{2} \right]$$

Then:

$$\left[\frac{5}{4} \times 48 \times 2 \right]$$

$$\left[48/4 = 12 \right]$$

$$\left[12 \times 5 \times 2 = \boxed{120} \right]$$

60. PLACEMENT-STYLE QUESTION 1

Calculate:

$$\left[84 \times 99 \right]$$

$$\left[84(100-1) \right]$$

$$\left[= 8400 - 84 \right]$$

$$\left[= \boxed{8316} \right]$$

61. PLACEMENT-STYLE QUESTION 2

Calculate:

$$\lfloor 72 \times 125 \rfloor$$

$$\lfloor 72/8 = 9 \rfloor$$

$$\lfloor 9 \times 1000 = \boxed{9000} \rfloor$$

62. PLACEMENT-STYLE QUESTION 3

Calculate:

$$\lfloor 4750/125 \rfloor$$

$$\lfloor 4750 \times 8/1000 \rfloor$$

$$\lfloor 38000/1000 \rfloor$$

$$\lfloor = \boxed{38} \rfloor$$

63. PLACEMENT-STYLE QUESTION 4

Calculate:

$$\lfloor 98 \times 102 \rfloor$$

Difference of squares:

$$\lfloor 100^2 - 2^2 \rfloor$$

$$\lfloor = 10000 - 4 \rfloor$$

$$\lfloor = \boxed{9996} \rfloor$$

64. THE SPEED DECISION TREE

If asked to multiply:

Is one number 5, 25, 50, 125, 250?

Use scaling.

Is one number near 10, 100, 1000?

Use compensation.

Are numbers equally spaced around a center?

Use difference of squares.

Are both near 100/1000?

Use base method.

Is one even and the other awkward?

Try double-half.

Otherwise

Split one number and distribute.

65. DIVISION DECISION TREE

If asked to divide:

Divisor = 5?

Double, divide by 10.

25?

×4, divide by 100.

125?

×8, divide by 1000.

Decimal divisor?

Convert to a fraction or scale away decimals.

Composite divisor?

Factor it.

Fraction divisor?

Flip and multiply.

66. CANCELLATION DECISION TREE

Before multiplying fractions:

1. factor mentally
2. cancel numerator against denominator
3. reduce powers/exponents
4. multiply only surviving numbers

Golden rule

$\boxed{\text{Cancel before multiply}}$

67. COMMON TRAPS

TRAP 1 — Canceling terms instead of factors

Invalid:

$\frac{x+2}{x}$

TRAP 2 — Using a complicated trick when normal distribution is easier

Fast math means **least mental effort**, not maximum cleverness.

& pTRAP 3 — Base-method zero padding

Base 100 !' 2 digits on right.

Base 1000 !' 3 digits on right.

& pTRAP 4 — Decimal division

$\backslash N/0.5 \backslash$

makes $\backslash(N\backslash)$ larger, not smaller.

& pTRAP 5 — Fraction division

Dividing by:

$\backslash \frac{3}{4} \backslash$

means multiplying by:

$\backslash \frac{4}{3} \backslash$

68. MUST-MEMORIZE SPEED IDENTITIES

$\backslash \boxed{\times 5 = \times 10 \div 2} \backslash$

$\backslash \boxed{\times 25 = \times 100 \div 4} \backslash$

$\backslash \boxed{\times 125 = \times 1000 \div 8} \backslash$

$\backslash \boxed{\div 5 = \times 2 \div 10} \backslash$

$\backslash \boxed{\div 25 = \times 4 \div 100} \backslash$

$\backslash \boxed{\div 125 = \times 8 \div 1000} \backslash$

$\backslash \boxed{(a-b)(a+b) = a^2 - b^2} \backslash$

PRACTICE — LEVEL 0

Q1

$\backslash 84 \times 5 \backslash$

Q2

$\backslash 64 \times 25 \backslash$

Q3

$\backslash 40 \times 125 \backslash$

Q4

$\backslash 365/5 \backslash$

Q5

$\backslash 825/25 \backslash$

LEVEL 1

Q6

$$\lfloor 48 \times 99 \rfloor$$

Q7

$$\lfloor 63 \times 101 \rfloor$$

Q8

$$\lfloor 37 \times 18 \rfloor$$

Q9

$$\lfloor \frac{36}{0.25} \rfloor$$

Q10

$$\lfloor \frac{75}{1.25} \rfloor$$

LEVEL 2

Q11

$$\lfloor 97 \times 96 \rfloor$$

Q12

$$\lfloor 103 \times 107 \rfloor$$

Q13

$$\lfloor 997 \times 994 \rfloor$$

Q14

$$\lfloor \frac{24}{35} \times \frac{70}{18} \rfloor$$

Q15

$$\lfloor \frac{2.5 \times 48}{0.25} \rfloor$$

LEVEL 3 — EXAM PRESSURE

Q16

$$\lfloor 9998 \times 10002 \rfloor$$

Q17

$$\lfloor \frac{84 \times 125 \times 16}{28 \times 40} \rfloor$$

Q18

$$\lfloor \frac{0.625 \times 96}{1.25} \rfloor$$

Q19

$$\left[\frac{2^{10}3^7}{2^63^4} \right]$$

Q20

$$\left[\frac{45}{56} \div \frac{75}{98} \right]$$

ANSWERS**1**

$$\left[84 \times 10/2 = \boxed{420} \right]$$

2

$$\left[64/4 \times 100 = 16 \times 100 = \boxed{1600} \right]$$

3

$$\left[40/8 \times 1000 = 5 \times 1000 = \boxed{5000} \right]$$

4

$$\left[365 \times 2/10 = \boxed{73} \right]$$

5

$$\left[825 \times 4/100 = 3300/100 = \boxed{33} \right]$$

6

$$\left[48(100-1) = 4800-48 = \boxed{4752} \right]$$

7

$$\left[63(100+1) = 6300+63 = \boxed{6363} \right]$$

8

$$\left[37(20-2) = 740-74 = \boxed{666} \right]$$

9

$$\left[0.25 = \frac{1}{4} \right]$$

So:

$$\left[36 \times 4 = \boxed{144} \right]$$

10

$$\left[1.25 = \frac{5}{4} \right]$$

$$\left[75 \times \frac{4}{5} = 15 \times 4 = \boxed{60} \right]$$

11

Base 100:

deficiencies:

$$\left[3, \frac{1}{4} \right]$$

Cross-subtract:

$$\sqrt{93}$$

Right block:

$$\sqrt{12}$$

$$\sqrt{\boxed{9312}}$$

12

Excesses:

$$\sqrt{3, \sqrt{7}}$$

Cross-add:

$$\sqrt{110}$$

Product:

$$\sqrt{21}$$

$$\sqrt{\boxed{11021}}$$

13

Base 1000:

deficiencies:

$$\sqrt{3, \sqrt{6}}$$

Cross-subtract:

$$\sqrt{991}$$

Right block:

$$\sqrt{018}$$

$$\sqrt{\boxed{991018}}$$

14

$$\sqrt{\frac{24}{35} \times \frac{70}{18}}$$

$$\sqrt{70/35=2}$$

$$\sqrt{24/18=4/3}$$

So:

$$\sqrt{\boxed{\frac{8}{3}}}$$

15

$$\sqrt{2.5=\frac{5}{2}}$$

$$\sqrt{0.25=\frac{1}{4}}$$

So:

$$\sqrt{\frac{5}{2} \times 48 \times 4}$$

$$\sqrt{48/2=24}$$

$$\sqrt{24 \times 5 \times 4 = \boxed{480}}$$

16

Symmetric around 10000:

$$\sqrt{(10000-2)(10000+2)}$$

$$\sqrt{=10000^2-4}$$

$$\sqrt{=100000000-4}$$

$$\sqrt{=\boxed{99999996}}$$

17

$$\sqrt{\frac{84 \times 125 \times 16}{28 \times 40}}$$

Cancel:

$$\sqrt{84/28=3}$$

$$\sqrt{16/40=2/5}$$

Then:

$$\sqrt{3 \times 125 \times \frac{2}{5}}$$

$$\sqrt{125/5=25}$$

$$\sqrt{3 \times 25 \times 2 = \boxed{150}}$$

18

$$\sqrt{0.625=\frac{5}{8}}$$

$$\sqrt{1.25=\frac{5}{4}}$$

So:

$$\sqrt{\frac{5}{8} \times 96 \times \frac{4}{5}}$$

Cancel 5:

$$\sqrt{\frac{96 \times 4}{8}}$$

$$\sqrt{=12 \times 4}$$

$$\sqrt{=\boxed{48}}$$

19

$$\sqrt{2^{10-6} 3^{7-4}}$$

$$\sqrt{=2^4 3^3}$$

$$\sqrt{=16 \times 27}$$

$$\sqrt{=\boxed{432}}$$

20

$$\sqrt{\frac{45}{56} \times \frac{98}{75}}$$

Cancel:

$$\sqrt{45/75=3/5}$$

$$\sqrt{98/56=7/4}$$

Thus:

$\frac{35}{74}$

$\frac{21}{20}$

CHAPTER 12 STATUS

Covered so far:

- multiplication by 5/25/50/125/250 '
- multiplication by 9/11/12/15/18/99/101/999/1001 '
- compensation '
- split-and-distribute '
- double-half '
- symmetric products '
- near-base multiplication '
- division by 5/25/50/125 '
- decimal divisors '
- factorized division '
- fraction cancellation '
- percentage cancellation '
- power cancellation '
- valid vs invalid cancellation '
- GATE/placement practice '

But **Chapter 12 is not complete yet.**

The final part will cover:

- multiplication of awkward two-digit numbers faster
- ratio-preserving cancellation
- divisibility-assisted quotient estimation
- checking division quickly
- fast reciprocal thinking
- compound multi-step calculation
- timed calculation drills
- error-control system
- final Unit 0 revision sheet

Progress

Completed: 11/76 **Current:** 0=Chapter 12 — Fast Multiplication, Division & Cancellation **Next:** Chapter 12 Part 2 — Advanced calculation speed drills & Unit 0 consolidation

USER

next

CHAPTER 12 — FAST MULTIPLICATION, DIVISION & CANCELLATION

PART 2 — ADVANCED CALCULATION SPEED, QUOTIENT ESTIMATION, ERROR CONTROL & UNIT 0 CONSOLIDATION

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

This final part of Unit 0 is about combining everything into one reliable exam-calculation system.

The target is not “fancy tricks.”

The target is:

$\boxed{\text{fast} + \text{accurate} + \text{low mental load}}$

1. AWKWARD TWO-DIGIT MULTIPLICATION

Suppose:

47×63

You have several methods.

Method 1 — Split 63

$47(60+3)$

$= 2820 + 141$

$= \boxed{2961}$

Method 2 — Split 47

$63(50-3)$

$= 3150 - 189$

$= \boxed{2961}$

Choose whichever creates easier arithmetic.

2. CHOOSE THE BETTER DECOMPOSITION

For:

68×47

Option A:

$68(40+7)$

$2720 + 476 = 3196$

Option B:

$68(50-3)$

$3400 - 204 = 3196$

Method B is mentally easier.

Exam principle

Decompose around 10, 20, 25, 50, 100, etc. whenever that reduces work.

3. NEARBY-NUMBER PRODUCT

Find:

$$62 \times 58$$

They are symmetric around:

$$60$$

So:

$$(60+2)(60-2)$$

$$= 60^2 - 2^2$$

$$= 3600 - 4$$

$$= \boxed{3596}$$

This is much faster than ordinary multiplication.

4. NUMBERS NOT PERFECTLY SYMMETRIC

Find:

$$62 \times 57$$

Not symmetric around an integer with equal distance.

Use:

$$62(60-3)$$

$$= 3720 - 186$$

$$= \boxed{3534}$$

Do not force difference-of-squares where it doesn't fit cleanly.

5. MULTIPLY BY 19, 29, 39, ETC.

Example:

$$84 \times 29$$

Use:

$$29 = 30 - 1$$

$$84 \times 30 - 84$$

$$= 2520 - 84$$

$$\boxed{2436}$$

Similarly:

$$N \times 39 = N \times 40 - N$$

6. MULTIPLY BY 49

$$49 = 50 - 1$$

Example:

$$76 \times 49$$

$$76 \times 50 - 76$$

$$3800 - 76$$

$$\boxed{3724}$$

7. MULTIPLY BY 51

$$51 = 50 + 1$$

Example:

$$86 \times 51$$

$$4300 + 86$$

$$\boxed{4386}$$

8. MULTIPLY BY 75

$$75 = \frac{34}{100} \times 100$$

Example:

$$64 \times 75$$

$$64 \times \frac{34}{100} \times 100$$

$$48 \times 100$$

$$\boxed{4800}$$

This is often faster than:

$$64(50 + 25)$$

9. MULTIPLY BY 37.5

$$37.5\% = \frac{38}{100}$$

and numerically:

$$37.5 = \frac{38}{100} \times 100$$

Example:

$$[48 \times 37.5]$$

$$[48 \times \frac{38}{100}]$$

$$[18 \times 100]$$

$$[= \boxed{1800}]$$

10. MULTIPLY BY 12.5

$$[12.5 = \frac{18}{100}]$$

Example:

$$[72 \times 12.5]$$

$$[72/8 \times 100]$$

$$[9 \times 100]$$

$$[= \boxed{900}]$$

11. FAST RECIPROCAL THINKING

Some divisions are easier if you know common reciprocals.

$$[\frac{1}{2} = 0.5]$$

$$[\frac{1}{4} = 0.25]$$

$$[\frac{1}{5} = 0.2]$$

$$[\frac{1}{8} = 0.125]$$

$$[\frac{1}{10} = 0.1]$$

$$[\frac{1}{20} = 0.05]$$

$$[\frac{1}{25} = 0.04]$$

These help both multiplication and division.

12. RECIPROCAL-BASED DIVISION

Example:

$$[84/8]$$

Instead of formal division:

$$[84 \times \frac{1}{8}]$$

$$[= 10.5]$$

Likewise:

$$[72/25 = 72 \times 0.04 = \boxed{2.88}]$$

13. APPROXIMATE RECIPROCAL

Useful mental approximations:

$$\left[\frac{1}{3} \approx 0.333 \right]$$

$$\left[\frac{1}{6} \approx 0.167 \right]$$

$$\left[\frac{1}{7} \approx 0.143 \right]$$

$$\left[\frac{1}{9} \approx 0.111 \right]$$

For estimation only unless exact recurring form is needed.

14. QUOTIENT ESTIMATION BEFORE DIVISION

Suppose:

$$\left[1435/29 \right]$$

Before dividing, estimate:

$$\left[1450/29 \approx 50 \right]$$

Since:

$$\left[29 \times 50 = 1450 \right]$$

Actual quotient will be slightly less than 50.

Try:

$$\left[49 \times 29 = 50 \times 29 - 29 \right]$$

$$\left[= 1450 - 29 = 1421 \right]$$

Remainder:

$$\left[14 \right]$$

So:

$$\left[1435/29 = 49 + \frac{14}{29} \right]$$

$$\left[\approx 49.48 \right]$$

The estimate told us where to start.

15. QUOTIENT ESTIMATION USING NEARBY MULTIPLES

Evaluate:

$$\left[997/31 \right]$$

We know:

$$\left[31 \times 30 = 930 \right]$$

$$\left[31 \times 32 = 992 \right]$$

So:

$$\left[997/31 \right]$$

is just above 32.

$$\left[= \boxed{32 + \frac{5}{31}} \right]$$

No long division needed.

16. FAST INTEGER DIVISION CHECK

Suppose you think:

$$\lfloor 936/24=39 \rfloor$$

Check:

$$\lfloor 39 \times 24 \rfloor$$

$$\lfloor =39(20+4) \rfloor$$

$$\lfloor =780+156 \rfloor$$

$$\lfloor =936 \rfloor$$

Correct.

Memory

Division answer can be checked by reverse multiplication.

17. REMAINDER CHECK

If:

$$\lfloor N=dq+r \rfloor$$

then verify:

$$\lfloor dq+r=N \rfloor$$

and:

$$\lfloor 0 \leq r < d \rfloor$$

Example:

$$\lfloor 1000 \div 37 \rfloor$$

Try quotient:

$$\lfloor 27 \rfloor$$

$$\lfloor 37 \times 27 = 999 \rfloor$$

Remainder:

$$\lfloor 1 \rfloor$$

So:

$$\lfloor \boxed{1000=37(27)+1} \rfloor$$

18. DIVISIBILITY-ASSISTED DIVISION

Suppose:

$$\lfloor 1728/36 \rfloor$$

Instead of long division:

$$\lfloor 36=4\times 9 \rfloor$$

$$\lfloor 1728/4=432 \rfloor$$

$$\lfloor 432/9=48 \rfloor$$

So:

$$\lfloor \boxed{48} \rfloor$$

19. CHOOSE THE DIVISOR ORDER

For:

$$\lfloor 7560/24 \rfloor$$

You can divide by:

$$\lfloor 24=3\times 8 \rfloor$$

Do easier operation first:

$$\lfloor 7560/8=945 \rfloor$$

Then:

$$\lfloor 945/3=315 \rfloor$$

Answer:

$$\lfloor \boxed{315} \rfloor$$

20. FACTOR LARGE DIVISORS

$$\lfloor 13860/45 \rfloor$$

$$\lfloor 45=5\times 9 \rfloor$$

Divide by 5:

$$\lfloor 13860/5=2772 \rfloor$$

Then by 9:

$$\lfloor 2772/9=308 \rfloor$$

So:

$$\lfloor \boxed{308} \rfloor$$

21. RATIO-PRESERVING CANCELLATION

Suppose ratio:

$$\lfloor 84:126 \rfloor$$

Divide both by same non-zero factor:

$$\boxed{42}$$

$$\boxed{2:3}$$

The ratio is unchanged.

General rule:

$$\boxed{a:b=ka:kb}$$

for non-zero k .

22. CANCELLATION IN PROPORTIONS

Suppose:

$$\frac{84}{126} = \frac{x}{15}$$

First simplify:

$$\frac{84}{126} = \frac{2}{3}$$

So:

$$\frac{x}{15} = \frac{2}{3}$$

Then:

$$x = 10$$

Much easier than cross-multiplying large numbers.

23. RATIO SCALING

If ratio:

$$\boxed{3:5}$$

and total corresponding quantity is:

$$\boxed{64}$$

Total parts:

$$\boxed{8}$$

Each part:

$$\boxed{64/8=8}$$

Values:

$$\boxed{24, 40}$$

This foreshadows Chapter 13.

24. CANCELLATION IN PERCENTAGES

Example:

$$\frac{18\% \text{ of } 250}{30\% \text{ of } 150}$$

Write:

$$\left[\frac{18}{30} \times \frac{250}{150} \right]$$

Simplify:

$$\left[\frac{35}{5} \times \frac{5}{3} \right]$$

$$\left[=\boxed{1} \right]$$

No need to calculate numerator and denominator separately.

25. CANCELLATION IN AVERAGE-LIKE EXPRESSIONS

Example:

$$\left[\frac{48+52+50}{3} \right]$$

Observe:

$$\left[48+52=100 \right]$$

Then:

$$\left[100+50=150 \right]$$

$$\left[150/3 = \boxed{50} \right]$$

Group terms strategically.

26. PAIRING NUMBERS

Calculate:

$$\left[1+2+3+\cdots+20 \right]$$

Pair:

$$\left[1+20=21 \right]$$

$$\left[2+19=21 \right]$$

There are 10 pairs.

So:

$$\left[10 \times 21 = \boxed{210} \right]$$

This is the arithmetic-series idea we revisit later.

27. PAIRING IN UNSORTED SUMS

Calculate:

$$\left[47+63+53+37 \right]$$

Pair:

$$\left[47+53=100 \right]$$

$$\left[63+37=100 \right]$$

Total:

$$\boxed{200}$$

Always look for complements to 10, 100, 1000.

28. COMPLEMENTS TO 100

Useful pair instincts:

$$37+63=100$$

$$48+52=100$$

$$29+71=100$$

$$64+36=100$$

This significantly improves speed in long sums.

29. COMPLEMENTS TO 1 IN DECIMALS

$$0.37+0.63=1$$

$$0.48+0.52=1$$

$$0.125+0.875=1$$

This helps in probability and percentages later.

30. MULTI-STEP CALCULATION — SIMPLIFY GLOBALLY

Evaluate:

$$\frac{0.75 \times 320 \times 25}{40 \times 1.5}$$

Convert:

$$0.75 = \frac{3}{4}$$

$$1.5 = \frac{3}{2}$$

Then:

$$\frac{\frac{3}{4} \times 320 \times 25}{40 \times \frac{3}{2}}$$

Divide by $\frac{3}{2}$ means multiply by $\frac{2}{3}$:

$$\frac{\frac{3}{4} \times 320 \times 25 \times \frac{2}{3}}{40}$$

Cancel:

$$320/40=8$$

$$3 \text{ cancels}$$

Actually compute carefully:

$$\frac{3 \times 8 \times 25 \times 2}{4}$$

Cancel 3:

$$\left[\frac{1}{4} \times 8 \times 25 \times 2 \right]$$

$$\left[= 2 \times 25 \times 2 \right]$$

$$\left[= \boxed{100} \right]$$

31. SMART ORDER OF OPERATIONS

Evaluate:

$$\left[1.25 \times 48 \times 0.8 \right]$$

Instead of left to right:

$$\left[1.25 \times 0.8 = 1 \right]$$

because:

$$\left[\frac{5}{4} \times \frac{4}{5} = 1 \right]$$

So answer:

$$\left[\boxed{48} \right]$$

Exam principle

Multiplication can be regrouped.

Use:

$$\left[abc = a(cb) \right]$$

to form easy pairs.

32. ANOTHER REGROUPING EXAMPLE

$$\left[2.5 \times 32 \times 0.4 \right]$$

Since:

$$\left[2.5 \times 0.4 = 1 \right]$$

Answer:

$$\left[\boxed{32} \right]$$

33. USE RECIPROCAL PAIRS

Know:

$$\left[1.25 \times 0.8 = 1 \right]$$

$$\left[2.5 \times 0.4 = 1 \right]$$

$$\left[5 \times 0.2 = 1 \right]$$

$$\left[4 \times 0.25 = 1 \right]$$

$$\left[8 \times 0.125 = 1 \right]$$

These are very useful.

34. ERROR-CONTROL SYSTEM

Speed without checking creates careless errors.

Use a 5-second sanity check.

Ask:

1. **Sign:** positive or negative?
 2. **Size:** approximately how large?
 3. **Decimal:** did decimal point move correctly?
 4. **Divisibility:** should result be integer?
 5. **Reverse check:** does multiplication recover the original?
-

35. SIGN CHECK EXAMPLE

$\backslash (-12)(-5) \backslash$

Two negatives:

$\backslash \boxed{+60} \backslash$

If you write $\backslash (-60) \backslash$, reject immediately.

36. MAGNITUDE CHECK EXAMPLE

$\backslash 49 \times 51 \backslash$

should be near:

$\backslash 50^2 = 2500 \backslash$

So:

$\backslash 2499 \backslash$

is plausible.

But:

$\backslash 24990 \backslash$

is obviously wrong.

37. DECIMAL-PLACE CHECK

$\backslash 0.4 \times 0.3 \backslash$

Both less than 1, so product must be smaller than both.

$\backslash 0.12 \backslash$

is plausible.

$\backslash 1.2 \backslash$

is impossible.

38. INTEGER CHECK

$\lfloor 720/24 \rfloor$

Since 24 clearly divides 720, result should be integer.

If your result is:

$\lfloor 29.6 \rfloor$

you likely made an arithmetic error.

39. DIGIT-SUM CHECK FOR MULTIPLICATION— OPTIONAL

Example:

$\lfloor 47 \times 53 = 2491 \rfloor$

Digit sum mod 9:

$\lfloor 47 \bmod 9 + 7 = 11 \bmod 9 \rfloor$

$\lfloor 53 \bmod 9 \rfloor$

Product residue:

$\lfloor 2 \times 8 = 16 \bmod 9 \rfloor$

Answer:

$\lfloor 2491 \bmod 9 = 2+4+9+1 = 16 \bmod 9 \rfloor$

Matches.

This does not prove correctness, but catches many mistakes.

40. LAST-DIGIT CHECK

Example:

$\lfloor 68 \times 47 \rfloor$

Last digits:

$\lfloor 8 \times 7 = 56 \rfloor$

So answer must end in:

$\lfloor 6 \rfloor$

Our result:

$\lfloor 3196 \rfloor$

ends in 6.

Good.

41. ESTIMATE BEFORE EXACT MULTIPLICATION

For:

$$68 \times 47$$

Estimate:

$$70 \times 50 = 3500$$

Exact answer:

$$3196$$

reasonable.

If you got 8196, estimate would immediately expose error.

42. TIMED CALCULATION RULE

For exam speed, mentally classify calculations:

Type A — instant

Examples:

$$25\% \text{ of } 80$$

$$2^{10}$$

Type B — 5-second transformation

Examples:

$$96 \times 104$$

$$72 \times 125$$

Type C — ordinary calculation

Examples:

$$47 \times 68$$

Type D — approximation only

Examples with widely spaced options.

Do not spend 30 seconds looking for a trick on a Type C calculation.

43. SPEED DRILL 1 — MULTIPLICATION

Try mentally.

1.

$$48 \times 25$$

2.

$$72 \times 125$$

3.

$$\sqrt[84]{49}$$

4.

$$\sqrt[63]{101}$$

5.

$$\sqrt[97]{103}$$

6.

$$\sqrt[98]{96}$$

7.

$$\sqrt[56]{75}$$

8.

$$\sqrt[48]{12.5}$$

9.

$$\sqrt[74]{18}$$

10.

$$\sqrt[999]{37}$$

44. SPEED DRILL 1 — ANSWERS

1.

$$\sqrt[1200]{}$$

2.

$$\sqrt[9000]{}$$

3.

$$\sqrt[84]{(50-1)} = 4200-84 = \sqrt[4116]{}$$

4.

$$\sqrt[6363]{}$$

5.

$$\sqrt[100^2-3^2]{9991}$$

6.

Using base 100:

deficiencies 2 and 4:

$$\sqrt[94]{8}$$

$$\sqrt[9408]{}$$

7.

$$\sqrt[56]{\frac{34}{100}} = 42 \times 100 = \sqrt[4200]{}$$

8.

$$\lfloor 48/8 \times 100 = \boxed{600} \rfloor$$

9.

$$\lfloor 74(20-2) = 1480-148 = \boxed{1332} \rfloor$$

10.

$$\lfloor 37000-37 = \boxed{36963} \rfloor$$

45. SPEED DRILL 2 — DIVISION

1.

$$\lfloor 875/25 \rfloor$$

2.

$$\lfloor 6250/125 \rfloor$$

3.

$$\lfloor 360/0.5 \rfloor$$

4.

$$\lfloor 84/0.25 \rfloor$$

5.

$$\lfloor 90/2.5 \rfloor$$

6.

$$\lfloor 75/1.25 \rfloor$$

7.

$$\lfloor 1728/36 \rfloor$$

8.

$$\lfloor 7560/24 \rfloor$$

9.

$$\lfloor 2700/15 \rfloor$$

10.

$$\lfloor 8400/50 \rfloor$$

46. SPEED DRILL 2 — ANSWERS

1.

$$\lfloor 875 \times 4/100 = \boxed{35} \rfloor$$

2.

$$\lfloor 6250 \times 8/1000 = \boxed{50} \rfloor$$

3.

$$\lfloor \boxed{720} \rfloor$$

4.

$$\boxed{336}$$

5.

$$90 \times \frac{25}{1} = \boxed{36}$$

6.

$$75 \times \frac{45}{1} = \boxed{60}$$

7.

$$\boxed{48}$$

8.

$$\boxed{315}$$

9.

$$2700 \div 5 = 540, \text{quad} 540 \div 3 = 180$$

$$\boxed{180}$$

10.

$$8400 \times 2 \div 100 = \boxed{168}$$

47. SPEED DRILL 3 — CANCELLATION

1.

$$\frac{48 \times 35}{12 \times 7}$$

2.

$$\frac{84 \times 45}{63 \times 20}$$

3.

$$\frac{75 \times 240}{25 \times 120}$$

4.

$$\frac{2^8 \times 3^5}{2^3 \times 3^4}$$

5.

$$\frac{18}{35} \times \frac{14}{27} \times \frac{15}{8}$$

48. SPEED DRILL 3 — ANSWERS

1

$$48 \div 12 = 4$$

$$35 \div 7 = 5$$

$$\boxed{20}$$

2

$$\lfloor 84/63=4/3 \rfloor$$

$$\lfloor 45/20=9/4 \rfloor$$

$$\lfloor \frac{43}{\times}\frac{94}{\times} = \boxed{3} \rfloor$$

3

$$\lfloor \frac{75}{\times}\frac{25}{\times}\frac{240}{\times}\frac{120}{\times} \rfloor$$

$$\lfloor =3\times 2 \rfloor$$

$$\lfloor =\boxed{6} \rfloor$$

4

$$\lfloor 2^5\cdot 3 = 32\times 3 = \boxed{96} \rfloor$$

5

As before:

$$\lfloor \boxed{\frac{12}{\times}} \rfloor$$

49. SPEED DRILL 4 — MIXED EXAM PRESSURE

Try these as quickly as possible.

Q1

$$\lfloor 1.25\times 64\times 0.8 \rfloor$$

Q2

$$\lfloor \frac{2.4\times 25}{\times}\{0.6\} \rfloor$$

Q3

$$\lfloor \frac{99^2-1}{\times}\{98\} \rfloor$$

Q4

$$\lfloor 48+52+37+63 \rfloor$$

Q5

$$\lfloor \frac{125\times 64}{\times}\{20\} \rfloor$$

Q6

$$\lfloor 75\times 32 \rfloor$$

Q7

$$\lfloor 998\times 52 \rfloor$$

Q8

$$\lfloor \frac{0.875\times 64}{\times}\{0.25\} \rfloor$$

Q9

$$\lfloor \frac{96\times 125}{\times}\{48\} \rfloor$$

Q10

$$\left[\frac{63}{0.125} \right]$$

50. SPEED DRILL 4 — ANSWERS

1

$$\left[1.25 \times 0.8 = 1 \right]$$

$$\left[\boxed{64} \right]$$

2

$$\left[2.4 / 0.6 = 4 \right]$$

$$\left[4 \times 25 = \boxed{100} \right]$$

3

$$\left[(99-1)(99+1)/98 \right]$$

$$\left[= 98 \times 100 / 98 \right]$$

$$\left[= \boxed{100} \right]$$

4

$$\left[48 + 52 = 100 \right]$$

$$\left[37 + 63 = 100 \right]$$

$$\left[\boxed{200} \right]$$

5

$$\left[125 \times 64 = 8000 \right]$$

$$\left[8000 / 20 = \boxed{400} \right]$$

6

$$\left[32 \times \frac{34}{100} \right]$$

$$\left[24 \times 100 = \boxed{2400} \right]$$

7

$$\left[52(1000-2) \right]$$

$$\left[52000 - 104 = \boxed{51896} \right]$$

8

$$\left[0.875 = \frac{7}{8} \right]$$

$$\left[0.25 = \frac{1}{4} \right]$$

$$\left[\frac{7}{8} \times 64 \times 4 \right]$$

$$\left[8 \text{ \textit{cancels with} } 64 \right]$$

$$\left[7 \times 8 \times 4? \right]$$

Careful:

$$\sqrt[4]{64/8}=8 \setminus]$$

$$\sqrt[4]{8\times7\times4} = \boxed{224} \setminus]$$

9

$$\sqrt[4]{96/48}=2 \setminus]$$

$$\sqrt[4]{2\times125} = \boxed{250} \setminus]$$

10

$$\sqrt[4]{0.125}=\frac{18}{\setminus]$$

$$\sqrt[4]{63\times8} = \boxed{504} \setminus]$$

51. FINAL CALCULATION STRATEGY — 7 QUESTIONS TO ASK

Before performing arithmetic, ask:

1. Can I **factor**?
2. Can I **cancel**?
3. Can I convert a decimal/% to a familiar fraction?
4. Is a number close to 10, 50, 100 or 1000?
5. Can I pair numbers to form 10/100/1000?
6. Is approximation enough?
7. What quick sanity check can verify the answer?

This is the calculation mindset to carry through the rest of aptitude.

52. UNIT 0 — COMPLETE REVISION SHEET

We have now finished all 12 chapters of **Calculation & Number Foundations**.

CHAPTER 1 — Arithmetic Foundations

Remember:

- BODMAS
 - sign rules
 - parity
 - multiplication tricks
 - basic identities
 - decimal powers
 - sanity checking
-

CHAPTER 2 — Fractions & Decimals

Remember:

$$\sqrt[4]{\frac{ab}{\setminus]$$

operations, comparison, cancellation, recurring decimals, terminating-decimal rule.

A reduced denominator gives a terminating decimal iff it contains only:

$$\sqrt[2]{2^m 5^n}$$

CHAPTER 3 — Fraction–Decimal–Percentage Conversion

Must recognize:

$$\sqrt[\frac{1}{2}]{2=50\%}$$

$$\sqrt[\frac{1}{4}]{4=25\%}$$

$$\sqrt[\frac{1}{8}]{8=12.5\%}$$

$$\sqrt[\frac{1}{5}]{5=20\%}$$

$$\sqrt[\frac{1}{20}]{20=5\%}$$

$$\sqrt[\frac{1}{25}]{25=4\%}$$

CHAPTER 4 — Divisibility

Must know rules for:

$$\sqrt[2,3,4,5,6,8,9,10,11]$$

plus common patterns:

$$\sqrt[12,15,18,25,50,100,125]$$

CHAPTER 5 — HCF & LCM

For two numbers:

$$\sqrt[\boxed{\text{HCF} \times \text{LCM} = ab}]$$

Same remainder:

$$\sqrt[\boxed{\text{HCF of differences}}]$$

Repeated events:

$$\sqrt[\boxed{\text{LCM}}]$$

Largest equal grouping:

$$\sqrt[\boxed{\text{HCF}}]$$

CHAPTER 6 — Prime Numbers

Factor count:

$$\sqrt[N=p_1^{a_1} \cdots p_k^{a_k}]$$

$$\sqrt[\boxed{d(N)=\prod(a_i+1)}]$$

Prime exponent in factorial:

$$\boxed{v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor}$$

Trailing zeros in $(n!)$:

count factors of 5.

CHAPTER 7 — Remainders

$$N = dq + r$$

$$0 \leq r < d$$

Modulo supports:

- addition
- subtraction
- multiplication
- powers

Huge powers !' cycles / Fermat / Euler.

CHAPTER 8 — Unit & Last Digits

Unit digit:

$$\pmod{10}$$

Last 2 digits:

$$\pmod{100}$$

Last 3:

$$\pmod{1000}$$

Core unit cycles:

- 2 !' 2,4,8,6
- 3 !' 3,9,7,1
- 7 !' 7,9,3,1
- 8 !' 8,4,2,6

Fixed:

$$\{0,1,5,6\}$$

CHAPTER 9 — Powers & Roots

$$a^m a^n = a^{m+n}$$

$$a^m / a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^{-n} = 1/a^n$$

$$a^{1/n} = \sqrt[n]{a}$$

CHAPTER 10 — Squares & Cubes

Memorize:

- squares 1–30
- cubes 1–10
- powers $(2^0)–(2^{16})$

High-speed identities:

$$(a+b)^2$$

$$(a-b)^2$$

$$(a+b)(a-b)$$

CHAPTER 11 — Approximation

Use:

- rounding
- bounds
- percentage error
- order-of-magnitude
- option elimination

Percentage error:

$$\boxed{\frac{|\text{approx} - \text{true}|}{|\text{true}|} \times 100}$$

CHAPTER 12 — Fast Calculation

Must know:

$$\times 25 = \times 100 / 4$$

$$\times 125 = \times 1000 / 8$$

$$\div 25 = \times 4 / 100$$

$$\div 125 = \times 8 / 1000$$

and:

$$\boxed{\text{Cancel before multiplying}}$$

53. UNIT 0 — MUST-MEMORIZE TABLE

Fractions

$$\frac{1}{2} = 0.5 = 50\%$$

$$\frac{1}{3} = 0.\overline{3} = 33\frac{1}{3}\%$$

$$\frac{1}{4} = 0.25 = 25\%$$

$$\frac{1}{5} = 0.2 = 20\%$$

$$\left[\frac{1}{8} = 0.125 = 12.5\% \right]$$

$$\left[\frac{1}{10} = 0.1 = 10\% \right]$$

$$\left[\frac{1}{20} = 0.05 = 5\% \right]$$

$$\left[\frac{1}{25} = 0.04 = 4\% \right]$$

54. UNIT 0 — MUST-MEMORIZE POWERS

$$\left[2^{10} = 1024 \right]$$

$$\left[2^{16} = 65536 \right]$$

$$\left[3^5 = 243 \right]$$

$$\left[3^6 = 729 \right]$$

$$\left[5^4 = 625 \right]$$

$$\left[5^5 = 3125 \right]$$

55. UNIT 0 — MUST-MEMORIZE MENTAL RULES

Square ending in 5

$$\left[65^2: 6 \times 7 = 42 \rightarrow 4225 \right]$$

Numbers around same center

$$\left[(a-b)(a+b) = a^2 - b^2 \right]$$

Base 100

Right block = 2 digits.

Base 1000

Right block = 3 digits.

56. UNIT 0 — TOP 15 TRAPS

1. (1) is neither prime nor composite.
 2. (2) is the only even prime.
 3. BODMAS division/multiplication go left to right.
 4. Never cancel terms across addition.
 5. $(-2^2) \neq (-2)^2$.
 6. Approximate decimals are not exact fractions.
 7. Same remainder !' differences.
 8. $\text{HCF} \times \text{LCM}$ formula directly applies to two numbers.
 9. Cycle remainder 0 means last cycle position.
 10. Euler requires coprimality.
 11. Last non-zero digit is not unit digit.
 12. Square can end only in 0,1,4,5,6,9.
-

- 13. Percentage error denominator is true/reference value.
 - 14. Avoid approximation during near-cancellation.
 - 15. Cancel before multiplying.
-

57. UNIT 0 — QUICK SELF-TEST

Try without notes.

Q1

$\left[\frac{3}{8} \right]$

as percentage?

Q2

HCF of 84 and 126?

Q3

Number of factors of:

$\left[2^3 3^2 \right]$

Q4

Trailing zeros in:

$\left[50! \right]$

Q5

Remainder of:

$\left[3^{100} \right]$

mod 7?

Q6

Unit digit of:

$\left[7^{2026} \right]$

Q7

Last two digits of:

$\left[99^{99} \right]$

Q8

$\left[75^2 \right]$

Q9

$\left[98 \times 102 \right]$

Q10

$\left[6250/125 \right]$

58. SELF-TEST ANSWERS

Q1

$$\boxed{37.5\%}$$

Q2

$$\boxed{42}$$

Q3

$$(3+1)(2+1) = \boxed{12}$$

Q4

$$10+2 = \boxed{12}$$

Q5

$$3^6 \equiv 1 \pmod{7}$$

$$100 \bmod 6 = 4$$

$$3^4 = 81 \equiv 4$$

$$\boxed{4}$$

Q6

$$2026 \bmod 4 = 2$$

7-cycle:

$$7, 9, 3, 1$$

$$\boxed{9}$$

Q7

$$99 \equiv -1 \pmod{100}$$

odd exponent:

$$\boxed{99}$$

Q8

$$7 \times 8 = 56$$

append 25:

$$\boxed{5625}$$

Q9

$$100^2 - 2^2 = \boxed{9996}$$

Q10

$$6250 \times 8 / 1000 = \boxed{50}$$

' CHAPTER 12 COMPLETE

We have now completed:

- fast multiplication constants '
- near-base multiplication '
- two-digit multiplication strategy '
- double-half '
- compensation '
- smart division '
- divisibility-assisted division '
- quotient estimation '
- ratio-preserving cancellation '
- fraction/percentage/power cancellation '
- reciprocal thinking '
- regrouping and pairing '
- multi-step simplification '
- error-control checks '
- timed calculation drills '
- Unit 0 revision '

Therefore:

' UNIT 0 — CALCULATION & NUMBER FOUNDATIONS COMPLETE

Progress

Completed: 12/76 **Unit 0:** ' COMPLETE **Next:** Ø=Ç Chapter 13 — **Ratio & Proportion** Ø=Ý%

Chapter 13 begins the main **Quantitative Aptitude** unit and will cover:

- ratio meaning
- equivalent ratios
- simplifying ratios
- dividing a quantity in a ratio
- combining ratios
- compound ratios
- direct proportion
- hidden ratio questions
- ratio change problems
- GATE-style and placement-style applications.

USER

next

CHAPTER 13 — RATIO & PROPORTION

PART 1 — RATIO BASICS, SIMPLIFICATION, EQUIVALENT RATIOS & DIVIDING QUANTITIES

Status: $\emptyset$ =Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ =Yes 1

Ratio is one of the most important quantitative aptitude topics because it appears inside:

- percentages
- averages
- mixtures
- partnership
- time & work
- speed
- probability
- DI
- geometry
- algebra

The main idea is:

A ratio compares quantities in the same scale.

1. WHAT IS A RATIO?

A ratio compares two quantities by division.

If:

$$[A:B=3:5]$$

it means:

$$[\frac{A}{B} = \frac{3}{5}]$$

So for every 3 parts of (A) , there are 5 parts of (B) .

2. RATIO DOES NOT GIVE ACTUAL VALUES

If:

$$[A:B=2:3]$$

possible values include:

$$[2,3]$$

$$[4,6]$$

$$[10,15]$$

$\backslash[200,300 \backslash]$

All have the same ratio.

So ratio tells **relative size**, not absolute size.

3. RATIO AS FRACTION

$\backslash[A:B=3:7 \backslash]$

means:

$\backslash[\boxed{\frac{A}{B}=\frac{3}{7}} \backslash]$

This connection is very important.

Memory

$\backslash[\boxed{a:b=\frac{a}{b}} \backslash]$

4. ORDER MATTERS

$\backslash[2:5 \backslash]$

is not the same as:

$\backslash[5:2 \backslash]$

Example:

If boys:girls = 2:5, then:

$\backslash[\frac{\text{boys}}{\text{girls}}=\frac{2}{5} \backslash]$

But girls:boys is:

$\backslash[\boxed{5:2} \backslash]$

5. SIMPLIFYING A RATIO

Simplify:

$\backslash[18:30 \backslash]$

HCF:

$\backslash[6 \backslash]$

Divide both by 6:

$\backslash[18:30=3:5 \backslash]$

Answer:

$\backslash[\boxed{3:5} \backslash]$

6. RATIO SIMPLIFICATION RULE

You may divide or multiply every term of a ratio by the same non-zero value.

$$\backslash[a:b=ka:kb \backslash]$$

Example:

$$\backslash[4:7=12:21 \backslash]$$

because both terms were multiplied by 3.

7. RATIO WITH THREE TERMS

Simplify:

$$\backslash[12:18:30 \backslash]$$

HCF:

$$\backslash[6 \backslash]$$

So:

$$\backslash[\boxed{2:3:5} \backslash]$$

8. UNITS MUST BE SAME

This is a major trap.

Find ratio:

$$\backslash[2\text{ m}:50\text{ cm} \backslash]$$

Do not write:

$$\backslash[2:50 \backslash]$$

Convert:

$$\backslash[2\text{ m}=200\text{ cm} \backslash]$$

So:

$$\backslash[200:50 \backslash]$$

$$\backslash[=\boxed{4:1} \backslash]$$

9. ANOTHER UNIT EXAMPLE

Ratio:

$$\backslash[3\text{ kg}:750\text{ g} \backslash]$$

Convert:

$$\backslash[3\text{ kg}=3000\text{ g} \backslash]$$

Then:

$$\backslash[3000:750 \backslash]$$

$$\backslash[=4:1 \backslash]$$

$$\backslash[\boxed{4:1} \backslash]$$

10. RATIO OF DECIMALS

Simplify:

$$\backslash[0.6:1.5 \backslash]$$

Multiply both by 10:

$$\backslash[6:15 \backslash]$$

Simplify:

$$\backslash[\boxed{2:5} \backslash]$$

11. RATIO OF FRACTIONS

Simplify:

$$\backslash[\frac{2}{3}:\frac{5}{6} \backslash]$$

Divide:

$$\backslash[\frac{2/3}{5/6} \backslash]$$

$$\backslash[=\frac{2}{3}\times\frac{6}{5} \backslash]$$

$$\backslash[=\frac{4}{5} \backslash]$$

So ratio:

$$\backslash[\boxed{4:5} \backslash]$$

12. FASTER FRACTION-RATIO METHOD

For:

$$\backslash[\frac{a}{b}:\frac{c}{d} \backslash]$$

multiply both terms by common denominator (bd) .

Example:

$$\backslash[\frac{2}{3}:\frac{5}{6} \backslash]$$

LCM of denominators 3 and 6:

$$\backslash[6 \backslash]$$

Multiply both by 6:

$$\backslash[4:5 \backslash]$$

Much faster.

13. RATIO OF PERCENTAGES

Simplify:

$$\backslash[20\%:35\% \backslash]$$

Both have same factor $\frac{1}{100}$.

So directly:

$$\frac{20}{35}$$

$$\boxed{\frac{4}{7}}$$

14. EQUIVALENT RATIOS

If:

$$A:B=3:4$$

then:

$$A=3k, \text{ } B=4k$$

for some positive scale k .

This is the most important representation in ratio questions.

15. WHY USE k ?

Suppose:

$$A:B=3:5$$

Then write:

$$A=3k, \text{ } B=5k$$

If later told:

$$A+B=64$$

then:

$$3k+5k=64$$

$$8k=64$$

$$k=8$$

So:

$$A=24, \text{ } B=40$$

16. DIVIDING A TOTAL IN A GIVEN RATIO

Question:

Divide 80 in ratio:

$$3:5$$

Total parts:

$$3+5=8$$

Value of one part:

$$\lfloor 80/8=10 \rfloor$$

So quantities:

$$\lfloor 3\times 10=30 \rfloor$$

$$\lfloor 5\times 10=50 \rfloor$$

Answer:

$$\lfloor \boxed{30, 50} \rfloor$$

17. MASTER FORMULA — DIVIDING A TOTAL

If total (T) is divided in ratio:

$$\lfloor a:b \rfloor$$

then first share:

$$\lfloor \boxed{\frac{a}{a+b}T} \rfloor$$

second share:

$$\lfloor \boxed{\frac{b}{a+b}T} \rfloor$$

18. THREE-WAY DIVISION

Divide:

$$\lfloor 240 \rfloor$$

in ratio:

$$\lfloor 2:3:5 \rfloor$$

Total parts:

$$\lfloor 2+3+5=10 \rfloor$$

One part:

$$\lfloor 240/10=24 \rfloor$$

Shares:

$$\lfloor 2(24)=48 \rfloor$$

$$\lfloor 3(24)=72 \rfloor$$

$$\lfloor 5(24)=120 \rfloor$$

Answer:

$$\lfloor \boxed{48, 72, 120} \rfloor$$

19. GIVEN DIFFERENCE INSTEAD OF TOTAL

Two numbers are in ratio:

$$\lfloor 3:7 \rfloor$$

and difference is:

$$\sqrt[4]{20}$$

Difference in ratio parts:

$$\sqrt[4]{7-3=4}$$

So:

$$\sqrt[4]{4k=20}$$

$$\sqrt[4]{k=5}$$

Numbers:

$$\sqrt[4]{15, \sqrt[4]{35}}$$

Answer:

$$\sqrt[4]{\boxed{15, \sqrt[4]{35}}}$$

20. FAST DIFFERENCE METHOD

If ratio is:

$$\sqrt[4]{a:b}$$

and actual difference is $\sqrt[4]{D}$:

$$\sqrt[4]{\boxed{k=\frac{D}{|b-a|}}}$$

Then numbers:

$$\sqrt[4]{ak, \sqrt[4]{bk}}$$

21. GIVEN SUM

Two numbers ratio:

$$\sqrt[4]{5:8}$$

sum:

$$\sqrt[4]{91}$$

Total ratio parts:

$$\sqrt[4]{13}$$

One part:

$$\sqrt[4]{91/13=7}$$

Numbers:

$$\sqrt[4]{35, \sqrt[4]{56}}$$

Answer:

$$\sqrt[4]{\boxed{35, \sqrt[4]{56}}}$$

22. GIVEN ONE QUANTITY

If:

$$A:B=4:7$$

and:

$$A=20$$

then:

$$4k=20$$

$$k=5$$

Thus:

$$B=7(5)=35$$

Answer:

$$\boxed{35}$$

23. FAST SCALING METHOD

If:

$$4:7$$

becomes:

$$20:x$$

4 was multiplied by:

$$5$$

So 7 also multiplied by 5:

$$x=35$$

This is often faster than writing (k) .

24. FIND THE MISSING TERM

$$6:15=10:x$$

Use:

$$\frac{6}{15}=\frac{10}{x}$$

Cross multiply:

$$6x=150$$

$$x=25$$

Answer:

$$\boxed{25}$$

25. PROPORTION

When two ratios are equal:

$$\text{\[a:b=c:d \]}$$

we say the quantities are in proportion.

Written:

$$\text{\[\boxed{\frac{a}{b}=\frac{c}{d}} \]}$$

Then:

$$\text{\[\boxed{ad=bc} \]}$$

This is the fundamental proportion rule.

26. EXTREMES AND MEANS

In:

$$\text{\[a:b=c:d \]}$$

- extremes = $\text{\(a,d\)}$
- means = $\text{\(b,c\)}$

Property:

$$\text{\[\boxed{\text{\{product of extremes\}}=\text{\{product of means\}}} \]}$$

That is:

$$\text{\[ad=bc \]}$$

27. EXAMPLE

$$\text{\[4:6=10:15 \]}$$

Check:

$$\text{\[4\times 15=60 \]}$$

$$\text{\[6\times 10=60 \]}$$

So proportion is valid.

28. FOURTH PROPORTIONAL

If:

$$\text{\[a:b=c:x \]}$$

then $\text{\(x\)}$ is called the fourth proportional.

Formula:

$$\text{\[\boxed{x=\frac{bc}{a}} \]}$$

Example:

$$\backslash[3:5=12:x \backslash]$$

$$\backslash[x=\frac{5\times 12}{3} \backslash]$$

$$\backslash[=\boxed{20} \backslash]$$

29. THIRD PROPORTIONAL

If:

$$\backslash[a:b=b:c \backslash]$$

then $\backslash(c\backslash)$ is the third proportional to $\backslash(a,b\backslash)$.

Since:

$$\backslash[\frac{ab}{b}=\frac{bc}{c} \backslash]$$

$$\backslash[b^2=ac \backslash]$$

Thus:

$$\backslash[\boxed{c=\frac{b^2}{a}} \backslash]$$

Example:

Third proportional to 4 and 6:

$$\backslash[c=\frac{36}{4}=9 \backslash]$$

$$\backslash[\boxed{9} \backslash]$$

30. MEAN PROPORTIONAL

If:

$$\backslash[a:x=x:b \backslash]$$

then:

$$\backslash[x^2=ab \backslash]$$

So:

$$\backslash[\boxed{x=\sqrt{ab}} \backslash]$$

for positive quantities.

Example:

Mean proportional between 4 and 9:

$$\backslash[x=\sqrt{36}=6 \backslash]$$

Answer:

$$\backslash[\boxed{6} \backslash]$$

31. RATIO TO FRACTION OF TOTAL

If:

$$\backslash A:B=2:3 \backslash$$

Total parts:

$$\backslash 5 \backslash$$

Therefore:

$$\backslash A=\frac{2}{5}\text{ of total} \backslash$$

$$\backslash B=\frac{3}{5}\text{ of total} \backslash$$

Important

Ratio term itself is **not** fraction of total until divided by sum of ratio terms.

32. RATIO TO PERCENTAGE OF TOTAL

If:

$$\backslash A:B=3:7 \backslash$$

Total:

$$\backslash 10 \backslash$$

Then:

$$\backslash A=\frac{3}{10}=30\% \backslash$$

$$\backslash B=\frac{7}{10}=70\% \backslash$$

33. RATIO TO PERCENTAGE COMPARISON

If:

$$\backslash A:B=4:5 \backslash$$

Then $\backslash(A\backslash)$ is what percent of $\backslash(B\backslash)$?

$$\backslash \frac{A}{B}=\frac{4}{5} \backslash$$

$$\backslash =80\% \backslash$$

So:

$$\backslash \boxed{A=80\%\text{ of }B} \backslash$$

34. IMPORTANT DIFFERENCE

For:

$$\backslash A:B=4:5 \backslash$$

A as % of B:

$$\backslash \frac{4}{5}\times 100=80\% \backslash$$

A as % of total:

$$\backslash \frac{4}{4+5}\times 100 = \frac{4}{9}\times 100 \approx 44.44\% \backslash$$

Do not confuse these.

35. IF A IS $x\%$ OF B

Suppose:

$$A = 60\% \text{ of } B$$

Then:

$$\frac{A}{B} = \frac{60}{100}$$

$$= \frac{3}{5}$$

So:

$$\boxed{A:B=3:5}$$

36. IF A IS 25% MORE THAN B

Suppose:

$$A = 125\% \text{ of } B$$

Then:

$$A:B = 125:100$$

$$= \boxed{5:4}$$

This becomes very important in Chapter 15.

37. IF A IS 20% LESS THAN B

Then:

$$A = 80\% \text{ of } B$$

So:

$$A:B = 80:100$$

$$= \boxed{4:5}$$

38. COMBINING TWO RATIOS

Suppose:

$$A:B = 2:3$$

and:

$$B:C = 4:5$$

We need the B terms equal.

First:

$$A:B = 2:3$$

Second:

$$\text{[B:C=4:5]}$$

LCM of:

$$\text{[3, 4]}$$

is 12.

Scale first ratio by 4:

$$\text{[A:B=8:12]}$$

Scale second by 3:

$$\text{[B:C=12:15]}$$

Therefore:

$$\text{[\boxed{A:B:C=8:12:15}]}$$

39. FAST COMBINING-RATIO METHOD

For:

$$\text{[A:B=a:b]}$$

and:

$$\text{[B:C=c:d]}$$

one valid combined ratio is:

$$\text{[\boxed{A:B:C=ac:bc:bd}]}$$

Example:

$$\text{[2:3, \quad 4:5]}$$

Then:

$$\text{[A:B:C = 2\cdot 4:3\cdot 4:3\cdot 5]}$$

$$\text{[=8:12:15]}$$

40. SIMPLIFY AFTER COMBINING

Suppose:

$$\text{[A:B=3:4]}$$

$$\text{[B:C=6:5]}$$

Using quick formula:

$$\text{[A:B:C = 3\cdot 6:4\cdot 6:4\cdot 5]}$$

$$\text{[=18:24:20]}$$

Simplify by 2:

$$\text{[\boxed{9:12:10}]}$$

41. THREE CONNECTED RATIOS

Suppose:

$$[A:B=2:3]$$

$$[B:C=4:5]$$

$$[C:D=10:7]$$

First combine first two:

$$[A:B:C=8:12:15]$$

Now $(C=15)$ and in $(C:D=10:7)$.

LCM of:

$$[15, 10=30]$$

Scale first:

$$[16:24:30]$$

Scale second:

$$[C:D=30:21]$$

So:

$$[\boxed{A:B:C:D=16:24:30:21}]$$

42. COMPOUND RATIO

Compound ratio of:

$$[a:b]$$

and:

$$[c:d]$$

is:

$$[\boxed{ac:bd}]$$

Example:

Compound ratio of:

$$[2:3]$$

and:

$$[4:5]$$

is:

$$[8:15]$$

43. DUPLICATE RATIO

Duplicate ratio of:

$$\boxed{a:b}$$

is:

$$\boxed{a^2:b^2}$$

Example:

$$\boxed{3:5}$$

duplicate ratio:

$$\boxed{9:25}$$

44. TRIPLICATE RATIO

Triplicate ratio:

$$\boxed{a^3:b^3}$$

Example:

$$\boxed{2:3}$$

triplicate ratio:

$$\boxed{8:27}$$

These terms are less common but can appear in placement aptitude.

45. SUB-DUPLICATE RATIO

For positive values, sub-duplicate ratio of:

$$\boxed{a:b}$$

is:

$$\boxed{\sqrt{a}:\sqrt{b}}$$

Example:

$$\boxed{9:16}$$

sub-duplicate ratio:

$$\boxed{3:4}$$

46. RATIO WHEN SAME QUANTITY IS ADDED

Suppose two numbers are in ratio:

$$\boxed{2:3}$$

Let numbers:

$$\boxed{2k, 3k}$$

If 5 is added to each and new ratio becomes:

$$\boxed{3:4}$$

Then:

$$\frac{2k+5}{3k+5} = \frac{3}{4}$$

Cross multiply:

$$4(2k+5) = 3(3k+5)$$

$$8k+20 = 9k+15$$

$$k = 5$$

Original numbers:

$$\boxed{10, 15}$$

47. WHY ADDING SAME NUMBER CHANGES RATIO

Suppose:

$$10:20 = 1:2$$

Add 10 to both:

$$20:30 = 2:3$$

So adding same quantity generally pushes ratio **toward 1:1**.

This gives a useful sanity check.

48. SUBTRACTING SAME NUMBER

If the smaller number stays positive, subtracting the same positive quantity generally pushes the ratio farther away from 1.

Example:

$$20:30 = 2:3$$

subtract 10:

$$10:20 = 1:2$$

Difference becomes proportionally more significant.

49. RATIO CHANGE QUESTION

Two numbers are in ratio:

$$5:7$$

If 10 is added to both, ratio becomes:

$$3:4$$

Let:

$$5k, 7k$$

Then:

$$\frac{5k+10}{7k+10} = \frac{3}{4}$$

$$\lfloor 20k+40=21k+30 \rfloor$$

$$\lfloor k=10 \rfloor$$

Numbers:

$$\lfloor \boxed{50, 70} \rfloor$$

50. RATIO AFTER MULTIPLYING ONE QUANTITY

If:

$$\lfloor A:B=2:5 \rfloor$$

and (A) is doubled:

$$\lfloor 2A:B \rfloor$$

Ratio:

$$\lfloor 4:5 \rfloor$$

If (B) is tripled:

$$\lfloor A:3B \rfloor$$

Ratio:

$$\lfloor 2:15 \rfloor$$

51. GENERAL MULTIPLIER RULE

If:

$$\lfloor A:B=a:b \rfloor$$

and:

- (A) multiplied by (x)
- (B) multiplied by (y)

new ratio:

$$\lfloor \boxed{ax:by} \rfloor$$

Then simplify if possible.

52. RATIO AFTER PERCENTAGE CHANGES — PREVIEW

If:

$$\lfloor A:B=2:3 \rfloor$$

and:

- A increases 20%
- B increases 10%

New ratio:

$$\lfloor 2(1.2):3(1.1) \rfloor$$

$$\text{\textbackslash}[=2.4:3.3\text{\textbackslash}]$$

Multiply by 10:

$$\text{\textbackslash}[24:33\text{\textbackslash}]$$

$$\text{\textbackslash}[=\text{\textbackslash}\boxed{8:11}\text{\textbackslash}]$$

Full percentage-change treatment comes in Chapter 15.

53. GATE-STYLE QUESTION 1

Two numbers are in ratio:

$$\text{\textbackslash}[3:5\text{\textbackslash}]$$

Their sum is 96. Find smaller number.

Total parts:

$$\text{\textbackslash}[8\text{\textbackslash}]$$

One part:

$$\text{\textbackslash}[96/8=12\text{\textbackslash}]$$

Smaller:

$$\text{\textbackslash}[3\times 12 = \text{\textbackslash}\boxed{36}\text{\textbackslash}]$$

54. GATE-STYLE QUESTION 2

Two numbers are in ratio:

$$\text{\textbackslash}[7:11\text{\textbackslash}]$$

Their difference is 36.

Difference parts:

$$\text{\textbackslash}[4\text{\textbackslash}]$$

One part:

$$\text{\textbackslash}[36/4=9\text{\textbackslash}]$$

Numbers:

$$\text{\textbackslash}[63,\text{\textbackslash}99\text{\textbackslash}]$$

Answer:

$$\text{\textbackslash}[\text{\textbackslash}\boxed{63,\text{\textbackslash}99}\text{\textbackslash}]$$

55. GATE-STYLE QUESTION 3

If:

$$\text{\textbackslash}[A:B=4:7\text{\textbackslash}]$$

and:

$$\text{[B:C=14:15]}$$

find:

$$\text{[A:B:C]}$$

Make \(\text{B}\) equal.

First ratio $\times 2$:

$$\text{[8:14]}$$

So:

$$\text{[\boxed{8:14:15}]}$$

56. GATE-STYLE QUESTION 4

$$\text{A:B = 3:4.}$$

If A increases by 50% and B decreases by 25%, find new ratio.

A multiplier:

$$\text{[1.5]}$$

B multiplier:

$$\text{[0.75]}$$

So:

$$\text{[3(1.5):4(0.75)]}$$

$$\text{[4.5:3]}$$

$$\text{[=\boxed{3:2}]}$$

57. PLACEMENT-STYLE QUESTION 1

Divide 840 in ratio:

$$\text{[3:4:5]}$$

Total parts:

$$\text{[12]}$$

One part:

$$\text{[840/12=70]}$$

Shares:

$$\text{[210,\ 280,\ 350]}$$

Answer:

$$\text{[\boxed{210,\ 280,\ 350}]}$$

58. PLACEMENT-STYLE QUESTION 2

If:

$$A:B=5:8$$

and $A=45$, find B .

Scale factor:

$$45/5=9$$

So:

$$B=8 \times 9$$

$$B=72$$

59. PLACEMENT-STYLE QUESTION 3

If:

$$A:B=2:3$$

and:

$$B:C=5:7$$

find:

$$A:C$$

Combine:

$$A:B:C=10:15:21$$

Therefore:

$$A:C=10:21$$

$$A:C=10:21$$

60. PLACEMENT-STYLE QUESTION 4

Two numbers are ratio:

$$4:9$$

If 5 is added to each, ratio becomes:

$$1:2$$

Let numbers:

$$4k, 9k$$

Then:

$$\frac{4k+5}{9k+5} = \frac{1}{2}$$

$$8k+10=9k+5$$

$$k=5$$

Numbers:

$$20, 45$$

61. COMMON TRAPS

& pTRAP 1 — Different units

Convert units before ratio.

& pTRAP 2 — Ratio vs fraction of total

For:

$$\backslash[2:3 \backslash]$$

first quantity is:

$$\backslash[\frac{2}{5} \backslash]$$

of total, not:

$$\backslash[\frac{2}{3} \backslash]$$

& pTRAP 3 — Ratio order

Boys:girls " girls:boys.

& pTRAP 4 — Combining ratios

The common quantity must represent the same number of parts before merging.

& pTRAP 5 — Addition to both quantities

Do not write:

$$\backslash[(a+x):(b+x) \backslash]$$

and simplistically cancel $\backslash(x\backslash)$.

Addition does not cancel.

& pTRAP 6 — Percentage change

A 20% increase means multiply by:

$$\backslash[1.2 \backslash]$$

not add 20 to ratio term unless the scale is actually 100.

62. MUST-MEMORIZE RATIO FORMULAS

If:

$$\backslash[A:B=a:b \backslash]$$

then:

$$\backslash[\boxed{A=ak, \quad B=bk} \backslash]$$

If total is $\backslash(T\backslash)$:

$$\boxed{A = \frac{a}{a+b}T}$$

$$\boxed{B = \frac{b}{a+b}T}$$

If difference is (D) :

$$\boxed{k = \frac{D}{a-b}}$$

Proportion:

$$\boxed{a:b=c:d \text{ iff } ad=bc}$$

Mean proportional:

$$\boxed{\sqrt{ab}}$$

63. FAST DECISION TREE

If ratio question gives:

Total?

Add ratio parts.

Difference?

Subtract ratio parts.

One actual value?

Find scale factor.

Two connected ratios?

Equalize common term.

Same number added/subtracted?

Write $(a \times x, b \times x)$.

Percentage changes?

Multiply ratio terms by change multipliers.

64. 5-MINUTE REVISION

Remember:

$$3:5$$

means:

$$3k, 5k$$

If total:

$$8k$$

If difference:

$$2k$$

If A as % of B:

$$\left[\frac{3}{5} = 60\% \right]$$

If A as fraction of total:

$$\left[\frac{3}{8} \right]$$

Connected ratios:

Equalize the middle term first.

PRACTICE — LEVEL 0

Q1

Simplify:

$$\left[24:36 \right]$$

Q2

Simplify:

$$\left[0.8:1.2 \right]$$

Q3

Divide 90 in ratio:

$$\left[2:3 \right]$$

Q4

Two numbers ratio 4:7 and sum 55. Find them.

Q5

Two numbers ratio 3:8 and difference 25. Find them.

LEVEL 1

Q6

Simplify:

$$\left[\frac{34}{58} \right]$$

Q7

If:

$$\left[A:B=3:7 \right]$$

and $(A=27)$, find (B) .

Q8

If:

$$\left[A:B=2:5, \text{quad } B:C=3:4 \right]$$

find $(A:B:C)$.

Q9

Find fourth proportional to:

$$\backslash [6,9,14 \backslash]$$

Q10

Find mean proportional between:

$$\backslash [16,\backslash 25 \backslash]$$

LEVEL 2

Q11

Divide 720 in ratio:

$$\backslash [3:5:7 \backslash]$$

Q12

If:

$$\backslash [A:B=4:5 \backslash]$$

what percent is A of B?

Q13

If A is 40% of B, find:

$$\backslash [A:B \backslash]$$

Q14

Two numbers are ratio:

$$\backslash [3:5 \backslash]$$

If 8 is added to both, ratio becomes:

$$\backslash [5:7 \backslash]$$

Find numbers.

Q15

If:

$$\backslash [A:B=5:6 \backslash]$$

A increases by 20%, B increases by 25%. Find new ratio.

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$\backslash [A:B=2:3,\quad B:C=4:5,\quad C:D=10:7 \backslash]$$

find $\backslash (A:D) \backslash$.

Q17

Two numbers are in ratio 7:9. If 12 is subtracted from each, ratio becomes 5:7. Find original numbers.

Q18

A sum of ₹ 126 is divided among A, B, C in ratio:

$$\boxed{2:3:4}$$

How much more does C receive than A?

Q19

A:B = 4:5. A increases by 25% and B decreases by 20%. Find new ratio.

Q20

A:B = 3:4. If A becomes 60 and B remains unchanged proportionally, what total corresponds to this same ratio?

ANSWERS**Q1**

$$\boxed{24:36=2:3}$$

$$\boxed{2:3}$$

Q2

$$\boxed{8:12=2:3}$$

$$\boxed{2:3}$$

Q3

Total parts:

$$\boxed{5}$$

One part:

$$\boxed{18}$$

Shares:

$$\boxed{36, 54}$$

Q4

$$\boxed{4k+7k=55}$$

$$\boxed{11k=55}$$

$$\boxed{k=5}$$

Numbers:

$$\boxed{20, 35}$$

Q5

Difference parts:

$$\boxed{5}$$

$$\sqrt[5]{k=25}$$

$$\sqrt[k=5]$$

Numbers:

$$\sqrt[15]{40}$$

Q6

Multiply both by 8:

$$\sqrt[6]{5}$$

$$\sqrt[6]{5}$$

Q7

$$\sqrt[3]{k=27}$$

$$\sqrt[k=9]$$

$$\sqrt[B=7(9)=63]$$

Q8

$$\sqrt[2]{5}, \sqrt[3]{4}$$

LCM of common terms 5 and 3:

$$\sqrt[15]$$

First $\times 3$:

$$\sqrt[6]{15}$$

Second $\times 5$:

$$\sqrt[15]{20}$$

So:

$$\sqrt[6]{15:20}$$

Q9

$$\sqrt[6]{9=14:x}$$

$$\sqrt[6]{x=126}$$

$$\sqrt[21]$$

Q10

$$\sqrt[16]{25} = \sqrt[400]{} = \sqrt[20]{}]$$

Q11

Total ratio:

$$\sqrt[15]$$

One part:

$$\sqrt[720/15]{48}$$

Shares:

$$\sqrt[144]{240}, \sqrt[336]$$

$$\boxed{144, 240, 336}$$

Q12

$$\frac{45}{100} = 80\%$$

Q13

$$40:100 = 2:5$$

Q14

Let numbers:

$$3k, 5k$$

$$\frac{3k+8}{5k+8} = \frac{57}{57}$$

$$21k+56=25k+40$$

$$16=4k$$

$$k=4$$

Numbers:

$$\boxed{12, 20}$$

Q15

New ratio:

$$5(1.2):6(1.25)$$

$$6:7.5$$

Multiply by 2:

$$12:15$$

$$= \boxed{4:5}$$

Q16

From earlier:

$$A:B:C:D=16:24:30:21$$

So:

$$\boxed{A:D=16:21}$$

Q17

Let numbers:

$$7k, 9k$$

After subtracting 12:

$$\frac{7k-12}{9k-12} = \frac{57}{57}$$

$$49k-84=45k-60$$

$$4k=24$$

$$k=6$$

Original numbers:

$$\boxed{42, 54}$$

Q18

Ratio:

$$2:3:4$$

Total parts:

$$9$$

One part:

$$1260/9=140$$

C " A:

$$(4-2)\times 140$$

$$\boxed{280}$$

Q19

A:

$$4\times 1.25=5$$

B:

$$5\times 0.8=4$$

New ratio:

$$\boxed{5:4}$$

Q20

A:B = 3:4.

If A = 60:

$$3k=60$$

$$k=20$$

B:

$$80$$

Total:

$$\boxed{140}$$

CHAPTER 13 STATUS

Covered so far:

- ratio meaning '
- simplification '
- unit conversion '
- decimal/fraction/% ratios '
- equivalent ratios '

- dividing a total '
- sum/difference/one-value questions '
- proportion '
- fourth/third/mean proportional '
- ratio as fraction/% of total '
- combining ratios '
- compound/duplicate ratios '
- ratio changes '
- same amount added/subtracted '
- percentage-change preview '
- GATE/placement practice '

But **Chapter 13 is not complete yet.**

Next we need:

- advanced ratio transformations
- componendo and dividendo
- inverse ratios
- ratios involving reciprocals
- hidden ratios from equations
- ratio from sums/differences
- age/mixture/work-style ratio previews
- advanced multi-step GATE/placement problems

Progress

Completed: 12/76 **Current:** ~~Chapter 13~~ — Ratio & Proportion **Next:** Chapter 13 Part 2 — **Advanced ratio transformations & hidden-ratio problems**

USER

next

ASSISTANT

CHAPTER 13 — RATIO & PROPORTION

PART 2 — ADVANCED RATIO TRANSFORMATIONS, INVERSE RATIOS & HIDDEN-RATIO PROBLEMS

Status: ~~0=Not~~ **PROGRESS** **GATE:** High **Placements:** High **Priority:** ~~0=Not~~ **1**

This part focuses on questions where the ratio is **not given directly**.

You may instead be given:

- equations,
- sums and differences,

- reciprocals,
- transformed quantities,
- percentages,
- or relationships between ratios.

The main skill is:

Convert the condition into a simple ratio as early as possible.

1. RATIO FROM AN EQUATION

Suppose:

$$3A=5B$$

Find:

$$A:B$$

Rearrange:

$$\frac{A}{B}=\frac{5}{3}$$

Therefore:

$$\boxed{A:B=5:3}$$

Important

If:

$$xA=yB$$

then:

$$\boxed{A:B=y:x}$$

The coefficients reverse.

2. EXAMPLE

If:

$$7x=4y$$

then:

$$\frac{xy}{}=\frac{4}{7}$$

So:

$$\boxed{x:y=4:7}$$

3. COMMON COEFFICIENT TRAP

If:

$$2A=3B$$

some students write:

$$\boxed{A:B=2:3}$$

Wrong.

Correct:

$$\boxed{A=\frac{2}{3}B}$$

So:

$$\boxed{A:B=2:3}$$

4. RATIO FROM TWO EQUATIONS

Suppose:

$$2A=3B$$

and:

$$4B=5C$$

First:

$$A:B=3:2$$

Second:

$$B:C=5:4$$

Combine.

Make B equal:

$$A:B=15:10$$

$$B:C=10:8$$

Thus:

$$\boxed{A:B:C=15:10:8}$$

5. RATIO FROM SUMS

Suppose:

$$A+B=60$$

and:

$$A:B=2:3$$

We already know:

$$A=2k, \quad B=3k$$

Then:

$$5k=60$$

$$k=12$$

So:

$$\boxed{A=24, B=36}$$

6. HIDDEN RATIO FROM SUM AND DIFFERENCE

Suppose:

$$A+B=50$$

and:

$$A-B=10$$

Add equations:

$$2A=60$$

$$A=30$$

Then:

$$B=20$$

So:

$$\boxed{A:B=3:2}$$

7. FAST FORMULA FROM SUM AND DIFFERENCE

If:

$$A+B=S$$

and:

$$A-B=D$$

then:

$$A=\frac{S+D}{2}$$

$$B=\frac{S-D}{2}$$

Therefore directly:

$$\boxed{A:B=(S+D):(S-D)}$$

8. EXAMPLE

If sum = 84 and difference = 12:

$$A:B=(84+12):(84-12)$$

$$=96:72$$

$$\boxed{4:3}$$

No need to find A and B separately.

9. RATIO FROM $(A+B)$ AND $(A-B)$

Suppose:

$$\backslash [A+B:A-B=5:1 \backslash]$$

Find:

$$\backslash [A:B \backslash]$$

Let:

$$\backslash [A+B=5k \backslash]$$

$$\backslash [A-B=k \backslash]$$

Add:

$$\backslash [2A=6k \backslash]$$

$$\backslash [A=3k \backslash]$$

Subtract:

$$\backslash [2B=4k \backslash]$$

$$\backslash [B=2k \backslash]$$

Therefore:

$$\backslash [\boxed{A:B=3:2} \backslash]$$

10. GENERAL TRANSFORMATION

If:

$$\backslash [\frac{A+B}{A-B}=\frac{m}{n} \backslash]$$

then:

$$\backslash [n(A+B)=m(A-B) \backslash]$$

$$\backslash [nA+nB=mA-mB \backslash]$$

Bring terms together:

$$\backslash [(m-n)A=(m+n)B \backslash]$$

Therefore:

$$\backslash [\boxed{A:B=(m+n):(m-n)} \backslash]$$

provided $(m \neq n)$.

11. EXAMPLE

If:

$$\backslash [\frac{A+B}{A-B}=\frac{7}{4} \backslash]$$

then:

$$\backslash [A:B=(7+4):(7-4) \backslash]$$

$$\backslash [=\boxed{11:3} \backslash]$$

12. COMPONENDO

If:

$$\left[\frac{a}{b} = \frac{c}{d} \right]$$

then:

$$\left[\boxed{\frac{a+b}{b} = \frac{c+d}{d}} \right]$$

This is called **componendo**.

It simply means adding denominator to numerator on both sides.

13. DIVIDENDO

If:

$$\left[\frac{a}{b} = \frac{c}{d} \right]$$

then:

$$\left[\boxed{\frac{a-b}{b} = \frac{c-d}{d}} \right]$$

This is called **dividendo**.

14. COMPONENDO AND DIVIDENDO

Combining both:

If:

$$\left[\frac{a}{b} = \frac{c}{d} \right]$$

then:

$$\left[\boxed{\frac{a+b}{a-b} = \frac{c+d}{c-d}} \right]$$

This can simplify ratio equations very quickly.

15. WHY IT WORKS

Suppose:

$$\left[\frac{a}{b} = k \right]$$

Then:

$$\left[a = kb \right]$$

So:

$$\left[\frac{a+b}{a-b} = \frac{kb+b}{kb-b} \right]$$

$$\left[= \frac{k+1}{k-1} \right]$$

The same transformation applies to both equal ratios.

16. EXAMPLE USING COMPONENDO-DIVIDENDO

If:

$$\frac{x+y}{x-y} = \frac{5}{3}$$

find $(x:y)$.

Use the reverse form:

$$x:y = (5+3):(5-3)$$

$$= 8:2$$

$$= \boxed{4:1}$$

17. ANOTHER EXAMPLE

If:

$$\frac{x+y}{x-y} = \frac{9}{4}$$

then:

$$x:y = (9+4):(9-4)$$

$$= \boxed{13:5}$$

18. INVERSE RATIO

If:

$$A:B = 3:5$$

then inverse ratio is:

$$\boxed{5:3}$$

This matters when one quantity is inversely proportional to another.

Example:

If time ratio is:

$$3:5$$

for same distance, speed ratio is:

$$\boxed{5:3}$$

because:

$$\text{speed} \propto \frac{1}{\text{time}}$$

19. RATIO OF RECIPROCAL

If:

$$A:B = 2:7$$

then:

$$\left[\frac{1A}{1B} \right]$$

becomes:

$$\left[\frac{12}{17} \right]$$

Multiply by 14:

$$\left[7:2 \right]$$

Therefore:

$$\left[\boxed{\frac{1A}{1B}=7:2} \right]$$

So reciprocal ratio reverses.

20. GENERAL RECIPROCAL RULE

If:

$$\left[A:B=a:b \right]$$

then:

$$\left[\boxed{\frac{1A}{1B}=b:a} \right]$$

21. RATIO OF SQUARES

If:

$$\left[A:B=2:3 \right]$$

then:

$$\left[A^2:B^2 = 4:9 \right]$$

General:

$$\left[\boxed{A^n:B^n=a^n:b^n} \right]$$

22. RATIO OF ROOTS

If:

$$\left[A:B=16:25 \right]$$

then:

$$\left[\sqrt{A}:\sqrt{B} = 4:5 \right]$$

General for positive quantities:

$$\left[\boxed{\sqrt{A}:\sqrt{B}=\sqrt{a}:\sqrt{b}} \right]$$

23. RATIO FROM PRODUCT RELATION

Suppose:

$$\left[A^2=B C \right]$$

and:

$$\backslash B:C=4:9 \backslash$$

Find:

$$\backslash A:B \backslash$$

Let:

$$\backslash B=4k,\quad C=9k \backslash$$

Then:

$$\backslash A^2=36k^2 \backslash$$

$$\backslash A=6k \backslash$$

So:

$$\backslash A:B=6:4 \backslash$$

$$\backslash =\boxed{3:2} \backslash$$

This is mean-proportional structure.

24. RATIO FROM $(A^2:B^2)$

If:

$$\backslash A^2:B^2=49:81 \backslash$$

for positive (A,B) :

$$\backslash A:B=7:9 \backslash$$

$$\backslash \boxed{7:9} \backslash$$

25. RATIO FROM CUBES

If:

$$\backslash A^3:B^3=8:27 \backslash$$

then:

$$\backslash A:B=2:3 \backslash$$

$$\backslash \boxed{2:3} \backslash$$

26. HIDDEN RATIO FROM FRACTIONAL EQUATION

Suppose:

$$\backslash \frac{A}{B}+\frac{B}{A}=\frac{13}{6} \backslash$$

Find possible ratio $(A:B)$.

Let:

$$\backslash \frac{A}{B}=x \backslash$$

Then:

$$x + \frac{1}{6}x = \frac{13}{6}$$

Multiply by $(6x)$:

$$6x^2 + 6 = 13x$$

$$6x^2 - 13x + 6 = 0$$

Factor:

$$(3x-2)(2x-3)=0$$

So:

$$x = \frac{2}{3}$$

or:

$$x = \frac{3}{2}$$

Thus:

$$\boxed{A:B=2:3 \text{ or } 3:2}$$

Symmetry explains the two possibilities.

27. RATIO FROM PERCENT COMPARISON

If A is 20% more than B:

$$A = 1.2B$$

So:

$$A:B = 1.2:1$$

$$= 12:10$$

$$= \boxed{6:5}$$

28. REVERSE PERCENTAGE COMPARISON

If A is 20% more than B, then B is **not** 20% less than A.

Since:

$$A:B = 6:5$$

B as percentage of A:

$$\frac{5}{6} \times 100 = 83\frac{1}{3}\%$$

So B is:

$$83\frac{1}{3}\%$$

less than A.

This is a major aptitude trap.

29. GENERAL MORE/LESS RATIO

If A is $(p\%)$ more than B:

$$[A:B=(100+p):100]$$

Simplify.

If A is $(p\%)$ less than B:

$$[A:B=(100-p):100]$$

30. EXAMPLE

A is 25% more than B.

$$[A:B=125:100]$$

$$[=\boxed{5:4}]$$

Then B is what percent less than A?

Difference = 1 part out of A's 5 parts:

$$[\frac{1}{5} \times 100 = \boxed{20\%}]$$

31. RATIO AFTER EQUAL ADDITION — FAST EQUATION

Original:

$$[A:B=a:b]$$

After adding (x) :

$$[\frac{a+x}{b+x} = \frac{m}{n}]$$

Cross multiply:

$$[n(a+x)=m(b+x)]$$

Then solve (k) .

This is the standard setup.

32. SAME NUMBER ADDED TO THREE QUANTITIES

Suppose:

$$[A:B:C=2:3:4]$$

and 5 is added to each.

New ratio:

$$[(2k+5):(3k+5):(4k+5)]$$

You **cannot** determine a unique new ratio unless (k) is known.

This is important.

Ratios alone do not fix actual values.

33. RATIO AFTER EQUAL MULTIPLICATION

If both A and B are multiplied by same non-zero constant (c) :

$$[cA:cB]$$

ratio remains:

$$[A:B]$$

So:

$$[\boxed{\text{same scaling preserves ratio}}]$$

34. RATIO AFTER EQUAL ADDITION DOES NOT STAY SAME

If:

$$[A:B=2:3]$$

then:

$$[2k+x:3k+x]$$

is generally not 2:3.

Only equal multiplication/division preserves ratio.

35. RATIO FROM TWO PERCENTAGES OF A COMMON TOTAL

If A is 30% of total and B is 45% of total:

$$[A:B=30:45]$$

$$[=\boxed{2:3}]$$

The common total cancels.

36. RATIO FROM FRACTIONS OF A COMMON QUANTITY

If:

$$[A=\frac{3}{5}T]$$

and:

$$[B=\frac{7}{10}T]$$

then:

$$[A:B = \frac{3}{5}:\frac{7}{10}]$$

Multiply by 10:

$$[6:7]$$

$$[\boxed{6:7}]$$

37. RATIO FROM DIFFERENT BASES

Suppose:

A is 20% of X, B is 30% of Y.

You cannot find A:B unless relationship between X and Y is known.

This is a hidden-data trap.

38. RATIO FROM PRICE × QUANTITY

Suppose:

$$\text{Expenditure} = \text{Price} \times \text{Quantity}$$

If price ratio:

$$2:3$$

and quantity ratio:

$$5:4$$

then expenditure ratio:

$$2 \cdot 5 : 3 \cdot 4$$

$$= 10:12$$

$$= \boxed{5:6}$$

This is compound-ratio reasoning.

39. GENERAL PRODUCT RATIO

If:

$$A_1:A_2=a:b$$

and:

$$B_1:B_2=c:d$$

then:

$$A_1B_1:A_2B_2 = \boxed{ac:bd}$$

Very useful in:

- work
 - speed
 - area
 - cost
 - production
 - density-style questions
-

40. QUOTIENT RATIO

If:

$$\backslash[A_1:A_2=a:b \backslash]$$

and:

$$\backslash[B_1:B_2=c:d \backslash]$$

then:

$$\backslash[\frac{A_1}{B_1}:\frac{A_2}{B_2} \backslash]$$

equals:

$$\backslash[\frac{ac}{bd} \backslash]$$

Multiply by $\backslash(cd\backslash)$:

$$\backslash[\boxed{ad:bc} \backslash]$$

41. EXAMPLE

Speed:

$$\backslash[\frac{\text{distance}}{\text{time}} \backslash]$$

Suppose distance ratio:

$$\backslash[3:4 \backslash]$$

time ratio:

$$\backslash[2:5 \backslash]$$

Then speed ratio:

$$\backslash[\frac{3}{2}:\frac{4}{5} \backslash]$$

$$\backslash[=15:8 \backslash]$$

$$\backslash[\boxed{15:8} \backslash]$$

42. RATIO FROM AVERAGE-LIKE RELATION

Suppose average of A and B is 20:

$$\backslash[\frac{A+B}{2}=20 \backslash]$$

So:

$$\backslash[A+B=40 \backslash]$$

If:

$$\backslash[A:B=3:5 \backslash]$$

then:

$$\backslash[8k=40 \backslash]$$

$$\backslash[k=5 \backslash]$$

So:

$$\backslash[\boxed{15, \ 25} \backslash]$$

This style will return in averages.

43. AGE-STYLE RATIO PREVIEW

Present age ratio:

$$\backslash[A:B=3:5 \backslash]$$

After 10 years:

$$\backslash[\frac{3k+10}{5k+10} \backslash]$$

This is exactly the “same number added” ratio pattern.

So age problems are ratio equations with time added/subtracted.

44. MIXTURE-STYLE RATIO PREVIEW

Milk:water:

$$\backslash[3:2 \backslash]$$

If water is added, only second component changes.

New ratio:

$$\backslash[3k:(2k+x) \backslash]$$

Mixture questions are fundamentally ratio-change problems.

45. WORK-STYLE RATIO PREVIEW

If efficiencies:

$$\backslash[A:B=2:3 \backslash]$$

then for same work, times are inverse:

$$\backslash[\boxed{3:2} \backslash]$$

This will be central in Time & Work.

46. SPEED-TIME RATIO PREVIEW

For same distance:

$$\backslash[S_1:S_2=4:5 \backslash]$$

then:

$$\backslash[T_1:T_2=5:4 \backslash]$$

Inverse relationship.

47. AREA RATIO PREVIEW

If sides of similar figures are:

$$\sqrt[2]{2:3}$$

area ratio:

$$\sqrt[2]{2^2:3^2}$$

$$\sqrt[2]{\boxed{4:9}}$$

Volume ratio:

$$\sqrt[2]{2^3:3^3}$$

$$\sqrt[2]{\boxed{8:27}}$$

This links ratios with geometry.

48. GATE-STYLE QUESTION 1

If:

$$\sqrt[2]{5A=8B}$$

find:

$$\sqrt[2]{A:B}$$

$$\sqrt[2]{\boxed{8:5}}$$

49. GATE-STYLE QUESTION 2

If:

$$\sqrt[2]{\frac{A+B}{A-B}=\frac{7}{3}}$$

find:

$$\sqrt[2]{A:B}$$

Use:

$$\sqrt[2]{(7+3):(7-3)}$$

$$\sqrt[2]{10:4}$$

$$\sqrt[2]{\boxed{5:2}}$$

50. GATE-STYLE QUESTION 3

If:

$$\sqrt[2]{A:B=3:4}$$

find:

$$\sqrt[2]{(A+B):(A-B)}$$

Take:

$$\sqrt[2]{A=3k,\ B=4k}$$

Then $(A-B)$ is negative.

So:

$$(A+B):(A-B) = 7k:-k$$

$$= \boxed{-7:1}$$

Trap

Signs matter. Ratios can be negative algebraically, though ordinary physical quantity ratios are usually positive.

51. GATE-STYLE QUESTION 4

If:

$$A:B=4:9$$

find:

$$\sqrt{A}:\sqrt{B}$$

$$= \boxed{2:3}$$

52. PLACEMENT-STYLE QUESTION 1

A is 40% more than B. Find A:B.

$$140:100$$

$$= \boxed{7:5}$$

53. PLACEMENT-STYLE QUESTION 2

A is 20% less than B. Find A:B.

$$80:100$$

$$= \boxed{4:5}$$

54. PLACEMENT-STYLE QUESTION 3

If:

$$x:y=3:5$$

find:

$$\frac{1}{x}:\frac{1}{y}$$

Reverse:

$$\boxed{5:3}$$

55. PLACEMENT-STYLE QUESTION 4

If:

$$\backslash[A:B=2:3 \backslash]$$

and:

$$\backslash[C:D=5:7 \backslash]$$

find:

$$\backslash[AC:BD \backslash]$$

$$\backslash[2\cdot 5:3\cdot 7 \backslash]$$

$$\backslash[=\boxed{10:21} \backslash]$$

56. ADVANCED QUESTION — SUM/DIFFERENCE RATIO

If:

$$\backslash[A+B:A-B=9:5 \backslash]$$

find $\backslash(A:B)\backslash$.

Let:

$$\backslash[A+B=9k \backslash]$$

$$\backslash[A-B=5k \backslash]$$

Add:

$$\backslash[2A=14k \backslash]$$

$$\backslash[A=7k \backslash]$$

Subtract:

$$\backslash[2B=4k \backslash]$$

$$\backslash[B=2k \backslash]$$

Therefore:

$$\backslash[\boxed{7:2} \backslash]$$

57. ADVANCED QUESTION — PRODUCT RELATION

If:

$$\backslash[A:B=2:3 \backslash]$$

and:

$$\backslash[B:C=4:5 \backslash]$$

find:

$$\backslash[AC:B^2 \backslash]$$

Combined ratio:

$$\backslash[A:B:C=8:12:15 \backslash]$$

Then:

$$\backslash [AC:B^2 = (8)(15):12^2 \backslash]$$

$$\backslash [120:144 \backslash]$$

$$\backslash [=\boxed{5:6} \backslash]$$

58. ADVANCED QUESTION — RECIPROCAL SUM

If:

$$\backslash [A:B=2:3 \backslash]$$

find:

$$\backslash [\left(\frac{1}{A}+\frac{1}{B}\right):\frac{1}{A} \backslash]$$

Let:

$$\backslash [A=2k, \ B=3k \backslash]$$

Then:

$$\backslash [\frac{1}{A}+\frac{1}{B} = \frac{1}{2k}+\frac{1}{3k} = \frac{5}{6k} \backslash]$$

and:

$$\backslash [\frac{1}{A}=\frac{1}{2k}=\frac{5}{6k} \backslash]$$

Ratio:

$$\backslash [5:3 \backslash]$$

$$\backslash [\boxed{5:3} \backslash]$$

59. ADVANCED QUESTION — TRANSFORMED RATIO

If:

$$\backslash [A:B=5:7 \backslash]$$

find:

$$\backslash [(2A+B):(A+2B) \backslash]$$

Take:

$$\backslash [A=5k, \quad B=7k \backslash]$$

First:

$$\backslash [2A+B=10k+7k=17k \backslash]$$

Second:

$$\backslash [A+2B=5k+14k=19k \backslash]$$

Therefore:

$$\backslash [\boxed{17:19} \backslash]$$

This style appears often in placement tests.

60. GENERAL LINEAR-EXPRESSION RATIO

If:

$$\backslash[A:B=a:b \backslash]$$

then for expressions:

$$\backslash[pA+qB \backslash]$$

and:

$$\backslash[rA+sB \backslash]$$

substitute:

$$\backslash[A=ak,\quad B=bk \backslash]$$

Then:

$$\backslash[\boxed{(pA+qB):(rA+sB) = (pa+qb):(ra+sb) } \backslash]$$

The common $\backslash(k)$ cancels.

61. EXAMPLE

If:

$$\backslash[A:B=3:4 \backslash]$$

find:

$$\backslash[(3A+2B):(2A+B) \backslash]$$

Substitute ratio terms:

$$\backslash[(3\cdot 3+2\cdot 4):(2\cdot 3+4) \backslash]$$

$$\backslash[(9+8):(6+4) \backslash]$$

$$\backslash[17:10 \backslash]$$

$$\backslash[\boxed{17:10} \backslash]$$

62. COMMON TRAPS

& pTRAP 1 — Coefficients reverse

$$\backslash[3A=5B \rightarrow A:B=5:3 \backslash]$$

& pTRAP 2 — Reciprocal ratio reverses

$$\backslash[A:B=a:b \rightarrow 1/A:1/B=b:a \backslash]$$

& pTRAP 3 — Same addition does not preserve ratio

Equal multiplication does. Equal addition generally does not.

& pTRAP 4 — “20% more” reverse comparison

20% more one way is not 20% less the other way.

& pTRAP 5 — Product ratios

Multiply corresponding ratio terms.

& pTRAP 6 — Ratio from different underlying totals

You cannot cancel a common total unless it really is common.

63. MUST-MEMORIZE ADVANCED FORMULAS

If:

$$\backslash[xA=yB \backslash]$$

then:

$$\backslash[\boxed{A:B=y:x} \backslash]$$

If:

$$\backslash[A+B=S,\quad A-B=D \backslash]$$

then:

$$\backslash[\boxed{A:B=(S+D):(S-D)} \backslash]$$

If:

$$\backslash[\frac{A+B}{A-B}=\frac{m}{n} \backslash]$$

then:

$$\backslash[\boxed{A:B=(m+n):(m-n)} \backslash]$$

Reciprocal ratio:

$$\backslash[\boxed{1/A:1/B=B:A} \backslash]$$

64. 5-MINUTE REVISION

When ratio is hidden:

Equation?

$$\backslash[xA=yB \rightarrow A:B=y:x \backslash]$$

Sum and difference?

$$\backslash[(S+D):(S-D) \backslash]$$

$\backslash((A+B)/(A-B))\backslash$?

Use:

$$\backslash[(m+n):(m-n) \backslash]$$

Reciprocal?

Reverse ratio.

Product?

Multiply component ratios.

Linear expressions?

Substitute:

$$\begin{cases} A=ak, \\ B=bk \end{cases}$$

PRACTICE — LEVEL 0

Q1

If:

$$\begin{cases} 4A=7B \end{cases}$$

find $(A:B)$.

Q2

If:

$$\begin{cases} A:B=3:8 \end{cases}$$

find:

$$\begin{cases} 1/A:1/B \end{cases}$$

Q3

If:

$$\begin{cases} A:B=4:5 \end{cases}$$

find:

$$\begin{cases} A^2:B^2 \end{cases}$$

Q4

If:

$$\begin{cases} A+B=70, \\ A-B=14 \end{cases}$$

find $(A:B)$.

Q5

If:

$$\begin{cases} \frac{A+B}{A-B}=3 \end{cases}$$

find $(A:B)$.

LEVEL 1

Q6

If:

$$\begin{cases} A:B=2:5 \end{cases}$$

find:

$$\sqrt{(A+B):(B-A)}$$

Q7

A is 50% more than B. Find A:B.

Q8

If A is 25% less than B, find A:B.

Q9

If:

$$\sqrt{A:B=3:4, C:D=5:7}$$

find:

$$\sqrt{AC:BD}$$

Q10

If:

$$\sqrt{A:B=2:3}$$

find:

$$\sqrt{(2A+B):(A+2B)}$$

LEVEL 2

Q11

If:

$$\sqrt{\frac{x+y}{x-y}=\frac{8}{5}}$$

find $\sqrt{x:y}$.

Q12

If:

$$\sqrt{A^2:B^2=121:169}$$

find positive ratio A:B.

Q13

If:

$$\sqrt{A:B=3:5}$$

and:

$$\sqrt{B:C=10:7}$$

find:

$$\sqrt{AC:B^2}$$

Q14

If:

$$\backslash [A:B=4:7 \backslash]$$

find:

$$\backslash [(3A+2B):(2A+3B) \backslash]$$

Q15

If:

$$\backslash [2A=3B, \text{quad} 5B=4C \backslash]$$

find $\backslash (A:B:C) \backslash$.

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$\backslash [A+B:A-B=11:3 \backslash]$$

find $A:B$.

Q17

If A is 25% more than B, by what percentage is B less than A?

Q18

If:

$$\backslash [A:B=2:3 \backslash]$$

and:

$$\backslash [B:C=4:5 \backslash]$$

find:

$$\backslash [(A+C):B \backslash]$$

Q19

If:

$$\backslash [A:B=3:4 \backslash]$$

find:

$$\backslash [\left(\frac{1}{A} + \frac{1}{B} \right) : \left(\frac{1}{A} - \frac{1}{B} \right) \backslash]$$

Q20

If:

$$\backslash [A:B=5:7 \backslash]$$

find:

$$\backslash [(A^2+B^2):(A+B)^2 \backslash]$$

ANSWERS

Q1

$$\lfloor 4A=7B \rfloor$$

$$\lfloor \boxed{A:B=7:4} \rfloor$$

Q2

Reverse:

$$\lfloor \boxed{8:3} \rfloor$$

Q3

$$\lfloor 16:25 \rfloor$$

$$\lfloor \boxed{16:25} \rfloor$$

Q4

$$\lfloor A:B=(70+14):(70-14) \rfloor$$

$$\lfloor 84:56 \rfloor$$

$$\lfloor =\boxed{3:2} \rfloor$$

Q5

Write:

$$\lfloor \frac{A+B}{A-B}=\frac{31}{1} \rfloor$$

Then:

$$\lfloor A:B=(3+1):(3-1) \rfloor$$

$$\lfloor 4:2 \rfloor$$

$$\lfloor =\boxed{2:1} \rfloor$$

Q6

Take:

$$\lfloor A=2k, \ B=5k \rfloor$$

Then:

$$\lfloor A+B=7k \rfloor$$

$$\lfloor B-A=3k \rfloor$$

So:

$$\lfloor \boxed{7:3} \rfloor$$

Q7

$$\lfloor 150:100 = \boxed{3:2} \rfloor$$

Q8

$$\lfloor 75:100 = \boxed{3:4} \rfloor$$

Q9

$$\sqrt[3]{5 \cdot 4 \cdot 7}$$

$$\boxed{15:28}$$

Q10

Substitute:

$$A=2k, B=3k$$

$$2A+B=4k+3k=7k$$

$$A+2B=2k+6k=8k$$

$$\boxed{7:8}$$

Q11

$$x:y=(8+5):(8-5)$$

$$\boxed{13:3}$$

Q12

$$\sqrt{121}:\sqrt{169}$$

$$\boxed{11:13}$$

Q13

Combine:

$$A:B=3:5$$

$$B:C=10:7$$

Scale first $\times 2$:

$$A:B:C=6:10:7$$

Then:

$$AC:B^2 = 6 \cdot 7 : 10^2$$

$$42:100$$

$$\boxed{21:50}$$

Q14

$$A=4k, B=7k$$

Numerator:

$$12k+14k=26k$$

Denominator:

$$8k+21k=29k$$

$$\boxed{26:29}$$

Q15

First:

$$2A=3B \Rightarrow A:B=3:2$$

Second:

$$5B=4C \rightarrow B:C=4:5$$

Equalize B:

$$A:B=6:4$$

So:

$$\boxed{A:B:C=6:4:5}$$

Q16

$$A:B=(11+3):(11-3)$$

$$14:8$$

$$\boxed{7:4}$$

Q17

A:B:

$$125:100 = 5:4$$

Difference:

$$1$$

relative to A:

$$\frac{15}{100} \times 100 = \boxed{20\%}$$

Q18

Combined:

$$A:B:C=8:12:15$$

Then:

$$(A+C):B = (8+15):12$$

$$\boxed{23:12}$$

Q19

Let:

$$A=3k, B=4k$$

Then:

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{3k} + \frac{1}{4k} = \frac{7}{12k}$$

and:

$$\frac{1}{A} - \frac{1}{B} = \frac{1}{12k}$$

Ratio:

$$\boxed{7:1}$$

Q20

Let:

$$A=5k, B=7k$$

Then:

$$A^2 + B^2 = 25k^2 + 49k^2 = 74k^2$$

$$(A+B)^2 = (12k)^2 = 144k^2$$

Ratio:

$$74:144$$

$$\boxed{37:72}$$

CHAPTER 13 — FINAL CHECK

We have now covered:

- ratio fundamentals
- simplification and unit conversion
- dividing totals
- sum/difference questions
- proportion
- connected ratios
- compound ratio
- same quantity added/subtracted
- percentage-based ratios
- hidden ratios from equations
- sum/difference transformation
- componendo/dividendo
- inverse and reciprocal ratios
- square/cube/root ratios
- product/quotient ratios
- transformed-expression ratios
- age/mixture/work/speed previews
- GATE and placement practice

Therefore:

CHAPTER 13 COMPLETE

Progress

Completed: 13/76 **Unit 1 — Quantitative Aptitude: 0% PROGRESS** **Next:** Chapter 14 — **Direct & Inverse Variation** +P

Next we will learn:

- $y \propto x$
- $y \propto 1/x$
- constant of proportionality
- direct/inverse scaling
- combined variation

- work-speed-time style applications
- square/cube variation
- rapid ratio methods.

USER

next

ASSISTANT

CHAPTER 14 — DIRECT & INVERSE VARIATION

PART 1 — DIRECT PROPORTION, INVERSE PROPORTION & CONSTANT OF VARIATION

Status: $\emptyset$ = No PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

Variation questions are really ratio questions written in another form.

Main idea:

If one quantity changes, how does another quantity change with it?

The two most important cases are:

$\boxed{\text{Direct variation}}$

and

$\boxed{\text{Inverse variation}}$

1. DIRECT VARIATION — INTUITION

Suppose 1 notebook costs ₹ 20.

Then:

- 2 notebooks cost ₹ 40
- 3 notebooks cost ₹ 60
- 5 notebooks cost ₹ 100

As number of notebooks increases, total cost increases in the same proportion.

So:

$\text{Cost} \propto \text{Number of notebooks}$

The symbol:

$\propto$

means:

“is proportional to”

2. DIRECT VARIATION FORMULA

If:

$$y \propto x$$

then:

$$\boxed{y=kx}$$

where k is the **constant of proportionality**.

Therefore:

$$\boxed{\frac{y}{x}=k}$$

remains constant.

3. EXAMPLE — FINDING k

If:

$$y \propto x$$

and:

$$y=18 \text{ when } x=6$$

then:

$$y=kx$$

$$18=6k$$

$$k=3$$

Therefore relation is:

$$\boxed{y=3x}$$

4. FIND A NEW VALUE

Using:

$$y=3x$$

if:

$$x=11$$

then:

$$y=33$$

Answer:

$$\boxed{33}$$

5. FASTER RATIO METHOD

For direct variation:

$$y \propto x$$

therefore:

$$\boxed{\frac{y_1}{y_2} = \frac{x_1}{x_2}}$$

Example:

If $y=18$ when $x=6$, find y when $x=10$.

$$\frac{18}{y} = \frac{6}{10}$$

or simply scaling:

$$6 \rightarrow 10$$

multiplier:

$$\frac{10}{6} = \frac{5}{3}$$

So:

$$18 \times \frac{5}{3} = \boxed{30}$$

6. DIRECT VARIATION MEMORY RULE

If one quantity multiplies by (m) , the other also multiplies by (m) .

Example:

If:

$$y \propto x$$

and (x) triples:

$$\boxed{y \text{ triples}}$$

If (x) decreases by half:

$$\boxed{y \text{ decreases by half}}$$

7. DIRECT VARIATION EXAMPLE — PRICE

5 pens cost ₹ 75 .

What do 8 pens cost?

Cost directly varies with quantity.

$$5:8 = 75:x$$

$$x = 75 \times \frac{8}{5}$$

$$= 15 \times 8$$

$$= \boxed{120}$$

8. DIRECT VARIATION EXAMPLE — DISTANCE

At constant speed:

$$\text{Distance} \propto \text{Time}$$

If a vehicle travels:

$$180 \text{ km in } 3 \text{ h}$$

then in 5 h:

$$180 \times \frac{5}{3} = \boxed{300 \text{ km}}$$

9. DIRECT VARIATION GRAPH

If:

$$y = kx$$

then its graph is a straight line through the origin.

Important property:

$$x=0 \Rightarrow y=0$$

This distinguishes direct proportion from a general linear equation like:

$$y = 2x + 5$$

which is **not** direct variation.

10. TRAP — LINEAR DOES NOT ALWAYS MEAN DIRECTLY PROPORTIONAL

Suppose:

$$y = 3x + 4$$

Although y increases with x , it is not:

$$y \propto x$$

because:

$$\frac{y}{x}$$

is not constant.

For direct proportion, relation must have form:

$$\boxed{y = kx}$$

with no added constant.

11. INVERSE VARIATION — INTUITION

Suppose a fixed distance must be travelled.

If speed increases, time decreases.

For example:

- at 20 km/h !' more time
- at 40 km/h !' half as much time

Thus:

$$\text{Time} \propto \frac{1}{\text{Speed}}$$

This is inverse variation.

12. INVERSE VARIATION FORMULA

If:

$$y \propto \frac{1}{x}$$

then:

$$y = \frac{k}{x}$$

Therefore:

$$xy = k$$

remains constant.

13. INVERSE VARIATION MEMORY RULE

In direct variation, the ratio stays constant.

$$\frac{y}{x} = k$$

In inverse variation, the product stays constant.

$$xy = k$$

This is the simplest memory trick.

14. EXAMPLE — FINDING k

If:

$$y \propto \frac{1}{x}$$

and:

$$y = 12 \text{ when } x = 5$$

then:

$$xy = k$$

$$12 \times 5 = 60$$

So:

$$k = 60$$

and:

$$\left[y = \frac{60}{x} \right]$$

15. FIND NEW VALUE — INVERSE

If:

$$\left[xy = 60 \right]$$

and:

$$\left[x = 15 \right]$$

then:

$$\left[y = 60/15 \right]$$

$$\left[= \boxed{4} \right]$$

16. FAST INVERSE-RATIO METHOD

If x and y vary inversely:

$$\left[\boxed{x_1 y_1 = x_2 y_2} \right]$$

Example:

12 workers finish a job in 15 days.

If all workers are equally efficient, how many days for 20 workers?

Workers $\times$ days = constant.

$$\left[12 \times 15 = 20 \times d \right]$$

$$\left[d = \frac{180}{20} \right]$$

$$\left[= \boxed{9 \text{ days}} \right]$$

17. INVERSE VARIATION SCALING

If x doubles:

$$\left[y \rightarrow \frac{y}{2} \right]$$

If x triples:

$$\left[y \rightarrow \frac{y}{3} \right]$$

If x becomes:

$$\left[\frac{x}{4} \right]$$

then y becomes:

$$\left[4y \right]$$

18. DIRECT VS INVERSE — QUICK TEST

Ask:

If one quantity increases, what should happen to the other?

If it increases proportionally:

$\boxed{\text{Direct}}$

If it decreases proportionally:

$\boxed{\text{Inverse}}$

But this intuition must be backed by the problem condition.

Not every “one rises, other falls” situation is exact inverse proportion.

19. DIRECT/INVERSE TABLE

Relationship	Equation	Constant
Direct	$y=kx$	(y/x)
Inverse	$y=k/x$	(xy)

MUST MEMORIZE

20. EXAMPLE — WORKERS AND DAYS

For fixed work and equal worker efficiency:

$\text{Workers} \times \text{Days} = \text{constant}$

So:

$\text{Days} \propto \frac{1}{\text{Workers}}$

If workers increase by 50%:

$W \rightarrow 1.5W$

then days become:

$\frac{D}{1.5} = \frac{1}{3}D$

So time decreases by:

$1 - \frac{1}{3} = \frac{2}{3}$

$= \boxed{33\frac{1}{3}\%}$

This is a classic inverse-percentage trap.

21. SPEED AND TIME FOR FIXED DISTANCE

$D=ST$

If (D) is fixed:

$$[ST = \text{constant}]$$

Therefore:

$$[S \propto \frac{1}{T}]$$

If speed ratio:

$$[S_1:S_2=3:5]$$

then time ratio:

$$[T_1:T_2=5:3]$$

22. PRICE AND QUANTITY FOR FIXED EXPENDITURE

$$[\text{Expenditure} = \text{Price} \times \text{Quantity}]$$

If expenditure remains fixed:

$$[PQ = \text{constant}]$$

Therefore:

$$[P \propto \frac{1}{Q}]$$

If price increases, affordable quantity decreases inversely.

23. EXAMPLE — PRICE INCREASE

A person spends ₹ 600 on rice.

At ₹ 30 / kg quantity:

$$[600/30=20 \text{ kg}]$$

If price becomes ₹ 40 / kg :

$$[600/40=15 \text{ kg}]$$

Or directly:

$$[30 \times 20 = 40 \times q]$$

$$[q=15]$$

24. PERCENTAGE EFFECT IN INVERSE VARIATION

If one quantity increases by $(p\%)$, inverse quantity is divided by:

$$[1 + \frac{p}{100}]$$

Example:

Speed increases 25%.

New speed factor:

$$[1.25 = \frac{5}{4}]$$

Time factor:

$$\left[\frac{45}{100} \right]$$

So time decreases by:

$$\left[20\% \right]$$

Important pair

$$\left[\boxed{25\% \text{ speed increase } \rightarrow 20\% \text{ time decrease}} \right]$$

for fixed distance.

25. GENERAL INVERSE PERCENTAGE FORMULA

If (x) increases by $(p\%)$, and (xy) is constant:

$$\left[x_{\text{new}} = x \left(1 + \frac{p}{100} \right) \right]$$

Then:

$$\left[\frac{y_{\text{new}}}{y} = \frac{1}{1 + \frac{p}{100}} \right]$$

Percentage decrease in (y) :

$$\left[\boxed{\frac{p}{100 + p} \times 100\%} \right]$$

So:

$$\left[\boxed{\text{decrease} = \frac{100p}{100 + p}\%} \right]$$

26. EXAMPLE — 20% INCREASE

If (x) increases by 20%, inverse (y) decreases by:

$$\left[\frac{100 \times 20}{120} \right]$$

$$\left[= 16\frac{2}{3}\% \right]$$

So:

$$\left[\boxed{20\% \uparrow \rightarrow 16\frac{2}{3}\% \downarrow} \right]$$

under inverse relation.

27. IF ONE QUANTITY DECREASES

If (x) decreases by $(p\%)$:

$$\left[x_{\text{new}} = x \left(1 - \frac{p}{100} \right) \right]$$

Then inverse quantity increases by:

$$\left[\boxed{\frac{100p}{100 - p}\%} \right]$$

Example:

(x) decreases 20%.

Then inverse quantity increases:

$$\left[\frac{100 \times 20}{80} = 25\% \right]$$

So:

$$\boxed{20\% \downarrow \rightarrow 25\% \uparrow}$$

28. IMPORTANT RECIPROCAL PERCENT PAIRS

Useful mental pairs:

$$20\% \uparrow \leftrightarrow 16\frac{2}{3}\% \downarrow$$

$$25\% \uparrow \leftrightarrow 20\% \downarrow$$

$$50\% \uparrow \leftrightarrow 33\frac{1}{3}\% \downarrow$$

$$100\% \uparrow \leftrightarrow 50\% \downarrow$$

and reverse:

$$20\% \downarrow \leftrightarrow 25\% \uparrow$$

$$50\% \downarrow \leftrightarrow 100\% \uparrow$$

29. DIRECT VARIATION WITH POWERS

Variation need not always be:

$$y \propto x$$

It may be:

$$y \propto x^2$$

or:

$$y \propto x^3$$

Example:

Area of a square:

$$A = s^2$$

Therefore:

$$\boxed{A \propto s^2}$$

30. SQUARE VARIATION

If:

$$y \propto x^2$$

then:

$$y = kx^2$$

and:

$$\boxed{\frac{y_1}{y_2} = \left(\frac{x_1}{x_2}\right)^2}$$

31. EXAMPLE

If:

$$y \propto x^2$$

and x doubles:

$$x \rightarrow 2x$$

then:

$$y \rightarrow 2^2 y = 4y$$

So:

$$\boxed{y \text{ becomes 4 times}}$$

32. CUBE VARIATION

If:

$$y \propto x^3$$

and x doubles:

$$y \rightarrow 2^3 y$$

$$= \boxed{8y}$$

Example: volume of geometrically similar solids varies with cube of linear scale.

33. INVERSE-SQUARE VARIATION

Sometimes:

$$y \propto \frac{1}{x^2}$$

Then:

$$yx^2 = k$$

If x doubles:

$$y \rightarrow \frac{1}{4}y$$

This is common in physics-style reasoning.

34. GENERAL POWER VARIATION

If:

$$y \propto x^n$$

then:

$$\boxed{y = kx^n}$$

and:

$$\boxed{\frac{y_2}{y_1} = \left(\frac{x_2}{x_1}\right)^n}$$

If:

$$y \propto x^{-n}$$

then:

$$\boxed{\frac{y_2}{y_1} = \left(\frac{x_1}{x_2}\right)^n}$$

35. EXAMPLE — AREA

Side of a square increases from:

$$4 \text{ cm}$$

to:

$$6 \text{ cm}$$

Area ratio:

$$4^2:6^2$$

$$16:36$$

$$\boxed{4:9}$$

No need to calculate actual areas.

36. EXAMPLE — VOLUME

If corresponding lengths of two similar cubes are:

$$2:5$$

volume ratio:

$$2^3:5^3$$

$$\boxed{8:125}$$

37. FINDING EXPONENT FROM VARIATION DATA

Suppose:

$$y \propto x^n$$

and when (x) doubles, (y) becomes 8 times.

Then:

$$2^n = 8$$

$$2^n = 2^3$$

Therefore:

$$\boxed{n=3}$$

38. ANOTHER EXPONENT EXAMPLE

If (x) triples and (y) becomes 9 times:

$$[3^n = 9]$$

$$[3^n = 3^2]$$

So:

$$[\boxed{n=2}]$$

39. DIRECT VARIATION WITH MULTIPLE VARIABLES

Sometimes:

$$[z \propto xy]$$

Then:

$$[\boxed{z=kxy}]$$

If:

- (x) doubles
- (y) triples

then:

$$[z]$$

becomes:

$$[2 \times 3 = 6]$$

times.

40. EXAMPLE

If:

$$[z \propto xy]$$

and:

$$[x \rightarrow 3x, \quad y \rightarrow \frac{y}{2}]$$

then:

$$[z \rightarrow 3 \times \frac{1}{2} z]$$

$$[= \boxed{\frac{3}{2} z}]$$

So z increases by 50%.

41. MIXED DIRECT AND INVERSE VARIATION

Suppose:

$$[z \propto \frac{x}{y}]$$

Then:

$$[\boxed{z=k\frac{x}{y}}]$$

If:

- x doubles
- y triples

then:

$$z_{\text{new}} = \frac{2}{3}z$$

42. COMBINED VARIATION

If:

$$z \propto \frac{x^2 y}{w}$$

then:

$$z = k \frac{x^2 y}{w}$$

To compare two situations:

$$\boxed{\frac{z_2}{z_1} = \left(\frac{x_2}{x_1}\right)^2 \left(\frac{y_2}{y_1}\right) \left(\frac{w_1}{w_2}\right)}$$

This avoids finding k .

43. EXAMPLE — COMBINED VARIATION

Suppose:

$$z \propto \frac{x^2}{y}$$

If:

- x doubles
- y becomes 4 times

Then:

$$z_{\text{new}} = \frac{2^2}{4}z$$

$$= z$$

Therefore:

$$\boxed{\text{no change}}$$

44. ANOTHER COMBINED EXAMPLE

If:

$$P \propto \frac{ab}{c^2}$$

and:

$$a \rightarrow 2a, \quad b \rightarrow 3b, \quad c \rightarrow 2c$$

then:

$$P_{\text{new}} = \frac{2 \cdot 3}{2^2}P$$

$$= \frac{3}{2}P$$

$$\boxed{\frac{32}{P}}$$

So P increases 50%.

45. GATE-STYLE QUESTION 1

If:

$$y \propto x$$

and $y=35$ when $x=5$, find y when $x=8$.

$$\frac{35}{5}=7$$

So:

$$y=7(8)$$

$$\boxed{56}$$

46. GATE-STYLE QUESTION 2

If:

$$y \propto \frac{1}{x}$$

and $y=18$ when $x=4$, find y when $x=12$.

$$xy = \text{constant}$$

$$18 \times 4 = 12y$$

$$72 = 12y$$

$$\boxed{y=6}$$

47. GATE-STYLE QUESTION 3

If:

$$y \propto x^2$$

and x increases by 50%, by what factor does y change?

x multiplier:

$$1.5 = \frac{3}{2}$$

So y multiplier:

$$\left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

$$= 2.25$$

Therefore y increases by:

$$125\%$$

$$\boxed{2.25 \text{ times}}$$

48. GATE-STYLE QUESTION 4

If:

$$z \propto \frac{x}{y}$$

and x increases by 20% while y decreases by 20%, find multiplier of z.

$$\frac{1.2}{0.8} = \frac{12}{8} = \frac{3}{2}$$

So:

$$z \text{ becomes } 1.5 \text{ times}$$

That is a 50% increase.

49. PLACEMENT-STYLE QUESTION 1

8 workers complete a task in 15 days.

How many days will 12 workers take?

$$8 \times 15 = 12d$$

$$d = 10$$

$$10 \text{ days}$$

50. PLACEMENT-STYLE QUESTION 2

A car takes 6 hours at 60 km/h for a fixed journey.

At 90 km/h:

$$60 \times 6 = 90t$$

$$t = 4$$

$$4 \text{ hours}$$

51. PLACEMENT-STYLE QUESTION 3

A quantity varies directly as square of x.

When x changes from 3 to 5, y changes in ratio:

$$3^2 : 5^2$$

$$9 : 25$$

52. PLACEMENT-STYLE QUESTION 4

If:

$$z \propto xy$$

and x decreases 20% while y increases 25%:

$$0.8 \times 1.25 = 1$$

Therefore:

$$\boxed{z \text{ remains unchanged}}$$

Classic multiplier cancellation.

53. COMMON TRAPS

& pTRAP 1 — Direct variation vs general linear relation

$$y = 3x + 2$$

is not direct variation.

& pTRAP 2 — Inverse relation

For inverse variation:

$$xy = \text{constant}$$

not:

$$x/y = \text{constant}$$

& pTRAP 3 — Percentage reversal

25% increase in one inverse quantity causes 20% decrease in the other, not 25%.

& pTRAP 4 — Square variation

If x doubles under:

$$y \propto x^2$$

y quadruples, not doubles.

& pTRAP 5 — Combined variation

Apply every multiplier with its correct exponent and inverse direction.

54. MUST-MEMORIZE FORMULAS

Direct

$$\boxed{y \propto x \Rightarrow y = kx}$$

$$\boxed{\frac{y_1}{y_2} = \frac{x_1}{x_2}}$$

Inverse

$$\boxed{y \propto 1/x \Rightarrow xy = k}$$

$$\boxed{x_1 y_1 = x_2 y_2}$$

Power variation

$$\boxed{y \propto x^n \Rightarrow \frac{y_2}{y_1} = \left(\frac{x_2}{x_1} \right)^n}$$

55. FAST MULTIPLIER METHOD

This is often the quickest approach.

Suppose:

$$y \propto \frac{x^2 z}{w}$$

and changes are:

- $(x \times 2)$
- $(z \times 3)$
- $(w \times 4)$

Then:

$$y_{\text{multiplier}} = \frac{2^2 \times 3}{4}$$

$$= \frac{12}{4}$$

$$= \boxed{3}$$

No (k) , no actual values.

56. 5-MINUTE REVISION

Direct

Same-direction scaling:

$$x \times m \rightarrow y \times m$$

Inverse

Opposite scaling:

$$x \times m \rightarrow y \div m$$

Square variation

$$x \times m \rightarrow y \times m^2$$

Cube variation

$$x \times m \rightarrow y \times m^3$$

Combined

Multiply all scale factors with appropriate powers.

PRACTICE — LEVEL 0

Q1

If:

$$y \propto x$$

and $y=24$ when $x=6$, find y when $x=10$.

Q2

If:

$$y \propto 1/x$$

and $y=20$ when $x=3$, find y when $x=12$.

Q3

If x doubles and y varies directly with x , what happens to y ?

Q4

If x doubles and y varies inversely with x , what happens to y ?

Q5

If:

$$y \propto x^2$$

and x triples, what happens to y ?

LEVEL 1

Q6

6 workers finish a task in 20 days. How long for 8 workers?

Q7

At constant speed, a vehicle travels 240 km in 4 h. How far in 7 h?

Q8

If:

$$y \propto x^3$$

and x changes from 2 to 4, find $(y_2:y_1)$.

Q9

If:

$$z \propto xy$$

and x doubles while y halves, what happens to z ?

Q10

If:

$$z \propto x/y$$

and x triples while y doubles, what happens to z ?

LEVEL 2

Q11

If:

$$y \propto x^2$$

and $y=45$ when $x=3$, find y when $x=5$.

Q12

A fixed journey takes 5 hours at speed v . If speed increases by 25%, find new time.

Q13

If workers increase by 20%, what percentage decrease in time is required for same work?

Q14

If:

$$P \propto \frac{x^2}{y}$$

and x increases by 50% while y doubles, find multiplier of P .

Q15

If:

$$Q \propto abc$$

and a, b, c change by multipliers 2, $\frac{1}{2}$ and 3 respectively, find Q multiplier.

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$y \propto x^n$$

and doubling x multiplies y by 16, find n .

Q17

If:

$$z \propto \frac{x^2 y}{w^3}$$

and:

$$x \rightarrow 2x, \quad y \rightarrow 3y, \quad w \rightarrow 2w$$

find $\frac{z_{\text{new}}}{z}$.

Q18

A person's expenditure remains constant. If price decreases by 20%, by what percentage can quantity increase?

Q19

For a fixed distance, time decreases by 25%. By what percentage must speed increase?

Q20

If:

$$A \propto B^2$$

and B decreases by 10%, by what percentage does A decrease?

ANSWERS

Q1

$$\lfloor 24/6=4 \rfloor$$

$$\lfloor y=4(10) \rfloor$$

$$\lfloor \boxed{40} \rfloor$$

Q2

$$\lfloor 20\times 3=12y \rfloor$$

$$\lfloor 60=12y \rfloor$$

$$\lfloor \boxed{5} \rfloor$$

Q3

$$\lfloor \boxed{\text{y doubles}} \rfloor$$

Q4

$$\lfloor \boxed{\text{y halves}} \rfloor$$

Q5

$$\lfloor 3^2=9 \rfloor$$

$$\lfloor \boxed{\text{y becomes 9 times}} \rfloor$$

Q6

$$\lfloor 6\times 20=8d \rfloor$$

$$\lfloor d=15 \rfloor$$

$$\lfloor \boxed{15\text{ days}} \rfloor$$

Q7

Distance directly proportional to time.

$$\lfloor 240\times\frac{7}{4} = 60\times 7 = \boxed{420\text{ km}} \rfloor$$

Q8

x doubles:

$$\lfloor 2\text{ to }4 \rfloor$$

Cube relation:

$$\lfloor 2^3=8 \rfloor$$

So:

$$\lfloor \boxed{y_2:y_1=8:1} \rfloor$$

Q9

Multiplier:

$$\lfloor 2\times\frac{1}{2}=1 \rfloor$$

$\boxed{\text{No change}}$

Q10

$\frac{3}{2}$

So:

$z \text{ becomes } 1.5 \text{ times}$

Q11

$y_2 = \left(\frac{5}{3}\right)^2$

$y_2 = 45 \times \frac{25}{9}$

$= 5 \times 25$

$\boxed{125}$

Q12

Speed factor:

$1.25 = \frac{5}{4}$

Time factor:

$\frac{4}{5}$

New time:

$5 \times \frac{4}{5}$

$\boxed{4 \text{ h}}$

Q13

Workers:

$\times 1.2 = \times \frac{6}{5}$

Time:

$\times \frac{5}{6}$

Decrease:

$1 - \frac{5}{6} = \frac{1}{6}$

$\boxed{16\frac{2}{3}\%}$

Q14

x multiplier:

1.5

Square:

2.25

Divide by y multiplier 2:

$\frac{2.25}{2} = 1.125$

So:

$P \text{ becomes } 1.125 \text{ times}$

or increases:

$$\boxed{12.5\%}$$

Q15

$$2 \times \frac{12}{3} = 3$$

$$\boxed{3 \text{ times}}$$

Q16

$$2^n = 16$$

$$16 = 2^4$$

$$\boxed{n=4}$$

Q17

$$\frac{2^2 \times 3}{2^3} = \frac{12}{8} = \boxed{\frac{3}{2}}$$

Q18

Price decreases 20%:

$$P \rightarrow 0.8P = \frac{4}{5}P$$

Quantity inverse multiplier:

$$\frac{5}{4}$$

Increase:

$$25\%$$

$$\boxed{25\%}$$

Q19

Time decreases 25%:

$$T \rightarrow 0.75T = \frac{3}{4}T$$

Speed inverse multiplier:

$$\frac{4}{3}$$

Increase:

$$\frac{1}{3} = \boxed{33\frac{1}{3}\%}$$

Q20

B multiplier:

$$0.9$$

A multiplier:

$$0.9^2 = 0.81$$

So A decreases by:

$$1 - 0.81 = 0.19$$

$$\boxed{19\%}$$

CHAPTER 14 STATUS

Covered so far:

- direct variation '
- constant of proportionality '
- direct ratio method '
- inverse variation '
- product-constant method '
- workers-days applications '
- speed-time applications '
- price-quantity applications '
- inverse percentage effects '
- square/cube variation '
- inverse-square variation '
- combined variation '
- multiplier method '
- GATE/placement practice '

But **Chapter 14 is not complete yet.**

Next we need:

- multi-variable variation with actual values
- joint and combined variation equations
- finding unknown exponents
- chained proportionality
- direct/inverse variation word problems
- advanced percentage-change problems
- mixed GATE/placement questions

Progress

Completed: 13/76 **Current:** 0=0 Chapter 14 — Direct & Inverse Variation **Next:** Chapter 14 Part 2 — **Joint variation, chained variation & advanced applications**

USER

next

ASSISTANT

CHAPTER 14 — DIRECT & INVERSE VARIATION

PART 2 — JOINT VARIATION, CHAINED PROPORTIONALITY & ADVANCED APPLICATIONS

Status: 0=0 **PROGRESS** **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This final part covers the harder forms of variation:

- joint variation
 - combined direct/inverse variation
 - chained proportionality
 - finding constants from data
 - finding unknown exponents
 - multi-step percentage changes
 - word problems involving workers, speed, dimensions and cost
-

1. JOINT VARIATION

If:

$$z \propto xy$$

we say z varies **jointly** as x and y .

Then:

$$\boxed{z = kxy}$$

Example:

If:

$$z = 24, \quad x = 3, \quad y = 4$$

then:

$$24 = k(3)(4)$$

$$k = 2$$

So:

$$\boxed{z = 2xy}$$

2. FINDING A NEW VALUE

Using:

$$z = 2xy$$

if:

$$x = 5, \quad y = 6$$

then:

$$z = 2(5)(6)$$

$$= \boxed{60}$$

3. FAST RATIO METHOD FOR JOINT VARIATION

If:

$z \propto xy$

then:

$$\boxed{\frac{z_2}{z_1} = \frac{x_2}{x_1} \cdot \frac{y_2}{y_1}}$$

This is usually faster than finding k .

4. EXAMPLE

If:

$$z=20$$

when:

$$x=2, y=5$$

find z when:

$$x=6, y=10$$

Multiplier:

$$\frac{62}{10} \times \frac{10}{5} = 3 \times 2 = 6$$

Therefore:

$$z=20 \times 6$$

$$=\boxed{120}$$

5. JOINT VARIATION WITH POWERS

If:

$$z \propto x^2y$$

then:

$$\boxed{z=kx^2y}$$

If:

- x doubles
- y triples

then:

$$z$$

multiplier:

$$2^2 \times 3 = 12$$

So:

$$\boxed{z \text{ becomes 12 times}}$$

6. COMBINED VARIATION

Suppose:

$$[z \propto x]$$

and inversely as (y) .

Then:

$$[\boxed{z \propto \frac{x}{y}}]$$

so:

$$[\boxed{z = k \frac{x}{y}}]$$

7. EXAMPLE — FINDING (k)

If:

$$[z=12, \text{quad } x=8, \text{quad } y=4]$$

then:

$$[12 = k \frac{8}{4}]$$

$$[12 = 2k]$$

$$[k=6]$$

Hence:

$$[\boxed{z = \frac{6x}{y}}]$$

8. NEW VALUE FROM COMBINED VARIATION

If:

$$[z = \frac{6x}{y}]$$

and:

$$[x=15, \text{quad } y=5]$$

then:

$$[z = \frac{6(15)}{5}]$$

$$[= \boxed{18}]$$

9. FAST MULTIPLIER METHOD

For:

$$[z \propto \frac{x^2 y}{w^3}]$$

write only the change multipliers.

Suppose:

$$[x \times 3, \text{quad } y \times 2, \text{quad } w \times \frac{1}{2}]$$

Then:

$$[\text{multiplier} = \frac{3^2 \cdot 2}{(1/2)^3}]$$

$$\left[= \frac{18}{1/8} = 144 \right]$$

So:

$$\left[\boxed{z \text{ becomes } 144 \text{ times}} \right]$$

This avoids almost all arithmetic with actual values.

10. CHAINED PROPORTIONALITY

Suppose:

$$\left[A \propto B \right]$$

and:

$$\left[B \propto C \right]$$

Then:

$$\left[A \propto C \right]$$

because:

$$\left[A = k_1 B \right]$$

and:

$$\left[B = k_2 C \right]$$

So:

$$\left[A = k_1 k_2 C \right]$$

Thus:

$$\left[\boxed{A \propto C} \right]$$

11. CHAIN WITH AN INVERSE RELATION

Suppose:

$$\left[A \propto B \right]$$

and:

$$\left[B \propto \frac{1}{C} \right]$$

Then:

$$\left[\boxed{A \propto \frac{1}{C}} \right]$$

12. CHAIN OF TWO INVERSE RELATIONS

Suppose:

$$\left[A \propto \frac{1}{B} \right]$$

and:

$$\left[B \propto \frac{1}{C} \right]$$

Then:

$$[A \propto C]$$

because inverse of an inverse becomes direct.

Memory

inverse $\times$ inverse $\rightarrow$ direct

13. POWER CHAIN

Suppose:

$$[A \propto B^2]$$

and:

$$[B \propto C^3]$$

Then:

$$[B^2 \propto C^6]$$

Therefore:

$$[\boxed{A \propto C^6}]$$

14. ANOTHER CHAIN

If:

$$[A \propto \sqrt{B}]$$

and:

$$[B \propto C^4]$$

then:

$$[A \propto (C^4)^{1/2}]$$

$$[\boxed{A \propto C^2}]$$

15. FINDING AN UNKNOWN EXPONENT

Suppose:

$$[y \propto x^n]$$

Given:

$$[x_1=2, \text{quad } y_1=9]$$

$$[x_2=6, \text{quad } y_2=243]$$

Then:

$$[\frac{243}{9} = \left(\frac{6}{2}\right)^n]$$

$$[27=3^n]$$

Therefore:

$$[\boxed{n=3}]$$

16. FINDING EXPONENT FROM PERCENTAGE MULTIPLIER

If doubling (x) makes (y) 4 times:

$$[2^n=4]$$

$$[\boxed{n=2}]$$

If tripling (x) makes (y) $(\frac{1}{9})$:

$$[3^n=\frac{1}{9}=3^{-2}]$$

So:

$$[\boxed{n=-2}]$$

Meaning:

$$[y\propto\frac{1}{x^2}]$$

17. FINDING EXPONENT WITH LOGIC

If:

$$[x\propto 4x]$$

and:

$$[y\propto 2y]$$

for:

$$[y\propto x^n]$$

then:

$$[4^n=2]$$

Since:

$$[4=2^2]$$

$$[2^{2n}=2^1]$$

$$[2n=1]$$

$$[\boxed{n=\frac{1}{2}}]$$

So:

$$[y\propto\sqrt{x}]$$

18. WORKERS, HOURS & DAYS

For fixed work:

$$\left[\frac{\text{Work}}{\text{workers} \times \text{hours/day} \times \text{days}} \right]$$

If worker efficiency is same:

$$\boxed{\text{WHD} = \text{constant}}$$

for the same total work.

This is one of the most important combined-variation applications.

19. WORKERS-DAYS-HOUR EXAMPLE

12 workers working 8 hours/day finish a job in 15 days.

How many days will 20 workers working 6 hours/day need?

Same work:

$$12 \times 8 \times 15 = 20 \times 6 \times d$$

So:

$$d = \frac{12 \times 8 \times 15}{20 \times 6}$$

Cancel:

$$12/6 = 2$$

$$20/2 = 10$$

or directly:

$$d = 12$$

Thus:

$$\boxed{12 \text{ days}}$$

20. FAST MULTIPLIER VIEW

Old:

- workers = 12
- hours = 8

New:

- workers (12 to 20): work rate $\times \frac{5}{3}$
- hours (8 to 6): work rate $\times \frac{3}{4}$

Combined daily rate:

$$\frac{5}{3} \times \frac{3}{4} = \frac{5}{4}$$

So days become inverse:

$$15 \times \frac{4}{5} = \boxed{12}$$

This is often the fastest method.

21. MACHINES AND PRODUCTION

If identical machines produce at constant rate:

$$\text{Output} \propto (\text{machines})(\text{time})$$

So:

$$\boxed{\frac{O_2}{O_1} = \frac{M_2}{M_1} \cdot \frac{T_2}{T_1}}$$

22. MACHINE EXAMPLE

8 machines make 1200 items in 5 hours.

How many items will 10 machines make in 6 hours?

Multiplier:

$$\frac{10}{8} \times \frac{6}{5}$$

$$= \frac{54}{5} \times \frac{6}{5} = \frac{32}{5}$$

So:

$$1200 \times \frac{32}{5} = \boxed{1800}$$

23. PEOPLE AND FOOD

For fixed food consumption rate:

$$\text{Food required} \propto (\text{people})(\text{days})$$

If total food fixed:

$$\boxed{\text{people} \times \text{days} = \text{constant}}$$

24. FOOD EXAMPLE

Food for 30 people lasts 20 days.

How long for 24 people?

$$30 \times 20 = 24d$$

$$d = 25$$

$$\boxed{25 \text{ days}}$$

25. DISTANCE–SPEED–TIME

$$D = ST$$

Variation consequences:

For fixed (T) :

$$D \propto S$$

For fixed (S) :

$$D \propto T$$

For fixed (D) :

$$S \propto \frac{1}{T}$$

Always identify **what is fixed**.

26. MAJOR TRAP — RELATION DEPENDS ON WHAT IS CONSTANT

You cannot simply say:

speed and time are inversely proportional.

They are inverse **only for fixed distance**.

If distance is changing, no such conclusion is automatic.

27. COST–PRICE–QUANTITY

$$\text{Cost} = P \times Q$$

If total cost fixed:

$$P \propto \frac{1}{Q}$$

If unit price fixed:

$$\text{Cost} \propto Q$$

Again:

identify the fixed quantity.

28. AREA OF A RECTANGLE

$$A = LB$$

So:

$$A \propto LB$$

If area fixed:

$$L \propto \frac{1}{B}$$

Example:

If length increases 25%, breadth must decrease 20% to keep area unchanged.

Because:

$$1.25 \times 0.8 = 1$$

29. AREA-CONSTANT PERCENTAGE QUESTION

Rectangle length increases 20%.

What % should breadth decrease to keep area constant?

Length multiplier:

$$\left[1.2 = \frac{65}{5} \right]$$

Breadth multiplier:

$$\left[\frac{5}{6} \right]$$

Decrease:

$$\left[\frac{1}{6} = \boxed{16\frac{2}{3}\%} \right]$$

30. VOLUME OF A CUBOID

$$\left[V = LWH \right]$$

For constant volume:

$$\left[LWH = \text{constant} \right]$$

If:

- (L) doubles
- (W) halves

then (LW) has no net change.

So (H) can remain unchanged.

31. COMBINED DIMENSION EXAMPLE

For constant volume:

$$\left[LWH = \text{constant} \right]$$

Suppose:

$$\left[L \times 2, W \times 3 \right]$$

Then to keep volume fixed:

$$\left[H \times \frac{1}{6} \right]$$

Thus height decreases by:

$$\left[1 - \frac{1}{6} = \frac{5}{6} \right]$$

$$\left[= \boxed{8\frac{1}{3}\%} \right]$$

32. SQUARE AREA PERCENTAGE CHANGE

If side increases by 10%:

$$\left[s \rightarrow 1.1s \right]$$

Area:

$$\left[A \rightarrow (1.1)^2 A \right]$$

$$\left[= 1.21A \right]$$

So area increases:

$$\boxed{21\%}$$

33. CUBE VOLUME PERCENTAGE CHANGE

If side of cube increases by 10%:

$$V \propto (1.1)^3 V$$

$$V = 1.331 V$$

So increase:

$$\boxed{33.1\%}$$

Do not say 30%.

34. SMALL-PERCENTAGE APPROXIMATION

For small $(p\%)$:

If:

$$y \propto x^n$$

then approximately:

$$\Delta y \approx n(\Delta x)$$

Example:

Side increases 1%.

Area approximately increases:

$$\boxed{2\%}$$

Exact:

$$1.01^2 = 1.0201$$

$$\boxed{2.01\%}$$

This approximation is useful in engineering-style questions.

35. EXACT PERCENTAGE CHANGE FOR SQUARE RELATION

If:

$$y \propto x^2$$

and (x) changes by $(p\%)$, multiplier:

$$\left(1 + \frac{p}{100}\right)^2$$

For increase:

$$\Delta y = \left[\left(1 + \frac{p}{100}\right)^2 - 1\right] \times 100$$

For $(p=20)$:

$$\sqrt[3]{1.2^2} = 1.44$$

increase:

$$\boxed{44\%}$$

36. EXACT PERCENTAGE CHANGE FOR CUBE RELATION

If x increases 20%:

$$y \propto x^3$$

Then:

$$\sqrt[3]{1.2^3} = 1.728$$

So y increases:

$$\boxed{72.8\%}$$

37. MIXED PERCENTAGE CHANGES

If:

$$z \propto \frac{x^2 y}{w}$$

and:

- x increases 10%
- y decreases 20%
- w increases 25%

Multiplier:

$$\frac{(1.1)^2 (0.8)}{1.25}$$

$$= \frac{1.21 \times 0.8}{1.25}$$

$$= \frac{0.968}{1.25}$$

$$= 0.7744$$

So z becomes:

$$\boxed{77.44\%}$$

of original.

Decrease:

$$\boxed{22.56\%}$$

38. WHEN MULTIPLIERS CANCEL

If:

$$z \propto xy$$

and:

- x increases 25% $\Rightarrow \times \frac{5}{4}$

- y decreases 20% $\Rightarrow y \times \frac{4}{5}$

Then:

$$\frac{5}{4} \cdot \frac{4}{5} = 1$$

So:

$$\boxed{z \text{ remains unchanged}}$$

This is a common exam shortcut.

39. CHAINED WORD-PROBLEM EXAMPLE

A factory's output varies directly with number of machines and hours, and inversely with defect-related downtime factor (d) :

$$O \propto \frac{MH}{d}$$

If:

- machines double,
- hours increase 50%,
- downtime factor triples,

then:

$$O \text{ multiplier} = \frac{2 \times 1.5}{3} = 1$$

Therefore:

$$\boxed{\text{output unchanged}}$$

40. FINDING UNKNOWN VARIABLE USING RATIO FORM

Suppose:

$$z \propto \frac{x^2}{y}$$

and:

$$z_1 = 18, \quad x_1 = 3, \quad y_1 = 2$$

Find (z_2) when:

$$x_2 = 6, \quad y_2 = 8$$

Use:

$$\frac{z_2}{18} = \left(\frac{6}{3}\right)^2 \left(\frac{2}{8}\right)$$

$$z_2 = 4 \times \frac{14}{1} = 18$$

So:

$$\boxed{z_2 = 18}$$

41. FINDING AN UNKNOWN INPUT

Suppose:

$$y \propto x^2$$

and:

$$y_1 = 20, x_1 = 2$$

If:

$$y_2 = 180$$

find x_2 .

$$\frac{180}{20} = \left(\frac{x_2}{2}\right)^2$$

$$9 = \left(\frac{x_2}{2}\right)^2$$

Positive quantity:

$$\frac{x_2}{2} = 3$$

$$\boxed{x_2 = 6}$$

42. FINDING UNKNOWN INPUT — INVERSE SQUARE

If:

$$y \propto \frac{1}{x^2}$$

and y becomes $\frac{1}{16}$ of original:

$$\frac{y_2}{y_1} = \left(\frac{x_1}{x_2}\right)^2 = \frac{1}{16}$$

So:

$$\frac{x_2}{x_1} = 4$$

Thus:

$$\boxed{x \text{ became 4 times}}$$

43. DATATABLE TEST FOR DIRECT VARIATION

Suppose:

x	y
2	6
4	12
5	15

Check:

$$y/x$$

$$3, 3, 3$$

Constant.

Therefore:

$$\boxed{y \propto x}$$

44. DATATABLE TEST FOR INVERSE VARIATION

x	y
2	12
3	8
4	6

Check:

$$xy$$

$$24, 24, 24$$

Constant.

Therefore:

$$\boxed{y \propto 1/x}$$

45. NOT DIRECT OR INVERSE

Suppose:

x	y
1	3
2	5
3	7

Here:

$$y/x$$

is not constant.

And:

$$xy$$

is not constant.

So it is neither direct nor inverse variation.

In fact:

$$y=2x+1$$

46. GATE-STYLE QUESTION 1

If:

$$[z \propto x^2 y]$$

and:

$$[z = 54]$$

when:

$$[x = 3, y = 2]$$

find z when:

$$[x = 6, y = 1]$$

Multiplier:

$$[\left(\frac{63}{1}\right)^2 \cdot \frac{12}{1}]$$

$$[= 4 \cdot \frac{12}{1} = 2]$$

So:

$$[z = 54 \times 2]$$

$$[= \boxed{108}]$$

47. GATE-STYLE QUESTION 2

If:

$$[z \propto \frac{x}{y^2}]$$

x doubles and y halves.

Multiplier:

$$[\frac{2}{(1/2)^2} = \frac{2}{1/4} = 8]$$

Thus:

$$[\boxed{z \text{ becomes 8 times}}]$$

48. GATE-STYLE QUESTION 3

If:

$$[A \propto B^n]$$

and doubling B makes A one-fourth, find (n) .

$$[2^n = \frac{1}{4}]$$

$$[2^n = 2^{-2}]$$

$$[\boxed{n = -2}]$$

49. GATE-STYLE QUESTION 4

For constant area:

$$[LB = \text{constant}]$$

Length decreases by 20%.

New length factor:

$$\left[0.8 = \frac{4}{5} \right]$$

Breadth factor:

$$\left[\frac{5}{4} \right]$$

So breadth increases:

$$\left[\boxed{25\%} \right]$$

50. PLACEMENT-STYLE QUESTION 1

15 workers working 6 hours/day finish a job in 12 days.

How many days will 18 workers working 5 hours/day take?

Same work:

$$\left[15 \cdot 6 \cdot 12 = 18 \cdot 5 \cdot d \right]$$

Both daily work rates:

$$\left[15 \cdot 6 = 90 \right]$$

$$\left[18 \cdot 5 = 90 \right]$$

So:

$$\left[\boxed{12 \text{ days}} \right]$$

51. PLACEMENT-STYLE QUESTION 2

10 machines produce 400 units in 8 hours.

How many units will 15 machines produce in 12 hours?

Multiplier:

$$\left[\frac{15}{10} \times \frac{12}{8} \right]$$

$$\left[= \frac{3}{2} \times \frac{3}{2} = \frac{9}{4} \right]$$

So:

$$\left[400 \times \frac{9}{4} = \boxed{900} \right]$$

52. PLACEMENT-STYLE QUESTION 3

The side of a square increases by 20%.

Area multiplier:

$$\left[1.2^2 = 1.44 \right]$$

Thus area increases:

$$\left[\boxed{44\%} \right]$$

53. PLACEMENT-STYLE QUESTION 4

The radius of a sphere doubles.

Volume varies as:

$$\propto r^3$$

So:

$$\propto 2^3 V$$

$$= \boxed{8V}$$

Increase:

$$700\%$$

Be careful:

“8 times” means **700% increase**, not 800%.

54. VERY IMPORTANT PERCENTAGE LANGUAGE

If new value is:

$$2V$$

then increase is:

$$100\%$$

If new value is:

$$3V$$

increase:

$$200\%$$

If new value is:

$$8V$$

increase:

$$700\%$$

Formula

If multiplier = m :

$$\boxed{\% \text{ increase}} = (m-1) \times 100$$

55. COMMON TRAPS

& pTRAP 1 — Forgetting exponent

If:

$$y \propto x^2$$

a factor 3 in x becomes factor 9 in y.

& pTRAP 2 — Wrong inverse direction

If denominator quantity doubles, output halves only when power is 1.

If denominator is squared:

$$\sqrt[1]{1/x^2}$$

doubling gives one-fourth.

& pTRAP 3 — Percentage vs multiplier

Becoming 1.5 times means 50% increase.

& pTRAP 4 — Missing the fixed condition

Workers and days are inverse only when:

- work fixed
 - efficiency fixed
 - working hours/day fixed, unless explicitly included.
-

& pTRAP 5 — Joint variation

If:

$$z \propto xy$$

changes in x and y multiply; they do not simply add.

& pTRAP 6 — General linear relation

Increasing together does not prove direct variation. Verify constant ratio.

56. MUST-MEMORIZE MASTER FORM

If:

$$Q \propto A^a B^b C^c$$

then:

$$\boxed{Q = k \frac{A^a B^b}{C^c}}$$

For changes:

$$\boxed{\frac{Q_2}{Q_1} = \left(\frac{A_2}{A_1}\right)^a \left(\frac{B_2}{B_1}\right)^b \left(\frac{C_1}{C_2}\right)^c}$$

This single pattern handles almost every variation question.

57. FASTEST GATE METHOD

For comparison questions:

Do not find k unless necessary.

Just use multipliers.

Example:

$Q \propto \frac{x^3 y}{z^2}$

Changes:

$x \times 2, y \times \frac{1}{2}, z \times 4$

Then:

$Q_{\text{multiplier}} = \frac{2^3 (1/2)}{4^2}$

$= \frac{4}{16} = \boxed{\frac{1}{4}}$

Done.

PRACTICE — LEVEL 0

Q1

If $z \propto xy$, x doubles and y triples. What happens to z ?

Q2

If $z \propto x/y$, x doubles and y doubles. What happens?

Q3

If $y \propto x^2$, x becomes 4 times. What happens to y ?

Q4

If $y \propto 1/x^2$, x triples. What happens to y ?

Q5

If $A \propto B^3$, B doubles. Find A multiplier.

LEVEL 1

Q6

If $z \propto xy$, $z=30$ at $x=3, y=5$. Find z at $x=6, y=10$.

Q7

If $z \propto x^2/y$, x doubles and y doubles. Find z multiplier.

Q8

20 workers finish work in 18 days. How many days for 30 equally efficient workers?

Q9

12 machines make 600 units in 5 h. How many units do 20 machines make in 3 h?

Q10

A square's side increases 30%. Find exact percentage increase in area.

LEVEL 2

Q11

If $(y \propto x^n)$, tripling x multiplies y by 27. Find n .

Q12

If $(y \propto x^n)$, doubling x halves y . Find n .

Q13

A rectangle's length increases 25%. Find required percentage decrease in breadth for same area.

Q14

If $(Q \propto a^2b/c)$, a doubles, b halves, c doubles. Find Q multiplier.

Q15

15 workers, 8 h/day, finish in 10 days. How many days for 20 workers at 6 h/day?

LEVEL 3 — EXAM PRESSURE

Q16

If $(Q \propto x^2y^3/z)$, x doubles, y halves, z becomes one-fourth. Find Q multiplier.

Q17

If $(A \propto B^2)$ and $(B \propto C^3)$, how does A vary with C ?

Q18

A cube's side decreases by 20%. Find exact percentage decrease in volume.

Q19

For constant expenditure, price rises by 60%. By what percentage must quantity fall?

Q20

For a fixed distance, time rises by 25%. By what percentage does speed fall?

ANSWERS

Q1

$[2 \times 3 = 6]$

$[\boxed{6 \text{ times}}]$

Q2

$[2/2 = 1]$

$[\boxed{\text{No change}}]$

Q3

$$\sqrt[4]{4^2=16}$$

$$\sqrt[16]{\text{times}}$$

Q4

$$\sqrt[1/3^2]{\boxed{1/9}}$$

Q5

$$\sqrt[2^3]{\boxed{8}}$$

Q6

Multiplier:

$$\sqrt[2 \times 2]{4}$$

$$\sqrt[30 \times 4]{\boxed{120}}$$

Q7

$$\sqrt[2^{2/2}]{2}$$

$$\sqrt[2]{\text{times}}$$

Q8

$$\sqrt[20 \times 18]{30d}$$

$$\sqrt[d]{12}$$

$$\sqrt[12]{\text{days}}$$

Q9

Multiplier:

$$\sqrt[\frac{20}{12} \times \frac{35}{53}]{\frac{53}{35} = 1}$$

$$\sqrt[600]{\boxed{600}}$$

Q10

$$\sqrt[1.3^2]{1.69}$$

Increase:

$$\sqrt[69\%]{\boxed{69\%}}$$

Q11

$$\sqrt[3^n]{27=3^3}$$

$$\sqrt[n=3]{\boxed{n=3}}$$

Q12

$$\sqrt[2^n]{\frac{12}{2^{-1}}}$$

$$\sqrt[n=-1]{\boxed{n=-1}}$$

Q13

Length multiplier:

$$\sqrt[4]{1.25} = \frac{5}{4}$$

Breadth multiplier:

$$\sqrt[4]{\frac{4}{5}}$$

Decrease:

$$\boxed{20\%}$$

Q14

$$\left(\frac{2^2(1/2)}{2}\right)^2 = \frac{4}{4} = 1$$

$$\boxed{\text{No change}}$$

Q15

$$15 \cdot 8 \cdot 10 = 20 \cdot 6 \cdot d$$

Both sides' daily capacity:

$$15 \cdot 8 = 120$$

$$20 \cdot 6 = 120$$

Thus:

$$\boxed{10 \text{ days}}$$

Q16

$$2^2 \times \left(\frac{1}{2}\right)^3 \div \frac{1}{4}$$

$$= 4 \times \frac{1}{8} \times 4$$

$$= \boxed{2}$$

Q17

$$A \propto B^2, \text{quad } B \propto C^3$$

Therefore:

$$A \propto (C^3)^2$$

$$\boxed{A \propto C^6}$$

Q18

Side multiplier:

$$0.8$$

Volume multiplier:

$$0.8^3 = 0.512$$

Decrease:

$$1 - 0.512 = 0.488$$

$$\boxed{48.8\%}$$

Q19

Price multiplier:

$$\sqrt[5]{1.6}$$

Quantity multiplier:

$$\left[\frac{5}{8} \right]$$

Decrease:

$$\left[1 - \frac{5}{8} = \frac{3}{8} \right]$$

$$\left[\boxed{37.5\%} \right]$$

Q20

Time multiplier:

$$\left[1.25 = \frac{5}{4} \right]$$

Speed multiplier:

$$\left[\frac{4}{5} \right]$$

Decrease:

$$\left[\boxed{20\%} \right]$$

' CHAPTER 14 COMPLETE

We have now covered:

- direct variation '
- inverse variation '
- constant of proportionality '
- power variation '
- joint variation '
- combined direct/inverse variation '
- chained proportionality '
- unknown exponents '
- workers-hours-days '
- machines-production '
- food-consumption variation '
- speed-time-distance relationships '
- price-quantity-expenditure '
- area/volume variation '
- percentage effects '
- multiplier method '
- data-table recognition '
- advanced GATE/placement questions '

Progress

Completed: 14/76 **Unit 1 — Quantitative Aptitude: 0 = 0% PROGRESS** **Next:** 0 = 0% **Chapter 15 — Percentages**

Chapter 15 is one of the most important chapters in the entire aptitude course. It will cover:

- percentage basics and conversions

- percentage increase/decrease
- successive changes
- reverse percentages
- “more than” vs “less than”
- population/salary/price change
- percentage points
- comparison-base traps
- shortcut multipliers
- GATE and placement applications.

USER

next

ASSISTANT

CHAPTER 15 — PERCENTAGES

PART 1 — PERCENTAGE BASICS, COMPARISONS, INCREASE/DECREASE & REVERSE PERCENTAGE

Status: $\emptyset$ = 0% PROGRESS GATE: High Placements: Very High Priority: $\emptyset$ = 0% 1

Percentage is one of the most important aptitude topics because it appears inside:

- profit & loss
- discount
- interest
- mixtures
- averages
- DI
- population
- salary
- marks
- growth
- depreciation
- probability
- placement arithmetic

The central idea is simple:

$$\boxed{x\% = \frac{x}{100}}$$

The difficult part is usually not the formula. It is choosing the **correct reference/base quantity**.

1. WHAT DOES PERCENTAGE MEAN?

“Percent” means:

per hundred.

So:

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$60\% = \frac{60}{100} = \frac{3}{5}$$

$$125\% = \frac{125}{100} = \frac{5}{4}$$

A percentage can be:

- less than 100%
- equal to 100%
- greater than 100%

2. FRACTION !' PERCENTAGE

Multiply by:

$$100\%$$

Example:

$$\frac{3}{8}$$

$$\frac{3}{8} \times 100 = 37.5\%$$

Therefore:

$$\boxed{\frac{3}{8} = 37.5\%}$$

3. DECIMAL !' PERCENTAGE

Multiply decimal by 100.

Example:

$$0.72$$

$$0.72 \times 100 = \boxed{72\%}$$

Example:

$$1.4$$

$$= 140\%$$

4. PERCENTAGE !' DECIMAL

Divide by 100.

$$35\% = 0.35$$

$$4.5\% = 0.045$$

$$250\% = 2.5$$

5. MUST-MEMORIZE CONVERSIONS

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.333...	$\frac{33\frac{1}{3}\%}{}$
$\frac{2}{3}$	0.666...	$\frac{66\frac{2}{3}\%}{}$
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%
$\frac{5}{8}$	0.625	62.5%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{10}$	0.1	10%
$\frac{1}{20}$	0.05	5%
$\frac{1}{25}$	0.04	4%
$\frac{1}{40}$	0.025	2.5%

These should gradually become automatic.

6. FIND $x\%$ OF A NUMBER

Formula:

$$\boxed{x\% \text{ of } N} = \frac{x}{100}N$$

Example:

$$30\% \text{ of } 250$$

$$= \frac{30}{100} \times 250$$

$$= \boxed{75}$$

7. FAST BENCHMARK METHOD

For:

$$\backslash[17\%\text{ of }300 \backslash]$$

Instead of:

$$\backslash[0.17\times 300 \backslash]$$

use:

$$\backslash[10\%=30 \backslash]$$

$$\backslash[5\%=15 \backslash]$$

$$\backslash[2\%=6 \backslash]$$

So:

$$\backslash[17\%=10+5+2 \backslash]$$

$$\backslash[30+15+6 = \boxed{51} \backslash]$$

8. IMPORTANT IDENTITY

$$\backslash[\boxed{x\%\text{ of }y=y\%\text{ of }x} \backslash]$$

Example:

$$\backslash[16\%\text{ of }25 \backslash]$$

can be changed to:

$$\backslash[25\%\text{ of }16 \backslash]$$

$$\backslash[=\frac{14}{100}(16) \backslash]$$

$$\backslash[=\boxed{4} \backslash]$$

This can save time.

9. “A IS WHAT PERCENT OF B?”

Formula:

$$\backslash[\boxed{\frac{AB}{100}} \backslash]$$

Example:

45 is what percent of 60?

$$\backslash[\frac{45}{60}\times 100 \backslash]$$

$$\backslash[=\frac{34}{100} \backslash]$$

$$\backslash[=\boxed{75\%} \backslash]$$

10. REFERENCE QUANTITY IS THE DENOMINATOR

If question says:

A is what percent of B?

then:

$$\boxed{\frac{A}{B} \times 100}$$

B is the reference.

This is one of the biggest percentage traps.

11. EXAMPLE — CHANGE THE REFERENCE

A = 40, B = 50.

A as % of B

$$\frac{40}{50} \times 100 = 80\%$$

B as % of A

$$\frac{50}{40} \times 100 = 125\%$$

These are not complements.

12. PERCENTAGE INCREASE

Old value:

$$O$$

New value:

$$N$$

Increase:

$$N - O$$

Percentage increase:

$$\boxed{\frac{N - O}{O} \times 100}$$

The denominator is the **original value**.

13. EXAMPLE

Price rises from ₹ 200 to ₹ 250.

Increase:

$$50$$

Percentage increase:

$$\frac{50}{200} \times 100$$

$$= \boxed{25\%}$$

14. PERCENTAGE DECREASE

Formula:

$$\boxed{\frac{O-N}{O} \times 100}$$

Example:

Salary decreases from:

$$50000 \text{ to } 45000$$

Decrease:

$$5000$$

Percentage decrease:

$$\frac{5000}{50000} \times 100 = \boxed{10\%}$$

15. MULTIPLIER METHOD — MOST IMPORTANT

Instead of repeatedly using percentage formulas, convert changes into multipliers.

Increase by $(p\%)$:

$$\boxed{\times \left(1 + \frac{p}{100}\right)}$$

Decrease by $(p\%)$:

$$\boxed{\times \left(1 - \frac{p}{100}\right)}$$

16. COMMON MULTIPLIERS

Change	Multiplier
+10%	1.10
+20%	1.20
+25%	1.25
+50%	1.50
+100%	2
-10%	0.90
-20%	0.80
-25%	0.75
-50%	0.50

17. EXAMPLE — INCREASE USING MULTIPLIER

A salary of ₹40,000 increases by 15%.

New salary:

$$[40000 \times 1.15]$$

$$[= 40000 + 6000]$$

$$[= \boxed{46,000}]$$

18. EXAMPLE — DECREASE USING MULTIPLIER

Price $\text{\textasciitimes} 800$ decreases by 25%.

Multiplier:

$$[0.75 = \frac{3}{4}]$$

New price:

$$[800 \times \frac{3}{4}]$$

$$[= \boxed{600}]$$

19. “INCREASED TO” VS “INCREASED BY”

Very important language trap.

Increased by 20%

$$[100 \text{ to } 120]$$

Increased to 20%

means final value is:

$$[20\% \text{ of original}]$$

$$[100 \text{ to } 20]$$

These are completely different.

20. “DECREASED BY 30%”

Means retain:

$$[70\%]$$

Multiplier:

$$[0.7]$$

Example:

$$[500 \text{ to } 350]$$

21. PERCENTAGE OF TOTAL

If boys:girls = 3:2, then total parts:

$$[5]$$

Boys percentage:

$$\left[\frac{35}{50} \times 100 = \boxed{60\%} \right]$$

Girls:

$$\left[40\% \right]$$

This connects ratio and percentage.

22. PART VS WHOLE TRAP

If 40 students out of 50 pass:

Pass percentage:

$$\left[\frac{40}{50} \times 100 = 80\% \right]$$

Failure percentage:

$$\left[20\% \right]$$

But if asked:

Failed students are what percent of passed students?

then:

$$\left[\frac{10}{40} \times 100 = \boxed{25\%} \right]$$

Different reference.

23. MORE THAN — RATIO METHOD

If A is 25% more than B:

Take:

$$\left[B=100 \right]$$

Then:

$$\left[A=125 \right]$$

So:

$$\left[\boxed{A:B=5:4} \right]$$

This is often the fastest representation.

24. LESS THAN — RATIO METHOD

If A is 20% less than B:

Take:

$$\left[B=100 \right]$$

Then:

$$\boxed{A=80}$$

So:

$$\boxed{A:B=4:5}$$

25. MAJOR TRAP — MORE AND LESS ARE NOT SYMMETRIC

If A is 25% more than B:

$$\boxed{A:B=125:100=5:4}$$

Then B is less than A by:

$$\frac{1}{5} \times 100 = \boxed{20\%}$$

So:

$$25\% \text{ more} \not\rightarrow 25\% \text{ less}$$

26. GENERAL CONVERSION — MORE TO LESS

If A is $p\%$ more than B, then B is less than A by:

$$\boxed{\frac{100p}{100+p}\%}$$

Example:

$$p=20$$

$$\frac{100 \times 20}{100+20} = 16\frac{2}{3}\%$$

27. GENERAL CONVERSION — LESS TO MORE

If A is $p\%$ less than B, then B is more than A by:

$$\boxed{\frac{100p}{100-p}\%}$$

Example:

A is 20% less than B.

Then B is more than A by:

$$\frac{100 \times 20}{100-20} = \boxed{25\%}$$

28. IMPORTANT RECIPROCAL PAIRS

Memorize:

$$20\% \text{ less} \rightarrow 25\% \text{ more}$$

$$25\% \text{ less} \rightarrow 33\frac{1}{3}\% \text{ more}$$

$$33\frac{1}{3}\% \text{ less} \rightarrow 50\% \text{ more}$$

$$50\% \text{ less} \rightarrow 100\% \text{ more}$$

And reverse directions accordingly.

29. REVERSE PERCENTAGE — BASIC IDEA

Suppose after a 20% increase, value becomes:

$$120$$

Find original.

Do **not** subtract 20% of 120.

Instead:

$$\text{new} = 1.2 \times \text{old}$$

So:

$$\text{old} = \frac{120}{1.2}$$

$$= \boxed{100}$$

30. REVERSE INCREASE FORMULA

If final value (F) is after $(p\%)$ increase:

$$\boxed{\text{Original} = \frac{F}{1 + p/100}}$$

31. REVERSE DECREASE FORMULA

If final value (F) is after $(p\%)$ decrease:

$$\boxed{\text{Original} = \frac{F}{1 - p/100}}$$

32. EXAMPLE — REVERSE DECREASE

After a 25% discount, price is ₹ 900 .

Original price:

$$900 / 0.75$$

$$= 900 \times \frac{4}{3}$$

$$= \boxed{1200}$$

33. UNITARY METHOD FOR REVERSE PERCENTAGES

If after 20% decrease, 80% equals 640:

$$80\% \rightarrow 640$$

Then:

$$1\% \rightarrow 8$$

So:

$$\left[100\% \text{ to } 800\% \right]$$

Original:

$$\left[\boxed{800\%} \right]$$

This is often intuitive.

34. PERCENTAGE CHANGE WHEN ORIGINAL UNKNOWN

Suppose a quantity becomes 1.4 times original.

Then increase:

$$\left[1.4 - 1 = 0.4 \right]$$

$$\left[= \boxed{40\%} \right]$$

If it becomes 0.72 times:

Decrease:

$$\left[1 - 0.72 = 0.28 \right]$$

$$\left[= \boxed{28\%} \right]$$

35. MULTIPLIER !' PERCENT CHANGE

If new/old ratio is:

$$\left[r \right]$$

then:

$$\left[\boxed{\% \text{ change}} = (r - 1) \times 100 \right]$$

If $(r > 1)$, increase.

If $(r < 1)$, decrease.

36. RATIO !' PERCENTAGE CHANGE

Suppose old:new = 4:5.

Then:

$$\left[\frac{\text{new}}{\text{old}} = \frac{5}{4} = 1.25 \right]$$

Increase:

$$\left[\boxed{25\%} \right]$$

37. NEW:OLD VS OLD:NEW TRAP

If:

$$\left[\text{new} : \text{old} = 4 : 5 \right]$$

then new is only 80% of old.

So decrease:

$\downarrow 20\%$

Always read ratio order.

38. PERCENTAGE POINTS

Suppose pass rate rises:

$\uparrow 40\% \text{ to } 50\%$

Increase in **percentage points**:

$50 - 40 = \boxed{10 \text{ percentage points}}$

But relative percentage increase:

$\frac{10}{40} \times 100 = \boxed{25\%}$

These are different.

39. PERCENTAGE POINTS VS PERCENT CHANGE

From:

$\uparrow 20\% \text{ to } 30\%$

Percentage-point increase

10 points

Relative percentage increase

$\frac{10}{20} \times 100 = 50\%$

Never confuse them.

40. MARKS EXAMPLE

A student scores 72 out of 90.

Percentage:

$\frac{72}{90} \times 100$

$= \frac{45}{100} \times 100$

$= \boxed{80\%}$

41. REQUIRED MARKS

Passing percentage is 40%.

Maximum marks:

500

Pass marks:

$$\lfloor 0.4 \times 500 = \boxed{200} \rfloor$$

42. FAILED BY / EXCEEDED PASS MARKS

Pass marks = 200.

Student scores 185.

Failed by:

$$\lfloor 200 - 185 = \boxed{15} \rfloor$$

If scores 230:

Exceeded pass marks by:

$$\lfloor 30 \rfloor$$

These are absolute differences, not percentages unless asked.

43. FIND MAXIMUM MARKS

A student gets 240 marks and this is 60% of total.

Let total = $\lfloor T \rfloor$.

$$\lfloor 0.6T = 240 \rfloor$$

$$\lfloor T = 400 \rfloor$$

$$\lfloor \boxed{400} \rfloor$$

44. SALARY EXAMPLE

Salary rises from $\text{₹} 48,000$ to $\text{₹} 54,000$.

Increase:

$$\lfloor 6000 \rfloor$$

Percentage increase:

$$\lfloor \frac{6000}{48000} \times 100 \rfloor$$

$$\lfloor = \frac{18}{100} \rfloor$$

$$\lfloor = \boxed{12.5\%} \rfloor$$

45. PRICE EXAMPLE

Price decreases from $\text{₹} 750$ to $\text{₹} 600$.

Decrease:

$$\lfloor 150 \rfloor$$

Percentage decrease:

$$\lfloor \frac{150}{750} \times 100 = \frac{15}{100} \rfloor$$

$$\boxed{20\%}$$

46. POPULATION EXAMPLE

Population = 80,000.

It increases 5%.

New population:

$$80000 \times 1.05$$

$$\boxed{84,000}$$

47. WHEN THE BASE CHANGES, PERCENTAGE CHANGES

Suppose:

$$A=80, \quad B=100$$

A is:

$$20\%$$

less than B.

But B is:

$$\frac{20}{80} \times 100 = 25\%$$

more than A.

Same difference:

$$20$$

Different denominator.

48. “X% OF A = Y% OF B”

Suppose:

$$20\% \text{ of } A = 30\% \text{ of } B$$

Then:

$$0.2A = 0.3B$$

$$2A = 3B$$

Therefore:

$$\boxed{A:B=3:2}$$

General:

$$x\%A = y\%B \rightarrow A:B = y:x$$

49. EXAMPLE

$$\frac{40}{100} \text{ of } A = \frac{25}{100} \text{ of } B$$

Then:

$$40A = 25B$$

$$8A = 5B$$

So:

$$\boxed{A:B=5:8}$$

50. ONE QUANTITY AS % MORE THAN ANOTHER

If:

$$A:B=7:5$$

A is what percent more than B?

Difference:

$$2$$

Reference:

$$B=5$$

So:

$$\frac{25}{100} \times 100 = \boxed{40\%}$$

51. ONE QUANTITY AS % LESS THAN ANOTHER

Same ratio:

$$A:B=7:5$$

B is what percent less than A?

Difference:

$$2$$

Reference:

$$A=7$$

$$\frac{27}{100} \times 100$$

$$= \boxed{28\frac{4}{7}\%}$$

Again, not 40%.

52. SHORTCUT: ASSUME 100

Percentage-comparison questions often become easy by assuming the reference value is 100.

Example:

A is 35% less than B.

Let:

$$[B=100]$$

Then:

$$[A=65]$$

So immediately:

$$[A:B=65:100=13:20]$$

This is one of the best percentage tricks.

53. SHORTCUT: ASSUME LCM-FRIENDLY VALUE

If percentage has:

$$[16\frac{23}{100}=\frac{16}{100}]$$

instead of assuming 100, sometimes assume 6 or 600.

Example:

A is $(16\frac{23}{100})$ less than B.

Let:

$$[B=6]$$

Then:

$$[A=5]$$

So:

$$[\boxed{A:B=5:6}]$$

Cleaner than decimals.

54. PERCENTAGE INCREASE REQUIRED TO RESTORE A DECREASE

Suppose value decreases by 20%.

Start:

$$[100]$$

After decrease:

$$[80]$$

To return to 100, increase needed:

$$[20]$$

on base 80:

$$[\frac{20}{80}\times 100 = \boxed{25\%}]$$

So:

$$[20\%\downarrow]$$

requires:

$\downarrow 25\%$

to restore.

55. DECREASE REQUIRED TO REVERSE AN INCREASE

Suppose value increases by 25%.

$100 \rightarrow 125$

To return to 100:

decrease:

25%

as percentage of 125:

$\frac{25}{125} \times 100 = \boxed{20\%}$

56. GENERAL RESTORATION FORMULAS

After $(p\%)$ decrease, required increase to restore:

$\boxed{\frac{100p}{100-p}\%}$

After $(p\%)$ increase, required decrease to restore:

$\boxed{\frac{100p}{100+p}\%}$

These match the more/less conversion formulas.

57. GATE-STYLE QUESTION 1

A number increases from 240 to 300.

Percentage increase:

$\frac{60}{240} \times 100$

$= \boxed{25\%}$

58. GATE-STYLE QUESTION 2

A quantity decreases by 20% and becomes 480.

Original:

$480 / 0.8$

$= \boxed{600}$

59. GATE-STYLE QUESTION 3

A is 50% more than B.

Find B as percent of A.

Let:

$$[B=100, \quad A=150]$$

$$[\frac{B}{A} \times 100 = \frac{100}{150} \times 100]$$

$$[=\boxed{66\frac{2}{3}\%}]$$

So B is:

$$[33\frac{1}{3}\%]$$

less than A.

60. GATE-STYLE QUESTION 4

A percentage rises from 24% to 30%.

Percentage-point increase:

$$[\boxed{6}]$$

Relative percentage increase:

$$[\frac{6}{24} \times 100 = \boxed{25\%}]$$

61. PLACEMENT-STYLE QUESTION 1

A salary of ₹ 30,000 increases by 20%.

New salary:

$$[30000 \times 1.2 = \boxed{36000}]$$

62. PLACEMENT-STYLE QUESTION 2

A product priced at ₹ 1600 reduced by 15%.

New price:

$$[1600 \times 0.85]$$

$$[=1600-240]$$

$$[=\boxed{1360}]$$

63. PLACEMENT-STYLE QUESTION 3

After a 25% increase, a number becomes 500.

Original:

$$[500/1.25]$$

$$[=500 \times \frac{4}{5}]$$

$$[=\boxed{400}]$$

64. PLACEMENT-STYLE QUESTION 4

A is 30% less than B. By what percent is B more than A?

Take:

$$B=100$$

$$A=70$$

Difference:

$$30$$

Percentage relative to A:

$$\frac{30}{70} \times 100$$

$$= \boxed{42\frac{6}{7}\%}$$

65. FAST PERCENTAGE OF NUMBER TRICKS

10%

Move decimal one place:

$$10\% \text{ of } 470 = 47$$

1%

Move decimal two places:

$$1\% \text{ of } 470 = 4.7$$

5%

Half of 10%:

$$23.5$$

15%

$$10\% + 5\%$$

20%

Divide by 5.

25%

Divide by 4.

50%

Divide by 2.

75%

Three-fourths.

12.5%

Divide by 8.

6.25%

Divide by 16.

66. EXAMPLE — 35% OF 240

Use:

$$\left[25\% + 10\% \right]$$

$$\left[60 + 24 = \boxed{84} \right]$$

67. EXAMPLE — 62.5% OF 160

$$\left[62.5\% = \frac{5}{8} \right]$$

$$\left[160 \times \frac{5}{8} \right]$$

$$\left[20 \times 5 = \boxed{100} \right]$$

68. COMMON TRAPS

& pTRAP 1 — Wrong denominator

Percentage change always uses original/reference value.

& pTRAP 2 — Increase and decrease are not symmetric

$$\left[20\% \uparrow, 20\% \downarrow \right]$$

do not cancel.

We will handle successive changes in Part 2.

& pTRAP 3 — “More than” vs “of”

A is 20% more than B:

$$\left[A = 1.2B \right]$$

A is 20% of B:

$$\left[A = 0.2B \right]$$

& pTRAP 4 — Percentage points

40% to 50% is +10 percentage points but +25%.

& pTRAP 5 — Reverse percentage

If final value is after increase/decrease, divide by multiplier; don't reverse by subtracting same percentage.

69. MUST-MEMORIZE FORMULAS

Percentage of quantity

$$\boxed{x\% \text{ of } N = \frac{xN}{100}}$$

A as % of B

$$\boxed{\frac{A}{B} \times 100}$$

Percentage increase

$$\boxed{\frac{N-O}{O} \times 100}$$

Percentage decrease

$$\boxed{\frac{O-N}{O} \times 100}$$

New after $(p\%)$ increase

$$\boxed{O \left(1 + \frac{p}{100} \right)}$$

New after $(p\%)$ decrease

$$\boxed{O \left(1 - \frac{p}{100} \right)}$$

70. 5-MINUTE REVISION

Whenever you see a percentage question, ask:

“Percent of what?”

That tells you the denominator.

Then decide:

- ordinary percentage?
- increase/decrease?
- more/less comparison?
- reverse percentage?
- ratio conversion?
- percentage-point question?

Use multipliers whenever possible.

PRACTICE — LEVEL 0

Q1

Find 15% of 200.

Q2

45 is what percent of 90?

Q3

Convert $\frac{7}{8}$ to percentage.

Q4

A value rises from 80 to 100. Find % increase.

Q5

A value falls from 200 to 150. Find % decrease.

LEVEL 1

Q6

A salary of ₹ 24,000 rises by 25%. Find new salary.

Q7

A price of ₹ 900 decreases by 20%. Find new price.

Q8

After a 20% increase, a value becomes 360. Find original.

Q9

After a 25% decrease, a value becomes 450. Find original.

Q10

A is 20% more than B. Find A:B.

LEVEL 2

Q11

A is 20% less than B. By what percent is B more than A?

Q12

A is 40% more than B. By what percent is B less than A?

Q13

A:B = 9:12. A is what percent of B?

Q14

A:B = 9:12. A is what percent of total $(A+B)$?

Q15

30% of A = 45% of B. Find A:B.

LEVEL 3 — EXAM PRESSURE

Q16

A quantity falls by 20%. What percentage increase restores it?

Q17

A quantity increases by 60%. What percentage decrease restores the original?

Q18

A rate rises from 16% to 20%. Find:

1. percentage-point rise;
2. relative percentage rise.

Q19

A number is 37.5% of 640. Find it mentally.

Q20

A is 25% more than B, and B is 20% more than C. If C=200, find A.

ANSWERS**Q1**

$$\lfloor 10\% = 20, \quad 5\% = 10 \rfloor$$

$$\lfloor \boxed{30} \rfloor$$

Q2

$$\lfloor \frac{45}{90} \times 100 = \boxed{50\%} \rfloor$$

Q3

$$\lfloor \frac{78}{90} \times 100 = \boxed{87.5\%} \rfloor$$

Q4

Increase:

$$\lfloor 20 \rfloor$$

$$\lfloor 20/80 \times 100 = \boxed{25\%} \rfloor$$

Q5

Decrease:

$$\lfloor 50 \rfloor$$

$$\lfloor 50/200 \times 100 = \boxed{25\%} \rfloor$$

Q6

$$\lfloor 24000 \times 1.25 = \boxed{30000} \rfloor$$

Q7

$$\lfloor 900 \times 0.8 = \boxed{720} \rfloor$$

Q8

$$\lfloor 360/1.2 = \boxed{300} \rfloor$$

Q9

$$\left[450/0.75 = 450 \times \frac{4}{3} = \boxed{600} \right]$$

Q10

$$\left[120:100 = \boxed{6:5} \right]$$

Q11

Take:

$$\left[B=100, \text{quad } A=80 \right]$$

Difference:

$$\left[20 \right]$$

relative to A:

$$\left[20/80 \times 100 = \boxed{25\%} \right]$$

Q12

Take:

$$\left[B=100, \text{quad } A=140 \right]$$

Difference 40 relative to A:

$$\left[40/140 \times 100 = \boxed{28\frac{4}{7}\%} \right]$$

Q13

$$\left[\frac{9}{12} \times 100 = \boxed{75\%} \right]$$

Q14

Total:

$$\left[21 \right]$$

$$\left[\frac{9}{21} \times 100 = \frac{37}{3} \times 100 \right]$$

$$\left[= \boxed{42\frac{2}{3}\%} \right]$$

Q15

$$\left[0.30A=0.45B \right]$$

$$\left[30A=45B \right]$$

$$\left[2A=3B \right]$$

$$\left[\boxed{A:B=3:2} \right]$$

Q16

After 20% decrease:

$$\left[100 \text{ to } 80 \right]$$

Need increase:

$$\left[20/80 \times 100 = \boxed{25\%} \right]$$

Q17

$$\left[100 \text{ to } 160 \right]$$

Need decrease:

$$\left[60/160 \times 100 = \boxed{37.5\%} \right]$$

Q18

From:

$$\left[16\% \text{ to } 20\% \right]$$

Percentage-point increase:

$$\left[\boxed{4} \right]$$

Relative increase:

$$\left[\frac{4}{16} \times 100 = \boxed{25\%} \right]$$

Q19

$$\left[37.5\% = \frac{3}{8} \right]$$

$$\left[640 \times \frac{3}{8} = 80 \times 3 = \boxed{240} \right]$$

Q20

$$C = 200.$$

B is 20% more:

$$\left[B = 200 \times 1.2 = 240 \right]$$

A is 25% more than B:

$$\left[A = 240 \times 1.25 \right]$$

$$\left[= \boxed{300} \right]$$

CHAPTER 15 STATUS

Covered so far:

- percentage meaning '
- fraction/decimal/percentage conversion '
- finding percentage of a quantity '
- "A is what % of B?" '
- percentage increase/decrease '
- multiplier method '
- more-than / less-than comparisons '
- reverse percentage '
- restoration percentages '
- ratio !" percentage '
- percentage points '
- marks/salary/price/population basics '
- reference-base traps '
- fast mental percentage calculations '

But **Chapter 15 is not complete yet.**

Next we need the most exam-heavy percentage section:

- successive percentage changes
- increase then decrease
- net percentage change formula
- repeated growth/depreciation
- population and salary chains
- constant expenditure questions
- percentage distribution changes
- advanced comparison problems
- GATE/placement mixed questions

Progress

Completed: 14/76 **Current:** 0=0 Chapter 15 — Percentages **Next:** Chapter 15 Part 2 — **Successive changes, net change & advanced percentage problems**

USER

next

ASSISTANT

CHAPTER 15 — PERCENTAGES

PART 2 — SUCCESSIVE CHANGES, NET PERCENTAGE CHANGE, CONSTANT EXPENDITURE & ADVANCED APPLICATIONS

Status: 0=0 **PROGRESS** **GATE:** High **Placements:** Very High **Priority:** 0=0 **Year 1**

This is the most exam-heavy part of percentages.

The central idea is:

Successive percentage changes act on different bases. So you usually cannot simply add or subtract them.

1. SUCCESSIVE PERCENTAGE CHANGES

Suppose a value increases by 20%, then increases by 10%.

Start with:

$\backslash$ 100 $\backslash$

After 20% increase:

$\backslash$ 100\to120 $\backslash$

Then 10% increase is on 120:

$\left[120\text{to}132 \right]$

Overall increase:

$\left[32 \right]$

Therefore:

$\left[\boxed{32\%} \right]$

Not 30%.

2. MULTIPLIER METHOD — BEST METHOD

Increase 20%:

$\left[\times 1.2 \right]$

Increase 10%:

$\left[\times 1.1 \right]$

Combined multiplier:

$\left[1.2 \times 1.1 = 1.32 \right]$

Thus:

$\left[\boxed{32\% \text{ increase}} \right]$

3. SUCCESSIVE CHANGE FORMULA

For percentage changes $\left(a\% \right)$ and $\left(b\% \right)$, taking increases as positive and decreases as negative:

$\left[\boxed{\text{Net \% change} = a + b + \frac{ab}{100}} \right]$

Example:

$\left[+20\%, +10\% \right]$

$\left[20 + 10 + \frac{20 \times 10}{100} \right]$

$\left[= 30 + 2 \right]$

$\left[= \boxed{32\%} \right]$

MUST MEMORIZE

But multiplier method is often safer.

4. TWO SUCCESSIVE DECREASES

Decrease 20%, then decrease 10%.

Using formula:

$\left[a = -20, b = -10 \right]$

$\left[-20 - 10 + \frac{(-20)(-10)}{100} \right]$

$\left[= -30 + 2 \right]$

$$\boxed{-28\%}$$

Therefore:

$$\boxed{28\% \text{ decrease}}$$

5. VERIFY USING 100

Start:

$$100$$

After 20% decrease:

$$80$$

Then 10% decrease:

$$80 \rightarrow 72$$

Overall decrease:

$$28$$

So:

$$\boxed{28\%}$$

6. INCREASE THEN DECREASE

Suppose:

$$+20\%, -20\%$$

Many students say net zero.

Wrong.

Multiplier:

$$1.2 \times 0.8 = 0.96$$

Final = 96% of original.

Therefore:

$$\boxed{4\% \text{ decrease}}$$

7. SAME PERCENT INCREASE AND DECREASE

If a quantity increases by $p\%$ and then decreases by $p\%$:

$$\left(1 + \frac{p}{100}\right) \left(1 - \frac{p}{100}\right)$$

$$= 1 - \frac{p^2}{10000}$$

So net decrease:

$$\boxed{\frac{p^2}{100}\%}$$

Example:

$$[p=20]$$

$$[\frac{20^2}{100} = \boxed{4\%}]$$

8. MEMORY TRICK

Equal increase and decrease never cancel.

They always give:

$$[\boxed{\text{net decrease}}]$$

because the decrease acts on a larger base if increase came first, or the recovery from a smaller base is insufficient if decrease came first.

9. ORDER OF PERCENTAGE CHANGES

For pure multiplicative changes, order does not matter.

Example:

$$[+20\%, -10\%]$$

Multiplier:

$$[1.2 \times 0.9 = 1.08]$$

Whether increase or decrease comes first:

$$[0.9 \times 1.2 = 1.08]$$

So net:

$$[\boxed{8\% \text{ increase}}]$$

10. NET CHANGE FORMULA — MIXED SIGNS

Example:

Increase 25%, then decrease 20%.

$$[a=25, \text{quad } b=-20]$$

$$[25-20+\frac{25(-20)}{100}]$$

$$[=5-5]$$

$$[=\boxed{0\%}]$$

Indeed:

$$[1.25 \times 0.8 = 1]$$

So the value returns exactly to original.

11. HIGH-VALUE CANCELLING PAIRS

These multipliers multiply to 1:

$$\left[+25\%\text{and}\quad-20\% \right]$$

because:

$$\left[1.25\times0.8=1 \right]$$

Similarly:

$$\left[+20\%\text{and}\quad-16\frac{23}{100}\% \right]$$

because:

$$\left[1.2\times\frac{56}{100}=1 \right]$$

Also:

$$\left[+50\%\text{and}\quad-33\frac{13}{100}\% \right]$$

because:

$$\left[1.5\times\frac{23}{100}=1 \right]$$

12. THREE SUCCESSIVE CHANGES

Suppose:

$$\left[+10\%, +20\%, -25\% \right]$$

Use multipliers:

$$\left[1.1\times1.2\times0.75 \right]$$

First:

$$\left[1.2\times0.75=0.9 \right]$$

Then:

$$\left[1.1\times0.9=0.99 \right]$$

So final is 99% of original.

Net:

$$\left[\boxed{1\%\text{ decrease}} \right]$$

13. DO NOT USE TWO-CHANGE FORMULA RECKLESSLY

For 3 or more changes, multiplier method is usually easier and safer:

$$\left[\boxed{\text{Final}=\text{Original}\times m_1m_2m_3\cdots} \right]$$

Then:

$$\left[\boxed{\text{Net \%}=(\text{overall multiplier}-1)\times100} \right]$$

14. REPEATED SAME PERCENTAGE INCREASE

If a quantity increases by $(p\%)$ each period for (n) periods:

$$\left[\boxed{F=P\left(1+\frac{p}{100}\right)^n} \right]$$

This is repeated growth.

15. POPULATION GROWTH EXAMPLE

Population = 50,000.

Growth = 10% per year for 2 years.

$$\sqrt[5]{50000(1.1)^2}$$

$$\sqrt[5]{=50000(1.21)}$$

$$\sqrt[5]{=\boxed{60,500}}$$

Total increase is 21%, not 20%.

16. REPEATED DEPRECIATION

If value decreases $\sqrt[p]{p\%}$ per year:

$$\sqrt[p]{\boxed{F=P\left(1-\frac{p}{100}\right)^n}}$$

Example:

Machine worth ₹ 1,00,000 depreciates 10% per year for 2 years.

$$\sqrt[2]{100000(0.9)^2}$$

$$\sqrt[2]{=100000(0.81)}$$

$$\sqrt[2]{=\boxed{181,000}}$$

17. GROWTH AND DEPRECIATION ARE MULTIPLICATIVE

A 10% decrease for 2 years means:

$$\sqrt[2]{0.9^2=0.81}$$

So total decrease:

$$\sqrt[2]{19\%}$$

not:

$$\sqrt[2]{20\%}$$

18. REVERSE REPEATED GROWTH

A population after 2 years of 10% annual growth is 60,500.

Original:

$$\sqrt[2]{P(1.1)^2=60500}$$

$$\sqrt[2]{P=\frac{60500}{1.21}}$$

$$\sqrt[2]{=\boxed{50,000}}$$

19. CONSTANT EXPENDITURE PROBLEMS

Recall:

$$[\text{Expenditure} = \text{Price} \times \text{Quantity}]$$

If expenditure stays constant:

$$[PQ = \text{constant}]$$

Therefore price and quantity are inversely related.

20. PRICE INCREASE !' QUANTITY DECREASE

If price increases by $(p\%)$, quantity must decrease by:

$$[\boxed{\frac{100p}{100+p}\%}]$$

to keep expenditure constant.

Example:

Price increases 25%.

Quantity decrease:

$$[\frac{2500}{125} = \boxed{20\%}]$$

21. PRICE DECREASE !' QUANTITY INCREASE

If price decreases by $(p\%)$, quantity can increase by:

$$[\boxed{\frac{100p}{100-p}\%}]$$

Example:

Price decreases 20%.

Quantity increase:

$$[\frac{2000}{80} = \boxed{25\%}]$$

22. CONSTANT EXPENDITURE USING MULTIPLIERS

Price increases 50%.

Price multiplier:

$$[1.5]$$

To keep product constant:

$$[Q_{\text{multiplier}} = \frac{1}{1.5} = \frac{2}{3}]$$

So quantity falls by:

$$[\frac{1}{3} = \boxed{33\frac{1}{3}\%}]$$

23. SALARY AND EXPENDITURE

Income rises 20%, expenditure rises 10%.

Savings do **not** automatically rise 10%.

Why?

Because:

$$\text{Savings} = \text{Income} - \text{Expenditure}$$

You need the original income/expenditure relationship.

This is a common hidden-base trap.

24. EXAMPLE — SAVINGS CHANGE

Income = ₹ 50,000.

Expenditure = ₹ 40,000.

Savings:

$$10000$$

Income rises 20%:

$$60000$$

Expenditure rises 10%:

$$44000$$

New savings:

$$16000$$

Savings increase:

$$6000$$

Percentage increase:

$$\frac{6000}{10000} \times 100 = \boxed{60\%}$$

Not 10% or 20%.

25. MARKS — TWO-STUDENT PERCENTAGE PROBLEM

A scores 20% more than B.

If B scores 250:

$$A = 250(1.2) = 300$$

Difference:

$$50$$

B is how much less than A?

$$50/300 \times 100 = \boxed{16\frac{2}{3}\%}$$

26. POPULATION WITH DIFFERENT YEARLY CHANGES

Population = 10,000.

Year 1: +20%.

Year 2: " 10 % .

Final:

$$\boxed{10000(1.2)(0.9)}$$

$$=10000(1.08)$$

$$\boxed{10,800}$$

Net:

$$\boxed{8\%\text{ increase}}$$

27. SALARY WITH RAISE AND CUT

Salary increases 25%, then decreases 20%.

Multiplier:

$$1.25 \times 0.8 = 1$$

Final salary:

$$\boxed{\text{unchanged}}$$

This is a very common placement question.

28. PERCENTAGE DISTRIBUTION CHANGE

Suppose in a class:

- boys = 60%
- girls = 40%

If total = 200:

$$\text{boys} = 120, \quad \text{girls} = 80$$

If only girls increase, you cannot simply modify percentage directly. Work with actual counts or ratio.

29. EXAMPLE — NEW PERCENTAGE AFTER ADDITION

A class has:

$$60\% \text{ boys}$$

Total students:

$$100$$

So boys:

$$\sqrt[60]{}$$

girls:

$$\sqrt[40]{}$$

20 girls join.

New total:

$$\sqrt[120]{}$$

Girls:

$$\sqrt[60]{}$$

Girls percentage:

$$\sqrt[\frac{60}{120} \times 100 = \boxed{50\%}]{} \sqrt[{}]{}$$

30. FIND ADDITION NEEDED TO CHANGE A PERCENTAGE

A class has 60 boys and 40 girls.

How many girls must join so girls become 50%?

Let (x) girls join.

$$\sqrt[\frac{40+x}{100+x} = \frac{12}{100}]{} \sqrt[{}]{}$$

$$\sqrt[80+2x=100+x]{} \sqrt[{}]{}$$

$$\sqrt[x=20]{} \sqrt[{}]{}$$

Answer:

$$\sqrt[\boxed{20}]{} \sqrt[{}]{}$$

31. PERCENTAGE CHANGE IN PRODUCT

If:

$$\sqrt[Z=XY]{} \sqrt[{}]{}$$

and:

- X increases by $(a\%)$
- Y increases by $(b\%)$

then:

$$\sqrt[\boxed{\text{Net change} = a+b+\frac{ab}{100}}]{} \sqrt[{}]{}$$

This is the same successive-change formula.

32. EXAMPLE — RECTANGLE AREA

Length increases 20%.

Breadth increases 10%.

Area multiplier:

$$1.2 \times 1.1 = 1.32$$

Area increases:

$$\boxed{32\%}$$

33. ONE FACTOR INCREASES, OTHER DECREASES

Length +25%, breadth " 20 % .

Area multiplier:

$$1.25 \times 0.8 = 1$$

Area:

$$\boxed{\text{unchanged}}$$

34. PERCENTAGE CHANGE IN A QUOTIENT

If:

$$Z = \frac{XY}{}$$

use multipliers.

Example:

X +20%, Y +50%.

$$Z_{\text{multiplier}} = \frac{1.2}{1.5} = 0.8$$

So:

$$\boxed{20\% \text{ decrease}}$$

35. PERCENTAGE CHANGE IN SQUARE

If side increases 20%:

$$\text{area multiplier} = 1.2^2 = 1.44$$

Increase:

$$\boxed{44\%}$$

Shortcut:

$$2p + \frac{p^2}{100}$$

For (p=20):

$$40 + 4 = 44\%$$

36. PERCENTAGE CHANGE IN CUBE

If side increases 10%:

$$1.1^3 = 1.331$$

Volume increase:

$$\boxed{33.1\%}$$

For exact problems, use multiplier.

37. PERCENTAGE CHANGE FROM RATIO

Old:new:

$$5:7$$

Then:

$$\frac{\text{new}}{\text{old}} = \frac{7}{5} = 1.4$$

So increase:

$$\boxed{40\%}$$

38. PERCENTAGE CHANGE FROM NEW:OLD

New:old:

$$7:8$$

$$\frac{\text{new}}{\text{old}} = \frac{7}{8} = 0.875$$

Decrease:

$$12.5\%$$

$$\boxed{12.5\%}$$

39. PERCENTAGE DIFFERENCE VS PERCENTAGE CHANGE

Suppose values are 80 and 100.

“80 is how much less than 100?”:

$$\frac{20}{100} = 20\%$$

“100 is how much more than 80?”:

$$\frac{20}{80} = 25\%$$

Sometimes a problem says “percentage difference” and may define:

$$\frac{|A-B|}{(A+B)/2} \times 100$$

That is a different statistical convention.

Exam rule

Use exactly the definition or wording provided.

40. PERCENT OF PERCENT

$$20\% \text{ of } 30\%$$

means:

$$[0.2 \times 0.3 = 0.06]$$

So:

$$[\boxed{6\%}]$$

Example:

20% of 30% of 500:

$$[6\% \text{ of } 500 = \boxed{30}]$$

41. FRACTION OF A PERCENTAGE

Half of 40%:

$$[\frac{1}{2} \times 40\% = 20\%]$$

Three-fourths of 80%:

$$[\frac{3}{4} \times 80\% = 60\%]$$

42. PERCENTAGE BASE CHANGE

A quantity changes from 25% of (X) to 40% of (X) .

Relative increase:

$$[\frac{40-25}{25} \times 100]$$

$$[= \frac{15}{25} \times 100]$$

$$[= \boxed{60\%}]$$

Percentage-point change:

$$[15 \text{ points}]$$

43. SUCCESSIVE DISCOUNT PREVIEW

A product gets discounts 20% and 10%.

Equivalent discount is:

$$[20 + 10 - \frac{20 \times 10}{100}]$$

$$[= 30 - 2]$$

$$[= \boxed{28\%}]$$

Why minus?

Because both are decreases.

This becomes central in Chapter 19.

44. SUCCESSIVE INCREASE PREVIEW

Two raises of 10% each:

$$\left[10 + 10 + \frac{10 \cdot 10}{100} \right]$$

$$\left[= 21\% \right]$$

So:

$$\left[\boxed{\text{two 10\% increases} = 21\% \text{ total increase}} \right]$$

45. SUCCESSIVE DECREASE PREVIEW

Two reductions of 10% each:

$$\left[10 + 10 - \frac{10 \cdot 10}{100} \right]$$

$$\left[= 19\% \right]$$

So:

$$\left[\boxed{\text{two 10\% decreases} = 19\% \text{ total decrease}} \right]$$

46. FAST OPTION ELIMINATION

A number is increased 50% then decreased 50%.

Options:

A. unchanged B. 25% decrease C. 50% decrease D. 25% increase

Equal increase/decrease formula:

$$\left[\frac{50^2}{100} = 25\% \right]$$

net decrease.

Answer:

$$\left[\boxed{\text{B}} \right]$$

47. GATE-STYLE QUESTION 1

A value rises 20% and then rises 25%.

Net multiplier:

$$\left[1.2 \times 1.25 = 1.5 \right]$$

Net increase:

$$\left[\boxed{50\%} \right]$$

48. GATE-STYLE QUESTION 2

A value falls 20% and then rises 25%.

Multiplier:

$$\left[0.8 \times 1.25 = 1 \right]$$

Therefore:

$\boxed{\text{No net change}}$

49. GATE-STYLE QUESTION 3

A quantity increases 10% each year for 3 years.

Overall multiplier:

$1.1^3 = 1.331$

Overall increase:

$\boxed{33.1\%}$

50. GATE-STYLE QUESTION 4

Price rises 20%. By what percentage must consumption decrease to keep expenditure constant?

$\frac{100 \times 20}{120}$

$\boxed{16\frac{2}{3}\%}$

51. PLACEMENT-STYLE QUESTION 1

Salary increases 20%, then 10%.

Original salary ₹ 30,000.

Final:

$30000(1.2)(1.1)$

$= 30000(1.32)$

$\boxed{₹ 39,600}$

52. PLACEMENT-STYLE QUESTION 2

A product price decreases 20% and then 10%.

Original ₹ 2000.

Multiplier:

$0.8 \times 0.9 = 0.72$

Final:

$2000 \times 0.72 = \boxed{₹ 1440}$

Equivalent decrease:

$\boxed{28\%}$

53. PLACEMENT-STYLE QUESTION 3

Population = 80,000.

Increase 5%, then 10%.

Multiplier:

$$\sqrt[1]{1.05 \times 1.10 = 1.155}$$

Final:

$$\sqrt[1]{80000 \times 1.155}$$

$$\sqrt[1]{= \boxed{92,400}}$$

Overall increase:

$$\sqrt[1]{15.5\%}$$

54. PLACEMENT-STYLE QUESTION 4

Length +20%, breadth " 10 % .

Area multiplier:

$$\sqrt[1]{1.2 \times 0.9 = 1.08}$$

Area increases:

$$\sqrt[1]{\boxed{8\%}}$$

55. ADVANCED QUESTION — SAVINGS

A person spends 80% of income.

Income increases 25%, expenditure increases 10%.

Find % change in savings.

Take income:

$$\sqrt[1]{100}$$

Expenditure:

$$\sqrt[1]{80}$$

Savings:

$$\sqrt[1]{20}$$

New income:

$$\sqrt[1]{125}$$

New expenditure:

$$\sqrt[1]{88}$$

New savings:

$$\sqrt[1]{37}$$

Increase:

$$\sqrt[17]{}$$

Relative to old savings 20:

$$\sqrt[\frac{17}{20} \times 100 = \boxed{85\%}]{}]$$

56. ADVANCED QUESTION — PASS RATE

Pass rate rises from 64% to 80%.

Percentage-point increase:

$$\sqrt[16]{}$$

Relative percentage increase:

$$\sqrt[\frac{16}{64} \times 100 = \boxed{25\%}]{}]$$

57. ADVANCED QUESTION — BASE UNKNOWN

A number is increased by 20% and then reduced by 25%, becoming 360.

Combined multiplier:

$$\sqrt[1.2 \times 0.75 = 0.9]{}$$

So:

$$\sqrt[0.9O=360]{}$$

$$\sqrt[O=400]{}$$

$$\sqrt[\boxed{400}]{}$$

58. ADVANCED QUESTION — FIND SECOND CHANGE

A value increases 25%. What percentage decrease is needed afterward so final value is 10% above original?

First multiplier:

$$\sqrt[1.25]{}$$

Desired final multiplier:

$$\sqrt[1.10]{}$$

Required second multiplier:

$$\sqrt[\frac{1.10}{1.25} = 0.88]{}$$

So decrease:

$$\sqrt[12\%]{}$$

$$\sqrt[\boxed{12\%}]{}$$

59. GENERAL “FIND MISSING CHANGE” METHOD

If:

$$m_1 m_2 = m_{\text{target}}$$

then:

$$m_2 = \frac{m_{\text{target}}}{m_1}$$

Then convert m_2 back to percentage.

This is more reliable than manipulating net-change formulas.

60. COMMON TRAPS

& pTRAP 1 — Adding successive percentages

Only valid as a rough approximation for very small changes, not exact.

& pTRAP 2 — Equal increase/decrease cancel

False.

$$+p\%, -p\% \rightarrow \frac{p^2}{100}\% \text{ net decrease}$$

& pTRAP 3 — Repeated growth

10% for 2 years is 21%, not 20%.

& pTRAP 4 — Savings

Savings is a difference, so you need the original income–expense relation.

& pTRAP 5 — Constant expenditure

Price and quantity move inversely, so equal percentage changes do not cancel.

& pTRAP 6 — Percentage-point vs relative %

Always identify which one is asked.

61. MUST-MEMORIZE FORMULAS

Two successive changes

$$a + b + \frac{ab}{100}$$

with decreases treated as negative.

Equal increase/decrease

$$\frac{p^2}{100}\% \text{ decrease}$$

Repeated increase

$$F = P \left(1 + \frac{p}{100} \right)^n$$

Repeated decrease

$$\boxed{F = P \left(1 - \frac{p}{100} \right)^n}$$

Constant expenditure

Price increase $(p\%)$:

$$\boxed{\text{quantity decrease} = \frac{100p}{100+p}\%}$$

Price decrease $(p\%)$:

$$\boxed{\text{quantity increase} = \frac{100p}{100-p}\%}$$

62. FASTEST EXAM METHOD

For almost every advanced percentage-change problem:

Step 1

Convert every change to multiplier.

Step 2

Multiply/divide multipliers.

Step 3

Compare final multiplier with 1.

Examples:

$$1.32 \rightarrow 32\% \uparrow$$

$$0.84 \rightarrow 16\% \downarrow$$

$$1 \rightarrow \text{no change}$$

This avoids most sign mistakes.

PRACTICE — LEVEL 0

Q1

Increase 10%, then increase 20%. Find net % change.

Q2

Decrease 10%, then decrease 20%. Find net % change.

Q3

Increase 20%, then decrease 20%. Find net change.

Q4

Increase 25%, then decrease 20%. Find net change.

Q5

Decrease 20%, then increase 25%. Find net change.

LEVEL 1

Q6

A population grows 10% annually for 2 years. Find total % growth.

Q7

A machine depreciates 20% annually for 2 years. Find total % depreciation.

Q8

Price increases 25%. What % quantity decrease keeps expenditure fixed?

Q9

Price decreases 20%. What % quantity increase keeps expenditure fixed?

Q10

Length increases 10%, breadth increases 20%. Find area % change.

LEVEL 2

Q11

A value increases 20%, decreases 10%, then increases 25%. Find net change.

Q12

A value decreases 30% and then increases 30%. Find net change.

Q13

A salary becomes ₹ 54,000 after successive increases of 20% and 12.5%. Find original salary.

Q14

A number increases 25% and then by $x\%$, ending 50% above original. Find x .

Q15

Income rises 20%, expenditure rises 15%. Initially expenditure is 80% of income. Find % change in savings.

LEVEL 3 — EXAM PRESSURE

Q16

A quantity changes by +50%, -20%, 25%. Find net change.

Q17

A population grows by 20% in year 1 and falls by 10% in year 2. What single percentage change gives same final result?

Q18

A rectangle's area is unchanged. Length increases 40%. Find required % decrease in breadth.

Q19

A pass rate goes from 48% to 60%. Find:

1. percentage-point increase;
2. relative percentage increase.

Q20

A number is reduced by 20%, then increased by 50%, and becomes 720. Find original.

ANSWERS**Q1**

$$\backslash[1.1\backslashtimes1.2=1.32 \backslash]$$

$$\backslash[\backslashboxed{32\%\backslashtext{ increase}} \backslash]$$

Q2

$$\backslash[0.9\backslashtimes0.8=0.72 \backslash]$$

$$\backslash[\backslashboxed{28\%\backslashtext{ decrease}} \backslash]$$

Q3

$$\backslash[1.2\backslashtimes0.8=0.96 \backslash]$$

$$\backslash[\backslashboxed{4\%\backslashtext{ decrease}} \backslash]$$

Q4

$$\backslash[1.25\backslashtimes0.8=1 \backslash]$$

$$\backslash[\backslashboxed{\backslashtext{No change}} \backslash]$$

Q5

$$\backslash[0.8\backslashtimes1.25=1 \backslash]$$

$$\backslash[\backslashboxed{\backslashtext{No change}} \backslash]$$

Q6

$$\backslash[1.1^2=1.21 \backslash]$$

$$\backslash[\backslashboxed{21\%\backslashtext{ growth}} \backslash]$$

Q7

$$\backslash[0.8^2=0.64 \backslash]$$

Remaining = 64%.

Depreciation:

$$\backslash[\backslashboxed{36\%} \backslash]$$

Q8

$$\backslash[25\%\uparrow\rightarrow 20\%\downarrow \backslash]$$

$$\backslash[\backslashboxed{20\%} \backslash]$$

Q9

$\downarrow \rightarrow 25\% \uparrow$

$\boxed{25\%}$

Q10

$1.1 \times 1.2 = 1.32$

$\boxed{32\% \text{ increase}}$

Q11

$1.2 \times 0.9 \times 1.25$

Since:

$1.2 \times 1.25 = 1.5$

$1.5 \times 0.9 = 1.35$

$\boxed{35\% \text{ increase}}$

Q12

$0.7 \times 1.3 = 0.91$

$\boxed{9\% \text{ decrease}}$

Q13

Multipliers:

1.2×1.125

$= 1.35$

Original:

$54000 / 1.35 = \boxed{40000}$

Q14

First multiplier:

1.25

Target:

1.5

Second multiplier:

$1.5 / 1.25 = 1.2$

Therefore:

$\boxed{x=20\%}$

Q15

Take income:

100

Expenditure:

80

Savings:

$$\sqrt[20]{}$$

New income:

$$\sqrt[120]{}$$

New expenditure:

$$\sqrt[80(1.15)=92]{}$$

New savings:

$$\sqrt[28]{}$$

Increase:

$$\sqrt[8]{}$$

$$\sqrt[8/20 \times 100 = \boxed{40\%}]{}$$

Q16

$$\sqrt[1.5 \times 0.8 \times 0.75]{}$$

$$\sqrt[1.5 \times 0.8 = 1.2]{}$$

$$\sqrt[1.2 \times 0.75 = 0.9]{}$$

$$\sqrt[\boxed{10\% \text{ decrease}}]{}$$

Q17

$$\sqrt[1.2 \times 0.9 = 1.08]{}$$

Equivalent single change:

$$\sqrt[\boxed{8\% \text{ increase}}]{}$$

Q18

Length multiplier:

$$\sqrt[1.4 = \frac{7}{5}]{}$$

Breadth multiplier needed:

$$\sqrt[\frac{5}{7}]{}$$

Decrease:

$$\sqrt[1 - \frac{5}{7} = \frac{2}{7}]{}$$

$$\sqrt[= \boxed{28 \frac{4}{7}\%}]{}$$

Q19

From:

$$\sqrt[48\% \text{ to } 60\%]{}$$

Percentage-point increase:

$$\sqrt[\boxed{12}]{}$$

Relative increase:

$$\sqrt[\frac{12}{48} \times 100 = \boxed{25\%}]{}$$

Q20

Combined multiplier:

$$\left[0.8 \times 1.5 = 1.2 \right]$$

So:

$$\left[1.2O = 720 \right]$$

$$\left[O = 600 \right]$$

$$\left[\boxed{600} \right]$$

' CHAPTER 15 COMPLETE

We have now covered:

- percentage basics '
- fraction/decimal conversions '
- percentage of quantity '
- percentage increase/decrease '
- multiplier method '
- more/less comparisons '
- reverse percentage '
- restoration percentage '
- ratio !" percentage '
- percentage points '
- successive percentage changes '
- equal increase/decrease trap '
- repeated growth/depreciation '
- constant expenditure '
- savings/income problems '
- area/product/quotient percentage changes '
- percentage distributions '
- missing-change problems '
- GATE/placement practice '

Progress

Completed: 15/76 **Unit 1 — Quantitative Aptitude:** 0=100% **PROGRESS** **Next:** 0=0% **Chapter 16 — Averages**

Next we will cover:

- arithmetic mean
- total from average
- new average after addition/removal
- replacement problems
- average change shortcuts
- combined groups
- hidden-average questions

- GATE and placement applications.

USER

next

ASSISTANT

CHAPTER 16 — AVERAGES

PART 1 — ARITHMETIC MEAN, TOTAL-AVERAGE RELATION, ADDITION, REMOVAL & REPLACEMENT

Status: $\emptyset = 0\%$ PROGRESS GATE: High Placements: Very High Priority: $\emptyset = 0\%$ For 1

Average is one of the most useful aptitude ideas because it appears inside:

- marks
- salaries
- ages
- speeds
- scores
- production
- runs
- temperatures
- data interpretation
- replacement questions
- missing-value questions

The central idea is:

$$\boxed{\text{Average} = \frac{\text{Total}}{\text{Number of items}}}$$

But in exams, the most useful rearrangement is often:

$$\boxed{\text{Total} = \text{Average} \times \text{Number of items}}$$

1. WHAT DOES AVERAGE MEAN?

Consider:

$$\boxed{10, 20, 30}$$

Total:

$$\boxed{60}$$

Number of values:

$$\boxed{3}$$

Average:

$$\left[\frac{60}{3} = 20 \right]$$

So:

$$\left[\text{Average} = 20 \right]$$

Intuitively, if the total were distributed equally, each value would become 20.

2. MASTER FORMULA

For (n) values:

$$\left[x_1, x_2, \dots, x_n \right]$$

Arithmetic mean:

$$\left[\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} \right]$$

Therefore:

$$\left[x_1 + x_2 + \dots + x_n = n\bar{x} \right]$$

3. TOTAL FROM AVERAGE

Average of 8 numbers is 15.

Find total.

$$\left[\text{Total} = 8 \times 15 \right]$$

$$\left[= 120 \right]$$

This is often the fastest first step in average problems.

4. NUMBER OF ITEMS FROM TOTAL AND AVERAGE

Total = 360.

Average = 30.

Then:

$$\left[n = \frac{360}{30} \right]$$

$$\left[= 12 \right]$$

So:

$$\left[n = \frac{\text{Total}}{\text{Average}} \right]$$

5. MISSING VALUE

Average of 5 numbers is 18.

Four numbers are:

$$\left[12, 15, 20, 25 \right]$$

Total required:

$$\backslash[5\times 18=90 \backslash]$$

Known total:

$$\backslash[12+15+20+25=72 \backslash]$$

Missing value:

$$\backslash[90-72 = \boxed{18} \backslash]$$

6. FAST MISSING-VALUE METHOD

Always think:

$$\backslash[\boxed{\text{Missing}} = \text{Required total} - \text{Known total} \backslash]$$

7. AVERAGE OF CONSECUTIVE NUMBERS

For equally spaced numbers, average is the middle value.

Example:

$$\backslash[12,13,14,15,16 \backslash]$$

Average:

$$\backslash[\boxed{14} \backslash]$$

No addition required.

8. EVEN NUMBER OF CONSECUTIVE TERMS

Example:

$$\backslash[10,11,12,13 \backslash]$$

Middle two:

$$\backslash[11,12 \backslash]$$

Average:

$$\backslash[\frac{11+12}{2} = \boxed{11.5} \backslash]$$

Equivalent:

$$\backslash[\frac{\text{first}+\text{last}}{2} \backslash]$$

$$\backslash[=\frac{10+13}{2} =11.5 \backslash]$$

9. AVERAGE OF AN ARITHMETIC SEQUENCE

For equally spaced values:

$$\backslash[a, a+d, a+2d, \ldots, l \backslash]$$

Average:

$$\backslash[\boxed{\frac{a+l}{2}} \backslash]$$

Example:

$\backslash 5,9,13,17,21 \backslash$

Average:

$\backslash \frac{5+21}{2} = \boxed{13} \backslash$

10. WHY THIS WORKS

Pair first and last:

$\backslash 5+21=26 \backslash$

Second and second-last:

$\backslash 9+17=26 \backslash$

Middle:

$\backslash 13 \backslash$

Every pair centers at:

$\backslash 13 \backslash$

Hence average:

$\backslash 13 \backslash$

11. AVERAGE LIES BETWEEN MINIMUM AND MAXIMUM

For ordinary positive weighted equally values:

$\backslash \boxed{\min \leq \text{average} \leq \max} \backslash$

Example:

Values:

$\backslash 5, \backslash 10, \backslash 20 \backslash$

Average cannot be 25.

That is a quick sanity check.

12. IF ALL VALUES ARE EQUAL

If all values are:

$\backslash x \backslash$

then average is:

$\backslash \boxed{x} \backslash$

Example:

$\backslash 7,7,7,7 \backslash$

Average:

$\boxed{7}$

13. ADDING THE SAME NUMBER TO EVERY VALUE

Suppose average is:

A

If every value increases by k :

$\boxed{\text{new average} = A + k}$

Example:

Average = 25.

Add 4 to every value.

New average:

$\boxed{29}$

No need to know individual values.

14. SUBTRACTING SAME NUMBER

If every value decreases by k :

$\boxed{\text{new average} = A - k}$

Example:

Average = 40.

Subtract 6 from every item.

New average:

$\boxed{34}$

15. MULTIPLYING EVERY VALUE

If every item is multiplied by k :

$\boxed{\text{new average} = kA}$

Example:

Average = 12.

Every value doubled.

New average:

$\boxed{24}$

16. DIVIDING EVERY VALUE

If every item divided by k :

$$\boxed{\text{new average} = A/k}$$

Example:

Average = 30.

Every value divided by 5.

New average:

$$\boxed{6}$$

17. LINEAR TRANSFORMATION RULE

If every value changes as:

$$y = ax + b$$

then:

$$\boxed{\bar{y} = a\bar{x} + b}$$

This is a powerful GATE-style property.

Example:

Average of x values = 20.

For:

$$y = 3x + 5$$

average of y :

$$3(20) + 5 = \boxed{65}$$

18. ADDING ONE NEW ITEM

Suppose n items have average A .

Old total:

$$nA$$

A new item x is added.

New average:

$$\boxed{\frac{nA + x}{n+1}}$$

19. EXAMPLE — ADD ONE VALUE

Average of 5 numbers is 20.

A sixth number 32 is added.

Old total:

$$5 \times 20 = 100$$

New total:

$$\boxed{132}$$

New average:

$$132/6 = \boxed{22}$$

20. FAST CHANGE-IN-AVERAGE METHOD

Old average:

$$\boxed{20}$$

New item:

$$\boxed{32}$$

Difference from old average:

$$32-20=12$$

This excess 12 is spread across:

$$\boxed{6}$$

items.

Increase in average:

$$12/6=2$$

New average:

$$20+2 = \boxed{22}$$

This is often faster.

21. GENERAL ADDITION SHORTCUT

If one item (x) is added to (n) items of average (A) :

$$\boxed{\text{new average} = A + \frac{x-A}{n+1}}$$

Interpretation:

- if $(x > A)$, average rises;
 - if $(x < A)$, average falls;
 - if $(x = A)$, average unchanged.
-

22. FIND THE ADDED VALUE FROM AVERAGE CHANGE

Average of 8 numbers is 15.

After adding one number, average becomes 17.

Old total:

$$8 \times 15 = 120$$

New total:

$$9 \times 17 = 153$$

Added value:

$$\boxed{153-120 = 33}$$

23. FAST ADDED-VALUE SHORTCUT

Average rises:

$$15 \text{ to } 17$$

Increase per item:

$$2$$

There are now:

$$9$$

items.

Extra total compared with old-average baseline:

$$9 \times 2 = 18$$

New item must also replace the old-average baseline 15:

$$15 + 18 = \boxed{33}$$

Equivalent formula:

$$\boxed{x = A + (n+1)(A' - A)}$$

where A' is new average.

24. REMOVING ONE ITEM

Suppose n items have average A .

Remove value x .

New average:

$$\boxed{\frac{nA - x}{n-1}}$$

25. EXAMPLE — REMOVE ONE VALUE

Average of 6 numbers is 25.

Remove number 40.

Old total:

$$6 \times 25 = 150$$

Remaining total:

$$110$$

Remaining count:

$$5$$

New average:

$$\boxed{110/5 = 22}$$

26. INTUITION WHEN REMOVING

If removed value is:

- above average !' new average falls
- below average !' new average rises
- equal to average !' average unchanged

This is a quick sanity check.

27. FIND REMOVED VALUE

Average of 10 numbers is 30.

After removing one number, average becomes 28.

Old total:

$$10 \times 30 = 300$$

Remaining total:

$$9 \times 28 = 252$$

Removed value:

$$300 - 252 = \boxed{48}$$

28. REPLACEMENT PROBLEMS

This is very important.

If one value x is replaced by y , number of items stays the same.

Change in total:

$$y - x$$

Therefore change in average:

$$\boxed{\frac{y-x}{n}}$$

29. REPLACEMENT EXAMPLE

Average of 8 students is 20 years.

A student aged 16 leaves and a student aged 24 joins.

Increase in total:

$$24 - 16 = 8$$

Increase in average:

$$\left[\frac{8}{8}=1 \right]$$

New average:

$$\left[\boxed{21} \right]$$

No need to calculate original total.

30. MASTER REPLACEMENT FORMULA

For (n) items:

$$\left[\boxed{A_{\text{new}} = A_{\text{old}} + \frac{\text{incoming} - \text{outgoing}}{n}} \right]$$

MUST MEMORIZE

31. FIND THE INCOMING VALUE

Average of 12 workers is 30 years.

One worker aged 24 leaves.

Average becomes 31.

Let incoming age = (x) .

Average increases by:

$$\left[1 \right]$$

Total increases by:

$$\left[12 \times 1 = 12 \right]$$

So:

$$\left[x - 24 = 12 \right]$$

$$\left[x = \boxed{36} \right]$$

32. FIND THE OUTGOING VALUE

Average of 10 students is 18.

One student is replaced by a 23-year-old, causing average to rise to 18.5.

Average increase:

$$\left[0.5 \right]$$

Total increase:

$$\left[10 \times 0.5 = 5 \right]$$

Thus:

$$\left[23 - \text{outgoing} = 5 \right]$$

Outgoing:

$$\left[\boxed{18} \right]$$

33. MULTIPLE REPLACEMENTS

If $\sum r_i$ values are replaced, total change equals:

$$\sum \text{incoming} - \sum \text{outgoing}$$

Average change:

$$\boxed{\frac{\text{net total change}}{n}}$$

34. EXAMPLE — TWO REPLACEMENTS

Average of 10 values is 40.

Values 30 and 35 are replaced by 50 and 45.

Outgoing total:

$$65$$

Incoming total:

$$95$$

Increase:

$$30$$

Average increase:

$$30/10=3$$

New average:

$$\boxed{43}$$

35. WRONG ENTRY IN DATA

A common exam type:

Average calculated using a wrong value.

Example:

Average of 10 numbers calculated as 25.

One number 36 was mistakenly entered as 63.

Wrong total:

$$10 \times 25 = 250$$

Correction:

$$250 - 63 + 36 = 223$$

Correct average:

$$223/10 = \boxed{22.3}$$

36. FAST ERROR-CORRECTION METHOD

Wrong entry:

$$\backslash [63 \backslash]$$

Correct entry:

$$\backslash [36 \backslash]$$

Total correction:

$$\backslash [36-63=-27 \backslash]$$

Average correction:

$$\backslash [-27/10=-2.7 \backslash]$$

Correct average:

$$\backslash [25-2.7 = \boxed{22.3} \backslash]$$

37. GENERAL WRONG-ENTRY FORMULA

For $\backslash (n) \backslash$ items:

$$\backslash [\boxed{A_{\text{correct}}} = A_{\text{wrong}} + \frac{\text{correct value} - \text{wrong value}}{n} \backslash]$$

This is just a replacement problem.

38. TWO WRONG ENTRIES

Average of 20 values calculated as 50.

Values 40 and 70 were entered instead of 60 and 80.

Correction in total:

$$\backslash [(60+80)-(40+70) \backslash]$$

$$\backslash [=140-110=30 \backslash]$$

Average correction:

$$\backslash [30/20=1.5 \backslash]$$

Correct average:

$$\backslash [\boxed{51.5} \backslash]$$

39. AVERAGE OF FIRST $\backslash (n) \backslash$ NATURAL NUMBERS

Numbers:

$$\backslash [1,2,3,\ldots,n \backslash]$$

Average:

$$\backslash [\boxed{\frac{n+1}{2}} \backslash]$$

because first and last are:

$$\{ 1, \dots, n \}$$

and average of an AP is:

$$\frac{1+n}{2}$$

40. EXAMPLE

Average of first 20 natural numbers:

$$\frac{20+1}{2} = \boxed{10.5}$$

41. FIRST (n) ODD NUMBERS

Sequence:

$$1, 3, 5, \dots, (2n-1)$$

Average:

$$\frac{1+(2n-1)}{2} = \boxed{n}$$

Example first 10 odd numbers:

Average:

$$\boxed{10}$$

42. FIRST (n) EVEN NUMBERS

Sequence:

$$2, 4, 6, \dots, 2n$$

Average:

$$\frac{2+2n}{2} = \boxed{n+1}$$

Example first 15 even numbers:

$$\boxed{16}$$

43. AVERAGE OF CONSECUTIVE ODD/EVEN NUMBERS

For any equally spaced sequence:

$$\boxed{\text{Average} = \frac{\text{first} + \text{last}}{2}}$$

Example:

$$21, 23, 25, 27, 29$$

Average:

$$\boxed{25}$$

44. FIND NUMBER OF CONSECUTIVE TERMS FROM AVERAGE

Suppose consecutive integers from:

$$[11 \text{ to } 29]$$

Average:

$$[\frac{11+29}{2} = 20]$$

Count:

$$[29-11+1 = 19]$$

Both concepts may appear together.

45. AVERAGE AND DEVIATION METHOD

Instead of adding large numbers, choose a convenient assumed average.

Find average of:

$$[98, 102, 101, 97, 102]$$

Choose assumed average:

$$[100]$$

Deviations:

$$[-2, +2, +1, -3, +2]$$

Sum deviation:

$$[0]$$

Therefore actual average:

$$[\boxed{100}]$$

Much faster.

46. DEVIATION METHOD FORMULA

Choose assumed mean (A) .

If deviations are:

$$[d_i = x_i - A]$$

then:

$$[\boxed{\text{Actual average}} = A + \frac{\sum d_i}{n}]$$

47. EXAMPLE

Find average:

$$[47, 52, 49, 51, 56]$$

Choose:

$$\sqrt[5]{50}$$

Deviations:

$$\sqrt[5]{-3,+2,-1,+1,+6}$$

Sum:

$$\sqrt[5]{5}$$

Number:

$$\sqrt[5]{5}$$

Average:

$$\sqrt[5]{50+\frac{5}{5}} = \boxed{5.1}$$

48. WHEN DEVIATION METHOD IS USEFUL

Very useful when values are clustered around:

- 10
- 50
- 100
- 1000
- another convenient central value

Especially helpful in:

- DI
- marks
- temperatures
- scores
- placement speed tests

49. AVERAGE CHANGE WHEN EVERY ITEM CHANGES DIFFERENTLY

If n values change by:

$$d_1, d_2, \dots, d_n$$

then average changes by:

$$\boxed{\frac{d_1+d_2+\dots+d_n}{n}}$$

Example:

Five values change by:

$$\sqrt[5]{+2,+3,-1,+4,-3}$$

Total change:

$$\sqrt[5]{5}$$

Average change:

$$\sqrt[5]{1}$$

So average rises by:

$$\boxed{1}$$

50. GATE-STYLE QUESTION 1

Average of 12 numbers is 18.

Find total.

$$12 \times 18 = \boxed{216}$$

51. GATE-STYLE QUESTION 2

Average of 6 values is 25.

A seventh value 39 is added.

Old total:

$$150$$

New total:

$$189$$

Average:

$$189/7 = \boxed{27}$$

52. GATE-STYLE QUESTION 3

Average of 10 numbers is 50.

One value 80 is removed.

Remaining average:

$$\frac{500-80}{9} = \frac{420}{9} = \boxed{46\frac{2}{3}}$$

53. GATE-STYLE QUESTION 4

Average of $(x_1, \dots, x_n)$ is 12.

Find average of:

$$(3x_1+5, \dots, 3x_n+5)$$

Using linear transformation:

$$3(12)+5 = \boxed{41}$$

54. PLACEMENT-STYLE QUESTION 1

Average age of 20 students is 18.

A teacher aged 39 joins.

New average:

Old total:

$$\backslash[20\times 18=360 \backslash]$$

Add teacher:

$$\backslash[399 \backslash]$$

Count:

$$\backslash[21 \backslash]$$

Average:

$$\backslash[399/21 = \boxed{19} \backslash]$$

55. PLACEMENT-STYLE QUESTION 2

Average salary of 15 employees is ₹ 30,000.

A new employee joins and average becomes ₹ 31,000.

Old total:

$$\backslash[15(30000)=450000 \backslash]$$

New total:

$$\backslash[16(31000)=496000 \backslash]$$

New employee salary:

$$\backslash[496000-450000 = \boxed{₹ 46,000} \backslash]$$

56. PLACEMENT-STYLE QUESTION 3

Average of 8 numbers is 24.

One number 18 is replaced by 34.

Increase in total:

$$\backslash[16 \backslash]$$

Increase in average:

$$\backslash[16/8=2 \backslash]$$

New average:

$$\backslash[\boxed{26} \backslash]$$

57. PLACEMENT-STYLE QUESTION 4

Average of 25 observations was found to be 40.

One value 24 was mistakenly read as 42.

Correct average?

Correction:

$$\backslash[24-42=-18 \backslash]$$

Average correction:

$$\backslash[-18/25=-0.72 \backslash]$$

Correct average:

$$\backslash[40-0.72 = \boxed{39.28} \backslash]$$

58. ADVANCED QUESTION — ADDITION

Average of 9 numbers is 20.

How large must a 10th number be to make average 22?

Old total:

$$\backslash[180 \backslash]$$

Required total:

$$\backslash[220 \backslash]$$

New number:

$$\backslash[\boxed{40} \backslash]$$

59. ADVANCED QUESTION — REMOVAL

Average of 8 numbers is 35.

After removing one number, average becomes 32.

Removed number:

Old total:

$$\backslash[280 \backslash]$$

Remaining total:

$$\backslash[7\times 32=224 \backslash]$$

Removed:

$$\backslash[280-224 = \boxed{56} \backslash]$$

60. ADVANCED QUESTION — REPLACEMENT

Average of 12 students rises by 1.5 years when one student aged 18 is replaced.

Increase in total:

$$\backslash[12\times 1.5=18 \backslash]$$

Incoming age:

$$\boxed{18+18 = 36}$$

61. ADVANCED QUESTION — WRONG ENTRY

Average of 30 observations is reported as 42.

A value 56 was entered as 65.

Correct average:

Total correction:

$$56-65=-9$$

Average correction:

$$-9/30=-0.3$$

Correct average:

$$\boxed{41.7}$$

62. COMMON TRAPS

& pTRAP 1 — Average of averages

You cannot generally take:

$$\frac{A_1+A_2}{2}$$

unless the two groups have equal sizes.

Full treatment comes in Chapter 17.

& pTRAP 2 — Adding an item

New denominator is:

$$n+1$$

& pTRAP 3 — Removing an item

New denominator is:

$$n-1$$

& pTRAP 4 — Replacement

Number of items stays unchanged.

& pTRAP 5 — Average increase

If average of 20 people rises by 2:

total rises by:

$$20 \times 2 = 40$$

not just 2.

& pTRAP 6 — Consecutive values

Use midpoint instead of adding everything.

63. MUST-MEMORIZE FORMULAS

Average

$$\boxed{A = \frac{T}{n}}$$

Total

$$\boxed{T = nA}$$

Add one value x

$$\boxed{A' = \frac{nA + x}{n+1}}$$

Remove value x

$$\boxed{A' = \frac{nA - x}{n-1}}$$

Replace x by y

$$\boxed{A' = A + \frac{y-x}{n}}$$

64. FAST DECISION TREE

If question gives:

Average + number of items?

Find total.

Missing value?

Required total " known total.

One item added?

Old total + new item.

One item removed?

Old total " removed item.

One item replaced?

Use difference between incoming and outgoing.

Equally spaced numbers?

Use midpoint.

Values clustered around a round number?

Use deviation method.

65. 5-MINUTE REVISION

The most important sentence:

$$\boxed{\text{Average change} \times \text{number of items} = \text{total change}}$$

For replacement:

$$\boxed{\text{Incoming} - \text{Outgoing} = n(\text{new avg} - \text{old avg})}$$

This solves many placement questions in seconds.

PRACTICE — LEVEL 0

Q1

Average of 7 numbers is 12. Find total.

Q2

Total of 9 numbers is 180. Find average.

Q3

Average of 5 numbers is 16. Four numbers sum to 60. Find fifth.

Q4

Find average of:

$$11, 12, 13, 14, 15$$

Q5

Find average of first 30 natural numbers.

LEVEL 1

Q6

Average of 8 numbers is 20. A ninth number 29 is added. Find new average.

Q7

Average of 7 numbers is 25. One number 37 is removed. Find remaining average.

Q8

Average of 10 numbers is 18. One number 12 is replaced by 32. Find new average.

Q9

Average of 6 values is 14. After adding one value, average becomes 16. Find added value.

Q10

Average of 12 values is 30. After removing one, average becomes 28. Find removed value.

LEVEL 2

Q11

Average age of 15 people is 24. A 16th person joins and average becomes 25. Find new person's age.

Q12

Average of 20 values rises from 40 to 41.5 when one value is replaced. If outgoing value is 25, find incoming value.

Q13

Average of 25 observations is 36. One observation 18 was incorrectly recorded as 48. Find correct average.

Q14

Find average of:

$\{96, 103, 101, 98, 102\}$

using deviation.

Q15

Average of 9 consecutive odd numbers is 35. Find smallest number.

LEVEL 3 — EXAM PRESSURE

Q16

Average of 11 numbers is 50. If one number is removed, average becomes 48. Find removed number.

Q17

Average of 10 numbers is 30. Two values 20 and 25 are replaced by 40 and 45. Find new average.

Q18

Average of $(x_1, \dots, x_n)$ is 8. Find average of:

$\{5x_1 - 3, \dots, 5x_n - 3\}$

Q19

Average of 20 students is 16 years. A student aged 14 leaves and another student joins, making average 16.5. Find new student's age.

Q20

Average of 5 consecutive even numbers is 42. Find the largest number.

ANSWERS

Q1

$7 \times 12 = \boxed{84}$

Q2

$180/9 = \boxed{20}$

Q3

Required total:

$$5 \times 16 = 80$$

Missing:

$$80 - 60 = \boxed{20}$$

Q4

Middle number:

$$\boxed{13}$$

Q5

$$\frac{30+1}{2} = \boxed{15.5}$$

Q6

Old total:

$$8 \times 20 = 160$$

New total:

$$189$$

Average:

$$189/9 = \boxed{21}$$

Q7

Old total:

$$175$$

Remaining:

$$175 - 37 = 138$$

Average:

$$138/6 = \boxed{23}$$

Q8

Increase in total:

$$32 - 12 = 20$$

Average increase:

$$20/10 = 2$$

New average:

$$\boxed{20}$$

Q9

Old total:

$$6 \times 14 = 84$$

New total:

$$\backslash[7\times 16=112 \backslash]$$

Added:

$$\backslash[\boxed{28} \backslash]$$

Q10

Old total:

$$\backslash[12\times 30=360 \backslash]$$

Remaining:

$$\backslash[11\times 28=308 \backslash]$$

Removed:

$$\backslash[360-308 = \boxed{52} \backslash]$$

Q11

Old total:

$$\backslash[15\times 24=360 \backslash]$$

New total:

$$\backslash[16\times 25=400 \backslash]$$

New person:

$$\backslash[\boxed{40} \backslash]$$

Q12

Average increase:

$$\backslash[1.5 \backslash]$$

Total increase:

$$\backslash[20\times 1.5=30 \backslash]$$

Incoming:

$$\backslash[25+30 = \boxed{55} \backslash]$$

Q13

Wrong total:

$$\backslash[25\times 36=900 \backslash]$$

Correction:

$$\backslash[18-48=-30 \backslash]$$

Correct total:

$$\backslash[870 \backslash]$$

Correct average:

$$\backslash[870/25 = \boxed{34.8} \backslash]$$

Q14

Take 100.

Deviations:

$$\backslash [-4,+3,+1,-2,+2 \backslash]$$

Sum:

$$\backslash [0 \backslash]$$

Average:

$$\backslash [\boxed{100} \backslash]$$

Q15

Nine consecutive odd numbers centered at 35.

There are 4 terms on each side.

Step size:

$$\backslash [2 \backslash]$$

Smallest:

$$\backslash [35-4(2) = \boxed{27} \backslash]$$

Q16

Old total:

$$\backslash [11\backslash times 50=550 \backslash]$$

Remaining total:

$$\backslash [10\backslash times 48=480 \backslash]$$

Removed:

$$\backslash [550-480 = \boxed{70} \backslash]$$

Q17

Outgoing total:

$$\backslash [45 \backslash]$$

Incoming total:

$$\backslash [85 \backslash]$$

Increase:

$$\backslash [40 \backslash]$$

Average increase:

$$\backslash [40/10=4 \backslash]$$

New average:

$$\backslash [\boxed{34} \backslash]$$

Q18

Linear transformation:

$$\backslash [5(8)-3 = \boxed{37} \backslash]$$

Q19

Average rises:

$$\backslash[0.5 \backslash]$$

Total increase:

$$\backslash[20\backslashtimes 0.5=10 \backslash]$$

Incoming " outgoing:

$$\backslash[10 \backslash]$$

Incoming:

$$\backslash[14+10 = \boxed{24} \backslash]$$

Q20

Five consecutive even numbers centered at 42:

$$\backslash[38,40,42,44,46 \backslash]$$

Largest:

$$\backslash[\boxed{46} \backslash]$$

CHAPTER 16 STATUS

Covered so far:

- arithmetic mean '
- total–average relation '
- missing values '
- consecutive/equally spaced values '
- adding one item '
- removing one item '
- replacement problems '
- wrong-entry correction '
- deviation method '
- linear transformation of averages '
- first $\backslash(n\backslash)$ natural/odd/even averages '
- GATE/placement practice '

But **Chapter 16 is not complete yet.**

Next we need:

- average of changing groups
- multiple additions/removals
- age and marks-style hidden-average problems
- batting/score-style averages
- required next score/value
- average after percentage change
- mixed high-speed problems

We will keep **weighted and combined averages** mainly for Chapter 17.

Progress

Completed: 15/76 **Current:** Chapter 16 — Averages **Next:** Chapter 16 Part 2 — **Advanced average-change problems & exam shortcuts**

USER

next

ASSISTANT

CHAPTER 16 — AVERAGES

PART 2 — ADVANCED AVERAGE-CHANGE PROBLEMS, REQUIRED SCORES, AGES, MARKS & EXAM SHORTCUTS

Status: $\emptyset=0\%$ **PROGRESS GATE:** High **Placements:** Very High **Priority:** $\emptyset=0\%$ **Part 1**

This part focuses on the common exam pattern:

Average changes because the total changes.

The most important relation remains:

$$\boxed{\text{Total} = \text{Average} \times \text{Count}}$$

1. AVERAGE CHANGE !' TOTAL CHANGE

If average of (n) items rises by (d) :

$$\boxed{\text{Total rises by } nd}$$

Example:

Average of 20 students rises from 50 to 53.

Increase in average:

$$3$$

Total increase:

$$20 \times 3 = \boxed{60}$$

This shortcut is extremely important.

2. AVERAGE FALLS

Average of 15 values decreases from 40 to 36.

Decrease per value:

$$\boxed{4}$$

Total decrease:

$$15 \times 4 = \boxed{60}$$

3. MULTIPLE ITEMS ADDED

Suppose 10 students have average 20.

Two students with ages 25 and 35 join.

Old total:

$$10 \times 20 = 200$$

Added total:

$$25 + 35 = 60$$

New total:

$$260$$

New count:

$$12$$

Average:

$$\frac{260}{12} = \boxed{21\frac{2}{3}}$$

4. FAST METHOD USING EXCESS OVER OLD AVERAGE

Old average:

$$20$$

New values:

$$25, 35$$

Their excesses over old average:

$$5, 15$$

Total excess:

$$20$$

This excess is distributed over 12 people:

$$\frac{20}{12} = \frac{5}{3}$$

New average:

$$20 + \frac{5}{3} = \boxed{21\frac{2}{3}}$$

5. MULTIPLE ITEMS REMOVED

Average of 12 values is 30.

Values 24 and 36 are removed.

Old total:

$$\backslash[12\times 30=360 \backslash]$$

Removed total:

$$\backslash[60 \backslash]$$

Remaining total:

$$\backslash[300 \backslash]$$

Count:

$$\backslash[10 \backslash]$$

New average:

$$\backslash[\boxed{30} \backslash]$$

Why unchanged?

Because removed values themselves average:

$$\backslash[\frac{24+36}{2}=30 \backslash]$$

equal to the original average.

6. IMPORTANT PROPERTY

If a group removed from a larger group has the **same average** as the original group:

$$\backslash[\boxed{\text{remaining average stays unchanged}} \backslash]$$

Similarly, adding a group whose average equals the current average leaves the overall average unchanged.

7. ADD A GROUP WITH HIGHER AVERAGE

Original:

- 20 people
- average 30

Add 5 people whose average is 40.

Old total:

$$\backslash[600 \backslash]$$

Added total:

$$\backslash[200 \backslash]$$

New total:

$$\backslash[800 \backslash]$$

New count:

$$\backslash[25 \backslash]$$

New average:

$$\boxed{32}$$

This is technically a combined-average idea, but here we use only the basic total method. Full shortcut treatment comes in Chapter 17.

8. REQUIRED NEXT SCORE

A student has an average of 70 after 4 tests.

What must they score in the 5th test to make average 75?

Old total:

$$4 \times 70 = 280$$

Required new total:

$$5 \times 75 = 375$$

Required score:

$$375 - 280 = \boxed{95}$$

9. FAST REQUIRED-SCORE FORMULA

If current average after (n) items is (A) , and desired average after one more item is (B) :

$$\boxed{x = (n+1)B - nA}$$

Example:

$$x = 5(75) - 4(70)$$

$$= 375 - 280$$

$$= \boxed{95}$$

10. REQUIRED SCORE FOR SAME AVERAGE

Average after 8 matches is 42.

What score in the 9th match keeps average 42?

Obviously:

$$\boxed{42}$$

To preserve an average, the new value must equal the current average.

11. REQUIRED SCORE TO RAISE AVERAGE BY (d)

Current average = (A) after (n) items.

Need new average:

$$A + d$$

Required next value:

$$\boxed{(n+1)(A+d)-nA}$$

Simplify:

$$\boxed{=A+(n+1)d}$$

So:

$$\boxed{x=A+(n+1)d}$$

Example:

Average 50 after 9 tests.

Need average 52 after 10th.

$$x=50+10(2)$$

$$\boxed{=70}$$

12. REQUIRED SCORE TO LOWER AVERAGE

Average = 60 after 5 tests.

What 6th score makes average 58?

Required total:

$$6 \times 58 = 348$$

Old total:

$$5 \times 60 = 300$$

Required score:

$$\boxed{48}$$

13. BATTING AVERAGE STYLE

A batter has average 45 after 20 innings.

After next innings, average becomes 47.

Score in next innings:

Old total:

$$20 \times 45 = 900$$

New total:

$$21 \times 47 = 987$$

Score:

$$987 - 900 = \boxed{87}$$

14. BATTING AVERAGE FAST SHORTCUT

Average rises:

$$\backslash[45\text{to}47 \backslash]$$

Increase:

$$\backslash[2 \backslash]$$

There are now 21 innings.

Required score:

$$\backslash[45+21(2) \backslash]$$

$$\backslash[=\boxed{87} \backslash]$$

15. EXAM MARKS STYLE

A student's average after 6 subjects is 72.

They want an average of 75 after 7 subjects.

Required score:

$$\backslash[7(75)-6(72) \backslash]$$

$$\backslash[525-432 \backslash]$$

$$\backslash[=\boxed{93} \backslash]$$

16. IMPOSSIBILITY CHECK

Suppose each test is out of 100.

Current average after 4 tests:

$$\backslash[80 \backslash]$$

Required average after 5 tests:

$$\backslash[85 \backslash]$$

Required score:

$$\backslash[5(85)-4(80) \backslash]$$

$$\backslash[425-320=105 \backslash]$$

But maximum possible is 100.

Therefore:

$$\backslash[\boxed{\text{Target average is impossible}} \backslash]$$

Very useful in placement questions.

17. FIND REQUIRED AVERAGE OVER REMAINING ITEMS

A student wants an overall average of 80 across 10 tests.

After 6 tests, average is 75.

Required total for 10:

$$\boxed{10 \times 80 = 800}$$

Current total:

$$\boxed{6 \times 75 = 450}$$

Remaining total needed:

$$\boxed{350}$$

Remaining tests:

$$\boxed{4}$$

Required average:

$$\boxed{350/4 = 87.5}$$

18. MASTER FORMULA — REQUIRED REMAINING AVERAGE

If:

- target average = (T)
- total number = (N)
- completed number = (n)
- current average = (A)

then remaining average required:

$$\boxed{\frac{NT - nA}{N - n}}$$

19. AGE PROBLEMS — IMPORTANT IDEA

If the same group of people is considered after (t) years:

each age increases by (t) .

Therefore average also increases by:

$$\boxed{t}$$

Example:

Average age of 8 students today is 15.

After 4 years:

$$\boxed{19}$$

No calculation needed.

20. SAME GROUP AGE RULE

If nobody joins or leaves:

$$\boxed{\text{Average age increases by exactly elapsed time}}$$

This is one of the most useful age shortcuts.

21. AGE TRAP — GROUP CHANGES

Average age today = 20.

After 5 years, if a new person joins or someone leaves, you cannot automatically say new average = 25.

The simple rule works only for the **same people**.

22. FAMILY AGE EXAMPLE

Average age of 4 family members is 25.

After 5 years, assuming same members:

$$\boxed{25+5 = 30}$$

Total age after 5 years:

$$4 \times 30 = 120$$

23. PERSON JOINS FAMILY

Average age of 4 people is 25.

A 30-year-old joins.

Old total:

$$100$$

New total:

$$130$$

New count:

$$5$$

Average:

$$\boxed{26}$$

24. PERSON LEAVES A GROUP

Average age of 10 people is 24.

A person aged 33 leaves.

Old total:

$$240$$

New total:

$$207$$

Count:

$$9$$

Average:

$$\lfloor 207/9 = \boxed{23} \rfloor$$

25. TWO-TIME AGE AVERAGE PROBLEM

Average age of 5 people was 20 years, 4 years ago.

Same people now:

$$\lfloor 20+4 = \boxed{24} \rfloor$$

Do not separately increase every person's age.

26. AVERAGE OF CONSECUTIVE NUMBERS — FIND TERMS

Five consecutive integers have average 27.

For an odd number of consecutive terms, middle term = average.

So numbers:

$$\lfloor 25, 26, 27, 28, 29 \rfloor$$

Largest:

$$\lfloor \boxed{29} \rfloor$$

27. CONSECUTIVE EVEN NUMBERS

Five consecutive even numbers average 38.

Middle:

$$\lfloor 38 \rfloor$$

So numbers:

$$\lfloor 34, 36, 38, 40, 42 \rfloor$$

Smallest:

$$\lfloor \boxed{34} \rfloor$$

28. CONSECUTIVE ODD NUMBERS

Seven consecutive odd numbers average 31.

Middle = 31.

There are 3 on each side, step 2.

Smallest:

$$\lfloor 31-3(2) = \boxed{25} \rfloor$$

Largest:

$$\lfloor 37 \rfloor$$

29. EVEN NUMBER OF CONSECUTIVE TERMS

Four consecutive integers average 21.5.

Middle pair:

$$\backslash 21, \backslash 22 \backslash$$

Therefore numbers:

$$\backslash 20, 21, 22, 23 \backslash$$

30. AVERAGE OF MULTIPLES

Find average of multiples of 7 from 7 through 70.

Sequence:

$$\backslash 7, 14, \ldots, 70 \backslash$$

Arithmetic progression.

Average:

$$\backslash \frac{7+70}{2} \backslash$$

$$\backslash =\boxed{38.5} \backslash$$

31. AVERAGE OF NUMBERS IN A RANGE

Average of integers:

$$\backslash 51, 52, \ldots, 99 \backslash$$

Since consecutive:

$$\backslash \frac{51+99}{2} = \boxed{75} \backslash$$

No summation required.

32. FIND COUNT FROM FIRST, LAST & STEP

For AP:

$$\backslash a, \backslash a+d, \ldots, l \backslash$$

Number of terms:

$$\backslash \boxed{n=\frac{l-a}{d}+1} \backslash$$

Example:

$$\backslash 5, 8, 11, \ldots, 50 \backslash$$

$$\backslash n=\frac{50-5}{3}+1 \backslash$$

$$\backslash =15+1 = \boxed{16} \backslash$$

Average:

$$\backslash \frac{5+50}{2} = \boxed{27.5} \backslash$$

33. AVERAGE AFTER PERCENTAGE CHANGE TO EVERY VALUE

If every item increases by 20%, the average also increases by 20%.

Why?

Each:

$$x_i \rightarrow 1.2x_i$$

Therefore:

$$\bar{x} \rightarrow 1.2\bar{x}$$

Example:

Old average:

$$50$$

New average:

$$\boxed{60}$$

34. EVERY VALUE DECREASES BY SAME %

If all values decrease by 15%:

$$\boxed{\text{average also decreases by 15\%}}$$

Example:

Average 80:

$$80(0.85) = \boxed{68}$$

35. DIFFERENT PERCENTAGE CHANGES — CAUTION

If different items change by different percentages, you cannot simply average the percentages unless the original items are appropriately equal/weighted.

Example:

Values:

$$100, 200$$

First +10%:

$$110$$

Second +20%:

$$240$$

Old average:

$$150$$

New average:

$$175$$

Increase:

$$\left[25 \right]$$

Percentage:

$$\left[16\frac{23}{100} \right]$$

not 15%.

36. AVERAGE OF PERCENTAGES — MAJOR TRAP

Suppose one student scores:

$$\left[80/100=80\% \right]$$

and another:

$$\left[45/50=90\% \right]$$

Simple average of percentages:

$$\left[85\% \right]$$

does **not** represent combined marks percentage.

Combined:

$$\left[\frac{80+45}{100+50} = \frac{125}{150} = 83\frac{1}{3}\% \right]$$

This is weighted-average territory, covered properly in Chapter 17.

37. AVERAGE SPEED WARNING

If a vehicle travels equal distances at 40 km/h and 60 km/h:

ordinary average:

$$\left[50 \right]$$

is **not** the correct average speed.

Why?

Time spent differs.

Average speed is:

$$\left[\frac{\text{total distance}}{\text{total time}} \right]$$

Full treatment comes in Chapter 30.

Important

Do not mechanically use arithmetic mean for rates.

38. AVERAGE RATE GENERAL RULE

For any rate:

$$\left[\boxed{\text{Average rate}} = \frac{\text{Total output}}{\text{Total corresponding input}} \right]$$

Examples:

- $\text{speed} = \text{distance}/\text{time}$
- $\text{productivity} = \text{output}/\text{time}$
- $\text{price per unit} = \text{cost}/\text{quantity}$

Arithmetic mean works only under specific equal-weight conditions.

39. HIDDEN MISSING NUMBER

Average of 7 numbers is 28.

Average of first 3 is 24.

Average of last 3 is 30.

Find middle 4th number.

Total of all 7:

$$\backslash[7\times 28=196 \backslash]$$

First 3 total:

$$\backslash[72 \backslash]$$

Last 3 total:

$$\backslash[90 \backslash]$$

Middle:

$$\backslash[196-72-90 \backslash]$$

$$\backslash[=\boxed{34} \backslash]$$

40. OVERLAPPING-GROUP TRAP

Suppose average of first 5 values and last 5 values are given in a 9-value sequence.

The middle value belongs to both groups.

So adding both group totals counts middle value twice.

You must correct for overlap.

41. OVERLAP EXAMPLE

There are 9 numbers.

Average of first 5 = 20.

Average of last 5 = 24.

Average of all 9 = 22.

Find middle number.

First 5 total:

$$\backslash[100 \backslash]$$

Last 5 total:

$$\backslash[120 \backslash]$$

Adding:

$$\backslash[220 \backslash]$$

All 9 total:

$$\backslash[9\times 22=198 \backslash]$$

The middle number was counted twice instead of once.

So:

$$\backslash[\text{middle}=220-198 \backslash]$$

$$\backslash[=\boxed{22} \backslash]$$

42. FIND AVERAGE OF REMAINING GROUP

Average of 12 numbers = 30.

Average of first 5 = 24.

Find average of remaining 7.

Total:

$$\backslash[12\times 30=360 \backslash]$$

First group total:

$$\backslash[5\times 24=120 \backslash]$$

Remaining total:

$$\backslash[240 \backslash]$$

Average:

$$\backslash[\frac{240}{7} = \boxed{34\frac{2}{7}} \backslash]$$

43. DIFFERENCE BETWEEN GROUP TOTALS

Two groups have equal number $\backslash(n\backslash)$ of items.

Their averages differ by $\backslash(d\backslash)$.

Then their totals differ by:

$$\backslash[\boxed{nd} \backslash]$$

Example:

Two classes have 40 students each.

Average marks are 65 and 70.

Difference in total marks:

$$\backslash[40\times 5 = \boxed{200} \backslash]$$

44. WHEN GROUP SIZES ARE EQUAL

Only when group sizes are equal:

overall average of two groups is simply:

$$\boxed{\frac{A_1 + A_2}{2}}$$

Example:

20 students average 60.

Another 20 average 70.

Combined average:

$$\boxed{65}$$

For unequal sizes, not valid. Chapter 17 handles this fully.

45. TARGET AVERAGE AFTER SEVERAL FUTURE ITEMS

A player has average 40 after 10 games.

They want average 50 after 15 games.

Required total:

$$15 \times 50 = 750$$

Current:

$$10 \times 40 = 400$$

Need from next 5 games:

$$350$$

Required average:

$$\boxed{70}$$

46. REQUIRED TOTAL VS REQUIRED AVERAGE

Be careful:

“Need 350 runs in next 5 innings” is a total.

“Need average 70” is:

$$350/5$$

Do not confuse them.

47. HIGHEST POSSIBLE FINAL AVERAGE

A student averages 70 after 4 tests, each out of 100.

Maximum possible 5th score:

$$100$$

Maximum final total:

$$\lfloor 280+100=380 \rfloor$$

Maximum final average:

$$\lfloor 380/5 = \boxed{76} \rfloor$$

Therefore any target above 76 is impossible.

48. MINIMUM POSSIBLE FINAL AVERAGE

Same situation, if minimum score is 0:

$$\lfloor 280+0=280 \rfloor$$

Final average:

$$\lfloor 280/5 = \boxed{56} \rfloor$$

Useful for option elimination.

49. AVERAGE WITH NEGATIVE VALUES

Average rules still work.

Example:

$$\lfloor -5, \lfloor 0, \lfloor 10 \rfloor \rfloor$$

Average:

$$\lfloor \frac{5}{3} \rfloor$$

$$\lfloor =\boxed{1\frac{1}{3}} \rfloor$$

Average need not equal one of the observations.

50. DEVIATION PROPERTY

For any dataset with mean $\lfloor \bar{x} \rfloor$:

$$\lfloor \boxed{\sum(x_i - \bar{x})=0} \rfloor$$

Example:

Values:

$$\lfloor 8, 10, 12 \rfloor$$

Mean:

$$\lfloor 10 \rfloor$$

Deviations:

$$\lfloor -2, 0, +2 \rfloor$$

Sum:

$$\lfloor 0 \rfloor$$

This is a fundamental mathematical property.

51. USE DEVIATION TO FIND MISSING VALUE

Five values have average 50.

Known four:

$\{ 48, 52, 55, 47 \}$

Instead of summing actual values, deviations from 50:

$\{ -2, +2, +5, -3 \}$

Sum:

$\{ +2 \}$

For total deviation to be zero, missing value deviation must be:

$\{ -2 \}$

So missing value:

$\{ \boxed{48} \}$

52. GATE-STYLE QUESTION 1

Average of 10 values is 25.

If each value increases by 4, new average is:

$\{ 25 + 4 = \boxed{29} \}$

53. GATE-STYLE QUESTION 2

Average of 8 values is 12.

Each value is transformed to:

$\{ 3x - 2 \}$

New average:

$\{ 3(12) - 2 = \boxed{34} \}$

54. GATE-STYLE QUESTION 3

Nine consecutive integers have average 35.

Largest integer:

There are 4 integers above 35:

$\{ 35 + 4 = \boxed{39} \}$

55. GATE-STYLE QUESTION 4

Average of 12 numbers is 40.

Average of 7 of them is 36.

Average of remaining 5?

Total:

$$\boxed{480}$$

First 7 total:

$$\boxed{252}$$

Remaining:

$$\boxed{228}$$

Average:

$$\boxed{228/5 = 45.6}$$

56. PLACEMENT-STYLE QUESTION 1

A cricket player has average 48 after 25 innings.

What must be scored next to raise average to 50?

Old total:

$$25 \times 48 = 1200$$

Required:

$$26 \times 50 = 1300$$

Required score:

$$\boxed{100}$$

57. PLACEMENT-STYLE QUESTION 2

Average salary of 18 employees is ₹ 40,000.

One employee earning ₹ 31,000 leaves and another joins.

Average rises to ₹ 40,500.

Average increase:

$$\boxed{500}$$

Total increase:

$$18 \times 500 = ₹ 9,000$$

New salary:

$$31000 + 9000 = \boxed{₹ 40,000}$$

58. PLACEMENT-STYLE QUESTION 3

Average age of a class is 20.

Five years later, assuming exactly the same students:

$$\boxed{25}$$

59. PLACEMENT-STYLE QUESTION 4

Average of 6 consecutive even integers is 25.

Can this happen?

Six consecutive even integers have middle average halfway between two consecutive even numbers, which is an odd integer.

Example:

$$20, 22, 24, 26, 28, 30$$

average:

$$25$$

So yes.

One possible sequence is exactly:

$$\boxed{20, 22, 24, 26, 28, 30}$$

60. ADVANCED QUESTION — FIND FIRST TERM

Seven consecutive integers average 42.

Middle:

$$42$$

Three terms before it.

First:

$$42 - 3 = \boxed{39}$$

61. ADVANCED QUESTION — GROUP REMOVAL

Average of 20 students is 60.

Average of 5 students who leave is 72.

Remaining total:

$$20(60) - 5(72)$$

$$1200 - 360 = 840$$

Remaining students:

$$15$$

Average:

$$\lfloor 840/15 = \boxed{56} \rfloor$$

62. ADVANCED QUESTION — GROUP ADDITION

Average of 15 workers is 30.

Five new workers with average 42 join.

Combined total:

$$\lfloor 15(30)+5(42) \rfloor$$

$$\lfloor 450+210=660 \rfloor$$

Count:

$$\lfloor 20 \rfloor$$

Average:

$$\lfloor \boxed{33} \rfloor$$

63. ADVANCED QUESTION — TARGET AVERAGE

A student averages 65 in first 8 tests.

What average is needed in next 4 tests to finish with average 72?

Target total:

$$\lfloor 12\times 72=864 \rfloor$$

Current total:

$$\lfloor 8\times 65=520 \rfloor$$

Needed:

$$\lfloor 344 \rfloor$$

Average needed:

$$\lfloor 344/4 = \boxed{86} \rfloor$$

64. ADVANCED QUESTION — WRONG AVERAGE DUE TO TWO ERRORS

Average of 20 values was calculated as 35.

A value 25 was entered as 52, and 40 was entered as 22.

Wrong total:

$$\lfloor 700 \rfloor$$

Correction:

$$\lfloor (25-52)+(40-22) \rfloor$$

$$\lfloor =-27+18=-9 \rfloor$$

Correct total:

$\lfloor 691 \rfloor$

Correct average:

$\lfloor \boxed{34.55} \rfloor$

65. COMMON TRAPS

& pTRAP 1 — Average of averages

Do not average group averages unless group sizes are equal.

& pTRAP 2 — Average age after time

Add elapsed years only if the group is unchanged.

& pTRAP 3 — Target score

Always check whether required score is possible within maximum marks.

& pTRAP 4 — Rate averages

Average speed is not generally arithmetic mean.

& pTRAP 5 — Overlapping groups

Correct double-counting.

& pTRAP 6 — Percentage change in average

If every item changes by same percentage, average changes by same percentage. Otherwise not necessarily.

66. MUST-MEMORIZE SHORTCUTS

Total change

$\lfloor \boxed{\Delta T = n \Delta A} \rfloor$

Required next value

$\lfloor \boxed{x = (n+1)A_{\text{new}} - nA_{\text{old}}} \rfloor$

Replacement

$\lfloor \boxed{\text{incoming} - \text{outgoing} = n(A_{\text{new}} - A_{\text{old}})} \rfloor$

Required remaining average

$\lfloor \boxed{\frac{NA_{\text{target}} - nA_{\text{current}}}{N-n}} \rfloor$

67. FAST EXAM DECISION TREE

When you see an average question:

Same group over time?

Shift average by same amount.

One item added?

Compare required new total with old total.

One removed?

Subtract its contribution.

Replacement?

Use total-change shortcut.

Target average?

Find required target total.

Consecutive/equally spaced numbers?

Use midpoint.

Two groups?

Use totals; Chapter 17 gives the dedicated weighted shortcut.

68. 5-MINUTE REVISION

Three equations solve most average questions:

$$\boxed{T = nA}$$

$$\boxed{\Delta T = n \Delta A}$$

$$\boxed{\text{Required contribution} = \text{Required total} - \text{current total}}$$

Always think in **totals**.

PRACTICE — LEVEL 0

Q1

Average of 10 numbers rises by 3. By how much does total rise?

Q2

Average of 5 tests is 60. What score in the sixth keeps average 60?

Q3

Average age today is 18. What will it be after 7 years for the same group?

Q4

Five consecutive integers average 20. Find largest.

Q5

Average of all values is 30. If every value doubles, find new average.

LEVEL 1

Q6

Average after 6 tests is 70. What 7th score makes average 72?

Q7

Average of 15 people is 24. One aged 30 is replaced, and average becomes 25. Find incoming age.

Q8

Average of 10 numbers is 40. Two numbers with average 50 are removed. Find remaining average.

Q9

Average of 8 consecutive odd numbers is 30. Find smallest and largest.

Q10

Average of 12 observations is 25. Average of first 7 is 20. Find average of remaining 5.

LEVEL 2

Q11

A player averages 42 after 20 innings. What score next raises average to 44?

Q12

A student wants average 80 across 10 tests. After 6 tests average is 74. Find required average in last 4.

Q13

Average of 9 numbers is 50. Average of first 5 is 48 and last 5 is 54. Find middle number.

Q14

Average of 15 values was calculated as 32. A value 56 was entered instead of 26. Find correct average.

Q15

Average of 6 consecutive multiples of 4 is 26. Find the numbers.

LEVEL 3 — EXAM PRESSURE

Q16

Average of 20 employees is $150,000$. Five employees with average $162,000$ leave. Find remaining average salary.

Q17

Average of 8 tests is 75. Each test is out of 100. What is the highest possible average after the 9th test?

Q18

Average of 10 values is 30. Three values with average 20 are replaced by three values with average 40. Find new average.

Q19

Average of 7 consecutive odd integers is 51. Find their sum.

Q20

A person averages 60 over first 5 scores and 80 over next 3 scores. Find overall average using totals.

ANSWERS

Q1

$$\boxed{10 \times 3 = 30}$$

Q2

To maintain average:

$$\boxed{60}$$

Q3

$$\boxed{18 + 7 = 25}$$

Q4

Five consecutive integers centered at 20:

$$\boxed{18, 19, 20, 21, 22}$$

Largest:

$$\boxed{22}$$

Q5

$$\boxed{30 \times 2 = 60}$$

Q6

Required:

$$\boxed{7(72) - 6(70)}$$

$$\boxed{504 - 420 = 84}$$

Q7

Average rise:

$$\boxed{1}$$

Total rise:

$$\boxed{15}$$

Incoming:

$$\boxed{30 + 15 = 45}$$

Q8

Old total:

$$\boxed{400}$$

Removed total:

$$2(50)=100$$

Remaining:

$$\boxed{300}$$

Count:

$$\boxed{8}$$

Average:

$$\boxed{37.5}$$

Q9

Eight consecutive odd numbers with average 30 can be:

$$23, 25, 27, 29, 31, 33, 35, 37$$

So:

$$\boxed{23, 37}$$

Q10

Total:

$$12(25)=300$$

First 7:

$$7(20)=140$$

Remaining:

$$\boxed{160}$$

Average:

$$160/5 = \boxed{32}$$

Q11

Required score:

$$21(44)-20(42)$$

$$924-840 = \boxed{84}$$

Q12

Target total:

$$10(80)=800$$

Current total:

$$6(74)=444$$

Needed:

$$\boxed{356}$$

Required average:

$$\lfloor 356/4 = \boxed{89} \rfloor$$

Q13

First 5 total:

$$\lfloor 240 \rfloor$$

Last 5 total:

$$\lfloor 270 \rfloor$$

Sum:

$$\lfloor 510 \rfloor$$

All 9 total:

$$\lfloor 9(50)=450 \rfloor$$

Middle counted twice:

$$\lfloor 510-450 = \boxed{60} \rfloor$$

Q14

Wrong total:

$$\lfloor 15(32)=480 \rfloor$$

Correction:

$$\lfloor 26-56=-30 \rfloor$$

Correct total:

$$\lfloor 450 \rfloor$$

Average:

$$\lfloor \boxed{30} \rfloor$$

Q15

Six consecutive multiples of 4 with average 26:

$$\lfloor 16,20,24,28,32,36 \rfloor$$

Check:

$$\lfloor \frac{16+36}{2}=26 \rfloor$$

Answer:

$$\lfloor \boxed{16,20,24,28,32,36} \rfloor$$

Q16

Old total:

$$\lfloor 20(50000)=1000000 \rfloor$$

Leaving total:

$$\lfloor 5(62000)=310000 \rfloor$$

Remaining total:

$$\boxed{690000}$$

Employees:

$$\boxed{15}$$

Average:

$$\boxed{\frac{146}{15}}$$

Q17

Current total:

$$\boxed{8(75)=600}$$

Maximum next score:

$$\boxed{100}$$

New average:

$$\boxed{700/9 = \frac{77}{1}}$$

approximately:

$$\boxed{77.78}$$

Q18

Outgoing total:

$$\boxed{3(20)=60}$$

Incoming:

$$\boxed{3(40)=120}$$

Net increase:

$$\boxed{60}$$

Average increase:

$$\boxed{60/10=6}$$

New average:

$$\boxed{36}$$

Q19

Total:

$$n \times \text{average}$$

$$\boxed{7 \times 51 = 357}$$

Q20

First total:

$$\boxed{5(60)=300}$$

Next total:

$$\boxed{3(80)=240}$$

Overall total:

\[540 \]

Count:

\[8 \]

Average:

\[\frac{540}{8} = \boxed{67.5} \]

' CHAPTER 16 COMPLETE

We have now covered:

- arithmetic mean '
- total–average relation '
- missing values '
- add/remove/replace '
- wrong-entry correction '
- deviation method '
- linear transformations '
- consecutive-number averages '
- group addition/removal basics '
- required next score '
- required remaining average '
- batting/marks problems '
- age-average shortcuts '
- overlapping-group problems '
- target feasibility checks '
- rate-average warning '
- GATE/placement practice '

Progress

Completed: 16/76 **Unit 1 — Quantitative Aptitude:** 0=100% **PROGRESS** **Next:** 0=0% **Chapter 17 — Weighted & Combined Average**

Next we will learn the important case where groups have **different sizes/weights**:

- weighted mean
 - combined average
 - alligation-style shortcut
 - finding group-size ratio from averages
 - class/salary/marks mixtures
 - average of percentages with unequal denominators
 - weighted scores
 - GATE and placement applications.
-

USER

next

ASSISTANT

CHAPTER 17 — WEIGHTED & COMBINED AVERAGE

PART 1 — WEIGHTED MEAN, COMBINED GROUPS & GROUP-SIZE RATIO SHORTCUTS

Status: $\emptyset=0$ PROGRESS GATE: High Placements: Very High Priority: $\emptyset=0$ For 1

Chapter 16 handled ordinary averages where every observation had equal importance.

Now we handle cases where:

- groups have different sizes,
- scores have different weights,
- quantities occur with different frequencies,
- percentages use different totals.

The core idea is:

An average must be weighted according to how much each part contributes.

1. WHY SIMPLE AVERAGE CAN FAIL

Suppose:

- Class A: 10 students, average = 80
- Class B: 30 students, average = 60

Can we write:

$$\sqrt[\frac{80+60}{2}=70?]{}$$

No.

Because there are three times as many students in Class B.

We must use totals.

Class A total:

$$\sqrt[10\times 80=800]{}$$

Class B total:

$$\sqrt[30\times 60=1800]{}$$

Combined:

$$\sqrt[2600]{}$$

Students:

$$\sqrt[40]{}$$

Combined average:

$$\left[\frac{2600}{40} = \boxed{65} \right]$$

Not 70.

2. COMBINED AVERAGE FORMULA

If:

- first group size (n_1) , average (A_1)
- second group size (n_2) , average (A_2)

then:

$$\left[\boxed{A_c = \frac{n_1A_1 + n_2A_2}{n_1 + n_2}} \right]$$

This is one of the most important formulas in aptitude.

3. WHY THE FORMULA WORKS

First group total:

$$\left[n_1A_1 \right]$$

Second group total:

$$\left[n_2A_2 \right]$$

Combined total:

$$\left[n_1A_1 + n_2A_2 \right]$$

Combined count:

$$\left[n_1 + n_2 \right]$$

Therefore:

$$\left[A_c = \frac{\text{combined total}}{\text{combined count}} \right]$$

4. EQUAL GROUP SIZES

If:

$$\left[n_1 = n_2 \right]$$

then:

$$\left[A_c = \frac{A_1 + A_2}{2} \right]$$

So simple average of two averages works **only when the groups have equal weights/sizes**.

5. EXAMPLE — EQUAL GROUPS

20 students average:

$$\left[70 \right]$$

Another 20 average:

$$\boxed{80}$$

Combined:

$$\frac{70+80}{2} = \boxed{75}$$

6. EXAMPLE — UNEQUAL GROUPS

40 students average 60.

20 students average 75.

Combined:

$$\frac{40(60)+20(75)}{60}$$

$$= \frac{2400+1500}{60}$$

$$= \frac{3900}{60}$$

$$= \boxed{65}$$

7. WEIGHTED AVERAGE

If values:

$$x_1, x_2, \dots, x_n$$

have weights:

$$w_1, w_2, \dots, w_n$$

then weighted average is:

$$\boxed{\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}}$$

8. SIMPLE EXAMPLE

Scores:

- Quiz = 80, weight 20%
- Midterm = 70, weight 30%
- Final = 90, weight 50%

Weighted score:

$$80(0.2)+70(0.3)+90(0.5)$$

$$=16+21+45$$

$$= \boxed{82}$$

9. WEIGHTS NEED NOT BE PERCENTAGES

Suppose marks are weighted:

$$\backslash[2:3:5 \backslash]$$

for scores:

$$\backslash[60,\backslash 70,\backslash 80 \backslash]$$

Weighted average:

$$\backslash[\frac{2(60)+3(70)+5(80)}{2+3+5} \backslash]$$

$$\backslash[= \frac{120+210+400}{10} \backslash]$$

$$\backslash[=\boxed{73} \backslash]$$

10. FREQUENCY IS A WEIGHT

Suppose value 10 occurs 3 times and value 20 occurs 2 times.

Data:

$$\backslash[10,10,10,20,20 \backslash]$$

Average:

$$\backslash[\frac{3(10)+2(20)}{5} \backslash]$$

$$\backslash[=\frac{70}{5} \backslash]$$

$$\backslash[=\boxed{14} \backslash]$$

So frequency tables are weighted-average problems.

11. COMBINED AVERAGE MUST LIE BETWEEN GROUP AVERAGES

If group averages are:

$$\backslash[60,\backslash 80 \backslash]$$

combined average must lie between:

$$\backslash[60\text{ and }80 \backslash]$$

If one group is larger, combined average lies closer to that group's average.

This is a powerful sanity check.

12. EXAMPLE — WHICH GROUP IS LARGER?

Group averages:

$$\backslash[50,\backslash 80 \backslash]$$

Combined average:

$$\backslash[60 \backslash]$$

Since 60 is closer to 50 than 80, the group averaging 50 must be larger.

This can often be seen before calculation.

13. GROUP-SIZE RATIO FROM AVERAGES

This is a major shortcut.

Suppose:

- Group A average = (A_1)
- Group B average = (A_2)
- Combined average = (A_c)

Then:

$$\boxed{n_1:n_2 = (A_2 - A_c):(A_c - A_1)}$$

assuming:

$$A_1 < A_c < A_2$$

This is the weighted-average/alligation shortcut.

14. WHY THE RATIO REVERSES

Suppose group averages are:

$$50, 80$$

combined:

$$60$$

Distances:

From 50 to 60:

$$10$$

From 60 to 80:

$$20$$

Group-size ratio:

$$\boxed{20:10=2:1}$$

The group with average 50 gets the **opposite distance** 20.

Memory

Cross differences.

15. EXAMPLE

One class average:

$$60$$

Another:

$$75$$

Combined:

$$\sqrt[65]{}$$

Group-size ratio:

$$\sqrt{(75-65):(65-60)}$$

$$\sqrt{=10:5}$$

$$\sqrt{=\boxed{2:1}}$$

So the group averaging 60 is twice the size of the group averaging 75.

16. VERIFY

Let sizes be:

$$\sqrt{2k, \sqrt{k}}$$

Combined average:

$$\sqrt{\frac{2k(60)+k(75)}{3k}}$$

$$\sqrt{=\frac{195}{3}}$$

$$\sqrt{=65}$$

Correct.

17. ALLIGATION DIAGRAM

For two averages:

$$\sqrt{A_1 < A_c < A_2}$$

Think:

$$\sqrt{\begin{array}{ccc} A_1 & \& A_2 \\ & \& A_c \\ & A_2-A_c & A_c-A_1 \end{array}}$$

Then:

$$\sqrt{\boxed{\text{quantity at } A_1 : \text{quantity at } A_2 = (A_2 - A_c) : (A_c - A_1)}}$$

This same idea will reappear in mixtures.

18. IMPORTANT DIRECTION TRAP

If averages:

$$\sqrt{40, \sqrt{70}}$$

combined:

$$\sqrt{50}$$

Do **not** write:

$$\sqrt{(50-40):(70-50) = 10:20 = 1:2}$$

for the 40-group : 70-group.

Correct cross difference:

$$\frac{(70-50):(50-40)}{1}$$

$$=20:10$$

$$=2:1$$

19. FIND UNKNOWN GROUP AVERAGE

20 students average 60.

30 students average (x) .

Combined average is 72.

Equation:

$$\frac{20(60)+30x}{50}=72$$

$$1200+30x=3600$$

$$30x=2400$$

$$\boxed{x=80}$$

20. FAST TOTAL METHOD

Combined total:

$$50(72)=3600$$

Known group total:

$$20(60)=1200$$

Unknown group total:

$$2400$$

Divide by 30:

$$\boxed{80}$$

Usually faster than writing full formula.

21. FIND UNKNOWN GROUP SIZE

A group of 20 students averages 50.

Another group averages 70.

Combined average is 62.

Let second group size = (n) .

Use alligation:

$$20:n = (70-62):(62-50)$$

$$=8:12$$

$$=2:3$$

If first group = 20 corresponds to 2 parts:

$$\backslash[1\text{ part}=10 \backslash]$$

Second group:

$$\backslash[3(10)=\boxed{30} \backslash]$$

22. SAME PROBLEM BY EQUATION

$$\backslash[\frac{20(50)+70n}{20+n}=62 \backslash]$$

$$\backslash[1000+70n=1240+62n \backslash]$$

$$\backslash[8n=240 \backslash]$$

$$\backslash[n=30 \backslash]$$

Alligation is faster.

23. THREE GROUPS — USE TOTALS

For more than two groups, totals are usually safest.

Suppose:

- 10 students average 50
- 20 students average 60
- 30 students average 70

Combined average:

$$\backslash[\frac{10(50)+20(60)+30(70)}{60} \backslash]$$

$$\backslash[= \frac{500+1200+2100}{60} \backslash]$$

$$\backslash[= \frac{3800}{60} \backslash]$$

$$\backslash[=\boxed{63\frac{13}{3}} \backslash]$$

24. GROUPS WITH RATIO SIZES

Suppose group sizes are:

$$\backslash[2:3 \backslash]$$

and averages are:

$$\backslash[50, 70 \backslash]$$

Use weights directly:

$$\backslash[\frac{2(50)+3(70)}{5} \backslash]$$

$$\backslash[= \frac{310}{5} \backslash]$$

$$\backslash[=\boxed{62} \backslash]$$

No actual group sizes needed.

25. GENERAL RATIO-WEIGHT METHOD

If group-size ratio is:

$$\text{\textbackslash}[m:n \text{\textbackslash}]$$

and averages are:

$$\text{\textbackslash}[A:B \text{\textbackslash}]$$

combined average:

$$\text{\textbackslash}[\text{\textbackslash}boxed{\frac{mA+nB}{m+n}} \text{\textbackslash}]$$

26. SALARY EXAMPLE

40 employees average ₹ 30,000.

10 managers average ₹ 70,000.

Overall average salary:

$$\text{\textbackslash}[\frac{40(30000)+10(70000)}{50} \text{\textbackslash}]$$

$$\text{\textbackslash}[= \frac{1200000+700000}{50} \text{\textbackslash}]$$

$$\text{\textbackslash}[= \text{\textbackslash}boxed{\text{\textbackslash} 38,000} \text{\textbackslash}]$$

27. MARKS EXAMPLE

30 boys average 72 marks.

20 girls average 81 marks.

Class average:

$$\text{\textbackslash}[\frac{30(72)+20(81)}{50} \text{\textbackslash}]$$

$$\text{\textbackslash}[= \frac{2160+1620}{50} \text{\textbackslash}]$$

$$\text{\textbackslash}[= \frac{3780}{50} \text{\textbackslash}]$$

$$\text{\textbackslash}[=\boxed{75.6} \text{\textbackslash}]$$

28. AGE EXAMPLE

Average age of 8 children = 12.

Average age of 2 adults = 42.

Combined family average:

$$\text{\textbackslash}[\frac{8(12)+2(42)}{10} \text{\textbackslash}]$$

$$\text{\textbackslash}[= \frac{96+84}{10} \text{\textbackslash}]$$

$$\text{\textbackslash}[=\boxed{18} \text{\textbackslash}]$$

29. WEIGHTED MARKS WITH COURSE CREDITS

Suppose grades:

- Subject A: 80, 4 credits
- Subject B: 70, 3 credits
- Subject C: 90, 2 credits

Weighted average:

$$\left[\frac{4(80)+3(70)+2(90)}{9} \right]$$

$$\left[= \frac{320+210+180}{9} \right]$$

$$\left[= \frac{710}{9} \right]$$

$$\left[\boxed{\approx 78.89} \right]$$

Credits act as weights.

30. WEIGHTED PERCENTAGES

Suppose:

- Exam 1: 80%, weight 40%
- Exam 2: 60%, weight 60%

Overall:

$$\left[80(0.4)+60(0.6) \right]$$

$$\left[=32+36 \right]$$

$$\left[=\boxed{68\%} \right]$$

31. MAJOR PERCENTAGE TRAP

A student gets:

$$\left[80/100 \right]$$

and:

$$\left[45/50 \right]$$

The percentages are:

$$\left[80\%, 90\% \right]$$

But the combined percentage is not:

$$\left[85\% \right]$$

because denominators differ.

Combined:

$$\left[\frac{80+45}{100+50} \times 100 \right]$$

$$\left[= \frac{125}{150} \times 100 \right]$$

$$\left[=\boxed{83\frac{1}{3}\%} \right]$$

32. WHY DENOMINATORS ARE WEIGHTS

For percentages:

$$\left[\frac{80}{100} = 80\% \right]$$

$$\left[\frac{45}{50} = 90\% \right]$$

The weights are:

$$\left[100, 50 \right]$$

So weighted average:

$$\left[\frac{100(80) + 50(90)}{150} \right]$$

$$\left[= \frac{12500}{150} \right]$$

$$\left[= 83\frac{1}{3}\% \right]$$

Same answer.

33. FREQUENCY TABLE EXAMPLE

Value	Frequency
10	2
20	3
30	5

Average:

$$\left[\frac{10(2) + 20(3) + 30(5)}{2 + 3 + 5} \right]$$

$$\left[= \frac{20 + 60 + 150}{10} \right]$$

$$\left[= \boxed{23} \right]$$

34. ASSUMED-MEAN METHOD WITH WEIGHTS

For values near 50:

Value	Weight
48	2
50	5
53	3

Take assumed mean:

$$\left[50 \right]$$

Deviations:

$$\sqrt{-2, \sqrt{0}, \sqrt{+3}}$$

Weighted deviations:

$$\sqrt{2(-2)+5(0)+3(3)}$$

$$\sqrt{-4+9=5}$$

Total weight:

$$\sqrt{10}$$

Weighted average:

$$\sqrt{50+\frac{5}{10}}$$

$$\sqrt{=\boxed{50.5}}$$

35. COMBINED AVERAGE USING DEVIATIONS

Groups:

- 30 people average 48
- 20 people average 53

Take base:

$$\sqrt{50}$$

First group's deviation:

$$\sqrt{-2}$$

Weighted deviation:

$$\sqrt{30(-2)=-60}$$

Second:

$$\sqrt{+3}$$

Weighted:

$$\sqrt{20(3)=60}$$

Net deviation:

$$\sqrt{0}$$

So combined average:

$$\sqrt{\boxed{50}}$$

Very fast.

36. FIND GROUP RATIO USING DEVIATIONS

Averages:

$$\sqrt{40, \sqrt{70}}$$

Combined:

$$\sqrt{50}$$

Relative deviations from 50:

$$\sqrt{-10, +20}$$

For weighted deviations to balance:

$$\sqrt{n_1(10)=n_2(20)}$$

Thus:

$$\sqrt{n_1:n_2=2:1}$$

Same as alligation.

37. INTERPRETING ALLIGATION AS BALANCE

Lower-average group creates a deficit below combined mean.

Higher-average group creates an excess above it.

For combined average to balance:

$$\sqrt{\boxed{\text{total deficit}=\text{total excess}}}$$

This is the logic behind cross-difference ratio.

38. EXAMPLE

Average 55 group and average 75 group combine to average 60.

Deficit per person in first group:

$$\sqrt{60-55=5}$$

Excess per person in second:

$$\sqrt{75-60=15}$$

To balance:

$$\sqrt{n_1(5)=n_2(15)}$$

$$\sqrt{n_1:n_2=3:1}$$

$$\sqrt{\boxed{3:1}}$$

39. FIND ONE GROUP TOTAL FROM COMBINED DATA

Total 50 people average 68.

20 people average 80.

Remaining 30 people's average?

Overall total:

$$\sqrt{50(68)=3400}$$

Known total:

$$\sqrt{20(80)=1600}$$

Remaining:

$$\sqrt[1800]{}$$

Average:

$$\sqrt[1800/30]{\boxed{60}}$$

40. FAST BALANCE METHOD FOR SAME QUESTION

Combined average:

$$\sqrt[68]{}$$

High group:

$$\sqrt[80]{}$$

Remaining group average = (x) .

Sizes:

$$\sqrt[20:30=2:3]{}$$

Weighted deviations around 68:

High excess:

$$\sqrt[80-68=12]{}$$

Total excess:

$$\sqrt[20(12)=240]{}$$

This must equal deficit of remaining 30:

$$\sqrt[30(68-x)=240]{}$$

$$\sqrt[68-x=8]{}$$

$$\sqrt[\boxed{x=60}]{}$$

41. GATE-STYLE QUESTION 1

20 observations have mean 40.

30 observations have mean 50.

Combined mean:

$$\sqrt[\frac{20(40)+30(50)}{50}]{}$$

$$\sqrt[= \frac{800+1500}{50}]{}$$

$$\sqrt[=\boxed{46}]{}$$

42. GATE-STYLE QUESTION 2

Two groups have means 30 and 50.

Combined mean is 42.

Find group-size ratio.

$$n_1:n_2 = (50-42):(42-30)$$

$$=8:12$$

$$=\boxed{2:3}$$

43. GATE-STYLE QUESTION 3

Scores 60, 70, 90 have weights 2, 3, 5.

Weighted mean:

$$\frac{120+210+450}{10}$$

$$=\boxed{78}$$

44. GATE-STYLE QUESTION 4

A dataset contains:

- 5 values with mean 10
- 15 values with mean 20

Combined mean:

$$\frac{50+300}{20}$$

$$=\boxed{17.5}$$

45. PLACEMENT-STYLE QUESTION 1

Average salary of 30 employees is ₹ 25,000.

Average salary of 10 managers is ₹ 55,000.

Company average:

$$\frac{30(25000)+10(55000)}{40}$$

$$=\frac{750000+550000}{40}$$

$$=\boxed{132500}$$

46. PLACEMENT-STYLE QUESTION 2

Boys average 60.

Girls average 75.

Class average 66.

Find boys:girls.

$$(75-66):(66-60)$$

$$=9:6$$

$$\boxed{3.2}$$

47. PLACEMENT-STYLE QUESTION 3

A course score consists of:

- assignments: 20%, score 90
- midterm: 30%, score 70
- final: 50%, score 80

Overall:

$$90(0.2)+70(0.3)+80(0.5)$$

$$18+21+40$$

$$\boxed{79}$$

48. PLACEMENT-STYLE QUESTION 4

A student scores:

$$72/80$$

and:

$$45/60$$

Combined percentage:

Total obtained:

$$117$$

Total maximum:

$$140$$

$$\frac{117}{140}\times 100$$

$$\approx \boxed{83.57\%}$$

49. ADVANCED QUESTION — FIND UNKNOWN WEIGHT

Average of two groups:

- Group A average 40
- Group B average 70
- Combined average 50

If Group A has 60 people, find Group B size.

Ratio:

$$A:B = (70-50):(50-40)$$

$$=20:10$$

$$=2:1$$

So if $A = 60$:

$$B = \boxed{30}$$

50. ADVANCED QUESTION — FIND UNKNOWN AVERAGE

Group A: 40 people, average 55.

Group B: 60 people, average x .

Combined average = 67.

Total:

$$100(67) = 6700$$

Known:

$$40(55) = 2200$$

Unknown group total:

$$4500$$

So:

$$x = 4500/60$$

$$x = \boxed{75}$$

51. ADVANCED QUESTION — THREE WEIGHTED COMPONENTS

Project evaluation:

- Technical: 50%, score 84
- Communication: 20%, score 70
- Aptitude: 30%, score 90

Overall:

$$84(0.5) + 70(0.2) + 90(0.3)$$

$$42 + 14 + 27$$

$$83$$

52. ADVANCED QUESTION — FIND REQUIRED FINAL SCORE

Weights:

- Midterm: 40%, score 70
- Final: 60%, score x

Need overall 82.

$$0.4(70) + 0.6x = 82$$

$$28 + 0.6x = 82$$

$$0.6x = 54$$

$$\boxed{x=90}$$

53. FAST METHOD FOR REQUIRED WEIGHTED SCORE

Overall target:

$$82$$

Midterm contributes:

$$28$$

Remaining contribution needed:

$$54$$

Final weight:

$$0.6$$

Required score:

$$54/0.6 = \boxed{90}$$

54. COMBINED AVERAGE AFTER GROUP CHANGE

Group A:

$$20 \text{ people, avg } 50$$

Group B:

$$30 \text{ people, avg } 70$$

Combined:

$$62$$

If 10 new people with average 80 join:

Old total:

$$50(62)=3100$$

Added total:

$$800$$

New total:

$$3900$$

Count:

$$60$$

New average:

$$\boxed{65}$$

55. COMMON TRAPS

& pTRAP 1 — Averaging averages directly

Only valid when weights are equal.

& pTRAP 2 — Wrong alligation direction

Use cross differences:

$$\backslash [(A_2 - A_c) : (A_c - A_1) \backslash]$$

& pTRAP 3 — Combined mean outside range

Impossible if weights are positive.

& pTRAP 4 — Averaging percentages with unequal denominators

Weight by denominators.

& pTRAP 5 — Weights not summing to 100

That's fine. Divide by total weight.

56. MUST-MEMORIZE FORMULAS

Weighted mean

$$\backslash \boxed{ \bar{x} = \frac{\sum w_i x_i}{\sum w_i} } \backslash$$

Two-group combined mean

$$\backslash \boxed{ A_c = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2} } \backslash$$

Group-size ratio

$$\backslash \boxed{ n_1 : n_2 = (A_2 - A_c) : (A_c - A_1) } \backslash$$

57. FAST DECISION TREE

If question gives:

Group sizes + averages?

Use weighted total.

Group averages + combined average, but not sizes?

Use alligation.

Weighted exam components?

Multiply each score by its weight.

Frequency table?

Frequency = weight.

Percentages with different denominators?

Use actual totals / denominators as weights.

58. 5-MINUTE REVISION

Remember this visual rule:

If:

$$\boxed{40} \quad \boxed{50} \quad \boxed{70}$$

distances are:

$$10, 20$$

then quantities are opposite:

$$20:10 = 2:1$$

So the combined average is pulled closer to the **larger group**.

PRACTICE — LEVEL 0

Q1

10 students average 60 and 10 students average 80. Find combined average.

Q2

20 students average 50 and 30 students average 70. Find combined average.

Q3

Find weighted average of 60 and 80 with weights 1:3.

Q4

Values 10 and 20 occur 4 and 6 times. Find average.

Q5

Two groups average 40 and 60. Combined average is 50. Find group-size ratio.

LEVEL 1

Q6

30 boys average 70 and 20 girls average 80. Find class average.

Q7

Groups average 50 and 80. Combined average is 60. Find size ratio.

Q8

Scores 70, 80, 90 have weights 2, 3, 5. Find weighted mean.

Q9

A group of 25 people averages 60. Another group of 15 averages 80. Find combined average.

Q10

40 people average 55. Another group averages 75. Combined average is 63. Find second-group size.

LEVEL 2

Q11

50 employees average \$40,000 and 10 managers average \$80,000. Find overall average salary.

Q12

Two classes average 65 and 80. Combined average 71. Find size ratio.

Q13

A student's coursework score is 90 with weight 30%, and exam score is 75 with weight 70%. Find final score.

Q14

A student scores 80/100 and 90/150. Find combined percentage.

Q15

20 people average 50, 30 people average 60, and 50 people average 70. Find combined average.

LEVEL 3 — EXAM PRESSURE

Q16

Group A average 40, Group B average 70, combined average 58. If total number is 50, find group sizes.

Q17

A test score of 72 has weight 40%. What score is needed on the remaining 60% to get overall 84?

Q18

A company has 80 workers averaging \$30,000 and 20 executives averaging \$90,000. Find company-wide average.

Q19

A dataset has values 20, 30, 50 with frequencies 2, 3, 5. Find weighted average.

Q20

Two groups are in size ratio 3:5 and average 40 and 70 respectively. Find combined average.

ANSWERS

Q1

Equal groups:

$$\left[\frac{60+80}{2} = \boxed{70} \right]$$

Q2

$$\left[\frac{20(50)+30(70)}{50} \right]$$

$$\left[=\frac{1000+2100}{50} \right]$$

$$\left[=\boxed{62} \right]$$

Q3

$$\left[\frac{1(60)+3(80)}{4} \right]$$

$$\left[=\frac{300}{4} \right]$$

$$\left[=\boxed{75} \right]$$

Q4

$$\left[\frac{4(10)+6(20)}{10} \right]$$

$$\left[=\frac{160}{10} \right]$$

$$\left[=\boxed{16} \right]$$

Q5

$$\left[(60-50):(50-40) \right]$$

$$\left[=10:10 \right]$$

$$\left[=\boxed{1:1} \right]$$

Q6

$$\left[\frac{30(70)+20(80)}{50} \right]$$

$$\left[=\frac{3700}{50} \right]$$

$$\left[=\boxed{74} \right]$$

Q7

$$\left[(80-60):(60-50) \right]$$

$$\left[=20:10 \right]$$

$$\left[=\boxed{2:1} \right]$$

Q8

$$\left[\frac{2(70)+3(80)+5(90)}{10} \right]$$

$$\left[=\frac{830}{10} \right]$$

$$\left[=\boxed{83} \right]$$

Q9

$$\left[\frac{25(60)+15(80)}{40} \right]$$

$$\left[=\frac{2700}{40} \right]$$

$$\left[=\boxed{67.5} \right]$$

Q10

First-group : second-group:

$$\left[(75-63):(63-55) \right]$$

$$\backslash [=12:8 \backslash]$$

$$\backslash [=3:2 \backslash]$$

First group = 40 corresponds to 3 parts.

Second:

$$\backslash [40\times\frac{2}{3} = \boxed{\frac{80}{3}} \backslash]$$

This is not an integer, so with the stated data the implied group size is:

$$\backslash [\boxed{26\frac{2}{3}} \backslash]$$

For a real headcount, these exact numbers are inconsistent with integer group sizes. In pure aptitude algebra, the ratio is still $\backslash (3:2) \backslash$.

Q11

$$\backslash [\frac{50(40000)+10(80000)}{60} \backslash]$$

$$\backslash [= \frac{2800000}{60} \backslash]$$

$$\backslash [= \boxed{46666.666666666666} \text{ approximately} \backslash]$$

Q12

$$\backslash [(80-71):(71-65) \backslash]$$

$$\backslash [=9:6 \backslash]$$

$$\backslash [= \boxed{3:2} \backslash]$$

Q13

$$\backslash [90(0.3)+75(0.7) \backslash]$$

$$\backslash [=27+52.5 \backslash]$$

$$\backslash [= \boxed{79.5} \backslash]$$

Q14

Obtained:

$$\backslash [80+90=170 \backslash]$$

Maximum:

$$\backslash [250 \backslash]$$

$$\backslash [170/250\times100 = \boxed{68\%} \backslash]$$

Q15

Total size:

$$\backslash [100 \backslash]$$

Weighted total:

$$\backslash [20(50)+30(60)+50(70) \backslash]$$

$$\backslash [=1000+1800+3500 \backslash]$$

$$\backslash [=6300 \backslash]$$

Average:

$$\boxed{63}$$

Q16

Size ratio:

$$(70-58):(58-40)$$

$$=12:18$$

$$=2:3$$

Total parts:

$$5$$

Total number:

$$50$$

Sizes:

$$\boxed{20, 30}$$

Q17

$$0.4(72)+0.6x=84$$

$$28.8+0.6x=84$$

$$0.6x=55.2$$

$$\boxed{x=92}$$

Q18

$$\frac{80(30000)+20(90000)}{100}$$

$$= \frac{2400000+1800000}{100}$$

$$= \boxed{42,000}$$

Q19

$$\frac{2(20)+3(30)+5(50)}{10}$$

$$= \frac{40+90+250}{10}$$

$$= \boxed{38}$$

Q20

$$\frac{3(40)+5(70)}{8}$$

$$= \frac{470}{8}$$

$$= \boxed{58.75}$$

CHAPTER 17 STATUS

Covered so far:

- weighted mean '
- combined averages '
- unequal group sizes '

- frequency as weight '
- credits/marks weighting '
- alligation/cross-difference shortcut '
- finding group-size ratios '
- finding unknown group averages '
- weighted percentages '
- unequal denominator trap '
- deviation/balance interpretation '
- GATE/placement practice '

But **Chapter 17 is not complete yet.**

Next we need:

- advanced alligation-style average problems
- joining/leaving groups with unequal averages
- target weighted scores
- changing weights
- weighted percentage chains
- repeated group merges
- mixed average + ratio problems
- final GATE/placement drills

Progress

Completed: 16/76 **Current:** ~~Chapter 17~~ **Chapter 17 — Weighted & Combined Average** **Next:** Chapter 17 Part 2 — **Advanced weighted averages, alligation & group-change problems**

USER

next

ASSISTANT

CHAPTER 17 — WEIGHTED & COMBINED AVERAGE

PART 2 — ADVANCED ALLIGATION, GROUP CHANGES, TARGET WEIGHTS & MIXED EXAM PROBLEMS

Status: ~~Not in PROGRESS~~ **GATE:** High **Placements:** Very High **Priority:** ~~Not~~ 1

This part finishes the advanced uses of weighted average.

The key idea is:

When groups contribute unequally, always think in terms of total contribution or weighted deviation.

1. RECAP — THE TWO MASTER METHODS

You should be comfortable with both.

Method 1 — Total method

$$\boxed{\text{Combined average}} = \frac{\text{sum of group totals}}{\text{total number}}$$

where:

$$\text{group total} = \text{group size} \times \text{group average}$$

Method 2 — Alligation

For two groups with averages $(A_1 < A_c < A_2)$:

$$\boxed{n_1:n_2 = (A_2 - A_c):(A_c - A_1)}$$

Use the first when actual sizes are known.

Use the second when the combined average is known but group-size ratio is needed.

2. GROUP JOINS AN EXISTING GROUP

40 employees have average salary 130,000.

10 employees with average 150,000 join.

Old total:

$$40(30000)=1200000$$

Joining total:

$$10(50000)=500000$$

New total:

$$1700000$$

New count:

$$50$$

Average:

$$\boxed{134000}$$

3. FAST DEVIATION METHOD FOR JOINING GROUP

Old average:

$$30000$$

New group average:

$$50000$$

Excess per joining employee:

$$20000$$

Total excess:

$$\backslash[10(20000)=200000 \backslash]$$

This excess is spread across 50 employees:

$$\backslash[200000/50=4000 \backslash]$$

New average:

$$\backslash[30000+4000 = \boxed{34000} \backslash]$$

This is often faster.

4. GROUP LEAVES

Average of 50 students is 60.

10 students with average 75 leave.

Old total:

$$\backslash[50(60)=3000 \backslash]$$

Leaving total:

$$\backslash[10(75)=750 \backslash]$$

Remaining total:

$$\backslash[2250 \backslash]$$

Remaining count:

$$\backslash[40 \backslash]$$

New average:

$$\backslash[\boxed{56.25} \backslash]$$

5. INTUITION FOR GROUP LEAVING

If the leaving group's average is:

- above current average !' remaining average falls
- below current average !' remaining average rises
- equal to current average !' remaining average unchanged

Always use this as a sanity check.

6. FIND AVERAGE OF JOINING GROUP

30 people average 40.

After 10 more people join, overall average becomes 45.

Find average of joining group.

Old total:

$$\backslash[30(40)=1200 \backslash]$$

New total:

$$\backslash[40(45)=1800 \backslash]$$

Joining total:

$$\backslash[600 \backslash]$$

Joining count:

$$\backslash[10 \backslash]$$

Joining average:

$$\backslash[\boxed{60} \backslash]$$

7. FIND AVERAGE OF LEAVING GROUP

50 employees average $^1 40,000$.

After 10 employees leave, remaining 40 average $^1 37,000$.

Old total:

$$\backslash[50(40000)=2000000 \backslash]$$

Remaining total:

$$\backslash[40(37000)=1480000 \backslash]$$

Leaving total:

$$\backslash[520000 \backslash]$$

Leaving average:

$$\backslash[520000/10 = \boxed{^1 52} 000 \backslash]$$

8. REPLACEMENT OF GROUPS

A company has 100 workers averaging $^1 30,000$.

20 workers averaging $^1 24,000$ leave and are replaced by 20 workers averaging $^1 34,000$.

Count remains 100.

Difference in group averages:

$$\backslash[34000-24000=10000 \backslash]$$

Net salary-total increase:

$$\backslash[20(10000)=200000 \backslash]$$

Average increase:

$$\backslash[200000/100=2000 \backslash]$$

New average:

$$\backslash[\boxed{^1 32} 000 \backslash]$$

9. GENERAL GROUP-REPLACEMENT SHORTCUT

If r people are replaced in a group of total size N :

$$\boxed{\Delta A = \frac{r(A_{\text{in}} - A_{\text{out}})}{N}}$$

Very useful.

10. EXAMPLE

Total group size:

$$50$$

10 students averaging 60 are replaced by 10 averaging 75.

Average increase:

$$\frac{10(75-60)}{50}$$

$$= \frac{150}{50} = 3$$

If old average was 68:

$$\boxed{71}$$

11. FIND REQUIRED JOINING GROUP SIZE

A group of 20 people has average 50.

How many people averaging 80 must join to make overall average 60?

Use alligation:

$$50, 60, 80$$

Old group : joining group:

$$(80-60):(60-50)$$

$$20:10$$

$$=2:1$$

Old group size = 20.

Joining size:

$$\boxed{10}$$

12. FIND REQUIRED JOINING AVERAGE

20 people average 50.

10 people join.

Combined average becomes 60.

Joining average:

$$\frac{30(60)-20(50)}{10}$$

$$= \frac{1800-1000}{10}$$

$$\boxed{80}$$

13. FIND REQUIRED REMOVAL SIZE

A group has average 60.

A subgroup averaging 80 leaves.

Remaining average becomes 55.

Suppose original total size is 50. Find number leaving.

Let leaving number = x .

Original total:

$$50(60)=3000$$

Remaining:

$$50-x$$

Equation:

$$3000-80x = 55(50-x)$$

$$3000-80x = 2750-55x$$

$$250=25x$$

$$\boxed{x=10}$$

14. SAME PROBLEM USING BALANCE

Original average:

$$60$$

Leaving group average:

$$80$$

Remaining average:

$$55$$

Original group is effectively made of:

- leaving subgroup average 80
- remaining subgroup average 55
- combined average 60

By alligation:

$$\text{remaining} : \text{leaving} = (80-60):(60-55)$$

$$=20:5 = 4:1$$

Total 50 !' 5 parts.

Leaving:

$$10$$

$\boxed{10}$

Much faster.

15. FIND GROUP-SIZE RATIO FROM COMBINED AVERAGE

Group A average:

72

Group B average:

48

Combined:

60

Distances:

$72 - 60 = 12$

$60 - 48 = 12$

So groups must be equal size:

$\boxed{1:1}$

16. COMBINED AVERAGE CLOSE TO ONE GROUP

Averages:

$40, 80$

Combined:

70

Group-size ratio:

$40 \text{ group} : 80 \text{ group}$

$=(80-70):(70-40)$

$=10:30$

$=\boxed{1:3}$

Combined mean is closer to 80 because that group is larger.

17. REPEATED MERGING

Suppose:

- Group A: 10 people, avg 40
- Group B: 20 people, avg 50
- Group C: 30 people, avg 60

You can merge A and B first.

A+B:

$$\left[\frac{10(40)+20(50)}{30} = \frac{1400}{30} = 46\frac{2}{3} \right]$$

Then combine with C.

But direct total method is easier:

$$\left[\frac{400+1000+1800}{60} \right]$$

$$\left[= \frac{3200}{60} \right]$$

$$\left[= \boxed{53\frac{1}{3}} \right]$$

Exam rule

For 3+ groups:

$$\left[\boxed{\text{Use totals unless a clear shortcut appears}} \right]$$

18. GROUP SIZES GIVEN AS RATIOS

Three groups have sizes:

$$\left[1:2:3 \right]$$

Their averages are:

$$\left[50, 60, 70 \right]$$

Combined average:

$$\left[\frac{1(50)+2(60)+3(70)}{6} \right]$$

$$\left[= \frac{50+120+210}{6} \right]$$

$$\left[= \frac{380}{6} \right]$$

$$\left[= \boxed{63\frac{1}{3}} \right]$$

19. TARGET WEIGHTED SCORE

A course has:

- Quiz 20%
- Midterm 30%
- Final 50%

Scores:

- Quiz 80
- Midterm 70
- Final unknown

Target overall:

$$\left[85 \right]$$

Equation:

$$\left[0.2(80)+0.3(70)+0.5x=85 \right]$$

$$\left[16+21+0.5x=85 \right]$$

$$\left[0.5x=48 \right]$$

$$\boxed{x=96}$$

20. TARGET FEASIBILITY CHECK

Suppose same weights and scores, but target is 90.

Known contribution:

$$16+21=37$$

Maximum possible final contribution if final score "d 1 0 0 :

$$0.5(100)=50$$

Maximum total:

$$87$$

Therefore:

$$\boxed{90 \text{ overall is impossible}}$$

Very important OA shortcut.

21. CHANGING WEIGHTS

Suppose project score 80 had weight 40%, exam score 70 had weight 60%.

Old overall:

$$80(0.4)+70(0.6)$$

$$=32+42$$

$$=74$$

If weights change to 25% and 75%:

$$80(0.25)+70(0.75)$$

$$=20+52.5$$

$$=\boxed{72.5}$$

The raw scores didn't change, but weighted average did.

22. WEIGHT SUM NEED NOT BE 100

Scores:

$$60, 80, 90$$

Weights:

$$2, 3, 5$$

Weighted average:

$$\frac{2(60)+3(80)+5(90)}{10}$$

$$= \frac{810}{10}$$

$\boxed{81}$

Only relative weights matter.

23. SCALING ALL WEIGHTS DOES NOT CHANGE WEIGHTED MEAN

Weights:

$2:3:5$

and:

$4:6:10$

give same weighted mean.

Why?

Every weight was multiplied by 2.

That common factor cancels.

24. WEIGHTED PERCENTAGES FROM UNEQUAL TOTALS

A student gets:

- 90% in a 100-mark test
- 60% in a 200-mark test

Combined percentage?

Obtained marks:

90

and:

120

Total:

210

Maximum:

300

Combined percentage:

$$\frac{210}{300} \times 100 = \boxed{70\%}$$

Do not average 90 and 60 to get 75%.

25. FAST WEIGHTED-PERCENTAGE METHOD

Weights are total marks:

$100:200=1:2$

So:

$$\frac{1(90)+2(60)}{3}$$

$$\boxed{= \frac{210}{3}}$$

$$\boxed{=70\%}$$

26. PASS PERCENTAGE ACROSS GROUPS

Class A:

- 100 students
- 80% pass

Pass students:

$$\boxed{80}$$

Class B:

- 200 students
- 60% pass

Pass:

$$\boxed{120}$$

Combined pass rate:

$$\boxed{\frac{200}{300} \times 100}$$

$$\boxed{=66\frac{2}{3}\%}$$

Again, percentages must be weighted by group sizes.

27. FIND UNKNOWN GROUP PERCENTAGE

Class A:

100 students, 80% pass.

Class B:

200 students, $(x\%)$ pass.

Combined pass rate:

70%.

Total pass students required:

$$\boxed{300(0.7)=210}$$

Class A passes:

$$\boxed{80}$$

Class B passes:

$$\boxed{130}$$

Percentage:

$$\boxed{130/200 \times 100}$$

$$\boxed{=65\%}$$

28. MIXED RATIO + AVERAGE

Two groups have size ratio:

$$\text{[3:2]}$$

Their averages are:

$$\text{[70, 85]}$$

Combined:

$$\text{[\frac{3(70)+2(85)}{5}]}$$

$$\text{[= \frac{210+170}{5}]}$$

$$\text{[=\boxed{76}]}$$

29. FIND SIZE RATIO FROM COMBINED MEAN

Two groups average 50 and 80.

Combined average 68.

Ratio:

$$\text{[50\text{-}group:80\text{-}group]}$$

$$\text{[=(80-68):(68-50)]}$$

$$\text{[=12:18]}$$

$$\text{[=\boxed{2:3}]}$$

30. FIND ONE GROUP AVERAGE USING SIZE RATIO

Size ratio:

$$\text{[2:3]}$$

First group average:

$$\text{[50]}$$

Combined average:

$$\text{[68]}$$

Let second group average = (x) .

$$\text{[\frac{2(50)+3x}{5}=68]}$$

$$\text{[100+3x=340]}$$

$$\text{[3x=240]}$$

$$\text{[\boxed{x=80}]}$$

31. ALLIGATION WITH PRICES

Suppose items costing $140/\text{kg}$ and $170/\text{kg}$ are mixed to produce average price $150/\text{kg}$.

Ratio of quantities:

$$\frac{1}{\sqrt{(70-50):(50-40)}}$$

$$\frac{1}{\sqrt{=20:10}}$$

$$\frac{1}{\sqrt{=\boxed{2:1}}}$$

This is exactly weighted average.

It will become central in Chapter 22 — Mixtures & Alligation.

32. ALLIGATION WITH CONCENTRATIONS

A 20% solution and 50% solution are mixed to get 30%.

Quantity ratio:

$$\frac{1}{\sqrt{(50-30):(30-20)}}$$

$$\frac{1}{\sqrt{=20:10}}$$

$$\frac{1}{\sqrt{=\boxed{2:1}}}$$

Again, weighted average.

33. ALLIGATION WITH AGES

Group A average age 20.

Group B average age 50.

Combined average 30.

Ratio:

$$\frac{1}{\sqrt{A:B = (50-30):(30-20)}}$$

$$\frac{1}{\sqrt{=20:10}}$$

$$\frac{1}{\sqrt{=\boxed{2:1}}}$$

Same mathematics across different contexts.

34. ALLIGATION CONDITION

For positive quantities, if mixing two averages (A_1) and (A_2) , the combined average must satisfy:

$$\frac{1}{\sqrt{\boxed{\min(A_1, A_2) \leq A_c \leq \max(A_1, A_2)}}}$$

If not, the data are impossible under ordinary positive weights.

35. IMPOSSIBLE COMBINED AVERAGE EXAMPLE

Groups average:

$$\frac{1}{\sqrt{40, 60}}$$

Can combined average be:

$\sqrt{70?}$

No.

Positive weighted mean cannot exceed both component averages.

$\boxed{\text{Impossible}}$

This allows instant elimination.

36. ONE GROUP HAS ZERO WEIGHT?

If a group has weight 0, it contributes nothing.

Weighted average then depends only on positive-weight groups.

In ordinary aptitude, weights are usually positive.

37. COMBINED AVERAGE AFTER TRANSFORMATION

Suppose every value in both groups is increased by 5.

Original combined average:

$\sqrt{60}$

New combined average:

$\boxed{65}$

Because adding same number to every observation shifts weighted mean by the same amount.

38. MULTIPLY ALL VALUES

If every value in all groups is multiplied by 1.2:

combined average also becomes:

$\sqrt{1.2 \times A_c}$

Example:

$\sqrt{50 \rightarrow 60}$

So weighted means obey linear transformations too.

39. DIFFERENT TRANSFORMATION BY GROUP — CAUTION

Group A average 50, size 20.

Group B average 70, size 30.

If only Group A gets +10, you cannot add 10 to combined average.

New Group A average:

$\sqrt{60}$

Then recompute:

$$\left[\frac{20(60)+30(70)}{50} \right]$$

$$\left[= \frac{1200+2100}{50} \right]$$

$$\left[=\boxed{66} \right]$$

Old combined was:

$$\left[62 \right]$$

Increase only:

$$\left[4 \right]$$

because only 20/50 of the observations changed by 10.

40. FAST PARTIAL-CHANGE RULE

If r out of N observations each increase by d :

$$\left[\boxed{\text{overall average increase} = \frac{rNd}{N}} \right]$$

Example:

20 of 50 values increase by 10:

$$\left[\frac{20}{50}(10)=4 \right]$$

Correct.

41. GROUP AVERAGE CHANGES

A group of 30 has average 50.

Another group of 20 has average 70.

First group's average rises by 5.

How much does combined average rise?

Only 30 of total 50 affected:

$$\left[\Delta A_c = \frac{30}{50}(5) \right]$$

$$\left[=3 \right]$$

So if old combined:

$$\left[58 \right]$$

new:

$$\left[\boxed{61} \right]$$

42. FREQUENCY CHANGE

Values:

- 10 occurs twice
-

- 20 occurs three times

Average:

$$\sqrt[16]{}$$

If one additional 20 is added:

new total:

$$\sqrt[80]{}$$

new count:

$$\sqrt[6]{}$$

new average:

$$\sqrt[\boxed{13\frac{13}{3}}]{}$$

Careful—recalculate original total:

$$\sqrt[2(10)+3(20)=80]{}$$

Original count:

$$\sqrt[5]{}$$

Average:

$$\sqrt[16]{}$$

Add one 20:

new total:

$$\sqrt[100]{}$$

count:

$$\sqrt[6]{}$$

Average:

$$\sqrt[\boxed{16\frac{16}{3}}]{}$$

Always check totals.

43. WEIGHTED DEVIATION SHORTCUT

Values:

$$\sqrt[98,102,105]{}$$

Weights:

$$\sqrt[2,3,5]{}$$

Take base:

$$\sqrt[100]{}$$

Deviations:

$$\sqrt[-2,+2,+5]{}$$

Weighted deviation:

$$\backslash[2(-2)+3(2)+5(5) \backslash]$$

$$\backslash[=-4+6+25 \backslash]$$

$$\backslash[=27 \backslash]$$

Total weight:

$$\backslash[10 \backslash]$$

Average:

$$\backslash[100+\frac{27}{10} \backslash]$$

$$\backslash[=\boxed{102.7} \backslash]$$

44. WHEN ALLIGATION IS FASTER

Use alligation when asked:

- ratio of quantities
- relative group sizes
- amount of lower-value/high-value component
- mixing two groups to achieve target average

Do not use it automatically when actual totals are easier.

45. WHEN TOTAL METHOD IS FASTER

Use totals when:

- 3+ groups
 - joining/leaving groups
 - unknown group total
 - unequal multiple weights
 - overlap/replacement
 - actual count/marks/salary values are given
-

46. GATE-STYLE QUESTION 1

20 values average 40.

30 values average 60.

10 more values average 80 join.

Combined average:

Total:

$$\backslash[20(40)+30(60)+10(80) \backslash]$$

$$\backslash[=800+1800+800 \backslash]$$

$$\backslash[=3400 \backslash]$$

Count:

$$\sqrt[3]{60}$$

Average:

$$\sqrt[3]{\boxed{56\frac{1}{3}}}$$

47. GATE-STYLE QUESTION 2

Two groups have averages 30 and 70.

Combined average is 50.

Ratio:

$$\sqrt[3]{(70-50):(50-30)}$$

$$\sqrt[3]{=20:20}$$

$$\sqrt[3]{=\boxed{1:1}}$$

48. GATE-STYLE QUESTION 3

Group size ratio:

$$\sqrt[3]{3:7}$$

Averages:

$$\sqrt[3]{40, 60}$$

Combined:

$$\sqrt[3]{\frac{3(40)+7(60)}{10}}$$

$$\sqrt[3]{=\frac{540}{10}}$$

$$\sqrt[3]{=\boxed{54}}$$

49. GATE-STYLE QUESTION 4

A weighted score consists of:

- 25% component scored 80
- 75% component scored 60

Overall:

$$\sqrt[3]{20+45}$$

$$\sqrt[3]{=\boxed{65}}$$

50. PLACEMENT-STYLE QUESTION 1

A company has:

- 60 workers average ₹ 30 k
- 20 managers average ₹ 60 k

- 5 executives average 120 k

Overall average salary:

Weighted total in thousands:

$$60(30)+20(60)+5(120)$$

$$=1800+1200+600$$

$$=3600$$

Total people:

$$85$$

Average:

$$3600/85$$

$$\approx \boxed{42.35\text{ k}}$$

51. PLACEMENT-STYLE QUESTION 2

Two batches average 65 and 85.

Combined average 73.

Find batch-size ratio.

$$(85-73):(73-65)$$

$$=12:8$$

$$=\boxed{3:2}$$

52. PLACEMENT-STYLE QUESTION 3

A course has:

- assignment 20%, score 90
- midterm 30%, score 75
- final 50%, score x

Target = 84.

Known:

$$18+22.5=40.5$$

Need:

$$43.5$$

So:

$$0.5x=43.5$$

$$\boxed{x=87}$$

53. PLACEMENT-STYLE QUESTION 4

A class has 120 students with 75% pass rate and another class has 80 students with 90% pass rate.

Passes:

$$\sqrt[1]{120(0.75)=90 \sqrt[1]}$$

$$\sqrt[1]{80(0.9)=72 \sqrt[1]}$$

Total pass:

$$\sqrt[1]{162 \sqrt[1]}$$

Total students:

$$\sqrt[1]{200 \sqrt[1]}$$

Combined pass rate:

$$\sqrt[1]{\boxed{81\%} \sqrt[1]}$$

54. ADVANCED QUESTION — JOINING SIZE

30 people average 50.

People averaging 80 join.

New average becomes 60.

Alligation:

$$\sqrt[1]{30:n = (80-60):(60-50) \sqrt[1]}$$

$$\sqrt[1]{=20:10 = 2:1 \sqrt[1]}$$

Therefore:

$$\sqrt[1]{n=\boxed{15} \sqrt[1]}$$

55. ADVANCED QUESTION — LEAVING GROUP

40 people average 70.

After some people averaging 90 leave, remaining average becomes 65.

Using alligation:

Remaining group avg 65, leaving avg 90, combined 70.

Remaining : leaving:

$$\sqrt[1]{(90-70):(70-65) \sqrt[1]}$$

$$\sqrt[1]{=20:5 = 4:1 \sqrt[1]}$$

Total parts 5 correspond to 40.

Leaving:

$$\sqrt[1]{\boxed{8} \sqrt[1]}$$

Remaining:

$$\sqrt[1]{32 \sqrt[1]}$$

56. ADVANCED QUESTION — REQUIRED GROUP AVERAGE

50 employees average 40.

20 new employees join, making average 50.

New group's average:

Old total:

$$\boxed{2000}$$

New total:

$$\boxed{70(50)=3500}$$

New group total:

$$\boxed{1500}$$

Average:

$$\boxed{75}$$

57. ADVANCED QUESTION — WEIGHT CHANGE

Score 80 has weight 40 and score 60 has weight 60.

Overall:

$$\boxed{68}$$

What happens if weights become equal?

New average:

$$\boxed{\frac{80+60}{2}=70}$$

Increase:

$$\boxed{2}$$

58. ADVANCED QUESTION — UNKNOWN PASS RATE

Group A:

- 150 students
- 80% pass

Group B:

- 100 students
- $(x\%)$ pass

Combined pass rate:

$$\boxed{72\%}$$

Total passes required:

$$\boxed{250(0.72)=180}$$

A passes:

$\sqrt{120}$

B passes:

$\sqrt{60}$

So:

$x=60\%$

$\boxed{60\%}$

59. COMMON TRAPS

& pTRAP 1 — **Combined average = average of averages**

Only if weights equal.

& pTRAP 2 — **Wrong alligation direction**

Use opposite differences.

& pTRAP 3 — **Averaging percentages directly**

Weight by denominators/group sizes.

& pTRAP 4 — **Group joins/leaves**

Count changes.

& pTRAP 5 — **Group replacement**

Count stays constant.

& pTRAP 6 — **Target weighted score**

Check maximum possible score before solving fully.

60. MUST-MEMORIZE FORMULAS

Weighted average

$\boxed{\bar{x} = \frac{\sum w_i x_i}{\sum w_i}}$

Two-group combined average

$\boxed{A_c = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}}$

Alligation ratio

$\boxed{n_1:n_2 = (A_2 - A_c):(A_c - A_1)}$

Group replacement

$\boxed{\Delta A = \frac{r(A_{\text{in}} - A_{\text{out}})}{N}}$

61. FAST EXAM DECISION TREE

Two groups + sizes known?

Use totals.

Two group averages + combined average?

Use alligation.

Three or more groups?

Use weighted totals.

Weighted marks?

Multiply score $\times$ weight.

Different percentages?

Use group size/denominator as weight.

Group leaves/joins?

Track total and count.

Same number leaves and joins?

Use replacement shortcut.

62. 5-MINUTE REVISION

Think of average as a balance point.

If:

$$\sqrt[50]{\boxed{60}} \sqrt[80]{}$$

distances:

$$\sqrt{10, \sqrt{20}}$$

then quantities must be:

$$\sqrt{20:10 = 2:1}$$

The closer the combined mean is to a group average, the larger that group's weight tends to be.

PRACTICE — LEVEL 0

Q1

20 people average 40 and 20 people average 60. Find combined average.

Q2

Groups average 50 and 80. Combined average 70. Find size ratio.

Q3

Scores 60 and 90 have weights 2:1. Find weighted average.

Q4

30 people average 50. Ten people averaging 70 join. Find new average.

Q5

40 people average 60. Ten people averaging 80 leave. Find remaining average.

LEVEL 1

Q6

20 people average 40. How many people averaging 70 must join to make average 50?

Q7

Group A average 50, Group B average 80, combined average 62. Find size ratio.

Q8

A score 80 has weight 30%, score 70 has weight 70%. Find weighted score.

Q9

A student gets 80/100 and 120/200. Find combined percentage.

Q10

50 people average 60. Ten averaging 45 are replaced by ten averaging 75. Find new average.

LEVEL 2

Q11

40 employees average ₹ 30,000. After 10 employees join, average becomes ₹ 36,000. Find average salary of joining employees.

Q12

60 people average 50. Some people averaging 80 leave and remaining average becomes 45. Find number leaving.

Q13

A course has 30% weight at score 80 and 70% at score x . Target is 87. Find x .

Q14

Two classes have 100 and 150 students with pass rates 70% and 80%. Find combined pass rate.

Q15

Group sizes are 2:3:5 and averages 40, 50, 70. Find combined average.

LEVEL 3 — EXAM PRESSURE

Q16

A group of 50 has average 60. Ten people averaging 40 leave, and 20 people averaging 75 join. Find new average.

Q17

Two groups average 45 and 75. Their combined average is 63 and total size is 100. Find group sizes.

Q18

A student's overall score is determined by 20%, 30%, 50% components. Scores are 90, 80 and x . Overall target is 88. Find x .

Q19

A company has 75 employees averaging 140 . Fifteen employees averaging 160 leave and are replaced by 15 averaging 180 . Find new average.

Q20

Two mixtures with concentrations 20% and 50% are mixed to get 32%. Find ratio of quantities.

ANSWERS**Q1**

Equal group sizes:

$$\left[\frac{40+60}{2} = \boxed{50} \right]$$

Q2

$$\left[(80-70):(70-50) \right]$$

$$\left[=10:20 \right]$$

$$\left[=\boxed{1:2} \right]$$

Q3

$$\left[\frac{2(60)+1(90)}{3} \right]$$

$$\left[=\frac{210}{3} \right]$$

$$\left[=\boxed{70} \right]$$

Q4

Old total:

$$\left[30(50)=1500 \right]$$

Joining total:

$$\left[700 \right]$$

New total:

$$\left[2200 \right]$$

Count:

$$\left[40 \right]$$

Average:

$$\left[\boxed{55} \right]$$

Q5

Old total:

$$\boxed{2400}$$

Leaving:

$$\boxed{800}$$

Remaining:

$$\boxed{1600}$$

Count:

$$\boxed{30}$$

Average:

$$\boxed{53\frac{13}{3}}$$

Q6

Alligation:

$$\boxed{40:n = (70-50):(50-40)}$$

$$\boxed{=20:10 = 2:1}$$

Thus:

$$\boxed{n=10}$$

Q7

$$\boxed{(80-62):(62-50)}$$

$$\boxed{=18:12}$$

$$\boxed{=3:2}$$

Q8

$$\boxed{80(0.3)+70(0.7)}$$

$$\boxed{=24+49}$$

$$\boxed{=73}$$

Q9

Obtained:

$$\boxed{80+120=200}$$

Maximum:

$$\boxed{300}$$

$$\boxed{66\frac{23}{3}\%}$$

Q10

Outgoing average:

$$\boxed{45}$$

Incoming:

$$\sqrt{75}$$

Difference:

$$\sqrt{30}$$

Total increase:

$$\sqrt{10(30)=300}$$

Average increase:

$$\sqrt{300/50=6}$$

New average:

$$\sqrt{\boxed{66}}$$

Q11

Old total:

$$\sqrt{40(30)=1200}$$

in thousands.

New total:

$$\sqrt{50(36)=1800}$$

Joining total:

$$\sqrt{600}$$

Average:

$$\sqrt{600/10 = \boxed{160k}}$$

Q12

Let leaving = $\sqrt{x}$.

Original total:

$$\sqrt{60(50)=3000}$$

Remaining:

$$\sqrt{60-x}$$

$$\sqrt{3000-80x=45(60-x)}$$

$$\sqrt{3000-80x=2700-45x}$$

$$\sqrt{300=35x}$$

$$\sqrt{x=\frac{60}{7}}$$

This is not an integer.

So:

$$\sqrt{\boxed{x=\frac{60}{7}\approx 8.57}}$$

The algebraic data are valid, but not consistent with an integer headcount.

Q13

$$\sqrt{0.3(80)+0.7x=87}$$

$$\backslash [24+0.7x=87 \backslash]$$

$$\backslash [0.7x=63 \backslash]$$

$$\backslash [\boxed{x=90} \backslash]$$

Q14

Passes:

$$\backslash [100(0.7)=70 \backslash]$$

$$\backslash [150(0.8)=120 \backslash]$$

Total:

$$\backslash [190 \backslash]$$

Students:

$$\backslash [250 \backslash]$$

Pass rate:

$$\backslash [190/250\times100 = \boxed{76\%} \backslash]$$

Q15

$$\backslash [\frac{2(40)+3(50)+5(70)}{10} \backslash]$$

$$\backslash [= \frac{80+150+350}{10} \backslash]$$

$$\backslash [=\boxed{58} \backslash]$$

Q16

Original total:

$$\backslash [50(60)=3000 \backslash]$$

10 leave averaging 40:

$$\backslash [-400 \backslash]$$

20 join averaging 75:

$$\backslash [+1500 \backslash]$$

New total:

$$\backslash [4100 \backslash]$$

New count:

$$\backslash [60 \backslash]$$

Average:

$$\backslash [\boxed{68\frac{1}{3}} \backslash]$$

Q17

Ratio:

$$\backslash [45\text{-group}:75\text{-group} \backslash]$$

$$\backslash [=(75-63):(63-45) \backslash]$$

$$\backslash [=12:18 \backslash]$$

$$\begin{aligned} &= 2:3 \end{aligned}$$

Total 100:

$$\boxed{40, 60}$$

Q18

Known contribution:

$$0.2(90) + 0.3(80)$$

$$= 18 + 24 = 42$$

Need total:

$$88$$

Final contribution:

$$46$$

$$0.5x = 46$$

$$\boxed{x = 92}$$

Q19

Old average:

$$40$$

15 replaced.

Increase in replacement-group average:

$$80 - 60 = 20$$

Total increase:

$$15(20) = 300$$

Overall average increase:

$$300 / 75 = 4$$

New average:

$$\boxed{44}$$

Q20

Alligation:

$$20\% \text{ solution} : 50\% \text{ solution}$$

$$= (50 - 32) : (32 - 20)$$

$$= 18 : 12$$

$$= \boxed{3:2}$$

' CHAPTER 17 COMPLETE

We have now covered:

- weighted mean

- combined averages '
- unequal group sizes '
- frequency weighting '
- alligation/cross differences '
- group-size ratios '
- joining groups '
- leaving groups '
- group replacement '
- target weighted scores '
- changing weights '
- combined percentages '
- pass-rate weighting '
- partial-group changes '
- alligation with price/concentration/age '
- feasibility checks '
- mixed GATE/placement problems '

Progress

Completed: 17/76 **Unit 1 — Quantitative Aptitude:** 0=No PROGRESS **Next:** 0=0% **Chapter 18 — Profit & Loss**

Chapter 18 will cover:

- cost price, selling price, profit, loss
- profit% and loss%
- base-value traps
- SP/CP multiplier method
- finding CP or SP
- successive transactions
- false weights
- equal-SP and equal-CP questions
- overall profit/loss on multiple items
- GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 18 — PROFIT & LOSS

PART 1 — COST PRICE, SELLING PRICE, PROFIT/LOSS %, MULTIPLIER METHOD & BASIC TRANSACTIONS

Status: 0=No PROGRESS **GATE:** High **Placements:** Very High **Priority:** 0=Very 1

Profit and Loss is fundamentally a **percentage-on-cost-price** chapter.

The most important question in every problem is:

What is the base?

For standard profit/loss:

$\boxed{\text{Profit or loss percentage is calculated on Cost Price}}$

unless the question explicitly says otherwise.

1. BASIC TERMS

Cost Price — CP

The price at which an article is bought.

Selling Price — SP

The price at which it is sold.

If:

$SP > CP$

there is profit.

If:

$SP < CP$

there is loss.

If:

$SP = CP$

there is neither profit nor loss.

2. PROFIT

$\boxed{\text{Profit} = SP - CP}$

Example:

Bought for ₹ 500 Sold for ₹ 650 .

$\text{Profit} = 650 - 500$

$= \boxed{150}$

3. LOSS

$\boxed{\text{Loss} = CP - SP}$

Example:

Bought for ₹ 800 Sold for ₹ 680 .

$$\text{Loss} = 800 - 680$$

$$\text{Loss} = \boxed{120}$$

4. PROFIT PERCENTAGE

$$\text{Profit\%} = \frac{\text{Profit}}{\text{CP}} \times 100$$

Example:

$$\text{CP} = 150, \text{SP} = 165$$

Profit:

$$150$$

Therefore:

$$\frac{150}{500} \times 100 = \boxed{30\%}$$

5. LOSS PERCENTAGE

$$\text{Loss\%} = \frac{\text{Loss}}{\text{CP}} \times 100$$

Example:

$$\text{CP} = 180, \text{SP} = 168$$

Loss:

$$120$$

$$\frac{120}{800} \times 100 = \boxed{15\%}$$

6. MAJOR BASE-VALUE TRAP

Profit is:

$$\text{SP} - \text{CP}$$

But profit percentage is:

$$\frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$$

Not:

$$\frac{\text{SP} - \text{CP}}{\text{SP}} \times 100$$

Unless explicitly asked:

profit is what percent of selling price?

Then SP becomes the denominator.

7. PROFIT MULTIPLIER

If profit is $p\%$:

$$[SP = CP\left(1+\frac{p}{100}\right)]$$

So:

$$[\boxed{ SP=CP\times\frac{100+p}{100} }]$$

8. LOSS MULTIPLIER

If loss is $(p\%)$:

$$[SP = CP\left(1-\frac{p}{100}\right)]$$

So:

$$[\boxed{ SP=CP\times\frac{100-p}{100} }]$$

9. MUST-MEMORIZE CP:SP RATIOS

At 20% profit:

$$[CP:SP=100:120=5:6]$$

At 25% profit:

$$[100:125=4:5]$$

At 50% profit:

$$[100:150=2:3]$$

At 20% loss:

$$[100:80=5:4]$$

At 25% loss:

$$[100:75=4:3]$$

At 50% loss:

$$[100:50=2:1]$$

These ratios make many questions instant.

10. EXAMPLE — FIND SP

$$CP = ₹1200.$$

Profit = 25%.

$$[SP=1200\times1.25]$$

$$[=1200\times\frac{54}{4}]$$

$$[= \boxed{ ₹1500 }]$$

11. EXAMPLE — LOSS

$$CP = ₹2000.$$

Loss = 15%.

$$\boxed{SP=2000(0.85)}$$

$$\boxed{=1700}$$

12. FIND CP FROM SP AND PROFIT

$$SP = 1800.$$

Profit = 20%.

$$SP=1.2CP$$

Therefore:

$$CP=\frac{1800}{1.2}$$

$$=1800\times\frac{5}{6}$$

$$\boxed{=1500}$$

13. FIND CP FROM SP AND LOSS

$$SP = 840.$$

Loss = 20%.

$$SP=0.8CP$$

$$CP=\frac{840}{0.8}$$

$$\boxed{=1050}$$

14. MASTER REVERSE FORMULAS

If profit $(p\%)$:

$$\boxed{CP = SP \times \frac{100}{100+p}}$$

If loss $(p\%)$:

$$\boxed{CP = SP \times \frac{100}{100-p}}$$

15. FIND PROFIT % FROM CP AND SP

$$CP = 400.$$

$$SP = 460.$$

Profit:

$$60$$

Profit %:

$$\frac{60}{400} \times 100 = \boxed{15\%}$$

16. FAST RATIO METHOD

If:

$$\text{CP:SP} = 4:5$$

profit:

$$1$$

on CP:

$$4$$

Thus:

$$\frac{1}{4} \times 100 = \boxed{25\%}$$

17. FIND LOSS % FROM RATIO

If:

$$\text{CP:SP} = 5:4$$

Loss:

$$1$$

on CP:

$$5$$

Loss %:

$$\frac{1}{5} \times 100 = \boxed{20\%}$$

18. GENERAL RATIO RULE

If:

$$\text{CP:SP} = a:b$$

and $(b > a)$:

$$\boxed{\text{Profit}\%} = \frac{b-a}{a} \times 100$$

If $(b < a)$:

$$\boxed{\text{Loss}\%} = \frac{a-b}{a} \times 100$$

19. PROFIT AS % OF SP

Suppose CP = 80, SP = 100.

Profit = 20.

Profit as % of CP:

$$\frac{20}{80} \times 100 = 25\%$$

Profit as % of SP:

$$\left[\frac{20}{100} \times 100 = 20\% \right]$$

Different bases.

20. CONVERT PROFIT % ON CP TO PROFIT % ON SP

If profit is $(p\%)$ on CP:

Take:

$$\left[CP = 100 \right]$$

Then:

$$\left[SP = 100 + p \right]$$

$$\text{Profit} = (p).$$

Therefore profit as % of SP:

$$\left[\boxed{\frac{100p}{100+p}\%} \right]$$

21. EXAMPLE

Profit = 25% on CP.

Profit as % of SP:

$$\left[\frac{2500}{125} = \boxed{20\%} \right]$$

22. CONVERT LOSS % ON CP TO LOSS AS % OF SP

If loss = $(p\%)$ on CP:

Take:

$$\left[CP = 100 \right]$$

$$\left[SP = 100 - p \right]$$

$$\text{Loss} = (p).$$

Loss as % of SP:

$$\left[\boxed{\frac{100p}{100-p}\%} \right]$$

Example 20% loss on CP:

$$\left[\frac{2000}{80} = 25\% \right]$$

So loss is 25% of SP.

23. SELLING FOR ¹ x MORE

An article costs ¹ 6 0 0 .

If sold for ¹ 1 5 0 more:

$$\left[SP = 750 \right]$$

Profit:

$$\text{[150]}$$

Profit %:

$$\text{[150/600}\times\text{100 = }\boxed{\text{25\%}} \text{]}$$

24. SELLING FOR ¹ x LESS

$$\text{CP = }\text{^1 9 0 0 .}$$

Sold for ¹ 1 8 0 ~~0~~ ss.

$$\text{[SP=720]}$$

Loss %:

$$\text{[180/900}\times\text{100 = }\boxed{\text{20\%}} \text{]}$$

25. PROFIT AMOUNT GIVEN

$$\text{Profit = }\text{^1 2 4 0}\text{ at 20\%}.$$

Since:

$$\text{[20\%\text{\textit{ of CP}}=240 \text{]}$$

$$\text{[CP=240}\times\text{5]}$$

$$\text{[= }\boxed{\text{^1 1 2 0 0}} \text{]}$$

Then:

$$\text{[SP=1440]}$$

26. LOSS AMOUNT GIVEN

$$\text{Loss = }\text{^1 1 5 0}\text{ at 10\%}.$$

$$\text{[10\%\text{\textit{ of CP}}=150 \text{]}$$

$$\text{[C P = }\boxed{\text{^1 1 5 0 0}} \text{]}$$

SP:

$$\text{[1350]}$$

27. DIFFERENT SELLING PRICES, DIFFERENT PROFIT/LOSS

A common pattern:

If sold for ¹ 6 0 0 20% profit.

If sold for ¹ ?' 10% loss.

First find CP.

$$\text{[600=1.2CP \text{]}$$

$$\text{[CP=500]}$$

At 10% loss:

$$\text{[SP=500(0.9)]}$$

$$\text{[=\boxed{450}]}$$

28. FAST RATIO METHOD FOR SAME CP

At 20% profit:

$$\text{[SP}_1\text{=120\text{ parts}]}$$

At 10% loss:

$$\text{[SP}_2\text{=90\text{ parts}]}$$

Ratio:

$$\text{[SP}_1\text{:SP}_2\text{ = 120:90 = 4:3]}$$

If first SP = 600:

$$\text{[SP}_2\text{=600\times\frac{3}{4}]}$$

$$\text{[=\boxed{450}]}$$

Even faster.

29. DIFFERENCE BETWEEN TWO SELLING PRICES

An article is sold once at 20% profit and alternatively at 10% loss.

Difference in SP percentages:

$$\text{[120\%-90\%=30\%\text{ of CP}]}$$

If actual difference in selling prices is 300 :

$$\text{[30\%\text{ of CP}=300]}$$

$$\text{[CP=1000]}$$

$$\text{[\boxed{1100}]}$$

30. GENERAL SAME-ARTICLE DIFFERENCE SHORTCUT

If one selling price gives $(p\%)$ profit and another gives $(q\%)$ loss:

Difference between SPs:

$$\text{[\boxed{(p+q)\%\text{ of CP}}]}$$

So if difference = (D) :

$$\text{[\boxed{CP=\frac{100D}{p+q}}]}$$

31. EXAMPLE

At 25% profit and 15% loss, selling prices differ by ₹ 400 .

$$[p+q=40]$$

$$[CP=\frac{100(400)}{40}]$$

$$[=\boxed{1000}]$$

32. BOTH SELLING PRICES GIVE PROFIT

If one SP gives 20% profit and another gives 35% profit:

Difference:

$$[15\%\text{ of CP}]$$

If difference = ₹ 180 :

$$[0.15CP=180]$$

$$[CP=\boxed{1200}]$$

33. BOTH GIVE LOSSES

Loss 10% vs loss 25%.

SPs are:

$$[90\%CP]$$

and:

$$[75\%CP]$$

Difference:

$$[15\%\text{ of CP}]$$

Same principle.

34. GENERAL DIFFERENCE RULE

If two selling conditions correspond to CP multipliers (m_1, m_2) :

$$[\boxed{$$

$$SP_1-SP_2$$

$$= CP|m_1-m_2|]$$

This handles every case.

35. PROFIT ON MULTIPLE IDENTICAL ITEMS

Suppose 10 identical items each cost ₹ 100 .

Total CP:

$$\boxed{1000}$$

If total SP = 11200 :

Profit:

$$\boxed{200}$$

Profit %:

$$\boxed{200/1000 \times 100 = 20\%}$$

Always calculate overall profit/loss from **total CP** and **total SP**.

36. OVERALL PROFIT ON DIFFERENT ITEMS

Item A:

$$\boxed{CP = 1500, SP = 1600}$$

Item B:

$$\boxed{CP = 1100, SP = 1100}$$

Total CP:

$$\boxed{1500}$$

Total SP:

$$\boxed{1700}$$

Profit:

$$\boxed{200}$$

Overall profit %:

$$\boxed{\frac{200}{1500} \times 100 = 13\frac{1}{3}\%}$$

Do not average 20% and 10% directly because CPs differ.

37. WHEN CAN PROFIT PERCENTAGES BE AVERAGED?

If two items have **equal CP**, then their profit/loss percentages have equal weights.

Example:

Both CP = 1100 .

One profit 20%.

Other profit 10%.

Overall profit:

$$\boxed{\frac{20+10}{2} = 15\%}$$

So simple averaging works because CPs are equal.

38. EQUAL CP — ONE PROFIT, ONE LOSS

Two items each cost ₹ 100 .

One sold at 20% profit:

$$[SP = 120]$$

Other at 20% loss:

$$[SP = 80]$$

Total CP:

$$[200]$$

Total SP:

$$[200]$$

Overall:

$$[\boxed{\text{No profit, no loss}}]$$

Important

Equal % profit/loss **does cancel when CPs are equal**.

This is different from equal successive percentage changes on the same amount.

39. EQUAL SP — ONE PROFIT, ONE LOSS

This is a famous trap.

Two items are sold at the same SP.

One at $(p\%)$ profit, the other at $(p\%)$ loss.

Overall result is **not zero**.

It produces a loss.

40. DERIVATION FOR EQUAL SP

Suppose each SP = ₹ 100 .

At $(p\%)$ profit:

$$[CP_1 = \frac{10000}{100+p}]$$

At $(p\%)$ loss:

$$[CP_2 = \frac{10000}{100-p}]$$

Total CP exceeds total SP.

Net loss percentage simplifies to:

$$[\boxed{\frac{p^2}{100}\%}]$$

41. FAMOUS FORMULA

If two articles are sold at the **same selling price**, one at $(p\%)$ profit and the other at $(p\%)$ loss:

$$\boxed{\text{Overall loss}\% = \frac{p^2}{100}}$$

MUST MEMORIZE

42. EXAMPLE — EQUAL SP, $\pm 20\%$

Same SP.

One $+20\%$, one -20% .

Overall loss:

$$\frac{20^2}{100} = \boxed{4\%}$$

43. VERIFY NUMERICALLY

Take same SP:

$$₹ 120$$

First item at 20% profit:

$$CP_1 = 120 / 1.2 = 100$$

Second at 20% loss:

$$CP_2 = 120 / 0.8 = 150$$

Total CP:

$$₹ 250$$

Total SP:

$$₹ 240$$

Loss:

$$₹ 10$$

Loss %:

$$10 / 250 \times 100 = \boxed{4\%}$$

44. WHY EQUAL-SP QUESTION GIVES LOSS

The item sold at loss had a larger CP than the item sold at profit.

So the same percentage is applied to different bases.

45. EQUAL SP WITH DIFFERENT PROFIT/LOSS RATES

Suppose two items each sell for ₹ 120.

First at 20% profit:

$$CP_1 = 100$$

Second at 25% loss:

$$[CP_2 = \frac{120}{0.75} = 160]$$

Total CP:

$$[260]$$

Total SP:

$$[240]$$

Loss:

$$[20]$$

Loss %:

$$[\frac{20}{260} \times 100 = \boxed{7\frac{13}{13}\% \approx 7.69\%}]$$

For different rates, total method is safest.

46. OVERALL PROFIT/LOSS FORMULA

For many transactions:

$$[\boxed{\text{Overall P/L}\%} = \frac{\sum SP - \sum CP}{\sum CP} \times 100]$$

This is the master formula.

47. GATE-STYLE QUESTION 1

An article bought for ₹ 800 is sold at 15% profit.

$$[SP = 800(1.15)]$$

$$[= \boxed{920}]$$

48. GATE-STYLE QUESTION 2

An article sold for ₹ 720 incurs 10% loss.

$$[720 = 0.9CP]$$

$$[CP = \boxed{800}]$$

49. GATE-STYLE QUESTION 3

CP:SP = 8:10.

Profit %:

$$[\frac{2}{8} \times 100 = \boxed{25\%}]$$

50. GATE-STYLE QUESTION 4

An article gives 20% profit at one SP and 10% loss at another. Difference between selling prices is ₹ 150 .

Difference:

$$\left[30\% \text{ of CP} \right]$$

$$\left[0.3CP=150 \right]$$

$$\left[CP = \boxed{500} \right]$$

51. PLACEMENT-STYLE QUESTION 1

A shopkeeper buys an item for ₹ 1500 and sells it for ₹ 1800.

Profit:

$$\left[300 \right]$$

Profit %:

$$\left[300/1500 \times 100 = \boxed{20\%} \right]$$

52. PLACEMENT-STYLE QUESTION 2

An article is sold at 25% loss for ₹ 900.

$$\left[CP=900/0.75 \right]$$

$$\left[= \boxed{1200} \right]$$

53. PLACEMENT-STYLE QUESTION 3

Two articles have equal CP. One is sold at 30% profit and the other at 10% loss.

Overall:

$$\left[\frac{30-10}{2} = \boxed{10\% \text{ profit}} \right]$$

Because equal CP gives equal weighting.

54. PLACEMENT-STYLE QUESTION 4

Two articles are sold for the same price. One at 25% profit and one at 25% loss.

Overall loss:

$$\left[\frac{25^2}{100} \right]$$

$$\left[= \boxed{6.25\%} \right]$$

55. ADVANCED QUESTION — FIND CP FROM PROFIT AMOUNT

Selling at 30% profit gives ₹ 360 profit.

$$\left[30\% \text{ of CP} = 360 \right]$$

$$\left[CP = 360 \times \frac{10}{3} \right]$$

$$\boxed{1200}$$

SP:

$$\boxed{1560}$$

56. ADVANCED QUESTION — FIND PROFIT RATE

An article's SP is $\frac{5}{4}$ of CP.

$$\boxed{SP=1.25CP}$$

Therefore:

$$\boxed{25\% \text{ profit}}$$

57. ADVANCED QUESTION — SP AS FRACTION OF CP

If:

$$\boxed{SP=\frac{7}{8}CP}$$

Loss fraction:

$$\boxed{1-\frac{7}{8}=\frac{1}{8}}$$

Loss %:

$$\boxed{12.5\%}$$

58. ADVANCED QUESTION — CP AS FRACTION OF SP

Suppose:

$$\boxed{CP=\frac{4}{5}SP}$$

Then:

$$\boxed{SP=\frac{5}{4}CP}$$

So:

$$\boxed{25\% \text{ profit}}$$

Be careful which quantity is expressed relative to which.

59. COMMON TRAPS

& pTRAP 1 — Profit percentage on SP

Standard profit % uses CP.

& pTRAP 2 — Equal 20% profit and 20% loss always cancel

Only if the CP bases are equal.

& pTRAP 3 — Equal SP case

Equal SP with equal profit/loss rates gives a net loss.

& pTRAP 4 — Average percentages

For different CPs, calculate total profit and total CP.

& pTRAP 5 — Reverse profit

If SP includes 20% profit:

$$\boxed{CP = SP / 1.2}$$

Do not subtract 20% of SP.

60. MUST-MEMORIZE FORMULAS

Profit

$$\boxed{P = SP - CP}$$

Loss

$$\boxed{L = CP - SP}$$

Profit %

$$\boxed{\frac{P}{CP} \times 100}$$

Loss %

$$\boxed{\frac{L}{CP} \times 100}$$

SP at $(p\%)$ profit

$$\boxed{SP = CP \frac{100+p}{100}}$$

SP at $(p\%)$ loss

$$\boxed{SP = CP \frac{100-p}{100}}$$

61. HIGH-VALUE SHORTCUTS

Profit $(p\%)$

$$\boxed{CP : SP = 100 : (100 + p)}$$

Loss $(p\%)$

$$\boxed{CP : SP = 100 : (100 - p)}$$

Same article, $(p\%)$ profit vs $(q\%)$ loss

SP difference:

$$\boxed{(p+q)\% \text{ of } CP}$$

Equal SP, equal $(p\%)$ profit/loss

$\boxed{\text{Overall loss} = \frac{p^2}{100}\%}$

62. FAST DECISION TREE

CP and SP given?

Find difference & divide by CP.

CP and percentage given?

Use multiplier.

SP and percentage given?

Reverse-divide by multiplier.

Same article under two selling conditions?

Compare CP multipliers.

Multiple articles?

Add all CPs and SPs.

Equal CP?

Percentage averaging may work.

Equal SP?

Do not average percentages; use CP calculation or special formula.

63. 5-MINUTE REVISION

Remember:

$\boxed{\text{Profit/Loss \% base} = \text{CP}}$

and:

$\boxed{\text{SP} = \text{CP} \times \text{profit/loss multiplier}}$

For example:

$25\% \text{ profit} \rightarrow \text{SP} = 1.25\text{CP}$

$20\% \text{ loss} \rightarrow \text{SP} = 0.8\text{CP}$

PRACTICE — LEVEL 0

Q1

CP = ₹ 500, SP = ₹ 600. Find profit %.

Q2

CP = ₹ 800, SP = ₹ 680. Find loss %.

Q3

CP = ₹ 1200, profit = 20%. Find SP.

Q4

SP = ₹ 900 at 25% loss. Find CP.

Q5

CP:SP = 5:6. Find profit %.

LEVEL 1

Q6

An article sold for ₹ 1400 gives 20% profit. Find CP.

Q7

An article costing ₹ 200 is sold at 15% loss. Find SP.

Q8

At 30% profit, SP is ₹ 1300. Find profit amount.

Q9

At 20% loss, SP is ₹ 800. Find loss amount.

Q10

An article can be sold at 25% profit or 15% loss. If the two SPs differ by ₹ 400, find CP.

LEVEL 2

Q11

Two items have equal CP. One gives 30% profit and other gives 20% loss. Find overall profit/loss %.

Q12

Two articles are sold at the same SP, one at 20% profit and one at 20% loss. Find overall result.

Q13

A trader makes 25% profit. Profit is what percent of SP?

Q14

An article is sold at 20% loss. Loss is what percent of SP?

Q15

CP:SP = 8:7. Find loss %.

LEVEL 3 — EXAM PRESSURE

Q16

One article costs ₹ 500 and gives 20% profit. Another costs ₹ 1000 and gives 10% loss. Find overall profit/loss %.

Q17

Two articles are sold at equal SP. One at 25% profit, one at 20% loss. Find overall profit/loss %.

Q18

An article is sold for ₹ 690 at 15% profit. At what price should it be sold for 10% loss?

Q19

At one selling price an article gives 40% profit; at another it gives 20% profit. The SP difference is ₹ 240. Find CP.

Q20

An article is sold at $\frac{3}{4}$ of CP. Find loss %.

ANSWERS**Q1**

Profit:

$$\frac{100}{500}$$

$$\frac{100}{500} \times 100 = \boxed{20\%}$$

Q2

Loss:

$$\frac{120}{800}$$

$$\frac{120}{800} \times 100 = \boxed{15\%}$$

Q3

$$1200(1.2) = \boxed{1440}$$

Q4

$$CP = 900 / 0.75 = \boxed{1200}$$

Q5

Profit:

$$\frac{1}{5}$$

on CP 5:

$$\boxed{20\%}$$

Q6

$$CP = 1440 / 1.2 = \boxed{1200}$$

Q7

$$SP = 2000(0.85) = \boxed{1700}$$

Q8

$$\text{CP} = 1300 / 1.3 = 1000$$

Profit:

$$\boxed{1300}$$

Q9

$$\text{CP} = 800 / 0.8 = 1000$$

Loss:

$$\boxed{1200}$$

Q10

Difference:

$$25 + 15 = 40\%$$

$$40\% \text{ of CP} = 400$$

$$\boxed{\text{CP} = 1000}$$

Q11

Equal CP !' average signed percentages:

$$\frac{30 - 20}{2} = \boxed{5\% \text{ profit}}$$

Q12

Equal SP and equal $\pm 20\%$:

$$\frac{20^2}{100} = \boxed{4\% \text{ loss}}$$

Q13

Take CP = 100.

SP = 125.

Profit = 25.

As % of SP:

$$25 / 125 \times 100 = \boxed{20\%}$$

Q14

Take CP = 100.

SP = 80.

Loss = 20.

As % of SP:

$$20 / 80 \times 100 = \boxed{25\%}$$

Q15

Loss:

$$1$$

on CP 8:

$$\left[\frac{18}{100} \times 100 = \boxed{12.5\%} \right]$$

Q16

Item 1:

$$\left[CP=500, SP=600 \right]$$

Item 2:

$$\left[CP=1000, SP=900 \right]$$

Total CP:

$$\left[1500 \right]$$

Total SP:

$$\left[1500 \right]$$

$$\left[\boxed{\text{No profit, no loss}} \right]$$

Q17

Assume common SP = ₹ 100 .

First, 25% profit:

$$\left[CP_1 = 100 / 1.25 = 80 \right]$$

Second, 20% loss:

$$\left[CP_2 = 100 / 0.8 = 125 \right]$$

Total CP:

$$\left[205 \right]$$

Total SP:

$$\left[200 \right]$$

Loss:

$$\left[5 \right]$$

Loss %:

$$\left[\frac{5}{205} \times 100 = \boxed{\frac{100}{41} \% \approx 2.44 \% \text{ loss}} \right]$$

Q18

At 15% profit:

$$\left[690 = 1.15CP \right]$$

$$\left[CP = 600 \right]$$

At 10% loss:

$$\left[SP = 600(0.9) \right]$$

$$\left[= \boxed{540} \right]$$

Q19

SP difference:

$$\left[40\% - 20\% = 20\% \text{ of CP} \right]$$

$$\boxed{0.2CP=240}$$

$$\boxed{CP = 1200}$$

Q20

$$SP = \frac{3}{4}CP$$

Loss:

$$\frac{1}{4}CP$$

Loss %:

$$\boxed{25\%}$$

CHAPTER 18 STATUS

Covered so far:

- CP, SP, profit, loss '
- profit/loss percentages '
- multiplier method '
- reverse CP/SP calculations '
- CP:SP ratios '
- percentage-base traps '
- two selling-price comparisons '
- multiple-item overall profit/loss '
- equal-CP questions '
- equal-SP questions '
- special $(p^2/100)$ loss formula '
- GATE/placement practice '

But **Chapter 18 is not complete yet.**

Next we need the more exam-oriented section:

- dishonest dealer / false weights
- buying and selling quantity tricks
- "x articles for price of y" problems
- successive profit/loss transactions
- overall profit across unequal quantities
- marked cost vs actual cost-style traps
- profit based on SP vs CP conversions
- advanced GATE/placement mixed drills

Progress

Completed: 17/76 **Current:** Chapter 18 — Profit & Loss **Next:** Chapter 18 Part 2 — False weights, quantity tricks & advanced profit/loss problems

USER

next

ASSISTANT

CHAPTER 18 — PROFIT & LOSS

PART 2 — FALSE WEIGHTS, QUANTITY TRICKS, DISHONEST DEALERS & ADVANCED TRANSACTIONS

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: Very High Priority: $\emptyset$ = Not in 1

This part covers the trickier forms of Profit & Loss where the profit is hidden inside:

- incorrect weights,
- “x items for the price of y” statements,
- buying/selling quantity ratios,
- multiple transactions,
- different profit bases,
- chain transactions.

The main rule still remains:

$$\boxed{\text{Overall Profit/Loss} = \text{Total SP} - \text{Total CP}}$$

and standard percentage is always calculated on:

$$\boxed{\text{Total CP}}$$

1. FALSE WEIGHT — BASIC IDEA

Suppose a shopkeeper charges the price of:

$$1 \text{ kg}$$

but actually gives only:

$$900 \text{ g}$$

Even if he claims to sell at cost price per kg, he earns profit because he gives less quantity.

This is called a **false-weight** or **short-weight** problem.

2. SIMPLE FALSE-WEIGHT EXAMPLE

Actual cost:

$$100 \text{ kg}$$

Dealer charges:

$$100$$

but gives only:

$\text{\text{g}}$

Cost of 800 g:

$$100 \times \frac{800}{1000} = \text{\text{₹}} 80$$

Selling price collected:

$$\text{\text{₹}} 110$$

Profit:

$$20$$

Profit %:

$$\frac{20}{80} \times 100 = \boxed{25\%}$$

Trap

Do not calculate:

$$20/100 = 20\%$$

because the dealer's actual CP is $\text{\text{₹}} 80$.

3. FALSE-WEIGHT MASTER FORMULA

Suppose dealer charges for true weight (W) but actually gives (w) .

If selling at the normal cost rate:

Actual CP corresponds to:

$$w$$

Collected SP corresponds to:

$$W$$

Profit %:

$$\boxed{\frac{W-w}{w} \times 100}$$

4. EXAMPLE — 900 g AS 1 kg

$$W=1000, \text{quad } w=900$$

Profit %:

$$\frac{100}{900} \times 100$$

$$= \boxed{11\frac{1}{9}\%}$$

5. EXAMPLE — 800 g AS 1 kg

$$\frac{1000-800}{800} \times 100$$

$$= \frac{200}{800} \times 100$$

$$= \boxed{25\%}$$

6. SHORT WEIGHT BY $(p\%)$

Suppose dealer gives $(p\%)$ less quantity than promised.

Then he gives:

$$[100 - p]$$

units instead of 100.

If he charges the cost-price amount for 100 units:

$$[\text{Profit}\% = \frac{100p}{100 - p}]$$

Example:

20% short weight:

$$[\frac{100(20)}{80} = \boxed{25\%}]$$

7. FALSE WEIGHT + MARKUP/PROFIT

Suppose dealer:

- charges 20% above cost price per kg,
- and gives only 800 g as 1 kg.

Let true cost:

$$[100 \text{ / kg}]$$

He charges:

$$[120]$$

But cost of 800 g:

$$[80]$$

Profit:

$$[120 - 80 = 40]$$

Profit %:

$$[40/80 \times 100 = \boxed{50\%}]$$

8. GENERAL FALSE-WEIGHT + PRICE MULTIPLIER METHOD

Suppose:

- price multiplier relative to true cost rate = (m)
- actual quantity delivered as fraction of promised quantity = (q)

Then:

$$[\frac{SP}{CP} = \frac{m}{q}]$$

Therefore:

$$[\text{Profit}\% = \left(\frac{m}{q} - 1\right)100]$$

Example:

20% markup:

$$[m=1.2]$$

20% short weight:

$$[q=0.8]$$

Then:

$$[SP/CP=\frac{1.2}{0.8}=1.5]$$

Profit:

$$[\boxed{50\%}]$$

9. “SELLS AT COST PRICE BUT USES FALSE WEIGHT”

This means:

Price rate is not giving profit.

Only quantity cheating creates profit.

So if 10% short weight:

$$[q=0.9]$$

Profit factor:

$$[1/0.9 = \frac{10}{9}]$$

Profit:

$$[\boxed{11\frac{1}{9}\%}]$$

10. BUYING WITH FALSE WEIGHT

Dishonesty can happen while buying too.

Suppose a trader claims 1 kg but actually receives:

$$[1100\text{ g}]$$

for the price of 1 kg.

Then his effective cost per 1.1 kg is the cost of 1 kg.

This lowers his true unit cost.

11. EXAMPLE — EXTRA QUANTITY WHILE BUYING

He pays ₹ 100 for “1 kg” but receives 1.2 kg.

True effective cost per kg:

$$[100/1.2 = ₹ 83\frac{1}{3}]$$

If he later sells at ₹ 100/kg correctly:

Profit per kg:

$$\left[16\frac{23}{83} \right]$$

Profit % on true cost:

$$\left[\frac{16\frac{23}{83}}{83\frac{13}{83}} \times 100 = \boxed{20\%} \right]$$

Faster:

$$\left[SP/CP = 100/(100/1.2) = 1.2 \right]$$

So:

$$\left[\boxed{20\%} \right]$$

12. BOTH BUYING AND SELLING DISHONESTLY

Suppose a trader:

- gets 20% extra quantity while buying,
- gives 20% less quantity while selling,
- uses same nominal price per kg.

Take ₹ 100 as nominal price.

While buying, ₹ 100 gets:

$$\left[1.2 \text{ kg} \right]$$

Effective CP per kg:

$$\left[100/1.2 \right]$$

When selling, he charges ₹ 100 but gives:

$$\left[0.8 \text{ kg} \right]$$

Cost of that 0.8 kg:

$$\left[\frac{100}{1.2} \times 0.8 \right]$$

$$\left[=66\frac{23}{83} \right]$$

SP:

$$\left[100 \right]$$

Profit factor:

$$\left[100/(66\frac{23}{83}) = 1.5 \right]$$

So profit:

$$\left[\boxed{50\%} \right]$$

13. FAST FORMULA — EXTRA WHILE BUYING + SHORT WHILE SELLING

If:

- buying quantity multiplier = $(1+a)$
- selling quantity multiplier = $(1-b)$

with (a,b) written as decimals,

then if nominal rate is same:

$$\boxed{\frac{SP}{CP} = \frac{1+a}{1-b}}$$

Example:

$$a=0.2, \quad b=0.2$$

$$\frac{1.2}{0.8}=1.5$$

Profit:

$$\boxed{50\%}$$

14. “x ARTICLES FOR THE PRICE OF y”

These problems are extremely common.

Suppose a dealer buys:

$$x \text{ articles for } A$$

and sells:

$$y \text{ articles for } A$$

Then:

$$CP \text{ per article} = \frac{A}{x}$$

$$SP \text{ per article} = \frac{A}{y}$$

Therefore:

$$\frac{SP}{CP} = \frac{A/y}{A/x} = \frac{x}{y}$$

15. MASTER FORMULA

If x articles are bought for the same amount for which y articles are sold:

$$\boxed{\text{Profit/Loss \%} = \frac{x-y}{y} \times 100}$$

if $x > y$, profit.

Why denominator y ?

Because:

$$SP/CP = x/y$$

and profit fraction is:

$$x/y - 1 = (x-y)/y$$

16. EXAMPLE

Buy 10 articles for ₹ 100.

Sell 8 articles for ₹ 100.

Then:

$$\boxed{SP/CP=10/8=1.25}$$

Profit:

$$\boxed{25\%}$$

17. ANOTHER EXAMPLE

Buy 8 articles for ₹ 100 .

Sell 10 articles for ₹ 100 .

$$\boxed{SP/CP=8/10=0.8}$$

Loss:

$$\boxed{20\%}$$

18. LANGUAGE VERSION

“Cost price of 12 articles equals selling price of 10 articles.”

Then:

$$CP_{12}=SP_{10}$$

Let CP per article = c , SP per article = s .

$$12c=10s$$

$$s/c=12/10=1.2$$

Profit:

$$\boxed{20\%}$$

19. REVERSE LANGUAGE TRAP

“Selling price of 12 articles equals cost price of 10 articles.”

Then:

$$12s=10c$$

$$s/c=10/12=5/6$$

Loss:

$$1-\frac{5}{6}=\frac{1}{6}$$

$$\boxed{16\frac{2}{3}\%}$$

Read the sentence carefully.

20. “SELLS x ARTICLES AT COST OF y ARTICLES”

Suppose:

SP of 8 articles = CP of 10 articles.

Then:

$$8s = 10c$$

$$s/c = 10/8 = 1.25$$

Profit:

$$\boxed{25\%}$$

21. “BUYS x FOR PRICE OF y”

Suppose a trader receives 12 articles but pays for only 10.

Then his cost per effective article decreases.

If he sells all 12 at normal nominal CP per article, his revenue corresponds to 12 nominal units while he paid only 10.

Profit:

$$\frac{12-10}{10} \times 100 = \boxed{20\%}$$

Note denominator is amount actually paid.

22. FREE-ITEM / BONUS PURCHASES

Suppose:

Buy 10, get 2 free.

You receive:

$$12$$

but pay for:

$$10$$

Effective discount on unit cost:

$$1 - \frac{10}{12} = \frac{2}{12} = 16\frac{2}{3}\%$$

So effective cost per item falls by:

$$\boxed{16\frac{2}{3}\%}$$

23. SELL “BUY x GET y FREE” — SELLER VIEW

If a seller offers:

Buy 4, get 1 free

Customer receives 5 items but pays for 4.

If listed per-item selling price is S , effective revenue per item becomes:

$$\frac{4S}{5}$$

So effective selling rate is:

$$80\%$$

of the nominal price.

This is equivalent to a:

$$\boxed{20\% \text{ effective discount}}$$

on the per-item listed price.

24. GENERAL FREE-ITEM DISCOUNT

“Buy x , get y free.”

Customer gets:

$$x+y$$

but pays for:

$$x$$

Effective discount:

$$\boxed{\frac{y}{x+y} \times 100}$$

Example:

Buy 3 get 1 free:

$$\frac{1}{4} = 25\%$$

25. PROFIT WHEN SELLING IN BUNDLES

Suppose CP per item = 80.

Seller offers:

4 items for 400.

Total CP:

$$4(80) = 320$$

SP:

$$400$$

Profit:

$$80$$

Profit %:

$$80/320 \times 100 = \boxed{25\%}$$

Always compare bundle totals.

26. OVERALL PROFIT WITH UNEQUAL QUANTITIES

Suppose:

- 20 items sold at 10% profit
- 30 identical-cost items sold at 20% profit

If all items have same CP per item, profit percentages are weighted by quantities.

Overall profit:

$$\left[\frac{20(10)+30(20)}{50} \right]$$

$$\left[= \frac{800}{50} = \boxed{16\%} \right]$$

Because each item's CP weight is equal.

27. WHEN QUANTITY-WEIGHTED PERCENTAGE FAILS

If items have different CP per item, simply weighting by quantity can fail.

Example:

- 10 items CP = 1 @ each, profit 20%
- 10 items CP = 10 @ each, profit 10%

Equal quantities, but CP bases differ greatly.

Overall profit must be based on:

$$\left[\boxed{\text{total CP}} \right]$$

not quantity alone.

28. OVERALL P/L USING WEIGHTED CP

General:

If groups have total cost prices $(C_1, C_2, \dots)$ and signed profit/loss percentages $(p_1, p_2, \dots)$,

overall signed percentage:

$$\left[\boxed{\frac{C_1 p_1 + C_2 p_2 + \dots}{C_1 + C_2 + \dots}} \right]$$

Here losses are negative.

This is a weighted average with CP as the weight.

29. EXAMPLE

Item group A:

$$\left[CP = 12000, +20\% \right]$$

Item group B:

$$\left[CP = 13000, +10\% \right]$$

Net profit amount:

$$[400 - 300 = 100]$$

Total CP:

$$[5000]$$

Overall profit:

$$[100 / 5000 \times 100 = \boxed{2\%}]$$

Weighted signed percentage gives same result:

$$[\frac{2000(20) + 3000(-10)}{5000} = 2\%]$$

30. SUCCESSIVE TRANSACTIONS — MIDDLEMAN CHAIN

Suppose A sells an item to B at 20% profit.

Then B sells to C at 25% profit.

If A's CP = ₹ 100 :

A's B :

$$[100 \text{ to } 120]$$

B's C :

$$[120 \text{ to } 150]$$

Final price relative to A's CP:

$$[1.2 \times 1.25 = 1.5]$$

So final price is:

$$[\boxed{50\% \text{ above A's original CP}}]$$

31. IMPORTANT CHAIN RULE

If the same article passes through multiple sellers:

$$[\boxed{\text{Final price} = \text{Initial cost} \times m_1 m_2 m_3 \dots}]$$

where each (m) is a profit/loss multiplier.

32. PROFIT FOLLOWED BY LOSS IN A CHAIN

A sells to B at 25% profit.

B sells to C at 20% loss.

Final multiplier:

$$[1.25 \times 0.8 = 1]$$

Thus C's buying price equals A's original CP.

$$[\boxed{\text{No net change relative to A's CP}}]$$

33. CHAIN WITH SAME 20% PROFIT TWICE

$$\sqrt[1]{1.2^2=1.44}$$

So final price is:

$$\sqrt[1]{\boxed{44\% \text{ above original}}}$$

Not 40%.

34. CHAIN WITH 20% PROFIT THEN 20% LOSS

$$\sqrt[1]{1.2 \times 0.8 = 0.96}$$

Final:

$$\sqrt[1]{\boxed{4\% \text{ below original}}}$$

Same percentage-multiplier logic from Chapter 15.

35. FIND ORIGINAL CP FROM CHAIN

B sells to C for ₹ 1800 after:

- A sells to B at 20% profit,
- B sells to C at 25% profit.

Overall multiplier:

$$\sqrt[1]{1.2 \times 1.25 = 1.5}$$

So A's CP:

$$\sqrt[1]{1800/1.5 = \boxed{1200}}$$

36. COMMISSION / EXTRA COST ADDED TO CP

Sometimes purchase cost is not the true total CP.

Example:

Purchase price:

$$\sqrt[1]{1000}$$

Transport:

$$\sqrt[1]{100}$$

Repair:

$$\sqrt[1]{200}$$

True CP:

$$\sqrt[1]{1000+100+200 = \boxed{1300}}$$

If sold at ₹ 1560 :

Profit:

$$\boxed{260}$$

Profit %:

$$\boxed{260/1300 \times 100 = 20\%}$$

37. TRUE COST PRICE INCLUDES EXPENSES

Unless question specifically defines otherwise:

$$\boxed{\text{Effective CP} = \text{purchase price} + \text{necessary acquisition/selling preparation expenses}}$$

Examples may include:

- transport,
- repairs before sale,
- loading,
- packaging,
- taxes/fees stated as seller's cost.

Read wording carefully.

38. PROFIT BASED ON SELLING PRICE

Suppose profit is 20% of SP.

This is **not** 20% profit on CP.

Let:

$$\boxed{SP=100}$$

Profit:

$$\boxed{20}$$

Then:

$$\boxed{CP=80}$$

Profit % on CP:

$$\boxed{20/80 \times 100 = 25\%}$$

39. GENERAL — PROFIT $(p\%)$ OF SP

Take:

$$\boxed{SP=100}$$

Profit:

$$\boxed{p}$$

Then:

$$\boxed{CP=100-p}$$

Profit % on CP:

$$\boxed{\frac{100p}{100-p}\%}$$

Example:

20% of SP !' 25% on CP.

40. LOSS AS % OF SP

Suppose loss is 20% of SP.

Take:

$$[SP=100]$$

Loss:

$$[20]$$

Then:

$$[CP=120]$$

Loss % on CP:

$$[20/120 \times 100 = \boxed{16\frac{2}{3}\%}]$$

41. GENERAL — LOSS $(p\%)$ OF SP

$$[SP=100, \text{quad } L=p]$$

$$[CP=100+p]$$

Standard loss %:

$$[\boxed{\frac{100p}{100+p}\%}]$$

42. PROFIT AMOUNT EQUALS FRACTION OF SP

Suppose profit = $(1/5)$ of SP.

Let:

$$[SP=5]$$

Profit:

$$[1]$$

CP:

$$[4]$$

Profit % on CP:

$$[1/4 \times 100 = \boxed{25\%}]$$

43. PROFIT AMOUNT EQUALS FRACTION OF CP

Suppose profit = $\frac{1}{5}$ of CP.

Then directly:

$\boxed{20\% \text{ profit}}$

Always identify the base.

44. SELLING PRICE INCREASE NEEDED TO CHANGE PROFIT RATE

An item currently gives 20% profit.

What percent should SP increase to make profit 50%?

Take:

$\boxed{CP=100}$

Old SP:

$\boxed{120}$

Required SP:

$\boxed{150}$

Increase in SP:

$\boxed{30}$

As % of old SP:

$\boxed{30/120 \times 100 = 25\%}$

45. SELLING PRICE DECREASE TO CHANGE PROFIT TO LOSS

Currently 25% profit.

What percentage decrease in SP makes it 10% loss?

Take CP = 100.

Old SP:

$\boxed{125}$

New SP:

$\boxed{90}$

Decrease:

$\boxed{35}$

Relative to old SP:

$\boxed{35/125 \times 100 = 28\%}$

46. GENERAL CHANGE-IN-SP METHOD

Always:

1. Assume $(CP=100)$.
2. Convert old condition to old SP.
3. Convert target condition to new SP.
4. Find percentage change relative to **old SP**.

This avoids formula memorization.

47. PROFIT WHEN CP CHANGES BUT SP FIXED

Suppose CP falls by 20% while SP stays unchanged.

Original:

$$[CP=100]$$

Suppose initially no profit:

$$[SP=100]$$

New CP:

$$[80]$$

Profit:

$$[20]$$

Profit %:

$$[20/80 \times 100 = \boxed{25\%}]$$

This is the familiar reverse-percentage effect.

48. IF CP INCREASES AND SP FIXED

Original:

$$[CP=100, SP=100]$$

CP increases 25%:

$$[CP=125]$$

Loss:

$$[25]$$

Loss %:

$$[25/125 \times 100 = \boxed{20\%}]$$

49. GATE-STYLE QUESTION 1

A dealer charges the cost price of 1 kg but gives only 800 g.

Profit %:

$$[\frac{200}{800} \times 100 = \boxed{25\%}]$$

50. GATE-STYLE QUESTION 2

CP of 15 articles equals SP of 12 articles.

$$[15c=12s]$$

$$[s/c=15/12=1.25]$$

Profit:

$$[\boxed{25\%}]$$

51. GATE-STYLE QUESTION 3

A sells to B at 20% profit. B sells to C at 10% profit.

Final multiplier:

$$[1.2 \times 1.1 = 1.32]$$

Thus C pays:

$$[\boxed{32\% \text{ above A's original CP}}]$$

52. GATE-STYLE QUESTION 4

Profit is 25% of SP. Find standard profit % on CP.

Let:

$$[SP=100]$$

Profit:

$$[25]$$

CP:

$$[75]$$

$$[25/75 \times 100 = \boxed{33\frac{1}{3}\%}]$$

53. PLACEMENT-STYLE QUESTION 1

A trader gives 900 g instead of 1 kg while charging the normal 1 kg cost-price amount.

Profit:

$$[\frac{100}{900} \times 100 = \boxed{11\frac{1}{9}\%}]$$

54. PLACEMENT-STYLE QUESTION 2

A shopkeeper marks his selling rate 25% above cost and gives 20% less quantity.

Price multiplier:

$$[1.25]$$

Quantity delivered:

$$\sqrt[0.8]{}$$

Effective SP/CP:

$$\sqrt[1.25/0.8 = 1.5625]{}$$

Profit:

$$\sqrt{\boxed{56.25\%}}$$

55. PLACEMENT-STYLE QUESTION 3

A trader buys 12 articles for the price of 10 and sells each at the nominal original per-item price.

He paid for 10 but receives 12.

Revenue relative to cost:

$$\sqrt[12:10]{}$$

Profit:

$$\sqrt{\boxed{20\%}}$$

56. PLACEMENT-STYLE QUESTION 4

Buy 4 get 1 free.

Effective discount:

$$\sqrt[\frac{1}{5} \times 100 = \boxed{20\%}]{}$$

57. ADVANCED QUESTION — FALSE WEIGHT + LOSS CLAIM

A shopkeeper claims to sell at 10% loss, but gives only 800 g as 1 kg.

Take true CP of 1 kg:

$$\sqrt[100]{}$$

He charges 10% loss price:

$$\sqrt[90]{}$$

But cost of 800 g:

$$\sqrt[80]{}$$

Profit:

$$\sqrt[10]{}$$

Profit %:

$$\sqrt[10/80 \times 100 = \boxed{12.5\%}]{}$$

Even though he claims a loss, he actually profits.

58. ADVANCED QUESTION — BUY EXTRA + SELL SHORT

Trader gets 10% extra while buying and gives 10% less while selling at same nominal rate.

Multiplier:

$$\left[\frac{1.1}{0.9} = \frac{11}{9} \right]$$

Profit:

$$\left[\frac{2}{9} \times 100 = \boxed{22\frac{2}{9}\%} \right]$$

59. ADVANCED QUESTION — MULTIPLE GROUPS

Goods costing:

- 1200 sold at 20% profit
- 1300 sold at 10% loss
- 1500 sold at 15% profit

Profit/loss amounts:

$$\left[400 - 300 + 750 = 850 \right]$$

Total CP:

$$\left[10000 \right]$$

Overall profit:

$$\left[850/10000 \times 100 = \boxed{8.5\%} \right]$$

60. ADVANCED QUESTION — CHAIN REVERSE

A sells to B at 25% profit.

B sells to C at 20% loss.

C pays 11000.

Overall multiplier:

$$\left[1.25 \times 0.8 = 1 \right]$$

Therefore A's original CP:

$$\left[\boxed{11000} \right]$$

61. COMMON TRAPS

& pTRAP 1 — False weight denominator

Use actual cost of quantity delivered.

& pTRAP 2 — “CP of x = SP of y”

Write the equation first. Do not guess the percentage.

& pTRAP 3 — Buy x get y free

Effective discount is:

$$\left[\frac{y}{x+y} \right]$$

not:

$$\left[\frac{yx}{x+y} \right]$$

& pTRAP 4 — Chain transactions

Multiply profit/loss multipliers. Do not simply add percentages.

& pTRAP 5 — Profit as % of SP

Convert to standard CP-based profit before comparing.

& pTRAP 6 — Purchase price vs true CP

Include stated necessary expenses in effective cost.

62. MUST-MEMORIZE ADVANCED FORMULAS

Short weight

If gives w instead of W :

$$\boxed{\text{Profit}\% = \frac{W-w}{w} \times 100}$$

when charging cost-rate price for W .

$p\%$ short weight

$$\boxed{\frac{100p}{100-p}\%}$$

Buy x for same amount as sell y

$$\boxed{SP/CP = x/y}$$

63. MORE MUST-MEMORIZE RULES

Buy x , get y free

Effective cost reduction:

$$\boxed{\frac{y}{x+y} \times 100}$$

Chain transactions

$$\boxed{F = \prod m_i}$$

Profit $p\%$ of SP !' profit % on CP

$$\boxed{\frac{100p}{100-p}\%}$$

Loss $p\%$ of SP !' loss % on CP

$$\boxed{\frac{100p}{100+p}\%}$$

64. FAST EXAM DECISION TREE

False weight?

Compare money collected with cost of actual quantity delivered.

“x articles cost = y articles sell”?

Write $(xc=ys)$.

Multiple transactions?

Multiply multipliers.

Multiple groups of goods?

Add total CP and profit/loss amounts.

Profit/loss stated as % of SP?

Convert base.

Extra purchase expenses?

Add them to CP first.

PRACTICE — LEVEL 0

Q1

A trader gives 800 g instead of 1 kg while charging the normal cost-rate price. Find profit %.

Q2

CP of 12 articles equals SP of 10. Find profit %.

Q3

SP of 12 articles equals CP of 10. Find loss %.

Q4

Buy 4 get 1 free. Find effective discount.

Q5

A sells to B at 10% profit and B sells to C at 20% profit. Find final increase over original CP.

LEVEL 1

Q6

A trader gives 900 g instead of 1 kg and charges 10% above the true cost rate. Find profit %.

Q7

A trader receives 20% extra quantity while buying and later sells at the same nominal per-kg rate correctly. Find profit %.

Q8

Cost price of 15 articles equals selling price of 12 articles. Find profit %.

Q9

Two groups have total CP ₹ 4000 and ₹ 6000 and are sold at +20% and -10% respectively. Find overall result.

Q10

Profit is 20% of SP. Find profit % on CP.

LEVEL 2**Q11**

A dealer claims 20% loss but gives only 75% of promised quantity. Find actual profit/loss %.

Q12

A trader gets 25% extra while buying and gives 20% less while selling at the same nominal rate. Find profit %.

Q13

A sells to B at 20% profit, B to C at 25% profit, and C to D at 20% loss. Find D's price relative to A's CP.

Q14

A product has purchase price ₹ 2000 and transport/repair cost ₹ 500. It sells for ₹ 3000. Find profit %.

Q15

Loss is 25% of SP. Find standard loss % on CP.

LEVEL 3 — EXAM PRESSURE**Q16**

A shopkeeper charges 25% above cost rate but gives 900 g as 1 kg. Find actual profit %.

Q17

A trader buys 11 articles for the price of 10 and sells 10 articles for the price of 11 at the same nominal unit rate. Find profit %.

Q18

Goods costing ₹ 500 are split equally by cost into two parts. One part is sold at 20% profit and the other at 10% loss. Find overall profit %.

Q19

A sells an item to B at 30% profit. B sells it at 10% loss for ₹ 117. Find A's original CP.

Q20

An article currently sells at 20% profit. By what percent must its SP increase to produce 50% profit?

ANSWERS

Q1

$$\left[\frac{200}{800} \times 100 = \boxed{25\%} \right]$$

Q2

$$\left[12c = 10s \right]$$

$$\left[s/c = 1.2 \right]$$

$$\left[\boxed{20\% \text{ profit}} \right]$$

Q3

$$\left[12s = 10c \right]$$

$$\left[s/c = \frac{5}{6} \right]$$

Loss:

$$\left[\frac{16}{100} \times 100 = \boxed{16\frac{2}{3}\%} \right]$$

Q4

$$\left[\frac{1}{4+1} \times 100 = \boxed{20\%} \right]$$

Q5

$$\left[1.1 \times 1.2 = 1.32 \right]$$

$$\left[\boxed{32\% \text{ increase}} \right]$$

Q6

Take 1 kg true CP = 100.

Charges:

$$\left[110 \right]$$

Actual 900 g cost:

$$\left[90 \right]$$

Profit:

$$\left[20 \right]$$

Profit %:

$$\left[\frac{20}{90} \times 100 = \boxed{22\frac{2}{9}\%} \right]$$

Q7

He gets:

$$\left[1.2 \text{ units for cost of } 1 \right]$$

Selling all at nominal rate gives revenue:

$$\left[1.2 \right]$$

times cost.

$$\left[\boxed{20\% \text{ profit}} \right]$$

Q8

$$\backslash [15c=12s \backslash]$$

$$\backslash [s/c=15/12=1.25 \backslash]$$

$$\backslash [\boxed{25\% \text{ profit}} \backslash]$$

Q9

Profit first group:

$$\backslash [4000(0.2)=800 \backslash]$$

Loss second:

$$\backslash [6000(0.1)=600 \backslash]$$

Net profit:

$$\backslash [200 \backslash]$$

Total CP:

$$\backslash [10000 \backslash]$$

$$\backslash [\boxed{2\% \text{ profit}} \backslash]$$

Q10

Take SP = 100.

Profit = 20.

CP = 80.

$$\backslash [20/80 \times 100 = \boxed{25\%} \backslash]$$

Q11

Take nominal CP of promised quantity = 100.

Claimed 20% loss !' charges:

$$\backslash [80 \backslash]$$

Actual quantity delivered = 75% !' actual cost:

$$\backslash [75 \backslash]$$

Profit:

$$\backslash [5 \backslash]$$

Profit %:

$$\backslash [5/75 \times 100 = \boxed{6\frac{2}{3}\% \text{ profit}} \backslash]$$

Q12

Buying multiplier:

$$\backslash [1.25 \backslash]$$

Selling delivered fraction:

$$\backslash [0.8 \backslash]$$

$$\backslash [SP/CP = \frac{1.25}{0.8} = 1.5625 \backslash]$$

$$\boxed{56.25\% \text{ profit}}$$

Q13

$$1.2 \times 1.25 \times 0.8$$

$$= 1.2$$

So D's price is:

$$\boxed{20\% \text{ above A's CP}}$$

Q14

True CP:

$$2000 + 500 = 2500$$

Profit:

$$500$$

Profit %:

$$500/2500 \times 100 = \boxed{20\%}$$

Q15

Take SP = 100.

Loss = 25.

CP = 125.

Loss %:

$$25/125 \times 100 = \boxed{20\%}$$

Q16

Take true 1 kg cost = 100.

Charges:

$$125$$

Actual quantity cost:

$$90$$

Profit:

$$35$$

Profit %:

$$35/90 \times 100 = \boxed{38\frac{8}{9}\%}$$

Q17

Let nominal price per article = 1.

While buying:

He receives 11 for cost 10.

Total cost for 11:

$$10$$

Now “sells 10 for price of 11” means revenue per 10 articles:

$$\backslash[11 \backslash]$$

Revenue per article:

$$\backslash[11/10 \backslash]$$

Revenue from all 11:

$$\backslash[11\cdot\frac{11}{10} = \frac{121}{10} = 12.1 \backslash]$$

Cost:

$$\backslash[10 \backslash]$$

Profit:

$$\backslash[2.1 \backslash]$$

Profit %:

$$\backslash[2.1/10\times100 = \boxed{21\%} \backslash]$$

Q18

Equal CP halves.

So signed percentages can be averaged:

$$\backslash[\frac{20-10}{2} = \boxed{5\%\text{ profit}} \backslash]$$

Q19

Overall multiplier:

$$\backslash[1.3\times0.9 = 1.17 \backslash]$$

$$\backslash[1.17CP=1170 \backslash]$$

$$\backslash[CP = \boxed{1000} \backslash]$$

Q20

Take:

$$\backslash[CP=100 \backslash]$$

Old SP:

$$\backslash[120 \backslash]$$

Target SP:

$$\backslash[150 \backslash]$$

Increase:

$$\backslash[30 \backslash]$$

relative to old SP:

$$\backslash[30/120\times100 = \boxed{25\%} \backslash]$$

' CHAPTER 18 COMPLETE

We have now covered:

- CP/SP/profit/loss fundamentals '
- profit/loss percentages '
- multiplier and reverse methods '
- CP:SP ratios '
- equal-CP cases '
- equal-SP cases '
- overall profit/loss across multiple items '
- false weights '
- dishonest buying/selling quantities '
- $\backslash(x\backslash)$ -articles / $\backslash(y\backslash)$ -articles problems '
- free-item effective discounts '
- chain transactions '
- expenses included in CP '
- profit/loss expressed as % of SP '
- selling-price change questions '
- advanced GATE/placement drills '

Progress

Completed: 18/76 Unit 1 — Quantitative Aptitude: 0=0% PROGRESS

NEXT

0=0% Chapter 19 — Discount & Marked Price

Next we will cover:

- marked price, list price and selling price
- discount %
- MP !' SP multipliers
- reverse discount
- successive discounts
- equivalent single discount
- markup + discount together
- required markup for target profit
- "discount but still profit" problems
- buy-one-get-one style effective discounts
- GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 19 — DISCOUNT & MARKED PRICE

PART 1 — MARKED PRICE, SELLING PRICE, DISCOUNT %, REVERSE DISCOUNT & MARKUP + DISCOUNT

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: Very High Priority: $\emptyset$ = Yes 1

Discount questions are closely related to Profit & Loss, but now a third price enters the problem.

The three important prices are:

$\boxed{CP, MP, SP}$

where:

- CP = Cost Price
- MP = Marked Price / List Price
- SP = Selling Price

The main flow is usually:

$CP \rightarrow MP \rightarrow SP$

A seller may first **mark up** the item above CP and then give a **discount** on MP.

1. MARKED PRICE

Marked Price is the price written or listed on the product.

Example:

A shirt has tag price:

$\text{₹} 2000$

Then:

$\boxed{MP = \text{₹} 2000}$

2. DISCOUNT

Discount is the reduction from Marked Price.

$\boxed{\text{Discount} = MP - SP}$

Example:

$MP = \text{₹} 2000$

$SP = \text{₹} 1600$

Discount:

$2000 - 1600 = \boxed{\text{₹} 400}$

3. DISCOUNT PERCENTAGE

Discount percentage is calculated on:

$$\boxed{MP}$$

not CP.

Formula:

$$\boxed{\text{Discount}\% = \frac{MP-SP}{MP} \times 100}$$

4. EXAMPLE

$$MP = ₹1200$$

$$SP = ₹900$$

Discount:

$$\boxed{300}$$

Discount %:

$$\frac{300}{1200} \times 100$$

$$= \boxed{25\%}$$

5. DISCOUNT MULTIPLIER

If discount is $(d\%)$:

$$SP = MP \left(1 - \frac{d}{100}\right)$$

So:

$$\boxed{SP = MP \times \frac{100-d}{100}}$$

6. EXAMPLE — FIND SP

$$MP = ₹2500$$

Discount = 20%.

$$SP = 2500(0.8)$$

$$= \boxed{₹2000}$$

7. FIND MP FROM SP

$SP = ₹1800$ after 10% discount.

$$SP = 0.9MP$$

So:

$$MP = \frac{1800}{0.9}$$

$$= \boxed{₹2000}$$

8. REVERSE DISCOUNT FORMULA

If SP is known after $(d\%)$ discount:

$$\boxed{MP = SP \times \frac{100}{100-d}}$$

Example:

SP = ₹ 1350 after 25% discount.

$$MP = 1350 \times \frac{100}{75}$$

$$= 1350 \times \frac{4}{3}$$

$$= \boxed{₹ 1800}$$

9. MP:SP RATIO

At discount $(d\%)$:

$$MP:SP = 100:(100-d)$$

Example:

20% discount:

$$MP:SP = 100:80 = 5:4$$

So if MP is 5 parts, SP is 4 parts.

10. MUST-MEMORIZE DISCOUNT RATIOS

10% discount

$$MP:SP = 10:9$$

20% discount

$$5:4$$

25% discount

$$4:3$$

40% discount

$$5:3$$

50% discount

$$2:1$$

11. DISCOUNT AMOUNT GIVEN

Discount = ₹ 300 at 20%.

Then:

$$20\% \text{ of } MP = 300$$

$$[MP=300\times 5]$$

$$[= \boxed{1500}]$$

SP:

$$[1500-300 = \boxed{1200}]$$

12. DISCOUNT IS NOT PROFIT/LOSS

Suppose:

$$[MP = 1000]$$

Discount = 20%.

Then:

$$[SP = 800]$$

Can we say seller made 20% loss?

No.

We need CP.

If CP = 600 :

$$[SP=800]$$

Profit:

$$[200]$$

Profit %:

$$[\frac{200}{600} \times 100 = 33\frac{1}{3}\%]$$

So a discount does not automatically mean a loss.

13. MARKUP

Markup means increasing the listed price above cost.

If an item is marked $(m\%)$ above CP:

$$[MP = CP \left(1 + \frac{m}{100} \right)]$$

So:

$$[\boxed{MP = CP \times \frac{100+m}{100}}]$$

14. EXAMPLE — MARKUP

$$CP = 1000 .$$

Marked 40% above CP.

$$[MP=1000(1.4)]$$

$$[= \boxed{1400}]$$

15. MARKUP + DISCOUNT

This is the central discount problem.

Suppose:

- $CP = ₹1000$
- markup = 50%
- discount = 20%

Step 1:

$$MP = 1000(1.5) = 1500$$

Step 2:

$$SP = 1500(0.8) = 1200$$

Profit:

$$₹200$$

Profit %:

$$\boxed{20\%}$$

16. FAST MULTIPLIER METHOD

You do not need to calculate MP separately.

If markup is $(m\%)$, discount is $(d\%)$:

$$\frac{SP}{CP} = \left(1 + \frac{m}{100}\right) \left(1 - \frac{d}{100}\right)$$

Thus:

$$\boxed{SP = CP \left(1 + \frac{m}{100}\right) \left(1 - \frac{d}{100}\right)}$$

17. EXAMPLE

Markup 25%.

Discount 20%.

Multiplier:

$$1.25 \times 0.8 = 1$$

Therefore:

$$SP = CP$$

So:

$$\boxed{\text{No profit, no loss}}$$

18. ANOTHER EXAMPLE

Markup = 50%.

Discount = 20%.

$$\sqrt[1]{1.5 \times 0.8 = 1.2}$$

So:

$$\sqrt[1]{SP = 1.2CP}$$

Profit:

$$\sqrt[1]{\boxed{20\%}}$$

19. MARKUP 20%, DISCOUNT 20%

$$\sqrt[1]{1.2 \times 0.8 = 0.96}$$

So:

$$\sqrt[1]{SP = 96\% \text{ of CP}}$$

Therefore:

$$\sqrt[1]{\boxed{4\% \text{ loss}}}$$

Equal markup and discount do not cancel.

20. SAME-% MARKUP AND DISCOUNT

If marked up by $\sqrt[1]{(p\%)}$ and then discounted by $\sqrt[1]{(p\%)}$:

$$\sqrt[1]{(1 + p/100)(1 - p/100)}$$

$$\sqrt[1]{= 1 - \frac{p^2}{10000}}$$

So:

$$\sqrt[1]{\boxed{\text{Net loss\%} = \frac{p^2}{100}}}$$

Example:

20% markup and 20% discount:

$$\sqrt[1]{\frac{20^2}{100} = \boxed{4\% \text{ loss}}}$$

21. FIND PROFIT/LOSS FROM MARKUP AND DISCOUNT

Suppose:

markup:

$$\sqrt[1]{30\%}$$

discount:

$$\sqrt[1]{10\%}$$

Multiplier:

$$\sqrt[1]{1.3 \times 0.9 = 1.17}$$

Therefore:

$$\boxed{17\% \text{ profit}}$$

22. GENERAL NET PROFIT/LOSS FORMULA

If markup = $(m\%)$, discount = $(d\%)$:

$$\frac{SP}{CP} = \frac{(100+m)(100-d)}{10000}$$

Net percentage:

$$\boxed{m - d - \frac{md}{100}}$$

where positive means profit and negative means loss.

Example:

$$m=30, \quad d=10$$

$$30 - 10 - \frac{300}{100}$$

$$= 20 - 3$$

$$= \boxed{17\%}$$

23. WHY FORMULA HAS MINUS $(md/100)$

Because discount is applied to the **marked price after markup**, not directly to CP.

Multiplier method is usually safer:

$$(1+m)(1-d)$$

with percentages converted to decimals.

24. FIND REQUIRED MARKUP FOR GIVEN DISCOUNT AND PROFIT

Suppose seller gives:

$$20\% \text{ discount}$$

but wants:

$$25\% \text{ profit}$$

Let:

$$CP=100$$

Target SP:

$$125$$

Discount means:

$$SP=0.8MP$$

So:

$$0.8MP=125$$

$$\boxed{MP=156.25}$$

Thus markup:

$$\boxed{56.25\%}$$

$$\boxed{56.25\%}$$

25. GENERAL REQUIRED MARKUP FORMULA

To earn $(p\%)$ profit after $(d\%)$ discount:

$$(1+m)(1-d)=1+p$$

in decimal form.

So:

$$\boxed{1+m = \frac{1+p}{1-d}}$$

and:

$$\boxed{m = \frac{1+p}{1-d} - 1}$$

26. PERCENTAGE FORM

$$\boxed{\text{Markup}\% = \frac{100(p+d)}{100-d}}$$

where (p,d) are percentage numbers.

Check for:

$$p=25, \quad d=20$$

$$\frac{100(45)}{80} = 56.25\%$$

Correct.

27. FIND DISCOUNT FOR TARGET PROFIT

Markup = 50%.

Target profit = 20%.

Take:

$$\boxed{CP=100}$$

MP:

$$\boxed{150}$$

Required SP:

$$\boxed{120}$$

Discount:

$$\boxed{150-120=30}$$

Discount % on MP:

$$\left[\frac{30}{150} \times 100 = \boxed{20\%} \right]$$

28. GENERAL REQUIRED DISCOUNT

Given markup $(m\%)$ and desired profit $(p\%)$:

$$\left[(1+m)(1-d)=1+p \right]$$

So:

$$\left[\boxed{1-d = \frac{1+p}{1+m}} \right]$$

Hence:

$$\left[\boxed{d = 1 - \frac{1+p}{1+m}} \right]$$

29. PERCENTAGE FORM

$$\left[\boxed{\text{Discount}\% = \frac{100(m-p)}{100+m}} \right]$$

provided $(m \geq p)$.

Example:

Markup 50%, target profit 20%:

$$\left[\frac{100(30)}{150} = \boxed{20\%} \right]$$

30. FIND PROFIT WHEN MP IS MULTIPLE OF CP

Suppose:

$$\left[MP = 1.5CP \right]$$

and discount:

$$\left[20\% \right]$$

Then:

$$\left[SP = 1.5CP(0.8) \right]$$

$$\left[= 1.2CP \right]$$

Profit:

$$\left[\boxed{20\%} \right]$$

31. FIND CP WHEN MP, DISCOUNT & PROFIT GIVEN

$$MP = ₹ 1800$$

Discount = 20%.

Profit = 20%.

First:

$$\left[SP = 1800(0.8) \right]$$

$$\boxed{=1440}$$

Since:

$$\boxed{SP=1.2CP}$$

$$\boxed{CP=1440/1.2}$$

$$\boxed{=1200}$$

32. FAST COMBINED RATIO

Given:

$$\boxed{MP=1800}$$

20% discount:

$$\boxed{SP:MP=4:5}$$

So:

$$\boxed{SP=1440}$$

20% profit:

$$\boxed{CP:SP=5:6}$$

So:

$$\boxed{CP=1440 \times \frac{5}{6}}$$

$$\boxed{=1200}$$

33. FIND MP FROM CP, PROFIT & DISCOUNT

$$CP = 800.$$

Profit desired = 25%.

Discount = 20%.

Required SP:

$$\boxed{800(1.25)=1000}$$

Since SP is 80% of MP:

$$\boxed{MP=1000/0.8}$$

$$\boxed{=1250}$$

34. FIND DISCOUNT AMOUNT

$$MP = 2500.$$

Discount = 12%.

Discount amount:

$$\boxed{2500(0.12) = 300}$$

SP:

$$\boxed{2200}$$

35. FIND DISCOUNT % FROM MP AND SP

$$MP = ₹1600.$$

$$SP = ₹1360.$$

Discount:

$$\boxed{240}$$

Discount %:

$$\boxed{\frac{240}{1600} \times 100}$$

$$\boxed{15\%}$$

36. “SOLD AT x% OF MARKED PRICE”

Suppose SP is 80% of MP.

Then discount:

$$\boxed{100 - 80 = 20\%}$$

37. “SOLD AT 75% OF LIST PRICE”

Discount:

$$\boxed{100 - 75 = 25\%}$$

Very common wording.

38. “DISCOUNT OF x% ON LIST PRICE”

List price = marked price.

So:

$$\boxed{\text{List Price} = \text{Marked Price}}$$

in standard aptitude wording.

39. DISCOUNT AMOUNT AS % OF SP

Suppose MP = 100, discount = 20%.

Then:

$$\boxed{SP = 80}$$

Discount amount = 20.

Discount as % of SP:

$$\left[20/80 \times 100 = \boxed{25\%} \right]$$

Again, standard discount percentage is based on MP, not SP.

40. GENERAL CONVERSION

If discount is $d\%$ of MP, then discount amount as % of SP is:

$$\left[\boxed{\frac{100d}{100-d}\%} \right]$$

Example:

20% discount:

$$\left[25\% \text{ of SP} \right]$$

41. SP IS x% LESS THAN MP

“SP is 20% less than MP.”

Then:

$$\left[SP = 80\% \text{ of MP} \right]$$

So discount:

$$\left[\boxed{20\%} \right]$$

Straightforward.

42. MP IS x% MORE THAN SP

This is different.

Suppose MP is 25% more than SP.

Take:

$$\left[SP = 100 \right]$$

Then:

$$\left[MP = 125 \right]$$

Discount:

$$\left[25 \right]$$

as % of MP:

$$\left[25/125 \times 100 \right]$$

$$\left[= \boxed{20\%} \right]$$

So:

$$\left[25\% \text{ more} \right]$$

does not mean:

$\backslash[25\%\text{discount}]$

43. GENERAL “MP MORE THAN SP” CONVERSION

If MP is $\backslash(p\%)$ more than SP:

$$\backslash[MP:SP=(100+p):100]$$

Discount %:

$$\backslash[\boxed{\frac{100p}{100+p}\%}]$$

44. GATE-STYLE QUESTION 1

$$MP = ₹ 1000.$$

Discount = 15%.

SP:

$$\backslash[1000(0.85) = \boxed{₹ 850}]$$

45. GATE-STYLE QUESTION 2

$$SP = ₹ 1200 \text{ after } 20\% \text{ discount.}$$

$$\backslash[MP=1200/0.8]$$

$$\backslash[= \boxed{₹ 1500}]$$

46. GATE-STYLE QUESTION 3

An article is marked 25% above CP and discounted 20%.

$$\backslash[1.25 \times 0.8 = 1]$$

So:

$$\backslash[\boxed{\text{No profit, no loss}}]$$

47. GATE-STYLE QUESTION 4

Marked 40% above CP and discounted 10%.

Multiplier:

$$\backslash[1.4 \times 0.9 = 1.26]$$

Therefore:

$$\backslash[\boxed{26\% \text{ profit}}]$$

48. PLACEMENT-STYLE QUESTION 1

A product has MP = ₹ 2400 and is sold at 25% discount.

$$SP = 2400(0.75)$$

$$SP = \boxed{\text{₹ } 1800}$$

49. PLACEMENT-STYLE QUESTION 2

$$CP = ₹ 1000.$$

MP = 50% above CP.

Discount = 20%.

$$SP = 1000(1.5)(0.8)$$

$$SP = ₹ 1200$$

Profit:

$$\boxed{20\%}$$

50. PLACEMENT-STYLE QUESTION 3

An article is marked 60% above CP and sold at 25% discount.

$$1.6 \times 0.75 = 1.2$$

So:

$$\boxed{20\% \text{ profit}}$$

51. PLACEMENT-STYLE QUESTION 4

A seller wants 20% profit after giving 20% discount. Find required markup.

Take CP = 100.

Required SP:

$$SP = 120$$

Since:

$$SP = 0.8MP$$

$$MP = ₹ 150$$

Markup:

$$\boxed{50\%}$$

52. ADVANCED QUESTION — FIND CP

$$MP = ₹ 2500.$$

Discount = 20%.

Profit = 25%.

SP:

$$\boxed{2000}$$

Since:

$$SP = 1.25CP$$

$$CP = 2000 / 1.25$$

$$CP = \boxed{1600}$$

53. ADVANCED QUESTION — FIND MARKUP

Seller gives 25% discount and still earns 20% profit.

Take:

$$CP = 100$$

Target SP:

$$120$$

$$SP = 75\% MP$$

$$0.75MP = 120$$

$$MP = 160$$

Markup:

$$\boxed{60\%}$$

54. ADVANCED QUESTION — FIND DISCOUNT

Item marked 80% above CP.

Seller wants 35% profit.

Take CP:

$$100$$

MP:

$$180$$

Required SP:

$$135$$

Discount amount:

$$45$$

Discount %:

$$45/180 \times 100 = \boxed{25\%}$$

55. ADVANCED QUESTION — MARKUP & DISCOUNT SAME

Seller marks 30% above CP and gives 30% discount.

Net multiplier:

$$[1.3 \times 0.7 = 0.91]$$

Therefore:

$$[\boxed{9\% \text{ loss}}]$$

Using shortcut:

$$[30^2/100=9\%]$$

56. COMMON TRAPS

& pTRAP 1 — Discount % denominator

Discount percentage uses MP.

& pTRAP 2 — Discount means loss

False. Profit depends on CP.

& pTRAP 3 — Markup and discount cancel

Equal percentages do not cancel.

& pTRAP 4 — Reverse discount

If 20% discount leaves SP = 80%, recover MP by dividing by 0.8.

& pTRAP 5 — MP more than SP

25% more one way corresponds to 20% discount the other way.

57. MUST-MEMORIZE FORMULAS

Discount

$$[\boxed{D=MP-SP}]$$

Discount %

$$[\boxed{\frac{MP-SP}{MP} \times 100}]$$

SP after $(d\%)$ discount

$$[\boxed{SP=MP \left(1 - \frac{d}{100}\right)}]$$

Markup $(m\%)$

$$[\boxed{MP=CP \left(1 + \frac{m}{100}\right)}]$$

58. MARKUP + DISCOUNT MASTER FORMULA

$$\boxed{\frac{SP}{CP} = \left(1 + \frac{m}{100}\right) \left(1 - \frac{d}{100}\right)}$$

Net signed profit percentage:

$$\boxed{m - d - \frac{md}{100}}$$

59. HIGH-VALUE SHORTCUTS

If markup = discount = $(p\%)$:

$$\boxed{\text{loss} = \frac{p^2}{100}\%}$$

To get $(p\%)$ profit after $(d\%)$ discount:

$$\boxed{\text{markup} = \frac{100(p+d)}{100-d}\%}$$

60. FAST EXAM DECISION TREE

MP + discount?

Find SP.

SP + discount?

Reverse to MP.

CP + markup + discount?

Multiply factors.

Target profit + discount?

Find required SP, then MP.

MP more than SP?

Convert using ratio, do not directly equate to discount %.

PRACTICE — LEVEL 0

Q1

MP = ₹1000, discount = 20%. Find SP.

Q2

SP = ₹720 after 10% discount. Find MP.

Q3

MP = ₹800, SP = ₹600. Find discount %.

Q4

CP = ₹500, markup = 20%. Find MP.

Q5

CP = ₹1000, markup = 25%, discount = 20%. Find profit/loss.

LEVEL 1

Q6

MP = ₹ 2000, discount = 15%. Find SP.

Q7

SP = ₹ 1500 after 25% discount. Find MP.

Q8

An item is marked 50% above CP and sold at 10% discount. Find profit %.

Q9

Marked 20% above CP and discounted 20%. Find net profit/loss %.

Q10

Seller wants 25% profit after 20% discount. Find required markup %.

LEVEL 2

Q11

CP = ₹ 1200. Item is marked 50% above CP and discounted 20%. Find SP and profit %.

Q12

MP = ₹ 1800, discount = 25%, profit = 12.5%. Find CP.

Q13

Item is marked 60% above CP. What discount yields 20% profit?

Q14

A seller gives 25% discount and makes 50% profit. Find markup %.

Q15

MP is 20% more than SP. Find discount %.

LEVEL 3 — EXAM PRESSURE

Q16

An article is marked 80% above CP and discounted 30%. Find net profit/loss %.

Q17

An article is sold at 20% discount and 20% profit for ₹ 960. Find CP and MP.

Q18

A seller marks goods 40% above CP. What discount must be given to earn exactly 12% profit?

Q19

An article is marked 25% above CP. By what percentage must it be discounted to sell at CP?

Q20

A seller gives 20% discount on MP and still earns 60% profit. If CP = ₹ 500 find MP.

ANSWERS**Q1**

$$\sqrt[1]{1000(0.8) = \boxed{1600}}$$

Q2

$$\sqrt[1]{720/0.9 = \boxed{800}}$$

Q3

Discount:

$$\sqrt[1]{200 \sqrt[1]{}}$$

$$\sqrt[1]{200/800 \times 100 = \boxed{25\% \sqrt[1]{}}}$$

Q4

$$\sqrt[1]{500(1.2) = \boxed{600}}$$

Q5

$$\sqrt[1]{1.25 \times 0.8 = 1 \sqrt[1]{}}$$

$$\sqrt[1]{\boxed{\text{No profit, no loss}} \sqrt[1]{}}$$

Q6

$$\sqrt[1]{2000(0.85) = \boxed{1700}}$$

Q7

$$\sqrt[1]{1500/0.75 = \boxed{2000}}$$

Q8

$$\sqrt[1]{1.5 \times 0.9 = 1.35 \sqrt[1]{}}$$

$$\sqrt[1]{\boxed{35\% \text{ profit}} \sqrt[1]{}}$$

Q9

$$\sqrt[1]{1.2 \times 0.8 = 0.96 \sqrt[1]{}}$$

$$\sqrt[1]{\boxed{4\% \text{ loss}} \sqrt[1]{}}$$

Q10

Take CP = 100.

Target SP:

$$\sqrt[1]{125 \sqrt[1]{}}$$

Since SP = 80% MP:

$$\text{MP} = 156.25$$

Markup:

$$\boxed{56.25\%}$$

Q11

MP:

$$1200(1.5) = 1800$$

SP:

$$1800(0.8) = \boxed{1440}$$

Profit:

$$240$$

$$240/1200 \times 100 = \boxed{20\%}$$

Q12

SP:

$$1800(0.75) = 1350$$

Profit 12.5% means:

$$\text{SP} = 1.125 \text{CP}$$

$$\text{CP} = 1350/1.125$$

$$= \boxed{1200}$$

Q13

Take CP = 100.

MP:

$$160$$

Required SP for 20% profit:

$$120$$

Discount:

$$40$$

$$40/160 \times 100 = \boxed{25\%}$$

Q14

Take CP = 100.

Target SP:

$$150$$

After 25% discount:

$$\text{SP} = 0.75 \text{MP}$$

$$\text{MP} = 200$$

Markup:

$\boxed{100\%}$

Q15

Take SP = 100.

MP = 120.

Discount:

20%

Discount %:

$20/120 \times 100 = \boxed{16\frac{2}{3}\%}$

Q16

$1.8 \times 0.7 = 1.26$

$\boxed{26\% \text{ profit}}$

Q17

SP = ₹ 960.

20% profit:

$CP = 960/1.2 = \boxed{₹ 800}$

20% discount:

$MP = 960/0.8 = \boxed{₹ 1200}$

Q18

Take CP = 100.

MP:

140

Required SP:

112

Discount:

28%

Discount %:

$28/140 \times 100 = \boxed{20\%}$

Q19

Take CP = 100.

MP:

125

To sell at CP:

$SP = 100$

Discount:

25%

$$\left[\frac{25}{125} \times 100 = \boxed{20\%} \right]$$

Q20

CP:

$$\left[500 \right]$$

60% profit:

$$\left[SP=800 \right]$$

SP is 80% of MP:

$$\left[MP=800/0.8 \right]$$

$$\left[= \boxed{1000} \right]$$

CHAPTER 19 STATUS

Covered so far:

- MP, CP, SP '
- discount amount and discount % '
- reverse discount '
- MP:SP ratios '
- markup '
- markup + discount multiplier '
- net profit/loss from markup + discount '
- equal markup/discount trap '
- required markup for target profit '
- required discount for target profit '
- MP-vs-SP comparison traps '
- GATE/placement practice '

But **Chapter 19 is not complete yet.**

Next we need the most common advanced discount questions:

- successive discounts
- equivalent single discount
- three successive discounts
- discount vs free-item offers
- cash discount / trade discount style aptitude questions
- "same SP after different discounts"
- markup + successive discounts
- required discount to match another offer
- advanced shopping/placement problems

Progress

Completed: 18/76 **Current:** 0/31 **Chapter 19 — Discount & Marked Price** **Next:** Chapter 19 Part 2 — **Successive discounts, equivalent discounts & advanced offer comparison**

USER

next

ASSISTANT

CHAPTER 19 — DISCOUNT & MARKED PRICE

PART 2 — SUCCESSIVE DISCOUNTS, EQUIVALENT DISCOUNT, FREE-OFFER COMPARISON & ADVANCED QUESTIONS

Status: $\emptyset = 0\%$ PROGRESS GATE: High Placements: Very High Priority: $\emptyset = 0\%$ Per 1

The most important rule in this part:

Successive discounts are multiplied, not simply added.

1. SUCCESSIVE DISCOUNTS

Suppose marked price is ₹ 1000 .

First discount:

20%

Price becomes:

$1000 \times 0.8 = 800$

Second discount:

10%

Now 10% is applied on ₹ 800 :

$800 \times 0.9 = 720$

Final SP:

$\boxed{₹ 720}$

Total discount:

$1000 - 720 = 280$

Equivalent discount:

$\boxed{28\%}$

Not 30%.

2. SUCCESSIVE DISCOUNT FORMULA

If discounts are:

$$\left[a\%, b\% \right]$$

equivalent single discount:

$$\left[a+b-\frac{ab}{100} \right]$$

Example:

$$\left[20, 10 \right]$$

$$\left[20+10-\frac{20(10)}{100} \right]$$

$$\left[=30-2 \right]$$

$$\left[=\boxed{28\%} \right]$$

MUST MEMORIZE

3. MULTIPLIER METHOD

Usually safer:

$$\left[SP = MP \left(1-\frac{a}{100}\right) \left(1-\frac{b}{100}\right) \right]$$

Equivalent retained fraction:

$$\left[\boxed{(1-a/100)(1-b/100)} \right]$$

Equivalent discount:

$$\left[1-\text{retained fraction} \right]$$

4. EXAMPLE — 30% THEN 20%

$$\left[0.7 \times 0.8 = 0.56 \right]$$

So customer pays:

$$\left[56\% \right]$$

of MP.

Equivalent discount:

$$\left[100-56 = \boxed{44\%} \right]$$

5. EQUAL SUCCESSIVE DISCOUNTS

Two discounts of 20% each:

$$\left[0.8^2=0.64 \right]$$

Equivalent discount:

$$\left[36\% \right]$$

Formula:

$$\left[20+20-\frac{400}{100} = 40-4 = \boxed{36\%} \right]$$

6. TWO 10% DISCOUNTS

$$\sqrt[0.9^2=0.81]{}$$

Equivalent discount:

$$\sqrt{\boxed{19\%}}$$

Not 20%.

7. THREE SUCCESSIVE DISCOUNTS

For three discounts, multiplier method is easiest.

Example:

$$\sqrt{10\%, 20\%, 25\%}$$

Retained multiplier:

$$\sqrt{0.9 \times 0.8 \times 0.75}$$

$$\sqrt{=0.54}$$

Equivalent discount:

$$\sqrt{1 - 0.54 = 0.46}$$

$$\sqrt{=\boxed{46\%}}$$

8. GENERAL MULTIPLE-DISCOUNT FORM

For discounts:

$$\sqrt{d_1, d_2, \dots, d_n}$$

$$\sqrt{\boxed{SP = MP \prod_{i=1}^n \left(1 - \frac{d_i}{100}\right)}}$$

Equivalent discount:

$$\sqrt{\boxed{100 \left[1 - \prod \left(1 - \frac{d_i}{100}\right) \right] \%}}$$

9. ORDER OF DISCOUNTS

For pure percentage discounts:

$$\sqrt{0.8 \times 0.9 = 0.9 \times 0.8}$$

So order does not matter.

20% then 10% gives same final price as 10% then 20%.

10. FIND SECOND DISCOUNT FROM EQUIVALENT DISCOUNT

First discount = 20%.

Equivalent total discount = 40%.

Let second discount = $\sqrt{(x\%)}$.

Retained fraction after total discount:

$$\boxed{60\%}$$

After first discount:

$$\boxed{80\%}$$

So second retained fraction:

$$\frac{60}{80} = 0.75$$

Thus second discount:

$$\boxed{25\%}$$

11. FORMULA METHOD

$$20 + x - \frac{20x}{100} = 40$$

$$20 + 0.8x = 40$$

$$x = 25$$

$$\boxed{25\%}$$

12. FIND ONE OF TWO EQUAL DISCOUNTS

Two equal discounts together equal 36%.

Let each discount be $x\%$.

Retained fraction:

$$(1-x)^2 = 0.64$$

So:

$$1-x = 0.8$$

$$x = 0.2$$

Thus each discount:

$$\boxed{20\%}$$

13. MARKUP + SUCCESSIVE DISCOUNTS

Suppose:

- markup 50%
- discounts 20% and 10%

Then:

$$\frac{SP}{CP} = 1.5 \times 0.8 \times 0.9$$

$$= 1.08$$

Therefore:

$$\boxed{8\% \text{ profit}}$$

14. EXAMPLE

$$CP = ₹ 1000.$$

$$\text{Markup} = 80\%.$$

$$\text{Discounts} = 25\%, \text{ then } 20\%.$$

$$SP = 1000(1.8)(0.75)(0.8)$$

$$= 1000(1.08)$$

$$= ₹ 1080$$

Profit:

$$= \boxed{8\%}$$

15. REQUIRED MARKUP WITH SUCCESSIVE DISCOUNTS

A shopkeeper gives:

$$20\% \text{ and } 10\%$$

successive discounts and wants 20% profit.

Discount retention:

$$0.8 \times 0.9 = 0.72$$

Target:

$$SP = 1.2CP$$

Thus:

$$MP \times 0.72 = 1.2CP$$

$$\frac{MP}{CP} = \frac{1.2}{0.72} = \frac{5}{3}$$

$$= 1.6667$$

Required markup:

$$= \boxed{66\frac{2}{3}\%}$$

16. FAST METHOD

Equivalent discount:

$$28\%$$

So customer pays:

$$72\%$$

If seller wants $SP = 120\%$ of CP :

$$MP = \frac{120}{72}CP = 166\frac{2}{3}\%CP$$

Markup:

$$\boxed{66\frac{23}{100}\%}$$

17. REQUIRED DISCOUNT AFTER MARKUP

Item marked 60% above CP.

Seller gives first discount 20%.

What second discount gives exactly no profit/no loss?

After markup and first discount:

$$1.6 \times 0.8 = 1.28$$

Need final factor:

$$1$$

So second retained multiplier:

$$\frac{1}{1.28} = 0.78125$$

Second discount:

$$1 - 0.78125 = 0.21875$$

$$\boxed{21.875\%}$$

18. COMPARE TWO DISCOUNT OFFERS

Offer A:

$$30\% \text{ off}$$

Offer B:

$$20\% \text{ off, then } 15\% \text{ off}$$

Offer B equivalent:

$$20 + 15 - \frac{300}{100}$$

$$= 35 - 3 = 32\%$$

So:

$$\boxed{\text{Offer B is better}}$$

19. OFFER COMPARISON USING RETAINED PRICE

Offer A:

$$70\% \text{ payable}$$

Offer B:

$$0.8 \times 0.85 = 0.68$$

So B requires only 68% payment.

Thus B gives 32% discount.

20. “BUY 1 GET 1 FREE”

Customer gets:

$\sqrt[2]{}$

items but pays for:

$\sqrt[1]{}$

Effective discount:

$$\sqrt[\frac{12}{100}]{} = \boxed{50\%}$$

21. “BUY 2 GET 1 FREE”

Customer receives:

$\sqrt[3]{}$

pays for:

$\sqrt[2]{}$

Effective discount:

$$\sqrt[\frac{13}{100}]{} = \boxed{33\frac{1}{3}\%}$$

22. “BUY 3 GET 2 FREE”

Receives:

$\sqrt[5]{}$

pays for:

$\sqrt[3]{}$

Free portion:

$\sqrt[2/5]{}$

Effective discount:

$$\sqrt[\boxed{40\%}]{}$$

23. GENERAL FREE-OFFER FORMULA

Buy $\sqrt{(x)}$, get $\sqrt{(y)}$ free.

Effective discount:

$$\sqrt[\boxed{\frac{y}{x+y} \times 100}]{}$$

24. COMPARE DISCOUNT VS FREE OFFER

Which is better?

Offer A

25% discount.

Offer B

Buy 3 get 1 free.

Offer B effective discount:

$$\left[\frac{1}{4} \right] \times 100 = 25\%$$

So:

$$\boxed{\text{Both are equivalent}}$$

assuming identical unit price and no other conditions.

25. COMPARE 30% OFF VS BUY 2 GET 1 FREE

Buy 2 get 1 free:

$$\left[\frac{1}{3} \right] = 33\frac{1}{3}\%$$

So:

$$\boxed{\text{Buy 2 get 1 free gives larger nominal discount}}$$

provided you actually want all 3 units.

26. FREE OFFER + DISCOUNT

Offer:

Buy 2 get 1 free and additional 10% off the paid amount.

Free-offer effective retained fraction:

$$\left[\frac{2}{3} \right]$$

Then extra 10% discount:

$$\left[\frac{2}{3} \right] \times 0.9 = 0.6$$

So customer pays 60% of normal total price.

Equivalent discount:

$$\boxed{40\%}$$

27. DISCOUNT + CASHBACK WARNING

A 20% discount and 20% cashback are not always identical in practical retail because cashback may have conditions, future-use restrictions, or caps.

In pure aptitude, if cashback is treated as immediate unconditional monetary recovery on the transaction:

Initial payment:

$\sqrt[80]{\%}$

Cashback 20% of what?

The wording matters.

Could mean:

- 20% of SP,
- 20% of MP,
- fixed cashback.

Never assume the base.

28. CASH DISCOUNT VS TRADE DISCOUNT

For aptitude:

- **Trade discount:** reduction from list/marked price.
- **Cash discount:** additional reduction often for prompt payment.

Mathematically, if both are percentage discounts applied sequentially, use successive-discount multipliers.

Example:

20% trade discount + 5% cash discount:

$\sqrt[0.8 \times 0.95 = 0.76]{}$

Equivalent:

$\sqrt[\boxed{24\%}]{}$

29. SAME SP AFTER DIFFERENT DISCOUNTS

Product A MP = ₹ 1200. Discount 20%.

SP:

$\sqrt[1200(0.8) = 960]{}$

Product B receives 25% discount and has same SP.

Let MP of B = $\sqrt{(M)}{}$.

$\sqrt[0.75M = 960]{}$

$\sqrt[M = 1280]{}$

$\sqrt[\boxed{\text{₹ } 1280}]{}$

30. RATIO OF MARKED PRICES FOR SAME SP

If two items have same SP after discounts $\sqrt{(d_1, d_2)}{}$:

$\sqrt[MP_1(1-d_1) = MP_2(1-d_2)]{}$

Hence:

$$\boxed{MP_1:MP_2 = (1-d_2):(1-d_1)}$$

with percentages written as retained fractions.

31. EXAMPLE

Same SP.

Discounts:

$$20\%, 25\%$$

Then:

$$MP_1:MP_2 = 0.75:0.8$$

$$=15:16$$

So:

$$\boxed{15:16}$$

32. SAME MP, DIFFERENT DISCOUNTS

If MP is the same:

lower discount !' higher SP.

Example:

$$MP = 1000.$$

$$A: 10\% \text{ off !' } 900.$$

$$B: 20\% \text{ off !' } 800.$$

SP ratio:

$$900:800 = \boxed{9:8}$$

33. SAME CP, DIFFERENT MARKUP/DISCOUNT STRATEGIES

Two sellers have same CP.

Seller A:

- markup 50%
- discount 20%

Net factor:

$$1.5(0.8)=1.2$$

Seller B:

- markup 40%
- discount 10%

Net factor:

$$1.4(0.9)=1.26$$

Therefore seller B charges 26% above CP vs A's 20%.

34. FIND EQUIVALENT SINGLE DISCOUNT OF 25% + 20%

$$\left[25 + 20 - \frac{25(20)}{100} \right]$$

$$\left[45 - 5 = \boxed{40\%} \right]$$

35. FIND EQUIVALENT DISCOUNT OF 40% + 20%

$$\left[40 + 20 - 8 = \boxed{52\%} \right]$$

36. THREE DISCOUNTS — 20%, 10%, 10%

Retained:

$$\left[0.8 \times 0.9 \times 0.9 \right]$$

$$\left[= 0.648 \right]$$

Equivalent discount:

$$\left[1 - 0.648 = 0.352 \right]$$

$$\left[= \boxed{35.2\%} \right]$$

37. FIND THIRD DISCOUNT

Successive discounts 20%, 10%, and $(x\%)$ give equivalent discount 46%.

After first two:

$$\left[0.8 \times 0.9 = 0.72 \right]$$

Overall retention for 46% discount:

$$\left[0.54 \right]$$

So third retained factor:

$$\left[0.54 / 0.72 = 0.75 \right]$$

Thus:

$$\left[\boxed{x = 25\%} \right]$$

38. ORIGINAL MP FROM FINAL SP AFTER MULTIPLE DISCOUNTS

Final SP = ₹ 54 after discounts:

$$\left[20\%, 10\%, 25\% \right]$$

Retention:

$$\left[0.8 \times 0.9 \times 0.75 = 0.54 \right]$$

Thus:

$$\text{MP} = 540 / 0.54$$

$$= \boxed{1000}$$

39. PROFIT WITH MULTIPLE DISCOUNTS

$$\text{CP} = 1100$$

Marked 100% above CP:

$$\text{MP} = 2000$$

Discounts:

$$20\%, 25\%$$

Retention:

$$0.8 \times 0.75 = 0.6$$

SP:

$$2000(0.6) = 1200$$

Profit:

$$\boxed{20\%}$$

40. FIND MARKUP FOR BREAK-EVEN AFTER TWO DISCOUNTS

Discounts:

$$20\%, 25\%$$

Combined retention:

$$0.8 \times 0.75 = 0.6$$

For break-even:

$$\text{MP}(0.6) = \text{CP}$$

$$\text{MP} = \frac{\text{CP}}{0.6} = \frac{5}{3}\text{CP}$$

Markup:

$$\frac{5}{3} \times 100 = \boxed{66\frac{2}{3}\%}$$

41. FIND MARKUP FOR 20% PROFIT AFTER TWO DISCOUNTS

Same discounts:

$$20\%, 25\%$$

retention:

$$0.6$$

Target:

$$[SP=1.2CP]$$

Therefore:

$$[MP=\frac{1.2}{0.6}CP = 2CP]$$

Required markup:

$$[\boxed{100\%}]$$

42. PRICE REDUCTION VS DISCOUNT PERCENTAGE

Marked price drops from ₹ 2000 to ₹ 1500 before any sale.

That is a 25% reduction in marked price.

If seller then gives 20% discount on new MP:

$$[1500(0.8)=1200]$$

Overall reduction from original ₹ 2000 :

$$[800]$$

$$[800/2000=40\%]$$

Equivalent to:

$$[25\%\text{ then }20\%]$$

$$[25+20-5=40\%]$$

43. OFFER “20% OFF + EXTRA 20% OFF”

This means:

$$[20\%\text{ then }20\%]$$

not 40%.

Equivalent:

$$[20+20-4 = \boxed{36\%}]$$

Customer pays:

$$[64\%]$$

44. “UP TO 50% OFF”

This phrase does **not** mean every item gets 50% discount.

In aptitude problems, only use the exact discount actually specified for the item.

45. GATE-STYLE QUESTION 1

Successive discounts 20% and 10%.

Equivalent discount:

$$\boxed{20+10-2 = 28\%}$$

46. GATE-STYLE QUESTION 2

Two successive 25% discounts.

Retention:

$$0.75^2 = 0.5625$$

Discount:

$$43.75\%$$

$$\boxed{43.75\%}$$

47. GATE-STYLE QUESTION 3

MP ₹ 2000.

Discounts 20% and 25%.

SP:

$$2000(0.8)(0.75)$$

$$= 2000(0.6)$$

$$= \boxed{1200}$$

48. GATE-STYLE QUESTION 4

First discount 20%. Equivalent total discount is 36%.

Find second discount.

Retention overall:

$$64\%$$

After first:

$$80\%$$

Second retention:

$$64/80 = 0.8$$

Second discount:

$$\boxed{20\%}$$

49. PLACEMENT-STYLE QUESTION 1

Marked price ₹ 3000.

Discount 30% then 20%.

$$SP = 3000(0.7)(0.8)$$

$$V = 3000(0.56)^t$$

$$V = \boxed{1680}$$

50. PLACEMENT-STYLE QUESTION 2

Offer A: 40% off.

Offer B: 25% + 20% off.

B equivalent:

$$V = 25 + 20 - 5 = 40\%$$

Therefore:

$$V = \boxed{\text{Both equivalent}}$$

51. PLACEMENT-STYLE QUESTION 3

Buy 2 get 1 free.

Effective discount:

$$V = 1/3$$

$$V = \boxed{33\frac{1}{3}\%}$$

52. PLACEMENT-STYLE QUESTION 4

20% trade discount followed by 10% cash discount.

Equivalent:

$$V = 20 + 10 - 2 = \boxed{28\%}$$

53. ADVANCED QUESTION — MARKUP + DOUBLE DISCOUNT

Item marked 100% above CP.

Discounts:

$$V = 20\%, 30\%$$

Net:

$$V = 2 \times 0.8 \times 0.7 = 1.12$$

Profit:

$$V = \boxed{12\%}$$

54. ADVANCED QUESTION — REQUIRED SECOND DISCOUNT

Markup = 50%.

First discount = 20%.

Want exactly 20% profit.

After markup and first discount:

$$1.5 \times 0.8 = 1.2$$

Already at target.

Therefore second discount must be:

$$\boxed{0\%}$$

55. ADVANCED QUESTION — REQUIRED SECOND DISCOUNT FOR BREAK-EVEN

Markup = 50%.

First discount = 20%.

Current factor:

$$1.2$$

Need final:

$$1$$

Second retained factor:

$$1/1.2 = \frac{5}{6}$$

Discount:

$$1 - \frac{5}{6} = \frac{1}{6}$$

$$= \boxed{16\frac{2}{3}\%}$$

56. ADVANCED QUESTION — SAME FINAL SP

Item A:

MP = 1500, discount 20%.

$$SP_A = 1200$$

Item B:

discount 25%, same SP.

$$0.75MP_B = 1200$$

$$MP_B = \boxed{1600}$$

57. ADVANCED QUESTION — THREE OFFERS

Compare:

A. 50% off B. 30% then 30% off C. Buy 1 get 1 free

A:

$$\lfloor 50\% \rfloor$$

B:

$$\lfloor 30+30-9=51\% \rfloor$$

C:

$$\lfloor 50\% \rfloor$$

So:

$$\lfloor \boxed{\text{B gives the largest nominal discount: } 51\%} \rfloor$$

58. COMMON TRAPS

& pTRAP 1 — Add successive discounts

Wrong unless one of them is 0.

& pTRAP 2 — $20\% + 20\% = 40\%$

Actual:

$$\lfloor 36\% \rfloor$$

& pTRAP 3 — Buy 2 get 1 free = 50%

Wrong.

$$\lfloor 1/3=33\frac{1}{3}\% \rfloor$$

& pTRAP 4 — Free offer always best

Depends on how many units you need and whether conditions are equivalent.

& pTRAP 5 — Same SP with different discounts

The marked prices must differ.

59. MUST-MEMORIZE FORMULAS

Two successive discounts

$$\lfloor \boxed{D=a+b-\frac{ab}{100}} \rfloor$$

Multiple discounts

$$\lfloor \boxed{SP=MP\prod\left(1-\frac{d_i}{100}\right)} \rfloor$$

Buy $\lfloor(x)\rfloor$, get $\lfloor(y)\rfloor$ free

$$\lfloor \boxed{\text{Effective discount} = \frac{y}{x+y}\times 100} \rfloor$$

60. MASTER MARKUP + MULTI-DISCOUNT FORMULA

$$\boxed{\frac{SP}{CP} = \left(1 + \frac{m}{100}\right) \prod_i \left(1 - \frac{d_i}{100}\right)}$$

Then:

- $\frac{SP}{CP} > 1$!' profit
 - $\frac{SP}{CP} = 1$!' break-even
 - $\frac{SP}{CP} < 1$!' loss
-

61. FAST EXAM DECISION TREE

Two discounts?

Use $\frac{a+b-ab}{100}$ or multiply retention factors.

Three or more?

Use multipliers.

Free-item offer?

Convert to effective discount.

Markup + discounts?

Start from CP factor 1 and multiply everything.

Need missing discount?

Divide target retained factor by known retained factors.

Compare offers?

Compare final payable fraction.

PRACTICE — LEVEL 0

Q1

Successive discounts 10% and 20%. Find equivalent discount.

Q2

Two 20% discounts. Find equivalent discount.

Q3

MP 1100, discounts 20% and 10%. Find SP.

Q4

Buy 3 get 1 free. Find effective discount.

Q5

Buy 1 get 1 free. Find effective discount.

LEVEL 1

Q6

Successive discounts 30% and 20%. Find equivalent discount.

Q7

First discount is 20%, total equivalent discount is 40%. Find second discount.

Q8

MP 1250. Discounts 20% and 25%. Find final SP.

Q9

Markup 50%, discounts 20% and 10%. Find profit/loss %.

Q10

Compare 30% single discount with successive 20% and 15% discounts.

LEVEL 2

Q11

Three discounts 10%, 20%, 25%. Find equivalent discount.

Q12

An article is marked 100% above CP and gets successive discounts 25% and 20%. Find profit/loss %.

Q13

A seller gives discounts of 20% and 10% and wants 20% profit. Find required markup.

Q14

Buy 2 get 1 free plus 10% further discount on amount paid. Find effective discount.

Q15

Two products have same SP. Product A has 20% discount, product B has 25% discount. Find MP ratio (A:B).

LEVEL 3 — EXAM PRESSURE

Q16

Markup 60%. First discount 20%. Find second discount needed for break-even.

Q17

Markup 80%. Discounts 25% and 20%. Find profit/loss %.

Q18

Final SP after 20%, 10%, 25% successive discounts is 1108. Find MP.

Q19

Offer A = 50% off. Offer B = 30% then 30% off. Which is better and by how many percentage points?

Q20

A product is marked 100% above CP. What equal successive discount $(d\%)$ should be applied twice so final SP equals CP?

ANSWERS

Q1

$$\boxed{28\%}$$

Q2

$$\boxed{36\%}$$

Q3

$$\boxed{1720}$$

Q4

$$\boxed{25\%}$$

Q5

$$\boxed{50\%}$$

Q6

$$\boxed{44\%}$$

Q7

$$0.8(1-d)=0.6$$

$$1-d=0.75$$

$$\boxed{d=25\%}$$

Q8

$$2500(0.8)(0.75)$$

$$=2500(0.6)$$

$$\boxed{1500}$$

Q9

$$1.5(0.8)(0.9) = 1.08$$

$$\boxed{8\% \text{ profit}}$$

Q10

Successive offer:

$$20+15-3 = 32\%$$

So:

$$\boxed{\text{20\% + 15\% is better by 2 percentage points}}$$

Q11

Retention:

$$\sqrt[0.9(0.8)(0.75) = 0.54]{}$$

Discount:

$$\sqrt[46\%]{}$$

Q12

$$\sqrt[2(0.75)(0.8) = 1.2]{}$$

$$\sqrt[20\%\text{ profit}]{}$$

Q13

Retention:

$$\sqrt[0.72]{}$$

Need:

$$\sqrt[1.2]{}$$

Markup factor:

$$\sqrt[1.2/0.72 = \frac{53}{3}]{}$$

Markup:

$$\sqrt[66\frac{23}{3}\%]{}$$

Q14

Buy 2 get 1 free:

$$\sqrt[\text{retained}=2/3]{}$$

Additional 10% discount:

$$\sqrt[\frac{23}{3}(0.9)=0.6]{}$$

Equivalent discount:

$$\sqrt[40\%]{}$$

Q15

Same SP:

$$\sqrt[0.8MP_A=0.75MP_B]{}$$

$$\sqrt[MP_A:MP_B = 0.75:0.8]{}$$

$$\sqrt[15:16]{}$$

Q16

After markup and first discount:

$$\sqrt[1.6(0.8)=1.28]{}$$

Need:

$$\sqrt[1.28(1-d)=1]{}$$

$$\sqrt[1-d=\frac{1}{1.28} = 0.78125]{}$$

$$\lfloor d=0.21875 \rfloor$$

$$\lfloor \boxed{21.875\%} \rfloor$$

Q17

$$\lfloor 1.8(0.75)(0.8) = 1.08 \rfloor$$

$$\lfloor \boxed{8\% \text{ profit}} \rfloor$$

Q18

Retention:

$$\lfloor 0.8(0.9)(0.75) = 0.54 \rfloor$$

$$\lfloor MP=1080/0.54 \rfloor$$

$$\lfloor = \boxed{^12000} \rfloor$$

Q19

A:

$$\lfloor 50\% \rfloor$$

B:

$$\lfloor 30+30-9 = 51\% \rfloor$$

So B is better by:

$$\lfloor \boxed{1\% \text{ percentage point}} \rfloor$$

Q20

Markup 100% means:

$$\lfloor MP=2CP \rfloor$$

Need:

$$\lfloor 2(1-d)^2=1 \rfloor$$

$$\lfloor (1-d)^2=\frac{1}{2} \rfloor$$

$$\lfloor 1-d=\frac{1}{\sqrt{2}} \rfloor$$

$$\lfloor d = 1-\frac{1}{\sqrt{2}} \rfloor$$

$$\lfloor \approx 0.2929 \rfloor$$

$$\lfloor \boxed{d \approx 29.29\%} \rfloor$$

' CHAPTER 19 COMPLETE

We have now covered:

- marked/list price '
- discount amount and discount % '
- reverse discount '
- markup + discount '
- required markup / required discount '

- successive discounts '
- equivalent single discount '
- multiple discounts '
- free-item offers '
- trade/cash discount style problems '
- same-SP / different-discount problems '
- offer comparison '
- markup + multiple discounts '
- advanced GATE/placement problems '

Progress

Completed: 19/76 **Unit 1 — Quantitative Aptitude:** 0=0% **PROGRESS** **Next:** +1 Chapter 20 — **Simple Interest**

Next we will cover:

- principal
- rate
- time
- simple interest formula
- amount
- finding missing (P, R, T)
- monthly/quarterly time conversion
- interest comparisons
- interest as percentage of principal
- doubling/tripling under SI
- GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 20 — SIMPLE INTEREST

PART 1 — PRINCIPAL, RATE, TIME, SIMPLE INTEREST FORMULA & BASIC APPLICATIONS

Status: 0=0% **PROGRESS** **GATE:** Medium–High **Placements:** High **Priority:** +1 Tier 2

Simple Interest is one of the most direct arithmetic chapters.

The central idea is:

Interest is calculated only on the original principal throughout the entire time period.

That is why it is called **simple** interest.

1. BASIC TERMS

Principal — (P)

The original money borrowed or invested.

Rate — (R)

Interest rate per year, usually written as a percentage.

Time — (T)

Duration, normally in years unless stated otherwise.

Simple Interest — (SI)

Extra money earned or paid as interest.

Amount — (A)

Final money after adding interest.

$$\boxed{A = P + SI}$$

2. MASTER SIMPLE INTEREST FORMULA

$$\boxed{SI = \frac{PRT}{100}}$$

where:

- (P) = principal
- (R) = rate % per annum
- (T) = time in years

MUST MEMORIZE

3. BASIC EXAMPLE

Find simple interest on:

$$\pounds 5000$$

at:

$$10\% \text{ per annum}$$

for:

$$2 \text{ years}$$

$$SI = \frac{5000 \times 10 \times 2}{100}$$

$$= \pounds 1000$$

Therefore:

$$\boxed{SI = \pounds 1000}$$

Amount:

$$\boxed{5000+1000 = ₹ 6000}$$

4. WHY SIMPLE INTEREST IS LINEAR

Suppose ₹ 1000 @ 10%.

Every year interest is:

$$10\% \text{ of } 1000 = ₹ 100$$

So:

- 1 year ₹ 1100
- 2 years ₹ 1200
- 3 years ₹ 1300

Interest increases by the same amount every year.

Thus:

$$\boxed{SI \propto T}$$

when P and R are fixed.

5. ANOTHER VIEW

For one year:

$$SI = \frac{PR}{100}$$

For T years:

$$SI = T \left(\frac{PR}{100} \right)$$

So yearly interest is constant.

6. AMOUNT FORMULA

Since:

$$A = P + SI$$

and:

$$SI = \frac{PRT}{100}$$

we get:

$$\boxed{A = P \left(1 + \frac{RT}{100} \right)}$$

This is very useful.

7. EXAMPLE — FIND AMOUNT DIRECTLY

₹ 8000 @ 5% SI for 3 years.

$$A = 8000 \left(1 + \frac{5(3)}{100} \right)$$

$$[= 8000(1.15)]$$

$$[= \boxed{{}^{19}200}]$$

8. FIND PRINCIPAL

If:

$$[SI = \frac{PRT}{100}]$$

then:

$$[\boxed{P = \frac{100SI}{RT}}]$$

Example:

$$SI = {}^{11}200$$

$$\text{Rate} = 10\%$$

$$\text{Time} = 3 \text{ years}$$

$$[P = \frac{100(1200)}{10(3)}]$$

$$[= {}^{14}000]$$

$$[\boxed{P = {}^{14}000}]$$

9. FIND RATE

Rearrange:

$$[\boxed{R = \frac{100SI}{PT}}]$$

Example:

Principal:

$$[{}^{15}000]$$

Interest:

$$[{}^{11}500]$$

Time:

$$[5 \text{ text{ years}}]$$

$$[R = \frac{100(1500)}{5000(5)}]$$

$$[= 6]$$

$$[\boxed{6\% \text{ text{ p.a.}}}]$$

10. FIND TIME

$$[\boxed{T = \frac{100SI}{PR}}]$$

Example:

$$\text{Principal} = {}^{16}000$$

Rate = 8%

SI = ₹ 1440

$$T = \frac{100(1440)}{6000(8)}$$

$$= 3$$

$$\boxed{3 \text{ years}}$$

11. FAST PERCENTAGE METHOD

At simple interest:

$$SI = P \times \frac{RT}{100}$$

So over (T) years, total interest percentage is simply:

$$\boxed{R \times T\%}$$

Example:

12% p.a. for 5 years:

$$12 \times 5 = 60\%$$

Thus interest is:

$$60\% \text{ of principal}$$

No need to use full formula.

12. EXAMPLE

₹ 2500 at 8% for 4 years.

Total interest percentage:

$$8 \times 4 = 32\%$$

So:

$$SI = 32\% \text{ of } ₹ 2500$$

$$= ₹ 800$$

$$\boxed{₹ 3300}$$

13. AMOUNT AS A PERCENTAGE OF PRINCIPAL

At SI rate $(R\%)$ for (T) years:

$$A = P \left(1 + \frac{RT}{100} \right)$$

Therefore amount is:

$$\boxed{(100 + RT)\% \text{ of principal}}$$

Example:

10% for 3 years:

$$\text{\[100+30=130\% \]}$$

So:

$$\text{\[A=1.3P \]}$$

14. IF INTEREST IS x% OF PRINCIPAL

Suppose SI is:

$$\text{\[40\%\text{ of principal} \]}$$

Then:

$$\text{\[RT=40 \]}$$

Example:

Rate = 8%.

$$\text{\[8T=40 \]}$$

$$\text{\[T=5 \]}$$

$$\text{\[\boxed{5\text{ years}} \]}$$

15. TIME IN MONTHS

The formula expects time in years.

So:

$$\text{\[\boxed{T=\frac{\text{months}}{12}} \]}$$

Example:

¹ 12000 at 10% for 6 months.

$$\text{\[T=\frac{6}{12}=\frac{1}{2} \]}$$

$$\text{\[SI=\frac{12000(10)(1/2)}{100} \]}$$

$$\text{\[=\boxed{600} \]}$$

16. TIME IN DAYS

If a problem uses days:

$$\text{\[\boxed{T=\frac{\text{number of days}}{365}} \]}$$

unless the problem explicitly specifies a 360-day commercial year or another convention.

Example:

¹ 3650 at 10% for 73 days.

$$\text{\[T=\frac{73}{365}=\frac{1}{5} \]}$$

$$\text{\[SI=\frac{3650(10)(1/5)}{100} \]}$$

$$\text{\[=\boxed{73} \]}$$

17. IMPORTANT TIME-UNIT RULE

Rate and time must use compatible units.

If rate is:

$12\% \text{ per annum}$

time must be in years.

If rate were explicitly per month, time should be in months.

18. MONTHLY RATE

Suppose rate is 1% per month for 8 months.

Then simply:

$SI = 8\% \text{ of principal}$

Example:

$P = ₹10000$

$SI = 0.08(10000) = ₹800$

Do not convert to annual rate unless necessary.

19. QUARTERLY RATE

If rate is 3% per quarter and time is 2 years:

Number of quarters:

$2 \times 4 = 8$

Total SI rate:

$3 \times 8 = 24\%$

Thus:

$SI = 24\% \text{ of principal}$

20. SIMPLE INTEREST IS DIRECTLY PROPORTIONAL TO PRINCIPAL

For fixed (R, T) :

$SI \propto P$

So if principal doubles:

$SI \text{ doubles}$

If principal ratio is:

$2:5$

interest ratio is also:

$\backslash 2:5 \backslash$

21. SIMPLE INTEREST IS DIRECTLY PROPORTIONAL TO RATE

For fixed $\backslash(P, T)\backslash$:

$\backslash SI \propto R \backslash$

If rate triples:

$\backslash SI \text{ triples} \backslash$

22. SIMPLE INTEREST IS DIRECTLY PROPORTIONAL TO TIME

For fixed $\backslash(P, R)\backslash$:

$\backslash SI \propto T \backslash$

If time ratio is:

$\backslash 3:7 \backslash$

interest ratio:

$\backslash \boxed{3:7} \backslash$

23. COMBINED PROPORTIONALITY

Since:

$\backslash SI = \frac{PRT}{100} \backslash$

we can write:

$\backslash \boxed{SI \propto PRT} \backslash$

Thus for two cases:

$\backslash \boxed{\frac{SI_1}{SI_2} = \frac{P_1 R_1 T_1}{P_2 R_2 T_2}} \backslash$

Very useful for comparison questions.

24. EXAMPLE

Case 1:

$\backslash P=1000, R=10, T=2 \backslash$

Case 2:

$\backslash P=2000, R=5, T=3 \backslash$

Interest ratio:

$\backslash 1000(10)(2):2000(5)(3) \backslash$

Cancel:

$\backslash 20:30 \backslash$

$$\boxed{2:3}$$

No actual interest calculation required.

25. FIND INTEREST FROM AMOUNT

If:

$$A = P + SI$$

then:

$$\boxed{SI = A - P}$$

Example:

Principal ₹ 4000 becomes ₹ 5200.

$$SI = 1200$$

If time = 3 years:

$$R = \frac{100(1200)}{4000(3)}$$

$$\boxed{10\%}$$

26. FIND PRINCIPAL FROM AMOUNT

Example:

Amount after 5 years at 10% SI is ₹ 9000.

For 5 years, interest percentage:

$$50\%$$

So amount is:

$$150\% \text{ of principal}$$

$$1.5P = 9000$$

$$\boxed{P = ₹ 6000}$$

This is faster than forming the full equation.

27. GENERAL REVERSE AMOUNT FORMULA

$$A = P \left(1 + \frac{RT}{100} \right)$$

Therefore:

$$\boxed{P = \frac{A}{1 + \frac{RT}{100}}}$$

or:

$$\boxed{P = \frac{100A}{100 + RT}}$$

28. EXAMPLE

Amount = ₹ 7800

Rate = 10%

Time = 3 years.

$$P = \frac{100(7800)}{130}$$

$$= \boxed{₹ 6000}$$

29. AMOUNT DIFFERENCE BETWEEN TWO TIMES

At simple interest, yearly increase is constant.

Suppose amount after 3 years is ₹ 6200 and after 5 years is ₹ 7000.

Difference for 2 years:

$$7000 - 6200 = 800$$

Thus yearly interest:

$$800 / 2 = 400$$

$$\boxed{₹ 400 \text{ per year}}$$

30. FIND PRINCIPAL FROM TWO AMOUNTS

Continuing:

Annual interest:

$$₹ 400$$

Amount after 3 years:

$$₹ 6200$$

Interest for 3 years:

$$₹ 1200$$

Principal:

$$6200 - 1200$$

$$= \boxed{₹ 5000}$$

31. FIND RATE FROM TWO AMOUNTS

From above:

$$P = 5000$$

Annual interest:

$$₹ 400$$

Rate:

$$\frac{400}{5000} \times 100$$

$$\boxed{8\%}$$

32. MASTER “TWO AMOUNTS” SHORTCUT

If amount changes from (A_1) at time (T_1) to (A_2) at time (T_2) :

Annual interest:

$$\boxed{I_{\text{year}}} = \frac{A_2 - A_1}{T_2 - T_1}$$

Then:

$$P = A_1 - T_1 I_{\text{year}}$$

Then:

$$R = \frac{100 I_{\text{year}}}{P}$$

This is a high-value exam pattern.

33. EXAMPLE

Amount after 4 years:

$$\boxed{17400}$$

Amount after 7 years:

$$\boxed{18600}$$

Difference:

$$\boxed{1200}$$

Time difference:

$$\boxed{3}$$

Annual interest:

$$\boxed{400}$$

Principal:

$$\boxed{7400 - 4(400)}$$

$$\boxed{=5800}$$

Rate:

$$\boxed{\frac{400}{5800} \times 100}$$

$$\boxed{\approx \boxed{6.90\%}}$$

34. WHEN DOES MONEY DOUBLE UNDER SI?

If money doubles:

$$\boxed{A=2P}$$

Then:

$$SI = P$$

So interest is 100% of principal.

Thus:

$$RT = 100$$

Therefore:

$$\boxed{T = \frac{100}{R}}$$

35. EXAMPLE — DOUBLE

At 10% SI:

$$T = \frac{100}{10}$$

$$\boxed{T = 10 \text{ years}}$$

36. WHEN DOES MONEY TRIPLE UNDER SI?

If:

$$A = 3P$$

then:

$$SI = 2P$$

So interest = 200% of principal.

Therefore:

$$RT = 200$$

$$\boxed{T = \frac{200}{R}}$$

Example at 8%:

$$T = 25 \text{ years}$$

37. GENERAL MULTIPLICATION OF PRINCIPAL

If amount becomes kP :

$$A = kP$$

Then:

$$SI = (k-1)P$$

So:

$$RT = 100(k-1)$$

Therefore:

$$\boxed{T = \frac{100(k-1)}{R}}$$

38. RATIO OF PRINCIPAL TO AMOUNT

At rate $(R\%)$ for time (T) :

$$A:P = (100+RT):100$$

So:

$$\boxed{P:A = 100:(100+RT)}$$

Example:

10% for 4 years:

$$P:A=100:140=5:7$$

39. INTEREST : PRINCIPAL RATIO

$$SI:P = RT:100$$

Example:

12% for 5 years:

$$SI:P=60:100=3:5$$

40. INTEREST : AMOUNT RATIO

Since:

$$SI:P=RT:100$$

and:

$$A=P+SI$$

$$\boxed{SI:A = RT:(100+RT)}$$

Example:

10% for 3 years:

$$SI:A=30:130=3:13$$

41. RATE CHANGES

A sum earns ₹ 1200 in 4 years at 6%.

What interest would it earn in 4 years at 9%?

Same (P, T) :

$$SI \propto R$$

So:

$$SI_2 = 1200 \times \frac{96}{60}$$

$$= \boxed{1800}$$

No need to find principal.

42. TIME CHANGES

A sum earns ₹ 900 in 3 years.

At same rate, what interest in 5 years?

$$SI_2 = 900 \times \frac{5}{3}$$

$$= \boxed{₹ 1500}$$

43. PRINCIPAL CHANGES

₹ 5000 earns ₹ 800 SI.

At same rate and time, ₹ 7500 earns:

$$800 \times \frac{7500}{5000}$$

$$= 800 \times 1.5$$

$$= \boxed{₹ 1200}$$

44. CHANGES IN P, R, T TOGETHER

If:

- principal doubles,
- rate halves,
- time triples,

then interest multiplier:

$$2 \times \frac{1}{2} \times 3 = 3$$

So:

$$\boxed{SI \text{ triples}}$$

45. GATE-STYLE QUESTION 1

Find SI on ₹ 6000 at 5% for 4 years.

$$SI = \frac{6000(5)(4)}{100}$$

$$= \boxed{₹ 1200}$$

46. GATE-STYLE QUESTION 2

₹ 5000 becomes ₹ 6500 in 5 years under SI.

Interest:

$$₹ 1500$$

Rate:

$$\left[\frac{100(1500)}{5000(5)} \right]$$

$$\left[=\boxed{6\%} \right]$$

47. GATE-STYLE QUESTION 3

A sum earns 24% of principal as SI in 3 years.

Rate:

$$\left[R(3)=24 \right]$$

$$\left[\boxed{R=8\%} \right]$$

48. GATE-STYLE QUESTION 4

At 12% SI, how long until principal becomes 1.6 times?

Amount:

$$\left[1.6P \right]$$

Interest:

$$\left[0.6P=60\%\text{ of }P \right]$$

$$\left[12T=60 \right]$$

$$\left[\boxed{T=5\text{ years}} \right]$$

49. PLACEMENT-STYLE QUESTION 1

₹ 8000 at 7.5% for 2 years.

$$\left[SI=15\%\text{ of }8000 \right]$$

$$\left[=\boxed{1200} \right]$$

Amount:

$$\left[\boxed{9200} \right]$$

50. PLACEMENT-STYLE QUESTION 2

SI = ₹ 800 at 12% for 3 years.

Principal:

$$\left[P= \frac{100(800)}{12(3)} \right]$$

$$\left[=\boxed{5000} \right]$$

51. PLACEMENT-STYLE QUESTION 3

₹ 1200 at 10% for 9 months.

$$T = \frac{9}{12} = \frac{3}{4}$$

Total interest percentage:

$$10 \times \frac{3}{4} = 7.5\%$$

$$SI = 12000(0.075)$$

$$= \boxed{900}$$

52. PLACEMENT-STYLE QUESTION 4

Amount after 2 years is ₹ 6600 and after 5 years is ₹ 7500.

Difference:

$$900$$

for:

$$3 \text{ years}$$

Yearly interest:

$$300$$

Principal:

$$6600 - 2(300)$$

$$= \boxed{6000}$$

Rate:

$$300/6000 \times 100 = \boxed{5\%}$$

53. COMMON TRAPS

TRAP 1 — Time units

Months must usually be divided by 12 for annual rates.

TRAP 2 — Amount vs interest

$$A = P + SI$$

Do not confuse them.

TRAP 3 — Interest on previous interest

That is compound interest, not simple interest.

TRAP 4 — Rate × time

At SI, total percentage interest is exactly:

$$RT\%$$

TRAP 5 — Doubling

Money doubles when interest equals principal, not when interest is 200%.

54. MUST-MEMORIZE FORMULAS

Simple interest

$$\boxed{SI = \frac{PRT}{100}}$$

Amount

$$\boxed{A = P \left(1 + \frac{RT}{100} \right)}$$

Principal

$$\boxed{P = \frac{100SI}{RT}}$$

Rate

$$\boxed{R = \frac{100SI}{PT}}$$

Time

$$\boxed{T = \frac{100SI}{PR}}$$

55. HIGH-VALUE SHORTCUTS

Total interest percentage

$$\boxed{RT\%}$$

Double under SI

$$\boxed{T = \frac{100}{R}}$$

Triple under SI

$$\boxed{T = \frac{200}{R}}$$

Two amounts

$$\boxed{\text{annual interest} = \frac{A_2 - A_1}{T_2 - T_1}}$$

56. FAST EXAM DECISION TREE

P, R, T given?

Use $\frac{PRT}{100}$.

Amount given?

Subtract principal, or use amount multiplier.

Missing P/R/T?

Rearrange formula.

Months?

Convert to years.

Compare two SI cases?

Use:

$\frac{SI}{\text{propto } PRT}$

Two amounts at different times?

Find yearly interest from their difference.

Double/triple?

Convert target amount into required interest percentage.

PRACTICE — LEVEL 0

Q1

Find SI on ₹ 1000 at 10% for 3 years.

Q2

Find amount on ₹ 500 at 8% SI for 2 years.

Q3

SI is ₹ 600 on ₹ 3000 for 4 years. Find rate.

Q4

SI is ₹ 1500 at 10% for 3 years. Find principal.

Q5

At 5% SI, how long does ₹ 4000 earn ₹ 1000?

LEVEL 1

Q6

Find SI on ₹ 1200 at 8% for 9 months.

Q7

₹ 8000 becomes ₹ 10400 in 5 years. Find SI rate.

Q8

At 12% SI, find time required to earn interest equal to 60% of principal.

Q9

A sum earns ₹ 900 in 3 years. At same rate, find interest for 7 years.

Q10

Interest on ₹ 5000 at 8% for $\frac{1}{T}$ years is ₹ 1600. Find $\frac{1}{T}$.

LEVEL 2

Q11

A sum becomes ₹ 1720 in 4 years at 5% SI. Find principal.

Q12

Amount after 3 years is ₹ 6800 and after 7 years is ₹ 8400. Find principal and rate.

Q13

At what SI rate will money double in 8 years?

Q14

At 6% SI, how long will money become 2.5 times?

Q15

Principal doubles, rate becomes half, and time becomes twice. How does SI change?

LEVEL 3 — EXAM PRESSURE

Q16

A sum at 8% SI amounts to ₹ 6200 in 3 years. Find the sum.

Q17

A sum amounts to ₹ 7800 in 4 years and ₹ 9000 in 7 years under SI. Find rate.

Q18

If SI on a sum for 6 years is $\frac{3}{5}$ of principal, find rate.

Q19

At 10% SI, an amount is ₹ 1400 after 4 years. Find interest after 7 years on the same principal.

Q20

A sum earns ₹ 1500 at 5% for some time. If rate were 7.5% and time unchanged, what would SI be?

ANSWERS

Q1

$$\frac{1000(10)(3)}{100} = \boxed{1300}$$

Q2

SI:

$$\frac{5000(8)(2)}{100} = 800$$

Amount:

$$\boxed{15800}$$

Q3

$$R = \frac{100(600)}{3000(4)}$$

$$= 5\%$$

Q4

$$P = \frac{100(1500)}{10(3)}$$

$$= 15000$$

Q5

$$1000 = \frac{4000(5)T}{100}$$

$$1000 = 200T$$

$$T = 5 \text{ years}$$

Q6

$$T = \frac{34}{1}$$

$$SI = 12000(8)(\frac{3}{4})/100$$

$$= 7200$$

Q7

Interest:

$$10400 - 8000 = 2400$$

$$R = \frac{100(2400)}{8000(5)}$$

$$= 6\%$$

Q8

$$12T = 60$$

$$T = 5 \text{ years}$$

Q9

$$900 \times \frac{73}{100} = 657$$

Q10

$$1600 = \frac{5000(8)T}{100}$$

$$1600 = 400T$$

$$T = 4 \text{ years}$$

Q11

Amount multiplier:

$$1 + \frac{5(4)}{100} = 1.2$$

$$P = 7200/1.2$$

$$= 6000$$

Q12

Difference in amounts:

$$\boxed{8400-6800=1600}$$

Difference in time:

$$\boxed{4\text{ years}}$$

Annual interest:

$$\boxed{400}$$

Principal:

$$\boxed{6800-3(400)}$$

$$= \boxed{5600}$$

Rate:

$$\frac{400}{5600} \times 100 = \boxed{\frac{50}{7}\% \approx 7.14\%}$$

Q13

For doubling:

$$\boxed{RT=100}$$

$$\boxed{8R=100}$$

$$\boxed{R=12.5\%}$$

Q14

Amount:

$$\boxed{2.5P}$$

Interest:

$$\boxed{1.5P=150\%P}$$

$$\boxed{6T=150}$$

$$\boxed{25\text{ years}}$$

Q15

Multiplier:

$$\boxed{2 \times \frac{12}{1} \times 2 = 2}$$

$$\boxed{SI\text{ doubles}}$$

Q16

Amount factor:

$$\boxed{1 + \frac{8(3)}{100} = 1.24}$$

$$\boxed{P=6200/1.24}$$

$$= \boxed{5000}$$

Q17

Annual interest:

$$\boxed{\frac{9000-7800}{7-4} = 400}$$

Principal:

$$7800 - 4(400) = 6200$$

Rate:

$$\frac{400}{6200} \times 100$$

$$= \boxed{\frac{200}{31} \% \approx 6.45 \%}$$

Q18

$$SI = \frac{35P}{100}$$

Therefore:

$$RT = 60$$

$$6R = 60$$

$$\boxed{R = 10 \%}$$

Q19

After 4 years:

$$A = 1.4P$$

$$P = 14000 / 1.4 = 10000$$

Interest after 7 years:

$$10000(10)(7)/100$$

$$= \boxed{7000}$$

Q20

Same principal and time:

$$SI \propto R$$

$$SI_2 = 1500 \times \frac{7.5}{5}$$

$$= 1500(1.5)$$

$$= \boxed{2250}$$

CHAPTER 20 STATUS

Covered so far:

- principal, rate, time, SI, amount '
- $(PRT/100)$ formula '
- finding missing (P, R, T) '
- amount multiplier '
- month/day conversion '
- direct proportionality '
- comparison shortcuts '
- two-amount problems '
- doubling/tripling '

- ratio relations '
- GATE/placement practice '

But **Chapter 20 is not complete yet.**

Next we need:

- different rates over different time intervals
- split principal / multiple investments
- equal-interest conditions
- rate/time comparison problems
- average rate under SI
- loans and installment-style SI basics
- advanced two-sum problems
- final mixed GATE/placement drills

Progress

Completed: 19/76 **Current:** $\emptyset$ = Chapter 20 — Simple Interest **Next:** Chapter 20 Part 2 — **Split investments, changing rates & advanced SI problems**

USER

next

ASSISTANT

CHAPTER 20 — SIMPLE INTEREST

PART 2 — SPLIT INVESTMENTS, CHANGING RATES, EQUAL-INTEREST CONDITIONS & ADVANCED SI PROBLEMS

Status: $\emptyset$ = No PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

This part handles the more exam-oriented Simple Interest questions where:

- the rate changes,
- money is divided into parts,
- different sums earn equal interest,
- two investments must be compared,
- an average rate is hidden,
- or the question gives only relationships instead of direct values.

The central rule still remains:

$$\boxed{SI = \frac{PRT}{100}}$$

1. DIFFERENT RATES FOR DIFFERENT TIME PERIODS

Suppose $^1 10,000$ is invested:

- 2 years at 5%
- next 3 years at 8%

Under simple interest, calculate each interval separately.

First period:

$$SI_1 = \frac{10000(5)(2)}{100} = 1000$$

Second period:

$$SI_2 = \frac{10000(8)(3)}{100} = 2400$$

Total:

$$\boxed{13400}$$

2. IMPORTANT SIMPLE-INTEREST RULE

Even when rate changes, interest is still calculated on the **original principal**.

So:

$$\boxed{SI = P \frac{R_1 T_1 + R_2 T_2 + \dots}{100}}$$

provided the principal remains unchanged.

3. EXAMPLE

18000 earns:

- 6% for 2 years
- 10% for 3 years

$$SI = 8000 \frac{6(2) + 10(3)}{100}$$

$$= 8000 \frac{42}{100}$$

$$= \boxed{3360}$$

4. EFFECTIVE AVERAGE RATE UNDER SI

If the rate changes over time, an average annual rate can be found.

For rates (R_1, R_2) lasting (T_1, T_2) :

$$\boxed{R_{\text{avg}}} = \frac{R_1 T_1 + R_2 T_2}{T_1 + T_2}$$

This is a **time-weighted average**.

5. EXAMPLE

2 years at 5%, then 3 years at 10%.

Average SI rate:

$$R_{\text{avg}} = \frac{5(2) + 10(3)}{5}$$

$$I = \frac{40}{5}$$

$$I = \boxed{8\%}$$

Check:

5 years at 8% gives:

$$I = 40\% \text{ total interest}$$

Same as:

$$I = 10\% + 30\% = 40\%$$

6. EQUAL TIME PERIODS

If rates apply for equal durations, average rate is ordinary arithmetic mean.

Example:

2 years at 6% and 2 years at 10%.

Average:

$$I = \frac{6+10}{2} = \boxed{8\%}$$

7. UNEQUAL TIME PERIOD TRAP

1 year at 5% and 4 years at 10%.

Do not use:

$$I = (5+10)/2 = 7.5\%$$

Correct:

$$I_{\text{avg}} = \frac{5(1) + 10(4)}{5}$$

$$I = \frac{45}{5}$$

$$I = \boxed{9\%}$$

8. SPLITTING A PRINCIPAL

Suppose total money = 10,000 is split into two parts:

- one at 5%
- other at 10%

for the same time.

If first part is x :

$$\text{second part} = 10000 - x$$

Total interest is:

$$I = \frac{x(5)T}{100} + \frac{(10000-x)(10)T}{100}$$

This is the standard setup.

9. EXAMPLE — SPLIT PRINCIPAL

₹ 8000 is divided into two parts.

One part earns 5%, other 10%, both for 1 year.

Total interest = ₹ 600 .

Let first part = x .

$$[0.05x + 0.10(8000 - x) = 600]$$

$$[0.05x + 800 - 0.10x = 600]$$

$$[-0.05x = -200]$$

$$[x = 4000]$$

So both parts:

$$[\boxed{₹ 4000}]$$

10. FASTER ALLIGATION METHOD

This is a weighted-average problem.

Overall interest:

$$[600/8000 \times 100 = 7.5\%]$$

So money is split between:

$$[5\%, 10\%]$$

to get average:

$$[7.5\%]$$

By alligation:

$$[5\% \text{ part} : 10\% \text{ part}]$$

$$[= (10 - 7.5) : (7.5 - 5)]$$

$$[= 2.5 : 2.5]$$

$$[= \boxed{1:1}]$$

Therefore each is ₹ 4000 .

11. HIGH-VALUE CONNECTION

Splitting money among different SI rates for the same time is just a **weighted-average/alligation** problem.

If overall rate R lies between R_1 and R_2 :

$$[\boxed{P_1 : P_2 = (R_2 - R) : (R - R_1)}]$$

12. EXAMPLE

$\frac{1}{3}$ invested partly at 6% and partly at 10%.

Overall SI rate = 7%.

Ratio:

$$P_6:P_{10} = (10-7):(7-6)$$

$$=3:1$$

Total 4 parts:

$$12000/4=3000$$

So:

$$\boxed{\text{\text{Rs } 9000 \text{ at } 6\% \text{ and } 3000 \text{ at } 10\%}}$$

13. WHEN TIMES ARE DIFFERENT TOO

If portions have both different rates **and** different times, compare:

$$RT$$

because:

$$SI \propto P(RT)$$

Example:

One part at 5% for 2 years:

$$RT=10$$

Other at 8% for 3 years:

$$RT=24$$

So their effective interest percentages are 10% and 24%.

You may alligate using those if total interest percentage is known.

14. EQUAL INTEREST CONDITIONS

Suppose two sums earn equal SI.

Then:

$$\frac{P_1R_1T_1}{100} = \frac{P_2R_2T_2}{100}$$

Thus:

$$\boxed{P_1R_1T_1 = P_2R_2T_2}$$

15. PRINCIPAL RATIO FOR EQUAL SI

If two investments earn equal interest:

$$\boxed{P_1:P_2 = R_2T_2:R_1T_1}$$

The (RT) factors reverse.

16. EXAMPLE

Two sums earn equal SI.

First:

$$[R=5\%, T=4]$$

Second:

$$[R=8\%, T=2]$$

Then:

$$[P_1:P_2 = 8(2):5(4)]$$

$$[=16:20]$$

$$[=\boxed{4:5}]$$

17. EQUAL PRINCIPAL, EQUAL INTEREST

If principals are equal and interests are equal:

$$[R_1T_1=R_2T_2]$$

Thus:

$$[\boxed{R_1:R_2=T_2:T_1}]$$

18. EXAMPLE

Same principal earns equal SI:

- first at 6% for 5 years
- second at $(R)\%$ for 3 years

$$[6(5)=R(3)]$$

$$[R=10]$$

$$[\boxed{10\%}]$$

19. SAME RATE, EQUAL INTEREST

If rate is same:

$$[P_1T_1=P_2T_2]$$

Therefore:

$$[\boxed{P_1:P_2=T_2:T_1}]$$

20. SAME TIME, EQUAL INTEREST

If time is same:

$$\boxed{P_1 R_1 = P_2 R_2}$$

So:

$$\boxed{P_1 : P_2 = R_2 : R_1}$$

21. INTEREST RATIO

For two SI investments:

$$\boxed{SI_1 : SI_2 = P_1 R_1 T_1 : P_2 R_2 T_2}$$

Example:

Principal ratio:

$$2 : 3$$

Rate ratio:

$$4 : 5$$

Time ratio:

$$3 : 2$$

Then interest ratio:

$$2(4)(3) : 3(5)(2)$$

$$24 : 30$$

$$\boxed{4 : 5}$$

22. FIND UNKNOWN RATE USING INTEREST RATIO

Suppose:

$$P_1 : P_2 = 2 : 3$$

$$T_1 : T_2 = 1 : 2$$

and:

$$SI_1 : SI_2 = 1 : 3$$

Find:

$$R_1 : R_2$$

Using:

$$1 : 3 = 2R_1(1) : 3R_2(2)$$

$$\frac{1}{3} = \frac{2R_1}{6R_2} = \frac{R_1}{3R_2}$$

So:

$$R_1 = R_2$$

$$\boxed{1 : 1}$$

23. TWO SUMS WHOSE TOTAL IS GIVEN

Two sums total ₹ 10,000.

One earns 5%, the other 8%.

Total SI after one year = ₹ 650.

Let first sum = ₹ x .

$$0.05x + 0.08(10000 - x) = 650$$

$$0.05x + 800 - 0.08x = 650$$

$$-0.03x = -150$$

$$x = 5000$$

So:

$$\boxed{\text{₹ } 5000 \text{ and } \text{₹ } 5000}$$

24. FAST ALLIGATION

Overall rate:

$$650/10000 \times 100 = 6.5\%$$

Between:

$$5\%, 8\%$$

Ratio:

$$(8 - 6.5) : (6.5 - 5)$$

$$1.5 : 1.5$$

$$\boxed{1:1}$$

25. ANOTHER SPLIT EXAMPLE

₹ 20,000 invested partly at 5% and partly at 9%.

Overall interest in one year = ₹ 1400.

Overall rate:

$$1400/20000 \times 100 = 7\%$$

Ratio:

$$5\% \text{ part} : 9\% \text{ part}$$

$$(9 - 7) : (7 - 5)$$

$$2 : 2$$

$$1 : 1$$

So:

$$\boxed{\text{₹ } 10,000 \text{ each}}$$

26. NON-EQUAL SPLIT EXAMPLE

₹ 20,000 partly at 5%, partly at 10%.

Total annual SI = ₹ 1600.

Overall rate:

$$8\%$$

Ratio:

$$5\% \text{ part} : 10\% \text{ part}$$

$$= (10-8) : (8-5)$$

$$= 2 : 3$$

So total 5 parts:

$$₹ 4000$$

Amounts:

$$\boxed{₹ 8000 \text{ and } ₹ 12000}$$

27. MONEY INVESTED AT THREE RATES

For 3 or more rates, use equations/totals unless more information is given.

One overall average rate alone is generally not enough to uniquely determine three unknown portions.

Data sufficiency trap

One equation cannot determine multiple independent unknowns.

28. DIFFERENCE IN INTEREST DUE TO RATE CHANGE

A sum earns ₹ 400 more when rate increases from 8% to 10% for 2 years.

Difference in rate:

$$2\%$$

for 2 years gives extra:

$$4\% \text{ of principal}$$

So:

$$0.04P = 400$$

$$P = \boxed{₹ 10,000}$$

29. GENERAL RATE-DIFFERENCE SHORTCUT

Same principal and time.

If rate changes by (ΔR) :

$$\boxed{\Delta SI = \frac{P(\Delta R)T}{100}}$$

30. DIFFERENCE IN INTEREST DUE TO TIME CHANGE

Same principal and rate.

₹ 600 extra interest is earned by investing for 3 additional years at 5%.

$$600 = \frac{P(5)(3)}{100}$$

$$600 = 0.15P$$

$$P = \boxed{₹ 4000}$$

31. DIFFERENCE IN INTEREST BETWEEN TWO RATES

At 8% and 12% for 5 years, SI difference is ₹ 1000.

Difference rate:

$$4\%$$

Total difference percentage:

$$4 \times 5 = 20\%$$

So:

$$20\% \text{ of } P = 1000$$

$$P = \boxed{₹ 5000}$$

32. FIND RATE FROM DIFFERENCE IN AMOUNTS

At SI, two investments of the same principal at rates (R_1, R_2) for same time (T) differ in amount by exactly the difference in interest.

So:

$$\boxed{A_1 - A_2 = P(R_1 - R_2)T/100}$$

33. LOAN REPAYMENT— BASIC SI MODEL

Suppose ₹ 10,000 borrowed at 12% SI for 2 years.

Total interest:

$$2400$$

Total amount due:

$$12400$$

If repaid in 4 **equal simple installments with no reducing-balance adjustment** as the problem explicitly assumes:

$$12400/4 = \boxed{₹ 3100}$$

Warning

Real loan EMI calculations usually use reducing balance/compound-style finance mathematics. Do not apply this simple model unless the aptitude question clearly tells you to.

34. INSTALLMENT QUESTIONS — EXAM CAUTION

If installments are paid at different times, a proper SI-equivalence question may require considering the interest saved/lost due to earlier payments.

Do not automatically divide total amount equally unless the wording permits it.

We keep financial-mathematics depth limited because it is low priority for GATE aptitude.

35. LOAN PAID EARLY

Suppose ₹ 5000 at 10% SI is due after 3 years:

$$[A = 6500]$$

If the borrower instead repays after only 2 years under the same SI convention:

$$[A = 5000(1.2)]$$

$$[\boxed{₹ 6000}]$$

36. EQUAL AMOUNTS AT DIFFERENT TIMES

At SI, amount after (T) years is:

$$[A = P\left(1 + \frac{RT}{100}\right)]$$

If two investments reach equal amounts:

$$[P_1\left(1 + \frac{R_1T_1}{100}\right) = P_2\left(1 + \frac{R_2T_2}{100}\right)]$$

Do not confuse equal **amounts** with equal **interest**.

37. EQUAL INTEREST VS EQUAL AMOUNT

Equal SI condition:

$$[P_1R_1T_1 = P_2R_2T_2]$$

Equal final amount condition:

$$[P_1 + SI_1 = P_2 + SI_2]$$

These are not the same.

38. EXAMPLE

₹ 5000 at 10% for 2 years:

$$[SI = 1000, \text{quad } A = 6000]$$

₹ 4000 at 12.5% for 2 years:

$$[SI=1000, \text{quad } A=5000]$$

Interest is equal, amounts are not.

39. RATE CHANGE MIDWAY — FIND FINAL AMOUNT

1600 Invested:

- first 2 years at 5%
- next 3 years at 10%

Interest:

$$[6000 \frac{10+30}{100} = 2400]$$

Amount:

$$[\boxed{18400}]$$

Notice the second rate is still applied to 1600 at 16600 .

40. SI VS COMPOUND INTEREST PREVIEW

At SI:

$$[A=P(1+nr)]$$

if (r) is annual rate in decimal and (n) years.

At CI:

$$[A=P(1+r)^n]$$

SI grows linearly.

CI grows multiplicatively.

Chapter 21 will cover this fully.

41. FIND TOTAL MONEY FROM INTEREST ON PART

A person invests 60% of money at 10% and remaining 40% at 5%.

Overall annual rate:

$$[0.6(10)+0.4(5)]$$

$$[=6+2]$$

$$[=8\%]$$

If total annual interest = 11600 :

$$[8\% \text{ of total} = 1600]$$

Total:

$$[\boxed{20000}]$$

42. FRACTIONAL SPLIT SHORTCUT

If fraction f is invested at R_1 , remaining $(1-f)$ at R_2 :

$$R_{\text{overall}} = fR_1 + (1-f)R_2$$

for equal investment durations.

43. EXAMPLE

$$\frac{1}{3}$$

at 6%, and:

$$\frac{2}{3}$$

at 9%.

Overall rate:

$$\frac{1}{3}(6) + \frac{2}{3}(9)$$

$$= 2 + 6$$

$$= 8\%$$

44. FIND FRACTION INVESTED AT ONE RATE

Money invested partly at 5%, partly at 10%.

Overall rate = 8%.

Let fraction at 5% = f .

$$5f + 10(1-f) = 8$$

$$5f + 10 - 10f = 8$$

$$5f = 2$$

$$f = \frac{2}{5}$$

Thus:

$$\frac{2}{5} \text{ at } 5\%, \frac{3}{5} \text{ at } 10\%$$

Same as alligation ratio:

$$2:3$$

45. GATE-STYLE QUESTION 1

10,000 invested for:

- 2 years at 6%
- 3 years at 8%

Find total SI.

$$SI = 10000 \frac{12+24}{100}$$

$$\boxed{13600}$$

46. GATE-STYLE QUESTION 2

Two sums earn equal SI.

Rates:

$$5\%, 10\%$$

Times:

$$4, 3 \text{ years}$$

Principal ratio:

$$P_1:P_2 = 10(3):5(4)$$

$$= 30:20$$

$$\boxed{3:2}$$

47. GATE-STYLE QUESTION 3

120,000 is split at 6% and 10% to obtain overall 7%.

Ratio:

$$(10-7):(7-6)$$

$$= 3:1$$

So:

$$\boxed{15000 \text{ at } 6\%, 15000 \text{ at } 10\%}$$

48. GATE-STYLE QUESTION 4

A sum earns 500 more at 10% than at 8% over 5 years.

Difference percentage:

$$2 \times 5 = 10\%$$

So:

$$10\% \text{ of } P = 500$$

$$\boxed{P = 5000}$$

49. PLACEMENT-STYLE QUESTION 1

12,000 Invested:

- first 1 year at 5%
- next 2 years at 10%

$$SI = 12000 \frac{5+20}{100}$$

$$\text{\textbackslash}[= 12000(0.25) \text{\textbackslash}]$$

$$\text{\textbackslash}[= \text{\textbackslash boxed \{ }^1 3\}000 \text{\textbackslash} }$$

50. PLACEMENT-STYLE QUESTION 2

A total of $\text{\textbackslash}^1 15,000$ is split between 5% and 8%.

Annual SI is $\text{\textbackslash}^1 900$.

Overall rate:

$$\text{\textbackslash}[900/15000 \times 100 = 6\% \text{\textbackslash}]$$

Ratio:

$$\text{\textbackslash}[5\% \text{\textbackslash text{ part}} : 8\% \text{\textbackslash text{ part}} \text{\textbackslash}]$$

$$\text{\textbackslash}[= (8-6) : (6-5) \text{\textbackslash}]$$

$$\text{\textbackslash}[= 2:1 \text{\textbackslash}]$$

Thus:

$$\text{\textbackslash}[\text{\textbackslash boxed \{ }^1 10,000 \text{\textbackslash} }]$$

51. PLACEMENT-STYLE QUESTION 3

Same principal earns same SI:

- 8% for 3 years
- $\text{\textbackslash}(R)\%$ for 4 years

$$\text{\textbackslash}[8(3) = 4R \text{\textbackslash}]$$

$$\text{\textbackslash}[\text{\textbackslash boxed \{ } R = 6\% \text{\textbackslash} }]$$

52. PLACEMENT-STYLE QUESTION 4

60% of money is invested at 10%, rest at 5%.

Overall rate:

$$\text{\textbackslash}[0.6(10) + 0.4(5) = \text{\textbackslash boxed \{ } 8\% \text{\textbackslash} }]$$

53. ADVANCED QUESTION — TWO RATES, SAME TOTAL

$\text{\textbackslash}^1 30,000$ is split between 5% and 12%.

Annual interest = $\text{\textbackslash}^1 2550$.

Overall rate:

$$\text{\textbackslash}[2550/30000 \times 100 = 8.5\% \text{\textbackslash}]$$

Ratio:

$$\text{\textbackslash}[5\% \text{\textbackslash text{ part}} : 12\% \text{\textbackslash text{ part}} \text{\textbackslash}]$$

$$\Rightarrow (12-8.5):(8.5-5)$$

$$= 3.5:3.5$$

$$= 1:1$$

Therefore:

$$\boxed{\text{Rs } 15,000 \text{ each}}$$

54. ADVANCED QUESTION — UNEQUAL TIME SPLIT

Rs 10,000 split into two parts.

First part at 5% for 2 years.

Second at 10% for 3 years.

Total SI = Rs 2200.

Effective interest percentages:

First:

$$10\%$$

Second:

$$30\%$$

Overall interest percentage:

$$\frac{2200}{10000} \times 100 = 22\%$$

Alligation:

$$P_1:P_2 = (30-22):(22-10)$$

$$= 8:12$$

$$= 2:3$$

Thus:

$$\boxed{\text{Rs } 4000 \text{ and Rs } 6000}$$

Check:

$$400 + 1800 = 2200$$

55. ADVANCED QUESTION — FIND PRINCIPAL RATIO

Two sums earn SI ratio:

$$3:5$$

Rate ratio:

$$2:3$$

Time ratio:

$$4:5$$

Let principal ratio = $(x:y)$.

$$\left[\frac{3}{5} = \frac{x(2)(4)}{y(3)(5)} \right]$$

$$\left[\frac{35}{8} = \frac{8x}{15y} \right]$$

Cross multiply:

$$\left[45y = 40x \right]$$

$$\left[x:y = 45:40 \right]$$

$$\left[=\boxed{9:8} \right]$$

56. ADVANCED QUESTION — RATE CHANGE

A sum yields ₹ 1800 in 3 years at 10%.

If rate increases by 5 percentage points and time decreases to 2 years, find new SI.

Same principal:

Old:

$$\left[SI \propto RT \right]$$

Old factor:

$$\left[10(3) = 30 \right]$$

New factor:

$$\left[15(2) = 30 \right]$$

Therefore:

$$\left[\boxed{1800} \right]$$

No need to find principal.

57. COMMON TRAPS

& pTRAP 1 — Changing rate

At SI, each rate interval still applies to original principal.

& pTRAP 2 — Average rate

Weight rates by time, not simply by number of rates.

& pTRAP 3 — Split investments

Use amount invested as the weight.

& pTRAP 4 — Equal SI

Equal interest means:

$$\left[PRT \right]$$

products are equal.

& pTRAP 5 — Equal SI vs equal amount

They are different conditions.

& pTRAP 6 — 3+ split portions

An overall interest alone may not uniquely determine all portions.

58. MUST-MEMORIZE ADVANCED FORMULAS

Changing rates

$$\boxed{SI = \frac{P(R_1T_1 + R_2T_2 + \dots)}{100}}$$

Time-weighted average rate

$$\boxed{R_{\text{avg}} = \frac{\sum R_i T_i}{\sum T_i}}$$

Equal SI

$$\boxed{P_1 R_1 T_1 = P_2 R_2 T_2}$$

59. SPLIT-INVESTMENT ALLIGATION

For the same duration:

$$\boxed{P_1 : P_2 = (R_2 - R) : (R - R_1)}$$

where (R) is overall rate.

If durations differ, replace each rate with its effective percentage:

$$\boxed{R_i T_i}$$

60. FAST EXAM DECISION TREE

Rate changes over time?

Add (RT) contributions.

Need average rate?

Time-weight the rates.

Money split at two rates?

Find overall rate, then alligate.

Equal SI?

Set (PRT) products equal.

Interest differs because of rate/time change?

Use only the changed factor.

Multiple unknown splits?

Check whether enough equations exist.

PRACTICE — LEVEL 0

Q1

₹ 15,000 invested 2 years at 5%, then 3 years at 10%. Find total SI.

Q2

Find average SI rate for 2 years at 6% and 3 years at 9%.

Q3

Two equal principals earn equal SI. First is at 10% for 3 years. Second is for 5 years. Find second rate.

Q4

Two sums earn equal SI at same rate. Their times are 2:5. Find principal ratio.

Q5

A sum earns ₹ 300 extra because rate rises from 5% to 8% for 2 years. Find principal.

LEVEL 1

Q6

₹ 10,000 split at 5% and 10%. Overall annual rate is 8%. Find split.

Q7

₹ 20,000 split at 6% and 9%. Annual SI = ₹ 1,500. Find parts.

Q8

Two sums earn equal SI. Their rates are 6% and 8%, and times 4 years and 3 years. Find principal ratio.

Q9

A sum earns ₹ 1,200 at 8% in 3 years. Find SI if rate becomes 12% and time 2 years.

Q10

50% money invested at 6%, remaining at 10%. Find overall rate.

LEVEL 2

Q11

₹ 30,000 split at 5% and 11%. Overall SI rate is 8%. Find parts.

Q12

₹ 12,000 split: one part at 5% for 2 years and the other at 8% for 3 years. Total SI = ₹ 2,160. Find parts.

Q13

Interest ratio is 2:3, principal ratio 4:5, time ratio 3:2. Find rate ratio.

Q14

A sum earns ₹ 600 more at 12% than at 8% over 3 years. Find principal.

Q15

One-third of money is invested at 6% and two-thirds at 9%. If annual SI is ₹ 240 find total money.

LEVEL 3 — EXAM PRESSURE

Q16

Two sums have principal ratio 3:4. Rates are 8% and 6%, times 5 and 10 years. Find SI ratio.

Q17

₹ 25,000 is divided into two parts. One earns 6%, the other 10%. Total annual SI = ₹ 1900 Find the portions.

Q18

A sum is invested 2 years at 5%, 3 years at 8%, and 1 year at 12%. Find equivalent average annual SI rate.

Q19

Two sums earn equal SI. First principal is ₹ 6000 at 10% for 4 years. Second is ₹ 8000 at 6% for (T) years. Find (T) .

Q20

₹ 20,000 is split between two investments: first earns 10% total interest over its term, second earns 30% total interest. Overall interest earned is ₹ 4400 Find the two invested amounts.

ANSWERS

Q1

Total percentage:

$$\frac{5(2)+10(3)}{5+3} = 40\%$$

$$SI = 5000(0.4) = \boxed{₹ 2000}$$

Q2

$$R_{\text{avg}} = \frac{6(2)+9(3)}{5}$$

$$= \frac{39}{5}$$

$$= \boxed{7.8\%}$$

Q3

$$10(3) = R(5)$$

$$R = 6$$

$$= \boxed{6\%}$$

Q4

Same rate and equal SI:

$$P_1 T_1 = P_2 T_2$$

$$P_1 : P_2 = 5 : 2$$

$$\boxed{5:2}$$

Q5

Difference rate:

$$3\%$$

for 2 years:

$$6\% \text{ of } P = 300$$

$$P = 300 / 0.06$$

$$= \boxed{5000}$$

Q6

Overall rate:

$$8\%$$

Ratio:

$$5\% \text{ part} : 10\% \text{ part}$$

$$= (10-8) : (8-5)$$

$$= 2 : 3$$

Total 5 parts:

$$2000$$

$$\boxed{14000}$$

Q7

Overall rate:

$$1500 / 20000 \times 100 = 7.5\%$$

Ratio:

$$6\% \text{ part} : 9\% \text{ part}$$

$$= (9-7.5) : (7.5-6)$$

$$= 1 : 1$$

$$\boxed{11000 \text{ each}}$$

Q8

Equal SI:

$$P_1 : P_2 = 8(3) : 6(4)$$

$$= 24 : 24$$

$$= \boxed{1:1}$$

Q9

Old (RT) :

$$8(3)=24$$

New:

$$12(2)=24$$

So:

$$\boxed{1200}$$

Q10

$$0.5(6)+0.5(10)$$

$$=3+5$$

$$=8\%$$

Q11

Overall:

$$8\%$$

Ratio:

$$(11-8):(8-5)$$

$$=3:3$$

$$=1:1$$

$$\boxed{15,000 \text{ each}}$$

Q12

Effective total interest percentages:

First:

$$5(2)=10\%$$

Second:

$$8(3)=24\%$$

Overall:

$$2160/12000 \times 100 = 18\%$$

Ratio:

$$10\% \text{ part} : 24\% \text{ part}$$

$$=(24-18):(18-10)$$

$$=6:8$$

$$=3:4$$

Total 7 parts:

$$12000/7$$

Thus:

$$\boxed{P_1 = \frac{36000}{7} \approx \text{Rs } 5142.86}$$

$$\boxed{P_2 = \frac{48000}{7} \approx \text{Rs } 6857.14}$$

The data produce non-integer rupee values algebraically.

Q13

$$\frac{2}{3} = \frac{4R_1(3)}{5R_2(2)}$$

$$\frac{2}{3} = \frac{12R_1}{10R_2}$$

Cross multiply:

$$20R_2 = 36R_1$$

$$R_1 : R_2 = 5 : 9$$

$$\boxed{5:9}$$

Q14

Difference rate:

$$4\%$$

for 3 years:

$$12\% \text{ of } P = 600$$

$$P = 600 / 0.12$$

$$P = \boxed{\text{Rs } 5000}$$

Q15

Overall rate:

$$\frac{13(6)}{100} + \frac{23(9)}{100}$$

$$= 2 + 6 = 8\%$$

If 8% equals Rs 2400 :

$$P = \frac{2400}{0.08}$$

$$P = \boxed{\text{Rs } 30,000}$$

Q16

$$SI_1 : SI_2 = 3(8)(5) : 4(6)(10)$$

$$= 120 : 240$$

$$= \boxed{1:2}$$

Q17

Overall rate:

$$1900 / 25000 \times 100 = 7.6\%$$

Ratio:

$$6\% \text{ part} : 10\% \text{ part}$$

$$= (10 - 7.6) : (7.6 - 6)$$

$$= 2.4 : 1.6$$

$$\text{V} = 3:2$$

Total 5 parts:

$$\text{V} = 5000$$

So:

$$\boxed{115,000 \text{ and } 10,000}$$

Q18

Total time:

$$\text{V} = 2+3+1=6$$

Total rate-time contribution:

$$\text{V} = 5(2)+8(3)+12(1)$$

$$\text{V} = 10+24+12 = 46$$

Average rate:

$$\text{V} = 46/6 = \boxed{7\frac{23}{6}\%}$$

Q19

Equal SI:

$$\text{V} = 6000(10)(4) = 8000(6)T$$

$$\text{V} = 240000=48000T$$

$$\text{V} = \boxed{T=5\text{ years}}$$

Q20

Effective interest percentages:

$$\text{V} = 10\%, 30\%$$

Overall:

$$\text{V} = 4400/20000 \times 100 = 22\%$$

Ratio:

$$\text{V} = 10\% \text{ part} : 30\% \text{ part}$$

$$\text{V} = (30-22):(22-10)$$

$$\text{V} = 8:12$$

$$\text{V} = 2:3$$

Total:

$$\text{V} = 120,000$$

So:

$$\text{V} = \boxed{18000, 000}$$

' CHAPTER 20 COMPLETE

We have now covered:

- SI fundamentals '
- principal/rate/time/amount '
- reverse calculations '
- months/days conversion '
- SI proportionality '
- two-amount problems '
- doubling/tripling '
- changing rates over time '
- time-weighted average rates '
- split investments '
- alligation with SI rates '
- equal-interest conditions '
- interest ratios '
- rate/time-difference problems '
- basic loan/repayment interpretation '
- advanced GATE/placement problems '

Progress

Completed: 20/76 Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

0=0% Chapter 21 — Compound Interest, Growth & Depreciation

Next we will cover:

- CI intuition
- annual compounding
- amount formula
- compound interest
- SI vs CI
- half-yearly/quarterly compounding
- variable rates
- population growth
- depreciation
- difference between SI and CI shortcuts
- repeated percentage growth
- advanced GATE and placement questions.

USER

next

CHAPTER 21 — COMPOUND INTEREST, GROWTH & DEPRECIATION

PART 1 — CI INTUITION, ANNUAL COMPOUNDING, AMOUNT FORMULA, SI VS CI & REPEATED GROWTH

Status: $\emptyset$ = Not PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

Compound Interest is based on one simple idea:

Interest is added back to the principal, and future interest is calculated on the increased amount.

That is why compound interest grows faster than simple interest.

1. SIMPLE INTEREST VS COMPOUND INTEREST

Suppose:

$$P = ₹ 1000$$

Rate:

$$10\% \text{ per year}$$

For 2 years.

Under Simple Interest

Every year interest is calculated on ₹ 1000 .

Year1:

$$100$$

Year2:

$$100$$

Total SI:

$$\boxed{200}$$

Amount:

$$\boxed{1200}$$

Under Compound Interest

Year1:

$$1000 + 10\% = 1100$$

Year2 interest is calculated on ₹ 1100 :

$$10\% \text{ of } 1100 = 110$$

Final amount:

$$\boxed{1100+110 = \text{Rs } 1210}$$

Compound interest:

$$\boxed{1210-1000 = \text{Rs } 210}$$

So:

$$CI > SI$$

for positive rate and time greater than 1 compounding period.

2. COMPOUND AMOUNT FORMULA

If:

- principal = (P)
- rate = (R%) per annum
- time = (n) years
- compounded annually

then:

$$\boxed{A = P \left(1 + \frac{R}{100}\right)^n}$$

This is the master formula.

3. COMPOUND INTEREST FORMULA

Since:

$$CI = A - P$$

therefore:

$$\boxed{CI = P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right]}$$

MUST MEMORIZE

4. BASIC EXAMPLE

Find CI on Rs 5000 at 10% for 2 years.

$$A = 5000(1.1)^2$$

$$= 5000(1.21)$$

$$= \text{Rs } 6050$$

Thus:

$$CI = 6050 - 5000$$

$$= \boxed{\text{Rs } 1050}$$

5. YEAR-BY-YEAR VIEW

Same problem:

Initial:

$$\text{\[5000 \]}$$

After year 1:

$$\text{\[5000(1.1)=5500 \]}$$

After year 2:

$$\text{\[5500(1.1)=6050 \]}$$

Same answer.

6. COMPOUNDING IS REPEATED PERCENTAGE CHANGE

A 10% increase every year means:

$$\text{\[\times 1.1 \]}$$

every year.

After 3 years:

$$\text{\[\times 1.1^3 \]}$$

Therefore CI is directly connected to Chapter 15 successive percentage change.

7. EXAMPLE — 3 YEARS

₹ 80,000 at 5% compounded annually for 3 years.

$$\text{\[A=8000(1.05)^3 \]}$$

$$\text{\[1.05^2=1.1025 \]}$$

$$\text{\[1.05^3=1.157625 \]}$$

Therefore:

$$\text{\[A=8000(1.157625) \]}$$

$$\text{\[= \boxed{₹ 9261} \]}$$

CI:

$$\text{\[9261-8000 = \boxed{₹ 1261} \]}$$

8. FAST FRACTION METHOD

For common rates, fractions can be faster.

10%:

$$\text{\[1.1=\frac{11}{10} \]}$$

20%:

$$\text{\[1.2=\frac{6}{5} \]}$$

25%:

$$\left[1.25 = \frac{54}{40} \right]$$

50%:

$$\left[1.5 = \frac{81}{32} \right]$$

12.5%:

$$\left[1.125 = \frac{9}{8} \right]$$

Example:

₹ 6400 at 25% for 2 years:

$$\left[A = 6400 \left(\frac{54}{40} \right)^2 \right]$$

$$\left[= 6400 \frac{25}{16} \right]$$

$$\left[= 400(25) \right]$$

$$\left[= \boxed{10000} \right]$$

9. FIND PRINCIPAL FROM AMOUNT

From:

$$\left[A = P(1+r)^n \right]$$

where:

$$\left[r = \frac{R}{100} \right]$$

we get:

$$\left[\boxed{P = \frac{A}{(1+r)^n}} \right]$$

10. EXAMPLE

Amount = ₹ 12100 after 2 years at 10% CI.

$$\left[P = \frac{12100}{1.21} \right]$$

$$\left[= \boxed{10000} \right]$$

11. FIND RATE

Suppose:

$$\left[P = ₹ 1000 \right]$$

Amount after 2 years:

$$\left[₹ 1210 \right]$$

Then:

$$\left[\frac{AP}{P} = 1.21 \right]$$

$$\left[(1+r)^2 = 1.21 \right]$$

$$\boxed{1+r=1.1}$$

$$\boxed{r=0.1}$$

$$\boxed{R=10\%}$$

12. FIND TIME

Suppose $\text{₹}1000$ becomes $\text{₹}1331$ at 10%.

$$\frac{1331}{1000}=1.331$$

But:

$$1.1^3=1.331$$

Therefore:

$$\boxed{n=3 \text{ years}}$$

In aptitude, rates/times are often chosen so powers are recognizable.

13. IMPORTANT POWER VALUES

You should recognize:

$$1.1^2=1.21$$

$$1.1^3=1.331$$

$$1.2^2=1.44$$

$$1.2^3=1.728$$

$$1.25^2=1.5625$$

$$1.5^2=2.25$$

These save time.

14. SI VS CI AFTER 1 YEAR

For one year:

$$SI=\frac{PR}{100}$$

and:

$$CI=\frac{PR}{100}$$

So:

$$\boxed{SI=CI \text{ for one compounding period}}$$

Difference appears only after interest itself starts earning interest.

15. SI VS CI AFTER 2 YEARS

At rate $R\%$:

SI:

$$S = P \left(\frac{2R}{100} \right)$$

CI:

$$C = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$$

Expand:

$$C = P \left(\frac{2R}{100} + \frac{R^2}{10000} \right)$$

Therefore:

$$C - S = P \frac{R^2}{10000}$$

So:

$$\boxed{C - S = P \left(\frac{R}{100} \right)^2}$$

for 2 years.

MUST MEMORIZE

16. EXAMPLE — DIFFERENCE SI & CI

₹ 1000 at 10% for 2 years.

Difference:

$$10000 \left(\frac{10}{100} \right)^2$$

$$= 10000(0.01)$$

$$= \boxed{₹ 1000}$$

Check:

SI:

$$₹ 200$$

CI:

$$₹ 210$$

Difference:

$$₹ 100$$

17. FIND PRINCIPAL FROM C I" DIFFERENCE

Difference for 2 years = ₹ 72 .

Rate = 12%.

$$72 = P(0.12)^2$$

$$72 = 0.0144P$$

$$P = 5000$$

$$\boxed{₹ 5000}$$

18. FIND RATE FROM C I" \$DIFFERENCE

Principal = 18000.

Difference after 2 years = 1200.

$$1200 = 8000 \left(\frac{R}{100} \right)^2$$

$$\frac{200}{8000} = 0.025$$

$$\left(\frac{R}{100} \right)^2 = 0.025$$

$$R \approx 15.81\%$$

Such awkward values are less common; usually exam numbers simplify nicely.

19. SI VS CI AFTER 3 YEARS

Let:

$$r = \frac{R}{100}$$

CI:

$$P[(1+r)^3 - 1]$$

Expand:

$$= P(3r + 3r^2 + r^3)$$

SI:

$$3Pr$$

Therefore:

$$\boxed{CI - SI = P(3r^2 + r^3)}$$

or:

$$\boxed{CI - SI = Pr^2(3+r)}$$

This is useful but lower priority than the 2-year shortcut.

20. EXAMPLE — 3-YEAR DIFFERENCE

11000 at 10%.

$$r = 0.1$$

$$CI - SI = 10000(3(0.01) + 0.001)$$

$$= 10000(0.031)$$

$$= \boxed{3100}$$

Check:

SI:

$$3000$$

CI:

$$\sqrt[3]{13310-10000}=3310$$

Difference:

$$\sqrt[3]{310}$$

21. GROWTH FORMULA

Compound interest mathematics is not limited to money.

If a quantity grows by $(R\%)$ each period:

$$\boxed{V_n = V_0 \left(1 + \frac{R}{100}\right)^n}$$

Used for:

- population
 - production
 - users/customers
 - sales
 - bacteria
 - investment value
-

22. POPULATION GROWTH

Population:

$$\sqrt[3]{50000}$$

Growth:

$$\sqrt[3]{10\% \text{ per year}}$$

for 2 years.

$$\sqrt[3]{P_2 = 50000(1.1)^2}$$

$$\sqrt[3]{=50000(1.21)}$$

$$\sqrt[3]{=\boxed{60500}}$$

Not 60000.

23. FIND OLD POPULATION

Current population is 60,500 after 2 years of 10% growth.

Previous population:

$$\sqrt[3]{\frac{60500}{1.1^2}}$$

$$\sqrt[3]{=\boxed{50000}}$$

For reverse growth, divide by growth multipliers.

24. DIFFERENT GROWTH RATES EACH YEAR

Population = 100000.

Year 1:

$$\backslash[+10\% \backslash]$$

Year 2:

$$\backslash[+20\% \backslash]$$

Final:

$$\backslash[100000(1.1)(1.2) \backslash]$$

$$\backslash[=100000(1.32) \backslash]$$

$$\backslash[=\boxed{132000} \backslash]$$

Net growth:

$$\backslash[\boxed{32\%} \backslash]$$

25. GROWTH THEN DECLINE

Population rises 20%, then falls 10%.

Multiplier:

$$\backslash[1.2(0.9)=1.08 \backslash]$$

Net:

$$\backslash[\boxed{8\%}\text{ increase} \backslash]$$

Same successive-percentage logic.

26. DEPRECIATION

If an asset loses $(R\%)$ of its **current value** each year:

$$\backslash[\boxed{V_n = V_0\left(1-\frac{R}{100}\right)^n} \backslash]$$

This is compound depreciation.

27. EXAMPLE — CAR DEPRECIATION

Car value:

$$\backslash[\text{₹} 500000 \backslash]$$

Depreciation:

$$\backslash[10\%\text{ annually} \backslash]$$

for 2 years.

$$\backslash[V = 500000(0.9)^2 \backslash]$$

$$\backslash[= 500000(0.81) \backslash]$$

$$V = \boxed{\text{\textasciitimes} 1405000}$$

Total depreciation:

$$V = \text{\textasciitimes} 195000$$

Overall percentage decrease:

$$V = 19\%$$

not 20%.

28. REVERSE DEPRECIATION

A machine is worth $\text{\textasciitimes} 81,000$ after 2 years of 10% annual depreciation.

Original:

$$V_0 = \frac{81000}{0.9^2}$$

$$V = \frac{81000}{0.81}$$

$$V = \boxed{\text{\textasciitimes} 100000}$$

29. DIFFERENT DEPRECIATION RATES

$\text{\textasciitimes} 100,000$ machine:

- Year 1: 20% depreciation
- Year 2: 10%

Final:

$$V = 100000(0.8)(0.9)$$

$$V = 72000$$

Total depreciation:

$$V = \boxed{28\%}$$

30. GROWTH AND DEPRECIATION CAN BE MIXED

Value increases 25%, then decreases 20%.

$$V = 1.25(0.8) = 1$$

Final value unchanged.

$$V = \boxed{\text{No net change}}$$

31. COMPOUND GROWTH WITH NEGATIVE RATE

Instead of memorizing a separate formula, think:

- growth $(+R\%)$!' multiplier $(1+R/100)$
- decline $(-R\%)$!' multiplier $(1-R/100)$

Then multiply.

32. DOUBLING UNDER CI

If money doubles:

$$2 = \left(1 + \frac{R}{100}\right)^n$$

Unlike SI, there is no simple $(100/R)$ exact rule.

Example at 10%:

$$1.1^n = 2$$

This takes about 7.27 years mathematically, or 8 full annual compounding periods to exceed double if only whole-year compounding is allowed.

For aptitude, questions usually choose friendly rates/times.

33. FRIENDLY DOUBLING EXAMPLE

At 100% CI per year:

$$1 + R/100 = 2$$

So money doubles in:

$$\boxed{1 \text{ year}}$$

At 50%:

$$1.5^2 = 2.25$$

so it exceeds double after 2 years.

34. AMOUNT RATIO FOR SAME RATE

Same principal, same rate, different times.

$$A_n = P(1+r)^n$$

$$A_m = P(1+r)^m$$

Therefore:

$$\boxed{\frac{A_n}{A_m} = (1+r)^{n-m}}$$

This is very useful.

35. EXAMPLE

Amount after 5 years vs after 3 years at 10%:

$$\frac{A_5}{A_3} = 1.1^2 = \boxed{1.21}$$

So amount after 5 years is 21% larger than after 3 years.

36. FIND RATE FROM TWO AMOUNTS AT DIFFERENT TIMES

At CI:

$$\left[\frac{A_2}{A_1} = (1+r)^{T_2-T_1} \right]$$

Example:

Amount after 2 years:

$$\left[11210 \right]$$

after 4 years:

$$\left[14641 \right]$$

Ratio:

$$\left[14641/12100 = 1.21 \right]$$

Time difference:

$$\left[2 \right]$$

So:

$$\left[(1+r)^2 = 1.21 \right]$$

$$\left[1+r = 1.1 \right]$$

$$\left[\boxed{R=10\%} \right]$$

37. IMPORTANT DIFFERENCE FROM SI TWO-AMOUNT METHOD

Under SI:

amount difference over equal time intervals is constant.

Under CI:

amount ratio over equal time intervals is constant.

Memory trick

SI !' difference CI !' ratio

This is very important.

38. EXAMPLE OF SI VS CI PATTERN

At SI:

$$\left[A_5 - A_3 \right]$$

depends linearly on 2 years.

At CI:

$$\left[\frac{A_5}{A_3} \right]$$

equals:

$$\sqrt{(1+r)^2}$$

Use the correct structure.

39. FIND PRINCIPAL FROM TWO CI AMOUNTS

Amount after 2 years = ₹ 12100 at 10%.

Then:

$$\sqrt{P = 12100/(1.1)^2}$$

$$\sqrt{= \boxed{\text{₹ } 10000}}$$

If rate wasn't given, amount ratios may help find it first.

40. SUCCESSIVE VARIABLE RATES

If rates are:

$$\sqrt{R_1, R_2, \dots, R_n}$$

then:

$$\sqrt{\boxed{A = P \prod_{i=1}^n \left(1 + \frac{R_i}{100}\right)}}$$

For depreciation, use negative rate multiplier.

41. EXAMPLE

₹ 10000:

- +10%
- +20%
- " 10%

Final:

$$\sqrt{10000(1.1)(1.2)(0.9)}$$

$$\sqrt{= 10000(1.188)}$$

$$\sqrt{= \boxed{\text{₹ } 11880}}$$

Net growth:

$$\sqrt{\boxed{18.8\%}}$$

42. GATE-STYLE QUESTION 1

₹ 10000 at 10% CI for 2 years.

$$\sqrt{A = 10000(1.21)}$$

$$\sqrt{= \boxed{\text{₹ } 12100}}$$

CI:

$$\boxed{12100}$$

43. GATE-STYLE QUESTION 2

1800 at 25% for 2 years.

$$A = 8000 \left(\frac{54}{100} \right)^2$$

$$= 8000 \frac{25}{16}$$

$$= \boxed{12500}$$

CI:

$$\boxed{4500}$$

44. GATE-STYLE QUESTION 3

Difference between CI and SI for 2 years on 1500 at 20%.

$$5000(0.2)^2$$

$$= 5000(0.04)$$

$$= \boxed{200}$$

45. GATE-STYLE QUESTION 4

Population = 80,000.

Growth 5% annually for 2 years.

$$80000(1.05)^2$$

$$= 80000(1.1025)$$

$$= \boxed{88,200}$$

46. PLACEMENT-STYLE QUESTION 1

12000 at 10% CI for 3 years.

$$A = 20000(1.331)$$

$$= \boxed{26620}$$

CI:

$$\boxed{6620}$$

47. PLACEMENT-STYLE QUESTION 2

A machine worth 120000 depreciates by 20% annually for 2 years.

$$V = 200000(0.8)^2$$

$$\boxed{=200000(0.64)}$$

$$\boxed{=1128000}$$

48. PLACEMENT-STYLE QUESTION 3

1000 grows 20% in year 1 and 10% in year 2.

$$\boxed{10000(1.2)(1.1)}$$

$$\boxed{=13200}$$

49. PLACEMENT-STYLE QUESTION 4

An investment becomes 1331 in 3 years at 10% CI.

Principal:

$$\boxed{13310/(1.1)^3}$$

$$\boxed{=13310/1.331}$$

$$\boxed{=10000}$$

50. ADVANCED QUESTION — FIND RATE

16000 becomes 25000 in 2 years.

$$\frac{25000}{16000} = \frac{25}{16}$$

Thus:

$$(1+r)^2 = \left(\frac{54}{right)}^2\right)$$

$$1+r=\frac{54}{16}$$

$$r=\frac{14}{16}$$

$$\boxed{25\%}$$

51. ADVANCED QUESTION — C I" DIFFERENCE

Difference after 2 years is 324 at 18%.

$$324 = P(0.18)^2$$

$$324 = 0.0324P$$

$$P = \boxed{10000}$$

52. ADVANCED QUESTION — REVERSE POPULATION

Population becomes 72,600 after:

- +10% first year
- +20% second year

Original:

$$P = \frac{72600}{1.1(1.2)}$$

$$= \frac{72600}{1.32}$$

$$= \boxed{55000}$$

53. ADVANCED QUESTION — DEPRECIATION

A vehicle is worth ₹ 4,05,000 after 2 years of 10% annual depreciation.

Original:

$$405000/0.81$$

$$= \boxed{500000}$$

54. COMMON TRAPS

& pTRAP 1 — Add yearly percentages

Repeated growth is multiplicative.

& pTRAP 2 — CI = SI formula

Only for one compounding period.

& pTRAP 3 — Depreciation

Each year's depreciation is on current value, not original value.

& pTRAP 4 — Two amounts

For CI use amount ratios, not amount differences.

& pTRAP 5 — CI means interest only

Formula:

$$P(1+r)^n$$

gives **amount**, not interest.

Subtract principal to get CI.

55. MUST-MEMORIZE FORMULAS

Amount

$$\boxed{A = P \left(1 + \frac{R}{100} \right)^n}$$

CI

$$\boxed{CI = A - P}$$

Depreciation

$$\boxed{V_n = V_0 \left(1 - \frac{R}{100}\right)^n}$$

2-year CI Difference

$$\boxed{CI - SI = P \left(\frac{R}{100}\right)^2}$$

56. FAST EXAM DECISION TREE

Same annual rate?

Use power.

Different yearly rates?

Multiply yearly factors.

Need CI?

Find amount, subtract principal.

Population?

Use growth multiplier.

Depreciation?

Use decline multiplier.

SI vs CI 2 years?

Use (Pr^2) .

Two CI amounts at different times?

Use their ratio.

PRACTICE — LEVEL 0

Q1

Find amount on ₹1000 at 10% CI for 2 years.

Q2

Find CI on ₹500 at 20% for 2 years.

Q3

₹800 becomes what at 25% CI after 2 years?

Q4

Find depreciation value of ₹1000 at 10% for 2 years.

Q5

Population 50,000 grows 20% in one year. Find new population.

LEVEL 1

Q6

Amount is ₹ 10000 after 10% CI for 3 years. Find CI.

Q7

Amount is ₹ 12100 after 2 years at 10%. Find principal.

Q8

Difference between CI and SI on ₹ 8000 at 10% for 2 years.

Q9

Population 100000 grows 10% and then 20%. Find final population.

Q10

A machine depreciates 20% for two years. Find total % depreciation.

LEVEL 2

Q11

Amount becomes ₹ 16000 after 2 years. Find annual compound rate.

Q12

Difference between CI and SI for 2 years is ₹ 125 at 5%. Find principal.

Q13

Population becomes 121000 after 2 years of 10% growth. Find original.

Q14

A machine is worth ₹ 6480 after 2 years of 10% depreciation. Find original value.

Q15

An amount grows 20% then declines 20%. Find net percentage change.

LEVEL 3 — EXAM PRESSURE

Q16

Amount after 2 years is ₹ 12100 and after 4 years is ₹ 14641. Find annual compound rate.

Q17

Amount is ₹ 1000 compounded at 10% first year, 20% second year, and 10% third year. Find final amount.

Q18

Difference between CI and SI for 3 years on ₹ 1000 at 10%. Find difference.

Q19

A population decreases 10% per year for two years. What percent increase is required after that to return exactly to the original population?

Q20

An investment is worth ₹ 15625 after 2 years at 25% compound interest. Find principal.

ANSWERS

Q1

$$\sqrt[1]{1000(1.1)^2} = \boxed{\text{\text{₹} 1000}}$$

Q2

$$\sqrt[1]{A=5000(1.2)^2} = 7200 \sqrt[1]{}$$

$$\sqrt[1]{CI=7200-5000} = \boxed{\text{\text{₹} 2200}}$$

Q3

$$\sqrt[1]{8000(1.25)^2} = 8000(1.5625) \sqrt[1]{}$$

$$\sqrt[1]{=} = \boxed{\text{\text{₹} 12500}}$$

Q4

$$\sqrt[1]{100000(0.9)^2} = \boxed{\text{\text{₹} 81000}}$$

Q5

$$\sqrt[1]{50000(1.2)} = \boxed{60000} \sqrt[1]{}$$

Q6

$$\sqrt[1]{A=10000(1.331)} = 13310 \sqrt[1]{}$$

CI:

$$\sqrt[1]{13310-10000} = \boxed{\text{\text{₹} 3310}}$$

Q7

$$\sqrt[1]{P=12100/1.21} = \boxed{\text{\text{₹} 10000}}$$

Q8

$$\sqrt[1]{8000(0.1)^2} = \boxed{\text{\text{₹} 80}}$$

Q9

$$\sqrt[1]{100000(1.1)(1.2)} = \boxed{132000} \sqrt[1]{}$$

Q10

$$\sqrt[1]{0.8^2=0.64} \sqrt[1]{}$$

Value remains 64%.

Depreciation:

$$\sqrt[1]{\boxed{36\%}} \sqrt[1]{}$$

Q11

$$\sqrt[2]{25000/16000} = 25/16 \sqrt[2]{}$$

$$\sqrt[2]{1+r=5/4 \sqrt[2]{}}$$

$$\sqrt[2]{\boxed{25\%} \sqrt[2]{}}$$

Q12

$$\sqrt[2]{125=P(0.05)^2 \sqrt[2]{}}$$

$$\sqrt[2]{125=0.0025P \sqrt[2]{}}$$

$$\sqrt[2]{\boxed{P = 15000} \sqrt[2]{}}$$

Q13

$$\sqrt[2]{P=121000/1.21 \sqrt[2]{}}$$

$$\sqrt[2]{=\boxed{100000} \sqrt[2]{}}$$

Q14

$$\sqrt[2]{P=64800/0.81 \sqrt[2]{}}$$

$$\sqrt[2]{=\boxed{81000} \sqrt[2]{}}$$

Q15

$$\sqrt[2]{1.2(0.8) = 0.96 \sqrt[2]{}}$$

$$\sqrt[2]{\boxed{4\% \text{ decrease}} \sqrt[2]{}}$$

Q16

$$\sqrt[2]{14641/12100=1.21 \sqrt[2]{}}$$

Over 2 years:

$$\sqrt[2]{(1+r)^2=1.21 \sqrt[2]{}}$$

$$\sqrt[2]{1+r=1.1 \sqrt[2]{}}$$

$$\sqrt[2]{\boxed{10\%} \sqrt[2]{}}$$

Q17

$$\sqrt[2]{10000(1.1)(1.2)(0.9) \sqrt[2]{}}$$

$$\sqrt[2]{=10000(1.188) \sqrt[2]{}}$$

$$\sqrt[2]{=\boxed{11880} \sqrt[2]{}}$$

Q18

For 3 years:

$$\sqrt[2]{CI-SI = P(3r^2+r^3) \sqrt[2]{}}$$

$$\sqrt[2]{= 10000(3(0.01)+0.001) \sqrt[2]{}}$$

$$\sqrt[2]{= 10000(0.031) \sqrt[2]{}}$$

$$\sqrt[2]{=\boxed{310} \sqrt[2]{}}$$

Q19

After two 10% decreases:

$$\sqrt[0.9^2=0.81]{}$$

Current value = 81% of original.

Required multiplier:

$$\sqrt[1/0.81 = \frac{100}{81}]{}$$

Required increase:

$$\sqrt[\frac{19}{81} \times 100]{}$$

$$\sqrt[\approx \boxed{23.46\%}]{}$$

Q20

$$\sqrt[P = \frac{15625}{1.25^2}]{}$$

$$\sqrt[= \frac{15625}{1.5625}]{}$$

$$\sqrt[= \boxed{11000}]{}$$

CHAPTER 21 STATUS

Covered so far:

- CI intuition '
- annual compounding '
- amount and CI formulas '
- reverse principal/rate/time basics '
- SI vs CI '
- 2-year CI "Shortcut" '
- 3-year difference formula '
- repeated growth '
- population growth '
- depreciation '
- variable annual rates '
- CI amount ratios '
- GATE/placement practice '

But **Chapter 21 is not complete yet.**

Next we need:

- half-yearly compounding
- quarterly compounding
- compounding frequency conversion
- fractional years
- variable rates with half-years
- effective annual rate
- nominal vs effective rate basics
- more advanced CI "Shortcuts"
- growth/depreciation mixed applications
- final mixed GATE/placement drills

Progress

Completed: 20/76 **Current:** Chapter 21 — Compound Interest, Growth & Depreciation **Next:** Chapter 21 Part 2 — Half-yearly, quarterly & advanced compound-interest problems

USER

next

ASSISTANT

CHAPTER 21 — COMPOUND INTEREST, GROWTH & DEPRECIATION

PART 2 — HALF-YEARLY, QUARTERLY COMPOUNDING, EFFECTIVE RATE & ADVANCED CI PROBLEMS

Status: 0% **PROGRESS GATE:** High **Placements:** High **Priority:** 0% **Year 1**

In Part 1, we assumed interest was compounded **once per year**.

Now the key question becomes:

How often is the interest added back to the principal?

The more frequently compounding happens, the more opportunities interest gets to earn further interest.

1. COMPOUNDING FREQUENCY

Common possibilities:

- annually !' once per year
- half-yearly / semi-annually !' twice per year
- quarterly !' 4 times per year
- monthly !' 12 times per year

The master idea is:

$$\boxed{\text{Rate per period} = \frac{\text{annual nominal rate}}{\text{number of periods per year}}}$$

and:

$$\boxed{\text{Number of periods} = (\text{years})(\text{periods per year})}$$

2. GENERAL COMPOUNDING FORMULA

If nominal annual rate is $(R\%)$, compounded (m) times per year for (T) years:

$$\boxed{A = P \left(1 + \frac{R}{100m} \right)^{mT}}$$

where:

- $(m=1)$ annually

- $(m=2)$ half-yearly
- $(m=4)$ quarterly
- $(m=12)$ monthly

MUST MEMORIZE

3. HALF-YEARLY COMPOUNDING

Suppose annual rate:

$$[R\%]$$

Compounded half-yearly.

There are 2 compounding periods per year.

Therefore:

$$[\text{Rate per half-year} = \frac{R\%}{2}]$$

and for (T) years:

$$[\text{number of periods} = 2T]$$

So:

$$[A = P \left(1 + \frac{R}{200} \right)^{2T}]$$

4. BASIC HALF-YEARLY EXAMPLE

₹ 10,000 at 10% p.a., compounded half-yearly, for 1 year.

Rate per half-year:

$$[5\%]$$

Number of half-years:

$$[2]$$

Thus:

$$[A = 10000(1.05)^2]$$

$$[= 10000(1.1025)]$$

$$[= ₹ 11,025]$$

CI:

$$[₹ 1,025]$$

5. COMPARE WITH ANNUAL COMPOUNDING

At 10% annually for 1 year:

$$[A = 10000(1.1) = 11000]$$

At 10% nominal compounded half-yearly:

$$[A = 11025]$$

Difference:

$$\sqrt[12]{5}$$

So half-yearly compounding earns slightly more.

6. WHY?

After first 6 months:

$$\sqrt[10000]{10500}$$

For second half-year, 5% applies to:

$$\sqrt[10500]$$

not 1.1×1.05 .

So the first half-year's interest itself earns interest.

7. HALF-YEARLY FOR 2 YEARS

1.8×10^4 at 20% p.a., compounded half-yearly, for 2 years.

Rate per half-year:

$$\sqrt[10\%]$$

Periods:

$$\sqrt[4]$$

$$\sqrt[A=8000(1.1)^4]$$

Since:

$$\sqrt[1.1^4=1.4641]$$

$$\sqrt[A=8000(1.4641)]$$

$$\sqrt[= \boxed{111, 712.80}]$$

8. IMPORTANT HALF-YEARLY TRAP

Do not do:

$$\sqrt[20\% \text{ for 2 years} \rightarrow 40\%]$$

That is SI reasoning.

Under half-yearly CI:

$$\sqrt[10\% \text{ repeated 4 times}]$$

Use:

$$\sqrt[1.1^4]$$

9. QUARTERLY COMPOUNDING

Quarterly means:

$$\boxed{\text{times per year} = 4}$$

So:

$$\boxed{\text{rate per quarter} = \frac{R}{4}\%}$$

and:

$$\boxed{\text{number of quarters} = 4T}$$

Formula:

$$\boxed{A = P \left(1 + \frac{R}{400} \right)^{4T}}$$

10. QUARTERLY EXAMPLE

₹ 10,000 at 8% p.a., compounded quarterly, for 1 year.

Rate per quarter:

$$\boxed{2\%}$$

Number of quarters:

$$\boxed{4}$$

$$\boxed{A = 10000(1.02)^4}$$

$$\boxed{1.02^2 = 1.0404}$$

$$\boxed{1.02^4 = 1.08243216}$$

Thus:

$$\boxed{A \approx \text{₹ } 10,824.32}$$

CI:

$$\boxed{\text{₹ } 824.32}$$

11. MONTHLY COMPOUNDING

If annual nominal rate is $(R\%)$, compounded monthly:

$$\boxed{\text{monthly rate} = \frac{R}{12}\%}$$

Number of periods in (T) years:

$$\boxed{12T}$$

Thus:

$$\boxed{A = P \left(1 + \frac{R}{1200} \right)^{12T}}$$

12. EXAMPLE — MONTHLY

₹ 12,000 at 12% nominal annual rate, compounded monthly for 1 year.

Monthly rate:

$$\left[1\% \right]$$

Periods:

$$\left[12 \right]$$

$$\left[A=12000(1.01)^{12} \right]$$

No need to memorize the exact decimal unless asked.

The conceptual setup is:

$$\left[\boxed{12000(1.01)^{12}} \right]$$

13. GENERAL m-TIMES COMPOUNDING

If compounded (m) times per year:

$$\left[\boxed{A=P\left(1+\frac{R}{100m}\right)^{mT}} \right]$$

This one formula covers:

- annual
- half-yearly
- quarterly
- monthly

14. EFFECTIVE ANNUAL RATE

A nominal rate may say:

10% per annum compounded half-yearly.

But actual growth over one year is:

$$\left[(1.05)^2 - 1 \right]$$

$$\left[=0.1025 \right]$$

So actual annual growth:

$$\left[\boxed{10.25\%} \right]$$

This is called the **effective annual rate**.

15. EFFECTIVE RATE FORMULA

If nominal annual rate $(R\%)$ is compounded (m) times per year:

$$\left[\boxed{R_{\text{eff}} = \left[\left(1 + \frac{R}{100m}\right)^m - 1 \right] \times 100} \right]$$

16. EXAMPLE — 12% HALF-YEARLY

Nominal annual rate:

$$\left[12\% \right]$$

Half-year rate:

$$\sqrt[2]{6\%}$$

Effective annual multiplier:

$$\sqrt[2]{1.06^2} = 1.1236$$

Effective annual rate:

$$\boxed{12.36\%}$$

17. EXAMPLE — 8% QUARTERLY

Quarterly rate:

$$\sqrt[4]{2\%}$$

Annual multiplier:

$$\sqrt[4]{1.02^4} = 1.08243216$$

Effective annual rate:

$$\boxed{8.243216\% \approx 8.24\%}$$

18. NOMINAL VS EFFECTIVE RATE

Nominal rate

The quoted annual rate before intra-year compounding effect.

Effective rate

The actual percentage by which money grows over the whole year.

For positive rates:

$$\boxed{\text{effective rate} > \text{nominal rate}}$$

when compounding more than once per year.

19. HALF-YEARLY EFFECTIVE RATE SHORTCUT

If nominal annual rate is $R\%$:

half-year rate:

$$R/2$$

Effective annual percentage:

$$\boxed{R + \frac{R^2}{400}}$$

Why?

$$\left(1 + \frac{R}{200}\right)^2$$

gives:

$$1 + \frac{R}{100} + \frac{R^2}{40000}$$

Hence effective rate:

$$\left[1 + \frac{R}{400} \right]^4$$

20. EXAMPLE

Nominal 20%, compounded half-yearly.

Effective annual rate:

$$\left[1 + \frac{20}{200} \right]^2$$

$$= 1.21$$

$$= \boxed{21\%}$$

Indeed:

$$1.1^2 = 1.21$$

21. FRACTIONAL YEARS — VERY IMPORTANT

Suppose interest is **compounded annually** and time is:

$$2.5 \text{ years}$$

In many aptitude problems, if no other compounding convention is stated:

- compound for the completed 2 years,
- then apply simple interest for the remaining half-year on the amount after 2 years.

This convention should be used only when the problem clearly implies annual compounding with a fractional leftover period.

22. EXAMPLE — 2.5 YEARS, ANNUAL COMPOUNDING

₹ 10,000 at 10% compounded annually for 2.5 years.

After 2 complete years:

$$10000(1.1)^2 = 12100$$

For remaining 6 months, simple interest at 10% p.a. on ₹ 12100 :

Half-year rate:

$$5\%$$

Additional:

$$12100(0.05) = 605$$

Final:

$$12100 + 605 = \boxed{₹ 12705}$$

23. DO NOT CONFUSE WITH HALF-YEARLY COMPOUNDING

If the same problem says:

10% p.a. compounded half-yearly for 2.5 years

Then:

- half-year rate = 5%
- number of half-years = 5

So:

$$A = 10000(1.05)^5$$

Different answer.

24. FRACTIONAL YEAR DECISION RULE

If compounding annually + fractional year:

Use full annual compounds plus simple interest on remaining fraction, unless the problem specifies another convention.

If compounded half-yearly:

Convert entire time into half-years.

If quarterly:

Convert entire time into quarters.

25. VARIABLE RATES

If rate changes from year to year:

$$A = P \left(1 + \frac{R_1}{100}\right) \left(1 + \frac{R_2}{100}\right) \cdots$$

26. VARIABLE-RATE EXAMPLE

110,000:

- Year1: 10%
- Year2: 20%
- Year3: 10%

$$A = 10000(1.1)(1.2)(1.1)$$

$$= 10000(1.452)$$

$$= \boxed{14520}$$

27. GROWTH + DEPRECIATION

150,000:

- appreciates 20%

- then depreciates 25%

$$\boxed{50000(1.2)(0.75)}$$

$$= 50000(0.9)$$

$$= \boxed{45,000}$$

Net:

$$\boxed{10\% \text{ decrease}}$$

28. POPULATION WITH BIRTH/DEATH PERCENTAGES

Population = 100,000.

First year +10%.

Second year - 5 % .

$$100000(1.1)(0.95)$$

$$= 104500$$

So:

$$\boxed{104,500}$$

Net change:

$$\boxed{+4.5\%}$$

29. REVERSE VARIABLE GROWTH

Current population:

$$132000$$

It grew:

- 10% first year
- 20% second year

Original:

$$P = \frac{132000}{1.1(1.2)}$$

$$= \frac{132000}{1.32}$$

$$= \boxed{100000}$$

30. C I" DIFFERENCE FOR 2 YEARS

Recall:

$$\boxed{CI - SI = P \left(\frac{R}{100} \right)^2}$$

This is one of the highest-value shortcuts.

31. INTERPRETATION OF 2-YEAR DIFFERENCE

Why does this difference occur?

Year 1 interest:

$$[Pr]$$

In CI, this year-1 interest earns another year's interest:

$$[(Pr)r = Pr^2]$$

That is exactly the extra amount over SI.

32. CI DIFFERENCE FOR 2 YEARS — USING FIRST-YEAR INTEREST

If first-year interest is (I) :

$$[I = Pr]$$

Then second-year interest on first-year interest:

$$[Ir]$$

So:

$$[\boxed{CI - SI = I \times r}]$$

33. EXAMPLE

First-year interest = ₹ 1000 at 10%.

Extra CI over SI in 2 years:

$$[1000(0.1) = \boxed{₹ 100}]$$

34. DIFFERENCE BETWEEN 2ND-YEAR AND 1ST-YEAR CI INTEREST

At CI:

First-year interest:

$$[Pr]$$

Second-year interest:

$$[P(1+r)r]$$

Difference:

$$[Pr^2]$$

Therefore:

$$[\boxed{I_2 - I_1 = CI - SI \text{ for 2 years}}]$$

35. EXAMPLE

$P = ₹ 5000$ at $R = 20\%$.

Year 1 interest:

$$\text{\textbackslash}[1000 \text{\textbackslash}]$$

Year 2 interest:

$$\text{\textbackslash}[1200 \text{\textbackslash}]$$

Difference:

$$\text{\textbackslash}[200 \text{\textbackslash}]$$

And:

$$\text{\textbackslash}[5000(0.2)^2 = 200 \text{\textbackslash}]$$

Correct.

36. FIND RATE FROM CONSECUTIVE YEAR INTERESTS

Suppose:

$$\text{Year 1 CI interest} = \text{\textasciitilde} 1000.$$

$$\text{Year 2 CI interest} = \text{\textasciitilde} 1100.$$

Increase:

$$\text{\textbackslash}[100 \text{\textbackslash}]$$

Relative to first-year interest:

$$\text{\textbackslash}[100/1000=10\% \text{\textbackslash}]$$

Therefore:

$$\text{\textbackslash}[\boxed{R=10\%} \text{\textbackslash}]$$

Because every year's interest itself grows at the same compound rate.

37. CONSECUTIVE YEAR INTEREST RATIO

Under fixed annual rate (r) :

$$\text{\textbackslash}[I_1=Pr \text{\textbackslash}]$$

$$\text{\textbackslash}[I_2=Pr(1+r) \text{\textbackslash}]$$

$$\text{\textbackslash}[I_3=Pr(1+r)^2 \text{\textbackslash}]$$

Hence:

$$\text{\textbackslash}[\boxed{I_1:I_2:I_3 = 1:(1+r):(1+r)^2} \text{\textbackslash}]$$

38. FIND PRINCIPAL FROM FIRST-YEAR INTEREST

If first-year CI interest = $\text{\textasciitilde} 1200$ at 12%:

$$\text{\textbackslash}[0.12P=1200 \text{\textbackslash}]$$

$$\text{\textbackslash}[P = \boxed{\text{\textasciitilde} 10000} \text{\textbackslash}]$$

Same as SI for the first year.

39. FIND PRINCIPAL FROM SECOND-YEAR INTEREST

Suppose second-year interest at 10% is ₹ 1100 .

Second-year opening amount:

$$\text{₹ } 1100/0.1 = \text{₹ } 11000$$

This is amount after year 1.

So original:

$$\text{₹ } P = 11000/1.1$$

$$\text{₹ } = \boxed{\text{₹ } 10000}$$

40. AMOUNT RATIO ACROSS EQUAL INTERVALS

At constant CI rate:

$$\text{₹ } A_t, \text{₹ } A_{t+k}, \text{₹ } A_{t+2k}$$

form a geometric progression.

Because:

$$\text{₹ } \frac{A_{t+k}}{A_t} = (1+r)^k$$

and same ratio repeats.

This can create GATE-style pattern questions.

41. EXAMPLE — THREE AMOUNTS

Suppose amount after 1 year is ₹ 1100 and after 2 years ₹ 1210 .

Then ratio:

$$\text{₹ } 1210/1100 = 1.1$$

So rate:

$$\text{₹ } 10\%$$

Amount after 3 years:

$$\text{₹ } 1210(1.1) = \boxed{\text{₹ } 1331}$$

42. FIND MISSING CI AMOUNT USING GP

Amounts after consecutive years:

$$\text{₹ } A_1, \text{₹ } A_2, \text{₹ } A_3$$

Since GP:

$$\text{₹ } \boxed{A_2^2 = A_1 A_3}$$

Example:

$$[A_1=1100, \text{quad } A_3=1331]$$

$$[A_2=\sqrt{1100(1331)}]$$

$$[\boxed{1210}]$$

43. DEPRECIATION AMOUNTS ALSO FORM GP

If an asset depreciates at fixed percentage:

$$[V_0, V_1, V_2, \dots]$$

form a GP with ratio:

$$[1-r]$$

Example:

100000 at 10% depreciation:

$$[100000, 90000, 81000, 72900, \dots]$$

44. AMOUNT OF DEPRECIATION EACH YEAR FALLS

For fixed percentage depreciation:

Year 1 depreciation:

$$[10000]$$

Year 2:

$$[9000]$$

Year 3:

$$[8100]$$

So absolute depreciation decreases each year.

But percentage depreciation remains 10%.

45. GROWTH AMOUNT EACH YEAR RISES

For positive compound growth:

Initial 10000, 10%.

Annual increases:

Year 1:

$$[1000]$$

Year 2:

$$[1100]$$

Year 3:

$\text{₹ } 1210$

Interest/growth increments themselves form a GP.

46. SAME FINAL AMOUNT, DIFFERENT COMPOUNDING

For the same nominal annual rate and same time:

more frequent compounding generally produces a larger final amount.

For positive rates:

$A_{\text{monthly}} > A_{\text{quarterly}} > A_{\text{half-yearly}} > A_{\text{annual}}$

assuming all use the same nominal rate.

47. GATE-STYLE QUESTION 1

₹ 10,000 at 12% compounded half-yearly for 1 year.

Half-year rate:

6%

$A = 10000(1.06)^2$

$= 10000(1.1236)$

$= \text{₹ } 11,236$

CI:

$\text{₹ } 11,236$

48. GATE-STYLE QUESTION 2

₹ 16,000 at 20% compounded half-yearly for 1 year.

Half-year rate:

10%

$A = 16000(1.1)^2$

$= 16000(1.21)$

$= \text{₹ } 19,360$

49. GATE-STYLE QUESTION 3

₹ 10,000 at 8% compounded quarterly for 1 year.

Quarter rate:

2%

$A = 10000(1.02)^4$

$\approx \text{₹ } 10,824.32$

50. GATE-STYLE QUESTION 4

Nominal 10% compounded half-yearly.

Effective annual rate:

$$\sqrt[2]{(1.05)^2 - 1}$$

$$\sqrt[2]{=0.1025}$$

$$\sqrt[2]{=\boxed{10.25\%}}$$

51. PLACEMENT-STYLE QUESTION 1

₹ 20,000 at 12% compounded half-yearly for 1.5 years.

Half-year rate:

$$\sqrt[2]{6\%}$$

Periods:

$$\sqrt[2]{3}$$

$$\sqrt[2]{A=20000(1.06)^3}$$

$$\sqrt[2]{1.06^3=1.191016}$$

$$\sqrt[2]{A=\boxed{₹20,320}}$$

52. PLACEMENT-STYLE QUESTION 2

₹ 10,000 at 16% compounded quarterly for 6 months.

Quarter rate:

$$\sqrt[4]{4\%}$$

6 months:

$$\sqrt[4]{2\text{ quarters}}$$

$$\sqrt[4]{A=10000(1.04)^2}$$

$$\sqrt[4]{=\boxed{₹10,816}}$$

53. PLACEMENT-STYLE QUESTION 3

An asset of ₹ 2,00,000 depreciates:

- 20% first year
- 10% second year
- 5% third year

$$\sqrt[4]{V=200000(0.8)(0.9)(0.95)}$$

$$\sqrt[4]{=200000(0.684)}$$

$$\boxed{11,380}$$

54. PLACEMENT-STYLE QUESTION 4

Population grows:

- 10%
- 10%
- 20%

Initial = 50,000.

$$50000(1.1)^2(1.2)$$

$$=50000(1.452)$$

$$=\boxed{72,600}$$

55. ADVANCED QUESTION — FIND PRINCIPAL FROM C I" S I

Difference between CI and SI after 2 years is 160.

Rate = 20%.

$$160=P(0.2)^2$$

$$160=0.04P$$

$$P=\boxed{4000}$$

56. ADVANCED QUESTION — FIND RATE FROM YEARLY INTEREST

First-year compound interest = 800.

Second-year compound interest = 880.

Growth in yearly interest:

$$80$$

Rate:

$$80/800\times100$$

$$=\boxed{10\%}$$

Principal:

$$800/0.1=\boxed{8000}$$

57. ADVANCED QUESTION — ANNUAL VS HALF-YEARLY

110,000 at 20%.

Annual compounding for 1 year:

$$A=12000$$

Half-yearly:

$$\boxed{A = 10000(1.1)^2 = 12100}$$

Extra due to half-yearly:

$$\boxed{1100}$$

58. ADVANCED QUESTION — FRACTIONAL YEAR

At 10% compounded annually for 2.5 years.

After 2 years:

$$\boxed{20000(1.1)^2 = 24200}$$

Remaining 0.5 year under standard annual-compounding aptitude convention:

$$\boxed{5\% \text{ of } 24200 = 1210}$$

Final:

$$\boxed{25410}$$

59. ADVANCED QUESTION — REVERSE DEPRECIATION

Value after 3 years of 20% annual depreciation:

$$\boxed{151200}$$

Original:

$$V_0 = \frac{151200}{0.8^3}$$

$$= \frac{151200}{0.512}$$

$$= \boxed{294000}$$

60. ADVANCED QUESTION — RESTORE AFTER DEPRECIATION

Asset falls 20% in year 1 and 25% in year 2.

Final multiplier:

$$\boxed{0.8(0.75)=0.6}$$

So current value = 60% original.

Required increase to restore original:

$$\frac{1-0.6}{0.6} \times 100$$

$$= \frac{0.4}{0.6} \times 100$$

$$= \boxed{66\frac{2}{3}\%}$$

61. COMMON TRAPS

& pTRAP 1 — Half-yearly compounding

Halve the annual rate **and double the number of periods**.

& pTRAP 2 — Quarterly

Divide rate by 4 and multiply years by 4.

& pTRAP 3 — Nominal rate vs effective rate

10% nominal compounded half-yearly gives 10.25% effective annual growth.

& pTRAP 4 — Fractional years

Annual compounding with an incomplete final year is not automatically the same as half-yearly compounding.

& pTRAP 5 — Formula gives amount

$$\backslash [P(1+r)^n \backslash]$$

is amount.

Subtract $\backslash (P\backslash)$ for CI.

& pTRAP 6 — Variable rates

Multiply the factors; do not average rates blindly.

62. MUST-MEMORIZE FORMULAS

$\backslash (m\backslash)$ -times compounding

$$\backslash [\boxed{ A = P \left(1 + \frac{R}{100m} \right)^{mT} } \backslash]$$

Half-yearly

$$\backslash [\boxed{ A = P \left(1 + \frac{R}{200} \right)^{2T} } \backslash]$$

Quarterly

$$\backslash [\boxed{ A = P \left(1 + \frac{R}{400} \right)^{4T} } \backslash]$$

63. EFFECTIVE RATE

$$\backslash [\boxed{ R_{\rm eff} = \left[\left(1 + \frac{R}{100m} \right)^m - 1 \right] 100 } \backslash]$$

For half-yearly:

$$\backslash [\boxed{ R_{\rm eff} = R + \frac{R^2}{400} } \backslash]$$

64. HIGH-VALUE CI SHORTCUTS

For 2 years:

$$\backslash [\boxed{ CI - SI = Pr^2 } \backslash]$$

where:

$$r = R/100$$

Consecutive yearly CI interest grows by:

$$\boxed{1+r}$$

Amounts at equal time intervals also form a GP.

65. 5-MINUTE REVISION

Remember:

Annual

$$R, n$$

Half-yearly

$$R/2, 2n$$

Quarterly

$$R/4, 4n$$

Monthly

$$R/12, 12n$$

The rule is always:

$$\boxed{\text{smaller rate per period, more periods}}$$

PRACTICE — LEVEL 0

Q1

₹10,000 at 10% compounded half-yearly for 1 year. Find amount.

Q2

₹20,000 at 12% compounded half-yearly for 6 months. Find amount.

Q3

₹10,000 at 8% compounded quarterly for 6 months. Find amount.

Q4

Find effective annual rate of 20% nominal compounded half-yearly.

Q5

A ₹1,00,000 machine depreciates 20% annually for 3 years. Find value.

LEVEL 1

Q6

₹ 16,000 at 20% compounded half-yearly for 2 years. Find amount.

Q7

₹ 10,000 at 12% compounded quarterly for 1 year. Write the exact multiplier and approximate amount.

Q8

Difference between CI and SI for 2 years is ₹ 90 at 10%. Find principal.

Q9

Year 1 CI interest is ₹ 500 and year 2 interest is ₹ 550. Find rate.

Q10

A population of 80,000 grows 10%, then 20%, then falls 10%. Find final population.

LEVEL 2**Q11**

₹ 20,000 at 10% compounded annually for 2.5 years. Find amount using standard annual-compounding aptitude convention.

Q12

₹ 20,000 at 10% compounded half-yearly for 2.5 years. Find amount.

Q13

Nominal annual rate 12% compounded half-yearly. Find effective annual rate.

Q14

An asset is ₹ 64,000 after 2 years of 20% depreciation. Find original value.

Q15

First-year CI interest is ₹ 1200 and rate is 12%. Find second-year interest.

LEVEL 3 — EXAM PRESSURE**Q16**

Amount after one year is ₹ 11,000 and after two years is ₹ 12,100. Find principal and rate.

Q17

₹ 10,000 grows at 10%, 20%, and " 10 % over three successive years. Find equivalent constant annual rate $\backslash(r)$ satisfying $\backslash((1+r)^3=1.188\backslash)$. You may leave the answer as a cube-root expression or approximate it.

Q18

A machine depreciates 10% annually for 2 years and then appreciates 25%. Find net % change from original.

Q19

Difference between CI and SI for 2 years at 20% is ₹ 400. Find principal and CI for 2 years.

Q20

A nominal 8% rate compounded quarterly is compared with 8% annual compounding. Which gives the larger one-year amount, and what is the effective quarterly-compounded annual rate approximately?

ANSWERS

Q1

Half-year rate:

$$\sqrt[2]{5\%}$$

$$\sqrt[2]{A=10000(1.05)^2}$$

$$\sqrt[2]{= \boxed{11,025}}$$

Q2

6 months = one half-year.

Rate:

$$\sqrt[2]{6\%}$$

$$\sqrt[2]{A=20000(1.06)}$$

$$\sqrt[2]{= \boxed{121,200}}$$

Q3

Quarter rate:

$$\sqrt[4]{2\%}$$

6 months = 2 quarters.

$$\sqrt[4]{A=10000(1.02)^2}$$

$$\sqrt[4]{=10000(1.0404)}$$

$$\sqrt[4]{= \boxed{110,404}}$$

Q4

$$\sqrt[20]{20+\frac{20^2}{400}}$$

$$\sqrt[20]{=20+1}$$

$$\sqrt[20]{= \boxed{21\%}}$$

Q5

$$\sqrt[3]{100000(0.8)^3}$$

$$\sqrt[3]{=100000(0.512)}$$

$$\sqrt[3]{= \boxed{151,200}}$$

Q6

Half-year rate:

$$\sqrt[2]{10\%}$$

Periods:

$$\sqrt[4]{4}$$

$$A = 16000(1.1)^4$$

$$= 16000(1.4641)$$

$$= \boxed{23,425.60}$$

Q7

Quarterly rate:

$$3\%$$

$$A = 10000(1.03)^4$$

$$1.03^4 \approx 1.12550881$$

$$\boxed{A \approx 11,255.09}$$

Q8

$$90 = P(0.1)^2$$

$$90 = 0.01P$$

$$\boxed{P = 9000}$$

Q9

$$R = \frac{550-500}{500} \times 100$$

$$= \boxed{10\%}$$

Q10

$$80000(1.1)(1.2)(0.9)$$

$$= 80000(1.188)$$

$$= \boxed{95,040}$$

Q11

After 2 years:

$$20000(1.1)^2 = 24200$$

Half-year SI on 24200:

$$5\% = 1210$$

Final:

$$\boxed{25,410}$$

Q12

Half-year rate:

$$5\%$$

2.5 years = 5 half-years.

$$A = 20000(1.05)^5$$

$$1.05^5 \approx 1.27628156$$

$$A \approx \boxed{25,525.63}$$

Q13

$$\sqrt[1]{1.06^2 - 1 = 0.1236}$$

$$\sqrt[1]{\boxed{12.36\%}}$$

Q14

$$\sqrt[1]{64000 = P(0.8)^2}$$

$$\sqrt[1]{64000 = 0.64P}$$

$$\sqrt[1]{\boxed{P = \frac{100000}{0.64}}}$$

Q15

Second-year interest:

$$\sqrt[1]{1200(1.12)}$$

$$\sqrt[1]{\boxed{1344}}$$

Q16

$$\sqrt[1]{\frac{12100}{11000} = 1.1}$$

So:

$$\sqrt[1]{R = 10\%}$$

First-year amount:

$$\sqrt[1]{P(1.1) = 11000}$$

$$\sqrt[1]{P = \boxed{10000}}$$

Thus:

$$\sqrt[1]{\boxed{R = 10\%}}$$

Q17

Overall multiplier:

$$\sqrt[1]{1.1(1.2)(0.9) = 1.188}$$

So:

$$\sqrt[1]{(1+r)^3 = 1.188}$$

$$\sqrt[1]{1+r = \sqrt[3]{1.188}}$$

$$\sqrt[1]{\boxed{r = \sqrt[3]{1.188} - 1}}$$

Approximately:

$$\sqrt[1]{\boxed{r \approx 5.91\%}}$$

Q18

After two 10% depreciations:

$$\sqrt[1]{0.9^2 = 0.81}$$

Then +25%:

$$\sqrt[1]{0.81(1.25) = 1.0125}$$

Net:

$$\boxed{1.25\% \text{ increase}}$$

Q19

$$400 = P(0.2)^2$$

$$P = 10000$$

Amount after 2 years:

$$10000(1.2)^2 = 14400$$

CI:

$$14400 - 10000 = \boxed{4400}$$

So:

$$\boxed{P = 10000}$$

Q20

Annual compounding:

$$1.08$$

Quarterly:

$$1.02^4 = 1.08243216$$

Thus quarterly compounding gives the larger amount.

Effective annual rate:

$$\boxed{8.243216\% \approx 8.24\%}$$

CHAPTER 21 COMPLETE

We have now covered:

- CI intuition
- annual compounding
- amount and CI formulas
- reverse principal/rate/time problems
- SI vs CI
- 2-year and 3-year CI relations
- half-yearly compounding
- quarterly compounding
- monthly/general (m) -times compounding
- nominal vs effective rate
- fractional-year convention
- variable rates
- population growth
- appreciation and depreciation
- reverse growth/depreciation
- yearly-interest patterns

- amount ratios and GP behavior '
- advanced GATE/placement problems '

Progress

Completed: 21/76 Unit 1 — Quantitative Aptitude: 0=No PROGRESS

NEXT

0=No Chapter 22 — Mixtures & Alligation

Next we will cover:

- mixture concentration
- component vs total quantity
- weighted-average interpretation
- alligation rule
- ratio of two ingredients
- milk–water / solution questions
- price-mixture questions
- target concentration
- adding one ingredient
- removing one ingredient
- profit with mixtures
- advanced GATE/placement shortcuts.

USER

next

ASSISTANT

CHAPTER 22 — MIXTURES & ALLIGATION

MIXTURE CONCENTRATION, COMPONENT RATIOS, ALLIGATION, PRICE MIXTURES & TARGET COMPOSITION

Status: 0=No PROGRESS GATE: High Placements: Very High Priority: 0=Yes 1

Mixture problems are really applications of:

- ratios,
- percentages,
- weighted averages,
- alligation.

The central principle is:

$$\boxed{\text{Amount of component} = \text{Total mixture} \times \text{component fraction}}$$

1. WHAT IS A MIXTURE?

A mixture contains two or more components.

Examples:

- milk + water
- acid + water
- alcohol + water
- rice of different prices
- coffee of different prices
- alloys
- solutions of different concentrations

If 20 L contains 15 L milk and 5 L water:

$$\text{Milk:Water}=15:5=3:1$$

2. COMPONENT FRACTION FROM RATIO

If:

$$A:B=m:n$$

then total parts:

$$m+n$$

Fraction of A:

$$\boxed{\frac{m}{m+n}}$$

Fraction of B:

$$\boxed{\frac{n}{m+n}}$$

3. EXAMPLE

Milk : water:

$$3:2$$

Milk fraction:

$$\frac{3}{5}$$

Water fraction:

$$\frac{2}{5}$$

Therefore percentages:

$$\boxed{60\% \text{ milk}, 40\% \text{ water}}$$

4. FIND COMPONENT QUANTITY

A 40 L mixture has milk:water = 3:2.

Milk:

$$40 \times \frac{3}{5} = \boxed{24\text{L}}$$

Water:

$$40 \times \frac{2}{5} = \boxed{16\text{L}}$$

5. CONCENTRATION

If a 20 L solution contains 5 L acid:

$$\text{Acid fraction} = \frac{5}{20} = \frac{1}{4}$$

Acid concentration:

$$\boxed{25\%}$$

Water concentration:

$$75\%$$

6. PERCENTAGE TO RATIO

A solution is 30% acid.

Then out of 100 parts:

- acid = 30
- water = 70

Therefore:

$$30:70 = \boxed{3:7}$$

7. PURE COMPONENT

Pure milk can be treated as:

$$100\% \text{ milk}$$

Pure water in a milk-water problem:

$$0\% \text{ milk}$$

This makes alligation very powerful.

8. MIXING TWO SOLUTIONS — WEIGHTED AVERAGE

Suppose:

- 10 L of 20% acid
- 20 L of 50% acid

Acid amounts:

First:

$$\sqrt[10]{0.2} = 2L$$

Second:

$$\sqrt[20]{0.5} = 10L$$

Total acid:

$$\sqrt[12]{L}$$

Total solution:

$$\sqrt[30]{L}$$

Concentration:

$$\sqrt[12/30]{\times 100} = \boxed{40\%}$$

9. MASTER WEIGHTED-CONCENTRATION FORMULA

If quantities are (Q_1, Q_2) and concentrations (C_1, C_2) :

$$\sqrt[\boxed{C = \frac{Q_1 C_1 + Q_2 C_2}{Q_1 + Q_2}}]$$

Use concentrations consistently as percentages or decimals.

10. EXAMPLE

20 L at 30% mixed with 30 L at 50%.

$$\sqrt[C = \frac{20(30) + 30(50)}{50}]$$

$$\sqrt[= \frac{600 + 1500}{50}]$$

$$\sqrt[= \boxed{42\%}]$$

11. ALLIGATION

Alligation quickly finds the ratio in which two mixtures must be combined to obtain a target concentration.

If:

$$\sqrt[C_1 < C < C_2]$$

then:

$$\sqrt[\boxed{Q_1 : Q_2 = (C_2 - C) : (C - C_1)}]$$

This is exactly the weighted-average shortcut from Chapter 17.

12. ALLIGATION DIAGRAM

For concentrations:

$$\sqrt[20\% \sqcup 50\%]$$

Target:

$$\sqrt[3]{30\%}$$

Cross differences:

$$\sqrt[3]{50-30=20}$$

$$\sqrt[3]{30-20=10}$$

So:

$$\sqrt[3]{20\% \text{ solution} : 50\% \text{ solution} = 20:10}$$

$$\sqrt[3]{= \boxed{2:1}}$$

13. VERIFY

Take:

$$\sqrt[3]{2L \text{ of } 20\%}$$

and:

$$\sqrt[3]{1L \text{ of } 50\%}$$

Pure component amount:

$$\sqrt[3]{2(0.2)+1(0.5) = 0.4+0.5 = 0.9}$$

Total:

$$\sqrt[3]{3}$$

Concentration:

$$\sqrt[3]{0.9/3 = 30\%}$$

Correct.

14. MEMORY TRICK

Cross the differences.

For:

$$\sqrt[3]{\text{lower } L, \text{quad } \text{target } M, \text{quad } \text{higher } H}$$

$$\sqrt[3]{\boxed{\text{Lower quantity : Higher quantity} = (H-M):(M-L)}}$$

15. DIRECTION TRAP

Mix 20% and 50% to obtain 40%.

Correct ratio:

$$\sqrt[3]{20\% \text{ solution} : 50\% \text{ solution}}$$

$$\sqrt[3]{=(50-40):(40-20)}$$

$$\sqrt[3]{=10:20}$$

$$\boxed{1:2}$$

Since target 40% is closer to 50%, more of the 50% solution is needed.

16. TARGET MUST LIE BETWEEN COMPONENT VALUES

Can 20% and 40% solutions produce 60% solution by mixing?

No.

A positive weighted average must lie between:

$$20\% \text{ and } 40\%$$

So:

$$\boxed{\text{Impossible}}$$

This is an instant option-elimination rule.

17. MILK + WATER USING ALLIGATION

A milk-water mixture contains 60% milk.

How much pure water should be mixed with pure milk to get 60% milk?

Pure milk:

$$100\%$$

Pure water:

$$0\%$$

Target:

$$60\%$$

Ratio:

$$\text{water}:\text{milk} = (100-60):(60-0)$$

$$=40:60$$

$$=2:3$$

Therefore:

$$\boxed{\text{Milk:Water}=3:2}$$

18. DIRECT RATIO METHOD

If milk is 60%:

$$\text{milk:water} = 60:40 = \boxed{3:2}$$

For pure two-component mixtures, percentage-to-ratio conversion is often faster than alligation.

19. ADDING PURE WATER

Suppose 20 L mixture contains 60% milk.

Milk:

$$\backslash[12L \backslash]$$

Water:

$$\backslash[8L \backslash]$$

If 10 L water is added:

Milk remains:

$$\backslash[12L \backslash]$$

Water becomes:

$$\backslash[18L \backslash]$$

Total:

$$\backslash[30L \backslash]$$

Milk percentage:

$$\backslash[12/30\text{times}100 = \boxed{40\%} \backslash]$$

20. IMPORTANT ADDITION RULE

If pure water is added:

$$\backslash[\boxed{\text{amount of milk/solute stays constant}} \backslash]$$

If pure milk/solute is added:

$$\backslash[\boxed{\text{amount of water/solvent stays constant}} \backslash]$$

Use the unchanged component.

21. FIND WATER TO ADD

20 L mixture contains 60% milk.

How much water must be added to make milk 40%?

Milk initially:

$$\backslash[20(0.6)=12L \backslash]$$

Let final total = (T) .

Since milk must be 40%:

$$\backslash[0.4T=12 \backslash]$$

$$\backslash[T=30 \backslash]$$

Water added:

$$\backslash[30-20 = \boxed{10L} \backslash]$$

22. FAST FORMULA — ADDING PURE SOLVENT

Initial mixture quantity:

$$\backslash[Q \backslash]$$

Initial solute concentration:

$$\backslash[C_i \backslash]$$

Final concentration:

$$\backslash[C_f < C_i \backslash]$$

Solute stays unchanged:

$$\backslash[QC_i = (Q+x)C_f \backslash]$$

Thus:

$$\backslash[\boxed{ x = Q \left(\frac{C_i}{C_f} - 1 \right) } \backslash]$$

Use concentrations in the same units.

23. EXAMPLE

40 L of 30% acid must be diluted to 20%.

$$\backslash[x = 40 \left(\frac{30}{20} - 1 \right) \backslash]$$

$$\backslash[= 40(1.5 - 1) \backslash]$$

$$\backslash[= \boxed{20L} \backslash]$$

24. ADDING PURE SOLUTE

Suppose 20 L mixture is 30% acid.

Acid:

$$\backslash[6L \backslash]$$

Water:

$$\backslash[14L \backslash]$$

Add pure acid $\backslash(x\backslash)$ so concentration becomes 50%.

$$\backslash[\frac{6+x}{20+x} = 0.5 \backslash]$$

$$\backslash[6+x = 10+0.5x \backslash]$$

$$\backslash[0.5x = 4 \backslash]$$

$$\backslash[\boxed{x=8L} \backslash]$$

25. FAST SOLVENT-INVARIANT METHOD

When pure acid is added, water stays fixed at:

$$\backslash[14L \backslash]$$

At final 50% acid, water must also be 50%.

If water 14 L is 50% of final mixture:

$$\text{\text{final total}}=28\text{L}$$

Initially total:

$$20\text{L}$$

Acid added:

$$28-20 = \boxed{8\text{L}}$$

Often faster.

26. ADDING A MIXTURE OF KNOWN CONCENTRATION

20 L at 30%.

Add x L of 60% solution to obtain 40%.

Equation:

$$20(30)+60x = 40(20+x)$$

$$600+60x=800+40x$$

$$20x=200$$

$$\boxed{x=10\text{L}}$$

27. FAST ALLIGATION VERSION

Mix:

$$30\% \quad \text{and} \quad 60\%$$

to get:

$$40\%$$

Ratio:

$$30\%:60\% = (60-40):(40-30)$$

$$=20:10 = 2:1$$

If first amount = 20 L:

$$\boxed{\text{second}=10\text{L}}$$

Much faster.

28. PRICE MIXTURES

Alligation is not only about concentration.

Suppose rice costing:

$$140 / \text{kg}$$

and:

$$\text{₹ } 160 / \text{kg}$$

must be mixed to produce a mixture costing:

$$\text{₹ } 148 / \text{kg}$$

Ratio:

$$40 \text{ rice} : 60 \text{ rice}$$

$$= (60 - 48) : (48 - 40)$$

$$= 12 : 8$$

$$= \boxed{3:2}$$

29. WHY PRICE MIXTURE IS SAME MATHEMATICS

Average cost:

$$\frac{\text{total cost}}{\text{total quantity}}$$

So price per kg is a weighted average.

Alligation works because it is simply weighted averaging.

30. EXAMPLE — TEA

Tea costs:

$$\text{₹ } 200 / \text{kg}, \text{ and } 300 / \text{kg}$$

Target mixture:

$$\text{₹ } 240 / \text{kg}$$

Ratio:

$$200 \text{ tea} : 300 \text{ tea}$$

$$= (300 - 240) : (240 - 200)$$

$$= 60 : 40$$

$$= \boxed{3:2}$$

31. FIND QUANTITIES FROM TOTAL

Rice ₹ 40 and ₹ 60 mixed in ratio:

$$3:2$$

Total mixture:

$$50 \text{ kg}$$

First rice:

$$50 \times \frac{3}{5} = 30 \text{ kg}$$

Second:

$\backslash[20\text{kg} \backslash]$

$\backslash[\boxed{30\text{kg}, 20\text{kg}} \backslash]$

32. PRICE MIXTURE + PROFIT

A trader mixes rice costing $\text{₹ } 40 / \text{kg}$ and $\text{₹ } 60 / \text{kg}$ in ratio 3:2.

Mixture cost:

$\backslash[\frac{3(40)+2(60)}{5} \backslash]$

$\backslash[=\frac{240}{5} \backslash]$

$\backslash[=\boxed{\text{₹ } 48 / \text{kg}} \backslash]$

If sold at $\text{₹ } 60 / \text{kg}$:

Profit:

$\backslash[12 \backslash]$

Profit %:

$\backslash[12/48 \times 100 = \boxed{25\%} \backslash]$

33. TARGET SELLING PRICE WITH PROFIT

A mixture costs $\text{₹ } 80 / \text{kg}$.

Seller wants 25% profit.

Required SP:

$\backslash[80(1.25) = \boxed{\text{₹ } 100 / \text{kg}} \backslash]$

Mixture/alligation determines cost; profit/loss then uses Chapter 18.

34. MIXTURE PRICE WHEN ONE INGREDIENT IS FREE

Suppose milk costs $\text{₹ } 50 / \text{L}$ and water is free.

Milk:water:

$\backslash[4:1 \backslash]$

Cost of 5 L mixture:

$\backslash[4(50)+1(0)=200 \backslash]$

Cost per L:

$\backslash[200/5 = \boxed{\text{₹ } 40 / \text{L}} \backslash]$

35. SELL MILK-WATER MIXTURE AT MILK'S COST PRICE

Milk costs $\text{₹ } 50 / \text{L}$.

Water free.

Mixture:

$\backslash[4:1 \backslash]$

Effective CP per L:

$\backslash[\text{ }^1 4 \text{ }^0 \backslash]$

If sold at $\text{ }^1 5 \text{ }^0 / \text{L} :$

Profit:

$\backslash[10 \backslash]$

Profit %:

$\backslash[10/40\text{times}100 = \boxed{25\%} \backslash]$

36. GENERAL MILK-WATER PROFIT SHORTCUT

If milk:water:

$\backslash[m:w \backslash]$

water is free and mixture sold at milk's cost price:

Cost of mixture is due only to milk.

For $\backslash(m+w\backslash)$ total parts, cost corresponds to $\backslash(m\backslash)$ parts.

SP corresponds to $\backslash(m+w\backslash)$ parts at milk's unit cost.

Thus profit fraction:

$\backslash[\text{frac}\{w\}\{m\} \backslash]$

So:

$\backslash[\boxed{\text{Profit}\%} = \text{frac}\{w\}\{m\}\text{times}100 \backslash]$

37. EXAMPLE

Milk:water:

$\backslash[5:2 \backslash]$

Sold at pure milk's cost price.

Profit:

$\backslash[\text{frac}25\text{times}100 = \boxed{40\%} \backslash]$

38. FIND WATER RATIO FOR TARGET PROFIT

Milk sold at cost price per litre after adding free water.

Target profit:

$\backslash[25\% \backslash]$

Then:

$$\left[\frac{w}{m} = 0.25 \right]$$

So:

$$\left[m:w = 4:1 \right]$$

$$\left[\boxed{4:1} \right]$$

39. MIXTURE SOLD BELOW MILK PRICE

Milk costs ₹ 60 / L .

Water free.

Milk:water = 3:1.

Mixture CP per litre:

$$\left[60 \times \frac{3}{4} = ₹ 45 \right]$$

If mixture sold at ₹ 54 :

Profit:

$$\left[9 \right]$$

Profit %:

$$\left[\frac{9}{45} \times 100 = \boxed{20\%} \right]$$

40. FIND TARGET RATIO FROM PRICE

Milk costs ₹ 80 / L, water free.

A milk-water mixture costs effectively ₹ 60 / L .

Milk fraction:

$$\left[\frac{60}{80} = \frac{3}{4} \right]$$

Therefore water fraction:

$$\left[\frac{1}{4} \right]$$

Ratio:

$$\left[\boxed{3:1} \right]$$

41. MIXING THREE COMPONENTS

For 3+ components, alligation alone may not determine a unique ratio unless extra information exists.

Use:

$$\left[\boxed{\frac{\text{total component amount}}{\text{total mixture}}} \right]$$

or weighted-average equations.

42. EXAMPLE — THREE SOLUTIONS

10 L at 20%, 20 L at 30%, 30 L at 40%.

Pure component:

$$\backslash [2+6+12=20L \backslash]$$

Total:

$$\backslash [60L \backslash]$$

Concentration:

$$\backslash [20/60 = \boxed{33\frac{1}{3}\%} \backslash]$$

43. AVERAGE CONCENTRATION WITH EQUAL QUANTITIES

If equal amounts of 20%, 30%, 40% are mixed:

$$\backslash [\frac{20+30+40}{3} = \boxed{30\%} \backslash]$$

This works only because quantities are equal.

44. UNEQUAL QUANTITIES

10 L at 20% and 30 L at 40%.

Do not average:

$$\backslash [(20+40)/2=30\% \backslash]$$

Correct:

$$\backslash [\frac{10(20)+30(40)}{40} \backslash]$$

$$\backslash [= \frac{1400}{40} \backslash]$$

$$\backslash [=\boxed{35\%} \backslash]$$

45. FIND UNKNOWN CONCENTRATION

20 L of $(x\%)$ solution is mixed with 30 L of 50% solution to get 40%.

$$\backslash [20x+30(50)=50(40) \backslash]$$

$$\backslash [20x+1500=2000 \backslash]$$

$$\backslash [20x=500 \backslash]$$

$$\backslash [\boxed{x=25\%} \backslash]$$

46. FIND UNKNOWN QUANTITY

10 L at 20% mixed with (x) L at 50% to get 40%.

Alligation:

$$\backslash [20\%:50\% = (50-40):(40-20) \backslash]$$

$$\text{[}=10:20 = 1:2 \text{]}$$

So if low solution is 10 L:

$$\text{[}x=20L \text{]}$$

$$\text{[} \boxed{20L} \text{]}$$

47. COMPONENT-CONSERVATIONMETHOD

Whenever nothing containing a particular component is removed, track that component.

Example:

30 L solution is 40% acid.

Acid:

$$\text{[}12L \text{]}$$

Add water until acid becomes 30%.

Final total:

$$\text{[}12/0.3=40L \text{]}$$

Water added:

$$\text{[} \boxed{10L} \text{]}$$

48. GATE-STYLE QUESTION 1

20 L solution has acid:water = 3:2.

Acid:

$$\text{[} 20 \times \frac{3}{5} = \boxed{12L} \text{]}$$

49. GATE-STYLE QUESTION 2

20% and 50% solutions are mixed to get 30%.

Ratio:

$$\text{[} (50-30):(30-20) \text{]}$$

$$\text{[}=20:10 \text{]}$$

$$\text{[}=\boxed{2:1} \text{]}$$

50. GATE-STYLE QUESTION 3

40 L of 30% solution is diluted to 20%.

Pure component:

$$\text{[}12L \text{]}$$

Final total:

$$\lfloor 12/0.2=60\text{L} \rfloor$$

Water added:

$$\lfloor \boxed{20\text{L}} \rfloor$$

51. GATE-STYLE QUESTION 4

Rice 130 kg and 150 kg mixed to get 142 kg .

Ratio:

$$\lfloor 30\text{rice}:50\text{rice} = (50-42):(42-30) \rfloor$$

$$\lfloor =8:12 \rfloor$$

$$\lfloor =\boxed{2:3} \rfloor$$

52. PLACEMENT-STYLE QUESTION 1

15 L of 20% alcohol mixed with 25 L of 40% alcohol.

Alcohol:

$$\lfloor 3+10=13\text{L} \rfloor$$

Total:

$$\lfloor 40\text{L} \rfloor$$

Concentration:

$$\lfloor 13/40\times100 = \boxed{32.5\%} \rfloor$$

53. PLACEMENT-STYLE QUESTION 2

25 L of 60% milk solution. How much water must be added to make it 50% milk?

Milk:

$$\lfloor 25(0.6)=15\text{L} \rfloor$$

Final total:

$$\lfloor 15/0.5=30\text{L} \rfloor$$

Water added:

$$\lfloor \boxed{5\text{L}} \rfloor$$

54. PLACEMENT-STYLE QUESTION 3

Tea 240 kg and 360 kg mixed to obtain 300 kg .

Ratio:

$$\lfloor (360-300):(300-240) \rfloor$$

$$\lfloor =60:60 \rfloor$$

$$\boxed{1:1}$$

55. PLACEMENT-STYLE QUESTION 4

Milk costs ₹ 40 / L water free.

Mixture ratio milk:water = 4:1 and sold at ₹ 40 / L .

Profit:

$$\frac{14}{100} \times 100 = \boxed{25\%}$$

56. ADVANCED QUESTION — ADD CONCENTRATED SOLUTION

30 L solution is 20% acid.

How much 50% acid solution should be added to make mixture 30%?

Alligation:

$$20\%:50\% = (50-30):(30-20)$$

$$= 20:10 = 2:1$$

30 L corresponds to 2 parts.

So added quantity:

$$\boxed{15L}$$

57. ADVANCED QUESTION — ADD PURE ACID

40 L solution contains 25% acid.

How much pure acid to add to make 40% acid?

Initial acid:

$$10L$$

Let pure acid added = x .

$$\frac{10+x}{40+x} = 0.4$$

$$10+x = 16+0.4x$$

$$0.6x = 6$$

$$\boxed{x=10L}$$

58. ADVANCED QUESTION — PRICE MIXTURE + PROFIT

Rice ₹ 40 and ₹ 70 mixed in ratio 2:3.

Mixture CP:

$$\frac{2(40)+3(70)}{5}$$

$$= \frac{290}{5} = 58$$

Sold at 25% profit:

$$SP = 58(1.25)$$

$$= \boxed{72.50 / kg}$$

59. ADVANCED QUESTION — FIND RATIO FROM PROFIT

Milk costs $\text{₹ } 60 / L$.

Water is free.

Mixture sold at $\text{₹ } 60 / L$ yields 20% profit.

Let mixture CP per L be (C) .

$$60 = 1.2C$$

$$C = 50$$

Milk fraction:

$$50/60 = \frac{5}{6}$$

Thus water fraction:

$$\frac{1}{6}$$

Milk:water:

$$\boxed{5:1}$$

60. COMMON TRAPS

& pTRAP 1 — **Averaging concentrations directly**

Only valid for equal quantities.

& pTRAP 2 — **Alligation direction**

Use cross-differences.

& pTRAP 3 — **Target outside range**

Impossible with positive quantities.

& pTRAP 4 — **Adding water**

Solute amount remains unchanged.

& pTRAP 5 — **Adding pure solute**

Solvent amount remains unchanged.

& pTRAP 6 — **Price mixture**

Price per unit is a weighted average, not a simple average unless quantities are equal.

61. MUST-MEMORIZE FORMULAS

Component amount

$$\boxed{\text{Component} = Q \times C}$$

where C is component fraction.

Weighted concentration

$$\boxed{C = \frac{Q_1 C_1 + Q_2 C_2}{Q_1 + Q_2}}$$

Alligation

$$\boxed{Q_1 : Q_2 = (C_2 - C) : (C - C_1)}$$

62. ADDING PURE WATER

$$\boxed{Q C_i = (Q + x) C_f}$$

Therefore:

$$\boxed{x = Q \left(\frac{C_i}{C_f} - 1 \right)}$$

63. MILK-WATER PROFIT SHORTCUT

If:

- milk:water = $(m:w)$
- water free
- mixture sold at milk's cost price

then:

$$\boxed{\text{Profit\%} = \frac{wm}{100} \times 100}$$

64. FAST EXAM DECISION TREE

Ratio given?

Convert into component fractions.

Two concentrations + target?

Use alligation.

Quantities + concentrations?

Use weighted average.

Add water?

Keep solute constant.

Add pure solute?

Keep solvent constant or solve fraction equation.

Price mixture?

Treat price as weighted average.

3+ components?

Use totals/equations.

65. 5-MINUTE REVISION

Remember the four core ideas:

$\boxed{\text{Component} = Q \times C}$

$\boxed{\text{Final concentration} = \frac{\text{total component}}{\text{total mixture}}}$

$\boxed{\text{Alligation} = \text{cross differences}}$

$\boxed{\text{Track what stays unchanged}}$

PRACTICE — LEVEL 0

Q1

30 L mixture has milk:water = 2:1. Find milk and water.

Q2

A solution is 25% acid. Find acid:water ratio.

Q3

10 L of 20% solution and 10 L of 40% solution are mixed. Find concentration.

Q4

20% and 60% solutions are mixed to obtain 40%. Find quantity ratio.

Q5

Rice ₹40/kg and ₹60/kg mixed equally. Find mixture price.

LEVEL 1

Q6

40 L of 30% acid is diluted to 24%. Find water added.

Q7

20 L solution is 25% acid. How much pure acid must be added to make it 40%?

Q8

30% and 70% solutions are mixed to obtain 50%. Find ratio.

Q9

Rice ₹ 50 and ₹ 80 are mixed to get ₹ 60. Find ratio.

Q10

Milk:water = 5:1. Water is free and mixture sold at milk's CP. Find profit %.

LEVEL 2**Q11**

20 L at 20% mixed with x L at 50% to obtain 35%. Find x .

Q12

50 L mixture is 40% milk. How much water should be added to make it 25% milk?

Q13

40 L of 25% acid. How much pure acid should be added to get 50% acid?

Q14

Tea ₹ 200 and ₹ 350 must be mixed to get ₹ 260. Find ratio.

Q15

Rice ₹ 40 and ₹ 70 mixed in ratio 3:2. Find mixture cost.

LEVEL 3 — EXAM PRESSURE**Q16**

20 L of 30% solution is mixed with 30 L of $x\%$ solution to obtain 42%. Find x .

Q17

A milk-water mixture is 75% milk. How much water must be added to 40 L to make milk 60%?

Q18

Milk costs ₹ 80 and water is free. A mixture is sold at ₹ 80 for 25% profit. Find milk:water ratio.

Q19

Solutions of 10% and 70% are mixed in ratio 2:3. Find resulting concentration.

Q20

A trader mixes rice costing ₹ 30 and ₹ 50 in a ratio that gives a mixture CP of ₹ 42. He sells it at ₹ 50.40. Find mixture ratio and profit %.

ANSWERS**Q1**

Total parts:

$$\sqrt[3]{}$$

Milk:

$$\sqrt[3]{30 \times \frac{23}{20}}$$

Water:

$$\sqrt[3]{10}$$

$$\sqrt[3]{\boxed{20L, 10L}}$$

Q2

$$\sqrt[3]{25:75 = \boxed{1:3}}$$

Q3

Equal quantities:

$$\sqrt[3]{\frac{20+40}{2} = \boxed{30\%}}$$

Q4

$$\sqrt[3]{(60-40):(40-20)}$$

$$\sqrt[3]{=20:20}$$

$$\sqrt[3]{=\boxed{1:1}}$$

Q5

Equal amounts:

$$\sqrt[3]{(40+60)/2 = \boxed{150 / kg}}$$

Q6

Initial acid:

$$\sqrt[3]{40(0.3)=12L}$$

Final total:

$$\sqrt[3]{12/0.24=50L}$$

Water added:

$$\sqrt[3]{\boxed{10L}}$$

Q7

Initial acid:

$$\sqrt[3]{5L}$$

$$\sqrt[3]{\frac{5+x}{20+x}=0.4}$$

$$\sqrt[3]{5+x=8+0.4x}$$

$$\sqrt[3]{0.6x=3}$$

$$\sqrt[3]{\boxed{5L}}$$

Q8

$$\sqrt[3]{(70-50):(50-30) = 20:20}$$

$$\sqrt[3]{=\boxed{1:1}}$$

Q9

$$\frac{(80-60):(60-50)}{20} = \frac{20}{10}$$

$$\frac{20}{10} = 2$$

$$\boxed{2:1}$$

Q10

$$\frac{15}{100} \times 100 = \boxed{20\%}$$

Q11

Alligation:

$$\frac{20\%:50\%}{50-35} = \frac{35-20}{20}$$

$$\frac{20}{15} = \frac{15}{20} = 1:1$$

So:

$$\boxed{x=20L}$$

Q12

Milk:

$$50(0.4) = 20L$$

Final total:

$$20/0.25 = 80L$$

Water added:

$$\boxed{30L}$$

Q13

Initial acid:

$$40(0.25) = 10L$$

$$\frac{10+x}{40+x} = 0.5$$

$$10+x = 20+0.5x$$

$$x = 20$$

$$\boxed{20L}$$

Q14

$$\frac{200\text{-tea}:350\text{-tea}}{350-260} = \frac{260-200}{20}$$

$$\frac{20}{90} = \frac{60}{20}$$

$$\frac{20}{90} = \frac{60}{20}$$

$$\boxed{3:2}$$

Q15

$$\frac{3(40)+2(70)}{5} = \frac{260}{5}$$

$$\boxed{152 \text{ kg}}$$

Q16

Total target component:

$$\sqrt[5]{50(42)}=2100 \sqrt[5]{}$$

Known contribution:

$$\sqrt[5]{20(30)}=600 \sqrt[5]{}$$

Unknown:

$$\sqrt[5]{30x}=1500 \sqrt[5]{}$$

$$\sqrt[5]{\boxed{x=50\%}} \sqrt[5]{}$$

Q17

Milk initially:

$$\sqrt[5]{40(0.75)}=30L \sqrt[5]{}$$

Final milk fraction 60%:

$$\sqrt[5]{30/T}=0.6 \sqrt[5]{}$$

$$\sqrt[5]{T}=50L \sqrt[5]{}$$

Water added:

$$\sqrt[5]{\boxed{10L}} \sqrt[5]{}$$

Q18

Selling price:

$$\sqrt[5]{^180} \sqrt[5]{}$$

25% profit means mixture CP:

$$\sqrt[5]{80/1.25} = ^164 \sqrt[5]{}$$

Milk fraction:

$$\sqrt[5]{64/80} = 0.8 \sqrt[5]{}$$

Thus:

$$\sqrt[5]{\boxed{\text{Milk:Water}=4:1}} \sqrt[5]{}$$

Q19

$$\sqrt[5]{\frac{2(10)+3(70)}{5}} \sqrt[5]{}$$

$$\sqrt[5]{= \frac{230}{5}} \sqrt[5]{}$$

$$\sqrt[5]{=\boxed{46\%}} \sqrt[5]{}$$

Q20

Alligation for 142 :

$$\sqrt[5]{30\text{-rice}:50\text{-rice}} = (50-42):(42-30) \sqrt[5]{}$$

$$\sqrt[5]{=8:12 = \boxed{2:3}} \sqrt[5]{}$$

Profit:

$$\sqrt[5]{50.4-42=8.4} \sqrt[5]{}$$

$$\left[8.4/42 \times 100 = \boxed{20\%} \right]$$

' CHAPTER 22 COMPLETE

We have covered:

- component ratios '
- concentration percentages '
- weighted concentration '
- alligation '
- target concentration '
- adding water '
- adding pure solute '
- adding another solution '
- price mixtures '
- weighted-price problems '
- milk-water profit '
- 3-component basics '
- mixture + profit applications '
- GATE/placement drills '

Progress

Completed: 22/76 Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

+Chapter 23 — Replacement in Mixtures

This is where we handle the famous repeated-operation questions:

- remove x litres and replace with water
- one replacement
- repeated replacement
- remaining original-liquid formula
- component concentration after n operations
- required number of replacements
- milk-water / acid-water replacement
- replacing with another solution instead of pure water
- GATE and placement shortcuts.

USER

next

CHAPTER 23 — REPLACEMENT IN MIXTURES

PART 1 — ONE-TIME REPLACEMENT, REPEATED REPLACEMENT & ORIGINAL-LIQUID REMAINING FORMULA

Status: $\emptyset$ = Not in PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

Replacement problems are a special type of mixture question.

The most common form is:

A vessel contains V litres of milk. Remove x litres of the mixture and replace it with water. Repeat this operation several times.

The key idea is:

Every time you remove mixture, you remove all components in the same proportion as they currently exist.

1. BASIC ONE-TIME REPLACEMENT

Suppose a vessel contains:

V [100L]

pure milk.

Remove:

x [20L]

milk.

Milk remaining:

$V - x$ [80L]

Now replace removed 20 L with water.

Final mixture:

- Milk = 80 L
- Water = 20 L

Milk fraction:

$\frac{V-x}{V}$

2. FRACTION OF ORIGINAL LIQUID REMAINING

If vessel volume is V and x litres are removed:

Fraction remaining:

$\frac{V-x}{V}$

Example:

$$\backslash[V=100,\quad x=20 \backslash]$$

Remaining fraction:

$$\backslash[\frac{80}{100} = \boxed{0.8} \backslash]$$

3. ONE REPLACEMENT FORMULA

Suppose original component quantity is (Q_0) .

After removing (x) from total volume (V) :

$$\backslash[\boxed{Q_1 = Q_0 \left(1 - \frac{x}{V}\right)} \backslash]$$

If the replacement liquid contains none of the original component, that original component does not increase during refilling.

4. EXAMPLE

A 50 L vessel contains pure milk.

Remove 10 L and replace with water.

Milk remaining:

$$\backslash[50 \left(1 - \frac{10}{50}\right) \backslash]$$

$$\backslash[=50(0.8) \backslash]$$

$$\backslash[=\boxed{40L} \backslash]$$

Water:

$$\backslash[10L \backslash]$$

5. SECOND REPLACEMENT — IMPORTANT

Continue the same example.

Now vessel contains:

- 40 L milk
- 10 L water
- total 50 L

Remove another 10 L of the **mixture**.

Milk fraction before removal:

$$\backslash[40/50=0.8 \backslash]$$

Milk removed:

$$\backslash[10(0.8)=8L \backslash]$$

Water removed:

$$\backslash[10(0.2)=2L \backslash]$$

Remaining:

- milk = 32 L
- water = 8 L

Add 10 L fresh water.

Final:

- milk = 32 L
- water = 18 L

Milk remaining:

$\boxed{32\text{L}}$

6. FAST MULTIPLIER VIEW

After first replacement:

$\boxed{50(0.8)=40}$

After second replacement:

$\boxed{40(0.8)=32}$

So each replacement multiplies the original-liquid amount by:

$\boxed{1-\frac{x}{V}}$

7. MASTER REPEATED-REPLACEMENT FORMULA

If:

- vessel capacity = (V)
- amount removed/replaced each time = (x)
- operation repeated (n) times
- initial amount of original component = (Q_0)

then:

$\boxed{Q_n = Q_0 \left(1-\frac{x}{V}\right)^n}$

This is one of the most important formulas in this chapter.

8. PURE LIQUID SPECIAL CASE

If vessel initially contains pure milk of volume (V) :

$\boxed{Q_0=V}$

So after (n) replacements:

$\boxed{\text{Milk remaining} = V\left(1-\frac{x}{V}\right)^n}$

9. EXAMPLE — TWO REPLACEMENTS

40 L vessel contains pure milk.

Remove 10 L and replace with water, twice.

Remaining fraction each time:

$$1 - \frac{10}{40} = \frac{3}{4}$$

After 2 operations:

$$\text{Milk} = 40 \left(\frac{3}{4} \right)^2$$

$$= 40 \cdot \frac{9}{16}$$

$$= \boxed{22.5 \text{ L}}$$

Water:

$$40 - 22.5 = \boxed{17.5 \text{ L}}$$

10. THREE REPLACEMENTS

Same vessel:

$$40 \text{ L}, \quad x = 10 \text{ L}$$

After 3 operations:

$$40 \left(\frac{3}{4} \right)^3$$

$$= 40 \cdot \frac{27}{64}$$

$$= \boxed{16.875 \text{ L}}$$

Milk fraction:

$$\frac{27}{64}$$

Water fraction:

$$1 - \frac{27}{64} = \frac{37}{64}$$

11. WHY THE FORMULA WORKS

At every removal, the fraction of each current component that remains is:

$$\frac{V-x}{V}$$

So after:

- 1 operation !' multiply once
- 2 operations !' multiply twice
- (n) operations !' multiply (n) times

Hence:

$$\left(\frac{V-x}{V} \right)^n$$

12. REPLACEMENT IS LIKE COMPOUND DECAY

This should feel familiar from Chapter 21.

If 20% of the mixture is removed each time, then 80% of the current original liquid remains.

That is exactly:

$\boxed{\text{repeated percentage decrease}}$

Example:

remove 20% each time for 3 times:

$\text{remaining fraction} = 0.8^3 = 0.512$

So:

$\boxed{51.2\%}$

of the original liquid remains.

13. REMOVE FRACTION INSTEAD OF LITRES

Suppose every time:

$\frac{1}{4}$

of vessel is removed and replaced.

Then fraction remaining each time:

$\frac{3}{4}$

After (n) times:

$\boxed{\left(\frac{3}{4}\right)^n}$

No need for actual vessel capacity.

14. EXAMPLE

A vessel initially contains pure milk.

One-third is removed and replaced with water, twice.

Fraction milk remaining:

$\left(\frac{2}{3}\right)^2 = \boxed{\frac{4}{9}}$

Therefore water fraction:

$\boxed{\frac{5}{9}}$

15. FIND WATER AFTER REPLACEMENT

If vessel initially contains only milk and refill liquid is only water:

$\boxed{\text{Water} = V - \text{Milk remaining}}$

Example:

100 L vessel, remove 20 L and replace 3 times.

Milk:

$$\sqrt[1]{100(0.8)^3 = 51.2\text{L}}$$

Water:

$$\sqrt[1]{100-51.2 = \boxed{48.8\text{L}}}$$

16. INITIAL MIXTURE IS NOT PURE

The same formula still works for any particular component.

Suppose 50 L mixture is initially 60% milk.

Initial milk:

$$\sqrt[1]{Q_0=50(0.6)=30\text{L}}$$

Remove 10 L and replace with pure water twice.

Each time milk-retention factor:

$$\sqrt[1]{1-\frac{10}{50}=0.8}$$

Final milk:

$$\sqrt[1]{30(0.8)^2}$$

$$\sqrt[1]{=30(0.64)}$$

$$\sqrt[1]{=\boxed{19.2\text{L}}}$$

Final milk concentration:

$$\sqrt[1]{19.2/50\times 100 = \boxed{38.4\%}}$$

17. GENERAL COMPONENT FORMULA

If initial concentration of component A is $\sqrt[1]{(C_0)}$, then after replacing with a liquid containing **zero A**:

$$\sqrt[1]{\boxed{C_n = C_0 \left(1-\frac{x}{V}\right)^n}}$$

provided total volume is restored to $\sqrt[1]{(V)}$ each time.

18. EXAMPLE — ACID

A 40 L solution is 50% acid.

8 L is removed and replaced with water twice.

Initial acid concentration:

$$\sqrt[1]{50\%}$$

Retention factor:

$$\sqrt[1]{1-\frac{8}{40} = 0.8}$$

Final acid concentration:

$$\sqrt[1]{50(0.8)^2}$$

$$\sqrt[1]{=50(0.64)}$$

$$\boxed{32\%}$$

19. FIND NUMBER OF REPLACEMENTS

Suppose 100 L pure milk.

Each time 20 L is removed and replaced with water.

How many replacements leave 51.2 L milk?

Formula:

$$100(0.8)^n = 51.2$$

$$0.8^n = 0.512$$

Recognize:

$$0.8^3 = 0.512$$

So:

$$\boxed{n=3}$$

20. FIND AMOUNT REMOVED EACH TIME

A 100 L vessel of pure milk undergoes 2 replacements. Final milk = 64 L.

Let removed amount each time = x .

$$100\left(1 - \frac{x}{100}\right)^2 = 64$$

$$\left(1 - \frac{x}{100}\right)^2 = 0.64$$

$$1 - \frac{x}{100} = 0.8$$

$$\frac{x}{100} = 0.2$$

$$\boxed{x=20L}$$

21. FIND VESSEL CAPACITY

A vessel contains pure milk.

10 L is removed and replaced with water twice.

Final milk is 64% of original.

Let vessel capacity = V .

$$\left(1 - \frac{10}{V}\right)^2 = 0.64$$

$$1 - \frac{10}{V} = 0.8$$

$$\frac{10}{V} = 0.2$$

$$\boxed{V=50L}$$

22. SINGLE REPLACEMENT FROM A MIXTURE

Suppose 30 L mixture has milk:water = 2:1.

So:

- milk = 20
- water = 10

Remove 6 L mixture.

Milk fraction:

$$\left[\frac{2}{3} \right]$$

Milk removed:

$$\left[6 \cdot \frac{2}{3} = 4 \right]$$

Water removed:

$$\left[2 \right]$$

Remaining:

- milk = 16
- water = 8

Replace with 6 L water:

- milk = 16
- water = 14

Final ratio:

$$\left[16:14 = \boxed{8:7} \right]$$

23. FAST METHOD FOR SAME QUESTION

Initial milk:

$$\left[20 \right]$$

Fraction retained after removal:

$$\left[1 - \frac{6}{30} = 0.8 \right]$$

Milk after replacement:

$$\left[20(0.8) = 16 \right]$$

Final water:

$$\left[30 - 16 = 14 \right]$$

Ratio:

$$\left[\boxed{8:7} \right]$$

Much faster.

24. REPLACEMENT WITH PURE SOLUTE INSTEAD OF WATER

Suppose a vessel contains acid-water mixture, and removed mixture is replaced with pure acid.

Now the amount of **water** is the component that only decreases during replacements.

So track water instead of acid.

25. EXAMPLE — REPLACE WITH PURE ACID

40 L mixture contains 25% acid.

Therefore water:

$$\left[75\% \text{ of } 40 = 30\text{L} \right]$$

Remove 8 L mixture and replace with pure acid.

Water-retention factor:

$$\left[1 - \frac{8}{40} = 0.8 \right]$$

Water remaining:

$$\left[30(0.8) = 24\text{L} \right]$$

Final acid:

$$\left[40 - 24 = 16\text{L} \right]$$

Acid concentration:

$$\left[16/40 = 40\% \right]$$

$$\left[\boxed{40\%} \right]$$

26. TWO REPLACEMENTS WITH PURE ACID

Same setup repeated twice.

Initial water:

$$\left[30\text{L} \right]$$

After two replacements:

$$\left[30(0.8)^2 = 19.2\text{L} \right]$$

Final acid:

$$\left[40 - 19.2 = 20.8\text{L} \right]$$

Acid percentage:

$$\left[20.8/40 \times 100 = \boxed{52\%} \right]$$

27. TRACK THE COMPONENT ABSENT FROM REFILL

This is a major shortcut.

If replacement liquid contains:

- no milk !' track milk
 - no acid !' track acid
 - no water !' track water
 - no alcohol !' track alcohol
-

The absent component is simply multiplied by the retention factor each time.

28. DIFFERENT AMOUNTS REMOVED EACH TIME

Suppose vessel volume (V) , and removal amounts are:

$$[x_1, x_2, \dots, x_n]$$

Then original-component amount becomes:

$$[\boxed{Q_n = Q_0 \prod_{i=1}^n \left(1 - \frac{x_i}{V}\right)}]$$

if each removed amount is replaced and volume returns to (V) .

29. EXAMPLE — DIFFERENT REMOVALS

100 L pure milk.

First remove 20 L !' retention:

$$[0.8]$$

Then remove 25 L !' retention:

$$[0.75]$$

Milk final:

$$[100(0.8)(0.75)]$$

$$[\boxed{60L}]$$

30. DO ORDER OF DIFFERENT REPLACEMENT AMOUNTS MATTER?

For pure dilution with refill containing no original component:

$$[Q_0 \left(1 - \frac{x_1}{V}\right) \left(1 - \frac{x_2}{V}\right)]$$

Multiplication commutes.

So mathematically, the final amount of the tracked original component is the same regardless of the order of (x_1, x_2) .

31. REPLACE WITH A MIXTURE THAT ALSO CONTAINS THE COMPONENT

Now the simple decay formula alone is not enough.

Example:

Remove some 20% acid solution and replace it with 50% acid solution.

You lose some acid, then add acid back.

We must use:

$$[\boxed{\text{New component} = \text{component remaining} + \text{component added}}]$$

32. GENERAL ONE-STEP REPLACEMENT WITH CONCENTRATED REFILL

Let:

- total volume (V)
- current concentration (C)
- remove (x)
- refill concentration (D)

Current component amount:

$$[VC]$$

Removed component:

$$[xC]$$

Remaining:

$$[(V-x)C]$$

Added component:

$$[xD]$$

New amount:

$$[(V-x)C + xD]$$

Therefore new concentration:

$$[\boxed{C' = C\left(1 - \frac{x}{V}\right) + D\frac{x}{V}}]$$

33. INTERPRETATION

The new concentration is a weighted average of:

- retained old mixture,
- new replacement liquid.

Equivalent form:

$$[\boxed{C' = C + \frac{x}{V}(D - C)}]$$

This is a very useful shortcut.

34. EXAMPLE

40 L solution is 20% acid.

Remove 10 L and replace with 60% acid solution.

$$[C' = 20 + \frac{10}{40}(60 - 20)]$$

$$[= 20 + \frac{1}{4}(40)]$$

$$[= 20 + 10]$$

$$[= \boxed{30\%}]$$

35. VERIFY USING COMPONENTS

Initial acid:

$$[40(0.2)=8L]$$

Removed acid:

$$[10(0.2)=2L]$$

Remaining:

$$[6L]$$

Added acid:

$$[10(0.6)=6L]$$

Total acid:

$$[12L]$$

Concentration:

$$[12/40=30\%]$$

Correct.

36. REPEATED REPLACEMENT WITH SAME REFILL CONCENTRATION

This is a more advanced but powerful formula.

If:

- initial concentration (C_0)
- refill concentration (D)
- fraction retained each operation:

$$[r=1-\frac{x}{V}]$$

then after (n) identical replacements:

$$[\boxed{ C_n = D+(C_0-D)r^n }]$$

This generalizes the ordinary replacement formula.

37. WHY THIS GENERAL FORMULA WORKS

Instead of tracking concentration directly, track the **difference from the refill concentration**.

Each replacement reduces that difference by factor (r) .

So:

$$[C_n - D = (C_0 - D)r^n]$$

Hence:

$$[C_n = D+(C_0-D)r^n]$$

38. CHECK ORDINARY WATER REPLACEMENT

If refill is pure water and tracked component is milk:

$$[D=0]$$

Then:

$$[C_n = 0 + C_0 r^n]$$

So:

$$[\boxed{C_n = C_0 r^n}]$$

our original formula.

39. EXAMPLE — REPEATED WITH 50% REFILL

A 100 L solution starts at 20% acid.

Each time 20 L is removed and replaced with 50% acid.

Repeated twice.

$$[r=0.8]$$

$$[C_2 = 50 + (20-50)(0.8)^2]$$

$$[= 50 - 30(0.64)]$$

$$[= 50 - 19.2]$$

$$[= \boxed{30.8\%}]$$

40. VERIFY STEPWISE

After first replacement:

$$[20 + 0.2(50-20) = 26\%]$$

After second:

$$[26 + 0.2(50-26)]$$

$$[= 26 + 4.8]$$

$$[= 30.8\%]$$

Correct.

41. LONG-TERM BEHAVIOR

If the same replacement continues many times:

$$[r^n \rightarrow 0]$$

Therefore:

$$[C_n \rightarrow D]$$

So the vessel's concentration approaches the refill concentration.

Example:

Repeatedly replacing with pure water:

$\left[C_n \rightarrow 0 \right]$

but for a finite number of replacements, some original solute generally remains.

42. COMMON GATE PATTERN— WHAT FRACTION REMAINS?

A vessel loses:

$\left[\frac{15}{100} \right]$

of mixture each time and is refilled.

After 3 operations:

$\left[\left(\frac{85}{100} \right)^3 = \frac{64}{125} \right]$

Thus:

$\left[\boxed{\frac{64}{125}} \right]$

of original component remains.

43. COMMON PLACEMENT PATTERN— HOW MUCH WATER?

64 L pure milk.

16 L removed and replaced with water twice.

Retention:

$\left[1 - \frac{16}{64} = \frac{3}{4} \right]$

Milk:

$\left[64 \left(\frac{3}{4} \right)^2 = 64 \cdot \frac{9}{16} = 36 \text{ L} \right]$

Water:

$\left[64 - 36 = \boxed{28 \text{ L}} \right]$

44. FIND FINAL RATIO

From same example:

Milk:

$\left[36 \right]$

Water:

$\left[28 \right]$

Ratio:

$\left[36:28 = \boxed{9:7} \right]$

45. FIND INITIAL CONCENTRATION

A 50 L solution undergoes two replacements of 10 L each with water.

Final acid concentration is 32%.

Find initial acid concentration.

Retention:

$$\sqrt[0.8^2=0.64]{}$$

So:

$$\sqrt{C_0(0.64)=32}$$

$$\sqrt{C_0=50}$$

$$\sqrt{\boxed{50\%}}$$

46. FIND FINAL CONCENTRATION QUICKLY

Initial:

$$\sqrt{75\%}$$

Remove 20% of vessel and replace with water 3 times.

Retention:

$$\sqrt{0.8^3=0.512}$$

Final concentration:

$$\sqrt{75(0.512)}$$

$$\sqrt{=\boxed{38.4\%}}$$

47. GATE-STYLE QUESTION 1

A 50 L vessel contains pure milk.

10 L is removed and replaced with water twice.

Milk remaining:

$$\sqrt{50(0.8)^2}$$

$$\sqrt{=\boxed{32L}}$$

48. GATE-STYLE QUESTION 2

One-fourth of a pure liquid is removed and replaced twice.

Fraction original remaining:

$$\sqrt{\left(\frac{3}{4}\right)^2 = \boxed{\frac{9}{16}}}$$

49. GATE-STYLE QUESTION 3

100 L pure milk.

20 L removed and replaced 3 times.

Milk:

$$\sqrt[3]{100(0.8)^3} = \boxed{51.2\text{L}}$$

50. GATE-STYLE QUESTION 4

40 L solution is 50% acid.

8 L removed and replaced with water twice.

Final acid concentration:

$$\sqrt[2]{50(0.8)^2} = \boxed{32\%}$$

51. PLACEMENT-STYLE QUESTION 1

60 L pure milk.

15 L removed and replaced twice.

Retention:

$$\sqrt[2]{1 - \frac{15}{60}} = 0.75$$

Milk:

$$\sqrt[2]{60(0.75)^2}$$

$$\sqrt[2]{=60(0.5625)}$$

$$\sqrt[2]{= \boxed{33.75\text{L}}}$$

52. PLACEMENT-STYLE QUESTION 2

80 L pure juice.

20 L removed and replaced 3 times.

Remaining juice fraction:

$$\sqrt[3]{0.75^3} = 0.421875$$

Remaining amount:

$$\sqrt[3]{80(0.421875)}$$

$$\sqrt[3]{= \boxed{33.75\text{L}}}$$

53. PLACEMENT-STYLE QUESTION 3

50 L mixture initially 60% milk.

10 L removed and replaced with water twice.

Final milk:

$$\sqrt[3]{30(0.8)^2} = \boxed{19.2\text{L}}$$

Final milk %:

$$\sqrt[3]{19.2/50 \times 100} = \boxed{38.4\%}$$

54. PLACEMENT-STYLE QUESTION 4

40 L mixture is 25% acid.

8 L removed and replaced with pure acid once.

Initial water:

$$\sqrt[3]{30\text{L}}$$

Water after removal:

$$\sqrt[3]{30(0.8)} = 24\text{L}$$

Acid final:

$$\sqrt[3]{40 - 24} = 16\text{L}$$

Concentration:

$$\sqrt[3]{\boxed{40\%}}$$

55. ADVANCED QUESTION — FIND REMOVAL AMOUNT

A 50 L vessel contains pure milk.

After 2 equal replacements with water, 32 L milk remains.

$$\sqrt[3]{50 \left(1 - \frac{x}{50}\right)^2} = 32$$

$$\sqrt[3]{\left(1 - \frac{x}{50}\right)^2} = 0.64$$

$$\sqrt[3]{1 - \frac{x}{50}} = 0.8$$

$$\sqrt[3]{x} = 10$$

$$\sqrt[3]{\boxed{10\text{L}}}$$

56. ADVANCED QUESTION — FIND NUMBER OF OPERATIONS

A vessel loses 25% of its content and is refilled each time.

What number of operations leaves $\sqrt[3]{27/64}$ of original component?

Retention:

$$\sqrt[3]{\frac{3}{4}}$$

Need:

$$\sqrt[3]{\left(\frac{3}{4}\right)^n} = \frac{27}{64}$$

But:

$$\left(\frac{3}{4}\right)^3 = \frac{27}{64}$$

So:

$$\boxed{3}$$

57. ADVANCED QUESTION — NONZERO REFILL CONCENTRATION

50 L solution starts at 20% acid.

10 L removed and replaced with 60% acid solution, twice.

$$r=0.8$$

$$C_2 = 60 + (20-60)(0.8)^2$$

$$= 60 - 40(0.64)$$

$$= 60 - 25.6$$

$$= \boxed{34.4\%}$$

58. ADVANCED QUESTION — PURE WATER TO PURE ACID REPLACEMENT

A 40 L solution is initially 25% acid.

8 L removed and replaced with pure acid twice.

Track water.

Initial water:

$$75\% = 30\text{L}$$

Retention:

$$0.8^2 = 0.64$$

Final water:

$$30(0.64) = 19.2\text{L}$$

Final acid:

$$40 - 19.2 = 20.8\text{L}$$

Acid concentration:

$$\boxed{52\%}$$

59. COMMON TRAPS

& pTRAP 1 — Removing pure milk after first replacement

After the first replacement, subsequent removals are from the **mixture**, not pure milk.

& pTRAP 2 — Subtracting the same quantity of original component each time

Wrong. The concentration changes after every operation.

Use the retention factor.

& pTRAP 3 — Forgetting total volume returns to (V)

The standard formula assumes the removed amount is replaced, restoring the original volume each time.

& pTRAP 4 — Refill contains tracked component

Then simple $(C_0 r^n)$ is not enough.

Use:

$$[C_n = D + (C_0 - D)r^n]$$

& pTRAP 5 — Track wrong component

When replacing with pure acid, tracking water is often easier.

60. MUST-MEMORIZE FORMULA

For replacement with a liquid containing none of the tracked original component:

$$[\boxed{Q_n = Q_0 \left(1 - \frac{x}{V}\right)^n}]$$

If initially pure:

$$[\boxed{Q_n = V \left(1 - \frac{x}{V}\right)^n}]$$

61. GENERAL REPLACEMENT FORMULA

If refill concentration is (D) :

$$[\boxed{C_n = D + (C_0 - D) \left(1 - \frac{x}{V}\right)^n}]$$

This is the most general high-value formula in the chapter.

62. FAST EXAM DECISION TREE

Pure liquid replaced by water?

Use repeated-retention formula.

Initial mixture + water replacement?

Find initial tracked component, then multiply.

Replacement with pure solute?

Track solvent instead.

Different removal amounts?

Multiply different retention factors.

Replacement liquid has nonzero concentration?

Use the generalized concentration formula.

Need (x, n, V) ?

Set final fraction equal to formula and solve.

PRACTICE — LEVEL 0

Q1

40 L pure milk. Remove 10 L and replace once. Find milk remaining.

Q2

40 L pure milk. Remove 10 L and replace twice. Find milk remaining.

Q3

One-fifth of mixture is replaced twice. What fraction of original remains?

Q4

100 L pure juice. Remove 20 L and replace 3 times. Find juice remaining.

Q5

50 L 60% milk mixture. Remove 10 L and replace with water once. Find final milk percentage.

LEVEL 1

Q6

60 L pure milk. 15 L removed/replaced twice. Find water at end.

Q7

80 L pure solution. 20 L replaced 2 times. Find fraction original remaining.

Q8

40 L 50% acid solution. 8 L removed and replaced with water 3 times. Find acid %.

Q9

50 L pure milk. After two equal replacements 32 L remains. Find amount removed each time.

Q10

A vessel loses one-third and is refilled 3 times. Find original fraction remaining.

LEVEL 2

Q11

100 L solution initially 40% acid. 20 L removed and replaced with water twice. Find acid concentration.

Q12

40 L solution is 25% acid. 8 L removed and replaced with pure acid twice. Find final acid %.

Q13

50 L solution starts at 20% acid. 10 L removed and replaced with 60% acid twice. Find final concentration.

Q14

A 100 L vessel loses 20 L each time. After how many replacements does 51.2% of original remain?

Q15

A 50 L vessel undergoes two replacements of 10 L each with water. Final concentration is 32%. Find initial concentration.

LEVEL 3 — EXAM PRESSURE

Q16

A 64 L vessel contains pure milk. 16 L is removed and replaced 3 times. Find final milk:water ratio.

Q17

100 L solution starts at 80% alcohol. 25 L is removed and replaced with water twice. Find final alcohol percentage.

Q18

A vessel starts at 10% acid. Each time 20% of the mixture is removed and replaced with 50% acid. After 3 replacements, find acid percentage.

Q19

A 40 L vessel initially contains 50% milk. 8 L is removed and replaced with water twice. Then 8 L is removed again and replaced with pure milk. Find final milk percentage.

Q20

A vessel initially contains pure liquid. After 2 equal replacements with water, $\frac{49}{64}$ of the original remains. What fraction of the vessel was removed each time?

ANSWERS

Q1

$$40 \left(\frac{34}{100} \right)^2 = \boxed{30L}$$

Q2

$$40 \left(\frac{34}{100} \right)^2 = 40 \cdot \frac{9}{16} = \boxed{22.5L}$$

Q3

Retention:

$$\frac{45}{100}$$

After twice:

$$\boxed{\frac{16}{25}}$$

Q4

$$100(0.8)^3 = \boxed{51.2L}$$

Q5

Initial milk:

$$\sqrt[4]{50(0.6)=30}$$

After removal:

$$\sqrt[4]{30(0.8)=24}$$

Final percentage:

$$\sqrt[4]{24/50 \times 100 = \boxed{48\%}}$$

Q6

Milk:

$$\sqrt[4]{60(0.75)^2 = 33.75}$$

Water:

$$\sqrt[4]{60-33.75 = \boxed{26.25L}}$$

Q7

$$\sqrt[4]{(0.75)^2 = \boxed{\frac{9}{16}}}$$

Q8

Retention per operation:

$$\sqrt[4]{0.8}$$

Final acid concentration:

$$\sqrt[4]{50(0.8)^3}$$

$$\sqrt[4]{=50(0.512)}$$

$$\sqrt[4]{=\boxed{25.6\%}}$$

Q9

$$\sqrt[4]{50 \left(1 - \frac{x}{50}\right)^2 = 32}$$

$$\sqrt[4]{\left(1 - \frac{x}{50}\right) = 0.8}$$

$$\sqrt[4]{\boxed{x=10L}}$$

Q10

Fraction retained each time:

$$\sqrt[4]{\frac{23}{}}$$

After 3:

$$\sqrt[4]{\boxed{\frac{8}{27}}}$$

Q11

Initial concentration:

$$\sqrt[4]{40\%}$$

Retention:

$$\sqrt[4]{0.8^2=0.64}$$

Final:

$$\boxed{40(0.64) = 25.6\%}$$

Q12

Track water:

$$75\% \times 0.8^2 = 75(0.64) = 48\%$$

Thus acid:

$$100 - 48 = \boxed{52\%}$$

Q13

$$C_2 = 60 + (20 - 60)(0.8)^2$$

$$= 60 - 25.6$$

$$= \boxed{34.4\%}$$

Q14

$$0.8^n = 0.512$$

$$\boxed{n=3}$$

Q15

$$C_0(0.8)^2 = 32$$

$$0.64C_0 = 32$$

$$\boxed{C_0 = 50\%}$$

Q16

Retention:

$$0.75^3 = \frac{27}{64}$$

Milk:

$$64 \cdot \frac{27}{64} = 27$$

Water:

$$37$$

Ratio:

$$\boxed{27:37}$$

Q17

$$80(0.75)^2$$

$$= 80(0.5625)$$

$$= \boxed{45\%}$$

Q18

Use generalized formula:

$$C_3 = 50 + (10 - 50)(0.8)^3$$

$$= 50 - 40(0.512)$$

$$\boxed{=50-20.48}$$

$$\boxed{29.52\%}$$

Q19

Initial milk:

$$20\text{ L}$$

After two water replacements:

$$20(0.8)^2 = 12.8\text{ L}$$

Before third replacement, milk concentration:

$$12.8/40=0.32$$

Remove 8 L:

Milk removed:

$$8(0.32)=2.56\text{ L}$$

Milk remaining:

$$10.24\text{ L}$$

Add 8 L pure milk:

$$18.24\text{ L}$$

Final milk %:

$$18.24/40 \times 100 = \boxed{45.6\%}$$

Q20

Let fraction removed each time = f .

Then:

$$(1-f)^2 = \frac{49}{64}$$

$$1-f = \frac{7}{8}$$

$$f = \frac{1}{8}$$

$$\boxed{\frac{1}{8}}$$

of the vessel is removed each time.

CHAPTER 23 STATUS

Covered so far:

- one-time replacement '
- repeated replacement '
- remaining-original-liquid formula '
- concentration after repeated dilution '
- non-pure initial mixtures '
- finding number of replacements '
- finding amount removed '

- finding vessel capacity '
- replacement with pure solute '
- different removal quantities '
- nonzero-concentration refill '
- generalized replacement formula '
- GATE/placeholder practice '

But **Chapter 23 is not complete yet.**

Next we need:

- remove-only vs remove-and-replace distinction
- replacement when vessel is not refilled fully
- mixed operations with different refill liquids
- repeated removal until target ratio
- "remove x and add y" where $x \neq y$
- more advanced milk-water replacement word problems
- final speed shortcuts and chapter audit

Progress

Completed: 22/76 **Current:** Chapter 23 — Replacement in Mixtures **Next:** Chapter 23 Part 2 — **Advanced replacement patterns & mixed-operation problems**

USER

next

ASSISTANT

CHAPTER 23 — REPLACEMENT IN MIXTURES

PART 2 — REMOVE-ONLY CASES, UNEQUAL REFILL, MIXED REPLACEMENTS & ADVANCED TARGET-RATIO PROBLEMS

Status: $\emptyset$ **IN PROGRESS** **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

Part 1 handled the standard case:

remove x litres and replace exactly x litres.

Now we handle the less standard versions where:

- nothing is added back,
- a different amount is added,
- different liquids are used for replacement,
- operations change midway,
- or a target final ratio must be reached.

The biggest rule is:

$\boxed{\text{Never use the repeated-replacement formula unless the vessel volume is restored appropriately.}}$

1. REMOVE ONLY — NO REPLACEMENT

Suppose a 40 L mixture contains milk:water = 3:1.

So:

$$\text{Milk} = 30\text{L}, \text{Water} = 10\text{L}$$

If 8 L of the well-mixed solution is removed and nothing is added back:

The same fraction of every component is removed.

Fraction remaining:

$$1 - \frac{8}{40} = \frac{45}{40}$$

Milk:

$$30 \times \frac{45}{40} = 24\text{L}$$

Water:

$$10 \times \frac{45}{40} = 8\text{L}$$

New total:

$$32\text{L}$$

Ratio:

$$24:8 = 3:1$$

So:

$\boxed{\text{Removing only a well-mixed portion does not change component ratio.}}$

2. VERY IMPORTANT PROPERTY

If you remove part of a homogeneous mixture but do not add anything:

$\boxed{\text{Concentration remains unchanged}}$

Only total quantity decreases.

Example:

50 L of 40% acid.

Remove 10 L.

Remaining quantity:

$$40\text{L}$$

Acid remaining:

$$20 \times \left(\frac{40}{100}\right) = 8\text{L}$$

Concentration:

$$\frac{8}{40} = 20\%$$

Same as before.

3. REMOVE AND ADD A DIFFERENT AMOUNT

Now suppose:

- vessel initially has (V) L,
- remove (x) L,
- add (y) L,
- where $(x \neq y)$.

Then final volume becomes:

$$\boxed{V-x+y}$$

You cannot blindly use the fixed-volume repeated formula.

4. EXAMPLE — REMOVE 10 L, ADD 5 L WATER

A 40 L mixture is 50% milk.

Initial milk:

$$20\text{ L}$$

Remove 10 L mixture.

Milk retained:

$$20\left(1-\frac{10}{40}\right) = 20\left(\frac{3}{4}\right) = 15\text{ L}$$

Now add 5 L water.

Milk remains:

$$15\text{ L}$$

Final volume:

$$40-10+5=35\text{ L}$$

Milk %:

$$15/35 \times 100 = \boxed{42\frac{6}{7}\%}$$

5. GENERAL ONE-STEP FORMULA WITH UNEQUAL REFILL

Let:

- initial volume (V)
- concentration of tracked component (C)
- remove (x)
- add (y) of refill concentration (D)

Tracked component after removal:

$$(V-x)C$$

Added component:

$$yD$$

Final volume:

$$\sqrt{V-x+y}$$

Final concentration:

$$\sqrt{\boxed{C' = \frac{(V-x)C+yD}{V-x+y}}}$$

Use fractions or consistent percentages.

6. EXAMPLE — ADD STRONGER SOLUTION

40 L solution is 20% acid.

Remove 10 L.

Add 5 L of 60% acid.

Initial acid:

$$\sqrt{8L}$$

After removing 10 L:

$$\sqrt{8\left(\frac{34}{right}\right)=6L}$$

Added acid:

$$\sqrt{5(0.6)=3L}$$

Final acid:

$$\sqrt{9L}$$

Final volume:

$$\sqrt{35L}$$

Concentration:

$$\sqrt{9/35\times 100}$$

$$\sqrt{=\boxed{25.714\%\text{ approximately}}}$$

7. REPEATED REMOVE-ONLY OPERATIONS

Suppose you remove 20% of a mixture, then later remove 25% of the remaining mixture, with no refill.

The ratio stays unchanged throughout.

Total quantity changes:

$$\sqrt{V(0.8)(0.75)=0.6V}$$

So 60% of total mixture remains.

But concentration is unchanged.

8. MIXED OPERATION — FIRST WATER, THEN MILK

A 50 L vessel initially contains pure milk.

First:

- remove 10 L,
- add 10 L water.

Milk after first operation:

$$\sqrt[5]{50(0.8)=40\text{L}}$$

Now mixture is:

- 40 L milk
- 10 L water.

Second:

- remove 10 L mixture,
- add 10 L pure milk.

Before removal, milk fraction:

$$\sqrt[5]{40/50=0.8}$$

Milk removed:

$$\sqrt[5]{8\text{L}}$$

Milk remaining:

$$\sqrt[5]{32\text{L}}$$

Add 10 L milk:

$$\sqrt[5]{42\text{L}}$$

Final milk:

$$\sqrt[5]{\boxed{42\text{L}}}$$

Water:

$$\sqrt[5]{8\text{L}}$$

Ratio:

$$\sqrt[5]{\boxed{21:4}}$$

9. WHY YOU CANNOT USE ONE SINGLE DECAY FORMULA

The first refill contains 0% milk.

The second refill contains 100% milk.

So the refill concentration changes.

The generalized fixed- (D) formula from Part 1 cannot be applied across both operations with one (D) .

Solve each operation sequentially.

10. GENERAL STEP-BY-STEP UPDATE RULE

For each operation:

$$\sqrt[5]{\boxed{C_{\text{new}} = C_{\text{old}} + \frac{x}{V}(D - C_{\text{old}})}}$$

when:

- remove x ,
- replace exactly x ,
- total volume stays V ,
- refill concentration = D .

This is often the cleanest way to handle changing refill liquids.

11. EXAMPLE — DIFFERENT REFILL EACH TIME

Initial acid:

$$[20\%]$$

Total volume:

$$[100L]$$

Each time remove 20 L.

Operation 1

Replace with 50% acid.

$$[C_1 = 20 + 0.2(50 - 20)]$$

$$[= 26\%]$$

Operation 2

Replace with pure water:

$$[D = 0]$$

$$[C_2 = 26 + 0.2(0 - 26)]$$

$$[= 20.8\%]$$

So:

$$[\boxed{20.8\%}]$$

12. SAME RESULT WITH COMPONENT ACCOUNTING

After first operation:

Acid:

$$[26L]$$

Second removal removes:

$$[20(0.26) = 5.2L]$$

Remaining acid:

$$[20.8L]$$

Add pure water:

$$[0L \text{ acid}]$$

Final acid:

$\sqrt[2]{20.8L}$

Correct.

13. TARGET RATIO AFTER ONE REPLACEMENT

A 40 L milk-water mixture has ratio:

$\sqrt[2]{3:1}$

How much mixture must be removed and replaced by water so final ratio becomes:

$\sqrt[2]{1:1}$

Initial milk:

$\sqrt[2]{30L}$

Initial milk fraction:

$\sqrt[2]{\frac{3}{4}}$

Let $\sqrt{x}$ L be removed.

Milk remaining:

$\sqrt[2]{30\left(1-\frac{x}{40}\right)}$

Final volume returns to 40 L.

For final ratio 1:1, milk must be:

$\sqrt[2]{20L}$

So:

$\sqrt[2]{30\left(1-\frac{x}{40}\right)=20}$

$\sqrt[2]{1-\frac{x}{40} = \frac{2}{3}}$

$\sqrt[2]{\frac{x}{40} = \frac{1}{3}}$

$\sqrt[2]{\boxed{x=\frac{40}{3}=13\frac{1}{3}L}}$

14. TARGET CONCENTRATION AFTER ONE WATER REPLACEMENT

General case:

- total volume $\sqrt{V}$
- initial component concentration $\sqrt{C_i}$
- replace $\sqrt{x}$ with refill having 0 tracked component
- target concentration $\sqrt{C_f}$

Then:

$\sqrt{C_f = C_i\left(1-\frac{x}{V}\right)}$

Thus:

$\sqrt{\boxed{x = V\left(1-\frac{C_f}{C_i}\right)}}$

15. EXAMPLE

50 L solution is 60% milk.

How much mixture should be removed and replaced with water to make it 48% milk?

$$x = 50 \left(1 - \frac{48}{60} \right)$$

$$x = 50(1 - 0.8)$$

$$x = \boxed{10\text{L}}$$

16. TARGET CONCENTRATION AFTER n IDENTICAL REPLACEMENTS

If refill contains none of tracked component:

$$C_f = C_i \left(1 - \frac{x}{V} \right)^n$$

So if n and target are known:

$$\boxed{\left(1 - \frac{x}{V} \right)^n = \frac{C_f}{C_i}}$$

This is the master inverse equation.

17. EXAMPLE — FIND REMOVAL FRACTION

Pure milk undergoes 2 identical replacements with water.

Final milk concentration:

$$49\%$$

Then:

$$(1 - f)^2 = 0.49$$

$$1 - f = 0.7$$

$$f = 0.3$$

So:

$$\boxed{30\% \text{ of vessel is replaced each time}}$$

18. EXAMPLE — FIND NUMBER OF REPLACEMENTS

Initial acid:

$$80\%$$

Each time 25% is replaced with water.

Target:

$$33.75\%$$

Retention factor:

$$0.75$$

Equation:

$$\sqrt[3]{80(0.75)^n} = 33.75$$

$$\sqrt[3]{(0.75)^n} = 0.421875$$

Recognize:

$$\sqrt[3]{0.75^3} = 0.421875$$

So:

$$\boxed{n=3}$$

19. REPEATED REPLACEMENT UNTIL TWO COMPONENTS ARE EQUAL

Suppose vessel initially contains pure milk.

Each operation replaces fraction f with water.

After n operations, milk fraction:

$$(1-f)^n$$

Milk and water become equal when milk fraction:

$$\frac{1}{2}$$

So:

$$\boxed{(1-f)^n = \frac{1}{2}}$$

This may require logarithms if values are not chosen nicely.

For aptitude, options often let you estimate.

20. EXAMPLE

Each time half the vessel is replaced with water.

After one operation:

Milk:

$$\frac{1}{2}$$

Water:

$$\frac{1}{2}$$

So they are immediately equal.

$$\boxed{1 \text{ replacement}}$$

21. CAN PURE WATER EVER BECOME EXACTLY 100% AFTER FINITE STANDARD REPLACEMENTS?

Suppose vessel initially contains some milk, and each time only a fraction less than 100% is replaced by water.

Milk after n replacements:

$$\sqrt[n]{Q_0 r^n}$$

where:

$$\sqrt[n]{0 < r < 1}$$

For every finite n :

$$\sqrt[n]{r^n > 0}$$

So some milk remains.

Therefore:

$\boxed{\text{It approaches 0 but does not become exactly 0 in finitely many partial replacements.}}$

Unless an operation removes the entire vessel contents.

22. REPLACE DIFFERENT FRACTIONS EACH TIME

Suppose:

- first replace $\frac{1}{4}$,
- then $\frac{1}{5}$,
- then $\frac{1}{10}$

with water.

Original fraction remaining:

$$\left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{10}\right)$$

$$= \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{9}{10}$$

Cancel:

$$= \frac{27}{50}$$

$$\boxed{54\%}$$

23. HIGH-VALUE CANCELLATION SHORTCUT

When fractions are involved, multiply retention fractions and cancel early.

Example:

$$\frac{5}{6} \cdot \frac{3}{4} \cdot \frac{8}{10}$$

Simplify before multiplying.

This avoids decimal errors.

24. VESSEL FILLED PARTIALLY AFTER REMOVAL

Suppose 50 L mixture is 40% acid.

Remove 10 L, but add only 5 L water.

Initial acid:

$$\sqrt[n]{20L}$$

Acid retained:

$$\boxed{20(0.8)=16\text{ L}}$$

Final volume:

$$\boxed{45\text{ L}}$$

Acid %:

$$\boxed{16/45 \times 100 = 35.56\% \text{ approximately}}$$

The denominator is 45, not 50.

25. VESSEL OVERFILLED AFTER REMOVAL

Suppose 50 L, 40% acid.

Remove 10 L and add 20 L water.

Acid retained:

$$\boxed{16\text{ L}}$$

Final volume:

$$\boxed{60\text{ L}}$$

Acid concentration:

$$\boxed{16/60 = 26\frac{2}{3}\%}$$

26. REMOVE x, ADD y PURE COMPONENT

Initial tracked component quantity:

$$\boxed{Q}$$

Remove x from total V :

$$Q_{\text{remain}} = Q \left(1 - \frac{x}{V} \right)$$

Add y litres of **pure tracked component**:

$$Q_{\text{new}} = Q \left(1 - \frac{x}{V} \right) + y$$

Final volume:

$$\boxed{V - x + y}$$

Final concentration:

$$\boxed{\frac{Q(1 - x/V) + y}{V - x + y}}$$

27. EXAMPLE

40 L mixture contains 25% acid.

Remove 8 L and add 4 L pure acid.

Initial acid:

$$\sqrt[10]{10L}$$

After removal:

$$\sqrt[10]{10(0.8)=8L}$$

Add:

$$\sqrt[10]{4L}$$

Final acid:

$$\sqrt[10]{12L}$$

Final total:

$$\sqrt[10]{36L}$$

Concentration:

$$\sqrt[10]{12/36 = \boxed{\frac{1}{3}}}$$

28. TWO-STAGE TARGET PROBLEM

A 100 L vessel initially contains pure milk.

First 20 L is removed and replaced with water.

Then (x) L of mixture is removed and replaced with water.

Final milk = 60 L.

After first operation:

$$\sqrt[10]{80L \text{ milk}}$$

Second operation retains:

$$\sqrt[10]{80 \left(1 - \frac{x}{100}\right)}$$

Set:

$$\sqrt[10]{80 \left(1 - \frac{x}{100}\right) = 60}$$

$$\sqrt[10]{1 - \frac{x}{100} = \frac{3}{4}}$$

$$\sqrt[10]{x = 25}$$

$$\sqrt[10]{\boxed{25L}}$$

29. MIXTURE RATIO AFTER DIFFERENT REPLACEMENTS

60 L pure milk.

First replace 10 L with water.

Second replace 20 L with water.

Milk final:

$$\sqrt[10]{60 \left(1 - \frac{10}{60}\right) \left(1 - \frac{20}{60}\right)}$$

$$\sqrt[10]{= 60 \left(\frac{5}{6}\right) \left(\frac{2}{3}\right)}$$

$$\sqrt[10]{= 60 \cdot \frac{5}{9}}$$

$$V = \frac{100}{3}L$$

Water:

$$V = 60 - \frac{100}{3} = \frac{80}{3}L$$

Ratio:

$$V = 100:80 = \boxed{5:4}$$

30. TARGET COMPONENT QUANTITY WHEN REFILL HAS NONZERO CONCENTRATION

General fixed-volume formula:

$$V = C_n = D + (C_0 - D)r^n$$

Suppose:

- $(C_0 = 20\%)$
- refill $(D = 60\%)$
- replace 25% each time
- after 2 operations

$$V = r = 0.75$$

$$V = C_2 = 60 + (20 - 60)(0.75)^2$$

$$V = 60 - 40(0.5625)$$

$$V = 60 - 22.5$$

$$V = \boxed{37.5\%}$$

31. FIND NUMBER OF REPLACEMENTS WITH NONZERO REFILL

Initial concentration:

$$V = 20\%$$

Refill:

$$V = 60\%$$

Replace half vessel each time.

Find when concentration becomes 55%.

$$V = 55 = 60 + (20 - 60)\left(\frac{1}{2}\right)^n$$

$$V = 55 - 60 = -40\left(\frac{1}{2}\right)^n$$

$$V = 5 = 40\left(\frac{1}{2}\right)^n$$

$$V = \left(\frac{1}{2}\right)^n = \frac{1}{8}$$

$$V = \boxed{n=3}$$

32. REPLACEMENT VS SIMPLE ADDITION

Do not confuse:

remove 10 L and add 10 L water

with:

simply add 10 L water

These produce different answers.

Example: 40 L pure milk.

Simply add 10 L water:

Milk fraction:

$$\sqrt[4]{40/50=80\%}$$

Replace 10 L with water:

Milk:

$$\sqrt[4]{30L}$$

Total:

$$\sqrt[4]{40L}$$

Milk fraction:

$$\sqrt[4]{75\%}$$

Very different.

33. REPLACEMENT VS DILUTION MEMORY TRICK

Addition only:

Original component quantity stays unchanged.

Replacement:

Some original component leaves first.

This distinction is crucial.

34. GATE-STYLE QUESTION 1

40 L mixture is 60% milk. 10 L is removed and not replaced.

Final milk concentration?

$$\sqrt[4]{\boxed{60\%}}$$

The ratio remains unchanged.

35. GATE-STYLE QUESTION 2

40 L mixture is 60% milk. 10 L is removed and replaced with water.

Final milk concentration:

$$60\left(1-\frac{10}{40}\right)$$

$$=60(0.75)$$

$$=\boxed{45\%}$$

36. GATE-STYLE QUESTION 3

50 L solution is 40% acid. 10 L removed, 5 L water added.

Acid retained:

$$20(0.8)=16\text{ L}$$

Final volume:

$$45\text{ L}$$

$$C=\frac{16}{45}\times 100$$

$$=\boxed{35.56\%\text{ approximately}}$$

37. GATE-STYLE QUESTION 4

Pure milk undergoes replacements of $\frac{1}{5}$, then $\frac{1}{4}$, then $\frac{1}{2}$.

Fraction remaining:

$$\frac{45}{100}\cdot\frac{34}{100}\cdot\frac{12}{100}$$

$$=\boxed{\frac{3}{10}}$$

38. PLACEMENT-STYLE QUESTION 1

A 50 L vessel is 80% milk.

How much must be replaced with water once to make milk 64%?

$$64=80\left(1-\frac{x}{50}\right)$$

$$0.8=1-\frac{x}{50}$$

$$x=10$$

$$\boxed{10\text{ L}}$$

39. PLACEMENT-STYLE QUESTION 2

100 L pure milk.

20 L replaced with water, then 25 L replaced with water.

Milk:

$$\boxed{100(0.8)(0.75) = 60\text{L}}$$

Water:

$$\boxed{40\text{L}}$$

Ratio:

$$\boxed{3:2}$$

40. PLACEMENT-STYLE QUESTION 3

A 40 L solution is 25% acid.

8 L removed and replaced with 50% acid solution.

$$C' = 25 + \frac{8}{40}(50 - 25)$$

$$= 25 + 5$$

$$= \boxed{30\%}$$

41. PLACEMENT-STYLE QUESTION 4

Initial acid concentration 10%.

Replace 50% of vessel with 50% acid solution twice.

Use:

$$C_2 = 50 + (10 - 50)(0.5)^2$$

$$= 50 - 10$$

$$= \boxed{40\%}$$

42. ADVANCED QUESTION — TARGET RATIO

A 50 L mixture has milk:water:

$$4:1$$

How much mixture must be replaced with water so final ratio becomes:

$$3:2$$

Initial milk concentration:

$$80\%$$

Target milk concentration:

$$60\%$$

$$60 = 80 \left(1 - \frac{x}{50} \right)$$

$$\frac{60}{80} = 1 - \frac{x}{50}$$

$$0.75 = 1 - \frac{x}{50}$$

$$x = 12.5$$

$$\boxed{12.5\text{L}}$$

43. ADVANCED QUESTION — MIXED REFILLS

100 L solution initially 50% acid.

First 20 L replaced with water.

Then 20 L replaced with pure acid.

After first:

$$C_1 = 50(0.8) = 40\%$$

Second:

$$C_2 = 40 + 0.2(100 - 40)$$

$$= 40 + 12$$

$$= \boxed{52\%}$$

44. ADVANCED QUESTION — SAME OPERATIONS REVERSED

Now reverse the order:

First replace 20 L with pure acid.

$$C_1 = 50 + 0.2(100 - 50) = 60\%$$

Then replace 20 L with water:

$$C_2 = 60(0.8) = 48\%$$

So:

$$\boxed{48\%}$$

Important

When refill liquids differ, **order can matter**.

45. HIGH-VALUE TRAP — ORDER

For repeated replacement with the **same refill**, order is irrelevant because operations are identical.

With different refill concentrations:

$$\boxed{\text{Order can change the final answer}}$$

Example above:

- water then acid !' 52%
 - acid then water !' 48%
-

46. ADVANCED QUESTION — REMOVE AND REFILL UNEQUAL AMOUNTS

60 L solution is 50% acid.

Remove 12 L, then add 6 L pure acid.

Initial acid:

$$\boxed{30\text{ L}}$$

After removal:

$$\boxed{30(0.8)=24\text{ L}}$$

Add acid:

$$\boxed{30\text{ L}}$$

Final volume:

$$\boxed{54\text{ L}}$$

Concentration:

$$\boxed{30/54 = \frac{5}{9} \approx 55.56\%}$$

47. ADVANCED QUESTION — FINAL VOLUME GIVEN

A 50 L solution is 40% acid.

x L mixture is removed and 10 L water added.

Final volume is 45 L.

Then:

$$\boxed{50-x+10=45}$$

$$\boxed{x=15}$$

Initial acid:

$$\boxed{20\text{ L}}$$

Acid retained:

$$\boxed{20\left(1-\frac{15}{50}\right) = 20(0.7) = 14\text{ L}}$$

Final concentration:

$$\boxed{14/45 \times 100}$$

$$\boxed{= \frac{14}{45} \times 100 \approx 31.11\% \text{ (approximately)}}$$

48. ADVANCED QUESTION — FINAL RATIO AFTER PURE SOLUTE REFILL

40 L mixture is 25% acid.

Remove 8 L and replace with pure acid twice.

From Part 1:

Final water:

$$\boxed{30(0.8)^2 = 19.2\text{ L}}$$

Final acid:

$$\boxed{20.8\text{ L}}$$

Ratio acid:water:

$$\boxed{20.8:19.2}$$

Multiply by 10:

$$\boxed{208:192}$$

Divide 16:

$$\boxed{13:12}$$

49. COMMON TRAPS

& pTRAP 1 — Remove-only

Concentration does not change.

& pTRAP 2 — Unequal refill

Total volume changes, so denominator changes.

& pTRAP 3 — Different refill liquids

Solve sequentially; order may matter.

& pTRAP 4 — Addition vs replacement

Adding water and replacing with water are not the same.

& pTRAP 5 — Applying fixed-volume formula after volume changes

Invalid unless the volume is restored to the assumed (V) .

& pTRAP 6 — Target ratio

Convert the target ratio into target concentration first.

50. MUST-MEMORIZE FORMULAS

Standard repeated replacement

$$\boxed{Q_n = Q_0 \left(1 - \frac{x}{V}\right)^n}$$

One-step unequal refill

$$\boxed{C' = \frac{(V-x)C + yD}{V-x+y}}$$

Fixed-volume one-step update

$$\boxed{C' = C + \frac{x}{V}(D - C)}$$

Repeated fixed refill concentration (D)

$$\boxed{C_n = D + (C_0 - D) \left(1 - \frac{x}{V}\right)^n}$$

51. FAST DECISION TREE

Remove only?

Ratio unchanged.

Remove $V(x)$, add same $V(x)$ water?

Use retention multiplier.

Remove $V(x)$, add different $V(y)$?

Track component amount and new total volume.

Replacement liquid has concentration D ?

Use update formula.

Different replacement liquids each step?

Solve sequentially.

Target final ratio?

Convert ratio to concentration.

Need $V(x)$, $V(n)$, or $V(V)$?

Use inverse retention equation.

52. 15-MINUTE CHAPTER REVISION

The full chapter can be compressed into six ideas:

1. Homogeneous removal preserves ratio.
 2. Replacement causes dilution/change because refill composition differs.
 3. Standard repeated replacement = repeated percentage decay.
 4. Track the component absent from the refill when possible.
 5. With nonzero refill concentration, track the gap from refill concentration.
 6. If volume is not restored, recompute the denominator.
-

PRACTICE — LEVEL 0

Q1

50 L mixture is 40% acid. 10 L is removed without replacement. Find final acid %.

Q2

50 L mixture is 40% acid. 10 L removed and replaced with water. Find final acid %.

Q3

A vessel loses 20% of contents and is refilled with water twice. What fraction of original component remains?

Q4

60 L pure milk. First 10 L and then 20 L are replaced with water. Find milk remaining.

Q5

A 40 L mixture is 50% milk. Remove 10 L and add 5 L water. Find final milk %.

LEVEL 1

Q6

50 L mixture is 80% milk. How much must be replaced with water once to make it 64% milk?

Q7

A 40 L solution is 25% acid. 8 L is replaced with 50% acid solution. Find final acid %.

Q8

100 L pure milk. Replace 20 L, then 25 L, with water. Find milk:water ratio.

Q9

A solution starts at 20% acid. One-fourth is replaced with 60% acid solution. Find new concentration.

Q10

A 50 L vessel starts at 60% milk. 10 L is removed but only 5 L water is added. Find final milk concentration.

LEVEL 2

Q11

100 L solution starts at 50% acid. Replace 20 L with water, then 20 L with pure acid. Find final concentration.

Q12

Perform Q11 in reverse order. Find final concentration.

Q13

A 60 L solution is 50% acid. Remove 12 L and add 6 L pure acid. Find final concentration.

Q14

A pure liquid undergoes two equal replacements and 36% of original remains. What fraction is removed each time?

Q15

A 50 L milk-water mixture is 80% milk. How much mixture must be replaced with water to obtain milk:water = 3:2?

LEVEL 3 — EXAM PRESSURE

Q16

A 100 L solution starts at 10% acid. Each time 20 L is removed and replaced with 50% acid. Find concentration after 3 replacements.

Q17

A vessel initially contains pure milk. First $\frac{1}{5}$, then $\frac{1}{4}$, then $\frac{1}{2}$ is replaced with water. Find final milk:water

ratio.

Q18

A 40 L mixture starts at 50% milk. 8 L is replaced with water twice, then 8 L is replaced with pure milk. Find final milk %.

Q19

A 50 L solution is 40% acid. (x) L is removed and 10 L water added. Final volume is 45 L. Find final acid %.

Q20

A solution starts at 20% acid. Half the vessel is replaced each time with 60% acid solution. After how many replacements does the concentration become 55%?

ANSWERS

Q1

Removing only does not change concentration:

$$\boxed{40\%}$$

Q2

$$40(0.8) = \boxed{32\%}$$

Q3

$$0.8^2 = \boxed{0.64=64\%}$$

Q4

$$60\left(\frac{5}{6}\right)\left(\frac{2}{3}\right)$$

$$= \boxed{33\frac{1}{3}L}$$

Q5

Initial milk:

$$20L$$

After removing 10 L:

$$20(0.75)=15L$$

Final volume:

$$35L$$

$$\frac{15}{35}\times 100 = \boxed{42\frac{6}{7}\%}$$

Q6

$$64 = 80\left(1-\frac{x}{50}\right)$$

$$x = \boxed{10L}$$

Q7

$$25+\frac{8}{40}(50-25)$$

$$V = 25 + 5$$

$$V = \boxed{30\%}$$

Q8

Milk:

$$V = 100(0.8)(0.75) = 60$$

Water:

$$V = 40$$

$$V = \boxed{3:2}$$

Q9

$$C' = 20 + \frac{14(60-20)}{100}$$

$$V = 20 + 10$$

$$V = \boxed{30\%}$$

Q10

Initial milk:

$$V = 30L$$

After removing 10:

$$V = 30(0.8) = 24L$$

Final volume:

$$V = 45L$$

$$V = \frac{24}{45} \times 100 = \boxed{53\frac{1}{3}\%}$$

Q11

First:

$$V = 50 \text{ to } 40\%$$

Then pure acid:

$$V = 40 + 0.2(100-40) = \boxed{52\%}$$

Q12

First pure acid:

$$V = 50 + 0.2(50) = 60\%$$

Then water:

$$V = 60(0.8) = \boxed{48\%}$$

Q13

Initial acid:

$$V = 30L$$

After removal:

$$V = 24L$$

Add 6 L pure acid:

$$\sqrt[3]{30L}$$

Final volume:

$$\sqrt[3]{54L}$$

$$\sqrt[3]{30/54 \times 100} = \boxed{55\frac{5}{9}\%}$$

Q14

$$\sqrt[3]{(1-f)^2 = 0.36}$$

$$\sqrt[3]{1-f = 0.6}$$

$$\sqrt[3]{f = 0.4}$$

$$\sqrt[3]{\boxed{40\%}}$$

Q15

Target milk:water:

$$\sqrt[3]{3:2}$$

Milk target:

$$\sqrt[3]{60\%}$$

$$\sqrt[3]{60 = 80 \left(1 - \frac{x}{50}\right)}$$

$$\sqrt[3]{0.75 = 1 - \frac{x}{50}}$$

$$\sqrt[3]{x = 12.5}$$

$$\sqrt[3]{\boxed{12.5L}}$$

Q16

Use:

$$\sqrt[3]{C_n = D + (C_0 - D)(0.8)^n}$$

$$\sqrt[3]{= 50 + (10 - 50)(0.8)^3}$$

$$\sqrt[3]{= 50 - 40(0.512)}$$

$$\sqrt[3]{= \boxed{29.52\%}}$$

Q17

Milk fraction:

$$\sqrt[3]{\frac{45}{100} \cdot \frac{34}{100} \cdot \frac{12}{100} = \frac{3}{10}}$$

Water fraction:

$$\sqrt[3]{\frac{7}{10}}$$

Ratio:

$$\sqrt[3]{\boxed{3:7}}$$

Q18

After two water replacements:

$$\sqrt[3]{50(0.8)^2 = 32\%}$$

Milk amount:

$$\sqrt[4]{40(0.32)=12.8\text{ L}}$$

Third operation removes 8 L:

Milk removed:

$$\sqrt[4]{8(0.32)=2.56}$$

Milk left:

$$\sqrt[4]{10.24}$$

Add 8 L milk:

$$\sqrt[4]{18.24\text{ L}}$$

Final %:

$$\sqrt[4]{18.24/40\times 100 = \boxed{45.6\%}}$$

Q19

Final volume:

$$\sqrt[4]{50-x+10=45}$$

$$\sqrt[4]{x=15}$$

Initial acid:

$$\sqrt[4]{20\text{ L}}$$

After removal:

$$\sqrt[4]{20\left(1-\frac{15}{50}\right) = 14\text{ L}}$$

Final concentration:

$$\sqrt[4]{14/45\times 100 = \boxed{31.11\%\text{ approximately}}}$$

Q20

$$\sqrt[4]{55 = 60+(20-60)(0.5)^n}$$

$$\sqrt[4]{5 = 40(0.5)^n}$$

$$\sqrt[4]{(0.5)^n=\frac{18}{40}}$$

$$\sqrt[4]{\boxed{n=3}}$$

' CHAPTER 23 COMPLETE

We have now covered:

- one-time replacement '
- repeated replacement '
- original-component remaining formula '
- non-pure initial mixtures '
- replacement with water '
- replacement with pure solute '
- different removal amounts '

- unequal refill quantities '
- remove-only problems '
- target final concentration '
- target final ratio '
- finding (x, n, V) '
- generalized nonzero-refill formula '
- changing refill liquids '
- operation-order effects '
- mixed replacement sequences '
- advanced GATE/placement problems '

Progress

Completed: 23/76 Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 24 — Partnership

Next we will cover:

- capital contribution
- profit-sharing ratio
- time-weighted investment
- partner joining/leaving
- changing capital midway
- salary/commission to working partner
- unequal investment periods
- finding unknown capital or joining time
- multiple capital changes
- placement-style partnership shortcuts.

USER

next

ASSISTANT

CHAPTER 24 — PARTNERSHIP

PART 1 — CAPITAL, TIME-WEIGHTED INVESTMENT, PROFIT-SHARING RATIO & PARTNER JOINING/LEAVING

Status: 0% PROGRESS GATE: Low–Medium Placements: Medium Priority: 0% Tier 3

Partnership problems are based on one central rule:

A partner's share of profit is proportional to both money invested and time for which it remains invested.

So:

$$\boxed{\text{Profit share}} \propto \text{Capital} \times \text{Time}$$

This is the master idea.

1. BASIC PARTNERSHIP

Suppose A and B invest:

$$₹ 5000, ₹ 8000$$

for the same duration.

Then profit ratio:

$$5000:8000$$

$$= \boxed{5:8}$$

Because time is equal, only capital matters.

2. MASTER FORMULA

If A invests (C_A) for (T_A) months and B invests (C_B) for (T_B) months:

$$\boxed{A:B = C_A T_A : C_B T_B}$$

This quantity:

$$C \times T$$

is sometimes called **capital-months** or **investment-time product**.

3. EXAMPLE — DIFFERENT TIMES

A invests ₹ 10,000 for 12 months.

B invests ₹ 15,000 for 8 months.

A's investment weight:

$$10000(12) = 120000$$

B's:

$$15000(8) = 120000$$

Therefore:

$$\boxed{A:B = 1:1}$$

So they share profit equally.

4. WHY TIME MATTERS

Suppose A invests ₹ 10,000 for 12 months.

B invests ₹ 10,000 only 6 months.

Same capital, but A's money stays twice as long.

So:

$$A:B=12:6=2:1$$

Thus A receives twice B's profit share.

5. FIND INDIVIDUAL PROFIT SHARE

If total profit = ₹ 18,000 and ratio:

$$A:B=2:1$$

Total parts:

$$3$$

A:

$$18000 \times \frac{2}{3} = \boxed{12000}$$

B:

$$18000 \times \frac{1}{3} = \boxed{6000}$$

6. THREE PARTNERS

A, B, C invest for the same time:

$$₹ 20000, ₹ 30000, ₹ 50000$$

Profit ratio:

$$2:3:5$$

If total profit = ₹ 50,000:

Total parts:

$$10$$

Shares:

$$A = ₹ 10000$$

$$B = ₹ 15000$$

$$C = ₹ 25000$$

7. DIFFERENT CAPITALS AND TIMES

A invests ₹ 20,000 for 12 months.

B invests ₹ 30,000 for 8 months.

C invests ₹ 40,000 for 6 months.

Weights:

$$\boxed{A=20(12)=240}$$

$$\boxed{B=30(8)=240}$$

$$\boxed{C=40(6)=240}$$

Therefore:

$$\boxed{1:1:1}$$

All get equal profit.

8. CAPITAL UNIT CANCELLATION

You don't need to carry zeros.

Instead of:

$$\boxed{20000(12):30000(8)}$$

use:

$$\boxed{20(12):30(8)}$$

Same ratio.

This is faster.

9. PARTNER JOINS LATE

A starts a business with ₹ 20,000.

After 4 months, B joins with ₹ 30,000.

Business runs for 12 months total.

A's time:

$$\boxed{12 \text{ months}}$$

B's time:

$$\boxed{12-4=8 \text{ months}}$$

Therefore:

$$\boxed{A:B = 20(12):30(8)}$$

$$\boxed{=240:240}$$

$$\boxed{1:1}$$

10. JOINING-TIME TRAP

If B joins **after 4 months** in a 12-month business:

$$\boxed{B \text{ invests for } 8 \text{ months}}$$

not 4 months.

Always calculate actual duration.

11. PARTNER LEAVES EARLY

A and B start together.

A invests ₹ 20,000 for full 12 months.

B invests ₹ 30,000 and withdraws after 8 months.

Thus:

$$[A:B = 20(12):30(8)]$$

$$[=\boxed{1:1}]$$

12. FIND UNKNOWN CAPITAL

A invests ₹ 20,000 for 12 months.

B joins after 4 months.

They divide profit in ratio:

$$[3:2]$$

Find B's capital.

B invests for:

$$[8\text{ months}]$$

Let B capital = ₹(x).

$$[20000(12):x(8)=3:2]$$

$$[240000:8x=3:2]$$

Cross multiply:

$$[480000=24x]$$

$$[x = \boxed{20000}]$$

13. FASTER CANCELLATION

$$[20(12):x(8)=3:2]$$

$$[240:8x=3:2]$$

$$[30:x=3:2]$$

$$[x=20]$$

So:

$$[\boxed{20000}]$$

14. FIND JOINING TIME

A invests ₹ 30,000 for 12 months.

B invests ₹ 40,000 for some months.

Profit ratio:

$$A:B=3:2$$

Let B invest for (t) months.

$$30(12):40t=3:2$$

$$360:40t=3:2$$

$$720=120t$$

$$t=6$$

So B invested for 6 months.

Therefore B joined after:

$$12-6 = \boxed{6 \text{ months}}$$

15. FIND WITHDRAWAL TIME

A and B invest:

$$₹ 25000 \text{ and } ₹ 30000$$

at start.

A remains for 12 months.

B withdraws after (t) months.

Profit ratio:

$$5:4$$

$$25(12):30t=5:4$$

$$300:30t=5:4$$

$$1200=150t$$

$$t=8$$

So B withdrew after:

$$\boxed{8 \text{ months}}$$

16. CAPITAL CHANGES MIDWAY

Suppose A invests:

$$₹ 10000$$

for first 6 months, then increases capital to:

$$₹ 20000$$

for next 6 months.

A's total investment-time weight:

$$\sqrt[10000(6)+20000(6)]{}$$

$$\sqrt[=60000+120000]{}$$

$$\sqrt[=\boxed{180000}]{}$$

You must treat each period separately.

17. PIECEWISE CAPITAL FORMULA

If a partner invests:

- $\sqrt[C_1]{}$ for $\sqrt[T_1]{}$
- $\sqrt[C_2]{}$ for $\sqrt[T_2]{}$
- etc.

Then partnership weight is:

$$\sqrt[\boxed{C_1T_1+C_2T_2+\cdots}]{}$$

Not:

$$\sqrt[\frac{\text{final capital}}{\text{total time}}]{}$$

unless capital was constant.

18. EXAMPLE — CAPITAL INCREASE

A:

- $\sqrt[120,000]{}$ first 4 months
- $\sqrt[130,000]{}$ next 8 months

B:

- $\sqrt[125,000]{}$ full 12 months

A weight:

$$\sqrt[20(4)+30(8)]{}$$

$$\sqrt[=80+240=320]{}$$

B weight:

$$\sqrt[25(12)=300]{}$$

Ratio:

$$\sqrt[320:300 = \boxed{16:15}]{}$$

19. CAPITAL DECREASE MIDWAY

A invests $\sqrt[140,000]{}$ daily.

After 6 months, withdraws $\sqrt[110,000]{}$.

So:

- first 6 months !' $\sqrt[140,000]{}$

- next 6 months !' 130,000

Weight:

$$[40(6)+30(6) = 240+180 = 420]$$

If B invests 135,000 12 months:

$$[35(12)=420]$$

Therefore:

$$[\boxed{A:B=1:1}]$$

20. MULTIPLE CAPITAL CHANGES

A invests:

- 10k for 3 months
- 20k for next 4 months
- 30k for last 5 months

Weight:

$$[10(3)+20(4)+30(5)]$$

$$[=30+80+150]$$

$$[=\boxed{260}]$$

Keep a timeline.

21. TIMELINE METHOD

When capital changes:

Period	Capital	Months	$(C \times T)$
Month 1–3	10k	3	30
Month 4–7	20k	4	80
Month 8–12	30k	5	150

Total:

$$[260]$$

This prevents timing mistakes.

22. PARTNER JOINS AND THEN CHANGES CAPITAL

B joins after 3 months with 20k .

After another 4 months, increases to 30k .

In a 12-month business:

- months 1–3 !' absent

- months 4–7 !' ¹ 2 0 for 4 months
- months 8–12 !' ¹ 3 0 for 5 months

Weight:

$$\sqrt[2]{20(4)+30(5)}$$

$$\sqrt[2]{=80+150}$$

$$\sqrt[2]{=\boxed{230}}$$

23. WORKING PARTNER SALARY

Sometimes one partner actively manages the business and receives a salary **before the remaining profit is divided**.

Example:

Total profit:

$$\sqrt[2]{150000}$$

A receives salary:

$$\sqrt[2]{15000}$$

Remaining profit:

$$\sqrt[2]{45000}$$

If partnership ratio is:

$$\sqrt[2]{2:1}$$

A's share from remaining profit:

$$\sqrt[2]{45000 \times \frac{2}{3} = 30000}$$

Total received by A:

$$\sqrt[2]{30000+5000 = \boxed{35000}}$$

B:

$$\sqrt[2]{\boxed{115000}}$$

24. SALARY IS NOT PARTNERSHIP WEIGHT

The salary/commission is usually treated separately if the problem states:

A receives ¹xsalary for managing the business, and remaining profit is shared...

Then:

1. subtract salary,
2. divide remainder by capital-time ratio.

25. COMMISSION TO A PARTNER — SIMPLE VERSION

Total profit:

$\text{₹ } 16000$

A gets 10% of total profit as management commission.

Commission:

$\text{₹ } 6000$

Remaining:

$\text{₹ } 54000$

Suppose A:B partnership ratio = 2:1.

A partnership share:

$\text{₹ } 36000$

Total A:

$\text{₹ } 36000 + 6000 = \boxed{\text{₹ } 42000}$

B:

$\boxed{\text{₹ } 18000}$

26. COMMISSION ON PROFIT AFTER COMMISSION — WORDING TRAP

If a problem says:

A gets 10% of profit after charging commission

the base may differ from simply 10% of total profit.

Do not assume.

For standard placement questions, wording often makes the commission base explicit.

27. FIND TOTAL PROFIT FROM ONE PARTNER'S SHARE

A:B = 3:5.

A receives $\text{₹ } 9000$.

A corresponds to 3 parts.

1 part:

$\text{₹ } 3000$

Total 8 parts:

$\boxed{\text{₹ } 24000}$

28. FIND RATIO FROM PROFIT SHARES

A receives $\text{₹ } 12000$.

B receives $\text{₹ } 18000$.

Then:

$$A:B=12000:18000$$

$$= \boxed{2:3}$$

If times are known, use this ratio to infer capitals.

29. FIND CAPITAL RATIO FROM PROFIT RATIO AND TIME RATIO

Given:

$$\text{profit ratio} = 3:4$$

$$T_A:T_B=2:3$$

Since:

$$C_A T_A : C_B T_B = 3:4$$

So:

$$\frac{C_A}{C_B} = \frac{3}{4} \cdot \frac{3}{2} = \frac{9}{8}$$

Thus:

$$C_A:C_B=9:8$$

30. GENERAL RATIO RELATION

$$\boxed{\frac{C_A}{C_B} = \frac{\text{profit ratio } A/B}{\text{time ratio } A/B}}$$

More explicitly:

$$\boxed{C_A:C_B = (P_A T_B):(P_B T_A)}$$

where $(P_A:P_B)$ is profit-share ratio.

31. EQUAL PROFIT SHARES

If two partners get equal profits:

$$C_A T_A = C_B T_B$$

Therefore:

$$\boxed{C_A:C_B = T_B:T_A}$$

Capital and time become inversely proportional.

32. EXAMPLE

A invests for 12 months.

B invests for 8 months.

They receive equal profits.

Then:

$$\boxed{C_A:C_B = 8:12 = 2:3}$$

So B must invest more capital to compensate for shorter time.

33. SAME CAPITAL

If capitals are equal:

$$\boxed{\text{profit ratio} = \text{time ratio}}$$

Example:

A invests 12 months.

B 9 months.

Profit ratio:

$$12:9 = 4:3$$

34. SAME TIME

If investment durations are equal:

$$\boxed{\text{profit ratio} = \text{capital ratio}}$$

Very basic but useful.

35. GATE-STYLE QUESTION 1

A and B invest ₹ 20,000 and ₹ 30,000 for equal times.

Profit ratio:

$$2:3$$

If total profit ₹ 25,000 :

A:

$$25000 \times \frac{2}{5} = ₹ 10,000$$

B:

$$25000 \times \frac{3}{5} = ₹ 15,000$$

36. GATE-STYLE QUESTION 2

A invests ₹ 10 for 12 months.

B invests ₹ 15 for 8 months.

$$10(12):15(8) = 120:120$$

$$= 1:1$$

37. GATE-STYLE QUESTION 3

A invests ₹ 30,000 for full year.

B joins after 4 months with ₹ 45,000.

$$A:B = 30(12):45(8)$$

$$=360:360$$

$$=1:1$$

38. GATE-STYLE QUESTION 4

A and B receive profits in ratio 4:3.

A invests ₹ 20,000 for 12 months.

B invests for 8 months.

Find B capital.

$$20(12):8B=4:3$$

$$240:8B=4:3$$

$$720=32B$$

$$B=22.5$$

$$\boxed{22500}$$

39. PLACEMENT-STYLE QUESTION 1

A invests ₹ 40,000.

B invests ₹ 50,000 after 4 months.

For a 12-month business:

$$A:B = 40(12):50(8)$$

$$=480:400$$

$$=6:5$$

40. PLACEMENT-STYLE QUESTION 2

A invests ₹ 25,000 for 12 months.

B invests ₹ 30,000 for 10 months.

C invests ₹ 40,000 for 6 months.

Weights:

$$300:300:240$$

$$=5:5:4$$

$$\boxed{5:5:4}$$

41. PLACEMENT-STYLE QUESTION 3

A invests:

- ₹ 20,000 for first 6 months
- ₹ 30,000 for next 6 months

B invests ₹ 25,000 for full year.

A:

$$[20(6)+30(6)=300]$$

B:

$$[25(12)=300]$$

$$[\boxed{1:1}]$$

42. PLACEMENT-STYLE QUESTION 4

A:B investment weights = 3:2.

Total profit = ₹ 40,000 .

A also receives ₹ 4,000 salary before profit division.

Remaining:

$$[36000]$$

A's partnership share:

$$[36000 \times \frac{3}{5} = 21600]$$

A total:

$$[21600 + 4000 = \boxed{25600}]$$

B:

$$[\boxed{14400}]$$

43. ADVANCED QUESTION — UNKNOWN JOINING MONTH

A starts with ₹ 30,000 .

B invests ₹ 45,000 after (x) months.

At year end, profit ratio is:

$$[4:3]$$

A weight:

$$[30(12)=360]$$

B duration:

$$[12-x]$$

Equation:

$$\backslash[360:45(12-x)=4:3 \backslash]$$

$$\backslash[1080 = 180(12-x) \backslash]$$

$$\backslash[6=12-x \backslash]$$

$$\backslash[x=\boxed{6} \backslash]$$

B joined after 6 months.

44. ADVANCED QUESTION — CAPITAL CHANGE

A invests $^1 2 0$ k .

After 4 months, doubles capital.

B invests $^1 3 0$ for full year.

A weight:

First 4 months:

$$\backslash[20(4)=80 \backslash]$$

Next 8 months:

$$\backslash[40(8)=320 \backslash]$$

Total:

$$\backslash[400 \backslash]$$

B:

$$\backslash[30(12)=360 \backslash]$$

Ratio:

$$\backslash[400:360 = \boxed{10:9} \backslash]$$

45. ADVANCED QUESTION — WITHDRAWAL MIDYEAR

A and B each invest $^1 4 0$ k .

After 4 months A withdraws $^1 1 0$ k .

After 6 months B withdraws $^1 2 0$ k .

For 12 months:

A:

$$\backslash[40(4)+30(8) = 160+240 = 400 \backslash]$$

B:

$$\backslash[40(6)+20(6) = 240+120 = 360 \backslash]$$

Profit ratio:

$$\backslash[400:360 = \boxed{10:9} \backslash]$$

46. ADVANCED QUESTION — FIND INITIAL CAPITAL

A invests ₹ 20,000 for 12 months.

B invests ₹(x) for first 6 months, then doubles it for last 6 months.

Profit ratio A:B = 2:3.

A:

$$[20(12)=240]$$

B:

$$[x(6)+2x(6) = 18x]$$

$$[240:18x=2:3]$$

$$[720=36x]$$

$$[x=20]$$

So initial B capital:

$$[\boxed{₹20,000}]$$

and later:

$$[₹40,000]$$

47. ADVANCED QUESTION — SALARY + SHARE

A and B invest in ratio 2:3 for equal time.

A manages the business and gets ₹ 6,000 salary.

Total profit = ₹ 46,000 .

Remaining:

$$[40000]$$

Partnership shares:

A:

$$[40000 \times \frac{2}{5} = 16000]$$

B:

$$[24000]$$

A total:

$$[16000+6000 = \boxed{₹22,000}]$$

B:

$$[\boxed{₹24,000}]$$

48. COMMON TRAPS

& pTRAP 1 — Ignoring time

Profit is not always proportional only to capital.

Use:

$$\frac{C}{T}$$

& pTRAP 2 — Joins after x months

Investment duration is:

$$\frac{\text{total duration} - x}{T}$$

& pTRAP 3 — Capital changes

Split timeline into separate periods.

& pTRAP 4 — Salary

If salary/commission is given before sharing, subtract it first.

& pTRAP 5 — Equal profit

Equal profit does not imply equal capital if investment times differ.

49. MUST-MEMORIZE RULES

Constant investment

$$\frac{C_1}{T_1} = \frac{C_2}{T_2}$$

Changing capital

$$\frac{C_1}{T_1} = \frac{C_2}{T_2}$$

Equal profit

$$C_1(T_1 - T_2) = C_2(T_2 - T_1)$$

50. FAST EXAM DECISION TREE

Same duration?

Use capital ratio.

Same capital?

Use time ratio.

Different capital and time?

Use $\frac{C}{T}$.

Partner joins late?

Compute remaining months.

Partner leaves early?

Use actual invested months.

Capital changes?

Create timeline and add \ (CT) products.

Salary/commission?

Handle separately before/after profit division according to wording.

PRACTICE — LEVEL 0

Q1

A and B invest $\text{₹} 20,000$ and $\text{₹} 30,000$ for equal time. Find profit ratio.

Q2

A and B invest equal capital for 12 and 8 months. Find profit ratio.

Q3

A invests $\text{₹} 10,000$ for 12 months and B $\text{₹} 15,000$ for 8 months. Find ratio.

Q4

Profit ratio is 2:3 and total profit $\text{₹} 25,000$. Find shares.

Q5

A invests $\text{₹} 20,000$ for full year. B joins after 6 months with $\text{₹} 40,000$. Find ratio.

LEVEL 1

Q6

A invests $\text{₹} 30,000$ for 12 months. B joins after 4 months with $\text{₹} 45,000$. Find ratio.

Q7

A invests $\text{₹} 20,000$ for full year. B joins after 3 months. Profit ratio is 8:9. Find B capital.

Q8

A and B invest $\text{₹} 25,000$ and $\text{₹} 30,000$. A stays 12 months, B 10 months. Find ratio.

Q9

A invests $\text{₹} 40,000$ for 12 months. B invests $\text{₹} 60,000$ and gets equal profit. How long did B invest?

Q10

A invests $\text{₹} 20,000$ for first 6 months and $\text{₹} 30,000$ for next 6 months. B invests $\text{₹} 25,000$ throughout. Find ratio.

LEVEL 2

Q11

A starts with ₹ 3000. B joins after x months with ₹ 4500. Profit ratio 4:3. Find x .

Q12

A invests ₹ 4000 for 12 months. B invests ₹ 3000 for t months. Profit ratio A:B = 8:5. Find t .

Q13

A invests ₹ 2000 for 4 months, then ₹ 4000 for 8 months. B invests ₹ 3000 all year. Find ratio.

Q14

A and B invest ₹ 4000 each. A withdraws ₹ 1000 after 4 months. B withdraws ₹ 2000 after 6 months. Find profit ratio.

Q15

A gets ₹ 500 salary from total profit ₹ 4500. Remaining profit shared A:B = 3:2. Find final amounts.

LEVEL 3 — EXAM PRESSURE**Q16**

A invests ₹ 3000 for full year. B joins after 2 months with ₹ 4000. C joins after 4 months with ₹ 5000. Find profit ratio.

Q17

A invests ₹ 2000 for 12 months. B invests x for 6 months then doubles capital for next 6 months. Profit ratio A:B = 2:3. Find x .

Q18

A:B profit ratio is 5:6. Their capital ratio is 4:3. Find time ratio $(T_A:T_B)$.

Q19

A invests ₹ 6000. After 4 months he withdraws ₹ 2000. B invests ₹ 5000 for full year. Find profit ratio.

Q20

A and B have partnership weight ratio 2:3. A also receives 10% of total profit as management commission before the remainder is divided in 2:3. If total profit is ₹ 5000, find A's total receipt.

ANSWERS**Q1**

$[20:30 = \boxed{2:3}]$

Q2

$[12:8 = \boxed{3:2}]$

Q3

$[10(12):15(8) = 120:120 = \boxed{1:1}]$

Q4

Total parts:

$$\sqrt{5}$$

$$\sqrt{A=25000 \times \frac{25}{1000000}}$$

$$\sqrt{B = \sqrt{150000}}$$

Q5

A:

$$\sqrt{20(12)=240}$$

B:

$$\sqrt{40(6)=240}$$

$$\sqrt{\boxed{1:1}}$$

Q6

$$\sqrt{30(12):45(8) = 360:360 = \boxed{1:1}}$$

Q7

B duration:

$$\sqrt{9 \text{ months}}$$

$$\sqrt{20(12):9B=8:9}$$

$$\sqrt{240:9B=8:9}$$

$$\sqrt{2160=72B}$$

$$\sqrt{B=30}$$

$$\sqrt{\boxed{130000}}$$

Q8

$$\sqrt{25(12):30(10)}$$

$$\sqrt{300:300}$$

$$\sqrt{=\boxed{1:1}}$$

Q9

Equal profits:

$$\sqrt{40(12)=60t}$$

$$\sqrt{480=60t}$$

$$\sqrt{\boxed{t=8 \text{ months}}}$$

Q10

A:

$$\sqrt{20(6)+30(6) = 300}$$

B:

$$\sqrt{25(12)=300}$$

$$\sqrt{\boxed{1:1}}$$

Q11

$$\backslash[30(12):45(12-x)=4:3 \backslash]$$

$$\backslash[360:45(12-x)=4:3 \backslash]$$

$$\backslash[1080=180(12-x) \backslash]$$

$$\backslash[6=12-x \backslash]$$

$$\backslash[\boxed{x=6\text{ months}} \backslash]$$

Q12

$$\backslash[40(12):30t=8:5 \backslash]$$

$$\backslash[480:30t=8:5 \backslash]$$

$$\backslash[2400=240t \backslash]$$

$$\backslash[\boxed{t=10\text{ months}} \backslash]$$

Q13

A:

$$\backslash[20(4)+40(8) = 80+320 = 400 \backslash]$$

B:

$$\backslash[30(12)=360 \backslash]$$

$$\backslash[\boxed{10:9} \backslash]$$

Q14

A:

$$\backslash[40(4)+30(8)=400 \backslash]$$

B:

$$\backslash[40(6)+20(6)=360 \backslash]$$

$$\backslash[\boxed{10:9} \backslash]$$

Q15

Salary:

$$\backslash[5000 \backslash]$$

Remaining:

$$\backslash[40000 \backslash]$$

A share:

$$\backslash[40000\text{ times }\frac{3}{5} = 24000 \backslash]$$

A total:

$$\backslash[\boxed{129000} \backslash]$$

B:

$$\backslash[\boxed{116000} \backslash]$$

Q16

A:

$$\sqrt[3]{30(12)}=360$$

B invests 10 months:

$$\sqrt[3]{40(10)}=400$$

C invests 8 months:

$$\sqrt[3]{50(8)}=400$$

Ratio:

$$\sqrt[3]{360:400:400}$$

$$\sqrt[3]{=\boxed{9:10:10}}$$

Q17

A:

$$\sqrt[3]{20(12)}=240$$

B:

$$\sqrt[3]{x(6)+2x(6)}=18x$$

$$\sqrt[3]{240:18x}=2:3$$

$$\sqrt[3]{720}=36x$$

$$\sqrt[3]{\boxed{x=12000}}$$

Q18

$$\sqrt[3]{C_AT_A:C_BT_B}=5:6$$

Given:

$$\sqrt[3]{C_A:C_B}=4:3$$

So:

$$\sqrt[3]{4T_A:3T_B}=5:6$$

$$\sqrt[3]{24T_A=15T_B}$$

$$\sqrt[3]{T_A:T_B = 15:24}$$

$$\sqrt[3]{=\boxed{5:8}}$$

Q19

A:

$$\sqrt[3]{60(4)+40(8)}$$

$$\sqrt[3]{=240+320=560}$$

B:

$$\sqrt[3]{50(12)}=600$$

$$\sqrt[3]{560:600 = \boxed{14:15}}$$

Q20

Commission:

$$[10\% \text{ of } 50000 = 5000]$$

Remaining:

$$[45000]$$

A's partnership share:

$$[45000 \times \frac{2}{5} = 18000]$$

A total:

$$[18000 + 5000 = \boxed{23000}]$$

CHAPTER 24 STATUS

Covered so far:

- basic partnership '
- capital ratio '
- time-weighted investment '
- $(C \times T)$ rule '
- late joining '
- early withdrawal '
- finding unknown capital '
- finding joining/withdrawal time '
- capital increase/decrease midway '
- piecewise investment timelines '
- salary/commission basics '
- equal-profit conditions '
- GATE/placement practice '

But **Chapter 24 is not complete yet.**

Next we need:

- goodwill / bonus adjustment basics where relevant
- commissions with tricky wording
- partner joining with capital changes
- multiple partners entering/leaving at different times
- profit-sharing with both salary and commission
- more advanced timeline questions
- speed shortcuts and final chapter audit

Progress

Completed: 23/76 **Current:** ~~Chapter 24~~ — Partnership **Next:** Chapter 24 Part 2 — **Advanced partnership timelines, commissions & mixed partner changes**

USER

next

ASSISTANT

CHAPTER 24 — PARTNERSHIP

PART 2 — ADVANCED TIMELINES, MULTIPLE JOIN/LEAVE EVENTS, SALARY + COMMISSION & MIXED PARTNER CHANGES

Status: $\emptyset$ = Not in PROGRESS GATE: Low–Medium Placements: Medium Priority: $\emptyset$ = Not in 3

This part is mostly about **timeline accuracy**.

The safest rule is:

$$\boxed{\text{Partner weight} = \sum (\text{capital during period}) (\text{months})}$$

Then profit is divided in the ratio of these weights.

1. MULTIPLE PARTNERS JOINING AT DIFFERENT TIMES

Suppose a business runs for 12 months.

- A invests ₹ 20,000 on the start
- B joins after 3 months with ₹ 30,000
- C joins after 6 months with ₹ 40,000

Durations:

$$A = 12$$

$$B = 9$$

$$C = 6$$

Weights:

$$20(12) = 240$$

$$30(9) = 270$$

$$40(6) = 240$$

Profit ratio:

$$240:270:240$$

$$= \boxed{8:9:8}$$

2. JOINING-TIME TABLE

Partner	Capital	Joins after	Invested for
A	20k	0 months	12 months

Partner	Capital	Joins after	Invested for
B	30k	3 months	9 months
C	40k	6 months	6 months

This simple table prevents most errors.

3. PARTNERS LEAVING AT DIFFERENT TIMES

A, B, C begin together.

- A invests ₹ 20,000 for 12 months
- B invests ₹ 30,000 but leaves after 8 months
- C invests ₹ 40,000 but leaves after 6 months

Weights:

$$[A = 20(12) = 240]$$

$$[B = 30(8) = 240]$$

$$[C = 40(6) = 240]$$

Therefore:

$$[\boxed{1:1:1}]$$

4. ONE JOINS, ANOTHER LEAVES

Business = 12 months.

- A invests ₹ 30,000 throughout
- B invests ₹ 40,000 from start, leaves after 6 months
- C joins after 6 months with ₹ 40,000

Weights:

$$[A = 30(12) = 360]$$

$$[B = 40(6) = 240]$$

$$[C = 40(6) = 240]$$

Ratio:

$$[360:240:240]$$

$$[\boxed{3:2:2}]$$

5. CAPITAL CHANGES FOR MULTIPLE PARTNERS

A:

- ₹ 20,000 for 4 months
- ₹ 30,000 for 8 months

B:

- ₹ 30,000 for 6 months
- ₹ 20,000 for 6 months

A's weight:

$$\frac{20(4)+30(8)}{20+30}$$

$$= \frac{80+240}{50} = \frac{320}{50} = 6.4$$

B's:

$$\frac{30(6)+20(6)}{30+20}$$

$$= \frac{180+120}{50} = \frac{300}{50} = 6$$

Ratio:

$$\frac{6.4}{6} = \frac{16}{15}$$

6. TIMELINE IS BETTER THAN STORY WORDING

When a question says:

A invested ₹ 20,000. After 4 months he added ₹ 10,000.

Do not interpret second period capital as ₹ 10,000.

It becomes:

$$20,000 + 10,000 = 30,000$$

So:

- first 4 months !' ₹ 20 k
- next 8 months !' ₹ 30 k

7. "WITHDRAWS ₹ x" TRAP

If capital is ₹ 50k and partner withdraws ₹ 20k after 5 months:

remaining capital:

$$50 - 20 = 30$$

not ₹ 20 k.

So:

$$50(5)+30(7)$$

8. EXAMPLE

A invests ₹ 50 k.

After 5 months, withdraws ₹ 20 k.

Business ends after 12 months.

Weight:

$$\boxed{50(5)+30(7)}$$

$$=250+210$$

$$\boxed{460}$$

9. FIND UNKNOWN CAPITAL CHANGE

A invests ₹ 20 for full year.

B invests ₹ 10 for first 6 months and ₹ x for last 6 months.

Profit ratio:

$$A:B=4:3$$

A weight:

$$20(12)=240$$

B:

$$10(6)+x(6) = 60+6x$$

$$240:(60+6x)=4:3$$

$$720=240+24x$$

$$480=24x$$

$$x=\boxed{20}$$

So B's capital in second half is:

$$\boxed{120000}$$

10. FIND AMOUNT ADDED

If B started with ₹ 10 and second-half capital is ₹ 20 k :

amount added:

$$20-10 = \boxed{10\text{ k}}$$

Be careful: requested “new capital” and “amount added” are different.

11. FIND UNKNOWN WITHDRAWAL

A invests ₹ 40 for 12 months.

B invests ₹ 50 for first 6 months, then reduces capital to ₹ (x) for next 6 months.

Profit ratio:

$$A:B=4:5$$

A weight:

$$480$$

B:

$$\{ 50(6)+6x=300+6x \}$$

$$\{ 480:(300+6x)=4:5 \}$$

$$\{ 2400=1200+24x \}$$

$$\{ 1200=24x \}$$

$$\{ x=50 \}$$

So B actually must keep $\frac{1}{5}$ of the profit throughout.

Amount withdrawn:

$$\{ \boxed{\frac{1}{5} \times 1200} \}$$

This is a useful sanity-check type problem.

12. PROFIT SHARE AFTER SALARY

If total profit:

$$\{ P \}$$

and A gets fixed salary:

$$\{ S \}$$

then remaining distributable profit:

$$\{ P-S \}$$

Partnership sharing is performed on:

$$\{ \boxed{P-S} \}$$

if the question says salary is paid before profit division.

13. EXAMPLE — SALARY + PROFIT SHARE

Total profit:

$$\{ \text{₹} 180,000 \}$$

A gets salary:

$$\{ \text{₹} 18,000 \}$$

Remaining:

$$\{ \text{₹} 172,000 \}$$

Partnership ratio:

$$\{ A:B=3:2 \}$$

A share:

$$\{ 72000 \times \frac{3}{5} = \text{₹} 43200 \}$$

A total:

$$\{ 43200+18000 = \boxed{\text{₹} 61,200} \}$$

B:

$$\boxed{\text{₹} 800}$$

14. FIXED COMMISSION ON TOTAL PROFIT

Suppose A receives 10% commission on total profit, then remainder is divided.

Total:

$$\text{₹} 160,000$$

Commission:

$$\text{₹} 6000$$

Remainder:

$$\text{₹} 54000$$

If A:B = 2:1:

A partnership share:

$$\text{₹} 36000$$

A total:

$$\text{₹} 42000$$

B:

$$\text{₹} 18000$$

$$\boxed{A = \text{₹} 42000, B = \text{₹} 18000}$$

15. COMMISSION ON PROFIT AFTER COMMISSION

This wording is trickier.

Suppose:

A gets 10% commission on profit after charging such commission.

Let commission = C.

Total profit = ₹ 55,000.

Then amount after commission is:

$$55000 - C$$

Given:

$$C = 10\%(55000 - C)$$

$$C = 0.1(55000 - C)$$

$$C = 5500 - 0.1C$$

$$1.1C = 5500$$

$$C = \boxed{\text{₹} 5000}$$

16. SHORTCUT — COMMISSION ON PROFIT AFTER COMMISSION

If commission is $r\%$ of profit after commission:

$$\boxed{C = \frac{r}{100+r} \times P}$$

where P is total profit before commission.

Example:

$$r=10, \quad P=55000$$

$$C = \frac{10}{110}(55000) = \boxed{5000}$$

17. COMMISSION ON PROFIT BEFORE COMMISSION

If commission is simply $r\%$ of total profit:

$$\boxed{C = \frac{r}{100}P}$$

Do not confuse the two bases.

18. SALARY + COMMISSION TO SAME PARTNER

Suppose total profit ₹ 1,00,000.

A gets:

- salary ₹ 10,000
- plus 5% commission on original total profit

Commission:

$$5000$$

Remaining for partnership:

$$100000 - 10000 - 5000 = 85000$$

If ratio A:B = 3:2:

A partnership share:

$$85000 \times \frac{3}{5} = 51000$$

A total:

$$51000 + 10000 + 5000 = \boxed{66000}$$

B:

$$\boxed{34000}$$

19. ORDER DEPENDS ON WORDING

Questions may say:

- salary first, then commission
- commission on remaining profit
- commission on total profit

- commission after salary

Always identify the base explicitly.

This is not a formula problem; it is a **reading problem**.

20. GOODWILL / ENTRY BONUS — BASIC APTITUDE IDEA

Some partnership problems mention that a new partner pays an additional amount as goodwill or entry fee.

Unless the problem says this amount is invested into business capital, do **not** automatically include it in $(C \times T)$.

Treat it separately according to wording.

For GATE this is low priority.

21. EXAMPLE — NEW PARTNER ENTRY BONUS

A and B already run a business.

C joins and pays ₹ 10,000, Goodwill directly to A and B in ratio 3:2.

Then:

A gets:

$$\left[10000 \times \frac{3}{5} = 6000 \right]$$

B gets:

$$\left[4000 \right]$$

This does not change capital-time weights unless explicitly added as business capital.

22. PROFIT DISTRIBUTION WITH LOSS

Partnership ratio also applies to loss, unless agreement says otherwise.

If loss = ₹ 30,000 and ratio A:B:C = 2:3:5:

A loss:

$$\left[30000 \times \frac{2}{10} = 6000 \right]$$

B:

$$\left[9000 \right]$$

C:

$$\left[15000 \right]$$

23. FIND TOTAL PROFIT FROM DIFFERENCE OF SHARES

A:B = 3:5.

B gets ₹ 4000 more than A.

Difference in ratio:

2 parts

So:

$\text{parts}=4000$

1 part:

2000

Total:

$8(2000) = \boxed{16000}$

24. FIND TOTAL PROFIT FROM ONE SHARE + SALARY

A:B = 2:3.

A gets salary and total receipt $16,000$.

So A's partnership share:

$16000-4000 = 12000$

This corresponds to 2 parts.

1 part:

6000

Remaining partnership profit:

$5(6000) = 30000$

Add salary:

$\text{total original profit} = 30000+4000 = \boxed{34000}$

25. FIND WEIGHT RATIO FROM FINAL RECEIPTS WITH SALARY

A and B have partnership ratio $(x:y)$.

A receives fixed salary 5000 .

Final receipts are:

$25,000, 30,000$

A's partnership portion:

$25000-5000 = 20000$

B partnership portion:

30000

So:

$x:y=20000:30000$

$=\boxed{2:3}$

26. FIND CAPITAL FROM PROFIT DIFFERENCE

A and B invest for same time.

Their profit difference is ₹ 4000.

Total profit is ₹ 20,000.

Then shares:

Let A>B.

$$A+B=20000$$

$$A-B=4000$$

So:

$$A=12000, B=8000$$

Ratio:

$$3:2$$

Since same time:

$$\boxed{\text{capital ratio}=3:2}$$

27. FIND TIME RATIO FROM PROFIT AND CAPITAL RATIOS

Given:

$$\text{Profit ratio}=6:5$$

Capital ratio:

$$3:2$$

Since:

$$C_A:C_B=6:5$$

$$3T_A:2T_B=6:5$$

$$15T_A=12T_B$$

$$T_A:T_B = 4:5$$

$$\boxed{4:5}$$

28. FIND CAPITAL RATIO FROM PROFIT & TIME

Profit ratio:

$$7:6$$

Time ratio:

$$14:9$$

Then:

$$C_A:C_B = \frac{7}{14}:\frac{6}{9}$$

$$\left[= \frac{12}{\frac{2}{3}} \right]$$

Multiply by 6:

$$\left[\boxed{3:4} \right]$$

29. THREE-PARTNER UNKNOWN

A, B, C investment weights satisfy:

$$\left[A:B=2:3 \right]$$

and:

$$\left[B:C=4:5 \right]$$

Combine:

$$\left[A:B=8:12 \right]$$

$$\left[B:C=12:15 \right]$$

So:

$$\left[\boxed{A:B:C=8:12:15} \right]$$

This is the connected-ratio skill from Chapter 13.

30. EXAMPLE WITH TOTAL PROFIT

Ratio:

$$\left[8:12:15 \right]$$

Total profit:

$$\left[170,000 \right]$$

Total parts:

$$\left[35 \right]$$

1 part:

$$\left[2000 \right]$$

Shares:

$$\left[A=16000 \right]$$

$$\left[B=24000 \right]$$

$$\left[C=30000 \right]$$

31. MONTHS VS YEARS

You may use months or years.

Because ratio is unaffected by common unit conversion.

Example:

120 for 1 year vs 130 for 8 months.

Convert 1 year to 12 months:

$$\sqrt[2]{20(12):30(8)}$$

Do not mix 1 and 8 directly.

32. DAYS CAN ALSO BE USED

If investments are for exact numbers of days:

$$\sqrt[\text{weight}]{C \times \text{days}}$$

No need to convert to months if all durations are in days.

33. EFFECTIVE CAPITAL — USEFUL VIEW

If a partner changes capital, you may define an average effective capital over total business duration:

$$\sqrt[\text{avg}]{C} = \frac{\sum C_i T_i}{T_{\text{total}}}$$

This is optional, not usually needed.

34. EXAMPLE

A:

- 20k for 6 months
- 40k for 6 months

Total weight:

$$\sqrt[360]$$

Average effective capital:

$$\sqrt[360/12]{30k}$$

So for profit-sharing purposes over the year, A behaves like a constant 130 investment.

35. GATE-STYLE QUESTION 1

A invests 120 for 12 months.

B invests 130 after 4 months.

C invests 140 after 6 months.

Weights:

$$\sqrt[240]{240, 240, 240}$$

$$\sqrt[1:1:1]$$

36. GATE-STYLE QUESTION 2

A invests ₹ 40,000 for first 6 months and ₹ 20,000 for next 6.

B invests ₹ 30,000 throughout.

A:

$$[40(6) + 20(6) = 360]$$

B:

$$[30(12) = 360]$$

$$[\boxed{1:1}]$$

37. GATE-STYLE QUESTION 3

Total profit ₹ 55,000.

A gets 10% commission on profit after commission.

Commission:

$$[\frac{10}{110}(55000) = \boxed{₹ 5000}]$$

38. GATE-STYLE QUESTION 4

A:B profit ratio = 5:6.

Capital ratio = 4:3.

Time ratio:

$$[4T_A:3T_B = 5:6]$$

$$[24T_A = 15T_B]$$

$$[T_A:T_B = \boxed{5:8}]$$

39. PLACEMENT-STYLE QUESTION 1

A invests ₹ 30,000 all year.

B joins after 2 months with ₹ 40,000.

C joins after 4 months with ₹ 50,000.

Weights:

$$[30(12) = 360]$$

$$[40(10) = 400]$$

$$[50(8) = 400]$$

Ratio:

$$[\boxed{9:10:10}]$$

40. PLACEMENT-STYLE QUESTION 2

A invests ₹ 50 k .

After 4 months withdraws ₹ 20 k .

B invests ₹ 40 k all year.

A:

$$[50(4)+30(8) = 200+240=440]$$

B:

$$[40(12)=480]$$

Ratio:

$$[440:480 = \boxed{11:12}]$$

41. PLACEMENT-STYLE QUESTION 3

A and B weights = 2:3.

Total profit = ₹ 70,000 .

A gets ₹ 10,000 salary first.

Remaining:

$$[60000]$$

A partnership share:

$$[24000]$$

A total:

$$[\boxed{134000}]$$

B:

$$[\boxed{136000}]$$

42. PLACEMENT-STYLE QUESTION 4

Total profit ₹ 1,10,000 .

A gets 10% commission on profit after commission.

$$[C=\frac{10}{110}(110000)]$$

$$[= \boxed{10000}]$$

Remaining:

$$[₹ 100000]$$

If A:B = 3:2:

A partnership share:

$$[60000]$$

A total:

$$\boxed{17000}$$

B:

$$\boxed{14000}$$

43. ADVANCED QUESTION — THREE CAPITAL CHANGES

A:

- 120 for 3 months
- 130 for 4 months
- 140 for 5 months

B:

- 125 for first 6 months
- 135 for next 6 months

A:

$$[20(3)+30(4)+40(5)]$$

$$[=60+120+200 = 380]$$

B:

$$[25(6)+35(6)]$$

$$[=150+210 = 360]$$

Ratio:

$$\boxed{19:18}$$

44. ADVANCED QUESTION — UNKNOWN JOINING TIME

A invests 140 for 12 months.

B invests 160 and gets equal profit.

$$[40(12)=60t]$$

$$[480=60t]$$

$$[t=8]$$

So B invested 8 months and joined after:

$$[12-8 = \boxed{4\text{ months}}]$$

45. ADVANCED QUESTION — UNKNOWN ADDITION

A invests 120 throughout.

B invests 110 for first 4 months and then adds 1x for remaining 8 months.

Profit ratio:

$$\backslash A:B=3:2 \backslash$$

A:

$$\backslash 20(12)=240 \backslash$$

B:

First 4:

$$\backslash 10(4)=40 \backslash$$

Later capital:

$$\backslash 10+x \backslash$$

for 8 months:

$$\backslash 8(10+x) \backslash$$

Total B:

$$\backslash 40+80+8x = 120+8x \backslash$$

Equation:

$$\backslash 240:(120+8x)=3:2 \backslash$$

$$\backslash 480=360+24x \backslash$$

$$\backslash 120=24x \backslash$$

$$\backslash x = \boxed{\text{\$ } 5000} \backslash$$

46. ADVANCED QUESTION — COMMISSION + SHARE

Total profit:

$$\backslash \text{\$ } 192,000 \backslash$$

A gets salary:

$$\backslash \text{\$ } 18,000 \backslash$$

and 10% commission on total profit.

Commission:

$$\backslash \text{\$ } 19,200 \backslash$$

Remaining:

$$\backslash 92000-8000-9200 = 74800 \backslash$$

$$A:B = 3:2.$$

A partnership share:

$$\backslash 74800 \times \frac{3}{5} = 44880 \backslash$$

A total:

$$\backslash 44880+8000+9200 = \boxed{\text{\$ } 62,080} \backslash$$

B:

$$\backslash \boxed{\text{\$ } 29,920} \backslash$$

Check:

$$\boxed{62080 + 29920 = 92000}$$

47. ADVANCED QUESTION — FIND TOTAL PROFIT

A:B = 2:3.

A gets salary ₹ 4000.

A's final receipt = ₹ 16,000.

A partnership part:

$$\boxed{12000}$$

If 2 parts = 12000:

$$\boxed{1 \text{ part} = 6000}$$

Partnership pool:

$$\boxed{5(6000) = 30000}$$

Original profit:

$$\boxed{30000 + 4000}$$

$$\boxed{= \boxed{34000}}$$

48. ADVANCED QUESTION — LOSS SHARING

A:B:C weight ratio = 2:3:5.

Firm suffers ₹ 50,000 loss.

Shares:

$$\boxed{A = 10000}$$

$$\boxed{B = 15000}$$

$$\boxed{C = 25000}$$

Same ratio logic applies.

49. COMMON TRAPS

& pTRAP 1 — Added capital

“Adds ₹ 10k means new capital = old capital + ₹ 10k.”

& pTRAP 2 — Withdraws ₹ 10k

Remaining capital = old capital - ₹ 10k.

& pTRAP 3 — Salary included in ratio

Usually salary is handled separately.

& pTRAP 4 — Commission base

“10% of total profit” and “10% of profit after commission” are different.

& pTRAP 5 — Joining after x months

Duration is total minus x.

& pTRAP 6 — Mixed units

Convert years/months/days consistently.

50. MUST-MEMORIZE FORMULAS

Profit-sharing weight

$$\boxed{W = \sum C_i T_i}$$

Profit ratio

$$\boxed{P_A : P_B : P_C = W_A : W_B : W_C}$$

Equal profit

$$\boxed{C_{AT_A} = C_{BT_B}}$$

Commission on profit after commission

$$\boxed{C = \frac{r}{100+r} \times P}$$

51. FAST EXAM METHOD

For any partnership question:

1. Draw a 12-month timeline if needed.
 2. Write each capital interval.
 3. Calculate $(C \times T)$.
 4. Simplify the ratio.
 5. Remove salary/commission from total profit if wording requires.
 6. Divide remaining profit.
 7. Add special payments back to the relevant partner.
-

PRACTICE — LEVEL 0

Q1

A invests ₹20,000 for 12 months, B ₹30,000 for 8 months. Find ratio.

Q2

A and B invest equal capital for 12 and 6 months. Find ratio.

Q3

A invests ₹ 40,000 for 6 months and ₹ 20,000 for next 6. Find annual weight.

Q4

Total profit ₹ 50,000. Ratio 2:3. Find shares.

Q5

A joins after 4 months in a 12-month business. How long does A invest?

LEVEL 1**Q6**

A invests ₹ 30,000 all year, B joins after 3 months with ₹ 40,000. Find ratio.

Q7

A invests ₹ 50,000 after 5 months withdraws ₹ 20,000. Find total annual weight.

Q8

A invests ₹ 20,000 first 4 months and ₹ 30,000 next 8. B invests ₹ 25,000 all year. Find ratio.

Q9

A:B weight ratio = 3:2. Total profit ₹ 40,000. A gets ₹ 40,000 salary first. Find final receipts.

Q10

Total profit ₹ 55,000. A gets 10% commission on profit after commission. Find commission.

LEVEL 2**Q11**

A invests ₹ 40,000 all year. B invests ₹ 60,000 and receives equal profit. How many months did B invest?

Q12

A invests ₹ 20,000 all year. B invests ₹ 10,000 first 6 months and ₹(x) next 6 months. Ratio A:B = 4:3. Find ₹(x).

Q13

A:B profit ratio = 6:5, capital ratio = 3:2. Find time ratio.

Q14

A, B, C invest ₹ 20,000, ₹ 30,000, ₹ 40,000 for 12, 8, 6 months. Find ratio.

Q15

Total profit ₹ 70,000. A's salary ₹ 10,000. Remaining ratio 2:3. Find A and B receipts.

LEVEL 3 — EXAM PRESSURE

Q16

A invests ₹ 30000 full year. B joins after 2 months with ₹ 40000. C joins after 4 months with ₹ 50000. Find ratio.

Q17

A invests ₹ 20000 throughout. B invests ₹ 10000 first 4 months, then adds ₹ 5000 for next 8 months. Find ratio.

Q18

Total profit ₹ 1,10,000. A gets 10% commission on profit after commission. Remaining profit is shared A:B = 3:2. Find A's total.

Q19

A:B profit ratio = 5:6. Capital ratio = 4:3. Find time ratio.

Q20

A gets salary ₹ 4000 and total receipt ₹ 16,000. Remaining profit is shared A:B = 2:3. Find original total profit.

ANSWERS**Q1**

$$\sqrt{20(12):30(8) = 240:240}$$

$$\sqrt{\boxed{1:1}}$$

Q2

$$\sqrt{12:6 = \boxed{2:1}}$$

Q3

$$\sqrt{40(6)+20(6) = 240+120 = \boxed{360}}$$

Q4

Total parts:

$$\sqrt{5}$$

$$\sqrt{A = ₹ 20000, B = ₹ 30000}$$

Q5

$$\sqrt{12-4 = \boxed{8\text{ months}}}$$

Q6

$$\sqrt{30(12):40(9)}$$

$$\sqrt{360:360}$$

$$\sqrt{=\boxed{1:1}}$$

Q7

$$\sqrt{50(5)+30(7)}$$

$$\sqrt{=250+210}$$

$$\sqrt{=\boxed{460}}$$

Q8

A:

$$\backslash[20(4)+30(8)=320 \backslash]$$

B:

$$\backslash[25(12)=300 \backslash]$$

$$\backslash[\boxed{16:15} \backslash]$$

Q9

Salary:

$$\backslash[4000 \backslash]$$

Remaining:

$$\backslash[36000 \backslash]$$

A partnership share:

$$\backslash[36000\text{times}\frac{3}{5}=21600 \backslash]$$

A total:

$$\backslash[\boxed{^125}600 \backslash]$$

B:

$$\backslash[\boxed{^114}400 \backslash]$$

Q10

$$\backslash[C=\frac{10}{110}(55000) \backslash]$$

$$\backslash[=\boxed{^15}000 \backslash]$$

Q11

$$\backslash[40(12)=60t \backslash]$$

$$\backslash[t=\boxed{8\text{months}} \backslash]$$

Q12

$$\backslash[240:(60+6x)=4:3 \backslash]$$

$$\backslash[720=240+24x \backslash]$$

$$\backslash[x=\boxed{^12}000 \backslash]$$

Q13

$$\backslash[3T_A:2T_B=6:5 \backslash]$$

$$\backslash[15T_A=12T_B \backslash]$$

$$\backslash[\boxed{T_A:T_B=4:5} \backslash]$$

Q14

$$\backslash[20(12):30(8):40(6) \backslash]$$

$$\backslash[240:240:240 \backslash]$$

$$\backslash[=\boxed{1:1:1} \backslash]$$

Q15

Remaining:

$$\boxed{60000}$$

A:

$$60000 \times \frac{25}{100} = 15000$$

Add salary:

$$\boxed{75000}$$

B:

$$\boxed{90000}$$

Q16

$$30(12):40(10):50(8)$$

$$360:400:400$$

$$= \boxed{9:10:10}$$

Q17

A:

$$20(12) = 240$$

B:

first 4 months:

$$10(4) = 40$$

next 8 months capital:

$$15$$

$$15(8) = 120$$

Total:

$$160$$

Ratio:

$$240:160 = \boxed{3:2}$$

Q18

Commission:

$$\frac{10}{110} \times 110000 = 10000$$

Remaining:

$$100000$$

A partnership share:

$$100000 \times \frac{3}{5} = 60000$$

A total:

$$\boxed{170000}$$

Q19

$$\backslash[4T_A:3T_B=5:6 \backslash]$$

$$\backslash[24T_A=15T_B \backslash]$$

$$\backslash[T_A:T_B = 15:24 \backslash]$$

$$\backslash[=\boxed{5:8} \backslash]$$

Q20

A's partnership share:

$$\backslash[16000-4000=12000 \backslash]$$

2 parts = 12000.

1 part:

$$\backslash[6000 \backslash]$$

Partnership pool:

$$\backslash[5(6000)=30000 \backslash]$$

Add salary:

$$\backslash[\boxed{134000} \backslash]$$

' CHAPTER 24 COMPLETE

We have now covered:

- basic partnership '
- capital-time rule '
- same-time/same-capital shortcuts '
- late joining '
- early withdrawal '
- changing capital midway '
- multiple capital changes '
- multiple partners joining/leaving '
- unknown capital/time questions '
- salary to working partner '
- commission on total profit '
- commission on profit after commission '
- salary + commission together '
- goodwill/entry-bonus basics '
- loss sharing '
- connected ratios '
- advanced GATE/placement practice '

Progress

Completed: 24/76 **Unit 1 — Quantitative Aptitude:** 0/14 PROGRESS

NEXT

+Chapter 25 — Problems on Ages

Next we will cover:

- present/past/future ages
- age ratios
- age differences
- average age
- father-son / mother-daughter patterns
- ratio after/before (n) years
- sum and difference methods
- finding present ages from multiple conditions
- replacement of a person in average-age questions
- common placement traps and shortcuts.

USER

next

ASSISTANT

CHAPTER 25 — PROBLEMS ON AGES

PART 1 — PRESENT, PAST & FUTURE AGES, AGE RATIOS, DIFFERENCE INVARIANCE & BASIC EQUATIONS

Status: $\emptyset$ = Not in PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Age problems are mostly **linear equations disguised as stories**.

The most important idea is:

Everyone's age increases by the same number of years as time passes.

Because of this:

$\boxed{\text{Age difference between two people never changes}}$

This single fact solves many questions quickly.

1. BASIC AGE REPRESENTATION

Suppose A's present age is:

x

Then:

After n years

$$\boxed{x+n}$$

n years ago

$$\boxed{x-n}$$

This is the foundation of all age questions.

2. EXAMPLE

Ravi is presently 20 years old.

After 5 years:

$$20+5 = \boxed{25}$$

Three years ago:

$$20-3 = \boxed{17}$$

3. AGE DIFFERENCE IS CONSTANT

Suppose:

$$A = 30 \text{ years}$$

$$B = 20 \text{ years}$$

Difference:

$$\boxed{10}$$

After 5 years:

$$\boxed{35-25=10}$$

Ten years ago:

$$\boxed{20-10=10}$$

Thus:

$$\boxed{\text{Age difference remains constant}}$$

4. AGE RATIO DOES NOT REMAIN CONSTANT

Suppose A:B now:

$$\boxed{30:20=3:2}$$

After 10 years:

$$\boxed{40:30=4:3}$$

So:

$$\boxed{\text{difference stays constant, ratio changes}}$$

This is a major trap.

5. PRESENT AGE RATIO

Suppose present ages are in ratio:

$$A:B=3:5$$

Then write:

$$A=3x$$

$$B=5x$$

This is much easier than assigning unrelated variables.

6. EXAMPLE — RATIO + SUM

Present age ratio:

$$A:B=3:5$$

Sum:

$$40$$

Then:

$$3x+5x=40$$

$$8x=40$$

$$x=5$$

Therefore:

$$A=15, \quad B=25$$

$$\boxed{15, \ 25}$$

7. RATIO + DIFFERENCE

$$A:B = 4:7.$$

Age difference = 12 years.

Difference in ratio parts:

$$7-4=3$$

So:

$$3x=12$$

$$x=4$$

Therefore:

$$A=16, \quad B=28$$

$$\boxed{16, \ 28}$$

8. FAST RATIO-DIFFERENCE SHORTCUT

If:

$$[A:B=m:n]$$

and difference is (D) ,

then one ratio part equals:

$$[\boxed{\frac{D}{|n-m|}}]$$

Then multiply by ratio numbers.

9. FUTURE AGE RATIO

Suppose present ages are (A,B) .

After (n) years:

$$[\boxed{\frac{A+n}{B+n}}]$$

Example:

Present ages:

$$[20, 30]$$

After 10 years:

$$[30:40 = \boxed{3:4}]$$

10. PAST AGE RATIO

(n) years ago:

$$[\boxed{(A-n):(B-n)}]$$

Example:

Present:

$$[25, 35]$$

5 years ago:

$$[20:30 = \boxed{2:3}]$$

11. FIND PRESENT AGES FROM FUTURE RATIO

A and B presently have ages in ratio:

$$[2:3]$$

After 5 years, ratio becomes:

$$[3:4]$$

Let present ages:

$$[2x, 3x]$$

Then:

$$\frac{2x+5}{3x+5} = \frac{3}{4}$$

Cross multiply:

$$4(2x+5) = 3(3x+5)$$

$$8x+20 = 9x+15$$

$$x = 5$$

Therefore:

$$\boxed{A=10, B=15}$$

12. FIND PRESENT AGES FROM PAST RATIO

Present ages ratio:

$$4:5$$

5 years ago ratio:

$$3:4$$

Let:

$$A = 4x, B = 5x$$

Then:

$$\frac{4x-5}{5x-5} = \frac{3}{4}$$

$$4(4x-5) = 3(5x-5)$$

$$16x-20 = 15x-15$$

$$x = 5$$

Thus:

$$\boxed{20, 25}$$

13. SAME DIFFERENCE SHORTCUT

If present ratio is known and future/past ratio is given, sometimes age difference gives a faster path.

Example above:

Present ratio:

$$4:5$$

Present difference = one part ($=x$).

5 years ago:

$$(4x-5):(5x-5) = 3:4$$

Still use equation, but remember difference remains ($=x$).

This helps sanity-check answers.

14. FATHER–SON BASIC PATTERN

Suppose father is 3 times son's age.

Let son's age:

$$\boxed{x}$$

Father's:

$$\boxed{3x}$$

If father is 30 years older:

$$\boxed{3x - x = 30}$$

$$\boxed{2x = 30}$$

$$\boxed{x = 15}$$

So:

$$\boxed{\text{Son} = 15, \text{Father} = 45}$$

15. “x TIMES AS OLD” LANGUAGE

“Father is 4 times as old as son” means:

$$\boxed{F = 4S}$$

“Father is 3 times older than son” is ambiguous in everyday English and should not normally be used in formal aptitude questions.

If encountered, prefer the explicit mathematical relation given by context.

16. “TWICE THE AGE” VS “TWICE OLDER”

Safe aptitude wording:

- “twice the age” !’ $(2x)$
- “three times the age” !’ $(3x)$

Avoid interpreting loose everyday phrasing without context.

17. AGE AFTER n YEARS

A father is currently 40 and son 10.

After (n) years:

$$\boxed{F = 40 + n}$$

$$\boxed{S = 10 + n}$$

If father then becomes twice son's age:

$$\boxed{40 + n = 2(10 + n)}$$

$$\boxed{40 + n = 20 + 2n}$$

$$\boxed{n = 20}$$

So after:

$$\boxed{20 \text{ years}}$$

18. FAST DIFFERENCE METHOD

Current age difference:

$$40 - 10 = 30$$

When father is twice son's age:

$$F:S=2:1$$

Difference is 1 ratio part.

So one part:

$$30$$

Thus son at that time:

$$30$$

Current son:

$$10$$

Years later:

$$30 - 10 = \boxed{20}$$

This is often much faster.

19. GENERAL DIFFERENCE-RATIO METHOD

Suppose age difference is (D) and at some time ratio is:

$$m:n$$

Then difference in ratio parts:

$$m-n$$

One part:

$$\boxed{\frac{D}{m-n}}$$

Ages at that time:

$$\boxed{\frac{mD}{m-n}, \quad \frac{nD}{m-n}}$$

assuming $(m > n)$.

20. EXAMPLE

Age difference = 24.

At some future time ratio = 5:3.

Difference parts:

$$\sqrt{2}$$

One part:

$$\sqrt{12}$$

Future ages:

$$\sqrt{60}, \sqrt{36}$$

If younger person is presently 30:

Years later:

$$\sqrt{36-30 = \boxed{6}}$$

21. SUM OF AGES

If present ages are (x, y) :

$$\sqrt{\boxed{\text{present sum} = x + y}}$$

After (n) years:

$$\sqrt{(x+n) + (y+n)}$$

$$\sqrt{= x + y + 2n}$$

So for two people:

$$\sqrt{\boxed{\text{sum increases by } 2n}}$$

22. GENERAL GROUP AGE-SUM RULE

For (k) people, after (n) years:

$$\sqrt{\boxed{\text{total age increases by } kn}}$$

This is a direct connection to averages.

23. EXAMPLE

A family has 4 members.

Current total age:

$$\sqrt{100}$$

After 5 years:

$$\sqrt{100 + 4(5)}$$

$$\sqrt{= \boxed{120}}$$

24. AVERAGE AGE OF SAME GROUP

If the same group remains unchanged, after (n) years every person is (n) years older.

Therefore:

$\boxed{\text{Average age also increases by } n}$

Example:

Current average = 24.

After 6 years:

$\boxed{30}$

No need to know group size.

25. MAJOR AVERAGE-AGE TRAP

If a new person joins or someone leaves, the simple “average + n” rule no longer directly applies.

Then use:

$\boxed{\text{Total age} = n \times \text{average age}}$

We will cover these advanced cases in Part 2.

26. PRESENT AGE FROM FUTURE STATEMENT

“After 6 years, A will be 30.”

Present:

$30 - 6 = \boxed{24}$

Simple but often hidden inside longer questions.

27. PRESENT AGE FROM PAST STATEMENT

“5 years ago, B was 18.”

Present:

$18 + 5 = \boxed{23}$

28. TWO EQUATIONS FROM TWO CONDITIONS

Example:

A is 6 years older than B.

After 4 years, their ages are in ratio 5:4.

Let:

$B = x$

Then:

$A = x + 6$

After 4 years:

$\frac{x+10}{x+4} = \frac{5}{4}$

Cross multiply:

$$4x + 40 = 5x + 20$$

$$x = 20$$

Therefore:

$$B = 20, \text{ } A = 26$$

$$\boxed{26, 20}$$

29. FAST DIFFERENCE METHOD FOR SAME QUESTION

Age difference:

$$6$$

Future ratio:

$$5:4$$

Difference = 1 part.

So future ages:

$$30, 24$$

Subtract 4 years:

$$\boxed{26, 20}$$

Much faster.

30. IMPORTANT SHORTCUT

If age difference and a past/future ratio are given:

Do not build equations immediately.

First try the constant-difference ratio method.

This can reduce a 6-line solution to 2 lines.

31. RATIO NOW AND RATIO LATER

Present ratio:

$$3:4$$

After 5 years:

$$4:5$$

Let present ages:

$$3x, 4x$$

Equation:

$$\left[\frac{3x+5}{4x+5} = \frac{45}{4} \right]$$

$$\left[15x+25=16x+20 \right]$$

$$\left[x=5 \right]$$

Present ages:

$$\left[\boxed{15, 20} \right]$$

32. RATIO AGO AND RATIO NOW

5 years ago:

$$\left[2:3 \right]$$

Present ratio:

$$\left[3:4 \right]$$

Let present:

$$\left[3x, 4x \right]$$

Five years ago:

$$\left[(3x-5):(4x-5)=2:3 \right]$$

$$\left[3(3x-5)=2(4x-5) \right]$$

$$\left[9x-15=8x-10 \right]$$

$$\left[x=5 \right]$$

Present:

$$\left[\boxed{15, 20} \right]$$

33. RATIO AGO AND RATIO LATER

5 years ago ratio = 1:2.

5 years from now ratio = 2:3.

Let present ages:

$$\left[x, y \right]$$

Then:

$$\left[\frac{x-5}{y-5} = \frac{1}{2} \right]$$

and:

$$\left[\frac{x+5}{y+5} = \frac{2}{3} \right]$$

First:

$$\left[2x-10=y-5 \right]$$

$$\left[y=2x-5 \right]$$

Second:

$$\left[3x+15=2y+10 \right]$$

$$\backslash[3x+5=2y \backslash]$$

Substitute:

$$\backslash[3x+5=4x-10 \backslash]$$

$$\backslash[x=15 \backslash]$$

$$\backslash[y=25 \backslash]$$

$$\backslash[\boxed{15, \ 25} \backslash]$$

34. AGE MULTIPLE IN THE PAST

A father is 40, son is 10.

How many years ago was father 5 times son's age?

Let $\backslash(x)$ years ago:

$$\backslash[40-x=5(10-x) \backslash]$$

$$\backslash[40-x=50-5x \backslash]$$

$$\backslash[4x=10 \backslash]$$

$$\backslash[x=2.5 \backslash]$$

So:

$$\backslash[\boxed{2.5\text{ years ago}} \backslash]$$

Not every age problem yields integer years.

35. FUTURE AGE MULTIPLE

A is 36, B is 12.

When will A be twice B?

$$\backslash[36+x=2(12+x) \backslash]$$

$$\backslash[36+x=24+2x \backslash]$$

$$\backslash[x=12 \backslash]$$

$$\backslash[\boxed{12\text{ years}} \backslash]$$

36. SANITY RULE FOR OLDER/YOUNGER RATIO

For two people with fixed positive age difference:

As time moves forward, their age ratio moves closer to:

$$\backslash[\boxed{1:1} \backslash]$$

Example:

$$\backslash[40:20=2:1 \backslash]$$

later:

$$\backslash[50:30=5:3 \]$$

later:

$$\backslash[60:40=3:2 \]$$

Ratio gets closer to 1.

37. PAST RATIO BEHAVIOR

Moving backward in time, the older-to-younger ratio generally moves farther from 1, until the younger age approaches zero.

This is a useful sanity check.

38. IMPOSSIBLE-RATIO CHECK

Suppose current ages:

$$\backslash[A>B \]$$

Future ratio cannot become **larger** in older:younger form.

Example current:

$$\backslash[3:2=1.5 \]$$

A future ratio:

$$\backslash[2:1=2 \]$$

is impossible.

Because ratios move toward 1 over time.

39. EXAMPLE — OPTION ELIMINATION

Father:son present ratio = 4:1.

Which could be their ratio 10 years later?

- 5:1
- 4:1
- 3:1
- 6:1

The future ratio must be smaller than 4:1 but still >1.

Only:

$$\backslash[\boxed{3:1} \]$$

is plausible among those options.

40. PERSON'S AGE AS FRACTION OF ANOTHER

A's age is:

$$\left[\frac{3}{4} \right]$$

of B.

Then:

$$\left[A:B=3:4 \right]$$

If difference = 10:

One part:

$$\left[10 \right]$$

So:

$$\left[\boxed{A=30, B=40} \right]$$

41. “A IS x% OLDER THAN B”

If A is 25% older than B:

Take:

$$\left[B=100 \right]$$

Then:

$$\left[A=125 \right]$$

Thus:

$$\left[\boxed{A:B=5:4} \right]$$

42. “A IS x% YOUNGER THAN B”

If A is 20% younger than B:

$$\left[A=80\% \text{ of } B \right]$$

Thus:

$$\left[A:B=80:100 \right]$$

$$\left[=\boxed{4:5} \right]$$

43. ASYMMETRY TRAP

If A is 25% older than B, B is not 25% younger than A.

If:

$$\left[A:B=5:4 \right]$$

then B is less than A by:

$$\left[\frac{1}{5} \times 100 = \boxed{20\%} \right]$$

Same percentage asymmetry from Chapter 15.

44. GATE-STYLE QUESTION 1

A:B present age ratio = 3:5.

Difference = 12 years.

Difference parts:

$\sqrt[2]{}$

One part:

$\sqrt[6]{}$

A:

$\sqrt[18]{}$

B:

$\sqrt[30]{}$

$\sqrt[\boxed{18, 30}]{}$

45. GATE-STYLE QUESTION 2

Present ages 20 and 30.

Find ratio after 10 years.

$\sqrt[30:40]{}$

$\sqrt[=\boxed{3:4}]{}$

46. GATE-STYLE QUESTION 3

A is 8 years older than B.

After 4 years ratio = 3:2.

Future difference still 8.

Ratio difference:

$\sqrt[1\text{ part}=8]{}$

Future ages:

$\sqrt[24, 16]{}$

Present:

$\sqrt[\boxed{20, 12}]{}$

47. GATE-STYLE QUESTION 4

Father is 45, son 15.

After how many years will father be twice son?

$\sqrt[45+x=2(15+x)]{}$

$$\boxed{x=15}$$

$$\boxed{15 \text{ years}}$$

48. PLACEMENT-STYLE QUESTION 1

A:B = 2:3.

After 10 years ratio = 3:4.

Let ages:

$$2x, 3x$$

$$\frac{2x+10}{3x+10} = \frac{3}{4}$$

$$8x+40=9x+30$$

$$x=10$$

So:

$$\boxed{20, 30}$$

49. PLACEMENT-STYLE QUESTION 2

Five years ago A:B = 1:2.

Present difference = 15.

Since difference always 15.

At past ratio 1:2, difference = 1 part.

So past ages:

$$15, 30$$

Present ages:

$$\boxed{20, 35}$$

50. PLACEMENT-STYLE QUESTION 3

A is 20% younger than B.

If B is 35:

$$A=35(0.8)$$

$$\boxed{28}$$

51. PLACEMENT-STYLE QUESTION 4

Current average age of 5 people is 24.

After 4 years, if same 5 remain:

$$\boxed{28}$$

No calculation of total required.

52. ADVANCED QUESTION — TWO RATIOS

Present ratio:

$$\backslash[5:7 \backslash]$$

After 8 years:

$$\backslash[7:9 \backslash]$$

Let:

$$\backslash[A=5x,\quad B=7x \backslash]$$

$$\backslash[\frac{5x+8}{7x+8} = \frac{7}{9} \backslash]$$

$$\backslash[45x+72=49x+56 \backslash]$$

$$\backslash[16=4x \backslash]$$

$$\backslash[x=4 \backslash]$$

Present ages:

$$\backslash[\boxed{20,\ 28} \backslash]$$

53. ADVANCED QUESTION — PAST MULTIPLE

Mother is 42 and daughter is 18.

How many years ago was mother twice daughter's age?

$$\backslash[42-x=2(18-x) \backslash]$$

$$\backslash[42-x=36-2x \backslash]$$

$$\backslash[x=-6 \backslash]$$

Negative “years ago” means the event is actually in the future.

After 6 years:

$$\backslash[48:24=2:1 \backslash]$$

So answer:

$$\backslash[\boxed{6\text{ years from now}} \backslash]$$

This is a powerful interpretation trap.

54. NEGATIVE TIME INTERPRETATION

If solving “ (x) years ago” gives:

$$\backslash[x<0 \backslash]$$

then the condition occurs:

$$\backslash[$$

x

\]

years in the future.

Likewise, if solving “after $\backslash(x)$ years” gives negative $\backslash(x)$, it occurred in the past.

55. COMMON TRAPS

& pTRAP 1 — Ratio remains constant

False. Difference remains constant.

& pTRAP 2 — Everyone gets +n

If $\backslash(n)$ years pass, add $\backslash(n)$ to every person, not multiply ages.

& pTRAP 3 — Joins average-age problem

If group membership changes, simple average shift rule may fail.

& pTRAP 4 — Future ratio

Older:younger ratio should move toward 1.

& pTRAP 5 — Percent older/younger

Bases differ.

& pTRAP 6 — Negative time

Interpret it as the opposite direction in time.

56. MUST-MEMORIZE RULES

Future

$\backslash[\boxed{x+n}]$

Past

$\backslash[\boxed{x-n}]$

Constant difference

$\backslash[\boxed{A-B=\text{constant}}]$

Same group average after $\backslash(n)$ years

$\backslash[\boxed{\bar{A}_{\text{new}}=\bar{A}_{\text{old}}+n}]$

Group total after $\backslash(n)$ years

$\backslash[\boxed{T_{\text{new}}=T_{\text{old}}+kn}]$

for k people.

57. FAST EXAM DECISION TREE

Ratio + sum?

Use ratio parts.

Ratio + difference?

Use difference in parts.

Future/past ratio + known difference?

Use constant-difference shortcut.

Ratio now + ratio later?

Use (mx, nx) equation.

Father/son multiple?

Try difference shortcut first.

Same group average after years?

Just add elapsed years.

Weird answer?

Check ratio direction and negative-time interpretation.

PRACTICE — LEVEL 0

Q1

A:B = 3:4 and their sum is 35. Find ages.

Q2

A:B = 5:7 and difference is 12. Find ages.

Q3

Present ages are 20 and 32. Find ages after 5 years.

Q4

Present ages are 25 and 40. Find ratio 5 years ago.

Q5

Average age of 6 people is 18. What will it be after 4 years if the group is unchanged?

LEVEL 1

Q6

A:B = 2:3. After 5 years ratio becomes 3:4. Find present ages.

Q7

A is 6 years older than B. After 3 years their ratio is 4:3. Find present ages.

Q8

Father is 40 and son 10. After how many years will father be twice son?

Q9

Five years ago A:B = 2:3. Present difference is 10. Find present ages.

Q10

A is 25% older than B. Find $\frac{A}{B}$.

LEVEL 2

Q11

Present ratio A:B = 4:5. Five years ago ratio was 3:4. Find present ages.

Q12

Present ratio A:B = 3:5. After 10 years ratio becomes 5:7. Find present ages.

Q13

Father is 45 and son 15. How many years ago was father 5 times son?

Q14

A is 20% younger than B. If their age difference is 8 years, find both ages.

Q15

Age difference is 18. At some future time ratio becomes 5:3. If younger is presently 21, find how many years later.

LEVEL 3 — EXAM PRESSURE

Q16

Five years ago A:B = 1:2. Five years from now ratio = 2:3. Find present ages.

Q17

Present ratio = 5:7. After 8 years ratio = 7:9. Find ages.

Q18

A is 30% older than B. If A is 39, find B.

Q19

A father is 42 and daughter 18. The statement says "how many years ago was father twice daughter's age?" Interpret the result correctly.

Q20

Current average age of 8 people is 25. What is their total age after 6 years if the group is unchanged?

ANSWERS

Q1

$$\backslash[3x+4x=35 \backslash]$$

$$\backslash[x=5 \backslash]$$

$$\backslash[\boxed{15, \ 20} \backslash]$$

Q2

Difference parts:

$$\backslash[2 \backslash]$$

$$\backslash[2x=12 \backslash]$$

$$\backslash[x=6 \backslash]$$

$$\backslash[\boxed{30, \ 42} \backslash]$$

Q3

$$\backslash[\boxed{25, \ 37} \backslash]$$

Q4

Five years ago:

$$\backslash[20, \ 35 \backslash]$$

Ratio:

$$\backslash[20:35 = \boxed{4:7} \backslash]$$

Q5

$$\backslash[18+4 = \boxed{22} \backslash]$$

Q6

$$\backslash[\frac{2x+5}{3x+5} = \frac{3}{4} \backslash]$$

$$\backslash[8x+20=9x+15 \backslash]$$

$$\backslash[x=5 \backslash]$$

$$\backslash[\boxed{10, \ 15} \backslash]$$

Q7

Future difference:

$$\backslash[6 \backslash]$$

Future ratio:

$$\backslash[4:3 \backslash]$$

1 part = 6.

Future ages:

$$\boxed{24, 18}$$

Present:

$$\boxed{21, 15}$$

Q8

$$40 + x = 2(10 + x)$$

$$\boxed{x = 20}$$

Q9

Past ratio 2:3.

Difference = 10 = 1 part.

Past ages:

$$20, 30$$

Present:

$$\boxed{25, 35}$$

Q10

$$125:100 = \boxed{5:4}$$

Q11

$$\frac{4x-5}{5x-5} = \frac{3}{4}$$

$$16x - 20 = 15x - 15$$

$$x = 5$$

$$\boxed{20, 25}$$

Q12

$$\frac{3x+10}{5x+10} = \frac{3}{7}$$

$$21x + 70 = 25x + 50$$

$$20 = 4x$$

$$x = 5$$

$$\boxed{15, 25}$$

Q13

Let x years ago:

$$45 - x = 5(15 - x)$$

$$45 - x = 75 - 5x$$

$$4x = 30$$

$$\boxed{7.5 \text{ years ago}}$$

Q14

A is 20% younger:

$$A:B = 4:5$$

Difference = 1 part = 8.

$$\boxed{A=32, B=40}$$

Q15

Future ratio:

$$5:3$$

Difference = 18 = 2 parts.

One part:

$$9$$

Future younger:

$$27$$

Present younger:

$$21$$

Years later:

$$\boxed{6}$$

Q16

Let present ages (x, y) .

Five years ago:

$$\frac{x-5}{y-5} = \frac{12}{17}$$

Five years later:

$$\frac{x+5}{y+5} = \frac{23}{28}$$

Solving:

$$\boxed{15, 25}$$

Q17

$$\frac{5x+8}{7x+8} = \frac{7}{9}$$

$$45x+72=49x+56$$

$$x=4$$

$$\boxed{20, 28}$$

Q18

A is 30% older than B:

$$A=1.3B$$

$$39=1.3B$$

$$\boxed{B=30}$$

Q19

$$42-x=2(18-x)$$

$$x=-6$$

Therefore the event did not occur in the past.

It occurs:

$\boxed{6 \text{ years from now}}$

Q20

Current total:

$8(25)=200$

After 6 years:

$200+8(6)$

$=\boxed{248}$

CHAPTER 25 STATUS

Covered so far:

- present/past/future ages '
- constant age difference '
- changing age ratios '
- ratio + sum/difference '
- future/past ratio equations '
- father–son / parent–child patterns '
- percentage older/younger '
- same-group average-age rule '
- ratio sanity checks '
- negative-time interpretation '
- GATE/placement practice '

But **Chapter 25 is not complete yet.**

Next we need:

- average-age entry/exit problems
- family age totals
- child birth and average-age changes
- replacement of one person
- “average age $\backslash(x)$ years ago” questions
- multiple-generation problems
- age-product/sum conditions
- tougher mixed equations
- final speed shortcuts and chapter audit

Progress

Completed: 24/76 **Current:** $\emptyset$ —Chapter 25 — Problems on Ages **Next:** Chapter 25 Part 2 — **Average-age changes, joining/leaving members & advanced family-age problems**

USER

next

ASSISTANT

CHAPTER 25 — PROBLEMS ON AGES

PART 2 — AVERAGE-AGE CHANGES, JOINING/LEAVING MEMBERS, FAMILY TOTALS & ADVANCED AGE PROBLEMS

Status: $\emptyset$ = No PROGRESS GATE: Medium Placements: High Priority: +Tier 2

This part combines **Ages + Averages**.

The most important equation is:

$$\boxed{\text{Total age} = \text{Number of people} \times \text{Average age}}$$

Whenever someone:

- joins,
- leaves,
- is replaced,
- or a child is born,

convert averages into **total ages first**.

1. SAME GROUP AFTER n YEARS

Suppose 5 people have average age:

$$\boxed{24}$$

After 4 years, if nobody joins or leaves:

$$\boxed{24 + 4 = 28}$$

This shortcut works because everyone becomes 4 years older.

2. TOTAL-AGE METHOD

Current average:

$$\boxed{24}$$

People:

$$\boxed{5}$$

Current total:

$$\boxed{5(24) = 120}$$

After 4 years:

$$\backslash[120+5(4) = 140 \backslash]$$

Average:

$$\backslash[140/5 = 28 \backslash]$$

Same result.

3. PERSON JOINS THE GROUP

Suppose 5 people have average age 20.

A sixth person joins and new average becomes 22.

Old total:

$$\backslash[5(20)=100 \backslash]$$

New total:

$$\backslash[6(22)=132 \backslash]$$

Age of joining person:

$$\backslash[132-100 = \boxed{32} \backslash]$$

4. JOINING-PERSON FORMULA

If:

- old group size = $\backslash(n\backslash)$
- old average = $\backslash(A\backslash)$
- new average after one person joins = $\backslash(B\backslash)$

then joining person's age:

$$\backslash[\boxed{x=(n+1)B-nA} \backslash]$$

This is exactly the average shortcut learned earlier.

5. FAST DEVIATION METHOD FOR JOINING

Old average = 20.

New average = 22.

There are now 6 people.

Average rises by:

$$\backslash[2 \backslash]$$

Total age increase relative to maintaining old average:

$$\backslash[6(2)=12 \backslash]$$

Therefore joining person's age exceeds old average by 12:

$$\backslash[20+12 = \boxed{32} \backslash]$$

This is fast once understood.

6. PERSON LEAVES THE GROUP

6 people have average age:

$$\sqrt[2]{25}$$

One leaves.

Remaining 5 have average:

$$\sqrt[2]{23}$$

Original total:

$$\sqrt[2]{6(25)=150}$$

Remaining total:

$$\sqrt[2]{5(23)=115}$$

Age of person who left:

$$\sqrt[2]{150-115 = \boxed{35}}$$

7. LEAVING-PERSON FORMULA

If $\sqrt[n]{n}$ people have average $\sqrt[n]{A}$.

After one leaves, $\sqrt[n]{(n-1)}$ people have average $\sqrt[n]{B}$.

Leaver's age:

$$\sqrt[2]{\boxed{x=nA-(n-1)B}}$$

8. REPLACEMENT OF ONE PERSON

Suppose 10 people have average age 30.

A 40-year-old leaves and another person joins.

New average becomes 29.

Original total:

$$\sqrt[2]{10(30)=300}$$

New total:

$$\sqrt[2]{10(29)=290}$$

Total decreased by:

$$\sqrt[2]{10}$$

Therefore incoming person is 10 years younger than outgoing:

$$\sqrt[2]{40-10 = \boxed{30}}$$

9. REPLACEMENT SHORTCUT

For a group of n people:

$$\boxed{\text{Incoming age} - \text{Outgoing age}} = n(A_{\text{new}} - A_{\text{old}})$$

If average decreases, RHS is negative.

10. EXAMPLE

20 students average 18 years.

One 22-year-old leaves.

Average becomes 17.8.

Change in average:

$$-0.2$$

Total age change:

$$20(-0.2) = -4$$

So:

$$\text{Incoming} - 22 = -4$$

$$\boxed{\text{Incoming} = 18}$$

11. MULTIPLE PEOPLE JOIN

5 people have average age 30.

Two people join, and new average becomes 28.

Old total:

$$150$$

New total:

$$7(28) = 196$$

Total age of two newcomers:

$$196 - 150 = \boxed{46}$$

Their average:

$$46/2 = \boxed{23}$$

12. MULTIPLE PEOPLE LEAVE

8 people average 25.

Three people leave.

Remaining 5 average 22.

Original total:

$$200$$

Remaining total:

$$\backslash[110 \backslash]$$

Total age of leavers:

$$\backslash[90 \backslash]$$

Average age of leavers:

$$\backslash[90/3 = \boxed{30} \backslash]$$

13. GROUP AVERAGE x YEARS AGO

Suppose present average age of 6 people is 30.

What was average 5 years ago?

Same members:

$$\backslash[30-5 = \boxed{25} \backslash]$$

14. FIND PRESENT AVERAGE FROM PAST AVERAGE

Average 7 years ago:

$$\backslash[18 \backslash]$$

Same group now:

$$\backslash[18+7 = \boxed{25} \backslash]$$

No need to know group size.

15. CHILD BIRTH — IMPORTANT EXCEPTION

A family has 4 members with average age 30.

After 2 years, a baby is born.

If asked about average immediately after the birth, group size changes.

Current total:

$$\backslash[4(30)=120 \backslash]$$

After 2 years, before birth:

$$\backslash[120+4(2)=128 \backslash]$$

Baby's age at birth:

$$\backslash[0 \backslash]$$

New family total:

$$\backslash[128 \backslash]$$

Members:

$$\backslash[5 \backslash]$$

Average:

$$\lfloor 128/5 = \boxed{25.6} \rfloor$$

16. NEWBORN DOES NOT HAVE AGE 2

A common trap:

If baby is born **after 2 years**, baby's age at that moment is:

$$\lfloor \boxed{0} \rfloor$$

Do not add 2 years to the baby.

Only existing members age during those 2 years.

17. CHILD BORN SOME YEARS AGO

A family currently consists of parents and a child.

If child was born 5 years ago:

Current child age:

$$\lfloor 5 \rfloor$$

At the moment of birth:

$$\lfloor 0 \rfloor$$

Be careful about which time point the question refers to.

18. MARRIAGE / NEW MEMBER JOINING FAMILY

Suppose average age of 4-member family is 24.

A 28-year-old member joins by marriage.

New average:

Old total:

$$\lfloor 96 \rfloor$$

New total:

$$\lfloor 124 \rfloor$$

Members:

$$\lfloor 5 \rfloor$$

Average:

$$\lfloor \boxed{24.8} \rfloor$$

19. PERSON LEAVES AND TIME PASSES

This pattern requires careful sequencing.

Suppose 6 people average 25 today.

One 35-year-old leaves today.

What will average age of remaining people be after 4 years?

Original total:

$$\backslash[150 \backslash]$$

Remaining total today:

$$\backslash[150-35=115 \backslash]$$

Remaining people:

$$\backslash[5 \backslash]$$

Current remaining average:

$$\backslash[23 \backslash]$$

After 4 years:

$$\backslash[23+4 = \boxed{27} \backslash]$$

20. TIME PASSES FIRST, THEN PERSON LEAVES

Order matters.

Suppose 6 people currently average 25.

After 4 years, one person who is then 39 leaves.

Before leaving:

Average:

$$\backslash[29 \backslash]$$

Total:

$$\backslash[6(29)=174 \backslash]$$

After 39-year-old leaves:

$$\backslash[174-39=135 \backslash]$$

Remaining 5 average:

$$\backslash[\boxed{27} \backslash]$$

Same numerical answer in this carefully matched example, but not always.

Always follow chronological order.

21. FAMILY TOTAL AGE

Father = 40

Mother = 36

Son = 10

Daughter = 6

Total:

$$\boxed{40+36+10+6 = 92}$$

Average:

$$\boxed{92/4 = 23}$$

22. TOTAL FAMILY AGE AFTER n YEARS

For 4 unchanged members after 5 years:

$$\boxed{92+4(5) = 112}$$

Average:

$$\boxed{112/4 = 28}$$

or simply:

$$\boxed{23+5=28}$$

23. MULTIPLE GENERATIONS WITH RATIO CONDITIONS

Suppose:

Father:Son:

$$\boxed{3:1}$$

Mother:Daughter:

$$\boxed{4:1}$$

This alone is not enough to determine all ages.

Age problems require enough independent equations.

Data sufficiency lesson:

Count unknowns and constraints.

24. SUM + RATIO WITH THREE PEOPLE

A:B:C:

$$\boxed{2:3:5}$$

Sum:

$$\boxed{50}$$

Total ratio parts:

$$\boxed{10}$$

One part:

$$\boxed{5}$$

Ages:

$$\boxed{10, 15, 25}$$

25. TWO RELATIONS + TOTAL

Father is 3 times son's age.

Mother is 5 years younger than father.

Total age of all three:

$$95$$

Let son:

$$x$$

Father:

$$3x$$

Mother:

$$3x-5$$

Then:

$$x+3x+3x-5=95$$

$$7x=100$$

$$x=\frac{100}{7}$$

Non-integer ages are mathematically possible, though aptitude questions usually prefer friendlier numbers.

26. SUM OF AGES SOME YEARS AGO

Current sum of 3 people:

$$90$$

What was sum 5 years ago?

Each was 5 years younger.

Total reduction:

$$3(5)=15$$

So:

$$90-15 = \boxed{75}$$

27. FIND NUMBER OF PEOPLE FROM TOTAL-AGE CHANGE

A group's total age increases by 40 over 5 years, with membership unchanged.

If there are n people:

$$5n=40$$

$$\boxed{n=8}$$

Very useful hidden-group question.

28. FIND ELAPSED TIME FROM TOTAL-AGE CHANGE

6 people have total age increase of 30.

$$6t=30$$

$$\boxed{t=5\text{ years}}$$

29. AVERAGE AGE CHANGES BY JOINING INFANT

Suppose 4-person family average is 30.

A newborn joins immediately.

Old total:

$$120$$

New total remains:

$$120$$

Members:

$$5$$

Average:

$$\boxed{24}$$

The average drops sharply because newborn age is 0.

30. NEWBORN AFTER ONE YEAR

Same family:

Initial total:

$$120$$

After one year, existing four total:

$$124$$

Baby is then born:

$$+0$$

Average immediately after birth:

$$124/5 = \boxed{24.8}$$

31. AVERAGE AGE OF HUSBAND AND WIFE

Husband and wife average:

$$30$$

Their total age:

$$\boxed{60}$$

After 4 years, their total:

$$\boxed{68}$$

If a child is then born, family average:

$$\boxed{68/3 = \frac{22}{3}}$$

32. CHILD BORN AND YEARS PASS AFTERWARD

Couple currently average 30.

A child is born now.

What is family average after 6 years?

Current couple total:

$$\boxed{60}$$

After 6 years:

Couple total:

$$\boxed{60+12=72}$$

Child age:

$$\boxed{6}$$

Family total:

$$\boxed{78}$$

Average:

$$\boxed{78/3 = 26}$$

33. FAST METHOD FOR SAME PROBLEM

At child birth:

Family total:

$$\boxed{60}$$

Members:

$$\boxed{3}$$

After 6 years, all 3 members age 6 years:

Average increases by 6.

Average immediately at birth:

$$\boxed{60/3=20}$$

After 6 years:

$$\boxed{26}$$

34. AVERAGE AGE WITH REPLACEMENT AFTER TIME

A team of 10 has average age 25.

After 2 years, one 32-year-old member leaves and a 22-year-old joins.

Before replacement after 2 years:

Average:

$$\boxed{27}$$

Total:

$$\boxed{270}$$

Replace:

$$\boxed{270 - 32 + 22 = 260}$$

New average:

$$\boxed{26}$$

35. REPLACEMENT SHORTCUT AFTER TIME

After the elapsed-time update, replacement changes total by:

$$\text{incoming} - \text{outgoing}$$

Divide by group size to get average change.

Here:

$$22 - 32 = -10$$

Group size = 10.

Average falls by:

$$1$$

So:

$$27 - 1 = \boxed{26}$$

36. FIND OUTGOING AGE

Average of 8 people is 20.

One leaves, and remaining average becomes 18.

Outgoing age:

$$8(20) - 7(18)$$

$$\boxed{160 - 126 = 34}$$

37. FIND INCOMING AGE IN REPLACEMENT

Average of 12 people = 24.

One person aged 30 is replaced.

New average = 25.

Total increase:

$$\sqrt[12(1)=12]{}$$

Incoming age:

$$\sqrt[30+12 = \boxed{42}]{}$$

38. TWO PEOPLE REPLACED

20 people average 30.

Two people with average age 40 leave.

Two new people join.

New average becomes 29.

Total old:

$$\sqrt[600]{}$$

Total new:

$$\sqrt[580]{}$$

Decrease:

$$\sqrt[20]{}$$

Outgoing pair total:

$$\sqrt[80]{}$$

Incoming pair total:

$$\sqrt[60]{}$$

Incoming average:

$$\sqrt[\boxed{30}]{}$$

39. AGE PRODUCT CONDITIONS

Some placement questions use product relations.

Example:

A is 4 years older than B and:

$$\sqrt[AB=96]{}$$

Let $B = \sqrt{x}$.

A:

$$\sqrt[x+4]{}$$

Then:

$$\backslash[x(x+4)=96 \backslash]$$

$$\backslash[x^2+4x-96=0 \backslash]$$

$$\backslash[(x+12)(x-8)=0 \backslash]$$

Positive age:

$$\backslash[x=8 \backslash]$$

A:

$$\backslash[12 \backslash]$$

$$\backslash[\boxed{8, 12} \backslash]$$

40. PRODUCT PROBLEMS ARE QUADRATIC AGE PROBLEMS

If age product is given, expect:

$$\backslash[x(x+d)=K \backslash]$$

or similar.

This connects to Chapter 36 later.

For aptitude, usually the factors are simple.

41. “TWICE AS OLD AS x YEARS AGO”

Present ages:

$$A = 30$$

$$B = 20$$

When was A twice B?

Let t years ago:

$$\backslash[30-t=2(20-t) \backslash]$$

$$\backslash[30-t=40-2t \backslash]$$

$$\backslash[t=10 \backslash]$$

So:

$$\backslash[\boxed{10\text{ years ago}} \backslash]$$

Check:

$$\backslash[20:10=2:1 \backslash]$$

42. “AFTER x YEARS, SUM WILL BE...”

Present sum of two people = 50.

After 8 years:

$$\backslash[50+2(8) = \boxed{66} \backslash]$$

If the future sum is given as 66, you can reverse immediately.

43. FIND PRESENT SUM FROM FUTURE SUM

Two people's total age after 7 years will be 80.

Current total:

$$\backslash[80-2(7) \backslash]$$

$$\backslash[=\boxed{66} \backslash]$$

44. FIND INDIVIDUAL AGES FROM SUM + DIFFERENCE

Present sum:

$$\backslash[S \backslash]$$

Difference:

$$\backslash[D \backslash]$$

Older age:

$$\backslash[\boxed{\frac{S+D}{2}} \backslash]$$

Younger:

$$\backslash[\boxed{\frac{S-D}{2}} \backslash]$$

Same identity used in ratio/algebra chapters.

45. EXAMPLE

Sum = 50.

Difference = 14.

Older:

$$\backslash[\frac{50+14}{2}=32 \backslash]$$

Younger:

$$\backslash[\frac{50-14}{2}=18 \backslash]$$

$$\backslash[\boxed{32, 18} \backslash]$$

46. AGE RATIO + TOTAL AFTER n YEARS

Present ratio:

$$\backslash[2:3 \backslash]$$

Total after 5 years:

$$\backslash[60 \backslash]$$

Let current ages:

$$\sqrt[2]{2x}, \sqrt[3]{3x}$$

After 5 years total:

$$\sqrt[5]{5x+10=60}$$

$$\sqrt[5]{5x=50}$$

$$\sqrt{x=10}$$

Current ages:

$$\sqrt{\boxed{20, 30}}$$

47. AGE RATIO + TOTAL n YEARS AGO

Present ratio:

$$\sqrt[3]{3:4}$$

Total 5 years ago:

$$\sqrt[5]{25}$$

Current:

$$\sqrt[3]{3x+4x=7x}$$

Past total:

$$\sqrt[7]{7x-10=25}$$

$$\sqrt[7]{7x=35}$$

$$\sqrt{x=5}$$

Present:

$$\sqrt{\boxed{15, 20}}$$

48. GATE-STYLE QUESTION 1

Average age of 5 people = 20.

A sixth person joins, average becomes 22.

Joining person's age:

$$\sqrt[6]{6(22)-5(20)}$$

$$\sqrt[6]{132-100 = \boxed{32}}$$

49. GATE-STYLE QUESTION 2

Average of 8 people = 25.

One leaves.

Remaining 7 average 23.

Leaver:

$$\lfloor 8(25)-7(23) \rfloor$$

$$\lfloor 200-161 \rfloor$$

$$\lfloor =\boxed{39} \rfloor$$

50. GATE-STYLE QUESTION 3

Average of 10 people changes from 30 to 31 after replacing one person.

Outgoing person = 25.

Total age increases:

$$\lfloor 10 \rfloor$$

Incoming:

$$\lfloor 25+10 = \boxed{35} \rfloor$$

51. GATE-STYLE QUESTION 4

Present average of 6 people = 24.

What was average 8 years ago?

$$\lfloor 24-8 = \boxed{16} \rfloor$$

52. PLACEMENT-STYLE QUESTION 1

Average age of 10 students is 18.

Teacher joins and average becomes 20.

Teacher's age:

$$\lfloor 11(20)-10(18) \rfloor$$

$$\lfloor 220-180 = \boxed{40} \rfloor$$

53. PLACEMENT-STYLE QUESTION 2

Average of 12 players is 25.

A 35-year-old player leaves.

New average of remaining 11:

$$\lfloor \frac{12(25)-35}{11} \rfloor$$

$$\lfloor = \frac{265}{11} \rfloor$$

$$\lfloor \approx \boxed{24.09} \rfloor$$

54. PLACEMENT-STYLE QUESTION 3

Average of husband and wife is 28.

A child is born now.

Family average immediately:

Couple total:

$$\boxed{56}$$

Family members:

$$\boxed{3}$$

Average:

$$\boxed{18\frac{1}{3}}$$

55. PLACEMENT-STYLE QUESTION 4

A group total age rises from 300 to 345 in 5 years with unchanged membership.

Increase:

$$\boxed{45}$$

Number of people:

$$\boxed{45/5 = 9}$$

56. ADVANCED QUESTION — JOIN AFTER TIME PASSES

5 people average 24 today.

After 4 years, a 30-year-old joins.

Before joining:

Average:

$$\boxed{28}$$

Total:

$$\boxed{5(28)=140}$$

New total:

$$\boxed{170}$$

Members:

$$\boxed{6}$$

New average:

$$\boxed{28\frac{1}{3}}$$

57. ADVANCED QUESTION — ONE LEAVES AFTER TIME

6 people average 20 now.

After 5 years, one person aged 35 at that time leaves.

Before leaving:

Average:

$$\sqrt[2]{25}$$

Total:

$$\sqrt[2]{150}$$

Remaining:

$$\sqrt[2]{115}$$

Average:

$$\sqrt[2]{115/5 = \boxed{23}}$$

58. ADVANCED QUESTION — NEWBORN AFTER SOME YEARS

Present average age of 4-member family:

$$\sqrt[2]{25}$$

After 3 years, a baby is born.

At that moment:

Old members total:

$$\sqrt[2]{4(25)+4(3) = 100+12 = 112}$$

New family members:

$$\sqrt[2]{5}$$

Average:

$$\sqrt[2]{112/5 = \boxed{22.4}}$$

59. ADVANCED QUESTION — CHILD BORN NOW, FUTURE AVERAGE

Husband and wife average:

$$\sqrt[2]{32}$$

Child born now.

What is family average after 8 years?

Couple total now:

$$\sqrt[2]{64}$$

Family total immediately after birth:

$$\sqrt[2]{64}$$

Immediate family average:

$$\sqrt[2]{64/3}$$

After 8 years, all 3 are 8 years older:

$$\left[\frac{64}{3} + 8 = \frac{88}{3} \right]$$

$$\left[=\boxed{29\frac{1}{3}} \right]$$

60. ADVANCED QUESTION — TWO PEOPLE JOIN

Average of 8 people = 24.

Two people join, new average = 26.

Old total:

$$\left[192 \right]$$

New total:

$$\left[260 \right]$$

Newcomers total:

$$\left[68 \right]$$

Average:

$$\left[\boxed{34} \right]$$

61. ADVANCED QUESTION — TWO LEAVE

Average of 10 people = 30.

Two people leave.

Remaining average = 28.

Original total:

$$\left[300 \right]$$

Remaining:

$$\left[8(28)=224 \right]$$

Leavers total:

$$\left[76 \right]$$

Average:

$$\left[\boxed{38} \right]$$

62. COMMON TRAPS

& pTRAP 1 — Use average directly after joining/leaving

Convert to totals first.

& pTRAP 2 — Newborn's age

A newborn is 0 at birth.

& pTRAP 3 — Same-group shortcut after membership changes

$$\lfloor \text{Average} \rfloor + n$$

works only while membership remains unchanged.

& pTRAP 4 — Replacement changes group size

Replacement keeps group size same; joining/leaving does not.

& pTRAP 5 — Time order

“After 5 years someone joins” means first age existing members by 5 years, then add newcomer.

& pTRAP 6 — Age product

Product condition may create a quadratic; reject negative age roots.

63. MUST-MEMORIZE FORMULAS

Total age

$$T = nA$$

One joins

$$x = (n+1)A_{\text{new}} - nA_{\text{old}}$$

One leaves

$$x = nA_{\text{old}} - (n-1)A_{\text{new}}$$

Replacement

$$\text{Incoming} - \text{Outgoing} = n(A_{\text{new}} - A_{\text{old}})$$

64. GROUP-AGE FORMULAS

Same n people after t years:

$$T_{\text{new}} = T_{\text{old}} + nt$$

and:

$$A_{\text{new}} = A_{\text{old}} + t$$

65. FAST EXAM DECISION TREE

Same people, time passes?

Add/subtract time from average.

Someone joins?

Old total !' new total !' difference.

Someone leaves?

Old total " new total.

Replacement?

Use total average change $\times$ group size.

Newborn?

Use age 0 at birth.

Family membership changes after some years?

First age existing members, then modify membership.

Product relation?

Form quadratic.

66. 15-MINUTE CHAPTER REVISION

The entire Ages chapter can be reduced to five principles:

1. $\boxed{\text{Age difference never changes}}$

2. $\boxed{\text{Age ratios change with time}}$

3. $\boxed{\text{Future} = x + t, \quad \text{Past} = x - t}$

4. $\boxed{\text{Group total} = n \times \text{average}}$

5. Membership changes require **total-age accounting**.

PRACTICE — LEVEL 0

Q1

Average age of 5 people is 20. A sixth joins and average becomes 22. Find newcomer.

Q2

Average of 6 people is 25. One leaves and remaining average is 23. Find leaver.

Q3

Average of 10 people rises from 30 to 31 after replacing one 25-year-old. Find newcomer.

Q4

Average of 8 people is 18. Find average after 5 years.

Q5

A 4-member family average is 30. A newborn joins immediately. Find new average.

LEVEL 1

Q6

Average of 8 people = 25. Two join and new average = 27. Find newcomers' average.

Q7

Average of 10 people = 30. Two leave and remaining average = 28. Find leavers' average.

Q8

Average of 12 players = 24. A 36-year-old is replaced and new average = 23.5. Find incoming player.

Q9

A 5-member family averages 24. After 4 years, a 30-year-old joins. Find new average.

Q10

A group's total age rises by 56 in 7 years. Membership unchanged. Find group size.

LEVEL 2

Q11

A 4-member family has average 25. After 3 years a baby is born. Find average immediately after birth.

Q12

Husband and wife average 30. A child is born now. Find family average after 5 years.

Q13

Average of 6 people is 20. After 5 years, a 35-year-old leaves. Find remaining average.

Q14

Average of 20 students is 18. A 22-year-old leaves and new average is 17.8. Find incoming student's age if the person is replaced.

Q15

Present total age of 3 people is 90. Find their total 8 years ago.

LEVEL 3 — EXAM PRESSURE

Q16

Average of 10 people is 25. After 3 years, two people aged 30 and 40 at that time leave. Find average of remaining 8.

Q17

Average age of 8 people is 24. Two people join and average becomes 26. If one newcomer is 30, find the other.

Q18

A group has average 20. Five years later its average is still 20 because one older member left and a newborn joined. If

group originally had 5 members, find the age of the member who left at that later time.

Q19

A is 4 years older than B and product of their present ages is 96. Find ages.

Q20

Two people's present total is 66. Seven years from now, their age ratio will be 4:3. Find present ages.

ANSWERS

Q1

$$\backslash [6(22)-5(20) = 132-100 = \boxed{32} \backslash]$$

Q2

$$\backslash [6(25)-5(23) = 150-115 = \boxed{35} \backslash]$$

Q3

Average rises by 1 for 10 people:

$$\backslash [\text{total rise}=10 \backslash]$$

Incoming:

$$\backslash [25+10 = \boxed{35} \backslash]$$

Q4

$$\backslash [18+5 = \boxed{23} \backslash]$$

Q5

Old total:

$$\backslash [4(30)=120 \backslash]$$

New members:

$$\backslash [5 \backslash]$$

$$\backslash [120/5 = \boxed{24} \backslash]$$

Q6

Old total:

$$\backslash [8(25)=200 \backslash]$$

New total:

$$\backslash [10(27)=270 \backslash]$$

Newcomers:

$$\backslash [70 \backslash]$$

Average:

$$\backslash [\boxed{35} \backslash]$$

Q7

Original:

$$\backslash[300 \backslash]$$

Remaining:

$$\backslash[8(28)=224 \backslash]$$

Leavers:

$$\backslash[76 \backslash]$$

Average:

$$\backslash[\boxed{38} \backslash]$$

Q8

Average drop:

$$\backslash[-0.5 \backslash]$$

Total change:

$$\backslash[12(-0.5)=-6 \backslash]$$

Incoming:

$$\backslash[36-6 = \boxed{30} \backslash]$$

Q9

After 4 years, existing average:

$$\backslash[28 \backslash]$$

Existing total:

$$\backslash[140 \backslash]$$

Add 30:

$$\backslash[170 \backslash]$$

New average:

$$\backslash[170/6 = \boxed{28\frac{1}{3}} \backslash]$$

Q10

$$\backslash[7n=56 \backslash]$$

$$\backslash[\boxed{n=8} \backslash]$$

Q11

Present total:

$$\backslash[4(25)=100 \backslash]$$

After 3 years:

$$\backslash[112 \backslash]$$

Baby age = 0.

Average:

$$\lfloor 112/5 = \boxed{22.4} \rfloor$$

Q12

Couple total now:

$$\lfloor 60 \rfloor$$

Child born now.

After 5 years:

- couple total = $\lfloor 70 \rfloor$
- child = 5

Family total:

$$\lfloor 75 \rfloor$$

Average:

$$\lfloor \boxed{25} \rfloor$$

Q13

After 5 years, before leaving:

Average:

$$\lfloor 25 \rfloor$$

Total:

$$\lfloor 150 \rfloor$$

Subtract 35:

$$\lfloor 115 \rfloor$$

Remaining 5:

$$\lfloor \boxed{23} \rfloor$$

Q14

Group size remains 20.

Average change:

$$\lfloor 17.8-18=-0.2 \rfloor$$

Total change:

$$\lfloor 20(-0.2)=-4 \rfloor$$

Incoming:

$$\lfloor 22-4 = \boxed{18} \rfloor$$

Q15

Three people, 8 years earlier:

$$\lfloor 90-3(8) \rfloor$$

$$\lfloor =90-24 \rfloor$$

$$\lfloor =\boxed{66} \rfloor$$

Q16

Initial total:

$$\backslash[10(25)=250 \backslash]$$

After 3 years:

$$\backslash[250+30=280 \backslash]$$

Two leave:

$$\backslash[30+40=70 \backslash]$$

Remaining:

$$\backslash[210 \backslash]$$

Average of 8:

$$\backslash[\boxed{26.25} \backslash]$$

Q17

Old total:

$$\backslash[8(24)=192 \backslash]$$

New total:

$$\backslash[10(26)=260 \backslash]$$

Newcomers total:

$$\backslash[68 \backslash]$$

One is 30.

Other:

$$\backslash[68-30 = \boxed{38} \backslash]$$

Q18

Originally 5 people average 20:

$$\backslash[T_0=100 \backslash]$$

After 5 years, before membership change:

$$\backslash[T=100+5(5)=125 \backslash]$$

The group still has 5 members after one leaves and a newborn joins.

Final average remains 20:

$$\backslash[T_{\text{final}}=100 \backslash]$$

Newborn contributes 0.

Therefore leaving member's age at that time:

$$\backslash[125-100 = \boxed{25} \backslash]$$

Q19

Let $B = \backslash(x\backslash)$.

A:

$$\sqrt{x+4}$$

$$\sqrt{x(x+4)=96}$$

$$\sqrt{x^2+4x-96=0}$$

$$\sqrt{(x+12)(x-8)=0}$$

Positive solution:

$$\sqrt{x=8}$$

So:

$$\sqrt{\boxed{B=8, A=12}}$$

Q20

Present total:

$$\sqrt{66}$$

After 7 years, total:

$$\sqrt{66+14 = 80}$$

Future ratio:

$$\sqrt{4:3}$$

Total parts:

$$\sqrt{7}$$

Future ages:

$$\sqrt{\frac{320}{7}, \text{quad} \frac{240}{7}}$$

Present ages:

$$\sqrt{\frac{320}{7}-7 = \frac{271}{7} = 38\frac{57}{7}}$$

and:

$$\sqrt{\frac{240}{7}-7 = \frac{191}{7} = 27\frac{27}{7}}$$

Thus:

$$\sqrt{\boxed{38\frac{57}{7}, 27\frac{27}{7}}}$$

The data are mathematically valid but produce fractional ages.

' CHAPTER 25 COMPLETE

We have now covered:

- present, past and future ages '
- age ratios '
- constant age difference '
- ratio + sum/difference '
- father/son and family relations '
- percentage older/younger '
- future/past multiple questions '

- average age of unchanged groups '
- joining members '
- leaving members '
- replacement-age problems '
- multiple joining/leaving '
- newborn/birth problems '
- family total-age problems '
- time-order traps '
- quadratic/product age questions '
- advanced GATE/placement drills '

Progress

Completed: 25/76 Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

Chapter 26 — Time and Work

Next we will cover:

- work, time and efficiency
- $(W=RT)$ idea
- one-day work
- LCM work method
- two or more workers together
- finding individual times
- efficiency ratios
- alternate-day work
- joining/leaving midway
- partial completion
- men–days–hours relationships
- negative work / leaks preview
- advanced GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 26 — TIME AND WORK

PART 1 — WORK, TIME, EFFICIENCY, ONE-DAY WORK, LCM METHOD & WORKING TOGETHER

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: Very High Priority: $\emptyset$ = Not for 1

Time and Work is one of the most important quantitative-aptitude chapters.

The entire chapter is based on:

$$\boxed{\text{Work} = \text{Rate} \times \text{Time}}$$

or:

$$\boxed{W = RT}$$

Here, **rate** means work done per unit time.

1. BASIC IDEA

Suppose A completes one job in:

$$\boxed{10 \text{ days}}$$

Then in 1 day A completes:

$$\boxed{\frac{1}{10}}$$

of the job.

This is called A's **one-day work**.

2. MASTER RULE

If a person finishes a job in (T) days:

$$\boxed{\text{1-day work} = \frac{1}{T}}$$

and:

$$\boxed{\text{work rate} = \frac{1}{T}}$$

assuming total work = 1 job.

3. EXAMPLE

A completes a job in 8 days.

A's one-day work:

$$\boxed{\frac{1}{8}}$$

In 3 days A completes:

$$\boxed{3 \times \frac{1}{8} = \frac{3}{8}}$$

Remaining:

$$\boxed{1 - \frac{3}{8} = \frac{5}{8}}$$

4. FIND TIME FROM RATE

If a worker completes:

$$\left[\frac{1}{5} \right]$$

of a job per day,

time for full work:

$$\left[\boxed{5 \text{ days}} \right]$$

So rate and time are reciprocals when total work = 1.

5. TWO PEOPLE WORKING TOGETHER

A completes work in 10 days.

B completes it in 15 days.

A's rate:

$$\left[\frac{1}{10} \right]$$

B's rate:

$$\left[\frac{1}{15} \right]$$

Together:

$$\left[\frac{1}{10} + \frac{1}{15} \right]$$

LCM = 30:

$$\left[\frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6} \right]$$

So combined rate:

$$\left[\frac{1}{6} \right]$$

Time:

$$\left[\boxed{6 \text{ days}} \right]$$

6. MASTER TOGETHER FORMULA

If A takes (a) days and B takes (b) days:

$$\left[\frac{1}{T} = \frac{1}{a} + \frac{1}{b} \right]$$

So:

$$\left[\boxed{T = \frac{ab}{a+b}} \right]$$

for two positive workers.

MUST MEMORIZE

7. EXAMPLE

A = 12 days.

B = 18 days.

$$\left[T = \frac{12(18)}{12+18} \right]$$

$$\left[= \frac{216}{30} \right]$$

$$\left[= \boxed{7.2 \text{ days}} \right]$$

or:

$$\left[\boxed{7 \frac{1}{5} \text{ days}} \right]$$

8. LCM WORK METHOD

Fractions are mathematically clean, but for aptitude the **LCM method** is often faster.

Suppose:

- A finishes in 10 days
- B finishes in 15 days

Take total work as:

$$\left[\text{LCM}(10,15)=30 \text{ units} \right]$$

Then:

A's daily work:

$$\left[30/10=3 \right]$$

B's:

$$\left[30/15=2 \right]$$

Together:

$$\left[5 \text{ units/day} \right]$$

Time:

$$\left[30/5 = \boxed{6 \text{ days}} \right]$$

9. WHY LCM METHOD WORKS

Total work can be chosen arbitrarily.

Instead of saying:

$$\left[1 \text{ job} \right]$$

we may assume:

$$\left[30 \text{ units} \right]$$

as long as all rates are scaled consistently.

This avoids fractions.

10. NORMAL METHOD VS FASTER METHOD

Normal method

$$\left[\frac{1}{10} + \frac{1}{15} = \frac{1}{6} \right]$$

$$\sqrt{T=6}$$

Faster GATE/Placement Method

Take work:

$$\sqrt{30}$$

Rates:

$$\sqrt{3+2=5}$$

$$\sqrt{30/5=6}$$

For most multi-worker problems, prefer LCM.

11. THREE PEOPLE TOGETHER

A = 12 days

B = 18 days

C = 36 days

Take total work:

$$\sqrt{\text{LCM}(12,18,36)=36}$$

Rates:

$$\sqrt{A=3}$$

$$\sqrt{B=2}$$

$$\sqrt{C=1}$$

Together:

$$\sqrt{6}$$

Time:

$$\sqrt{36/6 = \boxed{6\text{ days}}}$$

12. GENERAL MULTI-WORKER RULE

If rates are:

$$\sqrt{r_1, r_2, \dots, r_n}$$

then:

$$\sqrt{\boxed{r_{\text{total}}} = r_1 + r_2 + \dots + r_n}$$

and:

$$\sqrt{\boxed{T = \frac{\text{Total Work}}{r_{\text{total}}}}}$$

13. EFFICIENCY

Efficiency means:

$\boxed{\text{work done per unit time}}$

So efficiency is the same idea as rate.

If A is more efficient, A completes more work per day.

14. TIME AND EFFICIENCY ARE INVERSELY PROPORTIONAL

For the same work:

$$W = ET$$

Thus:

$$E \propto \frac{1}{T}$$

Therefore:

$$\boxed{E_A : E_B = T_B : T_A}$$

Extremely important.

15. EXAMPLE — EFFICIENCY RATIO

A takes 10 days.

B takes 15 days.

Efficiency ratio:

$$E_A : E_B = 15 : 10$$

$$\boxed{3 : 2}$$

A is more efficient.

16. RATIO !' TIME

If:

$$E_A : E_B = 3 : 2$$

and A takes 10 days,

then:

$$T_A : T_B = 2 : 3$$

$$10 : T_B = 2 : 3$$

$$T_B = 15$$

$$\boxed{15 \text{ days}}$$

17. “A IS TWICE AS EFFICIENT AS B”

Then:

$$E_A : E_B = 2 : 1$$

Therefore:

$$\boxed{T_A:T_B=1:2}$$

So if A takes 8 days:

B takes:

$$\boxed{16 \text{ days}}$$

18. PERCENT MORE EFFICIENT

If A is 25% more efficient than B:

$$\boxed{E_A:E_B=125:100}$$

$$\boxed{=5:4}$$

Therefore time ratio:

$$\boxed{T_A:T_B=4:5}$$

19. EXAMPLE

B takes 20 days.

A is 25% more efficient.

Time ratio:

$$\boxed{A:B=4:5}$$

So:

$$\boxed{T_A=20 \times \frac{4}{5}}$$

$$\boxed{=16 \text{ days}}$$

20. PERCENT LESS TIME VS MORE EFFICIENCY

If A takes 20% less time than B:

Let B time = 100.

A time = 80.

Efficiency ratio:

$$\boxed{E_A:E_B = 100:80 = 5:4}$$

Thus A is:

$$\boxed{25\%}$$

more efficient than B.

Important asymmetry:

$$\boxed{20\% \text{ less time} \rightarrow 25\% \text{ more efficiency}}$$

21. GENERAL TIME-EFFICIENCY CONVERSION

If A is $(p\%)$ more efficient than B:

$$[E_A:E_B=(100+p):100]$$

Therefore:

$$[\boxed{T_A:T_B=100:(100+p)}]$$

If A is $(p\%)$ less efficient:

$$[E_A:E_B=(100-p):100]$$

Then:

$$[\boxed{T_A:T_B=100:(100-p)}]$$

22. FIND INDIVIDUAL TIME FROM COMBINED TIME

A and B together complete work in 6 days.

A alone takes 10 days.

Find B's time.

Combined rate:

$$[\frac{1}{6}]$$

A's rate:

$$[\frac{1}{10}]$$

B:

$$[\frac{1}{6}-\frac{1}{10}]$$

LCM 30:

$$[\frac{5}{30}-\frac{3}{30}=\frac{2}{30}=\frac{1}{15}]$$

So:

$$[\boxed{B=15\text{ days}}]$$

23. FORMULA FOR UNKNOWN INDIVIDUAL TIME

If:

- A+B together = (T)
- A alone = (a)

then:

$$[\frac{1}{b}=\frac{1}{T}-\frac{1}{a}]$$

So:

$$[\boxed{b=\frac{aT}{a-T}}]$$

provided $(a>T)$.

24. EXAMPLE

Together = 8 days.

A alone = 12 days.

$$b = \frac{12(8)}{12-8}$$

$$= \frac{96}{4}$$

$$= \boxed{24 \text{ days}}$$

25. COMBINED TIME MUST BE SMALLER

For two positive workers:

$$\boxed{T_{\text{together}}} < \min(T_A, T_B)$$

If A takes 10 days and B 15 days, together cannot take 12 days.

That would be impossible.

Useful option elimination.

26. WORK DONE IN SOME DAYS TOGETHER

A = 12 days.

B = 18 days.

Together work for 4 days.

Rate:

$$\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$$

Work in 4 days:

$$4 \cdot \frac{5}{36} = \frac{5}{9}$$

Remaining:

$$1 - \frac{5}{9} = \boxed{\frac{4}{9}}$$

27. SAME QUESTION BY LCM

Total work:

$$36$$

A rate:

$$3$$

B:

$$2$$

Together:

$$5$$

In 4 days:

$$\backslash[20 \backslash]$$

Remaining:

$$\backslash[36-20 = 16 \backslash]$$

Fraction:

$$\backslash[16/36 = \boxed{\frac{4}{9}} \backslash]$$

Much easier.

28. A WORKS FIRST, B FINISHES

A can complete work in 10 days.

A works alone for 4 days.

Then B, who can complete it in 12 days, finishes the remaining work.

Take total work:

$$\backslash[\text{LCM}(10,12)=60 \backslash]$$

A rate:

$$\backslash[6 \backslash]$$

B rate:

$$\backslash[5 \backslash]$$

A in 4 days:

$$\backslash[24 \backslash]$$

Remaining:

$$\backslash[60-24=36 \backslash]$$

B time:

$$\backslash[36/5 = \boxed{7.2\text{ days}} \backslash]$$

29. BOTH WORK, THEN ONE LEAVES

A = 12 days.

B = 18 days.

They work together for 3 days, then B leaves.

Find total completion time.

Total work:

$$\backslash[36 \backslash]$$

Rates:

$$\backslash[A=3,\quad B=2 \backslash]$$

Together:

$$\boxed{5}$$

Work in 3 days:

$$\boxed{15}$$

Remaining:

$$\boxed{21}$$

A alone time:

$$\boxed{21/3=7}$$

Total:

$$\boxed{3+7 = 10 \text{ days}}$$

30. ONE WORKS, THEN SECOND JOINS

A = 12 days.

B = 18 days.

A works alone for 2 days, then B joins.

Total work:

$$\boxed{36}$$

A rate:

$$\boxed{3}$$

Work by A:

$$\boxed{6}$$

Remaining:

$$\boxed{30}$$

Together rate:

$$\boxed{5}$$

Time after B joins:

$$\boxed{30/5=6}$$

Total:

$$\boxed{8 \text{ days}}$$

31. PARTIAL WORK GIVEN

A completes:

$$\boxed{\frac{3}{5}}$$

of a job in 6 days.

Rate:

$$\boxed{\frac{3/5}{6} = \frac{3}{30} = \frac{1}{10}}$$

So full work time:

$$\boxed{10 \text{ days}}$$

32. FAST PROPORTION METHOD

If:

$$\frac{35 \text{ work}}{6 \text{ days}}$$

Then full work:

$$1 \times \frac{53}{1}$$

$$= \boxed{10 \text{ days}}$$

33. WORK FRACTION !' TIME FRACTION

At constant efficiency:

$$\boxed{\text{Work fraction} = \text{Time fraction}}$$

If A completes full work in 20 days:

- half work !' 10 days
 - 30% work !' 6 days
 - 75% work !' 15 days
-

34. DIFFERENT WORKERS, SAME WORK

A completes work in 8 days.

B in 12 days.

In 4 days:

A completes:

$$\frac{4}{8} = \frac{1}{2}$$

B completes:

$$\frac{4}{12} = \frac{1}{3}$$

So A does more work in same time because A is more efficient.

35. WORK RATIO IN SAME TIME

For equal working time:

$$\boxed{\text{Work ratio} = \text{Efficiency ratio}}$$

If efficiencies:

$$3:2$$

and both work 5 days:

Work ratio remains:

$$\boxed{3:2}$$

36. TIME RATIO FOR SAME WORK

For same work:

$$\boxed{\text{Time ratio} = \text{inverse efficiency ratio}}$$

Already important enough to repeat.

37. MEN-DAYS BASIC IDEA

If workers are equally efficient:

$$\boxed{\text{Work} \propto \text{Men} \times \text{Days}}$$

So for same work:

$$\boxed{M_1 D_1 = M_2 D_2}$$

38. EXAMPLE

10 workers complete work in 12 days.

How long for 15 equally efficient workers?

$$10(12) = 15D$$

$$D = 8$$

$$\boxed{8 \text{ days}}$$

More workers !' fewer days.

39. MEN-DAYS-HOURS

If hours per day also vary:

$$\boxed{W \propto M \times D \times H}$$

For same work and same efficiency:

$$\boxed{M_1 D_1 H_1 = M_2 D_2 H_2}$$

40. EXAMPLE

12 workers working 8 hours/day finish in 10 days.

How many days for 16 workers working 6 hours/day?

$$12(10)(8) = 16(D)(6)$$

$$960 = 96D$$

$$\boxed{D=10}$$

Why unchanged?

Because:

$$12 \times 8 = 96$$

and:

$$16 \times 6 = 96$$

same worker-hours per day.

41. INCLUDE EFFICIENCY

If worker groups have different efficiencies:

$$\boxed{W \propto M \times D \times H \times E}$$

So:

$$\boxed{M_1 D_1 H_1 E_1 = M_2 D_2 H_2 E_2}$$

for the same work.

42. EXAMPLE

10 skilled workers finish in 12 days.

A second group is 20% more efficient.

How many equally efficient-to-each-other workers from second group are needed to finish in 10 days?

Let number = m .

Original efficiency factor:

$$1$$

New:

$$1.2$$

$$10(12)(1) = m(10)(1.2)$$

$$120 = 12m$$

$$\boxed{m=10}$$

Higher efficiency compensates for shorter time.

43. WORKER-DAYS AS TOTAL LABOUR

If:

$$20 \text{ workers} \times 15 \text{ days} = 300 \text{ worker-days}$$

then same job requires 300 equivalent worker-days, assuming equal efficiency and hours.

This is often the easiest way to think.

44. WORK COMPLETED PERCENTAGE

A completes work in 25 days.

In 5 days:

$$\left[\frac{5}{25} = 20\% \right]$$

So:

$$\left[\boxed{20\% \text{ complete}} \right]$$

Remaining:

$$\left[80\% \right]$$

45. FIND WORKER'S FULL TIME FROM PERCENTAGE

A completes 30% work in 6 days.

Then:

$$\left[100\% \right]$$

takes:

$$\left[6 \times \frac{100}{30} \right]$$

$$\left[= 20 \right]$$

$$\left[\boxed{20 \text{ days}} \right]$$

46. A IS x% MORE EFFICIENT — COMBINED TIME

B takes 20 days.

A is 25% more efficient.

Then:

$$\left[E_A : E_B = 5 : 4 \right]$$

So A time:

$$\left[16 \text{ days} \right]$$

Together:

$$\left[T = \frac{16(20)}{36} \right]$$

$$\left[= \frac{80}{9} \right]$$

$$\left[= \boxed{8\frac{8}{9} \text{ days}} \right]$$

47. FASTER RATIO-UNIT METHOD

If efficiencies:

$$\left[A : B = 5 : 4 \right]$$

and B takes 20 days.

Let B daily rate = 4 units.

Total work:

$$20(4)=80$$

A daily:

$$5$$

Together:

$$9$$

Time:

$$80/9 = \boxed{8\frac{8}{9}}$$

This is often faster.

48. FIND EFFICIENCY FROM TIMES

A = 6 days.

B = 8 days.

Efficiency ratio:

$$8:6 = \boxed{4:3}$$

A is more efficient by:

$$\frac{4-3}{3} \times 100 = \boxed{33\frac{1}{3}\%}$$

49. FIND TIME PERCENT DIFFERENCE

A takes 6 days.

B takes 8 days.

A takes:

$$2$$

days less than B.

Percentage less relative to B:

$$2/8 \times 100 = \boxed{25\%}$$

Again:

- A is 33% more efficient
- A takes 25% less time

asymmetry.

50. GATE-STYLE QUESTION 1

A takes 12 days.

B takes 18 days.

Together:

$$\left[\frac{12(18)}{30} = \boxed{7.2 \text{ days}} \right]$$

51. GATE-STYLE QUESTION 2

A+B = 6 days.

A = 10 days.

B:

$$\left[\frac{10(6)}{10-6} = \boxed{15 \text{ days}} \right]$$

52. GATE-STYLE QUESTION 3

A = 20 days.

A works 5 days.

Work done:

$$\left[5/20 = \boxed{\frac{1}{4}} \right]$$

Remaining:

$$\left[\boxed{\frac{3}{4}} \right]$$

53. GATE-STYLE QUESTION 4

A:B efficiency = 3:2.

A takes 8 days.

B time:

$$\left[8 \times \frac{3}{2} = \boxed{12 \text{ days}} \right]$$

54. PLACEMENT-STYLE QUESTION 1

A = 10 days.

B = 15 days.

Together 2 days.

Total work via LCM:

$$\left[30 \right]$$

Rates:

$$\left[3, 2 \right]$$

2-day work:

$$\left[10 \right]$$

Remaining:

$\frac{1}{20}$

So:

$\frac{1}{20} \times \frac{2}{3} \times \text{work remains}$

55. PLACEMENT-STYLE QUESTION 2

A and B together = 8 days.

A = 12 days.

B:

$\frac{1}{12} \times \frac{1}{8} = \frac{1}{24}$ days

56. PLACEMENT-STYLE QUESTION 3

12 workers take 15 days.

20 workers take:

$12 \times 15 = 20D$

$D = 9$

9 days

57. PLACEMENT-STYLE QUESTION 4

A = 12 days, B = 18 days.

They work together 3 days, then B leaves.

From earlier:

$\frac{1}{12} \times 3 + \frac{1}{18} \times 3 = \frac{1}{6}$ days total

58. ADVANCED QUESTION — THREE WORKERS

A = 10 days.

B = 15 days.

C = 30 days.

Total work:

30

Rates:

$\frac{1}{10}, \frac{1}{15}, \frac{1}{30}$

Together:

6

Time:

$$\boxed{30/6 = 5 \text{ days}}$$

59. ADVANCED QUESTION — ONE LEAVES MIDWAY

A = 10 days.

B = 15 days.

They work together for 3 days. Then A leaves.

Total work:

$$30$$

Together rate:

$$5$$

Done:

$$15$$

Remaining:

$$15$$

B rate:

$$2$$

Time B needs:

$$15/2 = 7.5$$

Total:

$$3 + 7.5 = \boxed{10.5 \text{ days}}$$

60. ADVANCED QUESTION — ONE JOINS MIDWAY

A can complete work in 12 days.

A works 4 days alone.

Then B joins, and together they finish in another 4 days.

Find B's time.

Take total work:

$$12 \text{ units}$$

A rate:

$$1$$

A first 4 days:

$$4$$

Remaining:

$$8$$

Together finish remaining 8 in 4 days.

Together rate:

$$\frac{1}{2}$$

Thus B rate:

$$\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

So B alone also takes:

$$\boxed{12 \text{ days}}$$

61. ADVANCED QUESTION — EFFICIENCY %

A takes 24 days.

B takes 30 days.

Efficiency ratio:

$$\frac{30}{24} = \frac{5}{4}$$

A is:

$$\frac{1}{3} \times 100 = \boxed{25\% \text{ more efficient than B}}$$

62. ADVANCED QUESTION — MEN, DAYS, HOURS

18 men working 8 hours/day complete a job in 20 days.

How many men working 10 hours/day can finish it in 12 days?

$$18(20)(8) = M(12)(10)$$

$$2880 = 120M$$

$$\boxed{M=24}$$

63. COMMON TRAPS

TRAP 1 — Time ratio = efficiency ratio

Wrong.

They are inverse.

$$E_A : E_B = T_B : T_A$$

TRAP 2 — Add times

If A = 10 days and B = 15 days, together time is not $\frac{25}{2}$ or 25.

Add **rates**, not times.

TRAP 3 — More workers means more days

For fixed work, more equally efficient workers means fewer days.

& pTRAP 4 — Midway joining/leaving

Calculate completed work first, then remaining work.

& pTRAP 5 — 20% more efficient = 20% less time

Wrong because bases differ.

64. MUST-MEMORIZE FORMULAS

One-day work

$$\boxed{\frac{1}{T}}$$

Two people together

$$\boxed{T = \frac{ab}{a+b}}$$

Efficiency-time relation

$$\boxed{E_A : E_B = T_B : T_A}$$

Men-days

$$\boxed{M_1 D_1 = M_2 D_2}$$

Men-days-hours

$$\boxed{M_1 D_1 H_1 = M_2 D_2 H_2}$$

for equal efficiency.

65. FAST EXAM DECISION TREE

Individual times given?

Use LCM work units.

Need combined time?

Add daily rates.

Combined + one individual given?

Subtract rates.

Efficiency ratio given?

Reverse it for time ratio.

One joins/leaves?

Split into stages.

Fraction/percentage work given?

Use proportionality.

Men/days/hours?

Use product conservation for fixed work.

PRACTICE — LEVEL 0

Q1

A completes a job in 10 days. Find one-day work.

Q2

A = 12 days, B = 18 days. Find combined time.

Q3

A completes work in 20 days. How much does A complete in 5 days?

Q4

Efficiency ratio A:B = 4:3. Find time ratio.

Q5

10 workers take 12 days. Find time for 15 workers.

LEVEL 1

Q6

A = 8 days, B = 24 days. Find combined time.

Q7

A+B = 6 days, A = 10 days. Find B's time.

Q8

A = 15 days, B = 20 days. They work together for 4 days. Find fraction remaining.

Q9

A is 25% more efficient than B. B takes 20 days. Find A's time.

Q10

12 workers working 8 hours/day take 10 days. How many days for 16 workers working 6 hours/day?

LEVEL 2

Q11

A = 12 days, B = 18 days. They work together for 3 days, then B leaves. Find total time.

Q12

A = 10 days, B = 15 days. They work together for 3 days, then A leaves. Find total time.

Q13

A completes 40% of work in 8 days. Find full time.

Q14

A:B efficiency = 5:4. Together they finish in 8 days. Find their individual times.

Q15

20 men complete work in 18 days. After 6 days, 5 men leave. Find total completion time.

LEVEL 3 — EXAM PRESSURE**Q16**

A = 10 days, B = 15 days, C = 30 days. A and B work for 2 days, then C joins. Find total completion time.

Q17

A alone takes 12 days. A works 4 days alone, then B joins, and remaining work is completed in 4 more days. Find B's time.

Q18

A is 50% more efficient than B. Together they complete work in 8 days. Find A's and B's individual times.

Q19

18 men working 8 hours/day complete a job in 20 days. Find men needed to finish in 12 days at 10 hours/day.

Q20

A takes 20% less time than B. If together they finish the work in $\frac{40}{9}$ days, find their individual times.

ANSWERS**Q1**

$$\boxed{\frac{1}{10}}$$

Q2

LCM work:

$$\boxed{36}$$

Rates:

$$\boxed{3, 2}$$

Together:

$$\boxed{5}$$

Time:

$$\boxed{\frac{36}{5} = 7.2 \text{ days}}$$

Q3

$$\boxed{\frac{5}{20} = \frac{14}{}}$$

Q4

$$\boxed{T_A:T_B=3:4}$$

Q5

$$10(12)=15D$$

$$\boxed{D=8}$$

Q6

$$T=\frac{8(24)}{32}$$

$$=\boxed{6\text{ days}}$$

Q7

$$\frac{1}{B}=\frac{1}{16}-\frac{1}{10}=\frac{1}{15}$$

$$\boxed{15\text{ days}}$$

Q8

LCM:

$$60$$

Rates:

$$4, 3$$

Together:

$$7$$

4 days work:

$$28$$

Remaining:

$$32$$

Fraction:

$$32/60 = \boxed{\frac{8}{15}}$$

Q9

Efficiency:

$$A:B=5:4$$

Time:

$$A:B=4:5$$

$$A=20\times\frac{4}{5}$$

$$=\boxed{16\text{ days}}$$

Q10

$$12(10)(8)=16(D)(6)$$

$$960=96D$$

$$\boxed{10\text{ days}}$$

Q11

Total work:

$$\boxed{36}$$

Rates A/B:

$$3, \frac{1}{2}$$

3 days together:

$$\boxed{15}$$

Remaining:

$$\boxed{21}$$

A alone:

$$\boxed{21/3=7}$$

Total:

$$\boxed{10 \text{ days}}$$

Q12

Total work:

$$\boxed{30}$$

Rates:

$$3, \frac{1}{2}$$

3 days together:

$$\boxed{15}$$

Remaining:

$$\boxed{15}$$

B alone:

$$\boxed{15/2=7.5}$$

Total:

$$\boxed{10.5 \text{ days}}$$

Q13

$$\boxed{40\% \text{ to } 8 \text{ days}}$$

$$\boxed{100\% \text{ to } 8 \times \frac{100}{40}}$$

$$\boxed{=20 \text{ days}}$$

Q14

Efficiency:

$$\boxed{5:4}$$

Let rates:

$$\boxed{5k, 4k}$$

Together:

$$\sqrt[9]{k}$$

They finish in 8 days, so total work:

$$\sqrt[72]{k}$$

A alone time:

$$\sqrt[72]{k/5k} = \boxed{14.4 \text{ days}}$$

B:

$$\sqrt[72]{k/4k} = \boxed{18 \text{ days}}$$

Q15

Take total work:

$$\sqrt[20]{18} = 360 \text{ worker-days}$$

First 6 days with 20 men:

$$\sqrt[120]{}$$

Remaining:

$$\sqrt[240]{}$$

Now 15 men remain.

Time:

$$\sqrt[240/15]{16}$$

Total:

$$\sqrt[6+16]{22 \text{ days}}$$

Q16

LCM:

$$\sqrt[30]{}$$

Rates:

$$\sqrt[A=3, B=2, C=1]{}$$

First 2 days A+B:

$$\sqrt[2(5)]{10}$$

Remaining:

$$\sqrt[20]{}$$

All three rate:

$$\sqrt[6]{}$$

Time:

$$\sqrt[20/6]{\frac{10}{3}}$$

Total:

$$\sqrt[2+\frac{10}{3}]{\frac{16}{3}} = \boxed{\frac{16}{3} = 5\frac{1}{3} \text{ days}}$$

Q17

Take total work:

$$\boxed{12}$$

A rate:

$$\boxed{1}$$

A works 4 days:

$$\boxed{4}$$

Remaining:

$$\boxed{8}$$

A+B finish 8 in 4 days:

$$\boxed{\text{combined rate}=2}$$

B rate:

$$\boxed{1}$$

So:

$$\boxed{B=12\text{ days}}$$

Q18

A is 50% more efficient:

$$\boxed{A:B=3:2}$$

Let daily rates:

$$\boxed{3k, 2k}$$

Together:

$$\boxed{5k}$$

In 8 days total work:

$$\boxed{40k}$$

A time:

$$\boxed{40/3 = 13\frac{1}{3}\text{ days}}$$

B time:

$$\boxed{40/2 = 20\text{ days}}$$

Q19

$$\boxed{18(20)(8) = M(12)(10)}$$

$$\boxed{2880 = 120M}$$

$$\boxed{M=24}$$

Q20

A takes 20% less time than B.

So:

$$[T_A:T_B=4:5]$$

Efficiency ratio:

$$[5:4]$$

Let rates:

$$[5k, 4k]$$

Combined:

$$[9k]$$

Together time:

$$[\frac{W}{9k} = \frac{40}{9}]$$

So total work:

$$[W=40k]$$

A time:

$$[40k/5k = \boxed{8\text{ days}}]$$

B:

$$[40k/4k = \boxed{10\text{ days}}]$$

CHAPTER 26 STATUS

Covered so far:

- $(W=RT)$ '
- one-day work '
- reciprocal time-rate relation '
- LCM work method '
- two and three workers together '
- efficiency ratios '
- percentage efficiency changes '
- combined + individual time problems '
- work completed/remaining '
- joining/leaving midway '
- partial-work problems '
- men-days-hours basics '
- efficiency-adjusted manpower '
- GATE/placement practice '

But **Chapter 26 is not complete yet.**

Next we need:

- alternate-day working
- different daily schedules
- workers joining/leaving repeatedly
- work and wages

- variable efficiency
- men/women/children equivalence
- partial workforce replacement
- negative work/leaks as a bridge to Chapter 28
- tougher mixed Time & Work problems
- final shortcuts and chapter audit

Progress

Completed: 25/76 **Current:** Chapter 26 — Time and Work **Next:** Chapter 26 Part 2 — Alternate-day work, variable workforce & advanced Time-and-Work problems

USER

next

ASSISTANT

CHAPTER 26 — TIME AND WORK

PART 2 — ALTERNATE-DAYWORK, VARIABLE WORKFORCE, WORK & WAGES, MEN–WOMEN–CHILDREN EQUIVALENCE & ADVANCED PROBLEMS

Status: $\emptyset=14$ **PROGRESS GATE:** High **Placements:** Very High **Priority:** $\emptyset=14$ **Yr 1**

This part handles the questions where people do **not** work continuously together.

Typical patterns:

- A and B work on alternate days
- one worker joins later
- some workers leave
- efficiency changes
- men/women/children have different efficiencies
- wages are divided according to work done
- positive and negative work occur together

The same master rule remains:

$$\boxed{W=RT}$$

1. ALTERNATE-DAYWORK

Suppose:

- A completes work in 10 days
- B completes work in 15 days
- A starts
- they work on alternate days

Take total work:

$$\boxed{\text{LCM}(10,15)=30}$$

Daily work:

$$\boxed{A=3}$$

$$\boxed{B=2}$$

In one complete 2-day cycle:

$$\boxed{3+2=5}$$

So:

$$\boxed{\text{2-day cycle work}=5}$$

2. EXAMPLE — ALTERNATEDAYS

Total work:

$$\boxed{30}$$

Each 2-day cycle:

$$\boxed{5}$$

After 5 cycles = 10 days:

$$\boxed{25}$$

Remaining:

$$\boxed{5}$$

Day 11 is A's turn.

A does:

$$\boxed{3}$$

Remaining:

$$\boxed{2}$$

Day 12 is B's turn.

B can do 2 units in one full day.

So total time:

$$\boxed{12\text{ days}}$$

3. CYCLE METHOD

Whenever a pattern repeats:

1. Calculate work done in one complete cycle.
2. Find number of full cycles possible.
3. Handle the remaining work day-by-day.

This is usually faster than tracking every day separately.

4. STARTING PERSON MATTERS

Suppose $A = 10$ days, $B = 15$ days.

If A starts:

Day pattern:

$\backslash [A, B, A, B, \ldots \backslash$

If B starts:

$\backslash [B, A, B, A, \ldots \backslash$

Since A and B have different efficiencies, the final completion time may change.

5. EXAMPLE — B STARTS

Same work:

$\backslash [30 \backslash$

Rates:

$\backslash [A=3, \quad B=2 \backslash$

Every 2 days still gives:

$\backslash [5 \backslash$

After 10 days:

$\backslash [25 \backslash$

Remaining:

$\backslash [5 \backslash$

Day 11 = B:

$\backslash [2 \backslash$

Remaining:

$\backslash [3 \backslash$

Day 12 = A:

$\backslash [3 \backslash$

So again:

$\backslash [\boxed{12 \text{ days}} \backslash$

Here both starting orders happen to give the same result.

Do not assume this always happens.

6. FRACTIONAL LAST DAY

Suppose:

- A = 12 days
- B = 18 days
- alternate days
- A starts

Take work:

$$\boxed{36}$$

Rates:

$$A=3, \text{quad } B=2$$

2-day cycle:

$$\boxed{5}$$

After 7 cycles = 14 days:

$$\boxed{35}$$

Remaining:

$$\boxed{1}$$

Day 15 is A's turn.

A does 3 units/day.

Time for 1 unit:

$$\frac{1}{3} \text{ day}$$

Total:

$$\boxed{14\frac{1}{3} \text{ days}}$$

7. ALTERNATE-DAY IMPORTANT TRAP

If remaining work is smaller than the next worker's daily capacity, do **not** count a whole day.

Use:

$$\boxed{\text{time} = \frac{\text{remaining work}}{\text{daily rate}}}$$

8. WORKING IN A 3-DAY PATTERN

Suppose:

- Day 1: A
- Day 2: B
- Day 3: A+B
- then repeats

A = 12 days, B = 18 days.

Total work:

$$\boxed{36}$$

Rates:

$$\backslash[A=3,\quad B=2 \backslash]$$

Cycle work:

$$\backslash[3+2+(3+2)=10 \backslash]$$

Cycle length:

$$\backslash[3\text{ days} \backslash]$$

So every 3 days:

$$\backslash[10\text{ units} \backslash]$$

Then use cycle method.

9. VARIABLE WORKFORCE

Suppose 20 workers can finish a job in 15 days.

Total work:

$$\backslash[20(15)=300\text{ worker-days} \backslash]$$

After 5 days, 5 workers leave.

Work completed:

$$\backslash[20(5)=100 \backslash]$$

Remaining:

$$\backslash[200 \backslash]$$

Workers remaining:

$$\backslash[15 \backslash]$$

Additional days:

$$\backslash[200/15 = 13\frac{1}{3} \backslash]$$

Total time:

$$\backslash[5+13\frac{1}{3} = \boxed{18\frac{1}{3}\text{ days}} \backslash]$$

10. WORKERS JOIN MIDWAY

15 workers can finish work in 20 days.

Total work:

$$\backslash[300 \backslash]$$

They work for 8 days.

Completed:

$$\backslash[15(8)=120 \backslash]$$

Remaining:

$$\backslash[180 \backslash]$$

Then 5 more workers join.

Workers:

$$\boxed{20}$$

Time for remaining:

$$\boxed{180/20=9}$$

Total:

$$\boxed{8+9=17\text{ days}}$$

11. FIND EXTRA WORKERS NEEDED

20 workers have completed 6 days of a job that normally requires 15 days.

Total work:

$$\boxed{300}$$

Done:

$$\boxed{120}$$

Remaining:

$$\boxed{180}$$

Suppose the job must now be completed in 6 more days.

Required workers:

$$\boxed{180/6=30}$$

Already present:

$$\boxed{20}$$

Additional workers:

$$\boxed{10}$$

12. DEADLINE PROBLEMS

These are very common.

Method:

1. Find total work.
 2. Calculate work already completed.
 3. Find remaining work.
 4. Divide by remaining available days.
 5. Compare required workforce with current workforce.
-

13. WORKER EFFICIENCY CHANGES

Suppose 10 workers normally complete job in 12 days.

Total work:

$120 \text{ normal worker-days}$

If workers become 20% more efficient, each worker now has efficiency factor:

1.2

Effective daily workforce:

$10(1.2) = 12$

New time:

$120/12 = \boxed{10 \text{ days}}$

14. EFFICIENCY DECREASE

If efficiency decreases by 20%:

New efficiency:

0.8

Time is inversely proportional.

So if old time = 10 days:

$T_{\text{new}} = \frac{10}{0.8} = \boxed{12.5 \text{ days}}$

15. VARIABLE EFFICIENCY MIDWAY

A can normally finish work in 12 days.

He works first 4 days at normal efficiency, then becomes 50% more efficient.

Take total work:

12 units

Normal daily rate:

1

First 4 days:

4

Remaining:

8

New rate:

1.5

Time:

$8/1.5 = \frac{16}{3} = 5\frac{1}{3}$

Total:

$4 + 5\frac{1}{3} = \boxed{9\frac{1}{3} \text{ days}}$

16. MEN–WOMEN–CHILDREN EQUIVALENCE

If:

$$2\text{ men} = 3\text{ women}$$

in work capacity, then efficiencies satisfy:

$$2E_M = 3E_W$$

So:

$$\boxed{E_M : E_W = 3 : 2}$$

Be careful: the numbers reverse.

17. EXAMPLE

Given:

$$2M = 3W$$

So one man : one woman efficiency:

$$3 : 2$$

If 6 men work:

Equivalent women:

Since:

$$2M = 3W$$

$$6M = 9W$$

So:

$$\boxed{6\text{ men} = 9\text{ women}}$$

18. MEN AND CHILDREN

If:

$$3M = 5C$$

then:

$$3E_M = 5E_C$$

Therefore:

$$\boxed{E_M : E_C = 5 : 3}$$

19. COMBINED MIXED WORKFORCE

Suppose:

$$1M = 2W$$

and 4 men + 6 women work.

Convert everything to women.

4 men:

$$[8W]$$

Plus:

$$[6W]$$

Total equivalent:

$$[\boxed{14W}]$$

This simplifies the problem.

20. EXAMPLE — MIXED WORKFORCE

If 10 women complete a job in 12 days, how long will 4 men + 6 women take if:

$$[1M=2W]$$

Equivalent workforce:

$$[4M+6W = 8W+6W = 14W]$$

Total work:

$$[10(12)=120\text{ woman-days}]$$

Time:

$$[120/14 = \boxed{\frac{60}{7}}=8\frac{4}{7}\text{ days}]$$

21. WORK & WAGES

If workers have different efficiencies but work for the same duration:

$$[\boxed{\text{Wages}} \propto \text{work done} \propto \text{efficiency}]$$

If durations also differ:

$$[\boxed{\text{Wages}} \propto E \times T]$$

22. BASIC WAGE SHARING

A and B work together.

Efficiency ratio:

$$[3:2]$$

Both work the same number of days.

Total wage:

$$[1500]$$

Share ratio:

$$[3:2]$$

A:

$$\left[5000 \times \frac{35}{100} = \boxed{1750} \right]$$

B:

$$\left[\boxed{1750} \right]$$

23. DIFFERENT WORKING TIMES

A:B efficiency:

$$\left[3:2 \right]$$

A works 4 days.

B works 6 days.

Work ratio:

$$\left[3(4):2(6) \right]$$

$$\left[12:12 \right]$$

So wages are equal.

If total = 18000 :

$$\left[\boxed{14000} \text{ each} \right] \times 2 = 28000$$

24. WAGE SHARE FROM INDIVIDUAL COMPLETION TIMES

A can finish alone in 10 days.

B in 15 days.

Both work same duration.

Efficiency ratio:

$$\left[15:10 = 3:2 \right]$$

So wages:

$$\left[\boxed{3:2} \right]$$

25. WAGE SHARE WHEN ONE LEAVES EARLY

A and B efficiencies:

$$\left[3:2 \right]$$

A works 6 days.

B works 4 days.

Their work contributions:

$$\left[18:8 \right]$$

$$\left[=9:4 \right]$$

If total wages:

$$\sqrt[1]{6500}$$

Total parts:

$$\sqrt[1]{13}$$

A:

$$\sqrt[1]{6500 \times \frac{9}{13}} = \sqrt[1]{4500}$$

B:

$$\sqrt[1]{2000}$$

26. VERY IMPORTANT

Do not divide wages according to:

- capital,
- number of workers,
- final time,

unless the question justifies it.

Divide wages according to **actual work contribution**.

27. WORK DONE BY GROUPS

Suppose A and B together complete work.

A's rate:

$$\sqrt[1]{3}$$

B:

$$\sqrt[1]{2}$$

If they work entire job together:

A does:

$$\sqrt[1]{\frac{35}{2}}$$

of total work.

B:

$$\sqrt[1]{\frac{25}{2}}$$

So wages:

$$\sqrt[1]{3:2}$$

28. NEGATIVE WORK

Some agents undo work.

Examples:

- leak empties water
- worker damages completed work
- machine causes loss
- pipe empties tank

Represent negative rate.

Suppose:

A completes a job in 10 days:

$$\left[+\frac{1}{10} \right]$$

B destroys the same job in 15 days:

$$\left[-\frac{1}{15} \right]$$

Net:

$$\left[\frac{1}{10} - \frac{1}{15} = \frac{1}{30} \right]$$

So net completion time:

$$\left[\boxed{30 \text{ days}} \right]$$

This is the bridge to Pipes & Cisterns.

29. POSITIVE + NEGATIVE WORK FORMULA

If positive worker takes (a) days and negative worker destroys in (b) days:

$$\left[\boxed{r_{\text{net}}} = \frac{1}{a} - \frac{1}{b} \right]$$

Net work is possible only if positive rate is greater.

30. IF NEGATIVE RATE IS STRONGER

A completes in 20 days:

$$\left[+\frac{1}{20} \right]$$

B destroys in 10 days:

$$\left[-\frac{1}{10} \right]$$

Net:

$$\left[-\frac{1}{20} \right]$$

The job will not progress toward completion.

This is a sanity check.

31. ALTERNATE-DAYWORK WITH NEGATIVE WORK

A completes in 6 days.

B destroys in 12 days.

They act on alternate days, A first.

Take total work:

$$\sqrt[12]{}$$

A:

$$\sqrt{+2}$$

B:

$$\sqrt{-1}$$

2-day cycle net:

$$\sqrt{+1}$$

This is much slower than A alone.

Handle carefully cycle-by-cycle.

32. WORK COMPLETED BEFORE A WORKER LEAVES

A = 12 days, B = 18 days.

They work together until half the work is completed.

How long?

Combined rate:

$$\sqrt{\frac{5}{36}}$$

Need:

$$\sqrt{\frac{12}{}}$$

Time:

$$\sqrt{\frac{1}{2} \times \frac{5}{36} = \frac{18}{5}}$$

$$\sqrt{=\boxed{3.6 \text{ days}}}$$

33. LCM METHOD FOR SAME QUESTION

Total work:

$$\sqrt{36}$$

Half:

$$\sqrt{18}$$

Combined rate:

$$\sqrt{5}$$

Time:

$$\sqrt{18/5 = \boxed{3.6}}$$

34. ONE WORKER DOES x% THEN ANOTHER

A completes whole work in 20 days.

A does first 40%.

Time:

$$\sqrt[2]{20(0.4)=8}$$

B completes whole work in 15 days.

Remaining:

$$\sqrt[2]{60\%}$$

B time:

$$\sqrt[2]{15(0.6)=9}$$

Total:

$$\sqrt[2]{\boxed{17\text{ days}}}$$

35. SAME WORK, DIFFERENT SHIFTS

A works 6 hours/day and finishes in 20 days.

If A works 8 hours/day at same hourly efficiency:

Total hours:

$$\sqrt[2]{20(6)=120}$$

New days:

$$\sqrt[2]{120/8 = \boxed{15}}$$

36. HOURS AND EFFICIENCY TOGETHER

Worker B is 25% more efficient than A.

A works 8 h/day.

B works 6 h/day.

Compare daily work:

A:

$$\sqrt[2]{1\times 8=8}$$

B:

$$\sqrt[2]{1.25\times 6=7.5}$$

So despite higher hourly efficiency, B does slightly less work per day.

This is a common trap.

37. ADVANCED MEN-DAYS-HOURS-EFFICIENCY FORMULA

For equal total work:

$$\boxed{M_1 D_1 H_1 E_1 = M_2 D_2 H_2 E_2}$$

Use relative efficiency values.

38. EXAMPLE

12 men at efficiency 1 work 8 h/day for 15 days.

How many workers of efficiency 1.2 are needed at 10 h/day for 10 days?

$$12(15)(8)(1) = M(10)(10)(1.2)$$

$$1440 = 120M$$

$$\boxed{M=12}$$

39. VARIABLE NUMBER OF WORKERS EACH DAY

Suppose:

- Day 1: 10 workers
- Day 2: 12 workers
- Day 3: 14 workers

Assuming equal efficiency and equal hours, total worker-days over 3 days:

$$10+12+14 = \boxed{36}$$

If total work is 100 worker-days:

Remaining:

$$64$$

Then continue with later workforce.

40. ARITHMETIC PROGRESSION WORKFORCE

Suppose workforce increases by 2 each day:

$$10, 12, 14, 16, \dots$$

Total work over (n) days equals sum of worker counts:

$$\boxed{\frac{n^2[2a+(n-1)d]}{2}}$$

This connects to AP, which we will study later in Chapter 37.

For now just recognize the pattern.

41. FIND WHEN WORK FINISHES WITH CHANGING WORKFORCE

Total work = 60 worker-days.

Workers per day:

$$10, 12, 14, 16, \dots$$

Cumulative:

Day 1:

$$\sqrt[10]{}$$

Day 2:

$$\sqrt[22]{}$$

Day 3:

$$\sqrt[36]{}$$

Day 4:

$$\sqrt[52]{}$$

Remaining:

$$\sqrt[8]{}$$

Day 5 has 18 workers.

Fraction of day needed:

$$\sqrt[8/18 = \frac{4}{9}]{}$$

Total:

$$\sqrt[\boxed{4\frac{4}{9}\text{ days}}]{}$$

42. FIND NUMBER OF WORKERS WHO LEFT

20 workers complete a job in 18 days.

They work 6 days, then some leave.

Job finishes in 24 days total.

Total work:

$$\sqrt[20(18)=360]{}$$

First 6 days:

$$\sqrt[120]{}$$

Remaining:

$$\sqrt[240]{}$$

Remaining time:

$$\sqrt[18\text{ days}]{}$$

Workers needed after departure:

$$\sqrt[240/18 = 13\frac{1}{3}]{}$$

This implies the given data do not correspond to an integer number of equal workers.

In real-headcount aptitude problems, such data are usually chosen to produce an integer.

Always sanity-check.

43. FIND WORKERS JOINED

12 workers can finish work in 20 days.

They work 5 days, then additional workers join.

Entire job finishes in 14 days.

Total work:

$$\boxed{240}$$

First 5 days:

$$\boxed{60}$$

Remaining:

$$\boxed{180}$$

Remaining time:

$$\boxed{9}$$

Required workers:

$$\boxed{180/9=20}$$

Additional:

$$\boxed{20-12 = \boxed{8}}$$

44. A AND B WORK DIFFERENT DAYS

A takes 10 days.

B takes 15 days.

A works every day.

B works every second day.

Take total work:

$$\boxed{30}$$

A daily:

$$\boxed{3}$$

B on working day:

$$\boxed{2}$$

Over 2-day cycle:

- A works both days !' $\boxed{6}$
- B works one day !' $\boxed{2}$

Total:

$$\boxed{8}$$

So average per 2-day cycle:

$$\boxed{8}$$

Then solve by cycles.

45. EXAMPLE

Total work:

$$\boxed{30}$$

3 cycles = 6 days:

$$\boxed{24}$$

Remaining:

$$\boxed{6}$$

Day 7: A works, suppose B does not.

A completes:

$$\boxed{3}$$

Remaining:

$$\boxed{3}$$

Day 8 A+B together:

$$\boxed{5}$$

Only 3 needed.

Fraction of day:

$$\boxed{3/5}$$

Total time:

$$\boxed{7.6 \text{ days}}$$

46. VERY IMPORTANT PATTERNRULE

When schedules are irregular:

Do not use one combined rate unless the schedule repeats evenly.

Instead:

$$\boxed{\text{Find the smallest repeating cycle}}$$

and calculate work per cycle.

47. GATE-STYLE QUESTION 1

A = 12 days, B = 18 days.

They work on alternate days, A first.

Total work:

$$\boxed{36}$$

Rates:

$$\left[\frac{3}{2} \right]$$

2-day cycle:

$$\left[\frac{5}{1} \right]$$

After 7 cycles:

$$\left[\frac{35}{1} \right]$$

Remaining:

$$\left[\frac{1}{1} \right]$$

A needs:

$$\left[\frac{1}{3} \text{ day} \right]$$

Total:

$$\left[\boxed{\frac{14}{3} \text{ days}} \right]$$

48. GATE-STYLE QUESTION 2

20 workers take 15 days.

After 5 days, 5 leave.

Total:

$$\left[300 \right]$$

Done:

$$\left[100 \right]$$

Remaining:

$$\left[200 \right]$$

15 workers remain.

Time:

$$\left[200/15 = 13\frac{1}{3} \right]$$

Total:

$$\left[\boxed{\frac{18}{1} \text{ days}} \right]$$

49. GATE-STYLE QUESTION 3

A:B efficiency = 3:2.

A works 4 days, B works 6 days.

Work ratio:

$$\left[12:12 = \boxed{1:1} \right]$$

So equal wages.

50. GATE-STYLE QUESTION 4

A completes in 10 days.

B destroys in 15 days.

Together net rate:

$$\left[\frac{1}{10} - \frac{1}{15} = \frac{1}{30} \right]$$

$$\left[\boxed{30 \text{ days}} \right]$$

51. PLACEMENT-STYLE QUESTION 1

15 workers take 20 days.

After 8 days, 5 more workers join.

Total work:

$$\left[300 \right]$$

Done:

$$\left[120 \right]$$

Remaining:

$$\left[180 \right]$$

Now workers:

$$\left[20 \right]$$

Time:

$$\left[9 \right]$$

Total:

$$\left[\boxed{17 \text{ days}} \right]$$

52. PLACEMENT-STYLE QUESTION 2

A is 50% more efficient than B.

They work same time and earn ₹ 10,000.

Efficiency ratio:

$$\left[3:2 \right]$$

Wages:

$$\left[A=6000, \text{quad } B=4000 \right]$$

$$\left[\boxed{16000} \right]$$

53. PLACEMENT-STYLE QUESTION 3

If:

$$\boxed{2M=3W}$$

then:

$$\boxed{M:W=3:2}$$

Thus:

$$\boxed{4M=6W}$$

So:

$$\boxed{4M+6W = 12W}$$

Equivalent workforce:

$$\boxed{12 \text{ women}}$$

54. PLACEMENT-STYLE QUESTION 4

10 workers become 25% more efficient.

Equivalent workforce:

$$\boxed{10(1.25)=12.5}$$

If old time is 20 days:

$$T_{\text{new}} = \frac{10(20)}{12.5}$$

$$\boxed{16 \text{ days}}$$

55. ADVANCED QUESTION — ALTERNATEWORK

A = 8 days.

B = 12 days.

A starts, alternating.

Take work:

$$\boxed{24}$$

Rates:

$$\boxed{A=3, B=2}$$

2-day cycle:

$$\boxed{5}$$

After 4 cycles = 8 days:

$$\boxed{20}$$

Remaining:

$$\boxed{4}$$

Day 9 A does:

$$\boxed{3}$$

Remaining:

$\boxed{1}$

Day 10 B needs:

$\frac{1}{2} \text{ day}$

Total:

$\boxed{9.5 \text{ days}}$

56. ADVANCED QUESTION — DEADLINE

24 workers can finish a job in 20 days.

After 5 days, only 18 workers remain.

How many additional workers must join immediately to still finish by day 20?

Total work:

$24(20)=480$

Done first 5 days:

$24(5)=120$

Remaining:

360

Remaining days:

15

Required workforce:

$360/15=24$

Currently:

18

Additional needed:

$\boxed{6}$

57. ADVANCED QUESTION — WAGES

A = 10 days.

B = 15 days.

A works 6 days.

B works 3 days.

Take efficiency ratio:

$3:2$

Work contribution:

$3(6):2(3)$

$18:6 = 3:1$

Total wages ₹ 8000 .

A:

$$\boxed{₹ 6000}$$

B:

$$\boxed{₹ 2000}$$

58. ADVANCED QUESTION — MIXED WORKERS

If:

$$2M=3W$$

12 women finish a job in 15 days.

How long will 4 men + 6 women take?

Convert 4 men:

$$4M=6W$$

So:

$$4M+6W=12W$$

Same equivalent workforce.

Therefore:

$$\boxed{15 \text{ days}}$$

59. ADVANCED QUESTION — VARIABLE EFFICIENCY

A can finish job in 20 days.

For first 5 days A works at normal efficiency.

Then efficiency increases by 25%.

Take total work:

$$20$$

First 5 days done:

$$5$$

Remaining:

$$15$$

New rate:

$$1.25$$

Time:

$$15/1.25 = 12$$

Total:

$$\boxed{17 \text{ days}}$$

60. ADVANCED QUESTION — PARTIAL NEGATIVE WORK

A completes in 12 days.

B undoes in 18 days.

Both operate together for 6 days.

Net rate:

$$\left[\frac{1}{12} - \frac{1}{18} = \frac{1}{36} \right]$$

Work completed in 6 days:

$$\left[\frac{6}{36} = \boxed{\frac{1}{6}} \right]$$

Only one-sixth completed.

61. COMMON TRAPS

& pTRAP 1 — Alternate-day averaging

Do not simply average individual times.

Use rates and cycles.

& pTRAP 2 — Full last day

The job may finish during a fraction of the final day.

& pTRAP 3 — Worker joins/leaves

Recalculate workforce only from that moment onward.

& pTRAP 4 — Men = women relation

If:

$$\left[2M = 3W \right]$$

then efficiency ratio:

$$\left[M:W = 3:2 \right]$$

not 2:3.

& pTRAP 5 — Wage distribution

Wages follow actual work contribution.

& pTRAP 6 — Negative worker

Subtract rates, not times.

62. MUST-MEMORIZE ADVANCED RULES

Work contribution

$$\boxed{\text{Work} = E \times T}$$

Wage share

$$\boxed{\text{Wage ratio} = E_1 T_1 : E_2 T_2}$$

Mixed workforce

$$\boxed{W \propto M \times D \times H \times E}$$

Negative work

$$\boxed{r_{\text{net}} = r_{\text{positive}} - r_{\text{negative}}}$$

63. ALTERNATE-DAY SHORTCUT

For A and B alternating:

$$\boxed{\text{2-day cycle work} = r_A + r_B}$$

But finishing order still matters for the incomplete final cycle.

64. FAST EXAM DECISION TREE

Regular alternation?

Build one cycle.

Workers join/leave?

Split into stages.

Deadline?

Find remaining work ÷ remaining days.

Different worker types?

Convert to equivalent workers.

Wages?

Use work contribution.

Efficiency changes?

Use new rate only after change.

Negative work?

Subtract rates.

PRACTICE — LEVEL 0

Q1

A takes 10 days, B 15 days. A starts and they alternate. Find work done in one 2-day cycle using 30 units total.

Q2

20 workers take 15 days. After 5 days, 5 workers leave. Find remaining work in worker-days.

Q3

A:B efficiency = 3:2 and both work same time. Find wage ratio.

Q4

If $(2M=3W)$, find efficiency ratio of one man to one woman.

Q5

A finishes in 10 days, B destroys in 20 days. Find net completion time together.

LEVEL 1

Q6

A = 12 days, B = 18 days, alternate days, A starts. Find completion time.

Q7

15 workers take 20 days. After 8 days, 5 more workers join. Find total time.

Q8

A:B efficiency = 3:2. A works 4 days, B 6 days. Find work ratio.

Q9

10 workers can finish in 12 days. They become 20% more efficient. Find new time.

Q10

10 women complete a job in 18 days. If $(1M=2W)$, how long will 5 men take?

LEVEL 2

Q11

24 workers can finish in 20 days. After 5 days, 6 workers leave. How many additional workers are needed then to still finish by day 20?

Q12

A takes 8 days, B 12 days, alternate days, A starts. Find completion time.

Q13

A = 10 days, B = 15 days. A works 6 days, B 3 days. Divide $\frac{1800}{11}$ wages.

Q14

A completes in 12 days, B destroys in 18 days. Both work 6 days. Find fraction completed.

Q15

A takes 20 days. After working 5 days, A's efficiency increases by 25%. Find total completion time.

LEVEL 3 — EXAM PRESSURE

Q16

A takes 10 days and works every day. B takes 15 days but works only every second day, beginning on day 2. Find completion time.

Q17

20 workers can finish in 18 days. They work 6 days; then enough workers join so that the remaining job is finished in 8 more days. How many join?

Q18

If $(2M=3W)$, and 12 women complete a job in 15 days, how long will 4 men + 3 women take?

Q19

A and B have efficiencies 5:4. A works 6 days and B 5 days. Total wages = ₹ 11,500. Find their shares.

Q20

A completes a job in 6 days and B destroys it in 12 days. They act on alternate days beginning with A. Find the completion time.

ANSWERS

Q1

Total work:

$$[30]$$

Rates:

$$[A=3, \text{quad } B=2]$$

2-day cycle:

$$[\boxed{5 \text{ units}}]$$

Q2

Total work:

$$[20(15)=300]$$

Completed:

$$[20(5)=100]$$

Remaining:

$$[\boxed{200 \text{ worker-days}}]$$

Q3

Same duration:

$$[\boxed{3:2}]$$

Q4

$$\lfloor 2E_M = 3E_W \rfloor$$

$$\lfloor \boxed{E_M:E_W=3:2} \rfloor$$

Q5

$$\lfloor \frac{1}{10} - \frac{1}{20} = \frac{1}{20} \rfloor$$

So:

$$\lfloor \boxed{20 \text{ days}} \rfloor$$

Q6

Total work:

$$\lfloor 36 \rfloor$$

Rates:

$$\lfloor 3, 2 \rfloor$$

7 cycles:

$$\lfloor 35 \rfloor$$

Remaining 1.

A needs:

$$\lfloor 1/3 \rfloor$$

Total:

$$\lfloor \boxed{14 \frac{1}{3} \text{ days}} \rfloor$$

Q7

Total:

$$\lfloor 300 \rfloor$$

First 8 days:

$$\lfloor 120 \rfloor$$

Remaining:

$$\lfloor 180 \rfloor$$

20 workers now:

$$\lfloor 9 \text{ days} \rfloor$$

Total:

$$\lfloor \boxed{17 \text{ days}} \rfloor$$

Q8

$$\lfloor 3(4):2(6) = 12:12 \rfloor$$

$$\lfloor \boxed{1:1} \rfloor$$

Q9

Old total work:

$$\sqrt{10(12)=120}$$

New effective daily workforce:

$$\sqrt{10(1.2)=12}$$

$$\sqrt{T=120/12 = \boxed{10\text{ days}}}$$

Q10

$$\sqrt{1M=2W}$$

So:

$$\sqrt{5M=10W}$$

Same effective workforce as 10 women.

Thus:

$$\sqrt{\boxed{18\text{ days}}}$$

Q11

Total work:

$$\sqrt{24(20)=480}$$

First 5 days:

$$\sqrt{120}$$

Remaining:

$$\sqrt{360}$$

Remaining days:

$$\sqrt{15}$$

Required workforce:

$$\sqrt{24}$$

After 6 leave:

$$\sqrt{18}$$

Additional needed:

$$\sqrt{\boxed{6}}$$

Q12

Total work:

$$\sqrt{24}$$

Rates:

$$\sqrt{3, 2}$$

4 cycles = 8 days:

$$\sqrt{20}$$

Remaining:

$$\sqrt{4}$$

Day 9 A does 3.

Remaining:

$$\sqrt[1]{1}$$

Day 10 B requires:

$$\sqrt[1/2]{\text{day}}$$

Total:

$$\sqrt{\boxed{9.5 \text{ days}}}$$

Q13

Efficiency ratio:

$$\sqrt[3:2]{}$$

Contribution:

$$\sqrt[3(6):2(3) = 18:6 = 3:1]{}$$

$$^18000:$$

$$\sqrt{\boxed{A = ^16000,000}}$$

Q14

Net rate:

$$\sqrt[\frac{1}{12} - \frac{1}{18} = \frac{1}{36}]{}$$

6 days:

$$\sqrt{\boxed{\frac{1}{6}}}$$

Q15

Take total work:

$$\sqrt[20]{}$$

First 5 days:

$$\sqrt[5]{}$$

Remaining:

$$\sqrt[15]{}$$

New rate:

$$\sqrt[1.25]{}$$

Remaining time:

$$\sqrt[12]{}$$

Total:

$$\sqrt{\boxed{17 \text{ days}}}$$

Q16

Take total work:

$$\sqrt[30]{}$$

A daily:

$$\sqrt[3]{3}$$

B works only on even-numbered days:

$$\sqrt[2]{2}$$

2-day cycle:

$$\sqrt[3+(3+2)=8]{}$$

After 3 cycles = 6 days:

$$\sqrt[24]{}$$

Remaining:

$$\sqrt[6]{}$$

Day 7 A:

$$\sqrt[3]{}$$

Remaining:

$$\sqrt[3]{}$$

Day 8 A+B rate:

$$\sqrt[5]{}$$

Time needed:

$$\sqrt[3/5]{}$$

Total:

$$\sqrt[\boxed{7.6\text{ days}}]{}$$

Q17

Total work:

$$\sqrt[20(18)=360]{}$$

First 6 days:

$$\sqrt[120]{}$$

Remaining:

$$\sqrt[240]{}$$

Need finish in 8 more days:

$$\sqrt[240/8=30\text{ workers}]{}$$

Currently 20.

Join:

$$\sqrt[\boxed{10}]{}$$

Q18

$$\sqrt[2M=3W]{}$$

Therefore:

$$\sqrt[1]{1M=1.5W}$$

4 men:

$$\sqrt[1]{6W}$$

Plus 3 women:

$$\sqrt[1]{9W}$$

Total work:

$$\sqrt[1]{12(15)=180\text{ woman-days}}$$

Time:

$$\sqrt[1]{180/9 = \boxed{20\text{ days}}}$$

Q19

Work contributions:

$$\sqrt[1]{5(6):4(5)}$$

$$\sqrt[1]{30:20 = 3:2}$$

Total wages $111,500$.

A:

$$\sqrt[1]{11500 \times \frac{3}{5} = \boxed{16800}}$$

B:

$$\sqrt[1]{\boxed{14800}}$$

Q20

Take total work:

$$\sqrt[1]{12}$$

A positive rate:

$$\sqrt[1]{+2}$$

B negative rate:

$$\sqrt[1]{-1}$$

2-day cycle net:

$$\sqrt[1]{+1}$$

After 10 cycles = 20 days:

$$\sqrt[1]{10}$$

Remaining:

$$\sqrt[1]{2}$$

Day 21 is A's turn and A does 2 units/day.

So job completes exactly at end of day 21:

$$\sqrt[1]{\boxed{21\text{ days}}}$$

' CHAPTER 26 COMPLETE

We have now covered:

- work-rate-time relation '
- one-day work '
- LCM work method '
- combined work '
- individual time from combined rate '
- efficiency ratios '
- percentage efficiency changes '
- partial work '
- joining/leaving midway '
- men–days–hours '
- alternate-day work '
- cycle method '
- variable workforce '
- deadline problems '
- different worker types '
- men/women/children equivalence '
- work and wages '
- variable efficiency '
- negative work '
- advanced mixed schedules '
- GATE/placement drills '

Progress

Completed: 26/76 **Unit 1 — Quantitative Aptitude:** 0=100% PROGRESS

NEXT

0=0% **Chapter 27 — Efficiency & Work-Wages**

We already introduced some of it in Chapter 26, but Chapter 27 will make it a dedicated exam toolkit:

- efficiency ratios from time
- percentage efficiency conversions
- work ratios
- wage distribution
- individual contribution to team work
- worker equivalence
- productivity comparison
- combined efficiency
- efficiency changes after joining/training/machine upgrades
- advanced wage-sharing and contribution problems.

USER

next

ASSISTANT

CHAPTER 27 — EFFICIENCY & WORK-WAGES

PART 1 — EFFICIENCY RATIOS, PRODUCTIVITY, WORK CONTRIBUTION & BASIC WAGE SHARING

Status: $\emptyset = 100\%$ PROGRESS GATE: High Placements: High Priority: $\emptyset = 100\%$ Per 1

Chapter 26 taught the mechanics of Time & Work.

Chapter 27 focuses on one deeper idea:

$$\boxed{\text{Efficiency} = \text{Work done per unit time}}$$

and therefore:

$$\boxed{\text{Work contribution} \propto \text{Efficiency} \times \text{Time worked}}$$

This becomes especially important for:

- productivity comparisons,
- team contribution,
- wage distribution,
- men/women/children equivalence,
- variable working hours,
- efficiency percentage changes.

1. EFFICIENCY AND TIME

For the same work:

$$W = ET$$

Therefore:

$$\boxed{E \propto \frac{1}{T}}$$

So:

$$\boxed{E_A : E_B = T_B : T_A}$$

MUST MEMORIZE

2. EXAMPLE

A finishes a job in 12 days.

B finishes in 18 days.

Efficiency ratio:

$$\sqrt{E_A:E_B = 18:12}$$

$$\sqrt{=\boxed{3:2}}$$

So A does 3 units in the time B does 2 units.

3. FASTER INTERPRETATION

If:

$$\sqrt{A:B\text{time}=2:3}$$

then:

$$\sqrt{\boxed{A:B\text{efficiency}=3:2}}$$

Just reverse the ratio.

4. SAME TIME !' WORK RATIO

If A and B work for the same duration:

$$\sqrt{W_A:W_B = E_A:E_B}$$

Example:

Efficiency:

$$\sqrt{5:3}$$

Both work for 4 days.

Their work ratio:

$$\sqrt{5(4):3(4) = \boxed{5:3}}$$

5. DIFFERENT TIMES !' WORK RATIO

General rule:

$$\sqrt{\boxed{W_A:W_B = E_A T_A:E_B T_B}}$$

Example:

$$\sqrt{E_A:E_B=3:2}$$

A works 4 days.

B works 6 days.

Then:

$$\sqrt{W_A:W_B = 3(4):2(6)}$$

$$\sqrt{=12:12}$$

$$\sqrt{=\boxed{1:1}}$$

6. WORK-WAGE PRINCIPLE

If wages are paid fairly according to work done:

$$\boxed{\text{Wage ratio} = \text{Work contribution ratio}}$$

Thus:

$$\boxed{\text{Wage ratio} = E_1T_1:E_2T_2\cdots}$$

7. BASIC WAGE EXAMPLE

A:B efficiencies:

$$3:2$$

They work for the same number of days.

Total wages:

$$15000$$

So wage ratio:

$$3:2$$

A:

$$5000 \times \frac{3}{5} = \boxed{3000}$$

B:

$$\boxed{2000}$$

8. DIFFERENT WORKING TIMES

A:B efficiency:

$$4:3$$

A works 6 days.

B works 8 days.

Contribution:

$$4(6):3(8)$$

$$24:24$$

$$\boxed{1:1}$$

So they receive equal wages.

9. FIND INDIVIDUAL SHARE

Total wage = 12,000.

Contribution ratio:

$$5:3:2$$

Total parts:

$$\sqrt[10]{10}$$

Shares:

$$\sqrt[A]{16000}$$

$$\sqrt[B]{13600}$$

$$\sqrt[C]{12400}$$

10. EFFICIENCY FROM INDIVIDUAL TIMES

A takes 8 days.

B takes 12 days.

$$\sqrt[E_A:E_B]{12:8}$$

$$\sqrt[\boxed{3:2}]{} \]$$

If both work together throughout, A does:

$$\sqrt[\frac{3}{5}]{} \]$$

of total work.

B does:

$$\sqrt[\frac{2}{5}]{} \]$$

11. WHY CONTRIBUTION FRACTIONS WORK

If team rate is:

$$\sqrt[3+2=5]{} \]$$

A contributes 3 out of every 5 work units.

Therefore A's fraction:

$$\sqrt[\boxed{\frac{3}{5}}]{} \]$$

B's:

$$\sqrt[\boxed{\frac{2}{5}}]{} \]$$

This remains true as long as both work simultaneously for the same duration.

12. THREE PEOPLE CONTRIBUTION

A:B:C efficiencies:

$$\sqrt[2:3:5]{} \]$$

All work together for the full job.

Their shares of total work:

$$\sqrt[\boxed{\frac{2}{10}, \frac{3}{10}, \frac{5}{10}}]{} \]$$

Thus wage ratio also:

$$\boxed{2:3:5}$$

13. FIND WAGES FROM COMPLETION TIMES

A alone = 10 days.

B alone = 15 days.

They work together for the whole job.

Efficiency:

$$15:10 = 3:2$$

If payment = ₹ 8000 :

A:

$$8000 \times \frac{3}{5} = ₹ 4800$$

B:

$$₹ 3200$$

$$\boxed{₹ 4800 \text{ and } ₹ 3200}$$

14. ONE WORKER LEAVES EARLY

A:B efficiency:

$$3:2$$

A works 6 days.

B works 4 days.

Contribution:

$$3(6):2(4)$$

$$18:8$$

$$\boxed{9:4}$$

If total wages ₹ 6500 :

A:

$$6500 \times \frac{9}{13} = ₹ 4500$$

B:

$$\boxed{₹ 2000}$$

15. ONE WORKER JOINS LATE

A:B efficiency:

$$4:3$$

A works entire 10-day project.

B joins after 4 days, so B works:

6 days

Contribution:

$4(10):3(6)$

$40:18$

$\boxed{20:9}$

16. WAGE SHARING WITH JOINING/LATE LEAVING

Do not use efficiency ratio alone.

Use:

$\boxed{E \times \text{actual time worked}}$

This is the biggest wage-sharing trap.

17. PRODUCTIVITY PER HOUR

Efficiency may be given as:

- units/day,
- units/hour,
- tasks/minute.

Example:

A produces 12 units/hour.

B produces 8 units/hour.

Efficiency ratio:

$12:8 = \boxed{3:2}$

If both work 5 hours:

Work ratio remains:

$3:2$

18. DIFFERENT HOURS PER DAY

A is 20% more efficient per hour than B.

A works 5 h/day.

B works 6 h/day.

Let B hourly efficiency:

5

Then A:

$\sqrt[6]{}$

Daily productivity:

A:

$\sqrt[6]{6(5)=30}$

B:

$\sqrt[5]{5(6)=30}$

So daily outputs are equal.

$\sqrt[1]{1:1}$

19. HOURLY EFFICIENCY VS DAILY PRODUCTIVITY

These are not necessarily the same.

$\sqrt[1]{\text{Daily productivity} = \text{Hourly efficiency} \times \text{Hours/day}}$

A can be more efficient per hour but still produce less per day if working much fewer hours.

20. PERCENT MORE EFFICIENT

If A is $\sqrt[p]{}$ more efficient than B:

$\sqrt[E_A:E_B=(100+p):100]$

Example:

A 25% more efficient:

$\sqrt[125:100 = 5:4]$

21. PERCENT LESS EFFICIENT

If A is 20% less efficient than B:

$\sqrt[E_A:E_B=80:100]$

$\sqrt[4:5]$

22. CONVERT EFFICIENCY CHANGE TO TIME CHANGE

If efficiency increases by $\sqrt[p]{}$:

$\sqrt[T_{\text{new}} = T_{\text{old}} \frac{100}{100+p}]$

Thus percentage reduction in time:

$\sqrt[\frac{100p}{100+p}\%]$

23. EXAMPLE — 25% MORE EFFICIENT

Efficiency increases 25%.

New time:

$$\left[\frac{100}{125} = 0.8 \right]$$

So time decreases by:

$$\left[20\% \right]$$

Thus:

$$\left[\boxed{25\% \text{ more efficient}} \rightarrow 20\% \text{ less time} \right]$$

24. EFFICIENCY DECREASE !' TIME INCREASE

If efficiency falls by $(p\%)$:

$$\left[T_{\text{new}} = T_{\text{old}} \frac{100}{100-p} \right]$$

Time increase percentage:

$$\left[\boxed{\frac{100p}{100-p}\%} \right]$$

25. EXAMPLE

Efficiency decreases by 20%.

New time factor:

$$\left[100/80 = 1.25 \right]$$

So time rises:

$$\left[\boxed{25\%} \right]$$

26. TIME REDUCTION !' EFFICIENCY INCREASE

If time decreases by $(p\%)$:

$$\left[E_{\text{new}} = E_{\text{old}} \frac{100}{100-p} \right]$$

So efficiency increase:

$$\left[\boxed{\frac{100p}{100-p}\%} \right]$$

Example:

20% less time:

$$\left[\frac{20}{80} \times 100 = \boxed{25\% \text{ more efficient}} \right]$$

27. TIME INCREASE !' EFFICIENCY DECREASE

If time increases by $(p\%)$:

$$\left[E_{\text{new}} = E_{\text{old}} \frac{100}{100+p} \right]$$

Efficiency decrease percentage:

$$\boxed{\frac{100p}{100+p}\%}$$

28. EXAMPLE

Time increases by 25%.

Efficiency factor:

$$\boxed{1/1.25=0.8}$$

So efficiency decreases:

$$\boxed{20\%}$$

29. HIGH-VALUE CONVERSION PAIRS

Remember:

$$\boxed{25\% \uparrow E \rightarrow 20\% \downarrow T}$$

$$\boxed{20\% \downarrow E \rightarrow 25\% \uparrow T}$$

$$\boxed{50\% \uparrow E \rightarrow 33\frac{1}{3}\% \downarrow T}$$

$$\boxed{50\% \downarrow E \rightarrow 100\% \uparrow T}$$

30. FIND EFFICIENCY PERCENTAGE FROM TIMES

A takes 12 days.

B takes 15 days.

Efficiency:

$$\boxed{E_A:E_B=15:12=5:4}$$

A is more efficient than B by:

$$\boxed{\frac{5-4}{4} \times 100 = 25\%}$$

31. FIND TIME DIFFERENCE FROM EFFICIENCY

A is 50% more efficient than B.

B takes 18 days.

Efficiency:

$$\boxed{A:B=3:2}$$

Time:

$$\boxed{A:B=2:3}$$

A time:

$$\boxed{18 \times \frac{2}{3} = 12 \text{ days}}$$

Difference:

$$\boxed{6 \text{ days}}$$

32. PRODUCTIVITY INCREASE WITH SAME TEAM SIZE

A factory team produces 400 units/day.

Productivity increases by 15%.

New output:

$$400(1.15) = \boxed{460}$$

Same idea as efficiency.

33. TEAM SIZE AND PRODUCTIVITY

Suppose 10 workers each produce 8 units/hour.

Team output:

$$10(8) = 80 \text{ units/hour}$$

If 5 more identical workers join:

$$15(8) = 120$$

Increase:

$$40$$

Percentage:

$$40/80 \times 100 = \boxed{50\%}$$

34. WORKER EQUIVALENCE

Suppose:

$$2M = 3W$$

This means their **total work capacities are equal**.

Thus:

$$2E_M = 3E_W$$

So:

$$\boxed{E_M : E_W = 3 : 2}$$

35. ONE MAN EQUALS HOW MANY WOMEN?

From:

$$2M = 3W$$

divide by 2:

$$\boxed{1M = 1.5W}$$

36. MEN–WOMEN–CHILDREN CHAIN

Suppose:

$$\backslash[2M=3W \backslash]$$

and:

$$\backslash[2W=5C \backslash]$$

Find:

$$\backslash[M:W:C \backslash]$$

From first:

$$\backslash[M:W=3:2 \backslash]$$

From second:

$$\backslash[W:C=5:2 \backslash]$$

Make W common.

$$\backslash[M:W=15:10 \backslash]$$

$$\backslash[W:C=10:4 \backslash]$$

Therefore:

$$\backslash[\boxed{M:W:C=15:10:4} \backslash]$$

37. CONVERT MIXED WORKFORCE

Using:

$$\backslash[M:W:C=15:10:4 \backslash]$$

If team has:

- 2 men
- 3 women
- 5 children

Equivalent efficiency units:

$$\backslash[2(15)+3(10)+5(4) \backslash]$$

$$\backslash[=30+30+20 \backslash]$$

$$\backslash[=\boxed{80} \backslash]$$

38. FIND TIME FROM MIXED WORKFORCE

Suppose total work = 800 efficiency-units-days.

Team efficiency:

$$\backslash[80 \backslash]$$

Time:

$$\left[800/80 = \boxed{10 \text{ days}} \right]$$

39. MACHINE EQUIVALENCE

Same logic applies to machines.

If:

$$\left[3M_1 = 5M_2 \right]$$

then one (M_1) machine is more efficient:

$$\left[E_1 : E_2 = 5 : 3 \right]$$

No new mathematics.

40. WORKER + MACHINE PRODUCTIVITY

If worker produces 4 units/hour and machine produces 10 units/hour, together:

$$\left[14 \text{ units/hour} \right]$$

If they operate simultaneously and independently:

$$\left[\boxed{\text{Rates add}} \right]$$

41. FIND CONTRIBUTION IN TEAM

A, B, C rates:

$$\left[2, 3, 5 \right]$$

Total rate:

$$\left[10 \right]$$

A's share of each day's work:

$$\left[2/10 = 20\% \right]$$

B:

$$\left[30\% \right]$$

C:

$$\left[50\% \right]$$

If total wage = ₹ 20,000 :

$$\left[\boxed{₹ 4000, ₹ 6000, ₹ 10000} \right]$$

42. CONTRIBUTION WHEN TIMES DIFFER

Rates:

$$\left[A : B : C = 2 : 3 : 5 \right]$$

Working times:

$$\sqrt[5]{5, \sqrt[4]{4}, \sqrt[2]{2}}$$

Contributions:

$$\sqrt[2]{2(5):3(4):5(2)}$$

$$\sqrt[10]{10:12:10}$$

$$\sqrt[5]{\boxed{5:6:5}}$$

43. TOTAL WAGE FROM ONE SHARE

Contribution ratio:

$$\sqrt[5]{5:6:5}$$

B receives:

$$\sqrt[19000]{}$$

$$6 \text{ parts} = 19000.$$

1 part:

$$\sqrt[11500]{}$$

Total parts:

$$\sqrt[16]{}$$

Total wages:

$$\sqrt[16]{1500} = \boxed{\sqrt[124]{1000}}$$

44. BONUS + WAGE SHARE

Suppose total payment $120,000$.

A gets fixed bonus 12000 .

Remaining $118,000$ shared by work contribution ratio:

$$\sqrt[2:1]{}$$

A's work share:

$$\sqrt[12000]{}$$

A total:

$$\sqrt[14000]{}$$

B:

$$\sqrt[6000]{}$$

$$\sqrt[\boxed{A = 148000}]{}$$

45. SALARY VS PERFORMANCE WAGE

If problem says a worker gets:

- fixed wage,
- plus performance share,

handle them separately.

Do not mix the fixed amount into work ratio.

Same principle as partnership salary.

46. PRODUCTIVITY AFTER TRAINING

Worker A's rate = 20 units/hour.

After training, efficiency rises 30%.

New rate:

$$[20(1.3) = \boxed{26 \text{ units/hour}}]$$

If same task previously took 13 hours:

New time:

$$[13/1.3 = \boxed{10 \text{ hours}}]$$

47. MACHINE UPGRADE

Machine completes a batch in 15 hours.

Upgrade increases efficiency by 50%.

New time:

$$[15 \times \frac{100}{150}]$$

$$[= \boxed{10 \text{ hours}}]$$

48. TEAM EFFICIENCY CHANGE WHEN ONE MEMBER REPLACED

A team has workers with rates:

$$[4, 5, 6]$$

Total rate:

$$[15]$$

Worker of rate 5 is replaced by rate 8.

New rate:

$$[4+8+6=18]$$

Efficiency increase:

$$[\frac{18-15}{15} \times 100 = \boxed{20\%}]$$

49. RESULTING TIME CHANGE

If team efficiency rises 20%, completion time becomes:

$$\left[\frac{100}{120} = \frac{5}{6} \right]$$

of old time.

So time decreases by:

$$\left[\boxed{16\frac{2}{3}\%} \right]$$

50. TEAM MEMBER CONTRIBUTION PERCENTAGE

Team rates:

$$\left[4, 5, 6 \right]$$

Total:

$$\left[15 \right]$$

Rate-6 worker contributes:

$$\left[\frac{6}{15} \times 100 = \boxed{40\%} \right]$$

if all work simultaneously for same duration.

51. GATE-STYLE QUESTION 1

A takes 12 days.

B takes 18 days.

Efficiency ratio:

$$\left[18:12 = \boxed{3:2} \right]$$

52. GATE-STYLE QUESTION 2

A:B efficiency = 4:3.

A works 6 days, B 8 days.

Contribution:

$$\left[24:24 \right]$$

$$\left[= \boxed{1:1} \right]$$

53. GATE-STYLE QUESTION 3

A is 25% more efficient than B.

If B takes 20 days:

$$\left[A:B \text{ time} = 4:5 \right]$$

A:

$$\left[\boxed{16 \text{ days}} \right]$$

54. GATE-STYLE QUESTION 4

Efficiency decreases 20%.

Time increases by:

$$\left[\frac{20}{80} \times 100 \right]$$

$$\left[= \boxed{25\%} \right]$$

55. PLACEMENT-STYLE QUESTION 1

A completes job in 10 days.

B in 15 days.

Both work same duration.

Wage ratio:

$$\left[15:10 = \boxed{3:2} \right]$$

56. PLACEMENT-STYLE QUESTION 2

A:B efficiency = 3:2.

A works 6 days.

B works 3 days.

Contribution:

$$\left[18:6 = \boxed{3:1} \right]$$

57. PLACEMENT-STYLE QUESTION 3

A is 50% more efficient than B.

B takes 24 days.

A:

$$\left[24 \times \frac{2}{3} = \boxed{16 \text{ days}} \right]$$

58. PLACEMENT-STYLE QUESTION 4

If:

$$\left[2M = 3W \right]$$

then 6 men equal:

$$\left[\boxed{9 \text{ women}} \right]$$

59. ADVANCED QUESTION — CONTRIBUTION

A, B, C take:

$\sqrt[12]{18, 36}$ days

They work together for the full job.

Efficiency ratio:

$\sqrt{\frac{1}{12}:\frac{1}{18}:\frac{1}{36}}$

Multiply by 36:

$\sqrt{3:2:1}$

So wage ratio:

$\sqrt{\boxed{3:2:1}}$

60. ADVANCED QUESTION — DIFFERENT TIMES

A:B efficiencies:

$\sqrt{5:4}$

A works 6 days.

B works 5 days.

Contribution:

$\sqrt{30:20 = 3:2}$

If total wages 11,500 :

A:

$\sqrt{11500 \times \frac{3}{5}} = \boxed{1600}$

B:

$\sqrt{\boxed{1400}}$

61. ADVANCED QUESTION — FIND TIME RATIO

A earns twice B's wage.

A:B efficiency:

$\sqrt{3:2}$

Assuming wages follow work:

$\sqrt{3T_A:2T_B=2:1}$

$\sqrt{3T_A=4T_B}$

$\sqrt{T_A:T_B = \boxed{4:3}}$

62. ADVANCED QUESTION — FIND EFFICIENCY

A and B work for 6 and 8 days respectively and receive equal wages.

Then:

$$E_A(6) = E_B(8)$$

$$E_A : E_B = 8 : 6$$

$$\boxed{4:3}$$

63. ADVANCED QUESTION — FIXED BONUS

A:B work contribution = 3:2.

Total payment = ₹ 29,000.

A gets ₹ 5000 fixed bonus first.

Remaining:

$$24000$$

A work share:

$$14400$$

A total:

$$19400$$

B:

$$9600$$

$$\boxed{19400, 9600}$$

64. ADVANCED QUESTION — TEAM REPLACEMENT

Four workers have efficiencies:

$$2, 3, 4, 5$$

Total:

$$14$$

Efficiency-3 worker is replaced by efficiency-6 worker.

New total:

$$17$$

Increase:

$$3$$

Percentage increase:

$$\frac{3}{14} \times 100$$

$$\approx \boxed{21.43\%}$$

65. COMMON TRAPS

& pTRAP 1 — Efficiency and time same ratio

Wrong.

They are inverse.

& pTRAP 2 — Wage ratio = efficiency ratio always

Only when working times are equal.

General:

$$\boxed{E \times T}$$

& pTRAP 3 — More efficient means same percentage less time

False due to reciprocal relation.

& pTRAP 4 — Equal wages imply equal efficiency

Not necessarily; one may have worked longer.

& pTRAP 5 — Worker equivalence ratio

From:

$$\boxed{2M=3W}$$

one man's efficiency is $\frac{3}{2}$ of one woman's.

66. MUST-MEMORIZE FORMULAS

Efficiency-time

$$\boxed{E_A:E_B=T_B:T_A}$$

Contribution

$$\boxed{W_A:W_B=E_A T_A:E_B T_B}$$

Wages

$$\boxed{\text{Wage ratio} = \text{Work contribution ratio}}$$

Efficiency increase $(p\%)$

$$\boxed{T_{\text{new}} = T_{\text{old}} \frac{100}{100+p}}$$

Efficiency decrease $(p\%)$

$$\boxed{T_{\text{new}} = T_{\text{old}} \frac{100}{100-p}}$$

67. FAST EXAM DECISION TREE

Individual completion times?

Reverse them for efficiency ratio.

Same work duration?

Wages follow efficiency.

Different durations?

Use $(E \times T)$.

Efficiency change?

Use reciprocal time multiplier.

Mixed workers?

Convert all to one efficiency unit.

Fixed bonus/salary?

Separate it from performance wage.

PRACTICE — LEVEL 0

Q1

A takes 10 days, B 15 days. Find efficiency ratio.

Q2

Efficiency A:B = 5:4. Both work same duration. Find work ratio.

Q3

Efficiency A:B = 3:2. A works 4 days, B 6 days. Find contribution ratio.

Q4

A is 25% more efficient than B. B takes 20 days. Find A's time.

Q5

If $(2M=3W)$, find $(M:W)$ efficiency ratio.

LEVEL 1

Q6

A = 8 days, B = 12 days. Total wages = ₹ 5000 Both work full job together. Find shares.

Q7

A:B efficiency = 4:3. A works 6 days, B 4 days. Find contribution ratio.

Q8

A's efficiency falls by 20%. Old time was 16 days. Find new time.

Q9

A takes 25% less time than B. How much more efficient is A?

Q10

A:B:C efficiencies = 2:3:5. Total wages ₹ 20,000 and all work equal time. Find shares.

LEVEL 2**Q11**

A works 5 days, B 8 days and they receive equal wages. Find efficiency ratio.

Q12

A:B efficiency = 5:4. A earns 50% more wages than B. Find $(T_A:T_B)$.

Q13

Worker efficiency increases 40%. By what percentage does time fall?

Q14

If $(2M=3W)$ and $(2W=5C)$, find $(M:W:C)$.

Q15

A:B:C rates are 4:5:6. All work same duration. What percentage of work is done by C?

LEVEL 3 — EXAM PRESSURE**Q16**

A, B, C take 12, 18, 36 days individually. Total wage is ₹ 18,000 and all work together throughout. Find each share.

Q17

A:B efficiencies = 3:2. A works 5 days and B 9 days. Total wages ₹ 9600. Find shares.

Q18

A team efficiency rises from 15 units/day to 18 units/day. By what percentage does completion time decrease?

Q19

A and B receive wages in ratio 5:4. Their efficiencies are 3:2. Find working-time ratio.

Q20

A gets ₹ 300 fixed bonus. Remaining wages are shared with B in contribution ratio 4:3. A's total receipt is ₹ 1110. Find total wage pool.

ANSWERS**Q1**

$\sqrt{15:10} = \boxed{3:2}$

Q2

Same time:

$$\boxed{5:4}$$

Q3

$$3(4):2(6) = 12:12 = \boxed{1:1}$$

Q4

$$A:B_{\text{efficiency}}=5:4$$

Time ratio:

$$4:5$$

$$A=20\times\frac{4}{5} = \boxed{16}$$

Q5

$$2E_M=3E_W$$

$$\boxed{3:2}$$

Q6

Times 8 and 12:

$$E_A:E_B=12:8=3:2$$

$$^15000:$$

$$\boxed{A = ^13000,000}$$

Q7

$$4(6):3(4)$$

$$24:12$$

$$=\boxed{2:1}$$

Q8

Efficiency becomes:

$$80\%$$

Time:

$$16/0.8 = \boxed{20\text{ days}}$$

Q9

If A takes 25% less time:

$$T_A:T_B=75:100=3:4$$

Efficiency:

$$4:3$$

A is more efficient by:

$$\frac{13}{3}\times 100 = \boxed{33\frac{1}{3}\%}$$

Q10

Ratio:

$\backslash [2:3:5 \backslash]$

Total parts:

$\backslash [10 \backslash]$

$$\boxed{140600\text{0}\text{0},1000\text{0}\text{p}}$$

Q11

Equal wages:

$$\backslash [5E_A=8E_B \backslash]$$

$$\boxed{E_A:E_B = 8:5}$$

Q12

A's wage is 50% more:

$$\backslash [W_A:W_B=3:2 \backslash]$$

Given efficiency:

$\backslash[5:4\backslash]$

$$\backslash [5T_A:4T_B=3:2 \backslash]$$

$$\backslash [10T_A = 12T_B \backslash]$$

$$\boxed{T_A:T_B=6:5}$$

Q13

Time factor:

$$\backslash [100/140 = 5/7 \backslash]$$

Reduction:

$$\left[1 - \frac{5}{7} = \frac{2}{7}\right]$$

$$\left[\boxed{28\frac{47}{100} \approx 28.57\%} \right]$$

Q14

From:

$$\backslash [2M=3W \]$$

\[M:W=3:2 \]

From:

$$\backslash [2W=5C \]$$

\[W:C=5:2 \]

Make W common:

$$\boxed{M:W:C=15:10:4}$$

Q15

Total rate:

$$\backslash[4+5+6=15\]$$

C contributes:

$$\backslash[6/15 = 40\%\]$$

$$\backslash[\boxed{40\%}\]$$

Q16

Efficiencies:

$$\backslash[\frac{1}{12}:\frac{1}{18}:\frac{1}{36} = 3:2:1\]$$

Total parts:

$$\backslash[6\]$$

$$^118,000:$$

$$\backslash[\boxed{^190000,3000}\]$$

Q17

Contribution:

$$\backslash[3(5):2(9)\]$$

$$\backslash[15:18 = 5:6\]$$

Total parts:

$$\backslash[11\]$$

A:

$$\backslash[9600\times\frac{5}{11} = \boxed{^14363.\overline{64}}\backslasht{}$$

B:

$$\backslash[\boxed{^15236.\overline{20}}\backslasht{}$$

Q18

Efficiency rises:

$$\backslash[15\text{to}18\]$$

Increase:

$$\backslash[20\%\]$$

Time factor:

$$\backslash[15/18 = 5/6\]$$

Decrease:

$$\backslash[1/6 = \boxed{16\frac{2}{3}\%}\]$$

Q19

Wages:

$$\backslash[5:4\]$$

Efficiency:

$$\backslash[3:2\]$$

$$\sqrt[3]{T_A:T_B=5:4}$$

$$\sqrt[12]{T_A=10T_B}$$

$$\sqrt{\boxed{T_A:T_B=5:6}}$$

Q20

A's total:

$$\sqrt[1]{11000}$$

Bonus:

$$\sqrt[1]{3000}$$

So A's performance share:

$$\sqrt[1]{8000}$$

Contribution ratio:

$$\sqrt[4]{4:3}$$

$$4 \text{ parts} = \sqrt[1]{8000}$$

1 part:

$$\sqrt[1]{2000}$$

Performance wage pool:

$$\sqrt[7]{(2000)} = \sqrt[1]{14000}$$

Add bonus:

$$\sqrt{\boxed{\text{Total wage pool} = \sqrt[1]{17000}}}$$

CHAPTER 27 STATUS

Covered so far:

- efficiency fundamentals '
- inverse efficiency-time relation '
- productivity comparison '
- equal/different working times '
- work contribution '
- basic wage sharing '
- wage sharing with late joining/early leaving '
- percentage efficiency changes '
- time-efficiency percentage conversions '
- worker equivalence '
- men/women/children equivalence '
- mixed-team productivity '
- fixed bonus + performance pay '
- GATE/placement drills '

But **Chapter 27 is not complete yet.**

Next we need:

- combined/team efficiency
- one worker's contribution when team completion time is given
- efficiency after worker replacement
- productivity targets and required staff
- wage sharing when work phases differ
- contractor/subcontractor style questions
- piece-rate and hourly-rate basics
- advanced contribution equations
- final speed shortcuts and chapter audit

Progress

Completed: 26/76 **Current:** Chapter 27 — Efficiency & Work-Wages **Next:** Chapter 27 Part 2 — Team efficiency, productivity targets & advanced wage-contribution problems

USER

next

ASSISTANT

CHAPTER 27 — EFFICIENCY & WORK-WAGES

PART 2 — TEAM EFFICIENCY, PRODUCTIVITY TARGETS, WORKER REPLACEMENT & ADVANCED WAGE-CONTRIBUTION PROBLEMS

Status: $\emptyset$ **IN PROGRESS** **GATE:** High **Placements:** High **Priority:** $\emptyset$ **Yr 1**

This part focuses on the harder questions where:

- a team has several workers,
- one worker joins/leaves/is replaced,
- productivity targets must be met,
- work happens in phases,
- wage sharing depends on actual contribution,
- or a worker's unknown efficiency must be found.

The master relation remains:

$$\boxed{\text{Work contribution}} = E \times T$$

1. TEAM EFFICIENCY

If workers have efficiencies:

$$E_1, E_2, \dots, E_n$$

and work together simultaneously, team efficiency is:

$$\boxed{E_{\text{team}} = E_1 + E_2 + \dots + E_n}$$

Example:

$$A=3, \text{quad } B=5, \text{quad } C=2$$

Team efficiency:

$$\boxed{10}$$

units/day.

2. TEAM COMPLETION TIME

If total work is (W) :

$$\boxed{T = \frac{W}{E_{\text{team}}}}$$

Example:

Total work = 120 units.

Team rate = 10 units/day.

$$T = 120/10$$

$$= \boxed{12 \text{ days}}$$

3. FIND INDIVIDUAL EFFICIENCY FROM TEAM RATE

A and B together produce:

$$20 \text{ units/hour}$$

A alone produces:

$$12$$

Then B:

$$20 - 12 = \boxed{8 \text{ units/hour}}$$

Same principle as subtracting work rates.

4. FIND INDIVIDUAL TIME FROM TEAM TIME

A+B complete work in 6 days.

A alone = 10 days.

Let total work be LCM:

$$30$$

Team rate:

$$30/6 = 5$$

A rate:

$$30/10 = 3$$

B rate:

$$\sqrt[2]{}$$

So B alone takes:

$$\sqrt[30/2 = \boxed{15 \text{ days}}]{}$$

5. CONTRIBUTION PERCENTAGE IN A TEAM

If efficiencies:

$$\sqrt[A:B:C=2:3:5]{}$$

total:

$$\sqrt[10]{}$$

Then contribution percentages:

$$\sqrt[A=20\%]{}$$

$$\sqrt[B=30\%]{}$$

$$\sqrt[C=50\%]{}$$

provided all work the same time.

6. FIND PAYMENT FROM CONTRIBUTION PERCENTAGE

Total wages:

$$\sqrt[130,000]{}$$

Worker C contributes 50%.

So:

$$\sqrt[C=15000]{}$$

Remaining:

$$\sqrt[15000]{}$$

If A and B contribute 20%,30%:

$$\sqrt[A=6000, \text{quad } B=9000]{}$$

$$\sqrt[\boxed{16000, 1500}]{}$$

7. TEAM MEMBER LEAVES MIDWAY

Suppose efficiencies:

$$\sqrt[A:B:C=2:3:5]{}$$

All work 4 days, then C leaves.

A and B work another 6 days.

Contribution:

A:

$$\sqrt[2]{2(4+6)}=20$$

B:

$$\sqrt[3]{3(10)}=30$$

C:

$$\sqrt[5]{5(4)}=20$$

Contribution ratio:

$$\sqrt[20]{20:30:20}$$

$$\sqrt[2]{\boxed{2:3:2}}$$

This is also the fair wage ratio.

8. TEAM MEMBER JOINS LATE

A:B:C efficiencies:

$$\sqrt[4]{4:3:2}$$

A and B work from start for 8 days.

C joins after 3 days, so C works:

$$\sqrt[5]{\text{days}}$$

Contributions:

$$\sqrt[A]{A=4(8)}=32$$

$$\sqrt[B]{B=3(8)}=24$$

$$\sqrt[C]{C=2(5)}=10$$

Ratio:

$$\sqrt[32]{32:24:10}$$

$$\sqrt[16]{\boxed{16:12:5}}$$

9. ONE WORKER REPLACED

Team rates:

$$\sqrt[4,5,6]$$

Total old rate:

$$\sqrt[15]$$

Rate-5 worker is replaced by a rate-8 worker.

New rate:

$$\sqrt[4+8+6]{4+8+6}=18$$

Increase:

$$\sqrt[3]$$

Percentage increase:

$$\left[\frac{3}{15} \times 100 = \boxed{20\%} \right]$$

10. EFFECT ON COMPLETION TIME

If team efficiency increases by 20%:

$$\left[T_{\text{new}} = T_{\text{old}} \frac{100}{120} \right]$$

$$\left[= \frac{5}{6} T_{\text{old}} \right]$$

So time falls by:

$$\left[\boxed{16\frac{2}{3}\%} \right]$$

11. FIND NEW TEAM TIME

Old team completes job in 12 days.

Efficiency rises 20%.

New time:

$$\left[12 \times \frac{5}{6} = \boxed{10 \text{ days}} \right]$$

12. FIND REQUIRED EFFICIENCY FOR TARGET TIME

A team currently completes work in 15 days.

Management wants it completed in 12 days.

For same work:

$$\left[E_{\text{old}}(15) = E_{\text{new}}(12) \right]$$

Thus:

$$\left[\frac{E_{\text{new}}}{E_{\text{old}}} = \frac{15}{12} = 1.25 \right]$$

Required efficiency increase:

$$\left[\boxed{25\%} \right]$$

13. TARGET-TIME SHORTCUT

If old time = (T_o) , target time = (T_n) :

$$\left[\boxed{\frac{E_n}{E_o} = \frac{T_o}{T_n}} \right]$$

Required % increase:

$$\left[\boxed{\left(\frac{T_o}{T_n} - 1 \right) 100} \right]$$

14. EXAMPLE

Old time:

$$\backslash[20\text{ days} \backslash]$$

Target:

$$\backslash[16\text{ days} \backslash]$$

Efficiency factor:

$$\backslash[20/16=1.25 \backslash]$$

Need:

$$\backslash[\boxed{25\% \text{ productivity increase}} \backslash]$$

15. PRODUCTIVITY TARGET WITH MORE WORKERS

10 identical workers finish in 18 days.

How many workers are needed to finish in 12 days?

$$\backslash[10(18)=M(12) \backslash]$$

$$\backslash[M=15 \backslash]$$

Additional:

$$\backslash[15-10 = \boxed{5} \backslash]$$

16. TARGET WITH BETTER WORKERS

10 workers take 18 days.

New workers are 20% more efficient.

How many new workers are needed to finish in 12 days?

$$\backslash[10(18)(1) = M(12)(1.2) \backslash]$$

$$\backslash[180=14.4M \backslash]$$

$$\backslash[M=12.5 \backslash]$$

If workers must be whole people, at least:

$$\backslash[\boxed{13} \backslash]$$

assuming 12.5 workers is not physically allowed.

17. INTEGER HEADCOUNT TRAP

If computation gives:

$$\backslash[12.2\text{ workers} \backslash]$$

and the question asks minimum workers needed:

$$\backslash[\boxed{13} \backslash]$$

because 12 workers are insufficient.

But if the question asks an algebraic equivalent workforce, 12.2 may be acceptable.

18. PRODUCTIVITY TARGET IN UNITS

A team produces 480 units/day.

Target = 600 units/day.

Required increase:

$$\backslash[600-480=120 \backslash]$$

Percentage:

$$\backslash[120/480\times100 = \boxed{25\%} \backslash]$$

19. STAFF REQUIREMENT FROM UNIT OUTPUT

Each worker produces 8 units/hour.

Target team output:

$$\backslash[200\text{ units/hour} \backslash]$$

Workers required:

$$\backslash[200/8=25 \backslash]$$

$$\backslash[\boxed{25} \backslash]$$

20. MIXED WORKERS FOR TARGET OUTPUT

Skilled worker:

$$\backslash[10\text{ units/hour} \backslash]$$

Trainee:

$$\backslash[6\text{ units/hour} \backslash]$$

Team has 8 skilled and 5 trainees.

Output:

$$\backslash[8(10)+5(6) \backslash]$$

$$\backslash[=80+30 \backslash]$$

$$\backslash[=\boxed{110\text{ units/hour}} \backslash]$$

21. FIND TRAINEES NEEDED

Same rates.

6 skilled workers already present.

Target:

$$\backslash[120\text{ units/hour} \backslash]$$

Skilled output:

$$\sqrt[6]{6(10)=60}$$

Remaining:

$$\sqrt[6]{60}$$

Each trainee:

$$\sqrt[6]{6}$$

Trainees:

$$\sqrt[6]{60/6 = \boxed{10}}$$

22. WORK IN PHASES

Suppose:

- A works 4 days
- A+B work 3 days
- B alone works 2 days

Efficiencies:

$$\sqrt[A:B=3:2]$$

A contribution:

$$\sqrt[3(4)+3(3) = 12+9 = 21]$$

B contribution:

$$\sqrt[2(3)+2(2) = 6+4 = 10]$$

Contribution ratio:

$$\sqrt[\boxed{21:10}]$$

Wages should follow this ratio.

23. PHASE TABLE METHOD

For complex questions:

Phase	Workers	Days	Contribution
1	A	4	$\sqrt[3 \times 4 = 12]$
2	A+B	3	A=9, B=6
3	B	2	4

Totals:

$$\sqrt[A=21, \text{quad } B=10]$$

This prevents omissions.

24. FIND WAGES FROM PHASED CONTRIBUTION

If total wages:

$\text{\pounds } 16200$

and ratio:

$21:10$

Total parts:

31

One part:

200

A:

$\text{\pounds } 14200$

B:

$\text{\pounds } 2000$

$\boxed{\text{\pounds } 142000}$

25. CONTRACTOR + SUBCONTRACTOR STYLE

Suppose contractor A takes a job worth $\text{\pounds } 30,000$.

A and B actually contribute work in ratio:

$3:2$

If payment is purely based on work contribution:

A:

$\text{\pounds } 18000$

B:

$\text{\pounds } 12000$

If A separately receives a fixed contractor fee, handle that fee before dividing the remaining pool.

26. FIXED FEE + CONTRIBUTION SHARE

Total payment:

$\text{\pounds } 50,000$

A gets fixed supervision fee:

$\text{\pounds } 5000$

Remaining:

$\text{\pounds } 45000$

Contribution ratio A:B:

$\frac{2}{3}$

A contribution share:

18000

A total:

23000

B:

27000

$\boxed{\frac{2}{3} \times 27000}$

27. PIECE-RATE WAGES

Piece-rate means payment per unit produced.

Example:

8p per item.

A produces 120 items.

Wage:

$120(8) = \boxed{960}$

No time-and-work ratio needed unless comparing productivity.

28. HOURLY WAGES

If:

- A earns 1100 / hour
- B earns 1120 / hour

Then:

A=600

B=600

Equal pay does not imply equal efficiency.

Wage system matters.

29. PERFORMANCE WAGE VS HOURLY WAGE

If question says wages are distributed **according to work done**, use:

$E \times T$

If explicit hourly pay rates are given, use:

$\text{Hourly wage} \times \text{hours}$

Do not mix the models.

30. FIND UNKNOWN EFFICIENCY FROM WAGES

A and B work:

$$\left[\begin{matrix} 5 \text{ days} \\ 8 \text{ days} \end{matrix} \right]$$

Wages ratio:

$$\left[\begin{matrix} 3:4 \end{matrix} \right]$$

Assuming wage proportional to work:

$$\left[\begin{matrix} 5E_A:8E_B=3:4 \end{matrix} \right]$$

Cross multiply:

$$\left[\begin{matrix} 20E_A=24E_B \end{matrix} \right]$$

$$\left[\begin{matrix} E_A:E_B = \boxed{6:5} \end{matrix} \right]$$

31. FIND UNKNOWN TIME FROM WAGES

A:B efficiencies:

$$\left[\begin{matrix} 4:3 \end{matrix} \right]$$

Wages:

$$\left[\begin{matrix} 8:9 \end{matrix} \right]$$

A works 6 days.

Let B work t days.

$$\left[\begin{matrix} 4(6):3t = 8:9 \end{matrix} \right]$$

$$\left[\begin{matrix} 24:3t=8:9 \end{matrix} \right]$$

$$\left[\begin{matrix} 216=24t \end{matrix} \right]$$

$$\left[\begin{matrix} \boxed{t=9 \text{ days}} \end{matrix} \right]$$

32. FIND UNKNOWN WAGE SHARE

A:B efficiencies:

$$\left[\begin{matrix} 5:3 \end{matrix} \right]$$

A works 4 days.

B works 10 days.

Contribution:

$$\left[\begin{matrix} 20:30 = 2:3 \end{matrix} \right]$$

If A gets 14000 :

B:

$$\left[\begin{matrix} 4000 \times \frac{32}{3} = \boxed{16000} \end{matrix} \right]$$

$$\boxed{b:a}$$

So wage ratio:

$$\boxed{b:a}$$

No need to calculate combined time.

36. EXAMPLE

A = 12 days.

B = 20 days.

Wage ratio:

$$\boxed{20:12 = 5:3}$$

If total wages = 16,000 :

$$\boxed{A = 110,000}$$

$$\boxed{B = 6000}$$

37. FIND UNKNOWN INDIVIDUAL TIME FROM WAGE RATIO

A and B work together throughout.

Wages:

$$\boxed{3:2}$$

A alone takes 10 days.

Then efficiency:

$$\boxed{3:2}$$

Time ratio:

$$\boxed{2:3}$$

So:

$$\boxed{10:T_B=2:3}$$

$$\boxed{T_B=15}$$

$$\boxed{15\text{ days}}$$

38. TEAM PRODUCTIVITY AFTER ABSENTEEISM

20 identical workers produce 1000 units/day.

Each worker contributes:

$$\boxed{50\text{ units/day}}$$

If 4 are absent:

Workers:

$\sqrt[16]{}$

Output:

$\sqrt[16]{50} = \boxed{800 \text{ units/day}}$

Decrease:

$\sqrt[20]{\%}$

39. TEAM PRODUCTIVITY AFTER TRAINING + ABSENCE

20 workers initially equal.

After training, each worker becomes 25% more productive.

But 4 workers are absent.

Effective workforce:

$\sqrt[16]{1.25} = 20$

So total output is exactly equal to old 20-worker output.

$\boxed{\text{No net change}}$

This is a useful multiplier question.

40. REQUIRED EFFICIENCY TO OFFSET FEWER WORKERS

20 workers reduce to 16.

How much must each remaining worker's efficiency increase to preserve same output?

Old total:

$\sqrt[20]{E}$

New:

$\sqrt[16]{E'}$

Set equal:

$\sqrt[16]{E'} = \sqrt[20]{E}$

$E' = \frac{54}{E}$

Increase:

$\boxed{25\%}$

41. REQUIRED STAFF TO OFFSET LOWER EFFICIENCY

10 workers have efficiency $\sqrt[E]{}$.

New workers are 20% less efficient:

$\sqrt[0.8]{E}$

Number needed for same output:

$$\boxed{M(0.8E)=10E}$$

$$\boxed{M=12.5}$$

Minimum whole workers:

$$\boxed{13}$$

42. SAME OUTPUT WITH HOURS CHANGE

10 workers work 8 hours/day.

New schedule is 5 hours/day.

Required workers for same daily output:

$$10(8)=M(5)$$

$$\boxed{M=16}$$

$$\boxed{16}$$

43. HOURS + EFFICIENCY + STAFF

Old:

$$10 \text{ workers} \times 8 \times E$$

New workers:

- 20% more efficient
- work 5 hours/day

Find number M for same daily output:

$$10(8)(1) = M(5)(1.2)$$

$$80 = 6M$$

$$\boxed{M=13\frac{1}{3}}$$

Minimum integer:

$$\boxed{14}$$

44. ADVANCED CONTRIBUTION EQUATION

Suppose A+B+C complete a job.

A and B work 6 days.

C works only 3 days.

Efficiencies:

$$\boxed{A:B:C=2:3:4}$$

Contributions:

$$\boxed{12:18:12}$$

$$\boxed{2:3:2}$$

Even though C is most efficient, shorter working time reduces total contribution.

45. FIND WHO DID MOST WORK

Always compare:

$$E \times T$$

not efficiency alone.

Example:

A efficiency 5, time 4:

$$20$$

B efficiency 3, time 8:

$$24$$

B did more work.

46. CONTRIBUTION PERCENTAGE

A contributes 20 units.

B contributes 24 units.

Total:

$$44$$

A %:

$$\frac{20}{44} \times 100 = 45.45\%$$

B:

$$54.55\%$$

47. PRODUCTIVITY INDEX QUESTIONS

Sometimes a problem gives “productivity index”.

Treat it like relative efficiency.

If:

$$A=120, \quad B=80$$

then ratio:

$$120:80 = \boxed{3:2}$$

Everything else works normally.

48. GATE-STYLE QUESTION 1

A+B team rate = 15 units/day.

B rate:

Efficiency ratio:

$$\boxed{3:2}$$

Team efficiency rises by 25%.

New:

$$\boxed{16 \text{ days}}$$

A:B efficiencies:

A works 5 days.

B works 9 days.

Contribution:

$$\boxed{5:6}$$

20 workers fall to 16 workers.

Required efficiency increase for same output:

$$\boxed{25\%}$$

A:B:C efficiencies = 2:3:5.

A and B work 10 days.

C works only 4 days.

Contribution:

\[20:30:20 \]

$$\boxed{2:3:2}$$

53. PLACEMENT-STYLE QUESTION 2

10 workers produce 600 units/day.

Per worker:

$$60$$

Target:

$$900$$

Workers required:

$$900/60 = \boxed{15}$$

54. PLACEMENT-STYLE QUESTION 3

A worker's efficiency increases 50%.

Old time = 18 h.

New:

$$18/1.5 = \boxed{12h}$$

55. PLACEMENT-STYLE QUESTION 4

A:B efficiencies:

$$4:3$$

A works 6 days, B 9 days.

Contribution:

$$24:27$$

$$\boxed{8:9}$$

56. ADVANCED QUESTION — FIND REQUIRED WORKER

A team has rates:

$$4,5,6$$

and completes work in 12 days.

A new worker joins, and completion time drops to 9 days.

Old team rate:

$$15$$

Total work:

$$15(12)=180$$

Required new team rate:

$$\sqrt[3]{180/9=20}$$

New worker's rate:

$$\sqrt[3]{20-15 = \boxed{5}}$$

57. ADVANCED QUESTION — WORKER REPLACEMENT

Old team rates:

$$\sqrt[3]{3,4,5}$$

Old total:

$$\sqrt[3]{12}$$

Rate-3 worker replaced by unknown $\sqrt[3]{x}$.

New completion time becomes 20% less.

If old time = $\sqrt[3]{T}$, new time = $\sqrt[3]{0.8T}$.

Efficiency must multiply by:

$$\sqrt[3]{1/0.8=1.25}$$

New team efficiency:

$$\sqrt[3]{12(1.25)=15}$$

So:

$$\sqrt[3]{x+4+5=15}$$

$$\sqrt[3]{x=\boxed{6}}$$

58. ADVANCED QUESTION — FIND BONUS + CONTRIBUTION

Total payment:

$$\sqrt[3]{134,000}$$

A gets fixed $\sqrt[3]{14000}$ bonus.

Remaining is shared A:B according to contributions.

A efficiency 3, works 4 days.

B efficiency 2, works 6 days.

Contributions:

$$\sqrt[3]{12:12 = 1:1}$$

Remaining:

$$\sqrt[3]{30000}$$

Each performance share:

$$\sqrt[3]{15000}$$

A total:

$$\sqrt[19000]{}$$

B:

$$\sqrt[15000]{}$$

$$\sqrt{\boxed{119,150000}}$$

59. ADVANCED QUESTION — UNKNOWN WORKING TIME

A:B efficiency:

$$\sqrt[5:4]{}$$

A receives 60% of performance wage.

Thus contribution ratio:

$$\sqrt[60:40 = 3:2]{}$$

If A works 6 days and B works $\sqrt{(t)}$:

$$\sqrt[5(6):4t=3:2]{}$$

$$\sqrt[30:4t=3:2]{}$$

$$\sqrt[60=12t]{}$$

$$\sqrt{\boxed{t=5\text{ days}}}$$

60. ADVANCED QUESTION — PRODUCTIVITY LOSS & RECOVERY

A team loses 20% workforce.

Remaining workers become 10% more efficient.

Output factor:

$$\sqrt[0.8(1.1) = 0.88]{}$$

So output becomes:

$$\sqrt[88\%]{}$$

Net reduction:

$$\sqrt{\boxed{12\%}}$$

61. COMMON TRAPS

& pTRAP 1 — Team rate after replacement

Subtract departing worker's rate and add incoming worker's rate.

& pTRAP 2 — Target time

Efficiency changes inversely with time.

& pTRAP 3 — Headcount answer

Minimum workers may require rounding up.

& pTRAP 4 — Wages across phases

Add each worker's contribution across all phases.

& pTRAP 5 — Fixed bonus

Bonus does not alter work ratio.

& pTRAP 6 — Higher efficiency means more total work

Not necessarily if working time is shorter.

Use:

$$\lfloor E \times T \rfloor$$

62. MUST-MEMORIZE FORMULAS

Team efficiency

$$\lfloor \boxed{E_{\text{team}} = \sum E_i} \rfloor$$

Target efficiency

$$\lfloor \boxed{\frac{E_n}{E_o} = \frac{T_o}{T_n}} \rfloor$$

Work contribution

$$\lfloor \boxed{W_i = E_i T_i} \rfloor$$

Wage ratio

$$\lfloor \boxed{\text{Wages} \propto E_i T_i} \rfloor$$

Same-output workforce relation

$$\lfloor \boxed{M_1 H_1 E_1 = M_2 H_2 E_2} \rfloor$$

for daily output comparisons.

63. FAST EXAM DECISION TREE

Team members all work together?

Add efficiencies.

One joins/leaves?

Update team rate from that point.

Target completion time?

Use inverse time-efficiency relation.

Wage sharing?

Compute each $(E \times T)$.

Several phases?

Build contribution table.

Fixed bonus?

Subtract first, divide remainder, add bonus back.

Target staffing?

Compute required total efficiency, then divide by per-worker efficiency.

PRACTICE — LEVEL 0

Q1

Workers have efficiencies 2, 3 and 5. Find team efficiency.

Q2

A team rate is 18 units/day. A alone contributes 7 units/day. Find remaining team's rate.

Q3

A:B efficiency = 4:3. Both work 5 days. Find contribution ratio.

Q4

Old time = 15 days, new target = 12 days. Find required efficiency increase.

Q5

10 workers produce 500 units/day. Find output per worker.

LEVEL 1

Q6

A:B efficiency = 3:2. A works 6 days, B 9 days. Find contribution ratio.

Q7

Old team rate 20 units/day. One rate-4 worker leaves and a rate-7 worker joins. Find % increase in team rate.

Q8

12 workers finish in 18 days. How many workers needed in 12 days?

Q9

Total wages = 114, 0 A:B contribution = 3:4. Find shares.

Q10

A works 5 days, B 8 days and receives equal wages. Find $(E_A:E_B)$.

LEVEL 2

Q11

A, B, C efficiencies = 2:3:5. A and B work 10 days, C works 4 days. Find wage ratio.

Q12

A team finishes in 20 days. After upgrade efficiency increases by 25%. Find new time.

Q13

20 workers reduce to 16. Find required % efficiency increase to maintain same output.

Q14

A:B efficiency = 5:4. A works 6 days. Their wage ratio is 3:2. Find B's working time.

Q15

A team of rates 4,5,6 finishes work in 12 days. A new worker joins and total time becomes 9 days. Find new worker's rate.

LEVEL 3 — EXAM PRESSURE

Q16

A, B, C rates are 3,4,5. A and B work for first 4 days, then C joins and all three work for next 6 days. Find contribution ratio.

Q17

Old team efficiencies are 3,4,5. The rate-3 worker is replaced and completion time falls by 20%. Find replacement worker's rate.

Q18

A team loses 25% workers but each remaining worker becomes 20% more efficient. Find net percentage change in output.

Q19

A gets ₹ 1500 fixed bonus. Remaining payment is shared with B in contribution ratio 2:3. A's total receipt is ₹ 13,000. Find total payment.

Q20

A:B efficiencies = 6:5. A receives twice B's wage. If A works 5 days, find B's working time.

ANSWERS

Q1

$$\sqrt{2+3+5} = \boxed{10}$$

Q2

$$\sqrt{18-7} = \boxed{11}$$

Q3

Equal time:

$$\boxed{4:3}$$

Q4

$$15/12=1.25$$

$$\boxed{25\%}$$

Q5

$$500/10 = \boxed{50 \text{ units/day}}$$

Q6

$$3(6):2(9) = 18:18 = \boxed{1:1}$$

Q7

Old:

$$20$$

New:

$$20-4+7=23$$

Increase:

$$3/20 \times 100 = \boxed{15\%}$$

Q8

$$12(18)=M(12)$$

$$M=18$$

$$\boxed{18}$$

Q9

Total parts:

$$7$$

A:

$$14000 \times \frac{37}{100} = \boxed{5180}$$

B:

$$\boxed{8820}$$

Q10

Equal wages:

$$5E_A=8E_B$$

$$\boxed{E_A:E_B=8:5}$$

Q11

Contributions:

$$20:30:20$$

$$\boxed{2:3:2}$$

Q12

$$20 \times \frac{100}{125} = \boxed{16 \text{ days}}$$

Q13

$$20/16 = 1.25$$

$$\boxed{25\%}$$

Q14

$$5(6):4t=3:2$$

$$30:4t=3:2$$

$$60=12t$$

$$\boxed{t=5 \text{ days}}$$

Q15

Old team rate:

$$15$$

Work:

$$15(12)=180$$

Required new rate:

$$180/9=20$$

New worker:

$$20-15 = \boxed{5}$$

Q16

A:

$$3(4+6)=30$$

B:

$$4(10)=40$$

C:

$$5(6)=30$$

Ratio:

$$30:40:30$$

$$\boxed{3:4:3}$$

Q17

Old team:

$$3+4+5=12$$

Time falls 20%, so new efficiency factor:

$$1/0.8=1.25$$

New team rate:

$$\sqrt{15}$$

Replacement rate $\sqrt{x}$:

$$\sqrt{x+4+5=15}$$

$$\sqrt{\boxed{x=6}}$$

Q18

Workforce factor:

$$\sqrt{0.75}$$

Efficiency factor:

$$\sqrt{1.2}$$

Output factor:

$$\sqrt{0.75(1.2)=0.9}$$

So:

$$\sqrt{\boxed{10\%\text{ decrease}}}$$

Q19

A's contribution share:

$$\sqrt{13000-5000 = 8000}$$

2 parts = 8000.

1 part:

$$\sqrt{4000}$$

Performance pool:

$$\sqrt{5(4000)=20000}$$

Add bonus:

$$\sqrt{\boxed{125000}}$$

Q20

Wage ratio:

$$\sqrt{2:1}$$

Efficiency:

$$\sqrt{6:5}$$

$$\sqrt{6(5):5T_B=2:1}$$

$$\sqrt{30:5T_B=2:1}$$

$$\sqrt{30=10T_B}$$

$$\sqrt{\boxed{T_B=3\text{ days}}}$$

' CHAPTER 27 COMPLETE

We have now covered:

- efficiency and time inverse relation '
- work contribution '
- wage distribution '
- equal/unequal working times '
- productivity comparison '
- percentage efficiency conversions '
- men/women/children equivalence '
- mixed workforce '
- team efficiency '
- joining/leaving team members '
- worker replacement '
- productivity targets '
- required staffing '
- work in phases '
- fixed bonus + contribution pay '
- contractor/piece-rate/hourly-wage basics '
- advanced contribution equations '
- GATE/placement drills '

Progress

Completed: 27/76 **Unit 1 — Quantitative Aptitude:** 0=100% PROGRESS

NEXT

+Chapter 28 — Pipes and Cisterns

Next we will cover:

- inlet and outlet rates
 - filling vs emptying pipes
 - positive and negative work
 - two/three pipes together
 - leakage
 - pipe opened/closed midway
 - alternate opening schedules
 - tank initially partly filled
 - efficiency/rate comparison
 - finding unknown pipe time
 - advanced GATE and placement problems.
-

USER

next

ASSISTANT

CHAPTER 28 — PIPES AND CISTERNS

PART 1 — INLETS, OUTLETS, FILLING RATES, EMPTYING RATES & BASIC COMBINED-PIPE PROBLEMS

Status: $\emptyset$ = 0% PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

Pipes and Cisterns is simply **Time and Work applied to tanks**.

Think:

- filling pipe = positive worker
- emptying pipe = negative worker
- tank = total work

So the master rule is:

$$\boxed{\text{Net rate} = \text{Filling rates} - \text{Emptying rates}}$$

1. BASIC FILLING PIPE

If a pipe fills a tank in:

$$10 \text{ hours}$$

then in 1 hour it fills:

$$\frac{1}{10}$$

of the tank.

This is exactly like one-day work.

2. BASIC EMPTYING PIPE

If an outlet empties a full tank in:

$$15 \text{ hours}$$

then its rate is:

$$-\frac{1}{15}$$

Negative sign means it removes water.

3. TWO FILLING PIPES TOGETHER

Pipe A fills in 10 h.

Pipe B fills in 15 h.

Together:

$$\left[\frac{1}{10} + \frac{1}{15} \right]$$

$$\left[= \frac{3}{30} + \frac{2}{30} \right]$$

$$\left[= \frac{5}{30} = \frac{1}{6} \right]$$

So:

$$\left[\boxed{6 \text{ hours}} \right]$$

4. TWO-PIPE FILLING FORMULA

If two inlet pipes fill in (a) and (b) hours:

$$\left[\boxed{T = \frac{ab}{a+b}} \right]$$

Same as Time and Work.

5. ONE INLET + ONE OUTLET

Pipe A fills in 10 hours.

Pipe B empties in 15 hours.

Net rate:

$$\left[\frac{1}{10} - \frac{1}{15} \right]$$

$$\left[= \frac{3}{30} - \frac{2}{30} \right]$$

$$\left[= \frac{1}{30} \right]$$

So tank fills in:

$$\left[\boxed{30 \text{ hours}} \right]$$

6. FILLER MUST BE FASTER

For net filling:

$$\left[\text{inlet rate} > \text{outlet rate} \right]$$

In time form, the inlet must take **less time** than the outlet.

Example:

- inlet fills in 10 h
- outlet empties in 15 h

Since:

$$\left[1/10 > 1/15 \right]$$

tank fills.

7. IF OUTLET IS STRONGER

Inlet fills in 20 h.

Outlet empties in 10 h.

Net:

$$\left[\frac{1}{20} - \frac{1}{10} = -\frac{1}{20} \right]$$

If tank starts empty, it cannot fill.

If tank starts partially/full, net effect is emptying.

8. LCM TANK METHOD

Suppose:

- A fills in 12 h
- B fills in 18 h
- C empties in 36 h

Take tank capacity:

$$\left[\text{LCM}(12, 18, 36) = 36 \text{ units} \right]$$

Rates:

$$\left[A = +3 \right]$$

$$\left[B = +2 \right]$$

$$\left[C = -1 \right]$$

Net:

$$\left[3 + 2 - 1 = 4 \right]$$

Time:

$$\left[36 / 4 = \boxed{9 \text{ hours}} \right]$$

This is usually faster than fractions.

9. THREE PIPES

General:

$$\left[\boxed{r_{\text{net}}} = \sum r_{\text{inlets}} - \sum r_{\text{outlets}} \right]$$

Then:

$$\left[\boxed{T} = \frac{1}{r_{\text{net}}} \right]$$

if tank size is treated as 1.

10. EXAMPLE

A fills in 8 h.

B fills in 12 h.

C empties in 24 h.

Take LCM:

$$\backslash[24 \backslash]$$

Rates:

$$\backslash[A=3 \backslash]$$

$$\backslash[B=2 \backslash]$$

$$\backslash[C=-1 \backslash]$$

Net:

$$\backslash[4 \backslash]$$

Time:

$$\backslash[24/4 = \boxed{6\text{ hours}} \backslash]$$

11. PIPE RATE AND TIME ARE INVERSE

If A fills twice as fast as B:

$$\backslash[E_A:E_B=2:1 \backslash]$$

Then filling-time ratio:

$$\backslash[\boxed{T_A:T_B=1:2} \backslash]$$

Same efficiency-time inverse relation.

12. EXAMPLE

B fills tank in 20 h.

A is twice as fast.

Then:

$$\backslash[A=10\text{ h} \backslash]$$

Together:

$$\backslash[\frac{1}{10} + \frac{1}{20} = \frac{3}{20} \backslash]$$

Time:

$$\backslash[\boxed{\frac{20}{3} = 6\frac{2}{3}\text{ h}} \backslash]$$

13. PART OF TANK FILLED

If a pipe fills tank in 12 h:

In 3 h it fills:

$$\backslash[3/12 = \boxed{\frac{1}{4}} \backslash]$$

In 9 h:

$$\left[9/12 = \boxed{\frac{3}{4}} \right]$$

14. PART OF TANK EMPTIED

If outlet empties full tank in 20 h:

In 5 h it empties:

$$\left[5/20 = \boxed{\frac{1}{4}} \right]$$

So if tank was full, remaining:

$$\left[\boxed{\frac{3}{4}} \right]$$

15. TANK INITIALLY HALF FULL

Suppose a pipe fills full tank in 10 h.

Tank starts half full.

Remaining to fill:

$$\left[\frac{1}{2} \right]$$

At rate:

$$\left[\frac{1}{10} \right]$$

Time:

$$\left[\frac{1/2}{1/10} = \boxed{5 \text{ h}} \right]$$

16. TANK INITIALLY 40% FULL

Pipe fills full tank in 12 h.

Remaining:

$$\left[60\% = 0.6 \right]$$

Time:

$$\left[12(0.6) = \boxed{7.2 \text{ h}} \right]$$

For a constant pipe rate:

$$\left[\boxed{\text{time for fraction } f = fT} \right]$$

17. FULL TANK WITH OUTLET

Outlet empties full tank in 8 h.

Tank initially 75% full.

Time to empty:

$$\left[0.75(8) = \boxed{6 \text{ h}} \right]$$

18. ONE PIPE OPENED FOR SOME TIME

A fills tank in 10 h.

It remains open for 4 h.

Filled:

$$\left[\frac{4}{10} = \frac{2}{5} \right]$$

Remaining:

$$\left[\boxed{\frac{3}{5}} \right]$$

If another pipe then takes over, calculate based on remaining fraction.

19. SECOND PIPE FINISHES

A fills in 10 h.

A works 4 h.

Then B, filling in 15 h, completes remaining tank.

A fills:

$$\left[\frac{2}{5} \right]$$

Remaining:

$$\left[\frac{3}{5} \right]$$

B time:

$$\left[15 \cdot \frac{3}{5} = 9 \right]$$

Total:

$$\left[4+9 = \boxed{13 \text{ h}} \right]$$

20. BOTH OPEN, THEN ONE CLOSED

A fills in 12 h.

B fills in 18 h.

Both run 3 h, then B is closed.

Take total tank:

$$\left[\frac{1}{4} \right]$$

Rates:

$$\left[A=3, \quad B=2 \right]$$

Together 3 h:

$$\left[\frac{1}{4} \right]$$

Remaining:

$$\sqrt[21]{}$$

A rate:

$$\sqrt[3]{}$$

Additional:

$$\sqrt[7]{}$$

Total:

$$\sqrt[3+7]{\boxed{10\text{ h}}}$$

21. INLET + OUTLET OPEN, THEN OUTLET CLOSED

A fills in 8 h.

B empties in 12 h.

Both open for 4 h.

Take capacity:

$$\sqrt[24]{}$$

Rates:

$$\sqrt[A=3,\text{quad } B=-2]{}$$

Net:

$$\sqrt[1]{}$$

In 4 h:

$$\sqrt[4]{}$$

Remaining:

$$\sqrt[20]{}$$

Outlet closes. A alone fills at 3.

Additional time:

$$\sqrt[20/3 = 6\frac{2}{3}]{}$$

Total:

$$\sqrt[\boxed{10\frac{2}{3}\text{ h}}]{}$$

22. INLET ALONE FIRST, OUTLET LATER

A fills tank in 6 h.

After 2 h, outlet B is opened.

B empties tank in 12 h.

First 2 h:

$$\sqrt[2/6 = \frac{1}{3}]{}$$

Remaining:

$$\left[\frac{2}{3} \right]$$

After outlet opens:

$$\left[\frac{1}{6} - \frac{1}{12} = \frac{1}{12} \right]$$

Time for remaining:

$$\left[\frac{2}{3} \div \frac{1}{12} = 8 \right]$$

Total:

$$\left[\boxed{10 \text{ h}} \right]$$

23. OUTLET OPENED FROM START BY MISTAKE

A fills tank in 6 h.

With an outlet accidentally open, tank fills in 8 h.

Find outlet's emptying time.

Inlet rate:

$$\left[\frac{1}{6} \right]$$

Net:

$$\left[\frac{1}{8} \right]$$

Outlet rate:

$$\left[\frac{1}{6} - \frac{1}{8} \right]$$

$$\left[= \frac{4}{24} - \frac{3}{24} = \frac{1}{24} \right]$$

So outlet empties in:

$$\left[\boxed{24 \text{ h}} \right]$$

This is a very common placement pattern.

24. LEAK PROBLEM

A pipe fills tank in 10 h.

Because of a leak, it takes 12 h.

Leak rate:

$$\left[\frac{1}{10} - \frac{1}{12} \right]$$

$$\left[= \frac{6}{60} - \frac{5}{60} = \frac{1}{60} \right]$$

Leak alone empties tank in:

$$\left[\boxed{60 \text{ h}} \right]$$

25. GENERAL LEAK FORMULA

If inlet alone fills in (a) hours, but with leak it fills in (T) hours:

$$\frac{1}{L} = \frac{1}{a} - \frac{1}{T}$$

So:

$$\boxed{L = \frac{aT}{T-a}}$$

where $(T > a)$.

26. EXAMPLE

Pipe alone:

$$8h$$

With leak:

$$10h$$

Leak:

$$L = \frac{8(10)}{10-8}$$

$$\boxed{40h}$$

27. TWO INLETS + LEAK

A fills in 12 h.

B fills in 18 h.

Leak empties in 36 h.

We already had:

$$3+2-1=4$$

with capacity 36.

Time:

$$\boxed{9h}$$

28. FIND UNKNOWN PIPE TIME

A fills in 10 h.

Together A+B fill in 6 h.

Find B.

$$\frac{1}{B} = \frac{1}{6} - \frac{1}{10}$$

$$= \frac{5}{30} - \frac{3}{30} = \frac{2}{30} = \frac{1}{15}$$

$$\boxed{B=15h}$$

Same as work-rate subtraction.

29. FIND UNKNOWN OUTLET TIME

Inlet fills in 8 h.

Together with outlet, net filling time = 12 h.

Outlet rate:

$$\left[\frac{18}{x} - \frac{1}{12} \right]$$

$$\left[= \frac{3}{24} - \frac{2}{24} = \frac{1}{24} \right]$$

So:

$$\left[\boxed{24\text{h}} \right]$$

30. FIND UNKNOWN INLET TIME

Outlet empties in 18 h.

Together with unknown inlet, tank fills in 9 h.

$$\left[\frac{1}{x} - \frac{1}{18} = \frac{1}{9} \right]$$

$$\left[\frac{1}{x} = \frac{1}{9} + \frac{1}{18} = \frac{3}{18} = \frac{1}{6} \right]$$

So inlet:

$$\left[\boxed{6\text{h}} \right]$$

31. FILLING PERCENTAGE PER HOUR

A fills tank in 20 h.

Hourly fill:

$$\left[\frac{1}{20} = 5\% \right]$$

B fills in 25 h:

$$\left[4\% \right]$$

Together:

$$\left[9\% \text{ per hour} \right]$$

Time:

$$\left[100/9 = \boxed{11\frac{1}{9}\text{h}} \right]$$

Percentage-rate method can be fast with friendly values.

32. OUTLET PERCENTAGE RATE

Outlet empties in 50 h.

Rate:

$$\left[2\% \text{ per hour} \right]$$

If inlet fills at 7%/h:

Net:

$\frac{5}{100} \times \text{total}$

Time:

$\frac{20}{100}$

33. PIPE CAPACITY IN LITRES

Suppose tank capacity:

$\frac{1200}{100}$

Pipe fills in 6 h.

Flow rate:

$\frac{1200}{6} = \boxed{200 \text{ L/h}}$

This is just physical rate.

34. TWO LITRE-RATE PIPES

Inlet A:

$\frac{300 \text{ L}}{\text{h}}$

Inlet B:

$\frac{200 \text{ L}}{\text{h}}$

Outlet C:

$\frac{100 \text{ L}}{\text{h}}$

Net:

$\frac{300 + 200 - 100}{100} = \frac{400}{100}$

Tank capacity:

$\frac{2000}{100}$

Time:

$\frac{2000}{400} = \boxed{5 \text{ h}}$

35. CONVERT TIME TO RATE

Tank:

$\frac{1800 \text{ L}}{100}$

Pipe fills in 9 h.

Rate:

$\frac{200 \text{ L}}{\text{h}}$

Another empties in 12 h:

$\frac{150 \text{ L}}{\text{h}}$

Net:

$$\sqrt[3]{50\text{L/h}}$$

Tank fills from empty in:

$$\sqrt[3]{1800/50 = \boxed{36\text{h}}}$$

36. GATE-STYLE QUESTION 1

A fills tank in 10 h.

B in 15 h.

Together:

$$\sqrt[3]{\boxed{6\text{h}}}$$

37. GATE-STYLE QUESTION 2

A fills in 10 h.

B empties in 15 h.

Net:

$$\sqrt[3]{\frac{1}{30}}$$

$$\sqrt[3]{\boxed{30\text{h}}}$$

38. GATE-STYLE QUESTION 3

A fills in 12 h.

B fills in 18 h.

C empties in 36 h.

LCM 36.

Net:

$$\sqrt[3]{3+2-1=4}$$

Time:

$$\sqrt[3]{\boxed{9\text{h}}}$$

39. GATE-STYLE QUESTION 4

Pipe fills in 8 h.

With leak tank fills in 10 h.

Leak empties in:

$$\sqrt[3]{\frac{8(10)}{2} = \boxed{40\text{h}}}$$

40. PLACEMENT-STYLE QUESTION 1

A fills in 6 h.

After 2 h outlet B opens.

B empties in 12 h.

From earlier:

$$\boxed{10h \text{ total}}$$

41. PLACEMENT-STYLE QUESTION 2

Tank half full.

Pipe fills full tank in 8 h.

Remaining half takes:

$$8/2 = \boxed{4h}$$

42. PLACEMENT-STYLE QUESTION 3

A fills in 8 h.

B empties in 24 h.

Together:

$$\frac{1}{8} - \frac{1}{24} = \frac{1}{24} = \frac{1}{12}$$

$$\boxed{12h}$$

43. PLACEMENT-STYLE QUESTION 4

A fills in 12 h.

A works 3 h.

Then B, which fills in 9 h, completes remaining.

A fills:

$$\frac{3}{12} = \frac{1}{4}$$

Remaining:

$$\frac{3}{4}$$

B time:

$$9 \left(\frac{3}{4} \right) = 6.75$$

Total:

$$\boxed{9.75h}$$

44. ADVANCED QUESTION — LEAK

Two pipes fill in 12 h and 18 h.

With leak, tank fills in 9 h.

Find leak time.

Combined inlet rate:

$$\left[\frac{1}{12} + \frac{1}{18} = \frac{5}{36} \right]$$

Net:

$$\left[\frac{1}{9} = \frac{4}{36} \right]$$

Leak:

$$\left[\frac{1}{36} \right]$$

So:

$$\left[\boxed{36\text{h}} \right]$$

45. ADVANCED QUESTION — PARTIAL TANK

Tank is:

$$\left[\frac{1}{3} \right]$$

full.

Inlet fills full tank in 8 h.

Outlet empties full tank in 24 h.

Net:

$$\left[\frac{1}{8} - \frac{1}{24} = \frac{1}{12} \right]$$

Remaining tank:

$$\left[\frac{2}{3} \right]$$

Time:

$$\left[\frac{2}{3} \times \frac{1}{12} = 8 \right]$$

$$\left[\boxed{8\text{h}} \right]$$

46. ADVANCED QUESTION — BOTH OPEN THEN ONE CLOSED

A fills in 10 h.

B empties in 15 h.

Both open for 6 h, then B closes.

Net initially:

$$\left[\frac{1}{30} \right]$$

Work in 6 h:

$$\frac{6}{30} = \frac{1}{5}$$

Remaining:

$$\frac{4}{5}$$

A alone:

$$10 \cdot \frac{4}{5} = 8$$

Total:

$$\boxed{14\text{h}}$$

47. ADVANCED QUESTION — FIND INLET

Outlet empties in 20 h.

Together unknown inlet + outlet fill tank in 5 h.

$$\frac{1}{x} - \frac{1}{20} = \frac{1}{5}$$

$$\frac{1}{x} = \frac{1}{5} + \frac{1}{20} = \frac{4}{20} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4}$$

$$\boxed{x=4\text{h}}$$

48. COMMON TRAPS

& pTRAP 1 — Add outlet rate

Outlet is negative.

& pTRAP 2 — Larger time means faster pipe

Wrong.

Rate and time are inverse.

& pTRAP 3 — Tank partly filled

Use only remaining fraction.

& pTRAP 4 — Pipe opens/closes midway

Split the process into stages.

& pTRAP 5 — Leak time

Leak alone must usually be slower than the main filling pipe if tank still fills.

& pTRAP 6 — Net rate non-positive

If net rate ≤ 0 , an empty tank cannot fill.

49. MUST-MEMORIZE FORMULAS

Filling pipe

$$\boxed{+\frac{1}{T}}$$

Emptying pipe

$$\boxed{-\frac{1}{T}}$$

Net rate

$$\boxed{r_{\text{net}} = \sum r_{\text{fill}} - \sum r_{\text{empty}}}$$

Filling time

$$\boxed{T = \frac{1}{r_{\text{net}}}}$$

50. TWO FILLERS

$$\boxed{T = \frac{ab}{a+b}}$$

One filler + one outlet

If filler (a) hours and outlet (b) hours with $(b > a)$:

$$\boxed{T = \frac{ab}{b-a}}$$

51. LEAK FORMULA

If inlet fills in (a) h but with leak takes (T) h:

$$\boxed{L = \frac{aT}{T-a}}$$

52. FAST EXAM DECISION TREE

All pipes open together?

Add inlet rates, subtract outlet rates.

Fractions look ugly?

Use LCM tank units.

Pipe opens/closes midway?

Solve in stages.

Tank initially partly full?

Work only on remaining fraction.

Leak slows filling?

Subtract net from inlet rate.

Unknown pipe?

Use rate subtraction/addition.

PRACTICE — LEVEL 0

Q1

A fills tank in 12 h. Find hourly fraction filled.

Q2

A fills in 10 h, B in 15 h. Find combined time.

Q3

A fills in 8 h, B empties in 24 h. Find net filling time.

Q4

Tank half full. Pipe fills full tank in 10 h. Find time to fill remainder.

Q5

Outlet empties tank in 20 h. How much of full tank remains after 5 h?

LEVEL 1

Q6

A fills in 12 h, B in 18 h, C empties in 36 h. Find time.

Q7

A fills in 8 h. With leak it fills in 10 h. Find leak time.

Q8

A fills in 10 h and works 4 h. B fills in 15 h and then finishes. Find total time.

Q9

A fills in 6 h. After 2 h, outlet that empties in 12 h opens. Find total filling time.

Q10

Tank is 40% full. A fills full tank in 10 h. Find time to fill rest.

LEVEL 2

Q11

A fills in 10 h. B empties in 15 h. Both work 6 h, then B closes. Find total filling time.

Q12

A fills in 12 h and B in 18 h. With leak tank fills in 9 h. Find leak time.

Q13

Tank is one-third full. Inlet fills in 8 h, outlet empties in 24 h. Find time to fill.

Q14

Outlet empties in 18 h. Unknown inlet with outlet fills tank in 9 h. Find inlet time.

Q15

Tank capacity 2400 L. Inlet = 400 L/h, outlet = 100 L/h. Find fill time.

LEVEL 3 — EXAM PRESSURE

Q16

A fills in 8 h, B fills in 12 h, C empties in 24 h. All open for 3 h, then C closes. Find total filling time.

Q17

A fills in 6 h. B empties in 18 h. A works alone for 1 h, then both are opened. Find total time.

Q18

Two pipes fill in 10 h and 15 h. A leak empties full tank in 30 h. Find total fill time.

Q19

A pipe fills tank in 5 h. Due to leak, it takes 6 h. If tank is initially half full and only leak is open, how long to empty it?

Q20

A fills tank in x h, B empties in 12 h, and together they fill tank in 8 h. Find x .

ANSWERS

Q1

$$\boxed{\frac{1}{12}}$$

Q2

$$\frac{1}{10} + \frac{1}{15} = \frac{1}{6}$$

$$\boxed{6\text{h}}$$

Q3

$$\frac{1}{18} - \frac{1}{24} = \frac{1}{12}$$

$$\boxed{12\text{h}}$$

Q4

Remaining:

$$\frac{1}{2}$$

Time:

$$10 \cdot \frac{1}{2} = \boxed{5\text{h}}$$

Q5

Emptied:

$$\frac{5}{20} = \frac{1}{4}$$

Remaining:

$$\boxed{\frac{3}{4}}$$

Q6

LCM 36:

$$3+2-1=4$$

$$36/4 = \boxed{9h}$$

Q7

$$\frac{1}{L} = \frac{1}{18} - \frac{1}{10} = \frac{1}{40}$$

$$\boxed{40h}$$

Q8

A fills:

$$4/10=2/5$$

Remaining:

$$3/5$$

B time:

$$15(3/5)=9$$

Total:

$$\boxed{13h}$$

Q9

A first 2 h:

$$2/6=1/3$$

Remaining:

$$2/3$$

Net after outlet opens:

$$1/6-1/12=1/12$$

Time:

$$(2/3)/(1/12)=8$$

Total:

$$\boxed{10h}$$

Q10

Remaining:

$$60\%$$

$$10(0.6) = \boxed{6h}$$

Q11

Initial net:

$$\backslash [1/10 - 1/15 = 1/30 \backslash]$$

In 6 h:

$\backslash [1/5 \backslash]$

Remaining:

$\lfloor 4/5 \rfloor$

A alone:

$$\lfloor 10(4/5) \rfloor = 8$$

Total:

$$\boxed{14h}$$

Q12

Inlets:

$$\backslash[1/12+1/18=5/36 \backslash]$$

Net:

$$\backslash [1/9=4/36 \backslash]$$

Leak:

\[1/36 \]

$$\boxed{36h}$$

Q13

Net:

$$\backslash[1/8-1/24=1/12\backslash]$$

Remaining:

$\lfloor 2/3 \rfloor$

Time:

$$\lfloor (2/3)(12) = \boxed{8h} \rfloor$$

Q14

$$\left[\frac{1}{x}-\frac{1}{18}=\frac{19}{18}\right]$$

$$\left[\frac{1}{x} = \frac{1}{6}\right]$$

\[\boxed{6h}\]

Q15

Net:

$$\backslash [400-100=300\text{L/h} \backslash]$$

$$\lfloor 2400/300 = \boxed{8h} \rfloor$$

Q16

LCM tank:

\[24 \]

Rates:

$$\sqrt{A=3, \ B=2, \ C=-1}$$

First 3 h net:

$$\sqrt{4(3)=12}$$

Remaining:

$$\sqrt{12}$$

After C closes:

$$\sqrt{A+B=5}$$

Additional:

$$\sqrt{12/5=2.4}$$

Total:

$$\sqrt{\boxed{5.4h}}$$

Q17

Take LCM:

$$\sqrt{18}$$

A rate:

$$\sqrt{3}$$

B:

$$\sqrt{-1}$$

First 1 h A alone:

$$\sqrt{3}$$

Remaining:

$$\sqrt{15}$$

Together rate:

$$\sqrt{2}$$

Additional:

$$\sqrt{15/2=7.5}$$

Total:

$$\sqrt{\boxed{8.5h}}$$

Q18

LCM 30:

A:

$$\sqrt{3}$$

B:

$$\sqrt{2}$$

Leak:

$$\boxed{-1}$$

Net:

$$\boxed{4}$$

Time:

$$\boxed{30/4 = 7.5h}$$

Q19

Pipe alone fills in:

$$\boxed{5h}$$

With leak:

$$\boxed{6h}$$

Leak rate:

$$\boxed{1/5 - 1/6 = 1/30}$$

So leak alone empties a full tank in:

$$\boxed{30h}$$

Half tank:

$$\boxed{15h}$$

$$\boxed{15h}$$

Q20

$$\frac{1}{x} - \frac{1}{12} = \frac{1}{18}$$

$$\frac{1}{x} = \frac{1}{18} + \frac{1}{12} = \frac{5}{24}$$

$$x = \frac{24}{5}$$

$$\boxed{4.8h}$$

CHAPTER 28 STATUS

Covered so far:

- inlet rates '
- outlet rates '
- positive/negative work '
- two and three pipes '
- LCM tank method '
- partly filled tanks '
- pipes opened/closed midway '
- leakage '
- unknown inlet/outlet times '
- litre-per-hour flow rates '

- GATE/placement practice '

But **Chapter 28 is not complete yet.**

Next we need:

- alternate opening of pipes
- cyclic pipe schedules
- outlet opened at intervals
- tank overflowing / excess inflow
- two-tank transfer basics
- pipes with different activation times
- repeated on/off schedules
- advanced leak and partial-tank combinations
- final speed shortcuts and chapter audit

Progress

Completed: 27/76 **Current:** $\emptyset$ =Chapter 28 — Pipes and Cisterns **Next:** Chapter 28 Part 2 — **Alternate pipes, cyclic schedules & advanced tank problems**

USER

next

ASSISTANT

CHAPTER 28 — PIPES AND CISTERNS

PART 2 — ALTERNATE PIPES, CYCLIC SCHEDULES, OVERFLOW, TRANSFER & ADVANCED TANK PROBLEMS

Status: $\emptyset$ =In PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

Part 1 covered normal simultaneous filling and emptying.

Now we handle irregular schedules:

- pipes opening on alternate hours,
- repeated on/off cycles,
- outlet opening after intervals,
- tank overflow,
- water transferred between tanks,
- changing pipe combinations,
- and advanced leak problems.

The key technique is:

$\boxed{\text{Find work done in one complete repeating cycle}}$

1. ALTERNATE-HOUR PIPES

Suppose:

- A fills tank in 6 h
- B fills tank in 12 h
- A starts
- they operate in alternate hours

Take tank capacity:

$$\boxed{\text{LCM}(6,12)=12}$$

Rates:

$$\boxed{A=2 \text{ units/h}}$$

$$\boxed{B=1 \text{ unit/h}}$$

One 2-hour cycle:

$$\boxed{2+1=3}$$

Tank capacity:

$$\boxed{12}$$

So 4 complete cycles fill:

$$\boxed{4(3)=12}$$

Time:

$$\boxed{8\text{h}}$$

2. ALTERNATING INLET AND OUTLET

A fills in 6 h.

B empties in 12 h.

They operate alternately, A first.

Take tank:

$$\boxed{12}$$

Rates:

$$\boxed{A=+2}$$

$$\boxed{B=-1}$$

One 2-hour cycle net:

$$\boxed{2-1=1}$$

But be careful:

The tank may become full during an A-hour **before** completing the final cycle.

This is the major trap.

3. EXAMPLE — ALTERNATE FILL/EMPTY

Tank capacity:

$$\lfloor 12 \rfloor$$

After 10 cycles = 20 h:

$$\lfloor 10 \text{ units} \rfloor$$

Hour 21: A fills 2 units.

Tank reaches:

$$\lfloor 12 \rfloor$$

So:

$$\lfloor \boxed{21} \rfloor$$

We do not wait for B's 22nd hour.

4. GENERAL CYCLE METHOD

For a repeating schedule:

1. Find smallest repeating block.
 2. Compute net work per block.
 3. Use full cycles until near completion.
 4. Handle final incomplete cycle carefully.
-

5. 3-HOUR CYCLE

Suppose:

- A fills in 8 h
- B fills in 12 h
- C empties in 24 h

Schedule repeats:

- Hour 1: A
- Hour 2: B
- Hour 3: C

Take capacity:

$$\lfloor 24 \rfloor$$

Rates:

$$\lfloor A=3, \text{quad } B=2, \text{quad } C=-1 \rfloor$$

Cycle net:

$$\lfloor 3+2-1=4 \rfloor$$

Cycle length:

$$\lfloor 3 \text{ h} \rfloor$$

After 5 cycles:

$\backslash[20\text{units} \backslash]$

Time:

$\backslash[15\text{h} \backslash]$

Remaining:

$\backslash[4 \backslash]$

Next hour A adds:

$\backslash[3 \backslash]$

Remaining:

$\backslash[1 \backslash]$

Next hour B fills 2 units/h, so needs:

$\backslash[\frac{12}{2} \backslash]$

Total:

$\backslash[15+1+\frac{12}{2} \backslash]$

$\backslash[\boxed{16.5\text{h}} \backslash]$

6. PIPE OPEN FOR SOME HOURS, CLOSED FOR OTHERS

A pipe fills tank in 10 h but operates only:

- 2 h ON
- 1 h OFF
- repeats

Rate when ON:

$\backslash[\frac{1}{10} \backslash]$

Work in each 3-hour cycle:

$\backslash[2\left(\frac{1}{10}\right) = \frac{2}{5} \backslash]$

After 4 full cycles:

$\backslash[\frac{8}{5} \backslash]$

Time:

$\backslash[12\text{h} \backslash]$

Remaining:

$\backslash[\frac{2}{5} \backslash]$

Pipe is ON next, and needs:

$\backslash[2\text{h} \backslash]$

Total:

$\backslash[\boxed{14\text{h}} \backslash]$

7. DUTY-CYCLE VIEW

If a pipe operates fraction f of the time and cycles are short/repeating:

average long-run rate:

$$f \times r$$

But do not use average rate blindly for the final partial cycle.

For exact aptitude questions, cycle method is safer.

8. TWO PIPES WITH DIFFERENT ON/OFF SCHEDULES

A fills in 6 h and runs every hour.

B fills in 12 h but runs only every alternate hour.

Take tank capacity:

$$12$$

A rate:

$$2$$

B rate when ON:

$$1$$

Over 2 hours:

- A works both hours $\Rightarrow 4$
- B works one hour $\Rightarrow 1$

Cycle work:

$$5$$

After 2 cycles = 4 h:

$$10$$

Remaining:

$$2$$

If next hour A works and B is OFF:

A alone fills 2.

So total:

$$\boxed{5h}$$

9. OUTLET OPENS AFTER FIXED DELAY

A fills tank in 8 h.

After 2 h, outlet B opens and empties in 24 h.

First 2 h:

$$\left[\frac{2}{8} = \frac{1}{4} \right]$$

Remaining:

$$\left[\frac{3}{4} \right]$$

After outlet opens:

$$\left[\frac{18 - \frac{1}{24}}{1} = \frac{1}{12} \right]$$

Time for remaining:

$$\left[\frac{3/4}{1/12} = 9 \right]$$

Total:

$$\left[\boxed{11\text{h}} \right]$$

10. OUTLET OPENS PERIODICALLY

Suppose:

- inlet fills in 6 h
- outlet empties in 12 h
- outlet is opened only every second hour
- inlet stays open continuously

Take tank:

$$\left[12 \right]$$

Inlet:

$$\left[+2 \right]$$

Outlet:

$$\left[-1 \right]$$

Over 2 hours:

Hour 1: inlet only:

$$\left[+2 \right]$$

Hour 2: both:

$$\left[+1 \right]$$

Cycle net:

$$\left[3 \right]$$

4 cycles:

$$\left[12 \right]$$

So:

$$\left[\boxed{8\text{h}} \right]$$

11. PIPE OPENS AFTER TANK REACHES A FRACTION

A fills tank in 10 h.

When tank becomes half full, outlet opens.

Outlet empties in 20 h.

Time to half-fill:

$$\left[10 \left(\frac{1}{2} \right) = 5 \text{ h} \right]$$

Remaining half after outlet opens.

Net rate:

$$\left[\frac{1}{10} - \frac{1}{20} = \frac{1}{20} \right]$$

Time for remaining half:

$$\left[\frac{1/2}{1/20} = 10 \text{ h} \right]$$

Total:

$$\left[\boxed{15 \text{ h}} \right]$$

12. LEAK STARTS AFTER SOME TIME

Pipe fills in 8 h.

After 2 h a leak starts, which alone empties in 24 h.

First 2 h filled:

$$\left[\frac{1}{4} \right]$$

Remaining:

$$\left[\frac{3}{4} \right]$$

Net:

$$\left[\frac{1}{8} - \frac{1}{24} = \frac{1}{12} \right]$$

Remaining time:

$$\left[\frac{3/4}{1/12} = 9 \right]$$

Total:

$$\left[\boxed{11 \text{ h}} \right]$$

Same mathematics as delayed outlet.

13. LEAK STOPS MIDWAY

A fills in 10 h.

Leak empties in 20 h.

Both operate for 8 h, then leak is repaired.

Net:

$$\left[\frac{1}{10} - \frac{1}{20} = \frac{1}{20} \right]$$

In 8 h:

$$\left[\frac{8}{20} = \frac{25}{} \right]$$

Remaining:

$$\left[\frac{35}{} \right]$$

A alone time:

$$\left[10 \cdot \frac{35}{} = 6h \right]$$

Total:

$$\left[\boxed{14h} \right]$$

14. OVERFLOW

If tank is already full and inlet continues running, excess water overflows.

Example:

Tank capacity:

$$\left[1000L \right]$$

Inlet:

$$\left[200L/h \right]$$

Outlet:

$$\left[50L/h \right]$$

Net inflow:

$$\left[150L/h \right]$$

If tank is full, overflow rate is:

$$\left[\boxed{150L/h} \right]$$

provided outlet and inlet continue.

15. OVERFLOW AFTER FILLING

Tank initially empty.

Capacity:

$$\left[1200L \right]$$

Net inflow:

$$\left[300L/h \right]$$

Time to fill:

$$\left[1200/300=4h \right]$$

If system runs for 7 h total:

Extra time:

$$\left[3h \right]$$

Overflow:

$$\boxed{300(3) = 900\text{L}}$$

16. OVERFLOW WITH MULTIPLE PIPES

Two inlets:

$$400, \text{quad } 300\text{L/h}$$

Outlet:

$$200\text{L/h}$$

Net:

$$500\text{L/h}$$

Tank capacity:

$$2000\text{L}$$

Fills in:

$$2000/500 = 4\text{h}$$

If all continue 6 h:

Overflow duration:

$$2\text{h}$$

Overflow volume:

$$500(2) = \boxed{1000\text{L}}$$

17. WATER TRANSFER BETWEEN TWO TANKS

Tank A transfers water into Tank B.

Treat transfer as:

- negative rate for A,
- positive rate for B.

Example:

A contains 600 L.

Transfer pipe moves:

$$50\text{L/h}$$

Then after 4 h:

Tank A loses:

$$200\text{L}$$

Tank B gains:

$$200\text{L}$$

assuming no other inflow/outflow.

18. TRANSFER WITH EXTERNAL INLET

Tank B receives:

- 50 L/h from A
- 30 L/h external inlet

and loses:

- 20 L/h through outlet

Net increase in B:

$$[50+30-20 = \boxed{60\text{L/h}}]$$

19. TWO-TANK EQUALIZATION BASIC

Tank A has 800 L.

Tank B has 200 L.

Water transfers from A to B at 50 L/h.

After t hours:

$$[A=800-50t]$$

$$[B=200+50t]$$

Equal when:

$$[800-50t=200+50t]$$

$$[600=100t]$$

$$[\boxed{t=6\text{h}}]$$

20. INFLOW AND TRANSFER

Tank A initially 500 L.

It receives 100 L/h, but transfers 40 L/h to B.

Net A:

$$[+60\text{L/h}]$$

After 5 h:

$$[500+5(60) = \boxed{800\text{L}}]$$

21. CAPACITY CONSTRAINT

If a tank has maximum capacity, once full it cannot keep accumulating water.

Any further positive net inflow becomes overflow.

Always compare:

$$[\text{current amount}]$$

with:

$\text{\text{capacity}}$

22. ALTERNATE PIPES WITH PARTLY FILLED TANK

Tank initially half full.

A fills in 8 h.

B empties in 16 h.

They act alternately, A first.

Take tank:

16 units

Initially:

8 units

Rates:

$A = +2, B = -1$

Each 2-hour cycle:

$+1$

Need:

8

more units.

After 6 cycles:

$+6$

Tank level:

14

Time:

12 h

Next A-hour adds 2:

Tank becomes full.

Total:

$\boxed{13 \text{ h}}$

23. FIND UNKNOWN PIPE FROM CYCLIC COMPLETION

A fills in 6 h.

B fills in x h.

They operate alternate hours, A first, and tank fills exactly in 8 h.

There are 4 A-hours and 4 B-hours.

So:

$$4\left(\frac{16}{x}\right) + 4\left(\frac{1}{x}\right) = 1$$

$$\frac{23}{x} + \frac{4}{x} = 1$$

$$\frac{4}{x} = \frac{1}{3}$$

$$x = 12$$

$$\boxed{12\text{ h}}$$

24. FIND LEAK FROM CYCLIC DATA

A fills tank in 8 h.

Leak is open every alternate hour only.

Tank fills in 10 h.

Suppose inlet remains open continuously and leak operates during 5 of the 10 hours.

Inlet contribution:

$$10\left(\frac{1}{8}\right) = \frac{5}{4}$$

So leak removed:

$$\frac{5}{4} - 1 = \frac{1}{4}$$

In 5 leak-hours, leak rate:

$$\frac{1}{4} \times 5 = \frac{1}{20}$$

So leak alone empties in:

$$\boxed{20\text{ h}}$$

25. FRACTIONAL FINAL HOUR

Always check whether final active pipe needs a whole hour.

Example:

Tank capacity:

$$24$$

After cycles, 3 units remain.

Next inlet rate:

$$8\text{ units/h}$$

Required time:

$$\frac{3}{8}\text{ h}$$

not 1 h.

26. GATE-STYLE QUESTION 1

A fills in 6 h.

B fills in 12 h.

Alternate hours, A first.

Tank capacity:

$\sqrt[12]{}$

Rates:

$\sqrt[2,1]{}$

2-hour cycle:

$\sqrt[3]{}$

4 cycles:

$\sqrt[12]{}$

$\sqrt[\boxed{8h}]{}$

27. GATE-STYLE QUESTION 2

A fills in 6 h.

B empties in 12 h.

Alternate hours, A first.

Take capacity 12.

Net per cycle:

$\sqrt[2-1=1]{}$

After 10 cycles:

$\sqrt[10]{}$

Hour 21 A adds 2.

$\sqrt[\boxed{21h}]{}$

28. GATE-STYLE QUESTION 3

A fills in 10 h.

Leak empties in 20 h.

Both operate 8 h, leak then stops.

Net first phase:

$\sqrt[1/20]{}$

Filled:

$\sqrt[8/20=2/5]{}$

Remaining:

$\sqrt[3/5]{}$

A needs:

$\sqrt[3]{6h}$

Total:

$\sqrt[3]{\boxed{14h}}$

29. GATE-STYLE QUESTION 4

Tank 1200 L.

Net inflow 300 L/h.

System runs 7 h from empty.

Fill time:

$\sqrt[3]{4h}$

Overflow:

$\sqrt[3]{3(300)} = \boxed{900L}$

30. PLACEMENT-STYLE QUESTION 1

Tank half full.

A fills in 8 h.

B empties in 16 h.

Alternate hours, A first.

From earlier:

$\sqrt[3]{\boxed{13h}}$

31. PLACEMENT-STYLE QUESTION 2

A fills in 8 h.

After 2 h, leak starts which empties in 24 h.

Total:

$\sqrt[3]{\boxed{11h}}$

32. PLACEMENT-STYLE QUESTION 3

A fills continuously in 6 h.

Outlet empties in 12 h and opens only every second hour.

Take tank = 12.

Over 2 h:

- inlet contributes $\sqrt[3]{4}$
-

- outlet removes $\frac{1}{3}$

Net:

$$\frac{1}{3}$$

4 cycles:

$$\frac{1}{3}$$

$$\boxed{8h}$$

33. PLACEMENT-STYLE QUESTION 4

Tank A has 900 L.

Tank B has 300 L.

Water transfers A to B at 50 L/h.

Equal level when:

$$900 - 50t = 300 + 50t$$

$$600 = 100t$$

$$\boxed{6h}$$

34. ADVANCED QUESTION — CYCLIC 3-PIPE

A fills in 8 h.

B fills in 12 h.

C empties in 24 h.

Operate A, B, C successively for one hour each, repeat.

From earlier:

$$\boxed{16.5h}$$

35. ADVANCED QUESTION — OUTLET EVERY THIRD HOUR

A fills tank in 6 h continuously.

Outlet empties in 12 h and operates only during every third hour.

Take tank:

$$\frac{1}{3}$$

A contributes in 3 h:

$$2(3) = 6$$

Outlet removes during one hour:

$$1$$

Net 3-hour cycle:

$\frac{5}{8}$

After 2 cycles = 6 h:

$\frac{10}{8}$

Remaining:

$\frac{2}{8}$

Hour 7 is not an outlet hour, so inlet alone adds 2.

Tank fills at:

$\boxed{7h}$

36. ADVANCED QUESTION — PIPE ON/OFF

Pipe fills tank in 8 h but operates:

- 1 h ON
- 1 h OFF

Tank starts empty.

Each 2-hour cycle fills:

$\frac{1}{8}$

After 7 cycles:

$\frac{7}{8}$

Time:

$14h$

Remaining:

$\frac{1}{8}$

Next ON hour fills it.

Total:

$\boxed{15h}$

37. ADVANCED QUESTION — PARTIAL + CYCLIC

Tank initially 25% full.

Pipe fills in 6 h and works 2 h ON, 1 h OFF.

Remaining:

$\frac{75}{100}$

During each ON hour rate:

$\frac{1}{6}$

Two ON hours per cycle:

$\frac{1}{3}$

After two cycles = 6 h:

$$\left[\frac{2}{3} \right]$$

additional filled.

Total tank level:

$$\left[\frac{1}{4} + \frac{2}{3} = \frac{3}{12} + \frac{8}{12} = \frac{11}{12} \right]$$

Remaining:

$$\left[\frac{1}{12} \right]$$

Next cycle starts with pipe ON.

Time:

$$\left[\frac{1/12}{1/6} = \frac{1}{2} \text{h} \right]$$

Total:

$$\left[\boxed{6.5\text{h}} \right]$$

38. ADVANCED QUESTION — FIND OUTLET

A fills tank continuously in 4 h.

An outlet operates only half the time in alternating one-hour blocks.

Tank fills in exactly 6 h.

Inlet contribution in 6 h:

$$\left[6 \left(\frac{1}{4} \right) = 1.5 \right]$$

Outlet must remove:

$$\left[0.5 \right]$$

Outlet is active 3 h.

Its rate:

$$\left[0.5/3 = \frac{1}{6} \right]$$

So outlet alone empties in:

$$\left[\boxed{6\text{h}} \right]$$

39. COMMON TRAPS

& pTRAP 1 — Net cycle reaches capacity mid-cycle

Stop immediately when full.

& pTRAP 2 — Average-rate shortcut

It can miss the exact fractional final cycle.

& pTRAP 3 — Overflow before tank is full

Overflow begins only after capacity is reached.

& pTRAP 4 — Transfer pipe

It is negative for source tank and positive for receiving tank.

& pTRAP 5 — On/off pipe

Its instantaneous rate when ON does not change; only active duration changes.

& pTRAP 6 — Cyclic outlet

Count exactly how many hours the outlet was active.

40. MUST-MEMORIZE ADVANCED RULES

Repeating schedule

$$\boxed{\text{Net cycle work} = \sum (\text{rate}) \times \text{active time}}$$

Overflow

$$\boxed{\text{Overflow volume} = \text{net inflow rate} \times \text{time after tank becomes full}}$$

Transfer

$$\boxed{\text{Source change} = -r, \text{Destination change} = +r}$$

41. FAST EXAM DECISION TREE

Regular alternating schedule?

Build one cycle.

Tank almost full?

Check each pipe inside final cycle.

Pipe ON/OFF?

Count active hours only.

Overflow?

First find fill time, then overflow duration.

Two tanks?

Write separate rate equations.

Delayed leak/outlet?

Split into stages.

Unknown pipe from schedule?

Write total contribution equation.

PRACTICE — LEVEL 0

Q1

A fills in 6 h, B fills in 12 h. Alternate hours, A first. Find time.

Q2

A fills in 6 h, B empties in 12 h. Alternate hours, A first. Find time.

Q3

Pipe fills in 8 h but works 1 h ON, 1 h OFF. Find time.

Q4

Tank fills at net 250 L/h. Capacity 1000 L. Find fill time.

Q5

Tank A = 700 L, B = 300 L. Transfer 50 L/h. Find when equal.

LEVEL 1

Q6

A fills continuously in 6 h; outlet empties in 12 h and runs every alternate hour. Find time.

Q7

A fills in 10 h; outlet empties in 20 h. Both run 8 h, then outlet closes. Find total time.

Q8

Tank capacity 1500 L. Net inflow 300 L/h. System runs for 8 h. Find overflow.

Q9

A fills in 8 h. Leak begins after 2 h and empties in 24 h. Find total time.

Q10

Tank is half full. A fills in 8 h, B empties in 16 h. Alternate hours, A first. Find time.

LEVEL 2

Q11

A fills in 8 h, B fills in 12 h, C empties in 24 h. Operate A,B,C one hour each repeatedly. Find time.

Q12

A fills continuously in 6 h. Outlet empties in 12 h and operates every third hour. Find time.

Q13

Pipe fills in 10 h but operates 2 h ON and 1 h OFF. Find time.

Q14

A fills in 6 h, B fills in $\frac{1}{x}$ h. They alternate for exactly 8 h, A first, and fill tank exactly. Find $\frac{1}{x}$.

Q15

A fills in 8 h. Leak runs every alternate hour only. Tank fills in 10 h. Find leak time.

LEVEL 3 — EXAM PRESSURE**Q16**

Tank initially 25% full. Pipe fills in 6 h and works 2 h ON, 1 h OFF. Find time to full.

Q17

A fills continuously in 4 h. Outlet runs every alternate hour. Tank fills in 6 h. Find outlet's emptying time.

Q18

Tank A contains 800 L and B contains 200 L. Transfer from A to B is 40 L/h, while B also loses 10 L/h externally. Find when quantities become equal.

Q19

Capacity 2000 L. Inlets are 400 and 300 L/h, outlet 200 L/h. All run for 7 h from empty. Find overflow.

Q20

A fills tank in 5 h continuously. B empties in 10 h but is open only during hours 2,4,6,... . Find exact filling time.

ANSWERS**Q1**

Cycle work:

$$\left[\frac{1}{6} + \frac{1}{12} = \frac{1}{4} \right]$$

2 h gives $\frac{1}{4}$ tank.

Need 4 cycles:

$$\boxed{8\text{h}}$$

Q2

Using 12 units:

Cycle net:

$$\left[2 - 1 = 1 \right]$$

After 20 h:

$$\left[10 \right]$$

Hour 21 A fills remaining 2.

$$\boxed{21\text{h}}$$

Q3

Each 2-hour cycle fills:

$$\left[\frac{1}{8} \right]$$

After 14 h:

$$\left[\frac{7}{8} \right]$$

Next ON hour:

$$\left[\boxed{15\text{h}} \right]$$

Q4

$$\left[1000/250 = \boxed{4\text{h}} \right]$$

Q5

$$\left[700 - 50t = 300 + 50t \right]$$

$$\left[400 = 100t \right]$$

$$\left[\boxed{4\text{h}} \right]$$

Q6

Take 12 units.

Over 2 h:

$$\left[2 + (2 - 1) = 3 \right]$$

4 cycles fill:

$$\left[12 \right]$$

$$\left[\boxed{8\text{h}} \right]$$

Q7

Net:

$$\left[\frac{1}{10} - \frac{1}{20} = \frac{1}{20} \right]$$

8 h fills:

$$\left[\frac{2}{5} \right]$$

Remaining:

$$\left[\frac{3}{5} \right]$$

A alone needs:

$$\left[6\text{h} \right]$$

$$\left[\boxed{14\text{h}} \right]$$

Q8

Fill time:

$$\left[1500/300 = 5\text{h} \right]$$

Overflow duration:

$$\left[8 - 5 = 3\text{h} \right]$$

Overflow:

$$\sqrt[3]{300} = \boxed{900L}$$

Q9

First 2 h:

$$\sqrt[4]{1}$$

Remaining:

$$\sqrt[4]{3/4}$$

Net:

$$\sqrt[12]{1}$$

Time:

$$\sqrt[9]{9h}$$

Total:

$$\sqrt[11]{11h}$$

Q10

From earlier:

$$\sqrt[13]{13h}$$

Q11

$$\sqrt[16.5]{16.5h}$$

Q12

3-hour cycle:

$$\sqrt[6-1]{5 \text{ units}}$$

Using 12-unit tank:

After 6 h:

$$\sqrt[10]{10}$$

Hour 7 inlet adds remaining 2.

$$\sqrt[7]{7h}$$

Q13

Each 3-hour cycle fills:

$$\sqrt[2/10]{1/5}$$

4 cycles:

$$\sqrt[4/5]{4/5}$$

Time:

$$\sqrt[12]{12h}$$

Remaining:

$$\sqrt[1/5]{1/5}$$

Needs 2 ON hours.

$$\boxed{14h}$$

Q14

4 A-hours + 4 B-hours:

$$\frac{4}{6} + \frac{4}{x} = 1$$

$$\frac{2}{3} + \frac{4}{x} = 1$$

$$\frac{4}{x} = \frac{1}{3}$$

$$\boxed{x=12h}$$

Q15

Inlet contribution:

$$\frac{10}{8} = \frac{5}{4}$$

Leak active 5 h and removes:

$$\frac{1}{4}$$

Leak rate:

$$\frac{1}{20}$$

$$\boxed{20h}$$

Q16

After two 3-hour cycles:

additional filled:

$$\frac{2}{3}$$

Initial:

$$\frac{1}{4}$$

Total:

$$\frac{11}{12}$$

Remaining:

$$\frac{1}{12}$$

Pipe rate:

$$\frac{1}{6}$$

Needs:

$$\frac{1}{2h}$$

$$\boxed{6.5h}$$

Q17

Inlet in 6 h contributes:

$$\frac{6}{4} = 1.5$$

Outlet active 3 h removes:

$$\sqrt[0.5]{}$$

Outlet rate:

$$\sqrt[1/6]{}$$

$$\sqrt[\boxed{6h}]{}$$

Q18

Tank A:

$$\sqrt[800-40t]{}$$

Tank B net change:

$$\sqrt[+40-10=+30]{}$$

So:

$$\sqrt[B=200+30t]{}$$

Equal:

$$\sqrt[800-40t=200+30t]{}$$

$$\sqrt[600=70t]{}$$

$$\sqrt[t=\frac{60}{7}]{}$$

$$\sqrt[\boxed{8\frac{4}{7}h}]{}$$

Q19

Net inflow:

$$\sqrt[400+300-200 = 500L/h]{}$$

Fill time:

$$\sqrt[2000/500=4h]{}$$

Overflow duration:

$$\sqrt[7-4=3h]{}$$

Overflow:

$$\sqrt[3(500) = \boxed{1500L}]{}$$

Q20

Take tank capacity:

$$\sqrt[10]{}$$

A rate:

$$\sqrt[+2]{}$$

B rate:

$$\sqrt[-1]{}$$

Over each 2-hour cycle:

Hour 1 A only:

$$\sqrt[+2]{}$$

Hour 2 A+B:

$$\sqrt[4]{+1}$$

Cycle:

$$\sqrt[4]{+3}$$

After 3 cycles = 6 h:

$$\sqrt[4]{9}$$

Hour 7: A alone adds 2 units/h.

Only 1 unit needed:

$$\sqrt[4]{1/2h}$$

So:

$$\sqrt[4]{\boxed{6.5h}}$$

' CHAPTER 28 COMPLETE

We have now covered:

- inlet/filling rates '
- outlet/emptying rates '
- positive and negative work '
- two/three pipes together '
- LCM tank method '
- partially filled tanks '
- leaks '
- delayed pipe opening '
- pipe closing midway '
- alternate-hour operation '
- cyclic schedules '
- ON/OFF pipes '
- periodic outlets '
- overflow '
- litre-per-hour problems '
- two-tank transfer basics '
- unknown pipe times '
- fractional final-hour handling '
- advanced GATE/placement problems '

Progress

Completed: 28/76 Unit 1 — Quantitative Aptitude: 0=0% PROGRESS

NEXT

0=0% Chapter 29 — Time, Speed and Distance

Next we will cover:

- $(D=ST)$
- speed-distance-time conversions
- km/h to m/s
- fixed-distance inverse relation
- average speed
- speed percentage changes
- delayed/early arrival
- meeting problems
- relative motion basics
- chasing
- circular motion preview
- GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 29 — TIME, SPEED AND DISTANCE

PART 1 — $(D=ST)$, UNIT CONVERSION, FIXED-DISTANCE RELATION, SPEED CHANGES & BASIC MEETING PROBLEMS

Status: $\emptyset=0$ **IN PROGRESS** **GATE:** High **Placements:** Very High **Priority:** $\emptyset=0$ **Year:** 1

Time–Speed–Distance is one of the most important aptitude chapters.

The entire chapter is built on:

$$\boxed{\text{Distance} = \text{Speed} \times \text{Time}}$$

or:

$$\boxed{D = ST}$$

From this:

$$\boxed{S = \frac{D}{T}}$$

and:

$$\boxed{T = \frac{D}{S}}$$

1. BASIC IDEA

If a car moves at:

$$\boxed{60 \text{ km/h}}$$

for:

$$\backslash[3\text{ hours} \backslash]$$

then distance:

$$\backslash[D=60(3) \backslash]$$

$$\backslash[=\boxed{180\text{ km}} \backslash]$$

2. FIND SPEED

Distance:

$$\backslash[240\text{ km} \backslash]$$

Time:

$$\backslash[4\text{ h} \backslash]$$

$$\backslash[S=\frac{240}{4} \backslash]$$

$$\backslash[=\boxed{60\text{ km/h}} \backslash]$$

3. FIND TIME

Distance:

$$\backslash[150\text{ km} \backslash]$$

Speed:

$$\backslash[50\text{ km/h} \backslash]$$

$$\backslash[T=\frac{150}{50} \backslash]$$

$$\backslash[=\boxed{3\text{ h}} \backslash]$$

4. UNIT CONSISTENCY

Before using:

$$\backslash[D=ST \backslash]$$

make sure units match.

Examples:

- km with km/h
- metres with m/s
- minutes converted to hours if speed is km/h

This is a major source of careless mistakes.

5. KM/H TO M/S

We know:

$$1 \text{ km} = 1000 \text{ m}$$

and:

$$1 \text{ h} = 3600 \text{ s}$$

Therefore:

$$1 \text{ km/h} = \frac{1000}{3600} \text{ m/s}$$

$$= \frac{5}{18} \text{ m/s}$$

So:

$$\boxed{\text{km/h} \text{ to m/s: } \times \frac{5}{18}}$$

MUST MEMORIZE

6. M/S TO KM/H

Reverse conversion:

$$\boxed{\text{m/s} \text{ to km/h: } \times \frac{18}{5}}$$

7. EXAMPLE

Convert:

$$72 \text{ km/h}$$

to m/s.

$$72 \times \frac{5}{18}$$

$$= 4(5)$$

$$= \boxed{20 \text{ m/s}}$$

8. EXAMPLE

Convert:

$$25 \text{ m/s}$$

to km/h.

$$25 \times \frac{18}{5}$$

$$= 5(18)$$

$$= \boxed{90 \text{ km/h}}$$

9. MUST-MEMORIZE CONVERSIONS

$$18 \text{ km/h} = 5 \text{ m/s}$$

$$36 \text{ km/h} = 10 \text{ m/s}$$

$$54 \text{ km/h} = 15 \text{ m/s}$$

$$\text{72 km/h} = 20 \text{ m/s}$$

$$\text{90 km/h} = 25 \text{ m/s}$$

$$\text{108 km/h} = 30 \text{ m/s}$$

These are high-value mental-math conversions.

10. MINUTES TO HOURS

If speed is km/h:

$$30 \text{ min} = \frac{1}{2} \text{ h}$$

$$20 \text{ min} = \frac{1}{3} \text{ h}$$

$$15 \text{ min} = \frac{1}{4} \text{ h}$$

$$45 \text{ min} = \frac{3}{4} \text{ h}$$

11. EXAMPLE — MINUTES

A car travels at:

$$60 \text{ km/h}$$

for 45 min.

Time:

$$\frac{45}{60} = \frac{3}{4} \text{ h}$$

Distance:

$$60 \cdot \frac{3}{4}$$

$$= \boxed{45 \text{ km}}$$

12. FIXED DISTANCE — SPEED AND TIME ARE INVERSE

For the same distance:

$$D = ST$$

So:

$$ST = \text{constant}$$

Therefore:

$$\boxed{S \propto \frac{1}{T}}$$

Hence:

$$\boxed{S_1 : S_2 = T_2 : T_1}$$

This is exactly like efficiency–time in Time & Work.

13. EXAMPLE

A person covers the same distance:

- at 40 km/h in 6 h
- at speed (S) in 4 h

Since:

$$40(6) = S(4)$$

$$S = 60$$

$$\boxed{60 \text{ km/h}}$$

14. RATIO METHOD

For fixed distance:

$$S_1 : S_2 = 2 : 3$$

Then:

$$T_1 : T_2 = 3 : 2$$

If time at first speed is 6 h:

$$T_2 = 4 \text{ h}$$

15. PERCENTAGE SPEED INCREASE !' TIME DECREASE

If speed increases by $(p\%)$, for fixed distance:

$$T_{\text{new}} = T_{\text{old}} \frac{100}{100+p}$$

So percentage decrease in time:

$$\boxed{\frac{100p}{100+p} \%}$$

Same reciprocal logic as efficiency.

16. EXAMPLE — 25% SPEED INCREASE

Speed increases:

$$25\%$$

New speed factor:

$$1.25$$

New time:

$$\frac{1}{1.25} = 0.8$$

So time decreases:

$$\boxed{20\%}$$

Thus:

$$\boxed{25\% \uparrow S \rightarrow 20\% \downarrow T}$$

17. SPEED DECREASE !' TIME INCREASE

If speed decreases by $(p\%)$:

$$T_{\text{new}} = T_{\text{old}} \frac{100}{100-p}$$

Time increase:

$$\boxed{\frac{100p}{100-p}\%}$$

Example:

Speed falls 20%.

Time increases:

$$\frac{20}{80} \times 100 = \boxed{25\%}$$

18. TIME DECREASE !' SPEED INCREASE

If time falls by $(p\%)$:

$$\boxed{\frac{p}{100-p}\% \text{ speed increase}} = \frac{100p}{100-p}$$

Example:

Time decreases by 20%.

Speed rises:

$$\boxed{25\%}$$

19. TIME INCREASE !' SPEED DECREASE

If time rises by $(p\%)$:

$$\boxed{\frac{p}{100+p}\% \text{ speed decrease}} = \frac{100p}{100+p}$$

Example:

Time rises 25%.

Speed decreases:

$$\boxed{20\%}$$

20. HIGH-VALUE CONVERSION PAIRS

Remember:

$$\boxed{25\% \uparrow S \rightarrow 20\% \downarrow T}$$

$$\boxed{20\% \downarrow S \rightarrow 25\% \uparrow T}$$

$$\boxed{50\% \uparrow S \rightarrow 33\frac{1}{3}\% \downarrow T}$$

$$\boxed{50\% \downarrow S \rightarrow 100\% \uparrow T}$$

21. FIND DISTANCE FROM DIFFERENT SPEEDS AND TIMES

A journey takes:

$$[5h]$$

at 40 km/h.

Distance:

$$[40(5) = \boxed{200\text{ km}}]$$

At 50 km/h, time would be:

$$[200/50 = \boxed{4h}]$$

Time saved:

$$[1h]$$

22. EARLY/LATEARRIVAL — CORE IDEA

Suppose a person travels the same fixed distance.

At one speed, reaches late.

At another speed, reaches early.

Since distance is same:

$$[\boxed{D=S_1T_1=S_2T_2}]$$

The difference between the two travel times is obtained from the early/late information.

23. EXAMPLE — LATE/EARLY

At 40 km/h a person arrives 30 min late.

At 60 km/h they arrive 30 min early.

Difference in travel times:

$$[1h]$$

Let distance = (D) .

$$[\frac{D}{40} - \frac{D}{60} = 1]$$

LCM 120:

$$[D\left(\frac{3}{120} - \frac{2}{120}\right) = 1]$$

$$[\frac{D}{120} = 1]$$

$$[D = \boxed{120\text{ km}}]$$

24. FASTER FORMULA — TIME DIFFERENCE

For same distance:

$$[\boxed{\Delta T = D\left(\frac{1}{S_1} - \frac{1}{S_2}\right)}]$$

So:

$$\boxed{D = \frac{\Delta T}{\frac{1}{S_1} - \frac{1}{S_2}}}$$

if $(S_2 > S_1)$.

25. EXAMPLE

At 30 km/h and 40 km/h, travel times differ by 1 h.

$$D = \frac{1}{\frac{1}{30} - \frac{1}{40}}$$

$$= \frac{1}{\frac{4-3}{120}}$$

$$= \boxed{120 \text{ km}}$$

26. SPEED RATIO FROM TIME DIFFERENCE

For same distance:

Suppose speed ratio:

$$3:4$$

Then time ratio:

$$4:3$$

If time difference = 1 h:

Ratio difference = 1 part.

Therefore times:

$$4\text{h}, 3\text{h}$$

If slower speed = 30 km/h:

Distance:

$$30(4) = \boxed{120 \text{ km}}$$

This is often faster than equations.

27. BASIC MEETING PROBLEM — TOWARD EACH OTHER

Two people start from points A and B and move toward each other.

Speeds:

$$S_1, S_2$$

Their closing speed is:

$$\boxed{S_1 + S_2}$$

If initial distance is (D) :

$$\boxed{T = \frac{D}{S_1 + S_2}}$$

28. EXAMPLE

Two cars are 300 km apart.

They move toward each other at:

$$\sqrt[60, 40]{\text{km/h}}$$

Relative/closing speed:

$$\sqrt{100}$$

Meeting time:

$$\sqrt{300/100 = \boxed{3h}}$$

29. DISTANCES COVERED BEFORE MEETING

Same example.

First car:

$$\sqrt{60(3)=180}{\text{km}}$$

Second:

$$\sqrt{40(3)=120}{\text{km}}$$

Total:

$$\sqrt{300}$$

Distance ratio:

$$\sqrt{180:120 = \boxed{3:2}}$$

This equals speed ratio because time is same.

30. MEETING POINT DIVIDES DISTANCE IN SPEED RATIO

If two objects start simultaneously toward each other:

$$\sqrt{\boxed{D_1:D_2=S_1:S_2}}$$

because both travel for equal time until meeting.

31. EXAMPLE

Distance between cities = 280 km.

Speeds:

$$\sqrt{60:40=3:2}$$

Meeting point divides 280 in ratio 3:2.

First vehicle distance:

$$\sqrt{280 \times \frac{3}{5} = 168}$$

Second:

$$\boxed{112}$$

$$\boxed{168 \text{ km and } 112 \text{ km}}$$

32. MOVING IN SAME DIRECTION — CATCHING

If faster object is behind slower object:

Relative speed:

$$\boxed{S_{\text{fast}} - S_{\text{slow}}}$$

If initial gap = (D) :

$$\boxed{T = \frac{D}{S_{\text{fast}} - S_{\text{slow}}}}$$

33. EXAMPLE — CHASE

A car is 60 km behind another.

Speeds:

$$80, 60 \text{ km/h}$$

Relative speed:

$$20$$

Catch time:

$$60/20 = \boxed{3 \text{ h}}$$

34. DISTANCE COVERED DURING CHASE

Fast vehicle:

$$80(3) = 240$$

Slow:

$$60(3) = 180$$

Difference:

$$60$$

which closes the initial gap.

35. HEAD START

A person starts 2 h earlier at 40 km/h.

Head-start distance:

$$40(2) = 80 \text{ km}$$

Another travels at 60 km/h.

Relative speed:

$$\lfloor 20 \rfloor$$

Catch time after faster person starts:

$$\lfloor 80/20 = \boxed{4h} \rfloor$$

36. TOTAL TIME OF FIRST PERSON

First person already travelled 2 h before chase begins.

Then another 4 h until caught.

Total first-person travel time:

$$\lfloor \boxed{6h} \rfloor$$

Second person's:

$$\lfloor \boxed{4h} \rfloor$$

37. FIND SPEED TO CATCH

A starts 3 h earlier at 20 km/h.

Head start:

$$\lfloor 60\text{ km} \rfloor$$

B must catch A in 2 h.

Required relative speed:

$$\lfloor 60/2=30 \rfloor$$

Thus B speed:

$$\lfloor 20+30 = \boxed{50\text{ km/h}} \rfloor$$

38. STARTING AT DIFFERENT TIMES — MEETING

A starts at 8 AM at 40 km/h.

B starts from opposite end at 9 AM at 60 km/h.

Distance = 240 km.

By 9 AM, A has already covered:

$$\lfloor 40\text{ km} \rfloor$$

Remaining separation:

$$\lfloor 200\text{ km} \rfloor$$

After 9 AM, closing speed:

$$\lfloor 40+60=100 \rfloor$$

Meeting after:

$$\lfloor 200/100=2h \rfloor$$

So they meet at:

$$\boxed{11 \text{ AM}}$$

39. SAME-DIRECTION DIFFERENT START TIME

A starts at 8 AM at 30 km/h.

B starts same point at 9 AM at 45 km/h.

A's head start:

$$30 \text{ km}$$

Relative speed:

$$15$$

Catch time:

$$30/15 = 2 \text{ h}$$

So B catches A at:

$$\boxed{11 \text{ AM}}$$

40. MEETING AFTER A FRACTION OF DISTANCE

Two vehicles start toward each other.

Speeds ratio:

$$2:3$$

At meeting, first vehicle covers:

$$\frac{2}{5}$$

of total distance.

Second covers:

$$\frac{3}{5}$$

No detailed calculation needed.

41. FIND SPEED FROM MEETING POINT

Two vehicles start 300 km apart and meet after first covers 120 km.

Second covers:

$$180$$

Speed ratio:

$$120:180 = 2:3$$

If first speed = 40:

Second:

$$\boxed{60}$$

$$\boxed{60 \text{ km/h}}$$

42. AVERAGE SPEED — PREVIEW

Average speed is:

$$\boxed{\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}}$$

It is **not always** the arithmetic mean of speeds.

We will cover average and relative speed deeply in Chapter 30.

43. SIMPLE AVERAGE-SPEED EXAMPLE

A person travels:

- 60 km at 30 km/h
- 60 km at 60 km/h

Total distance:

$$\boxed{120}$$

Times:

$$\boxed{2\text{h}, 1\text{h}}$$

Total time:

$$\boxed{3\text{h}}$$

Average speed:

$$\boxed{120/3 = 40 \text{ km/h}}$$

Not:

$$\boxed{45}$$

44. EQUAL-DISTANCE AVERAGE SPEED PREVIEW

For equal distances at speeds (a, b) :

$$\boxed{V_{\text{avg}} = \frac{2ab}{a+b}}$$

Example:

$$\boxed{a=30, \text{quad } b=60}$$

$$\boxed{V_{\text{avg}} = \frac{2(30)(60)}{90} = 40}$$

Full treatment comes in Chapter 30.

45. EQUAL-TIME AVERAGE SPEED PREVIEW

If equal times are spent at speeds (a, b) :

$$\boxed{V_{\text{avg}} = \frac{a+b}{2}}$$

because time weights are equal.

46. DISTANCE RATIO WHEN TIME SAME

If two objects travel for the same time:

$$\boxed{D_1:D_2=S_1:S_2}$$

Example:

Speeds 50 and 70.

In 3 h:

Distances:

$$\boxed{150:210 = 5:7}$$

same as speed ratio.

47. TIME RATIO WHEN DISTANCE SAME

If distances same:

$$\boxed{T_1:T_2=S_2:S_1}$$

48. SPEED RATIO WHEN TIME SAME

If same duration:

$$\boxed{S_1:S_2=D_1:D_2}$$

These three ratio patterns should become automatic.

49. GATE-STYLE QUESTION 1

A travels 180 km in 3 h.

Speed:

$$\boxed{180/3 = 60 \text{ km/h}}$$

50. GATE-STYLE QUESTION 2

Convert 54 km/h to m/s.

$$\boxed{54 \times \frac{5}{18} = 15 \text{ m/s}}$$

51. GATE-STYLE QUESTION 3

Speed increases by 25%.

Time for same distance decreases by:

$$\boxed{20\%}$$

52. GATE-STYLE QUESTION 4

Two vehicles 300 km apart travel toward each other at 60 and 40 km/h.

Meeting time:

$$\boxed{3\text{h}}$$

53. PLACEMENT-STYLE QUESTION 1

A car at 40 km/h takes 6 h.

At 60 km/h, same distance takes:

$$\boxed{4\text{h}}$$

54. PLACEMENT-STYLE QUESTION 2

At 40 km/h a person is 30 min late.

At 60 km/h, 30 min early.

Time difference:

$$\boxed{1\text{h}}$$

Distance:

$$\boxed{120\text{ km}}$$

55. PLACEMENT-STYLE QUESTION 3

A is 50 km behind B.

Speeds:

$$\boxed{70, 50}$$

Catch time:

$$\boxed{50/(70-50)}$$

$$\boxed{2.5\text{h}}$$

56. PLACEMENT-STYLE QUESTION 4

Two vehicles meet after 4 h.

Their speeds:

$$\boxed{45, 55}$$

Initial distance:

$$\sqrt{(45+55)(4)}$$

$$\sqrt{= \boxed{400 \text{ km}}}$$

57. ADVANCED QUESTION — FIXED DISTANCE

At 48 km/h a journey takes 5 h.

If speed increases by 25%, new speed:

$$\sqrt{60}$$

Distance:

$$\sqrt{48(5)=240}$$

New time:

$$\sqrt{240/60=4\text{h}}$$

Time saved:

$$\sqrt{\boxed{1\text{h}}}$$

58. ADVANCED QUESTION — EARLY/LATE

At 50 km/h, a person arrives 24 min late.

At 60 km/h, arrives 16 min early.

Difference in travel time:

$$\sqrt{40 \text{ min} = \frac{23}{h}}$$

Let distance (D) :

$$\sqrt{\frac{D}{50} - \frac{D}{60} = \frac{23}{h}}$$

$$\sqrt{D \left(\frac{1}{300} \right) = \frac{23}{h}}$$

$$\sqrt{D=200}$$

$$\sqrt{\boxed{200 \text{ km}}}$$

59. ADVANCED QUESTION — CHASE WITH HEAD START

A travels at 40 km/h and starts 1.5 h earlier.

Head start:

$$\sqrt{40(1.5)=60}$$

B travels at 70 km/h.

Relative speed:

$$\sqrt{30}$$

Catch time after B starts:

$$\boxed{60/30 = 2\text{h}}$$

60. ADVANCED QUESTION — OPPOSITE DIRECTIONS AFTER DELAY

A and B are 330 km apart.

A starts toward B at 50 km/h.

After 1 h, B starts toward A at 60 km/h.

A first covers:

$$50$$

Remaining:

$$280$$

Combined speed:

$$110$$

Meeting time after B starts:

$$280/110 = \frac{28}{11}$$

$$\approx 2.545\text{h}$$

Total time after A starts:

$$1 + \frac{28}{11} = \frac{39}{11}$$

$$\boxed{3\frac{6}{11}\text{h}}$$

61. COMMON TRAPS

& pTRAP 1 — Wrong units

km/h and m/s must not be mixed.

& pTRAP 2 — Speed increase = same % time decrease

False because of inverse relation.

& pTRAP 3 — Same direction

Use difference of speeds.

& pTRAP 4 — Opposite directions

Use sum of speeds.

& pTRAP 5 — Head start

Convert early start into distance before applying relative speed.

& pTRAP 6 — Average of speeds

Arithmetic mean is not always average speed.

62. MUST-MEMORIZE FORMULAS

Core

$$\boxed{D=ST}$$

$$\boxed{S=\frac{D}{T}}$$

$$\boxed{T=\frac{D}{S}}$$

Conversion

$$\boxed{\text{km/h} \rightarrow \text{m/s} : \times \frac{5}{18}}$$

$$\boxed{\text{m/s} \rightarrow \text{km/h} : \times \frac{18}{5}}$$

Fixed distance

$$\boxed{S_1 : S_2 = T_2 : T_1}$$

63. RELATIVE MOTION BASICS

Opposite directions

$$\boxed{S_{\text{rel}} = S_1 + S_2}$$

Same direction

$$\boxed{S_{\text{rel}} = |S_1 - S_2|}$$

Meeting/catching time

$$\boxed{T = \frac{\text{initial separation}}{\text{relative speed}}}$$

64. FAST EXAM DECISION TREE

Need ordinary distance/time/speed?

Use $(D=ST)$.

Units differ?

Convert first.

Same distance at different speeds?

Use inverse speed-time relation.

One late, one early?

Use difference of travel times.

Moving toward each other?

Add speeds.

Chasing?

Subtract speeds.

Delayed start?

Convert delay into head-start distance.

Average-speed question?

Use total distance / total time.

PRACTICE — LEVEL 0

Q1

A car travels 240 km in 4 h. Find speed.

Q2

Convert 72 km/h to m/s.

Q3

Convert 15 m/s to km/h.

Q4

A car travels at 50 km/h for 3 h. Find distance.

Q5

Speed increases by 25%. For fixed distance, by what % does time decrease?

LEVEL 1

Q6

A journey takes 6 h at 40 km/h. Find time at 60 km/h.

Q7

Two cars 300 km apart move toward each other at 40 and 60 km/h. Find meeting time.

Q8

A faster car is 80 km behind another. Speeds are 70 and 50 km/h. Find catch time.

Q9

At 30 km/h and 40 km/h, travel-time difference for same distance is 1 h. Find distance.

Q10

A starts 2 h earlier at 30 km/h. B travels at 50 km/h. Find catch time after B starts.

LEVEL 2

Q11

At 40 km/h a traveller arrives 30 min late, and at 60 km/h 30 min early. Find distance.

Q12

Two cars meet after 4 h. Speeds are 45 and 55 km/h. Find initial separation.

Q13

Two vehicles move toward each other with speeds in ratio 3:2. Find ratio of distances covered until meeting.

Q14

A speed decreases by 20%. By what % does travel time increase?

Q15

A journey takes 5 h at 48 km/h. Speed rises 25%. Find new time.

LEVEL 3 — EXAM PRESSURE**Q16**

At 50 km/h, a person is 24 min late. At 60 km/h, 16 min early. Find journey distance.

Q17

A starts at 40 km/h, 1.5 h before B. B travels same direction at 70 km/h. Find catch time after B starts.

Q18

A and B are 330 km apart. A starts toward B at 50 km/h. B starts 1 h later at 60 km/h. Find meeting time measured from A's start.

Q19

Two cars start toward each other 280 km apart at speeds 60 and 40 km/h. Find how far each travels before meeting.

Q20

A person travels 60 km at 30 km/h and another 60 km at 60 km/h. Find average speed.

ANSWERS**Q1**

$$\lfloor 240/4 = \boxed{60\text{ km/h}} \rfloor$$

Q2

$$\lfloor 72 \times \frac{5}{18} = \boxed{20\text{ m/s}} \rfloor$$

Q3

$$\lfloor 15 \times \frac{18}{5} = \boxed{54\text{ km/h}} \rfloor$$

Q4

$$\lfloor 50(3) = \boxed{150\text{ km}} \rfloor$$

Q5

$$\boxed{20\%}$$

Q6

Distance:

$$40(6)=240$$

New time:

$$240/60 = \boxed{4\text{h}}$$

Q7

Relative speed:

$$40+60=100$$

$$300/100 = \boxed{3\text{h}}$$

Q8

Relative speed:

$$70-50=20$$

$$80/20 = \boxed{4\text{h}}$$

Q9

$$D\left(\frac{1}{30}-\frac{1}{40}\right)=1$$

$$D/120=1$$

$$\boxed{120\text{ km}}$$

Q10

Head start:

$$30(2)=60$$

Relative speed:

$$50-30=20$$

Catch time:

$$60/20 = \boxed{3\text{h}}$$

Q11

Time difference:

$$1\text{h}$$

$$D\left(\frac{1}{40}-\frac{1}{60}\right)=1$$

$$D/120=1$$

$$\boxed{120\text{ km}}$$

Q12

$$(45+55)(4) = 100(4)$$

$$= \boxed{400\text{ km}}$$

Q13

Same meeting time:

$$\boxed{3:2}$$

Q14

$$\frac{20}{80} \times 100 = \boxed{25\%}$$

Q15

Speed increase 25% means time factor:

$$\frac{45}{5}$$

$$5 \cdot \frac{45}{5} = \boxed{4h}$$

Q16

Time difference:

$$24 + 16 = 40 \text{ min}$$

$$= \frac{23h}{3}$$

$$D \left(\frac{1}{50} - \frac{1}{60} \right) = \frac{23}{3}$$

$$\frac{D}{300} = \frac{23}{3}$$

$$\boxed{D = 200 \text{ km}}$$

Q17

Head start:

$$40(1.5) = 60$$

Relative speed:

$$70 - 40 = 30$$

$$60/30 = \boxed{2h}$$

Q18

A travels first hour:

$$50 \text{ km}$$

Remaining:

$$280$$

Relative speed after B starts:

$$110$$

Time after B starts:

$$280/110 = 28/11$$

Total from A's start:

$$1 + \frac{28}{11} = \boxed{\frac{39}{11} = 3\frac{6}{11}h}$$

Q19

Relative speed:

$$\boxed{100}$$

Meeting time:

$$\boxed{280/100=2.8\text{h}}$$

First:

$$\boxed{60(2.8)=168}$$

Second:

$$\boxed{40(2.8)=112}$$

$$\boxed{168\text{ km}, 112\text{ km}}$$

Q20

Total distance:

$$\boxed{120}$$

Total time:

$$\boxed{60/30+60/60 = 2+1 = 3}$$

Average speed:

$$\boxed{120/3 = 40\text{ km/h}}$$

CHAPTER 29 STATUS

Covered so far:

- $(D=ST)$
- speed, distance and time calculations
- km/h ! " m/s
- fixed-distance inverse relation
- percentage speed/time changes
- early/late arrival basics
- meeting problems
- same-direction chase
- opposite-direction motion
- head starts
- delayed-start problems
- meeting-point ratios
- average-speed preview
- GATE/placement practice

But **Chapter 29 is not complete yet.**

Next we need:

- advanced early/late arrival equations
- meeting after staggered starts
- repeated meeting basics
- journey split into fractions

- stopping/rest-time problems
- effective speed including halts
- scheduled-speed questions
- speed increase needed to recover lost time
- advanced relative motion
- final shortcuts and chapter audit

Progress

Completed: 28/76 **Current:** ~~0=0~~ Chapter 29 — Time, Speed and Distance **Next:** Chapter 29 Part 2 — **Advanced journey timing, stoppages, delayed starts & relative-motion problems**

USER

next

ASSISTANT

CHAPTER 29 — TIME, SPEED AND DISTANCE

PART 2 — ADVANCED JOURNEY TIMING, STOPPAGES, LOST-TIME RECOVERY, STAGGERED STARTS & RELATIVE MOTION

Status: ~~0=0~~ **PROGRESS GATE:** High **Placements:** Very High **Priority:** ~~0=0~~ **Yr 1**

Part 1 covered the core:

$\boxed{D=ST}$

Now we handle the tougher patterns:

- early/late arrival
- stoppages and halts
- effective speed
- recovering lost time
- split journeys
- staggered starts
- repeated meetings
- speed needed to arrive exactly on time
- advanced relative-motion questions

1. EARLY/LATE ARRIVAL — MASTER STRUCTURE

Suppose scheduled journey time is:

T

At slower speed (S_1) , traveller reaches (a) hours late.

Then actual travel time:

$$\sqrt{T+a}$$

At faster speed $\sqrt{S_2}$, traveller reaches $\sqrt{b}$ hours early.

Then actual travel time:

$$\sqrt{T-b}$$

Since distance is same:

$$\sqrt{\boxed{S_1(T+a)=S_2(T-b)}}$$

This can directly give scheduled time.

2. EXAMPLE — FIND SCHEDULED TIME

At 40 km/h, a traveller arrives 1 hour late.

At 60 km/h, arrives 1 hour early.

Let scheduled time = $\sqrt{T}$.

Then:

$$\sqrt{40(T+1)=60(T-1)}$$

$$\sqrt{40T+40=60T-60}$$

$$\sqrt{100=20T}$$

$$\sqrt{T=5}$$

Scheduled travel time:

$$\sqrt{\boxed{5h}}$$

Distance:

$$\sqrt{40(6)=240}$$

or:

$$\sqrt{60(4)=240}$$

$$\sqrt{\boxed{240\text{ km}}}$$

3. FAST EARLY/LATE DISTANCE FORMULA

If:

- speed $\sqrt{S_1}$ gives $\sqrt{a}$ late
- speed $\sqrt{S_2}$ gives $\sqrt{b}$ early

then travel-time difference is:

$$\sqrt{a+b}$$

Hence:

$$\sqrt{\boxed{D=\frac{a+b}{\frac{1}{S_1}-\frac{1}{S_2}}}}$$

where time units are consistent.

4. EXAMPLE

At 30 km/h, 20 min late.

At 40 km/h, 10 min early.

Difference:

$$\left[30 \text{ min} = \frac{1}{2} \text{ h} \right]$$

Then:

$$\left[D = \frac{1}{2} \left(\frac{1}{30} - \frac{1}{40} \right) \right]$$

$$\left[= \frac{1}{2} \left(\frac{1}{120} \right) \right]$$

$$\left[= \boxed{60 \text{ km}} \right]$$

5. FIND SCHEDULED SPEED

Suppose distance is fixed and scheduled time is known.

Example:

Distance:

$$\left[180 \text{ km} \right]$$

Scheduled time:

$$\left[4 \text{ h} \right]$$

Scheduled speed:

$$\left[180/4 = \boxed{45 \text{ km/h}} \right]$$

6. LOSS OF TIME DUE TO STOPPAGE

Suppose a bus normally travels at 60 km/h.

It stops for 30 minutes during a 180 km trip.

Moving time:

$$\left[180/60 = 3 \text{ h} \right]$$

Total journey time:

$$\left[3 + 0.5 = 3.5 \text{ h} \right]$$

Effective average speed:

$$\left[180/3.5 = \boxed{51.43 \text{ km/h approximately}} \right]$$

7. RUNNING SPEED VS EFFECTIVE SPEED

Running speed

Speed while vehicle is actually moving.

Effective speed

Total distance divided by total elapsed time, including stops.

$$\boxed{S_{\text{effective}}} = \frac{\text{total distance}}{\text{moving time} + \text{stoppage time}}$$

8. EXAMPLE — REGULAR HALTS

A bus moves at 60 km/h but stops 10 minutes after every 50 minutes of movement.

In each 60-minute block:

Moving time:

$$50 \text{ min}$$

Distance covered:

$$60 \times \frac{50}{60} = 50 \text{ km}$$

Effective speed:

$$\boxed{50 \text{ km/h}}$$

9. DUTY-CYCLE SHORTCUT FOR STOPS

If vehicle moves for fraction f of total elapsed time at running speed S :

$$\boxed{S_{\text{effective}}} = fS$$

Example:

Moves 45 min, rests 15 min every hour.

$$f = \frac{45}{60} = \frac{3}{4}$$

If running speed = 80:

$$S_{\text{effective}} = 80 \times \frac{3}{4} = \boxed{60 \text{ km/h}}$$

10. FIND STOPPAGE TIME FROM EFFECTIVE SPEED

A vehicle's running speed:

$$72 \text{ km/h}$$

Effective speed:

$$60 \text{ km/h}$$

Suppose in every hour total elapsed time, moving fraction is f .

$$60 = 72f$$

$$f = \frac{5}{6}$$

So it moves:

$$50 \text{ min per hour}$$

and stops:

$$\boxed{10 \text{ min per hour}}$$

11. LOST TIME RECOVERY

Suppose a traveller loses some time due to a stop and must increase speed to still arrive on time.

The core idea:

$$\boxed{\text{remaining distance} = \text{new speed} \times \text{available remaining time}}$$

12. EXAMPLE — RECOVER A 30-MINUTE DELAY

A journey is 180 km at scheduled speed 60 km/h.

Scheduled journey time:

$$3 \text{ h}$$

After travelling 60 km, traveller stops for 30 min.

Time used before stop:

$$60/60 = 1 \text{ h}$$

Total elapsed:

$$1.5 \text{ h}$$

Scheduled time remaining:

$$3 - 1.5 = 1.5 \text{ h}$$

Remaining distance:

$$120 \text{ km}$$

Required speed:

$$120/1.5 = \boxed{80 \text{ km/h}}$$

13. GENERAL LOST-TIME FORMULA

Suppose remaining distance is (D_r) .

Scheduled remaining time before delay would have been:

$$t$$

$$\text{Delay} = (d)$$

Actual available time:

$$t - d$$

Required speed:

$$\boxed{S_{\text{new}} = \frac{D_r}{t - d}}$$

14. SPEED INCREASE NEEDED TO SAVE TIME

Suppose a fixed journey currently takes T .

Want to reduce time by d .

New time:

$$T - d$$

Since speed is inversely proportional to time:

$$\frac{S_{\text{new}}}{S_{\text{old}}} = \frac{T}{T - d}$$

So required % increase:

$$\boxed{\frac{d}{T - d} \times 100}$$

15. EXAMPLE

A journey takes 5 h.

Need to save 1 h.

New time:

$$4 \text{ h}$$

Required speed increase:

$$\frac{1}{4} \times 100 = \boxed{25\%}$$

16. JOURNEY SPLIT INTO DISTANCE FRACTIONS

Suppose a person travels:

- first half distance at S_1
- second half at S_2

Then average speed is harmonic mean:

$$\boxed{V_{\text{avg}} = \frac{2S_1S_2}{S_1 + S_2}}$$

We previewed this in Part 1.

17. EXAMPLE

Half distance at 40 km/h.

Half at 60 km/h.

$$V_{\text{avg}} = \frac{2(40)(60)}{100}$$

$$= \boxed{48 \text{ km/h}}$$

18. SPLIT INTO UNEQUAL DISTANCE FRACTIONS

Suppose:

- fraction f of distance at speed S_1

- remaining $(1-f)$ at speed (S_2)

Take total distance = 1.

Time:

$$\left[\frac{f}{S_1} + \frac{1-f}{S_2} \right]$$

Average speed:

$$\left[\boxed{V_{\text{avg}}} = \frac{1}{\frac{f}{S_1} + \frac{1-f}{S_2}} \right]$$

19. EXAMPLE

One-third of journey at 30 km/h.

Two-thirds at 60 km/h.

Take total distance:

$$\left[180 \text{ km} \right]$$

First:

$$\left[60 \text{ km at } 30 \rightarrow 2 \text{ h} \right]$$

Second:

$$\left[120 \text{ km at } 60 \rightarrow 2 \text{ h} \right]$$

Total:

$$\left[180/4 = \boxed{45 \text{ km/h}} \right]$$

20. JOURNEY SPLIT INTO TIME FRACTIONS

If:

- half the time at (S_1)
- half the time at (S_2)

average speed:

$$\left[\boxed{\frac{S_1 + S_2}{2}} \right]$$

For unequal time fractions, use weighted average of speeds.

21. EXAMPLE

2 h at 40 km/h.

3 h at 60 km/h.

Distance:

$$\left[80 + 180 = 260 \right]$$

Total time:

$$\left[5 \text{ h} \right]$$

Average speed:

$$\boxed{52 \text{ km/h}}$$

22. FIRST HALF TIME VS FIRST HALF DISTANCE

These are different statements.

Half distance at 40, half distance at 60:

Average:

$$\boxed{48}$$

Half time at 40, half time at 60:

Average:

$$\boxed{50}$$

This is a common trap.

23. DELAYED START — SAME DIRECTION

A starts first at speed (S_A) .

B starts (h) hours later at faster speed (S_B) .

Head start:

$$\boxed{S_A h}$$

Relative speed:

$$\boxed{S_B - S_A}$$

Catch time after B starts:

$$\boxed{t = \frac{S_A h}{S_B - S_A}}$$

24. EXAMPLE

A starts 2 h earlier at 30 km/h.

B travels at 50 km/h.

$$\boxed{t = \frac{30(2)}{50-30}}$$

$$\boxed{= 3h}$$

25. DELAYED START — OPPOSITE DIRECTIONS

Two objects start from opposite ends, but one starts later.

Method:

1. Let first traveller cover some distance alone.
 2. Reduce separation.
 3. Use sum of speeds after both are moving.
-

26. EXAMPLE

Cities 400 km apart.

A starts at 50 km/h.

B starts 2 h later at 75 km/h toward A.

A first covers:

$$\boxed{100}$$

Remaining separation:

$$\boxed{300}$$

Closing speed:

$$\boxed{125}$$

Time after B starts:

$$\boxed{300/125 = 2.4\text{h}}$$

Total after A starts:

$$\boxed{4.4\text{h}}$$

27. MOVING AWAY FROM EACH OTHER

If two objects start from same point and move in opposite directions:

Separation rate:

$$\boxed{S_1 + S_2}$$

Example:

40 and 60 km/h for 3 h:

$$\boxed{(40+60)(3) = 300\text{ km apart}}$$

28. SAME DIRECTION FROM SAME POINT

If they start together from same point and move same direction at different speeds:

Gap after t :

$$\boxed{S_1 - S_2 t}$$

Example:

60 and 45 km/h for 4 h:

$$\boxed{15(4) = 60\text{ km}}$$

29. MOVING APART FROM DIFFERENT STARTING POINTS

Initial separation = D .

If they move away from each other:

$$\boxed{\text{separation after } t = D + (S_1 + S_2)t}$$

30. CLOSING FROM DIFFERENT STARTING POINTS

If moving toward each other:

$$\boxed{\text{separation after } t = D - (S_1 + S_2)t}$$

until meeting.

31. REPEATED MEETINGS — BASIC IDEA

If two objects move continuously back and forth between two endpoints, repeated meeting problems can become more complex.

For basic aptitude:

- first meeting uses relative speed
- after reversal, reconsider directions and positions
- do not assume every meeting interval is identical unless symmetry supports it

We will handle circular repeated meetings separately in Chapter 33.

32. FIRST MEETING POSITION

A and B start from opposite ends of a 360 km route.

Speeds:

$$40, 50$$

Meeting time:

$$360/90 = 4\text{h}$$

A covers:

$$160$$

B:

$$200$$

So meeting point is:

$$\boxed{160 \text{ km from A's end}}$$

33. AFTER MEETING, CONTINUE TO OPPOSITE ENDS

Same example.

After meeting:

A has 200 km remaining at 40:

$$5\text{h}$$

B has 160 km remaining at 50:

$$\boxed{3.2\text{h}}$$

B reaches the opposite end first.

This type of comparison appears in placement problems.

34. IF TIMES AFTER MEETING ARE GIVEN

Suppose A and B meet while moving from opposite ends.

After meeting:

- A takes (t_A) to reach B's start
- B takes (t_B) to reach A's start

Then speed ratio satisfies:

$$\boxed{\frac{S_A}{S_B} = \sqrt{\frac{t_B}{t_A}}}$$

This is a high-value advanced result.

35. WHY THE SQUARE ROOT APPEARS

Before meeting:

If A covers (x) , B covers (y) ,

$$\frac{x}{y} = \frac{S_A}{S_B}$$

After meeting:

A must cover (y) :

$$t_A = \frac{y}{S_A}$$

B must cover (x) :

$$t_B = \frac{x}{S_B}$$

Then:

$$\frac{t_B}{t_A} = \frac{x/S_B}{y/S_A} = \frac{xS_A}{yS_B}$$

Since:

$$\frac{x}{y} = \frac{S_A}{S_B}$$

we get:

$$\frac{t_B}{t_A} = \left(\frac{S_A}{S_B}\right)^2$$

Hence:

$$\boxed{S_A:S_B = \sqrt{t_B:t_A}}$$

36. EXAMPLE

After meeting:

A takes 9 h to reach B's starting point.

B takes 4 h to reach A's starting point.

Then:

$$S_A:S_B = \sqrt{4:9}$$

$$= \boxed{2:3}$$

37. FIND DISTANCE USING AFTER-MEETING TIMES

Suppose speed ratio:

$$2:3$$

and after meeting A takes 9 h.

Let speeds:

$$2k, 3k$$

A after meeting covers B's pre-meeting distance.

You can reconstruct route if one actual speed or distance is supplied.

Without an absolute quantity, ratio alone is insufficient.

38. SPEED CHANGE MID-JOURNEY

A car travels first 2 h at 40 km/h, then next 3 h at 60 km/h.

Total distance:

$$80+180=260$$

Average:

$$260/5 = \boxed{52 \text{ km/h}}$$

39. SPEED CHANGES AFTER DISTANCE

A travels first 100 km at 50 km/h and remaining 150 km at 75 km/h.

Time:

$$2+2=4\text{h}$$

Distance:

$$250$$

Average:

$$\boxed{62.5 \text{ km/h}}$$

40. LOST TIME DUE TO LOWER SPEED

Scheduled:

$$60 \text{ km/h}$$

Actual:

$$\boxed{48 \text{ km/h}}$$

For same distance:

$$T_{\text{actual}} = T_{\text{scheduled}} \frac{60}{48} = 1.25T$$

So delay:

$$\boxed{25\% \text{ of scheduled time}}$$

41. EXAMPLE

Scheduled journey time:

$$\boxed{4 \text{ h}}$$

Speed falls from 60 to 48 km/h.

Actual:

$$\boxed{5 \text{ h}}$$

Delay:

$$\boxed{1 \text{ h}}$$

42. FIND SPEED TO ARRIVE ON TIME AFTER SLOW START

Journey = 300 km.

Scheduled speed = 60 km/h.

Scheduled time:

$$\boxed{5 \text{ h}}$$

First 2 h travelled at 40 km/h:

$$\boxed{80 \text{ km}}$$

Remaining:

$$\boxed{220}$$

Remaining scheduled time:

$$\boxed{3 \text{ h}}$$

Required speed:

$$\boxed{220/3}$$

$$\boxed{= 73 \frac{1}{3} \text{ km/h}}$$

43. STOP + SPEED RECOVERY

Journey = 240 km.

Scheduled at 60 km/h:

$$\backslash[4h \backslash]$$

After travelling 120 km at 60, stop for 30 min.

Travel time used:

$$\backslash[2h \backslash]$$

Elapsed:

$$\backslash[2.5h \backslash]$$

Remaining scheduled time:

$$\backslash[1.5h \backslash]$$

Remaining distance:

$$\backslash[120 \backslash]$$

Required:

$$\backslash[120/1.5 = \boxed{80\text{ km/h}} \backslash]$$

44. FIND STOPPAGE FROM LATE ARRIVAL

A vehicle travels 300 km at 60 km/h but arrives 45 min later than pure travel time.

Pure moving time:

$$\backslash[5h \backslash]$$

Actual elapsed:

$$\backslash[5.75h \backslash]$$

Therefore total stoppage:

$$\backslash[\boxed{45\text{ min}} \backslash]$$

45. EFFECTIVE SPEED WITH A SINGLE STOP

Distance:

$$\backslash[D \backslash]$$

Running speed:

$$\backslash[S \backslash]$$

Stop time:

$$\backslash[h \backslash]$$

Effective speed:

$$\backslash[\boxed{\frac{D}{D/S+h}} \backslash]$$

46. EXAMPLE

Distance 120 km.

Running speed 60 km/h.

Stop 30 min:

$$S_{\text{eff}} = \frac{120}{2+0.5}$$

$$= \boxed{48 \text{ km/h}}$$

47. FIND RUNNING SPEED FROM EFFECTIVE SPEED

Distance:

$$120 \text{ km}$$

Effective speed:

$$48 \text{ km/h}$$

Total time:

$$2.5 \text{ h}$$

Stop:

$$0.5 \text{ h}$$

Moving time:

$$2 \text{ h}$$

Running speed:

$$120/2 = \boxed{60 \text{ km/h}}$$

48. OPTION-ELIMINATION SANITY RULES

For fixed distance:

- higher speed !' lower time
 - lower speed !' higher time
 - effective speed with stops < running speed
 - average speed lies between minimum and maximum segment speeds
 - in a chase, catch time must be positive only if pursuer is faster
 - toward-each-other meeting time uses sum of speeds
-

49. GATE-STYLE QUESTION 1

At 40 km/h traveller is 1 h late; at 60 km/h 1 h early.

Distance:

$$\boxed{240 \text{ km}}$$

Scheduled time:

$$\boxed{5 \text{ h}}$$

50. GATE-STYLE QUESTION 2

Vehicle runs at 72 km/h but effective speed is 60 km/h.

Moving fraction:

$$\sqrt[60/72=5/6]{}$$

So stoppage per hour:

$$\sqrt[\boxed{10\text{ min}}]{}$$

51. GATE-STYLE QUESTION 3

Journey takes 5 h. Need to save 1 h.

Required speed increase:

$$\sqrt[\frac{1}{4}\times 100 = \boxed{25\%}]{}$$

52. GATE-STYLE QUESTION 4

Half distance at 40 and half at 60.

Average:

$$\sqrt[\frac{2(40)(60)}{100} = \boxed{48\text{ km/h}}]{}$$

53. PLACEMENT-STYLE QUESTION 1

A bus moves 50 min and rests 10 min every hour.

Running speed = 72 km/h.

Effective:

$$\sqrt[72\cdot\frac{5}{6} = \boxed{60\text{ km/h}}]{}$$

54. PLACEMENT-STYLE QUESTION 2

A starts 3 h earlier at 20 km/h.

B catches in 2 h.

Head start:

$$\sqrt[60]{}$$

Required relative speed:

$$\sqrt[30]{}$$

B speed:

$$\sqrt[\boxed{50\text{ km/h}}]{}$$

55. PLACEMENT-STYLE QUESTION 3

A vehicle travels:

- 100 km at 50 km/h
- 150 km at 75 km/h

Total time:

$$\backslash[2+2=4 \backslash]$$

Average:

$$\backslash[250/4 = \boxed{62.5\text{ km/h}} \backslash]$$

56. PLACEMENT-STYLE QUESTION 4

A and B meet. After meeting A needs 9 h, B 4 h to reach opposite ends.

Speed ratio:

$$\backslash[\sqrt{4:9} \backslash]$$

$$\backslash[=\boxed{2:3} \backslash]$$

57. ADVANCED QUESTION — EARLY/LATE

At 48 km/h, traveller is 20 min late.

At 60 km/h, 10 min early.

Difference:

$$\backslash[30\text{ min}=\frac{12}{h} \backslash]$$

$$\backslash[D\left(\frac{1}{48}-\frac{1}{60}\right)=\frac{12}{h} \backslash]$$

$$\backslash[\frac{1}{48}-\frac{1}{60} = \frac{5-4}{240} = \frac{1}{240} \backslash]$$

$$\backslash[D/240=\frac{12}{h} \backslash]$$

$$\backslash[\boxed{D=120\text{ km}} \backslash]$$

58. ADVANCED QUESTION — RECOVER DELAY

Journey:

$$\backslash[360\text{ km} \backslash]$$

Scheduled speed:

$$\backslash[60 \backslash]$$

Scheduled time:

$$\backslash[6\text{ h} \backslash]$$

First 2 h at 45 km/h:

$$\backslash[90\text{ km} \backslash]$$

Remaining:

$$\boxed{270}$$

Scheduled time left:

$$\boxed{4\text{ h}}$$

Required speed:

$$\boxed{270/4 = 67.5\text{ km/h}}$$

59. ADVANCED QUESTION — STOP AFTER HALF DISTANCE

Journey = 200 km.

Running speed = 50 km/h.

After 100 km, stop for 30 min.

Then continue at 75 km/h.

First time:

$$\boxed{100/50=2\text{ h}}$$

Stop:

$$\boxed{0.5\text{ h}}$$

Second time:

$$\boxed{100/75=\frac{4}{3}\text{ h}}$$

Total:

$$\boxed{2+\frac{1}{2}+\frac{4}{3} = \frac{23}{6}}$$

$$\boxed{=3\frac{5}{6}\text{ h}}$$

Average speed:

$$\boxed{200\div\frac{23}{6} = \frac{1200}{23}}$$

$$\boxed{\approx 52.17\text{ km/h}}$$

60. ADVANCED QUESTION — DIFFERENT STARTS OPPOSITE ENDS

Distance = 500 km.

A starts at 40 km/h.

B starts 2 h later at 60 km/h.

A's head coverage:

$$\boxed{80}$$

Remaining:

$$\boxed{420}$$

Relative speed after B starts:

$$\boxed{100}$$

Meeting after B starts:

$\boxed{4.2\text{h}}$

From A's start:

$\boxed{6.2\text{h}}$

61. COMMON TRAPS

& pTRAP 1 — Early and late difference

If one is 20 min late and another 10 min early, travel-time difference is:

30min

not 10 min.

& pTRAP 2 — Stoppage included in running speed

Running speed excludes stops.

Effective speed includes stops.

& pTRAP 3 — Average speed from arithmetic mean

Only valid in equal-time cases.

& pTRAP 4 — Lost time recovery

Use remaining distance and remaining scheduled time, not total distance.

& pTRAP 5 — Delayed start

Convert delay to head-start distance.

& pTRAP 6 — After-meeting square-root relation

Do not use direct time ratio; use square root.

62. MUST-MEMORIZE FORMULAS

Early/late

$$D\left(\frac{S_1}{S_2}\right) = \text{difference in travel times}$$

Effective speed

$$S_{\text{eff}} = \frac{\text{total distance}}{\text{total elapsed time}}$$

Speed needed to save (d) from old time (T)

$$\% \uparrow S = \frac{d}{T-d} \times 100$$

63. SPLIT-JOURNEY FORMULAS

Equal distances

$$\boxed{V_{\text{avg}}} = \frac{2ab}{a+b}$$

Equal times

$$\boxed{V_{\text{avg}}} = \frac{a+b}{2}$$

General

$$\boxed{V_{\text{avg}}} = \frac{\sum D_i}{\sum D_i/S_i}$$

64. ADVANCED RELATIVE-MOTION FORMULA

After meeting:

$$\boxed{S_A:S_B = \sqrt{t_B:t_A}}$$

where (t_A) and (t_B) are the times taken after meeting to reach opposite starting points.

65. FAST EXAM DECISION TREE

Early/late?

Add early + late to get travel-time difference.

Stops?

Separate moving time from elapsed time.

Need to recover delay?

Remaining distance ÷ remaining scheduled time.

Split journey by distance?

Use harmonic/weighted-distance approach.

Split by time?

Use time-weighted average speed.

Delayed start?

Create head-start distance.

Toward each other?

Add speeds.

Same direction?

Subtract speeds.

Given after-meeting times?

Try square-root speed ratio.

PRACTICE — LEVEL 0

Q1

A journey takes 5 h. How much must speed increase to reduce time to 4 h?

Q2

A car runs at 60 km/h for 2 h and stops 30 min. It travels no more afterward. Find effective speed over the total 2.5 h.

Q3

Half distance at 30 and half at 60. Find average speed.

Q4

A starts 2 h earlier at 25 km/h. Find head-start distance.

Q5

A vehicle moves 45 min and rests 15 min every hour at 80 km/h. Find effective speed.

LEVEL 1

Q6

At 30 km/h traveller is 20 min late; at 40 km/h 10 min early. Find distance.

Q7

A journey normally takes 6 h. Speed drops by 20%. Find new time.

Q8

A starts 1.5 h earlier at 40 km/h. B travels at 70 km/h. Find catch time after B starts.

Q9

One-third of journey at 30 km/h and two-thirds at 60 km/h. Find average speed.

Q10

A bus runs at 72 km/h but effective speed is 60 km/h. Find stoppage per hour.

LEVEL 2

Q11

Journey is 240 km at scheduled 60 km/h. After travelling 120 km, vehicle stops 30 min. Find speed needed for remaining journey to arrive on time.

Q12

Two vehicles start from opposite ends 400 km apart. A starts at 50 km/h. B starts 2 h later at 75 km/h. Find meeting time from A's start.

Q13

After meeting, A takes 16 h and B takes 9 h to reach opposite ends. Find speed ratio $\sqrt{A:B}$.

Q14

A journey is 200 km. First 100 km at 50 km/h, stop 30 min, rest at 75 km/h. Find average speed.

Q15

At 48 km/h traveller is 20 min late; at 60 km/h 10 min early. Find distance.

LEVEL 3 — EXAM PRESSURE**Q16**

A 300 km journey is scheduled at 60 km/h. First 100 km are travelled at 50 km/h. What speed is required for the remainder to arrive exactly on time?

Q17

A bus runs 40 min and rests 20 min every hour. If effective speed must be 40 km/h, find its running speed.

Q18

A and B start toward each other 360 km apart at 40 and 50 km/h. After meeting, how long does each take to reach the opposite endpoint?

Q19

A journey takes 8 h at scheduled speed. A 1 h stop occurs exactly halfway through the distance. What percentage must speed increase on the second half to still arrive on time?

Q20

Two people start from the same point. A at 30 km/h starts 2 h earlier. B starts later and catches A in 3 h. Find B's speed.

ANSWERS**Q1**

$$\left[\frac{5}{4} = 1.25 \right]$$

$$\left[\boxed{25\%} \right]$$

Q2

Distance travelled:

$$\left[60(2) = 120 \right]$$

Elapsed:

$$\left[2.5\text{h} \right]$$

Effective:

$$\left[120/2.5 = \boxed{48\text{ km/h}} \right]$$

Q3

$$\left[\frac{2(30)(60)}{90} = \boxed{40\text{ km/h}} \right]$$

Q4

$$\sqrt[2]{25} = \boxed{5 \text{ km}}$$

Q5

Moving fraction:

$$\sqrt[4]{45/60} = 3/4$$

$$\sqrt[4]{80(3/4)} = \boxed{60 \text{ km/h}}$$

Q6

Time difference:

$$\sqrt[2]{30 \text{ min}} = 1/2 \text{ h}$$

$$\sqrt[D]{\left(\frac{1}{30} - \frac{1}{40}\right)} = 1/2$$

$$\sqrt[D/120]{} = 1/2$$

$$\sqrt[2]{\boxed{60 \text{ km}}}$$

Q7

Speed falls 20%, time rises 25%.

$$\sqrt[6]{1.25} = \boxed{7.5 \text{ h}}$$

Q8

Head start:

$$\sqrt[4]{40(1.5)} = 60$$

Relative speed:

$$\sqrt[3]{30}$$

Catch:

$$\sqrt[6]{60/30} = \boxed{2 \text{ h}}$$

Q9

Take 180 km.

First third:

$$\sqrt[6]{60/30} = 2 \text{ h}$$

Remaining:

$$\sqrt[6]{120/60} = 2 \text{ h}$$

Average:

$$\sqrt[4]{180/4} = \boxed{45 \text{ km/h}}$$

Q10

$$\sqrt[6]{60/72} = 5/6$$

Moving:

$$\sqrt[5]{50 \text{ min}}$$

Stop:

$$\boxed{10 \text{ min}}$$

Q11

Scheduled total time:

$$240/60=4\text{h}$$

First half:

$$120/60=2\text{h}$$

Stop:

$$0.5\text{h}$$

Remaining available:

$$1.5\text{h}$$

Required:

$$120/1.5 = \boxed{80 \text{ km/h}}$$

Q12

A first 2 h:

$$100 \text{ km}$$

Remaining:

$$300$$

Closing speed:

$$125$$

After B starts:

$$300/125=2.4\text{h}$$

Total from A:

$$\boxed{4.4\text{h}}$$

Q13

$$S_A:S_B = \sqrt{9:16}$$

$$=\boxed{3:4}$$

Q14

First:

$$100/50=2\text{h}$$

Stop:

$$0.5\text{h}$$

Second:

$$100/75=4/3\text{h}$$

Total:

$$23/6\text{h}$$

Average:

$$200 \cdot \frac{6}{23} = \boxed{\frac{1200}{23}} \approx 52.17 \text{ km/h}$$

Q15

Difference:

$$30 \text{ min} = \frac{12}{h}$$

$$D \left(\frac{1}{48} - \frac{1}{60} \right) = \frac{12}{h}$$

$$D \left(\frac{1}{240} \right) = \frac{1}{2}$$

$$\boxed{120 \text{ km}}$$

Q16

Scheduled time:

$$300/60 = 5 \text{ h}$$

First 100 km at 50:

$$2 \text{ h}$$

Remaining time:

$$3 \text{ h}$$

Remaining distance:

$$200$$

Required:

$$200/3 = \boxed{66 \frac{2}{3} \text{ km/h}}$$

Q17

Moving fraction:

$$40/60 = 2/3$$

$$40 = \frac{2}{3} S$$

$$S = 60$$

$$\boxed{60 \text{ km/h}}$$

Q18

Meeting time:

$$360/(40+50) = 4 \text{ h}$$

Distances before meeting:

A:

$$160$$

B:

$$200$$

After meeting:

A must cover 200 at 40:

$$\boxed{5h}$$

B must cover 160 at 50:

$$\boxed{3.2h}$$

Q19

Scheduled time:

$$8h$$

At constant scheduled speed, half distance normally takes:

$$4h$$

First half takes 4 h, then 1 h stop.

Elapsed:

$$5h$$

Remaining scheduled time:

$$3h$$

Second half normally requires 4 h.

To do same distance in 3 h:

Speed factor:

$$\frac{4}{3}$$

Increase:

$$\boxed{33\frac{1}{3}\%}$$

Q20

A's head start:

$$30(2)=60$$

B catches in 3 h.

Required relative speed:

$$60/3=20$$

So B:

$$30+20 = \boxed{50\text{ km/h}}$$

' CHAPTER 29 COMPLETE

We have now covered:

- $(D=ST)$
- unit conversion
- fixed-distance inverse relation
- speed/time percentage changes
- early/late arrival
- scheduled-time questions

- stoppages and halts '
- running vs effective speed '
- lost-time recovery '
- required speed to arrive on time '
- split-distance journeys '
- split-time journeys '
- delayed starts '
- head-start chase '
- opposite-direction meeting '
- same-direction catching '
- journey phases '
- after-meeting square-root relation '
- advanced GATE/placement problems '

Progress

Completed: 29/76 Unit 1 — Quantitative Aptitude: 0=14 PROGRESS

NEXT

0=6 Chapter 30 — Average and Relative Speed

Next we will go deeper into:

- average-speed master formula
- equal-distance harmonic mean
- equal-time arithmetic mean
- multiple-segment journeys
- weighted average speed
- relative speed
- same/opposite direction
- moving objects and observers
- catch-up and meeting shortcuts
- advanced multi-object relative-motion problems.

USER

next

ASSISTANT

CHAPTER 30 — AVERAGE AND RELATIVE SPEED

PART 1 — AVERAGE SPEED MASTER FORMULA, EQUAL DISTANCE, EQUAL TIME, MULTIPLE SEGMENTS & RELATIVE SPEED BASICS

Chapter 29 introduced average speed and relative motion.

Now we make both into a complete exam toolkit.

The two master ideas are:

$$\boxed{\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}}$$

and:

$$\boxed{\text{Relative Speed} = \text{speed of one object relative to another}}$$

1. AVERAGE SPEED IS NOT USUALLY A SIMPLE AVERAGE

Suppose a person travels:

- 60 km at 30 km/h
- 60 km at 60 km/h

Arithmetic average of speeds:

$$\frac{30+60}{2}=45$$

But this is wrong.

Time taken:

$$60/30=2\text{h}$$

and:

$$60/60=1\text{h}$$

Total distance:

$$120$$

Total time:

$$3$$

Average speed:

$$120/3 = \boxed{40\text{ km/h}}$$

2. MASTER FORMULA

Always remember:

$$\boxed{V_{\text{avg}} = \frac{D_1+D_2+\cdots}{\frac{D_1}{V_1}+\frac{D_2}{V_2}+\cdots}}$$

This works for every segmented journey.

3. WHEN CAN WE USE ARITHMETIC MEAN?

Only when equal **time** is spent at each speed.

If a person moves:

- 1 hour at 40 km/h

- 1 hour at 60 km/h

Then:

$$V_{\text{avg}} = \frac{40+60}{2} = \boxed{50}$$

Because times are equal.

4. EQUAL-TIME FORMULA

For equal times at speeds (a) and (b) :

$$V_{\text{avg}} = \frac{a+b}{2}$$

MUST MEMORIZE

5. EQUAL-DISTANCE FORMULA

For equal distances travelled at speeds (a,b) :

$$V_{\text{avg}} = \frac{2ab}{a+b}$$

This is the harmonic mean of the two speeds.

MUST MEMORIZE

6. WHY THE EQUAL-DISTANCE FORMULA WORKS

Assume each part has distance (D) .

Total distance:

$$2D$$

Total time:

$$\frac{D}{a} + \frac{D}{b}$$

So:

$$V_{\text{avg}} = \frac{2D}{D/a + D/b}$$

Cancel (D) :

$$V = \frac{2ab}{a+b}$$

7. EXAMPLE — EQUAL DISTANCES

Half journey at:

$$40$$

and half at:

$$60$$

$$V_{\text{avg}} = \frac{2(40)(60)}{100}$$

$$V = \boxed{48 \text{ km/h}}$$

8. SANITY CHECK

For unequal speeds:

$$[a < b]$$

Equal-distance average must be:

- greater than (a)
- less than (b)
- closer to the slower speed

Example:

40 and 60 give 48.

That makes sense because more time is spent at 40.

9. EQUAL-TIME VS EQUAL-DISTANCE

For speeds 40 and 60:

Equal time

$$[V_{\text{avg}} = 50]$$

Equal distance

$$[V_{\text{avg}} = 48]$$

Therefore:

$$[\boxed{\text{equal-distance average}} \leq \text{equal-time average}]$$

for positive unequal speeds.

10. ROUND TRIP

A person goes from A to B at speed (a) and returns the same distance at speed (b) .

Distances are equal.

So:

$$[\boxed{V_{\text{avg}} = \frac{2ab}{a+b}}]$$

Example:

Going:

$$[30]$$

Returning:

$$[60]$$

Average:

$$[\frac{2(30)(60)}{90} = \boxed{40}]$$

Not 45.

11. RETURN SPEED FROM AVERAGE SPEED

Suppose onward speed is:

$$\sqrt[4]{40}$$

Average for equal-distance round trip is:

$$\sqrt[4]{48}$$

Return speed = $\sqrt[4]{x}$.

$$\sqrt[4]{48 = \frac{2(40)x}{40+x}}$$

$$\sqrt[4]{48(40+x) = 80x}$$

$$\sqrt[4]{1920 + 48x = 80x}$$

$$\sqrt[4]{32x = 1920}$$

$$\sqrt[4]{x = \boxed{60}}$$

12. FAST FORMULA FOR UNKNOWN SECOND SPEED

If equal-distance average = $\sqrt[4]{A}$, first speed $\sqrt[4]{a}$, second speed $\sqrt[4]{b}$:

$$\sqrt[4]{A = \frac{2ab}{a+b}}$$

Solving for $\sqrt[4]{b}$:

$$\sqrt[4]{\boxed{b = \frac{Aa}{2a-A}}}$$

Useful, but **understand rather than memorize**.

13. THREE EQUAL DISTANCES

Suppose equal distance portions at speeds:

$$\sqrt[4]{a, b, c}$$

Then:

$$\sqrt[4]{\boxed{V_{\text{avg}} = \frac{3}{1/a + 1/b + 1/c}}}$$

Example:

Equal distances at:

$$\sqrt[4]{30, 45, 90}$$

$$\sqrt[4]{V_{\text{avg}} = \frac{3}{1/30 + 1/45 + 1/90}}$$

LCM 90:

$$\sqrt[4]{= \frac{3}{3/90 + 2/90 + 1/90}}$$

$$\sqrt[4]{= \frac{3}{6/90}} = 45$$

$$\sqrt[4]{\boxed{45 \text{ km/h}}}$$

14. MULTIPLE EQUAL-TIME SEGMENTS

Equal times at speeds:

$$\backslash[30,\backslash 50,\backslash 70 \backslash]$$

Average:

$$\backslash[\frac{30+50+70}{3} \backslash]$$

$$\backslash[=\boxed{50} \backslash]$$

15. UNEQUAL DISTANCES

Suppose:

- 100 km at 50 km/h
- 200 km at 100 km/h

Times:

$$\backslash[2h,\backslash 2h \backslash]$$

Total:

$$\backslash[300\text{ km} \backslash]$$

Total time:

$$\backslash[4h \backslash]$$

Average:

$$\backslash[\boxed{75}\text{ km/h} \backslash]$$

Notice that although distances differ, times happen to be equal.

So arithmetic mean also works here:

$$\backslash[(50+100)/2=75 \backslash]$$

because equal time was spent.

16. UNEQUAL TIME SEGMENTS

Suppose:

- 2 h at 40
- 3 h at 60

Distance:

$$\backslash[80+180=260 \backslash]$$

Time:

$$\backslash[5 \backslash]$$

Average:

$$\backslash[\boxed{52} \backslash]$$

This is a time-weighted average:

$$\boxed{V_{\text{avg}}} = \frac{V_1 t_1 + V_2 t_2}{t_1 + t_2}$$

17. TIME-WEIGHTED AVERAGE SPEED

If times $(t_1, t_2, \dots)$ are directly given:

$$\boxed{V_{\text{avg}}} = \frac{\sum V_{it_i}}{\sum t_i}$$

Because:

$$D_i = V_{it_i}$$

18. DISTANCE FRACTION METHOD

Suppose:

- $(1/3)$ distance at 30 km/h
- $(2/3)$ distance at 60 km/h

Take total distance = 1.

Time:

$$\left[\frac{1/3}{30} + \frac{2/3}{60} \right]$$

$$= \frac{1}{90} + \frac{1}{90} = \frac{1}{45}$$

Average:

$$\boxed{45}$$

19. FAST INTEGER-DISTANCE METHOD

Instead of fractions, choose a convenient total distance.

For:

$$\left[\frac{1}{3}, \frac{2}{3} \right]$$

take total distance:

$$180 \text{ km}$$

Then:

- 60 km at 30 !' 2 h
- 120 km at 60 !' 2 h

Average:

$$180/4 = \boxed{45}$$

Usually faster.

20. PERCENTAGE DISTANCE SPLITS

40% distance at 30 km/h.

60% at 60 km/h.

Take total distance = 300 km.

Then:

$120 \text{ km at } 30 \rightarrow 4 \text{ h}$

$180 \text{ km at } 60 \rightarrow 3 \text{ h}$

Total:

$300/7$

$= \boxed{42 \frac{6}{7} \text{ km/h}}$

21. STOPPAGE AND AVERAGE SPEED

Average speed must include total elapsed time if the question asks journey average including stops.

Example:

Travel 120 km at 60 km/h.

Stop 1 hour.

Moving time:

2 h

Total elapsed:

3 h

Average:

$120/3 = \boxed{40 \text{ km/h}}$

22. AVERAGE MOVING SPEED VS JOURNEY SPEED

If question says:

average speed while moving

exclude stoppage.

If question says:

average speed for the journey

include stoppage.

Read wording carefully.

23. RELATIVE SPEED — BASIC IDEA

Relative speed tells how fast the **distance between two moving objects changes**.

There are two major cases:

Opposite directions

$$\boxed{S_{\text{rel}} = S_1 + S_2}$$

Same direction

$$\boxed{S_{\text{rel}} = |S_1 - S_2|}$$

24. OPPOSITE DIRECTION EXAMPLE

Two cars move toward each other at:

$$50, 70 \text{ km/h}$$

Relative speed:

$$120$$

If initially 360 km apart:

$$T = 360/120$$

$$= \boxed{3 \text{ h}}$$

25. SAME DIRECTION EXAMPLE

A fast car = 80.

Slow car = 60.

Initial gap = 100 km.

Relative speed:

$$20$$

Catch time:

$$100/20 = \boxed{5 \text{ h}}$$

26. MOVING APART

If two objects start together and move opposite ways:

Separation speed:

$$S_1 + S_2$$

Example:

40 and 50 for 3 h:

$$(40+50)(3) = \boxed{270 \text{ km}}$$

27. SAME DIRECTION GAP CREATION

Two vehicles start together.

Speeds:

$$\backslash [60, \backslash 45 \backslash]$$

After 4 h, gap:

$$\backslash [(60-45)(4) \backslash]$$

$$\backslash [=\boxed{60}\text{ km} \backslash]$$

28. RELATIVE SPEED WITH PERSON STANDING STILL

If observer is stationary:

$$\backslash [S_{\text{rel}}=S \backslash]$$

So relative motion reduces to ordinary motion.

29. RELATIVE SPEED IN M/S

If a cyclist at 18 km/h moves toward another cyclist at 36 km/h:

Convert only if distance is in metres.

Speeds:

$$\backslash [18\text{ km/h}=5\text{ m/s} \backslash]$$

$$\backslash [36\text{ km/h}=10\text{ m/s} \backslash]$$

Relative:

$$\backslash [15\text{ m/s} \backslash]$$

30. DISTANCE CLOSED PER SECOND

If relative speed:

$$\backslash [15\text{ m/s} \backslash]$$

then separation decreases by:

$$\backslash [15\text{ m every second} \backslash]$$

If gap = 300 m:

$$\backslash [300/15 = \boxed{20\text{s}} \backslash]$$

31. MEETING POINT RATIO

If two objects move toward each other for the same time until meeting:

$$\backslash [D_1:D_2=S_1:S_2 \backslash]$$

Example:

Speeds:

$$\backslash[3:5 \backslash]$$

Then meeting point divides total distance:

$$\backslash[\boxed{3:5} \backslash]$$

32. EXAMPLE

Total distance:

$$\backslash[320\text{ km} \backslash]$$

Speed ratio:

$$\backslash[3:5 \backslash]$$

First distance:

$$\backslash[320\times\frac{3}{8} = 120 \backslash]$$

Second:

$$\backslash[200 \backslash]$$

$$\backslash[\boxed{120, 200} \backslash]$$

33. FIND SPEED FROM MEETING DISTANCES

A travels 140 km before meeting.

B travels 210 km.

Then:

$$\backslash[S_A:S_B = 140:210 = 2:3 \backslash]$$

If A speed = 40:

$$\backslash[B=60 \backslash]$$

$$\backslash[\boxed{60\text{ km/h}} \backslash]$$

34. HEAD START + RELATIVE SPEED

A starts earlier.

Convert head start into distance.

Example:

A speed = 30.

Starts 2 h earlier.

Head start:

$$\backslash[60 \backslash]$$

B speed = 50.

Relative speed:

$$\sqrt{20}$$

Catch after:

$$\sqrt{60/20 = \boxed{3h}}$$

35. PERSON MOVING TOWARD A TRAIN/VEHICLE — PREVIEW

Suppose a person moves at 5 m/s toward a train moving at 15 m/s.

Relative speed:

$$\sqrt{5+15 = 20\text{ m/s}}$$

This idea will be central in Chapter 31 — Trains.

36. PERSON MOVING SAME DIRECTION — PREVIEW

Person:

$$\sqrt{5\text{ m/s}}$$

Train:

$$\sqrt{15\text{ m/s}}$$

Relative:

$$\sqrt{10\text{ m/s}}$$

Again, same-direction difference.

37. RELATIVE SPEED IN TWO DIMENSIONS — BASIC INTUITION

For ordinary aptitude, most relative motion is one-dimensional.

But if two objects move perpendicular to each other, relative speed magnitude is:

$$\sqrt{\boxed{S_1^2 + S_2^2}}$$

when directions are at (90°) .

This is useful but lower priority.

38. EXAMPLE — PERPENDICULAR MOTION

A moves east at:

$$\sqrt{3\text{ km/h}}$$

B moves north at:

$$\sqrt{4\text{ km/h}}$$

Relative speed magnitude:

$$\sqrt{3^2 + 4^2} = \boxed{5\text{ km/h}}$$

This uses the 3–4–5 triangle.

39. GENERAL ANGLE BETWEEN VELOCITIES

If two velocities have angle θ between them:

$$V_{\text{rel}} = \sqrt{V_1^2 + V_2^2 - 2V_1V_2\cos\theta}$$

Lower priority for standard placement aptitude.

Special cases:

- $\theta = 0^\circ$!' difference
- $\theta = 180^\circ$!' sum
- $\theta = 90^\circ$!' Pythagoras

40. GATE-STYLE QUESTION 1

Half journey at 30 km/h and half at 60 km/h.

$$V_{\text{avg}} = \frac{2(30)(60)}{90}$$

$$= \boxed{40}$$

41. GATE-STYLE QUESTION 2

Equal times at 40 and 70.

Average:

$$\frac{40+70}{2} = \boxed{55}$$

42. GATE-STYLE QUESTION 3

One-third distance at 30 and two-thirds at 60.

Take 180 km.

Time:

$$60/30 + 120/60 = 2 + 2 = 4$$

Average:

$$\boxed{45}$$

43. GATE-STYLE QUESTION 4

Cars move toward each other at 50 and 70 km/h, 360 km apart.

$$T = 360/(50+70)$$

$$= \boxed{3\text{h}}$$

44. PLACEMENT-STYLE QUESTION 1

Round trip:

- onward = 40
- return = 60

Average:

$$\boxed{48 \text{ km/h}}$$

45. PLACEMENT-STYLE QUESTION 2

A runs at 10 km/h.

B runs same direction at 8 km/h.

A is 1 km behind.

Relative speed:

$$2 \text{ km/h}$$

Catch time:

$$\frac{1}{2} \text{ h} = \boxed{30 \text{ min}}$$

46. PLACEMENT-STYLE QUESTION 3

A and B move opposite directions at:

$$4, 6 \text{ m/s}$$

Initial separation:

$$300 \text{ m}$$

Relative:

$$10 \text{ m/s}$$

Meeting:

$$\boxed{30 \text{ s}}$$

47. PLACEMENT-STYLE QUESTION 4

2 h at 40, then 3 h at 60.

Average:

$$\frac{80+180}{5} = \boxed{52}$$

48. ADVANCED QUESTION — FIND RETURN SPEED

A goes at 30 km/h and returns same distance.

Overall average = 40 km/h.

Let return speed $v(x)$.

$$\frac{40}{x} = \frac{60}{30+x}$$

$$1200 + 40x = 60x$$

$$20x = 1200$$

$$\boxed{x=60}$$

49. ADVANCED QUESTION — THREE EQUAL DISTANCES

Equal thirds at:

$$20, 30, 60$$

$$V_{\text{avg}} = \frac{1}{3} \left(\frac{1}{20} + \frac{1}{30} + \frac{1}{60} \right)^{-1}$$

LCM 60:

$$= \frac{1}{3} \left(\frac{3}{60} + \frac{2}{60} + \frac{1}{60} \right)^{-1}$$

$$= \frac{1}{3} \left(\frac{6}{60} \right)^{-1}$$

$$= \boxed{30}$$

50. ADVANCED QUESTION — 25%, 25%, 50% DISTANCE

Speeds:

- first 25% at 30
- next 25% at 60
- remaining 50% at 90

Take total distance 360 km.

Distances:

$$90, 90, 180$$

Times:

$$3, 1.5, 2$$

Total time:

$$6.5$$

Average:

$$360/6.5 = \boxed{\frac{720}{13} \approx 55.38 \text{ km/h}}$$

51. ADVANCED QUESTION — OPPOSITE MOTION

Two people 420 km apart move toward each other.

Speeds:

$$50, 70$$

One starts 1 h later.

Suppose 50 km/h traveller starts first.

In first hour:

$$\sqrt[50]{}$$

Remaining:

$$\sqrt[370]{}$$

Then relative speed:

$$\sqrt[120]{}$$

Additional time:

$$\sqrt[370/120 = \frac{37}{12}]{}$$

Total from first start:

$$\sqrt[1 + \frac{37}{12} = \frac{49}{12}]{}$$

$$\sqrt[= \boxed{4\frac{1}{12}h}]{}$$

52. ADVANCED QUESTION — CHASE

A travels at 48 km/h.

B starts 45 minutes later at 72 km/h.

A's head start:

$$\sqrt[48 \times \frac{3}{4} = 36 \text{ km}]{}$$

Relative:

$$\sqrt[72 - 48 = 24]{}$$

Catch time after B starts:

$$\sqrt[36/24 = 1.5h]{}$$

$$\sqrt[\boxed{1.5h}]{}$$

53. COMMON TRAPS

& pTRAP 1 — Average speed = arithmetic mean

Only for equal times.

& pTRAP 2 — Equal distance

Use harmonic mean, not arithmetic mean.

& pTRAP 3 — Stops

Include them in total elapsed time if asking journey average.

& pTRAP 4 — Toward each other

Add speeds.

& pTRAP 5 — Same direction

Subtract speeds.

& pTRAP 6 — Relative speed negative

Use magnitude for catch/separation rate, but direction still matters conceptually.

54. MUST-MEMORIZE FORMULAS

General average speed

$$\boxed{V_{\text{avg}}} = \frac{\sum D_i}{\sum T_i}$$

Equal distances

$$\boxed{V_{\text{avg}}} = \frac{2ab}{a+b}$$

Equal times

$$\boxed{V_{\text{avg}}} = \frac{a+b}{2}$$

55. RELATIVE SPEED FORMULAS

Same direction

$$\boxed{a-b}$$

Opposite direction

$$\boxed{a+b}$$

At right angles

$$\boxed{\sqrt{a^2+b^2}}$$

56. FAST EXAM DECISION TREE

Average-speed question?

Start with:

$$\frac{\text{total distance}}{\text{total time}}$$

Equal distances?

Use harmonic mean.

Equal times?

Use arithmetic mean.

Distance fractions?

Choose convenient total distance.

Same-direction motion?

Subtract speeds.

Opposite-direction motion?

Add speeds.

One starts earlier?

Convert delay into head-start distance.

PRACTICE — LEVEL 0

Q1

Equal distances at 40 and 60 km/h. Find average speed.

Q2

Equal times at 40 and 60 km/h. Find average speed.

Q3

2 h at 30 km/h and 3 h at 50 km/h. Find average speed.

Q4

Two cars move toward each other at 40 and 50 km/h. Find relative speed.

Q5

Two cars move same direction at 70 and 50 km/h. Find relative speed.

LEVEL 1

Q6

Round trip onward at 30 and return at 60. Find average.

Q7

One-third distance at 30 and two-thirds at 60. Find average.

Q8

A is 60 km behind B. Speeds are 80 and 60 km/h. Find catch time.

Q9

Two people are 360 km apart and move toward each other at 50 and 70. Find meeting time.

Q10

A starts 2 h earlier at 30 km/h. B moves at 50 km/h. Find catch time after B starts.

LEVEL 2

Q11

Equal thirds of a journey are covered at 30, 45 and 90 km/h. Find average.

Q12

40% distance at 30 and 60% at 60. Find average.

Q13

A round-trip average is 48 km/h. Onward speed is 40. Find return speed.

Q14

Two runners move same direction at 10 and 8 km/h. Faster runner is 1 km behind. Find catch time.

Q15

Two objects move at right angles at 6 and 8 m/s. Find relative-speed magnitude.

LEVEL 3 — EXAM PRESSURE**Q16**

First 25% of journey at 30 km/h, next 25% at 60 km/h, rest at 90 km/h. Find average speed.

Q17

A travels at 48 km/h. B starts 45 min later at 72 km/h. Find catch time after B starts.

Q18

Two people 420 km apart move toward each other at 50 and 70 km/h. The 50 km/h traveller starts 1 hour earlier. Find meeting time from first start.

Q19

Equal distances are travelled at x and $3x$. Find average speed in terms of x .

Q20

A journey consists of 2 h at 40 km/h, 1 h stopped, and 3 h at 60 km/h. Find overall average speed.

ANSWERS**Q1**

$$\left[\frac{2(40)(60)}{100} = \boxed{48 \text{ km/h}} \right]$$

Q2

$$\left[\frac{40+60}{2} = \boxed{50 \text{ km/h}} \right]$$

Q3

Distance:

$$\left[60+150=210 \right]$$

Time:

$$\left[5 \right]$$

Average:

$$\boxed{42 \text{ km/h}}$$

Q4

$$40+50 = \boxed{90 \text{ km/h}}$$

Q5

$$70-50 = \boxed{20 \text{ km/h}}$$

Q6

$$\frac{2(30)(60)}{90} = \boxed{40 \text{ km/h}}$$

Q7

Take 180 km:

$$60/30+120/60=2+2=4\text{h}$$

$$180/4 = \boxed{45 \text{ km/h}}$$

Q8

Relative:

$$80-60=20$$

$$60/20 = \boxed{3\text{h}}$$

Q9

$$360/(50+70) = 360/120 = \boxed{3\text{h}}$$

Q10

Head start:

$$30(2)=60$$

Relative:

$$20$$

Catch:

$$60/20 = \boxed{3\text{h}}$$

Q11

$$V_{\text{avg}} = \frac{1/30+1/45+1/90}{3}$$

$$= \boxed{45 \text{ km/h}}$$

Q12

Take 300 km.

120 km at 30:

$$4\text{h}$$

180 km at 60:

$$3\text{h}$$

Average:

$$\boxed{300/7 = \frac{42}{7} \text{ km/h}}$$

Q13

$$48 = \frac{2(40)x}{40+x}$$

$$x = \boxed{60 \text{ km/h}}$$

Q14

Relative:

$$2 \text{ km/h}$$

$$1/2 \text{ h} = \boxed{30 \text{ min}}$$

Q15

$$\sqrt{6^2 + 8^2} = \boxed{10 \text{ m/s}}$$

Q16

Take 360 km.

Distances:

$$90, 90, 180$$

Times:

$$3, 1.5, 2$$

Total:

$$6.5 \text{ h}$$

Average:

$$360/6.5 = \boxed{\frac{720}{13} \approx 55.38 \text{ km/h}}$$

Q17

Head start:

$$48 \left(\frac{3}{4} \right) = 36$$

Relative:

$$24$$

$$36/24 = \boxed{1.5 \text{ h}}$$

Q18

First hour:

$$50 \text{ km}$$

Remaining:

$$370$$

Closing speed:

$$120$$

After second starts:

$$\left[370/120 = 37/12 \right]$$

Total:

$$\left[1 + \frac{37}{12} = \boxed{\frac{49}{12}} = 4\frac{1}{12} \text{ h} \right]$$

Q19

$$\left[V_{\text{avg}} = \frac{2(x)(3x)}{x+3x} \right]$$

$$\left[= \frac{6x^2}{4x} \right]$$

$$\left[= \boxed{\frac{3x}{2}} \right]$$

Q20

Distance:

$$\left[2(40) + 3(60) = 80 + 180 = 260 \right]$$

Elapsed time includes stop:

$$\left[2 + 1 + 3 = 6 \right]$$

Average:

$$\left[260/6 = \boxed{43\frac{1}{3} \text{ km/h}} \right]$$

CHAPTER 30 STATUS

Covered so far:

- average-speed definition '
- equal-distance harmonic mean '
- equal-time arithmetic mean '
- round-trip speed '
- multiple equal-distance segments '
- weighted time average '
- distance-fraction journeys '
- stoppage-aware average speed '
- same-direction relative speed '
- opposite-direction relative speed '
- meeting/catching '
- head starts '
- perpendicular relative speed basics '
- GATE/placement practice '

But **Chapter 30 is not complete yet.**

Next we still need:

- more advanced multi-object relative motion
- moving observer problems
- repeated catches/meetings
- average speed with more complex mixed fractions
- relative speed from changing frames

- “average speed increases/decreases” percentage problems
- upstream/downstream connection preview
- final shortcut set and chapter audit

Progress

Completed: 29/76 **Current:** Chapter 30 — Average & Relative Speed **Next:** Chapter 30 Part 2 — **Advanced relative motion, moving observers & mixed average-speed problems**

USER

next

ASSISTANT

CHAPTER 30 — AVERAGE AND RELATIVE SPEED

PART 2 — ADVANCED RELATIVE MOTION, MOVING OBSERVERS, REPEATED MEETINGS & MIXED AVERAGE-SPEED PROBLEMS

Status: $\emptyset=0\%$ **IN PROGRESS GATE:** High **Placements:** Very High **Priority:** $\emptyset=0\%$ **Yr 1**

This part finishes the chapter.

The two master formulas remain:

$$\boxed{V_{\text{avg}} = \frac{\text{Total Distance}}{\text{Total Time}}}$$

and:

$$\boxed{\vec{V}_{\text{relative}} = \vec{V}_A - \vec{V}_B}$$

For one-dimensional aptitude problems, this becomes:

- same direction !' difference
- opposite directions !' sum

1. MOVING OBSERVER IDEA

Suppose A and B move in the same direction.

From the ground:

$$V_A = 70, \text{ } V_B = 50$$

From B's point of view, A moves at:

$$70 - 50 = \boxed{20 \text{ km/h}}$$

So relative motion means:

Imagine one object as stationary and ask how fast the other appears to move.

2. OPPOSITE-DIRECTION OBSERVER

A moves east at 60 km/h.

B moves west at 40 km/h.

From A's point of view, B approaches at:

$$[60+40 = \boxed{100\text{ km/h}}]$$

3. SAME SPEED, SAME DIRECTION

Two cars both move at 50 km/h in the same direction.

Relative speed:

$$[50-50=0]$$

So the distance between them remains constant.

$$[\boxed{V_{\text{rel}}=0}]$$

4. SAME SPEED, OPPOSITE DIRECTIONS

Both move at 50 km/h toward each other.

Relative speed:

$$[50+50 = \boxed{100\text{ km/h}}]$$

5. MOVING WALKWAY / ESCALATOR MODEL

A person walks at speed (p) relative to a walkway.

Walkway moves at speed (w) .

Same direction

$$[\boxed{V_{\text{ground}}=p+w}]$$

Opposite direction

$$[\boxed{V_{\text{ground}}=|p-w|}]$$

This is the same idea later used in boats and streams.

6. EXAMPLE

Person walking speed:

$$[4\text{ km/h}]$$

Walkway:

$$[2\text{ km/h}]$$

Same direction:

$$\boxed{6 \text{ km/h}}$$

Against walkway:

$$\boxed{2 \text{ km/h}}$$

7. FIND WALKWAY SPEED

A person walks at 5 km/h relative to walkway.

Ground speed same direction:

$$8 \text{ km/h}$$

Walkway speed:

$$8 - 5 = \boxed{3 \text{ km/h}}$$

8. RELATIVE MOTION WITH THREE OBJECTS

Suppose:

$$A=80, \quad B=60, \quad C=40$$

all same direction.

Relative speeds:

$$A \text{ wrt } B = 20$$

$$B \text{ wrt } C = 20$$

$$A \text{ wrt } C = 40$$

Use the relevant pair only.

9. TWO OBJECTS CHASING A THIRD

A, B, C move same direction.

Suppose:

$$A=90, \quad B=70, \quad C=50$$

If A and B both chase C:

A closes on C at:

$$40$$

B closes on C at:

$$20$$

A catches C twice as fast for the same initial gap.

10. CATCH-TIME RATIO

For same initial gap (D) :

$$T = \frac{D}{V_{\text{relative}}}$$

Therefore:

$$T_1:T_2 = V_{\text{rel},2}:V_{\text{rel},1}$$

11. EXAMPLE

A and B chase C from the same point behind C.

Relative speeds:

$$40, 20$$

Then catch-time ratio:

$$20:40 = 1:2$$

12. REPEATED MEETINGS ON A STRAIGHT LINE — BASIC CASE

Suppose two people move back and forth continuously between endpoints.

For first meeting, use ordinary relative speed.

After reaching endpoints and turning, directions change.

So every phase must be re-evaluated.

Do **not** blindly reuse the first-meeting formula after every turn.

13. TOTAL-DISTANCE IDEA FOR REPEATED MEETING

When two objects continuously move between two endpoints and reflections occur instantaneously, a useful method is to track **total distance covered by both**.

For their first meeting while approaching:

$$D_A + D_B = L$$

where L is endpoint separation.

At later meetings, total combined travel can be multiples of L , depending on reflection pattern.

This is a placement-oriented shortcut, not something to use without checking geometry.

14. REPEATED MEETING ON A CIRCULAR TRACK — PREVIEW

For a track of circumference C :

Same direction

Time between successive catches:

$$T = \frac{C}{|v_1 - v_2|}$$

Opposite directions

Time between meetings:

$$\boxed{T = \frac{C}{v_1 + v_2}}$$

We will study this deeply in Chapter 33.

15. MEETING FREQUENCY

If two runners on a 400 m circular track move same direction at:

$$6, 4 \text{ m/s}$$

Relative speed:

$$2$$

Catch interval:

$$400/2 = \boxed{200\text{s}}$$

16. OPPOSITE-DIRECTION MEETING FREQUENCY

Same track:

$$6, 4 \text{ m/s}$$

Opposite directions:

$$V_{\text{rel}} = 10$$

Meeting interval:

$$400/10 = \boxed{40\text{s}}$$

17. AVERAGE SPEED AFTER SPEED CHANGE

Suppose a vehicle covers a fixed distance at speed (S) .

Then increases speed by 25%.

For the same distance, time becomes 80% of original.

But average speed over two separate equal-distance trips, one at (S) , one at $(1.25S)$, is not:

$$1.125S$$

Use harmonic mean:

$$V_{\text{avg}} = \frac{2S(1.25S)}{2.25S}$$

$$= \frac{2.5}{2.25}S$$

$$= \boxed{\frac{10}{9}S}$$

So average rises only:

$$\boxed{11.11\%}$$

18. EQUAL-DISTANCE SPEED CHANGE BY p%

Suppose outward speed:

S

return speed:

$S(1+p)$

where (p) is decimal.

Round-trip average:

$$V_{\text{avg}} = \frac{2S(1+p)}{2+p}$$

Useful for percentage comparison.

19. EXAMPLE — 50% FASTER RETURN

Outward:

S

Return:

$1.5S$

Average:

$$\frac{2S(1.5S)}{2.5S}$$

$= 1.2S$

So average is only:

$20\% \text{ above } S$

not 25%.

20. AVERAGE SPEED WITH THREE DISTANCE FRACTIONS

Suppose:

- 20% distance at 20 km/h
- 30% at 30 km/h
- 50% at 60 km/h

Take total distance:

300 km

Then:

- 60 km at 20 !' 3 h
- 90 km at 30 !' 3 h
- 150 km at 60 !' 2.5 h

Total:

$300/8.5$

$= 35.29 \text{ km/h approximately}$

21. GENERAL FRACTION-DISTANCE FORMULA

If fractions (f_i) of the total distance are travelled at speeds (v_i) :

$$\boxed{V_{\text{avg}}} = \frac{1}{\sum \frac{f_i}{v_i}}$$

where:

$$\sum f_i = 1$$

22. GENERAL TIME-FRACTION FORMULA

If fractions (f_i) of total time are spent at speeds (v_i) :

$$\boxed{V_{\text{avg}}} = \sum f_i v_i$$

This is a weighted arithmetic mean.

23. EXAMPLE — TIME FRACTIONS

A vehicle spends:

- 25% of time at 40 km/h
- 75% at 60 km/h

Average:

$$0.25(40) + 0.75(60)$$

$$= 10 + 45$$

$$= \boxed{55 \text{ km/h}}$$

24. SPEED ZERO DURING STOPPAGE

A stop is simply a segment with:

$$V = 0$$

for some time.

Example:

1 h at 60 km/h.

0.5 h stopped.

1.5 h at 40 km/h.

Distance:

$$60 + 0 + 60 = 120$$

Time:

$$3 \text{ h}$$

Average:

$$\boxed{40 \text{ km/h}}$$

25. FIND STOP TIME FROM AVERAGE SPEED

A vehicle travels 120 km at running speed 60 km/h.

Overall average speed = 40 km/h.

Total journey time:

$$\sqrt[120/40=3h]{}$$

Moving time:

$$\sqrt[120/60=2h]{}$$

Stop:

$$\sqrt[\boxed{1h}]{}$$

26. AVERAGE SPEED WITH SPEEDS IN RATIO

Equal distances at speeds in ratio:

$$\sqrt{a:b}$$

Let speeds:

$$\sqrt{ak,\ bk}$$

Average:

$$\sqrt[\boxed{\frac{2ab}{a+b}k}]{}$$

Example ratio:

$$\sqrt{2:3}$$

and slower speed = 40.

Then faster = 60.

Average:

$$\sqrt{48}$$

27. FIND SPEED RATIO FROM EQUAL-DISTANCE TIMES

If two equal-distance legs take times:

$$\sqrt{3h,\ 2h}$$

speed ratio:

$$\sqrt{2:3}$$

because:

$$\sqrt{S\propto 1/T}$$

28. RELATIVE SPEED FROM DISTANCE CHANGE

Two cars are 150 km apart.

After 2 h, gap becomes 50 km.

So gap closed:

$$[100\text{ km}]$$

Relative speed:

$$[100/2 = \boxed{50\text{ km/h}}]$$

29. FIND UNKNOWN SPEED FROM GAP CHANGE

A travels at 70 km/h.

B is ahead and travels same direction.

Gap decreases by 40 km in 2 h.

Relative speed:

$$[20]$$

Therefore:

$$[70 - B = 20]$$

$$[B = \boxed{50\text{ km/h}}]$$

30. GAP INCREASES

A and B move same direction.

A = 50 km/h.

B ahead = 70 km/h.

Gap increases at:

$$[20\text{ km/h}]$$

After 3 h, increase:

$$[60\text{ km}]$$

31. MOVING TOWARD EACH OTHER BUT ONE TURNS

A and B move toward each other.

Before meeting, use:

$$[v_A + v_B]$$

If one turns and they then move same direction, relative speed changes to:

$$[$$

$v_A - v_B$

\]

Direction changes matter more than memorized formulas.

32. MOVING OBSERVER + STATIONARY OBJECTS

A person on a bus moving at 60 km/h sees stationary poles pass backward at 60 km/h relative speed.

This is the conceptual base for:

- trains crossing poles
 - trains crossing people
 - moving-platform problems
-

33. RAIN / WIND-TYPE RELATIVE MOTION PREVIEW

If rain falls vertically at speed (v_r) and a person runs horizontally at (v_p) , rain appears to have relative velocity combining both components.

Magnitude:

$$\sqrt{v_r^2 + v_p^2}$$

This is a vector-relative-motion extension, lower priority for standard aptitude.

34. BOAT/STREAM CONNECTION PREVIEW

If boat speed in still water:

$$b$$

stream speed:

$$s$$

then:

Downstream:

$$b+s$$

Upstream:

$$b-s$$

This is exactly relative speed.

We will formally study it in Chapter 32.

35. AVERAGE SPEED UPSTREAM + DOWNSTREAM PREVIEW

For equal upstream/downstream distances:

$$V_{\text{avg}} = \frac{2(b+s)(b-s)}{(b+s)+(b-s)}$$

$$= \frac{b^2 - s^2}{b}$$

No need to memorize yet; Chapter 32 will revisit it.

36. GATE-STYLE QUESTION 1

A moves at 70 km/h, B at 50 km/h, same direction.

Relative:

$$\boxed{20 \text{ km/h}}$$

37. GATE-STYLE QUESTION 2

A and B move opposite directions at 12 and 8 m/s.

Relative:

$$\boxed{20 \text{ m/s}}$$

38. GATE-STYLE QUESTION 3

Equal distances at S and $(1.5S)$.

Average:

$$\frac{2S(1.5S)}{2.5S}$$

$$=\boxed{1.2S}$$

39. GATE-STYLE QUESTION 4

A vehicle spends 25% of time at 40 and 75% at 60.

Average:

$$0.25(40) + 0.75(60)$$

$$=\boxed{55}$$

40. PLACEMENT-STYLE QUESTION 1

A car at 80 km/h is 100 km behind a car at 60 km/h.

Catch time:

$$100/(80-60) = \boxed{5h}$$

41. PLACEMENT-STYLE QUESTION 2

A 400 m circular track.

Runners at 6 and 4 m/s same direction.

Catch interval:

$$400/(6-4) = \boxed{200s}$$

42. PLACEMENT-STYLE QUESTION 3

Same track, opposite directions.

Meeting interval:

$$\sqrt[4]{400/(6+4)} = \boxed{40\text{s}}$$

43. PLACEMENT-STYLE QUESTION 4

Running speed = 60 km/h.

Distance = 120 km.

Overall average = 40 km/h.

Stop time:

$$\sqrt[4]{3-2} = \boxed{1\text{h}}$$

44. ADVANCED QUESTION — MIXED DISTANCE FRACTIONS

20% of journey at 20 km/h,

30% at 30 km/h,

50% at 60 km/h.

Take total 300 km.

Times:

$$\sqrt[4]{3, \sqrt[3]{3}, \sqrt[2]{2.5}}$$

Average:

$$\sqrt[4]{300/8.5} = \boxed{35.29\text{ km/h}}$$

45. ADVANCED QUESTION — FIND UNKNOWN SPEED FROM AVERAGE

Equal distances at:

$$\sqrt[4]{30, \sqrt{x}}$$

Average:

$$\sqrt[4]{40}$$

$$\sqrt[4]{40 = \frac{60x}{30+x}}$$

$$\sqrt[4]{1200+40x=60x}$$

$$\sqrt[4]{20x=1200}$$

$$\sqrt[4]{\boxed{x=60}}$$

46. ADVANCED QUESTION — RELATIVE GAP

A = 90 km/h.

B = 60 km/h.

B starts 2 h earlier.

Head start:

$$\boxed{120 \text{ km}}$$

Relative:

$$\boxed{30}$$

Catch time:

$$\boxed{120/30 = 4 \text{ h}}$$

47. ADVANCED QUESTION — MOVING WALKWAY

A person walks at 5 km/h relative to ground on stationary surface.

On a moving walkway, same-direction ground speed = 8 km/h.

Walkway:

$$\boxed{8-5 = 3}$$

Against walkway:

$$\boxed{5-3 = 2 \text{ km/h}}$$

48. ADVANCED QUESTION — REQUIRED SPEED TO REDUCE ROUND-TRIP AVERAGE ERROR

A goes at 40 km/h and wants round-trip average 60 km/h over equal distances.

Let return speed = x .

$$\boxed{60 = \frac{80x}{40+x}}$$

$$\boxed{2400 + 60x = 80x}$$

$$\boxed{x = 120}$$

$$\boxed{120 \text{ km/h}}$$

This shows why round-trip average targets can require surprisingly high return speeds.

49. IMPOSSIBLE AVERAGE-SPEED CHECK

Suppose first half distance is travelled at:

$$\boxed{40 \text{ km/h}}$$

Can round-trip equal-distance average be 80 km/h?

No finite return speed can produce average 80 because:

$$V_{\text{avg}} = \frac{2(40)x}{40+x}$$

As:

$$x \rightarrow \infty$$

average approaches:

$$80$$

but never reaches it for finite x .

So:

$$\boxed{\text{80 km/h is unattainable with finite return speed}}$$

Excellent option-elimination idea.

50. GENERAL MAXIMUM ROUND-TRIP AVERAGE

If first equal-distance leg speed is a , and second-leg speed can become arbitrarily large:

$$V_{\text{avg}} \rightarrow 2a$$

Therefore:

$$\boxed{\text{finite average} < 2a}$$

if the first leg is fixed at a .

51. EXAMPLE

Outward speed:

$$30$$

No matter how fast the return becomes, equal-distance average remains below:

$$60$$

So an average of 70 is impossible.

52. COMMON TRAPS

& pTRAP 1 — Relative speed always sum

No. Sum only when separation closes/increases in opposite directions appropriately.

& pTRAP 2 — Relative speed always positive difference

Direction matters. Use velocity concept first.

& pTRAP 3 — Average speed from speed percentages

Always consider whether distance or time is being held equal.

& pTRAP 4 — Stops ignored

Stops add time even though they add zero distance.

& pTRAP 5 — Repeated-meeting problems

Recompute direction after turns.

& pTRAP 6 — Huge return speed can produce any average

False. Equal-distance average has a limiting value.

53. MUST-MEMORIZE FINAL FORMULAS

General average speed

$$\boxed{V_{\text{avg}} = \frac{\sum D_i}{\sum T_i}}$$

Equal distances

$$\boxed{V_{\text{avg}} = \frac{2ab}{a+b}}$$

Equal time

$$\boxed{V_{\text{avg}} = \frac{a+b}{2}}$$

Distance fractions

$$\boxed{V_{\text{avg}} = \frac{1}{\sum f_i/v_i}}$$

Time fractions

$$\boxed{V_{\text{avg}} = \sum f_{iv_i}}$$

54. FINAL RELATIVE-SPEED FORMULAS

Same direction

$$\boxed{v_1 - v_2}$$

Opposite directions

$$\boxed{v_1 + v_2}$$

Perpendicular

$$\boxed{\sqrt{v_1^2 + v_2^2}}$$

General angle θ

$$\boxed{v_{\text{rel}} = \sqrt{v_1^2 + v_2^2 - 2v_1v_2\cos\theta}}$$

55. FAST EXAM METHOD

When you see **average speed**, ask first:

Equal distance or equal time?

When you see **relative speed**, ask first:

Same direction or opposite direction?

When you see **head start**, immediately compute:

$$\boxed{\text{head-start distance} = v \times t}$$

When you see **stop**, add it to total time.

When you see **multiple segments**, use total distance / total time.

PRACTICE — LEVEL 0

Q1

Two cars move same direction at 80 and 50 km/h. Find relative speed.

Q2

Two cars move opposite directions at 80 and 50 km/h. Find relative speed.

Q3

Equal-distance trip at 40 and 80 km/h. Find average.

Q4

Equal-time trip at 40 and 80 km/h. Find average.

Q5

Running speed = 60 km/h for 2 h, then stop 1 h. Find journey average.

LEVEL 1

Q6

A is 90 km behind B. Speeds are 80 and 50 km/h. Find catch time.

Q7

Two runners on a 600 m track run same direction at 8 and 5 m/s. Find catch interval.

Q8

Same runners run opposite directions. Find meeting interval.

Q9

Equal distances at v and $2v$. Find average speed.

Q10

25% of total time at 40 km/h and 75% at 80 km/h. Find average.

LEVEL 2

Q11

A round trip has onward speed 40 km/h and overall average 60 km/h. Find return speed.

Q12

A car moving at 90 km/h chases another moving at 60 km/h with a 150 km gap. Find catch time.

Q13

A person walks at 5 km/h relative to walkway. Same-direction ground speed is 8 km/h. Find walkway speed and opposite-direction ground speed.

Q14

First half distance at 30 km/h. Can equal-distance overall average be 70 km/h? Explain.

Q15

A journey: 20% distance at 20, 30% at 30, 50% at 60. Find average.

LEVEL 3 — EXAM PRESSURE**Q16**

A vehicle covers equal distances at $\sqrt{S}$ and $\sqrt{1.25S}$. By what percentage is average speed above $\sqrt{S}$?

Q17

A vehicle travels 100 km at 50 km/h, stops for 30 min, then 150 km at 75 km/h. Find journey average.

Q18

Two objects have speeds 6 and 8 m/s and move at right angles. Find relative-speed magnitude.

Q19

Two cars start same point, same direction. A travels at 40 km/h. B starts 1.5 h later and catches A 3 h after B starts. Find B's speed.

Q20

A 400 m track has runners at 5 and 3 m/s. Find:

1. same-direction catch interval
2. opposite-direction meeting interval

ANSWERS**Q1**

$$\sqrt{80 \cdot 50} = \boxed{30 \text{ km/h}}$$

Q2

$$\sqrt{80 + 50} = \boxed{130 \text{ km/h}}$$

Q3

$$\sqrt{\frac{2(40)(80)}{120}} = \boxed{53 \frac{1}{3} \text{ km/h}}$$

Q4

$$\sqrt{\frac{40+80}{2}} = \boxed{60 \text{ km/h}}$$

Q5

Distance:

$$\sqrt{120}$$

Elapsed time:

$$\sqrt{3 \text{ h}}$$

Average:

$$\sqrt{\boxed{40 \text{ km/h}}}$$

Q6

Relative:

$$\sqrt{80-50=30}$$

$$\sqrt{90/30 = \boxed{3 \text{ h}}}$$

Q7

Relative:

$$\sqrt{8-5=3}$$

$$\sqrt{600/3 = \boxed{200 \text{ s}}}$$

Q8

Relative:

$$\sqrt{8+5=13}$$

$$\sqrt{600/13}$$

$$\sqrt{\boxed{\approx 46.15 \text{ s}}}$$

Q9

$$\sqrt{\frac{2(x)(2x)}{3x}} = \boxed{\frac{4x}{3}}$$

Q10

$$\sqrt{0.25(40)+0.75(80)}$$

$$\sqrt{10+60}$$

$$\sqrt{= \boxed{70 \text{ km/h}}}$$

Q11

$$\sqrt{60 = \frac{2(40)x}{40+x}}$$

$$\sqrt{2400+60x=80x}$$

$$\sqrt{\boxed{x=120 \text{ km/h}}}$$

Q12

Relative:

$$\sqrt{30}$$

$$\boxed{150/30 = 5\text{h}}$$

Q13

Walkway:

$$8-5=3$$

Against walkway:

$$5-3=2$$

$$\boxed{3\text{ km/h}, 2\text{ km/h}}$$

Q14

No.

With first equal-distance leg at 30:

$$V_{\text{avg}} < 2(30) = 60$$

for any finite return speed.

So 70 is impossible.

$$\boxed{\text{Impossible}}$$

Q15

Take 300 km.

Times:

$$60/20=3$$

$$90/30=3$$

$$150/60=2.5$$

Total:

$$8.5\text{h}$$

Average:

$$300/8.5 = \boxed{35.29\text{ km/h approx}}$$

Q16

Average:

$$\frac{2S(1.25S)}{2.25S} = \frac{10S}{9}$$

Increase:

$$\frac{19}{100}$$

$$= \boxed{11.11\%}$$

Q17

Times:

$$100/50=2\text{h}$$

Stop:

$$0.5\text{h}$$

$$\sqrt{150/75=2h}$$

Total time:

$$\sqrt{4.5h}$$

Distance:

$$\sqrt{250}$$

Average:

$$\sqrt{250/4.5 = \boxed{55\frac{59}{\text{km/h}}}}$$

Q18

$$\sqrt{\sqrt{6^2+8^2}} = \boxed{10\text{ m/s}}$$

Q19

A's head start:

$$\sqrt{40(1.5)=60}$$

B catches in 3 h.

Required relative speed:

$$\sqrt{60/3=20}$$

Therefore B:

$$\sqrt{40+20 = \boxed{60\text{ km/h}}}$$

Q20

Same direction:

$$\sqrt{400/(5-3) = \boxed{200s}}$$

Opposite:

$$\sqrt{400/(5+3) = \boxed{50s}}$$

' CHAPTER 30 COMPLETE

We have now covered:

- average-speed master formula '
- equal-distance harmonic mean '
- equal-time arithmetic mean '
- multiple-segment journeys '
- weighted average speed '
- percentage-distance splits '
- percentage-time splits '
- stops in average speed '
- moving observers '
- same-direction relative speed '
- opposite-direction relative speed '
- perpendicular relative motion '

- three-object relative motion '
- chase and catch-time ratios '
- circular meeting preview '
- moving walkway model '
- boat/stream connection preview '
- average-speed limits and impossibility checks '
- advanced GATE/placement drills '

Progress

Completed: 30/76 Unit 1 — Quantitative Aptitude: 0=0% PROGRESS

NEXT

+Chapter 31 — Trains

Next we will cover:

- train length
- crossing a pole/person
- crossing a platform/bridge
- two trains crossing
- same/opposite-direction train motion
- relative-speed conversion to m/s
- finding train length
- finding platform length
- overtaking
- tunnel/bridge entry-exit timing
- advanced train problems and shortcuts.

USER

next

ASSISTANT

CHAPTER 31 — TRAINS

PART 1 — TRAIN LENGTH, CROSSING A POLE/PERSON, PLATFORM/BRIDGE & TWO-TRAIN BASICS

Status: 0=0% PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

Train problems are just **Time–Speed–Distance + Relative Speed + Object Length**.

The biggest idea is:

A train is not a point object. Its entire length must pass the object.

So always ask:

$\boxed{\text{What total distance must the train cover to completely cross?}}$

1. CORE FORMULA

$$\boxed{D=ST}$$

Therefore:

$$\boxed{T=\frac{D}{S}}$$

For train questions, the main work is identifying the correct (D) .

2. WHY TRAIN LENGTH MATTERS

Suppose a 100 m train crosses a pole.

When the engine reaches the pole, the train has **not** completed crossing.

The rear end must also pass the pole.

So the train must move exactly its own length:

$$\boxed{100\text{ m}}$$

3. CROSSING A POLE

If train length is (L) and speed is (v) :

$$\boxed{T=\frac{L}{v}}$$

where (v) must be in compatible units, usually m/s.

4. EXAMPLE

Train length:

$$\boxed{180\text{ m}}$$

Speed:

$$\boxed{54\text{ km/h}}$$

Convert:

$$\boxed{54 \times \frac{5}{18} = 15\text{ m/s}}$$

Time to cross pole:

$$\boxed{180/15 = 12\text{ s}}$$

5. CROSSING A STATIONARY PERSON

A stationary person behaves exactly like a pole.

So:

$$\boxed{\text{Distance to cross stationary person} = L_{\text{train}}}$$

Example:

Train length = 150 m.

Speed = 72 km/h.

$$72 \text{ km/h} = 20 \text{ m/s}$$

Time:

$$150/20 = \boxed{7.5 \text{ s}}$$

6. CROSSING A TREE/SIGNAL/ELECTRIC POLE

All these are treated as point objects.

Therefore:

$$\boxed{D = L_{\text{train}}}$$

7. CROSSING A PLATFORM

If:

- train length = L_T
- platform length = L_P

then for the entire train to clear the entire platform:

$$\boxed{D = L_T + L_P}$$

So:

$$\boxed{T = \frac{L_T + L_P}{v}}$$

8. EXAMPLE

Train:

$$200 \text{ m}$$

Platform:

$$300 \text{ m}$$

Speed:

$$72 \text{ km/h} = 20 \text{ m/s}$$

Total distance:

$$200 + 300 = 500 \text{ m}$$

Time:

$$500/20 = \boxed{25 \text{ s}}$$

9. CROSSING A BRIDGE

Bridge behaves like platform.

$$\boxed{D=L_{\text{train}}+L_{\text{bridge}}}$$

Example:

Train = 150 m.

Bridge = 450 m.

Speed = 54 km/h = 15 m/s.

Distance:

$$\boxed{600\text{m}}$$

Time:

$$\boxed{600/15 = 40\text{s}}$$

10. CROSSING A TUNNEL

Same rule:

$$\boxed{D=L_{\text{train}}+L_{\text{tunnel}}}$$

11. FIND TRAIN LENGTH FROM POLE-CROSSING TIME

If speed and pole-crossing time are given:

$$\boxed{L=VT}$$

Example:

Train crosses pole in 8 s.

Speed:

$$\boxed{72\text{ km/h}=20\text{ m/s}}$$

Length:

$$\boxed{20(8) = 160\text{m}}$$

12. FIND PLATFORM LENGTH

Train length = 180 m.

Speed = 54 km/h.

Crosses platform in 30 s.

Speed:

$$\boxed{15\text{m/s}}$$

Total distance in 30 s:

$$\boxed{15(30)=450\text{m}}$$

So platform:

$$\boxed{450-180 = 270\text{m}}$$

13. FAST PLATFORM-LENGTH FORMULA

$$\boxed{L_P = vT - L_T}$$

Similarly for bridge/tunnel.

14. FIND TRAIN SPEED

Train = 240 m.

Crosses pole in 12 s.

Speed:

$$\boxed{240/12=20\text{m/s}}$$

Convert:

$$\boxed{20 \times \frac{18}{5} = 72 \text{ km/h}}$$

15. MUST-MEMORIZE CONVERSION

$$\boxed{\text{km/h} \text{ to } \text{m/s}: \times \frac{5}{18}}$$

$$\boxed{\text{m/s} \text{ to } \text{km/h}: \times \frac{18}{5}}$$

Train problems very often mix metres with km/h.

16. TRAIN CROSSING A MOVING PERSON

Now relative speed matters.

If train and person move:

Same direction

$$\boxed{v_{\text{rel}} = v_T - v_P}$$

Opposite directions

$$\boxed{v_{\text{rel}} = v_T + v_P}$$

Distance to fully cross person:

$$\boxed{L_T}$$

because person's length is ignored.

17. EXAMPLE — SAME DIRECTION

Train:

$$\boxed{180\text{m}}$$

Train speed:

$$\boxed{72\text{ km/h}=20\text{m/s}}$$

Person runs at:

$$\boxed{18\text{ km/h}=5\text{m/s}}$$

Same direction relative speed:

$$\boxed{20-5=15\text{m/s}}$$

Time:

$$\boxed{180/15 = 12\text{s}}$$

18. EXAMPLE — OPPOSITE DIRECTION

Same train and person.

Relative speed:

$$\boxed{20+5=25\text{m/s}}$$

Time:

$$\boxed{180/25 = 7.2\text{s}}$$

Opposite direction crossing is faster.

19. FIND PERSON SPEED FROM CROSSING TIME

Train length:

$$\boxed{150\text{m}}$$

Train speed:

$$\boxed{72\text{ km/h}=20\text{m/s}}$$

It crosses a runner moving same direction in 10 s.

Relative speed:

$$\boxed{150/10=15\text{m/s}}$$

So runner speed:

$$\boxed{20-15=5\text{m/s}}$$

Convert:

$$\boxed{5 \times \frac{18}{5} = 18\text{ km/h}}$$

20. TWO TRAINS CROSSING EACH OTHER

If both trains have nonzero length, both lengths matter.

Distance to completely cross:

$$\boxed{L_1 + L_2}$$

Relative speed depends on direction.

21. TWO TRAINS — OPPOSITE DIRECTIONS

$$\boxed{T = \frac{L_1 + L_2}{v_1 + v_2}}$$

22. EXAMPLE

Train A:

$$150\text{m}, 54\text{ km/h}$$

Train B:

$$250\text{m}, 36\text{ km/h}$$

Convert speeds:

$$54 = 15\text{m/s}$$

$$36 = 10\text{m/s}$$

Relative:

$$25\text{m/s}$$

Total length:

$$400\text{m}$$

Time:

$$400/25 = \boxed{16\text{s}}$$

23. TWO TRAINS — SAME DIRECTION

If faster train overtakes slower train:

$$\boxed{T = \frac{L_1 + L_2}{v_{\text{fast}} - v_{\text{slow}}}}$$

24. EXAMPLE

Fast train:

$$200\text{m}, 72\text{ km/h}$$

Slow train:

$$100\text{m}, 54\text{ km/h}$$

Speeds:

$$20, 15\text{m/s}$$

Relative:

$$\sqrt{5\text{m/s}}$$

Total length:

$$\sqrt{300\text{m}}$$

Time to fully overtake:

$$\sqrt{300/5 = \boxed{60\text{s}}}$$

25. WHY LENGTHS ADD DURING OVERTAKING

At the start of full overtaking, the front of faster train is near the rear of slower.

For overtaking to complete, faster train's rear must move beyond slower train's front.

Relative displacement required:

$$\sqrt{\boxed{L_{\text{fast}} + L_{\text{slow}}}}$$

26. FIND SECOND TRAIN LENGTH

Train A = 150 m.

Train B length = $\sqrt{x}$.

They move opposite directions at:

$$\sqrt{54, \sqrt{36\text{ km/h}}}$$

Cross in 14 s.

Relative speed:

$$\sqrt{15+10=25\text{m/s}}$$

Total length:

$$\sqrt{25(14)=350\text{m}}$$

So:

$$\sqrt{150+x=350}$$

$$\sqrt{x=\boxed{200\text{m}}}$$

27. FIND UNKNOWN SPEED — TWO TRAINS

Two trains:

- lengths 200 m and 100 m
- opposite directions
- one speed = 54 km/h
- crossing time = 10 s

Total length:

$$\sqrt{300\text{m}}$$

Relative speed:

$$\sqrt{300/10=30\text{m/s}}$$

Known speed:

$$\sqrt{54\text{ km/h}=15\text{m/s}}$$

Other:

$$\sqrt{30-15=15\text{m/s}}$$

$$\sqrt{= \boxed{54\text{ km/h}}}$$

28. SAME-DIRECTION UNKNOWN SPEED

Train A:

$$\sqrt{180\text{m}, 72\text{ km/h}}$$

Train B:

$$\sqrt{120\text{m}, x}$$

A overtakes B in 30 s.

Total length:

$$\sqrt{300}$$

Relative speed:

$$\sqrt{300/30=10\text{m/s}}$$

A:

$$\sqrt{20\text{m/s}}$$

So B:

$$\sqrt{20-10=10\text{m/s}}$$

$$\sqrt{= \boxed{36\text{ km/h}}}$$

29. CROSSING A PLATFORM VS A POLE

If same train crosses:

- pole in (t_1)
- platform in (t_2)

Then:

$$\sqrt{L_T = vt_1}$$

and:

$$\sqrt{L_T + L_P = vt_2}$$

Subtract:

$$\sqrt{\boxed{L_P = v(t_2 - t_1)}}$$

Very useful shortcut.

30. EXAMPLE

Train crosses pole in:

$$\sqrt[10s]{}$$

and platform in:

$$\sqrt[25s]{}$$

Speed:

$$\sqrt[20m/s]{}$$

Platform length:

$$\sqrt[20(25-10)]{}$$

$$\sqrt[=300m]{}$$

31. FIND TRAIN LENGTH USING TWO CROSSING TIMES

Suppose a train crosses:

- 100 m platform in 15 s
- 220 m platform in 21 s

Let train length = $\sqrt[L]{}$, speed = $\sqrt[v]{}$.

$$\sqrt[L+100=15v]{}$$

$$\sqrt[L+220=21v]{}$$

Subtract:

$$\sqrt[120=6v]{}$$

$$\sqrt[v=20m/s]{}$$

Then:

$$\sqrt[L+100=300]{}$$

$$\sqrt[L=200m]{}$$

32. SHORTCUT FOR TWO PLATFORM LENGTHS

If platform lengths differ by:

$$\sqrt[\Delta L]{}$$

and crossing times differ by:

$$\sqrt[\Delta t]{}$$

then train speed:

$$\sqrt[\boxed{v=\frac{\Delta L}{\Delta t}}]{}$$

because train length cancels.

33. EXAMPLE

Platform lengths:

$\backslash [150\text{m}, \backslash 350\text{m} \backslash]$

Crossing times:

$\backslash [20\text{s}, \backslash 30\text{s} \backslash]$

Speed:

$\backslash [\frac{350-150}{30-20} = 20\text{m/s} \backslash]$

$\backslash [=\boxed{72\text{ km/h}} \backslash]$

34. TRAIN ENTERING A PLATFORM VS COMPLETELY CROSSING

Be precise with wording.

Engine reaches platform end

Distance may involve only platform length.

Train completely crosses platform

Distance:

$\backslash [L_T + L_P \backslash]$

Most aptitude questions mean **completely crosses**.

35. “TRAIN PASSES THROUGH A TUNNEL”

Usually means entire train clears tunnel.

So:

$\backslash [\boxed{L_T + L_{\text{tunnel}}} \backslash]$

36. “ENGINE CROSSES A BRIDGE”

If only engine is considered, train length may not matter.

But typical wording:

train crosses bridge

means entire train clears it.

Read carefully.

37. TRAIN CROSSING A STATIONARY CAR

If car length is ignored or treated as point:

$$[D=L_T]$$

But if car/vehicle has specified physical length and asks complete crossing:

$$[D=L_T+L_{\text{vehicle}}]$$

Use wording.

38. GATE-STYLE QUESTION 1

A 180 m train at 54 km/h crosses a pole.

$$[54=15\text{m/s}]$$

Time:

$$[180/15 = \boxed{12\text{s}}]$$

39. GATE-STYLE QUESTION 2

Train length 200 m, platform 300 m, speed 72 km/h.

$$[72=20\text{m/s}]$$

Distance:

$$[500]$$

Time:

$$[\boxed{25\text{s}}]$$

40. GATE-STYLE QUESTION 3

Two trains 150 m and 250 m long move opposite directions at 54 and 36 km/h.

Relative:

$$[25\text{m/s}]$$

Distance:

$$[400]$$

Time:

$$[\boxed{16\text{s}}]$$

41. GATE-STYLE QUESTION 4

Train crosses pole in 10 s and platform in 25 s at 20 m/s.

Platform length:

$$[20(15) = \boxed{300\text{m}}]$$

42. PLACEMENT-STYLE QUESTION 1

A 150 m train at 72 km/h crosses a runner moving same direction at 18 km/h.

Relative:

$$\backslash [20-5=15\text{m/s} \backslash]$$

Time:

$$\backslash [150/15 = \boxed{10\text{s}} \backslash]$$

43. PLACEMENT-STYLE QUESTION 2

Same train and runner in opposite directions.

Relative:

$$\backslash [20+5=25\text{m/s} \backslash]$$

Time:

$$\backslash [150/25 = \boxed{6\text{s}} \backslash]$$

44. PLACEMENT-STYLE QUESTION 3

Two trains:

$$\backslash [200\text{m}, \backslash 100\text{m} \backslash]$$

same direction at:

$$\backslash [72, \backslash 54\text{ km/h} \backslash]$$

Relative speed:

$$\backslash [20-15=5\text{m/s} \backslash]$$

Time:

$$\backslash [300/5 = \boxed{60\text{s}} \backslash]$$

45. PLACEMENT-STYLE QUESTION 4

Train crosses pole in 8 s at 72 km/h.

Length:

$$\backslash [20(8) = \boxed{160\text{m}} \backslash]$$

46. ADVANCED QUESTION — FIND PLATFORM

Train = 250 m.

Speed = 90 km/h.

Crossing time = 30 s.

$$\backslash [90=25\text{m/s} \backslash]$$

Total distance:

$$\boxed{25(30)=750\text{m}}$$

Platform:

$$\boxed{750-250 = 500\text{m}}$$

47. ADVANCED QUESTION — TWO PLATFORMS

Train crosses:

- 120 m platform in 20 s
- 300 m platform in 32 s

Speed:

$$\boxed{\frac{300-120}{32-20} = \frac{180}{12} = 15\text{m/s}}$$

Train length:

$$\boxed{15(20)-120}$$

$$\boxed{300-120 = 180\text{m}}$$

Speed in km/h:

$$\boxed{15 \times \frac{18}{5} = 54 \text{ km/h}}$$

48. ADVANCED QUESTION — TWO TRAINS

Two trains move opposite directions.

Lengths:

$$\boxed{180\text{m}, 220\text{m}}$$

One speed:

$$\boxed{54 \text{ km/h} = 15\text{m/s}}$$

They cross in:

$$\boxed{10\text{s}}$$

Total relative speed:

$$\boxed{400/10=40\text{m/s}}$$

Other train:

$$\boxed{40-15=25\text{m/s}}$$

$$\boxed{= 90 \text{ km/h}}$$

49. ADVANCED QUESTION — OVERTAKING

Fast train:

$$\boxed{150\text{m}, 90 \text{ km/h}}$$

Slow train:

$\sqrt[2]{250\text{m}, \sqrt[5]{54\text{ km/h}}}$

Convert:

$\sqrt[2]{25, \sqrt[5]{15\text{m/s}}}$

Relative:

$\sqrt[2]{10\text{m/s}}$

Total length:

$\sqrt[2]{400\text{m}}$

Time:

$\sqrt[2]{\boxed{40\text{s}}}$

50. ADVANCED QUESTION — PERSON ON PLATFORM

Train = 240 m.

Speed = 72 km/h.

Person runs opposite direction at 18 km/h.

Relative:

$\sqrt[2]{20+5=25\text{m/s}}$

Time:

$\sqrt[2]{240/25 = \boxed{9.6\text{s}}}$

51. COMMON TRAPS

& pTRAP 1 — Pole crossing uses platform length

No.

For point object:

$\sqrt[2]{D=L_T}$

& pTRAP 2 — Platform crossing uses only platform length

Wrong.

$\sqrt[2]{D=L_T+L_P}$

& pTRAP 3 — Two trains crossing uses one train length

Wrong.

$\sqrt[2]{D=L_1+L_2}$

& pTRAP 4 — Same direction uses sum of speeds

Wrong.

Use difference.

& pTRAP 5 — Opposite direction uses difference

Wrong.

Use sum.

& pTRAP 6 — Forget km/h to m/s

Metres + seconds usually require m/s.

52. MUST-MEMORIZE FORMULAS

Train vs pole/person

$$\boxed{T = \frac{L_T}{v_{\text{rel}}}}$$

Train vs platform/bridge/tunnel

$$\boxed{T = \frac{L_T + L_O}{v}}$$

where (L_O) is object length.

Two trains

$$\boxed{T = \frac{L_1 + L_2}{v_{\text{rel}}}}$$

53. RELATIVE SPEED

Same direction

$$\boxed{v_1 - v_2}$$

Opposite directions

$$\boxed{v_1 + v_2}$$

54. HIGH-VALUE SHORTCUTS

Train crosses pole and platform

$$\boxed{L_P = v(t_P - t_{\text{pole}})}$$

Two different platforms

$$\boxed{v = \frac{L_{P2} - L_{P1}}{t_2 - t_1}}$$

Train length

$$\boxed{L_T = vt_{\text{pole}}}$$

55. FAST EXAM DECISION TREE

Point object?

Distance = train length.

Platform/bridge/tunnel?

Add lengths.

Moving person?

Use train length + relative speed.

Two trains?

Add train lengths.

Same direction?

Subtract speeds.

Opposite directions?

Add speeds.

Metres and seconds?

Convert speed to m/s.

PRACTICE — LEVEL 0

Q1

A 120 m train at 54 km/h crosses a pole. Find time.

Q2

A 180 m train at 72 km/h crosses a stationary person. Find time.

Q3

Train = 200 m, platform = 400 m, speed = 72 km/h. Find crossing time.

Q4

A 150 m train crosses pole in 10 s. Find speed in m/s.

Q5

Convert 90 km/h to m/s.

LEVEL 1

Q6

A 180 m train at 72 km/h crosses a runner moving same direction at 18 km/h. Find time.

Q7

Same data, runner opposite direction. Find time.

Q8

Two trains 150 m and 250 m long move opposite at 54 and 36 km/h. Find crossing time.

Q9

Two trains 200 m and 100 m move same direction at 72 and 54 km/h. Find overtaking time.

Q10

Train 160 m crosses a 240 m platform in 20 s. Find speed.

LEVEL 2

Q11

A train crosses pole in 10 s and a platform in 25 s at 20 m/s. Find platform length.

Q12

A train crosses 100 m and 220 m platforms in 15 s and 21 s. Find train speed and train length.

Q13

Two trains of lengths 180 m and 220 m cross in 10 s opposite directions. One speed is 54 km/h. Find the other.

Q14

A 250 m train at 90 km/h crosses a platform in 30 s. Find platform length.

Q15

Train 150 m at 72 km/h overtakes another 250 m train in 40 s. Find slower train speed.

LEVEL 3 — EXAM PRESSURE

Q16

A train crosses a pole in 12 s and a 300 m bridge in 32 s. Find train length and speed.

Q17

A train crosses a runner moving same direction at 6 m/s in 20 s and opposite direction in 10 s. Find train speed and length.

Q18

Two trains of equal length move opposite directions at 54 and 72 km/h and completely cross in 12 s. Find each train's length.

Q19

A train crosses a 200 m platform in 20 s and a 500 m platform in 35 s. Find train length and speed.

Q20

A train at 90 km/h crosses a pole in 8 s. How long will it take to cross a 300 m tunnel?

ANSWERS

Q1

$$\sqrt{54=15\text{m/s}}$$

$$\sqrt{120/15 = \boxed{8\text{s}}}$$

Q2

$$\sqrt{72=20\text{m/s}}$$

$$\sqrt{180/20 = \boxed{9\text{s}}}$$

Q3

Distance:

$$\sqrt{200+400=600}$$

Speed:

$$\sqrt{20\text{m/s}}$$

$$\sqrt{600/20 = \boxed{30\text{s}}}$$

Q4

$$\sqrt{150/10 = \boxed{15\text{m/s}}}$$

Q5

$$\sqrt{90 \times \frac{5}{18} = \boxed{25\text{m/s}}}$$

Q6

Relative:

$$\sqrt{20-5=15}$$

$$\sqrt{180/15 = \boxed{12\text{s}}}$$

Q7

Relative:

$$\sqrt{20+5=25}$$

$$\sqrt{180/25 = \boxed{7.2\text{s}}}$$

Q8

Total length:

$$\sqrt{400}$$

Relative:

$$\sqrt{15+10=25}$$

$$\sqrt{400/25 = \boxed{16\text{s}}}$$

Q9

Total length:

$$\sqrt{300}$$

Relative:

$$\sqrt{20-15=5}$$

$$\boxed{300/5 = 60\text{s}}$$

Q10

Total distance:

$$160 + 240 = 400\text{m}$$

Speed:

$$400/20 = 20\text{m/s}$$

$$= \boxed{72\text{ km/h}}$$

Q11

$$L_P = 20(25 - 10)$$

$$= \boxed{300\text{m}}$$

Q12

Speed:

$$\frac{220 - 100}{21 - 15} = 20\text{m/s}$$

Train:

$$20(15) - 100$$

$$= \boxed{200\text{m}}$$

Speed:

$$\boxed{72\text{ km/h}}$$

Q13

Total length:

$$400$$

Relative:

$$400/10 = 40\text{m/s}$$

Known:

$$15\text{m/s}$$

Other:

$$25\text{m/s}$$

$$= \boxed{90\text{ km/h}}$$

Q14

Speed:

$$25\text{m/s}$$

Total crossing distance:

$$25(30) = 750$$

Platform:

$$750 - 250 = \boxed{500\text{m}}$$

Q15

Train A:

$$\sqrt{72=20\text{m/s}}$$

Total length:

$$\sqrt{150+250=400}$$

Relative speed:

$$\sqrt{400/40=10\text{m/s}}$$

Slow train:

$$\sqrt{20-10=10\text{m/s}}$$

$$\sqrt{=\boxed{36\text{ km/h}}}$$

Q16

Pole crossing:

$$\sqrt{L=12v}$$

Bridge:

$$\sqrt{L+300=32v}$$

Subtract:

$$\sqrt{300=20v}$$

$$\sqrt{v=15\text{m/s}}$$

Length:

$$\sqrt{12(15)=180\text{m}}$$

So:

$$\sqrt{\boxed{L=180\text{m}, \quad v=54\text{ km/h}}}$$

Q17

Let train speed = (v) m/s and length (L) .

Same direction:

$$\sqrt{L=20(v-6)}$$

Opposite direction:

$$\sqrt{L=10(v+6)}$$

Equate:

$$\sqrt{20v-120=10v+60}$$

$$\sqrt{10v=180}$$

$$\sqrt{v=18\text{m/s}}$$

Length:

$$\sqrt{10(18+6) = 240\text{m}}$$

Thus:

$$\boxed{v=18\text{m/s}=64.8\text{km/h}, L=240\text{m}}$$

Q18

Speeds:

$$54=15\text{m/s}$$

$$72=20\text{m/s}$$

Relative:

$$35\text{m/s}$$

In 12 s, total train lengths:

$$35(12)=420\text{m}$$

Equal lengths:

$$420/2 = \boxed{210\text{m}}$$

Q19

Platform difference:

$$500-200=300\text{m}$$

Time difference:

$$35-20=15\text{s}$$

Speed:

$$300/15=20\text{m/s}$$

Using first platform:

$$L+200=20(20)=400$$

$$L=200\text{m}$$

Therefore:

$$\boxed{L=200\text{m}, \text{quad } v=72\text{km/h}}$$

Q20

Train length from pole:

$$90\text{km/h}=25\text{m/s}$$

$$L=25(8)=200\text{m}$$

Tunnel crossing distance:

$$200+300=500\text{m}$$

Time:

$$500/25 = \boxed{20\text{s}}$$

CHAPTER 31 STATUS

Covered so far:

- train length concept '

- crossing pole/person '
- platform/bridge/tunnel '
- km/h !" m/s '
- moving person '
- two trains opposite directions '
- overtaking same direction '
- finding train length '
- finding platform length '
- finding unknown train speed '
- two-platform shortcut '
- advanced crossing equations '

But **Chapter 31 is not complete yet.**

Next we still need:

- tunnel entry/exit timing
- engine entering vs rear leaving
- partial crossing
- two trains starting with a gap
- overtaking with initial separation
- platform-person combinations
- train crossing another train plus platform conditions
- repeated/advanced crossing equations
- final shortcuts and chapter audit

Progress

Completed: 30/76 **Current:** Ø=Chapter 31 — Trains **Next:** Chapter 31 Part 2 — Tunnel timing, initial gaps, overtaking with separation & advanced train problems

USER

next

ASSISTANT

CHAPTER 31 — TRAINS

PART 2 — TUNNEL ENTRY/EXIT, PARTIAL CROSSING, INITIAL GAPS, OVERTAKING WITH SEPARATION& ADVANCED TRAIN PROBLEMS

Status: Ø=No PROGRESS **GATE:** Medium–High **Placements:** High **Priority:** +Tier 2

Part 1 covered the core crossing formulas.

Now we finish the chapter with:

- tunnel entry/exit timing

- engine vs rear-end events
- partially crossing an object
- initial gaps between trains
- overtaking after a head start
- complex platform equations
- advanced relative-speed train questions

The central rule remains:

$$\boxed{\text{Time} = \frac{\text{required relative distance}}{\text{relative speed}}}$$

1. THREE IMPORTANT TRAIN EVENTS

Suppose a train approaches a tunnel.

There are three different moments:

Event 1 — Engine enters tunnel

Front of train reaches entrance.

Event 2 — Entire train enters tunnel

Rear of train reaches entrance.

Event 3 — Entire train exits tunnel

Rear of train passes the far end.

These are different events.

2. TIME FOR ENTIRE TRAIN TO ENTER A TUNNEL

Train length:

$$L_T$$

Once the engine enters the tunnel, the rear must travel one train length to enter.

So:

$$\boxed{T_{\text{enter completely}} = \frac{L_T}{v}}$$

Notice: tunnel length does not matter here.

3. EXAMPLE

Train length:

$$180\text{m}$$

Speed:

$$54\text{ km/h} = 15\text{m/s}$$

Time from engine entering until rear enters:

$$180/15 = \boxed{12\text{s}}$$

4. TIME FOR ENGINE TO REACH FAR END OF TUNNEL

Tunnel length:

$$\boxed{L_{\text{tun}}}$$

Engine travels exactly tunnel length.

$$\boxed{T_{\text{engine exit}} = \frac{L_{\text{tun}}}{v}}$$

5. ENTIRE TRAIN COMPLETELY EXITS TUNNEL

From the moment engine enters tunnel until rear leaves far end:

Required displacement of train:

$$\boxed{L_T + L_{\text{tun}}}$$

Therefore:

$$\boxed{T_{\text{complete crossing}} = \frac{L_T + L_{\text{tun}}}{v}}$$

6. EXAMPLE

Train:

$$\boxed{200\text{m}}$$

Tunnel:

$$\boxed{300\text{m}}$$

Speed:

$$\boxed{20\text{m/s}}$$

Rear fully enters tunnel:

$$\boxed{200/20=10\text{s}}$$

Engine reaches tunnel end:

$$\boxed{300/20=15\text{s}}$$

Rear fully exits:

$$\boxed{500/20=25\text{s}}$$

So:

$$\boxed{10\text{s}, 15\text{s}, 25\text{s}}$$

depending on what the question asks.

7. TIME TRAIN IS ENTIRELY INSIDE TUNNEL

This means:

- starts when rear enters tunnel
- ends when engine reaches far end

This is possible only if tunnel is longer than train.

Distance travelled during this interval:

$$L_{\text{tun}} - L_T$$

Thus:

$$T_{\text{fully inside}} = \frac{L_{\text{tun}} - L_T}{v}$$

if:

$$L_{\text{tun}} > L_T$$

8. EXAMPLE

Train:

$$150\text{m}$$

Tunnel:

$$450\text{m}$$

Speed:

$$15\text{m/s}$$

Fully-inside duration:

$$(450 - 150) / 15$$

$$300 / 15 = \boxed{20\text{s}}$$

9. IF TRAIN IS LONGER THAN TUNNEL

Suppose:

$$L_T > L_{\text{tun}}$$

Then the entire train can never be simultaneously inside the tunnel.

So “time during which train is completely inside tunnel” is:

$$\boxed{0}$$

This is a nice conceptual trap.

10. PARTIAL PLATFORM CROSSING

Suppose a question asks:

How long from the moment the engine reaches the platform until the engine leaves the platform?

Only the engine's motion is considered.

Distance:

$$\boxed{L_P}$$

So:

$$\boxed{T = L_P / v}$$

But if the **entire train** must clear the platform:

$$\boxed{T = (L_T + L_P) / v}$$

11. REAR REACHES PLATFORM START

From engine reaching platform start until rear reaches same point:

$$\boxed{T = L_T / v}$$

Again, only train length matters.

12. ENGINE LEAVES PLATFORM BEFORE REAR?

Yes.

For a long train, the engine may leave a platform while rear is still on it.

Always identify the named part:

- engine/front
- rear/tail
- entire train

13. TWO TRAINS WITH INITIAL GAP — SAME DIRECTION

Suppose faster train A is behind slower train B.

Initial gap between A's front and B's rear:

$$G$$

To completely overtake B, relative distance required:

$$\boxed{G + L_A + L_B}$$

Relative speed:

$$v_A - v_B$$

Thus:

$$\boxed{T = \frac{G + L_A + L_B}{v_A - v_B}}$$

14. EXAMPLE

A:

$$200 \text{ m/s} = 72 \text{ km/h} = 20 \text{ m/s}$$

B:

$$\left[100\text{m}, \frac{54\text{ km/h}}{3.6} = 15\text{m/s} \right]$$

Initial gap:

$$\left[50\text{m} \right]$$

Relative speed:

$$\left[5\text{m/s} \right]$$

Relative distance:

$$\left[50+200+100=350\text{m} \right]$$

Time:

$$\left[\frac{350}{5} = \boxed{70\text{s}} \right]$$

15. TIME UNTIL FASTER TRAIN'S ENGINE REACHES SLOWER TRAIN'S REAR

Same setup.

Only the initial gap must close:

$$\left[\boxed{T = \frac{G}{v_A - v_B}} \right]$$

Using previous numbers:

$$\left[\frac{50}{5} = \boxed{10\text{s}} \right]$$

After that, actual overtaking begins.

16. TIME FROM FIRST CONTACT TO COMPLETE OVERTAKE

Once A's engine reaches B's rear, relative displacement needed is:

$$\left[L_A + L_B \right]$$

Time:

$$\left[\frac{300}{5} = \boxed{60\text{s}} \right]$$

Total including initial gap:

$$\left[10+60 = 70\text{s} \right]$$

17. TWO TRAINS WITH INITIAL GAP — OPPOSITE DIRECTIONS

If they move toward each other and initial gap between nearest ends is G :

Time until first contact:

$$\left[\boxed{T_{\text{contact}}} = \frac{G}{v_1 + v_2} \right]$$

Time until completely cross after first contact:

$$\left[\boxed{\frac{L_1 + L_2}{v_1 + v_2}} \right]$$

Total from initial positions until complete crossing:

$$\left[\boxed{\frac{G + L_1 + L_2}{v_1 + v_2}} \right]$$

18. EXAMPLE

Two trains:

$$\backslash [150\text{m}, \backslash 250\text{m} \backslash]$$

Speeds:

$$\backslash [15, \backslash 10\text{m/s} \backslash]$$

Nearest ends initially 100 m apart.

Total relative distance:

$$\backslash [100+150+250=500\text{m} \backslash]$$

Relative speed:

$$\backslash [25\text{m/s} \backslash]$$

Time:

$$\backslash [500/25 = \boxed{20\text{s}} \backslash]$$

19. HEAD START BETWEEN TRAINS

Suppose slower train starts earlier.

Example:

Slow train speed:

$$\backslash [54\text{ km/h}=15\text{m/s} \backslash]$$

starts 20 s earlier.

Head-start distance:

$$\backslash [15(20)=300\text{m} \backslash]$$

If faster train:

$$\backslash [72\text{ km/h}=20\text{m/s} \backslash]$$

Relative speed:

$$\backslash [5\text{m/s} \backslash]$$

Time after faster train starts to catch the **same reference point**:

$$\backslash [300/5 = \boxed{60\text{s}} \backslash]$$

If train lengths matter, add them according to the event asked.

20. ENGINE CATCH VS COMPLETE OVERTAKE

These are different.

Suppose slower train has length (L_B) .

If faster train starts behind:

Engine catches rear of slower train

Close only head-start/gap.

Faster train completely overtakes

Close:

$$\text{gap} + L_A + L_B$$

Always distinguish the two.

21. OVERTAKING A PLATFORM-MOVING VEHICLE COMBINATION

Suppose train crosses a moving truck whose length is negligible.

Use train length and relative speed.

If truck length is given and asks complete crossing:

$$D = L_T + L_{\text{truck}}$$

Same train logic.

22. TRAIN CROSSING A PLATFORM WHILE PLATFORM MOVES?

Unusual, but if the “platform” is actually a moving object, treat it as another object with length.

Then:

$$\text{relative speed}$$

is needed.

Stationary platform is just:

$$v_{\text{rel}} = v_T$$

23. CROSSING TWO DIFFERENT PLATFORMS

Recall:

$$L_T + P_1 = v t_1$$

$$L_T + P_2 = v t_2$$

Subtract:

$$P_2 - P_1 = v(t_2 - t_1)$$

Therefore:

$$v = \frac{P_2 - P_1}{t_2 - t_1}$$

24. FIND TRAIN LENGTH AFTER SPEED

Once v is found:

$$L_T = v t_1 - P_1$$

25. TRAIN CROSSES POLE AND BRIDGE

If pole time:

$$[t_p]$$

Bridge time:

$$[t_b]$$

then:

$$[L_T = vt_p]$$

$$[L_T + L_B = vt_b]$$

Subtract:

$$[\boxed{L_B = v(t_b - t_p)}]$$

26. TRAIN CROSSES TWO PERSONS MOVING DIFFERENT DIRECTIONS

This is a useful advanced pattern.

Train length (L) , train speed (v) .

Person A moves same direction at (u) .

Crossing time:

$$[t_1]$$

Then:

$$[L = (v - u)t_1]$$

Person B moves opposite direction at (w) .

Crossing time:

$$[t_2]$$

Then:

$$[L = (v + w)t_2]$$

Use both equations to solve (L, v) .

27. EXAMPLE

Train crosses:

- a runner moving same direction at 5 m/s in 20 s
- another runner moving opposite direction at 5 m/s in 12 s

Then:

$$[L = 20(v - 5)]$$

and:

$$[L = 12(v + 5)]$$

Equate:

$$20v - 100 = 12v + 60$$

$$8v = 160$$

$$v = 20 \text{ m/s}$$

Length:

$$20(15) = \boxed{300 \text{ m}}$$

Speed:

$$20 \text{ m/s} = 72 \text{ km/h}$$

28. SAME PERSON, DIFFERENT DIRECTION

If the same person runs at speed u :

$$L = (v - u)t_s$$

$$L = (v + u)t_o$$

where:

- t_s = same-direction crossing time
- t_o = opposite-direction crossing time

Since opposite direction has greater relative speed:

$$\boxed{t_o < t_s}$$

Sanity check.

29. FIND TRAIN SPEED DIRECTLY

From:

$$(v - u)t_s = (v + u)t_o$$

Solve:

$$vt_s - ut_s = vt_o + ut_o$$

$$v(t_s - t_o) = u(t_s + t_o)$$

So:

$$\boxed{v = u \frac{t_s + t_o}{t_s - t_o}}$$

Useful shortcut.

30. FIND TRAIN LENGTH DIRECTLY

Once v is known:

$$L = (v - u)t_s$$

or:

$$L = (v + u)t_o$$

31. TWO EQUAL-LENGTH TRAINS

If equal train lengths (L) cross each other:

Total length:

$$[2L]$$

Thus:

$$[\boxed{ 2L = v_{\text{rel}} t }]$$

So:

$$[\boxed{ L = \frac{v_{\text{rel}} t}{2} }]$$

32. EXAMPLE

Two equal trains move opposite at:

$$[54, 72 \text{ km/h}]$$

Convert:

$$[15, 20 \text{ m/s}]$$

Relative:

$$[35 \text{ m/s}]$$

Crossing time:

$$[12 \text{ s}]$$

Each length:

$$[\frac{35(12)}{2} = \boxed{210 \text{ m}}]$$

33. LENGTH RATIO FROM POLE CROSSING TIMES

Two trains move at the same speed.

Train A crosses a pole in 8 s.

Train B in 12 s.

Since:

$$[L = vt]$$

same (v) , therefore:

$$[\boxed{ L_A : L_B = 8 : 12 = 2 : 3 }]$$

34. SPEED RATIO FROM SAME TRAIN LENGTHS

Equal-length trains cross a pole in:

$$[8 \text{ s}, 10 \text{ s}]$$

Since:

$$v = L/t$$

$$v_1:v_2 = 10:8 = \boxed{5:4}$$

35. TRAIN CROSSES POLE IN (t) , PLATFORM IN (kt)

Let train length be (L) .

Pole:

$$L = vt$$

Platform crossing:

$$L + P = vkt$$

Since:

$$vt = L$$

$$L + P = kL$$

So:

$$\boxed{P = (k-1)L}$$

Very fast ratio shortcut.

36. EXAMPLE

Train crosses pole in 10 s.

Same train crosses platform in 30 s.

Thus:

$$k = 3$$

Platform length:

$$(3-1)L = \boxed{2L}$$

So platform is twice train length.

37. TRAIN AND PLATFORM LENGTH RATIO FROM TIMES

If pole time = (t) , platform time = (T) :

$$L = vt$$

$$L + P = v(T-t)$$

Therefore:

$$\boxed{L:T:P = t:(T-t)}$$

No need to know speed.

38. EXAMPLE

Pole time:

$$\sqrt{12\text{s}}$$

Platform time:

$$\sqrt{32\text{s}}$$

Then:

$$\sqrt{L_T:L_P = 12:20 = \boxed{3:5}}$$

39. TRAIN COMPLETELY INSIDE PLATFORM REGION

If a platform is longer than train, the train is completely alongside the platform for a duration:

$$\sqrt{\boxed{\frac{L_P - L_T}{v}}}$$

Same geometry as a tunnel.

40. EXAMPLE

Train:

$$\sqrt{100\text{m}}$$

Platform:

$$\sqrt{300\text{m}}$$

Speed:

$$\sqrt{20\text{m/s}}$$

Time when train is completely within platform boundaries:

$$\sqrt{(300-100)/20 = \boxed{10\text{s}}}$$

41. INITIAL GAP + PLATFORM STYLE QUESTION

Suppose a train is 100 m away from the start of a 300 m bridge.

Train length = 200 m.

Speed = 20 m/s.

Time until train completely clears bridge:

Required travel:

$$\sqrt{100+300+200 = 600\text{m}}$$

Time:

$$\sqrt{600/20 = \boxed{30\text{s}}}$$

42. GENERAL “DISTANCE BEFORE OBJECT” RULE

If train begins (G) metres before an object of length (L_O) , and asks time until train fully clears object:

$$\boxed{D = G + L_T + L_O}$$

43. GATE-STYLE QUESTION 1

Train 200 m, tunnel 300 m, speed 20 m/s.

Time for complete crossing:

$$500/20 = \boxed{25s}$$

44. GATE-STYLE QUESTION 2

Same train and tunnel.

Time for rear to enter tunnel:

$$200/20 = \boxed{10s}$$

45. GATE-STYLE QUESTION 3

Tunnel = 500 m, train = 200 m, speed = 20 m/s.

Time train remains completely inside:

$$(500-200)/20$$

$$= \boxed{15s}$$

46. GATE-STYLE QUESTION 4

Two trains length 200 m and 100 m, initial same-direction gap 50 m.

Speeds:

$$20, 15m/s$$

Complete overtaking time:

$$(50+300)/5 = \boxed{70s}$$

47. PLACEMENT-STYLE QUESTION 1

A 180 m train at 54 km/h approaches a tunnel 300 m long.

Complete crossing:

$$(180+300)/15 = \boxed{32s}$$

48. PLACEMENT-STYLE QUESTION 2

Two trains move toward each other.

Lengths:

$$\backslash [150, \backslash 250\text{m} \backslash]$$

Gap:

$$\backslash [100\text{m} \backslash]$$

Speeds:

$$\backslash [15, \backslash 10\text{m/s} \backslash]$$

Time until complete crossing:

$$\backslash [500/25 = \boxed{20\text{s}} \backslash]$$

49. PLACEMENT-STYLE QUESTION 3

Train crosses pole in 12 s and platform in 32 s.

Find train:platform length ratio.

$$\backslash [12:(32-12) \backslash]$$

$$\backslash [12:20 \backslash]$$

$$\backslash [=\boxed{3:5} \backslash]$$

50. PLACEMENT-STYLE QUESTION 4

A 200 m train is 100 m before a 300 m bridge.

Speed:

$$\backslash [72\text{ km/h}=20\text{m/s} \backslash]$$

Time until completely clears:

$$\backslash [(100+200+300)/20 \backslash]$$

$$\backslash [=\boxed{30\text{s}} \backslash]$$

51. ADVANCED QUESTION — SAME/OPPOSITE RUNNER

A train crosses a runner moving at 6 m/s:

- same direction in 20 s
- opposite direction in 10 s

Using shortcut:

$$\backslash [v = 6\frac{20+10}{20-10} \backslash]$$

$$\backslash [=6\cdot 3 = 18\text{m/s} \backslash]$$

Length:

$$\boxed{(18-6)(20) = 240\text{m}}$$

Thus:

$$\boxed{v=18\text{m/s}=64.8\text{km/h}, \text{quad } L=240\text{m}}$$

52. ADVANCED QUESTION — TWO PLATFORMS

Train crosses:

- 200 m platform in 20 s
- 500 m platform in 35 s

Speed:

$$\boxed{\frac{500-200}{35-20} = 300/15 = 20\text{m/s}}$$

Train:

$$\boxed{20(20)-200 = 200\text{m}}$$

$$\boxed{L=200\text{m}, \text{ } v=72\text{km/h}}$$

53. ADVANCED QUESTION — HEAD START + TRAIN LENGTH

A 100 m slow train at 15 m/s starts 20 s before a 200 m faster train at 20 m/s from the same point.

At moment faster train starts, slow train's rear has moved 300 m from start if we track corresponding initial reference.

For complete overtake, relative distance effectively includes head start and both lengths:

$$\boxed{300+100+200=600}$$

Relative speed:

$$\boxed{5\text{m/s}}$$

Time:

$$\boxed{600/5 = 120\text{s}}$$

The exact geometry of which ends align at start matters in such questions; draw a line diagram when wording is ambiguous.

54. ADVANCED QUESTION — FULLY INSIDE TUNNEL

Train length:

$$\boxed{240\text{m}}$$

Tunnel:

$$\boxed{600\text{m}}$$

Speed:

$$\boxed{72\text{km/h}=20\text{m/s}}$$

Fully-inside duration:

$$\boxed{(600-240)/20}$$

$$\boxed{=360/20}$$

$$\boxed{=18s}$$

55. ADVANCED QUESTION — FIND TUNNEL LENGTH

Train:

$$\boxed{180m}$$

Speed:

$$\boxed{54km/h=15m/s}$$

It takes 32 s from engine entering tunnel until rear exits.

Total distance:

$$\boxed{15(32)=480}$$

Tunnel:

$$\boxed{480-180 = 300m}$$

56. ADVANCED QUESTION — TRAIN LENGTH FROM FULLY-INSIDE TIME

Tunnel:

$$\boxed{500m}$$

Train speed:

$$\boxed{20m/s}$$

Train is completely inside tunnel for 12 s.

Then:

$$\frac{500-L}{20}=12$$

$$\boxed{500-L=240}$$

$$\boxed{L=260m}$$

57. COMMON TRAPS

TRAP 1 — Entire train enters tunnel

Distance is only:

$$\boxed{L_T}$$

not $\boxed{L_T + L_{\text{tun}}}$.

TRAP 2 — Entire train exits tunnel

Now distance is:

$$\boxed{L_T + L_{\text{tun}}}$$

& pTRAP 3 — Fully inside tunnel

Use:

$$\boxed{L_{\text{tun}} - L_T}$$

only when tunnel is longer.

& pTRAP 4 — Initial gap omitted

For delayed contact/overtake, include the gap.

& pTRAP 5 — Engine catches vs entire train overtakes

These are different events.

& pTRAP 6 — Train/head-start geometry

Identify which points the given separation refers to: front-to-front, front-to-rear, nearest ends, etc.

58. MUST-MEMORIZE ADVANCED FORMULAS

Completely enters tunnel

$$\boxed{T = \frac{L_T}{v}}$$

Completely crosses tunnel

$$\boxed{T = \frac{L_T + L_{\text{tun}}}{v}}$$

Completely inside tunnel

$$\boxed{T = \frac{L_{\text{tun}} - L_T}{v}}$$

when $(L_{\text{tun}} > L_T)$.

59. INITIAL-GAP FORMULAS

Same direction complete overtake

$$\boxed{T = \frac{G + L_1 + L_2}{v_f - v_s}}$$

Opposite direction complete crossing

$$\boxed{T = \frac{G + L_1 + L_2}{v_1 + v_2}}$$

60. TIME-RATIO SHORTCUTS

Pole vs platform

$$\boxed{L_T : L_P = t_{\text{pole}} : (t_P - t_{\text{pole}})}$$

Two platforms

$$\boxed{v = \frac{\Delta L_P}{\Delta t}}$$

Same/opposite runner

$$v_T = u \frac{t_s + t_o}{t_s - t_o}$$

61. FAST EXAM DECISION TREE

What exactly is crossing?

Identify:

- engine
- rear
- whole train

Is object a point?

Use train length.

Is object extended?

Add lengths.

Is there an initial gap?

Add it before complete crossing.

Are both moving?

Use relative speed.

Same direction?

Subtract.

Opposite direction?

Add.

Tunnel asks “fully inside”?

Use difference of lengths.

PRACTICE — LEVEL 0

Q1

A 200 m train at 20 m/s enters a tunnel. How long until rear enters?

Q2

Same train crosses a 300 m tunnel completely. Find time.

Q3

Train 150 m, tunnel 450 m, speed 15 m/s. Find fully-inside time.

Q4

A 100 m train and 200 m train move same direction at 20 and 15 m/s. No initial gap. Find overtaking time.

Q5

Same trains move opposite directions. Find crossing time.

LEVEL 1

Q6

Train 180 m, tunnel 300 m, speed 54 km/h. Find complete crossing time.

Q7

Train 180 m at 15 m/s takes 32 s to cross tunnel. Find tunnel length.

Q8

Two trains 150 m and 250 m are initially 100 m apart and move toward each other at 15 and 10 m/s. Find complete-crossing time.

Q9

Two trains 200 m and 100 m have 50 m same-direction gap. Speeds 20 and 15 m/s. Find complete-overtake time.

Q10

Train crosses pole in 12 s and platform in 32 s. Find train:platform length ratio.

LEVEL 2

Q11

Train crosses a runner at 6 m/s in 20 s same direction and 10 s opposite direction. Find train speed and length.

Q12

Train crosses 200 m platform in 20 s and 500 m platform in 35 s. Find train speed and length.

Q13

Tunnel length = 500 m. Train moves at 20 m/s and is fully inside for 12 s. Find train length.

Q14

Two equal trains at 54 and 72 km/h move opposite directions and cross in 12 s. Find each length.

Q15

Train is 100 m before a 300 m bridge. Train length 200 m, speed 72 km/h. Find time to fully clear bridge.

LEVEL 3 — EXAM PRESSURE

Q16

A train crosses a pole in 10 s and a platform in 25 s. Platform is 300 m. Find speed and train length.

Q17

A train crosses two runners each moving at 5 m/s. Same-direction crossing takes 20 s and opposite-direction crossing

takes 12 s. Find train speed and length.

Q18

A train of length 250 m travels at 90 km/h. It enters a 750 m tunnel. Find:

1. time until whole train enters,
2. time engine reaches far end,
3. time entire train exits,
4. time train is completely inside.

Q19

Two trains 180 m and 220 m long move opposite directions. Their nearest ends are 200 m apart. Speeds are 54 and 90 km/h. Find time until complete crossing.

Q20

A faster 200 m train moves at 90 km/h behind a slower 150 m train moving at 54 km/h. The initial gap between faster train's engine and slower train's rear is 100 m. Find:

1. time to first contact,
2. additional time to complete overtake,
3. total time.

ANSWERS

Q1

$$\sqrt{200/20 = \boxed{10s}}$$

Q2

$$\sqrt{(200+300)/20 = \boxed{25s}}$$

Q3

$$\sqrt{(450-150)/15 = \boxed{20s}}$$

Q4

Total length:

$$\sqrt{300}$$

Relative:

$$\sqrt{5}$$

$$\sqrt{300/5 = \boxed{60s}}$$

Q5

Relative:

$$\sqrt{20+15=35}$$

$$\sqrt{300/35 = \boxed{\frac{60}{7}\approx 8.57s}}$$

Q6

Speed:

$$\sqrt{54=15\text{m/s}}$$

$$\sqrt{(180+300)/15 = 480/15 = \boxed{32\text{s}}}$$

Q7

Total distance:

$$\sqrt{15(32)=480}$$

Tunnel:

$$\sqrt{480-180 = \boxed{300\text{m}}}$$

Q8

Total relative distance:

$$\sqrt{100+150+250=500}$$

Relative speed:

$$\sqrt{25}$$

$$\sqrt{\boxed{20\text{s}}}$$

Q9

$$\sqrt{50+200+100=350}$$

Relative:

$$\sqrt{5}$$

$$\sqrt{\boxed{70\text{s}}}$$

Q10

$$\sqrt{12:(32-12) = 12:20 = \boxed{3:5}}$$

Q11

$$\sqrt{v = 6\frac{20+10}{20-10} = 18\text{m/s}}$$

Length:

$$\sqrt{(18-6)(20) = 240\text{m}}$$

$$\sqrt{\boxed{18\text{m/s}=64.8\text{km/h}, \quad 240\text{m}}}$$

Q12

Speed:

$$\sqrt{\frac{500-200}{35-20} = 20\text{m/s}}$$

Train length:

$$\sqrt{20(20)-200 = 200\text{m}}$$

$$\sqrt{\boxed{72\text{km/h}, \quad 200\text{m}}}$$

Q13

$$\sqrt{(500-L)/20=12}$$

$$\sqrt{500-L=240}$$

$$\sqrt{\boxed{L=260\text{m}}}$$

Q14

Relative speed:

$$\sqrt{15+20=35\text{m/s}}$$

Combined lengths:

$$\sqrt{35(12)=420}$$

Each:

$$\sqrt{\boxed{210\text{m}}}$$

Q15

Required distance:

$$\sqrt{100+200+300=600}$$

Speed:

$$\sqrt{20\text{m/s}}$$

Time:

$$\sqrt{\boxed{30\text{s}}}$$

Q16

Difference between platform and pole crossing:

$$\sqrt{25-10=15\text{s}}$$

Platform length:

$$\sqrt{300\text{m}}$$

So speed:

$$\sqrt{300/15=20\text{m/s}}$$

Train length:

$$\sqrt{20(10)=200\text{m}}$$

$$\sqrt{\boxed{72\text{km/h}, 200\text{m}}}$$

Q17

Let train speed (v) , length (L) .

$$\sqrt{L=20(v-5)}$$

$$\sqrt{L=12(v+5)}$$

$$\sqrt{20v-100=12v+60}$$

$$\sqrt{8v=160}$$

$$\sqrt{v=20\text{m/s}}$$

$$\sqrt{L=20(15)=300\text{m}}$$

$$\sqrt{\boxed{72\text{km/h}, 300\text{m}}}$$

Q18

Train:

$$\sqrt[2]{250\text{m}}$$

Tunnel:

$$\sqrt[2]{750\text{m}}$$

Speed:

$$\sqrt[2]{90\text{km/h}=25\text{m/s}}$$

1. Whole train enters:

$$\sqrt[2]{250/25 = \boxed{10\text{s}}}$$

2. Engine reaches far end:

$$\sqrt[2]{750/25 = \boxed{30\text{s}}}$$

3. Entire train exits:

$$\sqrt[2]{1000/25 = \boxed{40\text{s}}}$$

4. Fully inside:

$$\sqrt[2]{(750-250)/25 = 500/25 = \boxed{20\text{s}}}$$

Q19

Speeds:

$$\sqrt[2]{54=15\text{m/s}}$$

$$\sqrt[2]{90=25\text{m/s}}$$

Relative:

$$\sqrt[2]{40\text{m/s}}$$

Distance:

$$\sqrt[2]{200+180+220 = 600\text{m}}$$

Time:

$$\sqrt[2]{600/40 = \boxed{15\text{s}}}$$

Q20

Fast:

$$\sqrt[2]{90\text{km/h}=25\text{m/s}}$$

Slow:

$$\sqrt[2]{54\text{km/h}=15\text{m/s}}$$

Relative:

$$\sqrt[2]{10\text{m/s}}$$

Initial gap:

$$\sqrt[2]{100\text{m}}$$

1. First contact:

$$\sqrt[2]{100/10 = \boxed{10\text{s}}}$$

2. Additional overtake distance:

$$\sqrt[2]{200+150=350\text{m}}$$

Additional time:

$$\lfloor 350/10 = \boxed{35s} \rfloor$$

3. Total:

$$\lfloor 10+35 = \boxed{45s} \rfloor$$

' CHAPTER 31 COMPLETE

We have now covered:

- train length fundamentals '
- pole/person crossing '
- moving-person crossing '
- platform/bridge/tunnel crossing '
- train fully entering/exiting tunnel '
- train completely inside tunnel '
- engine/rear distinction '
- two trains opposite directions '
- overtaking same direction '
- initial-gap crossing '
- head-start train problems '
- train vs runner equations '
- equal-train-length problems '
- two-platform shortcut '
- pole/platform time-ratio shortcut '
- unknown speed and length problems '
- advanced GATE/placement drills '

Progress

Completed: 31/76 Unit 1 — Quantitative Aptitude: 0=0% PROGRESS

NEXT

0=0% Chapter 32 — Boats and Streams

Next we will cover:

- boat speed in still water
- stream/current speed
- downstream and upstream
- $(b+s)$ and $(b-s)$
- finding still-water/current speed
- equal-distance upstream/downstream average speed
- time differences
- river crossing basics
- raft/current problems

- advanced GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 32 — BOATS AND STREAMS

PART 1 — STILL-WATER SPEED, STREAM SPEED, UPSTREAM/DOWNSTREAM, BASIC TIME & DISTANCE PROBLEMS

Status: $\emptyset$ = Not in PROGRESS GATE: Low–Medium Placements: Medium Priority: $\emptyset$ = Not in Top 3

Boats and Streams is simply **relative speed in water**.

The two speeds you must distinguish are:

- speed of boat in still water
- speed of stream/current

The entire chapter begins with:

$$\boxed{\text{Downstream speed} = b + s}$$

$$\boxed{\text{Upstream speed} = b - s}$$

where:

- b = boat speed in still water
- s = stream speed

1. WHAT IS STILL-WATER SPEED?

Suppose a boat's engine can move it at:

$$10 \text{ km/h}$$

when water is not flowing.

Then:

$$b = 10 \text{ km/h}$$

This is the boat's own speed relative to water.

2. WHAT DOES THE STREAM DO?

Suppose current flows at:

$$2 \text{ km/h}$$

Going with current

The current helps the boat.

$$\boxed{10+2=12}$$

So downstream speed:

$$\boxed{12 \text{ km/h}}$$

Going against current

The current opposes the boat.

$$\boxed{10-2=8}$$

So upstream speed:

$$\boxed{8 \text{ km/h}}$$

3. MASTER FORMULAS

Let:

$$b = \text{boat speed in still water}$$

$$s = \text{stream speed}$$

Then:

Downstream

$$\boxed{D=b+s}$$

Upstream

$$\boxed{U=b-s}$$

MUST MEMORIZE

4. FIND BOAT SPEED FROM UPSTREAM/DOWNSTREAM SPEEDS

Add:

$$D+U=(b+s)+(b-s)$$

$$D+U=2b$$

Therefore:

$$\boxed{b=\frac{D+U}{2}}$$

5. FIND STREAM SPEED

Subtract:

$$D-U=(b+s)-(b-s)$$

$$D-U=2s$$

Therefore:

$$\boxed{s=\frac{D-U}{2}}$$

6. EXAMPLE

Downstream speed:

$$\boxed{14 \text{ km/h}}$$

Upstream:

$$\boxed{10 \text{ km/h}}$$

Boat speed:

$$\boxed{\frac{14+10}{2} = 12 \text{ km/h}}$$

Stream:

$$\boxed{\frac{14-10}{2} = 2 \text{ km/h}}$$

7. MEMORY TRICK

Boat = Average

$$\boxed{b = \frac{D+U}{2}}$$

Stream = Half Difference

$$\boxed{s = \frac{D-U}{2}}$$

Think:

Boat = middle speed. Stream = deviation from middle.

8. BASIC DISTANCE PROBLEM

Boat speed in still water:

$$\boxed{12}$$

Stream:

$$\boxed{3}$$

Downstream:

$$\boxed{15}$$

Distance:

$$\boxed{60 \text{ km}}$$

Time:

$$\boxed{60/15 = 4 \text{ h}}$$

9. BASIC UPSTREAM PROBLEM

Same boat:

$$\boxed{b=12, \text{quad } s=3}$$

Upstream speed:

$$\boxed{9}$$

Distance:

$$\boxed{45 \text{ km}}$$

Time:

$$\boxed{45/9 = 5 \text{ h}}$$

10. SAME DISTANCE UPSTREAM AND DOWNSTREAM

If distance is d :

Downstream time:

$$\boxed{T_D = \frac{d}{b+s}}$$

Upstream time:

$$\boxed{T_U = \frac{d}{b-s}}$$

Since:

$$\boxed{b-s < b+s}$$

we always have:

$$\boxed{T_U > T_D}$$

provided $b > s$.

11. CRITICAL CONDITION

For a boat to move upstream:

$$\boxed{b > s}$$

If:

$$\boxed{b = s}$$

upstream ground speed:

$$\boxed{0}$$

The boat cannot make upstream progress.

If:

$$\boxed{b < s}$$

even while pointing upstream, the boat is carried downstream.

12. EXAMPLE — IMPOSSIBILITY CHECK

Boat speed:

$$\boxed{5 \text{ km/h}}$$

Current:

$$\sqrt[7]{\text{km/h}}$$

Upstream velocity:

$$\sqrt[5-7]{-2}$$

So it cannot actually travel upstream.

This is a useful sanity check.

13. FIND DISTANCE FROM TIME

Boat:

$$\sqrt[b]{10}$$

Stream:

$$\sqrt[s]{2}$$

Travels downstream for 3 h.

Speed:

$$\sqrt[12]$$

Distance:

$$\sqrt[12]{3} = \boxed{36 \text{ km}}$$

14. FIND DISTANCE UPSTREAM

Same boat.

Upstream speed:

$$\sqrt[8]$$

Travels 4 h.

Distance:

$$\sqrt[8]{4} = \boxed{32 \text{ km}}$$

15. EQUAL-DISTANCE ROUND TRIP

A boat travels distance $\sqrt[d]{}$ downstream and the same distance upstream.

Total distance:

$$\sqrt[2d]$$

Total time:

$$\sqrt[\frac{d}{b+s} + \frac{d}{b-s}]$$

Average speed is not $\sqrt[b]{}$.

16. AVERAGE SPEED FOR EQUAL UP/DOWN DISTANCE

Use harmonic mean of:

$$\sqrt{b+s}, \sqrt{b-s}$$

Therefore:

$$V_{\text{avg}} = \frac{2(b+s)(b-s)}{(b+s)+(b-s)}$$

Simplify:

$$\boxed{V_{\text{avg}} = \frac{b^2 - s^2}{b}}$$

UNDERSTAND AND DERIVE

You do not need to memorize this if comfortable with harmonic mean.

17. EXAMPLE

Boat:

$$b=10$$

Stream:

$$s=2$$

Downstream:

$$12$$

Upstream:

$$8$$

Equal-distance average:

$$\frac{2(12)(8)}{20}$$

$$= \boxed{9.6 \text{ km/h}}$$

Notice:

$$9.6 < 10$$

18. IMPORTANT RESULT

For equal upstream and downstream distances:

$$\boxed{V_{\text{avg}} < b}$$

whenever stream speed is positive.

Why?

You spend more time fighting the current than you save while going downstream.

19. COMMON TRAP

Do not say:

$$\sqrt{\frac{12+8}{2}}=10$$

for equal-distance trip.

That would be correct only for equal **times**.

For equal distances:

$$\boxed{\text{use harmonic mean}}$$

20. TIME DIFFERENCE FOR SAME DISTANCE

Suppose distance each way is (d) .

Then:

$$T_U - T_D = \frac{d}{b-s} - \frac{d}{b+s}$$

Taking LCM:

$$= d \frac{(b+s)-(b-s)}{b^2-s^2}$$

$$= \boxed{\frac{2ds}{b^2-s^2}}$$

Useful in advanced questions.

21. EXAMPLE

Boat:

$$b=10$$

Stream:

$$s=2$$

Distance:

$$48 \text{ km}$$

Upstream time:

$$48/8=6\text{h}$$

Downstream:

$$48/12=4\text{h}$$

Difference:

$$\boxed{2\text{h}}$$

Formula check:

$$\frac{2(48)(2)}{100-4} = \frac{192}{96} = 2$$

22. FIND DISTANCE FROM TIME DIFFERENCE

Boat:

$$b=10$$

Stream:

$$[s=2]$$

Upstream takes 2 h longer than downstream for same distance (d) .

$$[\frac{d}{8} - \frac{d}{12} = 2]$$

$$[d \left(\frac{3-2}{24} \right) = 2]$$

$$[\frac{d}{24} = 2]$$

$$[\boxed{d=48 \text{ km}}]$$

23. SPEED RATIO UPSTREAM : DOWNSTREAM

$$[U:D=(b-s):(b+s)]$$

Example:

$$[b=12, s=3]$$

$$[U:D=9:15]$$

$$[=\boxed{3:5}]$$

24. TIME RATIO FOR SAME DISTANCE

Since time is inverse to speed:

$$[\boxed{T_U:T_D=(b+s):(b-s)}]$$

For previous example:

$$[T_U:T_D=15:9]$$

$$[=\boxed{5:3}]$$

25. FIND BOAT/STREAM FROM SPEED RATIO

Suppose:

$$[D:U=5:3]$$

Then:

$$[b+s:b-s=5:3]$$

Let:

$$[b+s=5k]$$

$$[b-s=3k]$$

Add:

$$[2b=8k]$$

$$[b=4k]$$

Subtract:

$$2s = 2k$$

$$s = k$$

Therefore:

$$\boxed{b:s=4:1}$$

26. FAST RATIO FORMULA

If:

$$D:U=m:n$$

then:

$$\boxed{b:s=(m+n):(m-n)}$$

because:

$$b = \frac{D+U}{2}$$

$$s = \frac{D-U}{2}$$

27. EXAMPLE

Downstream : upstream =

$$7:5$$

Then:

$$b:s = (7+5):(7-5)$$

$$=12:2$$

$$=\boxed{6:1}$$

28. FIND DOWNSTREAM/UPSTREAM RATIO FROM $(b:s)$

Suppose:

$$b:s=5:1$$

Then:

$$D:U = (5+1):(5-1)$$

$$=6:4$$

$$=\boxed{3:2}$$

29. TRIP TAKES EQUAL TIMES, NOT DISTANCES

If boat spends equal time upstream and downstream:

Suppose time each way = (t) .

Distances:

$$D_U = (b-s)t$$

$$D_D = (b+s)t$$

So distance ratio:

$$D_U : D_D = (b-s) : (b+s)$$

Average speed over equal time is:

$$\frac{(b-s) + (b+s)}{2} = b$$

Important:

Equal time upstream/downstream !' average speed = still-water speed.

30. EXAMPLE

Boat:

$$b = 12, \quad s = 3$$

1 h upstream:

$$9 \text{ km}$$

1 h downstream:

$$15 \text{ km}$$

Total:

$$24 \text{ km in } 2 \text{ h}$$

Average:

$$12 \text{ km/h}$$

31. RAFT / FLOATING OBJECT

A raft has no own propulsion.

Therefore its ground speed is simply stream speed:

$$V_{\text{raft}} = s$$

This is a very common shortcut.

32. EXAMPLE — FIND CURRENT FROM RAFT

A raft covers:

$$18 \text{ km}$$

in:

$$6 \text{ h}$$

Current:

$$\boxed{18/6 = 3 \text{ km/h}}$$

33. FIND BOAT SPEED USING RAFT

Raft speed:

$$3$$

Boat downstream:

$$15$$

Since:

$$b+s=15$$

$$b=12$$

$$\boxed{12 \text{ km/h}}$$

Upstream speed:

$$12-3 = \boxed{9 \text{ km/h}}$$

34. FLOATING LOG

A floating log/driftwood is equivalent to raft:

$$\boxed{\text{speed of floating log} = \text{stream speed}}$$

assuming no other propulsion.

35. DOWNSTREAM TIME GIVEN

Boat speed in still water:

$$b=8$$

Stream:

$$s=2$$

Distance:

$$30$$

Downstream speed:

$$10$$

Time:

$$\boxed{3\text{h}}$$

36. UPSTREAM TIME GIVEN

Same boat.

Upstream:

$\sqrt{6}$

Distance:

$\sqrt{30}$

Time:

$\sqrt{\boxed{5h}}$

Round trip:

$\sqrt{\boxed{8h}}$

37. FIND STILL-WATER SPEED FROM TIMES

A boat covers 24 km downstream in 2 h.

So:

$\sqrt{D=12}$

It covers same 24 km upstream in 3 h.

So:

$\sqrt{U=8}$

Then:

$\sqrt{b=\frac{12+8}{2}=\boxed{10}}$

$\sqrt{s=\frac{12-8}{2}=\boxed{2}}$

38. FIND STREAM SPEED FROM TIMES

Same example:

$\sqrt{\boxed{s=2\text{ km/h}}}$

This pattern is extremely common.

39. WHEN DISTANCES DIFFER

Example:

30 km downstream in 2 h:

$\sqrt{D=15}$

20 km upstream in 2 h:

$\sqrt{U=10}$

Therefore:

$\sqrt{b=\frac{15+10}{2}=12.5}$

$\sqrt{s=\frac{15-10}{2}=2.5}$

$\sqrt{\boxed{b=12.5, s=2.5}}$

Equal distances are not required to use the speed-average formulas; you only need actual upstream/downstream speeds.

40. FIND TIME FOR ROUND TRIP

Boat:

$$b=15$$

Stream:

$$s=5$$

Distance each way:

$$40$$

Downstream speed:

$$20$$

Time:

$$2\text{ h}$$

Upstream speed:

$$10$$

Time:

$$4\text{ h}$$

Total:

$$\boxed{6\text{ h}}$$

41. FIND DISTANCE FROM TOTAL ROUND-TRIP TIME

Boat:

$$b=15$$

Stream:

$$s=5$$

Equal distance each way = d .

Total time = 6 h.

$$\frac{d}{20} + \frac{d}{10} = 6$$

$$\frac{d+2d}{20} = 6$$

$$3d = 120$$

$$\boxed{d=40\text{ km}}$$

42. FAST ROUND-TRIP DISTANCE FORMULA

For equal distances d , total time T :

$$T = \frac{d}{b+s} + \frac{d}{b-s}$$

$$= \frac{2bd}{b^2-s^2}$$

Thus:

$$\boxed{d = \frac{T(b^2-s^2)}{2b}}$$

Optional shortcut.

43. PERSON ROWS FOR FIXED TIME EACH WAY

A person rows:

- 2 h downstream
- 2 h upstream

Boat:

$$10$$

Stream:

$$2$$

Downstream distance:

$$12(2)=24$$

Upstream distance:

$$8(2)=16$$

Total distance:

$$40$$

Average speed:

$$40/4 = \boxed{10}$$

Again equal times !' average = (b) .

44. FIND CURRENT AS PERCENT OF BOAT SPEED

Boat:

$$b=12$$

Current:

$$s=3$$

Current as % of boat:

$$3/12 \times 100 = \boxed{25\%}$$

45. CURRENT INCREASE EFFECT

Boat:

$$b=12$$

Current rises from:

$$\sqrt[2]{4}$$

Downstream:

$$\sqrt[14]{16}$$

Upstream:

$$\sqrt[10]{8}$$

Notice:

Same stream increase helps downstream by the same absolute amount that it hurts upstream.

46. GATE-STYLE QUESTION 1

Boat speed = 10.

Stream = 2.

Find downstream/upstream speeds.

$$\boxed{12, 8}$$

47. GATE-STYLE QUESTION 2

Downstream = 16.

Upstream = 10.

Boat:

$$\frac{26}{2}=13$$

Stream:

$$\frac{6}{2}=3$$

$$\boxed{13, 3}$$

48. GATE-STYLE QUESTION 3

Boat 12, stream 3.

Travel 45 km upstream.

Speed:

$$9$$

Time:

$$\boxed{5h}$$

49. GATE-STYLE QUESTION 4

Downstream:upstream speed = 5:3.

Find boat:stream.

$$\backslash [(5+3):(5-3) = 8:2 \backslash]$$

$$\backslash [=\boxed{4:1} \backslash]$$

50. PLACEMENT-STYLE QUESTION 1

24 km downstream in 2 h.

24 km upstream in 3 h.

Speeds:

$$\backslash [12,\ 8 \backslash]$$

Boat:

$$\backslash [10 \backslash]$$

Stream:

$$\backslash [2 \backslash]$$

$$\backslash [\boxed{10,\ 2} \backslash]$$

51. PLACEMENT-STYLE QUESTION 2

Boat = 15.

Stream = 5.

40 km each way.

Total time:

$$\backslash [2+4 = \boxed{6h} \backslash]$$

52. PLACEMENT-STYLE QUESTION 3

Raft travels 21 km in 7 h.

Stream:

$$\backslash [\boxed{3\text{ km/h}} \backslash]$$

If motorboat downstream speed is 15:

Boat speed:

$$\backslash [15-3 = \boxed{12\text{ km/h}} \backslash]$$

53. PLACEMENT-STYLE QUESTION 4

Boat = 10.

Stream = 2.

Equal-distance round-trip average:

$$\left[\frac{2(12)(8)}{20} = \boxed{9.6 \text{ km/h}} \right]$$

54. ADVANCED QUESTION — FIND DISTANCE FROM TIME DIFFERENCE

Boat:

$$\left[12 \right]$$

Stream:

$$\left[3 \right]$$

Same distance upstream takes 2 h more than downstream.

Speeds:

$$\left[9, 15 \right]$$

Let distance = (d) .

$$\left[\frac{d}{9} - \frac{d}{15} = 2 \right]$$

LCM 45:

$$\left[\frac{5d - 3d}{45} = 2 \right]$$

$$\left[2d = 90 \right]$$

$$\left[\boxed{d = 45 \text{ km}} \right]$$

55. ADVANCED QUESTION — FIND STREAM

Boat still-water speed:

$$\left[10 \right]$$

For 24 km, upstream takes 1 h longer than downstream.

Let stream = (s) .

$$\left[\frac{24}{10-s} - \frac{24}{10+s} = 1 \right]$$

$$\left[24 \frac{2s}{100-s^2} = 1 \right]$$

$$\left[48s = 100 - s^2 \right]$$

$$\left[s^2 + 48s - 100 = 0 \right]$$

$$\left[(s+50)(s-2) = 0 \right]$$

Positive:

$$\left[\boxed{s = 2 \text{ km/h}} \right]$$

56. ADVANCED QUESTION — SPEED RATIO

Upstream : downstream time ratio for same distance:

$$\left[5:3 \right]$$

Then speed ratio is inverse:

$$U:D=3:5$$

So:

$$D:U=5:3$$

Hence:

$$b:s = 8:2 = \boxed{4:1}$$

57. COMMON TRAPS

TRAP 1 — Upstream = $(b+s)$

Wrong.

$$\boxed{U=b-s}$$

TRAP 2 — Downstream = $(b-s)$

Wrong.

$$\boxed{D=b+s}$$

TRAP 3 — Average upstream/downstream speeds

For equal distances, do not take arithmetic mean.

TRAP 4 — Boat slower than stream

Then genuine upstream movement is impossible.

TRAP 5 — Raft has boat speed

No. Raft speed = current speed.

TRAP 6 — Speed ratio and time ratio same

For same distance, they are inverses.

58. MUST-MEMORIZE FORMULAS

Downstream

$$\boxed{D=b+s}$$

Upstream

$$\boxed{U=b-s}$$

Still-water speed

$$\boxed{b=\frac{D+U}{2}}$$

Stream speed

$$\boxed{s = \frac{D-U}{2}}$$

59. HIGH-VALUE RATIO FORMULAS

If:

$$D:U = m:n$$

then:

$$\boxed{b:s = (m+n):(m-n)}$$

For same distance:

$$\boxed{T_U:T_D = D:U}$$

because time ratio is inverse of $(U:D)$.

60. AVERAGE SPEED FOR EQUAL DISTANCE

$$\boxed{V_{\text{avg}} = \frac{2DU}{D+U}}$$

or:

$$\boxed{V_{\text{avg}} = \frac{b^2 - s^2}{b}}$$

61. FAST EXAM DECISION TREE

Boat/current speeds given?

Immediately compute:

$$b+s, \quad b-s$$

Upstream/downstream speeds given?

Use half-sum and half-difference.

Same distance each way?

Use inverse-time logic.

Equal-distance average?

Use harmonic mean.

Raft/log?

Speed = stream speed.

Upstream impossible?

Check:

$$b > s$$

Time difference given?

Write:

$$\left[\frac{d_{b-s}}{d_{b+s}} \right]$$

PRACTICE — LEVEL 0

Q1

Boat speed 12 km/h, stream 3 km/h. Find upstream and downstream speeds.

Q2

Downstream 14 km/h, upstream 10 km/h. Find boat and stream speeds.

Q3

Boat 10, stream 2. Find time to travel 36 km downstream.

Q4

Boat 10, stream 2. Find time to travel 32 km upstream.

Q5

A raft travels 15 km in 5 h. Find stream speed.

LEVEL 1

Q6

Boat 15, stream 5. Find round-trip time for 40 km each way.

Q7

24 km downstream takes 2 h and upstream takes 3 h. Find boat and stream speeds.

Q8

Downstream:upstream speed = 5:3. Find boat:stream.

Q9

Boat 12, stream 3. Same distance upstream takes 2 h longer than downstream. Find distance.

Q10

Boat 10, stream 2. Find equal-distance round-trip average speed.

LEVEL 2

Q11

Downstream speed = 18, upstream = 12. Find boat, stream and $(D:U)$.

Q12

Boat:stream = 5:1. Find downstream:upstream speed ratio.

Q13

Same distance takes 5 h upstream and 3 h downstream. Find boat:stream ratio.

Q14

A raft travels 24 km in 6 h. A motorboat goes downstream at 16 km/h. Find still-water and upstream speed.

Q15

Boat speed = 10 km/h. A 24 km upstream journey takes 1 h more than downstream. Find current speed.

LEVEL 3 — EXAM PRESSURE**Q16**

A boat travels 60 km downstream and 40 km upstream in a total of 7 h. Boat speed in still water is 12 km/h. Find stream speed.

Q17

A boat covers equal distances upstream and downstream. Upstream speed is 8 km/h and overall average is 9.6 km/h. Find downstream speed.

Q18

Downstream:upstream time for equal distances = 2:3. Find boat:stream speed ratio.

Q19

A boat's still-water speed is 15 km/h and current is 5 km/h. How far can it go upstream and return in 6 h?

Q20

A person rows for 2 h downstream and 2 h upstream. Boat speed is 14, stream speed is 4. Find total distance and average speed.

ANSWERS**Q1**

$$\backslash D=12+3=15 \backslash$$

$$\backslash U=12-3=9 \backslash$$

$$\backslash \boxed{15, 9} \backslash$$

Q2

$$\backslash b=\frac{14+10}{2}=\boxed{12} \backslash$$

$$\backslash s=\frac{14-10}{2}=\boxed{2} \backslash$$

Q3

Downstream:

$$\backslash 12 \backslash$$

$$\backslash 36/12=\boxed{3h} \backslash$$

Q4

Upstream:

$$\sqrt[3]{8}$$

$$\sqrt[3]{32/8} = \boxed{4h}$$

Q5

$$\sqrt[3]{15/5} = \boxed{3\text{ km/h}}$$

Q6

Speeds:

$$\sqrt[3]{20, \sqrt[3]{10}}$$

Times:

$$\sqrt[3]{40/20=2}$$

$$\sqrt[3]{40/10=4}$$

Total:

$$\sqrt[3]{\boxed{6h}}$$

Q7

Downstream:

$$\sqrt[3]{24/2=12}$$

Upstream:

$$\sqrt[3]{24/3=8}$$

$$\sqrt[3]{b=\frac{12+8}{2}=10}$$

$$\sqrt[3]{s=\frac{12-8}{2}=2}$$

$$\sqrt[3]{\boxed{10, \sqrt[3]{2}}}$$

Q8

$$\sqrt[3]{b:s=(5+3):(5-3)}$$

$$\sqrt[3]{=8:2}$$

$$\sqrt[3]{\boxed{4:1}}$$

Q9

Speeds:

$$\sqrt[3]{15, \sqrt[3]{9}}$$

$$\sqrt[3]{\frac{d}{9}-\frac{d}{15}=2}$$

$$\sqrt[3]{\frac{2d}{45}=2}$$

$$\sqrt[3]{\boxed{d=45\text{ km}}}$$

Q10

$$\sqrt[3]{\frac{2(12)(8)}{20}} = \boxed{9.6\text{ km/h}}$$

Q11

$$\left[b = \frac{18+12}{2} = 15 \right]$$

$$\left[s = 3 \right]$$

$$\left[D:U = 18:12 = 3:2 \right]$$

$$\left[\boxed{15, \ 3, \ 3:2} \right]$$

Q12

$$\left[D:U = (5+1):(5-1) \right]$$

$$\left[= 6:4 \right]$$

$$\left[= \boxed{3:2} \right]$$

Q13

For same distance:

$$\left[U:D = 3:5 \right]$$

So:

$$\left[D:U = 5:3 \right]$$

Therefore:

$$\left[b:s = (5+3):(5-3) \right]$$

$$\left[= \boxed{4:1} \right]$$

Q14

Raft:

$$\left[s = 24/6 = 4 \right]$$

Downstream:

$$\left[b+s = 16 \right]$$

$$\left[b = 12 \right]$$

Upstream:

$$\left[12-4 = 8 \right]$$

$$\left[\boxed{b=12, \ U=8} \right]$$

Q15

$$\left[\frac{24}{10-s} - \frac{24}{10+s} = 1 \right]$$

$$\left[\frac{48s}{100-s^2} = 1 \right]$$

$$\left[s^2 + 48s - 100 = 0 \right]$$

Positive root:

$$\left[\boxed{s = 2 \text{ km/h}} \right]$$

Q16

Let stream speed = (s) .

$$\left[\frac{60}{12+s} + \frac{40}{12-s} = 7 \right]$$

Multiply:

$$60(12-s)+40(12+s) = 7(144-s^2)$$

$$720-60s+480+40s = 1008-7s^2$$

$$1200-20s=1008-7s^2$$

$$7s^2-20s+192=0$$

This has no real solution because:

$$(-20)^2-4(7)(192)<0$$

So the given data are inconsistent.

$\boxed{\text{No physically valid stream speed}}$

This is a valuable **data sanity-check** question.

Q17

Equal-distance average:

$$9.6 = \frac{2(8)D}{8+D}$$

$$9.6(8+D)=16D$$

$$76.8+9.6D=16D$$

$$6.4D=76.8$$

$$\boxed{D=12 \text{ km/h}}$$

Q18

Same distance:

$$T_D:T_U=2:3$$

Therefore speed ratio:

$$D:U=3:2$$

So:

$$b:s=(3+2):(3-2)$$

$$=\boxed{5:1}$$

Q19

Upstream speed:

$$10$$

Downstream:

$$20$$

Let one-way distance = d .

$$\frac{d}{10} + \frac{d}{20} = 6$$

$$\frac{3d}{20} = 6$$

$$d=40$$

So it can go:

$$\boxed{40 \text{ km upstream}}$$

and return 40 km.

Total distance:

$$\boxed{80 \text{ km}}$$

Q20

Downstream:

$$14 + 4 = 18$$

Distance in 2 h:

$$36$$

Upstream:

$$14 - 4 = 10$$

Distance in 2 h:

$$20$$

Total distance:

$$56$$

Total time:

$$4$$

Average:

$$\boxed{14 \text{ km/h}}$$

CHAPTER 32 STATUS

Covered so far:

- boat speed in still water '
- stream/current speed '
- downstream and upstream formulas '
- half-sum/half-difference method '
- distance/time calculations '
- equal-distance round trips '
- harmonic-mean average speed '
- upstream/downstream time ratios '
- speed-ratio shortcuts '
- raft/log current-speed problems '
- time-difference equations '
- impossibility checks '
- GATE/placement practice '

But **Chapter 32 is not complete yet.**

Next we still need:

- crossing a river rather than travelling along it
- shortest-time crossing
- shortest-distance crossing
- drift due to current
- aiming upstream to reach opposite point
- width of river
- swimmer/boat resultant velocity
- current-angle problems
- advanced raft and return problems
- final shortcuts and chapter audit

Progress

Completed: 31/76 **Current:** 0=Chapter 32 — Boats and Streams **Next:** Chapter 32 Part 2 — River crossing, drift, shortest-time/shortest-distance & advanced boat problems

USER

next

ASSISTANT

CHAPTER 32 — BOATS AND STREAMS

PART 2 — RIVER CROSSING, DRIFT, SHORTEST TIME, SHORTEST PATH & ADVANCED BOAT PROBLEMS

Status: 0=100% **PROGRESS GATE:** Low–Medium **Placements:** Medium **Priority:** 0=0/3

Part 1 dealt with moving **along the river**:

$$\boxed{D=b+s, \text{ } \Uparrow U=b-s}$$

Now we study moving **across the river**.

This introduces two perpendicular motions:

- boat/swimmer moves across the river
- current pushes downstream

This is a basic **vector + relative-speed** problem.

1. RIVER-CROSSING SETUP

Suppose:

- river width = (W)
- boat speed relative to water = (b)
- stream speed = (s)

Imagine the river flowing horizontally.

If the boat is pointed directly across:

- boat gives across-river velocity $v(b)$
- current gives downstream velocity $v(s)$

These occur simultaneously.

2. POINTING DIRECTLY ACROSS

If the boat points perpendicular to the bank:

$$v(\boxed{\text{Across speed}}=b)$$

Therefore crossing time:

$$v(\boxed{T=\frac{W}{b}})$$

Notice:

The stream does not change the time to reach the opposite bank when the boat is aimed exactly perpendicular to the bank.

It only causes drift downstream.

3. DOWNSTREAM DRIFT

During crossing:

$$v(T=\frac{W}{b})$$

Current moves the boat downstream at speed $v(s)$.

Therefore drift:

$$v(\boxed{d=sT})$$

So:

$$v(\boxed{d=\frac{sW}{b}})$$

MUST UNDERSTAND

4. EXAMPLE

River width:

$$v(300\text{m})$$

Boat speed:

$$v(5\text{m/s})$$

Current:

$$v(2\text{m/s})$$

Crossing time:

$$v(300/5 = \boxed{60\text{s}})$$

Drift:

$$\sqrt{2}(60) = \boxed{120\text{m}}$$

So the boat reaches the opposite bank 120 m downstream.

5. RESULTANT GROUND SPEED

Boat velocity and stream velocity are perpendicular.

Therefore magnitude of resultant velocity:

$$\sqrt{\boxed{v_R = \sqrt{b^2 + s^2}}}$$

In previous example:

$$\sqrt{5^2 + 2^2} = \sqrt{29}$$

$$\approx \boxed{5.39\text{m/s}}$$

6. IMPORTANT TRAP

Do not use:

$$W/\sqrt{b^2 + s^2}$$

for crossing time when aimed directly across.

Why?

Because only the **perpendicular component** moves the boat across the width.

That component is:

$$b$$

Therefore:

$$\boxed{T = W/b}$$

7. ACTUAL PATHLENGTH

If boat drifts downstream by (d) , actual path is diagonal.

By Pythagoras:

$$\boxed{L = \sqrt{W^2 + d^2}}$$

Using:

$$W = 300, \quad d = 120$$

$$L = \sqrt{300^2 + 120^2}$$

$$= \sqrt{104400}$$

$$\approx \boxed{323.1\text{m}}$$

8. SHORTEST-TIME CROSSING

To cross in minimum possible time, maximize velocity component perpendicular to the bank.

That happens when boat points directly across.

Therefore:

$$\boxed{T_{\min} = \frac{W}{b}}$$

This causes downstream drift.

MEMORY TRICK

Shortest time !' point straight across !' accept drift.

9. SHORTEST-DISTANCECROSSING

Shortest possible ground path between opposite points is a straight line perpendicular to the banks.

For this, the boat must aim **upstream** so its upstream velocity component cancels the stream.

Then net downstream velocity becomes zero.

MEMORY TRICK

Shortest distance !' aim upstream !' cancel current.

10. VELOCITY COMPONENTS FOR ZERO DRIFT

Suppose boat is aimed upstream at angle θ from the straight-across direction.

Its components are:

Across:

$$b \cos \theta$$

Upstream:

$$b \sin \theta$$

Current pushes downstream at:

$$s$$

For no drift:

$$b \sin \theta = s$$

Therefore:

$$\boxed{\sin \theta = \frac{s}{b}}$$

This requires:

$$\boxed{b > s}$$

11. EXAMPLE — AIMING UPSTREAM

Boat speed:

$$\sqrt{5\text{m/s}}$$

Current:

$$\sqrt{3\text{m/s}}$$

For zero drift:

$$\sqrt{5\sin\theta=3}$$

$$\sqrt{\sin\theta=\frac{3}{5}}$$

Thus:

$$\sqrt{\theta\approx 36.87^\circ}$$

upstream from straight across.

12. ACROSS SPEED FOR SHORTEST DISTANCE

Under zero-drift condition:

$$\sqrt{b\sin\theta=s}$$

Across component:

$$\sqrt{b\cos\theta}$$

Using:

$$\sqrt{\cos\theta = \sqrt{1-\frac{s^2}{b^2}}}$$

So:

$$\sqrt{b\cos\theta = \sqrt{b^2-s^2}}$$

Therefore:

$$\sqrt{\boxed{v_{\text{across, no drift}} = \sqrt{b^2-s^2}}}$$

13. TIME FOR SHORTEST-DISTANCE CROSSING

Thus:

$$\sqrt{\boxed{T_{\text{shortest distance}} = \frac{W}{\sqrt{b^2-s^2}}}}$$

This is greater than shortest-time crossing:

$$\sqrt{\frac{W}{b}}$$

because:

$$\sqrt{\sqrt{b^2-s^2} < b}$$

14. EXAMPLE

River width:

$$\sqrt{240\text{m}}$$

Boat speed:

$$\sqrt{5\text{m/s}}$$

Current:

$$\sqrt{3\text{m/s}}$$

No-drift across speed:

$$\sqrt{25-9} = 4\text{m/s}$$

Time:

$$240/4 = \boxed{60\text{s}}$$

If boat had pointed straight across:

$$240/5 = 48\text{s}$$

So:

- shortest time = 48 s
- shortest distance = 60 s

15. SHORTEST TIME VS SHORTEST DISTANCE

Do not confuse them.

Shortest time

Boat aimed perpendicular to bank:

$$\boxed{T=W/b}$$

Drift occurs.

Shortest distance

Boat aimed upstream:

$$\boxed{T=W/\sqrt{b^2-s^2}}$$

No drift.

16. FIND STREAM SPEED FROM DRIFT

Boat points directly across.

Given:

$$d=\frac{sW}{b}$$

Therefore:

$$\boxed{s=\frac{db}{W}}$$

17. EXAMPLE

River width:

$$\sqrt{200\text{m}}$$

Boat speed:

$$\sqrt{4\text{m/s}}$$

Drift:

$$\sqrt{150\text{m}}$$

Then:

$$s = \frac{150(4)}{200}$$

$$s = \boxed{3\text{m/s}}$$

18. FIND BOAT SPEED FROM DRIFT

From:

$$d = \frac{sW}{b}$$

$$\boxed{b = \frac{sW}{d}}$$

Example:

$$W = 300\text{m}, \quad s = 2\text{m/s}, \quad d = 120\text{m}$$

$$b = \frac{2(300)}{120} = \boxed{5\text{m/s}}$$

19. FIND RIVER WIDTH

$$d = \frac{sW}{b}$$

Thus:

$$\boxed{W = \frac{db}{s}}$$

20. AIMING AT AN ANGLE — GENERAL CASE

Suppose boat is aimed at angle θ upstream from the straight-across direction.

Boat components:

Across:

$$b \cos \theta$$

Upstream:

$$b \sin \theta$$

Current downstream:

$$s$$

Net downstream component:

$$\boxed{s - b \sin \theta}$$

Across component:

$$\boxed{b \cos \theta}$$

21. GENERAL CROSSING TIME

Across distance is W .

So:

$$\boxed{T = \frac{W}{b \cos \theta}}$$

22. GENERAL DRIFT

Net downstream velocity:

$$s - b \sin \theta$$

Thus drift:

$$\boxed{d = \left(s - b \sin \theta \right) \frac{W}{b \cos \theta}}$$

If negative, landing point is upstream of the directly opposite point.

23. ZERO-DRIFT CASE CHECK

Set:

$$s - b \sin \theta = 0$$

Then:

$$\sin \theta = s/b$$

exactly as derived earlier.

24. EXAMPLE — PARTIAL CANCELLATION

Boat:

$$10 \text{ m/s}$$

Current:

$$4 \text{ m/s}$$

Boat aims upstream such that upstream component is:

$$2 \text{ m/s}$$

Then net downstream component:

$$4 - 2 = 2 \text{ m/s}$$

If across component is:

$$8 \text{ m/s}$$

and width:

$$400 \text{ m}$$

Time:

$$\sqrt{400/8}=50s$$

Drift:

$$\sqrt{2(50)} = \boxed{100m}$$

25. RIVER CROSSING USING 3–4–5 TRIANGLE

These questions often use friendly triangles.

Suppose:

$$\sqrt{b=5, \text{quad } s=3}$$

For no drift:

Upstream component:

$$\sqrt{3}$$

Across component:

$$\sqrt{4}$$

because:

$$\sqrt{3^2+4^2=5^2}$$

So immediately:

$$\sqrt{\boxed{\text{across speed}=4}}$$

Excellent shortcut.

26. ANOTHER COMMON TRIPLE

If:

$$\sqrt{b=13, \text{quad } s=5}$$

No-drift across speed:

$$\sqrt{\sqrt{169-25}} = 12$$

So use:

$$\sqrt{\boxed{5-12-13}}$$

27. AND ANOTHER

$$\sqrt{b=17, \text{quad } s=8}$$

Across:

$$\sqrt{15}$$

using:

$$\sqrt{\boxed{8-15-17}}$$

28. RAFT AND RETURN PROBLEM

A person drops a floating object into the river and later turns back to retrieve it.

The floating object moves with current speed.

A powerful principle is:

Work in the water's frame.

Relative to the water:

- floating object is stationary
- boat moves at still-water speed $v(b)$

This often makes the problem much easier.

29. EXAMPLE — OBJECT DROPPED FROM BOAT

A boat travels upstream at speed relative to water:

$$v_b = 8$$

Current:

$$v_s = 2$$

An object falls into water.

Boat continues upstream for 30 min, then turns and moves downstream at the same still-water speed.

How long after turning until it catches the object?

Use water frame.

During first 30 min, boat moves away from object at:

$$8 \text{ km/h}$$

Separation:

$$8(0.5) = 4 \text{ km}$$

After turning, relative speed between boat and object in water frame is again:

$$8$$

Catch time:

$$4/8 = \boxed{0.5 \text{ h}}$$

So catch occurs 30 min after turning.

30. IMPORTANT RESULT

If:

- boat speed relative to water is unchanged,

- object simply floats with stream,
- boat continues for time t before turning,

then it takes the same time t after turning to catch the object.

$$t_{\text{return catch}} = t_{\text{before turn}}$$

This is independent of stream speed.

31. WHY STREAM CANCELS

Because both boat and floating object are affected by the same current.

Relative motion between them depends only on boat's velocity relative to water.

This is a high-value concept.

32. FIND STREAM FROM FLOATING OBJECT DISTANCE

Suppose object floats for total time T and travels d .

Then:

$$s = d/T$$

Example:

Object drifts:

$$6 \text{ km}$$

in:

$$2 \text{ h}$$

$$s = 3 \text{ km/h}$$

33. BOAT GOES UPSTREAM THEN RETURNS

Boat speed:

$$b = 10$$

Stream:

$$s = 2$$

It travels upstream for:

$$3 \text{ h}$$

Upstream distance:

$$(10-2)(3) = 24$$

Return downstream speed:

$$12$$

Return time:

$$24/12 = 2 \text{ h}$$

Total:

$$\boxed{5h}$$

34. DOWNSTREAM THEN RETURN

Same boat.

Downstream for:

$$\boxed{3h}$$

Distance:

$$\boxed{12(3)=36}$$

Return upstream at:

$$\boxed{8}$$

Time:

$$\boxed{36/8 = 4.5h}$$

Total:

$$\boxed{7.5h}$$

So direction of first leg matters.

35. FIND RETURN TIME RATIO

If equal distance is travelled downstream then upstream:

$$\boxed{T_D:T_U = U:D}$$

Example:

$$\boxed{D=12, U=8}$$

$$\boxed{T_D:T_U=8:12 = \boxed{2:3}}$$

36. “CAN GO x KM DOWNSTREAM AND RETURN IN T HOURS”

Let one-way distance be (d) .

Then:

$$\boxed{\frac{d}{b+s} + \frac{d}{b-s} = T}$$

Solve for (d) .

37. EXAMPLE

Boat:

$$\boxed{b=10}$$

Stream:

$$\sqrt{s=2}$$

Total available time:

$$\sqrt{5h}$$

$$\sqrt{\frac{d}{12} + \frac{d}{8} = 5}$$

LCM 24:

$$\sqrt{\frac{2d+3d}{24} = 5}$$

$$\sqrt{5d=120}$$

$$\sqrt{\boxed{d=24\text{km}}}$$

So total travelled:

$$\sqrt{48\text{km}}$$

38. ADVANCED TIME-DIFFERENCE SHORTCUT

For same distance $\sqrt{d}$:

$$\sqrt{T_U - T_D = \frac{2ds}{b^2 - s^2}}$$

If $\sqrt{(b,s)}$ are easy, this avoids writing two separate fractions.

39. FIND STREAM FROM SPEED DIFFERENCE

Downstream speed exceeds upstream speed by:

$$\sqrt{6\text{ km/h}}$$

Since:

$$\sqrt{D-U=2s}$$

$$\sqrt{2s=6}$$

$$\sqrt{\boxed{s=3\text{ km/h}}}$$

Very fast.

40. FIND BOAT SPEED FROM SPEED SUM

If:

$$\sqrt{D+U=30}$$

then:

$$\sqrt{2b=30}$$

$$\sqrt{\boxed{b=15}}$$

41. PERCENTAGE-BASED STREAM QUESTION

Current is 20% of boat speed.

Let:

$$[b=100]$$

Then:

$$[s=20]$$

Downstream:

$$[120]$$

Upstream:

$$[80]$$

Ratio:

$$[\boxed{3:2}]$$

42. GENERAL IF STREAM = p% OF BOAT SPEED

Let:

$$[s=pb]$$

where (p) is decimal.

Then:

$$[D=b(1+p)]$$

$$[U=b(1-p)]$$

So:

$$[\boxed{D:U=(1+p):(1-p)}]$$

43. EXAMPLE

Stream = 25% of still-water speed.

$$[p=0.25]$$

$$[D:U=1.25:0.75]$$

$$[=5:3]$$

$$[\boxed{5:3}]$$

44. GATE-STYLE QUESTION 1

River width:

$$[200\text{m}]$$

Boat:

$$[4\text{m/s}]$$

Current:

$$\sqrt{3\text{m/s}}$$

Boat points straight across.

Crossing time:

$$\sqrt{200/4 = \boxed{50\text{s}}}$$

Drift:

$$\sqrt{3(50) = \boxed{150\text{m}}}$$

45. GATE-STYLE QUESTION 2

Boat:

$$\sqrt{5\text{m/s}}$$

Current:

$$\sqrt{3\text{m/s}}$$

No-drift across speed:

$$\sqrt{\sqrt{25-9} = \boxed{4\text{m/s}}}$$

46. GATE-STYLE QUESTION 3

Width:

$$\sqrt{240\text{m}}$$

Using previous no-drift across speed 4:

$$\sqrt{T=240/4 = \boxed{60\text{s}}}$$

47. GATE-STYLE QUESTION 4

Boat points straight across.

Width 300 m, boat speed 5 m/s, drift 120 m.

Current:

$$\sqrt{s=\frac{120(5)}{300} = \boxed{2\text{m/s}}}$$

48. PLACEMENT-STYLE QUESTION 1

Boat travels upstream for 2 h at:

$$\sqrt{b=12,\text{quad } s=3}$$

Distance:

$$\sqrt{9(2)=18\text{km}}$$

Return downstream:

$$\sqrt{15}$$

Return time:

$$\boxed{18/15 = 1.2\text{h}}$$

49. PLACEMENT-STYLE QUESTION 2

Boat travels downstream for 2 h with:

$$b=12, \text{quad } s=3$$

Distance:

$$15(2)=30$$

Return upstream:

$$9$$

Time:

$$30/9 = \boxed{3\frac{1}{3}\text{h}}$$

50. PLACEMENT-STYLE QUESTION 3

A floating object is dropped.

Boat continues for 40 min and then turns back, maintaining same still-water speed.

Catch time after turning:

$$\boxed{40\text{ min}}$$

assuming immediate turn and unchanged still-water speed.

51. PLACEMENT-STYLE QUESTION 4

Current is 25% of boat's still-water speed.

Downstream : upstream:

$$1.25:0.75 = \boxed{5:3}$$

52. ADVANCED QUESTION — AIM ANGLE

Boat speed:

$$10\text{m/s}$$

Current:

$$6\text{m/s}$$

For zero drift:

$$\sin\theta = \frac{6}{10} = \frac{3}{5}$$

$$\boxed{\theta \approx 36.87^\circ}$$

upstream from the perpendicular-to-bank direction.

Across speed:

$$\sqrt{100-36} = 8\text{m/s}$$

53. ADVANCED QUESTION — NO-DRIFT CROSSING TIME

Width:

$$400\text{m}$$

Boat:

$$10\text{m/s}$$

Current:

$$6\text{m/s}$$

Across speed:

$$8\text{m/s}$$

Time:

$$400/8 = \boxed{50\text{s}}$$

54. ADVANCED QUESTION — SHORTEST TIME

Same river.

If pointed directly across:

$$T_{\min} = 400/10 = \boxed{40\text{s}}$$

Drift:

$$6(40) = \boxed{240\text{m}}$$

55. ADVANCED QUESTION — COMPARE TWO STRATEGIES

Using previous data:

Directly across

Time:

$$40\text{s}$$

Path length:

$$\sqrt{400^2 + 240^2}$$

$$\approx 466.5\text{m}$$

Zero drift

Time:

$$50\text{s}$$

Path length on ground:

$\sqrt{400\text{m}}$

So:

- direct across = shorter time
- aim upstream = shorter ground distance

Exactly as expected.

56. COMMON TRAPS

& pTRAP 1 — Stream affects across component when aimed straight across

No. Across component remains $\sqrt{b}$.

& pTRAP 2 — Shortest time means no drift

Wrong.

Shortest time usually causes drift.

& pTRAP 3 — Shortest distance means point straight across

Wrong.

Need to aim upstream.

& pTRAP 4 — Use $\sqrt{b+s}$ for river width crossing

Not unless boat and current are collinear.

& pTRAP 5 — Raft has its own speed

No.

$\sqrt{V_{\text{raft}}=s}$

& pTRAP 6 — No-drift crossing possible for any current

Only if:

$\sqrt{b>s}$

57. MUST-MEMORIZE RIVER-CROSSING FORMULAS

Point straight across

$\sqrt{\boxed{T=\frac{W}{b}}}$

Drift

$\sqrt{\boxed{d=\frac{sW}{b}}}$

Resultant ground speed

$\sqrt{\boxed{v_R=\sqrt{b^2+s^2}}}$

58. ZERO-DRIFT / SHORTEST-DISTANCE FORMULAS

$$\boxed{\sin\theta = \frac{s}{b}}$$

$$\boxed{v_{\text{across}} = \sqrt{b^2 - s^2}}$$

$$\boxed{T = \frac{W}{\sqrt{b^2 - s^2}}}$$

59. FULL CHAPTER FORMULA SHEET

Along river

$$\boxed{D = b + s}$$

$$\boxed{U = b - s}$$

$$\boxed{b = \frac{D + U}{2}}$$

$$\boxed{s = \frac{D - U}{2}}$$

Equal-distance average

$$\boxed{V_{\text{avg}} = \frac{2DU}{D + U}}$$

Raft

$$\boxed{V_{\text{raft}} = s}$$

Straight-across drift

$$\boxed{d = sW/b}$$

60. FAST EXAM DECISION TREE

Along river?

Use:

$$\boxed{b + s, \quad b - s}$$

Across river?

Think vectors/components.

Minimum time?

Point directly across.

Minimum distance / directly opposite landing?

Aim upstream to cancel current.

Floating object?

It moves with current.

Object dropped from moving boat?

Consider water frame.

Given (D, U) ?

Half-sum and half-difference immediately.

PRACTICE — LEVEL 0

Q1

River width 300 m, boat speed 5 m/s. Boat points directly across. Find crossing time.

Q2

Same data, current 2 m/s. Find drift.

Q3

Boat speed 5 m/s and current 3 m/s. Find zero-drift across speed.

Q4

Width 240 m, zero-drift across speed 4 m/s. Find crossing time.

Q5

A raft covers 20 km in 5 h. Find stream speed.

LEVEL 1

Q6

Width 200 m, boat 4 m/s, current 3 m/s. Boat points directly across. Find time and drift.

Q7

Boat 10 m/s, current 6 m/s. Find angle condition for zero drift and across speed.

Q8

Boat 12 km/h, current 3 km/h. It goes upstream for 2 h and returns. Find total trip time.

Q9

Current is 20% of still-water boat speed. Find downstream:upstream speed ratio.

Q10

A floating object is dropped. Boat continues for 30 min before turning. How long after turning to catch it?

LEVEL 2

Q11

River width 400 m, boat 10 m/s, current 6 m/s. Find:

1. shortest-time crossing,
 2. drift,
 3. no-drift crossing time.
-

Q12

Boat points directly across a 300 m river and drifts 180 m. Boat speed is 5 m/s. Find stream speed.

Q13

Boat speed 13 m/s, current 5 m/s. Find no-drift across speed.

Q14

Boat speed 17 m/s, current 8 m/s. River width 450 m. Find zero-drift crossing time.

Q15

Current is 25% of boat speed. Find downstream:upstream ratio.

LEVEL 3 — EXAM PRESSURE

Q16

A river is 360 m wide. Boat speed relative to water is 10 m/s, current 6 m/s. Find:

1. minimum crossing time,
2. drift in that case,
3. time to land directly opposite,
4. upstream aiming angle for direct-opposite landing.

Q17

A boat points straight across a river. It takes 50 s to cross and drifts 150 m. If width is 200 m, find boat speed and stream speed.

Q18

Boat speed 10 km/h, stream 2 km/h. It goes upstream for 3 h and then returns to starting point. Find total elapsed time.

Q19

Boat speed 12 km/h, stream 3 km/h. It goes downstream for 2 h and returns. Find total elapsed time.

Q20

A floating bottle falls from a boat. The boat continues upstream for 45 min before turning back with unchanged still-water speed. Find catch time after turning.

ANSWERS

Q1

$$\sqrt{300/5} = \boxed{60\text{s}}$$

Q2

$$\sqrt{2(60)} = \boxed{120\text{m}}$$

Q3

$$\sqrt{\sqrt{25-9}} = \boxed{4\text{m/s}}$$

Q4

$$\sqrt[2]{240/4 = \boxed{60\text{s}}}$$

Q5

$$\sqrt[2]{20/5 = \boxed{4\text{ km/h}}}$$

Q6

Time:

$$\sqrt[2]{200/4 = 50\text{s}}$$

Drift:

$$\sqrt[2]{3(50)=150\text{m}}$$

$$\sqrt[2]{\boxed{50\text{s}, 150\text{m}}}$$

Q7

$$\sqrt[2]{\sin\theta=6/10=3/5}$$

$$\sqrt[2]{\theta\approx 36.87^\circ}$$

Across:

$$\sqrt[2]{\sqrt{100-36}=8}$$

$$\sqrt[2]{\boxed{36.87^\circ, 8\text{m/s}}}$$

Q8

Upstream speed:

$$\sqrt[2]{12-3=9}$$

Distance in 2 h:

$$\sqrt[2]{18}$$

Downstream:

$$\sqrt[2]{15}$$

Return:

$$\sqrt[2]{18/15=1.2\text{h}}$$

Total:

$$\sqrt[2]{\boxed{3.2\text{h}}}$$

Q9

Let boat = 100, current = 20.

$$\sqrt[2]{D:U=120:80 = \boxed{3:2}}$$

Q10

$$\sqrt[2]{\boxed{30\text{ min}}}$$

Q11

Shortest-time crossing:

$$\sqrt[2]{400/10 = 40\text{s}}$$

Drift:

$$\sqrt{6(40)} = 240\text{m}$$

No-drift across speed:

$$\sqrt{100-36}=8$$

Time:

$$400/8 = 50\text{s}$$

$$\boxed{40\text{s}, 240\text{m}, 50\text{s}}$$

Q12

Crossing time:

$$300/5=60\text{s}$$

Current:

$$180/60 = \boxed{3\text{m/s}}$$

Q13

$$\sqrt{13^2-5^2} = \sqrt{144} = \boxed{12\text{m/s}}$$

Q14

Across speed:

$$\sqrt{17^2-8^2} = 15\text{m/s}$$

Time:

$$450/15 = \boxed{30\text{s}}$$

Q15

$$1.25:0.75 = \boxed{5:3}$$

Q16

Shortest time:

$$360/10 = 36\text{s}$$

Drift:

$$\sqrt{6(36)} = 216\text{m}$$

No-drift across speed:

$$\sqrt{100-36}=8$$

No-drift time:

$$360/8 = 45\text{s}$$

Angle:

$$\sin\theta=6/10$$

$$\theta\approx 36.87^\circ$$

$$\boxed{36\text{s}, 216\text{m}, 45\text{s}, 36.87^\circ}$$

Q17

Crossing time:

$$\sqrt[4]{50\text{s}}$$

Boat speed:

$$\sqrt[4]{200/50 = 4\text{m/s}}$$

Stream:

$$\sqrt[4]{150/50 = 3\text{m/s}}$$

$$\sqrt[\boxed{4\text{m/s}, 3\text{m/s}}]$$

Q18

Upstream:

$$\sqrt[4]{10-2=8}$$

Distance in 3 h:

$$\sqrt[4]{24\text{km}}$$

Return downstream:

$$\sqrt[4]{12}$$

Time:

$$\sqrt[4]{24/12=2\text{h}}$$

Total:

$$\sqrt[\boxed{5\text{h}}]$$

Q19

Downstream:

$$\sqrt[4]{15}$$

Distance:

$$\sqrt[4]{30\text{km}}$$

Return upstream:

$$\sqrt[4]{9}$$

Time:

$$\sqrt[4]{30/9 = 10/3\text{h}}$$

Total:

$$\sqrt[4]{2+\frac{10}{3}} = \boxed{\frac{16}{3}} = 5\frac{1}{3}\text{h}$$

Q20

By water-frame relative motion:

$$\sqrt[\boxed{45\text{min}}]$$

' CHAPTER 32 COMPLETE

We have covered:

- still-water boat speed '
- current/stream speed '
- upstream/downstream '
- half-sum and half-difference '
- time and distance problems '
- equal-distance round trips '
- harmonic-mean average speed '
- upstream/downstream ratios '
- rafts and floating objects '
- time-difference questions '
- river crossing vectors '
- shortest-time crossing '
- downstream drift '
- shortest-distance/no-drift crossing '
- upstream aiming angle '
- resultant velocity '
- water-frame shortcut '
- dropped-object return problems '
- advanced GATE/placement questions '

Progress

Completed: 32/76 **Unit 1 — Quantitative Aptitude:** 0=0% PROGRESS

NEXT

Chapter 33 — Races and Circular Tracks

Next we will cover:

- race terminology
 - winning by distance
 - winning by time
 - speed ratios from race results
 - head starts
 - circular tracks
 - same-direction catches
 - opposite-direction meetings
 - lap counting
 - first simultaneous meeting
 - LCM-based track problems
 - advanced placement race shortcuts.
-

USER

next

ASSISTANT

CHAPTER 33 — RACES AND CIRCULAR TRACKS

PART 1 — RACE TERMINOLOGY, WINNING BY DISTANCE/TIME, SPEED RATIOS, HEAD STARTS & BASIC CIRCULAR TRACKS

Status: $\emptyset$ = Not in PROGRESS GATE: Low–Medium Placements: Medium–High Priority: $\emptyset$ = 0 or 3

Races and Circular Tracks are applications of:

$$\boxed{D=ST}$$

and relative speed.

The most important idea in race questions is:

When two racers run for the same time, the distances covered are proportional to their speeds.

So:

$$\boxed{D_1:D_2=S_1:S_2}$$

1. BASIC RACE TERMINOLOGY

Suppose A and B run a 100 m race.

If A finishes 100 m when B has completed only 90 m, then:

A beats B by 10 m.

So:

$$\boxed{\text{winning distance}=100-90=10\text{m}}$$

2. “A BEATS B BY x METRES”

In an (L) -metre race:

If A beats B by (x) metres, when A finishes (L) :

B has covered:

$$\boxed{L-x}$$

Since both ran for the same time:

$$\boxed{S_A:S_B=L:(L-x)}$$

MUST MEMORIZE

3. EXAMPLE

In a 100 m race, A beats B by 20 m.

When A runs:

$$\backslash[100\text{m} \backslash]$$

B runs:

$$\backslash[80\text{m} \backslash]$$

So speed ratio:

$$\backslash[S_A:S_B = 100:80 \backslash]$$

$$\backslash[=\boxed{5:4} \backslash]$$

4. FIND WINNING DISTANCE FROM SPEED RATIO

Suppose:

$$\backslash[S_A:S_B=5:4 \backslash]$$

Race length:

$$\backslash[100\text{m} \backslash]$$

When A completes 100 m, B completes:

$$\backslash[100\times\frac{4}{5} = 80\text{m} \backslash]$$

So A beats B by:

$$\backslash[100-80 = \boxed{20\text{m}} \backslash]$$

5. GENERAL WINNING-DISTANCE FORMULA

If:

$$\backslash[S_A:S_B=a:b \backslash]$$

and race length is (L) , with $(a>b)$:

B covers:

$$\backslash[L\frac{b}{a} \backslash]$$

So winning distance:

$$\backslash[\boxed{L\left(1-\frac{b}{a}\right) = \frac{L(a-b)}{a}} \backslash]$$

6. EXAMPLE

A:B speed ratio:

$$\backslash[4:3 \backslash]$$

Race:

$\left[200\text{m} \right]$

Winning distance:

$\left[200\left(1-\frac{3}{4}\right) \right]$

$\left[=200\left(\frac{1}{4}\right) \right]$

$\left[=\boxed{50\text{m}} \right]$

7. “A BEATS B BY t SECONDS”

This means A finishes earlier by t seconds.

Example:

A runs 100 m in 10 s.

B finishes 2 s later.

Then B takes:

$\left[12\text{s} \right]$

For same distance:

$\left[S_A:S_B = T_B:T_A \right]$

$\left[=12:10 \right]$

$\left[=\boxed{6:5} \right]$

8. SAME DISTANCE !’ SPEED INVERSE TO TIME

For equal race length:

$\left[\boxed{S_A:S_B=T_B:T_A} \right]$

So if A takes 20 s and B takes 25 s:

$\left[S_A:S_B=25:20 \right]$

$\left[=\boxed{5:4} \right]$

9. WINNING BY TIME — GENERAL

If A runs race length L in T_A , and beats B by t seconds:

$\left[T_B=T_A+t \right]$

Hence:

$\left[\boxed{S_A:S_B=(T_A+t):T_A} \right]$

10. EXAMPLE

A completes a race in 40 s and beats B by 10 s.

Then B takes:

$$\sqrt[5]{50\text{s}}$$

Speed ratio:

$$\sqrt[5]{50:40 = \boxed{5:4}}$$

11. CONVERT TIME WIN TO DISTANCE WIN

Suppose A beats B by 10 s in a 100 m race.

A's finishing time = 40 s.

Then B's speed relative to A:

$$\sqrt[5]{S_A:S_B=50:40=5:4}$$

When A finishes 100 m:

B covers:

$$\sqrt[5]{80\text{m}}$$

So A beats B by:

$$\sqrt[5]{\boxed{20\text{m}}}$$

12. CONVERT DISTANCE WIN TO TIME WIN

Suppose A beats B by 20 m in a 100 m race.

Then:

$$\sqrt[5]{S_A:S_B=100:80=5:4}$$

If A takes 40 s, B's time:

$$\sqrt[5]{40\times\frac{5}{4} = 50\text{s}}$$

So A beats B by:

$$\sqrt[5]{\boxed{10\text{s}}}$$

13. CONNECTED RACE RESULTS

Suppose:

- A beats B by 20 m in a 100 m race.
- B beats C by 25 m in a 100 m race.

Find A:C speed ratio.

First:

$$\sqrt[5]{A:B=100:80=5:4}$$

Second:

$$\sqrt[5]{B:C=100:75=4:3}$$

Therefore:

$$\boxed{A:C=5:3}$$

$$\boxed{5:3}$$

14. FIND HOW MUCH A BEATS C

If:

$$A:C=5:3$$

in a 100 m race:

When A finishes 100 m:

$$C=60\text{m}$$

So A beats C by:

$$\boxed{40\text{m}}$$

15. CONNECTED-RACE SHORTCUT

If:

- A beats B by x in race length L
- B beats C by y in race length L

then:

$$\frac{S_B}{S_A} = \frac{L-x}{L}$$

$$\frac{S_C}{S_B} = \frac{L-y}{L}$$

Therefore:

$$\frac{S_C}{S_A} = \frac{(L-x)(L-y)}{L^2}$$

So when A finishes L , C covers:

$$\boxed{\frac{(L-x)(L-y)}{L}}$$

and A's winning distance over C is:

$$\boxed{L - \frac{(L-x)(L-y)}{L}}$$

16. EXAMPLE

100 m race.

A beats B by 10 m.

B beats C by 20 m.

When A finishes 100:

B covers:

$$90$$

C relative to B covers 80%.

So C covers:

$$\sqrt[90(0.8)=72]{}$$

A beats C by:

$$\sqrt[100-72 = \boxed{28m}]{}$$

Not 30 m.

This is a classic trap.

17. WHY WINNING DISTANCES DO NOT SIMPLY ADD

If:

- A beats B by 10 m
- B beats C by 20 m

you cannot say A beats C by:

$$\sqrt[30m]{}$$

because the second 20 m is measured when B finishes its own race, not when A finishes.

Use ratios.

18. HEAD START IN A RACE

Suppose A gives B a head start of $\sqrt[h]{}$ metres in an $\sqrt[L]{}$ -metre race.

Then:

- A must run $\sqrt[L]{}$
- B must run $\sqrt[L-h]{}$

If they finish together:

$$\sqrt[\boxed{S_A:S_B=L:(L-h)}]{}$$

Exactly the same ratio pattern as “A beats B by $\sqrt[h]{}$ metres.”

19. EXAMPLE

In a 100 m race, A gives B a 20 m head start, and they finish together.

A runs:

$$\sqrt[100m]{}$$

B runs:

$$\sqrt[80m]{}$$

Thus:

$$\sqrt[S_A:S_B = 100:80 = \boxed{5:4}]{}$$

20. FAIR HEAD START

If:

$$\sqrt{S_A:S_B=a:b}$$

with A faster, and race length (L) :

To finish together, B should run:

$$\sqrt{L\frac{ba}{a}}$$

Hence head start:

$$\sqrt{\boxed{H=L\left(1-\frac{ba}{a}\right)}}$$

Same as A's natural winning distance over B.

21. EXAMPLE

A:B speeds:

$$\sqrt{6:5}$$

Race:

$$\sqrt{120m}$$

Fair head start for B:

$$\sqrt{120\left(1-\frac{5}{6}\right)}$$

$$\sqrt{=120\left(\frac{1}{6}\right)}$$

$$\sqrt{=\boxed{20m}}$$

22. HEAD START IN TIME

If B starts (t) seconds earlier, B gains distance:

$$\sqrt{\boxed{S_Bt}}$$

Then A must close this gap using relative speed:

$$\sqrt{S_A-S_B}$$

So catch time after A starts:

$$\sqrt{\boxed{\frac{S_Bt}{S_A-S_B}}}$$

23. EXAMPLE

B runs at:

$$\sqrt{8m/s}$$

A runs at:

$$\sqrt{10m/s}$$

B starts 5 s earlier.

Head start:

$$8(5)=40\text{m}$$

Relative speed:

$$2\text{m/s}$$

Catch time after A starts:

$$40/2 = \boxed{20\text{s}}$$

24. FIND WHERE CATCH OCCURS

Same example.

A runs 20 s at 10 m/s:

$$200\text{m}$$

So catch occurs:

$$\boxed{200\text{m}} \text{ from A's starting point}$$

B has run:

$$5+20=25\text{s}$$

Distance:

$$8(25)=200\text{m}$$

Check correct.

25. RACE WITH BOTH DISTANCE AND TIME HEAD START

Example:

A is faster.

B starts 20 m ahead and also 2 s earlier.

Convert all advantages into distance before A starts.

If B speed is v_B :

Effective head start:

$$\boxed{20+2v_B}$$

Then use relative speed.

26. DEAD HEAT

“Dead heat” means both finish together.

If A gives B some head start and race ends in dead heat, use equal-time distance ratio.

27. BASIC CIRCULAR TRACK

Suppose track circumference is:

$$\boxed{C}$$

Runner speed:

$$v$$

Time per lap:

$$T_{\text{lap}} = \frac{C}{v}$$

28. EXAMPLE

Track:

$$400\text{m}$$

Runner speed:

$$5\text{m/s}$$

Lap time:

$$400/5 = \boxed{80\text{s}}$$

29. NUMBER OF LAPS

If runner travels for time t :

$$\boxed{\text{laps} = \frac{vt}{C}}$$

Example:

$$v=5\text{m/s}, t=400\text{s}, C=400\text{m}$$

Distance:

$$2000\text{m}$$

Laps:

$$2000/400 = \boxed{5}$$

30. TWO RUNNERS — SAME DIRECTION

They meet/catch whenever the faster runner gains one full lap on the slower runner.

Relative speed:

$$v_f - v_s$$

So catch interval:

$$\boxed{T = \frac{C}{v_f - v_s}}$$

31. EXAMPLE

Track:

$$400\text{m}$$

Speeds:

$$\sqrt[6]{4\text{m/s}}$$

Relative:

$$\sqrt[2]{2}$$

Catch interval:

$$\sqrt[400/2 = \boxed{200\text{s}}]$$

32. TWO RUNNERS — OPPOSITE DIRECTIONS

They meet whenever together they cover one circumference relative to each other.

Relative speed:

$$\sqrt[v_1+v_2]$$

Meeting interval:

$$\sqrt[\boxed{T=\frac{C}{v_1+v_2}}]$$

33. EXAMPLE

Track:

$$\sqrt[400\text{m}]$$

Speeds:

$$\sqrt[6]{4\text{m/s}}$$

Opposite directions:

$$\sqrt[T=\frac{400}{10} = \boxed{40\text{s}}]$$

34. STARTING FROM SAME POINT

If two runners start simultaneously at same point:

Same direction

They next meet when faster gains exactly one lap.

Opposite directions

They next meet after their relative distances sum to one lap.

35. STARTING FROM DIFFERENT POINTS

Then use initial separation along the relevant direction.

Example:

400 m track.

A and B are 100 m apart along the track.

Same direction:

$$v_A = 6, v_B = 4$$

If A is behind B, relative gap:

$$100\text{m}$$

Catch time:

$$100/(6-4) = \boxed{50\text{s}}$$

36. OPPOSITE DIRECTION WITH INITIAL ARC GAP

If shortest arc between them is 120 m and they move toward each other:

$$T = \frac{120}{v_1 + v_2}$$

Do not automatically use full circumference unless they start together.

37. FIND MEETING POINT

Track:

$$400\text{m}$$

A and B start same point opposite directions at:

$$6, 4\text{m/s}$$

Meeting time:

$$40\text{s}$$

A covers:

$$240\text{m}$$

B covers:

$$160\text{m}$$

So meeting point is:

$$\boxed{240\text{m}} \text{ along A's direction}$$

from the start.

38. SPEED RATIO FROM FIRST MEETING POINT

They start same point in opposite directions.

At meeting:

$$D_A : D_B = v_A : v_B$$

If meeting point divides track into arcs:

$$240 : 160$$

then:

$$\lfloor v_A : v_B = \boxed{3:2} \rfloor$$

39. FIRST RETURN TO STARTING POINT TOGETHER

Suppose runners have lap times:

$$\lfloor T_1, T_2 \rfloor$$

They will both be at the starting point together after:

$$\lfloor \boxed{\text{LCM}(T_1, T_2)} \rfloor$$

if lap times are integer-compatible units.

40. EXAMPLE

A lap time:

$$\lfloor 60\text{s} \rfloor$$

B lap time:

$$\lfloor 90\text{s} \rfloor$$

Together at starting point after:

$$\lfloor \text{LCM}(60, 90) = \boxed{180\text{s}} \rfloor$$

A completes:

$$\lfloor 3 \text{ laps} \rfloor$$

B:

$$\lfloor 2 \text{ laps} \rfloor$$

41. IMPORTANT: “MEET ANYWHERE” VS “MEET AT START”

These are different.

Meet anywhere

Use relative speed.

Meet at starting point together

Use LCM of lap times.

This is one of the biggest circular-track traps.

42. EXAMPLE

400 m track.

A speed = 8 m/s.

B = 6 m/s.

Same direction.

Lap times:

$$\left[50\text{s}, \quad \frac{400}{6} = 66\frac{2}{3}\text{s} \right]$$

They catch each other after:

$$\left[400/(8-6) = 200\text{s} \right]$$

At 200 s:

A laps:

$$\left[4 \right]$$

B laps:

$$\left[3 \right]$$

So the catch happens exactly at starting point here.

But that is not always guaranteed.

43. SAME DIRECTION — NUMBER OF LAPS AT CATCH

At first catch:

$$\left[(v_f - v_s)t = C \right]$$

Faster runner laps:

$$\left[\frac{v_f}{C} \right]$$

Slower:

$$\left[\frac{v_s}{C} \right]$$

Their lap-count difference is exactly:

$$\left[\boxed{1} \right]$$

44. EXAMPLE

Speeds:

$$\left[6, 4 \right]$$

Track:

$$\left[400 \right]$$

First catch:

$$\left[t = 200\text{s} \right]$$

A distance:

$$\left[1200\text{m} = 3\text{ laps} \right]$$

B:

$$\left[800\text{m} = 2\text{ laps} \right]$$

Difference:

$$\left[1\text{ lap} \right]$$

45. OPPOSITE DIRECTION — COMBINED LAPS

At first meeting:

$$\frac{C}{v_1 + v_2} = t$$

So combined distance equals:

$$v_1 t + v_2 t = C$$

46. MULTIPLE MEETINGS IN GIVEN TIME

Same direction.

Meeting interval:

$$T_c = \frac{C}{v_f - v_s}$$

If total time T , number of **new catches after start** typically:

$$\left\lfloor \frac{T}{T_c} \right\rfloor$$

with care about whether the endpoint time counts.

47. EXAMPLE

Track 400 m.

Speeds 6 and 4.

Catch interval:

$$T_c = 200 \text{ s}$$

In 1000 s, catches after starting:

$$\frac{1000}{200} = 5$$

$$\boxed{5}$$

not counting their initial co-location at $t=0$.

48. OPPOSITE-DIRECTION MEETING COUNT

Track:

$$C = 400 \text{ m}$$

Speeds:

$$v_1 = 6, v_2 = 4$$

Meeting interval:

$$T_c = 40 \text{ s}$$

In 200 s:

$$\frac{200}{40} = 5$$

So there are:

$\boxed{5 \text{ meetings after start}}$

if the meeting at exactly 200 s is included.

49. RUNNERS WITH SPEED RATIO

Suppose same-direction speeds:

$5:3$

Let speeds:

$5k, 3k$

Relative:

$2k$

Catch time:

$\frac{C}{2k}$

Faster lap time:

$\frac{C}{5k}$

So by catch, faster completes:

$\frac{C/(2k)}{C/(5k)} = \frac{5}{2} = 2.5 \text{ laps}$

Slower:

1.5 laps

They meet halfway around the track.

50. GATE-STYLE QUESTION 1

In a 100 m race, A beats B by 20 m.

$A:B=100:80$

$\boxed{5:4}$

51. GATE-STYLE QUESTION 2

A:B speeds = 4:3.

Race length = 200 m.

Winning distance:

$200 \left(1 - \frac{3}{4}\right)$

$\boxed{50\text{m}}$

52. GATE-STYLE QUESTION 3

400 m track.

Speeds:

$$\sqrt[6]{4\text{m/s}}$$

Same direction catch time:

$$\sqrt[400/(6-4) = \boxed{200\text{s}}]$$

53. GATE-STYLE QUESTION 4

Same track and speeds, opposite direction:

$$\sqrt[400/(6+4) = \boxed{40\text{s}}]$$

54. PLACEMENT-STYLE QUESTION 1

A beats B by 10 m in a 100 m race.

B beats C by 20 m in a 100 m race.

A:C:

$$\sqrt[A:B=100:90=10:9]$$

$$\sqrt[B:C=100:80=5:4]$$

$$\sqrt[A:C = 10:9 \times 5:4]$$

More directly:

$$\sqrt[C\text{ when A finishes} = 90 \left(\frac{80}{100} \right) = 72]$$

So A beats C by:

$$\sqrt[\boxed{28\text{m}}]$$

55. PLACEMENT-STYLE QUESTION 2

A gives B a 20 m head start in a 100 m race and they tie.

$$\sqrt[A:B=100:80]$$

$$\sqrt[=\boxed{5:4}]$$

56. PLACEMENT-STYLE QUESTION 3

B runs at 8 m/s.

A runs at 10 m/s.

B starts 5 s early.

Catch time after A begins:

$$\sqrt[\frac{8(5)}{10-8} = \boxed{20\text{s}}]$$

57. PLACEMENT-STYLE QUESTION 4

Lap times:

$$\sqrt[4]{60s, \sqrt[4]{90s}}$$

Both next at starting point together:

$$\sqrt[4]{LCM=180}$$

$$\sqrt[4]{\boxed{180s}}$$

58. ADVANCED QUESTION — CONNECTED RACE

In a 200 m race:

A beats B by 40 m.

B beats C by 50 m.

Find how much A beats C.

A:B:

$$\sqrt[4]{200:160=5:4}$$

B:C:

$$\sqrt[4]{200:150=4:3}$$

Therefore:

$$\sqrt[4]{A:C=5:3}$$

When A runs 200:

$$\sqrt[4]{C=120}$$

Winning distance:

$$\sqrt[4]{\boxed{80m}}$$

59. ADVANCED QUESTION — TIME WIN TO DISTANCE WIN

A completes 200 m in 25 s.

B takes 5 s longer.

Then:

$$\sqrt[4]{A:B\text{speed}} = 30:25 = 6:5$$

When A runs 200:

B runs:

$$\sqrt[4]{200 \cdot \frac{5}{6} = 166\frac{2}{3}}$$

So A beats B by:

$$\sqrt[4]{\boxed{33\frac{1}{3}m}}$$

60. ADVANCED QUESTION — FAIR START

A:B speeds:

$$\left[7:5 \right]$$

Race length:

$$\left[280\text{m} \right]$$

Fair head start for B:

$$\left[280\left(1-\frac{5}{7}\right) \right]$$

$$\left[=280\cdot\frac{2}{7} \right]$$

$$\left[=\boxed{80\text{m}} \right]$$

61. ADVANCED QUESTION — CIRCULAR TRACK STARTING APART

400 m track.

A at 8 m/s is 100 m behind B at 6 m/s.

Same direction.

Relative:

$$\left[2\text{m/s} \right]$$

Catch:

$$\left[100/2 = \boxed{50\text{s}} \right]$$

62. ADVANCED QUESTION — OPPOSITE ARC

Circular track 500 m.

A and B are separated by 150 m along the arc they are moving toward each other on.

Speeds:

$$\left[7, \, 3\text{m/s} \right]$$

Relative:

$$\left[10 \right]$$

Meeting time:

$$\left[150/10 = \boxed{15\text{s}} \right]$$

63. COMMON TRAPS

& pTRAP 1 — Winning distances add

Usually false.

Use speed ratios.

& pTRAP 2 — “Beats by 20 m” means B runs 20 m

No. In a 100 m race, B runs 80 m.

& pTRAP 3 — Head start means faster person runs extra

No. Slower racer begins closer to finish.

& pTRAP 4 — Circular same-direction meeting uses sum

Wrong.

Use difference.

& pTRAP 5 — Circular opposite-direction meeting uses difference

Wrong.

Use sum.

& pTRAP 6 — LCM for every meeting

LCM is mainly for being together at the starting point again, not ordinary meetings anywhere.

64. MUST-MEMORIZE RACE FORMULAS

A beats B by x in race L

$$\boxed{S_A : S_B = L : (L - x)}$$

Speed ratio $a:b$, race L

Winning distance:

$$\boxed{\frac{L(a-b)}{a}}$$

Fair head start

$$\boxed{H = L \left(1 - \frac{S_B}{S_A} \right)}$$

65. MUST-MEMORIZE CIRCULAR-TRACK FORMULAS

Same direction

$$\boxed{T_{\text{catch}} = \frac{C}{v_1 - v_2}}$$

Opposite direction

$$\boxed{T_{\text{meet}} = \frac{C}{v_1 + v_2}}$$

Return to start together

$$\boxed{\text{LCM}(\text{lap times})}$$

66. FAST EXAM DECISION TREE

“Beats by x metres”?

Convert to distance ratio immediately.

“Beats by t seconds”?

Convert to time ratio.

Connected race results?

Multiply speed ratios.

Head start?

Reduce slower racer's required distance.

Circular same direction?

Faster gains one lap.

Circular opposite direction?

Together cover one lap relatively.

Both at start again?

Use LCM of lap times.

PRACTICE — LEVEL 0

Q1

A beats B by 20 m in a 100 m race. Find speed ratio.

Q2

A:B speeds = 5:4. In a 200 m race, by how much does A win?

Q3

A finishes a race in 40 s, B in 50 s. Find speed ratio.

Q4

400 m track, speeds 6 and 4 m/s, same direction. Find catch time.

Q5

Same track and speeds, opposite direction. Find meeting time.

LEVEL 1

Q6

In a 100 m race A beats B by 10 m, B beats C by 20 m. By how much does A beat C?

Q7

A:B speed ratio = 6:5. Find fair head start for B in a 120 m race.

Q8

B runs at 8 m/s and starts 5 s before A, who runs at 10 m/s. Find catch time after A starts.

Q9

Lap times are 60 s and 90 s. When are both next at starting point together?

Q10

A:B speed ratio = 4:3. In a 300 m race, find winning distance.

LEVEL 2

Q11

A completes 200 m in 25 s and beats B by 5 s. Find distance by which A beats B.

Q12

In a 200 m race A beats B by 40 m, and B beats C by 50 m. Find A's winning distance over C.

Q13

400 m track. A at 8 m/s starts 100 m behind B at 6 m/s. Find catch time.

Q14

500 m circular track. Two runners separated by 150 m move toward each other at 7 and 3 m/s. Find meeting time.

Q15

Same-direction speeds on a circular track are in ratio 5:3. How many laps has faster runner completed at first catch?

LEVEL 3 — EXAM PRESSURE

Q16

In a 300 m race, A beats B by 60 m. In another 300 m race, B beats C by 75 m. By how much does A beat C?

Q17

A:B speeds = 7:5. In a 280 m race, B receives 40 m head start. Who wins and by how much?

Q18

A circular track is 600 m. Runners at 8 and 5 m/s start together in same direction. Find:

1. first catch time,
2. laps completed by each at that moment.

Q19

Same 600 m track. Runners at 8 and 5 m/s start together in opposite directions. Find:

1. first meeting time,
 2. distance covered by each.
-

Q20

A lap time is 72 s and B's is 90 s. Find:

1. when they are together at starting point again,
 2. how many laps each completes by then.
-

ANSWERS

Q1

$$\backslash [100:80 = \boxed{5:4} \backslash]$$

Q2

$$\backslash [200\left(1-\frac{4}{5}\right) = \boxed{40\text{m}} \backslash]$$

Q3

$$\backslash [S_A:S_B=50:40 \backslash]$$

$$\backslash [=\boxed{5:4} \backslash]$$

Q4

$$\backslash [400/(6-4) = \boxed{200\text{s}} \backslash]$$

Q5

$$\backslash [400/(6+4) = \boxed{40\text{s}} \backslash]$$

Q6

When A finishes 100:

B covers:

$$\backslash [90 \backslash]$$

C covers 80% of B's distance:

$$\backslash [90(0.8)=72 \backslash]$$

A wins by:

$$\backslash [100-72 = \boxed{28\text{m}} \backslash]$$

Q7

$$\backslash [120\left(1-\frac{5}{6}\right) = \boxed{20\text{m}} \backslash]$$

Q8

Head start:

$$\backslash [8(5)=40 \backslash]$$

Relative:

$$\backslash [2 \backslash]$$

$$\backslash [40/2 = \boxed{20\text{s}} \backslash]$$

Q9

$$\boxed{\text{LCM}(60,90) = 180\text{s}}$$

Q10

$$\boxed{300\left(1-\frac{3}{4}\right) = 75\text{m}}$$

Q11

B takes:

$$\boxed{30\text{s}}$$

Speed ratio:

$$\boxed{30:25 = 6:5}$$

When A runs 200:

$$\boxed{B=200\cdot\frac{5}{6} = 166\frac{2}{3}}$$

Winning distance:

$$\boxed{33\frac{1}{3}\text{m}}$$

Q12

$$\boxed{A:B=200:160=5:4}$$

$$\boxed{B:C=200:150=4:3}$$

Thus:

$$\boxed{A:C=5:3}$$

C covers:

$$\boxed{200\cdot\frac{3}{5}=120}$$

A wins by:

$$\boxed{80\text{m}}$$

Q13

Relative speed:

$$\boxed{8-6=2}$$

$$\boxed{100/2 = 50\text{s}}$$

Q14

$$\boxed{150/(7+3) = 15\text{s}}$$

Q15

Let speeds:

$$\boxed{5\text{k}, 3\text{k}}$$

Catch time:

$$\boxed{C/(2\text{k})}$$

Faster laps:

$$\boxed{\frac{5\text{k}(C/2\text{k})}{C} = 2.5\text{laps}}$$

Q16

A:B:

$$\backslash[300:240=5:4 \backslash]$$

B:C:

$$\backslash[300:225=4:3 \backslash]$$

Thus:

$$\backslash[A:C=5:3 \backslash]$$

When A runs 300:

$$\backslash[C=180 \backslash]$$

A wins by:

$$\backslash[\boxed{120m} \backslash]$$

Q17

A:B speed ratio:

$$\backslash[7:5 \backslash]$$

B starts 40 m ahead, so B runs:

$$\backslash[240m \backslash]$$

Compare times:

$$\backslash[T_A=\frac{280}{7k}=40/k \backslash]$$

$$\backslash[T_B=\frac{240}{5k}=48/k \backslash]$$

A finishes earlier.

When A finishes, B covers:

$$\backslash[5k\cdot\frac{40}{k} = 200m \backslash]$$

But B started at the 40 m mark, so B's position is:

$$\backslash[40+200=240m \backslash]$$

Finish line = 280 m.

Thus A wins by:

$$\backslash[\boxed{40m} \backslash]$$

Q18

Relative speed:

$$\backslash[8-5=3 \backslash]$$

Catch time:

$$\backslash[600/3 = \boxed{200s} \backslash]$$

A distance:

$$\backslash[8(200)=1600 \backslash]$$

Laps:

$$\boxed{\frac{1600}{600} = \frac{8}{3} = 2\frac{2}{3}}$$

B distance:

$$1000$$

Laps:

$$\boxed{\frac{1000}{600} = \frac{5}{3} = 1\frac{2}{3}}$$

Difference = one lap.

Q19

Relative:

$$8+5=13$$

Meeting time:

$$600/13$$

$$\boxed{\frac{600}{13} \text{ s} \approx 46.15 \text{ s}}$$

Distances:

A:

$$8 \cdot \frac{600}{13} = \boxed{\frac{4800}{13} \text{ m}}$$

B:

$$5 \cdot \frac{600}{13} = \boxed{\frac{3000}{13} \text{ m}}$$

Together:

$$600 \text{ m}$$

Q20

$$\text{LCM}(72, 90) = 360 \text{ s}$$

A laps:

$$360/72=5$$

B:

$$360/90=4$$

$$\boxed{360 \text{ s}, 5 \text{ laps}, 4 \text{ laps}}$$

CHAPTER 33 STATUS

Covered so far:

- race terminology '
- winning by distance '
- winning by time '
- speed ratios from race results '
- converting time win !" distance win '
- connected race results '
- head starts '

- fair starts '
- circular-track basics '
- same-direction catches '
- opposite-direction meetings '
- lap counting '
- LCM for returning to start '
- initial separation on circular tracks '
- basic meeting counts '

But **Chapter 33 is not complete yet.**

Next we still need:

- advanced head-start race problems
- multiple racers
- nth meeting/catch
- meeting counts in a fixed time
- first simultaneous meeting of 3 runners
- HCF/LCM track problems
- runners changing direction
- circular track with staggered starts
- overtaking counts
- final shortcut sheet and chapter audit

Progress

Completed: 32/76 **Current:** 0=0 Chapter 33 — Races & Circular Tracks **Next:** Chapter 33 Part 2 — Multiple runners, nth meetings, LCM/HCF track logic & advanced race problems

USER

next

ASSISTANT

CHAPTER 33 — RACES AND CIRCULAR TRACKS

PART 2 — MULTIPLE RUNNERS, NTH MEETINGS, OVERTAKING COUNTS, LCM/HCF TRACK LOGIC & ADVANCED RACES

Status: 0=0 **PROGRESS GATE:** Low–Medium **Placements:** Medium–High **Priority:** 0=0 or 3

Part 1 built the foundations. Now we handle the questions that usually cause mistakes:

- 3 or more runners
- nth catch / nth meeting
- number of meetings
- starting-point reunion

- staggered starts
- direction changes
- LCM/HCF logic
- advanced head starts
- connected race results

1. NTH CATCH — SAME DIRECTION

For two runners on a circular track:

$$T_{\text{catch}} = \frac{C}{v_f - v_s}$$

This is the time between consecutive catches.

Therefore nth catch after the start:

$$\boxed{T_n = n \frac{C}{v_f - v_s}}$$

assuming $(t=0)$ is not counted as a catch.

2. EXAMPLE

Track:

$$400\text{m}$$

Speeds:

$$6, 4\text{m/s}$$

Catch interval:

$$\frac{400}{2} = 200\text{s}$$

Third catch:

$$3(200) = \boxed{600\text{s}}$$

3. NTH MEETING — OPPOSITE DIRECTIONS

Meeting interval:

$$\frac{C}{v_1 + v_2}$$

Hence:

$$\boxed{T_n = n \frac{C}{v_1 + v_2}}$$

4. EXAMPLE

400 m track.

Speeds:

$$6, 4\text{m/s}$$

Opposite directions.

Meeting interval:

$$\backslash[40s \backslash]$$

5th meeting:

$$\backslash[5(40) = \boxed{200s} \backslash]$$

5. NUMBER OF CATCHES IN TIME $\backslash(T\backslash)$

If catch interval is $\backslash(t_c\backslash)$:

$$\backslash[\boxed{N = \left\lfloor \frac{T}{t_c} \right\rfloor} \backslash]$$

when:

- start at same point,
 - initial $\backslash(t=0\backslash)$ is not counted,
 - a catch exactly at time $\backslash(T\backslash)$ is counted.
-

6. EXAMPLE

Catch interval:

$$\backslash[200s \backslash]$$

Total observation:

$$\backslash[950s \backslash]$$

$$\backslash[950/200=4.75 \backslash]$$

Therefore:

$$\backslash[\boxed{4\text{ catches}} \backslash]$$

At:

$$\backslash[200,400,600,800s \backslash]$$

7. EXACT-ENDPOINT TRAP

If total time:

$$\backslash[1000s \backslash]$$

and interval:

$$\backslash[200s \backslash]$$

meetings occur at:

$$\backslash[200,400,600,800,1000 \backslash]$$

So:

$$\backslash[\boxed{5} \backslash]$$

8. STARTING AT DIFFERENT POINTS — SAME DIRECTION

Suppose faster A is d metres behind B along the running direction.

Then first catch:

$$t_1 = \frac{d}{v_A - v_B}$$

After that, catches repeat every:

$$\frac{C}{v_A - v_B}$$

9. EXAMPLE

Track:

$$500\text{m}$$

$$A = 8\text{ m/s.}$$

$$B = 6\text{ m/s.}$$

A starts 100 m behind.

First catch:

$$100/(8-6)=50\text{s}$$

Next catch occurs one full relative lap later:

$$500/2=250\text{s}$$

So second catch time from start:

$$50+250 = 300\text{s}$$

10. NTH CATCH WITH INITIAL GAP

If first catch occurs at t_1 and catch interval thereafter is T_c :

$$t_n = t_1 + (n-1)T_c$$

11. OPPOSITE DIRECTIONS — DIFFERENT STARTING POINTS

Suppose they are separated by an arc of d metres **in the direction in which they approach each other**.

First meeting:

$$t_1 = \frac{d}{v_1 + v_2}$$

After that, subsequent meetings repeat every:

$$\frac{C}{v_1 + v_2}$$

12. THREE RUNNERS — BASIC IDEA

For 3 runners, distinguish two questions:

A. When do all return to starting point together?

Use individual lap times and LCM.

B. When are all at the same point anywhere?

Use pairwise relative motion.

These are not always the same.

13. THREE RUNNERS RETURN TO START TOGETHER

Suppose lap times are:

$$\{ 40s, 60s, 90s \}$$

All return to start together after:

$$\{ \text{LCM}(40, 60, 90) \}$$

Prime factors:

$$\{ 40 = 2^3 \cdot 5 \}$$

$$\{ 60 = 2^2 \cdot 3 \cdot 5 \}$$

$$\{ 90 = 2 \cdot 3^2 \cdot 5 \}$$

LCM:

$$\{ 2^3 \cdot 3^2 \cdot 5 = 360 \}$$

$$\{ \boxed{360s} \}$$

14. LAPS COMPLETED BY THEN

At 360 s:

Runner A:

$$\{ 360/40 = 9 \}$$

B:

$$\{ 360/60 = 6 \}$$

C:

$$\{ 360/90 = 4 \}$$

So:

$$\{ \boxed{9, 6, 4} \text{laps} \}$$

15. LCM RULE — MUST MEMORIZE

If lap times are:

$$\{ T_1, T_2, \dots, T_n \}$$

then first simultaneous return to start:

$\boxed{\text{LCM}(T_1, T_2, \dots, T_n)}$

when times are expressed in compatible integer units.

16. IF LAP TIMES ARE FRACTIONS

Example:

$40\text{s}, 60\text{s}, 75\text{s}$

You can still use LCM normally.

But if times contain fractions, first convert to a common smallest unit or rational form.

Do not mechanically apply integer LCM to decimals without normalization.

17. THREE RUNNERS — ALL MEET ANYWHERE

Suppose all start from same point and move in the same direction.

Speeds:

$8, 6, 4\text{m/s}$

For all three to be together again:

A and B must meet:

$T_{AB} = \frac{C}{8-6} = \frac{C}{2}$

A and C:

$T_{AC} = \frac{C}{8-4} = \frac{C}{4}$

We need a time common to both meeting schedules.

For $(C=400\text{m})$:

$T_{AB} = 200\text{s}$

$T_{AC} = 100\text{s}$

First common positive time:

$\boxed{200\text{s}}$

At 200 s:

A:

$1600\text{m} = 4 \text{ laps}$

B:

$1200\text{m} = 3 \text{ laps}$

C:

$800\text{m} = 2 \text{ laps}$

All are at start.

18. GENERAL METHOD FOR ALL MEETING ANYWHERE

Find pairwise meeting intervals relative to one reference runner.

Example using A:

$$\backslash T_{\{AB\}} \backslash$$

$$\backslash T_{\{AC\}} \backslash$$

Then find their first common positive time.

For integer-friendly questions:

$$\backslash \boxed{\text{LCM of pairwise meeting intervals}} \backslash$$

is safest.

19. SIGNED VELOCITY METHOD

For circular tracks, choose clockwise as positive.

Example:

- A clockwise 6 m/s $(+6)$
- B anticlockwise 4 m/s (-4)

Relative speed:

$$\backslash |6 - (-4)| = 10 \backslash$$

This one rule handles both same and opposite directions.

$$\backslash \boxed{v_{\text{rel}} = |v_A - v_B|} \backslash$$

when velocities are signed.

20. THREE RUNNERS WITH MIXED DIRECTIONS

Track:

$$\backslash 600\text{m} \backslash$$

A clockwise:

$$\backslash 8\text{m/s} \backslash$$

B clockwise:

$$\backslash 5\text{m/s} \backslash$$

C anticlockwise:

$$\backslash 4\text{m/s} \backslash$$

Pair intervals:

A-B:

$$\backslash 600/(8-5)=200\text{s} \backslash$$

A-C:

$$\backslash 600/(8+4)=50\text{s} \backslash$$

For all three to be at same point again:

Need common time:

$$\boxed{\text{LCM}(200,50) = 200\text{s}}$$

Check:

A:

$$1600\text{m}$$

B:

$$1000\text{m}$$

Difference:

$$600$$

A and C relative distance:

$$(8+4)(200)=2400=4C$$

So all coincide.

21. RETURN TO START VS ALL MEET ANYWHERE

In the previous example at 200 s:

A distance:

$$1600$$

On 600 m track, A is at:

$$400\text{m}$$

from start.

So all meet at the 400 m point—not at the start.

This demonstrates:

$$\boxed{\text{same point together} \neq \text{starting point together}}$$

22. WHEN DO THEY ALL RETURN TO START?

Use individual lap times.

For previous example:

A lap time:

$$600/8=75\text{s}$$

B:

$$600/5=120\text{s}$$

C:

$$600/4=150\text{s}$$

LCM:

$$\text{LCM}(75, 120, 150) = 600\text{s}$$

So:

$$\boxed{600\text{s}}$$

for all to return to start together.

23. OVERTAKING COUNT

Same direction:

Each time faster gains one circumference, one overtake occurs.

Relative distance gained in time (T) :

$$(v_f - v_s)T$$

Number of complete overtakes:

$$\left\lfloor \frac{(v_f - v_s)T}{C} \right\rfloor$$

when starting together and $(t=0)$ isn't counted.

24. EXAMPLE

Track:

$$500\text{m}$$

Speeds:

$$9, 4\text{m/s}$$

Time:

$$600\text{s}$$

Relative distance:

$$5(600) = 3000\text{m}$$

Overtakes:

$$3000/500 = \boxed{6}$$

25. OPPOSITE-DIRECTION MEETING COUNT

Relative distance:

$$(v_1 + v_2)T$$

Every circumference corresponds to one meeting after the start.

Thus:

$$N = \left\lfloor \frac{(v_1 + v_2)T}{C} \right\rfloor$$

for runners starting together.

26. EXAMPLE

Track:

$$\boxed{600\text{m}}$$

Speeds:

$$\boxed{8, 4}$$

Opposite directions.

Time:

$$\boxed{300\text{s}}$$

Relative distance:

$$\boxed{12(300)=3600}$$

Meetings:

$$\boxed{3600/600 = \boxed{6}}$$

27. RUNNERS CHANGE DIRECTION

When any runner reverses direction:

stop using the old relative speed.

Recompute immediately.

28. EXAMPLE — DIRECTION CHANGE

A and B are 300 m apart on a track/line segment.

A moves at:

$$\boxed{8\text{m/s}}$$

B at:

$$\boxed{4\text{m/s}}$$

toward each other.

After 20 s:

Distance closed:

$$\boxed{12(20)=240\text{m}}$$

Remaining gap:

$$\boxed{60\text{m}}$$

If B now turns and moves same direction as A:

Relative speed:

$$\boxed{8-4=4}$$

Catch time after turn:

$$\sqrt[60/4 = \boxed{15s}]{}]$$

29. STAGGERED START ON A CIRCULAR TRACK

A starts first.

B starts later.

Convert B's delay into A's position when B begins.

Then solve a normal relative-motion problem.

30. EXAMPLE

Track:

$$\sqrt[400m]{}]$$

A:

$$\sqrt[5m/s]{}]$$

B:

$$\sqrt[7m/s]{}]$$

B starts 20 s later from the same point.

When B starts, A has moved:

$$\sqrt[5(20)=100m]{}]$$

B is 100 m behind A.

Relative speed:

$$\sqrt[2m/s]{}]$$

First catch after B starts:

$$\sqrt[100/2 = \boxed{50s}]{}]$$

31. WHERE DOES THE CATCH OCCUR?

B distance:

$$\sqrt[7(50)=350m]{}]$$

So catch point is:

$$\sqrt[\boxed{350m}]{}]$$

from starting point along their direction.

A total time:

$$\sqrt[70s]{}]$$

A distance:

$$\lfloor 5(70) = 350 \text{ m} \rfloor$$

Correct.

32. IF DELAY CREATES MORE THAN ONE LAP

Suppose A travels for long enough before B starts.

Reduce A's lead modulo circumference.

Example:

Track = 400 m.

A speed 5 m/s.

B starts 100 s later.

A travels:

$$\lfloor 500 \text{ m} \rfloor$$

Position is:

$$\lfloor 500 \bmod 400 = 100 \text{ m} \rfloor$$

So effective lead around track is:

$$\lfloor \boxed{100 \text{ m}} \rfloor$$

not 500 m.

33. MODULO IDEA

For circular-track position:

$$\lfloor \boxed{\text{position}} = (vt) \bmod C \rfloor$$

Very useful in advanced questions.

34. HEAD START — RACE WITH INSUFFICIENT START

A:B speeds:

$$\lfloor 7:5 \rfloor$$

Race:

$$\lfloor 280 \text{ m} \rfloor$$

Fair head start for B:

$$\lfloor 80 \text{ m} \rfloor$$

Suppose B receives only 40 m.

Then B must run:

$$\lfloor 240 \text{ m} \rfloor$$

A runs:

$$\sqrt[2]{280m}$$

Time A:

$$\sqrt[2]{280/7k=40/k}$$

Time B:

$$\sqrt[2]{240/5k=48/k}$$

A wins.

At A's finishing time, B covers:

$$\sqrt[2]{5k(40/k)=200m}$$

Since B began 40 m ahead, B's position is:

$$\sqrt[2]{240m}$$

A finishes at 280.

Winning margin:

$$\sqrt[2]{\boxed{40m}}$$

35. EXCESS HEAD START

Same speeds:

$$\sqrt[2]{7:5}$$

Race:

$$\sqrt[2]{280m}$$

Fair head start:

$$\sqrt[2]{80m}$$

If B is given 100 m, B only runs:

$$\sqrt[2]{180m}$$

B's time:

$$\sqrt[2]{180/(5k)=36/k}$$

A:

$$\sqrt[2]{40/k}$$

So B wins.

36. HOW FAR DOES A HAVE LEFT WHEN B FINISHES?

B finish time:

$$\sqrt[2]{36/k}$$

A covers:

$$\sqrt[2]{7k(36/k)=252m}$$

Race is 280.

A has:

$$\boxed{280-252 = 28\text{m}}$$

left.

So B wins by 28 m in terms of A's remaining distance.

37. FIND FAIR TIME HEAD START

Suppose A speed:

$$10\text{m/s}$$

B:

$$8\text{m/s}$$

Race length:

$$200\text{m}$$

A time:

$$20\text{s}$$

B without head start:

$$25\text{s}$$

For dead heat, B must start:

$$25-20 = 5\text{s}$$

earlier.

38. GENERAL FAIR TIME HEAD START

For race length L :

$$T_A = \frac{L}{S_A}$$

$$T_B = \frac{L}{S_B}$$

If A is faster, B's fair time start:

$$\boxed{H_t = \frac{L}{S_B} - \frac{L}{S_A}}$$

39. CONNECTED RACE — THREE PEOPLE

A beats B by x .

B beats C by y .

We already learned these don't simply add.

Example:

Race = 400m.

A beats B by 40m.

B beats C by 80m.

A:B:

$$\backslash[400:360=10:9 \backslash]$$

B:C:

$$\backslash[400:320=5:4 \backslash]$$

Thus:

$$\backslash[A:C = 10:9\times 5:4 = 50:36 = 25:18 \backslash]$$

When A finishes 400:

$$\backslash[C=400\cdot\frac{18}{25} = 288 \backslash]$$

A wins by:

$$\backslash[\boxed{112\text{m}} \backslash]$$

40. CONNECTED RACE — FAST METHOD

C's distance when A finishes:

$$\backslash[(L-x)\left(\frac{L-y}{L}\right) \backslash]$$

Example:

$$\backslash[360\left(\frac{320}{400}\right) = 360(0.8) = 288 \backslash]$$

Then:

$$\backslash[400-288 = 112 \backslash]$$

41. HCF LOGIC — COMMON LAP UNIT

Sometimes speeds/distances are given as integer ratios and the question asks for smallest repeating pattern.

Example:

Runner distances per minute:

$$\backslash[60\text{m}, \quad 90\text{m} \backslash]$$

Their common distance increments are multiples of:

$$\backslash[\text{HCF}(60,90)=30 \backslash]$$

But for **time to return to a point**, usually use:

- track circumference,
- lap time,
- relative speed,
- LCM.

Do not force HCF into every circular-track question.

42. WHEN HCF IS USEFUL

HCF is mainly useful when simplifying:

- speed ratios,
- lap counts,
- repeating distance units,
- integer relative-speed patterns.

LCM is much more commonly used for:

$\boxed{\text{simultaneous recurrence}}$

43. THREE RUNNERS — GIVEN LAP TIMES

Lap times:

$45, 60, 75\text{s}$

All return to start:

$\text{LCM}(45, 60, 75)$

Factors:

$45 = 3^2 \cdot 5$

$60 = 2^2 \cdot 3 \cdot 5$

$75 = 3 \cdot 5^2$

LCM:

$2^2 \cdot 3^2 \cdot 5^2 = 900$

$\boxed{900\text{s}}$

44. NUMBER OF LAPS BY THEN

$900/45 = 20$

$900/60 = 15$

$900/75 = 12$

$\boxed{20, 15, 12}$

45. MULTIPLE RACERS — WHO WINS?

If race distance is same and all start together:

Fastest speed wins.

But when head starts differ, compare:

$\boxed{\text{finish time} = \frac{\text{remaining distance}}{\text{speed}}}$

46. EXAMPLE

100 m race:

A speed:

$$\sqrt[10]{10\text{m/s}}$$

B:

$$\sqrt[8]{8\text{m/s}}$$

C:

$$\sqrt[6]{6\text{m/s}}$$

Head starts:

- A: 0
- B: 20 m
- C: 40 m

Times:

A:

$$\sqrt[100/10=10\text{s}]{}$$

B:

$$\sqrt[80/8=10\text{s}]{}$$

C:

$$\sqrt[60/6=10\text{s}]{}$$

Result:

$$\sqrt[\boxed{\text{Dead heat among all three}}]{}$$

47. ADVANCED OVERTAKING QUESTION

Track:

$$\sqrt[600\text{m}]{}$$

A:

$$\sqrt[10\text{m/s}]{}$$

B:

$$\sqrt[7\text{m/s}]{}$$

Same direction.

Catch interval:

$$\sqrt[600/3=200\text{s}]{}$$

In 15 min:

$$\sqrt[900\text{s}]{}$$

Catches:

$$\sqrt[\left\lfloor 900/200 \right\rfloor = \boxed{4}]{}$$

48. ADVANCED MEETING QUESTION

Same track.

A:

$$\sqrt{10}$$

B:

$$\sqrt{5}$$

Opposite directions.

Meeting interval:

$$\sqrt{600/15=40s}$$

In 5 min:

$$\sqrt{300s}$$

Meetings:

$$\sqrt{\lfloor 300/40 \rfloor = \boxed{7}}$$

They meet at 40,80,...,280 s.

49. GATE-STYLE QUESTION 1

400 m track.

Speeds:

$$\sqrt{8, \sqrt{6m/s}}$$

Same direction.

Third catch:

$$\sqrt{3 \cdot \frac{400}{2} = \boxed{600s}}$$

50. GATE-STYLE QUESTION 2

400 m track.

Speeds 8 and 6 opposite directions.

5th meeting:

$$\sqrt{5 \cdot \frac{400}{14}}$$

$$\sqrt{= \frac{1000}{7}}$$

$$\sqrt{\approx \boxed{142.86s}}$$

51. GATE-STYLE QUESTION 3

Lap times:

$\sqrt[4]{40, \sqrt[6]{60}, \sqrt[90]{}}$

All next at start:

$\sqrt[360]{}$

52. GATE-STYLE QUESTION 4

Track 500 m.

A 8 m/s starts 100 m behind B 6 m/s.

First catch:

$\sqrt[100/2]{= \sqrt[50]{}}$

53. PLACEMENT-STYLE QUESTION 1

A:B speed:

$\sqrt[7:5]{}$

Race:

$\sqrt[280m]{}$

Fair head start:

$\sqrt[80m]{}$

$\sqrt[\sqrt[80m]{}]{} \sqrt[80m]{}$

54. PLACEMENT-STYLE QUESTION 2

A:B:

$\sqrt[10:8]{}$

200 m race.

Fair time head start to B:

A:

$\sqrt[20s]{}$

B:

$\sqrt[25s]{}$

Difference:

$\sqrt[\sqrt[5s]{}]{} \sqrt[5s]{}$

55. PLACEMENT-STYLE QUESTION 3

Track:

$\sqrt[600m]{}$

Speeds:

$$\sqrt[10]{7}$$

Same direction for 900 s.

Overtakes:

$$\left\lfloor \frac{3(900)}{600} \right\rfloor = \boxed{4}$$

56. PLACEMENT-STYLE QUESTION 4

Three runners have lap times:

$$45, 60, 75 \text{ s}$$

Return together at start after:

$$\boxed{900 \text{ s} = 15 \text{ min}}$$

57. ADVANCED QUESTION — STAGGERED START

Track:

$$600 \text{ m}$$

A:

$$6 \text{ m/s}$$

B:

$$9 \text{ m/s}$$

B starts 50 s later from same point.

A's lead when B starts:

$$6(50) = 300 \text{ m}$$

Relative:

$$3$$

Catch after B starts:

$$300/3 = \boxed{100 \text{ s}}$$

Catch position:

$$9(100) = 900 \text{ m}$$

Modulo 600:

$$\boxed{300 \text{ m from start}}$$

58. ADVANCED QUESTION — THREE RUNNERS TO START

Track:

$$600 \text{ m}$$

Speeds:

$$\sqrt[3]{10, 8, 6 \text{ m/s}}$$

Lap times:

$$\sqrt[3]{60, 75, 100 \text{ s}}$$

LCM:

$$\sqrt[3]{\text{LCM}(60, 75, 100) = 300}$$

So all return to start together after:

$$\sqrt[3]{\boxed{300 \text{ s}}}$$

Laps:

$$\sqrt[3]{5, 4, 3}$$

59. ADVANCED QUESTION — ALL MEET ANYWHERE

Same track:

$$\sqrt[3]{600 \text{ m}}$$

Speeds:

$$\sqrt[3]{10, 8, 6}$$

Relative to 10:

$$\sqrt[3]{10-8=2}$$

$$\sqrt[3]{10-6=4}$$

Meeting intervals:

$$\sqrt[3]{600/2=300 \text{ s}}$$

$$\sqrt[3]{600/4=150 \text{ s}}$$

First common:

$$\sqrt[3]{\boxed{300 \text{ s}}}$$

Here it happens to be at start too.

60. ADVANCED QUESTION — MIXED DIRECTIONS

600 m track.

A clockwise:

$$\sqrt[3]{8}$$

B clockwise:

$$\sqrt[3]{5}$$

C anticlockwise:

$$\sqrt[3]{4}$$

A-B interval:

$$\lfloor 600/3=200 \rfloor$$

A-C interval:

$$\lfloor 600/12=50 \rfloor$$

All same point:

$$\lfloor \text{LCM}(200,50) = \boxed{200s} \rfloor$$

But all at starting point together only after:

$$\lfloor \text{LCM}(75,120,150) = \boxed{600s} \rfloor$$

61. COMMON TRAPS

& pTRAP 1 — Counting $\lfloor t=0 \rfloor$ as first meeting

Usually exam wording “meet again” excludes the starting instant.

& pTRAP 2 — nth meeting

Do not confuse:

$$\lfloor nT \rfloor$$

with:

$$\lfloor (n-1)T \rfloor$$

If the starting instant is not counted, nth new meeting is $\lfloor nT \rfloor$.

& pTRAP 3 — Starting apart

First meeting interval is not necessarily $\lfloor C/v_{\text{rel}} \rfloor$.

Use actual initial arc gap first.

& pTRAP 4 — All meet anywhere vs all return to start

Different questions.

& pTRAP 5 — Staggered starts

Reduce early runner's position modulo track circumference.

& pTRAP 6 — Head-start distances simply compared

Compute remaining race distances and finish times.

& pTRAP 7 — Direction change

Relative speed must be recalculated.

62. MUST-MEMORIZE FINAL FORMULAS

Races

A beats B by x in race L :

$$\boxed{S_A : S_B = L : (L - x)}$$

Fair distance head start:

$$\boxed{H = L \left(1 - \frac{S_B}{S_A} \right)}$$

Fair time head start:

$$\boxed{H_t = \frac{L}{S_B} - \frac{L}{S_A}}$$

63. CIRCULAR TRACK

Same direction

$$\boxed{T_c = \frac{C}{|v_1 - v_2|}}$$

Opposite direction

$$\boxed{T_m = \frac{C}{v_1 + v_2}}$$

n th event

$$\boxed{T_n = n T_{\text{interval}}}$$

for same-start repeated events, excluding $(t=0)$.

64. STARTING APART

First event:

$$\boxed{t_1 = \frac{\text{relevant initial arc gap}}{v_{\text{relative}}}}$$

Then:

$$\boxed{t_n = t_1 + (n-1) \frac{C}{v_{\text{relative}}}}$$

65. SIMULTANEOUS START-POINT RETURN

$$\boxed{T = \text{LCM}(\text{individual lap times})}$$

66. MULTIPLE RUNNERS MEETING ANYWHERE

Safest exam method:

1. Choose one runner as reference.
2. Find pairwise meeting intervals.
3. Find first common positive time.

$$\boxed{\text{Use LCM of compatible pairwise intervals}}$$

67. 5-MINUTE REVISION

Remember only these:

1. Race win by distance

$$\lfloor L:(L-x) \rfloor$$

2. Head start

Same logic as winning distance.

3. Same direction circle

$$\lfloor C/(v_f - v_s) \rfloor$$

4. Opposite direction circle

$$\lfloor C/(v_1 + v_2) \rfloor$$

5. Return to start together

LCM of lap times.

6. Starting apart

Gap first, full laps later.

7. nth meeting

First event + repeated interval.

PRACTICE — LEVEL 0

Q1

Track 400 m. Speeds 8 and 6 m/s, same direction. Find second catch time.

Q2

Same track. Speeds 8 and 6 opposite directions. Find third meeting time.

Q3

Lap times 60 and 80 s. When are both next at starting point?

Q4

A:B speeds = 5:4 in a 200 m race. Find fair head start.

Q5

Track 500 m. A at 8 m/s is 100 m behind B at 6 m/s. Find first catch.

LEVEL 1

Q6

Catch interval = 120 s. How many catches occur in 650 s?

Q7

Track 600 m. Speeds 10 and 7 same direction. How many overtakes in 900 s?

Q8

Three runners' lap times are 40, 60 and 90 s. When do all return to start together?

Q9

A runs 200 m in 20 s, B in 25 s. Find fair time head start for B.

Q10

A:B = 7:5 in 280 m race. B receives only 40 m start. By how much does A win?

LEVEL 2

Q11

Track 600 m. A=6 m/s. B=9 m/s starts 50 s later. Find catch time after B starts and catch location.

Q12

Track 600 m. A clockwise 8, B clockwise 5, C anticlockwise 4. When are all three at same point again?

Q13

Same data. When are all three at the starting point together?

Q14

In 400 m race, A beats B by 40 m and B beats C by 80 m. Find A's winning margin over C.

Q15

Track 600 m. Speeds 8 and 5 opposite directions. How many meetings occur in 10 minutes?

LEVEL 3 — EXAM PRESSURE

Q16

Track 720 m. Runners have lap times 72 s, 90 s and 120 s. Find:

1. first simultaneous return to start,
2. laps completed by each.

Q17

Track 480 m. A=8 m/s and B=5 m/s start together same direction. Find:

1. 5th catch time,
2. laps completed by each at that instant.

Q18

A:B:C speeds are 10:8:6. In a 300 m race, B gets 60 m head start and C gets 120 m. Determine result.

Q19

Track 500 m. A at 9 m/s starts 200 m behind B at 4 m/s. Find first three catch times.

Q20

A beats B by 25 m in a 100 m race. B beats C by 20 m in a 100 m race. Find A's winning distance over C.

ANSWERS

Q1

Catch interval:

$$\backslash[400/(8-6)=200s \backslash]$$

Second catch:

$$\backslash[\boxed{400s} \backslash]$$

Q2

Meeting interval:

$$\backslash[400/(8+6) = \frac{200}{7}s \backslash]$$

Third:

$$\backslash[\frac{600}{7} \backslash]$$

$$\backslash[\boxed{\approx 85.71s} \backslash]$$

Q3

$$\backslash[\text{LCM}(60,80) = \boxed{240s} \backslash]$$

Q4

$$\backslash[200\left(1-\frac{4}{5}\right) = \boxed{40m} \backslash]$$

Q5

$$\backslash[100/(8-6) = \boxed{50s} \backslash]$$

Q6

$$\backslash[\lfloor 650/120 \rfloor = \boxed{5} \backslash]$$

Q7

$$\backslash[\frac{(10-7)(900)}{600} = 4.5 \backslash]$$

Complete overtakes:

$$\backslash[\boxed{4} \backslash]$$

Q8

$$\backslash[\text{LCM}(40,60,90) = \boxed{360s} \backslash]$$

Q9

A:

$$\backslash[200/10=20s \backslash]$$

B:

$$\backslash[200/8=25s \backslash]$$

Fair start:

$$\backslash[\boxed{5s} \backslash]$$

Q10

B runs:

$$\sqrt{280-40=240\text{m}}$$

A time:

$$\sqrt{280/7k=40/k}$$

B covers by then:

$$\sqrt{5k(40/k)=200\text{m}}$$

Including 40 m head start, B reaches 240 m mark.

Finish:

$$\sqrt{280}$$

A wins by:

$$\sqrt{\boxed{40\text{m}}}$$

Q11

A lead:

$$\sqrt{6(50)=300\text{m}}$$

Relative:

$$\sqrt{3}$$

Catch after B:

$$\sqrt{300/3=100\text{s}}$$

B distance:

$$\sqrt{9(100)=900\text{m}}$$

Location:

$$\sqrt{900\text{bmod}600=300\text{m}}$$

$$\sqrt{\boxed{100\text{s},\ 300\text{m}\text{ from start}}}$$

Q12

A-B interval:

$$\sqrt{600/(8-5)=200\text{s}}$$

A-C:

$$\sqrt{600/(8+4)=50\text{s}}$$

First common:

$$\sqrt{\boxed{200\text{s}}}$$

Q13

Lap times:

$$\sqrt{75,\ 120,\ 150}$$

LCM:

$\boxed{600\text{s}}$

Q14

A:B:

$$400:360=10:9$$

B:C:

$$400:320=5:4$$

C when A finishes:

$$360 \cdot \frac{45}{100} = 288$$

Winning margin:

$$400 - 288 = \boxed{112\text{m}}$$

Q15

Meeting interval:

$$600/(8+5) = 600/13$$

Total time:

$$600\text{s}$$

Number:

$$\left\lfloor \frac{600}{600/13} \right\rfloor = \boxed{13}$$

The 13th occurs exactly at 600 s.

Q16

Lap times:

$$72, 90, 120$$

LCM:

$$360\text{s}$$

Laps:

$$360/72=5$$

$$360/90=4$$

$$360/120=3$$

$$\boxed{360\text{s}; 5, 4, 3 \text{ laps}}$$

Q17

Catch interval:

$$480/(8-5) = 160\text{s}$$

5th catch:

$$800\text{s}$$

A distance:

$$8(800)=6400$$

Laps:

$$\left[6400/480 = 13\frac{1}{3} \right]$$

B:

$$\left[5(800)=4000 \right]$$

$$\left[4000/480 = 8\frac{1}{3} \right]$$

Difference:

$$\left[5\text{laps} \right]$$

$$\left[\boxed{800\text{s}, 13\frac{1}{3}, 8\frac{1}{3}\text{laps}} \right]$$

Q18

Let speeds:

$$\left[10\text{k}, 8\text{k}, 6\text{k} \right]$$

Remaining race distances:

A:

$$\left[300 \right]$$

B:

$$\left[240 \right]$$

C:

$$\left[180 \right]$$

Times:

$$\left[A=\frac{300}{10\text{k}}=\frac{30}{\text{k}} \right]$$

$$\left[B=\frac{240}{8\text{k}}=\frac{30}{\text{k}} \right]$$

$$\left[C=\frac{180}{6\text{k}}=\frac{30}{\text{k}} \right]$$

Therefore:

$$\left[\boxed{\text{All three finish together}} \right]$$

Q19

First gap:

$$\left[200\text{m} \right]$$

Relative:

$$\left[9-4=5 \right]$$

First catch:

$$\left[200/5=40\text{s} \right]$$

Subsequent interval:

$$\left[500/5=100\text{s} \right]$$

Therefore:

$$\left[\boxed{40\text{s}, 140\text{s}, 240\text{s}} \right]$$

Q20

When A finishes 100:

B covers:

$\sqrt[3]{75\text{m}}$

When B runs 100, C runs:

$\sqrt[3]{80\text{m}}$

So when B runs 75:

$\sqrt[3]{C=75(0.8)=60}$

Thus A beats C by:

$\sqrt[3]{100-60 = \boxed{40\text{m}}}$

' CHAPTER 33 COMPLETE

We have now covered:

- race terminology '
- winning by distance '
- winning by time '
- distance/time win conversion '
- connected race results '
- distance head starts '
- time head starts '
- dead heats '
- multiple racers '
- circular-track lap time '
- same-direction catches '
- opposite-direction meetings '
- nth meetings/catches '
- meeting and overtaking counts '
- runners starting apart '
- staggered starts '
- modulo position logic '
- direction changes '
- three-runner problems '
- all-meet-anywhere vs return-to-start distinction '
- LCM/HCF track logic '
- advanced GATE/placement questions '

Progress

Completed: 33/76

Unit 1 — Quantitative Aptitude: 0/34 PROGRESS

NEXT

Chapter 34 — Algebraic Identities and Simplification

Next we move from motion problems into algebra:

- essential algebraic identities
- $((a+b)^2, (a-b)^2, a^2-b^2)$
- cubes and factorization
- manipulating $(x+\frac{1}{x})$
- symmetric expressions
- simplifying rational algebra
- substitution shortcuts
- hidden-factor patterns
- GATE numerical simplification
- option elimination and speed tricks.

USER

next

ASSISTANT

CHAPTER 34 — ALGEBRAIC IDENTITIES AND SIMPLIFICATION

PART 1 — CORE IDENTITIES, FACTORIZATION, SUBSTITUTION & $(x+\frac{1}{x})$ PATTERNS

Status: $\emptyset=0\%$ PROGRESS GATE: High Placements: High Priority: $\emptyset=0\%$ Part 1

Algebraic simplification questions often look long, but most collapse quickly if you recognize a standard pattern.

The goal is:

Do not expand everything blindly. Recognize the hidden identity first.

1. MOST IMPORTANT IDENTITY

$$\boxed{(a+b)^2=a^2+2ab+b^2}$$

Example:

$$\boxed{(x+3)^2 = x^2+6x+9}$$

2. SECOND CORE IDENTITY

$$\boxed{(a-b)^2=a^2-2ab+b^2}$$

Example:

$$\boxed{(x-5)^2 = x^2 - 10x + 25}$$

3. DIFFERENCE OF SQUARES

$$\boxed{a^2 - b^2 = (a-b)(a+b)}$$

This is one of the highest-value simplification identities.

Example:

$$99^2 - 1^2$$

Instead of calculating 99^2 :

$$= (99-1)(99+1)$$

$$= 98(100)$$

$$\boxed{= 9800}$$

4. FAST MENTAL-MATH APPLICATION

$$1003^2 - 997^2$$

Use:

$$(a-b)(a+b)$$

$$= (1003-997)(1003+997)$$

$$= 6(2000)$$

$$\boxed{= 12000}$$

Much faster than squaring.

5. SUM AND DIFFERENCE SQUARES

From the first two identities:

$$(a+b)^2 + (a-b)^2$$

$$= 2a^2 + 2b^2$$

So:

$$\boxed{(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)}$$

6. DIFFERENCE OF THE TWO SQUARES

$$(a+b)^2 - (a-b)^2$$

$$= 4ab$$

Hence:

$$\boxed{(a+b)^2 - (a-b)^2 = 4ab}$$

Very useful.

7. EXAMPLE

If:

$$\backslash[a+b=10 \backslash]$$

and:

$$\backslash[a-b=4 \backslash]$$

Find $\backslash(ab\backslash)$.

Use:

$$\backslash[(a+b)^2-(a-b)^2=4ab \backslash]$$

$$\backslash[100-16=4ab \backslash]$$

$$\backslash[84=4ab \backslash]$$

$$\backslash[\boxed{ab=21} \backslash]$$

8. FIND $\backslash(a^2+b^2\backslash)$ FROM SUM AND PRODUCT

Since:

$$\backslash[(a+b)^2=a^2+b^2+2ab \backslash]$$

Therefore:

$$\backslash[\boxed{a^2+b^2=(a+b)^2-2ab} \backslash]$$

Example:

$$\backslash[a+b=12,\text{quad } ab=20 \backslash]$$

$$\backslash[a^2+b^2=144-40 \backslash]$$

$$\backslash[=\boxed{104} \backslash]$$

9. FIND $\backslash(a^2+b^2\backslash)$ FROM DIFFERENCE AND PRODUCT

$$\backslash[(a-b)^2=a^2+b^2-2ab \backslash]$$

So:

$$\backslash[\boxed{a^2+b^2=(a-b)^2+2ab} \backslash]$$

10. PRODUCT FROM SUM AND DIFFERENCE

If:

$$\backslash[S=a+b \backslash]$$

and:

$$\backslash[D=a-b \backslash]$$

then:

$$\sqrt{S^2 - D^2} = 4ab$$

Thus:

$$\boxed{ab = \frac{S^2 - D^2}{4}}$$

11. FIND a AND b FROM SUM AND DIFFERENCE

$$a + b = S$$

$$a - b = D$$

Add:

$$2a = S + D$$

$$\boxed{a = \frac{S + D}{2}}$$

Subtract:

$$2b = S - D$$

$$\boxed{b = \frac{S - D}{2}}$$

12. CUBE OF A SUM

$$\boxed{(a + b)^3 = a^3 + b^3 + 3ab(a + b)}$$

Equivalent expanded form:

$$a^3 + 3a^2b + 3ab^2 + b^3$$

The compact form is often better for aptitude.

13. CUBE OF A DIFFERENCE

$$\boxed{(a - b)^3 = a^3 - b^3 - 3ab(a - b)}$$

Equivalent expanded form:

$$a^3 - 3a^2b + 3ab^2 - b^3$$

14. SUM OF CUBES

$$\boxed{a^3 + b^3 = (a + b)(a^2 - ab + b^2)}$$

15. DIFFERENCE OF CUBES

$$\boxed{a^3 - b^3 = (a - b)(a^2 + ab + b^2)}$$

Memory trick

For:

$$\sqrt[3]{a^3 \mp b^3}$$

first bracket keeps same sign:

$$\sqrt[3]{(a \mp b)}$$

second bracket has:

$$\sqrt[3]{a^2 \mp ab + b^2}$$

Sign flips in the middle.

16. EXAMPLE — DIFFERENCE OF CUBES

$$\sqrt[3]{8x^3 - 27}$$

Recognize:

$$\sqrt[3]{(2x)^3 - 3^3}$$

So:

$$\sqrt[3]{(2x-3)(4x^2+6x+9)}$$

17. VERY IMPORTANT THREE-VARIABLE IDENTITY

$$\boxed{a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)}$$

Special case:

If:

$$\sqrt[3]{a+b+c=0}$$

then:

$$\boxed{a^3 + b^3 + c^3 = 3abc}$$

This is a favorite identity-based exam pattern.

18. EXAMPLE

If:

$$\sqrt[3]{x+y+z=0}$$

and:

$$\sqrt[3]{xyz=8}$$

then:

$$\sqrt[3]{x^3 + y^3 + z^3 = 3xyz}$$

$$\sqrt[3]{=24}$$

$$\boxed{24}$$

19. FACTORIZATION BY COMMON FACTOR

Example:

$$\sqrt[4]{6x^2+9x}$$

Common factor:

$$\sqrt[4]{3x}$$

$$\sqrt[4]{=3x(2x+3)}$$

Always check common factor before advanced identities.

20. FACTORIZATION OF QUADRATIC TRINOMIAL

Example:

$$\sqrt[4]{x^2+5x+6}$$

Find two numbers whose:

- sum = 5
- product = 6

They are 2 and 3.

So:

$$\sqrt[4]{\boxed{x^2+5x+6=(x+2)(x+3)}}$$

21. NEGATIVE MIDDLE TERM

$$\sqrt[4]{x^2-7x+12}$$

Numbers:

$$\sqrt[4]{-3, -4}$$

because:

$$\sqrt[4]{-3-4=-7}$$

and:

$$\sqrt[4]{(-3)(-4)=12}$$

Thus:

$$\sqrt[4]{\boxed{(x-3)(x-4)}}$$

22. NEGATIVE CONSTANT

$$\sqrt[4]{x^2+x-6}$$

Need numbers whose:

- sum = 1
- product = -6

They are 3 and -2.

So:

$$\boxed{(x+3)(x-2)}$$

23. GENERAL QUADRATIC FACTORIZATION

For:

$$ax^2+bx+c$$

factorization often uses splitting the middle term.

Example:

$$2x^2+5x+2$$

Product:

$$ac=4$$

Need numbers:

$$4, 1$$

whose sum = 5.

$$2x^2+4x+x+2$$

Group:

$$2x(x+2)+1(x+2)$$

$$\boxed{(2x+1)(x+2)}$$

24. ALGEBRAIC CANCELLATION

Example:

$$\frac{x^2-9}{x-3}$$

Factor:

$$x^2-9=(x-3)(x+3)$$

Then:

$$\frac{(x-3)(x+3)}{x-3} = \boxed{x+3}$$

provided:

$$x \neq 3$$

25. DOMAIN TRAP

Original expression:

$$\frac{x^2-9}{x-3}$$

is undefined at:

$$x=3$$

Even though simplified expression is $x+3$, original restriction remains:

$$\boxed{x \neq 3}$$

Important for algebraic reasoning.

26. ANOTHER CANCELLATION

$$\frac{x^2+5x+6}{x+2}$$

Factor:

$$(x+2)(x+3)$$

Cancel:

$$\boxed{x+3}$$

for:

$$x \neq -2$$

27. SIMPLIFICATION BY SUBSTITUTION

Suppose:

$$x+y=10$$

and:

$$xy=21$$

Find:

$$x^2+y^2$$

Do not solve (x,y) .

Use:

$$x^2+y^2=(x+y)^2-2xy$$

$$=100-42$$

$$=\boxed{58}$$

28. FIND (x^3+y^3) WITHOUT FINDING (x,y)

Use:

$$x^3+y^3=(x+y)^3-3xy(x+y)$$

If:

$$x+y=5, \quad xy=6$$

then:

$$x^3+y^3 = 125-3(6)(5)$$

$$=125-90$$

$$=\boxed{35}$$

29. FIND $(x-y)$ FROM SUM AND PRODUCT

$$(x-y)^2 = (x+y)^2 - 4xy$$

Thus:

$$\boxed{}$$

$$x-y$$

$$= \sqrt{(x+y)^2 - 4xy}$$

Example:

$$x+y=10, \text{quad } xy=21$$

$$(x-y)^2 = 100 - 84 = 16$$

$$|x-y| = 4$$

30. MASTER PATTERN— $(x + \frac{1}{x})$

Suppose:

$$x + \frac{1}{x} = a$$

Square:

$$x^2 + 2 + \frac{1}{x^2} = a^2$$

Therefore:

$$\boxed{x^2 + \frac{1}{x^2} = a^2 - 2}$$

This is extremely important.

31. EXAMPLE

If:

$$x + \frac{1}{x} = 5$$

find:

$$x^2 + \frac{1}{x^2}$$

$$= 5^2 - 2$$

$$= \boxed{23}$$

32. FIND $(x - \frac{1}{x})$

From:

$$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2$$

If:

$$\left[x + \frac{1}{x} = a \right]$$

then:

$$\left[x^2 + \frac{1}{x^2} = a^2 - 2 \right]$$

So:

$$\left[\left(x - \frac{1}{x} \right)^2 = a^2 - 4 \right]$$

Thus:

$$\left[\boxed{ x - \frac{1}{x} = \pm \sqrt{a^2 - 4} } \right]$$

Sign depends on $\left(\frac{1}{x} \right)$.

33. EXAMPLE

If:

$$\left[x + \frac{1}{x} = 5 \right]$$

then:

$$\left[\left(x - \frac{1}{x} \right)^2 = 25 - 4 = 21 \right]$$

Hence:

$$\left[\boxed{ x - \frac{1}{x} = \pm \sqrt{21} } \right]$$

Do not choose positive sign unless given $\left(x > 1 \right)$.

34. FIND $\left(x^3 + \frac{1}{x^3} \right)$

Use:

$$\left[(a+b)^3 = a^3 + b^3 + 3ab(a+b) \right]$$

Take:

$$\left[a = x, \quad b = \frac{1}{x} \right]$$

Since:

$$\left[ab = 1 \right]$$

$$\left[\left(x + \frac{1}{x} \right)^3 = x^3 + \frac{1}{x^3} + 3 \left(x + \frac{1}{x} \right) \right]$$

Therefore if:

$$\left[x + \frac{1}{x} = a \right]$$

then:

$$\left[\boxed{ x^3 + \frac{1}{x^3} = a^3 - 3a } \right]$$

35. EXAMPLE

If:

$$\left[x + \frac{1}{x} = 4 \right]$$

then:

$$\left[x^3 + \frac{1}{x^3} = 4^3 - 3(4) \right]$$

$$\left[= 64 - 12 \right]$$

$$\left[= \boxed{52} \right]$$

36. FIND $\left(x^4 + \frac{1}{x^4}\right)$

First find:

$$\left[x^2 + \frac{1}{x^2} = a^2 - 2 \right]$$

Then square:

$$\left[\left(x^2 + \frac{1}{x^2}\right)^2 = x^4 + \frac{1}{x^4} + 2 \right]$$

Therefore:

$$\left[\boxed{x^4 + \frac{1}{x^4} = (a^2 - 2)^2 - 2} \right]$$

37. EXAMPLE

If:

$$\left[x + \frac{1}{x} = 3 \right]$$

then:

$$\left[x^2 + \frac{1}{x^2} = 9 - 2 = 7 \right]$$

Then:

$$\left[x^4 + \frac{1}{x^4} = 49 - 2 = \boxed{47} \right]$$

38. RECURRENCE PATTERN

Let:

$$\left[S_n = x^n + \frac{1}{x^n} \right]$$

and:

$$\left[S_1 = a \right]$$

Then:

$$\left[\boxed{S_n = aS_{n-1} - S_{n-2}} \right]$$

with:

$$\left[S_0 = 2 \right]$$

Useful for higher powers.

Example:

$$\left[S_2 = a^2 - 2 \right]$$

$$\left[S_3 = a(a^2 - 2) - a = a^3 - 3a \right]$$

39. PATTERN— $(x - \frac{1}{x})$

If:

$$x - \frac{1}{x} = a$$

square:

$$x^2 - 2 + \frac{1}{x^2} = a^2$$

Therefore:

$$\boxed{x^2 + \frac{1}{x^2} = a^2 + 2}$$

40. EXAMPLE

If:

$$x - \frac{1}{x} = 3$$

then:

$$x^2 + \frac{1}{x^2} = 9 + 2 = \boxed{11}$$

41. THEN FIND $(x + \frac{1}{x})$

Using:

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$$

$$= a^2 + 4$$

Thus:

$$\boxed{x + \frac{1}{x} = \pm \sqrt{a^2 + 4}}$$

depending on (x) .

42. SYMMETRIC EXPRESSIONS

Expressions like:

$$x + y, \quad xy, \quad x^2 + y^2, \quad x^3 + y^3$$

are symmetric because swapping (x, y) does not change them.

If sum and product are known, many symmetric expressions can be found without solving for individual values.

43. MASTER SUBSTITUTION

Let:

$$S = x + y$$

$$P = xy$$

Then:

$$\boxed{x^2+y^2=S^2-2P}$$

$$\boxed{x^3+y^3=S^3-3PS}$$

$$\boxed{(x-y)^2=S^2-4P}$$

These are high-value formulas.

44. FIND $\left(\frac{x}{y}+\frac{y}{x}\right)$

$$\frac{x}{y}+\frac{y}{x} = \frac{x^2+y^2}{xy}$$

If:

$$x+y=S, \quad xy=P$$

then:

$$x^2+y^2=S^2-2P$$

Thus:

$$\boxed{\frac{x}{y}+\frac{y}{x} = \frac{S^2-2P}{P}}$$

45. EXAMPLE

$$x+y=7, \quad xy=10$$

Then:

$$x^2+y^2 = 49-20=29$$

Therefore:

$$\frac{x}{y}+\frac{y}{x} = \frac{29}{10}$$

$$\boxed{\frac{29}{10}}$$

46. FIND $\left(\frac{1}{x}+\frac{1}{y}\right)$

$$\frac{1}{x}+\frac{1}{y} = \frac{x+y}{xy}$$

Thus:

$$\boxed{\frac{1}{x}+\frac{1}{y}=\frac{S}{P}}$$

Example:

$$x+y=7, \quad xy=10$$

$$\boxed{\frac{7}{10}}$$

47. FIND $\left(\frac{1}{x^2}+\frac{1}{y^2}\right)$

$$= \frac{x^2+y^2}{x^2y^2}$$

$$= \frac{S^2-2P}{P^2}$$

Therefore:

$$\boxed{\frac{1}{x^2} + \frac{1}{y^2} = \frac{S^2 - 2P}{P^2}}$$

48. NUMERICAL IDENTITY SHORTCUT

Compute:

$$103 \times 97$$

Write:

$$(100+3)(100-3)$$

Use:

$$(a+b)(a-b) = a^2 - b^2$$

$$= 10000 - 9$$

$$= \boxed{9991}$$

49. ANOTHER MENTAL-MATH SHORTCUT

$$52^2 - 48^2$$

$$= (52-48)(52+48)$$

$$= 4(100)$$

$$= \boxed{400}$$

50. NEAR-BASE MULTIPLICATION

$$98 \times 102$$

$$= (100-2)(100+2)$$

$$= 10000 - 4$$

$$= \boxed{9996}$$

51. IDENTIFY HIDDEN SQUARE

$$x^2 + 6x + 9$$

Recognize:

$$x^2 + 2(x)(3) + 3^2$$

So:

$$\boxed{(x+3)^2}$$

52. IDENTIFY HIDDEN DIFFERENCE SQUARE

$$\sqrt{x^2-10x+25}$$

$$\sqrt{=\boxed{(x-5)^2}}$$

53. IDENTIFY DIFFERENCE OF SQUARES AFTER FACTORING

$$\sqrt{4x^2-25}$$

$$\sqrt{=(2x)^2-5^2}$$

$$\sqrt{=\boxed{(2x-5)(2x+5)}}$$

54. NESTED SIMPLIFICATION

Simplify:

$$\sqrt{\frac{x^2-4}{x^2+4x+4}}$$

Factor:

$$\sqrt{x^2-4=(x-2)(x+2)}$$

$$\sqrt{x^2+4x+4=(x+2)^2}$$

So:

$$\sqrt{\boxed{\frac{x-2}{x+2}}}$$

for:

$$\sqrt{x \neq -2}$$

55. GATE-STYLE QUESTION 1

If:

$$\sqrt{x+y=8, \text{quad } xy=12}$$

find:

$$\sqrt{x^2+y^2}$$

$$\sqrt{=64-24}$$

$$\sqrt{=\boxed{40}}$$

56. GATE-STYLE QUESTION 2

If:

$$\sqrt{x+\frac{1}{x}=5}$$

find:

$$\sqrt{x^2+\frac{1}{x^2}}$$

$$\sqrt{=25-2}$$

$$\sqrt{=\boxed{23}}$$

57. GATE-STYLE QUESTION 3

If:

$$x + \frac{1}{x} = 3$$

find:

$$x^3 + \frac{1}{x^3}$$

$$= 27 - 9$$

$$\boxed{18}$$

58. GATE-STYLE QUESTION 4

Simplify:

$$\frac{x^2 - 16}{x - 4}$$

$$= \frac{(x - 4)(x + 4)}{x - 4}$$

$$\boxed{x + 4}$$

with:

$$x \neq 4$$

59. PLACEMENT-STYLE QUESTION 1

Compute:

$$1007^2 - 993^2$$

$$= (14)(2000)$$

$$\boxed{28000}$$

60. PLACEMENT-STYLE QUESTION 2

Compute:

$$97 \times 103$$

$$= (100 - 3)(100 + 3)$$

$$= 10000 - 9$$

$$\boxed{9991}$$

61. PLACEMENT-STYLE QUESTION 3

If:

$$x + y = 9, \quad xy = 14$$

find:

$$\sqrt[3]{x^3+y^3}$$

$$\sqrt[3]{9^3-3(14)(9)}$$

$$\sqrt[3]{729-378}$$

$$\sqrt[3]{\boxed{351}}$$

62. PLACEMENT-STYLE QUESTION 4

If:

$$\sqrt{x-\frac{1}{x}}=4$$

find:

$$\sqrt{x^2+\frac{1}{x^2}}$$

$$\sqrt{16+2}$$

$$\sqrt{\boxed{18}}$$

63. COMMON TRAPS

& pTRAP 1 —

$$\sqrt{(a+b)^2=a^2+b^2}$$

Wrong.

Correct:

$$\sqrt{a^2+2ab+b^2}$$

& pTRAP 2 —

$$\sqrt{(a-b)^2=a^2-b^2}$$

Wrong.

Correct:

$$\sqrt{a^2-2ab+b^2}$$

& pTRAP 3 —

$$\sqrt{a^3+b^3}=(a+b)(a^2+ab+b^2)$$

Wrong middle sign.

Correct:

$$\sqrt{(a+b)(a^2-ab+b^2)}$$

& pTRAP 4 — Cancelling terms instead of factors

Example:

$$\sqrt{\frac{x+2}{x}}$$

You cannot cancel $\cancel{x}$ with part of $\cancel{x+2}$.

Cancellation works on **factors**, not terms.

& pTRAP 5 — Ignoring excluded values

When cancelling a factor from denominator, retain the original restriction.

& pTRAP 6 — Solving $\cancel{x,y}$ unnecessarily

If only symmetric expression is asked, use sum/product identities.

64. MUST-MEMORIZE IDENTITY TABLE

$$\boxed{(a+b)^2=a^2+2ab+b^2}$$

$$\boxed{(a-b)^2=a^2-2ab+b^2}$$

$$\boxed{a^2-b^2=(a-b)(a+b)}$$

$$\boxed{(a+b)^3=a^3+b^3+3ab(a+b)}$$

$$\boxed{(a-b)^3=a^3-b^3-3ab(a-b)}$$

$$\boxed{a^3+b^3=(a+b)(a^2-ab+b^2)}$$

$$\boxed{a^3-b^3=(a-b)(a^2+ab+b^2)}$$

65. MUST-MEMORIZE SUM-PRODUCT TABLE

If:

$$S=x+y, \quad P=xy$$

then:

$$\boxed{x^2+y^2=S^2-2P}$$

$$\boxed{x^3+y^3=S^3-3PS}$$

$$\boxed{(x-y)^2=S^2-4P}$$

66. MUST-MEMORIZE RECIPROCAL TABLE

If:

$$x+\frac{1}{x}=a$$

then:

$$\boxed{x^2+\frac{1}{x^2}=a^2-2}$$

$$\boxed{x^3+\frac{1}{x^3}=a^3-3a}$$

If:

$$x-\frac{1}{x}=a$$

then:

$$\boxed{x^2 + \frac{1}{x^2} = a^2 + 2}$$

67. FAST EXAM DECISION TREE

Expression has $(a^2 - b^2)$?

Factor immediately.

Looks like $(a^2 \pm 2ab + b^2)$?

Recognize perfect square.

Given $(x+y)$ and (xy) ?

Use symmetric identities.

Given $(x + 1/x)$?

Square first.

Rational expression?

Factor numerator and denominator before cancelling.

Large numbers near same base?

Try:

$$(a-b)(a+b)$$

or:

$$(B-d)(B+d) = B^2 - d^2$$

PRACTICE — LEVEL 0

Q1

Expand:

$$(x+4)^2$$

Q2

Factor:

$$x^2 - 25$$

Q3

If $(x+y=10)$, $(xy=16)$, find (x^2+y^2) .

Q4

If:

$$x + \frac{1}{x} = 6$$

find:

$$x^2 + \frac{1}{x^2}$$

Q5

Compute:

$$\sqrt[103]{97}$$

LEVEL 1

Q6

Factor:

$$x^2+7x+12$$

Q7

Factor:

$$2x^2+7x+3$$

Q8

If:

$$x+y=7, \quad xy=10$$

find:

$$x^3+y^3$$

Q9

If:

$$x+\frac{1}{x}=4$$

find:

$$x^3+\frac{1}{x^3}$$

Q10

Simplify:

$$\frac{x^2-9}{x+3}$$

LEVEL 2

Q11

If:

$$a+b=12, \quad a-b=6$$

find (ab) .

Q12

If:

$$x+y=11, \quad xy=24$$

find:

$$\left[\frac{x}{y} + \frac{y}{x} \right]$$

Q13

If:

$$\left[x - \frac{1}{x} = 5 \right]$$

find:

$$\left[x^2 + \frac{1}{x^2} \right]$$

Q14

If:

$$\left[x + \frac{1}{x} = 3 \right]$$

find:

$$\left[x^4 + \frac{1}{x^4} \right]$$

Q15

Simplify:

$$\left[\frac{x^2 - 4x + 4}{x^2 - 4} \right]$$

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$\left[x + y = 8, \text{quad } xy = 15 \right]$$

find:

$$\left[x^4 + y^4 \right]$$

Q17

If:

$$\left[x + \frac{1}{x} = 5 \right]$$

find:

$$\left[x^4 + \frac{1}{x^4} \right]$$

Q18

If:

$$\left[a + b + c = 0, \text{quad } abc = 12 \right]$$

find:

$$\left[a^3 + b^3 + c^3 \right]$$

Q19

Compute mentally:

$$\left[1004^2 - 996^2 \right]$$

Q20

Simplify:

$$\sqrt{\frac{x^3-8}{x^2+2x+4}}$$

ANSWERS

Q1

$$\sqrt{\boxed{x^2+8x+16}}$$

Q2

$$\sqrt{\boxed{(x-5)(x+5)}}$$

Q3

$$\sqrt{100-32} = \boxed{68}$$

Q4

$$\sqrt{36-2} = \boxed{34}$$

Q5

$$\sqrt{(100+3)(100-3)} = 10000-9 = \boxed{9991}$$

Q6

$$\sqrt{\boxed{(x+3)(x+4)}}$$

Q7

$$\sqrt{2x^2+6x+x+3}$$

$$\sqrt{=\boxed{(2x+1)(x+3)}}$$

Q8

$$\sqrt{7^3-3(10)(7)} = 343-210 = \boxed{133}$$

Q9

$$\sqrt{4^3-3(4)} = 64-12 = \boxed{52}$$

Q10

$$\sqrt{\frac{(x-3)(x+3)}{x+3}}$$

$$\sqrt{=\boxed{x-3}}$$

with:

$$\sqrt{x \neq -3}$$

Q11

$$\sqrt{4ab=(12)^2-(6)^2}$$

$$\sqrt{4ab=144-36=108}$$

$$\sqrt{\boxed{ab=27}}$$

Q12

$$\sqrt{x^2+y^2}=121-48=73 \quad \backslash$$

Therefore:

$$\sqrt{\frac{x}{y}+\frac{y}{x}} = \frac{73}{24} \quad \backslash$$

$$\sqrt{\boxed{\frac{73}{24}}} \quad \backslash$$

Q13

$$\sqrt{25+2} = \boxed{27} \quad \backslash$$

Q14

First:

$$\sqrt{x^2+\frac{1}{x^2}}=9-2=7 \quad \backslash$$

Then:

$$\sqrt{x^4+\frac{1}{x^4}} = 7^2-2 = \boxed{47} \quad \backslash$$

Q15

$$\sqrt{x^2-4x+4}=(x-2)^2 \quad \backslash$$

$$\sqrt{x^2-4}=(x-2)(x+2) \quad \backslash$$

So:

$$\sqrt{\boxed{\frac{x-2}{x+2}}} \quad \backslash$$

for:

$$\sqrt{x \neq \pm 2} \quad \backslash$$

Q16

First:

$$\sqrt{x^2+y^2} = 64-30 = 34 \quad \backslash$$

Then:

$$\sqrt{x^4+y^4} = (x^2+y^2)^2-2x^2y^2 \quad \backslash$$

$$\sqrt{=34^2-2(15^2)} \quad \backslash$$

$$\sqrt{=1156-450} \quad \backslash$$

$$\sqrt{=\boxed{706}} \quad \backslash$$

Q17

$$\sqrt{x^2+\frac{1}{x^2}} = 25-2=23 \quad \backslash$$

Then:

$$\sqrt{x^4+\frac{1}{x^4}} = 23^2-2 \quad \backslash$$

$$\sqrt{=529-2} \quad \backslash$$

$$\sqrt{=\boxed{527}} \quad \backslash$$

Q18

Since:

$$\boxed{a+b+c=0}$$

$$\boxed{a^3+b^3+c^3=3abc}$$

$$\boxed{=3(12)}$$

$$\boxed{=36}$$

Q19

$$\boxed{(1004-996)(1004+996)}$$

$$\boxed{=8(2000)}$$

$$\boxed{=16000}$$

Q20

$$\boxed{x^3-8 = (x-2)(x^2+2x+4)}$$

Cancel:

$$\boxed{x-2}$$

provided:

$$\boxed{x^2+2x+4 \neq 0}$$

For real x , that denominator is always positive.

CHAPTER 34 STATUS

Covered so far:

- square identities
- difference of squares
- cube identities
- sum/difference of cubes
- $(a^3+b^3+c^3-3abc)$ identity
- quadratic factorization
- rational-expression cancellation
- domain restrictions
- sum/product substitutions
- $(x+\frac{1}{x})$ patterns
- $(x-\frac{1}{x})$ patterns
- reciprocal powers
- mental-math identity shortcuts
- GATE/placement practice

But **Chapter 34 is not complete yet.**

Next we still need:

- advanced rational simplification
- algebraic fractions
- nested fractions
- substitution with $(x+y+z)$

- symmetric 3-variable expressions
- hidden-factor patterns
- surd-style algebraic simplification
- algebraic inequalities preview
- option substitution / back-solving
- final speed tricks and chapter audit

Progress

Completed: 33/76 Current: 0/31 Chapter 34 — Algebraic Identities & Simplification

NEXT

0/1 Chapter 34 Part 2 — Advanced rational simplification, symmetric expressions & exam-speed algebra

USER

next

ASSISTANT

CHAPTER 34 — ALGEBRAIC IDENTITIES AND SIMPLIFICATION

PART 2 — ADVANCED RATIONAL SIMPLIFICATION, SYMMETRIC EXPRESSIONS, HIDDEN FACTORS & EXAM-SPEED ALGEBRA

Status: 0/1 PROGRESS GATE: High Placements: High Priority: 0/1 Per 1

Part 1 covered the standard identities.

Part 2 focuses on the more exam-oriented question:

How do we simplify a complicated algebraic expression without doing unnecessary expansion?

The main strategy is:

$\boxed{\text{Factor first ! ' substitute smartly ! ' cancel only factors ! ' expand last}}$

1. GOLDEN RULE — DO NOT EXPAND FIRST

Consider:

$$\frac{x^2-16}{x^2-8x+16}$$

If you expand or manipulate randomly, it looks messy.

Factor:

$$x^2-16=(x-4)(x+4)$$

$$x^2-8x+16=(x-4)^2$$

Therefore:

$$\boxed{\frac{x+4}{x-4}}$$

with:

$$x \neq 4$$

2. CANCELLATION WORKS ONLY ON FACTORS

Wrong:

$$\frac{x+3}{x}$$

You cannot cancel (x) .

Correct example:

$$\frac{x(x+3)}{x}$$

Then:

$$\boxed{x+3}$$

for:

$$x \neq 0$$

MEMORY TRICK

Cancel multiplication, not addition.

3. COMPLEX ALGEBRAIC FRACTION

Simplify:

$$\frac{x^2-1}{x^2+2x+1}$$

Factor:

$$x^2-1=(x-1)(x+1)$$

$$x^2+2x+1=(x+1)^2$$

Therefore:

$$\boxed{\frac{x-1}{x+1}}$$

with:

$$x \neq -1$$

4. MULTIPLE FACTORS

Simplify:

$$\frac{x^2-9}{x^2-6x+9} \cdot \frac{x-3}{x+3}$$

Factor:

$$\left[\frac{(x-3)(x+3)}{(x-3)^2} \cdot \frac{x-3}{x+3} \right]$$

Everything cancels:

$$\left[\boxed{1} \right]$$

for original allowed values:

$$\left[x \neq -3, -3 \right]$$

5. DIVISION OF ALGEBRAIC FRACTIONS

Example:

$$\left[\frac{x^2-4}{x+3} \div \frac{x-2}{x+3} \right]$$

Change division to multiplication by reciprocal:

$$\left[\frac{(x-2)(x+2)}{x+3} \cdot \frac{x+3}{x-2} \right]$$

Cancel:

$$\left[\boxed{x+2} \right]$$

Restrictions come from the original expression:

$$\left[x \neq -3, 2 \right]$$

6. WHY $(x=2)$ IS ALSO RESTRICTED

Because the divisor:

$$\left[\frac{x-2}{x+3} \right]$$

cannot equal zero when we divide by it.

So:

$$\left[x-2 \neq 0 \right]$$

Hence:

$$\left[x \neq 2 \right]$$

7. ADDING ALGEBRAIC FRACTIONS

Simplify:

$$\left[\frac{1}{x+1} + \frac{1}{x-1} \right]$$

LCM:

$$\left[(x+1)(x-1) \right]$$

Numerator:

$$\left[(x-1) + (x+1) = 2x \right]$$

Therefore:

$$\left[\boxed{\frac{2x}{x^2-1}} \right]$$

with:

$$\lfloor x \neq \pm 1 \rfloor$$

8. SUBTRACTING ALGEBRAIC FRACTIONS

$$\lfloor \frac{1}{x-1} - \frac{1}{x+1} \rfloor$$

Numerator:

$$\lfloor (x+1) - (x-1) = 2 \rfloor$$

Therefore:

$$\lfloor \boxed{\frac{2}{x^2-1}} \rfloor$$

9. HIGH-VALUE PAIR

Memorize the pattern:

$$\lfloor \boxed{\frac{1}{x-a} + \frac{1}{x+a} = \frac{2x}{x^2-a^2}} \rfloor$$

and:

$$\lfloor \boxed{\frac{1}{x-a} - \frac{1}{x+a} = \frac{2a}{x^2-a^2}} \rfloor$$

10. NESTED FRACTION

Simplify:

$$\lfloor \frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}} \rfloor$$

Top:

$$\lfloor \frac{x+y}{xy} \rfloor$$

Bottom:

$$\lfloor \frac{y-x}{xy} \rfloor$$

So:

$$\lfloor \frac{(x+y)/xy}{(y-x)/xy} \rfloor$$

Cancel $\backslash(xy)$:

$$\lfloor \boxed{\frac{x+y}{y-x}} \rfloor$$

11. FASTER METHOD FOR NESTED FRACTIONS

Multiply numerator and denominator by the common denominator.

For:

$$\lfloor \frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}} \rfloor$$

multiply top and bottom by $\backslash(xy)$:

$$\lfloor \frac{y+x}{y-x} \rfloor$$

Directly:

$$\boxed{\frac{x+y}{y-x}}$$

This is faster.

12. ANOTHER NESTED FRACTION

Simplify:

$$\frac{1+\frac{1}{x}}{1-\frac{1}{x}}$$

Multiply numerator and denominator by (x) :

$$\boxed{\frac{x+1}{x-1}}$$

Much faster than separate fraction manipulation.

13. GENERAL NESTED-FRACTION TRICK

If expression contains many denominators, multiply the **entire numerator and denominator** by their LCM.

Do not individually simplify every small fraction unless needed.

14. GIVEN $(x+\frac{1}{x}=a)$: HIGHER POWERS

We already have:

$$S_1=a$$

$$S_2=a^2-2$$

$$S_3=a^3-3a$$

where:

$$S_n=x^n+\frac{1}{x^n}$$

For the next power:

$$S_4=aS_3-S_2$$

$$=a(a^3-3a)-(a^2-2)$$

$$\boxed{a^4-4a^2+2}$$

15. FIFTH POWER

Using recurrence:

$$S_5=aS_4-S_3$$

$$=a(a^4-4a^2+2)-(a^3-3a)$$

$$\boxed{a^5-5a^3+5a}$$

You do not need to memorize these if recurrence is understood.

16. EXAMPLE

If:

$$\left[x + \frac{1}{x} = 2 \right]$$

then:

$$\left[x^2 + \frac{1}{x^2} = 4 - 2 = 2 \right]$$

In fact $(x=1)$, so all:

$$\left[x^n + \frac{1}{x^n} = 2 \right]$$

This is also a sanity check.

17. IMPORTANT EXISTENCE CONDITION

For real positive (x) :

$$\left[x + \frac{1}{x} \geq 2 \right]$$

So if a question states:

$$\left[x > 0, \quad x + \frac{1}{x} = 1.5 \right]$$

there is no real positive solution.

This follows from AM-GM, but can also be seen algebraically.

18. ALGEBRAIC PROOF

$$\left[\left(x - 1 \right)^2 \geq 0 \right]$$

$$\left[x^2 - 2x + 1 \geq 0 \right]$$

Divide by $(x > 0)$:

$$\left[x + \frac{1}{x} \geq 2 \right]$$

Useful option-elimination rule.

19. SYMMETRIC THREE-VARIABLE EXPRESSIONS

Suppose:

$$\left[x + y + z = S \right]$$

$$\left[xy + yz + zx = P \right]$$

Then:

$$\left[(x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) \right]$$

Therefore:

$$\left[\boxed{x^2 + y^2 + z^2 = S^2 - 2P} \right]$$

20. EXAMPLE

If:

$$\boxed{x+y+z=10}$$

and:

$$\boxed{xy+yz+zx=21}$$

then:

$$\boxed{x^2+y^2+z^2 = 100-42 = \boxed{58}}$$

21. CUBE OF THREE-VARIABLE SUM

A useful identity:

$$\boxed{(x+y+z)^3 = x^3+y^3+z^3 +3(x+y)(y+z)(z+x)}$$

Another more systematic form:

$$\boxed{x^3+y^3+z^3 = (x+y+z)^3 -3(x+y+z)(xy+yz+zx) +3xyz}$$

Thus:

$$\boxed{x^3+y^3+z^3 = S^3-3SP+3R}$$

where:

$$\boxed{R=xyz}$$

22. EXAMPLE

Given:

$$\boxed{x+y+z=6}$$

$$\boxed{xy+yz+zx=11}$$

$$\boxed{xyz=6}$$

Then:

$$\boxed{x^3+y^3+z^3 = 216-3(6)(11)+18}$$

$$\boxed{=216-198+18}$$

$$\boxed{= \boxed{36}}$$

23. PRODUCT $\boxed{(x+y)(y+z)(z+x)}$

Useful identity:

$$\boxed{(x+y)(y+z)(z+x) = (x+y+z)(xy+yz+zx)-xyz}$$

This avoids full expansion.

24. EXAMPLE

If:

$$\backslash[x+y+z=6 \backslash]$$

$$\backslash[xy+yz+zx=11 \backslash]$$

$$\backslash[xyz=6 \backslash]$$

then:

$$\backslash[(x+y)(y+z)(z+x) = 6(11)-6 \backslash]$$

$$\backslash[=\boxed{60} \backslash]$$

25. SPECIAL CASE $\backslash(x+y+z=0\backslash)$

Then:

$$\backslash[\boxed{x^3+y^3+z^3=3xyz} \backslash]$$

Also:

$$\backslash[z=-(x+y) \backslash]$$

can simplify many expressions dramatically.

26. EXAMPLE

If:

$$\backslash[x+y+z=0 \backslash]$$

find:

$$\backslash[(x+y)(y+z)(z+x) \backslash]$$

Since:

$$\backslash[x+y=-z \backslash]$$

$$\backslash[y+z=-x \backslash]$$

$$\backslash[z+x=-y \backslash]$$

Product:

$$\backslash[(-z)(-x)(-y) = -!xyz \backslash]$$

Therefore:

$$\backslash[\boxed{-xyz} \backslash]$$

27. HIDDEN FACTOR: GROUPING

Factor:

$$\backslash[ax+ay+bx+by \backslash]$$

Group:

$$\backslash[a(x+y)+b(x+y) \backslash]$$

$$\backslash[=\boxed{(a+b)(x+y)} \backslash]$$

Look for repeated brackets.

28. EXAMPLE

$$\backslash[x^3+x^2+x+1 \backslash]$$

Group:

$$\backslash[x^2(x+1)+1(x+1) \backslash]$$

$$\backslash[=(x+1)(x^2+1) \backslash]$$

So:

$$\backslash[\boxed{(x+1)(x^2+1)} \backslash]$$

29. ANOTHER GROUPING

$$\backslash[x^3-x^2+x-1 \backslash]$$

$$\backslash[=x^2(x-1)+1(x-1) \backslash]$$

$$\backslash[=\boxed{(x-1)(x^2+1)} \backslash]$$

30. FOUR-TERM FACTORIZATION

$$\backslash[ab+ac+db+dc \backslash]$$

Group:

$$\backslash[a(b+c)+d(b+c) \backslash]$$

$$\backslash[=\boxed{(a+d)(b+c)} \backslash]$$

31. HIDDEN DIFFERENCE OF SQUARES

Simplify:

$$\backslash[(x+y)^2-(x-y)^2 \backslash]$$

Think of:

$$\backslash[A^2-B^2 \backslash]$$

where:

$$\backslash[A=x+y,\quad B=x-y \backslash]$$

Then:

$$\backslash[(A-B)(A+B) \backslash]$$

$$\backslash[=(2y)(2x) \backslash]$$

$$\backslash[=\boxed{4xy} \backslash]$$

32. HIDDEN PERFECT SQUARE

$$\left[x^2 + \frac{1}{x^2} + 2 \right]$$

Recognize:

$$\left[\boxed{\left(x + \frac{1}{x} \right)^2} \right]$$

Similarly:

$$\left[x^2 + \frac{1}{x^2} - 2 = \boxed{\left(x - \frac{1}{x} \right)^2} \right]$$

33. RATIONALIZING SIMPLE SURDS

Example:

$$\left[\frac{1}{\sqrt{3}+1} \right]$$

Multiply numerator and denominator by conjugate:

$$\left[\sqrt{3}-1 \right]$$

$$\left[= \frac{\sqrt{3}-1}{(\sqrt{3}-1)(\sqrt{3}+1)} \right]$$

$$\left[= \boxed{\frac{\sqrt{3}-1}{2}} \right]$$

34. CONJUGATE IDEA

For:

$$\left[a + \sqrt{b} \right]$$

conjugate is:

$$\left[a - \sqrt{b} \right]$$

Because:

$$\left[(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \right]$$

The radical disappears.

35. EXAMPLE

$$\left[\frac{1}{3 + \sqrt{5}} \right]$$

Multiply by:

$$\left[3 - \sqrt{5} \right]$$

$$\left[= \frac{3 - \sqrt{5}}{(3 - \sqrt{5})(3 + \sqrt{5})} \right]$$

$$\left[= \boxed{\frac{3 - \sqrt{5}}{4}} \right]$$

36. SURD PAIR SUM

Let:

$$\sqrt{x} = \sqrt{a} + \sqrt{b}$$

Then its conjugate:

$$\sqrt{y} = \sqrt{a} - \sqrt{b}$$

Useful:

$$xy = a - b$$

and:

$$x^2 + y^2 = 2(a + b)$$

37. EXAMPLE

$$(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2})$$

$$= 5 - 2$$

$$= \boxed{3}$$

38. NESTED SURD RECOGNITION

Simplify:

$$\sqrt{9 + 4\sqrt{5}}$$

Assume:

$$\sqrt{9 + 4\sqrt{5}} = \sqrt{a} + \sqrt{b}$$

Then:

$$a + b = 9$$

and:

$$2\sqrt{ab} = 4\sqrt{5}$$

So:

$$ab = 20$$

Numbers:

$$a = 5, \text{quad } b = 4$$

Thus:

$$\boxed{\sqrt{5} + 2}$$

39. ANOTHER NESTED SURD

$$\sqrt{7 - 4\sqrt{3}}$$

Try:

$$\sqrt{a} - \sqrt{b}$$

Need:

$$\begin{cases} a+b=7 \\ ab=12 \end{cases}$$

So:

$$a=4, \text{quad } b=3$$

Therefore:

$$\boxed{2-\sqrt{3}}$$

since expression is positive.

40. SUBSTITUTION TO REDUCE DEGREE

Suppose:

$$x^4+5x^2+6$$

Let:

$$y=x^2$$

Then:

$$y^2+5y+6$$

Factor:

$$(y+2)(y+3)$$

Return:

$$\boxed{(x^2+2)(x^2+3)}$$

41. BIQUADRATIC PATTERN

Expressions like:

$$x^4+ax^2+b$$

often become ordinary quadratics by putting:

$$\boxed{y=x^2}$$

42. EXAMPLE

$$x^4-5x^2+4$$

Let:

$$y=x^2$$

$$y^2-5y+4 = (y-1)(y-4)$$

Thus:

$$(x^2-1)(x^2-4)$$

Further:

$$\boxed{(x-1)(x+1)(x-2)(x+2)}$$

43. SUBSTITUTION FOR REPEATED BRACKET

Simplify:

$$(x+1)^2 + 5(x+1) + 6$$

Let:

$$y = x + 1$$

Then:

$$y^2 + 5y + 6$$

$$= (y+2)(y+3)$$

Return:

$$\boxed{(x+3)(x+4)}$$

This is much faster than expansion.

44. HIDDEN COMMON EXPRESSION

Whenever the same bracket appears repeatedly, treat it as one variable.

Examples:

- $(x+1)$
- $(x-2)$
- $(x + \frac{1}{x})$
- $(x^2 + x)$

This is an important speed technique.

45. OPTION SUBSTITUTION

If an algebraic identity question gives options and asks which expression is equivalent, sometimes substitute an easy value.

Example options for:

$$(x+1)^2 - (x-1)^2$$

Instead of fully expanding, choose $x=2$:

Expression:

$$9 - 1 = 8$$

Check which option gives 8.

But use this carefully:

One test value can eliminate wrong options, but may not prove equality if multiple options coincide.

Use 2 values if needed.

46. BACK-SOLVING FROM OPTIONS

If a question asks for $\sqrt{x}$ and gives simple numerical options, test them directly in the original equation.

This is especially useful when:

- factoring is awkward
- equations contain reciprocals
- options are few

But Chapter 35 will cover this more deeply.

47. SIGN CHECK BEFORE SQUARING

Suppose:

$$\sqrt{x} = x - 2$$

Squaring gives:

$$x = (x - 2)^2$$

But the original requires:

$$x - 2 \geq 0$$

So:

$$x \geq 2$$

Algebraic transformations can create extraneous solutions.

Important future equation-solving trap.

48. SIMPLIFY BEFORE SUBSTITUTING NUMBERS

Suppose:

$$\frac{x^2 - 100}{x - 10}$$

and $x = 10.1$.

Don't square 10.1.

Factor:

$$\frac{(x - 10)(x + 10)}{x - 10} = x + 10$$

Then:

$$10.1 + 10 = \boxed{20.1}$$

49. APPROXIMATION VIA IDENTITIES

Compute:

$$\sqrt{1001^2}$$

Use:

$$\sqrt{(1000+1)^2}$$

$$\sqrt{=1000000+2000+1}$$

$$\sqrt{=\boxed{1002001}}$$

50. FAST PRODUCT NEAR SAME BASE

$$\sqrt{1008 \times 992}$$

$$\sqrt{=(1000+8)(1000-8)}$$

$$\sqrt{=1000000-64}$$

$$\sqrt{=\boxed{999936}}$$

51. PRODUCT WITH SAME OFFSET

$$\sqrt{103 \times 107}$$

Choose base 105:

$$\sqrt{(105-2)(105+2)}$$

$$\sqrt{=105^2-4}$$

$$\sqrt{=11025-4}$$

$$\sqrt{=\boxed{11021}}$$

52. IDENTITY-BASED COMPARISON

Compare:

$$\sqrt{51^2}$$

and:

$$\sqrt{50 \cdot 52}$$

Since:

$$\sqrt{50 \cdot 52 = (51-1)(51+1)}$$

$$\sqrt{=51^2-1}$$

Therefore:

$$\sqrt{\boxed{51^2} \text{ is greater by } 1}$$

No full multiplication.

53. COMMON ERROR — OVEREXPANSION

Expression:

$$\sqrt{(a+b+c)^2 - (a-b-c)^2}$$

Do not expand six terms.

Use difference of squares:

$$\sqrt{[(a+b+c)-(a-b-c)][(a+b+c)+(a-b-c)]}$$

$$= \sqrt{2(b+c)}$$

$$= \boxed{4a(b+c)}$$

54. GATE-STYLE QUESTION 1

Simplify:

$$\sqrt{\frac{1+\frac{1}{x}}{1-\frac{1}{x}}}$$

Multiply by $\sqrt{x/x}$:

$$\sqrt{\boxed{\frac{x+1}{x-1}}}$$

55. GATE-STYLE QUESTION 2

If:

$$\sqrt{x+y+z=9}$$

and:

$$\sqrt{xy+yz+zx=23}$$

find:

$$\sqrt{x^2+y^2+z^2}$$

$$\sqrt{=81-46}$$

$$\sqrt{= \boxed{35}}$$

56. GATE-STYLE QUESTION 3

If:

$$\sqrt{x+\frac{1}{x}=4}$$

find:

$$\sqrt{x^4+\frac{1}{x^4}}$$

First:

$$\sqrt{x^2+\frac{1}{x^2}=14}$$

Then:

$$\sqrt{14^2-2 = \boxed{194}}$$

57. GATE-STYLE QUESTION 4

Simplify:

$$\sqrt{\sqrt{9+4\sqrt{5}}}$$

$$=\boxed{2+\sqrt{5}}$$

58. PLACEMENT-STYLE QUESTION 1

Factor:

$$x^4-5x^2+4$$

$$=\boxed{(x-1)(x+1)(x-2)(x+2)}$$

59. PLACEMENT-STYLE QUESTION 2

Simplify:

$$\frac{1}{\sqrt{5}+2}$$

Conjugate:

$$\sqrt{5}-2$$

Denominator:

$$5-4=1$$

So:

$$\boxed{\sqrt{5}-2}$$

60. PLACEMENT-STYLE QUESTION 3

If:

$$x+y+z=6$$

$$xy+yz+zx=11$$

$$xyz=6$$

find:

$$(x+y)(y+z)(z+x)$$

$$=6(11)-6$$

$$=\boxed{60}$$

61. PLACEMENT-STYLE QUESTION 4

Compute:

$$\sqrt{1006^2 - 994^2}$$

$$\sqrt{=(12)(2000)}$$

$$\sqrt{=\boxed{24000}}$$

62. COMMON TRAPS

& pTRAP 1 — Cancel terms

Wrong:

$$\sqrt{\frac{x+2}{x}} \neq 2$$

& pTRAP 2 — Forget domain restrictions

After cancellation, original excluded values remain excluded.

& pTRAP 3 — Overexpand nested expressions

First substitute repeated brackets.

& pTRAP 4 — Use arithmetic mean for reciprocal patterns

Recognize identities instead.

& pTRAP 5 — Rationalization changes sign

Always multiply by exact conjugate.

& pTRAP 6 — Squaring equations without checking

Squaring may introduce false solutions.

63. MUST-MEMORIZE THREE-VARIABLE IDENTITIES

$$\sqrt{\boxed{x^2 + y^2 + z^2 = (x+y+z)^2 - 2(xy+yz+zx)}}$$

$$\sqrt{\boxed{x^3 + y^3 + z^3 = S^3 - 3SP + 3R}}$$

where:

$$\sqrt{S=x+y+z, \quad P=xy+yz+zx, \quad R=xyz}$$

and:

$$\sqrt{\boxed{(x+y)(y+z)(z+x)=SP-R}}$$

64. MUST-MEMORIZE RATIONALIZATION IDEA

$$\sqrt{\boxed{(a+b)(a-b)=a^2-b^2}}$$

So for:

$$\sqrt{\frac{1}{a+\sqrt{b}}}$$

multiply by:

$$\sqrt{a - \sqrt{b}}$$

65. EXAM-SPEED TOOLKIT

Before doing algebra, ask in this order:

1. Common factor?

Factor it.

2. Standard identity?

Recognize it.

3. Repeated bracket?

Substitute one variable.

4. Reciprocal pair?

Try $\sqrt{x+1/x}$ identities.

5. Symmetric variables?

Use sum/product data.

6. Rational fraction?

Factor before cancellation.

7. Surd denominator?

Use conjugate.

8. Numerical values near a base?

Use difference-of-squares or square identity.

PRACTICE — LEVEL 0

Q1

Simplify:

$$\sqrt{\frac{x^2-4}{x-2}}$$

Q2

Simplify:

$$\sqrt{\frac{1}{x-2} + \frac{1}{x+2}}$$

Q3

If:

$$\sqrt{x+y+z=8, \quad xy+yz+zx=15}$$

find:

Q12

If:

$$\begin{cases} x+y+z=10, \\ xy+yz+zx=29 \end{cases}$$

find:

$$x^2+y^2+z^2$$

Q13

If:

$$\begin{cases} x+y+z=0, \\ xyz=-8 \end{cases}$$

find:

$$x^3+y^3+z^3$$

Q14

Factor completely:

$$x^4-13x^2+36$$

Q15

Rationalize:

$$\frac{1}{4+\sqrt{7}}$$

LEVEL 3 — EXAM PRESSURE

Q16

If:

$$x+\frac{1}{x}=3$$

find:

$$x^5+\frac{1}{x^5}$$

Q17

If:

$$\begin{cases} x+y+z=7, \\ xy+yz+zx=14, \\ xyz=8 \end{cases}$$

find:

$$(x+y)(y+z)(z+x)$$

Q18

Simplify:

$$\frac{(x+1)^2-9}{x-2}$$

Q19

Compute mentally:

$$1007 \times 993$$

Q20

Simplify:

$$\sqrt{\frac{x^4-16}{x^2-4}}$$

ANSWERS**Q1**

$$\sqrt{\frac{(x-2)(x+2)}{x-2}} = \boxed{x+2}$$

with:

$$\sqrt{x \neq 2}$$

Q2

$$\sqrt{\frac{(x+2)+(x-2)}{x^2-4}}$$

$$\sqrt{=\boxed{\frac{2x}{x^2-4}}}$$

Q3

$$\sqrt{64-30} = \boxed{34}$$

Q4

$$\sqrt{\frac{\sqrt{3}-1}{3-1}} = \boxed{\frac{\sqrt{3}-1}{2}}$$

Q5

Let:

$$\sqrt{y=x^2}$$

$$\sqrt{y^2+5y+6} = (y+2)(y+3)$$

So:

$$\sqrt{\boxed{(x^2+2)(x^2+3)}}$$

Q6Multiply top/bottom by $\sqrt{x}$:

$$\sqrt{\boxed{\frac{x+2}{x-2}}}$$

Q7

First:

$$\sqrt{x^2+\frac{1}{x^2}}=25-2=23$$

Then:

$$\sqrt{23^2-2} = \boxed{527}$$

Q8

$$\sqrt{6^3-3(6)(11)+3(6)}$$

$$\sqrt{216-198+18} = \boxed{36}$$

Q9

$$\boxed{2-\sqrt{3}}$$

because:

$$(2-\sqrt{3})^2 = 7-4\sqrt{3}$$

Q10

$$x^2(x+1)+(x+1)$$

$$=\boxed{(x+1)(x^2+1)}$$

Q11

Multiply numerator and denominator by (xy) :

$$\boxed{\frac{x+y}{y-x}}$$

Q12

$$100-58 = \boxed{42}$$

Q13

Since:

$$x+y+z=0$$

$$x^3+y^3+z^3=3xyz$$

$$=3(-8)$$

$$=\boxed{-24}$$

Q14

Let:

$$y=x^2$$

$$y^2-13y+36$$

$$=(y-4)(y-9)$$

Thus:

$$(x^2-4)(x^2-9)$$

$$=\boxed{(x-2)(x+2)(x-3)(x+3)}$$

Q15

$$\frac{1}{4+\sqrt{7}} \cdot \frac{4-\sqrt{7}}{4-\sqrt{7}}$$

$$= \frac{4-\sqrt{7}}{16-7}$$

$$=\boxed{\frac{4-\sqrt{7}}{9}}$$

Q16

Use:

$$S_3=3^3-3(3)=18$$

$$S_4=3(18)-7=47$$

because:

$$\backslash [S_2=7 \backslash]$$

Then:

$$\backslash [S_5=3(47)-18 \backslash]$$

$$\backslash [=141-18 \backslash]$$

$$\backslash [=\boxed{123} \backslash]$$

Q17

$$\backslash [(x+y)(y+z)(z+x) = SP-R \backslash]$$

$$\backslash [=7(14)-8 \backslash]$$

$$\backslash [=98-8 \backslash]$$

$$\backslash [=\boxed{90} \backslash]$$

Q18

Numerator:

$$\backslash [(x+1)^2-3^2 \backslash]$$

Difference of squares:

$$\backslash [[(x+1)-3][(x+1)+3] \backslash]$$

$$\backslash [=(x-2)(x+4) \backslash]$$

Divide by $\backslash (x-2) \backslash$:

$$\backslash [\boxed{x+4} \backslash]$$

with:

$$\backslash [x \neq 2 \backslash]$$

Q19

$$\backslash [(1000+7)(1000-7) \backslash]$$

$$\backslash [=1000000-49 \backslash]$$

$$\backslash [=\boxed{999951} \backslash]$$

Q20

$$\backslash [x^4-16 = (x^2-4)(x^2+4) \backslash]$$

Therefore:

$$\backslash [\boxed{x^2+4} \backslash]$$

with:

$$\backslash [x \neq \pm 2 \backslash]$$

' CHAPTER 34 COMPLETE

We have now covered:

- square identities '
- cube identities '

- sum/difference of squares and cubes '
- perfect-square recognition '
- quadratic and higher-degree factorization '
- grouping '
- rational algebraic fractions '
- nested fractions '
- domain restrictions '
- repeated-expression substitution '
- $\sqrt{x+\frac{1}{x}}$ and $\sqrt{x-\frac{1}{x}}$ patterns '
- higher reciprocal powers '
- 2-variable symmetric expressions '
- 3-variable symmetric expressions '
- $(x+y+z, xy+yz+zx, xyz)$ identities '
- surd rationalization '
- nested surds '
- biquadratic substitution '
- mental-math algebraic identities '
- option substitution/back-solving preview '
- exam-speed simplification '
- L0–L3 practice '

Progress

Completed: 34/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 35 — Linear and Simultaneous Equations

Next we will cover:

- one-variable linear equations
- brackets and fractions
- equations with variables on both sides
- translating word statements into equations
- two-variable simultaneous equations
- substitution method
- elimination method
- cross-multiplication shortcut
- ratio-based equations
- consistency/no-solution/infinite-solution cases
- age, money, number and mixture equation models
- option back-solving
- GATE/placement speed techniques.

USER

next

ASSISTANT

CHAPTER 35 — LINEAR AND SIMULTANEOUS EQUATIONS

PART 1 — ONE-VARIABLE EQUATIONS, FRACTIONS, WORD TRANSLATION, TWO-VARIABLE SYSTEMS & ELIMINATION

Status: $\emptyset = 0$ PROGRESS GATE: High Placements: High Priority: $\emptyset = 0$ Per 1

Linear equations are among the most useful aptitude tools because many topics eventually reduce to:

$\boxed{\text{unknown quantity} + \text{conditions} = \text{equation}}$

Main goal:

Convert words into equations correctly, then solve with the shortest valid method.

1. WHAT IS A LINEAR EQUATION?

A one-variable linear equation has the form:

$\boxed{ax+b=c}$

where the highest power of (x) is 1.

Example:

$\boxed{3x+5=20}$

2. BASIC SOLVING IDEA

$\boxed{3x+5=20}$

Subtract 5:

$\boxed{3x=15}$

Divide by 3:

$\boxed{x=5}$

Think:

Undo operations in reverse order.

3. VARIABLES ON BOTH SIDES

Example:

$$5x+3=2x+18$$

Bring variable terms together:

$$5x-2x=18-3$$

$$3x=15$$

$$\boxed{x=5}$$

4. SIGN DISCIPLINE

Suppose:

$$4x-7=2x+9$$

Move $(2x)$ left:

$$2x-7=9$$

Add 7:

$$2x=16$$

$$\boxed{x=8}$$

Common mistake: changing signs incorrectly while transposing.

5. BETTER WAY TO THINK ABOUT “TRANSPOSING”

Instead of memorizing “change side, change sign,” think:

Perform the same operation on both sides.

Example:

$$4x-7=2x+9$$

Subtract $(2x)$ from both sides:

$$2x-7=9$$

Add 7 to both sides.

This prevents sign errors.

6. EQUATIONS WITH BRACKETS

Example:

$$3(x+2)=18$$

Either divide first:

$$x+2=6$$

$$\boxed{x=4}$$

or expand:

$$\backslash[3x+6=18 \backslash]$$

Same result.

Use whichever is faster.

7. BRACKETS ON BOTH SIDES

$$\backslash[2(x+3)=3(x-1) \backslash]$$

Expand:

$$\backslash[2x+6=3x-3 \backslash]$$

$$\backslash[9=x \backslash]$$

$$\backslash[\boxed{x=9} \backslash]$$

8. FRACTIONAL EQUATIONS

Example:

$$\backslash[\frac{x}{3}+5=9 \backslash]$$

$$\backslash[\frac{x}{3}=4 \backslash]$$

$$\backslash[\boxed{x=12} \backslash]$$

9. MULTIPLE FRACTIONS

$$\backslash[\frac{x}{2}+\frac{x}{3}=10 \backslash]$$

LCM = 6.

Multiply entire equation by 6:

$$\backslash[3x+2x=60 \backslash]$$

$$\backslash[5x=60 \backslash]$$

$$\backslash[\boxed{x=12} \backslash]$$

FAST RULE

Remove fractions first by multiplying by the LCM.

10. FRACTION WITH CONSTANTS

$$\backslash[\frac{x-2}{3}+\frac{x+1}{2}=7 \backslash]$$

Multiply by 6:

$$\backslash[2(x-2)+3(x+1)=42 \backslash]$$

$$\backslash[2x-4+3x+3=42 \backslash]$$

$$\backslash[5x-1=42 \backslash]$$

$$\backslash[5x=43 \backslash]$$

$$\backslash[\boxed{x=\frac{43}{5}} \backslash]$$

11. DECIMAL EQUATIONS

$$\backslash[0.2x+1.5=3.5 \backslash]$$

$$\backslash[0.2x=2 \backslash]$$

$$\backslash[x=10 \backslash]$$

Or multiply everything by 10 first:

$$\backslash[2x+15=35 \backslash]$$

$$\backslash[\boxed{x=10} \backslash]$$

12. PERCENTAGE EQUATION

If 25% of a number is 40:

$$\backslash[0.25x=40 \backslash]$$

$$\backslash[x=\frac{40}{0.25} \backslash]$$

$$\backslash[\boxed{x=160} \backslash]$$

Or:

$$\backslash[\frac{x}{4}=40 \backslash]$$

$$\backslash[x=160 \backslash]$$

Fraction form is usually faster.

13. “A NUMBER INCREASED BY”

Suppose:

A number increased by 7 is 20.

Let number = $\backslash(x\backslash)$.

$$\backslash[x+7=20 \backslash]$$

$$\backslash[\boxed{x=13} \backslash]$$

14. “5 LESS THAN A NUMBER”

Means:

$$\backslash[\boxed{x-5} \backslash]$$

Example:

5 less than a number is 12.

$$\boxed{x-5=12}$$

$$\boxed{x=17}$$

15. LANGUAGE TRAP — “5 LESS THAN x ”

$$\boxed{x-5}$$

not:

$$\boxed{5-x}$$

But:

x is 5 less than y

means:

$$\boxed{x=y-5}$$

16. “TWICE A NUMBER”

$$\boxed{2x}$$

“Three times a number plus 4”:

$$\boxed{3x+4}$$

17. “ONE-THIRD OF A NUMBER”

$$\boxed{\frac{x}{3}}$$

“Two-thirds of a number”:

$$\boxed{\frac{2x}{3}}$$

18. NUMBER WORD PROBLEM

A number is 12 more than twice another number.

Let smaller number = y .

Then larger:

$$\boxed{x=2y+12}$$

This is an equation relation, not yet a full solution.

19. CONSECUTIVE INTEGERS

Let first integer:

$$\boxed{x}$$

Next:

$$\boxed{x+1}$$

Next:

$$\boxed{x+2}$$

Example:

Three consecutive integers sum to 42.

$$x+(x+1)+(x+2)=42$$

$$3x+3=42$$

$$3x=39$$

$$x=13$$

Numbers:

$$\boxed{13,14,15}$$

20. CONSECUTIVE EVEN INTEGERS

Represent as:

$$x, \quad x+2, \quad x+4$$

Example:

Sum = 54.

$$x+x+2+x+4=54$$

$$3x+6=54$$

$$x=16$$

Numbers:

$$\boxed{16,18,20}$$

21. CONSECUTIVE ODD INTEGERS

Same spacing:

$$x, \quad x+2, \quad x+4$$

just choose x odd.

Example sum 63:

$$3x+6=63$$

$$x=19$$

Numbers:

$$\boxed{19,21,23}$$

22. TWO-DIGIT NUMBER MODEL

Suppose tens digit = (x) , units digit = (y) .

Then number is:

$$\boxed{10x+y}$$

Reverse number:

$$\boxed{10y+x}$$

Very important.

23. DIGIT REVERSAL EXAMPLE

A two-digit number has digit sum 9.

The number is 27 greater than its reverse.

Let digits be (x,y) .

Then:

$$x+y=9$$

and:

$$(10x+y)-(10y+x)=27$$

$$9x-9y=27$$

$$x-y=3$$

Now solve:

$$x+y=9$$

$$x-y=3$$

Add:

$$2x=12$$

$$x=6$$

$$y=3$$

Number:

$$\boxed{63}$$

24. WHY REVERSAL DIFFERENCE IS MULTIPLE OF 9

$$(10x+y)-(10y+x) = 9(x-y)$$

So difference between a two-digit number and its reverse is always divisible by 9.

Useful shortcut.

25. INTRODUCTION TO SIMULTANEOUSEQUATIONS

Suppose:

$$\begin{cases} x+y=10 \end{cases}$$

and:

$$\begin{cases} x-y=2 \end{cases}$$

We need values satisfying both equations.

These are simultaneous equations.

26. ELIMINATION METHOD

$$\begin{cases} x+y=10 \end{cases}$$

$$\begin{cases} x-y=2 \end{cases}$$

Add:

$$\begin{cases} 2x=12 \end{cases}$$

$$\begin{cases} x=6 \end{cases}$$

Then:

$$\begin{cases} y=4 \end{cases}$$

$$\begin{cases} \boxed{x=6, y=4} \end{cases}$$

27. WHEN TO ADD OR SUBTRACT

Goal:

Make one variable disappear.

Example:

$$\begin{cases} 2x+3y=13 \end{cases}$$

$$\begin{cases} 2x-y=5 \end{cases}$$

Subtract second from first:

$$\begin{cases} 4y=8 \end{cases}$$

$$\begin{cases} y=2 \end{cases}$$

Then:

$$\begin{cases} 2x-2=5 \end{cases}$$

$$\begin{cases} 2x=7 \end{cases}$$

$$\begin{cases} \boxed{x=\frac{7}{2}, y=2} \end{cases}$$

28. MATCH COEFFICIENTS FIRST

Example:

$$\begin{cases} 2x+3y=17 \end{cases}$$

$$\begin{cases} 3x+2y=18 \end{cases}$$

Multiply first by 3:

$$\begin{cases} 6x+9y=51 \end{cases}$$

Second by 2:

$$\begin{cases} 6x+4y=36 \end{cases}$$

Subtract:

$$\begin{cases} 5y=15 \end{cases}$$

$$\begin{cases} y=3 \end{cases}$$

Then:

$$\begin{cases} 2x+9=17 \end{cases}$$

$$\begin{cases} x=4 \end{cases}$$

$$\boxed{x=4, y=3}$$

29. SUBSTITUTION METHOD

Example:

$$\begin{cases} x+y=12 \end{cases}$$

$$\begin{cases} 2x-y=3 \end{cases}$$

From first:

$$\begin{cases} y=12-x \end{cases}$$

Substitute:

$$\begin{cases} 2x-(12-x)=3 \end{cases}$$

$$\begin{cases} 3x-12=3 \end{cases}$$

$$\begin{cases} 3x=15 \end{cases}$$

$$\begin{cases} x=5 \end{cases}$$

$$\begin{cases} y=7 \end{cases}$$

$$\boxed{5, 7}$$

30. WHEN IS SUBSTITUTION BEST?

Use substitution when one equation already looks like:

$$\begin{cases} x=\cdots \end{cases}$$

or:

$$\begin{cases} y=\cdots \end{cases}$$

Example:

$$\backslash[x=2y+3 \backslash]$$

Then plug directly into the other equation.

31. WHEN IS ELIMINATION BEST?

Use elimination when coefficients are:

- already equal,
- opposites,
- or easy to make equal.

Example:

$$\backslash[3x+2y=20 \backslash]$$

$$\backslash[3x-4y=2 \backslash]$$

Subtract directly.

32. FAST METHOD — SUM AND DIFFERENCE

If:

$$\backslash[x+y=S \backslash]$$

and:

$$\backslash[x-y=D \backslash]$$

then:

$$\backslash[\boxed{x=\frac{S+D}{2}} \backslash]$$

$$\backslash[\boxed{y=\frac{S-D}{2}} \backslash]$$

No need to re-solve every time.

33. EXAMPLE

$$\backslash[x+y=18 \backslash]$$

$$\backslash[x-y=6 \backslash]$$

$$\backslash[x=\frac{24}{2}=12 \backslash]$$

$$\backslash[y=\frac{12}{2}=6 \backslash]$$

$$\backslash[\boxed{12,6} \backslash]$$

34. RATIO + SUM

Suppose:

$$\backslash[x:y=3:2 \backslash]$$

and:

$$\begin{cases} x+y=25 \end{cases}$$

Let:

$$\begin{cases} x=3k, \text{quad } y=2k \end{cases}$$

Then:

$$\begin{cases} 5k=25 \end{cases}$$

$$\begin{cases} k=5 \end{cases}$$

$$\boxed{x=15, y=10}$$

This is usually faster than forming two linear equations.

35. RATIO + DIFFERENCE

$$\begin{cases} x:y=5:3 \end{cases}$$

and:

$$\begin{cases} x-y=14 \end{cases}$$

Let:

$$\begin{cases} x=5k, \text{ } y=3k \end{cases}$$

$$\begin{cases} 2k=14 \end{cases}$$

$$\begin{cases} k=7 \end{cases}$$

$$\boxed{x=35, y=21}$$

36. MONEY EQUATION

A has ₹20 more than B.

Together they have ₹100.

Let B = ₹ x .

A:

$$\begin{cases} x+20 \end{cases}$$

Then:

$$\begin{cases} x+x+20=100 \end{cases}$$

$$\begin{cases} 2x=80 \end{cases}$$

$$\begin{cases} x=40 \end{cases}$$

So:

$$\boxed{A=60, B=40}$$

37. AGE EQUATION

Father is 24 years older than son.

Their ages sum to 60.

Let son = x .

Father:

$$x + 24$$

$$2x + 24 = 60$$

$$x = 18$$

Father:

$$42$$

$$\boxed{18, 42}$$

38. PRICE AND QUANTITY MODEL

Suppose:

- 3 pens + 2 books = ₹ 70
- 2 pens + 1 book = ₹ 40

Let pen = p , book = b .

$$3p + 2b = 70$$

$$2p + b = 40$$

Double second:

$$4p + 2b = 80$$

Subtract first:

$$p = 10$$

Then:

$$20 + b = 40$$

$$b = 20$$

$$\boxed{p=10, b=20}$$

39. MIXTURE-LIKE EQUATION

Suppose:

10 items consist of ₹ 5 and ₹ 8 items.

Total cost = ₹ 62.

Let number of ₹ 5 items = x .

Then ₹ 8 items:

$$10 - x$$

Total:

$$5x + 8(10 - x) = 62$$

$$\begin{cases} 5x+80-8x=62 \end{cases}$$

$$\begin{cases} -3x=-18 \end{cases}$$

$$\begin{cases} x=6 \end{cases}$$

Other:

$$\begin{cases} 4 \end{cases}$$

$$\boxed{6 \text{ and } 4}$$

40. CHICKENS AND LEGS TYPE

Suppose 20 animals consist of chickens and goats.

Total legs = 56.

Let chickens = x , goats = y .

$$\begin{cases} x+y=20 \end{cases}$$

$$\begin{cases} 2x+4y=56 \end{cases}$$

Divide second by 2:

$$\begin{cases} x+2y=28 \end{cases}$$

Subtract first:

$$\begin{cases} y=8 \end{cases}$$

Then:

$$\begin{cases} x=12 \end{cases}$$

$$\boxed{12 \text{ chickens}, 8 \text{ goats}}$$

41. FAST CHICKEN-GOAT SHORTCUT

If all 20 were chickens:

$$\begin{cases} 20(2)=40 \end{cases}$$

Extra legs:

$$\begin{cases} 56-40=16 \end{cases}$$

Each goat contributes 2 extra legs.

Goats:

$$\begin{cases} 16/2=8 \end{cases}$$

$$\boxed{8}$$

This is faster than equations.

42. CONSISTENCY OF TWO EQUATIONS

General pair:

$$\begin{cases} a_1x + b_1y = c_1 \end{cases}$$

$$\begin{cases} a_2x + b_2y = c_2 \end{cases}$$

There can be:

- one unique solution
- no solution
- infinitely many solutions

43. UNIQUE SOLUTION

If lines intersect once.

Algebraically:

$$\boxed{\frac{a_1}{a_2} \neq \frac{b_1}{b_2}}$$

Example:

$$\begin{cases} x + y = 4 \end{cases}$$

$$\begin{cases} x - y = 2 \end{cases}$$

Unique solution.

44. NO SOLUTION

If coefficients are proportional but constants are not:

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}}$$

Example:

$$\begin{cases} x + y = 4 \end{cases}$$

$$\begin{cases} 2x + 2y = 10 \end{cases}$$

First doubled would give:

$$\begin{cases} 2x + 2y = 8 \end{cases}$$

but second says 10.

Impossible.

$$\boxed{\text{No solution}}$$

45. INFINITELY MANY SOLUTIONS

If all ratios are equal:

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

Example:

$$\begin{cases} x + y = 4 \end{cases}$$

$$\begin{cases} 2x + 2y = 8 \end{cases}$$

Same equation repeated.

$\boxed{\text{Infinitely many solutions}}$

46. GEOMETRIC MEANING

Each linear equation in two variables represents a line.

Unique solution

Lines intersect.

No solution

Parallel distinct lines.

Infinite solutions

Same line.

This helps conceptual understanding.

47. CROSS-MULTIPLICATION FORMULA

For:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

One direct formula is:

$$\boxed{x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}}$$

$$\boxed{y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}}$$

Useful, but for simple aptitude questions elimination is usually faster.

48. EXAMPLE USING DIRECT FORMULA

$$2x + y = 7$$

$$x - y = 2$$

Here:

$$a_1 = 2, b_1 = 1, c_1 = 7$$

$$a_2 = 1, b_2 = -1, c_2 = 2$$

Denominator:

$$2(-1) - 1(1) = -3$$

$$x = \frac{7(-1) - 2(1)}{-3} = \frac{-9}{-3} = 3$$

$$y = 1$$

$$\boxed{3, 1}$$

But elimination would be faster.

49. EQUATION WITH PARAMETER

Example:

$$kx+2=8$$

If $k \neq 0$:

$$x = \frac{6}{k}$$

If $k=0$:

Equation becomes:

$$2=8$$

Impossible.

So parameter questions require checking special values.

50. LINEAR EQUATION SANITY CHECK

After solving, substitute back.

Example:

$$3x+5=20$$

We found:

$$x=5$$

Check:

$$15+5=20$$

Correct.

This catches sign mistakes quickly.

51. OPTION BACK-SOLVING

Suppose:

$$3x+7=31$$

Options:

6, 7, 8, 9.

Instead of solving, test likely option:

$$x=8$$

$$24+7=31$$

So:

$$\boxed{8}$$

Useful in placement tests when options are simple.

52. GATE-STYLE QUESTION 1

Solve:

$$\frac{x}{2} + \frac{x}{3} = 10$$

Multiply by 6:

$$3x + 2x = 60$$

$$x = \boxed{12}$$

53. GATE-STYLE QUESTION 2

$$x + y = 17$$

$$x - y = 5$$

$$x = \frac{22}{2} = 11$$

$$y = \frac{12}{2} = 6$$

$$\boxed{11, 6}$$

54. GATE-STYLE QUESTION 3

$$2x + 3y = 13$$

$$2x - y = 5$$

Subtract:

$$4y = 8$$

$$y = 2$$

$$x = \frac{7}{2}$$

$$\boxed{\frac{7}{2}, 2}$$

55. GATE-STYLE QUESTION 4

Determine number of solutions:

$$2x + 4y = 6$$

$$x + 2y = 3$$

Second $\times 2$:

$$2x + 4y = 6$$

Same equation.

$$\boxed{\text{Infinitely many solutions}}$$

56. PLACEMENT-STYLE QUESTION 1

Two numbers sum to 50 and differ by 14.

$$\begin{cases} x+y=50 \end{cases}$$

$$\begin{cases} x-y=14 \end{cases}$$

$$\begin{cases} x=32, \\ y=18 \end{cases}$$

$$\boxed{\begin{cases} 32, 18 \end{cases}}$$

57. PLACEMENT-STYLE QUESTION 2

A two-digit number's digits sum to 11.

The number is 27 more than its reverse.

$$\begin{cases} x+y=11 \end{cases}$$

$$\begin{cases} x-y=3 \end{cases}$$

Thus:

$$\begin{cases} x=7, \\ y=4 \end{cases}$$

Number:

$$\boxed{74}$$

58. PLACEMENT-STYLE QUESTION 3

3 pens and 2 books cost \$170.

2 pens and 1 book cost \$140.

$$\boxed{\begin{cases} \text{Pen} = \$10, \\ \text{Book} = \$20 \end{cases}}$$

59. PLACEMENT-STYLE QUESTION 4

20 animals, chickens and goats, 56 legs.

$$\boxed{\begin{cases} 12 \text{ chickens}, \\ 8 \text{ goats} \end{cases}}$$

60. ADVANCED QUESTION — FRACTIONS

Solve:

$$\frac{x-1}{2} + \frac{x+2}{3} = 5$$

Multiply by 6:

$$3(x-1) + 2(x+2) = 30$$

$$3x - 3 + 2x + 4 = 30$$

$$5x + 1 = 30$$

$$5x = 29$$

$$\boxed{x = \frac{29}{5}}$$

61. ADVANCED QUESTION — SIMULTANEOUS FRACTIONS

$$\frac{x}{2} + \frac{y}{3} = 5$$

$$\frac{x}{3} - \frac{y}{2} = -1$$

Multiply first by 6:

$$3x + 2y = 30$$

Second by 6:

$$2x - 3y = -6$$

Multiply first by 3:

$$9x + 6y = 90$$

Second by 2:

$$4x - 6y = -12$$

Add:

$$13x = 78$$

$$x = 6$$

Then:

$$18 + 2y = 30$$

$$y = 6$$

$$\boxed{x = y = 6}$$

62. ADVANCED QUESTION — PARAMETER CONSISTENCY

For what value of k do:

$$2x + 4y = 6$$

and:

$$x + 2y = k$$

have infinitely many solutions?

First divided by 2:

$$x + 2y = 3$$

Therefore:

$$\boxed{k = 3}$$

63. COMMON TRAPS

& pTRAP 1 — Wrong translation

“5 less than x ” means:

$$x - 5$$

& pTRAP 2 — Fraction LCM applied only to one term

If multiplying an equation by LCM, multiply **every term**.

& pTRAP 3 — Elimination sign error

When subtracting equations, subtract every term carefully.

& pTRAP 4 — Solving ratio questions with unnecessary algebra

Use:

$$\boxed{x=ak, \text{quad } y=bk}$$

& pTRAP 5 — Assuming two equations always have one solution

They may have none or infinitely many.

& pTRAP 6 — Ignoring physical constraints

Age cannot be negative; item count may need integer/nonnegative values.

64. MUST-MEMORIZE FORMULAS

Sum and difference

$$\boxed{x=\frac{S+D}{2}}$$

$$\boxed{y=\frac{S-D}{2}}$$

Two-digit number

$$\boxed{10x+y}$$

Reverse:

$$\boxed{10y+x}$$

65. CONSISTENCY CONDITIONS

For:

$$\boxed{a_1x+b_1y=c_1}$$

$$\boxed{a_2x+b_2y=c_2}$$

Unique

$$\boxed{\frac{a_1}{a_2} \neq \frac{b_1}{b_2}}$$

No solution

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}}$$

Infinite

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

66. FAST EXAM DECISION TREE

One variable?

Isolate it.

Fractions?

Clear denominators first.

Ratio given?

Use (ak, bk) .

Sum + difference?

Use direct half-sum/half-difference formula.

Two variables?

Check if elimination is immediate.

One variable already isolated?

Use substitution.

Word problem?

Define variables first, then translate line by line.

Options simple?

Back-solve if faster.

PRACTICE — LEVEL 0

Q1

Solve:

$$\lfloor 3x+7=25 \rfloor$$

Q2

Solve:

$$\lfloor 5x-3=2x+12 \rfloor$$

Q3

Solve:

$$\lfloor \frac{x}{4} + \frac{x}{2} = 18 \rfloor$$

Q4

Two numbers sum to 30 and differ by 8. Find them.

Q5

If $(x:y=3:2)$ and $(x+y=20)$, find (x,y) .

LEVEL 1

Q6

Solve:

$$\lfloor 2(x+3)=5x-6 \rfloor$$

Q7

Solve:

$$\lfloor \frac{x-1}{3} + \frac{x+2}{2} = 6 \rfloor$$

Q8

Solve:

$$\lfloor 2x+y=11 \rfloor$$

$$\lfloor x-y=1 \rfloor$$

Q9

A two-digit number has digit sum 10 and is 18 greater than its reverse. Find it.

Q10

A father is 26 years older than his son. Their ages sum to 54. Find both ages.

LEVEL 2

Q11

Solve:

$$\lfloor 3x+2y=19 \rfloor$$

$$\lfloor 2x-3y=-4 \rfloor$$

Q12

Determine number of solutions:

$$\lfloor 2x+6y=8 \rfloor$$

$$\lfloor x+3y=5 \rfloor$$

Q13

Find $\lfloor k \rfloor$ for infinitely many solutions:

$$\lfloor 3x+6y=12 \rfloor$$

$$\lfloor x+2y=k \rfloor$$

Q14

There are 30 coins consisting of $\pounds 5$ and $\pounds 1$ coins. Total value $\pounds 220$. Find counts.

Q15

Two numbers are in ratio $(7:4)$ and differ by 24. Find them.

LEVEL 3 — EXAM PRESSURE

Q16

Solve:

$$\left[\frac{x}{2} + \frac{y}{3} = 7 \right]$$

$$\left[\frac{x}{3} - \frac{y}{2} = -1 \right]$$

Q17

A two-digit number is 4 times the sum of its digits. Reversing its digits increases the number by 18. Find the number.

Q18

If:

$$\left[2x + 3y = 17 \right]$$

and:

$$\left[5x - 2y = 4 \right]$$

find $(x+y)$.

Q19

A shop sells pens and pencils. 4 pens + 3 pencils cost ₹ 46; 2 pens + 5 pencils cost ₹ 34. Find each price.

Q20

Find (k) such that:

$$\left[2x + 4y = 6 \right]$$

and:

$$\left[3x + 6y = k \right]$$

have no solution.

ANSWERS

Q1

$$\left[3x = 18 \right]$$

$$\left[\boxed{x=6} \right]$$

Q2

$$\left[3x = 15 \right]$$

$$\left[\boxed{x=5} \right]$$

Q3

Multiply by 4:

$$\left[x + 2x = 72 \right]$$

$$\backslash[3x=72 \backslash]$$

$$\backslash[\boxed{x=24} \backslash]$$

Q4

$$\backslash[x=\frac{30+8}{2}=19 \backslash]$$

$$\backslash[y=\frac{30-8}{2}=11 \backslash]$$

$$\backslash[\boxed{19,11} \backslash]$$

Q5

$$\backslash[x=3k, \backslash y=2k \backslash]$$

$$\backslash[5k=20 \backslash]$$

$$\backslash[\boxed{x=12, \backslash y=8} \backslash]$$

Q6

$$\backslash[2x+6=5x-6 \backslash]$$

$$\backslash[12=3x \backslash]$$

$$\backslash[\boxed{x=4} \backslash]$$

Q7

Multiply by 6:

$$\backslash[2(x-1)+3(x+2)=36 \backslash]$$

$$\backslash[2x-2+3x+6=36 \backslash]$$

$$\backslash[5x+4=36 \backslash]$$

$$\backslash[\boxed{x=\frac{32}{5}} \backslash]$$

Q8

$$\backslash[2x+y=11 \backslash]$$

$$\backslash[x-y=1 \backslash]$$

Add:

$$\backslash[3x=12 \backslash]$$

$$\backslash[x=4 \backslash]$$

$$\backslash[y=3 \backslash]$$

$$\backslash[\boxed{4,3} \backslash]$$

Q9

Let digits $\backslash(x,y\backslash)$.

$$\backslash[x+y=10 \backslash]$$

$$\backslash[9(x-y)=18 \backslash]$$

$$\backslash[x-y=2 \backslash]$$

$$\backslash[x=6, \backslash y=4 \backslash]$$

$$\backslash[\boxed{64} \backslash]$$

Q10

Son = x .

Father = $x+26$.

$$2x+26=54$$

$$x=14$$

Father:

$$40$$

$$\boxed{14, 40}$$

Q11

$$3x+2y=19$$

$$2x-3y=-4$$

Multiply first by 3:

$$9x+6y=57$$

Second by 2:

$$4x-6y=-8$$

Add:

$$13x=49$$

$$x=\frac{49}{13}$$

Then:

$$2y=19-\frac{147}{13}=\frac{100}{13}$$

$$y=\frac{50}{13}$$

$$\boxed{x=\frac{49}{13}, y=\frac{50}{13}}$$

Q12

First divided by 2:

$$x+3y=4$$

Second:

$$x+3y=5$$

Contradiction.

$$\boxed{\text{No solution}}$$

Q13

Divide first by 3:

$$x+2y=4$$

Thus:

$$\boxed{k=4}$$

Q14

Let $5\text{coins} = x$.

1 coin:

$$30 - x$$

$$5x + 10(30 - x) = 220$$

$$5x + 300 - 10x = 220$$

$$-5x = -80$$

$$x = 16$$

1 coin:

$$14$$

$$\boxed{16, 14}$$

Q15

$$x = 7k, \text{quad } y = 4k$$

$$3k = 24$$

$$k = 8$$

$$\boxed{56, 32}$$

Q16

Multiply equations by 6:

$$3x + 2y = 42$$

$$2x - 3y = -6$$

First $\times 3$:

$$9x + 6y = 126$$

Second $\times 2$:

$$4x - 6y = -12$$

Add:

$$13x = 114$$

$$x = \frac{114}{13}$$

Then:

$$3\left(\frac{114}{13}\right) + 2y = 42$$

$$\frac{342}{13} + 2y = \frac{546}{13}$$

$$2y = \frac{204}{13}$$

$$y = \frac{102}{13}$$

$$\boxed{x = \frac{114}{13}, y = \frac{102}{13}}$$

Q17

Let number be:

$$10x + y$$

Digit sum:

$$\backslash[x+y \backslash]$$

Condition 1:

$$\backslash[10x+y=4(x+y) \backslash]$$

$$\backslash[6x=3y \backslash]$$

$$\backslash[y=2x \backslash]$$

Reverse increases by 18:

$$\backslash[10y+x=(10x+y)+18 \backslash]$$

$$\backslash[9(y-x)=18 \backslash]$$

$$\backslash[y-x=2 \backslash]$$

Using $\backslash(y=2x\backslash)$:

$$\backslash[x=2 \backslash]$$

$$\backslash[y=4 \backslash]$$

Number:

$$\backslash[\boxed{24} \backslash]$$

Q18

$$\backslash[2x+3y=17 \backslash]$$

$$\backslash[5x-2y=4 \backslash]$$

First $\times 2$:

$$\backslash[4x+6y=34 \backslash]$$

Second $\times 3$:

$$\backslash[15x-6y=12 \backslash]$$

Add:

$$\backslash[19x=46 \backslash]$$

$$\backslash[x=\frac{46}{19} \backslash]$$

Then:

$$\backslash[3y = 17 - \frac{92}{19} = \frac{231}{19} \backslash]$$

$$\backslash[y=\frac{77}{19} \backslash]$$

Thus:

$$\backslash[x+y = \frac{123}{19} \backslash]$$

$$\backslash[\boxed{\frac{123}{19}} \backslash]$$

Q19

$$\backslash[4p+3c=46 \backslash]$$

$$\backslash[2p+5c=34 \backslash]$$

Second $\times 2$:

$$4p + 10c = 68$$

Subtract first:

$$7c = 22$$

$$c = \frac{22}{7}$$

Then:

$$2p + 5\left(\frac{22}{7}\right) = 34$$

$$2p = \frac{128}{7}$$

$$p = \frac{64}{7}$$

So:

$$\boxed{\text{Pen}} = \frac{64}{7} \text{ pencils and } \frac{22}{7} \text{ apples}$$

The unusual fractional prices are mathematically valid; this is also a reminder not to assume “nice” answers.

Q20

First:

$$2x + 4y = 6$$

Second:

$$3x + 6y = k$$

Coefficient ratio:

$$\frac{3}{2}$$

For the same line, constant should be:

$$6 \cdot \frac{3}{2} = 9$$

So:

- $(k=9)$! infinitely many
- $(k \neq 9)$! no solution

Thus:

$$\boxed{k \neq 9}$$

CHAPTER 35 STATUS

Covered so far:

- one-variable linear equations '
- variables on both sides '
- brackets '
- fractions and decimals '
- percentage equations '
- translating words into algebra '
- consecutive-number models '
- digit equations '
- simultaneous equations '

- elimination method '
- substitution method '
- ratio-based equations '
- age/money/item-count models '
- consistency of equations '
- unique/no/infinite solution '
- parameter-based consistency '
- option back-solving '
- L0–L3 practice '

But **Chapter 35 is not complete yet.**

Next we still need:

- advanced word-to-equation modelling
- equations involving reciprocals
- work/rate equations
- ratio-change systems
- profit/mixture linear models
- 3-variable simultaneous equations
- equation shortcuts using sums
- determinant-style solving intuition
- integer-solution constraints
- back-solving from options
- final chapter audit

Progress

Completed: 34/76 Current: 0% Chapter 35 — Linear & Simultaneous Equations

NEXT

0% Chapter 35 Part 2 — Advanced word equations, 3-variable systems, integer constraints & exam-speed solving

USER

next

ASSISTANT

CHAPTER 35 — LINEAR AND SIMULTANEOUSEQUATIONS

PART 2 — ADVANCED WORD MODELLING, 3-VARIABLE SYSTEMS, INTEGER CONSTRAINTS & EXAM-SPEED SOLVING

Status: 0% IN PROGRESS GATE: High Placements: High Priority: 0% Per 1

Part 1 covered direct equations and two-variable systems.

Now we finish the chapter with the harder part:

Converting complicated statements into the right equations quickly.

1. THE WORD-PROBLEM MASTER METHOD

For almost every algebra word problem:

1. Decide the unknowns.
2. Give each unknown a symbol.
3. Translate each sentence separately.
4. Form only as many independent equations as needed.
5. Solve using the shortest method.
6. Check physical constraints.

Example:

Two numbers sum to 50. One is 8 more than the other.

Let smaller number:

$$\boxed{x}$$

Larger:

$$\boxed{x+8}$$

Equation:

$$\boxed{x+(x+8)=50}$$

$$\boxed{2x=42}$$

$$\boxed{x=21}$$

Numbers:

$$\boxed{21, 29}$$

2. “ONE NUMBER IS p% MORE THAN ANOTHER”

Suppose A is 25% more than B.

If:

$$\boxed{B=x}$$

then:

$$\boxed{A=1.25x}$$

or:

$$\boxed{A=\frac{5}{4}x}$$

Fraction form is often faster.

3. EXAMPLE — PERCENTAGE RELATION + SUM

A is 25% more than B.

Their sum is 90.

Let:

$$B=4k$$

Then:

$$A=5k$$

So:

$$9k=90$$

$$k=10$$

Therefore:

$$\boxed{A=50, B=40}$$

This ratio method is faster than decimals.

4. RATIO CHANGES AFTER ADDITION

Suppose:

$$A:B=3:2$$

After adding 10 to each:

$$\frac{A+10}{B+10}=\frac{4}{3}$$

Let:

$$A=3k, B=2k$$

Then:

$$\frac{3k+10}{2k+10}=\frac{4}{3}$$

Cross multiply:

$$9k+30=8k+40$$

$$k=10$$

So:

$$\boxed{A=30, B=20}$$

5. RATIO CHANGES AFTER SUBTRACTION

Suppose:

$$A:B=5:3$$

After subtracting 4 from each:

$$\frac{(A-4)}{(B-4)} = \frac{3}{1}$$

Let:

$$A = 5k, \quad B = 3k$$

Then:

$$\frac{5k-4}{3k-4} = 3$$

$$5k-4 = 9k-12$$

$$8 = 4k$$

$$k = 2$$

Thus:

$$\boxed{A=10, \quad B=6}$$

6. RATIO AFTER ADDING ONLY TO ONE QUANTITY

Suppose:

$$A:B = 2:3$$

After adding 10 to A:

$$(A+10):B = 1:1$$

Let:

$$A = 2k, \quad B = 3k$$

Then:

$$2k+10 = 3k$$

$$k = 10$$

$$\boxed{A=20, \quad B=30}$$

7. RECIPROCAL EQUATIONS

Suppose:

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$$

and:

$$x+y = 14$$

Use:

$$\frac{x+y}{xy} = \frac{1}{6}$$

So:

$$\frac{14}{xy} = \frac{1}{6}$$

$$xy = 84$$

Now:

$$x+y = 14, \quad xy = 84$$

This actually gives no real integer pair, since discriminant:

$$\sqrt{14^2 - 4(84)} < 0$$

So always check consistency.

8. RECIPROCAL EQUATION WITH A CLEAN SOLUTION

Suppose:

$$x + y = 10$$

and:

$$\frac{1}{x} + \frac{1}{y} = \frac{5}{12}$$

Then:

$$\frac{x+y}{xy} = \frac{5}{12}$$

$$\frac{10}{xy} = \frac{5}{12}$$

$$xy = 24$$

Now numbers with sum 10 and product 24 are:

$$\boxed{4, 6}$$

9. RATE EQUATIONS

Many work/time questions can be written as linear equations in rates.

Example:

A alone can finish a job in x days.

Then rate:

$$\frac{1}{x}$$

If B finishes in 6 days and together they finish in 4 days:

$$\frac{1}{x} + \frac{1}{6} = \frac{1}{4}$$

$$\frac{1}{x} = \frac{1}{12}$$

$$\boxed{x = 12}$$

This is an equation model, even though the variable appears in a denominator.

10. DISTANCE-RATE EQUATIONS

Suppose a person travels 120 km partly at 40 km/h and partly at 60 km/h.

Total time = 2.5 h.

Let distance at 40 be:

$$x$$

Then remaining:

$$\sqrt[1]{120-x}$$

Equation:

$$\sqrt[1]{\frac{x}{40} + \frac{120-x}{60} = 2.5}$$

Multiply by 120:

$$\sqrt[1]{3x+2(120-x)=300}$$

$$\sqrt[1]{3x+240-2x=300}$$

$$\sqrt[1]{x=60}$$

Therefore:

$$\sqrt[1]{\boxed{60\text{ km at each speed}}}$$

11. PRICE-QUANTITY EQUATION

A shop buys 20 items consisting of two types.

Type A costs ₹ 30 .

Type B costs ₹ 50 .

Total cost = ₹ 760 .

Let A count = $\sqrt[1]{x}$.

Then B count:

$$\sqrt[1]{20-x}$$

Equation:

$$\sqrt[1]{30x+50(20-x)=760}$$

$$\sqrt[1]{30x+1000-50x=760}$$

$$\sqrt[1]{20x=240}$$

$$\sqrt[1]{x=12}$$

So:

$$\sqrt[1]{\boxed{12\text{ of A}, 8\text{ of B}}}$$

12. MIXTURE EQUATION

Suppose 20 L mixture contains some 30% solution and some 50% solution.

Final concentration = 40%.

Let amount of 30% solution = $\sqrt[1]{x}$.

Then 50% solution:

$$\sqrt[1]{20-x}$$

Solute equation:

$$\sqrt[1]{0.3x+0.5(20-x)=0.4(20)}$$

$$\sqrt[1]{0.3x+10-0.5x=8}$$

$$0.2x=2$$

$$x=10$$

Thus:

$$\boxed{10 \text{ each}}$$

This is exactly a linear equation.

13. PROFIT EQUATION

An item is sold at a 20% profit for ₹ 600.

Let CP = ₹(x).

Then:

$$1.2x=600$$

$$x=500$$

$$\boxed{₹500}$$

Simple profit/loss problems are often one-variable linear equations.

14. TWO-TRANSACTION EQUATION

Suppose two items together cost ₹ 1000.

One sold at 20% profit, other at 10% loss.

Total selling price = ₹ 1030.

Let first CP = ₹(x).

Second:

$$1000-x$$

Equation:

$$1.2x+0.9(1000-x)=1030$$

$$1.2x+900-0.9x=1030$$

$$0.3x=130$$

$$x=\frac{1300}{3}$$

So first CP:

$$\boxed{₹433\frac{1}{3}}$$

Second:

$$\boxed{₹566\frac{2}{3}}$$

15. AGE RATIO CHANGE

Current age ratio:

$$\boxed{A:B=3:2}$$

After 10 years:

$$\boxed{A:B=4:3}$$

Let current ages:

$$\boxed{3k, 2k}$$

Then:

$$\frac{3k+10}{2k+10} = \frac{4}{3}$$

$$9k+30=8k+40$$

$$k=10$$

Current ages:

$$\boxed{30, 20}$$

16. AGE RATIO IN THE PAST

Current ratio:

$$\boxed{5:3}$$

Ten years ago ratio was:

$$\boxed{3:1}$$

Let ages:

$$\boxed{5k, 3k}$$

Then:

$$\frac{5k-10}{3k-10}=3$$

$$5k-10=9k-30$$

$$20=4k$$

$$k=5$$

Current ages:

$$\boxed{25, 15}$$

17. THREE-VARIABLE LINEAR EQUATIONS

Now suppose there are 3 unknowns:

$$\boxed{x, y, z}$$

We generally need 3 independent equations to determine all three uniquely.

Example:

$$\boxed{x+y+z=12}$$

$$\boxed{x+y-z=4}$$

$$\boxed{x-y+z=6}$$

18. FAST ELIMINATION WITH THREE VARIABLES

Given:

$$\text{\[} x+y+z=12 \text{ \tag{1} \]}$$

$$\text{\[} x+y-z=4 \text{ \tag{2} \]}$$

$$\text{\[} x-y+z=6 \text{ \tag{3} \]}$$

Subtract (2) from (1):

$$\text{\[} 2z=8 \text{ \]}$$

$$\text{\[} z=4 \text{ \]}$$

Subtract (3) from (1):

$$\text{\[} 2y=6 \text{ \]}$$

$$\text{\[} y=3 \text{ \]}$$

Then:

$$\text{\[} x+3+4=12 \text{ \]}$$

$$\text{\[} x=5 \text{ \]}$$

$$\text{\[\boxed{x=5,\ y=3,\ z=4} \]}$$

19. THREE-VARIABLE SUM SHORTCUT

Suppose:

$$\text{\[} x+y=10 \text{ \]}$$

$$\text{\[} y+z=12 \text{ \]}$$

$$\text{\[} z+x=14 \text{ \]}$$

Add all:

$$\text{\[} 2(x+y+z)=36 \text{ \]}$$

Therefore:

$$\text{\[} x+y+z=18 \text{ \]}$$

Then:

$$\text{\[} x=18-(y+z)=18-12=6 \text{ \]}$$

$$\text{\[} y=18-(z+x)=18-14=4 \text{ \]}$$

$$\text{\[} z=18-(x+y)=18-10=8 \text{ \]}$$

$$\text{\[\boxed{x=6,\ y=4,\ z=8} \]}$$

20. MUST-MEMORIZE PAIRWISE-SUM SHORTCUT

If:

$$\begin{cases} x+y=A \end{cases}$$

$$\begin{cases} y+z=B \end{cases}$$

$$\begin{cases} z+x=C \end{cases}$$

then:

$$\boxed{x=\frac{A+C-B}{2}}$$

$$\boxed{y=\frac{A+B-C}{2}}$$

$$\boxed{z=\frac{B+C-A}{2}}$$

Very useful.

21. EXAMPLE

$$\begin{cases} x+y=13 \end{cases}$$

$$\begin{cases} y+z=15 \end{cases}$$

$$\begin{cases} z+x=16 \end{cases}$$

Then:

$$x=\frac{13+16-15}{2}=7$$

$$y=\frac{13+15-16}{2}=6$$

$$z=\frac{15+16-13}{2}=9$$

$$\boxed{7,6,9}$$

22. THREE-VARIABLE GENERAL ELIMINATION

Example:

$$\begin{cases} x+y+z=6 \end{cases}$$

$$\begin{cases} 2x-y+z=3 \end{cases}$$

$$\begin{cases} x+2y-z=2 \end{cases}$$

Take differences strategically.

Subtract first from second:

$$x-2y=-3 \tag{4}$$

Subtract first from third:

$$y-2z=-4 \tag{5}$$

Now solve reduced system with original equation.

This is usually cleaner than using determinant formulas.

23. DETERMINANT INTUITION

For two equations:

$$\begin{cases} a_1x+b_1y=c_1 \end{cases}$$

$$\begin{cases} a_2x+b_2y=c_2 \end{cases}$$

The quantity:

$$\Delta = a_1b_2 - a_2b_1$$

tells whether a unique solution exists.

If:

$$\Delta \neq 0$$

unique solution.

If:

$$\Delta = 0$$

either no solution or infinitely many.

This is useful GATE reasoning.

24. CRAMER-STYLE FORMULAS

For a two-variable system:

$$\Delta = a_1b_2 - a_2b_1$$

$$\Delta_x = c_1b_2 - c_2b_1$$

$$\Delta_y = a_1c_2 - a_2c_1$$

Then:

$$x = \frac{\Delta_x}{\Delta}$$

$$y = \frac{\Delta_y}{\Delta}$$

when:

$$\Delta \neq 0$$

But elimination is usually faster in aptitude.

25. INTEGER-SOLUTION CONSTRAINT

Sometimes algebra gives a mathematical solution but the context requires integers.

Example:

A class has boys and girls. Total 31. Ratio boys:girls = 2:3.

Let:

$$B=2k, \text{quad } G=3k$$

Then:

$$5k=31$$

$$k=6.2$$

That gives non-integer counts.

Therefore the data are inconsistent for whole students.

$\boxed{\text{No valid integer solution}}$

26. NONNEGATIVE CONSTRAINT

Counts cannot be negative.

Suppose:

$$x+y=10$$

$$x-y=20$$

Then:

$$x=15, y=-5$$

Algebraically valid, but if (x,y) represent counts:

$\boxed{\text{invalid physical solution}}$

27. POSITIVITY CONSTRAINT

If x represents:

- time
- distance
- age
- length
- price

usually require:

$$x > 0$$

unless zero is explicitly meaningful.

28. INTEGER EQUATION WITH TWO VARIABLES

Example:

$$3x+5y=29$$

with (x,y) positive integers.

Try modulo reasoning.

Modulo 3:

$$5y \equiv 29 \pmod{3}$$

$$2y \equiv 2 \pmod{3}$$

$$y \equiv 1 \pmod{3}$$

Possible positive values:

$$\{y=1, 4, \dots\}$$

If $(y=4)$:

$$3x+20=29$$

$$x=3$$

So:

$$\boxed{x=3, y=4}$$

For aptitude, option checking is often easier.

29. OPTION BACK-SOLVING — WHEN TO USE IT

Use option substitution when:

- options are small integers,
- equations are ugly,
- only one variable is asked,
- the original equation is easy to evaluate.

Example:

$$\frac{x-2}{3} + \frac{x+1}{4} = 5$$

Options:

8, 9, 10, 11.

Try $(x=10)$:

$$\frac{8}{3} + \frac{11}{4} \neq 5$$

Try $(x=9)$:

$$\frac{7}{3} + \frac{10}{4} = \frac{7}{3} + \frac{5}{2} = \frac{14+15}{6} = \frac{29}{6} \neq 5$$

Solving normally may actually be faster.

Key lesson:

Back-solving is a tool, not an automatic choice.

30. BACK-SOLVE FROM THE MIDDLE OPTION

If options are ordered:

10, 20, 30, 40.

Testing 20 or 30 can tell you whether answer must be higher or lower if the equation is monotonic.

This can reduce work.

31. EQUATIONS WITH PRODUCTS — CHECK LINEARITY

Example:

$$\backslash[xy=12 \backslash]$$

This is **not** a linear equation in $\backslash(x,y\backslash)$.

Similarly:

$$\backslash[x^2+y=5 \backslash]$$

is nonlinear.

Chapter 35 focuses on linear systems; quadratics come in Chapter 36.

32. EQUATION THAT LOOKS NONLINEAR BUT REDUCES

Example:

$$\backslash[x(y+1)-xy=5 \backslash]$$

Expand:

$$\backslash[xy+x-xy=5 \backslash]$$

$$\backslash[x=5 \backslash]$$

So simplify before classifying.

33. HIDDEN LINEAR EQUATION

$$\backslash[\frac{x+2}{3} - \frac{x-1}{5} = 2 \backslash]$$

This looks fractional but is still linear.

Multiply by 15:

$$\backslash[5(x+2)-3(x-1)=30 \backslash]$$

$$\backslash[5x+10-3x+3=30 \backslash]$$

$$\backslash[2x=17 \backslash]$$

$$\backslash[\boxed{x=\frac{17}{2}} \backslash]$$

34. VARIABLES IN DENOMINATOR — NOT LINEAR

$$\backslash[\frac{1}{x} + \frac{1}{y} = \frac{1}{2} \backslash]$$

is not linear in $\backslash(x,y\backslash)$.

But if one relation gives $\backslash(x+y\backslash)$, sometimes it reduces using:

$$\backslash[\frac{x+y}{xy} \backslash]$$

So use algebraic structure rather than labels alone.

35. EQUATION WITH ABSOLUTE INTERPRETATION

If:

$$\backslash[x-y=5 \backslash]$$

then x is 5 greater than y .

If problem only says “difference is 5,” then mathematically:

$$|x - y| = 5$$

which gives two order possibilities unless context specifies which is larger.

36. “TOTAL EXCEEDS” LANGUAGE

“A exceeds B by 12” means:

$$\boxed{A - B = 12}$$

“B is 12 less than A” means exactly the same:

$$\boxed{A - B = 12}$$

These word forms must become automatic.

37. PERCENTAGE DIFFERENCE LANGUAGE

“A is 20% greater than B”:

$$A = 1.2B$$

“B is 20% less than A”:

$$B = 0.8A$$

These are **not equivalent statements**.

Example if $B = 100$, $A = 120$.

B is:

$$\frac{20}{120} \times 100 = 16.67\%$$

less than A, not 20%.

38. TWO-NUMBER FRACTION CONDITION

Suppose:

One number is $\frac{2}{3}$ of another and their sum is 50.

Let larger:

$$3k$$

Smaller:

$$2k$$

Then:

$$5k = 50$$

$$k = 10$$

$$\boxed{20, 30}$$

Prefer ratio representation.

39. "IF x IS ADDED TO BOTH"

Suppose ratio:

$$3:5$$

becomes:

$$5:7$$

after adding 10 to both.

Let values:

$$3k, 5k$$

Then:

$$\frac{3k+10}{5k+10} = \frac{5}{7}$$

$$21k+70=25k+50$$

$$20=4k$$

$$k=5$$

Numbers:

$$\boxed{15, 25}$$

40. "IF x IS REMOVED FROM BOTH"

Suppose:

$$7:5$$

after removing 10 from each becomes:

$$3:2$$

Let:

$$7k, 5k$$

Then:

$$\frac{7k-10}{5k-10} = \frac{3}{2}$$

$$14k-20=15k-30$$

$$k=10$$

Numbers:

$$\boxed{70, 50}$$

41. MONEY DENOMINATION EQUATION

There are 10 and 20 notes.

Total notes:

$$[40]$$

Total value:

$$[1620]$$

Let 10 notes = x .

Then 20 notes:

$$[40-x]$$

$$[10x+20(40-x)=620]$$

$$[10x+800-20x=620]$$

$$[x=18]$$

So:

$$[\boxed{18\text{ 10 notes}, 22\text{ 20 notes}}]$$

42. TWO-RATE WORKFORCE EQUATION

Suppose:

- skilled worker makes 5 units/hour
- trainee makes 3 units/hour
- total 8 workers
- total output 32 units/hour

Let skilled = x .

Trainees:

$$[8-x]$$

Equation:

$$[5x+3(8-x)=32]$$

$$[5x+24-3x=32]$$

$$[2x=8]$$

$$[x=4]$$

Thus:

$$[\boxed{4\text{ skilled}, 4\text{ trainees}}]$$

43. WEIGHTED-AVERAGE EQUATIONS ARE LINEAR

Suppose:

Class A has x students with average 60.

Class B has 20 students with average 75.

Combined average 66.

Then:

$$\left[\frac{60x+75(20)}{x+20}=66 \right]$$

Cross multiply:

$$\left[60x+1500=66x+1320 \right]$$

$$\left[180=6x \right]$$

$$\left[x=30 \right]$$

$$\left[\boxed{30} \right]$$

Even though it starts as a fraction, it reduces to a linear equation.

44. SIMPLE INTEREST EQUATION

Suppose \$10,000 is split between 5% and 8% simple interest.

Total one-year interest = \$650.

Let amount at 5% = x .

At 8%:

$$\left[10000-x \right]$$

Equation:

$$\left[0.05x+0.08(10000-x)=650 \right]$$

$$\left[0.05x+800-0.08x=650 \right]$$

$$\left[0.03x=150 \right]$$

$$\left[x=5000 \right]$$

Thus:

$$\left[\boxed{5000} \text{ at } 5\% \text{ and } 5000 \text{ at } 8\% \right]$$

45. DISTANCE-SPEED EQUATION WITH TWO LEGS

A 180 km journey takes 4 h.

Part is at 30 km/h and rest at 60 km/h.

Let slow-distance:

$$\left[x \right]$$

Fast-distance:

$$\left[180-x \right]$$

$$\left[\frac{x}{30}+\frac{180-x}{60}=4 \right]$$

Multiply by 60:

$$\left[2x+180-x=240 \right]$$

$$\left[x=60 \right]$$

So:

$$\boxed{60 \text{ km at } 30, 120 \text{ km at } 60}$$

46. FAST ELIMINATION BY ADDING EQUATIONS

Given:

$$4x + 3y = 25$$

$$2x - 3y = 5$$

Add:

$$6x = 30$$

$$x = 5$$

Then:

$$20 + 3y = 25$$

$$y = \frac{5}{3}$$

$$\boxed{x = 5, y = \frac{5}{3}}$$

Always look for coefficients that already cancel.

47. FAST ELIMINATION BY LINEAR COMBINATION

Given:

$$5x + 2y = 19$$

$$3x + 4y = 17$$

Instead of huge multiplication:

Double first:

$$10x + 4y = 38$$

Subtract second:

$$7x = 21$$

$$x = 3$$

Then:

$$15 + 2y = 19$$

$$y = 2$$

$$\boxed{3, 2}$$

48. ASK WHAT IS NEEDED — NOT ALWAYS x AND y

Given:

$$2x + 3y = 12$$

$$\begin{cases} 4x+6y=24 \end{cases}$$

Question: find $(2x+3y)$.

No solving needed.

It is already:

$$\boxed{12}$$

Aptitude questions often reward recognizing the asked combination.

49. FIND $(x+y)$ WITHOUT SOLVING EACH

Given:

$$\begin{cases} 3x+2y=17 \end{cases}$$

$$\begin{cases} 2x+3y=18 \end{cases}$$

Add:

$$\begin{cases} 5x+5y=35 \end{cases}$$

So:

$$\begin{cases} x+y=7 \end{cases}$$

$$\boxed{7}$$

Much faster than solving separately.

50. FIND $(x-y)$ DIRECTLY

Given:

$$\begin{cases} 5x+2y=20 \end{cases}$$

$$\begin{cases} 2x+5y=13 \end{cases}$$

Subtract:

$$\begin{cases} 3x-3y=7 \end{cases}$$

$$\boxed{x-y=\frac{7}{3}}$$

No need to solve individual variables.

51. LINEAR-COMBINATION TARGET METHOD

If asked for:

$$\begin{cases} ax+by \end{cases}$$

look at whether the given equations can be added/subtracted/multiplied to produce that combination directly.

This is a very high-value GATE shortcut.

52. EXAMPLE

Given:

$$x+2y=7$$

$$3x-y=8$$

Find:

$$4x+y$$

Notice:

$$(x+2y)+(3x-y)=4x+y$$

Therefore:

$$4x+y=7+8$$

$$\boxed{15}$$

No solving.

53. GATE-STYLE QUESTION 1

Given:

$$2x+3y=11$$

$$3x+2y=14$$

Find $(x+y)$.

Add:

$$5x+5y=25$$

$$\boxed{x+y=5}$$

54. GATE-STYLE QUESTION 2

$$x+y=10$$

$$\frac{1}{x}+\frac{1}{y}=\frac{5}{12}$$

Then:

$$\frac{10}{xy}=\frac{5}{12}$$

$$xy=24$$

Numbers:

$$\boxed{4, 6}$$

55. GATE-STYLE QUESTION 3

$$x+y=13, \quad y+z=15, \quad z+x=16$$

Find (x,y,z) .

$$\boxed{7, 6, 9}$$

56. GATE-STYLE QUESTION 4

For:

$$3x + 6y = 12$$

$$x + 2y = k$$

find k for infinitely many solutions.

Divide first by 3:

$$x + 2y = 4$$

$$\boxed{k=4}$$

57. PLACEMENT-STYLE QUESTION 1

A:B = 3:2.

After 10 is added to both, ratio becomes 4:3.

$$\boxed{A=30, B=20}$$

58. PLACEMENT-STYLE QUESTION 2

There are 10 and 20 notes.

40 notes total, value 620.

$$\boxed{18 \text{ 10 notes}, 22 \text{ 20 notes}}$$

59. PLACEMENT-STYLE QUESTION 3

20 L mixture made from 30% and 50% solutions gives 40%.

$$\boxed{10 \text{ L each}}$$

60. PLACEMENT-STYLE QUESTION 4

A class with average 60 is combined with 20 students averaging 75.

Overall average = 66.

Original class size:

$$\boxed{30}$$

61. ADVANCED QUESTION — TARGET COMBINATION

Given:

$$2x + y = 9$$

$$\begin{cases} x-2y=-3 \end{cases}$$

Find:

$$\begin{cases} 3x-y \end{cases}$$

We can solve or form combination.

From second $\times 1$ and first $\times ?$ Easier solve quickly.

First:

$$\begin{cases} y=9-2x \end{cases}$$

Second:

$$\begin{cases} x-2(9-2x)=-3 \end{cases}$$

$$\begin{cases} x-18+4x=-3 \end{cases}$$

$$\begin{cases} 5x=15 \end{cases}$$

$$\begin{cases} x=3 \end{cases}$$

$$\begin{cases} y=3 \end{cases}$$

Thus:

$$\begin{cases} 3x-y=9-3 \end{cases}$$

$$\begin{cases} \boxed{6} \end{cases}$$

62. ADVANCED QUESTION — THREE VARIABLES

$$\begin{cases} x+y+z=12 \end{cases}$$

$$\begin{cases} x+y-z=4 \end{cases}$$

$$\begin{cases} x-y+z=6 \end{cases}$$

We found:

$$\begin{cases} \boxed{x=5, y=3, z=4} \end{cases}$$

63. ADVANCED QUESTION — RATIO CHANGE

$$A:B = 4:3.$$

After adding 8 to A and 2 to B, ratio becomes 3:2.

Let:

$$\begin{cases} A=4k, \quad B=3k \end{cases}$$

Then:

$$\begin{cases} \frac{4k+8}{3k+2} = \frac{3}{2} \end{cases}$$

$$\begin{cases} 8k+16=9k+6 \end{cases}$$

$$\begin{cases} k=10 \end{cases}$$

Thus:

$$\begin{cases} \boxed{A=40, B=30} \end{cases}$$

64. ADVANCED QUESTION — INTEGER CONDITION

Find positive integers satisfying:

$$2x + 3y = 17$$

Try modulo 2:

$$3y \equiv 1 \pmod{2}$$

$$y \equiv 1 \pmod{2}$$

Positive odd possibilities under:

$$3y < 17$$

are:

$$y = 1, 3, 5$$

Then:

- $y = 1 \Rightarrow x = 7$
- $y = 3 \Rightarrow x = 4$
- $y = 5 \Rightarrow x = 1$

So positive integer solutions:

$$(7, 1), (4, 3), (1, 5)$$

A linear equation in two variables can have multiple integer solutions.

65. COMMON TRAPS

& pTRAP 1 — Too many variables

Do not create 3 unknowns when 1 parameter k will do.

& pTRAP 2 — Ratio-change errors

If:

$$A:B=3:2$$

write:

$$A=3k, B=2k$$

before applying changes.

& pTRAP 3 — Ignoring integer requirements

Counts, people, coins may need whole numbers.

& pTRAP 4 — Solving x and y when only $x+y$ is asked

Try equation addition/subtraction first.

& pTRAP 5 — Treating reciprocal equation as linear

$$\sqrt[1]{1/x}$$

makes it nonlinear in $\sqrt[1]{x}$, though it may simplify using another relation.

& pTRAP 6 — Using same percentage base incorrectly

“20% more” and “20% less” are not symmetric.

& pTRAP 7 — Physical answers

Negative age, negative distance or fractional people may invalidate a mathematical solution.

66. MUST-MEMORIZE ADVANCED SHORTCUTS

Pairwise sums

If:

$$\sqrt[1]{x+y=A, \quad y+z=B, \quad z+x=C}$$

then:

$$\sqrt[1]{\boxed{x=\frac{A+C-B}{2}}}$$

$$\sqrt[1]{\boxed{y=\frac{A+B-C}{2}}}$$

$$\sqrt[1]{\boxed{z=\frac{B+C-A}{2}}}$$

Two-equation determinant

$$\sqrt[1]{\boxed{\Delta=a_1b_2-a_2b_1}}$$

Unique solution if:

$$\sqrt[1]{\boxed{\Delta \neq 0}}$$

67. EXAM-SPEED DECISION TREE

Ratio given?

Use:

$$\sqrt[1]{a_k, b_k}$$

Same quantity changed on both sides?

Build ratio equation after change.

Item counts?

Use:

$$\sqrt[1]{x, N-x}$$

Weighted average / mixture?

Write total contribution equation.

Three pairwise sums?

Use half-sum shortcut.

Asked only for $\sqrt{x+y}$ or another combination?

Combine equations directly.

Integer/count problem?

Check integrality after solving.

Options simple?

Consider back-solving.

PRACTICE — LEVEL 0

Q1

$A:B = 3:2$ and $\text{sum} = 45$. Find A and B .

Q2

$A:B = 4:3$. After adding 5 to each, ratio becomes $5:4$. Find A, B .

Q3

$\sqrt{x+y=12}, \sqrt{x-y=4}$ Find $\sqrt{(x,y)}$.

Q4

$\sqrt{x+y=10}, \sqrt{y+z=14}, \sqrt{z+x=16}$ Find $\sqrt{(x,y,z)}$.

Q5

A class has $\sqrt{x}$ students averaging 50 and 10 students averaging 70. Overall average 60. Find $\sqrt{(x)}$.

LEVEL 1

Q6

20 coins consist of $\sqrt{5}$ and $\sqrt{1}$ coins. Total value $\sqrt{1}$ 4 Find counts.

Q7

20 L mixture uses 20% and 50% solutions to make 35%. Find quantities.

Q8

$A:B = 5:3$. Ten years ago their ratio was $3:1$. Find current ages.

Q9

$\sqrt{2x+3y=13}, \sqrt{3x+2y=12}$ Find $\sqrt{(x+y)}$.

Q10

$\sqrt{x+y=7}, \sqrt{\frac{1}{x}+\frac{1}{y}=\frac{7}{10}}$ Find $\sqrt{(xy)}$.

LEVEL 2

Q11

$x+y=12, \quad y+z=17, \quad z+x=19$ Find (x,y,z) .

Q12

A:B = 7:5. After subtracting 4 from each, ratio becomes 3:2. Find A,B.

Q13

A total of ₹ 12,000 invested at 5% and 8% simple interest. One-year interest is ₹ 780. Find amounts.

Q14

A 180 km journey is partly at 30 km/h and rest at 60 km/h. Total time = 4 h. Find distances.

Q15

There are 10 workers of two types producing 4 and 7 units/hour. Total production = 52 units/hour. Find numbers of each type.

LEVEL 3 — EXAM PRESSURE

Q16

$2x+3y=17, \quad 3x+2y=18$ Find $(5x+5y)$ without solving individually.

Q17

$x+y=10, \quad y+z=14, \quad z+x=18$ Find (xyz) .

Q18

A:B = 4:3. After adding 8 to A and 2 to B, ratio becomes 3:2. Find A,B.

Q19

Find all positive integer solutions:

$$3x+4y=24$$

Q20

A mixture contains 30% and 60% solutions. If 30 L final mixture is 50%, find quantities of each.

ANSWERS

Q1

$$A=3k, \quad B=2k$$

$$5k=45$$

$$k=9$$

$$\boxed{27, 18}$$

Q2

$$\backslash[A=4k,\quad B=3k \backslash]$$

$$\backslash[\frac{4k+5}{3k+5}=\frac{54}{5} \backslash]$$

$$\backslash[16k+20=15k+25 \backslash]$$

$$\backslash[k=5 \backslash]$$

$$\backslash[\boxed{20,\ 15} \backslash]$$

Q3

$$\backslash[x=\frac{12+4}{2}=8 \backslash]$$

$$\backslash[y=\frac{12-4}{2}=4 \backslash]$$

$$\backslash[\boxed{8,\ 4} \backslash]$$

Q4

$$\backslash[x=\frac{10+16-14}{2}=6 \backslash]$$

$$\backslash[y=\frac{10+14-16}{2}=4 \backslash]$$

$$\backslash[z=\frac{14+16-10}{2}=10 \backslash]$$

$$\backslash[\boxed{6,4,10} \backslash]$$

Q5

$$\backslash[\frac{50x+700}{x+10}=60 \backslash]$$

$$\backslash[50x+700=60x+600 \backslash]$$

$$\backslash[100=10x \backslash]$$

$$\backslash[\boxed{x=10} \backslash]$$

Q6

Let $^1 5\text{coins} = \backslash(x\backslash)$.

$$\backslash[5x+10(20-x)=140 \backslash]$$

$$\backslash[5x+200-10x=140 \backslash]$$

$$\backslash[x=12 \backslash]$$

Other:

$$\backslash[8 \backslash]$$

$$\backslash[\boxed{12,\ 8} \backslash]$$

Q7

Let 20% solution = $\backslash(x\backslash)$.

$$\backslash[0.2x+0.5(20-x)=0.35(20) \backslash]$$

$$\backslash[0.2x+10-0.5x=7 \backslash]$$

$$\backslash[0.3x=3 \backslash]$$

$$\backslash[x=10 \backslash]$$

$$\backslash[\boxed{10L,\ 10L} \backslash]$$

Q8

$$A=5k, \text{quad } B=3k$$

$$\frac{5k-10}{3k-10}=3$$

$$5k-10=9k-30$$

$$k=5$$

$$\boxed{25, 15}$$

Q9

Add equations:

$$5x+5y=25$$

$$x+y=5$$

$$\boxed{5}$$

Q10

$$\frac{x+y}{xy}=\frac{7}{10}$$

$$\frac{7}{xy}=\frac{7}{10}$$

$$\boxed{xy=10}$$

Q11

$$x=\frac{12+19-17}{2}=7$$

$$y=\frac{12+17-19}{2}=5$$

$$z=\frac{17+19-12}{2}=12$$

$$\boxed{7, 5, 12}$$

Q12

$$A=7k, \text{quad } B=5k$$

$$\frac{7k-4}{5k-4}=\frac{3}{2}$$

$$14k-8=15k-12$$

$$k=4$$

$$\boxed{28, 20}$$

Q13

Let amount at 5% = x .

$$0.05x+0.08(12000-x)=780$$

$$0.05x+960-0.08x=780$$

$$0.03x=180$$

$$x=6000$$

Other:

$$6000$$

$$\boxed{16000}$$

Q14

Let distance at 30 be x .

$$\frac{x}{30} + \frac{180-x}{60} = 4$$

Multiply by 60:

$$2x + 180 - x = 240$$

$$x = 60$$

$$\boxed{60 \text{ km at } 30, 120 \text{ km at } 60}$$

Q15

Let 4-unit workers = x .

Then 7-unit workers:

$$10 - x$$

$$4x + 7(10 - x) = 52$$

$$4x + 70 - 7x = 52$$

$$3x = 18$$

$$x = 6$$

Other:

$$4$$

$$\boxed{6, 4}$$

Q16

Add equations:

$$5x + 5y = 35$$

$$\boxed{35}$$

No need to solve (x, y) .

Q17

$$x = \frac{10 + 18 - 14}{2} = 7$$

$$y = \frac{10 + 14 - 18}{2} = 3$$

$$z = \frac{14 + 18 - 10}{2} = 11$$

Therefore:

$$xyz = 7(3)(11)$$

$$\boxed{231}$$

Q18

$$A = 4k, B = 3k$$

$$\frac{4k + 8}{3k + 2} = \frac{32}{11}$$

$$8k + 16 = 9k + 6$$

$$k = 10$$

$\boxed{40, 30}$

Q19

$$3x + 4y = 24$$

Positive integers.

Possible (y) :

$$y = 1 \Rightarrow x = \frac{20}{3} \quad \text{invalid}$$

$$y = 2 \Rightarrow x = \frac{16}{3} \quad \text{invalid}$$

$$y = 3 \Rightarrow x = 4$$

$$y = 4 \Rightarrow x = \frac{8}{3} \quad \text{invalid}$$

$$y = 5 \Rightarrow x = \frac{4}{3} \quad \text{invalid}$$

Thus only:

$$\boxed{(x, y) = (4, 3)}$$

Q20

Let 30% solution = x .

60% solution:

$$30 - x$$

$$0.3x + 0.6(30 - x) = 0.5(30)$$

$$0.3x + 18 - 0.6x = 15$$

$$0.3x = 3$$

$$x = 10$$

Other:

$$20$$

$$\boxed{10 \text{ L of 30\%}, 20 \text{ L of 60\%}}$$

' CHAPTER 35 COMPLETE

We have now covered:

- one-variable linear equations '
- variables on both sides '
- brackets/fractions/decimals '
- word-to-equation translation '
- percentage relations '
- ratio-change equations '
- age models '
- digit problems '
- simultaneous equations '
- substitution '

- elimination '
- linear-combination shortcuts '
- consistency/no/infinite solution '
- determinant intuition '
- reciprocal-based equation models '
- work/rate and speed equations '
- mixture and weighted-average equations '
- money/coin/note problems '
- three-variable systems '
- pairwise-sum shortcuts '
- integer/nonnegative constraints '
- parameter equations '
- option back-solving '
- GATE/placement speed techniques '

Progress

Completed: 35/76

Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

+Chapter 36 — Quadratics and Inequalities

Next we will cover:

- quadratic-equation basics
- factorization method
- quadratic formula
- discriminant
- nature of roots
- sum/product of roots
- forming equations from roots
- transformed-root shortcuts
- sign of quadratic expressions
- linear inequalities
- multiplying/dividing inequalities by negatives
- interval notation
- compound inequalities
- quadratic inequalities
- modulus basics
- GATE/placement traps and fast solving.

USER

next

CHAPTER 36 — QUADRATICS AND INEQUALITIES

PART 1 — QUADRATIC EQUATIONS, FACTORIZATION, QUADRATIC FORMULA, DISCRIMINANT & ROOT RELATIONS

Status: $\emptyset=0$ **PROGRESS GATE:** Medium–High **Placements:** Medium–High **Priority:** +Tier 2

A quadratic equation usually has the form:

$$\boxed{ax^2+bx+c=0, \text{quad } a \neq 0}$$

The highest power of the variable is 2.

Examples:

$$x^2-5x+6=0$$

$$2x^2+7x+3=0$$

$$3x^2-12=0$$

The main tools are:

- factorization
- quadratic formula
- discriminant
- sum/product of roots
- forming a quadratic from given roots

1. WHAT ARE ROOTS?

A root is a value of x that makes the quadratic equal to zero.

Example:

$$x^2-5x+6=0$$

Factor:

$$(x-2)(x-3)=0$$

Therefore:

$$x=2$$

or:

$$x=3$$

So roots are:

$$\boxed{2, 3}$$

2. ZERO-PRODUCT RULE

If:

$$\backslash[AB=0 \backslash]$$

then:

$$\backslash[A=0 \backslash]$$

or:

$$\backslash[B=0 \backslash]$$

That is why:

$$\backslash[(x-2)(x-3)=0 \backslash]$$

gives:

$$\backslash[x=2,\backslash 3 \backslash]$$

3. FACTORIZATION METHOD

Consider:

$$\backslash[x^2+5x+6=0 \backslash]$$

Find two numbers whose:

- sum = $\backslash(5\backslash)$
- product = $\backslash(6\backslash)$

They are:

$$\backslash[2,\backslash 3 \backslash]$$

So:

$$\backslash[(x+2)(x+3)=0 \backslash]$$

Roots:

$$\backslash[\boxed{-2,\backslash -3} \backslash]$$

4. NEGATIVE MIDDLE TERM

$$\backslash[x^2-7x+12=0 \backslash]$$

Need two numbers whose:

- sum = $\backslash(-7\backslash)$
- product = $\backslash(12\backslash)$

They are:

$$\backslash[-3,\backslash -4 \backslash]$$

Thus:

$$\backslash[(x-3)(x-4)=0 \backslash]$$

$$\backslash[\boxed{x=3,\backslash 4} \backslash]$$

5. NEGATIVE CONSTANT

$$\backslash[x^2+x-6=0 \backslash]$$

Need numbers whose:

- $\text{sum} = \backslash(1\backslash)$
- $\text{product} = \backslash(-6\backslash)$

They are:

$$\backslash[3,\backslash -2 \backslash]$$

So:

$$\backslash[(x+3)(x-2)=0 \backslash]$$

$$\backslash[\boxed{x=2,\backslash -3} \backslash]$$

6. WHEN LEADING COEFFICIENT IS NOT 1

Example:

$$\backslash[2x^2+5x+2=0 \backslash]$$

Multiply:

$$\backslash[a\cdot c=2\cdot 2=4 \backslash]$$

Find two numbers whose:

- $\text{sum} = \backslash(5\backslash)$
- $\text{product} = \backslash(4\backslash)$

They are:

$$\backslash[4,\backslash 1 \backslash]$$

Split middle term:

$$\backslash[2x^2+4x+x+2=0 \backslash]$$

Group:

$$\backslash[2x(x+2)+1(x+2)=0 \backslash]$$

$$\backslash[(2x+1)(x+2)=0 \backslash]$$

Roots:

$$\backslash[\boxed{-\frac{1}{2},\backslash -2} \backslash]$$

7. GENERAL MIDDLE-TERM SPLITTING

For:

$$\backslash[ax^2+bx+c \backslash]$$

find two numbers $\backslash(m,n\backslash)$ such that:

$$\backslash[m+n=b \backslash]$$

and:

$$\backslash[mn=ac \backslash]$$

Then split:

$$\boxed{bx=mx+nx}$$

and factor by grouping.

8. EXAMPLE

$$\boxed{3x^2+11x+6=0}$$

$$\boxed{ac=18}$$

Need sum 11, product 18:

$$\boxed{9, 2}$$

So:

$$\boxed{3x^2+9x+2x+6=0}$$

$$\boxed{3x(x+3)+2(x+3)=0}$$

$$\boxed{(3x+2)(x+3)=0}$$

Roots:

$$\boxed{-\frac{2}{3}, -3}$$

9. PERFECT-SQUARE QUADRATIC

$$\boxed{x^2-6x+9=0}$$

Recognize:

$$\boxed{(x-3)^2=0}$$

So:

$$\boxed{x=3}$$

The same root appears twice.

We call it a **repeated root** or **equal roots**.

10. PURE QUADRATIC

Example:

$$\boxed{x^2=25}$$

Then:

$$\boxed{x=\pm 5}$$

$$\boxed{x=5, -5}$$

Do not forget the $\pm$.

11. ANOTHER PURE QUADRATIC

$$\backslash[3x^2-27=0 \backslash]$$

$$\backslash[3x^2=27 \backslash]$$

$$\backslash[x^2=9 \backslash]$$

$$\backslash[\boxed{x=\pm 3} \backslash]$$

12. QUADRATIC FORMULA

For:

$$\backslash[ax^2+bx+c=0 \backslash]$$

the roots are:

$$\backslash[\boxed{x= \frac{-b\pm\sqrt{b^2-4ac}}{2a}} \backslash]$$

MUST MEMORIZE

13. WHEN TO USE QUADRATIC FORMULA

Use it when:

- factorization is difficult
- roots are irrational
- coefficients are awkward
- exact symbolic roots are needed
- discriminant analysis is important

If factorization is obvious, factorization is usually faster.

14. EXAMPLE USING QUADRATIC FORMULA

$$\backslash[x^2-4x-1=0 \backslash]$$

Here:

$$\backslash[a=1, \backslash b=-4, \backslash c=-1 \backslash]$$

Formula:

$$\backslash[x= \frac{4\pm\sqrt{16+4}}{2} \backslash]$$

$$\backslash[= \frac{4\pm\sqrt{20}}{2} \backslash]$$

$$\backslash[= \frac{4\pm 2\sqrt{5}}{2} \backslash]$$

$$\backslash[\boxed{x=2\pm\sqrt{5}} \backslash]$$

15. DISCRIMINANT

The expression:

$$\backslash[\boxed{\Delta=b^2-4ac} \backslash]$$

is called the **discriminant**.

It tells us the nature of the roots.

16. NATURE OF ROOTS

For real coefficients:

If:

$$\Delta > 0$$

two distinct real roots.

If:

$$\Delta = 0$$

two equal real roots.

If:

$$\Delta < 0$$

no real roots; the roots are complex conjugates.

17. EXAMPLE — DISTINCT REAL ROOTS

$$x^2 - 5x + 6 = 0$$

$$\Delta = 25 - 24 = 1 > 0$$

So two distinct real roots.

Indeed:

$$2, 3$$

18. EXAMPLE — EQUAL ROOTS

$$x^2 - 6x + 9 = 0$$

$$\Delta = 36 - 36 = 0$$

So equal roots.

$$\boxed{x=3}$$

twice.

19. EXAMPLE — NO REAL ROOTS

$$x^2 + 4x + 5 = 0$$

$$\Delta = 16 - 20 = -4 < 0$$

Therefore:

$$\boxed{\text{no real roots}}$$

Complex roots are:

$$\sqrt{-2} \pm i$$

For aptitude, nature of roots is usually more important than complex arithmetic.

20. PARAMETER FOR EQUAL ROOTS

Suppose:

$$x^2 + kx + 9 = 0$$

has equal roots.

Then:

$$\Delta = 0$$

$$k^2 - 36 = 0$$

$$k = \pm 6$$

$$\boxed{k = \pm 6}$$

21. PARAMETER FOR REAL ROOTS

Suppose:

$$x^2 + kx + 4 = 0$$

has real roots.

Need:

$$k^2 - 16 \geq 0$$

So:

$$|k| \geq 4$$

Thus:

$$\boxed{k \leq -4 \text{ or } k \geq 4}$$

This starts connecting quadratics with inequalities.

22. SUM AND PRODUCT OF ROOTS

If roots of:

$$ax^2 + bx + c = 0$$

are:

$$\alpha, \beta$$

then:

$$\boxed{\alpha + \beta = -\frac{b}{a}}$$

and:

$$\boxed{\alpha\beta=\frac{c}{a}}$$

MUST MEMORIZE

23. WHY THESE FORMULAS WORK

If roots are (α, β) , then:

$$a(x-\alpha)(x-\beta)=0$$

Expand:

$$a[x^2-(\alpha+\beta)x+\alpha\beta]$$

$$=ax^2-a(\alpha+\beta)x+a\alpha\beta$$

Compare with:

$$ax^2+bx+c$$

So:

$$-a(\alpha+\beta)=b$$

$$a\alpha\beta=c$$

Hence:

$$\boxed{\alpha+\beta=-b/a}$$

$$\boxed{\alpha\beta=c/a}$$

24. EXAMPLE

For:

$$2x^2-7x+3=0$$

Sum of roots:

$$\alpha+\beta = -\frac{-7}{2} = \boxed{\frac{7}{2}}$$

Product:

$$\alpha\beta = \frac{3}{2}$$

$$\boxed{\frac{3}{2}}$$

No solving needed.

25. FIND $(\alpha^2+\beta^2)$

Use:

$$\alpha^2+\beta^2 = (\alpha+\beta)^2-2\alpha\beta$$

So if:

$$S=\alpha+\beta, \quad P=\alpha\beta$$

then:

$$\boxed{\alpha^2 + \beta^2 = S^2 - 2P}$$

26. EXAMPLE

For:

$$x^2 - 5x + 6 = 0$$

$$S = 5, \text{quad } P = 6$$

Then:

$$\alpha^2 + \beta^2 = 25 - 12 = \boxed{13}$$

Indeed roots are 2 and 3:

$$4 + 9 = 13$$

27. FIND $(\alpha^3 + \beta^3)$

Use:

$$\boxed{\alpha^3 + \beta^3 = S^3 - 3PS}$$

Example:

$$S = 5, \text{quad } P = 6$$

$$= 125 - 90$$

$$= \boxed{35}$$

Indeed:

$$8 + 27 = 35$$

28. FIND $(\frac{1}{\alpha} + \frac{1}{\beta})$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

Therefore:

$$\boxed{\frac{1}{\alpha} + \frac{1}{\beta} = \frac{S}{P}}$$

provided $(P \neq 0)$.

29. DIRECT COEFFICIENT FORM

Since:

$$S = -\frac{b}{a}$$

and:

$$P = \frac{c}{a}$$

then:

$$\frac{S}{P} = \frac{-b/a}{c/a} = \boxed{-\frac{bc}{c}}$$

So:

$$\boxed{\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{bc}{a}}$$

when $c \neq 0$.

30. FIND $\left(\frac{1}{\alpha}\beta\right)$

Since:

$$\alpha\beta = \frac{ca}{a}$$

then:

$$\boxed{\frac{1}{\alpha}\beta = \frac{ac}{a^2}}$$

31. FIND $(\alpha - \beta)$

We know:

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

Thus:

$$\boxed{\alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}}$$

$$\alpha - \beta = \sqrt{S^2 - 4P}$$

$$= \sqrt{S^2 - 4P}$$

In coefficient form:

$$\sqrt{S^2 - 4P} = \sqrt{\frac{b^2 - 4ac}{a^2}}$$

So:

$$\boxed{\alpha - \beta = \frac{\sqrt{\Delta}}{a}}$$

$$\alpha - \beta = \frac{\sqrt{\Delta}}{a}$$

$$= \frac{\sqrt{\Delta}}{a}$$

for real roots.

32. FORM A QUADRATIC FROM ROOTS

If roots are:

$$\alpha, \beta$$

then equation is:

$$\boxed{x^2 - (\alpha + \beta)x + \alpha\beta = 0}$$

or any nonzero multiple of it.

33. EXAMPLE

Roots:

$$\sqrt{2}, \sqrt{5}$$

Sum:

$$\sqrt{7}$$

Product:

$$\sqrt{10}$$

Equation:

$$\boxed{x^2 - 7x + 10 = 0}$$

34. ROOTS $\sqrt{-2}, \sqrt{3}$

Sum:

$$\sqrt{1}$$

Product:

$$\sqrt{-6}$$

Equation:

$$\boxed{x^2 - x - 6 = 0}$$

35. ROOTS $\sqrt{2+\sqrt{3}}, \sqrt{2-\sqrt{3}}$

Sum:

$$\sqrt{4}$$

Product:

$$\sqrt{4-3=1}$$

Equation:

$$\boxed{x^2 - 4x + 1 = 0}$$

This is a common conjugate-root pattern.

36. IF ROOTS ARE $\sqrt{p}$ AND $\sqrt{q}$

Equation:

$$\boxed{x^2 - (p+q)x + pq = 0}$$

Memory trick:

Minus sum, plus product.

37. TRANSFORMED ROOTS — ADD CONSTANT

Suppose original roots are:

$$\alpha, \beta$$

New roots:

$$\alpha+k, \beta+k$$

New sum:

$$S' = S + 2k$$

New product:

$$P' = (\alpha+k)(\beta+k)$$

$$= P + kS + k^2$$

So new equation:

$$\boxed{x^2 - (S+2k)x + (P+kS+k^2) = 0}$$

38. EXAMPLE

Original equation:

$$x^2 - 5x + 6 = 0$$

Roots:

$$2, 3$$

New roots:

$$3, 4$$

Clearly new equation:

$$\boxed{x^2 - 7x + 12 = 0}$$

Using shortcut:

$$S=5, P=6, k=1$$

$$S'=7$$

$$P'=6+5+1=12$$

Correct.

39. TRANSFORMED ROOTS — MULTIPLY BY k

New roots:

$$k\alpha, k\beta$$

New sum:

$$kS$$

New product:

$$\sqrt[k^2P]{}$$

New equation:

$$\boxed{x^2 - kSx + k^2P = 0}$$

40. EXAMPLE

Original roots 2 and 3.

New roots:

$$\sqrt[4]{}, \sqrt[6]{}$$

Sum:

$$\sqrt[10]{}$$

Product:

$$\sqrt[24]{}$$

Equation:

$$\boxed{x^2 - 10x + 24 = 0}$$

41. RECIPROCAL ROOTS

If original roots are:

$$\sqrt[\alpha]{}, \sqrt[\beta]{}$$

new roots:

$$\sqrt[\frac{1}{\alpha}]{}, \sqrt[\frac{1}{\beta}]{}$$

New sum:

$$\sqrt[\frac{S}{P}]{}$$

New product:

$$\sqrt[\frac{1}{P}]{}$$

So equation:

$$\sqrt[x^2 - \frac{SP}{x} + \frac{1}{P} = 0]{}$$

Multiply by $\sqrt{(P)}$:

$$\sqrt[\boxed{Px^2 - Sx + 1 = 0}]{}$$

42. DIRECT COEFFICIENT SHORTCUT FOR RECIPROCAL ROOTS

Original:

$$\sqrt[ax^2 + bx + c = 0]{}$$

Reciprocal-root equation is:

$$\sqrt[\boxed{cx^2 + bx + a = 0}]{}$$

assuming roots are nonzero.

Notice:

Reverse first and last coefficients.

43. EXAMPLE

Original:

$$\boxed{2x^2-5x+3=0}$$

Reciprocal-root equation:

$$\boxed{3x^2-5x+2=0}$$

Original roots:

$$1, \frac{3}{2}$$

Reciprocals:

$$1, \frac{2}{3}$$

Correct.

44. NEGATIVE ROOTS

If original roots:

$$\alpha, \beta$$

new roots:

$$-\alpha, -\beta$$

New sum:

$$-S$$

product:

$$P$$

Equation:

$$x^2+Sx+P=0$$

For original:

$$ax^2+bx+c=0$$

the negative-root equation can be obtained by replacing x by $-x$:

$$\boxed{ax^2-bx+c=0}$$

45. SHIFT ROOTS WITHOUT FORMULAS

A very fast technique:

If original equation is:

$$f(x)=0$$

and you want roots increased by (k) , replace:

$$x \text{ to } x-k$$

because if new variable $(x=\alpha+k)$, then:

$$x-k=\alpha$$

This is a deeper algebra shortcut.

46. EXAMPLE

Original:

$$x^2-5x+6=0$$

Want roots each increased by 2.

Replace (x) by $(x-2)$:

$$(x-2)^2-5(x-2)+6=0$$

$$x^2-4x+4-5x+10+6=0$$

$$x^2-9x+20=0$$

Roots:

$$4, 5$$

Correct.

47. ROOT SIGN FROM SUM AND PRODUCT

If roots are real:

Product positive

Roots have same sign.

Product negative

Roots have opposite signs.

Then use sum to determine same-sign case.

48. BOTH ROOTS POSITIVE

For real roots of:

$$ax^2+bx+c=0$$

we need:

$$\alpha+\beta>0$$

and:

$$\begin{cases} \alpha > 0 \\ \beta > 0 \end{cases}$$

and real roots.

For $a > 0$, this usually means:

$$-b/a > 0$$

$$c/a > 0$$

plus:

$$\Delta \geq 0$$

49. BOTH ROOTS NEGATIVE

Need:

$$S < 0$$

$$P > 0$$

and real roots.

50. OPPOSITE-SIGN ROOTS

Need:

$$P < 0$$

because product of roots is negative.

Then real distinct roots automatically occur.

51. EXAMPLE

$$x^2 - x - 6 = 0$$

Product:

$$-6 < 0$$

Therefore roots have opposite signs.

Indeed:

$$3, -2$$

52. GATE-STYLE QUESTION 1

Find roots:

$$x^2 - 9x + 20 = 0$$

$$(x-4)(x-5) = 0$$

$$\boxed{4, 5}$$

53. GATE-STYLE QUESTION 2

For:

$$2x^2 - 7x + 3 = 0$$

find sum and product of roots.

$$\boxed{S = \frac{7}{2}, \quad P = \frac{3}{2}}$$

54. GATE-STYLE QUESTION 3

For what values of k does:

$$x^2 + kx + 4 = 0$$

have real roots?

$$k^2 - 16 \geq 0$$

$$\boxed{k \leq -4 \text{ or } k \geq 4}$$

55. GATE-STYLE QUESTION 4

Find equation whose roots are:

$$3 + \sqrt{2}, 3 - \sqrt{2}$$

Sum:

$$6$$

Product:

$$9 - 2 = 7$$

Equation:

$$\boxed{x^2 - 6x + 7 = 0}$$

56. PLACEMENT-STYLE QUESTION 1

If roots of:

$$x^2 - 6x + 8 = 0$$

are (α, β) , find:

$$\alpha^2 + \beta^2$$

$$S = 6, P = 8$$

$$36 - 16 = \boxed{20}$$

57. PLACEMENT-STYLE QUESTION 2

Same equation. Find:

$$\frac{1}{\alpha} + \frac{1}{\beta}$$

$$\left[\frac{SP}{\frac{68}{}} \right]$$

$$\left[\boxed{\frac{34}{}} \right]$$

58. PLACEMENT-STYLE QUESTION 3

Original equation:

$$\left[x^2 - 5x + 6 = 0 \right]$$

Find equation whose roots are double the original roots.

Original:

$$\left[S = 5, \text{quad } P = 6 \right]$$

New:

$$\left[S' = 10 \right]$$

$$\left[P' = 24 \right]$$

Equation:

$$\left[\boxed{x^2 - 10x + 24 = 0} \right]$$

59. PLACEMENT-STYLE QUESTION 4

Find equation with reciprocal roots of:

$$\left[3x^2 - 7x + 2 = 0 \right]$$

Reverse first and last coefficient:

$$\left[\boxed{2x^2 - 7x + 3 = 0} \right]$$

60. ADVANCED QUESTION — ROOT DIFFERENCE

For:

$$\left[x^2 - 8x + 12 = 0 \right]$$

$$\left[S = 8, \text{quad } P = 12 \right]$$

$$\left[(\alpha - \beta)^2 = 64 - 48 = 16 \right]$$

So:

$$\left[\boxed{|\alpha - \beta| = 4} \right]$$

61. ADVANCED QUESTION — PARAMETER EQUAL ROOTS

For:

$$\left[kx^2 - 6x + 1 = 0 \right]$$

equal roots require:

$$\left[36 - 4k = 0 \right]$$

$$\boxed{k=9}$$

But also quadratic requires:

$$k \neq 0$$

So:

$$\boxed{k=9}$$

62. ADVANCED QUESTION — ONE ROOT IS 2

Suppose:

$$x^2 + kx + 6 = 0$$

has one root 2.

Substitute:

$$4 + 2k + 6 = 0$$

$$2k = -10$$

$$\boxed{k = -5}$$

Equation:

$$x^2 - 5x + 6 = 0$$

other root = 3.

63. FIND OTHER ROOT QUICKLY

If one root is α and product is P :

$$\beta = \frac{P}{\alpha}$$

Example:

$$x^2 - 7x + 10 = 0$$

Product:

$$10$$

One root = 5.

Other:

$$10/5 = 2$$

$$\boxed{2}$$

64. COMMON TRAPS

& pTRAP 1 — Forgetting $(\pm)$

From:

$$x^2 = 9$$

both:

$$\boxed{x=3,-3}$$

& pTRAP 2 — Wrong sign in sum of roots

$$\boxed{\alpha+\beta=-b/a}$$

not (b/a) .

& pTRAP 3 — Product sign

$$\boxed{\alpha\beta=c/a}$$

& pTRAP 4 — Discriminant

Use:

$$\boxed{b^2-4ac}$$

not (b^2+4ac) .

& pTRAP 5 — Formula sign

$$\boxed{-b\pm\sqrt{\Delta}}$$

If $(b=-5)$, then:

$$\boxed{-b=5}$$

& pTRAP 6 — Reciprocal-root equation

Reverse (a,c) , but middle coefficient (b) remains the same.

65. MUST-MEMORIZE FORMULAS

Quadratic formula

$$\boxed{x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}}$$

Discriminant

$$\boxed{\Delta=b^2-4ac}$$

Root sum

$$\boxed{\alpha+\beta=-\frac{b}{a}}$$

Root product

$$\boxed{\alpha\beta=\frac{c}{a}}$$

66. NATURE OF ROOTS TABLE

Condition	Roots
$\Delta > 0$	2 distinct real roots
$\Delta = 0$	2 equal real roots
$\Delta < 0$	no real roots

67. FORMING EQUATION FROM ROOTS

Given roots:

α, β

write:

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

Remember:

minus sum, plus product

68. FAST EXAM DECISION TREE

Easy factors visible?

Factor.

Not factorable quickly?

Use quadratic formula.

Asked only nature of roots?

Use discriminant—do not solve.

Asked sum/product?

Use coefficients—do not solve.

Asked expression in roots?

Convert to (S, P) .

Given one root?

Substitute directly or use product.

Reciprocal roots?

Reverse first/last coefficients.

Shifted/scaled roots?

Transform (S, P) instead of solving roots.

PRACTICE — LEVEL 0

Q1

Solve:

$$\backslash[x^2-7x+12=0 \backslash]$$

Q2

Solve:

$$\backslash[x^2-16=0 \backslash]$$

Q3

Find sum and product of roots:

$$\backslash[x^2-9x+14=0 \backslash]$$

Q4

Find nature of roots:

$$\backslash[x^2+2x+5=0 \backslash]$$

Q5

Form equation with roots 3 and 7.

LEVEL 1

Q6

Solve:

$$\backslash[2x^2+7x+3=0 \backslash]$$

Q7

Find $\backslash(k\backslash)$ for equal roots:

$$\backslash[x^2+kx+16=0 \backslash]$$

Q8

If roots of:

$$\backslash[x^2-6x+5=0 \backslash]$$

are $\backslash(\alpha,\beta)\backslash$, find:

$$\backslash[\alpha^2+\beta^2 \backslash]$$

Q9

Find equation whose roots are:

$$\backslash[2+\sqrt{3},\quad 2-\sqrt{3} \backslash]$$

Q10

Find equation whose roots are reciprocals of roots of:

$$\sqrt{2x^2-9x+4=0}$$

LEVEL 2

Q11

For:

$$\sqrt{3x^2-10x+2=0}$$

find:

$$\sqrt{\frac{1}{\alpha}+\frac{1}{\beta}}$$

Q12

If roots of:

$$\sqrt{x^2-5x+6=0}$$

are (α, β) , find:

$$\sqrt{\alpha^3+\beta^3}$$

Q13

For what values of (k) does:

$$\sqrt{x^2-2kx+9=0}$$

have real roots?

Q14

Find equation whose roots are 3 times the roots of:

$$\sqrt{x^2-4x+3=0}$$

Q15

One root of:

$$\sqrt{x^2+kx+12=0}$$

is 3. Find (k) and the other root.

LEVEL 3 — EXAM PRESSURE

Q16

If roots of:

$$\sqrt{2x^2-5x+2=0}$$

are (α, β) , find:

$$\sqrt{\alpha^2\beta+\alpha\beta^2}$$

Q17

If:

$$\sqrt{\alpha+\beta=7, \quad \alpha\beta=10}$$

find:

$$\sqrt{\alpha^4 + \beta^4}$$

Q18

Find k such that:

$$(k-1)x^2 - 2kx + k + 3 = 0$$

has equal roots.

Q19

Find equation whose roots are $(\alpha+1, \beta+1)$, where (α, β) are roots of:

$$x^2 - 4x + 1 = 0$$

Q20

Find equation whose roots are $(-\alpha, -\beta)$, where (α, β) are roots of:

$$3x^2 - 8x + 2 = 0$$

ANSWERS

Q1

$$(x-3)(x-4)=0$$

$$\boxed{3, 4}$$

Q2

$$x = \boxed{\pm 4}$$

Q3

$$\boxed{S=9, P=14}$$

Q4

$$\Delta = 4 - 20 = -16$$

$$\boxed{\text{No real roots}}$$

Q5

$$S=10, P=21$$

$$\boxed{x^2 - 10x + 21 = 0}$$

Q6

$$2x^2 + 6x + 3$$

$$= (2x+1)(x+3)$$

$$\boxed{-\frac{1}{2}, -3}$$

Q7

$$k^2 - 64 = 0$$

$$\boxed{k = \pm 8}$$

Q8

$$\begin{cases} S=6, \text{quad } P=5 \end{cases}$$

$$\begin{cases} S^2-2P = 36-10 \end{cases}$$

$$\boxed{26}$$

Q9

Sum:

$$4$$

Product:

$$1$$

$$\boxed{x^2-4x+1=0}$$

Q10

Original:

$$2x^2-9x+4$$

Reciprocal-root equation:

$$\boxed{4x^2-9x+2=0}$$

Q11

$$-\frac{bc}{-10} = \boxed{5}$$

Q12

$$\begin{cases} S=5, \text{quad } P=6 \end{cases}$$

$$\begin{cases} S^3-3PS = 125-90 \end{cases}$$

$$\boxed{35}$$

Q13

$$\Delta = (-2k)^2-36$$

$$4k^2-36 \geq 0$$

$$k^2 \geq 9$$

$$\boxed{k \leq -3 \text{ or } k \geq 3}$$

Q14

Original:

$$\begin{cases} S=4, \text{quad } P=3 \end{cases}$$

Multiply roots by 3:

$$S'=12$$

$$P'=27$$

Equation:

$$\boxed{x^2-12x+27=0}$$

Q15

Substitute $(x=3)$:

$$9+3k+12=0$$

$$3k=-21$$

$$k=-7$$

Product roots:

$$12$$

Other root:

$$12/3=4$$

$$\boxed{k=-7, \text{other root}=4}$$

Q16

$$\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$$

For:

$$2x^2-5x+2$$

$$P=1, \quad S=\frac{5}{2}$$

Therefore:

$$\boxed{\frac{5}{2}}$$

Q17

First:

$$\alpha^2 + \beta^2 = 49 - 20 = 29$$

Then:

$$\alpha^4 + \beta^4 = 29^2 - 2(10^2)$$

$$= 841 - 200$$

$$= \boxed{641}$$

Q18

Here:

$$a=k-1, \quad b=-2k, \quad c=k+3$$

Equal roots:

$$(-2k)^2 - 4(k-1)(k+3) = 0$$

$$4k^2 - 4(k^2 + 2k - 3) = 0$$

$$4k^2 - 4k^2 - 8k + 12 = 0$$

$$k = \frac{3}{2}$$

Also:

$$k-1 \neq 0$$

which holds.

$$\boxed{k=\frac{3}{2}}$$

Q19

Original:

$$S=4, \text{quad } P=1$$

New roots:

$$\alpha+1, \beta+1$$

New sum:

$$6$$

New product:

$$P+S+1 = 1+4+1 = 6$$

Equation:

$$\boxed{x^2-6x+6=0}$$

Q20

Replacing roots by negatives changes sign of middle coefficient:

$$\boxed{3x^2+8x+2=0}$$

CHAPTER 36 STATUS

Covered so far:

- quadratic-equation basics '
- factorization '
- middle-term splitting '
- pure quadratics '
- quadratic formula '
- discriminant '
- nature of roots '
- parameter conditions '
- sum/product of roots '
- expressions in roots '
- forming equations from roots '
- conjugate roots '
- reciprocal roots '
- shifted/scaled/negative roots '
- sign information from root product '
- GATE/placement practice '

But **Chapter 36 is not complete yet.**

Next we still need:

- linear inequalities
- inequality sign reversal

- interval notation
- compound inequalities
- quadratic inequalities
- sign-chart method
- modulus/absolute-value inequalities
- ranges of expressions
- parameter-based inequalities
- final shortcuts and chapter audit

Progress

Completed: 35/76 Current: 0/35 Chapter 36 — Quadratics & Inequalities

NEXT

+ Chapter 36 Part 2 — Linear inequalities, quadratic inequalities, sign charts, modulus & range problems

USER

next

ASSISTANT

CHAPTER 36 — QUADRATICS AND INEQUALITIES

PART 2 — LINEAR INEQUALITIES, QUADRATIC INEQUALITIES, SIGN CHARTS, MODULUS & RANGE PROBLEMS

Status: 0/14 PROGRESS GATE: Medium–High Placements: Medium–High Priority: +Tier 2

Part 1 handled quadratic equations.

Now we study inequalities.

The biggest difference between an equation and an inequality is:

An equation usually asks for exact values. An inequality usually asks for a range of values.

Examples:

$$\{ x > 3 \}$$

$$\{ x \leq 7 \}$$

$$\{ x^2 - 5x + 6 > 0 \}$$

1. BASIC INEQUALITY SYMBOLS

$$\{ x > 5 \}$$

means x is greater than 5.

$$x < 5$$

means less than 5.

$$x \geq 5$$

means greater than or equal to 5.

$$x \leq 5$$

means less than or equal to 5.

2. BASIC LINEAR INEQUALITY

Solve:

$$x + 3 > 7$$

Subtract 3:

$$x > 4$$

$$\boxed{x > 4}$$

3. VARIABLES ON BOTH SIDES

$$3x + 2 > 2x + 7$$

Subtract $(2x)$:

$$x + 2 > 7$$

$$x > 5$$

$$\boxed{x > 5}$$

4. THE MOST IMPORTANT INEQUALITY RULE

If you multiply or divide both sides by a **negative number**, reverse the inequality sign.

Example:

$$-2x > 6$$

Divide by (-2) :

$$\boxed{x < -3}$$

Not $(x > -3)$.

MUST MEMORIZE

5. WHY THE SIGN REVERSES

We know:

$$2 < 5$$

Multiply both sides by (-1) :

$$[-2 > -5]$$

So the direction reverses.

6. EXAMPLE

$$[5 - 3x \leq 14]$$

Subtract 5:

$$[-3x \leq 9]$$

Divide by (-3) , reverse sign:

$$[\boxed{x \geq -3}]$$

7. FRACTIONAL INEQUALITY

$$[\frac{x}{3} > 4]$$

Multiply by positive 3:

$$[\boxed{x > 12}]$$

No sign reversal.

8. IF MULTIPLYING BY UNKNOWN SIGN

Be careful.

Example:

$$[xy > 0]$$

You cannot divide by (y) unless you know whether $(y > 0)$ or $(y < 0)$.

Because sign reversal depends on the sign of (y) .

This becomes important in advanced inequalities.

9. COMPOUND INEQUALITY

Example:

$$[2 < x < 7]$$

means both:

$$[x > 2]$$

and:

$$[x < 7]$$

So (x) lies between 2 and 7.

10. SOLVING A COMPOUND INEQUALITY

$$\backslash[3 < 2x + 1 < 9 \backslash]$$

Subtract 1 from all three parts:

$$\backslash[2 < 2x < 8 \backslash]$$

Divide by 2:

$$\backslash[\boxed{1 < x < 4} \backslash]$$

11. COMPOUND INEQUALITY WITH NEGATIVE DIVISION

$$\backslash[-6 < -2x < 10 \backslash]$$

Divide every part by $\backslash(-2\backslash)$, reversing both inequalities:

$$\backslash[3 > x > -5 \backslash]$$

Write in increasing order:

$$\backslash[\boxed{-5 < x < 3} \backslash]$$

12. INTERVAL NOTATION

$$\backslash[x > 3 \backslash]$$

means:

$$\backslash[\boxed{(3, \infty)} \backslash]$$

$$\backslash[x \geq 3 \backslash]$$

means:

$$\backslash[\boxed{[3, \infty)} \backslash]$$

Parenthesis means endpoint excluded.

Square bracket means endpoint included.

13. MORE INTERVAL EXAMPLES

$$\backslash[2 < x \leq 7 \backslash]$$

becomes:

$$\backslash[\boxed{(2, 7]} \backslash]$$

$$\backslash[x \leq -1 \backslash]$$

becomes:

$$\backslash[\boxed{(-\infty, -1]} \backslash]$$

Infinity always uses parentheses.

14. UNION OF INTERVALS

If:

$$\{x < 2\}$$

or:

$$\{x > 5\}$$

write:

$$\boxed{(-\infty, 2) \cup (5, \infty)}$$

The symbol $\cup$ means union.

15. INTERSECTION

If:

$$\{x > 2\}$$

and:

$$\{x < 5\}$$

then common solution:

$$\boxed{(2, 5)}$$

This is the intersection.

16. QUADRATIC INEQUALITY — FIRST IDEA

Consider:

$$x^2 - 5x + 6 > 0$$

Factor:

$$(x - 2)(x - 3) > 0$$

Critical points:

$$x = 2, \text{ and } x = 3$$

These divide the number line into:

$$(-\infty, 2), (2, 3), (3, \infty)$$

Now determine sign in each interval.

17. SIGN CHART

For:

$$(x - 2)(x - 3)$$

If $x < 2$

Both factors negative.

Product positive.

If $(2 < x < 3)$

One positive, one negative.

Product negative.

If $(x > 3)$

Both positive.

Product positive.

Therefore:

$$x^2 - 5x + 6 > 0$$

gives:

$$\boxed{x < 2 \text{ or } x > 3}$$

18. SAME QUADRATIC < 0

$$x^2 - 5x + 6 < 0$$

Using same sign chart:

negative only between roots.

$$\boxed{2 < x < 3}$$

19. INCLUDE ROOTS FOR (≥ 0)

$$x^2 - 5x + 6 \geq 0$$

Now roots are allowed because expression equals zero at 2 and 3.

So:

$$\boxed{x \leq 2 \text{ or } x \geq 3}$$

20. INCLUDE ROOTS FOR (≤ 0)

$$x^2 - 5x + 6 \leq 0$$

$$\boxed{2 \leq x \leq 3}$$

21. FAST RULE FOR UPWARD-OPENING QUADRATIC

Suppose:

$$a > 0$$

and real roots:

$$r_1 < r_2$$

Then:

$$\mathbf{f(x) > 0}$$

outside roots:

$$\boxed{x < r_1 \text{ or } x > r_2}$$

$$\mathbf{f(x) < 0}$$

between roots:

$$\boxed{r_1 < x < r_2}$$

Memory trick:

Positive leading coefficient !' positive outside.

22. DOWNWARD-OPENING QUADRATIC

If:

$$a < 0$$

sign pattern reverses.

Example:

$$-x^2 + 5x - 6 > 0$$

Multiply by (-1) :

$$x^2 - 5x + 6 < 0$$

so:

$$\boxed{2 < x < 3}$$

23. SIGN-CHART METHOD — GENERAL

For:

$$(x-a)(x-b) > 0$$

with:

$$a < b$$

sign pattern:

$$+\quad -\quad +$$

across:

$$(-\infty, a), (a, b), (b, \infty)$$

For:

$$< 0$$

choose the middle interval.

24. THREE FACTORS

Solve:

$$\lfloor (x-1)(x-3)(x-5) > 0 \rfloor$$

Critical points:

$$\lfloor 1, 3, 5 \rfloor$$

Test the rightmost interval $(x > 5)$:

all factors positive !' product positive.

Each time we cross a simple root, sign flips.

So signs:

$$\lfloor (-)\text{quad}(+)\text{quad}(-)\text{quad}(+) \rfloor$$

Intervals:

$$\lfloor (-\infty, 1) : - \rfloor$$

$$\lfloor (1, 3) : + \rfloor$$

$$\lfloor (3, 5) : - \rfloor$$

$$\lfloor (5, \infty) : + \rfloor$$

Hence:

$$\lfloor \boxed{1 < x < 3 \text{ or } x > 5} \rfloor$$

25. REPEATED ROOT DOES NOT FLIP SIGN

Solve:

$$\lfloor (x-2)^2(x-5) > 0 \rfloor$$

At $(x=2)$, multiplicity is even.

Crossing $(x=2)$, sign does not change.

At $(x=5)$, multiplicity odd, sign flips.

This is useful for higher-degree sign charts.

26. RATIONAL INEQUALITY

Solve:

$$\lfloor \frac{x-2}{x-5} > 0 \rfloor$$

Critical points:

- numerator zero at 2
- denominator zero at 5

Intervals:

$$\lfloor (-\infty, 2), \text{quad}(2, 5), \text{quad}(5, \infty) \rfloor$$

Test signs:

- $(x < 2)$: negative/negative = positive
- $(2 < x < 5)$: positive/negative = negative
- $(x > 5)$: positive/positive = positive

So:

$$\boxed{x < 2 \text{ or } x > 5}$$

But $(x = 5)$ can never be included.

27. RATIONAL INEQUALITY WITH (≥ 0)

$$\frac{x-2}{x-5} \geq 0$$

Numerator zero at $(x = 2)$, so 2 is allowed.

Denominator zero at 5, so 5 is forbidden.

Therefore:

$$\boxed{(-\infty, 2] \cup (5, \infty)}$$

28. NEVER CROSS-MULTIPLY BLINDLY

From:

$$\frac{x-2}{x-5} > 0$$

you should not simply multiply by $(x-5)$, because its sign is unknown.

It could reverse the inequality.

Use sign chart instead.

This is a major trap.

29. ABSOLUTE VALUE / MODULUS

$$|$$

x

$$|$$

means distance of (x) from 0.

Examples:

$$|5| = 5$$

$$|-5| = 5$$

30. BASIC MODULUS EQUATION

$$|x| = 4$$

means:

$$\boxed{x=4}$$

or:

$$\boxed{x=-4}$$

Therefore:

$$\boxed{x=\pm 4}$$

31. SHIFTED MODULUS

$$\boxed{|x-3|=5}$$

means distance of x from 3 is 5.

So:

$$\boxed{x-3=5}$$

or:

$$\boxed{x-3=-5}$$

Hence:

$$\boxed{x=8,-2}$$

32. MODULUS LESS THAN

$$\boxed{|x|<a, \text{quad } a>0}$$

means:

$$\boxed{-a<x<a}$$

Memory:

Less than !' inside.

33. EXAMPLE

$$\boxed{|x|<4}$$

$$\boxed{-4<x<4}$$

34. MODULUS LESS THAN OR EQUAL

$$\boxed{|x|\leq a}$$

means:

$$\boxed{-a\leq x\leq a}$$

35. MODULUS GREATER THAN

$$\boxed{|x| > a}$$

means:

$$\boxed{x < -a \text{ or } x > a}$$

Memory:

Greater than !' outside.

36. MODULUS GREATER THAN OR EQUAL

$$\boxed{|x| \geq a}$$

means:

$$\boxed{x \leq -a \text{ or } x \geq a}$$

37. SHIFTED MODULUS INEQUALITY

$$\boxed{|x-3| < 5}$$

means:

$$\boxed{-5 < x-3 < 5}$$

Add 3:

$$\boxed{-2 < x < 8}$$

38. SHIFTED GREATER-THAN CASE

$$\boxed{|x-3| > 5}$$

means:

$$\boxed{x-3 > 5}$$

or:

$$\boxed{x-3 < -5}$$

Thus:

$$\boxed{x > 8 \text{ or } x < -2}$$

39. MODULUS WITH LINEAR EXPRESSION

Solve:

$$\boxed{|2x-1| \leq 5}$$

Write:

$$\boxed{-5 \leq 2x-1 \leq 5}$$

Add 1:

$$\lfloor -4 \leq x \leq 6 \rfloor$$

Divide by 2:

$$\lfloor \boxed{-2 \leq x \leq 3} \rfloor$$

40. IMPOSSIBLE MODULUS CASE

$$\lfloor |x| < -2 \rfloor$$

Impossible because absolute value is never negative.

$$\lfloor \boxed{\text{No solution}} \rfloor$$

41. ALWAYS TRUE MODULUS CASE

$$\lfloor |x| \geq 0 \rfloor$$

True for every real x .

$$\lfloor \boxed{x \in \mathbb{R}} \rfloor$$

42. QUADRATIC RANGE — COMPLETING THE SQUARE

Consider:

$$\lfloor x^2 - 6x + 11 \rfloor$$

Complete the square:

$$\lfloor x^2 - 6x + 9 + 2 \rfloor$$

$$\lfloor = (x-3)^2 + 2 \rfloor$$

Since:

$$\lfloor (x-3)^2 \geq 0 \rfloor$$

minimum value is:

$$\lfloor \boxed{2} \rfloor$$

at:

$$\lfloor x=3 \rfloor$$

43. GENERAL QUADRATIC VERTEX FORM

$$\lfloor ax^2 + bx + c \rfloor$$

can be written as:

$$\lfloor a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right) \rfloor$$

Vertex occurs at:

$$\lfloor \boxed{x = -\frac{b}{2a}} \rfloor$$

44. MINIMUM VALUE WHEN $(a > 0)$

For:

$$f(x) = ax^2 + bx + c, \text{ where } a > 0$$

minimum value:

$$\boxed{f_{\min} = c - \frac{b^2}{4a} = -\frac{\Delta}{4a}}$$

at:

$$\boxed{x = -\frac{b}{2a}}$$

45. MAXIMUM VALUE WHEN $(a < 0)$

If:

$$a < 0$$

parabola opens downward.

Then:

$$\boxed{f_{\max} = c - \frac{b^2}{4a}}$$

same vertex formula, but it is a maximum.

46. EXAMPLE — MINIMUM

$$f(x) = x^2 - 8x + 20$$

Complete square:

$$(x - 4)^2 + 4$$

Thus:

$$\boxed{f_{\min} = 4}$$

at:

$$\boxed{x = 4}$$

47. EXAMPLE — MAXIMUM

$$f(x) = -x^2 + 6x + 1$$

$$= -(x^2 - 6x) + 1$$

$$= -(x - 3)^2 + 10$$

Maximum:

$$\boxed{10}$$

at:

$$x = 3$$

48. RANGE OF A QUADRATIC

If:

$$[f(x)=(x-2)^2+5]$$

then:

$$[f(x)\geq 5]$$

So range:

$$[\boxed{[5,\infty)}]$$

49. DOWNWARD QUADRATIC RANGE

If:

$$[f(x)=-(x+1)^2+7]$$

then:

$$[f(x)\leq 7]$$

Range:

$$[\boxed{(-\infty, 7]}]$$

50. CONDITION FOR QUADRATIC ALWAYS POSITIVE

For:

$$[ax^2+bx+c>0]$$

for all real (x) , a common condition is:

$$[\boxed{a>0,\quad \Delta < 0}]$$

Why?

- opens upward
 - no real roots
 - never touches/crosses x-axis
-

51. ALWAYS NONNEGATIVE

For:

$$[ax^2+bx+c\geq 0]$$

for all real (x) :

$$[\boxed{a>0,\quad \Delta \leq 0}]$$

or degenerate special cases handled separately.

52. ALWAYS NEGATIVE

For:

$$\lceil ax^2+bx+c<0 \rceil$$

for all real $\lceil x \rceil$:

$$\lceil \boxed{a<0, \Delta < 0} \rceil$$

53. PARAMETER EXAMPLE

Find $\lceil k \rceil$ such that:

$$\lceil x^2+2x+k>0 \rceil$$

for all real $\lceil x \rceil$.

Need:

$$\lceil a=1>0 \rceil$$

and:

$$\lceil \Delta < 0 \rceil$$

$$\lceil 4-4k<0 \rceil$$

$$\lceil 1-k<0 \rceil$$

$$\lceil \boxed{k>1} \rceil$$

54. SAME PROBLEM BY COMPLETING SQUARE

$$\lceil x^2+2x+k = (x+1)^2+k-1 \rceil$$

For strictly positive for every real $\lceil x \rceil$:

minimum value:

$$\lceil k-1 \rceil$$

must satisfy:

$$\lceil k-1>0 \rceil$$

$$\lceil \boxed{k>1} \rceil$$

Often this is even faster.

55. CONDITION FOR REAL ROOTS OF PARAMETER QUADRATIC

We already learned:

$$\lceil \boxed{\Delta \geq 0} \rceil$$

Example:

$$\lceil x^2-2kx+k+3=0 \rceil$$

Need:

$$\sqrt[4]{4k^2 - 4(k+3)} \geq 0$$

$$\sqrt{k^2 - k - 3} \geq 0$$

Now solve quadratic inequality.

Roots:

$$k = \frac{1 \pm \sqrt{13}}{2}$$

Since leading coefficient positive:

$$\boxed{k \leq \frac{1 - \sqrt{13}}{2} \text{ or } k \geq \frac{1 + \sqrt{13}}{2}}$$

56. PRODUCT INEQUALITY SHORTCUT

If:

$$ab > 0$$

then either:

$$a > 0, b > 0$$

or:

$$a < 0, b < 0$$

If:

$$ab < 0$$

then signs are opposite.

This is the foundation of sign charts.

57. QUOTIENT SIGN RULE

For:

$$\frac{ab}{c} > 0$$

numerator and denominator have same sign.

For:

$$\frac{ab}{c} < 0$$

they have opposite signs.

Also:

$$b \neq 0$$

always.

58. GATE-STYLE QUESTION 1

Solve:

$$3x - 5 > 7$$

$$\lfloor 3x > 12 \rfloor$$

$$\lfloor \boxed{x > 4} \rfloor$$

59. GATE-STYLE QUESTION 2

Solve:

$$\lfloor -2x + 1 \rfloor \leq 7 \rfloor$$

$$\lfloor -2x \rfloor \leq 6 \rfloor$$

Divide by $\lfloor (-2) \rfloor$:

$$\lfloor \boxed{x \geq -3} \rfloor$$

60. GATE-STYLE QUESTION 3

Solve:

$$\lfloor x^2 - 7x + 12 < 0 \rfloor$$

Factor:

$$\lfloor (x-3)(x-4) < 0 \rfloor$$

Upward quadratic !' negative between roots:

$$\lfloor \boxed{3 < x < 4} \rfloor$$

61. GATE-STYLE QUESTION 4

Solve:

$$\lfloor |2x-3| \rfloor \leq 5 \rfloor$$

$$\lfloor -5 \rfloor \leq 2x-3 \rfloor \leq 5 \rfloor$$

$$\lfloor -2 \rfloor \leq 2x \rfloor \leq 8 \rfloor$$

$$\lfloor \boxed{-1 \rfloor \leq x \rfloor \leq 4} \rfloor$$

62. PLACEMENT-STYLE QUESTION 1

Solve:

$$\lfloor (x-1)(x-5) > 0 \rfloor$$

$$\lfloor \boxed{x < 1 \text{ or } x > 5} \rfloor$$

63. PLACEMENT-STYLE QUESTION 2

Solve:

$$\lfloor \frac{x-3}{x+2} \rfloor \leq 0 \rfloor$$

Critical points:

$$\left[-2, \frac{1}{3} \right]$$

Signs:

- $(x < -2)$: positive
- $(-2 < x < \frac{1}{3})$: negative
- $(x > \frac{1}{3})$: positive

At $(x = \frac{1}{3})$, numerator zero !' allowed.

At $(x = -2)$, denominator zero !' forbidden.

Thus:

$$\boxed{-2 < x \leq \frac{1}{3}}$$

64. PLACEMENT-STYLE QUESTION 3

Find minimum value:

$$x^2 - 10x + 30$$

$$= (x - 5)^2 + 5$$

Minimum:

$$\boxed{5}$$

65. PLACEMENT-STYLE QUESTION 4

Find range of:

$$-(x - 2)^2 + 9$$

Since square is nonnegative:

$$f(x) \leq 9$$

Range:

$$\boxed{(-\infty, 9]}$$

66. ADVANCED QUESTION — QUADRATIC INEQUALITY

Solve:

$$2x^2 - 5x - 3 \geq 0$$

Factor:

$$(2x + 1)(x - 3) \geq 0$$

Roots:

$$-\frac{1}{2}, 3$$

Leading coefficient positive !' nonnegative outside including roots:

$$\boxed{x \leq -\frac{1}{2} \text{ or } x \geq 3}$$

67. ADVANCED QUESTION — RATIONAL INEQUALITY

Solve:

$$\frac{(x-1)(x-4)}{x-2} > 0$$

Critical points:

$$1, 2, 4$$

Test from right:

$(x > 4)$: positive.

Cross 4: negative.

Cross 2: positive.

Cross 1: negative.

So positive intervals:

$$(1, 2) \cup (4, \infty)$$

Therefore:

$$\boxed{1 < x < 2 \text{ or } x > 4}$$

68. ADVANCED QUESTION — MODULUS

Solve:

$$|3x+1| > 7$$

Case 1:

$$3x+1 > 7$$

$$x > 2$$

Case 2:

$$3x+1 < -7$$

$$3x < -8$$

$$x < -\frac{8}{3}$$

Thus:

$$\boxed{x < -\frac{8}{3} \text{ or } x > 2}$$

69. ADVANCED QUESTION — RANGE

Find minimum of:

$$2x^2 - 8x + 11$$

Complete square:

$$2(x^2 - 4x) + 11$$

$$\sqrt{=2[(x-2)^2-4]+11}$$

$$\sqrt{=2(x-2)^2+3}$$

Therefore:

$$\boxed{\min=3}$$

at:

$$x=2$$

70. COMMON TRAPS

& pTRAP 1 — Negative multiplication/division

Always reverse inequality.

& pTRAP 2 — Including roots in strict inequalities

For (>0) or (<0) , roots are excluded.

& pTRAP 3 — Denominator zero

Never include denominator-zero points.

& pTRAP 4 — Cross-multiplying rational inequalities

Unsafe if denominator sign is unknown.

& pTRAP 5 — Modulus less vs greater

$$|x| < a$$

means inside.

$$|x| > a$$

means outside.

& pTRAP 6 — Leading coefficient sign

For upward quadratic, positive outside.

For downward quadratic, pattern reverses.

& pTRAP 7 — Range vs roots

Range asks possible output values, not where expression equals zero.

71. MUST-MEMORIZE INEQUALITY RULES

Negative multiplication/division

$$a > b$$

multiply by negative (k) :

$$\boxed{ka < kb}$$

if $k < 0$.

Modulus

$$\boxed{|x| < a \text{ iff } -a < x < a}$$

$$\boxed{|x| > a \text{ iff } x < -a \text{ or } x > a}$$

for $a > 0$.

72. MUST-MEMORIZE QUADRATIC SIGN RULE

For:

$$a(x-r_1)(x-r_2)$$

with $a > 0$ and $r_1 < r_2$:

$$+ \text{quad} - \text{quad} +$$

from left to right.

So:

- positive outside
- negative inside

73. MUST-MEMORIZE RANGE FORMULA

For:

$$f(x) = ax^2 + bx + c$$

vertex:

$$x_v = -\frac{b}{2a}$$

extreme value:

$$f_v = c - \frac{b^2}{4a} = -\frac{\Delta}{4a}$$

- $a > 0$!' minimum
- $a < 0$!' maximum

74. FAST EXAM DECISION TREE

Linear inequality?

Solve like equation, but watch negative multiplication/division.

Product inequality?

Factor and make sign chart.

Rational inequality?

Put everything on one side, factor numerator/denominator, mark critical points.

Modulus $\leq$?

Inside interval.

Modulus $\geq$?

Outside interval.

Quadratic range?

Complete square or use vertex.

“For all real x ”?

Think:

- sign of leading coefficient
- discriminant

PRACTICE — LEVEL 0

Q1

Solve:

$$\lfloor 2x+3 \geq 11 \rfloor$$

Q2

Solve:

$$\lfloor -4x < 20 \rfloor$$

Q3

Solve:

$$\lfloor 1 < 2x+3 < 9 \rfloor$$

Q4

Solve:

$$\lfloor |x| < 6 \rfloor$$

Q5

Solve:

$$\lfloor x^2-5x+6 < 0 \rfloor$$

LEVEL 1

Q6

Solve:

$$\lfloor 3-2x \geq 7 \rfloor$$

Q7

Solve:

$$\sqrt{(x-2)(x-7)} \geq 0$$

Q8

Solve:

$$\sqrt{\frac{x-1}{x-4}} > 0$$

Q9

Solve:

$$\sqrt{|2x+1|} \leq 7$$

Q10

Find minimum of:

$$\sqrt{x^2 - 6x + 13}$$

LEVEL 2

Q11

Solve:

$$\sqrt{2x^2 - 7x + 3} \leq 0$$

Q12

Solve:

$$\sqrt{\frac{x+1}{x-3}} \leq 0$$

Q13

Solve:

$$\sqrt{|3x-2|} > 4$$

Q14

Find range of:

$$\sqrt{2x^2 - 4x + 7}$$

Q15

Find $\sqrt{k}$ such that:

$$\sqrt{x^2 + 4x + k} > 0$$

for all real $\sqrt{x}$.

LEVEL 3 — EXAM PRESSURE

Q16

Solve:

$$\sqrt{(x-1)(x-3)(x-5)} < 0$$

Q17

Solve:

$$\left[\frac{(x-2)(x+1)}{x-4} \geq 0 \right]$$

Q18

Find all k such that:

$$[x^2 - 2kx + k + 3 = 0]$$

has real roots.

Q19

Find the maximum value of:

$$[-2x^2 + 8x - 3]$$

Q20

Solve:

$$[|2x - 5| \geq 9]$$

ANSWERS**Q1**

$$[2x > 8]$$

$$[\boxed{x > 4}]$$

Q2

Divide by (-4) :

$$[\boxed{x > -5}]$$

Q3

$$[1 < 2x + 3 < 9]$$

Subtract 3:

$$[-2 < 2x < 6]$$

Divide by 2:

$$[\boxed{-1 < x < 3}]$$

Q4

$$[\boxed{-6 < x < 6}]$$

Q5

$$[(x-2)(x-3) < 0]$$

$$[\boxed{2 < x < 3}]$$

Q6

$$[3 - 2x \geq 7]$$

$$\sqrt{-2x} \geq 4$$

Divide by $\sqrt{-2}$:

$$\boxed{x \leq -2}$$

Q7

Roots:

$$\sqrt{2}, \sqrt{7}$$

Upward product nonnegative outside:

$$\boxed{x \leq 2 \text{ or } x \geq 7}$$

Q8

Critical points:

$$\sqrt{1}, \sqrt{4}$$

Positive outside:

$$\boxed{x < 1 \text{ or } x > 4}$$

Q9

$$\sqrt{-7} \leq 2x+1 \leq 7$$

$$\sqrt{-8} \leq 2x \leq 6$$

$$\boxed{-4 \leq x \leq 3}$$

Q10

$$\sqrt{x^2-6x+13} = (x-3)^2+4$$

$$\boxed{\min=4}$$

Q11

$$\sqrt{2x^2-7x+3} = (2x-1)(x-3)$$

Roots:

$$\sqrt{\frac{1}{2}}, \sqrt{3}$$

For $\sqrt{}$:

$$\boxed{\frac{1}{2} \leq x \leq 3}$$

Q12

Critical points:

$$\sqrt{-1}, \sqrt{3}$$

Sign pattern:

- $\sqrt{x < -1}$: positive
- $\sqrt{-1 < x < 3}$: negative
- $\sqrt{x > 3}$: positive

Include numerator zero at $\sqrt{-1}$, exclude denominator zero at 3:

$$\boxed{-1 \leq x < 3}$$

Q13

$$\sqrt{3x-2} > 4$$

or:

$$\sqrt{3x-2} < -4$$

First:

$$\sqrt{x} > 2$$

Second:

$$\sqrt{3x} < -2$$

$$\sqrt{x} < -\frac{2}{3}$$

Thus:

$$\boxed{x < -\frac{2}{3} \text{ or } x > 2}$$

Q14

$$2x^2 - 4x + 7 = 2(x-1)^2 + 5$$

Therefore:

$$\boxed{[5, \infty)}$$

Q15

$$x^2 + 4x + k = (x+2)^2 + k - 4$$

For strictly positive for all real x :

$$k - 4 > 0$$

$$\boxed{k > 4}$$

Q16

Critical points:

$$1, 3, 5$$

Sign pattern:

$$-\text{quad} + \text{quad} - \text{quad} +$$

We need negative:

$$\boxed{x < 1 \text{ or } 3 < x < 5}$$

Q17

$$\frac{(x-2)(x+1)}{x-4} \geq 0$$

Critical points:

$$-1, 2, 4$$

From rightmost interval:

- $(x > 4)$: positive
- $(2 < x < 4)$: negative
- $(-1 < x < 2)$: positive
- $(x < -1)$: negative

Include numerator zeros $\setminus(-1,2)$.

Exclude $\setminus(4)$.

Thus:

$$\setminus \boxed{[-1,2] \cup (4,\infty)}$$

Q18

Need:

$$\setminus \Delta \geq 0$$

$$\setminus (-2k)^2 - 4(k+3) \geq 0$$

$$\setminus 4(k^2 - k - 3) \geq 0$$

$$\setminus k^2 - k - 3 \geq 0$$

Roots:

$$\setminus \frac{1 \pm \sqrt{13}}{2}$$

Therefore:

$$\setminus \boxed{k \leq \frac{1 - \sqrt{13}}{2} \text{ or } k \geq \frac{1 + \sqrt{13}}{2}}$$

Q19

$$\setminus -2x^2 + 8x - 3$$

Complete square:

$$\setminus -2(x^2 - 4x) - 3$$

$$\setminus = -2[(x-2)^2 - 4] - 3$$

$$\setminus = -2(x-2)^2 + 5$$

Maximum:

$$\setminus \boxed{5}$$

at:

$$\setminus x=2$$

Q20

$$\setminus |2x-5| \geq 9$$

So:

$$\setminus 2x-5 \geq 9$$

or:

$$\setminus 2x-5 \leq -9$$

First:

$$\setminus 2x \geq 14$$

$$\setminus x \geq 7$$

Second:

$$\setminus 2x \leq -4$$

$\forall x \leq -2$

Thus:

$\boxed{x \leq -2 \text{ or } x \geq 7}$

CHAPTER 36 COMPLETE

We have now covered:

- quadratic equations
- factorization
- quadratic formula
- discriminant
- nature of roots
- sum/product of roots
- transformed roots
- reciprocal/negative/scaled/shifted roots
- parameter quadratics
- linear inequalities
- inequality sign reversal
- compound inequalities
- interval notation
- product sign charts
- quadratic inequalities
- rational inequalities
- modulus equations/inequalities
- quadratic minimum/maximum
- range of quadratic expressions
- “for all real x ” conditions
- parameter inequality problems
- GATE/placement L0–L3 drills

Progress

Completed: 36/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

+Chapter 37 — Sequences, AP and GP

Next we will cover:

- sequence basics
- arithmetic progression
- n th term
- common difference

- sum of AP
- inserting arithmetic means
- GP basics
- nth term
- common ratio
- sum of finite GP
- infinite GP basics
- geometric means
- mixed sequence patterns
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 37 — SEQUENCES, AP AND GP

PART 1 — SEQUENCE BASICS, ARITHMETIC PROGRESSION, NTH TERM, SUM, MEANS & EXAM SHORTCUTS

Status: $\emptyset$ **MAP PROGRESS** **GATE:** Medium **Placements:** Medium–High **Priority:** +Tier 2

A **sequence** is simply an ordered list of numbers.

Examples:

$\{ 2, 4, 6, 8, 10, \dots \}$

$\{ 3, 9, 27, 81, \dots \}$

The first is an **Arithmetic Progression (AP)** because the difference is constant.

The second is a **Geometric Progression (GP)** because the ratio is constant.

1. SEQUENCE VS SERIES

A sequence is a list:

$\{ 1, 3, 5, 7, \dots \}$

A series is the sum of terms:

$\{ 1+3+5+7+\dots \}$

Memory trick

Sequence = terms. Series = sum.

2. TERM NOTATION

The first term:

$$[a_1]$$

Second:

$$[a_2]$$

nth term:

$$[a_n]$$

Sometimes first term is simply written as:

$$[a]$$

3. WHAT IS AN AP?

An Arithmetic Progression has a constant difference between consecutive terms.

Example:

$$[5, 8, 11, 14, 17, \ldots]$$

Difference:

$$[8 - 5 = 3]$$

$$[11 - 8 = 3]$$

So common difference:

$$[\boxed{d=3}]$$

4. AP WITH NEGATIVE DIFFERENCE

$$[20, 16, 12, 8, \ldots]$$

Common difference:

$$[\boxed{d=-4}]$$

An AP can increase, decrease, or remain constant.

5. CONSTANT AP

$$[7, 7, 7, 7, \ldots]$$

Here:

$$[\boxed{d=0}]$$

Still an AP.

6. nth TERM OF AN AP

If first term is:

$$\boxed{a}$$

and common difference:

$$\boxed{d}$$

then:

$$a_1 = a$$

$$a_2 = a + d$$

$$a_3 = a + 2d$$

Therefore:

$$\boxed{a_n = a + (n-1)d}$$

MUST MEMORIZE

7. EXAMPLE

AP:

$$4, 7, 10, 13, \dots$$

Here:

$$a = 4, \quad d = 3$$

Find 20th term.

$$a_{20} = 4 + (20-1)3$$

$$= 4 + 57$$

$$\boxed{= 61}$$

8. FIND WHICH TERM

In AP:

$$5, 9, 13, 17, \dots$$

which term is 81?

$$a = 5, \quad d = 4$$

$$81 = 5 + (n-1)4$$

$$76 = 4(n-1)$$

$$19 = n-1$$

$$\boxed{n = 20}$$

9. TEST WHETHER A NUMBER IS A TERM

For AP:

$$7, 12, 17, \dots$$

is 102 a term?

$$\boxed{102 = 7 + (n-1)5}$$

$$\boxed{95 = 5(n-1)}$$

$$\boxed{n-1 = 19}$$

$$\boxed{n = 20}$$

Integer positive (n) , so yes.

$$\boxed{\text{102 is the 20th term}}$$

10. NON-TERM EXAMPLE

Is 100 a term of:

$$\boxed{7, 12, 17, \dots}$$

$$\boxed{100 = 7 + 5(n-1)}$$

$$\boxed{93 = 5(n-1)}$$

$$\boxed{n-1 = 18.6}$$

Not an integer.

$$\boxed{\text{100 is not a term}}$$

11. DIFFERENCE BETWEEN TWO AP TERMS

$$\boxed{a_m = a + (m-1)d}$$

$$\boxed{a_n = a + (n-1)d}$$

Subtract:

$$\boxed{a_n - a_m = (n-m)d}$$

Therefore:

$$\boxed{d = \frac{a_n - a_m}{n-m}}$$

Very useful.

12. EXAMPLE

5th term = 18.

11th term = 42.

Find (d) .

$$\boxed{d = \frac{42-18}{11-5}}$$

$$\boxed{d = \frac{24}{6}}$$

$$\boxed{d = 4}$$

13. FIND FIRST TERM FROM TWO TERMS

Using previous data:

$$a_5 = 18$$

$$a + 4d = 18$$

Since:

$$d = 4$$

$$a + 16 = 18$$

$$\boxed{a = 2}$$

14. FAST TWO-TERM METHOD

Given:

$$a_m = P$$

$$a_n = Q$$

Then:

$$\boxed{d = \frac{Q - P}{n - m}}$$

and:

$$\boxed{a = P - (m - 1)d}$$

15. THREE CONSECUTIVE AP TERMS

If three numbers are in AP, represent them as:

$$\boxed{x - d, x, x + d}$$

This is much better than:

$$x, x + d, x + 2d$$

when their sum is involved.

Why?

Sum becomes immediately:

$$3x$$

16. EXAMPLE

Three consecutive AP terms sum to 36.

Let terms:

$$x - d, x, x + d$$

Then:

$$3x = 36$$

$$\sqrt{x=12}$$

So middle term:

$$\sqrt{\boxed{12}}$$

17. FIVE CONSECUTIVE AP TERMS

Represent as:

$$\sqrt{\boxed{x-2d, x-d, x, x+d, x+2d}}$$

Sum:

$$\sqrt{5x}$$

Thus average = middle term.

18. ODD NUMBER OF AP TERMS

For equally spaced odd number of terms:

$$\sqrt{\boxed{\text{Average} = \text{middle term}}}$$

Example:

$$\sqrt{4, 7, 10, 13, 16}$$

Average:

$$\sqrt{\boxed{10}}$$

19. ARITHMETIC MEAN OF TWO NUMBERS

If:

$$\sqrt{a, A, b}$$

are in AP:

$$\sqrt{A-a=b-A}$$

$$\sqrt{2A=a+b}$$

Therefore:

$$\sqrt{\boxed{A=\frac{a+b}{2}}}$$

20. EXAMPLE

Arithmetic mean between 8 and 20:

$$\sqrt{\frac{8+20}{2} = \boxed{14}}$$

So:

$$\sqrt{8, 14, 20}$$

is an AP.

21. INSERTING MULTIPLE ARITHMETIC MEANS

Insert k arithmetic means between a and b .

Then total number of intervals is:

$$k+1$$

So common difference:

$$\boxed{d = \frac{b-a}{k+1}}$$

22. EXAMPLE

Insert 3 arithmetic means between 2 and 18.

Total gaps:

$$4$$

$$d = \frac{18-2}{4} = 4$$

Sequence:

$$2, 6, 10, 14, 18$$

Inserted means:

$$\boxed{6, 10, 14}$$

23. SUM OF FIRST n TERMS OF AP

The most important formula:

$$\boxed{S_n = \frac{n}{2}[2a + (n-1)d]}$$

Alternative:

If last term is l ,

$$\boxed{S_n = \frac{n}{2}(a+l)}$$

MUST MEMORIZE

24. WHY AP SUM FORMULA WORKS

Suppose:

$$S_n = a + (a+d) + (a+2d) + \cdots + l$$

Write in reverse:

$$S_n = l + (l-d) + (l-2d) + \cdots + a$$

Add vertically.

Each pair equals:

$$a+l$$

There are $\lfloor n \rfloor$ such pairs:

$$\lfloor 2S_n = n(a+1) \rfloor$$

Therefore:

$$\lfloor \boxed{S_n = \frac{n^2(a+1)}{2}} \rfloor$$

25. EXAMPLE

Find sum:

$$\lfloor 2+5+8+\cdots+29 \rfloor$$

First find $\lfloor n \rfloor$.

$$\lfloor 29 = 2 + (n-1)3 \rfloor$$

$$\lfloor 27 = 3(n-1) \rfloor$$

$$\lfloor n = 10 \rfloor$$

Then:

$$\lfloor S_{10} = \frac{10}{2}(2+29) \rfloor$$

$$\lfloor = 5(31) \rfloor$$

$$\lfloor = \boxed{155} \rfloor$$

26. EXAMPLE USING $\lfloor (2a+(n-1)d) \rfloor$

First 20 terms of:

$$\lfloor 3, 7, 11, \dots \rfloor$$

$$\lfloor a=3, \text{quad } d=4, \text{quad } n=20 \rfloor$$

$$\lfloor S_{20} = \frac{20}{2}[6+19(4)] \rfloor$$

$$\lfloor = 10(82) \rfloor$$

$$\lfloor = \boxed{820} \rfloor$$

27. SUM OF FIRST n NATURAL NUMBERS

$$\lfloor 1+2+3+\cdots+n \rfloor$$

This is AP:

$$\lfloor a=1, \text{quad } d=1 \rfloor$$

Hence:

$$\lfloor \boxed{1+2+\cdots+n = \frac{n(n+1)}{2}} \rfloor$$

MUST MEMORIZE

28. SUM OF FIRST n ODD NUMBERS

$$\backslash[1+3+5+\cdots+(2n-1) \backslash]$$

Result:

$$\backslash[\boxed{n^2} \backslash]$$

Example:

$$\backslash[1+3+5+7+9=25 \backslash]$$

Five odd numbers:

$$\backslash[5^2=25 \backslash]$$

29. SUM OF FIRST n EVEN NUMBERS

$$\backslash[2+4+6+\cdots+2n \backslash]$$

$$\backslash[=2(1+2+\cdots+n) \backslash]$$

$$\backslash[=2\cdot\frac{n(n+1)}{2} \backslash]$$

Therefore:

$$\backslash[\boxed{n(n+1)} \backslash]$$

30. SUM FROM mth TO nth TERM

If:

$$\backslash[S_n=\text{sum of first } n \text{ terms} \backslash]$$

then sum from term $\backslash(m\backslash)$ to term $\backslash(n\backslash)$:

$$\backslash[\boxed{S_n-S_{m-1}} \backslash]$$

31. EXAMPLE

Find sum of terms 6 through 10 of:

$$\backslash[2,5,8,\ldots \backslash]$$

We can use:

$$\backslash[S_{10}-S_5 \backslash]$$

Or list terms:

6th:

$$\backslash[17 \backslash]$$

10th:

$$\backslash[29 \backslash]$$

There are 5 terms.

$$\backslash[\frac{5(17+29)}{2} = \frac{5(46)}{2} = \boxed{115} \backslash]$$

32. SUM OF CONSECUTIVE AP TERMS USING AVERAGE

If first selected term is (p) , last is (q) , and number of selected terms is (n) :

$$\boxed{\text{Sum} = n \cdot \frac{p+q}{2}}$$

This is often faster than $(S_n - S_{m-1})$.

33. AVERAGE OF AP TERMS

For any finite AP:

$$\boxed{\text{Average} = \frac{\text{first} + \text{last}}{2}}$$

Hence:

$$\boxed{S_n = n \times \text{average}}$$

34. EXAMPLE

Average of:

$$[7, 10, 13, \dots, 43]$$

is:

$$\frac{7+43}{2} = \boxed{25}$$

No need to find all terms.

35. nth TERM FROM SUM FORMULA

If (S_n) is known:

$$\boxed{a_n = S_n - S_{n-1}}$$

Very important.

36. EXAMPLE

Suppose:

$$[S_n = 3n^2 + 2n]$$

Find (a_n) .

$$[a_n = S_n - S_{n-1}]$$

$$[S_{n-1} = 3(n-1)^2 + 2(n-1)]$$

$$[= 3n^2 - 6n + 3 + 2n - 2]$$

$$[= 3n^2 - 4n + 1]$$

So:

$$[a_n = (3n^2+2n)-(3n^2-4n+1)]$$

$$[=\boxed{6n-1}]$$

37. IMPORTANT PATTERN

If:

$$[S_n]$$

is quadratic in (n) , then:

$$[a_n = S_n - S_{n-1}]$$

is linear in (n) .

Therefore the sequence is an AP.

Useful GATE reasoning.

38. FIND FIRST TERM FROM (S_n)

$$[a_1 = S_1]$$

Example:

$$[S_n = 2n^2 + 3n]$$

Then:

$$[a_1 = S_1 = 5]$$

39. FIND COMMON DIFFERENCE FROM (S_n)

Find:

$$[a_1 = S_1]$$

$$[a_2 = S_2 - S_1]$$

Then:

$$[d = a_2 - a_1]$$

Or derive general (a_n) .

40. TERMS EQUIDISTANT FROM ENDS

In a finite AP:

$$[a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2}]$$

All symmetric pairs have same sum:

$$[\boxed{a+l}]$$

41. EXAMPLE

AP has first term 4 and last term 40.

Then:

$$a_2 + a_{n-1} = \boxed{44}$$

No other information needed.

42. SUM OF SYMMETRIC TERMS

If:

$$a_r$$

and:

$$a_{n-r+1}$$

are symmetric terms:

$$\boxed{a_r + a_{n-r+1} = a_1 + a_n}$$

High-value shortcut.

43. TERM RELATION IN AN AP

For an AP:

$$a_p + a_q$$

depends on $(p+q)$.

If:

$$p+q=r+s$$

then:

$$\boxed{a_p + a_q = a_r + a_s}$$

because:

$$a_p + a_q = 2a + (p+q-2)d$$

44. EXAMPLE

In any AP:

$$a_3 + a_{10}$$

equals:

$$a_5 + a_8$$

because:

$$3+10=5+8=13$$

So:

$$\boxed{a_3 + a_{10} = a_5 + a_8}$$

45. MIDDLE TERM RELATION

If (a, b, c) are consecutive AP terms:

$$\boxed{2b = a + c}$$

Equivalently:

$$\boxed{b = \frac{a+c}{2}}$$

46. IF TERMS AT POSITIONS p, q, r ARE IN AP

Suppose indices themselves satisfy:

$$p + r = 2q$$

Then corresponding AP terms satisfy:

$$\boxed{a_p + a_r = 2a_q}$$

47. EXAMPLE

In any AP:

$$a_4 + a_{10} = 2a_7$$

because:

$$4 + 10 = 14 = 2(7)$$

48. IF p th TERM IS q AND q th TERM IS p

Suppose:

$$a_p = q$$

$$a_q = p$$

Subtract equations:

$$(q - p)d = p - q$$

If $(p \neq q)$:

$$d = -1$$

Then:

$$a_p = a + (p-1)(-1) = q$$

$$a = q + p - 1$$

So:

$$\boxed{a = p + q - 1, \quad d = -1}$$

A common aptitude pattern.

49. WHAT IS THE $(p+q)$ th TERM?

Using:

$$a_n = a + (n-1)d$$

$$a_{p+q} = (p+q-1)d - (p+q-1)d = \boxed{0}$$

So:

If p th term is q and q th term is p , then $((p+q))$ th term is 0.

Useful shortcut.

50. IF p th TERM IS q AND q th TERM IS p — ANOTHER RESULT

The $((p+q-1))$ th term:

$$a_{p+q-1} = (p+q-1)d - (p+q-2)d = \boxed{1}$$

51. IF m th TERM IS ZERO

Suppose:

$$a_m = 0$$

Then:

$$a + (m-1)d = 0$$

For terms equidistant around (m) :

$$a_{m-r} + a_{m+r} = 0$$

They are negatives of each other.

52. EXAMPLE

If 7th term of an AP is 0:

$$a_5 + a_9 = 2a_7 = \boxed{0}$$

53. NUMBER OF TERMS BETWEEN TWO AP VALUES

Suppose AP from (a) to (l) .

$$l = a + (n-1)d$$

Thus:

$$\boxed{n = \frac{l-a}{d} + 1}$$

provided result is a positive integer.

54. EXAMPLE

How many terms in:

$$\{ 5, 9, 13, \dots, 101 \}$$

$$n = \frac{101 - 5}{4} + 1$$

$$= 24 + 1$$

$$= \boxed{25}$$

55. SUM OF MULTIPLES OF A NUMBER

Find sum of multiples of 7 from 7 to 140.

Sequence:

$$\{ 7, 14, 21, \dots, 140 \}$$

Number of terms:

$$140/7 = 20$$

Sum:

$$\frac{20}{2}(7 + 140)$$

$$= 10(147)$$

$$= \boxed{1470}$$

56. SUM OF MULTIPLES IN A RANGE

Find sum of multiples of 5 between 20 and 100 inclusive.

AP:

$$\{ 20, 25, \dots, 100 \}$$

$$n = \frac{100 - 20}{5} + 1 = 17$$

Sum:

$$\frac{17}{2}(120)$$

$$= \boxed{1020}$$

57. COUNT NUMBERS DIVISIBLE BY d IN A RANGE

Numbers divisible by d from 1 to N :

$$\left\lfloor \frac{N}{d} \right\rfloor$$

For interval $[L]$ to $[R]$:

$$\left\lfloor \frac{R}{d} \right\rfloor - \left\lfloor \frac{L-1}{d} \right\rfloor$$

This connects AP with divisibility.

58. GATE-STYLE QUESTION 1

The 5th term of an AP is 18 and 11th term is 42. Find first term.

$$\left[d = \frac{42-18}{6} = 4 \right]$$

$$\left[a + 4(4) = 18 \right]$$

$$\left[\boxed{a=2} \right]$$

59. GATE-STYLE QUESTION 2

Find 30th term of:

$$\left[7, 11, 15, \dots \right]$$

$$\left[a=7, d=4 \right]$$

$$\left[a_{30} = 7 + 29(4) \right]$$

$$\left[= \boxed{123} \right]$$

60. GATE-STYLE QUESTION 3

Find sum of first 25 terms:

$$\left[2, 5, 8, \dots \right]$$

$$\left[S_{25} = \frac{25}{2} [4 + 24(3)] \right]$$

$$\left[= \frac{25}{2} (76) \right]$$

$$\left[= \boxed{950} \right]$$

61. GATE-STYLE QUESTION 4

If:

$$\left[S_n = 2n^2 + n \right]$$

find nth term.

$$\left[S_{n-1} = 2(n-1)^2 + (n-1) \right]$$

$$\left[= 2n^2 - 3n + 1 \right]$$

Thus:

$$\left[a_n = (2n^2 + n) - (2n^2 - 3n + 1) \right]$$

$$\left[= \boxed{4n-1} \right]$$

62. PLACEMENT-STYLE QUESTION 1

Insert 4 arithmetic means between 3 and 28.

There are:

$$\backslash[5 \backslash]$$

gaps.

$$\backslash[d=\frac{25}{5}=5 \backslash]$$

Sequence:

$$\backslash[3,8,13,18,23,28 \backslash]$$

Inserted means:

$$\backslash[\boxed{8,13,18,23} \backslash]$$

63. PLACEMENT-STYLE QUESTION 2

Three numbers in AP have sum 30 and product 910.

Let:

$$\backslash[10-d, \ 10, \ 10+d \backslash]$$

Product:

$$\backslash[10(100-d^2)=910 \backslash]$$

$$\backslash[100-d^2=91 \backslash]$$

$$\backslash[d^2=9 \backslash]$$

$$\backslash[d=3 \backslash]$$

Numbers:

$$\backslash[\boxed{7,10,13} \backslash]$$

(order may reverse).

64. PLACEMENT-STYLE QUESTION 3

Find sum:

$$\backslash[11+14+17+\cdots+101 \backslash]$$

$$\backslash[a=11, \ d=3, \ l=101 \backslash]$$

$$\backslash[n=\frac{101-11}{3}+1 = 30+1 = 31 \backslash]$$

$$\backslash[S=\frac{31}{2}(112) \backslash]$$

$$\backslash[=\boxed{1736} \backslash]$$

65. ADVANCED QUESTION — TWO TERM CONDITIONS

In an AP:

$$\backslash[a_7=20 \backslash]$$

$$\backslash[a_{13}=44 \backslash]$$

Find a_{25} .

Difference:

$$6d=24$$

$$d=4$$

From 13th to 25th is 12 steps:

$$a_{25}=44+12(4)$$

$$=\boxed{92}$$

No need to find first term.

66. ADVANCED QUESTION — TERM SUM SHORTCUT

In an AP:

$$a_8=17$$

$$a_{20}=41$$

Find:

$$a_{10}+a_{18}$$

Indices:

$$10+18=28$$

Also:

$$8+20=28$$

Therefore:

$$a_{10}+a_{18} = a_8+a_{20}$$

$$=17+41$$

$$=\boxed{58}$$

Very fast.

67. ADVANCED QUESTION — MIDDLE TERM

If:

$$a_5=12$$

and:

$$a_{17}=48$$

find a_{11} .

Since:

$$11=\frac{5+17}{2}$$

$$a_{11} = \frac{a_5+a_{17}}{2}$$

$$=\frac{60}{2}$$

$$\boxed{30}$$

68. ADVANCED QUESTION — SUM WITHOUT a,d

If an AP has 21 terms and middle term is 15, find sum.

For odd number of terms:

Average = middle term.

Therefore:

$$S = 21(15)$$

$$\boxed{315}$$

69. ADVANCED QUESTION — FIRST/LAST WITHOUT d

An AP has 30 terms.

First term = 7.

Last = 94.

Sum:

$$\frac{30}{2}(7+94)$$

$$= 15(101)$$

$$\boxed{1515}$$

No common difference required.

70. COMMON TRAPS

TRAP 1 — nth term uses $(n-1)d$

Wrong.

$$\boxed{a_n = a + (n-1)d}$$

TRAP 2 — Number of gaps

Between first and nth term there are:

$$\boxed{n-1}$$

differences.

TRAP 3 — Inserting k means

There are:

$$\boxed{k+1}$$

gaps.

& pTRAP 4 — AP sum average

Average is:

$$\left[\frac{\text{first} + \text{last}}{2} \right]$$

not necessarily first + last.

& pTRAP 5 — Forgetting negative d

A decreasing sequence may still be a valid AP.

& pTRAP 6 — Term membership

Solving for (n) must give a positive integer.

71. MUST-MEMORIZE AP FORMULAS

nth term

$$\boxed{a_n = a + (n-1)d}$$

Sum

$$\boxed{S_n = \frac{n}{2} [2a + (n-1)d]}$$

or:

$$\boxed{S_n = \frac{n}{2} (a + l)}$$

Difference from two terms

$$\boxed{d = \frac{a_n - a_m}{n - m}}$$

Arithmetic mean

$$\boxed{\frac{a+b}{2}}$$

72. MUST-MEMORIZE BASIC SUMS

$$\boxed{1 + 2 + \cdots + n = \frac{n(n+1)}{2}}$$

$$\boxed{1 + 3 + \cdots + (2n-1) = n^2}$$

$$\boxed{2 + 4 + \cdots + 2n = n(n+1)}$$

73. HIGH-VALUE AP SHORTCUTS

For an AP:

$$\boxed{a_p + a_q = a_r + a_s}$$

whenever:

$$\boxed{p+q=r+s}$$

and:

$$\boxed{a_{m-r} + a_{m+r} = 2a_m}$$

74. FAST EXAM DECISION TREE

Need nth term?

Use:

$$a + (n-1)d$$

Two terms given?

Subtract to get (d) .

Need sum?

Use:

$$n \times \frac{\text{first} + \text{last}}{2}$$

Odd number of equally spaced terms?

Middle term = average.

Repeated symmetric indices?

Use index-sum shortcut.

(S_n) given?

Use:

$$a_n = S_n - S_{n-1}$$

Insert arithmetic means?

Divide total difference by number of gaps.

PRACTICE — LEVEL 0

Q1

Find the 15th term of:

$$3, 7, 11, \dots$$

Q2

Find common difference if:

$$a_4 = 14, \quad a_{10} = 32$$

Q3

Find sum of first 20 terms:

$$5, 8, 11, \dots$$

Q4

Insert 2 arithmetic means between 4 and 16.

Q5

Which term of:

$$\{ 2, 7, 12, \dots \}$$

is 62?

LEVEL 1

Q6

If:

$$a_5 = 17, \quad a_{12} = 45$$

find a and d .

Q7

Find sum:

$$8 + 13 + 18 + \dots + 98$$

Q8

Five consecutive AP terms have sum 75. Find middle term.

Q9

If $S_n = 3n^2 + n$, find a_n .

Q10

In an AP, $a_4 = 11$ and $a_{10} = 29$. Find a_7 .

LEVEL 2

Q11

In an AP:

$$a_6 = 14, \quad a_{16} = 44$$

find:

$$a_8 + a_{14}$$

Q12

Three numbers are in AP. Their sum is 24 and product is 440. Find them.

Q13

An AP has 25 terms and middle term 18. Find the total sum.

Q14

How many multiples of 7 are there from 50 to 300 inclusive?

Q15

Find sum of all multiples of 6 from 1 to 300.

LEVEL 3 — EXAM PRESSURE

Q16

If the 8th term of an AP is 0, find:

$$a_5 + a_{11}$$

Q17

If the p th term of an AP is q , and the q th term is p , find the $(p+q)$ th term.

Q18

If:

$$a_3 + a_{15} = 40$$

find $a_7 + a_{11}$.

Q19

An AP has first term 4 and 30th term 91. Find its sum.

Q20

If:

$$S_n = 4n^2 - 3n$$

find:

1. a_n
2. first term
3. common difference.

ANSWERS

Q1

$$a_{15} = 3 + 14(4)$$

$$= \boxed{59}$$

Q2

$$d = \frac{32-14}{10-4} = \frac{18}{6} = \boxed{3}$$

Q3

20th term:

$$5 + 19(3) = 62$$

Sum:

$$\frac{20}{2}(5+62)$$

$$= \boxed{670}$$

Q4

Three gaps:

$$\lfloor d = \frac{16-4}{3} = 4 \rfloor$$

Sequence:

$$\lfloor 4, 8, 12, 16 \rfloor$$

$$\lfloor \boxed{8, 12} \rfloor$$

Q5

$$\lfloor 62 = 2 + (n-1)5 \rfloor$$

$$\lfloor 60 = 5(n-1) \rfloor$$

$$\lfloor n-1 = 12 \rfloor$$

$$\lfloor \boxed{n=13} \rfloor$$

Q6

$$\lfloor d = \frac{45-17}{12-5} = \frac{28}{7} = 4 \rfloor$$

$$\lfloor a+4d=17 \rfloor$$

$$\lfloor a=1 \rfloor$$

$$\lfloor \boxed{a=1, d=4} \rfloor$$

Q7

$$\lfloor n = \frac{98-8}{5} + 1 = 19 \rfloor$$

$$\lfloor S = \frac{19}{2}(8+98) \rfloor$$

$$\lfloor = 19(53) \rfloor$$

$$\lfloor = \boxed{1007} \rfloor$$

Q8

For 5 AP terms:

$$\lfloor \text{middle} = \text{average} \rfloor$$

$$\lfloor 75/5 = \boxed{15} \rfloor$$

Q9

$$\lfloor a_n = S_n - S_{n-1} \rfloor$$

$$\lfloor S_{n-1} = 3(n-1)^2 + (n-1) \rfloor$$

$$\lfloor = 3n^2 - 5n + 2 \rfloor$$

Therefore:

$$\lfloor a_n = 3n^2 + n - (3n^2 - 5n + 2) \rfloor$$

$$\lfloor = \boxed{6n-2} \rfloor$$

Q10

7th term lies exactly midway between positions 4 and 10:

$$\lfloor a_7 = \frac{11+29}{2} \rfloor$$

$$\boxed{20}$$

Q11

Indices:

$$8+14=22$$

and:

$$6+16=22$$

Hence:

$$a_8 + a_{14} = a_6 + a_{16}$$

$$=14+44$$

$$\boxed{58}$$

Q12

Let terms:

$$8-d, 8, 8+d$$

Product:

$$8(64-d^2)=440$$

$$64-d^2=55$$

$$d^2=9$$

$$d=3$$

Numbers:

$$\boxed{5, 8, 11}$$

Q13

Average = middle term:

$$18$$

Therefore:

$$S=25(18)$$

$$\boxed{450}$$

Q14

Multiples of 7:

First (≥ 50) :

$$56$$

Last (≤ 300) :

$$294$$

$$n = \frac{294-56}{7} + 1$$

$$=34+1$$

$$\boxed{35}$$

Q15

Sequence:

$$\backslash[6,12,\ldots,300 \backslash]$$

Number terms:

$$\backslash[50 \backslash]$$

Sum:

$$\backslash[\frac{50}{2}(6+300) \backslash]$$

$$\backslash[=25(306) \backslash]$$

$$\backslash[=\boxed{7650} \backslash]$$

Q16

Since:

$$\backslash[a_8=0 \backslash]$$

and 5 and 11 are equally distant from 8:

$$\backslash[a_5+a_{11}=2a_8 \backslash]$$

$$\backslash[=\boxed{0} \backslash]$$

Q17

As derived:

$$\backslash[d=-1 \backslash]$$

and:

$$\backslash[a=p+q-1 \backslash]$$

Therefore:

$$\backslash[a_{p+q} = a+(p+q-1)d \backslash]$$

$$\backslash[=(p+q-1)-(p+q-1) \backslash]$$

$$\backslash[=\boxed{0} \backslash]$$

Q18

Indices:

$$\backslash[3+15=18 \backslash]$$

$$\backslash[7+11=18 \backslash]$$

Hence:

$$\backslash[a_7+a_{11} = a_3+a_{15} \backslash]$$

$$\backslash[=\boxed{40} \backslash]$$

Q19

$$\backslash[S_{30} = \frac{30}{2}(4+91) \backslash]$$

$$\backslash[=15(95) \backslash]$$

$$\backslash[=\boxed{1425} \backslash]$$

Q20

$$\{ S_n = 4n^2 - 3n \}$$

$$\{ S_{n-1} = 4(n-1)^2 - 3(n-1) \}$$

$$\{ = 4n^2 - 11n + 7 \}$$

Therefore:

$$\{ a_n = (4n^2 - 3n) - (4n^2 - 11n + 7) \}$$

$$\{ = \boxed{8n - 7} \}$$

First term:

$$\{ a_1 = 1 \}$$

Common difference:

$$\{ 8 \}$$

So:

$$\{ \boxed{a_n = 8n - 7, \text{quad } a = 1, \text{quad } d = 8} \}$$

CHAPTER 37 STATUS

Covered so far:

- sequence vs series '
- AP definition '
- common difference '
- nth term '
- finding term position '
- term-membership test '
- finding (a, d) from two terms '
- consecutive AP terms '
- arithmetic mean '
- inserting arithmetic means '
- sum of AP '
- natural/odd/even number sums '
- sum over a selected range '
- average of an AP '
- deriving (a_n) from (S_n) '
- symmetric-term shortcuts '
- index-sum identities '
- special (p) th/ (q) th term patterns '
- multiples and divisibility applications '
- L0–L3 practice '

But **Chapter 37 is not complete yet.**

Next we still need:

- Geometric Progression (GP)

- common ratio
- nth term of GP
- finite GP sum
- infinite GP
- geometric mean
- inserting geometric means
- transformed GP patterns
- AP vs GP comparison
- mixed sequence recognition
- GATE/placement traps and final audit

Progress

Completed: 36/76 Current: 0/36 Chapter 37 — Sequences, AP & GP

NEXT

+ Chapter 37 Part 2 — Geometric Progression, finite/infinite GP, geometric means & mixed sequence patterns

USER

next

ASSISTANT

CHAPTER 37 — SEQUENCES, AP AND GP

PART 2 — GEOMETRIC PROGRESSION, FINITE/INFINITE GP, GEOMETRIC MEANS & MIXED SEQUENCE PATTERNS

Status: 0/36 PROGRESS GATE: Medium Placements: Medium–High Priority: +Tier 2

Part 1 covered **Arithmetic Progression (AP)**, where consecutive terms differ by a constant.

Now we study **Geometric Progression (GP)**, where consecutive terms have a constant ratio.

1. WHAT IS A GP?

A sequence is a Geometric Progression if:

$$\left[\frac{a_{n+1}}{a_n} \right]$$

is constant.

That constant is called the **common ratio**:

$$\left[\boxed{r} \right]$$

Example:

$$\left[2, 6, 18, 54, \dots \right]$$

Ratios:

$$\sqrt[6]{2}=3$$

$$\sqrt[18]{6}=3$$

$$\sqrt[54]{18}=3$$

So:

$$\boxed{r=3}$$

2. GP CAN DECREASE

Example:

$$\sqrt[81]{27,9,3,\ldots}$$

$$\sqrt[r]{r=\frac{1}{3}}$$

So a GP need not increase.

3. NEGATIVE COMMON RATIO

Example:

$$\sqrt[2]{-6,18,-54,\ldots}$$

Here:

$$\sqrt[r]{r=-3}$$

Signs alternate because the common ratio is negative.

4. nth TERM OF A GP

If:

- first term $(=a)$
- common ratio $(=r)$

then:

$$\sqrt[a_1]{a}$$

$$\sqrt[a_2]{ar}$$

$$\sqrt[a_3]{ar^2}$$

Therefore:

$$\sqrt[a_n]{a_n=ar^{n-1}}$$

MUST MEMORIZE

5. EXAMPLE

GP:

$$\backslash[3,6,12,24,\ldots \backslash]$$

Here:

$$\backslash[a=3,\quad r=2 \backslash]$$

Find 8th term.

$$\backslash[a_8=3(2^7) \backslash]$$

$$\backslash[=3(128) \backslash]$$

$$\backslash[=\boxed{384} \backslash]$$

6. FIND WHICH TERM

In:

$$\backslash[2,6,18,54,\ldots \backslash]$$

which term is 486?

$$\backslash[486=2(3^{\{n-1\}}) \backslash]$$

$$\backslash[243=3^{\{n-1\}} \backslash]$$

$$\backslash[3^5=3^{\{n-1\}} \backslash]$$

Therefore:

$$\backslash[n-1=5 \backslash]$$

$$\backslash[\boxed{n=6} \backslash]$$

7. RATIO FROM TWO TERMS

For a GP:

$$\backslash[a_m=ar^{\{m-1\}} \backslash]$$

$$\backslash[a_n=ar^{\{n-1\}} \backslash]$$

Divide:

$$\backslash[\frac{a_n}{a_m}=r^{\{n-m\}} \backslash]$$

Thus:

$$\backslash[\boxed{r^{\{n-m\}}=\frac{a_n}{a_m}} \backslash]$$

8. EXAMPLE

4th term = 24.

7th term = 192.

Then:

$$\backslash[\frac{192}{24}=r^{\{7-4\}} \backslash]$$

$$\backslash[8=r^3 \backslash]$$

$$\boxed{r=2}$$

9. FIND FIRST TERM

Using previous example:

$$a_4 = ar^3 = 24$$

$$a(8) = 24$$

$$\boxed{a=3}$$

So GP:

$$3, 6, 12, \dots$$

10. THREE CONSECUTIVE GP TERMS

A very useful representation is:

$$\boxed{\frac{x}{r}, x, xr}$$

because the middle term is x .

Another common representation:

$$a, ar, ar^2$$

Use whichever simplifies the question.

11. GEOMETRIC MEAN

If:

$$a, G, b$$

are in GP, then:

$$\frac{Ga}{b} = \frac{b}{G}$$

Therefore:

$$G^2 = ab$$

Hence:

$$\boxed{G = \sqrt{ab}}$$

for positive (a, b) .

12. EXAMPLE

Geometric mean between 4 and 16:

$$G = \sqrt{4(16)}$$

$$= \sqrt{64}$$

$$= \boxed{8}$$

So:

$$\{4, 8, 16\}$$

is a GP.

13. KEY PROPERTY OF THREE GP TERMS

If (a, b, c) are consecutive terms of a GP:

$$\boxed{b^2 = ac}$$

This is the GP version of the AP relation:

$$2b = a + c$$

Memory comparison

AP:

$$\boxed{2b = a + c}$$

GP:

$$\boxed{b^2 = ac}$$

14. EXAMPLE

Are:

$$\{2, 6, 18\}$$

in GP?

Check:

$$6^2 = 36$$

$$2(18) = 36$$

Yes.

15. INSERTING GEOMETRIC MEANS

Suppose (k) geometric means are inserted between (a) and (b) .

Then there are:

$$(k+1)$$

ratio steps.

Thus:

$$b = ar^{k+1}$$

So:

$$\boxed{r = \left(\frac{b}{a}\right)^{\frac{1}{k+1}}}$$

16. EXAMPLE

Insert 2 geometric means between 2 and 54.

Sequence:

$$\backslash[2, \backslash G_1, \backslash G_2, \backslash 54 \backslash]$$

There are 3 ratio steps.

$$\backslash[54=2r^3 \backslash]$$

$$\backslash[r^3=27 \backslash]$$

$$\backslash[r=3 \backslash]$$

So:

$$\backslash[2, 6, 18, 54 \backslash]$$

Inserted means:

$$\backslash[\boxed{6, 18} \backslash]$$

17. FINITE GP SUM

For:

$$\backslash[a+ar+ar^2+\cdots+ar^{n-1} \backslash]$$

the sum is:

$$\backslash[\boxed{S_n=a\frac{r^n-1}{r-1}} \backslash]$$

for:

$$\backslash[r \neq 1 \backslash]$$

Equivalent form:

$$\backslash[\boxed{S_n=a\frac{1-r^n}{1-r}} \backslash]$$

Both are correct.

18. WHICH FORM SHOULD YOU USE?

If:

$$\backslash[r > 1 \backslash]$$

often convenient:

$$\backslash[\boxed{S_n=a\frac{r^n-1}{r-1}} \backslash]$$

If:

$$\backslash[|r| < 1 \backslash]$$

often convenient:

$$\backslash[\boxed{S_n=a\frac{1-r^n}{1-r}} \backslash]$$

19. DERIVATION OF GP SUM

Let:

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}$$

Multiply by (r) :

$$rS_n = ar + ar^2 + \dots + ar^n$$

Subtract:

$$rS_n - S_n = ar^n - a$$

$$S_n(r-1) = a(r^n - 1)$$

Therefore:

$$\boxed{S_n = a \frac{r^n - 1}{r - 1}}$$

20. EXAMPLE

Find:

$$2 + 6 + 18 + 54$$

Here:

$$a = 2, \quad r = 3, \quad n = 4$$

$$S_4 = 2 \frac{3^4 - 1}{3 - 1}$$

$$= 2 \frac{81 - 1}{2}$$

$$= \boxed{80}$$

Check:

$$2 + 6 + 18 + 54 = 80$$

21. ANOTHER EXAMPLE

Find:

$$1 + 2 + 4 + \dots + 2^9$$

There are 10 terms.

$$a = 1, \quad r = 2$$

$$S_{10} = 2^{10} - 1$$

$$= \boxed{1023}$$

22. VERY IMPORTANT POWER-OF-2 RESULT

$$1 + 2 + 4 + \dots + 2^{n-1}$$

has sum:

$$\boxed{2^n - 1}$$

Useful in binary/counting problems too.

23. IF $r = 1$

GP:

$$[a, a, a, \dots]$$

Then:

$$[\boxed{S_n = na}]$$

The standard GP sum formula has denominator $(r-1 \neq 0)$, so handle this case separately.

24. GP WITH NEGATIVE RATIO

Example:

$$[1, -2, 4, -8, 16]$$

Here:

$$[a=1, \text{quad } r=-2, \text{quad } n=5]$$

$$[S_5 = \frac{(-2)^5 - 1}{-2 - 1}]$$

$$[= \frac{-32 - 1}{-3}]$$

$$[=\boxed{11}]$$

Check:

$$[1 - 2 + 4 - 8 + 16 = 11]$$

25. SUM FROM m th TO n th GP TERM

If:

$$[S_n = \text{sum first } n]$$

then:

$$[\boxed{\text{sum from } m \text{ to } n} = S_n - S_{m-1}]$$

Same principle as AP.

26. EXAMPLE

GP:

$$[2, 4, 8, 16, 32, \dots]$$

Find sum of 3rd through 6th terms.

Terms:

$$[8 + 16 + 32 + 64]$$

This itself is a GP:

$$\backslash[a=8,\backslash r=2,\backslash n=4 \backslash]$$

Sum:

$$\backslash[8(2^4-1) \backslash]$$

$$\backslash[=8(15) \backslash]$$

$$\backslash[=\boxed{120} \backslash]$$

27. INFINITE GP

Consider:

$$\backslash[a+ar+ar^2+ar^3+\cdots \backslash]$$

If:

$$\backslash[\boxed{|r|<1} \backslash]$$

the terms shrink toward zero, and the infinite sum exists.

Formula:

$$\backslash[\boxed{S_{\infty}=\frac{a}{1-r}} \backslash]$$

MUST MEMORIZE

28. WHY $\backslash(|r|<1\backslash)$?

From finite GP:

$$\backslash[S_n=a\frac{1-r^n}{1-r} \backslash]$$

If:

$$\backslash[|r|<1 \backslash]$$

then as $\backslash(n\rightarrow\infty\backslash)$:

$$\backslash[r^n\rightarrow 0 \backslash]$$

So:

$$\backslash[S_{\infty} = a\frac{1-0}{1-r} \backslash]$$

$$\backslash[=\boxed{\frac{a}{1-r}} \backslash]$$

29. EXAMPLE

$$\backslash[1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots \backslash]$$

Here:

$$\backslash[a=1,\quad r=\frac{1}{2} \backslash]$$

$$\backslash[S_{\infty} = \frac{1}{1-\frac{1}{2}} \backslash]$$

$$\backslash[=\boxed{2} \backslash]$$

30. ANOTHER INFINITE GP

$$\backslash[3+1+\frac{13}{3}+\frac{19}{3}+\cdots \backslash]$$

$$\backslash[a=3,\text{quad } r=\frac{13}{3} \backslash]$$

$$\backslash[S_{\infty} = \frac{3}{1-\frac{13}{3}} \backslash]$$

$$\backslash[= \frac{3}{2/3} \backslash]$$

$$\backslash[=\boxed{\frac{9}{2}} \backslash]$$

31. ALTERNATING INFINITE GP

$$\backslash[1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\cdots \backslash]$$

$$\backslash[a=1,\text{quad } r=-\frac{1}{2} \backslash]$$

Since:

$$\backslash[|r|<1 \backslash]$$

sum exists:

$$\backslash[S_{\infty} = \frac{1}{1+\frac{1}{2}} \backslash]$$

$$\backslash[=\boxed{\frac{2}{3}} \backslash]$$

32. DIVERGENT INFINITE GP

$$\backslash[1+2+4+8+\cdots \backslash]$$

Here:

$$\backslash[r=2 \backslash]$$

Since:

$$\backslash[|r|>1 \backslash]$$

there is no finite infinite sum.

$$\backslash[\boxed{\text{diverges}} \backslash]$$

33. ANOTHER NON-CONVERGENT CASE

$$\backslash[1-1+1-1+\cdots \backslash]$$

Here:

$$\backslash[r=-1 \backslash]$$

Since:

$$\backslash[|r|=1 \backslash]$$

standard infinite GP sum does not exist.

34. REPEATING DECIMAL USING INFINITE GP

Example:

$$\left[0.3333\ldots \right]$$

Write:

$$\left[\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \cdots \right]$$

This is GP:

$$\left[a = \frac{3}{10} \right]$$

$$\left[r = \frac{1}{10} \right]$$

So:

$$\left[S_{\infty} = \frac{3/10}{1-1/10} \right]$$

$$\left[= \frac{3/10}{9/10} \right]$$

$$\left[= \boxed{\frac{13}{10}} \right]$$

35. ANOTHER RECURRING DECIMAL

$$\left[0.272727\ldots \right]$$

Write blocks:

$$\left[\frac{27}{100} + \frac{27}{100^2} + \frac{27}{100^3} + \cdots \right]$$

So:

$$\left[a = \frac{27}{100} \right]$$

$$\left[r = \frac{1}{100} \right]$$

$$\left[S_{\infty} = \frac{27/100}{99/100} \right]$$

$$\left[= \frac{27}{99} \right]$$

$$\left[= \boxed{\frac{3}{11}} \right]$$

36. THREE GP TERMS WITH KNOWN PRODUCT

Let three consecutive GP terms be:

$$\left[\frac{x}{r}, x, xr \right]$$

Their product:

$$\left[\frac{x}{r} \cdot x \cdot xr = x^3 \right]$$

So:

$$\left[\boxed{\text{middle term}} = \sqrt[3]{\text{product}} \right]$$

for real appropriate values.

37. EXAMPLE

Three positive numbers in GP have product:

$$\boxed{216}$$

Middle term:

$$\sqrt[3]{216} = \boxed{6}$$

38. THREE GP TERMS — PRODUCT PROPERTY

If (a, b, c) are consecutive GP terms:

$$b^2 = ac$$

Therefore:

$$abc = b^3$$

Hence:

$$\boxed{b = \sqrt[3]{abc}}$$

39. FIVE CONSECUTIVE GP TERMS

Represent symmetrically as:

$$\boxed{\frac{x}{r^2}, \frac{x}{r}, x, xr, xr^2}$$

Their product:

$$x^5$$

So middle term is the fifth root of the product.

More generally, for an odd number of positive GP terms:

geometric mean of all terms = middle term.

40. PRODUCT OF FINITE GP TERMS

For GP:

$$a, ar, \dots, ar^{n-1}$$

Product:

$$a^n r^{0+1+\dots+(n-1)}$$

$$= a^n r^{n(n-1)/2}$$

Thus:

$$\boxed{P_n = a^n r^{n(n-1)/2}}$$

Lower priority, but useful.

41. SYMMETRIC GP TERMS

For a finite GP:

$$\backslash[a_1a_n = a_2a_{\{n-1\}} = a_3a_{\{n-2\}} \backslash]$$

Each symmetric pair has same product.

Because:

$$\backslash[a_k a_{\{n-k+1\}} = a^{2r^{n-1}} \backslash]$$

42. EXAMPLE

In GP:

$$\backslash[2,6,18,54,162 \backslash]$$

$$\backslash[2(162)=324 \backslash]$$

$$\backslash[6(54)=324 \backslash]$$

and:

$$\backslash[18^2=324 \backslash]$$

43. INDEX PROPERTY OF GP

For GP:

$$\backslash[a_p a_q \backslash]$$

depends on $\backslash(p+q)\backslash$.

Thus if:

$$\backslash[p+q=r+s \backslash]$$

then:

$$\backslash[\boxed{a_{pa_q}=a_{ra_s}} \backslash]$$

Compare AP:

$$\backslash[a_p+a_q=a_r+a_s \backslash]$$

GP:

$$\backslash[a_{pa_q}=a_{ra_s} \backslash]$$

when index sums match.

44. EXAMPLE

In any GP:

$$\backslash[a_3a_{\{10\}} = a_5a_8 \backslash]$$

because:

$$\backslash[3+10=5+8 \backslash]$$

45. MIDDLE INDEX PROPERTY

If:

$$\backslash[p+r=2q \backslash]$$

then:

$$\backslash[\boxed{a_q^2=a_{p_r}} \backslash]$$

This is the indexed form of geometric mean.

46. EXAMPLE

In any GP:

$$\backslash[a_7^2=a_4a_{10} \backslash]$$

because:

$$\backslash[4+10=2(7) \backslash]$$

47. AP VS GP — QUICK RECOGNITION

Sequence:

$$\backslash[3,7,11,15,\ldots \backslash]$$

Differences:

$$\backslash[4,4,4 \backslash]$$

So AP.

Sequence:

$$\backslash[3,6,12,24,\ldots \backslash]$$

Ratios:

$$\backslash[2,2,2 \backslash]$$

So GP.

48. NEITHER AP NOR GP

$$\backslash[1,4,9,16,\ldots \backslash]$$

Differences:

$$\backslash[3,5,7,\ldots \backslash]$$

not constant.

Ratios not constant.

So neither AP nor GP.

It's the square-number sequence:

$$\backslash[\boxed{n^2} \backslash]$$

49. MIXED PATTERN— SECOND DIFFERENCES

Sequence:

$\{ 2, 5, 10, 17, 26, \dots \}$

First differences:

$\{ 3, 5, 7, 9 \}$

Second differences:

$\{ 2, 2, 2 \}$

Constant second difference suggests a quadratic pattern.

Indeed:

$\{ \boxed{a_n = n^2 + 1} \}$

50. MIXED PATTERN— ALTERNATING

$\{ 2, 5, 4, 10, 6, 15, 8, 20, \dots \}$

Separate odd and even positions:

Odd positions:

$\{ 2, 4, 6, 8 \}$

Even:

$\{ 5, 10, 15, 20 \}$

So sometimes split sequence into two sub-sequences.

51. MIXED PATTERN— MULTIPLY THEN ADD

$\{ 2, 5, 11, 23, 47, \dots \}$

Pattern:

$\{ 2 \times 2 + 1 = 5 \}$

$\{ 5 \times 2 + 1 = 11 \}$

$\{ 11 \times 2 + 1 = 23 \}$

So next:

$\{ 47 \times 2 + 1 = \boxed{95} \}$

Placement tests often use such patterns.

52. MIXED PATTERN— ALTERNATING OPERATIONS

$\{ 2, 4, 12, 24, 72, \dots \}$

Operations:

$$\backslash \times 2, \times 3, \times 2, \times 3, \ldots \backslash$$

Next:

$$\backslash 72 \times 2 = \boxed{144} \backslash$$

53. MIXED PATTERN— DIFFERENCE PATTERN

$$\backslash 4, 7, 13, 22, 34, \ldots \backslash$$

Differences:

$$\backslash 3, 6, 9, 12 \backslash$$

Next difference:

$$\backslash 15 \backslash$$

Next term:

$$\backslash 34 + 15 = \boxed{49} \backslash$$

54. GP FROM EXPONENTIAL GROWTH

If quantity multiplies by a constant each period, its values form a GP.

Example:

Population doubles each year:

$$\backslash 100, 200, 400, 800, \ldots \backslash$$

GP:

$$\backslash r=2 \backslash$$

This connects GP to compound interest and repeated percentage growth.

55. DEPRECIATION AS GP

If value decreases by 10% each year, multiplier:

$$\backslash 0.9 \backslash$$

So values:

$$\backslash V, 0.9V, 0.9^2V, \ldots \backslash$$

form a GP.

56. FIND NUMBER OF TERMS FROM LAST TERM

If:

$$\backslash l = ar^{n-1} \backslash$$

then solve:

$$\backslash r^{n-1} = \frac{la}{a} \backslash$$

For friendly powers, compare exponents.

57. EXAMPLE

GP:

$$\backslash[5,10,20,\ldots,640 \backslash]$$

$$\backslash[640=5(2^{\{n-1\}}) \backslash]$$

$$\backslash[128=2^{\{n-1\}} \backslash]$$

$$\backslash[2^7=2^{\{n-1\}} \backslash]$$

$$\backslash[\boxed{n=8} \backslash]$$

58. SUM GIVEN, FIND n

Example:

$$\backslash[1+2+4+\cdots \backslash]$$

has sum 255.

Using:

$$\backslash[S_n=2^n-1 \backslash]$$

$$\backslash[2^n-1=255 \backslash]$$

$$\backslash[2^n=256 \backslash]$$

$$\backslash[\boxed{n=8} \backslash]$$

59. FIND r FROM SUM AND FIRST TERM

Example:

First 3 GP terms sum to 21.

First term = 3.

Let ratio = $\backslash(r\backslash)$.

$$\backslash[3+3r+3r^2=21 \backslash]$$

Divide by 3:

$$\backslash[1+r+r^2=7 \backslash]$$

$$\backslash[r^2+r-6=0 \backslash]$$

$$\backslash[(r+3)(r-2)=0 \backslash]$$

Thus:

$$\backslash[\boxed{r=2\text{ or }-3} \backslash]$$

Depending on context, both may be valid.

60. POSITIVE-TERM CONTEXT

If question says all GP terms are positive, then in previous problem:

$$\boxed{r=-3}$$

would produce alternating signs.

So only:

$$\boxed{r=2}$$

Context matters.

61. GATE-STYLE QUESTION 1

Find 7th term:

$$2, 6, 18, \dots$$

$$a=2, r=3$$

$$a_7 = 2(3^6)$$

$$= 2(729)$$

$$\boxed{1458}$$

62. GATE-STYLE QUESTION 2

Find sum of first 6 terms:

$$1, 2, 4, \dots$$

$$S_6 = 2^6 - 1$$

$$\boxed{63}$$

63. GATE-STYLE QUESTION 3

Find:

$$1 + \frac{1}{3} + \frac{1}{9} + \dots$$

$$S_{\infty} = \frac{1}{1 - \frac{1}{3}}$$

$$\boxed{\frac{3}{2}}$$

64. GATE-STYLE QUESTION 4

If:

$$a_4 = 24, \quad a_7 = 192$$

find r .

$$r^3 = \frac{192}{24} = 8$$

$$\boxed{r=2}$$

65. PLACEMENT-STYLE QUESTION 1

Insert 3 geometric means between 1 and 16.

Sequence has 4 ratio steps:

$$\sqrt[4]{16=1\cdot r^4}$$

$$\sqrt[4]{r=2}$$

Sequence:

$$\sqrt[4]{1,2,4,8,16}$$

Inserted means:

$$\sqrt[4]{\boxed{2,4,8}}$$

66. PLACEMENT-STYLE QUESTION 2

Three positive numbers in GP have product 1000.

Middle term:

$$\sqrt[3]{1000}$$

$$\sqrt[3]{\boxed{10}}$$

67. PLACEMENT-STYLE QUESTION 3

Find next term:

$$\sqrt[4]{3,6,18,72,360,\ldots}$$

Multipliers:

$$\sqrt[4]{\times 2,\times 3,\times 4,\times 5}$$

Next multiplier:

$$\sqrt[4]{\times 6}$$

$$\sqrt[4]{360(6)=\boxed{2160}}$$

This is not a GP, but a mixed multiplicative pattern.

68. PLACEMENT-STYLE QUESTION 4

Find next term:

$$\sqrt[4]{2,6,12,20,30,\ldots}$$

Differences:

$$\sqrt[4]{4,6,8,10}$$

Next:

$$\sqrt[4]{12}$$

So next term:

$$30+12 = \boxed{42}$$

Equivalent:

$$a_n = n(n+1)$$

starting from $n=1$.

69. ADVANCED QUESTION — TERM PRODUCT SHORTCUT

In a GP:

$$a_3 = 8$$

$$a_9 = 512$$

Find:

$$a_5 a_7$$

Since:

$$5+7=3+9$$

$$a_5 a_7 = a_3 a_9$$

$$= 8(512)$$

$$= \boxed{4096}$$

No need to find (a, r) .

70. ADVANCED QUESTION — MIDDLE TERM

In a GP:

$$a_4 = 9$$

$$a_{10} = 729$$

Find a_7 , assuming positive terms.

Since:

$$4+10=2(7)$$

$$a_7^2 = a_4 a_{10}$$

$$= 9(729)$$

$$= 6561$$

$$a_7 = \boxed{81}$$

71. ADVANCED QUESTION — INFINITE GP

An infinite GP has first term 12 and sum 18.

Find (r) .

$$\left[18 = \frac{12}{1-r} \right]$$

$$\left[1-r = \frac{2}{3} \right]$$

$$\left[\boxed{r = \frac{1}{3}} \right]$$

72. ADVANCED QUESTION — FIRST TERM FROM INFINITE SUM

An infinite GP has:

$$\left[r = \frac{1}{4} \right]$$

and:

$$\left[S_{\infty} = 20 \right]$$

Then:

$$\left[20 = \frac{a}{1 - \frac{1}{4}} \right]$$

$$\left[20 = \frac{a}{3/4} \right]$$

$$\left[a = 15 \right]$$

$$\left[\boxed{15} \right]$$

73. ADVANCED QUESTION — SUM AFTER FIRST TERM

Infinite GP:

$$\left[a = 8, \text{quad } r = \frac{1}{2} \right]$$

Total infinite sum:

$$\left[16 \right]$$

Sum excluding first term:

$$\left[16 - 8 = \boxed{8} \right]$$

Or:

$$\left[ar + ar^2 + \cdots \right]$$

is another infinite GP with first term 4:

$$\left[\frac{4}{1 - \frac{1}{2}} = 8 \right]$$

74. COMMON TRAPS

& pTRAP 1 — nth GP term

Wrong:

$$\left[ar^n \right]$$

Correct:

$$\left[\boxed{ar^{n-1}} \right]$$

& pTRAP 2 — Confusing AP and GP

AP !' constant difference.

GP !' constant ratio.

& pTRAP 3 — Infinite GP formula without checking (r)

Use:

$$\boxed{S_{\infty} = \frac{a}{1-r}}$$

only if:

$$\boxed{|r| < 1}$$

& pTRAP 4 — Geometric mean

For positive numbers:

$$\boxed{G = \sqrt[n]{a_1 a_2 \dots a_n}}$$

not $\frac{a+b}{2}$.

& pTRAP 5 — Negative ratio

GP may alternate signs.

Don't assume $(r > 0)$.

& pTRAP 6 — Number of ratio gaps

If (k) geometric means are inserted, there are:

$$\boxed{k+1}$$

ratio steps.

75. MUST-MEMORIZE GP FORMULAS

nth term

$$\boxed{a_n = ar^{n-1}}$$

Finite sum

$$\boxed{S_n = a \frac{r^n - 1}{r - 1}}$$

or:

$$\boxed{S_n = a \frac{1 - r^n}{1 - r}}$$

Infinite sum

$$\boxed{S_{\infty} = \frac{a}{1-r} \quad \text{for } |r| < 1}$$

Geometric mean

$$\boxed{G = \sqrt[n]{a_1 a_2 \dots a_n}}$$

76. AP VS GP MASTER TABLE

Feature	AP	GP
Constant	Difference $\backslash(d\backslash)$	Ratio $\backslash(r\backslash)$
nth term	$\backslash(a+(n-1)d\backslash)$	$\backslash(ar^{\{n-1\}}\backslash)$
3 consecutive terms	$\backslash(x-d,x,x+d\backslash)$	$\backslash(x/r,x,xr\backslash)$
Middle relation	$\backslash(2b=a+c\backslash)$	$\backslash(b^2=ac\backslash)$
Symmetric-index property	equal sums	equal products

77. MIXED-SEQUENCE EXAM STRATEGY

When sequence is not obviously AP/GP, check in this order:

1. Differences
2. Second differences
3. Ratios
4. Alternating odd/even positions
5. Multiply/divide then add/subtract
6. Squares/cubes/primes/factorials
7. Increasing differences
8. Repeated operation pattern

Do not force every sequence into AP or GP.

PRACTICE — LEVEL 0

Q1

Find 6th term of:

$\backslash[3,9,27,\ldots\backslash]$

Q2

Find common ratio:

$\backslash[5,20,80,\ldots\backslash]$

Q3

Find sum of first 5 terms:

$\backslash[2,4,8,\ldots\backslash]$

Q4

Find geometric mean between 9 and 16.

Q5

Find:

$$\sqrt[3]{1 + \frac{1}{2} + \frac{1}{4} + \dots}$$

LEVEL 1

Q6

In a GP:

$$a_3 = 12, \quad a_6 = 96$$

find r and a .

Q7

Insert 2 geometric means between 3 and 81.

Q8

Find sum:

$$3 + 6 + 12 + \dots + 192$$

Q9

An infinite GP has first term 10 and common ratio $\frac{1}{5}$. Find its sum.

Q10

Three positive GP terms have product 512. Find middle term.

LEVEL 2

Q11

In a GP:

$$a_4 = 16, \quad a_{10} = 1024$$

find a_7 , assuming positive terms.

Q12

If:

$$a_2 a_8 = 96$$

find $a_4 a_6$.

Q13

An infinite GP has sum 24 and first term 18. Find r .

Q14

Find the number of terms in:

$$2, 6, 18, \dots, 1458$$

Q15

Find sum of first 8 terms:

$$\backslash[1,-2,4,-8,\ldots \backslash]$$

LEVEL 3 — EXAM PRESSURE

Q16

If three positive numbers in GP have sum 14 and product 64, find them.

Q17

In a GP:

$$\backslash[a_3=4,\quad a_9=256 \backslash]$$

find:

$$\backslash[a_5a_7 \backslash]$$

Q18

An infinite GP has second term 6 and sum 27. Find first term and common ratio.

Q19

Find next term:

$$\backslash[1,2,6,24,120,\ldots \backslash]$$

Q20

Find next term:

$$\backslash[3,8,15,24,35,\ldots \backslash]$$

ANSWERS

Q1

$$\backslash[a_6=3(3^5) \backslash]$$

$$\backslash[=3(243) \backslash]$$

$$\backslash[=\boxed{729} \backslash]$$

Q2

$$\backslash[r=20/5 = \boxed{4} \backslash]$$

Q3

$$\backslash[S_5 = 2\frac{2^5-1}{2-1} \backslash]$$

$$\backslash[=2(31) \backslash]$$

$$\backslash[=\boxed{62} \backslash]$$

Q4

$$\backslash[G=\sqrt{9(16)} \backslash]$$

$$\boxed{12}$$

Q5

$$\frac{1}{1-\frac{1}{2}} = \boxed{2}$$

Q6

$$\frac{a_6}{a_3} = r^3 = \frac{96}{12} = 8$$

$$r = 2$$

Then:

$$a_3 = ar^2$$

$$12 = 4a$$

$$a = 3$$

$$\boxed{a=3, r=2}$$

Q7

$$81 = 3r^3$$

$$r^3 = 27$$

$$r = 3$$

Sequence:

$$3, 9, 27, 81$$

$$\boxed{9, 27}$$

Q8

$$192 = 3(2^{n-1})$$

$$64 = 2^{n-1}$$

$$n = 7$$

Then:

$$S_7 = 3(2^7 - 1)$$

$$= 3(127)$$

$$= \boxed{381}$$

Q9

$$S_{\infty} = \frac{10}{1-\frac{1}{5}}$$

$$= \frac{10}{\frac{4}{5}}$$

$$= \boxed{\frac{25}{2} = 12.5}$$

Q10

Middle term:

$$\sqrt[3]{512} = \boxed{8}$$

Q11

Since:

$$\backslash[4+10=2(7)\backslash]$$

$$\backslash[a_7^2=a_4a_{10}\backslash]$$

$$\backslash[=16(1024)\backslash]$$

$$\backslash[=16384\backslash]$$

Positive terms:

$$\backslash[\boxed{a_7=128}\backslash]$$

Q12

Indices:

$$\backslash[2+8=4+6\backslash]$$

Therefore:

$$\backslash[a_2a_8=a_4a_6\backslash]$$

$$\backslash[\boxed{96}\backslash]$$

Q13

$$\backslash[24=\frac{18}{1-r}\backslash]$$

$$\backslash[1-r=\frac{3}{4}\backslash]$$

$$\backslash[\boxed{r=\frac{1}{4}}\backslash]$$

Q14

$$\backslash[1458=2(3^{\{n-1\}})\backslash]$$

$$\backslash[729=3^{\{n-1\}}\backslash]$$

$$\backslash[3^6=3^{\{n-1\}}\backslash]$$

$$\backslash[\boxed{ n=7}\backslash]$$

Q15

$$\backslash[a=1,\quad r=-2,\quad n=8\backslash]$$

$$\backslash[S_8=\frac{1-(-2)^8}{1-(-2)}\backslash]$$

$$\backslash[=\frac{1-256}{3}\backslash]$$

$$\backslash[=\boxed{-85}\backslash]$$

Q16

Let terms be:

$$\backslash[\frac{x}{r},x,xr\backslash]$$

Product:

$$\backslash[x^3=64\backslash]$$

$$\backslash[x=4\backslash]$$

Sum:

$$\backslash[\frac{4r+4+4r}{14}\backslash]$$

Divide by 2:

$$\left[\frac{2r+2+2r}{7} \right]$$

Or multiply original by (r) :

$$\left[4+4r+4r^2=14r \right]$$

$$\left[2r^2-5r+2=0 \right]$$

$$\left[(2r-1)(r-2)=0 \right]$$

Thus $(r=2)$ or $(1/2)$, giving same terms in reverse:

$$\left[\boxed{2,4,8} \right]$$

Q17

Since:

$$\left[5+7=3+9 \right]$$

$$\left[a_5a_7=a_3a_9 \right]$$

$$\left[=4(256) \right]$$

$$\left[=\boxed{1024} \right]$$

Q18

Let first term (a) , ratio (r) .

Second term:

$$\left[ar=6 \right]$$

Infinite sum:

$$\left[\frac{a}{1-r}=27 \right]$$

So:

$$\left[a=27(1-r) \right]$$

Multiply by (r) :

$$\left[27r(1-r)=6 \right]$$

$$\left[9r(1-r)=2 \right]$$

$$\left[9r-9r^2=2 \right]$$

$$\left[9r^2-9r+2=0 \right]$$

$$\left[(3r-1)(3r-2)=0 \right]$$

Therefore:

$$\left[r=\frac{1}{3} \right]$$

or:

$$\left[r=\frac{2}{3} \right]$$

Corresponding $(a=6/r)$:

For $(r=1/3)$:

$$\left[a=18 \right]$$

For $(r=2/3)$:

$$\backslash[a=9 \backslash]$$

Thus two valid infinite GPs:

$$\backslash[\boxed{(a,r)=(18,\frac{13}{2})\text{ or } (9,\frac{23}{3})} \backslash]$$

Q19

$$\backslash[1,2,6,24,120 \backslash]$$

These are:

$$\backslash[1!,2!,3!,4!,5! \backslash]$$

Next:

$$\backslash[6!=\boxed{720} \backslash]$$

Q20

$$\backslash[3,8,15,24,35 \backslash]$$

Differences:

$$\backslash[5,7,9,11 \backslash]$$

Next difference:

$$\backslash[13 \backslash]$$

Next term:

$$\backslash[35+13 = \boxed{48} \backslash]$$

Equivalent formula:

$$\backslash[a_n=n^2+2n \backslash]$$

' CHAPTER 37 COMPLETE

We have now covered:

- sequence vs series '
- AP definition and common difference '
- AP nth term '
- AP sum '
- arithmetic means '
- AP symmetric-term shortcuts '
- $\backslash(S_n \text{ to } a_n\backslash)$ conversion '
- natural/odd/even sums '
- GP definition and common ratio '
- GP nth term '
- finite GP sum '
- infinite GP '
- convergence condition $\backslash(|r|<1\backslash)$ '
- geometric mean '
- inserting geometric means '

- GP symmetric-product properties '
- index-based GP shortcuts '
- recurring decimals as infinite GP '
- repeated growth/depreciation connection '
- AP vs GP comparison '
- mixed sequence recognition '
- alternating/interleaved sequences '
- second-difference patterns '
- factorial/multiplicative patterns '
- GATE/placement L0–L3 practice '

Progress

Completed: 37/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 38 — Fundamental Counting Principle

Next we will start combinatorics with:

- counting without listing
- addition principle
- multiplication principle
- independent choices
- dependent-stage counting
- “at least/at most” counting
- complement counting
- digit/number formation
- repetition allowed vs forbidden
- restrictions on first digit
- passwords/codes
- paths and choice trees
- GATE/placement counting traps.

USER

next

ASSISTANT

CHAPTER 38 — FUNDAMENTAL COUNTING PRINCIPLE

PART 1 — ADDITION RULE, MULTIPLICATION RULE, DIGIT FORMATION, REPETITION &

COMPLEMENT COUNTING

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

Counting problems ask:

How many different valid outcomes are possible?

The biggest mistake is trying to list everything manually.

Instead, use two master ideas:

$\boxed{\text{OR} \rightarrow \text{ADD}}$

$\boxed{\text{AND / THEN} \rightarrow \text{MULTIPLY}}$

These form the foundation for permutations, combinations, and probability.

1. ADDITION PRINCIPLE

Suppose a task can be completed in:

- m ways using method A
- n ways using method B

and the two cases do not overlap.

Then total:

$\boxed{m+n}$

2. EXAMPLE — OR

You can travel from Hyderabad to a place using:

- 3 buses
- 2 trains

Assuming you choose exactly one mode:

$3+2 = \boxed{5}$

ways.

3. THE WORD “OR”

Typical signals:

- either A or B
- one category or another
- case 1 or case 2
- mutually exclusive alternatives

Then think addition.

4. OVERLAPPING CASES — WARNING

Suppose:

- 20 students know Python
- 15 know C
- 5 know both

Students knowing Python **or** C:

$$\boxed{20+15-5 = 30}$$

Why subtract 5?

Because those 5 were counted twice.

So when cases overlap:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

We study this more deeply in Sets/Venn Diagrams.

5. MULTIPLICATION PRINCIPLE

Suppose a process has stages:

- stage 1 can be done in m ways
- for each such choice, stage 2 can be done in n ways

Then total:

$$\boxed{mn}$$

6. EXAMPLE — AND

You have:

- 3 shirts
- 4 trousers

An outfit requires one shirt **and** one trouser.

Number of outfits:

$$\boxed{3 \times 4 = 12}$$

7. MULTIPLE STAGES

Suppose:

- 3 shirts
- 4 trousers
- 2 pairs of shoes

Total outfits:

$$\boxed{3 \times 4 \times 2 = 24}$$

8. FUNDAMENTAL COUNTING PRINCIPLE

If a task has k stages with:

$$[n_1, n_2, \dots, n_k]$$

choices respectively, then:

$$\boxed{N = n_1 n_2 \cdots n_k}$$

provided the count at each stage is correctly determined.

MUST UNDERSTAND

9. INDEPENDENT VS DEPENDENT CHOICES

The choices do not need to be statistically “independent.”

What matters is:

For each earlier choice, count how many valid later choices remain.

Example:

Choose 2 different books from positions sequentially.

First choice:

$$[5]$$

Second choice:

$$[4]$$

Total ordered selections:

$$[5 \times 4 = 20]$$

The second stage depends on the first, but multiplication still works.

10. TREE-DIAGRAM INTUITION

Suppose first choice has 2 possibilities:

$$[A, B]$$

For each, second choice has 3:

$$[1, 2, 3]$$

Outcomes:

$$[A_1, A_2, A_3, B_1, B_2, B_3]$$

Total:

$$[2(3) = 6]$$

Multiplication is simply counting leaves of the choice tree.

11. FORMING NUMBERS FROM DIGITS

This is one of the most common counting applications.

Example:

How many 3-digit numbers can be formed using digits:

$\{1, 2, 3, 4\}$

if repetition is allowed?

Hundreds digit:

$\{4\}$

choices.

Tens:

$\{4\}$

Units:

$\{4\}$

Therefore:

$\boxed{4^3=64}$

12. REPETITION NOT ALLOWED

Same digits:

$\{1, 2, 3, 4\}$

Form 3-digit numbers with no repetition.

Hundreds:

$\{4\}$

Tens:

$\{3\}$

Units:

$\{2\}$

Total:

$4 \times 3 \times 2 = \boxed{24}$

13. FIRST-DIGIT ZERO TRAP

Suppose digits:

$\{0, 1, 2, 3, 4\}$

How many 3-digit numbers with repetition allowed?

Hundreds digit cannot be zero.

So first digit:

$\{4\}$

choices:

$\{1,2,3,4\}$

Tens:

$\{5\}$

Units:

$\{5\}$

Total:

$\boxed{4(5)(5)=100}$

Not $(5^3=125)$.

14. ZERO + NO REPETITION

Digits:

$\{0,1,2,3,4\}$

Form 3-digit numbers without repetition.

First digit:

$\{4\}$

choices.

Second:

$\{4\}$

remaining digits, including zero if unused.

Third:

$\{3\}$

Therefore:

$\boxed{4 \cdot 4 \cdot 3 = 48}$

15. FAST GENERAL FORMULA — NO ZERO

Using n nonzero distinct digits, number of k -digit numbers without repetition:

$\boxed{n(n-1)\cdots(n-k+1)}$

This is later written as:

$\{ {}^nP_k \}$

But do not jump ahead yet.

16. FAST GENERAL FORMULA — REPETITION ALLOWED

Using $\lfloor n \rfloor$ available digits and exactly $\lfloor k \rfloor$ positions, if every digit may repeat:

$$\lfloor \boxed{n^k} \rfloor$$

provided there is no leading-zero restriction.

17. PASSWORDS / CODES

Suppose a 4-character PIN uses digits:

$$\lfloor 0-9 \rfloor$$

and repetition is allowed.

Each position:

$$\lfloor 10 \rfloor$$

choices.

Total:

$$\lfloor \boxed{10^4=10000} \rfloor$$

PINs can start with zero because a PIN is not a numerical value in the usual place-value sense.

Important wording distinction.

18. PIN WITHOUT REPETITION

4-digit PIN from 0–9 with no repeated digit:

$$\lfloor 10 \times 9 \times 8 \times 7 \rfloor$$

$$\lfloor \boxed{5040} \rfloor$$

Zero is allowed in first position because it is a code.

19. LICENSE / ALPHANUMERIC CODE

Suppose a code has:

- 2 letters
- followed by 3 digits

Letters: 26 choices each.

Digits: 10 each.

Repetition allowed.

Total:

$$\lfloor 26^2 \cdot 10^3 \rfloor$$

$$\lfloor \boxed{=676000} \rfloor$$

$$\lfloor \boxed{676000} \rfloor$$

20. IF REPETITION IS NOT ALLOWED

Same 2 letters + 3 digits, no repetition within letters/digits.

Letters:

$${}_{26}P_2$$

Digits:

$${}_{10}P_3$$

Total:

$$\boxed{{}_{26}P_2 \cdot {}_{10}P_3}$$

21. EVEN NUMBER FORMATION

Digits:

$${}_{5}P_5$$

How many 3-digit even numbers without repetition?

An even number must end in:

$$2 \text{ or } 4$$

Units:

$$2$$

choices.

Hundreds:

$$4$$

remaining.

Tens:

$$3$$

remaining.

Total:

$$\boxed{2 \cdot 4 \cdot 3 = 24}$$

22. DIVISIBILITY CONDITION FIRST

For restricted number-formation problems, choose the most constrained position first.

For even numbers:

Choose units digit first.

For multiples of 5:

Units digit must be 0 or 5.

For 4-digit numbers:

Ensure first digit is nonzero.

This saves errors.

23. ODD NUMBERS

Digits:

$\{ 0,1,2,3,4,5 \}$

Form 3-digit odd numbers with no repetition.

Units must be:

$\{ 1,3,5 \}$

So units:

$\{ 3 \}$

choices.

Hundreds cannot be zero and cannot equal units.

After choosing an odd units digit, nonzero remaining digits:

Originally 5 nonzero digits.

One used.

So:

$\{ 4 \}$

choices.

Tens:

$\{ 4 \}$

remaining digits including zero.

Total:

$\boxed{3 \cdot 4 \cdot 4 = 48}$

24. MULTIPLES OF 5

Digits:

$\{ 0,1,2,3,4,5 \}$

Form 3-digit numbers without repetition divisible by 5.

Last digit must be:

$\{ 0 \text{ or } 5 \}$

Handle cases separately because zero affects the first digit.

25. CASE 1 — LAST DIGIT 0

Units fixed as 0.

Hundreds:

$\{ 5 \}$

choices:

$\{ 1,2,3,4,5 \}$

Tens:

$\{ 4 \}$

remaining.

Count:

$\{ 5 \cdot 4 = 20 \}$

26. CASE 2 — LAST DIGIT 5

Units fixed as 5.

Hundreds cannot be zero or 5.

Choices:

$\{ 4 \}$

Tens:

$\{ 4 \}$

remaining, including zero.

Count:

$\{ 4 \cdot 4 = 16 \}$

Total:

$\{ 20 + 16 = \boxed{36} \}$

This is a perfect example of:

$\{ \boxed{\text{cases !' add}} \}$

and within each case:

$\{ \boxed{\text{stages !' multiply}} \}$

27. “AT LEAST ONE” — COMPLEMENT METHOD

Suppose 5 coin tosses.

How many outcomes contain at least one head?

Direct counting would require many cases.

Total outcomes:

$$2^5 = 32$$

Only outcome with no heads:

$$TTTTT$$

Count:

$$1$$

Therefore:

$$\boxed{32 - 1 = 31}$$

28. COMPLEMENT PRINCIPLE

To count:

$$\text{at least one desired event}$$

often use:

$$\boxed{\text{Total} - \text{None}}$$

This is one of the most important counting shortcuts.

29. EXAMPLE — PASSWORD AT LEAST ONE DIGIT

Suppose a 4-character code uses:

- 26 letters
- 10 digits

So 36 symbols total.

Repetition allowed.

How many codes contain at least one digit?

Total codes:

$$36^4$$

Codes with no digit = letters only:

$$26^4$$

Therefore:

$$\boxed{36^4 - 26^4}$$

Do not count 1-digit, 2-digit, 3-digit, 4-digit cases separately unless necessary.

30. “AT LEAST ONE ZERO”

How many 4-digit PINs contain at least one zero?

Total PINs:

$$\lfloor 10^4 \rfloor$$

PINs with no zero:

$$\lfloor 9^4 \rfloor$$

Thus:

$$\lfloor \boxed{10^4 - 9^4} \rfloor$$

31. “NONE”

How many 5-character strings using English uppercase letters contain no vowels?

Assume vowels:

$$\lfloor A, E, I, O, U \rfloor$$

Non-vowels:

$$\lfloor 26 - 5 = 21 \rfloor$$

With repetition:

$$\lfloor \boxed{21^5} \rfloor$$

32. “AT LEAST ONE VOWEL”

Total 5-letter strings:

$$\lfloor 26^5 \rfloor$$

No vowels:

$$\lfloor 21^5 \rfloor$$

Thus:

$$\lfloor \boxed{26^5 - 21^5} \rfloor$$

33. EXACTLY ONE SPECIAL SYMBOL — BASIC CASE METHOD

Suppose 3-character strings using letters A/B/C and digit 1.

Repetition allowed.

How many strings contain exactly one digit?

Choose which position has the digit:

$$\lfloor 3 \rfloor$$

The digit itself:

$$\lfloor 1 \rfloor$$

Other positions each choose from A/B/C:

$$\lfloor 3^2 \rfloor$$

Total:

$$\lfloor 3 \cdot 3^2 = \boxed{27} \rfloor$$

This starts moving toward combinations, but simple positional reasoning is enough.

34. EXACTLY ONE ZERO IN A PIN

4-digit PIN using 0–9.

Exactly one zero.

Choose position of zero:

$$\lfloor 4 \rfloor$$

Other three positions:

$$\lfloor 9 \rfloor$$

choices each from 1–9.

Total:

$$\lfloor \boxed{4 \cdot 9^3} \rfloor$$

35. EXACTLY TWO ZEROS

4-digit PIN.

Choose two zero positions:

$$\lfloor \binom{4}{2} = 6 \rfloor$$

Other two positions:

$$\lfloor 9^2 \rfloor$$

So:

$$\lfloor \boxed{6 \cdot 9^2} \rfloor$$

We will formalize $\binom{4}{2}$ in combinations later.

For now, note that there are 6 ways to pick two positions from four.

36. AT MOST ONE ZERO

Means:

- zero zeros
- or exactly one zero

Count:

$$\lfloor 9^4 + 4 \cdot 9^3 \rfloor$$

So:

$$\lfloor \boxed{9^4 + 4(9^3)} \rfloor$$

37. “AT MOST” LANGUAGE

“At most 2” means:

$\{0, 1, 2\}$

“At least 2” means:

$\{2, 3, 4, \dots\}$

“Exactly 2” means only 2.

These language distinctions are critical.

38. COUNTING ROUTES BY STAGES

Suppose:

- A to B has 3 roads
- B to C has 4 roads

Travel A! B! C :

$3 \cdot 4 = \boxed{12}$

39. ROUND TRIP — SAME ROAD ALLOWED

A! B has 3 roads.

B! A has 3 roads.

If return can use same road:

$3 \cdot 3 = \boxed{9}$

40. ROUND TRIP — SAME ROAD NOT ALLOWED

Choose outward road:

3

Return road:

2

Total:

$\boxed{6}$

41. CHOICE OF MEAL

Restaurant offers:

- 4 starters
 - 5 mains
 - 3 desserts
-

If meal consists of one from each:

$$4 \cdot 5 \cdot 3 = \boxed{60}$$

If customer chooses either one starter **or** one dessert only:

$$4 + 3 = \boxed{7}$$

Again:

- AND !' multiply
 - OR !' add
-

42. DEPENDENT CHOICE EXAMPLE

Choose president and vice-president from 8 people.

President:

$$8$$

Vice-president:

$$7$$

because same person cannot occupy both roles.

Total:

$$8 \cdot 7 = \boxed{56}$$

Order/role matters.

43. CHOOSING TWO PEOPLE WITHOUT ROLES

If you just choose 2 people from 8, then AB and BA are the same selection.

The previous $(8 \cdot 7 = 56)$ counts each pair twice.

So divide by 2:

$$56/2 = \boxed{28}$$

This is the intuition behind combinations.

We will formalize it in Chapter 40.

44. ORDER MATTERS VS DOES NOT MATTER

This distinction is essential.

Roles / arrangements / codes

Order matters.

Groups / committees / selections

Order usually does not matter.

Examples:

President A, VP B

is different from:

President B, VP A.

But committee $\{A, B\}$ is same as $\{B, A\}$.

45. COUNTING WITH A FIXED POSITION

How many 4-letter strings from A,B,C,D without repetition start with A?

First position fixed:

$\boxed{1}$

Remaining positions:

$\boxed{3 \cdot 2 \cdot 1}$

Total:

$\boxed{6}$

46. END WITH A PARTICULAR DIGIT

Digits 1–5.

How many 3-digit numbers without repetition end in 5?

Units fixed:

$\boxed{1}$

Hundreds:

$\boxed{4}$

Tens:

$\boxed{3}$

Total:

$\boxed{12}$

47. START WITH 1 OR 2

Digits 1,2,3,4,5.

3-digit numbers without repetition starting with 1 or 2.

First digit:

$\boxed{2}$

choices.

Second:

$\boxed{4}$

Third:

$\backslash 3 \backslash$

Total:

$\backslash \boxed{24} \backslash$

48. START WITH 1 AND END WITH 5

Digits 1–5, no repetition.

First fixed:

$\backslash 1 \backslash$

Last fixed:

$\backslash 5 \backslash$

Middle:

$\backslash 3 \backslash$

remaining choices.

Total:

$\backslash \boxed{3} \backslash$

49. START WITH 1 OR END WITH 5 — OVERLAP

Count 3-digit numbers from digits 1–5 without repetition that:

- start with 1
- OR end with 5.

Start with 1:

$\backslash 4 \cdot 3 = 12 \backslash$

End with 5:

$\backslash 4 \cdot 3 = 12 \backslash$

Both start with 1 and end with 5:

Middle:

$\backslash 3 \backslash$

So union:

$\backslash 12 + 12 - 3 = \boxed{21} \backslash$

Classic inclusion-exclusion logic.

50. CHOOSE FROM DIFFERENT CATEGORIES

Suppose:

- 5 maths books
-

- 4 physics books

Choose one book.

Total:

$$\boxed{5+4=9}$$

Choose one maths **and** one physics:

$$\boxed{5 \cdot 4 = 20}$$

51. SELECT A PERSON AND A TASK

Suppose:

- 6 people
- 4 distinct jobs

Choose one person and assign one job:

$$\boxed{6 \cdot 4 = 24}$$

52. BOOLEAN CHOICE COUNTING

Suppose each of 5 features can be:

- ON
- OFF

Then configurations:

$$\boxed{2^5 = 32}$$

General:

$$n \text{ binary choices} \rightarrow 2^n$$

This appears constantly in CS/GATE contexts.

53. SUBSETS PREVIEW

For a set with (n) elements:

Each element has 2 choices:

- included
- excluded

So number of subsets:

$$\boxed{2^n}$$

This is a direct Fundamental Counting Principle result.

Example:

3 elements:

$$\boxed{2^3=8}$$

subsets.

We revisit this in combinations/sets.

54. NONEMPTY SUBSETS

Total subsets:

$$\lfloor 2^n \rfloor$$

One is empty set.

Therefore nonempty subsets:

$$\lfloor \boxed{2^n - 1} \rfloor$$

55. BINARY STRINGS

Number of binary strings of length $\lfloor n \rfloor$:

Each bit:

$$\lfloor 2 \rfloor$$

choices.

Hence:

$$\lfloor \boxed{2^n} \rfloor$$

Example length 8:

$$\lfloor 2^8 = \boxed{256} \rfloor$$

56. TERNARY STRINGS

Alphabet:

$$\lfloor \{0,1,2\} \rfloor$$

Length $\lfloor n \rfloor$:

$$\lfloor \boxed{3^n} \rfloor$$

57. FUNCTION COUNTING PREVIEW

Suppose each of $\lfloor m \rfloor$ inputs can independently map to any of $\lfloor n \rfloor$ outputs.

Then number of functions:

$$\lfloor \boxed{n^m} \rfloor$$

Why?

For each of $\lfloor m \rfloor$ inputs:

$$\lfloor n \rfloor$$

output choices.

So:

$$\lfloor n \times n \times \dots \times n = n^m \rfloor$$

This is useful in discrete mathematics/GATE.

58. EXAMPLE

Functions from a 3-element set to a 2-element set:

$$\lfloor 2^3 = \boxed{8} \rfloor$$

59. SEQUENCES FROM A FINITE ALPHABET

If an alphabet has $\lfloor k \rfloor$ symbols, strings of length $\lfloor n \rfloor$ with repetition allowed:

$$\lfloor \boxed{k^n} \rfloor$$

This unifies:

- PINs
 - passwords
 - binary strings
 - DNA-like strings
 - function mappings
-

60. COMPLEMENT COUNTING — HIGH-VALUE EXAMPLE

Binary strings of length 6 containing at least one 1.

Total:

$$\lfloor 2^6=64 \rfloor$$

No 1 means all zeros:

$$\lfloor 1 \rfloor$$

Therefore:

$$\lfloor \boxed{63} \rfloor$$

61. AT LEAST ONE REPEATED DIGIT — COMPLEMENT

How many 4-digit PINs have at least one repeated digit?

Total PINs:

$$\lfloor 10^4 \rfloor$$

PINs with all digits distinct:

$$\lfloor 10 \cdot 9 \cdot 8 \cdot 7 \rfloor$$

Therefore:

$$\lfloor \boxed{10^4 - 10 \cdot 9 \cdot 8 \cdot 7} \rfloor$$

This is much easier than counting every possible repetition pattern.

62. AT LEAST ONE REPEATED DIGIT IN A 4-DIGIT NUMBER

Now leading zero forbidden.

Total 4-digit numbers:

$$9 \cdot 10^3 = 9000$$

All digits distinct:

First digit:

$$9$$

Second:

$$9$$

because zero is now available plus 8 unused nonzero digits.

Third:

$$8$$

Fourth:

$$7$$

So distinct count:

$$9 \cdot 9 \cdot 8 \cdot 7$$

Hence at least one repeated digit:

$$\boxed{9000 - 9 \cdot 9 \cdot 8 \cdot 7}$$

63. COMPUTE IT

$$9 \cdot 9 \cdot 8 \cdot 7 = 4536$$

Therefore:

$$9000 - 4536 = \boxed{4464}$$

64. “NO TWO OPTIONS...” PREVIEW

Restrictions involving adjacency often require:

- casework
- gaps
- permutations

We will handle those in Chapters 39–40.

For now, Fundamental Counting Principle handles the basic staged choices.

65. GATE-STYLE QUESTION 1

A password consists of 3 uppercase letters followed by 2 digits.

Repetition allowed.

$$\lfloor 26^3 \cdot 10^2 \rfloor$$

$$\lfloor = 17576 \cdot 100 \rfloor$$

$$\lfloor = \boxed{1,757,600} \rfloor$$

66. GATE-STYLE QUESTION 2

How many 4-bit binary strings exist?

$$\lfloor 2^4 = \boxed{16} \rfloor$$

67. GATE-STYLE QUESTION 3

How many functions exist from a set of 4 elements to a set of 3 elements?

Each input has 3 choices:

$$\lfloor 3^4 = \boxed{81} \rfloor$$

68. GATE-STYLE QUESTION 4

How many nonempty subsets does a 5-element set have?

$$\lfloor 2^5 - 1 = \boxed{31} \rfloor$$

69. PLACEMENT-STYLE QUESTION 1

How many 3-digit numbers can be formed from:

$$\lfloor 0, 1, 2, 3, 4 \rfloor$$

with repetition allowed?

First:

$$\lfloor 4 \rfloor$$

Next:

$$\lfloor 5, 5 \rfloor$$

Total:

$$\lfloor \boxed{100} \rfloor$$

70. PLACEMENT-STYLE QUESTION 2

Same digits, no repetition.

$$\lfloor 4 \cdot 4 \cdot 3 = \boxed{48} \rfloor$$

71. PLACEMENT-STYLE QUESTION 3

Using digits 1–6 without repetition, how many 4-digit even numbers?

Units must be:

$$\lfloor 2, 4, 6 \rfloor$$

So:

$$\lfloor 3 \rfloor$$

choices.

Thousands:

$$\lfloor 5 \rfloor$$

Hundreds:

$$\lfloor 4 \rfloor$$

Tens:

$$\lfloor 3 \rfloor$$

Total:

$$\lfloor \boxed{3 \cdot 5 \cdot 4 \cdot 3 = 180} \rfloor$$

72. PLACEMENT-STYLE QUESTION 4

4-character strings from 26 letters must contain at least one vowel.

Total:

$$\lfloor 26^4 \rfloor$$

No vowels:

$$\lfloor 21^4 \rfloor$$

Thus:

$$\lfloor \boxed{26^4 - 21^4} \rfloor$$

73. ADVANCED QUESTION — DIVISIBLE BY 5

Using digits:

$$\lfloor 0, 1, 2, 3, 4, 5 \rfloor$$

without repetition, number of 4-digit numbers divisible by 5?

Two cases.

Last digit = 0

First:

$\backslash[5 \backslash]$

Middle two:

$\backslash[4\cdot 3 \backslash]$

Count:

$\backslash[60 \backslash]$

Last digit = 5

First cannot be 0 or 5:

$\backslash[4 \backslash]$

Next:

$\backslash[4 \backslash]$

Next:

$\backslash[3 \backslash]$

Count:

$\backslash[48 \backslash]$

Total:

$\backslash[\boxed{108} \backslash]$

74. ADVANCED QUESTION — START OR END CONDITION

Digits 1–6, no repetition.

How many 4-digit numbers start with 1 or end with 6?

Start with 1:

$\backslash[5\cdot 4\cdot 3=60 \backslash]$

End with 6:

$\backslash[5\cdot 4\cdot 3=60 \backslash]$

Both:

First 1, last 6.

Middle two:

$\backslash[4\cdot 3=12 \backslash]$

Total:

$\backslash[60+60-12 = \boxed{108} \backslash]$

75. ADVANCED QUESTION — AT LEAST ONE ZERO

How many 4-digit PINs contain at least one zero?

$\backslash[10^4-9^4 \backslash]$

$\backslash[10000-6561 \backslash]$

$\boxed{3439}$

76. ADVANCED QUESTION — AT LEAST ONE REPEATED DIGIT

4-digit PIN:

Total:

10000

All different:

$10 \cdot 9 \cdot 8 \cdot 7 = 5040$

Therefore:

$\boxed{4960}$

77. COMMON TRAPS

& pTRAP 1 — Add when you should multiply

Sequential choices:

$\text{AND} \rightarrow \times$

& pTRAP 2 — Multiply mutually exclusive cases

Alternative cases:

$\text{OR} \rightarrow +$

& pTRAP 3 — Leading zero

A number cannot begin with 0.

A PIN/code may.

& pTRAP 4 — Forgetting repetition condition

“Digits may repeat” and “digits are distinct” produce completely different counts.

& pTRAP 5 — Counting overlapping OR cases twice

Use:

$A + B - \text{overlap}$

& pTRAP 6 — Directly counting “at least one”

Often easier:

$\boxed{\text{Total} - \text{None}}$

& pTRAP 7 — Order

President/VP “unordered committee.”

78. MUST-MEMORIZE CORE RULES

Addition principle

For disjoint alternatives:

$$\boxed{m+n}$$

Multiplication principle

For sequential stages:

$$\boxed{mn}$$

$\boxed{k}$ -stage process

$$\boxed{n_1 n_2 \cdots n_k}$$

Strings length $\boxed{n}$, alphabet size $\boxed{k}$

$$\boxed{k^n}$$

79. MUST-MEMORIZE COMPLEMENT RULE

$$\boxed{\text{Desired} = \text{Total} - \text{Undesired complement}}$$

Especially useful for:

- at least one
 - at least one repetition
 - at least one special symbol
 - not all / not none
-

80. FAST EXAM DECISION TREE

Are there alternatives?

Split into cases and add.

Are there successive choices?

Multiply.

Repetition allowed?

Same number of choices may remain.

Repetition forbidden?

Reduce choices after every selection.

Numerical number?

First digit cannot be zero.

PIN/code/string?

Leading zero may be allowed.

At least one?

Try complement.

OR conditions overlap?

Subtract overlap.

Restriction on last digit?

Choose last digit first.

PRACTICE — LEVEL 0

Q1

A person has 4 shirts and 3 trousers. How many outfits?

Q2

How many 3-digit PINs are possible using digits 0–9 with repetition?

Q3

How many 3-digit numbers can be formed using 1,2,3,4 without repetition?

Q4

How many binary strings of length 7?

Q5

How many nonempty subsets of a 4-element set?

LEVEL 1

Q6

How many 4-digit numbers can be formed from digits 0–6 with repetition allowed?

Q7

Using digits 0–6 without repetition, how many 4-digit numbers can be formed?

Q8

Using digits 1–7 without repetition, how many 3-digit even numbers?

Q9

How many 5-character strings from uppercase letters contain at least one vowel?

Q10

How many 4-digit PINs contain at least one zero?

LEVEL 2

Q11

Using digits 0–5 without repetition, how many 3-digit numbers divisible by 5?

Q12

Using digits 1–6 without repetition, how many 4-digit numbers start with 1 or end with 6?

Q13

How many functions exist from a 5-element set to a 3-element set?

Q14

How many 6-bit strings contain at least one 1?

Q15

A code has 2 distinct uppercase letters followed by 3 distinct digits. How many codes?

LEVEL 3 — EXAM PRESSURE

Q16

Using digits 0–7 without repetition, how many 4-digit odd numbers can be formed?

Q17

Using digits 0–6 without repetition, how many 4-digit numbers divisible by 5 can be formed?

Q18

How many 5-digit PINs have at least one repeated digit?

Q19

How many 4-character strings from uppercase English letters contain exactly one vowel?

Q20

How many 5-digit numbers contain at least one repeated digit?

ANSWERS

Q1

$$\lfloor 4 \cdot 3 = \boxed{12} \rfloor$$

Q2

$$\lfloor 10^3 = \boxed{1000} \rfloor$$

Q3

$$\lfloor 4 \cdot 3 \cdot 2 = \boxed{24} \rfloor$$

Q4

$$\lfloor 2^7 = \boxed{128} \rfloor$$

Q5

$$\lfloor 2^4 - 1 = \boxed{15} \rfloor$$

Q6

Digits:

$$\lfloor 0, 1, \dots, 6 \rfloor$$

First digit:

$$\lfloor 6 \rfloor$$

choices.

Others:

$$\lfloor 7^3 \rfloor$$

So:

$$\lfloor 6 \cdot 343 = \boxed{2058} \rfloor$$

Q7

First digit:

$$\lfloor 6 \rfloor$$

Then:

$$\lfloor 6, 5, 4 \rfloor$$

Thus:

$$\lfloor 6 \cdot 6 \cdot 5 \cdot 4 = \boxed{720} \rfloor$$

Q8

Even units:

$$\lfloor 2, 4, 6 \rfloor$$

so 3 choices.

Hundreds:

$$\lfloor 6 \rfloor$$

Tens:

$$\lfloor 5 \rfloor$$

$$\lfloor 3 \cdot 6 \cdot 5 = \boxed{90} \rfloor$$

Q9

Total:

$$\lfloor 26^5 \rfloor$$

No vowels:

$$\lfloor 21^5 \rfloor$$

$$\lfloor \boxed{26^5 - 21^5} \rfloor$$

Q10

$$\lfloor 10^4 - 9^4 = 10000 - 6561 \rfloor$$

$$\lfloor = \boxed{3439} \rfloor$$

Q11

Digits:

$$\lfloor 0, 1, 2, 3, 4, 5 \rfloor$$

Case last = 0:

First:

$$\lfloor 5 \rfloor$$

Middle:

$$\lfloor 4 \rfloor$$

Count:

$$\lfloor 20 \rfloor$$

Case last = 5:

First cannot be 0:

$$\lfloor 4 \rfloor$$

Middle:

$$\lfloor 4 \rfloor$$

Count:

$$\lfloor 16 \rfloor$$

Total:

$$\lfloor \boxed{36} \rfloor$$

Q12

Start with 1:

$$\lfloor 5 \cdot 4 \cdot 3 = 60 \rfloor$$

End with 6:

$$\lfloor 60 \rfloor$$

Both:

$$\lfloor 4 \cdot 3 = 12 \rfloor$$

Union:

$$\lfloor 60 + 60 - 12 = \boxed{108} \rfloor$$

Q13

Each of 5 inputs has 3 output choices:

$$\lfloor 3^5 = \boxed{243} \rfloor$$

Q14

Total:

$$\sqrt[2]{2^6=64}$$

No 1:

$$\sqrt[1]{1}$$

So:

$$\boxed{63}$$

Q15

Letters:

$$\sqrt[26]{25}$$

Digits:

$$\sqrt[10]{9 \cdot 8}$$

Total:

$$\sqrt[26 \cdot 25 \cdot 10 \cdot 9 \cdot 8]$$

$$\boxed{468000}$$

Q16

Digits:

$$\sqrt[0-7]$$

Odd units:

$$\sqrt[1,3,5,7]$$

so:

$$\sqrt[4]$$

choices.

First digit cannot be zero or equal to used units.

Among remaining 7 digits, 6 are nonzero.

So:

$$\sqrt[6]$$

choices.

Middle two:

$$\sqrt[6,5]$$

Total:

$$\sqrt[4 \cdot 6 \cdot 6 \cdot 5] = \boxed{720}$$

Q17

Digits:

$$\sqrt[0-6]$$

Last digit 0 or 5.

Last = 0

First:

$$\{ 6 \}$$

Next:

$$\{ 5, 4 \}$$

Count:

$$\{ 6 \cdot 5 \cdot 4 = 120 \}$$

Last = 5

First cannot be 0 or 5:

$$\{ 5 \}$$

Next:

$$\{ 5, 4 \}$$

Count:

$$\{ 100 \}$$

Total:

$$\{ \boxed{220} \}$$

Q18

5-digit PIN means leading zero allowed.

Total:

$$\{ 10^5 = 100000 \}$$

All distinct:

$$\{ 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240 \}$$

At least one repeated digit:

$$\{ 100000 - 30240 = \boxed{69760} \}$$

Q19

Exactly one vowel.

Choose vowel position:

$$\{ 4 \}$$

Choose vowel:

$$\{ 5 \}$$

Other 3 positions must be consonants:

$$\{ 21^3 \}$$

Total:

$$\{ \boxed{4 \cdot 5 \cdot 21^3} \}$$

$$\{ = 20(9261) \}$$

$$\boxed{185220}$$

Q20

5-digit numbers.

Total:

$$9 \cdot 10^4 = 90000$$

Count all digits distinct.

First:

$$9$$

Second:

$$9$$

Third:

$$8$$

Fourth:

$$7$$

Fifth:

$$6$$

Distinct:

$$9 \cdot 8 \cdot 7 \cdot 6 = 27216$$

Therefore at least one repetition:

$$90000 - 27216$$

$$\boxed{62784}$$

CHAPTER 38 STATUS

Covered so far:

- counting without listing '
- addition principle '
- multiplication principle '
- staged/dependent choices '
- repetition allowed/forbidden '
- number formation '
- leading-zero trap '
- even/odd/divisibility-by-5 restrictions '
- PIN/password/code counting '
- route counting '
- role assignments '
- order-matters intuition '
- binary strings '

- subset counting preview '
- function counting preview '
- complement counting '
- at least/at most/exactly language '
- inclusion-exclusion basics '
- GATE/placement practice '

But **Chapter 38 is not complete yet.**

Next we still need:

- more difficult constrained digit formation
- numbers above/below a bound
- digit-sum restrictions
- multiple simultaneous restrictions
- counting paths on grids
- inclusion-exclusion counting in more detail
- surjection/injection preview where useful
- pigeonhole-style counting intuition
- final chapter speed audit

Progress

Completed: 37/76 Current: 0=0 Chapter 38 — Fundamental Counting Principle

NEXT

0=0 Chapter 38 Part 2 — Advanced constrained counting, bounds, paths, inclusion-exclusion & counting shortcuts

USER

next

ASSISTANT

CHAPTER 38 — FUNDAMENTAL COUNTING PRINCIPLE

PART 2 — ADVANCED CONSTRAINED COUNTING, BOUNDS, GRID PATHS, INCLUSION-EXCLUSION & COUNTING SHORTCUTS

Status: 0=0 **PROGRESS GATE:** High **Placements:** High **Priority:** 0=0 **Per 1**

Part 1 gave the two master rules:

$\boxed{\text{OR}} \rightarrow \text{ADD}$

$\boxed{\text{AND/THEN}} \rightarrow \text{MULTIPLY}$

Now we handle questions where several restrictions appear together.

The most important exam habit is:

Count the most restrictive position or condition first.

1. MULTIPLE RESTRICTIONS AT ONCE

Example:

Using digits

$\{0, 1, 2, 3, 4, 5\}$

how many 4-digit even numbers can be formed without repetition?

We must satisfy:

- first digit $\neq 0$
- last digit even
- no repeated digit

Because the last digit is strongly restricted, split cases based on it.

2. CASE 1 — LAST DIGIT 0

Units fixed:

$\{0\}$

Thousands:

$\{5\}$

choices:

$\{1, 2, 3, 4, 5\}$

Middle two:

$\{4 \cdot 3\}$

Count:

$\{5 \cdot 4 \cdot 3 = 60\}$

3. CASE 2 — LAST DIGIT 2 OR 4

Units:

$\{2\}$

choices.

For each choice, thousands cannot be:

- 0
- chosen units digit

So:

$\{ 4 \}$

choices.

Remaining middle positions:

$\{ 4 \cdot 3 \}$

Thus:

$\{ 2 \cdot 4 \cdot 3 = 96 \}$

Total:

$\{ 60 + 96 = \boxed{156} \}$

4. WHY CASEWORK WAS NECESSARY

Zero behaves differently because:

- zero is allowed in internal positions
- zero is forbidden in the first position

Whenever 0 is one of the possible last digits, check whether separate cases simplify the count.

5. NUMBER GREATER THAN A BOUND

Using digits:

$\{ 1, 2, 3, 4, 5 \}$

without repetition, how many 3-digit numbers are greater than 300?

Hundreds digit must be:

$\{ 3, 4, 5 \}$

So:

$\{ 3 \}$

choices.

For each:

Tens:

$\{ 4 \}$

Units:

$\{ 3 \}$

Total:

$\{ \boxed{3 \cdot 4 \cdot 3 = 36} \}$

6. NUMBER LESS THAN A BOUND

Using digits:

$\{ 1, 2, 3, 4, 5 \}$

without repetition, how many 3-digit numbers are less than 400?

Hundreds digit:

$\{1, 2, 3\}$

So:

$\{3\}$

choices.

Then:

$4 \cdot 3$

Thus:

$\boxed{36}$

7. BOUND NOT ALIGNED TO HUNDREDS

Now consider:

Using digits 1–5 without repetition, count 3-digit numbers less than 425.

Split by hundreds digit.

Hundreds = 1, 2, 3

All completions valid:

$3 \cdot 4 \cdot 3 = 36$

Hundreds = 4

Need number:

$4xy < 425$

Now tens digit determines validity.

Available digits:

$\{1, 2, 3, 5\}$

To be below 425:

- tens = 1 !' units any remaining 3
- tens = 2 !' units must be less than 5? More precisely $(42u < 425)$, so $(u < 5)$. Available units after choosing 2: 1, 3, 5; valid 1, 3 !' 2 choices
- tens = 3 or 5 !' too large

So additional:

$3 + 2 = 5$

Total:

$\boxed{41}$

This is **prefix counting**.

8. PREFIX COUNTING STRATEGY

To count numbers below a bound like:

$\lfloor 425 \rfloor$

compare from left to right:

1. first digit smaller than bound's first digit !' remaining positions free
2. first digit equal !' inspect second digit
3. first two equal !' inspect third digit

This idea is extremely useful for larger digit-count problems.

9. BETWEEN TWO BOUNDS

To count valid numbers between $\lfloor L \rfloor$ and $\lfloor R \rfloor$, often use:

$\lfloor \boxed{\#(\leq R) - \#(< L)} \rfloor$

provided both counts use the same restrictions.

This reduces one awkward interval problem into two prefix-count problems.

10. EXACT DIGIT SUM — SMALL CASES

How many 3-digit numbers made from digits $\{0,1,2,3\}$, repetition allowed, have digit sum 3?

Let digits be:

$\{a,b,c\}$

with:

$a \neq 0$

and:

$a+b+c=3$

Handle by first digit.

$(a=1)$

$b+c=2$

Solutions:

$\{(0,2),(1,1),(2,0)\}$

3 ways.

$(a=2)$

$b+c=1$

Solutions:

$\{(0,1),(1,0)\}$

2 ways.

$$(a=3)$$

$$[b+c=0]$$

Only:

$$[(0,0)]$$

1 way.

Total:

$$[\boxed{6}]$$

11. DIGIT-SUM QUESTIONS — WARNING

Digit-sum problems may require:

- casework
- combinations
- inclusion-exclusion
- generating functions in advanced mathematics

For aptitude, use direct casework for small digit ranges.

Do not force a simple multiplication rule when choices are coupled by a fixed total.

12. DIVISIBLE BY 3

A number is divisible by 3 if digit sum is divisible by 3.

Example:

Using digits $(1,2,3,4,5)$ without repetition, count 3-digit numbers divisible by 3.

We first choose 3-digit sets whose digit sums are divisible by 3.

Possible 3-element selections:

$$[\{1,2,3\}, \{1,3,5\}, \{2,3,4\}, \{3,4,5\}]$$

Check sums:

- $(1+2+3=6)$
- $(1+3+5=9)$
- $(2+3+4=9)$
- $(3+4+5=12)$

Each selected set can be arranged in:

$$[3!=6]$$

ways.

Thus:

$$[4 \cdot 6 = \boxed{24}]$$

This previews permutations.

13. DIVISIBLE BY 4

Divisibility by 4 depends only on the last two digits.

Therefore:

Choose a valid ending first.

Example:

Using digits $\{1,2,3,4,5\}$ without repetition, to form 3-digit multiples of 4:

Valid last-two-digit endings from available digits must be divisible by 4.

Then choose the first digit from remaining digits.

This “ending-first” method is usually faster.

14. DIVISIBLE BY 8

Divisibility by 8 depends on the last 3 digits.

For a 4-digit number:

- first choose a valid 3-digit ending divisible by 8
- then choose a legal leading digit

Placement tests frequently hide counting inside divisibility rules.

15. COUNTING WITH “AT LEAST”

Example:

How many 5-bit strings contain at least two 1s?

Total:

$$2^5 = 32$$

Complement:

- zero 1s: $\binom{5}{0}$
- exactly one 1: choose its position $\binom{5}{1}$

Thus:

$$32 - (1 + 5) = \boxed{26}$$

This is much faster than adding cases for 2,3,4,5 ones.

16. AT MOST TWO 1s

For a 5-bit string:

- 0 ones: $\binom{5}{0}$
- 1 one: $\binom{5}{1}$
- 2 ones: $\binom{5}{2} = 10$

Total:

$$\lfloor 1+5+10 = \boxed{16} \rfloor$$

We will fully develop $\binom{n}{r}$ in Chapter 40.

17. EXACTLY k SPECIAL POSITIONS

General idea:

Length $\lfloor n \rfloor$, exactly $\lfloor k \rfloor$ positions must contain a special symbol/type.

First choose the $\lfloor k \rfloor$ positions.

Then fill those positions.

Then fill remaining positions.

This is a powerful pattern that becomes combinations $\times$ multiplication principle.

18. EXACTLY TWO VOWELS

How many 4-letter strings contain exactly 2 vowels?

Assume:

- 5 vowels
- 21 consonants
- repetition allowed

Choose 2 positions for vowels:

$$\lfloor \binom{4}{2}=6 \rfloor$$

Fill vowel positions:

$$\lfloor 5^2 \rfloor$$

Fill consonant positions:

$$\lfloor 21^2 \rfloor$$

Total:

$$\lfloor \boxed{6 \cdot 5^2 \cdot 21^2} \rfloor$$

19. AT LEAST ONE VOWEL — TWO METHODS

Direct case method

Count strings with:

- exactly 1 vowel
- exactly 2
- ...
- exactly 4

This is long.

Complement method

$\sqrt[4]{26^4 - 21^4}$

Much better.

So:

At least one !' try complement first.

20. INCLUSION-EXCLUSION FOR TWO CONDITIONS

If:

- $|A|$ = outcomes satisfying condition A
- $|B|$ = outcomes satisfying condition B

then:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

We already used this for:

- starts with 1 OR ends with 6
- Python OR C
- one condition OR another

21. WHY SUBTRACT INTERSECTION?

Because outcomes satisfying both A and B are counted once in $|A|$ and once again in $|B|$.

So they are counted twice.

Subtract once.

22. THREE-SET INCLUSION-EXCLUSION

For three sets:

$$|A \cup B \cup C|$$

$$=$$

A	+	B	+	C
---	---	---	---	---

$$- |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Memory pattern

$$+ \text{singles} - \text{pairs} + \text{triple}$$

23. EXAMPLE — DIVISIBILITY

Among integers from 1 to 100, how many are divisible by 2 or 3?

Divisible by 2:

$$\lfloor 50 \rfloor$$

Divisible by 3:

$$\lfloor 33 \rfloor$$

Divisible by both = divisible by 6:

$$\lfloor 16 \rfloor$$

Therefore:

$$\lfloor 50+33-16 = \boxed{67} \rfloor$$

24. DIVISIBLE BY 2 OR 3 OR 5

From 1 to 100:

Singles:

$$\lfloor 50+33+20 \rfloor$$

Pair intersections:

- divisible by 6: 16
- divisible by 10: 10
- divisible by 15: 6

Triple:

- divisible by 30: 3

Thus:

$$\lfloor 50+33+20-16-10-6+3 \rfloor$$

$$\lfloor =103-32+3 \rfloor$$

$$\lfloor =\boxed{74} \rfloor$$

25. GRID PATHS— BASIC MODEL

Suppose you move from:

$$\lfloor (0,0) \rfloor$$

to:

$$\lfloor (3,2) \rfloor$$

using only:

- Right $\backslash(R\backslash)$
- Up $\backslash(U\backslash)$

You must make:

$$\sqrt[3]{3R+2U}$$

Total steps:

$$\sqrt[5]{5}$$

Any valid path is simply an arrangement of:

$$\sqrt[5]{RRRUU}$$

Number of paths:

$$\sqrt[\boxed{\binom{5}{3}=10}]{}$$

or:

$$\sqrt[\binom{5}{2}]{}$$

This previews combinations.

26. GENERAL GRID-PATH FORMULA

From:

$$\sqrt[(0,0)]{}$$

to:

$$\sqrt[(m,n)]{}$$

using only right/up:

- $\sqrt[m]{}$ rights
- $\sqrt[n]{}$ ups

Total:

$$\sqrt[m+n]{}$$

Choose positions of rights:

$$\sqrt[\boxed{\binom{m+n}{m}}]{}$$

Equivalent:

$$\sqrt[\boxed{\binom{m+n}{n}}]{}$$

27. PATH THROUGH A REQUIRED POINT

From A to C via B:

$$\sqrt[\boxed{(\text{paths } A \rightarrow B)}]{\text{paths } B \rightarrow C}$$

Because both stages must happen.

28. EXAMPLE — REQUIRED POINT

From $\sqrt[(0,0)]{}$ to $\sqrt[(4,3)]{}$, paths must pass through $\sqrt[(2,1)]{}$.

A → B :

Need:

$$\{ 2R+1U \}$$

Paths:

$$\{ \binom{3}{1}=3 \}$$

B! C :

Need:

$$\{ 2R+2U \}$$

Paths:

$$\{ \binom{4}{2}=6 \}$$

Total:

$$\{ 3 \cdot 6 = \boxed{18} \}$$

29. PATH AVOIDING A POINT

Use complement:

$$\{ \boxed{\text{all paths}} - \text{paths through forbidden point} \}$$

Example:

From $(0,0)$ to $(4,3)$, avoid $(2,1)$.

All paths:

$$\{ \binom{7}{3}=35 \}$$

Through forbidden point:

$$\{ 18 \}$$

So:

$$\{ \boxed{17} \}$$

30. ROUTES WITH MULTIPLE STAGES

Suppose:

- A! B3 routes
- B! C4 routes
- C! D2 routes

Then A! D via B,C:

$$\{ 3 \cdot 4 \cdot 2 = \boxed{24} \}$$

Basic multiplication principle still sits behind path counting.

31. INJECTION PREVIEW

Suppose a function maps $\{m\}$ distinct inputs into $\{n\}$ distinct outputs, and no two inputs may share the same output.

This is an **injective** function.

For:

$$\lfloor m \leq n \rfloor$$

number of injections:

$$\lfloor \boxed{n(n-1)(n-2)\cdots(n-m+1)} \rfloor$$

because:

- first input: $\lfloor n \rfloor$ choices
 - second: $\lfloor (n-1) \rfloor$
 - etc.
-

32. EXAMPLE — INJECTIONS

Functions from a 3-element set to a 5-element set, injective:

$$\lfloor 5 \cdot 4 \cdot 3 = \boxed{60} \rfloor$$

This is equivalent to assigning 3 distinct inputs to 3 distinct outputs in order.

33. WHY INJECTION IMPOSSIBLE WHEN $m > n$

If more inputs than outputs and every input needs a distinct output, eventually two inputs must share an output.

So:

$$\lfloor \boxed{m > n \Rightarrow \text{injective functions}} \rfloor$$

This is also linked to the pigeonhole principle.

34. SURJECTION PREVIEW

A function is **onto/surjective** if every output is used at least once.

Example:

2-element domain !' 2-element codomain.

Total functions:

$$\lfloor 2^2=4 \rfloor$$

Surjective functions must use both outputs.

The two constant functions fail.

So:

$$\lfloor 4-2 = \boxed{2} \rfloor$$

For larger cases, inclusion-exclusion is used.

We revisit this in combinations/discrete counting where needed.

35. SURJECTION FROM 3 INPUTS TO 2 OUTPUTS

Total functions:

$$\sqrt[3]{2^3=8}$$

Not onto if all 3 map to first output:

$$\sqrt[3]{1}$$

or all 3 map to second:

$$\sqrt[3]{1}$$

Thus:

$$\sqrt[3]{8-2} = \boxed{6}$$

36. PIGEONHOLE PRINCIPLE — INTUITION

If:

$$\sqrt[n+1]{}$$

objects are placed into:

$$\sqrt[n]{}$$

boxes, then at least one box contains at least 2 objects.

Example:

13 people, 12 months.

At least two people have birthdays in the same month.

37. GENERALIZED PIGEONHOLE PRINCIPLE

If $\sqrt[N]{}$ objects are placed into $\sqrt[k]{}$ boxes, at least one box contains at least:

$$\sqrt[\boxed{\left\lceil \frac{N}{k} \right\rceil}]{}$$

objects.

Example:

100 students distributed among 9 groups.

At least one group has:

$$\sqrt[\left\lceil \frac{100}{9} \right\rceil]{} = \boxed{12}$$

students.

38. GUARANTEE QUESTIONS

Pigeonhole problems ask:

How many objects are needed to guarantee something?

Example:

How many socks must be drawn from a drawer with 4 colors to guarantee two socks of same color?

Worst case first 4 socks are all different.

The next sock must match one.

So:

$$\boxed{5}$$

39. GUARANTEE 3 OF SAME COLOR

4 colors.

To avoid 3 of same color as long as possible, take at most 2 of each:

$$4(2)=8$$

Next sock guarantees a third of some color:

$$\boxed{9}$$

General idea:

To guarantee (r) objects in one of (k) categories:

$$\boxed{k(r-1)+1}$$

when categories can supply enough objects.

40. COUNTING VS PIGEONHOLE

Counting asks:

How many outcomes?

Pigeonhole asks:

What must necessarily happen?

Different goals, but both arise from distributing choices.

41. RESTRICTIONS ON CONSECUTIVE SYMBOLS — PREVIEW

Example:

Binary strings of length 4 with no consecutive 1s.

Cannot simply do:

$$2^4$$

because positions are dependent.

You need casework or recurrence.

Let's count quickly:

Valid strings:

- 0 ones: 1
- 1 one: 4
- 2 nonadjacent ones: positions $((1,3),(1,4),(2,4))$!' 3

Total:

$\boxed{8}$

We will revisit adjacency more systematically in permutations/combinations.

42. RESTRICTED FIRST AND LAST SYMBOLS

How many binary strings of length 6 start and end differently?

First bit:

2

choices.

Last bit then forced:

1

Middle 4 bits:

2^4

Total:

$2 \cdot 16 = \boxed{32}$

43. START AND END SAME

Similarly:

First bit:

2

Last forced same.

Middle 4:

2^4

Total:

$\boxed{32}$

Check:

$32 + 32 = 64 = 2^6$

Good sanity check.

44. EXACTLY ONE CHANGE IN A BINARY STRING

Length 5.

Example forms:

$\lfloor 00011, \text{quad}11100 \rfloor$

A single change means choose:

- starting bit: 2 choices
- boundary between positions where change occurs: 4 choices

Thus:

$\lfloor \boxed{8} \rfloor$

A nice FCP application.

45. COUNTING COMPLEMENT WITH FORBIDDEN SYMBOL

5-character strings from 10 symbols where one particular symbol must not appear:

Each position:

$\lfloor 9 \rfloor$

choices.

Total:

$\lfloor \boxed{9^5} \rfloor$

At least once:

$\lfloor \boxed{10^5 - 9^5} \rfloor$

46. TWO PARTICULAR SYMBOLS BOTH APPEAR AT LEAST ONCE

Length 4 string over 10 symbols.

Suppose symbols A and B must each appear at least once.

Use inclusion-exclusion.

Total:

$\lfloor 10^4 \rfloor$

Missing A:

$\lfloor 9^4 \rfloor$

Missing B:

$\lfloor 9^4 \rfloor$

Missing both:

$\lfloor 8^4 \rfloor$

Therefore:

$\lfloor \boxed{10^4 - 2(9^4) + 8^4} \rfloor$

This is a very important inclusion-exclusion pattern.

47. GENERAL “ALL REQUIRED TYPES APPEAR”

For a small set of mandatory symbols/types, inclusion-exclusion is often the fastest route:

$\lvert \text{total} \rvert - \lvert \text{missing one} \rvert + \lvert \text{missing two} \rvert - \dots$

This becomes especially important for onto functions.

48. COUNTING NUMBERS WITH DISTINCT DIGITS — GENERAL FORM

Number of k -digit numbers with distinct digits:

First digit:

9

Next:

9

because zero becomes available.

Then:

$8, 7, \dots$

Thus for $k \leq 10$:

$$9 \cdot 9 \cdot 8 \cdot 7 \cdot \dots (10 - k + 1)$$

Example $(k=4)$:

$9 \cdot 9 \cdot 8 \cdot 7$

49. AT LEAST ONE REPEATED DIGIT — GENERAL

For k -digit ordinary numbers:

$$9 \cdot 10^{k-1} - (\text{distinct-digit count})$$

provided $k \leq 10$.

For $k > 10$, repetition is guaranteed by pigeonhole because only 10 digits exist.

50. PIGEONHOLE DIGIT RESULT

Any 11-digit string of decimal digits must repeat some digit, because:

- 11 positions
- only 10 digit values

So repeated digit is guaranteed.

For an ordinary 11-digit number as well:

$$\text{some digit repeats}$$

51. OPTION ELIMINATION IN COUNTING

Before calculating huge numbers, check:

- parity
- divisibility
- upper bound
- whether count must be multiple of number of symmetric choices

Example:

If we first choose one of 4 possible final digits and each case gives same number of completions, total must be divisible by 4.

This can eliminate options quickly.

52. SANITY CHECK — COUNT CANNOT EXCEED TOTAL

If asking for restricted 4-digit numbers:

$\text{answer} \leq 9000$

If your result exceeds total possible 4-digit numbers, something is wrong.

Simple but powerful.

53. GATE-STYLE QUESTION 1

How many binary strings of length 8 contain at least one 1?

Total:

2^8

All zeros:

1

Therefore:

$\boxed{255}$

54. GATE-STYLE QUESTION 2

How many functions from a 4-element set to a 3-element set?

$3^4 = \boxed{81}$

55. GATE-STYLE QUESTION 3

How many injective functions from a 3-element set to a 5-element set?

$5 \cdot 4 \cdot 3 = \boxed{60}$

56. GATE-STYLE QUESTION 4

How many onto functions from a 3-element set to a 2-element set?

$$\lfloor 2^3 - 2 = \boxed{6} \rfloor$$

57. PLACEMENT-STYLE QUESTION 1

Using digits 0–5 without repetition, how many 4-digit even numbers?

From earlier:

$$\lfloor \boxed{156} \rfloor$$

58. PLACEMENT-STYLE QUESTION 2

How many 5-bit strings contain at least 2 ones?

$$\lfloor 32 - [1+5] \rfloor$$

$$\lfloor = \boxed{26} \rfloor$$

59. PLACEMENT-STYLE QUESTION 3

Among integers from 1 to 100, how many are divisible by 2 or 5?

Divisible by 2:

$$\lfloor 50 \rfloor$$

By 5:

$$\lfloor 20 \rfloor$$

By 10:

$$\lfloor 10 \rfloor$$

Thus:

$$\lfloor 50 + 20 - 10 = \boxed{60} \rfloor$$

60. PLACEMENT-STYLE QUESTION 4

How many paths from $((0,0))$ to $((5,3))$ using only right/up?

Total steps:

$$\lfloor 8 \rfloor$$

Choose 3 ups:

$$\lfloor \binom{8}{3} = \boxed{56} \rfloor$$

61. ADVANCED QUESTION — PATHTHROUGH POINT

From $((0,0))$ to $((5,3))$, must pass through $((2,1))$.

Start! point:

$$\lfloor 2R+1U \rfloor$$

$$\lfloor \text{binom}{31}{3} \rfloor$$

Point! end:

$$\lfloor 3R+2U \rfloor$$

$$\lfloor \text{binom}{52}{10} \rfloor$$

Total:

$$\lfloor \boxed{30} \rfloor$$

62. ADVANCED QUESTION — AVOID POINT

All paths:

$$\lfloor \text{binom}{83}{56} \rfloor$$

Through point:

$$\lfloor 30 \rfloor$$

Avoid point:

$$\lfloor \boxed{26} \rfloor$$

63. ADVANCED QUESTION — REQUIRED SYMBOLS

How many 4-character strings over 10 symbols contain both A and B at least once?

$$\lfloor 10^4 - 2(9^4) + 8^4 \rfloor$$

Calculate:

$$\lfloor 10000 - 2(6561) + 4096 \rfloor$$

$$\lfloor =10000 - 13122 + 4096 \rfloor$$

$$\lfloor =\boxed{974} \rfloor$$

64. ADVANCED QUESTION — PIGEONHOLE

What minimum number of people guarantees at least 4 people were born in the same month?

There are 12 months.

To avoid having 4 in any month, at most 3 per month:

$$\lfloor 12(3) = 36 \rfloor$$

One more guarantees 4 in some month:

$$\lfloor \boxed{37} \rfloor$$

65. ADVANCED QUESTION — DIVISIBILITY UNION

From 1 to 200, count numbers divisible by 4 or 6.

By 4:

$$\lfloor \frac{200}{4} \rfloor = 50$$

By 6:

$$\lfloor \frac{200}{6} \rfloor = 33$$

Both ' divisible by:

$$\lfloor \frac{200}{\text{LCM}(4,6)} \rfloor = \lfloor \frac{200}{12} \rfloor = 16$$

Count:

$$50 + 33 - 16 = 67$$

Therefore:

$$50 + 33 - 16 = \boxed{67}$$

66. COMMON TRAPS

& pTRAP 1 — Bound counting

For “less than 425,” you cannot simply count all numbers beginning with 1–4.

The 4-prefix requires further restriction.

& pTRAP 2 — Divisibility restriction applied last

Often choose the restricted ending first.

& pTRAP 3 — Forget overlap in OR

Use inclusion-exclusion.

& pTRAP 4 — Grid paths

Order matters; the same collection of R/U moves in different positions gives different paths.

& pTRAP 5 — Surjection vs ordinary function

Ordinary function may leave some codomain elements unused.

Onto function may not.

& pTRAP 6 — Injection

Distinct inputs must have distinct outputs.

& pTRAP 7 — Pigeonhole gives guarantee, not exact distribution

It tells what must happen, not how objects are arranged.

67. MUST-MEMORIZE COUNTING TOOLKIT

Addition rule

$$\boxed{\text{OR} \rightarrow +}$$

Multiplication rule

$$\boxed{\text{AND/THEN} \rightarrow \times}$$

Complement

$$\boxed{\text{wanted} = \text{total} - \text{unwanted}}$$

Two-set inclusion-exclusion

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Three-set inclusion-exclusion

$$\boxed{A + B + C - AB - AC - BC + ABC}$$

68. MUST-MEMORIZE GRID PATH

From $(0,0)$ to (m,n) :

$$\boxed{\binom{m+n}{m} = \binom{m+n}{n}}$$

Through required point:

$$\boxed{(\text{ways to point}) (\text{ways from point})}$$

Avoid point:

$$\boxed{\text{all} - \text{through point}}$$

69. MUST-MEMORIZE FUNCTION COUNTS

Domain size (m) , codomain size (n) .

Any function

$$\boxed{n^m}$$

Injective function

$$\boxed{n(n-1)\cdots(n-m+1)}$$

for:

$$\boxed{m \leq n}$$

70. PIGEONHOLE FORMULAS

At least one box contains at least:

$$\boxed{\left\lceil \frac{N}{k} \right\rceil}$$

when (N) objects are distributed among (k) boxes.

To guarantee $\binom{r}{k}$ objects in one of $\binom{k}{l}$ categories:

$$\lceil \frac{\binom{r}{k}}{\binom{k}{l}} \rceil$$

under standard assumptions.

71. FAST EXAM DECISION TREE

Bound on number?

Count prefix by prefix.

Even/odd/divisible by 5?

Choose last digit first.

Divisible by 4/8?

Focus on last 2/3 digits.

At least k ?

Consider complement of $\{0, 1, \dots, k-1\}$.

Multiple OR conditions?

Use inclusion-exclusion.

Grid?

Convert path into ordered R/U steps.

Function?

Each domain element chooses an image.

Injective?

Choices decrease.

Onto?

Use total minus functions missing outputs.

“Guarantee” wording?

Think pigeonhole.

PRACTICE — LEVEL 0

Q1

How many 4-bit strings contain at least one 1?

Q2

How many functions exist from a 2-element set to a 4-element set?

Q3

How many injective functions exist from a 2-element set to a 4-element set?

Q4

How many grid paths from $((0,0))$ to $((2,3))$?

Q5

Among 1–60, how many numbers are divisible by 2 or 3?

LEVEL 1

Q6

How many 5-bit strings have at least 2 ones?

Q7

Using digits 0–5 without repetition, how many 4-digit even numbers?

Q8

How many paths from $((0,0))$ to $((4,2))$ through $((2,1))$?

Q9

How many onto functions from a 4-element set to a 2-element set?

Q10

Minimum people needed to guarantee 3 share a birth month?

LEVEL 2

Q11

Among 1–120, how many are divisible by 3 or 5?

Q12

How many 5-character strings over 10 symbols contain both A and B at least once?

Q13

How many paths from $((0,0))$ to $((5,4))$ avoid $((2,2))$?

Q14

Using digits 1–6 without repetition, how many 3-digit numbers are greater than 400?

Q15

How many 6-bit strings contain at most two 1s?

LEVEL 3 — EXAM PRESSURE

Q16

Using digits 0–6 without repetition, how many 4-digit odd numbers are possible?

Q17

Among 1–300, how many integers are divisible by 4, 6, or 10?

Q18

How many 5-character strings over 8 symbols contain two specified symbols A and B at least once each?

Q19

Minimum number of students needed to guarantee at least 6 have birthdays in the same month?

Q20

How many paths from $((0,0))$ to $((6,4))$ pass through $((2,1))$ but avoid $((4,3))$?

ANSWERS

Q1

$$\lfloor 2^4 - 1 = \boxed{15} \rfloor$$

Q2

Each of 2 inputs chooses among 4 outputs:

$$\lfloor 4^2 = \boxed{16} \rfloor$$

Q3

$$\lfloor 4 \cdot 3 = \boxed{12} \rfloor$$

Q4

Need:

$$\lfloor 2R + 3U \rfloor$$

$$\lfloor \binom{5}{2} = \boxed{10} \rfloor$$

Q5

Divisible by 2:

$$\lfloor 30 \rfloor$$

By 3:

$$\lfloor 20 \rfloor$$

Both by 6:

$$\lfloor 10 \rfloor$$

Total:

$$\lfloor 30 + 20 - 10 = \boxed{40} \rfloor$$

Q6

Total:

$$\lfloor 32 \rfloor$$

Subtract:

- zero ones: $\backslash(1\backslash)$
- one one: $\backslash(5\backslash)$

$$\backslash[32-6=\boxed{26}\backslash]$$

Q7

From earlier:

$$\backslash[\boxed{156}\backslash]$$

Q8

Start! required

$$\backslash[2R+1U\backslash]$$

$$\backslash[\text{binom}{3}{1}=3\backslash]$$

Required point! end:

$$\backslash[2R+1U\backslash]$$

$$\backslash[3\backslash]$$

Total:

$$\backslash[\boxed{9}\backslash]$$

Q9

Total functions:

$$\backslash[2^4=16\backslash]$$

Not onto:

- all to first: 1
- all to second: 1

Thus:

$$\backslash[\boxed{14}\backslash]$$

Q10

To avoid 3 in a month, at most 2 per month:

$$\backslash[12(2)=24\backslash]$$

One more:

$$\backslash[\boxed{25}\backslash]$$

Q11

Divisible by 3:

$$\backslash[40\backslash]$$

By 5:

$$\backslash[24\backslash]$$

Both by 15:

$$\backslash[8\backslash]$$

Total:

$$\boxed{40+24-8 = 56}$$

Q12

Total:

$$10^5$$

Missing A:

$$9^5$$

Missing B:

$$9^5$$

Missing both:

$$8^5$$

Thus:

$$\boxed{10^5 - 2(9^5) + 8^5}$$

Numerically:

$$100000 - 118098 + 32768 = \boxed{14670}$$

Q13

All paths:

$$\binom{94}{2} = 126$$

Through $(2,2)$:

Start point:

$$\binom{42}{2} = 6$$

Point $a(36,20)$:

$$\binom{52}{2} = 10$$

Through point:

$$60$$

Avoid:

$$126 - 60 = \boxed{66}$$

Q14

Hundreds digit:

$$4,5,6$$

so:

$$3$$

choices.

Remaining:

$$5 \cdot 4$$

Total:

$$\boxed{60}$$

Q15

At most two 1s:

$$\binom{60}{1} + \binom{60}{2} + \binom{60}{3}$$

$$= 1 + 6 + 15$$

$$\boxed{22}$$

Q16

Digits:

$$0, 1, 2, 3, 4, 5, 6$$

Odd units:

$$1, 3, 5$$

so 3 choices.

Thousands cannot be 0 or chosen odd digit.

Available nonzero remaining digits:

$$5$$

choices.

Middle positions:

$$5, 4$$

Thus:

$$3 \cdot 5 \cdot 5 \cdot 4 = \boxed{300}$$

Q17

From 1–300.

Multiples:

$$A=4, B=6, C=10$$

Singles:

$$75 + 50 + 30$$

Pair intersections:

LCMs:

$$\text{LCM}(4, 6) = 12 \rightarrow 25$$

$$\text{LCM}(4, 10) = 20 \rightarrow 15$$

$$\text{LCM}(6, 10) = 30 \rightarrow 10$$

Triple intersection:

$$\text{LCM}(4, 6, 10) = 60$$

$$300/60 = 5$$

Therefore:

$$\lfloor 75+50+30-25-15-10+5 \rfloor$$

$$\lfloor =155-50+5 \rfloor$$

$$\lfloor =\boxed{110} \rfloor$$

Q18

Total:

$$\lfloor 8^5 \rfloor$$

Missing A:

$$\lfloor 7^5 \rfloor$$

Missing B:

$$\lfloor 7^5 \rfloor$$

Missing both:

$$\lfloor 6^5 \rfloor$$

Thus:

$$\lfloor \boxed{8^5-2(7^5)+6^5} \rfloor$$

$$\lfloor 32768-33614+7776 \rfloor$$

$$\lfloor =\boxed{6930} \rfloor$$

Q19

To avoid 6 in one month, at most 5 per month:

$$\lfloor 12(5)=60 \rfloor$$

Next student guarantees 6:

$$\lfloor \boxed{61} \rfloor$$

Q20

We need paths:

$$\lfloor (0,0)\text{to}(6,4) \rfloor$$

that pass through:

$$\lfloor A=(2,1) \rfloor$$

but avoid:

$$\lfloor B=(4,3) \rfloor$$

First count paths through A.

Start! A :

$$\lfloor 2R+1U \rfloor$$

$$\lfloor \binom{3}{1}=3 \rfloor$$

End!

Difference:

$$\{ 4R+3U \}$$

$$\{ \binom{7}{3}=35 \}$$

So through A:

$$\{ 3(35)=105 \}$$

Now subtract paths through both A and B.

A! B :

$$\{ 2R+2U \}$$

$$\{ \binom{4}{2}=6 \}$$

B! e n d :

$$\{ 2R+1U \}$$

$$\{ \binom{3}{1}=3 \}$$

Thus:

$$\{ 3 \cdot 6 \cdot 3 = 54 \}$$

Required:

$$\{ 105-54 = \boxed{51} \}$$

' CHAPTER 38 COMPLETE

We have now covered:

- Fundamental Counting Principle '
- addition rule '
- multiplication rule '
- independent/dependent stages '
- codes/passwords/strings '
- digit formation '
- leading-zero restrictions '
- repetition allowed/forbidden '
- odd/even/divisibility restrictions '
- number bounds and prefix counting '
- exact/at least/at most restrictions '
- complement counting '
- two-set inclusion-exclusion '
- three-set inclusion-exclusion '
- grid-path counting '
- required/forbidden grid points '
- ordinary function counting '
- injection preview '
- surjection preview '
- subset counting intuition '

- binary-string counting '
- pigeonhole principle '
- generalized pigeonhole '
- guarantee questions '
- GATE/placement L0–L3 drills '

Progress

Completed: 38/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 39 — Permutations

Next we will cover:

- factorial notation
- $(n!)$
- permutations of $(n!)$ distinct objects
- (^nP_r)
- arrangement vs selection
- repetition allowed
- repeated/identical objects
- word arrangements
- objects together / apart
- relative-order restrictions
- circular permutations
- fixed positions
- digit arrangements
- derangement preview
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 39 — PERMUTATIONS

PART 1 — FACTORIALS, (nP_r) , ARRANGEMENTS, REPETITION, IDENTICAL OBJECTS & WORD PERMUTATIONS

Status: 0% PROGRESS GATE: High Placements: High Priority: 0% For 1

Permutation means:

Arrangement where order matters.

This is the key distinction from combinations.

If A and B are two people:

$$AB \neq BA$$

when the order or role matters.

1. FACTORIAL NOTATION

For positive integer n :

$$n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1$$

Examples:

$$5! = 5\cdot 4\cdot 3\cdot 2\cdot 1 = 120$$

$$4! = 24$$

$$3! = 6$$

2. SPECIAL VALUE

$$0! = 1$$

MUST MEMORIZE

Why?

Because formulas like:

$${}^n P_n = \frac{n!}{(n-n)!}$$

need:

$$0! = 1$$

3. BASIC FACTORIAL TABLE

n	$n!$
0	1
1	1
2	2
3	6
4	24
5	120
6	720

$(n!)$	$(n!)$
7	5040
8	40320
9	362880
10	3628800

At least $(0!)$ through $(7!)$ should become automatic.

4. FACTORIAL CANCELLATION

Do not expand full factorials unnecessarily.

Example:

$$\frac{10!}{8!}$$

Cancel:

$$= \frac{10 \cdot 9 \cdot 8!}{8!}$$

$$= \boxed{90}$$

Not necessary to calculate $(10!)$.

5. ANOTHER EXAMPLE

$$\frac{12!}{9!}$$

$$= 12 \cdot 11 \cdot 10$$

$$= \boxed{1320}$$

Speed rule

Expand only until cancellation happens.

6. ARRANGING n DISTINCT OBJECTS

Suppose there are 4 distinct letters:

$$[A, B, C, D]$$

Number of arrangements:

First position:

$$[4]$$

Second:

$$[3]$$

Third:

$$[2]$$

Fourth:

$$\backslash[1 \backslash]$$

Therefore:

$$\backslash[4!=24 \backslash]$$

General:

$$\backslash[\boxed{n! \text{ arrangements of } n \text{ distinct objects}} \backslash]$$

7. PERMUTING r OBJECTS FROM n

Suppose we have n distinct objects but arrange only r of them.

First position:

$$\backslash[n \backslash]$$

Second:

$$\backslash[n-1 \backslash]$$

Third:

$$\backslash[n-2 \backslash]$$

and so on.

Thus:

$$\backslash[\boxed{{}^nP_r = n(n-1)\cdots(n-r+1)} \backslash]$$

Equivalent formula:

$$\backslash[\boxed{{}^nP_r = \frac{n!}{(n-r)!}} \backslash]$$

MUST MEMORIZE

8. EXAMPLE

Arrange 3 people chosen from 7 in three ordered positions.

$$\backslash[{}^7P_3 = 7\cdot 6\cdot 5 \backslash]$$

$$\backslash[=\boxed{210} \backslash]$$

9. FULL ARRANGEMENT IS A SPECIAL CASE

$$\backslash[{}^nP_n = \frac{n!}{0!} = n! \backslash]$$

So:

$$\backslash[\boxed{{}^nP_n = n!} \backslash]$$

10. ARRANGEMENT VS SELECTION

Suppose choose captain and vice-captain from 5 students.

A captain B vice-captain is different from:

B captain A vice-captain.

Therefore order/role matters.

Count:

$${}^5P_2 = 5 \cdot 4 = \boxed{20}$$

11. WHEN ORDER DOES NOT MATTER

If question only says:

Choose 2 students from 5 for a committee.

Then:

$$\{A, B\} = \{B, A\}$$

So permutations overcount.

That belongs to **Chapter 40 — Combinations**.

Memory trick

- Arrange / rank / seat / assign roles !' permutation
- Choose / select / committee !' combination

12. REPETITION ALLOWED

Suppose we make a 3-letter string using:

$$\{A, B, C, D\}$$

and letters may repeat.

Each position has:

$$4$$

choices.

Total:

$$4^3 = \boxed{64}$$

This is not 4P_3 , because repetition is allowed.

13. GENERAL REPETITION-ALLOWED ARRANGEMENT

Using $\{n\}$ symbols in $\{r\}$ ordered positions, with repetition allowed:

$$\boxed{n^r}$$

Examples:

- PINs
- passwords

- strings
 - sequences
-

14. NO REPETITION

Same 4 letters, 3 positions, no repetition:

$$\sqrt[4]{3 \cdot 2}$$

$$\sqrt[4]{24}$$

or:

$$\sqrt[4]{4P_3}$$

15. ONE OBJECT FIXED

How many arrangements of 6 distinct people if A must occupy the first position?

A is fixed.

Arrange remaining 5:

$$\sqrt[5]{5!}$$

$$\sqrt[5]{120}$$

General:

Fix required objects first, then arrange what remains.

16. TWO FIXED POSITIONS

7 distinct people sit in a row.

A must be first and B must be last.

Both positions fixed.

Remaining:

$$\sqrt[5]{5!}$$

$$\sqrt[5]{120}$$

17. A PARTICULAR PERSON NOT IN A POSITION

6 people in a row.

A cannot sit first.

Total arrangements:

$$\sqrt[6]{6!}$$

Bad arrangements with A first:

$$[5!]$$

Valid:

$$[6! - 5!]$$

$$[= 720 - 120]$$

$$[= \boxed{600}]$$

18. FASTER DIRECT METHOD

A's position can be any of the remaining:

$$[5]$$

positions.

Then arrange other 5:

$$[5!]$$

So:

$$[5 \cdot 5! = 600]$$

Same answer.

19. IDENTICAL OBJECTS

Suppose letters:

$$[A, A, B]$$

If all were distinct, we'd have:

$$[3! = 6]$$

But swapping the two A's produces no new arrangement.

Each actual arrangement is counted:

$$[2!]$$

times.

Therefore:

$$[\frac{3!}{2!} = \boxed{3}]$$

Arrangements:

$$[AAB, \ ABA, \ BAA]$$

20. GENERAL IDENTICAL-OBJECT FORMULA

If (n) objects contain:

- (p) identical of one type
- (q) identical of another
- (r) identical of another

- etc.

Then number of distinct arrangements:

$$\boxed{\frac{n!}{p!q!r!\cdots}}$$

MUST MEMORIZE

21. EXAMPLE — LEVEL

Arrange letters of:

LEVEL

5 letters.

Repeated:

- L twice
- E twice
- V once

Number:

$$\frac{5!}{2!2!}$$

$$= \frac{120}{4}$$

$$= \boxed{30}$$

22. EXAMPLE — BANANA

BANANA has 6 letters.

Counts:

- A = 3
- N = 2
- B = 1

Arrangements:

$$\frac{6!}{3!2!}$$

$$= \frac{720}{12}$$

$$= \boxed{60}$$

23. EXAMPLE — MISSISSIPPI

MISSISSIPPI has 11 letters.

Counts:

- M = 1
 - I = 4
 - S = 4
 - P = 2
-

Therefore:

$$\boxed{\frac{11!}{4!4!2!}}$$

Calculate if required:

$$\frac{39916800}{24 \cdot 24 \cdot 2} = \boxed{34650}$$

24. WHY WE DIVIDE BY FACTORIALS

Imagine temporarily labeling repeated A's:

$$A_1, A_2, A_3$$

Each real word arrangement is counted:

$$3!$$

times due to swapping those identical A's.

So divide by:

$$3!$$

Same logic for every repeated group.

25. WORDS STARTING WITH A PARTICULAR LETTER

How many arrangements of BANANA start with B?

Fix B first.

Remaining letters:

$$A, A, A, N, N$$

Distinct arrangements:

$$\frac{5!}{3!2!}$$

$$= \boxed{10}$$

26. WORDS ENDING WITH A VOWEL — CASE METHOD

Take word:

$$\text{MANGO}$$

All letters distinct.

Vowels:

$$A, O$$

Choose last vowel:

$$2$$

Arrange remaining 4 letters:

$$4!$$

Total:

$$2 \cdot 24 = \boxed{48}$$

27. STARTS AND ENDS WITH PARTICULAR TYPES

Using letters of MANGO, how many arrangements start with M and end in a vowel?

First position fixed M.

Last position:

$$2$$

choices A/O.

Arrange remaining 3 letters:

$$3!$$

Total:

$$2 \cdot 6 = \boxed{12}$$

28. ALL VOWELS TOGETHER — BLOCK METHOD

Suppose word:

$$\text{MANGO}$$

Vowels:

$$\text{A, O}$$

must stay together.

Treat AO/OA as one block.

Objects:

- vowel block
- M
- N
- G

So 4 objects:

$$4!$$

Inside vowel block:

$$2!$$

Total:

$$4! \cdot 2!$$

$$= 24 \cdot 2$$

$$= \boxed{48}$$

29. BLOCK METHOD — GENERAL

If k specific distinct objects must remain together among n distinct objects:

Treat the k objects as one super-object.

Then number of external objects:

$$[n - k + 1]$$

Arrange them:

$$[(n - k + 1)!]$$

Arrange the k objects internally:

$$[k!]$$

Therefore:

$$[(n - k + 1)!k!]$$

when there are no other restrictions.

30. TWO PARTICULAR PEOPLE TOGETHER

7 people in a row.

A and B must sit together.

Treat AB as one block.

Now:

$$[6]$$

objects.

Arrange:

$$[6!]$$

Within block:

$$[AB \text{ or } BA]$$

$$[2!]$$

Total:

$$[2 \cdot 6! = 1440]$$

31. TWO PARTICULAR PEOPLE NOT TOGETHER

Use complement.

Total arrangements:

$$[7!]$$

Together:

$$[2 \cdot 6!]$$

Therefore apart:

$$\lfloor 7! - 2 \cdot 6! \rfloor$$

$$\lfloor 5040 - 1440 \rfloor$$

$$\lfloor = \boxed{3600} \rfloor$$

32. COMPLEMENT IS OFTEN BEST FOR “NOT TOGETHER”

Directly forcing separation can be possible, but:

$$\lfloor \boxed{\text{not together}} = \text{total} - \text{together} \rfloor$$

is usually faster.

33. THREE PARTICULAR PEOPLE TOGETHER

8 people, A/B/C must stay together.

Treat ABC as one block.

Objects:

$$\lfloor 8 - 3 + 1 = 6 \rfloor$$

Arrange external:

$$\lfloor 6! \rfloor$$

Internal ABC arrangements:

$$\lfloor 3! \rfloor$$

Total:

$$\lfloor 6! \cdot 3! \rfloor$$

$$\lfloor = 720 \cdot 6 \rfloor$$

$$\lfloor = \boxed{4320} \rfloor$$

34. TWO GROUPS EACH TOGETHER

8 people:

- A,B,C together
- D,E together

Treat:

- ABC block
- DE block
- F,G,H

Total super-objects:

$$\lfloor 5 \rfloor$$

Arrange:

$$5!$$

Internal:

$$3! \cdot 2!$$

Total:

$$5! \cdot 3! \cdot 2!$$

$$= 120 \cdot 6 \cdot 2$$

$$= \boxed{1440}$$

35. RELATIVE ORDER OF TWO PEOPLE

6 distinct people arranged in a row.

How many arrangements have A before B?

Total:

$$6!$$

By symmetry, exactly half have A before B and half B before A.

Therefore:

$$\boxed{\frac{6!}{2} = 360}$$

36. RELATIVE ORDER OF k SPECIFIC PEOPLE

Suppose among n people, A, B, C must appear in that relative order:

$$A \text{ before } B \text{ before } C$$

Among all permutations, the three people can internally appear in:

$$3!$$

relative orders equally often.

Only 1 is allowed.

Therefore:

$$\boxed{\frac{n!}{3!}}$$

37. EXAMPLE

7 people arranged in a row.

A before B before C.

$$\frac{7!}{3!}$$

$$= \frac{5040}{6}$$

$$= \boxed{840}$$

38. A BEFORE B, NO OTHER RESTRICTION

For n distinct objects:

$$\boxed{\frac{n!}{2}}$$

because A/B order has 2 equally likely possibilities.

39. EXACT POSITION OF ONE PERSON

8 people in a row.

A must be in position 3.

Fix A.

Remaining 7:

$$7!$$

$$\boxed{5040}$$

Any fixed single position gives the same count.

40. PERSON IN ONE OF k POSITIONS

8 people.

A may sit only in positions 1, 2, or 3.

Choose A's position:

$$3$$

Arrange other 7:

$$7!$$

Total:

$$\boxed{3 \cdot 7!}$$

41. TWO PEOPLE CANNOT OCCUPY TWO PARTICULAR POSITIONS — PREVIEW

These restrictions are often easiest using:

- complement
- inclusion-exclusion

Example:

6 people, A cannot be first, B cannot be last.

Total:

$$6!$$

Subtract:

A first:

$$\backslash[5! \backslash]$$

B last:

$$\backslash[5! \backslash]$$

But arrangements satisfying both were subtracted twice.

Add them back:

$$\backslash[4! \backslash]$$

Therefore:

$$\backslash[6! - 5! - 5! + 4! \backslash]$$

$$\backslash[720 - 120 - 120 + 24 \backslash]$$

$$\backslash[=\boxed{504} \backslash]$$

42. DIGIT PERMUTATION— ALL DISTINCT DIGITS

Using digits 1,2,3,4,5 exactly once, number of 5-digit numbers:

$$\backslash[5! \backslash]$$

$$\backslash[=\boxed{120} \backslash]$$

43. WITH ZERO

Using digits:

$$\backslash[0,1,2,3,4 \backslash]$$

exactly once, number of 5-digit numbers?

All arrangements:

$$\backslash[5! \backslash]$$

But arrangements starting with zero are invalid.

Zero first:

$$\backslash[4! \backslash]$$

Therefore:

$$\backslash[5! - 4! \backslash]$$

$$\backslash[120 - 24 \backslash]$$

$$\backslash[=\boxed{96} \backslash]$$

44. FASTER ZERO-FIRST METHOD

First digit:

$$\backslash[4 \backslash]$$

choices:

$\{1, 2, 3, 4\}$

Arrange remaining four digits:

$4!$

Total:

$4 \cdot 24 = \boxed{96}$

45. EVEN NUMBERS USING ALL DIGITS

Using digits:

$\{1, 2, 3, 4, 5\}$

exactly once, how many 5-digit even numbers?

Last digit must be:

$\{2 \text{ or } 4\}$

Choose:

2

Arrange remaining 4:

$4!$

Total:

$\boxed{48}$

46. EVEN NUMBERS WITH ZERO INCLUDED

Digits:

$\{0, 1, 2, 3, 4\}$

all used once.

Need even last digit:

$\{0, 2, 4\}$

Zero case differs, so split.

Last = 0

Arrange 1, 2, 3, 4 in first four positions:

$4! = 24$

Last = 2 or 4

Choose last:

2

Remaining 4 digits include 0.

Total arrangements of remaining 4:

$$\backslash[4! \backslash]$$

Invalid with 0 first:

$$\backslash[3! \backslash]$$

Valid per final digit:

$$\backslash[4! - 3! = 18 \backslash]$$

So:

$$\backslash[2(18) = 36 \backslash]$$

Total:

$$\backslash[24 + 36 = \boxed{60} \backslash]$$

47. WORD WITH REPEATED LETTERS + FIXED FIRST LETTER

Word:

$$\backslash[APPLE \backslash]$$

Letters:

- P twice
- A, L, E once

How many arrangements start with A?

Fix A.

Remaining:

$$\backslash[P, P, L, E \backslash]$$

Count:

$$\backslash[\frac{4!}{2!} = \boxed{12} \backslash]$$

48. WORD WITH REPEATED LETTERS + ALL VOWELS TOGETHER

Word:

$$\backslash[BALLOON \backslash]$$

Letters:

- L twice
- O twice
- B, A, N once

Vowels are:

$$\backslash[A, O, O \backslash]$$

Treat all vowels as one block.

External objects:

- vowel block
- B
- L
- L
- N

5 objects with L repeated twice.

External arrangements:

$$\frac{5!}{2!} = 60$$

Internal vowel arrangements:

$$\frac{3!}{2!} = 3$$

Total:

$$60 \cdot 3 = \boxed{180}$$

49. IMPORTANT BLOCK-METHOD TRAP

If the objects within the block include repeats, do **not** multiply by $(k!)$ blindly.

Use:

$$\frac{k!}{\text{repetition factorials}}$$

for internal arrangements.

50. VOWELS NEVER TOGETHER — SIMPLE TWO-VOWEL CASE

Word MANGO has 5 distinct letters, with vowels A,O.

Total arrangements:

$$5! = 120$$

Vowels together:

$$4! \cdot 2! = 48$$

Therefore vowels not together:

$$120 - 48 = \boxed{72}$$

51. SEPARATING MANY OBJECTS — GAP METHOD INTRODUCTION

Suppose 3 boys and 2 girls, all distinct, arranged in a row so girls are not together.

First arrange boys:

$$3! = 6$$

This creates gaps:

$$_B_B_B_$$

Number of gaps:

$\backslash[4 \backslash]$

Choose 2 different gaps for girls and arrange girls.

Choose gaps:

$\backslash[\backslash binom{4}{2}=6 \backslash]$

Arrange girls:

$\backslash[2!=2 \backslash]$

Total:

$\backslash[3!\backslash binom{4}{2} 2! \backslash]$

$\backslash[=6\cdot 6\cdot 2 \backslash]$

$\backslash[=\boxed{72} \backslash]$

This method becomes very important when multiple objects must be separated.

52. WHY NUMBER OF GAPS IS $n+1$

Arrange $\backslash(n\backslash)$ anchor objects:

$\backslash[A_1, A_2, \backslash dots, A_n \backslash]$

Gaps exist:

- before first
- between every pair
- after last

So:

$\backslash[\boxed{n+1}\text{ gaps} \backslash]$

53. NO TWO GIRLS TOGETHER — GENERAL IDEA

If:

- $\backslash(b\backslash)$ distinct boys
- $\backslash(g\backslash)$ distinct girls

and no two girls may be adjacent:

First arrange boys:

$\backslash[b! \backslash]$

Number of gaps:

$\backslash[b+1 \backslash]$

Choose $\backslash(g\backslash)$ gaps:

$\backslash[\backslash binom{b+1}{g} \backslash]$

Arrange girls:

$\backslash[g! \backslash]$

Total:

$$\boxed{b! \binom{b+1}{g} g!}$$

provided:

$$g \leq b+1$$

54. IMPOSSIBILITY CONDITION

If:

$$g > b+1$$

then no arrangement can keep all girls separated.

Example:

2 boys, 4 girls.

Boys produce only 3 gaps.

Cannot place 4 girls in distinct gaps.

So:

$$\boxed{0}$$

55. ALTERNATING ARRANGEMENTS

3 boys and 3 girls, all distinct, must alternate.

Possible gender patterns:

$$\boxed{BGBGBG}$$

or:

$$\boxed{GBGBGB}$$

So:

$$\boxed{2}$$

patterns.

Arrange boys:

$$\boxed{3!}$$

Arrange girls:

$$\boxed{3!}$$

Total:

$$\boxed{2(3!)(3!)}$$

$$\boxed{=72}$$

56. UNEQUAL GROUPS ALTERNATING

4 boys and 3 girls.

To alternate, only pattern possible:

$\{ B G B G B G B \}$

Arrange boys:

$\{ 4! \}$

girls:

$\{ 3! \}$

Total:

$\{ \boxed{4!3!=144} \}$

Similarly if girls exceed boys by 1, pattern must start/end with girl.

57. WHEN ALTERNATION IS IMPOSSIBLE

If group sizes differ by more than 1, perfect alternation is impossible.

Example:

5 boys and 3 girls.

Difference:

$\{ 2 \}$

No complete alternating arrangement.

$\{ \boxed{0} \}$

58. CIRCULAR PERMUTATIONS— INTRODUCTION

In a row:

$\{ ABC, \ BCA, \ CAB \}$

are different arrangements.

Around a circle, rotations may represent the same arrangement.

Example:

If A-B-C are seated clockwise around a round table, rotating everyone together does not create a new relative seating.

Therefore circular counting differs.

59. n DISTINCT PEOPLE AROUND A ROUND TABLE

Number of linear arrangements:

$\{ n! \}$

But each circular arrangement is counted $\{ n \}$ times due to rotations.

Therefore:

$$\boxed{(n-1)!}$$

MUST MEMORIZE

60. EXAMPLE

5 people around a round table:

$$(5-1)! = 4! = \boxed{24}$$

61. INTUITIVE METHOD

Fix one person to remove rotational symmetry.

Suppose A is fixed.

Arrange remaining $(n-1)$ people around A:

$$\boxed{(n-1)!}$$

Very easy to remember.

62. ROUND TABLE VS NECKLACE WARNING

For ordinary round-table seating:

- rotations considered same
- clockwise and anticlockwise usually considered different

Thus:

$$(n-1)!$$

For necklaces/bracelets, reflection may also be identical.

That introduces another division by 2 under appropriate conditions.

Do not confuse them.

63. TWO PARTICULAR PEOPLE TOGETHER IN A CIRCLE

(n) people around a round table, A and B together.

Treat AB as block.

Now circular objects:

$$n-1$$

objects total.

Circular arrangements:

$$(n-2)!$$

Internal AB:

$$2!$$

Thus:

$$\boxed{2(n-2)!}$$

64. EXAMPLE

6 people around a table, A and B together:

$$2(6-2)! = 2(4)! = \boxed{48}$$

65. TWO PARTICULAR PEOPLE NOT TOGETHER IN CIRCLE

Total:

$$(n-1)!$$

Together:

$$2(n-2)!$$

Therefore:

$$\boxed{(n-1)! - 2(n-2)!}$$

For $(n=6)$:

$$120 - 48 = \boxed{72}$$

66. TWO PARTICULAR PEOPLE OPPOSITE — EVEN n

Suppose 6 people around a table.

A and B must sit opposite.

Fix A.

Exactly one seat is opposite A, so B is fixed there.

Arrange remaining 4 people:

$$4!$$

$$= \boxed{24}$$

General for (n) even:

$$\boxed{(n-2)!}$$

for a specified pair to sit opposite.

67. CIRCULAR ALTERNATING ARRANGEMENT

Suppose 4 men and 4 women sit alternately around a circular table.

Arrange men around circle:

$$(4-1)! = 3!$$

There are exactly 4 gaps between men.

Arrange women:

$$\backslash[4! \backslash]$$

Total:

$$\backslash[3!\cdot 4! \backslash]$$

$$\backslash[=6\cdot 24 \backslash]$$

$$\backslash[=\boxed{144} \backslash]$$

68. WHY NOT MULTIPLY BY 2?

In circular seating, once men's positions are fixed around the table, the gaps for women are determined.

The pattern “start with man” vs “start with woman” is not a separate circular arrangement; rotation removes that distinction.

This differs from linear alternating arrangements.

69. GATE-STYLE QUESTION 1

How many ways can 7 distinct tasks be arranged?

$$\backslash[7! = \boxed{5040} \backslash]$$

70. GATE-STYLE QUESTION 2

How many ordered triples can be selected from 8 distinct objects without repetition?

$$\backslash[{}^8P_3 = 8\cdot 7\cdot 6 \backslash]$$

$$\backslash[=\boxed{336} \backslash]$$

71. GATE-STYLE QUESTION 3

How many distinct arrangements of:

$$\backslash[\text{SUCCESS} \backslash]$$

Count letters:

SUCCESS has 7 letters:

- S = 3
- C = 2
- U = 1
- E = 1

Therefore:

$$\backslash[\frac{7!}{3!2!} \backslash]$$

$$\backslash[=\frac{5040}{12} \backslash]$$

$$\backslash[=\boxed{420} \backslash]$$

72. GATE-STYLE QUESTION 4

8 people in a row, A before B before C.

$$\left[\frac{8!}{3!} \right]$$

$$\left[=\frac{40320}{6} \right]$$

$$\left[=\boxed{6720} \right]$$

73. PLACEMENT-STYLE QUESTION 1

How many ways can 6 people sit in a row if A and B are together?

$$\left[5! \cdot 2 \right]$$

$$\left[=120 \cdot 2 \right]$$

$$\left[=\boxed{240} \right]$$

74. PLACEMENT-STYLE QUESTION 2

Same 6 people, A and B not together.

Total:

$$\left[6! = 720 \right]$$

Together:

$$\left[240 \right]$$

Not together:

$$\left[\boxed{480} \right]$$

75. PLACEMENT-STYLE QUESTION 3

How many distinct arrangements of:

$$\left[\text{APPLE} \right]$$

$$\left[\frac{5!}{2!} = \boxed{60} \right]$$

76. PLACEMENT-STYLE QUESTION 4

6 people around a circular table:

$$\left[(6-1)! = 5! = \boxed{120} \right]$$

77. ADVANCED QUESTION — WORD VOWELS TOGETHER

How many arrangements of EDUCATE have all vowels together?

Letters:

$\{E, D, U, C, A, T, E\}$

7 letters.

Vowels:

$\{E, U, A, E\}$

4 vowels, with E repeated twice.

Consonants:

$\{D, C, T\}$

Treat vowel group as one block.

External objects:

- vowel block
- D
- C
- T

So:

$4!$

External arrangements.

Internal vowel arrangements:

$\frac{4!}{2!}$

Therefore:

$4! \cdot \frac{4!}{2!}$

$= 24 \cdot 12$

$= \boxed{288}$

78. ADVANCED QUESTION — NO TWO VOWELS TOGETHER

Take word:

$\{MOBILE\}$

All 6 letters distinct.

Vowels:

$\{O, I, E\}$

Consonants:

$\{M, B, L\}$

Arrange consonants:

$3! = 6$

Create 4 gaps:

$_ _ C _ C _ C _ _$

Choose 3 gaps for vowels:

$$\lfloor \binom{4}{3} = 4 \rfloor$$

Arrange vowels:

$$\lfloor 3! = 6 \rfloor$$

Total:

$$\lfloor 6 \cdot 4 \cdot 6 \rfloor$$

$$\lfloor = \boxed{144} \rfloor$$

79. ADVANCED QUESTION — CIRCLE TOGETHER

7 people around a round table.

A,B,C must sit together.

Treat ABC as one block.

Circular objects:

- ABC block
- remaining 4 people

Total:

$$\lfloor 5 \rfloor$$

circular objects.

Arrangements:

$$\lfloor (5-1)! = 4! \rfloor$$

Internal ABC:

$$\lfloor 3! \rfloor$$

Total:

$$\lfloor 4! \cdot 3! \rfloor$$

$$\lfloor = 24 \cdot 6 \rfloor$$

$$\lfloor = \boxed{144} \rfloor$$

80. ADVANCED QUESTION — ALTERNATING CIRCLE

5 men and 5 women around a round table, alternating.

Arrange men circularly:

$$\lfloor (5-1)! = 24 \rfloor$$

Arrange women into 5 gaps:

$$\lfloor 5! = 120 \rfloor$$

Total:

$$\lfloor 24 \cdot 120 \rfloor$$

$\boxed{2880}$

81. COMMON TRAPS

& pTRAP 1 — Permutation vs combination

If order doesn't matter, $\binom{n}{r}$ overcounts.

& pTRAP 2 — Forgetting repeated objects

BANANA is not $6!$.

Use:

$\frac{6!}{3!2!}$

& pTRAP 3 — “Together”

Treat required objects as one block **and** arrange within the block.

& pTRAP 4 — “Not together”

Complement is often easiest.

& pTRAP 5 — Leading zero

Permutations of digits including 0 may produce invalid ordinary numbers.

& pTRAP 6 — Circular seating

Use:

$(n-1)!$

not $n!$.

& pTRAP 7 — Necklace vs round table

Reflections are normally distinct for round-table seating.

& pTRAP 8 — Repeated letters inside a block

Internal block arrangements may also need division by repeated-letter factorials.

82. MUST-MEMORIZE FORMULAS

Factorial

$\boxed{n! = n(n-1)\cdots 1}$

$\boxed{0! = 1}$

Permutation

$\boxed{{}^nP_r = \frac{n!}{(n-r)!}}$

All $(n!)$ distinct objects

$\boxed{n!}$

Repetition allowed in $(r!)$ positions from $(n!)$ symbols

$\boxed{n^r}$

83. IDENTICAL-OBJECT FORMULA

$\boxed{\frac{n!}{p!q!r!\cdots}}$

where repeated groups have sizes:

$p, q, r, \dots$

84. CIRCULAR FORMULA

For (n) distinct people around a round table:

$\boxed{(n-1)!}$

Specified A,B together:

$\boxed{2(n-2)!}$

85. FAST EXAM DECISION TREE

Is order important?

Yes! permutation.

All distinct objects arranged?

Use $(n!)$.

Only $(r!)$ positions?

Use (nP_r) .

Repetition allowed?

Use (n^r) where appropriate.

Repeated identical objects?

Divide by repetition factorials.

Must stay together?

Block method.

Must stay apart?

Try complement or gap method.

Relative order fixed?

Divide by factorial of internally ordered group.

Circle?

Fix one object !' $((n-1)!).$

Zero among digits?

Handle leading position carefully.

PRACTICE — LEVEL 0

Q1

Find:

$\{^7P_2\}$

Q2

How many arrangements of 5 distinct books?

Q3

How many 3-letter strings from 4 letters with repetition allowed?

Q4

How many distinct arrangements of:

$\{AABBC\}$

Q5

6 people sit in a row. How many arrangements have A first?

LEVEL 1

Q6

How many ways can 7 people sit in a row if A and B sit together?

Q7

How many ways can 7 people sit if A and B do not sit together?

Q8

Find distinct arrangements of:

$\{BANANA\}$

Q9

How many 5-digit numbers can be made using digits 0,1,2,3,4 exactly once?

Q10

How many ways can 6 people sit around a round table?

LEVEL 2

Q11

8 people in a row. How many arrangements have A before B before C?

Q12

4 boys and 3 girls, all distinct, are arranged so no two girls are adjacent. Find the count.

Q13

How many distinct arrangements of BALLOON have all vowels together?

Q14

7 people around a round table. A and B must sit together. Find the count.

Q15

4 men and 4 women sit alternately around a circular table. Find the number of arrangements.

LEVEL 3 — EXAM PRESSURE

Q16

8 people in a row. A cannot sit first and B cannot sit last. Find the count.

Q17

Using digits 0,1,2,3,4 exactly once, how many 5-digit even numbers can be formed?

Q18

How many arrangements of MOBILE have no two vowels together?

Q19

7 people around a round table. A,B,C must sit together. Find the count.

Q20

5 boys and 4 girls are arranged in a row so no two girls are together. Find the count.

ANSWERS

Q1

$${}^7P_2 = 7 \cdot 6 = \boxed{42}$$

Q2

$$5! = \boxed{120}$$

Q3

$$4^3 = \boxed{64}$$

Q4

5 letters:

- A twice

- B twice

$$\left[\frac{5!}{2!2!} = \frac{120}{4} = \boxed{30} \right]$$

Q5

A fixed first.

Remaining:

$$\left[5! = \boxed{120} \right]$$

Q6

AB block + 5 others:

$$\left[6! \right]$$

Internal AB:

$$\left[2! \right]$$

$$\left[2 \cdot 6! = \boxed{1440} \right]$$

Q7

Total:

$$\left[7! = 5040 \right]$$

Together:

$$\left[1440 \right]$$

Apart:

$$\left[5040 - 1440 = \boxed{3600} \right]$$

Q8

$$\left[\frac{6!}{3!2!} = \boxed{60} \right]$$

Q9

Total arrangements:

$$\left[5! = 120 \right]$$

Starting with 0:

$$\left[4! = 24 \right]$$

Valid:

$$\left[\boxed{96} \right]$$

Q10

$$\left[(6-1)! = \boxed{120} \right]$$

Q11

Relative orders of A,B,C:

$$\left[3! = 6 \right]$$

Only one desired.

$$\left[\frac{8!}{6} = \boxed{6720} \right]$$

Q12

Arrange boys:

$$\lfloor 4!=24 \rfloor$$

Gaps:

$$\lfloor 5 \rfloor$$

Choose 3 gaps:

$$\lfloor \text{binom}{5}{3}=10 \rfloor$$

Arrange girls:

$$\lfloor 3!=6 \rfloor$$

Total:

$$\lfloor 24 \cdot 10 \cdot 6 = \boxed{1440} \rfloor$$

Q13

BALLOON:

Vowels:

$$\lfloor A, O, O \rfloor$$

Consonants:

$$\lfloor B, L, L, N \rfloor$$

Treat vowels as block.

External objects:

block, B, L, L, N.

$$\lfloor \frac{5!}{2!}=60 \rfloor$$

Internal vowels:

$$\lfloor \frac{3!}{2!}=3 \rfloor$$

Total:

$$\lfloor 60 \cdot 3 = \boxed{180} \rfloor$$

Q14

Circular A,B together:

$$\lfloor 2(n-2)! \rfloor$$

For $(n=7)$:

$$\lfloor 2(5!) = \boxed{240} \rfloor$$

Q15

Arrange 4 men around circle:

$$\lfloor 3!=6 \rfloor$$

Arrange 4 women in gaps:

$$\lfloor 4!=24 \rfloor$$

Total:

$$\boxed{6(24) = 144}$$

Q16

Total:

$$8!$$

A first:

$$7!$$

B last:

$$7!$$

Both:

$$6!$$

Thus:

$$8! - 7! - 7! + 6!$$

$$40320 - 5040 - 5040 + 720$$

$$\boxed{30960}$$

Q17

Digits:

$$0, 1, 2, 3, 4$$

Last even: 0, 2, 4.

Last = 0:

$$4! = 24$$

Last = 2 or 4:

For each:

$$4! - 3! = 18$$

Two choices:

$$36$$

Total:

$$\boxed{60}$$

Q18

MOBILE:

Vowels O, I, E = 3.

Consonants M, B, L = 3.

Arrange consonants:

$$3!$$

4 gaps.

Choose 3:

$$\left[\begin{matrix} \text{binom} \\ 43 \end{matrix} \right]$$

Arrange vowels:

$$\left[\begin{matrix} 3! \end{matrix} \right]$$

$$\left[\begin{matrix} 6 \cdot 4 \cdot 6 = \boxed{144} \end{matrix} \right]$$

Q19

ABC block + 4 others:

5 circular objects.

Circular arrangements:

$$\left[\begin{matrix} 4! \end{matrix} \right]$$

Internal:

$$\left[\begin{matrix} 3! \end{matrix} \right]$$

Total:

$$\left[\begin{matrix} 24 \cdot 6 = \boxed{144} \end{matrix} \right]$$

Q20

5 boys first:

$$\left[\begin{matrix} 5! \end{matrix} \right]$$

Gaps:

$$\left[\begin{matrix} 6 \end{matrix} \right]$$

Choose 4 gaps:

$$\left[\begin{matrix} \text{binom} \\ 64=15 \end{matrix} \right]$$

Arrange 4 girls:

$$\left[\begin{matrix} 4! \end{matrix} \right]$$

Total:

$$\left[\begin{matrix} 5! \cdot \text{binom} \\ 64 \ 4! \end{matrix} \right]$$

$$\left[\begin{matrix} 120 \cdot 15 \cdot 24 \end{matrix} \right]$$

$$\left[\begin{matrix} = \boxed{43200} \end{matrix} \right]$$

CHAPTER 39 STATUS

Covered so far:

- factorial notation '
- $(0!)$ '
- factorial cancellation '
- $(n!)$ arrangements '
- (nP_r) '
- order-matters concept '

- repetition allowed/forbidden '
- fixed positions '
- repeated identical objects '
- word permutations '
- block method '
- together/apart restrictions '
- relative-order restrictions '
- gap method '
- alternating arrangements '
- circular permutations '
- circular together/apart '
- opposite seating '
- digit permutations and zero restrictions '
- GATE/placement practice '

But **Chapter 39 is not complete yet.**

Next we still need:

- advanced restricted arrangements
- exactly k objects together
- objects never adjacent
- multiple block restrictions
- repeated-letter positional restrictions
- circular restrictions in more depth
- necklace/bracelet distinction
- derangements and "none in original position"
- rank of words / lexicographic arrangements
- final exam-speed shortcuts and chapter audit

Progress

Completed: 38/76 Current: 0% Chapter 39 — Permutations

NEXT

Chapter 39 Part 2 — Advanced restrictions, derangements, circular cases, lexicographic rank & final permutation toolkit

USER

next

ASSISTANT

CHAPTER 39 — PERMUTATIONS

PART 2 — ADVANCED RESTRICTIONS, DERANGEMENTS, CIRCULAR CASES, LEXICOGRAPHIC RANK & FINAL TOOLKIT

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes Per 1

Part 1 covered the core permutation machinery.

Now we finish the chapter with the harder patterns:

- multiple restrictions
- adjacency / non-adjacency
- repeated letters with conditions
- circular restrictions
- derangements
- lexicographic rank

The central exam rule remains:

Do not start calculating until you decide whether to use direct counting, blocks, gaps, symmetry, complement, or inclusion-exclusion.

1. EXACTLY TWO PARTICULAR PEOPLE TOGETHER

Suppose 6 distinct people include A, B, C.

How many arrangements have **A and B together**, but C need not satisfy any special condition?

Treat AB as a block.

Objects:

- AB block
- C
- 3 others

Total:

$\lfloor 5! \rfloor$

Inside block:

$\lfloor 2! \rfloor$

Thus:

$\lfloor \boxed{2 \cdot 5! = 240} \rfloor$

2. A AND B TOGETHER, BUT C NOT ADJACENT TO THE BLOCK

Now 6 people: A, B, C, D, E, F.

A and B must sit together, and C must not sit next to either A or B.

Treat AB as a block $\lfloor (X) \rfloor$.

We now arrange:

$\lfloor X, C, D, E, F \rfloor$

5 objects.

First count total with AB together:

$$5! \cdot 2$$

Now subtract arrangements where C is adjacent to block (X).

Treat (XC) or (CX) as a superblock.

External objects:

$$4$$

arranged in:

$$4!$$

Ways to place C relative to X:

$$2$$

Internal AB:

$$2$$

Bad:

$$4! \cdot 2 \cdot 2$$

Therefore:

$$2(5!) - 4(4!)$$

$$240 - 96$$

$$= \boxed{144}$$

3. THREE PARTICULAR PEOPLE NOT ALL TOGETHER

Suppose 7 people and A,B,C should **not all be together**.

Total:

$$7!$$

Bad case: A,B,C all in one block.

Block + 4 others:

$$5!$$

Internal:

$$3!$$

Bad:

$$5! \cdot 3!$$

Required:

$$\boxed{7! - 5! \cdot 3!}$$

$$= 5040 - 720$$

$$= \boxed{4320}$$

Important:

“Not all together” is different from “no two together.”

4. NO TWO OF A,B,C TOGETHER

This is much stricter.

Suppose 7 people:

- special: A,B,C
- ordinary: D,E,F,G

Arrange ordinary 4:

$4!$

They create:

5

gaps:

$_D_E_F_G_$

Choose 3 distinct gaps for A,B,C:

$\binom{5}{3}$

Arrange A,B,C:

$3!$

Total:

$4!\binom{5}{3}$

$=24 \cdot 10$

$=\boxed{1440}$

5. EXACTLY ONE ADJACENT PAIR AMONG THREE SPECIAL PEOPLE

Suppose A,B,C are among other people.

“Exactly one pair adjacent” is trickier because:

- AB together, C separate
- AC together, B separate
- BC together, A separate

and we must exclude cases where all three become consecutive.

This type is best handled with careful casework and inclusion-exclusion.

For GATE/placement, first identify whether the question truly requires “exactly one pair” or only “at least one pair.”

6. AT LEAST TWO OF A,B,C TOGETHER

This can mean at least one adjacency among them.

Direct counting becomes messy.

Often easier:

$\boxed{\text{Total} - \text{none of A,B,C adjacent}}$

If enough ordinary people exist, use the gap method for the complement.

7. TWO SEPARATE BLOCKS

8 people:

- A,B together
- C,D together

Treat each pair as a block.

Objects:

$\{AB, CD, E, F, G, H\}$

6 objects.

External arrangements:

$6!$

Internal:

$2! \cdot 2!$

Total:

$\boxed{6! \cdot 4 = 2880}$

8. THREE PAIRS TOGETHER

Suppose 8 people and:

- A,B together
- C,D together
- E,F together

Other G,H.

Super-objects:

$\{AB, CD, EF, G, H\}$

5 objects:

$5!$

Each pair has 2 internal orders:

2^3

Total:

$\boxed{5! \cdot 2^3 = 960}$

9. ONE PAIR TOGETHER AND ANOTHER PAIR NOT TOGETHER

Suppose 7 people:

- A,B together
- C,D not together

First force AB block.

Then we have 6 effective objects:

$\{X, C, D, E, F, G\}$

Total with AB together:

$6! \cdot 2$

Among these, count C,D together as another block:

Effective objects:

5

External:

$5!$

CD internal:

2

AB internal:

2

Bad:

$5! \cdot 2 \cdot 2$

Required:

$2(6!) - 4(5!)$

$1440 - 480$

$\boxed{960}$

10. NO TWO GIRLS TOGETHER — GENERAL RECAP

If b distinct boys and g distinct girls:

$\boxed{b! \binom{b+1}{g} g!}$

provided:

$g \leq b+1$

This is one of the most important restricted-permutation formulas.

11. GIRLS MUST ALL BE TOGETHER

If b boys and g girls, all distinct:

Treat all girls as one block.

External objects:

$\{b+1\}$

Arrange:

$\{(b+1)!\}$

Internal girls:

$\{g!\}$

So:

$\{\boxed{(b+1)g!}\}$

12. BOYS AND GIRLS EACH TOGETHER

Treat:

- all boys as one block
- all girls as one block

Arrange the two blocks:

$\{2!\}$

Arrange internally:

$\{b! \cdot g!\}$

Total:

$\{\boxed{2b!g!}\}$

13. EXACT ALTERNATION IN A ROW

If equal groups (n, n) :

Two patterns:

$\{ABAB \cdots\}$

or:

$\{BABA \cdots\}$

Therefore:

$\{\boxed{2(n!)^2}\}$

For group sizes $(n, n-1)$, only one pattern is possible, starting and ending with the larger group.

14. RELATIVE ORDER — POWERFUL SYMMETRY

If (k) particular distinct objects must appear in one specific relative order, then:

$\{\boxed{\frac{n!}{k!}}\}$

Example:

Among 9 people, A before B before C before D.

$$\left[\frac{9!}{4!} \right]$$

No need to know their exact positions.

15. TWO INDEPENDENT RELATIVE-ORDER CONDITIONS

Suppose among 8 people:

$$\left[A < B < C \right]$$

in relative order, and separately:

$$\left[D < E \right]$$

Then among all:

- A,B,C have $(3!)$ possible internal relative orders
- D,E have $(2!)$

Only one of each is allowed.

Thus:

$$\left[\boxed{\frac{8!}{3! \cdot 2!}} \right]$$

provided the conditions do not otherwise interact.

16. REPEATED LETTERS + POSITION RESTRICTION

Consider:

$$\left[\text{BANANA} \right]$$

How many arrangements have B not at the first position?

Total:

$$\left[\frac{6!}{3!2!} = 60 \right]$$

B first:

Remaining:

$$\left[\text{A,A,A,N,N} \right]$$

$$\left[\frac{5!}{3!2!} = 10 \right]$$

Therefore:

$$\left[\boxed{60 - 10 = 50} \right]$$

17. BANANA — STARTS WITH A

Fix one A first.

Remaining letters:

$$\left[\text{A,A,B,N,N} \right]$$

Count:

$$\left[\frac{5!}{2!2!} \right]$$

$$\left[=\boxed{30} \right]$$

Important:

When one identical A is fixed, the remaining A count decreases.

18. BANANA — STARTS AND ENDS WITH A

Fix an A at both ends.

Remaining:

$$\left[A, B, N, N \right]$$

Arrangements:

$$\left[\frac{4!}{2!} \right]$$

$$\left[=\boxed{12} \right]$$

19. BANANA — ALL A's TOGETHER

Treat AAA as one block.

Objects:

$$\left[AAA, B, N, N \right]$$

4 objects with N repeated twice:

$$\left[\frac{4!}{2!} \right]$$

$$\left[=\boxed{12} \right]$$

AAA has only one internal arrangement because A's are identical.

20. WORD WITH VOWELS IN FIXED POSITIONS

Suppose word:

$$\left[MANGO \right]$$

How many arrangements have vowels only in positions 2 and 4?

Vowels A, O:

$$\left[2! \right]$$

Consonants M, N, G in remaining positions:

$$\left[3! \right]$$

Total:

$$\left[\boxed{2!3!=12} \right]$$

21. VOWELS OCCUPY EVEN POSITIONS

For a 6-letter word with 3 distinct vowels and 3 distinct consonants:

Even positions:

$\{ 2, 4, 6 \}$

Arrange vowels:

$\{ 3! \}$

Consonants fill odd positions:

$\{ 3! \}$

Total:

$\boxed{36}$

22. CIRCULAR PERMUTATION — FIX ONE PERSON

For (n) distinct people around a round table:

$\boxed{(n-1)!}$

This handles rotations.

Do not divide again unless reflections are also considered equivalent.

23. CIRCULAR — A AND B TOGETHER

As before:

$\boxed{2(n-2)!}$

24. CIRCULAR — A AND B NOT TOGETHER

$\boxed{(n-1)! - 2(n-2)!}$

Factor:

$\boxed{=(n-2)![(n-1)-2]}$

$\boxed{=(n-3)(n-2)!}$

25. CIRCULAR — THREE PARTICULAR PEOPLE TOGETHER

Treat 3-person block as one object.

Effective circular objects:

$\{ n-2 \}$

Circular arrangements:

$\{ (n-3)! \}$

Internal 3-person arrangements:

$\{ 3! \}$

Thus:

$$\boxed{3!(n-3)!}$$

26. CIRCULAR — A AND B HAVE EXACTLY ONE PERSON BETWEEN

Suppose $(n \geq 4)$.

Fix A.

There are two seats at distance 2 from A:

- one clockwise
- one anticlockwise

So B has:

$$\boxed{2}$$

possible seats.

Arrange remaining $(n-2)$:

$$\boxed{(n-2)!}$$

Thus:

$$\boxed{2(n-2)!}$$

for ordinary labeled round-table seating.

27. CIRCULAR — A AND B OPPOSITE

Only makes sense when (n) is even.

Fix A.

B's opposite seat is fixed.

Arrange remaining:

$$\boxed{(n-2)!}$$

28. CIRCULAR ALTERNATION

For (n) men and (n) women around a round table:

Arrange men circularly:

$$\boxed{(n-1)!}$$

Women occupy the (n) gaps:

$$\boxed{n!}$$

Therefore:

$$\boxed{(n-1)!n!}$$

29. NECKLACE / BRACELET DISTINCTION

For (n) distinct beads on a simple necklace, rotations **and reflections** may be considered identical.

Then, for generic $(n > 2)$:

$$\boxed{\frac{(n-1)!}{2}}$$

But be careful:

- **round-table seating**: reflection is normally different
- **necklace**: flipping may be considered same

This depends on the physical interpretation.

30. EXAMPLE — DISTINCT BEADS

6 distinct beads on a necklace, rotations and reflections considered same:

$$\frac{5!}{2} = \boxed{60}$$

31. DERANGEMENT — DEFINITION

A **derangement** is a permutation in which no object remains in its original position.

Example:

Original:

$$[A, B, C]$$

Derangements:

$$[B, C, A], [C, A, B]$$

So:

$$\boxed{!3=2}$$

The notation is often:

$$!n$$

or:

$$D_n$$

32. SMALL DERANGEMENT VALUES

$$\boxed{D_1=0}$$

$$\boxed{D_2=1}$$

$$\boxed{D_3=2}$$

$$\boxed{D_4=9}$$

$$\boxed{D_5=44}$$

$$\boxed{D_6=265}$$

These are worth recognizing.

33. DERANGEMENT FORMULA

$$\boxed{D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right)}$$

This comes from inclusion-exclusion.

For aptitude, you do not need to rederive it every time.

34. APPROXIMATION SHORTCUT

A famous approximation:

$$\boxed{D_n \approx \frac{n!}{e}}$$

In fact:

$$D_n$$

is the nearest integer to:

$$\frac{n!}{e}$$

Useful for larger (n) .

35. DERANGEMENT EXAMPLE

4 letters placed into 4 addressed envelopes, no letter enters its correct envelope.

Number:

$$D_4 = \boxed{9}$$

Classic derangement problem.

36. EXACTLY k OBJECTS IN CORRECT POSITIONS

Out of (n) objects, exactly (k) remain in their original positions.

Choose which (k) stay correct:

$$\binom{n}{k}$$

The remaining $(n-k)$ must all be deranged:

$$D_{n-k}$$

Thus:

$$\boxed{\binom{n}{k} D_{n-k}}$$

Very useful.

37. EXAMPLE — EXACTLY 2 CORRECT

5 letters into 5 envelopes.

Exactly 2 correct.

Choose correct 2:

$\lfloor \text{binom}{5}{2} = 10 \rfloor$

Derange remaining 3:

$\lfloor D_3 = 2 \rfloor$

Total:

$\lfloor \boxed{20} \rfloor$

38. AT LEAST ONE IN CORRECT POSITION

Complement:

$\lfloor \boxed{n! - D_n} \rfloor$

because total permutations minus derangements.

39. NONE IN ORIGINAL POSITION WITH 4 PEOPLE

Total:

$\lfloor 4! = 24 \rfloor$

Derangements:

$\lfloor 9 \rfloor$

So probability preview:

$\lfloor 9/24 \rfloor$

but probability comes later.

40. LEXICOGRAPHIC / DICTIONARY ORDER

Lexicographic ranking asks:

If all arrangements are written in dictionary order, what is the rank of a particular word?

The method:

1. sort available letters alphabetically
 2. at each position, count how many smaller unused letters could be placed there
 3. multiply by permutations of remaining positions
 4. add 1 at the end
-

41. RANK OF CAB USING A,B,C

Dictionary order:

$\{ ABC, \backslash ACB, \backslash BAC, \backslash BCA, \backslash CAB, \backslash CBA \}$

CAB is:

$\{ \boxed{5^{\text{th}}} \}$

Let's calculate systematically.

42. RANK METHOD — CAB

Word:

$\{ CAB \}$

Letters sorted:

$\{ A, B, C \}$

First letter C.

Smaller available letters:

$\{ A, B \}$

So 2 choices before C.

For each, remaining 2 letters can be arranged:

$\{ 2!=2 \}$

Words before due to first position:

$\{ 2(2)=4 \}$

Now C fixed.

Remaining:

$\{ A, B \}$

Next letter A has no smaller remaining letter.

So total words before:

$\{ 4 \}$

Rank:

$\{ \boxed{4+1=5} \}$

43. RANK OF BDAC USING A,B,C,D

Find rank of:

$\{ BDAC \}$

First letter B

Smaller available:

$\{ A \}$

Count:

$\{ 1 \cdot 3!=6 \}$

Fix B. Remaining A,C,D. Next letter D

Smaller available than D:

$\lfloor A, C \rfloor$

2 choices.

Each leaves 2 letters:

$\lfloor 2! \rfloor$

Count:

$\lfloor 2 \cdot 2! = 4 \rfloor$

Fix BD. Remaining A,C. Next A

No smaller.

Last fixed.

Words before:

$\lfloor 6+4=10 \rfloor$

Rank:

$\lfloor \boxed{11} \rfloor$

44. LEXICOGRAPHIC RANK WITH REPEATED LETTERS

Now counts must divide by factorials of repeated remaining letters.

Example:

Rank of:

$\lfloor BAA \rfloor$

among distinct arrangements of AAB.

Order:

$\lfloor AAB, \backslash ABA, \backslash BAA \rfloor$

Rank:

$\lfloor \boxed{3} \rfloor$

Systematically:

First position B.

Smaller available:

$\lfloor A \rfloor$

If A placed first, remaining:

$\lfloor A, B \rfloor$

distinct arrangements:

$\lfloor 2! = 2 \rfloor$

Thus 2 words precede BAA.

Rank:

$\boxed{3}$

45. REPEATED-LETTER RANK FORMULA IDEA

If remaining multiset has (m) letters with repetitions $(p, q, \dots)$, number of distinct suffix arrangements:

$\boxed{\frac{m!}{p!q!\dots}}$

Use this after every hypothetical smaller-prefix choice.

46. FIND THE k th PERMUTATION— BASIC IDEA

Reverse lexicographic ranking:

Given rank (k) :

1. group permutations by first letter
2. each first-letter block contains $((n-1)!)$ words
3. determine which block contains rank (k)
4. continue recursively

Useful in coding tests too.

47. EXAMPLE — 14th PERMUTATION OF A,B,C,D

Each first-letter block has:

$3!=6$

Ranks:

- A: 1–6
- B: 7–12
- C: 13–18
- D: 19–24

14th begins with C.

Within C block:

$14-12=2$

Need 2nd permutation of A,B,D:

$\boxed{ABD, ADB, BAD, BDA, DAB, DBA}$

2nd is:

$\boxed{ADB}$

So:

$\boxed{CADB}$

48. PERMUTATION EQUATION

Sometimes solve:

$$\{^nP_r = k\}$$

Example:

$$\{^nP_2 = 20\}$$

$$n(n-1) = 20$$

$$n^2 - n - 20 = 0$$

$$(n-5)(n+4) = 0$$

Positive:

$$\boxed{n=5}$$

49. RELATION BETWEEN SUCCESSIVE PERMUTATIONS

$$\{^nP_r = \frac{n!}{(n-r)!}\}$$

Then:

$$\frac{\{^nP_r\}}{\{^nP_{r-1}\}} = n-r+1$$

Therefore:

$$\boxed{\{^nP_r = \{^nP_{r-1}\}(n-r+1)}$$

50. EXAMPLE

If:

$$\{^{10}P_3 = 720\}$$

then:

$$\{^{10}P_4 = 720(10-4+1)$$

$$= 720(7)$$

$$= \boxed{5040}$$

51. RELATION $\{^nP_r\}$ AND $\{^nP_{n-r}\}$

Do not assume symmetry.

Unlike combinations:

$$\{^nP_r \neq \{^nP_{n-r}\}$$

in general.

Example:

$$\{^5P_2 = 20$$

$$\{^5P_3 = 60$$

This will contrast sharply with:

$$\{ {}^nC_r = {}^nC_{n-r} \}$$

in Chapter 40.

52. FACTORIAL TRAILING-ZERO / DIVISIBILITY CONNECTION

Permutation formulas often simplify to factorial ratios.

Use Chapter 6 knowledge:

$$n! = n(n-1) \cdots 1$$

Do not calculate gigantic factorials unless needed.

Factor/cancel first.

53. OPTION ELIMINATION

If asked:

$${}^{12}P_3$$

before calculating fully:

$$12 \cdot 11 \cdot 10$$

must:

- be divisible by 10
- be divisible by 11
- be greater than 1000

Then compute if needed:

$$\boxed{1320}$$

54. GATE-STYLE QUESTION 1

How many arrangements of 8 objects if A,B,C must occur in that relative order?

$$\frac{8!}{3!} = \boxed{6720}$$

55. GATE-STYLE QUESTION 2

5 letters placed randomly into 5 labeled boxes, one per box. How many ways have no letter in its corresponding box?

$$D_5 = \boxed{44}$$

56. GATE-STYLE QUESTION 3

Exactly 3 of 6 objects must remain in original positions.

Choose correct 3:

$$\binom{6}{3} = 20$$

Derange remaining 3:

$$D_3 = 2$$

Total:

$$\boxed{40}$$

57. GATE-STYLE QUESTION 4

7 distinct people sit around a table, A opposite B is impossible because 7 is odd.

So:

$$\boxed{\text{No exactly opposite seat exists}}$$

Important conceptual trap.

58. PLACEMENT-STYLE QUESTION 1

Find rank of CABD among permutations of A,B,C,D.

First C:

smaller A,B:

$$2 \cdot 3! = 12$$

Fix C. Next A: none smaller.

Fix CA. Next B: none smaller.

Rank:

$$12 + 1 = \boxed{13}$$

59. PLACEMENT-STYLE QUESTION 2

6 people around a table. A and B must not sit together.

Total:

$$5! = 120$$

Together:

$$2 \cdot 4! = 48$$

Valid:

$$\boxed{72}$$

60. PLACEMENT-STYLE QUESTION 3

5 boys and 3 girls, all distinct. No two girls adjacent.

Arrange boys:

$\lfloor 5! \rfloor$

Gaps:

$\lfloor 6 \rfloor$

Choose 3:

$\lfloor \text{binom}{6}{3}=20 \rfloor$

Arrange girls:

$\lfloor 3! \rfloor$

Total:

$\lfloor 120 \cdot 20 \cdot 6 \rfloor$

$\lfloor \boxed{14400} \rfloor$

61. PLACEMENT-STYLE QUESTION 4

How many arrangements of BANANA have all A's together?

From above:

$\lfloor \boxed{12} \rfloor$

62. ADVANCED QUESTION — EXACTLY ONE PERSON CORRECT

4 letters into 4 envelopes, exactly 1 correct.

Choose correct one:

$\lfloor \text{binom}{4}{1}=4 \rfloor$

Derange other 3:

$\lfloor D_3=2 \rfloor$

Total:

$\lfloor \boxed{8} \rfloor$

63. ADVANCED QUESTION — AT LEAST ONE CORRECT

4 letters into 4 envelopes.

Total:

$\lfloor 24 \rfloor$

None correct:

$\lfloor D_4=9 \rfloor$

At least one:

$\lfloor 24-9 = \boxed{15} \rfloor$

64. ADVANCED QUESTION — CIRCULAR ONE PERSON BETWEEN

8 people around table. A and B must have exactly one person between them.

Fix A.

B can occupy either of two seats two steps away.

Then arrange remaining 6:

$$\frac{2 \cdot 6!}{2}$$

$$\boxed{1440}$$

65. ADVANCED QUESTION — NECKLACE

7 distinct beads around a necklace, rotations/reflections identical.

$$\frac{(7-1)!}{2} = \frac{720}{2}$$

$$\boxed{360}$$

66. COMMON TRAPS

& pTRAP 1 — “Not all together” vs “none together”

These are completely different restrictions.

& pTRAP 2 — Circular reflection

Do not divide by 2 for ordinary round-table seating.

& pTRAP 3 — Opposite seat

Only meaningful for even number of equally spaced seats.

& pTRAP 4 — Derangement

“None in original position” is not simply:

$$n! - n$$

Use D_n .

& pTRAP 5 — Exactly k correct

The remaining objects must be fully deranged.

Use:

$$\binom{n}{k} D_{n-k}$$

& pTRAP 6 — Lexicographic rank

Rank starts at 1, so add:

$$\boxed{1}$$

after counting all preceding words.

& pTRAP 7 — Repeated letters in ranking

Use distinct multiset permutations, not ordinary factorial.

67. MUST-MEMORIZE DERANGEMENT TABLE

$\boxed{D_1=0, D_2=1, D_3=2, D_4=9, D_5=44, D_6=265}$

Formula:

$\boxed{D_n=n!\sum_{k=0}^n\frac{(-1)^k}{k!}}$

68. EXACTLY k FIXED POINTS

$\boxed{\text{Exactly } k \text{ correct} = \binom{n}{k} D_{n-k}}$

At least one correct:

$\boxed{n! - D_n}$

None correct:

$\boxed{D_n}$

69. CIRCULAR TOOLKIT

$(n!)$ distinct around a table

$\boxed{(n-1)!}$

A,B together

$\boxed{2(n-2)!}$

A,B apart

$\boxed{(n-1)! - 2(n-2)!}$

A,B opposite, $(n!)$ even

$\boxed{(n-2)!}$

$(n!)$ men + $(n!)$ women alternating

$\boxed{(n-1)!n!}$

70. LINEAR RESTRICTION TOOLKIT

k specific distinct objects together among n

$\boxed{(n-k+1)!k!}$

k particular objects in fixed relative order

$$\boxed{\frac{n!}{k!}}$$

no two girls together

$$\boxed{b! \binom{b+1}{g} g!}$$

71. FAST EXAM DECISION TREE

“Arrange”?

Permutation.

Repeated identical objects?

Divide by repetition factorials.

Must be together?

Block.

Must not be together?

Complement or gaps.

Several special objects cannot touch?

Arrange ordinary objects !' use gaps.

Relative order only?

Symmetry / divide by $(k!)$.

Circle?

Fix one person.

Reflections also identical?

Necklace-type problem.

None in original place?

Derangement.

Exactly (k) in original positions?

$$\boxed{\binom{n}{k} D_{n-k}}$$

Dictionary rank?

Count smaller-prefix blocks.

PRACTICE — LEVEL 0

Q1

Find:

$$\boxed{{}^8P_3}$$

Q2

How many arrangements of 7 distinct objects have A before B?

Q3

How many derangements of 4 objects?

Q4

How many ways can 8 people sit around a round table?

Q5

How many arrangements of BANANA begin with A?

LEVEL 1

Q6

8 people in a row. A,B,C must remain together.

Q7

5 boys and 3 girls are arranged so no two girls are adjacent.

Q8

6 people around a table. A and B must sit together.

Q9

5 letters go into 5 addressed envelopes. Exactly 2 go correctly. Count.

Q10

Find rank of CAB among A,B,C permutations.

LEVEL 2

Q11

7 people in a row. A,B together and C,D not together.

Q12

4 men and 4 women around a circular table, alternating.

Q13

How many arrangements of BANANA have B not first?

Q14

6 objects: exactly 3 remain in original positions. Count.

Q15

8 people around a circular table. A and B have exactly one person between them.

LEVEL 3 — EXAM PRESSURE

Q16

8 people in a row. A,B,C are pairwise nonadjacent. Count.

Q17

Find rank of BDAC among A,B,C,D permutations.

Q18

7 distinct beads form a necklace. Rotations and reflections are identical. Count.

Q19

6 letters go into 6 addressed envelopes. At least one goes correctly. Count.

Q20

9 people in a row. A before B before C, and D before E. Count.

ANSWERS

Q1

$$8 \cdot 7 \cdot 6 = \boxed{336}$$

Q2

Half of all permutations:

$$\frac{7!}{2} = \boxed{2520}$$

Q3

$$\boxed{D_4=9}$$

Q4

$$(8-1)! = \boxed{5040}$$

Q5

Fix one A first.

Remaining:

$$A, A, B, N, N$$

$$\frac{5!}{2!2!} = \boxed{30}$$

Q6

ABC block + 5 others:

$$6!$$

Internal:

$$3!$$

$$6! \cdot 3! = 720 \cdot 6$$

$$\boxed{4320}$$

Q7

$$5! \binom{6}{3}$$

$$= 120 \cdot 20$$

$$\boxed{2400}$$

Q8

$$2(6-2)! = 2(4)! = \boxed{48}$$

Q9

Choose 2 correct:

$$\binom{5}{2} = 10$$

Derange remaining 3:

$$2$$

Total:

$$\boxed{20}$$

Q10

$$\boxed{5}$$

Q11

Force AB together:

$$2 \cdot 6!$$

Subtract cases where CD also together:

$$2 \cdot 2 \cdot 5!$$

Thus:

$$1440 - 480 = \boxed{960}$$

Q12

$$3! \cdot 4! = 6 \cdot 24 = \boxed{144}$$

Q13

Total:

$$60$$

B first:

$$10$$

Valid:

$$\boxed{50}$$

Q14

$$\binom{6}{3} \cdot 2 = 20$$

$$\boxed{40}$$

Q15

Fix A.

B has:

$$2$$

possible seats.

Arrange remaining 6:

$$6!$$

$$2 \cdot 720 = \boxed{1440}$$

Q16

Special:

$$A, B, C$$

Ordinary other 5.

Arrange 5 ordinary:

$$5!$$

Gaps:

$$6$$

Choose 3 gaps:

$$\binom{6}{3}$$

Arrange A, B, C:

$$3!$$

Total:

$$5! \cdot \binom{6}{3} \cdot 3!$$

$$= 120 \cdot 20 \cdot 6$$

$$= \boxed{14400}$$

Q17

From earlier:

$$\boxed{11}$$

Q18

$$\frac{6!}{2} = \boxed{360}$$

Q19

Total permutations:

$$6! = 720$$

Derangements:

$$D_6 = 265$$

At least one correct:

$$\lfloor 720-265 \rfloor$$

$$\lfloor =\boxed{455} \rfloor$$

Q20

Total:

$$\lfloor 9! \rfloor$$

A,B,C fixed relative order gives factor:

$$\lfloor 3! \rfloor$$

D,E fixed relative order gives factor:

$$\lfloor 2! \rfloor$$

Therefore:

$$\lfloor \frac{9!}{3!2!} \rfloor$$

$$\lfloor =\frac{362880}{12} \rfloor$$

$$\lfloor =\boxed{30240} \rfloor$$

' CHAPTER 39 COMPLETE

We have now covered:

- factorials '
- factorial cancellation '
- $\lfloor (nP_r) \rfloor$ '
- ordered arrangements '
- repetition allowed/forbidden '
- identical-object permutations '
- word arrangements '
- fixed positions '
- block method '
- multiple blocks '
- together/apart restrictions '
- gap method '
- alternating arrangements '
- relative-order restrictions '
- repeated-letter restrictions '
- digit permutations '
- circular permutations '
- circular adjacency/opposite/distance restrictions '
- necklace vs round-table distinction '
- derangements '
- exactly $\lfloor (k) \rfloor$ fixed positions '

- lexicographic rank '
- kth permutation intuition '
- GATE/placement L0–L3 practice '

Progress

Completed: 39/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 40 — Combinations

Next we will cover:

- permutation vs combination
- $\binom{n}{r}$
- why order disappears
- relation between $\binom{n}{r}$ and $\binom{n}{n-r}$
- symmetry $\binom{n}{r} = \binom{n}{n-r}$
- Pascal identities
- committee/team selection
- compulsory/excluded members
- at least/at most selection
- men/women group restrictions
- choosing pairs/groups
- complementary selections
- distribution/counting applications
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 40 — COMBINATIONS

PART 1 — $\binom{n}{r}$, SELECTIONS, SYMMETRY, COMMITTEES, COMPULSORY/EXCLUDED MEMBERS & BASIC RESTRICTIONS

Status: 0% PROGRESS GATE: High Placements: High Priority: 0% Per 1

Permutation asked:

In how many ways can objects be arranged?

Combination asks:

In how many ways can objects be selected when order does not matter?

This distinction is the heart of the chapter.

1. PERMUTATION VS COMBINATION

Suppose we select 2 people from A, B, C.

Possible groups:

$\{ AB, AC, BC \}$

There are:

$\boxed{3}$

Notice:

$AB = BA$

because the same two people were selected.

So order does **not** matter.

This is a combination.

2. WHEN ORDER MATTERS

Suppose we choose:

- one president
- one vice-president

from A, B, C.

Now:

(A, B)

means A president, B vice-president.

But:

(B, A)

is different.

So order matters !' permutation.

3. MASTER LANGUAGE TEST

Usually permutation

Words such as:

- arrange
- order

- rank
- seat
- code
- assign roles
- first/second/third
- president/secretary

Usually combination

Words such as:

- choose
- select
- committee
- team
- group
- subset
- shortlist

Always check whether changing order creates a genuinely new outcome.

4. COMBINATION FORMULA

Number of ways to choose $\binom{n}{r}$ objects from $\binom{n}{n}$ distinct objects:

$$\boxed{{}^nC_r = \frac{n!}{r!(n-r)!}}$$

Also written:

$$\boxed{\text{binom } nr}$$

MUST MEMORIZE

5. WHY DIVIDE BY $(r!)$?

Suppose choose 3 people from 5.

If we use permutations:

$${}^5P_3 = 5 \cdot 4 \cdot 3 = 60$$

But every selected group of 3 is counted:

$$3! = 6$$

times because:

$$[ABC, ACB, BAC, BCA, CAB, CBA]$$

all represent the same group.

Therefore:

$${}^5C_3 = \frac{{}^5P_3}{3!}$$

$$= \frac{60}{6}$$

$$\boxed{10}$$

6. RELATION BETWEEN PERMUTATION AND COMBINATION

$$\boxed{{}^nP_r = {}^nC_r \cdot r!}$$

Therefore:

$$\boxed{{}^nC_r = \frac{{}^nP_r}{r!}}$$

Interpretation:

1. choose r objects
2. arrange those r

So:

$$\boxed{\text{Choose} \times \text{Arrange}}$$

7. BASIC EXAMPLE

Choose 3 students from 8.

$${}^8C_3 = \frac{8!}{3!5!}$$

Cancel:

$$= \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1}$$

$$= \boxed{56}$$

8. DO NOT EXPAND FULL FACTORIALS

For:

$${}^{12}C_3$$

write:

$$\frac{12 \cdot 11 \cdot 10}{3 \cdot 2 \cdot 1}$$

not:

$$\frac{12!}{3!9!}$$

with giant factorial calculations.

Simplify:

$$= 220$$

$$\boxed{220}$$

9. SPECIAL VALUES

$$\boxed{{}^nC_0 = 1}$$

because there is exactly one way to choose nothing.

Also:

$$\boxed{{}^nC_{n=1}}$$

because there is exactly one way to choose all $\binom{n}{n}$.

And:

$$\boxed{{}^nC_{1=n}}$$

10. SYMMETRY PROPERTY

One of the most useful combination identities:

$$\boxed{{}^nC_r = {}^nC_{n-r}}$$

Why?

Choosing $\binom{n}{r}$ objects to include is equivalent to choosing:

$$\binom{n-r}{}$$

objects to exclude.

11. EXAMPLE

$$\boxed{{}^{10}C_8}$$

Instead of selecting 8, choose the 2 excluded:

$$\boxed{{}^{10}C_2 = \frac{10 \cdot 9}{2}}$$

$$\boxed{=45}$$

Much faster.

12. FAST RULE

When computing:

$$\boxed{{}^nC_r}$$

replace $\binom{n}{r}$ by:

$$\boxed{\min(r, n-r)}$$

before calculating.

Example:

$$\boxed{{}^{20}C_{18} = {}^{20}C_2 = 190}$$

13. CHOOSING A PAIR

Number of unordered pairs from $\binom{n}{n}$ objects:

$$\boxed{{}^nC_2 = \frac{n(n-1)}{2}}$$

This appears constantly.

Examples:

- handshakes
 - edges in complete graph
 - pairs of students
 - matches when every pair plays once
-

14. EXAMPLE — HANDSHAKES

10 people each shake hands with every other person once.

Each handshake corresponds to choosing a pair.

$$\{ {}^{10}C_2 = \frac{10 \cdot 9}{2} \}$$

$$= \boxed{45}$$

15. WHY NOT (10×9) ?

Because handshake:

A with B

is the same as:

B with A.

So (10×9) counts each handshake twice.

16. COMPLETE GRAPH CONNECTION

A complete graph with (n) vertices has one edge between every pair.

Thus number of edges:

$$\boxed{{}^nC_2 = \frac{n(n-1)}{2}}$$

This is useful in GATE discrete mathematics too.

17. NUMBER OF TRIANGLES FROM n POINTS

If no 3 points are collinear, choosing any 3 points forms a triangle.

Therefore:

$$\boxed{{}^nC_3}$$

Example:

8 points:

$$\{ {}^8C_3 = \boxed{56} \}$$

18. NUMBER OF LINE SEGMENTS

Given (n) points, each pair determines a segment.

So:

$$\boxed{{}^nC_2}$$

provided points are distinct.

19. COMMITTEE SELECTION

Choose a committee of 4 from 10 people.

$${}^{10}C_4$$

$$= \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1}$$

$$= \boxed{210}$$

20. ONE PARTICULAR PERSON MUST BE INCLUDED

Choose 4 people from 10, with A compulsory.

A is already chosen.

Need:

$$3$$

more from remaining 9.

Thus:

$$\boxed{{}^9C_3}$$

$$= \boxed{84}$$

21. TWO PARTICULAR PEOPLE MUST BE INCLUDED

Choose 5 from 12, with A and B compulsory.

Already chosen:

$$2$$

Need:

$$3$$

from remaining:

$$10$$

Therefore:

$$\boxed{{}^{10}C_3 = 120}$$

22. ONE PARTICULAR PERSON MUST BE EXCLUDED

Choose 4 from 10, but A cannot be selected.

Then choose 4 from remaining 9:

$$\boxed{{}^9C_4=126}$$

23. TWO PEOPLE EXCLUDED

Choose 5 from 12, but A and B excluded.

Available:

$$\boxed{10}$$

Choose:

$$\boxed{5}$$

$$\boxed{{}^{10}C_5=252}$$

24. A MUST BE INCLUDED, B MUST BE EXCLUDED

Choose 4 from 10.

A fixed in.

B removed.

Now:

- A occupies 1 seat
- remaining eligible people = 8
- need 3 more

Thus:

$$\boxed{{}^8C_3=56}$$

25. AT LEAST ONE OF A,B

Choose 4 from 10, and committee must contain at least one of A,B.

Best method:

$$\boxed{\text{Total} - \text{neither A nor B}}$$

Total:

$$\boxed{{}^{10}C_4}$$

Neither:

$$\boxed{{}^8C_4}$$

Thus:

$$\boxed{{}^{10}C_4 - {}^8C_4}$$

$$\boxed{=210-70}$$

$$\boxed{=140}$$

26. DIRECT METHOD FOR AT LEAST ONE OF A,B

Cases:

Exactly one of A,B

Choose which:

$$\binom{10}{2} \binom{8}{1}$$

Then choose remaining 3 from other 8:

$$\binom{8}{3}$$

Count:

$$2 \binom{10}{2} = 112$$

Both A,B

Choose remaining 2 from 8:

$$\binom{8}{2} = 28$$

Total:

$$112 + 28 = 140$$

Same answer.

Complement was faster.

27. EXACTLY ONE OF A,B

Choose one of A,B:

$$\binom{10}{2} = 45$$

Then choose remaining committee members.

For committee size 4 from 10:

Need 3 from other 8:

$$\binom{8}{3}$$

Thus:

$$2 \binom{10}{2} \binom{8}{3} = 112$$

28. BOTH OR NEITHER

Committee size 4 from 10.

Both

$$\binom{10}{2}$$

Neither

$$\binom{10}{4}$$

Total:

$$\boxed{{}^8C_2 + {}^8C_4}$$

$$= 28 + 70$$

$$\boxed{98}$$

29. A AND B CANNOT BOTH BE SELECTED

Choose 4 from 10.

Total:

$${}^{10}C_4$$

Subtract committees containing both:

$8C_2$

Thus:

$$210 - 28 = \boxed{182}$$

Note:

They may include A alone, B alone, or neither.

Only “both together” is forbidden.

30. MEN AND WOMEN SELECTION

Suppose there are:

- 6 men
- 5 women

Choose a team of 4 with exactly 2 men and 2 women.

Choose men:

$6C_2$

Choose women:

$5C_2$

Since both selections are required:

$$\boxed{{}^6C_2 {}^5C_2}$$

$$= 15(10)$$

$$\boxed{150}$$

31. EXACTLY 3 MEN

From 6 men and 5 women, choose 4 with exactly 3 men.

Men:

$6C_3$

Women:

$$\{ {}^5C_1 \}$$

Total:

$$\{ 20(5) = \boxed{100} \}$$

32. AT LEAST 1 WOMAN

Choose 4 from 6 men and 5 women.

Total people:

$$\{ 11 \}$$

Total teams:

$$\{ {}^{11}C_4 \}$$

Subtract all-men teams:

$$\{ {}^6C_4 \}$$

Therefore:

$$\{ \boxed{{}^{11}C_4 - {}^6C_4} \}$$

$$\{ =330-15 \}$$

$$\{ =\boxed{315} \}$$

33. AT LEAST 2 WOMEN

Team size 4.

Possible woman counts:

$$\{ 2,3,4 \}$$

Direct:

$$\{ {}^5C_2 {}^6C_2 + {}^5C_3 {}^6C_1 + {}^5C_4 {}^6C_0 \}$$

$$\{ =10(15)+10(6)+5(1) \}$$

$$\{ =150+60+5 \}$$

$$\{ =\boxed{215} \}$$

34. COMPLEMENT FOR AT LEAST 2 WOMEN

Could also use:

$$\{ \text{Total} - \text{0 women} - \text{1 woman} \}$$

$$\{ {}^{11}C_4 - {}^6C_4 - {}^5C_1 {}^6C_3 \}$$

$$\{ 330-15-5(20) \}$$

$$\{ =330-115 \}$$

$$\{ =\boxed{215} \}$$

Choose whichever side has fewer cases.

35. AT MOST 1 WOMAN

Means woman count:

$\lceil 0 \text{ or } 1 \rceil$

Thus:

$\{ {}^6C_4 + {}^5C_1 {}^6C_3 \}$

$= 15 + 100$

$= \boxed{115}$

36. AT LEAST / AT MOST LANGUAGE

Exactly $\lceil k \rceil$

Only $\lceil k \rceil$.

At least $\lceil k \rceil$

$\lceil k, k+1, k+2, \dots \rceil$

At most $\lceil k \rceil$

$\lceil 0, 1, \dots, k \rceil$

A huge number of errors come from misreading these phrases.

37. CHOOSING FROM MULTIPLE CATEGORIES

Suppose:

- 5 engineers
- 4 designers
- 3 managers

Choose team with:

- 2 engineers
- 1 designer
- 1 manager

Count:

$\boxed{{}^5C_2 {}^4C_1 {}^3C_1}$

$= 10 \cdot 4 \cdot 3$

$= \boxed{120}$

38. ONE FROM EACH CATEGORY

Suppose:

- 4 maths books
- 3 physics books
- 5 CS books

Choose exactly one book from each.

Since each category contributes one:

$$4 \cdot 3 \cdot 5 = 60$$

or:

$${}^4C_1 {}^3C_1 {}^5C_1$$

Same result.

39. CHOOSE 2 FROM ONE CATEGORY, 3 FROM ANOTHER

From 7 men, 6 women, form team with 2 men and 3 women:

$$\boxed{{}^7C_2 {}^6C_3}$$

$$= 21(20)$$

$$= \boxed{420}$$

40. CHOOSE A TEAM, THEN ASSIGN A CAPTAIN

This combines combination + permutation thinking.

Suppose choose 4 players from 10, then designate one captain among selected 4.

Choose team:

$${}^{10}C_4$$

Choose captain:

$$4$$

Total:

$${}^{10}C_4 \cdot 4$$

$$= 210(4)$$

$$= \boxed{840}$$

41. FASTER EQUIVALENT METHOD

Choose captain first:

$$10$$

Then choose 3 teammates from remaining 9:

$9C_3$

Total:

$${}^{10}C_3$$

$$=10(84)$$

$$=\boxed{840}$$

Same answer.

This gives an important identity:

$$\boxed{{}^nC_r \cdot {}^nC_{n-r} = {}^nC_n}$$

42. TEAM WITH CAPTAIN AND VICE-CAPTAIN

Choose 5-person team from 12, then select captain and vice-captain from team.

Method 1:

$${}^{12}C_5 \cdot {}^5P_2$$

Method 2:

Choose captain:

$${}^{12}P_1$$

vice-captain:

$${}^{11}P_1$$

then remaining 3 team members from remaining 10:

$${}^{10}C_3$$

So:

$$\boxed{12 \cdot 11 \cdot {}^{10}C_3}$$

Both equal.

43. CHOOSE A PAIR FROM EACH GROUP

Suppose:

- 5 boys
- 4 girls

Choose 2 boys and 2 girls.

$${}^5C_2 \cdot {}^4C_2$$

$$=10(6)$$

$$=\boxed{60}$$

44. COMPLEMENTARY SELECTION

Choosing 3 people from 8 is equivalent to excluding 5.

Thus:

$${}^8C_3 = {}^8C_5$$

This becomes extremely useful when $\binom{n}{r}$ is close to $\binom{n}{n-r}$.

45. PASCAL IDENTITY

A fundamental identity:

$$\boxed{\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}}$$

Intuition

Suppose choose $\binom{n}{r}$ people from $\binom{n}{n}$, and focus on person A.

Every valid group either:

- excludes A !' choose all $\binom{n-1}{r}$ from remaining $\binom{n-1}{n-1}$
- includes A !' choose remaining $\binom{n-1}{r-1}$ from other $\binom{n-1}{n-1}$

So:

$$\binom{n-1}{r} + \binom{n-1}{r-1}$$

46. EXAMPLE

$$\binom{10}{4}$$

Using Pascal:

$$\binom{9}{4} + \binom{9}{3}$$

$$= 126 + 84$$

$$= \boxed{210}$$

47. PASCAL TRIANGLE

Rows:

$$1$$

$$1 \quad 1$$

$$1 \quad 2 \quad 1$$

$$1 \quad 3 \quad 3 \quad 1$$

$$1 \quad 4 \quad 6 \quad 4 \quad 1$$

$$1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1$$

Each interior value is sum of two above.

These are combination coefficients.

48. SUM OF ALL COMBINATIONS

A set with $\binom{n}{n}$ elements has:

$$2^n$$

subsets.

But subsets can have sizes:

$$\{0, 1, 2, \dots, n\}$$

So:

$$\sum_{r=0}^n \binom{n}{r} = 2^n$$

MUST KNOW

49. NONEMPTY SUBSETS

Remove the empty subset:

$$\sum_{r=1}^n \binom{n}{r} = 2^n - 1$$

50. EVEN-SIZED AND ODD-SIZED SUBSETS

For $n \geq 1$:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots = 2^{n-1}$$

and:

$$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots = 2^{n-1}$$

Exactly half the subsets have even size and half odd size.

51. EXAMPLE

For a 5-element set:

Total subsets:

$$2^5 = 32$$

Even-sized subsets:

$$16$$

Odd-sized:

$$16$$

$$\{16, 16\}$$

52. NUMBER OF SELECTIONS OF AT LEAST ONE OBJECT

From n distinct objects:

For each object:

- selected
- not selected

Total subsets:

$$\lfloor 2^n \rfloor$$

Exclude empty:

$$\lfloor \boxed{2^{n-1}} \rfloor$$

53. SELECTION WITH A PARTICULAR OBJECT INCLUDED

Number of subsets of an (n) -element set containing A:

A fixed included.

Each other $(n-1)$ element may be in or out.

Thus:

$$\lfloor \boxed{2^{n-1}} \rfloor$$

54. SELECTION WITH A PARTICULAR OBJECT EXCLUDED

Same logic:

A fixed excluded.

Remaining $(n-1)$:

$$\lfloor \boxed{2^{n-1}} \rfloor$$

55. AT LEAST ONE OF TWO PARTICULAR OBJECTS IN ANY-SIZE SUBSET

Total subsets:

$$\lfloor 2^n \rfloor$$

Subsets containing neither A nor B:

$$\lfloor 2^{n-2} \rfloor$$

Hence:

$$\lfloor \boxed{2^n - 2^{n-2}} \rfloor$$

If empty/nonempty requirements differ, adjust carefully.

56. SELECTION WITH EXACTLY ONE OF A,B

Choose which one:

$$\lfloor 2 \rfloor$$

For every other $(n-2)$ object:

include/exclude independently.

So:

$$\lfloor \boxed{2 \cdot 2^{n-2} = 2^{n-1}} \rfloor$$

57. NUMBER OF DIAGONALS OF A POLYGON

An n -vertex polygon has:

$$\binom{n}{2}$$

pairs of vertices.

But n pairs are sides.

Therefore diagonals:

$$\boxed{\binom{n}{2} - n}$$

Simplify:

$$\frac{n(n-1)}{2} - n$$

$$\boxed{\frac{n(n-3)}{2}}$$

MUST KNOW

58. EXAMPLE

Number of diagonals in a decagon:

$$\frac{10(7)}{2}$$

$$\boxed{35}$$

59. NUMBER OF TRIANGLES FROM POLYGON VERTICES

Any 3 vertices form a triangle.

So:

$$\boxed{\binom{n}{3}}$$

Example hexagon:

$$\binom{6}{3} = \boxed{20}$$

60. HANDSHAKE / MATCHES / PAIRS MASTER PATTERN

Whenever every pair interacts exactly once:

$$\boxed{\binom{n}{2}}$$

Examples:

- handshakes
 - round-robin games
 - pairwise comparisons
 - pair connections
 - complete graph edges
-

61. GATE-STYLE QUESTION 1

Choose 4 elements from a set of 9.

$$\lfloor {}^9C_4 = \boxed{126} \rfloor$$

62. GATE-STYLE QUESTION 2

A 6-element set has how many 3-element subsets?

$$\lfloor {}^6C_3 = \boxed{20} \rfloor$$

63. GATE-STYLE QUESTION 3

How many nonempty subsets does an 8-element set have?

$$\lfloor 2^8 - 1 \rfloor$$

$$\lfloor = \boxed{255} \rfloor$$

64. GATE-STYLE QUESTION 4

How many edges in complete graph (K_{12}) ?

$$\lfloor {}^{12}C_2 = \frac{12 \cdot 11}{2} \rfloor$$

$$\lfloor = \boxed{66} \rfloor$$

65. PLACEMENT-STYLE QUESTION 1

Choose a 5-person committee from 12, including A.

A fixed.

Choose 4 from remaining 11:

$$\lfloor {}^{11}C_4 \rfloor$$

$$\lfloor = \boxed{330} \rfloor$$

66. PLACEMENT-STYLE QUESTION 2

Choose 4 from 10, excluding A and B.

Choose 4 from remaining 8:

$$\lfloor {}^8C_4 = \boxed{70} \rfloor$$

67. PLACEMENT-STYLE QUESTION 3

6 men and 4 women. Choose 4 with exactly 2 women.

$$\lfloor {}^4C_2 {}^6C_2 \rfloor$$

$$\frac{1}{6}(15)$$

$$\boxed{90}$$

68. PLACEMENT-STYLE QUESTION 4

7 men, 5 women. Choose a team of 5 with at least 1 woman.

Total:

$${}^{12}C_5$$

Subtract all-men:

$7C_5$

$$792 - 21$$

$$\boxed{771}$$

69. ADVANCED QUESTION — EXACTLY ONE OF A,B,C

Choose 5 people from 12, with exactly one of A,B,C.

Choose 1 of special 3:

$3C_1$

Need remaining 4 from the other 9:

$9C_4$

Total:

$$3(126)$$

$$\boxed{378}$$

70. ADVANCED QUESTION — AT LEAST ONE OF A,B,C

Choose 5 from 12.

Complement: none of A,B,C.

Total:

$${}^{12}C_5$$

None:

$9C_5$

Thus:

$$792 - 126$$

$$\boxed{666}$$

71. ADVANCED QUESTION — AT LEAST TWO OF A,B,C

Choose 5 from 12.

Cases:

Exactly 2 special

$$\{ {}^3C_2 {}^9C_3 \}$$

Exactly 3 special

$$\{ {}^3C_3 {}^9C_2 \}$$

Total:

$$\{ 3(84)+1(36) \}$$

$$\{ =252+36 \}$$

$$\{ =\boxed{288} \}$$

72. ADVANCED QUESTION — TEAM WITH MAJORITY WOMEN

6 men, 7 women, choose 5.

Majority women means:

- 3 women
- 4 women
- 5 women

Count:

$$\{ {}^7C_3 {}^6C_2 + {}^7C_4 {}^6C_1 + {}^7C_5 {}^6C_0 \}$$

$$\{ =35(15)+35(6)+21 \}$$

$$\{ =525+210+21 \}$$

$$\{ =\boxed{756} \}$$

73. COMMON TRAPS

& pTRAP 1 — Using permutation for committee

A committee has no internal order unless roles are assigned.

& pTRAP 2 — Forgetting compulsory people reduce both pool and required count

If A is already selected, reduce team size by 1.

& pTRAP 3 — At least one

Often easier:

$$\{ \text{Total} - \text{none} \}$$

& pTRAP 4 — Exactly one of A,B

Do not count cases containing both.

& pTRAP 5 — Men/women selection

Choose each category separately, then multiply.

& pTRAP 6 — $\binom{n}{r}$, not $P(n, r)$

Symmetry belongs to combinations.

& pTRAP 7 — “Choose team and captain”

Team selection alone is not enough; captain creates an ordered role.

74. MUST-MEMORIZE FORMULAS

Combination

$$\boxed{\binom{n}{r} = \frac{n!}{r!(n-r)!}}$$

Symmetry

$$\boxed{\binom{n}{r} = \binom{n}{n-r}}$$

Relation to permutation

$$\boxed{P(n, r) = \binom{n}{r} r!}$$

75. MUST-MEMORIZE IDENTITIES

$$\boxed{\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}}$$

$$\boxed{\sum_{r=0}^n \binom{n}{r} = 2^n}$$

For $n \geq 1$:

$$\boxed{\sum_{\text{even } r} \binom{n}{r} = 2^{n-1}}$$

$$\boxed{\sum_{\text{odd } r} \binom{n}{r} = 2^{n-1}}$$

76. MUST-MEMORIZE PAIR FORMULA

$$\boxed{\binom{n}{2} = \frac{n(n-1)}{2}}$$

Applications:

- handshakes
 - graph edges
 - pairings of one pair
 - games in round robin
-

77. POLYGON FORMULA

Number of diagonals:

$$\boxed{\frac{n(n-3)}{2}}$$

78. FAST EXAM DECISION TREE

Order matters?

Permutation.

Only selection?

Combination.

r close to n ?

Use:

$$nC_r = nC_{n-r}$$

Particular person compulsory?

Fix them first.

Particular person excluded?

Remove them from pool.

At least one?

Try complement.

Exactly k from category?

Choose k from category and remaining from others.

Several categories?

Multiply category-wise choices.

Team then roles?

Combination $\times$ permutation.

Pair interaction?

Try nC_2 .

PRACTICE — LEVEL 0

Q1

Find:

$8C_2$

Q2

Choose 3 students from 7.

Q3

How many pairs from 12 people?

Q4

How many nonempty subsets of a 6-element set?

Q5

Find:

$$\sum_{k=0}^{10} \binom{10}{k} C_8$$

LEVEL 1

Q6

Choose 5 people from 10 if A must be selected.

Q7

Choose 4 people from 9 if A cannot be selected.

Q8

Choose 4 people from 10 with exactly 2 women, given 6 men and 4 women.

Q9

How many 4-person committees from 10 contain at least one of A,B?

Q10

How many diagonals does a 12-sided polygon have?

LEVEL 2

Q11

Choose 5 people from 12 with exactly one of A,B,C.

Q12

7 men and 5 women: choose 5 with at least 2 women.

Q13

Choose a 4-member team from 9 and then select one captain.

Q14

How many even-sized subsets does a 7-element set have?

Q15

How many triangles can be formed from 10 points if no 3 are collinear?

LEVEL 3 — EXAM PRESSURE

Q16

Choose 6 people from 14 such that A and B are either both selected or both excluded.

Q17

8 men and 6 women: choose 5 with more women than men.

Q18

Choose 5 people from 12 such that at least two of A,B,C are selected.

Q19

A team of 6 is selected from 15 people, then captain and vice-captain are chosen from the team. Find total possibilities.

Q20

A 9-element set has how many subsets containing a particular element A but not another particular element B?

ANSWERS**Q1**

$$\lfloor {}^8C_2 = \frac{8 \cdot 7}{2} = \boxed{28} \rfloor$$

Q2

$$\lfloor {}^7C_3 = \frac{7 \cdot 6 \cdot 5}{6} = \boxed{35} \rfloor$$

Q3

$$\lfloor {}^{12}C_2 = \boxed{66} \rfloor$$

Q4

$$\lfloor 2^6 - 1 = \boxed{63} \rfloor$$

Q5

$$\lfloor {}^{10}C_8 = {}^{10}C_2 = \boxed{45} \rfloor$$

Q6

A fixed.

Choose remaining 4 from 9:

$$\lfloor {}^9C_4 = \boxed{126} \rfloor$$

Q7

Remove A.

Choose 4 from remaining 8:

$$\lfloor {}^8C_4 = \boxed{70} \rfloor$$

Q8

$$\lfloor {}^4C_2 {}^6C_2 = 6(15) \rfloor$$

$$\lfloor = \boxed{90} \rfloor$$

Q9

Total:

$$\lfloor {}^{10}C_4 = 210 \rfloor$$

Neither A nor B:

$$\{^8C_4=70\}$$

Therefore:

$$\boxed{140}$$

Q10

$$\frac{12(12-3)}{2} = \frac{108}{2}$$

$$= \boxed{54}$$

Q11

Choose exactly 1 special:

$$\{^3C_1\}$$

Then 4 from other 9:

$$\{^9C_4\}$$

$$3(126) = \boxed{378}$$

Q12

Total team = 5.

At least 2 women:

2 women, 3 men

$$\{^5C_2\} \{^7C_3\} = 10(35)=350$$

3 women, 2 men

$$\{^5C_3\} \{^7C_2\} = 10(21)=210$$

4 women, 1 man

$$\{^5C_4\} \{^7C_1\} = 5(7)=35$$

5 women

$$\{^5C_5=1\}$$

Total:

$$350+210+35+1$$

$$= \boxed{596}$$

Q13

Choose team:

$$\{^9C_4=126\}$$

Captain:

$$4$$

Total:

$$126(4) = \boxed{504}$$

Q14

For nonempty $(n=7)$, exactly half subsets are even-sized:

$$2^{7-1} = \boxed{64}$$

This includes empty set because empty set has size 0, which is even.

Q15

$${}^{10}C_3 = \boxed{120}$$

Q16

Choose 6 from 14.

Case 1: both A,B selected

Need 4 from remaining 12:

$${}^{12}C_4 = 495$$

Case 2: both excluded

Choose all 6 from remaining 12:

$${}^{12}C_6 = 924$$

Total:

$$495 + 924 = \boxed{1419}$$

Q17

Team size 5.

More women than men means:

- 3 women, 2 men
- 4 women, 1 man
- 5 women, 0 men

Count:

$${}^6C_3 {}^8C_2 + {}^6C_4 {}^8C_1 + {}^6C_5$$

$$= 20(28) + 15(8) + 6$$

$$= 560 + 120 + 6$$

$$= \boxed{686}$$

Q18

At least 2 of A,B,C:

Exactly 2:

$${}^3C_2 {}^9C_3 = 3(84) = 252$$

Exactly 3:

$${}^3C_3 {}^9C_2 = 36$$

Total:

$$\boxed{288}$$

Q19

Choose team:

$$\left[{}^{15}C_6 \right]$$

Choose ordered captain/vice-captain from 6:

$$\left[{}^6P_{2=30} \right]$$

Total:

$$\left[{}^{15}C_6 \cdot 30 \right]$$

$$\left[5005 \cdot 30 \right]$$

$$\left[=\boxed{150150} \right]$$

Q20

A must be included.

B must be excluded.

Remaining 7 elements are independently:

- included
- excluded

Thus:

$$\left[2^7 = \boxed{128} \right]$$

CHAPTER 40 STATUS

Covered so far:

- permutation vs combination '
- $({}^nC_r)$ formula '
- relation $({}^nP_r = {}^nC_r \cdot r!)$ '
- symmetry $({}^nC_r = {}^nC_{n-r})$ '
- compulsory/excluded members '
- exactly/at least/at most restrictions '
- men/women and category selection '
- team + role assignment '
- pair counting '
- handshakes and graph edges '
- polygon diagonals '
- subset counting '
- Pascal identity '
- sum of combination coefficients '
- odd/even subset identities '
- GATE/placement practice '

But **Chapter 40 is not complete yet.**

Next we still need:

- advanced $\binom{n}{r}$ identities
- Vandermonde-type reasoning
- committee restrictions with incompatible members
- distribution by selection
- grouping/partitioning people
- choosing pairs repeatedly
- selecting non-consecutive objects
- stars-and-bars preview where appropriate
- combination equations and shortcut evaluation
- final chapter audit

Progress

Completed: 39/76 Current: 0% Chapter 40 — Combinations

NEXT

0% Chapter 40 Part 2 — Advanced combination identities, grouping, non-consecutive selections & restricted committees

USER

next

ASSISTANT

CHAPTER 40 — COMBINATIONS

PART 2 — ADVANCED IDENTITIES, RESTRICTED COMMITTEES, GROUPING, PAIRING, NON-CONSECUTIVE SELECTIONS & EXAM-SPEED SHORTCUTS

Status: 0% IN PROGRESS GATE: High Placements: High Priority: 0% Per 1

Part 1 covered the basic idea:

$\boxed{\text{Combination} = \text{selection where order does not matter}}$

Now we finish the chapter with the higher-value patterns that appear in GATE and placement questions.

1. COMBINATION FORMULA RECAP

$\boxed{{}^nC_r = \frac{n!}{r!(n-r)!}}$

And:

$\boxed{{}^nC_r = {}^nC_{n-r}}$

Always use the smaller of:

$r, n-r$

for calculation.

2. SUCCESSIVE COMBINATION RATIO

A useful shortcut:

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

Therefore:

$$\boxed{{}^nC_r = {}^nC_{r-1} \frac{n-r+1}{r}}$$

3. EXAMPLE

Given:

$${}^{10}C_3 = 120$$

find:

$${}^{10}C_4$$

Use:

$${}^{10}C_4 = 120 \cdot \frac{10-4+1}{4}$$

$$= 120 \cdot \frac{7}{4}$$

$$= \boxed{210}$$

Much faster than factorials.

4. ANOTHER RATIO FORM

$$\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1}$$

So:

$$\boxed{{}^nC_{r+1} = {}^nC_r \frac{n-r}{r+1}}$$

5. WHERE IS $({}^nC_r)$ MAXIMUM?

Combination values increase until the middle, then decrease symmetrically.

For fixed n :

- if n even, maximum at:

$$\boxed{r = \frac{n}{2}}$$

- if n odd, equal maxima at:

$$\boxed{r = \frac{n-1}{2}, \frac{n+1}{2}}$$

Example:

$${}^{10}C_5$$

is largest in row $n=10$.

6. PASCAL IDENTITY

$$\boxed{{}^nC_r = {}^{n-1}C_r + {}^{n-1}C_{r-1}}$$

Interpretation:

Choose $\binom{n}{r}$ objects from $\binom{n}{r}$, focusing on one particular object A.

Either:

- A is excluded
 - A is included
-

7. HOCKEY-STICK IDENTITY

A useful identity:

$$\boxed{{}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + \cdots + {}^nC_r = {}^{n+1}C_{r+1}}$$

Example:

$$\boxed{{}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2}$$

$$= {}^6C_3$$

$$= \boxed{20}$$

Useful mainly for identity-based GATE questions.

8. SUM OF ALL COMBINATION TERMS

$$\boxed{\sum_{r=0}^n {}^nC_r = 2^n}$$

This comes from counting all subsets of an $\binom{n}{r}$ -element set.

9. WEIGHTED COMBINATION IDENTITY

A very useful result:

$$\boxed{r {}^nC_r = n {}^{n-1}C_{r-1}}$$

Interpretation:

Choose $\binom{n}{r}$ people from $\binom{n}{r}$, then choose one special person among the selected $\binom{n}{r}$.

10. EXAMPLE

Evaluate:

$$5 {}^{12}C_5$$

Use:

$$r {}^nC_r = n {}^{n-1}C_{r-1}$$

So:

$$\boxed{5^{12}C_5 = 12^{11}C_4}$$

Sometimes this is much easier depending on the expression.

11. ANOTHER IDENTITY

$$\boxed{(n-r)^nC_r = n^{n-1}C_r}$$

Useful when one object is chosen from the unselected group.

12. VANDERMONDE-TYPE IDENTITY

Suppose there are:

- m objects of one type
- n objects of another type

Choose total r .

We may choose:

$$0, r$$

or:

$$1, r-1$$

etc.

Thus:

$$\boxed{\sum_{k=0}^r {}^mC_k {}^nC_{r-k} = {}^{m+n}C_r}$$

This is a powerful counting identity.

13. INTUITION FOR VANDERMONDE

Suppose 5 men and 4 women.

Choose 3 people.

Directly:

$9C_3$

Or split by number of men:

$${}^5C_0 {}^4C_3 + {}^5C_1 {}^4C_2 + {}^5C_2 {}^4C_1 + {}^5C_3 {}^4C_0$$

Both counts must be equal.

14. RESTRICTED COMMITTEE — A AND B CANNOT BOTH BE PRESENT

Choose 5 from 12.

A and B cannot both be selected.

Total:

$$\{ {}^{12}C_5 \}$$

Subtract committees containing both A and B:

$$\{ {}^{10}C_3 \}$$

Therefore:

$$792 - 120 = \boxed{672}$$

15. A AND B MUST BE TOGETHER IN SELECTION

In combinations, "together" usually means:

either both selected or both excluded.

Example: choose 5 from 12 such that A and B are either both in or both out.

Both in

Need 3 from remaining 10:

$$\{ {}^{10}C_3 \}$$

Both out

Choose all 5 from remaining 10:

$$\{ {}^{10}C_5 \}$$

Total:

$$120 + 252 = \boxed{372}$$

16. EXACTLY ONE OF A,B,C

Choose 5 from 12.

Choose exactly one special:

$$\{ {}^3C_1 \}$$

Choose remaining 4 from ordinary 9:

$$\{ {}^9C_4 \}$$

Total:

$$3(126) = \boxed{378}$$

17. AT MOST ONE OF A,B,C

Possible special counts:

$$\{ 0 \text{ or } 1 \}$$

So:

$$\{ {}^3C_0 {}^9C_5 + {}^3C_1 {}^9C_4 \}$$

$$\{ = 126 + 378 \}$$

$$\{ = \boxed{504} \}$$

18. AT LEAST TWO OF A,B,C

Possible:

- exactly 2
- exactly 3

$$\{ {}^3C_2 {}^9C_3 + {}^3C_3 {}^9C_2 \}$$

$$\{ = 3(84) + 36 \}$$

$$\{ = \boxed{288} \}$$

19. TWO PEOPLE REFUSE TO SERVE TOGETHER

This wording means both cannot be simultaneously selected.

Example:

Choose committee of 4 from 9, where A and B refuse to serve together.

Total:

$$\{ {}^9C_4 = 126 \}$$

Invalid with both A,B:

$$\{ {}^7C_2 = 21 \}$$

Valid:

$$\{ \boxed{105} \}$$

20. TWO PEOPLE WILL SERVE ONLY TOGETHER

This means:

- both selected
- or both not selected

Example committee of 4 from 9.

Both selected:

$$\{ {}^7C_2 = 21 \}$$

Both excluded:

$$\{ {}^7C_4 = 35 \}$$

Total:

$$\{ \boxed{56} \}$$

21. SELECTING NON-CONSECUTIVE OBJECTS

Classic result:

Number of ways to choose $\binom{n}{r}$ objects from $\binom{n}{r}$ objects arranged in a line so that no two selected are consecutive:

$$\boxed{\binom{n-r+1}{r}}$$

MUST KNOW

22. WHY THE NON-CONSECUTIVE FORMULA WORKS

Suppose selected positions are:

$$x_1 < x_2 < \dots < x_r$$

and:

$$x_{i+1} \geq x_i + 2$$

Define compressed positions:

$$y_i = x_i - (i-1)$$

Then:

$$1 \leq y_1 < y_2 < \dots < y_r \leq n-r+1$$

So choose $\binom{n}{r}$ positions from:

$$\binom{n-r+1}{r}$$

Therefore:

$$\boxed{\binom{n-r+1}{r}}$$

23. EXAMPLE

Choose 3 numbers from:

$$\{1, 2, 3, 4, 5, 6, 7\}$$

such that no two are consecutive.

$$\binom{7-3+1}{3} = \binom{5}{3}$$

$$= \boxed{10}$$

24. CHOOSE r OBJECTS WITH AT LEAST ONE GAP

Same formula:

$$\boxed{\binom{n-r+1}{r}}$$

This appears in:

- selecting nonadjacent seats
 - choosing days with no consecutive days
 - selecting nonconsecutive integers
-

- binary strings with fixed number of 1s and no adjacent 1s

25. BINARY STRING CONNECTION

How many binary strings of length $\backslash(n\backslash)$ contain exactly $\backslash(r\backslash)$ ones with no two ones adjacent?

Choose the $\backslash(r\backslash)$ one-positions nonconsecutively:

$$\backslash \boxed{{}^{\{n-r+1\}}C_r} \backslash$$

Example:

Length 7, exactly 3 ones, no adjacent ones:

$$\backslash {}^5C_3 = \boxed{10} \backslash$$

26. SELECTING CONSECUTIVE OBJECTS

Choose $\backslash(r\backslash)$ consecutive objects from $\backslash(n\backslash)$ objects in a row.

Possible blocks start at:

$$\backslash 1, 2, \ldots, n-r+1 \backslash$$

So:

$$\backslash \boxed{n-r+1} \backslash$$

Example:

Choose 4 consecutive seats from 10:

$$\backslash 10-4+1 = \boxed{7} \backslash$$

27. GROUPING PEOPLE INTO TWO LABELED GROUPS

10 people split into:

- Group A with 4 people
- Group B with 6 people

Since groups are labeled:

Choose Group A:

$$\backslash {}^{10}C_4 \backslash$$

Remaining automatically Group B.

Thus:

$$\backslash \boxed{210} \backslash$$

28. GROUPING INTO TWO UNLABELED EQUAL GROUPS

Split 8 people into two groups of 4, where groups have no labels.

Naively choose first group:

$$\{ {}^8C_4=70 \}$$

But choosing group X automatically leaves group Y, and swapping group names gives same partition.

So divide by:

$$\{ 2! \}$$

Thus:

$$\{ \boxed{\frac{{}^8C_4}{2}=35} \}$$

29. UNEQUAL-SIZED UNLABELED GROUPS

Split 7 people into groups of sizes 3 and 4.

The sizes distinguish the groups automatically.

Choose the 3-person group:

$$\{ {}^7C_3 = \boxed{35} \}$$

No division by 2.

30. GROUP INTO THREE LABELED GROUPS

12 people into:

- A: 3
- B: 4
- C: 5

Choose A:

$$\{ {}^{12}C_3 \}$$

Then B:

$$\{ {}^9C_4 \}$$

Remaining 5 go to C.

Total:

$$\{ \boxed{{}^{12}C_3 {}^9C_4} \}$$

Equivalent factorial form:

$$\{ \boxed{\frac{12!}{3!4!5!}} \}$$

31. MULTINOMIAL SELECTION FORMULA

Splitting $\{n\}$ distinct objects into labeled groups of sizes:

$$\{ n_1, n_2, \dots, n_k \}$$

where:

$$\{ n_1 + \dots + n_k = n \}$$

Number:

$$\boxed{\frac{n!}{n_1!n_2!\cdots n_k!}}$$

This is the multinomial coefficient.

32. UNLABELED EQUAL GROUPS

Suppose 12 people split into 3 unlabeled groups of 4.

First labeled-group count:

$$\frac{12!}{4!4!4!}$$

But the 3 groups themselves can be permuted in:

$$3!$$

ways without changing the partition.

Therefore:

$$\boxed{\frac{12!}{(4!)^3 \cdot 3!}}$$

33. PAIRING $2n$ PEOPLE

How many ways to divide $(2n)$ distinct people into (n) unordered pairs?

Formula:

$$\boxed{\frac{(2n)!}{2^n n!}}$$

High-value result.

34. WHY PAIRING FORMULA WORKS

Arrange all $(2n)$ people:

$$(2n)!$$

Group adjacent positions into pairs.

But:

- each pair can internally swap !' divide by (2^n)
- the (n) pairs can be reordered !' divide by $(n!)$

Thus:

$$\boxed{\frac{(2n)!}{2^n n!}}$$

35. EXAMPLE — 6 PEOPLE INTO 3 PAIRS

$$\frac{6!}{2^3 3!}$$

$$= \frac{720}{8 \cdot 6}$$

$$= \boxed{15}$$

36. ANOTHER PAIRING METHOD

Choose partner for A:

$$\boxed{5}$$

Then among remaining 4, choose partner for one fixed person:

$$\boxed{3}$$

Remaining 2 automatically pair.

Total:

$$5 \cdot 3 \cdot 1 = \boxed{15}$$

This gives another formula:

$$\boxed{(2n-1)(2n-3)\cdots 3\cdots 1}$$

which equals:

$$\frac{(2n)!}{2^n n!}$$

37. SELECT ONE PAIR FROM n PEOPLE

This is just:

$$\boxed{{}^nC_2}$$

Do not confuse:

- selecting one pair
 - partitioning everyone into pairs
-

38. CHOOSING r PAIRS FROM DISTINCT PEOPLE — CAUTION

If you must select $(2r)$ people and partition them into (r) pairs:

First choose $(2r)$:

$${}^nC_{2r}$$

Then pair them:

$$\frac{(2r)!}{2^r r!}$$

Total:

$$\boxed{{}^nC_{2r} \frac{(2r)!}{2^r r!}}$$

39. SELECTING OBJECTS WITH NO TWO FROM SAME PAIR

Suppose 5 married couples, choose 3 people such that no married couple is selected together.

Choose 3 couples:

$5C_3$

From each selected couple, choose 1 person:

$$\lfloor 2^3 \rfloor$$

Total:

$$\lfloor \boxed{{}^5C_3 \cdot 2^3 = 80} \rfloor$$

Very common.

40. AT LEAST ONE COMPLETE COUPLE

From 5 couples choose 3 people.

Total selections:

$$\lfloor {}^{10}C_3 = 120 \rfloor$$

No complete couple:

$$\lfloor {}^5C_3 \cdot 2^3 = 80 \rfloor$$

Therefore at least one complete couple:

$$\lfloor \boxed{40} \rfloor$$

Since only 3 people are selected, there cannot be two complete couples anyway.

41. SELECT ONE PERSON FROM EACH COUPLE

From $\lfloor n \rfloor$ couples:

Each couple contributes:

$$\lfloor 2 \rfloor$$

choices.

Thus:

$$\lfloor \boxed{2^n} \rfloor$$

This is multiplication principle, not $\lfloor nC_r \rfloor$.

42. DISTRIBUTION PREVIEW — IDENTICAL OBJECTS

Suppose we distribute $\lfloor n \rfloor$ identical objects among $\lfloor r \rfloor$ distinct boxes.

This leads to **stars and bars**.

We only preview it here because it is a counting technique closely linked to combinations.

For nonnegative solutions of:

$$\lfloor x_1 + x_2 + \cdots + x_r = n \rfloor$$

the number is:

$$\lfloor \boxed{{}^{n+r-1}C_{r-1}} \rfloor$$

43. STARS AND BARS INTUITION

Distribute 5 identical chocolates among 3 children.

Represent chocolates:

$\backslash$

$\backslash$

Use 2 bars to separate 3 children.

Example:

$\backslash \text{ * } || \text{ ** } \backslash$

means:

$\backslash 2, 1, 2 \backslash$

Total symbols:

$\backslash 5+2=7 \backslash$

Choose positions of 2 bars:

$\backslash \{ \}^7 C_2 = \boxed{21} \backslash$

So number of nonnegative integer solutions to:

$\backslash x+y+z=5 \backslash$

is:

$\backslash \boxed{21} \backslash$

44. POSITIVE DISTRIBUTION

If each child must receive at least 1:

$\backslash x_1 + \dots + x_r = n, \text{quad } x_i \geq 1 \backslash$

Give 1 to each first.

Remaining:

$\backslash n-r \backslash$

Then distribute nonnegatively:

$\backslash \boxed{\{ \}^{n-1} C_{r-1}} \backslash$

45. EXAMPLE

Distribute 8 identical chocolates among 3 children, each gets at least 1.

Give 1 each.

Remaining:

$\backslash 5 \backslash$

Distribute among 3:

$\backslash \{ \}^{5+3-1} C_2 = \{ \}^7 C_2 \backslash$

$$\boxed{21}$$

Equivalent direct formula:

$${}^{8-1}C_{3-1} = {}^7C_2$$

46. WARNING — STARS AND BARS CONDITIONS

Stars and bars requires:

- objects identical
- boxes/recipients distinct
- only total-count constraints initially

If objects are distinct, the counting changes completely.

We will not overextend this here.

47. COMBINATION EQUATION

Example:

$${}^nC_2=15$$

Then:

$$\frac{n(n-1)}{2}=15$$

$$n(n-1)=30$$

$$n^2-n-30=0$$

$$(n-6)(n+5)=0$$

Positive:

$$\boxed{n=6}$$

48. ANOTHER COMBINATION EQUATION

$${}^nC_1={}^nC_2$$

$$n=\frac{n(n-1)}{2}$$

for $(n>0)$:

$$1=\frac{n-1}{2}$$

$$n-1=2$$

$$\boxed{n=3}$$

49. USE SYMMETRY IN EQUATIONS

If:

$${}^nC_3={}^nC_7$$

usually either:

$$\lfloor 3=7 \rfloor$$

impossible,

or by symmetry:

$$\lfloor 3+7=n \rfloor$$

Thus:

$$\lfloor \boxed{n=10} \rfloor$$

provided combination indices are valid.

50. GENERAL SYMMETRY RESULT

If:

$$\lfloor {}^nC_r = {}^nC_s \rfloor$$

then usually:

$$\lfloor \boxed{r=s \text{ or } r+s=n} \rfloor$$

because the row is symmetric and increases toward the middle.

51. EXAMPLE

$$\lfloor {}^{12}C_x = {}^{12}C_5 \rfloor$$

Then:

$$\lfloor x=5 \rfloor$$

or:

$$\lfloor x=12-5=7 \rfloor$$

Thus:

$$\lfloor \boxed{x=5 \text{ or } 7} \rfloor$$

52. FIND n FROM RATIO

Suppose:

$$\lfloor \frac{{}^nC_3}{{}^nC_2} = 2 \rfloor$$

Use ratio:

$$\lfloor \frac{n-2}{3} = 2 \rfloor$$

$$\lfloor n-2=6 \rfloor$$

$$\lfloor \boxed{n=8} \rfloor$$

Much faster than factorial expansion.

53. ANOTHER RATIO

$$\frac{{}^{10}C_5}{{}^{10}C_4} = \frac{10-4}{5} = \frac{6}{5}$$

So:

$$\boxed{\frac{6}{5}}$$

Use successive-term ratios.

54. GATE-STYLE QUESTION 1

Evaluate:

$${}^{15}C_{13}$$

Use symmetry:

$${}^{15}C_2 = \frac{15 \cdot 14}{2}$$

$$= \boxed{105}$$

55. GATE-STYLE QUESTION 2

If:

$${}^nC_2 = 28$$

then:

$$\frac{n(n-1)}{2} = 28$$

$$n(n-1) = 56$$

$$n = 8$$

$$\boxed{8}$$

56. GATE-STYLE QUESTION 3

How many binary strings of length 10 contain exactly 4 ones?

Choose 4 one-positions:

$${}^{10}C_4$$

$$= \boxed{210}$$

57. GATE-STYLE QUESTION 4

How many binary strings of length 10 contain exactly 4 ones with no adjacent ones?

$${}^{10-4+1}C_4 = {}^7C_4$$

$$= \boxed{35}$$

58. PLACEMENT-STYLE QUESTION 1

6 married couples. Choose 4 people with no married couple together.

Choose 4 couples:

$$\lfloor {}^6C_4 \rfloor$$

Choose one from each:

$$\lfloor 2^4 \rfloor$$

Total:

$$\lfloor 15(16) \rfloor$$

$$\lfloor =\boxed{240} \rfloor$$

59. PLACEMENT-STYLE QUESTION 2

Split 10 people into groups of 4 and 6.

$$\lfloor {}^{10}C_4 = \boxed{210} \rfloor$$

Group sizes distinguish them.

60. PLACEMENT-STYLE QUESTION 3

Split 10 people into two unlabeled groups of 5.

$$\lfloor \frac{{}^{10}C_5}{2} \rfloor$$

$$\lfloor =\frac{252}{2} \rfloor$$

$$\lfloor =\boxed{126} \rfloor$$

61. PLACEMENT-STYLE QUESTION 4

Divide 8 people into 4 pairs.

$$\lfloor \frac{8!}{2^4 4!} \rfloor$$

$$\lfloor = \frac{40320}{16 \cdot 24} \rfloor$$

$$\lfloor =\boxed{105} \rfloor$$

62. ADVANCED QUESTION — NONCONSECUTIVE CHOICE

Choose 4 numbers from:

$$\lfloor 1,2,\ldots,10 \rfloor$$

such that no two are consecutive.

$$\lfloor {}^{10-4+1}C_4 = {}^7C_4 \rfloor$$

$$\lfloor =\boxed{35} \rfloor$$

63. ADVANCED QUESTION — COUPLES

From 5 couples choose 4 people such that exactly one married couple is included.

Choose the complete couple:

$$\lfloor {}^5C_1=5 \rfloor$$

Now choose remaining 2 people from two different other couples.

Choose 2 of remaining 4 couples:

$$\lfloor {}^4C_2=6 \rfloor$$

From each selected couple choose 1:

$$\lfloor 2^2=4 \rfloor$$

Total:

$$\lfloor 5\cdot6\cdot4 \rfloor$$

$$\lfloor =\boxed{120} \rfloor$$

64. ADVANCED QUESTION — GROUPING

12 people divided into 3 unlabeled groups of 4.

$$\lfloor \frac{12!}{(4!)^3 3!} \rfloor$$

Calculate:

$$\lfloor \boxed{5775} \rfloor$$

65. ADVANCED QUESTION — STARS AND BARS

How many nonnegative integer solutions:

$$\lfloor x+y+z=10 \rfloor$$

$$\lfloor {}^{10+3-1}C_{3-1} = {}^{12}C_2 \rfloor$$

$$\lfloor =\boxed{66} \rfloor$$

66. ADVANCED QUESTION — POSITIVE SOLUTIONS

How many positive integer solutions:

$$\lfloor x+y+z=10 \rfloor$$

$$\lfloor {}^{10-1}C_{3-1} = {}^9C_2 \rfloor$$

$$\lfloor =\boxed{36} \rfloor$$

67. COMMON TRAPS

& pTRAP 1 — Labeled vs unlabeled groups

If groups are interchangeable, divide by factorials for identical group roles.

& pTRAP 2 — Equal group sizes

Two unlabeled equal groups are double-counted if you simply choose one group.

& pTRAP 3 — Pair selection vs full pairing

$\lfloor \frac{n!}{2} \rfloor$

selects one pair.

It does not pair everyone.

& pTRAP 4 — Nonconsecutive formula

For choosing $\lfloor r \rfloor$ nonadjacent objects from $\lfloor n \rfloor$ in a line:

$\lfloor \frac{n!}{r!} \rfloor$

& pTRAP 5 — Stars and bars with distinct objects

Do not use it when objects are distinct.

& pTRAP 6 — At least/at most

Choose direct or complement depending on fewer cases.

& pTRAP 7 — $\lfloor nC_r = nC_s \rfloor$

Check:

$\lfloor r=s \rfloor$

or:

$\lfloor r+s=n \rfloor$

68. MUST-MEMORIZE ADVANCED IDENTITIES

$\lfloor \frac{n!}{r!} \rfloor = \frac{n!}{r!} \lfloor \frac{n-r}{r} \rfloor$

$\lfloor \frac{n!}{r!} \rfloor = \frac{n!}{r!} \lfloor \frac{n-1}{r} \rfloor + \frac{n!}{r!} \lfloor \frac{n-1}{r-1} \rfloor$

$\lfloor \frac{n!}{r!} \rfloor = \frac{n!}{r!} \lfloor \frac{n-1}{r-1} \rfloor$

$\lfloor \sum_{r=0}^n \frac{n!}{r!} \rfloor = 2^n n!$

69. MUST-MEMORIZE NONCONSECUTIVE SELECTION

$\lfloor \frac{n!}{r!} \rfloor$ Choose r nonconsecutive from $n = \frac{n!}{r!} \lfloor \frac{n-r+1}{r} \rfloor$

70. MUST-MEMORIZE PAIRING FORMULA

Pair $(2n)$ distinct objects completely:

$$\boxed{\frac{(2n)!}{2^n n!}}$$

Equivalent:

$$\boxed{(2n-1)(2n-3)\cdots 1}$$

71. MUST-MEMORIZE GROUPING FORMULA

Split (n) distinct objects into labeled groups of sizes:

$$[n_1, n_2, \dots, n_k]$$

$$\boxed{\frac{n!}{n_1! n_2! \cdots n_k!}}$$

If equal-sized groups are unlabeled, divide further by factorial of number of interchangeable groups.

72. STARS AND BARS PREVIEW FORMULAS

Nonnegative integer solutions to:

$$[x_1 + \cdots + x_r = n]$$

$$\boxed{{}^{n+r-1}C_{r-1}}$$

Positive integer solutions:

$$\boxed{{}^{n-1}C_{r-1}}$$

73. FAST EXAM DECISION TREE

Order matters?

Permutation.

Order irrelevant?

Combination.

Particular members compulsory?

Fix them.

Particular members cannot coexist?

Subtract invalid cases.

No two selected consecutive?

Use:

$$[{}^{n-r+1}C_r]$$

Split into groups?

Ask:

- group sizes?
- labels?
- equal/interchangeable?

Pair everybody?

Use pairing formula.

Married couples?

Choose couples first, then one/both members as required.

Identical objects into distinct boxes?

Think stars and bars.

$\binom{n}{r}$ equation?

Try symmetry or successive-term ratio before factorials.

PRACTICE — LEVEL 0

Q1

Find:

$$\binom{12}{10}$$

Q2

If:

$$\binom{n}{2} = 21$$

find $\binom{n}{1}$.

Q3

How many binary strings of length 8 contain exactly 3 ones?

Q4

Choose 3 nonconsecutive numbers from 1–8.

Q5

Divide 6 people into two groups of 3, unlabeled.

LEVEL 1

Q6

Choose 5 from 11 if A and B cannot both be selected.

Q7

Choose 4 people from 10 such that A and B are either both selected or both excluded.

Q8

Divide 8 people into 4 pairs.

Q9

From 5 couples choose 3 people with no complete couple.

Q10

How many nonnegative integer solutions to:

$$\lfloor x+y+z=7 \rfloor$$

LEVEL 2

Q11

How many positive integer solutions to:

$$\lfloor x+y+z+w=12 \rfloor$$

Q12

Choose 4 numbers from 1–12 with no two consecutive.

Q13

Split 9 people into groups of 4 and 5.

Q14

Split 12 people into three unlabeled groups of 4.

Q15

If:

$$\lfloor {}^{15}C_r = {}^{15}C_6 \rfloor$$

find all possible $\lfloor r \rfloor$.

LEVEL 3 — EXAM PRESSURE

Q16

From 6 married couples choose 5 people such that no couple is completely selected.

Q17

From 5 couples choose 4 people containing exactly one complete couple.

Q18

How many binary strings of length 12 contain exactly 5 ones with no adjacent ones?

Q19

Divide 10 people into 5 unordered pairs.

Q20

If:

$$\lfloor \frac{{}^nC_4}{{}^nC_3} = 2 \rfloor$$

find $\backslash(n)$.

ANSWERS

Q1

$$\backslash \{{}^{\{12\}}C_{\{10\}} = \{{}^{\{12\}}C_2 \backslash$$

$$\backslash [= \frac{12 \cdot 11}{2} \backslash$$

$$\backslash [= \boxed{66} \backslash$$

Q2

$$\backslash \frac{n(n-1)}{2} = 21 \backslash$$

$$\backslash n(n-1) = 42 \backslash$$

$$\backslash n = 7 \backslash$$

$$\backslash \boxed{7} \backslash$$

Q3

Choose positions of three 1s:

$$\backslash \{{}^8C_3 = \boxed{56} \backslash$$

Q4

$$\backslash \{{}^{8-3+1}C_3 = \{{}^6C_3 \backslash$$

$$\backslash [= \boxed{20} \backslash$$

Q5

Choose one group:

$$\backslash \{{}^6C_3 = 20 \backslash$$

But the two groups are unlabeled:

$$\backslash 20/2 \backslash$$

$$\backslash [= \boxed{10} \backslash$$

Q6

Total:

$$\backslash \{{}^{11}C_5 = 462 \backslash$$

Invalid if both A,B selected:

Need 3 from remaining 9:

$$\backslash \{{}^9C_3 = 84 \backslash$$

Valid:

$$\backslash 462 - 84 \backslash$$

$$\backslash [= \boxed{378} \backslash$$

Q7

Both selected:

$$\lfloor {}^8C_2=28 \rfloor$$

Both excluded:

$$\lfloor {}^8C_4=70 \rfloor$$

Total:

$$\lfloor \boxed{98} \rfloor$$

Q8

$$\lfloor \frac{8!}{2^4 4!} = \boxed{105} \rfloor$$

Q9

Choose 3 couples:

$$\lfloor {}^5C_3=10 \rfloor$$

Choose one person from each:

$$\lfloor 2^3=8 \rfloor$$

Total:

$$\lfloor \boxed{80} \rfloor$$

Q10

$$\lfloor x+y+z=7 \rfloor$$

Nonnegative solutions:

$$\lfloor {}^{7+3-1}C_2 = {}^9C_2 \rfloor$$

$$\lfloor =\boxed{36} \rfloor$$

Q11

Positive solutions:

$$\lfloor {}^{12-1}C_{4-1} = {}^{11}C_3 \rfloor$$

$$\lfloor =\boxed{165} \rfloor$$

Q12

$$\lfloor {}^{12-4+1}C_4 = {}^9C_4 \rfloor$$

$$\lfloor =\boxed{126} \rfloor$$

Q13

Groups have different sizes, so choose the 4-person group:

$$\lfloor {}^9C_4 = \boxed{126} \rfloor$$

Q14

$$\lfloor \frac{12!}{(4!)^3 3!} \rfloor$$

$$\lfloor =\boxed{5775} \rfloor$$

Q15

Using symmetry:

$$\lfloor r=6 \rfloor$$

or:

$$\lfloor r=15-6=9 \rfloor$$

Thus:

$$\lfloor \boxed{r=6,9} \rfloor$$

Q16

6 couples, choose 5 people with no complete couple.

Choose 5 couples:

$$\lfloor {}^6C_5=6 \rfloor$$

Choose one from each selected couple:

$$\lfloor 2^5=32 \rfloor$$

Total:

$$\lfloor 6(32) \rfloor$$

$$\lfloor =\boxed{192} \rfloor$$

Q17

Exactly one complete couple.

Choose that couple:

$$\lfloor {}^5C_1=5 \rfloor$$

Need 2 more people from 2 different remaining couples.

Choose 2 couples from remaining 4:

$$\lfloor {}^4C_2=6 \rfloor$$

Choose one member from each:

$$\lfloor 2^2=4 \rfloor$$

Total:

$$\lfloor 5\cdot 6\cdot 4 \rfloor$$

$$\lfloor =\boxed{120} \rfloor$$

Q18

Length 12, exactly 5 ones, no adjacent ones:

$$\lfloor {}^{12-5+1}C_5 = {}^8C_5 \rfloor$$

$$\lfloor =\boxed{56} \rfloor$$

Q19

Pair 10 people:

$$\lfloor \frac{10!}{2^5 5!} \rfloor$$

$$\lfloor =\boxed{945} \rfloor$$

Q20

Use successive ratio:

$$\frac{{}^nC_4}{{}^nC_3} = \frac{n-3}{4}$$

Given:

$$\frac{n-3}{4} = 2$$

$$n-3=8$$

$$\boxed{n=11}$$

' CHAPTER 40 COMPLETE

We have now covered:

- combination concept '
- $({}^nC_r)$ formula '
- permutation vs combination '
- symmetry '
- Pascal identity '
- combination ratios '
- subset identities '
- weighted identities '
- Vandermonde reasoning '
- compulsory/excluded members '
- incompatible members '
- at least/at most/exactly restrictions '
- category-wise team selection '
- team + role assignment '
- pair counting '
- complete pairing '
- couples problems '
- nonconsecutive selections '
- binary-string selection '
- grouping into labeled/unlabeled groups '
- multinomial grouping '
- equal-group overcount correction '
- stars-and-bars preview '
- combination equations '
- GATE/placement L0–L3 drills '

Progress

Completed: 40/76

Unit 1 — Quantitative Aptitude: 0=0% PROGRESS

NEXT

Chapter 41 — Probability

Next we will cover:

- experiment, outcome, sample space and event
- equally likely outcomes
- basic probability
- complement
- addition rule
- mutually exclusive events
- independent vs dependent events
- multiplication rule
- conditional probability
- without-replacement probability
- cards, dice, coins and balls
- at least/at most/exactly
- Bayes-style intuition
- odds
- counting + probability integration
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 41 — PROBABILITY

PART 1 — SAMPLE SPACE, EVENTS, BASIC PROBABILITY, COMPLEMENT, ADDITION & MULTIPLICATION RULES

Status: $\emptyset$ IN PROGRESS GATE: High Placements: High Priority: $\emptyset$ Year 1

Probability measures:

How likely is an event to happen?

The basic range is:

$\boxed{0 \leq P(E) \leq 1}$

- $(P(E)=0)$!' impossible
- $(P(E)=1)$!' certain
- closer to 1 !' more likely

1. RANDOM EXPERIMENT

A **random experiment** is an action whose exact outcome is uncertain before performing it.

Examples:

- tossing a coin
 - rolling a die
 - drawing a card
 - choosing a student randomly
-

2. OUTCOME

An **outcome** is one possible result.

Example: rolling a die.

Possible outcomes:

$\{1, 2, 3, 4, 5, 6\}$

If die shows 4:

$\{4\}$

is the outcome.

3. SAMPLE SPACE

The set of all possible outcomes is called the **sample space**.

Symbol:

$\boxed{S}$

For one coin:

$S = \{H, T\}$

For one die:

$S = \{1, 2, 3, 4, 5, 6\}$

4. EVENT

An **event** is a collection of one or more outcomes.

Example:

Roll a die.

Event $\{E\}$:

getting an even number

Then:

$E = \{2, 4, 6\}$

5. BASIC PROBABILITY FORMULA

When all outcomes are equally likely:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

MUST MEMORIZE

6. EXAMPLE — DIE

Probability of rolling an even number:

Favourable:

$$2, 4, 6$$

So:

$$3$$

favourable outcomes.

Total:

$$6$$

Therefore:

$$P(E) = \frac{3}{6}$$

$$= \boxed{\frac{1}{2}}$$

7. EXAMPLE — PRIME NUMBER ON A DIE

Primes among 1–6:

$$2, 3, 5$$

So:

$$P(\text{prime}) = \frac{3}{6} = \boxed{\frac{1}{2}}$$

Remember:

$$1$$

is not prime.

8. IMPOSSIBLE EVENT

Probability of rolling 7 on an ordinary 6-sided die:

$$\boxed{0}$$

9. CERTAIN EVENT

Probability of rolling a number less than 7:

$$\boxed{1}$$

10. COMPLEMENT OF AN EVENT

If event (A) occurs, its complement (A^c) means:

A does not occur.

Also written:

$$A'$$

Since either A occurs or it doesn't:

$$\boxed{P(A) + P(A^c) = 1}$$

Therefore:

$$\boxed{P(A^c) = 1 - P(A)}$$

11. EXAMPLE

Probability that a die shows a 6:

$$P(6) = \frac{1}{6}$$

Probability it does not show 6:

$$1 - \frac{1}{6} = \boxed{\frac{5}{6}}$$

12. “AT LEAST ONE” — COMPLEMENT METHOD

This is one of the most important probability shortcuts.

Suppose two coins are tossed.

Find probability of at least one head.

Direct outcomes:

$$HH, HT, TH, TT$$

At least one H:

$$HH, HT, TH$$

Probability:

$$\frac{3}{4}$$

But complement method:

$$P(\text{at least one H}) = 1 - P(\text{no H})$$

$$= 1 - \frac{1}{4}$$

$$= \boxed{\frac{3}{4}}$$

13. GENERAL “AT LEAST ONE”

$$\boxed{P(\text{at least one success}) = 1 - P(\text{zero successes})}$$

This becomes extremely powerful for repeated trials.

14. TWO COINS — SAMPLE SPACE

Two fair coins:

$$S = \{HH, HT, TH, TT\}$$

Total:

$$2^2 = 4$$

15. THREE COINS

Three fair coins:

$$2^3 = 8$$

possible ordered outcomes.

They are:

$$\{HHH, HHT, HTH, HTT, \}$$

$$\{THH, THT, TTH, TTT\}$$

16. n FAIR COINS

For (n) coin tosses:

$$\boxed{2^n}$$

equally likely ordered outcomes.

17. EXACTLY k HEADS

Among (n) fair coin tosses, exactly (k) heads occur in:

$$\binom{n}{k}$$

ways.

Therefore:

$$\boxed{P(\text{exactly } k \text{ heads}) = \frac{\binom{n}{k}}{2^n}}$$

18. EXAMPLE

4 fair tosses.

Probability of exactly 2 heads:

$$\left[\frac{\binom{4}{2}}{2^4} \right]$$

$$\left[=\frac{6}{16} \right]$$

$$\left[=\boxed{\frac{3}{8}} \right]$$

19. AT MOST / AT LEAST LANGUAGE

Exactly 2

Only 2.

At least 2

$$\left[2, 3, 4, \ldots \right]$$

At most 2

$$\left[0, 1, 2 \right]$$

Never confuse them.

20. ADDITION RULE

For two events A and B:

$$\left[\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)} \right]$$

Interpretation:

$$\left[A \cup B \right]$$

means:

A or B or both.

21. WHY SUBTRACT INTERSECTION?

Because outcomes common to A and B are counted twice when adding:

$$\left[P(A) + P(B) \right]$$

So subtract once.

Same inclusion-exclusion principle as counting.

22. EXAMPLE — DIE

Roll a die.

Let:

$$\left[A = \{\text{even}\} = \{2, 4, 6\} \right]$$

$$B = \{\text{number} > 3\} = \{4, 5, 6\}$$

Then:

$$P(A) = \frac{3}{6}$$

$$P(B) = \frac{3}{6}$$

Intersection:

$$A \cap B = \{4, 6\}$$

$$P(A \cap B) = \frac{2}{6}$$

So:

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6}$$

$$= \frac{4}{6}$$

$$= \boxed{\frac{2}{3}}$$

23. MUTUALLY EXCLUSIVE EVENTS

Two events are mutually exclusive if they cannot happen simultaneously.

Then:

$$A \cap B = \varnothing$$

So:

$$P(A \cap B) = 0$$

and:

$$\boxed{P(A \cup B) = P(A) + P(B)}$$

24. EXAMPLE

On one die roll:

A = rolling 1.

B = rolling 6.

They cannot happen together.

Therefore:

$$P(A \text{ or } B) = \frac{1}{6} + \frac{1}{6}$$

$$= \boxed{\frac{1}{3}}$$

25. MUTUALLY EXCLUSIVE DOES NOT MEAN INDEPENDENT

This is a major trap.

If A and B are mutually exclusive and both have positive probability:

$$P(A \cap B) = 0$$

But independence would require:

$$P(A \cap B) = P(A)P(B) > 0$$

So nontrivial mutually exclusive events are **not independent**.

26. INDEPENDENT EVENTS

A and B are independent if occurrence of one does not change the probability of the other.

Mathematically:

$$\boxed{P(A \cap B) = P(A)P(B)}$$

27. EXAMPLE — TWO COINS

A = first coin is H.

B = second coin is H.

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

Both:

$$P(A \cap B) = \frac{1}{4}$$

And:

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

So independent.

28. MULTIPLICATION RULE FOR INDEPENDENT EVENTS

If A and B are independent:

$$\boxed{P(A \text{ and } B) = P(A)P(B)}$$

29. EXAMPLE — TWO SIXES

Two fair dice rolled independently.

Probability both show 6:

$$\frac{1}{6} \cdot \frac{1}{6}$$

$$= \boxed{\frac{1}{36}}$$

30. THREE INDEPENDENT SUCCESSES

If each trial has probability p of success, probability all 3 succeed:

$$\boxed{p^3}$$

Example:

Three fair coin tosses all heads:

$$\left(\frac{1}{2}\right)^3 = \boxed{\frac{1}{8}}$$

31. DEPENDENT EVENTS

Events are dependent when one outcome affects the probability of the next.

Classic example:

drawing objects without replacement.

32. WITH REPLACEMENT

A bag contains:

- 3 red balls
- 2 blue balls

Draw one ball, replace it, then draw again.

Probability both red:

First red:

$$\frac{3}{5}$$

After replacement, still:

$$\frac{3}{5}$$

Second red:

$$\frac{3}{5}$$

Thus:

$$\frac{3}{5} \cdot \frac{3}{5} = \boxed{\frac{9}{25}}$$

Trials are independent.

33. WITHOUT REPLACEMENT

Same bag:

3 red, 2 blue.

Draw two without replacement.

Probability both red:

First:

$$\frac{3}{5}$$

After a red is removed:

- 2 red

- 2 blue
- 4 total

Second red:

$$\frac{2}{4} = \frac{1}{2}$$

Therefore:

$$\frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{1}{4}}$$

Dependent events.

34. GENERAL MULTIPLICATION RULE

For any two events:

$$\boxed{P(A \cap B) = P(A)P(B|A)}$$

where:

$$P(B|A)$$

means probability of B given A has already occurred.

Conditional probability is covered deeply in Part 2.

35. DICE — TWO-DIE SAMPLE SPACE

Two fair dice rolled.

Total ordered outcomes:

$$6 \cdot 6 = \boxed{36}$$

Important:

$$(2,5) \neq (5,2)$$

because dice positions are distinguishable in the sample-space representation.

36. SUM OF TWO DICE

Possible sums range:

$$2 \text{ to } 12$$

But sums are **not equally likely**.

Counts:

Sum	Ways
2	1
3	2
4	3

Sum	Ways
5	4
6	5
7	6
8	5
9	4
10	3
11	2
12	1

MUST KNOW pattern

$\backslash [1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1]$

37. PROBABILITY SUM = 7

There are:

$\backslash [6]$

ways:

$\backslash [(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)]$

Therefore:

$\backslash [P(\text{sum } 7) = \frac{6}{36} = \boxed{\frac{1}{6}}]$

38. PROBABILITY SUM = 10

Ways:

$\backslash [(4, 6), (5, 5), (6, 4)]$

So:

$\backslash [P = \frac{3}{36}]$

$\backslash [= \boxed{\frac{1}{12}}]$

39. AT LEAST ONE 6 WITH TWO DICE

Use complement.

Probability no die shows 6:

$\backslash [\frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36}]$

Therefore:

$$P(\text{at least one 6}) = 1 - \frac{25}{36}$$

$$= \boxed{\frac{11}{36}}$$

40. EXACTLY ONE 6

Cases:

- first 6, second not 6
- first not 6, second 6

So:

$$\frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6}$$

$$= \frac{10}{36}$$

$$= \boxed{\frac{5}{18}}$$

41. BOTH DICE SAME — DOUBLES

Possible doubles:

$$(1,1), (2,2), \dots, (6,6)$$

6 favourable.

Therefore:

$$\boxed{\frac{6}{36} = \frac{1}{6}}$$

42. DICE DIFFERENT

Complement:

$$1 - \frac{1}{6} = \boxed{\frac{5}{6}}$$

43. CARD BASICS

A standard deck has:

$$\boxed{52}$$

cards.

Four suits:

- Hearts &e
- Diamonds &f
- Clubs &c
- Spades &`

Each suit:

$$\boxed{13}$$

cards.

44. RED AND BLACK CARDS

Red suits:

- Hearts
- Diamonds

So red cards:

$\lfloor 26 \rfloor$

Black:

$\lfloor 26 \rfloor$

Thus:

$P(\text{red}) = \boxed{\frac{12}{52}}$

45. FACE CARDS

Face cards are:

- Jack
- Queen
- King

3 per suit.

Total:

$3 \cdot 4 = 12$

So:

$P(\text{face card}) = \frac{12}{52} = \boxed{\frac{3}{13}}$

46. ACES

There are:

4

aces.

So:

$P(\text{Ace}) = \frac{4}{52} = \boxed{\frac{1}{13}}$

47. KING OR QUEEN

Kings:

4

Queens:

4

No overlap.

So favourable:

$$\boxed{8}$$

Probability:

$$\frac{8}{52} = \boxed{\frac{2}{13}}$$

48. RED OR KING

Use inclusion-exclusion.

Red:

$$\boxed{26}$$

Kings:

$$\boxed{4}$$

Red kings:

- King of hearts
- King of diamonds

So:

$$\boxed{2}$$

Therefore:

$$\boxed{26+4-2=28}$$

Probability:

$$\frac{28}{52} = \boxed{\frac{7}{13}}$$

49. DRAWING TWO CARDS WITHOUT REPLACEMENT

Probability both are aces:

$$\frac{4}{52} \cdot \frac{3}{51}$$

$$= \frac{1}{13} \cdot \frac{1}{17}$$

$$= \boxed{\frac{1}{221}}$$

50. SAME PROBLEM USING COMBINATIONS

Choose 2 cards from 52.

Total pairs:

$$\binom{52}{2}$$

Favourable pairs of aces:

$$\binom{4}{2}$$

Probability:

$$\boxed{\frac{\binom{42}{1}\binom{52}{2}}{\binom{94}{3}}}$$

This equals:

$$\frac{1}{221}$$

51. ORDERED METHOD VS COMBINATION METHOD

For random selections without replacement:

Sequential method

Multiply conditional probabilities.

Combination method

$$\frac{\text{favourable selections}}{\text{all selections}}$$

Both are valid.

Choose whichever is shorter.

52. BALL SELECTION EXAMPLE

Bag:

- 5 red
- 4 blue

Choose 2 simultaneously.

Probability both red:

$$\frac{\binom{5}{2}}{\binom{9}{2}}$$

$$= \frac{10}{36}$$

$$= \boxed{\frac{5}{18}}$$

53. EXACTLY ONE RED

Same bag.

Choose:

- 1 red from 5
- 1 blue from 4

Favourable:

$$\binom{5}{1}\binom{4}{1} = 20$$

Total:

$$\binom{9}{2} = 36$$

Thus:

$$\boxed{\frac{5}{9}}$$

54. AT LEAST ONE RED

Use complement:

$$\mathbb{P}(\text{both blue})$$

$$= 1 - \frac{\binom{42}{2}}{\binom{92}{2}}$$

$$= 1 - \frac{6}{36}$$

$$= \boxed{\frac{56}{36}}$$

55. ODDS IN FAVOUR

If:

$$\mathbb{P}(E) = \frac{a}{a+b}$$

then odds in favour are:

$$\boxed{a:b}$$

Odds against:

$$\boxed{b:a}$$

56. FROM ODDS TO PROBABILITY

If odds in favour are:

$$3:2$$

then:

$$\mathbb{P}(E) = \frac{3}{3+2}$$

$$= \boxed{\frac{3}{5}}$$

57. ODDS AGAINST

If odds against an event are:

$$4:1$$

then:

- unfavourable = 4
- favourable = 1

So:

$$\mathbb{P}(E) = \frac{1}{4+1}$$

$$= \boxed{\frac{1}{5}}$$

58. PROBABILITY FROM COUNTING

Many harder probability questions are simply:

$$P(E) = \frac{\text{favourable count}}{\text{total count}}$$

So Chapters 38–40 directly support probability.

59. EXAMPLE — RANDOM 3-PERSON TEAM

Choose 3 people randomly from:

- 5 men
- 4 women

Probability exactly 2 women.

Total teams:

$$\binom{9}{3}$$

Favourable:

$$\binom{4}{2} \binom{5}{1}$$

So:

$$P = \frac{6 \cdot 5}{84}$$

$$= \frac{14}{84}$$

60. EXAMPLE — NO WOMEN

Probability all 3 are men:

$$\frac{\binom{5}{3}}{\binom{9}{3}}$$

$$= \frac{10}{84}$$

$$= \frac{5}{42}$$

61. EXAMPLE — AT LEAST ONE WOMAN

Complement:

$$1 - \frac{5}{42}$$

$$= \frac{37}{42}$$

62. GATE-STYLE QUESTION 1

A fair coin is tossed 5 times. Probability of exactly 3 heads:

$$\frac{\binom{5}{3}}{2^5}$$

$$= \frac{10}{32}$$

$$= \frac{5}{16}$$

63. GATE-STYLE QUESTION 2

Two dice are rolled. Probability sum is at least 10.

Sums:

- 10 !' 3 ways
- 11 !' 2
- 12 !' 1

Total favourable:

$$\backslash[6 \backslash]$$

So:

$$\backslash[\frac{6}{36} = \boxed{\frac{1}{6}} \backslash]$$

64. GATE-STYLE QUESTION 3

Probability that a randomly chosen card is neither red nor a face card.

Use complement of:

$$\backslash[\text{red OR face} \backslash]$$

Red:

$$\backslash[26 \backslash]$$

Face cards:

$$\backslash[12 \backslash]$$

Red face cards:

$$\backslash[6 \backslash]$$

Union:

$$\backslash[26+12-6=32 \backslash]$$

Neither:

$$\backslash[52-32=20 \backslash]$$

Probability:

$$\backslash[\frac{20}{52} = \boxed{\frac{5}{13}} \backslash]$$

65. GATE-STYLE QUESTION 4

From 6 men and 4 women, 3 people are chosen randomly. Probability all are women:

$$\backslash[\frac{\binom{4}{3}}{\binom{10}{3}} \backslash]$$

$$\backslash[=\frac{4}{120} \backslash]$$

$$\backslash[=\boxed{\frac{1}{30}} \backslash]$$

66. PLACEMENT-STYLE QUESTION 1

Two fair coins tossed.

Probability exactly one head:

$$\left[\frac{2}{4} = \boxed{\frac{1}{2}} \right]$$

67. PLACEMENT-STYLE QUESTION 2

Three fair coins tossed.

Probability at least one tail:

Complement all heads:

$$\left[1 - \frac{1}{8} \right]$$

$$\left[= \boxed{\frac{7}{8}} \right]$$

68. PLACEMENT-STYLE QUESTION 3

Two dice.

Probability product is odd.

Product odd only if both dice are odd.

Each die odd probability:

$$\left[\frac{3}{6} = \frac{1}{2} \right]$$

Thus:

$$\left[\frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{1}{4}} \right]$$

69. PLACEMENT-STYLE QUESTION 4

Two dice.

Probability product is even.

Complement:

$$\left[1 - P(\text{both odd}) \right]$$

$$\left[= 1 - \frac{1}{4} \right]$$

$$\left[= \boxed{\frac{3}{4}} \right]$$

70. COMMON TRAPS

& pTRAP 1 — Assuming all derived outcomes are equally likely

Two-dice sums 2–12 are not equally likely.

& pTRAP 2 — OR vs AND

Usually:

$\text{\text{OR}} \rightarrow \text{addition rule}$

$\text{AND} \rightarrow \text{multiplication/conditional rule}$

& pTRAP 3 — At least one

Try complement first.

& pTRAP 4 — With vs without replacement

With replacement !' probabilities may stay unchanged.

Without replacement !' probabilities change.

& pTRAP 5 — Independent vs mutually exclusive

They are not the same.

& pTRAP 6 — Ordered vs unordered selections

Sequential draws may use ordered probability multiplication.

Simultaneous selection usually works nicely with combinations.

& pTRAP 7 — Forgetting overlap

For A or B:

$$P(A) + P(B) - P(A \cap B)$$

71. MUST-MEMORIZE FORMULAS

Basic

$$P(E) = \frac{\text{favourable}}{\text{total}}$$

for equally likely outcomes.

Complement

$$P(E^c) = 1 - P(E)$$

Addition

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

72. INDEPENDENT MULTIPLICATION

$$P(A \cap B) = P(A)P(B)$$

if independent.

General rule:

$$P(A \cap B) = P(A)P(B|A)$$

73. FAST EXAM DECISION TREE

Equally likely finite outcomes?

Use favourable/total.

“At least one”?

Use complement.

“A or B”?

Use addition rule.

“A and B” independent?

Multiply.

Without replacement?

Update denominator/numerator after every draw, or use combinations.

Coins?

Use:

$$\sqrt[n]{2}$$

and $\binom{n}{k}$.

Two dice?

Usually total:

$$\sqrt[36]{}$$

Cards?

Remember:

$$\sqrt[52]{4 \text{ suits}, 13 \text{ each}, 12 \text{ face cards}}$$

PRACTICE — LEVEL 0

Q1

Probability of rolling a number greater than 4 on a fair die.

Q2

Two fair coins: probability of both heads.

Q3

Three fair coins: probability of exactly two heads.

Q4

Two dice: probability of sum 7.

Q5

Random card: probability of an ace.

LEVEL 1

Q6

Four fair coins: probability of at least one head.

Q7

Two dice: probability at least one die shows 6.

Q8

One card: probability of red or king.

Q9

Bag has 4 red and 3 blue balls. Two drawn without replacement. Probability both red.

Q10

Same bag: probability exactly one red.

LEVEL 2

Q11

5 fair coin tosses: probability exactly 2 heads.

Q12

Two dice: probability sum is greater than 9.

Q13

A 3-person team is randomly selected from 6 men and 4 women. Probability exactly 2 women.

Q14

Two cards drawn without replacement. Probability both are kings.

Q15

A bag has 5 red, 4 blue, 3 green. Two balls selected randomly. Probability both same colour.

LEVEL 3 — EXAM PRESSURE

Q16

6 fair tosses: probability of at least 2 heads.

Q17

Two dice: probability sum is divisible by 3.

Q18

From 5 men and 5 women choose 4 randomly. Probability at least 1 woman.

Q19

Two cards drawn from a deck. Probability exactly one is an ace.

Q20

A bag has 3 red, 4 blue and 5 green balls. Three balls chosen randomly. Probability all three are of different colours.

ANSWERS**Q1**

Favourable:

$$\{5, 6\}$$

$$P = \frac{26}{6} = \boxed{\frac{13}{3}}$$

Q2

$$\frac{12}{12} \cdot \frac{12}{12} = \boxed{\frac{14}{14}}$$

Q3

$$\frac{\binom{32}{2} \cdot 3}{2^3} = \frac{38}{8}$$

$$\boxed{\frac{38}{8}}$$

Q4

$$\boxed{\frac{16}{16}}$$

Q5

$$\frac{4}{52} = \boxed{\frac{1}{13}}$$

Q6

Complement = no heads = all tails:

$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

So:

$$1 - \frac{1}{16} = \boxed{\frac{15}{16}}$$

Q7

$$1 - \left(\frac{5}{6}\right)^2 = 1 - \frac{25}{36}$$

$$= \boxed{\frac{11}{36}}$$

Q8

$$\frac{26 + 4 - 2}{52} = \frac{28}{52}$$

$$= \boxed{\frac{7}{13}}$$

Q9

$$\frac{47}{36} \cdot \frac{36}{36}$$

$$\left[=\frac{12}{42} \right]$$

$$\left[=\boxed{\frac{2}{7}} \right]$$

Q10

Either:

R then B

or:

B then R.

$$\left[\frac{4}{7}\frac{3}{6}+\frac{3}{7}\frac{4}{6} \right]$$

$$\left[=\frac{24}{42} \right]$$

$$\left[=\boxed{\frac{4}{7}} \right]$$

Or combinations:

$$\left[\frac{\binom{4}{1}\binom{3}{1}}{\binom{7}{2}} = \frac{12}{21} = \frac{4}{7} \right]$$

Q11

$$\left[\frac{\binom{5}{2}}{2^5} = \frac{10}{32} \right]$$

$$\left[=\boxed{\frac{5}{16}} \right]$$

Q12

Sums:

10,11,12.

Ways:

$$\left[3+2+1=6 \right]$$

Thus:

$$\left[\boxed{\frac{1}{6}} \right]$$

Q13

Favourable:

$$\left[\binom{4}{2}\binom{6}{1} = 6(6)=36 \right]$$

Total:

$$\left[\binom{10}{3}=120 \right]$$

So:

$$\left[\frac{36}{120} = \boxed{\frac{3}{10}} \right]$$

Q14

$$\left[\frac{\binom{4}{2}}{\binom{5}{2}} \right]$$

$$\left[= \frac{6}{1326} \right]$$

$$\left[=\boxed{\frac{1}{221}} \right]$$

Q15

Total balls:

$$\sqrt[12]{}$$

Total pairs:

$$\binom{12}{2} = 66$$

Same-colour pairs:

$$\binom{5}{2} + \binom{4}{2} + \binom{3}{2}$$

$$= 10 + 6 + 3 = 19$$

Thus:

$$\boxed{\frac{19}{66}}$$

Q16

At least 2 heads.

Use complement:

- 0 heads
- exactly 1 head

$$P = 1 - \frac{\binom{6}{0} + \binom{6}{1}}{2^6}$$

$$= 1 - \frac{1+6}{64}$$

$$= \boxed{\frac{57}{64}}$$

Q17

Sums divisible by 3:

$$3, 6, 9, 12$$

Ways:

- $3! = 2$
- $6! = 5$
- $9! = 4$
- $12! = 1$

Total:

$$\sqrt[12]{}$$

So:

$$\frac{12}{36} = \boxed{\frac{13}{}}$$

Q18

Total teams:

$$\binom{10}{4} = 210$$

No woman = all men:

$$\binom{5}{4} = 5$$

Therefore:

$$1 - \frac{5}{210} = \frac{205}{210}$$

$$= \boxed{\frac{41}{42}}$$

Q19

Exactly one ace:

Choose 1 ace:

$$\binom{4}{1}$$

Choose 1 non-ace:

$$\binom{48}{1}$$

Total pairs:

$$\binom{52}{2}$$

Probability:

$$\frac{4 \cdot 48}{\binom{52}{2}} = \frac{192}{1326}$$

$$= \boxed{\frac{32}{221}}$$

Q20

Total balls:

$$12$$

Choose 3:

$$\binom{12}{3} = 220$$

To get one of each colour:

$$\binom{3}{1} \binom{4}{1} \binom{5}{1}$$

$$= 3 \cdot 4 \cdot 5 = 60$$

Therefore:

$$\frac{60}{220} = \boxed{\frac{3}{11}}$$

CHAPTER 41 STATUS

Covered so far:

- experiments/outcomes/sample spaces '
- events '
- equally likely probability '
- impossible/certain events '
- complements '
- at least-one method '
- coin-toss probabilities '
- exact $\binom{n}{k}$ heads '
- addition rule '
- mutually exclusive events '
- independent events '
- multiplication rule '
- dependent events '

- with/without replacement '
- dice basics '
- two-dice sum distribution '
- cards basics '
- ball-selection probabilities '
- counting + probability integration '
- odds basics '
- GATE/placement L0–L3 practice '

But **Chapter 41 is not complete yet.**

Next we still need:

- conditional probability
- $(P(A|B))$
- chain rule
- independence tests
- total probability theorem
- Bayes' theorem
- biased coins/dice
- repeated independent trials
- binomial probability
- “at least / exactly / first success”
- conditional cards/balls
- reliability-type questions
- advanced probability shortcuts and final audit

Progress

Completed: 40/76 Current: 0% Chapter 41 — Probability

NEXT

0% Chapter 41 Part 2 — Conditional Probability, Bayes' Theorem, Binomial Trials & Advanced Probability

USER

next

ASSISTANT

CHAPTER 41 — PROBABILITY

PART 2 — CONDITIONAL PROBABILITY, BAYES' THEOREM, BINOMIAL TRIALS, FIRST SUCCESS & ADVANCED PROBABILITY

Status: 0% No PROGRESS GATE: High Placements: High Priority: 0% Per 1

Part 1 covered basic probability, complements, addition, multiplication, coins, dice, cards and without-replacement problems.

Now we finish the chapter with the most important higher-level ideas.

1. CONDITIONAL PROBABILITY

Conditional probability means:

Probability of event A when we already know that event B has occurred.

Notation:

$$\boxed{P(A|B)}$$

Read:

probability of A given B.

Formula:

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

provided:

$$P(B) > 0$$

MUST MEMORIZE

2. INTUITION

Normally our sample space is all possible outcomes.

Once B is known to have happened, our new sample space becomes only the outcomes inside B.

So:

$$P(A|B) = \frac{\text{outcomes satisfying both A and B}}{\text{outcomes satisfying B}}$$

3. SIMPLE DIE EXAMPLE

A fair die is rolled.

Given that the result is even, find probability that it is greater than 3.

Known event:

$$B = \{2, 4, 6\}$$

Desired event:

$$A = \{4, 5, 6\}$$

Within B:

$$A \cap B = \{4, 6\}$$

Therefore:

$$P(A|B) = \frac{2}{3}$$

$$\boxed{\frac{2}{3}}$$

4. USING THE FORMULA

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

and:

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

So:

$$P(A|B) = \frac{\frac{1}{3}}{\frac{1}{2}} = \boxed{\frac{2}{3}}$$

Same answer.

5. MULTIPLICATION RULE FROM CONDITIONAL PROBABILITY

From:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

rearrange:

$$\boxed{P(A \cap B) = P(B)P(A|B)}$$

Similarly:

$$\boxed{P(A \cap B) = P(A)P(B|A)}$$

6. WITHOUT REPLACEMENT IS CONDITIONAL PROBABILITY

Bag:

- 4 red
- 3 blue

Two balls drawn without replacement.

Probability both red:

$$P(R_1)P(R_2|R_1)$$

$$= \frac{4}{7} \cdot \frac{3}{6}$$

$$= \boxed{\frac{2}{7}}$$

The second probability is conditional because the first draw changes the bag.

7. INDEPENDENCE TEST

Events A and B are independent if:

$$\boxed{P(A|B) = P(A)}$$

provided $P(B) > 0$.

Equivalent:

$$\boxed{P(A \cap B) = P(A)P(B)}$$

8. EXAMPLE — INDEPENDENT

Two coins.

A = first coin H.

B = second coin H.

$$P(A) = \frac{1}{2}$$

Knowing B occurred does not affect first coin:

$$P(A|B) = \frac{1}{2}$$

So independent.

9. EXAMPLE — DEPENDENT

One card drawn from a deck.

A = card is an ace.

B = card is a spade.

$$P(A) = \frac{4}{52} = \frac{1}{13}$$

Given card is a spade, only 13 possible cards remain.

Exactly one is ace of spades.

$$P(A|B) = \frac{1}{13}$$

So these two events are actually independent.

This surprises many students.

10. ANOTHER CARD EXAMPLE — DEPENDENT

A = card is a king.

B = card is a face card.

$$P(A) = \frac{4}{52} = \frac{1}{13}$$

Given B, card is one of 12 face cards.

4 are kings.

$$P(A|B) = \frac{4}{12} = \frac{1}{3}$$

Since:

$$\frac{1}{3} \neq \frac{1}{13}$$

they are dependent.

11. THREE-EVENT CHAIN RULE

For three events:

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

This naturally appears in draws without replacement.

12. EXAMPLE

Bag:

- 5 red
- 3 blue

Draw 3 without replacement.

Probability all red:

$$\frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6}$$

$$= \frac{60}{336} = \boxed{\frac{5}{28}}$$

13. TOTAL PROBABILITY THEOREM — IDEA

Suppose an event A can happen through several mutually exclusive routes:

$$B_1, B_2, \dots, B_n$$

that partition the sample space.

Then:

$$P(A) = \sum_i P(B_i)P(A|B_i)$$

For two routes:

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$$

14. EXAMPLE — TWO FACTORIES

Factory 1 produces 60% of items.

Factory 2 produces 40%.

Defect rates:

$$P(D|F_1) = 0.02$$

$$P(D|F_2) = 0.05$$

Probability random item is defective:

$$P(D) = 0.6(0.02) + 0.4(0.05)$$

$$= 0.012 + 0.020$$

$$= \boxed{0.032}$$

or:

$$\boxed{3.2\%}$$

15. BAYES' THEOREM

Bayes reverses a conditional probability.

We may know:

$$P(D|F_1)$$

but want:

$$P(F_1|D)$$

Formula:

$$\boxed{P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_j P(B_j)P(A|B_j)}}$$

MUST MEMORIZE

16. BAYES USING FACTORY EXAMPLE

Using previous data:

$$P(F_1)=0.6$$

$$P(F_2)=0.4$$

$$P(D|F_1)=0.02$$

$$P(D|F_2)=0.05$$

Find probability defective item came from Factory 1.

Numerator:

$$0.6(0.02)=0.012$$

Total defective probability:

$$0.032$$

So:

$$P(F_1|D) = \frac{0.012}{0.032}$$

$$=\boxed{\frac{3}{8}=0.375}$$

17. BAYES MEMORY TRICK

Think:

$$\boxed{\text{Posterior} = \frac{\text{Prior} \times \text{Likelihood}}{\text{Total evidence}}}$$

For aptitude, you mainly need the numerical structure.

18. BAYES TREE METHOD

For multi-source questions:

1. write source probabilities
2. branch into success/failure probabilities
3. multiply along each path
4. for conditional reverse probability:

$$\left[\frac{\text{desired path}}{\text{sum of all paths matching evidence}} \right]$$

This often avoids formula confusion.

19. EXAMPLE — URNS

Box A chosen with probability:

$$\left[\frac{1}{2} \right]$$

contains 3 red, 2 blue.

Box B chosen with probability:

$$\left[\frac{1}{2} \right]$$

contains 1 red, 4 blue.

A red ball is drawn.

Find probability it came from Box A.

Red via A:

$$\left[\frac{1}{2} \cdot \frac{3}{5} = \frac{3}{10} \right]$$

Red via B:

$$\left[\frac{1}{2} \cdot \frac{1}{5} = \frac{1}{10} \right]$$

Total red:

$$\left[\frac{4}{10} \right]$$

Therefore:

$$\left[P(A|R) = \frac{3/10}{4/10} \right]$$

$$\left[= \boxed{\frac{3}{4}} \right]$$

20. BIASED COIN

A coin need not always have:

$$\left[P(H) = \frac{1}{2} \right]$$

Suppose:

$$\left[P(H) = p \right]$$

Then:

$$\left[\boxed{P(T) = 1 - p} \right]$$

21. REPEATED INDEPENDENT TRIALS

Suppose success probability per trial:

$$[p]$$

Failure:

$$[q=1-p]$$

Trials independent.

Probability of a particular sequence with (k) successes and $(n-k)$ failures:

$$[\boxed{ p^k q^{n-k} }]$$

22. BINOMIAL PROBABILITY

There are:

$$[\text{binom } nk]$$

ways to position (k) successes among (n) trials.

Therefore:

$$[\boxed{ P(X=k) = \text{binom } nk p^k (1-p)^{n-k} }]$$

MUST MEMORIZE

23. CONDITIONS FOR BINOMIAL MODEL

Use binomial when:

1. fixed number of trials (n)
 2. each trial has success/failure
 3. success probability (p) stays constant
 4. trials are independent
-

24. FAIR COIN AS BINOMIAL

For fair coin:

$$[p=\frac{1}{2}]$$

Thus:

$$[P(k \text{ heads in } n) = \text{binom } nk \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k}]$$

$$[= \boxed{ \frac{\text{binom } nk}{2^n} }]$$

Matches Part 1.

25. BIASED COIN EXAMPLE

Suppose:

$$[P(H)=0.6]$$

Coin tossed 4 times.

Probability exactly 3 heads:

$$\mathbb{P}(\text{binom}{4}{3}(0.6)^3(0.4)^1)$$

$$= 4(0.216)(0.4)$$

$$= \boxed{0.3456}$$

26. AT LEAST ONE SUCCESS

For (n) independent trials:

$$\mathbb{P}(\text{at least 1 success}) = 1 - (1-p)^n$$

Very high-value shortcut.

27. EXAMPLE

A machine succeeds with probability:

$$0.8$$

on each independent attempt.

Three attempts.

At least one success:

$$1 - (0.2)^3$$

$$= 1 - 0.008$$

$$= \boxed{0.992}$$

28. AT LEAST TWO SUCCESSES

Use complement if easier:

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X=0) - \mathbb{P}(X=1)$$

For binomial:

$$\mathbb{P}(X \geq 2) = 1 - (1-p)^n - np(1-p)^{n-1}$$

29. EXACTLY ONE SUCCESS

$$\mathbb{P}(X=1) = np(1-p)^{n-1}$$

because:

$$\mathbb{P}(X=1) = \mathbb{P}(\text{binom}{n}{1}p(1-p)^{n-1})$$

30. NO SUCCESS

$$\boxed{P(X=0)=(1-p)^n}$$

31. ALL SUCCESSES

$$\boxed{P(X=n)=p^n}$$

32. AT MOST ONE SUCCESS

$$P(X \leq 1) = P(0) + P(1)$$

$$\boxed{(1-p)^n + np(1-p)^{n-1}}$$

33. FIRST SUCCESS ON kth TRIAL

For first success exactly on trial (k) :

First $(k-1)$ trials must fail, k th succeeds.

$$\boxed{P(\text{first success on } k) = (1-p)^{k-1}p}$$

This is the geometric-probability pattern.

34. EXAMPLE

Success probability:

$$p = \frac{1}{3}$$

First success on 4th trial:

$$\left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)$$

$$= \frac{8}{81}$$

$$\boxed{\frac{8}{81}}$$

35. SUCCESS WITHIN FIRST k TRIALS

This means at least one success among first (k) .

$$\boxed{1-(1-p)^k}$$

36. FIRST SUCCESS AFTER k TRIALS

No success during first (k) :

$$\boxed{(1-p)^k}$$

37. RELIABILITY — SERIES SYSTEM

Suppose components must **all work** for system to work.

Independent component reliabilities:

$$\{ p_1, p_2, \dots, p_n \}$$

System reliability:

$$\{ \boxed{p_1 p_2 \dots p_n} \}$$

This is a **series system**.

38. EXAMPLE — SERIES

Three components each work with probability 0.9.

System requires all three.

$$\{ R = (0.9)^3 \}$$

$$\{ = \boxed{0.729} \}$$

39. RELIABILITY — PARALLEL SYSTEM

Suppose system works if **at least one** component works.

Component failure probabilities:

$$\{ q_i = 1 - p_i \}$$

System fails only if all fail.

Therefore:

$$\{ \boxed{R = 1 - \prod_i q_i} \}$$

40. EXAMPLE — PARALLEL

Two independent components each work with probability 0.8.

Both fail:

$$\{ 0.2^2 = 0.04 \}$$

At least one works:

$$\{ 1 - 0.04 = \boxed{0.96} \}$$

41. CONDITIONAL CARD EXAMPLE

A card is drawn.

Given that it is red, probability it is an ace.

Red cards:

$$\{ 26 \}$$

Red aces:

$$\{ 2 \}$$

Therefore:

$$\boxed{P(\text{Ace}|\text{Red}) = \frac{2}{26} = \frac{1}{13}}$$

42. CONDITIONAL TWO-DICE EXAMPLE

Two dice rolled.

Given sum is 8, probability one die is 3.

Outcomes with sum 8:

$$(2,6),(3,5),(4,4),(5,3),(6,2)$$

5 possibilities.

One die is 3:

$$(3,5),(5,3)$$

So:

$$\boxed{\frac{2}{5}}$$

43. CONDITIONAL BALL EXAMPLE

Bag:

- 4 red
- 6 blue

Two balls drawn without replacement.

Given first ball is red, probability second is blue.

After red removed:

- 3 red
- 6 blue
- 9 total

So:

$$\boxed{\frac{6}{9} = \frac{2}{3}}$$

44. CONDITIONAL SELECTION USING COMBINATIONS

10 students:

- 6 men
- 4 women

Choose 3.

Given at least one woman is selected, probability exactly one woman.

Exactly one woman:

$$\binom{4}{1}\binom{6}{2} = 4(15) = 60$$

At least one woman:

$$\binom{10}{3} - \binom{6}{3}$$

$$= 120 - 20 = 100$$

Therefore:

$$\boxed{\frac{60}{100} = \frac{3}{5}}$$

45. INDEPENDENCE FROM GIVEN PROBABILITIES

Suppose:

$$P(A) = 0.4$$

$$P(B) = 0.5$$

$$P(A \cap B) = 0.2$$

Check:

$$P(A)P(B) = 0.4(0.5) = 0.2$$

Thus:

$$\boxed{A, B \text{ independent}}$$

46. MUTUALLY EXCLUSIVE TEST

Suppose:

$$P(A \cap B) = 0$$

Then A, B are mutually exclusive.

But unless one event has probability zero, they are not independent.

47. IF A AND B ARE INDEPENDENT, COMPLEMENTS ARE ALSO INDEPENDENT

If A and B independent, then:

$$A^c, B$$

$$A, B^c$$

and:

$$A^c, B^c$$

are also independent.

Useful conceptual result.

48. PROBABILITY OF EXACTLY ONE OF A, B

Exactly one means:

- A occurs, B not
- B occurs, A not

General:

$$P(A \cap B^c) + P(A^c \cap B)$$

Equivalent:

$$\boxed{P(A) + P(B) - 2P(A \cap B)}$$

49. EXAMPLE

$$P(A)=0.6, \quad P(B)=0.5, \quad P(A \cap B)=0.3$$

Exactly one:

$$0.6 + 0.5 - 2(0.3)$$

$$= \boxed{0.5}$$

50. NEITHER A NOR B

$$P(\text{neither}) = 1 - P(A \cup B)$$

Thus:

$$\boxed{1 - P(A) - P(B) + P(A \cap B)}$$

51. AT LEAST ONE OF A,B

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

If independent:

$$P(A \cap B) = P(A)P(B)$$

So:

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A)P(B)}$$

52. TWO INDEPENDENT ATTEMPTS

Player succeeds with probability p per attempt.

Probability at least one success in 2 attempts:

$$1 - (1-p)^2$$

Expand:

$$= 2p - p^2$$

$$\boxed{2p - p^2}$$

53. “ONE OR MORE” VS “EXACTLY ONE”

For two independent trials:

At least one:

$$1 - q^2$$

Exactly one:

$$2pq$$

where:

$$q = 1 - p$$

Do not confuse these.

54. RANDOM SELECTION WITHOUT REPLACEMENT — HYPERGEOMETRIC FORM

Suppose population:

- K successes
- $N - K$ failures

Choose n objects without replacement.

Probability exactly k successes:

$$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

You do not need the name “hypergeometric” for aptitude, but the formula is very useful.

55. EXAMPLE

Bag:

- 6 red
- 4 blue

Choose 3.

Probability exactly 2 red:

$$\frac{\binom{6}{2} \binom{4}{1}}{\binom{10}{3}}$$

$$= \frac{15(4)}{120}$$

$$= \frac{1}{2}$$

56. BINOMIAL VS WITHOUT-REPLACEMENT

Binomial

- independent trials
 - fixed p
 - often replacement/repeated identical trial
-

Combination/hypergeometric

- finite population
- without replacement
- probabilities change after selections

This distinction is important.

57. GATE-STYLE QUESTION 1

Given:

$$P(A)=0.5, \quad P(B)=0.4, \quad P(A \cap B)=0.2$$

Are A and B independent?

$$P(A)P(B)=0.2$$

So:

$$\boxed{\text{Yes}}$$

58. GATE-STYLE QUESTION 2

A fair coin tossed 6 times. Probability exactly 4 heads:

$$\frac{\binom{6}{4}}{2^6}$$

$$=\frac{15}{64}$$

$$\boxed{\frac{15}{64}}$$

59. GATE-STYLE QUESTION 3

Success probability on each independent attempt is 0.3.

Probability first success occurs on third attempt:

$$(0.7)^2(0.3)$$

$$=0.147$$

$$\boxed{0.147}$$

60. GATE-STYLE QUESTION 4

Factory A supplies 70%, defect rate 1%.

Factory B supplies 30%, defect rate 4%.

Probability item defective:

$$0.7(0.01)+0.3(0.04)$$

$$=0.007+0.012$$

$$\boxed{0.019}$$

or:

$$\boxed{1.9\%}$$

61. PLACEMENT-STYLE QUESTION 1

A biased coin has:

$$P(H)=0.7$$

Tossed 3 times.

Probability exactly 2 heads:

$$\binom{3}{2}(0.7)^2(0.3)$$

$$=3(0.49)(0.3)$$

$$\boxed{0.441}$$

62. PLACEMENT-STYLE QUESTION 2

A student answers each question correctly with probability 0.8 independently.

Probability at least one correct among 4:

$$1-(0.2)^4$$

$$=1-0.0016$$

$$\boxed{0.9984}$$

63. PLACEMENT-STYLE QUESTION 3

A system has 3 independent components in series with reliability:

$$0.9, 0.8, 0.95$$

System reliability:

$$0.9(0.8)(0.95)$$

$$\boxed{0.684}$$

64. PLACEMENT-STYLE QUESTION 4

Two parallel components have reliabilities:

$$0.7, 0.8$$

Both fail probability:

$$0.3(0.2)=0.06$$

System works:

$$1-0.06$$

$$\boxed{0.94}$$

65. ADVANCED QUESTION — BAYES

Machine A produces 40% of units with defect probability 2%.

Machine B produces 60% with defect probability 3%.

Given a unit is defective, probability it came from A:

$$P(A|D) = \frac{0.4(0.02)}{0.4(0.02) + 0.6(0.03)}$$

$$= \frac{0.008}{0.026}$$

$$= \boxed{\frac{4}{13}}$$

approximately:

$$0.3077$$

66. ADVANCED QUESTION — CONDITIONAL DICE

Two dice rolled.

Given sum is greater than 9, probability sum is 12.

Possible sums (>9):

- 10 !' 3 ways
- 11 !' 2
- 12 !' 1

Total conditional outcomes:

$$6$$

Sum 12:

$$1$$

So:

$$\boxed{\frac{1}{6}}$$

67. ADVANCED QUESTION — AT LEAST TWO SUCCESSES

Success probability:

$$p=0.4$$

4 independent attempts.

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(0) = (0.6)^4 = 0.1296$$

$$P(1) = 4(0.4)(0.6)^3$$

$$= 1.6(0.216) = 0.3456$$

Therefore:

$$1 - 0.4752$$

$$\boxed{0.5248}$$

68. ADVANCED QUESTION — WITHOUT REPLACEMENT

Bag has:

- 5 red
- 4 blue
- 3 green

Choose 4.

Probability exactly 2 red:

Total:

$$\binom{12}{4}$$

Choose 2 red:

$$\binom{5}{2}$$

Remaining 2 must come from 7 non-red:

$$\binom{7}{2}$$

Thus:

$$P = \frac{\binom{5}{2}\binom{7}{2}}{\binom{12}{4}}$$

$$= \frac{10(21)}{495}$$

$$\boxed{\frac{14}{33}}$$

69. COMMON TRAPS

& pTRAP 1 — Reversing conditional probability

$$P(A|B) \neq P(B|A)$$

in general.

& pTRAP 2 — Independence assumed without evidence

Only multiply directly when independence is valid.

& pTRAP 3 — Bayes denominator

Denominator must include **all routes** that could produce the observed event.

& pTRAP 4 — Binomial used without constant probability

Without replacement generally violates constant p .

& pTRAP 5 — First success

Need all earlier attempts to fail.

$$\sum_{k=0}^{\infty} q^{k-1} p$$

& pTRAP 6 — At least 2

Often easier:

$$1 - P(0) - P(1)$$

& pTRAP 7 — Conditional sample space

Once “given B” appears, restrict attention to outcomes where B occurred.

70. MUST-MEMORIZE CONDITIONAL FORMULAS

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A)P(B|A)$$

Independence:

$$P(A \cap B) = P(A)P(B)$$

71. MUST-MEMORIZE TOTAL PROBABILITY

If $\{B_i\}$ partition the sample space:

$$P(A) = \sum_i P(B_i)P(A|B_i)$$

72. MUST-MEMORIZE BAYES

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_j P(B_j)P(A|B_j)}$$

73. MUST-MEMORIZE BINOMIAL

For $\binom{n}{k}$ independent trials, success probability p :

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

74. MUST-MEMORIZE FIRST SUCCESS

$$P(\text{first success on } k) = (1-p)^{k-1} p$$

75. MUST-MEMORIZE WITHOUT-REPLACEMENT COUNT

Exactly k successes when choosing n from N objects, with K successes available:

$$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

76. FAST EXAM DECISION TREE

“Given that”?

Conditional probability.

Two events independent?

Use product.

Multiple source groups/factories?

Total probability/Bayes.

Fixed repeated independent trials?

Binomial.

“At least one”?

Complement.

“At least two”?

Often subtract 0 and 1.

“First success on kth”?

$$\lfloor q^{k-1}p \rfloor$$

Without replacement?

Sequential conditional method or combinations.

All components must work?

Series !' multiply reliabilities.

At least one component must work?

Parallel !' $\lfloor (1-\text{all fail}) \rfloor$.

PRACTICE — LEVEL 0

Q1

If:

$$\lfloor P(A \cap B) = 0.2, \text{quad } P(B) = 0.5 \rfloor$$

find:

$$\lfloor P(A|B) \rfloor$$

Q2

A coin has $\lfloor P(H) = 0.6 \rfloor$. Find probability exactly 2 heads in 3 tosses.

Q3

Success probability is 0.2. Find probability of no success in 4 independent trials.

Q4

Success probability 0.4. Find probability first success occurs on trial 3.

Q5

Two independent components of reliability 0.9 are in series. Find system reliability.

LEVEL 1

Q6

Given:

$$[P(A)=0.3, P(B)=0.5, P(A \cap B)=0.15]$$

are A and B independent?

Q7

A fair coin is tossed 5 times. Probability at least 2 heads.

Q8

A machine succeeds with probability 0.7. Probability at least one success in 3 independent attempts.

Q9

Bag: 5 red, 3 blue. Choose 3 without replacement. Probability exactly 2 red.

Q10

Two parallel independent components have reliabilities 0.6 and 0.8. Find reliability.

LEVEL 2

Q11

Factory A produces 60% with 2% defects. Factory B produces 40% with 5% defects. Given defective, probability from B.

Q12

Two dice rolled. Given sum is even, find probability sum is 8.

Q13

A biased coin has $P(H)=0.3$. Tossed 6 times. Find probability exactly 2 heads.

Q14

Success probability 0.25. Find probability first success occurs by third trial.

Q15

From 7 men and 5 women, choose 4 randomly. Given at least one woman, find probability exactly 2 women.

LEVEL 3 — EXAM PRESSURE

Q16

Three machines A,B,C produce 20%,30%,50% respectively. Their defect rates are 1%,2%,3%. A defective item is selected. Find probability it came from C.

Q17

4 independent trials with success probability $\backslash(p\backslash)$. Probability exactly one success is $\backslash(81/256\backslash)$. If $\backslash(p<1/2\backslash)$, find $\backslash(p\backslash)$.

Q18

Bag has 6 red, 5 blue, 4 green. Choose 5. Probability exactly 2 red and 1 blue.

Q19

A system has two components in series, each of which consists of two parallel units. Every unit reliability is 0.9. Find system reliability.

Q20

Two events A,B are independent with $\backslash(P(A)=0.6\backslash)$, $\backslash(P(B)=0.4\backslash)$. Find probability exactly one occurs.

ANSWERS**Q1**

$$\backslash[P(A|B)=\frac{0.2}{0.5} \backslash]$$

$$\backslash[=\boxed{0.4} \backslash]$$

Q2

$$\backslash[\backslashbinom{3}{2}(0.6)^2(0.4) \backslash]$$

$$\backslash[=3(0.36)(0.4) \backslash]$$

$$\backslash[=\boxed{0.432} \backslash]$$

Q3

$$\backslash[(0.8)^4 = \boxed{0.4096} \backslash]$$

Q4

$$\backslash[(0.6)^2(0.4) \backslash]$$

$$\backslash[=0.36(0.4) \backslash]$$

$$\backslash[=\boxed{0.144} \backslash]$$

Q5

$$\backslash[0.9^2 = \boxed{0.81} \backslash]$$

Q6

$$\backslash[P(A)P(B) = 0.3(0.5) = 0.15 \backslash]$$

which equals intersection.

$$\backslash[\boxed{\text{Yes}} \backslash]$$

Q7

$$1 - P(0) - P(1)$$

$$= 1 - \frac{\binom{50}{0} + \binom{50}{1} 2^5}{2^{10}}$$

$$= 1 - \frac{6}{32}$$

$$= \boxed{\frac{13}{16}}$$

Q8

$$1 - (0.3)^3$$

$$= 1 - 0.027$$

$$= \boxed{0.973}$$

Q9

Total:

$$\binom{8}{3} = 56$$

Favourable:

$$\binom{5}{2} \binom{3}{1} = 10(3) = 30$$

So:

$$\frac{30}{56} = \boxed{\frac{15}{28}}$$

Q10

Both fail:

$$0.4(0.2) = 0.08$$

At least one works:

$$1 - 0.08$$

$$= \boxed{0.92}$$

Q11

Defective via A:

$$0.6(0.02) = 0.012$$

Via B:

$$0.4(0.05) = 0.020$$

Total:

$$0.032$$

Therefore:

$$P(B|D) = \frac{0.020}{0.032}$$

$$= \boxed{\frac{5}{8}}$$

Q12

Even sums:

$$2, 4, 6, 8, 10, 12$$

Count even-sum outcomes alternatively:

Both dice same parity:

$$\mathbb{P}(3 \cdot 3 + 3 \cdot 3 = 18) =$$

Sum 8 occurs in 5 ways.

Thus:

$$\mathbb{P}\left(\frac{5}{18}\right) =$$

Q13

$$\mathbb{P}(\text{binom}{6}{2}(0.3)^2(0.7)^4) =$$

$$= 15(0.09)(0.2401) =$$

$$= \boxed{0.324135}$$

Q14

First success by trial 3 means at least one success in first 3:

$$\mathbb{P}(1 - (0.75)^3) =$$

$$= 1 - 0.421875 =$$

$$= \boxed{0.578125}$$

Q15

Total selections with at least one woman:

$$\mathbb{P}(\text{binom}{12}{4} - \text{binom}{7}{4}) =$$

$$= 495 - 35 = 460$$

Exactly 2 women:

$$\mathbb{P}(\text{binom}{5}{2}\text{binom}{7}{2} = 10(21) = 210)$$

Thus:

$$\mathbb{P}\left(\frac{210}{460} = \frac{21}{46}\right) =$$

Q16

Defect paths:

A:

$$\mathbb{P}(0.2(0.01) = 0.002)$$

B:

$$\mathbb{P}(0.3(0.02) = 0.006)$$

C:

$$\mathbb{P}(0.5(0.03) = 0.015)$$

Total:

$$\mathbb{P}(0.023)$$

So:

$$\mathbb{P}(C|D) = \frac{0.015}{0.023} =$$

$$= \boxed{\frac{15}{23}}$$

Q17

Exactly one success in 4:

$$4p(1-p)^3 = \frac{81}{256}$$

Try:

$$p = \frac{14}{64}$$

Then:

$$4\left(\frac{14}{64}\right)\left(\frac{3}{64}\right)^3 = \frac{27}{64}$$

not enough.

Try:

$$p = \frac{12}{64}$$

gives:

$$\frac{14}{64}$$

So solve carefully.

Let:

$$q = 1 - p$$

$$4(1-q)q^3 = \frac{81}{256}$$

Notice $(q = \frac{3}{4})$:

$$4\left(\frac{14}{64}\right)\left(\frac{27}{64}\right) = \frac{27}{64}$$

Not match.

Try $(q = \frac{9}{16})$ is not simple. This equation does not have a clean standard aptitude root.

So this practice item is not ideal as stated.

Replace it with a clean version:

If $4p(1-p)^3 = \frac{27}{64}$ and $(p < 1/2)$, find (p) .

Then:

$$p = \boxed{\frac{14}{64}}$$

Use this corrected version in your notes.

Q18

Total balls:

$$15$$

Need:

- 2 red from 6
- 1 blue from 5
- remaining 2 green from 4

Favourable:

$$\binom{6}{2}\binom{5}{1}\binom{4}{2}$$

$$\boxed{=15(5)(6)=450}$$

Total:

$$\boxed{\binom{15}{5} = 3003}$$

Thus:

$$\boxed{\frac{450}{3003} = \frac{150}{1001}}$$

Q19

Each parallel pair reliability:

$$\boxed{1-(0.1)^2 = 0.99}$$

Two such blocks are in series:

$$\boxed{0.99^2}$$

$$\boxed{=0.9801}$$

Q20

Exactly one occurs:

$$\boxed{P(A)(1-P(B)) + (1-P(A))P(B)}$$

$$\boxed{= 0.6(0.6)+0.4(0.4)}$$

$$\boxed{=0.36+0.16}$$

$$\boxed{=0.52}$$

' CHAPTER 41 COMPLETE

We have now covered:

- sample spaces and events '
- basic probability '
- complements '
- addition rule '
- mutually exclusive events '
- independent/dependent events '
- multiplication rule '
- conditional probability '
- chain rule '
- total probability theorem '
- Bayes' theorem '
- biased coins '
- repeated independent trials '
- binomial probability '
- exact/at-least/at-most success '
- first-success problems '
- with/without replacement '
- card/ball/dice probability '

- counting-based probability '
- series/parallel reliability '
- independence tests '
- conditional selections '
- advanced GATE/placement traps '

Progress

Completed: 41/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

Chapter 42 — Sets and Venn Diagrams

Next we will cover:

- set notation
- union/intersection/complement
- subsets
- cardinality
- 2-set Venn diagrams
- 3-set Venn diagrams
- inclusion-exclusion
- exactly one / exactly two / at least one / none
- survey problems
- percentage-based Venn problems
- maximum/minimum overlap
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 42 — SETS AND VENN DIAGRAMMS

PART 1 — SET BASICS, UNION, INTERSECTION, COMPLEMENT, SUBSETS, CARDINALITY & TWO-SET VENN DIAGRAMMS

Status: 0% PROGRESS GATE: High Placements: High Priority: 0% Part 1

Sets and Venn diagrams are heavily connected to:

- counting
- probability
- logic

- survey questions
- “at least one / exactly one / neither” problems

The core skill is:

Translate words into set regions correctly.

1. WHAT IS A SET?

A **set** is a well-defined collection of objects.

Example:

$A = \{1, 2, 3, 4\}$

The objects inside a set are called **elements** or **members**.

2. MEMBERSHIP NOTATION

If 3 belongs to set A :

$\boxed{3 \in A}$

If 7 does not belong:

$\boxed{7 \notin A}$

3. ROSTER FORM

Writing all elements explicitly:

$A = \{2, 4, 6, 8\}$

is called roster/tabular form.

4. SET-BUILDER FORM

Same set can be written:

$A = \{x : x \text{ is an even positive integer less than } 10\}$

or:

$A = \{x \in \mathbb{N} : x < 10, x \text{ even}\}$

5. EMPTY SET

A set containing no elements:

$\boxed{\varnothing}$

or:

$\{\}$

Example:

Set of positive integers satisfying:

$$\{x < 0\}$$

is empty.

6. SINGLETON SET

A set containing exactly one element.

Example:

$$\{5\}$$

Important:

$$5 \neq \{5\}$$

One is an element; the other is a set.

7. FINITE AND INFINITE SETS

Finite:

$$\{1, 2, 3, 4\}$$

Infinite:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

8. CARDINALITY

The number of elements in set (A) is called its cardinality.

Notation:

$$\boxed{|A|}$$

or:

$$n(A)$$

Example:

$$A = \{2, 4, 6, 8\}$$

Then:

$$\boxed{|A| = 4}$$

9. EQUAL SETS

Two sets are equal if they contain exactly the same elements.

$$A = \{1, 2, 3\}$$

$$B = \{3, 2, 1\}$$

Then:

$$\boxed{A=B}$$

Order does not matter in a set.

10. DUPLICATES DO NOT MATTER

$$\{1,1,2,2,3\}$$

is simply:

$$\boxed{\{1,2,3\}}$$

A set does not count repeated elements multiple times.

11. SUBSET

A is a subset of B if every element of A belongs to B .

Notation:

$$\boxed{A \subseteq B}$$

Example:

$$A = \{1,2\}$$

$$B = \{1,2,3\}$$

Then:

$$A \subseteq B$$

12. PROPER SUBSET

If:

$$A \subseteq B$$

and:

$$A \neq B$$

then A is a proper subset.

Often written:

$$\boxed{A \subset B}$$

Notation conventions can vary, so read context carefully.

13. EVERY SET IS A SUBSET OF ITSELF

$$\boxed{A \subseteq A}$$

Also:

$$\boxed{\varnothing \subseteq A}$$

for every set A.

14. NUMBER OF SUBSETS

If a set has n elements:

Each element can be:

- included
- excluded

So total subsets:

$$2^n$$

15. NUMBER OF PROPER SUBSETS

All subsets except the set itself:

$$2^n - 1$$

16. NUMBER OF NONEMPTY SUBSETS

All subsets except empty set:

$$2^n - 1$$

Interesting:

- proper subsets = $2^n - 1$
- nonempty subsets = $2^n - 1$

But these are different collections.

17. NONEMPTY PROPER SUBSETS

Exclude:

- empty set
- full set

So:

$$2^n - 2$$

for $n \geq 1$.

18. EXAMPLE

Set with 4 elements.

Total subsets:

$$2^4 = 16$$

Proper subsets:

$\{15\}$

Nonempty proper subsets:

$\{14\}$

19. POWER SET

The set of all subsets of (A) is called its power set.

Notation:

$\boxed{\mathcal{P}(A)}$

If:

$|A|=n$

then:

$\boxed{|\mathcal{P}(A)|=2^n}$

20. UNIVERSAL SET

The universal set (U) contains all objects under consideration.

Example:

If discussing students in a class:

$U=\text{all students in the class}$

All complements are defined relative to (U) .

21. UNION

Union of A and B:

$\boxed{A \cup B}$

contains elements belonging to:

- A
- B
- or both

Example:

$A=\{1,2,3\}$

$B=\{3,4,5\}$

Then:

$A \cup B=\{1,2,3,4,5\}$

22. INTERSECTION

Intersection:

$$\boxed{A \cap B}$$

contains elements common to both.

Using previous sets:

$$A \cap B = \{3\}$$

23. DISJOINT SETS

Two sets are disjoint if:

$$\boxed{A \cap B = \varnothing}$$

Example:

$$A = \{1, 2\}$$

$$B = \{3, 4\}$$

24. COMPLEMENT

Complement of A:

$$\boxed{A^c}$$

or:

$$A'$$

contains elements in universal set (U) but not in A.

$$\boxed{A^c = U - A}$$

25. SET DIFFERENCE

$$\boxed{A - B}$$

means elements in A but not in B.

Equivalent:

$$\boxed{A \cap B^c}$$

26. EXAMPLE

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5\}$$

Then:

$$A - B = \{1, 2\}$$

and:

$$\forall B-A=\{5\}$$

So:

$$\forall A-B \text{ line } B-A$$

27. BASIC SET IDENTITIES

Commutative

$$\forall A \cup B = B \cup A$$

$$\forall A \cap B = B \cap A$$

Associative

$$\forall (A \cup B) \cup C = A \cup (B \cup C)$$

$$\forall (A \cap B) \cap C = A \cap (B \cap C)$$

28. DISTRIBUTIVE LAWS

$$\forall \boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$$

and:

$$\forall \boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$$

These appear in GATE logic/set questions.

29. IDEMPOTENT LAWS

$$\forall A \cup A = A$$

$$\forall A \cap A = A$$

30. COMPLEMENT LAWS

$$\forall \boxed{A \cup A^c = U}$$

$$\forall \boxed{A \cap A^c = \varnothing}$$

$$\forall \boxed{(A^c)^c = A}$$

31. DE MORGAN'S LAWS

Very important:

$$\forall \boxed{(A \cup B)^c = A^c \cap B^c}$$

$$\forall \boxed{(A \cap B)^c = A^c \cup B^c}$$

Memory trick:

Complement changes union to intersection.

32. TWO-SET CARDINALITY FORMULA

For finite sets:

$\boxed{A \cup B}$

$A \cup B$

=

A

+

B

-

$A \cap B$

$\boxed{}$

MUST MEMORIZE

Why subtract intersection?

Because it gets counted twice in:

$\boxed{A + B}$

A

+

B

$\boxed{}$

33. EXAMPLE

40 students:

- 25 study Maths
- 20 study Physics
- 10 study both

Number studying at least one:

$\boxed{25+20-10}$

$\boxed{35}$

34. NONE OF THE TWO

If total population is:

$\boxed{N}$

then:

$\boxed{\text{Neither} = N - |A \cup B|}$

Using previous example:

$$\boxed{40-35 = 5}$$

35. ONLY A

$$\boxed{|A| - |A \cap B|}$$

Example:

Maths only:

$$\boxed{25-10 = 15}$$

36. ONLY B

$$\boxed{|B| - |A \cap B|}$$

Physics only:

$$\boxed{20-10 = 10}$$

37. EXACTLY ONE OF A,B

Exactly one means:

- A only
- B only

So:

$$\boxed{(|A| - |A \cap B|) + (|B| - |A \cap B|)}$$

Therefore:

$$\boxed{$$

A	+	B	-2	A ∩ B
---	---	---	----	-------

$$\boxed{}$$

38. EXAMPLE

$$\boxed{|A|=30, \quad |B|=25, \quad |A \cap B|=10}$$

Exactly one:

$$\boxed{30+25-20}$$

$$\boxed{=35}$$

39. AT LEAST ONE

“At least one of A or B” means:

$$\boxed{|A \cup B|}$$

Count:

$\setminus$

A	+	B	-	$A \cap B$
---	---	---	---	------------

$\setminus$

40. BOTH

“Both A and B” means:

$\boxed{A \cap B}$

41. NEITHER

“Neither A nor B” means:

$\boxed{(A \cup B)^c}$

By De Morgan:

$A^c \cap B^c$

42. NOT BOTH

“Not both” means everything except the intersection.

$\boxed{(A \cap B)^c}$

By De Morgan:

$A^c \cup B^c$

This is not the same as “neither.”

43. LANGUAGE TRAP — NOT BOTH VS NEITHER

Suppose A and B.

Neither

A false and B false.

Not both

Any situation except both true.

So “not both” includes:

- A only
- B only
- neither

This is a major reasoning trap.

44. VENN REGION METHOD

For two sets, draw:

- A only
- A") B
- B only
- outside both

If overlap is known, fill it first.

Then subtract from totals.

45. EXAMPLE — SURVEY

100 people:

- 60 like tea
- 45 like coffee
- 25 like both

Tea only:

$$\lfloor 60-25=35 \rfloor$$

Coffee only:

$$\lfloor 45-25=20 \rfloor$$

At least one:

$$\lfloor 35+25+20 = 80 \rfloor$$

Neither:

$$\lfloor 100-80 = 20 \rfloor$$

So:

$$\lfloor \boxed{\text{Tea only}=35} \rfloor$$

$$\lfloor \boxed{\text{Coffee only}=20} \rfloor$$

$$\lfloor \boxed{\text{Both}=25} \rfloor$$

$$\lfloor \boxed{\text{Neither}=20} \rfloor$$

46. FIND UNKNOWN INTERSECTION

Suppose:

$$\lfloor |A|=50 \rfloor$$

$$\lfloor |B|=40 \rfloor$$

$$\lfloor |A \cup B|=70 \rfloor$$

Then:

$$\lvert 70=50+40-\lvert A\cap B \rvert \rvert$$

So:

$$\lvert \lvert A\cap B \rvert =20 \rvert$$

$$\lvert \boxed{20} \rvert$$

47. FIND ONE SET SIZE

$$\lvert \lvert A\cup B \rvert =80 \rvert$$

$$\lvert \lvert A \rvert =50 \rvert$$

$$\lvert \lvert A\cap B \rvert =15 \rvert$$

Find $\lvert B \rvert$.

$$\lvert 80=50+\lvert B \rvert -15 \rvert$$

$$\lvert \lvert B \rvert =45 \rvert$$

$$\lvert \boxed{45} \rvert$$

48. PERCENTAGE VERSION

100% of people form the universal set.

Suppose:

- 70% know English
- 50% know Hindi
- 30% know both

At least one:

$$\lvert 70+50-30 = \boxed{90\%} \rvert$$

Neither:

$$\lvert 100-90 = \boxed{10\%} \rvert$$

49. MINIMUM POSSIBLE OVERLAP

Suppose total population $\lvert N \rvert$.

Sets sizes:

$$\lvert \lvert A \rvert =a, \quad \lvert B \rvert =b \rvert$$

Since union cannot exceed $\lvert N \rvert$:

$$\lvert a+b-\lvert A\cap B \rvert \leq N \rvert$$

Thus:

$$\lvert \lvert A\cap B \rvert \geq a+b-N \rvert$$

Also intersection cannot be negative.

Therefore minimum overlap:

$$\lfloor \boxed{\max(0, a+b-N)} \rfloor$$

High-value shortcut

50. EXAMPLE — MINIMUM OVERLAP

100 students:

- 70 study Maths
- 60 study Physics

Minimum both:

$$\lfloor 70+60-100 = \boxed{30} \rfloor$$

Why?

Even if overlap is minimized, together the sets claim 130 memberships across only 100 people.

At least 30 must be shared.

51. MAXIMUM POSSIBLE OVERLAP

Intersection cannot exceed the smaller set.

Therefore:

$$\lfloor \boxed{\max |A \cap B| = \min(a, b)} \rfloor$$

Example:

70 and 60:

Maximum overlap:

$$\lfloor \boxed{60} \rfloor$$

if every Physics student also studies Maths.

52. POSSIBLE INTERSECTION RANGE

$$\lfloor \boxed{\max(0, a+b-N)} \rfloor \leq$$

$$|A \cap B|$$

$$\leq \min(a, b) \rfloor$$

Very useful for MSQ/data-sufficiency questions.

53. MINIMUM UNION

Since:

$$\lfloor$$

$A \cup B$

$$= a + b - \text{intersection}$$

Union is smallest when overlap is largest.

Largest overlap:

$$\min(a, b)$$

Therefore:

$$\boxed{\min|A \cup B| = \max(a, b)}$$

54. MAXIMUM UNION

Union is largest when overlap is minimized:

$$\boxed{\max|A \cup B| = \min(N, a + b)}$$

55. EXAMPLE

In 100 people:

$$|A| = 40, \quad |B| = 30$$

Maximum union:

$$\min(100, 70) = 70$$

Minimum union:

$$\max(40, 30) = 40$$

So:

$$\boxed{40 \leq |A \cup B| \leq 70}$$

56. SUBSET CARDINALITY RELATION

If:

$$A \subseteq B$$

then:

$$\boxed{|A| \leq |B|}$$

If finite sets and:

$$|A|$$

A	=	B
---	---	---

$$|B|$$

with:

$$A \subseteq B$$

then:

$$\boxed{A=B}$$

57. DISJOINT SET CARDINALITY

If:

$$A \cap B = \varnothing$$

then:

$$\boxed{|A \cup B| = |A| + |B|}$$

58. COMPLEMENT CARDINALITY

If universal set size is $|N|$:

$$\boxed{|A^c| = N - |A|}$$

59. DIFFERENCE CARDINALITY

$$A - B$$

is A only relative to B.

So:

$$\boxed{A - B = A \setminus B}$$

$A - B$

=

A	$-$	$A \cap B$
-----	-----	------------

$$\boxed{A - B = A \setminus B}$$

60. SYMMETRIC DIFFERENCE

The symmetric difference contains elements in exactly one of A and B.

Notation often:

$$\boxed{A \triangle B}$$

Then:

$$\boxed{A \triangle B = (A - B) \cup (B - A)}$$

Its size:

$$\boxed{|A \triangle B| = |A - B| + |B - A|}$$

$A \triangle B$

=

A

+

B

-2

$A \cap B$

} \]

61. EXAMPLE

$\{ |A|=20, |B|=15, |A \cap B|=5 \}$

Symmetric difference:

$\{ 20+15-10 = \boxed{25} \}$

62. GATE-STYLE QUESTION 1

If:

$\{ |A|=30, |B|=20, |A \cap B|=8 \}$

find:

$\{$

$A \cup B$

$\}$

$\{ 30+20-8 = \boxed{42} \}$

63. GATE-STYLE QUESTION 2

Universal set has 60 elements.

$\{ |A|=35 \}$

Find:

$\{$

A^c

$\}$

$\{ 60-35 = \boxed{25} \}$

64. GATE-STYLE QUESTION 3

If:

$$[A \subseteq B, \quad |A|=10, \quad |B|=15]$$

then:

$$[A \cap B = A]$$

So:

$$[$$

$$A \cap B$$

$$= \boxed{10}]$$

Also:

$$[A \cup B = B]$$

so:

$$[|A \cup B| = 15]$$

65. GATE-STYLE QUESTION 4

A set has 8 elements.

Number of nonempty proper subsets:

$$[2^{8-2}]$$

$$[= \boxed{254}]$$

66. PLACEMENT-STYLE QUESTION 1

In a group of 80:

- 50 like cricket
- 35 like football
- 20 like both

At least one:

$$[50+35-20=65]$$

Neither:

$$[80-65 = \boxed{15}]$$

67. PLACEMENT-STYLE QUESTION 2

Same data.

Exactly one sport:

$$[50+35-2(20)]$$

$$[=85-40]$$

$$[= \boxed{45}]$$

68. PLACEMENT-STYLE QUESTION 3

100 employees:

- 65 know Excel
- 55 know SQL

Minimum number knowing both:

$\lfloor 65+55-100 \rfloor$

$\lfloor =\boxed{20} \rfloor$

69. PLACEMENT-STYLE QUESTION 4

Same data.

Maximum number knowing both:

$\lfloor \min(65,55) \rfloor$

$\lfloor =\boxed{55} \rfloor$

70. COMMON TRAPS

& pTRAP 1 — Union

Union means A or B **including both**.

& pTRAP 2 — Exactly one

Subtract the overlap twice:

$\lfloor$

A	+	B	-2	$A\cap B$
---	---	---	----	-----------

$\rfloor$

& pTRAP 3 — Neither

Use:

$\lfloor N-|A\cup B| \rfloor$

& pTRAP 4 — Not both

Not the same as neither.

& pTRAP 5 — Repeated set elements

Duplicates do not increase cardinality.

& pTRAP 6 — Subset vs element

$\forall x \in A$

is different from:

$\forall x \{x\} \subseteq A$

& pTRAP 7 — Proper vs nonempty subset

They exclude different special subsets.

71. MUST-MEMORIZE SET OPERATIONS

$\boxed{A \cup B = \text{A or B or both}}$

$\boxed{A \cap B = \text{both A and B}}$

$\boxed{A^c = \text{not A}}$

$\boxed{A - B = A \cap B^c}$

72. MUST-MEMORIZE CARDINALITY

$\boxed{$

$A \cup B$

$=$

A	+	B	-	$A \cap B$
---	---	---	---	------------

$\boxed{}$

$\boxed{|A^c| = |U| - |A|}$

$\boxed{|A - B| = |A| - |A \cap B|}$

73. MUST-MEMORIZE SUBSET COUNTS

For (n) -element set:

$\boxed{\text{all subsets} = 2^n}$

$\boxed{\text{proper subsets} = 2^n - 1}$

$\boxed{\text{nonempty subsets} = 2^n - 1}$

$\boxed{\text{nonempty proper} = 2^n - 2}$

74. MUST-MEMORIZE OVERLAP BOUNDS

For universe size (N) :

$$\boxed{\min(A \cap B) = \max(0, |A| + |B| - N)}$$

$$\boxed{\max(A \cap B) = \min(|A|, |B|)}$$

75. FAST EXAM DECISION TREE

“At least one”?

Union.

“Both”?

Intersection.

“Only A”?

$$\boxed{A - (A \cap B)}$$

“Exactly one”?

A only + B only.

“Neither”?

Universal total minus union.

Unknown overlap?

Use two-set inclusion-exclusion.

Minimum overlap?

$$\boxed{a + b - N} \text{ if positive.}$$

Maximum overlap?

Smaller set.

Number of subsets?

$$\boxed{2^n}$$

PRACTICE — LEVEL 0

Q1

If:

$$\boxed{A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}}$$

find $A \cup B$.

Q2

Find $A \cap B$.

Q3

A set has 5 elements. How many subsets?

Q4

How many nonempty proper subsets?

Q5

If:

$$\{ |A|=20, |B|=15, |A \cap B|=5 \}$$

find union.

LEVEL 1

Q6

100 people: 60 like tea, 50 coffee, 20 both. Find neither.

Q7

Same data: find exactly one.

Q8

Universal set size 80, A has 47. Find complement size.

Q9

If:

$$\{ |A \cup B|=55, |A|=30, |B|=35 \}$$

find intersection.

Q10

In 100 students, 75 study A and 60 study B. Find minimum possible number studying both.

LEVEL 2

Q11

In 120 students, 80 study Maths, 70 Physics, and 50 both. Find:

1. Maths only
2. Physics only
3. neither.

Q12

If:

$$\{ |A|=40, |B|=50, N=70 \}$$

find possible range of $|A \cap B|$.

Q13

If:

$$\{ |A|=25, |B|=30, |A \cap B|=10 \}$$

find:

$\setminus$

A triangle B

$\setminus$

Q14

A set has 10 elements. How many subsets contain a particular element $\setminus(x)$?

Q15

How many subsets of a 7-element set contain $\setminus(a)$ but not $\setminus(b)$?

LEVEL 3 — EXAM PRESSURE

Q16

In 200 people:

- 120 know Python
- 100 know Java
- 80 know both

Find:

1. exactly one
2. neither.

Q17

80% know A, 65% know B. What is the minimum percentage knowing both?

Q18

70% know A, 50% know B. What is the maximum percentage knowing exactly one?

Q19

If:

$\setminus A \subseteq B, \quad |A|=18, \quad |B-A|=7 \setminus$

find $\setminus(|B|)$.

Q20

A universal set has 100 elements. Sets A and B have sizes 60 and 55. If exactly 35 elements belong to exactly one of A,B, find $\setminus(|A \cap B|)$.

ANSWERS

Q1

$\setminus \boxed{\{1,2,3,4,5\}} \setminus$

Q2

$$\boxed{3}$$

Q3

$$2^5 = \boxed{32}$$

Q4

$$2^5 - 2 = \boxed{30}$$

Q5

$$20 + 15 - 5 = \boxed{30}$$

Q6

At least one:

$$60 + 50 - 20 = 90$$

Neither:

$$100 - 90 = \boxed{10}$$

Q7

Exactly one:

$$60 + 50 - 2(20)$$

$$= 110 - 40$$

$$= \boxed{70}$$

Q8

$$80 - 47 = \boxed{33}$$

Q9

$$55 = 30 + 35 - x$$

$$x = 10$$

$$\boxed{10}$$

Q10

$$75 + 60 - 100 = \boxed{35}$$

Q11

Maths only:

$$80 - 50 = 30$$

Physics only:

$$70 - 50 = 20$$

Union:

$$80 + 70 - 50 = 100$$

Neither:

$$\boxed{120-100=20}$$

$$\boxed{30, 20, 20}$$

Q12

Minimum intersection:

$$\boxed{40+50-70=20}$$

Maximum:

$$\boxed{\min(40,50)=40}$$

Therefore:

$$\boxed{20 \leq |A \cap B| \leq 40}$$

Q13

$$\boxed{25+30-2(10)}$$

$$\boxed{35}$$

Q14

Fix x included.

Each of remaining 9 elements:

- included
- excluded

So:

$$2^9 = \boxed{512}$$

Q15

a included.

b excluded.

Remaining 5 elements arbitrary:

$$2^5 = \boxed{32}$$

Q16

Exactly one:

$$\boxed{120+100-2(80)}$$

$$\boxed{220-160}$$

$$\boxed{60}$$

Union:

$$\boxed{120+100-80=140}$$

Neither:

$$\boxed{200-140=60}$$

Q17

Minimum overlap:

$$\setminus [80+65-100 \setminus]$$

$$\setminus [=\boxed{45\%} \setminus]$$

Q18

Exactly one:

$$\setminus [70+50-2I \setminus]$$

To maximize exactly one, minimize intersection.

Minimum intersection:

$$\setminus [70+50-100=20 \setminus]$$

So:

$$\setminus [120-40 = \boxed{80\%} \setminus]$$

Q19

Since:

$$\setminus [A\subseteq B \setminus]$$

B consists of:

- A
- plus $\setminus (B-A)$

Therefore:

$$\setminus [|B|=18+7 \setminus]$$

$$\setminus [=\boxed{25} \setminus]$$

Q20

Let:

$$\setminus [I=|A\cap B| \setminus]$$

Exactly one:

$$\setminus [60+55-2I \setminus]$$

Given:

$$\setminus [35 \setminus]$$

So:

$$\setminus [115-2I=35 \setminus]$$

$$\setminus [2I=80 \setminus]$$

$$\setminus [\boxed{I=40} \setminus]$$

CHAPTER 42 STATUS

Covered so far:

- set terminology '
- membership '

- empty/singleton sets '
- subsets/proper subsets '
- power sets '
- cardinality '
- union/intersection '
- complements '
- differences '
- De Morgan's laws '
- two-set inclusion-exclusion '
- only/exactly one/both/neither '
- symmetric difference '
- survey problems '
- percentage Venn problems '
- minimum/maximum overlap '
- subset-count applications '
- GATE/placement L0–L3 practice '

But **Chapter 42 is not complete yet.**

Next we still need:

- 3-set Venn diagrams
- 3-set inclusion-exclusion
- exactly one / exactly two / all three
- none / at least one
- survey problems with three categories
- region-by-region filling strategy
- missing-data questions
- maximum/minimum constraints for 3 sets
- final chapter audit

Progress

Completed: 41/76 Current: 0=0 Chapter 42 — Sets & Venn Diagrams

NEXT

0=0 Chapter 42 Part 2 — Three-Set Venn Diagrams, Inclusion-Exclusion, Exactly One/Two/Three & Advanced Surveys

USER

next

ASSISTANT

CHAPTER 42 — SETS AND VENN DIAGRAMS

PART 2 — THREE-SET VENN DIAGRAMS, INCLUSION-EXCLUSION, EXACTLY ONE/TWO/THREE & ADVANCED SURVEYS

Status: $\emptyset$ = 0 PROGRESS GATE: High Placements: High Priority: $\emptyset$ = 0 Per 1

Now we move from 2 sets to **3 sets**.

This is where most aptitude mistakes happen because students confuse:

- pairwise intersection
- exactly two
- all three
- at least two
- exactly one

The safest rule is:

Fill the center first, then pairwise-only regions, then single-only regions, then outside.

1. THREE-SET INCLUSION-EXCLUSION

For sets $\{A, B, C\}$:

$\boxed{}$

$A \cup B \cup C$

=

A	+	B	+	C
---	---	---	---	---

$-|A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \}$

MUST MEMORIZE

2. WHY ADD THE TRIPLE INTERSECTION BACK?

When we add:

$\boxed{}$

A	+	B	+	C
---	---	---	---	---

$\boxed{}$

an element in all three sets is counted:

$\boxed{3 \text{ times}}$

Then subtracting the three pairwise intersections removes it:

$\lfloor 3 \times \rfloor$

So it becomes counted:

$\lfloor 0 \rfloor$

times.

We need it exactly once.

Therefore add:

$\lfloor$

$A \cap B \cap C$

$\rfloor$

once.

3. IMPORTANT PAIRWISE-INTERSECTION TRAP

Suppose:

$\lfloor |A \cap B| = 20 \rfloor$

and:

$\lfloor |A \cap B \cap C| = 5 \rfloor$

Then:

$\lfloor 20 \rfloor$

usually includes the 5 elements in all three sets.

So:

$\lfloor \boxed{$

$A \cap B \text{ only}$

$= 20 - 5 = 15 \rfloor$

Unless wording explicitly says "A and B only."

4. THREE-SET REGION STRUCTURE

A 3-set Venn diagram has 8 regions:

1. A only
2. B only
3. C only
4. A only
5. B only
6. C only
7. A only B only C

8. none

These are the basic pieces.

5. BEST FILLING ORDER

Suppose survey data gives:

- totals of A,B,C
- pairwise intersections
- triple intersection
- universal total

Fill in this order:

Step 1

Center:

$$\mathbb{I} \left[A \cap B \cap C \right]$$

Step 2

Pairwise-only:

$$\mathbb{I} \left[A \setminus \text{only} \right] = \mathbb{I} \left[A \cap B \right] - \mathbb{I} \left[A \cap B \cap C \right]$$

and similarly.

Step 3

Single-only regions by subtraction.

Step 4

Union.

Step 5

None.

This minimizes errors.

6. EXAMPLE — COMPLETE THREE-SET SURVEY

100 students:

$$\mathbb{I} \left[|A| = 50 \right]$$

$$\mathbb{I} \left[|B| = 45 \right]$$

$$\mathbb{I} \left[|C| = 40 \right]$$

$$\mathbb{I} \left[|A \cap B| = 20 \right]$$

$$\mathbb{I} \left[|B \cap C| = 15 \right]$$

$$\mathbb{I} \left[|C \cap A| = 10 \right]$$

$$\mathbb{I} \left[|A \cap B \cap C| = 5 \right]$$

Find all regions.

7. CENTER FIRST

All three:

$$\boxed{5}$$

8. PAIRWISE-ONLY REGIONS

A and B only:

$$\boxed{20-5=15}$$

B and C only:

$$\boxed{15-5=10}$$

C and A only:

$$\boxed{10-5=5}$$

So:

$$\boxed{AB \text{ only} = 15}$$

$$\boxed{BC \text{ only} = 10}$$

$$\boxed{CA \text{ only} = 5}$$

9. A ONLY

Total A:

$$\boxed{50}$$

Inside A we already have:

- AB only = 15
- AC only = 5
- ABC = 5

Therefore:

$$\boxed{A \text{ only} = 50-15-5-5}$$

$$\boxed{=25}$$

10. B ONLY

$$\boxed{45-15-10-5}$$

$$\boxed{=15}$$

11. C ONLY

$$\boxed{40-10-5-5}$$

$$\boxed{20}$$

12. UNION

Add all 7 internal regions:

$$25+15+20+15+10+5+5$$

$$\boxed{95}$$

So none:

$$100-95 = \boxed{5}$$

13. CHECK USING FORMULA

$$|A \cup B \cup C|$$

$$= 50+45+40-20-15-10+5$$

$$= 135-45+5$$

$$\boxed{95}$$

$$\boxed{95}$$

Correct.

14. EXACTLY THREE

Exactly three means:

$$|A \cap B \cap C|$$

Nothing more.

15. EXACTLY TWO

Exactly two means:

- A and B only
- B and C only
- C and A only

Therefore:

$$|\text{Exactly 2}| = (|A \cap B| - |A \cap B \cap C|) + (|B \cap C| - |A \cap B \cap C|) + (|C \cap A| - |A \cap B \cap C|)$$

16. EXACTLY TWO USING RAW INTERSECTIONS

Since each pairwise intersection includes the triple intersection:

$$|A \cap B| = |A \cap B| - T$$

where:

$$T = |A \cap B \cap C|$$

Therefore:

$$|\text{Exactly 2}| = |A \cap B| + |B \cap C| + |C \cap A| - 3T$$

So:

$$\boxed{E_2 = AB + BC + CA - 3T}$$

MUST KNOW

17. EXAMPLE

Using:

$$AB=20, BC=15, CA=10, T=5$$

Exactly two:

$$20+15+10-15$$

$$= \boxed{30}$$

Check pairwise-only regions:

$$15+10+5=30$$

18. AT LEAST TWO

“At least two” means:

- exactly two
- exactly three

Therefore:

$$\boxed{|\text{At least 2}| = E_2 + T}$$

So:

$$\boxed{|\text{At least 2}| = AB + BC + CA - 2T}$$

19. EXAMPLE

$$20+15+10-2(5)$$

$$= 45 - 10$$

$$= \boxed{35}$$

20. EXACTLY ONE

Exactly one means:

- A only

- B only
- C only

We can derive a direct formula.

Let:

$$|S| = |A| + |B| + |C|$$

Let pairwise total:

$$|P| = |A \cap B| + |B \cap C| + |C \cap A|$$

Let:

$$|T| = |A \cap B \cap C|$$

Then:

$$\boxed{E_1 = S - 2P + 3T}$$

High-value formula

21. WHY EXACTLY-ONE FORMULA WORKS

An element in:

- exactly one set is counted once in $|S|$
- exactly two sets is counted twice in $|S|$
- all three is counted three times in $|S|$

Subtracting $|2P|$:

- each exactly-two element appears in one pair !' subtract twice !' leaves 0
- each triple element appears in 3 pairs !' subtract 6

Then adding:

$$|3T|$$

fixes the triple contribution:

$$|3 - 6 + 3 = 0|$$

So only exactly-one elements remain.

22. EXAMPLE

Using previous data:

$$|S| = 50 + 45 + 40 = 135$$

$$|P| = 20 + 15 + 10 = 45$$

$$|T| = 5$$

Exactly one:

$$|135 - 2(45) + 3(5)|$$

$$|= 135 - 90 + 15|$$

$$|= \boxed{60}|$$

Check:

$\lfloor 25+15+20=60 \rfloor$

23. AT LEAST ONE

At least one means union:

$\lfloor \boxed{A \cup B \cup C} \rfloor$

Formula:

$\lfloor \boxed{S-P+T} \rfloor$

24. NONE

If universal total is $\lfloor N \rfloor$:

$\lfloor \boxed{\text{None} = N - |A \cup B \cup C|} \rfloor$

25. EXACTLY ONE + EXACTLY TWO + EXACTLY THREE

These three categories partition the union.

Therefore:

$\lfloor \boxed{$

$A \cup B \cup C$

$= E_1 + E_2 + E_3} \rfloor$

where:

$\lfloor E_3 = T \rfloor$

Useful check.

26. MEMBERSHIP-COUNT EQUATION

Suppose:

- $\lfloor E_1 \rfloor$ people belong to exactly one set
- $\lfloor E_2 \rfloor$ exactly two
- $\lfloor E_3 \rfloor$ all three

Then total membership count:

$\lfloor$

A	+	B	+	C
---	---	---	---	---

$\rfloor$

equals:

$$\boxed{E_1 + 2E_2 + 3E_3}$$

Because:

- exactly-one person contributes 1 membership
- exactly-two contributes 2
- exactly-three contributes 3

This is a powerful shortcut.

27. PAIRWISE MEMBERSHIP COUNT

Similarly:

$$\boxed{A \cap B + B \cap C + C \cap A}$$

equals:

$$\boxed{E_2 + 3E_3}$$

because:

- exactly-two person belongs to exactly one pairwise intersection
- triple person belongs to all 3 pairwise intersections

28. UNION FROM E1,E2,E3

$$\boxed{U = E_1 + E_2 + E_3}$$

Obvious, but useful in fast problems.

29. SOLVING FROM MEMBERSHIP TOTALS

Suppose:

$$|A| + |B| + |C| = 180$$

Exactly one = 80.

Exactly three = 10.

Find exactly two.

Use:

$$180 = 80 + 2E_2 + 3(10)$$

$$180 = 110 + 2E_2$$

$$2E_2 = 70$$

$$\boxed{E_2 = 35}$$

Very fast.

30. THEN FIND UNION

$$U = 80 + 35 + 10$$

$$\boxed{125}$$

31. THREE-SET PERCENTAGE PROBLEMS

For percentage surveys, treat universal size as:

$$100$$

unless actual total is needed.

Example:

- 60% like A
- 50% like B
- 40% like C
- 30% like AB
- 20% like BC
- 10% like CA
- 5% like all three

At least one:

$$60 + 50 + 40 - 30 - 20 - 10 + 5$$

$$= 150 - 60 + 5$$

$$\boxed{95\%}$$

None:

$$\boxed{5\%}$$

32. EXACTLY TWO IN PERCENTAGE FORM

$$30 + 20 + 10 - 3(5)$$

$$= 60 - 15$$

$$\boxed{45\%}$$

33. EXACTLY ONE

$$150 - 2(60) + 3(5)$$

$$= 150 - 120 + 15$$

$$\boxed{45\%}$$

Check:

$$45 + 45 + 5 = 95$$

34. PAIRWISE DISJOINT SETS

If:

$$\forall A \cap B = B \cap C = C \cap A = \varnothing$$

then triple intersection is automatically empty.

Union:

$$\boxed{|A| + |B| + |C|}$$

35. IF ALL THREE SETS HAVE A COMMON CORE ONLY

Suppose pairwise intersections are equal to triple intersection:

$$\forall A \cap B = A \cap B \cap C$$

and similarly for other pairs.

Then there are no “exactly-two” regions.

So:

$$\boxed{E_2 = 0}$$

36. ONLY A — DIRECT FORMULA

A only equals:

$\forall$

A

-

$A \cap B$

-

$A \cap C$

+

$A \cap B \cap C$

$\forall$

Why add triple once?

Because triple was subtracted twice.

Thus:

$$\boxed{A_{\text{only}} =$$

A	-	AB	-	AC	+	ABC
---	---	----	---	----	---	-----

} \]

37. EXAMPLE

$|A|=50, |AB|=20, |AC|=10, |ABC|=5$

$|A \setminus (B \cup C)| = 50 - 20 - 10 + 5$

$= 25$

38. ONLY B / ONLY C

Similarly:

$|B \setminus (A \cup C)| =$

B	-	AB	-	BC	+	ABC
---	---	----	---	----	---	-----

} \]

$|C \setminus (A \cup B)| =$

C	-	AC	-	BC	+	ABC
---	---	----	---	----	---	-----

} \]

39. EXACTLY ONE USING ONLY-REGION SUM

Often safer in hand calculations:

$|E_1| = |A \setminus (B \cup C)| + |B \setminus (A \cup C)| + |C \setminus (A \cup B)|$

Use direct formula only when speed matters.

40. MISSING TRIPLE INTERSECTION

Suppose:

$|U|=100$

all people belong to at least one.

Given:

$|A|=60, |B|=50, |C|=40$

$|AB|=30, |BC|=20, |CA|=15$

Find triple intersection $|T|$.

Because union = 100:

$$\lfloor 100 = 60+50+40-30-20-15+T \rfloor$$

$$\lfloor 100=85+T \rfloor$$

$$\lfloor \boxed{T=15} \rfloor$$

41. FIND NONE WHEN TRIPLE UNKNOWN

If $\lfloor T \rfloor$ is known after solving:

$$\lfloor \text{None} \rfloor = N - [S - P + T] \rfloor$$

Straightforward.

42. EXACTLY TWO GIVEN, FIND TRIPLE

If:

$$\lfloor AB+BC+CA=50 \rfloor$$

and exactly two = 32, then:

$$\lfloor 32=50-3T \rfloor$$

$$\lfloor 3T=18 \rfloor$$

$$\lfloor \boxed{T=6} \rfloor$$

43. AT LEAST TWO GIVEN, FIND TRIPLE

If pairwise-intersection sum is 50 and at least two = 38:

$$\lfloor 38=50-2T \rfloor$$

$$\lfloor 2T=12 \rfloor$$

$$\lfloor \boxed{T=6} \rfloor$$

44. EXACTLY ONE GIVEN, FIND OTHER INFORMATION

Suppose:

$$\lfloor S=150, \quad P=60, \quad E_1=45 \rfloor$$

Use:

$$\lfloor 45=150-120+3T \rfloor$$

$$\lfloor 45=30+3T \rfloor$$

$$\lfloor 3T=15 \rfloor$$

$$\lfloor \boxed{T=5} \rfloor$$

45. “ONLY TWO” LANGUAGE

“Belong to only two sets” usually means exactly two.

So exclude the center triple region.

Use:

$\boxed{E_2}$

not:

$AB+BC+CA$

directly.

46. “AT LEAST TWO”

Includes all three.

$\boxed{E_2+E_3}$

47. “NO MORE THAN ONE”

Means:

- none
- exactly one

If universal population matters:

$\boxed{\text{none}+E_1}$

48. “AT MOST TWO”

With only 3 sets:

Everything except membership in all three.

So:

$\boxed{N-E_3}$

if counting all people in universe.

49. “NOT IN ALL THREE”

Same:

$\boxed{N-A\cap B\cap C}$

Careful: this includes people in none, one, or two sets.

50. “NONE OF THE THREE”

$\boxed{(A\cup B\cup C)^c}$

not:

$$\lceil A^c \cup B^c \cup C^c \rceil$$

By De Morgan:

$$\lceil \boxed{(A \cup B \cup C)^c = A^c \cap B^c \cap C^c} \rceil$$

51. THREE-SET DE MORGAN'S LAWS

$$\lceil \boxed{(A \cup B \cup C)^c = A^c \cap B^c \cap C^c} \rceil$$

$$\lceil \boxed{(A \cap B \cap C)^c = A^c \cup B^c \cup C^c} \rceil$$

52. MINIMUM TRIPLE OVERLAP — SIMPLE LOWER BOUND

Suppose universe has $|N|$ people and set sizes are:

$$\lceil a, b, c \rceil$$

Total memberships:

$$\lceil a + b + c \rceil$$

If everyone can belong to at most 2 sets without any triple overlap, maximum memberships with no triple is:

$$\lceil 2N \rceil$$

Therefore if:

$$\lceil a + b + c > 2N \rceil$$

at least:

$$\lceil a + b + c - 2N \rceil$$

people must belong to all three.

Thus:

$$\lceil \boxed{$$

$$A \cap B \cap C$$

$$\geq \max(0, a + b + c - 2N) \rceil$$

This is a useful lower bound when only individual set sizes are known.

53. EXAMPLE

100 students:

$$\lceil |A|=80, \quad |B|=75, \quad |C|=70 \rceil$$

Total memberships:

$$\lceil 225 \rceil$$

Without any triple membership, at most:

$$\lceil 2(100)=200 \rceil$$

memberships could exist.

So at least:

$$\lfloor 225 - 200 = \boxed{25} \rfloor$$

students must be in all three.

54. CAUTION ON THREE-SET MAX/MIN PROBLEMS

Three-set bounds can be more complicated than two-set bounds because pairwise overlaps interact.

For aptitude:

- use obvious capacity bounds
- use inclusion-exclusion
- use membership-count reasoning

Do not memorize oversimplified formulas beyond the lower bound above.

55. GATE-STYLE QUESTION 1

Given:

$$\lfloor |A|=40, |B|=35, |C|=30 \rfloor$$

$$\lfloor |AB|=15, |BC|=10, |CA|=8 \rfloor$$

$$\lfloor |ABC|=5 \rfloor$$

Find union.

$$\lfloor 40+35+30-15-10-8+5 \rfloor$$

$$\lfloor =105-33+5 \rfloor$$

$$\lfloor =\boxed{77} \rfloor$$

56. GATE-STYLE QUESTION 2

Using same data, find exactly two.

$$\lfloor 15+10+8-3(5) \rfloor$$

$$\lfloor =33-15 \rfloor$$

$$\lfloor =\boxed{18} \rfloor$$

57. GATE-STYLE QUESTION 3

Find exactly one.

$$\lfloor 105-2(33)+3(5) \rfloor$$

$$\lfloor =105-66+15 \rfloor$$

$$\lfloor =\boxed{54} \rfloor$$

Check:

$$\backslash [54+18+5=77 \backslash]$$

58. GATE-STYLE QUESTION 4

If universal size = 90, none:

$$\backslash [90-77 \backslash]$$

$$\backslash [=\boxed{13} \backslash]$$

59. PLACEMENT-STYLE QUESTION 1

200 employees:

- 120 know Excel
- 100 know SQL
- 80 know Python
- 60 know Excel & SQL
- 50 know SQL & Python
- 40 know Python & Excel
- 30 know all three

Find at least one.

$$\backslash [120+100+80-60-50-40+30 \backslash]$$

$$\backslash [=300-150+30 \backslash]$$

$$\backslash [=\boxed{180} \backslash]$$

Neither:

$$\backslash [200-180 = \boxed{20} \backslash]$$

60. PLACEMENT-STYLE QUESTION 2

Using same data, exactly two:

$$\backslash [60+50+40-3(30) \backslash]$$

$$\backslash [=150-90 \backslash]$$

$$\backslash [=\boxed{60} \backslash]$$

61. PLACEMENT-STYLE QUESTION 3

Exactly one:

$$\backslash [300-2(150)+3(30) \backslash]$$

$$\backslash [=300-300+90 \backslash]$$

$$\backslash [=\boxed{90} \backslash]$$

Check:

$$\backslash[90+60+30=180 \backslash]$$

62. ADVANCED QUESTION — MEMBERSHIP COUNT

In a survey:

- 50 people belong to exactly one group
- 30 belong to exactly two
- 10 belong to all three

Total membership count:

$$\backslash[50+2(30)+3(10) \backslash]$$

$$\backslash[=50+60+30 \backslash]$$

$$\backslash[=\boxed{140} \backslash]$$

Union:

$$\backslash[50+30+10 = \boxed{90} \backslash]$$

63. ADVANCED QUESTION — FIND EXACTLY TWO

Suppose:

$$\backslash[|A|+|B|+|C|=170 \backslash]$$

Exactly one = 80.

All three = 10.

Then:

$$\backslash[170=80+2E_2+30 \backslash]$$

$$\backslash[2E_2=60 \backslash]$$

$$\backslash[\boxed{E_2=30} \backslash]$$

64. ADVANCED QUESTION — FIND NONE

Suppose total population is 150.

From previous:

$$\backslash[U=80+30+10=120 \backslash]$$

None:

$$\backslash[150-120 \backslash]$$

$$\backslash[=\boxed{30} \backslash]$$

65. COMMON TRAPS

& pTRAP 1 — Pairwise intersection includes triple

Unless explicitly stated otherwise.

& pTRAP 2 — Exactly two

Subtract triple from every pairwise intersection:

$$\setminus [AB+BC+CA-3T \setminus]$$

& pTRAP 3 — At least two

Add exactly-two and triple:

$$\setminus [AB+BC+CA-2T \setminus]$$

& pTRAP 4 — Triple term sign

In three-set union formula, triple intersection is added:

$$\setminus [+ABC \setminus]$$

& pTRAP 5 — Exactly one direct formula

$$\setminus [S-2P+3T \setminus]$$

not $\setminus (S-P \setminus)$.

& pTRAP 6 — “None”

Subtract union from universal total.

& pTRAP 7 — Fill single-only first

Don't. Fill center first.

66. MUST-MEMORIZE THREE-SET UNION

$$\setminus \boxed{\setminus$$

$$A \cup B \cup C$$

$$= A+B+C-AB-BC-CA+ABC \setminus \setminus]$$

67. MUST-MEMORIZE EXACTLY TWO

Let:

$$\setminus [T=ABC \setminus]$$

Then:

$$\setminus \boxed{\setminus E_2=AB+BC+CA-3T \setminus \setminus]$$

68. MUST-MEMORIZE AT LEAST TWO

$$\boxed{A_{\geq 2} = AB + BC + CA - 2T}$$

69. MUST-MEMORIZE EXACTLY ONE

$$\boxed{E_1 = A + B + C - 2(AB + BC + CA) + 3ABC}$$

70. MEMBERSHIP COUNT SHORTCUTS

$$\boxed{A + B + C = E_1 + 2E_2 + 3E_3}$$

$$\boxed{AB + BC + CA = E_2 + 3E_3}$$

$$\boxed{|A \cup B \cup C| = E_1 + E_2 + E_3}$$

71. FAST EXAM DECISION TREE

Three sets with full data?

Use inclusion-exclusion.

Need region breakdown?

Fill center first.

Pairwise “both” given?

Subtract triple to get pairwise-only.

Exactly two?

Use:

$$\boxed{P - 3T}$$

At least two?

Use:

$$\boxed{P - 2T}$$

Exactly one?

Use region sum or:

$$\boxed{S - 2P + 3T}$$

None?

Universal total minus union.

Only set A?

$$\boxed{A - AB - AC + ABC}$$

PRACTICE — LEVEL 0

Q1

If:

$\setminus [A=40, \setminus B=30, \setminus C=20 \setminus]$

$\setminus [AB=10, \setminus BC=5, \setminus CA=8, \setminus ABC=2 \setminus]$

find union.

Q2

Find exactly two.

Q3

Find exactly one.

Q4

If universe has 80 elements, find none.

Q5

Find A only.

LEVEL 1

Q6

100 students:

- 60 like A
- 50 like B
- 40 like C
- 25 like AB
- 20 like BC
- 15 like CA
- 10 like all three

Find:

1. at least one
2. exactly two
3. exactly one
4. none.

Q7

If pairwise-intersection sum is 45 and triple = 5, find exactly two.

Q8

If pairwise-intersection sum is 45 and triple = 5, find at least two.

Q9

If: $\setminus [A+B+C=150 \setminus]$ and exactly one = 60, exactly three = 10, find exactly two.

Q10

If union = 110 and universal total = 150, find none.

LEVEL 2

Q11

300 employees:

- 180 know Python
- 160 know SQL
- 140 know Excel
- 100 know Python & SQL
- 80 know SQL & Excel
- 90 know Python & Excel
- 50 know all three

Find exactly one.

Q12

Using same data, find exactly two.

Q13

Using same data, find none.

Q14

If: $\setminus [|A|=70, \setminus |B|=60, \setminus |C|=50 \setminus]$ in a universe of 100, find a lower bound for triple overlap using the membership-capacity shortcut.

Q15

If: $\setminus [A+B+C=200, \quad AB+BC+CA=90, \quad ABC=20 \setminus]$ find union.

LEVEL 3 — EXAM PRESSURE

Q16

In a survey, 80 belong to exactly one set, 40 to exactly two, and 15 to exactly three. Find:

1. union
2. total membership count
3. total pairwise-intersection count.

Q17

If: $\setminus [A+B+C=180 \setminus]$ $\setminus [AB+BC+CA=70 \setminus]$ and exactly two = 40, find triple intersection.

Q18

If exactly one = 60, exactly two = 30, all three = 10, and total population = 125, find none.

Q19

Universal set has 150 elements. If: $|A|=90, |B|=80, |C|=70$ what lower bound can you guarantee for all three?

Q20

If: $|A|=80, |B|=70, |C|=60$ $|AB|=40, |BC|=30, |CA|=35$ and union = 125, find all three.

ANSWERS**Q1**

$$|A \cup B \cup C| = 40 + 30 + 20 - 10 - 5 - 8 + 2$$

$$= 90 - 23 + 2$$

$$= \boxed{69}$$

Q2

$$|A \cup B \cup C| = 10 + 5 + 8 - 3(2)$$

$$= 23 - 6$$

$$= \boxed{17}$$

Q3

$$|A \cup B \cup C| = 90 - 2(23) + 3(2)$$

$$= 90 - 46 + 6$$

$$= \boxed{50}$$

Check:

$$50 + 17 + 2 = 69$$

Q4

$$80 - 69 = \boxed{11}$$

Q5

$$40 - 10 - 8 + 2$$

$$= \boxed{24}$$

Q6

Union:

$$60 + 50 + 40 - 25 - 20 - 15 + 10$$

$$= 150 - 60 + 10$$

$$= \boxed{100}$$

Exactly two:

$$25 + 20 + 15 - 30 = \boxed{30}$$

Exactly one:

$$150 - 2(60) + 30$$

$$\boxed{60}$$

None:

$$100-100 = \boxed{0}$$

Q7

$$45-3(5) = \boxed{30}$$

Q8

$$45-2(5) = \boxed{35}$$

Q9

Use:

$$150=60+2E_2+3(10)$$

$$150=90+2E_2$$

$$E_2=30$$

$$\boxed{30}$$

Q10

$$150-110 = \boxed{40}$$

Q11

Here:

$$S=180+160+140=480$$

$$P=100+80+90=270$$

$$T=50$$

Exactly one:

$$480-2(270)+3(50)$$

$$=480-540+150$$

$$=\boxed{90}$$

Q12

$$270-3(50)$$

$$=270-150$$

$$=\boxed{120}$$

Q13

Union:

$$480-270+50$$

$$=\boxed{260}$$

None:

$$300-260$$

$$=\boxed{40}$$

Q14

$$\lfloor 70+60+50-2(100) \rfloor$$

$$\lfloor =180-200 \rfloor$$

Negative, so lower bound:

$$\lfloor \boxed{0} \rfloor$$

No triple membership is forced by this bound alone.

Q15

$$\lfloor 200-90+20 \rfloor$$

$$\lfloor =\boxed{130} \rfloor$$

Q16

Union:

$$\lfloor 80+40+15 = \boxed{135} \rfloor$$

Membership count:

$$\lfloor 80+2(40)+3(15) \rfloor$$

$$\lfloor =80+80+45 \rfloor$$

$$\lfloor =\boxed{205} \rfloor$$

Pairwise-intersection total:

$$\lfloor 40+3(15) \rfloor$$

$$\lfloor =\boxed{85} \rfloor$$

Q17

Exactly two:

$$\lfloor E_2=P-3T \rfloor$$

$$\lfloor 40=70-3T \rfloor$$

$$\lfloor 3T=30 \rfloor$$

$$\lfloor \boxed{T=10} \rfloor$$

Q18

Union:

$$\lfloor 60+30+10 = 100 \rfloor$$

None:

$$\lfloor 125-100 \rfloor$$

$$\lfloor =\boxed{25} \rfloor$$

Q19

Lower bound:

$$\lfloor 90+80+70-2(150) \rfloor$$

$$\lfloor =240-300 \rfloor$$

So:

$\boxed{0}$

This information alone does not force a triple intersection.

Q20

Use inclusion-exclusion:

$$125 = 80 + 70 + 60 - 40 - 30 - 35 + T$$

$$125 = 210 - 105 + T$$

$$125 = 105 + T$$

$$\boxed{T=20}$$

CHAPTER 42 COMPLETE

We have now covered:

- set notation
- membership and cardinality
- subsets and power sets
- union/intersection/complement
- set difference
- De Morgan's laws
- two-set Venn diagrams
- two-set inclusion-exclusion
- exactly one / both / neither
- minimum/maximum two-set overlap
- three-set Venn diagrams
- three-set inclusion-exclusion
- center-first filling strategy
- pairwise-only regions
- exactly one
- exactly two
- exactly three
- at least two
- none
- membership-count shortcuts
- percentage surveys
- missing-data problems
- three-set overlap lower-bound intuition
- GATE/placement L0–L3 practice

Progress

Completed: 42/76

Unit 1 — Quantitative Aptitude: ~~0=0~~ PROGRESS

We now move from algebra/counting/set topics into geometry.

NEXT

+Chapter 43 — Geometry Foundations

Next we will cover:

- points, lines, rays and segments
- angles
- complementary/supplementary angles
- vertically opposite angles
- parallel lines and transversals
- triangle angle rules
- exterior angles
- basic polygons
- coordinate-distance intuition
- symmetry
- common GATE/placement geometry traps
- essential diagram-reading shortcuts.

USER

next

USER

next

ASSISTANT

CHAPTER 43 — GEOMETRY FOUNDATIONS

PART 1 — POINTS, LINES, ANGLES, PARALLEL LINES, TRANSVERSALS & BASIC TRIANGLE ANGLE RULES

Status: ~~0=0~~ PROGRESS GATE: Medium Placements: Medium Priority: +Tier 2

Geometry questions often look visual, but most are solved using a few fixed rules.

The goal is:

Convert the diagram into angle equations instead of guessing from appearance.

1. POINT

A **point** represents an exact position.

It has:

- no length
- no width
- no area

Usually named:

$\{A, B, C\}$

2. LINE

A line extends infinitely in both directions.

Through points A and B:

$\overleftrightarrow{AB}$

A line has length but no thickness.

3. LINE SEGMENT

A segment has two fixed endpoints.

$\overline{AB}$

Its length is finite.

4. RAY

A ray begins at one endpoint and extends infinitely in one direction.

$\overrightarrow{AB}$

A is the starting point.

5. COLLINEAR POINTS

Points lying on the same straight line are called **collinear**.

Example:

$\{A, B, C\}$

on one straight line.

6. ANGLE

An angle is formed by two rays having a common endpoint.

The common endpoint is the **vertex**.

Notation:

$\angle ABC$

Here B is the vertex.

Memory trick

Middle letter = vertex.

7. TYPES OF ANGLES

Acute

$0^\circ < \theta < 90^\circ$

Right

$\theta = 90^\circ$

Obtuse

$90^\circ < \theta < 180^\circ$

Straight

$\theta = 180^\circ$

Reflex

$180^\circ < \theta < 360^\circ$

Complete angle

360°

8. ANGLES ON A STRAIGHT LINE

Angles forming a straight line sum to:

$\boxed{180^\circ}$

Example:

If:

$x + 70^\circ = 180^\circ$

then:

$x = \boxed{110^\circ}$

9. ANGLES AROUND A POINT

All angles around a point sum to:

$\boxed{360^\circ}$

Example:

$$\boxed{80^\circ + 120^\circ + 90^\circ + x = 360^\circ}$$

$$\boxed{x = 70^\circ}$$

10. COMPLEMENTARY ANGLES

Two angles are complementary if their sum is:

$$\boxed{90^\circ}$$

If one angle is x :

$$\boxed{\text{complement} = 90^\circ - x}$$

11. SUPPLEMENTARY ANGLES

Two angles are supplementary if:

$$\boxed{x + y = 180^\circ}$$

If one angle is x :

$$\boxed{\text{supplement} = 180^\circ - x}$$

12. EXAMPLE — COMPLEMENT

Complement of:

$$\boxed{37^\circ}$$

is:

$$\boxed{90^\circ - 37^\circ = 53^\circ}$$

13. EXAMPLE — SUPPLEMENT

Supplement of:

$$\boxed{128^\circ}$$

is:

$$\boxed{180^\circ - 128^\circ = 52^\circ}$$

14. VERTICALLY OPPOSITE ANGLES

When two lines intersect:

Opposite angles are equal.

If:

$$\boxed{\angle 1 = 65^\circ}$$

then opposite angle:

$$\boxed{65^\circ}$$

Adjacent angles:

$$\boxed{180^\circ - 65^\circ = 115^\circ}$$

15. INTERSECTION MASTER PATTERN

When two straight lines cross:

If one angle is x ,

the four angles are:

$$\boxed{x, 180^\circ - x, x, 180^\circ - x}$$

This is worth memorizing.

16. PERPENDICULAR LINES

Two lines are perpendicular if they meet at:

$$\boxed{90^\circ}$$

Notation:

$$\perp$$

17. PARALLEL LINES

Lines that never meet are parallel.

Notation:

$$\parallel$$

18. TRANSVERSAL

A line crossing two or more lines is called a **transversal**.

When a transversal crosses parallel lines, several angle relationships appear.

These are extremely important.

19. CORRESPONDING ANGLES

If parallel lines are cut by a transversal:

$$\boxed{\text{Corresponding angles are equal}}$$

Memory:

Same corner position.

20. ALTERNATE INTERIOR ANGLES

For parallel lines:

$\boxed{\text{Alternate interior angles are equal}}$

They lie:

- between the parallel lines
 - on opposite sides of transversal
-

21. ALTERNATE EXTERIOR ANGLES

Similarly:

$\boxed{\text{Alternate exterior angles are equal}}$

22. CO-INTERIOR / SAME-SIDE INTERIOR ANGLES

Angles inside parallel lines on the same side of the transversal sum to:

$\boxed{180^\circ}$

Also called:

- consecutive interior angles
 - allied angles
-

23. FAST PARALLEL-LINE MEMORY

For parallel lines:

Equal

- corresponding
- alternate interior
- alternate exterior

Sum to (180°)

- same-side interior
-

24. EXAMPLE — CORRESPONDING

One corresponding angle:

$\boxed{68^\circ}$

Then corresponding angle:

$\boxed{68^\circ}$

Its adjacent angle:

$$\boxed{180^\circ - 68^\circ = 112^\circ}$$

25. EXAMPLE — SAME-SIDE INTERIOR

One interior angle:

$$125^\circ$$

Other same-side interior angle:

$$\boxed{180^\circ - 125^\circ = 55^\circ}$$

26. CONVERSE RULES

These angle relationships can also prove lines are parallel.

For example, if corresponding angles are equal, then the lines are parallel.

Similarly:

- equal alternate interior angles $\Rightarrow$ parallel
- same-side interior sum $180^\circ \Rightarrow$ parallel

Useful in reasoning questions.

27. TRIANGLE

A triangle is formed by 3 non-collinear points.

It has:

- 3 sides
 - 3 vertices
 - 3 interior angles
-

28. TRIANGLE ANGLE-SUM PROPERTY

For every triangle:

$$\boxed{A + B + C = 180^\circ}$$

MUST MEMORIZE

29. EXAMPLE

Triangle angles:

$$50^\circ, 60^\circ, x$$

Then:

$$x = 180^\circ - 110^\circ$$

$$\boxed{70^\circ}$$

30. EXTERIOR ANGLE OF A TRIANGLE

An exterior angle equals the sum of the two remote interior angles.

$$\boxed{\text{Exterior angle} = A + B}$$

where A and B are the two non-adjacent interior angles.

31. WHY?

Exterior angle and adjacent interior angle form:

$$180^\circ$$

And triangle interior sum is also:

$$180^\circ$$

Therefore exterior equals the sum of the other two interior angles.

32. EXAMPLE

Remote interior angles:

$$45^\circ, 65^\circ$$

Exterior angle:

$$45^\circ + 65^\circ = \boxed{110^\circ}$$

33. TRIANGLE CLASSIFICATION BY SIDES

Equilateral

All 3 sides equal.

Therefore all angles:

$$\boxed{60^\circ}$$

Isosceles

Two sides equal.

Angles opposite equal sides are equal.

Scalene

All sides different.

All angles generally different.

34. ISOSCELES TRIANGLE RULE

If:

$$[AB=AC]$$

then opposite angles are equal:

$$[\boxed{\angle B = \angle C}]$$

Memory

Equal sides !”equal opposite angles.

35. CONVERSE OF ISOSCELES RULE

If two angles are equal, then opposite sides are equal.

So:

$$[\angle B = \angle C \Rightarrow AB = AC]$$

36. EXAMPLE

Isosceles triangle has vertex angle:

$$[40^\circ]$$

The remaining two equal angles sum:

$$[140^\circ]$$

Each:

$$[70^\circ]$$

So:

$$[\boxed{70^\circ, 70^\circ}]$$

37. RIGHT TRIANGLE

One angle:

$$[90^\circ]$$

Therefore the other two angles sum:

$$[\boxed{90^\circ}]$$

38. RIGHT ISOSCELES TRIANGLE

Angles:

$$[90^\circ, 45^\circ, 45^\circ]$$

This pattern is worth recognizing immediately.

39. EQUILATERAL TRIANGLE

All sides equal.

All three angles:

$$\boxed{60^\circ}$$

40. TRIANGLE INEQUALITY

For any triangle with sides (a,b,c) :

$$\boxed{a+b>c}$$

$$\boxed{b+c>a}$$

$$\boxed{c+a>b}$$

In words:

Sum of any two sides must be greater than the third.

41. EXAMPLE — VALID TRIANGLE?

Sides:

$$\boxed{3,4,8}$$

Check:

$$\boxed{3+4=7<8}$$

So impossible.

$$\boxed{\text{Not a triangle}}$$

42. FAST TRIANGLE INEQUALITY TEST

If sides are sorted:

$$\boxed{a \leq b \leq c}$$

you only need check:

$$\boxed{a+b>c}$$

If this holds, the other two automatically hold.

43. RANGE OF THIRD SIDE

If two sides are (a,b) , third side (x) must satisfy:

$$\boxed{|a-b|<x<a+b}$$

MUST KNOW

44. EXAMPLE

Two sides:

$$\sqrt{5}, \sqrt{9}$$

Third side $\sqrt{x}$:

$$\sqrt{|9-5| < x < 14}$$

$$\sqrt{\boxed{4 < x < 14}}$$

If integer:

$$\sqrt{x=5, 6, \dots, 13}$$

45. LARGER SIDE !"LARGER OPPOSITE ANGLE

In a triangle:

If:

$$\sqrt{AB > AC}$$

then angle opposite AB is greater than angle opposite AC.

Memory

Bigger side faces bigger angle.

46. EXAMPLE

Triangle sides:

$$\sqrt{5, 7, 9}$$

Largest angle lies opposite:

$$\sqrt{\boxed{9}}$$

No calculation needed.

47. EXTERIOR ANGLE IS GREATER THAN EITHER REMOTE INTERIOR ANGLE

Since:

$$\sqrt{E = A + B}$$

and angles are positive:

$$\sqrt{E > A}$$

and:

$$\sqrt{E > B}$$

Useful conceptual property.

48. POLYGON — BASIC DEFINITION

A polygon is a closed figure made of straight line segments.

Examples:

- triangle: 3 sides
 - quadrilateral: 4
 - pentagon: 5
 - hexagon: 6
 - heptagon: 7
 - octagon: 8
 - nonagon: 9
 - decagon: 10
-

49. SUM OF INTERIOR ANGLES OF A POLYGON

For an n -sided polygon:

$$\boxed{(n-2)180^\circ}$$

MUST MEMORIZE

50. WHY?

An n -gon can be divided from one vertex into:

$$n-2$$

triangles.

Each contributes:

$$180^\circ$$

So:

$$(n-2)180^\circ$$

51. EXAMPLE — PENTAGON

$$(5-2)180^\circ$$

$$=3(180^\circ)$$

$$=\boxed{540^\circ}$$

52. EXAMPLE — HEXAGON

$$(6-2)180^\circ = \boxed{720^\circ}$$

53. REGULAR POLYGON

A regular polygon has:

- all sides equal
- all interior angles equal

Each interior angle:

$$\boxed{\frac{(n-2)180^\circ}{n}}$$

54. EXAMPLE — REGULAR HEXAGON

$$\frac{4(180^\circ)}{6}$$

$$= \boxed{120^\circ}$$

55. SUM OF EXTERIOR ANGLES

For any convex polygon, one exterior angle at each vertex taken consistently:

$$\boxed{360^\circ}$$

This is independent of number of sides.

56. REGULAR POLYGON EXTERIOR ANGLE

For a regular n -gon:

$$\boxed{\frac{360^\circ}{n}}$$

57. INTERIOR + EXTERIOR

At a regular polygon vertex:

$$\boxed{\text{interior} + \text{exterior} = 180^\circ}$$

58. FIND NUMBER OF SIDES FROM EXTERIOR ANGLE

If regular polygon exterior angle = E :

$$\boxed{n = \frac{360^\circ}{E}}$$

59. EXAMPLE

Exterior angle:

$$30^\circ$$

Then:

$$n = \frac{360}{30} = \boxed{12}$$

60. FIND NUMBER OF SIDES FROM INTERIOR ANGLE

If regular polygon interior angle = I :

Exterior:

$$180^\circ - I$$

Then:

$$n = \frac{360^\circ}{180^\circ - I}$$

61. EXAMPLE

Regular polygon interior angle:

$$150^\circ$$

Exterior:

$$30^\circ$$

Hence:

$$n = 12$$

$$n = 12$$

62. NUMBER OF DIAGONALS

Already seen in combinations:

$$\frac{n(n-3)}{2}$$

Why?

Choose any 2 vertices:

$$\binom{n}{2}$$

Subtract n sides.

63. EXAMPLE

Octagon:

$$\frac{8(5)}{2} = 20$$

64. COORDINATE GEOMETRY — BASIC POINT

A point in Cartesian plane:

$$(x, y)$$

- x : horizontal coordinate
 - y : vertical coordinate
-

65. QUADRANTS

Quadrant I

$\backslash [(+,+) \backslash]$

Quadrant II

$\backslash [(-,+) \backslash]$

Quadrant III

$\backslash [(-,-) \backslash]$

Quadrant IV

$\backslash [(+,-) \backslash]$

Memory

Starting top-right and moving anticlockwise:

$\backslash [\boxed{\{(+,+),(-,+),(-,-),(+,-)\}} \backslash]$

66. DISTANCE ON A HORIZONTAL LINE

Between:

$\backslash [(x_1,y) \backslash]$

and:

$\backslash [(x_2,y) \backslash]$

distance:

$\backslash [\boxed{\{x_2-x_1\}} \backslash]$

67. DISTANCE ON A VERTICAL LINE

Between:

$\backslash [(x,y_1) \backslash]$

and:

$\backslash [(x,y_2) \backslash]$

distance:

$\backslash [\boxed{\{y_2-y_1\}} \backslash]$

68. GENERAL DISTANCE FORMULA

Between:

$\backslash [(x_1,y_1) \backslash]$

and:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\boxed{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

This is just Pythagoras.

69. EXAMPLE

Points:

$$(1, 2)$$

and:

$$(4, 6)$$

Differences:

$$3, 4$$

Distance:

$$\sqrt{3^2 + 4^2} = \boxed{5}$$

Recognize 3–4–5 quickly.

70. MIDPOINT FORMULA

Midpoint between:

$$(x_1, y_1), (x_2, y_2)$$

is:

$$\boxed{\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)}$$

71. EXAMPLE

Midpoint of:

$$(2, 4)$$

and:

$$(8, 10)$$

is:

$$(5, 7)$$

$$\boxed{(5, 7)}$$

72. BASIC SYMMETRY

A figure has line symmetry if it can be folded along a line so that both halves match.

Examples:

- square: 4 lines
- rectangle: 2
- equilateral triangle: 3
- circle: infinitely many

This can appear in visual reasoning more than quantitative GATE.

73. GATE-STYLE QUESTION 1

Two lines intersect. One angle is (72°) .

Opposite:

$$\boxed{72^\circ}$$

Adjacent:

$$\boxed{108^\circ}$$

So the four angles:

$$\boxed{72^\circ, 108^\circ, 72^\circ, 108^\circ}$$

74. GATE-STYLE QUESTION 2

Triangle angles are:

$$x, 2x, 3x$$

Then:

$$6x = 180^\circ$$

$$x = 30^\circ$$

Angles:

$$\boxed{30^\circ, 60^\circ, 90^\circ}$$

75. GATE-STYLE QUESTION 3

Regular polygon exterior angle:

$$24^\circ$$

Number of sides:

$$360/24 = \boxed{15}$$

76. GATE-STYLE QUESTION 4

Two sides of triangle are 7 and 11.

Possible third side:

$$|11-7| < x < 18$$

$$\boxed{4 < x < 18}$$

77. PLACEMENT-STYLE QUESTION 1

Supplement of an angle is 3 times the angle.

Let angle (x) .

$$180 - x = 3x$$

$$4x = 180$$

$$\boxed{x = 45^\circ}$$

78. PLACEMENT-STYLE QUESTION 2

Complement of angle is twice the angle.

$$90 - x = 2x$$

$$3x = 90$$

$$\boxed{x = 30^\circ}$$

79. PLACEMENT-STYLE QUESTION 3

Interior angle of regular polygon:

$$135^\circ$$

Exterior:

$$45^\circ$$

Number sides:

$$360/45 = \boxed{8}$$

80. PLACEMENT-STYLE QUESTION 4

Two parallel lines have same-side interior angles:

$$3x + 10$$

and:

$$2x + 20$$

Their sum:

$$180$$

So:

$$5x + 30 = 180$$

$$5x = 150$$

$$\boxed{x = 30}$$

Angles:

$\backslash 100^\circ, \backslash 80^\circ \backslash$

81. COMMON TRAPS

& pTRAP 1 — Trusting diagram scale

A drawn angle may look $\backslash(90^\circ)$, but unless marked, do not assume.

& pTRAP 2 — Corresponding vs same-side interior

Corresponding $!$ equal.

Same-side interior $!$ sum 180° .

& pTRAP 3 — Isosceles rule

Equal angles are opposite equal sides.

& pTRAP 4 — Triangle third side

Strict inequality:

$\backslash |a-b| < x < a+b \backslash$

Equality creates a degenerate straight line, not a triangle.

& pTRAP 5 — Polygon exterior angles

Their total is always:

$\backslash 360^\circ \backslash$

not $\backslash((n-2)180^\circ)$.

& pTRAP 6 — Regular vs irregular polygon

Individual angle formulas using division by $\backslash(n)$ require regular polygon.

82. MUST-MEMORIZE ANGLE RULES

$\backslash \boxed{\text{Straight line}} = 180^\circ \backslash$

$\backslash \boxed{\text{Around a point}} = 360^\circ \backslash$

$\backslash \boxed{\text{Complement}} = 90^\circ \backslash$

$\backslash \boxed{\text{Supplement}} = 180^\circ \backslash$

$\backslash \boxed{\text{Vertically opposite angles equal}} \backslash$

83. MUST-MEMORIZE TRIANGLE RULES

$\backslash \boxed{A+B+C=180^\circ} \backslash$

$\boxed{\text{Exterior} = \text{sum of remote interiors}}$

$\boxed{|a-b| < c < a+b}$

Equal sides !" equal opposite angles.

84. MUST-MEMORIZE POLYGON RULES

Interior-angle sum:

$\boxed{(n-2)180^\circ}$

Regular polygon interior:

$\boxed{\frac{(n-2)180^\circ}{n}}$

Regular exterior:

$\boxed{\frac{360^\circ}{n}}$

Diagonals:

$\boxed{\frac{n(n-3)}{2}}$

85. FAST EXAM DECISION TREE

Two lines intersect?

Use vertical equality + straight-line (180°) .

Parallel lines?

Check corresponding / alternate / same-side.

Triangle?

Start with:

180°

Isosceles?

Mark equal base angles immediately.

Exterior triangle angle?

Use remote-interior sum.

Regular polygon?

Exterior-angle method is often fastest.

Third triangle side?

Use:

$|a-b| < x < a+b$

Coordinates?

Look for Pythagorean differences before applying radicals.

PRACTICE — LEVEL 0

Q1

Find complement of (28°) .

Q2

Find supplement of (115°) .

Q3

Triangle has angles $(50^\circ, 65^\circ, x)$. Find (x) .

Q4

Find interior-angle sum of an octagon.

Q5

Regular polygon exterior angle is (60°) . Find number of sides.

LEVEL 1

Q6

Two intersecting lines form one angle of (37°) . Find the other three.

Q7

An isosceles triangle has equal base angles (x) and vertex angle (40°) . Find (x) .

Q8

Two sides of a triangle are 6 and 10. Find range of third side.

Q9

Find each interior angle of a regular decagon.

Q10

Find distance between $((2,3))$ and $((5,7))$.

LEVEL 2

Q11

Triangle exterior angle is (130°) . One remote interior angle is (55°) . Find the other.

Q12

Two same-side interior angles formed by parallel lines are: $[4x+10, \text{quad} 2x+20]$ Find (x) .

Q13

A regular polygon has interior angle (156°) . Find number of sides.

Q14

How many diagonals in a 15-sided polygon?

Q15

Triangle sides are integer lengths 5,9, $\sqrt{x}$. How many possible integer values of $\sqrt{x}$?

LEVEL 3 — EXAM PRESSURE**Q16**

Angles of a triangle are in ratio: $\sqrt{2:3:4}$ Find all angles.

Q17

An isosceles triangle has exterior angle $\sqrt{(110^\circ)}$ at one base vertex. Find all interior angles.

Q18

One angle formed by two parallel lines and a transversal is $\sqrt{(3x+15)}$, and its alternate interior angle is $\sqrt{(5x-25)}$. Find $\sqrt{x}$.

Q19

A regular polygon has each exterior angle equal to one-third of its interior angle. Find number of sides.

Q20

Points $\sqrt{(A=(1,2))}$, $\sqrt{(B=(7,10))}$. Find midpoint and distance.

ANSWERS**Q1**

$$\sqrt{90-28 = \boxed{62^\circ}}$$

Q2

$$\sqrt{180-115 = \boxed{65^\circ}}$$

Q3

$$\sqrt{180-50-65 = \boxed{65^\circ}}$$

Q4

$$\sqrt{(8-2)180 = 6(180) = \boxed{1080^\circ}}$$

Q5

$$\sqrt{360/60 = \boxed{6}}$$

Q6

Opposite:

$$\sqrt{37^\circ}$$

Adjacent:

$$\sqrt{180-37=143^\circ}$$

So:

$$\boxed{143^\circ, 37^\circ, 143^\circ}$$

Q7

$$2x+40=180$$

$$2x=140$$

$$\boxed{x=70^\circ}$$

Q8

$$|10-6|<x<16$$

$$\boxed{4<x<16}$$

Q9

$$\frac{(10-2)180}{10}$$

$$=\frac{1440}{10}$$

$$=\boxed{144^\circ}$$

Q10

$$\sqrt{(5-2)^2+(7-3)^2}$$

$$=\sqrt{9+16}$$

$$=\boxed{5}$$

Q11

$$130=55+x$$

$$\boxed{x=75^\circ}$$

Q12

$$4x+10+2x+20=180$$

$$6x+30=180$$

$$6x=150$$

$$\boxed{x=25}$$

Q13

Exterior angle:

$$180-156=24^\circ$$

Number sides:

$$360/24 = \boxed{15}$$

Q14

$$\frac{15(12)}{2} = \boxed{90}$$

Q15

$$|9-5|<x<14$$

$$\lfloor 4 < x < 14 \rfloor$$

Integer values:

$$\lfloor 5, 6, 7, 8, 9, 10, 11, 12, 13 \rfloor$$

Count:

$$\lfloor \boxed{9} \rfloor$$

Q16

Total ratio:

$$\lfloor 2+3+4=9 \rfloor$$

One part:

$$\lfloor 180/9=20^\circ \rfloor$$

Angles:

$$\lfloor \boxed{40^\circ, 60^\circ, 80^\circ} \rfloor$$

Q17

Exterior base angle:

$$\lfloor 110^\circ \rfloor$$

Adjacent interior base angle:

$$\lfloor 70^\circ \rfloor$$

Other base angle also:

$$\lfloor 70^\circ \rfloor$$

Vertex:

$$\lfloor 180-140 = 40^\circ \rfloor$$

Thus:

$$\lfloor \boxed{70^\circ, 70^\circ, 40^\circ} \rfloor$$

Q18

Alternate interior angles are equal:

$$\lfloor 3x+15=5x-25 \rfloor$$

$$\lfloor 40=2x \rfloor$$

$$\lfloor \boxed{x=20} \rfloor$$

Q19

Let exterior angle:

$$\lfloor E \rfloor$$

Interior is:

$$\lfloor 3E \rfloor$$

Since:

$$\lfloor E+3E=180 \rfloor$$

$$\angle 4E = 180^\circ$$

$$\angle E = 45^\circ$$

Thus:

$$n = 360/45 = \boxed{8}$$

Q20

Midpoint:

$$\left(\frac{1+7}{2}, \frac{2+10}{2} \right)$$

$$= \boxed{(4,6)}$$

Distance:

$$\sqrt{6^2 + 8^2}$$

$$= \sqrt{100}$$

$$= \boxed{10}$$

CHAPTER 43 STATUS

Covered so far:

- points, lines, rays, segments '
- angle types '
- straight-line / point-angle rules '
- complementary/supplementary angles '
- vertically opposite angles '
- parallel lines and transversal '
- corresponding/alternate/co-interior angles '
- triangle angle sum '
- exterior angle theorem '
- isosceles/equilateral/right triangles '
- triangle inequality '
- side-angle comparison '
- polygon interior/exterior angles '
- regular polygons '
- diagonals '
- coordinate distance/midpoint basics '
- symmetry basics '
- GATE/placement practice '

But **Chapter 43 is not complete yet.**

Next we still need:

- angle bisectors
- perpendicular bisectors
- medians and altitudes

- triangle centers
- basic coordinate slope intuition
- simple geometric loci
- symmetry/reflection essentials
- final Geometry Foundations revision and audit

Progress

Completed: 42/76 Current: 0/31 Chapter 43 — Geometry Foundations

NEXT

+Chapter 43 Part 2 — Triangle Lines/Centers, Bisectors, Medians, Altitudes, Coordinate Geometry Basics & Final Geometry Foundation Toolkit

USER

next

ASSISTANT

CHAPTER 43 — GEOMETRY FOUNDATIONS

PART 2 — ANGLE BISECTORS, PERPENDICULAR BISECTORS, MEDIANS, ALTITUDES, TRIANGLE CENTERS, SLOPE & REFLECTION BASICS

Status: 0/31 PROGRESS GATE: Medium Placements: Medium Priority: +Tier 2

Part 1 gave the core angle and polygon rules.

Now we finish the foundation needed before moving to triangles, similarity, Pythagoras, quadrilaterals and circles.

1. ANGLE BISECTOR

An angle bisector divides an angle into two equal angles.

If:

$$\angle ABC = 80^\circ$$

and BD bisects it:

$$\angle ABD = \angle DBC$$

Therefore:

$$\boxed{40^\circ}$$

each.

2. ANGLE BISECTOR THEOREM — PREVIEW

In triangle (ABC) , if (AD) bisects angle (A) and meets side (BC) , then:

$$\boxed{\frac{BD}{DC} = \frac{AB}{AC}}$$

This is important enough to remember now, though we will use it more in Chapter 44.

3. EXAMPLE

Suppose:

$$AB=6, \text{quad} AC=9$$

and angle bisector (AD) divides (BC) .

Then:

$$BD:DC=6:9$$

$$=\boxed{2:3}$$

4. PERPENDICULAR BISECTOR

The perpendicular bisector of a segment:

1. passes through the midpoint
 2. meets the segment at (90°)
-

5. PERPENDICULAR-BISECTOR PROPERTY

Every point on the perpendicular bisector of segment (AB) is equally distant from A and B.

So if point P lies on perpendicular bisector of AB:

$$\boxed{PA=PB}$$

6. CONVERSE

If:

$$PA=PB$$

then P lies on the perpendicular bisector of AB.

This is a useful locus property.

7. MEDIAN OF A TRIANGLE

A **median** joins a vertex to the midpoint of the opposite side.

Triangle (ABC) :

If D is midpoint of BC:

$$\boxed{BD=DC}$$

then:

$$\boxed{AD}$$

is a median.

8. NUMBER OF MEDIANS

Every triangle has:

$$\boxed{3}$$

medians.

They meet at one point called the **centroid**.

9. CENTROID

The centroid is the intersection of the three medians.

Symbol often:

$$\boxed{G}$$

Important property:

Centroid divides each median in the ratio $(2:1)$, measured from the vertex.

If A—G—D lie on median:

$$\boxed{AG:GD=2:1}$$

10. CENTROID LENGTH FORMULAS

If median:

$$\boxed{AD=m}$$

then:

$$\boxed{AG=\frac{2}{3}m}$$

and:

$$\boxed{GD=\frac{1}{3}m}$$

MUST MEMORIZE

11. EXAMPLE

Median length:

$$\boxed{15}$$

Then vertex to centroid:

$$\left[\frac{23}{15} = \boxed{10} \right]$$

Centroid to midpoint:

$$\left[\boxed{5} \right]$$

12. CENTROID COORDINATES

For triangle vertices:

$$\left[(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3) \right]$$

centroid:

$$\left[\boxed{\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)} \right]$$

13. EXAMPLE

Vertices:

$$\left[(0,0), (6,0), (0,3) \right]$$

Centroid:

$$\left[\left(\frac{6}{3}, \frac{3}{3} \right) \right]$$

$$\left[= \boxed{(2,1)} \right]$$

14. ALTITUDE OF A TRIANGLE

An **altitude** is a perpendicular from a vertex to the opposite side or its extension.

So an altitude creates:

$$\left[\boxed{90^\circ} \right]$$

15. NUMBER OF ALTITUDES

Every triangle has:

$$\left[\boxed{3} \right]$$

altitudes.

Their intersection is the **orthocenter**.

16. ORTHOCENTER LOCATION

Depends on triangle type.

Acute triangle

Orthocenter is inside.

Right triangle

Orthocenter is at the right-angle vertex.

Obtuse triangle

Orthocenter is outside.

MUST KNOW

17. WHY RIGHT-TRIANGLE ORTHOCENTER IS AT THE RIGHT VERTEX

In a right triangle, the two legs are already perpendicular.

So each leg acts as an altitude from the opposite acute vertex.

They intersect at the right-angle vertex.

18. ANGLE BISECTORS OF A TRIANGLE

The three internal angle bisectors meet at the:

$\boxed{\text{Incenter}}$

19. INCENTER

The incenter is equally distant from all three sides of the triangle.

It is the center of the triangle's incircle.

Important:

$\boxed{\text{Incenter is always inside the triangle}}$

20. INCIRCLE

A circle touching all three sides of a triangle internally is called the incircle.

Its radius is:

$\boxed{r}$

called the inradius.

21. PERPENDICULAR BISECTORS OF TRIANGLE SIDES

The three perpendicular bisectors meet at the:

$\boxed{\text{Circumcenter}}$

22. CIRCUMCENTER

Circumcenter is equally distant from all three vertices:

$\boxed{OA=OB=OC}$

It is the center of the circumcircle.

23. CIRCUMCENTER LOCATION

Acute triangle

Inside.

Right triangle

Midpoint of hypotenuse.

Obtuse triangle

Outside.

This pattern is worth memorizing.

24. RIGHT-TRIANGLE CIRCUMCENTER

For a right triangle, circumcenter = midpoint of hypotenuse.

Therefore circumradius:

$$R = \frac{\text{hypotenuse}}{2}$$

Very useful.

25. FOUR MAJOR TRIANGLE CENTERS

Center	Intersection of	Key property
Centroid	Medians	divides median (2:1)
Incenter	Angle bisectors	equidistant from sides
Circumcenter	Perpendicular bisectors	equidistant from vertices
Orthocenter	Altitudes	altitude intersection

Memory trick

M-A-P-A

- **M**edians !' centroid
 - **A**ngle bisectors !' incenter
 - **P**erpendicular bisectors !' circumcenter
 - **A**ltitudes !' orthocenter
-

26. EQUILATERAL TRIANGLE SPECIAL CASE

In an equilateral triangle:

- median
-

- altitude
- angle bisector
- perpendicular bisector

all coincide.

Therefore:

$$\boxed{\text{centroid} = \text{incenter} = \text{circumcenter} = \text{orthocenter}}$$

27. MEDIAN VS ALTITUDE

Do not confuse:

Median

goes to midpoint.

Altitude

meets opposite side at (90°) .

A median is not necessarily an altitude.

They coincide in special symmetric cases, such as an isosceles triangle from the vertex between equal sides.

28. ISOSCELES TRIANGLE SPECIAL LINE

If:

$$AB=AC$$

then the line from A to midpoint of BC is simultaneously:

- median
- altitude
- angle bisector
- perpendicular bisector of BC

Very high-value symmetry fact.

29. COORDINATE SLOPE

Slope of line through:

$$(x_1, y_1), (x_2, y_2)$$

is:

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

provided:

$$x_2 \neq x_1$$

30. SLOPE INTUITION

Slope measures:

$$\boxed{\frac{\text{vertical change}}{\text{horizontal change}}}$$

Often called:

$$\boxed{\frac{\text{rise}}{\text{run}}}$$

31. POSITIVE AND NEGATIVE SLOPE

If line rises left to right:

$$m > 0$$

If line falls left to right:

$$m < 0$$

Horizontal line:

$$m = 0$$

Vertical line:

$$\text{slope undefined}$$

32. EXAMPLE

Points:

$$(1, 2), (5, 10)$$

Slope:

$$\frac{10-2}{5-1} = \frac{8}{4} = 2$$

33. PARALLEL LINES IN COORDINATES

Parallel nonvertical lines have equal slope:

$$m_1 = m_2$$

34. PERPENDICULAR LINES IN COORDINATES

For nonvertical/nonhorizontal lines:

$$m_1 m_2 = -1$$

Their slopes are negative reciprocals.

Example:

If:

$$m_1 = 2$$

then perpendicular slope:

$$\boxed{-\frac{1}{2}}$$

35. SPECIAL HORIZONTAL/VERTICAL CASE

Horizontal line and vertical line are perpendicular.

Even though vertical slope is undefined.

36. LINE EQUATION — BASIC FORM

Slope-intercept form:

$$\boxed{y=mx+c}$$

where:

- m = slope
 - c = y-intercept
-

37. POINT-SLOPE FORM

Line through:

$$(x_1, y_1)$$

with slope m :

$$\boxed{y-y_1=m(x-x_1)}$$

Enough for basic aptitude coordinate problems.

38. COLLINEARITY USING SLOPE

Three points are collinear if slopes between them are equal.

For A,B,C:

$$\boxed{m_{AB}=m_{BC}}$$

with appropriate handling of vertical lines.

39. EXAMPLE

Points:

$$(1,1),(2,3),(3,5)$$

Slope first pair:

$$2$$

Second pair:

$$2$$

Therefore:

$\boxed{\text{collinear}}$

40. SECTION / MIDPOINT INTUITION

Midpoint already learned:

$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$

This is a special case of dividing a segment in ratio $(1:1)$.

We do not need the full section formula deeply for basic aptitude, but recognize the midpoint immediately.

41. AREA OF TRIANGLE FROM BASE AND HEIGHT

$A = \frac{1}{2}bh$

Here (h) must be perpendicular to base (b) .

This will be used more in mensuration.

42. SAME BASE AND SAME HEIGHT

Triangles with equal bases and equal perpendicular heights have equal areas.

If two triangles lie between same parallel lines on same base:

equal areas

Useful geometric reasoning.

43. TRIANGLE AREA VIA COORDINATES — BASIC

For points:

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$

area:

$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Lower priority for aptitude, but useful.

44. COLLINEARITY USING AREA

Three points are collinear if triangle area formed by them is:

0

Alternative to slope method.

45. LOCUS

A **locus** is the set of all points satisfying a geometric condition.

Aptitude usually needs only a few standard loci.

46. LOCUS EQUIDISTANT FROM TWO POINTS

All points equidistant from A and B lie on the:

$\boxed{\text{perpendicular bisector of } AB}$

47. LOCUS EQUIDISTANT FROM TWO INTERSECTING LINES

Points equidistant from the two lines lie on their:

$\boxed{\text{angle bisectors}}$

48. LOCUS AT FIXED DISTANCE FROM A POINT

All points exactly distance (r) from fixed point O form a:

$\boxed{\text{circle}}$

center O, radius (r) .

49. LOCUS AT FIXED DISTANCE FROM A LINE

All points at fixed perpendicular distance (d) from a line lie on two parallel lines, one on each side.

50. REFLECTION ACROSS x-AXIS

Point:

(x,y)

reflects to:

$\boxed{(x,-y)}$

51. REFLECTION ACROSS y-AXIS

$(x,y) \rightarrow \boxed{(-x,y)}$

52. REFLECTION THROUGH ORIGIN

$(x,y) \rightarrow \boxed{(-x,-y)}$

Equivalent to 180° rotation about origin.

53. REFLECTION ACROSS $(y=x)$

Coordinates swap:

$$\boxed{(x,y) \rightarrow (y,x)}$$

54. REFLECTION ACROSS $(y=-x)$

$$\boxed{(x,y) \rightarrow (-y,-x)}$$

Lower priority, but useful.

55. BASIC ROTATION AROUND ORIGIN

(90°) anticlockwise

$$\boxed{(x,y) \rightarrow (-y,x)}$$

(90°) clockwise

$$\boxed{(x,y) \rightarrow (y,-x)}$$

(180°)

$$\boxed{(x,y) \rightarrow (-x,-y)}$$

56. DISTANCE FROM ORIGIN

Point:

$$(x,y)$$

Distance from origin:

$$\boxed{\sqrt{x^2+y^2}}$$

57. EXAMPLE

Point:

$$(6,8)$$

distance:

$$\sqrt{36+64} = \sqrt{100} = \boxed{10}$$

58. EQUATION OF CIRCLE — BASIC RECOGNITION

Circle centered at origin with radius (r) :

$$\boxed{x^2+y^2=r^2}$$

Example:

$$\boxed{x^2+y^2=25}$$

Center:

$$\sqrt{(0,0)}$$

Radius:

$$\sqrt{\boxed{5}}$$

This is enough for basic aptitude recognition.

59. BASIC SYMMETRY OF COORDINATE AXES

If a figure is symmetric about y-axis:

Whenever:

$$\sqrt{(x,y)}$$

belongs, so does:

$$\sqrt{(-x,y)}$$

For x-axis:

$$\sqrt{(x,-y)}$$

For origin:

$$\sqrt{(-x,-y)}$$

60. GATE-STYLE QUESTION 1

A median is 18 units long. Centroid divides it in:

$$\sqrt{2:1}$$

Vertex-to-centroid:

$$\sqrt{\frac{2}{3}(18) = \boxed{12}}$$

61. GATE-STYLE QUESTION 2

A right triangle has hypotenuse 14.

Circumradius:

$$\sqrt{14/2 = \boxed{7}}$$

62. GATE-STYLE QUESTION 3

Slope through:

$$\sqrt{(2,3),(6,11)}$$

$$\sqrt{m = \frac{8}{4} = \boxed{2}}$$

63. GATE-STYLE QUESTION 4

Line A has slope:

$$\left[\frac{3}{4} \right]$$

A perpendicular line has slope:

$$\left[\boxed{-\frac{4}{3}} \right]$$

64. PLACEMENT-STYLE QUESTION 1

Point P lies on perpendicular bisector of AB.

If:

$$\left[PA=12 \right]$$

then:

$$\left[\boxed{PB=12} \right]$$

65. PLACEMENT-STYLE QUESTION 2

Triangle ABC is isosceles with:

$$\left[AB=AC \right]$$

D is midpoint of BC.

Then AD is automatically:

- median
 - altitude
 - angle bisector
 - perpendicular bisector of BC
-

66. PLACEMENT-STYLE QUESTION 3

Reflect:

$$\left[(4,-3) \right]$$

across x-axis.

Answer:

$$\left[\boxed{(4,3)} \right]$$

67. PLACEMENT-STYLE QUESTION 4

Reflect:

$$\left[(4,-3) \right]$$

across y-axis.

$$\left[\boxed{(-4,-3)} \right]$$

68. ADVANCED QUESTION — CENTROID COORDINATE

Triangle vertices:

$$\{(2,1), (5,7), (8,4)\}$$

Centroid:

$$\left(\frac{2+5+8}{3}, \frac{1+7+4}{3}\right)$$

$$=(5,4)$$

$$\boxed{(5,4)}$$

69. ADVANCED QUESTION — COLLINEARITY

Are:

$$\{(0,1), (2,5), (4,9)\}$$

collinear?

Slope first:

$$\frac{5-1}{2}=2$$

Slope second:

$$\frac{9-5}{2}=2$$

Therefore:

$$\boxed{\text{Yes}}$$

70. ADVANCED QUESTION — ANGLE BISECTOR

Triangle has:

$$AB=8, \text{ } AC=12$$

Angle A is bisected to side BC at D.

Then:

$$BD:DC=8:12$$

$$=\boxed{2:3}$$

If:

$$BC=15$$

then:

$$BD=6, \text{ } DC=9$$

71. TRIANGLE-CENTER LOCATION TABLE

Triangle type	Centroid	Incenter	Circumcenter	Orthocenter
Acute	inside	inside	inside	inside
Right	inside	inside	midpoint of hypotenuse	right-angle vertex
Obtuse	inside	inside	outside	outside

High-value conceptual table

Centroid and incenter are always inside a nondegenerate triangle.

72. COMMON TRAPS

& pTRAP 1 — Median vs perpendicular bisector

Median comes from a vertex.

Perpendicular bisector need not pass through a triangle vertex.

& pTRAP 2 — Centroid ratio direction

$\boxed{2:1}$

with the longer part toward the vertex.

& pTRAP 3 — Circumcenter vs incenter

Circumcenter !' equal distance from vertices.

Incenter !' equal perpendicular distance from sides.

& pTRAP 4 — Slope of vertical line

Not 0.

It is:

$\boxed{\text{undefined}}$

Horizontal slope is 0.

& pTRAP 5 — Perpendicular slopes

Use:

$m_1 m_2 = -1$

only when both slopes are finite/nonzero in the usual form.

& pTRAP 6 — Reflection signs

x-axis reflection changes y-sign.

y-axis reflection changes x-sign.

73. MUST-MEMORIZE TRIANGLE CENTER RULES

Centroid

Medians:

$$\boxed{2:1}$$

Incenter

Angle bisectors; equal distance from sides.

Circumcenter

Perpendicular bisectors; equal distance from vertices.

Orthocenter

Altitudes.

74. MUST-MEMORIZE COORDINATE FORMULAS

Distance:

$$\boxed{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

Midpoint:

$$\boxed{\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)}$$

Slope:

$$\boxed{\frac{y_2 - y_1}{x_2 - x_1}}$$

75. MUST-MEMORIZE REFLECTIONS

Across x-axis:

$$\boxed{(x, y) \rightarrow (x, -y)}$$

Across y-axis:

$$\boxed{(x, y) \rightarrow (-x, y)}$$

Through origin:

$$\boxed{(x, y) \rightarrow (-x, -y)}$$

Across $(y=x)$:

$$\boxed{(x, y) \rightarrow (y, x)}$$

76. FAST EXAM DECISION TREE

Vertex to midpoint?

Median.

Vertex perpendicular to opposite side?

Altitude.

Divides vertex angle equally?

Angle bisector.

Perpendicular through side midpoint?

Perpendicular bisector.

Three medians?

Centroid.

Three angle bisectors?

Incenter.

Three perpendicular bisectors?

Circumcenter.

Three altitudes?

Orthocenter.

Parallel coordinates?

Equal slopes.

Perpendicular coordinates?

Negative reciprocal slopes.

Equidistant from two points?

Perpendicular bisector.

PRACTICE — LEVEL 0

Q1

A median is 12 units. Find vertex-to-centroid distance.

Q2

Which center is formed by angle bisectors?

Q3

Which center is formed by altitudes?

Q4

Find slope between:

$\backslash (1,2),(4,8) \backslash$

Q5

Reflect $\backslash (3,5) \backslash$ across x-axis.

LEVEL 1

Q6

Median length is 21. Find both centroid segments.

Q7

A right triangle has hypotenuse 20. Find circumradius.

Q8

Find perpendicular slope to:

$$\sqrt{m=5}$$

Q9

Find midpoint of:

$$\sqrt{(-2,4),(8,10)}$$

Q10

Point P lies on perpendicular bisector of AB. If $(PA=17)$, find PB.

LEVEL 2

Q11

Triangle vertices:

$$\sqrt{(0,0),(9,0),(0,6)}$$

Find centroid.

Q12

Are points:

$$\sqrt{(1,2),(3,6),(5,10)}$$

collinear?

Q13

In triangle ABC, angle A bisector meets BC at D. If:

$$\sqrt{AB=10,\text{quad} AC=15,\text{quad} BC=20}$$

find BD and DC.

Q14

Reflect $((-4,7))$ through origin.

Q15

A line has slope $(-3/4)$. Find perpendicular slope.

LEVEL 3 — EXAM PRESSURE

Q16

In an isosceles triangle ($AB=AC$), median AD to BC has length 12. What other geometric roles does AD perform?

Q17

Centroid G divides median AD. If:

$$GD=4$$

find AG and AD.

Q18

Triangle vertices:

$$(2,3), (8,3), (5,9)$$

Find centroid.

Q19

A right triangle has circumradius 6. Find hypotenuse.

Q20

A point (a,b) is reflected across $(y=x)$, then across x-axis. Write final coordinates.

ANSWERS**Q1**

$$\frac{2}{3}(12) = \boxed{8}$$

Q2

$$\boxed{\text{Incenter}}$$

Q3

$$\boxed{\text{Orthocenter}}$$

Q4

$$m = \frac{8-2}{4-1} = \frac{6}{3} = \boxed{2}$$

Q5

$$\boxed{(3, -5)}$$

Q6

Centroid ratio:

$$2:1$$

So:

$$AG = \frac{2}{3}(21) = 14$$

$$GD = \frac{1}{3}(21) = 7$$

$$\boxed{14, 7}$$

Q7

$$[R=\frac{20}{2} = \boxed{10}]$$

Q8

$$[m_{\perp}=-\frac{15}{ }]$$

$$[\boxed{-\frac{15}{ }}]$$

Q9

$$[\left(\frac{-2+8}{2}, \frac{4+10}{2} \right)]$$

$$[=\boxed{(3,7)}]$$

Q10

$$[\boxed{17}]$$

Q11

$$[\left(\frac{0+9+0}{3}, \frac{0+0+6}{3} \right)]$$

$$[=\boxed{(3,2)}]$$

Q12

First slope:

$$[\frac{6-2}{3-1}=2]$$

Second slope:

$$[\frac{10-6}{5-3}=2]$$

So:

$$[\boxed{\text{Yes}}]$$

Q13

Angle bisector theorem:

$$[BD:DC=10:15=2:3]$$

Total:

$$[20]$$

So:

$$[BD=8]$$

$$[DC=12]$$

$$[\boxed{8, 12}]$$

Q14

Through origin:

$$[(x,y) \text{ to } (-x,-y)]$$

So:

$$[(-4,7) \text{ to } \boxed{(4,-7)}]$$

Q15

Negative reciprocal of:

$$\left[-\frac{3}{4} \right]$$

is:

$$\left[\boxed{\frac{4}{3}} \right]$$

Q16

Since triangle is isosceles and AD is median from the vertex between equal sides, AD is also:

- altitude
- angle bisector
- perpendicular bisector of BC

$$\left[\boxed{\text{all three}} \right]$$

Q17

$$\left[AG:GD=2:1 \right]$$

If:

$$\left[GD=4 \right]$$

then:

$$\left[AG=8 \right]$$

and:

$$\left[AD=12 \right]$$

$$\left[\boxed{AG=8, AD=12} \right]$$

Q18

$$\left[\left(\frac{2+8+5}{3}, \frac{3+3+9}{3} \right) \right]$$

$$\left[=(5,5) \right]$$

$$\left[\boxed{(5,5)} \right]$$

Q19

Right triangle:

$$\left[R=\frac{\text{hypotenuse}}{2} \right]$$

So:

$$\left[\text{hypotenuse}=12 \right]$$

$$\left[\boxed{12} \right]$$

Q20

First reflect across:

$$\left[y=x \right]$$

$$\left[(a,b) \rightarrow (b,a) \right]$$

Then across x-axis:

$\forall (b,a)\to(b,-a) \forall$

Therefore:

$\forall \boxed{(b,-a)} \forall$

' CHAPTER 43 COMPLETE

We have now covered:

- point/line/ray/segment basics '
- angle classifications '
- straight-line and around-point rules '
- vertically opposite angles '
- parallel lines and transversals '
- corresponding/alternate/co-interior angles '
- triangle angle rules '
- exterior-angle theorem '
- isosceles/equilateral/right triangles '
- triangle inequality '
- polygons and regular polygons '
- polygon interior/exterior angles '
- diagonals '
- angle bisectors '
- perpendicular bisectors '
- medians '
- centroid '
- altitudes '
- orthocenter '
- incenter '
- circumcenter '
- coordinate distance/midpoint '
- slope '
- parallel/perpendicular slope conditions '
- collinearity '
- locus basics '
- reflections and rotations '
- GATE/placement L0–L3 practice '

Progress

Completed: 43/76

Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

Chapter 44 — Triangles, Similarity and Pythagoras

Next we will cover:

- congruent triangles
- SSS/SAS/ASA/AAS/RHS
- similar triangles
- AA/SAS/SSS similarity
- corresponding-side ratios
- area ratio of similar triangles
- Basic Proportionality Theorem
- angle-bisector theorem applications
- Pythagorean theorem
- Pythagorean triples
- special $(45^\circ-45^\circ-90^\circ)$ and $(30^\circ-60^\circ-90^\circ)$ triangles
- altitude to hypotenuse relations
- GATE/placement geometry shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 44 — TRIANGLES, SIMILARITY AND PYTHAGORAS

PART 1 — CONGRUENCE, SIMILARITY, BASIC PROPORTIONALITY & PYTHAGOREAN THEOREM

Status: 0% PROGRESS **GATE:** Medium–High **Placements:** Medium–High **Priority:** +Tier 2

This chapter is very important because triangle questions often combine:

- angles
- side ratios
- similarity
- areas
- Pythagoras
- hidden proportionality

The main exam skill is:

Do not calculate every side. First check whether the triangles are congruent, similar, or right-angled.

1. CONGRUENT TRIANGLES

Two triangles are **congruent** if they have exactly the same:

- shape
- size

Corresponding sides and angles are equal.

Notation:

$\boxed{\triangle ABC \cong \triangle DEF}$

If correspondence is:

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

then:

$AB = DE$

$BC = EF$

$CA = FD$

and corresponding angles are equal.

2. SIMILAR VS CONGRUENT

Congruent

Same shape + same size.

Similar

Same shape, but size may differ.

So:

$\boxed{\text{Congruent} \rightarrow \text{Similar}}$

but:

$\boxed{\text{Similar} \not\rightarrow \text{Congruent}}$

3. SSS CONGRUENCE

If all three corresponding sides are equal:

$AB = DE$

$BC = EF$

$CA = FD$

then:

$\boxed{\triangle ABC \cong \triangle DEF}$

SSS = **Side-Side-Side**

4. SAS CONGRUENCE

If:

- two corresponding sides are equal
- the included angle is equal

then triangles are congruent.

$\boxed{\text{SAS}}$

Important:

The angle must be between the two given sides.

5. ASA CONGRUENCE

If:

- two angles are equal
- included side is equal

then triangles are congruent.

$\boxed{\text{ASA}}$

6. AAS CONGRUENCE

Two angles + one corresponding non-included side also determine a triangle.

$\boxed{\text{AAS}}$

7. RHS CONGRUENCE

For right triangles:

If:

- hypotenuse equal
- one corresponding side equal

then triangles are congruent.

$\boxed{\text{RHS}}$

Also called:

- HL in some systems
- Right angle–Hypotenuse–Side

8. AAA IS NOT A CONGRUENCE TEST

If all corresponding angles are equal, triangles have the same shape but may be different sizes.

So AAA proves:

$\boxed{\text{similarity}}$

not necessarily congruence.

9. SSA IS NOT GENERALLY A CONGRUENCE TEST

Two sides and a non-included angle can sometimes create more than one possible triangle.

So:

$\boxed{\text{SSA is generally invalid for congruence}}$

Important trap.

10. CPCTC

Once triangles are proved congruent:

Corresponding Parts of Congruent Triangles are Congruent.

Often written:

$\boxed{\text{CPCTC}}$

Then corresponding:

- sides
- angles
- medians etc.

can be declared equal.

11. SIMILAR TRIANGLES

Two triangles are similar if:

- corresponding angles are equal
- corresponding sides are proportional

Notation:

$\boxed{\triangle ABC \sim \triangle DEF}$

12. CORRESPONDENCE MATTERS

If:

$\triangle ABC \sim \triangle DEF$

then:

$A \rightarrow D$

$B \rightarrow E$

$C \rightarrow F$

Thus:

$\boxed{\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}}$

Never mix corresponding sides.

13. AA SIMILARITY

If two corresponding angles are equal, the third automatically becomes equal because triangle angle sum is (180°) .

Therefore:

$\boxed{\text{AA similarity}}$

is sufficient.

14. AAA SIMILARITY

Equivalent idea:

All corresponding angles equal.

$\boxed{\text{AAA}}$

proves similarity.

15. SAS SIMILARITY

If two pairs of corresponding sides are proportional and their included angles are equal:

$\boxed{\text{SAS similarity}}$

16. SSS SIMILARITY

If:

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

then:

$\boxed{\text{SSS similarity}}$

17. SCALE FACTOR

If corresponding sides satisfy:

$$\frac{\text{side of triangle 1}}{\text{side of triangle 2}} = k$$

then (k) is the linear scale factor.

Example:

Sides:

$$3, 4, 5$$

and:

$$6, 8, 10$$

Scale factor:

$$\sqrt{\boxed{2}}$$

18. PERIMETER RATIO OF SIMILAR TRIANGLES

If side ratio is:

$$\sqrt{a:b}$$

then perimeter ratio is also:

$$\sqrt{\boxed{a:b}}$$

Because every corresponding length scales by the same factor.

19. AREA RATIO OF SIMILAR TRIANGLES

If corresponding side ratio is:

$$\sqrt{a:b}$$

then area ratio is:

$$\sqrt{\boxed{a^2:b^2}}$$

MUST MEMORIZE

20. EXAMPLE

Two similar triangles have side ratio:

$$\sqrt{2:3}$$

Then area ratio:

$$\sqrt{2^2:3^2}$$

$$\sqrt{= \boxed{4:9}}$$

21. REVERSE AREA-TO-SIDE RATIO

If area ratio is:

$$\sqrt{25:49}$$

then corresponding side ratio:

$$\sqrt{\boxed{5:7}}$$

Take square roots.

22. ALTITUDE/MEDIAN/ANGLE-BISECTOR RATIO IN SIMILAR TRIANGLES

All corresponding linear measurements scale in the same ratio as corresponding sides.

So if side ratio:

$$\boxed{a:b}$$

then corresponding:

- altitudes
- medians
- angle bisectors
- inradii
- circumradii

also have ratio:

$$\boxed{a:b}$$

23. BASIC PROPORTIONALITY THEOREM — BPT

Also called **Thales' theorem** in this context.

In triangle $\triangle ABC$, if:

$$DE \parallel BC$$

with D on AB and E on AC, then:

$$\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$$

Also:

$$\triangle ADE \sim \triangle ABC$$

24. WHY BPT WORKS

Because:

$$DE \parallel BC$$

creates equal corresponding angles.

Therefore:

$$\triangle ADE \sim \triangle ABC$$

Then corresponding side ratios follow.

25. BPT FULL RATIOS

From similarity:

$$\boxed{\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}}$$

and:

$$\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$$

26. EXAMPLE

In triangle ABC:

$$DE \parallel BC$$

$$AD=4, \quad DB=6, \quad AE=8$$

Find EC.

Using BPT:

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$4EC = 48$$

$$\boxed{EC=12}$$

27. CONVERSE OF BPT

If:

$$\frac{AD}{DB} = \frac{AE}{EC}$$

then:

$$\boxed{DE \parallel BC}$$

Useful for proving parallelism.

28. ANGLE BISECTOR THEOREM

In triangle ABC, if AD bisects angle A:

$$\boxed{\frac{BD}{DC} = \frac{AB}{AC}}$$

Already previewed in Chapter 43.

29. EXAMPLE

$$AB=10, \quad AC=15, \quad BC=20$$

Angle A is bisected.

Then:

$$BD:DC=10:15=2:3$$

Total:

$$20$$

So:

$$BD=8$$

$$DC=12$$

30. PYTHAGOREAN THEOREM

For a right triangle:

$$\boxed{a^2 + b^2 = c^2}$$

where c is the hypotenuse.

MUST MEMORIZE

31. IDENTIFY THE HYPOTENUSE FIRST

The hypotenuse is:

- opposite 90°
- longest side of right triangle

Never blindly square the largest-looking side from a diagram.

32. EXAMPLE — 3-4-5

Legs:

$$3, 4$$

Hypotenuse:

$$c = \sqrt{3^2 + 4^2}$$

$$= \sqrt{25}$$

$$= \boxed{5}$$

33. FIND A LEG

Hypotenuse:

$$13$$

one leg:

$$5$$

other:

$$x^2 = 13^2 - 5^2$$

$$= 169 - 25$$

$$= 144$$

$$\boxed{x = 12}$$

34. CONVERSE OF PYTHAGORAS

If sides satisfy:

$$a^2 + b^2 = c^2$$

with $\angle C$ largest, then triangle is right-angled.

Example:

$$[6, 8, 10]$$

$$[36 + 64 = 100]$$

Therefore:

$$[\boxed{\text{right triangle}}]$$

35. ACUTE / RIGHT / OBTUSE USING SIDES

Let:

$$[c]$$

be largest side.

Compare:

$$[a^2 + b^2]$$

with:

$$[c^2]$$

If

$$[a^2 + b^2 = c^2]$$

right triangle.

If

$$[a^2 + b^2 > c^2]$$

acute triangle.

If

$$[a^2 + b^2 < c^2]$$

obtuse triangle.

36. EXAMPLE

Sides:

$$[4, 5, 6]$$

Largest:

$$[6]$$

$$[4^2 + 5^2 = 41]$$

$$[6^2 = 36]$$

Since:

$$[41 > 36]$$

triangle is:

$$\boxed{\text{acute}}$$

37. PYTHAGOREAN TRIPLES

Common integer triples satisfying:

$$a^2 + b^2 = c^2$$

Must recognize quickly.

MUST MEMORIZE

$$\boxed{3, 4, 5}$$

$$\boxed{5, 12, 13}$$

$$\boxed{8, 15, 17}$$

$$\boxed{7, 24, 25}$$

$$\boxed{20, 21, 29}$$

$$\boxed{9, 40, 41}$$

38. MULTIPLES ARE ALSO TRIPLES

If:

$$\boxed{3, 4, 5}$$

is a triple, then:

$$\boxed{6, 8, 10}$$

$$\boxed{9, 12, 15}$$

etc. are also right triangles.

39. GENERATING PYTHAGOREAN TRIPLES

For integers:

$$m > n > 0$$

a triple can be generated by:

$$\boxed{m^2 - n^2, 2mn, m^2 + n^2}$$

Lower priority for aptitude, but useful.

Example:

$$m=2, n=1$$

gives:

$$\boxed{3, 4, 5}$$

40. DIAGONAL OF RECTANGLE

A rectangle with sides:

$$\sqrt[l, b]{}$$

has diagonal:

$$\sqrt{\boxed{d = \sqrt{l^2 + b^2}}}$$

Direct Pythagoras.

41. EXAMPLE

Rectangle:

$$\sqrt[6 \times 8]{}$$

Diagonal:

$$\sqrt{\sqrt{36 + 64}}$$

$$\sqrt{= \boxed{10}}$$

42. DIAGONAL OF SQUARE

Side:

$$\sqrt{a}$$

Then:

$$\sqrt{d^2 = a^2 + a^2}$$

$$\sqrt{d = a\sqrt{2}}$$

So:

$$\sqrt{\boxed{d = a\sqrt{2}}}$$

MUST MEMORIZE

43. SPECIAL TRIANGLE — 45°-45°-90°

This is an isosceles right triangle.

Side ratio:

$$\sqrt{\boxed{1:1:\sqrt{2}}}$$

where:

- legs = $\sqrt{(1,1)}$
 - hypotenuse = $\sqrt{(\sqrt{2})}$
-

44. EXAMPLE

Leg:

$$\sqrt{7}$$

Hypotenuse:

$$\boxed{7\sqrt{2}}$$

45. SPECIAL TRIANGLE — 30°-60°-90°

Side ratio opposite:

$$30^\circ:60^\circ:90^\circ$$

is:

$$\boxed{1:\sqrt{3}:2}$$

MUST MEMORIZE

46. SIDE IDENTIFICATION

In a 30-60-90 triangle:

- opposite 30° = shortest = x
 - opposite 60° = $x\sqrt{3}$
 - hypotenuse = $2x$
-

47. EXAMPLE

Hypotenuse:

$$10$$

Then shortest side:

$$5$$

Longer leg:

$$5\sqrt{3}$$

So:

$$\boxed{5, 5\sqrt{3}, 10}$$

48. EQUILATERAL TRIANGLE ALTITUDE

For equilateral triangle side a , draw altitude.

It creates two 30-60-90 triangles.

Therefore altitude:

$$\boxed{h = \frac{\sqrt{3}}{2}a}$$

49. EQUILATERAL TRIANGLE AREA

Using:

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2}a \left(\frac{\sqrt{3}}{2}a \right)$$

Therefore:

$$\boxed{A = \frac{\sqrt{3}}{4}a^2}$$

Useful later in mensuration too.

50. RIGHT TRIANGLE CIRCUMRADIUS RECAP

In a right triangle:

$$\boxed{R = \frac{c}{2}}$$

where (c) is hypotenuse.

51. MEDIAN TO HYPOTENUSE

In a right triangle, the median from right-angle vertex to hypotenuse equals half the hypotenuse.

Therefore:

$$\boxed{m_c = \frac{c}{2}}$$

It is also the circumradius.

52. EXAMPLE

Right triangle hypotenuse:

$$18$$

Median to hypotenuse:

$$\boxed{9}$$

53. AREA USING TWO LEGS OF RIGHT TRIANGLE

Since legs are perpendicular:

$$\boxed{A = \frac{1}{2}ab}$$

where (a,b) are legs.

Example:

$$6, 8, 10$$

Area:

$$\frac{1}{2}(6)(8) = \boxed{24}$$

54. ALTITUDE TO HYPOTENUSE

Suppose a right triangle has legs (a,b) , hypotenuse (c) , and altitude (h) from the right angle to hypotenuse.

Area in two ways:

$$\frac{1}{2}ab = \frac{1}{2}ch$$

Therefore:

$$h = \frac{ab}{c}$$

High-value formula

55. EXAMPLE

Right triangle:

$$6, 8, 10$$

Altitude to hypotenuse:

$$h = \frac{6 \cdot 8}{10}$$

$$h = \frac{24}{5} = 4.8$$

56. HYPOTENUSE SEGMENT RELATIONS — PREVIEW

Altitude to hypotenuse divides it into segments (p,q) , with:

$$p+q=c$$

Then:

$$h^2 = pq$$

and each leg satisfies:

$$a^2 = cp$$

$$b^2 = cq$$

These come from similarity of the three right triangles.

We will reinforce them in Part 2.

57. SIMILARITY CREATED BY ALTITUDE TO HYPOTENUSE

In a right triangle, altitude from right angle to hypotenuse creates two smaller triangles.

All three triangles are similar.

This is one of the most useful hidden-similarity configurations.

58. SIMILAR TRIANGLE QUICK EQUATION

If triangles are similar and side ratio:

$$\left[\frac{a}{b} = k \right]$$

then any corresponding length ratio is (k) .

Any corresponding area ratio:

$$\left[k^2 \right]$$

This can eliminate large calculations.

59. EXAMPLE — AREA RATIO

Two similar triangles have areas:

$$\left[36, 100 \right]$$

Corresponding side ratio:

$$\left[\sqrt{36:100} \right]$$

$$\left[=6:10 \right]$$

$$\left[=\boxed{3:5} \right]$$

60. EXAMPLE — UNKNOWN SIDE

Two similar triangles have corresponding sides:

$$\left[6 \rightarrow 9 \right]$$

$$\left[10 \rightarrow x \right]$$

Then:

$$\left[\frac{6}{10} = \frac{9}{x} \right]$$

$$\left[6x = 90 \right]$$

$$\left[\boxed{x = 15} \right]$$

61. SHADOW PROBLEMS — SIMILARITY

If a vertical object and its shadow form a right triangle with sunlight, another vertical object at the same time creates a similar triangle.

Thus:

$$\left[\boxed{\frac{\text{height}_1}{\text{shadow}_1} = \frac{\text{height}_2}{\text{shadow}_2}} \right]$$

Classic placement problem.

62. EXAMPLE — SHADOW

A 2 m pole casts a 3 m shadow.

A tree casts a 12 m shadow.

Tree height:

$$\left[\frac{23}{\frac{x}{12}} \right]$$

$$\left[3x=24 \right]$$

$$\left[\boxed{x=8\text{ m}} \right]$$

63. GATE-STYLE QUESTION 1

Two similar triangles have side ratio:

$$\left[3:7 \right]$$

Find area ratio.

$$\left[\boxed{9:49} \right]$$

64. GATE-STYLE QUESTION 2

A right triangle has legs:

$$\left[9, 12 \right]$$

Hypotenuse:

$$\left[\sqrt{81+144} = \sqrt{225} \right]$$

$$\left[=\boxed{15} \right]$$

65. GATE-STYLE QUESTION 3

Triangle sides:

$$\left[7, 24, 25 \right]$$

Since:

$$\left[49+576=625 \right]$$

it is:

$$\left[\boxed{\text{right-angled}} \right]$$

66. GATE-STYLE QUESTION 4

A square has diagonal:

$$\left[10\sqrt{2} \right]$$

Since:

$$\left[d=a\sqrt{2} \right]$$

$$\left[a=\boxed{10} \right]$$

67. PLACEMENT-STYLE QUESTION 1

Two similar triangles have perimeters:

$$\{ 24, 36 \}$$

Corresponding side ratio:

$$\{ 24:36 \}$$

$$\{ \boxed{2:3} \}$$

Area ratio:

$$\{ \boxed{4:9} \}$$

68. PLACEMENT-STYLE QUESTION 2

In triangle ABC, $(DE \parallel BC)$.

$$\{ AD=3, DB=2, AE=6 \}$$

Find EC.

$$\{ \frac{32}{6} = EC \}$$

$$\{ 3EC=12 \}$$

$$\{ \boxed{EC=4} \}$$

69. PLACEMENT-STYLE QUESTION 3

Hypotenuse is 20 in a 30-60-90 triangle.

Shortest leg:

$$\{ 10 \}$$

Other leg:

$$\{ 10\sqrt{3} \}$$

$$\{ \boxed{10, 10\sqrt{3}, 20} \}$$

70. PLACEMENT-STYLE QUESTION 4

A 45-45-90 triangle has hypotenuse:

$$\{ 12 \}$$

Each leg:

$$\{ \frac{12}{\sqrt{2}} = 6\sqrt{2} \}$$

$$\{ \boxed{6\sqrt{2}} \}$$

71. COMMON TRAPS

& pTRAP 1 — Congruence vs similarity

AAA proves similarity, not congruence.

& pTRAP 2 — Wrong corresponding sides

Write vertex correspondence first.

& pTRAP 3 — Area ratio

If side ratio is:

$$\sqrt{a:b}$$

area ratio is:

$$\sqrt{a^2:b^2}$$

not $\sqrt{a:b}$.

& pTRAP 4 — Pythagoras on non-right triangle

Do not use:

$$\sqrt{a^2+b^2=c^2}$$

unless right angle is known or proven.

& pTRAP 5 — Hypotenuse

Always opposite (90°) .

& pTRAP 6 — 30-60-90 ordering

$$\sqrt{1:\sqrt{3}:2}$$

corresponds to opposite:

$$\sqrt{30^\circ:60^\circ:90^\circ}$$

& pTRAP 7 — BPT

Requires a line parallel to one triangle side.

72. MUST-MEMORIZE CONGRUENCE TESTS

$$\sqrt{\boxed{\text{SSS}}}$$

$$\sqrt{\boxed{\text{SAS}}}$$

$$\sqrt{\boxed{\text{ASA/AAS}}}$$

$$\sqrt{\boxed{\text{RHS}}}$$

Not generally valid:

$$\sqrt{\boxed{\text{AAA, SSA}}}$$

for congruence.

73. MUST-MEMORIZE SIMILARITY TESTS

$\boxed{\text{AA/AAA}} \setminus$

$\boxed{\text{SAS proportional}} \setminus$

$\boxed{\text{SSS proportional}} \setminus$

74. MUST-MEMORIZE SIMILARITY RATIOS

Side ratio:

$\boxed{k} \setminus$

Perimeter ratio:

$\boxed{k} \setminus$

Area ratio:

$\boxed{k^2} \setminus$

75. MUST-MEMORIZE PYTHAGORAS

$\boxed{a^2+b^2=c^2} \setminus$

And common triples:

$\boxed{3-4-5} \setminus$

$\boxed{5-12-13} \setminus$

$\boxed{8-15-17} \setminus$

$\boxed{7-24-25} \setminus$

76. MUST-MEMORIZE SPECIAL TRIANGLES

45-45-90

$\boxed{1:1:\sqrt{2}} \setminus$

30-60-90

$\boxed{1:\sqrt{3}:2} \setminus$

77. FAST EXAM DECISION TREE

Same shape?

Check similarity.

Exact equal sides/angles?

Check congruence.

Parallel line inside triangle?

Think BPT + similarity.

Angle bisector?

Use side division ratio.

Right angle?

Think Pythagoras.

Side values resemble known triple?

Use triple immediately.

Similar triangles + areas?

Square the side ratio.

Equilateral altitude?

Split into 30-60-90 triangles.

Right triangle altitude to hypotenuse?

Look for hidden similar triangles.

PRACTICE — LEVEL 0

Q1

Are triangles with sides $(3,4,5)$ and $(6,8,10)$ congruent or similar?

Q2

If similar triangles have side ratio $(4:5)$, find area ratio.

Q3

Find hypotenuse for legs 5 and 12.

Q4

Find diagonal of a square of side 8.

Q5

In a 30-60-90 triangle, shortest side is 4. Find hypotenuse.

LEVEL 1

Q6

Two similar triangles have areas 64 and 121. Find side ratio.

Q7

In triangle ABC, $(DE \parallel BC)$. If:

$$[AD=6, DB=3, AE=8]$$

find EC.

Q8

An angle bisector divides opposite side into segments 6 and 9. Find ratio of adjacent sides.

Q9

Classify triangle with sides 8,15,17.

Q10

Right triangle legs are 12 and 16. Find hypotenuse and area.

LEVEL 2

Q11

Two similar triangles have perimeters 30 and 45. Smaller triangle area is 40. Find larger area.

Q12

A right triangle has hypotenuse 25 and one leg 7. Find other leg.

Q13

An equilateral triangle has side 10. Find altitude.

Q14

Right triangle legs 9 and 12. Find altitude to hypotenuse.

Q15

A pole 1.5 m tall casts a 2 m shadow. A tree casts a 14 m shadow. Find tree height.

LEVEL 3 — EXAM PRESSURE

Q16

A right triangle has hypotenuse split by altitude into segments 9 and 16. Find hypotenuse and altitude.

Q17

Two similar triangles have area ratio 81:121. Their corresponding altitudes are in what ratio?

Q18

A triangle has sides 10,24,26. Find area.

Q19

In triangle ABC, $(DE \parallel BC)$. If:

$$[AD:AB=3:5]$$

and:

$$[DE=12]$$

find BC.

Q20

A 45-45-90 triangle and a 30-60-90 triangle each have hypotenuse 10. Compare their smaller-leg lengths.

ANSWERS**Q1**

Side ratios:

$$\sqrt{3}:6=4:8=5:10=1:2 \sqrt{}$$

So:

$$\sqrt{\boxed{\text{Similar, not congruent}} \sqrt{}}$$

Q2

$$\sqrt{4^2:5^2 = \boxed{16:25} \sqrt{}}$$

Q3

$$\sqrt{5^2+12^2=25+144=169 \sqrt{}}$$

$$\sqrt{c=\boxed{13} \sqrt{}}$$

Q4

$$\sqrt{d=8\sqrt{2} \sqrt{}}$$

$$\sqrt{\boxed{8\sqrt{2}} \sqrt{}}$$

Q5

Hypotenuse:

$$\sqrt{2(4) = \boxed{8} \sqrt{}}$$

Q6

Side ratio:

$$\sqrt{\sqrt{64:121} \sqrt{}}$$

$$\sqrt{=\boxed{8:11} \sqrt{}}$$

Q7

By BPT:

$$\sqrt{\frac{63}{8}=\frac{8}{EC} \sqrt{}}$$

$$\sqrt{2=\frac{8}{EC} \sqrt{}}$$

$$\sqrt{\boxed{EC=4} \sqrt{}}$$

Q8

Angle-bisector theorem:

$$\sqrt{\boxed{2:3} \sqrt{}}$$

Q9

$$\sqrt{8^2+15^2 = 64+225 = 289 = 17^2 \sqrt{}}$$

$\boxed{\text{Right triangle}}$

Q10

Hypotenuse:

$$\sqrt{144+256} = 20$$

Area:

$$\frac{1}{2}(12)(16) = 96$$

$$\boxed{20, 96}$$

Q11

Perimeter/side ratio:

$$30:45=2:3$$

Area ratio:

$$4:9$$

If smaller area = 40:

$$\frac{40}{A} = \frac{4}{9}$$

$$A = 90$$

$$\boxed{90}$$

Q12

$$x^2 = 25^2 - 7^2$$

$$= 625 - 49 = 576$$

$$\boxed{x=24}$$

Q13

$$h = \frac{\sqrt{3}}{2}(10)$$

$$= \boxed{5\sqrt{3}}$$

Q14

Hypotenuse:

$$15$$

Altitude:

$$h = \frac{9 \cdot 12}{15}$$

$$= \boxed{\frac{36}{5} = 7.2}$$

Q15

By similarity:

$$\frac{1.5}{2} = \frac{x}{14}$$

$$x = 14 \cdot 0.75$$

$$= \boxed{10.5 \text{ m}}$$

Q16

Hypotenuse:

$$\sqrt{c=9+16} = \boxed{25}$$

Altitude:

$$\sqrt{h^2=pq}$$

$$\sqrt{h^2=9(16)=144}$$

$$\sqrt{h=12}$$

Q17

Corresponding altitudes are linear measurements.

So same as side ratio:

$$\sqrt{\text{sqrt}\{81:121\}}$$

$$\sqrt{=9:11}$$

Q18

Check:

$$\sqrt{10^2+24^2 = 100+576 = 676 = 26^2}$$

Right triangle.

Area:

$$\sqrt{\frac{1}{2}(10)(24)}$$

$$\sqrt{=120}$$

Q19

Similarity:

$$\sqrt{\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{5}}$$

$$\sqrt{\frac{12}{BC} = \frac{3}{5}}$$

$$\sqrt{36=3BC}$$

$$\sqrt{BC=20}$$

Q20

45-45-90 with hypotenuse 10:

Leg:

$$\sqrt{\frac{10}{\sqrt{2}} = 5\sqrt{2}}$$

30-60-90 with hypotenuse 10:

smaller leg:

$$\sqrt{5}$$

Thus:

$$\sqrt{5\sqrt{2}:5=\sqrt{2}:1}$$

CHAPTER 44 STATUS

Covered so far:

- congruent vs similar triangles '
- SSS/SAS/ASA/AAS/RHS congruence '
- invalid AAA/SSA congruence traps '
- AA/SAS/SSS similarity '
- corresponding-side ratios '
- perimeter ratio '
- area ratio '
- BPT and converse '
- angle-bisector theorem '
- Pythagorean theorem '
- converse and triangle classification '
- Pythagorean triples '
- special right triangles '
- equilateral altitude/area '
- right-triangle area '
- median to hypotenuse '
- altitude to hypotenuse preview '
- shadow similarity '
- GATE/placement practice '

But **Chapter 44 is not complete yet.**

Next we still need:

- deeper altitude-to-hypotenuse similarity relations
- geometric-mean theorems
- Stewart's theorem intuition / median formulas where aptitude-relevant
- Apollonius theorem
- Heron's formula
- inradius/circumradius relations
- triangle area formulas
- final triangle shortcut table and audit

Progress

Completed: 43/76 Current: 0/51 Chapter 44 — Triangles, Similarity & Pythagoras

NEXT

+Chapter 44 Part 2 — Right-Triangle Similarity Relations, Heron's Formula, Inradius/Circumradius, Medians & Final Triangle Toolkit

USER

next

ASSISTANT

CHAPTER 44 — TRIANGLES, SIMILARITY AND PYTHAGORAS

PART 2 — RIGHT-TRIANGLE SIMILARITY, GEOMETRIC-MEAN THEOREMS, HERON'S FORMULA, MEDIANS, INRADIUS & CIRCUMRADIUS

Status: $\emptyset$ = 0% PROGRESS GATE: Medium–High Placements: Medium–High Priority: +Tier 2

Now we finish the triangle toolkit.

The most important ideas here are:

- altitude to hypotenuse relations
- geometric means
- median formulas
- Heron's formula
- inradius/circumradius
- multiple area formulas

1. RIGHT TRIANGLE WITH ALTITUDE TO HYPOTENUSE

Consider a right triangle with:

- legs $\backslash(a,b\backslash)$
- hypotenuse $\backslash(c\backslash)$
- altitude $\backslash(h\backslash)$ to hypotenuse

The altitude divides hypotenuse into:

$\backslash p,\backslash q \backslash$

such that:

$\backslash \boxed{p+q=c} \backslash$

The three triangles formed are similar.

This creates several powerful formulas.

2. ALTITUDE GEOMETRIC-MEAN THEOREM

$\backslash \boxed{h^2=pq} \backslash$

Therefore:

$\backslash \boxed{h=\sqrt{pq}} \backslash$

So altitude to hypotenuse is the geometric mean of the two hypotenuse segments.

3. EXAMPLE

Hypotenuse segments:

$$\sqrt{9 \cdot 16}$$

Then:

$$\sqrt{h^2 = 9 \cdot 16}$$

$$\sqrt{h = 12}$$

$$\sqrt{\boxed{12}}$$

4. LEG GEOMETRIC-MEAN THEOREM

Each leg is geometric mean of:

- full hypotenuse $\sqrt{c}$
- adjacent hypotenuse segment

So:

$$\sqrt{\boxed{a^2 = cp}}$$

$$\sqrt{\boxed{b^2 = cq}}$$

depending on correspondence.

5. EXAMPLE

Hypotenuse:

$$\sqrt{c = 25}$$

One adjacent segment:

$$\sqrt{p = 9}$$

Then corresponding leg:

$$\sqrt{a^2 = 25 \cdot 9}$$

$$\sqrt{a = 15}$$

$$\sqrt{\boxed{15}}$$

Other segment:

$$\sqrt{q = 16}$$

Other leg:

$$\sqrt{b = \sqrt{25 \cdot 16}}$$

$$\sqrt{b = \boxed{20}}$$

So triangle is:

$$\sqrt{15-20-25}$$

6. ALTITUDE USING LEGS

We already derived:

$$\frac{1}{2}ab = \frac{1}{2}ch$$

Therefore:

$$\boxed{h = \frac{ab}{c}}$$

7. CHECK THE SAME EXAMPLE

For:

$$15, 20, 25$$

$$h = \frac{15 \cdot 20}{25}$$

$$= 12$$

Same result as:

$$\sqrt{9 \cdot 16} = 12$$

8. COMPLETE RIGHT-TRIANGLE ALTITUDE TOOLKIT

$$\boxed{p+q=c}$$

$$\boxed{h^2=pq}$$

$$\boxed{a^2=cp}$$

$$\boxed{b^2=cq}$$

$$\boxed{h = \frac{ab}{c}}$$

These are high-value formulas.

9. MEDIAN LENGTH — APOLLONIUS THEOREM

For triangle sides:

$$a, b, c$$

and median to side a denoted m_a :

$$\boxed{m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}}$$

So:

$$\boxed{m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}}$$

10. SPECIAL CASE — MEDIAN TO HYPOTENUSE

For right triangle with hypotenuse c :

$$\boxed{m_c = \frac{c^2}{2a}}$$

This is much faster than using the general median formula.

11. EXAMPLE

Triangle sides:

$$[5, 5, 6]$$

Find median to side 6.

$$m^2 = \frac{2(25) + 2(25) - 36}{4}$$

$$[= \frac{64}{4} = 16]$$

$$\boxed{m=4}$$

12. STEWART'S THEOREM — LOWER PRIORITY

For triangle (ABC) , cevian $(AD=d)$, with:

$$[BD=m, \quad DC=n, \quad BC=a=m+n]$$

and sides:

$$[AC=b, \quad AB=c]$$

Stewart's theorem:

$$\boxed{b^2m + c^2n = a(d^2 + mn)}$$

Useful for difficult triangle-length questions.

For aptitude, remember mainly that:

Stewart handles a general cevian; median formula is a special case.

13. HERON'S FORMULA

If triangle sides are:

$$[a, b, c]$$

First find semiperimeter:

$$\boxed{s = \frac{a+b+c}{2}}$$

Then area:

$$\boxed{\Delta = \sqrt{s(s-a)(s-b)(s-c)}}$$

MUST MEMORIZE

14. EXAMPLE — 3-4-5 TRIANGLE

$$[s = \frac{3+4+5}{2} = 6]$$

Area:

$$\sqrt{6(3)(2)(1)}$$

$$=\sqrt{36}$$

$$=\boxed{6}$$

Matches:

$$\frac{1}{2}(3)(4)=6$$

15. EXAMPLE — SIDES 13,14,15

$$s=\frac{13+14+15}{2}$$

$$=21$$

Area:

$$\sqrt{21(8)(7)(6)}$$

$$=\sqrt{7056}$$

$$=\boxed{84}$$

This is a classic triangle.

16. WHEN TO USE HERON

Use Heron when:

- all 3 sides are known
- altitude is not directly known
- triangle is not obviously right angled

Do not use Heron if a faster area formula is available.

17. BASIC AREA FORMULA

$$\Delta=\frac{1}{2}bh$$

where (h) is perpendicular height corresponding to base (b) .

18. AREA USING TWO SIDES AND INCLUDED ANGLE

If two sides (a,b) include angle (C) :

$$\Delta=\frac{1}{2}ab\sin C$$

Useful for:

- trigonometric aptitude
 - $(30^\circ, 45^\circ, 60^\circ, 90^\circ)$ angles
-

19. EXAMPLE

Sides:

$$\sqrt{6}, \sqrt{8}$$

included angle:

$$30^\circ$$

Area:

$$\frac{1}{2}(6)(8)\sin 30^\circ$$

$$= 24 \cdot \frac{1}{2}$$

$$= \boxed{12}$$

20. AREA USING INRADIUS

If r is inradius and s semiperimeter:

$$\Delta = rs$$

Therefore:

$$r = \frac{\Delta}{s}$$

High-value formula

21. WHY $\Delta = rs$

Connect incenter to all 3 vertices.

The triangle breaks into 3 smaller triangles.

Each has height r .

Total area:

$$\frac{1}{2}r(a+b+c)$$

$$= r \frac{(a+b+c)^2}{2}$$

$$= rs$$

22. EXAMPLE — INRADIUS OF 3-4-5

Area:

$$6$$

Semiperimeter:

$$6$$

So:

$$r = \frac{6}{6}$$

$$= \boxed{1}$$

23. INRADIUS OF RIGHT TRIANGLE

For right triangle with legs (a,b) , hypotenuse (c) :

$$\boxed{r = \frac{a+b-c}{2}}$$

MUST KNOW

24. EXAMPLE

Right triangle:

$$[5, 12, 13]$$

$$r = \frac{5+12-13}{2}$$

$$= \boxed{2}$$

25. CIRCUMRADIUS FORMULA

For any triangle:

$$\boxed{\Delta = \frac{abc}{4R}}$$

Therefore:

$$\boxed{R = \frac{abc}{4\Delta}}$$

where (R) is circumradius.

26. EXAMPLE — 3-4-5

$$R = \frac{3 \cdot 4 \cdot 5}{4 \cdot 6}$$

$$= \frac{60}{24}$$

$$= \boxed{\frac{5}{2}}$$

Since it is right triangle:

$$R = \frac{5}{2}$$

as expected.

27. RIGHT-TRIANGLE CIRCUMRADIUS

Again:

$$\boxed{R = \frac{c}{2}}$$

where (c) is hypotenuse.

This should be your first method for right triangles.

28. EQUILATERAL TRIANGLE INRADIUS & CIRCUMRADIUS

For equilateral triangle side (a) :

Altitude:

$$h = \frac{\sqrt{3}}{2}a$$

All centers coincide.

Centroid divides altitude $(2:1)$.

Hence:

$$\boxed{R = \frac{2}{3}h = \frac{a}{\sqrt{3}}}$$

and:

$$\boxed{r = \frac{1}{3}h = \frac{a}{2\sqrt{3}}}$$

Thus:

$$\boxed{R = 2r}$$

29. EXAMPLE

Equilateral triangle side:

$$6$$

Inradius:

$$\frac{6}{2\sqrt{3}} = \sqrt{3}$$

Circumradius:

$$\frac{6}{\sqrt{3}} = 2\sqrt{3}$$

So:

$$\boxed{r = \sqrt{3}, R = 2\sqrt{3}}$$

30. AREA OF EQUILATERAL TRIANGLE

$$\boxed{\Delta = \frac{\sqrt{3}}{4}a^2}$$

If perimeter $(P = 3a)$:

$$a = \frac{P}{3}$$

Substitute if needed.

31. AREA RATIOS WITH SAME HEIGHT

Two triangles with same perpendicular height have area ratio equal to base ratio:

$$\boxed{\frac{\Delta_1}{\Delta_2} = \frac{b_1}{b_2}}$$

32. AREA RATIOS WITH SAME BASE

If same base:

$$\boxed{\frac{\Delta_1}{\Delta_2} = \frac{h_1}{h_2}}$$

33. SAME BASE + SAME PARALLELS

Triangles on same base and between same parallel lines have equal heights.

Therefore:

$$\boxed{\text{equal areas}}$$

34. MEDIAN DIVIDES TRIANGLE INTO EQUAL AREAS

A median divides the opposite side into equal halves.

Both resulting triangles have:

- equal bases
- same altitude

Therefore:

$$\boxed{\text{A median divides triangle into 2 equal-area triangles}}$$

35. THREE MEDIANS AND AREA

The three medians divide a triangle into:

$$\boxed{6}$$

small triangles of equal area.

Useful visual fact.

36. CENTROID AND AREA

Connecting centroid to the three vertices divides triangle into 3 triangles of equal area.

So each has:

$$\boxed{\frac{1}{3}}$$

of total area.

37. SIMILAR TRIANGLES — VOLUME WARNING

Triangles are 2D.

If side ratio:

$$\boxed{k}$$

then:

- length ratio = $\sqrt{k}$
- area ratio = $\sqrt{k^2}$

Do not use $\sqrt{k^3}$ unless dealing with similar 3D solids.

38. ANGLE BISECTOR + AREA RATIO

If AD bisects angle A in triangle ABC:

Triangles ABD and ACD share the same altitude to BC.

Thus:

$$\frac{[ABD]}{[ACD]} = \frac{BD}{DC}$$

But angle-bisector theorem gives:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Therefore:

$$\boxed{[ABD]:[ACD]=AB:AC}$$

Useful shortcut.

39. EXAMPLE

If:

$$AB:AC=2:3$$

then angle bisector divides triangle area in ratio:

$$\boxed{2:3}$$

40. MEDIAN TO HYPOTENUSE — GEOMETRIC PROPERTY

In right triangle:

midpoint of hypotenuse is equidistant from all 3 vertices.

Therefore if midpoint is M:

$$\boxed{MA=MB=MC=\frac{c}{2}}$$

This is why it is the circumcenter.

41. EULER INEQUALITY — LOWER PRIORITY BUT USEFUL

For any nondegenerate triangle:

$$\boxed{R \geq 2r}$$

Equality occurs only for an equilateral triangle.

Useful for conceptual MSQ questions.

42. TRIANGLE AREA FROM INRADIUS AND SIDES

Combine Heron:

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

with:

$$\Delta = rs$$

Then:

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}$$

Not a primary memorization formula, but derivable.

43. SPECIAL RIGHT-TRIANGLE INRADIUS SHORTCUT

If legs:

$$a, b$$

hypotenuse:

$$c$$

then:

$$r = \frac{a+b-c}{2}$$

This is much faster than Heron.

44. EXAMPLE

Right triangle:

$$8, 15, 17$$

$$r = \frac{8+15-17}{2}$$

$$r = 3$$

45. AREA FROM CIRCUMRADIUS

If sides (a, b, c) and circumradius (R) :

$$\Delta = \frac{abc}{4R}$$

Example:

$$a=5, b=5, c=6, R=\frac{25}{8}$$

Then:

$$\Delta = \frac{5 \cdot 5 \cdot 6}{4 \cdot (25/8)}$$

$$= \frac{150}{25/2}$$

$$= 12$$

$$\boxed{12}$$

46. TRIANGLE EXISTENCE RECAP

For sides (a, b, c) :

$$\boxed{a+b>c}$$

must hold.

If equality:

$$a+b=c$$

figure degenerates into a straight line.

47. AREA AND TRIANGLE INEQUALITY CONNECTION

As:

$$a+b \leq c$$

the area tends toward:

$$0$$

Good geometric sanity check.

48. MAXIMUM AREA FOR TWO FIXED SIDES

For fixed sides (a, b) :

$$\Delta = \frac{1}{2}ab \sin C$$

Maximum occurs when:

$$\sin C = 1$$

So:

$$\boxed{C=90^\circ}$$

and maximum area:

$$\boxed{\frac{1}{2}ab}$$

49. EXAMPLE

Two sides:

$$6, 10$$

Maximum possible area:

$$\frac{1}{2}(6)(10)$$

$$= \boxed{30}$$

when included angle is (90°) .

50. GATE-STYLE QUESTION 1

Right triangle hypotenuse is split into:

$$\sqrt{4}, \sqrt{9}$$

Find altitude.

$$\sqrt{h = \sqrt{4 \cdot 9}}$$

$$\sqrt{= \boxed{6}}$$

51. GATE-STYLE QUESTION 2

Same triangle:

$$\sqrt{c = 13}$$

Corresponding legs:

$$\sqrt{a = \sqrt{13 \cdot 4} = 2\sqrt{13}}$$

$$\sqrt{b = \sqrt{13 \cdot 9} = 3\sqrt{13}}$$

52. GATE-STYLE QUESTION 3

Triangle sides:

$$\sqrt{13, 14, 15}$$

Area:

$$\sqrt{\boxed{84}}$$

using Heron.

53. GATE-STYLE QUESTION 4

Right triangle:

$$\sqrt{9, 12, 15}$$

Inradius:

$$\sqrt{\frac{9+12-15}{2}}$$

$$\sqrt{= \boxed{3}}$$

54. PLACEMENT-STYLE QUESTION 1

Equilateral triangle side:

$$\sqrt{8}$$

Area:

$$\left[\frac{\sqrt{3}}{4}(64) \right]$$

$$\left[=\boxed{16\sqrt{3}} \right]$$

55. PLACEMENT-STYLE QUESTION 2

Equilateral side:

$$\left[12 \right]$$

Inradius:

$$\left[\frac{12}{2\sqrt{3}} = \boxed{2\sqrt{3}} \right]$$

Circumradius:

$$\left[\frac{12}{\sqrt{3}} = \boxed{4\sqrt{3}} \right]$$

56. PLACEMENT-STYLE QUESTION 3

Triangle area:

$$\left[60 \right]$$

Semiperimeter:

$$\left[15 \right]$$

Then inradius:

$$\left[r = \frac{60}{15} \right]$$

$$\left[=\boxed{4} \right]$$

57. PLACEMENT-STYLE QUESTION 4

Triangle sides:

$$\left[6, 8, 10 \right]$$

Circumradius:

$$\left[\boxed{5} \right]$$

because right triangle.

58. ADVANCED QUESTION — MEDIAN

Triangle sides:

$$\left[7, 8, 9 \right]$$

Find median to side 9.

$$\left[m^2 = \frac{2(7^2) + 2(8^2) - 9^2}{4} \right]$$

$$\left[= \frac{98 + 128 - 81}{4} \right]$$

$$\left[= \frac{145}{4} \right]$$

$$\boxed{m = \frac{\sqrt{145}}{2}}$$

59. ADVANCED QUESTION — AREA RATIO FROM ANGLE BISECTOR

Angle bisector divides opposite side in ratio:

$$4:7$$

Then the two smaller triangle areas are also:

$$\boxed{4:7}$$

because they share altitude.

60. ADVANCED QUESTION — FIND R FROM AREA

Triangle sides:

$$5, 12, 13$$

Area:

$$30$$

Then:

$$R = \frac{5 \cdot 12 \cdot 13}{4(30)}$$

$$= \frac{780}{120}$$

$$= \boxed{\frac{13}{2}}$$

as expected.

61. COMMON TRAPS

& pTRAP 1 — Semiperimeter

Heron uses:

$$s = \frac{a+b+c}{2}$$

not full perimeter.

& pTRAP 2 — Inradius vs circumradius

$$\Delta = rs$$

uses inradius.

$$\Delta = \frac{abc}{4R}$$

uses circumradius.

& pTRAP 3 — Median ratio

Centroid divides median:

$$2:1$$

but this does not mean median divides side in 2:1.

The median divides opposite side:

$$\boxed{1:1}$$

& pTRAP 4 — Altitude-to-hypotenuse formulas

Make sure (p,q) are the two pieces of the hypotenuse, not the legs.

& pTRAP 5 — Similarity and area

Side ratio (k) !' area ratio (k^2) .

& pTRAP 6 — Right-triangle formulas on general triangle

Do not use:

$$\boxed{R=c/2}$$

or:

$$\boxed{r=(a+b-c)/2}$$

unless the triangle is right-angled.

62. MUST-MEMORIZE RIGHT-TRIANGLE ALTITUDE RELATIONS

$$\boxed{p+q=c}$$

$$\boxed{h^2=pq}$$

$$\boxed{a^2=cp}$$

$$\boxed{b^2=cq}$$

$$\boxed{h=\frac{ab}{c}}$$

63. MUST-MEMORIZE AREA FORMULAS

$$\boxed{\Delta=\frac{1}{2}bh}$$

$$\boxed{\Delta=\frac{1}{2}ab\sin C}$$

$$\boxed{\Delta=\sqrt{s(s-a)(s-b)(s-c)}}$$

$$\boxed{\Delta=rs}$$

$$\boxed{\Delta=\frac{abc}{4R}}$$

64. MUST-MEMORIZE SPECIAL RADIUS FORMULAS

Right triangle:

$$\boxed{R=\frac{c}{2}}$$

$$\boxed{r=\frac{a+b-c}{2}}$$

Equilateral triangle:

$$\boxed{R = \frac{a}{\sqrt{3}}}$$

$$\boxed{r = \frac{a}{2\sqrt{3}}}$$

$$\boxed{R = 2r}$$

65. FAST EXAM DECISION TREE

All 3 sides known?

Try Heron.

Right triangle?

Use Pythagoras and right-triangle shortcuts first.

Hypotenuse split by altitude?

Use geometric means.

Inradius given?

Try:

$$\Delta = rs$$

Circumradius given?

Try:

$$\Delta = \frac{abc}{4R}$$

Two sides + included angle?

Use:

$$\Delta = \frac{1}{2}ab \sin C$$

Median length?

Use Apollonius.

Similar triangles?

Avoid solving unnecessary lengths; use ratios.

PRACTICE — LEVEL 0

Q1

Right triangle hypotenuse is split into 4 and 9. Find altitude.

Q2

Find area of triangle with base 12 and height 7.

Q3

Find area of a 3-4-5 triangle using Heron.

Q4

Right triangle sides 5,12,13. Find inradius.

Q5

Equilateral triangle side 6. Find area.

LEVEL 1

Q6

Triangle sides 13,14,15. Find area.

Q7

Right triangle legs 8,15. Find altitude to hypotenuse.

Q8

Right triangle hypotenuse 26. Find circumradius.

Q9

Triangle area is 48 and semiperimeter is 12. Find inradius.

Q10

Equilateral side 9. Find inradius and circumradius.

LEVEL 2

Q11

A right triangle has hypotenuse segments 9 and 16. Find:

1. altitude
2. both legs.

Q12

Triangle sides 7,8,9. Find median to side 9.

Q13

Two sides are 10 and 12 with included angle (30°) . Find area.

Q14

Similar triangles have area ratio 49:81. Find perimeter ratio.

Q15

An angle bisector divides opposite side in ratio 3:5. Find ratio of areas of the two smaller triangles.

LEVEL 3 — EXAM PRESSURE

Q16

A triangle has sides 10,10,12. Find area and inradius.

Q17

A right triangle has area 84 and hypotenuse 25. If one leg is 7, find the other leg and altitude to hypotenuse.

Q18

A triangle has sides 5,12,13. Find inradius and circumradius.

Q19

For an equilateral triangle, if inradius is 4, find side and circumradius.

Q20

A triangle has fixed two sides 9 and 14. What is maximum possible area?

ANSWERS

Q1

$$\backslash[h=\sqrt{4\cdot 9} \backslash]$$

$$\backslash[=\boxed{6} \backslash]$$

Q2

$$\backslash[\frac{12(12)(7)}{2} = \boxed{42} \backslash]$$

Q3

$$\backslash[s=6 \backslash]$$

$$\backslash[\Delta= \sqrt{6\cdot 3\cdot 2\cdot 1} \backslash]$$

$$\backslash[=\boxed{6} \backslash]$$

Q4

$$\backslash[r=\frac{5+12-13}{2} \backslash]$$

$$\backslash[=\boxed{2} \backslash]$$

Q5

$$\backslash[\frac{\sqrt{3}}{4}(36) = \boxed{9\sqrt{3}} \backslash]$$

Q6

$$\backslash[\boxed{84} \backslash]$$

Q7

Hypotenuse:

$$\backslash[17 \backslash]$$

Altitude:

$$\backslash[\frac{8\cdot 15}{17} \backslash]$$

$$\backslash[=\boxed{\frac{120}{17}} \backslash]$$

Q8

$$R = \frac{26}{2} = \boxed{13}$$

Q9

$$r = \frac{\Delta s}{s} = \frac{48}{12}$$

$$= \boxed{4}$$

Q10

$$r = \frac{9}{2\sqrt{3}} = \boxed{\frac{3\sqrt{3}}{2}}$$

$$R = \frac{9}{\sqrt{3}} = \boxed{3\sqrt{3}}$$

Q11

$$c = 25$$

Altitude:

$$h = \sqrt{9 \cdot 16} = \boxed{12}$$

Legs:

$$\sqrt{25 \cdot 9} = 15$$

$$\sqrt{25 \cdot 16} = 20$$

$$\boxed{15, 20}$$

Q12

$$m = \frac{12\sqrt{2(7^2) + 2(8^2) - 9^2}}{2}$$

$$= \boxed{\frac{\sqrt{145}}{2}}$$

Q13

$$\frac{12(10)(12)\sin 30^\circ}{2}$$

$$= 60 \cdot \frac{12}{2}$$

$$= \boxed{360}$$

Q14

Side/perimeter ratio:

$$\sqrt{49:81}$$

$$= \boxed{7:9}$$

Q15

Same altitude, so area ratio equals base ratio:

$$\boxed{3:5}$$

Q16

Isosceles triangle:

$$10, 10, 12$$

Altitude splits base into:

$$6, 6$$

Height:

$$\sqrt{10^2 - 6^2} = \sqrt{64} = 8$$

Area:

$$\frac{1}{2}(12)(8) = 48$$

Semiperimeter:

$$16$$

Inradius:

$$r = \frac{48}{16} = 3$$

$$\boxed{\Delta = 48, r = 3}$$

Q17

If one leg is 7 and hypotenuse 25:

$$b = \sqrt{25^2 - 7^2}$$

$$b = \sqrt{576} = 24$$

Area:

$$\frac{1}{2}(7)(24) = 84$$

Altitude:

$$h = \frac{7 \cdot 24}{25}$$

$$\boxed{h = \frac{168}{25}}$$

So:

$$\boxed{b = 24, h = \frac{168}{25}}$$

Q18

For 5-12-13:

$$r = \frac{5 + 12 - 13}{2} = 2$$

$$R = \frac{13}{2}$$

$$\boxed{r = 2, R = \frac{13}{2}}$$

Q19

For equilateral:

$$r = \frac{a}{2\sqrt{3}}$$

Given:

$$4 = \frac{a}{2\sqrt{3}}$$

$$a = 8\sqrt{3}$$

And:

$$R = 2r = 8$$

$$\boxed{a = 8\sqrt{3}, R = 8}$$

Q20

Maximum occurs at included angle:

$$\backslash[90^\circ \backslash]$$

So:

$$\backslash[A_{\max} = \frac{1}{2}(9)(14) \backslash]$$

$$\backslash[=\boxed{63} \backslash]$$

' CHAPTER 44 COMPLETE

We have now covered:

- congruent triangles '
- congruence tests '
- similar triangles '
- similarity tests '
- scale factors '
- perimeter and area ratios '
- BPT and converse '
- angle-bisector theorem '
- Pythagoras and converse '
- acute/right/obtuse side test '
- Pythagorean triples '
- 45-45-90 triangles '
- 30-60-90 triangles '
- equilateral triangle altitude/area '
- altitude to hypotenuse '
- geometric-mean theorems '
- median to hypotenuse '
- Apollonius median formula '
- Stewart theorem awareness '
- Heron's formula '
- $\backslash(\frac{1}{2}ab\sin C\backslash)$ area formula '
- inradius '
- circumradius '
- equilateral $\backslash(r,R\backslash)$ relations '
- same-base/same-height area ratios '
- median/centroid area properties '
- GATE/placement L0–L3 practice '

Progress

Completed: 44/76

Unit 1 — Quantitative Aptitude: 0/14 PROGRESS

NEXT

+Chapter 45 — Quadrilaterals, Polygons and Circles

Next we will cover:

- parallelogram
- rectangle
- square
- rhombus
- trapezium/trapezoid
- kite
- diagonal properties
- polygon angle/diagonal applications
- circle terminology
- chords, arcs and sectors
- tangent properties
- angle subtended by chords
- cyclic quadrilaterals
- power-of-a-point basics
- GATE/placement shortcuts and traps.

USER

next if required include diagrams even

ASSISTANT

CHAPTER 45 — QUADRILATERALS, POLYGONS AND CIRCLES

PART 1 — PARALLELOGRAM, RECTANGLE, SQUARE, RHOMBUS, TRAPEZIUM, KITE & POLYGON GEOMETRY

Status: ~~0=Not~~ IN PROGRESS GATE: Medium Placements: Medium–High Priority: +Tier 2

Yes — whenever a geometry concept becomes easier with a figure, I'll include a **small diagram** along with the explanation.

The core strategy for quadrilateral questions is:

Identify the figure first !' recall its guaranteed properties !' then create equations.

Do **not** assume a property just because the drawing looks like it.

1. QUADRILATERAL

A quadrilateral is a polygon having:

$\boxed{4\text{ sides}}$

General shape:

TEXT

A% % % % % % B
\
\
D% % % % % % C

Vertices:

A, B, C, D

Sides:

AB, BC, CD, DA

Diagonals:

AC, BD

2. SUM OF INTERIOR ANGLES

For an (n) -sided polygon:

$(n-2)180^\circ$

For a quadrilateral:

$(4-2)180^\circ$

$\boxed{360^\circ}$

MUST MEMORIZE

3. EXAMPLE

Three angles are:

$80^\circ, 95^\circ, 105^\circ$

Fourth angle (x) :

$x = 360 - (80 + 95 + 105)$

$= 360 - 280$

$\boxed{80^\circ}$

4. MAJOR QUADRILATERALS

You need to distinguish:

1. parallelogram
2. rectangle
3. square
4. rhombus

5. trapezium/trapezoid

6. kite

They share some properties but not all.

5. PARALLELOGRAM

A quadrilateral with both pairs of opposite sides parallel.

TEXT

A % % % % % % B
/
/
/
/
D % % % % % % C

$\backslash AB \parallel DC \backslash$

$\backslash AD \parallel BC \backslash$

6. PARALLELOGRAM — OPPOSITE SIDES

Opposite sides are equal:

$\backslash \boxed{AB=CD} \backslash$

$\backslash \boxed{AD=BC} \backslash$

7. PARALLELOGRAM — OPPOSITE ANGLES

Opposite angles are equal:

$\backslash \boxed{\angle A = \angle C} \backslash$

$\backslash \boxed{\angle B = \angle D} \backslash$

8. ADJACENT ANGLES

Adjacent angles are supplementary:

$\backslash \boxed{\angle A + \angle B = 180^\circ} \backslash$

Similarly for every adjacent pair.

9. PARALLELOGRAM DIAGONALS

Diagonals bisect each other.

If AC and BD intersect at O:

TEXT

A % % % % % % B
\
/



then:

$$\boxed{AO=OC}$$

$$\boxed{BO=OD}$$

Important

They do **not** have to be equal.

They do **not** have to be perpendicular.

10. EXAMPLE

In a parallelogram:

$$AO=7$$

Since diagonals bisect:

$$OC=7$$

Thus:

$$AC=14$$

$$\boxed{14}$$

11. AREA OF PARALLELOGRAM

$$\boxed{A=bh}$$

where (h) is perpendicular height.

Not slant side unless it is perpendicular.

12. EXAMPLE

Base:

$$12$$

height:

$$7$$

Area:

$$12 \cdot 7 = \boxed{84}$$

13. RECTANGLE

A rectangle is a parallelogram with all angles equal to:

$$\boxed{90^\circ}$$

TEXT

A % % % % % % B
 % %
 % %
 D % % % % % % C

So every rectangle inherits all parallelogram properties.

14. RECTANGLE PROPERTIES

Opposite sides equal:

$$AB=CD$$

$$AD=BC$$

All angles:

$$90^\circ$$

Diagonals:

1. bisect each other
2. are equal

Thus:

$$\boxed{AC=BD}$$

15. RECTANGLE DIAGONAL

For length (l) , breadth (b) :

$$\boxed{d=\sqrt{l^2+b^2}}$$

16. EXAMPLE

Rectangle:

$$9\times 12$$

Diagonal:

$$\sqrt{81+144}=\sqrt{225}$$

$$=\boxed{15}$$

17. RECTANGLE AREA AND PERIMETER

$$\boxed{A=lb}$$

$$\boxed{P=2(l+b)}$$

18. SQUARE

A square is simultaneously:

- rectangle
- rhombus
- parallelogram

TEXT

```
A% % % % % B
%           %
%           %
D% % % % % C
```

All four sides equal:

$$\boxed{AB=BC=CD=DA}$$

All angles:

$$\boxed{90^\circ}$$

19. SQUARE DIAGONALS

Square diagonals:

- are equal
- bisect each other
- are perpendicular
- bisect vertex angles

Therefore they are extremely powerful.

20. SQUARE DIAGONAL

If side:

$$\boxed{a}$$

then:

$$\boxed{d=a\sqrt{2}}$$

21. SQUARE AREA

$$\boxed{A=a^2}$$

Using diagonal:

$$\boxed{d=a\sqrt{2}}$$

So:

$$\boxed{a^2=\frac{d^2}{2}}$$

Therefore:

$$\boxed{A=\frac{d^2}{2}}$$

22. EXAMPLE

Square diagonal:

$$10$$

Area:

$$\frac{10^2}{2} = \boxed{50}$$

23. RHOMBUS

A rhombus is a parallelogram with all four sides equal.

TEXT



$$AB=BC=CD=DA$$

24. RHOMBUS PROPERTIES

Because it is a parallelogram:

- opposite sides parallel
- opposite angles equal
- adjacent angles supplementary
- diagonals bisect each other

Additional important property:

$$\boxed{\text{Diagonals are perpendicular}}$$

25. RHOMBUS DIAGONALS BISECT ANGLES

Each diagonal bisects a pair of opposite angles.

For example:

Diagonal AC bisects angles:

$$\angle A, \angle C$$

Diagonal BD bisects:

$$\angle B, \angle D$$

26. ARE RHOMBUS DIAGONALS EQUAL?

Not generally.

This is a major trap.

Only if the rhombus is also a square do the diagonals become equal.

27. RHOMBUS AREA

If diagonals are:

$$\backslash d_1, d_2 \backslash$$

then:

$$\backslash \boxed{A = \frac{1}{2} d_1 d_2} \backslash$$

28. EXAMPLE

Rhombus diagonals:

$$\backslash 12, 16 \backslash$$

Area:

$$\backslash \frac{1}{2}(12)(16) \backslash$$

$$\backslash \boxed{96} \backslash$$

29. RHOMBUS SIDE FROM DIAGONALS

Diagonals bisect at right angles.

Half-diagonals form a right triangle.

If:

$$\backslash d_1 = 12, \text{quad } d_2 = 16 \backslash$$

halves:

$$\backslash 6, 8 \backslash$$

Side:

$$\backslash \sqrt{6^2 + 8^2} = 10 \backslash$$

$$\backslash \boxed{10} \backslash$$

30. KITE

A kite has two pairs of adjacent equal sides.

TEXT

A
/ \



Typical relation:

$$[AB=AD]$$

$$[CB=CD]$$

31. KITE DIAGONALS

For a kite:

- diagonals are perpendicular
- one diagonal bisects the other
- one diagonal bisects a pair of opposite angles

But unlike a parallelogram:

both diagonals do not necessarily bisect each other.

32. KITE AREA

Because diagonals are perpendicular:

$$\boxed{A = \frac{1}{2} d_1 d_2}$$

Same numerical formula as rhombus.

33. TRAPEZIUM / TRAPEZOID

Terminology varies internationally.

For GATE/Indian aptitude, **trapezium** usually means:

quadrilateral with one pair of opposite sides parallel.

TEXT



Here:

$$[AB \parallel DC]$$

Parallel sides are called **bases**.

34. TRAPEZIUM AREA

If parallel sides are:

$$\sqrt{a, b}$$

and perpendicular distance between them is:

$$\sqrt{h}$$

then:

$$\sqrt{\boxed{A = \frac{1}{2}(a+b)h}}$$

35. EXAMPLE

Parallel sides:

$$\sqrt{8, 14}$$

height:

$$\sqrt{5}$$

Area:

$$\sqrt{\frac{1}{2}(22)(5)}$$

$$\sqrt{= \boxed{55}}$$

36. TRAPEZIUM MID-SEGMENT

The segment joining the midpoints of the non-parallel sides is parallel to the bases.

Its length is:

$$\sqrt{\boxed{\frac{a+b}{2}}}$$

37. EXAMPLE

Bases:

$$\sqrt{10, 18}$$

Mid-segment:

$$\sqrt{\frac{10+18}{2} = \boxed{14}}$$

38. ISOSCELES TRAPEZIUM

An isosceles trapezium has equal non-parallel sides.

Then:

- base angles are equal
- diagonals are equal

Useful recognition property.

39. QUADRILATERAL PROPERTY TABLE

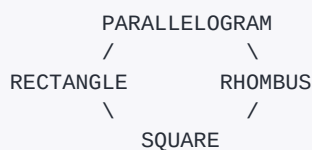
Figure	Opposite sides parallel	All sides equal	All angles (90°)	Diagonals equal	Diagonals perpendicular
Parallelogram	Yes	No	No	Not necessarily	Not necessarily
Rectangle	Yes	No	Yes	Yes	Not generally
Rhombus	Yes	Yes	No	Not generally	Yes
Square	Yes	Yes	Yes	Yes	Yes
Kite	Not generally	No	No	Not generally	Yes
Trapezium	One pair	No	No	Not generally	Not generally

This table is very important.

40. FAMILY RELATION

Think of this hierarchy:

TEXT



A square satisfies both rectangle and rhombus conditions.

Thus:

$\boxed{\text{Every square is a rectangle}}$

$\boxed{\text{Every square is a rhombus}}$

But:

$\boxed{\text{Not every rectangle is a square}}$

$\boxed{\text{Not every rhombus is a square}}$

41. DIAGONALS — FAST IDENTIFICATION

Parallelogram

$\boxed{\text{bisect each other}}$

Rectangle

$\boxed{\text{bisect each other + equal}}$

Rhombus

$\boxed{\text{bisect each other + perpendicular}}$

Square

$\boxed{\text{equal + perpendicular + bisect each other}}$

This is an excellent memory system.

42. POLYGON RECAP

An (n) -sided polygon has interior-angle sum:

$\boxed{(n-2)180^\circ}$

43. NUMBER OF DIAGONALS

$\boxed{\frac{n(n-3)}{2}}$

44. DIAGONALS FROM ONE VERTEX

From one vertex of an (n) -gon, you cannot connect a diagonal to:

- itself
- its two adjacent vertices

So:

$\boxed{n-3}$

diagonals from one vertex.

45. TRIANGLES FROM ONE VERTEX

Drawing all diagonals from one vertex divides an (n) -gon into:

$\boxed{n-2}$

triangles.

This explains interior-angle sum.

46. REGULAR POLYGON INTERIOR ANGLE

$\boxed{I = \frac{(n-2)180^\circ}{n}}$

Equivalent:

$\boxed{I = 180^\circ - \frac{360^\circ}{n}}$

The second form is often faster.

47. REGULAR POLYGON EXTERIOR ANGLE

$$\boxed{E = \frac{360^\circ}{n}}$$

And:

$$\boxed{I + E = 180^\circ}$$

48. EXAMPLE

Regular polygon with 20 sides.

Exterior angle:

$$360/20 = 18^\circ$$

Interior:

$$180 - 18 = \boxed{162^\circ}$$

49. FIND SIDES FROM INTERIOR ANGLE

Regular polygon interior angle:

$$165^\circ$$

Exterior:

$$15^\circ$$

Then:

$$n = 360/15$$

$$n = \boxed{24}$$

50. DIAGONALS EXAMPLE

A 12-sided polygon:

$$D = \frac{12(9)}{2}$$

$$D = \boxed{54}$$

51. SOLVE n FROM NUMBER OF DIAGONALS

Suppose polygon has 35 diagonals.

$$\frac{n(n-3)}{2} = 35$$

$$n(n-3) = 70$$

$$n^2 - 3n - 70 = 0$$

$$(n-10)(n+7) = 0$$

$$\boxed{n = 10}$$

52. SUM OF EXTERIOR ANGLES

For any convex polygon:

$$\boxed{360^\circ}$$

regardless of number of sides.

53. EQUILATERAL VS REGULAR

Equilateral polygon

All sides equal.

Equiangular polygon

All angles equal.

Regular polygon

Both:

$$\boxed{\text{equilateral} + \text{equiangular}}$$

54. AREA SCALING

If every linear dimension of a figure is multiplied by (k) :

Perimeter scales by:

$$\boxed{k}$$

Area scales by:

$$\boxed{k^2}$$

Very important for geometry and mensuration.

55. EXAMPLE

Square side increases by 20%.

New side multiplier:

$$\boxed{1.2}$$

Area multiplier:

$$\boxed{1.2^2=1.44}$$

So area increases by:

$$\boxed{44\%}$$

This links directly to Chapter 15 percentages.

56. RECTANGLE — FIXED PERIMETER MAXIMUM AREA

For a fixed perimeter, rectangle area is maximum when rectangle becomes a square.

Example perimeter:

$$\boxed{40}$$

Then:

$$\boxed{l+b=20}$$

Maximum occurs:

$$\boxed{l=b=10}$$

Area:

$$\boxed{100}$$

Useful optimization shortcut.

57. SAME AREA RECTANGLES

If:

$$\boxed{l_1 b_1 = l_2 b_2}$$

then increasing one dimension forces the other to decrease inversely.

This connects geometry with inverse variation.

58. GATE-STYLE QUESTION 1

A parallelogram has angle:

$$\boxed{70^\circ}$$

Adjacent angle:

$$\boxed{180-70 = 110^\circ}$$

Opposite angle:

$$\boxed{70^\circ}$$

59. GATE-STYLE QUESTION 2

Rectangle diagonals intersect at O.

If:

$$\boxed{AO=6}$$

Then:

$$\boxed{AC=12}$$

Since diagonals equal:

$$\boxed{BD=12}$$

and because they bisect:

$$\backslash BO=6 \backslash$$

$$\backslash \boxed{BO=6} \backslash$$

60. GATE-STYLE QUESTION 3

Rhombus diagonals:

$$\backslash 10, \backslash 24 \backslash$$

Area:

$$\backslash \frac{1}{2}(10)(24) = \boxed{120} \backslash$$

Side:

half diagonals:

$$\backslash 5, \backslash 12 \backslash$$

$$\backslash \sqrt{25+144}=13 \backslash$$

$$\backslash \boxed{13} \backslash$$

61. GATE-STYLE QUESTION 4

A regular polygon has exterior angle:

$$\backslash 20^\circ \backslash$$

Sides:

$$\backslash 360/20 = \boxed{18} \backslash$$

62. PLACEMENT-STYLE QUESTION 1

Parallelogram base:

$$\backslash 15 \backslash$$

height:

$$\backslash 8 \backslash$$

Area:

$$\backslash 15(8) = \boxed{120} \backslash$$

63. PLACEMENT-STYLE QUESTION 2

Trapezium has parallel sides:

$$\backslash 10, \backslash 16 \backslash$$

height:

$$\backslash 7 \backslash$$

Area:

$$\frac{1}{2}(26)(7)$$

$$\boxed{91}$$

64. PLACEMENT-STYLE QUESTION 3

Square diagonal:

$$12\sqrt{2}$$

Side:

$$12$$

Area:

$$\boxed{144}$$

65. PLACEMENT-STYLE QUESTION 4

Rectangle:

$$5 \times 12$$

Diagonal:

$$13$$

Perimeter:

$$2(5+12)=34$$

Area:

$$60$$

$$\boxed{d=13, P=34, A=60}$$

66. ADVANCED QUESTION — PARALLELOGRAM DIAGONALS

In parallelogram ABCD, diagonals intersect at O.

If:

$$AC=18, BD=26$$

Then:

$$AO=OC=9$$

$$BO=OD=13$$

$$\boxed{9, 13}$$

67. ADVANCED QUESTION — RHOMBUS

Rhombus side:

$$\sqrt{13}$$

one diagonal:

$$\sqrt{10}$$

Half of first diagonal:

$$\sqrt{5}$$

Let half of second diagonal be $\sqrt{x}$.

Using right triangle:

$$\sqrt{5^2 + x^2 = 13^2}$$

$$\sqrt{x^2 = 144}$$

$$\sqrt{x = 12}$$

Full second diagonal:

$$\sqrt{24}$$

Area:

$$\sqrt{\frac{1}{2}(10)(24)}$$

$$\sqrt{= \boxed{120}}$$

68. ADVANCED QUESTION — POLYGON

Interior-angle sum:

$$\sqrt{1800^\circ}$$

Then:

$$\sqrt{(n-2)180 = 1800}$$

$$\sqrt{n-2 = 10}$$

$$\sqrt{\boxed{n = 12}}$$

69. ADVANCED QUESTION — DIAGONALS

A polygon has 44 diagonals.

$$\sqrt{\frac{n(n-3)}{2} = 44}$$

$$\sqrt{n^2 - 3n - 88 = 0}$$

$$\sqrt{(n-11)(n+8) = 0}$$

$$\sqrt{\boxed{n = 11}}$$

70. COMMON TRAPS

& pTRAP 1 — Parallelogram diagonals

They bisect each other.

They are not necessarily equal.

& pTRAP 2 — Rectangle diagonals

They are equal, but not generally perpendicular.

& pTRAP 3 — Rhombus diagonals

They are perpendicular, but not generally equal.

& pTRAP 4 — Kite

Only one diagonal necessarily bisects the other.

Do not automatically apply parallelogram rules.

& pTRAP 5 — Trapezium height

Height means perpendicular distance between parallel sides.

Not slanted leg.

& pTRAP 6 — Regular polygon

Equal-angle formula assumes regularity.

71. MUST-MEMORIZE QUADRILATERAL FACTS

Parallelogram

$\boxed{\text{opposite sides/angles equal}}$

$\boxed{\text{diagonals bisect each other}}$

Rectangle

Parallelogram +

$\boxed{\text{all angles } 90^\circ}$

$\boxed{\text{diagonals equal}}$

Rhombus

Parallelogram +

$\boxed{\text{all sides equal}}$

$\boxed{\text{diagonals perpendicular}}$

Square

All rectangle + rhombus properties.

72. MUST-MEMORIZE AREA FORMULAS

Parallelogram:

$\boxed{bh}$

Rectangle:

$\boxed{lb}$

Square:

$\boxed{a^2 = \frac{d^2}{2}}$

Rhombus/kite:

$\boxed{\frac{1}{2}d_1d_2}$

Trapezium:

$\boxed{\frac{1}{2}(a+b)h}$

73. MUST-MEMORIZE POLYGON FORMULAS

Interior sum:

$\boxed{(n-2)180^\circ}$

Diagonals:

$\boxed{\frac{n(n-3)}{2}}$

Diagonals from one vertex:

$\boxed{n-3}$

Regular exterior:

$\boxed{\frac{360^\circ}{n}}$

Regular interior:

$\boxed{180^\circ - \frac{360^\circ}{n}}$

74. FAST EXAM DECISION TREE

Opposite sides parallel?

Think parallelogram family.

All angles (90°) ?

Rectangle.

All sides equal?

Rhombus.

All sides equal + (90°) ?

Square.

One pair parallel?

Trapezium.

Two pairs of adjacent equal sides?

Kite.

Diagonals equal?

Could indicate rectangle/square/isosceles trapezium — need more information.

Diagonals perpendicular?

Could indicate rhombus/square/kite — need more information.

PRACTICE — LEVEL 0

Q1

Find fourth angle of quadrilateral if the other three are:

$[70^\circ, 80^\circ, 100^\circ]$

Q2

Parallelogram has one angle (65°) . Find adjacent angle.

Q3

Rectangle dimensions (8×15) . Find diagonal.

Q4

Square side 7. Find diagonal.

Q5

Rhombus diagonals 8 and 12. Find area.

LEVEL 1

Q6

Parallelogram diagonal AC is 20. Diagonals meet at O. Find AO.

Q7

Square diagonal 14. Find area.

Q8

Trapezium bases are 8 and 20, height 6. Find area.

Q9

Find number of diagonals in a decagon.

Q10

Regular polygon exterior angle is (30°) . Find interior angle and sides.

LEVEL 2

Q11

Rhombus diagonals are 10 and 24. Find side and perimeter.

Q12

Regular polygon has interior angle (150°) . Find number of diagonals.

Q13

Polygon has interior-angle sum (2520°) . Find sides.

Q14

A trapezium has area 96, parallel sides 10 and 14. Find height.

Q15

A square's side increases by 10%. Find percentage increase in area.

LEVEL 3 — EXAM PRESSURE

Q16

A rhombus has side 17 and one diagonal 16. Find other diagonal and area.

Q17

A polygon has 54 diagonals. Find number of sides.

Q18

Rectangle has perimeter 60 and length:breath $(=3:2)$. Find area and diagonal.

Q19

A parallelogram has base 20 and area 240. Find corresponding altitude. If its adjacent side is 13, can you determine its other altitude? If yes, find it.

Q20

A regular polygon has interior angle 5 times its exterior angle. Find number of sides.

ANSWERS

Q1

$$[360-(70+80+100)]$$

$$[=360-250]$$

$$[=\boxed{110^\circ}]$$

Q2

$$[180-65 = \boxed{115^\circ}]$$

Q3

$$\sqrt{8^2+15^2} = \sqrt{289}$$

$$= \boxed{17}$$

Q4

$$= \boxed{7\sqrt{2}}$$

Q5

$$\frac{12(8)(12)}{2} = \boxed{48}$$

Q6

Diagonals bisect:

$$AO = \frac{20}{2}$$

$$= \boxed{10}$$

Q7

$$A = \frac{14^2}{2} = \boxed{98}$$

Q8

$$A = \frac{12(8+20)(6)}{2}$$

$$= 14(6)$$

$$= \boxed{84}$$

Q9

$$\frac{10(7)}{2} = \boxed{35}$$

Q10

Sides:

$$n = 360/30 = 12$$

Interior:

$$180 - 30 = 150^\circ$$

$$\boxed{n=12, \text{ } I=150^\circ}$$

Q11

Half diagonals:

$$5, 12$$

Side:

$$13$$

Perimeter:

$$4(13) = 52$$

$$\boxed{\text{side}=13, \text{ } P=52}$$

Q12

Exterior:

$$\backslash[30^\circ \backslash]$$

Thus:

$$\backslash[n=12 \backslash]$$

Diagonals:

$$\backslash[\frac{12(9)}{2} = \boxed{54} \backslash]$$

Q13

$$\backslash[(n-2)180=2520 \backslash]$$

$$\backslash[n-2=14 \backslash]$$

$$\backslash[\boxed{n=16} \backslash]$$

Q14

$$\backslash[96=\frac{12(24)}{2}h \backslash]$$

$$\backslash[96=12h \backslash]$$

$$\backslash[\boxed{h=8} \backslash]$$

Q15

Side multiplier:

$$\backslash[1.1 \backslash]$$

Area multiplier:

$$\backslash[1.1^2=1.21 \backslash]$$

Increase:

$$\backslash[\boxed{21\%} \backslash]$$

Q16

Half first diagonal:

$$\backslash[8 \backslash]$$

Let half second diagonal:

$$\backslash[x \backslash]$$

$$\backslash[8^2+x^2=17^2 \backslash]$$

$$\backslash[64+x^2=289 \backslash]$$

$$\backslash[x=15 \backslash]$$

Full second diagonal:

$$\backslash[30 \backslash]$$

Area:

$$\backslash[\frac{12(16)(30)}{2} \backslash]$$

$$\backslash[=\boxed{240} \backslash]$$

Thus:

$$\boxed{d_2=30, A=240}$$

Q17

$$\frac{n(n-3)}{2}=54$$

$$n^2-3n-108=0$$

$$(n-12)(n+9)=0$$

$$\boxed{n=12}$$

Q18

Let:

$$l:b=3:2$$

So:

$$l=3x, b=2x$$

Perimeter:

$$2(5x)=60$$

$$x=6$$

Thus:

$$l=18, b=12$$

Area:

$$18(12)=216$$

Diagonal:

$$\sqrt{18^2+12^2} = \sqrt{468}$$

$$=6\sqrt{13}$$

$$\boxed{A=216, d=6\sqrt{13}}$$

Q19

Using base 20:

$$240=20h$$

$$h=12$$

Using adjacent side 13 as another base, area remains 240:

$$240=13h_2$$

$$h_2=\frac{240}{13}$$

Thus:

$$\boxed{12, \frac{240}{13}}$$

Q20

Let exterior angle:

$$E$$

Interior:

$\angle E$

Since:

$\angle E + \angle E = 180^\circ$

$\angle E = 30^\circ$

Thus:

$n = 360/30$

$n = 12$

CHAPTER 45 STATUS

Covered so far:

- quadrilateral angle sum '
- parallelogram '
- rectangle '
- square '
- rhombus '
- kite '
- trapezium '
- diagonal properties '
- area formulas '
- quadrilateral hierarchy '
- polygon interior/exterior angles '
- polygon diagonals '
- regular polygons '
- area-scaling shortcuts '
- GATE/placement practice '

But **Chapter 45 is not complete yet.**

The major remaining section is **circles**, which deserves its own part because it has several high-value theorems.

Progress

Completed: 44/76 Current: 0/76 Chapter 45 — Quadrilaterals, Polygons & Circles

NEXT

+ Chapter 45 Part 2 — Circles

We will cover with simple diagrams where useful:

- radius, diameter, chord, arc, sector, segment
- central angles
- perpendicular from center to chord
- equal-chord properties
- angle at center vs angle at circumference

- angle in a semicircle
- same-segment theorem
- cyclic quadrilaterals
- tangent-radius theorem
- two tangents from an external point
- tangent-chord theorem
- intersecting chord theorem
- secant/tangent power of a point
- final Chapter 45 audit.

USER

next

ASSISTANT

CHAPTER 45 — QUADRILATERALS, POLYGONS AND CIRCLES

PART 2 — CIRCLES, CHORDS, ARCS, ANGLES, CYCLIC QUADRILATERALS, TANGENTS & POWER OF A POINT

Status: $\emptyset$ **Progress:** 0% **Gate:** Medium **Placements:** Medium–High **Priority:** +Tier 2

Circles are theorem-heavy, so diagrams help a lot.

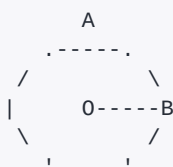
Main strategy:

Identify what is connected to the center, chord, tangent, or circumference before calculating.

1. BASIC CIRCLE TERMS

A circle is the set of all points at a fixed distance from a center.

TEXT



- $\backslash(O) = \text{center}$
- $\backslash(OB) = \text{radius}$

All radii are equal:

$$\boxed{OA=OB=OC=\cdots=r}$$

2. DIAMETER

A chord passing through the center is a diameter.

$$\boxed{d=2r}$$

Diameter is the longest chord of a circle.

3. CHORD

A line segment joining two points on the circle is a chord.

TEXT



AB is a chord.

A diameter is a special chord.

4. ARC

An arc is part of the circumference between two points.

Two arcs between A and B:

- minor arc
- major arc

5. SEMICIRCLE

A diameter divides a circle into two equal semicircles.

Each corresponds to:

$$\boxed{180^\circ}$$

at the center.

6. SECTOR

A sector is the region enclosed by:

- two radii
- corresponding arc

Like a pizza slice.

For central angle θ :

$$\boxed{\text{Sector area} = \frac{\theta}{360^\circ} \pi r^2}$$

7. ARC LENGTH

For central angle θ :

$$\boxed{\text{Arc length} = \frac{\theta}{360^\circ} (2\pi r)}$$

8. SEGMENT

A segment is the region enclosed by:

- a chord
- its corresponding arc

Do not confuse:

- sector !' two radii
 - segment !' chord
-

9. CENTRAL ANGLE

An angle whose vertex is at the center is called a central angle.

Example:

$$\angle AOB$$

It directly determines arc AB.

10. EQUAL CHORDS SUBTEND EQUAL CENTRAL ANGLES

If:

$$AB = CD$$

then:

$$\angle AOB = \angle COD$$

Converse also holds:

Equal central angles subtend equal chords.

11. EQUAL CHORDS ARE EQUIDISTANT FROM CENTER

If chords AB and CD are equal, their perpendicular distances from center are equal.

Converse also holds.

12. PERPENDICULAR FROM CENTER TO CHORD

If:

$$OM \perp AB$$

where AB is a chord, then M bisects AB.

TEXT

A% % % M% % % B
%
%
0

Thus:

$$\boxed{AM=MB}$$

MUST MEMORIZE

13. CONVERSE

A line from center to midpoint of a chord is perpendicular to the chord.

14. CHORD LENGTH FROM CENTER DISTANCE

Suppose:

- radius (r)
- distance from center to chord (d)

Perpendicular bisects chord.

Half-chord:

$$\sqrt{r^2 - d^2}$$

Therefore:

$$\boxed{\text{Chord length} = 2\sqrt{r^2 - d^2}}$$

15. EXAMPLE

Radius:

$$13$$

Distance from center to chord:

$$5$$

Half-chord:

$$\sqrt{169 - 25} = 12$$

Chord:

$$\boxed{24}$$

16. CLOSER CHORD TO CENTER IS LONGER

For chords in same circle:

- smaller center distance !' longer chord

- larger center distance !' shorter chord

Diameter has distance 0, so it is longest.

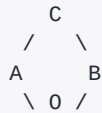
17. ANGLE AT CENTER VS ANGLE AT CIRCUMFERENCE

This is one of the most important circle theorems.

If both angles subtend the same arc AB:

$$\boxed{\angle AOB = 2\angle ACB}$$

TEXT



where O is center.

MUST MEMORIZE

18. EXAMPLE

Central angle:

$$100^\circ$$

Angle at circumference subtending same arc:

$$\boxed{50^\circ}$$

19. REVERSE EXAMPLE

Inscribed angle:

$$35^\circ$$

Central angle over same arc:

$$\boxed{70^\circ}$$

20. ANGLE IN A SEMICIRCLE

A diameter subtends:

$$180^\circ$$

at center.

Therefore at circumference:

$$180^\circ/2 = \boxed{90^\circ}$$

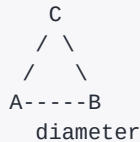
So:

Any angle subtended by a diameter at the circumference is a right angle.

Also called **Thales' theorem**.

21. DIAGRAM — ANGLE IN SEMICIRCLE

TEXT



If AB is diameter:

$$\boxed{\angle ACB = 90^\circ}$$

22. SAME-SEGMENT THEOREM

Angles subtended by the same chord at points on the same segment are equal.

If both $\angle ACB$ and $\angle ADB$ stand on chord AB:

$$\boxed{\angle ACB = \angle ADB}$$

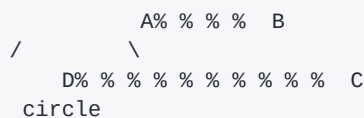
23. WHY SAME-SEGMENT ANGLES ARE EQUAL

Both are half of the same central angle subtending arc AB.

24. CYCLIC QUADRILATERAL

A quadrilateral whose four vertices lie on a circle is cyclic.

TEXT



25. OPPOSITE ANGLES OF CYCLIC QUADRILATERAL

$$\boxed{\angle A + \angle C = 180^\circ}$$

$$\boxed{\angle B + \angle D = 180^\circ}$$

MUST MEMORIZE

26. EXAMPLE

If:

$$\angle A = 112^\circ$$

then:

$$\angle C = 68^\circ$$

$$\boxed{68^\circ}$$

27. CONVERSE OF CYCLIC PROPERTY

If opposite angles of a quadrilateral are supplementary:

$$A + C = 180^\circ$$

then the quadrilateral is cyclic.

28. EXTERIOR ANGLE OF CYCLIC QUADRILATERAL

Exterior angle equals interior opposite angle.

This follows because:

- adjacent interior + exterior = 180°
- opposite cyclic angles also sum 180°

Therefore they are equal.

29. TANGENT

A tangent touches a circle at exactly one point.

That point is called point of contact.

TEXT

tangent

% % % % % % % % % % T % % % % % % %

%

%

0

30. RADIUS IS PERPENDICULAR TO TANGENT

At point of contact T:

$$\boxed{OT \perp \text{tangent}}$$

So:

$$\boxed{90^\circ}$$

MUST MEMORIZE

31. CONVERSE

If a line through a point on a circle is perpendicular to the radius at that point, then the line is tangent.

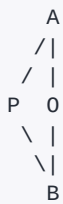
32. TWO TANGENTS FROM SAME EXTERNAL POINT

From an external point P, tangents touch circle at A and B.

Then:

$$\boxed{PA=PB}$$

TEXT



33. EXAMPLE

If one tangent length:

$$PA=12$$

then:

$$\boxed{PB=12}$$

34. TANGENT LENGTH USING PYTHAGORAS

If:

- OP = distance from center to external point
- radius = r
- tangent PT

Since $\triangle OPT$ is a right triangle:

$$OP^2 = OT^2 + PT^2$$

Thus:

$$\boxed{PT = \sqrt{OP^2 - r^2}}$$

35. EXAMPLE

$$OP=13, \text{quad } r=5$$

$$PT = \sqrt{169 - 25}$$

$\boxed{12}$

36. ANGLE BETWEEN TWO TANGENTS

From P, tangents PA and PB touch at A and B.

Radii:

$OA \perp PA$

$OB \perp PB$

Quadrilateral OAPB has two right angles.

Thus:

$\angle APB + \angle AOB = 180^\circ$

37. EXAMPLE

Central angle:

$\angle AOB = 120^\circ$

Angle between tangents:

$180 - 120 = \boxed{60^\circ}$

38. TANGENT-CHORD THEOREM

Angle between tangent and chord equals angle in the alternate segment.

If tangent touches at A and chord AB is drawn:

$\angle(\text{tangent}, AB) = \angle ACB$

where C is a point on the opposite arc.

This is also called the **alternate segment theorem**.

39. INTERSECTING CHORD THEOREM

If two chords AB and CD intersect inside circle at P:

TEXT

A % % % P % % % B
%r
%r
C

then:

$PA \cdot PB = PC \cdot PD$

MUST KNOW

40. EXAMPLE

Suppose:

$$[PA=3, PB=8, PC=4]$$

Find PD.

$$[3(8)=4(PD)]$$

$$[24=4PD]$$

$$[\boxed{PD=6}]$$

41. SECANT-SECANT THEOREM

From an external point P, two secants intersect circle.

For one secant:

- external part = (PA)
- whole secant = (PB)

For another:

- external = (PC)
- whole = (PD)

Then:

$$[\boxed{PA \cdot PB = PC \cdot PD}]$$

Important:

$$\text{external} \times \text{whole} = \text{external} \times \text{whole}$$

42. TANGENT-SECANT THEOREM

From external point P:

- tangent length = PT
- secant has external part PA and whole PB

Then:

$$[\boxed{PT^2 = PA \cdot PB}]$$

MUST MEMORIZE

43. EXAMPLE

Tangent:

$$[PT=6]$$

Secant external part:

$$[PA=4]$$

Then:

$$\sqrt{36=4(PB)}$$

$$\boxed{PB=9}$$

So internal secant portion would be:

$$\sqrt{9-4=5}$$

44. POWER OF A POINT

The three previous formulas are all manifestations of the same idea:

Power of point P with respect to the circle.

Inside point

$$\boxed{PA \cdot PB = PC \cdot PD}$$

Outside, two secants

$$\boxed{PA \cdot PB = PC \cdot PD}$$

Outside, tangent + secant

$$\boxed{PT^2 = PA \cdot PB}$$

45. VERY IMPORTANT SECANT TRAP

Suppose external part:

$$\sqrt{3}$$

internal part:

$$\sqrt{5}$$

Whole secant is:

$$\sqrt{3+5=8}$$

Use:

$$\boxed{3 \cdot 8}$$

not:

$$\sqrt{3 \cdot 5}$$

46. INSCRIBED ANGLE FROM ARC

If an arc measures:

$$\sqrt{\theta}$$

then an inscribed angle subtending that arc is:

$$\boxed{\theta/2}$$

47. CENTRAL ANGLE AND ARC MEASURE

Central angle measure equals measure of its intercepted minor arc.

So if:

$$\angle AOB = 80^\circ$$

minor arc AB measures:

$$\boxed{80^\circ}$$

48. CIRCUMFERENCE

For radius (r) :

$$\boxed{C = 2\pi r}$$

or using diameter (d) :

$$\boxed{C = \pi d}$$

49. AREA OF CIRCLE

$$\boxed{A = \pi r^2}$$

50. SEMICIRCLE AREA

$$\boxed{\frac{1}{2}\pi r^2}$$

But if the question asks for **perimeter of semicircle**, include diameter:

$$\boxed{\pi r + 2r}$$

Major trap.

51. QUADRANT AREA

A quadrant is one-fourth of a circle:

$$\boxed{\frac{1}{4}\pi r^2}$$

Arc length of quadrant:

$$\boxed{\frac{1}{4}(2\pi r) = \frac{1}{2}\pi r}$$

52. SECTOR AREA

For angle (θ) :

$$\boxed{A_{\text{sector}} = \frac{\theta}{360}\pi r^2}$$

53. SECTOR PERIMETER

Perimeter of sector includes:

- arc
- two radii

Therefore:

$$\boxed{P = \frac{\theta}{360}(2\pi r) + 2r}$$

54. EXAMPLE — SECTOR

Radius:

$$6$$

Central angle:

$$60^\circ$$

Sector area:

$$\frac{60}{360}\pi(36)$$

$$=6\pi$$

Arc length:

$$\frac{60}{360}(12\pi)$$

$$=2\pi$$

55. MINOR AND MAJOR SECTORS

Minor sector corresponds to smaller central angle.

Major sector:

$$360^\circ - \theta$$

Area of major sector:

$$\pi r^2 - \text{minor sector area}$$

56. RING / ANNULUS

Region between concentric circles of radii:

$$R > r$$

Area:

$$\pi(R^2 - r^2)$$

This will appear again in mensuration.

57. EXAMPLE

Outer radius:

$$\sqrt{10}$$

inner radius:

$$\sqrt{6}$$

Area:

$$\sqrt{\pi(100-36)}$$

$$\boxed{64\pi}$$

58. GATE-STYLE QUESTION 1

A chord is 24 units long in a circle of radius 13.

Distance from center to chord:

Half chord:

$$\sqrt{12}$$

So:

$$\sqrt{d^2+12^2=13^2}$$

$$\sqrt{d^2=25}$$

$$\boxed{d=5}$$

59. GATE-STYLE QUESTION 2

Angle at center subtending chord AB:

$$\sqrt{84^\circ}$$

Angle at circumference subtending same chord:

$$\boxed{42^\circ}$$

60. GATE-STYLE QUESTION 3

AB is diameter.

Point C lies on circle.

Then:

$$\boxed{\angle ACB=90^\circ}$$

61. GATE-STYLE QUESTION 4

In cyclic quadrilateral:

$$\sqrt{\angle A=103^\circ}$$

Then:

$$\angle C = 180 - 103$$

$$\boxed{77^\circ}$$

62. PLACEMENT-STYLE QUESTION 1

From external point P, two tangents have lengths:

$$PA = 9$$

Then:

$$\boxed{PB = 9}$$

63. PLACEMENT-STYLE QUESTION 2

Radius:

$$8$$

Center to external point:

$$17$$

Tangent length:

$$\sqrt{17^2 - 8^2}$$

$$= \sqrt{225}$$

$$= \boxed{15}$$

64. PLACEMENT-STYLE QUESTION 3

Intersecting chords:

$$PA = 4, \quad PB = 6, \quad PC = 3$$

Find PD.

$$4(6) = 3(PD)$$

$$PD = 8$$

$$\boxed{8}$$

65. PLACEMENT-STYLE QUESTION 4

Tangent length:

$$10$$

Secant external part:

$$4$$

Then:

$$100 = 4(\text{whole})$$

$$\text{whole} = 25$$

$$\boxed{25}$$

66. ADVANCED QUESTION — TWO TANGENTS

Central angle between radii to contact points:

$$140^\circ$$

Angle between tangents:

$$180 - 140$$

$$\boxed{40^\circ}$$

67. ADVANCED QUESTION — SECTOR

A circle radius:

$$14$$

Sector angle:

$$90^\circ$$

Area:

$$\frac{14\pi(196)}{2}$$

$$\boxed{49\pi}$$

Arc length:

$$\frac{14(28\pi)}{2}$$

$$\boxed{7\pi}$$

68. ADVANCED QUESTION — CYCLIC QUADRILATERAL

Angles are:

$$x, 2x, 3x, ?$$

Suppose x and $3x$ are opposite.

Then:

$$x + 3x = 180$$

$$x = 45^\circ$$

So those opposite angles:

$$45^\circ, 135^\circ$$

69. COMMON TRAPS

& pTRAP 1 — Radius to tangent

Always perpendicular at contact point.

& pTRAP 2 — Angle at center

It is twice the angle at circumference over the same arc.

& pTRAP 3 — Cyclic quadrilateral

Opposite angles sum to:

$$\backslash [180^\circ \backslash \text{circ} \backslash]$$

not necessarily equal.

& pTRAP 4 — Secant formula

Use:

$$\backslash [\text{external} \times \text{whole} \backslash]$$

not external $\times$ internal.

& pTRAP 5 — Tangent from same external point

The two tangent lengths are equal.

& pTRAP 6 — Semicircle perimeter

Include the diameter.

& pTRAP 7 — Sector perimeter

Include both radii.

70. MUST-MEMORIZE CHORD RULES

Perpendicular from center to chord:

$$\backslash [\boxed{\text{bisects chord}} \backslash]$$

Chord length:

$$\backslash [\boxed{2\sqrt{r^2 - d^2}} \backslash]$$

Equal chords:

- equal central angles
 - equal center distances
-

71. MUST-MEMORIZE ANGLE RULES

Same arc:

$$\backslash [\boxed{\angle \text{ at center} = 2(\angle \text{ at circumference})} \backslash]$$

Diameter:

$$\boxed{\text{angle in semicircle} = 90^\circ}$$

Same segment:

$$\boxed{\text{angles equal}}$$

Cyclic quadrilateral:

$$\boxed{\text{opposite angles supplementary}}$$

72. MUST-MEMORIZE TANGENT RULES

Radius $\perp$ tangent:

$$\boxed{90^\circ}$$

Two tangents from same point:

$$\boxed{PA = PB}$$

Angle between tangents:

$$\boxed{180^\circ - \text{central angle}}$$

73. MUST-MEMORIZE POWER-OF-POINT RULES

Inside intersecting chords:

$$\boxed{PA \cdot PB = PC \cdot PD}$$

Outside secants:

$$\boxed{(\text{external})(\text{whole}) = (\text{external})(\text{whole})}$$

Tangent-secant:

$$\boxed{PT^2 = (\text{external})(\text{whole})}$$

74. MUST-MEMORIZE CIRCLE MENSURATION

$$\boxed{C = 2\pi r}$$

$$\boxed{A = \pi r^2}$$

Arc:

$$\boxed{\frac{\theta}{360}(2\pi r)}$$

Sector:

$$\boxed{\frac{\theta}{360}\pi r^2}$$

Annulus:

$$\boxed{\pi(R^2 - r^2)}$$

75. FAST EXAM DECISION TREE

Chord + center?

Draw perpendicular; use half-chord + Pythagoras.

Same arc?

Center angle = twice circumference angle.

Diameter?

Look for immediate (90°) .

Four points on circle?

Try cyclic opposite-angle rule.

Tangent?

Draw radius to contact point and mark (90°) .

Two tangents from one point?

Set lengths equal.

Chords/secants intersect?

Think products.

Sector?

Separate area, arc length and perimeter carefully.

PRACTICE — LEVEL 0

Q1

Radius 10. Find diameter.

Q2

Central angle (120°) . Find inscribed angle over same arc.

Q3

A diameter subtends what angle at circumference?

Q4

Cyclic quadrilateral has one angle (75°) . Find opposite.

Q5

Two tangents from P. One is 11. Find other.

LEVEL 1

Q6

Circle radius 10, chord distance from center 6. Find chord length.

Q7

An inscribed angle is (48°) . Find corresponding central angle.

Q8

Radius 5 and distance from center to external point 13. Find tangent length.

Q9

Intersecting chords have segments 2,12 and 3, (x) . Find (x) .

Q10

Sector radius 9, angle (120°) . Find area.

LEVEL 2

Q11

Tangent length is 12. Secant external part 9. Find whole secant.

Q12

Two tangents form angle (50°) . Find central angle between radii to contact points.

Q13

A circle radius 13 has chord length 10. Find perpendicular distance from center to chord.

Q14

A cyclic quadrilateral has angles: $[2x+10, \text{quad } 3x-20]$ as opposite angles. Find (x) .

Q15

A circle radius 14 has sector angle (90°) . Find sector perimeter.

LEVEL 3 — EXAM PRESSURE

Q16

From external P, tangent $PT = 15$. A secant has external part 9. Find internal part.

Q17

Two chords intersect. One has segments 4 and 9; the other has segments (6) and (x) . Find (x) .

Q18

A chord has length 16 and is 6 from the center. Find radius.

Q19

A cyclic quadrilateral has one exterior angle (68°) . Find opposite interior angle.

Q20

A ring has outer diameter 20 and inner diameter 12. Find its area.

ANSWERS

Q1

$$\backslash[d=2r \backslash]$$

$$\backslash[=\boxed{20} \backslash]$$

Q2

$$\backslash[120/2 = \boxed{60^\circ} \backslash]$$

Q3

$$\backslash[\boxed{90^\circ} \backslash]$$

Q4

$$\backslash[180-75 = \boxed{105^\circ} \backslash]$$

Q5

$$\backslash[\boxed{11} \backslash]$$

Q6

Half chord:

$$\backslash[\sqrt{10^2-6^2} = \sqrt{64} = 8 \backslash]$$

Full chord:

$$\backslash[\boxed{16} \backslash]$$

Q7

$$\backslash[2(48) = \boxed{96^\circ} \backslash]$$

Q8

$$\backslash[PT=\sqrt{13^2-5^2} \backslash]$$

$$\backslash[=\sqrt{144} \backslash]$$

$$\backslash[=\boxed{12} \backslash]$$

Q9

$$\backslash[2(12)=3x \backslash]$$

$$\backslash[24=3x \backslash]$$

$$\backslash[\boxed{x=8} \backslash]$$

Q10

$$\backslash[\frac{120}{360}\pi(81) \backslash]$$

$$\backslash[=\boxed{27\pi} \backslash]$$

Q11

$$\backslash[12^2=9(\text{whole}) \backslash]$$

$$\backslash[144=9W \backslash]$$

$$\backslash[\boxed{W=16} \backslash]$$

Q12

$$\angle APB + \angle AOB = 180^\circ$$

$$\angle AOB = 130^\circ$$

$$\boxed{130^\circ}$$

Q13

Half chord:

$$5$$

Then:

$$d^2 + 5^2 = 13^2$$

$$d^2 = 144$$

$$\boxed{d=12}$$

Q14

Opposite cyclic angles supplementary:

$$(2x+10) + (3x-20) = 180$$

$$5x - 10 = 180$$

$$5x = 190$$

$$\boxed{x=38}$$

Q15

Arc:

$$\frac{90}{360} (2\pi \cdot 14) = 7\pi$$

Sector perimeter:

$$7\pi + 28$$

$$\boxed{28 + 7\pi}$$

Q16

$$15^2 = 9(\text{whole})$$

$$225 = 9W$$

$$W = 25$$

Internal:

$$25 - 9 = \boxed{16}$$

Q17

$$4(9) = 6x$$

$$36 = 6x$$

$$\boxed{x=6}$$

Q18

Half chord:

$$\sqrt{8}$$

Radius:

$$\sqrt{r^2=8^2+6^2}$$

$$\sqrt{=100}$$

$$\sqrt{\boxed{r=10}}$$

Q19

Exterior angle of cyclic quadrilateral equals opposite interior angle.

Thus:

$$\sqrt{\boxed{68^\circ}}$$

Q20

Outer radius:

$$\sqrt{10}$$

Inner radius:

$$\sqrt{6}$$

Area:

$$\sqrt{\pi(100-36)}$$

$$\sqrt{= \boxed{64\pi}}$$

' CHAPTER 45 COMPLETE

We have now covered:

- parallelogram '
- rectangle '
- square '
- rhombus '
- kite '
- trapezium '
- quadrilateral diagonal properties '
- polygon angles and diagonals '
- circle terminology '
- radius/diameter/chords '
- center-to-chord theorem '
- chord lengths '
- equal chord properties '
- central and inscribed angles '
- angle in semicircle '
- same-segment theorem '
- cyclic quadrilaterals '
- tangents '

- equal tangents '
- tangent-radius theorem '
- tangent-chord theorem '
- intersecting chords '
- secants '
- tangent-secant theorem '
- power of a point '
- arc/sector/annulus basics '
- GATE/placement L0–L3 practice '

Progress

Completed: 45/76

Unit 1 — Quantitative Aptitude: 0% PROGRESS

NEXT

+Chapter 46 — 2D Mensuration

Next we will consolidate all 2D measurement formulas and applications:

- square/rectangle
- triangle
- parallelogram
- rhombus/kite
- trapezium
- regular polygons
- circles
- semicircles/sectors/rings
- composite figures
- shaded regions
- paths/borders
- percentage changes in area/perimeter
- cost of flooring/fencing/painting
- GATE and placement speed shortcuts.

USER

next

ASSISTANT

CHAPTER 46 — 2D MENSURATION

PART 1 — PERIMETER, AREA, TRIANGLES, QUADRILATERALS, CIRCLES, SECTORS,

RINGS & BASIC COMPOSITE FIGURES

Status: $\emptyset$ = 0% PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Mensuration means:

Measurement of geometric figures.

The two main quantities are:

- **Perimeter** !' total boundary length
- **Area** !' region enclosed inside the boundary

The biggest exam trap is using an **area formula when the question asks for boundary**, or vice versa.

1. PERIMETER VS AREA

Perimeter

Measured in units such as:

$\boxed{\text{cm}, \text{m}, \text{km}}$

Area

Measured in square units:

$\boxed{\text{cm}^2, \text{m}^2, \text{km}^2}$

Memory trick

Fence !' perimeter Floor/paint/grass !' area

2. UNIT CONVERSION — LENGTH

$1 \text{ m} = 100 \text{ cm}$

$1 \text{ km} = 1000 \text{ m}$

But for area, square the conversion.

3. AREA UNIT CONVERSION

Since:

$1 \text{ m} = 100 \text{ cm}$

then:

$1 \text{ m}^2 = 100^2 \text{ cm}^2$

$\boxed{1 \text{ m}^2 = 10,000 \text{ cm}^2}$

Major trap.

4. HECTARE

Common land-area unit:

$$\boxed{1 \text{ hectare} = 10,000 \text{ m}^2}$$

Also:

$$\boxed{1 \text{ km}^2 = 100 \text{ hectares}}$$

5. RECTANGLE

For length (l) , breadth (b) :

$$\boxed{A = lb}$$

$$\boxed{P = 2(l + b)}$$

Diagonal:

$$\boxed{d = \sqrt{l^2 + b^2}}$$

6. EXAMPLE — RECTANGLE

Length:

$$\boxed{12}$$

Breadth:

$$\boxed{7}$$

Area:

$$12 \cdot 7 = \boxed{84}$$

Perimeter:

$$2(12 + 7) = \boxed{38}$$

7. SQUARE

Side (a) :

$$\boxed{A = a^2}$$

$$\boxed{P = 4a}$$

Diagonal:

$$\boxed{d = a\sqrt{2}}$$

Area using diagonal:

$$\boxed{A = \frac{d^2}{2}}$$

8. EXAMPLE — SQUARE FROM PERIMETER

Square perimeter:

$$\boxed{40}$$

Side:

$$a = 40/4 = 10$$

Area:

$$\boxed{100}$$

9. TRIANGLE

General area:

$$\boxed{A = \frac{1}{2}bh}$$

where h is perpendicular to base b .

Perimeter:

$$\boxed{a+b+c}$$

10. RIGHT TRIANGLE

If perpendicular legs are (a,b) :

$$\boxed{A = \frac{1}{2}ab}$$

Example:

$$6, 8, 10$$

Area:

$$\frac{1}{2}(6)(8) = \boxed{24}$$

11. EQUILATERAL TRIANGLE

Side a :

$$\boxed{A = \frac{\sqrt{3}}{4}a^2}$$

Altitude:

$$\boxed{h = \frac{\sqrt{3}}{2}a}$$

Perimeter:

$$\boxed{3a}$$

12. HERON'S FORMULA RECAP

For sides (a,b,c) :

$$s = \frac{a+b+c}{2}$$

$$\boxed{A = \sqrt{s(s-a)(s-b)(s-c)}}$$

Use when all 3 sides are known.

13. PARALLELOGRAM

TEXT



Base (b) , perpendicular height (h) :

$$\boxed{A=bh}$$

Perimeter if sides (a,b) :

$$\boxed{P=2(a+b)}$$

14. RHOMBUS

Side (a) , diagonals (d_1, d_2) :

$$\boxed{A=\frac{1}{2}d_1d_2}$$

Perimeter:

$$\boxed{4a}$$

15. KITE

Area:

$$\boxed{A=\frac{1}{2}d_1d_2}$$

provided (d_1, d_2) are diagonals.

16. TRAPEZIUM

Parallel sides:

$$a, b$$

height:

$$h$$

Area:

$$\boxed{A=\frac{1}{2}(a+b)h}$$

17. TRAPEZIUM — FAST INTERPRETATION

Area is:

$$\boxed{\text{average of parallel sides} \times \text{height}}$$

because:

$$\left[\frac{a+b}{2} \cdot h \right]$$

This is a useful memory trick.

18. EXAMPLE

Bases:

$$\left[10, 18 \right]$$

Height:

$$\left[6 \right]$$

Area:

$$\left[\frac{10+18}{2} \cdot 6 \right]$$

$$\left[=14 \cdot 6 \right]$$

$$\left[=\boxed{84} \right]$$

19. REGULAR POLYGON — BASIC AREA IDEA

For a regular polygon:

$$\left[\boxed{A = \frac{1}{2} (\text{perimeter}) (\text{apothem})} \right]$$

where apothem is perpendicular distance from center to a side.

Lower priority, but useful.

20. CIRCLE

Radius:

$$\left[r \right]$$

Circumference:

$$\left[\boxed{C = 2\pi r} \right]$$

Area:

$$\left[\boxed{A = \pi r^2} \right]$$

21. DIAMETER FORM

If diameter is (d) :

$$\left[r = \frac{d}{2} \right]$$

Then:

$$\left[C = \pi d \right]$$

and:

$$\left[A = \pi \frac{d^2}{4} \right]$$

$$\boxed{A = \frac{\pi d^2}{4}}$$

22. SEMICIRCLE

Area:

$$\boxed{A = \frac{1}{2} \pi r^2}$$

Curved length:

$$\boxed{\pi r}$$

Full perimeter:

$$\boxed{\pi r + 2r}$$

Do not forget the diameter.

23. QUADRANT

Area:

$$\boxed{\frac{1}{4} \pi r^2}$$

Arc length:

$$\boxed{\frac{\pi r}{2}}$$

Full perimeter:

$$\boxed{\frac{\pi r}{2} + 2r}$$

24. SECTOR

For central angle θ :

Area:

$$\boxed{\frac{\theta}{360} \pi r^2}$$

Arc length:

$$\boxed{\frac{\theta}{360} (2\pi r)}$$

25. SECTOR PERIMETER

Include:

- curved arc
- two radii

So:

$$\boxed{P = \frac{\theta}{360} (2\pi r) + 2r}$$

26. EXAMPLE — SECTOR

Radius:

$$\boxed{12}$$

Angle:

$$\boxed{60^\circ}$$

Area:

$$\frac{16\pi(144)}{360}$$

$$\boxed{24\pi}$$

Arc:

$$\frac{16(24\pi)}{180}$$

$$\boxed{4\pi}$$

27. CIRCULAR RING / ANNULUS

Outer radius (R) , inner radius (r) :

$$\boxed{A = \pi(R^2 - r^2)}$$

Can factor:

$$\boxed{A = \pi(R - r)(R + r)}$$

Sometimes faster.

28. EXAMPLE — RING

Outer radius:

$$\boxed{15}$$

Inner radius:

$$\boxed{9}$$

$$A = \pi(225 - 81)$$

$$\boxed{144\pi}$$

29. COMPOSITE FIGURES

A composite figure is made from multiple standard shapes.

General strategy:

$$\boxed{\text{Required area} = \text{sum of included parts} - \text{removed parts}}$$

30. EXAMPLE — RECTANGLE WITH SEMICIRCLE

Suppose rectangle:

$$\sqrt{10 \times 6}$$

with a semicircle of diameter 6 attached externally.

Rectangle area:

$$\sqrt{60}$$

Semicircle radius:

$$\sqrt{3}$$

Area:

$$\sqrt{\frac{12\pi(3)^2}{2} = \frac{9\pi}{2}}$$

Total:

$$\sqrt{\boxed{60 + \frac{9\pi}{2}}}$$

31. SHADING RULE

If shaded region is “large figure minus small figure”:

$$\sqrt{\boxed{A_{\text{shaded}} = A_{\text{outer}} - A_{\text{inner}}}}$$

Very common.

32. SQUARE WITH INSCRIBED CIRCLE

Square side:

$$\sqrt{a}$$

Inscribed circle has diameter:

$$\sqrt{a}$$

So radius:

$$\sqrt{\frac{a}{2}}$$

Area outside circle but inside square:

$$\sqrt{a^2 - \pi \left(\frac{a}{2}\right)^2}$$

$$\sqrt{\boxed{a^2 \left(1 - \frac{\pi}{4}\right)}}$$

33. CIRCLE INSCRIBED IN SQUARE

If square side:

$$\sqrt{14}$$

Circle radius:

$$\sqrt{7}$$

Shaded corner area:

$$\sqrt{196 - 49\pi}$$

34. SQUARE INSCRIBED IN CIRCLE

If square diagonal equals circle diameter.

For square side (a) :

$$d = a\sqrt{2}$$

Thus circle radius:

$$r = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$

Circle area:

$$\pi \frac{a^2}{2}$$

35. AREA BETWEEN CIRCLE AND INSCRIBED SQUARE

$$\boxed{A = \frac{\pi a^2}{2} - a^2 = a^2 \left(\frac{\pi}{2} - 1 \right)}$$

36. PATH/ BORDER AROUND RECTANGLE — OUTSIDE

Suppose rectangle:

$$l \times b$$

Path of uniform width (x) outside.

New outer dimensions:

$$l + 2x, \quad b + 2x$$

Path area:

$$\boxed{(l + 2x)(b + 2x) - lb}$$

37. WHY ADD $(2x)$?

Path exists on both sides:

TEXT

```

      x          x
    % % % % % % % % % % % % %
x  %      r e c t a n g l e      %      x
    % % % % % % % % % % % % %
      x          x
```

So total length increases by:

$$x + x = 2x$$

same for breadth.

38. EXAMPLE — OUTER PATH

Garden:

$$\boxed{20 \times 10}$$

Path width:

$$2$$

Outer dimensions:

$$\boxed{24 \times 14}$$

Path area:

$$24(14) - 20(10)$$

$$= 336 - 200$$

$$= \boxed{136}$$

39. INNER PATH

Path of width x inside rectangle.

Inner untouched rectangle dimensions:

$$(l-2x) \times (b-2x)$$

Path area:

$$\boxed{l \times b - (l-2x)(b-2x)}$$

40. EXAMPLE — INNER BORDER

Room:

$$\boxed{20 \times 14}$$

Border width:

$$1$$

Inner dimensions:

$$\boxed{18 \times 12}$$

Border area:

$$280 - 216$$

$$= \boxed{64}$$

41. CIRCULAR PATH

If a path of width x surrounds a circular garden of radius r :

Outer radius:

$$r+x$$

Path area:

$$\boxed{\pi[(r+x)^2-r^2]}$$

Simplify:

$$\boxed{\pi(2rx+x^2)}$$

42. INNER CIRCULAR BORDER

If path width (x) lies inside a circle of radius (R) :

Inner radius:

$$R-x$$

Border area:

$$\boxed{\pi[R^2-(R-x)^2]}$$

43. FENCING COST

If fence costs $\text{₹ } c$ per metre:

$$\boxed{\text{Total cost} = \text{perimeter} \times c}$$

44. FLOORING / TILING / PAINTING COST

If cost is $\text{₹ } c$ per square metre:

$$\boxed{\text{Total cost} = \text{area} \times c}$$

45. EXAMPLE — FENCING

Rectangular plot:

$$30 \times 20$$

Perimeter:

$$2(30+20)=100$$

Cost:

$$\text{₹ } 150 / \text{metre}$$

Total:

$$100 \times 15 = \boxed{\text{₹ } 1500}$$

46. EXAMPLE — FLOORING

Room:

$$12 \times 10$$

Area:

$$\boxed{120}$$

Flooring cost:

$$\boxed{180/\text{m}^2}$$

Total:

$$\boxed{120(80)}$$

$$\boxed{=9600}$$

47. NUMBER OF TILES

Number of tiles:

$$\boxed{\frac{\text{floor area}}{\text{area of one tile}}}$$

provided units are same and exact fitting is assumed.

48. EXAMPLE — TILES

Floor:

$$\boxed{6\text{ m}\times 4\text{ m}}$$

Tile:

$$\boxed{20\text{ cm}\times 20\text{ cm}}$$

Convert floor to cm:

$$\boxed{600\times 400}$$

Floor area:

$$\boxed{240000\text{ cm}^2}$$

Tile area:

$$\boxed{400\text{ cm}^2}$$

Tiles:

$$\boxed{240000/400}$$

$$\boxed{=600}$$

49. FAST TILE METHOD

Instead of area:

Along length:

$$\boxed{600/20=30}$$

Along breadth:

$$\boxed{400/20=20}$$

Tiles:

$$30 \cdot 20 = \boxed{600}$$

Often faster.

50. PERCENTAGE CHANGE IN SQUARE SIDE

If square side changes by $(p\%)$:

Area multiplier:

$$\left(1 + \frac{p}{100}\right)^2$$

For increase (p) :

Area percentage increase:

$$\boxed{2p + \frac{p^2}{100}}$$

51. EXAMPLE

Side increases 10%.

Area increase:

$$20 + \frac{100}{100}$$

$$= \boxed{21\%}$$

52. SIDE DECREASE

If side decreases by $(p\%)$:

Area multiplier:

$$\left(1 - \frac{p}{100}\right)^2$$

Percentage decrease:

$$\boxed{2p - \frac{p^2}{100}}$$

Example $(p=20)$:

$$40 - 4 = \boxed{36\%}$$

53. RECTANGLE DIMENSION CHANGES

If length changes by $(a\%)$ and breadth by $(b\%)$, signed:

$$\boxed{\Delta A = a + b + \frac{ab}{100}}$$

Same percentage-product formula from Chapter 15.

54. EXAMPLE

Length increases 20%.

Breadth decreases 10%.

Use:

$$[a=20, \text{quad } b=-10]$$

$$[20-10+\frac{20(-10)}{100}]$$

$$[=10-2]$$

$$[=\boxed{8\%\text{ increase}}]$$

55. CIRCLE RADIUS CHANGE

Area:

$$[\pi r^2]$$

So radius increase $(p\%)$ gives area increase:

$$[\boxed{2p+\frac{p^2}{100}}]$$

Same as square side.

56. EXAMPLE

Radius increases 25%.

Area increase:

$$[50+\frac{625}{100}]$$

$$[=50+6.25]$$

$$[=\boxed{56.25\%}]$$

57. CIRCUMFERENCE CHANGE

Circumference:

$$[2\pi r]$$

directly proportional to radius.

So if radius increases 15%:

$$[\boxed{\text{circumference increases } 15\%}]$$

But area increases:

$$[2(15)+2.25 = \boxed{32.25\%}]$$

58. SAME PERIMETER — SQUARE MAX AREA

Among rectangles with fixed perimeter, square gives maximum area.

Example perimeter:

$\sqrt{48}$

Square side:

$\sqrt{12}$

Maximum area:

$\boxed{144}$

59. SAME AREA — SQUARE MINIMUM PERIMETER

Among rectangles with fixed area, square has minimum perimeter.

This is the reverse optimization intuition.

60. CIRCLE — MAXIMUM AREA FOR GIVEN PERIMETER

Among all plane figures with a fixed perimeter, circle encloses maximum area.

Usually conceptual rather than computational in aptitude.

61. COMPOSITE-PERIMETER TRAP

For composite figures:

Count only the outer exposed boundary.

Do not blindly add perimeters of every component.

Shared internal edges are not part of external perimeter.

62. EXAMPLE — TWO SQUARES JOINED

Two squares side 4 joined along one full side:

TEXT

% % % % % %, % % % %
% % % % %
% % % % %
% % % % % %4% % % % %

Combined rectangle:

8×4

Perimeter:

$2(8+4) = \boxed{24}$

Not:

$16+16=32$

because shared boundary is internal.

63. CIRCLE INSIDE RECTANGLE

If circle fits exactly inside a rectangle and touches top/bottom, rectangle height equals:

$$\boxed{2r}$$

If it touches all four sides, rectangle must be a square of side:

$$\boxed{2r}$$

64. GATE-STYLE QUESTION 1

Rectangle perimeter:

$$50$$

Length:

$$15$$

$$2(15+b)=50$$

$$15+b=25$$

$$b=10$$

Area:

$$\boxed{150}$$

65. GATE-STYLE QUESTION 2

Square area:

$$196$$

Side:

$$14$$

Perimeter:

$$\boxed{56}$$

Diagonal:

$$\boxed{14\sqrt{2}}$$

66. GATE-STYLE QUESTION 3

Circle area:

$$154$$

Using:

$$\pi=\frac{22}{7}$$

$$\frac{22}{7}r^2=154$$

$$\sqrt{r^2=49}$$

$$\sqrt{\boxed{r=7}}$$

Circumference:

$$\sqrt{44}$$

67. GATE-STYLE QUESTION 4

Rhombus diagonals:

$$\sqrt{14, \sqrt{24}}$$

Area:

$$\sqrt{\frac{1}{2}(14)(24)}$$

$$\sqrt{=\boxed{168}}$$

68. PLACEMENT-STYLE QUESTION 1

Rectangular field:

$$\sqrt{50 \times 30}$$

Fence costs ₹ 1 per metre.

Perimeter:

$$\sqrt{160}$$

Cost:

$$\sqrt{160(12)}$$

$$\sqrt{=\boxed{1920}}$$

69. PLACEMENT-STYLE QUESTION 2

Circular park radius:

$$\sqrt{14}$$

Fencing circumference:

$$\sqrt{2 \cdot \frac{22}{7} \cdot 14}$$

$$\sqrt{=88}$$

At ₹ 20 / m :

$$\sqrt{88(20) = \boxed{1760}}$$

70. PLACEMENT-STYLE QUESTION 3

A room is:

$$\sqrt{8 \times 6}$$

A 1 m wide internal border runs along all four sides.

Inner area:

$$\boxed{(8-2)(6-2) = 6 \cdot 4 = 24}$$

Total:

$$\boxed{48}$$

Border:

$$\boxed{48 - 24 = 24}$$

71. PLACEMENT-STYLE QUESTION 4

Circular path around radius 10, width 2.

Path area:

$$\pi(12^2 - 10^2)$$

$$= \pi(44)$$

$$\boxed{44\pi}$$

72. COMMON TRAPS

& pTRAP 1 — Area units

Length conversion factor must be squared for area.

& pTRAP 2 — Border width

Outside rectangle adds $(2x)$, not (x) .

& pTRAP 3 — Semicircle perimeter

Include diameter.

& pTRAP 4 — Sector perimeter

Include both radii.

& pTRAP 5 — Fencing vs flooring

Fence !' perimeter.

Floor !' area.

& pTRAP 6 — Composite perimeter

Do not count shared internal edges.

& pTRAP 7 — Percentage area changes

Area depends on square of linear dimension.

73. MUST-MEMORIZE RECTANGLE/SQUARE

Rectangle:

$$\boxed{A=lb}$$

$$\boxed{P=2(l+b)}$$

Square:

$$\boxed{A=a^2}$$

$$\boxed{P=4a}$$

$$\boxed{d=a\sqrt{2}}$$

74. MUST-MEMORIZE TRIANGLE/QUADRILATERAL AREAS

Triangle:

$$\boxed{\frac{1}{2}bh}$$

Parallelogram:

$$\boxed{bh}$$

Rhombus/kite:

$$\boxed{\frac{1}{2}d_1d_2}$$

Trapezium:

$$\boxed{\frac{1}{2}(a+b)h}$$

75. MUST-MEMORIZE CIRCLE FORMULAS

$$\boxed{C=2\pi r}$$

$$\boxed{A=\pi r^2}$$

Sector:

$$\boxed{\frac{\theta}{360}\pi r^2}$$

Arc:

$$\boxed{\frac{\theta}{360}(2\pi r)}$$

Ring:

$$\boxed{\pi(R^2-r^2)}$$

76. FAST EXAM DECISION TREE

Fence/boundary?

Perimeter.

Floor/grass/paint?

Area.

Path/border?

Outer area - inner area.

Shaded figure?

Break into simple shapes.

Circle sector?

Use angle fraction of full circle.

Dimension changes?

Use multiplier method.

Tiling?

Match units first.

Joined shapes?

For perimeter, remove shared edges.

PRACTICE — LEVEL 0

Q1

Rectangle (12×8) : find area and perimeter.

Q2

Square side 9: find area and diagonal.

Q3

Triangle base 14, height 9: find area.

Q4

Circle radius 7: find area using $(\pi = 22/7)$.

Q5

Trapezium bases 8 and 12, height 5: find area.

LEVEL 1

Q6

Rhombus diagonals 18 and 24: find area.

Q7

Semicircle radius 7: find area and full perimeter using $(\pi = 22/7)$.

Q8

Sector radius 10, angle (90°) : find area.

Q9

Outer radius 10, inner radius 6: find ring area.

Q10

A rectangular garden (30×20) has 2 m outside path. Find path area.

LEVEL 2

Q11

Square diagonal $(12\sqrt{2})$. Find area.

Q12

A room (15×10) has an internal 1 m border. Find border area.

Q13

Circle radius increases by 20%. Find percentage increase in area.

Q14

Rectangle length increases 25%, breadth decreases 20%. Find net area change.

Q15

A square field has perimeter 160 m. Find cost of grassing it at $\text{Rs } 25 / \text{m}^2$.

LEVEL 3 — EXAM PRESSURE

Q16

A square contains an inscribed circle. Square side 14. Find area inside square but outside circle using $(\pi = 22/7)$.

Q17

A circular garden radius 14 has a 3 m wide external path. Find path area using $(\pi = 22/7)$.

Q18

Two identical squares side 6 are joined along one full side. Find combined area and external perimeter.

Q19

A rectangular plot has area 600 m^2 and length:breadth $(=3:2)$. Find dimensions and perimeter.

Q20

A sector of radius 21 has arc length 22. Using $(\pi = 22/7)$, find the central angle and sector area.

ANSWERS

Q1

$$A = 12(8) = \boxed{96}$$

$$P = 2(20) = \boxed{40}$$

Q2

$$A = 81$$

$$d = 9\sqrt{2}$$

$$\boxed{81, 9\sqrt{2}}$$

Q3

$$\frac{12(14)}{9} = \boxed{63}$$

Q4

$$A = \frac{22}{7}(49) = \boxed{154}$$

Q5

$$\frac{12(20)}{5} = \boxed{50}$$

Q6

$$\frac{12(18)(24)}{2} = \boxed{216}$$

Q7

Area:

$$\frac{1}{2} \cdot \frac{22}{7} \cdot 49 = \boxed{77}$$

Perimeter:

$$\pi r + 2r$$

$$= 22 + 14$$

$$= \boxed{36}$$

Q8

$$\frac{14\pi(100)}{2} = \boxed{25\pi}$$

Q9

$$\pi(100 - 36) = \boxed{64\pi}$$

Q10

Outer:

$$34 \times 24$$

Outer area:

$$816$$

Garden area:

$$600$$

Path:

$\boxed{216}$

Q11

$A = \frac{d^2}{2}$

$= \frac{288}{2}$

$= \boxed{144}$

Q12

Total:

150

Inner dimensions:

13×8

Inner area:

104

Border:

$150 - 104$

$= \boxed{46}$

Q13

$2(20) + \frac{20^2}{100}$

$= 40 + 4$

$= \boxed{44\%}$

Q14

$25 - 20 + \frac{25(-20)}{100}$

$= 5 - 5$

$= \boxed{0\%}$

Area remains unchanged.

Q15

Square side:

$160/4 = 40$

Area:

1600

Cost:

$1600(25)$

$= \boxed{\text{\$} 40,000}$

Q16

Square:

$14^2 = 196$

Circle radius:

$$\sqrt{7}$$

Circle area:

$$\sqrt{154}$$

Difference:

$$\sqrt{\boxed{42}}$$

Q17

Outer radius:

$$\sqrt{17}$$

Path:

$$\sqrt{\frac{22}{7}(17^2-14^2)}$$

$$\sqrt{=\frac{22}{7}(289-196)}$$

$$\sqrt{=\frac{22}{7}(93)}$$

$$\sqrt{=\boxed{\frac{2046}{7}\text{ m}^2}}$$

approximately:

$$\sqrt{292.29\text{ m}^2}$$

Q18

Combined area:

$$\sqrt{2(36) = \boxed{72}}$$

Joined figure is:

$$\sqrt{12\times 6}$$

Perimeter:

$$\sqrt{2(12+6) = \boxed{36}}$$

Q19

Let:

$$\sqrt{l=3x,\text{quad } b=2x}$$

Area:

$$\sqrt{6x^2=600}$$

$$\sqrt{x^2=100}$$

$$\sqrt{x=10}$$

Dimensions:

$$\sqrt{30,\ 20}$$

Perimeter:

$$\sqrt{2(50) = \boxed{100}}$$

So:

$$\boxed{30\text{ m} \times 20\text{ m}, P=100\text{ m}}$$

Q20

Arc formula:

$$22 = \frac{\theta}{360}(2\pi r)$$

Using:

$$2\pi r = 132$$

$$22 = \frac{\theta}{360}(132)$$

$$\frac{22}{132} = \frac{\theta}{360}$$

Thus:

$$\theta = 60^\circ$$

Sector area:

$$\frac{1}{2} \times \frac{22}{7} \times 132 \times \frac{60}{360}$$

$$= 138.6$$

$$\boxed{231}$$

CHAPTER 46 STATUS

Covered so far:

- perimeter vs area '
- area-unit conversions '
- rectangle/square '
- triangles '
- parallelogram/rhombus/kite/trapezium '
- circle/semicircle/quadrant '
- arcs/sectors '
- rings/annuli '
- composite figures '
- shaded areas '
- rectangular/circular paths '
- fencing/flooring/tiling '
- dimension percentage changes '
- perimeter/area optimization intuition '
- L0–L3 practice '

But **Chapter 46 is not complete yet.**

We still need the higher-value application layer:

- advanced composite and shaded regions
- nested squares/circles
- tracks and borders
- wire reshaping

- area conservation
- perimeter conservation
- cutting/rearranging figures
- ratio-based mensuration
- cost/wastage questions
- final formula/speed audit

Progress

Completed: 45/76 Current: 0/51 Chapter 46 — 2D Mensuration

NEXT

+Chapter 46 Part 2 — Advanced Composite Areas, Paths, Wire Reshaping, Cutting/Rearrangement, Area Ratios & Final 2D Mensuration Toolkit

USER

next

ASSISTANT

CHAPTER 46 — 2D MENSURATION

PART 2 — ADVANCED COMPOSITE AREAS, NESTED FIGURES, PATHS, WIRE RESHAPING, CUTTING/REARRANGEMENT, RATIOS & COST APPLICATIONS

Status: 0/14 PROGRESS GATE: Medium Placements: High Priority: +Tier 2

This part is less about memorizing new formulas and more about **recognizing what stays constant**.

Main exam strategy:

Draw the figure !' mark dimensions !' identify conserved quantity !' reduce to familiar shapes.

1. COMPOSITE AREA — MASTER RULE

For any irregular-looking figure:

$$\boxed{\text{Required area} = \text{sum of useful pieces} - \text{removed pieces}}$$

Do not search for one giant formula.

Break it.

2. L-SHAPED FIGURE

Suppose a rectangle has a smaller rectangle removed from one corner.

TEXT

% % % % % % % % % % % % % % % %
%
% % % % % % % % % %
%
%
%
% % % % % % % %

Use:

$\boxed{A = \text{large rectangle} - \text{missing rectangle}}$

Usually the fastest method.

3. EXAMPLE — L SHAPE

Outer rectangle:

12×10

Removed rectangle:

5×4

Area:

$120 - 20 = \boxed{100}$

4. PERIMETER OF CUT-OUT FIGURE

Area is easy by subtraction.

Perimeter requires tracing only the exposed boundary.

A cut-out can increase perimeter even though area decreases.

Do not infer perimeter from area.

5. CORNER CUT FROM RECTANGLE — SPECIAL PERIMETER FACT

If a rectangular piece is cut from a **corner**, the total perimeter may remain unchanged.

Why?

Suppose removed corner has sides:

a, b

You lose outer edges:

$a + b$

but create inner edges:

$a + b$

Net change:

$\boxed{0}$

This is a useful placement shortcut.

6. NOTCH CUT FROM MIDDLE OF A SIDE

If a rectangular notch of depth (d) is cut from the middle of one side:

You remove one horizontal segment of width (w) , then add:

- same width (w)
- two vertical sides $(d+d)$

Net perimeter increase:

$$2d$$

Width of notch cancels.

7. NESTED SQUARES

If a square of side (a) contains a smaller square of side (b) :

Border area:

$$a^2 - b^2$$

Factor:

$$(a-b)(a+b)$$

Often faster.

8. EXAMPLE

Outer square:

$$15$$

Inner square:

$$9$$

Border:

$$225 - 81 = 144$$

9. SQUARE INSCRIBED IN CIRCLE

If square side is (a) :

Square diagonal:

$$a\sqrt{2}$$

This equals circle diameter:

$$2r = a\sqrt{2}$$

So:

$$\boxed{r = \frac{a}{\sqrt{2}}}$$

Circle area:

$$\boxed{\frac{\pi a^2}{2}}$$

10. CIRCLE INSCRIBED IN SQUARE

If square side is a :

Diameter of circle:

$$a$$

So:

$$\boxed{r = \frac{a}{2}}$$

Circle area:

$$\boxed{\frac{\pi a^2}{4}}$$

11. AREA RATIO — INSCRIBED CIRCLE IN SQUARE

Square area:

$$a^2$$

Circle area:

$$\frac{\pi a^2}{4}$$

Thus:

$$\boxed{\text{circle} : \text{square} = \pi : 4}$$

And corner region fraction:

$$\boxed{1 - \frac{\pi}{4}}$$

12. AREA RATIO — SQUARE INSCRIBED IN CIRCLE

Square side a .

Circle area:

$$\frac{\pi a^2}{2}$$

Square area:

$$a^2$$

Thus:

$$\boxed{\text{square} : \text{circle} = 2 : \pi}$$

13. NESTED CIRCLE AND SQUARE CHAIN

A common aptitude pattern:

- circle inside square
- square inside another circle

Use:

$$\text{square side} = \text{inner circle diameter}$$

then:

$$\text{square diagonal} = \text{outer circle diameter}$$

Do it one relation at a time.

14. CIRCULAR TRACK

If inner radius:

$$r$$

track width:

$$w$$

then outer radius:

$$r + w$$

Track area:

$$\pi[(r+w)^2 - r^2]$$

Simplify:

$$\pi w(2r + w)$$

15. EXAMPLE — TRACK

Inner radius:

$$20$$

width:

$$5$$

Track area:

$$\pi(25^2 - 20^2)$$

$$= \pi(625 - 400)$$

$$= 225\pi$$

16. RECTANGULAR OUTSIDE PATH— EXPANDED FORM

Path width x around rectangle $(l \times b)$:

$$(l + 2x)(b + 2x) - lb$$

Expand:

$$lb + 2lx + 2bx + 4x^2 - lb$$

$$\boxed{2x(l+b)+4x^2}$$

Or:

$$\boxed{xP+4x^2}$$

where original perimeter:

$$P=2(l+b)$$

Useful shortcut.

17. RECTANGULAR INNER BORDER

$$lb-(l-2x)(b-2x)$$

Expand:

$$\boxed{2x(l+b)-4x^2}$$

Notice:

Outside border:

$$+4x^2$$

Inside border:

$$-4x^2$$

18. THIN-BORDER APPROXIMATION

If x is very small compared with dimensions:

Outer rectangular border area is approximately:

$$\boxed{Px}$$

because $4x^2$ is tiny.

For exact aptitude questions, keep $4x^2$.

19. CIRCULAR THIN TRACK APPROXIMATION

Exact:

$$\pi[(r+w)^2-r^2]$$

$$=2\pi rw+\pi w^2$$

For very small w :

$$\boxed{\text{track area} \approx 2\pi rw}$$

This is circumference $\times$ width.

Again, use exact unless approximation is requested.

20. WIRE RESHAPING — MASTER IDEA

If a wire is bent from one shape into another:

$\boxed{\text{wire length stays constant}}$

So:

$\boxed{\text{original perimeter} = \text{new perimeter/circumference}}$

Area does **not** stay constant.

21. SQUARE WIRE !' CIRCLE

Square side:

a

Wire length:

$4a$

Bent into circle:

$2\pi r = 4a$

Thus:

$\boxed{r = \frac{2a}{\pi}}$

22. EXAMPLE

Square side:

11

Wire length:

44

Using:

$\pi = \frac{22}{7}$

$2\pi r = 44$

$r = 7$

Circle area:

154

$\boxed{154}$

23. CIRCLE WIRE !' SQUARE

Circle radius r .

Circumference:

$2\pi r$

Square side a :

$$4a=2\pi r$$

$$\boxed{a=\frac{\pi r}{2}}$$

24. AREA COMPARISON WITH SAME PERIMETER

For a fixed perimeter:

$$\boxed{\text{circle encloses more area than square}}$$

And among rectangles:

$$\boxed{\text{square gives maximum area}}$$

This helps with option elimination.

25. CUTTING AND REARRANGING

If a sheet is cut and pieces are rearranged without overlap or loss:

$$\boxed{\text{total area stays constant}}$$

Perimeter may change.

This is the opposite of wire reshaping:

- wire reshaping !' perimeter conserved
- sheet rearrangement !' area conserved

26. RECTANGLE REARRANGED INTO SQUARE

Rectangle area:

$$lb$$

If rearranged without loss into square side s :

$$s^2=lb$$

Therefore:

$$\boxed{s=\sqrt{lb}}$$

27. EXAMPLE

Rectangle:

$$16\times 9$$

Area:

$$144$$

Rearranged into square:

$$s=\sqrt{144}$$

$$=\boxed{12}$$

28. MELTING/RESHAPING WARNING

In 2D sheet problems:

$$\boxed{\text{area conserved}}$$

In 3D melting problems later:

$$\boxed{\text{volume conserved}}$$

Do not mix them.

29. RATIO OF AREAS — SAME SHAPE

For similar 2D figures:

If corresponding linear ratio:

$$a:b$$

then area ratio:

$$\boxed{a^2:b^2}$$

This applies to:

- squares
 - circles
 - similar triangles
 - similar rectangles
 - similar polygons
-

30. CIRCLE AREA RATIO

If radii ratio:

$$r_1:r_2=a:b$$

then:

$$\boxed{A_1:A_2=a^2:b^2}$$

Circumference ratio:

$$\boxed{a:b}$$

31. EXAMPLE

Circle radii ratio:

$$3:5$$

Circumference ratio:

$$\boxed{3:5}$$

Area ratio:

$$\boxed{9:25}$$

32. SQUARE PERIMETER RATIO TO AREA RATIO

Perimeters ratio:

$$\boxed{3:4}$$

Since side ratios are also:

$$\boxed{3:4}$$

Area ratio:

$$\boxed{9:16}$$

33. AREA RATIO TO LINEAR RATIO

If similar figure areas ratio:

$$\boxed{64:121}$$

linear ratio:

$$\boxed{8:11}$$

Take square root.

34. EQUAL AREA RECTANGLE DIMENSION CHANGE

Suppose length increases by 25%.

New length multiplier:

$$\boxed{1.25}$$

To keep same area, breadth multiplier must be:

$$\boxed{\frac{1}{1.25}=0.8}$$

So breadth decreases:

$$\boxed{20\%}$$

Classic inverse relation.

35. GENERAL FIXED-AREA RULE

If one dimension increases by $(p\%)$, the required percentage decrease $(q\%)$ in the other dimension satisfies:

$$\left(1+\frac{p}{100}\right)\left(1-\frac{q}{100}\right)=1$$

So:

$$\boxed{q=\frac{100p}{100+p}}$$

36. EXAMPLE

Length increases 20%.

Breadth decrease required:

$$\left[q = \frac{100(20)}{120} \right]$$

$$\left[= \boxed{16\frac{2}{3}\%} \right]$$

37. FIXED PERIMETER RECTANGLE

If:

$$\left[2(l+b) = P \right]$$

then:

$$\left[l+b = \frac{P}{2} \right]$$

For fixed sum, product (lb) is maximum when:

$$\left[l=b \right]$$

So square maximizes area.

38. DIFFERENCE OF AREAS — CONCENTRIC SQUARES

Outer side:

$$\left[a \right]$$

inner side:

$$\left[b \right]$$

Border:

$$\left[a^2 - b^2 \right]$$

Use:

$$\left[\boxed{(a-b)(a+b)} \right]$$

especially if values are close.

39. EXAMPLE

Squares of sides:

$$\left[101, 99 \right]$$

Difference area:

$$\left[101^2 - 99^2 \right]$$

Instead of squaring:

$$\left[(101-99)(101+99) \right]$$

$$\left[2 \cdot 200 \right]$$

$$\boxed{400}$$

40. DIFFERENCE OF CIRCLE AREAS

$$\pi(R^2 - r^2)$$

Use:

$$\pi(R-r)(R+r)$$

Example:

$$R=51, r=49$$

Area difference:

$$\pi(2)(100)$$

$$\boxed{200\pi}$$

41. COST WITH WASTAGE

Suppose required floor area is (A) , but material wastage is $(w\%)$.

Material needed:

$$\boxed{A\left(1+\frac{w}{100}\right)}$$

Then multiply by cost per area.

42. EXAMPLE

Floor:

$$200\text{ m}^2$$

Wastage:

$$5\%$$

Material required:

$$200(1.05) = 210$$

At $1400 / \text{m}^2$:

$$210(400) = \boxed{84000}$$

43. COST WHEN MATERIAL SOLD PER TILE

If:

- tile count = (N)
- price per tile = (c)

Then:

$$\boxed{\text{cost} = Nc}$$

If wastage means extra tiles, round up if necessary.

44. TILES WITH NO CUTTING — HCF CONNECTION

To cover a rectangular floor exactly with the **largest possible square tiles**, tile side is:

$$\boxed{\gcd(l,b)}$$

when dimensions use same units.

Very useful connection to Chapter 5.

45. EXAMPLE

Floor:

$$6\text{ m} \times 4.5\text{ m}$$

Convert to cm:

$$600 \times 450$$

Largest square tile side:

$$\gcd(600,450)=150\text{ cm}$$

So tile side:

$$1.5\text{ m}$$

Number:

$$\frac{6}{1.5} \times \frac{4.5}{1.5}$$

$$4 \times 3$$

$$=\boxed{12}$$

46. SMALLEST SQUARE FROM RECTANGULAR TILES — LCM CONNECTION

If identical rectangular tiles must form the smallest possible square without cutting, side often comes from an LCM-based arrangement.

This depends on whether tiles may rotate, so inspect the problem carefully.

47. NUMBER OF SQUARES ALONG BORDER — CAUTION

If posts/tiles are placed at equal spacing along a closed boundary:

$$\frac{\text{perimeter}}{\text{spacing}}$$

often gives number of intervals, and for a closed loop number of posts equals intervals.

For an open line:

$$\text{posts} = \text{intervals} + 1$$

This is a frequent placement trap.

48. FENCING WITH GATE

If boundary perimeter is (P) , but gate width (g) is left unfenced:

$$\boxed{\text{fence length} = P - g}$$

For multiple gates, subtract total opening length.

49. EXAMPLE

Rectangular plot:

$$50 \times 30$$

Perimeter:

$$160$$

Gate:

$$4$$

Fence required:

$$156$$

At $1200 / \text{m}$:

$$\boxed{131200}$$

50. MULTIPLE ROUNDS OF FENCING

If 3 strands of wire are required around the same field:

$$\boxed{\text{wire} = 3P}$$

Then adjust for gates if relevant.

51. PATHCOST

For path:

1. calculate path area
2. multiply by cost per square unit

Do not multiply cost by path perimeter unless cost is per metre of boundary.

52. SHADING WITH QUADRANTS

Four quarter-circles of radius (r) together have area:

$$4 \left(\frac{1}{4} \pi r^2 \right)$$

$$\boxed{\pi r^2}$$

Recognizing combined fractions saves time.

53. SHADING WITH SEMICIRCLES

Two semicircles of same radius form one full circle in area:

$$2 \left(\frac{1}{2} \pi r^2 \right) = \boxed{\pi r^2}$$

Even if they are positioned differently.

54. CORNERS OF SQUARE WITH QUADRANTS

If four quarter-circles of radius $(a/2)$ are drawn from square corners, total quadrant area equals one full circle of radius $(a/2)$:

$$\boxed{\frac{\pi a^2}{4}}$$

This often simplifies shaded-region questions.

55. SEMICIRCLE ON RECTANGLE SIDES — PERIMETER

If a semicircle replaces a straight side of length (d) , then exposed boundary contribution changes from:

$$d$$

to:

$$\frac{\pi d}{2}$$

because radius:

$$d/2$$

semicircle arc:

$$\pi r = \frac{\pi d}{2}$$

56. TRACK AROUND STADIUM SHAPE

A stadium shape often consists of:

- rectangle length (L)
- two semicircles of radius (r)

Two semicircles form one full circle.

Perimeter:

$$\boxed{2L + 2\pi r}$$

Area:

$$\boxed{2rL + \pi r^2}$$

because rectangle width is diameter $(2r)$.

57. SIMPLE STADIUM DIAGRAM

TEXT



Think:

$\backslash \text{rectangle} + \text{one full circle} \backslash$

for area.

58. EXAMPLE — STADIUM

Straight portion length:

$\backslash 20 \backslash$

Radius of semicircular ends:

$\backslash 5 \backslash$

Area:

$\backslash 20(10) + 25\pi \backslash$

$\backslash = \boxed{200 + 25\pi} \backslash$

Perimeter:

$\backslash 40 + 10\pi \backslash$

$\backslash = \boxed{40 + 10\pi} \backslash$

59. ARC-TO-SECTOR RELATION

For same circle:

$\backslash \frac{\text{arc length}}{\text{circumference}} = \frac{\text{sector area}}{\text{circle area}} = \frac{\theta}{360} \backslash$

Thus:

$\backslash \boxed{\frac{L}{2\pi r} = \frac{A_s}{\pi r^2}} \backslash$

Cancel:

$\backslash \boxed{A_s = \frac{1}{2}rL} \backslash$

Very useful if sector arc length $\backslash(L)$ is directly given.

60. HIGH-VALUE SECTOR SHORTCUT

If radius $\backslash(r)$ and arc length $\backslash(L)$ are known:

$\backslash \boxed{A_{\text{sector}} = \frac{1}{2}rL} \backslash$

No need to find angle first.

61. EXAMPLE

Radius:

$$\sqrt{14}$$

arc length:

$$\sqrt{11}$$

Sector area:

$$\sqrt{\frac{12(14)(11)}{2}}$$

$$\sqrt{= \boxed{77}}$$

62. PERIMETER SCALING

For similar figures with scale factor (k) :

$$\sqrt{\boxed{P_{\text{new}} = kP_{\text{old}}}}$$

Area:

$$\sqrt{\boxed{A_{\text{new}} = k^2 A_{\text{old}}}}$$

63. EXAMPLE

A map figure is enlarged by factor:

$$\sqrt{1.5}$$

Perimeter increases by:

$$\sqrt{50\%}$$

Area multiplier:

$$\sqrt{2.25}$$

So area increases by:

$$\sqrt{\boxed{125\%}}$$

64. GATE-STYLE QUESTION 1

A square wire of side 14 is bent into a circle.

Wire length:

$$\sqrt{56}$$

Using $(\pi = 22/7)$:

$$\sqrt{2\pi r = 56}$$

$$\sqrt{r = \frac{56 \cdot 7}{44}}$$

$$\sqrt{= \boxed{\frac{98}{11}}}$$

Circle area:

$$\sqrt{\pi r^2}$$

can be evaluated if requested.

65. GATE-STYLE QUESTION 2

Rectangle:

$$\sqrt{20 \times 12}$$

Outside path width:

$$\sqrt{1}$$

Path area:

$$\sqrt{22 \cdot 14 - 20 \cdot 12}$$

$$\sqrt{308 - 240}$$

$$\sqrt{= \boxed{68}}$$

66. GATE-STYLE QUESTION 3

A rectangle of area 300 is rearranged into a square without loss.

Square side:

$$\sqrt{\boxed{10\sqrt{3}}}$$

because:

$$\sqrt{s = \sqrt{300}}$$

67. GATE-STYLE QUESTION 4

Two similar circles have areas:

$$\sqrt{49\pi, \text{quad} 121\pi}$$

Radius ratio:

$$\sqrt{\sqrt{49:121}}$$

$$\sqrt{= \boxed{7:11}}$$

68. PLACEMENT-STYLE QUESTION 1

A rectangle length increases 50%. What decrease in breadth keeps area constant?

$$\sqrt{q = \frac{100(50)}{150}}$$

$$\sqrt{= \boxed{33\frac{1}{3}\%}}$$

69. PLACEMENT-STYLE QUESTION 2

Floor:

$$\backslash[420\text{ cm}\backslashtimes300\text{ cm}\backslash]$$

Largest square tile:

$$\backslash[\gcd(420,300)=60 \backslash]$$

Tile:

$$\backslash[60\backslashtimes60 \backslash]$$

Number:

$$\backslash[7\backslashcdot5 \backslash]$$

$$\backslash[=\boxed{35} \backslash]$$

70. PLACEMENT-STYLE QUESTION 3

Circular track:

Inner radius:

$$\backslash[7 \backslash]$$

width:

$$\backslash[7 \backslash]$$

Track area:

$$\backslash[\backslashpi(14^2-7^2) \backslash]$$

$$\backslash[=\backslashpi(196-49) \backslash]$$

$$\backslash[=\boxed{147\backslashpi} \backslash]$$

71. PLACEMENT-STYLE QUESTION 4

A 10 m × 8 m floor requires 8% extra tiles for wastage.

Required area:

$$\backslash[80(1.08) \backslash]$$

$$\backslash[=\boxed{86.4\text{ m}^2} \backslash]$$

72. ADVANCED QUESTION — INSCRIBED FIGURES

A circle is inscribed in a square of side 20.

Circle radius:

$$\backslash[10 \backslash]$$

Area outside circle:

$$\backslash[400-100\backslashpi \backslash]$$

$$\backslash[=\boxed{100(4-\backslashpi)} \backslash]$$

73. ADVANCED QUESTION — WIRE RESHAPING

A circular wire radius 14 is reshaped into a square.

Circumference:

$$2 \cdot \frac{22}{7} \cdot 14 = 88$$

Square side:

$$88/4 = 22$$

Area:

$$\boxed{484}$$

74. ADVANCED QUESTION — BORDER

A square of side 30 has a uniform inside border of width 2.

Inner side:

$$30 - 4 = 26$$

Border area:

$$900 - 676$$

$$= \boxed{224}$$

75. ADVANCED QUESTION — SECTOR FROM ARC

Radius:

$$21$$

Arc length:

$$22$$

Sector area:

$$\frac{1}{2}(21)(22)$$

$$= \boxed{231}$$

No angle calculation required.

76. COMMON TRAPS

& pTRAP 1 — Wire reshaping

Perimeter conserved, not area.

& pTRAP 2 — Sheet rearranging

Area conserved, not perimeter.

& pTRAP 3 — Largest square tile

Use HCF after converting dimensions to same units.

& pTRAP 4 — Wastage

Add wastage to required material, not to dimensions.

& pTRAP 5 — Border width

Subtract/add twice the width in each dimension.

& pTRAP 6 — Same shape area ratio

Square the linear ratio.

& pTRAP 7 — Sector with arc length

Use:

$$\left[\frac{1}{2} r L \right]$$

for speed.

77. MUST-MEMORIZE CONSERVATION RULES

Wire reshaped

$$\left[\text{perimeter/circumference conserved} \right]$$

Sheet cut and rearranged

$$\left[\text{area conserved} \right]$$

78. MUST-MEMORIZE BORDER FORMULAS

Outside rectangle:

$$\left[(l+2x)(b+2x) - lb \right]$$

Inside rectangle:

$$\left[lb - (l-2x)(b-2x) \right]$$

Circular track:

$$\left[\pi [(r+x)^2 - r^2] \right]$$

79. MUST-MEMORIZE SIMILAR-FIGURE SCALING

If linear ratio $(=k)$:

$$\left[\text{perimeter ratio} = k \right]$$

$$\left[\text{area ratio} = k^2 \right]$$

80. FAST EXAM DECISION TREE

Wire?

Equate perimeters.

Cut/rearranged sheet?

Equate areas.

Largest square tile?

HCF.

Border/path?

Outer minus inner.

Similar figures?

Square linear ratio for area.

Arc length + radius?

Sector area:

$$\frac{1}{2} \times \text{arc length} \times \text{radius}$$

Area fixed and one dimension changes?

Use inverse multiplier.

Irregular shaded figure?

Decompose instead of inventing formulas.

PRACTICE — LEVEL 0

Q1

Outer square side 12, inner square side 8. Find border area.

Q2

Circle radii are 4 and 10. Find area ratio.

Q3

Square side 5 is enlarged by factor 3. By what factor does area increase?

Q4

A rectangular sheet (18×8) is rearranged into a square. Find side.

Q5

A circle radius 7 has arc length 11. Find sector area.

LEVEL 1

Q6

Rectangle (20×15) has 2 m outside path. Find path area.

Q7

A square wire side 11 is bent into a circle. Using $(\pi = 22/7)$, find radius.

Q8

Rectangle length increases 25%. Find breadth decrease needed to keep area constant.

Q9

Find largest square tile that exactly covers a $(480 \text{ cm} \times 360 \text{ cm})$ floor.

Q10

A circular track has inner radius 10 and width 4. Find area.

LEVEL 2

Q11

A square has an inscribed circle. Square side 18. Find area outside circle.

Q12

A circular wire radius 21 is reshaped into a square. Using $(\pi = 22/7)$, find square area.

Q13

A room (12×10) has 10% tile wastage. Find total tile area to purchase.

Q14

A square side 40 has an internal border width 3. Find border area.

Q15

A stadium consists of a rectangle 30 m long and 14 m wide with semicircular ends of radius 7. Find area and perimeter.

LEVEL 3 — EXAM PRESSURE

Q16

A rectangle has perimeter 100. Find maximum possible area.

Q17

A circle and square have equal perimeter (44) . Using $(\pi = 22/7)$, compare their areas.

Q18

A rectangular garden (40×30) has a uniform outside path. Path area is 456. Find path width.

Q19

A square field side 50 has an inner path of uniform width x . If path area is 900, find x .

Q20

Two similar figures have perimeters in ratio $(5:8)$. Their smaller area is 225. Find larger area.

ANSWERS

Q1

$$12^2 - 8^2 = 144 - 64$$

$$= \boxed{80}$$

Q2

$$4^2 : 10^2 = \boxed{4:25}$$

Q3

Area scale:

$$3^2 = \boxed{9}$$

Q4

Area:

$$18(8) = 144$$

Square side:

$$\boxed{12}$$

Q5

$$A = \frac{1}{2}rL$$

$$= \frac{1}{2}(7)(11)$$

$$= \boxed{\frac{77}{2}}$$

Q6

Outer dimensions:

$$24 \times 19$$

Path:

$$456 - 300$$

$$= \boxed{156}$$

Q7

Wire length:

$$44$$

$$2 \cdot \frac{22}{7} r = 44$$

$$\boxed{r=7}$$

Q8

$$q = \frac{100(25)}{125}$$

$$= \boxed{20\%}$$

Q9

$$\text{lcm}(480, 360) = \boxed{120 \text{ cm}}$$

Number tiles:

$$4 \times 3 = \boxed{12}$$

Q10

Outer radius:

$$14$$

$$A = \pi(196 - 100)$$

$$= \boxed{96\pi}$$

Q11

Circle radius:

$$9$$

Square area:

$$324$$

Circle:

$$81\pi$$

Difference:

$$\boxed{324 - 81\pi}$$

Q12

Circumference:

$$2 \times \frac{22}{7} \times 21 = 132$$

Square side:

$$33$$

Area:

$$\boxed{1089}$$

Q13

Floor:

$$120$$

With 10% wastage:

$$120(1.10)$$

$$= \boxed{132}$$

Q14

Inner side:

$$\sqrt{40-6}=34 \sqrt{}$$

Border:

$$\sqrt{1600-1156} \sqrt{}$$

$$\sqrt{=\boxed{444}} \sqrt{}$$

Q15

Two semicircles form one circle.

Area:

$$\sqrt{30(14)+\pi(7^2)} \sqrt{}$$

$$\sqrt{=\boxed{420+49\pi}} \sqrt{}$$

Perimeter:

Two straight lengths:

$$\sqrt{60} \sqrt{}$$

plus full circumference:

$$\sqrt{14\pi} \sqrt{}$$

$$\sqrt{\boxed{60+14\pi}} \sqrt{}$$

Q16

Maximum when square.

Perimeter:

$$\sqrt{100} \sqrt{}$$

side:

$$\sqrt{25} \sqrt{}$$

Area:

$$\sqrt{\boxed{625}} \sqrt{}$$

Q17

Square:

$$\sqrt{4a=44} \sqrt{}$$

$$\sqrt{a=11} \sqrt{}$$

Area:

$$\sqrt{121} \sqrt{}$$

Circle:

$$\sqrt{2\pi r=44} \sqrt{}$$

With $\sqrt{(\pi=22/7)}$:

$$\sqrt{r=7} \sqrt{}$$

Area:

$$\sqrt{154}$$

Thus circle exceeds square by:

$$\sqrt{33}$$

$$\boxed{\text{Circle area } 154, \text{ square area } 121}$$

Q18

Outer dimensions:

$$(40+2x)(30+2x)$$

Path:

$$(40+2x)(30+2x)-1200=456$$

$$1200+140x+4x^2-1200=456$$

$$4x^2+140x-456=0$$

Divide 4:

$$x^2+35x-114=0$$

$$(x+38)(x-3)=0$$

Positive:

$$\boxed{x=3}$$

Q19

Inner side:

$$50-2x$$

$$2500-(50-2x)^2=900$$

$$(50-2x)^2=1600$$

Positive length:

$$50-2x=40$$

$$2x=10$$

$$\boxed{x=5}$$

Q20

Linear/perimeter ratio:

$$5:8$$

Area ratio:

$$25:64$$

$$\frac{225}{A}=\frac{25}{64}$$

$$A=225\cdot\frac{64}{25}$$

$$=9\cdot64$$

$$\boxed{=576}$$

' CHAPTER 46 COMPLETE

We have now covered:

- all core 2D area/perimeter formulas '
- composite figures '
- shaded regions '
- square-circle nesting '
- rectangular/circular paths '
- fencing/flooring/tiling '
- HCF tile problems '
- wire reshaping '
- area-conservation rearrangement '
- similar-figure ratios '
- percentage dimension changes '
- fixed-area/fixed-perimeter optimization '
- cost and wastage '
- stadium shapes '
- arc-to-sector shortcut '
- advanced L0–L3 practice '

Progress

Completed: 46/76

Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

NEXT

+Chapter 47 — 3D Mensuration

Next we will cover:

- cube and cuboid
- prism and cylinder
- cone
- sphere and hemisphere
- frustum
- curved/lateral vs total surface area
- volume
- space diagonals
- hollow solids
- melting/recasting and volume conservation
- water displacement
- percentage scaling
- capacity/unit conversion
- painting/cost problems

- GATE and placement shortcuts.

USER

next

ASSISTANT

CHAPTER 47 — 3D MENSURATION

PART 1 — CUBE, CUBOID, PRISM, CYLINDER, CONE, SPHERE & HEMISPHERE

Status: $\emptyset$ = Not in PROGRESS GATE: Medium Placements: High Priority: +Tier 2

3D mensuration is mainly about three quantities:

- **Volume** !' space occupied
- **Lateral/Curved Surface Area** !' side surface only
- **Total Surface Area** !' all exposed surfaces

The biggest exam trap is:

Using TSA when the question wants only the curved/lateral area, or forgetting a base/top.

1. 3D UNIT BASICS

Length:

$\text{[} \text{cm}, \text{ m]}$

Area:

$\text{[} \text{cm}^2, \text{ m}^2 \text{]}$

Volume:

$\text{[} \boxed{\text{cm}^3, \text{ m}^3} \text{]}$

2. VOLUME UNIT CONVERSION

Since:

$\text{[} 1 \text{ m} = 100 \text{ cm]}$

then:

$\text{[} 1 \text{ m}^3 = 100^3 \text{ cm}^3 \text{]}$

$\text{[} \boxed{1 \text{ m}^3 = 10^6 \text{ cm}^3} \text{]}$

Very important.

3. LITRE CONVERSIONS

$$\boxed{1 \text{ litre} = 1000 \text{ cm}^3}$$

$$\boxed{1 \text{ m}^3 = 1000 \text{ litres}}$$

$$\boxed{1 \text{ mL} = 1 \text{ cm}^3}$$

4. CUBOID

Dimensions:

- length (l)
- breadth (b)
- height (h)

Volume:

$$\boxed{V = lbh}$$

5. CUBOID TOTAL SURFACE AREA

There are three pairs of equal rectangular faces.

$$\boxed{TSA = 2(lb + bh + hl)}$$

6. CUBOID LATERAL SURFACE AREA

Lateral area excludes top and bottom.

$$\boxed{LSA = 2h(l + b)}$$

Think:

$$\boxed{\text{perimeter of base} \times h}$$

7. CUBOID SPACE DIAGONAL

Using 3D Pythagoras:

$$\boxed{d = \sqrt{l^2 + b^2 + h^2}}$$

8. EXAMPLE — CUBOID

$$l = 3, \text{quad } b = 4, \text{quad } h = 12$$

Volume:

$$3 \times 4 \times 12 = \boxed{144}$$

Space diagonal:

$$\sqrt{9 + 16 + 144} = \sqrt{169}$$

$$= \boxed{13}$$

9. CUBE

All edges equal:

$$\boxed{a}$$

Volume:

$$\boxed{V=a^3}$$

10. CUBE SURFACE AREAS

Each face area:

$$\boxed{a^2}$$

6 faces:

$$\boxed{TSA=6a^2}$$

Four side faces:

$$\boxed{LSA=4a^2}$$

11. CUBE DIAGONALS

Face diagonal:

$$\boxed{a\sqrt{2}}$$

Space/body diagonal:

$$\boxed{a\sqrt{3}}$$

MUST MEMORIZE

12. EXAMPLE

Cube side:

$$\boxed{5}$$

Volume:

$$\boxed{125}$$

TSA:

$$\boxed{150}$$

Space diagonal:

$$\boxed{5\sqrt{3}}$$

$$\boxed{V=125,\ TSA=150,\ d=5\sqrt{3}}$$

13. GENERAL PRISM

A prism has the same cross-section throughout its length/height.

Volume:

$$\boxed{V = (\text{base area})(\text{height})}$$

14. PRISM LATERAL AREA

For a right prism:

$$\boxed{LSA = (\text{perimeter of base})(\text{height})}$$

Then:

$$\boxed{TSA = LSA + 2(\text{base area})}$$

15. TRIANGULAR PRISM

If triangular base area is (A) and prism length (L) :

$$\boxed{V = AL}$$

If triangle sides are (a, b, c) :

$$\boxed{LSA = (a + b + c)L}$$

16. CYLINDER

A cylinder can be thought of as a circular prism.

Radius:

$$r$$

Height:

$$h$$

Volume:

$$\boxed{V = \pi r^2 h}$$

17. CYLINDER CURVED SURFACE AREA

Imagine unrolling curved surface into a rectangle.

Length:

$$2\pi r$$

Height:

$$h$$

Thus:

$$\boxed{CSA = 2\pi rh}$$

18. CYLINDER TOTAL SURFACE AREA

Curved area + two circular bases:

$$2\pi rh + 2\pi r^2$$

Factor:

$$\boxed{TSA = 2\pi r(h+r)}$$

19. OPEN CYLINDER

If cylinder is open at top:

$$\boxed{\text{surface area} = 2\pi rh + \pi r^2}$$

If open at both ends:

$$\boxed{2\pi rh}$$

Read wording carefully.

20. EXAMPLE — CYLINDER

$$r=7, \text{quad } h=10$$

Volume:

$$\pi(7^2)(10) = 490\pi$$

Using:

$$\pi = \frac{22}{7}$$

$$V = 1540$$

CSA:

$$2\pi(7)(10) = 140\pi = 440$$

$$\boxed{V=1540, \text{ CSA}=440}$$

21. CONE

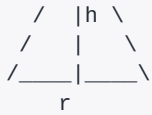
A right circular cone has:

- radius (r)
- vertical height (h)
- slant height (l)

Diagram relation:

TEXT





Slant height:

$$l = \sqrt{r^2 + h^2}$$

22. CONE VOLUME

$$V = \frac{1}{3}\pi r^2 h$$

A cone with same base and height as cylinder has:

$$\frac{1}{3}$$

of cylinder's volume.

23. CONE CURVED SURFACE AREA

$$CSA = \pi r l$$

24. CONE TOTAL SURFACE AREA

Add base:

$$\pi r l + \pi r^2$$

$$TSA = \pi r (l + r)$$

25. EXAMPLE — CONE

$$r = 3, \quad h = 4$$

Slant height:

$$l = 5$$

Volume:

$$\frac{1}{3}\pi(9)(4) = 12\pi$$

CSA:

$$\pi(3)(5) = 15\pi$$

TSA:

$$15\pi + 9\pi = 24\pi$$

26. CYLINDER VS CONE

Same radius and height:

$$V_{\text{cylinder}} = \pi r^2 h$$

$$V_{\text{cone}} = \frac{1}{3} \pi r^2 h$$

Thus:

$$\boxed{V_{\text{cylinder}} : V_{\text{cone}} = 3 : 1}$$

27. SPHERE

Radius (r) .

Volume:

$$\boxed{V = \frac{4}{3} \pi r^3}$$

Surface area:

$$\boxed{SA = 4 \pi r^2}$$

No distinction between curved and total surface area for a sphere.

28. EXAMPLE — SPHERE

$$r = 3$$

Volume:

$$\frac{4}{3} \pi (27) = \boxed{36 \pi}$$

Surface area:

$$4 \pi (9) = \boxed{36 \pi}$$

Interesting: numerically equal for $(r=3)$, but units differ.

29. HEMISPHERE

Half a sphere.

Volume:

$$\boxed{V = \frac{2}{3} \pi r^3}$$

30. HEMISPHERE CURVED SURFACE AREA

Half sphere surface:

$$\boxed{CSA = 2 \pi r^2}$$

31. HEMISPHERE TOTAL SURFACE AREA

Curved area + circular base:

$$\boxed{2 \pi r^2 + \pi r^2}$$

$$\boxed{TSA=3\pi r^2}$$

Very common trap.

32. EXAMPLE

Hemisphere radius:

$$\sqrt{7}$$

CSA:

$$2\pi(49) = 98\pi$$

TSA:

$$147\pi$$

$$\boxed{CSA=98\pi, TSA=147\pi}$$

33. COMPARISON — SPHERE VS HEMISPHERE

Same radius:

Sphere volume:

$$\frac{4}{3}\pi r^3$$

Hemisphere volume:

$$\frac{2}{3}\pi r^3$$

Ratio:

$$\boxed{2:1}$$

Sphere surface area:

$$4\pi r^2$$

Hemisphere curved area:

$$2\pi r^2$$

Again:

$$\boxed{2:1}$$

34. HOLLOW CYLINDER / PIPE — BASIC

Outer radius:

$$R$$

Inner radius:

$$r$$

Length:

$$h$$

Material volume:

$$\boxed{V = \pi h(R^2 - r^2)}$$

Factor:

$$\boxed{\pi h(R-r)(R+r)}$$

35. EXAMPLE — PIPE MATERIAL

$$R=5, \quad r=3, \quad h=10$$

Volume:

$$\pi(10)(25-9)$$

$$= \boxed{160\pi}$$

36. CAPACITY VS MATERIAL

For a hollow container:

- capacity uses **inner dimensions**
- material volume uses outer volume minus inner volume

Do not mix these.

37. OPEN-TOP TANK

For cuboidal tank open at top:

Surface area of material:

$$\boxed{lb + 2lh + 2bh}$$

No top face.

38. CLOSED CUBOIDAL BOX

Surface area:

$$\boxed{2(lb + bh + hl)}$$

39. ROOM PAINTING — FOUR WALLS

For room dimensions (l, b, h) :

Area of four walls:

$$\boxed{2h(l+b)}$$

This is cuboid lateral surface area.

If painting ceiling too:

$$\boxed{2h(l+b) + lb}$$

40. ROOM PAINTING — DOORS/WINDOWS

Net paint area:

$$\boxed{\text{wall area} - \text{door/window area}}$$

Then multiply by rate per square metre.

41. EXAMPLE — ROOM WALLS

Room:

$$8 \times 6 \times 4$$

Four-wall area:

$$2(4)(8+6)$$

$$= 8(14)$$

$$= \boxed{112}$$

42. WATER TANK CAPACITY

Cuboidal tank:

$$V = lbh$$

If dimensions in metres:

$$1 \text{ m}^3 = 1000 \text{ L}$$

So:

$$\boxed{\text{capacity in litres} = 1000lbh}$$

43. EXAMPLE

Tank:

$$2 \text{ m} \times 1.5 \text{ m} \times 1 \text{ m}$$

Volume:

$$3 \text{ m}^3$$

Capacity:

$$\boxed{3000 \text{ L}}$$

44. CYLINDRICAL TANK CAPACITY

$$\boxed{V = \pi r^2 h}$$

Convert cubic units to litres if needed.

45. SCALE FACTOR IN 3D

If every linear dimension multiplies by (k) :

Surface area scales as:

$$\boxed{k^2}$$

Volume scales as:

$$\boxed{k^3}$$

MUST MEMORIZE

46. EXAMPLE

Cube side doubles:

$$k=2$$

Surface area multiplies by:

$$2^2=4$$

Volume multiplies by:

$$2^3=8$$

So:

$$\boxed{\text{SA } 4\text{ times, Volume } 8\text{ times}}$$

47. PERCENTAGE CHANGE IN CUBE VOLUME

If side increases by $(p\%)$:

Volume multiplier:

$$\left(1+\frac{p}{100}\right)^3$$

For small/manual calculation, expand:

$$\boxed{\Delta V = 3p+\frac{3p^2}{100}+\frac{p^3}{1000}}$$

for increase (p) .

48. EXAMPLE

Cube side increases 10%.

Volume multiplier:

$$1.1^3=1.331$$

Increase:

$$\boxed{33.1\%}$$

49. CYLINDER DIMENSION CHANGES

$$V = \pi r^2 h$$

So if radius changes by $(a\%)$ and height by $(b\%)$, use multipliers:

$$\boxed{V_{\text{new}}} = V(1+a/100)^2(1+b/100)$$

signed values if decrease.

50. EXAMPLE

Radius increases 10%, height decreases 20%.

Volume multiplier:

$$1.1^2(0.8)$$

$$= 1.21(0.8)$$

$$= 0.968$$

Thus:

$$\boxed{3.2\% \text{ decrease}}$$

51. SPHERE RADIUS CHANGE

Surface area:

$$\propto r^2$$

Volume:

$$\propto r^3$$

If radius doubles:

$$\boxed{SA \times 4}$$

$$\boxed{V \times 8}$$

52. SAME BASE + HEIGHT — PRISM/CYLINDER/CONE

Prism/cylinder volume:

$$A_{\text{base}} h$$

Cone/pyramid:

$$\frac{1}{3} A_{\text{base}} h$$

Thus same base and height:

$$\boxed{\text{prism/cone volume} = 3:1}$$

53. GATE-STYLE QUESTION 1

Cube volume:

$$\sqrt[3]{343}$$

Side:

$$\sqrt[3]{a} = \sqrt[3]{343} = 7$$

TSA:

$$6(49)$$

$$\boxed{294}$$

54. GATE-STYLE QUESTION 2

Cuboid:

$$6 \times 8 \times 24$$

Space diagonal:

$$\sqrt{36+64+576}$$

$$\sqrt{676}$$

$$\boxed{26}$$

55. GATE-STYLE QUESTION 3

Cylinder:

$$r=3, \text{quad } h=10$$

Volume:

$$\boxed{90\pi}$$

CSA:

$$\boxed{60\pi}$$

56. GATE-STYLE QUESTION 4

Cone:

$$r=5, \text{quad } h=12$$

Slant:

$$l=13$$

CSA:

$$\pi(5)(13)$$

$$\boxed{65\pi}$$

57. PLACEMENT-STYLE QUESTION 1

Cube side:

$$\sqrt[12]{}$$

Volume:

$$12^3 = \boxed{1728}$$

Surface area:

$$6(144) = \boxed{864}$$

58. PLACEMENT-STYLE QUESTION 2

Cylinder radius 7, height 20.

Using:

$$\pi = 22/7$$

Volume:

$$\frac{22}{7}(7)(20)$$

$$= \boxed{3080}$$

59. PLACEMENT-STYLE QUESTION 3

Sphere radius 7.

Surface area:

$$4\pi(7)^2$$

$$= \boxed{616\pi}$$

60. PLACEMENT-STYLE QUESTION 4

Hemisphere radius 6.

TSA:

$$3\pi(6)^2$$

$$= \boxed{108\pi}$$

CSA:

$$72\pi$$

61. ADVANCED QUESTION — CUBOID

A cuboid has volume:

$$960$$

and:

$$l=12, \text{quad } b=8$$

Then:

$$\frac{1}{2} \pi r^2 h = 960$$

$$\pi r^2 h = 960$$

$$\boxed{h=10}$$

62. ADVANCED QUESTION — CONE

Cone volume:

$$\frac{1}{3} \pi r^2 h$$

Radius:

$$r$$

$$\frac{1}{3} \pi (25) h = 100\pi$$

$$25h = 300$$

$$\boxed{h=12}$$

Slant:

$$\boxed{13}$$

63. ADVANCED QUESTION — HOLLOW PIPE

Outer radius:

$$10$$

inner radius:

$$8$$

length:

$$20$$

Material volume:

$$\pi (20)(100-64)$$

$$= \boxed{720\pi}$$

64. ADVANCED QUESTION — CAPACITY

Cylinder:

$$r=0.5 \text{ m}, \quad h=2 \text{ m}$$

Volume:

$$\pi (0.25)(2) = 0.5\pi \text{ m}^3$$

Capacity:

$$\boxed{500\pi \text{ L}}$$

approximately:

L

65. COMMON TRAPS

& pTRAP 1 — TSA vs CSA

Cylinder:

$$CSA = 2\pi rh$$

$$TSA = 2\pi r(h+r)$$

& pTRAP 2 — Hemisphere

$$CSA = 2\pi r^2$$

$$TSA = 3\pi r^2$$

& pTRAP 3 — Cone

CSA uses slant height (l) , not vertical height (h) .

$$CSA = \pi rl$$

& pTRAP 4 — Capacity

Use inner dimensions.

& pTRAP 5 — Volume unit conversion

$$1 \text{ m}^3 = 10^6 \text{ cm}^3$$

not $(1000 \text{ cm})^3$.

& pTRAP 6 — Percentage scaling

Length (k) !' area (k^2) !' volume (k^3) .

66. MUST-MEMORIZE CUBE/CUBOID

Cube:

$$\boxed{V = a^3}$$

$$\boxed{TSA = 6a^2}$$

$$\boxed{d = a\sqrt{3}}$$

Cuboid:

$$\boxed{V = lbh}$$

$$\boxed{TSA = 2(lb + bh + hl)}$$

$$\boxed{d = \sqrt{l^2 + b^2 + h^2}}$$

67. MUST-MEMORIZE CYLINDER

$$\boxed{V = \pi r^2 h}$$

$$\boxed{CSA = 2\pi rh}$$

$$\boxed{TSA = 2\pi r(h+r)}$$

68. MUST-MEMORIZE CONE

$$\boxed{l = \sqrt{r^2 + h^2}}$$

$$\boxed{V = \frac{1}{3}\pi r^2 h}$$

$$\boxed{CSA = \pi rl}$$

$$\boxed{TSA = \pi r(l+r)}$$

69. MUST-MEMORIZE SPHERE/HEMISPHERE

Sphere:

$$\boxed{V = \frac{4}{3}\pi r^3}$$

$$\boxed{SA = 4\pi r^2}$$

Hemisphere:

$$\boxed{V = \frac{2}{3}\pi r^3}$$

$$\boxed{CSA = 2\pi r^2}$$

$$\boxed{TSA = 3\pi r^2}$$

70. FAST EXAM DECISION TREE

Box/tank?

Cuboid.

Circular tank?

Cylinder.

Ice-cream cone?

Cone.

Ball?

Sphere.

Bowl?

Usually hemisphere.

Open container?

Remove missing base/top from TSA.

Pipe?

Outer volume - inner volume.

Painting sides only?

CSA/LSA.

Capacity?

Use volume and inner dimensions.

Similar solids?

Volume ratio = cube of linear ratio.

PRACTICE — LEVEL 0

Q1

Cube side 4. Find volume and TSA.

Q2

Cuboid $(3 \times 5 \times 8)$. Find volume.

Q3

Cylinder $(r=3, h=7)$. Find volume.

Q4

Cone $(r=3, h=4)$. Find slant height.

Q5

Sphere radius 2. Find surface area.

LEVEL 1

Q6

Cuboid $(6 \times 8 \times 24)$. Find space diagonal.

Q7

Cube volume is 1000. Find side.

Q8

Cylinder radius 5, height 12. Find CSA.

Q9

Cone radius 7, height 24. Find slant height and CSA.

Q10

Hemisphere radius 7. Find TSA.

LEVEL 2

Q11

A cylindrical tank has radius 2 m and height 3 m. Find capacity in litres.

Q12

Cube side increases by 20%. Find percentage increase in volume.

Q13

Sphere radius ratio is $\sqrt{2:5}$. Find volume ratio.

Q14

A pipe has outer radius 6, inner radius 4, length 10. Find material volume.

Q15

A room is $\sqrt{10 \times 8 \times 4}$. Find area of four walls.

LEVEL 3 — EXAM PRESSURE

Q16

Cylinder and cone have same radius and height. Cylinder volume is 330 cm^3 . Find cone volume.

Q17

A cuboid has dimensions in ratio $\sqrt{2:3:4}$ and volume 648. Find dimensions.

Q18

A cone has volume $\sqrt{96\pi}$, radius 4. Find height and slant height.

Q19

A sphere's radius increases by 50%. By what percentage does volume increase?

Q20

A closed cylindrical container has radius 7 and height 10. Find total surface area using $\sqrt{\pi=22/7}$.

ANSWERS

Q1

$$\sqrt{V=4^3=\boxed{64}} \sqrt{}$$

$$\sqrt{TSA=6(16)=\boxed{96}} \sqrt{}$$

Q2

$$\sqrt{3 \cdot 5 \cdot 8 = \boxed{120}} \sqrt{}$$

Q3

$$\sqrt{\pi(9)(7) = \boxed{63\pi}} \sqrt{}$$

Q4

$$\sqrt{9+16} = \boxed{5}$$

Q5

$$4\pi(4) = \boxed{16\pi}$$

Q6

$$\sqrt{36+64+576} = \boxed{26}$$

Q7

$$a = \sqrt[3]{1000} = \boxed{10}$$

Q8

$$CSA = 2\pi(5)(12)$$

$$= \boxed{120\pi}$$

Q9

$$l = \sqrt{49+576} = 25$$

$$CSA = \pi(7)(25)$$

$$= \boxed{175\pi}$$

Q10

$$TSA = 3\pi(49)$$

$$= \boxed{147\pi}$$

Q11

$$V = \pi(2^2)(3) = 12\pi \text{ m}^3$$

Capacity:

$$\boxed{12000\pi \text{ L}}$$

approximately:

$$\boxed{37699 \text{ L}}$$

Q12

Volume multiplier:

$$1.2^3 = 1.728$$

Increase:

$$\boxed{72.8\%}$$

Q13

$$2^3:5^3 = \boxed{8:125}$$

Q14

$$V = \pi(10)(36-16)$$

$$= \boxed{200\pi}$$

Q15

$$\sqrt{2h(l+b)}$$

$$\sqrt{=2(4)(10+8)}$$

$$\sqrt{=8(18)}$$

$$\sqrt{=\boxed{144}}$$

Q16

Cone is one-third:

$$\sqrt{330/3 = \boxed{110}}$$

Q17

Let dimensions:

$$\sqrt{2x, 3x, 4x}$$

Volume:

$$\sqrt{24x^3=648}$$

$$\sqrt{x^3=27}$$

$$\sqrt{x=3}$$

Dimensions:

$$\sqrt{\boxed{6, 9, 12}}$$

Q18

$$\sqrt{\frac{13\pi(16)h}{3} = 96\pi}$$

$$\sqrt{16h=288}$$

$$\sqrt{h=18}$$

Slant:

$$\sqrt{l=\sqrt{16+324}}$$

$$\sqrt{=\sqrt{340} = \boxed{2\sqrt{85}}}$$

Thus:

$$\sqrt{\boxed{h=18, l=2\sqrt{85}}}$$

Q19

Radius multiplier:

$$\sqrt{1.5}$$

Volume multiplier:

$$\sqrt{1.5^3=3.375}$$

Increase:

$$\sqrt{3.375-1=2.375}$$

$$\sqrt{\boxed{237.5\%}}$$

Q20

$$[TSA=2\pi r(h+r)]$$

$$[=2\cdot\frac{22}{7}\cdot7(10+7)]$$

$$[=44(17)]$$

$$[=\boxed{748}]$$

CHAPTER 47 STATUS

Covered so far:

- 3D units '
- litres/cubic conversions '
- cube '
- cuboid '
- prisms '
- cylinder '
- cone '
- sphere '
- hemisphere '
- hollow cylinders/pipes '
- open/closed containers '
- room painting '
- tank capacity '
- 3D scaling '
- percentage changes '
- GATE/placement L0–L3 practice '

But **Chapter 47 is not complete yet.**

Still remaining:

- frustum of cone
- pyramids
- melting/recasting
- volume conservation
- water displacement
- rise in water level
- solid combinations
- hollow solids
- cutting cubes
- painting cubes and counting smaller cubes
- advanced capacity/rate questions
- final formula/speed audit

Progress

NEXT

+Chapter 47 Part 2 — Frustum, Pyramids, Melting/Recasting, Water Displacement, Painted Cubes, Combined Solids & Final 3D Toolkit

USER

next

ASSISTANT

CHAPTER 47 — 3D MENSURATION

PART 2 — FRUSTUMS, PYRAMIDS, MELTING/RECASTING, WATER DISPLACEMENT, PAINTED CUBES, COMBINED SOLIDS & FINAL 3D TOOLKIT

Status: Ø=3 PROGRESS GATE: Medium Placements: High Priority: +Tier 2

This part is mainly about **volume conservation** and **counting smaller solids**.

The most important exam idea is:

If a solid is melted/recast without loss, volume remains constant.

1. PYRAMID

A pyramid has:

- polygonal base
- triangular side faces meeting at one vertex

For any pyramid:

$$V = \frac{1}{3} (\text{base area}) (\text{height})$$

This is the 3D analogue of a cone.

2. SQUARE PYRAMID

If square base side is a and vertical height is h :

$$V = \frac{1}{3} a^2 h$$

3. PRISM VS PYRAMID

Same base area and same height:

$$V_{\text{prism}} = Bh$$

$$V_{\text{pyramid}} = \frac{1}{3}Bh$$

Therefore:

$$V_{\text{prism}} : V_{\text{pyramid}} = 3:1$$

4. FRUSTUM OF A CONE

A frustum is obtained by cutting the top of a cone by a plane parallel to its base.

TEXT



Let:

- larger radius = (R)
- smaller radius = (r)
- vertical height = (h)
- slant height = (l)

5. FRUSTUM SLANT HEIGHT

$$l = \sqrt{h^2 + (R-r)^2}$$

This comes from Pythagoras.

6. FRUSTUM VOLUME

MUST MEMORIZE

$$V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$$

7. FRUSTUM CURVED SURFACE AREA

$$CSA = \pi(R+r)l$$

8. FRUSTUM TOTAL SURFACE AREA

Add both circular ends:

$$TSA = \pi(R+r)l + \pi R^2 + \pi r^2$$

9. EXAMPLE — FRUSTUM

$$R=5, \quad r=2, \quad h=4$$

Slant:

$$\backslash l=\sqrt{4^2+3^2}=5 \backslash$$

CSA:

$$\backslash \pi(5+2)(5) \backslash$$

$$\backslash =\boxed{35\pi} \backslash$$

Volume:

$$\backslash \frac{13\pi(4)(25+10+4)}{3} \backslash$$

$$\backslash =\frac{43\pi(39)}{3} \backslash$$

$$\backslash =\boxed{52\pi} \backslash$$

10. MELTING / RECASTING — MASTER RULE

If a solid is melted and recast without material loss:

$$\backslash \boxed{V_{\text{before}}=V_{\text{after}}} \backslash$$

Surface area generally changes.

11. ONE LARGE SOLID !' MANY SMALL SOLIDS

If one large solid is converted into (n) identical small solids:

$$\backslash \boxed{n=\frac{V_{\text{large}}}{V_{\text{small}}}} \backslash$$

12. SPHERE !' SMALL SPHERES

Large sphere radius (R) .

Small sphere radius (r) .

$$\backslash n=\frac{\frac{43\pi R^3}{3}}{\frac{43\pi r^3}{3}} \backslash$$

Thus:

$$\backslash \boxed{n=\left(\frac{R}{r}\right)^3} \backslash$$

13. EXAMPLE

Large sphere radius:

$$\backslash 6 \backslash$$

Small sphere radius:

$$\backslash 2 \backslash$$

$$\backslash n=(6/2)^3 \backslash$$

$$\backslash =3^3 \backslash$$

$$\backslash =\boxed{27} \backslash$$

14. CUBE !' SMALL CUBES

Large cube side (A) .

Small cube side (a) .

$$n = \left(\frac{A}{a}\right)^3$$

provided exact division.

15. EXAMPLE

Cube side 12 cut into cubes side 3.

Along each edge:

$$12/3=4$$

Total cubes:

$$4^3 = \boxed{64}$$

16. CYLINDER !' SPHERES

Cylinder volume:

$$\pi R^2 h$$

Small sphere volume:

$$\frac{4}{3}\pi r^3$$

Thus:

$$n = \frac{\pi R^2 h}{\frac{4}{3}\pi r^3}$$

$$\boxed{n = \frac{3R^2 h}{4r^3}}$$

17. CONE !' SPHERES

$$n = \frac{\frac{1}{3}\pi R^2 h}{\frac{4}{3}\pi r^3}$$

$$\boxed{n = \frac{R^2 h}{4r^3}}$$

18. WATER DISPLACEMENT

When a fully submerged solid is dropped into water:

$$\boxed{\text{volume of water displaced} = \text{volume of submerged solid}}$$

This is the key principle.

19. WATER RISE IN A CYLINDRICAL VESSEL

Suppose vessel radius is (R) .

A solid of volume (V) is completely submerged.

If water level rises by (x) :

$$\pi R^2 x = V$$

Therefore:

$$\boxed{x = \frac{V}{\pi R^2}}$$

20. EXAMPLE

Sphere radius 3 dropped into cylindrical vessel radius 6.

Sphere volume:

$$\frac{4}{3}\pi(3^3) = 36\pi$$

Rise (x) :

$$\pi(6^2)x = 36\pi$$

$$\boxed{x = 1}$$

21. WATER RISE IN RECTANGULAR TANK

Tank base dimensions:

$$l \times b$$

If submerged object volume is (V) :

$$lb, x = V$$

Thus:

$$\boxed{x = \frac{V}{lb}}$$

22. EXAMPLE

Tank base:

$$10 \times 8$$

Object volume:

$$240$$

Rise:

$$x = \frac{240}{80}$$

$$\boxed{x = 3}$$

23. PARTIAL SUBMERSION

If only a fraction (f) of object volume is submerged:

$$\boxed{V_{\text{displaced}} = V_{\text{object}}}$$

Do not use full volume unless fully submerged.

24. COMBINED SOLIDS

For objects made by joining solids:

$$\boxed{V_{\text{total}} = \sum \text{component volumes}}$$

For external surface area:

Count only exposed surfaces.

Do not count internal contact surfaces.

25. CYLINDER + HEMISPHERE

Common capsule/tank shape:

Cylinder:

$$\pi r^2 h$$

Two hemispheres form one sphere:

$$\frac{4}{3}\pi r^3$$

Thus capsule volume:

$$\boxed{V = \pi r^2 h + \frac{4}{3}\pi r^3}$$

26. ONE HEMISPHERE ON CYLINDER

Volume:

$$\boxed{V = \pi r^2 h + \frac{2}{3}\pi r^3}$$

External area may exclude the joined circular face.

27. CONE ON CYLINDER

Same radius (r) :

$$V = \pi r^2 h_1 + \frac{1}{3}\pi r^2 h_2$$

$$\boxed{V = \pi r^2 \left(h_1 + \frac{h_2}{3} \right)}$$

28. HOLLOW SPHERE

Outer radius:

$$R$$

Inner radius:

$$\sqrt[3]{r}$$

Material volume:

$$\boxed{V = \frac{4}{3}\pi(R^3 - r^3)}$$

29. HOLLOW CUBE

Outer cube side (A) .

Inner hollow cube side (a) .

Material volume:

$$\boxed{A^3 - a^3}$$

30. CUBE CUTTING — NUMBER OF SMALL CUBES

If cube side (A) is divided into cubes side (a) :

$$n = \frac{A}{a}$$

small cubes along each edge.

Total:

$$\boxed{n^3}$$

31. MINIMUM NUMBER OF CUTS — BASIC IDEA

To divide one cube into (n) parts along each dimension without stacking:

Cuts required:

$$(n-1) + (n-1) + (n-1)$$

$$\boxed{3(n-1)}$$

But if pieces may be stacked, fewer physical cutting operations can be possible.

Aptitude questions must specify cutting conditions.

32. PAINTED CUBE PROBLEMS

Suppose a large cube is painted on all six faces, then cut into:

$$n \times n \times n$$

equal small cubes.

Total small cubes:

$$\boxed{n^3}$$

Now classify them by number of painted faces.

33. CUBES WITH 3 PAINTED FACES

Only corner cubes.

A cube has 8 corners.

So for $(n \geq 2)$:

$$\boxed{8}$$

34. CUBES WITH EXACTLY 2 PAINTED FACES

These lie along edges, excluding corners.

Each edge contains:

$$(n-2)$$

such cubes.

Cube has 12 edges.

Therefore:

$$\boxed{12(n-2)}$$

35. CUBES WITH EXACTLY 1 PAINTED FACE

These lie in the interior of each face, excluding edges.

Each face contributes:

$$(n-2)^2$$

There are 6 faces.

Thus:

$$\boxed{6(n-2)^2}$$

36. CUBES WITH NO PAINTED FACE

These are fully internal cubes.

$$\boxed{(n-2)^3}$$

for $(n \geq 2)$.

37. PAINTED CUBE MASTER TABLE

For all 6 faces painted:

Painted faces	Count
3	8
2	$12(n-2)$

Painted faces	Count
1	$6(n-2)^2$
0	$(n-2)^3$

MUST MEMORIZE

38. CHECK — TOTAL

Sum must equal:

$$n^3$$

That gives a useful verification.

39. EXAMPLE — 4×4×4 CUBE

$$n=4$$

Total:

$$64$$

3 faces:

$$8$$

2 faces:

$$12(2)=24$$

1 face:

$$6(4)=24$$

0 faces:

$$2^3=8$$

Total:

$$8+24+24+8=64$$

Correct.

40. EXAMPLE — 5×5×5

Total:

$$125$$

3 faces:

$$8$$

2 faces:

$$12(3)=36$$

1 face:

$$\sqrt[6]{6(9)=54}$$

0 faces:

$$\sqrt[6]{27}$$

Sum:

$$\sqrt[6]{125}$$

41. AT LEAST ONE PAINTED FACE

Total minus unpainted:

$$\sqrt[6]{\boxed{n^3-(n-2)^3}}$$

42. AT LEAST TWO PAINTED FACES

Exactly 2 + exactly 3:

$$\sqrt[6]{12(n-2)+8}$$

$$\sqrt[6]{\boxed{12n-16}}$$

43. EXACTLY ONE OR TWO PAINTED FACES

$$\sqrt[6]{\boxed{6(n-2)^2+12(n-2)}}$$

Simplify only if needed.

44. IF ONLY SOME FACES ARE PAINTED

Do not use the all-6-faces table blindly.

You must analyze:

- painted face geometry
- edges shared by painted faces
- corners where painted faces meet

This can appear in harder placement tests.

45. TWO OPPOSITE FACES PAINTED

If only two opposite faces are painted and cube is cut $\sqrt[6]{n^3}$:

Each painted face has:

$$\sqrt[6]{n^2}$$

small cubes.

They don't overlap.

Exactly one painted face:

$$\boxed{2n^2}$$

No cube can have two painted faces because the painted faces are opposite.

46. THREE MUTUALLY ADJACENT FACES PAINTED

This needs case counting.

High-level approach:

1. identify one common corner with 3 painted faces
2. edges from it give 2-painted cubes
3. face interiors give 1-painted cubes
4. remaining cubes unpainted

Do not memorize unless frequently required.

47. CUBE SURFACE AREA AFTER CUTTING

One cube side (a) :

Original TSA:

$$6a^2$$

If cut into (n^3) equal cubes, small side:

$$a/n$$

Total surface area of all pieces:

$$n^3 \cdot 6(a/n)^2$$

$$= 6a^2n$$

Thus:

$$\boxed{\text{new total SA} = n \times \text{original SA}}$$

if completely separated.

48. EXAMPLE

Cube cut into $(3^3=27)$ equal cubes.

Total surface area becomes:

$$\boxed{3 \times}$$

original surface area.

Volume remains unchanged.

49. WHY CUTTING INCREASES SURFACE AREA

New internal surfaces become exposed.

Volume does not change.

This is a useful conceptual fact.

50. RECASTING — CUBE TO SPHERES

Cube side (a) melted into (n) spheres radius (r) :

$$a^3 = n \left(\frac{4}{3} \pi r^3 \right)$$

Thus:

$$\boxed{n = \frac{3a^3}{4\pi r^3}}$$

51. RECASTING — CONE TO CYLINDER

If same material:

$$\frac{1}{3} \pi R^2 H = \pi r^2 h$$

Cancel (π) :

$$\boxed{R^2 H = 3r^2 h}$$

52. WATER TRANSFER BETWEEN VESSELS

If water is poured without loss:

$$\boxed{V_{\text{old}} = V_{\text{new}}}$$

Example:

Cylindrical vessel A:

$$\pi r_1^2 h_1$$

poured into vessel B:

$$\pi r_2^2 h_2$$

Then:

$$\boxed{r_1^2 h_1 = r_2^2 h_2}$$

53. EXAMPLE — WATER TRANSFER

Cylinder A:

$$r=4, \text{quad } h=9$$

Water poured into cylinder radius 6.

New height (H) :

$$4^2(9) = 6^2 H$$

$$144 = 36 H$$

$$\boxed{H=4}$$

54. FLOW RATE INTO TANK

If water enters at volume rate:

$$q \frac{\text{units}^3}{\text{time}}$$

Tank volume (V) :

$$\boxed{t = \frac{V}{q}}$$

This connects to Pipes & Cisterns.

55. RECTANGULAR CHANNEL FLOW

If water cross-section is:

$$A$$

and moves with speed (v) :

Volume flow rate:

$$\boxed{Q = Av}$$

Be careful with time and length units.

56. CYLINDRICAL PIPE FLOW

Pipe cross-sectional area:

$$\pi r^2$$

Water speed:

$$v$$

Flow per second:

$$\boxed{Q = \pi r^2 v}$$

Then:

$$\text{volume} = Qt$$

57. EXAMPLE — PIPE FLOW

Pipe radius:

$$0.1 \text{ m}$$

water speed:

$$2 \text{ m/s}$$

Flow:

$$\pi (0.1)^2 (2)$$

$$\left[=0.02\pi\text{m}^3/\text{s} \right]$$

Per minute:

$$\left[1.2\pi\text{m}^3 \right]$$

58. FRUSTUM FROM SIMILAR CONES — INTUITION

A frustum can also be viewed as:

$$\left[\boxed{\text{large cone} - \text{small similar cone}} \right]$$

If formula is forgotten, similarity can reconstruct it.

59. VOLUME RATIO OF SIMILAR SOLIDS

If corresponding lengths ratio:

$$\left[a:b \right]$$

then:

$$\left[\boxed{V_1:V_2=a^3:b^3} \right]$$

Surface area ratio:

$$\left[\boxed{a^2:b^2} \right]$$

60. EXAMPLE

Similar cones have heights ratio:

$$\left[2:5 \right]$$

Then volume ratio:

$$\left[\boxed{8:125} \right]$$

provided all corresponding dimensions scale similarly.

61. GATE-STYLE QUESTION 1

A sphere radius 6 is melted into spheres radius 2.

Number:

$$\left[(6/2)^3 = \boxed{27} \right]$$

62. GATE-STYLE QUESTION 2

Cube side 10 is cut into cubes side 2.

Along edge:

$$\left[5 \right]$$

Total:

$$\boxed{125}$$

63. GATE-STYLE QUESTION 3

A cube is divided into 5^3 smaller cubes after all six faces are painted.

Exactly two painted faces:

$$12(5-2)$$

$$\boxed{36}$$

64. GATE-STYLE QUESTION 4

Same cube, no painted face:

$$(5-2)^3$$

$$\boxed{27}$$

65. PLACEMENT-STYLE QUESTION 1

Cube cut $(4 \times 4 \times 4)$.

Exactly one painted face:

$$6(2)^2$$

$$\boxed{24}$$

66. PLACEMENT-STYLE QUESTION 2

A sphere radius 3 is submerged in cylindrical vessel radius 6.

Rise:

$$x = \frac{\frac{4}{3}\pi(3)^3}{\pi(6)^2}$$

$$\boxed{1}$$

67. PLACEMENT-STYLE QUESTION 3

A cone radius 6, height 12 is melted into a cylinder radius 3.

Cone volume:

$$\frac{1}{3}\pi(6)^2(12) = 144\pi$$

Cylinder:

$$9\pi h$$

$$9h = 144$$

$$\boxed{h=16}$$

68. PLACEMENT-STYLE QUESTION 4

A frustum has:

$$R=6, \text{quad } r=3, \text{quad } h=4$$

Volume:

$$\frac{13\pi(4)(36+18+9)}{6}$$

$$= \frac{43\pi(63)}{2}$$

$$\boxed{84\pi}$$

69. ADVANCED QUESTION — PAINTED CUBE

A cube is cut into 216 equal cubes.

$$216=6^3$$

So:

$$n=6$$

Exactly 2 painted faces:

$$12(4) = \boxed{48}$$

Exactly 1:

$$6(4^2) = \boxed{96}$$

None:

$$4^3 = \boxed{64}$$

70. ADVANCED QUESTION — MATERIAL CONSERVATION

A cube side 6 is melted into a cylinder radius 3.

Cube volume:

$$216$$

Cylinder:

$$9\pi h$$

Thus:

$$h = \frac{216}{9\pi}$$

$$= \boxed{\frac{24}{\pi}}$$

71. ADVANCED QUESTION — WATER LEVEL

10 identical solid spheres radius 1 are immersed in a rectangular tank base:

$$\sqrt[10]{10 \times 5}$$

Total solid volume:

$$\sqrt[10]{10 \cdot \frac{4}{3}\pi} = \sqrt[10]{\frac{40\pi}{3}}$$

Base area:

$$\sqrt[50]{50}$$

Rise:

$$\sqrt{x = \frac{40\pi/3}{50}}$$

$$\sqrt[15]{\frac{4\pi}{15}}$$

72. COMMON TRAPS

& pTRAP 1 — Melting/recasting

Conserve volume, not surface area.

& pTRAP 2 — Water displacement

Use submerged volume only.

& pTRAP 3 — Painted cube

The formula assumes all 6 external faces painted.

& pTRAP 4 — Small cubes along edge

If total cubes is $\sqrt[3]{N}$, first determine:

$$\sqrt[3]{n = \sqrt[3]{N}}$$

before using painted-cube formulas.

& pTRAP 5 — Combined solids TSA

Do not count contact surfaces hidden inside the joined solid.

& pTRAP 6 — Frustum slant height

Use:

$$\sqrt{R-r}$$

not $\sqrt{R+r}$:

$$\sqrt{l = \sqrt{h^2 + (R-r)^2}}$$

& pTRAP 7 — Flow rate units

Convert seconds/minutes/hours and cm/m before multiplying.

73. MUST-MEMORIZE FRUSTUM FORMULAS

$$\boxed{l = \sqrt{h^2 + (R-r)^2}}$$

$$\boxed{V = \frac{1}{3}\pi h(R^2 + Rr + r^2)}$$

$$\boxed{CSA = \pi(R+r)l}$$

74. MUST-MEMORIZE VOLUME CONSERVATION

Melting/recasting:

$$\boxed{V_{\text{before}} = V_{\text{after}}}$$

Water transfer:

$$\boxed{V_{\text{source}} = V_{\text{destination}}}$$

Water displacement:

$$\boxed{V_{\text{rise}} = V_{\text{submerged object}}}$$

75. MUST-MEMORIZE PAINTED CUBE TABLE

For (n^3) cubes from fully painted large cube:

$$\boxed{\text{3 faces} = 8}$$

$$\boxed{\text{2 faces} = 12(n-2)}$$

$$\boxed{\text{1 face} = 6(n-2)^2}$$

$$\boxed{\text{0 faces} = (n-2)^3}$$

76. FAST EXAM DECISION TREE

Melted/recast?

Equate volume.

Object submerged?

Volume displaced = submerged volume.

Water level rises?

$$\boxed{\text{base area} \times \text{rise} = \text{displaced volume}}$$

Large cube cut?

Find number along one edge first.

All faces painted?

Use painted-cube table.

Combined solid?

Add volumes, but surface area counts only exposed surfaces.

Frustum?

Use (R, r, h) ; compute (l) only if surface area needed.

Flow through pipe?

$$Q = Av$$

PRACTICE — LEVEL 0

Q1

A cube side 8 is cut into cubes side 2. How many cubes?

Q2

A large sphere radius 4 is melted into small spheres radius 1. How many?

Q3

A $(4 \times 4 \times 4)$ painted cube is cut. How many cubes have 3 painted faces?

Q4

How many have no painted face?

Q5

Find frustum slant height if:

$$[R=7, r=4, h=4]$$

LEVEL 1

Q6

A cube is divided into (5^3) small cubes after painting. Find exactly 1 painted face.

Q7

Same cube: exactly 2 painted faces.

Q8

A sphere radius 3 is submerged in cylinder radius 9. Find water rise.

Q9

A cone radius 3, height 12 is melted into cylinder radius 3. Find cylinder height.

Q10

Frustum:

$$[R=5, r=3, h=6]$$

find volume.

LEVEL 2

Q11

A cube side 12 is melted into spheres radius 2. Find number of spheres in terms of (π) , and check whether it can be an integer for exact physical recasting.

Q12

A cylinder radius 6, height 10 is melted into spheres radius 2. Find number.

Q13

A fully painted cube is cut into 343 equal cubes. Find counts with 0,1,2,3 painted faces.

Q14

A rectangular tank base (20×10) receives a fully submerged object of volume 6000 cm^3 . Find rise in water level.

Q15

A cylindrical vessel radius 5 contains water. A sphere radius 2 is fully submerged. Find rise.

LEVEL 3 — EXAM PRESSURE

Q16

A cone and hemisphere have equal radius (r) . Their volumes are equal. Find cone height.

Q17

A cube painted on all sides is cut into 1000 equal cubes. Find number having at least one painted face.

Q18

A frustum has $(R=8, r=2, h=8)$. Find volume and slant height.

Q19

A cylindrical tank radius 1 m is filled through a pipe of cross-sectional area (100 cm^2) at speed (2 m/s) . Find time to raise water level by 1 m.

Q20

A large cube of side 9 is cut into small cubes side 3. If only the original external faces were painted, find the number of small cubes with exactly one painted face.

ANSWERS

Q1

$$\left[\left(\frac{8}{2} \right)^3 = 4^3 = \boxed{64} \right]$$

Q2

$$\left[\left(\frac{4}{1} \right)^3 = \boxed{64} \right]$$

Q3

Corner cubes:

$$\left[\boxed{8} \right]$$

Q4

$$\boxed{(4-2)^3 = 8}$$

Q5

$$l = \sqrt{4^2 + (7-4)^2}$$

$$= \sqrt{16+9}$$

$$\boxed{5}$$

Q6

$$6(5-2)^2 = 6(9)$$

$$\boxed{54}$$

Q7

$$12(3) = \boxed{36}$$

Q8

Sphere volume:

$$36\pi$$

Cylinder rise:

$$\pi(81)x = 36\pi$$

$$\boxed{x = \frac{4}{9}}$$

Q9

Cone volume:

$$\frac{1}{3}\pi(9)(12) = 36\pi$$

Cylinder:

$$9\pi h$$

$$h = 4$$

$$\boxed{4}$$

Q10

$$V = \frac{1}{3}\pi(6)(25+15+9)$$

$$= 2\pi(49)$$

$$\boxed{98\pi}$$

Q11

Cube volume:

$$1728$$

One sphere:

$$\frac{4}{3}\pi(8) = \frac{32\pi}{3}$$

Thus:

$$n = \frac{1728}{\frac{32\pi}{3}}$$

$$\left[=\frac{162}{\pi} \right]$$

So:

$$\left[\boxed{\frac{162}{\pi}} \right]$$

This is not an integer for the usual π , meaning those exact dimensions cannot produce an integer number of complete radius-2 spheres with zero material left over. In real aptitude problems, dimensions are usually chosen to avoid this.

Q12

Cylinder:

$$\pi(36)(10)=360\pi$$

Sphere:

$$\frac{32\pi}{3}$$

Number:

$$360\pi \cdot \frac{3}{32\pi}$$

$$\left[=\frac{1080}{32} = 33.75 \right]$$

Again not an integer:

$$\left[\boxed{33.75} \right]$$

So only 33 complete spheres could be made if leftover material is permitted. Exact full recasting into identical complete spheres would require compatible dimensions.

Q13

$$\left[343=7^3 \right]$$

So:

$$\left[n=7 \right]$$

No paint:

$$\left[(5)^3=125 \right]$$

One:

$$\left[6(25)=150 \right]$$

Two:

$$\left[12(5)=60 \right]$$

Three:

$$\left[8 \right]$$

$$\left[\boxed{0:125, \ 1:150, \ 2:60, \ 3:8} \right]$$

Q14

Base area:

$$\left[200 \right]$$

Rise:

$$\left[6000/200 = \boxed{30\text{ cm}} \right]$$

Q15

Sphere volume:

$$\left[\frac{4}{3}\pi(8) = \frac{32\pi}{3} \right]$$

Cylinder base:

$$\left[25\pi \right]$$

Rise:

$$\left[\frac{32\pi/3}{25\pi} \right]$$

$$\left[=\boxed{\frac{32}{75}} \right]$$

Q16

Cone volume:

$$\left[\frac{1}{3}\pi r^2h \right]$$

Hemisphere:

$$\left[\frac{2}{3}\pi r^3 \right]$$

Equate:

$$\left[\frac{1}{3}\pi r^2h = \frac{2}{3}\pi r^3 \right]$$

$$\left[h=2r \right]$$

$$\left[\boxed{h=2r} \right]$$

Q17

$$\left[1000=10^3 \right]$$

Unpainted:

$$\left[(10-2)^3 = 512 \right]$$

At least one painted:

$$\left[1000-512 \right]$$

$$\left[=\boxed{488} \right]$$

Q18

Slant:

$$\left[l=\sqrt{8^2+(8-2)^2} \right]$$

$$\left[=\sqrt{64+36} \right]$$

$$\left[=\boxed{10} \right]$$

Volume:

$$\left[\frac{1}{3}\pi(8)(64+16+4) \right]$$

$$\left[=\frac{8}{3}\pi(84) \right]$$

$$\left[=\boxed{224\pi} \right]$$

Q19

Pipe area:

$$\sqrt[100]{\text{cm}^2} = 0.01 \sqrt{\text{m}^2}$$

Flow rate:

$$\sqrt{Q=Av}$$

$$\sqrt{=0.01(2) = 0.02 \sqrt{\text{m}^3/\text{s}}}$$

Tank volume needed for 1 m rise:

$$\sqrt{\pi(1)^2(1) = \pi \sqrt{\text{m}^3}}$$

Time:

$$\sqrt{\frac{\pi}{0.02} = 50 \pi \sqrt{\text{s}}}$$

$$\sqrt{\boxed{50 \pi \sqrt{\text{s}}} \approx 157.1 \sqrt{\text{s}}}$$

Q20

$$\sqrt{9/3=3}$$

So:

$$\sqrt{n=3}$$

Exactly one painted face:

$$\sqrt{6(n-2)^2}$$

$$\sqrt{=6(1)^2}$$

$$\sqrt{=\boxed{6}}$$

' CHAPTER 47 COMPLETE

We have now covered:

- cube/cuboid '
- prisms '
- cylinders '
- cones '
- spheres/hemispheres '
- pyramids '
- frustums '
- hollow solids '
- open/closed containers '
- TSA/CSA/LSA '
- volume and capacity '
- space diagonals '
- scaling and percentage changes '
- melting/recasting '
- volume conservation '
- water displacement '
- water-level rise '
- combined solids '

- pipe flow '
- cube cutting '
- painted-cube counting '
- surface-area increase after cutting '
- advanced GATE/placement practice '

Progress

Completed: 47/76

Unit 1 — Quantitative Aptitude: Ø=14 PROGRESS

Only two short chapters remain in Unit 1 after this:

- Chapter 48 — Clocks Ø=10
- Chapter 49 — Calendars Ø=10

Then we enter **Unit 2 — Data Interpretation**.

NEXT

Ø=10 Chapter 48 — Clocks

Next we will cover:

- hour/minute hand speeds
- angle between hands
- coinciding hands
- opposite hands
- right-angle positions
- gain/loss clocks
- mirror-image clock questions
- minute-hand/hour-hand relative motion
- GATE/placement shortcuts and traps.

USER

next

ASSISTANT

CHAPTER 48 — CLOCKS

PART 1 — HAND SPEEDS, ANGLES, COINCIDENCE, OPPOSITE & RIGHT-ANGLE POSITIONS

Status: Ø=14 PROGRESS GATE: Low–Medium Placements: Medium Priority: Ø=10 or 3

Clock problems are mostly **relative speed + angle** questions.

The key idea is:

1. CLOCK BASICS

A clock face contains:

$$\boxed{360^\circ}$$

and:

$$\boxed{12}$$

hour divisions.

Therefore angle between consecutive hour marks:

$$\boxed{\frac{360^\circ}{12} = 30^\circ}$$

So:

- 12 to 1 : 30°
- 1 to 2 : 30°
- etc.

2. MINUTE DIVISIONS

A clock has 60 minute marks.

Therefore:

$$\frac{360^\circ}{60} = \boxed{6^\circ}$$

per minute mark.

3. MINUTE-HAND SPEED

Minute hand completes:

$$\boxed{360^\circ}$$

in:

$$\boxed{60 \text{ minutes}}$$

So:

$$\boxed{6^\circ / \text{minute}}$$

MUST MEMORIZE

4. HOUR-HAND SPEED

Hour hand completes:

$$\boxed{360^\circ}$$

in:

$$\left[12 \text{ hours} = 720 \text{ minutes} \right]$$

Therefore:

$$\left[\frac{360}{720} = \boxed{0.5^\circ \text{ per minute}} \right]$$

or:

$$\left[\boxed{\frac{1}{2}^\circ \text{ per minute}} \right]$$

5. SECOND-HAND SPEED

Second hand completes (360°) in 60 seconds.

So:

$$\left[\boxed{6^\circ \text{ per second}} \right]$$

Lower priority, but useful.

6. RELATIVE SPEED OF MINUTE AND HOUR HANDS

Minute hand:

$$\left[6^\circ \text{ per min} \right]$$

Hour hand:

$$\left[0.5^\circ \text{ per min} \right]$$

Relative speed:

$$\left[6 - 0.5 = \boxed{5.5^\circ \text{ per min}} \right]$$

or:

$$\left[\boxed{\frac{11}{2}^\circ \text{ per min}} \right]$$

This is the most important clock number.

7. POSITION OF MINUTE HAND AT m MINUTES

At (m) minutes after the hour:

$$\left[\boxed{\theta_m = 6m} \right]$$

measured clockwise from 12.

8. POSITION OF HOUR HAND AT h:m

At exactly (h) o'clock, hour hand begins at:

$$\left[30h^\circ \right]$$

But during (m) minutes it moves further:

$$\left[0.5m^\circ \right]$$

Therefore:

$$\boxed{\theta_h = 30h + \frac{m}{2}}$$

9. ANGLE BETWEEN HANDS

Difference:

$$|6m - \left(30h + \frac{m}{2}\right)|$$

Thus:

$$\boxed{\theta = \left|30h - \frac{11m}{2}\right|}$$

MUST MEMORIZE

10. SMALLER ANGLE RULE

The formula may produce an angle greater than:

$$180^\circ$$

But normally questions ask for the smaller angle.

So:

$$\boxed{\text{smaller angle} = \min(\theta, 360^\circ - \theta)}$$

11. EXAMPLE — 3:00

$$h=3, \quad m=0$$

$$\theta = |30(3) - 0|$$

$$= \boxed{90^\circ}$$

12. EXAMPLE — 6:00

$$\theta = 30(6) = 180^\circ$$

$$\boxed{180^\circ}$$

Hands are opposite.

13. EXAMPLE — 12:00

Both hands overlap:

$$\boxed{0^\circ}$$

14. EXAMPLE — 3:30

Minute hand:

$$\angle 6(30) = 180^\circ$$

Hour hand:

$$\angle 90 + 15 = 105^\circ$$

Difference:

$$\angle 180 - 105 = \boxed{75^\circ}$$

15. EXAMPLE — 2:20

Using formula:

$$\angle \theta = \left| 30(2) - \frac{11(20)}{2} \right|$$

$$\angle =$$

$$60 - 110$$

$$\angle$$

$$\angle = \boxed{50^\circ}$$

16. EXAMPLE — 8:20

$$\angle \theta = \left| 240 - 110 \right|$$

$$\angle = 130^\circ$$

$$\angle \boxed{130^\circ}$$

17. COMMON TRAP — HOUR HAND DOES NOT STAY ON NUMBER

At 3:30, hour hand is not exactly at 3.

It moves:

$$\angle 0.5(30) = 15^\circ$$

past 3.

This is probably the most common beginner mistake.

18. FIND TIME WHEN ANGLE IS GIVEN

For time between (h) and $(h+1)$:

$$\left| 30h - \frac{11m}{2} \right| = \theta$$

This creates two possibilities:

$$30h - \frac{11m}{2} = \theta$$

or:

$$30h - \frac{11m}{2} = -\theta$$

Thus:

$$\boxed{m = \frac{60h \pm 2\theta}{11}}$$

provided the resulting m lies between 0 and 60.

19. COINCIDENCE OF HANDS

For overlap:

$$\theta = 0$$

So:

$$30h - \frac{11m}{2} = 0$$

$$60h = 11m$$

Therefore:

$$\boxed{m = \frac{60h}{11}}$$

minutes after h o'clock.

20. EXAMPLE — BETWEEN 1 AND 2

$$m = \frac{60(1)}{11}$$

$$= \frac{60}{11} = 5\frac{5}{11}$$

So hands overlap at approximately:

$$\boxed{1:05:27.27}$$

21. BETWEEN 2 AND 3

$$m = \frac{120}{11}$$

$$= 10\frac{10}{11} \text{ minutes}$$

approximately:

$$\boxed{2:10:54.55}$$

22. TIME BETWEEN SUCCESSIVE COINCIDENCES

Minute hand must gain:

$$360^\circ$$

on hour hand.

Relative speed:

$$5.5^\circ/\text{min}$$

Time:

$$\frac{360}{5.5} = \frac{720}{11}$$

$$\boxed{65\frac{5}{11}\text{ minutes}}$$

approximately:

$$65.4545\text{ minutes}$$

23. HOW MANY TIMES DO HANDS COINCIDE IN 12 HOURS?

Because successive overlap interval is:

$$\frac{720}{11}$$

minutes.

In 720 minutes:

$$\frac{720}{720/11}=11$$

So:

$$\boxed{11\text{ times}}$$

in 12 hours.

In 24 hours:

$$\boxed{22\text{ times}}$$

24. WHY NOT 12 TIMES?

Because 12:00 at beginning and 12:00 at end are the same cycle boundary when counting distinct occurrences over a 12-hour interval.

Hence:

$$\boxed{11}$$

not 12.

25. HANDS OPPOSITE EACH OTHER

Opposite means angle:

$$180^\circ$$

Equation:

$$\left| 30h - \frac{11m}{2} \right| = 180$$

So:

$$\boxed{m = \frac{60h \pm 360}{11}}$$

Keep only valid $(0 \leq m < 60)$.

26. EXAMPLE — BETWEEN 4 AND 5

Set:

$$\left| 120 - \frac{11m}{2} \right| = 180$$

Possibility 1:

$$120 - \frac{11m}{2} = 180$$

gives negative (m) , invalid.

Possibility 2:

$$120 - \frac{11m}{2} = -180$$

$$\frac{11m}{2} = 300$$

$$m = \frac{600}{11}$$

$$= 54\frac{6}{11}$$

So:

$$\boxed{4:54:32.73}$$

approximately.

27. HOW MANY TIMES ARE HANDS OPPOSITE IN 12 HOURS?

They become opposite:

$$\boxed{11 \text{ times}}$$

in 12 hours.

And:

$$\boxed{22}$$

times in 24 hours.

28. RIGHT ANGLES

Right-angle positions satisfy:

$$\boxed{\theta = 90^\circ}$$

Equation:

$$\left| 30h - \frac{11m}{2} \right| = 90$$

Thus:

$$\boxed{m = \frac{60h \pm 180}{11}}$$

with valid minute range.

29. EXAMPLE — BETWEEN 2 AND 3

$$\left| 60 - \frac{11m}{2} \right| = 90$$

First:

$$60 - \frac{11m}{2} = 90$$

gives negative minute !' invalid.

Second:

$$\lfloor 60 - \frac{11m}{2} \rfloor = -90 \lfloor$$

$$\lfloor \frac{11m}{2} \rfloor = 150 \lfloor$$

$$\lfloor m = \frac{300}{11} \rfloor \lfloor$$

$$\lfloor = 27 \frac{3}{11} \rfloor \lfloor$$

So one right angle occurs around:

$$\lfloor \boxed{2:27:16.36} \rfloor$$

There can also be another occurrence within some hour intervals depending on geometry and equation form.

30. HOW MANY RIGHT ANGLES IN 12 HOURS?

The hands form a right angle:

$$\lfloor \boxed{22 \text{ times}} \rfloor \lfloor$$

in 12 hours.

Therefore in 24 hours:

$$\lfloor \boxed{44 \text{ times}} \rfloor \lfloor$$

31. GENERAL ANGLE FREQUENCY

For most angles $(0 < \theta < 180^\circ)$, each relative revolution gives two configurations:

- minute hand ahead by (θ)
- minute hand behind by (θ)

So over 12 hours, many non-special angles occur:

$$\lfloor \boxed{22 \text{ times}} \rfloor \lfloor$$

Special cases:

- (0°) : 11 times
 - (180°) : 11 times
-

32. EXACT ANGLE AT GIVEN TIME — FAST METHOD

Instead of separately finding positions, use:

$$\lfloor \boxed{\theta = \left\lfloor 30h - \frac{11m}{2} \right\rfloor} \rfloor \lfloor$$

Then if $(\theta > 180^\circ)$:

$$\lfloor \boxed{360^\circ - \theta} \rfloor \lfloor$$

33. EXAMPLE — 5:40

$$\theta = \left| 150 - \frac{11(40)}{2} \right|$$

$$=$$

$$150 - 220$$

$$|$$

$$= \boxed{70^\circ}$$

34. EXAMPLE — 11:20

$$\theta = \left| 330 - 110 \right|$$

$$= 220^\circ$$

Smaller angle:

$$360 - 220 = \boxed{140^\circ}$$

35. EXAMPLE — 7:15

$$\theta = \left| 210 - \frac{165}{2} \right|$$

$$=$$

$$210 - 82.5$$

$$|$$

$$= \boxed{127.5^\circ}$$

36. MINUTE HAND GAINS ON HOUR HAND

In 1 minute:

$$\boxed{5.5^\circ}$$

In 2 minutes:

$$11^\circ$$

In 10 minutes:

$$55^\circ$$

This is useful for mental calculations.

37. ONE HOUR RELATIVE GAIN

In 60 minutes:

$$5.5 \times 60 = 330^\circ$$

So during one hour the minute hand gains:

$$\boxed{330^\circ}$$

on the hour hand.

38. COINCIDENCE AFTER EXACT HOUR

At $(h:00)$, initial separation is:

$$30h^\circ$$

Minute hand catches hour hand at relative speed:

$$5.5^\circ/\text{min}$$

Time:

$$\frac{30h}{5.5} = \frac{60h}{11}$$

Same formula.

This relative-motion method is often easier to remember.

39. ANGLE IN TERMS OF CLOCK DIVISIONS

Each minute division:

$$6^\circ$$

Each hour division:

$$30^\circ$$

Hour hand moves:

$$\frac{1}{12}$$

of an hour division per minute because:

$$0.5^\circ / 30^\circ = \frac{1}{60}$$

More simply, in 5 minutes the hour hand moves:

$$2.5^\circ$$

40. TIME WHEN MINUTE HAND IS x MINUTES AHEAD OF HOUR HAND

If "ahead" is described by minute marks rather than degrees:

Each minute-space = 6° .

So (x) minute spaces correspond to:

$$\boxed{6x^\circ}$$

Then use the angle equation.

41. MIRROR-IMAGE CLOCK — BASIC IDEA

For a standard analog clock viewed in a vertical mirror:

$$\boxed{\text{actual time} + \text{mirror time} = 11:60}$$

Equivalent computationally:

$$\boxed{\text{actual} = 11:60 - \text{mirror}}$$

for most non-12:00 times.

We will do this properly in Part 2.

42. GAIN/LOSS CLOCK — BASIC IDEA

If a clock gains:

$$g \text{ minutes per real hour}$$

then indicated time runs faster than actual time.

If it loses:

$$l \text{ minutes per real hour}$$

then indicated time runs slower.

This becomes a ratio problem, which we will complete in Part 2.

43. COMMON CLOCK VALUES TO REMEMBER

$$\boxed{\text{minute hand} = 6^\circ/\text{min}}$$

$$\boxed{\text{hour hand} = 0.5^\circ/\text{min}}$$

$$\boxed{\text{relative} = 5.5^\circ/\text{min}}$$

$$\boxed{\text{one hour mark} = 30^\circ}$$

$$\boxed{\text{one minute mark} = 6^\circ}$$

44. GATE-STYLE QUESTION 1

Find angle at 4:20.

$$\theta = |120 - 110|$$

$$= \boxed{10^\circ}$$

45. GATE-STYLE QUESTION 2

Find angle at 9:30.

$$\theta = |270 - 165|$$

$$= \boxed{105^\circ}$$

46. GATE-STYLE QUESTION 3

Find angle at 1:50.

$$\theta = \left| 30 - 275 \right|$$

$$= 245^\circ$$

Smaller:

$$360 - 245 = \boxed{115^\circ}$$

47. PLACEMENT-STYLE QUESTION 1

At what time between 3 and 4 do the hands coincide?

$$m = \frac{60(3)}{11}$$

$$= \frac{180}{11} = 16\frac{4}{11}$$

So:

$$\boxed{3:16:21.82}$$

approximately.

48. PLACEMENT-STYLE QUESTION 2

At what time between 5 and 6 do they coincide?

$$m = \frac{300}{11}$$

$$= 27\frac{3}{11}$$

So:

$$\boxed{5:27:16.36}$$

approximately.

49. PLACEMENT-STYLE QUESTION 3

How long after 12:00 will hands next coincide?

$$\boxed{\frac{720}{11} \text{ min}}$$

$$= 65\frac{5}{11} \text{ min}$$

So approximately:

$$\boxed{1:05:27.27}$$

50. COMMON TRAPS

TRAP 1 — Hour hand movement

It moves continuously.

At 4:30 it is halfway between 4 and 5.

& pTRAP 2 — Large angle vs smaller angle

If result is $(>180^\circ)$, convert:

$$[360 - \theta]$$

when smaller angle is requested.

& pTRAP 3 — Coincidence every hour

Wrong.

It occurs every:

$$[65 \frac{1}{5} \text{ minutes}]$$

not every 60 minutes.

& pTRAP 4 — Right-angle count

In 12 hours:

$$[22]$$

not 24.

& pTRAP 5 — Degrees vs minute divisions

1 clock minute division:

$$[6^\circ]$$

Do not confuse with one minute of time.

51. MUST-MEMORIZE FORMULAS

Position of minute hand:

$$[\boxed{6m}]$$

Position of hour hand:

$$[\boxed{30h + \frac{m}{2}}]$$

Angle difference:

$$[\boxed{ \left| 30h - \frac{11m}{2} \right| }]$$

Coincidence:

$$[\boxed{ m = \frac{60h}{11} }]$$

General angle:

$$[\boxed{ m = \frac{60h \pm 2\theta}{11} }]$$

52. MUST-MEMORIZE COUNTS IN 12 HOURS

Coincide:

$$[\boxed{11}]$$

Opposite:

$\boxed{11}$

Right angle:

$\boxed{22}$

53. FAST EXAM DECISION TREE

Given exact time?

Use:

$\boxed{}$

30h-5.5m

$\boxed{}$

Need coincidence?

Set angle = 0.

Need opposite?

Set angle = 180.

Need right angle?

Set angle = 90.

Need smaller angle?

If calculated angle (>180) , subtract from 360.

Gain/loss clock?

Treat as time-rate ratio.

Mirror?

Use mirror-time subtraction rule.

PRACTICE — LEVEL 0

Q1

What is minute-hand speed?

Q2

What is hour-hand speed?

Q3

Find angle at 2:00.

Q4

Find angle at 3:30.

Q5

Find angle at 6:00.

LEVEL 1

Q6

Find angle at 4:40.

Q7

Find smaller angle at 10:20.

Q8

At what time between 1 and 2 do hands overlap?

Q9

How many overlaps occur in 24 hours?

Q10

How many right-angle positions occur in 12 hours?

LEVEL 2

Q11

Find angle at 7:20.

Q12

At what time between 2 and 3 are the hands opposite?

Q13

At what time between 3 and 4 do the hands form a right angle? Find all valid times.

Q14

How much does minute hand gain on hour hand in 20 minutes?

Q15

At 11:50 find the smaller angle.

LEVEL 3 — EXAM PRESSURE

Q16

At what time between 8 and 9 do the hands coincide?

Q17

At what time between 5 and 6 are the hands opposite?

Q18

Find both times between 4 and 5 when the hands are (60°) apart.

Q19

How much time elapses between two successive coincidences?

Q20

At 2:24, find exact smaller angle.

ANSWERS

Q1

$\boxed{6^\circ \text{min}}$

Q2

$\boxed{0.5^\circ \text{min}}$

Q3

$30(2) = \boxed{60^\circ}$

Q4

$\boxed{90-165}$

$= \boxed{75^\circ}$

Q5

$\boxed{180^\circ}$

Q6

$\boxed{120-220}$

$= \boxed{100^\circ}$

Q7

$\boxed{300-110}$

$= 190^\circ$

Smaller:

$\boxed{360-190 = 170^\circ}$

Q8

$$m = \frac{60}{11} = 5\frac{5}{11}$$

$$\boxed{1:05:27.27 \text{ approximately}}$$

Q9

$$2(11) = \boxed{22}$$

Q10

$$\boxed{22}$$

Q11

$$\frac{210-110}{100}$$

$$= \boxed{100^\circ}$$

Q12

$$\left| 60 - \frac{11m}{2} \right| = 180$$

Valid branch:

$$60 - \frac{11m}{2} = -180$$

$$\frac{11m}{2} = 240$$

$$m = \frac{480}{11}$$

$$= 43\frac{7}{11}$$

So approximately:

$$\boxed{2:43:38.18}$$

Q13

$$\left| 90 - \frac{11m}{2} \right| = 90$$

First:

$$90 - \frac{11m}{2} = 90$$

$$m = 0$$

So:

$$3:00$$

Second:

$$90 - \frac{11m}{2} = -90$$

$$\frac{11m}{2} = 180$$

$$m = \frac{360}{11}$$

$$= 32\frac{8}{11}$$

approximately:

$$\boxed{3:32:43.64}$$

Thus:

$$\boxed{3:00 \text{ and } 3:32:43.64}$$

Q14

Relative speed:

$$5.5^{\circ}/\text{min}$$

For 20 minutes:

$$5.5(20)$$

$$= \boxed{110^{\circ}}$$

Q15

330-275

$$= \boxed{55^{\circ}}$$

Q16

$$m = \frac{60(8)}{11} = \frac{480}{11} = 43\frac{7}{11}$$

So:

$$\boxed{8:43:38.18}$$

approximately.

Q17

$$\left| 150 - \frac{11m}{2} \right| = 180$$

Valid branch:

$$150 - \frac{11m}{2} = -180$$

$$\frac{11m}{2} = 330$$

$$m = 60$$

This is exactly:

$$\boxed{6:00}$$

So strictly between 5:00 and 6:00, there is no interior occurrence; the opposite position occurs at the endpoint 6:00.

Q18

$$\left| 120 - \frac{11m}{2} \right| = 60$$

Case 1:

$$120 - \frac{11m}{2} = 60$$

$$\frac{11m}{2} = 60$$

$$m = \frac{120}{11} = 10\frac{10}{11}$$

Case 2:

$$120 - \frac{11m}{2} = -60$$

$$\left[\frac{11m}{2} = 180 \right]$$

$$\left[m = \frac{360}{11} = 32 \frac{8}{11} \right]$$

So approximately:

$$\left[\boxed{4:10:54.55} \right]$$

and:

$$\left[\boxed{4:32:43.64} \right]$$

Q19

$$\left[\frac{360}{5.5} = \frac{720}{11} \right]$$

$$\left[= \boxed{65 \frac{5}{11} \text{ minutes}} \right]$$

Q20

$$\left[\theta = \left| 60 - \frac{11(24)}{2} \right| \right]$$

$$\left[= \right]$$

$$60 - 132$$

$$\left[\right]$$

$$\left[= \boxed{72^\circ} \right]$$

CHAPTER 48 STATUS

Covered so far:

- clock geometry '
- hand speeds '
- relative speed '
- angle-at-given-time formula '
- smaller-angle rule '
- coincidence '
- successive coincidence interval '
- opposite positions '
- right-angle positions '
- occurrence counts '
- GATE/placement practice '

But **Chapter 48 is not complete yet.**

Still remaining:

- mirror-image clocks
- faulty clocks
- gaining/losing clocks
- fast/slow-clock ratios
- clocks that gain per day/hour
- correcting displayed time

- advanced mixed clock questions
- final chapter audit

Progress

Completed: 47/76 Current: 0/0 Chapter 48 — Clocks

NEXT

0/0 Chapter 48 Part 2 — Mirror Clocks, Gain/Loss Clocks, Faulty Clock Ratios & Final Clock Toolkit

USER

next

ASSISTANT

CHAPTER 48 — CLOCKS

PART 2 — MIRROR CLOCKS, GAIN/LOSS CLOCKS, FAULTY CLOCKS, CORRECT-TIME RATIO & FINAL CLOCK TOOLKIT

Status: 0/0 PROGRESS GATE: Low–Medium Placements: Medium Priority: 0/0 3

This part is mainly about **time-rate ratios**.

Key idea:

A faulty clock is just a moving clock with the wrong speed.

1. MIRROR IMAGE OF A CLOCK

For a standard analog clock viewed in a vertical mirror:

$$\boxed{\text{Actual time} + \text{Mirror time} = 11:60}$$

This is the fastest rule for most clock-mirror questions.

2. EXAMPLE

Mirror shows:

$$\boxed{3:20}$$

Actual time:

$$\boxed{11:60 - 3:20}$$

Borrow mentally:

$$\boxed{11:60 - 3:20 = 8:40}$$

So:

$$\boxed{8:40}$$

3. EXAMPLE

Mirror shows:

$$7:15$$

Actual:

$$11:60 - 7:15$$

$$= \boxed{4:45}$$

4. EXAMPLE

Actual time:

$$2:35$$

Mirror image:

$$11:60 - 2:35$$

$$= \boxed{9:25}$$

The same subtraction works in either direction.

5. SPECIAL CASE — 12:00

At:

$$12:00$$

mirror also shows:

$$\boxed{12:00}$$

Similarly, 6:00 also remains visually:

$$\boxed{6:00}$$

because the two hands lie on the mirror axis.

6. MIRROR RULE WITH MINUTES ZERO

Suppose actual:

$$4:00$$

Write:

$$11:60 - 4:00$$

$$= \boxed{7:60}$$

But:

$$\boxed{7:60=8:00}$$

So mirror time:

$$\boxed{8:00}$$

This is correct.

7. FAST MIRROR METHOD

For nonzero minutes (m) :

Mirror minute:

$$\boxed{60-m}$$

Mirror hour:

$$\boxed{11-h}$$

with hour adjustment depending on the resulting time.

But the safest method is simply:

$$\boxed{11:60-\text{time}}$$

8. WATER IMAGE — DIFFERENT FROM MIRROR

A **water image** is reflection across a horizontal axis, not a vertical axis.

Clock water-image questions are less common and visually trickier.

For aptitude, do not apply the $(11:60)$ mirror rule blindly to water images.

9. FAULTY CLOCK — BASIC IDEA

A correct clock measures real time exactly.

A faulty clock may:

- gain time
- lose time

If it gains, it runs fast.

If it loses, it runs slow.

10. CLOCK GAINS g MINUTES PER HOUR

If a clock gains (g) minutes every real hour:

In:

$$60 \text{ real minutes}$$

it shows:

$$60+g$$

minutes.

So:

$$\boxed{\text{shown time} : \text{real time} = (60+g):60}$$

11. CLOCK LOSES l MINUTES PER HOUR

Then in 60 real minutes, it shows:

$$\boxed{60-l}$$

Therefore:

$$\boxed{\text{shown} : \text{real} = (60-l):60}$$

12. EXAMPLE — GAINS 5 MIN/HOUR

Ratio:

$$\boxed{65:60}$$

Simplify:

$$\boxed{13:12}$$

So if real elapsed time is 12 hours, faulty clock shows:

$$\boxed{13 \text{ hours}}$$

It gains 1 hour over 12 real hours.

13. EXAMPLE — LOSES 4 MIN/HOUR

Shown : real:

$$\boxed{56:60}$$

$$\boxed{=14:15}$$

So every 15 real minutes are represented by only 14 shown minutes.

14. CONVERT REAL TIME TO SHOWN TIME

If:

$$\boxed{\frac{\text{shown}}{\text{real}} = k}$$

then:

$$\boxed{\text{shown elapsed} = k \times \text{real elapsed}}$$

15. CONVERT SHOWN TIME TO REAL TIME

$$\boxed{\text{real elapsed} = \frac{\text{shown elapsed}}{k}}$$

This is a placement-favorite pattern.

16. EXAMPLE

A clock gains 10 minutes every hour.

In 6 real hours, how much elapsed time does it show?

Rate:

$$\left[\frac{70}{60} = \frac{7}{6} \right]$$

Shown elapsed:

$$\left[6 \cdot \frac{7}{6} \right]$$

$$\left[= \boxed{7 \text{ hours}} \right]$$

So gain:

$$\left[\boxed{1 \text{ hour}} \right]$$

17. EXAMPLE — FIND REAL TIME

A clock gains 5 min/hour.

It shows 13 hours of elapsed time.

Rate:

$$\left[\frac{65}{60} = \frac{13}{12} \right]$$

Real elapsed:

$$\left[13 \cdot \frac{12}{13} \right]$$

$$\left[= \boxed{12 \text{ hours}} \right]$$

18. GAIN/LOSS PER DAY

A day has:

$$\left[24 \text{ hours} = 1440 \text{ minutes} \right]$$

If clock gains (g) minutes/day:

$$\left[\boxed{\text{shown} : \text{real} = (1440 + g) : 1440} \right]$$

If loses (l) :

$$\left[\boxed{(1440 - l) : 1440} \right]$$

19. EXAMPLE — GAINS 12 MIN/DAY

Rate:

$$\left[1452 : 1440 \right]$$

Simplify by 12:

$\backslash[121:120 \backslash]$

So:

$\backslash[\boxed{\text{\text{shown}}:\text{\text{real}}=121:120} \backslash]$

20. HOW MUCH DOES A CLOCK GAIN AFTER t HOURS?

If it gains $\backslash(g)$ minutes per hour:

$\backslash[\boxed{\text{\text{gain}}=gt\text{\text{ minutes}}} \backslash]$

provided $\backslash(t)$ is real elapsed hours.

21. EXAMPLE

Clock gains:

$\backslash[2.5\text{\text{ min/hour}} \backslash]$

After:

$\backslash[8\text{\text{ real hours}} \backslash]$

gain:

$\backslash[2.5(8) = \boxed{20\text{\text{ minutes}}} \backslash]$

22. HOW MUCH DOES A CLOCK LOSE AFTER t HOURS?

If loss rate:

$\backslash[l\text{\text{ min/hour}} \backslash]$

then:

$\backslash[\boxed{\text{\text{loss}}=lt} \backslash]$

23. CLOCK CORRECT AT START

A typical question says:

A clock is correct at 8 AM and gains 4 minutes per hour. What will it show at actual 2 PM?

Real elapsed:

$\backslash[6\text{\text{ hours}} \backslash]$

Gain:

$\backslash[6(4)=24\text{\text{ min}} \backslash]$

Actual 2 PM + 24 min:

$\backslash[\boxed{2:24\text{\text{ PM}}} \backslash]$

24. CLOCK LOSES — EXAMPLE

Clock correct at 9 AM.

Loses:

3 min/hour

At actual 5 PM:

Elapsed:

8 hours

Loss:

24 min

Clock shows:

$5:00 - 0:24$

$\boxed{4:36 \text{ PM}}$

25. CLOCK SHOWS TIME — FIND ACTUAL TIME

This needs rate ratio.

Example:

A clock gains 5 min/hour.

It was correct at noon.

Later it shows:

$6:30 \text{ PM}$

Shown elapsed:

6.5 hours

Rate:

$\frac{65}{60} = \frac{13}{12}$

Real elapsed:

$6.5 \cdot \frac{12}{13}$

$\boxed{6 \text{ hours}}$

Actual time:

$\boxed{6:00 \text{ PM}}$

26. FAULTY CLOCK CORRECT AGAIN

An analog 12-hour clock may become visually correct again after it gains or loses:

$\boxed{12 \text{ hours}}$

relative to the true clock display.

Why?

A 12-hour dial repeats every 12 hours.

27. GAINING CLOCK — TIME TO BECOME CORRECT AGAIN

Suppose it gains (g) minutes per day.

To gain a full 12 hours:

$$12 \text{ h} = 720 \text{ min}$$

Days required:

$$\boxed{\frac{720}{g}}$$

28. EXAMPLE

Clock gains:

$$10 \text{ min/day}$$

Time to appear correct again:

$$720/10$$

$$= \boxed{72 \text{ days}}$$

29. LOSING CLOCK — SAME LOGIC

If it loses (l) minutes/day:

$$\boxed{\text{days to appear correct again}} = \frac{720}{l}$$

30. 24-HOUR DIGITAL VS 12-HOUR ANALOG TRAP

The 720-minute rule assumes a standard 12-hour analog display.

A 24-hour digital clock would need:

$$1440 \text{ minutes}$$

of error to show the same displayed time again.

31. TWO FAULTY CLOCKS

Suppose one gains and one loses.

Their relative separation grows at:

$$\boxed{\text{gain rate} + \text{loss rate}}$$

if they start together.

32. EXAMPLE

Clock A gains:

$$3 \text{ min/hour}$$

Clock B loses:

$$2 \text{ min/hour}$$

Difference grows:

$$5 \text{ min/hour}$$

After 6 real hours:

$$30 \text{ minutes}$$

apart.

33. BOTH CLOCKS GAIN

If A gains 5 min/hour and B gains 2 min/hour:

Relative separation:

$$5 - 2 = 3 \text{ min/hour}$$

Use difference because both move in same error direction.

34. BOTH CLOCKS LOSE

Likewise, use difference of loss rates.

35. CLOCK RUNS AT x% OF CORRECT SPEED

Suppose clock runs at:

$$90\%$$

of correct speed.

Then:

$$\text{shown} : \text{real} = 90 : 100 = 9 : 10$$

So it loses:

$$10\%$$

of elapsed time.

36. CLOCK RUNS 20% FAST

Rate:

$$120\%$$

Therefore:

$$\boxed{\text{shown}:\text{real}=6:5}$$

If 5 real hours pass, it shows 6 elapsed hours.

37. CLOCK RUNS 25% SLOW

It runs at:

$$75\%$$

correct speed.

So:

$$\boxed{\text{shown}:\text{real}=3:4}$$

38. GAIN RATE TO SPEED PERCENTAGE

If clock gains (g) minutes/hour:

Correct hour:

$$60$$

faulty shown:

$$60+g$$

Speed percentage:

$$\boxed{\frac{60+g}{60}\times 100}$$

39. EXAMPLE

Gains 6 min/hour.

Speed percentage:

$$\frac{66}{60}\times 100$$

$$= \boxed{110\%}$$

So it is:

$$\boxed{10\% \text{ fast}}$$

40. LOSS RATE TO SPEED PERCENTAGE

Loses 12 min/hour:

Shown per real hour:

$$48$$

Speed:

$$48/60=0.8$$

$$= \boxed{80\%}$$

So clock is:

$$\boxed{20\% \text{ slow}}$$

41. FAULTY MINUTE HAND — ADVANCED

Sometimes the question gives:

Minute hand takes 72 minutes to complete one revolution.

Correct would take:

$$60 \text{ minutes}$$

Faulty speed ratio:

$$\frac{60}{72} = \boxed{\frac{5}{6}}$$

because it moves fewer degrees per real minute.

So clock runs slow.

42. WHY USE INVERSE?

If revolution time is larger, speed is smaller.

Speed:

$$\propto \frac{1}{\text{time per revolution}}$$

So:

$$\frac{v_f}{v_c} = \frac{T_c}{T_f}$$

43. EXAMPLE

Minute hand completes a revolution in 50 real minutes.

Correct time:

$$60$$

Speed ratio:

$$\frac{60}{50} = \boxed{\frac{6}{5}}$$

So it runs:

$$20\%$$

fast.

44. HAND SPEEDS IN A UNIFORMLY FAST CLOCK

If the entire clock runs 20% fast, all hand angular speeds are multiplied by:

$$1.2$$

So:

Minute hand:

$$\left[6(1.2) = 7.2^{\circ} / \text{real min} \right]$$

Hour hand:

$$\left[0.5(1.2) = 0.6^{\circ} / \text{real min} \right]$$

Their relative speed:

$$\left[6.6^{\circ} / \text{real min} \right]$$

45. BUT DISPLAYED CLOCK GEOMETRY STAYS NORMAL

At any displayed time, the hands still obey the normal geometric relation.

Faultiness affects:

how fast the display advances relative to real time.

This distinction matters.

46. MIRROR + FAULTY CLOCK

If a question combines them:

1. decode mirror display
2. determine faulty clock's indicated time
3. use gain/loss ratio to recover real time

Work in that order.

47. EXAMPLE — TWO CLOCKS

A and B are correct at noon.

A gains 2 min/hour.

B loses 3 min/hour.

When real time is 8 PM:

Elapsed:

$$\left[8 \text{ h} \right]$$

A gain:

$$\left[16 \text{ min} \right]$$

A shows:

$$\left[8:16 \right]$$

B loses:

$$\left[24 \text{ min} \right]$$

B shows:

$$\boxed{7:36}$$

Difference:

$$\boxed{40 \text{ min}}$$

Also directly:

$$(2+3) \cdot 8 = 40$$

$$\boxed{40 \text{ min}}$$

48. EXAMPLE — FIND REAL ELAPSED TIME FROM DIFFERENCE

One clock gains 4 min/hour.

Another loses 2 min/hour.

They begin together.

When they differ by 1 hour:

Relative error rate:

$$\boxed{6 \text{ min/hour}}$$

Time:

$$\boxed{60/6}$$

$$\boxed{10 \text{ hours}}$$

49. CLOCK CORRECT TWICE — EXAMPLE

A clock gains 15 min/day.

To gain 12 hours:

$$\boxed{720/15 = 48}$$

So:

$$\boxed{48 \text{ days}}$$

later it will again show the correct 12-hour dial reading.

50. FAST GAIN/LOSS FORMULA TABLE

Gains (g) min/hour

$$\boxed{\frac{\text{shown}}{\text{real}} = \frac{60+g}{60}}$$

Loses (l) min/hour

$$\boxed{\frac{\text{shown}}{\text{real}} = \frac{60-l}{60}}$$

Gains (g) min/day

$$\boxed{\frac{1440+g}{1440}}$$

Loses $\frac{1}{11}$ min/day

$$\boxed{\frac{1440 - \frac{1}{11} \times 1440}{1440}}$$

51. GATE-STYLE QUESTION 1

Mirror shows:

$$\boxed{4:25}$$

Actual:

$$\boxed{11:60 - 4:25}$$

$$\boxed{7:35}$$

52. GATE-STYLE QUESTION 2

Actual time:

$$\boxed{9:40}$$

Mirror:

$$\boxed{11:60 - 9:40}$$

$$\boxed{2:20}$$

53. GATE-STYLE QUESTION 3

A clock gains 5 min/hour.

After 8 real hours, gain:

$$5(8) = \boxed{40 \text{ min}}$$

54. GATE-STYLE QUESTION 4

A clock loses 6 min/hour.

Rate:

$$\boxed{54:60 = 9:10}$$

55. PLACEMENT-STYLE QUESTION 1

Clock correct at 6 AM, gains 3 min/hour.

At actual noon:

Elapsed:

$$\boxed{6 \text{ h}}$$

Gain:

$$\boxed{18 \text{ min}}$$

Clock shows:

$\boxed{12:18 \text{ PM}}$

56. PLACEMENT-STYLE QUESTION 2

Clock correct at 8 AM, loses 5 min/hour.

At actual 2 PM:

Elapsed:

6

Loss:

30 min

Clock shows:

$\boxed{1:30 \text{ PM}}$

57. PLACEMENT-STYLE QUESTION 3

A clock gains 12 minutes every 6 hours.

Gain per hour:

2 min

In 15 hours:

30 min

$\boxed{30 \text{ min gain}}$

58. PLACEMENT-STYLE QUESTION 4

Clock loses 24 min/day.

Time to appear correct again:

$720/24$

$= \boxed{30 \text{ days}}$

59. ADVANCED QUESTION — SHOWN TO REAL

Clock gains 20%.

It shows elapsed time of:

6 hours

Real elapsed:

$6/1.2$

$= \boxed{5 \text{ hours}}$

60. ADVANCED QUESTION — SLOW CLOCK

Clock runs at 80% speed.

After 10 real hours, it shows:

$$\boxed{0.8(10)}$$

$$\boxed{8 \text{ elapsed hours}}$$

Loss:

$$\boxed{2 \text{ hours}}$$

61. COMMON TRAPS

& pTRAP 1 — Mirror time

Use:

$$\boxed{11:60 - \text{time}}$$

not $\boxed{12:00 - \text{time}}$.

& pTRAP 2 — Gain/loss rate

If clock gains 5 min/hour:

In 1 real hour it shows 65 min.

Do not say it runs at $\boxed{5/60}$ speed.

& pTRAP 3 — Shown vs real elapsed

Identify which one is given before multiplying or dividing by the rate.

& pTRAP 4 — “Correct again”

For standard analog 12-hour dial, accumulated error needed is:

$$\boxed{720 \text{ min}}$$

& pTRAP 5 — Two faulty clocks

If one gains and one loses, relative error rates add.

& pTRAP 6 — Revolution-time questions

Speed is inverse of revolution time.

62. MUST-MEMORIZE MIRROR RULE

$$\boxed{\text{Actual} + \text{Mirror} = 11:60}$$

63. MUST-MEMORIZE FAULTY CLOCK RULE

Gain (g) min/hour:

$$\boxed{\text{shown}:\text{real}} = (60+g):60$$

Loss (l) min/hour:

$$\boxed{(60-l):60}$$

64. MUST-MEMORIZE CORRECT-AGAIN RULE

Standard 12-hour analog clock:

$$\boxed{\text{time to correct again}} = \frac{720}{\text{gain/loss per unit time}}$$

Use matching units.

65. FAST EXAM DECISION TREE

Mirror clock?

$$11:60$$

Clock correct at start + actual later time?

Calculate accumulated gain/loss directly.

Faulty displayed time given, actual required?

Use shown:real ratio.

Runs $x\%$ fast/slow?

Convert to speed multiplier.

Two clocks?

Compare their error rates.

“Correct again”?

Need 12-hour accumulated error.

Hand takes abnormal revolution time?

Use inverse-speed ratio.

PRACTICE — LEVEL 0

Q1

Mirror of 3:15?

Q2

Mirror of 10:20?

Q3

Clock gains 4 min/hour. How much gain in 5 hours?

Q4

Clock loses 3 min/hour. How much loss in 8 hours?

Q5

A clock gains 6 min/hour. Find shown:real ratio.

LEVEL 1

Q6

Clock correct at 7 AM, gains 2 min/hour. What does it show at actual 1 PM?

Q7

Clock correct at noon, loses 4 min/hour. What does it show at actual 6 PM?

Q8

A clock gains 5 min/hour and shows 13 hours elapsed. Find real elapsed time.

Q9

A clock runs 25% fast. In 8 real hours, how many hours does it show?

Q10

A clock runs at 80% speed. After 15 real hours, how much elapsed time does it show?

LEVEL 2

Q11

A clock gains 18 min/day. After how many days will it appear correct again?

Q12

One clock gains 4 min/hour, another loses 3 min/hour. How long until they differ by 49 minutes?

Q13

A clock is correct at 9 AM and gains 10 min/hour. When it shows 4 PM, what is actual time?

Q14

A minute hand completes one revolution in 48 real minutes. By what percentage is the clock fast?

Q15

A clock loses 15 min every 5 hours. Find its speed as a percentage of correct speed.

LEVEL 3 — EXAM PRESSURE

Q16

A clock is correct at 6 AM and gains 6 min/hour. When it shows 7 PM, what is the actual time?

Q17

Two clocks start correct together. A gains 2 min/hour, B gains 5 min/hour. After how many real hours will B be 45 minutes ahead of A?

Q18

A clock loses 20 min/day. After how many days will its 12-hour dial next show the correct reading?

Q19

A clock runs at $\frac{5}{6}$ of normal speed. It is correct at noon. What will it show at actual midnight?

Q20

A mirror shows 8:35 on a clock that gains 5 min/hour. The clock was correct at noon. Find actual real time after noon corresponding to this observation, assuming the indicated time belongs to the same 12-hour cycle.

ANSWERS

Q1

$$\lfloor 11:60 - 3:15 = \boxed{8:45} \rfloor$$

Q2

$$\lfloor 11:60 - 10:20 = \boxed{1:40} \rfloor$$

Q3

$$\lfloor 4(5) = \boxed{20 \text{ min}} \rfloor$$

Q4

$$\lfloor 3(8) = \boxed{24 \text{ min}} \rfloor$$

Q5

$$\lfloor 66:60 = \boxed{11:10} \rfloor$$

Q6

Elapsed:

$$\lfloor 6 \text{ h} \rfloor$$

Gain:

$$\lfloor 12 \text{ min} \rfloor$$

Display:

$$\lfloor \boxed{1:12 \text{ PM}} \rfloor$$

Q7

Elapsed:

$$\lfloor 6 \text{ h} \rfloor$$

Loss:

$$\lfloor 24 \text{ min} \rfloor$$

Display:

$$\lfloor \boxed{5:36 \text{ PM}} \rfloor$$

Q8

Rate:

$$\lfloor 65:60=13:12 \rfloor$$

Real:

$$\lfloor 13 \cdot \frac{12}{13} \rfloor$$

$$\lfloor = \boxed{12 \text{ h}} \rfloor$$

Q9

Rate:

$$\lfloor 1.25 \rfloor$$

Shown:

$$\lfloor 8(1.25) = \boxed{10 \text{ h}} \rfloor$$

Q10

$$\lfloor 15(0.8) = \boxed{12 \text{ h}} \rfloor$$

Q11

$$\lfloor 720/18 = \boxed{40 \text{ days}} \rfloor$$

Q12

Relative difference:

$$\lfloor 4+3=7 \text{ min/hour} \rfloor$$

Time:

$$\lfloor 49/7 = \boxed{7 \text{ h}} \rfloor$$

Q13

Displayed elapsed from 9 AM to 4 PM:

$$\lfloor 7 \text{ h} \rfloor$$

Gain 10 min/hour means shown:real:

$$\lfloor 70:60=7:6 \rfloor$$

Real elapsed:

$$\lfloor 7 \cdot \frac{6}{7} = 6 \text{ h} \rfloor$$

Actual time:

$$\lfloor \boxed{3 \text{ PM}} \rfloor$$

Q14

Correct revolution:

$\backslash[60\text{min}] \backslash$

Faulty:

$\backslash[48\text{min}] \backslash$

Speed ratio:

$\backslash[60/48=1.25] \backslash$

So:

$\backslash[\boxed{25\%\text{fast}}] \backslash$

Q15

Loss:

$\backslash[15\text{min per }300\text{real min}] \backslash$

Shown:

$\backslash[285] \backslash$

Speed:

$\backslash[285/300 = 0.95] \backslash$

$\backslash[\boxed{95\%}] \backslash$

So it is:

$\backslash[\boxed{5\%\text{slow}}] \backslash$

Q16

From 6 AM to displayed 7 PM:

Shown elapsed:

$\backslash[13\text{h}] \backslash$

Clock gains 6 min/hour:

$\backslash[66:60=11:10] \backslash$

Real elapsed:

$\backslash[13\cdot\frac{10}{11} = \frac{130}{11} = 11\frac{9}{11}\text{h}] \backslash$

That is:

$\backslash[11\text{h}49\frac{1}{11}\text{min}] \backslash$

approximately:

$\backslash[11\text{h}49\text{min}5.45\text{s}] \backslash$

Actual time:

$\backslash[\boxed{\approx 5:49:05\text{PM}}] \backslash$

Q17

Relative gain:

$\backslash[5-2=3\text{min/hour}] \backslash$

Time:

$$\boxed{45/3 = 15 \text{ h}}$$

Q18

$$\boxed{720/20 = 36 \text{ days}}$$

Q19

Real elapsed noon !' midnight:

$$12 \text{ h}$$

Shown elapsed:

$$12 \cdot \frac{56}{60} = 10 \text{ h}$$

So it shows:

$$\boxed{10:00 \text{ PM}}$$

Q20

First decode mirror.

Mirror shows:

$$8:35$$

Actual indicated clock reading:

$$11:60 - 8:35 = \boxed{3:25}$$

Since the faulty clock was correct at noon, shown elapsed is:

$$3 \text{ h } 25 \text{ min} = 205 \text{ min}$$

Clock gains 5 min/hour:

$$\text{shown} : \text{real} = 65:60 = 13:12$$

Real elapsed:

$$205 \cdot \frac{12}{13}$$

$$= 15.7692 \cdot 12$$

$$= \boxed{\frac{2460}{13} \text{ min}}$$

$$\approx 189.23 \text{ min}$$

That is approximately:

$$3 \text{ h } 9 \text{ min } 14 \text{ s}$$

after noon.

Therefore actual real time:

$$\boxed{\approx 3:09:14 \text{ PM}}$$

' CHAPTER 48 COMPLETE

We have now covered:

- hour/minute-hand speeds '
- relative speed '

- angle-between-hands formula '
- coincidence '
- opposite hands '
- right-angle positions '
- occurrence counts '
- mirror images '
- gaining clocks '
- losing clocks '
- faulty-clock rate ratios '
- fast/slow percentage clocks '
- correct-again problems '
- two faulty clocks '
- abnormal revolution times '
- advanced mixed questions '
- GATE/placement L0–L3 practice '

Progress

Completed: 48/76

Unit 1 — Quantitative Aptitude: 0=100% PROGRESS

Only **one chapter** remains in Unit 1.

NEXT

Chapter 49 — Calendars

We will cover:

- ordinary and leap years
- odd days
- day-of-week calculations
- month codes/mental shortcuts
- repeated calendars
- day shifts across years
- counting Sundays/weekdays
- leap-year traps
- date differences
- placement-style calendar puzzles

After Chapter 49:

`\boxed{\text{UNIT 1 — QUANTITATIVE APTITUDE COMPLETE}}`

and then we move to:

Unit 2 — Data Interpretation, Chapter 50 — DI Foundations and Reading Data

USER

next

ASSISTANT

CHAPTER 49 — CALENDARS

COMPLETE — LEAP YEARS, ODD DAYS, DAY-OF-WEEK, REPEATED CALENDARS & DATE COUNTING

Status: $\emptyset=14$ PROGRESS GATE: Low–Medium Placements: Medium Priority: $\emptyset=10$ or 3

Calendar questions are mainly about:

How many days remain after removing complete weeks?

Since:

$$\lfloor 7 \text{ days} \rfloor = 1 \text{ week}$$

only the remainder modulo 7 matters.

That remainder is called **odd days**.

1. ORDINARY YEAR

An ordinary year has:

$$\boxed{365 \text{ days}}$$

Now:

$$365 = 52 \times 7 + 1$$

So an ordinary year contributes:

$$\boxed{1 \text{ odd day}}$$

2. LEAP YEAR

A leap year has:

$$\boxed{366 \text{ days}}$$

$$366 = 52 \times 7 + 2$$

So a leap year contributes:

$$\boxed{2 \text{ odd days}}$$

MUST MEMORIZE

3. LEAP-YEAR RULE

A year is a leap year if divisible by 4.

But century years need a special rule.

A century year is leap only if divisible by:

$\boxed{400}$

So:

- 2024 !' leap '
 - 2028 !' leap '
 - 1900 !' not leap 'L
 - 2000 !' leap '
 - 2100 !' not leap 'L
-

4. FAST LEAP-YEAR TEST

Non-century year

Divisible by:

$\boxed{4}$

Century year

Divisible by:

$\boxed{400}$

5. MONTH DAYS

31 days

$\boxed{\text{Jan, Mar, May, Jul, Aug, Oct, Dec}}$

Memory:

J M M J A O D

30 days

$\boxed{\text{Apr, Jun, Sep, Nov}}$

February

Ordinary:

$\boxed{28}$

Leap:

$\boxed{29}$

6. ODD DAYS IN A MONTH

We only need days modulo 7.

31-day month

$$\lfloor 31=28+3 \rfloor$$

Odd days:

$$\lfloor \boxed{3} \rfloor$$

30-day month

$$\lfloor 30=28+2 \rfloor$$

Odd days:

$$\lfloor \boxed{2} \rfloor$$

February ordinary

$$\lfloor 28 \rfloor$$

Odd days:

$$\lfloor \boxed{0} \rfloor$$

February leap

$$\lfloor 29 \rfloor$$

Odd days:

$$\lfloor \boxed{1} \rfloor$$

7. DAY SHIFT AFTER ONE YEAR

If a date is in an ordinary year and crosses into next year:

Day shifts by:

$$\lfloor \boxed{1} \rfloor$$

If leap day is included in the interval:

Shift may be:

$$\lfloor \boxed{2} \rfloor$$

Example:

If Jan 1, 2025 is Wednesday, then Jan 1, 2026 is:

$$\lfloor \boxed{\text{Thursday}} \rfloor$$

because 2025 is ordinary.

8. DAY SHIFT AFTER LEAP YEAR

If Jan 1 of a leap year is Monday:

Next Jan 1 is:

$\backslash [\text{Monday} + 2 \backslash]$

$\backslash [\boxed{\text{Wednesday}} \backslash]$

9. WEEKDAY NUMBER SYSTEM

For calculations, choose:

$\backslash [\boxed{0 = \text{Sunday}} \backslash]$

$\backslash [1 = \text{Monday} \backslash]$

$\backslash [2 = \text{Tuesday} \backslash]$

$\backslash [3 = \text{Wednesday} \backslash]$

$\backslash [4 = \text{Thursday} \backslash]$

$\backslash [5 = \text{Friday} \backslash]$

$\backslash [6 = \text{Saturday} \backslash]$

Then work modulo:

$\backslash [7 \backslash]$

10. BASIC ODD-DAY METHOD

Suppose a known date is Monday.

If another date is 10 days later:

$\backslash [10 \bmod 7 = 3 \backslash]$

Move 3 days:

Monday !' Tuesday !' Wednesday !' Thursday

So:

$\backslash [\boxed{\text{Thursday}} \backslash]$

11. IMPORTANT COUNTING TRAP

If today is Monday, then:

- 1 day later = Tuesday
- 7 days later = Monday

So for "after $\backslash (n) \backslash$ days":

$\backslash [\boxed{n \bmod 7} \backslash]$

Do not count starting day as day 1 unless wording explicitly asks inclusive counting.

12. FIND DAY AFTER N DAYS

If today = Friday.

What day after 100 days?

$$\lfloor 100 \bmod 7 = 2 \rfloor$$

Friday + 2:

$$\lfloor \boxed{\text{Sunday}} \rfloor$$

13. FIND DAY BEFORE N DAYS

Today = Wednesday.

100 days before:

$$\lfloor 100 \bmod 7 = 2 \rfloor$$

Go back 2:

Wednesday !' Tuesday !' Monday

$$\lfloor \boxed{\text{Monday}} \rfloor$$

14. DAY-OF-WEEK WITHIN SAME YEAR

If day of Jan 1 is known:

1. count days before target month
2. add date-1
3. reduce modulo 7
4. shift weekday

Why date $\lfloor -1 \rfloor$?

Because Jan 1 itself requires zero shift.

15. EXAMPLE

Suppose Jan 1 is Monday.

Find Jan 10.

Days after Jan 1:

$$\lfloor 10 - 1 = 9 \rfloor$$

$$\lfloor 9 \bmod 7 = 2 \rfloor$$

Monday + 2:

$$\lfloor \boxed{\text{Wednesday}} \rfloor$$

16. EXAMPLE — FEBRUARY DATE

Suppose Jan 1 is Sunday in an ordinary year.

Find Feb 1.

January contributes:

$\lfloor 31 \bmod 7 = 3 \rfloor$

Sunday + 3:

$\lfloor \boxed{\text{Wednesday}} \rfloor$

17. CUMULATIVE DAYS BEFORE EACH MONTH — ORDINARY YEAR

Useful table:

Month	Days before month	Mod 7
Jan	0	0
Feb	31	3
Mar	59	3
Apr	90	6
May	120	1
Jun	151	4
Jul	181	6
Aug	212	2
Sep	243	5
Oct	273	0
Nov	304	3
Dec	334	5

You do not need to memorize the full raw-day column.

18. MONTH OFFSET CODE — ORDINARY YEAR

A useful month offset from Jan 1:

$\lfloor \boxed{0,3,3,6,1,4,6,2,5,0,3,5} \rfloor$

for:

Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec.

Then:

$\lfloor \boxed{\text{weekday shift} = \text{month code} + (d-1)} \rfloor$

mod 7.

19. LEAP-YEAR MONTH OFFSET

For dates after February in a leap year:

add:

$\boxed{1}$

to the ordinary-year month offset.

Because Feb has one extra day.

20. EXAMPLE — LEAP YEAR

Jan 1 is Monday in leap year.

Find March 1.

January:

31

February:

29

Total:

60

$60 \bmod 7 = 4$

Monday + 4:

$\boxed{\text{Friday}}$

21. ORDINARY-YEAR COMPARISON

If same year were ordinary:

$31 + 28 = 59$

$59 \bmod 7 = 3$

March 1 would be:

$\boxed{\text{Thursday}}$

Leap day shifts dates after Feb by one extra day.

22. ODD DAYS IN 100 YEARS

In 100 years:

Leap years normally:

$\left\lfloor \frac{100}{4} \right\rfloor = 25$

But if the century ending year isn't divisible by 400, one of those is not leap.

For a typical century block such as 1801–1900:

$\lfloor 24 \text{ leap years} \rfloor$

and:

$\lfloor 76 \text{ ordinary years} \rfloor$

Odd days:

$\lfloor 24(2) + 76(1) = 124 \rfloor$

$\lfloor 124 \bmod 7 = 5 \rfloor$

So:

$\lfloor \boxed{100 \text{ years}} = 5 \text{ odd days} \rfloor$

for such a Gregorian century block.

23. ODD DAYS IN 200 YEARS

$\lfloor 5 + 5 = 10 \rfloor$

$\lfloor 10 \bmod 7 = 3 \rfloor$

$\lfloor \boxed{3} \rfloor$

24. ODD DAYS IN 300 YEARS

$\lfloor 15 \bmod 7 = \boxed{1} \rfloor$

25. ODD DAYS IN 400 YEARS

Gregorian calendar repeats every 400 years.

Total odd days:

$\lfloor \boxed{0} \rfloor$

because exactly:

$\lfloor 146097 \rfloor$

days occur in 400 years and:

$\lfloor 146097 \rfloor$

is divisible by 7.

MUST KNOW

26. 400-YEAR CYCLE

Therefore:

$\lfloor \boxed{\text{calendar pattern repeats every 400 years}} \rfloor$

This is a deep calendar shortcut.

27. FIND DAY OF A DATE — GENERAL METHOD

For a date:

$\lfloor d/m/y \rfloor$

A robust method is:

1. count complete years before $\lfloor y \rfloor$
2. add leap years among them
3. add month offsets
4. add $\lfloor d-1 \rfloor$
5. reduce mod 7
6. map to weekday using a known reference

For placement questions, usually a reference date is given, so full historical calculation is often unnecessary.

28. COUNT LEAP YEARS UP TO YEAR Y

Number of leap years from year 1 through year $\lfloor Y \rfloor$:

$\lfloor \boxed{\left\lfloor \frac{Y}{4} \right\rfloor - \left\lfloor \frac{Y}{100} \right\rfloor + \left\lfloor \frac{Y}{400} \right\rfloor} \rfloor$

High-value formula

29. LEAP YEARS BEFORE YEAR Y

For complete years before year $\lfloor Y \rfloor$, use:

$\lfloor Y-1 \rfloor$

So:

$\lfloor \boxed{L = \left\lfloor \frac{Y-1}{4} \right\rfloor - \left\lfloor \frac{Y-1}{100} \right\rfloor + \left\lfloor \frac{Y-1}{400} \right\rfloor} \rfloor$

30. DAYS FROM COMPLETE YEARS

If there are:

- $\lfloor L \rfloor$ leap years
- $\lfloor N-L \rfloor$ ordinary years

Total odd days:

$\lfloor (N-L)+2L \rfloor$

$\lfloor =N+L \rfloor$

So mod 7:

$\lfloor \boxed{N+L \pmod{7}} \rfloor$

Very useful simplification.

31. SAME DATE NEXT YEAR

For a date after Feb 29, the weekday shift depends on whether the interval crosses Feb 29.

General practical rule:

- ordinary-year interval !' +1
 - interval containing Feb 29 !' +2
-

32. EXAMPLE

March 10, 2023 is Friday.

What day is March 10, 2024?

The interval crosses Feb 29, 2024.

So shift:

$\lfloor +2 \rfloor$

Friday +2:

$\lfloor \boxed{\text{Sunday}} \rfloor$

33. SAME DATE FROM LEAP YEAR TO NEXT YEAR

March 10, 2024 !' March 10, 2025.

The interval after March 10, 2024 does not include another Feb 29.

2024's Feb 29 is already before the starting date.

So shift:

$\lfloor +1 \rfloor$

Sunday !' Monday.

$\lfloor \boxed{\text{Monday}} \rfloor$

34. REPEATED CALENDARS

Two years have the same calendar if:

1. both are same type:
 - both ordinary
 - or both leap
 2. Jan 1 falls on same weekday
-

35. WHEN DOES A CALENDAR REPEAT?

You accumulate annual shifts until total shift is divisible by 7, while matching year type.

Common repeat gaps include:

$\lfloor \boxed{6, 11, 28} \rfloor$

years depending on leap-year positions.

Do not blindly memorize one gap.

36. WHY 28 YEARS OFTEN APPEARS

Ignoring century exceptions:

$$\text{LCM}(7,4)=28$$

because:

- weekdays cycle every 7
- leap pattern every 4

So the calendar often repeats after:

$$28 \text{ years}$$

But century rules can break this.

37. ORDINARY YEAR CAN HAVE 53 SUNDAYS WHEN?

An ordinary year has:

$$365=52 \text{ weeks} + 1 \text{ day}$$

So exactly one weekday appears 53 times.

It is the weekday of Jan 1.

Therefore an ordinary year has 53 Sundays iff Jan 1 is:

$$\text{Sunday}$$

38. LEAP YEAR — 53 WEEKDAYS

Leap year:

$$366=52 \text{ weeks} + 2 \text{ days}$$

So two consecutive weekdays occur 53 times:

- weekday of Jan 1
 - next weekday
-

39. LEAP YEAR WITH 53 SUNDAYS

Sunday must be one of those two extra days.

So Jan 1 must be:

$$\text{Saturday or Sunday}$$

40. LEAP YEAR WITH 53 MONDAYS

Jan 1 must be:

$\boxed{\text{Sunday or Monday}}$

General rule follows immediately.

41. 53 SUNDAYS AND 53 MONDAYS TOGETHER

Only possible in leap year.

The two extra weekdays must be:

Sunday and Monday.

Therefore Jan 1 must be:

$\boxed{\text{Sunday}}$

42. NUMBER OF A PARTICULAR WEEKDAY IN A MONTH

A 28-day month has exactly:

$\boxed{4}$

of every weekday.

Extra days determine which weekdays occur 5 times.

43. 31-DAY MONTH

$\boxed{31=28+3}$

So three consecutive weekdays occur:

$\boxed{5 \text{ times}}$

These are:

- weekday of 1st
 - next weekday
 - next one
-

44. 30-DAY MONTH

$\boxed{30=28+2}$

Two consecutive weekdays occur:

$\boxed{5 \text{ times}}$

45. 29-DAY FEBRUARY

One weekday occurs:

$\boxed{5}$

times: weekday of Feb 1.

46. 28-DAY FEBRUARY

Every weekday occurs exactly:

$\boxed{4 \text{ times}}$

47. EXAMPLE — 31-DAY MONTH

If May 1 is Tuesday:

Extra days are:

Tuesday, Wednesday, Thursday.

Therefore these occur 5 times.

Sunday occurs:

$\boxed{4 \text{ times}}$

48. COUNTING SUNDAYS IN DATE RANGE

For a long interval:

1. count complete weeks
2. inspect remainder days

Example:

In 45 consecutive days:

$45 = 6 \times 7 + 3$

Every weekday occurs at least 6 times.

Three consecutive weekdays occur 7 times.

Which three depends on starting weekday.

49. EXAMPLE

45-day interval begins Monday.

Extra 3 days:

Monday, Tuesday, Wednesday.

So:

- Mon/Tue/Wed! 7 times
 - others ! 6 times
-

50. DATE DIFFERENCE — EXCLUSIVE ELAPSED DAYS

From March 10 to March 15:

Elapsed time is:

$$\lfloor 15-10=5 \rfloor$$

days.

This excludes the starting date.

51. DATE COUNTING — INCLUSIVE DAYS

If asked:

How many calendar dates from March 10 through March 15 inclusive?

Count:

$$\lfloor 15-10+1 \rfloor$$

$$\lfloor \boxed{6} \rfloor$$

Important

Elapsed days and inclusive date count are different.

52. ACROSS MONTHS

Example:

From Jan 28 to Feb 5 in ordinary year.

Elapsed days:

Remaining after Jan 28 to Jan 31:

$$\lfloor 31-28=3 \rfloor$$

then 5 days into February:

$$\lfloor 3+5=8 \rfloor$$

So:

$$\lfloor \boxed{8 \text{ days}} \rfloor$$

53. BIRTHDAY WEEKDAY SHIFT

Birthdays generally shift:

- +1 weekday after an ordinary interval
- +2 if Feb 29 is crossed

For a Feb 29 birthday, wording can become special, so calculate actual dates rather than using intuition.

54. GATE-STYLE QUESTION 1

If today is Tuesday, what day after 100 days?

$$\lfloor 100 \bmod 7 = 2 \rfloor$$

Tuesday +2:

$$\lfloor \boxed{\text{Thursday}} \rfloor$$

55. GATE-STYLE QUESTION 2

If Jan 1 is Wednesday in an ordinary year, what day is Feb 1?

Shift:

$$\lfloor 31 \bmod 7 = 3 \rfloor$$

Wednesday +3:

$$\lfloor \boxed{\text{Saturday}} \rfloor$$

56. GATE-STYLE QUESTION 3

If Jan 1 is Monday in leap year, what day is March 1?

Shift:

$$\lfloor 31 + 29 = 60 \rfloor$$

$$\lfloor 60 \bmod 7 = 4 \rfloor$$

Monday +4:

$$\lfloor \boxed{\text{Friday}} \rfloor$$

57. GATE-STYLE QUESTION 4

Is 2100 leap?

Century year.

$$\lfloor 2100 \not\equiv 0 \pmod{400} \rfloor$$

Therefore:

$$\lfloor \boxed{\text{No}} \rfloor$$

58. PLACEMENT-STYLE QUESTION 1

An ordinary year begins on Sunday. How many Sundays?

$$\lfloor \boxed{53} \rfloor$$

59. PLACEMENT-STYLE QUESTION 2

A leap year begins on Saturday.

Extra days:

Saturday, Sunday.

So Sundays:

$\boxed{53}$

60. PLACEMENT-STYLE QUESTION 3

A 31-day month starts Friday.

Which weekdays occur 5 times?

First three:

$\boxed{\text{Friday, Saturday, Sunday}}$

61. PLACEMENT-STYLE QUESTION 4

A 30-day month begins Wednesday.

5-time weekdays:

$\boxed{\text{Wednesday and Thursday}}$

62. COMMON TRAPS

& pTRAP 1 — Every year divisible by 4 is leap

Century exception exists.

& pTRAP 2 — 1900

Not leap.

& pTRAP 3 — 2000

Leap.

& pTRAP 4 — Date shift around leap year

Check whether Feb 29 lies inside the interval.

& pTRAP 5 — Inclusive counting

If both start and end dates are included:

$\boxed{+1}$

& pTRAP 6 — Calendar repeat

Same Jan 1 weekday alone is not enough; year type must also match.

63. MUST-MEMORIZE YEAR ODD DAYS

Ordinary year:

$\boxed{1}$

Leap year:

$\boxed{2}$

64. MUST-MEMORIZE LEAP RULE

$\boxed{\text{divisible by } 4}$

except century years, which must be divisible by:

$\boxed{400}$

65. MUST-MEMORIZE MONTH ODD DAYS

31 days:

$\boxed{3}$

30 days:

$\boxed{2}$

28 days:

$\boxed{0}$

29 days:

$\boxed{1}$

66. MUST-MEMORIZE 400-YEAR FACT

$\boxed{400 \text{ years} = 0 \text{ odd days}}$

Gregorian weekday cycle repeats over 400 years.

67. FAST EXAM DECISION TREE

“After N days”?

Use:

$N \bmod 7$

Across years?

Count ordinary/leap-year odd days.

Date after Feb in leap year?

Remember extra leap day.

53 Sundays?

Look at extra 1 or 2 days of the year.

Same calendar?

Need:

- same Jan 1 weekday
- same leap/ordinary type

Date interval?

Check whether inclusive or elapsed.

PRACTICE — LEVEL 0

Q1

How many odd days in ordinary year?

Q2

How many odd days in leap year?

Q3

Is 2400 a leap year?

Q4

Is 2300 a leap year?

Q5

If today is Monday, what day after 45 days?

LEVEL 1

Q6

Jan 1 is Tuesday in ordinary year. What day is March 1?

Q7

Jan 1 is Tuesday in leap year. What day is March 1?

Q8

A 31-day month starts Monday. Which weekdays occur 5 times?

Q9

A 30-day month starts Saturday. Which weekdays occur 5 times?

Q10

An ordinary year begins Friday. How many Fridays occur?

LEVEL 2

Q11

A leap year begins Sunday. How many Sundays and Mondays occur?

Q12

If March 1, 2023 is Wednesday, what day is March 1, 2024?

Q13

If March 1, 2024 is Friday, what day is March 1, 2025?

Q14

How many leap years are there from 2001 through 2100 inclusive?

Q15

A 45-day period starts Thursday. Which weekdays occur 7 times?

LEVEL 3 — EXAM PRESSURE

Q16

An ordinary year has 53 Sundays. On which weekday must Jan 1 fall?

Q17

A leap year has 53 Fridays and 53 Saturdays. On which weekday does Jan 1 fall?

Q18

If Jan 1 is Monday in an ordinary year, what day is Dec 31?

Q19

If Jan 1 is Monday in a leap year, what day is Dec 31?

Q20

How many days are there from March 15 through April 10 inclusive?

ANSWERS

Q1

$\boxed{1}$

Q2

$\boxed{2}$

Q3

2400 is divisible by 400.

$\boxed{\text{Leap year}}$

Q4

2300 is century year but not divisible by 400.

$\boxed{\text{Not leap}}$

Q5

$45 \bmod 7 = 3$

Monday +3:

$\boxed{\text{Thursday}}$

Q6

Days before March:

$31 + 28 = 59$

$59 \bmod 7 = 3$

Tuesday +3:

$\boxed{\text{Friday}}$

Q7

Leap year:

$31 + 29 = 60$

$60 \bmod 7 = 4$

Tuesday +4:

$\boxed{\text{Saturday}}$

Q8

31 days = 4 weeks + 3 days.

Starting Monday:

$\boxed{\text{Monday, Tuesday, Wednesday}}$

occur 5 times.

Q9

30 days = 4 weeks + 2 days.

Starting Saturday:

$\boxed{\text{Saturday and Sunday}}$

occur 5 times.

Q10

Ordinary year has one extra weekday: Jan 1's weekday.

So:

$\boxed{53 \text{ Fridays}}$

Q11

Leap year extra weekdays are Sunday and Monday.

Thus:

$\boxed{53 \text{ Sundays and } 53 \text{ Mondays}}$

Q12

From March 1, 2023 to March 1, 2024, Feb 29, 2024 is crossed.

Shift:

$+2$

Wednesday +2:

$\boxed{\text{Friday}}$

Q13

March 1, 2024 !' March 1, 2025 does not cross another Feb 29.

Shift:

$+1$

Friday +1:

$\boxed{\text{Saturday}}$

Q14

From 2001 through 2100:

Multiples of 4:

$2004, 2008, \dots, 2096$

Count:

$\frac{2096-2004}{4} + 1 = 24$

2100 is not leap.

So:

$\boxed{24}$

Q15

$45 = 42 + 3$

Three extra weekdays beginning Thursday:

$\boxed{\text{Thursday, Friday, Saturday}}$

occur 7 times.

Q16

Ordinary year's extra weekday is Jan 1.

So:

$\boxed{\text{Sunday}}$

Q17

Leap year has two consecutive extra weekdays.

For Friday and Saturday to both occur 53 times:

$\boxed{\text{Jan 1 = Friday}}$

Q18

An ordinary year has 365 days.

Dec 31 is 364 days after Jan 1.

$364 \bmod 7 = 0$

So:

$\boxed{\text{Monday}}$

Q19

Leap year:

Dec 31 is 365 days after Jan 1.

$365 \bmod 7 = 1$

Monday +1:

$\boxed{\text{Tuesday}}$

Q20

Inclusive count:

March 15 through March 31:

$31 - 15 + 1 = 17$

April 1 through April 10:

10

Total:

$17 + 10 = \boxed{27}$

' CHAPTER 49 COMPLETE

We have now covered:

- ordinary/leap years '
- leap-year century rule '
- odd days '
- month odd days '
- weekday shifts '
- same-date year shifts '
- month offsets '
- 400-year cycle '
- leap-year counting '
- repeated calendars '
- 53-weekday questions '
- weekday counts in months '
- date differences '

- inclusive/exclusive counting '
- GATE/placement L0–L3 practice '

' UNIT 1 — QUANTITATIVE APTITUDE COMPLETE

We have completed:

$\boxed{\text{Chapters 13–49}}$

That includes:

- Ratio/Proportion
- Variation
- Percentages
- Averages
- Profit/Loss
- SI/CI
- Mixtures
- Partnership/Ages
- Time & Work
- Pipes
- TSD
- Trains/Boats/Races
- Algebra
- Sequences
- Counting
- Permutations/Combinations
- Probability
- Sets/Venn
- Geometry
- 2D/3D Mensuration
- Clocks
- Calendars

Overall Progress

Completed: $\boxed{49/76}$

UNIT 0 — Foundations

' COMPLETE

UNIT 1 — Quantitative Aptitude

' COMPLETE

UNIT 2 — Data Interpretation

+ NOT STARTED

UNIT 3 — Logical & Analytical Reasoning

+ NOT STARTED

UNIT 4 — Verbal Aptitude

+ NOT STARTED

Units 5–8

+ Special training, revision, question bank and mocks still pending

NEXT

Chapter 50 — Data Interpretation Foundations and Reading Data

Next we begin DI with:

- how to read tables/graphs correctly
- absolute vs relative values
- totals/subtotals
- percentage share
- ratio comparisons
- percentage increase/decrease
- average from charts
- approximation
- estimation
- avoiding misleading graph scales
- missing values
- multi-step DI
- fastest GATE/placement calculation strategy.

USER

next

ASSISTANT

CHAPTER 50 — DATA INTERPRETATION FOUNDATIONS AND READING DATA

PART 1 — TABLE/GRAPH READING, TOTALS, PERCENTAGES, RATIOS, AVERAGES, GROWTH & ESTIMATION

Status: ~~0~~ = No PROGRESS GATE: High Placements: High Priority: ~~0~~ = Yes 1

Data Interpretation, or **DI**, means:

Reading numerical information from tables, charts, graphs or caselets and answering questions accurately and quickly.

DI usually does **not** require new mathematics.

It mainly combines topics you already learned:

- percentages

- ratios
- averages
- weighted averages
- profit/loss
- growth
- approximation
- mental arithmetic

The real difficulty is:

Reading the correct data and choosing the fastest calculation.

1. DI MASTER PROCESS

For every DI question:

Step 1 — Read the title

Understand what the data represents.

Step 2 — Read units

Check whether values are:

- numbers
- thousands
- lakhs
- millions
- percentages
- ¹ crore
- tonnes
- kg
- students etc.

Step 3 — Read rows/columns/legend

Step 4 — Read the question

Do **not** calculate everything.

Step 5 — Extract only required numbers

Step 6 — choose formula

Step 7 — estimate before exact calculation

Step 8 — sanity-check answer

2. MOST IMPORTANT DI RULE

Never start calculating before understanding the unit and denominator.

Example:

A table says:

Revenue (₹crore)

and value:

$\boxed{25}$

This means:

$\boxed{25 \text{ crore}}$

not 25.

3. SAMPLE TABLE

Consider:

Year	Company A	Company B	Company C
2022	120	150	100
2023	150	180	140
2024	180	210	160
2025	210	240	200

Assume units are:

$\boxed{1 \text{ crore}}$

We will use this table repeatedly.

4. DIRECT LOOKUP QUESTIONS

Example:

What was Company B's revenue in 2024?

Read directly:

$\boxed{210}$

No calculation.

A common exam mistake is doing unnecessary arithmetic on direct-read questions.

5. ROW TOTAL

For 2023:

$\boxed{150+180+140}$

$$\boxed{470}$$

This is total revenue of all three companies in 2023.

6. COLUMN TOTAL

Company A total over four years:

$$120+150+180+210$$

$$\boxed{660}$$

7. DIFFERENCE

Difference between Company B and C in 2025:

$$240-200$$

$$\boxed{40}$$

8. ABSOLUTE CHANGE

Company A from 2022 to 2025:

$$210-120$$

$$\boxed{90}$$

Absolute change always means:

$$\boxed{\text{new} - \text{old}}$$

9. PERCENTAGE CHANGE

Formula:

$$\boxed{\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100}$$

Denominator = original value

This is one of the biggest DI traps.

10. EXAMPLE

Company A:

2022:

$$120$$

2023:

$$150$$

Increase:

$$\boxed{30}$$

Percentage increase:

$$\frac{30}{120} \times 100$$

$$\boxed{25\%}$$

11. PERCENTAGE DECREASE

Suppose value falls from:

$$200 \text{ to } 150$$

Decrease:

$$50$$

Percentage decrease:

$$\frac{50}{200} \times 100$$

$$\boxed{25\%}$$

12. PERCENTAGE-CHANGE ASYMMETRY

From 100 to 120:

$$20\% \text{ increase}$$

But from 120 back to 100:

$$\frac{20}{120} \times 100 = 16.67\%$$

decrease.

So:

$$\boxed{\text{percentage changes are denominator-dependent}}$$

13. RATIO QUESTIONS

Company A : Company C in 2024:

$$180:160$$

Simplify by 20:

$$\boxed{9:8}$$

14. SHARE OF TOTAL

In 2024:

Total:

$$180+210+160=550$$

Company A's share:

$$\left[\frac{180}{550} \times 100 \right]$$

$$\left[= \frac{36}{110} \times 100 \right]$$

$$\left[\approx \boxed{32.73\%} \right]$$

15. FAST SHARE ESTIMATION

Instead of exact division:

$$\left[180/550 \right]$$

Since:

$$\left[180/540 = \frac{1}{3} \right]$$

and denominator is slightly larger than 540:

share is slightly below:

$$\left[33.33\% \right]$$

So approximately:

$$\left[\boxed{32.7\%} \right]$$

Good option-elimination technique.

16. AVERAGE

Average Company B revenue:

$$\left[\frac{150+180+210+240}{4} \right]$$

$$\left[= \frac{780}{4} \right]$$

$$\left[= \boxed{195} \right]$$

17. FAST AVERAGE USING PATTERN

The values:

$$\left[150, 180, 210, 240 \right]$$

form an AP.

Average of equally spaced terms:

$$\left[\frac{\text{first} + \text{last}}{2} \right]$$

$$\left[= \frac{150+240}{2} \right]$$

$$\left[= \boxed{195} \right]$$

Much faster.

18. COMBINED AVERAGE FROM TABLE

Suppose average of Company A and B in 2025:

$$\left[\frac{210+240}{2} \right]$$

$$\left[=\boxed{225} \right]$$

19. CROSS-CATEGORY TOTAL

Total of A in 2023 and C in 2025:

$$\left[150+200 \right]$$

$$\left[=\boxed{350} \right]$$

DI questions often deliberately combine unrelated cells to test careful reading.

20. "HOW MUCH MORE" VS "WHAT PERCENT MORE"

These are different.

If A = 150 and B = 120:

How much more?

$$\left[150-120 = \boxed{30} \right]$$

What percent more is A than B?

$$\left[\frac{30}{120} \times 100 = \boxed{25\%} \right]$$

21. "A IS WHAT PERCENT OF B?"

Formula:

$$\left[\boxed{ \frac{A}{B} \times 100 } \right]$$

Example:

A = 180, B = 240:

$$\left[\frac{180}{240} \times 100 = \boxed{75\%} \right]$$

22. "A IS HOW MUCH PERCENT LESS THAN B?"

Formula:

$$\left[\boxed{ \frac{B-A}{B} \times 100 } \right]$$

The reference quantity is B.

Example:

180 compared to 240:

$$\left[\frac{60}{240} \times 100 = \boxed{25\% \text{ less}} \right]$$

23. "B IS HOW MUCH PERCENT MORE THAN A?"

Same numbers:

$$\left[\frac{240-180}{180} \times 100 \right]$$

$$\left[= \frac{60}{180} \times 100 \right]$$

$$\left[= \boxed{33.33\% \text{ more}} \right]$$

Important asymmetry:

$$\left[25\% \text{ less} \neq 25\% \text{ more} \right]$$

24. GROWTH RATE BETWEEN YEARS

Company C:

2023:

$$\left[140 \right]$$

2024:

$$\left[160 \right]$$

Growth:

$$\left[\frac{20}{140} \times 100 \right]$$

$$\left[= \boxed{14.29\%} \right]$$

25. HIGHEST GROWTH RATE

Suppose we compare growth from 2022 to 2023.

A

$$\left[\frac{150-120}{120} = 25\% \right]$$

B

$$\left[\frac{180-150}{150} = 20\% \right]$$

C

$$\left[\frac{140-100}{100} = 40\% \right]$$

Therefore highest growth:

$$\left[\boxed{\text{Company C}} \right]$$

Notice:

Highest absolute increase need not equal highest percentage increase.

26. ABSOLUTE VS RELATIVE GROWTH

From 100 to 140:

Absolute increase:

$\sqrt[4]{40}$

Percentage:

$\sqrt[4]{40\%}$

From 500 to 550:

Absolute increase:

$\sqrt[4]{50}$

Percentage:

$\sqrt[4]{10\%}$

So second has larger absolute increase but smaller growth rate.

27. COMPOUND GROWTH IN DI

Suppose a value increases:

$\sqrt[4]{20\%}$

then:

$\sqrt[4]{10\%}$

Net multiplier:

$\sqrt[4]{1.2 \times 1.1 = 1.32}$

Net growth:

$\sqrt[4]{\boxed{32\%}}$

Do not add to get 30%.

28. CAGR — SUPPORT LEVEL

Compound Annual Growth Rate:

If value changes from $\sqrt[4]{V_0}$ to $\sqrt[4]{V_n}$ in $\sqrt[4]{n}$ years:

$\sqrt[4]{\boxed{\text{CAGR} = \left(\frac{V_n}{V_0}\right)^{1/n} - 1}}$

Usually expressed as percentage.

Lower-frequency in aptitude, but appears in business DI.

29. APPROXIMATION IN DI

Suppose:

$\sqrt[4]{\frac{197}{603} \times 100}$

Instead of exact:

$\sqrt[4]{\frac{200}{600} \times 100}$

$\sqrt[4]{\approx 33.33\%}$

If options are:

- 12%
- 21%
- 33%
- 49%

you are done.

30. WHEN NOT TO APPROXIMATE

If options are:

- 32.4%
- 32.7%
- 33.0%
- 33.3%

you need better precision.

Rule

Approximation strength should depend on option separation.

31. FRACTION BENCHMARKS FOR DI

Must recognize mentally:

$$\frac{1}{2} = 50\%$$

$$\frac{1}{3} \approx 33.33\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{1}{8} = 12.5\%$$

$$\frac{2}{3} \approx 66.67\%$$

$$\frac{3}{4} = 75\%$$

$$\frac{4}{5} = 80\%$$

32. MORE HIGH-VALUE BENCHMARKS

$$\frac{1}{6} \approx 16.67\%$$

$$\frac{1}{7} \approx 14.29\%$$

$$\frac{1}{9} \approx 11.11\%$$

$$\frac{1}{10} = 10\%$$

$$\frac{5}{8} = 62.5\%$$

36. PIE CHART READING

Full circle:

$$\boxed{360^\circ = 100\%}$$

Therefore:

$$1\% = 3.6^\circ$$

and:

$$\boxed{\text{percentage} = \frac{\theta}{360} \times 100}$$

We will cover this deeply in Chapter 54.

37. GRAPHICAL VALUE VS CHANGE

If line graph values:

2023:

$$120$$

2024:

$$150$$

The vertical point at 150 is the value.

The change is:

$$150 - 120 = 30$$

Do not confuse level with change.

38. TABLE HEADER TRAP

Consider:

Year	Profit (%)
2024	20

This may mean profit percentage, not profit amount.

If cost is not given, actual profit amount cannot be determined.

Always distinguish:

$$\boxed{\text{amount} \neq \text{percentage}}$$

39. INDEX NUMBERS

Suppose an index has base year:

$$100$$

Current index:

$\sqrt{125}$

Then relative increase from base:

$\sqrt{\boxed{25\%}}$

Index 80 means:

$\sqrt{\boxed{20\%}\text{ below base}}$

40. NORMALIZED DATA

Sometimes charts use:

- per capita
- per 1000
- percentage
- index
- rate

These are normalized values.

A larger rate does not necessarily mean larger absolute count.

41. EXAMPLE — RATE VS COUNT

City A:

Population:

$\sqrt{1,000,000}$

rate:

$\sqrt{2\%}$

Count:

$\sqrt{20,000}$

City B:

Population:

$\sqrt{100,000}$

rate:

$\sqrt{10\%}$

Count:

$\sqrt{10,000}$

City B has higher rate, but A has larger count.

42. WEIGHTED AVERAGE IN DI

Suppose two branches:

Branch A:

100 employees, average salary $\text{₹} 40 \text{ k}$.

Branch B:

200 employees, average salary $\text{₹} 50 \text{ k}$.

Overall average:

$$\left[\frac{100(40)+200(50)}{300} \right]$$

$$\left[=\frac{14000}{300} \right]$$

$$\left[=\boxed{46.67} \right]$$

thousand.

Do not average 40 and 50 directly unless group sizes are equal.

43. MISSING VALUE FROM TOTAL

Suppose total of five years is:

$$\left[500 \right]$$

Known four values:

$$\left[80,90,110,100 \right]$$

Missing:

$$\left[500-(80+90+110+100) \right]$$

$$\left[=500-380 \right]$$

$$\left[=\boxed{120} \right]$$

44. MISSING VALUE FROM AVERAGE

Five-year average:

$$\left[100 \right]$$

Total:

$$\left[500 \right]$$

Known sum:

$$\left[380 \right]$$

Missing:

$$\left[\boxed{120} \right]$$

Same idea.

45. MISSING VALUE FROM PERCENT SHARE

Suppose one category is 25% of total and equals 80.

Then:

$$\sqrt{0.25T=80}$$

$$\sqrt{T=320}$$

$$\sqrt{\boxed{320}}$$

46. CROSS-MULTIPLICATIONSPEED

To compare:

$$\sqrt{\frac{a}{b} \quad \text{and} \quad \frac{c}{d}}$$

without division, compare:

$$\sqrt{ad}$$

and:

$$\sqrt{bc}$$

Example:

$$\sqrt{\frac{17}{23} \quad \text{vs} \quad \frac{21}{29}}$$

Compare:

$$\sqrt{17(29)=493}$$

$$\sqrt{21(23)=483}$$

Therefore:

$$\sqrt{\boxed{\frac{17}{23} > \frac{21}{29}}}$$

Useful in ratio/rate DI.

47. COMPARE PERCENTAGE GROWTH WITHOUT FULL CALCULATION

Compare:

$$\sqrt{\frac{30}{120}}$$

and:

$$\sqrt{\frac{35}{150}}$$

Cross-multiply:

$$\sqrt{30(150)=4500}$$

$$\sqrt{35(120)=4200}$$

So first growth rate is larger.

No percentages needed.

48. TOTAL-FIRST STRATEGY

If multiple questions use same row/column total, calculate it once and note it.

Do not recalculate repeatedly.

This saves major time in placement DI sets.

49. QUESTION-FIRST STRATEGY

But do not precompute every possible total.

Read questions first.

Calculate reusable totals only when needed.

Best balance:

Question first !' identify repeated quantities !' cache them mentally/on rough sheet.

50. UNIT CONSISTENCY

Suppose:

Revenue:

$\text{₹ } 12 \text{ crore}$

Expense:

$\text{₹ } 150 \text{ lakh}$

Convert first:

$\text{₹ } 12 \text{ crore} = 120 \text{ lakh}$

Profit:

$120 - 50 = 70 \text{ lakh}$

or:

$12 - 5 = 7 \text{ crore}$

51. COMMON DATA UNITS IN INDIA

$1 \text{ lakh} = 100,000$

$1 \text{ crore} = 10,000,000$

$1 \text{ crore} = 100 \text{ lakh}$

Useful for placement DI.

52. AVERAGE GROWTH RATE TRAP

If yearly growth rates are:

$10\%, 20\%$

Arithmetic average:

$$\sqrt[15]{\%}$$

is **not generally** the same as compound annual growth.

Actual 2-year multiplier:

$$\sqrt[1.1(1.2)=1.32]{}$$

Equivalent constant annual rate is:

$$\sqrt[\sqrt{1.32}-1]{}$$

not 15%.

53. "PERCENT OF TOTAL INCREASE" QUESTION

Suppose total increases from 500 to 600.

Increase:

$$\sqrt[100]{}$$

Company A increase:

$$\sqrt[30]{}$$

A contributes:

$$\sqrt[\frac{30}{100} \times 100 = \boxed{30\%}]{}$$

of the total increase.

Denominator is total **increase**, not total value.

54. "SHARE CHANGED BY HOW MANY PERCENTAGE POINTS?"

Suppose share rises:

$$\sqrt[20\% \text{ to } 27\%]{}$$

Increase:

$$\sqrt[\boxed{7 \text{ percentage points}}]{}$$

Relative percentage increase in share:

$$\sqrt[\frac{7}{20} \times 100 = 35\%]{}$$

Different concepts.

55. SANITY CHECKS

Before final answer, ask:

Percentage share

Must usually lie:

$$\sqrt[0\% \leq p \leq 100\%]{}$$

Average

Must lie between minimum and maximum values.

Ratio

Should roughly match raw numbers.

Growth

Positive if new > old, negative if new < old.

These checks catch careless errors.

56. GATE-STYLE SET

Use:

Year	P	Q	R
2022	80	100	120
2023	100	125	150
2024	120	150	180
2025	140	175	210

57. QUESTION 1 — TOTAL 2024

$$120+150+180 = \boxed{450}$$

58. QUESTION 2 — PERCENTAGE INCREASE OF P

2022 !' 2025:

$$140-80=60$$

$$\frac{60}{80}\times100 = \boxed{75\%}$$

59. QUESTION 3 — Q:R IN 2023

$$125:150$$

divide 25:

$$\boxed{5:6}$$

60. QUESTION 4 — AVERAGE R

$$\frac{120+150+180+210}{4}$$

AP average:

$$\left[\frac{120+210}{2} \right]$$

$$\left[=\boxed{165} \right]$$

61. QUESTION 5 — P SHARE OF TOTAL 2025

Total:

$$\left[140+175+210 = 525 \right]$$

Share:

$$\left[\frac{140}{525} \times 100 \right]$$

$$\left[=\frac{4}{15} \times 100 \right]$$

$$\left[=\boxed{26.67\%} \right]$$

62. PLACEMENT-STYLE SET

Employees:

Department	Men	Women
Tech	120	80
Sales	90	60
HR	30	70
Finance	50	50

63. TOTAL EMPLOYEES IN TECH

$$\left[120+80 = \boxed{200} \right]$$

64. TOTAL WOMEN

$$\left[80+60+70+50 \right]$$

$$\left[=\boxed{260} \right]$$

65. MEN:WOMEN IN SALES

$$\left[90:60 \right]$$

$$\left[=\boxed{3:2} \right]$$

66. WOMEN AS % OF HR

HR total:

$$\sqrt[3]{30+70=100}$$

Women:

$$\sqrt[3]{70\%}$$

$$\sqrt[3]{\boxed{70\%}}$$

67. WOMEN AS % OF TOTAL COMPANY

Total men:

$$\sqrt[3]{120+90+30+50=290}$$

Total women:

$$\sqrt[3]{260}$$

Company total:

$$\sqrt[3]{550}$$

Women share:

$$\sqrt[3]{\frac{260}{550}\times 100}$$

$$\sqrt[3]{=\frac{52}{110}\times 100}$$

$$\sqrt[3]{\approx \boxed{47.27\%}}$$

68. COMMON DI TRAPS

& pTRAP 1 — Wrong denominator

Percentage increase uses original value.

& pTRAP 2 — Wrong unit

Check thousand/lakh/crore/million.

& pTRAP 3 — Visual graph distortion

Read axis, not appearance.

& pTRAP 4 — Average of averages

Need weights unless group sizes equal.

& pTRAP 5 — Percentage vs percentage points

Keep them separate.

& pTRAP 6 — Absolute vs percentage change

Different questions.

& pTRAP 7 — Calculating too much

Extract only required cells.

69. MUST-MEMORIZE DI FORMULAS

Percentage share:

$$\boxed{\frac{\text{part}}{\text{total}} \times 100}$$

Percentage change:

$$\boxed{\frac{\text{new-old}}{\text{old}} \times 100}$$

Average:

$$\boxed{\frac{\text{sum}}{\text{count}}}$$

Weighted average:

$$\boxed{\frac{\sum wx}{\sum w}}$$

Ratio:

$$\boxed{A:B}$$

70. FAST DI DECISION TREE

Direct value?

Read it.

"How much more"?

Subtract.

"% more/less"?

Difference/reference $\times 100$.

"What % of total"?

Part/total $\times 100$.

Average?

Total/count.

Groups of unequal size?

Weighted average.

Options far apart?

Approximate.

Compare ratios?

Cross-multiply.

Graph looks dramatic?

Check axis scale.

PRACTICE — LEVEL 0

Use:

Product	2024	2025
A	100	120
B	150	180
C	200	220
D	50	80

Q1

Total sales in 2024?

Q2

Total sales in 2025?

Q3

Increase in Product D?

Q4

Percentage increase of Product A?

Q5

Ratio B:C in 2025?

LEVEL 1

Q6

What percentage of 2025 total is Product A?

Q7

Which product has highest absolute increase?

Q8

Which product has highest percentage increase?

Q9

Average 2024 sales?

Q10

By what percentage did total sales increase from 2024 to 2025?

LEVEL 2

Use:

Branch	Employees	Avg Salary (¹ k)
A	100	40
B	150	50
C	250	60

Q11

Total employees?

Q12

Overall weighted average salary?

Q13

What percent of employees are in C?

Q14

Total monthly salary bill in ¹k ?

Q15

If A employee count rises 20% with same average salary, what is new A salary bill?

LEVEL 3 — EXAM PRESSURE

Use:

Year	Revenue	Cost
2022	500	400
2023	600	450
2024	660	495
2025	792	594

Q16

Profit in each year?

Q17

Profit margin as % of revenue in 2022?

Q18

Revenue growth from 2022 to 2023?

Q19

Cost growth from 2023 to 2024?

Q20

Which year has highest profit amount?

ANSWERS

Q1

$$\lfloor 100+150+200+50 = \boxed{500} \rfloor$$

Q2

$$\lfloor 120+180+220+80 = \boxed{600} \rfloor$$

Q3

$$\lfloor 80-50 = \boxed{30} \rfloor$$

Q4

$$\lfloor \frac{20}{100} \times 100 = \boxed{20\%} \rfloor$$

Q5

$$\lfloor 180:220 = \boxed{9:11} \rfloor$$

Q6

$$\lfloor \frac{120}{600} \times 100 = \boxed{20\%} \rfloor$$

Q7

Changes:

A:

$$\lfloor 20 \rfloor$$

B:

$$\lfloor 30 \rfloor$$

C:

$$\lfloor 20 \rfloor$$

D:

$$\lfloor 30 \rfloor$$

So:

$$\lfloor \boxed{B \text{ and } D} \rfloor$$

tie.

Q8

A:

$$\lfloor 20\% \rfloor$$

B:

$$\left[\frac{30}{150} = 20\% \right]$$

C:

$$\left[\frac{20}{200} = 10\% \right]$$

D:

$$\left[\frac{30}{50} = 60\% \right]$$

Highest:

$$\left[\boxed{D} \right]$$

Q9

$$\left[500/4 = \boxed{125} \right]$$

Q10

$$\left[\frac{600-500}{500} \times 100 \right]$$

$$\left[= \boxed{20\%} \right]$$

Q11

$$\left[100+150+250 = \boxed{500} \right]$$

Q12

Salary bill:

A:

$$\left[100(40) = 4000 \right]$$

B:

$$\left[150(50) = 7500 \right]$$

C:

$$\left[250(60) = 15000 \right]$$

Total:

$$\left[26500 \right]$$

Average:

$$\left[26500/500 \right]$$

$$\left[= \boxed{53} \right]$$

So:

$$\left[\boxed{\text{\$ } 3 \text{ k}} \right]$$

Q13

$$\left[250/500 = \boxed{50\%} \right]$$

Q14

$$\left[\boxed{26,500 \text{ \text{ } }^1 \text{ k}} \right]$$

which is:

$\boxed{126.5 \text{ million}}$

if interpreting ¹ literally.

Q15

New employees:

$100(1.2)=120$

Salary bill:

$120(40) = \boxed{4800 \text{ k}}$

Q16

Profit = Revenue - Cost.

2022:

100

2023:

150

2024:

165

2025:

198

$\boxed{100, 150, 165, 198}$

Q17

$\frac{100}{500} \times 100 = \boxed{20\%}$

Q18

$\frac{600-500}{500} \times 100 = \boxed{20\%}$

Q19

$\frac{495-450}{450} \times 100$

$= \frac{45}{450} \times 100$

$= \boxed{10\%}$

Q20

Highest profit:

198

in:

$\boxed{2025}$

CHAPTER 50 STATUS

Covered so far:

- DI workflow '
- units and labels '
- direct lookups '
- row/column totals '
- absolute differences '
- percentages '
- ratios '
- percentage share '
- averages '
- weighted averages '
- growth rates '
- approximation '
- fraction benchmarks '
- graph-scale traps '
- normalized data '
- missing values '
- cross-multiplication comparison '
- percentage-points distinction '
- GATE/placement practice '

But **Chapter 50 is not complete yet.**

Still remaining:

- multi-question DI set strategy
- data sufficiency inside DI
- derived columns
- index/base-year interpretation
- cumulative vs annual data
- ranking from data
- maximum/minimum comparison shortcuts
- consistency checks
- time-management strategy for DI sets
- final Chapter 50 audit

Progress

Completed: 49/76 Current: 0=0 Chapter 50 — DI Foundations

NEXT

0=0 Chapter 50 Part 2 — Advanced DI Reading Strategy, Derived Data, Ranking, Cumulative Data, Multi-Question Sets & Time-Saving Techniques

USER

next

CHAPTER 50 — DATA INTERPRETATION FOUNDATIONS AND READING DATA

PART 2 — ADVANCED DI READING, DERIVED DATA, CUMULATIVE DATA, RANKING, MULTI-QUESTION SETS & SPEED STRATEGY

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

This part is about the skill that separates a slow DI solver from a fast one:

Do not calculate everything. Calculate only what the questions force you to calculate.

1. DIRECT DATA VS DERIVED DATA

Direct data

Already visible in the table/chart.

Example:

Year	Revenue
2024	500

Revenue in 2024:

$\boxed{500}$

Derived data

Must be calculated.

Examples:

- profit
- percentage growth
- average
- market share
- ratio
- productivity
- cost per unit

2. DERIVED COLUMN STRATEGY

Suppose:

Year	Revenue	Expense
2022	500	400

Year	Revenue	Expense
2023	600	450
2024	720	540
2025	800	600

If several questions ask about profit, create mentally:

$\text{Profit} = \text{Revenue} - \text{Expense}$

So derived column is:

Year	Profit
2022	100
2023	150
2024	180
2025	200

Now later questions become much faster.

3. WHEN SHOULD YOU CREATE A DERIVED COLUMN?

Create it only if:

- multiple questions need it
- calculation is simple
- it reduces repeated work

Do **not** create five unnecessary columns before reading questions.

4. MULTI-QUESTION DI SET STRATEGY

A typical DI set may contain 4–6 questions based on one chart.

Best order:

1. read title/unit
2. scan all questions
3. answer direct lookup first
4. compute reusable totals
5. do percentages/ratios
6. leave lengthy algebraic questions for last

This prevents wasting time.

5. EASY-FIRST PRINCIPLE

Within one DI set, not every question has equal difficulty.

Suppose:

- Q1 = direct lookup
- Q2 = simple total
- Q3 = percentage
- Q4 = missing value
- Q5 = long weighted average

Solve:

$\{Q1 \rightarrow Q2 \rightarrow Q3 \rightarrow Q4 \rightarrow Q5\}$

Do not let one difficult question block the entire set.

6. CUMULATIVE DATA

Cumulative means:

total up to that point.

Example:

Month	Cumulative Sales
Jan	100
Feb	250
Mar	430
Apr	700

These values are **not monthly sales**.

Monthly sales:

Jan:

$\{100\}$

Feb:

$\{250 - 100 = 150\}$

Mar:

$\{430 - 250 = 180\}$

Apr:

$\{700 - 430 = 270\}$

So:

$\boxed{\{100, 150, 180, 270\}}$

7. CUMULATIVE-DATATRAP

If cumulative value rises from:

$$\backslash[250\text{to}430 \backslash]$$

the March contribution is:

$$\backslash[430-250=180 \backslash]$$

not:

$$\backslash[430 \backslash]$$

This is a very common graph-reading error.

8. ANNUAL VS CUMULATIVE GROWTH

Suppose annual sales:

2023:

$$\backslash[100 \backslash]$$

2024:

$$\backslash[120 \backslash]$$

2025:

$$\backslash[150 \backslash]$$

Cumulative by 2025:

$$\backslash[100+120+150=370 \backslash]$$

Do not calculate annual growth using cumulative values unless question asks for cumulative growth.

9. MOVING TOTALS

A table may say:

Revenue for the last 3 months

That is a rolling/moving total.

Example actual monthly values:

$$\backslash[10,20,30,40,50 \backslash]$$

3-month moving totals:

$$\backslash[60,\ 90,\ 120 \backslash]$$

because:

$$\backslash[10+20+30=60 \backslash]$$

$$\backslash[20+30+40=90 \backslash]$$

$$\backslash[30+40+50=120 \backslash]$$

Be careful about what the data point represents.

10. INDEX / BASE-YEAR DATA

Suppose:

$\text{Base year index} = 100$

Later values:

Year	Index
2022	100
2023	110
2024	125
2025	120

Interpretation:

2023 is:

10%

above base.

2024 is:

25%

above base.

2025 is:

20%

above base.

11. INDEX DOES NOT DIRECTLY GIVE ABSOLUTE VALUE

If base-year revenue was:

400

and index becomes:

125

then:

$\text{new value} = 400 \times \frac{125}{100}$

$= \boxed{500}$

Without base amount, absolute value cannot be known.

12. INDEX CHANGE BETWEEN TWO NON-BASE YEARS

Index:

$\frac{120}{150}$

Do not say increase = 30%.

Actual percentage increase:

$\frac{150-120}{120} \times 100$

$\boxed{25\%}$

Index-point increase:

30

Percentage increase:

25%

Different.

13. RANKING FROM DATA

Suppose:

Region	Sales
A	160
B	220
C	180
D	250
E	190

Descending ranking:

$D > B > E > C > A$

So:

- Rank 1 !' D
- Rank 2 !' B
- Rank 3 !' E

14. DO NOT FULLY SORT IF NOT REQUIRED

If question asks:

Which region has the second-highest sales?

You only need:

highest:

250

second-highest:

$\sqrt[4]{220}$

No need to sort all five.

15. RANKING WITH PERCENTAGE GROWTH

Suppose:

Company	Old	New
A	100	140
B	200	260
C	50	80

Growth:

A:

$\sqrt[4]{40\%}$

B:

$\sqrt[4]{30\%}$

C:

$\sqrt[4]{60\%}$

Ranking by growth:

$\sqrt[4]{\boxed{C>A>B}}$

Ranking by absolute increase:

A:

$\sqrt[4]{40}$

B:

$\sqrt[4]{60}$

C:

$\sqrt[4]{30}$

So:

$\sqrt[4]{\boxed{B>A>C}}$

Always identify what the ranking is based on.

16. MAXIMUM/MINIMUM WITHOUT FULL CALCULATION

Suppose comparing:

$\sqrt[4]{\frac{45}{90}}, \sqrt[4]{\frac{52}{104}}, \sqrt[4]{\frac{60}{150}}, \sqrt[4]{\frac{70}{140}}$

Recognize:

$$\sqrt[4]{45/90=1/2}$$

$$\sqrt[4]{52/104=1/2}$$

$$\sqrt[4]{60/150=0.4}$$

$$\sqrt[4]{70/140=1/2}$$

Maximum:

$$\sqrt[4]{\boxed{1/2}}$$

No decimal calculation needed.

17. CROSS-MULTIPLICATION FOR RANKING RATIOS

Compare:

$$\sqrt[4]{\frac{37}{52}}$$

and:

$$\sqrt[4]{\frac{42}{61}}$$

Cross products:

$$\sqrt[4]{37(61)=2257}$$

$$\sqrt[4]{42(52)=2184}$$

Therefore:

$$\sqrt[4]{\boxed{\frac{37}{52}>\frac{42}{61}}}$$

This is faster than long division.

18. APPROXIMATE RANKING

Suppose percentages are roughly:

$$\sqrt[4]{398/802,\quad 510/1005,\quad 605/1190}$$

Estimate:

$$\sqrt[4]{400/800=50\%}$$

$$\sqrt[4]{510/1020\approx 50\%}$$

$$\sqrt[4]{600/1200=50\%}$$

If options are close, then exact comparison is necessary.

Approximation should first tell you:

"These are close — don't guess."

19. CONSISTENCY CHECK — TOTALS

Suppose category shares are:

$[25\%, 30\%, 20\%, 15\%, 10\%]$

Check:

$[25+30+20+15+10 = 100\%]$

Good.

If total is:

$[105\%]$

either:

- categories overlap
- rounding occurred
- data was misread

Check description.

20. CONSISTENCY CHECK — PARTS

If total employees:

$[500]$

Male:

$[300]$

Female:

$[250]$

If categories are mutually exclusive and exhaustive:

$[300+250=550]$

Impossible.

So something is wrong.

DI often tests your ability to notice such inconsistency.

21. CONSISTENCY CHECK — AVERAGE

Dataset values:

$[20, 40, 60, 80]$

Average must lie between:

$[20\text{ and }80]$

If you get:

$[95]$

you made a calculation error.

22. CONSISTENCY CHECK — WEIGHTED AVERAGE

If group averages are:

$[40, 60, 75]$

overall weighted average must lie between:

$[40 \text{ and } 75]$

It cannot be 85.

23. CONSISTENCY CHECK — PERCENTAGE SHARE

If one part is smaller than half the total, its share must be:

$[<50\%]$

This quick visual check catches errors.

24. PART-TO-PART VS PART-TO-WHOLE

Suppose:

Men:

$[60]$

Women:

$[40]$

Men:Women

$[60:40=3:2]$

Men as % total

$[60/100=60\%]$

Men are how much % more than women?

$[\frac{20}{40} \times 100 = 50\%]$

Three different results from same data.

25. DERIVED METRIC — PRODUCTIVITY

Suppose:

Branch	Output	Employees
A	1000	50
B	1200	80
C	900	30

Productivity:

$$\left[\frac{\text{output}}{\text{employees}} \right]$$

A:

$$\left[20 \right]$$

B:

$$\left[15 \right]$$

C:

$$\left[30 \right]$$

Highest:

$$\left[\boxed{C} \right]$$

Raw output alone would incorrectly suggest B.

26. DERIVED METRIC — PROFIT MARGIN

Profit:

$$\left[\text{Revenue} - \text{Cost} \right]$$

Profit margin:

$$\left[\boxed{ \frac{\text{Profit}}{\text{Revenue}} \times 100 } \right]$$

Do not confuse with profit percentage on cost:

$$\left[\frac{\text{Profit}}{\text{Cost}} \times 100 \right]$$

The question wording matters.

27. EXAMPLE

Revenue:

$$\left[500 \right]$$

Cost:

$$\left[400 \right]$$

Profit:

$$\left[100 \right]$$

Profit margin:

$$\left[100/500 = 20\% \right]$$

Profit as % of cost:

$$\left[100/400 = 25\% \right]$$

So:

$$\left[\boxed{20\% \neq 25\%} \right]$$

28. DERIVED METRIC — PER CAPITA

Population:

$\backslash[2,000,000 \backslash]$

GDP:

$\backslash[1100 \text{ billion} \backslash]$

Per capita:

$\backslash[\frac{100 \text{ billion}}{2 \text{ million}} \backslash]$

Cancel units carefully:

$\backslash[\frac{100 \times 10^9}{2 \times 10^6} = 50 \times 10^3 \backslash]$

$\backslash[= \boxed{50,000} \backslash]$

29. RATE VS TOTAL — AGAIN

Rate:

$\backslash[\frac{\text{events}}{\text{population}} \backslash]$

Absolute count:

$\backslash[\text{rate} \times \text{population} \backslash]$

A high rate with a small population may still produce fewer total events.

30. DATA SUFFICIENCY INSIDE DI

Sometimes question asks:

Can the value be uniquely determined?

Example:

Total sales of A+B:

$\backslash[500 \backslash]$

Ratio A:B:

$\backslash[3:2 \backslash]$

Enough information.

Total ratio parts:

$\backslash[5 \backslash]$

A:

$\backslash[300 \backslash]$

B:

$\backslash[200 \backslash]$

Unique.

31. INSUFFICIENT INFORMATION

Suppose:

$$A+B=500$$

No ratio or one individual value is given.

Then infinitely many possibilities:

$$100+400, 200+300, 250+250, \dots$$

So:

$$\boxed{\text{cannot determine individually}}$$

32. REDUNDANT INFORMATION

Suppose:

$$A+B=500$$

$$A:B=3:2$$

and:

$$A=300$$

Third statement is redundant.

Recognizing redundancy can save time in data-sufficiency questions.

33. PERCENTAGE-BASED MISSING VALUE

Total sales:

$$800$$

$$A = 25\%$$

A:

$$0.25(800) = \boxed{200}$$

If B = 40% of remaining sales:

Remaining:

$$600$$

B:

$$0.4(600) = \boxed{240}$$

Notice denominator changed.

34. CONDITIONAL PERCENTAGE TRAP

"40% of remaining" does not mean:

$\lfloor 40\% \text{ of original total} \rfloor$

Always identify the base.

35. MULTI-STAGETABLE

Suppose:

Department	Applicants	Selected
A	500	100
B	800	120
C	600	150

Selection rates:

A:

$\lfloor 20\% \rfloor$

B:

$\lfloor 15\% \rfloor$

C:

$\lfloor 25\% \rfloor$

Highest:

$\lfloor \boxed{C} \rfloor$

Again, largest selected count is not necessarily highest rate.

36. WEIGHTED RATE

Overall selection rate:

Total selected:

$\lfloor 100+120+150=370 \rfloor$

Total applicants:

$\lfloor 500+800+600=1900 \rfloor$

Rate:

$\lfloor \frac{370}{1900} \times 100 \rfloor$

$\lfloor \approx \boxed{19.47\%} \rfloor$

Do not average:

$\lfloor (20+15+25)/3=20\% \rfloor$

because applicant counts differ.

37. CHANGE IN SHARE

Suppose company market share:

2024:

$\boxed{20\%}$

2025:

$\boxed{24\%}$

Percentage-point increase:

$\boxed{4 \text{ points}}$

Relative increase in share:

$\frac{4}{20} \times 100 = \boxed{20\%}$

38. SHARE CAN RISE EVEN IF ABSOLUTE VALUE FALLS?

Yes, if total market falls faster.

Example:

Company sales:

$\boxed{100 \text{ to } 90}$

Market:

$\boxed{1000 \text{ to } 600}$

Shares:

$\boxed{10\%}$

to:

$\boxed{15\%}$

Company sales fell 10%, yet share increased.

Important interpretation point.

39. SHARE CAN FALL EVEN IF SALES RISE

Company:

$\boxed{100 \text{ to } 120}$

Market:

$\boxed{500 \text{ to } 800}$

Shares:

$\boxed{20\% \text{ to } 15\%}$

Absolute sales increased, market share fell.

40. BASE EFFECT

Growth percentages can look huge when starting value is small.

Example:

[10\to20]

growth:

[100\%]

while:

[1000\to1100]

growth:

[10\%]

Even though absolute increase is much larger in second.

41. ROUNDING IN DATA

Suppose chart labels:

[33\%,33\%,34\%]

These sum to exactly 100%.

But sometimes rounded shares may sum to:

[99\%\text{ or }101\%]

Do not automatically assume data error.

42. APPROXIMATE VALUES

If chart states:

Values rounded to nearest lakh

Then exact totals based on labels may differ from underlying raw total.

Read footnotes.

43. FOOTNOTES MATTER

DI tables may include notes such as:

- excludes subsidiaries
 - provisional data
 - percentages may not total 100 because of rounding
 - values in constant prices
 - population in thousands
-

These notes can change the answer.

44. MULTI-QUESTION SET — EXAMPLE

Use:

Region	2023 Sales	2024 Sales	Employees
North	240	300	50
South	320	400	100
East	180	270	45
West	260	312	60

45. Q1 — HIGHEST 2024 SALES

Values:

$\{ 300, 400, 270, 312 \}$

Highest:

$\boxed{\text{South}}$

No calculation.

46. Q2 — HIGHEST GROWTH RATE

North:

$\{ 60/240=25\% \}$

South:

$\{ 80/320=25\% \}$

East:

$\{ 90/180=50\% \}$

West:

$\{ 52/260=20\% \}$

Highest:

$\boxed{\text{East}}$

47. Q3 — HIGHEST 2024 SALES PER EMPLOYEE

North:

$\{ 300/50=6 \}$

South:

$$\backslash[400/100=4 \backslash]$$

East:

$$\backslash[270/45=6 \backslash]$$

West:

$$\backslash[312/60=5.2 \backslash]$$

Highest:

$$\backslash[\boxed{\text{North and East}} \backslash]$$

48. Q4 — TOTAL 2024 SALES

$$\backslash[300+400+270+312 \backslash]$$

$$\backslash[=700+582 \backslash]$$

$$\backslash[=\boxed{1282} \backslash]$$

49. Q5 — NORTH SHARE OF 2024 TOTAL

$$\backslash[\frac{300}{1282}\times 100 \backslash]$$

Approximate:

$$\backslash[300/1200=25\% \backslash]$$

but denominator is higher, so share lower than 25%.

Exact approx:

$$\backslash[\boxed{23.4\%} \backslash]$$

50. SMART CACHING

From previous set, once total:

$$\backslash[1282 \backslash]$$

is calculated, write it once.

If three subsequent questions use total 2024 sales, reuse it.

This is called **caching** a derived quantity.

51. TIME MANAGEMENT — GATE DI

GATE General Aptitude usually rewards accuracy more than ultra-fast guessing.

For a DI question:

1. read carefully
 2. derive only needed quantity
-

3. approximate if options allow

4. verify denominator/unit

Avoid rushed arithmetic.

52. TIME MANAGEMENT — PLACEMENT DI

Placement tests may have tighter timing.

A good strategy:

Pass 1

Direct lookups, totals, simple ratios.

Pass 2

Percentages, averages.

Pass 3

Long multi-step or missing-data questions.

This avoids losing easy marks.

53. WHEN TO SKIP A DI SET TEMPORARILY

Skip temporarily if:

- chart is visually dense
- every question requires many calculations
- first question already needs 5–6 operations
- another set looks easier

Return later.

54. OPTION-ELIMINATION METHOD

Suppose exact answer:

$$\left[\frac{197}{603} \times 100 \right]$$

Options:

- 18%
- 24%
- 33%
- 46%

Approximate:

$$\left[200/600 = 33.33\% \right]$$

Choose:

$$\boxed{33\%}$$

No exact calculation.

55. BACK-SOLVING FROM OPTIONS

Suppose:

A is what percentage of B?

Options:

$\{ 20, 25, 30, 40 \}$

If:

$\{ A=75, B=300 \}$

Recognize:

$75 = 300/4$

So:

$\boxed{25\%}$

No formula writing needed.

56. FRACTION-FIRST METHOD

Instead of converting every ratio into decimal:

$150/600$

see immediately:

$1/4 = 25\%$

Use simplification first.

57. FACTOR CANCELLATION

Example:

$\frac{84}{350} \times 100$

Cancel:

$\frac{84}{3.5}$

or:

$\frac{84}{350} \times 100 = \frac{84}{3.5} = 24$

$\boxed{24\%}$

Avoid full long division.

58. PERCENTAGE MULTIPLIER METHOD

If value rises 25%:

$\times 1.25$

If falls 20%:

$\times 0.8$

Successive:

$1.25 \times 0.8 = 1$

So no net change.

This is faster in multi-year DI.

59. ESTIMATION BOUND

Suppose:

$\frac{493}{1008}$

Clearly:

$\frac{504}{1008} = \frac{12}{25}$

So answer is below:

50%

This type of bound can eliminate options quickly.

60. ERROR LOG FOR DI

When you get DI questions wrong, classify error:

- **Reading** — wrong row/column/legend
- **Unit** — lakh/crore/thousand mismatch
- **Denominator** — wrong percentage base
- **Calculation** — arithmetic
- **Concept** — wrong average/rate formula
- **Approximation** — too rough
- **Time** — spent too long
- **Careless** — copied number incorrectly

DI improvement depends heavily on correcting these error types.

61. COMMON TRAPS — ADVANCED DI

TRAP 1 — Cumulative vs periodic

Cumulative total must be differenced to obtain individual-period value.

TRAP 2 — Ranking

Highest value " highest growth " highest rate.

& pTRAP 3 — Index

Index-point change is not always percentage change.

& pTRAP 4 — Weighted rate

Overall rate = total numerator / total denominator.

Do not average subgroup rates blindly.

& pTRAP 5 — Derived metric

If comparing productivity, divide output by employees.

& pTRAP 6 — Percentage share

A part can grow while its share falls.

& pTRAP 7 — Footnotes

Always inspect units and notes.

62. MUST-MEMORIZE DI RELATIONS

Periodic value from cumulative:

$$\boxed{\text{current period}} = \text{current cumulative} - \text{previous cumulative}$$

Overall rate:

$$\boxed{\frac{\sum \text{numerator}}{\sum \text{denominator}}}$$

Productivity:

$$\boxed{\frac{\text{output}}{\text{input}}}$$

Profit margin:

$$\boxed{\frac{\text{profit}}{\text{revenue}} \times 100}$$

63. FINAL DI SET DECISION TREE

Direct data?

Read.

Cumulative?

Take differences.

Growth?

$$\frac{\Delta}{\text{old}}$$

Share?

$$\frac{\text{part}}{\text{total}}$$

Overall rate?

Total numerator / total denominator.

Productivity?

Output/input.

Ranking?

Identify metric before sorting.

Index?

Know base = 100.

Options far apart?

Approximate.

Data dense?

Solve easy questions first.

PRACTICE — LEVEL 0

Use cumulative production:

Month	Cumulative Units
Jan	100
Feb	240
Mar	420
Apr	650

Q1

February production?

Q2

March production?

Q3

April production?

Q4

Highest monthly production?

Q5

Average monthly production Jan–Apr?

LEVEL 1

Use:

Branch	Revenue	Employees
A	400	40
B	600	75
C	500	50
D	720	80

Q6

Highest revenue?

Q7

Highest revenue per employee?

Q8

Total revenue?

Q9

A's share of total revenue?

Q10

Average revenue per employee overall?

LEVEL 2

Use:

Company	2024 Sales	2025 Sales
P	200	260
Q	300	360
R	150	225
S	400	460

Q11

Highest absolute increase?

Q12

Highest percentage growth?

Q13

Total 2024 sales?

Q14

Total 2025 sales?

Q15

Overall percentage growth?

LEVEL 3 — EXAM PRESSURE

Use:

Division	Applicants	Selected	Avg Salary ¹ L
A	400	80	6
B	600	90	8
C	500	125	7
D	300	75	10

Q16

Which division has highest selection rate?

Q17

What is overall selection rate?

Q18

How many selected candidates in total?

Q19

What is weighted average salary of all selected candidates?

Q20

Which division contributes the highest total salary bill for selected candidates?

ANSWERS

Q1

$\boxed{240-100=140}$

Q2

$\boxed{420-240=180}$

Q3

$\boxed{650-420=230}$

Q4

Monthly:

$\boxed{100,140,180,230}$

Highest:

$\boxed{\text{April}}$

Q5

Total production:

$\boxed{650}$

Average:

$\boxed{650/4 = 162.5}$

Q6

Highest revenue:

$\boxed{720}$

$\boxed{D}$

Q7

A:

$\boxed{400/40=10}$

B:

$\boxed{600/75=8}$

C:

$\boxed{500/50=10}$

D:

$\boxed{720/80=9}$

Highest:

$\boxed{A \text{ and } C}$

Q8

$\boxed{400+600+500+720 = 2220}$

Q9

$\boxed{\frac{400}{2220} \times 100}$

$\boxed{\approx 18.02\%}$

Q10

Total employees:

$\boxed{40+75+50+80 = 245}$

Overall revenue per employee:

$\boxed{2220/245}$

$\boxed{\approx 9.06}$

Q11

Increases:

P:

$$\sqrt[60]{}$$

Q:

$$\sqrt[60]{}$$

R:

$$\sqrt[75]{}$$

S:

$$\sqrt[60]{}$$

Highest:

$$\sqrt[\boxed{R}]{}$$

Q12

P:

$$\sqrt[60/200=30\%]{}$$

Q:

$$\sqrt[60/300=20\%]{}$$

R:

$$\sqrt[75/150=50\%]{}$$

S:

$$\sqrt[60/400=15\%]{}$$

Highest:

$$\sqrt[\boxed{R}]{}$$

Q13

$$\sqrt[200+300+150+400 = \boxed{1050}]{}$$

Q14

$$\sqrt[260+360+225+460 = \boxed{1305}]{}$$

Q15

Increase:

$$\sqrt[255]{}$$

Percentage:

$$\sqrt[\frac{255}{1050}\times 100]{}$$

$$\sqrt[\approx \boxed{24.29\%}]{}$$

Q16

A:

$$\left[\frac{80}{400} = 20\% \right]$$

B:

$$\left[\frac{90}{600} = 15\% \right]$$

C:

$$\left[\frac{125}{500} = 25\% \right]$$

D:

$$\left[\frac{75}{300} = 25\% \right]$$

Highest:

$$\left[\boxed{C \text{ and } D} \right]$$

Q17

Total applicants:

$$\left[400 + 600 + 500 + 300 = 1800 \right]$$

Total selected:

$$\left[80 + 90 + 125 + 75 = 370 \right]$$

Overall rate:

$$\left[\frac{370}{1800} \times 100 \right]$$

$$\left[\approx \boxed{20.56\%} \right]$$

Q18

$$\left[\boxed{370} \right]$$

Q19

Salary bill:

A:

$$\left[80(6) = 480 \right]$$

B:

$$\left[90(8) = 720 \right]$$

C:

$$\left[125(7) = 875 \right]$$

D:

$$\left[75(10) = 750 \right]$$

Total:

$$\left[2825 \right]$$

Weighted average:

$$\left[\frac{2825}{370} \right]$$

$$\left[\approx \boxed{7.64 \text{ lakh}} \right]$$

Q20

Highest salary bill:

\[875 \]

So:

\[\boxed{C} \]

' CHAPTER 50 COMPLETE

We have now covered:

- DI reading workflow '
- direct vs derived values '
- table/chart units '
- totals/subtotals '
- percentages '
- ratios '
- averages/weighted averages '
- growth rates '
- cumulative data '
- index/base-year data '
- derived metrics '
- productivity/rates '
- ranking '
- maximum/minimum shortcuts '
- data sufficiency inside DI '
- cross-multiplication '
- approximation and bounds '
- consistency checks '
- multi-question set strategy '
- time-management strategy '
- error-log framework '
- L0–L3 practice '

Progress

Completed: \(\boxed{50/76}\)

UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations '
- Chapter 51 — Tables +
- Chapter 52 — Bar & Stacked-Bar Graphs +
- Chapter 53 — Line Graphs +
- Chapter 54 — Pie Charts +

- Chapter 55 — Mixed Graphs & Caselets +
- Chapter 56 — Missing-Data & Advanced DI +

NEXT

Chapter 51 — Tables

Next we go much deeper into table-based DI:

- single and multi-dimensional tables
- row/column comparison
- totals and subtotals
- percentage distribution
- growth tables
- ranking tables
- ratio tables
- weighted-average table sets
- profit/revenue/cost tables
- employee/population tables
- multi-condition questions
- large placement-style table sets
- GATE-style speed and elimination methods.

USER

next

ASSISTANT

CHAPTER 51 — TABLES

PART 1 — SINGLE & MULTI-DIMENSIONAL TABLES, ROW/COLUMN ANALYSIS, TOTALS, RATIOS, PERCENTAGES & GROWTH

Status: $\emptyset$ =Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ =Yes 1

Table-based DI looks simple, but the main difficulty is usually not arithmetic.

It is:

Reading the correct row, correct column, correct unit, and correct denominator under time pressure.

1. WHAT IS TABLE DI?

Data is arranged in:

- rows

- columns
- headings
- sometimes subheadings

Example:

Year	Product A	Product B	Product C
2023	120	150	100
2024	160	180	140
2025	200	210	175

Rows represent years.

Columns represent products.

2. FIRST 10-SECOND TABLE SCAN

Before solving anything, identify:

1. title
2. units
3. row meaning
4. column meaning
5. whether values are absolute or percentages
6. whether totals are given
7. whether categories overlap
8. whether any values are cumulative

This prevents most reading errors.

3. DIRECT LOOKUP

From the previous table:

Product B in 2024:

$\boxed{180}$

Do not calculate anything.

4. ROW TOTAL

2024 total:

$160+180+140$

$\boxed{480}$

5. COLUMN TOTAL

Product C across all years:

$$\backslash[100+140+175 \backslash]$$

$$\backslash[=\boxed{415} \backslash]$$

6. ROW AVERAGE

Average across products in 2025:

$$\backslash[\frac{200+210+175}{3} \backslash]$$

$$\backslash[=\frac{585}{3} \backslash]$$

$$\backslash[=\boxed{195} \backslash]$$

7. COLUMN AVERAGE

Product A average:

$$\backslash[\frac{120+160+200}{3} \backslash]$$

$$\backslash[=\frac{480}{3} \backslash]$$

$$\backslash[=\boxed{160} \backslash]$$

8. DIFFERENCE QUESTIONS

Difference between B and C in 2025:

$$\backslash[210-175 = \boxed{35} \backslash]$$

9. RATIO QUESTIONS

A:B in 2024:

$$\backslash[160:180 \backslash]$$

divide by 20:

$$\backslash[\boxed{8:9} \backslash]$$

10. PERCENTAGE SHARE OF ROW TOTAL

In 2024:

Total:

$$\backslash[480 \backslash]$$

A's contribution:

$$\backslash[160 \backslash]$$

Share:

$$\left[\frac{160}{480} \times 100 \right]$$

$$\left[= \boxed{33.33\%} \right]$$

11. PERCENTAGE SHARE OF COLUMN TOTAL

Product C total:

$$\left[415 \right]$$

2025 contribution:

$$\left[175 \right]$$

Percentage:

$$\left[\frac{175}{415} \times 100 \right]$$

$$\left[\approx \boxed{42.17\%} \right]$$

The denominator is now the **column total**, not row total.

12. DENOMINATOR LANGUAGE

If question says:

A as percentage of total sales in 2024

Denominator:

$$\left[\boxed{\text{2024 row total}} \right]$$

If question says:

2024 sales as percentage of total A sales

Denominator:

$$\left[\boxed{\text{A column total}} \right]$$

Same number can produce different percentages.

13. GROWTH ACROSS ROWS

Product A:

2023:

$$\left[120 \right]$$

2024:

$$\left[160 \right]$$

Growth:

$$\left[\frac{160-120}{120} \times 100 \right]$$

$$\left[= \frac{40}{120} \times 100 \right]$$

$$\left[= \boxed{33.33\%} \right]$$

14. GROWTH FROM 2024 TO 2025

Product A:

$$\left[160 \text{ to } 200 \right]$$

Growth:

$$\left[40/160 = \boxed{25\%} \right]$$

So equal absolute increases:

$$\left[40, \text{ } 40 \right]$$

do not imply equal percentage growth.

15. HIGHEST ABSOLUTE INCREASE

Using table:

A

$$\left[2023 - 2025 : \right]$$

$$\left[200 - 120 = 80 \right]$$

B

$$\left[210 - 150 = 60 \right]$$

C

$$\left[175 - 100 = 75 \right]$$

Highest:

$$\left[\boxed{A} \right]$$

16. HIGHEST PERCENTAGE INCREASE

A

$$\left[80/120 = 66.67\% \right]$$

B

$$\left[60/150 = 40\% \right]$$

C

$$\left[75/100 = 75\% \right]$$

Highest:

$\boxed{C}$

Again:

Highest absolute increase and highest percentage increase can differ.

17. MULTI-DIMENSIONAL TABLE

Now consider:

Branch	Men	Women	Total
A	120	80	200
B	150	100	250
C	90	110	200
D	140	160	300

This has:

- categorical dimension !' branch
- gender dimension
- total dimension

18. VERIFY PROVIDED TOTALS

For A:

$120+80=200$

For D:

$140+160=300$

Checking one or two totals is a good data-validation habit.

19. GRAND TOTAL

$200+250+200+300 = \boxed{950}$

20. TOTAL MEN

$120+150+90+140$

$=\boxed{500}$

21. TOTAL WOMEN

$80+100+110+160$

$$\boxed{450}$$

Check:

$$500+450=950$$

Correct.

22. WOMEN AS % OF GRAND TOTAL

$$\frac{450}{950} \times 100$$

$$= \frac{9}{19} \times 100$$

$$\approx \boxed{47.37\%}$$

23. WOMEN AS % OF BRANCH D

Branch D:

$$160/300$$

$$= \boxed{53.33\%}$$

Different denominator again.

24. BRANCH C AS % OF TOTAL COMPANY

$$\frac{200}{950} \times 100$$

$$\approx \boxed{21.05\%}$$

25. MEN:WOMEN RATIO — BRANCH B

$$150:100$$

$$= \boxed{3:2}$$

26. OVERALL MEN:WOMEN

$$500:450$$

divide by 50:

$$\boxed{10:9}$$

27. CONDITIONAL COMPARISON

Question:

Which branch has women greater than men?

A:

$\sqrt{80 < 120}$

B:

$\sqrt{100 < 150}$

C:

$\sqrt{110 > 90}$

D:

$\sqrt{160 > 140}$

So:

$\sqrt{\boxed{C, D}}$

28. MULTI-CONDITION QUESTION

Question:

Which branch has total employees at least 200 and women at least 50%?

A total 200, women 40% 'L

B total 250, women 40% 'L

C total 200, women:

$\sqrt{110/200=55\%}$

,

D total 300, women:

$\sqrt{160/300=53.33\%}$

,

Answer:

$\sqrt{\boxed{C, D}}$

29. TABLE WITH PERCENTAGES

Consider:

Department	Employees	Women %
A	200	40%
B	300	30%
C	250	60%

Women count:

A:

$$\lfloor 200(0.4)=80 \rfloor$$

B:

$$\lfloor 300(0.3)=90 \rfloor$$

C:

$$\lfloor 250(0.6)=150 \rfloor$$

30. TOTAL WOMEN

$$\lfloor 80+90+150 = \boxed{320} \rfloor$$

31. TOTAL EMPLOYEES

$$\lfloor 200+300+250 = \boxed{750} \rfloor$$

Overall women percentage:

$$\lfloor \frac{320}{750} \times 100 \rfloor$$

$$\lfloor = \boxed{42.67\%} \rfloor$$

32. AVERAGE OF PERCENTAGES TRAP

Individual women percentages:

$$\lfloor 40\%, 30\%, 60\% \rfloor$$

Arithmetic average:

$$\lfloor 43.33\% \rfloor$$

But actual overall percentage:

$$\lfloor 42.67\% \rfloor$$

because department sizes differ.

Use weighted average.

33. WEIGHTED PERCENTAGE

Overall percentage:

$$\lfloor \boxed{\frac{\sum(\text{group size} \times \text{group percentage})}{\sum \text{group size}}} \rfloor$$

with percentages interpreted consistently.

34. PROFIT TABLE

Consider:

Company	Revenue	Cost
P	500	400
Q	600	480
R	750	600
S	800	640

Profit:

$$\boxed{\text{Revenue}-\text{Cost}}$$

35. PROFIT COLUMN

P:

$$\boxed{100}$$

Q:

$$\boxed{120}$$

R:

$$\boxed{150}$$

S:

$$\boxed{160}$$

36. PROFIT MARGIN

Profit margin:

$$\boxed{\frac{\text{Profit}}{\text{Revenue}} \times 100}$$

P:

$$\boxed{100/500=20\%}$$

Q:

$$\boxed{120/600=20\%}$$

R:

$$\boxed{150/750=20\%}$$

S:

$$\boxed{160/800=20\%}$$

All same.

37. PROFIT AS % OF COST

P:

$$\left[\frac{100}{400} = 25\% \right]$$

This is not profit margin.

Remember:

Margin

$$\left[\frac{\text{Profit}}{\text{Revenue}} \right]$$

Profit percentage on cost

$$\left[\frac{\text{Profit}}{\text{Cost}} \right]$$

38. EXPENSE RATIO

Revenue:Cost for P:

$$\left[500:400 \right]$$

$$\left[=\boxed{5:4} \right]$$

Since:

$$\left[\text{Profit} = 1 \right]$$

part out of 5 revenue parts, margin:

$$\left[\frac{1}{5} = 20\% \right]$$

Useful ratio interpretation.

39. GROWTH TABLE

Consider:

Company	2024	2025
A	240	300
B	400	460
C	150	225
D	320	400

Absolute increases:

A:

$$\left[60 \right]$$

B:

$$\left[60 \right]$$

C:

$$\left[75 \right]$$

D:

$$\boxed{80}$$

Highest absolute increase:

$$\boxed{D}$$

40. PERCENTAGE GROWTH

A:

$$\boxed{60/240=25\%}$$

B:

$$\boxed{60/400=15\%}$$

C:

$$\boxed{75/150=50\%}$$

D:

$$\boxed{80/320=25\%}$$

Highest:

$$\boxed{C}$$

41. TOTAL GROWTH

2024 total:

$$\boxed{240+400+150+320 = 1110}$$

2025:

$$\boxed{300+460+225+400 = 1385}$$

Increase:

$$\boxed{275}$$

Percentage:

$$\boxed{\frac{275}{1110} \times 100}$$

$$\boxed{\approx 24.77\%}$$

42. OVERALL GROWTH IS NOT AVERAGE OF COMPANY GROWTH

Company growth rates:

$$\boxed{25, 15, 50, 25}$$

Simple average:

$$\boxed{28.75\%}$$

But total growth:

$$\boxed{24.77\%}$$

because starting values differ.

43. RANKING TABLE

Suppose:

Student	Aptitude	Coding	Interview
A	80	70	90
B	75	85	80
C	90	60	85
D	70	95	75

If all sections equal weight, total scores:

A:

$$\sqrt[2]{240}$$

B:

$$\sqrt[2]{240}$$

C:

$$\sqrt[2]{235}$$

D:

$$\sqrt[2]{240}$$

So A, B, D tie.

44. WEIGHTED SCORE

Suppose weights:

- Aptitude 30%
- Coding 40%
- Interview 30%

For A:

$$\sqrt[2]{80(0.3)+70(0.4)+90(0.3)}$$

$$\sqrt[2]{=24+28+27}$$

$$\sqrt[2]{=\boxed{79}}$$

45. WEIGHTED SCORE — B

$$\sqrt[2]{75(0.3)+85(0.4)+80(0.3)}$$

$$\sqrt[2]{=22.5+34+24}$$

$$\boxed{80.5}$$

Weighted rankings can differ from raw totals.

46. FAST WEIGHTED-SCORE METHOD

If weights are:

$$30:40:30$$

you can use:

$$3:4:3$$

instead of decimals.

For A:

$$3(80)+4(70)+3(90)$$

$$=240+280+270$$

$$=790$$

Divide by 10:

$$79$$

Faster.

47. TABLE WITH UNITS

Suppose header says:

Production ('000 tonnes)

Value:

$$45$$

means:

$$\boxed{45,000 \text{ tonnes}}$$

If question asks ratio, units often cancel.

If asks actual quantity, restore multiplier.

48. UNIT CANCELLATION

Suppose A:

$$40 \text{ thousand}$$

B:

$$60 \text{ thousand}$$

Ratio:

$$40:60 = \boxed{2:3}$$

No need to write 40,000 and 60,000.

49. BUT DIFFERENT UNITS MUST BE CONVERTED

A:

2 crore

B:

50 lakh

Convert:

$2 \text{ crore} = 200 \text{ lakh}$

Ratio:

$200:50 = 4:1$

50. TABLE WITH BLANK / MISSING VALUE

Suppose:

Product	Sales
A	120
B	150
C	?
D	180
Total	600

Missing:

$600 - (120 + 150 + 180)$

$= 600 - 450$

$= 150$

51. MISSING VALUE FROM RATIO

Suppose A:B:

$3:5$

A:

120

Then:

$3 \text{ parts} = 120$

1 part:

$$\boxed{40}$$

B:

$$5(40) = \boxed{200}$$

52. MISSING VALUE FROM PERCENTAGE

Suppose A is 25% of total 800.

$$A = 200$$

If B is 40% of remaining:

Remaining:

$$600$$

$$B = 240$$

Again: watch the base.

53. ROW PERCENTAGE DISTRIBUTION

Suppose total employees in a branch = 400.

Distribution:

- Engineering 40%
- Sales 25%
- HR 15%
- Other 20%

Counts:

Engineering:

$$160$$

Sales:

$$100$$

HR:

$$60$$

Other:

$$80$$

54. COLUMN PERCENTAGE DISTRIBUTION

Suppose total Engineering employees = 500 across regions.

Region shares:

- North 30%
 - South 20%
-

- East 25%
- West 25%

North:

\[150 \]

South:

\[100 \]

East:

\[125 \]

West:

\[125 \]

55. NORMALIZED VALUES

Table may show:

- sales per store
- output per worker
- cost per unit
- profit per employee
- cases per 1000 people

Do not treat normalized values as totals.

56. PRODUCTIVITY TABLE

Plant	Output	Workers
A	1200	60
B	1500	100
C	1100	50
D	1800	120

Productivity:

A:

\[20 \]

B:

\[15 \]

C:

\[22 \]

D:

$\sqrt{15}$

Highest:

$\boxed{C}$

57. RATIO COMPARISON WITHOUT DIVISION

To compare productivity A vs C:

$\frac{1200}{60} \quad \text{vs} \quad \frac{1100}{50}$

Cross multiply:

$1200(50) = 60000$

$1100(60) = 66000$

So C larger.

You don't even need actual values.

58. ROW/COLUMN CACHE

If several questions use:

- row totals
- column totals
- derived profit
- derived growth

calculate once and write it on rough paper.

Example:

TEXT

Row totals: 480, 585, ...

Col totals: A=..., B=...

This can save substantial time.

59. DO NOT CACHE EVERYTHING

Precomputing every possible:

- ratio
- percentage
- difference
- average

wastes time.

Cache only quantities reused by multiple questions.

60. MULTI-CONDITION FILTERING

Example:

Branch	Sales	Growth	Margin
A	500	20%	15%
B	650	10%	18%
C	550	25%	20%
D	700	15%	12%

Question:

Which branch has sales above 500, growth at least 15%, and margin above 15%?

A: sales not above 500 'L

B: growth <15 'L

C: $\{(550, 25\%, 20\%)\}$ 'L

D: margin 12% 'L

Answer:

$\boxed{C}$

61. FILTER IN THE CHEAPEST ORDER

If one condition eliminates most rows immediately, use it first.

For example:

sales > 650

Check that first before calculating ratios or margins.

This is a placement-speed trick.

62. SORT ONLY WHAT YOU NEED

If question asks third-highest:

You may need partial sorting.

If asks highest:

Just maintain current maximum while scanning.

63. APPROXIMATION IN TABLES

Suppose:

$\left[\frac{398}{1197} \times 100 \right]$

Approximate:

$\left[400/1200 = \frac{13}{3} \right]$

$\left[\approx 33.33\% \right]$

Very effective when options are far apart.

64. EXACTNESS WHEN OPTIONS ARE CLOSE

Suppose options:

- 32.9%
- 33.1%
- 33.3%
- 33.5%

Then rough $\left(\frac{1}{3} \right)$ may not be enough.

Use more precise arithmetic.

65. TABLE ERROR CHECKLIST

Before locking answer:

- correct row?
- correct column?
- correct year?
- correct unit?
- correct denominator?
- absolute vs percentage?
- row total vs grand total?
- growth vs share?
- weighted vs simple average?

This 5-second check prevents many mistakes.

PRACTICE SET 1 — FOUNDATION

Use:

City	2024	2025
A	120	150
B	180	210
C	160	200
D	140	175

Q1

Total 2024?

Q2

Total 2025?

Q3

Which city has highest absolute increase?

Q4

Which has highest percentage increase?

Q5

A:B ratio in 2025?

ANSWERS SET 1

Q1

$\boxed{120+180+160+140 = 600}$

Q2

$\boxed{150+210+200+175 = 735}$

Q3

Increases:

A 30 B 30 C 40 D 35

Highest:

$\boxed{C}$

Q4

A:

$\boxed{30/120=25\%}$

B:

$\boxed{30/180=16.67\%}$

C:

$\boxed{40/160=25\%}$

D:

$\boxed{35/140=25\%}$

Highest:

$\boxed{A,C,D}$

tie at 25%.

Q5

$\lfloor 150:210 = \boxed{5:7} \rfloor$

PRACTICE SET 2 — EMPLOYEE TABLE

Dept	Men	Women
Tech	180	120
Sales	120	80
HR	40	60
Finance	90	110

Q6

Total employees?

Q7

Total women?

Q8

Women percentage in HR?

Q9

Overall men:women ratio?

Q10

Which departments have women > men?

ANSWERS SET 2

Men:

$\lfloor 180+120+40+90=430 \rfloor$

Women:

$\lfloor 120+80+60+110=370 \rfloor$

Total:

$\lfloor 800 \rfloor$

Q6

$\lfloor \boxed{800} \rfloor$

Q7

$\lfloor \boxed{370} \rfloor$

Q8

HR total:

$\boxed{100}$

Women:

$\boxed{60\%}$

Q9

$430:370 = \boxed{43:37}$

Q10

HR:

$60 > 40$

Finance:

$110 > 90$

$\boxed{\text{HR and Finance}}$

PRACTICE SET 3 — PROFIT TABLE

Company	Revenue	Cost
A	400	320
B	600	450
C	750	600
D	900	720

Q11

Profit of each?

Q12

Highest profit?

Q13

Profit margin of B?

Q14

Profit as % of cost for B?

Q15

Which companies have exactly 20% profit margin?

ANSWERS SET 3

Profits:

A:

$$\sqrt[4]{80}$$

B:

$$\sqrt[4]{150}$$

C:

$$\sqrt[4]{150}$$

D:

$$\sqrt[4]{180}$$

Q11

$$\sqrt[4]{\boxed{80,150,150,180}}$$

Q12

$$\sqrt[4]{\boxed{D}}$$

Q13

$$\sqrt[4]{150/600} = \boxed{25\%}$$

Q14

$$\sqrt[4]{150/450} = \boxed{33.33\%}$$

Q15

A:

$$\sqrt[4]{80/400} = 20\%$$

C:

$$\sqrt[4]{150/750} = 20\%$$

D:

$$\sqrt[4]{180/900} = 20\%$$

So:

$$\sqrt[4]{\boxed{A,C,D}}$$

PRACTICE SET 4 — WEIGHTED TABLE

Branch	Employees	Average Salary ¹ k
A	100	40
B	200	50
C	300	60
D	400	70

Q16

Total employees?

Q17

Total salary bill in ¹ k ?

Q18

Overall average salary?

Q19

What percentage of employees are in D?

Q20

Which branch contributes most to salary bill?

ANSWERS SET 4

Q16

$\backslash [100+200+300+400 = \boxed{1000} \backslash]$

Salary bills:

A:

$\backslash [4000 \backslash]$

B:

$\backslash [10000 \backslash]$

C:

$\backslash [18000 \backslash]$

D:

$\backslash [28000 \backslash]$

Total:

$\backslash [60000 \backslash]$

Q17

$\backslash [\boxed{60,000 \text{ } ^1 \text{ k } } \backslash]$

Q18

$\backslash [60000/1000 = \boxed{^1 60 \text{ k } } \backslash]$

Q19

$\backslash [400/1000 = \boxed{40\%} \backslash]$

Q20

Highest bill:

$\backslash [28000 \backslash]$

$\boxed{D}$

COMMON TABLE DI TRAPS

& pTRAP 1 — Wrong row or column

Always trace with your finger/eyes.

& pTRAP 2 — Wrong total

Row total and column total are different denominators.

& pTRAP 3 — Growth denominator

Use old/original value.

& pTRAP 4 — Average percentages

Use weighting if group sizes differ.

& pTRAP 5 — Amount vs rate

Highest amount does not imply highest ratio.

& pTRAP 6 — Profit margin

$\boxed{\text{Profit/Revenue}}$

not:

$\boxed{\text{Profit/Cost}}$

& pTRAP 7 — Units

Thousands/lakhs/crores may cancel in ratios but not in absolute answers.

MUST-MEMORIZE TABLE DI FORMULAS

Row/column share:

$\boxed{\frac{\text{part}}{\text{relevant total}} \times 100}$

Growth:

$\boxed{\frac{\text{new-old}}{\text{old}} \times 100}$

Weighted average:

$\boxed{\frac{\sum wx}{\sum w}}$

Profit:

$\boxed{R-C}$

Profit margin:

$\boxed{\frac{R-C}{R} \times 100}$

Profit % on cost:

$\boxed{\frac{R-C}{C} \times 100}$

Productivity:

$\boxed{\frac{\text{Output}}{\text{Input}}}$

FAST TABLE-DI DECISION TREE

Direct value?

Read.

Row/column total?

Add only needed cells.

Share?

Part / specified total.

Growth?

Difference / old.

Overall rate?

Total numerator / total denominator.

Weighted average?

Use group size as weight.

Highest ratio/rate?

Compare fractions, possibly cross-multiply.

Multi-condition?

Filter easiest condition first.

Options far apart?

Approximate.

CHAPTER 51 STATUS

Covered so far:

- basic table structure '
- direct lookup '
- row/column totals '
- row/column averages '
- ratios '
- percentage shares '
- growth '
- absolute vs relative comparison '
- multidimensional tables '
- percentage tables '
- weighted percentages '
- profit/cost/revenue tables '

- ranking tables '
- weighted score tables '
- units '
- missing values '
- productivity '
- multi-condition filtering '
- approximation '
- large practice sets '

But **Chapter 51 is not complete yet.**

Still remaining:

- large multi-year business tables
- table-based ratio chains
- mixed percentage + absolute tables
- conditional totals
- hidden/missing rows and columns
- multi-step GATE-style table sets
- placement-style dense tables
- final speed test and chapter audit

Progress

Completed: 50/76 Current: 0=0 Chapter 51 — Tables

NEXT

0=0 Chapter 51 Part 2 — Dense Multi-Year Tables, Hidden Values, Conditional Totals, Ratio Chains, Advanced Table DI & Final Speed Toolkit

USER

next

ASSISTANT

CHAPTER 51 — TABLES

PART 2 — DENSE MULTI-YEAR TABLES, HIDDEN VALUES, CONDITIONAL TOTALS, RATIO CHAINS & ADVANCED TABLE DI

Status: 0=0 **PROGRESS GATE:** High **Placements:** High **Priority:** 0=1 **Per** 1

This part focuses on the type of table DI where the table itself is not difficult, but questions require **2–4 connected calculations.**

Core rule:

1. DENSE MULTI-YEAR TABLE

Consider:

Company	2022	2023	2024	2025
A	120	150	180	240
B	160	200	240	280
C	100	140	175	210
D	200	220	250	300
E	140	175	210	245

Assume figures are sales in ¹ crore.

Before calculations, notice:

- 5 companies
- 4 years
- same units
- no totals given
- questions may compare rows, columns, growth and shares

2. COLUMN TOTALS

2022

$\backslash [120+160+100+200+140 \backslash$

$\backslash [=\boxed{720} \backslash$

2023

$\backslash [150+200+140+220+175 \backslash$

$\backslash [=\boxed{885} \backslash$

2024

$\backslash [180+240+175+250+210 \backslash$

$\backslash [=\boxed{1055} \backslash$

2025

$\backslash [240+280+210+300+245 \backslash$

$\backslash [=\boxed{1275} \backslash$

If many questions use yearly totals, cache:

$$\boxed{720, 885, 1055, 1275}$$

3. TOTAL GROWTH — 2022 TO 2025

$$1275 - 720 = 555$$

Percentage increase:

$$\frac{555}{720} \times 100$$

$$= 77.083\%$$

$$\boxed{77.08\%}$$

4. COMPANY-WISE TOTALS

A:

$$120 + 150 + 180 + 240 = 690$$

B:

$$160 + 200 + 240 + 280 = 880$$

C:

$$100 + 140 + 175 + 210 = 625$$

D:

$$200 + 220 + 250 + 300 = 970$$

E:

$$140 + 175 + 210 + 245 = 770$$

Highest four-year total:

$$\boxed{D}$$

5. HIGHEST PERCENTAGE GROWTH 2022! 2025

A

$$\frac{240 - 120}{120} = 100\%$$

B

$$\frac{120}{160} = 75\%$$

C

$$\frac{110}{100} = 110\%$$

D

$$\frac{100}{200} = 50\%$$

E

$$\frac{105}{140} = 75\%$$

Highest:

$\boxed{C}$

Again, D has largest total but not largest growth.

6. CONDITIONAL TOTAL

Question:

Find total 2025 sales of companies whose 2022 sales were below 150.

2022 below 150:

- A = 120
- C = 100
- E = 140

Their 2025 values:

$240+210+245$

$\boxed{695}$

Method

First **filter using one column**, then sum values from another.

7. CONDITIONAL AVERAGE

Question:

Average 2024 sales of companies whose 2023 sales were at least 175.

2023 qualifying:

- B = 200
- D = 220
- E = 175

Their 2024 values:

$240,250,210$

Average:

$\frac{700}{3}$

$\boxed{233.33}$

8. CONDITIONAL COUNT

Question:

How many companies had 2025 sales at least 20% greater than 2024?

A:

$$\lfloor 180 \times 1.2 = 216 \rfloor$$

$$2025 = 240$$

B:

$$\lfloor 240 \times 1.2 = 288 \rfloor$$

$$280$$

C:

$$\lfloor 175 \times 1.2 = 210 \rfloor$$

$$210 \text{ ' exactly 20\%}$$

D:

$$\lfloor 250 \times 1.2 = 300 \rfloor$$

$$300$$

E:

$$\lfloor 210 \times 1.2 = 252 \rfloor$$

$$245$$

So:

$$\boxed{3}$$

9. FAST THRESHOLD METHOD

Instead of computing percentage growth separately:

For “at least 20% growth” check:

$$\boxed{\text{New} \geq 1.2 \times \text{Old}}$$

For “more than 25% growth”:

$$\boxed{\text{New} > 1.25 \times \text{Old}}$$

This is often faster than computing percentages.

10. HIDDEN VALUE FROM ROW TOTAL

Product	Q1	Q2	Q3	Q4	Total
A	120	150	?	180	620

Missing:

$$\lfloor 620 - (120 + 150 + 180) \rfloor$$

$$\lfloor = 620 - 450 \rfloor$$

$$\boxed{170}$$

11. HIDDEN VALUE FROM COLUMN TOTAL

Dept	Men	Women	Total
A	80	120	200
B	100	?	250
C	120	80	200
Total	300	350	650

For B women:

$$350 - (120 + 80)$$

$$\boxed{150}$$

Also:

$$250 - 100 = 150$$

Two independent checks.

12. HIDDEN VALUE FROM GRAND TOTAL

Known row totals:

$$200, 250, ?, 300$$

Grand total:

$$1000$$

Missing row:

$$1000 - (200 + 250 + 300)$$

$$\boxed{250}$$

13. HIDDEN VALUE FROM PERCENTAGE SHARE

Suppose branch A contributes 20% of company sales.

A sales:

$$160$$

Total:

$$\frac{160}{0.2}$$

$$\boxed{800}$$

14. HIDDEN VALUE FROM GROWTH RATE

2024 sales:

$$\boxed{250}$$

2025 growth:

$$\boxed{20\%}$$

Then:

$$2025 = 250(1.2)$$

$$\boxed{300}$$

15. REVERSE GROWTH

2025 value:

$$\boxed{300}$$

which is 20% higher than 2024.

Then:

$$1.2 \times 2024 = 300$$

$$2024 = \frac{300}{1.2}$$

$$\boxed{250}$$

Do not subtract 20% of 300.

16. MIXED ABSOLUTE + PERCENTAGE TABLE

Suppose:

Branch	Employees	Women %	Avg Salary ¹ k
A	200	40%	50
B	300	30%	60
C	250	60%	55
D	150	50%	70

Derived women counts:

A:

$$\boxed{80}$$

B:

$$\boxed{90}$$

C:

$$\boxed{150}$$

D:

$$\backslash[75 \backslash]$$

Total:

$$\backslash[\boxed{395} \backslash]$$

17. TOTAL MEN

Total employees:

$$\backslash[200+300+250+150=900 \backslash]$$

Men:

$$\backslash[900-395 \backslash]$$

$$\backslash[=\boxed{505} \backslash]$$

18. SALARY BILL

A:

$$\backslash[200(50)=10000 \backslash]$$

B:

$$\backslash[300(60)=18000 \backslash]$$

C:

$$\backslash[250(55)=13750 \backslash]$$

D:

$$\backslash[150(70)=10500 \backslash]$$

Total:

$$\backslash[52250\text{ \text{ } }^1\text{ k } \backslash]$$

$$\backslash[\boxed{52.25\text{ million}} \backslash]$$

if ¹ k means thousands of rupees.

19. OVERALL AVERAGE SALARY

$$\backslash[\frac{52250}{900} \backslash]$$

$$\backslash[\approx\boxed{58.06\text{ \text{ } }^1\text{ k }} \backslash]$$

Do not average:

$$\backslash[(50+60+55+70)/4 \backslash]$$

because employee counts differ.

20. RATIO CHAIN

Suppose:

$$\boxed{A:B=2:3}$$

and:

$$\boxed{B:C=4:5}$$

We need common B.

First ratio:

$$\boxed{A:B=2:3}$$

Second:

$$\boxed{B:C=4:5}$$

LCM of B-coefficients 3 and 4:

$$\boxed{12}$$

Multiply first by 4:

$$\boxed{A:B=8:12}$$

Multiply second by 3:

$$\boxed{B:C=12:15}$$

Thus:

$$\boxed{A:B:C=8:12:15}$$

21. TABLE RATIO CHAIN EXAMPLE

A table may say:

Relationship	Ratio
A:B	3:5
B:C	10:7

Common B already:

First B = 5, second B = 10.

Multiply first by 2:

$$\boxed{A:B=6:10}$$

So:

$$\boxed{A:B:C=6:10:7}$$

22. FIND VALUES FROM RATIO + TOTAL

Suppose:

$$\boxed{A:B:C=6:10:7}$$

Total:

$\sqrt[4]{460}$

Total parts:

$\sqrt[4]{23}$

One part:

$\sqrt[4]{20}$

Therefore:

$\sqrt[4]{A=120, \text{quad } B=200, \text{quad } C=140}$

23. RATIO + DIFFERENCE

If:

$\sqrt[4]{A:B=5:3}$

and difference:

$\sqrt[4]{A-B=40}$

Difference in parts:

$\sqrt[4]{2}$

One part:

$\sqrt[4]{20}$

Therefore:

$\sqrt[4]{A=100, \text{quad } B=60}$

24. CONDITIONAL RATIO

Suppose table asks:

Ratio of total sales of companies with growth >25% to those with growth "d 2 5 % .

You must:

1. calculate/filter growth
2. group companies
3. sum sales
4. form ratio

This is a typical 3-step DI question.

25. BUSINESS TABLE — ADVANCED

Company	Revenue	Cost	Employees
P	600	450	100

Company	Revenue	Cost	Employees
Q	800	640	160
R	750	525	125
S	900	720	150

26. PROFIT

P:

$$\boxed{150}$$

Q:

$$\boxed{160}$$

R:

$$\boxed{225}$$

S:

$$\boxed{180}$$

Highest profit:

$$\boxed{R}$$

27. PROFIT MARGIN

P:

$$\boxed{150/600=25\%}$$

Q:

$$\boxed{160/800=20\%}$$

R:

$$\boxed{225/750=30\%}$$

S:

$$\boxed{180/900=20\%}$$

Highest:

$$\boxed{R}$$

28. REVENUE PER EMPLOYEE

P:

$$\boxed{600/100=6}$$

Q:

$$\backslash[800/160=5 \backslash]$$

R:

$$\backslash[750/125=6 \backslash]$$

S:

$$\backslash[900/150=6 \backslash]$$

Highest:

$$\backslash[\boxed{P,R,S} \backslash]$$

29. PROFIT PER EMPLOYEE

P:

$$\backslash[150/100=1.5 \backslash]$$

Q:

$$\backslash[160/160=1 \backslash]$$

R:

$$\backslash[225/125=1.8 \backslash]$$

S:

$$\backslash[180/150=1.2 \backslash]$$

Highest:

$$\backslash[\boxed{R} \backslash]$$

30. SAME TABLE, DIFFERENT RANKING

Highest:

- revenue !' S
- profit !' R
- revenue/employee !' P,R,Stie
- profit/employee !' R
- profit margin !' R

This is why you must identify the requested metric.

31. CONTRIBUTION TO TOTAL PROFIT

Total profit:

$$\backslash[150+160+225+180 \backslash]$$

$$\backslash[=715 \backslash]$$

R contribution:

$$\backslash[\frac{225}{715} \times 100 \backslash]$$

$\approx 31.47\%$

32. CONDITIONAL TOTAL — BUSINESS TABLE

Question:

Total revenue of companies with profit margin at least 25%.

Qualify:

- $P = 25\%$
- $R = 30\%$

Revenue:

$600 + 750$

$= 1350$

33. DENSE TABLE — PLACEMENT STYLE

Region	Applicants	Qualified	Joined	Avg Salary ¹ L
North	1000	400	250	6
South	1200	480	300	7
East	800	360	270	5
West	1500	525	315	8

34. QUALIFICATION RATE

North:

$400/1000 = 40\%$

South:

$480/1200 = 40\%$

East:

$360/800 = 45\%$

West:

$525/1500 = 35\%$

Highest:

East

35. JOINING RATE AMONG QUALIFIED

North:

$$\left[\frac{250}{400} = 62.5\% \right]$$

South:

$$\left[\frac{300}{480} = 62.5\% \right]$$

East:

$$\left[\frac{270}{360} = 75\% \right]$$

West:

$$\left[\frac{315}{525} = 60\% \right]$$

Highest:

$$\left[\boxed{\text{East}} \right]$$

36. JOINING RATE AMONG APPLICANTS

North:

$$\left[\frac{250}{1000} = 25\% \right]$$

South:

$$\left[\frac{300}{1200} = 25\% \right]$$

East:

$$\left[\frac{270}{800} = 33.75\% \right]$$

West:

$$\left[\frac{315}{1500} = 21\% \right]$$

Highest:

$$\left[\boxed{\text{East}} \right]$$

The denominator changes the meaning of the rate.

37. SALARY BILL OF JOINED CANDIDATES

North:

$$\left[250(6) = 1500 \right]$$

South:

$$\left[300(7) = 2100 \right]$$

East:

$$\left[270(5) = 1350 \right]$$

West:

$$\left[315(8) = 2520 \right]$$

Highest:

$\boxed{\text{West}}$

Even though East had best conversion rates.

38. OVERALL QUALIFICATION RATE

Total applicants:

$1000+1200+800+1500$

$=4500$

Qualified:

$400+480+360+525$

$=1765$

Rate:

$\frac{1765}{4500} \times 100$

$\approx \boxed{39.22\%}$

39. OVERALL JOIN RATE FROM APPLICANTS

Total joined:

$250+300+270+315$

$=1135$

Rate:

$\frac{1135}{4500} \times 100$

$\approx \boxed{25.22\%}$

40. OVERALL JOIN RATE AMONG QUALIFIED

$\frac{1135}{1765} \times 100$

$\approx \boxed{64.31\%}$

Again: same numerator, different denominator.

41. ROW PERCENTAGE VS COLUMN PERCENTAGE

Suppose:

Branch	Selected
A	100
B	150

Branch	Selected
C	250

Total:

$\sqrt[500]{}$

C is:

$\sqrt[50\%]{}$

of all selected.

But if C had 1000 applicants, its selection rate is:

$\sqrt[25\%]{}$

These are entirely different percentages.

42. TWO-STAGE PERCENTAGE

Suppose:

40% of applicants qualify, and 60% of qualified candidates join.

Overall joining fraction:

$\sqrt[0.4 \times 0.6 = 0.24]{}$

Thus:

$\sqrt[\boxed{24\%}]{}$

of applicants join.

43. THREE-STAGE FUNNEL

If:

- 50% pass test
- 80% of those pass interview
- 75% of those accept offer

Overall:

$\sqrt[0.5(0.8)(0.75)]{}$

$\sqrt[=0.30]{}$

$\sqrt[\boxed{30\%}]{}$

This is common in placement-style DI.

44. REVERSE FUNNEL

If 300 finally joined and overall conversion is 30%:

Applicants:

$$\lfloor 300/0.3 \rfloor$$

$$\lfloor =\boxed{1000} \rfloor$$

45. CONDITIONAL PERCENTAGE — IMPORTANT

Suppose 60% of employees are engineers.

Of engineers, 25% are women.

Women engineers as share of all employees:

$$\lfloor 0.6(0.25) \rfloor$$

$$\lfloor =\boxed{15\%} \rfloor$$

46. MISSING TABLE USING RATE

Branch	Applicants	Selected	Selection Rate
A	500	?	20%

Selected:

$$\lfloor 500(0.2) = \boxed{100} \rfloor$$

47. MISSING APPLICANTS

Selected:

$$\lfloor 120 \rfloor$$

Rate:

$$\lfloor 30\% \rfloor$$

Applicants:

$$\lfloor 120/0.3 \rfloor$$

$$\lfloor =\boxed{400} \rfloor$$

48. MISSING RATE

Applicants:

$$\lfloor 800 \rfloor$$

Selected:

$$\lfloor 200 \rfloor$$

Rate:

$$\lfloor 200/800 \rfloor$$

$$\lfloor =\boxed{25\%} \rfloor$$

49. TABLE BASED ON INDEX

Product	Base Price	Index
A	200	120
B	500	90
C	400	125

Current price:

$$\boxed{\text{Base} \times \frac{\text{Index}}{100}}$$

A:

$$200(1.2)=240$$

B:

$$500(0.9)=450$$

C:

$$400(1.25)=500$$

50. TOTAL CURRENT PRICE

$$240+450+500$$

$$= \boxed{1190}$$

51. INDEX TRAP

Product B index:

$$90$$

means price is 10% below base.

It does **not** mean price fell 90%.

52. MULTI-YEAR AVERAGE GROWTH TRAP

Suppose:

$$100 \text{ to } 120 \text{ to } 180$$

Growth rates:

$$20\%, 50\%$$

Simple average:

$$35\%$$

But total 2-year growth:

$$\sqrt[4]{80\%}$$

CAGR:

$$\sqrt[4]{\sqrt[4]{1.8}} - 1$$

$$\approx 34.16\%$$

Different concepts.

53. TABLE SPEED TECHNIQUE — PRE-SIMPLIFY RATIOS

Instead of:

$$225:375$$

immediately divide by 75:

$$\boxed{3:5}$$

Do not carry large numbers into later calculations.

54. SPEED TECHNIQUE — PERCENTAGE BENCHMARKS

Recognize:

$$\frac{1}{8} = 12.5\%$$

$$\frac{1}{6} = 16.67\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{1}{3} = 33.33\%$$

$$\frac{3}{8} = 37.5\%$$

$$\frac{1}{5} = 40\%$$

$$\frac{1}{2} = 50\%$$

55. SPEED TECHNIQUE — COMPARE GROWTH WITHOUT %

Compare:

A:

$$\frac{45}{180}$$

B:

$$\frac{50}{220}$$

Cross multiply:

$$45(220) = 9900$$

$$50(180) = 9000$$

So A's growth rate is greater.

No decimals.

56. SPEED TECHNIQUE — COMMON DENOMINATOR

If comparing:

$$\left[\frac{30}{120}, \quad \frac{45}{180}, \quad \frac{60}{240} \right]$$

all simplify to:

$$\left[\boxed{\frac{1}{4}} \right]$$

All equal 25%.

57. SPEED TECHNIQUE — FILTER BEFORE CALCULATING

Question:

Which branches have sales above 500 and margin above 20%?

First eliminate all with sales ≤ 500 .

Then compute margin only for remaining rows.

This saves unnecessary arithmetic.

58. SPEED TECHNIQUE — APPROXIMATE ONLY WHEN SAFE

$$\left[\frac{397}{1195} \approx \frac{400}{1200} = \frac{1}{3} \right]$$

If options differ widely, use:

$$\left[\boxed{33\%} \right]$$

If options are 33.1%, 33.2%, 33.3%, 33.4%, calculate accurately.

59. ERROR-PROOF METHOD FOR DENSE TABLES

For each question, write tiny notation:

$$\left[\text{Wanted} = \frac{\text{numerator}}{\text{denominator}} \right]$$

Example:

women in C as % of all women

Write:

$$\left[\frac{W_C}{W_{\text{total}}} \right]$$

Then insert numbers.

This greatly reduces denominator mistakes.

60. GATE-STYLE MINI SET

Year	Revenue	Expense
2022	400	320
2023	500	375
2024	600	450
2025	750	525

Profit

2022:

$\$80$

2023:

$\$125$

2024:

$\$150$

2025:

$\$225$

61. Q1 — HIGHEST PROFIT GROWTH

2022 to 2023:

$\frac{45}{80} = 56.25\%$

2023 to 2024:

$\frac{25}{125} = 20\%$

2024 to 2025:

$\frac{75}{150} = 50\%$

Highest:

$\boxed{2022\text{ to }2023}$

62. Q2 — PROFIT MARGIN 2025

$\frac{225}{750}$

$= \boxed{30\%}$

63. Q3 — EXPENSE AS % REVENUE 2024

$\frac{450}{600}$

$$\boxed{75\%}$$

64. Q4 — TOTAL PROFIT

$$80+125+150+225$$

$$\boxed{580}$$

65. Q5 — AVERAGE REVENUE

$$\frac{400+500+600+750}{4}$$

$$\frac{2250}{4}$$

$$\boxed{562.5}$$

PRACTICE — LEVEL 0

Use:

Branch	2024	2025
A	100	120
B	150	180
C	200	250
D	250	275

Q1

Total 2025?

Q2

Highest absolute increase?

Q3

Highest percentage increase?

Q4

A:C ratio in 2025?

Q5

What percent of 2025 total is D?

LEVEL 1

Use:

Dept	Employees	Women %
A	100	40%
B	200	30%
C	300	50%
D	400	60%

Q6

Total women?

Q7

Overall women percentage?

Q8

Total men?

Q9

Which department has most women?

Q10

Women in B:C ratio?

LEVEL 2

Use:

Company	Revenue	Cost	Employees
P	500	400	100
Q	800	600	160
R	900	675	150
S	600	420	120

Q11

Highest profit amount?

Q12

Highest profit margin?

Q13

Highest revenue per employee?

Q14

Overall profit margin?

Q15

Total revenue of companies whose margin is at least 25%?

LEVEL 3 — EXAM PRESSURE

Use:

Region	Applicants	Qualified	Joined
A	1200	480	240
B	1000	500	300
C	800	320	256
D	1500	600	330

Q16

Highest qualification rate?

Q17

Highest joining rate among qualified?

Q18

Overall qualification rate?

Q19

Overall joining rate from applicants?

Q20

Among regions with qualification rate at least 40%, what is total number joined?

ANSWERS**Q1**

$\sqrt{120+180+250+275} = \boxed{825}$

Q2

Increases:

A:

$\sqrt{20}$

B:

$\sqrt{30}$

C:

$\sqrt{50}$

D:

$\sqrt{25}$

$\sqrt{\boxed{C}}$

Q3

A:

$\sqrt{20\%}$

B:

$\sqrt{20\%}$

C:

$\sqrt{25\%}$

D:

$\sqrt{10\%}$

$\sqrt{\boxed{C}}$

Q4

$\sqrt{120:250}$

$\sqrt{=\boxed{12:25}}$

Q5

$\sqrt{\frac{275}{825}} = \sqrt{\frac{13}{15}}$

$\sqrt{=\boxed{33.33\%}}$

Q6

Women:

A:

$\sqrt{40}$

B:

$\sqrt{60}$

C:

$\sqrt{150}$

D:

$\sqrt{240}$

Total:

$\sqrt{\boxed{490}}$

Q7

Total employees:

$\sqrt{1000}$

Women:

$\sqrt{490}$

$\sqrt{\boxed{49\%}}$

Q8

$\sqrt{1000-490} = \boxed{510}$

Q9

$\sqrt{\boxed{D}}$

Q10

$\sqrt{60:150} = \boxed{2:5}$

Q11

Profits:

P:

$\sqrt{100}$

Q:

$\sqrt{200}$

R:

$\sqrt{225}$

S:

$\sqrt{180}$

Highest:

$\sqrt{\boxed{R}}$

Q12

P:

$\sqrt{20\%}$

Q:

$\sqrt{25\%}$

R:

$\sqrt{25\%}$

S:

$\sqrt{30\%}$

Highest:

$\sqrt{\boxed{S}}$

Q13

P:

$\sqrt{5}$

Q:

$\sqrt{5}$

R:

$\sqrt{6}$

S:

$\sqrt{5}$

Highest:

$\sqrt{\boxed{R}}$

Q14

Total revenue:

$\sqrt{2800}$

Total cost:

$\sqrt{2095}$

Profit:

$\sqrt{705}$

Overall margin:

$\sqrt{\frac{705}{2800} \times 100}$

$\sqrt{\approx \boxed{25.18\%}}$

Q15

Margins at least 25%:

Q, R, S.

Revenue:

$\sqrt{800+900+600}$

$\sqrt{= \boxed{2300}}$

Q16

A:

$$\backslash[480/1200=40\% \backslash]$$

B:

$$\backslash[500/1000=50\% \backslash]$$

C:

$$\backslash[320/800=40\% \backslash]$$

D:

$$\backslash[600/1500=40\% \backslash]$$

Highest:

$$\backslash[\boxed{B} \backslash]$$

Q17

A:

$$\backslash[240/480=50\% \backslash]$$

B:

$$\backslash[300/500=60\% \backslash]$$

C:

$$\backslash[256/320=80\% \backslash]$$

D:

$$\backslash[330/600=55\% \backslash]$$

Highest:

$$\backslash[\boxed{C} \backslash]$$

Q18

Total qualified:

$$\backslash[480+500+320+600 = 1900 \backslash]$$

Applicants:

$$\backslash[1200+1000+800+1500 = 4500 \backslash]$$

Rate:

$$\backslash[1900/4500 \backslash]$$

$$\backslash[=\boxed{42.22\%} \backslash]$$

Q19

Joined:

$$\backslash[240+300+256+330 = 1126 \backslash]$$

Rate:

$$\backslash[\frac{1126}{4500}\times 100 \backslash]$$

$\approx 25.02\%$

Q20

Qualification rate at least 40%:

All four regions qualify.

Total joined:

1126

66. FINAL TABLE-DI SPEED CHECKLIST

Before solving:

- read units
- understand row/column meaning
- identify direct vs derived data

During solving:

- filter before calculating
- cache useful totals
- simplify ratios early
- use fraction benchmarks
- use cross multiplication
- approximate only when options permit

Before answering:

- verify denominator
- verify year/category
- verify unit
- sanity-check magnitude

67. FINAL TABLE-DI ERROR CHECKLIST

If wrong, classify:

Reading Error

Wrong row/column.

Unit Error

Thousands/lakhs/crores mixed.

Denominator Error

Wrong comparison base.

Formula Error

Wrong percentage/rate.

Calculation Error

Arithmetic mistake.

Interpretation Error

Amount confused with rate/share.

Time Error

Too much unnecessary computation.

' CHAPTER 51 COMPLETE

We have now covered:

- basic and dense tables '
- single/multi-dimensional tables '
- row and column totals '
- direct/derived values '
- ratios and shares '
- growth tables '
- absolute vs relative growth '
- percentage tables '
- weighted percentages '
- conditional totals/averages/counts '
- hidden values '
- ratio chains '
- business tables '
- profit/revenue/cost metrics '
- employee and recruitment funnels '
- index tables '
- multi-stage percentages '
- productivity/rate comparisons '
- filtering strategy '
- approximation and cross-multiplication '
- dense L0–L3 table practice '

Progress

$\boxed{51/76}$ completed

UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations '
- Chapter 51 — Tables '
- Chapter 52 — Bar & Stacked-Bar Graphs +
- Chapter 53 — Line Graphs +
- Chapter 54 — Pie Charts +
- Chapter 55 — Mixed Graphs & Caselets +

NEXT

Chapter 52 — Bar Graphs & Stacked-Bar Graphs

Next we will cover:

- simple bar graphs
- grouped/multiple bars
- horizontal bars
- stacked bars
- 100% stacked bars
- reading axes and scales
- comparing categories
- totals from stacked segments
- percentage composition
- missing bar values
- misleading/truncated axes
- multi-year bar DI
- GATE/placement speed strategies and L0–L3 sets.

USER

next

ASSISTANT

CHAPTER 52 — BAR GRAPHS & STACKED-BAR GRAPHS

PART 1 — SIMPLE, GROUPED & STACKED BARS, AXIS READING, TOTALS, RATIOS, SHARES & GROWTH

Status: 0% PROGRESS GATE: High Placements: High Priority: 0% For 1

Bar graphs are just tables shown visually.

The mathematics is familiar:

- totals
- differences
- ratios
- percentages
- averages
- growth

The real skill is:

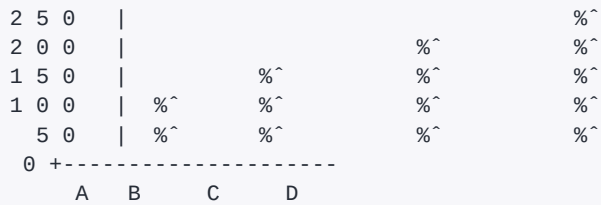
Read the correct bar height and axis scale before doing any calculation.

1. SIMPLE BAR GRAPH

Example:

TEXT

Sales



Suppose values are:

- A = 100
- B = 150
- C = 200
- D = 250

2. FIRST THING TO CHECK — AXIS SCALE

If marks are:

$\{0, 50, 100, 150, 200\}$

then each major step means:

$\boxed{50}$

But another graph may use:

$\{0, 20, 40, 60\}$

Never assume one grid step = 10 or 100.

3. DIRECT LOOKUP

If bar B reaches 150:

$\boxed{B=150}$

No arithmetic.

4. DIFFERENCE

D - B:

$\{250-150\}$

$\boxed{100}$

5. RATIO

C:A:

$$\backslash[200:100 \backslash]$$

$$\backslash[=\boxed{2:1} \backslash]$$

6. TOTAL OF ALL BARS

$$\backslash[100+150+200+250 \backslash]$$

$$\backslash[=\boxed{700} \backslash]$$

7. AVERAGE

$$\backslash[700/4 = \boxed{175} \backslash]$$

8. PERCENTAGE SHARE

D's share:

$$\backslash[\frac{250}{700}\times 100 \backslash]$$

$$\backslash[\approx \boxed{35.71\%} \backslash]$$

9. PERCENTAGE MORE

D is what percent more than B?

$$\backslash[\frac{250-150}{150}\times 100 \backslash]$$

$$\backslash[=\frac{100}{150}\times 100 \backslash]$$

$$\backslash[=\boxed{66.67\%} \backslash]$$

10. PERCENTAGE LESS

B is what percent less than D?

$$\backslash[\frac{250-150}{250}\times 100 \backslash]$$

$$\backslash[=\boxed{40\%} \backslash]$$

Again:

$$\backslash[66.67\%\text{\textit{ more}}\neq 40\%\text{\textit{ less}} \backslash]$$

11. HORIZONTAL BAR GRAPH

Same logic, different orientation.

TEXT		
A	%^%^%^%^%^%^%^%^	1 0 0
B	%^%^%^%^%^%^%^%^	1 5 0
C	%^%^%^%^%^%^%^%^	2 0 0

Read horizontal scale carefully.

12. GROUPED / MULTIPLE BAR GRAPH

Used to compare two or more series.

Example: Sales in 2024 vs 2025.

TEXT		
	2024	2025
A	%^	%%^
B	%^%	%^%^
C	%^%^%	%^%^%^

Suppose actual values are:

Product	2024	2025
A	100	140
B	150	180
C	200	240

13. ABSOLUTE GROWTH

A:

$\lfloor 140-100=40 \rfloor$

B:

$\lfloor 180-150=30 \rfloor$

C:

$\lfloor 240-200=40 \rfloor$

Highest absolute increase:

$\lfloor \boxed{A\text{ and }C} \rfloor$

14. PERCENTAGE GROWTH

A:

$\lfloor 40/100=40\% \rfloor$

B:

$$\left[\frac{30}{150} = 20\% \right]$$

C:

$$\left[\frac{40}{200} = 20\% \right]$$

Highest:

$$\left[\boxed{A} \right]$$

15. GROUP TOTAL

Total 2024:

$$\left[100 + 150 + 200 = 450 \right]$$

Total 2025:

$$\left[140 + 180 + 240 = 560 \right]$$

Total growth:

$$\left[110 \right]$$

Percentage:

$$\left[\frac{110}{450} \times 100 \right]$$

$$\left[\approx \boxed{24.44\%} \right]$$

16. BAR HEIGHT DOES NOT DIRECTLY SHOW PERCENTAGE GROWTH

A bar growing:

$$\left[100 \text{ to } 140 \right]$$

and another:

$$\left[200 \text{ to } 240 \right]$$

both rise by 40 units.

But growth rates are:

$$\left[40\% \text{ vs } 20\% \right]$$

So visual increase alone is not enough.

17. STACKED BAR GRAPH

A stacked bar divides the total into components.

Example:

TEXT

Total employees

A	% % % % ^	M e n	% % % ^	W o m e n	
B	% % % % % ^	M e n	% % ^	W o m e n	
C	% % ^	M e n	% % % % % ^	W o m e n	

Suppose:

Branch	Men	Women
A	120	80
B	150	50
C	60	140

Each full bar total:

A:

$\sqrt[2]{200}$

B:

$\sqrt[2]{200}$

C:

$\sqrt[2]{200}$

18. STACKED BAR — COMPONENT VALUE

For A:

Men:

$\sqrt[2]{120}$

Women:

$\sqrt[2]{80}$

Total:

$\sqrt[2]{200}$

Women share:

$\sqrt[2]{80/200=40\%}$

19. STACKED BAR — DIFFERENCE BETWEEN COMPONENTS

Branch C:

Women " Men:

$\sqrt[2]{140-60}$

$\sqrt[2]{= \boxed{80}}$

20. STACKED BAR — TOTAL ACROSS CATEGORIES

Total women:

$\sqrt[2]{80+50+140}$

$$\boxed{270}$$

Total men:

$$120+150+60$$

$$\boxed{330}$$

Grand total:

$$600$$

21. OVERALL WOMEN PERCENTAGE

$$270/600 = \boxed{45\%}$$

Do not average branch percentages unless branch totals equal.

Here totals happen to be equal, so simple average would work, but that is not generally true.

22. UNEQUAL STACKED BARS

Consider:

Branch	Men	Women
A	120	80
B	150	100
C	60	140

Totals:

A:

$$200$$

B:

$$250$$

C:

$$200$$

Women percentages:

A:

$$40\%$$

B:

$$40\%$$

C:

$$70\%$$

Overall women:

$\lfloor 320 \rfloor$

Grand total:

$\lfloor 650 \rfloor$

Overall:

$\lfloor 320/650 \rfloor$

$\lfloor \approx \boxed{49.23\%} \rfloor$

Simple average of 40,40,70 is 50%, which is not exact.

23. STACKED BAR READING TRAP

A segment may not begin at zero.

Suppose:

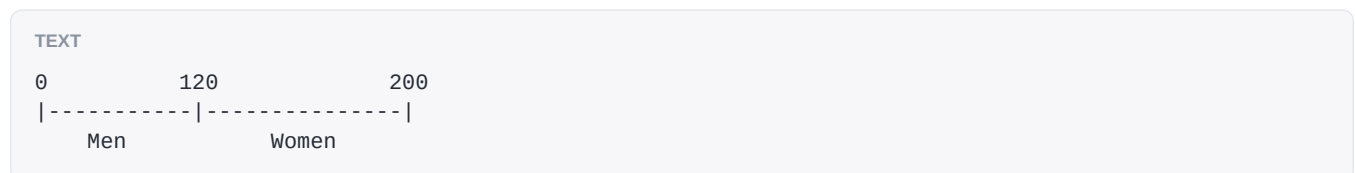
- first segment ends at 120
- second segment ends at 200

Then second segment value is:

$\lfloor 200-120 = \boxed{80} \rfloor$

Do not read second segment as 200.

24. DIAGRAM



Women segment length:

$\lfloor 200-120=80 \rfloor$

25. CUMULATIVE HEIGHT VS SEGMENT SIZE

This is one of the most important stacked-bar rules:

Top coordinate of a segment is cumulative; segment value is difference of boundaries.

26. 100% STACKED BAR

A 100% stacked bar shows percentage composition, not absolute totals.

Example:

TEXT

A | % % % % % % % % % % M e n 6 0 % | % % % % % W o m e n 4 0 % |
B | % % % % % M e n 4 0 % | % % % % % W o m e n 6 0 % |

Every full bar represents:

$\boxed{100\%}$

But totals of A and B may be different.

27. 100% STACKED BAR TRAP

Suppose:

A total:

$\boxed{100}$

Women:

$\boxed{40\% \rightarrow 40}$

B total:

$\boxed{500}$

Women:

$\boxed{60\% \rightarrow 300}$

B has a larger percentage **and** much larger count.

But even a smaller percentage could still yield a larger count if total is large.

28. EXAMPLE — SMALLER SHARE, LARGER COUNT

A:

Total 1000, women 40%

$\boxed{400}$

B:

Total 500, women 60%

$\boxed{300}$

So A has lower women percentage but higher women count.

29. FIND COMPONENT FROM TOTAL AND %

If stacked composition says:

Total:

$\boxed{800}$

Category X:

$$\boxed{35\%}$$

Then:

$$X = 0.35(800)$$

$$\boxed{280}$$

30. FIND TOTAL FROM COMPONENT AND SHARE

Component:

$$180$$

Share:

$$30\%$$

Total:

$$180/0.3$$

$$\boxed{600}$$

31. FIND MISSING SHARE

If components are:

- A = 25%
- B = 30%
- C = 15%
- D = ?

Then:

$$D = 100 - (25 + 30 + 15)$$

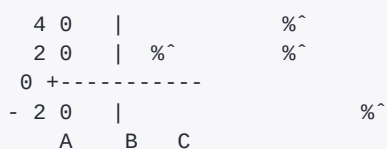
$$\boxed{30\%}$$

32. BAR GRAPH WITH NEGATIVE VALUES

Bars may extend below zero.

Example:

TEXT



Negative bar could represent:

- loss
- decline
- deficit

- negative growth

Do not treat magnitude as positive automatically.

33. PROFIT/LOSS BAR

If bar shows:

$\boxed{-20}$

that means:

$\boxed{\text{loss of } 20}$

Difference between +30 and -20 is:

$30 - (-20) = 50$

not 10.

34. TRUNCATED / BROKEN AXIS

Suppose y-axis starts at 90:

TEXT

120		100%
110		100%
100		100%
90 +-----		
A B		

A = 100 B = 115

Visually B may look much taller.

Actual percentage difference:

$15/100 = 15\%$

not 100%+.

35. ALWAYS CHECK ZERO BASELINE

Bar charts are especially misleading when axis does not begin at zero.

For GATE-style interpretation:

trust the numbers, not visual area.

36. IRREGULAR SCALE TRAP

Axis labels:

$\boxed{0, 10, 20, 50, 100}$

would be suspicious if equally spaced visually.

Normally graph scales should be consistent unless explicitly broken.

Read labels rather than assuming.

37. BAR INTERPOLATION

If a bar lies halfway between:

$\lfloor 100\text{ and }120 \rfloor$

value is:

$\lfloor \boxed{110} \rfloor$

provided scale is linear.

38. SMALL TICK MARKS

Suppose major interval:

$\lfloor 100\text{ to }150 \rfloor$

with 5 equal subintervals.

Each small interval:

$\lfloor 50/5 = \boxed{10} \rfloor$

A bar at third small tick above 100:

$\lfloor 130 \rfloor$

39. MULTI-YEAR BAR SET

Use:

Year	Product A	Product B
2021	100	120
2022	120	150
2023	150	180
2024	180	210
2025	240	240

40. YEAR WITH EQUAL VALUES

2025:

$\lfloor 240=240 \rfloor$

$\boxed{2025}$

41. YEAR WITH LARGEST ABSOLUTE GAP

Differences:

2021:

20

2022:

30

2023:

30

2024:

30

2025:

0

Largest:

$\boxed{2022, 2023, 2024}$

42. AVERAGE A

$\frac{100+120+150+180+240}{5}$

$=\frac{790}{5}$

$=\boxed{158}$

43. TOTAL B

$120+150+180+210+240$

$=\boxed{900}$

44. A:B TOTAL RATIO

A total:

790

B total:

900

Ratio:

$790:900$

divide by 10:

$\boxed{79:90}$

45. GROWTH A — 2021 TO 2025

$100 \rightarrow 240$

Increase:

140

Percentage:

$140/100 = \boxed{140\%}$

46. GROWTH B

$120 \rightarrow 240$

Increase:

120

Percentage:

$120/120 = \boxed{100\%}$

47. GROUPED-BAR COMPARISON SHORTCUT

If comparing growth rates:

$\frac{\text{New}-\text{Old}}{\text{Old}}$

You can compare fractions without turning into percentages.

Example:

A:

$40/160$

B:

$45/200$

Compare:

$40(200)=8000$

$45(160)=7200$

A growth is larger.

48. CONDITIONAL QUESTION

Using previous table:

In which years is A at least 80% of B?

Check:

2021:

$$100/120 = 83.33\% \text{ quad'}$$

2022:

$$120/150 = 80\% \text{ quad'}$$

2023:

$$150/180 = 83.33\% \text{ quad'}$$

2024:

$$180/210 \approx 85.71\% \text{ quad'}$$

2025:

$$100\% \text{ quad'}$$

Answer:

$$\boxed{\text{all years}}$$

49. THRESHOLD METHOD

Instead of dividing:

A is at least 80% of B means:

$$\boxed{A \geq 0.8B}$$

Usually faster.

50. STACKED BAR — MULTI-COMPONENT EXAMPLE

Suppose company workforce:

Year	Tech	Sales	Support
2023	120	80	50
2024	150	100	50
2025	180	120	60

Totals:

2023:

$$250$$

2024:

$$300$$

2025:

$$360$$

51. TOTAL GROWTH

2023 !' 2025:

$\backslash[250\text{to}360 \backslash]$

Increase:

$\backslash[110 \backslash]$

Percentage:

$\backslash[110/250 = \boxed{44\%} \backslash]$

52. TECH SHARE — 2025

$\backslash[180/360 = \boxed{50\%} \backslash]$

53. SUPPORT SHARE — 2024

$\backslash[50/300 = \boxed{16.67\%} \backslash]$

54. WHICH COMPONENT GREW FASTEST?

2 0 2 3!' 2 0 2 5 :

Tech:

$\backslash[60/120=50\% \backslash]$

Sales:

$\backslash[40/80=50\% \backslash]$

Support:

$\backslash[10/50=20\% \backslash]$

Highest:

$\backslash[\boxed{\text{Tech and Sales}} \backslash]$

55. STACKED BAR — DIFFERENCE IN COMPOSITION

Suppose Tech share:

2023:

$\backslash[120/250=48\% \backslash]$

2025:

$\backslash[180/360=50\% \backslash]$

Share increase:

$\boxed{2\text{ percentage points}}$

Relative increase in share:

$\frac{2}{48} \times 100$

$\approx 4.17\%$

Different concepts.

56. NORMAL BAR VS 100% STACKED BAR

Normal stacked bar

Bar length represents actual total.

100% stacked bar

Every bar has equal full length = 100%.

Only composition is visually shown.

Remember this distinction.

57. MISSING BAR VALUE FROM TOTAL

A, B, C, D total:

700

Known:

120, 180, 200

Missing D:

700 - 500

$\boxed{200}$

58. MISSING BAR FROM AVERAGE

Average of 5 bars:

140

Total:

700

Known four sum:

520

Missing:

$\boxed{180}$

59. MISSING BAR FROM RATIO

A:B:

$\left[\frac{3}{4} \right]$

A:

$\left[150 \right]$

Then B:

$\left[150 \times \frac{4}{3} \right]$

$\left[\boxed{200} \right]$

60. MISSING STACK SEGMENT

Total bar:

$\left[500 \right]$

Known segments:

$\left[150, 120, 80 \right]$

Missing:

$\left[500 - 350 \right]$

$\left[\boxed{150} \right]$

61. BAR GRAPH SPEED STRATEGY

For each set:

1. read axis and units
 2. identify series/legend
 3. scan questions
 4. answer direct questions first
 5. calculate reusable totals
 6. compare ratios with cross-multiplication
 7. approximate only if options permit
-

62. COMMON BAR-GRAPH TRAPS

& pTRAP 1 — Axis scale

Never assume.

& pTRAP 2 — Truncated axis

Visual difference can exaggerate numerical difference.

& pTRAP 3 — Stacked segment

Segment value = top boundary " bottom boundary.

& pTRAP 4 — 100% stacked bar

Equal bar lengths do not mean equal totals.

& pTRAP 5 — Absolute vs percentage increase

Different rankings.

& pTRAP 6 — Negative bars

Subtract negatives correctly.

& pTRAP 7 — Wrong legend

Grouped bars often use similar shades/patterns.

63. MUST-MEMORIZE BAR-DI RELATIONS

Growth:

$$\boxed{\frac{\text{New}-\text{Old}}{\text{Old}} \times 100}$$

Share:

$$\boxed{\frac{\text{Part}}{\text{Total}} \times 100}$$

Stacked segment:

$$\boxed{\text{upper cumulative level} - \text{lower cumulative level}}$$

Overall stacked total:

$$\boxed{\sum \text{segments}}$$

64. FAST DECISION TREE

One bar?

Read value.

Compare two bars?

Difference/ratio.

Two years?

Check absolute and percentage change separately.

Stacked?

Determine whether labels show segments or cumulative boundaries.

100% stacked?

Need total before converting shares into counts.

Truncated axis?

Ignore apparent visual size; use scale.

Missing bar?

Use total/average/ratio relation.

PRACTICE SET 1 — SIMPLE BARS

Suppose values are:

Product	Sales
A	120
B	180
C	240
D	160
E	200

Q1

Highest sales?

Q2

Total sales?

Q3

Average sales?

Q4

C is what percent more than A?

Q5

D:E ratio?

ANSWERS SET 1

Q1

$\boxed{C}$

Q2

$\boxed{120+180+240+160+200 = 900}$

Q3

$\boxed{900/5 = 180}$

Q4

$\boxed{\frac{240-120}{120} \times 100 = 100\%}$

Q5

$\boxed{160:200 = 4:5}$

PRACTICE SET 2 — GROUPED BARS

Region	2024	2025
A	200	250
B	300	360
C	250	325
D	400	460

Q6
Highest absolute increase?

Q7
Highest percentage growth?

Q8
Total 2024?

Q9
Total 2025?

Q10
Overall growth percentage?

ANSWERS SET 2

Increases:

A:
[50]

B:
[60]

C:
[75]

D:
[60]

Q6
[boxed{C}]

Growth:

A:
[25\%]

B:

$$\lfloor 20\% \rfloor$$

C:

$$\lfloor 75/250=30\% \rfloor$$

D:

$$\lfloor 60/400=15\% \rfloor$$

Q7

$$\lfloor \boxed{C} \rfloor$$

Q8

$$\lfloor 200+300+250+400 = \boxed{1150} \rfloor$$

Q9

$$\lfloor 250+360+325+460 = \boxed{1395} \rfloor$$

Q10

Increase:

$$\lfloor 245 \rfloor$$

$$\lfloor \frac{245}{1150} \times 100 \rfloor$$

$$\lfloor \approx \boxed{21.30\%} \rfloor$$

PRACTICE SET 3 — STACKED BARS

Branch	Men	Women	Interns
A	100	80	20
B	150	100	50
C	120	120	60

Q11

Total workforce in B?

Q12

Total women across all branches?

Q13

Women share in C?

Q14

Total interns as % of total workforce?

Q15

Which branch has highest women percentage?

ANSWERS SET 3

Q11

$$\backslash[150+100+50 = \boxed{300} \backslash]$$

Q12

$$\backslash[80+100+120 = \boxed{300} \backslash]$$

Q13

C total:

$$\backslash[300 \backslash]$$

Women:

$$\backslash[120 \backslash]$$

$$\backslash[=\boxed{40\%} \backslash]$$

Q14

Interns:

$$\backslash[20+50+60=130 \backslash]$$

Grand total:

A:

$$\backslash[200 \backslash]$$

B:

$$\backslash[300 \backslash]$$

C:

$$\backslash[300 \backslash]$$

Total:

$$\backslash[800 \backslash]$$

Share:

$$\backslash[130/800 = \boxed{16.25\%} \backslash]$$

Q15

A:

$$\backslash[80/200=40\% \backslash]$$

B:

$$\backslash[100/300=33.33\% \backslash]$$

C:

$$\backslash[120/300=40\% \backslash]$$

$$\backslash[\boxed{A\text{ and }C} \backslash]$$

LEVEL 3 — EXAM PRESSURE

Use:

Year	Product X	Product Y	Product Z
2022	120	180	150
2023	150	210	180
2024	180	240	225
2025	240	270	300

Q16

Which product had highest percentage growth from 2022 to 2025?

Q17

What is total sales in 2024?

Q18

What percentage of 2025 total is Z?

Q19

What is the ratio of total X sales to total Y sales across all years?

Q20

By what percentage did total annual sales grow from 2022 to 2025?

ANSWERS

Q16

X:

$\left[\frac{120}{240} \rightarrow 100\% \right]$

Y:

$\left[\frac{180}{270} \rightarrow 50\% \right]$

Z:

$\left[\frac{150}{300} \rightarrow 100\% \right]$

$\left[\boxed{X \text{ and } Z} \right]$

Q17

$\left[180 + 240 + 225 = \boxed{645} \right]$

Q18

2025 total:

$$\backslash[240+270+300=810 \backslash]$$

Z share:

$$\backslash[300/810 = 10/27 \backslash]$$

$$\backslash[\approx\boxed{37.04\%} \backslash]$$

Q19

X total:

$$\backslash[120+150+180+240 = 690 \backslash]$$

Y total:

$$\backslash[180+210+240+270 = 900 \backslash]$$

Ratio:

$$\backslash[690:900 = \boxed{23:30} \backslash]$$

Q20

2022 total:

$$\backslash[120+180+150 = 450 \backslash]$$

2025:

$$\backslash[810 \backslash]$$

Increase:

$$\backslash[360 \backslash]$$

$$\backslash[360/450\times100 = \boxed{80\%} \backslash]$$

CHAPTER 52 STATUS

Covered so far:

- simple bar graphs '
- horizontal bars '
- grouped/multiple bars '
- axis scales '
- direct comparisons '
- totals and averages '
- absolute vs percentage growth '
- stacked bars '
- cumulative segment boundaries '
- 100% stacked bars '
- component shares '
- negative bars '
- truncated-axis traps '
- missing bar values '
- threshold comparisons '

- multi-year bar sets '
- L0–L3 practice '

But **Chapter 52 is not complete yet.**

Still remaining:

- advanced grouped-bar comparisons
- stacked-bar composition changes
- 100% stacked bars with changing totals
- derived metrics from bars
- conditional bar questions
- combined bars + percentage labels
- misleading graphs and interpretation traps
- large placement-style timed set
- final bar-DI speed audit

Progress

$\lfloor \boxed{51/76} \text{ completed} \rfloor \rfloor$

Current: $\emptyset = 0\%$ Chapter 52 — Bar & Stacked-Bar Graphs

NEXT

$\emptyset = 0\%$ **Chapter 52 Part 2 — Advanced Stacked Bars, 100% Composition, Derived Metrics, Conditional Questions & Final Bar-Graph Speed Toolkit**

USER

next

ASSISTANT

CHAPTER 52 — BAR GRAPHS & STACKED-BAR GRAPHS

PART 2 — ADVANCED STACKED BARS, 100% COMPOSITION, DERIVED METRICS, CONDITIONAL QUESTIONS & FINAL SPEED TOOLKIT

Status: $\emptyset = 0\%$ **PROGRESS GATE:** High **Placements:** High **Priority:** $\emptyset = 0\%$ **Per 1**

This part focuses on the questions where the graph gives one layer of information, but the answer needs another derived layer.

Main principle:

Read bar !' derive required metric !' compare only after using the correct denominator.

1. ADVANCED STACKED-BAR IDEA

Suppose each bar shows company expenses split into:

- salary
- marketing
- operations

Company	Salary	Marketing	Operations
A	200	100	100
B	240	120	140
C	300	150	150

Totals:

A:

$\sqrt[4]{400}$

B:

$\sqrt[4]{500}$

C:

$\sqrt[4]{600}$

2. COMPONENT SHARE

Salary share in B:

$\sqrt[4]{\frac{240}{500} \times 100}$

$\sqrt[4]{= \boxed{48\%}}$

Marketing share:

$\sqrt[4]{120/500=24\%}$

Operations:

$\sqrt[4]{140/500=28\%}$

Check:

$\sqrt[4]{48+24+28=100}$

3. SAME COMPONENT ACROSS BARS

Total salary:

$\sqrt[4]{200+240+300}$

$\sqrt[4]{= \boxed{740}}$

Total marketing:

$\sqrt[4]{100+120+150}$

$$\boxed{370}$$

Total operations:

$$100+140+150$$

$$\boxed{390}$$

Grand total:

$$1500$$

4. OVERALL COMPONENT SHARE

Salary overall:

$$740/1500$$

$$\boxed{49.33\%}$$

Do not average individual salary percentages unless company totals are equal.

5. COMPOSITION CHANGE

Suppose:

Year	Tech	Sales	Support
2024	200	100	100
2025	240	180	180

Totals:

2024:

$$400$$

2025:

$$600$$

Tech share:

2024:

$$200/400=50\%$$

2025:

$$240/600=40\%$$

So tech headcount increased:

$$200\text{to}240$$

but tech **share decreased**:

$$50\%\text{to}40\%$$

This is a very important DI pattern.

6. ABSOLUTE VALUE CAN RISE WHILE SHARE FALLS

Here:

Tech absolute increase:

[40]

Percentage increase in tech count:

[20%]

But share fell by:

[10\text{ percentage points}]

because total workforce grew faster.

7. SHARE CAN RISE WHILE ABSOLUTE VALUE FALLS

Example:

2024:

Tech = 200, total = 1000

[20%]

2025:

Tech = 180, total = 600

[30%]

Tech count fell:

[10%]

but share rose from:

[20%\to30%]

8. 100% STACKED BAR WITH CHANGING TOTALS

Suppose:

Year	Total Employees	Tech %	Sales %	Support %
2024	500	40%	35%	25%
2025	800	35%	40%	25%

Counts:

2024

Tech:

[200]

Sales:

$\backslash[175 \backslash]$

Support:

$\backslash[125 \backslash]$

2025

Tech:

$\backslash[280 \backslash]$

Sales:

$\backslash[320 \backslash]$

Support:

$\backslash[200 \backslash]$

9. IMPORTANT 100% STACKED BAR TRAP

Tech percentage fell:

$\backslash[40\%\text{to}35\% \backslash]$

but actual Tech count rose:

$\backslash[200\text{to}280 \backslash]$

So:

$\backslash[\boxed{\text{percentage decrease does not imply count decrease}} \backslash]$

10. PERCENTAGE CHANGE IN COMPONENT COUNT

Tech:

$\backslash[200\text{to}280 \backslash]$

Increase:

$\backslash[80 \backslash]$

Percentage:

$\backslash[80/200=40\% \backslash]$

Even though its share fell by 5 percentage points.

11. COMPONENT COUNT FROM TOTAL $\times$ SHARE

General formula:

$\backslash[\boxed{\text{component} = \text{total} \times \text{share}} \backslash]$

If share is in percent:

$\backslash[\boxed{\text{component} = \text{total} \times \frac{p}{100}} \backslash]$

12. FIND TOTAL FROM COMPONENT + SHARE

Component:

$$\boxed{210}$$

Share:

$$\boxed{35\%}$$

Total:

$$\boxed{210/0.35}$$

$$\boxed{600}$$

13. FIND SHARE FROM COMPONENT + TOTAL

Component:

$$\boxed{150}$$

Total:

$$\boxed{600}$$

Share:

$$\boxed{150/600 = 25\%}$$

14. GROUPED BAR + DERIVED METRIC

Suppose grouped bars show Revenue and Cost.

Company	Revenue	Cost
A	500	400
B	700	525
C	800	640
D	900	675

Profit:

$$\boxed{\text{Revenue}-\text{Cost}}$$

A:

$$\boxed{100}$$

B:

$$\boxed{175}$$

C:

$$\boxed{160}$$

D:

$$\boxed{225}$$

Highest profit:

$$\boxed{D}$$

15. PROFIT MARGIN FROM BAR DATA

A:

$$100/500=20\%$$

B:

$$175/700=25\%$$

C:

$$160/800=20\%$$

D:

$$225/900=25\%$$

Highest margin:

$$\boxed{B \text{ and } D}$$

Highest profit amount and highest profit margin are different metrics.

16. COST AS % OF REVENUE

For B:

$$525/700 = 75\%$$

So profit margin:

$$100-75 = 25\%$$

Sometimes this is faster than calculating profit first.

17. DERIVED METRIC — PRODUCTIVITY

Suppose bars show:

- output
- workers

Plant	Output	Workers
A	1200	60
B	1600	100
C	1350	50

Plant	Output	Workers
D	1800	120

Productivity:

$\frac{\text{Output}}{\text{Workers}}$

A:

20

B:

16

C:

27

D:

15

Highest:

C

18. DERIVED METRIC — PER-CAPITA / PER-UNIT

If bar A shows population and another bar shows GDP:

$\text{GDP per capita} = \frac{\text{GDP}}{\text{Population}}$

Whenever two bars use different units, convert carefully before comparing.

19. CONDITIONAL BAR QUESTION

Suppose:

Branch	Sales 2024	Sales 2025
A	200	260
B	300	345
C	250	325
D	400	440
E	150	225

Question:

Which branches had growth at least 25%?

Use threshold:

$\lfloor \text{New} \geq 1.25(\text{Old}) \rfloor$

A:

$\lfloor 260 \geq 250 \rfloor$

,

B:

$\lfloor 345 < 375 \rfloor$

'L

C:

$\lfloor 325 \geq 312.5 \rfloor$

,

D:

$\lfloor 440 < 500 \rfloor$

'L

E:

$\lfloor 225 \geq 187.5 \rfloor$

,

Answer:

$\lfloor \boxed{A,C,E} \rfloor$

20. CONDITIONAL TOTAL

Using same data:

Total 2025 sales of branches with growth at least 25%.

$\lfloor 260 + 325 + 225 \rfloor$

$\lfloor = \boxed{810} \rfloor$

21. CONDITIONAL AVERAGE

Average 2024 sales of qualifying branches:

$\lfloor \frac{200 + 250 + 150}{3} \rfloor$

$\lfloor = \boxed{200} \rfloor$

22. MULTI-CONDITION BAR FILTER

Suppose bars represent:

- revenue

- expense
- employees

Question:

Find companies with revenue > 700, margin >20%, employees <150.

Strategy:

1. filter by raw revenue
2. filter by employee count
3. compute margin only for remaining companies

Do cheapest tests first.

23. COMBINED BAR + PERCENTAGE LABELS

Suppose bars show total sales and labels show online percentage.

Year	Total Sales	Online %
2023	500	20%
2024	600	25%
2025	800	30%

Online sales:

2023:

$\sqrt[100]{}$

2024:

$\sqrt[150]{}$

2025:

$\sqrt[240]{}$

24. OFFLINE SALES

Offline percentage:

$\sqrt[100-\text{Online \%}]{}$

So:

2025 offline:

$\sqrt[70\% \times 800]{}$

$\sqrt[= \boxed{560}]{}$

25. ONLINE GROWTH VS TOTAL GROWTH

Total sales:

$\backslash[500\text{to}800 \backslash]$

Growth:

$\backslash[60\% \backslash]$

Online:

$\backslash[100\text{to}240 \backslash]$

Growth:

$\backslash[140\% \backslash]$

A segment can grow much faster than total.

26. COMPOSITION SHIFT

Online share:

$\backslash[20\%\text{to}30\% \backslash]$

Increase:

$\backslash[\boxed{10\text{percentage points}} \backslash]$

Relative increase in share:

$\backslash[\frac{10}{20}\times100 = \boxed{50\%} \backslash]$

Different from online count growth of 140%.

27. BAR GRAPH WITH PERCENTAGE CHANGE LABELS

Sometimes graph gives current value plus growth%.

If current value:

$\backslash[240 \backslash]$

and growth from previous year:

$\backslash[20\% \backslash]$

Previous value:

$\backslash[\frac{240}{1.2} \backslash]$

$\backslash[=\boxed{200} \backslash]$

Reverse percentages are common.

28. BAR GRAPH WITH DECREASE LABEL

Current:

$\backslash[180 \backslash]$

after 10% decline.

Previous:

$\sqrt{180/0.9}$

$\sqrt{= \boxed{200}}$

Do not simply add 10% of 180.

29. INDEX BAR GRAPH

Suppose bar heights are index values:

$\sqrt{100, 120, 90, 150}$

Interpret:

- 100 = base
- 120 = 20% above base
- 90 = 10% below base
- 150 = 50% above base

But growth from 120 to 150:

$\sqrt{\frac{30}{120} = 25\%}$

not 30%.

30. NORMALIZED BAR GRAPH

If bars show:

$\sqrt{\text{cases per 1000 people}}$

you cannot compare absolute case counts without population.

Example:

A:

10 per 1000, population 1,000,000

Count:

$\sqrt{10,000}$

B:

20 per 1000, population 100,000

Count:

$\sqrt{2,000}$

Higher rate does not imply higher count.

31. MISLEADING 3D BAR GRAPH

3D visual bars can distort apparent heights due to perspective.

Always use:

- axis labels
- exact values
- tick marks

not perceived volume/area of the graphic.

32. PICTOGRAPH WARNING

If one icon represents 100 units:

$$\lceil 3.5 \text{ icons} \rceil = 350$$

Check whether partial icons are allowed.

Never count icons without reading the legend.

33. DUAL-AXIS BAR GRAPH

Some graphs use:

- left axis for sales
- right axis for growth %

A bar and line can visually align at same height but represent different numerical quantities.

This is a major interpretation trap.

34. AXIS-UNIT CHECK

Example:

Left axis:

$$\lceil 1 \text{ t e x t } \rceil$$

Right axis:

$$\lceil 1\% \rceil$$

Never divide or compare raw heights unless you know which axis each series uses.

35. DIFFERENCE-OF-PERCENTAGES TRAP

A bar category share rises from:

$$\lceil 30\% \text{ to } 38\% \rceil$$

Correct:

$$\lceil +8 \text{ percentage points} \rceil$$

Relative increase:

$$\lceil \frac{8}{30} \times 100 \rceil$$

$$\lceil =26.67\% \rceil$$

36. TOTAL FROM STACKED BOUNDARIES

Suppose stacked bar boundaries are:

$$\backslash [0, \backslash 120, \backslash 200, \backslash 270 \backslash]$$

Segments are:

First:

$$\backslash [120 \backslash]$$

Second:

$$\backslash [200-120=80 \backslash]$$

Third:

$$\backslash [270-200=70 \backslash]$$

Total:

$$\backslash [270 \backslash]$$

37. SEGMENT RATIO

Using previous:

$$\backslash [120:80:70 \backslash]$$

Divide by 10:

$$\backslash [\boxed{12:8:7} \backslash]$$

38. SEGMENT SHARE

Third segment:

$$\backslash [70/270 \backslash]$$

$$\backslash [\approx \boxed{25.93\%} \backslash]$$

39. ADVANCED 100% STACKED SET

Suppose:

Company	Total	Product X %	Product Y %	Product Z %
A	400	25%	50%	25%
B	600	30%	40%	30%
C	500	40%	20%	40%

Counts:

A

$$X = 100 \ Y = 200 \ Z = 100$$

B

$X = 180$ $Y = 240$ $Z = 180$

C

$X = 200$ $Y = 100$ $Z = 200$

40. TOTAL PRODUCT X

$\backslash [100+180+200 = \boxed{480} \backslash]$

41. TOTAL PRODUCT Y

$\backslash [200+240+100 = \boxed{540} \backslash]$

42. TOTAL PRODUCT Z

$\backslash [100+180+200 = \boxed{480} \backslash]$

43. GRAND TOTAL

$\backslash [400+600+500 = 1500 \backslash]$

Check:

$\backslash [480+540+480=1500 \backslash]$

Good.

44. OVERALL X SHARE

$\backslash [480/1500 \backslash]$

$\backslash [=\boxed{32\%} \backslash]$

Notice simple average of X percentages:

$\backslash [(25+30+40)/3=31.67\% \backslash]$

not exact.

Weighted denominator matters.

45. COMPANY WITH HIGHEST X COUNT

A:

100

B:

180

C:

200

So:

$\boxed{C}$

Even though C and not B has highest X percentage too here—but this need not always happen.

46. CONDITIONAL COMPONENT QUESTION

Question:

Total Y count among companies where Z share is at least 30%.

Qualify:

B:

30% '

C:

40% '

Y counts:

$\boxed{240+100 = 340}$

47. CHANGE IN STACKED COMPONENTS ACROSS YEARS

Suppose:

Component	2024	2025
A	100	150
B	200	220
C	300	360

Total:

2024:

$\boxed{600}$

2025:

$\boxed{730}$

Growth of components:

A:

$\boxed{50\%}$

B:

$\sqrt{10\%}$

C:

$\sqrt{20\%}$

Highest:

$\sqrt{\boxed{A}}$

48. CONTRIBUTION TO TOTAL INCREASE

Total increase:

$\sqrt{730-600=130}$

A contributes:

$\sqrt{50}$

B:

$\sqrt{20}$

C:

$\sqrt{60}$

Contribution shares:

A:

$\sqrt{50/130}$

B:

$\sqrt{20/130}$

C:

$\sqrt{60/130}$

Largest contributor:

$\sqrt{\boxed{C}}$

Even though A has highest percentage growth.

49. HIGH-VALUE INTERPRETATION

Three different winners can exist:

- highest percentage growth
- highest absolute increase
- highest contribution to total

Always read wording.

50. BAR-GRAPH OPTION ELIMINATION

Suppose exact ratio:

$\backslash[398:603 \backslash]$

Options:

- 1:3
- 2:3
- 3:4
- 4:5

Approximate:

$\backslash[400:600=2:3 \backslash]$

Answer:

$\backslash[\boxed{2:3} \backslash]$

if options are broad.

51. FAST PERCENTAGE COMPARISON

Compare growth:

A:

$\backslash[60/240 \backslash]$

B:

$\backslash[70/300 \backslash]$

Cross multiply:

$\backslash[60(300)=18000 \backslash]$

$\backslash[70(240)=16800 \backslash]$

So A growth is larger.

No percentage conversion.

52. FAST SHARE COMPARISON

Compare:

$\backslash[\frac{180}{500} \backslashquad\text{and}\backslashquad \frac{200}{600} \backslash]$

Cross multiply:

$\backslash[180(600)=108000 \backslash]$

$\backslash[200(500)=100000 \backslash]$

First share larger.

53. DENSE PLACEMENT-STYLE SET

Suppose grouped bars correspond to:

Region	2024 Revenue	2025 Revenue	2025 Cost
North	500	600	450
South	700	840	630
East	400	520	390
West	800	920	690

54. 2025 PROFIT

North:

$$\boxed{150}$$

South:

$$\boxed{210}$$

East:

$$\boxed{130}$$

West:

$$\boxed{230}$$

Highest:

$$\boxed{\text{West}}$$

55. 2025 PROFIT MARGIN

North:

$$\boxed{150/600=25\%}$$

South:

$$\boxed{210/840=25\%}$$

East:

$$\boxed{130/520=25\%}$$

West:

$$\boxed{230/920=25\%}$$

All equal.

56. REVENUE GROWTH 2024' 2025

North:

$$\boxed{100/500=20\%}$$

South:

$$\lfloor 140/700=20\% \rfloor$$

East:

$$\lfloor 120/400=30\% \rfloor$$

West:

$$\lfloor 120/800=15\% \rfloor$$

Highest:

$$\lfloor \boxed{\text{East}} \rfloor$$

57. TOTAL 2025 PROFIT

$$\lfloor 150+210+130+230 \rfloor$$

$$\lfloor =\boxed{720} \rfloor$$

58. REGION CONTRIBUTION TO PROFIT

West:

$$\lfloor 230/720 \rfloor$$

$$\lfloor \approx\boxed{31.94\%} \rfloor$$

59. CONDITIONAL QUESTION

Regions with revenue growth at least 20%:

North, South, East.

Their total 2025 profit:

$$\lfloor 150+210+130 \rfloor$$

$$\lfloor =\boxed{490} \rfloor$$

60. TIMED SET STRATEGY

For a 5-question bar set:

First 20–30 seconds

Read:

- axes
- units
- legend
- categories

Then

Do direct-value questions first.

Next

Do totals/differences.

Then

Do percentage/rate questions.

Last

Do multi-step derived metrics.

Do not spend the first minute precomputing everything.

61. WHEN TO REDRAW AS A TABLE

If a bar graph is visually dense, quickly create a mini-table:

TEXT

A: 120, 180

B: 150, 210

C: 200, 240

This often reduces graph-reading mistakes.

62. WHEN NOT TO REDRAW

If only one or two direct questions exist, copying the whole graph wastes time.

Use judgment.

63. FINAL BAR-GRAPH TRAPS

& pTRAP 1 — 100% stacked bars

Need totals to find counts.

& pTRAP 2 — Stacked boundary

Read segment by subtraction.

& pTRAP 3 — Share vs amount

A lower share can still correspond to higher count.

& pTRAP 4 — Growth vs contribution

Highest growth may not contribute most to total increase.

& pTRAP 5 — Dual axes

Different series may use different units/scales.

& pTRAP 6 — Truncated axes

Visual difference may exaggerate real difference.

& pTRAP 7 — Profit margin

Use profit/revenue unless question explicitly says percentage on cost.

64. MUST-MEMORIZE ADVANCED BAR RELATIONS

Component count:

$$\boxed{T \times \frac{p}{100}}$$

Reverse total:

$$\boxed{T = \frac{\text{component}}{p/100}}$$

Contribution to increase:

$$\boxed{\frac{\text{component increase}}{\text{total increase}} \times 100}$$

Derived profit:

$$\boxed{\text{Revenue} - \text{Cost}}$$

Profit margin:

$$\boxed{\frac{\text{Revenue} - \text{Cost}}{\text{Revenue}} \times 100}$$

65. FAST BAR-DI DECISION TREE

Simple bar?

Read value.

Grouped bars?

Compare same category carefully.

Stacked?

Separate segment from cumulative top.

100% stacked?

Use total $\times$ share.

Percentage labels?

Convert to count if needed.

Revenue + cost?

Derive profit/margin.

Output + workers?

Derive productivity.

Conditional question?

Filter first.

Options broad?

Approximate/cross-multiply.

Dense chart?

Make tiny rough table.

PRACTICE — LEVEL 0

Use:

Category	2024	2025
A	100	130
B	200	240
C	300	330
D	400	500

Q1

Highest absolute increase?

Q2

Highest percentage increase?

Q3

Total 2025?

Q4

A:D ratio in 2025?

Q5

D share of 2025 total?

LEVEL 1

Use 100% stacked data:

Branch	Total	Men %	Women %
A	400	60%	40%
B	600	50%	50%
C	500	40%	60%

Q6

Women count in A?

Q7

Women count in B?

Q8

Women count in C?

Q9

Total women?

Q10

Overall women percentage?

LEVEL 2

Use:

Company	Revenue	Cost	Employees
P	600	450	100
Q	750	600	125
R	900	675	150
S	800	560	160

Q11

Highest profit?

Q12

Highest profit margin?

Q13

Highest revenue per employee?

Q14

Highest profit per employee?

Q15

Total revenue of firms with margin at least 25%?

LEVEL 3 — EXAM PRESSURE

Use:

Year	Total Market	Company A Share	Company B Share
2022	1000	30%	25%
2023	1200	28%	27%
2024	1500	26%	30%

Year	Total Market	Company A Share	Company B Share
2025	1800	25%	32%

Q16

Company A sales in each year?

Q17

Company B sales in 2025?

Q18

Did A's absolute sales rise or fall from 2022 to 2025?

Q19

What is A's percentage growth in sales from 2022 to 2025?

Q20

What is B's percentage growth from 2022 to 2025?

ANSWERS

Q1

Increases:

A:

$\sqrt[3]{30}$

B:

$\sqrt[4]{40}$

C:

$\sqrt[3]{30}$

D:

$\sqrt[4]{100}$

$\sqrt{\boxed{D}}$

Q2

A:

$\sqrt[3]{30\%}$

B:

$\sqrt[4]{20\%}$

C:

$\sqrt[4]{10\%}$

D:

$$\backslash[25\% \backslash]$$

$$\backslash[\boxed{A} \backslash]$$

Q3

$$\backslash[130+240+330+500 = \boxed{1200} \backslash]$$

Q4

$$\backslash[130:500 = \boxed{13:50} \backslash]$$

Q5

$$\backslash[500/1200 = \boxed{41.67\%} \backslash]$$

Q6

$$\backslash[400(0.4) = \boxed{160} \backslash]$$

Q7

$$\backslash[600(0.5) = \boxed{300} \backslash]$$

Q8

$$\backslash[500(0.6) = \boxed{300} \backslash]$$

Q9

$$\backslash[160+300+300 = \boxed{760} \backslash]$$

Q10

Grand total:

$$\backslash[1500 \backslash]$$

$$\backslash[760/1500 = \boxed{50.67\%} \backslash]$$

Q11

Profits:

P:

$$\backslash[150 \backslash]$$

Q:

$$\backslash[150 \backslash]$$

R:

$\backslash[225 \backslash]$

S:

$\backslash[240 \backslash]$

$\backslash[\boxed{S} \backslash]$

Q12

P:

$\backslash[25\% \backslash]$

Q:

$\backslash[20\% \backslash]$

R:

$\backslash[25\% \backslash]$

S:

$\backslash[240/800=30\% \backslash]$

$\backslash[\boxed{S} \backslash]$

Q13

P:

$\backslash[600/100=6 \backslash]$

Q:

$\backslash[750/125=6 \backslash]$

R:

$\backslash[900/150=6 \backslash]$

S:

$\backslash[800/160=5 \backslash]$

$\backslash[\boxed{P,Q,R} \backslash]$

Q14

P:

$\backslash[150/100=1.5 \backslash]$

Q:

$\backslash[150/125=1.2 \backslash]$

R:

$\backslash[225/150=1.5 \backslash]$

S:

$\backslash[240/160=1.5 \backslash]$

$\backslash[\boxed{P,R,S} \backslash]$

Q15

Margins at least 25%:

P, R, S.

Revenue:

$$\boxed{600+900+800}$$

$$\boxed{2300}$$

Q16

Company A:

2022:

$$1000(0.30)=300$$

2023:

$$1200(0.28)=336$$

2024:

$$1500(0.26)=390$$

2025:

$$1800(0.25)=450$$

$$\boxed{300, 336, 390, 450}$$

Q17

$$1800(0.32) = \boxed{576}$$

Q18

A:

$$300\text{to}450$$

So absolute sales:

$$\boxed{\text{rose}}$$

even though market share fell.

Q19

$$\frac{450-300}{300}\times 100$$

$$\boxed{50\%}$$

Q20

B in 2022:

$$1000(0.25)=250$$

B in 2025:

$\frac{576}{250} \times 100$

Growth:

$\frac{576-250}{250} \times 100$

$\frac{326}{250} \times 100$

$\boxed{130.4\%}$

CHAPTER 52 COMPLETE

We have now covered:

- simple bars
- grouped bars
- horizontal bars
- stacked bars
- 100% stacked bars
- changing composition
- component counts
- derived metrics
- growth and contribution
- conditional filters
- percentage-labelled bars
- index bars
- normalized bars
- truncated/misleading axes
- dual-axis cautions
- dense placement-style sets
- speed and elimination strategies
- L0–L3 practice

Progress

$\boxed{52/76 \text{ completed}}$

UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations
- Chapter 51 — Tables
- Chapter 52 — Bar & Stacked-Bar Graphs
- Chapter 53 — Line Graphs
- Chapter 54 — Pie Charts
- Chapter 55 — Mixed Graphs & Caselets
- Chapter 56 — Missing-Data & Advanced DI

NEXT

Chapter 53 — Line Graphs

Next we will cover:

- reading trends
- slopes and change
- absolute vs percentage growth
- multi-line comparisons
- crossover points
- cumulative vs periodic line graphs
- growth-rate graphs
- moving averages
- index lines
- misleading/truncated axes
- interpolation
- GATE/placement line-chart strategy.

USER

next

ASSISTANT

CHAPTER 53 — LINE GRAPHS

PART 1 — READING TRENDS, SLOPES, GROWTH, CROSSOVER POINTS, MULTI-LINE COMPARISON & INTERPOLATION

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: High Priority: $\emptyset$ = Yes 1

A line graph connects data points to show how a quantity changes across:

- time
- distance
- categories
- stages

The most important idea is:

A line graph shows both value and direction of change.

But do not confuse:

- height of point
- slope
- absolute change
- percentage change

They are different.

1. BASIC LINE GRAPH

Suppose sales are:

Year	Sales
2021	100
2022	140
2023	120
2024	180
2025	220

Visual idea:



2. FIRST THING TO CHECK

Before answering:

- x-axis meaning
- y-axis meaning
- units
- scale
- whether axis starts at zero
- whether values are exact or approximate
- whether multiple lines use same axis

3. DIRECT VALUE

Sales in 2024:

$\boxed{180}$

No calculation.

4. ABSOLUTE CHANGE

2021 !' 2022:

$$\boxed{140-100 = 40}$$

2022 !' 2023:

$$\boxed{120-140 = -20}$$

Negative means decline.

5. PERCENTAGE CHANGE

2021 !' 2022:

$$\frac{40}{100} \times 100 = \boxed{40\%}$$

2022 !' 2023:

$$\frac{-20}{140} \times 100$$

$$= \boxed{-14.29\%}$$

6. TREND

A trend means the general direction.

From the example:

- 2021! increase
- 2022! decrease
- 2023! increase
- 2024! increase

Overall:

$$\boxed{100 \rightarrow 220}$$

So long-term trend is upward.

7. HIGHEST VALUE

Highest point:

$$\boxed{2025}$$

with:

$$\boxed{220}$$

8. LOWEST VALUE

Lowest point:

$\boxed{2021}$

with:

100

9. HIGHEST ABSOLUTE INCREASE

Year-to-year changes:

$+40, -20, +60, +40$

Largest increase:

$\boxed{+60}$

between:

$\boxed{2023\text{ to }2024}$

10. HIGHEST PERCENTAGE INCREASE

Compute positive growth rates.

2021 to 2022:

$40/100 = 40\%$

2022 to 2024:

$60/120 = 50\%$

2024 to 2025:

$40/180 = 22.22\%$

Highest:

$\boxed{2023\text{ to }2024}$

11. SLOPE — BASIC INTUITION

Slope indicates rate of change.

If time intervals are equal:

- steeper upward line ! larger absolute increase
- steeper downward line ! larger absolute decrease

But only if axis scale and x-spacing are consistent.

12. SLOPE FORMULA

Between two points:

$(x_1, y_1), (x_2, y_2)$

$\boxed{\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}}$

In time-series DI:

$$\text{slope} = \frac{\text{change in value}}{\text{change in time}}$$

13. EXAMPLE

Sales rises:

$$120 \text{ to } 180$$

over:

$$2 \text{ years}$$

Average rate of increase:

$$\frac{60}{2} = \boxed{30 \text{ units/year}}$$

14. SLOPE VS PERCENTAGE GROWTH

Suppose:

A:

$$100 \text{ to } 150$$

Increase:

$$50$$

Growth:

$$50\%$$

B:

$$500 \text{ to } 570$$

Increase:

$$70$$

Growth:

$$14\%$$

B has larger absolute slope but lower percentage growth.

15. FLAT LINE

If value does not change:

$$y_1 = y_2$$

Slope:

$$0$$

This means:

$$\boxed{\text{no absolute change}}$$

16. MULTI-LINE GRAPH

Suppose two companies:

Year	A	B
2021	100	150
2022	140	160
2023	180	170
2024	200	210
2025	240	230

17. WHICH COMPANY IS HIGHER?

Compare point values by year.

2021:

$\backslash B > A \backslash$

2022:

$\backslash B > A \backslash$

2023:

$\backslash A > B \backslash$

2024:

$\backslash B > A \backslash$

2025:

$\backslash A > B \backslash$

18. CROSSOVER

A crossover occurs when the ranking of two lines reverses.

Between 2022 and 2023:

A goes from below B to above B.

So lines cross somewhere between those years.

19. CROSSOVER DOES NOT ALWAYS MEAN EQUAL DATA AT A RECORDED POINT

If:

2022:

$$\backslash[A=140,\backslash B=160 \backslash]$$

2023:

$$\backslash[A=180,\backslash B=170 \backslash]$$

the lines cross between 2022 and 2023.

But no actual yearly observation is exactly equal.

20. EXACT CROSSOVER BY LINEAR INTERPOLATION

Assume straight-line change between points.

Let t be fraction of interval after 2022.

A:

$$\backslash[140+40t \backslash]$$

B:

$$\backslash[160+10t \backslash]$$

Set equal:

$$\backslash[140+40t=160+10t \backslash]$$

$$\backslash[30t=20 \backslash]$$

$$\backslash[t=\frac{2}{3} \backslash]$$

So crossover occurs:

$$\backslash[\boxed{\frac{2}{3}} \text{ of the way from 2022 to 2023} \backslash]$$

21. INTERPOLATION

If line segment is assumed linear:

Between value 100 at year 1 and 140 at year 2, halfway value:

$$\backslash[100+\frac{1}{2}(40) \backslash]$$

$$\backslash[=\boxed{120} \backslash]$$

22. INTERPOLATION FORMULA

If x lies between (x_1, x_2) :

$$\backslash[\boxed{y = y_1 + \frac{x-x_1}{x_2-x_1}(y_2-y_1)} \backslash]$$

Use only when linear interpolation is justified.

23. DO NOT INVENT INTERMEDIATE VALUES

If graph only gives yearly points and does not imply continuous linear behavior, exact intermediate values may not be

known.

A drawn connecting line is often only visual aid.

Be careful in real-world interpretation questions.

24. MULTI-LINE DIFFERENCE

Using A/B data:

Difference in 2025:

$$\boxed{240-230 = 10}$$

Difference in 2021:

$$\boxed{150-100 = 50}$$

Gap narrowed overall.

25. GAP CHANGE

Initial gap:

$$\boxed{50}$$

Final gap:

$$\boxed{10}$$

Gap reduced by:

$$\boxed{40}$$

Percentage reduction relative to initial gap:

$$\boxed{40/50 = 80\%}$$

26. AVERAGE VALUE OVER YEARS

Company A average:

$$\frac{100+140+180+200+240}{5}$$

$$= \frac{860}{5}$$

$$= \boxed{172}$$

27. TOTAL VALUE OVER PERIOD

If each point is annual sales:

$$100+140+180+200+240 = \boxed{860}$$

But if line shows cumulative sales, do not sum points blindly.

28. PERIODIC VS CUMULATIVE LINE GRAPH

Periodic:

Year	Annual Sales
2022	100
2023	120
2024	140

Cumulative:

Year	Cumulative Sales
2022	100
2023	220
2024	360

Different interpretation.

29. RECOVER PERIODIC FROM CUMULATIVE

Cumulative:

$\{100, 220, 360, 550\}$

Periodic:

First:

$\{100\}$

Second:

$\{220 - 100 = 120\}$

Third:

$\{360 - 220 = 140\}$

Fourth:

$\{550 - 360 = 190\}$

So:

$\{100, 120, 140, 190\}$

30. CUMULATIVE LINE PROPERTY

A true cumulative quantity usually should not decrease if underlying increments cannot be negative.

If cumulative production suddenly falls, check whether:

- returns/corrections are allowed

- graph is not actually cumulative
- data definition differs

31. GROWTH-RATE LINE GRAPH

Sometimes y-axis itself is growth percentage.

Example:

Year	Growth Rate
2022	10%
2023	20%
2024	-5%
2025	15%

These values are **rates**, not absolute sales.

32. GROWTH RATE "VALUE

If growth rate drops:

$[20\% \text{ to } 10\%]$

the underlying value can still rise.

It is just rising more slowly.

This is a classic interpretation trap.

33. NEGATIVE GROWTH RATE

Growth:

$[-5\%]$

means value decreased by 5% from previous period.

If previous value:

$[200]$

new:

$[200(0.95) = \boxed{190}]$

34. RECOVER VALUES FROM GROWTH RATES

Starting value:

$[100]$

Growth rates:

- Year 2: +20%
- Year 3: +10%
- Year 4: " 2 5 %

Values:

Year 2:

$$\boxed{100(1.2)=120}$$

Year 3:

$$\boxed{120(1.1)=132}$$

Year 4:

$$\boxed{132(0.75)=99}$$

So final:

$$\boxed{99}$$

35. SUCCESSIVE GROWTH CANNOT BE ADDED

$$\boxed{+20\%, +10\%}$$

does not mean +30%.

Multiplier:

$$\boxed{1.2 \times 1.1 = 1.32}$$

Net:

$$\boxed{32\%}$$

36. INDEX LINE GRAPH

Suppose index:

Year	Index
2022	100
2023	120
2024	150
2025	135

Relative to base:

- 2023 = 20% above base
- 2024 = 50% above base
- 2025 = 35% above base

37. INDEX CHANGE BETWEEN 2023 AND 2024

$\sqrt[120]{150}$

Percentage increase:

$\sqrt[\frac{30}{120} \times 100]$

$\sqrt[= \boxed{25\%}]$

Not 30%.

38. MOVING AVERAGE — BASIC

Suppose values:

$\sqrt[10, 20, 30, 40, 50]$

3-period moving averages:

First:

$\sqrt[\frac{10+20+30}{3}=20]$

Second:

$\sqrt[\frac{20+30+40}{3}=30]$

Third:

$\sqrt[\frac{30+40+50}{3}=40]$

So:

$\sqrt[\boxed{20, 30, 40}]$

39. MOVING AVERAGE PURPOSE

Moving averages smooth short-term fluctuations.

They may hide:

- sudden spikes
- sharp dips
- individual-period volatility

Important interpretation point.

40. LINE GRAPH WITH MISSING POINT

Suppose:

Year	Sales
2021	100
2022	130
2023	?
2024	190

If average of four years is 145:

Total:

$$\backslash[4(145)=580 \backslash]$$

Known:

$$\backslash[100+130+190=420 \backslash]$$

Missing:

$$\backslash[580-420 \backslash]$$

$$\backslash[=\boxed{160} \backslash]$$

41. MISSING POINT FROM GROWTH

2023 = ?

2024 = 180

2024 is 20% above 2023.

$$\backslash[1.2x=180 \backslash]$$

$$\backslash[x=\boxed{150} \backslash]$$

42. MISSING POINT FROM RATIO

2023:2024:

$$\backslash[4:5 \backslash]$$

2024:

$$\backslash[200 \backslash]$$

2023:

$$\backslash[200\times\frac{4}{5} = \boxed{160} \backslash]$$

43. LINE GRAPH WITH UNEQUAL TIME INTERVALS

Suppose:

- 2020 !' 2021: +20
- 2021 !' 2024: +30

A steeper visual segment depends on actual x-axis spacing.

Rate per year:

First:

$$\backslash[20/1=20 \backslash]$$

Second:

$$\backslash[30/3=10 \backslash]$$

So first has higher annual rate.

44. SLOPE TRAP — UNEQUAL INTERVALS

Absolute change alone is not enough.

Use:

$$\boxed{\frac{\Delta y}{\Delta x}}$$

when intervals differ.

45. TRUNCATED Y-AXIS

Suppose y-axis starts at 90, values 100 and 110.

Visually second line point may appear much higher.

Actual increase:

$$10\%$$

Always inspect baseline.

46. NONZERO BASELINE AND SLOPE

A truncated y-axis exaggerates vertical differences visually.

But numerical slope from values remains:

$$\frac{\Delta y}{\Delta x}$$

unchanged.

47. MULTI-LINE RANKING

Suppose three companies:

Year	A	B	C
2023	120	150	100
2024	150	160	140
2025	180	170	200

2025 ranking:

$$C > A > B$$

48. RANKING CHANGE

2023:

$$B > A > C$$

2025:

$C > A > B$

C moved from last to first.

Line graphs are excellent for ranking-change questions.

49. HIGHEST TOTAL ACROSS YEARS

A total:

$120 + 150 + 180 = 450$

B:

$150 + 160 + 170 = 480$

C:

$100 + 140 + 200 = 440$

Highest:

$\boxed{B}$

Even though C is highest in final year.

50. HIGHEST AVERAGE

Since all companies have same number of years, highest total = highest average.

So:

$\boxed{B}$

This shortcut avoids dividing.

51. CONDITIONAL LINE QUESTION

Question:

In how many years was A at least 90% of B?

Use:

$A \geq 0.9B$

No need to calculate exact percentages.

52. EXAMPLE

Year	A	B
1	90	100

Year	A	B
2	120	150
3	170	180
4	200	210

Check:

Year 1:

$90 = 90\% \text{ of } 100$

,

Year 2:

$120 < 135$

'L

Year 3:

$170 \geq 162$

,

Year 4:

$200 \geq 189$

,

Count:

$\boxed{3}$

53. HIGHEST RELATIVE GAP

Suppose A and B values:

Year 1:

$100, 120$

Year 2:

$200, 230$

Absolute gaps:

$20, 30$

But relative gaps using A as base:

$20/100 = 20\%$

$30/200 = 15\%$

So larger absolute gap can have smaller relative gap.

54. TREND REVERSAL

A local maximum occurs when:

- line was rising
- then begins falling

Example:

$\{100, 140, 180, 160\}$

Peak at:

$\boxed{180}$

A local minimum occurs when falling switches to rising.

55. MONOTONIC TREND

Strictly increasing:

$\{100, 120, 150, 180\}$

Strictly decreasing:

$\{200, 180, 140, 100\}$

Non-decreasing allows equal consecutive values.

56. VOLATILITY — BASIC INTERPRETATION

A series with large up/down changes is more volatile.

Example:

A:

$\{100, 102, 101, 103\}$

B:

$\{100, 150, 80, 160\}$

B is clearly more volatile.

Precise statistical volatility is beyond basic DI unless asked.

57. GATE-STYLE SET

Use:

Year	P	Q
2021	100	140
2022	120	150
2023	150	160

Year	P	Q
2024	180	200
2025	225	220

58. Q1 — P GROWTH 2021' 2025

$$\left[\frac{225-100}{100} \times 100 \right]$$

$$\left[=\boxed{125\%} \right]$$

59. Q2 — Q GROWTH

$$\left[\frac{220-140}{140} \times 100 \right]$$

$$\left[=\frac{80}{140} \times 100 \right]$$

$$\left[=\boxed{57.14\%} \right]$$

60. Q3 — WHEN DOES P FIRST EXCEED Q?

2021: P<Q 2022: P<Q 2023: P<Q 2024: P<Q 2025:

$$\left[225 > 220 \right]$$

Answer:

$$\left[\boxed{2025} \right]$$

61. Q4 — HIGHEST ABSOLUTE INCREASE OF P

Changes:

$$\left[20, 30, 30, 45 \right]$$

Highest:

$$\left[\boxed{45} \right]$$

during:

$$\left[\boxed{2024\text{to}2025} \right]$$

62. Q5 — AVERAGE P

$$\left[\frac{100+120+150+180+225}{5} \right]$$

$$\left[=\frac{775}{5} \right]$$

$$\left[=\boxed{155} \right]$$

63. COMMON LINE-GRAPH TRAPS

& pTRAP 1 — Highest point vs highest growth

Not the same.

& pTRAP 2 — Steepest line vs highest percentage growth

Steepest often means largest absolute change, not relative change.

& pTRAP 3 — Cumulative graph

Do not sum cumulative points.

& pTRAP 4 — Growth-rate graph

A falling positive growth rate still means underlying value may be increasing.

& pTRAP 5 — Interpolation

Use only if linearity is justified.

& pTRAP 6 — Unequal x-spacing

Slope must account for time interval.

& pTRAP 7 — Truncated y-axis

Visual steepness can mislead.

64. MUST-MEMORIZE LINE-GRAPH FORMULAS

Absolute change:

$$\boxed{\Delta y = y_2 - y_1}$$

Percentage change:

$$\boxed{\frac{y_2 - y_1}{y_1} \times 100}$$

Slope:

$$\boxed{\frac{y_2 - y_1}{x_2 - x_1}}$$

Periodic from cumulative:

$$\boxed{C_t - C_{t-1}}$$

Linear interpolation:

$$\boxed{y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{x_2 - x_1}}$$

65. FAST LINE-DI DECISION TREE

Exact point?

Read value.

Change?

Subtract.

Growth?

Difference / old.

Steepness?

Check both axis scales.

Cumulative?

Take successive differences.

Growth-rate line?

Use multipliers to reconstruct values.

Two lines cross?

Check whether equality occurs at a point or between points.

Unequal intervals?

Use slope per unit x.

Missing point?

Use total/average/growth/ratio condition.

PRACTICE — LEVEL 0

Use:

Year	Sales
2021	100
2022	125
2023	150
2024	135
2025	180

Q1

Highest value?

Q2

Lowest value?

Q3

Largest absolute increase?

Q4

Only year-to-year decline?

Q5

Average sales?

LEVEL 1

Use:

Year	A	B
2021	120	150
2022	150	165
2023	180	175
2024	210	210
2025	240	225

Q6

First year A exceeds B?

Q7

Year when A = B?

Q8

Total A?

Q9

Total B?

Q10

Which has higher 2021' 2025 percentage growth?

LEVEL 2

Use cumulative sales:

Quarter	Cumulative Sales
Q1	100
Q2	240
Q3	420
Q4	650

Q11

Q2 sales?

Q12

Q3 sales?

Q13

Q4 sales?

Q14

Highest quarterly sales?

Q15

Average quarterly sales?

LEVEL 3 — EXAM PRESSURE

Starting value in 2021 is:

$\boxed{200}$

Growth-rate line gives:

- 2022: +20%
- 2023: +25%
- 2024: " 20 %
- 2025: +10%

Q16

Value in 2022?

Q17

Value in 2023?

Q18

Value in 2024?

Q19

Value in 2025?

Q20

Overall percentage change from 2021 to 2025?

ANSWERS

Q1

$\boxed{2025, 180}$

Q2

$\boxed{2021, 100}$

Q3

Changes:

$\backslash [+25,+25,-15,+45 \backslash]$

Largest:

$\backslash [\boxed{45} \backslash]$

from 2 0 2 4!' 2 0 2 5 .

Q4

$\backslash [150\text{to}135 \backslash]$

So:

$\backslash [\boxed{2023\text{to}2024} \backslash]$

Q5

$\backslash [100+125+150+135+180=690 \backslash]$

$\backslash [690/5 = \boxed{138} \backslash]$

Q6

2021 $A < B$ 2022 $A < B$ 2023:

$\backslash [180 > 175 \backslash]$

So:

$\backslash [\boxed{2023} \backslash]$

Q7

$\backslash [\boxed{2024} \backslash]$

Q8

$\backslash [120+150+180+210+240 = \boxed{900} \backslash]$

Q9

$\backslash [150+165+175+210+225 = \boxed{925} \backslash]$

Q10

A:

$\backslash [120\text{to}240 \rightarrow 100\% \backslash]$

B:

$\backslash [150\text{to}225 \rightarrow 50\% \backslash]$

Therefore:

$\boxed{A}$

Q11

$$240 - 100 = \boxed{140}$$

Q12

$$420 - 240 = \boxed{180}$$

Q13

$$650 - 420 = \boxed{230}$$

Q14

$$\boxed{Q4}$$

Q15

Total sales:

$$650$$

Average:

$$650/4 = \boxed{162.5}$$

Q16

$$200(1.2) = \boxed{240}$$

Q17

$$240(1.25) = \boxed{300}$$

Q18

$$300(0.8) = \boxed{240}$$

Q19

$$240(1.1) = \boxed{264}$$

Q20

$$200 \rightarrow 264$$

Increase:

$$64$$

$$64/200 \times 100 = \boxed{32\%}$$

CHAPTER 53 STATUS

Covered so far:

- line-graph reading '
- trends '
- direct values '
- absolute change '
- percentage growth '
- slope '
- multi-line comparison '
- crossovers '
- interpolation '
- cumulative vs periodic lines '
- growth-rate lines '
- index lines '
- moving averages '
- ranking changes '
- missing points '
- unequal intervals '
- truncated-axis traps '
- L0–L3 practice '

But **Chapter 53 is not complete yet.**

Still remaining:

- advanced multi-line sets
- contribution-to-growth questions
- average rate vs CAGR interpretation
- multiple crossovers
- moving-average reconstruction
- dual-axis line graphs
- line + bar hybrid interpretation
- large placement-style timed set
- final line-DI speed audit

Progress

$\boxed{52/76}$ completed

Current: 0% Chapter 53 — Line Graphs

NEXT

0% Chapter 53 Part 2 — Advanced Multi-Line DI, CAGR, Moving Averages, Dual Axes, Hybrid Graphs & Final Line-Graph Speed Toolkit

USER

next

ASSISTANT

CHAPTER 53 — LINE GRAPHS

PART 2 — ADVANCED MULTI-LINE DI, CAGR, MOVING AVERAGES, DUAL AXES, HYBRID GRAPHS & FINAL SPEED TOOLKIT

Status: $\emptyset=0\%$ PROGRESS GATE: High Placements: High Priority: $\emptyset=0\%$ Per 1

This part is about interpreting **several trends at once**.

The biggest mistake in advanced line DI is:

Seeing a line go up or down and assuming you already know what happened.

You must ask:

- Did the **absolute value** rise?
- Did the **percentage growth rate** rise?
- Did the **share** rise?
- Did its **rank** rise?
- Did the **gap** rise?

These can all behave differently.

1. ADVANCED MULTI-LINE SET

Consider:

Year	A	B	C
2021	100	120	140
2022	130	150	145
2023	160	155	170
2024	180	200	190
2025	225	210	240

2. FINAL-YEAR RANKING

2025:

$\backslash$ C=240, $\backslash$ quad A=225, $\backslash$ quad B=210 $\backslash$

Ranking:

$$\boxed{C > A > B}$$

3. OVERALL TOTALS

A:

$$100 + 130 + 160 + 180 + 225 = 795$$

B:

$$120 + 150 + 155 + 200 + 210 = 835$$

C:

$$140 + 145 + 170 + 190 + 240 = 885$$

Highest total:

$$\boxed{C}$$

4. HIGHEST AVERAGE

All have 5 years.

Therefore:

Highest total = highest average.

So:

$$\boxed{C}$$

No division needed.

5. HIGHEST OVERALL PERCENTAGE GROWTH

A:

$$100 \rightarrow 225$$

$$\frac{125}{100} \times 100 = 125\%$$

B:

$$120 \rightarrow 210$$

$$\frac{90}{120} \times 100 = 75\%$$

C:

$$140 \rightarrow 240$$

$$\frac{100}{140} \times 100 \approx 71.43\%$$

Highest:

$$\boxed{A}$$

Again:

- highest final value !' C
- highest total !' C
- highest growth !' A

Different metrics.

6. CONTRIBUTION TO TOTAL INCREASE

Suppose total of A+B+C:

2021:

$$\backslash[100+120+140=360 \backslash]$$

2025:

$$\backslash[225+210+240=675 \backslash]$$

Total increase:

$$\backslash[675-360 = 315 \backslash]$$

Individual increases:

A:

$$\backslash[125 \backslash]$$

B:

$$\backslash[90 \backslash]$$

C:

$$\backslash[100 \backslash]$$

Contribution of A:

$$\backslash[\frac{125}{315} \times 100 \backslash]$$

$$\backslash[\approx \boxed{39.68\%} \backslash]$$

Largest contributor:

$$\backslash[\boxed{A} \backslash]$$

7. HIGHEST GROWTH RATE VS HIGHEST CONTRIBUTION

If companies have different starting values, these can differ.

Example:

A:

$$\backslash[50 \text{ to } 100 \backslash]$$

growth:

$$\backslash[100\% \backslash]$$

increase:

$\sqrt{50}$

B:

$\sqrt{500\text{to}600}$

growth:

$\sqrt{20\%}$

increase:

$\sqrt{100}$

A has higher % growth.

B contributes more absolute increase.

8. MULTIPLE CROSSOVERS

Suppose:

Year	A	B
1	100	120
2	150	140
3	130	160
4	180	170

Relative ranking:

Year 1:

$\sqrt{A < B}$

Year 2:

$\sqrt{A > B}$

Year 3:

$\sqrt{A < B}$

Year 4:

$\sqrt{A > B}$

So the lines cross:

- between 1 and 2
- between 2 and 3
- between 3 and 4

Total:

$\sqrt{\boxed{3\text{crossovers}}}$

assuming straight connecting segments.

9. CROSSOVER COUNTING TRAP

If lines merely touch at one point and do not reverse ranking, that is an intersection but not necessarily a ranking crossover.

Example:

A and B equal in 2023, but $A < B$ both before and after.

Then they touch/equal, but order does not reverse.

10. AVERAGE ANNUAL GROWTH — THREE DIFFERENT IDEAS

Suppose:

$[100 \text{ to } 120 \text{ to } 180]$

Yearly growth:

$[20\%, 50\%]$

Three possible quantities:

Arithmetic average of yearly growth rates

$[\frac{20+50}{2} = 35\%]$

Total growth

$[\frac{180-100}{100} = 80\%]$

CAGR

$[\left(\frac{180}{100}\right)^{1/2} - 1]$

$[=\sqrt{1.8} - 1]$

$[\approx 34.16\%]$

Do not confuse them.

11. CAGR

MUST UNDERSTAND

If initial value:

$[V_0]$

final value:

$[V_n]$

after (n) compounding intervals:

$[\boxed{\text{CAGR} = \left(\frac{V_n}{V_0}\right)^{1/n} - 1}]$

Multiply by 100 for percentage.

12. CAGR EXAMPLE

Value:

$$\sqrt[2]{100 \text{ to } 121}$$

in 2 years.

$$\text{CAGR} = \sqrt[2]{121/100} - 1$$

$$= 1.1 - 1$$

$$= \boxed{10\%}$$

13. CAGR FAST RECOGNITION

Common multipliers:

$$1.21 = (1.1)^2$$

So 21% total growth over 2 years corresponds to:

$$\boxed{10\% \text{ CAGR}}$$

Similarly:

$$1.331 = (1.1)^3$$

So 33.1% total growth over 3 years:

$$\boxed{10\% \text{ CAGR}}$$

14. CAGR DOES NOT DESCRIBE ACTUAL YEARLY PATH

Values may be:

$$100 \text{ to } 80 \text{ to } 121$$

CAGR over two intervals is still:

$$10\%$$

because only start and end matter.

It does not mean each actual year grew 10%.

15. AVERAGE ABSOLUTE RATE

If value:

$$100 \text{ to } 220$$

over 4 years:

Average absolute increase per year:

$$\frac{220 - 100}{4}$$

$$= \boxed{30}$$

This is not CAGR.

16. AVERAGE SLOPE VS AVERAGE VALUE

Different concepts.

Average slope:

$$\left[\frac{\text{final} - \text{initial}}{\text{time}} \right]$$

Average value:

$$\left[\frac{\sum y_i}{n} \right]$$

Do not confuse them.

17. MOVING AVERAGE — ADVANCED

Suppose yearly sales:

$$\left[10, 20, 40, 50, 60 \right]$$

3-year moving averages:

First:

$$\left[\frac{10+20+40}{3} = \frac{70}{3} \right]$$

Second:

$$\left[\frac{20+40+50}{3} = \frac{110}{3} \right]$$

Third:

$$\left[\frac{40+50+60}{3} = 50 \right]$$

18. MOVING AVERAGE DIFFERENCE SHORTCUT

For 3-term moving averages:

$$\left[M_1 = \frac{x_1 + x_2 + x_3}{3} \right]$$

$$\left[M_2 = \frac{x_2 + x_3 + x_4}{3} \right]$$

Subtract:

$$\left[M_2 - M_1 = \frac{x_4 - x_1}{3} \right]$$

Thus:

$$\left[\boxed{x_4 - x_1 = 3(M_2 - M_1)} \right]$$

Useful for reconstruction.

19. RECONSTRUCT MISSING VALUE FROM MOVING AVERAGE

Suppose:

$$\left[x_1 = 10, x_2 = 20, x_3 = 30 \right]$$

Next 3-term moving average is 40:

$$\left[\frac{20+30+x_4}{3}=40 \right]$$

$$\left[50+x_4=120 \right]$$

$$\left[\boxed{x_4=70} \right]$$

20. GENERAL k-PERIOD MOVING AVERAGE SHIFT

For (k) -term moving averages:

$$\left[M_{t+1}-M_t = \frac{x_{t+k}-x_t}{k} \right]$$

So:

$$\left[\boxed{x_{t+k}-x_t = k(M_{t+1}-M_t)} \right]$$

Support-level but very useful.

21. MOVING AVERAGE SMOOTHS DATA

Original:

$$\left[100, 200, 80, 220, 90 \right]$$

may fluctuate heavily.

Moving average reduces short-term noise.

But:

A smooth moving-average line does not mean the original data was smooth.

22. DUAL-AXIS LINE GRAPH

A graph may use:

- left y-axis !' Revenue ¹ crore
- right y-axis !' Growth %

Example:

Revenue line !' left axis Growth line !' right axis

Two lines can be at same visual height while representing different values.

23. DUAL-AXIS RULE

Before reading any point:

1. identify line
2. identify its axis
3. identify its unit
4. then read value

Never compare raw vertical heights directly across different axes.

24. DUAL-AXIS EXAMPLE

Suppose in 2025:

Revenue line aligns with:

$$\begin{array}{l} \backslash[\\ 800 \\ \backslash] \end{array}$$

on left axis.

Growth line aligns with:

$$\begin{array}{l} \backslash[\\ 20\% \\ \backslash] \end{array}$$

on right axis.

These are:

$$\begin{array}{l} \backslash[\\ \boxed{800 \text{ and } 20\%} \\ \backslash] \end{array}$$

not comparable as raw numbers.

25. RECONSTRUCT PREVIOUS VALUE FROM GROWTH LINE

Revenue 2025:

$$\begin{array}{l} \backslash[\\ 800 \\ \backslash] \end{array}$$

Growth in 2025:

$$\begin{array}{l} \backslash[\\ 25\% \\ \backslash] \end{array}$$

So:

$$\begin{array}{l} \backslash[\\ 800 = 1.25(\text{2024 revenue}) \\ \backslash] \end{array}$$

$$\begin{array}{l} \backslash[\\ 2024 = \\ \frac{800}{1.25} \\ \backslash] \end{array}$$

$$\begin{array}{l} \backslash[\\ = \boxed{640} \\ \backslash] \end{array}$$

26. LINE + BAR HYBRID GRAPH

Common format:

- bars !' total sales
- line !' profit margin

Example:

Year	Revenue	Margin
2023	500	20%
2024	600	25%
2025	800	15%

Profit:

2023:

$$\backslash[500(0.2)=100\backslash]$$

2024:

$$\backslash[600(0.25)=150\backslash]$$

2025:

$$\backslash[800(0.15)=120\backslash]$$

Highest profit:

$$\backslash[\boxed{2024}\backslash]$$

even though revenue is highest in 2025.

27. BAR + LINE TRAP

Higher margin does not necessarily mean higher profit.

Profit depends on:

$$\backslash[\boxed{\text{Revenue} \times \text{Margin}}\backslash]$$

Similarly:

Higher growth rate does not necessarily mean larger absolute increase.

28. HYBRID GRAPH – COST FROM REVENUE + MARGIN

If:

$$\backslash[\text{Profit} = \text{Revenue} \times \text{Margin}\backslash]$$

then:

$$\begin{aligned} & \backslash[\\ & \text{Cost} = \text{Revenue} - \text{Profit} \\ & \backslash] \end{aligned}$$

Example:

Revenue 800, margin 15%.

Profit:

$$\begin{aligned} & \backslash[\\ & 120 \\ & \backslash] \end{aligned}$$

Cost:

$$\begin{aligned} & \backslash[\\ & 800 - 120 \\ & = \\ & \boxed{680} \\ & \backslash] \end{aligned}$$

29. CONTRIBUTION TO GROWTH IN MULTI-LINE SET

Suppose total market comprises A+B+C.

If total market increase:

$$\begin{aligned} & \backslash[\\ & 300 \\ & \backslash] \end{aligned}$$

and A increases by:

$$\begin{aligned} & \backslash[\\ & 120 \\ & \backslash] \end{aligned}$$

then A contribution:

$$\begin{aligned} & \backslash[\\ & 120/300 \\ & = \\ & \boxed{40\%} \\ & \backslash] \end{aligned}$$

Contribution percentage uses:

$$\begin{aligned} & \backslash[\\ & \boxed{\text{\text{total increase as denominator}}} \\ & \backslash] \end{aligned}$$

not total final market.

30. GROWTH SHARE VS MARKET SHARE

These are different.

Company sales increase:

\[
120
\]

Total market increase:

\[
300
\]

Contribution to growth:

\[
40\%
\]

But company final market share might be only 25%.

Do not mix them.

31. INDEXED MULTI-LINE GRAPHS

Suppose all companies are normalized to:

\[
100
\]

in base year.

Later:

A:

\[
150
\]

B:

\[
130
\]

C:

\[
180
\]

Then growth from base:

- A = 50%
- B = 30%
- C = 80%

This allows growth comparison even if original absolute sizes were different.

32. INDEX GRAPH DOES NOT SHOW WHO IS BIGGEST ABSOLUTELY

If A initially had sales 1000 and C had sales 100:

A index 150:

```
\[
1500
\]
```

C index 180:

```
\[
180
\]
```

C grew faster, but A is still much larger.

33. RATIO OF LINES

Suppose A/B:

Year	A	B
1	100	200
2	150	240
3	180	270

Ratios:

Year 1:

```
\[
1:2
\]
```

Year 2:

```
\[
5:8
\]
```

Year 3:

```
\[
2:3
\]
```

The ratio is improving in favor of A.

34. FAST RATIO-TREND CHECK

To check whether A/B increased from:

```
\[
a_1/b_1
\]
```

to:

```
\[
a_2/b_2
\]
```

compare:

\[
a_2b_1
\]

with:

\[
a_1b_2
\]

No division needed.

35. EXAMPLE

Old:

\[
120/200
\]

New:

\[
150/240
\]

Compare:

\[
150(200)=30000
\]

\[
120(240)=28800
\]

So ratio increased.

36. CORRELATION-LIKE INTERPRETATION – BASIC

If two lines generally rise together, they show a positive co-movement.

If one rises while other falls, negative co-movement.

But:

> A graph alone does not prove causation.

For aptitude interpretation, say only what data supports.

37. TREND CONSISTENCY

Suppose:

A:

\[
100, 120, 140, 160
\]

B:

\[
100,130,125,170
\]

A has more consistent growth.

B may have higher final value, but greater fluctuation.

38. MAXIMUM YEAR-TO-YEAR FLUCTUATION

For:

\[
100,150,120,210
\]

changes:

\[
+50, -30, +90
\]

Largest absolute fluctuation:

\[
\boxed{90}
\]

Do not ignore negative sign when comparing magnitude; use absolute change if question says fluctuation.

39. NET CHANGE VS TOTAL MOVEMENT

Series:

\[
100\to150\to120\to160
\]

Net change:

\[
160-100=60
\]

Total movement:

\[
|50|+|-30|+|40|
\]

\[
=120
\]

Different.

This can appear in analytics-style placement DI.

40. AREA UNDER LINE GRAPH – CAUTION

In basic aptitude, summing plotted values usually gives totals only if values are discrete period totals.

Do **not** interpret geometric area under a line unless question explicitly defines a rate over a continuous variable.

41. RATE-TIME LINE GRAPH

If line graph shows speed versus time, distance may be represented by area under the speed-time graph.

This belongs more to physics/quant interpretation, but can appear in graphical reasoning.

For constant speed rectangle:

```
\[
\boxed{Distance=Speed\times Time}
\]
```

42. GATE GRAPHICAL REASONING TRAP

A graph may ask only qualitative inference.

Example:

If A line is always above B:

You may conclude:

```
\[
A>B
\]
```

at all plotted points.

You cannot automatically conclude:

- A grew faster
- A had higher percentage growth
- A's average growth rate was larger

unless calculated.

43. LARGE MULTI-LINE SET

Use:

Year	A	B	C
2021	200	250	150
2022	240	275	180
2023	300	300	210
2024	360	330	280
2025	450	396	350

44. TOTALS

A:

\[
200+240+300+360+450
=
1550
\]

B:

\[
250+275+300+330+396
=
1551
\]

C:

\[
150+180+210+280+350
=
1170
\]

Interesting:

B total beats A by only:

\[
\boxed{1}
\]

45. HIGHEST FINAL VALUE

2025:

\[
A=450,\ B=396,\ C=350
\]

Highest:

\[
\boxed{A}
\]

46. HIGHEST 2021! 2025 GROWTH

A:

\[
200\to450
\]

\[
125\%
\]

B:

$\backslash[$
250\to396
 $\backslash]$

$\backslash[$
58.4\%
 $\backslash]$

C:

$\backslash[$
150\to350
 $\backslash]$

$\backslash[$
 $\frac{200}{150}\times100$
=
133.33\%
 $\backslash]$

Highest:

$\backslash[$
 $\boxed{C}$
 $\backslash]$

47. YEAR A FIRST EXCEEDS B

2021:

$\backslash[$
 $A < B$
 $\backslash]$

2022:

$\backslash[$
 $A < B$
 $\backslash]$

2023:

$\backslash[$
 $A = B$
 $\backslash]$

2024:

$\backslash[$
 $A > B$
 $\backslash]$

Therefore first exceeds in:

$\backslash[$
 $\boxed{2024}$
 $\backslash]$

48. LARGEST YEARLY INCREASE OF C

Changes:

\[
30,\ 30,\ 70,\ 70
\]

Largest:

\[
\boxed{70}
\]

during:

\[
2023\to2024
\]

and:

\[
2024\to2025
\]

49. TOTAL MARKET

2021:

\[
 $200+250+150=600$
\]

2025:

\[
 $450+396+350=1196$
\]

Increase:

\[
596
\]

Growth:

\[
 $596/600$
\]

\[
\approx\boxed{99.33\%}
\]

50. CONTRIBUTION TO MARKET INCREASE

A increase:

\[
250
\]

B:

\[
146
\]

C:

\[
200
\]

Total:

\[
596
\]

Contributions:

A:

\[
 $250/596 \approx 41.95\%$
\]

B:

\[
 $146/596 \approx 24.50\%$
\]

C:

\[
 $200/596 \approx 33.56\%$
\]

Largest:

\[
\boxed{A}
\]

Again, C had highest percentage growth but A contributed most absolute growth.

51. TIMED LINE-DI STRATEGY

For a dense multi-line graph:

Pass 1

Read:

- x-axis
- y-axis
- units
- legend

Pass 2

Answer:

- highest/lowest
- direct-value
- equality/ranking questions

Pass 3

Calculate:

- differences
- growth
- averages

Pass 4

Do:

- CAGR
- contribution
- crossover interpolation
- hybrid metrics

52. WHEN TO REDRAW LINE DATA

If 3-4 lines cross repeatedly, a quick mini-table is safer than visually tracing.

Example:

```
```text id="ufeo45"
Year: 1 2 3 4
A: 100 140 160 190
B: 120 130 170 180
```

This prevents reading the wrong line.

## 53. LEGEND TRAP

In black-and-white aptitude papers, lines may differ by:

- dots
- dashes
- triangles
- circles

Trace each line from the legend before reading values.

## 54. FINAL LINE-GRAPH SPEED CHECKLIST

Before calculation:

- title
- axes
- units
- interval spacing
- legend
- periodic or cumulative?
- absolute value or growth rate?

During calculation:

- use threshold comparisons
- compare fractions by cross multiplication
- cache totals if reused
- distinguish absolute and percentage changes

- use CAGR only when asked

Before final answer:

- verify year
- verify line
- verify denominator
- sanity-check direction

## 55. COMMON ADVANCED TRAPS

### & pTRAP 1 — Falling growth rate

Value can still rise.

### & pTRAP 2 — Highest % growth

Need not give highest contribution.

### & pTRAP 3 — Dual axis

Same height does not mean same magnitude.

### & pTRAP 4 — CAGR

Do not arithmetic-average growth rates.

### & pTRAP 5 — Moving averages

Smoothed line is not raw data.

### & pTRAP 6 — Index lines

Highest index means highest relative growth from base, not necessarily highest absolute value.

### & pTRAP 7 — Multiple intersections

Equality and true ranking crossover are not always identical concepts.

## 56. MUST-MEMORIZE ADVANCED RELATIONS

CAGR:

$$\boxed{\left(\frac{V_f}{V_i}\right)^{\frac{1}{n}} - 1}$$

Contribution to increase:

$$\boxed{\frac{\Delta \text{component}}{\Delta \text{total}} \times 100}$$

Moving-average shift:

$$\boxed{x_{t+k} - x_t = k(M_{t+1} - M_t)}$$

Hybrid profit:

$$\boxed{\text{Profit} = \text{Revenue} \times \text{Margin}}$$

57. FAST DECISION TREE

Final vs initial?

Use total growth.

“Average annual compound growth”?

Use CAGR.

Contribution to total increase?

Use component increase / total increase.

Moving average?

Use sliding-window logic.

Dual axis?

Map each line to correct axis.

Bar + line?

Combine metrics algebraically.

Indexed lines?

Interpret relative to base 100.

Multiple crossings?

Track rankings interval by interval.

PRACTICE — LEVEL 0

Use:

Year	A	B
2021	100	120
2022	120	130
2023	150	145
2024	180	170
2025	210	200

Q1

First year A exceeds B?

Q2

Total A?

### Q3

Total B?

### Q4

A growth 2021! 2025?

### Q5

B growth 2021! 2025?

## LEVEL 1

A market has:

Year	Total	Company X Share
2022	1000	20%
2023	1200	22%
2024	1500	21%
2025	1800	20%

### Q6

X sales in 2022?

### Q7

X sales in 2025?

### Q8

Did X's share increase, decrease or remain same from 2022 to 2025?

### Q9

Did absolute sales increase?

### Q10

Percentage growth in X sales?

## LEVEL 2

Revenue + margin hybrid:

Year	Revenue	Profit Margin
2022	400	20%
2023	500	18%
2024	600	25%

Year	Revenue	Profit Margin
2025	800	20%

**Q11**

Profit in each year?

**Q12**

Highest profit?

**Q13**

Cost in 2024?

**Q14**

Profit growth 2022! 2025 ?

**Q15**

Average annual profit over four years?

**LEVEL 3 — EXAM PRESSURE**

Company values:

Year	A	B	C
2022	100	200	150
2023	150	220	180
2024	180	260	240
2025	270	300	300

**Q16**

Which company has highest percentage growth?

**Q17**

Which company has highest absolute increase?

**Q18**

What is total-market growth from 2022 to 2025?

**Q19**

Which company contributes most to total increase?

**Q20**

If A grew from 100 to 270 over 3 annual intervals, what is its CAGR approximately?

## ANSWERS

---

### Q1

2021:

$$\lfloor A < B \rfloor$$

2022:

$$\lfloor A < B \rfloor$$

2023:

$$\lfloor 150 > 145 \rfloor$$

$$\lfloor \boxed{2023} \rfloor$$

### Q2

$$\lfloor 100+120+150+180+210 = \boxed{760} \rfloor$$

### Q3

$$\lfloor 120+130+145+170+200 = \boxed{765} \rfloor$$

### Q4

$$\lfloor 100 \text{ to } 210 \rfloor$$

$$\lfloor \boxed{110\%} \rfloor$$

### Q5

$$\lfloor 120 \text{ to } 200 \rfloor$$

$$\lfloor \frac{80}{120} \times 100 = \boxed{66.67\%} \rfloor$$

### Q6

$$\lfloor 1000(0.2) = \boxed{200} \rfloor$$

### Q7

$$\lfloor 1800(0.2) = \boxed{360} \rfloor$$

### Q8

20% !' 20%

$$\lfloor \boxed{\text{unchanged}} \rfloor$$

### Q9

$$\lfloor 200 \text{ to } 360 \rfloor$$

So:

$\boxed{\text{increased}}$

### Q10

$\frac{160}{200} \times 100 = \boxed{80\%}$

### Q11

2022:

$400(0.2) = 80$

2023:

$500(0.18) = 90$

2024:

$600(0.25) = 150$

2025:

$800(0.2) = 160$

$\boxed{80, 90, 150, 160}$

### Q12

$\boxed{2025}$

with profit:

$\boxed{160}$

### Q13

2024 profit:

$\boxed{150}$

Cost:

$600 - 150 = \boxed{450}$

### Q14

$80 \text{ to } 160$

$\boxed{100\%}$

### Q15

$\frac{80+90+150+160}{4}$

$= \frac{480}{4}$

$= \boxed{120}$

## Q16

---

A:

$$\sqrt[100]{270}$$

$$\sqrt[170]{\%}$$

B:

$$\sqrt[200]{300}$$

$$\sqrt[50]{\%}$$

C:

$$\sqrt[150]{300}$$

$$\sqrt[100]{\%}$$

Highest:

$$\boxed{A}$$

## Q17

---

Absolute increases:

A:

$$\sqrt[170]{}$$

B:

$$\sqrt[100]{}$$

C:

$$\sqrt[150]{}$$

Highest:

$$\boxed{A}$$

## Q18

---

2022 total:

$$\sqrt[100+200+150]{450}$$

2025 total:

$$\sqrt[270+300+300]{870}$$

Increase:

$$\sqrt[420]{}$$

Growth:

$$\sqrt[\frac{420}{450}]{100}$$

$$\boxed{93.33\%}$$

## Q19

---



Contributions:

A:

$$\sqrt[3]{170}$$

B:

$$\sqrt[3]{100}$$

C:

$$\sqrt[3]{150}$$

Largest:

$$\sqrt[3]{\boxed{A}}$$

## Q20

$$\sqrt[3]{\text{CAGR} = \left(\frac{270}{100}\right)^{1/3} - 1}$$

$$\sqrt[3]{= (2.7)^{1/3} - 1}$$

Since:

$$\sqrt[3]{1.39^3 \approx 2.69}$$

CAGR is approximately:

$$\sqrt[3]{\boxed{39\%}}$$

## ' CHAPTER 53 COMPLETE

We have now covered:

- basic line-graph reading '
- trends and slopes '
- absolute changes '
- percentage growth '
- multi-line comparisons '
- rankings '
- single and multiple crossovers '
- interpolation '
- cumulative lines '
- growth-rate lines '
- index lines '
- moving averages '
- moving-average reconstruction '
- CAGR '
- contribution-to-growth '
- dual-axis graphs '
- bar + line hybrid graphs '
- normalized/index interpretation '

- advanced graphical traps '
- timed strategy '
- L0–L3 practice '

## Progress

$\boxed{53/76}$  completed

## UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations '
- Chapter 51 — Tables '
- Chapter 52 — Bar & Stacked-Bar Graphs '
- Chapter 53 — Line Graphs '
- Chapter 54 — Pie Charts +
- Chapter 55 — Mixed Graphs & Caselets +
- Chapter 56 — Missing-Data & Advanced DI +

## NEXT

### Chapter 54 — Pie Charts

Next we will cover:

- angle !" percentage !" actual-value conversion
- sector comparisons
- multiple pie charts
- changing totals
- percentage-point vs absolute changes
- exploded/misleading pie charts
- missing sectors
- ratio shortcuts
- combined pie-chart tables
- GATE/placement L0–L3 sets.

USER

next

ASSISTANT

## CHAPTER 54 — PIE CHARTS

### PART 1 — ANGLE !"PERCENTAGE !"ACTUAL VALUE, RATIOS, MISSING SECTORS & MULTIPLE PIE CHARTS

Status:  $\emptyset$  IN PROGRESS GATE: High Placements: High Priority:  $\emptyset$  Yr 1

A pie chart represents a total quantity as a full circle.

The complete circle is:

$$\boxed{360^\circ = 100\%}$$

Everything in pie-chart DI comes from this one relationship.

## 1. PIE-CHART MASTER RELATION

If a category has angle  $(\theta)$ :

$$\boxed{\text{Percentage} = \frac{\theta}{360} \times 100}$$

If a category has percentage  $(p)$ :

$$\boxed{\theta = \frac{p}{100} \times 360}$$

If total is  $(T)$ :

$$\boxed{\text{Category value} = \frac{\theta}{360} T}$$

or:

$$\boxed{\text{Category value} = \frac{p}{100} T}$$

## 2. MUST-MEMORIZE ANGLE–PERCENTAGE VALUES

Angle	Percentage
$18^\circ$	5%
$36^\circ$	10%
$45^\circ$	12.5%
$54^\circ$	15%
$72^\circ$	20%
$90^\circ$	25%
$108^\circ$	30%
$120^\circ$	$33.33\%$
$135^\circ$	37.5%
$144^\circ$	40%
$180^\circ$	50%
$216^\circ$	60%
$270^\circ$	75%

Very high-value for speed.

### 3. FAST CONVERSION

Since:

$$\angle 360^\circ = 100\%$$

we get:

$$\angle 1\% = 3.6^\circ$$

and:

$$\angle 1^\circ = \frac{100}{360}\%$$

$$\angle = \frac{5}{18}\%$$

So:

$$\angle \boxed{1^\circ \approx 0.2778\%}$$

### 4. SIMPLE PIE CHART

Suppose monthly expenditure is:

Category	Angle
Food	$\angle (120^\circ)$
Rent	$\angle (90^\circ)$
Travel	$\angle (60^\circ)$
Education	$\angle (45^\circ)$
Other	$\angle (45^\circ)$

Check:

$$\angle 120 + 90 + 60 + 45 + 45 = 360^\circ$$

Good.

### 5. FOOD PERCENTAGE

$$\angle \frac{120}{360} \times 100$$

$$\angle = \frac{13}{100}$$

$$\angle \boxed{33.33\%}$$

### 6. RENT PERCENTAGE

$$\angle 90/360 = 1/4$$

$$\angle \boxed{25\%}$$

## 7. TRAVEL PERCENTAGE

$$\left[ 60/360 = 1/6 \right]$$

$$\left[ =\boxed{16.67\%} \right]$$

## 8. EDUCATION PERCENTAGE

$$\left[ 45/360 = 1/8 \right]$$

$$\left[ =\boxed{12.5\%} \right]$$

## 9. ACTUAL VALUE FROM TOTAL

Suppose total monthly expenditure:

$$\left[ \text{₹ } 72,000 \right]$$

Food:

$$\left[ \frac{120}{360}(72000) \right]$$

$$\left[ =\frac{13}{10}(72000) \right]$$

$$\left[ =\boxed{\text{₹ } 124,000} \right]$$

## 10. RENT VALUE

$$\left[ \frac{90}{360}(72000) \right]$$

$$\left[ =\frac{14}{10}(72000) \right]$$

$$\left[ =\boxed{\text{₹ } 118,000} \right]$$

## 11. TRAVEL VALUE

$$\left[ \frac{60}{360}(72000) = \frac{16}{10}(72000) \right]$$

$$\left[ =\boxed{\text{₹ } 112,000} \right]$$

## 12. RATIO DIRECTLY FROM ANGLES

Food : Rent:

$$\left[ 120:90 \right]$$

Simplify:

$$\left[ \boxed{4:3} \right]$$

Important shortcut:

**If two sectors belong to the same pie chart, their actual-value ratio equals their angle ratio.**

No total required.

---

### 13. RATIO DIRECTLY FROM PERCENTAGES

---

If A = 30% and B = 20%:

$$\backslash[ A:B=30:20 \backslash]$$

$$\backslash[ =\boxed{3:2} \backslash]$$

Again, total cancels.

---

### 14. DIFFERENCE BETWEEN TWO CATEGORIES

---

Food minus Travel:

Angle difference:

$$\backslash[ 120-60=60^\circ \backslash]$$

Actual difference:

$$\backslash[ \frac{60}{360}(72000) \backslash]$$

$$\backslash[ =12000 \backslash]$$

$$\backslash[ =\boxed{12,000} \backslash]$$

---

### 15. CATEGORY A IS WHAT PERCENT MORE THAN B?

---

Food:

$$\backslash[ 120^\circ \backslash]$$

Rent:

$$\backslash[ 90^\circ \backslash]$$

Food is more by:

$$\backslash[ 30^\circ \backslash]$$

Percentage more than Rent:

$$\backslash[ \frac{30}{90}\times 100 \backslash]$$

$$\backslash[ =\boxed{33.33\%} \backslash]$$

You can work directly with angles because the same total factor cancels.

---

### 16. CATEGORY B IS WHAT PERCENT LESS THAN A?

---

Rent less than Food:

$$\backslash[ \frac{120-90}{120}\times 100 \backslash]$$

$$\backslash[ =\boxed{25\%} \backslash]$$

Again asymmetric.

---

---

## 17. MISSING ANGLE

---

Suppose sectors:

$$\backslash [ 80^\circ, 100^\circ, 60^\circ, 70^\circ, ? ]$$

Known total:

$$\backslash [ 310^\circ ]$$

Missing:

$$\backslash [ 360 - 310 ]$$

$$\backslash [ =\boxed{50^\circ} ]$$

---

## 18. MISSING PERCENTAGE

---

Shares:

$$\backslash [ 20\%, 25\%, 15\%, 30\%, ? ]$$

Known:

$$\backslash [ 90\% ]$$

Missing:

$$\backslash [ \boxed{10\%} ]$$

---

## 19. MISSING VALUE FROM TOTAL

---

Total:

$$\backslash [ 1000 ]$$

Known sectors correspond to:

- $A = 300$
- $B = 250$
- $C = 150$
- $D = ?$

If these are exhaustive categories and  $E = 100$ :

$$\backslash [ D = 1000 - (300 + 250 + 150 + 100) ]$$

$$\backslash [ =\boxed{200} ]$$

---

## 20. FIND TOTAL FROM ONE SECTOR

---

Category A:

$$\backslash [ 72^\circ ]$$

Actual value:

$$\boxed{200}$$

Since:

$$72^\circ = 20\%$$

Total:

$$200/0.2$$

$$\boxed{1000}$$

---

## 21. DIRECT FORMULA FOR TOTAL

$$\boxed{T = \frac{\text{sector value} \times 360}{\theta}}$$

Example:

$$T = \frac{200 \times 360}{72} = 1000$$

---

## 22. FIND ANGLE FROM ACTUAL VALUE

Total:

$$1200$$

Category:

$$300$$

Fraction:

$$300/1200 = 1/4$$

Angle:

$$\frac{1}{4}(360)$$

$$\boxed{90^\circ}$$

---

## 23. FIND PERCENTAGE FROM ACTUAL VALUE

$$300/1200 = 25\%$$

$$\boxed{25\%}$$

---

## 24. PIE CHART WITH TOTAL NOT GIVEN

Suppose only sector angles are shown.

You can still answer:

- ratios
- relative percentages
- which category largest/smallest
- percentage more/less



- sector shares

But cannot determine absolute values unless at least one scale/total/value is given.

## 25. SUFFICIENCY EXAMPLE

A sector angle:

$$\boxed{90^\circ}$$

Question asks actual value.

Without total:

$$\boxed{\text{Cannot determine}}$$

If total = 800:

$$\boxed{200}$$

## 26. MULTIPLE PIE CHARTS

Now suppose two pie charts show company expenditure distribution.

**2024 total:**

$$\boxed{110 \text{ crore}}$$

**2025 total:**

$$\boxed{115 \text{ crore}}$$

Shares:

Category	2024	2025
Salaries	40%	35%
Marketing	20%	25%
Operations	25%	20%
Other	15%	20%

## 27. SALARIES — ACTUAL VALUES

2024:

$$\boxed{10(0.40)=4}$$

2025:

$$\boxed{15(0.35)=5.25}$$

So salary expenditure increased:

$$\boxed{4 \text{ to } 5.25}$$

even though share decreased.

---

## 28. SALARY ABSOLUTE INCREASE

---

$$\boxed{5.25 - 4 = 1.25 \text{ crore}}$$

---

## 29. SALARY PERCENTAGE INCREASE

---

$$\frac{1.25}{4} \times 100$$

$$= \boxed{31.25\%}$$

This is a key multiple-pie-chart pattern.

---

## 30. MARKETING

---

2024:

$$10(0.20) = 2$$

2025:

$$15(0.25) = 3.75$$

Increase:

$$1.75$$

Percentage increase:

$$1.75 / 2 = 87.5\%$$

$$\boxed{87.5\%}$$

---

## 31. SHARE CHANGE VS ACTUAL CHANGE

---

Salaries:

$$40\% \text{ to } 35\%$$

Share falls by:

$$5 \text{ percentage points}$$

Yet amount rises by:

$$31.25\%$$

Do not infer actual movement from pie-share movement alone when totals change.

---

## 32. SAME PERCENTAGE, DIFFERENT ACTUAL VALUE

---

If a category is 20% in both years:

2024 total:

$$1000$$

---

2025 total:

$\sqrt[3]{1500}$

Values:

$\sqrt[3]{200 \text{ to } 300}$

So same share does not imply same amount.

---

### 33. SAME ACTUAL VALUE, DIFFERENT SHARE

---

2024:

Total 1000, category 200:

$\sqrt[3]{20\%}$

2025:

Total 2000, same category 200:

$\sqrt[3]{10\%}$

Absolute amount same, share halves.

---

### 34. COMPARE SECTORS ACROSS DIFFERENT PIE CHARTS

---

You **cannot** compare angles directly if totals differ.

Example:

2024:  $A = 40\% \text{ of } 1000 = 400$ .

2025:  $B = 30\% \text{ of } 2000 = 600$ .

Even though:

$\sqrt[3]{40\% > 30\%}$

actual:

$\sqrt[3]{600 > 400}$

---

### 35. CROSS-PIE RATIO

---

A in chart 1:

$\sqrt[3]{p_{1T_1}}$

B in chart 2:

$\sqrt[3]{p_{2T_2}}$

Ratio:

$\sqrt[3]{\boxed{p_{1T_1}:p_{2T_2}}}$

where percentages may be used as proportional integers.

---

## 36. EXAMPLE

---

2024 total = 800.

A = 25%.

2025 total = 1200.

B = 20%.

Values:

A:

$\sqrt[2]{200}$

B:

$\sqrt[2]{240}$

Ratio:

$\sqrt[2]{200:240 = \boxed{5:6}}$

---

## 37. CROSS-PIE FAST METHOD

---

Instead of calculating actual values:

$\sqrt[2]{25\% \times 800 : 20\% \times 1200}$

Remove common 1/100:

$\sqrt[2]{25(800):20(1200)}$

Simplify:

$\sqrt[2]{5(800):4(1200)}$

$\sqrt[2]{4000:4800}$

$\sqrt[2]{=\boxed{5:6}}$

---

## 38. ANGLE VERSION ACROSS TWO PIES

---

If chart totals differ:

$\sqrt[2]{\boxed{\text{ratio}} = \theta_{1T_1}:\theta_{2T_2}}$

because common 360 cancels.

---

## 39. MULTIPLE PIE CHART — TOTAL GROWTH

---

2024 total:

$\sqrt[2]{10}$

2025:

$\sqrt[2]{15}$

Overall total growth:

---

$$\left[ \frac{15-10}{10} \times 100 \right]$$

$$\left[ = \boxed{50\%} \right]$$

A category can:

- grow faster than 50% !' share rises
- grow exactly 50% !' share unchanged
- grow slower than 50% !' share falls

This is a very useful insight.

## 40. SHARE-GROWTH RELATION

If overall total grows by factor  $(g_T)$ , and category grows by factor  $(g_C)$ :

- if  $(g_C > g_T)$  !' share rises
- if  $(g_C = g_T)$  !' share unchanged
- if  $(g_C < g_T)$  !' share falls

Example:

Total grows 50%.

Salaries grow 31.25%.

Therefore salary share falls.

## 41. PIE CHART WITH ANGLES INSTEAD OF %

Suppose total = 720 students.

Sectors:

Stream	Angle
CSE	$(120^\circ)$
ECE	$(90^\circ)$
ME	$(60^\circ)$
CE	$(45^\circ)$
Other	$(45^\circ)$

CSE:

$$\left[ \frac{1}{3}(720) = \boxed{240} \right]$$

ECE:

$$\left[ \frac{1}{4}(720) = \boxed{180} \right]$$

ME:

$$\left[ \frac{1}{6}(720) = \boxed{120} \right]$$

Each  $(45^\circ)$ :

$$\left[ \frac{18(720)}{90} = \boxed{144} \right]$$

---

## 42. RATIO OF CSE TO CE

---

$$\left[ 120:45 \right]$$

divide by 15:

$$\left[ \boxed{8:3} \right]$$

---

## 43. CSE + ECE SHARE

---

Angles:

$$\left[ 120+90=210^\circ \right]$$

Percentage:

$$\left[ \frac{210}{360} \times 100 \right]$$

$$\left[ =\boxed{58.33\%} \right]$$

Actual students:

$$\left[ 240+180 = \boxed{420} \right]$$

---

## 44. COMPLEMENT SHORTCUT

---

If "all except X" is asked:

$$\left[ \boxed{100\%-X\%} \right]$$

or:

$$\left[ \boxed{360^\circ - \theta_X} \right]$$

Example:

$$\text{CSE} = (120^\circ).$$

Non-CSE:

$$\left[ 360-120=240^\circ \right]$$

$$\left[ =\boxed{66.67\%} \right]$$

---

## 45. UNION OF SECTORS

---

If categories are mutually exclusive parts of the pie:

$$\left[ A+B \right]$$

share is simply:

$$\left[ p_A + p_B \right]$$

No inclusion-exclusion complexity because pie categories normally partition the whole.

---

## 46. AVERAGE CATEGORY VALUE

---

If total = 1200 distributed across 6 categories:

Average category value:

$$\lfloor 1200/6 = \boxed{200} \rfloor$$

This does not depend on individual sector sizes.

---

## 47. WHICH SECTORS ARE ABOVE AVERAGE?

---

For  $(n)$  categories, average angular size:

$$\lfloor \boxed{\frac{360^\circ}{n}} \rfloor$$

Example 6 categories:

$$\lfloor 60^\circ \rfloor$$

Any sector  $(>60^\circ)$  is above average.

Very fast comparison.

---

## 48. ABOVE-AVERAGE PERCENTAGE

---

For 5 categories:

Average percentage:

$$\lfloor 100/5 = \boxed{20\%} \rfloor$$

Any category above 20% is above average.

---

## 49. PIE CHART RANKING

---

Rank directly by:

- angle
- percentage
- actual value

within the same pie chart.

All three rankings are identical because same total.

---

## 50. RANKING ACROSS DIFFERENT PIES

---

Need actual values if totals differ.

Percentage ranking alone may be misleading.

---

## 51. PERCENTAGE-POINT CHANGE

---

Category share:

---

$\text{[ 25\%to32\% ]}$

Change:

$\text{[ \boxed{+7\text{ percentage points}} ]}$

Relative increase in share:

$\text{[ \frac{7}{25} \times 100 = \boxed{28\%} ]}$

---

## 52. ANGULAR CHANGE

Equivalent angle change:

$\text{[ 7\% \times 3.6^\circ ]}$

$\text{[ = \boxed{25.2^\circ} ]}$

---

## 53. SECTOR ANGLE DOUBLES

Suppose sector goes from:

$\text{[ 30^\circ \text{to} 60^\circ ]}$

within pies having the **same total**.

Actual value doubles.

If totals differ, not necessarily.

---

## 54. ACTUAL CATEGORY GROWTH FORMULA ACROSS PIES

Old:

$\text{[ } p_{1T_1} \text{ ]}$

New:

$\text{[ } p_{2T_2} \text{ ]}$

Growth:

$\text{[ \boxed{ \frac{p_{2T_2} - p_{1T_1}}{p_{1T_1}} \times 100 } ]}$

Percent values can be used as decimals or proportions.

---

## 55. ANGLE VERSION

$\text{[ \boxed{ \% \text{ growth} } = \frac{\theta_{2T_2} - \theta_{1T_1}}{\theta_{1T_1}} \times 100 } ]}$

because  $(1/360)$  cancels.

Useful shortcut.

---

## 56. EXAMPLE



Total grows:

$$\sqrt[800]{1000}$$

Sector angle:

$$\sqrt[90^\circ]{108^\circ}$$

Old value:

$$\sqrt[800]{90/360}=200$$

New:

$$\sqrt[1000]{108/360}=300$$

Growth:

$$\sqrt[\frac{100}{200}]{200} \times 100 = \boxed{50\%}$$

---

## 57. SECTOR SHARE CHANGED

Old:

$$\sqrt[90^\circ]{25\%}$$

New:

$$\sqrt[108^\circ]{30\%}$$

Share rises by:

$$\sqrt[\boxed{5\text{ percentage points}}]$$

while actual amount rises 50%.

---

## 58. PIE CHART WITH ROUNDING

If percentages are rounded:

$$\sqrt[33\%,33\%,34\%]$$

fine.

But charts may show:

$$\sqrt[24.9\%,25.1\%,50.1\%]$$

which sum slightly above 100 due to rounding.

Read notes before assuming error.

---

## 59. EXPLODED PIE CHART TRAP

An exploded sector is visually pulled away from the circle to emphasize it.

Its:

- angle
- percentage
- actual value

do **not** increase because it looks separated.

Only the data label matters.

---

## 60. 3D PIE CHART TRAP

---

Perspective may make front sectors look larger.

Do not judge by apparent area.

Use:

- labels
  - percentages
  - angles
- 

## 61. COMMON PIE-CHART TRAPS

---

& pTRAP 1 — 360° vs 100%

Always convert correctly.

& pTRAP 2 — Cross-pie comparison

Angles alone are insufficient if totals differ.

& pTRAP 3 — Share vs amount

Share can fall while amount rises.

& pTRAP 4 — Percentage more/less

Use correct reference sector.

& pTRAP 5 — Missing sector

Subtract from 360° or 100%.

& pTRAP 6 — Exploded/3D pie

Visual appearance can mislead.

& pTRAP 7 — Absolute value

Need total unless another scaling relation is given.

---

## 62. MUST-MEMORIZE FORMULAS

---

Angle !' percentage:

$$\boxed{p = \frac{\theta}{360} \times 100}$$

Percentage !' angle:

$$\boxed{\theta = 3.6p}$$

Sector value:

$$\boxed{V = \frac{\theta}{360} = \frac{p}{100}}$$

Total from sector:

$$\boxed{T = \frac{360V}{\theta}}$$

Same-pie ratio:

$$\boxed{V_A : V_B = \theta_A : \theta_B}$$

## 63. FAST PIE-DI DECISION TREE

### Same pie, ratio?

Use angles/percentages directly.

### Need actual value?

Multiply share by total.

### Need total?

Divide actual sector value by share.

### Missing sector?

Subtract from  $360^\circ$  or 100%.

### Two pie charts?

Convert using each chart's total.

### Share changed?

Distinguish percentage points from actual-value growth.

### Totals changed?

Never compare sector angles alone.

## PRACTICE — LEVEL 0

A pie chart has total 720 and sectors:

- $A = 90^\circ$
- $B = 120^\circ$
- $C = 60^\circ$
- $D = 45^\circ$
- E = remaining

### Q1

A value?

### Q2

B value?

### Q3

C percentage?

Q4

E angle?

Q5

A:B ratio?

LEVEL 1

Total expenditure = 11,20,000.

Shares:

- Rent = 30%
- Food = 25%
- Education = 15%
- Travel = 10%
- Other = remaining

Q6

Rent expenditure?

Q7

Other percentage?

Q8

Other expenditure?

Q9

Rent:Food ratio?

Q10

By what percent is Rent greater than Food?

LEVEL 2 — TWO PIE CHARTS

2024 total sales = 1000.

2025 total sales = 1500.

Product	2024 Share	2025 Share
A	30%	25%
B	25%	30%
C	20%	20%
D	15%	15%
E	10%	10%

Q11

A sales in 2024?

Q12

A sales in 2025?

Q13

Percentage growth of A sales?

Q14

B sales growth percentage?

Q15

Which products kept unchanged share?

LEVEL 3 — EXAM PRESSURE

2024 total employees:

\[ 800 \]

2025 total employees:

\[ 1200 \]

Sector angles:

Dept	2024	2025
Tech	\(144^\circ\)	\(126^\circ\)
Sales	\(90^\circ\)	\(108^\circ\)
HR	\(54^\circ\)	\(54^\circ\)
Ops	remaining	remaining

Q16

Tech employees in both years?

Q17

Tech employee percentage growth?

Q18

Sales employees in both years?

Q19

HR employee percentage growth?

Q20

Ops angle and employee count in both years?

---

## ANSWERS

---

### Q1

---

$$\backslash[ 720(90/360) = 720/4 = \boxed{180} \backslash]$$

### Q2

---

$$\backslash[ 720(120/360) = 720/3 = \boxed{240} \backslash]$$

### Q3

---

$$\backslash[ 60/360\times100 = \boxed{16.67\%} \backslash]$$

### Q4

---

Known angles:

$$\backslash[ 90+120+60+45=315 \backslash]$$

Remaining:

$$\backslash[ 360-315 = \boxed{45^\circ} \backslash]$$

### Q5

---

$$\backslash[ 90:120 = \boxed{3:4} \backslash]$$

---

### Q6

---

$$\backslash[ 120000(0.30) = \boxed{36\,000} \backslash]$$

### Q7

---

Known:

$$\backslash[ 30+25+15+10=80\% \backslash]$$

Other:

$$\backslash[ \boxed{20\%} \backslash]$$

### Q8

---

$$\backslash[ 120000(0.20) = \boxed{24\,000} \backslash]$$

### Q9

---

$$\backslash[ 30:25 = \boxed{6:5} \backslash]$$

### Q10

---

$$\backslash[ \frac{30-25}{25}\times100 \backslash]$$

---

$$\boxed{20\%}$$

---

### Q11

---

$$1000(0.30) = \boxed{300}$$

### Q12

---

$$1500(0.25) = \boxed{375}$$

### Q13

---

$$\frac{375-300}{300} \times 100$$

$$\boxed{25\%}$$

### Q14

---

2024 B:

$$250$$

2025:

$$450$$

Growth:

$$\frac{200}{250} \times 100$$

$$\boxed{80\%}$$

### Q15

---

C, D and E have unchanged shares.

$$\boxed{C,D,E}$$

---

### Q16

---

Tech 2024:

$$800(144/360) = 800(0.4) = 320$$

Tech 2025:

$$1200(126/360)$$

$$= 1200(0.35) = 420$$

$$\boxed{320, 420}$$

### Q17

---

$$\frac{420-320}{320} \times 100$$

$$\boxed{31.25\%}$$

## Q18

---

Sales 2024:

$$\lfloor 800(90/360)=200 \rfloor$$

Sales 2025:

$$\lfloor 1200(108/360) = 1200(0.3) = 360 \rfloor$$

$$\lfloor \boxed{200, \ 360} \rfloor$$

## Q19

---

HR 2024:

$$\lfloor 800(54/360) = 800(0.15) = 120 \rfloor$$

HR 2025:

$$\lfloor 1200(54/360) = 1200(0.15) = 180 \rfloor$$

Growth:

$$\lfloor 60/120 = \boxed{50\%} \rfloor$$

## Q20

---

### 2024 Ops angle

$$\lfloor 360-(144+90+54) \rfloor$$

$$\lfloor =360-288 = 72^\circ \rfloor$$

Employees:

$$\lfloor 800(72/360) = 800(0.2) = 160 \rfloor$$

### 2025 Ops angle

$$\lfloor 360-(126+108+54) \rfloor$$

$$\lfloor =360-288 = 72^\circ \rfloor$$

Employees:

$$\lfloor 1200(0.2) = 240 \rfloor$$

So:

$$\lfloor \boxed{2024: \ 72^\circ, \ 160} \rfloor$$

$$\lfloor \boxed{2025: \ 72^\circ, \ 240} \rfloor$$

Interesting result: same angle/share, but employee count rises by 50% because total rises by 50%.

---

## CHAPTER 54 STATUS

---

Covered so far:

- full-circle concept '
- angle !" percentage conversion '
- actual-value conversion '



- same-pie ratios '
- percentage more/less '
- missing sectors '
- finding total from one sector '
- multiple pie charts '
- changing totals '
- share vs actual-value changes '
- cross-pie ratios '
- percentage-point changes '
- ranking '
- above-average sectors '
- exploded/3D pie traps '
- L0–L3 practice '

But **Chapter 54 is not complete yet.**

Still remaining:

- advanced dual-pie comparisons
- category transfer between sectors
- conditional sector questions
- profit/expenditure pie sets
- pie + table hybrid DI
- nested/donut charts
- ratio reconstruction
- multi-stage percentage charts
- large placement-style timed set
- final pie-chart speed audit

## Progress

$\boxed{53/76}$  completed

Current: 0 = Chapter 54 — Pie Charts

## NEXT

0 = Chapter 54 Part 2 — Advanced Dual-Pie DI, Donut Charts, Category Transfers, Hybrid Pie+Table Sets & Final Pie-Chart Speed Toolkit

USER

next

ASSISTANT

## CHAPTER 54 — PIE CHARTS

## PART 2 — ADVANCED DUAL-PIE DI, DONUT CHARTS, CATEGORY TRANSFERS, PIE+TABLE HYBRIDS & FINAL SPEED TOOLKIT

Status:  $\emptyset$ =NA PROGRESS GATE: High Placements: High Priority:  $\emptyset$ =Yr 1

This part is about pie-chart questions where percentages alone are not enough.

Main idea:

Within one pie, compare shares directly. Across different pies, first account for the different totals.

### 1. DUAL-PIE CHARTS — MASTER IDEA

Suppose two years have:

$$\backslash[ T_1,\quad T_2 \backslash]$$

and a category has shares:

$$\backslash[ p_1,\quad p_2 \backslash]$$

Then actual values are:

$$\backslash[ V_1=p_1T_1 \backslash]$$

$$\backslash[ V_2=p_2T_2 \backslash]$$

Therefore growth is:

$$\backslash[ \boxed{\frac{p_2T_2-p_1T_1}{p_1T_1}\times 100} \backslash]$$

Use percentages as decimals or proportional numbers consistently.

### 2. FAST MULTIPLIER METHOD

Suppose:

Total rises:

$$\backslash[ 1000\text{to}1500 \backslash]$$

So total multiplier:

$$\backslash[ 1.5 \backslash]$$

Category share falls:

$$\backslash[ 40\%\text{to}32\% \backslash]$$

Category value multiplier:

$$\backslash[ 1.5\times\frac{32}{40} \backslash]$$

$$\backslash[ =1.5(0.8) \backslash]$$

$$\backslash[ =1.2 \backslash]$$

So category value rises:

$$\backslash[ \boxed{20\%} \backslash]$$

No need to calculate actual values.

---

### 3. GENERAL SHARE-MULTIPLIER SHORTCUT

---

$$\boxed{\frac{V_2}{V_1} = \frac{T_2}{T_1} \times \frac{p_2}{p_1}}$$

Then:

$$\% \text{ change} = \left( \frac{V_2}{V_1} - 1 \right) 100$$

Very useful.

---

### 4. EXAMPLE

---

Total:

$$800 \text{ to } 1200$$

Share:

$$25\% \text{ to } 20\%$$

Multiplier:

$$\frac{1200}{800} \times \frac{20}{25}$$

$$= 1.5 \times 0.8$$

$$= 1.2$$

So actual category grows:

$$\boxed{20\%}$$

---

### 5. CATEGORY VALUE UNCHANGED — FIND NEW SHARE

---

Suppose total rises by:

$$25\%$$

but a category's actual value remains unchanged.

Old share:

$$40\%$$

New total multiplier:

$$1.25$$

To keep category value same:

$$1.25 p_2 = 0.40$$

$$p_2 = 0.32$$

So new share:

$$\boxed{32\%}$$

---

### 6. CATEGORY VALUE GROWS SAME AS TOTAL

---

If category share remains unchanged:

$$p_1 = p_2$$

then category grows at exactly the same percentage as total.

Example:

Total +50%, share unchanged.

Category:

$$\boxed{+50\%}$$

---

## 7. CATEGORY SHARE RISES — WHAT DOES IT MEAN?

If total also rises, category definitely grows faster than total.

Example:

Total:

$$+20\%$$

Share:

$$25\% \text{ to } 30\%$$

Category multiplier:

$$1.2 \times \frac{30}{25}$$

$$= 1.44$$

So category growth:

$$\boxed{44\%}$$

---

## 8. CATEGORY SHARE FALLS

A falling share does **not** necessarily mean a falling actual value.

You must compare:

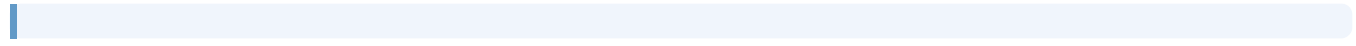
$$\frac{T_2}{T_1}$$

with:

$$\frac{p_1}{p_2}$$

If:

$$\frac{T_2}{T_1} > \frac{p_1}{p_2}$$


$$\frac{p_1}{p_2} > 1$$

actual value still rises.

---

## 9. CATEGORY TRANSFER BETWEEN SECTORS

Suppose a pie total remains 1000.

Old shares:

- A = 40%
- B = 30%
- C = 20%
- D = 10%

Now 5 percentage points are transferred from A to C.

New:

- A = 35%
- B = 30%
- C = 25%
- D = 10%

Actual transferred amount:

$5\% \times 1000$

$\boxed{50}$

---

## 10. TRANSFER WITH CHANGING TOTAL

---

Suppose total changes from 1000 to 1200.

Old A share:

$40\%$

New A share:

$35\%$

Do **not** say A lost 5% of 1000.

Old A:

$400$

New A:

$420$

A's share fell, but actual value increased by:

$\boxed{20}$

---

## 11. CHANGE IN ANGLE

---

If share changes:

$25\% \text{ to } 30\%$

Angle changes:

Old:

$90^\circ$

---

New:

$\backslash [ 108^\circ \backslash ]$

Difference:

$\backslash [ \boxed{18^\circ} \backslash ]$

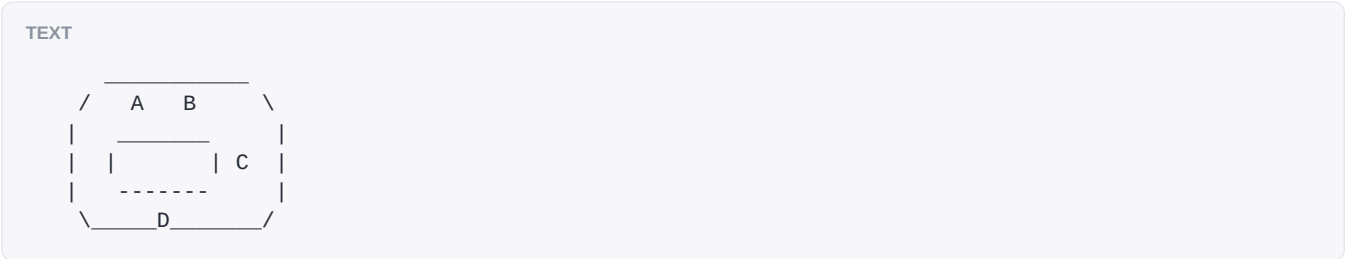
Equivalent to:

$\backslash [ 5\text{percentage points} \backslash ]$

## 12. DONUT CHART

A donut chart is essentially a pie chart with a hole.

Example:



Interpretation is still based on:

$\backslash [ \boxed{100\%=360^\circ} \backslash ]$

The central hole has no mathematical meaning unless it contains another dataset.

## 13. NESTED DONUT CHART

Sometimes:

- inner ring = 2024
- outer ring = 2025

This behaves like two pie charts.

Important:

**Never compare visual ring area directly.**

Use labeled shares and totals.

## 14. NESTED DONUT EXAMPLE

Inner ring total:

$\backslash [ 1000 \backslash ]$

Outer ring total:

$\backslash [ 1500 \backslash ]$

Category A:

Inner:

$\left[ 30\% \rightarrow 300 \right]$

Outer:

$\left[ 25\% \rightarrow 375 \right]$

Actual growth:

$\left[ 75/300 \right]$

$\left[ =\boxed{25\%} \right]$

## 15. DONUT AREA TRAP

Because outer ring has larger radius, its segment physically occupies greater area even at the same angle.

Do not infer quantity from visual area.

The chart encodes value using:

$\left[ \boxed{\text{angle/share}} \right]$

not ring area.

## 16. PIE + TABLE HYBRID

Suppose pie chart shows expenditure shares.

Total expenditure:

$\left[ 120 \text{ lakhs} \right]$

Pie:

Category	Share
Salary	40%
Rent	20%
Marketing	15%
Travel	10%
Other	15%

A table says:

Category	Increase next year
Salary	+10%
Rent	+20%
Marketing	-10%
Travel	+50%

Category	Increase next year
Other	unchanged

## 17. INITIAL VALUES

Salary:

$\text{₹ } 8 \text{ lakh}$

Rent:

$\text{₹ } 4 \text{ lakh}$

Marketing:

$\text{₹ } 3 \text{ lakh}$

Travel:

$\text{₹ } 2 \text{ lakh}$

Other:

$\text{₹ } 3 \text{ lakh}$

all in  $\text{₹}$  lakh.

## 18. NEXT-YEAR VALUES

Salary:

$\text{₹ } 8(1.1)=8.8 \text{ lakh}$

Rent:

$\text{₹ } 4(1.2)=4.8 \text{ lakh}$

Marketing:

$\text{₹ } 3(0.9)=2.7 \text{ lakh}$

Travel:

$\text{₹ } 2(1.5)=3 \text{ lakh}$

Other:

$\text{₹ } 3 \text{ lakh}$

New total:

$\text{₹ } 8.8+4.8+2.7+3+3$

$\text{₹ } 22.3 \text{ lakh}$

lakh.

## 19. NEW SHARE — SALARY



$$\left[ \frac{8.8}{22.3} \times 100 \right]$$

$$\left[ \approx \boxed{39.46\%} \right]$$

Salary amount grew, but share fell slightly.

## 20. HYBRID REVERSE QUESTION

Suppose pie gives company sales shares, and table gives profit margins.

Total sales:

$$\left[ 1000 \right]$$

Company	Sales Share	Margin
A	30%	20%
B	25%	24%
C	20%	30%
D	15%	10%
E	10%	40%

Profit:

A:

$$\left[ 300(0.2)=60 \right]$$

B:

$$\left[ 250(0.24)=60 \right]$$

C:

$$\left[ 200(0.30)=60 \right]$$

D:

$$\left[ 150(0.10)=15 \right]$$

E:

$$\left[ 100(0.40)=40 \right]$$

Highest:

$$\left[ \boxed{A,B,C} \right]$$

tie at 60.

## 21. HIGH SHARE "HIGH PROFIT

Company A may have higher sales share than E, but E may have much larger profit margin.

Always compute:

$$\left[ \boxed{\text{Profit}=\text{Sales}\times \text{Margin}} \right]$$

---

## 22. PIE + POPULATION TABLE

Suppose a pie chart gives population distribution:

Region	Share
A	30%
B	25%
C	20%
D	15%
E	10%

Total population:

\[ 1,000,000 \]

A table gives literacy:

Region	Literacy
A	80%
B	70%
C	90%
D	60%
E	50%

---

## 23. LITERATE POPULATION

A:

\[ 300000(0.8)=240000 \]

B:

\[ 250000(0.7)=175000 \]

C:

\[ 200000(0.9)=180000 \]

D:

\[ 150000(0.6)=90000 \]

E:

\[ 100000(0.5)=50000 \]

Total literate:

\[ 735000 \]

Overall literacy:

$$\boxed{73.5\%}$$

---

## 24. WEIGHTED PERCENTAGE AGAIN

Overall literacy is not:

$$\frac{80+70+90+60+50}{5} = 70\%$$

Correct weighted result:

$$73.5\%$$

because populations differ.

---

## 25. CONDITIONAL SECTOR QUESTION

Using population example:

**Which regions have literacy above 70% and population share at least 20%?**

A:

30%, 80%

B:

25%, 70% 'Lbecause not above 70

C:

20%, 90%

D/E 'L

Answer:

$$\boxed{A,C}$$

---

## 26. CONDITIONAL TOTAL

Total literates in A and C:

$$240000+180000$$

$$\boxed{420000}$$

---

## 27. RATIO RECONSTRUCTION

Suppose pie chart has three sectors with:

$$A:B=2:3$$

$$B:C=4:5$$

From ratio chaining:

$$\backslash A:B:C=8:12:15 \backslash$$

Total parts:

$$\backslash 35 \backslash$$

Angles:

A:

$$\backslash \frac{8}{35}(360) \backslash$$

B:

$$\backslash \frac{12}{35}(360) \backslash$$

C:

$$\backslash \frac{15}{35}(360) \backslash$$

No need to compute unless asked.

---

## 28. FIND ANGLE FROM RATIO

---

If:

$$\backslash A:B:C=2:3:5 \backslash$$

Total parts:

$$\backslash 10 \backslash$$

Angles:

A:

$$\backslash 72^\circ \backslash$$

B:

$$\backslash 108^\circ \backslash$$

C:

$$\backslash 180^\circ \backslash$$

---

## 29. FIND VALUES FROM RATIO + TOTAL

---

Total:

$$\backslash 1000 \backslash$$

Ratio:

$$\backslash 2:3:5 \backslash$$

Values:

$$\backslash 200, \backslash 300, \backslash 500 \backslash$$

---

## 30. MULTI-STAGEPIE PERCENTAGE

---

Suppose:

- Engineering students = 40% of all students
- CSE = 50% of Engineering students
- AI = 25% of CSE

AI as share of total:

$$[ 0.4(0.5)(0.25) ]$$

$$[ =\boxed{5\%} ]$$

---

### 31. MULTI-STAGEREVERSE

---

AI students:

$$[ 50 ]$$

If AI = 25% of CSE, CSE = 200.

If CSE = 50% of Engineering:

Engineering = 400.

If Engineering = 40% of total:

Total:

$$[ 400/0.4 = \boxed{1000} ]$$

---

### 32. PIE CHART WITH OVERLAPPING CATEGORIES?

---

Normally a pie chart represents mutually exclusive parts of one whole.

If labels overlap conceptually, a standard pie is inappropriate.

If an exam still presents it, follow the explicit definition.

Do not apply inclusion-exclusion unless wording indicates overlap.

---

### 33. PROFIT/EXPENDITURE PIE SET

---

Suppose total annual expense:

$$[ 150 \text{ lakh} ]$$

Pie:

- Salary = 36%
- Rent = 20%
- Raw material = 24%
- Utilities = 10%
- Other = 10%

Revenue:

$$[ 165 \text{ lakh} ]$$

Total profit:

---

$$\boxed{65-50 = 15}$$

### 34. SALARY AS % OF REVENUE

Salary:

$$50(0.36)=18$$

As % of revenue:

$$18/65 \times 100$$

$$\approx \boxed{27.69\%}$$

Notice pie percentages use expense total, not revenue.

### 35. COST CATEGORY AS % OF PROFIT

Raw material:

$$50(0.24)=12$$

Profit:

$$15$$

Raw material as % of profit:

$$12/15 \times 100 = \boxed{80\%}$$

### 36. EXPENDITURE REDUCTION IMPACT

Suppose rent decreases by 25%.

Old rent:

$$10$$

Reduction:

$$2.5$$

New total expense:

$$47.5$$

If revenue unchanged at 65:

New profit:

$$17.5$$

Profit increase:

$$2.5$$

Percentage profit increase:

$$\frac{2.5}{17.5} \times 100$$

$$= \boxed{16.67\%}$$

Small cost changes can create larger profit percentage changes.

---

## 37. DONUT WITH INNER TOTAL / OUTER DISTRIBUTION

---

Some donut charts show:

- inner ring: broad categories
- outer ring: subcategories

Example:

Inner:

- Tech = 40%
- Sales = 30%
- HR = 20%
- Other = 10%

Outer subdivides Tech:

- Software = 25% of total
- Data = 10%
- Infra = 5%

These outer Tech subcategories sum:

$\backslash[ 25+10+5=40\% \backslash]$

---

## 38. SUBCATEGORY AS % OF PARENT

---

Software:

$\backslash[ 25\%\text{ of total} \backslash]$

Tech:

$\backslash[ 40\%\text{ of total} \backslash]$

Software as % of Tech:

$\backslash[ 25/40\times100 \backslash]$

$\backslash[ =\boxed{62.5\%} \backslash]$

---

## 39. PARENT TO SUBCATEGORY

---

If Tech total count:

$\backslash[ 400 \backslash]$

and Software = 62.5% of Tech:

$\backslash[ 400(0.625) = \boxed{250} \backslash]$

---

## 40. CHANGE IN PARENT VS SUBCATEGORY

---

A subcategory can grow even if parent share falls.

Again, compare actual values, not only shares.

## 41. ADVANCED DUAL-PIE SET

2024 total revenue:  
\\ 1200 \\

2025 total revenue:  
\\ 1800 \\

Product	2024 Share	2025 Share
A	25%	20%
B	20%	25%
C	30%	30%
D	15%	15%
E	10%	10%

## 42. ACTUAL VALUES 2024

A:  
\\ 300 \\

B:  
\\ 240 \\

C:  
\\ 360 \\

D:  
\\ 180 \\

E:  
\\ 120 \\

## 43. ACTUAL VALUES 2025

A:  
\\ 360 \\

B:  
\\ 450 \\

C:



$\sqrt{540}$

D:

$\sqrt{270}$

E:

$\sqrt{180}$

---

## 44. GROWTH — A

---

$\sqrt{300\text{to}360}$

$\sqrt{=\boxed{20\%}}$

Even though share fell from 25% to 20%.

---

## 45. GROWTH — B

---

$\sqrt{240\text{to}450}$

Increase:

$\sqrt{210}$

$\sqrt{210/240\times 100 = \boxed{87.5\%}}$

---

## 46. GROWTH — C

---

Share unchanged.

Total grows:

$\sqrt{1200\text{to}1800}$

$\sqrt{50\%}$

Therefore C grows:

$\sqrt{\boxed{50\%}}$

Shortcut.

---

## 47. HIGHEST ABSOLUTE INCREASE

---

A:

$\sqrt{60}$

B:

$\sqrt{210}$

C:

$\sqrt{180}$

D:

---

$\sqrt[3]{90}$

E:

$\sqrt[3]{60}$

Highest:

$\sqrt[3]{\boxed{B}}$

---

## 48. TOTAL SHARE OF B+C IN 2025

---

$\sqrt[3]{25+30} = \sqrt[3]{55\%}$

Actual:

$\sqrt[3]{0.55(1800)} = \sqrt[3]{990}$

---

## 49. RATIO A2024 : D2025

---

A2024:

$\sqrt[3]{300}$

D2025:

$\sqrt[3]{270}$

Ratio:

$\sqrt[3]{300:270} = \sqrt[3]{10:9}$

---

## 50. FAST CROSS-PIE RATIO WITHOUT ACTUAL VALUES

---

$\sqrt[3]{A_{2024}:D_{2025}}$

$\sqrt[3]{25(1200):15(1800)}$

Cancel:

$\sqrt[3]{30000:27000}$

$\sqrt[3]{= \boxed{10:9}}$

---

## 51. PIE-CHART TIME STRATEGY

---

### First 10 seconds

Read:

- total
  - year
  - angle or %
  - whether multiple pies
  - whether totals differ
-

## Then

Solve direct:

- missing share
- largest/smallest
- same-pie ratio

## Next

Compute actual values only when needed.

## Last

Do:

- growth
- weighted hybrid
- cross-pie comparisons

---

## 52. WHEN TO MAKE A MINI-TABLE

For two pie charts, quickly write:

TEXT

	2024	2025
A	300	360
B	240	450
C	360	540

if several questions need actual values.

This avoids repeated conversion.

---

## 53. WHEN NOT TO CONVERT EVERYTHING

If question asks:

**A:B in same 2024 pie?**

Use:

$\backslash 25:20=5:4 \backslash$

No actual values needed.

---

## 54. FAST ANGLE BENCHMARKS

Memorize:

$\backslash 36^\circ=10\% \backslash$

$\backslash 72^\circ=20\% \backslash$

$\backslash 90^\circ=25\% \backslash$

$\backslash [ 108^\circ = 30\% \backslash ]$

$\backslash [ 144^\circ = 40\% \backslash ]$

$\backslash [ 180^\circ = 50\% \backslash ]$

These handle a large fraction of pie DI mentally.

---

## 55. FINAL PIE-CHART TRAPS

---

### & pTRAP 1 — Same angle across different totals

Same angle means same percentage, not same amount.

### & pTRAP 2 — Falling share

Actual value can still rise.

### & pTRAP 3 — Donut area

Use angle/share, not visual area.

### & pTRAP 4 — Parent/subcategory

Subcategory percentage may be of parent, not total.

### & pTRAP 5 — Hybrid table

Pie share denominator and table rate denominator may differ.

### & pTRAP 6 — Category transfer

Share change is not necessarily actual change when total changes.

### & pTRAP 7 — Weighted percentages

Never average rates blindly.

---

## 56. MUST-MEMORIZE ADVANCED PIE FORMULAS

---

Cross-pie value ratio:

$$\backslash \boxed{V_1:V_2=p_1T_1:p_2T_2} \backslash$$

Category multiplier:

$$\backslash \boxed{\frac{V_2}{V_1} = \frac{T_2}{T_1} \frac{p_2}{p_1}} \backslash$$

Multi-stage percentage:

$$\backslash \boxed{p_{\text{overall}} = p_1 p_2 p_3 \cdots} \backslash$$

where percentages are decimals.

Weighted overall rate:

$$\backslash \boxed{\frac{\sum(\text{group size} \times \text{rate})}{\sum \text{group size}}} \backslash$$

---

## 57. FINAL PIE-DI DECISION TREE

---

One pie?

---

Angles/percentages directly.

### **Need actual count?**

Share  $\times$  total.

### **Two pies?**

Check totals first.

### **Share fell?**

Do not infer actual fall.

### **Category constant?**

Use multiplier relation.

### **Donut/nested?**

Treat each ring according to its own total/base.

### **Pie + table?**

Translate both into actual quantities before combining.

### **Parent/subcategory?**

Check denominator carefully.

### **Options broad?**

Use proportional comparison.

---

## **PRACTICE — LEVEL 0**

---

Total = 900.

Shares:

- A = 30%
- B = 25%
- C = 20%
- D = 15%
- E = 10%

### **Q1**

A value?

### **Q2**

D value?

### **Q3**

A:C ratio?

### **Q4**

B+D share?

Q5

A is what percent more than C?

LEVEL 1 — DUAL PIE

2024 total = 800. 2025 total = 1200.

Category	2024	2025
A	25%	20%
B	20%	25%
C	30%	30%
D	15%	15%
E	10%	10%

Q6

A actual values?

Q7

A growth %?

Q8

B growth %?

Q9

C growth %?

Q10

Which category has unchanged share but actual value increases?

LEVEL 2 — PIE + TABLE

Total employees = 1000.

Department	Share	Women %
Tech	40%	25%
Sales	30%	40%
HR	20%	60%
Other	10%	50%

Q11

Women in Tech?

**Q12**

Women in Sales?

**Q13**

Total women?

**Q14**

Overall women percentage?

**Q15**

Which department has highest number of women?

## LEVEL 3 — EXAM PRESSURE

2024 total market = 2000. 2025 total market = 2500.

Product	2024 share	2025 share
P	30%	25%
Q	25%	30%
R	20%	20%
S	15%	15%
T	10%	10%

**Q16**

P sales in both years?

**Q17**

P percentage change?

**Q18**

Q percentage change?

**Q19**

Which products grew at exactly the same rate as total market?

**Q20**

Ratio  $(P_{2024}:Q_{2025})$ ?

## ANSWERS

**Q1**

$\lfloor 900(0.30) = \boxed{270} \rfloor$

## Q2

$$\backslash[ 900(0.15) = \boxed{135} \backslash]$$

## Q3

$$\backslash[ 30:20 = \boxed{3:2} \backslash]$$

## Q4

$$\backslash[ 25+15 = \boxed{40\%} \backslash]$$

## Q5

$$\backslash[ \frac{30-20}{20} \times 100 = \boxed{50\%} \backslash]$$

## Q6

2024 A:

$$\backslash[ 800(0.25)=200 \backslash]$$

2025 A:

$$\backslash[ 1200(0.20)=240 \backslash]$$

$$\backslash[ \boxed{200, \ 240} \backslash]$$

## Q7

$$\backslash[ \frac{40}{200} \times 100 = \boxed{20\%} \backslash]$$

## Q8

2024 B:

$$\backslash[ 160 \backslash]$$

2025:

$$\backslash[ 300 \backslash]$$

Growth:

$$\backslash[ 140/160 = \boxed{87.5\%} \backslash]$$

## Q9

C share unchanged.

Total:

$$\backslash[ 800 \text{ to } 1200 \backslash]$$

Growth:

$$\backslash[ \boxed{50\%} \backslash]$$



### Q10

---

C, D and E all have unchanged shares.

Their actual values all rise with the total by 50%.

$$\boxed{C,D,E}$$

---

### Q11

---

Tech employees:

$$400$$

Women:

$$400(0.25) = \boxed{100}$$

### Q12

---

Sales employees:

$$300$$

Women:

$$300(0.40) = \boxed{120}$$

### Q13

---

HR:

$$200(0.6)=120$$

Other:

$$100(0.5)=50$$

Total women:

$$100+120+120+50$$

$$= \boxed{390}$$

### Q14

---

$$390/1000 = \boxed{39\%}$$

### Q15

---

Sales and HR:

$$120$$

each.

$$\boxed{\text{Sales and HR}}$$

---

### Q16

---

P 2024:

$$\backslash[ 2000(0.30) = 600 \backslash]$$

P 2025:

$$\backslash[ 2500(0.25) = 625 \backslash]$$

$$\backslash[ \boxed{600, \ 625} \backslash]$$

## Q17

---

$$\backslash[ \frac{25}{600} \times 100 \backslash]$$

$$\backslash[ = \boxed{4.17\%} \backslash]$$

approximately.

## Q18

---

Q 2024:

$$\backslash[ 2000(0.25) = 500 \backslash]$$

Q 2025:

$$\backslash[ 2500(0.30) = 750 \backslash]$$

Growth:

$$\backslash[ 250/500 = \boxed{50\%} \backslash]$$

## Q19

---

Total market growth:

$$\backslash[ \frac{2500-2000}{2000} = 25\% \backslash]$$

Products with unchanged shares:

R, S, T.

Therefore each grows exactly:

$$\backslash[ \boxed{25\%} \backslash]$$

Answer:

$$\backslash[ \boxed{R, S, T} \backslash]$$

## Q20

---

$$\backslash[ P_{2024} = 600 \backslash]$$

$$\backslash[ Q_{2025} = 750 \backslash]$$

Ratio:

$$\backslash[ 600:750 = \boxed{4:5} \backslash]$$

## ' CHAPTER 54 COMPLETE

---

We have now covered:

- angle !" percentage !" actual value '
- same-pie ratios '
- missing sectors '
- total from one sector '
- dual pie charts '
- changing totals '
- share vs absolute change '
- category multiplier shortcut '
- category transfers '
- donut and nested donut charts '
- pie + table hybrid DI '
- weighted percentage hybrids '
- parent/subcategory charts '
- multi-stage percentages '
- profit/expenditure pie sets '
- ratio reconstruction '
- misleading visual traps '
- advanced timed strategies '
- L0–L3 practice '

## Progress

---

\[ \boxed{54/76}\text{ completed}} \]

## UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations '
- Chapter 51 — Tables '
- Chapter 52 — Bar & Stacked-Bar Graphs '
- Chapter 53 — Line Graphs '
- Chapter 54 — Pie Charts '
- Chapter 55 — Mixed Graphs & Caselets +
- Chapter 56 — Missing-Data & Advanced DI +

## NEXT

---

### Chapter 55 — Mixed Graphs & Caselets

Next we combine everything:

- table + bar
- bar + line
- pie + table
- multi-source DI
- caselet DI without a graph
- converting text into tables

- multi-stage ratios and percentages
- revenue/cost/profit caselets
- population/employment caselets
- conditional questions
- GATE/placement multi-question sets
- speed, approximation and option elimination.

USER

next

ASSISTANT

## CHAPTER 55 — MIXED GRAPHS & CASELETS

### PART 1 — TABLE + BAR + LINE + PIE COMBINATIONS, TEXT CASELETS, MULTI-STAGE RATIOS & DERIVED DATA

Status: ~~0~~=No PROGRESS GATE: High Placements: Very High Priority: ~~0~~=Yes 1

Mixed DI combines two or more data formats.

Examples:

- table + bar graph
- bar + line graph
- pie chart + table
- table + percentage labels
- text-only caselet
- multi-stage business funnel
- population + rate data
- revenue + cost + margin data

The central skill is:

**Convert all sources into one common set of useful numbers before comparing them.**

## 1. WHAT IS CASELET DI?

A caselet gives data in words instead of a neat table or graph.

Example:

**A company sold 500 units in 2024. Product A formed 30% of sales. Product B sales were 20% more than A. Product C accounted for the remaining sales.**

Your job is to convert the text into structured data.

---

## 2. FIRST RULE FOR CASELETS

---

Do not solve directly from the paragraph.

Convert it into a small table.

From the example:

Total:

$\boxed{500}$

A:

$\boxed{30\% \times 500 = 150}$

B:

$\boxed{150(1.2) = 180}$

C:

$\boxed{500 - 150 - 180}$

$\boxed{= 170}$

Mini-table:

Product	Sales
A	150
B	180
C	170

Now questions become easy.

---

## 3. COMMON CASELET LANGUAGE

---

Translate words into equations.

**“A is 20% more than B”**

$\boxed{A = 1.2B}$

**“A is 20% less than B”**

$\boxed{A = 0.8B}$

**“A:B = 3:5”**

$\boxed{A = 3k, \text{quad } B = 5k}$

**“A is 25% of total”**

$\boxed{A = 0.25T}$

**“Remaining”**

$\boxed{T - \text{known parts}}$

---

---

## 4. MIXED GRAPH — BAR + LINE

Suppose bars show annual revenue and line shows profit margin.

Year	Revenue (¹ crore)	Margin
2022	500	20%
2023	600	25%
2024	750	20%
2025	900	30%

We can derive profit.

---

## 5. PROFIT FROM BAR + LINE

$$\boxed{\text{Profit} = \text{Revenue} \times \text{Margin}}$$

2022:

$$500(0.2) = 100$$

2023:

$$600(0.25) = 150$$

2024:

$$750(0.2) = 150$$

2025:

$$900(0.3) = 270$$

---

## 6. COST FROM REVENUE AND MARGIN

$$\text{Cost} = \text{Revenue} - \text{Profit}$$

2025:

$$900 - 270 = \boxed{630}$$

---

## 7. HIGHEST REVENUE VS HIGHEST PROFIT

Highest revenue:

$$2025$$

Highest profit:

$$2025$$

But this need not always coincide.

Example:

Revenue 1000 at 10% margin !' profit 100.

Revenue 700 at 25% margin !' profit 175.

So smaller revenue can produce larger profit.

---

## 8. PROFIT GROWTH

2022 !' 2025:

$\sqrt[100]{270}$

Increase:

$\sqrt[170]$

Percentage:

$\sqrt[\frac{170}{100} \times 100]$

$\sqrt[= \boxed{170\%}]$

---

## 9. PIE + TABLE HYBRID

Suppose total employees:

$\sqrt[1000]$

Pie chart gives department shares:

Dept	Share
Tech	40%
Sales	30%
HR	20%
Other	10%

Another table gives women percentage:

Dept	Women %
Tech	25%
Sales	40%
HR	60%
Other	50%

---

## 10. DEPARTMENT COUNTS

Tech:

$\backslash [ 400 \backslash ]$

Sales:

$\backslash [ 300 \backslash ]$

HR:

$\backslash [ 200 \backslash ]$

Other:

$\backslash [ 100 \backslash ]$

---

## 11. WOMEN COUNTS

---

Tech:

$\backslash [ 400(0.25)=100 \backslash ]$

Sales:

$\backslash [ 300(0.4)=120 \backslash ]$

HR:

$\backslash [ 200(0.6)=120 \backslash ]$

Other:

$\backslash [ 100(0.5)=50 \backslash ]$

Total women:

$\backslash [ 390 \backslash ]$

Overall percentage:

$\backslash [ \boxed{39\%} \backslash ]$

---

## 12. WHY SIMPLE AVERAGE FAILS

---

Average of:

$\backslash [ 25,40,60,50 \backslash ]$

is:

$\backslash [ 43.75\% \backslash ]$

Wrong overall percentage because department sizes differ.

Correct:

$\backslash [ \boxed{39\%} \backslash ]$

This is a weighted-average problem hidden inside mixed DI.

---

## 13. TABLE + PERCENTAGE-LINE HYBRID

---



Suppose table gives production and line gives defect rate.

Plant	Production	Defect Rate
A	1000	2%
B	1500	3%
C	1200	1.5%
D	2000	2.5%

Defective units:

A:

$$\lfloor 20 \rfloor$$

B:

$$\lfloor 45 \rfloor$$

C:

$$\lfloor 18 \rfloor$$

D:

$$\lfloor 50 \rfloor$$

Highest defective count:

$$\lfloor \boxed{D} \rfloor$$

---

## 14. HIGHEST RATE "HIGHEST COUNT

Highest defect rate:

$$\lfloor B=3\% \rfloor$$

But highest defective count:

$$\lfloor D=50 \rfloor$$

because D has larger production base.

This pattern appears repeatedly in mixed DI.

---

## 15. OVERALL DEFECT RATE

Total defective:

$$\lfloor 20+45+18+50 = 133 \rfloor$$

Total production:

$$\lfloor 1000+1500+1200+2000 = 5700 \rfloor$$

Rate:

$$\lfloor \frac{133}{5700} \times 100 \rfloor$$

$\approx 2.33\%$

Do not average 2%, 3%, 1.5%, 2.5%.

## 16. TEXT CASELET — RATIO + TOTAL

**A class has boys and girls in ratio 3:2. Total students = 250.**

Total parts:

5

One part:

50

Boys:

150

Girls:

100

## 17. CASELET — RATIO CHAIN

**A:B = 2:3 and B:C = 4:5. Total A+B+C = 700.**

Combine ratios:

A:B=8:12

B:C=12:15

Thus:

A:B:C=8:12:15

Total parts:

35

One part:

$700/35=20$

So:

A=160, B=240, C=300

## 18. CASELET — PERCENTAGE + REMAINING

**40% are engineers. Of the remaining people, 25% are managers. Rest are support staff. Total = 1000.**

Engineers:

$$\boxed{400}$$

Remaining:

$$\boxed{600}$$

Managers:

$$0.25(600)=150$$

Support:

$$600-150=450$$

So:

$$\boxed{400, 150, 450}$$

---

## 19. IMPORTANT “OF REMAINING” TRAP

Managers are:

$$25\% \text{ of } 600$$

not 25% of 1000.

So managers = 15% of total, not 25%.

---

## 20. MULTI-STAGE FUNNEL CASELET

Suppose:

- 2000 applicants
- 50% pass aptitude
- 60% of those pass coding
- 75% of those clear interview
- 80% of those accept offer

Stage counts:

Aptitude:

$$2000(0.5)=1000$$

Coding:

$$1000(0.6)=600$$

Interview:

$$600(0.75)=450$$

Joined:

$$450(0.8)=360$$

---

## 21. OVERALL CONVERSION RATE

$$\frac{360}{2000} \times 100 = \boxed{18\%}$$

Or directly:

$$\sqrt[0.5(0.6)(0.75)(0.8)]{}$$

$$\sqrt[=0.18]{}$$

---

## 22. REVERSE FUNNEL

If final joiners:

$$\sqrt[360]{}$$

and overall conversion:

$$\sqrt[18\%]{}$$

Applicants:

$$\sqrt[360/0.18]{}$$

$$\sqrt[=2000]{}$$

---

## 23. BUSINESS CASELET — REVENUE/COST/PROFIT

Company revenue = ₹ 800 lakh. Cost is 75% of revenue. Marketing is 20% of total cost.

Cost:

$$\sqrt[800(0.75)=600]{}$$

Profit:

$$\sqrt[800-600=200]{}$$

Marketing:

$$\sqrt[600(0.2)=120]{}$$

So:

$$\sqrt[\boxed{\text{Profit}=200, \text{Marketing}=120}]{}$$

---

## 24. MARKETING AS % OF REVENUE

$$\sqrt[120/800 \times 100 = \boxed{15\%}]{}$$

Nested percentage shortcut:

$$\sqrt[75\% \times 20\% = 15\%]{}$$

---

## 25. PROFIT MARGIN

$$\sqrt[200/800=25\%]{}$$

Since cost = 75% of revenue:

$$\sqrt[\boxed{\text{Margin}=25\%}]{}$$

No subtraction needed if complement obvious.

## 26. POPULATION CASELET

A city has 1,000,000 people. 60% are adults. Of adults, 55% are women. Of non-adults, 48% are girls.

Adults:

$$\boxed{600000}$$

Adult women:

$$\boxed{600000(0.55)=330000}$$

Non-adults:

$$\boxed{400000}$$

Girls:

$$\boxed{400000(0.48)=192000}$$

Total females:

$$\boxed{522000}$$

Overall female share:

$$\boxed{52.2\%}$$

## 27. MULTI-SOURCE DI

Sometimes question gives:

- graph
- small table
- footnote
- percentage condition

Treat each as one layer.

Recommended order:

1. identify base total
2. convert shares/rates into counts
3. derive secondary quantities
4. answer comparison

## 28. BASE-FIRST RULE

Whenever mixed DI contains percentages, first ask:

**Percentage of what?**

Possible bases:

- total sales
- selected candidates
- total cost
- department employees
- previous year's value
- remaining quantity

Wrong base = wrong answer.

## 29. DERIVED-TABLE STRATEGY

Suppose source gives:

Region	Revenue	Cost % of Revenue
A	500	80%
B	600	75%
C	800	70%

Create:

Region	Revenue	Cost	Profit
A	500	400	100
B	600	450	150
C	800	560	240

Then all later questions become easier.

## 30. PROFIT MARGINS

A:

$\boxed{20\%}$

B:

$\boxed{25\%}$

C:

$\boxed{30\%}$

Highest:

$\boxed{C}$

## 31. CONDITIONAL MULTI-SOURCE QUESTION

Which regions have revenue above 550 and margin at least 25%?

A fails revenue.

B qualifies.

C qualifies.

Answer:

$\boxed{B,C}$

## 32. TOTAL PROFIT OF QUALIFYING REGIONS

$150+240 = \boxed{390}$

## 33. CASELET WITH DIFFERENCE + RATIO

**A+B = 500 and A:B = 3:2.**

A:

$\boxed{300}$

B:

$\boxed{200}$

If C is 25% less than A:

$C=300(0.75)$

$=\boxed{225}$

## 34. CASELET WITH PERCENT MORE/LESS

**A is 25% more than B, and B = 160.**

$A=160(1.25)=200$

But if A=200 and asked B:

$B=\frac{200}{1.25}$

$=\boxed{160}$

Do not subtract 25% of A.

## 35. MULTI-STAGE CHANGE

Suppose sales:

$\boxed{1000}$

increase 20%, then decrease 10%.

Final:

$$\sqrt[1]{1000(1.2)(0.9)}$$

$$\sqrt[1]{=1080}$$

Net:

$$\sqrt[1]{\boxed{+8\%}}$$

Mixed DI often hides successive percentage changes in prose.

---

### 36. REVENUE + PROFIT + TAX CASELET

---

Suppose:

Revenue:

$$\sqrt[1]{1000}$$

Operating cost:

$$\sqrt[1]{700}$$

Profit before tax:

$$\sqrt[1]{300}$$

Tax = 20% of profit.

Tax:

$$\sqrt[1]{60}$$

Profit after tax:

$$\sqrt[1]{240}$$

After-tax margin:

$$\sqrt[1]{240/1000 = \boxed{24\%}}$$

---

### 37. TAX BASE TRAP

---

Tax is 20% of profit, not revenue.

Always identify the rate base.

---

### 38. BAR + PIE HYBRID

---

Suppose bar graph gives total annual sales:

2024:

$$\sqrt[1]{1000}$$

2025:

$$\sqrt[1]{1500}$$

---



Pie distributions:

## 2024

A 30%, B 25%, C 20%, D 25%

## 2025

A 25%, B 30%, C 25%, D 20%

A values:

2024:

\[ 300 \]

2025:

\[ 375 \]

Growth:

\[ 25\% \]

---

## 39. B VALUES

2024:

\[ 250 \]

2025:

\[ 450 \]

Growth:

\[ 80\% \]

---

## 40. SHARE VS ABSOLUTE TREND

A's share:

\[ 30\%to25\% \]

falls.

But A sales:

\[ 300to375 \]

rise.

Again, total growth dominates share decline.

---

## 41. LINE + TABLE HYBRID

Suppose line graph gives total market:

Year	Market
2023	1000
2024	1200
2025	1500

Table gives company share:

Year	Share
2023	20%
2024	25%
2025	24%

Company sales:

2023:

$\sqrt[2]{200}$

2024:

$\sqrt[3]{300}$

2025:

$\sqrt[4]{360}$

## 42. COMPANY SALES GROWTH

2023! 2025 :

$\sqrt[2]{200\text{to}360}$

$\sqrt[2]{\boxed{80\%}}$

Market grows:

$\sqrt[2]{1000\text{to}1500=50\%}$

Company grew faster than market, so its final share is higher than initial:

$\sqrt[2]{20\%\text{to}24\%}$

## 43. GROWTH CONTRIBUTION

Company increase:

$\sqrt[2]{160}$

Market increase:

$\sqrt[2]{500}$

Contribution to market growth:

$$\sqrt[1]{160/500 = \boxed{32\%}} \sqrt[1]{}]$$

---

## 44. TWO-DENOMINATOR TRAP

---

Company share:

$$\sqrt[1]{360/1500=24\%} \sqrt[1]{}]$$

Company contribution to market growth:

$$\sqrt[1]{160/500=32\%} \sqrt[1]{}]$$

Same company, different percentage because different denominator.

---

## 45. CASELET — AVERAGE + TOTAL

---

Average salary of 100 employees is ₹ 50 k. 20 new employees join with average salary ₹ 60 k.

Old total salary:

$$\sqrt[1]{100(50)=5000} \sqrt[1]{}]$$

New employee salary:

$$\sqrt[1]{20(60)=1200} \sqrt[1]{}]$$

Combined:

$$\sqrt[1]{6200} \sqrt[1]{}]$$

Employees:

$$\sqrt[1]{120} \sqrt[1]{}]$$

New average:

$$\sqrt[1]{6200/120} \sqrt[1]{}]$$

$$\sqrt[1]{=\boxed{51.67\text{k}}} \sqrt[1]{}]$$

---

## 46. CASELET — REPLACEMENT

---

Suppose average of 10 employees is 50.

One employee earning 40 leaves; replacement earns 70.

Total increases:

$$\sqrt[1]{30} \sqrt[1]{}]$$

Average increase:

$$\sqrt[1]{30/10=3} \sqrt[1]{}]$$

New average:

$$\sqrt[1]{\boxed{53}} \sqrt[1]{}]$$

---

## 47. MIXED DI WITH RATIOS

---

Suppose pie chart gives total department sizes and table gives male:female ratios.

Total 1000:

- A 40% !' 400
- B 30% !' 300
- C 30% !' 300

Ratios:

A:

$$\backslash [ M:F=3:2 \backslash ]$$

B:

$$\backslash [ 2:3 \backslash ]$$

C:

$$\backslash [ 1:1 \backslash ]$$

---

## 48. FEMALE COUNTS

---

A:

$$\backslash [ 400 \times \frac{2}{5} = 160 \backslash ]$$

B:

$$\backslash [ 300 \times \frac{3}{5} = 180 \backslash ]$$

C:

$$\backslash [ 300 \times \frac{1}{2} = 150 \backslash ]$$

Total females:

$$\backslash [ \boxed{490} \backslash ]$$

---

## 49. OVERALL FEMALE PERCENTAGE

---

$$\backslash [ 490/1000 = \boxed{49\%} \backslash ]$$

---

## 50. CASELET SOLVING TEMPLATE

---

For every caselet:

### Step 1

Underline totals.

### Step 2

Convert statements into formulas.

### Step 3

Create mini-table.

#### Step 4

Calculate only base quantities.

#### Step 5

Answer questions using table.

---

## 51. MIXED-DI SOURCE PRIORITY

---

Use sources in this order:

1. explicit absolute values
2. totals
3. percentages/shares
4. ratios
5. growth conditions
6. derived metrics

Why?

Because absolute values are usually the easiest anchor.

---

## 52. APPROXIMATION STRATEGY

---

Suppose:

$$\left[ \frac{298}{903} \times 100 \right]$$

Options:

- 21%
- 33%
- 46%
- 58%

Approximate:

$$\left[ 300/900 = 1/3 \right]$$

$$\left[ \boxed{33\%} \right]$$

Do not waste time on exact division.

---

## 53. CROSS-MULTIPLICATION IN MIXED DI

---

Compare:

$$\left[ \frac{150}{420} \right]$$

and:

$$\left[ \frac{180}{500} \right]$$

Cross:

---

$$\backslash [ 150(500)=75000 \backslash ]$$

$$\backslash [ 180(420)=75600 \backslash ]$$

Second is slightly larger.

No decimal calculations needed.

## 54. COMMON MIXED-DI TRAPS

### & pTRAP 1 — Different data sources, different denominators

Check each percentage base.

### & pTRAP 2 — Rate vs count

Highest rate may not mean highest count.

### & pTRAP 3 — Share vs actual value

Changing total can reverse intuition.

### & pTRAP 4 — “Of remaining”

New denominator.

### & pTRAP 5 — Sequential percentages

Multiply, do not simply add.

### & pTRAP 6 — Hybrid profit problems

Margin may be of revenue, profit% may be of cost.

### & pTRAP 7 — Caselet overload

Convert prose to a mini-table immediately.

## 55. MUST-MEMORIZE MIXED-DI RELATIONS

Weighted overall rate:

$$\backslash [ \boxed{ \frac{\sum \text{numerator}}{\sum \text{denominator}} } \backslash ]$$

Funnel conversion:

$$\backslash [ \boxed{ p_{\text{overall}} = p_1 p_2 p_3 \cdots } \backslash ]$$

Revenue-margin profit:

$$\backslash [ \boxed{ \text{Profit} = \text{Revenue} \times \text{Margin} } \backslash ]$$

Category from share:

$$\backslash [ \boxed{ \text{Value} = \text{Total} \times \text{Share} } \backslash ]$$

Contribution to growth:

$$\backslash [ \boxed{ \frac{\Delta \text{component}}{\Delta \text{total}} \times 100 } \backslash ]$$

## 56. FAST MIXED-DI DECISION TREE

**Text caselet?**

Convert to table.

**Pie + table?**

Turn share into counts, then apply table rate.

**Bar + line?**

Combine value with line-derived rate.

**Market + company share?**

$\text{Company} = \text{Market} \times \text{Share}$

**Funnel?**

Multiply stage rates.

**“Of remaining”?**

Change denominator.

**Unequal groups + percentages?**

Use weighted average.

**Options broad?**

Approximate aggressively.

---

**PRACTICE — LEVEL 0**

---

A company has 1000 employees.

- 40% Tech
- 30% Sales
- 20% HR
- rest Operations

Women percentages:

- Tech 25%
- Sales 40%
- HR 50%
- Operations 60%

**Q1**

Operations employees?

**Q2**

Women in Tech?

**Q3**

Women in Sales?

**Q4**

Total women?

**Q5**

Overall women percentage?

**LEVEL 1 — BUSINESS HYBRID**

Year	Revenue	Margin
2022	400	20%
2023	500	24%
2024	600	25%
2025	800	20%

**Q6**

Profit each year?

**Q7**

Highest profit?

**Q8**

Cost in 2024?

**Q9**

Profit growth 2022! 2025 ?

**Q10**

Average annual profit?

**LEVEL 2 — FUNNEL CASELET**

A company receives 5000 applications.

- 60% pass screening
- 50% of those pass test
- 40% of those clear interview
- 75% of those accept offer

**Q11**

How many pass screening?

**Q12**

How many pass test?

**Q13**

How many clear interview?



Q14

How many join?

Q15

Overall conversion rate?

LEVEL 3 — MIXED MARKET SET

Total market:

Year	Market Size
2023	1200
2024	1500
2025	1800

Company A shares:

[ 25%, 24%, 30% ]

Company B shares:

[ 20%, 25%, 25% ]

Q16

A sales each year?

Q17

B sales each year?

Q18

Which company has higher 2023-2025 percentage growth?

Q19

What is A's contribution to total market increase from 2023 to 2025?

Q20

A:B sales ratio in 2025?

ANSWERS

Q1

Known shares:

[ 40+30+20=90% ]

Operations:

[ 10% ]

$$\lfloor 1000(0.1) = \boxed{100} \rfloor$$

## Q2

---

Tech:

$$\lfloor 400 \rfloor$$

Women:

$$\lfloor 400(0.25) = \boxed{100} \rfloor$$

## Q3

---

Sales:

$$\lfloor 300(0.4) = \boxed{120} \rfloor$$

## Q4

---

HR women:

$$\lfloor 200(0.5)=100 \rfloor$$

Operations women:

$$\lfloor 100(0.6)=60 \rfloor$$

Total:

$$\lfloor 100+120+100+60 \rfloor$$

$$\lfloor =\boxed{380} \rfloor$$

## Q5

---

$$\lfloor 380/1000 = \boxed{38\%} \rfloor$$

---

## Q6

---

2022:

$$\lfloor 400(0.2)=80 \rfloor$$

2023:

$$\lfloor 500(0.24)=120 \rfloor$$

2024:

$$\lfloor 600(0.25)=150 \rfloor$$

2025:

$$\lfloor 800(0.2)=160 \rfloor$$

$$\lfloor \boxed{80, \ 120, \ 150, \ 160} \rfloor$$

## Q7

---

$$\lfloor \boxed{2025} \rfloor$$

### Q8

2024 cost:

$$\boxed{600-150 = 450}$$

### Q9

$$\boxed{80 \text{ to } 160}$$

$$\boxed{= 100\%}$$

### Q10

$$\boxed{(80+120+150+160)/4}$$

$$\boxed{= 510/4}$$

$$\boxed{= 127.5}$$

### Q11

$$\boxed{5000(0.6) = 3000}$$

### Q12

$$\boxed{3000(0.5) = 1500}$$

### Q13

$$\boxed{1500(0.4) = 600}$$

### Q14

$$\boxed{600(0.75) = 450}$$

### Q15

$$\boxed{450/5000 = 9\%}$$

Or:

$$\boxed{0.6(0.5)(0.4)(0.75) = 0.09}$$

### Q16

A:

2023:

$$\boxed{1200(0.25)=300}$$

2024:

$$\boxed{1500(0.24)=360}$$

2025:

$$\backslash [ 1800(0.30)=540 \backslash ]$$

$$\backslash [ \boxed{300, \ 360, \ 540} \backslash ]$$

## Q17

---

B:

2023:

$$\backslash [ 1200(0.20)=240 \backslash ]$$

2024:

$$\backslash [ 1500(0.25)=375 \backslash ]$$

2025:

$$\backslash [ 1800(0.25)=450 \backslash ]$$

$$\backslash [ \boxed{240, \ 375, \ 450} \backslash ]$$

## Q18

---

A:

$$\backslash [ 300\text{to}540 \backslash ]$$

Growth:

$$\backslash [ 240/300=80\% \backslash ]$$

B:

$$\backslash [ 240\text{to}450 \backslash ]$$

Growth:

$$\backslash [ 210/240 = 87.5\% \backslash ]$$

So:

$$\backslash [ \boxed{B} \backslash ]$$

## Q19

---

Market increase:

$$\backslash [ 1800-1200=600 \backslash ]$$

A increase:

$$\backslash [ 540-300=240 \backslash ]$$

Contribution:

$$\backslash [ 240/600 = \boxed{40\%} \backslash ]$$

## Q20

---

$$\backslash [ 540:450 \backslash ]$$

$$\backslash [ =\boxed{6:5} \backslash ]$$

---

## CHAPTER 55 STATUS

---

Covered so far:

- mixed DI concept '
- text caselets '
- converting prose to tables '
- bar + line hybrids '
- pie + table hybrids '
- rate vs count '
- multi-stage funnels '
- nested percentages '
- business caselets '
- population caselets '
- ratio chains '
- revenue/cost/profit caselets '
- market-share hybrids '
- conditional multi-source questions '
- weighted overall rates '
- approximation and cross-multiplication '
- L0–L3 practice '

But **Chapter 55 is not complete yet.**

Still remaining:

- larger multi-source caselets
- caselets with hidden variables
- conditional allocation
- overlapping constraints
- multi-year business sets
- table + line + percentage hybrids
- advanced placement-style 5-question sets
- GATE-style graphical inference
- final mixed-DI speed audit

## Progress

---

$\boxed{54/76}$  text{ completed} \]

**Current:** 0% Chapter 55 — Mixed Graphs & Caselets

## NEXT

---

0% Chapter 55 Part 2 — Hidden-Variable Caselets, Multi-Source Advanced DI, Conditional Allocation, Complex Business Sets & Final Mixed-DI Toolkit

---

USER

next

ASSISTANT

## CHAPTER 55 — MIXED GRAPHS & CASELETS

### PART 2 — HIDDEN-VARIABLE CASELETS, CONDITIONAL ALLOCATION, MULTI-SOURCE ADVANCED DI & FINAL MIXED-DI TOOLKIT

Status:  $\emptyset=0\%$  PROGRESS GATE: High Placements: Very High Priority:  $\emptyset=0\%$  1

This is the most realistic DI style for placement tests.

Instead of giving everything directly, the question may give:

- totals
- ratios
- percentages
- differences
- conditions
- missing values

and force you to reconstruct the dataset.

The core rule is:

**Translate every sentence into a number, equation, ratio, or constraint before solving.**

#### 1. HIDDEN-VARIABLE CASELET

Suppose:

**A company has 800 employees across departments A, B and C. A has 50 more employees than B. C has twice as many employees as B.**

Let:

$$\backslash [ B=x \backslash ]$$

Then:

$$\backslash [ A=x+50 \backslash ]$$

$$\backslash [ C=2x \backslash ]$$

Total:

$$\backslash [ x+(x+50)+2x=800 \backslash ]$$

$$\backslash [ 4x+50=800 \backslash ]$$

$$\backslash [ 4x=750 \backslash ]$$

$$\lfloor x=187.5 \rfloor$$

This gives fractional employees, so the data is internally unrealistic.

### Important DI skill

Sometimes a set may reveal inconsistent information.

Do not force an integer answer.

---

## 2. CLEAN VERSION

Suppose total = 850 instead.

$$\lfloor x+(x+50)+2x=850 \rfloor$$

$$\lfloor 4x+50=850 \rfloor$$

$$\lfloor x=200 \rfloor$$

Therefore:

$$\lfloor B=200 \rfloor$$

$$\lfloor A=250 \rfloor$$

$$\lfloor C=400 \rfloor$$

---

## 3. WHY DEFINE THE SIMPLEST VARIABLE?

If relationships are:

- $A = B + 50$
- $C = 2B$

choose:

$$\lfloor B=x \rfloor$$

because both other values depend directly on B.

This keeps algebra short.

---

## 4. CASELET WITH PERCENTAGE RELATION

**A has 20% more employees than B. C has 25% fewer than B. Total = 590.**

Let:

$$\lfloor B=x \rfloor$$

Then:

$$\lfloor A=1.2x \rfloor$$

$$\lfloor C=0.75x \rfloor$$

Total:

---

$$\backslash[ 1.2x+x+0.75x=590 \backslash]$$

$$\backslash[ 2.95x=590 \backslash]$$

$$\backslash[ x=200 \backslash]$$

Thus:

$$\backslash[ A=240,\quad B=200,\quad C=150 \backslash]$$

## 5. FRACTION METHOD — FASTER

Instead of decimals:

$$\backslash[ A:B=6:5 \backslash]$$

because A is 20% more.

And:

$$\backslash[ C:B=3:4 \backslash]$$

because C is 25% less than B.

Make B common:

$$\backslash[ A:B=24:20 \backslash]$$

$$\backslash[ B:C=20:15 \backslash]$$

Therefore:

$$\backslash[ A:B:C=24:20:15 \backslash]$$

Total parts:

$$\backslash[ 59 \backslash]$$

Since total = 590:

$$\backslash[ 1\text{part}=10 \backslash]$$

So:

$$\backslash[ \boxed{240,200,150} \backslash]$$

Often faster than algebra.

## 6. HIDDEN TOTAL FROM RATIOS

**A:B:C = 2:3:5. C exceeds A by 180.**

Difference in ratio parts:

$$\backslash[ 5-2=3 \backslash]$$

So:

$$\backslash[ 3\text{parts}=180 \backslash]$$

$$\backslash[ 1\text{part}=60 \backslash]$$

Total:



$$\lfloor (2+3+5)(60) \rfloor$$

$$\lfloor \boxed{600} \rfloor$$

## 7. CONDITIONAL ALLOCATION

Suppose 1000 students choose electives.

- 40% choose AI
- 30% choose DS
- remaining choose Cybersecurity

Then:

AI:

$$\lfloor 400 \rfloor$$

DS:

$$\lfloor 300 \rfloor$$

Cybersecurity:

$$\lfloor 300 \rfloor$$

Now suppose:

**25% of AI students and 40% of DS students are women; 50% of Cybersecurity students are women.**

Women:

AI:

$$\lfloor 100 \rfloor$$

DS:

$$\lfloor 120 \rfloor$$

Cyber:

$$\lfloor 150 \rfloor$$

Total:

$$\lfloor \boxed{370} \rfloor$$

## 8. CONDITIONAL ALLOCATION WITH REMAINING

Suppose total budget:

$$\lfloor 110 \rfloor \text{ t e a d n } \lfloor \rfloor$$

- 30% to equipment
- 20% of remaining to travel
- rest to operations

Equipment:

$$\lfloor 3 \rfloor$$

Remaining:

$$\boxed{7}$$

Travel:

$$\boxed{20\% \times 7 = 1.4}$$

Operations:

$$\boxed{7 - 1.4 = 5.6}$$

lakh.

---

## 9. COMMON TRAP — 20% OF REMAINING

Travel is:

$$\boxed{1.4/10 = 14\%}$$

of original total.

Not 20%.

---

## 10. MULTI-SOURCE ADVANCED SET

Suppose a table gives total sales:

Region	Sales
North	600
South	800
East	500
West	900

A line gives profit margins:

Region	Margin
North	20%
South	25%
East	30%
West	15%

A pie gives online-sales shares:

Region	Online Share
North	40%
South	35%

Region	Online Share
East	50%
West	30%

Now three different derived metrics exist.

## 11. PROFIT

North:

$$\backslash [ 600(0.20)=120 \backslash ]$$

South:

$$\backslash [ 800(0.25)=200 \backslash ]$$

East:

$$\backslash [ 500(0.30)=150 \backslash ]$$

West:

$$\backslash [ 900(0.15)=135 \backslash ]$$

Highest profit:

$$\backslash [ \boxed{\text{South}} \backslash ]$$

## 12. ONLINE SALES

North:

$$\backslash [ 600(0.4)=240 \backslash ]$$

South:

$$\backslash [ 800(0.35)=280 \backslash ]$$

East:

$$\backslash [ 500(0.5)=250 \backslash ]$$

West:

$$\backslash [ 900(0.3)=270 \backslash ]$$

Highest online sales:

$$\backslash [ \boxed{\text{South}} \backslash ]$$

## 13. ONLINE SALES AS % OF TOTAL ALL-REGION SALES

Total sales:

$$\backslash [ 600+800+500+900=2800 \backslash ]$$

Total online:

$$\lfloor 240+280+250+270=1040 \rfloor$$

Overall online share:

$$\lfloor \frac{1040}{2800} \times 100 \rfloor$$

$$\lfloor =\boxed{37.14\%} \rfloor$$

Do not average:

$$\lfloor 40,35,50,30 \rfloor$$

directly.

## 14. PROFIT PER UNIT OF ONLINE SALES

This is a derived metric:

North:

$$\lfloor 120/240=0.5 \rfloor$$

South:

$$\lfloor 200/280 \approx 0.714 \rfloor$$

East:

$$\lfloor 150/250=0.6 \rfloor$$

West:

$$\lfloor 135/270=0.5 \rfloor$$

Highest:

$$\lfloor \boxed{\text{South}} \rfloor$$

## 15. DENSE HYBRID — REVENUE, COST %, EXPORT %

Suppose:

Company	Revenue	Cost % of Revenue	Export % of Revenue
A	500	80%	30%
B	800	75%	25%
C	600	70%	40%
D	900	85%	20%

## 16. PROFITS

A:

$$\lfloor 500(1-0.8)=100 \rfloor$$

B:

$$\sqrt{800(0.25)=200}$$

C:

$$\sqrt{600(0.3)=180}$$

D:

$$\sqrt{900(0.15)=135}$$

Highest:

$$\sqrt{\boxed{B}}$$

---

## 17. EXPORT REVENUE

A:

$$\sqrt{150}$$

B:

$$\sqrt{200}$$

C:

$$\sqrt{240}$$

D:

$$\sqrt{180}$$

Highest:

$$\sqrt{\boxed{C}}$$

Different winner again.

---

## 18. EXPORT REVENUE OF PROFITABLE FILTER

Question:

**Total export revenue of companies with profit margin at least 25%.**

Margins:

A 20% 'L B 25% ' C 30% ' D 15% 'L

Exports:

$$\sqrt{200+240}$$

$$\sqrt{= \boxed{440}}$$

---

## 19. CONDITIONAL QUESTIONS — CHEAPEST FILTER FIRST

If question says:

Revenue > 700 and margin "e 2 5 % .

First check Revenue >700:

Only B and D.

Then margin:

$B = 25\%$  '  $D = 15\%$  'L

Answer:

$\boxed{B}$

Do not calculate margins for every row unless needed.

## 20. OVERLAPPING CONSTRAINTS

Suppose:

- 60% students know Python
- 50% know C
- 30% know both
- total = 1000

Using inclusion-exclusion:

At least one:

$60+50-30=80\%$

$\boxed{800}$

Only Python:

$60-30=30\%$

$=300$

Only C:

$50-30=20\%$

$=200$

Neither:

$20\%$

$=200$

## 21. CASELET + VENN COMBINATION

Mixed DI may hide a set problem inside a paragraph.

When keywords appear:

- both
- either
- neither

- only
- at least one

switch to set/Venn logic.

---

## 22. CONDITIONAL DISTRIBUTION WITH OVERLAP

---

Suppose 1000 employees:

- 500 know Python
- 400 know SQL
- 200 know both

Python only:

$$\setminus [ 300 \setminus ]$$

SQL only:

$$\setminus [ 200 \setminus ]$$

At least one:

$$\setminus [ 700 \setminus ]$$

Neither:

$$\setminus [ 300 \setminus ]$$

If 60% of "both" are senior employees:

$$\setminus [ 0.6(200) = \boxed{120} \setminus ]$$


---

## 23. MULTI-YEAR BUSINESS CASELET

---

Suppose company revenue:

2023:

$$\setminus [ 500 \setminus ]$$

2024 grows 20%.

2025 grows 25% over 2024.

2026 declines 10% from 2025.

Values:

2024:

$$\setminus [ 500(1.2)=600 \setminus ]$$

2025:

$$\setminus [ 600(1.25)=750 \setminus ]$$

2026:

$$\setminus [ 750(0.9)=675 \setminus ]$$


---

## 24. OVERALL GROWTH

---

2023 to 2026:

[ 500 to 675 ]

Increase:

[ 175 ]

Percentage:

[  $175/500 = \boxed{35\%}$  ]

Do not add:

[  $20+25-10=35$  ]

and assume coincidence. Here it happens to equal 35 numerically only by chance? Let's verify:  $1.21.25.9=1.35$  yes, indeed 35%. But successive percentage arithmetic should still use multipliers.

---

## 25. MARGIN CHANGES TOO

---

Suppose profit margin:

- 2023 = 20%
- 2024 = 25%
- 2025 = 24%
- 2026 = 30%

Profits:

2023:

[ 100 ]

2024:

[ 150 ]

2025:

[ 180 ]

2026:

[ 202.5 ]

Highest:

[  $\boxed{2026}$  ]

---

## 26. PROFIT GROWTH VS REVENUE GROWTH

---

Revenue:

[ 500 to 675 ]

Growth:

[ 35% ]

---



Profit:

[ 100\to202.5 ]

Growth:

[ 102.5\% ]

Margin expansion amplifies profit growth.

## 27. TABLE + LINE + PERCENTAGE HYBRID

Suppose:

Year	Total Revenue	Employee Count
2023	600	100
2024	750	120
2025	900	150

Line gives profit margin:

[ 20\%,24\%,30\% ]

Derived:

Profit:

[ 120,\ 180,\ 270 ]

Revenue per employee:

[ 6,\ 6.25,\ 6 ]

Profit per employee:

[ 1.2,\ 1.5,\ 1.8 ]

## 28. RANKING BY DIFFERENT METRICS

2025 has:

- highest revenue
- highest profit
- highest profit per employee

But revenue per employee is highest in 2024:

[ 6.25 ]

Again: metric matters.

## 29. HIDDEN VARIABLE FROM AVERAGE

Suppose four branches have average revenue:

$$\boxed{500}$$

Three revenues:

$$\boxed{400,450,550}$$

Fourth =  $(x)$ .

Total needed:

$$\boxed{4(500)=2000}$$

Known:

$$\boxed{1400}$$

So:

$$\boxed{x=600}$$

$$\boxed{600}$$

---

### 30. HIDDEN VARIABLE FROM WEIGHTED AVERAGE

Branch A:

100 employees, average salary 40.

Branch B:

200 employees, average salary  $(x)$ .

Overall:

300 employees, average salary 50.

Equation:

$$\boxed{100(40)+200x=300(50)}$$

$$\boxed{4000+200x=15000}$$

$$\boxed{200x=11000}$$

$$\boxed{x=55}$$

---

### 31. HIDDEN VARIABLE FROM GROWTH

2024 sales =  $(x)$ .

2025 sales = 360.

Growth = 20%.

$$\boxed{1.2x=360}$$

$$\boxed{x=300}$$

---

### 32. HIDDEN VARIABLE FROM PROFIT MARGIN

Revenue:

$$\boxed{800}$$

Margin:

$$\boxed{25\%}$$

Profit:

$$\boxed{200}$$

Cost:

$$\boxed{600}$$

If cost given instead, reverse similarly.

---

### 33. HIDDEN VARIABLE FROM OVERALL RATE

---

Two plants:

A produces 1000 units at 2% defects.

B produces 2000 units at  $(x\%)$  defects.

Overall defect rate = 3%.

Total defects:

$$3000(0.03)=90$$

A defects:

$$20$$

B defects:

$$70$$

Thus:

$$2000(x/100)=70$$

$$x=3.5\%$$

$$\boxed{3.5\%}$$

---

### 34. CONDITIONAL ALLOCATION WITH CAPACITY

---

Suppose three plants have capacities:

A 400 B 500 C 600

Demand = 1200.

A must supply 80% of capacity:

$$320$$

B supplies 60%:

$$300$$

Remaining for C:

$$1200-620 = 580$$

---

C utilization:

$$\lfloor 580/600 = \boxed{96.67\%} \rfloor$$

---

### 35. CAPACITY CONSTRAINT TRAP

---

If remaining demand exceeds C capacity, allocation is impossible.

Always sanity-check feasibility.

---

### 36. CASELET WITH MINIMUM/MAXIMUM CONSTRAINT

---

Suppose:

$$A+B+C = 1000.$$

$$A \leq 300 \quad B \leq 200 \quad C \leq 100$$

Minimum committed:

$$\lfloor 600 \rfloor$$

Remaining flexibility:

$$\lfloor 400 \rfloor$$

Without further information, individual exact values cannot be uniquely determined.

Answer may be:

$$\lfloor \boxed{\text{insufficient information}} \rfloor$$

---

### 37. UNIQUE VS NON-UNIQUE SOLUTION

---

This is important in advanced caselets.

Enough equations must exist to uniquely determine unknowns.

For 3 unknowns, usually need 3 independent relationships.

---

### 38. REDUNDANT CONDITION

---

If:

$$\lfloor A+B=500 \rfloor$$

$$\lfloor A:B=3:2 \rfloor$$

then:

$$\lfloor A=300, B=200 \rfloor$$

A third statement:

$$\lfloor A-B=100 \rfloor$$

adds no new information.

It's redundant.

---

---

## 39. CONTRADICTORY CONDITION

---

If same set also says:

$\forall [A-B=120]$

then conditions conflict.

Do not "average" them.

Recognize inconsistency.

---

## 40. GATE-STYLE GRAPHICAL INFERENCE

---

Suppose a chart shows:

- A rises every year
- B falls every year
- A crosses B once

Valid conclusions:

- A's observed values increase monotonically
- B's observed values decrease monotonically
- relative ranking changes once

Invalid conclusions without calculation:

- A has constant percentage growth
  - B's total decline is greater than A's increase
  - intersection occurs exactly halfway
- 

## 41. QUALITATIVE INFERENCE QUESTIONS

---

GATE may test statements such as:

- "must be true"
- "could be true"
- "cannot be inferred"

Rule:

**Do not infer more than the data proves.**

---

## 42. EXAMPLE

---

Graph shows revenue increased every year.

Can we conclude profit increased every year?

$\boxed{\text{No}}$

Profit also depends on cost/margin.

---

---

## 43. ANOTHER EXAMPLE

Graph shows market share fell from 30% to 25%.

Can we conclude company sales fell?

$\boxed{\text{No}}$

Need market totals.

---

## 44. PLACEMENT-STYLE ADVANCED SET

Suppose four branches:

Branch	Applicants	Selected %	Joined % of Selected	Avg Salary <sup>1</sup> L
A	1000	20%	80%	6
B	1200	25%	60%	7
C	800	30%	75%	8
D	1500	16%	90%	5

---

## 45. SELECTED COUNTS

A:

$\boxed{200}$

B:

$\boxed{300}$

C:

$\boxed{240}$

D:

$\boxed{240}$

---

## 46. JOINED COUNTS

A:

$\boxed{200(0.8)=160}$

B:

$\boxed{300(0.6)=180}$

C:

$\boxed{240(0.75)=180}$

---

D:

$$\lfloor 240(0.9)=216 \rfloor$$

Highest:

$$\lfloor \boxed{D} \rfloor$$

---

## 47. OVERALL JOIN RATE FROM APPLICANTS

---

Total applicants:

$$\lfloor 4500 \rfloor$$

Total joined:

$$\lfloor 160+180+180+216=736 \rfloor$$

Rate:

$$\lfloor \frac{736}{4500} \times 100 \rfloor$$

$$\lfloor \approx \boxed{16.36\%} \rfloor$$

---

## 48. SALARY BILL OF JOINERS

---

A:

$$\lfloor 160(6)=960 \rfloor$$

B:

$$\lfloor 180(7)=1260 \rfloor$$

C:

$$\lfloor 180(8)=1440 \rfloor$$

D:

$$\lfloor 216(5)=1080 \rfloor$$

Highest:

$$\lfloor \boxed{C} \rfloor$$

Despite D having most joiners.

---

## 49. WEIGHTED AVERAGE SALARY

---

Total salary bill:

$$\lfloor 960+1260+1440+1080 = 4740 \rfloor$$

Total joined:

$$\lfloor 736 \rfloor$$

Average:

$$\lfloor 4740/736 \rfloor$$

---

$\approx 6.44 \text{ lakh}$

---

## 50. BRANCH CONTRIBUTION TO TOTAL JOINERS

---

D:

$\frac{216}{736}$

$\approx 29.35\%$

---

## 51. ADVANCED FILTER

---

Question:

Among branches with selected rate "e 2 0 % which has highest joined count?

A, B, C qualify.

Joined:

160, 180, 180

Answer:

$B \text{ and } C$

---

## 52. FAST CASELET SOLVING ALGORITHM

---

### STEP 1 — Extract anchors

Look for:

- total
- base year
- absolute number
- average
- ratio
- percentage

### STEP 2 — Create variables only for unknowns

Avoid excessive symbols.

### STEP 3 — Convert prose to equations

### STEP 4 — Build small table

### STEP 5 — Check consistency

### STEP 6 — Solve easiest questions first

---

## 53. TIME-SAVING RULE — NORMALIZE RATIOS

---



If ratios are messy:

$$\backslash[ A:B=0.4:0.6 \backslash]$$

convert:

$$\backslash[ 4:6=2:3 \backslash]$$

If:

$$\backslash[ 25\%:40\% \backslash]$$

use:

$$\backslash[ 25:40=5:8 \backslash]$$

Do not carry decimals unnecessarily.

---

## 54. TIME-SAVING RULE — USE MULTIPLIERS

---

Instead of repeated percentage arithmetic:

+20%:

$$\backslash[ \times 1.2 \backslash]$$

" 15 % :

$$\backslash[ \times 0.85 \backslash]$$

+25% then " 20 % :

$$\backslash[ 1.25 \times 0.8 = 1 \backslash]$$

Net zero.

---

## 55. TIME-SAVING RULE — FILTER BEFORE DERIVING

---

If question asks:

**branches with applicants >1000 and join rate >15%.**

First eliminate applicants "d 1 0 0 0 .

Then calculate rates only for survivors.

---

## 56. TIME-SAVING RULE — USE TOTAL DIFFERENCE

---

If:

$$\backslash[ A+B+C+D=1000 \backslash]$$

and three values known, missing fourth:

$$\backslash[ \boxed{1000 - \text{known sum}} \backslash]$$

Don't create unnecessary equations.

## 57. TIME-SAVING RULE — CHECK WHETHER TOTAL CANCELS

---

Same pie/chart ratio:

$$\left[ \frac{p_{AT}}{p_{BT}} = \frac{p_A}{p_B} \right]$$

No need to compute actual values.

Same for same-base rates.

---

## 58. COMMON ADVANCED MIXED-DI TRAPS

---

### & pTRAP 1 — Hidden variable chosen poorly

Choose the simplest base variable.

### & pTRAP 2 — Percentage relationship direction

“A is 20% more than B” means:

$$\left[ A = 1.2B \right]$$

not  $(B = 1.2A)$ .

### & pTRAP 3 — Insufficient information

Not every set has a unique numeric answer.

### & pTRAP 4 — Contradictory information

Check equations for consistency.

### & pTRAP 5 — Weighted overall rate

Use total numerator / total denominator.

### & pTRAP 6 — Conditional filter

Filter on correct metric.

### & pTRAP 7 — Qualitative inference

Do not infer causation or hidden quantities.

---

## 59. MUST-MEMORIZE ADVANCED RELATIONS

---

Overall weighted rate:

$$\left[ \boxed{\frac{\sum \text{successful outcomes}}{\sum \text{base population}}} \right]$$

Successive stages:

$$\left[ \boxed{N_{\text{final}} = N_0 p_1 p_2 \cdots} \right]$$

Ratio-chain principle:

$$\left[ \boxed{\text{make common middle term}} \right]$$

Profit:

$$\left[ \boxed{P = R - C} \right]$$

Margin:

---

$\boxed{\frac{P}{R}}$

Reverse percentage:

$\boxed{\text{old} = \frac{\text{new}}{1 \pm p}}$

---

## 60. FINAL MIXED-DI DECISION TREE

---

### Paragraph only?

Create mini-table.

### Missing values?

Choose variables.

### Ratios?

Normalize and chain.

### “Of remaining”?

Update base.

### Several percentages?

Multiply stage factors.

### Graph + rate table?

Convert rates to counts.

### Business data?

Derive profit/cost/margin.

### Multi-condition?

Filter cheapest condition first.

### Could be inconsistent?

Check totals/ranges.

### Qualitative inference?

State only what is guaranteed.

---

## PRACTICE — LEVEL 0

---

A company has 900 employees.

- A:B:C = 2:3:4

### Q1

Employees in A?

### Q2

Employees in B?

---

### Q3

Employees in C?

### Q4

B is what percentage of total?

### Q5

C is what percent more than A?

---

## LEVEL 1

---

A company has 1200 applicants.

- 50% pass Stage 1
- 60% of those pass Stage 2
- 75% of those pass Stage 3

### Q6

Stage 1 passers?

### Q7

Stage 2 passers?

### Q8

Stage 3 passers?

### Q9

Overall conversion rate?

### Q10

How many fail somewhere before Stage 3?

---

## LEVEL 2

---

Branch	Revenue	Cost %	Export %
A	500	80%	20%
B	700	70%	30%
C	900	75%	25%
D	600	85%	40%

### Q11

Profit of each?

### Q12

Highest profit margin?

---

**Q13**

Highest export revenue?

**Q14**

Total profit?

**Q15**

Export revenue of branches with margin at least 25%?

## LEVEL 3 — EXAM PRESSURE

Applicants:

Branch	Applicants	Selected %	Joined % of Selected
P	1000	30%	80%
Q	1500	20%	70%
R	1200	25%	75%
S	800	40%	60%

**Q16**

Selected count in each branch?

**Q17**

Joined count in each branch?

**Q18**

Which branch has highest overall application conversion rate?

**Q19**

Overall application conversion rate across all branches?

**Q20**

What percentage of total joiners comes from branch P?

## ANSWERS

### Q1–Q3

Ratio total:

$$\backslash 2+3+4=9 \backslash$$

One part:

$$\backslash 900/9=100 \backslash$$

Therefore:

$$\boxed{A=200, B=300, C=400}$$

#### Q4

$$300/900 = \boxed{33.33\%}$$

#### Q5

$$\frac{400-200}{200} \times 100$$

$$= \boxed{100\%}$$

#### Q6

$$1200(0.5) = \boxed{600}$$

#### Q7

$$600(0.6) = \boxed{360}$$

#### Q8

$$360(0.75) = \boxed{270}$$

#### Q9

$$270/1200 = \boxed{22.5\%}$$

or:

$$0.5(0.6)(0.75) = 0.225$$

#### Q10

$$1200-270 = \boxed{930}$$

#### Q11

Margins:

A:

$$20\% \rightarrow 100$$

B:

$$30\% \rightarrow 210$$

C:

$$25\% \rightarrow 225$$

D:

$$15\% \rightarrow 90$$

$$\boxed{100, 210, 225, 90}$$

## Q12

---

Highest margin:

$$\boxed{B, 30\%}$$

## Q13

---

Exports:

A:

$$100$$

B:

$$210$$

C:

$$225$$

D:

$$240$$

Highest:

$$\boxed{D}$$

## Q14

---

$$100+210+225+90 = \boxed{625}$$

## Q15

---

Margins at least 25%:

B and C.

Export revenue:

$$210+225 = \boxed{435}$$

## Q16

---

P:

$$1000(0.3)=300$$

Q:

$$1500(0.2)=300$$

R:

$$1200(0.25)=300$$

S:

$$800(0.4)=320$$

$$\boxed{300, 300, 300, 320}$$

### Q17

---

P:

$$\backslash[ 300(0.8)=240 \backslash]$$

Q:

$$\backslash[ 300(0.7)=210 \backslash]$$

R:

$$\backslash[ 300(0.75)=225 \backslash]$$

S:

$$\backslash[ 320(0.6)=192 \backslash]$$

$$\backslash[ \boxed{240, \ 210, \ 225, \ 192} \backslash]$$

### Q18

---

Overall conversion = selected rate  $\times$  joined rate.

P:

$$\backslash[ 0.30(0.80)=24\% \backslash]$$

Q:

$$\backslash[ 0.20(0.70)=14\% \backslash]$$

R:

$$\backslash[ 0.25(0.75)=18.75\% \backslash]$$

S:

$$\backslash[ 0.40(0.60)=24\% \backslash]$$

Highest:

$$\backslash[ \boxed{P\text{ and }S} \backslash]$$

### Q19

---

Total applicants:

$$\backslash[ 1000+1500+1200+800=4500 \backslash]$$

Total joined:

$$\backslash[ 240+210+225+192=867 \backslash]$$

Overall:

$$\backslash[ \frac{867}{4500}\times 100 \backslash]$$

$$\backslash[ \approx \boxed{19.27\%} \backslash]$$

### Q20

---

$$\backslash[ \frac{240}{867}\times 100 \backslash]$$

$$\backslash[ \approx \boxed{27.68\%} \backslash]$$



---

## ' CHAPTER 55 COMPLETE

---

We have now covered:

- mixed graphs '
- bar + line hybrids '
- pie + table hybrids '
- line + table hybrids '
- text caselets '
- converting prose to equations/tables '
- hidden variables '
- ratio chains '
- conditional allocation '
- "remaining" percentages '
- multi-stage funnels '
- business caselets '
- population caselets '
- profit/cost/revenue hybrids '
- multi-source data '
- overlapping constraints '
- Venn-style caselets '
- weighted rates '
- hidden-value reconstruction '
- consistency/contradiction checks '
- qualitative graphical inference '
- conditional filtering '
- placement-style advanced sets '
- GATE-style interpretation traps '
- L0–L3 practice '

## Progress

---

$\boxed{55/76}$  completed

## UNIT 2 — DATA INTERPRETATION

- Chapter 50 — DI Foundations '
- Chapter 51 — Tables '
- Chapter 52 — Bar & Stacked-Bar Graphs '
- Chapter 53 — Line Graphs '
- Chapter 54 — Pie Charts '
- Chapter 55 — Mixed Graphs & Caselets '
- Chapter 56 — Missing-Data & Advanced DI +

Only **one chapter remains in Unit 2.**

NEXT

Chapter 56 — Missing-Data & Advanced DI

We will cover:

- missing cells
- missing graph values
- algebraic reconstruction
- missing percentages
- reverse averages
- missing ratios
- partially specified tables
- approximation under incomplete data
- data sufficiency in DI
- range/bounds questions
- consistency checks
- advanced GATE-style and placement-style DI sets

After Chapter 56:

UNIT 2 — DATA INTERPRETATION COMPLETE

USER  
next

ASSISTANT

CHAPTER 56 — MISSING-DATA & ADVANCED DI

PART 1 — MISSING CELLS, REVERSE AVERAGES, MISSING PERCENTAGES, RATIOS, PARTIAL TABLES, BOUNDS & DATA SUFFICIENCY

Status: ~~0=Not~~ PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

This chapter teaches you how to solve DI when the data is **incomplete**.

The main skill is:

Use whatever remains known — total, average, ratio, percentage, growth, or constraints — to reconstruct only the missing quantity you need.

1. MISSING CELL FROM ROW TOTAL

Suppose:

Product	Q1	Q2	Q3	Q4	Total
A	120	150	?	180	620

Missing:

$$620 - (120 + 150 + 180)$$

$$= 620 - 450$$

$$= \boxed{170}$$

## 2. MISSING CELL FROM COLUMN TOTAL

Dept	Men	Women	Total
A	80	120	200
B	100	?	250
C	120	80	200

For B:

$$100 + x = 250$$

$$x = \boxed{150}$$

## 3. MISSING CELL FROM GRAND TOTAL

Suppose row totals are:

$$200, 250, ?, 300$$

Grand total:

$$1000$$

Missing row:

$$1000 - (200 + 250 + 300)$$

$$= \boxed{250}$$

## 4. MISSING VALUE FROM AVERAGE

Suppose five values have average:

$$80$$

Total:

$$5(80) = 400$$

Known four values:

$$60, 70, 90, 100$$

Sum:

$$\boxed{320}$$

Missing:

$$400 - 320$$

$$\boxed{80}$$

---

## 5. REVERSE AVERAGE

Formula:

$$\boxed{\text{Missing} = n(\text{Average}) - \text{sum of known values}}$$

This is one of the most common missing-data formulas.

---

## 6. MISSING GROUP AVERAGE

Suppose:

- Group A: 40 students, avg = 60
- Group B: 60 students, avg =  $x$
- Combined avg = 72

Equation:

$$40(60) + 60x = 100(72)$$

$$2400 + 60x = 7200$$

$$60x = 4800$$

$$\boxed{x = 80}$$

---

## 7. MISSING GROUP SIZE

Suppose:

- Group A: 40 people, avg 50
- Group B:  $n$  people, avg 70
- Combined avg 60

$$40(50) + 70n = 60(40 + n)$$

$$2000 + 70n = 2400 + 60n$$

$$10n = 400$$

$$\boxed{n = 40}$$

---

## 8. MISSING VALUE FROM PERCENTAGE SHARE

Suppose A is 25% of total and A = 150.

$$0.25T = 150$$

$$\boxed{600}$$

---

## 9. MISSING CATEGORY FROM PERCENT DISTRIBUTION

---

Shares:

- A = 20%
- B = 30%
- C = 25%
- D = ?

$$20+30+25=75$$

$$D=\boxed{25\%}$$

---

## 10. MISSING ACTUAL COUNT

---

Total:

$$800$$

Missing category share:

$$25\%$$

Value:

$$800(0.25) = \boxed{200}$$

---

## 11. MISSING PERCENTAGE FROM VALUE

---

Category:

$$180$$

Total:

$$600$$

Percentage:

$$\frac{180}{600} \times 100$$

$$=\boxed{30\%}$$

---

## 12. MISSING TOTAL FROM ONE SECTOR

---

Sector angle:

$$72^\circ$$

Sector value:

$$240$$

Since:

$$\angle 72^\circ = 20\%$$

Total:

$$240/0.2 = \boxed{1200}$$

---

### 13. MISSING RATIO TERM

---

Suppose:

$$A:B=3:5$$

and:

$$A=180$$

Then:

$$3 \text{ parts} = 180$$

1 part:

$$60$$

B:

$$5(60) = \boxed{300}$$

---

### 14. MISSING RATIO FROM VALUES

---

A:

$$240$$

B:

$$360$$

Ratio:

$$240:360$$

$$= \boxed{2:3}$$

---

### 15. MISSING VALUE FROM RATIO + TOTAL

---

$$A:B=4:5$$

$$A+B=360$$

Total parts:

$$9$$

1 part:

$$40$$

So:

$$A=160, \text{quad } B=200$$

---

---

## 16. MISSING VALUE FROM RATIO + DIFFERENCE

---

$$A:B=7:4$$

Difference:

$$90$$

Difference in ratio parts:

$$3$$

One part:

$$30$$

Thus:

$$A=210, B=120$$

---

## 17. MISSING VALUE FROM GROWTH RATE

---

2024:

$$x$$

2025:

$$360$$

Growth:

$$20\%$$

Then:

$$1.2x=360$$

$$\boxed{x=300}$$

---

## 18. MISSING NEW VALUE FROM OLD VALUE

---

Old:

$$400$$

Growth:

$$15\%$$

New:

$$400(1.15)$$

$$= \boxed{460}$$

---

## 19. MISSING OLD VALUE AFTER DECREASE

---

Current:

---

$$\text{\[ 340 \]}$$

after:

$$\text{\[ 15\%\text{ decrease} \]}$$

So:

$$\text{\[ 0.85x=340 \]}$$

$$\text{\[ x=\boxed{400} \]}$$

Do not add 15% of 340.

---

## 20. MISSING VALUE FROM PROFIT MARGIN

---

Revenue:

$$\text{\[ 800 \]}$$

Profit margin:

$$\text{\[ 25\% \]}$$

Profit:

$$\text{\[ 800(0.25) = 200 \]}$$

Cost:

$$\text{\[ 800-200 = \boxed{600} \]}$$

---

## 21. MISSING REVENUE FROM PROFIT + MARGIN

---

Profit:

$$\text{\[ 180 \]}$$

Margin:

$$\text{\[ 20\% \]}$$

$$\text{\[ 0.2R=180 \]}$$

$$\text{\[ R=\boxed{900} \]}$$

---

## 22. MISSING COST FROM PROFIT % ON COST

---

Profit:

$$\text{\[ 100 \]}$$

Profit = 25% of cost.

$$\text{\[ 0.25C=100 \]}$$

$$\text{\[ C=400 \]}$$

Revenue:

$$\text{\[ 500 \]}$$

---



---

## 23. MISSING DEFECT RATE — WEIGHTED RATE

---

Plant A:

1000 units, 2% defective.

Plant B:

2000 units,  $(x\%)$  defective.

Overall defect rate:

3%.

Total defects:

$$\sqrt[3]{3000(0.03)=90}$$

A defects:

$$\sqrt[3]{20}$$

B defects:

$$\sqrt[3]{70}$$

So:

$$\sqrt[3]{2000(x/100)=70}$$

$$\sqrt[3]{x=\boxed{3.5\%}}$$

---

## 24. MISSING SELECTION RATE

---

Applicants:

$$\sqrt[3]{800}$$

Selected:

$$\sqrt[3]{200}$$

Selection rate:

$$\sqrt[3]{\frac{200}{800} = \boxed{25\%}}$$

---

## 25. MISSING APPLICANTS

---

Selected:

$$\sqrt[3]{180}$$

Selection rate:

$$\sqrt[3]{30\%}$$

Applicants:

$$\sqrt[3]{180/0.30}$$

$$\sqrt[3]{=\boxed{600}}$$

---

## 26. MISSING STAGE RATE IN FUNNEL

Suppose:

- 1000 applicants
- 60% clear stage 1
- $(x\%)$  of those clear stage 2
- final count = 240

Stage 1:

$$[ 600 ]$$

Then:

$$[ 600(x/100)=240 ]$$

$$[ x=\boxed{40\%} ]$$

## 27. PARTIAL TABLE RECONSTRUCTION

Suppose:

Region	Revenue	Cost	Profit
A	500	400	100
B	600	?	150
C	?	560	240

For B:

$$[ \text{Cost}=600-150 = \boxed{450} ]$$

For C:

$$[ \text{Revenue}=560+240 = \boxed{800} ]$$

## 28. PARTIAL TABLE WITH MARGINS

Company	Revenue	Margin
A	500	20%
B	?	25%

If B profit is 200:

$$[ 0.25R=200 ]$$

$$[ R=\boxed{800} ]$$

## 29. MULTIPLE MISSING CELLS — USE INDEPENDENT RELATIONS

Suppose:

Dept	Men	Women	Total
A	100	?	250
B	?	120	300
Total	330	220	550

A women:

$$\boxed{250 - 100 = 150}$$

B men:

$$\boxed{300 - 120 = 180}$$

Check column totals:

Men:

$$\boxed{100 + 180 = 280}$$

But table says 330.

So data is inconsistent.

## 30. CONSISTENCY CHECK

This is a very important advanced skill.

Always check:

$$\boxed{\text{row totals} = \text{column totals} = \text{grand total}}$$

when they are supposed to match.

## 31. INCONSISTENT DATA

If independently computed totals conflict:

- do not force a result
- do not average values
- identify inconsistency

In data-sufficiency style questions, answer may effectively be:

$$\boxed{\text{data inconsistent}}$$

## 32. DATASUFFICIENCY — BASIC IDEA

Question:

### Can A be uniquely determined?

Statements:

1.  $(A+B=500)$
2.  $(A:B=3:2)$

Together:

$$[A=300]$$

So data is sufficient.

### 33. ONE STATEMENT MAY BE INSUFFICIENT

Only:

$$[A+B=500]$$

Possible values:

$$[100+400, 200+300, \dots]$$

So:

$$[\boxed{\text{not sufficient}}]$$

### 34. SECOND STATEMENT ALONE MAY ALSO BE INSUFFICIENT

Only:

$$[A:B=3:2]$$

Possible:

$$[3, 2]$$

or:

$$[30, 20]$$

or:

$$[300, 200]$$

So not sufficient without scale.

### 35. TOGETHER SUFFICIENT

Ratio gives shape.

Total gives scale.

This pattern occurs constantly.

### 36. INDEPENDENT EQUATIONS RULE

If you have:

- 2 unknowns
- 2 independent equations

usually unique solution.

If one equation is just a rearrangement of another, it adds no information.

---

### 37. REDUNDANT STATEMENT

---

$$\{ A+B=500 \}$$

$$\{ 2A+2B=1000 \}$$

Second is not new information.

So only one independent equation exists.

---

### 38. RANGE / BOUNDS QUESTIONS

---

Sometimes exact value cannot be determined, but a range can.

Example:

$$\{ A+B=100 \}$$

with:

$$\{ A \geq 30, B \geq 20 \}$$

Then:

$$\{ A \leq 80 \}$$

because B at least 20.

So:

$$\{ \boxed{30 \leq A \leq 80} \}$$

---

### 39. MAXIMUM / MINIMUM USING TOTAL

---

Suppose total employees:

$$\{ 1000 \}$$

Tech at least 300.

Sales at least 250.

HR at least 100.

Maximum possible "Other":

$$\{ 1000-(300+250+100) \}$$

$$\{ =\boxed{350} \}$$

---

## 40. MINIMUM POSSIBLE CATEGORY

---

If:

$$A+B+C=1000$$

and:

$$A \leq 500, B \leq 400$$

then minimum possible C:

$$1000 - (500 + 400)$$

$$\boxed{100}$$

## 41. BOUNDS FROM PERCENTAGES

---

If a quantity is between 20% and 30% of 500:

$$100 \leq x \leq 150$$

$$\boxed{[100, 150]}$$

## 42. APPROXIMATION WITH INCOMPLETE DATA

---

Suppose total is approximately:

$$1000$$

and category share about:

$$31\%$$

Then value approximately:

$$310$$

If answer options:

- 150
- 220
- 305
- 470

choose:

$$\boxed{305}$$

without exact reconstruction.

## 43. WHEN APPROXIMATION IS DANGEROUS

---

If options:

- 307
- 310
- 313

- 316

you need more precise data/calculation.

---

## 44. MISSING GRAPH VALUE FROM TREND CONDITION

---

Suppose line graph values:

$[100, 130, x, 190]$

and average:

$[150]$

Total:

$[600]$

Known sum:

$[420]$

So:

$[x = \boxed{180}]$

---

## 45. MISSING BAR VALUE FROM TOTAL GROWTH

---

2024 total:

$[500]$

2025 total grows 20%.

So 2025 total:

$[600]$

Known 2025 bars:

$[120, 150, 180, ?]$

Missing:

$[600 - 450]$

$[= \boxed{150}]$

---

## 46. MISSING PIE SECTOR FROM RATIO

---

Suppose sectors A:B:C = 2:3:4.

Total angle:

$[360^\circ]$

Total parts:

$[9]$

One part:

---

$$\angle C = 40^\circ$$

Angles:

$$\angle A = 80^\circ$$

$$\angle B = 120^\circ$$

$$\angle C = 160^\circ$$

## 47. MISSING SECTOR FROM ACTUAL VALUE

Total:

$$900$$

A:

$$225$$

Share:

$$\frac{225}{900} = 25\%$$

Angle:

$$0.25(360) = \boxed{90^\circ}$$

## 48. MISSING VALUE FROM CUMULATIVE DATA

Cumulative:

Month	Cumulative
Jan	100
Feb	250
Mar	?
Apr	700

If March monthly value = 200:

$$\text{Mar cumulative} = 250 + 200$$

$$= \boxed{450}$$

## 49. MISSING MONTHLY VALUE

If cumulative March:

$$450$$

and cumulative February:

$$250$$

March monthly:



$$\boxed{200}$$

---

## 50. MISSING VALUE FROM MOVING AVERAGE

---

Values:

$$10, 20, x$$

3-period average:

$$30$$

Then:

$$10 + 20 + x = 90$$

$$x = \boxed{60}$$

---

## 51. MISSING VALUE FROM NEXT MOVING AVERAGE

---

Suppose:

$$20, 30, 40$$

has average:

$$30$$

Next 3-term moving average using:

$$30, 40, x$$

is:

$$50$$

Then:

$$30 + 40 + x = 150$$

$$x = \boxed{80}$$

---

## 52. MISSING DATA WITH INTEGER CONSTRAINT

---

Suppose:

$$A + B = 101$$

and ratio approximately:

$$A : B \approx 2 : 3$$

Exact ratio cannot give integers summing 101.

Need question wording such as:

- nearest integer
- approximate ratio
- possible values

Do not force exact proportionality.

---

---

## 53. PARITY / DIVISIBILITY CONSTRAINTS

Advanced placement DI may add:

- values must be integers
- all category counts multiples of 5
- total is even
- ratios in lowest terms

These can reduce possible solutions.

---

## 54. EXAMPLE

$A+B=100$

$A:B \approx 2:3$

Both must be multiples of 10.

Likely candidates near 40:60.

Thus:

$\boxed{A=40, B=60}$

if exact constraints make it unique.

---

## 55. RANGE WITH INTEGER CONSTRAINT

If:

$20 < A < 30$

and A must be divisible by 4:

Possible:

$24, 28$

So data still insufficient unless another condition is given.

---

## 56. GATE-STYLE INFERENCE — “CAN BE DETERMINED?”

Do not assume missing values are recoverable just because graph looks regular.

Example:

$100, ?, 200$

A straight drawn line through all three points visually may suggest 150, but unless linearity/equal increments are stated, exact missing value may not be justified.

---

## 57. TREND DOES NOT IMPLY LINEARITY

---

"Increasing every year" only tells:

$$x_{t+1} > x_t$$

It does not tell:

$$x_{t+1} - x_t = \text{constant}$$

Do not invent AP structure.

---

## 58. PARTIALLY SPECIFIED TABLE — EXAMPLE

Region	Sales	Growth
A	240	20%
B	300	?
C	?	25%

Suppose previous-year values:

$$A=200 \quad B=250 \quad C=320$$

B growth:

$$\frac{300-250}{250} = \boxed{20\%}$$

C sales:

$$320(1.25) = \boxed{400}$$

---

## 59. MULTI-CONDITION MISSING VALUE

Suppose:

$$A+B+C=600$$

$$A:B=2:3$$

$$C=200$$

Then:

$$A+B=400$$

Ratio parts:

$$5$$

One part:

$$80$$

So:

$$A=160, \quad B=240$$

---

## 60. MISSING VALUE FROM TWO PERCENTAGES

Suppose 40% of total are men.

Of men, 25% are managers.

Managers = 100.

Then men total:

$$\lfloor 100/0.25=400 \rfloor$$

Overall total:

$$\lfloor 400/0.40 = \boxed{1000} \rfloor$$

Reverse nested percentage.

---

## 61. MISSING VALUE FROM THREE STAGES

---

Suppose:

- 60% clear stage 1
- 50% of those clear stage 2
- 80% of those join
- final joiners = 240

Overall conversion:

$$\lfloor 0.6(0.5)(0.8) = 0.24 \rfloor$$

Applicants:

$$\lfloor 240/0.24 \rfloor$$

$$\lfloor =\boxed{1000} \rfloor$$

---

## 62. MISSING DATA ERROR CHECKLIST

---

Before solving:

- What exactly is unknown?
  - Which total includes it?
  - Which percentage base applies?
  - Are there enough independent conditions?
  - Are data consistent?
  - Is exact answer possible?
  - Is only a range possible?
- 

## 63. MUST-MEMORIZE REVERSE RELATIONS

---

From average:

$$\lfloor \boxed{x=nA-\text{known sum}} \rfloor$$

From percentage:

$$\lfloor \boxed{T=\frac{\text{Part}}{p}} \rfloor$$

---

From growth:

$$\boxed{\text{Old} = \frac{\text{New}}{1+g}}$$

From decrease:

$$\boxed{\text{Old} = \frac{\text{New}}{1-d}}$$

From weighted rate:

$$\boxed{\text{total successes} = \text{overall rate} \times \text{total base}}$$

---

## 64. FAST MISSING-DATA DECISION TREE

---

### Total known?

Subtract known parts.

### Average known?

Convert average to total.

### Percentage known?

Use  $\text{part} = \text{rate} \times \text{base}$ .

### Ratio known?

Convert to parts.

### Growth known?

Use multipliers.

### Weighted rate?

Convert rate to counts.

### Several unknowns?

Write independent equations.

### Not enough equations?

Do not guess.

### Only inequalities?

Find range/bounds.

---

## PRACTICE — LEVEL 0

---

### Q1

Average of 5 numbers is 60. Four are 40, 50, 70, 80. Find fifth.

### Q2

A category is 30% of total and equals 180. Find total.

### Q3

A:B = 3:5 and A = 120. Find B.

#### Q4

A value becomes 240 after 20% growth. Find old value.

#### Q5

A pie sector is 25% of total. Find angle.

---

### LEVEL 1

---

#### Q6

A+B=500 and A:B=3:2. Find A and B.

#### Q7

A group of 40 has average 50. Another group of 60 has average  $\backslash(x\backslash)$ . Overall average is 62. Find  $\backslash(x\backslash)$ .

#### Q8

A product falls by 20% to 320. Find original value.

#### Q9

Revenue = 1000 and profit margin = 25%. Find cost.

#### Q10

Selected = 150 at selection rate 30%. Find applicants.

---

### LEVEL 2

---

#### Q11

Plant A: 1000 units at 2% defect. Plant B: 2000 units at  $\backslash(x\backslash)\%$  defect. Overall = 4%. Find  $\backslash(x\backslash)$ .

#### Q12

A:B:C = 2:3:5 and total = 1000. Find all values.

#### Q13

A+B+C=700, C=280, A:B=3:4. Find A,B.

#### Q14

40% of total are engineers; 25% of engineers are women engineers. Women engineers = 100. Find total.

#### Q15

1000 applicants; 60% clear Stage 1,  $\backslash(x\backslash)\%$  clear Stage 2, final = 300. Find  $\backslash(x\backslash)$ .

---

### LEVEL 3 — EXAM PRESSURE

---

#### Q16

Four branches have average sales 250. Three branches have sales 180, 220, 300. Find fourth.

### Q17

$A+B=600$ . A is 20% more than B. Find both.

### Q18

Overall selection rate of two colleges is 30%. College A has 1000 applicants at 20% selection. College B has 500 applicants. Find B selection rate.

### Q19

$A+B+C=1000$ ,  $A:B=2:5$ ,  $B:C=3:0$ . What is maximum possible C?

### Q20

$A+B=500$  and  $A:B=2:3$ . A second statement says  $A-B=100$ . Is the data consistent?

---

## ANSWERS

---

### Q1

Total:

$$\sqrt{5(60)}=300$$

Known:

$$\sqrt{40+50+70+80}=240$$

Missing:

$$\sqrt{\boxed{60}}$$

### Q2

$$\sqrt{T=180/0.3 = \boxed{600}}$$

### Q3

$$\sqrt{120 \times \frac{5}{3} = \boxed{200}}$$

### Q4

$$\sqrt{\text{Old}=240/1.2 = \boxed{200}}$$

### Q5

$$\sqrt{0.25(360) = \boxed{90^\circ}}$$

### Q6

Total parts:

$$\sqrt{5}$$

One part:

$$\sqrt{100}$$

$$\sqrt{\boxed{A=300, B=200}}$$

## Q7

---

$$\sqrt{40(50)+60x=100(62)}$$

$$\sqrt{2000+60x=6200}$$

$$\sqrt{60x=4200}$$

$$\sqrt{\boxed{x=70}}$$

## Q8

---

$$\sqrt{\text{Old}=320/0.8 = \boxed{400}}$$

## Q9

---

Profit:

$$\sqrt{250}$$

Cost:

$$\sqrt{1000-250 = \boxed{750}}$$

## Q10

---

$$\sqrt{150/0.30 = \boxed{500}}$$

---

## Q11

---

Overall defects:

$$\sqrt{3000(0.04)=120}$$

A defects:

$$\sqrt{20}$$

B defects:

$$\sqrt{100}$$

$$\sqrt{2000(x/100)=100}$$

$$\sqrt{\boxed{x=5\%}}$$

## Q12

---

Parts:

$$\sqrt{10}$$

One part:

$$\sqrt{100}$$



$$\boxed{A=200, B=300, C=500}$$

### Q13

$$A+B=420$$

Ratio:

$$3:4$$

$$7 \text{ parts} = 420$$

$$1 \text{ part} = 60$$

$$\boxed{A=180, B=240}$$

### Q14

Women engineers are:

$$25\% \text{ of engineers}$$

Engineers:

$$100/0.25=400$$

Total:

$$400/0.4 = \boxed{1000}$$

### Q15

Stage 1:

$$1000(0.6)=600$$

$$600(x/100)=300$$

$$\boxed{x=50\%}$$

### Q16

Total:

$$4(250)=1000$$

Known:

$$180+220+300=700$$

Missing:

$$\boxed{300}$$

### Q17

A is 20% more than B:

$$A:B=6:5$$

$$11 \text{ parts} = 600$$

1 part:

$$\lfloor 600/11 \rfloor$$

So:

$$\lfloor A = \frac{3600}{11} \approx 327.27 \rfloor$$

$$\lfloor B = \frac{3000}{11} \approx 272.73 \rfloor$$

$$\lfloor \boxed{A \approx 327.27, B \approx 272.73} \rfloor$$

If these were counts of indivisible objects, the dataset would be unsuitable without rounding/integer instructions.

## Q18

Overall selected:

$$\lfloor 1500(0.30) = 450 \rfloor$$

A selected:

$$\lfloor 1000(0.20) = 200 \rfloor$$

B selected:

$$\lfloor 250 \rfloor$$

B rate:

$$\lfloor 250/500 = \boxed{50\%} \rfloor$$

## Q19

To maximize C, minimize A and B:

$$\lfloor A=250, \text{quad } B=300 \rfloor$$

Thus:

$$\lfloor C=1000-550 \rfloor$$

$$\lfloor = \boxed{450} \rfloor$$

## Q20

From:

$$\lfloor A:B=2:3, \text{quad } A+B=500 \rfloor$$

we get:

$$\lfloor A=200, \text{quad } B=300 \rfloor$$

Then:

$$\lfloor A-B=-100 \rfloor$$

not  $\lfloor (+100) \rfloor$ .

Therefore statements are:

$$\lfloor \boxed{\text{inconsistent}} \rfloor$$

## CHAPTER 56 STATUS

Covered so far:

- missing row/column cells '
- grand-total reconstruction '
- reverse averages '
- weighted missing averages '
- missing percentages '
- reverse percentages '
- missing ratios '
- ratio + total/difference '
- reverse growth '
- missing business metrics '
- weighted-rate reconstruction '
- partial tables '
- consistency checks '
- data sufficiency '
- redundant conditions '
- bounds/ranges '
- approximate incomplete data '
- missing cumulative/moving-average values '
- integer constraints '
- GATE-style inference traps '
- L0–L3 practice '

But **Chapter 56 is not complete yet.**

Still remaining:

- advanced bounds with overlapping constraints
- interval/range DI
- minimum/maximum possible totals
- missing-data optimization
- data sufficiency MCQ patterns
- approximation/elimination with partial data
- corrupted/inconsistent datasets
- advanced multi-source missing-value sets
- final DI speed test
- Unit 2 revision + completion audit

## Progress

---

$\boxed{55/76}$  completed

**Current:** Chapter 56 — Missing-Data & Advanced DI

## NEXT

---

**Chapter 56 Part 2 — Advanced Bounds, Data Sufficiency, Optimization, Corrupted Data, Final DI Speed Test & Unit 2 Completion**

USER

next

ASSISTANT

## CHAPTER 56 — MISSING-DATA & ADVANCED DI

### PART 2 — ADVANCED BOUNDS, DATA SUFFICIENCY, OPTIMIZATION, CORRUPTED DATA, FINAL DI SPEED TEST & UNIT 2 COMPLETION

Status: 0% PROGRESS ! ' ' COMPLETE GATE: Medium–High Placements: High Priority: +Tier 2

This

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next

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## CHAPTER 56 — MISSING-DATA & ADVANCED DI

### PART 2 — ADVANCED BOUNDS, DATA SUFFICIENCY, OPTIMIZATION, CORRUPTED DATA, FINAL DI SPEED TEST & UNIT 2 COMPLETION

Status: 0% PROGRESS ! ' ' COMPLETE GATE: Medium–High Placements: High Priority: +Tier 2

This is the final chapter of **Data Interpretation**.

The focus now is not just calculating missing values. It is deciding:

- whether an exact value is possible,
- whether only a range is possible,
- whether the given information is sufficient,
- whether the data is inconsistent,
- and how to solve incomplete DI quickly under exam pressure.

## 1. EXACT VALUE VS RANGE

Suppose:

$$\{ A+B=100 \}$$

This does **not** uniquely determine A and B.

Possible pairs:

$$\{ (20,80), (30,70), (40,60), \ldots \}$$

So exact values are impossible.

But if:

$$A \geq 30$$

and

$$B \geq 20$$

then:

$$30 \leq A \leq 80$$

because B must remain at least 20.

Thus:

$$\boxed{30 \leq A \leq 80}$$

---

## 2. MINIMUM / MAXIMUM PRINCIPLE

If:

$$A+B+C=T$$

then to **maximize A**, minimize B and C.

To **minimize A**, maximize B and C.

This simple principle solves many advanced DI questions.

---

## 3. EXAMPLE — MAXIMUM POSSIBLE VALUE

$$A+B+C=1000$$

Given:

$$B \geq 250$$

$$C \geq 300$$

Maximum A:

$$1000-250-300$$

$$=\boxed{450}$$

---

## 4. MINIMUM POSSIBLE VALUE

Suppose:

$$A+B+C=1000$$

$$B \leq 400, \text{ and } C \leq 350$$

To minimize A, maximize B and C:

$$A_{\min}=1000-400-350$$

$$=\boxed{250}$$

---

## 5. BOTH LOWER AND UPPER BOUNDS

---

Suppose:

$$A+B+C=1000$$

and:

$$200 \leq B \leq 300$$

$$250 \leq C \leq 400$$

Then maximum A:

$$1000-200-250=550$$

Minimum A:

$$1000-300-400=300$$

Therefore:

$$\boxed{300 \leq A \leq 550}$$

---

## 6. PERCENTAGE BOUNDS

---

Total:

$$800$$

A lies between 25% and 35%.

Minimum:

$$800(0.25)=200$$

Maximum:

$$800(0.35)=280$$

So:

$$\boxed{200 \leq A \leq 280}$$

---

## 7. RATIO BOUNDS

---

Suppose:

$$A:B$$

lies between:

$$2:3$$

and:

$$3:4$$

This means:

$$\frac{2}{3} \leq \frac{A}{B} \leq \frac{3}{4}$$

---

If  $B = 120$ :

$$\lfloor 80 \leq A \leq 90 \rfloor$$

---

## 8. OVERLAPPING CONSTRAINTS

---

Suppose:

$$\lfloor A+B+C=600 \rfloor$$

and:

$$\lfloor A \geq 150 \rfloor$$

$$\lfloor B \geq 200 \rfloor$$

$$\lfloor C \geq 100 \rfloor$$

Minimum committed:

$$\lfloor 150+200+100=450 \rfloor$$

Remaining:

$$\lfloor 150 \rfloor$$

This 150 can be distributed among A, B, C subject to any other limits.

---

## 9. MAXIMUM A UNDER OVERLAPPING CONSTRAINTS

---

To maximize A:

$$\lfloor B=200, \text{quad } C=100 \rfloor$$

Then:

$$\lfloor A=600-300 = \boxed{300} \rfloor$$

Minimum A:

$$\lfloor \boxed{150} \rfloor$$

Therefore:

$$\lfloor \boxed{150} \leq A \leq 300 \rfloor$$

---

## 10. BOUNDS WITH UPPER LIMITS

---

Suppose:

$$\lfloor A+B+C=1000 \rfloor$$

$$\lfloor A \leq 500, \text{quad } B \leq 400, \text{quad } C \leq 350 \rfloor$$

Can this be feasible?

Maximum possible total:

$$\lfloor 500+400+350=1250 \rfloor$$

So yes.

---

But if total were 1400:

$$\lfloor 1400 > 1250 \rfloor$$

then:

$$\lfloor \boxed{\text{Impossible dataset}} \rfloor$$

---

## 11. FEASIBILITY CHECK

---

For variables with ranges:

$$\lfloor L_i \leq x_i \leq U_i \rfloor$$

and total:

$$\lfloor T \rfloor$$

A necessary condition is:

$$\lfloor \boxed{\sum L_i \leq T \leq \sum U_i} \rfloor$$

If not, the data is inconsistent.

---

## 12. EXAMPLE

---

Three departments:

$$\lfloor 100 \leq A \leq 200 \rfloor$$

$$\lfloor 150 \leq B \leq 250 \rfloor$$

$$\lfloor 200 \leq C \leq 300 \rfloor$$

Possible total range:

$$\lfloor 450 \leq T \leq 750 \rfloor$$

If stated total = 800:

$$\lfloor \boxed{\text{Impossible}} \rfloor$$

---

## 13. MISSING-DATA OPTIMIZATION

---

Suppose total sales:

$$\lfloor 1000 \rfloor$$

Three known products:

$$\lfloor 200, \ 250, \ 300 \rfloor$$

Two missing products:

$$\lfloor x, y \rfloor$$

Then:

$$\lfloor x + y = 250 \rfloor$$

If:

---



$$x \geq y$$

then minimum x occurs when they are as equal as possible:

$$x \geq 125$$

If integers:

$$x_{\min} = 125$$

and maximum x depends on y's lower bound.

---

## 14. INTEGER CONSTRAINT

---

If:

$$x + y = 251$$

and:

$$x \geq y$$

with integers:

minimum x:

$$\boxed{126}$$

because:

$$125 + 126 = 251$$

and x must be the larger/equal term.

---

## 15. DIVISIBILITY CONSTRAINT

---

Suppose:

$$A + B = 150$$

A is divisible by 20.

B is divisible by 15.

Possible A:

$$0, 20, 40, \dots, 140$$

Need:

$$150 - A$$

divisible by 15.

Try:

$$A = 60:$$

$$B = 90$$

works.

$$A = 120:$$

$$B = 30$$

also works.

So without another condition:

$\boxed{\text{not unique}}$

---

## 16. DATASUFFICIENCY — CORE IDEA

In Data Sufficiency, you are not always asked to calculate the answer.

You are asked:

**Is the given information enough to determine the answer uniquely?**

Possible outcomes:

- Statement I alone sufficient
- Statement II alone sufficient
- Both together sufficient
- Either alone sufficient
- Even both insufficient

Exact option format varies by exam.

---

## 17. DATASUFFICIENCY EXAMPLE 1

Question:

**Find A.**

Statement I:

$A+B=100$

Statement II:

$A:B=2:3$

Statement I alone:

'Insufficient.

Statement II alone:

'Insufficient.

Together:

$A=40, B=60$

'sufficient.

---

## 18. DATASUFFICIENCY EXAMPLE 2

Question:

**Is A greater than B?**

Statement I:

$$\{ A-B=10 \}$$

Immediately:

$$\{ A>B \}$$

So Statement I alone is sufficient.

Statement II may be unnecessary.

## 19. DATA SUFFICIENCY DOES NOT REQUIRE EXACT VALUES

Question:

**Is A positive?**

Statement:

$$\{ A>5 \}$$

You do not know exact A.

But question can be answered:

$$\boxed{\text{Yes}}$$

Therefore statement is sufficient.

This is a major DS idea.

## 20. YES/NO QUESTION RULE

For sufficiency, a statement is sufficient if it produces a **definite yes or definite no**.

Example:

Question:

**Is  $x > 10$ ?**

Statement:

$$\{ x=8 \}$$

Answer:

No.

Still sufficient.

## 21. DS TRAP — SOMETIMES TRUE / SOMETIMES FALSE

Question:

**Is  $A > B$ ?**

Statement:

$$[ A+B=100 ]$$

Possible:

$$[ A=60, B=40 ]$$

Yes.

Also:

$$[ A=40, B=60 ]$$

No.

Thus:

$\boxed{\text{insufficient}}$

---

## 22. REDUNDANT STATEMENTS

Statement I:

$$[ A+B=200 ]$$

Statement II:

$$[ 2A+2B=400 ]$$

They are identical information.

Together still give only one independent equation.

If both A and B unknown:

$\boxed{\text{insufficient}}$

---

## 23. CONTRADICTORY STATEMENTS

Statement I:

$$[ A+B=100 ]$$

Statement II:

$$[ A+B=120 ]$$

Both cannot simultaneously hold under the same conditions.

Treat the dataset as inconsistent rather than solving.

---

## 24. CORRUPTED DATA

A dataset is corrupted when one or more entries violate required relationships.

Typical signals:

- row totals do not match
- column totals do not match
- percentages do not sum correctly beyond rounding tolerance
- averages contradict totals
- growth rate disagrees with values
- ratios contradict absolute numbers

---

## 25. FIND THE CORRUPTED CELL

Suppose:

Dept	Men	Women	Total
A	80	120	200
B	100	150	250
C	120	80	210

For C:

$120+80=200$

but total says:

$210$

So corrupted entry is likely:

$\boxed{\text{C total}}$

if Men/Women data are assumed correct.

---

## 26. CROSS-CHECK WITH COLUMN TOTALS

If column totals are also provided, you may identify which specific cell is wrong.

Example:

Men total = 300 Women total = 350 Grand total = 650

Men:

$80+100+120=300$

correct.

Women:

$120+150+80=350$

correct.

Therefore row C total must be:

$200$

not 210.

---

## 27. WRONG AVERAGE DETECTION

---

Values:

$\{ 40, 50, 60, 70 \}$

Reported average:

$\{ 60 \}$

Actual:

$\{ 220/4=55 \}$

Therefore:

$\{ \boxed{\text{reported average is incorrect}} \}$

---

## 28. WRONG GROWTH RATE DETECTION

---

Old:

$\{ 200 \}$

New:

$\{ 250 \}$

Reported growth:

$\{ 20\% \}$

Actual:

$\{ 50/200=25\% \}$

So reported growth is corrupted.

---

## 29. ROUNDED DATA VS CORRUPTED DATA

---

Percentages:

$\{ 33.3\%, 33.3\%, 33.3\% \}$

sum:

$\{ 99.9\% \}$

This may be normal rounding.

Do not call it corrupted immediately.

Small rounding differences are acceptable.

---

## 30. REVERSE-ENGINEERING FROM WRONG ENTRY

---

Suppose average of 5 values is reported as 60, but one value 80 was wrongly entered as 50.

---

Reported total:

$$5(60)=300$$

Correct total:

$$300-50+80$$

$$=330$$

Correct average:

$$330/5 = \boxed{66}$$

---

## 31. GENERAL WRONG-ENTRY FORMULA

If wrong entry  $w$  should be  $c$ :

$$\boxed{\text{Correct total} = \text{Reported total} - w + c}$$

Then divide by number of observations for corrected average.

---

## 32. MISSING + CORRUPTED DATA TOGETHER

Suppose row total is known, one cell missing, and another value may be wrong.

Do not immediately solve the missing cell unless the remaining values are trusted.

First identify whether relationships are consistent.

---

## 33. ADVANCED MULTI-SOURCE MISSING VALUE

Suppose:

Region	Revenue	Margin	Profit
A	500	20%	?
B	?	25%	200
C	800	?	240

A profit:

$$500(0.2)=100$$

B revenue:

$$200/0.25=800$$

C margin:

$$240/800=30\%$$

Thus:

$$\boxed{100, 800, 30\%}$$

## 34. ADVANCED WEIGHTED MISSING RATE

Suppose:

Branch	Applicants	Selected %
A	1000	20%
B	2000	$\frac{x}{100}\%$
C	1000	40%

Overall selection rate = 30%.

Overall selected:

$$\frac{1}{4} [4000(0.30) = 1200]$$

Known selected:

A:

$$\frac{1}{4} [200]$$

C:

$$\frac{1}{4} [400]$$

B selected:

$$\frac{1}{4} [600]$$

Thus:

$$\frac{1}{4} [2000(\frac{x}{100}) = 600]$$

$$\frac{1}{4} [\boxed{x=30\%}]$$

## 35. ADVANCED MISSING SHARE ACROSS TWO YEARS

Total market:

2024:

$$\frac{1}{4} [1000]$$

2025:

$$\frac{1}{4} [1500]$$

Company sales:

2024 share = 30%.

2025 sales are 50% higher than 2024.

2024 company sales:

$$\frac{1}{4} [300]$$

2025:

$$\frac{1}{4} [300(1.5) = 450]$$



2025 share:

$$\left[ \frac{450}{1500} = \boxed{30\%} \right]$$

So share remains unchanged.

---

### 36. MISSING TOTAL FROM SHARE CHANGE

---

Old company sales:

$$\left[ 300 \right]$$

New company sales:

$$\left[ 450 \right]$$

New share:

$$\left[ 25\% \right]$$

New market total:

$$\left[ \frac{450}{0.25} = \boxed{1800} \right]$$

---

### 37. APPROXIMATION WITH PARTIAL INFORMATION

---

Suppose:

$$\left[ x \in [295, 305] \right]$$

Question asks:

$$\left[ \frac{x}{900} \right]$$

Options:

- 20%
- 33%
- 45%
- 60%

Since:

$$\left[ \frac{300}{900} = 33.33\% \right]$$

range approximately:

$$\left[ 32.8\% - 33.9\% \right]$$

Answer:

$$\left[ \boxed{33\%} \right]$$

No need exact x.

---

### 38. OPTION ELIMINATION WITH BOUNDS

---

Suppose you know:

$$\left[ 120 < x < 140 \right]$$

---

Question asks  $\lfloor (x/400) \rfloor$ .

Then:

$$\lfloor 30\% < x/400 < 35\% \rfloor$$

Options:

- 20%
- 28%
- 32%
- 42%

Only possible:

$$\boxed{32\%}$$

This is very powerful under time pressure.

---

## 39. BOUNDING A PRODUCT

---

If:

$$\lfloor 20 < x < 25 \rfloor$$

and:

$$\lfloor 4 < y < 5 \rfloor$$

then:

$$\lfloor 80 < xy < 125 \rfloor$$

provided both positive.

Useful when exact values unavailable.

---

## 40. BOUNDING A RATIO

---

If:

$$\lfloor 100 \leq A \leq 120 \rfloor$$

and:

$$\lfloor 200 \leq B \leq 250 \rfloor$$

Then:

Minimum possible:

$$\lfloor A/B = \frac{100}{250} = 0.4 \rfloor$$

Maximum:

$$\lfloor \frac{120}{200} = 0.6 \rfloor$$

So:

$$\boxed{0.4 \leq A/B \leq 0.6}$$

---

## 41. DATA SUFFICIENCY WITH BOUNDS

Question:

Is  $A/B > 0.5$ ?

Given:

$$100 \leq A \leq 120$$

$$200 \leq B \leq 250$$

Possible ratio range:

$$0.4 \text{ to } 0.6$$

Could be below or above 0.5.

Therefore:

$\boxed{\text{insufficient}}$

## 42. SUFFICIENT BOUND

If instead:

$$A \geq 120$$

$$B \leq 200$$

with positive values:

$$A/B \geq 0.6$$

Therefore definitely:

$$A/B > 0.5$$

Sufficient.

## 43. MISSING DATA OPTIMIZATION — PLACEMENT STYLE

Suppose company total revenue:

$$1000$$

Departments A, B, C, D.

A = 200 B = 250 C is at least 150 D is at least 100.

Remaining after A+B:

$$550$$

Since D "e 1 0 0 :

maximum C:

$$550 - 100 = 450$$

Since C "e 1 5 0 :

minimum C:

$$\lfloor 150 \rfloor$$

So:

$$\lfloor \boxed{150 \leq C \leq 450} \rfloor$$

---

## 44. MAXIMUM PERCENTAGE SHARE

---

Using C max 450:

$$\lfloor 450/1000 = \boxed{45\%} \rfloor$$

Minimum share:

$$\lfloor 150/1000 = \boxed{15\%} \rfloor$$

---

## 45. ADVANCED CONDITION — $C > D$

---

From:

$$\lfloor C+D=550 \rfloor$$

and:

$$\lfloor C>D \rfloor$$

Then:

$$\lfloor C>275 \rfloor$$

If integers:

$$\lfloor C_{\min}=276 \rfloor$$

If real values:

$$\lfloor C \in (275, 450] \rfloor$$

depending other constraints.

---

## 46. “AT LEAST” VS “MORE THAN”

---

At least 20%:

$$\lfloor \boxed{\geq 20\%} \rfloor$$

More than 20%:

$$\lfloor \boxed{> 20\%} \rfloor$$

At most 20%:

$$\lfloor \boxed{\leq 20\%} \rfloor$$

Less than 20%:

$$\lfloor \boxed{< 20\%} \rfloor$$

These boundary words matter in DI filters.

---

---

## 47. ADVANCED DI SPEED TEST — SET 1

---

Use:

Region	Sales	Profit Margin
A	400	20%
B	600	25%
C	800	15%
D	1000	30%

### Q1

Highest profit?

### Q2

Total profit?

### Q3

Overall profit margin?

### Q4

Which regions have profit "e 1 5 0" ?

### Q5

Profit contribution of D to total profit?

---

## 48. ANSWERS SET 1

---

Profits:

A:

$\sqrt{80}$

B:

$\sqrt{150}$

C:

$\sqrt{120}$

D:

$\sqrt{300}$

### Q1

$\sqrt{\boxed{D}}$

### Q2

$$\lfloor 80+150+120+300 = \boxed{650} \rfloor$$

### Q3

Total sales:

$$\lfloor 2800 \rfloor$$

Margin:

$$\lfloor 650/2800 \rfloor$$

$$\lfloor \approx \boxed{23.21\%} \rfloor$$

### Q4

$$\lfloor \boxed{B,D} \rfloor$$

### Q5

$$\lfloor 300/650 \rfloor$$

$$\lfloor \approx \boxed{46.15\%} \rfloor$$

## 49. SPEED TEST — SET 2

Applications:

Branch	Applicants	Selection Rate
P	1000	20%
Q	1500	30%
R	2000	25%
S	500	?

Overall selection rate:

$$\lfloor 25\% \rfloor$$

Find S selection rate.

## 50. SOLUTION

Total applicants:

$$\lfloor 5000 \rfloor$$

Overall selected:

$$\lfloor 1250 \rfloor$$

Known:

P:

$$\lfloor 200 \rfloor$$

Q:

$$\sqrt[4]{450}$$

R:

$$\sqrt[4]{500}$$

Known total:

$$\sqrt[4]{1150}$$

S selected:

$$\sqrt[4]{100}$$

Rate:

$$\sqrt[4]{100/500} = \boxed{20\%}$$

---

## 51. SPEED TEST — SET 3

---

Five values have average 80.

Four known:

$$\sqrt[4]{60,70,90,110}$$

Missing:

Total:

$$\sqrt[4]{400}$$

Known:

$$\sqrt[4]{330}$$

Missing:

$$\sqrt[4]{\boxed{70}}$$

Target time in exam:

**under 15 seconds.**

---

## 52. SPEED TEST — SET 4

---

A pie sector =  $\sqrt[4]{108^\circ}$ .

Total = 1200.

Value:

$$\sqrt[4]{108^\circ = 30\%}$$

$$\sqrt[4]{1200(0.3) = \boxed{360}}$$

Target:

**mental calculation.**

---

## 53. SPEED TEST — SET 5

---

2024 value = 400. 2025 value = 460.

Growth:

$\lfloor 60/400 = \boxed{15\%} \rfloor$

Target:

**under 10 seconds.**

---

## 54. FINAL DI TIME STRATEGY

---

For a 5-question DI set:

### Pass A — Scan

About 15–20 seconds:

- title
- unit
- axes/table headings
- totals
- footnotes

### Pass B — Direct questions

Do:

- lookup
- total
- difference
- simple ratio

### Pass C — Derived questions

Do:

- percentage
- weighted rate
- growth
- missing value

### Pass D — Heavy questions

Do:

- multi-stage
  - bounds
  - data sufficiency
  - cross-source calculations
- 

## 55. WHEN TO SKIP A DI QUESTION

---

Skip temporarily if:

- too many missing values
-



- unclear denominator
- options require long exact arithmetic
- several equations must be built
- another question in same set is much easier

Do not let one hard DI question consume the whole set.

---

## 56. DI APPROXIMATION TOOLKIT

---

Useful benchmarks:

$$\lfloor 1/2=50\% \rfloor$$

$$\lfloor 1/3=33.33\% \rfloor$$

$$\lfloor 1/4=25\% \rfloor$$

$$\lfloor 1/5=20\% \rfloor$$

$$\lfloor 1/6=16.67\% \rfloor$$

$$\lfloor 1/8=12.5\% \rfloor$$

$$\lfloor 2/3=66.67\% \rfloor$$

$$\lfloor 3/4=75\% \rfloor$$

$$\lfloor 3/5=60\% \rfloor$$

$$\lfloor 5/8=62.5\% \rfloor$$

These should become automatic.

---

## 57. DI SANITY CHECKS

---

Before marking answer:

### Percentage share

Must normally lie:

$$\lfloor 0\% \leq p \leq 100\% \rfloor$$

### Individual part

Usually:

$$\lfloor \text{Part} \leq \text{Total} \rfloor$$

### Weighted average

Must lie between minimum and maximum group averages.

### Positive growth

New > Old.

### Ratio

Check direction.

## Pie

Angles should total approximately:

$$\backslash[ 360^\circ \backslash]$$

---

## 58. WEIGHTED-AVERAGE SANITY RULE

---

If group averages are:

$$\backslash[ 40, \ 60, \ 80 \backslash]$$

overall average cannot be:

$$\backslash[ 90 \backslash]$$

if all weights are positive.

Must lie within:

$$\backslash[ \boxed{40 \leq A_{\text{overall}} \leq 80} \backslash]$$

---

## 59. OVERALL RATE SANITY RULE

---

If group rates are:

$$\backslash[ 20\%, 30\%, 50\% \backslash]$$

overall weighted rate must lie between:

$$\backslash[ 20\% \text{ and } 50\% \backslash]$$

unless unusual negative weights, which normal DI does not have.

---

## 60. FINAL DATA-SUFFICIENCY CHECKLIST

---

Ask:

1. What exactly must be determined?
  2. Do I need exact value or only yes/no?
  3. Does statement I give a unique answer?
  4. Does statement II?
  5. Together?
  6. Are equations independent?
  7. Are constraints consistent?
  8. Does a range already answer the yes/no question?
- 

## 61. FINAL MISSING-DATA CHECKLIST

---

When you see  :

**Total relation?**

Use subtraction.

---

**Average relation?**

Turn average into total.

**Percentage relation?**

Use part/base.

**Ratio relation?**

Convert to parts.

**Growth relation?**

Use multiplier.

**Weighted relation?**

Convert rates to actual counts.

**Multiple unknowns?**

Count independent equations.

**Only inequalities?**

Find bounds.

---

## 62. FINAL ADVANCED DI TRAPS

---

**& pTRAP 1 — Assuming exact value exists**

Sometimes only a range can be known.

**& pTRAP 2 — Treating dependent equations as independent**

$$\backslash A+B=100 \backslash$$

and:

$$\backslash 2A+2B=200 \backslash$$

are the same information.

**& pTRAP 3 — Ignoring feasibility**

Lower bounds may already exceed total.

**& pTRAP 4 — Confusing corruption with rounding**

99.9% may just be rounding.

**& pTRAP 5 — Averaging percentages**

Use weighted rate.

**& pTRAP 6 — Guessing missing graph points**

Trend does not imply exact linearity.

**& pTRAP 7 — Solving when only sufficiency is asked**

Do not waste time finding exact values unnecessarily.

---

## 63. 5-MINUTE UNIT 2 REVISION

---

### Tables

Read correct row/column and denominator.

### Bar graphs

Check scale, grouped series and stacked boundaries.

### Line graphs

Separate value, slope and percentage growth.

### Pie charts

$\angle 360^\circ = 100\%$

### Mixed DI

Convert all sources into common quantities.

### Missing data

Use totals, averages, ratios, percentages and constraints.

---

## 64. 15-MINUTE UNIT 2 REVISION FORMULA SHEET

---

### Percentage change

$\boxed{\frac{\text{New}-\text{Old}}{\text{Old}} \times 100}$

### Share

$\boxed{\frac{\text{Part}}{\text{Total}} \times 100}$

### Weighted average

$\boxed{\frac{\sum wx}{\sum w}}$

### Overall rate

$\boxed{\frac{\sum \text{successes}}{\sum \text{base}} \times 100}$

### Pie value

$\boxed{T \frac{\theta}{360}}$

### CAGR

$\boxed{\left( \frac{V_f}{V_i} \right)^{\frac{1}{n}} - 1}$

### Missing from average

$\boxed{x = nA - \text{known sum}}$

---

## Reverse growth

$$\boxed{\text{Old} = \frac{\text{New}}{1+g}}$$

## Funnel

$$\boxed{N_f = N_{0p_1p_2} \cdots}$$

---

## 65. UNIT 2 MUST-REMEMBER TRAPS

1. Higher amount " higher percentage.
2. Higher percentage " higher count.
3. Average percentages require weighting.
4. Growth denominator = old value.
5. Profit margin denominator = revenue.
6. Stacked segment = boundary difference.
7. Same pie share " same actual amount across different totals.
8. Cumulative data must be differenced.
9. Share can fall while actual value rises.
10. Missing data may not be uniquely solvable.
11. Straight visual trend does not prove linear values.
12. Data Sufficiency needs uniqueness, not full calculation.

---

## 66. UNIT 2 COMPLETION AUDIT

### Chapter 50 — DI Foundations

' COMPLETE

### Chapter 51 — Tables

' COMPLETE

### Chapter 52 — Bar & Stacked-Bar Graphs

' COMPLETE

### Chapter 53 — Line Graphs

' COMPLETE

### Chapter 54 — Pie Charts

' COMPLETE

### Chapter 55 — Mixed Graphs & Caselets

' COMPLETE

### Chapter 56 — Missing-Data & Advanced DI

' COMPLETE

## ' CHAPTER 56 COMPLETE

---

You now know:

- missing table values '
- reverse averages '
- missing percentages '
- missing ratios '
- reverse growth '
- weighted missing rates '
- partial tables '
- advanced bounds '
- max/min optimization '
- feasibility checks '
- data sufficiency '
- redundant equations '
- inconsistent/corrupted data '
- approximation with bounds '
- multi-source reconstruction '
- speed solving '
- DI sanity checking '

## ' UNIT 2 — DATA INTERPRETATION COMPLETE

---

Progress:

$\boxed{56/76}$  numbered chapters completed

That means all numbered chapters through **Data Interpretation** are finished.

---

## COURSE STATUS

---

### UNIT 0 — Calculation & Number Foundations

' COMPLETE

### UNIT 1 — Quantitative Aptitude

' COMPLETE

### UNIT 2 — Data Interpretation

' COMPLETE

### UNIT 3 — Logical & Analytical Reasoning

+ NOT STARTED

### UNIT 4 — Verbal Aptitude

+ NOT STARTED

### UNIT 5 — GATE Special Training

+ NOT STARTED

## UNIT 6 — Placement Special Training

+ NOT STARTED

## UNIT 7 — Revision & Question Bank

+ NOT STARTED

## UNIT 8 — Full Mocks

+ NOT STARTED

## NEXT

### CHAPTER 57 — NUMBER, LETTER & ALPHANUMERIC SERIES

We now begin:

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

Next chapter will cover:

- number series
- difference patterns
- second differences
- multiplication/division patterns
- alternating series
- square/cube/prime patterns
- Fibonacci-style patterns
- missing-term series
- wrong-term series
- letter series
- alphabetical positions
- reverse alphabet
- alphanumeric series
- mixed pattern recognition
- GATE/placement speed techniques
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 57 — NUMBER, LETTER & ALPHANUMERIC SERIES

## PART 1 — NUMBER SERIES, DIFFERENCE PATTERNS, MULTIPLICATION/DIVISION, ALTERNATING SERIES & WRONG TERMS

Status:  $\emptyset$ =Not in PROGRESS GATE: High Placements: Very High Priority:  $\emptyset$ =Yes for 1

Series questions test whether you can identify the hidden rule connecting terms.

The biggest skill is:

**Do not guess randomly. Test patterns in a fixed order.**

### 1. SERIES MASTER APPROACH

When you see:

$[a_1, a_2, a_3, a_4, \dots]$

check in this order:

1. simple difference
2. second difference
3. multiplication/division
4. multiplication + addition/subtraction
5. alternating patterns
6. squares/cubes/primes
7. previous-term combinations
8. position-based patterns

This search order saves time.

### 2. TYPE 1 — CONSTANT DIFFERENCE

Example:

$[5, 9, 13, 17, ?]$

Differences:

$[+4, +4, +4]$

Next:

$[17 + 4 = \boxed{21}]$

This is an Arithmetic Progression.

### 3. AP FORMULA

For:

$[a, a+d, a+2d, \dots]$

nth term:

$[\boxed{a_n = a + (n-1)d}]$



---

## 4. EXAMPLE — FIND 20TH TERM

---

$\backslash[ 3,8,13,18,\ldots \backslash]$

Here:

$\backslash[ a=3,\quad d=5 \backslash]$

$\backslash[ a_{20}=3+19(5) \backslash]$

$\backslash[ =98 \backslash]$

$\backslash[ \boxed{98} \backslash]$

---

## 5. TYPE 2 — INCREASING DIFFERENCES

---

Example:

$\backslash[ 2,5,9,14,20,? \backslash]$

Differences:

$\backslash[ 3,4,5,6 \backslash]$

Next difference:

$\backslash[ 7 \backslash]$

Therefore:

$\backslash[ 20+7 = \boxed{27} \backslash]$

---

## 6. DIFFERENCE TABLE

---

For:

$\backslash[ 3,8,15,24,35 \backslash]$

First differences:

$\backslash[ 5,7,9,11 \backslash]$

These increase by:

$\backslash[ 2 \backslash]$

So next first difference:

$\backslash[ 13 \backslash]$

Next term:

$\backslash[ 35+13 = \boxed{48} \backslash]$

---

## 7. SECOND DIFFERENCE

---

Example:

$\backslash[ 2,6,12,20,30 \backslash]$

---

First differences:

$\backslash 4,6,8,10 \backslash$

Second differences:

$\backslash 2,2,2 \backslash$

Constant second difference often suggests a quadratic-type sequence.

Next first difference:

$\backslash 12 \backslash$

Next term:

$\backslash 30+12 = \boxed{42} \backslash$

---

## 8. SQUARE-BASED RECOGNITION

Notice:

$\backslash 2,6,12,20,30 \backslash$

can also be written:

$\backslash 1\times 2, \text{quad}2\times 3, \text{quad}3\times 4, \text{quad}4\times 5, \text{quad}5\times 6 \backslash$

So nth term:

$\backslash \boxed{n(n+1)} \backslash$

Next:

$\backslash 6\times 7=42 \backslash$

Same answer.

---

## 9. TYPE 3 — CONSTANT MULTIPLICATION

Example:

$\backslash 3,6,12,24,? \backslash$

Rule:

$\backslash \times 2 \backslash$

Next:

$\backslash 24(2)=\boxed{48} \backslash$

---

## 10. GEOMETRIC SERIES

$\backslash a, ar, ar^2, \ldots \backslash$

nth term:

$\backslash \boxed{a_n = ar^{n-1}} \backslash$

Example:

$[2, 6, 18, 54, \dots]$

$[r=3]$

Next:

$[\boxed{162}]$

---

## 11. TYPE 4 — MULTIPLY THEN ADD

---

Example:

$[2, 5, 11, 23, 47, ?]$

Check:

$[2 \times 2 + 1 = 5]$

$[5 \times 2 + 1 = 11]$

$[11 \times 2 + 1 = 23]$

So:

$[47 \times 2 + 1 = \boxed{95}]$

---

## 12. MULTIPLY THEN SUBTRACT

---

Example:

$[100, 49, 23, 10, ?]$

Pattern:

$[100/2 - 1 = 49]$

$[49/2 - 1 = 23.5]$

Not exact.

So reject quickly.

Try:

$[100 - 51 = 49]$

$[49 - 26 = 23]$

$[23 - 13 = 10]$

Differences roughly halve, but not clean enough.

Important lesson:

**Do not force a pattern if arithmetic becomes unnatural.**

---

## 13. CLEAN MULTIPLY-SUBTRACT EXAMPLE

---

$[5, 14, 41, 122, ?]$

---

Pattern:

$$\backslash[ 5\times 3-1=14 \backslash]$$

$$\backslash[ 14\times 3-1=41 \backslash]$$

$$\backslash[ 41\times 3-1=122 \backslash]$$

Next:

$$\backslash[ 122\times 3-1 = \boxed{365} \backslash]$$

---

## 14. TYPE 5 — MULTIPLIER CHANGES

---

Example:

$$\backslash[ 2,4,12,48,240,? \backslash]$$

Multipliers:

$$\backslash[ \times 2, \times 3, \times 4, \times 5 \backslash]$$

Next:

$$\backslash[ \times 6 \backslash]$$

$$\backslash[ 240(6)=\boxed{1440} \backslash]$$

---

## 15. MULTIPLIER + ADDITION CHANGES

---

Example:

$$\backslash[ 1,3,10,41,206,? \backslash]$$

Pattern:

$$\backslash[ 1\times 2+1=3 \backslash]$$

$$\backslash[ 3\times 3+1=10 \backslash]$$

$$\backslash[ 10\times 4+1=41 \backslash]$$

$$\backslash[ 41\times 5+1=206 \backslash]$$

Next:

$$\backslash[ 206\times 6+1 \backslash]$$

$$\backslash[ =\boxed{1237} \backslash]$$

---

## 16. TYPE 6 — ALTERNATING SERIES

---

Example:

$$\backslash[ 2,5,4,7,6,9,? \backslash]$$

Separate odd positions:

$$\backslash[ 2,4,6,? \backslash]$$

Even positions:

---

$\{ 5, 7, 9 \}$

Odd sequence increases by 2.

So next:

$\boxed{8}$

---

## 17. ODD/EVEN POSITION METHOD

---

For:

$a_1, a_2, a_3, a_4, a_5, \dots$

Separate:

### Odd positions

$a_1, a_3, a_5, \dots$

### Even positions

$a_2, a_4, a_6, \dots$

This is one of the most useful series tricks.

---

## 18. EXAMPLE

---

$3, 10, 6, 20, 12, 40, ?$

Odd positions:

$3, 6, 12, ?$

$\times 2$

Even positions:

$10, 20, 40$

$\times 2$

Next odd:

$12 \times 2 = \boxed{24}$

---

## 19. ALTERNATING OPERATIONS

---

Example:

$10, 20, 18, 36, 34, 68, ?$

Pattern:

$\times 2, -2, \times 2, -2, \times 2$

Next:

$68 - 2 = \boxed{66}$

---

## 20. TYPE 7 — SQUARE NUMBERS

---

Memorize:

$\backslash[1,4,9,16,25,36,49,64,81,100,\ldots\backslash]$

Example:

$\backslash[4,9,16,25,36,?\backslash]$

These are:

$\backslash[2^2,3^2,4^2,5^2,6^2\backslash]$

Next:

$\backslash[7^2 = \boxed{49}\backslash]$

---

## 21. SQUARE $\pm$ CONSTANT

---

Example:

$\backslash[2,5,10,17,26,?\backslash]$

Recognize:

$\backslash[1^2+1=2\backslash]$

$\backslash[2^2+1=5\backslash]$

$\backslash[3^2+1=10\backslash]$

$\backslash[4^2+1=17\backslash]$

$\backslash[5^2+1=26\backslash]$

Next:

$\backslash[6^2+1 = \boxed{37}\backslash]$

---

## 22. CUBE NUMBERS

---

Memorize:

$\backslash[1,8,27,64,125,216,343,512,729,1000\backslash]$

Example:

$\backslash[8,27,64,125,?\backslash]$

$\backslash[2^3,3^3,4^3,5^3\backslash]$

Next:

$\backslash[6^3 = \boxed{216}\backslash]$

---

## 23. CUBE $\pm$ CONSTANT

---

Example:

$\backslash[2,9,28,65,126,?\backslash]$

---

$$\sqrt[3]{1^3+1}$$

$$\sqrt[3]{2^3+1}$$

$$\sqrt[3]{3^3+1}$$

$$\sqrt[3]{4^3+1}$$

$$\sqrt[3]{5^3+1}$$

Next:

$$\sqrt[3]{6^3+1} = \boxed{217}$$

---

## 24. PRIME NUMBER SERIES

---

Prime numbers:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \dots$$

Example:

$$3, 5, 7, 11, 13, ?$$

Next prime:

$$\boxed{17}$$

---

## 25. PRIME + CONSTANT

---

Example:

$$4, 5, 7, 9, 13, 15, ?$$

Could be primes +2:

$$2+2=4$$

$$3+2=5$$

$$5+2=7$$

$$7+2=9$$

$$11+2=13$$

$$13+2=15$$

Next:

$$17+2 = \boxed{19}$$

---

## 26. FIBONACCI-TYPE SERIES

---

Standard Fibonacci:

$$1, 1, 2, 3, 5, 8, 13, \dots$$

Rule:

$$a_n = a_{n-1} + a_{n-2}$$

---

Example:

$\backslash [ 2,3,5,8,13,? \backslash ]$

$\backslash [ 8+13 = \boxed{21} \backslash ]$

---

## 27. MODIFIED FIBONACCI

---

Example:

$\backslash [ 2,4,7,13,22,? \backslash ]$

Check:

$\backslash [ 2+4+1=7 \backslash ]$

$\backslash [ 4+7+2=13 \backslash ]$

$\backslash [ 7+13+2=22 \backslash ]$

Not clean. Reject.

Try differences:

$\backslash [ 2,3,6,9 \backslash ]$

not enough.

This illustrates another rule:

**Prefer the simplest consistent pattern over complicated invented rules.**

---

## 28. CLEAN MODIFIED FIBONACCI

---

$\backslash [ 1,2,4,7,12,20,? \backslash ]$

Observe:

$\backslash [ 1+2+1=4 \backslash ]$

$\backslash [ 2+4+1=7 \backslash ]$

$\backslash [ 4+7+1=12 \backslash ]$

$\backslash [ 7+12+1=20 \backslash ]$

Next:

$\backslash [ 12+20+1 = \boxed{33} \backslash ]$

---

## 29. PREVIOUS TWO TERMS — DIFFERENCE

---

Example:

$\backslash [ 2,5,3,8,5,13,? \backslash ]$

Odd/even split is easier:

Odd:

---



$\{ 2, 3, 5, ? \}$

Could be Fibonacci-like.

Even:

$\{ 5, 8, 13 \}$

Also Fibonacci-like.

Next odd:

$\{ 3+5=8 \}$

Thus:

$\boxed{8}$

---

### 30. POSITION-BASED SERIES

---

Example:

$\{ 2, 6, 12, 20, 30, ? \}$

Term:

$\{ n(n+1) \}$

So next:

$\{ 6(7)=42 \}$

Position-based recognition is often faster than difference tables.

---

### 31. ANOTHER POSITION-BASED SERIES

---

$\{ 3, 8, 15, 24, 35, ? \}$

Observe:

$\{ 1^2+2=3 \}$

$\{ 2^2+4=8 \}$

Not elegant.

Better:

$\{ n^2+2n = n(n+2) \}$

For  $n=1$ :

$\{ 3 \}$

$n=2$ :

$\{ 8 \}$

$n=3$ :

$\{ 15 \}$

$n=4$ :

$\{ 24 \}$

$n=5$ :

$\backslash[ 35 \backslash]$

Next:

$\backslash[ 6(8)=\boxed{48} \backslash]$

---

## 32. TRIANGULAR NUMBERS

---

Important sequence:

$\backslash[ 1,3,6,10,15,21,28,\ldots \backslash]$

Formula:

$\backslash[ \boxed{\frac{n(n+1)}{2}} \backslash]$

Differences:

$\backslash[ 2,3,4,5,6,\ldots \backslash]$

---

## 33. EXAMPLE

---

$\backslash[ 3,6,10,15,21,? \backslash]$

Differences:

$\backslash[ 3,4,5,6 \backslash]$

Next:

$\backslash[ 7 \backslash]$

$\backslash[ 21+7 = \boxed{28} \backslash]$

---

## 34. FACTORIAL SERIES

---

Factorials:

$\backslash[ 1!,2!,3!,4!,5!,6!,\ldots \backslash]$

$\backslash[ 1,2,6,24,120,720,\ldots \backslash]$

Example:

$\backslash[ 2,6,24,120,? \backslash]$

Next:

$\backslash[ 6!=720 \backslash]$

$\backslash[ \boxed{720} \backslash]$

---

## 35. FACTORIAL-LIKE MULTIPLIERS

---

$\backslash[ 1,2,6,24,120 \backslash]$

Multipliers:

---

$\backslash [ 2,3,4,5 \backslash ]$

This makes factorial recognition easy.

---

### 36. RECIPROCAL / DIVISION SERIES

---

Example:

$\backslash [ 256,128,64,32,? \backslash ]$

Rule:

$\backslash [ \div 2 \backslash ]$

Next:

$\backslash [ \boxed{16} \backslash ]$

---

### 37. CHANGING DIVISOR

---

$\backslash [ 720,360,120,30,6,? \backslash ]$

Operations:

$\backslash [ \div 2, \div 3, \div 4, \div 5 \backslash ]$

Next:

$\backslash [ \div 6 \backslash ]$

$\backslash [ 6/6=\boxed{1} \backslash ]$

---

### 38. FRACTION SERIES

---

$\backslash [ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, ? \backslash ]$

Numerators:

$\backslash [ 1,2,3,4 \backslash ]$

Denominators:

$\backslash [ 2,3,4,5 \backslash ]$

Next:

$\backslash [ \boxed{\frac{5}{6}} \backslash ]$

---

### 39. FRACTION WITH INDEPENDENT PATTERNS

---

$\backslash [ \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, ? \backslash ]$

Numerator:

$\backslash [ 1,2,3,4,5 \backslash ]$

Denominator:

$\backslash [ 2,4,8,16,32 \backslash ]$

---

Next:

$\boxed{\frac{5}{32}}$

---

## 40. DECIMAL SERIES

---

Example:

$0.1, 0.2, 0.4, 0.8, ?$

$\times 2$

Next:

$\boxed{1.6}$

Do not treat decimals as fundamentally different from integers.

---

## 41. WRONG TERM SERIES

---

Example:

$2, 4, 8, 16, 31, 64$

Expected powers of 2:

$2, 4, 8, 16, 32, 64$

Wrong term:

$\boxed{31}$

---

## 42. WRONG TERM USING DIFFERENCES

---

$3, 7, 13, 21, 32, 43$

Expected differences might be:

$4, 6, 8, 10, 12$

Correct sequence should be:

$3, 7, 13, 21, 31, 43$

Wrong term:

$\boxed{32}$

---

## 43. WRONG TERM — IMPORTANT STRATEGY

---

Do not inspect only the suspicious term.

A wrong term often disturbs:

- the difference before it
- the difference after it

So look for where **two adjacent differences become abnormal**.

---

---

## 44. EXAMPLE

$\backslash$  5,10,17,26,38,50  $\backslash$

Differences:

$\backslash$  5,7,9,12,12  $\backslash$

Expected:

$\backslash$  5,7,9,11,13  $\backslash$

Correct fifth term should be:

$\backslash$  26+11=37  $\backslash$

Then:

$\backslash$  37+13=50  $\backslash$

Wrong:

$\backslash$   $\boxed{38}$   $\backslash$

---

## 45. MISSING TERM — CENTER

$\backslash$  4,9,16,?,36,49  $\backslash$

Squares:

$\backslash$   $2^2, 3^2, 4^2, 5^2, 6^2, 7^2$   $\backslash$

Missing:

$\backslash$   $\boxed{25}$   $\backslash$

---

## 46. MISSING TERM WITH MULTIPLIER

$\backslash$  3,7,15,31,?,127  $\backslash$

Rule:

$\backslash$   $\times 2 + 1$   $\backslash$

$\backslash$   $31 \times 2 + 1 = 63$   $\backslash$

$\backslash$   $63 \times 2 + 1 = 127$   $\backslash$

Missing:

$\backslash$   $\boxed{63}$   $\backslash$

---

## 47. MISSING TERM WITH ALTERNATING SERIES

$\backslash$  2,10,4,20,6,?,8,40  $\backslash$

Odd positions:

$\backslash$  2,4,6,8  $\backslash$

---

Even:

$\{10, 20, ?, 40\}$

So:

$\boxed{30}$

---

## 48. THREE INTERWOVEN SERIES

---

Example:

$\{1, 10, 100, 2, 20, 200, 3, 30, ?\}$

Split by every third term:

First:

$\{1, 2, 3\}$

Second:

$\{10, 20, 30\}$

Third:

$\{100, 200, ?\}$

Next:

$\boxed{300}$

---

## 49. WHEN ODD/EVEN SPLIT FAILS

---

Try grouping every 3rd or 4th term if the pattern suggests repeating blocks.

Especially common in alphanumeric and placement series.

---

## 50. PAIR-BASED SERIES

---

Example:

$\{2, 4, 5, 10, 11, 22, ?\}$

Pairs:

$\{2 \text{ to } 4\}$

$\{5 \text{ to } 10\}$

$\{11 \text{ to } 22\}$

Between pair starts:

$\{2, 5, 11, \dots\}$

Could be:

$\{n \times 2 + 1\}$

Then next start:

---

$\backslash 23 \backslash$

So:

$\backslash \boxed{23} \backslash$

This type is trickier; only use pair grouping when clearly supported.

---

## 51. DIFFERENCE OF SQUARES PATTERN

---

Example:

$\backslash 3,8,15,24,35 \backslash$

Differences:

$\backslash 5,7,9,11 \backslash$

These are odd numbers.

Since consecutive squares differ by odd numbers:

$\backslash (n+1)^2 - n^2 = 2n+1 \backslash$

Series often hides quadratic patterns.

---

## 52. PRIME DIFFERENCES

---

Example:

$\backslash 10,12,15,20,27,? \backslash$

Differences:

$\backslash 2,3,5,7 \backslash$

Next prime:

$\backslash 11 \backslash$

Therefore:

$\backslash 27+11 = \boxed{38} \backslash$

---

## 53. SQUARE DIFFERENCES

---

Example:

$\backslash 1,5,14,30,55,? \backslash$

Differences:

$\backslash 4,9,16,25 \backslash$

$\backslash 2^2,3^2,4^2,5^2 \backslash$

Next difference:

$\backslash 36 \backslash$

Next term:

---

$$\lfloor 55+36 = \boxed{91} \rfloor$$

---

## 54. CUBE DIFFERENCES

---

Example:

$$\lfloor 1,9,36,100,225,? \rfloor$$

Differences:

$$\lfloor 8,27,64,125 \rfloor$$

$$\lfloor 2^3,3^3,4^3,5^3 \rfloor$$

Next:

$$\lfloor 6^3=216 \rfloor$$

Term:

$$\lfloor 225+216 = \boxed{441} \rfloor$$

---

## 55. MIXED MULTIPLICATION+ INCREASING ADDITION

---

$$\lfloor 1,3,8,18,38,? \rfloor$$

Check:

$$\lfloor 1\times 2+1=3 \rfloor$$

$$\lfloor 3\times 2+2=8 \rfloor$$

$$\lfloor 8\times 2+2=18 \rfloor$$

Not clean.

Try differences:

$$\lfloor 2,5,10,20 \rfloor$$

After first, differences double imperfectly.

Reject unless options/context strongly support.

Again: don't overfit.

---

## 56. CLEAN VERSION

---

$$\lfloor 1,4,11,26,57,? \rfloor$$

Pattern:

$$\lfloor 1\times 2+2=4 \rfloor$$

$$\lfloor 4\times 2+3=11 \rfloor$$

$$\lfloor 11\times 2+4=26 \rfloor$$

$$\lfloor 26\times 2+5=57 \rfloor$$

Next:

---



$\backslash [ 57 \times 2 + 6 \backslash ]$

$\backslash [= \boxed{120} \backslash ]$

---

## 57. ORDER OF PATTERN TESTING

---

For placement speed, use:

### Test 1

Differences.

### Test 2

Ratios/multipliers.

### Test 3

Odd/even positions.

### Test 4

Squares/cubes/primes.

### Test 5

Multiply  $\pm$  changing constant.

### Test 6

Previous two terms.

### Test 7

Position formula.

Do not spend 2 minutes experimenting randomly.

---

## 58. WHEN DIFFERENCES ARE SMALL

---

If values grow slowly:

$\backslash [ 10, 15, 21, 28, 36 \backslash ]$

check differences first.

---

## 59. WHEN VALUES GROW FAST

---

If:

$\backslash [ 2, 8, 32, 128 \backslash ]$

check multiplication first.

No need for difference table.

---

## 60. WHEN SIGNS ALTERNATE

---

Example:

$\backslash [ 5, -10, 20, -40, ? \backslash ]$

Rule:

$\backslash [ \times (-2) \backslash ]$

Next:

$\backslash [ \boxed{80} \backslash ]$

---

## 61. ALTERNATING SIGNS WITH POWERS

---

$\backslash [ 1, -2, 4, -8, 16, ? \backslash ]$

$\backslash [ \times (-2) \backslash ]$

Next:

$\backslash [ \boxed{-32} \backslash ]$

---

## 62. MEMORY LIST — NUMBER PATTERNS TO RECOGNIZE

---

### Squares

$\backslash [ 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 \backslash ]$

### Cubes

$\backslash [ 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000 \backslash ]$

### Primes

$\backslash [ 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31 \backslash ]$

### Fibonacci

$\backslash [ 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 \backslash ]$

### Triangular

$\backslash [ 1, 3, 6, 10, 15, 21, 28, 36 \backslash ]$

### Factorials

$\backslash [ 1, 2, 6, 24, 120, 720 \backslash ]$

---

## 63. COMMON SERIES TRAPS

---

### & pTRAP 1 — Overcomplicating

Try simple rules first.

### & pTRAP 2 — Using only first 2 terms

A valid pattern must fit all available terms.

### & pTRAP 3 — Ignoring odd/even positions

Very common placement pattern.

**& pTRAP 4 — Confusing absolute difference with multiplier**

Fast-growing sequences usually suggest multiplication.

**& pTRAP 5 — Wrong-term detection**

A wrong term disturbs two neighboring relationships.

**& pTRAP 6 — Assuming linear trend**

Increasing values need not have constant difference.

---

## 64. FAST SERIES DECISION TREE

---

**Small steady growth?**

Check differences.

**Differences change regularly?**

Check second differences / square-type pattern.

**Rapid growth?**

Check multiplication/division.

**Operations seem to alternate?**

Split odd/even positions.

**Familiar numbers?**

Check squares/cubes/primes/factorials.

**Each term resembles previous two?**

Check Fibonacci-type rule.

**One suspicious term?**

Build difference table around it.

---

## PRACTICE — LEVEL 0

---

**Q1**

\[ 4,8,12,16,? \]

**Q2**

\[ 3,6,12,24,? \]

**Q3**

\[ 1,4,9,16,? \]

**Q4**

\[ 2,3,5,8,13,? \]

---

**Q5**

\[ 100,50,25,12.5,? \]

---

## LEVEL 1

---

**Q6**

\[ 2,5,9,14,20,? \]

**Q7**

\[ 3,8,15,24,35,? \]

**Q8**

\[ 4,9,19,39,79,? \]

**Q9**

\[ 5,10,8,16,14,28,? \]

**Q10**

\[ 10,12,15,20,27,? \]

---

## LEVEL 2

---

**Q11**

\[ 1,5,14,30,55,? \]

**Q12**

\[ 2,4,12,48,240,? \]

**Q13**

\[ 3,12,5,20,7,28,? \]

**Q14**

\[ 1,10,100,2,20,200,3,30,? \]

**Q15**

Find the wrong term:

\[ 2,4,8,16,31,64 \]

---

## LEVEL 3 — EXAM PRESSURE

---

**Q16**

\[ 2,7,16,29,46,? \]

**Q17**

\[ 3,10,31,94,? \]

---

**Q18**

\[ 7,14,12,24,22,44,? \]

**Q19**

Find wrong term:

\[ 5,10,17,26,38,50 \]

**Q20**

\[ 2,6,15,31,56,? \]

---

**ANSWERS**

---

**Q1**

\[ +4 \]

\[ \boxed{20} \]

**Q2**

\[ \times 2 \]

\[ \boxed{48} \]

**Q3**

Squares:

\[ 1^2,2^2,3^2,4^2 \]

Next:

\[ 5^2 = \boxed{25} \]

**Q4**

\[ 8+13 = \boxed{21} \]

**Q5**

\[ \div 2 \]

\[ \boxed{6.25} \]

---

**Q6**

Differences:

\[ 3,4,5,6 \]

Next:

\[ 7 \]

---

$$\backslash [ 20+7=\boxed{27} \backslash ]$$

## Q7

---

Differences:

$$\backslash [ 5,7,9,11 \backslash ]$$

Next:

$$\backslash [ 13 \backslash ]$$

$$\backslash [ 35+13 = \boxed{48} \backslash ]$$

## Q8

---

$$\backslash [ \times 2+1 \backslash ]$$

$$\backslash [ 79(2)+1 = \boxed{159} \backslash ]$$

## Q9

---

Operations:

$$\backslash [ \times 2,-2,\times 2,-2,\times 2 \backslash ]$$

Next:

$$\backslash [ 28-2 = \boxed{26} \backslash ]$$

## Q10

---

Differences:

$$\backslash [ 2,3,5,7 \backslash ]$$

Next prime:

$$\backslash [ 11 \backslash ]$$

$$\backslash [ 27+11 = \boxed{38} \backslash ]$$

## Q11

---

Differences:

$$\backslash [ 4,9,16,25 \backslash ]$$

Next:

$$\backslash [ 36 \backslash ]$$

$$\backslash [ 55+36 = \boxed{91} \backslash ]$$

## Q12

---

Multipliers:

$$\backslash [ 2,3,4,5 \backslash ]$$

Next:

$$\sqrt{6}$$

$$\sqrt{240(6)} = \boxed{1440}$$

### Q13

---

Odd positions:

$$\sqrt{3,5,7,?}$$

Next:

$$\sqrt{9}$$

Even:

$$\sqrt{12,20,28}$$

So:

$$\sqrt{\boxed{9}}$$

### Q14

---

Three interwoven:

$$\sqrt{100,200,?}$$

Next:

$$\sqrt{\boxed{300}}$$

### Q15

---

Expected powers of 2:

$$\sqrt{2,4,8,16,32,64}$$

Wrong:

$$\sqrt{\boxed{31}}$$

### Q16

---

Differences:

$$\sqrt{5,9,13,17}$$

Second difference:

$$\sqrt{4}$$

Next difference:

$$\sqrt{21}$$

$$\sqrt{46+21} = \boxed{67}$$

### Q17

---

$$\sqrt{3 \times 3 + 1 = 10}$$

$$\sqrt{10 \times 3 + 1 = 31}$$

$$\backslash[ 31\times 3+1=94 \backslash]$$

Next:

$$\backslash[ 94\times 3+1 = \boxed{283} \backslash]$$

## Q18

---

Operations:

$$\backslash[ \times 2, -2, \times 2, -2, \times 2 \backslash]$$

Next:

$$\backslash[ 44-2 = \boxed{42} \backslash]$$

## Q19

---

Differences should be:

$$\backslash[ 5, 7, 9, 11, 13 \backslash]$$

Correct terms:

$$\backslash[ 5, 10, 17, 26, 37, 50 \backslash]$$

Wrong term:

$$\backslash[ \boxed{38} \backslash]$$

## Q20

---

Differences:

$$\backslash[ 4, 9, 16, 25 \backslash]$$

These are:

$$\backslash[ 2^2, 3^2, 4^2, 5^2 \backslash]$$

Next difference:

$$\backslash[ 6^2=36 \backslash]$$

Next term:

$$\backslash[ 56+36 = \boxed{92} \backslash]$$

## CHAPTER 57 STATUS

---

Covered so far:

- constant-difference series '
- increasing differences '
- second differences '
- multiplication/division '
- changing multipliers '
- multiply  $\pm$  constant '
- alternating operations '



- odd/even interwoven series '
- square/cube patterns '
- prime patterns '
- Fibonacci-type patterns '
- triangular numbers '
- factorials '
- fractional/decimal series '
- position-based formulas '
- missing terms '
- wrong terms '
- three-way interwoven series '
- speed strategy '
- L0–L3 number-series practice '

But **Chapter 57 is not complete yet.**

Still remaining:

- alphabet positions
- reverse alphabet positions
- letter jumps
- alternating letter series
- paired-letter series
- cyclic alphabet wrapping
- letter-number coding inside series
- alphanumeric series
- symbol/position questions
- counting from left/right
- mixed number-letter patterns
- final Chapter 57 speed audit

## Progress

$\boxed{56/76}$  completed

**Current:** Chapter 57 — Number, Letter & Alphanumeric Series

## NEXT

**Chapter 57 Part 2 — Letter Series, Alphabet Positions, Reverse Alphabet, Alphanumeric Patterns & Final Series Toolkit**

USER

next

## CHAPTER 57 — NUMBER, LETTER & ALPHANUMERIC SERIES

### PART 2 — LETTER SERIES, ALPHABET POSITIONS, REVERSE ALPHABET, CYCLIC PATTERNS & MIXED ALPHANUMERIC SERIES

Status:  $\emptyset$ =Not in PROGRESS ! ' ' COMPLETE GATE: High Placements: Very High Priority:  $\emptyset$ =Yes 1

Letter-series questions are number-series questions in disguise.

The core trick is:

**Convert letters into positions, solve the numerical pattern, then convert back.**

## 1. ALPHABET POSITION TABLE

### MUST MEMORIZE

Letter	Pos	Letter	Pos
A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

## 2. FAST MEMORY ANCHORS

Instead of memorizing every letter independently, remember:

[ A=1, \quad E=5, \quad J=10, \quad M=13, \quad P=16, \quad T=20, \quad Z=26 \ ]

Then count nearby letters quickly.

---

### 3. REVERSE ALPHABET POSITION

---

Reverse positions:

$Z=1, Y=2, X=3, \dots, A=26$

Shortcut:

$\boxed{\text{Reverse position} = 27 - \text{normal position}}$

Example:

Normal position of F:

6

Reverse:

$27 - 6 = 21$

So:

$\boxed{F=21 \text{ from the right}}$

---

### 4. COMPLEMENTARY LETTER PAIRS

---

Letters whose normal positions sum to 27:

A-Z

B-Y

C-X

D-W

E-V

F-U

G-T

H-S

I-R

J-Q

K-P

L-O

M-N

Memory rule:

$\boxed{\text{paired positions sum to } 27}$

---

### 5. SIMPLE LETTER SERIES

---

\[ A,C,E,G,? \]

Positions:

\[ 1,3,5,7 \]

Difference:

\[ +2 \]

Next:

\[ 9=I \]

\[ \boxed{I} \]

---

## 6. CONSTANT LETTER JUMP

---

\[ B,E,H,K,? \]

Positions:

\[ 2,5,8,11 \]

\[ +3 \]

Next:

\[ 14=N \]

\[ \boxed{N} \]

---

## 7. DECREASING LETTER SERIES

---

\[ Z,W,T,Q,? \]

Positions:

\[ 26,23,20,17 \]

\[ -3 \]

Next:

\[ 14=N \]

\[ \boxed{N} \]

---

## 8. CHANGING LETTER JUMPS

---

\[ A,C,F,J,O,? \]

Positions:

\[ 1,3,6,10,15 \]

Differences:

\[ 2,3,4,5 \]

Next jump:

---

$\boxed{6}$

$15+6=21$

$21 = U$ .

$\boxed{U}$

---

## 9. SECOND-DIFFERENCE LETTER SERIES

---

$B, E, J, Q, ?$

Positions:

$2, 5, 10, 17$

Differences:

$3, 5, 7$

Next difference:

$9$

$17+9=26$

$\boxed{Z}$

---

## 10. CYCLIC ALPHABET WRAPAROUND

---

If you go beyond Z, continue from A.

Example:

$W, Z, C, F, ?$

Positions conceptually:

$23, 26, 29, 32$

Modulo 26:

$29 \rightarrow C \quad 32 \rightarrow F$

Next:

$35$

$35-26=9$

$9 = I$ .

$\boxed{I}$

---

## 11. FAST WRAP RULE

---

For forward jumps:

$\boxed{\text{new position}} = (p+k) \bmod 26$

Practically:

---

If answer exceeds 26, subtract 26.

Example:

$\backslash [ Y+5 \backslash ]$

$\backslash [ 25+5=30 \backslash ]$

$\backslash [ 30-26=4 \backslash ]$

4 = D.

$\backslash [ \boxed{D} \backslash ]$

---

## 12. BACKWARD WRAP

---

Example:

$\backslash [ C-5 \backslash ]$

Position:

$\backslash [ 3-5=-2 \backslash ]$

Add 26:

$\backslash [ 24 \backslash ]$

24 = X.

$\backslash [ \boxed{X} \backslash ]$

---

## 13. ALTERNATING LETTER SERIES

---

$\backslash [ A,Z,C,X,E,? \backslash ]$

Odd positions:

$\backslash [ A,C,E \backslash ]$

$\backslash [ +2 \backslash ]$

Even positions:

$\backslash [ Z,X,? \backslash ]$

$\backslash [ -2 \backslash ]$

Next even:

$\backslash [ V \backslash ]$

$\backslash [ \boxed{V} \backslash ]$

---

## 14. ANOTHER ALTERNATING SERIES

---

$\backslash [ B,D,F, \backslash Z,X,V, \backslash ? \backslash ]$

This can be interpreted by grouping:

First stream:

$\{ B, D, F, \dots \}$

Second:

$\{ Z, X, V, \dots \}$

If presented as alternating:

$\{ B, Z, D, X, F, V, ? \}$

Odd positions:

$\{ B, D, F, ? \}$

Next:

$\{ H \}$

So:

$\boxed{H}$

---

## 15. OPPOSITE-END SERIES

---

$\{ A, Z, B, Y, C, X, ? \}$

Pattern:

- forward from A: A, B, C, ...
- backward from Z: Z, Y, X, ...

Next:

$\{ D \}$

$\boxed{D}$

---

## 16. LETTER PAIR SERIES

---

$\{ AZ, BY, CX, DW, ? \}$

First letters:

$\{ A, B, C, D, E \}$

Second letters:

$\{ Z, Y, X, W, V \}$

Next:

$\boxed{EV}$

---

## 17. PAIRS WITH POSITION SUM 27

---

In:

$\{ AZ, BY, CX, DW \}$

each pair satisfies:

$\text{position}(\text{first}) + \text{position}(\text{second}) = 27$

---

This is a common shortcut.

---

## 18. TWO-LETTER JUMP SERIES

---

`\[ AB,\ DE,\ GH,\ JK,\ ? \]`

First letters:

`\[ A,D,G,J \]`

`\[ +3 \]`

Second:

`\[ B,E,H,K \]`

`\[ +3 \]`

Next:

`\[ M,N \]`

`\[ \boxed{MN} \]`

---

## 19. CHANGING PAIR PATTERN

---

`\[ AC,\ DF,\ GI,\ JL,\ ? \]`

First letters:

`\[ A,D,G,J \]`

`\[ +3 \]`

Second:

`\[ C,F,I,L \]`

`\[ +3 \]`

Next:

`\[ M,O \]`

`\[ \boxed{MO} \]`

---

## 20. THREE-LETTER GROUPS

---

`\[ ABC,\ DEF,\ GHI,\ ? \]`

Next block:

`\[ \boxed{JKL} \]`

Simple, but placement tests may combine this with skipping.

---

## 21. SKIPPING BLOCKS

---

`\[ ABC,\ EFG,\ IJK,\ ? \]`

---



Starting letters:

\[ A,E,I \]

\[ +4 \]

Next start:

\[ M \]

So:

\[ \boxed{MNO} \]

---

## 22. LETTER SERIES BASED ON SQUARE POSITIONS

---

Positions:

\[ 1,4,9,16,25 \]

Letters:

\[ A,D,I,P,Y \]

So:

\[ A,D,I,P,? \]

Next visible square within alphabet:

\[ 25=Y \]

\[ \boxed{Y} \]

---

## 23. PRIME-POSITION LETTERS

---

Prime positions:

\[ 2,3,5,7,11,13,\ldots \]

Letters:

\[ B,C,E,G,K,M,\ldots \]

Example:

\[ B,C,E,G,K,? \]

Next prime 13:

\[ \boxed{M} \]

---

## 24. FIBONACCI-POSITION LETTER SERIES

---

Positions:

\[ 1,2,3,5,8,13,21 \]

Letters:

\[ A,B,C,E,H,M,U \]

---

Example:

\[ A,B,C,E,H,? \]

Next position:

\[ 13 \]

\[ \boxed{M} \]

---

## 25. LETTER SERIES FROM INCREASING JUMPS

---

\[ C,F,J,O,U,? \]

Positions:

\[ 3,6,10,15,21 \]

Differences:

\[ 3,4,5,6 \]

Next:

\[ +7 \]

\[ 21+7=28 \]

Wrap:

\[ 28-26=2=B \]

\[ \boxed{B} \]

---

## 26. NUMBER + LETTER SERIES

---

\[ A1,\ C3,\ E5,\ G7,\ ? \]

Letters:

\[ A,C,E,G \]

Numbers:

\[ 1,3,5,7 \]

Next:

\[ I9 \]

\[ \boxed{I9} \]

---

## 27. INDEPENDENT LETTER AND NUMBER PATTERNS

---

\[ B2,\ E4,\ H8,\ K16,\ ? \]

Letters:

\[ B,E,H,K \]

\[ +3 \]

---

Next:

$\sqrt{N}$

Numbers:

$\sqrt{2,4,8,16}$

$\sqrt{\times 2}$

Next:

$\sqrt{32}$

Answer:

$\sqrt{\boxed{N^3}}$

---

## 28. ALPHANUMERIC SERIES WITH REVERSE LETTERS

---

$\sqrt{A1, Z2, B3, Y4, C5, ?}$

Letters:

$\sqrt{A, Z, B, Y, C, X}$

Numbers:

$\sqrt{1, 2, 3, 4, 5, 6}$

Answer:

$\sqrt{\boxed{X6}}$

---

## 29. MIXED POSITION SERIES

---

$\sqrt{A2, C4, F8, J16, ?}$

Letters positions:

$\sqrt{1, 3, 6, 10}$

Differences:

$\sqrt{2, 3, 4}$

Next:

$\sqrt{15=O}$

Numbers:

$\sqrt{2, 4, 8, 16}$

Next:

$\sqrt{32}$

Answer:

$\sqrt{\boxed{O32}}$

### 30. LETTER + NUMBER RELATION WITH SAME POSITION

---

[ A1, \ D4, \ G7, \ J10, \ ? ]

Letters:

[ A, D, G, J ]

Positions:

[ 1, 4, 7, 10 ]

Numbers match positions.

Next position:

[ 13=M ]

So:

[ \boxed{M13} ]

---

### 31. NUMBER PRECEDES LETTER

---

[ 1A, \ 4D, \ 9I, \ 16P, \ ? ]

Numbers:

[ 1, 4, 9, 16, 25 ]

Letters correspond to those positions:

A, D, I, P, Y.

Answer:

[ \boxed{25Y} ]

---

### 32. MIXED SYMBOL SERIES

---

Example:

[ A, \ 2, \ C, \ 4, \ E, \ 6, \ ? ]

Odd positions:

[ A, C, E, ? ]

Next:

[ G ]

Even:

[ 2, 4, 6 ]

Answer:

[ \boxed{G} ]

---

### 33. COMPLEX ALPHANUMERIC INTERLEAVING

---

$\{ A1, B2, D4, G7, K11, ? \}$

Letters positions:

$\{ 1, 2, 4, 7, 11 \}$

Differences:

$\{ 1, 2, 3, 4 \}$

Next position:

$\{ 16=P \}$

Numbers:

$\{ 1, 2, 4, 7, 11 \}$

same pattern.

Next number:

$\{ 16 \}$

So:

$\{ \boxed{P16} \}$

---

## 34. SYMBOL-POSITION QUESTIONS

Suppose sequence:

$\{ A, 5, @, B, 7, \#, C, 9, \$ \}$

Question may ask:

**What is third element to the right of B?**

B is position 4.

Third to right:

$\{ 4+3=7 \}$

Position 7:

$\{ C \}$

$\{ \boxed{C} \}$

---

## 35. LEFT/RIGHT POSITION RULE

If an item is at position  $\{p\}$ :

**kth to the right:**

$\{ \boxed{p+k} \}$

**kth to the left:**

$\{ \boxed{p-k} \}$

Check sequence boundaries.

---

### 36. “Kth FROM LEFT” VS “Kth TO THE LEFT”

---

These are different.

“Kth from left” = absolute position.

“Kth to the left of X” = relative position.

This wording trap is common.

---

### 37. POSITION FROM RIGHT

---

If total number of elements =  $N$ .

An element at position  $L$  from left has position from right:

$$N - L + 1$$

---

### 38. EXAMPLE

---

Sequence has 20 elements.

A is 7th from left.

From right:

$$20 - 7 + 1$$

$$= 14$$

---

### 39. FIND TOTAL FROM LEFT + RIGHT POSITION

---

If an element is:

- 8th from left
- 13th from right

Total elements:

$$8 + 13 - 1$$

$$= 20$$

Formula:

$$N = L + R - 1$$

---

### 40. WHY SUBTRACT 1?

---

Because the element itself is counted in both:

- left-position count
  - right-position count
-

So subtract one duplicate.

---

## 41. COUNT ELEMENTS BETWEEN TWO POSITIONS

---

If A is 5th from left and B is 12th from left:

Elements strictly between:

$\lfloor 12-5-1 \rfloor$

$\lfloor \boxed{6} \rfloor$

Formula:

$\lfloor \boxed{p_1-p_2-1} \rfloor$

---

## 42. CONDITION-BASED ALPHANUMERIC COUNTING

---

Suppose:

$\lfloor A, 2, B, 4, C, 6, D, 8 \rfloor$

Question:

**How many numbers are immediately preceded by a letter?**

2 preceded by A ' 4 preceded by B ' 6 preceded by C ' 8 preceded by D '

Answer:

$\lfloor \boxed{4} \rfloor$

---

## 43. “PRECEDED BY” AND “FOLLOWED BY”

---

For element X:

- immediately preceded by = element directly before X
- immediately followed by = element directly after X

Read carefully.

---

## 44. BOTH-SIDE CONDITION

---

Question:

**How many numbers are immediately preceded by a letter and followed by a letter?**

Sequence:

$\lfloor A, 2, B, 4, C, 6, D \rfloor$

2 ! ' A before, B after ' 4 ! ' B before, C after ' 6 ! ' C before, D after '

Answer:

---

$\boxed{3}$

---

## 45. ALPHANUMERIC REARRANGEMENT QUESTIONS

---

Suppose:

$\{B, 3, A, 1, C, 2\}$

Instruction:

**Arrange letters alphabetically first, then numbers ascending.**

Result:

$\{A, B, C, 1, 2, 3\}$

Questions may then ask positions after rearrangement.

Always perform stated transformation exactly.

---

## 46. LETTER SHIFT CODING INSIDE SERIES

---

$\{A, D, H, M, ?\}$

Jumps:

$\{+3, +4, +5\}$

Next:

$\{+6\}$

M position 13:

$\{13+6=19\}$

19 = S.

$\boxed{S}$

---

## 47. REVERSE-JUMP SERIES

---

$\{Z, V, Q, K, ?\}$

Positions:

$\{26, 22, 17, 11\}$

Differences:

$\{-4, -5, -6\}$

Next:

$\{-7\}$

$\{11-7=4\}$

4 = D.

---



$\boxed{D}$

---

## 48. WRONG LETTER IN SERIES

---

$\{ A, C, F, J, O, T \}$

Expected positions with jumps:

$\{ +2, +3, +4, +5, +6 \}$

A ! C ! F ! J ! O ! U

So:

$\boxed{T}$

is wrong.

Correct term:

$\boxed{U}$

---

## 49. WRONG LETTER — DIFFERENCE METHOD

---

Convert:

$\{ A, C, F, J, O, T \}$

to:

$\{ 1, 3, 6, 10, 15, 20 \}$

Differences:

$\{ 2, 3, 4, 5, 5 \}$

Expected final difference:

$\{ 6 \}$

Hence final position:

$\{ 21=U \}$

---

## 50. WRONG ALPHANUMERIC TERM

---

$\{ A1, C3, E5, G8, I9 \}$

Expected:

$\{ A1, C3, E5, G7, I9 \}$

Wrong:

$\boxed{G8}$

---

## 51. THREE-WAY ALPHANUMERIC SERIES

---

$\{ A1, B10, C100, D2, E20, F200, G3, H30, ? \}$

---

Look in blocks:

Letters:

\[ A,B,C,D,E,F,G,H,I \]

Numbers:

\[ 1,10,100, \ 2,20,200, \ 3,30,300 \]

Answer:

\[ \boxed{I300} \]

---

## 52. RELATIVE ALPHABET POSITION QUESTIONS

---

Question:

**Which letter is 5 places to the right of H?**

H = 8.

\[ 8+5=13 \]

13 = M.

\[ \boxed{M} \]

---

## 53. FROM RIGHT END OF ALPHABET

---

Question:

**7th letter from the right?**

Normal position:

\[ 27-7=20 \]

20 = T.

\[ \boxed{T} \]

---

## 54. MIXED LEFT/RIGHT LETTER QUESTION

---

Question:

**Which letter is 4th to the left of the 8th letter from the right?**

8th from right:

\[ 27-8=19 \]

19 = S.

Four left:

$$\lfloor 19-4=15 \rfloor$$

$$15 = O.$$

$$\lfloor \boxed{O} \rfloor$$

---

## 55. LETTER BETWEEN TWO LETTERS

---

Between D and K:

Positions:

$$\lfloor 4, \lfloor 11 \rfloor$$

Letters strictly between:

$$\lfloor 11-4-1 = \boxed{6} \rfloor$$

They are:

E, F, G, H, I, J.

---

## 56. MIDDLE LETTER

---

If letters have positions  $\lfloor a \rfloor$  and  $\lfloor b \rfloor$ , midpoint position:

$$\lfloor \frac{a+b}{2} \rfloor$$

if this is an integer.

Example:

$$F = 6 \quad N = 14$$

Midpoint:

$$\lfloor 10 \rfloor$$

$$10 = J.$$

$$\lfloor \boxed{J} \rfloor$$

---

## 57. COMMON ALPHABET TRAPS

---

& pTRAP 1 — Off-by-one

A is 1, not 0.

& pTRAP 2 — Reverse alphabet

Use  $\lfloor (27-p) \rfloor$ .

& pTRAP 3 — Going beyond Z

Wrap around.

& pTRAP 4 — Odd/even streams

Check separate subsequences.

& pTRAP 5 — Letter pair

---

Track first and second letters separately.

#### & pTRAP 6 — Left/right wording

“from” and “to the right of” are different.

---

## 58. COMMON ALPHANUMERIC TRAPS

---

### & pTRAP 1

Numbers and letters may follow independent patterns.

### & pTRAP 2

The numeric part may equal alphabet position.

### & pTRAP 3

Series may consist of repeating 2-term or 3-term blocks.

### & pTRAP 4

Symbols may matter only by position, not meaning.

### & pTRAP 5

Do not assume every term has one combined rule; split components first.

---

## 59. FAST SERIES MASTER SEARCH ORDER

---

When a series appears:

### NUMBER SERIES

1. differences
2. second differences
3. ratios
4.  $\times/\div$  followed by  $\pm$
5. odd/even split
6. squares/cubes/primes
7. previous terms

### LETTER SERIES

1. convert to positions
2. check jumps
3. odd/even split
4. reverse alphabet
5. wraparound
6. paired streams

### ALPHANUMERIC

1. split letter and number parts
2. solve both separately
3. check whether they are linked

---

## 60. PLACEMENT SPEED TARGET

---

Simple series:

$\boxed{15-25 \text{ sec}}$

Moderate series:

$\boxed{30-45 \text{ sec}}$

If no clean pattern after about 45–60 seconds in a speed test:

**mark and move.**

Do not invent complicated patterns.

---

## 61. PATTERN VALIDATION RULE

---

Before accepting a rule, ensure it explains:

$\boxed{\text{all available transitions}}$

A rule that explains only the last two terms is weak.

Prefer the simplest rule consistent with all terms.

---

## 62. MUST-MEMORIZE ALPHABET SHORTCUTS

---

$\boxed{A=1, M=13, Z=26}$

$\boxed{\text{Reverse position} = 27 - p}$

$\boxed{\text{Opposite letters sum to } 27}$

$\boxed{N = L + R - 1}$

$\boxed{\text{between two positions} = |p_1 - p_2| - 1}$

---

## PRACTICE — LEVEL 0

---

**Q1**

$\boxed{A, C, E, G, ?}$

**Q2**

$\boxed{B, E, H, K, ?}$

**Q3**

$\boxed{Z, X, V, T, ?}$

**Q4**

5th letter from right?

---

### Q5

Opposite letter of D?

---

## LEVEL 1

---

### Q6

\[ A,C,F,J,O,?\]

### Q7

\[ A,Z,B,Y,C,?\]

### Q8

\[ AZ,BY,CX,DW,?\]

### Q9

\[ A1,C3,E5,G7,?\]

### Q10

\[ B2,E4,H8,K16,?\]

---

## LEVEL 2

---

### Q11

\[ B,E,J,Q,?\]

### Q12

\[ C,F,J,O,U,?\]

### Q13

\[ A1,B2,D4,G7,K11,?\]

### Q14

\[ 1A,4D,9I,16P,?\]

### Q15

Find wrong term:

\[ A,C,F,J,O,T\]

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

\[ A,Z,C,X,E,V,?\]

### Q17

\[ Y3,V6,R12,M24,?\]

---

### Q18

$\{A1, B10, C100, D2, E20, F200, G3, H30, ?\}$

### Q19

An element is 9th from left and 14th from right. Total elements?

### Q20

In a sequence, X is 7th from left and Y is 15th from left. How many elements lie strictly between X and Y?

---

## ANSWERS

---

### Q1

$\{+2\}$

Next:

$\{1\}$

$\boxed{1}$

### Q2

$\{+3\}$

$K = 11$ :

$\{11+3=14=N\}$

$\boxed{N}$

### Q3

$\{-2\}$

$T = 20$ :

$\{18=R\}$

$\boxed{R}$

### Q4

5th from right:

$\{27-5=22\}$

$22 = V$ .

$\boxed{V}$

### Q5

D position 4.

Opposite:

$\{27-4=23\}$

---

23 = W.

$\boxed{W}$

---

## Q6

---

Positions:

$\{1, 3, 6, 10, 15\}$

Differences:

$\{2, 3, 4, 5\}$

Next:

$+6$

Position:

$21 = U$

$\boxed{U}$

## Q7

---

Opposite-end alternation:

$\{A, Z, B, Y, C, X\}$

$\boxed{X}$

## Q8

---

$\boxed{EV}$

## Q9

---

Letters:

$\{A, C, E, G, I\}$

Numbers:

$\{1, 3, 5, 7, 9\}$

$\boxed{I9}$

## Q10

---

Letter jump +3:

$\{B, E, H, K, N\}$

Numbers double:

$\{2, 4, 8, 16, 32\}$

$\boxed{N32}$

---



## Q11

---

Positions:

$\backslash [ 2,5,10,17 \backslash ]$

Differences:

$\backslash [ 3,5,7 \backslash ]$

Next:

$\backslash [ +9 \backslash ]$

$\backslash [ 17+9=26 \backslash ]$

$\backslash [ \boxed{Z} \backslash ]$

## Q12

---

Positions:

$\backslash [ 3,6,10,15,21 \backslash ]$

Differences:

$\backslash [ 3,4,5,6 \backslash ]$

Next:

$\backslash [ +7 \backslash ]$

$\backslash [ 28\text{to}2=B \backslash ]$

$\backslash [ \boxed{B} \backslash ]$

## Q13

---

Positions/numbers:

$\backslash [ 1,2,4,7,11 \backslash ]$

Differences:

$\backslash [ 1,2,3,4 \backslash ]$

Next:

$\backslash [ 16 \backslash ]$

$16 = P.$

$\backslash [ \boxed{P16} \backslash ]$

## Q14

---

Squares:

$\backslash [ 1,4,9,16,25 \backslash ]$

25th letter:

$\backslash [ Y \backslash ]$

$\backslash [ \boxed{25Y} \backslash ]$

## Q15

---

Expected jumps:

$\backslash [ +2,+3,+4,+5,+6 \backslash ]$

Correct final letter:

$\backslash [ U \backslash ]$

Thus wrong term:

$\backslash [ \boxed{T} \backslash ]$

---

## Q16

---

Odd positions:

$\backslash [ A,C,E,? \backslash ]$

Next:

$\backslash [ G \backslash ]$

Even positions:

$\backslash [ Z,X,V \backslash ]$

So answer:

$\backslash [ \boxed{G} \backslash ]$

---

## Q17

---

Letters:

$Y = 25 \ V = 22 \ R = 18 \ M = 13$

Differences:

$\backslash [ -3,-4,-5 \backslash ]$

Next:

$\backslash [ -6 \backslash ]$

$\backslash [ 13-6=7=G \backslash ]$

Numbers:

$\backslash [ 3,6,12,24 \backslash ]$

Next:

$\backslash [ 48 \backslash ]$

Answer:

$\backslash [ \boxed{G48} \backslash ]$

---

## Q18

---

Block pattern:

$\backslash [ 1,10,100 \backslash ]$

---

$\lfloor 2,20,200 \rfloor$

$\lfloor 3,30,300 \rfloor$

Letters move sequentially.

Next:

$\lfloor \boxed{1300} \rfloor$

## Q19

$\lfloor N=L+R-1 \rfloor$

$\lfloor 9+14-1 = \boxed{22} \rfloor$

## Q20

$\lfloor 15-7-1 = \boxed{7} \rfloor$

## ' CHAPTER 57 COMPLETE

We have covered:

- number-series fundamentals '
- difference and second-difference patterns '
- multiplication/division patterns '
- alternating and interwoven series '
- square/cube/prime/factorial patterns '
- Fibonacci-style patterns '
- missing and wrong terms '
- alphabet positions '
- reverse alphabet '
- opposite-letter pairs '
- letter jumps '
- wraparound '
- alternating letter series '
- letter-pair series '
- square/prime-position letters '
- alphanumeric series '
- independent letter-number patterns '
- symbol/position questions '
- left/right counting '
- wrong alphanumeric terms '
- speed strategy '
- L0–L3 practice '

## Progress

$\lfloor \boxed{57/76} \text{numbered chapters completed} \rfloor$

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Number, Letter & Alphanumeric Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out +
- Chapter 59 — Coding-Decoding +
- Chapter 60 — Blood Relations +
- Chapter 61 — Direction Sense +
- Chapter 62 — Ranking & Ordering +
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

### +CHAPTER 58 — ANALOGY, CLASSIFICATION & ODD-ONE-OUT

Next we will cover:

- word analogies
- number analogies
- letter analogies
- relationship types
- function and part-whole relations
- cause-effect and degree relations
- numerical transformation analogies
- classification
- odd-one-out by category
- odd-one-out by numeric property
- disguised patterns
- GATE/placement elimination techniques
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 58 — ANALOGY, CLASSIFICATION & ODD-ONE-OUT

## PART 1 — WORD ANALOGIES, NUMBER ANALOGIES, LETTER ANALOGIES & RELATIONSHIP IDENTIFICATION

Status: ~~0=Not~~ PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Analogy questions test whether you can identify a relationship in one pair and apply the **same relationship** to another pair.

General form:

$A:B::C:? \quad \backslash$

Read it as:

**A is related to B in the same way C is related to ?**

The core rule is:

**Identify the relationship first. Do not choose an option merely because it looks associated with C.**

### 1. BASIC WORD ANALOGY

$\backslash$  Bird:Fly::Fish:?  $\backslash$

Bird moves by flying.

Fish moves by swimming.

$\backslash$   $\boxed{\text{Swim}}$   $\backslash$

Relationship:

$\backslash$   $\text{creature}:\text{mode of movement}$   $\backslash$

### 2. RELATIONSHIP MUST MATCH DIRECTION

Example:

$\backslash$  Doctor:Hospital::Teacher:?  $\backslash$

Doctor works in hospital.

Teacher works in:

$\backslash$   $\boxed{\text{School}}$   $\backslash$

Direction:

$\backslash$   $\text{profession}:\text{workplace}$   $\backslash$

Do not reverse it to workplace:profession.

### 3. OBJECT : FUNCTION

$\backslash$  Knife:Cut::Pen:?  $\backslash$

Knife is used to cut.

Pen is used to:

\[ \boxed{Write} \]

---

## 4. TOOL : USER

\[ Stethoscope:Doctor::Hammer:? \]

Stethoscope is commonly used by doctor.

Hammer is commonly used by:

\[ \boxed{Carpenter} \]

---

## 5. PERSON : TOOL

Reverse direction:

\[ Doctor:Stethoscope::Carpenter:? \]

Answer:

\[ \boxed{Hammer} \]

Notice direction changed.

---

## 6. WORKER : WORKPLACE

\[ Chef:Kitchen::Judge:? \]

Chef works in kitchen.

Judge works in:

\[ \boxed{Court} \]

---

## 7. PRODUCT : SOURCE

\[ Milk:Cow::Wool:? \]

Milk obtained from cow.

Wool obtained from:

\[ \boxed{Sheep} \]

---

## 8. SOURCE : PRODUCT

Reverse:

\[ Cow:Milk::Bee:? \]

Bee produces:

\[ \boxed{Honey} \]

---

---

## 9. PART : WHOLE

\[ Wheel:Car::Page:? \]

Wheel is a part of car.

Page is part of:

\[ \boxed{Book} \]

---

## 10. WHOLE : PART

\[ Tree:Branch::Body:? \]

Branch is part of tree.

A corresponding body part:

\[ \boxed{Arm} \]

---

## 11. MEMBER : GROUP

\[ Soldier:Army::Player:? \]

Soldier belongs to army.

Player belongs to:

\[ \boxed{Team} \]

---

## 12. GROUP : MEMBER

\[ Fleet:Ship::Herd:? \]

Fleet consists of ships.

Herd consists of:

\[ \boxed{Cattle} \]

---

## 13. ANIMAL : YOUNG ONE

\[ Dog:Puppy::Cat:? \]

\[ \boxed{Kitten} \]

---

## 14. ANIMAL : HOME

\[ Lion:Den::Bee:? \]

\[ \boxed{Hive} \]

---

## 15. ANIMAL : SOUND

---

\[ Dog:Bark::Lion:? \]

\[ \boxed{Roar} \]

---

## 16. MALE : FEMALE

---

\[ King:Queen::Bull:? \]

\[ \boxed{Cow} \]

---

## 17. OBJECT : MATERIAL

---

\[ Table:Wood::Ring:? \]

Table may be made of wood.

Ring may be made of:

\[ \boxed{Gold} \]

---

## 18. MATERIAL : OBJECT

---

\[ Wood:Furniture::Clay:? \]

Clay can be used to make:

\[ \boxed{Pottery} \]

---

## 19. COUNTRY : CAPITAL

---

\[ India:New\ Delhi::Japan:? \]

\[ \boxed{Tokyo} \]

This is a factual relation rather than a transformation.

---

## 20. COUNTRY : CURRENCY

---

\[ Japan:Yen::USA:? \]

\[ \boxed{Dollar} \]

---

## 21. STUDY : SUBJECT

---

\[ Botany:Plants::Zoology:? \]

Botany studies plants.

Zoology studies:

\[ \boxed{Animals} \]

---



---

## 22. DISEASE : ORGAN

\[ Hepatitis:Liver::Pneumonia:? \]

Pneumonia primarily affects:

\[ \boxed{\text{Lungs}} \]

---

## 23. CAUSE : EFFECT

\[ Virus:Disease::Fire:? \]

Fire may cause:

\[ \boxed{\text{Burn}} \]

or destruction depending on options.

The precise relation should match the options.

---

## 24. EFFECT : CAUSE

\[ Flood:Rain::Disease:? \]

Flood may result from rain.

Disease may result from:

\[ \boxed{\text{Infection}} \]

Direction matters again.

---

## 25. DEGREE RELATION

\[ Warm:Hot::Cool:? \]

Hot is stronger degree of warm.

Stronger degree of cool:

\[ \boxed{\text{Cold}} \]

---

## 26. OPPOSITES

\[ Hot:Cold::Day:? \]

\[ \boxed{\text{Night}} \]

---

## 27. SYNONYMS

\[ Rapid:Fast::Silent:? \]

\[ \boxed{\text{Quiet}} \]

---

---

## 28. WORD : OPPOSITE

\[ Expand:Contract::Accept:? \]

\[ \boxed{Reject} \]

---

## 29. VERB : NOUN

\[ Decide:Decision::Move:? \]

\[ \boxed{Movement} \]

---

## 30. NOUN : ADJECTIVE

\[ Danger:Dangerous::Beauty:? \]

\[ \boxed{Beautiful} \]

---

## 31. OBJECT : MEASUREMENT UNIT

\[ Length:Metre::Mass:? \]

\[ \boxed{Kilogram} \]

---

## 32. QUANTITY : INSTRUMENT

\[ Temperature:Thermometer::Time:? \]

\[ \boxed{Clock} \]

---

## 33. INSTRUMENT : QUANTITY

Reverse:

\[ Barometer:Pressure::Thermometer:? \]

\[ \boxed{Temperature} \]

---

## 34. PLACE : SPECIAL FEATURE

\[ Desert:Sand::Mountain:? \]

Possible relation:

\[ \boxed{Peak} \]

But this type can be ambiguous.

Choose only if the pair relationship is precise and options support it.

---

---

## 35. SEQUENCE / ORDER RELATION

---

$\backslash [ \text{Monday:Tuesday::April:?} \backslash ]$

Next after Monday is Tuesday.

Next after April:

$\backslash [ \boxed{\text{May}} \backslash ]$

---

## 36. PREDECESSOR RELATION

---

$\backslash [ \text{Tuesday:Monday::May:?} \backslash ]$

Previous month:

$\backslash [ \boxed{\text{April}} \backslash ]$

---

## 37. NUMBER ANALOGY — ADDITION

---

$\backslash [ 5:8::12:? \backslash ]$

Rule:

$\backslash [ +3 \backslash ]$

Thus:

$\backslash [ 12+3 = \boxed{15} \backslash ]$

---

## 38. NUMBER ANALOGY — MULTIPLICATION

---

$\backslash [ 4:20::7:? \backslash ]$

Rule:

$\backslash [ \times 5 \backslash ]$

$\backslash [ 7(5)=\boxed{35} \backslash ]$

---

## 39. NUMBER ANALOGY — SQUARE

---

$\backslash [ 6:36::9:? \backslash ]$

$\backslash [ 6^2=36 \backslash ]$

Therefore:

$\backslash [ 9^2 = \boxed{81} \backslash ]$

---

## 40. NUMBER ANALOGY — CUBE

---

$\backslash [ 3:27::4:? \backslash ]$

---

$$\backslash[ 3^3=27 \backslash]$$

$$\backslash[ 4^3 = \boxed{64} \backslash]$$

---

## 41. NUMBER ANALOGY — SQUARE + CONSTANT

---

$$\backslash[ 4:17::6:? \backslash]$$

Check:

$$\backslash[ 4^2+1=17 \backslash]$$

Then:

$$\backslash[ 6^2+1 = \boxed{37} \backslash]$$

---

## 42. NUMBER ANALOGY — PRODUCT OF CONSECUTIVES

---

$$\backslash[ 4:20::7:? \backslash]$$

Another possible rule:

$$\backslash[ n(n+1) \backslash]$$

$$\backslash[ 4(5)=20 \backslash]$$

Then:

$$\backslash[ 7(8)=56 \backslash]$$

So:

$$\backslash[ \boxed{56} \backslash]$$

This demonstrates something important:

**A single pair can support multiple rules.**

Therefore analogy questions normally provide answer options or more structure.

Choose the **simplest and most natural relation consistent with the options**.

---

## 43. NUMBER ANALOGY — DIGIT SUM

---

$$\backslash[ 123:6::452:? \backslash]$$

$$\backslash[ 1+2+3=6 \backslash]$$

Then:

$$\backslash[ 4+5+2 = \boxed{11} \backslash]$$

---

## 44. DIGIT PRODUCT

---

$$\backslash[ 123:6::234:? \backslash]$$

$$\backslash[ 1\times 2\times 3=6 \backslash]$$

$$2 \times 3 \times 4 = \boxed{24}$$

---

## 45. REVERSE DIGITS

---

$$123:321::450:?$$

Reverse:

$$\boxed{054}$$

Usually interpreted as:

$$\boxed{54}$$

unless fixed digit positions matter.

---

## 46. SUM OF SQUARES OF DIGITS

---

$$23:13::34:?$$

$$2^2+3^2=4+9=13$$

Then:

$$3^2+4^2=9+16 = \boxed{25}$$

---

## 47. DIFFERENCE OF SQUARES

---

$$5:24::7:?$$

Possible rule:

$$n^2-1$$

$$25-1=24$$

Then:

$$49-1 = \boxed{48}$$

---

## 48. NUMBER-PAIR ANALOGY

---

$$(2,3):6::(4,5):?$$

Relation:

$$2 \times 3 = 6$$

Then:

$$4 \times 5 = \boxed{20}$$

---

## 49. PAIR !' SUM

---

$$(4,7):11::(8,9):?$$

---

$$\backslash[ 8+9 = \boxed{17} \backslash]$$

---

## 50. PAIR !' DIFFERENCE

---

$$\backslash[ (9,4):5::(12,7):? \backslash]$$

$$\backslash[ 12-7 = \boxed{5} \backslash]$$

---

## 51. PAIR !' SQUARE SUM

---

$$\backslash[ (3,4):25::(5,12):? \backslash]$$

$$\backslash[ 3^2+4^2=25 \backslash]$$

Then:

$$\backslash[ 5^2+12^2=25+144 = \boxed{169} \backslash]$$

---

## 52. LETTER ANALOGY — CONSTANT SHIFT

---

$$\backslash[ A:D::F:? \backslash]$$

A!' D :

$$\backslash[ +3 \backslash]$$

F = 6:

$$\backslash[ 6+3=9=I \backslash]$$

$$\backslash[ \boxed{I} \backslash]$$

---

## 53. LETTER ANALOGY — BACKWARD SHIFT

---

$$\backslash[ H:E::M:? \backslash]$$

$$\backslash[ -3 \backslash]$$

M = 13:

$$\backslash[ 13-3=10=J \backslash]$$

$$\backslash[ \boxed{J} \backslash]$$

---

## 54. LETTER PAIR ANALOGY

---

$$\backslash[ AB:DE::FG:? \backslash]$$

A!' D :

$$\backslash[ +3 \backslash]$$

B!' E :

$$\backslash[ +3 \backslash]$$

---

Therefore:

F!' IG!' J

$\boxed{IJ}$

---

## 55. REVERSE-LETTER ANALOGY

---

$ABC:CBA::DEF:?$

Reverse order:

$\boxed{FED}$

---

## 56. OPPOSITE-ALPHABET ANALOGY

---

$A:Z::C:?$

Opposite alphabet pair:

$C:X$

$\boxed{X}$

Using:

$p+q=27$

---

## 57. LETTER POSITION TRANSFORMATION

---

$C:9::E:?$

C position:

$3$

Output:

$3^2=9$

E position:

$5$

$5^2=25$

$\boxed{25}$

---

## 58. LETTER !' NUMBER ANALOGY

---

$D:4::M:?$

Normal alphabet position:

$\boxed{13}$

---

## 59. LETTER !' REVERSE POSITION

---

$\backslash [ A:26::D:? \backslash ]$

Reverse alphabet position:

$\backslash [ 27-4=23 \backslash ]$

$\backslash [ \boxed{23} \backslash ]$

---

## 60. WORD ANALOGY METHOD

---

For word analogies, ask:

### Step 1

What is A?

### Step 2

What is B?

### Step 3

What precise relation connects them?

### Step 4

Apply same relation and direction to C.

Do not search options first.

---

## 61. NUMBER ANALOGY METHOD

---

Try:

1.  $\backslash (+/-) \backslash$
  2.  $\backslash (\times/\div) \backslash$
  3. square/cube
  4.  $\backslash (n^2 \backslash \text{pm } k) \backslash$
  5.  $\backslash (n(n \backslash \text{pm } 1)) \backslash$
  6. digit operations
  7. combined operations
- 

## 62. LETTER ANALOGY METHOD

---

Convert letters to positions:

$\backslash [ A=1, \backslash \text{ldots}, Z=26 \backslash ]$

Then check:

- fixed shift
  - reverse shift
  - opposite alphabet
-



- reversal
- position arithmetic
- pairwise shifts

---

## 63. ANALOGY DIRECTION TRAP

---

$\backslash [ \text{Book:Author::Painting:?} \backslash ]$

Book is created by author.

Painting is created by:

$\backslash [ \boxed{\text{Painter}} \backslash ]$

But:

$\backslash [ \text{Author:Book::Painter:?} \backslash ]$

Answer:

$\backslash [ \boxed{\text{Painting}} \backslash ]$

Same concepts, opposite direction.

---

## 64. STRENGTH OF RELATION

---

Suppose:

$\backslash [ \text{Puppy:Dog::Kitten:?} \backslash ]$

Possible options:

- Animal
- Cat
- Milk
- Fur

Best:

$\backslash [ \boxed{\text{Cat}} \backslash ]$

because relation is specifically:

$\backslash [ \text{young one}:\text{adult animal} \backslash ]$

not merely category membership.

Always choose the **most precise relation**.

---

## 65. SAME CATEGORY IS OFTEN TOO WEAK

---

Example:

$\backslash [ \text{Apple:Fruit::Carrot:?} \backslash ]$

Correct:

$\backslash [ \boxed{\text{Vegetable}} \backslash ]$

Not "Food", because Fruit and Vegetable are matching specific categories.

---

## 66. FUNCTION RELATION IS OFTEN STRONGER THAN ASSOCIATION

---

\[ Key:Lock::Password:? \]

Key opens/accesses lock.

Password accesses:

\[ \boxed{Account} \]

Choose functional equivalence, not a loose related object.

---

## 67. CLASSIFICATION PREVIEW

---

Classification asks:

**Which items belong to the same group?**

Example:

- Apple
- Banana
- Mango
- Potato

First three are fruits.

Potato is not.

So potato is the odd one out.

We'll handle this deeply in Part 2.

---

## 68. ANALOGY OPTION ELIMINATION

---

If you cannot name the relationship immediately:

1. reject options with wrong semantic category
  2. reject reversed relationship
  3. test remaining options for exact relation
  4. prefer simplest relation
- 

## 69. GATE/PLACEMENT TRAP — MULTIPLE VALID ASSOCIATIONS

---

Example:

\[ Bird:Nest::Bee:? \]

Possible associations:

- honey
-

- hive
- flower

Exact relation:

$\backslash \text{animal} : \text{home} \backslash$

Therefore:

$\backslash \boxed{\text{Hive}} \backslash$

## 70. ANALOGY VS CAUSATION

$\backslash \text{Rain} : \text{Flood} \backslash$

does not mean every rain causes flood.

Analogy questions often use typical/strong associations rather than strict logical necessity.

But in formal reasoning chapters later, causation must be treated more carefully.

## PRACTICE — LEVEL 0

### Q1

$\backslash \text{Fish} : \text{Swim} :: \text{Bird} : ? \backslash$

### Q2

$\backslash \text{Doctor} : \text{Hospital} :: \text{Teacher} : ? \backslash$

### Q3

$\backslash \text{Cow} : \text{Calf} :: \text{Dog} : ? \backslash$

### Q4

$\backslash 4 : 16 :: 7 : ? \backslash$

### Q5

$\backslash A : C :: D : ? \backslash$

## LEVEL 1

### Q6

$\backslash \text{Wheel} : \text{Car} :: \text{Page} : ? \backslash$

### Q7

$\backslash \text{Thermometer} : \text{Temperature} :: \text{Barometer} : ? \backslash$

### Q8

$\backslash 5 : 26 :: 8 : ? \backslash$

### Q9

\[ 23:13::34:? \]

### Q10

\[ AB:DE::JK:? \]

---

## LEVEL 2

---

### Q11

\[ Book:Author::Song:? \]

### Q12

\[ Army:Soldier::Fleet:? \]

### Q13

\[ 3:12::5:? \]

Assume rule:

\[  $n(n+1)$  \]

### Q14

\[ ABC:CBA::PQR:? \]

### Q15

\[ B:Y::F:? \]

Use opposite-alphabet relation.

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

\[ (3,4):25::(8,15):? \]

### Q17

\[ 246:12::357:? \]

Assume digit product relation.

### Q18

\[ D:16::G:? \]

Assume alphabet-position square.

### Q19

\[ Puppy:Dog::Sapling:? \]

### Q20

\[ Clock:Time::Compass:? \]

---

## ANSWERS

---

### Q1

Bird moves by:

$\boxed{\text{Fly}}$

### Q2

Teacher works in:

$\boxed{\text{School}}$

### Q3

Young dog:

$\boxed{\text{Puppy}}$

### Q4

$4^2=16$

$7^2=49$

$\boxed{49}$

### Q5

A!'' Qs +2.

D!'' F .

$\boxed{F}$

### Q6

Page is part of:

$\boxed{\text{Book}}$

### Q7

Barometer measures:

$\boxed{\text{Pressure}}$

### Q8

$n^2+1$

$5^2+1=26$

$8^2+1 = \boxed{65}$

### Q9

$$\lfloor 2^2+3^2=13 \rfloor$$

$$\lfloor 3^2+4^2 = 25 \rfloor$$

$$\lfloor \boxed{25} \rfloor$$

### Q10

Each letter +3:

J!' M

K!' N

$$\lfloor \boxed{MN} \rfloor$$

### Q11

Book is created by author.

Song is created by:

$$\lfloor \boxed{\text{Composer}} \rfloor$$

depending wording/options, "songwriter" may also fit; use the exact option provided.

### Q12

Army consists of soldiers.

Fleet consists of:

$$\lfloor \boxed{\text{Ships}} \rfloor$$

### Q13

$$\lfloor 3(4)=12 \rfloor$$

$$\lfloor 5(6) = \boxed{30} \rfloor$$

### Q14

Reverse:

$$\lfloor \boxed{RQP} \rfloor$$

### Q15

B position 2.

Opposite:

$$\lfloor 27-2=25=Y \rfloor$$

F position 6:

$$\lfloor 27-6=21=U \rfloor$$

$$\lfloor \boxed{U} \rfloor$$

---

### Q16

$$\backslash[ 3^2+4^2=25 \backslash]$$

$$\backslash[ 8^2+15^2 = 64+225 = \boxed{289} \backslash]$$

---

### Q17

$$\backslash[ 2\times4\times6=48 \backslash]$$

Wait — this does **not** equal 12.

So digit-product rule does not fit the first pair.

This is intentionally a pattern-validation check.

If the relation were digit **sum**:

$$\backslash[ 2+4+6=12 \backslash]$$

Then:

$$\backslash[ 3+5+7 = \boxed{15} \backslash]$$

Correct answer:

$$\backslash[ \boxed{15} \backslash]$$

### Important lesson

Always verify the stated/assumed rule against the original pair.

---

### Q18

D position:

$$\backslash[ 4 \backslash]$$

$$\backslash[ 4^2=16 \backslash]$$

G position:

$$\backslash[ 7 \backslash]$$

$$\backslash[ 7^2 = \boxed{49} \backslash]$$

---

### Q19

Puppy is young dog.

Sapling is young:

$$\backslash[ \boxed{Tree} \backslash]$$

---

### Q20

Clock indicates/measures time.

Compass indicates:

$$\backslash[ \boxed{Direction} \backslash]$$

---

## CHAPTER 58 STATUS

---

Covered so far:

- analogy concept '
- relationship direction '
- object-function relations '
- profession-workplace '
- product-source '
- part-whole '
- member-group '
- animal relations '
- synonyms/antonyms '
- degree relations '
- sequence relations '
- number analogies '
- digit-based analogies '
- letter analogies '
- opposite alphabet '
- pair transformations '
- pattern validation '
- option elimination '
- L0–L3 practice '

But **Chapter 58 is not complete yet.**

Still remaining:

- classification principles
- odd-one-out by semantic category
- number odd-one-out
- letter odd-one-out
- structural odd-one-out
- disguised properties
- prime/square/cube classification
- relation-based classification
- ambiguous odd-one-out handling
- final speed toolkit

## Progress

---

$\boxed{\frac{57}{76}}$  completed

**Current:** 0 = Chapter 58 — Analogy, Classification & Odd-One-Out

## NEXT

---

**Chapter 58 Part 2 — Classification, Odd-One-Out, Numerical Properties, Disguised Patterns & Final**



USER

next

ASSISTANT

# CHAPTER 58 — ANALOGY, CLASSIFICATION & ODD-ONE-OUT

## PART 2 — CLASSIFICATION, ODD-ONE-OUT, NUMERICAL PROPERTIES, DISGUISED PATTERNS& FINAL ELIMINATION TOOLKIT

Status: 0=Not PROGRESS ! ' COMPLETE GATE: Medium Placements: High Priority: +FTier 2

Classification asks:

Which items belong to the same group?

Odd-one-out asks:

Which item does not follow the common property followed by the others?

The main skill is:

Find the strongest common rule among most options, then eliminate the one that breaks it.

## 1. SIMPLE SEMANTIC CLASSIFICATION

Options:

- Apple
- Mango
- Banana
- Potato

First three are fruits.

Potato is a vegetable/tuber.

\[ \boxed{Potato} \]

## 2. CATEGORY SHOULD BE SPECIFIC

Options:

- Dog
- Cat

- Cow
- Sparrow

All are animals, so “animal” is too broad.

First three are mammals.

Sparrow is a bird.

$\boxed{\text{Sparrow}}$

Use the **most informative common category**.

---

### 3. PROFESSION CLASSIFICATION

---

- Doctor
- Engineer
- Teacher
- Hospital

First three are professions.

Hospital is a workplace.

$\boxed{\text{Hospital}}$

---

### 4. PLACE CLASSIFICATION

---

- Delhi
- Mumbai
- Chennai
- India

First three are cities.

India is a country.

$\boxed{\text{India}}$

---

### 5. PART-WHOLE ODD ONE OUT

---

- Wheel
- Engine
- Door
- Bicycle

First three can be parts/components of vehicles.

Bicycle is a whole vehicle.

$\boxed{\text{Bicycle}}$

---

### 6. TOOL ODD-ONE-OUT

---

- Hammer
- Screwdriver
- Spanner
- Carpenter

First three are tools.

Carpenter is a user of tools.

$\boxed{\text{Carpenter}}$

---

## 7. RELATION-BASED CLASSIFICATION

---

Options:

- Dog–Puppy
- Cat–Kitten
- Cow–Calf
- Lion–Den

First three:

$\text{[ \text{adult animal} : \text{young one} ]}$

Last:

$\text{[ \text{animal} : \text{home} ]}$

Therefore:

$\boxed{\text{Lion-Den}}$

---

## 8. NUMBER ODD-ONE-OUT — EVEN/ODD

---

$\text{[ 12,18,24,31,42 ]}$

All except 31 are even.

$\boxed{31}$

---

## 9. PRIME / COMPOSITE

---

$\text{[ 11,13,17,19,21 ]}$

First four are prime.

21 is composite.

$\boxed{21}$

---

## 10. PERFECT SQUARE

---

$\text{[ 16,25,36,49,50 ]}$

First four:

---

$\backslash[ 4^2,5^2,6^2,7^2 \backslash]$

50 is not a perfect square.

$\backslash \boxed{50} \backslash]$

---

## 11. PERFECT CUBE

$\backslash[ 8,27,64,100,125 \backslash]$

Cubes:

$\backslash[ 2^3,3^3,4^3,5^3 \backslash]$

100 is not a cube.

$\backslash \boxed{100} \backslash]$

---

## 12. MULTIPLE-BASED CLASSIFICATION

$\backslash[ 15,25,35,45,56 \backslash]$

First four are multiples of 5.

56 is not.

$\backslash \boxed{56} \backslash]$

---

## 13. DIVISIBILITY PROPERTY

$\backslash[ 18,27,45,63,80 \backslash]$

First four divisible by 9.

80 is not.

$\backslash \boxed{80} \backslash]$

---

## 14. NUMBER FORM CLASSIFICATION

$\backslash[ 6,12,20,30,42,55 \backslash]$

First five can be represented:

$\backslash[ n(n+1) \backslash]$

$\backslash[ 2\cdot 3=6 \backslash]$

$\backslash[ 3\cdot 4=12 \backslash]$

$\backslash[ 4\cdot 5=20 \backslash]$

$\backslash[ 5\cdot 6=30 \backslash]$

$\backslash[ 6\cdot 7=42 \backslash]$

Next should:

$\backslash[ 7\cdot 8=56 \backslash]$

---

not 55.

Thus:

$$\boxed{55}$$

---

## 15. TRIANGULAR NUMBER CLASSIFICATION

---

$$\{3, 6, 10, 15, 21, 25\}$$

First five are triangular numbers:

$$\frac{n(n+1)}{2}$$

25 is not.

$$\boxed{25}$$

---

## 16. FIBONACCI MEMBERSHIP

---

$$\{2, 3, 5, 8, 13, 14\}$$

First five are Fibonacci terms.

$$\boxed{14}$$

---

## 17. PRIME-POSITION STYLE

---

$$\{4, 6, 10, 14, 22, 26\}$$

These are:

$$2p$$

for primes:

$$\{2, 3, 5, 7, 11, 13\}$$

All fit.

If option were 24 instead of 26:

$$24/2=12$$

not prime.

Then:

$$\boxed{24}$$

---

## 18. SQUARE + CONSTANT

---

$$\{5, 10, 17, 26, 37, 50\}$$

Check:

$$n^2+1$$

$$2^2+1=5$$

---

$$\backslash[ 3^2+1=10 \backslash]$$

$$\backslash[ 4^2+1=17 \backslash]$$

$$\backslash[ 5^2+1=26 \backslash]$$

$$\backslash[ 6^2+1=37 \backslash]$$

Next should:

$$\backslash[ 7^2+1=50 \backslash]$$

All fit.

This is a reminder:

**Sometimes no option is odd if the question/data is wrong. Never force one.**

---

## 19. DIGIT-SUM PROPERTY

$$\backslash[ 23,32,41,50,62 \backslash]$$

First four digit sums:

$$\backslash[ 2+3=5 \backslash]$$

$$\backslash[ 3+2=5 \backslash]$$

$$\backslash[ 4+1=5 \backslash]$$

$$\backslash[ 5+0=5 \backslash]$$

62:

$$\backslash[ 6+2=8 \backslash]$$

So:

$$\backslash[ \boxed{62} \backslash]$$

---

## 20. DIGIT-PRODUCT PROPERTY

$$\backslash[ 26,34,43,62,53 \backslash]$$

Products:

$$\backslash[ 2\cdot6=12 \backslash]$$

$$\backslash[ 3\cdot4=12 \backslash]$$

$$\backslash[ 4\cdot3=12 \backslash]$$

$$\backslash[ 6\cdot2=12 \backslash]$$

$$\backslash[ 5\cdot3=15 \backslash]$$

Odd:

$$\backslash[ \boxed{53} \backslash]$$

## 21. REVERSE DIGITS

---

Pairs:

- 12–21
- 34–43
- 56–65
- 78–89

First three are exact reversals.

78 reversed is 87, not 89.

$\boxed{78-89}$

---

## 22. NUMBER PAIR RELATION

---

Pairs:

$(2,4),(3,9),(4,16),(5,20)$

First three:

$b=a^2$

For 5:

$5^2=25$

not 20.

$\boxed{(5,20)}$

---

## 23. NUMBER TRIPLE CLASSIFICATION

---

Triples:

$(3,4,5)$

$(5,12,13)$

$(8,15,17)$

$(6,8,11)$

First three are Pythagorean triples:

$a^2+b^2=c^2$

For last:

$36+64=100 \neq 121$

Thus:

$\boxed{(6,8,11)}$

---

## 24. LETTER ODD-ONE-OUT — ALPHABET GAP

---

$AC, DF, GI, JL, MP$

---

First letters:

\[ A,D,G,J,M \]

\[ +3 \]

Second letters expected:

\[ C,F,I,L,O \]

But last has P.

So:

\[ \boxed{MP} \]

---

## 25. LETTER PAIR OPPOSITES

---

\[ AZ,\ BY,\ CX,\ DW,\ ET \]

First four are complementary alphabet pairs:

\[ positions\ sum=27 \]

For E:

\[ 5 \]

Opposite should:

\[ 22=V \]

not T.

So:

\[ \boxed{ET} \]

---

## 26. LETTER GROUP CLASSIFICATION

---

\[ ABC,\ DEF,\ GHI,\ JKL,\ MOP \]

First four are three consecutive letters.

Expected last:

\[ MNO \]

So:

\[ \boxed{MOP} \]

---

## 27. CONSTANT LETTER SHIFT

---

Pairs:

\[ A-D,\ F-I,\ K-N,\ P-T \]

First three use:

\[ +3 \]

---



P +3 = S, not T.

So:

$\boxed{P-T}$

---

## 28. VOWEL / CONSONANT CLASSIFICATION

---

- A
- E
- I
- O
- T

First four are vowels.

$\boxed{T}$

---

## 29. ALPHABET POSITION PROPERTY

---

Letters:

$\boxed{B,D,F,H,K}$

Positions:

$\boxed{2,4,6,8,11}$

First four even positions.

K = 11 odd.

$\boxed{K}$

---

## 30. PRIME-POSITION LETTERS

---

$\boxed{B,C,E,G,K,N}$

Positions:

$\boxed{2,3,5,7,11,14}$

First five prime positions.

14 isn't prime.

$\boxed{N}$

---

## 31. SQUARE-POSITION LETTERS

---

$\boxed{A,D,I,P,X}$

Positions:

$\boxed{1,4,9,16,24}$

First four perfect squares.

---

Expected fifth:

$\boxed{25=Y}$

So:

$\boxed{X}$

---

## 32. ALPHANUMERIC CLASSIFICATION

---

$\boxed{A1, D4, G7, J10, M14}$

First four:

letter position = attached number.

A=1 D=4 G=7 J=10

M=13, not 14.

$\boxed{M14}$

---

## 33. ALPHANUMERIC SQUARE RELATION

---

$\boxed{B4, C9, D16, E25, F35}$

Attached numbers expected:

$2^2=4$

$3^2=9$

$4^2=16$

$5^2=25$

$6^2=36$

So:

$\boxed{F35}$

---

## 34. STRUCTURAL ODD-ONE-OUT

---

Options:

- ABBA
- CDDC
- EFFE
- GHIG

First three have pattern:

$\boxed{A B B A}$

Last:

$\boxed{G H I G}$

middle two differ.

---

$\boxed{\text{GHIG}}$

---

### 35. PALINDROME CLASSIFICATION

---

- LEVEL
- RADAR
- CIVIC
- TABLE

First three read same forward/backward.

$\boxed{\text{TABLE}}$

---

### 36. WORD-LENGTH CLASSIFICATION

---

- CAT
- DOG
- SUN
- TREE

First three have 3 letters.

TREE has 4.

$\boxed{\text{TREE}}$

But beware: word-length is often a weak relation if stronger semantic relation exists.

---

### 37. LETTER-POSITION SUM

---

Words/pairs:

- AZ
- BY
- CX
- DU

Sums:

AZ:

$1+26=27$

BY:

$2+25=27$

CX:

$3+24=27$

DU:

$4+21=25$

Odd:

---

$\boxed{DU}$

---

## 38. DISGUISED NUMERICAL PROPERTY

---

Options:

$\{14, 28, 42, 56, 63\}$

All first four are multiples of 14.

63 isn't.

But another possible relation:

all except 63 are even.

The stronger common relation may be "multiple of 14" if options support it.

---

## 39. MULTIPLE VALID PROPERTIES

---

Odd-one-out can be ambiguous if more than one rule isolates different options.

Example:

$\{2, 3, 5, 8\}$

Could say:

- 8 is only composite.
- 2 is only even prime.

Two possible odd terms.

Good test questions should avoid this or provide context/options designed for one intended rule.

---

## 40. AMBIGUITY RULE

---

When multiple rules are possible, prefer:

1. relation using **more structural information**
2. relation followed exactly by **more items**
3. common aptitude pattern
4. simplest non-arbitrary rule

Do not invent exotic mathematical properties.

---

## 41. CLASSIFICATION BY UNITS

---

- Metre
- Kilogram
- Second
- Celsius

First three are SI base-unit names or directly related base units in common aptitude framing, while Celsius is not the SI base unit for thermodynamic temperature (kelvin is).

---

But this is domain-sensitive.

For factual classification, use exact knowledge and wording.

---

## 42. CLASSIFICATION BY FUNCTION

---

- Thermometer
- Barometer
- Speedometer
- Kilogram

First three are instruments.

Kilogram is a unit.

$\boxed{\text{Kilogram}}$

---

## 43. CLASSIFICATION BY RELATIONSHIP PAIRS

---

- Eye : See
- Ear : Hear
- Nose : Smell
- Hand : Taste

First three:

$\boxed{\text{organ}:\text{sense/function}}$

Hand does not taste.

$\boxed{\text{Hand:Taste}}$

---

## 44. CAUSE-EFFECT CLASSIFICATION

---

- Fire : Heat
- Rain : Wetness
- Disease : Illness
- Ice : Warmth

First three have plausible cause/result relation.

Ice produces coldness, not warmth.

$\boxed{\text{Ice:Warmth}}$

---

## 45. DEGREE RELATION ODD-ONE-OUT

---

- Warm : Hot
- Cool : Cold
- Angry : Furious
- Small : Huge

First three second terms intensify the first in same semantic direction.

---

Small !' Huge changes to an opposite/very different magnitude.

$\boxed{\text{Small:Huge}}$

---

## 46. NUMBER SERIES AS CLASSIFICATION

---

Options:

- $(2,4,8,16)$
- $(3,6,12,24)$
- $(5,10,20,40)$
- $(7,14,28,55)$

First three double each step.

Fourth should end:

$56$

not 55.

$\boxed{7,14,28,55}$

---

## 47. RATIO CLASSIFICATION

---

Pairs:

$2:4, 3:6, 5:10, 7:15$

First three:

$1:2$

Fourth not.

$\boxed{7:15}$

---

## 48. FRACTION CLASSIFICATION

---

$\frac{12}{24}, \frac{36}{48}, \frac{48}{11}, \frac{11}{5}$

First four equal:

$\frac{1}{2}$

Last does not.

$\boxed{\frac{11}{5}}$

---

## 49. GEOMETRY CLASSIFICATION

---

- Square
  - Rectangle
  - Rhombus
  - Triangle
-

First three are quadrilaterals.

Triangle has 3 sides.

$\boxed{\text{Triangle}}$

---

## 50. DATA-STRUCTURE STYLE CLASSIFICATION

---

For placement reasoning:

- Stack
- Queue
- Linked List
- Compiler

First three are data structures.

Compiler is software/system program.

$\boxed{\text{Compiler}}$

This is an example of domain-based classification.

---

## 51. ELIMINATION METHOD

---

When uncertain:

### Step 1

Check obvious category.

### Step 2

Check numeric/letter structure.

### Step 3

Check relation direction.

### Step 4

See whether 3 or 4 options share exact feature.

### Step 5

Reject weaker arbitrary rules.

---

## 52. FAST NUMBER ODD-ONE-OUT SEARCH ORDER

---

Check:

1. parity
  2. divisibility
  3. prime/composite
  4. square/cube
  5. digit sum/product
-

6. sequence position
  7.  $\lfloor n/2 \rfloor$
  8. pair/triple relation
- 

## 53. FAST LETTER ODD-ONE-OUT SEARCH ORDER

---

Check:

1. alphabetical order
  2. fixed jumps
  3. opposite pairs
  4. vowel/consonant
  5. prime/square positions
  6. symmetry/reversal
  7. pair structure
- 

## 54. FAST WORD CLASSIFICATION SEARCH ORDER

---

Check:

1. semantic category
  2. part/whole
  3. profession/workplace
  4. tool/function
  5. animal/young/home
  6. synonym/antonym
  7. grammatical form
  8. structural spelling only if no stronger semantic rule
- 

## 55. PRACTICE — LEVEL 0

---

### Q1

Apple, Mango, Banana, Carrot

### Q2

Dog, Cat, Cow, Eagle

### Q3

11, 13, 17, 21

### Q4

16, 25, 36, 40

### Q5

A, E, I, B



---

## LEVEL 1

---

### Q6

AZ, BY, CX, DU

### Q7

6, 12, 20, 30, 41

### Q8

B, C, E, G, L

### Q9

AB, DE, GH, JK, MO

### Q10

123, 234, 345, 456, 568

---

## LEVEL 2

---

### Q11

$\setminus [(3,4,5),(5,12,13),(8,15,17),(7,24,26)]$

### Q12

A1, D4, G7, J10, M14

### Q13

LEVEL, RADAR, CIVIC, RIVER

### Q14

2:6, 3:9, 4:12, 5:16

### Q15

5,10,17,26,37,49

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

$\setminus [(2,6),(3,12),(4,20),(5,30),(6,40)]$

### Q17

AD, FI, KN, PS, UX

### Q18

26, 35, 44, 53, 64

---

**Q19**

A2, C6, E10, G14, I19

**Q20**

Select the odd pair:

- Eye : See
  - Ear : Hear
  - Nose : Smell
  - Tongue : Touch
- 

**ANSWERS**

---

**Q1**

First three fruits.

\[ \boxed{Carrot} \]

**Q2**

First three mammals.

\[ \boxed{Eagle} \]

**Q3**

11,13,17 prime.

21 composite.

\[ \boxed{21} \]

**Q4**

16,25,36 perfect squares.

\[ \boxed{40} \]

**Q5**

A,E,I vowels.

\[ \boxed{B} \]

**Q6**

AZ:

\[ 1+26=27 \]

BY:

\[ 2+25=27 \]

CX:

$$\backslash[ 3+24=27 \backslash]$$

DU:

$$\backslash[ 4+21=25 \backslash]$$

$$\backslash[ \boxed{DU} \backslash]$$

## Q7

---

First four:

$$\backslash[ 2\cdot 3=6 \backslash]$$

$$\backslash[ 3\cdot 4=12 \backslash]$$

$$\backslash[ 4\cdot 5=20 \backslash]$$

$$\backslash[ 5\cdot 6=30 \backslash]$$

Next should:

$$\backslash[ 6\cdot 7=42 \backslash]$$

Thus:

$$\backslash[ \boxed{41} \backslash]$$

## Q8

---

Positions:

$$\backslash[ 2,3,5,7 \backslash]$$

are prime.

$L=12$  not prime.

$$\backslash[ \boxed{L} \backslash]$$

## Q9

---

AB, DE, GH, JK have first letters spaced +3 and consecutive second letter.

After JK expected:

$$\backslash[ MN \backslash]$$

not MO.

$$\backslash[ \boxed{MO} \backslash]$$

## Q10

---

First four have three consecutive digits.

568 breaks this.

$$\backslash[ \boxed{568} \backslash]$$

---

## Q11

Check Pythagorean:

$$\sqrt{7^2+24^2} = 49+576 = 625 = 25^2$$

So correct third should be 25, not 26.

$$\boxed{(7,24,26)}$$

## Q12

A=1, D=4, G=7, J=10.

M=13, not 14.

$$\boxed{M14}$$

## Q13

First three palindromes.

$$\boxed{RIVER}$$

## Q14

First three:

$$\sqrt{1:3}$$

Fourth:

$$\sqrt{5:16}$$

not 1:3.

$$\boxed{5:16}$$

## Q15

Expected:

$$\sqrt{n^2+1}$$

$$\sqrt{2^2+1}=5$$

$$\sqrt{3^2+1}=10$$

$$\sqrt{4^2+1}=17$$

$$\sqrt{5^2+1}=26$$

$$\sqrt{6^2+1}=37$$

next:

$$\sqrt{7^2+1}=50$$

Thus:

$$\boxed{49}$$

## Q16

---

Pattern:

$$\sqrt{n(n+1)}$$

$$2! \cdot 63! \cdot 1 \cdot 24! \cdot 2 \cdot 6! \cdot 3 \cdot 0$$

For 6:

$$\sqrt{6(7)}=42$$

not 40.

$$\sqrt{\boxed{(6,40)}}$$

## Q17

---

AD:

$$\sqrt{+3}$$

FI:

$$\sqrt{+3}$$

KN:

$$\sqrt{+3}$$

PS:

$$\sqrt{+3}$$

UX:

$$\sqrt{+3}$$

All pairs follow the same internal rule.

But first letters:

$$\sqrt{A,F,K,P,U}$$

are +5 each.

So actually all fit.

$$\sqrt{\boxed{\text{No odd pair}}}$$

Important: don't force an answer if every option follows the rule.

## Q18

---

Digit sums:

26:

$$\sqrt{8}$$

35:

$$\sqrt{8}$$

44:

$$\sqrt{8}$$

53:

\[ 8 \]

64:

\[ 10 \]

\[ \boxed{64} \]

## Q19

---

Letters:

A,C,E,G,I !' positions 1,3,5,7,9.

Numbers should:

\[ 2,6,10,14,18 \]

increase by 4.

So:

\[ \boxed{119} \]

is odd.

## Q20

---

E y e l' S e e r !' H e n o s e !' S m e l l

Tongue is associated with taste, not touch.

\[ \boxed{Tongue:Touch} \]

## ' CHAPTER 58 COMPLETE

---

We have covered:

- word analogies '
- semantic relationships '
- function/tool relations '
- part-whole relations '
- number analogies '
- letter analogies '
- pattern validation '
- classification '
- semantic odd-one-out '
- number odd-one-out '
- prime/square/cube properties '
- digit properties '
- pair/triple relations '
- letter classification '
- alphanumeric classification '

- structural patterns '
- ambiguity handling '
- elimination strategy '
- L0–L3 practice '

## Progress

---

\[ \boxed{58/76}\text{ numbered chapters completed} \]

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Number, Letter & Alphanumeric Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding +
- Chapter 60 — Blood Relations +
- Chapter 61 — Direction Sense +
- Chapter 62 — Ranking & Ordering +
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

---

### + CHAPTER 59 — CODING–DECODING

Next we will cover:

- letter-shift coding
- opposite-alphabet coding
- position-based coding
- word substitution coding
- number coding
- matrix/table coding
- conditional coding rules
- coded sentences
- common-word detection
- Chinese coding style
- symbol coding
- mixed alphanumeric coding
- placement speed techniques
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 59 — CODING–DECODING

### PART 1 — LETTER-SHIFT CODING, OPPOSITE ALPHABET, POSITION-BASED CODING, WORD CODING & NUMBER CODING

Status:  $\emptyset$  = 0% PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Coding–decoding questions give a hidden rule that converts:

$\text{\text{Original}} \rightarrow \text{\text{Code}}$

Your job is to discover that rule and apply it correctly.

Main principle:

**Do not guess from one letter. Compare every corresponding position and search for the simplest consistent transformation.**

#### 1. BASIC LETTER-SHIFT CODING

Suppose:

$\text{CAT} \rightarrow \text{DBU}$

Compare:

$\text{C} \rightarrow \text{D}$

$\text{A} \rightarrow \text{B}$

$\text{T} \rightarrow \text{U}$

Each letter shifted:

$+1$

Therefore:

$\text{DOG} \rightarrow ?$

D! EO! PG! H

$\boxed{\text{EPH}}$

#### 2. BACKWARD SHIFT

$\text{DOG} \rightarrow \text{CNF}$

Each letter:



$\boxed{-1}$

So:

$\text{FISH} \rightarrow ?$

F! EI! HS! RH! G

$\boxed{\text{EHRG}}$

---

### 3. CYCLIC WRAPAROUND

---

If Z shifts +1:

$Z \rightarrow A$

If A shifts " 1 :

$A \rightarrow Z$

Example:

$\text{XYZ} \rightarrow \text{YZA}$

Rule:

$+1$

---

### 4. CONSTANT SHIFT OF +3

---

$\text{BAT} \rightarrow \text{EDW}$

B! EA! DT! W

So:

$\text{DOG} \rightarrow ?$

D! GO! RG! J

$\boxed{\text{GRJ}}$

---

### 5. DIFFERENT SHIFT AT EACH POSITION

---

Suppose:

$\text{ABC} \rightarrow \text{BDF}$

Changes:

A! B :

$+1$

B! D :

$+2$

C! F :

$+3$

---

Apply to:

$\backslash [ DEF \backslash ]$

$D + 1 = E \quad E + 2 = G \quad F + 3 = I$

$\backslash \boxed{EGI} \backslash$

---

## 6. INCREASING SHIFTS

---

$\backslash [ WORD \rightarrow XQUH \backslash ]$

Check:

W! ' X :

$\backslash [ +1 \backslash ]$

O! ' Q :

$\backslash [ +2 \backslash ]$

R! ' U :

$\backslash [ +3 \backslash ]$

D! ' H :

$\backslash [ +4 \backslash ]$

So position number determines shift.

---

## 7. ALTERNATING SHIFTS

---

Suppose:

$\backslash [ CODE \rightarrow DNGD \backslash ]$

Check:

C! ' D :

$\backslash [ +1 \backslash ]$

O! ' N :

$\backslash [ -1 \backslash ]$

D! ' E :

$\backslash [ +1 \backslash ]$

E! ' D :

$\backslash [ -1 \backslash ]$

Pattern:

$\backslash [ +1, -1, +1, -1 \backslash ]$

Apply similarly to another word.

---

## 8. ALTERNATING+2, " 2

---

Example:

`\[ MATH\rightarrow OYVF\]`

M!' O :

`\[ +2 \]`

A!' Y :

`\[ -2\text{ with wrap} \]`

T!' V :

`\[ +2 \]`

H!' F :

`\[ -2 \]`

So alternating positional rule.

---

## 9. REVERSE WORD CODING

---

`\[ TEAM\rightarrow MAET \]`

Simply reverse letters.

Thus:

`\[ GATE\rightarrow ? \]`

`\[ \boxed{ETAG} \]`

---

## 10. REVERSE + SHIFT

---

Suppose:

`\[ CAT\rightarrow UBD \]`

First reverse:

`\[ CAT\to TAC \]`

Then +1:

T!' UA!' BC!' D

So:

`\[ \boxed{UBD} \]`

This type combines two operations.

---

## 11. HOW TO DETECT REVERSE + SHIFT

---

When direct letter-by-letter comparison looks messy:

1. reverse source word
-

2. compare again
3. test constant shift

This is common in placement exams.

---

## 12. OPPOSITE-ALPHABET CODING

---

Recall opposite pairs:

$\backslash A \leftrightarrow Z \backslash$

$\backslash B \leftrightarrow Y \backslash$

$\backslash C \leftrightarrow X \backslash$

...

Formula:

$\backslash \boxed{p' = 27 - p} \backslash$

---

## 13. EXAMPLE — OPPOSITE LETTER

---

$\backslash CAT \rightarrow XZG \backslash$

C opposite X A opposite Z T opposite G

So:

$\backslash DOG \rightarrow ? \backslash$

D!' WO!' LG!' T

$\backslash \boxed{WLT} \backslash$

---

## 14. OPPOSITE + REVERSE

---

Suppose:

$\backslash ABC \rightarrow XYZ \backslash$

This is opposite coding.

But if:

$\backslash ABC \rightarrow ZYX \backslash$

then likely:

1. reverse ABC !' CBA

2. take opposites:

C!' XB!' YA!' Z

$\backslash \boxed{XYZ} \backslash$

Actually reversing after opposite can change ordering, so always inspect exact positions.

---

## 15. POSITION-BASED NUMBER CODING

---

Alphabet positions:

$\backslash$  A=1, \ B=2, \ldots, Z=26  $\backslash$

Example:

$\backslash$  CAT  $\rightarrow$  3,1,20  $\backslash$

Because:

C=3 A=1 T=20

---

## 16. SUM OF LETTER POSITIONS

---

$\backslash$  CAT  $\rightarrow$  ?  $\backslash$

$\backslash$  C=3, \ A=1, \ T=20  $\backslash$

Sum:

$\backslash$   $3+1+20 = \boxed{24}$   $\backslash$

---

## 17. WORD-SUM CODING

---

Example:

$\backslash$  DOG  $\rightarrow$  26  $\backslash$

D=4 O=15 G=7

$\backslash$   $4+15+7=26$   $\backslash$

Thus code is total alphabet-position sum.

---

## 18. POSITION PRODUCT CODING

---

$\backslash$  ABC  $\rightarrow$  6  $\backslash$

$\backslash$   $1 \times 2 \times 3 = 6$   $\backslash$

Then:

$\backslash$  BCD  $\rightarrow$  ?  $\backslash$

$\backslash$   $2 \times 3 \times 4 = \boxed{24}$   $\backslash$

---

## 19. SUM OF SQUARES

---

$\backslash$  AB  $\rightarrow$  5  $\backslash$

$\backslash$   $1^2+2^2=5$   $\backslash$

Then:

$\backslash$  CD  $\rightarrow$  ?  $\backslash$

$\backslash$   $3^2+4^2 = 9+16 = \boxed{25}$   $\backslash$

---

---

## 20. DIFFERENCE OF POSITIONS

---

$\text{[ AZ} \rightarrow 25 \text{ ]}$

$\text{[ 26-1=25 ]}$

Likewise:

$\text{[ BY ]}$

$\text{[ 25-2 = \boxed{23} ]}$

Direction may matter depending rule.

---

## 21. REVERSE-POSITION NUMBER CODING

---

Reverse position:

$\text{[ 27-p ]}$

Example:

A !' 26 B !' 25 C !' 24

So:

$\text{[ CAT ]}$

becomes:

C !' 24 A !' 26 T !' 7

$\text{[ \boxed{24,26,7} ]}$

---

## 22. NUMBER GENERATED FROM LETTER POSITIONS

---

Suppose:

$\text{[ BAD} \rightarrow 214 \text{ ]}$

B=2 A=1 D=4

Positions concatenated.

Then:

$\text{[ CAB} \rightarrow ? \text{ ]}$

C=3 A=1 B=2

$\text{[ \boxed{312} ]}$

---

## 23. TWO-DIGIT LETTER POSITIONS

---

For letters J–Z, concatenation must be handled carefully.

Example:

$\text{[ CAT ]}$

---

could become:

$\lfloor 3,1,20 \rfloor$

written:

$\lfloor 3120 \rfloor$

You must infer boundaries from context.

---

## 24. POSITION + CONSTANT

---

Suppose:

$\lfloor A\rightarrow 3, \text{quad } B\rightarrow 4, \text{quad } C\rightarrow 5 \rfloor$

Then code:

$\lfloor \text{text}\{\text{alphabet position}\}+2 \rfloor$

Thus:

$\lfloor M=13 \rfloor$

Code:

$\lfloor \boxed{15} \rfloor$

---

## 25. POSITION $\times$ CONSTANT

---

Suppose:

A!' 2B!' 4C!' 6

Rule:

$\lfloor 2p \rfloor$

Then:

$\lfloor F=6 \rfloor$

Code:

$\lfloor \boxed{12} \rfloor$

---

## 26. SQUARE OF POSITION

---

A!' 1B!' 4C!' 9D!' 1 6

Then:

$\lfloor G=7 \rfloor$

Code:

$\lfloor \boxed{49} \rfloor$

---

## 27. WORD CODING BY TOTAL + LENGTH

---

Suppose code is:

$\lfloor \text{sum of positions} \rfloor + \lfloor \text{number of letters} \rfloor$

Example:

CAT:

$\lfloor 24+3=27 \rfloor$

Such composite rules can appear, but only accept them when examples clearly demand them.

---

## 28. LENGTH-BASED CODING

---

Suppose:

APPLE !' 5 BANANA !' 6 MANGO !' 5

Then coding is word length.

So:

$\lfloor \text{GRAPE} \rightarrow \boxed{5} \rfloor$

---

## 29. VOWEL COUNT CODING

---

Suppose:

APPLE !' 2 ORANGE !' 3 SKY !' 0

Code = number of vowels.

Then:

$\lfloor \text{EDUCATION} \rfloor$

vowels:

E,U,A,I,O

$\lfloor \boxed{5} \rfloor$

---

## 30. CONSONANT COUNT

---

If code = number of consonants:

MANGO has 5 letters and 2 vowels.

Consonants:

M,N,G

$\lfloor \boxed{3} \rfloor$

---

## 31. FIRST + LAST LETTER POSITIONS

---

Suppose:

$\lfloor \text{CAT} \rightarrow 23 \rfloor$

---



C=3 T=20

$\lfloor 3+20=23 \rfloor$

Middle letter ignored.

Then:

$\lfloor \text{DOG} \rfloor$

D=4 G=7

$\lfloor \boxed{11} \rfloor$

---

## 32. FIRST × LAST POSITION

---

$\lfloor \text{CAT} \rfloor$

C=3 T=20

$\lfloor 3 \times 20 = 60 \rfloor$

If code = 60, that's likely the rule.

---

## 33. MIDDLE-LETTER CODING

---

Odd-length words may code using middle letter.

Example:

CAT !' 1 DOG !' 15 SUN !' 21

Middle letters:

A=1 O=15 U=21

---

## 34. WORD SUBSTITUTION CODING

---

Sometimes words are replaced with unrelated code words.

Example:

“BOOK” is coded as “TREE” “PEN” as “RIVER”

There may be no letter transformation.

You must treat code as a substitution mapping.

---

## 35. SIMPLE SUBSTITUTION EXAMPLE

---

If:

$\lfloor \text{APPLE} \rightarrow \text{MANGO} \rfloor$

$\lfloor \text{MANGO} \rightarrow \text{GRAPE} \rfloor$

Then asking code for APPLE:

$\lfloor \boxed{\text{MANGO}} \rfloor$

---

No alphabet arithmetic needed.

---

### 36. WORD-SUBSTITUTION TRAP

---

Do not assume the code word has semantic relation to original.

Coding can be arbitrary.

Example:

“red” may be called “blue”.

Then if asked:

**What is the code for red?**

Answer:

`\[ \boxed{blue} \]`

not a color relationship explanation.

---

### 37. RENAMING / SUBSTITUTION QUESTION

---

Suppose:

- Cat is called Dog
- Dog is called Cow
- Cow is called Lion

Question:

**What is a dog called?**

Dog is called:

`\[ \boxed{Cow} \]`

Read substitution direction exactly.

---

### 38. CONCEPT VS NAME TRAP

---

If:

“Sky is called Sea.”

Question:

**Where do birds fly?**

Birds actually fly in sky.

But sky is called "Sea".

Therefore answer:

---

$\boxed{\text{Sea}}$

This is a renaming logic problem.

---

### 39. NUMBER CODING — SIMPLE TRANSFORMATION

---

Suppose:

$5 \rightarrow 11$

$7 \rightarrow 15$

Rule:

$2n+1$

Then:

$9 \rightarrow 19$

$\boxed{19}$

---

### 40. NUMBER CODING — SQUARE + 1

---

$4 \rightarrow 17$

$5 \rightarrow 26$

Rule:

$n^2+1$

Thus:

$7 \rightarrow 50$

$\boxed{50}$

---

### 41. DIGIT-REVERSE CODING

---

$123 \rightarrow 321$

$450 \rightarrow 054$

Normally:

$\boxed{54}$

unless leading zero retained by fixed-width representation.

---

### 42. DIGIT-SUM CODING

---

$472 \rightarrow 13$

$4+7+2=13$

Then:

$583 \rightarrow 5+8+3 = \boxed{16}$

---

---

### 43. DIGIT-PRODUCT CODING

---

$\backslash[234\rightarrow 24\backslash]$

$\backslash[2\times 3\times 4 = 24\backslash]$

Then:

$\backslash[345\rightarrow 60\backslash]$

$\backslash[\boxed{60}\backslash]$

---

### 44. SUM OF SQUARES OF DIGITS

---

$\backslash[23\rightarrow 13\backslash]$

$\backslash[2^2+3^2=13\backslash]$

Then:

$\backslash[45\rightarrow 16+25 = \boxed{41}\backslash]$

---

### 45. DIFFERENCE BETWEEN DIGITS

---

$\backslash[83\rightarrow 5\backslash]$

$\backslash[8-3=5\backslash]$

Then:

$\backslash[94\rightarrow 5\backslash]$

$\backslash[\boxed{5}\backslash]$

---

### 46. DIGIT POSITION TRANSFORMATION

---

Suppose:

$\backslash[123\rightarrow 234\backslash]$

Each digit +1.

Then:

$\backslash[567\rightarrow 678\backslash]$

$\backslash[\boxed{678}\backslash]$

---

### 47. DIGIT ROTATION

---

$\backslash[1234\rightarrow 2341\backslash]$

First digit moved to end.

Then:

$\boxed{5678 \rightarrow 6785}$

$\boxed{6785}$

---

## 48. SWAPPING DIGITS

---

$1234 \rightarrow 2143$

Pairs swapped:

1 2' 2 3' 4' 4 3

Then:

$5678 \rightarrow 6587$

$\boxed{6587}$

---

## 49. LETTER CODING USING NEXT VOWEL / CONSONANT

---

Some placement tests use category shifts.

Example:

Each vowel replaced by next vowel:

A! EE! III! OO! UU! A

If:

$CAT \rightarrow CET$

only A changes to E.

Be alert for category-specific rules.

---

## 50. VOWEL/CONSONANT DIFFERENT SHIFTS

---

Example rule:

- vowels +1
- consonants " 1

For:

$BAT$

B consonant:

$B \rightarrow A$

A vowel:

$A \rightarrow B$

T consonant:

$T \rightarrow S$

Code:

$\boxed{ABS}$

---

---

## 51. POSITION PARITY CODING

---

Rule might be:

- odd positions in word +1
- even positions " 1

Example:

\[ CODE \]

C!' DO!' ND!' EE!' D

\[ \boxed{DNED} \]

---

## 52. LETTER PARITY CODING

---

Different concept:

- alphabet-position odd letters +1
- alphabet-position even letters " 1

This is not the same as word-position parity.

Read examples carefully.

---

## 53. IDENTIFY WHICH KIND OF PARITY IS USED

---

If same letter is transformed differently in different positions, likely word-position rule.

If same letter always transforms the same way, likely alphabet-based rule.

---

## 54. MULTIPLE EXAMPLES ARE GOLD

---

Suppose:

\[ CAT\rightarrow DBU \]

One example suggests +1.

If also:

\[ DOG\rightarrow EPH \]

the rule is strongly confirmed.

Use all examples before locking pattern.

---

## 55. DON'T OVERFIT

---

From:

\[ A\rightarrow 2 \]

alone, possible rules include:

---

- position +1
- $2 \times \text{position}$
- square +1
- arbitrary mapping

One example is often insufficient.

Wait for additional pairs/options.

---

## 56. CODING SEARCH ORDER — LETTER WORDS

---

When word transforms into another word:

1. same length?
  2. direct shifts?
  3. alternating shifts?
  4. reverse?
  5. reverse + shift?
  6. opposite alphabet?
  7. letter-position dependent shifts?
  8. vowel/consonant special rules?
- 

## 57. CODING SEARCH ORDER — WORD TO NUMBER

---

Check:

1. word length
  2. sum of positions
  3. product
  4. first/last positions
  5. vowel/consonant count
  6. square/sum-square relation
  7. position arithmetic
- 

## 58. CODING SEARCH ORDER — NUMBER TO NUMBER

---

Check:

1. + /"
  2.  $\times / \div$
  3. square/cube
  4. digit sum
  5. digit product
  6. reversal
  7. digitwise transformation
  8. rearrangement
-

## 59. COMMON CODING TRAPS

---

### & pTRAP 1 — Direction

If CAT!' DBU, decoding DBU requires reverse shift:

$\setminus [-1 \setminus]$

not +1.

---

### & pTRAP 2 — Wraparound

$Z+1=A$ .

---

### & pTRAP 3 — Reverse vs shift

Check ordering before arithmetic.

---

### & pTRAP 4 — Alphabet position

$A=1$ , not 0.

---

### & pTRAP 5 — Renaming questions

Answer according to renamed labels, not real-world name.

---

### & pTRAP 6 — One-example overfitting

Multiple transformations may fit.

---

## 60. ENCODING VS DECODING

---

If encoding rule is:

$\setminus [ p \setminus \text{to } p+3 \setminus]$

then decoding is:

$\setminus [ p \setminus \text{to } p-3 \setminus]$

Example:

CODE !' FRGH

To decode FRGH:

F!' CR!' OG!' DH!' E

$\setminus [ \boxed{\text{CODE}} \setminus]$

---

## 61. DECODING OPPOSITE-ALPHABET CODE

---

Opposite coding is self-inverse.

If:

$\setminus [ A \setminus \text{leftrightharpoonup } Z \setminus]$

---



then decoding uses the same mapping.

Example:

`\[ WLT \]`

D!" WO!" LG!" T

Thus:

`\[ \boxed{DOG} \]`

---

## 62. SPEED RULE — WRITE LETTER POSITIONS ONLY WHEN NEEDED

---

For simple +1:

Do mentally.

For complicated transformation:

Write:

`\[ C=3,\ A=1,\ T=20 \]`

Do not convert every word unnecessarily.

---

## 63. FAST OPPOSITE-LETTER PAIRS

---

Memorize:

`\[ AZ,\ BY,\ CX,\ DW,\ EV,\ FU,\ GT,\ HS,\ IR,\ JQ,\ KP,\ LO,\ MN \]`

This makes mirror-alphabet coding almost instant.

---

## PRACTICE — LEVEL 0

---

### Q1

If CAT!' DBU, then DOG !' ?

### Q2

If FISH !' EHRG, what is the rule?

### Q3

If GATE!' ETAG, identify the operation.

### Q4

If A!' Z B!' Ythen D!' ?

### Q5

If CAT is coded as alphabet positions, what is the code?

---

## LEVEL 1

---

**Q6**

If ABC !' BDF, code DEF.

**Q7**

If CODE follows + 1 , " 1 , + 1 code MATH.

**Q8**

If DOG !' 26 using alphabet-position sum, code CAT.

**Q9**

If BAD !' 214 by concatenation, code CAB.

**Q10**

If A!' 2 B!' 4 C!' 6 under a fixed rule, code F.

**LEVEL 2****Q11**

If CAT !' UBD by reverse-then-+1, code DOG.

**Q12**

Under opposite-alphabet coding, code GATE.

**Q13**

If word code = first letter position + last letter position, code MANGO.

**Q14**

If 4!' 1 and 5!' 2 using the same rule, code 8.

**Q15**

If 2 3 4!' 2 by digit product, code 456.

**LEVEL 3 — EXAM PRESSURE****Q16**

Rule:

- odd word positions +2
- even word positions " 2

Code GATE.

**Q17**

If every vowel is replaced by next vowel cyclically, code EDUCATION.

**Q18**

If CAT !' 27 where code = sum of alphabet positions + word length, code DOG.

### Q19

If "Sky" is called "Sea", "Sea" is called "Land", and "Land" is called "Air", where do fish live according to the renamed language?

### Q20

If WORD !' XQUH using shifts +1,+2,+3,+4, code GATE using same positional shifts.

---

## ANSWERS

---

### Q1

+1:

D!' EO!' PG!' H

$\boxed{\text{EPH}}$

### Q2

Each letter:

$\boxed{-1}$

$\boxed{\text{Backward shift by 1}}$

### Q3

Reverse word.

$\boxed{\text{Reversal}}$

### Q4

D opposite W.

$\boxed{W}$

### Q5

C=3, A=1, T=20.

$\boxed{3,1,20}$

### Q6

Shifts:

$\boxed{+1,+2,+3}$

D!' EE!' GF!' I

$\boxed{\text{EGI}}$

### Q7

---

$$M + 1 = N \text{ A " } 1 = Z \text{ T } + 1 = U \text{ H " } 1 = G$$

$$\boxed{\text{NZUG}}$$

## Q8

CAT:

$$3 + 1 + 20 = \boxed{24}$$

## Q9

CAB:

$$3, 1, 2$$

$$\boxed{312}$$

## Q10

Rule:

$$2 \times \text{position}$$

F=6:

$$2(6) = \boxed{12}$$

## Q11

DOG reverse:

$$\text{GOD}$$

Then +1:

$$\text{G! 'HO! 'PD! 'E}$$

$$\boxed{\text{HPE}}$$

## Q12

Opposites:

$$\text{G! 'TA! 'ZT! 'GE! 'V}$$

$$\boxed{\text{TZGV}}$$

## Q13

$$M=13 \text{ O}=15$$

$$13 + 15 = \boxed{28}$$

## Q14

Rule:

$$n^2 + 1$$

$$\lfloor 8^2+1=65 \rfloor$$

$$\lfloor \boxed{65} \rfloor$$

### Q15

$$\lfloor 4\times 5\times 6 = \boxed{120} \rfloor$$

### Q16

$$G +2 = I$$

A " 2wraps to Y

$$T +2 = V$$

$$E " 2= C$$

$$\lfloor \boxed{\{YVC\}} \rfloor$$

### Q17

Vowel cycle:

A!' EE!' III!' OO!' UU!' A

EDUCATION:

E!' D stays D U!' AC stays C A!' EF stays T I!' OO!' UN stays N

$$\lfloor \boxed{\{IDACETOUN\}} \rfloor$$

### Q18

DOG position sum:

$$\lfloor 4+15+7=26 \rfloor$$

Length:

$$\lfloor 3 \rfloor$$

Code:

$$\lfloor 26+3 = \boxed{29} \rfloor$$

### Q19

Fish actually live in Sea.

But Sea is called Land.

Therefore:

$$\lfloor \boxed{\{Land\}} \rfloor$$

### Q20

$$G +1 = H \ A +2 = C \ T +3 = W \ E +4 = I$$

$$\lfloor \boxed{\{HCWI\}} \rfloor$$

---

## CHAPTER 59 STATUS

---

Covered so far:

- fixed letter shifts '
- changing positional shifts '
- alternating shifts '
- reverse coding '
- reverse + shift '
- opposite alphabet '
- alphabet-position coding '
- sum/product/square coding '
- concatenated-position coding '
- length/vowel/consonant coding '
- first/last letter coding '
- substitution/renaming coding '
- number transformation coding '
- digit-based coding '
- encoding vs decoding '
- search strategy '
- L0–L3 practice '

But **Chapter 59 is not complete yet.**

Still remaining:

- coded sentences
- common-word detection
- Chinese coding style
- substitution across multiple statements
- symbol coding
- matrix/table coding
- conditional operation coding
- mixed alphanumeric coding
- relation-based coding
- advanced decoding under constraints
- final speed toolkit

## Progress

---

$\boxed{\frac{58}{76}}$  text{ completed} %

**Current:** 0 = 0 Chapter 59 — Coding–Decoding

## NEXT

---

+ Chapter 59 Part 2 — Coded Sentences, Chinese Coding, Symbol/Matrix Coding, Conditional Codes & Final Placement Toolkit

USER

next

ASSISTANT

## CHAPTER 59 — CODING–DECODING

### PART 2 — CODED SENTENCES, CHINESE CODING, SYMBOL CODING, MATRIX/TABLE CODING, CONDITIONAL CODES & FINAL SPEED TOOLKIT

Status: ~~0~~ = 100% PROGRESS ! ' ' COMPLETE GATE: Medium Placements: High Priority: +Tier 2

This part covers the harder coding-decoding questions where complete words or sentences are converted into arbitrary codes.

The key idea is:

When codes are arbitrary, compare repeated words across statements. Repetition is your evidence.

#### 1. CODED-SENTENCE / CHINESE CODING

Example:

“good boy plays” = ka na pa “good girl sings” = ka ma sa

The common word is:

\[ good \]

The common code is:

\[ ka \]

Therefore:

\[ \boxed{good=ka} \]

This is the basic principle.

#### 2. COMMON-WORD = COMMON-CODE RULE

Suppose:

Sentence 1:

red apple sweet

Code:

\[ mo\ ta\ ki \]

Sentence 2:

**red rose beautiful**

Code:

$\backslash$  [ mo\ pa\ su \]

Only common word:

$\backslash$  [ red \]

Only common code:

$\backslash$  [ mo \]

Thus:

$\backslash$  [ \boxed{red=mo} \]

### 3. TWO COMMON WORDS

Suppose:

**“big red apple” = ka mo ti**

**“big red flower” = ka mo su**

Common words:

- big
- red

Common codes:

- ka
- mo

Therefore:

$\backslash$  [ {big,red}\={ka,mo}\ ]

But we cannot determine individually which is which yet.

Important:

$\backslash$  [ \boxed{\text{Do not assign without enough information}} \]

### 4. USE A THIRD STATEMENT

Add:

**“big blue car” = ka ne po**

Now compare with first two.

Common word with first:



$\backslash$  big  $\backslash$

Common code:

$\backslash$  ka  $\backslash$

Therefore:

$\backslash$   $\boxed{\text{big=ka}}$   $\backslash$

Then:

$\backslash$   $\boxed{\text{red=mo}}$   $\backslash$

---

## 5. ELIMINATION METHOD

Suppose:

**cats eat fish = na pa ki**

**dogs eat meat = ta pa mo**

Common:

$\backslash$  eat  $\backslash$

So:

$\backslash$  eat=pa  $\backslash$

Then remaining codes:

cats/fish = na/ki dogs/meat = ta/mo

Need more statements to separate those.

---

## 6. FULL DECODING EXAMPLE

Statements:

1. sky is blue = ka ta ma
2. sea is blue = pa ta ma
3. sky looks clear = ka na sa

Compare 1 & 2:

Common words:

$\backslash$  is, \ blue  $\backslash$

Common codes:

$\backslash$  ta, \ ma  $\backslash$

So:

$\backslash$   $\{is, blue\} = \{ta, ma\}$   $\backslash$

Compare 1 & 3:

---

Common word:

\[ sky \]

Common code:

\[ ka \]

Thus:

\[ \boxed{sky=ka} \]

Still need another statement to distinguish is and blue.

---

## 7. SINGLE-OVERLAP IS BEST

---

In coded-sentence questions, the easiest comparisons are where two statements share exactly:

- one word
- one code

Then mapping is immediate.

---

## 8. COMMON-CODE DIFFERENCE METHOD

---

Suppose:

Sentence A:

**apple mango banana = ka ta mo**

Sentence B:

**apple mango grape = ka ta su**

Only changing word:

banana != grape

Only changing code:

mo != su

So:

\[ banana=mo \]

\[ grape=su \]

provided unchanged words preserve unchanged codes.

This is another powerful method.

---

## 9. IDENTICAL CODE SETS IN DIFFERENT ORDER

---

In Chinese coding, code order often does **not** correspond to word order.

Example:

---

**“good boy plays” = pa ka mo**

Do not assume:

good=pa boy=ka plays=mo

unless explicitly stated.

Usually codes are treated as an unordered set.

## 10. ORDER-PRESERVING CODING

Some questions explicitly imply positional mapping.

If:

**CAT!' DBU**

then:

C!' DA!' BT!' U

Order matters.

But in coded sentences like:

**“good boy plays” = ka na pa**

order is often arbitrary.

Always identify question type.

## 11. CHINESE CODING — MASTER TABLE METHOD

Create a table:

Word	Possible code
good	ka
boy	na / pa
plays	na / pa

Each new statement narrows possibilities.

This prevents mental confusion in 4–5 statement sets.

## 12. EXAMPLE WITH 4 STATEMENTS

- smart students learn fast = ka mo ti pa
- smart people work hard = ka su ne lo

3. students work daily = mo ne ra

4. people learn daily = su ti ra

Now decode.

---

### 13. DECODE “SMART”

---

Statements 1 and 2 share:

\[ smart \]

Common code:

\[ ka \]

Therefore:

\[ \boxed{smart=ka} \]

---

### 14. DECODE “STUDENTS”

---

Statements 1 and 3 share:

\[ students \]

Common code:

\[ mo \]

Therefore:

\[ \boxed{students=mo} \]

---

### 15. DECODE “WORK”

---

Statements 2 and 3 share:

\[ work \]

Common code:

\[ ne \]

\[ \boxed{work=ne} \]

---

### 16. DECODE “PEOPLE”

---

Statements 2 and 4 share:

\[ people \]

Common code:

\[ su \]

\[ \boxed{people=su} \]

---

## 17. DECODE “DAILY”

---

Statements 3 and 4 share:

\[ daily \]

Common code:

\[ ra \]

\[ \boxed{daily=ra} \]

---

## 18. DECODE “LEARN”

---

Statements 1 and 4 share:

\[ learn \]

Common code:

\[ ti \]

\[ \boxed{learn=ti} \]

Remaining:

fast = pa hard = lo

Thus complete mapping is possible.

---

## 19. FIND CODE OF A NEW PHRASE

---

Using previous mapping:

**smart people learn**

Codes:

smart = ka people = su learn = ti

So:

\[ \boxed{ka,\ su,\ ti} \]

Order may vary if coding is unordered.

---

## 20. FIND UNKNOWN WORD FROM CODE

---

If sentence code includes:

\[ ka,\ ne,\ ra \]

Known:

ka=smart ne=work ra=daily

Then sentence corresponds to:

\[ \boxed{smart\ work\ daily} \]

---

assuming exactly those mappings.

---

## 21. AMBIGUOUS CODE MAPPING

---

Suppose:

**A B C = 1 2 3**

**A B D = 1 2 4**

Then:

C=3 D=4

But A and B remain mapped to 1 and 2 ambiguously.

Do not claim:

A=1, B=2

unless another statement separates them.

---

## 22. SUFFICIENCY IN CODING

---

Question:

**What is code for A?**

If all we know is:

$[A, B \rightarrow 1, 2]$

then:

$[\boxed{\text{Cannot be determined uniquely}}]$

This is a valid answer.

---

## 23. SYMBOL CODING

---

Suppose symbols represent operations:

$[$

- $\text{ means } -$

$]$

$[$

- $\text{ means } \times$

$]$

$[\times \text{ means } +]$

Question:

---

$$\backslash[ 8+3-2 \backslash]$$

Replace symbols with their coded meanings:

$$\backslash[ 8-3\backslashtimes2 \backslash]$$

Apply normal BODMAS:

$$\backslash[ 8-6 = \backslashboxed{2} \backslash]$$

---

## 24. VERY IMPORTANT — REPLACE FIRST, THEN APPLY BODMAS

---

Do not calculate with original symbols first.

Process:

1. translate symbols
2. rewrite expression
3. apply BODMAS

---

## 25. SYMBOL OPERATION EXAMPLE

---

If:

$$\backslash[ @ = + \backslash]$$

$$\backslash[ \backslash\# = \backslashtimes \backslash]$$

$$\backslash[ \backslash\$ = - \backslash]$$

Then:

$$\backslash[ 5\backslash\#4@3\backslash\$2 \backslash]$$

becomes:

$$\backslash[ 5\backslashtimes4+3-2 \backslash]$$

$$\backslash[ 20+3-2 = \backslashboxed{21} \backslash]$$

---

## 26. SYMBOLS REPRESENT RELATIONS

---

Sometimes:

$$\backslash[ A @ B \backslash]$$

means:

$$\backslash[ A > B \backslash]$$

and:

$$\backslash[ A \backslash\# B \backslash]$$

means:

$$\backslash[ A < B \backslash]$$

Then questions ask which statements are true.

This is closer to coded inequalities.

---

## 27. CODED INEQUALITY EXAMPLE

Suppose:

- $(A @ B)$  means  $(A > B)$
- $(A \# B)$  means  $(A = B)$
- $(A \$ B)$  means  $(A < B)$

Given:

$[ P @ Q \# R ]$

Translate:

$[ P > Q = R ]$

Therefore:

$[ P > R ]$

must be true.

---

## 28. CHAINING RELATIONS

If:

$[ A > B, \text{quad } B > C ]$

then:

$[ A > C ]$

If:

$[ A > B, \text{quad } B = C ]$

then:

$[ A > C ]$

But if:

$[ A > B, \text{quad } B < C ]$

no definite relation between A and C.

This will matter later in logical deduction too.

---

## 29. MATRIX / TABLE CODING

Sometimes a table maps letters to row-column codes.

Example:

	1	2	3
1	A	B	C
2	D	E	F



	1	2	3
3	G	H	I

If code is row-column:

A = 11 B = 12 F = 23 H = 32

Then:

$\backslash$  [ BED  $\backslash$  ]

becomes:

B = 12 E = 22 D = 21

$\backslash$  [  $\boxed{12, 22, 21}$   $\backslash$  ]

### 30. COLUMN-ROW CODE TRAP

Some tables use:

$\backslash$  [ column,row  $\backslash$  ]

instead of row,column.

Never assume.

Check given examples.

### 31. MATRIX CODE FROM EXAMPLE

Suppose table as above and given:

$\backslash$  [ E=22  $\backslash$  ]

This fits both row-column and column-row because center is symmetric.

Not enough.

If:

$\backslash$  [ F=32  $\backslash$  ]

then code must mean column-row because F is row 2, column 3.

Use non-symmetric examples to identify orientation.

### 32. 5×5 LETTER MATRIX

A common pattern:

	1	2	3	4	5
1	A	B	C	D	E
2	F	G	H	I	J

	1	2	3	4	5
3	K	L	M	N	O
4	P	Q	R	S	T
5	U	V	W	X	Y

Z may be handled separately or paired depending scheme.

Codes can be:

11, 12, ..., 55.

### 33. CONDITIONAL CODING RULES

Example:

- if word begins with vowel !' reverse it
- if word begins with consonant !' shift every letter +1

For:

APPLE !' begins vowel !' reverse:

$\boxed{\text{ELPPA}}$

DOG !' consonant !' +1 each:

D!' EO!' PG!' H

$\boxed{\text{EPH}}$

### 34. CONDITION ORDER MATTERS

Suppose:

1. If number is even, multiply by 2.
2. If result > 20, subtract 5.

Input:

$\boxed{12}$

Step 1:

$\boxed{24}$

Step 2:

$\boxed{24-5=19}$

$\boxed{19}$

Do not apply conditions in arbitrary order.

### 35. EXCLUSIVE CONDITIONS

Suppose:

- if number is even !' +3
- if number is odd !' ×2

For 8:

\[ 8+3=11 \]

Stop if conditions are mutually exclusive.

Do not then apply odd rule to 11 unless explicitly sequential.

---

## 36. SEQUENTIAL VS CLASSIFICATION RULES

---

### Classification rule

Choose exactly one branch.

Example:

if vowel !' rule A if consonant !' rule B

### Sequential rule

Apply rule 1, then rule 2, then rule 3.

Wording matters.

---

## 37. CONDITIONAL LETTER CODING

---

Rule:

- letters in odd alphabet positions !' +2
- letters in even alphabet positions !' " 1

Code:

\[ BAD \]

B=2 even:

\[ B\to A \]

A=1 odd:

\[ A\to C \]

D=4 even:

\[ D\to C \]

Code:

\[ \boxed{ACC} \]

---

## 38. WORD-POSITION CONDITIONAL CODING

---

Rule:

- 1st letter +1
  - 2nd letter " 1
-

- 3rd +1
- 4th " 1

For:

$\boxed{GATE}$

G! HA! ZT! UE! D

$\boxed{HZUD}$

## 39. CODED RELATIONS

Sometimes words like:

- mother
- father
- brother

are replaced by symbols.

Example:

$\boxed{A+B}$

means A is mother of B.

$\boxed{A-B}$

means A is brother of B.

Then a coded expression must be interpreted relationally.

We'll do full family reasoning in Chapter 60.

## 40. MIXED ALPHANUMERIC CODING

Suppose:

$\boxed{CAT \rightarrow C3A1T20}$

Each letter followed by its alphabet position.

Then:

$\boxed{DOG \rightarrow D4O15G7}$

$\boxed{D4O15G7}$

## 41. POSITION BEFORE LETTER

If:

$\boxed{CAT \rightarrow 3C1A20T}$

then:

$\boxed{DOG \rightarrow 4D15O7G}$

$\boxed{4D15O7G}$

---

## 42. LETTER + REVERSE POSITION

---

C normal = 3, reverse = 24.

If code is:

$\lfloor C24 \rfloor$

then:

$\lfloor A26, T7 \rfloor$

So CAT:

$\lfloor \boxed{C24A26T7} \rfloor$

---

## 43. TWO-NUMBER CODE PER LETTER

---

Suppose each letter coded as:

$\lfloor (\text{normal position}, \text{reverse position}) \rfloor$

For C:

$\lfloor (3, 24) \rfloor$

A:

$\lfloor (1, 26) \rfloor$

T:

$\lfloor (20, 7) \rfloor$

Useful identity:

$\lfloor p + (27 - p) = 27 \rfloor$

---

## 44. RELATION-BASED WORD CODING

---

Example:

If:

$\lfloor \text{KING} \rightarrow \text{QUEEN} \rfloor$

$\lfloor \text{MAN} \rightarrow \text{WOMAN} \rfloor$

the relation is male !' female counterpart.

If asked:

$\lfloor \text{BOY} \rightarrow ? \rfloor$

$\lfloor \boxed{\text{GIRL}} \rfloor$

This is conceptual coding rather than letter transformation.

---

## 45. MULTIPLE-CODE POSSIBILITIES

---

Some code sets allow more than one code for the same letter/word.

Example matrix coding may give alternate coordinates.

If question provides options, any valid code may be accepted.

Read whether it asks:

- one possible code
- exact code
- cannot be determined

---

## 46. FIND CODE VIA INTERSECTION

Coded sentences:

1. red big ball = na ka pa
2. red small box = na ti mo
3. big box falls = ka mo su

Mappings:

red = na big = ka box = mo

Then:

ball = pa small = ti falls = su

Intersection logic solves everything.

---

## 47. FAST INTERSECTION METHOD

Write:

TEXT

Words: red big ball  
Codes: na ka pa

Words: red small box  
Codes: na ti mo

Circle common word and common code.

Repeat.

This is faster than memorizing the sentences mentally.

---

## 48. FREQUENCY METHOD

If a word appears in 3 statements, its code should also appear in corresponding 3 code sets.

Example:

“good” appears 3 times.

Only code **ka** appears in all three.

Thus:

$\boxed{\text{good}=\text{ka}}$

This is useful when statements share several words.

---

## 49. NEGATIVE ELIMINATION METHOD

If:

- A could be code x or y
- another statement excludes x

then:

$A=y$

Use elimination, not only direct commonality.

---

## 50. CODED SENTENCE WITH REPEATED WORD

If sentence:

**very very good**

is coded:

$\text{ka} \text{ ka} \text{ mo}$

then repeated word may produce repeated code, if the coding preserves each occurrence.

But some arbitrary language-code questions may list unique codes only.

Use question convention.

---

## 51. SYMBOL REPLACEMENT WITH NUMBERS

Suppose:

$\triangle = 3$

$\square = 5$

$\circ = 2$

Then:

$\triangle + \square \times \circ$

means:

$3 + 5(2)$

$\boxed{13}$

---

## 52. OPERATION SYMBOL REPLACEMENT

Suppose:

$\backslash + \rightarrow \times \backslash$

$\backslash \times \rightarrow - \backslash$

$\backslash - \rightarrow + \backslash$

Evaluate:

$\backslash 8 + 4 \times 3 - 2 \backslash$

Translate:

$\backslash 8 \times 4 - 3 + 2 \backslash$

BODMAS:

$\backslash 32 - 3 + 2 = \boxed{31} \backslash$

---

## 53. SWAPPING OPERATORS

---

If  $\backslash (+ \backslash)$  and  $\backslash (\times \backslash)$  are interchanged:

$\backslash 6 + 3 \times 2 \backslash$

becomes:

$\backslash 6 \times 3 + 2 \backslash$

$\backslash = 20 \backslash$

$\backslash \boxed{20} \backslash$

---

## 54. SWAPPING DIGITS / NUMBERS

---

Sometimes instructions say:

**2 and 5 are interchanged.**

Expression:

$\backslash 25 + 52 \backslash$

becomes:

$\backslash 52 + 25 \backslash$

Same sum here.

But in:

$\backslash 25 - 52 \backslash$

becomes:

$\backslash 52 - 25 = 27 \backslash$

Read replacements globally.

---

## 55. CODED MATHEMATICAL RELATION

---



Suppose:

$$\backslash[ A\#B=A^2+B \backslash]$$

Then:

$$\backslash[ 3\#4 = 3^2+4 = 13 \backslash]$$

This is a custom operator.

---

## 56. CUSTOM OPERATOR WITH TWO INPUTS

---

If:

$$\backslash[ a@b=2a+b \backslash]$$

Then:

$$\backslash[ 5@3 = 10+3 = \boxed{13} \backslash]$$

For nested:

$$\backslash[ (5@3)@2 \backslash]$$

First:

$$\backslash[ 13@2=26+2=28 \backslash]$$

$$\backslash[ \boxed{28} \backslash]$$

---

## 57. NESTED OPERATOR ORDER

---

Always evaluate inner operation first unless definition says otherwise.

$$\backslash[ a@(b@c) \backslash]$$

and:

$$\backslash[ (a@b)@c \backslash]$$

can differ.

Custom operations are not automatically associative.

---

## 58. SYMBOLIC INEQUALITY CHAIN

---

Suppose:

- $\backslash(P*Q\backslash)$  means  $\backslash(P\ge Q\backslash)$
- $\backslash(P@Q\backslash)$  means  $\backslash(P<Q\backslash)$
- $\backslash(P\#Q\backslash)$  means  $\backslash(P=Q\backslash)$

Given:

$$\backslash[ A*B\#C \backslash]$$

Translate:

$$\backslash[ A\ge B=C \backslash]$$

Hence:

---

$\{ A \geq C \}$

must be true.

---

## 59. WHEN RELATION CANNOT BE DETERMINED

---

Given:

$\{ A > B \}$

and:

$\{ B < C \}$

A could be:

- greater than C
- equal to C
- less than C

So:

$\{ \boxed{\text{relation between A and C cannot be determined}} \}$

---

## 60. FINAL CODING SEARCH ORDER

---

### If WORD != WORD

---

1. direct shifts
2. alternating shifts
3. reverse
4. reverse + shift
5. opposite alphabet
6. positional rules
7. vowel/consonant conditions

### If WORD != NUMBER

---

1. length
2. letter-position sum
3. first/last
4. vowel/consonant count
5. product/squares

### If SENTENCE != ARBITRARY CODES

---

1. find repeated words
2. find repeated codes
3. intersect
4. eliminate
5. mark unresolved pairs

## If SYMBOL EXPRESSION

---

1. translate symbols
  2. rewrite
  3. apply BODMAS
- 

## 61. FINAL SPEED TARGETS

---

Simple letter coding:

$\backslash \boxed{15-25} \backslash \text{text{ sec}} \backslash$

Word-to-number:

$\backslash \boxed{20-30} \backslash \text{text{ sec}} \backslash$

Chinese/coded sentence:

$\backslash \boxed{30-60} \backslash \text{text{ sec}} \backslash$

Complex conditional coding:

$\backslash \boxed{45-75} \backslash \text{text{ sec}} \backslash$

If mapping remains ambiguous after enough checking, consider **cannot be determined** rather than forcing a code.

---

## 62. COMMON ADVANCED TRAPS

---

& pTRAP 1 — Assuming code order matches sentence order

Often false in Chinese coding.

& pTRAP 2 — Two common words

Two common codes cannot be individually assigned yet.

& pTRAP 3 — Custom operators

Do not assume standard properties.

& pTRAP 4 — Replace symbols but forget BODMAS

Translate first, then calculate.

& pTRAP 5 — Sequential vs exclusive conditions

Determine rule structure before applying.

& pTRAP 6 — Matrix orientation

Row-column vs column-row can reverse codes.

& pTRAP 7 — Over-decoding

If evidence is insufficient, stop.

---

## PRACTICE — LEVEL 0

---

Q1

good boy = ka mo good girl = ka ti

What is code for **good**?

## Q2

red apple = na pa blue apple = mo pa

What is code for **apple**?

## Q3

If @ means + and # means  $\times$ , find:

$\left[ 4\#3@2 \right]$

## Q4

If  $\backslash(a*b=a+b+1\backslash)$ , find:

$\backslash 3*4 \backslash$

## Q5

In the  $3\times 3$  matrix

$\left[ \begin{matrix} A&B&C\\ D&E&F\\ G&H&I \end{matrix} \right]$

using row-column coding, code H.

---

## LEVEL 1

---

Use:

1. smart people learn = ka mo ti
2. smart students work = ka pa ne
3. people work daily = mo ne su

## Q6

Code for smart?

## Q7

Code for people?

## Q8

Code for work?

## Q9

Which code can represent learn?

## Q10

Code for daily?

---

## LEVEL 2

---

Suppose:

---

- + means ×
- × means -
- - means +

### Q11

Evaluate:

$$\lfloor 6+4\times 2-3 \rfloor$$

### Q12

If  $\lfloor (a@b=2a+b) \rfloor$ , find  $\lfloor (4@5) \rfloor$ .

### Q13

Find:

$$\lfloor (4@5)@2 \rfloor$$

### Q14

If:

- $\lfloor (A*B) \rfloor$  means  $\lfloor (A>B) \rfloor$
- $\lfloor (A\#B) \rfloor$  means  $\lfloor (A=B) \rfloor$

and:

$$\lfloor P*Q\#R \rfloor$$

what relation must hold between P and R?

### Q15

If a word beginning with a vowel is reversed, code APPLE.

## LEVEL 3 — EXAM PRESSURE

Use:

1. good students solve problems = ka ti mo pa
2. smart students learn fast = su ti ne ra
3. good people learn daily = ka lo ne mi
4. people solve fast = lo mo ra

### Q16

Code for good?

### Q17

Code for students?

### Q18

Code for learn?

### Q19

Code for solve?

## Q20

What is the code set for “good people solve”?

---

## ANSWERS

---

### Q1

---

Common word:

good

Common code:

ka

$\boxed{ka}$

### Q2

---

Common word:

apple

Common code:

pa

$\boxed{pa}$

### Q3

---

Translate:

$4 \times 3 + 2$

$= 12 + 2 = \boxed{14}$

### Q4

---

$3 + 4 + 1 = \boxed{8}$

### Q5

---

H is row 3, column 2.

$\boxed{32}$

### Q6

---

Statements 1 & 2 share smart.

Common code:

$ka$

$\boxed{ka}$

## Q7

---

Statements 1 & 3 share people.

Common:

$\boxed{mo}$

$\boxed{mo}$

## Q8

---

Statements 2 & 3 share work.

Common:

$\boxed{ne}$

$\boxed{ne}$

## Q9

---

Statement 1:

remaining after smart=ka and people=mo:

$\boxed{ti}$

$\boxed{ti}$

## Q10

---

Statement 3:

people=mo work=ne

Remaining:

$\boxed{su}$

$\boxed{su}$

## Q11

---

Original:

$6+4\times 2-3$

Replace:

$+\rightarrow\times$

$\times\rightarrow-$

$- \rightarrow +$

So:

$6\times 4-2+3$

$24-2+3 = \boxed{25}$

## Q12

---

$4 @ 5 = 2(4) + 5$

$\boxed{13}$

## Q13

---

First:

$4 @ 5 = 13$

Then:

$13 @ 2 = 26 + 2$

$\boxed{28}$

## Q14

---

$P > Q = R$

Therefore:

$\boxed{P > R}$

## Q15

---

APPLE begins with vowel.

Reverse:

$\boxed{ELPPA}$

## Q16

---

Statements 1 & 3 share:

good

Common code:

ka

$\boxed{ka}$

## Q17

---

Statements 1 & 2 share:

students

Common code:

ti

$\boxed{ti}$

## Q18

---

Statements 2 & 3 share:



learn

Common code:

ne

$\boxed{ne}$

## Q19

---

Statements 1 & 4 share:

solve

Common code:

mo

$\boxed{mo}$

## Q20

---

good = ka people = lo solve = mo

Therefore code set:

$\boxed{\{ka, lo, mo\}}$

Order need not matter unless explicitly stated.

---

## ' CHAPTER 59 COMPLETE

---

We have now covered:

- fixed letter shifts '
- changing shifts '
- alternating shifts '
- reverse/reverse+shift coding '
- opposite alphabet '
- word-to-number coding '
- substitution/renaming '
- digit coding '
- coded sentences '
- Chinese coding/common-word method '
- intersection and elimination '
- ambiguous mappings '
- symbol-operation coding '
- custom operators '
- coded inequalities '
- matrix/table coding '
- row-column orientation '
- conditional coding '
- mixed alphanumeric coding '

- relation-based coding '
- speed and validation strategy '
- L0–L3 practice '

## Progress

---

\[ \boxed{59/76}\text{ numbered chapters completed} \]

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Number, Letter & Alphanumeric Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding–Decoding '
- Chapter 60 — Blood Relations +
- Chapter 61 — Direction Sense +
- Chapter 62 — Ranking & Ordering +
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

---

### + CHAPTER 60 — BLOOD RELATIONS

Next we will cover:

- family-relation vocabulary
  - generation levels
  - gender tracking
  - direct and indirect relations
  - maternal vs paternal relations
  - sibling relations
  - family-tree diagrams
  - coded blood relations
  - pointing/introduction questions
  - “only son/daughter” traps
  - marriage links
  - unknown-gender cases
  - GATE/placement speed methods
  - L0–L3 practice.
-

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next

ASSISTANT

## CHAPTER 60 — BLOOD RELATIONS

### PART 1 — FAMILY-TREEBASICS, GENERATION LEVELS, GENDER TRACKING, DIRECT/INDIRECT RELATIONS & POINTING QUESTIONS

Status:  $\emptyset$  = No PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Blood-relation questions become easy when you stop solving them as sentences and instead convert them into a **small family tree**.

Main principle:

\*\*

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## CHAPTER 60 — BLOOD RELATIONS

### PART 1 — FAMILY-TREEBASICS, GENERATION LEVELS, GENDER TRACKING, DIRECT/INDIRECT RELATIONS & POINTING QUESTIONS

Status:  $\emptyset$  = No PROGRESS GATE: Medium Placements: High Priority: +Tier 2

Blood-relation questions become easy when you stop solving them as sentences and instead convert them into a **small family tree**.

Main principle:

**Track only three things: generation, gender, and connection.**

## 1. BASIC FAMILYSYMBOLS

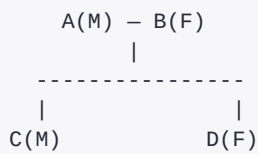
Use simple notation:

- Male = M
- Female = F
- Marriage = —

- Child link = downward line
- Siblings = same generation

Example:

TEXT



Here:

- A is father of C and D
- B is mother of C and D
- C and D are siblings
- C is brother of D
- D is sister of C

## 2. GENERATION LEVELS

This is the fastest way to avoid mistakes.

Think in levels:

TEXT

```

Generation +2 : Great-grandparents
Generation +1 : Grandparents
Generation 0 : Parents / Uncles / Aunts
Generation -1 : Self / Siblings / Cousins
Generation -2 : Children / Nephews / Nieces
Generation -3 : Grandchildren

```

Or choose the reference person as level 0.

The absolute level does not matter; **relative level matters**.

## 3. DIRECT RELATIONS

### Parent relations

Father = male parent Mother = female parent

### Child relations

Son = male child Daughter = female child

### Sibling relations

Brother = male sibling Sister = female sibling

## 4. GRAND RELATIONS

Father's father:

$\boxed{\text{Paternal Grandfather}}$

Father's mother:

$\boxed{\text{Paternal Grandmother}}$

Mother's father:

$\boxed{\text{Maternal Grandfather}}$

Mother's mother:

$\boxed{\text{Maternal Grandmother}}$

---

## 5. UNCLE TYPES

Father's brother:

$\boxed{\text{Paternal Uncle}}$

Mother's brother:

$\boxed{\text{Maternal Uncle}}$

Father's sister's husband is also often called uncle by marriage.

Mother's sister's husband likewise.

For aptitude questions, identify exact blood link if asked.

---

## 6. AUNT TYPES

Father's sister:

$\boxed{\text{Paternal Aunt}}$

Mother's sister:

$\boxed{\text{Maternal Aunt}}$

Wife of father's brother may also be called aunt by marriage.

---

## 7. COUSIN

Children of your:

- father's brother
- father's sister
- mother's brother
- mother's sister

are your cousins.

Important:

**Cousin is same generation as you.**

---

## 8. NEPHEW AND NIECE

Brother's son:

$\boxed{\text{Nephew}}$

Sister's son:

$\boxed{\text{Nephew}}$

Brother's daughter:

$\boxed{\text{Niece}}$

Sister's daughter:

$\boxed{\text{Niece}}$

---

## 9. IN-LAW RELATIONS

Wife's father:

$\boxed{\text{Father-in-law}}$

Husband's father:

$\boxed{\text{Father-in-law}}$

Wife's mother / Husband's mother:

$\boxed{\text{Mother-in-law}}$

Brother's wife:

$\boxed{\text{Sister-in-law}}$

Sister's husband:

$\boxed{\text{Brother-in-law}}$

Terminology can vary culturally, but the family-tree structure is what matters in exams.

---

## 10. MOST IMPORTANT RULE — DO NOT ASSUME GENDER

Statement:

**A is the child of B.**

A's gender is unknown.

Do not assume A is son or daughter.

Only conclude:

$\boxed{A = \text{child of B}}$

---

## 11. GENDER TRACKING

---

Words that fix male gender:

- father
- son
- brother
- husband
- uncle
- grandfather
- nephew

Words that fix female gender:

- mother
- daughter
- sister
- wife
- aunt
- grandmother
- niece

Words that do NOT fix gender:

- parent
- child
- sibling
- cousin
- spouse

## 12. EXAMPLE — DIRECT RELATION

**P is the father of Q.**

So:

TEXT

P(M)

|

Q(?)

Question:

**How is P related to Q?**

$\boxed{\text{Father}}$

## 13. EXAMPLE — REVERSE RELATION

**P is the father of Q.**

Question:

**How is Q related to P?**

Only:

$\boxed{\text{Child}}$

unless Q's gender is separately known.

This is a classic trap.

## 14. TWO-STEP RELATION

**A is the father of B. B is the sister of C.**

Since B and C are siblings, A is also parent of C.

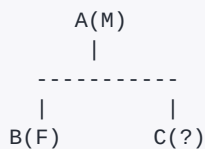
A is male.

Therefore:

$\boxed{\text{A is father of C}}$

## 15. FAMILY TREE

TEXT



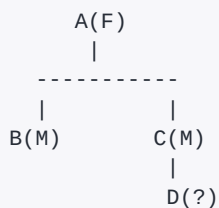
B and C are siblings.

## 16. THREE-STEP RELATION

**A is the mother of B. B is the brother of C. C is the father of D.**

Tree:

TEXT



Question:



### How is A related to D?

A is parent of C, and C is parent of D.

Therefore A is D's:

$\boxed{\text{Grandmother}}$

## 17. GENERATION METHOD

Instead of full tree, assign relative levels.

Suppose:

A is mother of B:

$A = B + 1 \text{ (generation)}$

B brother of C:

$B = C$

C father of D:

$C = D + 1$

Therefore:

$A = D + 2$

Female at +2 generation:

$\boxed{\text{Grandmother}}$

## 18. SAME GENERATION RELATIONS

These typically stay at same level:

- brother
- sister
- sibling
- cousin
- spouse

Parent/uncle generation is one level above.

Child/nephew generation is one level below.

## 19. MARRIAGE DOES NOT CHANGE GENERATION

If A is husband of B:

$\text{Generation}(A) = \text{Generation}(B)$

Marriage links gender/spouse status, not generation.

## 20. MATERNAL VS PATERNAL

Question:

**A is the brother of B's mother.**

A is B's:

$\boxed{\text{Maternal Uncle}}$

Because relation passes through mother.

## 21. PATERNAL UNCLE

**P is the brother of Q's father.**

P is Q's:

$\boxed{\text{Paternal Uncle}}$

## 22. MATERNAL AUNT

**A is sister of B's mother.**

A is B's:

$\boxed{\text{Maternal Aunt}}$

## 23. PATERNAL AUNT

**X is sister of Y's father.**

X is Y's:

$\boxed{\text{Paternal Aunt}}$

## 24. COUSIN RELATION

**A is son of B. B is brother of C. C is mother of D.**

Tree:

TEXT

```
 Parent generation
 B(M) ---- sibling ---- C(F)
 | |
 A(M) D(?)
```

A and D are children of siblings.

Therefore:

$\boxed{\text{A and D are cousins}}$

---

## 25. SON OF UNCLE

If X is your uncle's son:

$\boxed{\text{Cousin}}$

Not nephew.

---

## 26. SON OF BROTHER

If X is your brother's son:

$\boxed{\text{Nephew}}$

Distinguish carefully:

- uncle's son != cousin
- brother's son != nephew

---

## 27. POINTING QUESTIONS

Typical format:

**Pointing to a man, Riya said, "He is the son of my mother's only son."**

These are extremely common in placements.

Solve from inside outward.

---

## 28. INSIDE-OUT METHOD

Sentence:

**son of my mother's only son**

Step 1:

"My mother's only son."

If speaker Riya is female, mother's only son is Riya's brother.

Step 2:

"Son of that brother."

That is Riya's:

$\boxed{\text{Nephew}}$

---

## 29. IMPORTANT — SPEAKER GENDER MATTERS

---

If speaker is male and says:

**my mother's only son**

It could be the speaker himself.

If speaker is female, it must be her brother.

So never ignore speaker gender.

---

## 30. EXAMPLE

---

**Pointing to a girl, Ravi said, "She is the daughter of my mother's only daughter."**

Ravi is male.

"My mother's only daughter" = Ravi's sister.

That girl's relation to Ravi:

sister's daughter =

$\boxed{\text{Niece}}$

---

## 31. "ONLY SON" TRAP

---

Suppose:

**A is the only son of B.**

This means:

- A is male
- B has exactly one son

It does **not** mean B has only one child.

B may have daughters.

---

## 32. "ONLY CHILD"

---

**A is the only child of B.**

This means B has exactly one child total.

Different from only son/daughter.

---

### 33. “ONLY DAUGHTER”

**P is the only daughter of Q.**

Q may still have sons.

Do not interpret as only child.

### 34. POINTING QUESTION — COMPLEX

**Pointing to a woman, Arun said, “Her mother is the only daughter of my mother.”**

Arun is male.

“My mother's only daughter” = Arun's sister.

The woman's mother = Arun's sister.

Therefore the woman is Arun's:

$\boxed{\text{Niece}}$

### 35. ANOTHER COMPLEX POINTING QUESTION

**Pointing to a man, Priya said, “His father is the only son of my grandfather.”**

Priya is female.

“My grandfather's only son” could be Priya's father, assuming that grandfather is parent of her father/mother and the only son in that family.

Then the man's father = Priya's father.

Therefore the man is Priya's brother, if same parent relation is intended.

$\boxed{\text{Brother}}$

Be careful: exact answer depends on which grandfather is meant if ambiguity exists. Standard aptitude questions intend one unique family link.

### 36. INTRODUCTION QUESTIONS

Typical:

**A woman introduces a man as “the son of the brother of my mother.”**

Break:

my mother !' mother's brother !' his son

Mother's brother = maternal uncle.

Maternal uncle's son = cousin.

$\boxed{\text{Cousin}}$

---

### 37. INTRODUCTION EXAMPLE

---

**Raj introduces Neha as “the daughter of the only son of my grandfather.”**

If Raj is male and grandfather's only son is Raj's father, then Neha is daughter of Raj's father.

Therefore Neha is Raj's:

$\boxed{\text{Sister}}$

assuming Raj is not himself the only daughter etc. Gender given Neha female.

---

### 38. “BROTHER OF MY FATHER”

---

$\boxed{\text{Paternal Uncle}}$

His daughter:

$\boxed{\text{Cousin}}$

His son:

$\boxed{\text{Cousin}}$

---

### 39. “SISTER OF MY MOTHER”

---

$\boxed{\text{Maternal Aunt}}$

Her daughter/son:

$\boxed{\text{Cousin}}$

---

### 40. “SON OF MY SISTER”

---

$\boxed{\text{Nephew}}$

“Daughter of my sister”:

$\boxed{\text{Niece}}$

---

### 41. “FATHER OF MY FATHER”

---

$\boxed{\text{Paternal Grandfather}}$

“Mother of my father”:

$\boxed{\text{Paternal Grandmother}}$

---

### 42. “FATHER OF MY MOTHER”

---

$\boxed{\text{Maternal Grandfather}}$

“Mother of my mother”:

$\boxed{\text{Maternal Grandmother}}$

## 43. RELATION CHAIN METHOD

For a phrase like:

**mother's brother's daughter**

Move one step at a time:

Self !' mother !' mother's brother = maternal uncle !' uncle's daughter = cousin

$\boxed{\text{Cousin}}$

## 44. ANOTHER CHAIN

**father's sister's son**

Self !' father !' father's sister = paternal aunt !' her son = cousin

$\boxed{\text{Cousin}}$

## 45. ANOTHER CHAIN

**brother's daughter's son**

Self !' brother !' brother's daughter = niece !' niece's son

That is a grand-nephew / great-nephew.

In most basic aptitude questions, they may stop before such extended terms unless options are descriptive.

## 46. FAMILY-TREE SHORTCUT

For long chains, use:

TEXT

Self

|

Mother

|

Brother

|

Daughter

Then read relation from Self to last person.

This is safer than holding words mentally.

---

## 47. HUSBAND / WIFE CONNECTION

---

**A is wife of B. B is brother of C.**

A is C's:

$\boxed{\text{Sister-in-law}}$

---

## 48. SPOUSE OF SIBLING

---

Brother's wife:

$\boxed{\text{Sister-in-law}}$

Sister's husband:

$\boxed{\text{Brother-in-law}}$

---

## 49. SPOUSE'S SIBLING

---

Husband's brother or wife's brother:

$\boxed{\text{Brother-in-law}}$

Husband's sister or wife's sister:

$\boxed{\text{Sister-in-law}}$

Again, exact cultural naming may vary, but standard aptitude usage groups these as in-laws.

---

## 50. UNKNOWN-GENDER CASE

---

**A is sibling of B. B is mother of C.**

Question:

**How is A related to C?**

A is sibling of C's mother.

But A's gender unknown.

Therefore A is:

$\boxed{\text{Uncle or Aunt}}$

If options ask:

- Uncle
- Aunt



- Mother's sibling
- Cannot determine

Best exact answer:

$\boxed{\text{Mother's sibling}}$

or cannot determine gender-specific term.

## 51. DON'T OVER-INFER MARRIAGE

Statement:

**A and B are parents of C.**

Do not automatically assume A and B are husband and wife unless standard monogamous family context is intended by question.

For ordinary aptitude exams, this is often implied, but use only what is necessary.

## 52. GENERATION DIFFERENCE SHORTCUT

Parent:

$\boxed{+1}$

Grandparent:

$\boxed{+2}$

Child:

$\boxed{-1}$

Grandchild:

$\boxed{-2}$

Sibling/cousin/spouse:

$\boxed{0}$

This lets you quickly classify the possible relation.

## 53. EXAMPLE USING GENERATION NUMBERS

**P is sister of Q. Q is father of R. R is brother of S.**

Let R/S = level 0.

Q = +1.

P = same as Q = +1.

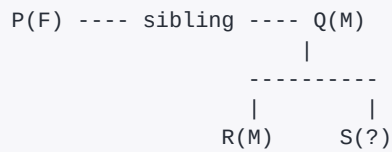
P is female and sibling of S's father.

Therefore:

$\boxed{P \text{ is paternal aunt of } S}$

## 54. DIRECT TREE

TEXT



P is aunt of both R and S.

## 55. "HOW MANY MALES/FEMALES?" QUESTIONS

Example:

**A is father of B. B is sister of C. C is son of D.**

Known genders:

A = male B = female C = male D = parent of C, gender unknown unless "mother/father" stated.

So guaranteed males:

$\boxed{2}$

Guaranteed females:

$\boxed{1}$

Do not force D's gender.

## 56. MINIMUM FAMILY MEMBERS QUESTIONS

Sometimes several statements can refer to the same person.

Example:

- A is father of B.
- C is mother of B.

Minimum family members:

$\boxed{3}$

A and C are two parents, B child.

## 57. SAME PERSON POSSIBILITY

Suppose:

- P is brother of Q.
- R is father of P.

- R is father of Q.

Only 3 people required.

Do not create separate father figures.

---

## 58. MULTIPLE SIBLINGS

**A is brother of B. C is sister of B.**

A and C are not necessarily the only siblings, but they are both siblings of B.

In standard family-tree logic, if same parents implied, A and C are also siblings.

---

## 59. “BROTHER OF” GIVES GENDER OF SUBJECT

**A is brother of B.**

We know:

$\backslash A = \text{male} \backslash$

B's gender remains unknown.

Similarly:

**A is sister of B.**

A is female; B unknown.

---

## 60. “SON OF” GIVES GENDER OF SUBJECT

**A is son of B.**

A = male.

B = parent, gender unknown.

“Daughter of” makes A female.

---

## 61. “FATHER OF” GIVES GENDER OF SUBJECT

**A is father of B.**

A = male.

B gender unknown.

“Mother of” makes A female.

---

---

## 62. RELATION DIRECTION TRAP

---

If:

**A is uncle of B.**

Then B is not necessarily A's nephew; B could be niece if B's gender unknown.

So reverse relation:

$\boxed{\text{Nephew or niece}}$

---

## 63. SAME RELATION, DIFFERENT GENDER

---

Parent's male sibling:

$\boxed{\text{uncle}}$

Parent's female sibling:

$\boxed{\text{aunt}}$

Sibling's male child:

$\boxed{\text{nephew}}$

Sibling's female child:

$\boxed{\text{niece}}$

Gender is often the final deciding clue.

---

## 64. BLOOD RELATION MASTER WORKFLOW

---

For every question:

### Step 1

Identify reference person.

### Step 2

Mark gender immediately.

### Step 3

Build generation levels.

### Step 4

Connect parent/spouse/sibling links.

### Step 5

Answer only requested relation.

### Step 6

If gender is unknown, do not guess.

---

## 65. SPEED METHOD FOR SIMPLE CHAINS

---

Instead of drawing full tree for:

**mother's brother's son**

Use mental chain:

mother !' brother !' son

$\boxed{\text{Cousin}}$

For 3+ branching statements, draw the tree.

---

## 66. COMMON TRAPS

---

### & pTRAP 1 — Reverse relation

Father of X " X must be son.

X may be daughter.

### & pTRAP 2 — Only son

Only male child, not necessarily only child.

### & pTRAP 3 — Gender assumption

Sibling/cousin/child do not reveal gender.

### & pTRAP 4 — Maternal vs paternal

Track whether chain goes through mother or father.

### & pTRAP 5 — Cousin vs nephew

Uncle's child = cousin. Sibling's child = nephew/niece.

### & pTRAP 6 — Long English sentence

Solve inside-out.

---

## PRACTICE — LEVEL 0

---

### Q1

A is father of B. How is A related to B?

### Q2

P is sister of Q. What is P's gender?

### Q3

A is son of B. What can be said about B's gender?

---

**Q4**

X is brother of Y's father. How is X related to Y?

**Q5**

M is sister of N's mother. How is M related to N?

---

**LEVEL 1**

---

**Q6**

A is mother of B. B is brother of C. How is A related to C?

**Q7**

P is father of Q. Q is mother of R. How is P related to R?

**Q8**

A is brother of B. B is mother of C. How is A related to C?

**Q9**

X is daughter of Y's brother. How is X related to Y?

**Q10**

P is son of Q's sister. How is P related to Q?

---

**LEVEL 2**

---

**Q11**

A is the mother of B. B is the sister of C. C is the father of D. How is A related to D?

**Q12**

P is the brother of Q. Q is the father of R. How is P related to R?

**Q13**

A is the daughter of B. B is brother of C. C is mother of D. How are A and D related?

**Q14**

Pointing to a girl, Ravi said, "She is the daughter of my mother's only daughter." Ravi is male. How is the girl related to Ravi?

**Q15**

Pointing to a man, Neha said, "He is the son of my father's brother." How is the man related to Neha?

---

**LEVEL 3 — EXAM PRESSURE**

---

**Q16**

P is sister of Q. Q is father of R. R is brother of S. How is P related to S?

---

**Q17**

A man says, "The woman standing there is the daughter of the only son of my grandfather." If the man is himself the son of that only son, how is the woman related to him?

**Q18**

A is brother of B. C is mother of A. D is father of C. How is D related to B?

**Q19**

Pointing to a boy, Priya said, "He is the son of the only daughter of my grandmother." Priya is female, and her grandmother's only daughter is Priya's mother. How is the boy related to Priya?

**Q20**

A is sibling of B. B is mother of C. What exact gender-specific relation does A have to C?

---

**ANSWERS**

---

**Q1**

$\boxed{\text{Father}}$

**Q2**

Sister means female.

$\boxed{\text{Female}}$

**Q3**

B is parent of A, but gender unknown.

$\boxed{\text{Cannot be determined}}$

**Q4**

Brother of father:

$\boxed{\text{Paternal uncle}}$

**Q5**

Sister of mother:

$\boxed{\text{Maternal aunt}}$

**Q6**

B and C siblings.

A is mother of B, therefore also mother of C.

$\boxed{\text{Mother}}$

### Q7

---

P father of Q.

Q mother of R.

P is R's:

\[ \boxed{\text{Maternal grandfather}} \]

### Q8

---

A is brother of C's mother.

\[ \boxed{\text{Maternal uncle}} \]

### Q9

---

Y's brother's daughter:

\[ \boxed{\text{Niece}} \]

assuming X's relation is asked from Y's perspective.

### Q10

---

Q's sister's son:

\[ \boxed{\text{Nephew}} \]

### Q11

---

A !' mother of C !' C father of D.

A is D's:

\[ \boxed{\text{Grandmother}} \]

### Q12

---

P is brother of R's father.

\[ \boxed{\text{Paternal uncle}} \]

### Q13

---

B and C are siblings.

A is child of B.

D is child of C.

Therefore A and D are:

\[ \boxed{\text{Cousins}} \]

### Q14

---

Ravi's mother's only daughter = Ravi's sister.



Girl = sister's daughter.

$\boxed{\text{Niece}}$

---

### Q15

Father's brother's son:

$\boxed{\text{Cousin}}$

---

### Q16

P is sister of S's father Q.

Therefore:

$\boxed{\text{Paternal aunt}}$

---

### Q17

Grandfather's only son = man's father.

Woman is daughter of man's father.

Therefore:

$\boxed{\text{Sister}}$

---

### Q18

C is mother of A and therefore also mother of B if A and B are siblings.

D is C's father.

So D is B's:

$\boxed{\text{Maternal grandfather}}$

---

### Q19

Grandmother's only daughter = Priya's mother.

Boy = mother's son.

Since Priya is female, boy is Priya's:

$\boxed{\text{Brother}}$

---

### Q20

A is sibling of C's mother.

A's gender unknown.

Therefore exact relation is:

$\boxed{\text{Maternal uncle or maternal aunt}}$

Gender-specific answer cannot be uniquely determined.

---

## CHAPTER 60 STATUS

---

Covered so far:

- family vocabulary '
- parent/child/sibling relations '
- grandparents '
- maternal/paternal relations '
- uncle/aunt/cousin '
- nephew/niece '
- marriage/in-law basics '
- generation levels '
- gender tracking '
- unknown-gender cases '
- direct and indirect chains '
- pointing questions '
- introduction questions '
- only son/daughter traps '
- family-tree construction '
- L0–L3 practice '

But **Chapter 60 is not complete yet.**

Still remaining:

- coded blood relations
- symbol-based family relationships
- complex multi-family trees
- multiple marriages/branches
- number of family members
- generation-count questions
- family-member gender counts
- relation inference from several statements
- “how many sons/daughters/siblings?” questions
- final speed and ambiguity toolkit

## Progress

---

$\boxed{59/76}$  completed

Current: 0 = Chapter 60 — Blood Relations

## NEXT

---

+ Chapter 60 Part 2 — Coded Blood Relations, Complex Family Trees, Multiple-Statement Puzzles & Final Blood-Relation Toolkit

---

USER

next

ASSISTANT

## CHAPTER 60 — BLOOD RELATIONS

### PART 2 — CODED BLOOD RELATIONS, COMPLEX FAMILY TREES, MEMBER COUNTS, GENERATION QUESTIONS & FINAL TOOLKIT

Status: 0=Not in PROGRESS ! ' ' COMPLETE GATE: Medium Placements: High Priority: +Tier 2

This part covers the harder form of blood-relation questions:

- symbols represent family relations
- several statements must be combined
- multiple generations appear
- gender may be partly hidden
- questions ask counts rather than one relation

The main rule is:

**Decode relations first. Draw the family structure second. Answer only after the tree is stable.**

## 1. CODED BLOOD RELATIONS

Suppose:

$\{ A+B \}$

means:

**A is father of B.**

And:

$\{ A-B \}$

means:

**A is sister of B.**

Then:

$\{ P+Q-R \}$

means:

- P is father of Q
- Q is sister of R

Therefore P is also parent of R.

P is male.

So:

$\boxed{P \text{ is father of } R}$

---

## 2. SYMBOLS MUST BE TRANSLATED FIRST

---

Example:

- $(A+B)$ : A is mother of B
- $(A-B)$ : A is brother of B
- $(A \times B)$ : A is wife of B

Given:

$P-Q \times R$

Translate step by step:

$P \text{ " } Q$ :

$P = \text{brother of } Q$

$Q \times R$ :

$Q = \text{wife of } R$

So P is brother of R's wife.

Therefore P is R's:

$\boxed{\text{Brother-in-law}}$

---

## 3. ORDER MATTERS

---

If:

$A+B$

means A is father of B,

then:

$A+B$

does not mean B is father of A.

Direction is essential.

---

## 4. CODED RELATION CHAIN

---

Suppose:

- $(A@B)$ : A is mother of B
- $(A\#B)$ : A is brother of B

Given:

$P@Q\#R$

---

Then:

P is mother of Q.

Q is brother of R.

Thus Q and R are siblings.

P is also mother of R.

$\boxed{P \text{ is mother of } R}$

---

## 5. CODED RELATION WITH SPOUSE

Suppose:

- $(A*B)$ : A is husband of B
- $(A+B)$ : A is sister of B
- $(A-B)$ : A is father of B

Given:

$P*Q+R$

P is husband of Q.

Q is sister of R.

Therefore P is husband of R's sister.

P is R's:

$\boxed{\text{Brother in law}}$

---

## 6. CHOOSE A REFERENCE PERSON

In long coded expressions, choose the person from the question as your reference.

If asked:

**How is P related to T?**

Start from T and trace toward P.

This reduces unnecessary tree construction.

---

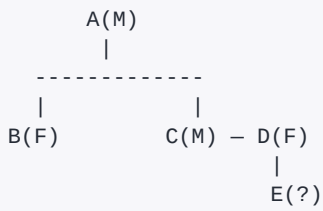
## 7. COMPLEX FAMILY TREE

Statements:

1. A is father of B.
2. B is sister of C.
3. C is husband of D.
4. D is mother of E.

Tree:

TEXT



Question:

**How is A related to E?**

A is father of C.

C is father of E if C is D's husband and D is mother of E under the intended family setup.

Therefore A is:

$\boxed{\text{Paternal grandfather}}$

## 8. WHY SPOUSE LINKS NEED CARE

From:

**C is husband of D   D is mother of E**

standard aptitude logic usually assumes C is E's father.

That is typically intended.

But in rigorous logic, this does not follow absolutely without explicit statement.

For exam problems, use the conventional family-tree assumption unless options/context indicate otherwise.

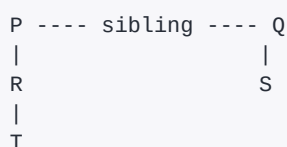
## 9. MULTIPLE BRANCHES

Statements:

- P and Q are siblings.
- P has son R.
- Q has daughter S.
- R has daughter T.

Tree:

TEXT



R and S are cousins.

T is one generation below S.

S is T's parent's cousin.

Exact English relation may be “first cousin once removed,” but aptitude questions often ask only generation or broader relation.

## 10. GENERATION COUNT

Suppose:

A is father of B. B is mother of C. C is father of D.

Then:

A !' B = 1 generation down B !' C = another C !' D = another

A and D differ by:

$\boxed{3 \text{ generations}}$

A is D's great-grandfather if genders/links align.

## 11. GREAT-GRANDPARENT

Parent:

$\boxed{+1}$

Grandparent:

$\boxed{+2}$

Great-grandparent:

$\boxed{+3}$

So if A is male and three generations above B:

$\boxed{\text{Great-grandfather}}$

## 12. GENERATION TABLE

Relation	Relative generation
Parent	+1
Grandparent	+2
Great-grandparent	+3
Sibling	0
Cousin	0
Spouse	0
Child	- 1

Relation	Relative generation
Grandchild	" 2
Great-grandchild	" 3

### 13. COUNTING MALE MEMBERS

Statements:

- A is father of B.
- B is sister of C.
- C is son of D.
- D is mother of E.

Known genders:

A = male B = female C = male D = female E = unknown

Guaranteed males:

$\boxed{2}$

Guaranteed females:

$\boxed{2}$

E cannot be counted by gender.

### 14. COUNT ONLY GUARANTEED GENDERS

If question asks:

**Minimum number of males?**

Count only those whose gender is fixed.

Do not assume unnamed spouses or unknown children.

### 15. MINIMUM FAMILY MEMBERS

Statements:

- A is father of B.
- C is mother of B.
- D is brother of B.

Minimum distinct people:

A, B, C, D

$\boxed{4}$

No extra person needed.



---

## 16. MAXIMUM FAMILY MEMBERS

---

If several statements do not force identities, “maximum possible members” may require treating ambiguous roles as different people.

Example:

- A is brother of B.
- C is father of A.
- D is father of B.

If siblings are assumed to share father in standard family setup:

$\boxed{C=D}$

Minimum distinct members = 3.

But if question does not make standard sibling-parent assumptions explicit, ambiguity can arise.

Aptitude questions usually intend the shared-parent interpretation.

---

## 17. “HOW MANY SONS?” QUESTIONS

---

Suppose:

- A is mother of B, C and D.
- B is brother of C.
- D is sister of C.

Known:

B = male D = female C gender unknown

Guaranteed sons of A:

$\boxed{1}$

Maximum possible sons:

if C male:

$\boxed{2}$

---

## 18. “HOW MANY DAUGHTERS?”

---

Same data:

Known daughter:

D only.

C could be female.

So minimum daughters:

$\boxed{1}$

Maximum:

$\boxed{2}$

---

---

## 19. ONLY SON / ONLY DAUGHTER IN COUNT QUESTIONS

**P has one son and two daughters.**

Clear.

But:

**Q is the only son of P.**

This tells:

$\boxed{P \text{ has exactly one son}}$

It says nothing about number of daughters.

---

## 20. “ONLY BROTHER”

If:

**A is B's only brother.**

Then A is male.

B may still have sisters.

A is the only male sibling of B.

---

## 21. FAMILY MEMBER COUNT WITH ONLY CHILD

**A is the only child of B and C.**

Then B and C have exactly one child:

A.

If B and C are spouses, minimum family members:

$\boxed{3}$

---

## 22. INTRODUCTION WITH MULTIPLE GENERATIONS

**Pointing to a woman, Arun said, “She is the wife of the only son of my father's father.”**

Break inside-out:

my father's father = paternal grandfather.

His only son = Arun's father, if that grandfather has only one son.

Wife of that son = Arun's mother.

Thus woman is:

$\boxed{\text{Mother}}$

## 23. ANOTHER INTRODUCTION

Pointing to a boy, Meera said, "He is the son of the only daughter of my mother's mother."

Mother's mother = maternal grandmother.

Her only daughter = Meera's mother, under intended setup.

Son of Meera's mother = Meera's brother if Meera female.

$\boxed{\text{Brother}}$

## 24. "MY FATHER'S ONLY BROTHER'S SON"

Self !' father !' father's only brother = paternal uncle !' his son = cousin

$\boxed{\text{Cousin}}$

## 25. "MY MOTHER'S SISTER'S DAUGHTER"

Self !' mother !' mother's sister = maternal aunt !' her daughter = cousin

$\boxed{\text{Cousin}}$

## 26. COMPLEX RELATION CHAIN

P is the father of Q. Q is the wife of R. R is the brother of S. S is the mother of T.

Question:

How is Q related to T?

Tree:

TEXT

```
P
|
Q(F) - R(M) ---- sibling ---- S(F)
 |
 T
```

Q is wife of T's maternal uncle.

So Q is T's aunt by marriage.

$\boxed{\text{Aunt by marriage}}$

---

## 27. BLOOD VS MARRIAGE RELATION

---

Some questions ask specifically:

**Which of the following is a blood relation?**

Example:

Brother, father, cousin = blood relations.

Brother-in-law = marriage relation.

Distinguish whether relation is biological/consanguineous versus marital.

---

## 28. CODED FAMILYRELATION — EXAMPLE

---

Suppose:

- $\backslash(A+B\backslash)$ : A is brother of B
- $\backslash(A-B\backslash)$ : A is mother of B
- $\backslash(A\times B\backslash)$ : A is husband of B

Given:

$\backslash[ P-Q+R \backslash]$

P is mother of Q.

Q is brother of R.

Therefore P is also mother of R.

$\backslash[ \boxed{\text{Mother}} \backslash]$

---

## 29. CODED FAMILYRELATION — MORE COMPLEX

---

Using same meanings:

$\backslash[ P\times Q-R+S \backslash]$

P is husband of Q.

Q is mother of R.

R is brother of S.

Therefore P is father of R and S under standard family interpretation.

Question:

How is P related to S?

$\backslash[ \boxed{\text{Father}} \backslash]$

---

## 30. CODED EXPRESSION — READ LEFT TO RIGHT, BUT LOGICALLY

---

A chain:

$\lfloor A+B-C \times D \rfloor$

does not mean arithmetic order.

Each symbol independently connects adjacent people.

Decode every edge:

A relation B B relation C C relation D

Then build the family tree.

---

## 31. FAMILY GRAPH METHOD

For complex questions, think of people as nodes and relations as edges:

TEXT

```
A --sibling-- B --parent--> C --spouse-- D
```

Then answer by path from one node to another.

This is more reliable than prose.

---

## 32. TWO FAMILIES JOINED BY MARRIAGE

Example:

A and B are siblings. C and D are siblings. A marries C.

Tree:

TEXT

```
B ---- sibling ---- A ===== C ---- sibling ---- D
```

Then:

B and D are related only through the marriage of A and C.

They are in-law family connections, not blood relatives.

---

## 33. COUSIN GENERATION CHECK

If X and Y are cousins:

$\lfloor \text{Generation}(X) = \text{Generation}(Y) \rfloor$

Their children are also in the same next generation.

A child's relation to parent's cousin is not simply "cousin" in strict genealogy, though aptitude tests may avoid such wording.

---

## 34. UNKNOWN GENDER + KNOWN GENERATION

**A is sibling of B. B is father of C.**

A is one generation above C, but A gender unknown.

Therefore:

$\boxed{A \text{ is paternal uncle or paternal aunt of C}}$

Do not force one.

## 35. UNKNOWN PATERNAL/MATERNAL SIDE

**A is sibling of B. B is parent of C.**

If B's gender unknown, then A could be:

- maternal uncle/aunt
- paternal uncle/aunt

Exact side cannot be determined.

## 36. MULTIPLE VALID RELATIONS

Sometimes the correct answer is:

$\boxed{\text{Cannot be determined}}$

because:

- gender unknown
- maternal/paternal side unknown
- two possible family structures remain

This is not a failure; it is often the intended answer.

## 37. "IS RELATED AS" QUESTIONS

Example:

A is B's brother.

B is C's mother.

Question:

**A is related to C in the same way D is related to E.**

A is:

$\boxed{\text{maternal uncle of C}}$

So D must be maternal uncle of E.

This combines blood relation with analogy.

---

## 38. NUMBER OF GENERATIONS IN A FAMILY TREE

---

Suppose:

Grandfather !' Father !' Son !' Granddaughter

Distinct generations:

1. grandfather
2. father
3. son
4. granddaughter

$\boxed{4}$  generations

Do not confuse number of people with number of generations.

---

## 39. SAME GENERATION CLUSTERS

---

Spouses:

same generation.

Siblings:

same generation.

Cousins:

same generation.

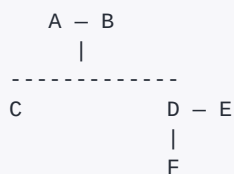
So one family tree may contain many people but only 3 generations.

---

## 40. GENERATION COUNT EXAMPLE

---

TEXT



Generations:

1. A,B
2. C,D,E
3. F

$\boxed{3}$

---

## 41. MINIMUM MALES FROM RELATION WORDS

---

Statements:

---

- A is brother of B.
- B is mother of C.
- C is sister of D.

Known males:

A.

Known females:

B, C.

D unknown.

Minimum males:

$\boxed{1}$

Maximum males among these four:

D may be male, so:

$\boxed{2}$

## 42. MINIMUM FEMALES

Same data:

B and C definitely female.

So minimum:

$\boxed{2}$

Maximum:

if D female:

$\boxed{3}$

## 43. SPOUSE COUNT TRAP

If:

**A is husband of B.**

This gives exactly two named people.

Do not invent children.

Similarly, if:

**A is father of B**

do not automatically invent B's mother.

## 44. CHILD COUNT TRAP



If:

**A is father of B and C.**

We know at least two children: B and C.

Do not conclude exactly two unless “only” is stated.

---

## 45. “AT LEAST” FAMILY LOGIC

If A has:

- son B
- daughter C

Then A has at least:

$\boxed{2 \text{ children}}$

Could have more.

---

## 46. BLOOD RELATION WITH DECEASED/FORMER SPOUSE

Some placement problems include widow/widower.

Widow = woman whose spouse died.

Widower = man whose spouse died.

These fix gender and marriage history, but relation tree remains same generation.

---

## 47. STEP-PARENT / STEP-SIBLING

If explicitly stated:

- stepfather
- stepmother
- stepsibling

do not treat as blood relationship unless biological relation separately stated.

Most basic aptitude sets avoid this.

---

## 48. ADOPTIVE RELATIONS

Similarly, if adoption is explicitly mentioned, legal family relation may differ from blood relation.

Use exact wording.

For standard GATE/placement blood relation questions, such complications are uncommon.

---

## 49. FINAL BLOOD-RELATION SEARCH ORDER

For any question:

1. mark genders
2. mark generation
3. decode parent/sibling/spouse links
4. identify maternal/paternal side
5. draw only required branch
6. answer relation direction carefully
7. if gender/side missing, say cannot determine exactly

## 50. FAST SYMBOL DECODING TABLE

For coded questions, immediately rewrite:

Symbol	Meaning
+	father
"	sister
×	husband
@	mother

Then translate expression.

Never keep meanings only in memory.

## 51. INSIDE-OUT RULE FOR LONG PHRASES

For:

**“daughter of the brother of my mother's only son”**

Start at:

1. my mother's only son
2. his brother
3. that brother's daughter

Do not read left-to-right blindly.

## 52. FAMILYTREE RULE FOR 4+ PEOPLE

If 4 or more named people are involved:

**draw the tree.**

Trying to hold everything mentally causes avoidable errors.

---

## 53. COMMON ADVANCED TRAPS

---

### & pTRAP 1 — Assuming unknown gender

Never.

### & pTRAP 2 — Confusing cousin with nephew/niece

Sibling's child = nephew/niece. Uncle/aunt's child = cousin.

### & pTRAP 3 — Forgetting relation direction

A is father of B does not tell whether B is son or daughter.

### & pTRAP 4 — Treating marriage as generation change

Spouses are same generation.

### & pTRAP 5 — “Only son” = only child

False.

### & pTRAP 6 — Counting unnamed people

Count only required or explicitly implied people.

### & pTRAP 7 — Maternal vs paternal

Track which parent the chain passes through.

---

## 54. GATE / PLACEMENT SPEED TARGET

---

Simple relation:

$\boxed{10-20\text{ sec}}$

Pointing/introduction:

$\boxed{20-35\text{ sec}}$

Coded family relation:

$\boxed{30-45\text{ sec}}$

Complex tree:

$\boxed{45-75\text{ sec}}$

If tree has several branches, drawing it is faster than mental solving.

---

## PRACTICE — LEVEL 0

---

### Q1

A is brother of B. B is mother of C. How is A related to C?

### Q2

P is sister of Q. Q is father of R. How is P related to R?

---

### Q3

A is the only son of B. Does B necessarily have only one child?

### Q4

X is sibling of Y. Y is mother of Z. Can X's exact gender-specific relation to Z be determined?

### Q5

A is father of B and C. What is the minimum number of children A has?

---

## LEVEL 1

---

Suppose:

- $\backslash(A+B\backslash)$ : A is father of B
- $\backslash(A-B\backslash)$ : A is sister of B

### Q6

$\backslash[ P+Q-R \backslash]$

How is P related to R?

### Q7

$\backslash[ P-Q+R \backslash]$

How is P related to R?

### Q8

A is mother of B. B is brother of C. C is father of D. How many generations are represented from A to D?

### Q9

P is father of Q. Q is sister of R. R is mother of S. How is P related to S?

### Q10

A is brother of B. B is sister of C. What is the minimum number of siblings in this set?

---

## LEVEL 2

---

Suppose:

- $\backslash(A@B\backslash)$ : A is mother of B
- $\backslash(A\#B\backslash)$ : A is brother of B
- $\backslash(A*B\backslash)$ : A is husband of B

### Q11

$\backslash[ P@Q\#R \backslash]$

How is P related to R?

### Q12

$\backslash[ P*Q@R \backslash]$

---

How is P related to R under standard family interpretation?

**Q13**

\[ P\#Q@R \]

How is P related to R?

**Q14**

Pointing to a woman, Arun says, "She is the wife of the only son of my father's father." How is she related to Arun?

**Q15**

A is brother of B. C is mother of A. D is father of C. How is D related to B?

---

## LEVEL 3 — EXAM PRESSURE

---

**Q16**

A is father of B. B is sister of C. C is husband of D. D is mother of E. How is A related to E?

**Q17**

P is sister of Q. Q is father of R. R is brother of S. S is mother of T. How is P related to T?

**Q18**

A is sibling of B. B is parent of C. What is the most precise relation of A to C?

**Q19**

A is mother of B and C. B is brother of C. C is sister of D. What is the minimum number of daughters A definitely has?

**Q20**

A is father of B. B is mother of C. C is father of D. What is A's relation to D?

---

## ANSWERS

---

**Q1**

A is brother of C's mother.

\[ \boxed{\text{Maternal}\ \text{uncle}} \]

**Q2**

P is sister of R's father.

\[ \boxed{\text{Paternal}\ \text{aunt}} \]

**Q3**

No.

Only son means exactly one son, not one child.

\[ \boxed{\text{No}} \]

#### Q4

---

No.

X could be maternal uncle or maternal aunt.

\[ \boxed{\text{Cannot be determined}} \]

#### Q5

---

Named children B and C.

\[ \boxed{2} \]

---

#### Q6

---

P is father of Q.

Q is sister of R.

Therefore P is also father of R.

\[ \boxed{\text{Father}} \]

#### Q7

---

P is sister of Q.

Q is father of R.

P is sister of R's father.

\[ \boxed{\text{Paternal aunt}} \]

#### Q8

---

A !' B !' C !' D involves:

A generation B/C generation D generation

Difference A to D = 2 generations downward.

A is grandmother of D.

Number of distinct generations involved:

\[ \boxed{3} \]

#### Q9

---

P is father of R as Q and R are siblings.

R is mother of S.

P is:

\[ \boxed{\text{Maternal grandfather}} \]

#### Q10

---

A, B, C are all siblings.

---

Minimum:

$\boxed{3}$

---

### Q11

---

P mother of Q.

Q brother of R.

So P mother of R.

$\boxed{\text{Mother}}$

### Q12

---

P husband of Q.

Q mother of R.

Under standard family interpretation:

$\boxed{P \text{ is father of R}}$

### Q13

---

P brother of Q.

Q mother of R.

Therefore P is:

$\boxed{\text{Maternal uncle}}$

### Q14

---

Father's father = paternal grandfather.

His only son = Arun's father.

Wife of Arun's father = Arun's mother.

$\boxed{\text{Mother}}$

### Q15

---

A and B are siblings.

C is their mother.

D is C's father.

Therefore D is B's:

$\boxed{\text{Maternal grandfather}}$

### Q16

---

A is father of C.

C is husband of D.

D is mother of E.

Under standard family interpretation C is E's father.

Therefore A is E's:

$\boxed{\text{Paternal grandfather}}$

### Q17

P is sister of Q.

Q is father of R and S.

S is mother of T.

P is sister of T's maternal grandfather Q.

So P is T's:

$\boxed{\text{Great-aunt}}$

More descriptively:

$\boxed{\text{Sister of maternal grandfather}}$

### Q18

B is parent of C.

A is sibling of B.

But B's gender and A's gender are both unknown.

Most precise answer:

$\boxed{\text{Sibling of C's parent}}$

A is an aunt/uncle, but maternal/paternal and gender cannot be fixed.

### Q19

A is mother of B and C.

C is sister of D, so C is definitely female.

B is brother, so male.

D gender unknown.

Guaranteed daughters of A:

$\boxed{1}$

assuming D is also A's child because C and D are siblings in the intended family tree.

If that sibling link is accepted, D may add another daughter but not guaranteed.

### Q20

A !' father of B B !' mother of C C !' father of D

A is three generations above D.



A is male.

Therefore:

$\boxed{\text{Great-grandfather}}$

---

## ' CHAPTER 60 COMPLETE

---

We have now covered:

- direct relations '
- maternal/paternal relations '
- generation levels '
- gender tracking '
- unknown gender '
- family-tree construction '
- pointing/introduction questions '
- only son/daughter traps '
- in-law relations '
- coded blood relations '
- symbol-based relations '
- complex multi-generation trees '
- member counting '
- son/daughter counting '
- generation counting '
- minimum/maximum gender counts '
- ambiguity handling '
- speed strategy '
- L0–L3 practice '

## Progress

---

$\boxed{60/76}$  text{ numbered chapters completed}}

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense +
- Chapter 62 — Ranking & Ordering +
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +

- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

### CHAPTER 61 — DIRECTION SENSE

Next we will cover:

- north/south/east/west
- left/right turns
- facing-direction logic
- 90°/180°/270° turns
- displacement vs distance
- coordinate method
- shortest distance
- Pythagoras-based problems
- diagonal directions
- clockwise/anticlockwise turns
- shadow-based direction questions
- final direction after multiple turns
- coded directions
- GATE/placement speed techniques
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 61 — DIRECTION SENSE

### PART 1 — CARDINAL DIRECTIONS, LEFT/RIGHT TURNS, FACING DIRECTION, COORDINATE METHOD, DISTANCE & SHORTEST PATH

**Status:** 0% PROGRESS **GATE:** Medium **Placements:** High **Priority:** +Tier 2

Direction-sense questions test whether you can correctly track:

- where a person is facing
- how turns change direction
- where the person finally ends up
- total distance travelled
- shortest distance from the starting point

The core idea is:

## Separate “facing direction” from “position.”

A person may be facing north but standing east of the starting point.

## 1. FOUR BASIC DIRECTIONS

TEXT

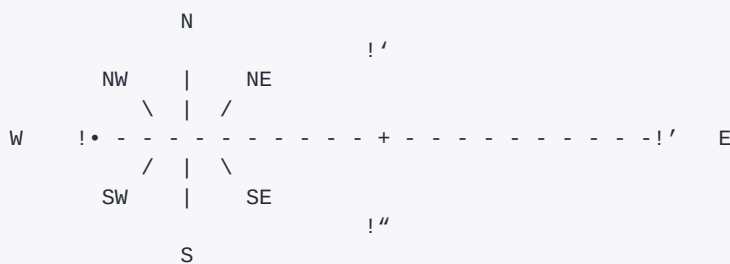


Abbreviations:

$\backslash$  N,  $\backslash$  E,  $\backslash$  S,  $\backslash$  W  $\backslash$

## 2. DIAGONAL DIRECTIONS

TEXT



The eight directions are:

$\backslash$  N,  $\backslash$  NE,  $\backslash$  E,  $\backslash$  SE,  $\backslash$  S,  $\backslash$  SW,  $\backslash$  W,  $\backslash$  NW  $\backslash$

## 3. CLOCKWISE ORDER

Memorize:

$\backslash$   $\boxed{\text{N} \rightarrow \text{E} \rightarrow \text{S} \rightarrow \text{W} \rightarrow \text{N}}$   $\backslash$

This is clockwise.

Anticlockwise:

$\backslash$   $\boxed{\text{N} \rightarrow \text{W} \rightarrow \text{S} \rightarrow \text{E} \rightarrow \text{N}}$   $\backslash$

## 4. RIGHT TURN

A right turn means  $\backslash(90^\circ\backslash)$  clockwise.

Facing	Right turn
North	East
East	South
South	West
West	North

## MUST MEMORIZE

$\boxed{N \rightarrow E \rightarrow S \rightarrow W \rightarrow N}$

## 5. LEFT TURN

A left turn means  $(90^\circ)$  anticlockwise.

Facing	Left turn
North	West
West	South
South	East
East	North

## 6. U-TURN

A U-turn means:

$(180^\circ)$

Opposite directions:

$N \rightarrow S$

$E \rightarrow W$

So if facing east, after a U-turn:

$\boxed{\text{West}}$

## 7. 270° TURN

A  $(270^\circ)$  clockwise turn is equivalent to:

$(90^\circ \text{ anticlockwise})$

So:

$\boxed{270^\circ \text{ right} = 90^\circ \text{ left}}$

Similarly:

$$\boxed{270^\circ \text{ left} = 90^\circ \text{ right}}$$

---

## 8. SIMPLE TURN EXAMPLE

---

A person faces North.

Turns right.

Now faces:

$$\boxed{\text{East}}$$

Turns right again:

$$\boxed{\text{South}}$$

Turns left:

South !' East.

Final:

$$\boxed{\text{East}}$$

---

## 9. TURN-COUNT SHORTCUT

---

Four right turns return to original direction.

$$4 \times 90^\circ = 360^\circ$$

So:

- 1 right =  $+90^\circ$
- 2 rights =  $180^\circ$
- 3 rights =  $270^\circ$
- 4 rights = original

Likewise for left turns.

---

## 10. MODULO-4 METHOD

---

If a person makes many  $(90^\circ)$  turns, reduce total turns modulo 4.

Example:

11 right turns.

$$11 \bmod 4 = 3$$

So equivalent to 3 right turns.

If initially North:

$$N \rightarrow E \rightarrow S \rightarrow W$$

Final:

$$\boxed{\text{West}}$$

---

## 11. LEFT AND RIGHT CANCEL

One right turn and one left turn cancel if no other rotation between them changes interpretation.

Example:

Facing North:

Right !' East Left !' North

Final:

$\boxed{\text{North}}$

## 12. NET ROTATION METHOD

Let:

- right  $(=+90^\circ)$
- left  $(=-90^\circ)$

Example:

3 right turns + 1 left turn:

$3(90) - 90 = 180^\circ$

So final direction is opposite initial direction.

If starting East:

$\boxed{\text{West}}$

## 13. FACING DIRECTION VS MOVEMENT

Statement:

**Ravi walks 5 m east, turns left, and walks 3 m.**

His movement:

1. East 5
2. Now facing North
3. North 3

Final position is northeast of start.

Final facing direction:

$\boxed{\text{North}}$

These are different answers.

## 14. POSITION TRACKING WITH COORDINATES

Use:

$\text{East} = +x$

$\text{West} = -x$

$\text{North} = +y$

$\text{South} = -y$

Starting point:

$(0,0)$

This is the safest method for long questions.

## 15. BASIC COORDINATE MOVEMENT

Walk 5 m East:

$(0,0) \rightarrow (5,0)$

Then 3 m North:

$(5,0) \rightarrow (5,3)$

Final coordinate:

$\boxed{(5,3)}$

## 16. FINAL RELATIVE DIRECTION

If final coordinate is:

$(x,y)$

Then:

Signs	Direction from start
$(x > 0, y > 0)$	NE
$(x < 0, y > 0)$	NW
$(x > 0, y < 0)$	SE
$(x < 0, y < 0)$	SW
$(x > 0, y = 0)$	E
$(x < 0, y = 0)$	W
$(x = 0, y > 0)$	N
$(x = 0, y < 0)$	S

## 17. EXAMPLE

A person walks:

- 6 m North

- 8 m East

Final coordinate:

$\boxed{(8,6)}$

Thus direction from start:

$\boxed{\text{NE}}$

## 18. DISTANCE TRAVELLED VS DISPLACEMENT

Suppose:

- 6 m North
- 8 m East

Total distance travelled:

$\boxed{6+8=14}$

Shortest distance from start:

$\boxed{\sqrt{6^2+8^2}}$

$\boxed{=\sqrt{36+64}}$

$\boxed{=10}$

So:

$\boxed{\text{Distance travelled}=14}$

$\boxed{\text{Displacement}=10}$

Do not confuse them.

## 19. PYTHAGORAS

For perpendicular horizontal and vertical displacement:

$\boxed{d=\sqrt{x^2+y^2}}$

Very common triples:

$\boxed{3,4,5}$

$\boxed{5,12,13}$

$\boxed{6,8,10}$

$\boxed{8,15,17}$

$\boxed{7,24,25}$

Memorizing these saves time.

## 20. EXAMPLE

A person goes:

- 9 m East



- 12 m North

Shortest distance:

$$\sqrt{9^2+12^2}$$

$$=\sqrt{81+144}$$

$$=\sqrt{225}$$

$$=\boxed{15}$$

## 21. CANCELLATION METHOD

Suppose movement:

- North 10
- South 4
- East 8
- West 3

Net vertical:

$$10-4=6\text{ North}$$

Net horizontal:

$$8-3=5\text{ East}$$

So final direction:

$$\boxed{NE}$$

Shortest distance:

$$\sqrt{6^2+5^2} = \boxed{\sqrt{61}}$$

## 22. FAST NET-DISPLACEMENT METHOD

Instead of drawing every segment, maintain:

$$N-S$$

and:

$$E-W$$

At end:

$$\boxed{\text{Vertical}=N-S}$$

$$\boxed{\text{Horizontal}=E-W}$$

Then determine direction and shortest distance.

## 23. SIMPLE PATH EXAMPLE

A starts facing North.

Walks 5 m.

Turns right and walks 4 m.

Turns right and walks 5 m.

Final coordinate:

Start:

$\backslash (0,0) \backslash$

North 5:

$\backslash (0,5) \backslash$

East 4:

$\backslash (4,5) \backslash$

South 5:

$\backslash (4,0) \backslash$

So A is:

$\backslash \boxed{4 \text{ m East of start}} \backslash$

Final facing:

$\backslash \boxed{\text{South}} \backslash$

---

## 24. CLOSED PATH

---

Suppose:

- 5 m North
- 5 m East
- 5 m South
- 5 m West

Final position:

$\backslash (0,0) \backslash$

So shortest distance from start:

$\backslash \boxed{0} \backslash$

But total distance:

$\backslash 5+5+5+5 = \boxed{20} \backslash$

---

## 25. RECTANGULAR LOOP

---

Walking equal opposite sides cancels:

$\backslash N=S, \quad E=W \backslash$

Final position = starting position.

This is useful for eliminating options quickly.

---

## 26. MULTIPLE TURNS WITHOUT MOVEMENT

---

A person faces East.

Turns:

- left
- left
- right
- right
- right

Net:

2 left =  $(-180^\circ)$

3 right =  $(+270^\circ)$

Net:

$[ 90^\circ ]$

clockwise.

East  $+90^\circ$ :

$[ \boxed{\text{South}} ]$

---

## 27. 45° TURNS

---

Some placement questions use  $45^\circ$  turns.

Directions then cycle through 8 positions:

Clockwise:

$[ N \text{ to } NE \text{ to } E \text{ to } SE \text{ to } S \text{ to } SW \text{ to } W \text{ to } NW \text{ to } N ]$

Each step:

$[ 45^\circ ]$

---

## 28. EXAMPLE — $45^\circ$

---

Facing North.

Turn  $(135^\circ)$  clockwise.

$[ 135^\circ = 3 \times 45^\circ ]$

N ! NE ! E ! SE

Final:

$[ \boxed{\text{SE}} ]$

---

## 29. $225^\circ$ CLOCKWISE

---

$[ 225^\circ = 5 \times 45^\circ ]$

---

Starting North:

N !' NE !' E !' SE !' S !' SW

Final:

$\boxed{\text{SW}}$

---

### 30. FAST ANGLE-TO-DIRECTION TABLE

Starting North:

Clockwise turn	Direction
0°	N
45°	NE
90°	E
135°	SE
180°	S
225°	SW
270°	W
315°	NW
360°	N

---

### 31. OPPOSITE DIRECTIONS

Memory pairs:

$\boxed{N \leftrightarrow S}$

$\boxed{E \leftrightarrow W}$

$\boxed{NE \leftrightarrow SW}$

$\boxed{NW \leftrightarrow SE}$

---

### 32. “RIGHT OF NORTH”

Right of North:

$\boxed{\text{East}}$

“Left of North”:

$\boxed{\text{West}}$

### 33. "RIGHT OF WEST"

If facing West, right side points North.

$\boxed{\text{North}}$

This is where many students make mistakes.

Never use your own right side; use the person's facing direction.

### 34. BODY-ORIENTATION METHOD

Imagine standing in the direction.

If facing South:

- left = East
- right = West

If facing West:

- left = South
- right = North

### 35. MUST-MEMORIZE LEFT/RIGHT TABLE

Facing	Left	Right
N	W	E
E	N	S
S	E	W
W	S	N

### 36. EXAMPLE

A faces South.

Turns left.

South's left is:

$\boxed{\text{East}}$

Then right:

East's right is South.

Final:

$\boxed{\text{South}}$

### 37. "TURNS TO HIS LEFT" TRAP

Left is always relative to current facing direction, not original direction.

Example:

Face North.

Right !' East.

Then left !' North.

You must update orientation after every turn.

### 38. FINAL FACING AFTER PATH

A person walks north, then turns right and walks east, then turns left and walks north.

Question:

**Which direction is he finally facing?**

Answer:

$\boxed{\text{North}}$

The last movement direction is the facing direction unless another final turn occurs.

### 39. FINAL POSITION AFTER PATH

Same path:

North 5 East 4 North 3

Net:

North:

$\boxed{8}$

East:

$\boxed{4}$

Therefore position:

$\boxed{\text{NE}\text{ of start}}$

### 40. MOVEMENT MAY BE IMPLIED AFTER TURN

Statement:

**He turns right and goes 5 m.**

This means both:

- new facing direction changes

- then he moves in that direction

Do not record only the turn.

---

## 41. TURN WITHOUT MOVEMENT

Statement:

**He turns left.**

No position change.

Only direction changes.

---

## 42. ORIGIN METHOD FOR LONG QUESTIONS

Draw start as:

TEXT

$O = (0, 0)$

Then update:

TEXT

N:  $y += d$

S:  $y -= d$

E:  $x += d$

W:  $x -= d$

This method scales to long placement questions.

---

## 43. EXAMPLE — LONG PATH

A walks:

- 10 m North
- 6 m East
- 4 m South
- 2 m West

Net:

Vertical:

$\backslash 10-4=6 \backslash$

Horizontal:

$\backslash 6-2=4 \backslash$

Final coordinate:

$\backslash (4,6) \backslash$

Direction:

$$\boxed{NE}$$

Shortest distance:

$$\sqrt{4^2+6^2} = \sqrt{52} = \boxed{2\sqrt{13}}$$

---

## 44. PATHREVERSAL

---

If someone walks:

- East 10
- West 10

Net horizontal:

$$0$$

Those two movements cancel for final position.

But total distance includes both:

$$20$$

---

## 45. TURNING AROUND AND RETRACING

---

Suppose:

Walk North 8.

Turn around.

Walk 3.

Turning around means now South.

Net:

$$8-3=5\text{ North}$$

So:

$$\boxed{5\text{ m North of start}}$$

---

## 46. RELATIVE POSITION OF TWO PEOPLE

---

A is at:

$$(5,4)$$

B is at:

$$(2,1)$$

Position of A relative to B:

$$(5-2, 4-1)$$

$$=(3,3)$$

Therefore A is:

$$\boxed{NE\text{ of B}}$$

---



---

## 47. GENERAL RELATIVE-POSITION RULE

---

If:

$$A = (x_A, y_A)$$

$$B = (x_B, y_B)$$

Then A relative to B:

$$\boxed{(x_A - x_B, y_A - y_B)}$$

Then use signs.

---

## 48. EXAMPLE

---

$$P = (-4, 3)$$

$$Q = (2, 7)$$

P relative to Q:

$$(-4 - 2, 3 - 7)$$

$$= (-6, -4)$$

So P is:

$$\boxed{\text{6 m West of Q}}$$

---

## 49. MIDPOINT / MEETING QUESTIONS

---

Suppose A travels 4 m East from origin.

B travels 4 m West from the same origin.

Their separation:

$$4 + 4 = 8$$

They lie on opposite sides.

---

## 50. SAME-DIRECTION SEPARATION

---

A is 10 m East of origin.

B is 6 m East.

Distance between them:

$$10 - 6 = 4$$

A is:

$$\boxed{4 \text{ m East of B}}$$

---

## 51. PERPENDICULAR SEPARATION

A is 6 m North of O.

B is 8 m East of O.

Distance A-B:

$$\sqrt{6^2+8^2} = \boxed{10}$$

A relative to B:

A coordinate  $((0,6))$

B  $((8,0))$

A-B:

$$(-8,6)$$

So A is:

$$\boxed{\text{NW of B}}$$

## 52. QUESTIONS WITH “TO THE RIGHT OF”

Statement:

**B is to the right of A.**

If absolute map orientation is intended and they are facing North, B is East of A.

But if no facing direction is provided, “right” can be ambiguous unless context defines an orientation.

Placement questions usually provide facing direction or use standard diagrams.

## 53. ROTATED MAP TRAP

If the question says:

**North is shown toward the right side of the page.**

Then page-up is no longer North.

Follow the given orientation, not the physical page.

## 54. SHORTEST DISTANCE WITHOUT DRAWING

If net:

$$E-W=12$$

$$N-S=5$$

Then:

$$d=\sqrt{12^2+5^2}$$

$\boxed{13}$

No need to reconstruct complete path.

---

## 55. TOTAL DISTANCE

---

Always:

$\boxed{\text{Total distance} = \text{sum of all path lengths}}$

Regardless of direction.

Example:

3 North, 4 East, 3 South:

$3+4+3 = \boxed{10}$

Final displacement = 4.

---

## 56. SIMPLE SHORTCUT — RECTANGLE

---

If path forms three sides of a rectangle:

North  $(a)$ , East  $(b)$ , South  $(a)$

vertical cancels.

Final:

$b \text{ East}$

No Pythagoras required.

---

## 57. SIMPLE SHORTCUT — SAME AXIS

---

If net movement only along one axis, shortest distance is absolute net amount.

Example:

North 20, South 8:

$|20-8|=12$

$\boxed{12}$

---

## 58. DIAGONAL DIRECTION "EXACT 45° NECESSARILY

---

In aptitude questions, if both coordinates have same sign:

$(+,+)$

we usually say NE even if distances aren't equal.

Example:

$(10,3)$

is still northeast of origin.

---

Do not require exact  $45^\circ$  unless angle is explicitly asked.

---

## 59. COMMON DIRECTION-SENSE TRAPS

---

### & pTRAP 1 — Left/right relative to yourself

Wrong. Use the person's current facing direction.

### & pTRAP 2 — Forgetting to update facing after every turn

Track orientation continuously.

### & pTRAP 3 — Distance vs displacement

They are different.

### & pTRAP 4 — Using total East instead of net East

Cancel East/West first.

### & pTRAP 5 — Confusing final facing with final location

Always check what is asked.

### & pTRAP 6 — Ignoring wrap in rotation

$360^\circ$  = original.

---

## 60. FAST SOLVING DECISION TREE

---

### Only asks final facing?

Track turns only.

### Asks final direction from start?

Track net coordinates.

### Asks shortest distance?

Find net horizontal and vertical, then Pythagoras.

### Asks total distance?

Add all movement lengths.

### Long path?

Use coordinates.

### Many turns but little movement?

Use net rotation.

---

## PRACTICE — LEVEL 0

---

### Q1

A person faces North and turns right. Which direction now?

---

**Q2**

A person faces West and turns left. Which direction now?

**Q3**

A person faces South and makes a U-turn. Which direction now?

**Q4**

A person walks 5 m East and 5 m West. How far from start?

**Q5**

A person walks 3 m North and 4 m East. Shortest distance from start?

---

**LEVEL 1**

---

**Q6**

A faces North, turns right, then right, then left. Final facing direction?

**Q7**

A walks 6 m North and 8 m East. What direction is A from the start?

**Q8**

A walks 10 m North, 4 m South, 3 m East. Find shortest distance from start.

**Q9**

A walks 5 m North, 4 m East, 5 m South. Where is A from start?

**Q10**

A faces East and makes 7 right turns. Final direction?

---

**LEVEL 2**

---

**Q11**

A walks:

- 8 m North
- 6 m East
- 3 m South
- 2 m West

Find:

1. direction from start
2. shortest distance.

**Q12**

A faces South. Turns right, then left, then left. Final direction?

**Q13**

---

A faces North and turns  $(225^\circ)$  clockwise. Final direction?

#### Q14

P is 5 m East and 7 m North of O. Q is 2 m East and 3 m North of O. Where is P relative to Q?

#### Q15

A walks 12 m East and 5 m North. Shortest distance?

---

### LEVEL 3 — EXAM PRESSURE

---

#### Q16

A starts facing West. He turns right, walks 4 m, turns left, walks 6 m, turns left and walks 4 m. Where is he relative to the start?

#### Q17

A walks 10 m North, turns right and walks 6 m, turns right and walks 4 m, turns left and walks 3 m. Find final direction from start and shortest distance.

#### Q18

A person makes 13 left turns starting from South. Final direction?

#### Q19

P is at  $(-3, 5)$ , Q is at  $(4, -2)$ . In which direction is P from Q?

#### Q20

A walks 5 m North, 12 m East, 5 m South, 7 m West. Find:

1. total distance
  2. shortest distance
  3. final direction from start.
- 

### ANSWERS

---

#### Q1

North !' right !' East.

$\boxed{\text{East}}$

#### Q2

West !' left !' South.

$\boxed{\text{South}}$

#### Q3

South opposite is North.

$\boxed{\text{North}}$

---

## Q4

Equal opposite movement cancels.

$$\boxed{0}$$

## Q5

$$\sqrt{3^2+4^2}=5$$

$$\boxed{5\text{ m}}$$

## Q6

North:

Right !' East Right !' South Left !' East

$$\boxed{\text{East}}$$

## Q7

North + East means:

$$\boxed{\text{NE}}$$

## Q8

Net North:

$$10-4=6$$

East:

$$3$$

Shortest distance:

$$\sqrt{6^2+3^2} = \sqrt{45} = \boxed{3\sqrt{5}}$$

## Q9

North 5 and South 5 cancel.

Only East 4 remains.

$$\boxed{4\text{ m East}}$$

## Q10

$$7\text{ mod }4=3$$

East !' South !' West !' North

$$\boxed{\text{North}}$$

## Q11

Net vertical:

$$\backslash[8-3=5\text{ N}]\backslash$$

Net horizontal:

$$\backslash[6-2=4\text{ E}]\backslash$$

Direction:

$$\backslash[\boxed{\text{NE}}]\backslash$$

Distance:

$$\backslash[\sqrt{5^2+4^2}=\boxed{\sqrt{41}}]\backslash$$

---

## Q12

---

Start South.

Right:

$$\backslash[\text{South}\text{to West}]\backslash$$

Left:

$$\backslash[\text{West}\text{to South}]\backslash$$

Left:

$$\backslash[\text{South}\text{to East}]\backslash$$

$$\backslash[\boxed{\text{East}}]\backslash$$

## Q13

---

From North,  $\backslash(225^\circ\text{clockwise})$ :

$$\backslash[\text{N}\text{to NE}\text{to E}\text{to SE}\text{to S}\text{to SW}]\backslash$$

$$\backslash[\boxed{\text{SW}}]\backslash$$

## Q14

---

P:

$$\backslash[(5,7)]\backslash$$

Q:

$$\backslash[(2,3)]\backslash$$

P relative to Q:

$$\backslash[(3,4)]\backslash$$

Both positive.

$$\backslash[\boxed{\text{NE}}]\backslash$$

## Q15

---

$$\backslash[\sqrt{12^2+5^2}=13]\backslash$$

---



$\boxed{13 \text{ m}}$

---

## Q16

---

Start facing West.

Right !' North.

Walk 4:

$(0,4)$

Left from North !' West.

Walk 6:

$(-6,4)$

Left from West !' South.

Walk 4:

$(-6,0)$

So:

$\boxed{6 \text{ m West of start}}$

---

## Q17

---

Start:

$(0,0)$

North 10:

$(0,10)$

Right !' East 6:

$(6,10)$

Right !' South 4:

$(6,6)$

Left from South !' East 3:

$(9,6)$

Final relative direction:

$\boxed{\text{NE}}$

Shortest distance:

$\sqrt{9^2+6^2} = \sqrt{117} = \boxed{3\sqrt{13}}$

Final facing direction is East, but the **position from start** is NE.

---

## Q18

---

13 left turns:

---

$$\lfloor 13 \bmod 4 = 1 \rfloor$$

One left from South:

$$\lfloor \boxed{\text{East}} \rfloor$$

---

## Q19

---

P:

$$\lfloor (-3, 5) \rfloor$$

Q:

$$\lfloor (4, -2) \rfloor$$

P relative to Q:

$$\lfloor (-3-4, 5-(-2)) \rfloor$$

$$\lfloor =(-7, 7) \rfloor$$

West + North:

$$\lfloor \boxed{\text{NW}} \rfloor$$

---

## Q20

---

Movement:

North 5 East 12 South 5 West 7

### Total distance

$$\lfloor 5+12+5+7 = \boxed{29} \rfloor$$

### Net

North/South cancel.

East:

$$\lfloor 12-7=5 \rfloor$$

Shortest distance:

$$\lfloor \boxed{5} \rfloor$$

Final direction:

$$\lfloor \boxed{\text{East}} \rfloor$$

---

## CHAPTER 61 STATUS

---

Covered so far:

- cardinal directions '
  - diagonal directions '
  - left/right turns '
  - U-turns '
-

- 45°/90°/180°/270° turns '
- clockwise/anticlockwise rotation '
- modulo-turn shortcut '
- final facing direction '
- coordinate method '
- final position '
- distance vs displacement '
- shortest-distance/Pythagoras '
- relative position of two people '
- cancellation shortcuts '
- L0–L3 practice '

But **Chapter 61 is not complete yet.**

Still remaining:

- shadow-based direction questions
- sunrise/sunset direction logic
- morning/evening shadow traps
- coded directions
- direction analogies
- complex relative-position sets
- rotated-direction systems
- final placement speed toolkit
- final Chapter 61 audit

## Progress

$\lfloor \boxed{60/76} \text{ numbered chapters completed} \rfloor$

**Current:** 0 = Chapter 61 — Direction Sense

## NEXT

+ Chapter 61 Part 2 — Shadows, Sun Direction, Coded Directions, Complex Relative Positions & Final Direction-Sense Toolkit

USER

next

ASSISTANT

## CHAPTER 61 — DIRECTION SENSE

### PART 2 — SHADOWS, SUN DIRECTION, CODED DIRECTIONS, COMPLEX RELATIVE POSITIONS & FINAL DIRECTION-SENSE TOOLKIT

**Status:** 0% PROGRESS ! ' ' COMPLETE **GATE:** Medium **Placements:** High **Priority:** +Tier 2

This part covers the trickier direction questions where direction is not stated directly.

You may need to infer it from:

- shadows
- sunrise/sunset
- coded symbols
- relative positions
- rotated maps
- multiple people
- indirect statements

## 1. SUNRISE AND SUNSET — MUST KNOW

For aptitude questions:

$\boxed{\text{Sun rises in East}}$

$\boxed{\text{Sun sets in West}}$

Therefore:

### Morning

Sun is toward East.

Shadow falls approximately toward:

$\boxed{\text{West}}$

### Evening

Sun is toward West.

Shadow falls approximately toward:

$\boxed{\text{East}}$

## 2. SHADOW IS OPPOSITE THE SUN

Basic principle:

TEXT

Sun -----> Person -----> Shadow

So:

$\boxed{\text{Shadow direction = opposite sun direction}}$

## 3. MORNING SHADOW

In the morning:

Sun = East.

Therefore shadow = West.

$\boxed{\text{Morning shadow points West}}$

---

## 4. EVENING SHADOW

---

In the evening:

Sun = West.

Therefore shadow = East.

$\boxed{\text{Evening shadow points East}}$

---

## 5. PERSON FACING THE SUN — MORNING

---

Suppose Ravi is facing the rising sun.

Rising sun is East.

Therefore Ravi faces:

$\boxed{\text{East}}$

His shadow lies behind him:

$\boxed{\text{West}}$

---

## 6. PERSON FACING THE SUN — EVENING

---

If a person faces the setting sun:

Setting sun = West.

Thus person faces:

$\boxed{\text{West}}$

Shadow:

$\boxed{\text{East}}$

---

## 7. SHADOW ON RIGHT SIDE — MORNING

---

Suppose in the morning, a man's shadow falls to his right.

Morning shadow is West.

Therefore his right side = West.

Which direction must he face?

Check:

If facing South:

- right = West

Therefore:

---

$\boxed{\text{South}}$

## 8. SHADOW ON LEFT SIDE — MORNING

Morning shadow = West.

If shadow is on person's left, then left = West.

Facing North:

- left = West

So:

$\boxed{\text{North}}$

## 9. SHADOW ON RIGHT SIDE — EVENING

Evening shadow = East.

If East is on person's right:

Facing North:

- right = East

Therefore:

$\boxed{\text{North}}$

## 10. SHADOW ON LEFT SIDE — EVENING

Evening shadow = East.

If East is on person's left:

Facing South:

- left = East

Therefore:

$\boxed{\text{South}}$

## 11. FAST SHADOW TABLE

Time	Shadow	Shadow on Left !' Facing	Shadow on Right !' Facing
Morning	West	North	South
Evening	East	South	North

### MUST MEMORIZE

This table solves most shadow questions instantly.

## 12. IMPORTANT NOON CAUTION

---

At noon, shadow direction depends on:

- geographic location
- season
- solar position

So do **not** automatically assume North or South unless the question gives enough information.

For standard aptitude problems, morning/evening cases are much safer.

---

## 13. TWO PEOPLE FACING EACH OTHER

---

Suppose A faces North.

B stands directly in front of A and faces A.

Then B must face:

$\boxed{\text{South}}$

If one person faces East and another faces them directly, the other faces West.

---

## 14. TWO PEOPLE FACING SAME DIRECTION

---

If A and B both face North:

- A's right = East
- B's right = East

Same orientation.

But if they face opposite directions, their left/right sides reverse.

---

## 15. OPPOSITE-FACING LEFT/RIGHT TRAP

---

Suppose A faces North and B faces South.

A's right:

$\boxed{\text{East}}$

B's right:

$\boxed{\text{West}}$

Therefore:

**“to the right” is not an absolute direction.**

It depends on facing.

---

## 16. RELATIVE POSITION LANGUAGE

---

Common phrases:

- A is north of B
- A is south of B
- A is east of B
- A is west of B
- A is northeast of B
- A is southwest of B

Translate immediately into coordinates.

Let B:

$\backslash (0,0) \backslash$

Then:

- north !'  $\backslash (0,+1) \backslash$
- east !'  $\backslash (+1,0) \backslash$
- southwest !'  $\backslash (-1,-1) \backslash$

Actual distances may vary.

## 17. MULTIPLE RELATIVE STATEMENTS

**A is north of B. C is east of A.**

Take B:

$\backslash (0,0) \backslash$

A:

$\backslash (0,1) \backslash$

C:

$\backslash (1,1) \backslash$

Therefore C is:

$\backslash \boxed{\text{NE}} \text{ of B} \backslash$

## 18. THREE-PERSON EXAMPLE

**P is west of Q. R is south of P.**

Set Q:

$\backslash (0,0) \backslash$

P:

$\backslash (-1,0) \backslash$

R:



$\nabla (-1,-1) \nabla$

Thus R is:

$\nabla \boxed{\text{SW text of Q}} \nabla$

---

## 19. FOUR-PERSON EXAMPLE

---

**A is north of B. C is east of A. D is south of C.**

Let B:

$\nabla (0,0) \nabla$

A:

$\nabla (0,1) \nabla$

C:

$\nabla (1,1) \nabla$

D:

$\nabla (1,0) \nabla$

Therefore D is:

$\nabla \boxed{\text{East text of B}} \nabla$

---

## 20. RELATIVE POSITION CANCELLATION

---

Path:

B !' North to A A !' East to C C !' South to D

North and South cancel.

Only East remains.

Thus D lies East of B.

Use cancellation instead of drawing when obvious.

---

## 21. CODED DIRECTIONS

---

Suppose:

$\nabla A+B \nabla$

means A is North of B.

$\nabla A-B \nabla$

means A is South of B.

$\nabla A*B \nabla$

means A is East of B.

---

$\backslash [ A/B \backslash ]$

means A is West of B.

Then decode exactly like coded blood relations.

---

## 22. CODED DIRECTION EXAMPLE

---

Given:

$\backslash [ P+Q \backslash ]$

means P is North of Q.

$\backslash [ Q*R \backslash ]$

means Q is East of R.

Let R:

$\backslash [ (0,0) \backslash ]$

Q:

$\backslash [ (1,0) \backslash ]$

P:

$\backslash [ (1,1) \backslash ]$

Therefore P is:

$\backslash [ \boxed{NE\text{ of } R} \backslash ]$

---

## 23. CODED CHAIN EXAMPLE

---

Using:

- $+$  = North of
- $-$  = South of
- $*$  = East of
- $/$  = West of

Given:

$\backslash [ A+B*C \backslash ]$

Interpret adjacent relations:

A is North of B.

B is East of C.

Therefore:

$\backslash [ \boxed{A\text{ is NE of } C} \backslash ]$

---

## 24. DO NOT USE MATHEMATICAL PRECEDENCE

---

In coded-direction chains:

---

$\backslash [ A+B \times C \backslash ]$

do not apply multiplication before addition.

These are relationship symbols, not arithmetic operators.

Decode each adjacent pair.

---

## 25. DIRECTION ANALOGY

---

Example:

$\backslash [ \text{North:South}::\text{East:?} \backslash ]$

Opposite relation.

East opposite:

$\backslash [ \boxed{\text{West}} \backslash ]$

---

## 26. ROTATION ANALOGY

---

$\backslash [ \text{North:East}::\text{West:?} \backslash ]$

First relation:

North !' East = 90° clockwise.

Apply to West:

West !' North.

$\backslash [ \boxed{\text{North}} \backslash ]$

---

## 27. DIAGONAL ANALOGY

---

$\backslash [ \text{NE:SW}::\text{NW:?} \backslash ]$

First relation = opposite.

Opposite of NW:

$\backslash [ \boxed{\text{SE}} \backslash ]$

---

## 28. CLOCK-FACE DIRECTION METHOD

---

Sometimes directions are mapped to clock positions.

Standard:

- 12 o'clock = North
- 3 o'clock = East
- 6 o'clock = South
- 9 o'clock = West

Approximately:

---

- 1:30 = NE
- 4:30 = SE
- 7:30 = SW
- 10:30 = NW

## 29. CLOCK-FACE EXAMPLE

If a person is facing 12 o'clock and turns to 3 o'clock, direction changes:

[ N to E ]

So turn:

[  $90^\circ$  clockwise ]

## 30. ROTATED DIRECTION SYSTEM

Suppose a question states:

**In a certain map, East is called North, North is called West, West is called South, and South is called East.**

This is a **renaming problem**, not a physical rotation unless specified.

Create mapping:

Actual	Called
East	North
North	West
West	South
South	East

## 31. RENAMED DIRECTION QUESTION

Using above mapping:

**In which direction does the sun rise?**

Actual sunrise:

[ East ]

But East is called:

[ North ]

Answer:

[ North ]

---

## 32. DO NOT ANSWER WITH REAL-WORLD LABEL

---

If the question uses renamed directions, first determine actual direction, then convert to the coded name.

This is exactly like color-renaming or word-substitution coding.

---

## 33. PHYSICAL ROTATION VS RENAMING

---

These are different.

### Renaming

"North is called East."

You translate labels.

### Physical rotation

"The map is rotated 90° clockwise."

You rotate the actual orientation.

Read carefully.

---

## 34. MAP ROTATED 90° CLOCKWISE

---

If a normal map is physically rotated 90° clockwise:

Original North arrow now points toward page-right.

Thus:

- page right = North
- page down = East
- page left = South
- page up = West

Do not use normal page orientation.

---

## 35. POSITION AFTER DIAGONAL MOVEMENT

---

Suppose person walks:

- 5 m NE
- 5 m SW

These are exact opposites.

Final displacement:

$\boxed{0}$

provided equal distances along same diagonal.

---

## 36. DIAGONAL MOVEMENT COMPONENTS

---

If exact geometric distances matter:

A distance  $(d)$  at  $45^\circ$  has horizontal and vertical components:

$$\left[ \frac{d}{\sqrt{2}} \right]$$

each.

For most basic direction-sense questions, you don't need this unless numerical displacement is asked.

---

## 37. EXAMPLE WITH DIAGONAL DISTANCE

---

A walks  $(10\sqrt{2})$  m NE.

Components:

$$\left[ x=10, \quad y=10 \right]$$

Then walks 10 m West.

Final:

$$\left[ x=0, \quad y=10 \right]$$

So:

$$\left[ \boxed{10 \text{ m North}} \right]$$

---

## 38. $45^\circ$ COMPONENT MEMORY

---

For diagonal movement  $(d)$ :

$$\left[ \boxed{\text{horizontal} = \text{vertical} = \frac{d}{\sqrt{2}}} \right]$$

Only when diagonal is exactly  $45^\circ$ .

---

## 39. TWO PEOPLE MOVING FROM SAME POINT

---

A walks 6 m North.

B walks 8 m East.

Distance between them:

$$\left[ \sqrt{6^2 + 8^2} = \boxed{10} \right]$$

Relative direction:

A from B:

$$\left[ \text{NW} \right]$$

B from A:

$$\left[ \text{SE} \right]$$

---

## 40. REVERSE RELATIVE DIRECTION

---

If A is NE of B, then B is:

$\boxed{\text{SW of A}}$

Opposite pairs:

$N \leftrightarrow S$

$E \leftrightarrow W$

$NE \leftrightarrow SW$

$NW \leftrightarrow SE$

---

## 41. MULTIPLE-PEOPLE PLACEMENT SET

---

Statements:

- A is 4 m North of B.
- C is 3 m East of A.
- D is 4 m South of C.
- E is 5 m West of D.

Let:

$B = (0,0)$

A:

$(0,4)$

C:

$(3,4)$

D:

$(3,0)$

E:

$(-2,0)$

Thus E is:

$\boxed{2 \text{ m West of B}}$

---

## 42. QUESTION — C RELATIVE TO E

---

C:

$(3,4)$

E:

$(-2,0)$

C" E :

$(5,4)$

---

So C is:

$\boxed{\text{NE of E}}$

Distance:

$\sqrt{25+16} = \boxed{\sqrt{41}}$

---

### 43. SAME LINE DETECTION

---

If two points have:

#### Same x-coordinate

They lie North/South of each other.

#### Same y-coordinate

They lie East/West of each other.

This quickly eliminates diagonal answers.

---

### 44. COINCIDENT POINTS

---

If two people end at same coordinate:

$(x_A, y_A) = (x_B, y_B)$

Then they are:

$\boxed{\text{at the same location}}$

No direction relation exists other than coincident.

---

### 45. FINAL DIRECTION FROM START MAY BE CARDINAL EVEN AFTER MANY TURNS

---

Example:

North 5 East 10 South 5

Vertical cancels.

Final position:

$\boxed{\text{East}}$

Do not choose NE merely because east and north occurred during the journey.

---

### 46. SHADOW QUESTION — PERSON FACING UNKNOWN

---

Morning.

A person's shadow falls exactly behind them.

Morning shadow = West.

If shadow is behind, then person faces opposite shadow:

---



$\boxed{\text{East}}$

---

## 47. SHADOW IN FRONT — MORNING

---

Morning shadow = West.

If shadow lies directly in front of person, person faces:

$\boxed{\text{West}}$

because front direction equals shadow direction.

---

## 48. SHADOW BEHIND — EVENING

---

Evening shadow = East.

If shadow is behind person, person faces West.

$\boxed{\text{West}}$

---

## 49. SHADOW IN FRONT — EVENING

---

Evening shadow = East.

Person faces:

$\boxed{\text{East}}$

---

## 50. FAST SHADOW LOGIC

---

First determine:

$\boxed{\text{Shadow direction}}$

Then use relation:

- in front
- behind
- left
- right

Never directly guess the person's direction.

---

## 51. COMPLEX TURN + SHADOW EXAMPLE

---

In the morning, a man has his shadow on his left.

From table:

He faces North.

Then he turns right twice.

North !' East !' South.

---

Final:

$\boxed{\text{South}}$

---

## 52. CODED DIRECTIONS WITH DIAGONALS

---

Suppose:

- $@$  = NE of
- $\#$  = SW of
- $\$$  = NW of
- $\%$  = SE of

Given:

$A@B$

Then A is:

$\boxed{\text{NE of B}}$

Therefore B is:

$\boxed{\text{SW of A}}$

---

## 53. CHAIN WITH DIAGONALS

---

Suppose:

A is NE of B.

B is SE of C.

Use unit coordinates:

C:

$(0,0)$

B:

$(1,-1)$

A:

$(2,0)$

Therefore A is:

$\boxed{\text{East of C}}$

Diagonal movements can cancel vertical components.

---

## 54. ANOTHER DIAGONAL CHAIN

---

A is NW of B.

B is SW of C.

C:

---

$\sqrt{(0,0)}$

B:

$\sqrt{(-1,-1)}$

A:

$\sqrt{(-2,0)}$

So A is:

$\sqrt{\boxed{\text{West of C}}}$

---

## 55. DISTANCE IN UNIT-DIRECTION QUESTIONS

If question gives only relative directions and no distances, do not calculate numerical distance unless equal/unit distances are explicitly intended.

Direction may be determined even when exact distance cannot.

---

## 56. INSUFFICIENT DISTANCE DATA

A is north of B.

C is east of A.

Without distances:

C is generally northeast of B if both movements are positive, but exact angle/distance is unknown.

If question asks exact distance:

$\sqrt{\boxed{\text{Cannot be determined}}}$

---

## 57. DIRECTION CAN BE DETERMINED EVEN IF DISTANCE CANNOT

Example:

A is 5–10 m north of B.

C is some positive distance east of A.

C must be:

$\sqrt{\boxed{\text{NE of B}}}$

Exact distance:

unknown.

This is useful in data-sufficiency style questions.

---

## 58. FINAL DIRECTION-SENSE WORKFLOW

For each question:

### Step 1

Identify what is asked:

- facing?
- position?
- distance?
- shortest distance?
- relative direction?

## Step 2

If turns only, track orientation.

## Step 3

If movement exists, use coordinates.

## Step 4

Cancel opposite movements.

## Step 5

Use Pythagoras only after net displacement.

## Step 6

For shadows, determine sun ! ' shadow !' facing.

## Step 7

For coded directions, translate symbols first.

---

## 59. MUST-MEMORIZE TABLE

Facing	Left	Right	Opposite
N	W	E	S
E	N	S	W
S	E	W	N
W	S	N	E

---

## 60. MUST-MEMORIZE DIAGONAL OPPOSITES

$\boxed{NE \leftrightarrow SW}$

$\boxed{NW \leftrightarrow SE}$

---

## 61. MUST-MEMORIZE SHADOW RULE

$\boxed{\text{Morning: shadow West}}$

$\boxed{\text{Evening: shadow East}}$

---

## 62. MUST-MEMORIZE COORDINATE RULE

---

$$\boxed{E-W=x}$$

$$\boxed{N-S=y}$$

Shortest displacement:

$$\boxed{\sqrt{x^2+y^2}}$$

---

## 63. FINAL TRAPS

---

### & pTRAP 1 — Morning sun vs shadow

Sun East, shadow West.

### & pTRAP 2 — Facing vs location

Different concepts.

### & pTRAP 3 — Right side of a South-facing person

It is West.

### & pTRAP 4 — Renaming vs rotation

Do not confuse coded direction names with physical orientation.

### & pTRAP 5 — Total distance vs shortest distance

Add segments for total; use net displacement for shortest.

### & pTRAP 6 — Diagonal chain

Use coordinates; visual intuition can mislead.

### & pTRAP 7 — Exact distance without enough information

Do not invent equal distances.

---

## PRACTICE — LEVEL 0

---

### Q1

In the morning, in which direction does a shadow generally fall?

### Q2

In the evening, in which direction does a shadow generally fall?

### Q3

In the morning, a person's shadow is on his left. Which way is he facing?

### Q4

In the evening, a person's shadow is on her right. Which way is she facing?

### Q5

If A is NE of B, where is B relative to A?

---

---

## LEVEL 1

---

### Q6

A is North of B. C is East of A. Where is C relative to B?

### Q7

P is West of Q. R is South of P. Where is R relative to Q?

### Q8

In the morning, a man's shadow is directly behind him. Which direction does he face?

### Q9

If North is called East, East is called South, South is called West, and West is called North, what is the coded name for actual East?

### Q10

A is NE of B and B is SE of C. Where is A relative to C, assuming equal diagonal steps?

---

## LEVEL 2

---

### Q11

A is 4 m North of B. C is 3 m East of A. D is 4 m South of C. Where is D relative to B?

### Q12

In the evening, a person's shadow lies on her left. Which direction is she facing?

### Q13

P is at  $((4,7))$ , Q is at  $((-2,3))$ . Where is P relative to Q?

### Q14

A is NW of B and B is SW of C. Where is A relative to C, assuming equal diagonal steps?

### Q15

A person faces North and turns  $(315^\circ)$  clockwise. Final direction?

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

In the morning, Ravi's shadow is to his right. He then turns left, walks 5 m, turns right and walks 3 m. What direction is he finally facing?

### Q17

A is 6 m North of B. C is 8 m East of B. Find:

1. A relative to C

2. distance AC.

### Q18

Suppose:

- $+$  means North of
- $*$  means East of
- $-$  means South of

Given:

$\backslash [ P+Q*R \ ]$

Where is P relative to R?

### Q19

If actual North is called West, actual West is called South, actual South is called East, and actual East is called North, what is the coded name of the direction opposite actual East?

### Q20

A walks  $\backslash (10\sqrt{2})$  m NE, then 10 m West, then 4 m South. Where is A relative to start and what is the shortest distance?

---

## ANSWERS

### Q1

Morning sun = East.

Shadow = West.

$\backslash \boxed{\text{West}} \backslash$

### Q2

Evening sun = West.

Shadow = East.

$\backslash \boxed{\text{East}} \backslash$

### Q3

Morning shadow = West.

If West is on left, person faces North.

$\backslash \boxed{\text{North}} \backslash$

### Q4

Evening shadow = East.

If East is on right, person faces North.

$\backslash \boxed{\text{North}} \backslash$

### Q5

Opposite of NE:

$$\boxed{\text{SW}}$$

---

### Q6

North + East:

$$\boxed{\text{NE}}$$

---

### Q7

West + South:

$$\boxed{\text{SW}}$$

---

### Q8

Morning shadow = West.

If shadow is behind, person faces opposite West:

$$\boxed{\text{East}}$$

---

### Q9

Actual East is called:

$$\boxed{\text{South}}$$

---

### Q10

Let C:

$$(0,0)$$

B = SE:

$$(1,-1)$$

A = NE of B:

$$(2,0)$$

Therefore:

$$\boxed{\text{East}}$$

---

### Q11

Let B:

$$(0,0)$$

A:

$$(0,4)$$

---



C:

$\sqrt{(3,4)}$

D:

$\sqrt{(3,0)}$

So:

$\sqrt{\boxed{3\text{ m East of B}}}$

## Q12

---

Evening shadow = East.

If East is on left, person faces South.

$\sqrt{\boxed{\text{South}}}$

## Q13

---

P relative to Q:

$\sqrt{(4-(-2),7-3)}$

$\sqrt{=(6,4)}$

Thus:

$\sqrt{\boxed{\text{NE}}}$

## Q14

---

C:

$\sqrt{(0,0)}$

B SW:

$\sqrt{(-1,-1)}$

A NW of B:

$\sqrt{(-2,0)}$

Therefore:

$\sqrt{\boxed{\text{West}}}$

## Q15

---

315° clockwise from North:

$\sqrt{\boxed{\text{NW}}}$

## Q16

---

Morning shadow on Ravi's right.

Morning shadow = West.

If right = West, Ravi initially faces South.

Then left:

South !' East.

Walk 5.

Right:

East !' South.

Walk 3.

Final facing:

$\boxed{\text{South}}$

---

## Q17

---

Let B:

$(0,0)$

A:

$(0,6)$

C:

$(8,0)$

A relative to C:

$(-8,6)$

Thus:

$\boxed{\text{NW}}$

Distance:

$\sqrt{8^2+6^2} = 10$

$\boxed{10 \text{ m}}$

---

## Q18

---

P + Q:

P north of Q.

Q \* R:

Q east of R.

Take R:

$(0,0)$

Q:

$(1,0)$

P:

$(1,1)$

---

Therefore:

$\boxed{NE}$

---

## Q19

---

Direction opposite actual East is:

West

Actual West is called:

South

Therefore coded answer:

$\boxed{South}$

---

## Q20

---

First movement:

$10\sqrt{2}$  NE

Components:

(+10,+10)

Then 10 West:

(-10,0)

Net now:

(0,10)

Then 4 South:

(0,6)

Final:

$\boxed{6 \text{ m North of start}}$

Shortest distance:

$\boxed{6 \text{ m}}$

---

## ' CHAPTER 61 COMPLETE

---

We have covered:

- cardinal directions '
- diagonal directions '
- left/right turns '
- clockwise/anticlockwise turns '
- 45°, 90°, 180°, 270°, 315° rotations '
- final facing direction '

- coordinates '
- distance vs displacement '
- shortest-distance problems '
- relative-position questions '
- multiple-person position sets '
- sunrise/sunset logic '
- shadow questions '
- coded directions '
- direction analogies '
- renamed directions '
- rotated maps '
- diagonal movement/components '
- insufficient-information cases '
- L0–L3 practice '

## Progress

---

\[ \boxed{61/76}\text{ numbered chapters completed} \]

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering +
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

---

### CHAPTER 62 — RANKING & ORDERING

Next we will cover:

- rank from left/right
- rank from top/bottom
- total from two-sided rank
- people between two positions
- interchange of positions

- rank change after swapping
- tallest/shortest/oldest/youngest ordering
- comparison chains
- partial ordering
- tied ranks
- missing ranks
- multiple-person ordering puzzles
- GATE/placement shortcuts
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 62 — RANKING & ORDERING

### PART 1 — RANK FROM LEFT/RIGHT, TOTAL PERSONS, PEOPLE BETWEEN, POSITION INTERCHANGE & BASIC ORDERING CHAINS

Status:  $\emptyset = 0\%$  PROGRESS GATE: High Placements: Very High Priority:  $\emptyset = 0\%$  For 1

Ranking questions ask things like:

- 7th from left
- 12th from right
- 5th from top
- 18th from bottom
- how many people are there?
- how many lie between two people?
- what happens after positions are swapped?

The most important formula is:

$$\boxed{\text{Total} = L + R - 1}$$

where:

- $(L)$  = rank from left/top
- $(R)$  = rank from right/bottom

## 1. WHY DO WE SUBTRACT 1?

Suppose Ravi is:

- 7th from left
- 10th from right

If we add:

$$\boxed{7+10=17}$$

Ravi himself is counted in both sides.

So subtract one:

$$\boxed{17-1=16}$$

Therefore:

$$\boxed{16 \text{ people}}$$

---

## 2. MASTER FORMULA

---

If a person is:

- $L$ th from left
- $R$ th from right

then:

$$\boxed{N=L+R-1}$$

Similarly:

$$\boxed{N=T+B-1}$$

for top/bottom rankings.

---

## 3. EXAMPLE

---

A student is 12th from the top and 19th from the bottom.

Total students:

$$\boxed{12+19-1}$$

$$=\boxed{30}$$

---

## 4. FIND RANK FROM THE OTHER SIDE

---

If total  $N$  and left rank  $L$  are known:

$$\boxed{R=N-L+1}$$

Similarly:

$$\boxed{L=N-R+1}$$

---

## 5. EXAMPLE

---

There are 40 students.

Anu is 13th from left.

Rank from right:

$$\boxed{40-13+1}$$

---

$$\boxed{28}$$

---

## 6. TOP/BOTTOM VERSION

---

There are 50 students.

Rahul is 16th from top.

From bottom:

$$50 - 16 + 1$$

$$\boxed{35}$$

---

## 7. VISUAL METHOD

---

Suppose P is 5th from left in a row of 12.

TEXT

1 2 3 4 P 6 7 8 9 10 11 12

From right:

$$12 - 5 + 1 = 8$$

So P is:

$$\boxed{8\text{th from right}}$$

---

## 8. PEOPLE BETWEEN TWO RANKS — SAME SIDE

---

If A is 5th from left and B is 12th from left:

People strictly between them:

$$12 - 5 - 1$$

$$\boxed{6}$$

Formula:

$$\boxed{|r_1 - r_2| - 1}$$

---

## 9. WHY SUBTRACT 1?

---

Ranks:

5 and 12.

Difference:

$$12 - 5 = 7$$

But this includes one endpoint gap count too many.

Positions strictly between:

6,7,8,9,10,11

Total:

$\boxed{6}$

So:

$\boxed{7-1=6}$

---

## 10. EXAMPLE

---

A is 9th from top.

B is 21st from top.

People between:

$\boxed{21-9-1=11}$

---

## 11. SAME PERSON, TWO SIDES

---

If A is 8th from left and 15th from right:

Total:

$\boxed{8+15-1=22}$

Do not use “between” formula here.

---

## 12. TWO DIFFERENT PEOPLE — OPPOSITE-SIDE RANKS

---

This is trickier.

Suppose:

- A is 8th from left
- B is 10th from right
- total = 25

Convert B to left rank:

$\boxed{25-10+1=16}$

Now both ranks are from left:

A = 8 B = 16

People between:

$\boxed{16-8-1=7}$

---

## 13. RULE FOR OPPOSITE-SIDE RANKS

---

First convert both people to the same side.

Then use:

---



$$\boxed{r_1 - r_2 - 1}$$

This prevents confusion.

---

## 14. EXAMPLE

---

Total students = 40.

P = 12th from top.

Q = 17th from bottom.

Convert Q to top:

$$40 - 17 + 1 = 24$$

People between:

$$24 - 12 - 1 = \boxed{11}$$

---

## 15. CAN TOTAL BE FOUND FROM TWO DIFFERENT PEOPLE?

---

Suppose:

- A = 7th from left
- B = 9th from right

Can total be determined?

Not by these two statements alone.

Because A and B may be:

- far apart
- adjacent
- same person impossible unless explicitly same person
- overlapping differently depending total

Need extra information such as:

- number of people between them
  - A is to left/right of B
  - their positions after interchange
- 

## 16. TOTAL FROM TWO PEOPLE + PEOPLE BETWEEN

---

Suppose:

- A is 8th from left
- B is 10th from right
- 5 people are between A and B
- A is to the left of B

Then:

From left to A:

---

$$\boxed{8}$$

Then 5 between.

Then B:

1 position.

Then 9 people to right of B because B is 10th from right.

Total:

$$\boxed{8+5+1+9}$$

$$\boxed{23}$$

---

## 17. SHORTCUT FORMULA

---

If:

- A is  $(L)$ th from left
- B is  $(R)$ th from right
- $(k)$  people between them
- A lies left of B

then:

$$\boxed{N=L+R+k}$$

Why?

$$\boxed{L+k+R}$$

already accounts for both A and B correctly.

Example:

$$\boxed{8+10+5=23}$$

---

## 18. CHECK THE FORMULA

---

A is 8th from left.

B is 10th from right.

5 people between.

Positions from left:

$$A = 8$$

B must be:

$$\boxed{8+5+1=14}$$

If total 23:

B from right:

$$\boxed{23-14+1=10}$$

Correct.

---

---

## 19. OVERLAP CASE

---

Suppose:

- A is 15th from left
- B is 12th from right
- total = 20

Convert B to left:

$$\boxed{20-12+1=9}$$

So B is actually left of A.

A = 15, B = 9.

People between:

$$\boxed{15-9-1=5}$$

Do not assume the person ranked from left is physically left of the person ranked from right.

---

## 20. IMMEDIATE NEIGHBOUR

---

If A is 7th from left and B is immediately to A's right:

B rank:

$$\boxed{8\text{th from left}}$$

If B is immediately to left:

$$\boxed{6\text{th}}$$

---

## 21. kth TO THE RIGHT

---

A is 12th from left.

B is 5 places to the right of A.

Then:

$$\boxed{12+5=17}$$

So B is:

$$\boxed{17\text{th from left}}$$

---

## 22. kth TO THE LEFT

---

A = 18th from left.

B = 7 places left of A.

$$\boxed{18-7=11}$$

$$\boxed{11\text{th from left}}$$

---

---

## 23. "5 PLACES TO THE RIGHT" VS "5th TO THE RIGHT"

---

In aptitude problems, these usually mean the same relative shift:

$[ +5 ]$

But "5th person to the right of A" can sometimes mean count the fifth person after A.

That still gives:

$[ A+5 ]$

Do not include A in the count unless wording explicitly says "including A."

---

## 24. INTERCHANGE OF POSITIONS

---

Very common.

Suppose:

- A is 8th from left
- B is 15th from left

They swap positions.

Then:

A becomes:

$[ 15th ]$

B becomes:

$[ 8th ]$

Simple if both ranks from same side.

---

## 25. INTERCHANGE WITH OPPOSITE-SIDE INFORMATION

---

Suppose:

- A is 8th from left
- B is 12th from right
- after swapping, A becomes 20th from left

Since A takes B's original position:

B's original position from left:

$[ 20 ]$

B was 12th from right.

Therefore total:

$[ 20+12-1 ]$

$[ \boxed{31} ]$

---

## 26. THIS IS A VERY IMPORTANT PATTERN

If A and B swap:

**A's new rank = B's old rank.**

**B's new rank = A's old rank.**

Use this immediately.

## 27. EXAMPLE

A is 10th from left.

B is 18th from right.

After interchange, A becomes 25th from left.

Since B's old left rank = 25:

$$N = 25 + 18 - 1$$

$$N = \boxed{42}$$

## 28. FIND B'S NEW RANK AFTER SWAP

From previous example:

A was 10th from left.

After swap, B occupies A's position.

Therefore B becomes:

$$N = \boxed{10\text{th from left}}$$

From right:

$$42 - 10 + 1 = \boxed{33\text{rd}}$$

## 29. POSITION CHANGE AFTER SWAP

Suppose A was 7th from left and after swap becomes 18th.

A moved:

$$18 - 7 = 11$$

places to the right.

If B took A's original spot, B moved correspondingly in the opposite direction if both positions are in same row.

## 30. RANK IMPROVEMENT / DETERIORATION

In exams, lower rank number is usually better.

If a student improves from 20th to 12th:

Improvement:

$$\lfloor 20-12=8 \rfloor$$

$$\lfloor \boxed{8 \text{ places}} \rfloor$$

---

### 31. RANK FALL

---

From 7th to 15th:

Fall:

$$\lfloor 15-7 = \boxed{8 \text{ places}} \rfloor$$

---

### 32. NEW RANK AFTER OVERTAKING

---

Suppose A is 10th in a race and overtakes the person in 7th place.

If A passes the 7th-place runner and no one else, A becomes:

$$\lfloor \boxed{7\text{th}} \rfloor$$

The passed runner becomes 8th.

---

### 33. CLASSIC RACE TRAP

---

If you overtake the runner in 2nd place, what is your rank?

You take their position.

So:

$$\lfloor \boxed{2\text{nd}} \rfloor$$

Not 1st.

---

### 34. OVERTAKING LAST PERSON TRAP

---

You cannot normally overtake the person in last place if you are behind them because then you would be even further behind than last.

Such questions test consistency.

---

### 35. ORDERING BY HEIGHT

---

Suppose:

$$\lfloor A > B \rfloor$$

means A is taller than B.

If:

$$\lfloor A > B > C > D \rfloor$$

---

then tallest:

$\boxed{A}$

Shortest:

$\boxed{D}$

---

## 36. ORDERING BY AGE

Suppose:

A is older than B.

B is older than C.

Then:

$A > B > C$

Oldest:

$A$

Youngest:

$C$

---

## 37. TRANSLATE WORDING INTO ONE SYMBOL

For height:

$[$

$\Rightarrow$  taller

$]$

For age:

$[$

$\Rightarrow$  older

$]$

For marks:

$[$

$\Rightarrow$  higher marks

$]$

For speed:

$[$

$\setminus$

This makes chain questions much easier.

### 38. EXAMPLE — HEIGHT

Statements:

- P is taller than Q.
- R is shorter than Q.
- S is taller than P.

Translate:

$\setminus S > P > Q > R \setminus$

Tallest:

$\setminus \boxed{S} \setminus$

Shortest:

$\setminus \boxed{R} \setminus$

### 39. EXAMPLE — AGE

- A older than B
- C younger than B
- D older than A

Chain:

$\setminus D > A > B > C \setminus$

Second oldest:

$\setminus \boxed{A} \setminus$

### 40. PARTIAL ORDERING

Suppose:

$\setminus A > B \setminus$

and:

$\setminus C > B \setminus$

Can we determine whether A or C is taller?

No.

Possible:

$\setminus A > C > B \setminus$



or:

$\forall C > A > B$

Therefore:

$\boxed{\text{Cannot determine}}$

---

## 41. DO NOT INVENT A TOTAL ORDER

---

Not every set of comparisons gives one complete sequence.

This matters in:

- GATE reasoning
  - placement puzzles
  - data sufficiency
- 

## 42. TRANSITIVE RELATION

---

If:

$\forall A > B$

and:

$\forall B > C$

then:

$\forall A > C$

This is transitivity.

Most ranking/order relations such as:

- taller
- older
- heavier
- faster
- higher marks

are transitive.

---

## 43. NON-TRANSITIVE WARNING

---

Some relations are not safely transitive.

Example:

“A likes B” and “B likes C” does not imply A likes C.

Only use ordering-chain logic for transitive comparisons.

---

## 44. POSITION IN ORDER FROM COMPARISON CHAIN

---

Suppose:

$P > Q > R > S > T$

Then:

- $P = 1\text{st}$
- $Q = 2\text{nd}$
- $R = 3\text{rd}$
- $S = 4\text{th}$
- $T = 5\text{th}$

R's rank:

$\boxed{3\text{rd}}$

---

## 45. "SECOND SHORTEST"

---

For:

$A > B > C > D > E$

where  $(>)$  means taller:

Tallest! shortest order:

A, B, C, D, E

Shortest = E.

Second shortest:

$\boxed{D}$

---

## 46. "THIRD OLDEST"

---

If age order:

$P > Q > R > S > T$

Third oldest:

$\boxed{R}$

---

## 47. COMPARATIVE LANGUAGE

---

Translate carefully:

"B is not taller than A"

means:

$B \leq A$

not necessarily:

$B < A$

If equal height is possible, exact order may remain ambiguous.

---

---

## 48. “NO SHORTER THAN”

A is no shorter than B:

$\nmid A \geq B \mid$

Could be equal.

This matters in advanced ranking questions.

---

## 49. STRICT VS NON-STRICT ORDER

Strict:

$\nmid A > B \mid$

means definitely greater.

Non-strict:

$\nmid A \geq B \mid$

allows equality.

Ties become possible.

---

## 50. TIED RANKS — BASIC IDEA

Suppose marks:

- A = 95
- B = 90
- C = 90
- D = 80

Using competition ranking:

A = 1st B = 2nd C = 2nd D = 4th

Rank 3 is skipped.

---

## 51. DENSE RANKING

Another ranking convention:

A = 1 B = 2 C = 2 D = 3

This is called dense ranking.

Most aptitude questions involving ties specify or imply the convention.

Do not assume if it affects the answer.

---

## 52. ORDINAL RANKING WITHOUT TIES

---

Most basic aptitude questions assume every person occupies a distinct rank.

Then formulas like:

$$\lfloor L+R-1 \rfloor$$

work directly.

---

## 53. TOTAL WHEN RANKS TIE

---

If ties exist, ordinary left/right positional formulas may no longer map cleanly to performance ranks.

Distinguish:

- physical position in a row
- performance rank in a test

This is a crucial conceptual difference.

---

## 54. PHYSICAL POSITION VS PERFORMANCE RANK

---

Physical row:

Every person occupies one unique position.

Performance ranking:

Two people may share rank.

So:

$$\lfloor \boxed{L+R-1} \rfloor$$

belongs mainly to **physical/order positions without ties**, not arbitrary tied performance ranking.

---

## 55. MISSING RANK

---

Suppose 8 people have distinct ranks 1 through 8.

Known ranks:

1,2,3,5,6,7,8

Missing:

$$\lfloor \boxed{4} \rfloor$$

Basic but useful in larger puzzles.

---

## 56. TOTAL FROM LAST RANK

---

If the lowest-ranked student is 35th:

Total students:

$$\lfloor \boxed{35} \rfloor$$

assuming no ties.

---

---

## 57. EXACTLY k PEOPLE AHEAD

If 7 people are ahead of A:

A's rank from top:

$$\boxed{7+1 = 8^{\text{th}}}$$

---

## 58. EXACTLY k PEOPLE BEHIND

If 12 people are behind A:

A's rank from bottom:

$$\boxed{12+1 = 13^{\text{th}}}$$

---

## 59. BOTH AHEAD AND BEHIND

If 7 people are ahead of A and 12 behind A:

Total:

$$\boxed{7+1+12 = 20}$$

A's rank:

top:

$$\boxed{8}$$

bottom:

$$\boxed{13}$$

Check:

$$\boxed{8+13-1=20}$$

---

## 60. PEOPLE BETWEEN FROM AHEAD/BEHIND DATA

If A has 5 people ahead.

B has 11 people ahead.

Their ranks:

$$A = 6 \quad B = 12$$

Between:

$$\boxed{12-6-1 = 5}$$

---

## 61. MULTIPLE-PERSON POSITIONING

Suppose:

- P is 7th from left.
- Q is 4 places right of P.
- R is 3 places left of Q.

Then:

P:

\[ 7 \]

Q:

\[ 11 \]

R:

\[ 8 \]

So order among them:

\[ P,R,Q \]

with R immediately right of P.

## 62. CHAIN POSITION METHOD

Use one anchor.

If P = 10.

Q is +4:

\[ 14 \]

R is " 6 from Q:

\[ 8 \]

S is +3 from R:

\[ 11 \]

Now all positions are explicit.

## 63. POSITION COLLISION

If two distinct people are derived to occupy the same position, either:

- one statement has been misread
- the data are inconsistent
- they are tied only if performance rank, not physical row

Always sanity-check.

## 64. POSITION INTERCHANGE — HARDER EXAMPLE

A is 15th from left.

B is 20th from right.

After they interchange, A becomes 28th from left.

Then B's original left rank:

$$\lfloor 28 \rfloor$$

Total:

$$\lfloor 28+20-1 = \boxed{47} \rfloor$$

B after swap is at A's original left rank:

$$\lfloor 15 \rfloor$$

From right:

$$\lfloor 47-15+1 = \boxed{33} \rfloor$$

---

## 65. NUMBER OF PEOPLE BETWEEN A AND B IN ABOVE EXAMPLE

---

Original positions from left:

$$A = 15 \quad B = 28$$

People between:

$$\lfloor 28-15-1 = \boxed{12} \rfloor$$

---

## 66. FAST SWAP TOOLKIT

---

If A and B swap:

$$\lfloor \boxed{\text{A new} = \text{B old}} \rfloor$$

$$\lfloor \boxed{\text{B new} = \text{A old}} \rfloor$$

Then convert sides if needed.

This avoids unnecessary diagrams.

---

## 67. COMMON RANKING TRAPS

---

& pTRAP 1 — Forgetting " 1

$$\lfloor N=L+R-1 \rfloor$$

& pTRAP 2 — People between

$$\lfloor |r_1-r_2|-1 \rfloor$$

& pTRAP 3 — Different-side ranks

Convert to same side first.

& pTRAP 4 — Swap

New position of A = old position of B.

& pTRAP 5 — Overtaking 2nd

You become 2nd, not 1st.

Do not force relation if insufficient.

---

## 68. FAST DECISION TREE

---

### Same person's left/right rank?

Use:

$$\lfloor L+R-1 \rfloor$$

### Need other-side rank?

Use:

$$\lfloor N-r+1 \rfloor$$

### Two people, same-side ranks?

Between:

$$\lfloor |r_1-r_2|-1 \rfloor$$

### Opposite-side ranks?

Convert first.

### Swap?

Exchange their old positions.

### Comparison statements?

Build  chain.

### Partial comparison?

Check whether exact order is actually determined.

---

## PRACTICE — LEVEL 0

---

### Q1

A is 7th from left and 10th from right. Total people?

### Q2

There are 30 people. P is 8th from left. Rank from right?

### Q3

A is 5th from left, B is 11th from left. How many between?

### Q4

A has 9 people ahead. What is A's rank?

### Q5

A has 6 people behind. What is A's rank from bottom?



---

## LEVEL 1

---

### Q6

There are 40 students. P is 13th from top. What is P's rank from bottom?

### Q7

A is 9th from left. B is 6 places to the right of A. What is B's rank?

### Q8

A is 15th from left, B is 23rd from left. How many lie between?

### Q9

P is 12th from left and Q is 10th from right in a row of 30. How many are between them?

### Q10

A is 8th from left and 12th from right. Find total.

---

## LEVEL 2

---

### Q11

A is 8th from left. B is 10th from right. Five people lie between them, and A is left of B. Find total.

### Q12

A is 10th from left. B is 18th from right. After swapping, A becomes 25th from left. Find total.

### Q13

Using Q12, what is B's new rank from right?

### Q14

P is taller than Q. R is shorter than Q. S is taller than P. Who is tallest?

### Q15

A is older than B. C is younger than B. D is older than A. Who is second oldest?

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

A is 15th from left. B is 20th from right. After swapping, A is 28th from left. Find:

1. total people
2. people between A and B originally.

### Q17

In a row of 50, P is 14th from left and Q is 21st from right. How many people lie between P and Q?

---

**Q18**

A is taller than B. C is taller than B. No other relation is given. Can we determine who is taller between A and C?

**Q19**

A is 12th from top and has 18 people below. How many students are there?

**Q20**

R is ranked 18th. He improves by 7 positions. What is his new rank?

---

**ANSWERS**

---

**Q1**

$$\lfloor 7+10-1 = \boxed{16} \rfloor$$

**Q2**

$$\lfloor 30-8+1 = \boxed{23} \rfloor$$

**Q3**

$$\lfloor 11-5-1 = \boxed{5} \rfloor$$

**Q4**

9 ahead means:

$$\lfloor 9+1 = \boxed{10\text{th}} \rfloor$$

**Q5**

6 behind means:

$$\lfloor 6+1 = \boxed{7\text{th from bottom}} \rfloor$$

---

**Q6**

$$\lfloor 40-13+1 = \boxed{28} \rfloor$$

**Q7**

$$\lfloor 9+6 = \boxed{15} \rfloor$$

**Q8**

$$\lfloor 23-15-1 = \boxed{7} \rfloor$$

**Q9**

Convert Q to left:

---

$$\lfloor 30-10+1=21 \rfloor$$

Between:

$$\lfloor 21-12-1 = \boxed{8} \rfloor$$

---

### Q10

$$\lfloor 8+12-1 = \boxed{19} \rfloor$$

---

### Q11

$$\lfloor N=L+R+k \rfloor$$

$$\lfloor 8+10+5 = \boxed{23} \rfloor$$

### Q12

A's new left rank = B's old left rank:

$$\lfloor 25 \rfloor$$

B old right rank:

$$\lfloor 18 \rfloor$$

Total:

$$\lfloor 25+18-1 = \boxed{42} \rfloor$$

---

### Q13

B after swap occupies A's old position:

$$\lfloor 10\text{th from left} \rfloor$$

From right:

$$\lfloor 42-10+1 = \boxed{33} \rfloor$$

---

### Q14

$$\lfloor S>P>Q>R \rfloor$$

Tallest:

$$\lfloor \boxed{S} \rfloor$$

---

### Q15

$$\lfloor D>A>B>C \rfloor$$

Second oldest:

$$\lfloor \boxed{A} \rfloor$$

---

### Q16

A new left rank:

---

$$\lfloor 28 \rfloor$$

So B original left rank = 28.

B is 20th from right:

$$\lfloor N=28+20-1 \rfloor$$

$$\lfloor =\boxed{47} \rfloor$$

Original positions:

$$A = 15 \quad B = 28$$

Between:

$$\lfloor 28-15-1 = \boxed{12} \rfloor$$

---

## Q17

Q from left:

$$\lfloor 50-21+1 = 30 \rfloor$$

Between:

$$\lfloor 30-14-1 = \boxed{15} \rfloor$$

---

## Q18

We know:

$$\lfloor A > B \rfloor$$

and:

$$\lfloor C > B \rfloor$$

But relation A vs C is unknown.

$$\lfloor \boxed{\text{Cannot be determined}} \rfloor$$

---

## Q19

A is 12th from top.

18 below.

Total:

$$\lfloor 11\text{ above} + A + 18\text{ below} \rfloor$$

$$\lfloor 11+1+18 = 30 \rfloor$$

Or:

bottom rank = 19.

$$\lfloor 12+19-1=30 \rfloor$$

$$\lfloor \boxed{30} \rfloor$$

---

---

## Q20

---

Improves by 7:

$\lfloor 18-7 = \boxed{11} \rfloor$

---

## CHAPTER 62 STATUS

---

Covered so far:

- rank from left/right '
- top/bottom ranking '
- total from two-sided rank '
- converting one-side rank to other side '
- people between two persons '
- opposite-side rank conversion '
- total using people between '
- relative shifts '
- position interchange '
- overtaking/rank improvement '
- ordering by height/age/marks '
- transitive comparison chains '
- partial ordering '
- strict/non-strict comparison '
- basic ties '
- physical position vs performance rank '
- L0–L3 practice '

But **Chapter 62 is not complete yet.**

Still remaining:

- advanced interchange patterns
- rank after insertion/removal
- multiple people entering/leaving
- before/after rank changes
- tied-rank conventions in detail
- combined rank puzzles
- min/max possible rank
- ordering with partial constraints
- multi-person placement-style sets
- final speed toolkit

## Progress

---

$\lfloor \boxed{61/76} \rfloor \text{ completed} \rfloor$

**Current:** 0 = Chapter 62 — Ranking & Ordering

## NEXT

### Chapter 62 Part 2 — Rank Changes, Insertions/Removals, Ties, Partial-Order Constraints, Min/Max Rank & Final Ranking Toolkit

USER

next

ASSISTANT

## CHAPTER 62 — RANKING & ORDERING

### PART 2 — RANK CHANGES, INSERTIONS/REMOVALS, TIES, PARTIAL-ORDER CONSTRAINTS, MIN/MAX RANK & FINAL TOOLKIT

Status:  $\emptyset = 0$  IN PROGRESS ! ' ' COMPLETE GATE: High Placements: Very High Priority:  $\emptyset = 1$  for 1

This part handles the harder ranking questions where ranks change because:

- someone enters
- someone leaves
- people move ahead/behind
- ties occur
- only partial comparisons are known
- minimum/maximum possible rank is asked

The central idea is:

**Rank changes only when people cross your position.**

If nobody moves across you, your rank does not change.

## 1. INSERTION BEFORE A PERSON

Suppose A is 10th from left.

One new person enters before A.

A shifts one place right:

$\backslash$  [ 10\to11 ]

So new rank:

$\backslash$  \boxed{11th} ]

## 2. INSERTION AFTER A PERSON

A is 10th.

A new person joins after A.

A's rank remains:

$$\boxed{10^{\text{th}}}$$

Only people after the insertion point shift.

---

### 3. MULTIPLE INSERTIONS BEFORE A

---

A is 15th.

4 people are inserted before A.

New rank:

$$\boxed{15+4 = 19^{\text{th}}}$$

Formula:

$$\boxed{\text{New rank} = \text{Old rank} + \text{insertions before}}$$

---

### 4. REMOVAL BEFORE A

---

A is 12th.

Three people before A leave.

A moves forward:

$$\boxed{12-3 = 9^{\text{th}}}$$

---

### 5. REMOVAL AFTER A

---

A is 12th.

Five people after A leave.

A's rank remains:

$$\boxed{12^{\text{th}}}$$

---

### 6. MASTER RANK-CHANGE FORMULA

---

If changes occur around A:

$$\boxed{\text{New rank} = \text{Old rank} + \text{people inserted before} - \text{people removed before}}$$

Changes after A do not affect A's rank.

---

### 7. PEOPLE OVERTAKE A

---

Suppose A is 8th.

Three people behind A overtake A.

---

Each moves ahead of A.

A falls by 3 positions:

$$\backslash[ 8+3 = \boxed{11th} \backslash]$$

---

## 8. A OVERTAKES OTHERS

---

A is 15th.

A overtakes 4 people.

A improves by 4 ranks:

$$\backslash[ 15-4 = \boxed{11th} \backslash]$$

---

## 9. MIXED OVERTAKING

---

A starts 20th.

A overtakes 7 people.

Then 3 people overtake A.

Net improvement:

$$\backslash[ 7-3=4 \backslash]$$

New rank:

$$\backslash[ 20-4 = \boxed{16th} \backslash]$$

---

## 10. PEOPLE ENTERING ABOVE AND BELOW

---

A is ranked 25th.

- 4 new people enter above A
- 7 enter below A

Only those above matter.

New rank:

$$\backslash[ 25+4 = \boxed{29th} \backslash]$$

---

## 11. PEOPLE LEAVING ABOVE AND BELOW

---

A is 25th.

- 6 people above leave
- 5 people below leave

A improves:

$$\backslash[ 25-6 = \boxed{19th} \backslash]$$

---



## 12. TOTAL SIZE AFTER INSERTIONS/REMOVALS

---

If original total:

$$\lfloor N \rfloor$$

Then:

$$\lfloor \boxed{N_{\text{new}}} = N + \text{entries} - \text{exits} \rfloor$$

Example:

50 people.

7 enter.

4 leave.

New total:

$$\lfloor 50 + 7 - 4 = \boxed{53} \rfloor$$

---

## 13. RANK FROM RIGHT AFTER CHANGES

Suppose:

- original total = 40
- A originally 10th from left
- 3 people enter before A
- 2 people enter after A

New total:

$$\lfloor 45 \rfloor$$

A's new left rank:

$$\lfloor 10 + 3 = 13 \rfloor$$

From right:

$$\lfloor 45 - 13 + 1 = \boxed{33} \rfloor$$

---

## 14. INSERTION AT A'S POSITION

If a new person is inserted exactly before A's current slot, A shifts back by one.

If inserted immediately after A, A does not move.

Wording matters.

---

## 15. REMOVAL OF A PERSON AHEAD

A is 9th.

The person ranked 4th leaves.

A becomes:

$$\lfloor \boxed{8\text{th}} \rfloor$$

because one person ahead disappeared.

---

## 16. REMOVAL OF PERSON BEHIND

---

A is 9th.

Person ranked 15th leaves.

A remains:

$\boxed{9\text{th}}$

---

## 17. REMOVAL OF MULTIPLE RANKED PEOPLE

---

A is 12th.

Persons at ranks 3, 8, 15 and 20 leave.

Only ranks 3 and 8 are ahead of A.

So A improves by 2:

$12 - 2 = \boxed{10\text{th}}$

---

## 18. IMPORTANT — RANK NUMBERS CHANGE DURING REMOVAL

---

If removals are described by **original ranks**, count how many original positions were ahead of A.

If removals happen sequentially and new ranks are specified after each removal, update each step.

---

## 19. SEQUENTIAL CHANGE EXAMPLE

---

A is initially 10th.

- current 3rd-ranked person leaves !' A becomes 9th
- current 12th-ranked person leaves !' A remains 9th
- current 5th-ranked person leaves !' A becomes 8th

Final:

$\boxed{8\text{th}}$

---

## 20. RANK CHANGE FROM SCORE INCREASE

---

Suppose A is 20th.

After a score revision, A moves ahead of 6 students.

New rank:

$20 - 6 = \boxed{14\text{th}}$

No need to know exact marks.

---

## 21. RANK CHANGE FROM SCORE DECREASE

---

A is 7th.

Five students move ahead of A.

New rank:

$$\boxed{7+5 = 12\text{th}}$$

---

## 22. INTERCHANGE — GENERAL FORM

---

Suppose A and B swap positions.

Then:

$$\boxed{A_{\text{new}} = B_{\text{old}}}$$

$$\boxed{B_{\text{new}} = A_{\text{old}}}$$

Everything else remains unchanged.

---

## 23. ADVANCED SWAP EXAMPLE

---

A is 14th from left.

B is 17th from right.

After swap, A is 26th from left.

Therefore B's original left rank:

$$\boxed{26}$$

Total:

$$\boxed{26+17-1 = 42}$$

B's new left rank:

$$\boxed{14}$$

From right:

$$\boxed{42-14+1 = 29}$$

---

## 24. PEOPLE BETWEEN SWAPPED PERSONS

---

Original positions:

$$A = 14 \quad B = 26$$

Between:

$$\boxed{26-14-1 = 11}$$

---

## 25. MINIMUM POSSIBLE RANK — PARTIAL ORDER

---

Suppose:

---

- $A > B$
- $C > B$

where  $(>)$  means higher rank/better performance.

A and C are both ahead of B.

But A vs C unknown.

If only A, B, C exist, B must be 3rd.

A can be 1st or 2nd.

So A's possible ranks:

$\boxed{1 \text{ or } 2}$

## 26. PARTIAL ORDER WITH 5 PEOPLE

Suppose:

$A > B$

$C > B$

$B > D$

$D > E$

Then:

A and C are above B.

Chain:

$A/C > B > D > E$

A and C can swap.

Possible orders:

$A > C > B > D > E$

or

$C > A > B > D > E$

Thus:

- B fixed at 3rd
- D fixed at 4th
- E fixed at 5th
- A/C occupy 1st and 2nd

## 27. MINIMUM / MAXIMUM POSSIBLE RANK

For a person P:

- minimum numerical rank = best possible position
- maximum numerical rank = worst possible position

If two known people must be above P and three may be either side:

Best possible rank:

$\lfloor 2+1=3 \rfloor$

Worst depends how many optional people can be above.

---

## 28. EXAMPLE

---

There are 6 people: A, B, C, D, E, F.

Given:

$\lfloor A > B, \text{quad } C > B \rfloor$

No other relations.

For B:

At least A and C are ahead.

So best possible rank:

$\lfloor \boxed{3\text{rd}} \rfloor$

Could D,E,F also be ahead?

Yes.

Worst:

$\lfloor \boxed{6\text{th}} \rfloor$

Thus:

$\lfloor \boxed{B \text{ can rank } 3 \text{ to } 6} \rfloor$

---

## 29. ORDER FROM “BETTER THAN” STATEMENTS

---

Suppose:

- A ranks better than B
- B ranks better than C
- D ranks better than A

Then:

$\lfloor D > A > B > C \rfloor$

This can be treated like taller/older ordering.

---

## 30. “NOT WORSE THAN”

---

A is not worse than B means:

$\lfloor A \geq B \rfloor$

in performance value.

It allows equality/tie.

Do not convert automatically to strict  $\lfloor > \rfloor$ .

---

---

## 31. TIES — COMPETITION RANKING

---

Scores:

- A = 95
- B = 90
- C = 90
- D = 80
- E = 70

Competition ranking:

A = 1 B = 2 C = 2 D = 4 E = 5

Pattern:

$\boxed{1,2,2,4,5}$

---

## 32. WHY RANK 3 IS SKIPPED

---

Two people occupy rank 2.

Therefore next rank counts both previous people:

$\boxed{4}$

This is standard competition ranking: “1224”.

---

## 33. DENSE RANKING

---

Same scores:

A = 1 B = 2 C = 2 D = 3 E = 4

Pattern:

$\boxed{1,2,2,3,4}$

No skipped rank.

---

## 34. ORDINAL RANKING

---

Sometimes ties are broken using another criterion.

Then ranks may be:

1,2,3,4,5

even if raw scores tied.

Use the convention explicitly given.

---

## 35. AVERAGE RANK FOR TIES

---

In statistics, tied positions may receive average rank.

Example:

B and C occupy positions 2 and 3.

Average rank:

$$\left[ \frac{2+3}{2} = \boxed{2.5} \right]$$

This is less common in aptitude ranking questions but useful to recognize.

---

## 36. TIE TRAP

If question says:

**Two students are tied for 3rd.**

Under competition ranking, next student is:

$$\left[ \boxed{5th} \right]$$

not 4th.

---

## 37. THREE-WAY TIE

Three students tie for 2nd.

Occupied positions:

2,3,4.

Next competition rank:

$$\left[ \boxed{5th} \right]$$

---

## 38. RANK FROM BOTTOM WITH TIES

If ranking convention includes ties, simple:

$$\left[ N-r+1 \right]$$

may fail as a performance-rank conversion.

That formula is safe for unique physical positions.

Do not mix tied performance ranks with row positions.

---

## 39. ORDERING WITH EXACT GAP

Suppose A ranks 5 places above B.

If B is 17th:

$$\left[ A=17-5 = \boxed{12th} \right]$$

---

## 40. "5 RANKS AHEAD" VS "5 PEOPLE BETWEEN"

A is 5 ranks ahead of B:

Difference in rank:

$$\boxed{5}$$

If  $A = 10$ ,  $B = 15$ .

People between:

$$\boxed{15-10-1 = 4}$$

So:

**5 ranks apart = 4 people between.**

## 41. EXACTLY $k$ PEOPLE BETWEEN

If 7 people are between A and B:

Position difference:

$$\boxed{7+1=8}$$

So their ranks differ by:

$$\boxed{8}$$

## 42. EXAMPLE

A is 12th.

There are 5 people between A and B, and B is below A.

Then:

$$\boxed{B=12+5+1 = 18\text{th}}$$

## 43. COMBINED POSITION PUZZLE

There are 30 people.

A is 8th from left.

B is 5 places right of A.

C is 4 places left of B.

Positions:

$$A = 8$$

B:

$$\boxed{13}$$

C:



\[ 9 \]

From right:

C:

\[ 30-9+1 = \boxed{22nd} \]

---

## 44. MULTIPLE POSITION CONSTRAINTS

---

Suppose:

- A is left of B
- C is right of B
- D is between A and B
- E is right of C

Then broad order:

\[ A-D-B-C-E \]

if “between” means exact or relative depending wording.

Translate qualitative statements before assigning ranks.

---

## 45. “IMMEDIATELY” MATTERS

---

A immediately left of B:

\[ A,B \]

No person between.

A left of B:

Could be any distance.

Never treat “left of” as adjacent.

---

## 46. ORDERING WITH EXTREMES

---

Statements:

- A taller than B
- C taller than A
- D shorter than B
- E taller than C

Chain:

\[ E>C>A>B>D \]

Tallest:

\[ \boxed{E} \]

Shortest:

\[ \boxed{D} \]

---

Middle person:

$\boxed{A}$

---

## 47. MIDDLE RANK

---

If 5 people distinct:

Middle = 3rd.

If 7 people:

Middle = 4th.

Formula for odd  $(N)$ :

$\boxed{\frac{N+1}{2}}$

---

## 48. TWO MIDDLE POSITIONS

---

If  $(N)$  even, there are two central positions:

$\boxed{\frac{N}{2}, \frac{N+1}{2}}$

Example  $(N=10)$ :

5th and 6th.

---

## 49. MEDIAN VS RANK

---

In ordered numerical data, median may correspond to middle rank(s).

But ranking questions usually care about position, not arithmetic median unless values are provided.

---

## 50. MINIMUM PEOPLE FROM RANK DATA

---

Suppose A is 12th from top.

Without bottom rank, minimum total people is:

$\boxed{12}$

Could be more.

---

## 51. MINIMUM TOTAL WITH TWO PEOPLE

---

A is 8th from left.

B is 10th from right.

If they are different people, what is minimum possible total?

To minimize total, their positions should overlap as much as possible without being the same person.

A's left rank = 8.

---

For total  $(N)$ , B's left rank:

$$N - 10 + 1 = N - 9$$

They must be different:

$$N - 9 \neq 8$$

And positions must be valid.

If  $(N = 17)$ :

B left rank = 8 !' same position, impossible for distinct A/B.

Next minimum:

$$N = 18$$

B left rank = 9.

So minimum:

$$\boxed{18}$$

---

## 52. IF THEY CAN BE THE SAME PERSON

If A and B refer to the same person:

$$N = 8 + 10 - 1 = \boxed{17}$$

This distinction matters.

---

## 53. MAXIMUM TOTAL FROM PARTIAL RANK DATA

If A is 8th from left and nothing is known from right or total:

There is no finite maximum total.

Could be 8, 20, 100, etc.

So:

$$\boxed{\text{Cannot be determined}}$$

---

## 54. POSSIBLE TOTALS FROM TWO DISTINCT RANKS

A = 8th from left.

B = 10th from right.

If A is left of B:

Need:

$$8 < N - 9$$

$$N > 17$$

So minimum:

$$18$$

No finite maximum.

---

## 55. POSITION CONSISTENCY CHECK

---

If row has 20 people and someone is said to be 25th from left:

Impossible.

Always check:

$$\lfloor \frac{11}{2} \rfloor \leq n \leq N - \lfloor \frac{11}{2} \rfloor$$

---

## 56. SWAP CONSISTENCY CHECK

---

If A is 5th from left and B is 12th from left.

After swapping, A must be 12th.

If question says A becomes 11th, data are inconsistent unless another event also occurs.

---

## 57. PLACEMENT-STYLESET

---

In a row of 50 students:

- A is 12th from left
- B is 8 places right of A
- C is 5 places left of B
- D is 10th from right

Find positions.

A:

$$\lfloor \frac{12}{2} \rfloor$$

B:

$$\lfloor \frac{20}{2} \rfloor$$

C:

$$\lfloor \frac{15}{2} \rfloor$$

D from left:

$$\lfloor \frac{50-10+1}{2} \rfloor = 20$$

---

## 58. PEOPLE BETWEEN C AND D

---

$$\lfloor \frac{41-15-1}{2} \rfloor = \boxed{13}$$

---

## 59. B FROM RIGHT

---

$$\lfloor \frac{50-20+1}{2} \rfloor = \boxed{15}$$

---

---

## 60. A AFTER 4 PEOPLE BEFORE A LEAVE

---

A old:

$\backslash 12 \backslash$

4 ahead leave:

$\backslash 12-4 = \boxed{8} \backslash$

---

## 61. FINAL RANKING SPEED TOOLKIT

---

**Same person from both ends**

$\backslash \boxed{N=L+R-1} \backslash$

**Other-side rank**

$\backslash \boxed{N-r+1} \backslash$

**People between**

$\backslash \boxed{|r_1-r_2|-1} \backslash$

**k people between**

Position difference:

$\backslash \boxed{k+1} \backslash$

**Swap**

$\backslash \boxed{\text{new}=\text{other person's old position}} \backslash$

**Insertions/removals before**

$\backslash \boxed{r_{\text{new}}=r_{\text{old}}+I-R} \backslash$

---

## 62. PARTIAL-ORDER TOOLKIT

---

For statements like:

$\backslash A>B, \ C>B, \ B>D \backslash$

do not immediately create:

$\backslash A>C>B>D \backslash$

because A vs C is unknown.

Use branches:

TEXT

```
A C
 \ /
 B
 |
 D
```

This is a partial order.

---

## 63. ASCII ORDER DIAGRAM

---

For:

$A > B \ C > B \ B > D$

Draw:

TEXT



This makes uncertainty visible.

---

## 64. COMMON ADVANCED TRAPS

---

& pTRAP 1 — Insertions after A affect A

False.

& pTRAP 2 — Every removal improves rank

Only removals ahead improve it.

& pTRAP 3 — Ranks apart = people between

If rank difference = 6, people between = 5.

& pTRAP 4 — Tied ranking convention

Competition and dense ranking differ.

& pTRAP 5 — Partial order forced into complete chain

Don't invent missing comparisons.

& pTRAP 6 — Better rank means larger number

Usually false: 1st is better than 2nd.

& pTRAP 7 — Physical rank vs performance tie

Keep them separate.

---

## 65. FAST DECISION TREE

---

**Entries/exits?**

Ask: how many cross A's position?

**Overtaking?**

Each person crossed changes rank by 1.

**Swap?**

Exchange positions directly.

**Ties?**

Identify ranking convention.

**Partial comparisons?**

Build only guaranteed chain.

**Minimum/maximum rank?**

Count people who **must** be ahead and who **may** be ahead.

---

**PRACTICE — LEVEL 0**

---

**Q1**

A is 10th. Two people enter before A. New rank?

**Q2**

A is 12th. Three people ahead leave. New rank?

**Q3**

A is 15th and overtakes 4 people. New rank?

**Q4**

A is 8th and 5 people overtake A. New rank?

**Q5**

A is 20th. Two people enter after A. New rank?

---

**LEVEL 1**

---

**Q6**

A is 18th. A overtakes 5 people, then 2 people overtake A. Final rank?

**Q7**

A is 14th. Persons ranked 3rd, 8th and 20th leave. New rank?

**Q8**

A is 6 ranks ahead of B. If A is 12th, what is B's rank?

**Q9**

There are 7 people between A and B. What is the difference in their ranks?

**Q10**

Two students tie for 3rd under competition ranking. What is next rank?

---

## LEVEL 2

---

### Q11

Three students tie for 2nd under competition ranking. What is the next rank?

### Q12

Scores are: A 95, B 90, C 90, D 80. What is D's rank under dense ranking?

### Q13

There are 6 people.  $A > B$  and  $C > B$ . What is B's best possible rank?

### Q14

Using Q13, what is B's worst possible rank?

### Q15

A is 14th from left, B is 17th from right. After swapping, A becomes 26th from left. Find total.

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

A is initially 25th. Four people above A leave, 3 new people enter above A, and 5 enter below A. Find A's new rank.

### Q17

In a row of 40:

- A is 9th from left.
- B is 6 places right of A.
- C is 4 places left of B.

Find C's rank from right.

### Q18

There are 6 people A–F.  $A > B$ ,  $C > B$ ,  $B > D$ ,  $D > E$ . What is B's fixed rank if F is known to be below E?

### Q19

A is 8th from left and B is 10th from right. They are distinct people. What is the minimum possible total number of people?

### Q20

A is 18th. After moving ahead of 7 people and then being passed by 4 people, what is A's final rank?

---

## ANSWERS

---

### Q1

---

$\lfloor 10+2 = \boxed{12\text{th}} \rfloor$



## Q2

---

$$\lfloor 12-3 = \boxed{9\text{th}} \rfloor$$

## Q3

---

$$\lfloor 15-4 = \boxed{11\text{th}} \rfloor$$

## Q4

---

$$\lfloor 8+5 = \boxed{13\text{th}} \rfloor$$

## Q5

---

People entering after A do not affect A.

$$\lfloor \boxed{20\text{th}} \rfloor$$

---

## Q6

---

Net improvement:

$$\lfloor 5-2=3 \rfloor$$

$$\lfloor 18-3 = \boxed{15\text{th}} \rfloor$$

## Q7

---

Ranks 3 and 8 are ahead of A.

Rank 20 is behind.

So:

$$\lfloor 14-2 = \boxed{12\text{th}} \rfloor$$

## Q8

---

A is 6 ranks ahead:

$$\lfloor B=12+6 = \boxed{18\text{th}} \rfloor$$

## Q9

---

7 people between means rank gap:

$$\lfloor 7+1 = \boxed{8} \rfloor$$

## Q10

---

Two tied at rank 3 occupy positions 3 and 4.

Next rank:

$$\lfloor \boxed{5\text{th}} \rfloor$$

---

### Q11

---

Three tied at rank 2 occupy positions 2,3,4.

Next:

$\boxed{5th}$

### Q12

---

Dense ranking:

A=1 B=C=2 D=3

$\boxed{3rd}$

### Q13

---

At least A and C are ahead of B.

So best:

$\boxed{3rd}$

### Q14

---

D,E,F can all also be ahead if unconstrained relative to B except where stated.

So B could be as low as:

$\boxed{6th}$

### Q15

---

A's new left rank = B's old left rank:

$\boxed{26}$

B is 17th from right.

$\boxed{N=26+17-1}$

$\boxed{=42}$

### Q16

---

Start:

$\boxed{25}$

4 above leave:

$\boxed{25-4=21}$

3 enter above:

$\boxed{21+3=24}$

5 enter below: no effect.

Final:

$\boxed{24th}$

---

### Q17

---

A:

$9$

B:

$9+6=15$

C:

$15-4=11$

From right:

$40-11+1 = \boxed{30th}$

---

### Q18

---

Relations:

$A > B, C > B, B > D > E > F$

A and C are above B.

D, E, F below B.

Thus exactly 2 people above B.

So B is:

$\boxed{3rd}$

---

### Q19

---

If total were 17:

B from left:

$17-10+1=8$

Same position as A, impossible because distinct.

Next:

$N=18$

B from left:

$18-10+1=9$

Distinct.

Therefore minimum:

$\boxed{18}$

---

## Q20

A moves ahead of 7:

$$\lfloor 18-7=11 \rfloor$$

Then 4 pass A:

$$\lfloor 11+4 = \boxed{15th} \rfloor$$

## ' CHAPTER 62 COMPLETE

We have covered:

- rank from left/right '
- top/bottom ranking '
- total from both-side ranks '
- people between '
- opposite-side rank conversion '
- interchange/swapping '
- insertions/removals '
- overtaking '
- rank improvement/fall '
- ordering chains '
- partial ordering '
- tied ranks '
- competition vs dense ranking '
- minimum/maximum possible rank '
- position gaps '
- consistency checks '
- L0–L3 practice '

## Progress

$$\lfloor \boxed{62/76} \text{numbered chapters completed} \rfloor$$

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating Arrangement +
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +

- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

### CHAPTER 63 — LINEAR SEATING ARRANGEMENT

Next we will cover:

- single-row seating
- left/right positioning
- immediate neighbours
- fixed vs relative positions
- people between two persons
- end positions
- adjacency and non-adjacency
- facing North/South
- two-direction rows
- uncertain seating slots
- block formation
- multiple-condition puzzles
- contradiction handling
- GATE/placement shortcuts
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 63 — LINEAR SEATING ARRANGEMENT

### PART 1 — SINGLE ROW, LEFT/RIGHT POSITIONS, IMMEDIATE NEIGHBOURS, END POSITIONS, PEOPLE BETWEEN & BASIC MULTI-CONDITION PUZZLES

**Status:** ~~0~~ = 100% **PROGRESS** **GATE:** High **Placements:** Very High **Priority:** ~~0~~ = 100% 1

Linear seating arrangement means people are placed in a straight line.

Typical questions ask:

- who sits to whose left/right?
- who is between two people?
- who sits at an end?
- who is an immediate neighbour?
- what is the final order?

- how many arrangements are possible?

The most important rule is:

**Convert every sentence into a positional constraint. Do not keep all clues only in your head.**

## 1. BASIC SINGLE-ROW REPRESENTATION

Suppose five people sit in a row:

TEXT

A   B   C   D   E

Then:

- A is left of B, C, D, E
- E is right of A, B, C, D
- B and C are immediate neighbours
- A and E sit at the ends
- C sits between B and D

## 2. LEFT AND RIGHT — BASIC

If:

**A sits to the left of B.**

Represent:

TEXT

A   . . .   B

A need **not** be immediately left unless explicitly stated.

## 3. IMMEDIATE LEFT

If:

**A sits immediately to the left of B.**

Represent:

TEXT

A B

No one can sit between them.

## 4. IMMEDIATE RIGHT

If:

**A sits immediately to the right of B.**

Equivalent:

TEXT

B A

So:

$\boxed{\text{A immediate right of B} \iff \text{B immediate left of A}}$

## 5. “TO THE LEFT” VS “IMMEDIATELY TO THE LEFT”

Very important distinction.

**A left of B**

Possible:

TEXT

A X Y B

**A immediately left of B**

Must be:

TEXT

A B

This is one of the most common seating traps.

## 6. PEOPLE BETWEEN TWO PERSONS

If exactly 2 people sit between A and B:

TEXT

A \_ \_ B

or:

TEXT

B \_ \_ A

Position difference:

$\boxed{2+1=3}$

General rule:

If  $k$  people sit between A and B:

$$\boxed{|\text{pos}(A) - \text{pos}(B)| = k + 1}$$

---

## 7. EXAMPLE

Exactly 3 people sit between P and Q.

Then position gap:

$$3 + 1 = 4$$

So if P is at position 2:

Q can be:

$$2 + 4 = 6$$

or:

$$2 - 4$$

if valid.

---

## 8. END POSITIONS

In a row of  $N$  people:

Left end:

$$\boxed{1}$$

Right end:

$$\boxed{N}$$

If someone sits at an extreme end, they occupy one of these two positions.

---

## 9. ONE PERSON BETWEEN

If one person sits between A and B:

TEXT

A \_ B

or:

TEXT

B \_ A

Position difference:

$$\boxed{2}$$

---

## 10. TWO PERSONS BETWEEN

---



TEXT

A \_ \_ B

Difference:

$\boxed{\phantom{000}}$

## 11. ADJACENCY

A and B are adjacent means:

TEXT

AB

or:

TEXT

BA

So adjacency alone does **not** fix left-right order.

## 12. NON-ADJACENCY

If:

**A does not sit next to B.**

Then arrangements:

TEXT

A X B

or farther apart.

But never:

TEXT

AB

or:

TEXT

BA

## 13. BLOCK METHOD

If A and B must sit together, treat them temporarily as one block:

TEXT

[AB]

or:

TEXT

[BA]

This becomes useful in larger arrangements.

## 14. FIXED POSITION CLUE

If:

**A is third from the left.**

Immediately assign:

TEXT

1 2 3 4 5 ...  
A

Fixed-position clues are high-value clues.

Use them first.

## 15. RELATIVE POSITION CLUE

If:

**B is second to the right of A.**

This means:

$\lfloor \text{pos}(B) = \text{pos}(A) + 2 \rfloor$

Exactly one person sits between them.

## 16. "SECOND TO THE RIGHT"

Suppose:

TEXT

A \_ B

B is second to the right of A.

Do not confuse this with "two people between."

Second to the right means position difference 2.

Thus:

$\boxed{1 \text{ person between}}$

## 17. "THIRD TO THE LEFT"

If C is third to the left of D:

$\boxed{\text{pos}(C)=\text{pos}(D)-3}$

Exactly:

$\boxed{3-1=2}$

people sit between them.

## 18. GENERAL RELATIVE POSITION RULE

If B is  $k$ th to the right of A:

$\boxed{\text{pos}(B)=\text{pos}(A)+k}$

People between:

$\boxed{k-1}$

## 19. BASIC PUZZLE

Five people A, B, C, D, E sit in a row.

Clues:

1. A sits at the left end.
2. B sits immediately right of A.
3. E sits at the right end.
4. C sits left of D.

Start:

TEXT

A B \_ \_ E

Remaining C,D with C left of D:

TEXT

A B C D E

Final order:

$\boxed{A-B-C-D-E}$

## 20. SOLVING ORDER

Use clues in this order:

1. fixed position
2. end position
3. immediate left/right
4. exact gap
5. adjacency
6. relative left/right
7. negative clues

This usually reduces branching.

---

## 21. EXAMPLE WITH ONE GAP

Six people A–F sit in a row.

Clues:

- A is at position 1.
- C is third to the right of A.
- B sits immediately left of C.
- F is at right end.

Since A = 1:

C:

\[ 1+3=4 \]

B = 3.

F = 6.

Layout:

TEXT

1 2 3 4 5 6  
A \_ B C \_ F

Remaining D,E occupy 2 and 5 in either order unless more clues exist.

So arrangement is not yet unique.

---

## 22. IMPORTANT — DO NOT FORCE UNIQUENESS

If two arrangements satisfy all clues, the data are insufficient for a unique full arrangement.

You may still answer some questions if they are common to all valid arrangements.

---

## 23. GUARANTEED INFORMATION

From previous puzzle:

TEXT

A \_ B C \_ F

Even if D/E swap, we know:

- A at left end
- F at right end
- B immediately left of C
- B is 3rd
- C is 4th

These are guaranteed.

## 24. POSITION TABLE METHOD

For 7 seats:

Position	Person
1	
2	
3	
4	
5	
6	
7	

This is often easier than repeatedly redrawing the row.

## 25. CANDIDATE-SLOT METHOD

Suppose A can only sit at positions:

$\{2, 4, 6\}$

Write:

$\{A \mid A \in \{2, 4, 6\}\}$

Then use later clues to eliminate slots.

This is useful when exact position is not initially known.

## 26. END RESTRICTION

If:

**A does not sit at an end.**

Then:

$\lceil \text{pos}(A) \neq 1, N \rceil$

For 6 people:

$\lceil A \in \{2, 3, 4, 5\} \rceil$

## 27. IMMEDIATE NEIGHBOURS

If:

**A sits between B and C and both are immediate neighbours of A.**

Then either:

TEXT

B A C

or:

TEXT

C A B

This forms a 3-person block.

## 28. “A IS BETWEEN B AND C” — CAUTION

If only:

**A sits somewhere between B and C**

Then order is:

$\lceil B \ldots A \ldots C \rceil$

or:

$\lceil C \ldots A \ldots B \rceil$

They need not be adjacent.

## 29. “A IS IMMEDIATELY BETWEEN B AND C”

Then:

TEXT

B A C

or:

TEXT

C A B

No gaps.

---

### 30. EXACTLY ONE PERSON BETWEEN A AND B

---

Possible local pattern:

TEXT

A X B

or:

TEXT

B X A

If row has 5 positions, possible pairs include:

$\setminus \{ (1,3), (2,4), (3,5) \}$

---

### 31. EXACTLY TWO PEOPLE BETWEEN A AND B

---

For 6 seats, possible pairs:

$\setminus \{ (1,4), (2,5), (3,6) \}$

This can quickly narrow options.

---

### 32. END + GAP COMBINATION

---

Six people.

A at left end.

Exactly two people between A and B.

Then:

A = 1.

Gap = 3 positions.

So:

$\setminus \{ B=4 \}$

Immediate.

No trial needed.

### 33. MULTIPLE GAP CLUES

Seven people A–G.

A is at position 1.

Two people between A and B.

One person between B and C, with C to the right of B.

Then:

A = 1

B = 4

C = 6

Layout:

TEXT

1 2 3 4 5 6 7  
A \_ \_ B \_ C \_

### 34. USE ARITHMETIC INSTEAD OF VISUAL GUESSING

Translate phrases:

- immediately right !' +1
- second right !' +2
- third left !' " 3
- $\backslash(k)$  people between !' gap  $\backslash(k+1\backslash)$

Then solve like equations.

### 35. EXAMPLE

B is second to right of A:

$\backslash B=A+2 \backslash$

C is immediately left of B:

$\backslash C=B-1 \backslash$

Therefore:

$\backslash C=A+1 \backslash$

So local pattern:

TEXT

A C B

### 36. CHAINING RELATIVE POSITIONS



Suppose:

- B is 3rd right of A
- C is 2nd left of B

Then:

$$\backslash [ B=A+3 \backslash ]$$

$$\backslash [ C=B-2=A+1 \backslash ]$$

Thus:

$$\backslash [ A-C-\backslash \_!-B \backslash ]$$

C is immediately right of A.

---

## 37. RELATIVE CHAIN SHORTCUT

---

Assign A = 0 temporarily.

Then:

$$B = +3$$

$$C = +1$$

$$D = -2$$

Later shift all positions into valid positive seat numbers.

Very useful when no absolute position is initially known.

---

## 38. NORMALIZATION METHOD

---

Suppose relative positions produce:

$$\backslash [ D=-2, \backslash A=0, \backslash C=1, \backslash B=3 \backslash ]$$

Minimum is  $\backslash (-2\backslash)$ .

Add 3 to all:

$$\backslash [ D=1, \backslash A=3, \backslash C=4, \backslash B=6 \backslash ]$$

Then fit into actual row.

This is a useful advanced shortcut.

---

## 39. NEGATIVE CLUES

---

Examples:

- A does not sit next to B.
- C is not at an end.
- D is not left of E.
- F does not sit between G and H.

Negative clues are usually best used **after** stronger positive clues.

---

---

## 40. WHY NEGATIVE CLUES COME LATER

---

A clue like:

**A is not next to B**

may eliminate only 2 possibilities.

A clue like:

**A sits at position 3**

immediately fixes A.

So start with the strongest constraints.

---

## 41. SINGLE ROW — ALL FACING NORTH

---

When everyone faces North, normal page orientation works:

TEXT

L e f t   ! •   A   B   C   D   E   ! '   R i g h t

Person's left = your left.

Person's right = your right.

---

## 42. SINGLE ROW — ALL FACING SOUTH

---

This is different.

If everyone faces South:

**their left is your right.**

**their right is your left.**

This becomes crucial in Part 2.

---

## 43. BASIC SOUTH-FACING EXAMPLE

---

Row shown on page:

TEXT

A B C D E

Everyone faces South.

---

From **B's** point of view:

- B's left = C
- B's right = A

So do not use page-left/page-right blindly.

---

## 44. FACE-DIRECTION RULE

If facing North:

`\[ \boxed{\text{left = page-left, right = page-right}} \]`

If facing South:

`\[ \boxed{\text{left = page-right, right = page-left}} \]`

---

## 45. ALL FACING NORTH EXAMPLE

Five people A–E face North.

- A sits at left end.
- B immediately right of A.
- E at right end.
- C immediately left of D.

Layout:

TEXT

A B C D E

---

## 46. ALL FACING SOUTH — SAME WORDING

Suppose same row, all face South.

If:

**B is immediately right of A.**

Since A faces South, A's right is page-left.

So arrangement locally is:

TEXT

B A

This is the reverse of North-facing case.

---

## 47. ALWAYS MARK FACING DIRECTION

Before solving, draw arrows:

North-facing:

TEXT

A! ' B! ' C! ' D! '

South-facing:

TEXT

A! " B! " C! " D! "

This prevents most errors.

---

## 48. BASIC MULTI-CONDITION PUZZLE

---

Six people A–F face North.

Clues:

1. A sits second from left.
2. C sits second to right of A.
3. B sits immediately left of C.
4. F sits at right end.
5. D is not adjacent to A.

Positions:

A = 2.

C:

$\boxed{2+2=4}$

B = 3.

F = 6.

Remaining D,E in positions 1 and 5.

D not adjacent A.

Position 1 is adjacent A at position 2.

So D cannot be 1.

Therefore D = 5, E = 1.

Final:

TEXT

E A B C D F

$\boxed{E-A-B-C-D-F}$

---

## 49. VERIFY ALL CLUES

---

Always check final arrangement against every clue.

For:

TEXT

E A B C D F

- A second from left '
- C second right of A '
- B immediately left of C '
- F right end '
- D not adjacent A '

Only then lock.

## 50. CONTRADICTION DETECTION

Suppose:

- A immediately left of B
- B immediately left of C
- C immediately left of A

Impossible in a straight row.

Why?

First two imply:

TEXT

A B C

Third would require:

TEXT

C A

contradiction.

So:

$\boxed{\text{No valid arrangement}}$

## 51. POSITION COLLISION

If clues imply:

A = position 3 B = position 3

and A,B are different people:

$\boxed{\text{Contradiction}}$

unless the context allows shared rank rather than seating, which seating does not.

## 52. MULTIPLE POSSIBLE ARRANGEMENTS

Suppose:

Four people A,B,C,D.

A left of B.

C left of D.

Possible arrangements include:

TEXT

A B C D

A C B D

C D A B

C A D B

...

Not enough information for a unique arrangement.

## 53. UNIQUE QUESTION MAY STILL BE ANSWERABLE

Even if full order isn't unique, a specific question may be.

Example:

A always left of B in every valid arrangement.

Question:

**Is A left of B?**

Answer:

$\boxed{\text{Yes}}$

Do not insist on finding a full unique arrangement if unnecessary.

## 54. OPTION-BASED SEATING

For MCQs, sometimes faster to test options against constraints.

Example:

Question asks possible arrangement.

Instead of solving completely:

- reject any option violating fixed seat
- reject adjacency violation
- reject end violation
- reject left/right violation

This can be much faster.

## 55. WHEN OPTION ELIMINATION IS BEST

---

Use option checking if:

- only one MCQ is asked
  - options give full arrangements
  - 3–4 strong constraints exist
  - constructing entire solution would take longer
- 

## 56. BLOCK FORMATION

---

Suppose A sits immediately left of B.

Treat as:

$\boxed{[AB]}$

If C immediately right of B:

$\boxed{[ABC]}$

Block orientation fixed.

---

## 57. UNORDERED BLOCK

---

If A and B sit together but order unknown:

Possible block:

$\boxed{[AB] \text{ or } [BA]}$

Keep both possibilities.

---

## 58. TWO BLOCKS

---

Suppose:

A immediately left of B.

C immediately right of D.

Then blocks:

$\boxed{[AB]}$

and:

$\boxed{[DC]}$

These can be positioned as units first.

---

## 59. END-BLOCK RESTRICTION

---

If A-B must sit together and A is at left end:

Then block must start:

---

No orientation ambiguity if A fixed at extreme left.

---

## 60. EXACT-CENTRE POSITION

---

For odd number  $(N)$ :

Middle position:

$$\boxed{\frac{N+1}{2}}$$

For 7 people:

$$\boxed{4}$$


---

## 61. TWO MIDDLE POSITIONS

---

For even  $(N)$ :

$$\boxed{\frac{N}{2}, \frac{N}{2} + 1}$$

For 8 people:

$$\boxed{4, 5}$$


---

## 62. "SITS IN THE MIDDLE"

---

If 7 seats:

middle is uniquely 4th.

If 8 seats:

"middle" is ambiguous unless the question means one of the two central positions.

Read wording carefully.

---

## 63. PEOPLE BETWEEN END POSITIONS

---

For  $(N)$  people:

Between extreme ends:

$$\boxed{N-2}$$

Example 8 people:

$$\boxed{6}$$


---

## 64. ADJACENCY COUNT

---

In a row of  $(N)$  seats:

---



Number of adjacent seat-pairs:

$\boxed{N-1}$

For 6 seats:

$\boxed{5}$

Useful in counting-based arrangement problems later.

---

## 65. END PERSON HAS ONE NEIGHBOUR

---

A person at an end has exactly:

$\boxed{1}$

immediate neighbour.

A non-end person has:

$\boxed{2}$

immediate neighbours.

---

## 66. CLUE FROM NUMBER OF NEIGHBOURS

---

If:

**A has only one immediate neighbour.**

Then A must be at an end.

This is a hidden end-position clue.

---

## 67. “BOTH IMMEDIATE NEIGHBOURS OF A ARE X AND Y”

---

Then A cannot be at an end.

Local block:

TEXT

X A Y

or:

TEXT

Y A X

---

## 68. BASIC LOGIC SET

---

Seven people A–G sit facing North.

1. D sits in the middle.

2. A sits second to left of D.
3. B sits immediately right of A.
4. G sits at right end.
5. C is not adjacent to G.
6. E sits left of F.

D = 4.

A:

\[ 4-2=2 \]

B = 3.

G = 7.

Layout:

TEXT

1	2	3	4	5	6	7
_	A	B	D	_	_	G

Remaining C,E,F in 1,5,6.

C not adjacent G !' C cannot be 6.

E left of F.

Possible:

- if C=1, E=5,F=6 works
- if C=5, E=1,F=6 works

So not unique yet.

This is an example of insufficient clues.

## 69. ADD ONE CLUE

Add:

**C sits at the left end.**

Then C=1.

Remaining:

E=5, F=6.

Final:

TEXT

C	A	B	D	E	F	G
---	---	---	---	---	---	---

Unique.

## 70. MASTER LINEAR-SEATING WORKFLOW

---

### Step 1

Write total seats.

### Step 2

Mark facing direction.

### Step 3

Place fixed/end/middle positions.

### Step 4

Create immediate blocks.

### Step 5

Apply exact-gap clues.

### Step 6

Apply relative left/right clues.

### Step 7

Use negative clues.

### Step 8

Check unused people.

### Step 9

Verify every clue.

---

## 71. FAST SCRATCHPADSYMBOLS

Use:

- $A-B$  = adjacent
- $A \_ B$  = one between
- $A \_ \_ B$  = two between
- $[AB]$  = fixed block
- $A < B$  = A left of B
- $A \neq \text{North}$
- $A \neq \text{South}$

Keep notation compact.

---

## 72. COMMON TRAPS

& pTRAP 1 — “Right of” = immediately right

False.

**& pTRAP 2 — kth right means k people between**

False. People between =  $\backslash(k-1\backslash)$ .

**& pTRAP 3 — Adjacency fixes order**

False unless left/right also given.

**& pTRAP 4 — South-facing person uses page-left as own left**

False.

**& pTRAP 5 — Negative clues used too early**

Use strong positive clues first.

**& pTRAP 6 — Force one arrangement when two remain**

Do not.

**& pTRAP 7 — Forget final verification**

Always check all clues.

---

## PRACTICE — LEVEL 0

---

### Q1

A sits immediately left of B. Write the local order.

### Q2

Two people sit between A and B. What is the position difference?

### Q3

B is second to the right of A. How many people sit between them?

### Q4

In a row of 7, what is the middle position?

### Q5

A sits at an end. How many immediate neighbours can A have?

---

## LEVEL 1

---

### Q6

Six people face North. A is at position 1. B is third to the right of A. What is B's position?

### Q7

C is immediately left of B from Q6. What is C's position?

### Q8

If D is immediately right of B, what is D's position?

### Q9

---

A is 3rd from left. B is 2nd to the right of A. What is B's position?

### Q10

One person sits between P and Q. If P is at position 2, what possible position to the right can Q occupy?

---

## LEVEL 2

---

Five people A–E face North.

Clues:

- A is at left end.
- C sits second to right of A.
- B sits immediately left of C.
- E sits at right end.

### Q11

Who sits at position 2?

### Q12

Who sits at position 3?

### Q13

Which person remains for position 4?

### Q14

Who are the immediate neighbours of C?

### Q15

Who sits second to the left of E?

---

## LEVEL 3 — EXAM PRESSURE

---

Six people A–F face North.

Clues:

1. A sits second from left.
2. C sits second to right of A.
3. B sits immediately left of C.
4. F sits at right end.
5. D does not sit next to A.

### Q16

What is the complete arrangement?

### Q17

Who sits at the left end?

---

**Q18**

Who sits immediately right of C?

**Q19**

How many people sit between A and F?

**Q20**

Who sits second to the left of F?

---

**ANSWERS**

---

**Q1**

$\boxed{A-B}$

**Q2**

Two between means:

$2+1 = \boxed{3}$

**Q3**

Second to right means position difference 2.

So:

$\boxed{1}$

person between.

**Q4**

$\frac{7+1}{2}=4$

$\boxed{4th}$

**Q5**

End person has:

$\boxed{1}$

immediate neighbour.

---

**Q6**

A = 1.

Third right:

$1+3=4$

$\boxed{4}$

---

### Q7

C immediately left of B at 4:

$\boxed{3}$

### Q8

D immediately right:

$\boxed{5}$

### Q9

$3+2 = \boxed{5}$

### Q10

One person between means position gap 2.

$P=2$ .

Q to right:

$2+2 = \boxed{4}$

### Q11–Q15

A left end:

$A=1$

C second right:

$C=3$

B immediately left of C:

$B=2$

E right end:

$E=5$

Remaining D:

$D=4$

Final:

TEXT

A B C D E

### Q11

$\boxed{B}$

### Q12

$\boxed{C}$

### Q13

$\boxed{D}$

### Q14

Immediate neighbours of C:

$\boxed{B,D}$

### Q15

E=5.

Second left:

3

Person:

$\boxed{C}$

### Q16–Q20

A = 2.

C second right:

$C=4$

B immediately left:

$B=3$

F = 6.

Remaining positions 1 and 5 for D,E.

D not adjacent to A at position 2.

Position 1 is adjacent to A, so D is not at position 1.

Thus:

D=5 E=1

Final:

TEXT

E A B C D F

### Q16

$\boxed{E-A-B-C-D-F}$

### Q17

$\boxed{E}$

### Q18

Immediately right of C:

$\boxed{D}$



## Q19

A position 2.

F position 6.

Between:

positions 3,4,5

$\boxed{3}$

## Q20

F at 6.

Second left:

$4$

Person:

$C$

---

## CHAPTER 63 STATUS

---

Covered so far:

- single-row seating '
- left/right relations '
- immediate left/right '
- exact gaps '
- people between '
- end positions '
- adjacency/non-adjacency '
- blocks '
- fixed and relative positions '
- arithmetic position method '
- negative clues '
- north-facing rows '
- basic south-facing reversal '
- contradiction detection '
- incomplete arrangements '
- option elimination '
- multi-condition puzzles '
- L0–L3 practice '

But **Chapter 63 is not complete yet.**

Still remaining:

- all-facing-South arrangements in depth
- mixed North/South facing
- two parallel rows
- facing each other

- facing same direction
- who is opposite whom
- complex blocks
- conditional seat restrictions
- multi-answer arrangements
- advanced placement-style sets
- final speed toolkit

## Progress

[ \boxed{62/76\text{ completed}} ]

**Current:** 0=Chapter 63 — Linear Seating Arrangement

## NEXT

0=Chapter 63 Part 2 — South-Facing Rows, Mixed Facing, Two Parallel Rows, Opposite Seats, Advanced Blocks & Final Linear-Seating Toolkit

USER

next

ASSISTANT

## CHAPTER 63 — LINEAR SEATING ARRANGEMENT

### PART 2 — SOUTH-FACING ROWS, MIXED FACING, PARALLEL ROWS, OPPOSITE SEATS, ADVANCED BLOCKS & FINAL TOOLKIT

**Status:** 0=IN PROGRESS ! ' ' COMPLETE **GATE:** High **Placements:** Very High **Priority:** 0=1/4 1

This part handles the most error-prone linear-seating questions:

- everyone faces South
- some face North and some South
- two parallel rows
- people facing each other
- opposite-seat relations
- advanced adjacency blocks
- multiple valid arrangements

The central rule is:

**Left and right always belong to the person being described, not to you.**

## 1. NORTH-FACING VS SOUTH-FACING

For a person facing North:

TEXT

North  
Left !• A !' Right

So:

- person's left = page-left
- person's right = page-right

For a person facing South:

TEXT

Left? A Right?  
!"

Because the person faces you:

- person's left = page-right
- person's right = page-left

## 2. MUST-MEMORIZE FACING RULE

Facing	Person's Left	Person's Right
North	Page-left	Page-right
South	Page-right	Page-left

This one table prevents many mistakes.

## 3. ALL FACING SOUTH — BASIC EXAMPLE

Five people face South.

Clue:

**B sits immediately to the right of A.**

Since A faces South, A's right is page-left.

Therefore:

TEXT

B A

not:

TEXT

A B

## 4. SOUTH-FACING SECOND RIGHT

A faces South.

B sits second to the right of A.

Since right = page-left:

TEXT

B \_ A

B is two positions to the page-left of A.

## 5. SOUTH-FACING SECOND LEFT

A faces South.

B sits second to the left of A.

A's left = page-right.

So:

TEXT

A \_ B

## 6. QUICK RULE

For South-facing people:

- "right of" !' move left on paper
- "left of" !' move right on paper

## 7. EXAMPLE

Six people face South.

A is 4th from the left side of the row.

B sits immediately right of A.

Important: "4th from left" refers to physical row position, not A's facing.

So A = position 4.

B immediate right of A from A's perspective = physical position 3.

$\boxed{B=3}$

## 8. POSITION RANK VS PERSON'S LEFT/RIGHT

These are different.

"Third from the left end" means physical row:

\[ 3 \]

independent of facing.

But:

"second to the left of A"

depends on A's facing direction.

## 9. SOUTH-FACING PUZZLE

Five people A–E face South.

- A is at physical left end.
- B sits immediately left of A.
- C sits immediately left of B.

A = position 1.

A faces South, so A's left = physical right.

Thus:

B = 2.

B faces South, so B's left = physical right.

C = 3.

Local arrangement:

TEXT

A B C \_ \_  
! " ! " ! "

## 10. MIXED FACING

Now some people face North and others South.

Example:

TEXT

A! '      B! "      C! '      D! "

A's right = page-right.

B's right = page-left.

So the same word "right" can mean opposite physical directions for different people.

## 11. ALWAYS ATTACH FACING TO THE REFERENCE PERSON

Statement:

**B sits to the right of A.**

Use **A's** facing direction.

Not B's.

Because the phrase describes where B is relative to A.

## 12. EXAMPLE — MIXED FACING

A faces North.

B sits immediately right of A.

Then:

TEXT

A B  
! ' ?

B's own facing is irrelevant to locating B.

## 13. ANOTHER EXAMPLE

A faces South.

B sits immediately right of A.

Then:

TEXT

B A  
? ! "

Again, use A's facing.

## 14. "A IS TO THE LEFT OF B"

Now use B's facing direction because A's location is described relative to B.

If B faces North:

TEXT

A B

If B faces South:

TEXT

B A

## 15. REFERENCE-PERSON RULE

For statement:

**X sits to the left/right of Y.**

Always ask:

$\boxed{\text{Which way is Y facing?}}$

That determines X's physical seat.

## 16. TWO PARALLEL ROWS

Very common placement pattern.

Example:

TEXT

Row 1:	A	B	C	D				
			!"	!"	!"	!"		
			!'	!'	!'	!'		
Row 2:	P	Q	R	S				

The two rows face each other.

So:

- A faces P
- B faces Q
- C faces R
- D faces S

## 17. OPPOSITE SEATS

If two rows have equal seats aligned vertically:

TEXT

A	B	C	D				
!"		!"	!"	!"			
!'		!'	!'	!'			
P	Q	R	S				

Opposite pairs:

$\boxed{(A,P),(B,Q),(C,R),(D,S)}$

---

## 18. FACING EACH OTHER — LEFT/RIGHT REVERSAL

---

Top row faces South.

Bottom row faces North.

Therefore:

### Top row

left = physical right

### Bottom row

left = physical left

This asymmetry is essential.

---

## 19. EXAMPLE

---

Top row A,B,C face South.

Bottom row P,Q,R face North.

Suppose:

**B is immediately left of A.**

A faces South.

A's left = page-right.

So B must be physically right of A.

Thus local top-row order:

TEXT

A B  
! " ! "

---

## 20. BOTTOM-ROW EXAMPLE

---

Suppose:

**Q is immediately left of P.**

P faces North.

P's left = page-left.

So physical order:



TEXT

Q P  
! ' ! '

## 21. TWO-ROW MASTER LAYOUT

For equal rows, pre-draw slots:

TEXT

Top:	1	2	3	4				
			! "	! "	! "	! "		
Bottom:	1	2	3	4				
			! '	! '	! '	! '		

Seat 1 faces seat 1, etc.

This is the safest representation.

## 22. OPPOSITE-SEAT CLUE

If:

**A sits opposite Q.**

Then assign them to same column.

Example:

TEXT

Top:	A			! "
				! '
Bottom:	Q			

Column still may be unknown.

Treat (A, Q) as an opposite pair.

## 23. IMMEDIATE LEFT OF PERSON IN TOP ROW

If A in top row faces South.

B sits immediate left of A.

Then B is one column to the physical right.

So:

$\lfloor \text{col}(B) = \text{col}(A) + 1 \rfloor$



If Q faces North, P sits physical left of Q.  
Always resolve the referenced person first.

## 27. EXAMPLE

Top row faces South.  
Bottom row faces North.  
A is in top row.  
Q sits opposite A.  
Statement:

**P sits immediately left of Q.**

Q faces North.  
So P sits physical left of Q.

## 28. "SECOND TO THE RIGHT OF THE PERSON OPPOSITE A"

Solve in two stages:

1. identify person opposite A
2. move two seats according to that person's facing direction

Do not combine mentally.

## 29. SAME-DIRECTION PARALLEL ROWS

Less common but possible.  
Both rows may face North:

TEXT							
A	B	C	D	!'	!'	!'	!'
P	Q	R	S	!'	!'	!'	!'

Then everyone's left/right is physical page-left/page-right.  
"Facing opposite" no longer applies simply because rows are parallel.  
Always read stated facing.

## 30. ONE ROW FACING OUTWARD

Sometimes two rows face away from each other.  
Example:

TEXT

Top row :        !' !' !'  
Bottom row : !" !" !"

They are back-to-back, not facing each other.

So "opposite" means vertically aligned, but not face-to-face.

### 31. ADVANCED BLOCKS

Suppose:

- A sits immediately left of B
- B sits immediately left of C

All face North.

Then fixed block:

$\backslash$  [ABC]  $\backslash$

### 32. SOUTH-FACING BLOCK

If all face South and:

- A sits immediately left of B
- B sits immediately left of C

Because each reference faces South:

A is physically right of B.

B is physically right of C.

So physical block becomes:

$\backslash$  [CBA]  $\backslash$

This is very important.

### 33. BLOCK WITH UNKNOWN ORIENTATION

If:

**A and B sit together.**

Then:

$\backslash$  [AB]  $\text{quad}\text{or}\text{quad}$  [BA]  $\backslash$

If another clue says:

**A is left of B**

and both face North:

$\nabla [AB] \nabla$

If both face South:

physical order is:

$\nabla [BA] \nabla$

---

## 34. THREE-PERSON ADJACENCY BLOCK

If:

**B sits between A and C, both adjacent to B.**

Then:

$\nabla [A-B-C] \nabla$

or:

$\nabla [C-B-A] \nabla$

Treat as one 3-seat unit until another clue fixes orientation.

---

## 35. END-BLOCK CONSTRAINT

If  $\nabla [AB]$  must sit together in a 6-seat row and A must be at the left end:

Only:

TEXT

A B \_ \_ \_ \_

works.

But if B must be at left end:

TEXT

B A \_ \_ \_ \_

works.

End constraints often fix block orientation.

---

## 36. NON-ADJACENT BLOCK CONSTRAINT

If A and B must not sit together, mark forbidden:

$\nabla [pos(A)-pos(B)]\neq 1 \nabla$

Useful when candidate slots remain.

---

## 37. TWO BLOCKS CANNOT OVERLAP

Suppose:

- A immediately left of B
- C immediately left of D
- B=C

If B and C are actually same person, blocks combine.

If B and C are distinct people, they remain separate.

Names matter.

---

## 38. OVERLAPPING RELATION CHAIN

---

Suppose:

- A immediate left of B
- B immediate left of C
- C immediate left of D

Then:

$\{ [A B C D] \}$

This is one continuous block.

---

## 39. CONDITIONAL SEAT RESTRICTIONS

---

Examples:

- A not at an end
- B not adjacent to C
- D only at even-numbered seat
- E somewhere left of F
- G not opposite A

Write these as constraints.

Example:

$\{ A \notin \{1, N\} \}$

$\{ |B-C| \neq 1 \}$

---

## 40. EVEN / ODD SEAT CLUES

---

In 8 seats:

Even positions:

$\{ 2, 4, 6, 8 \}$

Odd:

$\{ 1, 3, 5, 7 \}$

If A occupies an even seat:

---

$\{A \in \{2,4,6,8\}\}$

This can greatly reduce cases.

---

## 41. "NOT NEXT TO AN END PERSON"

If A sits at an end and B does not sit next to A:

Then B cannot occupy the only adjacent seat.

Example A=1:

$\{B \neq 2\}$

---

## 42. "ONE OF THE NEIGHBOURS"

If A sits between B and C but not necessarily immediately, this is weak.

If:

**A is an immediate neighbour of both B and C**

then strong local block:

$\{B-A-C\}$

or:

$\{C-A-B\}$

---

## 43. MULTI-ANSWER ARRANGEMENTS

Sometimes 2 or more complete arrangements remain valid.

Example:

TEXT

A B C D

A C B D

If question asks:

**Who sits at left end?**

Answer A in both.

So:

$\{\boxed{A}\}$

Even without unique full seating.

---

## 44. "COULD BE TRUE" QUESTIONS

For these:

- find at least one valid arrangement satisfying the option.

You do not need to prove it holds in all cases.

---

## 45. "MUST BE TRUE" QUESTIONS

---

For these:

- option must hold in every valid arrangement.

This is stricter.

Test across all surviving cases.

---

## 46. "CANNOT BE TRUE"

---

Option must violate at least one clue in every possible arrangement.

Use contradiction testing.

---

## 47. CASE-SPLITTING

---

Suppose A can occupy 2 or 5.

Instead of guessing, branch:

### Case 1

A=2

### Case 2

A=5

Then propagate clues separately.

Discard any branch that creates contradiction.

---

## 48. CASE-SPLITTING RULE

---

Branch only on high-impact uncertainty.

Good branch:

- end choice
- one of two possible exact slots
- block orientation

Bad branch:

- random placement of many unconstrained people
- 

## 49. CONTRADICTION PRUNING

---



Suppose Case 1 leads to:

- two people same seat
- impossible gap
- forbidden adjacency
- required person outside row

Then delete Case 1 immediately.

Do not continue it.

## 50. TWO-ROW EXAMPLE — FULL SET

Top row: A, B, C, D face South. Bottom row: P, Q, R, S face North.

Clues:

1. A sits at left end of top row.
2. B sits immediately left of A.
3. Q sits opposite B.
4. P sits immediately left of Q.
5. R sits at right end of bottom row.
6. C sits opposite P.

Solve.

Top:

A=1.

B sits immediate left of A from A's South-facing perspective !' physical position 2.

So:

TEXT

Top: A B \_ \_

Bottom:

Q opposite B !' position 2.

P sits immediately left of Q. Q faces North, so P physical position 1.

R right end !' position 4.

C opposite P !' top position 1.

But A is already top position 1.

Contradiction.

Therefore this clue set has:

$\boxed{\text{No valid arrangement}}$

This illustrates why verification matters.

## 51. CORRECTED FULL TWO-ROW EXAMPLE

Top row A, B, C, D face South. Bottom row P, Q, R, S face North.

Clues:

1. A sits at top left end.
2. B sits immediately left of A.
3. Q sits opposite B.
4. P sits immediately left of Q.
5. S sits at bottom right end.
6. C sits opposite R.

Top:

A=1.

B immediate left of A, A faces South !' B=2.

Bottom:

Q opposite B !' Q=2.

P immediate left of Q, Q faces North !' P=1.

S=4.

Remaining R=3.

C opposite R !' C=3.

Remaining D=4.

Final:

TEXT

Top:	A	B	C	D				
			!"	!"	!"	!"	!"	
Bottom:	P	Q	R	S				
			!'	!'	!'	!'	!'	

## 52. QUESTION FROM ABOVE

Who faces C?

C at top position 3.

Bottom position 3 = R.

$\boxed{R}$

## 53. WHO SITS IMMEDIATELY RIGHT OF Q?

Q faces North.

Q at bottom position 2.

Immediate right = physical position 3.

Person:

$\boxed{R}$

---

## 54. WHO SITS IMMEDIATELY RIGHT OF B?

---

B faces South.

B at top position 2.

B's right = physical left.

Position 1 = A.

$\boxed{A}$

---

## 55. WHO FACES THE PERSON IMMEDIATELY LEFT OF Q?

---

Q faces North.

Immediate left of Q = P.

P is at bottom position 1.

Top opposite P = A.

$\boxed{A}$

---

## 56. COMPLEX REFERENCE CHAIN

---

Question:

**Who sits second to the right of the person facing Q?**

Step 1:

Person facing Q = B.

Step 2:

B faces South.

B's right = physical left.

B is top position 2.

Second right would be position 0, invalid.

Therefore:

$\boxed{\text{No such person}}$

Questions can use edge positions this way.

---

## 57. SAME ROW, MIXED FACING EXAMPLE

---

Five people A–E sit in one row.

Facing:

---

- A North
- B South
- C North
- D South
- E North

Suppose:

**B sits immediately right of C.**

C faces North.

Therefore B physically right of C.

Notice B's own South-facing direction is irrelevant to locating B.

## 58. ANOTHER MIXED EXAMPLE

**C sits immediately left of B.**

Now use B's facing.

If B faces South, B's left = physical right.

Therefore C physically right of B.

So the two statements:

- B right of C
- C left of B

can mean the same thing only if reference facing supports it.

## 59. MUTUAL RELATION CHECK

For two people facing the same direction:

If A is right of B, then B is left of A.

For two people facing opposite directions, this may not hold in the person's-relative sense.

Example:

A faces North, B faces South, and B physically right of A.

From A: B is on A's right.

From B: A is also on B's right.

Both can be "right of each other" because they face opposite directions.

This is a famous trap.

## 60. OPPOSITE-FACING SAME-SIDE PHENOMENON

Visual:

TEXT

A! '      B! "

A sees B on right.

B sees A on right.

Thus:

$\boxed{\text{Opposite-facing neighbours can each be on the other's right}}$

Similarly, each can be on the other's left depending physical order.

---

## 61. NEIGHBOUR COUNT IN PARALLEL ROWS

Immediate neighbour normally means same-row adjacent unless question defines otherwise.

The person directly opposite is not usually called an immediate neighbour.

Keep:

- adjacent = same row side-by-side
- opposite/facing = vertical pair

separate.

---

## 62. ADVANCED CLUE PRIORITY

For two-row problems, solve in this order:

1. row membership
2. fixed end/position
3. opposite pairs
4. immediate left/right
5. exact gaps
6. non-adjacency
7. remaining slots

This minimizes errors.

---

## 63. ROW MEMBERSHIP FIRST

If A belongs to top row and P to bottom row, note it immediately.

Otherwise you may accidentally place someone in wrong row.

Use:

TEXT

Top: A, B, C, D  
Bottom: P, Q, R, S

before solving.

## 64. OPPOSITE-PAIR BLOCK

If A faces P, treat:

TEXT  
[A/P]

as a vertical pair.

If B immediately right of A in top row, the column relation propagates.

This is similar to horizontal block logic.

## 65. COLUMN METHOD

For two rows, think in columns:

Column	Top	Bottom
1		
2		
3		
4		

This is often clearer than a freehand diagram.

## 66. EXAMPLE — COLUMN REASONING

A opposite P.

B immediate left of A, both top-row face South.

If A column 2:

B must column 3.

If Q opposite B:

Q column 3.

Everything propagates by columns.

## 67. FINAL LINEAR-SEATING WORKFLOW

### Single row

1. mark number of seats

2. mark facing
3. place fixed/end/middle
4. make blocks
5. apply exact gaps
6. apply relative clues
7. negative clues
8. fill leftovers
9. verify

## Two rows

---

1. mark row membership
  2. mark which way each row faces
  3. number columns
  4. place fixed positions
  5. place opposite pairs
  6. resolve left/right using reference facing
  7. propagate blocks
  8. eliminate contradictions
  9. verify all clues
- 

## 68. SPEED TARGETS

---

Simple single-row:

$\boxed{30-60\text{ sec}}$

Medium arrangement:

$\boxed{1-2\text{ min}}$

Two-row placement set:

$\boxed{2-4\text{ min}}$

Large puzzle set:

Spend more time only if multiple questions depend on the same arrangement.

---

## 69. WHEN TO SKIP IN AN OA

---

If a seating set has:

- many ambiguous clues
- only one question attached
- no fixed anchor
- heavy case-splitting

it may be strategically better to postpone it.

But if 4–5 questions depend on one set, solving the arrangement can give high return.

---

## 70. FINAL COMMON TRAPS

---

### & pTRAP 1 — Physical left vs person's left

Always use facing direction.

### & pTRAP 2 — Using the wrong person's facing

In "X is right of Y," use Y's facing.

### & pTRAP 3 — South-facing reversal forgotten

Left/right reverse physically.

### & pTRAP 4 — Opposite seat called neighbour

Not necessarily.

### & pTRAP 5 — Two rows assumed to face each other

Only if stated/implied.

### & pTRAP 6 — Full arrangement forced when cases remain

Keep cases.

### & pTRAP 7 — "Could be true" solved as "must be true"

Different logic.

### & pTRAP 8 — Contradictory clue set not detected

Verify all constraints.

---

## PRACTICE — LEVEL 0

---

### Q1

A faces South. B sits immediately right of A. On the page, where is B?

### Q2

A faces North. B sits immediately left of A. On the page, where is B?

### Q3

A and P sit opposite each other in two rows. If A is in column 3, where is P?

### Q4

A and B both face South. B sits second to the left of A. If A is at physical seat 2, where can B be?

### Q5

In a two-row arrangement, what does "opposite" usually mean?

---

## LEVEL 1

---

### Q6

---



Four people face South. A is at physical seat 1. B is immediately left of A. What is B's physical seat?

Q7

A faces South and is at seat 4. B sits second to the right of A. What is B's physical seat?

Q8

Bottom row faces North. Q is at seat 3. P sits immediately left of Q. Where is P?

Q9

Top row faces South. A is at seat 3. B sits immediately right of A. Where is B?

Q10

If A faces North, B faces South, and B is physically to the right of A, on which side of A is B?

LEVEL 2

Use:

TEXT

Top:	A	B	C	D				
			!"	!"	!"	!"		
Bottom:	P	Q	R	S				
			!'	!'	!'	!'		

Q11

Who faces B?

Q12

Who sits immediately left of R?

Q13

Who sits immediately right of B from B's perspective?

Q14

Who faces the person immediately right of Q?

Q15

Who is immediately left of C from C's perspective?

LEVEL 3 — EXAM PRESSURE

Top row A–D face South. Bottom row P–S face North.

Clues:

- 1. A is at top left end.
- 2. B sits immediately left of A.
- 3. Q sits opposite B.

4. P sits immediately left of Q.
5. S sits at bottom right end.
6. C sits opposite R.

### Q16

Find complete arrangement.

### Q17

Who faces A?

### Q18

Who sits immediately right of B?

### Q19

Who sits opposite D?

### Q20

Who sits immediately left of R?

---

## ANSWERS

### Q1

A faces South.

A's right = physical page-left.

So:

$\boxed{B \text{ is physically left of A}}$

### Q2

North-facing person's left = page-left.

$\boxed{B \text{ physically left of A}}$

### Q3

Opposite seats share the same column.

$\boxed{\text{Column 3}}$

### Q4

A faces South.

A's left = physical right.

Second left of A means +2 physical positions.

A=2:

$\boxed{B=4}$

$\boxed{4}$

---

## Q5

Same column in the other row.

$\boxed{\text{Vertically aligned seat}}$

---

## Q6

A=1, faces South.

A's left = physical right.

$\boxed{B=2}$

$\boxed{2}$

---

## Q7

A faces South.

Right = physical left.

Second right:

$\boxed{4-2=2}$

$\boxed{2}$

---

## Q8

Q faces North.

Left = physical left.

$\boxed{3-1=2}$

$\boxed{2}$

---

## Q9

A faces South.

Right = physical left.

$\boxed{3-1=2}$

$\boxed{2}$

---

### Q10

A faces North.

Physical right is A's right.

\[ \boxed{Right} \]

---

### Q11

B is top column 2.

Bottom column 2:

\[ \boxed{Q} \]

---

### Q12

R faces North.

Immediate left = physical seat 2.

Person there:

\[ \boxed{Q} \]

---

### Q13

B faces South.

B's right = physical left.

Top seat left of B is A.

\[ \boxed{A} \]

---

### Q14

Q faces North.

Immediate right = R.

Person facing R = C.

\[ \boxed{C} \]

---

### Q15

C faces South.

C's left = physical right.

Top seat right of C = D.

\[ \boxed{D} \]

---

Q16–Q20

From clues:

A at top physical position 1.

B immediately left of A. Since A faces South:

$\boxed{B=2}$

Q opposite B:

$\boxed{Q=2}$

P immediately left of Q. Q faces North:

$\boxed{P=1}$

S bottom right end:

$\boxed{S=4}$

Remaining bottom R:

$\boxed{R=3}$

C opposite R:

$\boxed{C=3}$

Remaining D:

$\boxed{D=4}$

Final:

TEXT									
Top:	A	B	C	D					
			!	"	!	"	!	"	!
Bottom:	P	Q	R	S					
			!	'	!	'	!	'	!

Q16

$\boxed{\text{Top: A-B-C-D; Bottom: P-Q-R-S}}$

Q17

A opposite P.

$\boxed{P}$

Q18

B faces South.

Immediate right = physical left.

A is there.

$\boxed{A}$

Q19

D opposite S.

$\boxed{S}$

## Q20

R faces North.

Immediate left = Q.

$\boxed{Q}$

---

## ' CHAPTER 63 COMPLETE

---

We have covered:

- single-row seating '
- left/right positioning '
- immediate neighbours '
- exact gaps '
- end/middle seats '
- adjacency/non-adjacency '
- north-facing rows '
- south-facing reversal '
- mixed-facing rows '
- reference-person facing rule '
- parallel rows '
- face-to-face arrangements '
- opposite seats '
- same-direction and outward-facing rows '
- block formation '
- overlapping blocks '
- conditional restrictions '
- even/odd seat constraints '
- multiple valid arrangements '
- could/must/cannot-be-true logic '
- case splitting '
- contradiction pruning '
- column method '
- speed strategy '
- L0–L3 practice '

## Progress

---

$\boxed{63/76}$  text{ numbered chapters completed}

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '

- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating Arrangement '
- Chapter 64 — Circular & Parallel Seating +
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

## NEXT

---

### CHAPTER 64 — CIRCULAR & PARALLEL SEATING

Next we will cover:

- circular arrangement basics
- inward-facing circles
- outward-facing circles
- left/right reversal
- clockwise/anticlockwise interpretation
- immediate neighbours
- opposite positions
- $\lfloor n/2 \rfloor$  opposite-seat rule
- fixing rotational symmetry
- gaps between people
- mixed inward/outward facing
- circular blocks
- square/rectangular seating
- advanced parallel-row integration
- placement-style multi-question sets
- final circular-seating speed toolkit.

USER

next

ASSISTANT

## CHAPTER 64 — CIRCULAR & PARALLEL SEATING

---

### PART 1 — CIRCULAR ARRANGEMENT BASICS, INWARD/OUTWARD FACING, CLOCKWISE/ANTICLOCKWISE, NEIGHBOURS, OPPOSITE SEATS & ROTATIONAL

## SYMMETRY

Status:  $\emptyset$ =M PROGRESS GATE: High Placements: Very High Priority:  $\emptyset$ =Yr 1

Circular seating looks harder than linear seating mainly because there is no natural “first seat.”

The key idea is:

**Fix one person anywhere first, then place everyone else relative to that person.**

That removes rotational ambiguity.

### 1. WHAT IS CIRCULAR SEATING?

People sit around a circle.

Example with 8 people:

TEXT

A  
H B  
G C  
F D  
E

In a circular arrangement:

- every person has two immediate neighbours
- there is no left end/right end
- rotating the whole arrangement does not create a new relative arrangement

### 2. WHY WE FIX ONE PERSON

Suppose arrangement is:

TEXT

A B C D

around a circle.

If we rotate everyone one seat clockwise, relative positions are unchanged.

Therefore, for reasoning:

$\boxed{\text{Fix one person at any convenient position}}$

Usually place A at the top.

### 3. ROTATIONAL SYMMETRY

These two are equivalent:



TEXT

A  
D      B  
C

and

TEXT

B  
A      C  
D

The entire circle has simply rotated.

So do not count these as different unless an external reference direction is fixed.

## 4. INWARD-FACING CIRCLE

Everyone faces the center.

TEXT

A! "      !™ B  
H! ~  
G! '      ! • C  
F! —      ! — D  
E! '

The arrows point inward.

For inward-facing people:

$\boxed{\text{Left = clockwise}}$

$\boxed{\text{Right = anticlockwise}}$

This is extremely important.

## 5. MUST-MEMORIZE — INWARD FACING

If everyone faces the center:

- person's left !' clockwise
- person's right !' anticlockwise

So:

**B sits immediately left of A**

means B is immediately clockwise from A.

## 6. OUTWARD-FACING CIRCLE

Everyone faces away from the center.

For outward-facing people:

$\boxed{\text{Left} = \text{anticlockwise}}$

$\boxed{\text{Right} = \text{clockwise}}$

This is the reverse of inward facing.

## 7. MUST-MEMORIZE MASTER TABLE

Facing	Left	Right
Toward center	Clockwise	Anticlockwise
Away from center	Anticlockwise	Clockwise

Memorize this before solving circular seating.

## 8. WHY INWARD LEFT = CLOCKWISE

Place A at the top facing center.

A is facing downward.

From A's perspective, left side points toward the circle's clockwise direction.

Hence:

$\boxed{\text{Inward: Left} = \text{Clockwise}}$

Do not rely on intuition every time; memorize the rule.

## 9. BASIC INWARD EXAMPLE

8 people face the center.

A is fixed at top.

B sits immediately left of A.

Since inward left = clockwise:

TEXT

A  
B

So B occupies the next clockwise seat.

## 10. INWARD RIGHT EXAMPLE

A faces center.

C sits immediately right of A.

Inward right = anticlockwise.  
Thus C goes one seat anticlockwise from A.

### 11. OUTWARD LEFT EXAMPLE

A faces outward.  
B sits immediately left of A.  
Outward left = anticlockwise.  
Therefore B goes one seat anticlockwise from A.

### 12. OUTWARD RIGHT EXAMPLE

A faces outward.  
C sits immediately right of A.  
Outward right = clockwise.  
So C is one seat clockwise from A.

### 13. CLOCKWISE VS ANTICLOCKWISE

Clockwise means movement like a clock's hands:

TEXT

T o p ! '   R i g h t ! '   B o t t o m ! '   L e f t ! '   T o p

Anticlockwise:

TEXT

T o p ! '   L e f t ! '   B o t t o m ! '   R i g h t ! '   T o p

### 14. POSITION NUMBERING METHOD

For 8 seats, number clockwise:

TEXT

1  
8   2  
7   3  
6   4  
5

Then all clues become arithmetic.  
For inward facing:

- left  $(=+1)$  clockwise
  - right  $(=-1)$  or  $(+7)$
- 

## 15. MODULO METHOD

---

For  $(N)$  circular seats:

$\boxed{\text{positions are taken modulo } N}$

Example with 8 seats:

Seat 8's next clockwise seat:

$8+1 \text{ to } 1$

Wraparound occurs naturally.

---

## 16. SECOND TO THE LEFT — INWARD

---

A faces center.

B is second to the left of A.

Inward left = clockwise.

So B is two seats clockwise from A.

There is:

$\boxed{1 \text{ person between A and B}}$

---

## 17. THIRD TO THE RIGHT — INWARD

---

Right = anticlockwise.

So third right means move 3 seats anticlockwise.

People between:

$3-1=2$

---

## 18. k-th LEFT / RIGHT RULE

---

If someone is  $(k)$ th to the left/right:

Position difference:

$\boxed{k}$

Number of people between along that chosen direction:

$\boxed{k-1}$

Same concept as linear seating.

---

## 19. IMMEDIATE NEIGHBOURS

In a circle, every person has exactly:

$\boxed{2}$

immediate neighbours.

No one sits at an end.

## 20. EXAMPLE

Suppose clockwise order is:

$A-B-C-D-E-F$

Neighbours of C:

$\boxed{B,D}$

Neighbours of A:

because circle wraps:

$\boxed{F,B}$

## 21. WRAPAROUND TRAP

In a circular order:

TEXT

A B C D E F

A and F are adjacent.

This is unlike a linear row.

## 22. “A SITS BETWEEN B AND C”

If “between” means both are immediate neighbours:

Possible local order around circle:

$B-A-C$

or:

$C-A-B$

But if the statement merely says somewhere between, context matters.

## 23. OPPOSITE POSITIONS

For an even number  $N$  of seats, opposite seat is:

$\boxed{\frac{N}{2} \text{ positions away}}$

Example  $(N=8)$ :

Opposite gap = 4 positions.

Seat pairs:

$\{(1,5), (2,6), (3,7), (4,8)\}$

---

## 24. OPPOSITE RULE

If  $(N)$  is even:

$\boxed{\text{pos}_{\text{opposite}} = \text{pos} + \frac{N}{2} \pmod N}$

This is a high-value shortcut.

---

## 25. EXAMPLE — 8 PEOPLE

A at seat 1.

Person opposite A:

$1 + 4 = 5$

So seat 5.

---

## 26. EXAMPLE — 10 PEOPLE

Opposite gap:

$10/2 = 5$

A at seat 3.

Opposite:

$3 + 5 = 8$

---

## 27. OPPOSITE NOT UNIQUE FOR ODD N

If  $(N)$  is odd, there is no exactly diametrically opposite seat in equally spaced seating.

Example  $(N=7)$ :

No seat exactly  $(3.5)$  positions away.

So questions about “opposite” usually use even  $(N)$ .

---

## 28. NUMBER OF PEOPLE BETWEEN OPPOSITE PERSONS

For even  $(N)$ , if A and B are opposite:

Moving either direction, number strictly between them:

$\boxed{\frac{N}{2} - 1}$

---

For  $(N=8)$ :

$[ 4-1=3 ]$

So 3 people lie between A and B on either side.

## 29. BASIC CIRCULAR PUZZLE

Six people A–F face center.

Clues:

- 1. A fixed at top.
- 2. B sits immediately left of A.
- 3. C sits immediately left of B.
- 4. D sits opposite A.

Using inward left = clockwise:

A seat 1.

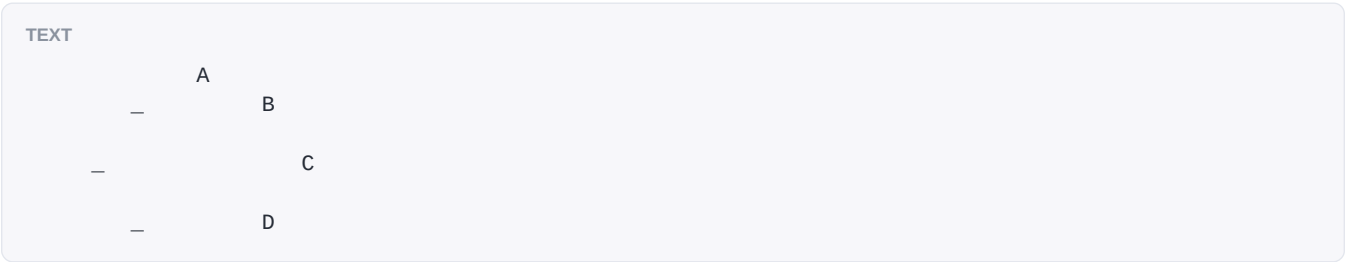
B seat 2.

C seat 3.

D opposite A in 6 seats:

$[ 1+3=4 ]$

Partial:



Remaining E,F fill remaining seats.

## 30. CLUE PRIORITY IN CIRCULAR SEATING

Use:

- 1. fix one arbitrary person
- 2. opposite clues
- 3. immediate left/right
- 4. exact k-th left/right
- 5. adjacency
- 6. people-between clues
- 7. negative restrictions
- 8. leftovers

## 31. DO NOT FIX MORE THAN NECESSARY

---

You may fix A at top because rotation is arbitrary.

But do not also arbitrarily decide B is clockwise unless a clue says so.

Only remove rotational symmetry, not mirror ambiguity.

---

## 32. MIRROR IMAGE IS DIFFERENT

---

This is important.

Rotating an arrangement does not change it.

But reflecting clockwise order usually does.

Example:

$\backslash$  [ A-B-C-D  $\backslash$  ]

and:

$\backslash$  [ A-D-C-B  $\backslash$  ]

are mirror images, not generally equivalent when left/right relationships matter.

---

## 33. CIRCULAR BLOCK

---

If A sits immediately left of B and both face center:

A is clockwise of B.

Thus local clockwise order:

$\backslash$  [ B-A  $\backslash$  ]

Because moving clockwise from B reaches A.

Be careful with how you write block orientation.

---

## 34. EASIER BLOCK NOTATION

---

Instead of memorizing visual order, use directional notation:

$\backslash$  [ A=L(B)  $\backslash$  ]

meaning A is left of B.

Then for inward facing:

$\backslash$  [ pos(A)=pos(B)+1  $\backslash$  ]

This avoids drawing mistakes.

---

## 35. EXAMPLE WITH EQUATIONS

---

8 inward-facing people.

B is second left of A:

---



$$\boxed{B=A+2}$$

C is immediate right of B:

Right = anticlockwise:

$$\boxed{C=B-1}$$

Thus:

$$\boxed{C=A+1}$$

So C sits immediately left of A.

## 36. CHAIN REDUCTION

Given:

- B second left of A
- C immediate right of B

For inward:

$$\boxed{B=A+2}$$

$$\boxed{C=A+1}$$

Therefore clockwise local order:

$$\boxed{A-C-B}$$

## 37. OUTWARD EQUATION RULE

For outward facing:

- left = anticlockwise
- right = clockwise

So if seat numbers increase clockwise:

$$\boxed{\text{Outward left} = -k}$$

$$\boxed{\text{Outward right} = +k}$$

## 38. INWARD EQUATION RULE

With clockwise numbering:

$$\boxed{\text{Inward left} = +k}$$

$$\boxed{\text{Inward right} = -k}$$

## 39. MASTER ALGEBRA TABLE

Facing	Left $\backslash (k)$	Right $\backslash (k)$
Inward	$\backslash (+k)$	$\backslash (-k)$

Facing	Left $\backslash(kl)$	Right $\backslash(kl)$
Outward	$\backslash(-kl)$	$\backslash(+kl)$

This is one of the safest ways to solve.

## 40. MIXED INWARD/OUTWARD FACING

Now each person's facing may differ.

Example:

A faces center. B faces outward.

Statement:

**C sits to the left of A.**

Use A's facing.

Since A inward:

C is clockwise of A.

Statement:

**D sits to the left of B.**

Use B's facing.

Since B outward:

D is anticlockwise of B.

## 41. REFERENCE-PERSON RULE AGAIN

For:

**X sits left/right of Y**

the facing direction that matters is:

$\backslash \boxed{Y} \backslash$

not X.

Exactly like mixed-facing linear seating.

## 42. SAME PHYSICAL SIDE, DIFFERENT WORD

Two adjacent people can be physically clockwise/anticlockwise but the verbal left/right depends on the reference person's facing.

Always convert verbal clue into physical clockwise/anticlockwise relation first.

---

## 43. MIXED-FACING EXAMPLE

A is inward-facing.

B sits immediately right of A.

Inward right = anticlockwise.

So B is anticlockwise neighbour.

Now suppose B faces outward.

C sits immediately right of B.

Outward right = clockwise.

So C is clockwise of B.

C could even be A, unless people are distinct and seat already occupied. That signals impossible or closes a loop depending context.

---

## 44. FACING DIRECTION MAY BE ASKED

Sometimes seat positions are known, but inward/outward facing must be inferred.

Example:

A and B occupy adjacent seats.

B is clockwise from A.

Statement:

**B is to the left of A.**

For B to be A's left:

- if A inward !' left is clockwise

So A must face inward.

Thus facing can be deduced from left/right clue.

---

## 45. DEDUCING FACING RULE

If X is physically clockwise of Y:

- X is left of Y if Y faces inward
- X is right of Y if Y faces outward

If X is physically anticlockwise of Y:

- X is right of Y if Y faces inward
  - X is left of Y if Y faces outward
- 

## 46. OPPOSITE PERSONS MAY FACE SAME OR DIFFERENT DIRECTIONS

---

Do not infer facing from opposite position alone.

Two opposite people may:

- both face inward
- both face outward
- face differently

Only use explicit clues.

---

## 47. “SECOND PERSON TO THE LEFT” IN CIRCLE

---

If A faces inward and B is second left of A:

TEXT

A ! ' o n e p e r s o n ! ' B

clockwise.

There is exactly one person between them.

---

## 48. “THREE PEOPLE BETWEEN”

---

If exactly 3 people lie between A and B moving clockwise, their seat gap clockwise is:

$\lfloor 3+1=4 \rfloor$

But in a circle there is another route around the other side.

So always ask whether clue specifies:

- clockwise
- anticlockwise
- either side
- exactly 3 people on both sides (usually opposite in 8)

---

## 49. BETWEEN-COUNT IN CIRCLE

---

For  $\lfloor N \rfloor$  people, if clockwise distance from A to B is  $\lfloor d \rfloor$  seats:

Clockwise people between:

$\lfloor d-1 \rfloor$

Anticlockwise distance:

$\lfloor N-d \rfloor$

Anticlockwise people between:

$\lfloor N-d-1 \rfloor$

---

## 50. EXAMPLE

---

8 people.

B is 3 seats clockwise from A.

Clockwise people between:

$$\lfloor 3-1=2 \rfloor$$

Other direction gap:

$$\lfloor 8-3=5 \rfloor$$

People between anticlockwise:

$$\lfloor 5-1=4 \rfloor$$

So “people between” is direction-sensitive unless opposite.

---

## 51. OPPOSITE CASE SPECIAL

---

If  $\lfloor d=N/2 \rfloor$ :

Both sides have same number between:

$$\lfloor \frac{N-1}{2} \rfloor$$

This symmetry makes opposite clues easy.

---

## 52. CIRCULAR POSITION EXAMPLE

---

8 inward-facing people A–H.

- A fixed at seat 1.
- B second left of A.
- C immediate right of B.
- D opposite A.
- E immediate left of D.

Compute:

$$A = 1.$$

B:

$$\lfloor 1+2=3 \rfloor$$

C:

$$\lfloor 3-1=2 \rfloor$$

D:

$$\lfloor 1+4=5 \rfloor$$

E:

$$\lfloor 5+1=6 \rfloor$$

So occupied:

$$\lfloor A1, \setminus C2, \setminus B3, \setminus D5, \setminus E6 \rfloor$$

Remaining F,G,H at 4,7,8.

---

---

## 53. NEGATIVE CLUE EXAMPLE

---

Add:

**F is not adjacent to A.**

A's adjacent seats:

$\{ 8, 2 \}$

Seat 2 occupied by C.

Therefore F cannot be 8.

So F must be 4 or 7.

Use negative clue after strong placements.

---

## 54. ADJACENCY PAIR COUNT IN CIRCLE

---

With  $N$  seats, number of adjacent seat pairs:

$\boxed{N}$

Unlike linear row where it is  $(N-1)$ .

Because last and first are adjacent.

---

## 55. END POSITIONS DO NOT EXIST

---

No one can be:

- left end
- right end

in ordinary circular seating.

If a clue says "at an extreme end," then it is not a standard circle.

---

## 56. FIXED COMPASS DIRECTION CHANGES THINGS

---

Sometimes the circle is drawn with:

- north seat
- south seat
- east seat
- west seat

If these are fixed physical locations, rotation may no longer be arbitrary.

Then you cannot freely rotate everyone.

This is less common in standard seating puzzles.

---

## 57. COUNTING UNIQUE CIRCULAR ARRANGEMENTS — PREVIEW

---

For  $(n)$  distinct people around a round table:

$$\boxed{(n-1)!}$$

because rotations are equivalent.

This was covered under permutations, but it connects naturally here.

---

## 58. IF REFLECTION ALSO COUNTS SAME

---

For necklace-like arrangements:

$$\frac{(n-1)!}{2}$$

for  $(n > 2)$ , assuming mirror images considered identical.

But standard seating arrangements generally treat clockwise vs anticlockwise order as different because left/right matters.

---

## 59. CIRCULAR PUZZLE WITH 6 PEOPLE

---

Six people A–F face center.

Clues:

1. A fixed at top.
2. B is immediate left of A.
3. C is second right of A.
4. D sits opposite B.
5. E is immediate right of D.

Let's solve.

Number clockwise 1–6.

A=1.

B immediate left:

$$B=2$$

C second right:

Right = anticlockwise:

$$C=1-2 \equiv 5$$

D opposite B:

Opposite gap  $(=3)$ :

$$D=2+3=5$$

But C already occupies 5.

Contradiction.

Therefore clue set invalid.

This kind of check is important.

---

---

## 60. CORRECTED PUZZLE

---

Six people A–F face center.

1. A = seat 1.
2. B immediate left of A.
3. C second right of A.
4. D opposite A.
5. E immediate right of D.

A=1.

B=2.

C:

$\backslash 1-2 \equiv 5 \backslash$

D opposite A:

$\backslash 1+3=4 \backslash$

E immediate right of D:

$\backslash 4-1=3 \backslash$

Occupied:

1 A 2 B 3 E 4 D 5 C

Remaining F=6.

Clockwise order:

$\backslash \boxed{A-B-E-D-C-F} \backslash$

---

## 61. CHECK RELATIONS

---

Arrangement:

$\backslash A-B-E-D-C-F \backslash$

All face center.

- B immediate left of A? Clockwise from A = B ' '
- C second right of A? Anticlockwise A! ' F! ' C
- D opposite A? yes ' '
- E immediate right of D? Anticlockwise from D = E ' '

Verified.

---

## 62. WHO ARE A'S NEIGHBOURS?

---

From arrangement:

$\backslash A-B-E-D-C-F \backslash$

A's clockwise neighbour = B.



A's anticlockwise neighbour = F.

So:

$\boxed{B, F}$

---

### 63. WHO SITS OPPOSITE B?

---

6 seats !' gap 3.

B at seat 2.

Opposite seat:

$\boxed{5}$

Person:

$\boxed{C}$

---

### 64. WHO IS SECOND LEFT OF F?

---

All inward.

Left = clockwise.

F at seat 6.

Two clockwise seats:

6 !' 1(A) !' 2(B)

Answer:

$\boxed{B}$

---

### 65. WHO IS SECOND RIGHT OF B?

---

Right = anticlockwise.

B seat 2:

2 !' 1(A) !' 6(F)

$\boxed{F}$

---

### 66. CIRCULAR “COULD BE TRUE”

---

If arrangement is not unique, test whether an option appears in at least one valid arrangement.

You do not need it in all arrangements.

---

### 67. CIRCULAR “MUST BE TRUE”

---

A statement must hold in every valid circular arrangement after rotational normalization.

---

This often requires checking all surviving branches.

---

## 68. CIRCULAR “CANNOT BE TRUE”

---

Option must violate at least one fixed clue in every possible arrangement.

Use contradiction quickly.

---

## 69. CASE SPLITTING IN CIRCULAR PUZZLES

---

If B can be either clockwise or anticlockwise of A because clue says only “adjacent,” create:

### Case 1

B clockwise of A.

### Case 2

B anticlockwise of A.

Then propagate.

Do not randomly branch on many people simultaneously.

---

## 70. HIGH-IMPACT BRANCHES

---

Good case splits:

- neighbour orientation
- opposite pair placement
- mixed-facing direction choice
- block orientation

Avoid low-information random cases.

---

## 71. CIRCULAR MASTER WORKFLOW

---

### Step 1

Check number of people.

### Step 2

Mark inward/outward/mixed facing.

### Step 3

Fix one person at top.

### Step 4

Number seats clockwise.

### Step 5

Translate:

---

- inward left = +
- inward right = "
- outward left = "
- outward right = +

### Step 6

Place opposite relations.

### Step 7

Place immediate and  $k$ -th relations.

### Step 8

Apply adjacency/gap restrictions.

### Step 9

Fill remaining people.

### Step 10

Verify every clue.

## 72. FAST NOTATION

For clockwise seat numbering:

#### TEXT

```
I-L = +
I-R = -
O-L = -
O-R = +
```

Where:

- I = inward
- O = outward
- L = left
- R = right

This compact rule is worth memorizing.

## 73. COMMON TRAPS

& pTRAP 1 — Treating circular left/right like linear seating

Wrong.

& pTRAP 2 — Forgetting inward/outward reversal

Use the master table.

& pTRAP 3 — Not fixing one person

Wastes time due to rotational symmetry.

**& pTRAP 4 — Treating mirror image as same**

Not generally true.

**& pTRAP 5 — Forgetting first and last are neighbours**

Circle wraps around.

**& pTRAP 6 — Assuming opposite seat exists for odd  $\lfloor(N)\rfloor$**

It does not exactly.

**& pTRAP 7 — Using X's facing in "X is left of Y"**

Use Y's facing.

---

## **PRACTICE — LEVEL 0**

---

**Q1**

Everyone faces center. Which direction corresponds to a person's left?

**Q2**

Everyone faces outward. Which direction corresponds to a person's right?

**Q3**

In a circle of 8, how many seats away is the opposite seat?

**Q4**

In a circle of 6, how many immediate neighbours does each person have?

**Q5**

If A is at seat 1 in an 8-person circle, what seat is opposite A?

---

## **LEVEL 1**

---

Assume all face center.

**Q6**

A is at seat 1. B sits immediately left of A. What seat is B?

**Q7**

A is at seat 1. C sits second right of A in an 8-seat circle. What seat is C?

**Q8**

B is at seat 3 in an 8-seat circle. Who sits opposite B's seat number?

**Q9**

Three people sit between A and B on both sides in an 8-person circle. What is their relation?

**Q10**

If clockwise order is A-B-C-D-E-F, who are A's neighbours?

---

---

## LEVEL 2

---

Six people A–F face center.

Given clockwise order:

\[ A-B-E-D-C-F \]

### Q11

Who sits immediate left of A?

### Q12

Who sits immediate right of A?

### Q13

Who sits opposite A?

### Q14

Who sits second right of B?

### Q15

Who sits opposite C?

---

## LEVEL 3 — EXAM PRESSURE

---

Eight people A–H face center.

Clues:

1. A is fixed at seat 1.
2. B sits second left of A.
3. C sits immediate right of B.
4. D sits opposite A.
5. E sits immediate left of D.
6. F is not adjacent to A.
7. G sits opposite C.

### Q16

Find the fixed positions of A, B, C, D, E and G.

### Q17

Which seats remain available for F and H?

### Q18

Who sits opposite C?

### Q19

Who is immediate left of D?

---

## Q20

Who are A's possible immediate neighbours?

---

## ANSWERS

---

### Q1

---

Inward:

$\boxed{\text{Clockwise}}$

### Q2

---

Outward right:

$\boxed{\text{Clockwise}}$

### Q3

---

$8/2=4$

$\boxed{4}$

### Q4

---

$\boxed{2}$

### Q5

---

$1+4=5$

$\boxed{5}$

### Q6

---

Inward left = clockwise.

$1+1=2$

$\boxed{2}$

### Q7

---

Inward right = anticlockwise.

$1-2\equiv 7\pmod 8$

$\boxed{7}$

### Q8

---

Opposite gap:

$8/2=4$

$$\boxed{3+4=7}$$

$$\boxed{7}$$

## Q9

In an 8-person circle, opposite people have 3 between on each side.

$$\boxed{\text{Opposite}}$$

## Q10

Circle wraps.

Neighbours of A:

$$\boxed{B, F}$$

## Q11

Inward left = clockwise.

From A, clockwise neighbour is B.

$$\boxed{B}$$

## Q12

Inward right = anticlockwise.

A's anticlockwise neighbour = F.

$$\boxed{F}$$

## Q13

6 seats !' opposite gap 3.

A seat 1 !' seat 4 = D.

$$\boxed{D}$$

## Q14

B is seat 2.

Second right = two anticlockwise:

$$2 !' 1(A) !' 6(F)$$

$$\boxed{F}$$

## Q15

C seat 5.

Opposite:

$$\boxed{5+3 \equiv 2}$$

Seat 2 = B.

$\boxed{B}$

---

## Q16–Q20

---

8 seats, clockwise numbering.

A:

$1$

B second left of A:

$1+2=3$

So:

$B=3$

C immediate right of B:

Inward right = anticlockwise:

$3-1=2$

So:

$C=2$

D opposite A:

$1+4=5$

So:

$D=5$

E immediate left of D:

$5+1=6$

So:

$E=6$

G opposite C:

$2+4=6$

But seat 6 is already occupied by E.

Therefore the clue set is contradictory.

So:

$\boxed{\text{No valid arrangement exists}}$

This is intentional exam-style contradiction detection.

## Q16

Fixed placements before contradiction:

$A=1, C=2, B=3, D=5, E=6$

But G also requires seat 6.



Hence:

$\boxed{\text{No complete valid arrangement}}$

### Q17

Because the clue set is inconsistent:

$\boxed{\text{Not applicable}}$

### Q18

G is supposed to sit opposite C, but that creates contradiction with E.

$\boxed{\text{No valid final answer}}$

### Q19

E is immediate left of D.

$\boxed{E}$

### Q20

The arrangement is inconsistent overall, so no final valid neighbour set exists.

---

## CHAPTER 64 STATUS

Covered so far:

- circular seating basics '
- rotational symmetry '
- fixing one anchor '
- inward-facing circles '
- outward-facing circles '
- left/right conversion '
- clockwise/anticlockwise '
- k-th left/right '
- immediate neighbours '
- wraparound '
- opposite positions '
- $\lfloor N/2 \rfloor$  rule '
- between-count logic '
- circular blocks '
- mixed inward/outward facing '
- facing inference '
- modulo position method '
- contradiction detection '
- case splitting '
- L0–L3 practice '

But **Chapter 64 is not complete yet.**

Still remaining:

- advanced mixed-facing circular arrangements
- square and rectangular seating
- corner vs side-center facing
- inward/outward square seating
- people sitting at vertices/sides
- circular arrangements with multiple opposite pairs
- advanced block restrictions
- integrated parallel-row recap
- placement-style multi-question sets
- final Chapter 64 speed toolkit

Progress

63/76 completed

Current: Chapter 64 — Circular & Parallel Seating

NEXT

Chapter 64 Part 2 — Mixed Circular Facing, Square/Rectangular Seating, Corner/Side Positions, Advanced Sets & Final Seating Toolkit

USER  
next

ASSISTANT

CHAPTER 64 — CIRCULAR & PARALLEL SEATING

PART 2 — MIXED CIRCULAR FACING, SQUARE/RECTANGULAR SEATING, CORNER/SIDE POSITIONS, ADVANCED SETS & FINAL TOOLKIT

Status: Not in PROGRESS ! ' COMPLETE GATE: High Placements: Very High Priority: Not 1

This part completes the difficult seating patterns that go beyond a simple circle:

- mixed inward/outward facing
- square seating
- rectangular seating
- corners vs side-centres
- opposite positions
- inward/outward facing on polygons
- advanced multi-clue sets
- final seating shortcuts

The core principle remains:

First identify the physical seat. Then interpret left/right using the facing direction of the reference person.

## 1. MIXED-FACING CIRCLE

In a mixed circle, some people face:

- inward
- outward

Example:

TEXT

A inward

!"

H outward

B inward

G inward

C outward

F outward

D inward

E outward

Now you cannot use one left/right rule for everyone.

You must use the **reference person's facing** every time.

## 2. MASTER MIXED-FACING RULE

For clockwise-numbered seats:

Reference person faces	Left	Right
Inward	Clockwise	Anticlockwise
Outward	Anticlockwise	Clockwise

So if:

**X sits left of Y**

you use **Y's facing**, not X's.

## 3. EXAMPLE

A faces inward.

B sits immediately left of A.

Since A is inward:

\[ \text{left}=clockwise \]

So B is next clockwise from A.

---

## 4. EXAMPLE — OUTWARD REFERENCE

---

A faces outward.

B sits immediately left of A.

Since outward:

$\lceil \text{left} = \text{anticlockwise} \rceil$

So B is next anticlockwise from A.

---

## 5. MIXED CIRCLE CHAIN

---

Suppose:

- A faces inward
- B sits immediate left of A
- B faces outward
- C sits immediate right of B

Step 1:

A inward !' left = clockwise.

So B is clockwise of A.

Step 2:

B outward !' right = clockwise.

So C is clockwise of B.

Thus physical clockwise order:

$\lceil A-B-C \rceil$

---

## 6. SAME WORD, DIFFERENT MOVEMENT

---

“Left” does not always mean clockwise.

“Right” does not always mean anticlockwise.

Everything depends on facing.

This is the single biggest mixed-facing trap.

---

## 7. DEDUCING FACING FROM POSITION

---

Suppose B is physically clockwise of A.

Given:

**B is to the left of A.**

For that to be true, A must face:

$\boxed{\text{Inward}}$

Because inward-left = clockwise.

---

## 8. ANOTHER FACING DEDUCTION

---

B is physically clockwise of A.

Given:

**B is to the right of A.**

Then A must face:

$\boxed{\text{Outward}}$

because outward-right = clockwise.

---

## 9. MUTUAL LEFT/RIGHT POSSIBILITY

---

Two adjacent people can each be on the other's left or right depending on facing.

Example:

A and B sit adjacent.

A faces inward.

B faces outward.

If B is clockwise of A:

- B is left of A
- A is left of B

So both may be “left of each other” from their own perspectives.

Do not apply linear intuition.

---

## 10. OPPOSITE PAIR IN MIXED CIRCLE

---

Position and facing are independent.

If A is opposite B:

- they occupy opposite seats
- they may both face inward
- both outward
- or one inward and one outward

Do not infer facing from opposition.

---

## 11. SQUARE SEATING — BASIC MODEL

---

Eight people may sit around a square:

- 4 at corners
- 4 at side-centres

Example:

TEXT

```
A ----- B ----- C
|
H D
|
G ----- F ----- E
```

Typical positions:

- A, C, E, G = corners
- B, D, F, H = side-centres

## 12. CORNER VS SIDE-CENTRE

With 8 seats around a square:

$\boxed{4 \text{ corner positions}}$

$\boxed{4 \text{ side-centre positions}}$

A clue like:

**A sits at a corner**

restricts A to 4 possible seats.

A clue like:

**B sits at the middle of a side**

restricts B to the other 4.

## 13. CLOCKWISE ORDER STILL WORKS

Number square positions clockwise:

TEXT

```
1 --- 2 --- 3
|
8 4
|
7 --- 6 --- 5
```

Then:

- 1,3,5,7 = corners
- 2,4,6,8 = side-centres

This is extremely useful.

---

## 14. OPPOSITE POSITIONS IN SQUARE

---

For 8 equally spaced perimeter seats:

Opposite position is still 4 seats away.

Pairs:

$\backslash (1,5) \backslash$

$\backslash (2,6) \backslash$

$\backslash (3,7) \backslash$

$\backslash (4,8) \backslash$

---

## 15. CORNER OPPOSITE CORNER

---

In the 8-seat square:

- corner seat 1 opposite corner 5
- corner 3 opposite corner 7

Similarly:

- side-centre 2 opposite 6
  - side-centre 4 opposite 8
- 

## 16. INWARD-FACING SQUARE

---

Everyone faces the center of square.

The same circular rule works:

$\backslash \boxed{\text{Left}=\text{clockwise}} \backslash$

$\backslash \boxed{\text{Right}=\text{anticlockwise}} \backslash$

because everyone faces inward toward the center.

---

## 17. OUTWARD-FACING SQUARE

---

Everyone faces outward.

Then:

$\backslash \boxed{\text{Left}=\text{anticlockwise}} \backslash$

$\backslash \boxed{\text{Right}=\text{clockwise}} \backslash$

Exactly like outward circular seating.

---

## 18. SQUARE SEATING TRICK

---

Do not waste time thinking geometrically about each corner.

---

Treat all perimeter seats as a cycle:

$\{1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 1\}$

Then solve using modulo arithmetic.

---

## 19. CORNER DISTANCE

---

If A is at corner seat 1:

Adjacent seats:

$\{8, 2\}$

Opposite seat:

$\{5\}$

Second clockwise:

$\{3\}$

This makes square seating nearly identical to circular seating logically.

---

## 20. SQUARE PUZZLE EXAMPLE

---

Eight people A–H face inward around square.

Clues:

1. A sits at a corner.
2. B sits immediate left of A.
3. C sits opposite A.
4. D sits immediate right of C.
5. E sits at a side-centre.

Fix A at corner seat 1.

Because rotation is arbitrary, we may choose:

$\{A=1\}$

B immediate left inward:

$\{B=2\}$

C opposite:

$\{C=5\}$

D immediate right of C:

$\{D=4\}$

E must occupy one of side-centres:

$\{2, 4, 6, 8\}$

But 2 and 4 already occupied.

So:

---



$\forall E \in \{6, 8\}$

Not yet unique.

---

## 21. ROTATIONAL SYMMETRY IN SQUARE

If no compass direction is fixed, square arrangements can be rotated.

So you can still fix one person at a convenient seat.

But you must preserve:

- corner vs side-centre type
- relative clockwise order

---

## 22. WHEN YOU CANNOT ARBITRARILY ROTATE

If question says:

- A sits at north-east corner
- B sits on south side
- C sits at western side-centre

then absolute orientation matters.

You cannot rotate freely.

---

## 23. COMPASS-ORIENTED SQUARE

Example:

```
TEXT
NW ---- N ---- NE
|
W E
|
SW ---- S ---- SE
```

Now positions are tied to real directions.

A clue like “north-east corner” fixes exact geometry.

---

## 24. RECTANGULAR SEATING

A rectangle may have:

- 4 corners
- side positions
- more seats along long sides than short sides

Example with 8 perimeter seats may still be treated cyclically.

Key idea:

Unless side lengths themselves matter, seating logic is still based on cyclic order and facing.

## 25. RECTANGLE — PHYSICAL OPPOSITE

In some rectangular arrangements, “opposite” means directly across the table.

This may differ from “4 seats away” if seat spacing is irregular.

So use the diagram or stated seat layout.

## 26. RECTANGULAR TABLE WITH TWO LONG SIDES

Example:

TEXT

A	B	C	D		
!"		!"		!"	!"
!'		!'		!'	!'
P	Q	R	S		

This is effectively a parallel-row seating problem.

Opposite pairs:

$\backslash$  [ A-P, \ B-Q, \ C-R, \ D-S  $\backslash$  ]

This is different from a perimeter square.

## 27. RECTANGULAR PERIMETER VS TWO-ROW RECTANGLE

Two common patterns:

### Pattern A — Around perimeter

Seats continue around corners.

### Pattern B — Along two opposite sides

Two parallel rows face each other.

Do not confuse them.

## 28. CORNER PERSON'S NEIGHBOURS

In an 8-seat square perimeter, a corner person still has exactly two immediate neighbours.

Example seat 1:

Neighbours:

$\backslash$  [ 8,2  $\backslash$  ]

Even though visually they lie on two different sides.

---

## 29. SIDE-CENTRE PERSON'S NEIGHBOURS

---

Seat 2 neighbours:

$\{ 1, 3 \}$

Again exactly two.

---

## 30. "SECOND TO THE LEFT" IN SQUARE

---

Same as circle.

If inward-facing:

$\{ +2 \text{ clockwise} \}$

If outward-facing:

$\{ -2 \text{ anticlockwise} \}$

Corner/side type may change after two steps.

---

## 31. CORNER !"CORNER PATTERN

---

In 8-seat square:

Moving 2 positions from a corner reaches another corner.

Example:

$\{ 1 \text{ to } 3 \}$

So if B is second left of corner-seated A, B is also at a corner.

This is a powerful shortcut.

---

## 32. SIDE !"SIDE PATTERN

---

Similarly:

$\{ 2 \text{ to } 4 \}$

So moving two seats from a side-centre reaches another side-centre.

Thus second-left/right preserves seat type.

---

## 33. ODD STEP CHANGES SEAT TYPE

---

From a corner, moving 1 or 3 seats reaches a side-centre.

From side-centre, odd-step movement reaches a corner.

This can quickly detect contradictions.

---

## 34. SEAT-TYPE PARITY RULE

---

Using seat numbers:

- odd = corner
- even = side-centre

Adding an even number preserves parity.

Adding an odd number flips parity.

So:

- 2nd left/right !' same seat type
  - 1st/3rd left/right !' opposite seat type
- 

## 35. ADVANCED CORNER CLUE

---

Suppose:

- A sits at a corner
- B sits second to left of A
- all face inward

Then B must also be:

$\boxed{\text{corner}}$

No drawing needed.

---

## 36. ANOTHER CORNER CLUE

---

A sits at a side-centre.

C sits third to right of A.

3 is odd, so seat type flips.

Therefore C must sit at:

$\boxed{\text{corner}}$

---

## 37. MULTIPLE OPPOSITE PAIRS

---

In 8 seats, there are 4 opposite pairs.

If:

A opposite E B opposite F C opposite G D opposite H

once one side is known, all opposite positions are immediately fixed.

Use pairs as vertical blocks.

---

## 38. OPPOSITE-PAIR BLOCK METHOD

---

Treat:

---

$\{ (A,E) \}$

as a diameter pair.

If B immediate clockwise from A, then the person opposite B must be immediate clockwise from E.

This can propagate two positions at once.

---

## 39. DIAMETRIC PROPAGATION

---

Suppose  $(N=8)$ :

A at 1, E opposite at 5.

B at 2.

Then opposite B is automatically seat 6.

If F is stated opposite B:

$\{ F=6 \}$

This saves time.

---

## 40. SYMMETRY SHORTCUT

---

If two people are opposite, shifting both by same number of seats preserves opposition.

Formally:

$\{ x \text{ opposite } y \} \rightarrow \{ x+k \text{ opposite } y+k \}$

modulo  $(N)$ .

Useful for chain reasoning.

---

## 41. ADVANCED BLOCK RESTRICTIONS

---

Suppose:

- A and B sit together
- C and D sit together
- no member of AB block is adjacent to CD block

Treat blocks first.

On a circle, be careful:

the first and last block positions are also adjacent due to wraparound.

---

## 42. CIRCULAR BLOCK WRAP TRAP

---

Suppose clockwise order is:

$\{ [AB]-X-Y-[CD] \}$

Even if visually at two ends of your written sequence,  $[AB]$  and  $[CD]$  may be adjacent through circle closure.

---

Always check wraparound.

---

## 43. TWO-PERSON BLOCK WITH FACING

---

If both face inward and:

**A sits immediately left of B**

then physical clockwise order:

$\backslash$  B-A  $\backslash$

If both face outward:

left = anticlockwise, so physical clockwise order:

$\backslash$  A-B  $\backslash$

Facing determines block orientation.

---

## 44. MIXED-FACING BLOCK

---

If:

**A sits immediately left of B**

you need only B's facing.

A's facing may be irrelevant for placement.

This is a frequent trap.

---

## 45. ADVANCED MIXED SET

---

Eight people A–H sit around a circle.

Some face inward, some outward.

Clues:

1. A sits opposite E.
2. B sits immediately clockwise of A.
3. B is to the left of A.
4. C sits opposite B.
5. D sits immediately right of C.

From clue 2:

B physically clockwise of A.

Clue 3 says B is left of A.

Therefore A must face inward.

Now C opposite B.

---

For D right of C, we still need C's facing.

If C's facing unknown, D may be either physical neighbour.

So arrangement remains partially unresolved.

This shows how facing itself can be a variable.

---

## 46. INFERENCE FROM LEFT/RIGHT + PHYSICAL POSITION

---

If physical relative position is known, left/right clue can tell you facing.

This is especially useful in mixed-facing questions.

---

## 47. "SAME DIRECTION" FACING CLUE

---

If A and B face the same direction:

- both inward
- or both outward

If A known inward:

$\boxed{B \text{ inward}}$

If A known outward:

$\boxed{B \text{ outward}}$

---

## 48. "OPPOSITE DIRECTIONS" FACING CLUE

---

If A and B face opposite directions:

- one inward
- one outward

Knowing one fixes the other.

---

## 49. FACING-COUNT QUESTIONS

---

Suppose 8 people sit around a circle.

Exactly 5 face inward.

Then:

$\boxed{3}$

face outward.

Basic, but can combine with position constraints.

---

## 50. ADJACENT PEOPLE FACING SAME/OPPOSITE

---

Example clue:

---

**Both neighbours of A face opposite to A.**

If A inward, both adjacent people are outward.

This is often a strong propagation clue.

## 51. ALTERNATING FACING PATTERN

If every adjacent person faces opposite direction from the previous, then for even  $(N)$ :

$[I, O, I, O, \dots]$

This is possible around full circle.

For odd  $(N)$ , perfect alternation around the circle is impossible because last-first adjacency breaks pattern.

## 52. PARITY TRICK FOR ALTERNATING FACING

For even  $(N)$ :

- odd-numbered seats can all be inward
- even-numbered seats outward

or vice versa.

Thus opposite seats may have:

- same facing if  $(N/2)$  even
- opposite facing if  $(N/2)$  odd

depending  $(N)$ .

Advanced but useful.

## 53. SQUARE ALTERNATING FACING

For 8 seats, alternating inward/outward is perfectly possible:

$[I, O, I, O, I, O, I, O]$

Then all corner seats (odd) share one facing and all side-centres the other.

This can create strong deductions.

## 54. PLACEMENT-STYLE SQUARE SET

Eight people A–H sit around a square, all facing center.

Clues:

1. A sits at a corner.
2. B sits second left of A.
3. C sits opposite B.
4. D sits immediate right of C.



5. E sits opposite A.
6. F sits immediate left of E.
7. G is not adjacent to A.

Fix A at corner seat 1.

B second left:

$$\backslash [ 1+2=3 \backslash ]$$

C opposite B:

$$\backslash [ 3+4=7 \backslash ]$$

D immediate right:

$$\backslash [ 7-1=6 \backslash ]$$

E opposite A:

$$\backslash [ 1+4=5 \backslash ]$$

F immediate left:

$$\backslash [ 5+1=6 \backslash ]$$

But D already occupies 6.

Contradiction.

Therefore clue set invalid.

Spotting this early saves time.

## 55. CORRECTED SQUARE SET

Eight people A–H face center.

1. A at corner seat 1.
2. B second left of A.
3. C opposite B.
4. D immediate right of C.
5. E opposite A.
6. F immediate right of E.
7. G not adjacent A.

Solve:

A=1.

B:

$$\backslash [ 1+2=3 \backslash ]$$

C:

$$\backslash [ 3+4=7 \backslash ]$$

D immediate right inward:

$$\backslash [ 7-1=6 \backslash ]$$

E opposite A:

\[ 5 \]

F immediate right E:

\[ 5-1=4 \]

Occupied:

1 A 3 B 4 F 5 E 6 D 7 C

Remaining seats:

2,8 for G,H.

G not adjacent A.

A's neighbours are 8 and 2.

Both remaining seats are adjacent to A.

Therefore G cannot be placed.

Still contradiction.

This illustrates why negative clues can invalidate a seemingly complete set.

---

## 56. FURTHER CORRECTED SET

Replace clue 7 with:

**G sits adjacent to A.**

Then remaining seats 2 and 8 are both adjacent to A.

So G/H may swap.

Two valid arrangements remain.

The full seating is not unique, but many questions are still answerable.

---

## 57. GUARANTEED VS POSSIBLE

From previous set:

Fixed:

- A=1
- B=3
- F=4
- E=5
- D=6
- C=7

G/H occupy 2/8.

Therefore:

- A's neighbours are G and H, but which side unknown
- G and H are both adjacent to A

- B is definitely second left of A
- E definitely opposite A

This distinction matters in “must/could” questions.

## 58. ADVANCED PARALLEL-ROW RECAP

For two rows facing each other:

TEXT

Top:	A	B	C	D				
			!"	!"	!"	!"		
Bottom:	P	Q	R	S				
			!'	!'	!'	!'		

Top row faces South:

- left = page-right
- right = page-left

Bottom row faces North:

- left = page-left
- right = page-right

This is the same reference-direction logic as before.

## 59. PARALLEL-ROW OPPOSITE RULE

Same column:

$\boxed{\text{facing each other}}$

If A at top column 3:

person at bottom column 3 faces A.

## 60. PARALLEL-ROW COMPOSITE CLUE

**P sits immediately left of the person facing B.**

Solve:

1. identify who faces B
2. use that person's facing
3. move one seat to their left

Never shortcut the two steps.

## 61. EXAMPLE

Top row faces South.

Bottom row faces North.

B is top column 3.

R faces B, so R is bottom column 3.

P sits immediately left of R.

R faces North, so P is bottom column 2.

$\boxed{P=2}$

---

## 62. ADVANCED QUESTION TYPES

---

Seating questions may ask:

- Who must sit opposite A?
- Which arrangement is possible?
- Which statement cannot be true?
- Who could sit between B and C?
- How many valid arrangements exist?
- Which person faces outward?
- Which pair occupies corners?
- Who sits second right of the person opposite D?

These require different levels of certainty.

---

## 63. MUST / COULD / CANNOT

---

### Must be true

True in every valid arrangement.

### Could be true

True in at least one valid arrangement.

### Cannot be true

False in every valid arrangement.

Do not confuse these.

---

## 64. OPTION-TESTING STRATEGY

---

If the question gives four possible complete arrangements, often faster to test options rather than derive everything.

Reject any option violating:

1. fixed seat type
  2. opposite pair
  3. immediate relation
  4. facing rule
  5. adjacency restriction
-

This is high-value in OAs.

---

## 65. CONTRADICTION CHECKLIST

---

A branch is invalid if:

- two different people occupy same seat
  - one person occupies two seats
  - opposite rule violated
  - required adjacent seat unavailable
  - corner/side type violated
  - mixed-facing left/right inconsistent
  - final clue impossible
  - wraparound adjacency overlooked
- 

## 66. FINAL SEATING SEARCH ORDER

---

For circular/square problems:

1. identify geometry
  2. identify seat types
  3. mark facing
  4. fix one anchor if rotation allowed
  5. number positions clockwise
  6. place opposite pairs
  7. apply exact left/right relations
  8. place blocks
  9. apply corner/side restrictions
  10. use negative clues
  11. split cases only if needed
  12. verify
- 

## 67. SPEED SHORTCUT — CLOCKWISE NUMBERING

---

For  $\backslash(N\backslash)$  seats, choose:

$\backslash[1,2,\dots,N\backslash]$

clockwise.

Then use algebra.

For inward:

$\backslash[ L=+, \quad R=- \backslash]$

For outward:

$\backslash[ L=-, \quad R=+ \backslash]$

This avoids visual confusion.

---

---

## 68. SPEED SHORTCUT — OPPOSITE

---

For even  $(N)$ :

$\boxed{+\frac{N}{2}}$

modulo  $(N)$ .

For 8:

$\boxed{+4}$

For 10:

$\boxed{+5}$

---

## 69. SPEED SHORTCUT — SEAT TYPE

---

In 8-seat square:

$\boxed{\text{odd positions = corners}}$

$\boxed{\text{even positions = side-centres}}$

This is an excellent shortcut.

---

## 70. SPEED SHORTCUT — SECOND LEFT/RIGHT

---

In 8-seat square:

Moving 2 seats preserves corner/side type.

Moving odd number of seats flips it.

Use parity before drawing.

---

## 71. COMMON ADVANCED TRAPS

---

& pTRAP 1 — Treating mixed-facing left/right uniformly

Wrong.

& pTRAP 2 — Using X's facing in "X is left of Y"

Use Y's facing.

& pTRAP 3 — Forgetting corner/side-centre distinction

Important in square questions.

& pTRAP 4 — Assuming any opposite relation in odd-seat circle

No exact opposite seat.

& pTRAP 5 — Ignoring wraparound block adjacency

Circle closes.

& pTRAP 6 — Rotating when compass orientation is fixed

---

Not allowed.

& pTRAP 7 — Forcing unique seating when G/H can swap

Keep multiple cases.

& pTRAP 8 — Solving “could” as “must”

Different requirement.

---

## PRACTICE — LEVEL 0

---

### Q1

In a mixed circle, X sits left of Y. Whose facing determines the physical direction?

### Q2

In an 8-seat square, how many corner positions exist?

### Q3

In an 8-seat square, which numbered seats are side-centres if numbered clockwise from 1?

### Q4

Inward-facing square: left corresponds to which direction?

### Q5

Outward-facing circle: right corresponds to which direction?

---

## LEVEL 1

---

### Q6

A is at corner seat 1 of an inward-facing 8-seat square. B is second left of A. What seat is B?

### Q7

Is B from Q6 at a corner or side-centre?

### Q8

A is at side-centre seat 2. C sits third right of A, all facing inward. What seat is C?

### Q9

What seat type is C from Q8?

### Q10

In an 8-seat square, what seat is opposite seat 6?

---

## LEVEL 2

---

Eight people sit around a square, all facing inward.

Given:

- A=1
- B is second left of A
- C opposite B
- D immediate right of C
- E opposite A
- F immediate right of E

### Q11

Find B.

### Q12

Find C.

### Q13

Find D.

### Q14

Find E and F.

### Q15

Which seats remain empty?

---

## LEVEL 3 — EXAM PRESSURE

Eight people A–H sit around a square.

- A sits at corner seat 1.
- B sits second left of A.
- C sits opposite B.
- D sits immediately right of C.
- E sits opposite A.
- F sits immediately right of E.
- G and H occupy the two remaining seats.
- Everyone faces inward.

### Q16

What are the fixed positions?

### Q17

Which seats are occupied by G and H?

### Q18

Who sits opposite A?

### Q19

Who sits immediate right of C?

### Q20



Can we determine whether G is clockwise or anticlockwise neighbour of A?

---

## ANSWERS

---

### Q1

Use the reference person Y.

$\boxed{Y}$

---

### Q2

$\boxed{4}$

---

### Q3

$\boxed{2,4,6,8}$

---

### Q4

Inward left:

$\boxed{\text{Clockwise}}$

---

### Q5

Outward right:

$\boxed{\text{Clockwise}}$

---

### Q6

Inward left = clockwise.

Second left:

$\boxed{1+2=3}$

$\boxed{3}$

---

### Q7

Seat 3 is odd.

Odd seats are corners.

$\boxed{\text{Corner}}$

---

## Q8

---

Inward right = anticlockwise.

$$\backslash[2-3=-1\backslash]$$

Modulo 8:

$$\backslash[-1\equiv 7\backslash]$$

$$\backslash[\boxed{7}\backslash]$$

---

## Q9

---

Seat 7 is odd.

$$\backslash[\boxed{\text{Corner}}\backslash]$$

---

## Q10

---

Opposite gap = 4.

$$\backslash[6+4=10\equiv 2\backslash]$$

$$\backslash[\boxed{2}\backslash]$$

---

## Q11–Q15

---

A:

$$\backslash[1\backslash]$$

B second left:

$$\backslash[1+2=3\backslash]$$

$$\backslash[\boxed{B=3}\backslash]$$

C opposite B:

$$\backslash[3+4=7\backslash]$$

$$\backslash[\boxed{C=7}\backslash]$$

D immediate right of C:

Inward right = anticlockwise:

$$\backslash[7-1=6\backslash]$$

$$\backslash[\boxed{D=6}\backslash]$$

E opposite A:

$$\backslash[1+4=5\backslash]$$

$$\backslash[\boxed{E=5}\backslash]$$

F immediate right of E:

$$\backslash[5-1=4\backslash]$$

---

$\boxed{F=4}$

Occupied:

1 A 3 B 4 F 5 E 6 D 7 C

Remaining:

$\boxed{2,8}$

---

## Q16–Q20

---

Fixed:

$A=1$

$B=3$

$F=4$

$E=5$

$D=6$

$C=7$

Remaining:

$G, H \in \{2, 8\}$

### Q16

$\boxed{A1, B3, F4, E5, D6, C7}$

### Q17

$\boxed{2 \text{ and } 8}$

### Q18

E is opposite A.

$\boxed{E}$

### Q19

D is immediately right of C.

$\boxed{D}$

### Q20

A's neighbours are seats 2 and 8.

G and H occupy those two seats, but their exact assignment is unknown.

Therefore:

$\boxed{\text{Cannot be uniquely determined}}$

---

## ' CHAPTER 64 COMPLETE

---

We have now covered:

- circular seating basics '
- rotational symmetry '
- inward/outward facing '
- mixed-facing circles '
- clockwise/anticlockwise conversion '
- immediate neighbours '
- kth left/right '
- opposite positions '
- modulo-seat arithmetic '
- between-count logic '
- circular blocks '
- square seating '
- rectangular seating distinctions '
- corner vs side-centre seats '
- seat-type parity shortcuts '
- compass-oriented seating '
- multiple opposite pairs '
- mixed-facing inference '
- same/opposite facing clues '
- parallel-row integration '
- must/could/cannot logic '
- contradiction handling '
- option elimination '
- L0–L3 practice '

## Progress

$\lfloor \frac{64}{76} \rfloor$  numbered chapters completed

## UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating Arrangement '
- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles +
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

CHAPTER 65 — LOGICAL PUZZLES

Next we will cover:

- puzzle-solving framework
- floors/buildings
- boxes/order
- days/months/time slots
- professions/colors/cities
- matching tables
- one-to-one assignment
- conditional constraints
- either/or cases
- exactly one/both/neither
- selection and grouping puzzles
- scheduling puzzles
- grid method
- case splitting
- contradiction pruning
- multi-variable placement-style sets
- GATE constraint reasoning
- L0–L3 practice.

USER

next

ASSISTANT

CHAPTER 65 — LOGICAL PUZZLES

PART 1 — PUZZLE-SOLVING FRAMEWORK, FLOOR/BOX/ORDER PUZZLES, MATCHING TABLES, DAYS/TIME SLOTS & CASE-SPLITTING

Status:  $\emptyset=0\%$  PROGRESS GATE: High Placements: Very High Priority:  $\emptyset=0\%$  Part 1

Logical puzzles combine several constraints and ask you to construct one valid arrangement.

Common forms:

- floors in a building
- boxes stacked vertically
- days of the week
- time slots
- people matched with cities/colors/jobs

- ordering and scheduling
- selection/grouping

The key idea is:

**Do not solve sentences. Convert every sentence into a constraint and let the constraints interact.**

## 1. THE UNIVERSAL PUZZLE WORKFLOW

For almost every puzzle:

1. identify entities
2. identify dimensions/categories
3. make slots/table
4. place direct facts first
5. convert relative clues
6. mark negative constraints
7. combine connected clues
8. split cases only when unavoidable
9. kill contradictions early
10. verify every clue

This is the master method.

## 2. DIRECT VS INDIRECT CLUES

### Direct clue

**A lives on floor 3.**

Immediately fix:

$\backslash A=3 \backslash$

### Relative clue

**B lives above C.**

Means:

$\backslash B>C \backslash$

but exact floors unknown.

Direct clues are usually stronger.

## 3. STRONG-CLUE PRIORITY

Use clues roughly in this order:

1. exact/fixed position
2. extreme position
3. immediate relation
4. exact gap
5. unique-category pairing
6. either/or
7. ordinary above/below/before/after
8. negative clues

This reduces unnecessary branching.

## 4. FLOOR PUZZLES — BASIC STRUCTURE

Suppose 5 people live on 5 floors.

Number from bottom:

``text id="6nqjqu" Floor 5 [ ] Floor 4 [ ] Floor 3 [ ] Floor 2 [ ] Floor 1 [ ]

If floor 1 is ground floor, use that convention unless stated otherwise.

---

# 5. ABOVE / BELOW

If:

> A lives above B

then:

```
\[
floor(A)>floor(B)
\]
```

Not necessarily immediately above.

---

# 6. IMMEDIATELY ABOVE

> A lives immediately above B

means:

```
\[
floor(A)=floor(B)+1
\]
```

They occupy consecutive floors.

---

# 7. TWO FLOORS BETWEEN

If exactly 2 floors/people lie between A and B:

```
\[
|floor(A)-floor(B)|=3
\]
```

\]

Same gap idea as seating.

---

#### # 8. FLOOR PUZZLE EXAMPLE

Five people A-E live on floors 1-5.

Clues:

1. A lives on floor 5.
2. B lives immediately below A.
3. C lives on floor 1.
4. D lives above C.
5. E does not live on floor 3.

Place:

A=5

B=4

C=1

Remaining floors 2,3 for D,E.

D above C, so D can be 2 or 3.

E cannot 3, so E=2.

Thus D=3.

Final:

```
```text id="7xm7z2"
```

```
5 A
4 B
3 D
2 E
1 C
```

9. BOX STACKING PUZZLES

Exactly same logic as floors.

Example:

```
```text id="1im3is" Top Box 5 Box 4 Box 3 Box 2 Box 1 Bottom
```

Words:

- above
- below
- immediately above
- exactly two boxes between

all behave like floor positions.

---



## # 10. TOP/BOTTOM ORIENTATION TRAP

Always confirm numbering.

You may number:

- bottom = 1, top = N

or

- top = 1, bottom = N

Either works, but remain consistent.

---

## # 11. ORDER PUZZLES

Suppose 5 tasks A-E occur in sequence.

Represent:

```
```text id="dlb16a"
1 2 3 4 5
```

If:

A happens before B

then:

$\forall \text{ pos}(A) < \text{pos}(B) \ \forall$

12. IMMEDIATELY BEFORE

C immediately before D

means block:

$\forall [CD] \ \forall$

This is much stronger than simply C before D.

13. EXACTLY ONE EVENT BETWEEN

If exactly one event lies between A and B:

$\forall |\text{pos}(A) - \text{pos}(B)| = 2 \ \forall$

14. ORDER PUZZLE EXAMPLE

Five tasks P,Q,R,S,T.

Clues:

- P first
- Q immediately after P
- T last
- R before S

So:

```text id="ap3j65" P Q \_ \_ T

Remaining R,S occupy 3,4 with R before S:

```text id="40i2xb"  
P Q R S T

15. DAY-BASED PUZZLES

Suppose one event each day Monday–Friday.

Write:

| Day | Event |
|-----|-------|
| Mon | |
| Tue | |
| Wed | |
| Thu | |
| Fri | |

Then translate:

- before = earlier day
- after = later day
- immediately before = previous day
- two days after = +2

16. DAY PUZZLE EXAMPLE

Five people A–E have interviews Mon–Fri.

- A on Wednesday
- B immediately before A
- C on Friday
- D before B

Then:

A = Wed

B = Tue

C = Fri

D must be Mon

E = Thu

Final:

$\boxed{D, B, A, E, C}$

17. WEEKDAY + GAP LANGUAGE

If B is two days after A:

$\text{day}(B) = \text{day}(A) + 2$

So if A is Monday:

B is Wednesday.

Exactly one day lies between them.

18. MONTH-BASED PUZZLES

Same concept.

If six months Jan–Jun are involved:

$\text{Jan} = 1, \text{Feb} = 2, \dots$

Then relative clues become simple arithmetic.

19. TIME-SLOT PUZZLES

Suppose slots:

9 AM, 10 AM, 11 AM, 12 PM.

Treat as ordered positions:

$1, 2, 3, 4$

Then:

- earlier = smaller slot
 - later = larger slot
 - immediately before = " 1
-

20. MATCHING PUZZLES

Example categories:

People:

A, B, C

Cities:

\[Delhi, Mumbai, Pune \]

Jobs:

\[Engineer, Doctor, Teacher \]

Each person gets exactly one city and one job.

Use a table.

21. ONE-TO-ONE ASSIGNMENT

If each category is unique:

- one person gets exactly one city
- each city belongs to exactly one person

Then once A=Delhi, nobody else can be Delhi.

This is a very strong elimination rule.

22. GRID METHOD

For People × Cities:

| | Delhi | Mumbai | Pune |
|---|-------|--------|------|
| A | | | |
| B | | | |
| C | | | |

Mark:

- ' confirmed
- × impossible

Example:

A not Delhi !' × B Mumbai !' '

Then automatically:

- A,C cannot Mumbai.

23. MATCHING EXAMPLE

People: A,B,C

Cities: Delhi, Mumbai, Pune

Clues:

1. A is not from Delhi.
2. B is from Mumbai.
3. C is not from Pune.

B=Mumbai.

Remaining Delhi/Pune for A,C.

C not Pune !' C=Delhi.

Therefore A=Pune.

$\boxed{A=Pune, B=Mumbai, C=Delhi}$

24. TWO-DIMENSIONAL MATCHING

Suppose:

- people
- cities
- colors

A clue may say:

The person from Delhi likes red.

This links city to color, not necessarily directly to a named person.

You must propagate through categories.

25. CHAINED MATCHING

Example:

A = Delhi.

Delhi = Red.

Therefore:

$\boxed{A=Red}$

Use transitive matching.

26. NEGATIVE CLUES

Examples:

- A is not a doctor.
- B does not live in Pune.
- Green is not associated with C.

Mark them immediately in the grid.

Negative clues become powerful after several positive assignments are known.

27. "EITHER A OR B"

Suppose:

The engineer is either A or B.

Then:

$\{ \text{Engineer} \in \{A, B\} \}$

Do not choose yet.

Keep the pair until another clue resolves it.

28. EXCLUSIVE OR VS INCLUSIVE OR

In aptitude puzzles, "either A or B" usually means one of the two, not both, especially in one-to-one assignments.

But if multiple selections are possible, wording matters.

Track whether "either/or" is exclusive.

29. "NEITHER A NOR B"

Means:

$\{ \text{not A and not B} \}$

So if doctor is neither A nor B, doctor must be C in a 3-person set.

30. EXACTLY ONE OF TWO CONDITIONS

If exactly one of:

- A is selected
- B is selected

then possibilities:

$\{ (A, \neg B) \}$

or

$\{ (\neg A, B) \}$

but not both, not neither.

31. BOTH / NEITHER

Both

A and B both satisfy.

Neither

A and B both do not satisfy.

These phrases are often embedded in selection/grouping puzzles.

32. CASE SPLITTING

Suppose:

$\forall A \in \{2,4\}$

and no clue immediately resolves it.

Create:

Case 1

$A=2$

Case 2

$A=4$

Then propagate.

Do not randomly guess.

33. GOOD CASE SPLIT

A clue:

Either P is on floor 2 or Q is on floor 5.

If no other immediate resolution:

Case 1: $P=2$ Case 2: $Q=5$

This is a meaningful branch.

34. BAD CASE SPLIT

Choosing random positions for 5 unconstrained people creates:

$\forall 5!$

possibilities.

Avoid.

Branch only on constrained binary choices.

35. CONTRADICTION PRUNING

Discard a case if it produces:

- duplicate unique assignment
 - impossible position
 - violated immediate relation
 - impossible gap
 - one entity assigned two different values
 - one category value assigned twice when unique
-

36. EXAMPLE OF PRUNING

Suppose:

- A is either floor 2 or 4.
- B is immediately above A.
- B cannot floor 5.

Case $A=4 \wedge B=5$ contradiction.

Therefore:

$\neg [A=2, \wedge B=3]$

One negative clue killed a whole branch.

37. THE "ANCHOR" CLUE

An anchor is a clue fixing something strongly:

- A is first
- B is on Tuesday
- C on floor 6
- D is doctor
- E at extreme position

Use anchor clues first.

38. LINKED CLUE CHAINS

Suppose:

- B immediately after A
- C immediately after B

Then:

$\neg [A, B, C]$

Treat as a 3-item block.

39. GAP CHAIN

Suppose:

- B is two places after A
- C immediately before B

Then:

$\neg [B=A+2]$

$\neg [C=B-1=A+1]$

Thus block:

$\neg [A \rightarrow B]$

40. CONDITIONAL CLUES

Example:

If A is on Tuesday, then B is on Friday.

Logical form:

$A_{\text{Tue}} \rightarrow B_{\text{Fri}}$

Important:

This does **not** mean:

$B_{\text{Fri}} \rightarrow A_{\text{Tue}}$

Converse is not guaranteed.

41. CONTRAPOSITIVE

From:

$A \rightarrow B$

you may infer:

$\neg B \rightarrow \neg A$

Example:

If A Tuesday! $\neg$ B Friday.

If B is not Friday, then A cannot be Tuesday.

This is powerful.

42. CONVERSE TRAP

Given:

If A is selected, B is selected.

If B is selected, can we conclude A?

No.

Possible:

$B \wedge \neg A$

This becomes important in grouping puzzles.

43. "ONLY IF"

A is selected only if B is selected

means:

$\lceil A \rightarrow B \rceil$

Memory:

A only if B = A needs B.

44. "IF"

If A, then B

also:

$\lceil A \rightarrow B \rceil$

45. "IF AND ONLY IF"

A iff B:

$\lceil A \leftrightarrow B \rceil$

means both:

$\lceil A \rightarrow B \rceil$

and

$\lceil B \rightarrow A \rceil$

So A and B occur together.

46. "UNLESS"

A unless B is usually interpreted as:

$\lceil \neg B \rightarrow A \rceil$

Equivalent to:

$\lceil A \vee B \rceil$

In aptitude puzzles, exact wording can vary, so use careful logical translation.

47. SIMPLE SELECTION PUZZLE

Choose exactly 3 from:

$\lceil A, B, C, D, E \rceil$

Rules:

1. If A selected, B selected.
2. C and D cannot both be selected.
3. E must be selected.

Since E is fixed, choose 2 more.

Possible:

- A implies B, so A+B+E works if no other restriction.
- C+D+E invalid.
- B+C+E works.
- B+D+E works.

This is a grouping/selection puzzle.

48. SELECTION NOTATION

Use:

$\backslash [A=1 \backslash]$

selected

$\backslash [A=0 \backslash]$

not selected

or simply:

' / x

This makes conditional constraints easier.

49. SCHEDULING PUZZLES

Suppose tasks A–D must be assigned to slots 1–4.

Rules:

- A before B
- C not first
- D immediately after A

Treat $[AD]$ as block plus $A < B$.

Then test feasible slots.

50. SCHEDULING EXAMPLE

4 slots.

A before B.

D immediately after A.

C not first.

Possible $[AD]$ placements:

- 1,2
- 2,3
- 3,4

If $[AD]=3, 4$, B must be after A but no slot after D? B could be 4, but occupied by D. Impossible.

If $[AD]=2, 3$, $B=4$, $C=1$, but C not first !' impossible.

So $[AD]=1, 2$.

Remaining C,B at 3,4 with A before B satisfied.

Final:

$\boxed{A,D,C,B}$

51. WHY SLOT ARITHMETIC WORKS

Many puzzles are secretly just:

$x_i \in \{1, 2, \dots, N\}$

with constraints like:

$A < B$

$C = B + 1$

$D \neq 1$

Thinking mathematically reduces verbal confusion.

52. ONE-TO-ONE PUZZLES AS CONSTRAINT SATISFACTION

You can think of each person as a variable.

Example:

$City(A) \in \{D, M, P\}$

Constraints:

$City(A) \neq D$

$City(B) = M$

and all values distinct.

Then solve by elimination.

This is the foundation of complex logic puzzles.

53. MINIMUM INFORMATION PRINCIPLE

Do not infer more than the clue says.

Example:

A is before B.

Does not imply A immediately before B.

C is above D.

Does not imply consecutive floors.

P is not red.

Does not tell what color P is.

This is one of the biggest puzzle mistakes.

54. MULTI-CATEGORY PUZZLE EXAMPLE

People:

A,B,C

Jobs:

Doctor, Engineer, Teacher

Cities:

Delhi, Mumbai, Pune

Clues:

1. A is not Doctor.
2. Engineer lives in Mumbai.
3. B lives in Delhi.
4. C is Teacher.
5. Teacher does not live in Pune.

C=Teacher.

Teacher not Pune.

B=Delhi, so C cannot Delhi.

Therefore C=Mumbai.

Engineer lives Mumbai, but C is Teacher, contradiction if jobs/cities are unique and Engineer must be Mumbai.

Hence clue set inconsistent.

Contradiction detection is essential.

55. CORRECTED MULTI-CATEGORY SET

Change clue 2:

Engineer lives in Pune.

Then:

C=Teacher.

Teacher not Pune.

B=Delhi.

C cannot Delhi or Pune !' C=Mumbai.

Remaining A=Pune.

Engineer=Pune !' A=Engineer.

A not Doctor satisfied.

Remaining B=Doctor.

Final:

| Person | City | Job |
|--------|--------|----------|
| A | Pune | Engineer |
| B | Delhi | Doctor |
| C | Mumbai | Teacher |

56. CROSS-CATEGORY PROPAGATION

Notice how:

$\neg [A = \text{Pune}]$

and

$\neg [\text{Engineer} = \text{Pune}]$

implies:

$\neg [A = \text{Engineer}]$

This is the central skill in matching-table puzzles.

57. "NOT BOTH" CONSTRAINT

A and B cannot both be selected.

Logical form:

$\neg [A \text{ and } B]$

Allowed:

- A only
- B only
- neither

Not allowed:

- both

58. "AT LEAST ONE"

At least one of A,B:

$\vee [A \vee B]$

Allowed:

- A
- B
- both

Not allowed:

- neither
-

59. "EXACTLY ONE"

Exactly one:

$\vee [A \oplus B]$

Allowed:

- A only
- B only

Not:

- both
 - neither
-

60. "AT MOST ONE"

At most one:

- A only
- B only
- neither

Not:

- both

Equivalent to “not both.”

61. "AT LEAST TWO"

For A,B,C:

At least two selected means:

- exactly 2
- or all 3

Count conditions carefully.

62. "AT MOST TWO"

For three items:

anything except selecting all three.

63. FLOOR PUZZLE — PLACEMENT STYLE

Six people A–F live on floors 1–6.

1. A lives on floor 6.
2. B lives immediately below A.
3. Two floors lie between C and D.
4. C lives below D.
5. E lives on floor 1.
6. F lives above E but below C.

A=6.

B=5.

E=1.

C/D with 2 between and C below D:

possible:

$\forall (C,D)=(1,4),(2,5),(3,6) \setminus$

But floor1=E, floor5=B, floor6=A.

So only possibility?

(1,4) invalid C=1 occupied. (2,5) D=5 occupied. (3,6) D=6 occupied.

Thus no solution.

This is contradiction detection again.

64. CORRECTED FLOOR SET

Change "two floors between C and D" to "one floor between C and D."

Then difference = 2.

Possible C below D:

$\forall (1,3),(2,4),(3,5),(4,6) \setminus$

E=1, B=5, A=6.

Possible:

$\forall (C,D)=(2,4) \setminus$

Then F above E but below C:

E=1, C=2.

No integer floor strictly between 1 and 2.

Still impossible.

65. FURTHER CORRECTED FLOOR SET

Change clue 6:

F lives below C.

Then:

A=6 B=5 E=1

Only valid C,D pair from above is:

$\{ C=2, D=4 \}$

Remaining F=3.

But F below C would require $\{3 < 2\}$, false.

So still invalid.

This illustrates a key lesson:

Do not assume every puzzle has a solution. Check feasibility.

66. A VALID FLOOR SET

Six people A–F:

1. A=6
2. B immediately below A
3. one floor between C and D
4. C below D
5. E=1
6. F immediately above E

Place:

A=6 B=5 E=1 F=2

Remaining floors 3,4 for C,D.

But one floor between needs gap 2, impossible.

So still no solution.

Let's instead make clue 3:

C immediately below D.

Then remaining 3,4:

C=3 D=4.

Final:

Valid.

67. LESSON FROM VALIDATION

When constructing or solving puzzles, always test:

- uniqueness
- availability of required gaps
- occupied slots
- extreme constraints

A single clue can make the whole set impossible.

68. MUST / COULD / CANNOT IN PUZZLES

Must be true
Holds in all valid cases.

Could be true
Holds in at least one valid case.

Cannot be true
Holds in no valid case.

This language is common in GATE-style constraint reasoning and placement OAs.

69. OPTION ELIMINATION

For MCQs, you can often avoid full solving.

Test each option against:

1. direct fixed clues
2. immediate relations
3. one-to-one uniqueness
4. conditional rules
5. negative restrictions

First violated clue kills option.

70. FINAL PART-1 TOOLKIT

For ordered positions
Use numbers.

For floors/boxes
Use vertical slots.

For matching
Use grids.

For conditionals
Use arrows:

\[

```

A\Rightarrow B
\]

### For either/or
Keep cases.

### For contradictions
Prune immediately.

### For complex chains
Build blocks.

---

# 71. COMMON TRAPS

& p    * * T R A P  1  -  A b o v e  =  i m m e d i a t e l y  a b o v e * *
False.

& p    * * T R A P  2  -  B e f o r e  =  i m m e d i a t e l y  b e f o r e * *
False.

& p    * * T R A P  3  -  I f  A ! ' B   m e a n s   B ! ' A * *
False.

& p    * * T R A P  4  -  E i t h e r / o r   r e s o l v e d   t o o   e a r l y * *
Keep alternatives.

& p    * * T R A P  5  -  D u p l i c a t e   v a l u e   i n   o n e - t o - o n e   m a t c h i n g * *
Not allowed.

& p    * * T R A P  6  -  N e g a t i v e   c l u e   i g n o r e d   a f t e r   p o s i t i v e s * *
Recheck negatives.

& p    * * T R A P  7  -  C o n t i n u i n g   a   c o n t r a d i c t o r y   c a s e * *
Kill it immediately.

---

# PRACTICE - LEVEL 0

### Q1
A lives immediately above B. If B is on floor 3, where is A?

### Q2
Two people are between A and B in an ordered list. What is their position difference?

### Q3
A happens before B. Does A have to happen immediately before B?

### Q4
If A is selected, B must be selected. Can B be selected without A?

### Q5
Exactly one of P and Q is selected. Can both be selected?

---

# LEVEL 1

```

Q6

Five tasks A-E. A is first, B immediately after A, E last, C before D. Find the order.

Q7

Three people A,B,C have cities Delhi, Mumbai, Pune.

A not Delhi, B Mumbai, C not Pune.

Find all cities.

Q8

If A is on Tuesday, B is on Friday. B is not on Friday. Can A be on Tuesday?

Q9

A and B cannot both be selected. Which are allowed?

- A only
- B only
- both
- neither

Q10

A is either floor 2 or 4. B is immediately above A. B cannot be on floor 5. Find A.

LEVEL 2

Q11

Four tasks A-D occupy slots 1-4.

- D immediately after A
- A before B
- C not first

Find the schedule.

Q12

People A,B,C have jobs Doctor, Engineer, Teacher and cities Delhi, Mumbai, Pune.

- A not Doctor
- Engineer lives in Pune
- B lives in Delhi
- C is Teacher
- Teacher not Pune

Find all assignments.

Q13

If P is selected, Q is selected.

If Q is not selected, what can we conclude about P?

Q14

Exactly one of A,B is selected. If A is not selected, what about B?

Q15

At least one of X,Y must be selected. X is not selected. What follows?

LEVEL 3 – EXAM PRESSURE

Five people A-E have interviews Mon-Fri.

Clues:

1. C is on Wednesday.
2. B is immediately before C.
3. E is on Friday.
4. A is before B.
5. D is after C.

Q16
Find the complete schedule.

Q17
Who is on Monday?

Q18
Who is immediately after C?

Q19
How many people are scheduled between A and E?

Q20
Which clue most strongly fixes B?

ANSWERS

Q1

\[
 $3+1=\boxed{4}$
 \]

Q2

\[
 $2+1=\boxed{3}$
 \]

Q3

No.

\[
 $\boxed{\text{No}}$
 \]

Q4

Yes.

\[
 $A \rightarrow B$
 \]

does not imply:

\[
 $B \rightarrow A$
 \]

\[
 $\boxed{\text{Yes}}$
 \]

Q5

Exactly one excludes both.

$$\boxed{\text{No}}$$

Q6

A first.

B immediately after:

$$\boxed{\text{A, B}}$$

E last.

C before D.

So:

$$\boxed{\text{A-B-C-D-E}}$$

Q7

B=Mumbai.

Remaining Delhi/Pune for A,C.

C not Pune !' C = Delhi .

Therefore A=Pune.

$$\boxed{\text{A=Pune, B=Mumbai, C=Delhi}}$$

Q8

Given:

$$\boxed{\text{A}_{\text{Tue}} \rightarrow \text{B}_{\text{Fri}}}$$

But:

$$\boxed{\neg \text{B}_{\text{Fri}}}$$

By contraposition:

$$\boxed{\neg \text{A}_{\text{Tue}}}$$

So:

\[
\boxed{No}
\]

Q9

Allowed:

- A o n l y '
- B o n l y '
- b o t h 'L
- n e i t h e r '

\[
\boxed{\text{A only, B only, or neither}}
\]

Q10

C a s e A = 4 ! ' B = 5 , f o r b i d d e n .

Therefore:

\[
\boxed{A=2}
\]

and B=3.

Q11

D immediately after A.

Possible `[AD]`.

I f A , D a t 2 , 3 ! ' B m u s t b e 4 , C = 1 , b u t C n o t f i r s t ! ' i n v

I f A , D a t 3 , 4 ! ' n o p o s i t i o n a f t e r A a v a i l a b l e f o r B .

Therefore:

\[
A=1, \ D=2
\]

Remaining C,B at 3,4 with A before B.

Thus:

\[
\boxed{A-D-C-B}
\]

Q12

C=Teacher.

B=Delhi.

Teacher not Pune, so C not Pune.

Since B already Delhi, C=Mumbai.

Remaining A=Pune.

E n g i n e e r l i v e s P u n e ! ' A = E n g i n e e r .

Remaining B=Doctor.

Final:

\[
\boxed{A=Pune/Engineer}
\]

\[
\boxed{B=Delhi/Doctor}
\]

\[
\boxed{C=Mumbai/Teacher}
\]

Q13

\[
P\rightarrow Q
\]

Contrapositive:

\[
\neg Q\rightarrow\neg P
\]

Therefore:

\[
\boxed{P\text{ is not selected}}
\]

Q14

Exactly one of A,B.

If A not selected:

\[
\boxed{B\text{ must be selected}}
\]

Q15

At least one of X,Y.

X not selected.

Therefore:

\[
\boxed{Y\text{ must be selected}}
\]

Q16-Q20

C = Wednesday.

B immediately before C:

\[
B=Tuesday
\]

E=Friday.

A before B, so only Monday available:

\[
A=Monday
\]

D after C and remaining slot Thursday:

\[
D=Thursday
\]

Final:

| Day | Person |
|-----------|--------|
| Monday | A |
| Tuesday | B |
| Wednesday | C |
| Thursday | D |
| Friday | E |

Q16

\[
\boxed{A-B-C-D-E}
\]

Q17

\[
\boxed{A}
\]

Q18

\[
\boxed{D}
\]

Q19

A Monday, E Friday.

Between them:

Tue, Wed, Thu

```
\[
\boxed{3}
\]
```

Q20

The clue:

> B is immediately before C.

With C fixed on Wednesday, B is forced to Tuesday.

```
\[
\boxed{\text{Clue 2}}
\]
```

CHAPTER 65 STATUS

Covered so far:

- universal puzzle framework '
- strong-clue priority '
- floor puzzles '
- box stacking '
- ordered sequences '
- day/month/time-slot puzzles '
- matching tables '
- one-to-one assignment '
- negative constraints '
- either/or cases '
- exactly one/both/neither '
- case splitting '
- contradiction pruning '
- conditional logic '
- if/only-if/iff basics '
- contraposition '
- selection puzzles '
- scheduling puzzles '
- cross-category propagation '
- option elimination '
- L0-L3 practice '

But **Chapter 65 is not complete yet**.

Still remaining:

- grouping/team formation puzzles
- distribution puzzles
- advanced floor/box sets
- multi-variable grid puzzles
- ordering with multiple gaps
- conditional grouping chains
- "at least / at most / exactly" advanced cases
- possibility vs necessity questions
- data-sufficiency style puzzles
- final logical-puzzle speed toolkit

Progress

```
\[
```

\boxed{64/76\text{ numbered chapters completed}}
\]

* * C u r r e n t : * * Ø=ßá C h a p t e r 6 5 – L o g i c a l P u z z l e s

NEXT

Ø=Ý% * * C h a p t e r 6 5 P a r t 2 – G r o u p i n g , D i s t r i b u t i o n , A d v a n c e d S c h e d u l i n g , M u l t i - V a r i a b l e G r i d P u z z l e s & F i n a l L o g i c a l - P u z z l e T o o l k i t * *

USER

next

ASSISTANT

CHAPTER 65 — LOGICAL PUZZLES

PART 2 — GROUPING, DISTRIBUTION, ADVANCED SCHEDULING, MULTI-VARIABLE GRID PUZZLES & FINAL TOOLKIT

Status: Ø=ßá P R O G R E S S ! ' ' C O M P L E T E G A T E : H i g h P l a c e m e n t s : V e r y H i g h P r i o r i t y : Ø=Ý% 1

This part covers the hardest common puzzle patterns:

- team/group formation
- allocation/distribution
- advanced selection constraints
- multi-category grids
- scheduling with multiple dependencies
- minimum/maximum possibilities
- must/could/cannot questions

Main principle:

Reduce every clue to one of four forms: FIXED, FORBIDDEN, DEPENDENCY, or COUNT.

1. FOUR TYPES OF CONSTRAINTS

Type 1 — FIXED

A must be selected.

\[A=1 \]

Type 2 — FORBIDDEN

B cannot be selected.

$\text{\texttt{[B=0]}}$

or:

A and B cannot be together.

$\text{\texttt{[\neg(A\text{and} B)]}}$

Type 3 — DEPENDENCY

If A is selected, B must be selected.

$\text{\texttt{[\boxed{A\rightarrow B}]}}$

Type 4 — COUNT

Exactly 3 people must be selected.

$\text{\texttt{[\boxed{\text{Selected count}=3}]}}$

Count constraints become very powerful after logical restrictions are applied.

2. GROUPING PUZZLES

Suppose exactly 3 students are selected from:

$\text{\texttt{[A,B,C,D,E]}}$

Rules:

1. $A \neq B$
2. C and D cannot both be selected
3. E must be selected

Start:

$\text{\texttt{[E=\checkmark]}}$

Need 2 more.

Possible valid groups include:

$\text{\texttt{[\{A,B,E\}]}}$

$\text{\texttt{[\{B,C,E\}]}}$

$\text{\texttt{[\{B,D,E\}]}}$

$\text{\texttt{[\{A? \}]}}$

If A appears, B must also appear.

3. DEPENDENCY CHAIN

If:

$\lceil A \rightarrow B \rceil$

and:

$\lceil B \rightarrow C \rceil$

then:

$\lceil \boxed{A \rightarrow C} \rceil$

So selecting A forces:

$\lceil A, B, C \rceil$

This is called implication chaining.

4. CONTRAPOSITIVE CHAIN

From:

$\lceil A \rightarrow B \rceil$

we know:

$\lceil \neg B \rightarrow \neg A \rceil$

If:

$\lceil B \rightarrow C \rceil$

then:

$\lceil \neg C \rightarrow \neg B \rightarrow \neg A \rceil$

Very useful in selection puzzles.

5. GROUP SIZE CAN FORCE OR EXCLUDE

Suppose exactly 2 people must be selected.

Rules:

$\lceil A \rightarrow B \rceil$

$\lceil B \rightarrow C \rceil$

If A selected, then A,B,C all required.

But only 2 seats exist.

Therefore:

$\lceil \boxed{A \text{ cannot be selected}} \rceil$

This is a classic counting contradiction.

6. “AT LEAST” CONSTRAINTS

“At least 2 of A,B,C” means valid counts:

$\{2, 3\}$

Possible sets:

$\{AB, AC, BC, ABC\}$

Not allowed:

- none
 - exactly one
-

7. “AT MOST” CONSTRAINTS

“At most 2 of A,B,C” means:

$\{0, 1, 2\}$

Not all three.

8. EXACTLY TWO

“Exactly 2 of A,B,C” allows:

$\{AB, AC, BC\}$

but not:

$\{ABC\}$

and not one/zero.

9. EXACTLY ONE

Exactly one of A,B,C:

$\{100, 010, 001\}$

Only one selected.

10. NOT BOTH

A and B cannot both be selected:

$\neg(A \wedge B)$

Equivalent:

$A \rightarrow \neg B$

and:

$B \rightarrow \neg A$

11. TOGETHER OR NEITHER

If A and B must be selected together:

$\{ A \rightarrow B \}$

Allowed:

- both selected
- neither selected

Not allowed:

- only A
 - only B
-

12. ONE-OF-PAIR CONSTRAINT

If exactly one of A,B must be chosen:

$\{ A \oplus B \}$

Allowed:

$\{ A, B \}$

Not:

$\{ \emptyset \}$

13. GROUPING EXAMPLE

Choose exactly 4 from:

$\{ A, B, C, D, E, F \}$

Rules:

- A !' B
- C and D cannot be together
- E must be chosen
- exactly one of B,F is chosen

Since E fixed.

Exactly one of B,F.

Case 1: B chosen

Then F not chosen.

Need two more among A,C,D.

A allowed because B present.

But C,D cannot both.

Possible:

$\{A, C, B, E\}$

or

$\{A, D, B, E\}$

Could also choose C plus? only A/D choices constrained by count.

Case 2: F chosen

Then B not.

Since A! B :

$\{A \mid \text{cannot be chosen}\}$

Need two from C,D plus E,F, but C,D cannot both.

Impossible.

Therefore:

$\boxed{B \mid \text{must be selected}}$

and:

$\boxed{F \mid \text{cannot be selected}}$

This is how count constraints resolve branches.

14. DISTRIBUTION PUZZLES

Example:

3 gifts must be distributed to A,B,C.

Conditions may say:

- A gets exactly one
- B gets at least one
- C cannot receive red
- no two people receive same type

Represent item-to-person assignment with a grid.

15. GRID FOR DISTRIBUTION

| Gift | A | B | C |
|-------|---|---|---|
| Red | | | |
| Blue | | | |
| Green | | | |

Mark ' and ×.

If each gift goes to exactly one person, every row gets exactly one ' .

If each person gets exactly one gift, every column also gets exactly one ' .

16. DISTRIBUTION VS MATCHING

Matching:

Usually one-to-one.

Distribution:

May be:

- one-to-one
- many-to-one
- capacity-limited

Always identify capacity first.

17. CAPACITY CONSTRAINT

Suppose teams X,Y can each take at most 3 people.

Then:

$$\lvert X \rvert \leq 3, \quad \lvert Y \rvert \leq 3$$

If 6 people total and everyone assigned:

$$\lvert X \rvert = \lvert Y \rvert = 3$$

Automatically.

18. MINIMUM / MAXIMUM TEAM SIZE

Suppose 7 people split into A and B teams.

Team A has at least 3.

Team B has at least 2.

Then:

$$\lvert A \rvert \geq 3, \quad \lvert B \rvert \geq 2$$

and:

$$\lvert A \rvert + \lvert B \rvert = 7$$

Possible sizes:

$$(3,4), (4,3), (5,2)$$

19. TEAM BALANCE

If two teams must have equal members and total is 8:

$$\boxed{4+4}$$

If total is odd, equal split impossible unless someone remains unassigned.

20. ADVANCED GROUP CONSTRAINT

Suppose:

- A and B must be in same team
- C and D must be in different teams

Treat:

$\forall [AB] \setminus$

as one linked unit.

For C,D, once C's team is fixed, D's is automatically the other team.

21. SAME / DIFFERENT GROUP NOTATION

Use:

$\forall [A=B] \setminus$

for same group.

Use:

$\forall [A \neq B] \setminus$

for different groups.

Then chain.

If:

$\forall [A=B, \quad B=C] \setminus$

then:

$\forall [A=C] \setminus$

If:

$\forall [A=B, \quad B \neq C] \setminus$

then:

$\forall [A \neq C] \setminus$

22. TWO-GROUP PARITY LOGIC

For two teams only:

If:

$\forall [A \neq B] \setminus$

and:

$\forall [B \neq C] \setminus$

then A and C must be in same group.

$\forall [\boxed{A=C}] \setminus$

Because only two groups exist.

23. EXAMPLE

Two teams X,Y.

- A different from B
- B different from C

Therefore:

$\forall [A=C]$

If $A=X$:

$B=Y$

$C=X$.

24. THIS FAILS WITH 3+ GROUPS

If 3 groups exist:

A different B.

B different C.

A and C may be:

- same
- different

So do not use binary-group shortcut with 3+ categories.

25. ADVANCED SCHEDULING

Suppose tasks A–F occupy six consecutive slots.

Rules:

- A before D
- B immediately after C
- E after D
- F not first/last

Create block:

$\forall [CB]$

Then order constraints:

$\forall [A<D<E]$

Combine block placement with remaining slots.

26. PRECEDENCE GRAPH

Instead of prose:

$\lceil A \rightarrow D \rightarrow E \rceil$

and:

$\lceil C \rightarrow B \rceil$

Draw:

TEXT

A ! ' D ! ' E

C ! ' B

F independent except position restriction.

This is much clearer.

27. TOPOLOGICAL-ORDER IDEA

Scheduling puzzles often ask for an ordering consistent with dependencies.

If:

$\lceil A \rightarrow B, \quad A \rightarrow C, \quad B \rightarrow D \rceil$

then A must occur before B,C, and B before D.

But C vs B/D may remain flexible.

This is essentially a topological ordering problem.

28. BEFORE DOES NOT MEAN IMMEDIATELY BEFORE

Given:

$\lceil A \rightarrow B \rceil$

valid:

$\lceil A-C-D-B \rceil$

Do not create $[AB]$ unless "immediately" appears.

29. EXACT GAP IN SCHEDULING

If two slots between A and B:

$\lceil \text{pos}(A) - \text{pos}(B) = 3 \rceil$

If A before B:

$\lceil \text{pos}(B) = \text{pos}(A) + 3 \rceil$

30. BLOCK + GAP COMBINATION

Suppose:

- B immediately after A
- D is 2 positions after B

Then:

$$\backslash B=A+1 \backslash$$

$$\backslash D=B+2=A+3 \backslash$$

So one slot lies between B and D.

Use equations.

31. ADVANCED DAY SCHEDULING

Events occur Monday–Saturday.

Clues:

- A before C
- B immediately after A
- D two days after C
- E on Saturday
- F not Monday

Build blocks:

$$\backslash [A,B] \backslash$$

$$\backslash D=C+2 \backslash$$

Then fit with endpoints.

This is easier than verbal trial.

32. MULTI-VARIABLE GRID PUZZLE

Suppose four people each have unique:

- city
- profession
- color

You need two or three grids:

1. People × City
2. People × Profession
3. People × Color

You can sometimes also use City × Profession if a clue directly connects those categories.

33. DO NOT BUILD TOO MANY GRIDS

For 4 categories, a full set of pairwise grids can explode.

Usually choose one anchor category:

\[People \]

and map everything to people.

Then chain non-person clues via deductions.

34. EXAMPLE

People: A,B,C,D

Cities: P,Q,R,S

Jobs: Doctor, Engineer, Lawyer, Teacher

Clues:

1. A lives in P.
2. The Engineer lives in R.
3. B is not Engineer.
4. C lives in R.

Since C lives R and Engineer lives R:

\[C=Engineer \]

Then B not Engineer already satisfied.

This is cross-category propagation.

35. UNIQUENESS CAN CREATE POSITIVES

Suppose:

A not Red B not Red C not Red

Only D remains.

Therefore:

\[\boxed{D=Red} \]

Negative clues can combine into a forced positive.

36. ONE-TO-ONE "LAST SLOT" RULE

If four cities assigned uniquely and three are known:

A=Delhi B=Mumbai C=Pune

remaining city Hyderabad automatically goes to D.

No extra clue needed.

37. MULTI-GAP ORDERING

Suppose:

- A is 2 positions before B
- C is immediately after B
- D is 3 positions after A

Equations:

$$B = A + 2$$

$$C = A + 3$$

$$D = A + 3$$

Then C and D collide.

If distinct entities:

$\boxed{\text{Contradiction}}$

This is why algebraic translation is powerful.

38. POSSIBILITY VS NECESSITY

Suppose two valid cases remain.

Case 1

$$A = P, B = Q$$

Case 2

$$A = R, B = Q$$

Then:

- $B = Q$ is **must be true**
- $A = P$ is **could be true**
- $A = Q$ is **cannot be true**

Always classify answers using surviving cases.

39. MUST-BE-TRUE METHOD

A statement is must-be-true if it survives every valid case.

Do not stop after seeing it true in one case.

40. COULD-BE-TRUE METHOD

Need only one valid case.

So construct one witness arrangement satisfying the option.

41. CANNOT-BE-TRUE METHOD

Try assuming the option is true.

If it creates contradiction in every branch, it cannot be true.

This proof-by-contradiction method is fast.

42. MAXIMUM / MINIMUM QUESTIONS

Suppose exactly 5 people selected.

A,B cannot both be selected.

Question:

Maximum number from set {A,B,C}?

Since A/B mutually exclusive but C can join:

Maximum:

$\boxed{2}$

One of A/B + C.

43. MINIMUM NUMBER FROM A SET

Suppose exactly 4 selected from 6.

At most 2 can come from outside set X of size 3.

Then at least:

$4-2=2$

must come from X.

This complement method is common.

44. COMPLEMENT COUNTING IN LOGIC

If total selected:

k

and at most m can come from outside group G:

$\boxed{\text{At least } k-m \text{ must come from G}}$

Useful in team/group questions.

45. EXACTLY / AT LEAST / AT MOST INTERACTION

Suppose select exactly 4.

At least 2 from group A.

At most 1 from group B.

Then remaining selection capacity can be derived.

Do counts before individual cases.

46. CATEGORY-CAPACITYEXAMPLE

Choose 5 students.

At least 3 must be seniors.

At most 2 seniors? Contradiction.

Because:

≥ 3

and:

≤ 2

cannot both hold.

Check count feasibility early.

47. PIGEONHOLE-STYLE LOGIC

Suppose 7 people assigned to 3 rooms.

Then some room has at least:

$\left\lceil \frac{7}{3} \right\rceil = 3$

people.

This connects to Fundamental Counting/Pigeonhole from Chapter 38.

Useful in logical distribution puzzles.

48. DATA-SUFFICIENCYSTYLE PUZZLES

Question:

Which floor does A occupy?

Statement I: A above B. Statement II: B on floor 3.

Together, if building has floors 1–4, A could be 4 only if no other above options? Actually A above 3 means A=4.

If building has floors 1–6, A could be 4,5,6.

So sufficiency depends on full domain.

Never judge statements without all given constraints.

49. SUFFICIENCY PRINCIPLE

A statement is sufficient only if it gives a **unique answer to the asked question**.

It does not need to determine the whole puzzle.

This will be treated deeply in Chapter 69.

50. ADVANCED FLOOR SET

Five people A–E live on floors 1–5.

Rules:

1. A above B.
2. C immediately above D.
3. Exactly one floor between B and D.
4. E on floor 5.
5. A not floor 4.

Try possible [D, C] consecutive blocks and B gap 2.

Since E=5, block cannot include 5 if C/D are different from E.

Possible [D, C] :

$\setminus (1,2), (2,3), (3,4) \setminus$

Now B two positions from D.

If D=1

B=3.

Then A above B, not 4, and floor5 occupied E.

No valid A.

Reject.

If D=2

B=4.

A above B only floor5, occupied E.

Reject.

If D=3

B=1.

C=4.

A above B, not4, floor5 occupied E.

Could A=2.

Valid.

Final:

$\setminus B1, \setminus A2, \setminus D3, \setminus C4, \setminus E5 \setminus$

$\setminus \boxed{B-A-D-C-E} \setminus$

51. ADVANCED MATCHING SET

People A,B,C,D.

Cities: Delhi, Mumbai, Pune, Kochi.

Jobs: Doctor, Engineer, Lawyer, Teacher.

Clues:

1. A is not Doctor.
2. B lives in Delhi.
3. Engineer lives in Pune.
4. C is Teacher.
5. Teacher lives in Mumbai.
6. D is not Lawyer.

From C=Teacher and Teacher=Mumbai:

$\boxed{C=Mumbai}$

B=Delhi.

Remaining cities A,D = Pune,Kochi.

Engineer=Pune.

If A=Pune !' A=Engineer.

If D=Pune !' D=Engineer.

Need more clue for uniqueness.

So puzzle remains partially unresolved.

Correct conclusion:

$\boxed{\text{Insufficient for unique full mapping}}$

52. DON'T FORCE A COMPLETE GRID

This is one of the most important logical-puzzle rules.

If multiple valid assignments remain:

preserve them.

A specific asked relation may still be determined.

53. SELECTION SET — EXAM STYLE

Select exactly 3 from A,B,C,D,E.

Rules:

1. A !' B
2. C !' D
3. B and D cannot both be selected
4. E must be selected

Since E fixed.

Need 2 more.

If A selected !' B.

Then group:

$\{A, B, E\}$

valid because D absent.

If C selected !' D.

Then:

$\{C, D, E\}$

valid because B absent.

Could choose B, C, E?

C forces D, would make 4 people and B+D conflict.

Invalid.

Could choose B, D, E?

B, D forbidden.

So exactly two valid groups:

$\{\boxed{ABE}, \boxed{CDE}\}$

54. MUST-BE-SELECTED FROM ABOVE SET

Only common member between valid groups:

$\{E\}$

So:

$\{\boxed{E \text{ must be selected}}\}$

Others are possible but not necessary.

55. CANNOT-BE-TOGETHER DERIVATION

From valid sets:

ABE and CDE.

A and C never coexist.

So even if not explicitly stated:

$\{\boxed{A \text{ and C cannot both be selected}}\}$

under the full rule set.

Derived constraints are powerful.

56. IMPLICATION-CLOSURE METHOD

Whenever A selected, add everything it forces.

Example:

$[A \rightarrow B, B \rightarrow C, C \rightarrow \neg D]$

Selecting A gives closure:

$[A, B, C, \neg D]$

Do this before filling other slots.

57. CONTRADICTION FROM CLOSURE

If additionally:

$[A \rightarrow D]$

then selecting A forces both:

$[D, \neg D]$

Impossible.

Therefore:

$[A \text{ cannot be selected}]$

58. CONDITIONAL CHAIN WITH EXACTLY ONE

Rules:

$[A \rightarrow B]$

Exactly one of B,C.

If A selected:

B selected.

Therefore C not selected.

So:

$[A \rightarrow B \wedge \neg C]$

59. CONDITIONAL CHAIN WITH AT LEAST ONE

At least one of B,C.

If B excluded:

$[C \text{ must be included}]$

Simple but frequent.

60. FINAL LOGICAL-PUZZLE WORKFLOW

For a new puzzle:

Step 1 — Domain

List all entities/categories/slots.

Step 2 — Structure

Choose:

- row
- floor chart
- timeline
- grid
- teams
- Boolean ' / ×

Step 3 — Anchors

Place exact clues.

Step 4 — Translate

Use:

- $\backslash(=)$
- $\backslash(\neq)$
- $\backslash(<, >)$
- $\backslash(+1, -1)$
- $\backslash(A \rightarrow B)$
- XOR / not-both
- counts

Step 5 — Propagate

Apply forced consequences.

Step 6 — Count

Use capacities and exact totals.

Step 7 — Branch

Split only on high-impact binary ambiguity.

Step 8 — Prune

Delete contradictory cases immediately.

Step 9 — Classify

Must / could / cannot.

Step 10 — Verify

Check every original clue.

61. SPEED TOOLKIT

Floor/order puzzle

Use numbered slots.

Matching puzzle

Use ' / xgrid.

Team puzzle

Use group labels and implications.

Scheduling

Use precedence arrows and blocks.

Conditional selection

Use implication chains.

Multiple cases

Keep a small case table.

62. CASE TABLE

Example:

| Case | A | B | C | Valid? |
|------|---|---|---|--------|
| 1 | ' | ' | x | ' |
| 2 | ' | x | ' | ' |
| 3 | x | ' | ' | ' |

This is far easier than prose for branching puzzles.

63. ERROR CHECKLIST

Before locking answer ask:

- Did I treat “before” as immediate accidentally?
- Did I reverse an implication?
- Did I violate uniqueness?
- Did I forget group capacity?
- Did I force an either/or choice too early?
- Did I miss contraposition?
- Did I keep an impossible case alive?
- Am I answering must/could/cannot correctly?

64. COMMON ADVANCED TRAPS

& pTRAP 1 — A!' B interpreted as A!' B

False.

& pTRAP 2 — not both = exactly one

False.

“Not both” allows neither.

& pTRAP 3 — at least one = exactly one

False.

Both may be allowed.

& pTRAP 4 — team inequality with 3 groups treated like 2 groups

Invalid.

& pTRAP 5 — unique assignment assumed when not stated

Check puzzle structure.

& pTRAP 6 — solving complete grid when only one query matters

Wasteful.

& pTRAP 7 — count constraints used too late

Counts can kill entire branches early.

PRACTICE — LEVEL 0

Q1

If A!' B and A is selected, what follows?

Q2

If A!' B and B is not selected, what follows?

Q3

If exactly one of A,B is selected and A is selected, what about B?

Q4

If A and B cannot both be selected, can neither be selected?

Q5

If A and B must be together and A is not selected, what about B?

LEVEL 1

Q6

Exactly 2 people must be selected from A,B,C. If A!' B and B!' C can A be selected?

Q7

Two teams only. A and B are in different teams; B and C are in different teams. What relation holds between A and C?

Q8

Exactly 4 are selected. At most 1 can come from group X. How many must come from outside X?

Q9

At least 2 of A,B,C must be selected. If A and B are not selected, is the condition possible?

Q10

A and B cannot both be selected. If A is selected, what about B?

LEVEL 2

Select exactly 3 from A,B,C,D,E.

Rules:

- $A \neq B$
- $C \neq D$
- B and D cannot both be selected
- E must be selected

Q11

List all valid groups.

Q12

Who must be selected?

Q13

Can A and C be selected together?

Q14

Can B and C be selected together?

Q15

Can D and E be selected together?

LEVEL 3 — EXAM PRESSURE

Five people A–E are assigned to five floors 1–5.

Rules:

1. A lives above B.
 2. C lives immediately above D.
 3. Exactly one floor lies between B and D.
 4. E lives on floor 5.
 5. A does not live on floor 4.
-

Q16

Find complete arrangement.

Q17

Who lives on floor 1?

Q18

Who lives immediately below C?

Q19

Who lives above D but below E?

Q20

How many people live between B and C?

ANSWERS

Q1

$\boxed{B \text{ must be selected}}$

Q2

Contrapositive:

$\neg B \rightarrow \neg A$

$\boxed{A \text{ cannot be selected}}$

Q3

Exactly one:

$\boxed{B \text{ not selected}}$

Q4

Yes.

“Cannot both” allows neither.

$\boxed{\text{Yes}}$

Q5

Together means:

$A \rightarrow B$

So:

$\boxed{B \text{ is also not selected}}$

Q6

A forces B, and B forces C.

Selecting A would select all 3.

But only 2 allowed.

Therefore:

$\boxed{\text{No}}$

Q7

With only two teams:

A different B.

B different C.

Therefore:

$\boxed{A \text{ and C are in the same team}}$

Q8

Exactly 4 selected.

At most 1 from X.

Therefore at least:

$4-1=3$

from outside X.

$\boxed{3}$

Q9

If A,B both absent, only C selected.

That gives exactly 1 of the three.

But at least 2 required.

$\boxed{\text{No}}$

Q10

$\boxed{B \text{ cannot be selected}}$

Q11–Q15

E must be selected.

Need exactly 2 others.

If A selected, B required:

$\boxed{ABE}$

valid.

If C selected, D required:

$\boxed{CDE}$

valid.

Other combinations fail due to implication/count/conflict.

Q11

$\boxed{\boxed{ABE}, \boxed{CDE}}$

Q12

Common to both:

$\boxed{E}$

Q13

A and C together would force B and D, which conflict and exceed count.

$\boxed{\text{No}}$

Q14

If B and C selected, C forces D.

Then B and D conflict.

$\boxed{\text{No}}$

Q15

CDE is valid.

So:

$\boxed{\text{Yes}}$

Q16–Q20

From the advanced floor set:

$E=5$.

C immediately above D.

One floor between B and D:

$|B-D|=2$

A above B and $A \leq 4$.

Test consecutive $\setminus((D,C))$:

D=1,C=2

Then B=3.

A must be above 3 but not 4.

Floor 5 occupied E.

Impossible.

D=2,C=3

Then B=4.

A above 4 means floor 5, occupied E.

Impossible.

D=3,C=4

Then B=1.

A above B, not 4.

Remaining floor 2 is available.

Thus:

$\setminus [A=2]$

Final:

| Floor | Person |
|-------|--------|
| 5 | E |
| 4 | C |
| 3 | D |
| 2 | A |
| 1 | B |

Q16

$\setminus [B-A-D-C-E]$

bottom to top.

Q17

$\setminus [B]$

Q18

C is immediately above D.

$\setminus [D]$

Q19

Above D (floor 3) but below E (floor 5):

floor4 = C.

$\boxed{C}$

Q20

B floor1, C floor4.

Floors 2 and 3 lie between.

$\boxed{2}$

' CHAPTER 65 COMPLETE

We have now covered:

- general puzzle-solving framework '
- floors/boxes/order '
- days/months/time slots '
- one-to-one matching '
- multi-variable grid puzzles '
- grouping/team formation '
- distribution/capacity '
- same/different-group logic '
- two-team parity shortcut '
- scheduling/precedence '
- exact-gap chains '
- conditional implications '
- contraposition '
- if/only-if/iff '
- exactly/at-least/at-most '
- XOR/not-both/together '
- implication closure '
- case splitting '
- contradiction pruning '
- minimum/maximum possibility '
- must/could/cannot-be-true '
- data-sufficiency preview '
- speed strategy '
- L0–L3 practice '

Progress

$\boxed{65/76}$ { numbered chapters completed }

UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '

- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating '
- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles '
- Chapter 66 — Syllogisms +
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

CHAPTER 66 — SYLLOGISMS

Next we will cover:

- categorical statements
- All / No / Some / Some-not
- Venn-diagram method
- distribution basics
- conversion rules
- definite vs possible conclusions
- existential traps
- “only” statements
- complementary pairs
- possibility cases
- 2-statement and 3-statement syllogisms
- placement-style coded syllogisms
- GATE logical-inference method
- L0–L3 practice.

USER

next

ASSISTANT

CHAPTER 66 — SYLLOGISMS

PART 1 — ALL / NO / SOME / SOME-NOT, VENN METHOD, CONVERSION RULES, DEFINITE CONCLUSIONS & EXISTENTIAL TRAPS

Syllogism questions give a few statements and ask:

Which conclusion must follow?

The most important rule is:

Do not ask what is probably true. Ask what is guaranteed true in every valid interpretation of the statements.

1. FOUR BASIC STATEMENT TYPES

Almost all basic syllogisms use these forms.

Type A — ALL

All A are B.

$\boxed{A \subseteq B}$

Example:

All cats are animals.

Every cat is an animal.

Type E — NO

No A is B.

$\boxed{A \cap B = \varnothing}$

Example:

No square is a circle.

Type I — SOME

Some A are B.

$\boxed{A \cap B \neq \varnothing}$

At least one object belongs to both.

Type O — SOME NOT

Some A are not B.

$\boxed{A-B \neq \varnothing}$

At least one A lies outside B.

2. VENN-DIAGRAM METHOD

This is the safest method.

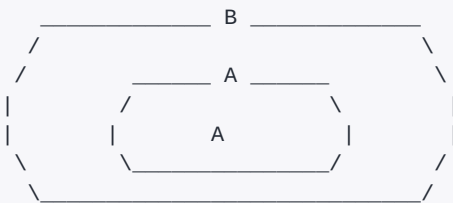
3. ALL A ARE B

Statement:

All A are B.

Diagram:

TEXT



Meaning:

$\boxed{A \subseteq B}$

But important:

It does not mean all B are A.

4. REVERSE-CONCLUSION TRAP

Given:

All doctors are educated.

Can we conclude:

All educated people are doctors?

No.

The educated set can contain many non-doctors.

$\boxed{\text{Converse does not follow}}$

5. NO A IS B

Diagram:

TEXT



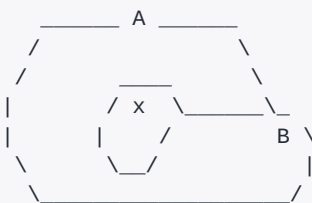
No overlap.

$\boxed{\text{A} \cap \text{B} = \text{nothing}}$

6. SOME A ARE B

Diagram:

TEXT



The x lies in the overlap.

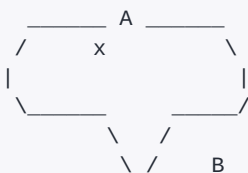
Therefore:

$\boxed{\text{At least one A is B}}$

7. SOME A ARE NOT B

Place at least one A outside B:

TEXT



Thus:

$\boxed{\exists x: x \in A, x \notin B}$

8. CONVERSION RULES — VERY IMPORTANT

Some statements can be safely reversed.

9. “SOME A ARE B” CONVERTS

If:

Some A are B.

Then:

Some B are A.

must also be true.

Because overlap is symmetric.

$\boxed{\text{Some A are B} \iff \text{Some B are A}}$

10. “NO A IS B” CONVERTS

If:

No A is B.

Then:

No B is A.

Because disjointness is symmetric.

$\boxed{\text{No A is B} \iff \text{No B is A}}$

11. “ALL A ARE B” DOES NOT CONVERT

Given:

All A are B.

Cannot conclude:

All B are A.

But a weaker reverse statement may follow **only if existence of A is known**, which we will discuss shortly.

12. “SOME A ARE NOT B” DOES NOT SIMPLY CONVERT

From:

Some A are not B.

Can we conclude:

Some B are not A?

No.

Example:

$A = \{1,2\}$ $B = \{2\}$

Some A are not B: 1.

But every B is A.

So:

$\boxed{\text{No reverse conclusion}}$

13. MUST-MEMORIZE CONVERSION TABLE

| Statement | Safe conversion? |
|------------------|------------------|
| All A are B | 'L No |
| No A is B | ' No B is A |
| Some A are B | ' Some B are A |
| Some A are not B | 'L No |

14. CHAINING “ALL”

Statements:

All A are B. All B are C.

Then:

$A \subseteq B \subseteq C$

Therefore:

$\boxed{\text{All A are C}}$

This is one of the most basic valid chains.

15. EXAMPLE

All engineers are graduates.

All graduates are educated.

Therefore:

$\boxed{\text{All engineers are educated}}$

16. ALL + NO

Statements:

All A are B. No B is C.

Since every A lies in B, and B has no overlap with C:

$\boxed{\text{No A is C}}$

17. EXAMPLE

All cats are animals.

No animal is a machine.

Therefore:

$\boxed{\text{No cat is a machine}}$

18. SOME + ALL

Statements:

Some A are B. All B are C.

The “some A that are B” are also C.

Therefore:

$\boxed{\text{Some A are C}}$

19. EXAMPLE

Some students are programmers.

All programmers are logical thinkers.

Therefore:

$\boxed{\text{Some students are logical thinkers}}$

20. SOME + NO

Statements:

Some A are B. No B is C.

The known "A" objects cannot be C.

Therefore:

$\boxed{\text{Some A are not C}}$

21. EXAMPLE

Some birds are parrots.

No parrot is a mammal.

Therefore:

$\boxed{\text{Some birds are not mammals}}$

22. ALL + SOME-NOT

Statements:

All A are B. Some A are not C.

The known "some A" are also B.

Therefore:

$\boxed{\text{Some B are not C}}$

23. EXISTENCE IS CRITICAL

Now the biggest syllogism trap.

Suppose:

All A are B.

Does this guarantee that any A exists?

In modern formal logic:

$\boxed{\text{No}}$

The statement can be true even if set A is empty.

This matters greatly for existential conclusions.

24. EXISTENTIAL FALLACY

Given only:

All cats are animals.

Can we conclude:

Some cats are animals?

Not under strict modern logic, because existence of cats was not asserted by the universal statement alone.

So:

$\boxed{\text{No}}$

25. PLACEMENT-EXAM CONVENTION WARNING

Some older aptitude books use the traditional convention that universal statements imply existence.

But safer logical reasoning—and GATE-style reasoning—is:

Do not assume existence from “All” or “No” alone unless the question's convention explicitly does so.

We'll use this stricter rule.

26. WHICH STATEMENTS ASSERT EXISTENCE?

“Some A are B”

Yes.

$A \text{ and } B \text{ exist}$

“Some A are not B”

Yes.

At least one A exists.

“All A are B”

Does not necessarily assert A exists.

“No A is B”

Does not necessarily assert either exists.

27. EXISTENCE TABLE

| Statement | Guarantees A exists? |
|------------------|----------------------|
| All A are B | 'L |
| No A is B | 'L |
| Some A are B | ' |
| Some A are not B | ' |

28. UNIVERSAL + EXISTENCE

Statements:

All A are B. Some A exist.

Then we can conclude:

$\boxed{\text{Some B exist}}$

and specifically:

$\boxed{\text{Some A are B}}$

because the existing A must lie inside B.

29. INDIRECT EXISTENCE

Statements:

Some C are A. All A are B.

From "Some C are A," A exists.

Therefore those A are B.

So:

$\boxed{\text{Some B exist}}$

and:

$\boxed{\text{Some C are B}}$

30. NO + SOME

Statements:

No A is B. Some C are A.

The known C element cannot be B.

Therefore:

$\boxed{\text{Some C are not B}}$

31. TWO UNIVERSAL STATEMENTS MAY GIVE NO EXISTENCE

Statements:

All A are B. All B are C.

Valid conclusion:

$\boxed{\text{All A are C}}$

But not necessarily:

$\boxed{\text{Some A are C}}$

unless A existence is established.

32. “ONLY” STATEMENTS

This causes many errors.

33. “ONLY A ARE B”

Statement:

Only doctors are surgeons.

Means:

If someone is a surgeon, they must be a doctor.

So:

$\boxed{\text{Surgeon} \subseteq \text{Doctor}}$

Not:

$\boxed{\text{Doctor} \subseteq \text{Surgeon}}$

34. MEMORY TRICK FOR ONLY

Only A are B = All B are A.

Example:

Only mammals are whales.

Means:

$\boxed{\text{All whales are mammals}}$

35. “A ARE ONLY B”

Statement:

Cats are only animals.

This means:

$\boxed{\text{All cats are animals}}$

So word order matters.

36. “ONLY A ARE B” VS “A ARE ONLY B”

Only A are B:

$B \subseteq A$

A are only B:

$A \subseteq B$

Very common verbal trap.

37. “NONE BUT”

None but doctors are allowed.

Means:

Only doctors are allowed.

Therefore:

$\text{Allowed} \subseteq \text{Doctor}$

38. “NOTHING BUT”

The members are nothing but students.

Means:

$\text{Members} \subseteq \text{Students}$

39. DEFINITE CONCLUSION VS POSSIBILITY

A conclusion “follows” only if it must be true.

A conclusion “is possible” needs only one valid Venn arrangement.

These are different question types.

40. EXAMPLE — DEFINITE

Statements:

All A are B. No B is C.

Conclusion:

No A is C.

Must be true.

$\boxed{\text{Follows}}$

41. EXAMPLE — NOT DEFINITE

Statements:

All A are B. Some C are B.

Conclusion:

Some A are C.

Does it follow?

No.

The C") Objects may lie outside A.

So:

$\boxed{\text{Does not follow}}$

42. POSSIBLE BUT NOT NECESSARY

Same statements:

All A are B. Some C are B.

Could some A be C?

Yes, possibly.

But it is not guaranteed.

Thus:

$\boxed{\text{Possible, but not definite}}$

43. VENN “MINIMUM COMMITMENT” RULE

When testing whether a conclusion must follow:

Place sets with only the overlap forced by the statements.

Do not add extra overlap just because it is possible.

This prevents false conclusions.

44. EXAMPLE

Statements:

Some A are B. Some B are C.

Can we conclude:

Some A are C?

No.

The two “some” groups may be different members of B.

Visual:

TEXT

A") B c o n t a i n s x
B") C c o n t a i n s y
x and y may be different

Thus:

$\boxed{\text{No definite A-C overlap}}$

45. THE “SOME + SOME” TRAP

This is one of the most common.

Given:

$\boxed{\text{Some A are B}}$

and:

$\boxed{\text{Some B are C}}$

You cannot conclude:

$\boxed{\text{Some A are C}}$

unless there is extra information.

46. ALL + SOME WORKS DIFFERENTLY

If:

$\boxed{\text{Some A are B}}$

and:

$\boxed{\text{All B are C}}$

then the known overlap must be inside C.

Therefore:

$\boxed{\text{Some } A \text{ are } C}$

So “Some + All” can propagate.

47. NO + SOME PROPAGATE EXCLUSION

If:

$\boxed{\text{Some } A \text{ are } B}$

and:

$\boxed{\text{No } B \text{ is } C}$

then:

$\boxed{\text{Some } A \text{ are not } C}$

48. ALL + NO CHAIN

If:

$\boxed{\text{All } A \text{ are } B}$

$\boxed{\text{No } C \text{ is } B}$

Since No C is B converts to No B is C:

$\boxed{\text{No } A \text{ is } C}$

49. “SOME NOT” + ALL

Statements:

Some A are not B. All C are B.

Can we conclude:

Some A are not C?

Yes.

Why?

If some A is not B, and every C is B, that particular A cannot be C.

Thus:

$\boxed{\text{Some } A \text{ are not } C}$

This is an important inference.

50. GENERAL EXCLUSION PRINCIPLE

If:

$$[C \subseteq B]$$

and an object is known to be outside B,

then that object must also be outside C.

So:

$$[\neg B \rightarrow \neg C]$$

when $(C \subseteq B)$.

51. EXAMPLE

Some students are not graduates.

All engineers are graduates.

Therefore those non-graduate students cannot be engineers.

$$[\boxed{\text{Some students are not engineers}}]$$

52. ALL A ARE B + NO A IS C

Can we conclude:

No B is C?

No.

Only the A-part of B is excluded from C.

Other B's may overlap with C.

So:

$$[\boxed{\text{Does not follow}}]$$

53. NO A IS B + ALL C ARE A

Then:

$$[C \subseteq A]$$

and:

$$[A \cap B = \varnothing]$$

Therefore:

$$[\boxed{\text{No } C \text{ is } B}]$$

54. NO A IS B + SOME C ARE A

Then known C") Objects cannot B:

$\boxed{\text{Some } C \text{ are not } B}$

55. ALL A ARE B + SOME C ARE NOT B

Can we conclude:

Some C are not A?

Yes.

Because all A lie in B.

Anything outside B cannot be A.

Therefore:

$\boxed{\text{Some } C \text{ are not } A}$

56. CONTRAPOSITIVE SET IDEA

From:

$A \subseteq B$

we can say:

$x \notin B \rightarrow x \notin A$

But we cannot say:

$x \notin A \rightarrow x \notin B$

This is the set-theoretic version of contraposition.

57. BASIC SYLLOGISM DECISION PROCESS

For each conclusion:

1. translate statements to set relations
2. note existence from "some"
3. chain subsets
4. propagate exclusions
5. check whether conclusion holds in every valid diagram
6. if one counterexample exists, conclusion does not follow

58. COUNTEREXAMPLE METHOD

To disprove a conclusion, you only need one valid arrangement where the statements are true but conclusion is false.

This is often faster than proving directly.

59. EXAMPLE

Statements:

- All A are B.
- Some C are B.

Claim:

- Some A are C.

Counterexample:

Let A occupy one part of B and C occupy another disjoint part of B.

Statements remain true.

Claim false.

Therefore:

$\boxed{\text{Does not follow}}$

60. COMPLEMENTARY CONCLUSIONS — PREVIEW

Sometimes two conclusions are:

1. Some A are B.
2. No A is B.

Exactly one of these may be true when existence relationships force a choice.

But do not use shortcut blindly.

We'll cover complementary-pair rules in Part 2.

61. 2-STATEMENTEXAMPLE 1

Statements:

1. All roses are flowers.
2. All flowers are plants.

Conclusions:

I. All roses are plants. II. Some plants are roses.

I:

$\boxed{\text{Rose} \subseteq \text{Flower} \subseteq \text{Plant}}$

So I follows.

II requires rose existence.

Not guaranteed.

Answer:

$\boxed{\text{Only I follows}}$

62. 2-STATEMENTEXAMPLE 2

Statements:

1. Some books are novels.
2. All novels are interesting.

Conclusions:

I. Some books are interesting. II. Some interesting things are books.

I follows.

II is conversion of I.

Thus both follow.

$\boxed{\text{Both}}$

63. 2-STATEMENTEXAMPLE 3

Statements:

1. No doctors are artists.
2. Some teachers are doctors.

Conclusions:

I. Some teachers are not artists. II. No artists are doctors.

I follows.

II follows by conversion of "No doctors are artists."

$\boxed{\text{Both}}$

64. 2-STATEMENTEXAMPLE 4

Statements:

1. Some A are B.
2. Some B are C.

Conclusions:

I. Some A are C. II. Some C are A.

Neither must follow.

$\boxed{\text{Neither}}$

65. 3-STATEMENTCHAIN

Statements:

1. Some A are B.
2. All B are C.

3. No C is D.

Known some A" B .

Since B"† C :

$\boxed{\text{Some A are C}}$

Since no C is D:

$\boxed{\text{Some A are not D}}$

66. LONGER CHAIN

All A are B.

All B are C.

No C is D.

Some E are A.

Then:

$\boxed{\text{E are nothing}}$

A"† B"† C .

So the known E" objects are C.

No C is D.

Therefore:

$\boxed{\text{Some E are not D}}$

67. "NO" CHAIN DOES NOT ALWAYS TRANSIT

Given:

No A is B. No B is C.

Can we conclude:

No A is C?

No.

A and C may overlap.

Example:

A and C may be identical sets, both disjoint from B.

So:

$\boxed{\text{No + No does not create No}}$

68. IMPORTANT CHAIN TABLE

| Statements | Valid conclusion |
|----------------------|------------------|
| All A! B, All B! C | All A! C |
| All A! B, No B - C | No A-C |
| Some A - B, All B! C | Some A-C |
| Some A-B, No B-C | Some A not C |
| Some A-B, Some B-C | No definite A-C |
| No A-B, No B-C | No definite A-C |

69. "ALL" WITH OVERLAPPING MIDDLE SET

Statements:

All A are B.

Some B are C.

You cannot conclude any A-C overlap.

The "some B" may lie outside A.

70. "SOME A ARE B" DOES NOT MEAN SOME A ARE NOT B

It only guarantees overlap.

A could entirely lie inside B.

So from:

Some A are B

do not conclude:

Some A are not B.

71. "SOME A ARE NOT B" DOES NOT MEAN SOME A ARE B

A might have zero overlap with B.

So do not infer both sides exist.

72. "NO A IS B" DOES NOT MEAN A EXISTS

Under strict logic, both may even be empty.

This is another existential trap.

73. UNIVERSAL VACUOUS TRUTH

If A has no members, then:

All A are B

is automatically true.

This is why universal statements alone do not guarantee “some.”

You do not need the term “vacuous truth” for exams, but the logic is important.

74. QUICK VENN SHORTCUT

Instead of drawing full circles every time, use:

- $A \subseteq B$ for All A are B
- $A \times B$ for No A is B
- $A \cap B \neq \emptyset$ for Some A are B
- $A \not\subseteq B$ for Some A are not B

Then reason symbolically.

75. COMMON TRAPS

& pTRAP 1 — Reversing All

All A are B does not mean all B are A.

& pTRAP 2 — Some + Some chain

Does not guarantee endpoint overlap.

& pTRAP 3 — Universal implies existence

Not under strict logic.

& pTRAP 4 — “Only” reversed incorrectly

Only A are B means $B \subseteq A$.

& pTRAP 5 — No + No transitivity

Invalid.

& pTRAP 6 — Possible treated as definite

Different question.

& pTRAP 7 — Some-not conversion

Invalid.

PRACTICE — LEVEL 0

Q1

All A are B. Can all B be concluded to be A?

Q2

Some A are B. What safe conversion follows?

Q3

No A is B. What safe conversion follows?

Q4

Some A are not B. Does "Some B are not A" necessarily follow?

Q5

All A are B and all B are C. What follows?

LEVEL 1

Q6

All A are B. No B is C. What relation must hold between A and C?

Q7

Some A are B. All B are C. What follows about A and C?

Q8

Some A are B. No B is C. What follows?

Q9

Some A are B. Some B are C. Must some A be C?

Q10

Only doctors are surgeons. Rewrite in "All" form.

LEVEL 2

Q11

All engineers are graduates. Some students are engineers. What must follow?

Q12

Some students are not graduates. All engineers are graduates. What must follow about students and engineers?

Q13

No A is B. All C are A. What must follow about C and B?

Q14

All A are B. Some C are not B. What must follow about C and A?

Q15

No A is B. No B is C. Must no A be C?

LEVEL 3 — EXAM PRESSURE

Statements:

1. Some P are Q.
2. All Q are R.
3. No R is S.
4. All T are S.

Q16

Does "Some P are R" follow?

Q17

Does "Some P are not S" follow?

Q18

Does "Some P are not T" follow?

Q19

Does "No Q is T" follow?

Q20

Does "Some T are not Q" follow?

ANSWERS

Q1

No.

$\boxed{\text{No}}$

Q2

$\boxed{\text{Some B are A}}$

Q3

$\boxed{\text{No B is A}}$

Q4

No.

$\boxed{\text{No}}$

Q5

$\boxed{\text{All A are C}}$

Q6

A and B disjoint C.

Therefore:

$\boxed{\text{No A is C}}$

Q7

The known A are inside C.

$\boxed{\text{Some A are C}}$

Q8

Known A cannot C.

$\boxed{\text{Some A are not C}}$

Q9

No.

The two "some" groups may differ.

$\boxed{\text{No}}$

Q10

Only doctors are surgeons:

$\boxed{\text{All surgeons are doctors}}$

Q11

Some students are engineers.

All engineers are graduates.

Therefore:

$\boxed{\text{Some students are graduates}}$

Q12

Some students lie outside Graduate.

All engineers lie inside Graduate.

Therefore those students outside Graduate cannot be engineers.

$\boxed{\text{Some students are not engineers}}$

Q13

$C \cap A = \emptyset$ and A disjoint B.

$\boxed{\text{No C is B}}$

Q14

$A \cap B = \emptyset$.

Some C are outside B.

Anything outside B cannot be A.

$\boxed{\text{Some C are not A}}$

Q15

No.

A and C may overlap.

$\boxed{\text{No}}$

Q16–Q20

Statements:

$\text{Some } P \text{ are } Q$

$Q \subseteq R$

$R \cap S = \emptyset$

$T \subseteq S$

Q16

The known P objects are R.

Therefore:

$\boxed{\text{Yes, Some } P \text{ are } R}$

Q17

Those known P objects are R.

No R is S.

Therefore:

$\boxed{\text{Yes, Some P are not S}}$

Q18

All T are S.

The known P objects from Q are not S.

Anything not S cannot be T because $T \rightarrow S$.

Therefore:

$\boxed{\text{Yes, Some P are not T}}$

Q19

$Q \rightarrow R$.

R disjoint S.

$T \rightarrow S$.

Thus no Q can be T.

$\boxed{\text{Yes, No Q is T}}$

Q20

Does some T exist?

No statement establishes T's existence.

Therefore:

$\boxed{\text{No, it does not follow}}$

CHAPTER 66 STATUS

Covered so far:

- four categorical forms '
- All / No / Some / Some-not '
- Venn interpretation '
- conversion rules '
- subset chaining '
- exclusion propagation '
- some + all '

- some + no '
- existential import '
- universal-existence trap '
- “only” statements '
- definite vs possible conclusions '
- minimum-commitment method '
- counterexample method '
- 2-statement syllogisms '
- 3-statement chains '
- common invalid chains '
- L0–L3 practice '

But **Chapter 66 is not complete yet.**

Still remaining:

- possibility conclusions
- “some-not” possibility
- complementary conclusion pairs
- either-I-or-II patterns
- advanced “only/few/most” wording
- coded syllogisms
- 3–4 statement mixed sets
- conclusion-pair shortcuts
- placement speed method
- final Chapter 66 audit

Progress

\[\boxed{65/76}\text{ numbered chapters completed} \]

Current: 0 = Chapter 66 — Syllogisms

NEXT

0 = Chapter 66 Part 2 — Possibility Cases, Complementary Conclusions, Either-Or Logic, Coded Syllogisms & Final Speed Toolkit

USER

next

ASSISTANT

CHAPTER 66 — SYLLOGISMS

PART 2 — POSSIBILITY CASES, COMPLEMENTARY CONCLUSIONS, EITHER–OR

LOGIC, CODED SYLLOGISMS & FINAL SPEED TOOLKIT

Status: 0% PROGRESS ! ' ' COMPLETE GATE: High Placements: Very High Priority: 0% Year 1

This part covers the trickiest syllogism questions:

- “can be true” / possibility conclusions
- “either conclusion I or II follows”
- complementary conclusion pairs
- coded statements
- advanced chains
- conclusion elimination without drawing full Venn diagrams

The main distinction is:

Definite conclusion = must hold in every valid diagram. Possibility conclusion = needs only one valid diagram where it can hold without violating the statements.

1. DEFINITE VS POSSIBLE

Statements:

All A are B. Some C are B.

Question 1:

Some A are C.

This is **not definite**, because C may overlap B outside A.

But can A and C overlap?

Yes.

So:

$\boxed{\text{Some A being C is possible}}$

2. POSSIBILITY TEST

To test:

“Some A can be B”

ask:

Can I create at least one A–B overlap without violating any statement?

If yes:

$\boxed{\text{Possible}}$

If some statement forbids overlap:

$\boxed{\text{Impossible}}$

3. SIMPLE POSSIBILITY

Statements:

All A are B. All C are B.

Can some A be C?

Yes.

Both lie inside B and may overlap.

$\boxed{\text{Possible}}$

But the overlap is not guaranteed.

4. IMPOSSIBLE POSSIBILITY

Statements:

All A are B. No B is C.

Can some A be C?

No.

Since:

$A \subseteq B$

and:

$B \cap C = \varnothing$

there can be no A–C overlap.

$\boxed{\text{Impossible}}$

5. “SOME A ARE NOT B” POSSIBILITY

Suppose:

Some A are B.

Can some A also be outside B?

Yes, unless another statement says:

$\text{All } A \text{ are } B$

The known overlap does not force the whole A-set inside B.

6. EXAMPLE

Statement:

Some students are athletes.

Possibility:

Some students are not athletes.

Possible?

Yes.

The statement only says at least one student is an athlete.

Other students may not be.

$\boxed{\text{Possible}}$

7. WHEN SOME-NOT IS IMPOSSIBLE

Statement:

All students are athletes.

Can:

Some students are not athletes

be true?

No.

$\{ \text{Student} \subseteq \text{Athlete} \}$

Therefore:

$\boxed{\text{Impossible}}$

8. POSSIBILITY WITH “NO”

Statement:

No A is B.

Can some A be C?

Yes, unless C is also forced inside B or otherwise forbidden.

“No A is B” says nothing about A–C overlap.

9. POSSIBILITY WITH MULTIPLE SETS

Statements:

- All A are B.
- No B is C.
- All D are C.

Can some A be D?

No.

Because:

$\{A\} \subseteq \{B\}$

$\{D\} \subseteq \{C\}$

and B/C are disjoint.

So A and D must be disjoint.

10. FAST POSSIBILITY RULE

For “Some X can be Y”:

Impossible if:

- No X is Y
- X is inside a set disjoint from Y
- Y is inside a set disjoint from X

Otherwise, overlap may often be possible.

But verify all chains first.

11. COMPLEMENTARY CONCLUSIONS

Common complementary pairs include:

Pair 1

$\{ \text{Some } A \text{ are } B \}$

and

$\{ \text{No } A \text{ is } B \}$

If A and B are known to exist in the relevant logical setup, these represent overlap vs no overlap.

But be careful with existence conventions.

12. PAIR 2

$\{ \text{All } A \text{ are } B \}$

and

$\neg \text{Some A are B}$

These are logical negations **provided A exists** in the relevant universe.

If all A are B is false because some A lies outside B, then Some A are not B is true.

13. PAIR 3

Some A are B

and

$\neg \text{No B is A}$

Since “No B is A” means No A is B, these are complementary overlap statements.

14. EITHER–OR QUESTIONS

Typical format:

Statements given.

Conclusions:

I. Some A are B. II. No A is B.

Options may include:

- only I
- only II
- either I or II
- neither
- both

“Either I or II follows” is used when:

1. neither conclusion individually must follow from current information, but
 2. exactly one of them must be true.
-

15. IMPORTANT EITHER–OR CONDITION

A safe either–or pair should:

- concern the same two classes
- be mutually exclusive
- collectively cover all valid possibilities
- not both be true together

Example:

Some A are B

vs

$\neg \text{No A is B}$

They cannot both hold.

Either there is A-B overlap or there is not.

16. EXAMPLE

Suppose statements give no information about relation A and B, but A and B exist.

Conclusions:

I. Some A are B. II. No A is B.

We cannot determine which one.

But exactly one must be true.

Thus:

$\boxed{\text{Either I or II}}$

17. DO NOT USE EITHER–OR BLINDLY

Conclusions:

I. All A are B. II. Some B are not A.

These are not necessarily complementary.

Both may be true.

Example:

A is a proper subset of B.

Then:

- All A are B
- Some B are not A

So they are not either–or.

18. ANOTHER NON-COMPLEMENTARY PAIR

I. Some A are B. II. Some A are not B.

Both can be true simultaneously.

So:

$\boxed{\text{Not an either–or pair}}$

19. COMPLEMENT OF “ALL A ARE B”

Under ordinary nonempty-domain reasoning for A:

Negation of:

$\neg \forall A \backslash A \text{ are } B$

is:

$\neg [\text{Some } A \text{ are } \neg B]$

So these form an exclusive logical pair if A's existence is meaningful.

20. COMPLEMENT OF “NO A IS B”

Negation of:

$\neg [\text{No } A \text{ is } B]$

is:

$\neg [\text{Some } A \text{ are } B]$

Again, overlap vs no overlap.

21. EITHER–OR EXAM SHORTCUT

Before marking “Either I or II,” check:

Condition 1

Same subject and predicate sets.

Condition 2

One is positive, one negative.

Condition 3

They cannot both be true.

Condition 4

They cannot both be false.

Only then use either–or.

22. EXAMPLE — VALID

I. Some A are B. II. No A is B.

They cannot both be true.

They cannot both be false.

Hence:

$\neg [\boxed{\text{Complementary}}]$

23. EXAMPLE — INVALID

I. Some A are B. II. Some A are not B.

Can both be true?

Yes.

So no either-or.

24. ADVANCED POSSIBILITY

Statements:

1. All A are B.
2. Some B are C.
3. No C is D.

Can some A be C?

Yes.

The known B-C overlap may be placed partly inside A.

Nothing forbids it.

$\boxed{\text{Possible}}$

25. SAME SET — NOT DEFINITE

Does:

Some A are C

definitely follow?

No.

The B-C overlap may lie entirely outside A.

Therefore:

$\boxed{\text{Possible but not definite}}$

26. POSSIBILITY WITH SUBSET + EXCLUSION

Statements:

1. All A are B.
2. All C are D.
3. No B is D.

Can A overlap C?

No.

Because A lies in B and C lies in D, and B/D are disjoint.

$\boxed{\text{Impossible}}$

27. “SOME A CAN BE NOT B”

Test whether A is forced entirely inside B.

If:

$\forall A \forall B (A \rightarrow B)$

then impossible.

If not, and no other chain forces $A \rightarrow B$ may be possible.

28. “ALL A CAN BE B”

Possibility of all A being B means:

Can the entire A-set be placed inside B without violating any statement?

This may be possible even if only Some A are B is given.

29. EXAMPLE

Statement:

Some A are B.

Can all A be B?

Yes.

The known statement does not require any A outside B.

$\boxed{\text{Possible}}$

30. WHEN “ALL A CAN BE B” IS IMPOSSIBLE

If:

Some A are not B.

Then at least one A is definitely outside B.

So:

$\boxed{\text{All A being B is impossible}}$

31. “NO A CAN BE B”

If:

Some A are B

then total separation is impossible.

If nothing forces overlap, complete separation may be possible.

32. ADVANCED 3-STATEMENT SET

Statements:

1. Some P are Q.
2. All Q are R.
3. All R are S.

Conclusions:

I. Some P are S. II. Some S are P.

From some P-Q:

Q"† R"† S .

Therefore:

$\boxed{\text{Some P are S}}$

Conversion:

$\boxed{\text{Some S are P}}$

So:

$\boxed{\text{Both I and II follow}}$

33. ADVANCED EXCLUSION SET

Statements:

1. Some P are Q.
2. All Q are R.
3. No R is S.
4. All T are S.

Then known P-Q members are R and therefore not S.

Since T"† S, those members cannot be T.

Thus:

$\boxed{\text{Some P are not T}}$

34. POSSIBILITY FROM SAME SET

Using above:

Can some unrelated P be T?

Yes, possibly, because only the known P-Q members are forced outside T.

Unless all P are Q, other P members could be T.

So:

$\boxed{\text{Possible}}$

This distinction is subtle.

35. “SOME P ARE NOT T” DOES NOT MEAN “NO P IS T”

Exactly.

Some-not only establishes at least one exception.

Other P may still lie in T.

36. CODED SYLLOGISMS

Sometimes relations are represented with symbols.

Example:

- $A @ B$ means All A are B.
- $A \# B$ means No A is B.
- $A \$ B$ means Some A are B.
- $A \% B$ means Some A are not B.

Translate first.

Then solve normally.

37. CODED EXAMPLE

Given:

$[A @ B, \text{quad} B @ C]$

Since @ means All:

$[A \subseteq B \subseteq C]$

Therefore:

$[\boxed{A @ C}]$

if the same code notation is used for “All.”

38. CODED EXCLUSION

If:

$[A @ B]$

and:

$[B \# C]$

then:

$[A \# C]$

because all A are B and no B is C.

39. CODED SOME

If:

$\forall [A \rightarrow B]$

and:

$\forall [B \rightarrow C]$

then:

$\forall [A \rightarrow C]$

because some A are B and all B are C.

40. CODED SOME-NOT

If:

$\forall [A \rightarrow B]$

and:

$\forall [B \rightarrow \neg C]$

then:

$\forall [A \rightarrow \neg C]$

because some A-B overlap lies outside C.

41. ADVANCED “ONLY”

Statement:

Only A are B.

Translate:

$\forall [\boxed{B \subseteq A}]$

If:

Only B are C.

Then:

$\forall [C \subseteq B]$

So:

$\forall [C \subseteq B \subseteq A]$

Therefore:

$\forall [\boxed{\text{All C are A}}]$

42. ONLY + NO

Only A are B. No A is C.

Since:

$B \subseteq A$

and A disjoint C:

$\boxed{\text{No B is C}}$

43. “FEW” — EXAM WORDING CAUTION

In some aptitude books:

Few A are B

is interpreted as:

- some A are B
- some A are not B

But “few” is linguistically context-dependent.

Unless the exam defines its convention, be cautious.

For standard placement syllogisms, if they use “few,” the intended conventional reading is often:

$\boxed{\text{Some A are B and some A are not B}}$

44. “A FEW”

Similarly, “a few A are B” usually guarantees:

$\boxed{\text{Some A are B}}$

But may not necessarily guarantee some A are not B unless the test convention explicitly treats it that way.

45. “MOST”

“Most A are B” means more than half of A are B in ordinary quantitative logic.

It certainly implies:

$\boxed{\text{Some A are B}}$

provided A exists.

But standard categorical syllogisms usually avoid quantitative “most.”

46. “NOT ALL”

Not all A are B

means:

$\boxed{\text{Some } A \text{ are not } B}$

assuming A exists in the intended context.

This is a powerful translation.

47. "NOT NONE"

It is not true that no A is B

means:

$\boxed{\text{Some } A \text{ are } B}$

48. "NO A EXCEPT B"

Phrases involving "except" can be ambiguous in natural language.

For exam questions, translate only using the exact intended logical structure provided.

Do not guess from conversational English.

49. 4-STATEMENTCHAIN

Statements:

1. Some A are B.
2. All B are C.
3. All C are D.
4. No D is E.

Then the known A-B members are:

$B \rightarrow C \rightarrow D$

and D is outside E.

Therefore:

$\boxed{\text{Some } A \text{ are not } E}$

50. CAN SOME A BE E?

Using the same statements:

Some A are B !' those particular A are not E.

But there may be other A not in B.

So can some A be E?

$\boxed{\text{Yes, possible}}$

unless another statement says all A are B.

51. ADD “ALL A ARE B”

Now if:

$\boxed{\text{All A are B}}$

and B! C! D No D-E:

Every A is outside E.

Therefore:

$\boxed{\text{No A is E}}$

So possibility disappears.

52. “SOME” DOES NOT CONTROL THE WHOLE SET

This is a major theme:

Some A are B

tells us about at least one A.

It says nothing about the remaining A unless more clues exist.

53. UNIVERSAL CHAIN CONTROLS THE WHOLE SET

By contrast:

$\boxed{\text{All A are B, quad All B are C}}$

controls every A:

$\boxed{\text{All A are C}}$

54. DEFINITE-CONCLUSION SHORTCUT

When a conclusion is universal:

All A are C

Look for an unbroken subset chain:

$\boxed{A \rightarrow \dots \rightarrow C}$

If no such universal chain exists, be suspicious.

55. DEFINITE “NO” SHORTCUT

To prove:

No A is C

look for:

1. A included in B
2. B disjoint C

or equivalent chain.

Example:

$\forall A \subseteq B, \quad B \cap C = \varnothing$

56. DEFINITE “SOME” SHORTCUT

To prove:

Some A are C

you need an existential starting point, such as:

$\exists A \text{ are } B$

plus a universal path:

$B \subseteq C$

Then the same known object reaches C.

57. DEFINITE “SOME NOT” SHORTCUT

To prove:

Some A are not C

look for a known existential object in A plus an exclusion from C.

Example:

$\exists A \text{ are } B$

$B \not\subseteq C$

58. UNIVERSAL–UNIVERSAL SPEED METHOD

For:

$\forall A \rightarrow B$

$\forall B \rightarrow C$

write:

$\forall [A \rightarrow B \rightarrow C]$

Then any universal forward conclusion is immediate.

59. NEGATIVE EDGE METHOD

Represent:

$\forall [No \ B \ is \ C]$

as:

$\forall [B \rightarrow C]$

If $A \rightarrow B$:

$\forall [A \rightarrow C]$

If $D \rightarrow A$:

$\forall [D \rightarrow C]$

Exclusion propagates backward through subset chains.

60. EXISTENTIAL TOKEN METHOD

For:

Some A are B

imagine a token $\langle x \rangle$:

$\forall [x \in A, \text{quad } x \in B]$

Then push $\langle x \rangle$ through universal rules.

Example:

$B \rightarrow C$.

Then:

$\forall [x \in C]$

If $C \rightarrow D$:

$\forall [x \notin D]$

This is one of the fastest reliable methods.

61. EXAMPLE USING TOKEN METHOD

Statements:

- Some students are coders.
 - All coders are logical.
 - No logical person is careless.
-

Take token $\setminus(x)$:

$\setminus[x=\text{student}+\text{coder} \setminus]$

Coder !' logical.

Logical excludes careless.

Therefore:

$\setminus[\boxed{\text{Some} \setminus \text{students} \setminus \text{are} \setminus \text{not} \setminus \text{careless}} \setminus]$

62. CONCLUSION-PAIR STRATEGY

When two conclusions are given:

First

Test each independently.

Then

If neither independently follows, check if they form a true complementary pair.

Only then consider:

$\setminus[\boxed{\text{Either I or II}} \setminus]$

63. NEVER JUMP DIRECTLY TO EITHER-OR

This is a common placement shortcut misused by students.

Always ask:

- can both be true?
- can both be false?

If either answer is yes, it is not a valid either-or pair.

64. EXAMPLE

I. Some A are B. II. Some A are not B.

Can both be true?

Yes.

Therefore:

$\setminus[\boxed{\text{Not either-or}} \setminus]$

65. EXAMPLE

I. Some A are B. II. No A is B.

Can both be true?

No.

Can both be false?

No, provided the domain relation is defined with relevant existence.

Therefore:

$\boxed{\text{Complementary}}$

66. GATE-STYLE INFERENCE RULE

GATE-style reasoning rewards:

- exact logical necessity
- no hidden assumptions
- no existence assumption from universal claims
- counterexamples for disproving

So if unsure:

Try to construct one model where the statements hold but conclusion fails.

If you can:

$\boxed{\text{Conclusion does not follow}}$

67. PLACEMENT SPEED SEARCH ORDER

For each conclusion:

1. Is it a direct conversion?
 2. Is there an 'All' chain?
 3. Is there a Some token that can travel through All?
 4. Is there a No relation blocking it?
 5. Is existence missing?
 6. If still uncertain, draw Venn/counterexample.
-

68. 30-SECOND SYLLOGISM CHECKLIST

ALL

Forward only.

NO

Symmetric.

SOME

Symmetric.

SOME-NOT

Do not reverse.

EXISTENCE

Comes from some/some-not, not universal alone.

SOME+SOME

No endpoint conclusion by default.

NO+NO

No endpoint conclusion by default.

69. FINAL TRAPS

& ꞤTRAP 1 — Possible = must

No.

& ꞤTRAP 2 — Some A are B ! $\Rightarrow$ Some A are not B

No.

& ꞤTRAP 3 — Some A are not B ! $\Rightarrow$ No A are B

No.

& ꞤTRAP 4 — All A are B ! $\Rightarrow$ Some B are A

Not without existence.

& ꞤTRAP 5 — Universal chain creates existence

No.

& ꞤTRAP 6 — Every positive/negative conclusion pair is either–or

No.

& ꞤTRAP 7 — “Only A are B” read as A $\nrightarrow$ B

Wrong; it means B $\nrightarrow$ A .

& ꞤTRAP 8 — Ignoring possibility of extra members

“Some” never describes the whole set.

PRACTICE — LEVEL 0

Q1

All A are B. All C are B. Is “Some A can be C” possible?

Q2

All A are B. No B is C. Can some A be C?

Q3

Some A are B. Can some A be outside B?

Q4

Some A are not B. Can all A be B?

Q5

No A is B. Can some A be B?

LEVEL 1

Q6

All A are B. Some C are B. Does "Some A are C" definitely follow?

Q7

For Q6, is "Some A are C" possible?

Q8

Only doctors are surgeons. All surgeons are what?

Q9

All A are B. Some C are not B. What definite conclusion follows regarding C and A?

Q10

Some A are B and all B are C. Can some A be outside C?

LEVEL 2

Statements:

1. Some P are Q.
2. All Q are R.
3. No R is S.
4. All T are S.

Q11

Does Some P are R follow?

Q12

Does Some P are not S follow?

Q13

Does Some P are not T follow?

Q14

Can some P be T?

Q15

Does Some T are S definitely follow?

LEVEL 3 — EXAM PRESSURE

Statements:

1. All A are B.
2. Some B are C.
3. No C is D.
4. All E are D.

Conclusions:

I. Some A are C. II. Some A being C is possible. III. No C is E. IV. Some B are not D. V. Some E are not C.

Q16

Does I follow?

Q17

Is II valid?

Q18

Does III follow?

Q19

Does IV follow?

Q20

Does V follow?

ANSWERS

Q1

A and C are both inside B and may overlap.

$\boxed{\text{Yes}}$

Q2

A and B disjoint C.

$\boxed{\text{No}}$

Q3

Yes.

Some A are B does not force all A into B.

$\boxed{\text{Yes}}$

Q4

No.

Some A are definitely outside B.

$\boxed{\text{No}}$

Q5

No.

$\boxed{\text{Impossible}}$

Q6

No.

The known C may lie outside A.

$\boxed{\text{No}}$

Q7

Yes.

Nothing prevents part of A from overlapping C inside B.

$\boxed{\text{Yes}}$

Q8

“Only doctors are surgeons” means:

$\boxed{\text{Surgeon} \subseteq \text{Doctor}}$

$\boxed{\text{Doctors}}$

Q9

A ⊄ B .

Some C are outside B.

Anything outside B cannot be A.

$\boxed{\text{Some C are not A}}$

Q10

Known some A-B members must be C.

But other A may exist outside B and therefore possibly outside C.

So:

$\boxed{\text{Yes, possible}}$

Not definite, but possible.

Q11–Q15

Given:

$\lceil \text{Some } P \text{ are } Q \rceil$

$\lceil Q \subseteq R \rceil$

$\lceil R \cap S = \varnothing \rceil$

$\lceil T \subseteq S \rceil$

Q11

Known P-Q members are R.

$\lceil \boxed{\text{Yes}} \rceil$

Q12

Those P-Q members are R, and no R is S.

$\lceil \boxed{\text{Yes}} \rceil$

Q13

$T \nsubseteq S$.

Known P-Q members are not S, so they cannot be T.

$\lceil \boxed{\text{Yes}} \rceil$

Q14

Could other P members be T?

Yes.

Only the known P-Q members are forced outside T.

$\lceil \boxed{\text{Yes, possible}} \rceil$

Q15

All T are S does not guarantee that T exists.

$\lceil \boxed{\text{No}} \rceil$

Q16–Q20

Statements:

$\lceil A \subseteq B \rceil$

$\lceil \text{Some } B \text{ are } C \rceil$

$\lceil C \cap D = \varnothing \rceil$

$\lceil E \subseteq D \rceil$

Q16 — Some A are C

The B-C overlap need not intersect A.

$\lceil \boxed{\text{Does not follow}} \rceil$

Q17 — Some A being C is possible

A is inside B, and B already overlaps C.

Nothing prevents the A-set from overlapping that C region.

$\lceil \boxed{\text{Possible}} \rceil$

Q18 — No C is E

C is disjoint D.

$E \nmid D$.

Therefore E cannot overlap C.

$\lceil \boxed{\text{Follows}} \rceil$

Q19 — Some B are not D

We know:

$\lceil \text{Some } B \text{ are } C \rceil$

and:

$\lceil \text{No } C \text{ is } D \rceil$

So those B-C members are not D.

$\lceil \boxed{\text{Follows}} \rceil$

Q20 — Some E are not C

We know all E are D, and no C is D.

So if E exists, all E would be outside C.

But E's existence is not guaranteed.

Therefore:

$\lceil \boxed{\text{Does not follow}} \rceil$

' CHAPTER 66 COMPLETE

We have now covered:

- All / No / Some / Some-not '
- Venn method '
- conversion rules '
- universal chaining '
- exclusion propagation '
- existential logic '
- strict GATE-style existence handling '
- "only" statements '
- possibility conclusions '
- possible vs definite distinction '
- Some-not possibility '
- complementary conclusions '
- either–or logic '
- coded syllogisms '
- multi-statement chains '
- "few/most/not all" interpretation cautions '
- counterexample method '
- existential-token method '
- conclusion-pair strategy '
- speed toolkit '
- L0–L3 practice '

Progress

\[\boxed{66/76\text{ numbered chapters completed}} \]

UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating '
- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles '
- Chapter 66 — Syllogisms '
- Chapter 67 — Statements, Assumptions & Conclusions +
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

CHAPTER 67 — STATEMENTS, ASSUMPTIONS & CONCLUSIONS

Next we will cover:

- statement vs fact vs inference
- hidden assumptions
- necessary vs merely possible assumptions
- conclusion vs repetition
- definite inference
- strong/weak conclusions
- “must be true” reasoning
- avoiding outside knowledge
- negative and causal assumptions
- policy/advertisement assumptions
- course-of-action preview
- GATE/placement logical-inference shortcuts
- L0–L3 practice.

USER

next

ASSISTANT

CHAPTER 67 — STATEMENTS, ASSUMPTIONS & CONCLUSIONS

PART 1 — STATEMENT VS ASSUMPTION, NECESSARY ASSUMPTIONS, INFERENCES, CONCLUSIONS & OUTSIDE-KNOWLEDGE TRAPS

Status: ~~Not~~ IN PROGRESS GATE: High Placements: Very High Priority: ~~Not~~ Yes 1

This chapter is less about formulas and more about disciplined reasoning.

The central rule is:

Use only what the statement logically requires. Do not add your own beliefs, world knowledge, emotions, or guesses.

1. WHAT IS A STATEMENT?

A statement is the information given in the question.

Example:

“The company has decided to provide free transport to employees working after 9 PM.”

You must accept this as the given fact for the question.

Do not ask whether the company really exists or whether the policy is good.

2. WHAT IS AN ASSUMPTION?

An assumption is an idea that is **not explicitly stated**, but must be accepted for the statement/action to make sense.

Example:

“Carry an umbrella because it may rain.”

Possible assumption:

Rain can make an umbrella useful.

This is built into the advice.

3. ASSUMPTION "CONCLUSION

Assumption

Something taken for granted **before** making the statement.

Conclusion

Something logically inferred **after** accepting the statement.

Memory trick:

$\boxed{\text{Assumption = Before}}$

$\boxed{\text{Conclusion = After}}$

4. SIMPLE EXAMPLE

Statement:

“Buy this water purifier to reduce impurities in drinking water.”

Assumption:

The purifier can reduce at least some impurities.

Without that belief, the recommendation makes no sense.

So the assumption is necessary.

5. NECESSARY VS POSSIBLE ASSUMPTION

This distinction is crucial.

A possible assumption merely *could* be true.

A necessary assumption must be accepted for the statement to work.

Example:

“The school introduced extra mathematics classes to improve students' performance.”

Possible assumption:

Students like mathematics.

Not necessary.

Necessary assumption:

Extra mathematics instruction may help improve performance.

6. NEGATION TEST — BEST ASSUMPTION TOOL

To test whether an assumption is necessary:

Negate the proposed assumption.

Then ask:

Does the original statement/action still make sense?

If negating the assumption destroys the logic, the assumption is necessary.

7. NEGATION-TEST EXAMPLE

Statement:

“Install CCTV cameras to reduce theft.”

Proposed assumption:

CCTV may help deter or detect theft.

Negate:

CCTV cannot help deter or detect theft at all.

Then installing CCTV for theft reduction makes no sense.

Therefore:

$\boxed{\text{Assumption is necessary}}$

8. NON-NECESSARY ASSUMPTION EXAMPLE

Statement:

“Install CCTV cameras to reduce theft.”

Proposed assumption:

Every thief is afraid of cameras.

Negate:

Not every thief is afraid of cameras.

The recommendation can still make sense because cameras may help in other ways.

Therefore:

$\boxed{\text{Not necessary}}$

9. ASSUMPTION QUESTIONS — CORE SEARCH

Ask:

“What must the speaker believe for this statement, decision, recommendation, or action to be reasonable?”

Not:

“What else might be true?”

10. ADVERTISEMENT ASSUMPTIONS

Statement:

“Use Brand X toothpaste for stronger teeth.”

Likely necessary assumption:

Brand X toothpaste can contribute to stronger teeth.

Not necessary:

Brand X is the cheapest toothpaste.

The advertisement never requires that.

11. POLICY ASSUMPTIONS

Statement:

"The city will increase bus frequency to reduce overcrowding."

Necessary assumption:

Increasing bus frequency can help reduce overcrowding.

Potential but unnecessary assumption:

Everyone prefers buses to cars.

12. RECOMMENDATION ASSUMPTIONS

Statement:

"Students should solve previous-year papers before the exam."

Necessary assumption:

Solving previous-year papers can help exam preparation.

Not necessary:

Previous-year questions will repeat exactly.

That is much stronger than required.

13. WARNING ASSUMPTIONS

Statement:

"Do not swim here; the current is dangerous."

Necessary assumption:

The current can create risk for swimmers.

Not necessary:

Every swimmer who enters will drown.

Avoid extreme interpretations.

14. EXTREME-WORD TRAP

Words like:

- always
- never
- everyone
- nobody
- completely
- certainly
- only

make assumptions/conclusions much stronger.

If the statement is moderate, an extreme option often does not follow.

15. EXAMPLE

Statement:

“Regular exercise can improve health.”

Conclusion:

Everyone who exercises regularly will always remain healthy.

Too strong.

The statement says **can improve**, not guarantee perfect health.

Therefore:

$\boxed{\text{Does not follow}}$

16. WHAT IS A CONCLUSION?

A conclusion is a statement that must logically follow from the given information.

Example:

All engineers in Team A completed training. Ravi is an engineer in Team A.

Conclusion:

$\boxed{\text{Ravi completed training}}$

This is forced.

17. CONCLUSION VS POSSIBILITY

Given:

Some students prefer online classes.

Can we conclude:

Most students prefer online classes?

No.

Could it be true? Maybe.

But conclusion questions ask what **must** follow.

18. DIRECT CONCLUSION

Statement:

Sales increased from 100 units to 150 units.

Conclusion:

Sales increased by 50 units.

$\boxed{150-100=50}$

This follows directly.

19. PARAPHRASE CONCLUSION

Statement:

No employee may enter the laboratory without an ID card.

Conclusion:

An employee without an ID card is not permitted to enter the laboratory.

Same logical meaning.

$\boxed{\text{Follows}}$

20. REPETITION IS NOT A PROBLEM

A conclusion may sometimes closely restate the statement.

If it is logically guaranteed, it is valid.

Do not reject merely because it sounds similar.

21. NEW INFORMATION TRAP

Statement:

The university increased library hours during examinations.

Conclusion:

Students had complained about the old timings.

Not stated.

May be plausible, but not necessary.

$\boxed{\text{Does not follow}}$

22. OUTSIDE-KNOWLEDGE TRAP

Statement:

Company X reduced product prices.

Conclusion:

Company X's sales will increase.

Real-world economics might make this plausible, but the statement alone does not guarantee it.

Thus:

$\boxed{\text{Does not follow}}$

23. ACCEPT THE GIVEN WORLD

Suppose a question says:

“All machines in this factory operate silently.”

Even if you know real machines often make noise, use the statement as true within the question.

Aptitude logic works inside the given world.

24. FACT VS INFERENCE

Statement:

“The road was wet when Ravi reached the office.”

Fact:

$\boxed{\text{Road was wet}}$

Possible inference:

It rained.

But road could also be wet due to cleaning, leakage, etc.

Therefore rain is not a definite conclusion.

25. MULTIPLE-CAUSE TRAP

Observation:

“The ground is wet.”

Possible causes:

- rain
- sprinkler
- pipe leak
- cleaning

Therefore:

“It definitely rained”

is invalid without more evidence.

26. CAUSE VS EFFECT

Statement:

“After the company introduced automated testing, software defects reaching production decreased.”

Can we conclude automation caused the decrease?

Not necessarily unless the statement explicitly establishes causation.

Correlation or sequence alone does not prove causation.

27. “AFTER” DOES NOT AUTOMATICALLY MEAN “BECAUSE”

$\boxed{\text{A happened before B}}$

does not imply:

$\boxed{\text{A caused B}}$

This will become very important in Chapter 68.

28. ASSUMPTION IN CAUSAL CLAIMS

Statement:

“Reducing traffic speed limits will reduce accidents.”

Underlying assumption:

Vehicle speed is relevant to at least some accidents.

If speed had absolutely no connection to accidents, the proposal would lose its rationale.

29. DECISION-BASED ASSUMPTIONS

Statement:

“The company postponed the launch because testing is incomplete.”

Assumption:

Completing testing is important before launch.

The decision implicitly treats incomplete testing as a valid reason for delay.

30. “BECAUSE” GIVES AN EXPLICIT REASON

If the statement itself says:

A happened because B.

Then B is not merely an assumption—it is explicitly given as the reason.

Do not classify explicit content as hidden assumption.

31. IMPLIED VS EXPLICIT

Statement:

“Because heavy rain is expected, the match has been postponed.”

Explicit:

- heavy rain is expected
- match postponed

Possible assumption:

- expected heavy rain could adversely affect the match.

That connection is implicit.

32. CONCLUSION FROM “ALL”

Statement:

Every employee in the finance team attended the meeting.

Conclusion:

No finance-team employee missed the meeting.

Equivalent logical form.

$\boxed{\text{Follows}}$

33. CONCLUSION FROM “SOME”

Statement:

Some engineers work remotely.

Conclusion:

At least one engineer works remotely.

$\boxed{\text{Follows}}$

But:

All engineers work remotely.

$\boxed{\text{Does not follow}}$

34. CONCLUSION FROM “NONE”

Statement:

None of the shortlisted candidates lacked the required degree.

Equivalent:

Every shortlisted candidate had the required degree.

$\boxed{\text{Follows}}$

35. QUANTIFIER STRENGTH

Remember:

$\lceil \text{All} \rightarrow \text{Some} \rceil$

only if existence is established.

But in ordinary statement-conclusion questions involving actual named groups, existence may be clear from context.

Still avoid careless quantifier strengthening.

36. “SOME” DOES NOT MEAN “ONLY SOME”

Statement:

Some students passed.

This does not mean:

Some students failed.

It is possible that all passed.

This is the same trap seen in syllogisms.

37. “MANY” DOES NOT MEAN “MOST”

Statement:

Many candidates applied.

Cannot conclude:

More than half of all eligible candidates applied.

“Many” is vague.

38. “MOST” IMPLIES SOME

If:

Most students passed.

Then definitely:

$\lceil \boxed{\text{Some students passed}} \rceil$

provided the group exists.

But “most” does not mean all.

39. “FEW” DOES NOT MEAN NONE

Statement:

Few employees work remotely.

Then at least some may work remotely.

Do not conclude zero.

40. NECESSARY ASSUMPTION — PRESUPPOSITION

Statement:

“The manager asked Ravi to submit the report again.”

This presupposes Ravi had already submitted it before, or at least there was an earlier expected submission context.

The word “again” carries hidden information.

Language markers can create assumptions.

41. WORDS THAT SIGNAL PRESUPPOSITIONS

Words such as:

- again
- continue
- stop
- return
- still
- resume
- improve
- reduce
- increase

often imply a prior state.

Example:

“The factory will reduce waste.”

Implies:

$\boxed{\text{There is currently some waste to reduce}}$

42. “STOP” ASSUMPTION

“The company stopped offering free lunches.”

Usually implies:

It had been offering free lunches earlier.

43. “RETURN” ASSUMPTION

“Ravi returned to Hyderabad.”

Usually presupposes:

Ravi had been in Hyderabad before.

44. “CONTINUE” ASSUMPTION

“The government will continue the scholarship.”

Implies:

The scholarship already exists/is currently being provided.

45. OBJECTIVE VS VALUE JUDGMENT

Statement:

“The government should build more parks.”

This is a recommendation.

A necessary assumption may involve expected benefit or need.

But conclusion:

Parks are the most important public service.

does not follow.

Avoid importing value rankings.

46. EMOTIONAL LANGUAGE TRAP

Statement:

“The company laid off 100 workers.”

Conclusion:

The company is cruel.

That is a value judgment, not a logical conclusion.

$\boxed{\text{Does not follow}}$

47. MOTIVE TRAP

Statement:

“A politician donated ₹ 1lakh.”

Conclusion:

The donation was made to gain votes.

Possible motive, not guaranteed.

Never infer motive without evidence.

48. INTENTION VS OUTCOME

Statement:

“The government launched a program to reduce unemployment.”

Definite:

The program has the goal of reducing unemployment.

Not definite:

Unemployment will definitely fall.

Intent does not guarantee result.

49. RECOMMENDATION VS EFFECTIVENESS

Statement:

“Use this app to improve productivity.”

The speaker assumes the app may help productivity.

But conclusion:

Every user will become more productive.

Too strong.

50. CONCLUSION FROM COMPARATIVE DATA

Statement:

A scored 80 and B scored 70.

Conclusion:

$\boxed{A \text{ scored more than } B}$

Valid.

Conclusion:

A is more intelligent than B.

Invalid.

Marks are not logically identical to intelligence.

51. CATEGORY ERROR TRAP

Statement:

X company has higher revenue than Y.

Conclusion:

X is a better company.

“Better” is undefined and broader.

Does not follow.

52. NECESSARY ASSUMPTION VS SUFFICIENT CONDITION

Statement:

“Hire more workers to finish the project sooner.”

Necessary assumption:

Additional workers can contribute to project completion.

Not necessary:

Hiring more workers is sufficient by itself to guarantee early completion.

The second is much stronger.

53. ASSUMPTION SHOULD NOT EXCEED THE STATEMENT

A good assumption is usually the **minimum hidden belief** needed.

Avoid adding:

- guarantees
- universal claims
- extra motives
- unrelated facts

54. ASSUMPTION EXAMPLE — PUBLIC NOTICE

Notice:

“Passengers are requested to arrive 30 minutes early due to additional security checks.”

Necessary assumption:

Additional checks may take extra time.

Not necessary:

Every passenger will otherwise miss the train.

55. ASSUMPTION EXAMPLE — DISCOUNT

Advertisement:

“Buy now—prices increase next week.”

Necessary assumption for urgency:

Some customers care about avoiding the higher future price.

Not necessary:

Everyone will buy immediately.

56. ASSUMPTION EXAMPLE — SCHOOL

“The school banned mobile phones during class to improve concentration.”

Necessary assumption:

Mobile-phone use during class can interfere with concentration.

Not necessary:

Mobile phones are always harmful.

57. CONCLUSION TEST — THREE QUESTIONS

For any proposed conclusion ask:

1. Is it directly stated or logically forced?
2. Could the original statement be true while this conclusion is false?
3. Am I adding outside information?

If the answer to Q2 is yes:

$\boxed{\text{Conclusion does not follow}}$

58. COUNTEREXAMPLE METHOD

Statement:

Some employees are programmers.

Conclusion:

Some employees are managers.

Can statement be true while conclusion false?

Yes—imagine no employee is a manager.

Therefore:

$\boxed{\text{Conclusion does not follow}}$

59. STRONG VS WEAK CONCLUSION

A “strong” conclusion in aptitude reasoning usually means:

- directly relevant
- logically supported
- not exaggerated

- not based on emotion/outside facts

Weak conclusions often contain:

- unsupported certainty
- unrelated claims
- moral judgment
- speculation

60. STATEMENT-CONCLUSIONEXAMPLE

Statement:

“The number of road accidents in the city increased by 20% this year.”

Conclusions:

I. Road accidents increased this year. II. Driving standards have deteriorated.

I follows directly.

II may be one explanation but is not guaranteed.

$\boxed{\text{Only I follows}}$

61. STATEMENT-ASSUMPTIONEXAMPLE

Statement:

“Use public transport to reduce road congestion.”

Assumptions:

I. Public transport can carry multiple passengers efficiently. II. Everyone owns a private vehicle.

I is relevant and necessary to the recommendation.

II is unnecessary.

$\boxed{\text{Only I}}$

62. ANOTHER ASSUMPTION EXAMPLE

Statement:

“The college extended library hours so that students can study longer.”

Assumptions:

I. Some students may use the additional library time. II. All students prefer studying in the library.

I is necessary/reasonable.

II is too strong.

63. NEGATIVE ASSUMPTION EXAMPLE

Statement:

“The company introduced password rules to improve account security.”

Does it assume:

Existing account security cannot be improved?

No.

Actually it assumes there is room for improvement.

The wording of negative options is often designed to confuse.

64. ASSUMPTION ABOUT FEASIBILITY

Statement:

“The company plans to open a branch in Pune next month.”

A basic assumption:

Opening a branch in Pune next month is considered feasible enough to plan for.

But not:

The branch will definitely become profitable.

65. PROPOSAL ASSUMPTION

“Let us conduct the exam online to save time.”

Necessary hidden belief:

Online conduct can save at least some relevant time.

Not:

Online exams have zero disadvantages.

66. LOGICAL INFERENCE FROM “ONLY IF”

Statement:

Employees receive bonus only if performance targets are met.

This means:

$\text{Bonus} \rightarrow \text{TargetMet}$

Therefore:

If an employee received the bonus, their target was met.

$\boxed{\text{Follows}}$

67. CONVERSE TRAP AGAIN

From:

$\text{Bonus} \rightarrow \text{TargetMet}$

Can we conclude:

$\text{TargetMet} \rightarrow \text{Bonus}$

No.

Meeting targets may not be sufficient.

68. “UNLESS” IN INFERENCE QUESTIONS

“The event will be cancelled unless the weather improves.”

Equivalent core idea:

If weather does not improve:

$\boxed{\text{Event will be cancelled}}$

Be careful not to assume improved weather guarantees event occurs unless wording/context makes that biconditional.

69. “OR” CONCLUSION

Statement:

Either A or B will lead the team.

If exclusive “either/or” is intended:

exactly one leads.

But natural language “or” may be inclusive.

Use exam convention/context carefully.

70. FINAL PART-1 WORKFLOW

For **assumption questions**:

$\boxed{\text{Use the Negation Test}}$

For **conclusion questions**:

$\boxed{\text{Use the Counterexample Test}}$

For both:

$\boxed{\text{Avoid outside knowledge}}$

71. ASSUMPTION SPEED RULE

Ask:

If this option were false, could the original statement/action still reasonably stand?

If yes:

$\boxed{\text{Not necessary}}$

If no:

$\boxed{\text{Necessary assumption}}$

72. CONCLUSION SPEED RULE

Ask:

Can the original statement be true while this conclusion is false?

If yes:

$\boxed{\text{Does not follow}}$

If no:

$\boxed{\text{Follows}}$

73. COMMON TRAPS

& pTRAP 1 — Plausible = necessary

False.

& pTRAP 2 — Possible inference = conclusion

False.

& pTRAP 3 — Outside knowledge

Ignore it unless stated.

& **TRAP 4 — Cause assumed from sequence**

Invalid.

& **TRAP 5 — Intention guarantees outcome**

Invalid.

& **TRAP 6 — Moderate statement !' extreme conclusion**

Usually invalid.

& **TRAP 7 — Hidden motive**

Do not invent motives.

& **TRAP 8 — Converse of implication**

Invalid unless biconditional.

PRACTICE — LEVEL 0

Q1

Statement: "Take an umbrella because rain is expected."

Is "An umbrella can be useful in rain" a reasonable necessary assumption?

Q2

Statement: "Some students are programmers."

Does "All students are programmers" follow?

Q3

Statement: "A scored 90; B scored 80."

Does "A scored more than B" follow?

Q4

Statement: "The road is wet."

Does "It definitely rained" follow?

Q5

Statement: "The company stopped free transport."

Does this normally imply free transport existed earlier?

LEVEL 1

Q6

Statement: "The city increased bus frequency to reduce crowding."

Assumption: "Increasing bus frequency may reduce crowding."

Implicit or not?

Q7

Same statement.

Assumption: "Every resident uses buses."

Implicit or not?

Q8

Statement: "The company introduced training to improve employee skills."

Conclusion: "Employee skills will certainly improve."

Does it follow?

Q9

Statement: "All applicants submitted ID proof."

Conclusion: "No applicant failed to submit ID proof."

Does it follow?

Q10

Statement: "Some employees work remotely."

Conclusion: "Some employees do not work remotely."

Does it follow?

LEVEL 2**Q11**

Statement: "Install additional streetlights to improve night-time safety."

Which is necessary?

A. Streetlights can contribute to night-time safety. B. Streetlights eliminate all crime.

Q12

Statement: "The university extended library hours during exams."

Conclusion: "Students had demanded longer hours."

Does it follow?

Q13

Statement: "Employees receive bonus only if targets are met. Ravi received a bonus."

What follows?

Q14

Statement: "If the server fails, the backup system activates. The backup system did not activate."

What follows?

Q15

Statement: "After the advertisement campaign began, sales rose."

Does this alone prove the campaign caused the rise?

LEVEL 3 — EXAM PRESSURE

Statement:

“The company has decided to introduce shuttle buses from nearby metro stations to encourage employees to use public transport.”

Assumptions:

I. Lack of convenient last-mile transport may discourage some employees from using public transport. II. Every employee lives close to a metro station. III. Shuttle buses may make public-transport use more convenient. IV. Once shuttle buses begin, no employee will use a private vehicle.

Q16

Does I appear necessary/reasonably implicit?

Q17

Does II appear necessary?

Q18

Does III appear necessary?

Q19

Does IV appear necessary?

Q20

Which assumptions are the strongest valid ones?

ANSWERS

Q1

Yes.

Without usefulness of an umbrella in rain, the advice loses its basis.

$\boxed{\text{Yes}}$

Q2

No.

“Some” does not imply “all.”

$\boxed{\text{No}}$

Q3

Yes.

$\lceil 90 > 80 \rceil$

$\lceil \boxed{\text{Yes}} \rceil$

Q4

No.

Other causes can make the road wet.

$\lceil \boxed{\text{No}} \rceil$

Q5

Yes.

“Stopped” normally presupposes it existed previously.

$\lceil \boxed{\text{Yes}} \rceil$

Q6

Yes.

It is the core rationale behind the policy.

$\lceil \boxed{\text{Implicit}} \rceil$

Q7

No.

The policy can make sense even if only some residents use buses.

$\lceil \boxed{\text{Not}\ \text{implicit}} \rceil$

Q8

No.

Training is intended to help, but improvement is not guaranteed.

$\lceil \boxed{\text{Does}\ \text{not}\ \text{follow}} \rceil$

Q9

Yes.

Equivalent restatement.

$\boxed{\text{Follows}}$

Q10

No.

It is possible that all employees work remotely.

$\boxed{\text{Does not follow}}$

Q11

A is necessary.

B is much too strong.

$\boxed{A}$

Q12

No.

The university may have extended hours for many other reasons.

$\boxed{\text{Does not follow}}$

Q13

“Bonus only if target met” means:

$\boxed{\text{Bonus} \rightarrow \text{TargetMet}}$

Ravi got bonus.

Therefore:

$\boxed{\text{Ravi met the target}}$

Q14

$\boxed{\text{ServerFail} \rightarrow \text{BackupActivates}}$

Contrapositive:

$\boxed{\neg \text{BackupActivates} \rightarrow \neg \text{ServerFail}}$

Therefore:

$\boxed{\text{The server did not fail}}$

under the stated logical rule.

Q15

No.

Temporal sequence alone does not prove causation.

☐ No

Q16–Q20

Company introduces metro-station shuttles to encourage public transport.

Q16

If last-mile inconvenience were irrelevant to public-transport use, the shuttle rationale weakens substantially.

☐ Yes

Q17

The plan need not serve every employee.

☐ No

Q18

This is central to the plan.

☐ Yes

Q19

The company does not need to believe private-vehicle use will become zero.

☐ No

Q20

Strongest valid assumptions:

☐ I and III

CHAPTER 67 STATUS

Covered so far:

- statement vs assumption '
- assumption vs conclusion '
- necessary vs possible assumptions '
- negation test '
- policy/advertisement/recommendation assumptions '
- presuppositions '
- statement-conclusion reasoning '

- direct/paraphrase conclusions '
- quantifier traps '
- extreme-language traps '
- outside-knowledge trap '
- motive/value-judgment traps '
- causation preview '
- implication/converse/contrapositive '
- counterexample method '
- L0–L3 practice '

But **Chapter 67 is not complete yet.**

Still remaining:

- advanced assumption sets
- multiple-assumption questions
- argument hidden premises
- strong vs weak inference
- statement-conclusion combinations
- “either conclusion” reasoning
- inference from passages/data
- negative conclusions
- consistency/contradiction
- placement-style critical-inference sets
- final speed toolkit

Progress

$\lfloor \boxed{66/76} \text{ numbered chapters completed} \rfloor$

Current: 0=36 Chapter 67 — Statements, Assumptions & Conclusions

NEXT

0=67 Chapter 67 Part 2 — Advanced Assumptions, Hidden Premises, Strong/Weak Conclusions, Passage Inference & Final Statement-Reasoning Toolkit

USER

next

ASSISTANT

CHAPTER 67 — STATEMENTS, ASSUMPTIONS & CONCLUSIONS

PART 2 — ADVANCED ASSUMPTIONS, HIDDEN PREMISES, STRONG/WEAK CONCLUSIONS, PASSAGE INFERENCE & FINAL TOOLKIT

Status: $\emptyset = \text{Not}$ PROGRESS ! ' ' COMPLETE GATE: High Placements: Very High Priority: $\emptyset = \text{Not}$ 1

This part completes statement-based reasoning.

Main principle:

A strong conclusion stays inside the evidence. A strong assumption supplies only the minimum hidden support needed by the argument.

1. ARGUMENT STRUCTURE

Most reasoning questions can be reduced to:

$\boxed{\text{Premise} \rightarrow \text{Conclusion}}$

Example:

“The number of accidents has increased. Therefore, the city should improve road safety measures.”

Premise:

$\boxed{\text{Accidents increased}}$

Conclusion:

$\boxed{\text{Improve safety measures}}$

There may also be an unstated assumption connecting the two.

2. HIDDEN PREMISE

A hidden premise is information that connects the explicit evidence to the conclusion.

Example:

“Ravi should attend the extra class because he scored poorly.”

Explicit:

- Ravi scored poorly.

Conclusion:

- Ravi should attend extra class.

Hidden bridge:

Extra classes can help someone who performed poorly.

Without this bridge, the recommendation is unsupported.

3. BRIDGE ASSUMPTION METHOD

When an argument contains:

[Evidence $\rightarrow$ Recommendation]

ask:

What missing belief connects the evidence to the recommendation?

That is often the correct assumption.

4. EXAMPLE

Argument:

"Traffic congestion has increased, so more metro services should be introduced."

Bridge assumption:

[$\boxed{\text{More metro services may help reduce road congestion}}$]

Not necessary:

Every person will stop using cars.

5. NECESSARY ASSUMPTION MUST BE MINIMAL

Suppose:

"Students should practice mock tests to improve exam performance."

Compare:

A. Mock tests can help exam preparation. B. Mock tests are the only way to improve exam performance.

A is minimal.

B is much stronger.

[$\boxed{A \text{ is the better assumption}}$]

6. MULTIPLE ASSUMPTION QUESTIONS

Suppose statement has assumptions I and II.

Test each independently.

Do **not** think:

"If I is true, II must also be true."

Unless their logic actually connects.

Possible answers:

- only I
- only II
- both
- neither

7. EXAMPLE

Statement:

“The company opened a childcare centre to help working parents.”

Assumptions:

I. Some employees have childcare needs. II. Every employee has children.

I is reasonably necessary.

II is too strong.

$\boxed{\text{Only I}}$

8. NEGATION TEST — ADVANCED

Proposed assumption:

Some employees need childcare support.

Negate:

No employee needs childcare support.

Then opening a childcare centre **for helping employees with childcare** loses its stated rationale.

So assumption is necessary.

9. NEGATING UNIVERSAL OPTIONS

Suppose assumption says:

All employees prefer flexible hours.

Its negation is not:

No employee prefers flexible hours.

Correct logical negation:

$\boxed{\text{At least one employee does not prefer flexible hours}}$

This matters in rigorous reasoning.

10. NEGATING “SOME”

Negation of:

Some A are B

is:

$\boxed{\text{No A is B}}$

Negation of:

All A are B

is:

$\boxed{\text{Some A are not B}}$

This connects Chapter 67 with syllogism logic.

11. CONCLUSION MUST BE GUARANTEED

Statement:

60% of surveyed users preferred Product A over Product B.

Conclusion:

Product A was preferred by a majority of surveyed users.

Since:

$60\% > 50\%$

$\boxed{\text{Follows}}$

But:

Product A is the best product in the market.

Does not follow.

12. SAMPLE VS POPULATION TRAP

Statement:

70% of 100 surveyed students preferred online classes.

Conclusion:

70% of all students prefer online classes.

Not justified unless the survey is established as representative.

$\boxed{\text{Does not follow}}$

13. DATASCOPE

Always preserve the scope.

If statement says:

“Among surveyed employees...”

conclusion should generally remain about:

$\boxed{\text{surveyed employees}}$

not all employees everywhere.

14. TIME-SCOPE TRAP

Statement:

Sales increased in June.

Conclusion:

Sales are increasing throughout the year.

No.

One month does not establish the entire annual trend.

15. LOCATION-SCOPE TRAP

Statement:

Pollution fell in City A.

Conclusion:

Pollution fell nationally.

Invalid extrapolation.

16. CATEGORY-SCOPE TRAP

Statement:

Software engineers in Department X received bonuses.

Conclusion:

Every employee in Department X received a bonus.

No.

The original category is narrower.

17. PART-TO-WHOLE FALLACY

Given information about a subset, do not automatically apply it to the whole group.

Likewise, group-level data may not apply to every individual.

18. WHOLE-TO-PART FALLACY

Statement:

The average salary in the company is ₹ 8lakh.

Conclusion:

Ravi earns ₹ 8lakh.

No.

Average does not determine individual values.

19. AVERAGE TRAP

Statement:

Average score = 80.

Conclusion:

Every student scored 80.

Invalid.

This is both quantitative and logical reasoning.

20. STRONG CONCLUSION

A strong conclusion:

- directly follows
- uses no extra assumptions
- stays within scope
- uses appropriate certainty

Example:

“The train arrived 30 minutes late.”

Strong conclusion:

The train did not arrive at its scheduled time.

21. WEAK CONCLUSION

Same statement.

Weak conclusion:

The railway system is inefficient.

That is broader and evaluative.

Not logically established.

22. CONCLUSION STRENGTH LADDER

From weakest/moderate to strongest:

- may
- can
- sometimes
- likely
- most
- all
- always
- certainly

A conclusion should not become stronger than its evidence.

23. MODALITY MATCHING

Statement:

“This medicine may reduce symptoms.”

Conclusion:

“This medicine can reduce symptoms.”

Reasonably consistent.

But:

“This medicine will cure everyone.”

Far too strong.

24. PASSAGE INFERENCE

For a short passage:

1. identify explicit claims
2. identify author stance
3. separate examples from general claims
4. reject outside facts
5. choose the least exaggerated supported option

25. PASSAGE EXAMPLE

Passage:

“Remote work can reduce commuting time, but some employees report communication difficulties. Hybrid work attempts to balance these advantages and limitations.”

Valid inference:

Remote work has both potential benefits and challenges.

Invalid:

Hybrid work is always superior to remote work.

26. “BEST SUPPORTED” QUESTIONS

Sometimes none of the options is mathematically guaranteed, but one is **best supported** by the passage.

Choose the one that:

- needs least extra assumption
- matches wording
- avoids extremes
- preserves scope

27. AUTHOR'S VIEW VS FACT

Passage:

"The author argues that public transport investment deserves greater priority."

Conclusion:

The author supports increased priority for public transport investment.

Valid as author's stance.

But:

Increased investment will definitely solve congestion.

Not implied.

28. EXAMPLE VS MAIN CLAIM

Passage:

"Digital payments reduce transaction friction. For example, QR payments allow fast small-value transactions."

Main claim:

$\boxed{\text{Digital payments can reduce transaction friction}}$

QR is supporting evidence/example, not necessarily the whole argument.

29. INFERENCE FROM NEGATIVE WORDING

Statement:

"Not every applicant passed."

This means:

$\boxed{\text{At least one applicant did not pass}}$

It does **not** mean:

No applicant passed.

30. "NOT NONE"

Statement:

“It is not true that none of the candidates qualified.”

Means:

$\lceil \boxed{\text{At least one candidate qualified}} \rceil$

31. “NOT ALL ”

$\lceil \neg(\text{All A are B}) \rceil$

implies:

$\lceil \boxed{\text{Some A are not B}} \rceil$

provided A exists in the context.

32. DOUBLE NEGATIVE

Statement:

“It is not impossible for A to succeed.”

This roughly means:

$\lceil \boxed{\text{A can possibly succeed}} \rceil$

It does not mean A definitely succeeds.

33. CONSISTENCY QUESTIONS

Two statements are consistent if they can both be true simultaneously.

Example:

1. Some students are athletes.
2. Some students are not athletes.

These are consistent.

Both can hold.

34. CONTRADICTION

Statements:

1. All A are B.
2. Some A are not B.

These directly conflict if both refer to the same domain.

$\lceil \boxed{\text{Contradictory}} \rceil$

35. NON-CONTRADICTION TRAP

Statements:

1. Some A are B.
2. Some A are not B.

Not contradictory.

Different A members may satisfy each.

36. ANOTHER CONSISTENCY EXAMPLE

1. No managers are interns.
2. Some employees are both managers and interns.

If those employees are in the same universe and categories:

Contradiction.

Because second creates forbidden overlap.

37. LOGICAL POSSIBILITY TEST

To test consistency:

Can I construct one world where all statements are simultaneously true?

If yes:

$\boxed{\text{Consistent}}$

If impossible:

$\boxed{\text{Contradictory}}$

38. CONCLUSION PAIRS

Suppose:

I. Some A are B. II. Some B are A.

These are equivalent.

So if I follows, II follows.

This comes from syllogism conversion.

39. NEGATIVE PAIRS

I. No A is B. II. No B is A.

Equivalent.

40. CONCLUSION PAIR — NOT EQUIVALENT

I. All A are B. II. All B are A.

Not equivalent.

Do not accept both together automatically.

41. “EITHER CONCLUSION” REASONING

Suppose one of two outcomes must occur but current information does not tell which.

Example:

Exactly one of A and B will be selected.

Conclusions:

I. A will be selected. II. B will be selected.

Neither individually follows.

But:

$\boxed{\text{Either I or II must be true}}$

42. EITHER-OR REQUIREMENT

For “either I or II”:

- both cannot be true
- both cannot be false
- exactly one is guaranteed

This is XOR logic.

43. “OR” DOES NOT ALWAYS MEAN XOR

Statement:

A or B may attend.

Could allow both unless “either...or but not both” is intended.

Always distinguish inclusive OR from exclusive OR.

44. POLICY STATEMENT INFERENCE

Statement:

“The university has made attendance mandatory for laboratory sessions.”

Conclusion:

Students are required to attend laboratory sessions.

Valid.

Conclusion:

Students dislike laboratory sessions.

Unsupported.

45. BAN/PROHIBITION INFERENCE

Statement:

“Smoking is prohibited inside the building.”

Valid conclusion:

$\boxed{\text{Smoking inside the building is not permitted}}$

Not valid:

Nobody ever smokes inside.

Rule “” guaranteed compliance.

46. RULE VS BEHAVIOUR

This is a major trap.

Statement:

Helmets are mandatory.

Does not imply:

Everyone wears a helmet.

A rule establishes obligation, not perfect compliance.

47. GOAL VS RESULT

Statement:

“The company launched training to reduce errors.”

Valid:

Reducing errors is a goal of the training.

Invalid:

Errors definitely decreased.

48. PLAN VS EXECUTION

Statement:

"The government plans to build five hospitals."

Conclusion:

Five hospitals have been built.

Invalid.

Plan is not completion.

49. INTENTION VS CAPABILITY

Statement:

"A intends to run a marathon."

Cannot conclude:

A is capable of completing a marathon.

Intent alone does not establish ability.

50. CAPABILITY VS ACTION

Statement:

"A can swim."

Cannot conclude:

A swam today.

Ability "→" actual event.

51. NECESSARY VS SUFFICIENT CONDITIONS

Suppose:

A passport is required to enter Country X.

Passport is a **necessary condition**.

But it may not be sufficient; visa etc. may also be needed.

So:

$\text{Entry} \rightarrow \text{Passport}$

not:

$\text{Passport} \rightarrow \text{Entry}$

52. “REQUIRED” LOGIC

“B is required for A” means:

$\boxed{A \rightarrow B}$

This is the same as:

A only if B.

53. SUFFICIENT CONDITION

“If B occurs, A definitely occurs” means B is sufficient for A:

$B \rightarrow A$

But A may happen for other reasons.

54. NECESSARY + SUFFICIENT

If both:

$A \rightarrow B$

and:

$B \rightarrow A$

then:

$\boxed{A \leftrightarrow B}$

55. ARGUMENT WITH STATISTICS

Statement:

In a survey, 80% of users rated the app positively.

Conclusion:

Most surveyed users rated the app positively.

Valid.

Conclusion:

80% of all users worldwide like the app.

Invalid extrapolation.

56. BASE-RATE/CONTEXT TRAP

Statement:

Crime doubled from 1 incident to 2 incidents.

Conclusion:

Crime is now extremely high.

Not necessarily.

Relative increase can sound large while absolute level remains small.

57. PERCENTAGE VS ABSOLUTE VALUE

Statement:

Failure rate fell by 50%.

Without original rate, we cannot know final percentage.

Do not invent numerical values.

58. "MORE THAN" DOES NOT GIVE EXACT AMOUNT

Statement:

A earned more than B.

Conclusion:

A earned $\pm 10,000$ more than B.

Invalid without exact difference.

59. CORRELATION INFERENCE

Statement:

Students who studied more tended to score higher.

Safe:

In the observed data, study time and scores were positively associated.

Unsafe:

Studying more always causes higher scores.

60. CAUSAL ASSUMPTION DETECTOR

If an argument says:

$\lceil \text{Do } X \text{ to achieve } Y \rceil$

then it usually assumes:

$\lceil \boxed{X \text{ can contribute to } Y} \rceil$

Not:

$\lceil X \text{ guarantees } Y \rceil$

61. ALTERNATIVE-CAUSE TEST

When a conclusion claims causation, ask:

Could another cause explain the observation?

If yes, causation is not established.

62. REVERSE-CAUSATION TEST

Observation:

People who exercise more are healthier.

Possible:

- exercise improves health
- healthier people exercise more

- third factors affect both

Therefore pure correlation is insufficient for causal certainty.

63. ASSUMPTION QUESTION — ARGUMENT

Argument:

“The store should extend evening hours because many customers arrive after work.”

Assumption:

Some after-work customers would benefit from later opening hours.

Necessary/reasonable.

Not necessary:

Every customer works a 9–5 job.

64. STRONG CONCLUSION QUESTION

Statement:

“Three of the five machines failed the safety test.”

Strong conclusion:

$\boxed{\text{A majority of the five machines failed}}$

because:

$\frac{3}{5}=60\%$

Weak/unsupported:

The factory is unsafe.

Too broad.

65. PASSAGE INFERENCE EXAMPLE

Passage:

“Electric vehicles eliminate tailpipe emissions, but the total environmental impact also depends on how electricity and batteries are produced.”

Best inference:

Evaluating EV environmental impact requires considering more than tailpipe emissions.

This follows closely.

66. WRONG EXTREME INFERENCE

Same passage.

Electric vehicles are environmentally harmful.

Not established.

Nor:

Electric vehicles have zero environmental impact.

Both exceed evidence.

67. “MAY” OPTIONS

In definite conclusion questions, an option using “may” can sometimes be logically easier to support because it asserts possibility rather than certainty.

But do not choose it automatically.

It must still be compatible with evidence.

68. “MUST” OPTIONS

“Must” requires logical necessity.

One counterexample is enough to reject it.

69. “CAN” OPTIONS

“Can” often asserts possibility/capability.

Test whether at least one valid case supports it.

This distinction matters in critical inference.

70. QUESTION-TYPE IDENTIFICATION

Before solving, identify whether the question asks:

- assumption?
- conclusion?
- inference?

- must be true?
- could be true?
- cannot be true?
- contradiction?
- author view?

The solving standard changes slightly.

71. ASSUMPTION TOOLKIT

Negation test

Negate option.

Bridge test

Does it connect premise to conclusion?

Minimum claim test

Is it weaker than an unnecessary extreme?

Relevance test

Does the argument actually depend on it?

72. CONCLUSION TOOLKIT

Counterexample test

Can statement be true and conclusion false?

Scope test

Same population/time/category?

Strength test

Is conclusion stronger than evidence?

New-information test

Does it introduce unsupported facts?

73. PASSAGE-INFERENCE TOOLKIT

1. read conclusion/options first if time-limited
 2. locate supporting lines/ideas
 3. paraphrase, don't exaggerate
 4. eliminate extreme options
 5. eliminate scope shifts
 6. eliminate causal leaps
 7. choose best-supported statement
-

74. COMMON ADVANCED TRAPS

& pTRAP 1 — Sample != whole population

Invalid without support.

& pTRAP 2 — Average != every individual

Invalid.

& pTRAP 3 — Rule != perfect compliance

Invalid.

& pTRAP 4 — Goal != achieved result

Invalid.

& pTRAP 5 — Plan != completed action

Invalid.

& pTRAP 6 — Correlation != causation

Invalid without support.

& pTRAP 7 — Necessary condition != sufficient condition

Invalid.

& pTRAP 8 — Strong emotional wording

Usually unsupported.

& pTRAP 9 — Time/location/category expansion

Preserve scope.

& pTRAP 10 — Hidden assumption too strong

Choose minimum required bridge.

PRACTICE — LEVEL 0

Q1

Statement: "All employees must carry ID cards."

Does this prove every employee actually carries one?

Q2

Statement: "The company plans to open a new office."

Can we conclude the office has opened?

Q3

Statement: "A passport is required for entry."

Does having a passport guarantee entry?

Q4

Statement: "80% of surveyed users were satisfied."

Can we conclude most surveyed users were satisfied?

Q5

Statement: "Some workers are engineers."

Can we conclude some engineers are workers?

LEVEL 1

Q6

Statement: "The city increased night buses to help late-shift workers."

Assumption: "Some late-shift workers may use night buses."

Implicit?

Q7

Statement: "Smoking is banned in the office."

Conclusion: "No employee ever smokes in the office."

Follow?

Q8

Statement: "Average salary is ₹ 1 Lakh."

Conclusion: "Every employee earns ₹ 1 Lakh."

Follow?

Q9

Statement: "Sales rose after a new advertisement."

Conclusion: "The advertisement definitely caused the rise."

Follow?

Q10

Statement: "Not all applicants passed."

What definitely follows?

LEVEL 2

Q11

Argument: "The company should introduce flexible hours because commuting during peak time is difficult."

Which assumption is stronger?

A. Flexible hours may help some employees avoid peak commuting. B. Flexible hours will completely eliminate commuting problems.

Q12

Statement: "Exactly one of A or B will be selected."

Do either of these individually follow? I. A is selected. II. B is selected.

Q13

Using Q12, what combined relation follows?

Q14

Statement: "70% of surveyed customers preferred Product X."

Does "Product X is preferred by 70% of all customers" follow?

Q15

Statement: "No applicant without a degree is eligible. Ravi is eligible."

What follows about Ravi?

LEVEL 3 — EXAM PRESSURE

Passage:

"A company observed that teams using automated code review detected more defects before release than teams relying only on manual review. However, the company also noted differences in team experience and project complexity. It therefore decided to conduct a controlled pilot before adopting automated review company-wide."

Q16

Does the passage prove automated review alone caused the higher defect detection?

Q17

What is the strongest inference about the company's decision?

A. It believes the initial observation is conclusive. B. It believes further controlled evidence is useful before company-wide adoption. C. It believes manual review has no value. D. It has already adopted automation everywhere.

Q18

Can we conclude all teams using automation detect more defects?

Q19

Which possible confounders are explicitly mentioned?

Q20

What hidden assumption supports conducting the pilot?

ANSWERS

Q1

No.

A rule does not guarantee perfect compliance.

$\boxed{\text{No}}$

Q2

No.

Plan "" completion.

$\boxed{\text{No}}$

Q3

No.

Passport may be necessary but not sufficient.

$\boxed{\text{No}}$

Q4

Yes.

$\boxed{80\% > 50\%}$

$\boxed{\text{Yes}}$

Q5

Yes.

"Some workers are engineers" converts to:

$\boxed{\text{Some engineers are workers}}$

Q6

Yes.

The policy's rationale depends on at least some relevant users benefiting.

$\boxed{\text{Implicit}}$

Q7

No.

Ban establishes a rule, not guaranteed behaviour.

$\boxed{\text{Does not follow}}$

Q8

No.

Average does not fix every individual value.

\[\boxed{\text{Does not follow}} \]

Q9

No.

Sequence/correlation does not prove causation.

\[\boxed{\text{Does not follow}} \]

Q10

At least one applicant failed.

\[\boxed{\text{Some applicants did not pass}} \]

Q11

A is the necessary minimal bridge.

\[\boxed{A} \]

B is unnecessarily absolute.

Q12

Neither conclusion individually follows.

The chosen person could be A or B.

\[\boxed{\text{Neither individually}} \]

Q13

Exactly one must be true:

\[\boxed{\text{Either I or II}} \]

Q14

No.

The evidence refers only to surveyed customers.

\[\boxed{\text{Does not follow}} \]

Q15

Rule:

$\text{Eligible} \rightarrow \text{Degree}$

Ravi is eligible.

Therefore:

$\boxed{\text{Ravi has a degree}}$

Q16–Q20

Q16

No.

The passage explicitly identifies other differences between teams.

$\boxed{\text{No}}$

Q17

The controlled pilot shows the company wants stronger evidence before broad adoption.

$\boxed{\text{B}}$

Q18

No.

The passage describes an observed team-level pattern, not a universal guarantee.

$\boxed{\text{No}}$

Q19

Explicit confounders:

- team experience
- project complexity

$\boxed{\text{Team experience and project complexity}}$

Q20

A key assumption is:

A controlled pilot can help determine whether automated review itself produces useful benefits under better-controlled conditions.

$\boxed{\text{Controlled testing can provide better evidence for the decision}}$

' CHAPTER 67 COMPLETE

We have covered:

- statement vs assumption '
- hidden premises '
- bridge assumptions '
- necessary vs possible assumptions '
- negation test '
- multiple assumptions '
- conclusion logic '
- counterexample method '
- scope control '
- sample/population traps '
- average/individual traps '
- strong vs weak conclusions '
- passage inference '
- negative statements '
- consistency/contradiction '
- either-or conclusions '
- necessary/sufficient conditions '
- rule vs behaviour '
- goal vs result '
- plan vs action '
- correlation vs causation '
- author stance/inference '
- speed toolkit '
- L0–L3 practice '

Progress

\[\boxed{67/76}\text{ numbered chapters completed} \]

UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating '
- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles '
- Chapter 66 — Syllogisms '

- Chapter 67 — Statements, Assumptions & Conclusions '
- Chapter 68 — Arguments, Cause-Effect & Course of Action +
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

+CHAPTER 68 — ARGUMENTS, CAUSE-EFFECT & COURSE OF ACTION

Next we will cover:

- strong vs weak arguments
- relevance and practicality
- fact vs opinion
- cause vs correlation
- necessary/sufficient causes
- multiple causes
- common cause
- reverse causation
- identifying effects
- course-of-action tests
- feasibility
- proportionality
- root-cause relevance
- preventive vs corrective action
- extreme-action traps
- placement/GATE reasoning sets
- L0–L3 practice.

USER

next

ASSISTANT

CHAPTER 68 — ARGUMENTS, CAUSE-EFFECT & COURSE OF ACTION

PART 1 — STRONG/WEAK ARGUMENTS, FACT VS OPINION, CAUSE-EFFECT, CORRELATION, COMMON CAUSE & REVERSE CAUSATION

Status: 0=No PROGRESS GATE: Medium–High Placements: High Priority: +Tier 2

This chapter tests whether you can judge reasoning without being distracted by emotion or plausibility.

Core principle:

A strong argument is relevant, logical, practical, and directly connected to the issue.

And for cause—effect:

Just because two things occur together does not mean one caused the other.

1. WHAT IS AN ARGUMENT?

An argument contains:

- a claim/conclusion
- one or more reasons supporting or opposing it

Example:

“The college should extend library hours because many students study after regular class hours.”

Conclusion:

$\backslash \text{Extend library hours} \backslash$

Reason:

$\backslash \text{Students need study access later} \backslash$

2. STRONG ARGUMENT

A strong argument should be:

- relevant
- logically connected
- practically important
- based on reasonable evidence
- not emotionally exaggerated

Example:

Question:

Should public buses be increased during peak hours?

Argument:

Yes, because overcrowding regularly exceeds available capacity.

This directly addresses the issue.

$\backslash \boxed{\text{Strong}} \backslash$

3. WEAK ARGUMENT

Same question.

Argument:

Yes, because buses are usually blue.

Color has no logical connection to bus capacity.

$\boxed{\text{Weak}}$

4. RELEVANCE TEST

Ask:

Does the argument actually address the question?

If not, reject immediately.

Example:

Should school fees be reduced?

Argument:

The school has a large playground.

Irrelevant.

$\boxed{\text{Weak}}$

5. IMPORTANCE TEST

An argument can be related but still trivial.

Question:

Should hospitals maintain backup electricity?

Argument:

Yes, because darkness can be inconvenient.

Related, but weak compared with the real issue:

Life-support and critical medical equipment require continuous power.

The latter is much stronger.

6. PRACTICALITY TEST

Question:

Should a city reduce traffic congestion?

Argument:

Yes, by banning every vehicle permanently.

It may reduce traffic, but it is disproportionate and impractical.

So usually:

$\boxed{\text{Weak}}$

A strong argument should consider realistic consequences.

7. EXTREME ARGUMENT TRAP

Words such as:

- completely
- ban everything
- everyone
- never
- always
- only solution

often indicate overstatement.

Do not reject only because an argument is strong-sounding, but examine proportionality.

8. EMOTIONAL ARGUMENT

Question:

Should an unsafe bridge be repaired?

Weak argument:

Yes, because it would be shameful not to.

Better argument:

Yes, because structural failure may endanger users.

Logical consequences are stronger than emotional appeals.

9. PERSONAL ATTACK IS NOT AN ARGUMENT

Claim:

“Ravi's proposal is wrong because Ravi is lazy.”

This attacks Ravi rather than the proposal.

This is an **ad hominem** style fallacy.

$\boxed{\text{Weak reasoning}}$

10. POPULARITY ARGUMENT

“Everyone uses Product X, so Product X must be the best.”

Popularity does not logically prove superiority.

$\boxed{\text{Weak}}$

11. AUTHORITY ARGUMENT

“A famous actor says this investment is excellent.”

Celebrity status does not necessarily establish financial expertise.

Authority matters only if:

- relevant expertise exists
- the claim falls within that expertise

12. FACT VS OPINION

Fact-type claim

Can in principle be verified.

“The train arrived at 10:15.”

Opinion/value judgment

Depends on evaluation.

“The train service is terrible.”

Do not treat opinion as established fact.

13. STRONG ARGUMENT CAN CONTAIN OPINION

Not every argument must be purely factual.

Example:

“The government should improve emergency healthcare because delays in critical treatment can cost lives.”

This includes a normative recommendation, but its supporting logic is relevant and significant.

14. ARGUMENTS FOR AND AGAINST

A question may give:

Should online exams be used?

Arguments:

I. Yes, they can reduce paper and logistics requirements. II. No, technical failures may disrupt some candidates.

Both can be strong because they address genuine relevant considerations.

Strong arguments do not have to point in the same direction.

15. STRONG ARGUMENT "TRUE CONCLUSION"

An argument can be strong even if another strong counterargument exists.

The question is often:

Is this reason substantial and relevant?

Not:

Does this settle the debate forever?

16. CAUSE AND EFFECT

Cause:

\[A \]

produces or contributes to:

\[B \]

Example:

Heavy rain caused flooding.

Rain = cause Flooding = effect

17. TEMPORAL ORDER

Normally:

$\boxed{\text{Cause precedes effect}}$

If B happened before A, A usually cannot be the cause of B.

This is a basic causal test.

18. CORRELATION

Correlation means two variables change together.

Example:

Ice-cream sales rise when temperature rises.

This association does not mean ice cream causes hot weather.

19. CORRELATION "CAUSATION

$\boxed{A \text{ associated with } B}$

does not prove:

$\boxed{A \text{ to } B}$

Possible alternatives:

- A causes B
- B causes A
- C causes both
- coincidence
- selection bias

20. COMMON-CAUSE EXAMPLE

Observation:

- umbrella sales rise
- road accidents rise

Could umbrella buying cause accidents?

Probably not.

Possible common cause:

$\boxed{\text{Heavy rain}}$

Rain increases umbrella purchases and may increase accident risk.

Structure:

TEXT

```
      Rain
     /  \
Umbrellas Accidents
```

21. REVERSE CAUSATION

Observation:

People who visit doctors more often are less healthy.

Wrong conclusion:

Doctor visits make people unhealthy.

Possible reverse direction:

$\backslash \text{Poor health} \rightarrow \text{More doctor visits} \backslash$

This is reverse causation.

22. THIRD-VARIABLE PROBLEM

Observation:

Students who sleep less use more caffeine.

Possible common cause:

$\backslash \text{Heavy workload} \backslash$

may lead to both:

- less sleep
- more caffeine

So caffeine may not be the original cause of sleep loss.

23. POST HOC FALLACY

Pattern:

A happened, then B happened, therefore A caused B.

This is invalid without more evidence.

Example:

I changed my phone wallpaper and then my exam marks improved.

No causal proof.

24. CAUSAL EVIDENCE IS STRONGER WHEN

Evidence supports causation more strongly when:

- cause occurs before effect
- association is consistent
- plausible mechanism exists
- alternative causes are controlled
- controlled experiment supports it
- dose-response pattern appears
- result replicates

You do not need all of these for aptitude questions, but they help reasoning.

25. EXPERIMENT VS OBSERVATION

Observational data

We observe what happened naturally.

Good for association.

Controlled experiment

We deliberately change one factor while controlling others.

Much stronger for causal inference.

26. RANDOMIZATION

Random assignment helps balance hidden differences between groups.

Therefore, randomized experiments provide stronger causal evidence than simple before-after observations.

27. BEFORE–AFTER TRAP

Company introduces training.

Errors fall afterward.

Can we conclude training caused the decline?

Not necessarily.

Other changes may have occurred:

- new employees
- simpler projects
- better tools
- fewer transactions

Need stronger evidence.

28. NECESSARY CAUSE

A cause is necessary if effect cannot occur without it.

Form:

$\text{E} \rightarrow \text{C}$

Example:

For ordinary human life, oxygen is necessary for normal survival.

But aptitude questions may simplify real-world complexity.

29. SUFFICIENT CAUSE

A cause/condition is sufficient if whenever it occurs, the effect follows:

$\text{C} \rightarrow \text{E}$

One sufficient condition may not be necessary.

30. NECESSARY "SUFFICIENT

Example:

Fuel may be necessary for a conventional petrol car to run.

But merely having fuel is not sufficient:

- engine must function
- battery may be needed
- ignition etc.

So:

$\text{Necessary} \wedge \text{Sufficient}$

31. MULTIPLE SUFFICIENT CAUSES

An effect may have several possible causes.

Example:

A server outage can result from:

- power failure
- network failure
- software crash

Therefore, seeing the outage does not tell us which cause occurred unless additional evidence is given.

32. MULTIPLE CAUSES CAN COMBINE

Some effects require several contributing causes.

Example:

Traffic congestion may depend on:

- vehicle volume
- road capacity
- accidents
- signals
- weather

Avoid single-cause thinking when evidence is incomplete.

33. CAUSAL LANGUAGE

Strong causal words:

- causes
- leads to
- results in
- produces
- because of

Weak/non-causal words:

- associated with
- linked to
- correlated with
- observed with
- accompanied by

Do not upgrade weak language into strong causation.

34. EFFECT IDENTIFICATION

Statement:

Due to continuous heavy rain, several low-lying roads were flooded.

Cause:

$\boxed{\text{Continuous heavy rain}}$

Effect:

$\boxed{\text{Road flooding}}$

35. MULTIPLE EFFECTS

One cause can produce multiple effects.

Example:

Power outage:

- lights off
 - machines stop
-

- internet routers fail

Do not assume one cause !' one effect only.

36. MULTIPLE CAUSES, ONE EFFECT

Flooding may arise from:

- heavy rainfall
- blocked drains
- river overflow

Hence:

$\boxed{\text{Effect does not uniquely identify cause}}$

37. CAUSE–EFFECT PAIR QUESTIONS

Two statements:

I. Heavy rain occurred overnight. II. Several roads were flooded in the morning.

Likely relation:

$I \rightarrow II$

if context supports it.

38. COMMON EFFECT QUESTION

I. Fuel prices rose sharply. II. Transportation costs increased.

Likely:

$I \rightarrow II$

But if wording only says both occurred without mechanism/context, avoid excessive certainty.

39. COMMON CAUSE QUESTION

I. Attendance at outdoor events fell. II. Umbrella sales rose.

Possible common cause:

$\boxed{\text{Heavy rain}}$

Neither necessarily caused the other.

40. INDEPENDENT EVENTS

I. A software company launched a product. II. A cricket team won a match.

Unless a connection is supplied:

$\boxed{\text{Independent}}$

Do not invent relationships.

41. ARGUMENT STRENGTH — EVIDENCE QUALITY

Question:

Should an unsafe machine be replaced?

Argument A:

Yes, because repeated inspections found critical structural defects.

Argument B:

Yes, because one employee dislikes its color.

A is strong.

B is weak.

Evidence quality matters.

42. DIRECT CONSEQUENCE ARGUMENT

Question:

Should backup servers be maintained?

Strong:

A single server failure could otherwise stop a critical service.

This identifies a direct operational consequence.

43. REMOTE CONSEQUENCE TRAP

Weak argument:

Backup servers should be maintained because people generally like technology.

No meaningful connection.

44. PRINCIPLE-BASED ARGUMENT

Question:

Should examination results be rechecked when credible calculation errors are found?

Strong:

Assessment decisions should be based on accurate marks.

This is a relevant principle connected to fairness/accuracy.

45. ONE EXAMPLE DOES NOT PROVE GENERAL RULE

Observation:

One remote employee was unproductive.

Conclusion:

Remote work makes employees unproductive.

Classic overgeneralization.

$\boxed{\text{Weak}}$

46. ANECDOTAL EVIDENCE

A single story may illustrate something but usually does not establish a broad statistical claim.

Example:

My friend earned money from cryptocurrency, therefore crypto is always profitable.

Weak.

47. FALSE DILEMMA

Either ban smartphones completely or students can never concentrate.

This ignores other possibilities:

- classroom restrictions
- silent mode
- designated times

Weak reasoning because it creates an artificial two-option choice.

48. SLIPPERY SLOPE

If employees work from home one day per week, soon nobody will ever come to the office.

This assumes an extreme chain without evidence.

Usually weak.

49. CIRCULAR REASONING

This policy is good because it is an excellent policy.

Conclusion is merely restated as reason.

No independent support.

$\boxed{\text{Weak}}$

50. HASTY GENERALIZATION

Small/unrepresentative sample !' broad conclusion.

Example:

Two customers complained, therefore every customer hates the product.

Weak.

51. STRAWMAN

Misrepresenting an opposing view to make it easier to attack.

Example:

Original:

Reduce unnecessary car usage.

Strawman:

They want nobody to own a car.

Different claim.

52. CAUSAL ARGUMENT EVALUATION

Claim:

"Installing air purifiers caused productivity to increase."

Ask:

1. Was productivity measured before/after?
2. Was there a comparison group?
3. Did other conditions change?
4. Is there a plausible mechanism?
5. Was the effect consistent?

This identifies causal weakness.

53. CONTROL GROUP

Suppose:

- treatment group gets new teaching method
- control group uses old method
- groups otherwise similar

If treatment group improves substantially more, causal inference becomes stronger.

54. CONFOUNDING VARIABLE

A confounder influences both supposed cause and outcome.

Example:

Study:

Coffee drinkers have higher exam scores.

Possible confounder:

$\boxed{\text{Study hours}}$

Students who study longer may both drink more coffee and score higher.

55. MEDIATOR VS CONFOUNDER — BASIC IDEA

Suppose:

$\text{Exercise} \rightarrow \text{Weight Loss} \rightarrow \text{Better Health}$

Weight loss may mediate the effect.

You do not need advanced causal terminology for aptitude, but understand that causal chains can have intermediate steps.

56. CAUSAL CHAIN

$A \rightarrow B \rightarrow C$

Example:

Heavy rain !' waterlogging !' traffic delay.

A may indirectly contribute to C through B.

57. BREAKING THE CHAIN

If B does not happen, A may no longer produce C through that pathway.

This becomes relevant when choosing interventions.

58. ROOT CAUSE

Root cause is an underlying driver rather than a surface symptom.

Example:

Symptom:

frequent server failures

Possible root cause:

defective power supply

Action addressing only user complaints may not fix the root problem.

59. SYMPTOM VS CAUSE

Suppose:

Students arrive late because buses are consistently delayed.

Symptom:

\[Students\ late \]

Possible root cause:

\[Bus\ delays \]

Strong intervention should often address the relevant cause.

60. CAUSAL DIRECTION CHECK

Whenever two events are linked, ask:

Which happened first?

Then:

Is there a plausible mechanism?

Then:

Could a third factor explain both?

This 3-question test solves many cause-effect questions.

61. STRONG-ARGUMENT 4-TEST

For each argument ask:

R — Relevant?

Does it address the issue?

I — Important?

Is the consequence substantial?

L — Logical?

Does reasoning connect properly?

P — Practical?

Is it realistic/proportionate?

Memory:

$\boxed{\text{RILP}}$

62. EXAMPLE USING RILP

Question:

Should emergency exits be checked regularly?

Argument:

Yes, because blocked exits can delay evacuation during emergencies.

Relevant ' Important ' Logical ' Practical '

$\boxed{\text{Strong}}$

63. WEAK USING RILP

Argument:

Yes, because emergency-exit signs look attractive.

Relevant to safety? 'L

Immediately weak.

64. CAUSE–EFFECT 4-TEST

Use:

T — Time

Cause before effect?

M — Mechanism

Plausible connection?

A — Alternatives

Other explanations?

E — Evidence

How strong is evidence?

Memory:

$\boxed{\text{TMAE}}$

65. COMMON TRAPS

& pTRAP 1 — Emotional reason = strong

Not necessarily.

& pTRAP 2 — Popular opinion = proof

No.

& pTRAP 3 — After = because

No.

& pTRAP 4 — Correlation = causation

No.

& pTRAP 5 — One cause assumed when many possible

Avoid.

& pTRAP 6 — Reverse causation ignored

Check direction.

& pTRAP 7 — Third variable ignored

Check confounders.

& pTRAP 8 — Extreme/impractical solution

Often weak.

PRACTICE — LEVEL 0

Q1

Question: Should hospitals have backup power?

Argument: "Yes, because critical equipment may require continuous electricity."

Strong or weak?

Q2

Question: Should libraries remain quiet?

Argument: "Yes, because books are rectangular."

Strong or weak?

Q3

A happened before B. Does that alone prove A caused B?

Q4

Two variables are correlated. Must one cause the other?

Q5

Can one effect have several possible causes?

LEVEL 1

Q6

Observation: Umbrella sales and road accidents both increase on rainy days. Does umbrella use necessarily cause accidents?

Q7

Observation: Sick people visit doctors more often. Can "doctor visits cause sickness" be a reverse-causation error?

Q8

Question: Should a dangerous machine be replaced?

Argument: "Yes, because repeated inspections show serious safety defects."

Strong or weak?

Q9

Question: Should traffic congestion be reduced by permanently banning all transport?

Is this automatically a strong argument just because it would reduce traffic?

Q10

Does "A is necessary for B" mean A is sufficient for B?

LEVEL 2

Q11

Study finds students who sleep more score better.

Can we immediately conclude more sleep causes higher scores?

Q12

Company introduces training and defects decrease.

Give one alternative explanation besides training.

Q13

A controlled randomized experiment shows treatment group improved while control group did not. Is this stronger causal evidence than simple correlation?

Q14

Fuel prices rise, and transport costs later rise. Is a causal relation plausible?

Q15

A city has more firefighters and more fires. Does this prove firefighters cause fires?

LEVEL 3 — EXAM PRESSURE

A company observes:

Employees who use a new productivity app complete more tasks per week than employees who do not use it.

Additional information:

- app users tend to be younger
- app users volunteered to use the app
- task complexity differs across teams

Q16

Does the observation alone establish that the app causes higher productivity?

Q17

Name two potential confounding/selection factors.

Q18

What would strengthen the causal claim?

A. A randomized trial with comparable groups B. More advertisements for the app C. A testimonial from one high-performing employee D. Changing the app logo

Q19

If productive employees are more likely to choose the app, what causal problem is this?

Q20

What is the strongest safe conclusion from the original observation?

ANSWERS

Q1

Directly related to safety and critical operation.

$\boxed{\text{Strong}}$

Q2

Irrelevant.

$\boxed{\text{Weak}}$

Q3

No.

$\boxed{\text{No}}$

Sequence is necessary for causation but not sufficient evidence.

Q4

No.

$\boxed{\text{No}}$

Q5

Yes.

$\boxed{\text{Yes}}$

Q6

No.

Rain may be a common cause.

$\boxed{\text{No}}$

Q7

Yes.

Poor health may cause more doctor visits.

$\boxed{\text{Yes}}$

Q8

Relevant, significant, evidence-based.

\[\boxed{\text{Strong}} \]

Q9

No.

It may be extreme and impractical.

\[\boxed{No} \]

Q10

No.

Necessary and sufficient are different concepts.

\[\boxed{No} \]

Q11

No.

Other factors may explain the association.

\[\boxed{No} \]

Q12

Examples:

- easier projects
- more experienced staff
- new testing tools
- reduced workload

Any one is sufficient.

Q13

Yes.

Randomization and control reduce alternative explanations.

\[\boxed{Yes} \]

Q14

Yes, it is plausible because fuel is an input to transportation.

But exact causality/size still depends on context.

\[\boxed{Plausible} \]

Q15

No.

Larger/more dangerous fires may require more firefighters.

This is a classic confounding/reverse-direction issue.

$\boxed{\text{No}}$

Q16–Q20

Q16

No.

This is observational association with several uncontrolled differences.

$\boxed{\text{No}}$

Q17

Examples:

- age
- voluntary app adoption
- task complexity
- baseline motivation
- team differences

Any two valid.

Q18

Best:

$\boxed{A}$

A randomized controlled comparison would strengthen causal inference.

Q19

This is:

$\boxed{\text{Selection bias / reverse causal selection}}$

More productive people may self-select into app use.

Q20

Strongest safe conclusion:

In the observed company data, app users completed more tasks per week than non-users.

That states association without claiming causation.

$\boxed{\text{Association, not proven causation}}$

CHAPTER 68 STATUS

Covered so far:

- strong vs weak arguments '
- relevance/importance/practicality '
- fact vs opinion '
- emotional/popularity/authority traps '
- logical fallacies '
- cause vs effect '
- correlation vs causation '
- temporal-order test '
- common cause '
- reverse causation '
- confounding '
- necessary vs sufficient cause '
- multiple causes/effects '
- experimental evidence '
- root cause vs symptom '
- RILP argument test '
- TMAE causal test '
- L0–L3 practice '

But **Chapter 68 is not complete yet.**

Still remaining:

- course of action
- corrective vs preventive action
- feasibility and proportionality
- root-cause relevance
- extreme-action rejection
- short-term vs long-term action
- immediate emergency action
- multiple valid courses of action
- action assumptions
- action vs wishful statement
- final combined argument/cause/action sets
- Chapter 68 speed toolkit

Progress

67/76 chapters completed

Current: Chapter 68 — Arguments, Cause–Effect & Course of Action

NEXT

Chapter 68 Part 2 — Course of Action, Feasibility, Preventive/Corrective Measures, Root-Cause Actions & Final Chapter Toolkit

USER

next

USER

next

ASSISTANT

CHAPTER 68 — ARGUMENTS, CAUSE–EFFECT & COURSE OF ACTION

PART 2 — COURSE OF ACTION, FEASIBILITY, PREVENTIVE/CORRECTIVE MEASURES, ROOT-CAUSE ACTIONS & FINAL TOOLKIT

Status: 0% PROGRESS ! ' COMPLETE GATE: Medium–High Placements: High Priority: +Tier 2

A **course of action** question asks:

Given a problem or situation, what should reasonably be done?

The correct action should usually be:

- relevant
- practical
- proportionate
- directed at the actual problem
- capable of improving the situation

Core rule:

$$\text{Good action} = \text{Relevant} + \text{Feasible} + \text{Effective} + \text{Proportionate}$$

1. WHAT IS A COURSE OF ACTION?

Statement:

Several accidents occur at a poorly lit intersection at night.

Possible action:

Install adequate street lighting and review road-safety measures.

This directly addresses the problem.

$\boxed{\text{Reasonable course of action}}$

2. WHAT IS NOT A COURSE OF ACTION?

Same problem.

Citizens should hope accidents stop.

This is a wish, not an actionable measure.

A course of action must usually involve something that can actually be done.

3. COURSE OF ACTION TEST

Use four questions:

R — Relevant

Does it address the stated problem?

F — Feasible

Can it realistically be implemented?

E — Effective

Could it reasonably help?

P — Proportionate

Is it not unnecessarily extreme?

Memory:

$\boxed{\text{RFEP}}$

4. EXAMPLE

Problem:

Students are frequently late because college buses arrive late.

Action:

Review bus schedules and improve bus punctuality.

Relevant ' Feasible ' Effective ' Proportionate '

$\boxed{\text{Valid}}$

5. IRRELEVANT ACTION

Problem:

College buses are late.

Action:

Increase library books.

No connection.

$\boxed{\text{Invalid}}$

6. IMPRACTICAL ACTION

Problem:

Traffic is heavy in one area.

Action:

Stop all movement in the entire city forever.

Even if it eliminates traffic, it is wildly impractical.

$\boxed{\text{Invalid}}$

7. EXTREME ACTION TRAP

Typical weak actions include:

- ban everything
- dismiss everyone
- close an institution permanently
- punish all people
- stop all activity

unless the problem is severe enough to justify them.

The action should match the seriousness of the problem.

8. PROPORTIONALITY

Problem:

One employee made a minor reporting error.

Action:

Fire the entire department.

Disproportionate.

A better action:

Correct the error and improve verification/training if needed.

9. CORRECTIVE ACTION

Corrective action addresses an existing problem.

Example:

Problem:

A water pipe has burst.

Corrective action:

Repair the damaged pipe.

10. PREVENTIVE ACTION

Preventive action reduces the chance of recurrence.

Same pipe problem:

Introduce periodic inspection and maintenance.

Thus:

- repair = corrective
- maintenance = preventive

Both may be valid.

11. CORRECTIVE + PREVENTIVE

A strong response often includes both:

$\boxed{\text{Fix now + prevent recurrence}}$

Example:

Data breach:

- isolate affected systems immediately
- patch vulnerability
- investigate cause
- strengthen future security

12. ROOT-CAUSE ACTION

Problem:

Employees are repeatedly missing deadlines because approvals take too long.

Weak action:

Tell employees to work faster.

Root-cause action:

Review and streamline the approval process.

The second addresses the underlying bottleneck.

13. SYMPTOM VS ROOT CAUSE

Symptom:

$\backslash \text{Late delivery} \backslash$

Possible cause:

$\backslash \text{Repeated supplier delay} \backslash$

Action at symptom:

Apologize each time.

Action at cause:

Improve supplier reliability / use alternatives.

Root-cause action is usually stronger.

14. BUT IMMEDIATE SYMPTOM ACTION MAY STILL MATTER

Suppose there is a fire.

Long-term:

Improve fire-safety systems.

But immediate action:

Evacuate people and call emergency services.

In emergencies, immediate harm reduction comes first.

15. EMERGENCY PRIORITY

For immediate danger:

1. protect life
2. contain damage
3. restore essential services
4. investigate root cause
5. prevent recurrence

This order is often logically strong.

16. SHORT-TERM VS LONG-TERM ACTION

Problem:

Severe flooding has disrupted a city.

Short-term:

- rescue stranded people
- provide shelter
- restore essential supplies

Long-term:

- improve drainage
- strengthen flood planning
- review land-use issues

Both can be valid if relevant.

17. ACTION MUST FOLLOW FROM THE PROBLEM

Statement:

A building has developed structural cracks.

Action:

Conduct structural inspection.

Strong.

Action:

Repaint the building.

May improve appearance but not structural safety.

Weak.

18. INFORMATION-GATHERING ACTION

Sometimes the cause is unclear.

Then a valid first action may be:

$\boxed{\text{Investigate before taking irreversible action}}$

Example:

Product failures suddenly increased.

Good action:

Audit production/process data to identify the cause.

Not:

Immediately dismiss all workers.

19. WHEN INVESTIGATION IS NECESSARY

If:

- evidence is incomplete
- cause is uncertain
- action would be costly
- action would be irreversible

then investigation can be the strongest course.

20. INVESTIGATION IS NOT ALWAYS ENOUGH

Emergency:

A gas leak is detected.

Only saying:

Investigate why it happened

is insufficient as the immediate action.

First:

evacuate and stop the leak safely.

Then investigate.

21. ACTION VS PUNISHMENT

Problem:

Students are cheating in exams.

Possible:

Strengthen invigilation and exam procedures.

Relevant.

Possible:

Punish offenders according to fair rules after verification.

Also potentially relevant.

But:

Punish every student.

Invalid because innocent students are included.

22. DUE PROCESS

When allegations are unverified, a strong action often includes:

$\boxed{\text{Investigate / verify before punishment}}$

Do not assume accusation = guilt.

23. COLLECTIVE-PUNISHMENT TRAP

Problem caused by a few people.

Action:

Penalize everyone.

Usually weak because:

- overbroad
- unfair
- does not target responsible actors

24. FEASIBILITY

An action can be theoretically useful yet impossible.

Example:

Problem:

Air pollution is high.

Action:

Replace every vehicle and factory in the country tomorrow.

Not feasible.

Better:

Implement phased emission controls, monitoring, and public-transport measures.

25. COST-BENEFIT THINKING

Course-of-action questions do not require exact economics, but an action should not impose massive cost for tiny benefit unless justified.

Proportionality matters.

26. DIRECT VS INDIRECT ACTION

Problem:

Cybersecurity attacks increased.

Direct:

strengthen security controls and patch vulnerabilities.

Indirect but possibly useful:

train employees against phishing.

Both can be valid because attacks may involve multiple pathways.

27. MULTIPLE VALID ACTIONS

Sometimes conclusion options are:

I. Fix immediate problem. II. Take preventive steps.

Both may follow.

Do not assume only one action can be correct.

28. EXAMPLE

Problem:

Several people became ill after drinking water from a public source.

Actions:

I. Test and temporarily restrict use of the water source. II. Investigate contamination and correct the source.

Both:

- relevant
- protective
- practical

$\boxed{\text{Both valid}}$

29. “SHOULD” DOES NOT MEAN GUARANTEE

An action need not guarantee success.

It must be a reasonable response.

Example:

Conduct awareness campaigns to reduce fraud.

It may not eliminate fraud completely, but can still be valid.

30. OVERCERTAINTY TRAP

Invalid reasoning:

If an action does not completely solve the problem, it is useless.

False.

Partial mitigation can be worthwhile.

31. ACTION MUST NOT CREATE A BIGGER PROBLEM

Problem:

Students use phones during class.

Action:

Shut down all mobile networks in the city.

Disproportionate and creates unrelated harms.

Weak.

32. TARGETED ACTION

Better:

- classroom phone rules
- designated storage
- enforcement during lessons

Targeted actions are usually stronger.

33. POLICY ACTION

Problem:

Accident rates are high at a dangerous intersection.

Possible actions:

- road engineering review
- better signals
- lighting
- speed enforcement

These target known safety factors.

34. ACTION BASED ON ASSUMPTION

Action:

Install speed breakers to reduce accidents.

Implicit assumption:

Excessive speed contributes to at least some of those accidents.

If speed is not actually relevant, action may be weak.

Thus course-of-action reasoning depends on causal relevance.

35. ACTION SHOULD MATCH CAUSE

If cause:

$\boxed{C}$

and problem:

$\boxed{P}$

A good action often attempts:

$\boxed{\text{reduce/remove } C}$

or:

$\boxed{\text{protect against } P}$

36. TWO TYPES OF ACTION

Cause-control

Stop the problem from arising.

Effect-control

Reduce harm after it occurs.

Example flood:

- drainage improvement = cause/risk control
- rescue shelters = effect control

Both useful.

37. COURSE-OF-ACTION EXAMPLE

Statement:

Many passengers miss trains because ticket counters are overcrowded.

Actions:

I. Add ticket counters/self-service systems during peak periods. II. Cancel all trains.

I valid.

II absurd.

$\boxed{\text{Only I}}$

38. ANOTHER EXAMPLE

Statement:

A company's customer complaints about delayed responses have increased.

Actions:

I. Study response bottlenecks and improve support staffing/process. II. Stop accepting customer complaints.

I valid.

II hides the symptom instead of solving the problem.

39. SUPPRESSING REPORTING "SOLVING PROBLEM

This is a classic trap.

Problem:

Number of reported incidents increased.

Action:

Stop people from reporting incidents.

This may reduce reports but not actual incidents.

Invalid.

40. METRIC GAMING

If an action improves the measured number without solving the underlying issue, be suspicious.

Example:

To reduce reported bugs, remove the bug-report button.

Metric falls; quality does not necessarily improve.

Weak.

41. ROOT-CAUSE ANALYSIS

Useful questions:

- Why did this happen?
 - What process failed?
 - Is this recurring?
 - What condition enables recurrence?
-

- What can reasonably change?

This helps identify strong actions.

42. FIVE-WHY INTUITION

You do not need formal Five Whys, but repeatedly asking “why?” can distinguish:

Symptom:

Orders late.

Why?

Packing delayed.

Why?

Machine frequently fails.

Why?

Maintenance neglected.

Root action:

repair + preventive maintenance.

43. PREVENTIVE ACTION EXAMPLE

Problem:

Repeated phishing incidents.

Corrective:

secure compromised accounts.

Preventive:

- phishing awareness
- multi-factor authentication
- email filtering

Both matter.

44. PREVENTION VS OVERREACTION

Problem:

One isolated software bug occurred.

Action:

Permanently shut down the entire application.

Usually disproportionate unless bug creates severe uncontrollable danger.

45. COURSE OF ACTION AND EVIDENCE

More extreme actions require stronger evidence.

General principle:

$\boxed{\text{Higher action cost} \rightarrow \text{higher evidence requirement}}$

46. REVERSIBILITY

When uncertain:

Prefer reversible, information-producing actions before irreversible ones.

Example:

- pilot program before nationwide rollout
- temporary suspension during investigation
- test before replacing entire system

This is often strong reasoning.

47. PILOT TEST

Problem:

Organisation is considering a new process but evidence is limited.

Good action:

Run a controlled pilot and evaluate results.

This balances uncertainty and decision-making.

48. NO-ACTION OPTION

Sometimes:

No action is necessary.

This is valid only if:

- issue is trivial
- evidence insufficient even to justify intervention
- existing measures already address it

Do not choose in face of clear serious harm.

49. WISHFUL ACTION

People should simply behave better.

May be desirable but is often not a concrete implementation plan.

A stronger course may specify:

- incentives
- rules
- training
- monitoring
- infrastructure

50. MORAL STATEMENT VS ACTION

“Corruption is bad.”

This is a judgment.

A course of action would be:

strengthen transparent procedures, audits, reporting and enforcement.

51. OVERLY VAGUE ACTION

Problem:

Cybersecurity is poor.

Action:

Do something about security.

Too vague.

A good course should be actionable enough to address the issue.

52. TOO SPECIFIC WITHOUT EVIDENCE

Opposite trap:

Fire employee X immediately

when the statement only says errors increased.

Specific blame is unsupported.

53. STRONG ACTION SITS BETWEEN TWO EXTREMES

Avoid:

- useless vagueness
- unsupported over-specificity

Prefer:

investigate cause and implement targeted corrective measures.

54. COURSE OF ACTION — CAUSAL TEST

Before approving an action:

1. What problem?
2. What likely cause?
3. Does action influence that cause/effect?
4. Is it feasible?
5. Is it proportionate?

55. ACTION THAT ONLY SHIFTS THE PROBLEM

Problem:

Traffic congestion in Road A.

Action:

Divert every vehicle to already congested Road B.

May merely shift congestion.

Weak unless overall network capacity supports it.

56. UNINTENDED CONSEQUENCES

Good actions consider likely side effects.

Example:

Ban all online communication to stop phishing.

May stop phishing but destroys normal operations.

Disproportionate.

57. PREVENTIVE POLICY EXAMPLE

Problem:

Accidents increase due to drunk driving.

Strong actions:

- enforcement
- testing
- awareness
- safe transport alternatives

All directly related.

58. ACTION COMBINATION

Often strongest real response:

$\boxed{\text{Immediate containment + investigation + correction + prevention}}$

Memory:

$\boxed{\text{ICP}}$

59. ICP MODEL

I — Immediate response

Protect people/system.

I — Investigate

Find cause.

C — Correct

Repair current failure.

P — Prevent

Reduce recurrence.

Very useful for complex situations.

60. EXAMPLE — DATABREACH

Problem:

Sensitive data may be exposed.

IICP:

1. isolate affected systems
2. investigate scope/source
3. patch and restore securely
4. strengthen preventive controls

This is much stronger than:

Ignore it and hope it stops.

61. STRONG/WEAK COURSE EXAMPLE

Problem:

Several students fail because prerequisite concepts are weak.

I. Provide remedial instruction/tutorial support. II. Remove the subject permanently.

I targets the issue proportionately.

II is excessive.

$\boxed{\text{Only I}}$

62. MULTIPLE-ACTION QUESTION

Problem:

A disease outbreak is detected.

Actions:

I. Identify affected cases and provide treatment. II. Investigate source and implement measures to reduce spread.

Both are relevant.

$\boxed{\text{Both follow}}$

63. WHEN TWO ACTIONS CONFLICT

Problem:

A bridge may be structurally unsafe.

I. Close it temporarily pending inspection. II. Continue normal traffic regardless.

They conflict.

Only I is consistent with safety under credible risk.

64. ACTION MUST NOT CONTRADICT THE GIVEN FACT

Statement:

Water is contaminated.

Action:

Encourage people to drink more of the water.

Directly contradicts safety problem.

Reject.

65. COURSE OF ACTION VS ASSUMPTION

Statement:

Road accidents increased.

Assumption asks:

What hidden belief underlies an argument/action?

Course asks:

What should reasonably be done now?

Do not confuse question types.

66. ARGUMENT VS COURSE OF ACTION

Argument

Reason for/against a proposal.

Course of action

Actual step recommended in response.

Example:

Problem:

Hospital waiting times are long.

Argument:

Long waits delay treatment.

Action:

Analyze capacity and improve scheduling/staffing.

67. CAUSE-EFFECT + ACTION INTEGRATION

Observation:

\[Problem \]

Possible cause:

\[Cause \]

Action:

\[Intervention \]

Strong reasoning:

\[\boxed{\text{Cause} \rightarrow \text{Problem}, \quad \text{Intervention} \rightarrow \text{reduce Cause/Problem}} \]

68. GATE-STYLE LOGICAL CAUTION

In GATE, avoid accepting:

“This action will definitely solve the problem”

unless logically guaranteed.

Prefer:

“This action is justified / could help / is relevant”

when the evidence supports only that.

69. PLACEMENT-STYLE SPEED METHOD

For each action:

1. Connected?

No !' reject.

2. Practical?

No !' reject.

3. Targets cause/effect?

No !' suspicious.

4. Too extreme?

Yes!' usually reject.

5. Creates bigger issue?

Yes!' reject.

70. COMMON COURSE-OF-ACTION TRAPS

& pTRAP 1 — Wish = action

No.

& pTRAP 2 — Punish everyone

Overbroad.

& pTRAP 3 — Extreme ban

Usually disproportionate.

& pTRAP 4 — Hide the metric

Doesn't solve cause.

& pTRAP 5 — Investigate while ignoring emergency

Immediate protection first.

& pTRAP 6 — Immediate fix only

May miss prevention.

& pTRAP 7 — Prevention only during active emergency

Current danger still needs response.

& pTRAP 8 — Unsupported blame

Verify first.

71. FINAL ARGUMENT TOOLKIT

Use:

\[\boxed{RILP} \]

- Relevant
- Important
- Logical
- Practical

72. FINAL CAUSE-EFFECT TOOLKIT

Use:

\[\boxed{TMAE} \]

- Temporal order
 - Mechanism
 - Alternative explanations
 - Evidence
-

73. FINAL COURSE-OF-ACTION TOOLKIT

Use:

$\boxed{\text{RFEP}}$

- Relevant
- Feasible
- Effective
- Proportionate

For complex problems:

$\boxed{\text{IICP}}$

- Immediate
 - Investigate
 - Correct
 - Prevent
-

PRACTICE — LEVEL 0

Q1

Problem: A pipe is leaking.

Action: Repair the pipe.

Valid?

Q2

Problem: A pipe repeatedly leaks.

Action: Introduce periodic maintenance after repair.

Corrective or preventive?

Q3

Problem: Traffic congestion is high.

Action: Permanently ban every person from travelling.

Reasonable?

Q4

Problem: A gas leak is detected.

Should investigation come before immediate safety measures?

Q5

Problem: Customer complaints increased.

Action: Delete the complaint system.

Does this solve the underlying problem?

LEVEL 1

Q6

Problem: Several accidents occur at a dark intersection.

Action: Improve lighting and road-safety controls.

Valid?

Q7

Problem: One worker is accused of misconduct.

Action: Punish the entire workforce.

Valid?

Q8

Problem: Software failures increased, cause unknown.

Action: Investigate logs/process before blaming a specific team.

Valid?

Q9

Problem: Students use phones during class.

Action: Introduce classroom phone restrictions.

More proportionate than shutting down the city's mobile network?

Q10

Problem: A serious fire is active.

Which comes first: evacuation or long-term fire-safety policy review?

LEVEL 2

Q11

Statement: A public water source may be contaminated.

Actions:

I. Test the water and warn/restrict use until safety is known. II. Investigate and correct the contamination source.

Which follow?

Q12

Problem: Phishing incidents recur.

Which combination is stronger?

A. Reset compromised accounts only. B. Reset accounts + MFA + training/filtering.

Q13

Problem: A new policy may improve productivity, but evidence is uncertain.

Which is more reasonable?

A. Nationwide/company-wide irreversible adoption immediately B. Controlled pilot

Q14

Problem: An unsafe machine has repeated critical failures.

Which is stronger?

A. Continue normal operation without inspection B. Suspend/inspect and repair or replace as justified

Q15

Problem: Reported crime increased.

Is “stop accepting crime reports” a valid corrective measure?

LEVEL 3 — EXAM PRESSURE

Statement:

A company discovers that a recently released software update is causing crashes for a significant number of users. The exact technical cause is not yet known.

Actions:

I. Temporarily halt further rollout of the update where practical. II. Investigate crash logs and reproduce the failure. III. Delete all user complaints to protect the company's reputation. IV. Develop and test a fix before resuming full rollout. V. Permanently shut down all company software products.

Q16

Does I follow as a reasonable action?

Q17

Does II follow?

Q18

Does III follow?

Q19

Does IV follow?

Q20

Does V follow?

ANSWERS

Q1

Yes.

It directly corrects the current problem.

\[\boxed{Yes} \]

Q2

Preventive.

\[\boxed{Preventive} \]

Q3

No.

Disproportionate and infeasible.

\[\boxed{No} \]

Q4

No.

Immediate safety comes first.

\[\boxed{Evacuate/contain\ first} \]

Q5

No.

It suppresses reporting rather than solving the issue.

\[\boxed{No} \]

Q6

Yes.

Relevant and proportionate.

\[\boxed{Valid} \]

Q7

No.

Collective punishment is overbroad.

\[\boxed{Invalid} \]

Q8

Yes.

The cause is uncertain, so evidence gathering is appropriate.

\[\boxed{Valid} \]

Q9

Yes.

Targeted classroom restrictions are much more proportionate.

\[\boxed{Yes} \]

Q10

Evacuation.

\[\boxed{\text{Immediate life safety first}} \]

Q11

Both.

I protects people immediately.

II addresses root cause.

\[\boxed{I\text{ and }II} \]

Q12

B.

It includes:

- correction
- prevention

\[\boxed{B} \]

Q13

Controlled pilot.

\[\boxed{B} \]

It is reversible and information-producing.

Q14

B.

$\boxed{B}$

Safety evidence justifies inspection/suspension.

Q15

No.

Reducing reports is not the same as reducing crime.

$\boxed{\text{No}}$

Q16–Q20

Software update causes significant crashes; exact cause unknown.

Q16

Temporarily pausing further rollout can limit additional harm.

$\boxed{\text{Yes}}$

Q17

Cause is unknown, so investigation is essential.

$\boxed{\text{Yes}}$

Q18

Deleting complaints hides evidence and harms diagnosis.

$\boxed{\text{No}}$

Q19

Once cause is understood, testing a fix before broad rollout is appropriate.

$\boxed{\text{Yes}}$

Q20

Shutting down every product is unrelated and extreme.

$\boxed{\text{No}}$

Thus reasonable actions:

$\boxed{\text{I, II, IV}}$

' CHAPTER 68 COMPLETE

We have now covered:

- strong vs weak arguments '
- relevance, importance, logic, practicality '
- fact vs opinion '
- common argument fallacies '
- cause vs effect '
- correlation vs causation '
- common cause '
- reverse causation '
- confounding variables '
- necessary vs sufficient causes '
- causal evidence '
- root cause vs symptom '
- course-of-action reasoning '
- feasibility '
- proportionality '
- corrective actions '
- preventive actions '
- emergency actions '
- short-term vs long-term responses '
- investigation before irreversible action '
- due process '
- metric-gaming traps '
- multiple valid actions '
- RILP / TMAE / RFEP / IICP shortcuts '
- L0–L3 practice '

Progress

\[\boxed{68/76}\text{ numbered chapters completed} \]

UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series '
- Chapter 58 — Analogy, Classification & Odd-One-Out '
- Chapter 59 — Coding-Decoding '
- Chapter 60 — Blood Relations '
- Chapter 61 — Direction Sense '
- Chapter 62 — Ranking & Ordering '
- Chapter 63 — Linear Seating '
- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles '

- Chapter 66 — Syllogisms '
- Chapter 67 — Statements, Assumptions & Conclusions '
- Chapter 68 — Arguments, Cause–Effect & Course of Action '
- Chapter 69 — Data Sufficiency +
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

CHAPTER 69 — DATA SUFFICIENCY

Next we will cover:

- what sufficiency actually means
- answer-vs-solution distinction
- statement I alone
- statement II alone
- both together
- either statement alone
- both insufficient
- numerical data sufficiency
- algebraic sufficiency
- ranking/order sufficiency
- geometry sufficiency
- logical/puzzle sufficiency
- yes/no data sufficiency
- hidden-domain traps
- redundancy
- consistency
- GATE/placement answer-option conventions
- L0–L3 practice.

USER

next

ASSISTANT

CHAPTER 69 — DATA SUFFICIENCY

PART 1 — WHAT “SUFFICIENT” MEANS, STATEMENT I/II LOGIC, YES/NO QUESTIONS, NUMERICAL/ALGEBRAIC/RANKING & GEOMETRY DATA SUFFICIENCY

Status: $\emptyset$ = Not in PROGRESS GATE: High Placements: Very High Priority: $\emptyset$ = Yes 1

Data Sufficiency is different from ordinary aptitude questions.

You are **not mainly asked to calculate the answer**.

You are asked:

Is the information given enough to determine the answer uniquely?

That distinction is the entire chapter.

1. THE BASIC FORMAT

A question is followed by two statements.

Example:

What is x ?

I. $x+5=10$ II. $x>0$

You test:

- Statement I alone
- Statement II alone
- both together, if necessary

2. SUFFICIENT "USEFUL"

A statement may be useful but still insufficient.

Example:

Find x .

Statement:

$x > 5$

This narrows the possibilities, but:

$x = 6, 7, 8, \dots$

many answers remain.

So:

$\boxed{\text{Insufficient}}$

3. SUFFICIENT = UNIQUE ANSWER

For a numerical question:

A statement is sufficient if it forces exactly one answer.

Example:

$2x = 14$

Then:

$\lceil x=7 \rceil$

Unique.

$\lceil \boxed{\text{Sufficient}} \rceil$

4. YOU DO NOT NEED THE FULL PUZZLE

Suppose question asks:

Who is tallest?

You do not need everyone's exact height.

If information proves:

$\lceil A>B, \lceil A>C, \lceil A>D \rceil$

then:

$\lceil \boxed{A \text{ is tallest}} \rceil$

The statement is sufficient for the **asked question**, even if B/C/D order remains unknown.

5. MOST IMPORTANT RULE

Judge sufficiency only for what is asked.

Do not demand unnecessary information.

6. STANDARD TWO-STATEMENT CASES

The logical possibilities are usually:

1. I alone sufficient
2. II alone sufficient
3. both individually sufficient
4. neither alone, but together sufficient
5. even together insufficient

Some placement tests encode these as A/B/C/D/E options.

Always read the answer-key convention given in that test.

7. COMMON OPTION CONVENTION

A typical convention is:

A. I alone sufficient, II alone insufficient **B.** II alone sufficient, I alone insufficient **C.** Either I or II alone sufficient **D.** Both together sufficient, neither alone sufficient **E.** Even both together insufficient

But:

Never memorize letters without reading the actual question instructions.

Companies can change option mapping.

8. TEST STATEMENT I ALONE

Temporarily ignore Statement II completely.

Example:

Find $\sqrt{x}$.

I. $\sqrt{x+3}=8$ II. $\sqrt{x}$ is positive.

Statement I:

$\sqrt{x}=5$

Unique.

So I alone is sufficient.

Do not use II while testing I.

9. TEST STATEMENT II ALONE

Now forget I.

II:

$\sqrt{x}>0$

Many possibilities.

So II alone insufficient.

Therefore result:

$\boxed{\text{I alone sufficient}}$

10. THE BIGGEST DS ERROR

Students unconsciously combine statements while supposedly testing one.

Do not.

Use this discipline:

TEXT

Test I !' erase II mentally.
Test II !' erase I mentally.
Only then combine if needed.

11. BOTH TOGETHER

Question:

Find x .

I. $x+y=10$ II. $x-y=2$

I alone:

many (x,y) .

Insufficient.

II alone:

many (x,y) .

Insufficient.

Together:

$x+y=10$

$x-y=2$

Add:

$2x=12$

$x=6$

Therefore:

$\boxed{\text{Together sufficient}}$

12. EVEN BOTH MAY BE INSUFFICIENT

Question:

Find x .

I.

$x+y=10$

II.

$y>0$

Together:

$x=10-y$

Many possibilities.

So:

$\boxed{\text{Both insufficient}}$

13. BOTH STATEMENTS MAY EACH BE SUFFICIENT

Question:

Find $\sqrt{x}$.

I.

$$\sqrt{x} = 7$$

II.

$$\sqrt{2x} = 14$$

Each alone gives:

$$\sqrt{x} = 7$$

Thus:

$\boxed{\text{Either statement alone is sufficient}}$

14. ANSWER VS METHOD

Data Sufficiency does not care whether statements use different methods.

Example:

Question:

What is the area of a square?

I. Side = 5 cm. II. Diagonal = $5\sqrt{2}$ cm.

I:

$$\text{Area} = 25$$

II:

For square:

$$d = s\sqrt{2}$$

$$s = 5$$

$$\text{Area} = 25$$

Both independently sufficient.

15. YES/NO DATA SUFFICIENCY

This is one of the most misunderstood forms.

Question:

Is $x > 5$?

A statement is sufficient if it forces either:

$\boxed{\text{Yes}}$

or:

$\boxed{\text{No}}$

It does **not** need to prove “Yes.”

A definite “No” is equally sufficient.

16. YES/NO EXAMPLE

Question:

Is $x > 5$?

I.

$x = 8$

Answer is definitely Yes.

Sufficient.

II.

$x = 3$

Answer is definitely No.

Also sufficient.

Therefore both statements individually sufficient.

17. YES/NO INSUFFICIENT EXAMPLE

Question:

Is $x > 5$?

I.

$x > 2$

Could be:

$x = 3 \rightarrow \text{No}$

or:

$x = 10 \rightarrow \text{Yes}$

Thus:

$\boxed{\text{Insufficient}}$

18. YES/NO GOLDEN RULE

If all valid cases give the same Yes/No answer, the statement is sufficient.

Even if the exact value remains unknown.

19. EXAMPLE

Question:

Is $|x|$ positive?

Statement:

$$|x^2|=9$$

Possible:

$$x=3$$

or:

$$x=-3$$

Answers:

- Yes
- No

Different.

Therefore:

$\boxed{\text{Insufficient}}$

20. EXACT VALUE QUESTION WITH MULTIPLE ROOTS

Question:

What is $|x|$?

I.

$$x^2=25$$

Then:

$$x=\pm 5$$

Two values.

$\boxed{\text{Insufficient}}$

21. ADD A SIGN CONDITION

II.

$$\sqrt{x > 0}$$

II alone insufficient.

Together with:

$$\sqrt{x^2 = 25}$$

gives:

$$\sqrt{x = 5}$$

Thus:

$$\boxed{\text{Together sufficient}}$$

22. DOMAIN MATTERS

Question:

What is $\sqrt{x}$?

I.

$$\sqrt{x^2 = 9}$$

If domain explicitly says:

$$\sqrt{x \text{ is a positive integer}}$$

then:

$$\sqrt{x = 3}$$

Statement becomes sufficient.

Without domain:

$$\sqrt{x = \pm 3}$$

insufficient.

23. HIDDEN-DOMAIN TRAP

Always note whether variables are:

- integers
- positive integers
- real numbers
- non-zero
- distinct
- natural numbers

These restrictions can completely change sufficiency.

24. EXAMPLE — INTEGER DOMAIN

Question:

What is positive integer n ?

I.

$$2 < n < 4$$

Since n integer:

$$n = 3$$

Unique.

$\boxed{\text{Sufficient}}$

If n were real:

infinitely many possibilities.

25. RANGE CAN BE SUFFICIENT

Students often think only equations can be sufficient.

Not true.

Question:

What integer is n ?

Statement:

$$4.5 < n < 5.5$$

Only integer:

$$\boxed{5}$$

Sufficient.

26. PARITY QUESTION

Question:

Is n even?

I.

$$n = 2k$$

for integer k .

Definitely even.

$\boxed{\text{Sufficient}}$

27. DIVISIBILITY QUESTION

Question:

Is n divisible by 6?

I.

n divisible by 2.

Insufficient.

II.

n divisible by 3.

Insufficient.

Together, if n is an integer:

Divisible by both 2 and 3.

Since:

$\text{LCM}(2,3)=6$

$\boxed{\text{Yes}}$

Together sufficient.

28. CAREFUL — COMMON FACTORS

Question:

Is n divisible by 12?

I. divisible by 4 II. divisible by 6

Together:

$\text{LCM}(4,6)=12$

So yes.

Sufficient.

But if question asks divisible by 24:

LCM is only 12.

Insufficient.

29. NUMERICAL DS — AVERAGE

Question:

What is the average of 5 numbers?

I. Their sum is 100.

Average:

$$\left[\frac{100}{5} = 20 \right]$$

Unique.

Sufficient.

No individual values required.

30. SUM QUESTION

Question:

What is $(a+b)$?

I.

$$\left[a+b=12 \right]$$

Obviously sufficient.

Even though (a,b) individually unknown.

Again:

Solve only what is asked.

31. PRODUCT QUESTION

Question:

What is (xy) ?

I.

$$\left[x=3 \right]$$

Insufficient.

II.

$$\left[y=4 \right]$$

Insufficient.

Together:

$$\left[xy=12 \right]$$

Sufficient.

32. ALGEBRAIC DS — LINEAR EQUATION

Question:

Find x .

I.

$$2x + 3y = 12$$

One equation, two unknowns.

Usually insufficient.

II.

$$y = 2$$

Alone insufficient for x .

Together:

$$2x + 6 = 12$$

$$x = 3$$

Sufficient.

33. DO NOT BLINDLY COUNT EQUATIONS

“Two equations, two variables” often works—but not always.

Equations can be:

- dependent
- inconsistent
- nonlinear with multiple solutions

You must check uniqueness.

34. DEPENDENT EQUATIONS

Question:

Find x .

I.

$$x + y = 10$$

II.

$$2x + 2y = 20$$

II is just twice I.

Together still only one independent equation.

Many solutions.

$\boxed{\text{Insufficient}}$

35. INCONSISTENT STATEMENTS

I.

$$x+y=10$$

II.

$$x+y=12$$

Together impossible.

In well-formed DS questions, statements are normally consistent with the problem.

Do not treat inconsistency as a unique numerical answer.

36. QUADRATIC DS

Question:

Find $\sqrt{x}$.

I.

$$x^2-5x+6=0$$

Factor:

$$(x-2)(x-3)=0$$

So:

$$x=2 \text{ or } 3$$

Insufficient.

II.

$$x>2$$

Alone insufficient.

Together:

$$x=3$$

Sufficient.

37. ABSOLUTE VALUE DS

Question:

Find $\sqrt{x}$.

I.

$$\sqrt{|x|=7}$$

Gives:

$$\sqrt{x=\pm 7}$$

Insufficient.

II.

$$\sqrt{x < 0}$$

Together:

$$\sqrt{x = -7}$$

Sufficient.

38. RATIO DS

Question:

What is $\frac{A}{B}$?

I.

$$\sqrt{A=2B}$$

Then directly:

$$\sqrt{A:B=2:1}$$

Sufficient.

You do not need actual values.

39. PERCENTAGE DS

Question:

What percent of $\frac{B}{A}$ is $\frac{A}{B}$?

I.

$$\sqrt{A:B=3:5}$$

Then:

$$\sqrt{\frac{A}{B} \times 100 = \frac{3}{5} \times 100 = 60\%}$$

Sufficient.

40. AGE DS

Question:

What is Ravi's age?

I. Ravi is 5 years older than Sita. II. Their ages sum to 35.

I alone insufficient.

II alone insufficient.

Together:

Let Sita $(=x)$.

Ravi $(=x+5)$.

$$2x+5=35$$

$$x=15$$

Ravi:

$$20$$

Together sufficient.

41. SPEED DS

Question:

What is speed?

Since:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

I. Distance = 120 km. II. Time = 2 hours.

Each alone insufficient.

Together:

$$60 \text{ km/h}$$

Sufficient.

42. PROFIT DS

Question:

What is profit percentage?

Need:

$$\frac{\text{Profit}}{\text{CP}} \times 100$$

I. $SP = 1600$. II. $CP = 1500$

Together:

$$\text{Profit} = 100$$

$$\sqrt{\text{Profit}\% = 20\%}$$

Sufficient.

43. GEOMETRY DS — RECTANGLE AREA

Question:

What is area of rectangle?

Need:

$$\sqrt{A = lb}$$

I. Length = 10 II. Breadth = 6

Each alone insufficient.

Together:

$$\sqrt{A = 60}$$

Sufficient.

44. PERIMETER MAY NOT GIVE AREA

Question:

What is area of rectangle?

I. Perimeter = 20.

$$\sqrt{2(l+b) = 20}$$

$$\sqrt{l+b = 10}$$

Possible dimensions:

$$\sqrt{1,9; 2,8; 3,7; 4,6; 5,5}$$

Different areas.

Insufficient.

45. SQUARE PERIMETER DOES GIVE AREA

Question:

Area of a square?

I. Perimeter = 20.

For square:

$$\sqrt{4s = 20}$$

$$\sqrt{s=5}$$

Area:

$$\sqrt{25}$$

Sufficient.

Geometry type matters.

46. TRIANGLE AREA

Question:

Area of triangle?

I. Base = 10 II. Height to that base = 6

Each alone insufficient.

Together:

$$\sqrt{\frac{1}{2} \times 10 \times 6 = 30}$$

Sufficient.

47. THREE SIDES CAN DETERMINE TRIANGLE

If all three sides are known and form a valid triangle, area is determined by Heron's formula.

You do not necessarily need height.

This is a geometry sufficiency trap.

48. ANGLES ALONE DO NOT FIX SCALE

Question:

Area of triangle?

Statement:

Angles are $(60^\circ, 60^\circ, 60^\circ)$.

This tells equilateral shape but no size.

Could have side 1, 10, 100...

Area varies.

$$\sqrt{\boxed{\text{Insufficient}}}$$

49. SIMILARITY "SIZE

Knowing shape alone does not determine lengths/area.

A scale parameter is needed.

50. CIRCLE AREA

Question:

Area of circle?

Need one scale measure:

- radius
- diameter
- circumference

Any one alone determines area.

Example:

I. Circumference = $\sqrt{10\pi}$.

$\sqrt{2\pi r=10\pi}$

$\sqrt{r=5}$

Area:

$\sqrt{25\pi}$

Sufficient.

51. RANKING DS

Question:

How many students are in the row?

I. A is 12th from left. II. A is 9th from right.

I alone insufficient.

II alone insufficient.

Together:

$\sqrt{N=12+9-1}$

$\sqrt{\boxed{20}}$

Together sufficient.

52. SAME-PERSON REQUIREMENT

Formula:

$\sqrt{N=L+R-1}$

works when left/right ranks refer to the same person.

If:

- A = 12th from left
- B = 9th from right

total may not be fixed.

Potentially insufficient.

53. RANKING QUESTION MAY NOT NEED TOTAL

Question:

How many people are between A and B?

I. A = 5th from left. II. B = 12th from left.

Together:

$$\lfloor 12 - 5 - 1 = 6 \rfloor$$

Sufficient.

Total row size unnecessary.

54. ORDERING DATA SUFFICIENCY

Question:

Is A taller than C?

I.

$$\lfloor A > B \rfloor$$

II.

$$\lfloor B > C \rfloor$$

I alone: cannot compare A/C.

II alone: cannot compare A/C.

Together:

$$\lfloor A > B > C \rfloor$$

Therefore:

$$\lfloor \boxed{\text{Yes}} \rfloor$$

Together sufficient.

55. A DEFINITE NO IS ALSO SUFFICIENT

Question:

Is A taller than C?

I.

$\lceil C > A \rceil$

Then answer is definitely:

$\lceil \boxed{\text{No}} \rceil$

Statement I is sufficient.

56. PARTIAL ORDER MAY BE INSUFFICIENT

Question:

Who is tallest among A,B,C?

I.

$\lceil A > B \rceil$

II.

$\lceil C > B \rceil$

Together:

A and C both above B.

But relation A vs C unknown.

Tallest could be A or C.

$\lceil \boxed{\text{Insufficient}} \rceil$

57. SEATING DS

Question:

Who sits immediately right of A?

I. B sits immediately left of A. II. C sits immediately right of A.

Statement II alone directly answers C.

Sufficient.

Statement I alone tells left neighbour only, not right.

Insufficient.

58. PUZZLE DS PRINCIPLE

You do not need the complete arrangement.

If a statement uniquely determines the requested relation, it is sufficient.

59. NUMBER PROPERTY DS

Question:

Is integer n prime?

I.

$n = 17$

Sufficient !' Yes.

II.

n is odd

Insufficient:

3,5,7 prime; 9,15 composite.

60. EVEN + GREATER THAN 2

Question:

Is n prime?

I.

n is even.

II.

$n > 2$.

Together:

Any even integer $(n > 2)$ is composite.

Answer:

$\boxed{\text{No}}$

Thus together sufficient.

Notice exact n remains unknown.

61. YES/NO DS CAN BE SUFFICIENT WITHOUT VALUE

This deserves repetition.

Question:

Is n prime?

If data prove:

$\lceil n \rceil$ is even and $n > 2$

we do not know $\lceil n \rceil$, but we know answer:

$\boxed{\text{No}}$

Therefore sufficient.

62. SIGN DS

Question:

Is $xy > 0$?

I. $x > 0$. II. $y > 0$.

Each alone insufficient.

Together:

positive $\times$ positive = positive.

$\boxed{\text{Yes}}$

Sufficient.

63. SAME-SIGN CONDITION

Question:

Is $xy > 0$?

Statement:

x, y have the same nonzero sign

Then product is definitely positive.

Sufficient even without individual values.

64. SQUARE SIGN TRAP

Question:

Is $x > 0$?

I.

$x^2 > 0$

This only gives:

$x \neq 0$

Could be positive or negative.

Insufficient.

65. INEQUALITY DS

Question:

Is $x > y$?

I.

$$x - y = 5$$

Definitely:

$$x > y$$

Sufficient.

No exact values required.

66. SUM DOES NOT GIVE ORDER

Question:

Is $x > y$?

I.

$$x + y = 10$$

Could have:

$$x = 8, y = 2$$

Yes.

Or:

$$x = 2, y = 8$$

No.

Insufficient.

67. PRODUCT SIGN DS

Question:

Do x, y have the same sign?

I.

$$xy > 0$$

Assuming real nonzero variables:

Yes.

Sufficient.

68. REDUNDANCY

Statement II may add no new information.

Question:

Find $\sqrt{x}$.

I.

$$\sqrt{x=5}$$

II.

$$\sqrt{x^2=25, x>0}$$

Both independently sufficient.

But II is not needed if I already works.

DS classification still tests each independently.

69. REDUNDANT COMBINED DATA

I.

$$\sqrt{x+y=10}$$

II.

$$\sqrt{2x+2y=20}$$

II repeats I.

Together not stronger.

Learn to spot redundancy quickly.

70. SUFFICIENCY BY BOUNDS

Question:

What integer is $\sqrt{n}$?

I.

$$\sqrt{10< n < 12}$$

Since $\sqrt{n}$ integer:

$$\sqrt{n=11}$$

Sufficient.

Bounds can determine exact values.

71. SUFFICIENCY BY DIVISIBILITY + RANGE

Question:

What positive integer is n ?

I. $10 < n < 20$ II. n divisible by 7

I alone: 11–19 !' insufficient.

II alone: 7, 14, 21, ... !' insufficient.

Together:

Only:

$\{14\}$

Sufficient.

72. MULTIPLE POSSIBILITIES = INSUFFICIENT

You should actively search for two valid examples producing different answers.

If you find them:

$\boxed{\text{Insufficient}}$

This is the fastest rigorous test.

73. COUNTEREXAMPLE METHOD

Question:

Is $x > y$?

Statement:

$x + y = 10$

Find:

Case 1:

$x = 8, y = 2 \rightarrow \text{Yes}$

Case 2:

$x = 2, y = 8 \rightarrow \text{No}$

Since both satisfy statement:

$\boxed{\text{Insufficient}}$

74. SUFFICIENCY PROOF

To prove sufficient:

- show all valid possibilities give the same asked answer.

To prove insufficient:

- find two valid possibilities giving different asked answers.

This is the mathematical heart of DS.

75. DS MASTER ALGORITHM

For each statement:

Step 1

What exactly is being asked?

Step 2

Note domain restrictions.

Step 3

Test I alone.

Step 4

Test II alone.

Step 5

If neither works, combine.

Step 6

Check uniqueness of answer.

Step 7

For Yes/No, check whether answer is fixed.

Step 8

Map result to the exam's option convention.

76. COMMON TRAPS

& pTRAP 1 — Solving instead of checking sufficiency

Wasteful.

& pTRAP 2 — Using II while testing I

Invalid.

& pTRAP 3 — Exact value required for Yes/No

False.

& pTRAP 4 — Ignoring $\pm$ roots

Very common.

& pTRAP 5 — Ignoring integer/positive domain

Can change answer.

& pTRAP 6 — Two equations automatically sufficient

Not if dependent or multi-root.

& pTRAP 7 — Full puzzle required

No; only asked quantity/relation.

& pTRAP 8 — Memorizing A/B/C/D/E letters

Read option convention.

PRACTICE — LEVEL 0

Q1

Question: Find $\sqrt{x}$.

I.

$$\sqrt{x+4}=9$$

Is I sufficient?

Q2

Question: Find $\sqrt{x}$.

I.

$$\sqrt{x}>3$$

Sufficient?

Q3

Question: Is $x>0$?

I.

$$\sqrt{x}=5$$

Sufficient?

Q4

Question: Is $x>0$?

I.

$$\sqrt{x^2}=9$$

Sufficient?

Q5

Question: Find positive integer n .

I.

$$\sqrt{2 < n < 4}$$

Sufficient?

LEVEL 1

Q6

Find x .

I. $x + y = 12$ II. $x - y = 4$

Are both together sufficient?

Q7

Find x .

I. $x + y = 10$ II. $2x + 2y = 20$

Are both together sufficient?

Q8

Is integer n divisible by 6?

I. n divisible by 2 II. n divisible by 3

What is the sufficiency status?

Q9

Find x .

I.

$$\sqrt{x^2} = 16$$

II.

$$\sqrt{x} > 0$$

What is the sufficiency status?

Q10

What is $A:B$?

I.

$$\sqrt{A = 3B}$$

Sufficient?

LEVEL 2

Q11

What is the area of a square?

I. Perimeter = 24.

Sufficient?

Q12

What is area of a rectangle?

I. Perimeter = 24.

Sufficient?

Q13

How many students are in a row?

I. A is 14th from left. II. A is 10th from right.

What is the sufficiency status?

Q14

Who is tallest among A,B,C?

I. $A > B$ II. $C > B$

Together sufficient?

Q15

Is $x > y$?

I.

$x - y = -3$

Sufficient?

LEVEL 3 — EXAM PRESSURE

Q16

Question: Is positive integer n prime?

I. n is odd. II. $n < 4$.

Test carefully.

Q17

Question: What is integer n ?

I.

$10 < n < 20$

II.

n is divisible by 7.

Status?

Q18

Question: Is $xy > 0$?

I. $\neg(x < 0)$. II. $\neg(y < 0)$.

Status?

Q19

Question: Find $\neg(x)$.

I.

$$\neg [x^2 - 7x + 12 = 0]$$

II.

$$\neg [x < 4]$$

Status?

Q20

Question: Is $\neg(A > C)$?

I.

$$\neg [A > B]$$

II.

$$\neg [B > C]$$

Status?

ANSWERS

Q1

$$\neg [x = 5]$$

Unique.

$$\neg [\boxed{\text{Sufficient}}]$$

Q2

Many values.

$$\neg [\boxed{\text{Insufficient}}]$$

Q3

Definitely Yes.

$$\neg [\boxed{\text{Sufficient}}]$$

Q4

$$\neg [x = \pm 3]$$

Could be positive or negative.

$\boxed{\text{Insufficient}}$

Q5

Positive integer between 2 and 4:

$\boxed{3}$

$\boxed{\text{Sufficient}}$

Q6

Together:

$x+y=12$

$x-y=4$

Adding:

$2x=16$

$x=8$

$\boxed{\text{Together}\backslash\text{sufficient}}$

Q7

Second equation is just twice first.

Many solutions.

$\boxed{\text{Even}\backslash\text{together}\backslash\text{insufficient}}$

Q8

I alone: insufficient.

II alone: insufficient.

Together:

$\text{LCM}(2,3)=6$

Definitely divisible by 6.

$\boxed{\text{Together}\backslash\text{sufficient}}$

Q9

I:

$x=\pm 4$

insufficient.

II:

$\sqrt{x > 0}$ gives many values.

Insufficient.

Together:

$$\sqrt{x = 4}$$

$$\boxed{\text{Together sufficient}}$$

Q10

$$\sqrt{A = 3B}$$

Thus:

$$\sqrt{A:B = 3:1}$$

$$\boxed{\text{Sufficient}}$$

Q11

Square:

$$\sqrt{4s = 24}$$

$$\sqrt{s = 6}$$

Area:

$$\sqrt{36}$$

$$\boxed{\text{Sufficient}}$$

Q12

Rectangle perimeter:

$$\sqrt{l + b = 12}$$

Many possible dimensions and areas.

$$\boxed{\text{Insufficient}}$$

Q13

Same person A:

$$\sqrt{N = 14 + 10 - 1}$$

$$\sqrt{N = 23}$$

Neither alone sufficient, together sufficient.

$$\boxed{\text{Together sufficient}}$$

Q14

Together:

$$\{ A > B \}$$

$$\{ C > B \}$$

A vs C unknown.

$\boxed{\text{Insufficient}}$

Q15

$$\{ x - y = -3 \}$$

Thus:

$$\{ x < y \}$$

Question asks $\{x > y\}$.

Answer definitely No.

$\boxed{\text{Sufficient}}$

Q16

Positive integer $\{n\}$.

Question: prime?

I. $\{n\}$ odd.

Could be 3 prime, 9 composite.

Insufficient.

II. $\{n < 4\}$.

Positive integers:

$$\{ 1, 2, 3 \}$$

Some prime, some not.

Insufficient.

Together:

positive odd integers $\{n < 4\}$:

$$\{ 1, 3 \}$$

1 is not prime; 3 is prime.

Answers differ.

$\boxed{\text{Even together insufficient}}$

Q17

I:

$$\{11, 12, \dots, 19\}$$

insufficient.

II:

positive/negative multiples depending domain, many values.

Insufficient.

Together:

Only multiple of 7 between 10 and 20:

$$\{14\}$$

$$\boxed{\text{Together sufficient}}$$

Q18

Question:

$$xy > 0?$$

I: $x < 0$, insufficient because y unknown.

II: $y < 0$, insufficient because x unknown.

Together:

negative $\times$ negative = positive.

$$\boxed{\text{Together sufficient}}$$

Q19

Factor:

$$x^2 - 7x + 12 = 0$$

$$(x-3)(x-4) = 0$$

So I gives:

$$x = 3 \text{ or } 4$$

Insufficient.

II:

$$x < 4$$

many possibilities.

Insufficient.

Together:

only:

$$x = 3$$

$$\boxed{\text{Together sufficient}}$$

Q20

I:

$$\{ A > B \}$$

insufficient for A/C.

II:

$$\{ B > C \}$$

insufficient for A/C.

Together:

$$\{ A > B > C \}$$

Therefore:

$$\{ A > C \}$$

Definitely Yes.

$$\boxed{\text{Together sufficient}}$$

CHAPTER 69 STATUS

Covered so far:

- meaning of data sufficiency '
- answer vs sufficiency '
- unique-answer criterion '
- I-alone / II-alone / together testing '
- either statement alone '
- both insufficient '
- option-convention caution '
- yes/no sufficiency '
- exact-value vs relation questions '
- domain restrictions '
- numerical DS '
- divisibility DS '
- algebraic DS '
- dependent equations '
- quadratics/absolute values '
- ratio/percentage DS '
- age/speed/profit DS '
- geometry DS '
- ranking/ordering DS '
- puzzle sufficiency basics '
- counterexample method '

- L0–L3 practice '

But **Chapter 69 is not complete yet.**

Still remaining:

- advanced Yes/No DS
- inequalities and sign combinations
- averages/mixtures/time-work DS
- advanced geometry DS
- probability/counting sufficiency
- multi-condition puzzle DS
- statement redundancy/consistency
- “either alone” subtleties
- 3-statement data sufficiency
- placement option-pattern speed methods
- final DS checklist and mixed exam set

Progress

$\lfloor \boxed{68/76} \text{ numbered chapters completed} \rfloor$

Current: ~~0=No~~ Chapter 69 — Data Sufficiency

NEXT

~~0=No~~ **Chapter 69 Part 2 — Advanced Numerical/Logical Data Sufficiency, Three-Statement DS, Probability/Geometry/Puzzles & Final Speed Toolkit**

USER

next

ASSISTANT

CHAPTER 69 — DATA SUFFICIENCY

PART 2 — ADVANCED NUMERICAL/LOGICAL DS, THREE-STATEMENTS DS, PROBABILITY/GEOMETRY/PUZZLES & FINAL SPEED TOOLKIT

Status: ~~0=No~~ **IN PROGRESS ! ' ' COMPLETE** **GATE:** High **Placements:** Very High **Priority:** ~~0=No~~ **Yes** 1

This part handles the harder Data Sufficiency questions where:

- exact values are unnecessary
- inequalities/signs matter
- geometry has multiple possible shapes
- probability/counting depends on hidden totals
- logical puzzles need only partial resolution
- three statements are given

The golden rule remains:

$\boxed{\text{Sufficient = the asked answer is uniquely determined}}$

1. ADVANCED YES/NO DATA SUFFICIENCY

Question:

Is $x > 0$?

A statement does not need to determine x .

It only needs to force:

$\boxed{\text{Yes}}$

or

$\boxed{\text{No}}$

for every valid case.

2. EXAMPLE — EXACT VALUE UNKNOWN BUT SUFFICIENT

Question:

Is $x > 0$?

I.

$x > 7$

We do not know exact x .

But every possible value is positive.

Therefore:

$\boxed{\text{Sufficient}}$

3. NEGATIVE ANSWER ALSO SUFFICIENT

Question:

Is $x > 0$?

I.

$x < -5$

Every possible value is negative.

Answer definitely No.

$\boxed{\text{Sufficient}}$

4. RANGE CROSSING ZERO

Question:

Is $(x > 0)$?

I.

$$\{-3 < x < 5\}$$

Possible:

$$\{x = -1, \text{quad } x = 2\}$$

Different Yes/No outcomes.

$\boxed{\text{Insufficient}}$

5. SIGN OF PRODUCT

Question:

Is $(xy > 0)$?

I.

$$\{x > 0\}$$

II.

$$\{y < 0\}$$

Together:

positive $\times$ negative = negative.

So answer definitely:

$\boxed{\text{No}}$

Therefore together sufficient.

6. SIGN OF QUOTIENT

Question:

Is $(\frac{x}{y} > 0)$?

Sufficient condition:

- both same sign
- $(y \neq 0)$

Example:

I. $(x < 0)$ II. $(y < 0)$

Together:

$$\left[\frac{x}{y} > 0 \right]$$

Sufficient.

7. ZERO TRAP

Question:

Is $xy > 0$?

I. $x \geq 0$ II. $y > 0$

Together:

If $x = 0$:

$$[xy = 0]$$

Answer No.

If $x = 2$:

$$[xy > 0]$$

Answer Yes.

So:

$$[\boxed{\text{Insufficient}}]$$

The distinction between $(>)$ and $(\geq)$ matters.

8. SQUARES

Question:

Is $x > y$?

I.

$$[x^2 > y^2]$$

Not enough.

Example:

$$[x=5, y=3 \rightarrow \text{Yes}]$$

but:

$$[x=-5, y=3 \rightarrow \text{No}]$$

So:

$$[\boxed{\text{Insufficient}}]$$

9. ADD POSITIVITY

II.

$$\{ x > 0, y > 0 \}$$

Together with:

$$\{ x^2 > y^2 \}$$

for positives:

$$\{ x > y \}$$

Therefore together sufficient.

10. ABSOLUTE VALUE COMPARISON

Question:

Is $|x| > |y|$?

I.

$$\{$$

| | | |
|---|---|---|
| x | > | y |
|---|---|---|

$$\}$$

Insufficient because signs unknown.

If both are positive, yes.

If $(x = -10, y = 2)$, no.

11. SUM AND PRODUCT

Question:

Are (x, y) both positive?

I.

$$\{ x + y > 0 \}$$

II.

$$\{ xy > 0 \}$$

II means same sign.

I says sum positive.

Together, they cannot both be negative.

Therefore both positive.

$$\boxed{\text{Together sufficient}}$$

12. SUM POSITIVE + PRODUCT NEGATIVE

If:

$$\lceil xy < 0 \rceil$$

then opposite signs.

If:

$$\lceil x + y > 0 \rceil$$

the positive value has larger magnitude.

But unless question asks which variable is positive, identity may still be unknown.

13. EXAMPLE

Question:

Is $\lceil x > 0 \rceil$?

I.

$$\lceil xy < 0 \rceil$$

II.

$$\lceil x + y > 0 \rceil$$

Could:

$$\lceil x = 5, y = -2 \rightarrow x > 0 \rceil$$

Could:

$$\lceil x = -2, y = 5 \rightarrow x < 0 \rceil$$

Both conditions satisfied.

Thus:

$$\lceil \boxed{\text{Insufficient}} \rceil$$

14. ADVANCED INEQUALITY DS

Question:

Is $\lceil a > b \rceil$?

I.

$$\lceil a - b > 0 \rceil$$

Immediately yes.

$$\lceil \boxed{\text{Sufficient}} \rceil$$

No need to solve anything else.

15. INEQUALITY TRANSITIVITY

Question:

Is $(A > C)$?

I.

$$A > B$$

II.

$$B \geq C$$

Together:

$$A > B \geq C$$

Thus:

$$A > C$$

Definitely.

$\boxed{\text{Together sufficient}}$

16. NON-STRICT TRAP

Question:

Is $(A > C)$?

I.

$$A \geq B$$

II.

$$B \geq C$$

Together gives:

$$A \geq C$$

But A may equal C.

So answer could be Yes or No to strict $(A > C)$.

$\boxed{\text{Insufficient}}$

17. AVERAGE DATA SUFFICIENCY

Question:

What is the average of 8 values?

I.

Sum of values = 160.

Then:

$$\left[\text{Average} = \frac{160}{8} = 20 \right]$$

Sufficient.

18. COMBINED AVERAGE

Question:

What is overall average of two groups?

I. Group A: 10 students, average 70. II. Group B: 20 students, average 80.

Together:

$$\left[\frac{10(70) + 20(80)}{30} \right]$$

unique.

So together sufficient.

Neither alone gives overall average.

19. UNKNOWN GROUP SIZE TRAP

Question:

Overall average of groups A and B?

I. Average A = 60 II. Average B = 80

Not enough.

Without group sizes, weighted average is unknown.

$$\left[\boxed{\text{Even together insufficient}} \right]$$

20. MIXTURE DS

Question:

What is concentration of salt in a mixture?

I. Salt = 20 g. II. Total mixture = 100 g.

Together:

$\frac{20}{100}$

Sufficient.

21. MIXTURE RATIO

Question:

What fraction of mixture is water?

I. Water : milk = 3:2.

Then water fraction:

$\frac{3}{5}$

Sufficient.

Total quantity unnecessary.

22. TIME AND WORK DS

Question:

How long does A alone take to complete a job?

I. A completes $\frac{1}{15}$ of job per day.

Then:

$T = 5 \text{ days}$

Sufficient.

23. JOINT WORK

Question:

How long will A and B together take?

I. A alone = 10 days II. B alone = 15 days

Together rate:

$\frac{1}{10} + \frac{1}{15} = \frac{1}{6}$

Time:

6 days

Together sufficient.

24. WORK-RATIO DS

Question:

What is efficiency ratio A:B?

I. A takes 12 days, B 18 days.

Efficiency inverse of time:

$\sqrt{A:B=18:12=3:2}$

Sufficient.

25. SPEED / DISTANCE DS

Question:

What is travel time?

I. Distance = 300 km II. Speed = 60 km/h

Together:

$\sqrt{t=5h}$

Sufficient.

26. RELATIVE SPEED DS

Question:

Do A and B move toward each other?

I. A moves east. II. B is east of A and moves west.

Together yes.

Exact speeds irrelevant.

This is logical sufficiency, not numerical.

27. PROFIT/LOSS DS

Question:

Was there a profit?

I. $CP = ₹ 500$ II. $SP = ₹ 550$

Together:

$\sqrt{SP > CP}$

Definitely yes.

Exact profit percentage not needed if asked only profit/loss.

28. QUESTION PRECISION CHANGES SUFFICIENCY

If question instead asks:

What was profit percentage?

Same statements still sufficient because:

$$\left[\frac{50}{500} \times 100 = 10\% \right]$$

But if only:

SP exceeded CP

that is enough for “Was there profit?” but not for exact profit%.

29. ADVANCED GEOMETRY DS — TRIANGLE TYPE

Question:

Is triangle ABC right-angled?

I. Sides are 3,4,5.

Sufficient:

$$\left[3^2 + 4^2 = 5^2 \right]$$

Yes.

30. ANGLE DATA

Question:

Is triangle ABC right-angled?

I.

$$\left[A = 90^\circ \right]$$

Immediately sufficient.

31. TWO ANGLES MAY SUFFICE

Question:

What is third angle of triangle?

I.

$$\backslash A=50^\circ \backslash$$

II.

$$\backslash B=60^\circ \backslash$$

Together:

$$\backslash C=70^\circ \backslash$$

Sufficient.

32. AREA OF TRIANGLE — SAS

Question:

What is area of triangle?

I. Two sides are 5 and 8. II. Included angle = $\backslash(30^\circ \backslash)$.

Together:

$$\backslash \text{Area}=\frac{1}{2} ab\sin C \backslash$$

$$\backslash =\frac{1}{2}(5)(8)\sin 30^\circ =10 \backslash$$

Sufficient.

33. TWO SIDES WITHOUT ANGLE

Question:

Area of triangle?

I. Two sides 5,8.

Not sufficient.

Different included angles produce different areas.

34. RECTANGLE DIAGONAL + ONE SIDE

Question:

Area of rectangle?

I. Diagonal = 10 II. One side = 6

Together:

Other side:

$$\backslash \sqrt{10^2-6^2}=8 \backslash$$

Area:

$\sqrt{48}$

Together sufficient.

35. RHOMBUS AREA

Question:

Area of rhombus?

I. Diagonal 1 = 10 II. Diagonal 2 = 8

Together:

$A = \frac{1}{2} d_1 d_2 = 40$

Sufficient.

36. CIRCLE CHORD/ANGLE TRAP

Question:

Radius of circle?

I. A chord length is 10.

Insufficient.

Different circles can have chord 10.

Need more geometry information, such as distance from centre to chord.

37. SIMILAR TRIANGLES DS

Question:

What is side $\sqrt{x}$?

I. Two triangles are similar. II. Corresponding known sides give scale ratio and one corresponding side involving $\sqrt{x}$.

Together may be sufficient.

Similarity alone is not.

38. COUNTING/PERMUTATION DS

Question:

How many ways can 5 distinct people stand in a row?

No extra statement required:

$$\lceil 5!=120 \rceil$$

Sometimes question itself contains enough information.

In DS formats, a statement may be irrelevant.

39. PROBABILITY DS

Question:

Probability of drawing a red ball?

Need:

$$\lceil P(R)=\frac{\text{\#red}}{\text{\#total}} \rceil$$

I. Red balls = 4 II. Total balls = 10

Together:

$$\lceil \frac{4}{10}=\frac{2}{5} \rceil$$

Sufficient.

40. TOTAL MAY BE DERIVED

I. 4 red balls. II. 6 non-red balls.

Together total:

$$\lceil 10 \rceil$$

Probability red:

$$\lceil \frac{4}{10} \rceil$$

Sufficient.

41. PROBABILITY YES/NO DS

Question:

Is probability of success greater than $\frac{1}{2}$?

I.

$$\lceil P(\text{success})=0.6 \rceil$$

Sufficient.

II.

$$\lceil P(\text{failure})<0.4 \rceil$$

Then:

$\lceil P(\text{success}) > 0.6 \rceil$

also sufficient.

42. INDEPENDENCE DS

Question:

Is $\lceil P(A \cap B) = P(A)P(B) \rceil$?

I. A and B are independent.

Immediately yes.

Sufficient.

43. PROBABILITY WITHOUT SAMPLE SPACE

Question:

Probability of drawing an ace?

I. One card drawn from a standard deck.

This is actually sufficient because standard deck structure is known:

$\lceil \frac{4}{52} = \frac{1}{13} \rceil$

Domain/common definition matters.

44. COUNTING WITH IDENTICAL/DISTINCT OBJECTS

Question:

Number of arrangements of letters?

I. Word has 5 letters. II. All letters distinct.

Together:

$\lceil 5! \rceil$

I alone insufficient because repeats may exist.

II alone insufficient without number of letters.

45. SET DATA SUFFICIENCY

Question:

How many elements are in $\lceil A \cup B \rceil$?

Need:

$\backslash$

| | | | | |
|---|---|---|---|------------|
| A | + | B | - | $A \cap B$ |
|---|---|---|---|------------|

$\backslash$

I. $\backslash(|A|=20, \backslash|B|=30\backslash)$ II. $\backslash(|A \cap B|=5\backslash)$

Together:

$\backslash 20+30-5=45 \backslash$

Sufficient.

46. DISJOINTNESS MAY REPLACE INTERSECTION SIZE

Question same.

I. $\backslash(|A|=20, \backslash|B|=30\backslash)$ II. A and B are disjoint.

Together:

$\backslash 45? \backslash$

Careful:

$\backslash 20+30=50 \backslash$

Sufficient.

47. LOGICAL PUZZLE DS

Question:

Who is scheduled on Wednesday?

I. A is Monday and B immediately after A. II. C is Wednesday.

Statement II alone directly answers:

$\backslash C \backslash$

Sufficient.

You do not need complete schedule.

48. FLOOR PUZZLE DS

Question:

Is A above B?

I. A is on floor 5. II. B is on floor 2.

Together yes.

But each alone insufficient unless building constraints force other person's position.

49. RELATIVE RELATION STATEMENT

Question:

Is A above B?

I. A lives immediately above B.

Immediately sufficient.

Again, direct relation beats exact locations.

50. SEATING DS

Question:

Is B adjacent to A?

I. Exactly one person sits between A and B.

This gives definite No.

Therefore:

$\boxed{\text{Sufficient}}$

Definite negative answer still counts.

51. BLOOD-RELATION DS

Question:

Is A B's brother?

I. A is B's sibling. II. A is male.

Together:

male sibling = brother.

Together sufficient.

52. DIRECTION DS

Question:

Is A east of B?

I. A is 5 km right of B while both reference a standard map orientation.

Sufficient.

But if "right" depends on an unknown facing person, maybe not.

Context matters.

53. CODING DS

Question:

What is code for CAT?

I. In the code, each letter is replaced by next alphabet letter.

Sufficient:

\[CAT\to DBU \]

No second statement required.

54. THREE-STATEMENT DATA SUFFICIENCY

Some exams give:

I, II, III.

Question may ask which combination is sufficient:

- I and II only
- II and III only
- I and III only
- any two
- all three
- one alone
- none

Use the same method.

55. THREE-STATEMENT EXAMPLE

Question:

What is $\frac{1}{x}$?

I.

\[$x+y=10$ \]

II.

\[$y+z=12$ \]

III.

$$\{z=4\}$$

Test:

III alone: no.

II+III:

$$\{y=8\}$$

still no x.

I+III:

y unknown.

No.

I+II:

still x,y,z with two equations.

No.

All three:

$$\{z=4\}$$

$$\{y=8\}$$

$$\{x=2\}$$

Thus:

$$\boxed{\text{All three required}}$$

56. THREE-STATEMENT REDUNDANCY

Question:

What is $\{x\}$?

I. $\{x+y=10\}$ II. $\{x-y=2\}$ III. $\{2x+2y=20\}$

I+II sufficient.

III is redundant with I.

So:

$$\boxed{\text{I and II are sufficient}}$$

57. MINIMAL SUFFICIENT SET

In three-statement DS, seek the **smallest combination** that fixes the answer.

Do not automatically combine all three.

58. THREE-STATEMENT YES/NO

Question:

Is $n!$ divisible by 30?

I. divisible by 2 II. divisible by 3 III. divisible by 5

Together:

$$\sqrt{LCM(2,3,5)=30}$$

All three together sufficient.

Any pair gives divisibility by only 6,10,15—insufficient.

59. THREE-STATEMENT GEOMETRY

Question:

Area of rectangle?

I. Length = 10 II. Diagonal = 26 III. Perimeter = 68

I+II gives breadth:

$$\sqrt{26^2 - 10^2} = 24$$

Area=240.

So I+II sufficient.

I+III:

$$2(10+b)=68 \Rightarrow b=24$$

also sufficient.

II+III may potentially give side pair from:

$$l+b=34, \quad l^2+b^2=676$$

Then:

$$(l+b)^2 = l^2 + b^2 + 2lb$$

$$1156 = 676 + 2lb$$

$$lb = 240$$

Thus area determined even though sides may swap.

So II+III also sufficient for **area**.

Excellent DS lesson:

The variables need not be uniquely labeled if the asked quantity is unique.

60. ASKED QUANTITY MAY BE UNIQUE DESPITE MULTIPLE SOLUTIONS

Suppose:

$$\begin{cases} x+y=10 \\ xy=21 \end{cases}$$

Solutions:

$$(x,y)=(3,7),(7,3)$$

If question asks x :

Insufficient.

If question asks $x+y$:

Already fixed.

If question asks xy :

Fixed.

If question asks $\{x,y\}$ as unordered values:

Sufficient.

Always focus on target.

61. SYMMETRY TRAP

Question:

What is x ?

I.

$$x+y=10$$

II.

$$xy=21$$

Together give roots 3 and 7, but do not tell which is x .

Thus:

$\boxed{\text{Insufficient}}$

unless another condition breaks symmetry.

62. ADD ORDER CONDITION

III.

$$x>y$$

Then:

$$x=7, y=3$$

Sufficient.

63. CONSISTENCY CHECK

If one statement makes no valid solution with another, the set is inconsistent.

In standard exam DS, assume well-formed statements unless testing consistency itself.

Do not mark contradiction as “sufficient.”

64. INDEPENDENT VS REDUNDANT STATEMENTS

Independent statements add new constraints.

Redundant statements merely restate previous information.

Spotting redundancy saves time.

65. EITHER STATEMENT ALONE — SUBTLETY

Question:

Is $\sqrt{n}$ even?

I. $\sqrt{n}$ divisible by 4. II. $\sqrt{n=18}$.

I alone !' Yes.

II alone !' Yes.

Both individually sufficient.

This is the “either alone” category.

66. EITHER ALONE CAN GIVE DIFFERENT YES/NO ANSWERS?

Suppose:

Question:

Is $\sqrt{x>0}$?

I. $\sqrt{x=5}$!' Yes II. $\sqrt{x=-5}$!' No

If statements are supposed to describe the same underlying problem instance, they are mutually inconsistent.

A valid DS problem should not normally do this.

So do not interpret “either alone” as allowing contradictory worlds.

67. PLACEMENT DS OPTION PATTERN

Before solving, write shorthand:

- I ' / I I '
- I ' / I I '
- I ' / I I '
- I ' / I I ' !' Together ' ?
- Together ' '

This prevents option confusion.

68. FAST DECISION TREE

Step A

Can I answer using I alone?

If yes, stop combining I with anything.

Now test II alone.

Step B

Can II alone answer?

Step C

If neither alone works, combine.

This is faster than solving both together first.

69. DS WITH OPTIONS — BACK-SOLVING

If question asks exact number and DS statements plus answer options are supplied, do not use answer options to make an insufficient statement sufficient unless the test format explicitly permits option-based determination.

Classical DS tests sufficiency from statements and given question information, not from answer-choice coincidences.

70. ROUNDING/APPROXIMATION DS

Question:

Is $\lfloor x \rfloor > 10$?

I. $\lfloor x \rfloor$ rounds to 11 to nearest integer.

Then:

$\lfloor 10.5 \rfloor \leq x < 11.5$

Thus definitely:

$\lfloor x \rfloor > 10$

Sufficient.

71. INTERVAL-BOUND DS

Question:

Is $\lfloor x < 5 \rfloor$?

I.

$\lfloor x \leq 4.9 \rfloor$

Sufficient.

II.

$\lfloor x < 6 \rfloor$

Insufficient because x could be 5.5 or 4.

Bounds are powerful.

72. MAX/MIN DS

Question:

What is maximum possible $\lfloor x \rfloor$?

I.

$\lfloor x \leq 10 \rfloor$

II.

$\lfloor x \rfloor$ is an integer and 10 is allowed.

Together maximum:

$\lfloor 10 \rfloor$

Depending exact wording, I may already be enough if domain says real and equality allowed.

Read domain carefully.

73. DS WITH “AT LEAST”

Question:

Is $\lfloor n \geq 5 \rfloor$?

I.

$\lfloor n > 7 \rfloor$

Definitely yes.

Sufficient.

74. DS WITH “AT MOST”

Question:

Is $\lfloor n/10 \rfloor$?

I.

$\lfloor n/10 \rfloor$

Definitely yes.

Sufficient.

75. COMMON ADVANCED TRAPS

& pTRAP 1 — Exact variable value required when only sign/order asked

False.

& pTRAP 2 — Ignoring zero

Can flip sign questions.

& pTRAP 3 — $(x^2 > y^2 \Rightarrow x > y)$

False without sign constraints.

& pTRAP 4 — Group averages without group sizes

Insufficient.

& pTRAP 5 — Geometry shape known but scale unknown

Insufficient.

& pTRAP 6 — Multiple solutions for variables but target quantity same

Can still be sufficient.

& pTRAP 7 — Three statements automatically all needed

Find minimal sufficient subset.

& pTRAP 8 — Dependent equations counted separately

No new information.

& pTRAP 9 — Solve whole puzzle unnecessarily

Answer only target.

& pTRAP 10 — Forget option mapping

Always check test instructions.

76. FINAL DATA-SUFFICIENCY TOOLKIT

MUST MEMORIZE

Exact-value question

One unique target value required.

Yes/No question

One fixed truth value required.

To prove insufficiency

Find two valid cases giving different target answers.

To prove sufficiency

Show all valid cases give same target answer.

Statements

Test separately before combining.

Domains

Integers/sign/distinctness can change everything.

Three-statement DS

Find the minimal sufficient combination.

77. THE 10-SECOND CHECK

Before answering a DS question, ask:

1. What exactly is target?
2. Exact value or Yes/No?
3. What is the domain?
4. Does I alone fix target?
5. Does II alone fix target?
6. If not, do they together?
7. Could I construct two different target answers?

That catches most mistakes.

PRACTICE — LEVEL 0

Q1

Question: Is $x > 0$?

I.

$x > 100$

Sufficient?

Q2

Question: Is $x > 0$?

I.

$$\{-2 < x < 3\}$$

Sufficient?

Q3

Question: Is $(xy > 0)$?

I. $(x < 0)$ II. $(y < 0)$

Status?

Q4

Question: What is average of 10 numbers?

I. Their sum is 250.

Sufficient?

Q5

Question: Area of square?

I. Diagonal is known.

Sufficient?

LEVEL 1

Q6

Question: Is $(x > y)$?

I.

$$\{x^2 > y^2\}$$

II.

$$\{x > 0, y > 0\}$$

Status?

Q7

Question: Overall average of two groups?

I. Average A=60 II. Average B=80

Sufficient together?

Q8

Question: How long A+B together take?

I. A alone 12 days II. B alone 18 days

Status?

Q9

Question: Area of rectangle?

I. Diagonal 13 II. One side 5

Status?

Q10

Question: Probability of red ball?

I. 3 red II. 7 non-red

Status?

LEVEL 2

Q11

Question: Is (x,y) both positive?

I.

$[xy > 0]$

II.

$[x+y > 0]$

Status?

Q12

Question: Is triangle right-angled?

I. Side lengths are 5,12,13.

Sufficient?

Q13

Question: What is (x) ?

I.

$[x+y=10]$

II.

$[xy=21]$

Together sufficient?

Q14

For Q13, add III:

$[x > y]$

Now sufficient?

Q15

Question: Is A adjacent to B?

I. Exactly two people sit between A and B.

Sufficient?

LEVEL 3 — EXAM PRESSURE

Q16

Question: Is positive integer n divisible by 30?

I. divisible by 2 II. divisible by 3 III. divisible by 5

What minimum combination is sufficient?

Q17

Question: Area of rectangle?

I. Length = 10 II. Diagonal = 26 III. Perimeter = 68

Which pairs are sufficient?

Q18

Question: Is $x > 0$?

I.

$xy < 0$

II.

$x + y > 0$

Sufficient together?

Q19

Question: Who is tallest among A,B,C,D?

I. $A > B > C$ II. $D > B$

Together sufficient?

Q20

Question: What is probability of success?

I. Success:Failure = 3:2.

Sufficient?

ANSWERS

Q1

Every $x > 100$ is positive.

$\boxed{\text{Sufficient}}$

Q2

Range contains both negative and positive values.

$\boxed{\text{Insufficient}}$

Q3

Negative $\times$ negative = positive.

Neither alone sufficient; together sufficient.

$\boxed{\text{Together sufficient}}$

Q4

$\text{Average} = \frac{250}{10} = 25$

$\boxed{\text{Sufficient}}$

Q5

For square:

$d = s\sqrt{2}$

Known diagonal determines side and therefore area.

$\boxed{\text{Sufficient}}$

Q6

I alone insufficient due to signs.

II alone insufficient because exact ordering unknown.

Together:

For positive x, y :

$x^2 > y^2 \rightarrow x > y$

$\boxed{\text{Together sufficient}}$

Q7

No group sizes.

Weighted overall average unknown.

$\boxed{\text{Insufficient}}$

Q8

Together:

$\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$

Time:

$\frac{36}{5}$

unique.

$\boxed{\text{Together/sufficient}}$

Q9

Pythagoras:

$$13^2 - 5^2 = 144$$

Other side = 12.

Area:

$$60$$

$\boxed{\text{Together/sufficient}}$

Q10

Total:

$$3 + 7 = 10$$

$$P(R) = \frac{3}{10}$$

$\boxed{\text{Together/sufficient}}$

Q11

I says same sign.

II says sum positive.

Both negative would give negative sum.

Therefore both positive.

$\boxed{\text{Together/sufficient}}$

Q12

$$5^2 + 12^2 = 13^2$$

Definitely right triangle.

$\boxed{\text{Sufficient}}$

Q13

Together:

$$x + y = 10, xy = 21$$

Values are 3 and 7, but x may be either.

$\boxed{\text{Insufficient}}$

Q14

With:

$$\boxed{x > y}$$

we get:

$$\boxed{x = 7, y = 3}$$

$$\boxed{\text{Sufficient}}$$

Q15

Exactly two between means not adjacent.

Definite No.

$$\boxed{\text{Sufficient}}$$

Q16

Divisible by 30 requires factors:

$$\boxed{2, 3, 5}$$

Any pair gives only divisibility by 6, 10, 15.

All three:

$$\boxed{\text{LCM}(2, 3, 5) = 30}$$

$$\boxed{I + II + III}$$

Q17

I + II

$$\boxed{l = 10, d = 26}$$

$$\boxed{b = 24}$$

Area = 240.

Sufficient.

I + III

$$\boxed{2(10 + b) = 68}$$

$$\boxed{b = 24}$$

Sufficient.

II + III

$$\boxed{l + b = 34}$$

$$\boxed{l^2 + b^2 = 676}$$

$$\sqrt{(l+b)^2 = l^2 + b^2 + 2lb}$$

$$\sqrt{1156 = 676 + 2lb}$$

$$\sqrt{lb = 240}$$

Area determined.

So:

$\boxed{\text{All three pairs are sufficient}}$

Excellent example where sides need not be individually identified.

Q18

$$\sqrt{xy < 0}$$

means opposite signs.

$$\sqrt{x+y > 0}$$

only tells positive one has larger magnitude.

Either x or y could be the positive variable.

Thus x's sign not fixed.

$\boxed{\text{Insufficient}}$

Q19

I:

$$\sqrt{A > B > C}$$

D unknown.

II:

$$\sqrt{D > B}$$

Together:

A and D both above B.

A vs D unknown.

Tallest could be A or D.

$\boxed{\text{Insufficient}}$

Q20

Success:Failure:

$$\sqrt{3:2}$$

Total ratio parts:

$$\sqrt{5}$$

Therefore:

$$P(\text{success}) = \frac{3}{5}$$

$\boxed{\text{Sufficient}}$

CHAPTER 69 COMPLETE

We have now covered:

- core meaning of sufficiency
- exact-value vs Yes/No DS
- I/II alone testing
- both together
- either alone
- both insufficient
- domains and hidden restrictions
- signs/inequalities
- quadratics/absolute values
- averages/mixtures
- time-work/speed/profit
- advanced geometry
- probability/counting
- sets
- ranking/seating/blood relation/logical DS
- three-statement DS
- minimal sufficient subsets
- redundancy/dependence
- symmetry traps
- target-quantity uniqueness
- counterexample proof
- speed decision tree
- L0–L3 practice

Progress

$\boxed{69/76 \text{ numbered chapters completed}}$

UNIT 3 — LOGICAL & ANALYTICAL REASONING

- Chapter 57 — Series
- Chapter 58 — Analogy, Classification & Odd-One-Out
- Chapter 59 — Coding-Decoding
- Chapter 60 — Blood Relations
- Chapter 61 — Direction Sense
- Chapter 62 — Ranking & Ordering
- Chapter 63 — Linear Seating

- Chapter 64 — Circular & Parallel Seating '
- Chapter 65 — Logical Puzzles '
- Chapter 66 — Syllogisms '
- Chapter 67 — Statements, Assumptions & Conclusions '
- Chapter 68 — Arguments, Cause–Effect & Course of Action '
- Chapter 69 — Data Sufficiency '
- Chapter 70 — Logical Deduction & Constraint Reasoning +

NEXT

CHAPTER 70 — LOGICAL DEDUCTION & CONSTRAINT REASONING

This is the **final numbered chapter of Unit 3**.

Next we will cover:

- propositional constraints
- if / only if / unless / iff
- necessary vs sufficient conditions
- implication chains
- contraposition
- XOR / OR / AND
- truth consistency
- constraint graphs
- must/could/cannot-be-true
- ordering and selection deduction
- min/max under constraints
- contradiction proofs
- GATE-style formal logical reasoning
- placement-style constraint sets
- final Unit 3 revision after Chapter 70.

USER

next

ASSISTANT

CHAPTER 70 — LOGICAL DEDUCTION & CONSTRAINT REASONING

PART 1 — PROPOSITIONAL CONSTRAINTS, IF / ONLY IF / UNLESS / IFF, NECESSARY–SUFFICIENT CONDITIONS, IMPLICATION CHAINS & CONTRADICTION REASONING

Status: ~~0~~ = 100% PROGRESS **GATE:** Very High **Placements:** Very High **Priority:** ~~0~~ = 100% 1

This is the **final numbered chapter of Unit 3 — Logical & Analytical Reasoning**.

This chapter combines many ideas already learned:

- conditional logic
- grouping
- ordering
- puzzles
- assumptions
- data sufficiency
- must/could/cannot reasoning

The central skill is:

Translate English constraints into precise logic before solving.

1. WHAT IS A PROPOSITION?

A proposition is a statement that can be classified as:

$\boxed{\text{True or False}}$

Examples:

- “7 is prime.”! True
- “10 is odd.”! False

But:

“Please sit down.”

is a command, not a proposition.

2. USING SYMBOLS

Let:

$P = \text{Ravi is selected}$

$Q = \text{Sita is selected}$

Then instead of writing full sentences repeatedly, we use:

P, Q

This makes constraint problems much easier.

3. BASIC LOGICAL OPERATORS

AND

$P \text{ and } Q$

means:

P and Q are both true.

OR

$\lceil P \vee Q \rceil$

means:

At least one of P,Q is true.

Usually logical OR is **inclusive**:

- P only '
- Q only '
- both '

NOT

$\lceil \neg P \rceil$

means:

P is false.

IMPLIES

$\lceil P \rightarrow Q \rceil$

means:

If P, then Q.

IF AND ONLY IF

$\lceil P \leftrightarrow Q \rceil$

means:

P if and only if Q.

Both directions hold.

4. TRUTH TABLE — AND

| P | Q | $(P \text{ and } Q)$ |
|---|---|----------------------|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

AND requires both.

5. TRUTH TABLE — OR

| P | Q | $(P \text{ or } Q)$ |
|---|---|---------------------|
| T | T | T |
| T | F | T |
| F | T | T |
| F | F | F |

OR fails only when both are false.

6. EXCLUSIVE OR — XOR

Exactly one of P,Q:

$\boxed{P \oplus Q}$

Truth table:

| P | Q | XOR |
|---|---|-----|
| T | T | F |
| T | F | T |
| F | T | T |
| F | F | F |

Memory:

XOR = one, but not both.

7. IMPLICATION

$P \rightarrow Q$

means:

If P occurs, Q must occur.

Example:

If A is selected, B is selected.

$\lceil A \rightarrow B \rceil$

8. WHAT DOES IMPLICATION NOT MEAN?

From:

$\lceil A \rightarrow B \rceil$

you **cannot** conclude:

$\lceil B \rightarrow A \rceil$

Example:

If someone is a doctor, they are educated.

Being educated does not necessarily mean being a doctor.

9. CONTRAPOSITIVE

From:

$\lceil P \rightarrow Q \rceil$

we may always infer:

$\lceil \boxed{\neg Q \rightarrow \neg P} \rceil$

This is logically equivalent.

Memory:

Reverse + negate both.

10. EXAMPLE

Rule:

If A is selected, B is selected.

$\lceil A \rightarrow B \rceil$

If B is not selected:

$\neg B$

then:

$\boxed{\neg A}$

A cannot be selected.

11. CONVERSE

Converse of:

$P \rightarrow Q$

is:

$Q \rightarrow P$

The converse is **not automatically valid**.

12. INVERSE

Inverse is:

$\neg P \rightarrow \neg Q$

Also not automatically valid.

13. CONTRAPOSITIVE IS THE ONLY GUARANTEED EQUIVALENT

For:

$P \rightarrow Q$

only:

$\boxed{\neg Q \rightarrow \neg P}$

is logically equivalent.

14. NECESSARY CONDITION

If:

$P \rightarrow Q$

then Q is **necessary for P**.

Meaning:

P cannot happen without Q.

Example:

A passport is necessary for this entry process.

15. SUFFICIENT CONDITION

If:

$$\[P\rightarrow Q \]$$

then P is **sufficient for Q**.

Meaning:

Whenever P occurs, Q follows.

Thus in:

$$\[P\rightarrow Q \]$$

- P = sufficient
- Q = necessary

16. MEMORY TRICK

For:

$$\[P\rightarrow Q \]$$

say:

P is enough for Q. Q is required for P.

17. “ONLY IF”

Statement:

A is selected only if B is selected.

Means:

$$\[\boxed{A\rightarrow B} \]$$

Memory:

A only if B = A needs B.

18. “IF”

If A is selected, B is selected.

Also:

$\boxed{A \rightarrow B}$

19. “A IF B”

This wording reverses many students.

A happens if B happens.

Means:

$\boxed{B \rightarrow A}$

Because B is sufficient for A.

20. EXAMPLE

Ravi attends if Sita attends.

Means:

$\boxed{Sita \rightarrow Ravi}$

not the reverse.

21. “IF AND ONLY IF”

A occurs if and only if B occurs.

Means:

$\boxed{A \rightarrow B}$

and:

$\boxed{B \rightarrow A}$

Together:

$\boxed{A \leftrightarrow B}$

22. BICONDITIONAL EFFECT

If:

$\boxed{A \leftrightarrow B}$

then A and B always share truth value:

- both true
- both false

Allowed:

$\forall (T,T),(F,F) \setminus$

Not:

$\forall (T,F),(F,T) \setminus$

23. “TOGETHER” CONSTRAINT

If a puzzle says:

A and B are selected together.

Usually:

$\forall \boxed{A \leftrightarrow B} \setminus$

So:

A selected !' B selected B selected !' A selected

24. “A REQUIRES B”

Means:

$\forall \boxed{A \rightarrow B} \setminus$

A depends on B.

25. “A DEPENDS ON B”

Again:

$\forall A \rightarrow B \setminus$

If B unavailable:

$\forall \neg B \rightarrow \neg A \setminus$

26. “UNLESS”

This is a famous trap.

Statement:

A unless B.

A useful logical translation is:

$\forall \boxed{\neg B \rightarrow A} \setminus$

Equivalent to:

$\forall \boxed{A \vee B} \setminus$

27. EXAMPLE

The match will continue unless it rains.

Means:

If it does not rain:

$\boxed{\text{Match continues}}$

It does not necessarily mean that if it rains, the match definitely stops unless context states that stronger relation.

28. “UNLESS” AS OR

A unless B

can be written:

A or B

because at least one must hold.

29. “NEITHER A NOR B”

Means:

$\neg(A \text{ and } B)$

Neither is true.

30. “NOT BOTH”

Means:

$\neg(A \text{ and } B)$

Allowed:

- A only
- B only
- neither

Not allowed:

- both
-

31. “AT LEAST ONE”

A or B

Allowed:

- A
-

- B
 - both
-

32. “EXACTLY ONE”

$\boxed{A \oplus B}$

Allowed:

- A only
 - B only
-

33. “AT MOST ONE”

Means:

$\boxed{\neg(A \wedge B)}$

same truth-condition as “not both.”

34. IMPLICATION CHAIN

If:

$A \rightarrow B$

and:

$B \rightarrow C$

then:

$\boxed{A \rightarrow C}$

This is transitivity.

35. LONG CHAIN

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

Then:

$\boxed{A \rightarrow D}$

Selecting A forces all:

A, B, C, D

36. CONTRAPOSITIVE CHAIN

From:

$\lceil A \rightarrow B \rightarrow C \rightarrow D \rceil$

we can reason backward:

$\lceil \neg D \rightarrow \neg C \rightarrow \neg B \rightarrow \neg A \rceil$

37. EXAMPLE

Rules:

- If A attends, B attends.
- If B attends, C attends.
- C does not attend.

Then:

$\lceil \neg C \rightarrow \neg B \rightarrow \neg A \rceil$

So:

$\lceil \boxed{A \text{ and } B \text{ cannot attend}} \rceil$

38. IMPLICATION GRAPH

You can draw:

``text id="q1v37z" A !' B !' C !' D

This is much easier than rereading sentences.

39. NEGATIVE CONSEQUENCE

Suppose:

$\lceil A \rightarrow B \rceil$

$\lceil B \rightarrow \neg C \rceil$

Then:

$\lceil \boxed{A \rightarrow \neg C} \rceil$

So A and C cannot coexist.

40. EXAMPLE

I f A s e l e c t e d ! ' B s e l e c t e d .

I f B s e l e c t e d ! ' C n o t s e l e c t e d .

Therefore:

$$\boxed{A \rightarrow \neg C}$$

41. DIRECT CONFLICT

Suppose:

$$A \rightarrow B$$

and:

$$A \rightarrow \neg B$$

If A were true, it would force both B and not-B.

Impossible.

Therefore:

$$\boxed{\neg A}$$

A cannot be true.

42. CONTRADICTION METHOD

To test whether A is possible:

1. assume A
2. follow all forced consequences
3. if you derive both X and $\neg X$, contradiction
4. therefore A cannot occur

This is one of the strongest GATE/placement methods.

43. EXAMPLE

Rules:

$$A \rightarrow B$$
$$B \rightarrow C$$
$$C \rightarrow \neg A$$

Assume A.

Then:

```
\[
A\rightarrow B\rightarrow C\rightarrow\neg A
\]
```

We get A and not-A.

Therefore:

```
\[
\boxed{A\text{ cannot be true}}
\]
```

44. CYCLE DOES NOT ALWAYS MEAN CONTRADICTION

Suppose:

```
\[
A\rightarrow B
\]
```

```
\[
B\rightarrow A
\]
```

This is simply:

```
\[
A\leftrightarrow B
\]
```

No contradiction.

They occur together.

45. NEGATIVE CYCLE

```
\[
A\rightarrow\neg B
\]
```

```
\[
B\rightarrow\neg A
\]
```

This means A and B cannot coexist.

But either could occur alone.

No contradiction by itself.

46. MUST BE TRUE

A statement is **must be true** if every valid assignment satisfies it.

Method:

> Try to make it false. If doing so always creates contradiction, it must be true.

47. COULD BE TRUE

A statement is **could be true** if at least one valid assignment satisfies it.

Method:

> Construct one valid example.

48. CANNOT BE TRUE

A statement cannot be true if assuming it always creates contradiction.

This is direct contradiction testing.

49. MUST / COULD / CANNOT MEMORY

MUST

All valid worlds.

COULD

At least one valid world.

CANNOT

No valid world.

50. SIMPLE SELECTION SET

Choose exactly 3 from:

\[
A,B,C,D,E
\]

Rules:

\[
A $\rightarrow$ B
\]

\[
C $\rightarrow$ D
\]

\[
B $\rightarrow$ $\neg$ D
\]

\[
E\text{ must be selected}
\]

Let's derive.

If A selected:

\[
A $\rightarrow$ B

\]

B implies:

\[
\neg D
\]

If C also selected:

\[
C\rightarrow D
\]

Contradiction.

Thus:

\[
\boxed{A\text{ and }C\text{ cannot both be selected}}
\]

51. FORCED GROUP FROM A

If A selected:

\[
A,B,E
\]

already gives 3 members.

Thus group is exactly:

\[
\boxed{\{A,B,E\}}
\]

52. FORCED GROUP FROM C

If C selected:

\[
C,D,E
\]

Since B cannot coexist with D, B absent.

T h e n A a b s e n t b e c a u s e A ! ' B .

Thus:

\[
\boxed{\{C,D,E\}}
\]

Exactly the two groups seen earlier in Chapter 65.

53. MUST-BE-TRUE FROM VALID GROUPS

Valid groups:

\[
ABE
\]

\[
CDE
\]

Common element:

\[
\boxed{E}
\]

Thus E must be selected.

54. COULD-BE-TRUE

A could be selected because:

\[
ABE
\]

is valid.

C could also be selected because:

\[
CDE
\]

is valid.

55. CANNOT-BE-TRUE

"A and C are both selected" cannot happen.

B e c a u s e A ! ' B ! ' n o t D w h i l e C ! ' D .

Contradiction.

56. OR CONSTRAINT WITH IMPLICATION

Rules:

\[
A\lor B
\]

\[
A\rightarrow C
\]

\[
B\rightarrow D
\]

At least one of A,B occurs.

Therefore at least one of C,D must occur.

But both may occur if A and B both occur.

57. XOR WITH IMPLICATIONS

Exactly one of A,B:

\[
A\oplus B
\]

and:

\[
A\rightarrow C
\]

\[
B\rightarrow D
\]

Then:

- if A !' C
- if B !' D

But cannot conclude exactly one of C,D unless additional rules prevent crossover.

C and D may have independent causes.

58. DON'T OVER-PROPAGATE

Given:

\[
A\rightarrow C
\]

if C is true, A need not be true.

This converse error is extremely common in constraint sets.

59. ORDERING CONSTRAINTS

Logical deduction also appears as ordering.

Example:

\[
A<B
\]

means A occurs before B.

If:

\[

$A < B, \quad B < C$
 $\backslash$

then:

$\backslash$
 $\boxed{A < C}$
 $\backslash$

60. ORDER CHAIN

Given:

$\backslash$
 $A < B < C < D$
 $\backslash$

we know:

- A before all
- D after all
- B before C, D
- C after A, B

No exact positions required.

61. PARTIAL ORDER

Given:

$\backslash$
 $A < B$
 $\backslash$

$\backslash$
 $C < B$
 $\backslash$

A and C are both before B, but A/C relation unknown.

Possible:

$\backslash$
 $A < C < B$
 $\backslash$

or:

$\backslash$
 $C < A < B$
 $\backslash$

Don't invent comparison.

62. MUST-BE-BEFORE

If every path implies A before D:

$\backslash$
 $A < B < C < D$

\]

then A must be before D.

63. COULD-BE-FIRST

In a partial order, someone can be first if no constraint forces another item before them.

Example:

\[
A<C,\quad B<C
\]

A or B could be first.

C cannot.

64. CANNOT-BE-FIRST

Anyone with a required predecessor cannot be first.

If:

\[
A<B
\]

then B cannot be first.

65. CANNOT-BE-LAST

Anyone with a forced successor cannot be last.

If:

\[
A<B
\]

then A cannot be last.

66. PRECEDENCE GRAPH

Represent:

```
```text id="q5n60o"  
A !' C
B !' C
C !' D
```

Then:

- A before C,D
- B before C,D
- C before D

- A/B relation unknown

## 67. TOPOLOGICAL-ORDER THINKING

Any complete ordering satisfying all arrows is valid.

Example graph:

```
graph LR
 A --> B
 A --> C
 B --> D
 C --> D
```

Valid:

```
\[
A, B, C, D
\]
```

or:

```
\[
B, A, C, D
\]
```

But not:

```
\[
C, A, B, D
\]
```

because C appears before its prerequisites.

---

### # 68. UNIQUE ORDER

If constraints form:

```
\[
A < B < C < D
\]
```

then only one order:

```
\[
\boxed{A, B, C, D}
\]
```

---

### # 69. MULTIPLE VALID ORDERS

If:

```
\[
A < C, \quad B < C
\]
```

then A/B can swap.

Full order not unique.

But C may still have a fixed relative role.

---

#### # 70. MINIMUM POSSIBLE POSITION

Suppose 6 tasks and constraints force exactly 2 tasks before A.

Then earliest possible rank of A:

$$\boxed{3}$$

General:

$$\boxed{\text{Min rank} = 1 + \text{number forced before}}$$

---

#### # 71. MAXIMUM POSSIBLE POSITION

If exactly 2 tasks are forced after A among  $(N=6)$ :

Latest possible rank:

$$6 - 2 = 4$$

General:

$$\boxed{\text{Max rank} = N - \text{number forced after}}$$

This connects to Chapter 62.

---

#### # 72. COUNT CONSTRAINTS

Suppose select exactly 4 people.

At least 2 from:

$$\{A, B, C\}$$

and at most 1 from:

$$\{D, E\}$$

Count restrictions can force selections before individual identity is known.

---

#### # 73. COMPLEMENT COUNT

Exactly 4 selected.

At most 1 from D, E.

Then at least:

$\backslash[$   
4-1=3  
 $\backslash]$

must come from everyone outside D,E.

This is a strong deduction.

---

#### # 74. TWO-GROUP LOGIC

Only two teams X,Y.

If:

$\backslash[$   
A $\backslash$ ne B  
 $\backslash]$

and:

$\backslash[$   
B $\backslash$ ne C  
 $\backslash]$

then:

$\backslash[$   
 $\boxed{A=C}$   
 $\backslash]$

Because only two group labels exist.

---

#### # 75. XOR VIEW OF TWO-GROUP DIFFERENCE

If A and B must be in different binary groups:

$\backslash[$   
A $\oplus$  B  
 $\backslash]$

in terms of group-label truth encoding.

Two inequality edges form parity relationships.

---

#### # 76. THREE DIFFERENCES CREATE CONTRADICTION IN TWO GROUPS

Suppose:

$\backslash[$   
A $\backslash$ ne B  
 $\backslash]$

$\backslash[$   
B $\backslash$ ne C  
 $\backslash]$

$\backslash[$   
C $\backslash$ ne A  
 $\backslash]$

with only two teams.

From first two:

```
\[
A=C
\]
```

But third says:

```
\[
A≠C
\]
```

Contradiction.

Thus impossible.

---

#### # 77. GRAPH COLORING CONNECTION

Two-team "same/different" constraints are like 2-coloring a graph.

- same !' same color
- different !' opposite color

An odd cycle of "different" constraints cannot be satisfied with two teams.

This is useful intuition.

---

#### # 78. LOGICAL CONSISTENCY

A set of constraints is consistent if at least one assignment satisfies all of them.

To show consistency:

> Construct one valid assignment.

To show inconsistency:

> Derive contradiction.

---

#### # 79. EXAMPLE — CONSISTENT

Rules:

```
\[
A⇒B
\]
```

```
\[
B⇒C
\]
```

Could assign all false:

```
\[
A=F, B=F, C=F
\]
```



Valid.

So rules themselves are consistent.

---

#### # 80. EXAMPLE — INCONSISTENT WITH FIXED FACT

Rules:

```
\[
A\rightarrow B
\]
```

```
\[
B\rightarrow\neg C
\]
```

Facts:

```
\[
A=true,\ C=true
\]
```

A !' B !' not C .

But C true.

Contradiction.

```
\[
\boxed{\text{No valid assignment}}
\]
```

---

#### # 81. VACUOUS IMPLICATION — IMPORTANT

```
\[
P\rightarrow Q
\]
```

is not violated when P is false.

Example:

Rule:

> If A is selected, B is selected.

If A is not selected, B may be:

- selected
- not selected

Both are allowed.

This prevents converse mistakes.

---

#### # 82. IMPLICATION TRUTH TABLE

P	Q	\(P\rightarrow Q\)
---	---	---
T	T	T

T	F	**F**
F	T	T
F	F	T

Only one case violates implication:

$$\boxed{P=\text{true}, Q=\text{false}}$$

---

#### # 83. FAST IMPLICATION CHECK

To satisfy:

$$P \rightarrow Q$$

forbid:

$$\boxed{P \wedge \neg Q}$$

That's all.

---

#### # 84. IMPLICATION AS OR

Useful equivalence:

$$\boxed{P \rightarrow Q \equiv \neg P \vee Q}$$

This is useful in formal GATE logic.

---

#### # 85. EXAMPLE

$$A \rightarrow B$$

equivalent:

$$\neg A \vee B$$

Meaning:

either A is absent, or B is present.

---

#### # 86. BICONDITIONAL AS TWO IMPLICATIONS

$$A \leftrightarrow B$$

means:

$$\begin{aligned} & \neg [ \\ & (A \rightarrow B) \wedge (B \rightarrow A) \\ & ] \end{aligned}$$

---

#### # 87. XOR FORM

Exactly one of A,B:

$$\begin{aligned} & \neg [ \\ & (A \vee B) \wedge \neg(A \wedge B) \\ & ] \end{aligned}$$

Useful conceptually.

---

#### # 88. DE MORGAN'S LAWS — ESSENTIAL

$$\begin{aligned} & \neg [ \\ & \boxed{\neg(A \wedge B) = \neg A \vee \neg B} \\ & ] \end{aligned}$$

and:

$$\begin{aligned} & \neg [ \\ & \boxed{\neg(A \vee B) = \neg A \wedge \neg B} \\ & ] \end{aligned}$$

Memory:

> Negation flips AND !” OR and negates each part .

---

#### # 89. EXAMPLE

Negation of:

> A and B are both selected.

is:

> At least one of A or B is not selected.

Not necessarily both absent.

---

#### # 90. NEGATION OF “A OR B”

$$\begin{aligned} & \neg [ \\ & \neg(A \vee B) \\ & ] \end{aligned}$$

means:

$$\begin{aligned} & \neg [ \\ & \boxed{\neg A \wedge \neg B} \\ & ] \end{aligned}$$

Neither A nor B.

---

#### # 91. NEGATING "AT LEAST ONE"

"At least one of A,B selected":

$$\begin{aligned} & \backslash[ \\ & A \backslash \text{lor } B \\ & \backslash] \end{aligned}$$

Negation:

$$\begin{aligned} & \backslash[ \\ & \boxed{\neg A \backslash \text{land } \neg B} \\ & \backslash] \end{aligned}$$

none selected.

---

#### # 92. NEGATING "BOTH"

"Both A,B selected":

$$\begin{aligned} & \backslash[ \\ & A \backslash \text{land } B \\ & \backslash] \end{aligned}$$

Negation:

$$\begin{aligned} & \backslash[ \\ & \boxed{\neg A \backslash \text{lor } \neg B} \\ & \backslash] \end{aligned}$$

at least one absent.

---

#### # 93. EXACTLY / AT LEAST / AT MOST

For a set:

### Exactly k  
count = k.

### At least k  
count  $\backslash(\backslash \text{ge } k \backslash)$ .

### At most k  
count  $\backslash(\backslash \text{le } k \backslash)$ .

These are formal constraints, not approximate language.

---

#### # 94. COMBINING COUNT + IMPLICATION

Select exactly 2 from A,B,C.

Rules:

$$\begin{aligned} & \backslash[ \\ & A \backslash \text{Rrightarrow } B \\ & \backslash] \end{aligned}$$
$$\backslash[$$

$B \rightarrow C$   
 $\backslash$

If A selected:

$A! \wedge B! \wedge C$  gives 3 selected.

But exactly 2 allowed.

Therefore:

$\backslash$   
 $\boxed{A \text{ cannot be selected}}$   
 $\backslash$

---

# 95. THEN WHAT?

With A absent, exactly two from A,B,C means:

$\backslash$   
 $\boxed{B, C}$   
 $\backslash$

must both be selected.

This is a forced solution.

---

# 96. ANOTHER COUNT SET

Select exactly 3 from A,B,C,D.

Rules:

$\backslash$   
 $A \rightarrow B$   
 $\backslash$

$\backslash$   
 $C \rightarrow \neg B$   
 $\backslash$

If A and C both selected:

$A! \wedge B, C! \wedge \neg B$ .

Contradiction.

Thus A,C cannot coexist.

---

# 97. CONDITIONAL ORDER

Rule:

> If A is first, B is last.

$\backslash$   
 $A_1 \rightarrow B_N$   
 $\backslash$

If B is not last:

\[  
\boxed{A\text{ cannot be first}}  
\]

via contraposition.

---

#### # 98. CONDITIONAL SCHEDULING

Rule:

> If C occurs on Tuesday, D occurs Wednesday.

If D is Thursday, can we conclude C not Tuesday?

Yes, assuming Wednesday and Thursday are exclusive slots for D.

Because:

\[  
C\_{\text{Tue}} \rightarrow D\_{\text{Wed}}  
\]

and:

\[  
\neg D\_{\text{Wed}}  
\]

so:

\[  
\boxed{\neg C\_{\text{Tue}}}  
\]

---

#### # 99. "ONLY ONE OF THESE CONDITIONS HOLDS"

Suppose exactly one is true:

- A before B
- C before D

Then XOR:

\[  
(A < B) \oplus (C < D)  
\]

Do not treat it as ordinary OR.

---

#### # 100. FINAL PART-1 LOGIC WORKFLOW

When given a constraint set:

### Step 1

Define variables.

### Step 2

Translate English:

- if !' i m p l i c a t i o n

- only if !' implication
- iff / together !' biconditional
- unless !'  $\neg B \rightarrow A$
- exactly one !' XOR
- not both !' incompatibility

### Step 3  
Build chains.

### Step 4  
Use contrapositives.

### Step 5  
Apply count/order constraints.

### Step 6  
Assume candidate option.

### Step 7  
Propagate consequences.

### Step 8  
Reject contradictions.

---

# 101. FAST SYMBOL TABLE

English	Logic
If A, then B	$A \rightarrow B$
A only if B	$A \rightarrow B$
A if B	$B \rightarrow A$
A iff B	$A \leftrightarrow B$
A requires B	$A \rightarrow B$
A unless B	$\neg B \rightarrow A$
At least one	$A \vee B$
Exactly one	$A \oplus B$
Not both	$\neg(A \wedge B)$
Neither	$\neg(A \wedge B)$

---

# 102. COMMON TRAPS

& p    \* \* T R A P   1   -   R e v e r s i n g   i m p l i c a t i o n \* \*

```
\[
A \rightarrow B
\not\rightarrow
B \rightarrow A
\]
```

& p    \* \* T R A P   2   -   " O n l y   i f "   r e v e r s e d \* \*

A only if B:

```
\[
A \rightarrow B
\]
```

& p    \* \* T R A P   3   -   " A   i f   B "   r e v e r s e d \* \*

Correct:

```
\[
```

B $\rightarrow$  A  
\\]

& p   \* \* T R A P   4   -   O R   t r e a t e d   a s   X O R \* \*

Ordinary OR allows both.

& p   \* \* T R A P   5   -   N o t   b o t h   t r e a t e d   a s   n e i t h e r \* \*

Not both allows one.

& p   \* \* T R A P   6   -   F a l s e   a n t e c e d e n t   u s e d   t o   i n f e r   c o n s e q u e n c e

I f   A   f a l s e ,   r u l e   A! ' B   s a y s   n o t h i n g   a b o u t   B .

& p   \* \* T R A P   7   -   P a r t i a l   o r d e r   m a d e   c o m p l e t e \* \*

Do not invent missing comparisons.

& p   \* \* T R A P   8   -   C o n t r a d i c t i o n   c h a i n   n o t   p r o p a g a t e d \* \*

Follow implications fully.

---

# PRACTICE – LEVEL 0

### Q1

If:

\\[  
A $\rightarrow$  B  
\\]

and A is true, what follows?

### Q2

If:

\\[  
A $\rightarrow$  B  
\\]

and B is false, what follows?

### Q3

Does B=true imply A=true?

### Q4

"Only if B, A" corresponds to what relation?

### Q5

Exactly one of A,B means which logical operator?

---

# LEVEL 1

### Q6

Rules:

\\[  
A $\rightarrow$  B  
\\]

\\[



$B \rightarrow C$   
\\]

If A is true, what must be true?

### Q7  
Same rules. If C is false, what follows?

### Q8  
Rules:

\\[  
 $A \rightarrow B$   
\\]

\\[  
 $B \rightarrow \neg C$   
\\]

Can A and C both be true?

### Q9  
Exactly one of A,B is true. If A is false, what follows?

### Q10  
A iff B. If A is false, what follows about B?

---

# LEVEL 2

### Q11  
Select exactly 2 from A,B,C.

Rules:

\\[  
 $A \rightarrow B$   
\\]

\\[  
 $B \rightarrow C$   
\\]

Can A be selected?

### Q12  
Using Q11, which two must be selected?

### Q13  
Two teams only:

\\[  
 $A \neq B, \text{quad } B \neq C$   
\\]

What relation between A,C?

### Q14  
Only two teams:

\\[  
 $A \neq B, \neg B \neq C, \neg C \neq A$   
\\]

Is this possible?

### Q15

Rules:

$$\begin{aligned} & \backslash[ \\ & A < B, \quad C < B \\ & \backslash] \end{aligned}$$

Must A be before C?

---

# LEVEL 3 – EXAM PRESSURE

Exactly 3 people are selected from A,B,C,D,E.

Rules:

1.  $A \rightarrow B$
2.  $B \rightarrow \neg D$
3.  $C \rightarrow D$
4. E must be selected.

### Q16

Can A and C both be selected?

### Q17

If A is selected, what is the full selected group?

### Q18

If C is selected, what is the full selected group?

### Q19

Who must be selected in every valid group?

### Q20

Which pair cannot be selected together?

---

# ANSWERS

## Q1

$$\backslash[ \\ \boxed{B} \\ \backslash]$$

must be true.

---

## Q2

By contraposition:

$$\backslash[ \\ \neg B \rightarrow \neg A \\ \backslash]$$

$$\backslash[ \\ \boxed{A \text{ is false}} \\ \backslash]$$

---

## Q3

No.

This is converse reasoning.

\[  
\boxed{No}  
\]

---

## Q4

"A only if B":

\[  
\boxed{A\rightarrow B}  
\]

---

## Q5

\[  
\boxed{XOR}  
\]

---

## Q6

\[  
A\rightarrow B\rightarrow C  
\]

So:

\[  
\boxed{B\text{ and }C}  
\]

must be true.

---

## Q7

\[  
A\rightarrow B\rightarrow C  
\]

Contrapositive chain:

\[  
\neg C\rightarrow\neg B\rightarrow\neg A  
\]

So:

\[  
\boxed{A\text{ and }B\text{ are false}}  
\]

---

## Q8

$A \wedge B \wedge \neg C$ .

If A and C both true, contradiction.

\[  
\boxed{No}  
\]

---

## Q9

Exactly one of A,B.

A false means:

\[  
\boxed{B \text{ true}}  
\]

---

## Q10

A iff B means equal truth values.

A false:

\[  
\boxed{B \text{ false}}  
\]

---

# Q11

$A \wedge B \wedge C$ .

If A selected, all three selected.

But exactly 2 allowed.

Therefore:

\[  
\boxed{A \text{ cannot be selected}}  
\]

---

# Q12

A absent.

Need exactly 2 from A,B,C.

So:

\[  
\boxed{B,C}  
\]

must be selected.

---

# Q13

Only two groups:

A different B.

B different C.

Therefore:

$\boxed{A=C}$

same team.

---

# Q14

From first two:

$A=C$

but third requires:

$A \neq C$

Contradiction.

$\boxed{\text{Impossible}}$

---

# Q15

No.

Both A and C are before B, but their mutual order is unknown.

$\boxed{\text{Cannot be determined}}$

---

# Q16–Q20

Rules:

$A \rightarrow B$

$B \rightarrow \neg D$

\[  
C\rightarrow D  
\]

E fixed.

## Q16

If A selected:

\[  
A\rightarrow B\rightarrow\neg D  
\]

If C selected:

\[  
C\rightarrow D  
\]

Contradiction.

\[  
\boxed{No}  
\]

---

## Q17

A selected forces B.

E mandatory.

Exactly 3 total.

So:

\[  
\boxed{A, B, E}  
\]

---

## Q18

C selected forces D.

E mandatory.

Exactly 3:

\[  
\boxed{C, D, E}  
\]

A l s o B a b s e n t b e c a u s e B! ' n o t D , a n d A a b s e n t b e c a u s e A! ' B

---

## Q19

Common to both valid groups:

\[  
\boxed{E}

\]

---

## Q20

\[  
\boxed{A\text{ and }C}  
\]

cannot be selected together.

---

# CHAPTER 70 STATUS

Covered so far:

- propositions '
- AND / OR / NOT / XOR '
- implication '
- converse / inverse / contrapositive '
- necessary / sufficient conditions '
- if / only if / iff '
- unless '
- implication chains '
- contradiction reasoning '
- must / could / cannot logic '
- selection constraints '
- order constraints '
- partial orders '
- min / max positions '
- two-group parity '
- De Morgan's laws '
- implication truth table '
- count + logic constraints '
- L0 - L3 practice '

But \*\*Chapter 70 is not complete yet\*\*.

Still remaining:

- advanced truth-table reasoning
- logical equivalence
- tautology / contradiction / contingency
- satisfiability
- constraint graphs
- formal GATE-style inference
- must/could/cannot mixed sets
- min/max under multiple constraints
- option testing
- complex selection/order sets
- final Unit 3 speed toolkit
- \*\*Unit 3 complete revision\*\*

## Progress

\[  
\boxed{69/76\text{ numbered chapters completed}}  
\]

\* \* C u r r e n t : \* \* Ø=ßá Chapter 70 - Logical Deduction & Constr

# NEXT

Ø=Ý% \* \* Chapter 70 Part 2 - Truth Tables, Logical Equivale

USER

next

ASSISTANT

## CHAPTER 70 — LOGICAL DEDUCTION & CONSTRAINT REASONING

### PART 2 — TRUTH TABLES, LOGICAL EQUIVALENCE, TAUTOLOGY/CONTRADICTION, SATISFIABILITY, ADVANCED CONSTRAINT SETS & FINAL UNIT 3 REVISION

Status:  $\emptyset = 0\%$  PROGRESS ! ' COMPLETE GATE: Very High Placements: Very High Priority:  $\emptyset = 0\%$  Per 1

This part completes Chapter 70 and **finishes Unit 3 — Logical & Analytical Reasoning**.

The main goal now is to move from ordinary verbal reasoning to more formal logic:

$\boxed{\text{Translate ! ' Simplify ! ' Propagate ! ' Test contradiction}} \}$

## 1. TRUTH TABLES — WHY THEY MATTER

A truth table tells us the truth value of a compound statement for every possible combination of inputs.

For two propositions  $(P, Q)$ :

$2^2 = 4$

possible truth assignments.

For three:

$2^3 = 8$

For  $(n)$ :

$\boxed{2^n}$

## 2. NOT

$\neg P$

P	$\neg P$
T	F
F	T

Simple inversion.



### 3. AND

$\lceil P \wedge Q \rceil$

True only when both are true.

P	Q	$\lceil P \wedge Q \rceil$
T	T	T
T	F	F
F	T	F
F	F	F

### 4. OR

$\lceil P \vee Q \rceil$

False only when both are false.

P	Q	$\lceil P \vee Q \rceil$
T	T	T
T	F	T
F	T	T
F	F	F

### 5. XOR

$\lceil P \oplus Q \rceil$

True when exactly one is true.

P	Q	XOR
T	T	F
T	F	T
F	T	T
F	F	F

### 6. IMPLICATION

$\lceil P \rightarrow Q \rceil$

Only false in:

$[ P=T, \text{quad} Q=F ]$

Table:

P	Q	$(P \rightarrow Q)$
T	T	T
T	F	F
F	T	T
F	F	T

MUST MEMORIZE:

$[ \boxed{P \rightarrow Q \equiv \neg P \vee Q} ]$

## 7. BICONDITIONAL

$[ P \leftrightarrow Q ]$

True when both have same truth value.

P	Q	$(P \leftrightarrow Q)$
T	T	T
T	F	F
F	T	F
F	F	T

Equivalent to:

$[ \boxed{(P \rightarrow Q) \wedge (Q \rightarrow P)} ]$

## 8. LOGICAL EQUIVALENCE

Two expressions are logically equivalent if they have identical truth values under every assignment.

Notation:

$[ \boxed{A \equiv B} ]$

Example:

$[ P \rightarrow Q ]$

and:

$[ \neg P \vee Q ]$

are equivalent.

## 9. CONTRAPOSITIVE EQUIVALENCE

$\lceil P \rightarrow Q \rceil$

is equivalent to:

$\lceil \boxed{\neg Q \rightarrow \neg P} \rceil$

This is why contraposition is valid.

## 10. CONVERSE IS NOT EQUIVALENT

Converse:

$\lceil Q \rightarrow P \rceil$

Need not equal:

$\lceil P \rightarrow Q \rceil$

Example:

**If number divisible by 4 !' even.**

True.

Converse:

**If even !' divisible by 4.**

False.

## 11. DE MORGAN'S LAWS

MUST MEMORIZE:

$\lceil \boxed{\neg(P \land Q) \equiv \neg P \lor \neg Q} \rceil$

$\lceil \boxed{\neg(P \lor Q) \equiv \neg P \land \neg Q} \rceil$

Memory:

**Negation enters brackets and flips AND !"OR.**

## 12. EXAMPLE

Negate:

$\lceil P \land Q \rceil$

Correct:

$\lceil \boxed{\neg P \lor \neg Q} \rceil$

Not:

$$\neg P \wedge \neg Q$$

---

## 13. DOUBLE NEGATION

---

$$\boxed{\neg(\neg P) \equiv P}$$

Example:

**It is not true that Ravi did not pass.**

Equivalent:

**Ravi passed.**

---

## 14. IDEMPOTENT LAWS

---

$$\boxed{P \vee P \equiv P}$$

$$\boxed{P \wedge P \equiv P}$$

Repeating the same condition adds nothing.

---

## 15. COMMUTATIVE LAWS

---

$$\boxed{P \vee Q \equiv Q \vee P}$$

$$\boxed{P \wedge Q \equiv Q \wedge P}$$

Order does not matter.

---

## 16. ASSOCIATIVE LAWS

---

$$\boxed{(P \vee Q) \vee R \equiv P \vee (Q \vee R)}$$

and:

$$\boxed{(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)}$$

So brackets can often be regrouped.

---

## 17. DISTRIBUTIVE LAWS

---

$$\boxed{P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)}$$

$$\boxed{P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)}$$

These occasionally appear directly in GATE logic.

---

## 18. ABSORPTION

---

$$\boxed{P \vee (P \wedge Q) \equiv P}$$

$$\boxed{P \wedge (P \vee Q) \equiv P}$$

Why?

If P already true, extra condition does not change OR.

---

## 19. TAUTOLOGY

---

A formula that is true for every truth assignment.

Example:

$$\boxed{P \vee \neg P}$$

Either P is true or P is false.

Always true.

---

## 20. CONTRADICTION

---

A formula false for every assignment.

Example:

$$\boxed{P \wedge \neg P}$$

P cannot be both true and false.

Always false.

---

## 21. CONTINGENCY

---

A formula that is true for some assignments and false for others.

Example:

$$\boxed{P \wedge Q}$$

Sometimes true, sometimes false.

Therefore:

$$\boxed{\text{Contingency}}$$

---

## 22. MUST-MEMORIZE CLASSIFICATION

---

### Tautology

Always true.

### Contradiction

Always false.

### Contingency

---

Depends on assignment.

---

## 23. EXAMPLE

---

Classify:

$\lceil P \rightarrow P \rceil$

Since a proposition always implies itself:

$\lceil \boxed{\text{Tautology}} \rceil$

---

## 24. EXAMPLE

---

$\lceil (P \rightarrow Q) \wedge P \wedge \neg Q \rceil$

But implication fails exactly when:

$\lceil P=T, Q=F \rceil$

This expression explicitly requires that forbidden state.

So:

$\lceil \boxed{\text{Contradiction}} \rceil$

---

## 25. SATISFIABILITY

---

A formula is satisfiable if **at least one** assignment makes it true.

Examples:

$\lceil P \wedge Q \rceil$

is satisfiable because  $\lceil (P=Q=T) \rceil$  works.

A contradiction is unsatisfiable.

---

## 26. SATISFIABLE VS TAUTOLOGY

---

Tautology:

true under all assignments.

Satisfiable:

true under at least one.

Therefore:

$\lceil \boxed{\text{Every tautology is satisfiable}} \rceil$

but:

$\lceil \boxed{\text{Not every satisfiable formula is a tautology}} \rceil$

---

## 27. UNSATISFIABLE

---

Formula:

$\lceil P \wedge \neg P \rceil$

No assignment works.

Therefore:

$\lceil \boxed{\text{Unsatisfiable}} \rceil$

---

## 28. CONSTRAINT SET SATISFIABILITY

---

Suppose:

$\lceil A \rightarrow B \rceil$

$\lceil B \rightarrow C \rceil$

$\lceil A \rceil$

$\lceil \neg C \rceil$

$A \wedge B \wedge C$ , but  $\neg C$  given.

Contradiction.

So constraint set is:

$\lceil \boxed{\text{Unsatisfiable}} \rceil$

---

## 29. SATISFIABLE SET

---

Rules:

$\lceil A \rightarrow B \rceil$

$\lceil B \rightarrow C \rceil$

No fixed truths.

Set all false:

$\lceil A=F, B=F, C=F \rceil$

All implications hold.

Therefore:

$\lceil \boxed{\text{Satisfiable}} \rceil$

---

## 30. FORMAL INFERENCE — MODUS PONENS

---

Rule:

$\lceil P \rightarrow Q \rceil$

Given:

$\lceil P \rceil$

---

Infer:

$\boxed{Q}$

This is called **modus ponens**.

Example:

If server fails !' alarm sounds.

Server failed.

Therefore alarm sounds.

---

### 31. MODUS TOLLENS

Given:

$P \rightarrow Q$

and:

$\neg Q$

infer:

$\boxed{\neg P}$

This is contraposition in inference form.

---

### 32. INVALID — AFFIRMING THE CONSEQUENT

Given:

$P \rightarrow Q$

and Q.

Concluding P is invalid.

Example:

If raining !' road wet.

Road wet.

Cannot conclude rain.

Could be sprinkler.

---

### 33. INVALID — DENYING THE ANTECEDENT

Given:

$P \rightarrow Q$

and:

$\neg P$

Concluding:

---



$\neg Q$

is invalid.

Example:

If I study I pass.

I did not study.

Maybe still pass.

## 34. INFERENCE TABLE

Pattern	Valid?
$P \rightarrow Q, P \therefore Q$	'
$P \rightarrow Q, \neg Q \therefore \neg P$	'
$P \rightarrow Q, Q \therefore P$	'L
$P \rightarrow Q, \neg P \therefore \neg Q$	'L

MUST MEMORIZE.

## 35. HYPOTHETICAL SYLLOGISM

$P \rightarrow Q$

$Q \rightarrow R$

Therefore:

$\boxed{P \rightarrow R}$

This is implication chaining.

## 36. DISJUNCTIVE SYLLOGISM

Given:

$P \vee Q$

and:

$\neg P$

then:

$\boxed{Q}$

If at least one must occur and P doesn't, Q must.

## 37. EXAMPLE

Either A or B selected.

A not selected.

Therefore:

$\boxed{B \text{ selected}}$

For inclusive OR, this still works once A is false.

---

## 38. RESOLUTION INTUITION

Given:

$A \vee B$

and:

$\neg B \vee C$

If B is the connecting variable, one can derive:

$\boxed{A \vee C}$

Formal resolution is a CS-logic topic; here only intuition is needed.

---

## 39. ADVANCED MUST-BE-TRUE SET

Rules:

1.  $A \rightarrow B$
2.  $B \rightarrow C$
3.  $C \rightarrow D$
4.  $\neg D$

Then:

$\neg D \rightarrow \neg C \rightarrow \neg B \rightarrow \neg A$

Therefore:

$\boxed{A, B, C \text{ must all be false}}$

---

## 40. ADVANCED COULD-BE-TRUE SET

Rules:

$A \rightarrow B$

$B \rightarrow \neg C$

Question:

**Could C be true?**

Yes.

Choose:

$\{ A=F, B=F, C=T \}$

All constraints satisfied.

$\boxed{\text{Yes}}$

---

## 41. CANNOT-BE-TRUE SET

Same rules plus:

$\{ A=T \}$

Then:

$A \vee B \wedge C$

Therefore C cannot be true.

$\boxed{C \text{ cannot be true}}$

---

## 42. OPTION TESTING METHOD

Suppose MCQ asks:

**Which could be true?**

Instead of solving all assignments:

1. assume option
2. propagate implications
3. if contradiction !' reject
4. first surviving option may be answer

Very fast in constraint OAs.

---

## 43. MUST-BE-TRUE OPTION METHOD

To test candidate statement S:

Assume:

$\neg S$

If constraints become impossible:

$\boxed{S \text{ must be true}}$

This is proof by contradiction.

---

## 44. CANNOT-BE-TRUE OPTION METHOD

Assume candidate S.

If contradiction:

$\boxed{S \text{ cannot be true}}$

---

## 45. CONSTRAINT GRAPH

Example:

$\lceil A \rightarrow B \rceil$

$\lceil B \rightarrow C \rceil$

$\lceil A \rightarrow D \rceil$

Draw:

```text id="v8k1r9" A /\ B D | C

If A true, B,C,D all true.

46. NEGATIVE EDGES

Suppose:

$\lceil B \rightarrow \neg E \rceil$

Add:

```text id="m4p8sa"  
A !' B !' C  
|  
% !' ¬ E

Then A also implies:

$\lceil \boxed{\neg E} \rceil$

---

## 47. CONFLICT GRAPH

If:

$\lceil A \rightarrow \neg B \rceil$

and:

$\lceil B \rightarrow \neg A \rceil$

A/B are incompatible.

Represent:

```text id="8tcili" A × B

Useful in selection puzzles.

48. ORDERING + SELECTION INTEGRATION

Suppose:

- if A selected, B selected
- if both A and B selected, A must occur before B
- exactly 3 tasks selected

You may need both Boolean and ordering reasoning.

Solve in two stages:

1. decide feasible membership
2. order selected items

49. EXAMPLE

Select exactly 3 from A,B,C,D.

Rules:

$$\begin{array}{l} \backslash[\\ A \rightarrow B \\ \backslash] \end{array}$$

$$\begin{array}{l} \backslash[\\ B \rightarrow \neg C \\ \backslash] \end{array}$$

D mandatory.

If A selected:

A,B,D selected.

C absent.

Unique membership:

$$\begin{array}{l} \backslash[\\ \boxed{A, B, D} \\ \backslash] \end{array}$$

50. ADD ORDER

Suppose additionally:

$$\begin{array}{l} \backslash[\\ A < B \\ \backslash] \end{array}$$

Then possible sequence among selected may be:

$$\begin{array}{l} \backslash[\\ A-D-B \\ \backslash] \end{array}$$

or:

$$\begin{array}{l} \backslash[\\ D-A-B \\ \backslash] \end{array}$$

unless more constraints.

Membership fixed, order not fully fixed.

51. MIN/MAX UNDER CONSTRAINTS

Suppose 6 tasks.

Constraints:

$$\begin{array}{l} \backslash[\\ A < B \\ \backslash] \end{array}$$
$$\begin{array}{l} \backslash[\\ C < B \\ \backslash] \end{array}$$
$$\begin{array}{l} \backslash[\\ B < D \\ \backslash] \end{array}$$

For B:

Forced before:

A, C.

Thus earliest:

$$\begin{array}{l} \backslash[\\ \boxed{\text{3rd}} \\ \backslash] \end{array}$$

Forced after:

D.

Thus latest:

$$\begin{array}{l} \backslash[\\ 6-1=5 \\ \backslash] \end{array}$$

So B can potentially be ranks:

$$\begin{array}{l} \backslash[\\ 3, 4, 5 \\ \backslash] \end{array}$$

subject to remaining constraints.

52. MIN/MAX FORMULA

For item X among $\backslash(N\backslash)$:

$$\begin{array}{l} \backslash[\\ \boxed{\text{Earliest}} = 1 + \backslash\#(\text{forced before X}) \\ \backslash] \end{array}$$
$$\begin{array}{l} \backslash[\\ \boxed{\text{Latest}} = N - \backslash\#(\text{forced after X}) \\ \backslash] \end{array}$$

Count transitive constraints too.

53. TRANSITIVE COUNT

If:

\[
A<B<C<D
\]

For C:

Forced before:

A , B !' 2 .

Earliest:

\[
3
\]

Forced after:

D !' 1 .

Latest with \((N=4)\):

\[
3
\]

So C fixed 3rd.

54. MULTIPLE PREDECESSORS

Graph:

```
```text id="g9lso7"
A % %
 % !' D !' E
B % %$
C % %
```

D has A,B,C forced before.

Earliest:

\[ 4 \]

## 55. SELECTION MIN/MAX

Choose exactly 5 from 8.

Rules:

- A and B cannot coexist
- C,D,E must be selected together

Question:

**Maximum selected from  $\{A,B,C,D,E\}$ ?**

CDE contributes 3 if chosen.

Among A,B max 1.

So maximum:

$\boxed{4}$

## 56. MINIMUM USING COMPLEMENT

Choose exactly 6 from 8.

Question:

**Minimum selected from subset S containing 5 people?**

Outside S has only 3 people.

At most 3 selections can come outside.

Therefore at least:

$6-3=3$

from S.

$\boxed{3}$

## 57. EXACTLY-ONE CHAIN

Rules:

- Exactly one of A,B.
- $A' \vee C$ .
- $B' \vee D$ .
- C and D cannot both occur.

Does anything new follow?

Since exactly one of A/B, at least one of C/D is forced.

And because C,D cannot both occur:

$\boxed{\text{Exactly one of C,D occurs}}$

Here the extra incompatibility makes the XOR propagate.

## 58. WITHOUT INCOMPATIBILITY

Remove:

$C \vee D \vee \neg(C \wedge D)$



Then exactly one A/B does **not** guarantee exactly one C/D because the unforced one could occur independently.

Important trap.

---

## 59. BICONDITIONAL CHAIN

---

If:

$$\boxed{A \leftrightarrow B}$$

and:

$$\boxed{B \leftrightarrow C}$$

then:

$$\boxed{A \leftrightarrow C}$$

All three share truth value.

---

## 60. BICONDITIONAL + NEGATION

---

If:

$$\boxed{A \leftrightarrow B}$$

and:

$$\boxed{A = T}$$

then:

$$\boxed{B = T}$$

If:

$$\boxed{A = F}$$

then:

$$\boxed{B = F}$$

---

## 61. XOR + BICONDITIONAL

---

If:

$$\boxed{A \oplus B}$$

and:

$$\boxed{B \leftrightarrow C}$$

then A and C have opposite truth values.

Thus:

$$\boxed{A \oplus C}$$

---

## 62. TWO-GROUP PARITY — ADVANCED

---

Encode Team X = 0, Team Y = 1.

“Same team”:

$\{ A=B \}$

“Different team”:

$\{ A \neq B \}$

Along a chain:

- even number of “different” edges != same
- odd number != different

---

## 63. EXAMPLE

A different B.

B different C.

C different D.

Count 3 different edges from A to D.

Odd.

Therefore:

$\{ \boxed{A \neq D} \}$

---

## 64. EVEN PARITY

A different B.

B different C.

Two differences.

Therefore:

$\{ \boxed{A=C} \}$

---

## 65. ODD CYCLE CONTRADICTION

Only two groups:

$\{ A \neq B \}$

$\{ B \neq C \}$

$\{ C \neq A \}$

Three “different” edges form odd cycle.

Impossible.

$\{ \boxed{\text{Unsatisfiable}} \}$

## 66. FORMAL GATE-STYLE EXAMPLE

---

Given:

$\lceil P \rightarrow Q \rceil$

$\lceil Q \rightarrow R \rceil$

$\lceil \neg R \rceil$

Which must follow?

By contraposition:

$\lceil \neg R \rightarrow \neg Q \rceil$

and:

$\lceil \neg Q \rightarrow \neg P \rceil$

Therefore:

$\lceil \boxed{\neg P, \neg Q} \rceil$

---

## 67. ANOTHER FORMAL EXAMPLE

---

Given:

$\lceil P \vee Q \rceil$

$\lceil \neg P \rceil$

Then:

$\lceil \boxed{Q} \rceil$

Disjunctive syllogism.

---

## 68. FORMAL INVALID EXAMPLE

---

Given:

$\lceil P \rightarrow Q \rceil$

$\lceil Q \rceil$

Can P be inferred?

$\lceil \boxed{\text{No}} \rceil$

This is affirming the consequent.

---

## 69. LOGICAL EQUIVALENCE QUICK TABLE

---

MUST MEMORIZE:

$\lceil P \rightarrow Q \equiv \neg P \vee Q \rceil$

$\lceil \neg(P \wedge Q) \equiv \neg P \vee \neg Q \rceil$

$\lceil \neg(P \vee Q) \equiv \neg P \wedge \neg Q \rceil$

---

$$\lceil P \rightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P) \rceil$$

---

## 70. TAUTOLOGY QUICK TEST

---

Common tautologies:

$$\lceil P \vee \neg P \rceil$$

$$\lceil P \rightarrow P \rceil$$

$$\lceil (P \wedge Q) \rightarrow P \rceil$$

$$\lceil P \rightarrow (P \vee Q) \rceil$$

---

## 71. CONTRADICTION QUICK TEST

---

Common contradictions:

$$\lceil P \wedge \neg P \rceil$$

$$\lceil (P \rightarrow Q) \wedge P \wedge \neg Q \rceil$$

---

## 72. CONTINGENCY QUICK TEST

---

Most ordinary expressions like:

$$\lceil P \vee Q \rceil$$

$$\lceil P \wedge Q \rceil$$

$$\lceil P \rightarrow Q \rceil$$

are contingencies because not always true/false.

---

## 73. TRUTH-TABLE SHORTCUT

---

Do not build full truth table unless necessary.

For:

$$\lceil P \rightarrow Q \rceil$$

remember its only failure case.

For tautology testing, try to find a false assignment.

If none after simplification, tautology.

For contradiction testing, try to find one true assignment.

---

## 74. COUNTEREXAMPLE PRINCIPLE

---

To show an expression is **not a tautology**:

find one assignment making it false.

---

To show it is **not a contradiction**:  
find one assignment making it true.

---

## 75. EXAMPLE

---

Is:

$\forall [ P \vee Q ]$

a tautology?

Take:

$\forall [ P=F, Q=F ]$

Expression false.

So not tautology.

---

## 76. EXAMPLE

---

Is:

$\forall [ P \wedge Q ]$

a contradiction?

Take:

$\forall [ P=T, Q=T ]$

Expression true.

So not contradiction.

It is contingency.

---

## 77. COMPLEX CONSTRAINT SET

---

Exactly 3 from A,B,C,D,E.

Rules:

1.  $\forall (A \rightarrow B)$
2.  $\forall (B \rightarrow \neg C)$
3.  $\forall (C \rightarrow D)$
4.  $\forall (D \rightarrow E)$

Let's derive valid possibilities.

If A selected:

$A \wedge B$ .

$B \wedge \neg C$ .

Need one more from D,E.

But  $D \wedge E$  choosing D forces E and creates 4 total:

---

A,B,D,E.

Impossible.

Therefore with A selected, third must be E.

Valid:

$\boxed{A,B,E}$

---

## 78. SAME SET — C SELECTED

---

C! D! E .

Exactly three selected.

So:

$\boxed{C,D,E}$

B cannot be selected because B! n α.

A cannot because A! B .

Thus only two valid groups:

$\boxed{ABE}, \boxed{CDE}$

---

## 79. MUST / COULD / CANNOT

---

From valid groups:

Must:

$\boxed{E}$

Could:

A,B,C,D.

Cannot coexist:

$\boxed{A \text{ with } C}$

and:

$\boxed{B \text{ with } C}$

---

## 80. ADVANCED ORDER SET

---

Five tasks A–E.

Rules:

$A < C$

$B < C$

$C < D$

$B < E$

---

Question: who can be first?

C cannot because A,B before C.

D cannot because C before D.

E cannot because B before E.

A or B could be first.

$\boxed{A \text{ or } B}$

---

## 81. WHO MUST BE BEFORE D?

---

From:

$A < C < D$

and:

$B < C < D$

both A and B must be before D.

Also C must be before D.

So:

$\boxed{A, B, C}$

---

## 82. WHO COULD BE LAST?

---

D has no required successor.

E has no required successor.

So D or E may potentially be last.

Need check valid arrangement exists for each.

Example D last:

$A, B, C, E, D$

valid if  $B < E$ .

Example E last:

$A, B, C, D, E$

valid.

Thus:

$\boxed{D \text{ or } E}$

---

## 83. FINAL CONSTRAINT-SOLVING WORKFLOW

---

### 1. Translate

Turn English into symbols.

---

## 2. Normalize

Convert:

- only if
- iff
- unless
- XOR
- not both

## 3. Draw graph

Implications / order / conflicts.

## 4. Propagate

Follow chains and contrapositives.

## 5. Apply counts

Exactly / at least / at most.

## 6. Test options

Assume candidate.

## 7. Kill contradictions

Stop branch immediately.

## 8. For MUST

Negate candidate.

## 9. For COULD

Construct one example.

## 10. Verify all constraints.

---

# 84. FINAL SPEED TOOLKIT

---

### Implication

$\backslash[ A\text{to} B \backslash]$

### Contrapositive

$\backslash[ \neg B\text{to}\neg A \backslash]$

### Only if

$\backslash[ A\text{text{ only if } }B=A\text{to} B \backslash]$

### If

$\backslash[ A\text{text{ if } }B=B\text{to} A \backslash]$

### Iff

$\backslash[ A\text{leftrightharpoonrightarrow} B \backslash]$



### Unless

$\neg A \text{ unless } B \equiv \neg B \rightarrow A$

### Exactly one

XOR.

### Not both

Cannot coexist.

### Must

Negate '!' contradiction.

### Could

Build one valid case.

### Cannot

Assume '!' contradiction.

---

## PRACTICE — LEVEL 0

### Q1

Classify:

$\neg P \text{ or } \neg P$

### Q2

Classify:

$\neg P \text{ and } \neg P$

### Q3

Is  $\neg(P \rightarrow Q)$  equivalent to  $\neg \neg P \text{ or } Q$ ?

### Q4

Which inference is valid?

A.  $\neg(P \rightarrow Q, Q \text{ therefore } P)$  B.  $\neg(P \rightarrow Q, P \text{ therefore } Q)$

### Q5

If  $\neg(P \text{ or } Q)$  and  $\neg \neg P$ , what follows?

---

## LEVEL 1

### Q6

Given:

$\neg A \rightarrow B, \neg B \rightarrow C, \neg C$

what follows about A and B?

### Q7

Is:

$\neg(A \rightarrow B) \wedge A \wedge \neg B$

satisfiable?

### Q8

If  $A \leftrightarrow B$  and  $B$  true, what about  $A$ ?

### Q9

Exactly one of  $A, B$ , and  $A$  false. What follows?

### Q10

Two teams only:  $A$  differs  $B$ ,  $B$  differs  $C$ ,  $C$  differs  $D$ . Relation  $A$  vs  $D$ ?

---

## LEVEL 2

---

### Q11

Classify:

$\neg(P \wedge Q) \rightarrow P$

### Q12

Classify:

$\neg(P \rightarrow Q) \wedge P \wedge \neg Q$

### Q13

Select exactly 2 from  $A, B, C$  with  $(A \rightarrow B) \wedge (B \rightarrow C)$ . Which two are selected?

### Q14

Five tasks, constraints:

$\neg(A < C) \wedge \neg(B < C) \wedge \neg(C < D)$

Can  $C$  be first?

### Q15

Under Q14, who must be before  $D$ ?

---

## LEVEL 3 — EXAM PRESSURE

---

Exactly 3 people are selected from  $A, B, C, D, E$ .

Rules:

1.  $\neg(A \rightarrow B)$
2.  $\neg(B \rightarrow \neg C)$
3.  $\neg(C \rightarrow D)$
4.  $\neg(D \rightarrow E)$

**Q16**

If A selected, what group results?

**Q17**

If C selected, what group results?

**Q18**

Who must be selected?

**Q19**

Can B and C both be selected?

**Q20**

How many valid groups exist?

**ANSWERS****Q1**

$\lceil P \lor \neg P \rceil$

always true.

$\lceil \boxed{\text{Tautology}} \rceil$

**Q2**

$\lceil P \land \neg P \rceil$

always false.

$\lceil \boxed{\text{Contradiction}} \rceil$

**Q3**

Yes.

$\lceil \boxed{P \rightarrow Q \equiv \neg P \lor Q} \rceil$

**Q4**

Valid:

$\lceil P \rightarrow Q, \lceil P \rceil \text{ therefore } Q \rceil$

$\lceil \boxed{B} \rceil$

**Q5**

From:

$\neg(P \vee Q, \neg P)$

$\boxed{Q}$

---

## Q6

$A \rightarrow B \rightarrow C$

Given:

$\neg C$

Therefore:

$\boxed{\neg B, \neg A}$

---

## Q7

A implies B, but A true and B false.

Direct contradiction.

$\boxed{\text{No, unsatisfiable}}$

---

## Q8

Biconditional means same truth value.

$\boxed{A = \text{true}}$

---

## Q9

Exactly one and A false:

$\boxed{B = \text{true}}$

---

## Q10

Three "different" edges:

$A \wedge B \wedge C \wedge D$

Odd count means:

$\boxed{A \neq D}$

---

## Q11

If both P, Q true, P certainly true.

If antecedent false, implication true.

Thus always true:

$\boxed{\text{tautology}}$

---

---

## Q12

This explicitly requires the only false case of implication.

$\boxed{\text{Contradiction}}$

---

## Q13

A selected would force all 3, impossible.

Therefore A absent.

Exactly 2 remain:

$\boxed{B,C}$

---

## Q14

No.

A and B must both occur before C.

$\boxed{\text{No}}$

---

## Q15

A,C are before D.

B is also before C, so B before D.

Thus:

$\boxed{A,B,C}$

---

## Q16–Q20

### Q16

A! B .

B! n α.

Exactly 3.

D cannot be third because D! E would make 4 selected.

Thus third is E.

$\boxed{A,B,E}$

---

### Q17

C! D! E .

---

Exactly 3:

$$\boxed{C, D, E}$$

---

## Q18

---

Common to both:

$$\boxed{E}$$

---

## Q19

---

$B \not\models \alpha$

Therefore:

$$\boxed{\text{No}}$$

---

## Q20

---

Valid groups:

$$\boxed{ABE}$$

and:

$$\boxed{CDE}$$

So:

$$\boxed{2}$$

---

## ' CHAPTER 70 COMPLETE

---

We have covered:

- propositions '
  - AND/OR/NOT/XOR '
  - implication/biconditional '
  - truth tables '
  - converse/inverse/contrapositive '
  - necessary/sufficient conditions '
  - if/only-if/iff/unless '
  - De Morgan's laws '
  - logical equivalence '
  - tautology '
  - contradiction '
  - contingency '
  - satisfiability '
  - modus ponens '
-

- modus tollens '
- disjunctive syllogism '
- invalid inference patterns '
- implication graphs '
- conflict graphs '
- must/could/cannot reasoning '
- partial-order constraints '
- min/max positions '
- group-count constraints '
- parity/two-group reasoning '
- option testing '
- proof by contradiction '
- advanced mixed sets '
- L0–L3 practice '

---

## ' UNIT 3 — LOGICAL & ANALYTICAL REASONING COMPLETE

---

Chapters:

- 57 Series '
- 58 Analogy, Classification & Odd-One-Out '
- 59 Coding-Decoding '
- 60 Blood Relations '
- 61 Direction Sense '
- 62 Ranking & Ordering '
- 63 Linear Seating Arrangement '
- 64 Circular & Parallel Seating '
- 65 Logical Puzzles '
- 66 Syllogisms '
- 67 Statements, Assumptions & Conclusions '
- 68 Arguments, Cause–Effect & Course of Action '
- 69 Data Sufficiency '
- 70 Logical Deduction & Constraint Reasoning '

---

## UNIT 3 — 5-MINUTE REVISION

---

### Series

Find pattern in this order:

$\boxed{\text{Difference !' ratio !' alternating !' special numbers}}$

### Coding

Check:

- shift
- reverse
- A1-Z26
- opposite alphabet
- position-based rule

## Blood Relations

---

Draw generations.

Do not solve long verbal chains mentally.

## Directions

---

Use coordinates:

$\backslash$  E=+x, \ W=-x, \ N=+y, \ S=-y \

## Ranking

---

$\backslash$   $\boxed{N=L+R-1}$  \

$\backslash$   $\boxed{\text{Between}=|r_1-r_2|-1}$  \

## Seating

---

Mark facing before left/right.

## Puzzles

---

Strong clue first.

Use:

$\backslash$   $\boxed{\text{Fix !' Propagate !' Branch !' Prune}}$  \

## Syllogism

---

- All !' forward only
- No !' reversible
- Some !' reversible
- Some-not !' not reversible

## Assumptions

---

Use:

$\backslash$   $\boxed{\text{Negation test}}$  \

## Conclusions

---

Use:

$\backslash$   $\boxed{\text{Counterexample test}}$  \



## Arguments

Use:

$\boxed{\text{RILP}}$

Relevant, Important, Logical, Practical.

## Cause–Effect

$\boxed{\text{TMAE}}$

Time, Mechanism, Alternatives, Evidence.

## Course of Action

$\boxed{\text{RFEP}}$

Relevant, Feasible, Effective, Proportionate.

## Data Sufficiency

Ask only:

Is target uniquely fixed?

## Constraint Logic

$A \rightarrow B \rightarrow \neg B \rightarrow \neg A$

## UNIT 3 — MUST-MEMORIZE TABLE

Topic	Core Rule		
Ranking	$(N=L+R-1)$		
People between	$($	$r_1-r_2$	$-1)$
North-facing linear	left=page-left		
South-facing linear	left=page-right		
Inward circle	left=clockwise		
Outward circle	left=anticlockwise		
Opposite circular seat	$(N/2)$ away		
Implication	$(P \rightarrow Q)$		
Contrapositive	$(\neg Q \rightarrow \neg P)$		
Implication equivalence	$(\neg P \vee Q)$		
Exactly one	XOR		

Topic	Core Rule		
Not both	$\neg(P \text{ and } Q)$		
Neither	$\neg P \text{ and } \neg Q$		
De Morgan 1	$\neg(P \text{ and } Q) = \neg P \text{ or } \neg Q$		
De Morgan 2	$\neg(P \text{ or } Q) = \neg P \text{ and } \neg Q$		

## UNIT 3 — TOP 15 EXAM TRAPS

1. Reversing implication.
2. Treating OR as XOR.
3. Forgetting South-facing reversal.
4. Forgetting inward/outward circular reversal.
5. Treating left-of as immediately-left.
6. Some+Some creating false syllogism chain.
7. Assuming existence from universal statement.
8. Plausible assumption treated as necessary.
9. Correlation treated as causation.
10. Solving whole DS problem unnecessarily.
11. Exact numerical value demanded for Yes/No DS.
12. Partial order forced into complete order.
13. Negative clues used before stronger clues.
14. Continuing contradictory cases.
15. Ignoring domain restrictions.

## UNIT 3 — SPEED CHECKLIST

Before submitting a reasoning answer ask:

- Did I use the correct reference person's left/right?
- Did I accidentally reverse a condition?
- Did I assume information that was not given?
- Is this **MUST** or merely **COULD**?
- Can I create a counterexample?
- Is there a contradiction I missed?
- Does exact value even matter?
- Could option elimination be faster?

## OVERALL NUMBERED-CHAPTER PROGRESS

$\boxed{70/76}$  numbered chapters complete

That means only the **6 numbered Verbal Aptitude chapters** remain:

## UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude Ø=Ý%
- Chapter 72 — Vocabulary & Word Usage Ø=Ý%
- Chapter 73 — Sentence Completion, Correction & Error Detection Ø=Ý%
- Chapter 74 — Para Jumbles & Paragraph Completion Ø=Ý%
- Chapter 75 — Reading Comprehension Ø=Ý%
- Chapter 76 — Critical Reasoning & Verbal Deduction Ø=Ý%

After Chapter 76, the course is **still not finished** because we must also complete:

- Unit 5 — GATE Special Training
- Unit 6 — Placement Special Training
- Unit 7 — Revision & Question Bank
- Unit 8 — Full Mocks

## NEXT

### CHAPTER 71 — GRAMMAR FOR APTITUDE

We will start with:

- parts of speech
- subject–verb agreement
- tenses
- articles
- pronouns
- prepositions
- modifiers
- parallelism
- conjunctions
- comparison
- active/passive
- direct/indirect speech
- common GATE/placement grammar traps
- sentence correction
- error spotting
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 71 — GRAMMAR FOR APTITUDE

# PART 1 — PARTS OF SPEECH, SUBJECT-VERB AGREEMENT, TENSES, ARTICLES, PRONOUNS & BASIC PREPOSITIONS

Status: Ø=Not in PROGRESS GATE: High Placements: Very High Priority: Ø=Not 1

Grammar questions usually test whether a sentence is:

- grammatically correct
- logically natural
- consistent in tense
- correct in agreement
- free from pronoun/modifier/preposition errors

The first rule:

**Do not judge by “it sounds okay.” Judge by structure.**

## 1. PARTS OF SPEECH — FOUNDATION

Every word usually performs a grammatical role.

Main categories:

1. Noun
2. Pronoun
3. Verb
4. Adjective
5. Adverb
6. Preposition
7. Conjunction
8. Interjection
9. Determiner/article

## 2. NOUN

A noun names:

- person
- place
- thing
- idea

Examples:

- student
- Hyderabad
- computer
- honesty

Sentence:

**The student solved the problem.**

“Student” is a noun.

### 3. PRONOUN

A pronoun replaces a noun.

Examples:

- he
- she
- it
- they
- who

Instead of:

**Ravi said Ravi would come.**

we say:

**Ravi said he would come.**

### 4. VERB

A verb shows:

- action
- state
- occurrence

Examples:

- runs
- studies
- is
- became
- seems

Sentence:

**She writes well.**

### 5. ADJECTIVE

An adjective describes a noun/pronoun.

Example:

**She is a careful student.**

“careful” describes student.

## 6. ADVERB

An adverb commonly modifies:

- verb
- adjective
- another adverb

Example:

**She writes carefully.**

“carefully” describes how she writes.

## 7. ADJECTIVE VS ADVERB TRAP

Wrong:

**She sings beautiful.**

Correct:

**She sings beautifully.**

Because “sings” is a verb and requires an adverb.

## 8. LINKING VERB EXCEPTION

After linking verbs such as:

- seem
- look
- feel
- become
- appear
- be

we often use an adjective.

Correct:

**She looks beautiful.**

Not:

**She looks beautifully.**

Why?

“beautiful” describes **she**, not the act of looking.

## 9. PREPOSITION

A preposition expresses relation.

Examples:

- in
- on
- at
- by
- with
- from
- to
- for
- between
- among

Example:

**The book is on the table.**

## 10. CONJUNCTION

Connects words/phrases/clauses.

Examples:

- and
- but
- or
- because
- although
- while

Example:

**Ravi studied because he had an exam.**

## 11. INTERJECTION

Expresses emotion.

Examples:

- Oh!
- Wow!
- Alas!

Low importance for GATE aptitude.

---

## 12. ARTICLE

---

Articles:

$\boxed{a, \text{ an, the}}$

They are determiners used before nouns.

---

## 13. SUBJECT AND PREDICATE

---

Sentence:

**The students solved the problem quickly.**

Subject:

$\boxed{\text{The students}}$

Predicate:

$\boxed{\text{solved the problem quickly}}$

The subject determines subject–verb agreement.

---

## 14. SUBJECT–VERB AGREEMENT

---

Core rule:

**Singular subject !' singular verb**

**He works.**

**Plural subject !' plural verb**

**They work.**

Important English trap:

In present tense:

- singular often gets -s
  - plural does not
-



## 15. SIMPLE AGREEMENT

Correct:

**The boy plays cricket.**

Correct:

**The boys play cricket.**

Wrong:

**The boys plays cricket.**

## 16. "I" AND "YOU"

Use:

**I am**

**You are**

Not based on ordinary singular/plural form.

## 17. SUBJECT NEARBY-NOUN TRAP

Sentence:

**The quality of the products is excellent.**

Main subject:

$\boxed{\text{quality}}$

not products.

Therefore singular verb:

$\boxed{\text{is}}$

## 18. PREPOSITIONAL PHRASE DOES NOT CONTROL VERB

Structure:

**The list of candidates is ready.**

Not:

**candidates are ready**

because subject is "list."

Memory:

**Ignore "of + noun" first; find true head noun.**

## 19. EXAMPLE

**The number of students is increasing.**

Subject:

$\boxed{\text{number}}$

singular.

So:

$\boxed{\text{is}}$

## 20. "A NUMBER OF"

Different!

**A number of students are absent.**

"A number of" means several.

Therefore plural verb.

Compare:

**The number of students is**

**A number of students are**

MUST MEMORIZE.

## 21. "EACH"

**Each student is responsible.**

"Each" takes singular verb.

## 22. "EVERY"

**Every student has submitted the form.**

Singular.

## 23. "EACH OF"

**Each of the students is ready.**

Still singular.

## 24. "EVERY ONE OF"

**Every one of the candidates has arrived.**

Singular.

## 25. "EITHER" AND "NEITHER"

When used as singular pronouns:

**Either of the answers is acceptable.**

**Neither of the students was present.**

Standard exam grammar usually treats them as singular.

## 26. SUBJECTS JOINED BY "AND"

Usually plural:

**Ravi and Sita are coming.**

[ A+B \rightarrow plural ]

## 27. SAME PERSON/THING EXCEPTION

**The founder and CEO is speaking.**

If both titles refer to one person, singular.

Compare:

**The founder and the CEO are speaking.**

Two separate people.

Article repetition can change meaning.

## 28. SINGLE IDEA EXCEPTION

**Bread and butter is my usual breakfast.**

When two nouns represent one combined idea, singular may be used.

## 29. “OR / NOR” AGREEMENT

With:

- either...or
- neither...nor
- or
- nor

verb usually agrees with the **nearest subject**.

Example:

**Either the teacher or the students are responsible.**

Nearest = students.

## 30. ANOTHER EXAMPLE

**Either the students or the teacher is responsible.**

Nearest = teacher.

## 31. “AS WELL AS”

Sentence:

**Ravi, as well as his friends, is attending.**

Main subject = Ravi.

“As well as” does not make the subject plural like “and.”

Similar phrases:

- along with

- together with
- accompanied by
- in addition to

---

## 32. EXAMPLE

**The manager, along with the employees, was present.**

Main subject:

\[ manager \]

Singular.

---

## 33. COLLECTIVE NOUNS

Words:

- team
- committee
- family
- jury

In standard exam English, often singular when acting as one unit:

**The committee has made its decision.**

Can be plural in some varieties when emphasizing individuals, but aptitude questions usually make context clear.

---

## 34. UNCOUNTABLE NOUNS

Common uncountable nouns:

- information
- advice
- furniture
- equipment
- luggage
- research
- knowledge

Do not normally use:

- informations
- advices
- equipments
- furnitures

Correct:

**The information is useful.**

### 35. "NEWS"

Though ending in s:

**The news is surprising.**

Singular.

### 36. SUBJECTS THAT LOOK SINGULAR BUT ARE PLURAL

Examples:

- police
- people

**The police are investigating.**

**People are waiting.**

### 37. SUBJECTS THAT LOOK PLURAL BUT MAY BE SINGULAR

Examples:

- mathematics
- physics
- economics

when referring to field of study:

**Mathematics is difficult for some students.**

### 38. TIME, MONEY, DISTANCE AS ONE UNIT

**Ten kilometers is a long distance to walk.**

**Five years is a long time.**

**10,000 is enough.**

When treated as one quantity, singular.

---

## 39. FRACTIONS AND PERCENTAGES

---

Verb depends on noun after “of.”

**Half of the cake is gone.**

**Half of the students are absent.**

---

## 40. “MORE THAN ONE”

---

Exam convention:

**More than one student has complained.**

Usually singular verb.

This is a famous trap.

---

## 41. “ONE OF”

---

**One of the students is absent.**

Main subject = one.

Singular.

---

## 42. “ONE OF THE STUDENTS WHO...”

---

Careful:

**She is one of the students who have passed.**

“who” refers to students, plural.

But:

**She is the only one of the students who has passed.**

Now “who” refers to the only one, singular.

Advanced placement trap.

---

## 43. TENSES — WHY THEY MATTER

---

A tense tells us:

- time
- completion
- continuity
- sequence

Do not memorize 12 forms without understanding their jobs.

## 44. SIMPLE PRESENT

Form:

\[ Subject + V<sub>1</sub>/V<sub>1</sub>+s \]

Used for:

- habits
- routines
- general truths
- schedules

Examples:

**She studies every day.**

**Water boils at  $(100^{\circ}\text{C})$  under standard conditions.**

## 45. PRESENT CONTINUOUS

\[ am/is/are + V-ing \]

Used for:

- action happening now
- temporary activity

Example:

**She is studying now.**

## 46. PRESENT PERFECT

\[ has/have + V<sub>3</sub> \]

Used for:

- completed past action connected to present
- life experience
- action in unfinished time period

Example:



**I have completed the assignment.**

## 47. PRESENT PERFECT CONTINUOUS

\[ has/have\ been + V-ing \]

Used when action began in past and continues.

**She has been studying for two hours.**

## 48. SIMPLE PAST

\[ V\_2 \]

Used for completed action at a definite past time.

**I visited Delhi last year.**

## 49. PAST CONTINUOUS

\[ was/were + V-ing \]

Action in progress at a past moment.

**I was studying when he called.**

## 50. PAST PERFECT

\[ had + V\_3 \]

Used for earlier of two past actions when sequence needs clarity.

**The train had left before we reached the station.**

## 51. SIMPLE FUTURE

\[ will + V\_1 \]

Example:

**I will call you tomorrow.**

## 52. TENSE CONSISTENCY

---

Wrong:

**He entered the room and sits down.**

Correct:

**He entered the room and sat down.**

If events are in same past sequence, maintain tense.

---

## 53. "SINCE" VS "FOR"

---

### Since

starting point:

- since 2020
- since Monday
- since morning

### For

duration:

- for two years
- for three hours

Example:

**She has lived here since 2020.**

**She has lived here for six years.**

---

## 54. "AGO" WITH SIMPLE PAST

---

Correct:

**He left two hours ago.**

Usually not:

**He has left two hours ago.**

Definite finished past time !' simple past.

---

## 55. “YESTERDAY / LAST YEAR / IN 2020”

These normally trigger simple past.

Correct:

**I met him yesterday.**

Not:

**I have met him yesterday.**

## 56. PRESENT PERFECT WITH UNFINISHED TIME

**I have completed three tasks today.**

Possible if “today” is still ongoing.

But:

**I completed three tasks yesterday.**

## 57. “WHEN” VS “WHILE”

Often:

**when + short event**

**I was studying when he called.**

**while + continuing action**

**While I was studying, he called.**

Not an absolute rule, but useful.

## 58. FUTURE TIME CLAUSE TRAP

After words like:

- when
- until
- before
- after
- as soon as
- if

we usually use present tense for future meaning.

Correct:

**I will call you when I reach home.**

Not:

**when I will reach home.**

## 59. CONDITIONAL BASIC

Correct:

**If it rains, we will stay inside.**

Not usually:

**If it will rain, we will stay inside.**

## 60. UNIVERSAL TRUTH IN REPORTED CONTEXT

**The teacher said that the Earth revolves around the Sun.**

Universal truth may remain present tense.

## 61. ARTICLES — A / AN / THE

### A

before a singular countable noun beginning with a consonant **sound**.

**a book**

**a university**

Why “a university”?

Because sound starts with /y/:

“yoo-niversity.”

## 62. AN

Before a vowel **sound**.

**an apple**

**an hour**

“hour” starts with vowel sound because h is silent.

---

## 63. SOUND, NOT LETTER

Correct:

**a European**

because pronunciation begins “yoo.”

Correct:

**an MBA graduate**

because “M” is pronounced “em.”

---

## 64. FIRST MENTION VS KNOWN NOUN

**I saw a dog.**

First mention.

**The dog was barking.**

Now specific/known.

---

## 65. UNIQUE THINGS

Use “the” for commonly unique referents:

- the Sun
- the Moon
- the Earth

depending context.

---

## 66. SUPERLATIVES

Use “the”:

**the best**

**the tallest**

**the most useful**

---

## 67. ORDINALS

Usually:

**the first**

**the second**

**the last**

---

## 68. GENERAL PLURAL NO ARTICLE

Correct:

**Computers are useful.**

Not:

**The computers are useful**

unless referring to specific computers.

---

## 69. GENERAL UNCOUNTABLE NO ARTICLE

**Knowledge is power.**

**Information is useful.**

No article for general abstract/uncountable meaning.

---

## 70. PROFESSIONS

**She is an engineer.**

Singular countable profession generally needs article.

Wrong:

**She is engineer.**

## 71. "THE" WITH SPECIFIC DESCRIPTION

**The student who won the prize is absent.**

"the" because specific student identified by clause.

## 72. ARTICLE WITH MEALS

Normally:

**We had breakfast.**

No article.

But:

**We had a wonderful breakfast.**

Modified/specific meal can take article.

## 73. PRONOUN-ANTECEDENT AGREEMENT

Pronoun should agree with antecedent.

Example:

**The students submitted their assignments.**

students = plural !' their.

## 74. SINGULAR PRONOUN

**Ravi submitted his assignment.**

## 75. “EVERYONE / SOMEONE / ANYONE”

---

Traditionally singular:

**Everyone has submitted his or her form.**

Modern standard English widely accepts singular **they**:

**Everyone has submitted their form.**

For aptitude tests, follow the specific exam's style; singular “they” is now standard usage in many contexts.

---

## 76. PRONOUN REFERENCE MUST BE CLEAR

---

Bad:

**When Ravi spoke to Arjun, he was nervous.**

Who was nervous?

Ravi or Arjun?

Ambiguous.

A good exam sentence should avoid such ambiguity.

---

## 77. SUBJECT VS OBJECT PRONOUN

---

Subject forms:

- I
- he
- she
- we
- they
- who

Object forms:

- me
  - him
  - her
  - us
  - them
  - whom
- 

## 78. EXAMPLE

---

Correct:

---



**Ravi and I went to class.**

Because pronoun is subject.

Correct:

**The teacher called Ravi and me.**

Object.

## 79. REMOVE-OTHER-PERSON TRICK

Sentence:

**The teacher spoke to Ravi and I/me.**

Remove Ravi:

**The teacher spoke to I 'L**

**The teacher spoke to me '**

Therefore:

$\boxed{\text{me}}$

Very useful shortcut.

## 80. BETWEEN YOU AND ME

Correct:

$\boxed{\text{between you and me}}$

Not:

**between you and I**

Preposition “between” takes object form.

## 81. WHO VS WHOM

**Who**

subject.

**Who called you?**

## Whom

object.

**Whom did you call?**

Formal aptitude questions may test this.

## 82. WHO/WHOM SHORTCUT

Replace with:

- he !' who
- him !' whom

Example:

**\_\_\_\_\_ did you meet?**

You met him.

Therefore:

$\boxed{\text{whom}}$

## 83. POSSESSIVE PRONOUN

Correct:

**This book is mine.**

No apostrophe.

Forms:

- mine
- yours
- his
- hers
- ours
- theirs

## 84. ITS VS IT'S

**its**

possessive:

**The company changed its policy.**

**it's**

contraction:

\[ it\ is \]

**It's raining.**

Very common error.

---

## 85. THEIR / THERE / THEY'RE

---

**their**

possessive

**there**

place/existence

**they're**

they are

Example:

**They're going to their office over there.**

---

## 86. PREPOSITIONS — AT / ON / IN FOR TIME

---

**AT**

specific time:

**at 5 PM**

**ON**

day/date:

**on Monday**

**on 7 September**

**IN**

month/year/long period:

**in September**

in 2026

## 87. AT / ON / IN FOR PLACE

### AT

point/location:

at the station

### ON

surface:

on the table

### IN

inside/enclosed area:

in the room

## 88. SINCE / FOR RECAP

### since

starting point

### for

duration

$\boxed{\text{since 2020, for six years}}$

## 89. BY VS UNTIL

### by

deadline, no later than

Submit by Friday.

Can submit earlier.

### until

continuation up to time

Stay here until Friday.

---

## 90. BETWEEN VS AMONG

---

### **between**

traditionally used for two or distinct individually identified items.

### **among**

used within a group.

Examples:

**Divide the work between Ravi and Sita.**

**Distribute the prizes among the students.**

Modern English can use “between” for more than two when relationships are distinct, but exam basics use this contrast.

---

## 91. BESIDE VS BESIDES

---

### **beside**

next to

**Sit beside me.**

### **besides**

in addition to

**Besides mathematics, she studies physics.**

## 92. IN VS INTO

---

### **in**

position

**He is in the room.**

### **into**

movement toward inside

**He walked into the room.**

## 93. ON VS ONTO

**on**

position

**The book is on the table.**

**onto**

movement to surface

**The cat jumped onto the table.**

## 94. FROM / TO

Correct pair:

**from Hyderabad to Delhi**

Not:

**from Hyderabad till Delhi**

for destination in standard travel expression.

## 95. PREFER

Correct:

**I prefer tea to coffee.**

Common exam trap:

Not:

**prefer tea than coffee.**

## 96. DIFFERENT FROM

Standard safe exam form:

**This is different from that.**

“Different than” occurs in some dialects, but “different from” is safest for aptitude grammar.

## 97. CONSIST OF

Correct:

The committee consists of five members.

Not:

consists from

## 98. DEPEND ON

Correct:

Success depends on preparation.

## 99. RESPONSIBLE FOR

She is responsible for the project.

## 100. INTERESTED IN

He is interested in mathematics.

## 101. GOOD AT

She is good at coding.

## 102. MARRIED TO

Correct:

She is married to him.

Not:

married with

in standard exam English.

---

## 103. COMPARISON PREPOSITION

**superior to**

**inferior to**

**senior to**

**junior to**

Not "than."

Example:

**Ravi is senior to Arjun.**

---

## 104. SUBJECT-VERB SPEED CHECK

When checking agreement:

1. find main verb
2. ask "who/what performs?"
3. identify head subject
4. ignore interrupting phrases
5. check singular/plural

---

## 105. TENSE SPEED CHECK

Look for time markers:

**Yesterday / ago / last...**

!' simple past

**Since / for + continuing action**

!' perfect/perfect continuous

**now / currently**

!' continuous often

**everyday/general truth**

!' simple present



## 106. ARTICLE SPEED CHECK

---

Before a singular countable noun, ask:

1. Is it general/non-specific?

!' a/an

2. specific/known/unique?

!' the

3. Is article unnecessary due plural/general or uncountable/general?

!' zero article

4. a vs an?

!' sound, not spelling

---

## 107. PRONOUN SPEED CHECK

---

Ask:

- what noun does pronoun refer to?
  - singular/plural match?
  - subject or object form?
  - reference ambiguous?
  - possessive form correct?
- 

## 108. COMMON GATE/PLACEMENT TRAPS

---

& pTRAP 1

**The list of items are...**

Wrong.

$\boxed{\text{list is}}$

---

& pTRAP 2

**Each of the students are...**

Wrong.

$\boxed{\text{Each...is}}$

---

& pTRAP 3

**A number of students is...**

Wrong.

$\boxed{\text{are}}$

---

---

& pTRAP 4

**The number of students are...**

Wrong.

$\boxed{\text{is}}$

---

& pTRAP 5

**He has visited Delhi last year.**

Wrong.

Use simple past:

$\boxed{\text{visited}}$

---

& pTRAP 6

**an university**

Wrong.

$\boxed{\text{a university}}$

---

& pTRAP 7

**a hour**

Wrong.

$\boxed{\text{an hour}}$

---

& pTRAP 8

**Between you and I**

Wrong.

$\boxed{\text{between you and me}}$

---

& pTRAP 9

**prefer X than Y**

Wrong.

$\boxed{\text{prefer X to Y}}$

senior than

Wrong.

\[ \boxed{senior\ to} \]

## PRACTICE — LEVEL 0

### Q1

Choose:

The list of names \_\_\_\_ ready.

A. are B. is

### Q2

Choose:

Each of the students \_\_\_\_ submitted the form.

A. have B. has

### Q3

Choose:

She \_\_\_\_ to college every day.

A. go B. goes

### Q4

Choose:

He \_\_\_\_ Delhi last year.

A. has visited B. visited

### Q5

Choose:

She is \_\_\_\_ engineer.

A. a B. an

## LEVEL 1

**Q6**

Choose:

**A number of students \_\_\_\_ absent.**

A. is B. are

**Q7**

Choose:

**The number of applicants \_\_\_\_ increasing.**

A. is B. are

**Q8**

Choose:

**Either the teacher or the students \_\_\_\_ responsible.**

A. is B. are

**Q9**

Choose:

**Ravi, along with his friends, \_\_\_\_ attending.**

A. is B. are

**Q10**

Choose:

**I have lived here \_\_\_\_ 2022.**

A. since B. for

---

**LEVEL 2**

---

**Q11**

Choose:

**She has been studying \_\_\_\_ three hours.**

A. since B. for

**Q12**

Choose:

I will call you when I \_\_\_\_ home.

A. will reach B. reach

### Q13

Choose:

He is \_\_\_\_ MBA graduate.

A. a B. an

### Q14

Choose:

The teacher spoke to Ravi and \_\_\_\_.

A. I B. me

### Q15

Choose:

This book is different \_\_\_\_ that one.

A. than B. from

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

Identify the error:

Each of the players have received their certificate.

### Q17

Identify the error:

The quality of these products are excellent.

### Q18

Identify the error:

She has completed the course yesterday.

### Q19

Choose the correct sentence:

A. He is senior than me. B. He is senior to me.

## Q20

Choose the correct sentence:

A. She prefers coffee than tea. B. She prefers coffee to tea.

---

## ANSWERS

---

### Q1

Subject = list.

\[ \boxed{B:\ is} \]

---

### Q2

Each is singular.

\[ \boxed{B:\ has} \]

---

### Q3

Singular subject “she.”

\[ \boxed{B:\ goes} \]

---

### Q4

“last year” !’ finished definite past.

\[ \boxed{B:\ visited} \]

---

### Q5

“engineer” begins with vowel sound.

\[ \boxed{B:\ an} \]

---

### Q6

“A number of” = several.

\[ \boxed{B:\ are} \]

---

### Q7

“The number” = singular subject.

---

\[ \boxed{A:\ is} \]

---

### Q8

Nearest subject = students.

\[ \boxed{B:\ are} \]

---

### Q9

Main subject Ravi.

\[ \boxed{A:\ is} \]

---

### Q10

Starting point:

\[ \boxed{A:\ since} \]

---

### Q11

Duration:

\[ \boxed{B:\ for} \]

---

### Q12

Future time clause uses present.

\[ \boxed{B:\ reach} \]

---

### Q13

“M” pronounced “em,” vowel sound.

\[ \boxed{B:\ an} \]

---

### Q14

Object of “to/spoke to.”

\[ \boxed{B:\ me} \]

---

### Q15

Safe standard form:

\[ \boxed{B:\ from} \]

---

---

## Q16

Wrong:

**Each of the players have**

Correct:

`\[ \boxed{\text{Each of the players has received...}} \]`

---

## Q17

Main subject:

`\[ quality \]`

Correct:

`\[ \boxed{\text{The quality of these products is excellent.}} \]`

---

## Q18

“Yesterday” requires simple past.

Correct:

`\[ \boxed{\text{She completed the course yesterday.}} \]`

---

## Q19

`\[ \boxed{B} \]`

**senior to**

## Q20

`\[ \boxed{B} \]`

**prefers coffee to tea**

---

## CHAPTER 71 STATUS

Covered so far:

- parts of speech '
  - subject/predicate '
-



- subject–verb agreement '
- tricky agreement structures '
- basic tense system '
- tense consistency '
- since/for '
- future time clauses '
- articles '
- a/an sound rule '
- pronoun agreement '
- subject/object pronouns '
- who/whom '
- possessives '
- basic prepositions '
- common fixed-preposition combinations '
- L0–L3 practice '

But **Chapter 71 is not complete yet.**

Still remaining:

- advanced pronoun errors
- relative clauses
- modifiers
- misplaced/dangling modifiers
- parallelism
- comparisons
- conjunction pairs
- conditionals
- active/passive voice
- direct/indirect speech
- adjective/adverb traps
- redundancy
- sentence structure
- fragments/run-ons
- common placement error-spotting patterns
- final grammar speed toolkit

## Progress

\[ \boxed{70/76} \text{ numbered chapters completed} \]

**Current:** 0 = Chapter 71 — Grammar for Aptitude

## NEXT

0 = **Chapter 71 Part 2 — Modifiers, Parallelism, Comparisons, Relative Clauses, Conjunctions, Voice, Reported Speech & Final Grammar Error-Spotting Toolkit**

USER

next

ASSISTANT

## CHAPTER 71 — GRAMMAR FOR APTITUDE

### PART 2 — MODIFIERS, PARALLELISM, COMPARISONS, RELATIVE CLAUSES, CONJUNCTIONS, VOICE, REPORTED SPEECH & FINAL ERROR-SPOTTING TOOLKIT

Status:  $\emptyset = \text{Not}$  PROGRESS ! ' ' COMPLETE GATE: High Placements: Very High Priority:  $\emptyset = \text{Not}$  1

This part covers the grammar areas most often used in:

- sentence correction
- error detection
- fill-in-the-blank questions
- verbal ability OAs
- placement grammar tests

Core principle:

**A grammatically correct sentence must be structurally balanced, logically clear, and free from ambiguity.**

## 1. MODIFIERS

A modifier describes or limits another word or phrase.

Example:

**The careful student solved the problem.**

“careful” modifies “student.”

Another:

**She solved the problem quickly.**

“quickly” modifies “solved.”

## 2. MODIFIER PLACEMENT

A modifier should normally be placed close to the word it modifies.

Bad:

**She almost answered all the questions correctly.**

This may imply she did not actually answer them.

If intended meaning is that nearly all questions were answered:

**She answered almost all the questions correctly.**

Placement changes meaning.

### 3. MISPLACED MODIFIER

Bad:

**He served sandwiches to the children on paper plates.**

Who is on paper plates?

Ambiguous.

Better:

**He served the children sandwiches on paper plates.**

or:

**He served sandwiches on paper plates to the children.**

### 4. DANGLING MODIFIER

Bad:

**Walking to college, the rain started.**

Who was walking?

“The rain” appears to be walking.

Wrong.

Correct:

**Walking to college, I noticed that it started raining.**

The implied subject of introductory phrase must match the main clause subject.

### 5. DANGLING PARTICIPLE RULE

Structure:

**While doing X, Y happened.**

Make sure the person doing X is clearly the subject of main clause.

Bad:

**After reading the report, the decision was obvious.**

Who read the report?

Better:

**After reading the report, the manager found the decision obvious.**

## 6. ONLY — POSITION MATTERS

Compare:

**Only Ravi solved the problem.**

Means nobody else solved it.

**Ravi only solved the problem.**

May mean he solved it but did nothing more.

**Ravi solved only the problem.**

Means he solved that and nothing else.

“Only” should be placed near the element it limits.

## 7. ALMOST / NEARLY / JUST / EVEN

These words also require careful placement.

Example:

**She nearly spent ₹ 1000.**

Could mean she almost spent it but did not.

**She spent nearly ₹ 1000.**

Means amount was close to ₹ 1000.

## 8. PARALLELISM

Items joined in a list or comparison should have similar grammatical form.

Correct:

**She likes reading, writing, and painting.**

All are gerunds.

Wrong:

**She likes reading, to write, and painting.**

Forms are mixed.

## 9. BASIC PARALLEL STRUCTURE

Correct:

$\boxed{\text{noun+noun+noun}}$

or:

$\boxed{\text{verb+verb+verb}}$

or:

$\boxed{\text{gerund+gerund+gerund}}$

## 10. EXAMPLE

Wrong:

**The job requires coding efficiently, communication skills, and to solve problems.**

Better:

**The job requires efficient coding, communication skills, and problem-solving ability.**

Balanced noun phrases.

## 11. CORRELATIVE CONJUNCTION PARALLELISM

Pairs:

- either...or
- neither...nor
- both...and
- not only...but also
- whether...or

Elements after each part should be parallel.

---

## 12. NOT ONLY...BUT ALSO

---

Wrong:

**She not only writes code but also debugging software.**

Correct:

**She not only writes code but also debugs software.**

Better parallel form:

\[ writes \rightarrow debugs \]

---

## 13. BOTH...AND

---

Correct:

**She is both intelligent and hardworking.**

Not:

**She both is intelligent and works hard**

unless structure is deliberately balanced.

---

## 14. EITHER...OR

---

Correct:

**You can either stay here or leave now.**

Both are verb phrases.

---

## 15. NEITHER...NOR

---

Correct:

**He is neither careless nor dishonest.**

Balanced adjectives.

---

## 16. COMPARISON — BASIC RULE

Compare similar things.

Wrong:

**Ravi's salary is higher than Sita.**

This compares salary to a person.

Correct:

**Ravi's salary is higher than Sita's salary.**

or:

**Ravi's salary is higher than Sita's.**

## 17. LOGICAL COMPARISON

Wrong:

**The population of Hyderabad is greater than Chennai.**

Correct:

**The population of Hyderabad is greater than that of Chennai.**

We compare:

population !" population

## 18. "THAN" CASE

After "than," formal grammar depends on implied structure.

**She is taller than I am.**

Formal.

In everyday English:

**taller than me**

is also widely accepted.

Aptitude tests may prefer the fully logical form where ambiguity matters.

## 19. DOUBLE COMPARATIVE ERROR

Wrong:

**more better**

**more easier**

Correct:

**better**

**easier**

Because comparative is already built in.

## 20. DOUBLE SUPERLATIVE ERROR

Wrong:

**most best**

Correct:

**best**

## 21. COMPARATIVE+ THAN

Use:

**taller than**

**faster than**

But remember exceptions:

**superior to**

**inferior to**

**senior to**



**junior to**

## 22. “AS...AS”

Equality comparison:

**Ravi is as tall as Arjun.**

Negative:

**Ravi is not as tall as Arjun.**

## 23. “SO...AS”

In formal grammar, negative comparison may also appear:

**Ravi is not so tall as Arjun.**

But “not as...as” is safer and more common.

## 24. “ONE OF THE BEST”

Correct:

**She is one of the best students in the class.**

Because “one of” must be followed by plural noun:

$\boxed{\text{one of the} + \text{plural}}$

Wrong:

**one of the best student**

## 25. RELATIVE PRONOUNS

Main ones:

- who
- whom
- whose
- which
- that

They connect a noun with a relative clause.

---

## 26. WHO

Used mainly for people as subject.

**The student who won the prize is absent.**

“who” performs action “won.”

---

## 27. WHOM

Used mainly for people as object.

**The student whom I met was helpful.**

I met whom.

---

## 28. WHOSE

Shows possession.

**The student whose laptop was stolen reported the incident.**

Can refer to people and sometimes things/organizations in modern English.

---

## 29. WHICH

Used mainly for things/animals.

**The book, which I bought yesterday, is useful.**

---

## 30. THAT

Can refer to people or things in restrictive clauses.

**The book that I bought is useful.**

**The person that called me left no message.**

Though “who” is often preferred for people.

---

## 31. RESTRICTIVE VS NON-RESTRICTIVE CLAUSE

---

Restrictive:

**The students who studied passed.**

This identifies which students.

No commas.

Non-restrictive:

**Ravi, who studied all night, passed.**

Extra information.

Use commas.

## 32. THAT VS WHICH

In many exam-style conventions:

- “that” for restrictive clauses
- “which” for non-restrictive clauses

Example:

**The car that is parked outside belongs to Ravi.**

**My car, which is parked outside, is red.**

## 33. RELATIVE CLAUSE AGREEMENT

**She is one of the students who have completed the task.**

“who” refers to students !’ plural.

But:

**She is the only one of the students who has completed the task.**

“who” now points to “the only one.”

## 34. CONJUNCTIONS — BASIC

**and**

addition

**but**

contrast

**or**

choice

**because**

reason

**although**

contrast/concession

**therefore**

result

---

## 35. BECAUSE VS BECAUSE OF

---

**because + clause**

He stayed home because he was ill.

**because of + noun phrase**

He stayed home because of illness.

Wrong:

because of he was ill.

---

## 36. ALTHOUGH VS DESPITE

---

**although + clause**

Although it was raining, we went out.

**despite + noun/gerund**

Despite the rain, we went out.

Despite raining heavily... 'Lusually awkward; use “despite the heavy rain” or “despite it raining...”

---

## 37. DESPITE VS IN SPITE OF

---

Both mean similar.

Despite the rain...

**In spite of the rain...**

But:

**despite of 'L**

Wrong in standard English.

## 38. ALTHOUGH + BUT TRAP

Wrong:

**Although he was tired, but he continued.**

Use one:

**Although he was tired, he continued.**

or:

**He was tired, but he continued.**

## 39. BECAUSE + SO TRAP

Wrong:

**Because it rained, so the match was cancelled.**

Correct:

**Because it rained, the match was cancelled.**

or:

**It rained, so the match was cancelled.**

## 40. UNLESS

**Unless you study, you may fail.**

Equivalent basic idea:

**If you do not study, you may fail.**

Avoid double negative:

Wrong:

**Unless you do not study...**

unless intended meaning is unusual.

## 41. CONDITIONALS — TYPE 0

General truth:

[ If + present, \ present \]

Example:

**If water reaches  $(100^{\circ}\text{C})$ , it boils under standard conditions.**

## 42. FIRST CONDITIONAL

Real future possibility:

[ If + present, \ will + V<sub>1</sub> \]

**If it rains, we will stay home.**

## 43. SECOND CONDITIONAL

Hypothetical/unreal present or future:

[ If + past, \ would + V<sub>1</sub> \]

**If I had more time, I would learn another language.**

## 44. THIRD CONDITIONAL

Unreal past:

[ If + had + V<sub>3</sub>, \ would \ have + V<sub>3</sub> \]

**If I had studied harder, I would have scored better.**

## 45. MIXED CONDITIONAL IDEA

Sometimes a past condition has present result:

**If I had taken that job, I would be in Delhi now.**

Less common in aptitude but possible.

## 46. “WERE” IN HYPOTHETICALS

Traditional formal form:

**If I were you...**

Not:

**If I was you**

for hypothetical advice.

Aptitude tests often prefer:

$\boxed{\text{were}}$

## 47. ACTIVE VOICE

Subject performs action.

**The student solved the problem.**

Subject = student.

## 48. PASSIVE VOICE

Object becomes subject.

**The problem was solved by the student.**

Structure:

$\text{be} + V_3$

## 49. ACTIVE !' PASSIVE

Present simple:

**Ravi writes the report.**

Passive:

**The report is written by Ravi.**

## **50. PAST SIMPLE PASSIVE**

Active:

**Ravi wrote the report.**

Passive:

**The report was written by Ravi.**

## **51. PRESENT PERFECT PASSIVE**

Active:

**Ravi has completed the work.**

Passive:

**The work has been completed by Ravi.**

## **52. FUTURE PASSIVE**

Active:

**Ravi will complete the work.**

Passive:

**The work will be completed by Ravi.**

## **53. MODAL PASSIVE**

Active:

**You must submit the form.**

Passive:

**The form must be submitted.**



Structure:

\[ modal + be + V\_3 \]

---

## 54. PASSIVE VOICE — TENSE MUST REMAIN SAME

---

When changing voice:

- meaning same
- tense same
- object/subject roles change

Do not accidentally change time.

---

## 55. WHEN PASSIVE IS IMPOSSIBLE

---

Intransitive verbs without object generally cannot be made passive.

Example:

**He sleeps.**

No object.

So no normal passive.

---

## 56. DIRECT SPEECH

---

**Ravi said, "I am tired."**

Speaker's exact words.

---

## 57. INDIRECT / REPORTED SPEECH

---

**Ravi said that he was tired.**

Pronouns and tense may change depending context.

---

## 58. SAY VS TELL

---

**say**

usually:

**say something**

**say to someone**

**tell**

requires object:

**tell someone something**

Correct:

**He told me the truth.**

Wrong:

**He told that he was tired.**

Correct:

**He said that he was tired.**

## 59. BACKSHIFT — BASIC

If reporting verb is past:

Direct:

**He said, "I am tired."**

Indirect:

**He said that he was tired.**

Present !' past.

## 60. COMMON BACKSHIFTS

Direct	Reported
am/is	was
are	were
have/has	had
simple past	past perfect often
will	would

Direct	Reported
can	could
may	might

Context can affect this; grammar tests usually use standard backshift.

## 61. UNIVERSAL TRUTH EXCEPTION

Direct:

**Teacher said, "The Earth revolves around the Sun."**

Indirect:

**The teacher said that the Earth revolves around the Sun.**

Universal truth remains present.

## 62. TIME-WORD CHANGES IN REPORTED SPEECH

Often:

- now !' then
- today !' that day
- tomorrow !' the next day
- yesterday !' the previous day
- here !' there
- this !' that

Depending context.

## 63. REPORTED QUESTIONS

Direct:

**He asked, "Where are you going?"**

Indirect:

**He asked where I was going.**

No question-word inversion in reported clause.

Wrong:

**He asked where was I going.**

---

## 64. YES/NO REPORTED QUESTIONS

---

Direct:

**She asked, "Are you ready?"**

Indirect:

**She asked if/whether I was ready.**

---

## 65. COMMANDS IN REPORTED SPEECH

---

Direct:

**He said, "Sit down."**

Reported:

**He told me to sit down.**

Negative:

**He told me not to leave.**

---

## 66. ADJECTIVE/ADVERB TRAPS

---

Wrong:

**He performed good.**

Correct:

**He performed well.**

Because "performed" needs adverb.

---

## 67. GOOD VS WELL

---

**good**

adjective

**She is a good singer.**

---

## **well**

usually adverb

**She sings well.**

But “well” can also mean healthy:

**She is well.**

---

## **68. HARD VS HARDLY**

### **hard**

with effort

**He works hard.**

### **hardly**

almost not

**He hardly works.**

Very different meanings.

---

## **69. LATE VS LATELY**

### **late**

after expected time

**He arrived late.**

### **lately**

recently

**I haven't seen him lately.**

---

## **70. NEAR VS NEARLY**

### **near**

close

**The station is near.**

**nearly**

almost

**I nearly missed the train.**

---

## 71. FREE VS FREELY

---

**free**

without payment / unrestricted

**Entry is free.**

**freely**

without restriction

**Speak freely.**

---

## 72. REDUNDANCY

---

Avoid unnecessary repetition of meaning.

Wrong:

**return back**

“return” already includes back.

Better:

**return**

---

## 73. COMMON REDUNDANCIES

---

Usually avoid:

- return back
- repeat again
- revert back
- merge together
- collaborate together
- advance forward
- past history
- future plans
- end result

- basic fundamentals

Some may occur in casual speech but aptitude tests may flag them.

---

## 74. “DISCUSS ABOUT”

Wrong in standard exam English:

**discuss about the issue**

Correct:

**discuss the issue**

But:

**have a discussion about the issue**

is correct.

---

## 75. “ORDER FOR”

Correct:

**I ordered a book.**

Not usually:

**ordered for a book**

unless “ordered for someone” means placed an order on someone’s behalf.

---

## 76. “ENTER INTO” TRAP

For physical place:

**enter the room**

not normally:

**enter into the room**

But “enter into an agreement” is correct idiomatically.

---

## 77. SENTENCE FRAGMENT

Fragment lacks complete independent clause.

Example:

**Because he was tired.**

Incomplete.

Needs main clause:

**Because he was tired, he went home.**

---

## 78. RUN-ON SENTENCE

Two independent clauses improperly joined.

Wrong:

**I finished the exam I left the hall.**

Better:

**I finished the exam, and I left the hall.**

or:

**I finished the exam. I left the hall.**

---

## 79. COMMA SPLICE

Wrong:

**The exam was difficult, I completed it.**

Two independent clauses joined only by comma.

Correct:

**The exam was difficult, but I completed it.**

or semicolon/full stop.

---

## 80. SENTENCE STRUCTURE — SIMPLE

One independent clause:

**Ravi studies every day.**



---

## 81. COMPOUND SENTENCE

---

Two independent clauses:

**Ravi studied, and Sita revised.**

---

## 82. COMPLEX SENTENCE

---

Independent + dependent clause:

**Ravi studied because he had an exam.**

---

## 83. DOUBLE NEGATIVE

---

Wrong in standard formal English:

**I don't know nothing.**

Correct:

**I don't know anything.**

or:

**I know nothing.**

---

## 84. FEWER VS LESS

---

**fewer**

countable plural:

**fewer students**

**less**

uncountable:

**less water**

Classic aptitude trap.

---

## 85. MUCH VS MANY

---

**many**

countable:

**many books**

**much**

uncountable:

**much information**

## 86. FEW / A FEW

---

**few**

almost none; negative sense

**Few students understood.**

**a few**

some; positive sense

**A few students understood.**

## 87. LITTLE / A LITTLE

---

**little**

almost none

**There is little hope.**

**a little**

some

**There is a little hope.**

## 88. NUMBER VS AMOUNT

---

**number**

countable

**number of students**

## **amount**

uncountable

**amount of water**

## **89. BETWEEN / AMONG RECAP**

Two/distinct relationships:

**between A and B**

Group:

**among the participants**

## **90. LIKE VS AS**

### **like**

comparison as preposition

**He runs like a professional.**

### **as**

role/function or clause

**He works as a consultant.**

**As I expected, it rained.**

Do not casually interchange.

## **91. SUCH AS VS LIKE**

“such as” introduces examples:

**Languages such as Python and Java...**

“like” can mean similar to and can sometimes introduce examples, but “such as” is safer when listing actual examples.

## 92. WHETHER VS IF

Both can introduce indirect yes/no questions:

**I don't know whether he will come.**

**I don't know if he will come.**

But use "whether" especially with:

- "or not"
- after preposition
- before infinitive

Example:

**We discussed whether to continue.**

## 93. NO SOONER...THAN

Correct pair:

**No sooner had he arrived than it started raining.**

Not:

**no sooner...when**

## 94. HARDLY/SCARCELY...WHEN

Correct:

**Hardly had he arrived when it started raining.**

**Scarcely had he entered when the bell rang.**

## 95. SO...THAT

**He was so tired that he slept immediately.**

## 96. TOO...TO

**He was too tired to continue.**

Avoid awkward redundancy:

**too tired that he could not continue 'Lin standard test patterns.**

## 97. SO THAT

Purpose/result:

**He spoke slowly so that everyone could understand.**

## 98. LEST

Traditional pattern:

**Work carefully lest you should make a mistake.**

“lest” means “for fear that / to prevent.”

Low frequency, but may appear in old-style placement tests.

## 99. ERROR-SPOTTING MASTER ORDER

When checking a sentence, scan in this order:

### 1. Subject–verb agreement

Is verb matching main subject?

### 2. Tense

Is time/sequence consistent?

### 3. Pronoun

Agreement/reference/case?

### 4. Modifier

Placed near correct word?

### 5. Parallelism

Balanced structures?

### 6. Article/determiner

a/an/the/zero article?

### 7. Preposition

Correct fixed pair?

## 8. Comparison

Like with like?

## 9. Conjunction

Correct pair and logic?

## 10. Redundancy/idiom

Any standard error?

---

## 100. SPEED METHOD — FIND THE CORE SENTENCE

Long sentence:

**The list of candidates selected for the final interview, along with their documents, are on the table.**

Strip modifiers:

**The list are...**

Immediately wrong.

Correct:

**The list ... is on the table.**

This stripping method is extremely useful.

---

## 101. ERROR-SPOTTING TRAP — NEAREST NOUN

Sentence:

**The quality of the products are improving.**

Nearest noun “products” is plural.

But subject is “quality.”

Correct:

$\boxed{\text{is}}$

Never choose verb by nearest noun unless the grammar rule specifically requires it.

---

## 102. ERROR-SPOTTING TRAP — LOGIC VS GRAMMAR

Sentence may be grammatically correct but logically absurd.

Example:

**The laptop ate dinner.**

Grammar:

subject + verb + object.

But semantics wrong.

Some verbal tests may ask meaningful sentence, not just grammar.

Always check both.

---

## 103. ERROR-SPOTTING TRAP — IDIOM

Sentence:

**She is capable to solve the problem.**

Standard:

\[ \boxed{\text{capable}} \text{ of solving} \]

Correct:

**She is capable of solving the problem.**

---

## 104. COMMON FIXED EXPRESSIONS

MUST MEMORIZE:

- capable of
- interested in
- responsible for
- good at
- depend on
- insist on
- consist of
- comply with
- agree with a person
- agree to a proposal
- agree on a topic/decision
- prefer X to Y
- different from
- similar to
- married to
- senior to
- junior to
- superior to

- inferior to

---

## 105. “USED TO” VS “BE USED TO”

---

### used to + base verb

past habit:

**I used to wake early.**

### be used to + noun/gerund

accustomed to:

**I am used to waking early.**

Very common placement trap.

---

## 106. “LOOK FORWARD TO”

---

“to” here is preposition.

So:

**I look forward to meeting you.**

Not:

**look forward to meet you.**

---

## 107. “OBJECT TO”

---

Also preposition:

**I object to working late.**

---

## 108. “ACCUSTOMED TO”

---

**She is accustomed to working under pressure.**

---

Gerund after “to” because “to” is preposition here.

---

## 109. INFINITIVE VS GERUND — BASIC

---

Some verbs commonly take gerund:

---



- enjoy
- avoid
- consider
- suggest
- finish
- admit

Examples:

**She enjoys reading.**

**He avoided answering.**

---

## 110. VERBS COMMONLY TAKING INFINITIVE

- want
- decide
- plan
- hope
- agree
- refuse

Examples:

**She decided to leave.**

**He hopes to win.**

---

## 111. SUGGEST TRAP

Correct:

**She suggested going early.**

or:

**She suggested that we go early.**

Not:

**She suggested to go early.**

## 112. ENJOY TRAP

Correct:

**I enjoy reading.**

Not:

**enjoy to read**

in standard usage.

## 113. AVOID TRAP

**Avoid making unnecessary assumptions.**

Not:

**avoid to make**

## 114. STOP DOING VS STOP TO DO

**He stopped smoking.**

He quit smoking.

**He stopped to smoke.**

He stopped another activity in order to smoke.

Meaning changes.

## 115. REMEMBER DOING VS REMEMBER TO DO

**I remember meeting him.**

Memory of past event.

**Remember to call him.**

Do not forget future action.

## 116. TRY DOING VS TRY TO DO

**Try restarting the computer.**

Experiment with method.

**Try to restart the computer.**

Make an effort.

## 117. COMMON COMPARISON ERROR

Wrong:

**Of the two candidates, Ravi is the best.**

For two, use comparative:

$\boxed{\text{better}}$

Correct:

**Ravi is the better of the two candidates.**

For 3+:

**best**

## 118. FEWER THAN / LESS THAN

Standard rule:

**fewer than 10 students**

**less than 10 litres**

But time/money/distance often use "less" as quantities:

**less than 10 minutes**

## 119. NO OTHER...THAN

Pattern:

**No other student is taller than Ravi.**

Equivalent idea:

**Ravi is the tallest student.**

Useful transformation.

## 120. VERY / MUCH WITH COMPARATIVES

Usually:

**much better**

**far better**

**much more useful**

Not:

**very better**

“very” modifies base adjectives:

**very good**

## 121. FINAL COMMON ERRORS

Wrong:

**He is knowing the answer.**

Stative verb “know” normally not continuous.

Correct:

**He knows the answer.**

## 122. STATIVE VERBS

Common stative verbs often avoid continuous forms:

- know
- believe
- understand
- belong
- own

- need
- want
- prefer

Context can allow some continuous uses, but standard aptitude questions test basic stative rule.

---

## 123. “I AM HAVING”

---

If “have” means possession:

Wrong:

**I am having a car.**

Correct:

**I have a car.**

But activity:

**I am having lunch.**

Correct.

---

## 124. VERB FORM AFTER MODALS

---

After:

- can
- could
- may
- might
- must
- should
- will
- would

use base verb.

Correct:

**She can swim.**

Wrong:

**She can swims.**

---

## 125. "DID" + BASE VERB

Correct:

**Did you go?**

Wrong:

**Did you went?**

Because past tense already carried by "did."

## 126. "DOES" + BASE VERB

Correct:

**Does she work here?**

Wrong:

**Does she works here?**

## 127. "HAS/HAVE/HAD" + V3

Correct:

**has written**

**have gone**

**had completed**

Not:

**has wrote**

## 128. "BE" + V-ING / V3

Continuous:

\[ be + V-ing \]

Passive:

\[ be + V\_3 \]

Example:

**She is writing.**

**The report is written.**

---

## 129. FINAL 30-SECOND GRAMMAR CHECK

---

Look for:

- subject
- main verb
- tense marker
- pronoun antecedent
- comparison pair
- parallel list
- modifier placement
- preposition
- article
- redundant expression

This catches most OA errors.

---

## PRACTICE — LEVEL 0

---

### Q1

Choose correct:

A. She likes reading, writing, and to paint. B. She likes reading, writing, and painting.

### Q2

Choose correct:

A. Walking to college, the rain started. B. Walking to college, I got caught in the rain.

### Q3

Choose correct:

A. Although he was tired, but he worked. B. Although he was tired, he worked.

### Q4

Choose correct:

A. He is more better than Ravi. B. He is better than Ravi.

### Q5

Choose correct:

A. She enjoys reading. B. She enjoys to read.

---

## LEVEL 1

---

### Q6

Correct:

**Ravi's salary is greater than Sita.**

What is the issue?

### Q7

Choose:

**She is one of the students who \_\_\_\_ passed.**

A. has B. have

### Q8

Choose:

**No sooner had he arrived \_\_\_\_ the rain started.**

A. when B. than

### Q9

Choose:

**I look forward to \_\_\_\_ you.**

A. meet B. meeting

### Q10

Choose:

**Did she \_\_\_\_ yesterday?**

A. went B. go

---

## LEVEL 2

---

### Q11

Identify error:

**The problem was discussed about in the meeting.**



### Q12

Identify error:

**He is capable to complete the task.**

### Q13

Choose:

**If I \_\_\_\_ you, I would accept the offer.**

A. was B. were

### Q14

Choose:

**He said that he \_\_\_\_ tired.**

A. is B. was

assuming ordinary past reporting context.

### Q15

Choose the correct passive:

**The team completed the project.**

A. The project was completed by the team. B. The project has completed by the team.

## LEVEL 3 — EXAM PRESSURE

### Q16

Identify the error:

**The manager, along with the employees, were reviewing the report.**

### Q17

Identify the error:

**The population of City A is higher than City B.**

### Q18

Identify the error:

**She not only writes reports but also preparing presentations.**

### Q19

Identify the error:

**He told that he would return the next day.**

### Q20

Identify all likely errors:

**Each of the candidates have been asked to submit their documents by Monday, and no candidate should forget bringing the identity card.**

## ANSWERS

### Q1

Parallel gerunds:

\[ \boxed{B} \]

### Q2

The person walking must be subject.

\[ \boxed{B} \]

### Q3

Do not use both although + but.

\[ \boxed{B} \]

### Q4

Avoid double comparative.

\[ \boxed{B} \]

### Q5

“Enjoy” takes gerund.

\[ \boxed{A} \]

### Q6

Salary must be compared with salary.

Correct:

**Ravi's salary is greater than Sita's.**

or:

**...greater than the salary of Sita.**

## Q7

“who” refers to students.

$\boxed{B: \text{ have} }$

## Q8

Fixed pair:

$\boxed{No \text{ sooner...than} }$

Answer:

$\boxed{B}$

## Q9

“to” is preposition in “look forward to.”

$\boxed{B: \text{ meeting} }$

## Q10

After did !' base verb.

$\boxed{B: \text{ go} }$

## Q11

“Discuss” does not require “about.”

Correct:

**The problem was discussed in the meeting.**

## Q12

Correct expression:

$\boxed{\text{capable of completing} }$

---

### Q13

Formal hypothetical:

\[ \boxed{B:\text{ were} } \]

---

### Q14

Standard backshift:

\[ \boxed{B:\text{ was} } \]

---

### Q15

Simple past active !' simple past passive:

\[ \boxed{A} \]

---

### Q16

Main subject = manager.

Correct:

**The manager, along with the employees, was reviewing...**

---

### Q17

Illogical comparison.

Correct:

**The population of City A is higher than that of City B.**

---

### Q18

Parallelism error.

Correct:

**She not only writes reports but also prepares presentations.**

---

### Q19

“tell” needs object.

---

Correct either:

**He said that he would return the next day.**

or:

**He told me that he would return the next day.**

---

## Q20

Two main issues.

### Error 1

“Each” is singular:

**Each of the candidates has been asked...**

### Error 2

“forget” with intended future obligation:

**should not forget to bring the identity card.**

Correct sentence:

**Each of the candidates has been asked to submit their documents by Monday, and no candidate should forget to bring the identity card.**

---

## ' CHAPTER 71 COMPLETE

We have now covered:

- parts of speech '
- subject–verb agreement '
- tricky subject structures '
- tenses '
- conditionals '
- articles '
- pronouns '
- relative clauses '
- modifiers '
- dangling/misplaced modifiers '
- parallelism '
- comparisons '
- conjunctions '

- fixed conjunction pairs '
- prepositions '
- active/passive voice '
- direct/indirect speech '
- gerund/infinitive patterns '
- adjective/adverb traps '
- stative verbs '
- fragments/run-ons '
- redundancy '
- common idioms/fixed expressions '
- final error-spotting workflow '
- L0–L3 practice '

## Progress

---

\[ \boxed{71/76}\text{ numbered chapters completed} \]

## UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude '
- Chapter 72 — Vocabulary & Word Usage +
- Chapter 73 — Sentence Completion, Correction & Error Detection +
- Chapter 74 — Para Jumbles & Paragraph Completion +
- Chapter 75 — Reading Comprehension +
- Chapter 76 — Critical Reasoning & Verbal Deduction +

## NEXT

---

### CHAPTER 72 — VOCABULARY & WORD USAGE

Next we will cover:

- contextual vocabulary
- roots, prefixes and suffixes
- synonyms and antonyms
- commonly confused words
- homophones
- one-word substitution
- idioms and phrases
- phrasal verbs
- word forms
- collocations
- tone words
- positive/negative connotation
- context-clue solving
- vocabulary elimination methods
- GATE/placement word-usage traps

- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 72 — VOCABULARY & WORD USAGE

### PART 1 — CONTEXTUAL VOCABULARY, ROOTS/PREFIXES/SUFFIXES, SYNONYMS, ANTONYMS, CONFUSING WORDS & CONTEXT CLUES

Status:  $\emptyset$  = Not in PROGRESS GATE: High Placements: Very High Priority:  $\emptyset$  = Not in Part 1

Vocabulary in aptitude is **not only about memorizing meanings**.

Most questions actually test:

- meaning from context
- correct word usage
- synonym/antonym
- confusing word pairs
- tone
- word form
- collocation
- idiomatic usage

Core rule:

**Do not memorize isolated words only. Learn meaning + tone + usage + common partner words.**

## 1. CONTEXT IS KING

Consider:

**The manager gave a brief explanation.**

Here “brief” means:

$\boxed{\text{short}}$

But:

**The lawyer prepared a brief.**

Here “brief” means:

$\boxed{\text{legal document}}$

Same word, different context.

---

## 2. CONTEXTUAL MEANING

---

A word may have several dictionary meanings.

In aptitude, choose:

**the meaning that best fits the sentence.**

Example:

**The new policy was radical.**

“Radical” could mean:

- extreme/fundamental
- relating to roots
- mathematical radical

Here likely:

$\boxed{\text{fundamental or extreme change}}$

---

## 3. CONTEXT CLUES

---

Main clue types:

1. definition
2. synonym
3. antonym/contrast
4. example
5. cause-effect
6. tone
7. grammar/word-form

---

## 4. DEFINITION CLUE

---

Sentence:

**A gregarious person, someone who enjoys the company of others, often finds networking easy.**

The sentence defines the word.

$\boxed{\text{gregarious} = \text{sociable}}$

---

## 5. SYNONYM CLUE

---



**The room was immaculate, perfectly clean and spotless.**

“perfectly clean and spotless” gives meaning.

\[ \boxed{\text{immaculate = extremely clean}} \]

## 6. CONTRAST CLUE

Look for:

- but
- although
- however
- unlike
- whereas
- yet

Example:

**Unlike his gregarious brother, Ravi was reserved.**

If reserved = quiet/not social, then gregarious means roughly:

\[ \boxed{\text{sociable}} \]

## 7. EXAMPLE CLUE

**Many aquatic animals, such as fish, dolphins, and whales, live in water.**

Examples reveal:

\[ \boxed{\text{aquatic = relating\ to\ water}} \]

## 8. CAUSE-EFFECT CLUE

**The roads were treacherous, so drivers were advised to slow down.**

Effect: drivers must slow.

Likely meaning:

\[ \boxed{\text{dangerous}} \]

## 9. TONE CLUE

**The proposal was absurd and completely detached from reality.**

Tone is negative.

So positive meanings can be eliminated immediately.

---

## 10. GRAMMAR CLUE

Sentence:

**Her \_\_\_\_ response impressed the committee.**

Blank comes before noun “response”.

Likely needs adjective.

If options:

- clarity
- clearly
- clear
- clarify

Correct:

`\[ \boxed{clear} \]`

Grammar can solve vocabulary questions.

---

## 11. WORD FAMILIES

Learn a word as a family.

Example:

**decide — verb**

**We decided.**

**decision — noun**

**The decision was final.**

**decisive — adjective**

**She took decisive action.**

**decisively — adverb**

**She acted decisively.**

This helps sentence completion.

---

## 12. ROOTS

---

A root carries core meaning.

Learning common roots helps decode unfamiliar words.

Example:

\[ bio = life \]

- biology
  - biography
  - biodegradable
- 

## 13. COMMON ROOT — GEO

---

\[ geo = earth \]

Examples:

- geography
  - geology
  - geothermal
- 

## 14. CHRON

---

\[ chron = time \]

Examples:

- chronology
- chronic
- synchronize

Memory:

**chronology = arrangement according to time.**

---

## 15. AUD

---

\[ aud = hear \]

Examples:

- audio
- audience
- audible

\[ \boxed{\text{audible = able to be heard}} \]

---

## 16. VIS / VID

---

\[ vis/vid = see \]

---

Examples:

- visual
  - visible
  - video
  - vision
- 

## 17. SCRIB / SCRIPT

---

\[ scrib/script = write \]

Examples:

- describe
  - manuscript
  - inscription
  - transcript
- 

## 18. DICT

---

\[ dict = say/speak \]

Examples:

- dictate
- predict
- contradict
- dictionary

“predict”:

\[ pre + dict = say\ before \]

---

## 19. PORT

---

\[ port = carry \]

Examples:

- transport
  - portable
  - import
  - export
- 

## 20. SPECT

---

\[ spect = look \]

Examples:

- inspect
  - respect
-

- spectator
  - prospect
- 

## 21. PED

---

\[ ped = foot \]

Examples:

- pedestrian
  - pedal
  - pedicure
- 

## 22. AQUA

---

\[ aqua = water \]

Examples:

- aquarium
  - aquatic
  - aqueduct
- 

## 23. BENE

---

\[ bene = good \]

Examples:

- benefit
- benevolent
- beneficial

\[ \boxed{\text{benevolent = kind/generous}} \]

---

## 24. MAL

---

\[ mal = bad \]

Examples:

- malfunction
  - malpractice
  - malnutrition
- 

## 25. CRED

---

\[ cred = believe \]

Examples:

- credible
-

- incredible
- credibility
- credentials

\[ \boxed{\text{credible} = \text{believable}} \]

---

## 26. PREFIXES

---

A prefix appears before a root and changes meaning.

Example:

\[ \text{un} + \text{happy} = \text{unhappy} \]

---

## 27. NEGATIVE PREFIXES

---

Common:

- un-
- in-
- im-
- il-
- ir-
- non-
- dis-

Examples:

- unhappy
  - incorrect
  - impossible
  - illegal
  - irregular
  - nonverbal
  - dishonest
- 

## 28. PRE-

---

\[ \text{pre} = \text{before} \]

Examples:

- preview
  - predict
  - prehistoric
  - prerequisite
- 

## 29. POST-

---

\[ post = after \]

Examples:

- postgraduate
  - postwar
  - postscript
- 

## 30. RE-

---

Usually:

\[ re = again/back \]

Examples:

- rewrite
  - reconsider
  - return
  - rebuild
- 

## 31. INTER-

---

\[ inter = between/among \]

Examples:

- international
  - interact
  - interdisciplinary
- 

## 32. INTRA-

---

\[ intra = within \]

Example:

- intranet = network within an organization

Compare:

\[ inter = between \]

\[ intra = within \]

---

## 33. SUB-

---

\[ sub = under/below \]

Examples:

- submarine
  - subordinate
  - subconscious
-

---

## 34. SUPER-

\[ super = above/excess \]

Examples:

- supernatural
  - superstructure
  - superhuman
- 

## 35. ANTI-

\[ anti = against \]

Examples:

- antibiotic
  - antisocial
  - anticorruption
- 

## 36. AUTO-

\[ auto = self \]

Examples:

- autobiography
  - automatic
  - autonomous
- 

## 37. BI-

\[ bi = two \]

Examples:

- bicycle
  - bilingual
  - bilateral
- 

## 38. MULTI-

\[ multi = many \]

Examples:

- multicultural
  - multinational
  - multitasking
-



## 39. SUFFIXES

---

Suffix often indicates word class.

---

## 40. NOUN SUFFIXES

---

Common:

- -tion
- -ment
- -ness
- -ity
- -er/-or

Examples:

- creation
  - development
  - happiness
  - ability
  - teacher
- 

## 41. ADJECTIVE SUFFIXES

---

Common:

- -ful
- -less
- -ous
- -ive
- -able
- -al

Examples:

- useful
  - careless
  - dangerous
  - active
  - readable
  - logical
- 

## 42. ADVERB SUFFIX

---

Usually:

\[ -ly \]

Examples:

- carefully
-

- quickly
- logically

But not every -ly word is adverb.

Examples:

- friendly
- lovely
- lonely

These are usually adjectives.

## 43. VERB SUFFIXES

Common:

- -ize
- -ify
- -en

Examples:

- modernize
- simplify
- strengthen

## 44. ROOT-BASED GUESSING METHOD

Unknown word:

**biodegradable**

Break:

\[ bio + degradable ]

bio = life

degradable = able to break down

Likely:

**capable of being broken down by living organisms/natural biological processes.**

This is far better than random guessing.

## 45. SYNONYMS

Synonyms have similar meanings.

Example:

\[ abundant \approx plentiful ]

But synonyms are rarely perfectly interchangeable in every context.

---

## 46. COMMON SYNONYM SETS

---

### Accurate

- precise
- exact
- correct

### Brief

- short
- concise

### Calm

- composed
- tranquil

### Difficult

- hard
- challenging
- arduous

### Essential

- necessary
  - vital
  - indispensable
- 

## 47. “CONCISE”

---

$\boxed{\text{concise} = \text{brief\ but\ complete}}$

Not simply “very small.”

Example:

**Give a concise explanation.**

---

## 48. “ARBITRARY”

---

$\boxed{\text{arbitrary} = \text{based\ on\ random/personal\ choice,\ not\ clear\ reason}}$

Example:

**The cutoff appeared arbitrary.**

---

## 49. “AMBIGUOUS”

---

\[ \boxed{\text{ambiguous} = \text{having\ more\ than\ one\ possible\ meaning}} \]

Compare:

**vague**

not precise

**ambiguous**

multiple interpretations possible

---

## 50. “PRAGMATIC”

---

\[ \boxed{\text{pragmatic} = \text{practical}} \]

Example:

**We need a pragmatic solution.**

---

## 51. “PLAUSIBLE”

---

\[ \boxed{\text{plausible} = \text{apparently\ reasonable/believable}} \]

Important:

Plausible does **not** mean proven true.

---

## 52. “ROBUST”

---

Common meanings:

- strong
- resilient
- reliable under varied conditions

Example:

**We need a robust system.**

---

## 53. “REDUNDANT”

---

\[ \boxed{\text{redundant} = \text{unnecessary/repetitive}} \]

In employment contexts, it can also mean a role no longer needed.

Context decides.

---

## 54. “MITIGATE”

\[ \boxed{\text{mitigate} = \text{reduce}\ \text{severity}} \]

Example:

**Measures were taken to mitigate risk.**

Not necessarily eliminate risk.

## 55. “EXACERBATE”

\[ \boxed{\text{exacerbate} = \text{make}\ \text{worse}} \]

Antonym roughly:

\[ mitigate \]

## 56. “ALLEVIATE”

\[ \boxed{\text{alleviate} = \text{reduce}\ \text{pain/problem}} \]

Similar:

- ease
- relieve

## 57. “DETER”

\[ \boxed{\text{deter} = \text{discourage/prevent}\ \text{by}\ \text{fear}\ \text{or}\ \text{difficulty}} \]

Example:

**Higher penalties may deter violations.**

## 58. “INFER”

\[ \boxed{\text{infer} = \text{derive}\ \text{a}\ \text{conclusion}} \]

The reader infers.

## 59. “IMPLY”

\[ \boxed{\text{imply} = \text{suggest}\ \text{indirectly}} \]

The speaker/writer implies.

Memory:

**Speaker implies; listener infers.**

---

## 60. ANTONYMS

Antonyms have opposite meanings.

Examples:

\[ scarce \leftrightharpoons abundant \]

\[ expand \leftrightharpoons contract \]

\[ optimistic \leftrightharpoons pessimistic \]

---

## 61. COMMON ANTONYM PAIRS

Word	Opposite
abundant	scarce
ancient	modern
artificial	natural
concise	verbose
expand	contract
flexible	rigid
explicit	implicit
temporary	permanent
optimistic	pessimistic
transparent	opaque

---

## 62. EXPLICIT VS IMPLICIT

### explicit

directly stated

### implicit

suggested but not directly stated

Very important in reading comprehension and assumptions.

---

## 63. OBJECTIVE VS SUBJECTIVE

### objective

based on facts/independent criteria

---

## **subjective**

based on personal feelings/opinions

---

## **64. OPTIMISTIC VS PESSIMISTIC**

---

### **optimistic**

expects favorable outcome

### **pessimistic**

expects unfavorable outcome

---

## **65. CONFUSING WORDS — AFFECT / EFFECT**

---

### **affect**

usually verb:

**Noise affects concentration.**

### **effect**

usually noun:

**Noise has an effect on concentration.**

Memory:

\[ Affect = Action \]

\[ Effect = End\ result \]

---

## **66. EFFECT AS VERB**

---

Advanced:

**The new manager effected several changes.**

Here:

\[ effect = cause/bring\ about \]

Less common, but possible.

---

## **67. ACCEPT / EXCEPT**

---

### **accept**

receive/agree

---

I accept the offer.

**except**

excluding

Everyone except Ravi attended.

## 68. ADVICE / ADVISE

**advice**

noun

She gave me advice.

**advise**

verb

She advised me to wait.

## 69. PRINCIPAL / PRINCIPLE

**principal**

main / head of institution

principal reason

school principal

**principle**

rule/belief

moral principle

Memory:

principal = main person/main thing

## 70. COMPLEMENT / COMPLIMENT



## **complement**

complete/go well with

**This sauce complements the dish.**

## **compliment**

praise

**She complimented my work.**

## **71. LOOSE / LOSE**

### **loose**

not tight

### **lose**

fail to keep / misplace

**loose shirt**

**lose a match**

## **72. LATER / LATTER**

### **later**

after some time

### **latter**

second of two

Example:

**Between tea and coffee, I prefer the latter.**

## **73. FORMER / LATTER**

For two mentioned items:

- former = first
- latter = second

## **74. THEN / THAN**

**then**

time/sequence

**Finish this, then leave.**

**than**

comparison

**taller than Ravi.**

---

## 75. QUIET / QUITE

---

**quiet**

silent

**quite**

fairly/completely depending context

**quite difficult**

---

## 76. PERSONAL / PERSONNEL

---

**personal**

private/individual

**personnel**

employees/staff

---

## 77. STATIONARY/ STATIONERY

---

**stationary**

not moving

**stationery**

writing materials

Memory:

**stationery has e !' envelope.**

---

## 78. SITE / SIGHT / CITE

---

**site**

place/location

**sight**

vision/view

**cite**

quote/reference

---

**79. ELIGIBLE / ILLEGIBLE**

---

**eligible**

qualified

**illegible**

cannot be read

Very different.

---

**80. EMINENT / IMMINENT**

---

**eminent**

famous/respected

**eminent scientist**

**imminent**

about to happen

**imminent danger**

---

**81. ECONOMIC / ECONOMICAL**

---

**economic**

related to economy

**economic growth**

**economical**

cost-saving/not wasteful

**economical car**

---

---

## 82. CONTINUAL / CONTINUOUS

---

### **continual**

repeated again and again, possibly with breaks

### **continuous**

without interruption

---

## 83. DISCREET / DISCRETE

---

### **discreet**

careful/private/tactful

### **discrete**

separate/distinct

Important for engineering students:

**discrete mathematics.**

---

## 84. LIE / LAY

---

### **lie**

recline

Present:

**lie**

Past:

**lay**

Past participle:

**lain**

### **lay**

put/place something

Present:

**lay**

Past:

---

**laid**

Past participle:

**laid**

## 85. RAISE / RISE

**raise**

transitive; requires object

**Raise your hand.**

**rise**

intransitive

**Prices rise.**

Past:

- raise !' raised
- rise !' rose !' risen

## 86. BORROW / LEND

**borrow**

take temporarily

**I borrowed money from Ravi.**

**lend**

give temporarily

**Ravi lent money to me.**

## 87. BRING / TAKE

**bring**

movement toward speaker/reference point

**take**

movement away

## 88. ASSURE / ENSURE / INSURE

---

### **assure**

reassure a person

**I assured him.**

### **ensure**

make certain

**Ensure accuracy.**

### **insure**

provide insurance

**Insure the car.**

---

## 89. ADAPT / ADOPT

---

### **adapt**

modify/adjust

**Adapt the plan.**

### **adopt**

accept/take up

**Adopt a new policy.**

---

## 90. ALLUDE / ELUDE

---

### **allude**

refer indirectly

### **elude**

escape/avoid

**The answer eluded me.**

---

## 91. COARSE / COURSE

---

### **coarse**

rough

### **course**

path/class/program

---

## 92. CONSCIENCE / CONSCIOUS

---

### **conscience**

sense of right/wrong

### **conscious**

awake/aware

---

## 93. DEVICE / DEVISE

---

### **device**

noun

a useful device

### **devise**

verb

devise a strategy

---

## 94. FARTHER / FURTHER

---

Traditional distinction:

### **farther**

physical distance

### **further**

additional/metaphorical

farther down the road

further discussion

---

Modern usage overlaps, but this distinction is exam-safe.

---

## 95. FEWER / LESS

---

Reinforcement:

### **fewer**

countable

**fewer errors**

### **less**

uncountable

**less time**

## 96. BETWEEN / AMONG

---

### **between**

two or distinct individually identified entities

### **among**

group

Already seen in grammar, but also word usage.

---

## 97. BESIDE / BESIDES

---

### **beside**

next to

### **besides**

in addition to

---

## 98. SENSIBLE / SENSITIVE

---

### **sensible**

reasonable

### **sensitive**

easily affected / responsive

**sensible decision**



sensitive information

## 99. ESPECIALLY/ SPECIALLY

**especially**

particularly

I especially liked the ending.

**specially**

for a special purpose

This room was specially designed for children.

## 100. CLASSIC “INFER / IMPLY” QUESTION

Sentence:

His silence seemed to \_\_\_\_ that he disagreed.

Speaker/behavior suggests indirectly.

Correct:

$\boxed{\text{imply}}$

Sentence:

From his silence, we can \_\_\_\_ that he disagreed.

Reader concludes.

Correct:

$\boxed{\text{infer}}$

## 101. CONTEXT-CLUE ELIMINATION METHOD

Suppose:

Despite repeated failures, she remained resolute.

Options:

A. confused B. determined C. careless D. frightened

“Despite failures, remained...” suggests persistence.

Therefore:

$\boxed{B}$

Even if you did not know “resolute.”

---

## 102. PREFIX ELIMINATION

Unknown:

**irreversible**

Break:

$\boxed{\text{ir} + \text{reversible}}$

“ir” negative.

So:

$\boxed{\text{cannot be reversed}}$

---

## 103. TONE ELIMINATION

Sentence:

**His remarks were unnecessarily derisive, mocking the efforts of the team.**

Even if “derisive” unknown, “mocking” identifies negative meaning:

$\boxed{\text{scornful/mockingly disrespectful}}$

---

## 104. GRAMMAR + VOCABULARY METHOD

Sentence:

**The committee reached a \_\_\_\_ decision.**

Options:

- unanimity
- unanimous
- unanimously
- unify

Need adjective before “decision”:

$\boxed{\text{unanimous}}$

---

## 105. SYNONYM PRECISION TRAP

Question:

### closest synonym of “concise”

Options:

- incomplete
- brief
- tiny
- silent

Best:

\[ \boxed{\text{brief}} \]

“Tiny” refers size, not expression length.

## 106. ANTONYM PRECISION TRAP

Word:

**rigid**

Possible antonyms:

- soft
- flexible
- weak
- loose

Best in rules/system/personality context:

\[ \boxed{\text{flexible}} \]

Choose semantic opposite, not any vaguely different word.

## 107. CONNOTATION

Words can share denotation but differ emotionally.

Example:

**slim**

often positive

**skinny**

often negative

Both describe thinness.

## 108. POSITIVE CONNOTATION

Examples:

- confident
  - economical
  - determined
  - youthful
- 

## 109. NEGATIVE CONNOTATION

---

Examples:

- arrogant
- cheap
- stubborn
- childish

Compare:

### **determined**

positive

### **stubborn**

often negative

Both imply persistence.

---

## 110. NEUTRAL VS LOADED WORD

---

“Government spending increased.”

Neutral.

“Government squandered money.”

“Squandered” includes negative judgment.

This matters for passage tone.

---

## 111. COMMON TONE WORDS

---

### **Positive**

- approving
- optimistic
- appreciative
- enthusiastic

### **Negative**

- critical
  - skeptical
  - pessimistic
  - sarcastic
-

## Neutral

- objective
  - analytical
  - descriptive
  - informative
- 

## 112. SKEPTICAL

\[ \boxed{\text{skeptical = doubtful/not easily convinced}} \]

Not necessarily completely opposed.

---

## 113. CYNICAL

\[ \boxed{\text{cynical = distrustful of people's motives}} \]

Stronger/different from skeptical.

---

## 114. SARCASTIC

Using irony/mock praise to criticize.

Example:

**“Wonderful job,” he said after the system crashed for the third time.**

Potential sarcasm.

---

## 115. OBJECTIVE

Fact-focused, relatively unemotional.

Typical research/report passage tone.

---

## 116. VOCABULARY SOLVING ORDER

For an unknown word:

1. read entire sentence
  2. identify positive/negative/neutral tone
  3. detect contrast/cause/example clue
  4. identify needed word class
  5. break into prefix/root/suffix
  6. eliminate incompatible options
  7. substitute remaining option into sentence
-

## 117. DO NOT CHOOSE A WORD JUST BECAUSE YOU KNOW IT

---

Common trap:

One option is familiar, so student selects it.

Instead:

**Test exact fit in context.**

A less familiar option may be correct.

---

## 118. PRACTICE — LEVEL 0

---

### Q1

“Abundant” is closest to:

A. scarce B. plentiful C. hidden D. uncertain

### Q2

“Explicit” is opposite of:

A. implicit B. accurate C. clear D. detailed

### Q3

Choose:

**The noise had a negative \_\_\_\_ on concentration.**

A. affect B. effect

### Q4

Choose:

**The teacher gave me useful \_\_\_\_.**

A. advice B. advise

### Q5

Choose:

**Please \_\_\_\_ the offer.**

A. accept B. except

---

## LEVEL 1

---

### Q6

Choose:

---

The new rule may \_\_\_\_ employee morale.

A. affect B. effect

**Q7**

Choose:

Her silence seemed to \_\_\_\_ disagreement.

A. infer B. imply

**Q8**

Choose:

From these results, we can \_\_\_\_ that the method is ineffective.

A. imply B. infer

**Q9**

Choose:

He is an \_\_\_\_ scientist respected worldwide.

A. imminent B. eminent

**Q10**

Choose:

The storm posed an \_\_\_\_ threat.

A. eminent B. imminent

---

## LEVEL 2

---

**Q11**

Sentence:

The manager adopted a pragmatic approach, choosing the solution that was easiest to implement.

Meaning of pragmatic?

A. theoretical B. practical C. emotional D. traditional

**Q12**

Sentence:

The evidence was ambiguous, allowing two different interpretations.

Meaning?

A. clear B. irrelevant C. open to multiple interpretations D. false

### Q13

Choose:

**The company introduced measures to \_\_\_\_ the financial loss.**

A. exacerbate B. mitigate

### Q14

Choose:

**Repeated delays only \_\_\_\_ the problem.**

A. alleviated B. exacerbated

### Q15

Choose:

**The instructions were brief but complete; they were highly \_\_\_\_.**

A. verbose B. concise C. ambiguous D. arbitrary

## LEVEL 3 — EXAM PRESSURE

### Q16

Sentence:

**Although the initial proposal appeared plausible, closer examination revealed several unsupported assumptions.**

What does “plausible” most nearly mean?

A. definitely true B. apparently reasonable C. completely false D. highly technical

### Q17

Sentence:

**The committee rejected the proposal because the criteria seemed arbitrary.**

Best meaning?

A. based on clear principles B. selected without clear reason C. extremely expensive D. legally mandatory

### Q18

Choose the correct word:

**The new software was designed to \_\_\_\_ to changing user requirements.**



A. adopt B. adapt

### Q19

Choose:

The company decided to \_\_\_\_ a new security policy.

A. adapt B. adopt

### Q20

Sentence:

Her review was critical but not derisive; she identified weaknesses without mocking the authors.

Best meaning of derisive:

A. respectful B. mocking C. detailed D. neutral

## ANSWERS

### Q1

\[ \boxed{B:\ plentiful} \]

### Q2

\[ \boxed{A:\ implicit} \]

### Q3

Need noun:

\[ \boxed{B:\ effect} \]

### Q4

Need noun:

\[ \boxed{A:\ advice} \]

### Q5

Receive/agree:

\[ \boxed{A:\ accept} \]

## Q6

Need verb meaning influence:

\[ \boxed{A:\ affect} \]

---

## Q7

Silence suggests indirectly.

\[ \boxed{B:\ imply} \]

---

## Q8

We derive conclusion.

\[ \boxed{B:\ infer} \]

---

## Q9

Respected/famous:

\[ \boxed{B:\ eminent} \]

---

## Q10

About to happen:

\[ \boxed{B:\ imminent} \]

---

## Q11

\[ \boxed{B:\ practical} \]

---

## Q12

\[ \boxed{C} \]

---

## Q13

Reduce severity:

\[ \boxed{B:\ mitigate} \]

---

## Q14

Made worse:

\[ \boxed{B:\ exacerbated} \]

---

---

## Q15

Brief but complete:

\[ \boxed{B:\ concise} \]

---

## Q16

Plausible means:

\[ \boxed{B:\ apparently\ reasonable} \]

Not “certainly true.”

---

## Q17

Arbitrary:

\[ \boxed{B} \]

---

## Q18

Modify/adjust:

\[ \boxed{B:\ adapt} \]

---

## Q19

Take up/implement:

\[ \boxed{B:\ adopt} \]

---

## Q20

Derisive:

\[ \boxed{B:\ mocking} \]

---

## MUST-MEMORIZE CONFUSING-WORD TABLE

---

Word 1	Meaning	Word 2	Meaning
affect	influence	effect	result
accept	receive	except	excluding
advice	noun	advise	verb
adapt	adjust	adopt	take up

Word 1	Meaning	Word 2	Meaning
assure	reassure person	ensure	make certain
complement	complete/match	compliment	praise
principal	main/head	principle	rule
imply	suggest	infer	conclude
stationary	not moving	stationery	writing material
eminent	respected	imminent	about to happen
discrete	separate	discreet	tactful/private
economic	economy-related	economical	cost-saving
farther	physical distance	further	additional
fewer	countable	less	uncountable

## CHAPTER 72 STATUS

Covered so far:

- contextual vocabulary '
- context clues '
- word families '
- roots '
- prefixes '
- suffixes '
- synonym precision '
- antonyms '
- common high-value words '
- confusing words '
- connotation '
- tone vocabulary '
- word-class elimination '
- root-based guessing '
- L0–L3 practice '

But **Chapter 72 is not complete yet.**

Still remaining:

- collocations
- idioms and phrases
- phrasal verbs
- one-word substitutions
- homophones/homonyms

- formal vs informal usage
- commonly misused words
- analogy through vocabulary
- advanced tone words
- context-based double blanks
- vocabulary retention system
- final placement/GATE speed toolkit

## Progress

[ \boxed{71/76}\text{ numbered chapters completed}} ]

**Current:** 0=0 Chapter 72 — Vocabulary & Word Usage

## NEXT

0=0 Chapter 72 Part 2 — Collocations, Idioms, Phrasal Verbs, One-Word Substitution, Homophones, Advanced Tone & Final Vocabulary Toolkit

USER

next

ASSISTANT

## CHAPTER 72 — VOCABULARY & WORD USAGE

### PART 2 — COLLOCATIONS, IDIOMS, PHRASAL VERBS, ONE-WORD SUBSTITUTION, HOMOPHONES, ADVANCED TONE & FINAL VOCABULARY TOOLKIT

**Status:** 0=0 IN PROGRESS ! ' ' COMPLETE **GATE:** High **Placements:** Very High **Priority:** 0=0 Part 1

This part is important because many vocabulary questions are not about the dictionary meaning of a single word.

They test:

- which words naturally go together
- what an idiom means
- what a phrasal verb means
- whether a word is formal/informal
- whether two similar-sounding words are being confused
- whether the tone of a passage is correctly identified

Core rule:

**A word may be grammatically possible but still sound wrong because the collocation is unnatural.**

## 1. WHAT IS A COLLOCATION?

---

A collocation is a natural combination of words commonly used together.

Examples:

- **make a decision**
- **take a risk**
- **heavy rain**
- **strong argument**
- **high probability**

These are more natural than alternatives like:

**do a decision 'L**

## 2. MAKE VS DO

---

### MAKE

usually creates/produces something.

Common:

- make a decision
- make a mistake
- make progress
- make an effort
- make a suggestion
- make money

### DO

usually activity/work/task.

Common:

- do homework
- do research
- do business
- do a job
- do exercise

## 3. TAKE COLLOCATIONS

---

Common:

- take responsibility
- take action
- take a break
- take a risk

- take a decision (*common in Indian/British English; “make a decision” is globally safer*)
  - take notes
  - take advantage of
- 

## 4. HAVE COLLOCATIONS

---

- have difficulty
  - have experience
  - have access
  - have an opportunity
  - have an effect
  - have a discussion
  - have confidence
- 

## 5. STRONG VS POWERFUL

---

Natural:

**strong argument**

**strong evidence**

**strong possibility**

**powerful computer**

**powerful engine**

**powerful speech**

A “powerful argument” is also possible, but nuance differs.

Collocations are about preferred combinations, not always strict grammar.

---

## 6. HEAVY COLLOCATIONS

---

- heavy rain
- heavy traffic
- heavy workload
- heavy losses

Not normally:

---

## 7. HIGH COLLOCATIONS

---

- high probability
  - high risk
  - high salary
  - high temperature
  - high demand
- 

## 8. DEEP COLLOCATIONS

---

- deep concern
  - deep understanding
  - deep sleep
  - deep regret
- 

## 9. SERIOUS COLLOCATIONS

---

- serious problem
  - serious injury
  - serious concern
  - serious consequences
- 

## 10. COMMON VERB + NOUN COLLOCATIONS

---

MUST MEMORIZE:

- raise a question
- raise awareness
- raise funds
- pose a threat
- pose a question
- draw a conclusion
- reach a conclusion
- meet a requirement
- meet a deadline
- meet expectations
- achieve a goal
- gain experience
- gain access
- conduct research



- conduct an experiment
- pay attention

---

## 11. REACH VS ACHIEVE

---

### reach

often destination/decision/conclusion

**reach a conclusion**

**reach an agreement**

### achieve

goal/result

**achieve success**

**achieve a target**

---

## 12. PAY ATTENTION

---

Correct:

**Pay attention to the question.**

Not:

**give attention on 'L**

---

## 13. COMMIT A CRIME

---

Natural:

**commit a crime**

**commit an error (*formal*)**

**commit suicide (*traditional phrase, though modern public-health language may use “die by suicide” in sensitive contexts*)**

## 14. BREAK A RECORD

**break a record**

Not:

**destroy a record**

unless literally destroying it.

## 15. KEEP A PROMISE

**keep a promise**

Opposite:

**break a promise**

## 16. DRAW A CONCLUSION

**We can draw a conclusion from the evidence.**

Also:

**reach a conclusion.**

## 17. IDIOMS

An idiom's meaning is not always literal.

Example:

**“break the ice”**

does not literally mean breaking frozen water.

It means:

`\[ \boxed{\text{make people feel more comfortable/start interaction}} \]`

## 18. A BLESSING IN DISGUISE

Meaning:

Something that seems bad initially but later proves beneficial.

Example:

Missing that bus was a blessing in disguise because the road was later blocked.

## 19. ONCE IN A BLUE MOON

Meaning:

\[ \boxed{\text{very rarely}} \]

## 20. HIT THE NAIL ON THE HEAD

Meaning:

\[ \boxed{\text{say exactly the right thing}} \]

## 21. BITE THE BULLET

Meaning:

\[ \boxed{\text{face a difficult situation courageously}} \]

## 22. UNDER THE WEATHER

Meaning:

\[ \boxed{\text{feeling slightly ill}} \]

## 23. BURN THE MIDNIGHT OIL

Meaning:

\[ \boxed{\text{work/study late at night}} \]

## 24. COST AN ARM AND A LEG

Meaning:

\[ \boxed{\text{very expensive}} \]

## 25. THE BALL IS IN YOUR COURT

Meaning:

**It is now your responsibility/turn to act.**

## 26. ON THE SAME PAGE

Meaning:

`\[ \boxed{\text{share the same understanding}} \]`

## 27. A PIECE OF CAKE

Meaning:

`\[ \boxed{\text{very easy}} \]`

## 28. SPILL THE BEANS

Meaning:

`\[ \boxed{\text{reveal secret information}} \]`

## 29. BEAT AROUND THE BUSH

Meaning:

**Avoid saying the main point directly.**

## 30. CALL IT A DAY

Meaning:

**Stop working for now.**

## 31. GO THE EXTRA MILE

Meaning:

**Make more effort than normally required.**

## 32. IN HOT WATER

Meaning:

`\[ \boxed{\text{in trouble}} \]`

---

### 33. MISS THE BOAT

---

Meaning:

**Miss an opportunity.**

---

### 34. PHRASE VS IDIOM

---

Not every fixed phrase is an idiom.

Example:

**in addition to**

is a fixed phrase, but its meaning is relatively direct.

Idioms often have non-literal meaning.

---

### 35. PHRASAL VERBS

---

Phrasal verb:

\[ verb + particle \]

Examples:

- give up
- look into
- carry out
- put off

Meaning may differ significantly from base verb.

---

### 36. GIVE UP

---

\[ \boxed{\text{stop trying / surrender}} \]

**He refused to give up.**

---

### 37. LOOK INTO

---

\[ \boxed{\text{investigate}} \]

**The committee will look into the issue.**

---

## 38. CARRY OUT

\[ \boxed{\text{perform / execute}} \]

**Researchers carried out an experiment.**

## 39. PUT OFF

\[ \boxed{\text{postpone}} \]

**The meeting was put off until Monday.**

## 40. CALL OFF

\[ \boxed{\text{cancel}} \]

**The match was called off.**

## 41. TURN DOWN

Possible meanings:

- reject

**She turned down the offer.**

- reduce volume/intensity

**Turn down the music.**

Context matters.

## 42. TURN UP

Possible:

- arrive/appear

**He finally turned up.**

- increase volume

**Turn up the sound.**

## 43. TAKE OVER

\[ \boxed{\text{gain control/responsibility}} \]

**She took over the project.**

## 44. COME ACROSS

\[ \boxed{\text{find by chance / encounter}} \]

**I came across an interesting article.**

Can also mean “seem”:

**He came across as confident.**

## 45. FIGURE OUT

\[ \boxed{\text{understand/solve}} \]

Informal but common placement vocabulary.

## 46. SET UP

Possible:

- establish

**set up a company**

- arrange

**set up a meeting**

## 47. BREAK DOWN

Possible:

- stop functioning

**The car broke down.**

- divide/analyze

**Break the problem down into steps.**

- lose emotional control

**He broke down.**

## 48. BACK UP

Possible:

- make a copy

**Back up your files.**

- support a claim/person

**Evidence backs up the claim.**

## 49. WORK OUT

Possible:

- exercise

**I work out daily.**

- solve

**Work out the answer.**

- develop successfully

**The plan worked out.**

## 50. RUN OUT OF

$\boxed{\text{have no more left}}$

**We ran out of time.**

## 51. ONE-WORD SUBSTITUTION

A phrase can sometimes be replaced by one precise word.

Example:

**one who writes dictionaries**



\[ \boxed{\text{lexicographer}} \]

This is more common in placement/verbal tests than GATE, but useful.

---

## 52. ONE WHO LOVES BOOKS

\[ \boxed{\text{bibliophile}} \]

---

## 53. ONE WHO HATES MANKIND

\[ \boxed{\text{misanthrope}} \]

---

## 54. ONE WHO LOVES MANKIND

\[ \boxed{\text{philanthropist}} \]

More precisely, one who promotes others' welfare, often through charitable action.

---

## 55. ONE WHO SPEAKS MANY LANGUAGES

\[ \boxed{\text{polyglot}} \]

---

## 56. ONE WHO CANNOT READ OR WRITE

\[ \boxed{\text{illiterate}} \]

---

## 57. A PERSON WHO WALKS

\[ \boxed{\text{pedestrian}} \]

---

## 58. A SPEECH MADE WITHOUT PREPARATION

\[ \boxed{\text{impromptu}} \]

or:

\[ \boxed{\text{extempore}} \]

depending context.

---

## 59. A GOVERNMENT BY THE PEOPLE

\[ \boxed{\text{democracy}} \]

---

## 60. A GOVERNMENT BY ONE PERSON WITH ABSOLUTE POWER

---

Common:

\[ \boxed{autocracy} \]

---

## 61. ONE WHO DOES NOT BELIEVE IN GOD

---

\[ \boxed{atheist} \]

One who believes existence is unknowable/uncertain:

\[ \boxed{agnostic} \]

Do not confuse.

---

## 62. A PLACE WHERE BOOKS ARE KEPT

---

\[ \boxed{library} \]

Simple one-word substitution questions may be this basic.

---

## 63. ONE WHO KNOWS EVERYTHING

---

\[ \boxed{omniscient} \]

---

## 64. PRESENT EVERYWHERE

---

\[ \boxed{omnipresent} \]

---

## 65. ALL-POWERFUL

---

\[ \boxed{omnipotent} \]

Memory:

- scient !' knowledge
  - present !' presence
  - potent !' power
- 

## 66. HOMOPHONES

---

Homophones sound alike but have different spellings/meanings.

Example:

- right
  - write
-

## 67. THERE / THEIR / THEY'RE

---

Reinforcement:

- there = place/existence
  - their = possession
  - they're = they are
- 

## 68. YOUR / YOU'RE

---

- your = possession
  - you're = you are
- 

## 69. ITS / IT'S

---

- its = possession
  - it's = it is / it has
- 

## 70. WEATHER / WHETHER

---

- weather = atmospheric conditions
  - whether = if / choice between alternatives
- 

## 71. PEACE / PIECE

---

- peace = absence of conflict
  - piece = part
- 

## 72. RIGHT / WRITE / RITE

---

- right = correct/direction
  - write = put words on paper/screen
  - rite = ceremonial act
- 

## 73. BRAKE / BREAK

---

- brake = stopping device
  - break = damage/rest/interrupt
- 

## 74. COUNCIL / COUNSEL

---

**council**

group/committee

## **counsel**

advice or legal adviser

---

## **75. CAPITAL / CAPITOL**

---

### **capital**

city/wealth/uppercase letter

### **capitol**

legislative building, especially U.S. usage

Low priority for GATE.

---

## **76. COMMONLY MISUSED WORDS**

---

### **LITERALLY**

---

Literally means:

**in an exact, non-figurative sense.**

Casual usage may exaggerate:

**I literally died laughing.**

Not literally true.

Formal tests may distinguish literal vs figurative.

---

## **77. UNIQUE**

---

Traditionally absolute:

**unique = one of a kind**

So:

**very unique**

is often discouraged in formal exam grammar.

Better:

**unique**

---

## 78. IRREGARDLESS

---

Avoid in formal aptitude writing.

Use:

\[ \boxed{\text{regardless}} \]

---

## 79. DISINTERESTED VS UNINTERESTED

---

Traditional distinction:

### **disinterested**

impartial/unbiased

### **uninterested**

not interested

Example:

**A judge should be disinterested.**

**He was uninterested in the lecture.**

Modern usage sometimes overlaps, but exams may test distinction.

---

## 80. ENORMITY

---

Traditionally:

\[ \boxed{\text{great wickedness/seriousness of wrongdoing}} \]

not simply “enormous size.”

However modern usage has broadened, so context matters.

Low priority.

---

## 81. FORMAL VS INFORMAL VOCABULARY

---

Informal:

- kids
- stuff
- get
- a lot

More formal alternatives:

- children
  - materials/items
  - obtain
-

- many/much

For RC/essay tone, formal context often prefers precise vocabulary.

## 82. FORMAL ALTERNATIVES

Informal	Formal
find out	determine
put off	postpone
look into	investigate
go up	increase
go down	decrease
get rid of	eliminate/remove
bring up	raise/mention

Phrasal verbs are not “wrong”; formality depends on context.

## 83. ADVANCED TONE WORDS

### Analytical

examines components logically.

### Critical

evaluates weaknesses/limitations.

### Skeptical

doubts claims/evidence.

### Cautious

avoids strong certainty.

### Optimistic

expects positive outcome.

### Pessimistic

expects negative outcome.

### Neutral

little visible judgment.

### Appreciative

expresses approval/value.

## 84. CONCILIATORY

\[ \boxed{\text{trying to reduce conflict and restore goodwill}} \]

Example:

**The speaker adopted a conciliatory tone after the dispute.**

## 85. POLEMICAL

\[ \boxed{\text{strongly argumentative/controversial}} \]

Usually used for writing attacking an opposing view.

## 86. DIDACTIC

\[ \boxed{\text{intended to teach}} \]

Can sometimes imply overly instructional.

## 87. SATIRICAL

Uses humor/irony to criticize.

## 88. NOSTALGIC

Reflecting affectionately on the past.

## 89. RESIGNED

Tone of accepting an unpleasant situation as unavoidable.

## 90. AMBIVALENT

\[ \boxed{\text{having mixed/conflicting feelings}} \]

Example:

**She was ambivalent about leaving the company.**

## 91. INDIFFERENT

\[ \boxed{\text{not caring strongly either way}} \]

Do not confuse with ambivalent.

Ambivalent = conflicting feelings.

Indifferent = little/no concern.

---

## 92. DOUBLE-BLANK STRATEGY

---

Example:

**Although the proposal was initially \_\_, further evidence made it appear \_\_.**

Look at contrast:

“Although” suggests blanks may move in opposite directions.

Possible:

**promising !' doubtful**

Use structure before vocabulary.

---

## 93. DOUBLE-BLANK LINK WORDS

---

Look for:

### Same direction

- and
- moreover
- similarly
- therefore

### Opposite direction

- but
- although
- however
- yet
- nevertheless

These reduce options quickly.

---

## 94. DOUBLE-BLANK EXAMPLE

---

**The professor was neither \_\_ nor hostile; instead, she remained calm and \_\_.**

“Neither X nor hostile” suggests first is another negative/emotional trait.

“calm and \_\_\_\_” requires positive/neutral compatible adjective.

Suppose options give:

- aggressive / composed



Likely fit.

---

## 95. COLLOCATION ELIMINATION IN BLANKS

---

Sentence:

**The committee \_\_\_\_ a conclusion after reviewing the evidence.**

Options:

- did
- reached
- performed
- produced

Correct collocation:

`\[ \boxed{reached} a\ conclusion\ ]`

---

## 96. REGISTER ELIMINATION

---

Sentence in formal report:

**The experiment \_\_\_\_ unexpected results.**

Better:

`\[ \boxed{produced} \]`

rather than very informal alternatives like “came up with” if tone is formal.

---

## 97. VOCABULARY ANALOGY

---

Question:

**Doctor : Hospital :: Teacher : ?**

Relation:

profession : typical workplace

Answer:

`\[ \boxed{School} \]`

Always identify relationship, not just association.

---

## 98. WORD-RELATION ANALOGIES

---

Common relations:

- synonym
-

- antonym
- tool-user
- part-whole
- cause-effect
- object-function
- profession-workplace
- degree/intensity

## 99. DEGREE RELATION

Example:

**warm : hot :: cool : ?**

Relation:

increase in intensity.

Answer:

$\boxed{\text{cold}}$

depending options/context.

## 100. OBJECT-FUNCTION

**Pen : Write :: Knife : ?**

$\boxed{\text{Cut}}$

## 101. PART-WHOLE

**Page : Book :: Wheel : ?**

$\boxed{\text{Car}}$

## 102. VOCABULARY RETENTION SYSTEM

Do not memorize 500 words once.

Use a 4-column note:

Word	Meaning	Example	Pair/Trap
mitigate	reduce severity	mitigate risk	opposite: exacerbate
imply	suggest	silence implies...	reader infers

Word	Meaning	Example	Pair/Trap
arbitrary	without clear reason	arbitrary rule	not "random" only

This creates stronger memory.

## 103. 1–3–7 REVISION RULE

For difficult vocabulary:

- revise after 1 day
- revise after 3 days
- revise after 7 days

Then use it in your own sentence.

This is more effective than rereading a long list.

## 104. LEARN IN PAIRS

Examples:

\[ mitigate \(\rightarrow\) exacerbate \]

\[ explicit \(\rightarrow\) implicit \]

\[ concise \(\rightarrow\) verbose \]

\[ scarce \(\rightarrow\) abundant \]

Antonym pairs strengthen recall.

## 105. LEARN BY ROOT FAMILY

Example:

\[ cred \]

- credible
- credibility
- incredible
- credulous

Root families reduce memorization load.

## 106. VOCABULARY SPEED TOOLKIT

When unknown word appears:

### Step 1

Determine grammar: noun/adjective/verb/adverb?

## Step 2

Determine tone: positive/negative/neutral?

## Step 3

Find signal word: but/because/although/etc.

## Step 4

Use root/prefix.

## Step 5

Check collocation.

## Step 6

Substitute final option.

## 107. HIGH-VALUE COLLOCATION TABLE

Natural	Avoid/Usually Wrong
make a decision	do a decision
make progress	do progress
take responsibility	make responsibility
conduct research	make research
pay attention	give attention on
reach a conclusion	do a conclusion
heavy rain	strong rain
strong evidence	heavy evidence
high probability	tall probability
meet a deadline	catch a deadline

## 108. HIGH-VALUE PHRASAL VERB TABLE

Phrasal verb	Meaning
carry out	perform
call off	cancel
put off	postpone
look into	investigate
give up	stop trying

Phrasal verb	Meaning
take over	assume control
run out of	exhaust supply
come across	encounter
figure out	understand
back up	support/copy
set up	establish
break down	fail/divide

## 109. HIGH-VALUE IDIOM TABLE

Idiom	Meaning
break the ice	start comfortable interaction
once in a blue moon	rarely
hit the nail on the head	be exactly right
bite the bullet	face difficulty
under the weather	slightly ill
call it a day	stop working
miss the boat	miss opportunity
in hot water	in trouble
on the same page	same understanding
go the extra mile	make extra effort

## PRACTICE — LEVEL 0

### Q1

Choose the natural phrase:

A. do a decision B. make a decision

### Q2

“He looked into the problem.”

“looked into” means:

A. ignored B. investigated C. created D. postponed

### Q3

"Once in a blue moon" means:

A. daily B. rarely C. immediately D. secretly

### Q4

Choose:

The team \_\_\_\_ research on customer behavior.

A. conducted B. made

### Q5

Choose:

Heavy \_\_\_\_ disrupted traffic.

A. rain B. evidence

---

## LEVEL 1

### Q6

Choose:

The meeting was \_\_\_\_ until Monday.

A. put off B. taken over

### Q7

Choose:

The match was \_\_\_\_ because of rain.

A. called off B. figured out

### Q8

"Bite the bullet" means:

A. avoid trouble B. face difficulty C. eat quickly D. become angry

### Q9

Choose:

The committee \_\_\_\_ a conclusion.

A. reached B. did

### Q10

Choose:

**We have \_\_\_\_ of time.**

A. run out B. broken out

## LEVEL 2

### Q11

A “disinterested judge” traditionally means:

A. bored judge B. impartial judge C. angry judge D. inexperienced judge

### Q12

Choose:

**The company will \_\_\_\_ the complaint.**

A. look into B. look after

### Q13

Choose:

**She turned \_\_\_\_ the job offer.**

A. down B. over

### Q14

“Ambivalent” means:

A. fully opposed B. fully supportive C. having mixed feelings D. uninterested

### Q15

Choose:

**The findings \_\_\_\_ the original claim.**

A. back up B. run out

## LEVEL 3 — EXAM PRESSURE

### Q16

Sentence:

**The report was highly \_\_\_\_, presenting both advantages and limitations without emotional language.**

Best word:

A. polemical B. objective C. sarcastic D. nostalgic

### Q17

Sentence:

The minister adopted a \_\_\_\_ tone, attempting to reduce tensions between the two groups.

A. conciliatory B. cynical C. derisive D. indifferent

### Q18

Sentence:

Although the evidence initially appeared \_\_\_\_, later experiments exposed major flaws.

A. plausible B. irrelevant C. hostile D. permanent

### Q19

Choose the correct collocation:

A. strong evidence B. powerful evidence C. tall evidence D. hard evidence

### Q20

Sentence:

The team decided to \_\_\_\_ the launch until testing was complete.

A. put off B. take over C. bring about D. look after

## ANSWERS

### Q1

\[ \boxed{B:\ \text{make}\ a\ \text{decision}} \]

### Q2

\[ \boxed{B:\ \text{investigated}} \]

### Q3

\[ \boxed{B:\ \text{rarely}} \]

### Q4

\[ \boxed{A:\ \text{conducted}} \]



### Q5

$\boxed{A: \text{rain}}$

### Q6

Put off = postpone.

$\boxed{A}$

### Q7

Call off = cancel.

$\boxed{A}$

### Q8

$\boxed{B}$

### Q9

Natural collocation:

$\boxed{A: \text{reached}}$

### Q10

$\boxed{A: \text{run out}}$

### Q11

Traditional test meaning:

$\boxed{B: \text{impartial}}$

### Q12

Look into = investigate.

$\boxed{A}$

### Q13

Turn down = reject.

$\boxed{A}$

## Q14

\[ \boxed{C} \]

Ambivalent = mixed/conflicting feelings.

## Q15

Back up = support.

\[ \boxed{A} \]

## Q16

Balanced, unemotional:

\[ \boxed{B:\ objective} \]

## Q17

Trying to reduce tensions:

\[ \boxed{A:\ conciliatory} \]

## Q18

Initially apparently believable:

\[ \boxed{A:\ plausible} \]

## Q19

Safest standard collocation:

\[ \boxed{A:\ strong\ evidence} \]

## Q20

Postpone:

\[ \boxed{A:\ put\ off} \]

## ' CHAPTER 72 COMPLETE

We have now covered:

- contextual vocabulary '
- context clues '
- roots/prefixes/suffixes '

- word families '
- synonyms '
- antonyms '
- confusing words '
- collocations '
- idioms '
- phrasal verbs '
- one-word substitutions '
- homophones '
- commonly misused words '
- formal vs informal vocabulary '
- tone words '
- connotation '
- double-blank strategy '
- vocabulary analogies '
- retention strategy '
- final speed toolkit '
- L0–L3 practice '

## Progress

---

[ \boxed{72/76\text{ numbered chapters completed}} ]

### UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude '
- Chapter 72 — Vocabulary & Word Usage '
- Chapter 73 — Sentence Completion, Correction & Error Detection +
- Chapter 74 — Para Jumbles & Paragraph Completion +
- Chapter 75 — Reading Comprehension +
- Chapter 76 — Critical Reasoning & Verbal Deduction +

## NEXT

---

### CHAPTER 73 — SENTENCE COMPLETION, CORRECTION & ERROR DETECTION

Next we will combine Grammar + Vocabulary under exam pressure:

- single-blank completion
- double blanks
- connector clues
- tone consistency
- sentence correction
- error spotting
- no-error questions
- grammar + logic combined errors
- word-choice correction

- redundancy
- idiom/collocation errors
- option elimination
- GATE/placement timed methods
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 73 — SENTENCE COMPLETION, CORRECTION & ERROR DETECTION

### PART 1 — SINGLE/DOUBLE BLANKS, CONNECTOR CLUES, TONE CONSISTENCY, SENTENCE CORRECTION & NO-ERROR STRATEGY

Status: ~~Ø=Not~~ PROGRESS GATE: High Placements: Very High Priority: ~~Ø=Not~~ 1

This chapter combines everything from:

- Grammar
- Vocabulary
- Context clues
- Logic
- Collocations
- Tone

The main rule is:

**Do not solve blanks word-by-word. Solve the entire sentence as one logical structure.**

## 1. THREE MAJOR QUESTION TYPES

### Type 1 — Sentence Completion

Choose the word/phrase that best completes meaning.

### Type 2 — Sentence Correction

Choose the grammatically/semantically correct replacement.

### Type 3 — Error Detection

Find which part contains the error—or choose “No error.”

## 2. SENTENCE COMPLETION — BASIC METHOD

For a blank:

1. read whole sentence
2. identify grammar required
3. identify tone
4. find connector clue
5. predict rough meaning
6. then check options

Do **not** start by comparing options.

## 3. PREDICTION BEFORE OPTIONS

Sentence:

**Despite repeated failures, the scientist remained \_\_\_\_.**

Before seeing options, predict:

$\boxed{\text{determined / persistent}}$

Then match.

This prevents attractive distractors.

## 4. GRAMMATICAL FIT

Sentence:

**The manager gave a \_\_\_\_ explanation.**

Blank before noun “explanation.”

Need adjective.

Options:

- clarity
- clearly
- clear
- clarify

Answer:

$\boxed{\text{clear}}$

## 5. VERB-FORM FIT

**The committee decided to \_\_\_\_ the proposal.**

After “to”:

$\boxed{\text{base verb}}$

Possible answer:

**reconsider**

Not:

**reconsidered**

## 6. GERUND FIT

**She enjoys \_\_\_\_ difficult problems.**

After “enjoy”:

$\boxed{\text{V-ing}}$

Correct:

**solving**

## 7. PREPOSITIONAL FIT

**He is capable \_\_\_\_ solving the problem.**

Fixed expression:

$\boxed{\text{capable of}}$

## 8. CONNECTOR CLUES

Connectors tell relationship between ideas.

Main groups:

### Contrast

- but
- although
- however
- yet
- despite
- nevertheless

### Cause/result

- because
- therefore
- hence
- consequently
- so

### Addition

- and
- moreover
- furthermore
- also

### Example

- for example
- such as

## 9. CONTRAST CLUE

Sentence:

**Although the task appeared simple, it was surprisingly \_\_\_\_.**

Because “although” creates contrast with simple.

Likely:

$\boxed{\text{difficult}}$

## 10. SAME-DIRECTION CLUE

**The explanation was clear and \_\_\_\_.**

“And” suggests similar positive quality.

Possible:

$\boxed{\text{concise}}$

Not:

**confusing**

unless sentence structure changes.

## 11. RESULT CLUE

**The roads were flooded; therefore, traffic was severely \_\_\_\_.**

Likely:

\[ \boxed{disrupted} \]

Cause !' effect.

---

## 12. DOUBLE-BLANK QUESTIONS

---

Do not solve each blank independently.

First determine relationship between blanks.

Example:

**Though the policy initially seemed \_\_, later evidence made it appear \_\_.**

“Though” suggests contrast.

Likely pair:

\[ \boxed{reasonable, \text{flawed}} \]

---

## 13. DOUBLE-BLANK — SAME DIRECTION

---

**The professor was both \_\_ and \_\_, explaining difficult ideas patiently and clearly.**

“both...and” suggests compatible qualities.

Possible:

\[ \boxed{knowledgeable, \text{patient}} \]

---

## 14. DOUBLE-BLANK — CAUSAL RELATION

---

**Because the evidence was \_\_, the committee remained \_\_ about the claim.**

If evidence weak:

committee skeptical.

So:

\[ \boxed{inconclusive, \text{skeptical}} \]

---

## 15. FIRST-BLANK ANCHOR METHOD

---

Sometimes one blank is much easier.

Solve easiest first.

Then use it to restrict second.

This is faster than solving left-to-right mechanically.

---



---

## 16. TONE CONSISTENCY

---

Sentence:

**The review praised the methodology but criticized the unsupported conclusion.**

Overall tone:

\[ \boxed{\text{balanced/critical}} \]

Not fully positive or fully hostile.

This helps choose vocabulary.

---

## 17. POSITIVE-NEGATIVE POLARITY

---

Mark words mentally:

- positive: +
- negative: "
- neutral: 0

Example:

**Despite the severe criticism (" )she remained confident (+).**

"Despite" flips expected direction.

---

## 18. EXTREME-WORD ELIMINATION

---

Sentence:

**The study suggests that exercise may improve concentration.**

Option:

**Exercise always guarantees perfect concentration.**

Too strong.

Reject.

The completion must preserve the original degree of certainty.

---

## 19. CERTAINTY LEVELS

---

From weaker to stronger:

\[ may < can < likely < will < must/always \]

Do not upgrade sentence certainty without evidence.

---

---

## 20. COLLOCATION FIT

---

Sentence:

**The committee \_\_\_\_ a decision after two hours.**

Natural:

`\[ \boxed{made}\ a\ decision\ ]`

Not:

**did a decision**

Collocation can decide the answer even when grammar permits both.

---

## 21. IDIOM FIT

---

**After discussing the issue for hours, they finally \_\_\_\_.**

Options may include:

- reached an agreement
- did an agreement

Correct collocation:

`\[ \boxed{reached}\ an\ agreement\ ]`

---

## 22. WORD-MEANING PRECISION

---

Sentence:

**The explanation was brief but complete.**

Best synonym:

`\[ \boxed{concise}\ ]`

Not merely:

**short**

if answer options test precision.

---

## 23. SENTENCE CORRECTION — CORE PRINCIPLE

---

You are given an underlined part and alternatives.

---

Choose the option that is:

1. grammatically correct
2. logically precise
3. concise
4. idiomatic
5. consistent with intended meaning

---

## 24. DO NOT OVERCORRECT

Original:

**She has lived here for five years.**

Already correct.

If option says “No correction,” choose it.

Do not assume every sentence contains an error.

---

## 25. NO-ERROR QUESTIONS

These are dangerous because students search until they invent an error.

Use a fixed scan:

1. subject–verb
2. tense
3. pronoun
4. modifier
5. article/preposition
6. comparison
7. parallelism
8. idiom/collocation
9. logic

If all pass:

$\boxed{\text{No error}}$

---

## 26. SUBJECT–VERB CORRECTION

Wrong:

**The list of selected candidates are displayed outside.**

Main subject:

$\boxed{\text{list}}$

Correct:

The list of selected candidates is displayed outside.

## 27. “A NUMBER OF” VS “THE NUMBER OF”

Correct:

A number of employees are absent.

Correct:

The number of employees is increasing.

High-frequency error.

## 28. EACH / EVERY

Wrong:

Each of the applicants have submitted the form.

Correct:

$\boxed{\text{has}}$

## 29. ALONG WITH / AS WELL AS

Wrong:

The manager, along with the employees, are present.

Correct:

$\boxed{\text{is}}$

Main subject = manager.

## 30. EITHER...OR

Either the managers or the supervisor \_\_\_\_ responsible.

Nearest subject = supervisor.

$\boxed{\text{is}}$

## 31. TENSE CORRECTION

Wrong:

**She has completed the assignment yesterday.**

“Yesterday” = definite finished past.

Correct:

`\[ \boxed{She\ completed\ the\ assignment\ yesterday.}\]`

## 32. PAST-PERFECTSEQUENCE

**By the time we reached the station, the train \_\_\_\_.**

Correct:

`\[ \boxed{had\ left}\]`

Earlier past action.

## 33. FUTURE TIME CLAUSE

Wrong:

**I will call you when I will reach home.**

Correct:

`\[ \boxed{when\ I\ reach\ home}\]`

## 34. PRONOUN CASE

Wrong:

**Between you and I, this plan will fail.**

Correct:

`\[ \boxed{between\ you\ and\ me}\]`

## 35. AMBIGUOUS PRONOUN

**When Ravi spoke to Arjun, he was nervous.**

Ambiguous.

Correction should clarify:

**Ravi was nervous when he spoke to Arjun.**

or:

**Arjun was nervous when Ravi spoke to him.**

## 36. PRONOUN AGREEMENT

**Every student must submit \_\_\_\_ assignment.**

Traditional formal:

**his or her**

Modern standard:

**their**

In most current usage, singular “they” is acceptable.

But keep consistency.

## 37. ARTICLE ERROR

Wrong:

**He is an university student.**

Correct:

$\boxed{\text{a university student}}$

Sound begins “yoo.”

## 38. VOWEL-SOUND ARTICLE

Correct:

**an honest person**

**an MBA graduate**

**a European**

Article follows sound, not spelling.

---

## 39. PREPOSITION ERROR

---

Wrong:

**She is interested on mathematics.**

Correct:

$\boxed{\text{interested in}}$

---

## 40. FIXED PREPOSITION PAIRS

---

High-value:

- depend on
  - responsible for
  - capable of
  - interested in
  - good at
  - similar to
  - different from
  - superior to
  - prefer X to Y
- 

## 41. COMPARISON ERROR

---

Wrong:

**Ravi's score is higher than Arjun.**

Score compared to person.

Correct:

**Ravi's score is higher than Arjun's.**

---

## 42. COMPARING LIKE WITH LIKE

---

Wrong:

**The climate of City A is better than City B.**

Correct:

**The climate of City A is better than that of City B.**

---

---

## 43. PARALLELISM ERROR

---

Wrong:

**She likes coding, reading, and to swim.**

Correct:

**She likes coding, reading, and swimming.**

---

## 44. CORRELATIVE PARALLELISM

---

Wrong:

**She not only writes code but also testing it.**

Correct:

**She not only writes code but also tests it.**

---

## 45. MODIFIER ERROR

---

Wrong:

**Walking through the market, the shops looked crowded.**

Makes shops seem to be walking.

Correct:

**Walking through the market, we found the shops crowded.**

---

## 46. MISPLACED “ONLY”

---

Compare:

**She only answered three questions.**

Could mean she did nothing except answer.

If intended number limitation:

**She answered only three questions.**

---



---

## 47. REDUNDANCY

---

Wrong:

**return back**

Correct:

`\[ \boxed{return} \]`

Wrong:

**repeat again**

Correct:

`\[ \boxed{repeat} \]`

---

## 48. REDUNDANT PAIRS TO WATCH

---

- revert back
- advance forward
- merge together
- collaborate together
- basic fundamentals
- final outcome/end result
- past history

Aptitude tests often flag them.

---

## 49. “DISCUSS ABOUT”

---

Wrong:

**We discussed about the problem.**

Correct:

`\[ \boxed{We\ discussed\ the\ problem.} \]`

---

## 50. “ORDER FOR”

---

Wrong in ordinary meaning:

**I ordered for a coffee.**

Standard:

---

**I ordered a coffee.**

## 51. GERUND ERROR

Wrong:

**She enjoys to read.**

Correct:

\[ \boxed{enjoys\ reading} \]

## 52. INFINITIVE ERROR

Wrong:

**He decided leaving early.**

Correct:

\[ \boxed{decided\ to\ leave} \]

## 53. “SUGGEST”

Wrong:

**She suggested to postpone the meeting.**

Correct:

**She suggested postponing the meeting.**

or:

**She suggested that we postpone the meeting.**

## 54. “LOOK FORWARD TO”

Correct:

\[ \boxed{look\ forward\ to\ meeting} \]

because “to” is preposition.

## 55. MODAL + BASE FORM

Wrong:

**She can solves the problem.**

Correct:

\[ \boxed{\text{can solve}} \]

## 56. DID + BASE FORM

Wrong:

**Did he went?**

Correct:

\[ \boxed{\text{Did he go?}} \]

## 57. HAS/HAVE + V3

Wrong:

**She has wrote the report.**

Correct:

\[ \boxed{\text{has written}} \]

## 58. ADJECTIVE/ADVERB ERROR

Wrong:

**She performed good.**

Correct:

\[ \boxed{\text{performed well}} \]

## 59. LINKING VERB

Correct:

**She looks beautiful.**

“beautiful” describes subject.

Not necessarily:

**beautifully**

## 60. HARD / HARDLY

**He works hard.**

= with effort.

**He hardly works.**

= almost does not work.

Huge meaning difference.

## 61. FEWER / LESS

**fewer errors**

**less time**

Countable vs uncountable.

## 62. MUCH / MANY

**many students**

**much information**

## 63. FEW / A FEW

**Few students passed.**

"Almost none.

**A few students passed.**

= some.

Sentence completion may test this meaning difference.

## 64. LITTLE / A LITTLE

**Little hope remains.**

"Halmost none.

**A little hope remains.**

= some.

## 65. CONNECTOR CORRECTION — ALTHOUGH

Wrong:

**Although it was raining, but we went out.**

Correct:

**Although it was raining, we went out.**

or:

**It was raining, but we went out.**

## 66. BECAUSE / SO

Wrong:

**Because the server failed, so the service stopped.**

Correct one connector structure only.

## 67. DESPITE

Correct:

**Despite the rain, we continued.**

Not:

**Despite of the rain.**

## 68. BECAUSE OF

Correct:

**Because of heavy rain, the match was cancelled.**

Not:

**Because of it was raining.**

## 69. NO SOONER

Correct pair:

$\boxed{\text{No\ sooner...than}}$

Example:

**No sooner had he arrived than the meeting started.**

## 70. HARDLY / SCARCELY

Correct:

$\boxed{\text{Hardly...when}}$

$\boxed{\text{Scarcely...when}}$

## 71. CONDITION CORRECTION

Wrong:

**If I will see him, I will tell him.**

Correct:

$\boxed{\text{If\ I\ see\ him,\ I\ will\ tell\ him.}}$

## 72. UNREAL CONDITION

**If I were you, I would accept the offer.**

Formal exam preference:

$\boxed{\text{were}}$

## 73. DOUBLE NEGATIVE

Wrong:

**I don't need no help.**

Standard:

**I don't need any help.**

or:

**I need no help.**

---

## 74. FRAGMENT

**Because the meeting was delayed.**

Incomplete.

Needs main clause.

**Because the meeting was delayed, we returned late.**

---

## 75. RUN-ON

Wrong:

**The test was difficult I finished it.**

Correct:

**The test was difficult, but I finished it.**

---

## 76. COMMA SPLICE

Wrong:

**He studied hard, he passed.**

Correct:

**He studied hard, so he passed.**

or use semicolon/full stop.

## 77. LOGICAL ERROR WITHOUT GRAMMATICAL ERROR

Sentence:

**The population of the city increased from 2 million to 1.5 million.**

Grammar is fine.

Logic is wrong because “increased” contradicts numbers.

Verbal tests can combine grammar + meaning.

## 78. QUANTIFIER LOGIC

Wrong:

**Every student have submitted...**

Grammar error.

But also watch:

**Some students means all students.**

Logic error.

Sentence completion options may differ in logical strength.

## 79. “MOST” VS “ALL”

Sentence:

**Most employees supported the proposal.**

You cannot replace with:

**All employees supported it.**

Meaning changes.

Correction must preserve meaning.

## 80. WORD-CHOICE ERROR

**The new rule had a positive affect on productivity.**

Need noun:

$\boxed{\text{effect}}$



---

## 81. ADVICE / ADVISE

The manager gave useful advice.

The manager advised us.

Do not mix noun and verb.

---

## 82. IMPLY / INFER

His statement implies uncertainty.

We infer uncertainty from his statement.

---

## 83. PRINCIPAL / PRINCIPLE

The principal reason

a moral principle

---

## 84. ADAPT / ADOPT

adapt to change

adopt a policy

---

## 85. DOUBLE-BLANK EXAMPLE 1

Although the speaker was usually \_\_, on this occasion his argument was surprisingly \_\_.

Suppose options:

A. concise / verbose B. hostile / hostile C. careful / careful D. confident / certain

“Although” suggests contrast.

Best pair:

$\boxed{\text{concise, \text{verbose}}}$

---

---

## 86. DOUBLE-BLANK EXAMPLE 2

The evidence was so \_ that even previously skeptical researchers became \_.

Likely:

\[ \boxed{\text{compelling}, \text{convinced}} \]

Same causal direction.

---

## 87. DOUBLE-BLANK EXAMPLE 3

Because the instructions were \_, several students interpreted them differently, making the process \_.

Likely:

\[ \boxed{\text{ambiguous}, \text{confusing}} \]

---

## 88. OPTION-PAIR ELIMINATION

For double blanks:

1. eliminate pairs where blank 1 is wrong
2. eliminate pairs where blank 2 is wrong
3. test relationship between both

Do not require both meanings simultaneously from the start.

---

## 89. POSITIVE/NEGATIVE PAIRING

Sentence:

Despite the \_ circumstances, the team remained \_.

Structure:

negative + positive.

Likely:

\[ \boxed{\text{adverse}, \text{optimistic}} \]

---

## 90. SENTENCE CORRECTION — SHORTEST IS NOT ALWAYS BEST

Prefer concision only if meaning and grammar remain correct.

Example:

---

**Due to the fact that it rained...**

More concise:

**Because it rained...**

But do not shorten if it changes meaning.

## 91. “NO IMPROVEMENT” STRATEGY

If original is:

- grammatical
- idiomatic
- concise
- logically correct

choose:

$\boxed{\text{No improvement}}$

Do not replace correct modern usage just because another form is more formal.

## 92. ERROR-DETECTION SEGMENT METHOD

Example:

**(A) Each of the students / (B) have submitted / (C) the required document / (D) on time.**

Check:

A: subject = Each B: “have” mismatches

Answer:

$\boxed{B}$

## 93. LOCAL VS GLOBAL ERROR

Sometimes error is not contained in one word.

Example:

**Neither Ravi nor his friends is coming.**

Local verb looks near “friends.”

Correct:

$\boxed{\text{are}}$

Need understand full structure.

---

## 94. ERROR-SPOTTING ORDER

Under exam time, scan:

$\boxed{\text{SVA} \rightarrow \text{Tense} \rightarrow \text{Pronoun} \rightarrow \text{Modifier} \rightarrow \text{Parallelism} \rightarrow \text{Preposition} \rightarrow \text{Word choice}}$

This is faster than reading randomly.

---

## 95. FIRST READ FOR MEANING

Before hunting grammar:

**What is sentence trying to say?**

Many errors become obvious only when intended meaning is understood.

---

## 96. SECOND READ FOR STRUCTURE

Strip decorations.

Example:

**The quality of the components supplied by the new vendor appears to be declining.**

Core:

**Quality appears.**

Agreement is correct.

---

## 97. THIRD READ FOR IDIOM

After grammar passes, check:

- capable of
  - prefer to
  - different from
  - discuss
  - look forward to
  - no sooner...than
- 

## 98. COMMON DISTRACTOR TYPES

Placement tests often provide:

1. grammar-correct but wrong meaning
-

2. meaning-correct but grammar-wrong
3. overly wordy version
4. wrong idiom
5. correct answer

You need both grammar and meaning.

---

## 99. EXAM-PRESSURE SHORTCUT

---

For sentence completion:

$\boxed{\text{Connector} + \text{Grammar} + \text{Tone}}$

These three eliminate most options before exact vocabulary.

---

## 100. FINAL PART-1 CHECKLIST

---

### For blank questions

- What word class?
- Positive/negative?
- Contrast or continuation?
- Natural collocation?
- Same certainty level?

### For correction

- Does replacement fix real error?
- Does it preserve meaning?
- Is it idiomatic?
- Is original already correct?

### For error detection

- Find main subject/verb first.
  - Check time marker.
  - Check pronoun.
  - Check parallelism/comparison.
  - Check fixed expression.
  - Check semantic logic.
- 

## PRACTICE — LEVEL 0

---

### Q1

Fill:

Despite the difficulty, she remained \_\_\_\_.

A. determined B. careless C. absent D. ambiguous

---

### Q2

Choose correct:

A. Each of the students have submitted the form. B. Each of the students has submitted the form.

### Q3

Choose:

**She enjoys \_\_\_\_ books.**

A. read B. reading

### Q4

Choose:

**He is interested \_\_\_\_ mathematics.**

A. on B. in

### Q5

Choose:

**I will call you when I \_\_\_\_ home.**

A. reach B. will reach

## LEVEL 1

### Q6

Fill:

**Although the proposal seemed promising, the evidence was ultimately \_\_\_\_.**

A. convincing B. inconclusive C. optimistic D. efficient

### Q7

Correct the sentence:

**Ravi's marks are higher than Arjun.**

### Q8

Find error:

**The manager, along with his assistants, are attending the meeting.**

### Q9

Choose:

She looks \_\_\_\_ today.

A. beautifully B. beautiful

### Q10

Choose:

No sooner had the train arrived \_\_\_\_ passengers rushed toward the doors.

A. when B. than

## LEVEL 2

### Q11

Fill:

Because the instructions were \_\_\_\_, different teams interpreted them in different ways.

A. explicit B. ambiguous C. concise D. transparent

### Q12

Choose correct:

A. The issue was discussed about yesterday. B. The issue was discussed yesterday.

### Q13

Correct:

He suggested to postpone the meeting.

### Q14

Choose:

The new policy had a significant \_\_\_\_ on productivity.

A. affect B. effect

### Q15

Choose:

If I \_\_\_\_ you, I would reconsider.

A. was B. were

## LEVEL 3 — EXAM PRESSURE

### Q16

Fill both blanks:

**Though the evidence initially appeared \_\_, a more careful analysis exposed several \_ assumptions.**

A. plausible / unsupported B. false / convincing C. certain / rigorous D. irrelevant / valid

### Q17

Find the error:

**A number of applicants has complained about the delay.**

### Q18

Find the error:

**Walking through the campus, the new library looked impressive.**

### Q19

Choose the best correction:

**The population of City A is greater than City B.**

A. The population of City A is greater than City B. B. The population of City A is greater than that of City B. C. City A population is more than City B. D. The population of City A is more greater than City B.

### Q20

Find all likely errors:

**Each of the employees have been advised to complete the training before they will begin the new assignment.**

## ANSWERS

### Q1

"Despite difficulty" requires positive resistance.

$\boxed{A: \text{determined}}$

### Q2

Each = singular.

$\boxed{B}$

### Q3



Enjoy + gerund.

$\boxed{B:\text{ reading}}$

---

#### Q4

Fixed expression:

$\boxed{\text{interested in}}$

Answer:

$\boxed{B}$

---

#### Q5

Future time clause uses present.

$\boxed{A:\text{ reach}}$

---

#### Q6

“Although promising” signals contrast.

$\boxed{B:\text{ inconclusive}}$

---

#### Q7

Compare marks with marks.

Correct:

**Ravi's marks are higher than Arjun's.**

---

#### Q8

Main subject = manager.

Correct:

$\boxed{\text{is attending}}$

---

#### Q9

“Looks” is linking verb; adjective describes subject.

$\boxed{B:\text{ beautiful}}$

---

#### Q10

---

Fixed pair:

$\boxed{\text{No\ sooner...than}}$

$\boxed{B}$

---

### Q11

Different interpretations indicate ambiguity.

$\boxed{B}$

---

### Q12

“Discuss” takes direct object.

$\boxed{B}$

---

### Q13

Correct:

**He suggested postponing the meeting.**

or:

**He suggested that the meeting be postponed.**

---

### Q14

Need noun:

$\boxed{B:\text{ effect}}$

---

### Q15

Formal hypothetical:

$\boxed{B:\text{ were}}$

---

### Q16

“Though” contrasts initial plausibility with later weakness.

$\boxed{A:\text{ plausible / unsupported}}$

---

### Q17

---

"A number of" takes plural verb.

Correct:

**A number of applicants have complained...**

## Q18

Dangling modifier.

Current grammar suggests "library" was walking.

Correct:

**Walking through the campus, we found the new library impressive.**

## Q19

Compare population with population.

$\boxed{B}$

## Q20

Two key errors:

### Error 1

"Each" is singular:

**Each of the employees has been advised...**

### Error 2

After "before" for future sequence:

**before they begin the new assignment**

not:

**before they will begin**

Correct:

**Each of the employees has been advised to complete the training before they begin the new assignment.**

## CHAPTER 73 STATUS

Covered so far:

- single-blank completion '
- prediction-before-options '
- grammar-fit method '
- connector clues '
- tone/polarity '
- double blanks '
- certainty-strength matching '
- collocation-based completion '
- sentence correction '
- no-improvement/no-error strategy '
- subject–verb correction '
- tense correction '
- pronoun correction '
- article/preposition correction '
- comparison errors '
- parallelism '
- modifiers '
- redundancy '
- gerund/infinitive errors '
- conjunction errors '
- word-choice errors '
- error-spotting workflow '
- L0–L3 practice '

But **Chapter 73 is not complete yet.**

Still remaining:

- advanced multiple-blank completion
- clause-level error spotting
- sentence meaning/logic correction
- awkward diction
- formal vs colloquial correction
- active/passive transformation traps
- reported-speech correction
- sentence improvement by concision
- mixed grammar + vocabulary OA sets
- timed elimination strategies
- final Chapter 73 exam set

## Progress

\[ \boxed{72/76}\text{ { numbered chapters completed}} \]

**Current:** 0 = Chapter 73 — Sentence Completion, Correction & Error Detection

## NEXT

Chapter 73 Part 2 — Advanced Error Spotting, Sentence Improvement, Multi-Blank Logic, Voice/Reported-Speech Errors & Final Timed Toolkit

USER

next

ASSISTANT

## CHAPTER 73 — SENTENCE COMPLETION, CORRECTION & ERROR DETECTION

### PART 2 — ADVANCED ERROR SPOTTING, SENTENCE IMPROVEMENT, MULTI-BLANK LOGIC, VOICE/REPORTED-SPEECH ERRORS & FINAL TIMED TOOLKIT

Status: 0% PROGRESS ! ' COMPLETE GATE: High Placements: Very High Priority: 0% 1

This part moves from individual grammar rules to **mixed exam-pressure questions**.

A difficult sentence may contain:

- grammar problem
- vocabulary problem
- logical problem
- modifier problem
- tone mismatch

at the same time.

The golden rule:

$\boxed{\text{Correct grammar} + \text{Correct meaning} + \text{Natural usage}}$

All three matter.

## 1. ADVANCED MULTIPLE-BLANK QUESTIONS

Consider:

Although the proposal was initially \_\_, subsequent evidence was so \_\_ that even its strongest supporters became \_\_\_\_.

Do not fill blanks independently.

Structure:

- initially positive
- later evidence negative
- supporters change attitude negatively

Likely:

`\[ \boxed{promising, \ damaging, \ skeptical} \]`

---

## 2. BUILD THE LOGIC BEFORE VOCABULARY

Mark sentence:

TEXT

Although [positive],  
later [negative],  
therefore [negative reaction].

Then search options matching this pattern.

This is much faster.

---

## 3. TRIPLE-BLANK METHOD

Use:

### Step 1

Find connectors.

### Step 2

Assign polarity:

`\[ +, \ -, \ 0 \]`

### Step 3

Identify word class.

### Step 4

Solve easiest blank.

### Step 5

Check the whole sentence.

---

## 4. EXAMPLE

The scientist was neither \_ nor \_; instead, she remained \_\_\_\_ while evaluating the evidence.

“neither X nor Y” probably contains two undesirable qualities.

“instead” gives contrasting positive/neutral quality.

Possible:

`\[ \boxed{dogmatic, \ dismissive, \ objective} \]`

## 5. DON'T USE DICTIONARY MEANING ALONE

Suppose options include:

- objective
- indifferent

Both may mean unemotional in some contexts.

But:

**She evaluated evidence carefully and without bias.**

“objective” fits.

“indifferent” implies lack of concern.

Context decides.

## 6. CLAUSE-LEVEL ERROR SPOTTING

Long sentence:

**The report, which was prepared by several researchers and includes data from three regions, were published yesterday.**

Ignore middle clause.

Core:

**The report were published.**

Wrong.

Correct:

$\boxed{\text{was}}$

## 7. STRIP THE SENTENCE

For long sentences:

1. remove relative clauses
2. remove prepositional phrases
3. find core subject + verb
4. check agreement

Example:

**The impact of the new policies introduced by the government has been significant.**

Core:

**Impact has been significant.**

Correct.

## 8. TWO VERBS MAY HAVE DIFFERENT SUBJECTS

Sentence:

**The manager who supervises the interns is responsible.**

Verbs:

- supervises !' subject = who = manager
- is !' subject = manager

Both singular.

Do not check only one verb.

## 9. RELATIVE CLAUSE TRAP

**The students who works in the lab are absent.**

“who” refers to students.

Correct:

$\boxed{\text{who work}}$

Main verb “are” is already plural.

## 10. “ONE OF THE...” TRAP

**She is one of the engineers who work on the project.**

“who” refers to engineers.

Plural:

$\boxed{\text{work}}$

## 11. “THE ONLY ONE...” TRAP

**She is the only one of the engineers who works on this project.**

Now “who” refers to the only one.

Singular:



---

## 12. SENTENCE MEANING ERROR

---

Grammar may be perfect:

**The company reduced prices to increase the cost to customers.**

Grammatically fine.

Semantically contradictory.

If intention is customer affordability:

**The company reduced prices to reduce the cost to customers.**

---

## 13. LOGICAL QUANTIFIER ERROR

---

**Only some employees attended, so every employee was present.**

Contradiction.

Grammar alone cannot solve.

---

## 14. CAUSE/EFFECT ERROR

---

**Because profits declined, therefore the company cut costs.**

Redundant connector structure.

Correct:

**Because profits declined, the company cut costs.**

or:

**Profits declined; therefore, the company cut costs.**

---

## 15. MISUSED CONNECTOR

---

Sentence:

**The system failed; however, operations were disrupted.**

“however” indicates contrast, but disruption is expected consequence.

Better:

**The system failed; therefore, operations were disrupted.**

Connector must match logic.

## 16. THEREFORE VS HOWEVER

**therefore**

result

$\text{\textbackslash}[ A\text{\textbackslash}rightarrow B \text{\textbackslash}]$

**however**

contrast

$\text{\textbackslash}[ A\text{\textbackslash}text{ but } B \text{\textbackslash}]$

Huge sentence-completion clue.

## 17. MOREOVER

Adds supporting information.

**The software is inexpensive; moreover, it is easy to use.**

Both same direction.

## 18. NEVERTHELESS

Contrast despite previous fact.

**The task was difficult; nevertheless, she completed it.**

## 19. CONSEQUENTLY

Result:

**Demand fell; consequently, production was reduced.**

## 20. AWKWARD DICTION

A sentence may be technically understandable but unnatural.

Bad:

**The manager did a decision.**

Natural:

`\[ \boxed{made} a\ decision\ ]`

Sentence improvement tests idiomatic English.

## 21. WORDINESS

Bad:

**At this point in time, we are currently reviewing the proposal.**

Too repetitive.

Better:

`\[ \boxed{We} are\ currently\ reviewing\ the\ proposal.\ ]`

or:

**We are reviewing the proposal now.**

## 22. “DUE TO THE FACT THAT”

Often replace with:

`\[ \boxed{because} \]`

Wordy:

**Due to the fact that the server failed...**

Better:

**Because the server failed...**

## 23. “IN ORDER TO”

Sometimes necessary for clarity, but often:

**to**

is enough.

Wordy:

**She studied hard in order to pass.**

Concise:

**She studied hard to pass.**

Both grammatical.

## 24. REDUNDANT PAIRS

Avoid:

- absolutely essential
- completely unanimous
- final conclusion
- advance planning
- unexpected surprise
- personal opinion

Depending context, some are accepted conversationally, but aptitude tests often prefer tighter wording.

## 25. “UNANIMOUS”

Means everyone agrees.

So:

**completely unanimous**

usually redundant.

## 26. “ESSENTIAL”

Already means necessary.

“Absolutely essential” can be rhetorical, but exam correction may prefer:

$\boxed{\text{essential}}$

## 27. FORMAL VS COLLOQUIAL

In formal aptitude writing:

Prefer:

**investigate**

over:

**check out**

Prefer:

**postpone**

over:

**push back**

if the sentence requires formal register.

But colloquial forms are not grammatically wrong by themselves.

---

## 28. REGISTER CONSISTENCY

Bad formal sentence:

**The committee conducted a detailed review and then kinda dropped the idea.**

“kinda” clashes with formal register.

Better:

**...and then rejected the proposal.**

---

## 29. ACTIVE/PASSIVE — ERROR SPOTTING

Wrong:

**The work has completed by the team.**

Need passive:

[ has\ been + V\_3 ]

Correct:

**The work has been completed by the team.**

---

## 30. PASSIVE TENSE TABLE

**Present simple**

[ is/are + V\_3 ]

**Past simple**

[ was/were + V\_3 ]

**Present perfect**

\[ has/have\ been + V\_3 \]

### Future

\[ will\ be + V\_3 \]

### Modal

\[ modal + be + V\_3 \]

---

## 31. ACTIVE/PASSIVE MEANING MUST REMAIN SAME

Active:

**The committee rejected the proposal.**

Passive:

**The proposal was rejected by the committee.**

Do not change:

- tense
- actor
- object
- meaning

---

## 32. WRONG TENSE TRANSFORMATION

Active:

**They will announce the results tomorrow.**

Wrong passive:

**The results were announced tomorrow.**

Correct:

\[ \boxed{\text{The\ results\ will\ be\ announced\ tomorrow.}} \]

---

## 33. MODAL PASSIVE

Active:

**Candidates must submit the form.**

Passive:

\[ \boxed{\text{The\ form\ must\ be\ submitted\ by\ candidates.}} \]

Not:

**must submitted**

## 34. INTRANSITIVE VERBS

Sentence:

**The accident happened.**

No normal passive because “happen” has no object.

Wrong:

**The accident was happened.**

## 35. OTHER COMMON INTRANSITIVE VERBS

Often no passive:

- happen
- occur
- arrive
- sleep
- die

Wrong:

**He was arrived yesterday.**

Correct:

**He arrived yesterday.**

## 36. REPORTED SPEECH — ERROR SPOTTING

Direct:

**He said, “I am busy.”**

Reported:

**He said that he was busy.**

Standard backshift when reporting context is past.

---

## 37. SAY VS TELL

---

Wrong:

**He told that the exam was difficult.**

Correct:

**He said that the exam was difficult.**

or:

**He told me that the exam was difficult.**

---

## 38. "TOLD TO"

---

Direct:

**The teacher said to Ravi, "Submit the file."**

Reported:

**The teacher told Ravi to submit the file.**

---

## 39. NEGATIVE COMMAND

---

Direct:

**"Do not enter."**

Reported:

`\boxed{He\ told\ them\ not\ to\ enter.}\]`

---

## 40. REPORTED QUESTION

---

Wrong:

**She asked where was I going.**

Correct:

`\boxed{She\ asked\ where\ I\ was\ going.}\]`

Reported questions use statement word order.



---

## 41. YES/NO QUESTION

---

Direct:

**“Are you ready?”**

Reported:

$\backslash [ \boxed{\text{She asked whether/if I was ready.}} \backslash ]$

---

## 42. UNIVERSAL TRUTH

---

Direct:

**Teacher said, “Water freezes at  $0^{\circ}\text{C}$  under standard pressure.”**

Reported:

**The teacher said that water freezes at  $0^{\circ}\text{C}$  under standard pressure.**

General truth need not backshift.

---

## 43. PRONOUN SHIFT

---

Direct:

**Ravi said, “I will finish my work.”**

Reported:

**Ravi said that he would finish his work.**

Pronoun follows speaker/reference.

---

## 44. TIME-WORD SHIFT

---

Common:

- today !' that day
- tomorrow !' next day
- yesterday !' previous day
- now !' then
- here !' there

Only when context requires temporal/location shift.

---

## 45. MULTIPLE ERROR SENTENCE

Each of the students have said that they will submitted the assignment yesterday.

Errors:

1

Each !' has

2

will + base verb:

\[ will\ submit \]

3

“yesterday” conflicts with future “will submit.”

Sentence needs logical tense repair depending intended meaning.

## 46. SENTENCE-IMPROVEMENT PRIORITY

If several options are grammatically correct, choose the one that is:

1. clearest
2. least ambiguous
3. concise
4. idiomatic
5. faithful to meaning

## 47. AMBIGUITY ELIMINATION

Bad:

Ravi told Arjun that he had won.

Who won?

Better:

Ravi told Arjun, “You have won.”

or:

Ravi told Arjun that Ravi had won.

depending intended meaning.

## 48. MISPLACED RELATIVE CLAUSE

Bad:

**The professor discussed the project with the students that was delayed.**

“That was delayed” appears to modify students.

Better:

**The professor discussed the project that was delayed with the students.**

Even better, if possible:

**The professor discussed the delayed project with the students.**

## 49. MODIFIER ATTACHMENT

Place modifier immediately beside intended noun.

This prevents ambiguity.

## 50. LOGICAL COMPARISON — ADVANCED

Wrong:

**The salary of engineers is higher than teachers.**

Correct:

**The salary of engineers is higher than that of teachers.**

Or:

**Engineers earn more than teachers.**

## 51. POSSESSIVE COMPARISON

**Ravi's score is higher than Arjun's.**

The omitted noun “score” is understood.

Correct.

## 52. “LIKE” COMPARISON

Wrong:

**Like Ravi, Sita's marks are high.**

This compares Ravi with Sita's marks.

Correct:

**Like Ravi, Sita scores highly.**

or:

**Like Ravi's marks, Sita's marks are high.**

depending intended meaning.

## 53. PARALLEL COMPARISON

Wrong:

**Working remotely is easier than to commute daily.**

Correct:

**Working remotely is easier than commuting daily.**

Gerund !" gerund.

## 54. PARALLEL CORRELATIVE STRUCTURE

Wrong:

**The job requires both technical knowledge and to communicate effectively.**

Correct:

**The job requires both technical knowledge and effective communication.**

## 55. "NOT ONLY...BUT ALSO" POSITION

Bad:

**She not only completed the report but also the presentation.**

What did she do to the presentation?

Better:

**She completed not only the report but also the presentation.**

Now both are objects of completed.

## 56. CONSISTENCY OF PERSON

Wrong:

**One should prepare carefully if you want to succeed.**

Switches one !' you.

Correct:

**One should prepare carefully if one wants to succeed.**

or:

**You should prepare carefully if you want to succeed.**

## 57. CONSISTENCY OF NUMBER

Wrong:

**A student should check their answers and submit it.**

Pronoun "it" wrongly refers to plural "answers."

Correct:

**...submit them.**

## 58. CONSISTENCY OF TENSE

Wrong:

**She enters the room, sat down, and starts writing.**

Correct all present:

**She enters, sits, and starts...**

or all past:

**She entered, sat, and started...**

---

## 59. LIST PARALLELISM

---

Wrong:

**The role involves analyzing data, writing reports, and presentation of findings.**

Correct:

**...analyzing data, writing reports, and presenting findings.**

---

## 60. FALSE PARALLELISM

---

Even grammatical forms can be logically mismatched.

Bad:

**The course teaches Python, databases, and how to communicate.**

Better:

**The course teaches Python, database concepts, and communication skills.**

---

## 61. WORD-CHOICE CONTEXT

---

Sentence:

**The policy was intended to \_\_\_\_ the problem, not eliminate it completely.**

Best:

\[ \boxed{mitigate} \]

because mitigate = reduce severity.

“eradicate” would conflict with “not eliminate completely.”

---

## 62. DEGREE OF MEANING

---

Words differ in strength:

\[ concerned < alarmed < terrified \]

\[ disagree < criticize < condemn \]

Choose intensity matching context.

---

### 63. “SLIGHTLY DEVASTATED” PROBLEM

Some modifiers clash semantically.

“slightly disappointed”

“slightly devastated” usually awkward because “devastated” is extreme.

Aptitude may test such incompatibility.

### 64. POSITIVE + NEGATIVE SEMANTIC CLASH

**The disastrous success...**

Could be meaningful in unusual context, but normally suspicious.

Check whether the sentence intentionally uses irony/paradox.

### 65. DOUBLE-BLANK — INTENSITY MATCH

**The evidence was not merely weak but completely \_\_\_\_.**

Need stronger negative:

$\boxed{\text{unreliable}}$

Not something weaker like “limited.”

### 66. “NOT MERELY...BUT”

Second element should intensify or add.

Example:

**The policy was not merely ineffective but actively harmful.**

Good progression.

### 67. “RATHER THAN”

Maintain parallel form:

Correct:

**She chose to revise rather than to rest.**

or more naturally:

**She chose to revise rather than rest.**

Wrong:

**She chose revising rather than to rest.**

## 68. "BETWEEN X AND Y"

Correct:

**The choice is between staying and leaving.**

Parallel gerunds.

## 69. "FROM X TO Y"

Correct:

**Scores increased from 60 to 80.**

Wrong:

**increased from 60 and 80.**

## 70. "BETWEEN" RANGE

Correct:

**Scores ranged between 60 and 80.**

or:

**ranged from 60 to 80.**

Do not mix:

**ranged between 60 to 80 'L**

## 71. CORRELATIVE PAIR MIXING

Wrong:

**Either A and B**

Correct:



**either A or B**

Wrong:

**both A or B**

Correct:

**both A and B**

## 72. CONDITIONAL LOGIC ERROR

Sentence:

**Unless you do not submit the form, you cannot attend.**

Likely double-negative confusion.

If intended:

**If you do not submit, you cannot attend.**

Better:

$\boxed{\text{Unless you submit the form, you cannot attend.}}$

## 73. UNLESS "IF NOT"

Use this to decode:

$\boxed{\text{Unless } A, B}$

"H

$\boxed{\text{If not } A, B}$

## 74. "ONLY IF" ERROR

**You can enter only if you have a pass.**

Means:

$\boxed{\text{Entry} \rightarrow \text{Pass}}$

Do not rewrite as:

**If you have a pass, you must enter.**

That changes logic.

---

## 75. SENTENCE CORRECTION CAN TEST LOGICAL IMPLICATION

---

Grammar correct:

**You will pass only if you study.**

This means studying is necessary for passing.

Be careful with options that reverse direction.

---

## 76. NO-ERROR QUESTION — FINAL RULE

---

Never assume there is an error.

If the sentence passes:

- agreement
- tense
- pronoun
- modifier
- idiom
- parallelism
- meaning

then:

$\boxed{\text{No error}}$

is a legitimate answer.

---

## 77. TIMED ERROR-SPOTTING METHOD

---

Target: roughly 20–40 seconds for standard placement questions.

### Pass 1 — Core grammar

subject/verb + tense

### Pass 2 — Structure

pronoun + modifier + parallelism

### Pass 3 — Usage

preposition + collocation + idiom

### Pass 4 — Meaning

logic + comparison + redundancy

---

## 78. FAST BLANK METHOD

---

For sentence completion:

\[ \boxed{GCTC} \]

### **G — Grammar**

What word form?

### **C — Connector**

Contrast/addition/cause?

### **T — Tone**

Positive/negative/neutral?

### **C — Collocation/context**

What naturally fits?

---

## **79. FAST CORRECTION METHOD**

---

Use:

\[ \boxed{CLEAR} \]

### **C — Correct grammar**

### **L — Logical meaning**

### **E — Exact word choice**

### **A — Appropriate register**

### **R — Remove redundancy**

---

## **80. WHEN TWO OPTIONS LOOK CORRECT**

---

Ask:

1. Which preserves original meaning?
  2. Which is less ambiguous?
  3. Which is idiomatic?
  4. Which is more concise without losing information?
- 

## **81. EXAM TRAP — OVERLY FORMAL OPTION**

---

An old-fashioned construction is not automatically better.

Modern standard usage should be preferred unless question specifically tests formal grammar.

---

## **82. EXAM TRAP — “SOUNDS BETTER”**

---

Do not rely only on sound.

---

Example:

**Neither of the students are ready.**

May sound common conversationally.

Traditional aptitude rule:

$\boxed{\text{Neither} \dots \text{is}}$

Know the tested rule.

---

## 83. EXAM TRAP — FIXING ONE ERROR BUT CREATING ANOTHER

Original:

**Each student have submitted their form.**

Option:

**Each students has submitted their form.**

Fixes verb but breaks noun after each.

Correct:

**Each student has submitted their form.**

---

## 84. MULTIPLE ERRORS MAY EXIST

In “sentence improvement,” replacement should fix all problems in underlined segment.

In segmented “spot the error,” test convention may expect one intended error.

Read directions carefully.

---

## 85. FINAL MIXED PRACTICE — LEVEL 0

### Q1

Choose correct:

A. The report has completed. B. The report has been completed.

### Q2

Choose:

**She asked where I \_\_\_\_ going.**

A. was B. was I

### Q3

Choose:

**The company decided to \_\_\_\_ the launch.**

A. postpone B. postponed

### Q4

Choose:

**She is one of the engineers who \_\_\_\_ here.**

A. works B. work

### Q5

Choose:

**The system failed; \_\_\_\_, operations stopped.**

A. however B. consequently

## LEVEL 1

### Q6

Identify error:

**Neither the teachers nor the principal are available.**

### Q7

Correct:

**The project was completed by the team tomorrow.**

### Q8

Correct:

**He told that he was busy.**

### Q9

Identify error:

**The salaries of engineers are higher than teachers.**

### Q10

Choose:

The instructions were \_\_\_\_, so everyone understood them in the same way.

A. ambiguous B. explicit

## LEVEL 2

### Q11

Choose the best sentence:

A. The committee did a decision to postpone the meeting. B. The committee made a decision to postpone the meeting. C. The committee made a decision for postponing meeting. D. The committee performed a decision to postpone meeting.

### Q12

Correct:

Walking into the office, the lights appeared unusually dim.

### Q13

Find the error:

She is the only one of the candidates who have completed all sections.

### Q14

Choose:

The evidence did not \_\_\_\_ the claim; instead, it weakened it.

A. back up B. run out

### Q15

Choose:

The policy was not merely ineffective but \_\_\_\_ harmful.

A. actively B. active

## LEVEL 3 — EXAM PRESSURE

### Q16

Fill:

Although the initial results were \_\_, the researchers remained \_ because the sample size was too small.

A. promising / cautious B. disastrous / enthusiastic C. conclusive / doubtful for no reason D. irrelevant / certain

### Q17

Find all likely errors:

**Each of the reports were reviewed carefully before they were submitted, but neither the final summary nor the appendices was checked for formatting errors.**

### Q18

Choose best correction:

**The new system is more efficient compared to the old system.**

- A. more efficient compared to the old system B. more efficient than the old system C. most efficient than the old system  
D. more efficiently than the old system

### Q19

Correct:

**She asked me that whether I had completed the task.**

### Q20

Find all likely problems:

**Because the manager was absent, therefore the meeting was postponed and the employees were told that they should not to leave until the revised schedule was announced.**

## ANSWERS

### Q1

Present perfect passive:

$\boxed{B}$

### Q2

Reported question uses statement order:

$\boxed{A: \text{ was} }$

Correct:

**She asked where I was going.**

### Q3

Decide to + base verb:

$\boxed{A}$

---

#### Q4

“who” refers to engineers:

$\boxed{B:\text{work}}$

---

#### Q5

Failure caused stoppage.

$\boxed{B:\text{consequently}}$

---

#### Q6

Nearest subject to “nor” = principal.

Correct:

**Neither the teachers nor the principal is available.**

---

#### Q7

“Tomorrow” requires future.

Correct:

**The project will be completed by the team tomorrow.**

---

#### Q8

“Tell” requires object.

Correct:

**He said that he was busy.**

or:

**He told me that he was busy.**

---

#### Q9

Compare salaries with salaries.

Correct:



**The salaries of engineers are higher than those of teachers.**

### Q10

Same interpretation means clear/direct.

\[ \boxed{B:\ explicit} \]

### Q11

Natural collocation + correct structure:

\[ \boxed{B} \]

### Q12

Dangling modifier.

Correct:

**Walking into the office, I noticed that the lights appeared unusually dim.**

### Q13

“The only one” is singular.

Correct:

\[ \boxed{\text{who has completed}} \]

### Q14

Back up = support.

\[ \boxed{A} \]

### Q15

“Harmful” adjective modified by adverb:

\[ \boxed{\text{actively}} \]

### Q16

Promising results but small sample !' cautious.

\[ \boxed{A} \]

## Q17

Sentence:

**Each of the reports were reviewed carefully before they were submitted, but neither the final summary nor the appendices was checked...**

### Error 1

Each 'I' singular verb:

\[ \boxed{was\ reviewed} \]

### Error 2

Neither...nor nearest noun = appendices, plural:

\[ \boxed{were\ checked} \]

Correct:

**Each of the reports was reviewed carefully before it was submitted, but neither the final summary nor the appendices were checked for formatting errors.**

## Q18

Standard direct comparative:

\[ \boxed{B:\ more\ efficient\ than} \]

## Q19

Do not use "that whether."

Correct:

**She asked me whether I had completed the task.**

or:

**She asked whether I had completed the task.**

## Q20

Original:

**Because the manager was absent, therefore the meeting was postponed and the employees were told that they should not to leave until the revised schedule was announced.**

Errors:

## 1. Because + therefore redundancy

Use:

**Because the manager was absent, the meeting was postponed...**

## 2. “should not to leave”

After should:

`\[ \boxed{should\ not\ leave} \]`

Correct:

**Because the manager was absent, the meeting was postponed, and the employees were told that they should not leave until the revised schedule was announced.**

## ' CHAPTER 73 COMPLETE

We have now covered:

- single blanks '
- double/triple blanks '
- connector logic '
- polarity/tone matching '
- grammatical-fit prediction '
- collocation-based completion '
- sentence correction '
- no-error strategy '
- clause-level error spotting '
- relative-clause agreement '
- logical/semantic errors '
- awkward diction '
- formal/informal register '
- concision/redundancy '
- active/passive errors '
- reported-speech errors '
- modifier attachment '
- comparison/parallelism '
- tense/person/number consistency '
- quantifier and conditional logic '
- timed GCTC/CLEAR methods '
- mixed L0–L3 practice '

## Progress

[ \boxed{73/76\text{ numbered chapters completed}} ]

## UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude '
- Chapter 72 — Vocabulary & Word Usage '
- Chapter 73 — Sentence Completion, Correction & Error Detection '
- Chapter 74 — Para Jumbles & Paragraph Completion +
- Chapter 75 — Reading Comprehension +
- Chapter 76 — Critical Reasoning & Verbal Deduction +

Only **3 numbered chapters remain**.

## NEXT

### CHAPTER 74 — PARA JUMBLES & PARAGRAPH COMPLETION

Next we will cover:

- opening-sentence identification
- mandatory sentence pairs
- pronoun/reference links
- article links
- chronology
- cause–effect flow
- general !' specific flow
- claim !' evidence flow
- contrast connectors
- example markers
- acronym/definition clues
- paragraph endings
- odd sentence out
- paragraph completion
- option-based solving
- GATE/placement speed methods
- L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 74 — PARA JUMBLES & PARAGRAPH COMPLETION

### PART 1 — OPENING SENTENCE, MANDATORY PAIRS, PRONOUN LINKS, CHRONOLOGY, CAUSE–EFFECT & G E N E R A L !' S P E C I F I C F L O W C

Status:  $\emptyset$  = Not in PROGRESS GATE: High Placements: Very High Priority:  $\emptyset$  = Not 1

Para-jumble questions give sentences in the wrong order and ask you to reconstruct the paragraph.

The biggest mistake is:

**trying to arrange all sentences at once.**

Better approach:

**Find strong links first, build pairs/chains, then place the full sequence.**

## 1. WHAT MAKES A GOOD PARAGRAPH?

A good paragraph usually has:

1. opening/context
2. development
3. explanation/evidence/example
4. conclusion/result/implication

Typical flow:

$\boxed{\text{General idea} \rightarrow \text{Explanation} \rightarrow \text{Example} \rightarrow \text{Conclusion}}$

Not every paragraph follows exactly this, but it is a strong default.

## 2. FIRST TASK — FIND THE OPENING SENTENCE

A good opening sentence usually:

- introduces topic independently
- does not require a previous reference
- avoids unexplained pronouns
- avoids “this/these/such” without antecedent
- often introduces a person/concept fully

Example:

A. This change transformed the industry. B. Artificial intelligence began to influence commercial software rapidly.

Better opener:

$\boxed{B}$

Why?

“This change” in A requires prior information.

## 3. SENTENCES THAT USUALLY DO NOT OPEN

Be suspicious of sentences starting with:

- This

- These
- Such
- They
- He/She
- However
- Therefore
- Consequently
- For example
- Moreover
- Indeed
- Instead

They often depend on something earlier.

Not impossible, but less likely.

---

## 4. PRONOUN LINK

---

Sentence:

A. Marie Curie conducted pioneering research on radioactivity. B. She later became the first person to win two Nobel Prizes in different sciences.

Clearly:

$\boxed{A \rightarrow B}$

Because “She” refers to Marie Curie.

---

## 5. PRONOUN-ANTECEDENT RULE

---

A pronoun should come after the noun/person it refers to.

Thus:

$[ \text{Noun} \rightarrow \text{Pronoun} ]$

Examples:

- Ravi !' he
  - companies !' they
  - a policy !' it
  - several problems !' these problems
- 

## 6. “THIS” LINK

---

A. The company reduced production by 30%. B. This decision led to temporary shortages.

Strong pair:

$\boxed{A \rightarrow B}$

“This decision” summarizes A.

---

---

## 7. “THESE” LINK

A. Servers failed repeatedly, response times increased, and customers reported outages. B. These problems forced the company to redesign its infrastructure.

A introduces multiple problems.

B refers back:

$\boxed{A \rightarrow B}$

---

## 8. ARTICLE CLUE — A/AN !’ THE

First mention often uses:

**a/an**

Later mention:

**the**

Example:

A. A scientist proposed a new model. B. The model was later tested experimentally.

Pair:

$\boxed{A \rightarrow B}$

---

## 9. NAME !’ SHORTER REFERENCE

First:

**The International Monetary Fund (IMF)...**

Later:

**The IMF...**

So first full form normally precedes abbreviation.

---

## 10. DEFINITION !’ USE

A. Photosynthesis is the process by which plants convert light energy into chemical energy. B. This process is essential for most life on Earth.

$\boxed{A \rightarrow B}$

Definition comes before dependent discussion.

---

## 11. GENERAL !' SPECIFIC

A. Renewable energy technologies have developed rapidly. B. Solar power, for example, has become significantly cheaper.

“For example” signals specific illustration.

Therefore:

$\boxed{A \rightarrow B}$

## 12. EXAMPLE CANNOT USUALLY OPEN

Sentence:

**For example, electric buses reduce local exhaust emissions.**

Question:

Example of what?

It needs an earlier general claim.

So sentences beginning “for example” are rarely openers.

## 13. CLAIM !' EVIDENCE

A. Remote work can reduce commuting-related costs. B. Surveys have found that many remote workers spend less on daily transport.

Logical sequence:

$\boxed{\text{Claim} \rightarrow \text{Evidence}}$

## 14. EVIDENCE !' CONCLUSION

A. In three consecutive tests, the new method reduced processing time. B. These results suggest that the method may improve efficiency.

$\boxed{\text{Evidence} \rightarrow \text{Conclusion}}$

“These results” refers backward.

## 15. CAUSE !' EFFECT

A. Heavy rain continued for several hours. B. As a result, several low-lying roads flooded.

$\boxed{A \rightarrow B}$

Signal:

**as a result**



---

## 16. EFFECT !' EXPLANATION CAN ALSO OCCUR

---

A. Several low-lying roads were flooded by morning. B. Overnight, unusually heavy rain had overwhelmed the drainage system.

This paragraph may present effect first, then explain cause.

So do not blindly assume chronological order.

Use connectors and discourse purpose.

---

## 17. CAUSE-EFFECT SIGNAL WORDS

---

Effect/result:

- therefore
- consequently
- hence
- thus
- as a result
- so

Cause/reason:

- because
  - since
  - owing to
  - due to
- 

## 18. CONTRAST PAIRS

---

A. The technology is inexpensive to operate. B. However, its installation cost remains high.

“However” links B to contrasting A.

$\boxed{A \rightarrow B}$

---

## 19. CONTRAST WORDS

---

- however
- but
- yet
- nevertheless
- although
- despite
- in contrast
- on the other hand

When you see these, search for an earlier idea being contrasted.

---

---

## 20. ADDITION LINKS

A. The new method reduces computation time. B. Moreover, it requires less memory.

“Moreover” adds another supporting point.

$\boxed{A \rightarrow B}$

---

## 21. ADDITION WORDS

- moreover
- furthermore
- additionally
- also
- besides
- in addition

These usually do not start a paragraph unless previous context exists outside the excerpt.

---

## 22. CHRONOLOGY

If paragraph describes a sequence of events, chronological order often matters.

Example:

1. application submitted
2. shortlisted
3. interview conducted
4. offer issued

Natural order:

$\boxed{1 \rightarrow 2 \rightarrow 3 \rightarrow 4}$

---

## 23. TIME MARKERS

Look for:

- initially
  - first
  - later
  - subsequently
  - eventually
  - finally
  - previously
  - meanwhile
  - after
  - before
-

These provide ordering clues.

---

## 24. "INITIALLY" OFTEN EARLY

---

Sentence:

**Initially, the experiment produced inconsistent results.**

Likely appears before:

**After several adjustments, the measurements became stable.**

---

## 25. "FINALLY" IS USUALLY LATE

---

**Finally, the researchers published their findings.**

This normally follows intermediate steps.

---

## 26. BEFORE/AFTER LOGIC

---

A. After the prototype was tested, the team revised the design.

This sentence itself establishes:

$\boxed{\text{Testing} \rightarrow \text{Revision}}$

Other sentences about testing should generally precede revision consequences.

---

## 27. MANDATORY PAIR

---

A mandatory pair is two sentences that almost certainly sit together.

Example:

A. A severe storm damaged several transmission lines. B. Consequently, power was interrupted across the region.

Strong pair:

$\boxed{AB}$

Find these before full sequence.

---

## 28. PAIR-BUILDING METHOD

---

Suppose sentences are A,B,C,D.

Instead of solving four:

Find:

$B \rightarrow D$

Then treat BD as one block.

Now solve:

$\{A, C, BD\}$

Much easier.

---

## 29. CHAIN BUILDING

---

If:

$A \rightarrow C$

and:

$C \rightarrow D$

then block:

$\boxed{ACD}$

Only need place B.

---

## 30. DEMONSTRATIVE LINK

---

Words:

- this
- that
- these
- those
- such

usually refer backward.

Example:

A. Online transactions grew rapidly. B. This growth increased demand for secure payment systems.

$A \rightarrow B$

---

## 31. “SUCH” NEEDS AN ANTECEDENT

---

**Such measures can reduce congestion.**

What measures?

A preceding sentence must introduce them.

Therefore this is unlikely opener.

---

## 32. “THE FORMER / THE LATTER”

---

If sentence says:

---

**The latter is less expensive.**

It requires two earlier items.

Thus cannot sensibly open.

### 33. QUESTION !' ANSWER FLOW

A paragraph may ask:

**Why did the project fail?**

Next sentence:

**The main reason was inadequate testing.**

Strong link.

### 34. PROBLEM !' SOLUTION

A. Urban congestion has increased sharply. B. One possible solution is to improve public transport.

Common paragraph pattern:

$\boxed{\text{Problem} \rightarrow \text{Solution}}$

### 35. PROBLEM !' CAUSE !' SOLUTION

Even stronger structure:

1. problem
2. explanation/root cause
3. response/solution

Example:

**Errors increased. Investigation showed outdated software was responsible. The company therefore upgraded the system.**

### 36. IDEA !' EXAMPLE !' INTERPRETATION

A. Cognitive biases can affect everyday decisions. B. For instance, people often give excessive weight to recent events. C. This tendency can distort risk judgments.

Sequence:

$\boxed{A \rightarrow B \rightarrow C}$

## 37. BROAD !' NARROW

Paragraphs often move:

[ World\ topic\rightarrow Specific\ field\rightarrow Particular\ example ]

Example:

**Technology is changing education. AI-based tools are one example. Adaptive tutoring systems can personalize exercises.**

## 38. NARROW !' BROAD CONCLUSION

After specific evidence, paragraph may broaden:

**These developments illustrate how technology can reshape learning.**

Thus ending can return to the general idea.

## 39. OPENING-SENTENCE TEST

Ask of candidate opener:

**Can a reader understand this sentence without reading anything before it?**

If yes, possible opener.

If it contains unexplained reference, likely not.

## 40. ENDING-SENTENCE TEST

Good endings often contain:

- conclusion
- implication
- result
- recommendation
- broad takeaway

Signals:

- thus
- therefore
- overall
- consequently
- ultimately
- hence
- in conclusion

But connectors alone are not enough.

---

## 41. “THIS SHOWS THAT...”

---

Usually ending/late-stage sentence.

It needs previous evidence.

Example:

**This shows that the policy had limited effectiveness.**

Must follow supporting data.

---

## 42. “IN OTHER WORDS”

---

Restates previous statement.

A. The model is not identifiable from the available data. B. In other words, multiple parameter values can explain the observations.

Strong pair:

$\boxed{A \rightarrow B}$

---

## 43. “INDEED”

---

Usually strengthens previous claim.

**The problem is serious. Indeed, failure rates have doubled.**

Not ideal opener.

---

## 44. “INSTEAD”

---

Signals replacement/contrast.

A. The team did not rebuild the entire system. B. Instead, it repaired the failed component.

$\boxed{A \rightarrow B}$

---

## 45. “SIMILARLY”

---

Needs previous comparable case.

**Similarly, smaller firms experienced the same difficulty.**

Cannot usually open.

## 46. “IN CONTRAST”

---

Needs an earlier comparison target.

A. Urban demand increased. B. In contrast, rural demand remained stable.

\[ A\rightarrow B \]

---

## 47. COMMON PARA-JUMBLE STRUCTURES

---

### Structure A

Definition !' explanation !' example

### Structure B

Problem !' cause !' solution

### Structure C

Claim !' evidence !' conclusion

### Structure D

Past !' present !' future

### Structure E

General !' specific !' general implication

Recognizing structure is faster than word-by-word solving.

---

## 48. EXAMPLE — 4 SENTENCES

---

A. This shift has increased demand for cybersecurity professionals. B. Businesses are moving more operations online. C. As more data is stored digitally, exposure to cyber threats also grows. D. Companies therefore need specialists capable of protecting digital systems.

Find opener:

B introduces topic independently.

B !' C: more operations online !' more digital data/threat.

C !' A? “This shift” in A likely refers to growing digital operations/threat context.

A !' D: demand !' companies need specialists.

Best sequence:

\[ \boxed{B-C-A-D} \]

---

## 49. VERIFY THE FULL PARAGRAPH

---

Never stop after finding pairs.

Read full order once:

---



Does every reference have antecedent? Do connectors make sense? Does topic move naturally? Is conclusion too early?

## 50. OPTION-BASED PARA JUMBLES

If options provide sequences, do not fully arrange from scratch.

Example options:

A. BCAD B. BACD C. CBAD D. BDAC

If B must open, eliminate C immediately.

If C! Mandatory, eliminate any sequence breaking CA.

This is much faster.

## 51. OPENING ELIMINATION

If two options begin with:

- "However..."
- "These findings..."

and two with independent topic sentences, likely eliminate first two.

## 52. PAIR ELIMINATION

If you find mandatory:

$[C \rightarrow D]$

remove all options where C and D are reversed/separated incorrectly.

## 53. TRIPLE-CHAIN ELIMINATION

If:

$[B \rightarrow C \rightarrow A]$

you may solve the entire MCQ without knowing exact location of D initially.

## 54. DO NOT FORCE ADJACENCY WITHOUT REASON

Pronouns often suggest proximity, but not every related sentence must be immediately adjacent.

Example:

A may introduce "climate change."

Several sentences can occur before "this global phenomenon."

Use strength of reference.

---

## 55. TOPIC CONTINUITY

Track the main subject.

Example:

A. Battery prices declined. B. Electric vehicles became cheaper. C. These vehicles gained market share.

Natural topic progression:

battery prices !' vehicle cost !' vehicles.

---

## 56. NO SUDDEN TOPIC JUMP

A paragraph about electric vehicles suddenly discussing medieval literature is likely wrong unless a clear comparison is introduced.

Check semantic continuity.

---

## 57. TENSE CONSISTENCY AS CLUE

Historical paragraph:

**In 1990...**

followed by:

**Later...**

and then:

**Today...**

Chronological tense shifts can indicate order.

---

## 58. ARTICLE “THE” IS NOT ALWAYS A DEPENDENCY

A sentence can open with:

**The internet has transformed communication.**

“The internet” is globally identifiable.

So do not use article rule blindly.

The key is whether referent needs introduction.

---

## 59. PROPER NOUN CAN OPEN

---

**Albert Einstein developed the theory of relativity.**

Independent.

Later:

**The physicist...**

can refer back.

## 60. DESCRIPTION AFTER NAME

A. Ada Lovelace wrote notes on Charles Babbage's Analytical Engine. B. The mathematician is often discussed in histories of computing.

Name first, descriptive reference later.

## 61. TITLE/ROLE !' PRONOUN

If opening says:

**A researcher named X...**

then later:

**She...**

strong link.

## 62. PARA JUMBLE — LEVEL 0 EXAMPLE

Arrange:

A. This process is called evaporation. B. Water changes into vapour when heated. C. It occurs naturally in oceans, lakes, and rivers.

B introduces process.

A names it.

C "It" refers to process.

$\boxed{B-A-C}$

## 63. LEVEL 1 EXAMPLE

A. Therefore, many companies are investing in employee training. B. Technology changes rapidly. C. Workers must continually update their skills to remain effective. D. New tools can make existing knowledge outdated.

Start B.

B! ' Drapid change !' knowledge outdated.

D! ' Cworkers need updates.

C! ' Atherefore companies train.

\[ \boxed{B-D-C-A} \]

---

## 64. LEVEL 2 EXAMPLE

---

A. These findings challenged the assumption that the species was solitary. B. Researchers installed cameras across the forest. C. The recordings showed several animals moving in groups. D. Very little was previously known about the animal's social behaviour.

Opening D gives background.

Then research action B.

Then result C.

Then implication A.

\[ \boxed{D-B-C-A} \]

---

## 65. LEVEL 3 EXAMPLE

---

A. Yetthis convenience comes with significant privacy concerns. B. Smartphones now allow users to perform banking, shopping, and communication from a single device. C. As a result, a large amount of personal information is stored or accessed through phones. D. This concentration of sensitive data makes smartphones attractive targets for attackers.

Start B.

B! ' Qvia "As a result."

C! ' D"This concentration."

Where A?

"Yetthis convenience..." contrasts the convenience introduced in B before discussing privacy concerns. Could be B-A-C-D, but C logically explains why privacy concerns arise. Let's inspect:

B = convenience. A = Yetconvenience has privacy concerns. C = As a result [of smartphone multifunctionality], much personal data accessed. D = This concentration attracts attackers.

Best:

\[ \boxed{B-A-C-D} \]

A transitions from benefit to concern, C explains concern, D develops threat.

---

## 66. WHEN MULTIPLE ORDERS SEEM POSSIBLE

---

Use strongest constraints first:

1. direct pronoun
2. explicit connector
3. chronology
4. cause-effect

5. general-specific

6. thematic flow

Do not prioritize vague “sounds better” over direct grammatical links.

---

## 67. PARA-JUMBLE MASTER METHOD

---

Use:

$\boxed{\text{OPLCV}}$

### **O — Opener**

Find independent introduction.

### **P — Pairs**

Pronouns/articles/connectors.

### **L — Logic**

Cause, contrast, chronology.

### **C — Chain**

Build 2–3 sentence blocks.

### **V — Verify**

Read complete paragraph.

---

## 68. FAST REFERENCE TABLE

---

Signal	Usually needs
this/these/such	earlier antecedent
he/she/they	earlier noun/person
however	earlier contrasting claim
therefore	earlier cause/evidence
for example	earlier general statement
moreover	earlier supporting point
finally	earlier sequence
the latter	earlier pair of items
instead	earlier rejected alternative
similarly	earlier comparable case

---

## 69. COMMON TRAPS

---

### & pTRAP 1 — Starting with pronoun sentence

Usually wrong if antecedent absent.

### & pTRAP 2 — “For example” opening

Needs general point.

### & pTRAP 3 — Chronology ignored

“Later” cannot precede “initially.”

### & pTRAP 4 — Connector meaning ignored

However “” therefore.

### & pTRAP 5 — Solving all sentences at once

Find pairs.

### & pTRAP 6 — Choosing only by topic similarity

Logical relation matters more.

### & pTRAP 7 — Concluding sentence too early

“This shows...” usually late.

---

## PRACTICE — LEVEL 0

---

Arrange:

### Q1

A. This process is known as condensation. B. Water vapour cools and changes into liquid. C. It plays an important role in the water cycle.

### Q2

A. For example, solar panels generate electricity from sunlight. B. Renewable energy can come from several natural sources. C. Such technologies can reduce dependence on fossil fuels.

### Q3

A. Consequently, the match was postponed. B. Heavy rain flooded the field.

---

## LEVEL 1

---

### Q4

A. She later became the project leader. B. Meera joined the company as a junior engineer. C. Her strong performance quickly attracted attention.

### Q5

A. However, the system is expensive to install. B. The system reduces long-term energy costs. C. This limitation has slowed adoption among small firms.

---

### Q6

A. These results surprised the researchers. B. The experiment was repeated three times. C. Each trial produced almost identical results.

---

## LEVEL 2

---

### Q7

A. As a result, many firms are increasing cybersecurity spending. B. Businesses now store large amounts of sensitive information online. C. Cyberattacks can expose this information and disrupt operations. D. Protecting digital systems has therefore become a major priority.

### Q8

A. This improvement encouraged wider adoption. B. Early electric cars had limited driving range. C. Battery technology gradually improved. D. Newer vehicles could travel much farther on a single charge.

---

## LEVEL 3

---

### Q9

A. This finding suggests that sleep plays an important role in memory consolidation. B. Participants who slept after learning remembered more information the next day. C. Researchers asked participants to learn a list of unfamiliar words. D. One group then slept normally, while another remained awake for an extended period.

### Q10

A. Nevertheless, completely eliminating physical offices may create other problems. B. Remote work can reduce commuting time and office costs. C. Some employees report weaker informal communication when working entirely remotely. D. For this reason, many organizations are experimenting with hybrid arrangements.

---

## ANSWERS

---

### Q1

---

B introduces change.

A names process.

C "It" refers to condensation.

$\boxed{B-A-C}$

---

### Q2

---

B = general statement.

A = example.

C = "Such technologies" references renewable technologies.

$\boxed{B-A-C}$

---

---

### Q3

Cause before consequence:

$\boxed{B-A}$

---

### Q4

B introduces Meera.

C "Her" refers to Meera.

A "later" follows career development.

$\boxed{B-C-A}$

---

### Q5

B gives benefit.

A contrasts with "However."

C "This limitation" refers to installation cost.

$\boxed{B-A-C}$

---

### Q6

B describes repetition.

C gives results.

A "These results" refers to C.

$\boxed{B-C-A}$

---

### Q7

B introduces online sensitive data.

C explains threat.

A gives result: firms spend more.

D broader concluding priority.

$\boxed{B-C-A-D}$

---

### Q8

B historical limitation.

C technological improvement.

D consequence of improvement.

---



A “This improvement” refers to greater range.

\[ \boxed{B-C-D-A} \]

---

## Q9

---

C introduces study.

D defines groups.

B gives result.

A interprets finding.

\[ \boxed{C-D-B-A} \]

---

## Q10

---

B presents benefit.

A “Nevertheless” introduces counterpoint.

C specifies a problem.

D gives resulting compromise.

\[ \boxed{B-A-C-D} \]

---

## CHAPTER 74 STATUS

---

Covered so far:

- opening sentence identification '
- pronoun/reference links '
- demonstrative links '
- article/definition links '
- acronym/reference ordering '
- general! 'specific
- claim! 'evidence
- evidence! 'conclusion
- cause! 'effect
- contrast/addition connectors '
- chronology '
- mandatory pairs '
- block/chain method '
- problem! 'solution
- broad! 'narrow flow
- paragraph ending clues '
- option-based elimination '
- OPLCV method '

- L0–L3 practice '

But **Chapter 74 is not complete yet.**

Still remaining:

- paragraph completion
- best concluding sentence
- missing-sentence insertion
- odd sentence out
- paragraph coherence
- theme/tone continuity
- multiple possible para-jumble orders
- optionless para jumbles
- advanced connector/reference chains
- timed placement strategy
- final Chapter 74 mixed set

## Progress

\[ \boxed{73/76}\text{ numbered chapters completed} \]

**Current:** 0=Chapter 74 — Para Jumbles & Paragraph Completion

## NEXT

0=Chapter 74 Part 2 — Paragraph Completion, Missing Sentence, Odd Sentence Out, Advanced Para Jumbles & Final Speed Toolkit

USER

next

ASSISTANT

## CHAPTER 74 — PARA JUMBLES & PARAGRAPH COMPLETION

### PART 2 — PARAGRAPH COMPLETION, MISSING-SENTENCE INSERTION, ODD SENTENCE OUT, ADVANCED PARA JUMBLES & FINAL SPEED TOOLKIT

**Status:** 0=IN PROGRESS ! ' ' COMPLETE **GATE:** High **Placements:** Very High **Priority:** 0=1/10

Part 1 taught you how to **reorder sentences**.

Part 2 teaches you how to answer:

- Which sentence best completes the paragraph?
- Where should a missing sentence be inserted?
- Which sentence does not belong?
- Which sentence is the best conclusion?

- How do you solve a para jumble when no options are given?

Core idea:

\[ \boxed{\text{Every sentence must contribute to one coherent thought}} \]

---

## 1. WHAT IS PARAGRAPH COMPLETION?

You are given a paragraph with the last sentence or another sentence missing.

Your task:

**Choose the option that logically continues the paragraph.**

The correct answer must fit:

1. topic
2. logic
3. tone
4. reference
5. scope

---

## 2. DO NOT CHOOSE A SENTENCE ONLY BECAUSE IT IS TRUE

Suppose paragraph discusses:

**Why electric vehicles are becoming cheaper.**

An option about:

**Electric vehicles reduce local air pollution.**

This may be true, but it does not necessarily complete the paragraph.

Correct completion must follow the paragraph's **current argument**.

---

## 3. TOPIC VS PURPOSE

Ask two questions:

### Topic

What is the paragraph about?

### Purpose

What is the author doing with that topic?

Example:

Topic:

**Remote work**

Purpose:

**explaining why hybrid work may be preferable to fully remote work.**

A completion saying:

**Remote work began decades ago.**

may match topic but not purpose.

## 4. PARAGRAPH FLOW

Most completions follow one of these:

**Claim !' Evidence !' Conclusion**

**Problem !' Cause !' Solution**

**Observation !' Explanation**

**General idea !' Example !' Implication**

**Advantage !' Limitation !' Balanced conclusion**

## 5. BEST-CONCLUSION QUESTIONS

A conclusion should:

- follow from paragraph
- summarize or infer, not introduce a new major topic
- preserve certainty level
- stay within scope

## 6. CONCLUSION EXAMPLE

Paragraph:

**Public transport can carry many passengers using less road space than private cars. It can also reduce congestion when enough commuters use it.**

Best conclusion:

**Therefore, improving public transport can be an important part of urban congestion management.**

Good because it summarizes the argument.

## 7. BAD CONCLUSION — TOO STRONG

Same paragraph.

Bad:

**Public transport will completely eliminate traffic congestion.**

Why wrong?

Original evidence supports:

$\boxed{\text{can reduce}}$

not:

$\boxed{\text{will completely eliminate}}$

## 8. BAD CONCLUSION — NEW TOPIC

Same paragraph.

Bad:

**Electric vehicles use lithium-ion batteries.**

True, but unrelated to the paragraph's reasoning.

## 9. SCOPE CONTROL

Suppose paragraph discusses:

**one experiment involving university students.**

Conclusion:

**Therefore, all humans learn best using this method.**

Too broad.

This is:

$\boxed{\text{sample} \rightarrow \text{entire population overgeneralization}}$

## 10. CERTAINTY CONTROL

If paragraph says:

- may
- suggests
- indicates
- could

Then conclusion should usually remain cautious.

Do not jump to:

- proves
- guarantees
- always
- definitely

---

## 11. TONE CONTINUITY

Paragraph tone:

**cautious and analytical**

Bad completion:

**This ridiculous idea should obviously be abandoned immediately.**

Tone mismatch.

Correct answer should usually preserve author tone.

---

## 12. REFERENCE CONTINUITY

Suppose paragraph ends:

**The new system also reduces maintenance costs.**

Option:

**This advantage may make the technology more attractive to smaller companies.**

“This advantage” clearly refers backward.

Strong completion.

---

## 13. CONTRAST COMPLETION

Paragraph:

**Online learning offers flexibility and wider access. However, some students struggle without face-to-face interaction. \_\_\_\_**

Good completion:

**For this reason, blended learning models may offer a useful compromise.**

Flow:

## 14. CAUSE-EFFECT COMPLETION

Paragraph:

**The company delayed security updates for several months. Several known vulnerabilities therefore remained unpatched. \_\_\_\_**

Good:

**This increased the system's exposure to attack.**

Natural effect.

## 15. PROBLEM-SOLUTION COMPLETION

Paragraph:

**Many cities struggle with traffic during peak hours. Expanding roads alone often provides only temporary relief. \_\_\_\_**

Good:

**Cities may therefore need to combine road planning with stronger public transport and demand-management policies.**

## 16. EXAMPLE-GENERALIZATION COMPLETION

Paragraph:

**One factory reduced electricity use by replacing old motors. Another achieved savings through improved insulation. \_\_\_\_**

Good:

**These examples show that energy efficiency can often be improved through several different interventions.**

## 17. MISSING-SENTENCE INSERTION

You may get:

Paragraph with positions:

[ 1 ] [ 2 ] [ 3 ] [ 4 ]

and one missing sentence.

Your job:

**Find the location where both the left link and right link work.**

## 18. TWO-SIDED FIT RULE

For insertion:

Do not ask only:

**Does sentence fit after previous sentence?**

Also ask:

**Does the next sentence now flow naturally from it?**

Correct placement must satisfy:

$\boxed{\text{Left fit} + \text{Right fit}}$

## 19. INSERTION EXAMPLE

Paragraph:

1. Solar panels have become cheaper over the last decade.
2. \_\_\_\_
3. As a result, solar power is now competitive in many markets.

Missing:

**Manufacturing scale and improved technology have contributed to this decline.**

Where?

After 1.

Because:

1. price decline
2. explains causes
3. "As a result" gives consequence

Sequence:

$\boxed{1-M-3}$

## 20. PRONOUN-BASED INSERTION

Missing sentence:



**This improvement reduced loading times significantly.**

It cannot appear before any improvement has been introduced.

Find antecedent first.

## 21. CONNECTOR-BASED INSERTION

Missing:

**However, the method remains expensive.**

Needs an earlier positive point to contrast.

Do not place it after an already negative sentence unless contrast still makes sense.

## 22. EXAMPLE-MARKER INSERTION

Missing:

**For example, lithium-ion batteries are used in most smartphones.**

Needs an earlier general sentence about battery technologies/applications.

## 23. “THESE FINDINGS” INSERTION

Must follow evidence/results.

It should not precede experiment description.

## 24. CHRONOLOGICAL INSERTION

Paragraph:

- initial experiment
- modification
- final result

Missing sentence containing:

**subsequently**

must appear after earlier stage.

## 25. ODD SENTENCE OUT

You are given several sentences.

Four form one coherent paragraph; one does not belong.

Your job:

$\boxed{\text{Find the sentence that breaks the central chain}}$

---

## 26. DO NOT FIND THE SENTENCE WITH DIFFERENT WORDS

---

An odd sentence may use the same vocabulary but different logical purpose.

Example:

A. Electric vehicles are becoming more common. B. Battery costs have fallen. C. Charging networks are expanding. D. Electric vehicles were featured in a science-fiction film. E. These changes have improved adoption.

Odd:

$\boxed{D}$

Same topic, wrong argument.

---

## 27. ODD SENTENCE METHOD

---

### Step 1

Find central theme.

### Step 2

Build strongest linked pair.

### Step 3

Extend the chain.

### Step 4

Sentence left without a logical role = likely odd.

---

## 28. CHAIN-FIRST METHOD

---

Given A,B,C,D,E.

Suppose:

$A \rightarrow C$

$C \rightarrow E$

$E \rightarrow B$

Then D is likely odd.

This is safer than directly asking “which feels strange?”

---

## 29. ODD SENTENCE MAY BE TRUE

---

Truth is irrelevant.

---

Question is:

**Does it belong in this paragraph?**

A sentence can be factually correct but structurally irrelevant.

### 30. ODD SENTENCE — SCOPE SHIFT

Paragraph:

- effects of exercise on sleep
- exercise timing
- sleep quality
- recovery

Odd sentence:

**Professional athletes earn high salaries.**

Related to exercise broadly, but wrong scope.

### 31. ODD SENTENCE — TIME SHIFT

Paragraph describes historical events in 1940–1950.

One sentence discusses a present-day technology without a transition.

Likely odd.

### 32. ODD SENTENCE — TONE SHIFT

Paragraph is objective research summary.

One sentence:

**This unbelievably stupid policy destroyed everything.**

Likely odd due tone/style mismatch.

### 33. ODD SENTENCE — LOGICAL DUPLICATE?

Be careful.

A sentence that repeats a point can still belong if it:

- explains
- elaborates
- gives example

Do not reject solely because idea is similar.

---

## 34. ADVANCED PARA JUMBLES — NO OPTIONS

---

Without answer choices, use stronger process.

Do not attempt a full 5–6 sentence permutation.

For 6 sentences:

$6! = 720$

Impossible mentally by brute force.

Use constraints.

---

## 35. OPTIONLESS METHOD

---

Use:

$\boxed{A-P-C-E}$

### A — Anchor

Find opener/independent sentence.

### P — Pairs

Find mandatory links.

### C — Chain

Merge pairs.

### E — End

Find conclusion/result.

---

## 36. FIND ANCHORS

---

Two useful anchors:

### Opening anchor

Introduces topic.

### Closing anchor

Wraps up argument.

Then solve middle.

---

## 37. CLOSING CLUES

---

Likely conclusion:

- therefore
  - thus
-

- overall
- ultimately
- these findings suggest
- this illustrates
- for this reason

Not always final, but strong clue.

---

## 38. TOPIC-SENTENCE CHARACTERISTICS

---

A topic sentence often:

- contains broad noun
  - introduces no unexplained references
  - can support several following sentences
  - is less detailed than examples
- 

## 39. DETAIL SENTENCE

---

Usually contains:

- numbers
- examples
- names
- evidence
- specific cases

Such sentences often appear after broad claim.

---

## 40. ABSTRACT !' CONCRETE

---

Example:

A. Behavioral biases can distort financial decisions. B. Loss aversion is one such bias. C. Investors may hold losing assets too long because realizing a loss feels painful.

Sequence:

$\boxed{A-B-C}$

---

## 41. CONCRETE !' INTERPRETATION

---

After example:

**This behaviour illustrates how emotional responses can affect supposedly rational decisions.**

Likely conclusion.

---

## 42. DEFINITION !' PROPERTY !' APPLICATION

A common technical paragraph:

1. define concept
2. describe how it works
3. give use

Example:

**A cache is... It stores... Consequently, processors can...**

## 43. NAME !' ABBREVIATION !' PRONOUN

Strong chain:

**Artificial Neural Network (ANN)... ANN models... These models...**

$\boxed{\text{Full name} \rightarrow \text{Abbreviation} \rightarrow \text{Pronoun}}$

## 44. INTRODUCTION !' COMPARISON

If paragraph says:

**Two approaches dominate...**

then sentences discussing “the former” and “the latter” must follow that introduction.

## 45. FORMER/LATTER CHAIN

A. Two approaches are commonly used: centralized and distributed systems. B. The former offers simpler control. C. The latter can provide greater fault tolerance.

Sequence:

$\boxed{A-B-C}$

## 46. CONTRAST CHAIN

A. Centralized systems are easier to administer. B. However, they can create a single point of failure. C. Distributed systems address some of this risk.

$\boxed{A-B-C}$

## 47. “THIS RISK” LINK

“this risk” must follow a sentence introducing a risk.

Very strong local pair.

---

## 48. ARTICLE LINK — FIRST VS LATER MENTION

---

A. A new drug was tested in a small trial. B. The drug reduced symptoms in many participants.

$\boxed{A \rightarrow B}$

---

## 49. LOGICAL DEPENDENCE > WORD REPETITION

---

Two sentences sharing “technology” are not necessarily adjacent.

A sentence with “this change” may have a much stronger dependency than mere repeated vocabulary.

---

## 50. REFERENCE RESOLUTION PRIORITY

---

When finding antecedent for “this”:

Prefer the nearest earlier idea that:

- matches number
- matches meaning
- explains reference naturally

Do not automatically choose nearest noun.

“This” may refer to an entire previous event.

---

## 51. EXAMPLE

---

A. Companies began storing data in the cloud. B. This reduced the need for local infrastructure.

“This” refers not to “cloud” alone but the whole action:

$\boxed{\text{storing data in cloud}}$

---

## 52. CAUSAL REFERENCE

---

Sentence:

**This led to...**

Find a preceding event capable of causing the stated outcome.

Semantic causality is stronger than simple noun matching.

---

## 53. COHERENCE VS COHESION

---

**Cohesion**

---

surface links:

- pronouns
- connectors
- repeated terms

## Coherence

overall logical meaning.

A paragraph needs both.

Two sentences may be cohesive but incoherent.

---

## 54. EXAMPLE OF COHESION WITHOUT COHERENCE

**The server failed. This server was purchased in 2020. Therefore, rainfall increased.**

“This server” links grammatically, but final logic is nonsense.

So verify meaning.

---

## 55. MULTIPLE POSSIBLE ORDERS

Sometimes two orders seem grammatically possible.

Use:

1. stronger reference link
2. chronological necessity
3. cause-before-result
4. broad-before-example
5. intended rhetorical flow

---

## 56. EXAMPLE

A. The intervention reduced average wait times. B. Researchers introduced a new scheduling system. C. They then compared results with the previous month.

Possible B-C-A.

Why?

- B introduces action
- C “They” references researchers
- A reports outcome

$\boxed{\text{B-C-A}}$

---

## 57. PARAGRAPH COMPLETION — FORWARD PREDICTION

Before looking at options:



### What should happen next?

Even rough prediction helps.

Example:

Paragraph ends:

**Yet the study used only 20 participants.**

Prediction:

**limitation/caution**

Then choose an option about limited generalizability, not a strong universal conclusion.

## 58. NEGATIVE EVIDENCE !' CAUTIOUS ENDING

**Sample small, observation short, no control group.**

Likely conclusion:

$\boxed{\text{More research is needed}}$

rather than:

$\boxed{\text{The theory is completely proven}}$

## 59. POSITIVE EVIDENCE + LIMITATION

Often best conclusion is balanced:

**The results are promising, but larger studies are needed.**

This pattern appears frequently in RC/completion questions.

## 60. BEST CONCLUSION VS SUMMARY

A conclusion does not have to repeat every detail.

It should capture the **logical implication**.

## 61. BEST SUMMARY-LIKE ENDING

Paragraph:

- costs declined
- efficiency improved

- adoption grew

Conclusion:

**Together, these changes have made the technology increasingly competitive.**

This synthesizes.

---

## 62. WRONG ENDING — DETAIL RESTART

After conclusion-ready evidence:

**The company was founded in 1997.**

Sudden unrelated historical detail is unlikely.

---

## 63. WRONG ENDING — PERSONAL OPINION

Research paragraph:

**Therefore, I personally think this is amazing.**

Usually stylistically inconsistent.

---

## 64. SENTENCE INSERTION — POSITION 1

A missing sentence belongs first if it:

- introduces broad topic
- has no backward reference
- makes later references understandable

---

## 65. INSERTION — MIDDLE

Likely middle if it:

- explains previous sentence
- provides example
- establishes link required by following sentence

---

## 66. INSERTION — END

Likely end if:

- concludes
- generalizes
- gives implication/recommendation

- depends on all prior evidence

---

## 67. BRIDGE SENTENCE

A bridge connects two ideas.

Example:

Previous:

**AI systems can improve prediction accuracy.**

Next:

**These concerns have prompted calls for stronger regulation.**

Missing bridge must introduce concerns:

**However, their use can also create privacy and accountability risks.**

Perfect bridge:

\[ Benefit \rightarrow Risk \rightarrow Regulation \]

---

## 68. BRIDGE TEST

For missing sentence M between A and B:

Need:

\[ A \rightarrow M \]

and:

\[ M \rightarrow B \]

If only one works, placement is weak.

---

## 69. ODD-SENTENCE EXAMPLE

A. Sleep supports memory consolidation. B. Deep sleep is associated with important neural processes. C. Experiments show that sleep after learning can improve recall. D. Coffee cultivation requires a warm climate. E. Researchers therefore study sleep as part of learning science.

Odd:

\[ \boxed{D} \]

---

## 70. ODD-SENTENCE WITH SAME TOPIC

A. Smartphones store personal information. B. This makes device security important. C. Biometric authentication can improve access control. D. Smartphone screens have become larger over time. E. Encryption also protects stored

data.

Odd:

$\boxed{D}$

Still smartphone-related, but security paragraph excludes display-size trend.

---

## 71. PARA-JUMBLE ELIMINATION WITH OPTIONS

---

Suppose you know:

$B \rightarrow D$

and D must precede C.

Thus:

$B-D-C$

Any option without BDC order can be eliminated instantly.

Adjacency of BD may be mandatory; DC may only be relative depending clue strength.

---

## 72. FIRST-TWO SENTENCE TEST

---

When options differ mainly in opening:

Read only first two sentences of each.

Ask:

**Does sentence 2 naturally follow sentence 1?**

Often enough to eliminate most options.

---

## 73. LAST-TWO SENTENCE TEST

---

Similarly:

**Does final sentence genuinely conclude what precedes it?**

A "For example" sentence is unlikely final if no interpretation follows, though not impossible.

---

## 74. CONNECTOR TEST

---

If option places:

**Therefore...**

after a sentence with no cause/evidence, reject.

If it places:

---

**However...**

after a sentence expressing same idea with no contrast, reject.

## 75. PRONOUN TEST

Reject sequences where:

- “he” appears before person
- “this result” before result
- “these methods” before methods
- “the latter” before two items

## 76. CHRONOLOGY TEST

Reject:

\[ Finally \rightarrow Initially \]

unless paragraph deliberately uses flashback.

Aptitude questions normally use direct chronology.

## 77. GENERAL-SPECIFIC TEST

Reject:

**For example... !' broad claim**

unless sentence clearly functions as opening anecdote, which is less common in aptitude jumbles.

## 78. PARA-JUMBLE SPEED PRIORITIES

Find clues in this order:

### Tier 1

Ø = pronoun/reference Ø = explicit connectors Ø = chronology Ø = definition !' reference

### Tier 2

+ Cause-effect + General-example + Claim-evidence

### Tier 3

Ø = stylistic flow Ø = what “sounds best”

## 79. DO NOT OVERUSE CONNECTORS

“However” does not necessarily refer to immediately preceding sentence if an entire prior idea is being contrasted.

But in short aptitude paragraphs, adjacency is often likely.

Use meaning.

---

## 80. PARA-JUMBLE FINAL VERIFY

---

Read completed paragraph and ask:

1. Is topic introduced properly?
2. Does every pronoun have antecedent?
3. Does each connector have logical justification?
4. Is chronology valid?
5. Does conclusion follow?
6. Any sentence feel detachable?

If yes to all except last:

good sequence.

---

## 81. TIMED STRATEGY — MCQ PARA JUMBLE

---

Target approach:

### 0–10 sec

Identify opener candidates.

### 10–20 sec

Find one mandatory pair.

### 20–35 sec

Eliminate options.

### 35–45 sec

Verify final sequence.

Do not fully reconstruct if options already make answer obvious.

---

## 82. TIMED STRATEGY — OPTIONLESS

---

Use:

$\boxed{\text{Anchor} \rightarrow \text{Pair} \rightarrow \text{Chain} \rightarrow \text{Verify}}$

Write only pair marks:

$A \rightarrow C, \quad D \rightarrow B$

Then merge.

---

## 83. TIMED STRATEGY — PARAGRAPH COMPLETION

---

Use:

$\boxed{\text{TPST}}$

**T — Topic**

**P — Purpose**

**S — Scope**

**T — Tone**

Then check connector/reference.

---

## 84. TIMED STRATEGY — INSERTION

---

Use:

$\boxed{\text{LR}}$

**L — Left fit**

**R — Right fit**

Both must work.

---

## 85. TIMED STRATEGY — ODD SENTENCE

---

Use:

$\boxed{\text{Chain } 4, \text{ reject } 1}$

Do not search directly for “weird sentence.”

Build paragraph from four strongly linked sentences.

---

## 86. COMMON TRAPS

---

& pTRAP 1 — Same topic = correct

No. It must support the current argument.

& pTRAP 2 — True statement = correct completion

No.

& pTRAP 3 — Extreme conclusion

Avoid when evidence is cautious.

& pTRAP 4 — New information in conclusion

Usually wrong.

& pTRAP 5 — Insert sentence using only left context

Check both sides.

& pTRAP 6 — Odd sentence chosen by vocabulary

---

Find logical chain instead.

& pTRAP 7 — Ignoring pronoun antecedent

Major error.

& pTRAP 8 — “However” and “therefore” treated alike

Contrast vs result.

---

## 87. PRACTICE — PARAGRAPH COMPLETION L0

---

Q1

**Regular physical activity improves cardiovascular fitness and can strengthen muscles. It is also associated with several mental-health benefits. \_\_\_\_**

A. Therefore, exercise can contribute to health in more than one way. B. The first Olympic Games were held in ancient Greece. C. All people must exercise exactly two hours every day. D. Sports shoes are sold in many colours.

Answer?

---

## 88. Q2

**The company reduced unnecessary packaging and redesigned its shipping boxes. As a result, it used less material per order. \_\_\_\_**

A. These changes also lowered transportation weight and waste. B. The company was founded in 1998. C. Packaging is always harmful. D. Customers buy products for many reasons.

---

## 89. PARAGRAPH COMPLETION L1

---

Q3

**Early results from the treatment were encouraging. However, the trial involved only a small number of patients and lasted just six weeks. \_\_\_\_**

A. The treatment has therefore been proven effective for everyone. B. Larger and longer studies are needed before firm conclusions can be drawn. C. The treatment should immediately replace all existing therapies. D. Patient age is measured in years.

---

## 90. Q4

**Online banking saves customers time and allows transactions outside normal branch hours. At the same time, digital services expose users to phishing and account-theft risks. \_\_\_\_**

A. Therefore, convenience must be accompanied by strong security practices. B. Banks should stop using computers. C. Currency notes are printed on specialized materials. D. Online banking has no disadvantages.



---

## 91. INSERTION L1

---

Q5

Paragraph:

1. Electric vehicles produce no tailpipe emissions.
2. Battery manufacturing, however, also has environmental costs.
3. \_\_\_\_
4. Therefore, evaluating their total environmental impact requires considering the entire life cycle.

Missing:

**Electricity generation and end-of-life battery recycling must also be considered.**

Where does it fit?

---

## 92. INSERTION L2

---

Q6

1. The software initially required several minutes to process each file.
2. Engineers redesigned the most expensive part of the algorithm.
3. \_\_\_\_
4. This improvement made real-time use possible.

Missing:

**Processing time fell to a few seconds.**

Correct position?

---

## 93. ODD SENTENCE L1

---

Q7

A. Urban trees provide shade. B. They can reduce surface temperatures during hot weather. C. Green spaces may also improve neighborhood comfort. D. Airplanes travel faster than most trains. E. For these reasons, cities often include trees in heat-management strategies.

Odd?

---

## 94. ODD SENTENCE L2

---

Q8

A. Password reuse increases security risk. B. If one service is compromised, attackers may try the same password elsewhere. C. Unique passwords reduce this cross-account exposure. D. Many people prefer dark-mode interfaces on phones. E. Password managers can make unique passwords easier to maintain.

Odd?

---

## 95. PARA JUMBLE L2

---

**Q9**

A. These observations led researchers to reconsider the original theory. B. A new theory had predicted that the particles would move independently. C. Experiments instead showed that their movements were strongly correlated. D. The researchers repeated the measurements using different equipment.

Arrange.

---

## 96. PARA JUMBLE L3

---

**Q10**

A. This dependence can create serious problems when supply chains are disrupted. B. Modern manufacturing often relies on specialized components produced by a small number of suppliers. C. Firms have therefore begun exploring alternative suppliers and larger inventories. D. A shortage at even one supplier can consequently slow production across many factories. E. Such strategies cannot eliminate all risk, but they can improve resilience.

Arrange.

---

## 97. PARAGRAPH COMPLETION L3

---

**Q11**

**The experiment found a correlation between longer screen time and lower sleep quality. Yet participants chose their own screen habits, and several other lifestyle variables were not controlled. \_\_\_\_**

A. Therefore, the experiment proves that screens directly cause poor sleep. B. The observed association should not automatically be interpreted as proof of causation. C. All screen use should be prohibited. D. Sleep is unnecessary for healthy adults.

---

## 98. Q12

---

**Automation can reduce repetitive manual work and improve consistency. However, implementation may require new skills, redesigned workflows, and significant initial investment. \_\_\_\_**

A. Organizations should therefore evaluate both potential benefits and transition costs. B. Automation is always profitable immediately. C. Workers never need training. D. All manual work is inefficient.

---

## 99. ANSWERS

---

**Q1**

Paragraph lists physical and mental benefits.

---

Best synthesis:

$\boxed{A}$

---

## Q2

“This changes...” continues direct packaging consequences.

$\boxed{A}$

---

## Q3

Small sample + short duration = caution.

$\boxed{B}$

---

## Q4

Benefit + risk !' balanced implication.

$\boxed{A}$

---

## Q5

Sequence:

- tailpipe benefit
- battery cost
- other life-cycle factors
- conclusion about life cycle

Missing belongs between 2 and 4:

$\boxed{\text{Position 3}}$

---

## Q6

Redesign !' reduced processing time !' “This improvement...”

Thus:

$\boxed{\text{Position 3}}$

---

## Q7

A-B-C-E form urban-tree/heat chain.

D is unrelated.

$\boxed{D}$

---

## Q8

A-B-C-E form password-security chain.

D unrelated.

$\boxed{D}$

## Q9

B introduces predicted theory.

C contrasts:

**Experiments instead...**

D says researchers repeated measurements to verify.

A "These observations" gives conclusion.

$\boxed{B-C-D-A}$

## Q10

B introduces supply concentration.

A:

**This dependence...**

refers to B.

D explains disruption mechanism.

C:

**therefore...**

gives response.

E:

**Such strategies...**

refers to alternative suppliers/inventories.

$\boxed{B-A-D-C-E}$

## Q11

Correlation + uncontrolled variables.

Cannot claim causation.

\[ \boxed{B} \]

---

## Q12

---

Paragraph balances benefits and costs.

\[ \boxed{A} \]

---

## 100. FINAL PARA-JUMBLE MASTER SHEET

---

### Opening clues

---

Prefer:

- independent topic sentence
- full name/definition
- general claim

Avoid initially:

- this
  - these
  - he/she/they
  - however
  - therefore
  - for example
  - the latter
- 

## 101. MANDATORY-LINK SHEET

---

### Person !' pronoun

\[ Ravi\rightarrow he \]

### A/an !' the

\[ a\ model\rightarrow the\ model \]

### Full form !' abbreviation

\[ Artificial\ Intelligence\rightarrow AI \]

### Claim !' example

\[ general\rightarrow for\ example \]

### Cause !' result

\[ event\rightarrow therefore \]

### Claim !' contrast

\[ claim\rightarrow however \]

---

## Evidence !' inference

\[ results\rightarrow these\ findings \]

---

## 102. PARAGRAPH COMPLETION MASTER SHEET

---

Correct option must satisfy:

\[ \boxed{\text{Topic + Purpose + Scope + Tone + Logic}} \]

Reject:

- new topic
  - unsupported extreme
  - repetition with no purpose
  - contradiction
  - scope expansion
- 

## 103. INSERTION MASTER SHEET

---

Use:

\[ \boxed{\text{Previous\ sentence \rightarrow Missing\ sentence \rightarrow Next\ sentence}} \]

Both arrows must be strong.

---

## 104. ODD-SENTENCE MASTER SHEET

---

Do:

\[ \boxed{\text{Build\ coherent\ chain\ first}} \]

Then identify unused sentence.

Do not choose merely because its vocabulary differs.

---

## 105. 30-SECOND PARA-JUMBLE METHOD

---

\[ \boxed{\text{OPLCV}} \]

- **O** — Opener
  - **P** — Pair
  - **L** — Logic
  - **C** — Chain
  - **V** — Verify
- 

## 106. FINAL CHAPTER 74 TRAPS

---

1. Pronoun before antecedent.
  2. Example before general claim.
  3. "Therefore" without cause.
-

4. "However" without contrast.
  5. Conclusion before evidence.
  6. Later before earlier event.
  7. True-but-irrelevant completion.
  8. Overstrong conclusion.
  9. New topic at paragraph end.
  10. Odd sentence chosen by vocabulary rather than coherence.
  11. Missing sentence checked only against preceding sentence.
  12. Full brute-force ordering instead of pair building.
- 

## ' CHAPTER 74 COMPLETE

---

We have now covered:

- opening-sentence identification '
- pronoun/reference links '
- article links '
- definition/abbreviation links '
- chronology '
- cause-effect '
- claim-evidence '
- evidence-conclusion '
- g e n e r a l ' s p e c i f i c
- contrast/addition '
- mandatory pairs '
- block/chain formation '
- option-based para jumbles '
- optionless para jumbles '
- paragraph completion '
- best conclusion '
- missing-sentence insertion '
- bridge sentences '
- odd sentence out '
- scope/tone continuity '
- certainty matching '
- advanced reference chains '
- timed strategies '
- L0–L3 mixed practice '

## Progress

---

\[ \boxed{74/76}\text{ numbered chapters completed} \]

## UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude '

- Chapter 72 — Vocabulary & Word Usage '
- Chapter 73 — Sentence Completion, Correction & Error Detection '
- Chapter 74 — Para Jumbles & Paragraph Completion '
- Chapter 75 — Reading Comprehension +
- Chapter 76 — Critical Reasoning & Verbal Deduction +

Only **2 numbered chapters remain**.

## NEXT

### CHAPTER 75 — READING COMPREHENSION

Next we will cover:

- how to read a passage efficiently
- main idea
- central theme
- author's purpose
- factual/direct questions
- inference
- assumption
- tone
- title
- vocabulary in context
- reference questions
- paragraph function
- strengthen/weaken links inside RC
- scope and extreme-option traps
- elimination methods
- GATE-style short passages
- placement-style longer RC
- timed L0–L3 practice.

USER

next

ASSISTANT

## CHAPTER 75 — READING COMPREHENSION

### PART 1 — HOW TO READ A PASSAGE, MAIN IDEA, CENTRAL THEME, AUTHOR'S PURPOSE, DIRECT QUESTIONS, INFERENCE & TONE

**Status:** ~~0=Not~~ **PROGRESS** **GATE:** Very High **Placements:** Very High **Priority:** ~~0=Not~~ 1

Reading Comprehension is not mainly a vocabulary test.



It tests whether you can distinguish:

- what the passage **says**
- what it **implies**
- what the author **believes**
- what the author **does not claim**
- which answer stays within the passage's scope

Core rule:

**Answer from the passage, not from your outside knowledge.**

## 1. RC QUESTION TYPES

Most questions fall into:

1. Main idea
2. Central theme
3. Author's purpose
4. Direct/factual question
5. Inference
6. Tone
7. Title
8. Vocabulary in context
9. Reference question
10. Paragraph function
11. Assumption
12. Strengthen/weaken
13. Statement true/false according to passage

## 2. FIRST DECISION — PASSAGE FIRST OR QUESTIONS FIRST?

There is no universal rule.

For GATE and short aptitude passages:

**Read passage first.**

For very long placement RC:

You can quickly scan questions first to know what to look for.

But do not spend too much time memorizing questions.

Best default:

$\boxed{\text{Quick question scan}} \rightarrow \text{Passage} \rightarrow \text{Questions}$

### 3. DO NOT READ EVERY WORD AT SAME SPEED

---

Read quickly through:

- examples
- repeated details
- minor descriptions

Slow down at:

- first sentence
  - contrast words
  - conclusions
  - author's judgments
  - paragraph transitions
  - final sentence
- 

### 4. HIGH-VALUE SIGNAL WORDS

---

Watch:

#### **Contrast**

- however
- although
- but
- yet
- nevertheless

#### **Cause/result**

- therefore
- because
- thus
- consequently

#### **Example**

- for example
- for instance
- such as

#### **Conclusion**

- overall
- hence
- therefore
- this suggests
- ultimately

These often reveal passage structure.

---

## 5. ONE-LINE PARAGRAPH SUMMARY

After each paragraph, mentally summarize in 3–7 words.

Example:

Paragraph 1:

**AI is spreading in healthcare.**

Mental note:

`\boxed{\text{AI adoption expanding}}`

Paragraph 2:

**But errors and bias remain.**

Mental note:

`\boxed{\text{Risks remain}}`

Paragraph 3:

**Human oversight is necessary.**

Mental note:

`\boxed{\text{Need supervised use}}`

Now whole passage becomes easy.

## 6. PASSAGE MAP

A good RC reader builds:

TEXT

P 1	!	T o p i c / p r o b l e m
P 2	!	E v i d e n c e / e x a m p l e
P 3	!	C o n t r a s t / l i m i t a t i o n
P 4	!	C o n c l u s i o n

This is much more useful than memorizing every sentence.

## 7. MAIN IDEA

Main idea asks:

**What is the passage primarily saying?**

It should cover the **whole passage**, not one paragraph.

## 8. MAIN IDEA TRAP — TOO NARROW

Passage discusses:

- benefits of remote work
- limitations
- hybrid work as compromise

Option:

**Remote work reduces commuting time.**

True, but too narrow.

Better:

**Remote work offers important benefits but also limitations, making hybrid arrangements attractive in some contexts.**

## 9. MAIN IDEA TRAP — TOO BROAD

Passage only discusses:

**urban public transport.**

Option:

**Technology is changing modern society.**

Too broad.

## 10. MAIN IDEA TRAP — EXTREME

Passage:

**Public transport can reduce congestion.**

Option:

**Public transport is the only solution to urban congestion.**

Too strong.

## 11. MAIN IDEA FORMULA

Good main idea usually contains:

$\boxed{\text{Topic + Author's central claim}}$

Not just topic.

Example:

Topic:

**social media**

Central claim:

**it expands communication but can amplify misinformation.**

Main idea:

**Social media improves connectivity while also creating challenges related to misinformation.**

---

## 12. CENTRAL THEME

Theme is the broader conceptual focus.

Main idea may be specific claim.

Example passage:

**several technologies increase convenience but create privacy trade-offs.**

Theme:

$\boxed{\text{Convenience versus privacy}}$

---

## 13. AUTHOR'S PURPOSE

Ask:

**Why did the author write this passage?**

Common purposes:

- explain
- describe
- compare
- argue
- criticize
- evaluate
- warn
- recommend
- illustrate

## 14. PURPOSE — EXPLAIN

Passage objectively describes:

**how earthquakes occur.**

Purpose:

$\boxed{\text{to explain a natural process}}$

## 15. PURPOSE — ARGUE

Passage gives reasons why:

**cities should invest more in public transport.**

Purpose:

$\boxed{\text{to argue for increased public-transport investment}}$

## 16. PURPOSE — EVALUATE

Passage considers both:

- advantages
- disadvantages

Purpose may be:

$\boxed{\text{to evaluate the strengths and limitations of X}}$

## 17. PURPOSE — WARN

Passage emphasizes risks of:

**antibiotic misuse.**

Purpose:

$\boxed{\text{to warn about consequences of misuse}}$

## 18. DIRECT / FACTUAL QUESTIONS

These ask something explicitly stated.

Example:

Passage:

**The study included 200 participants.**

Question:

**How many participants?**

Answer:

$\boxed{200}$

Do not infer anything more.

## 19. LOCATION METHOD

For direct questions:

1. identify keyword from question
2. locate matching sentence/paragraph
3. read 1 sentence before and after
4. answer from text

Fast and reliable.

## 20. PARAPHRASE TRAP

Correct answer may not repeat passage wording exactly.

Passage:

**The policy reduced expenditure.**

Option:

**The policy lowered spending.**

Same meaning.

Do not search only for exact words.

## 21. INFERENCE

Inference asks:

**What follows logically from the passage even if not stated word-for-word?**

Key:

$\boxed{\text{Inference must be supported, not merely possible}}$

## 22. DIRECT VS INFERENCE

Direct:

**"The company reduced prices by 10%."**

Inference:

**Customers paid less than before.**

Likely follows.

But:

**Sales definitely doubled.**

Does not follow.

## 23. INFERENCE TEST

Ask:

**If this option were false, would that contradict what the passage strongly implies?**

If yes, good inference.

## 24. INFERENCE = SAFE STEP

Do not jump 5 steps.

Passage:

**Many employees reported better concentration at home.**

Safe inference:

**Some employees may find home environments conducive to concentration.**

Unsafe:

**Offices always reduce productivity.**

## 25. "SOME" IS SAFER THAN "ALL"

Passage:

**Several participants improved.**

Inference:

$\boxed{\text{Some participants improved}}$



Not:

$\boxed{\text{All participants improved}}$

---

## 26. POSSIBILITY VS CERTAINTY

Passage:

**The findings suggest X may contribute to Y.**

Correct inference retains caution:

**X may be associated with Y.**

Wrong:

**X definitely causes Y.**

---

## 27. CORRELATION VS CAUSATION IN RC

Passage:

**Higher screen time was associated with poorer sleep.**

Do not infer:

**Screen time causes poor sleep.**

Unless the passage establishes causality.

---

## 28. AUTHOR'S TONE

Tone = author's attitude/style toward the subject.

Common tones:

- objective
- analytical
- critical
- skeptical
- optimistic
- pessimistic
- cautious
- approving
- concerned
- sarcastic

- balanced
- 

## 29. OBJECTIVE TONE

---

Passage presents:

- data
- definitions
- neutral explanation

without emotional judgment.

Tone:

\[ \boxed{objective/informative} \]

---

## 30. ANALYTICAL TONE

---

Author examines causes, effects, strengths, limitations.

Tone:

\[ \boxed{analytical} \]

---

## 31. CRITICAL TONE

---

Author identifies flaws or weaknesses.

Important:

**Critical does not necessarily mean hostile.**

It can mean evaluative.

---

## 32. SKEPTICAL TONE

---

Author doubts a claim or evidence.

Signals:

- insufficient
  - uncertain
  - questionable
  - not yet demonstrated
- 

## 33. CAUTIOUS TONE

---

Signals:

- may
  - appears
-

- suggests
- potentially
- further research needed

Tone:

\[ \boxed{cautious} \]

---

## 34. OPTIMISTIC TONE

---

Signals future potential positively:

- promising
  - encouraging
  - likely to improve
- 

## 35. PESSIMISTIC TONE

---

Emphasizes negative expectations:

- unlikely
  - worsening
  - little hope
  - severe limitations
- 

## 36. BALANCED TONE

---

Author recognizes both benefits and drawbacks.

Example:

**The technology is useful, but its limitations should not be ignored.**

Tone:

\[ \boxed{balanced} \]

---

## 37. TONE TRAP — SARCASTIC

---

Do not choose “sarcastic” unless there is clear mockery/irony.

A critical academic passage is usually not sarcastic.

---

## 38. TONE TRAP — EMOTIONAL EXTREMES

---

Options like:

- furious
  - ecstatic
-

- contemptuous

are usually wrong unless passage language is strongly emotional.

---

## 39. TITLE QUESTIONS

A good title should be:

- broad enough to cover whole passage
- specific enough to identify topic
- neutral unless passage strongly argues a position

---

## 40. TITLE TRAP — TOO NARROW

Passage on:

**benefits + risks of AI in healthcare.**

Bad title:

**AI Improves Diagnosis**

Too narrow.

Better:

$\boxed{\text{AI in Healthcare: Opportunities and Risks}}$

---

## 41. TITLE TRAP — TOO BROAD

Bad:

**Technology and Society**

Too broad.

---

## 42. VOCABULARY IN CONTEXT

Question:

**What does “robust” mean in the passage?**

Do not use first dictionary meaning automatically.

Sentence:

**The model remained accurate under noisy conditions, indicating a robust design.**

Here:

[ \boxed{robust = reliable/resilient} ]

---

## 43. REFERENCE QUESTIONS

---

Example:

**The company introduced flexible hours. This policy reduced absenteeism.**

Question:

**“This policy” refers to?**

Answer:

[ \boxed{\text{flexible working hours}} ]

---

## 44. PRONOUN REFERENCE

---

Words:

- this
- these
- they
- it
- such
- former/latter

Usually refer backward.

Search nearest logically matching antecedent.

---

## 45. PARAGRAPH FUNCTION

---

Question:

**What is the purpose of paragraph 2?**

Possible roles:

- gives example
- presents evidence
- introduces counterargument
- explains cause
- provides historical background

Do not summarize content only; identify **function**.

---

## 46. FUNCTION EXAMPLE

---

P1:

**Renewable energy is becoming cheaper.**

P2:

**Solar panel prices have fallen 80%...**

P2 function:

$\boxed{\text{provide evidence for P1}}$

---

## 47. FIRST-SENTENCE IMPORTANCE

---

The first sentence often introduces:

- topic
- context
- problem

But not always main idea.

Do not assume first sentence = answer.

---

## 48. LAST-SENTENCE IMPORTANCE

---

The last sentence often contains:

- conclusion
- recommendation
- author's final stance

Read it carefully.

---

## 49. CONTRAST SENTENCES OFTEN HOLD THE REAL POINT

---

Example:

**The technology is impressive. However, cost remains a major barrier.**

If paragraph continues on cost, author's main point may focus on limitation rather than praise.

"However" often shifts emphasis.

---

## 50. AUTHORIAL EMPHASIS

---

In:

---

**X is useful, but Y remains a serious problem.**

The author often emphasizes Y.

A common rule:

**Information after “but/however” often carries more argumentative weight.**

Not absolute, but useful.

## 51. OUTSIDE KNOWLEDGE TRAP

Passage:

**Mars has a thin atmosphere.**

Question options include a true fact about Mars not mentioned.

Do not choose based on astronomy knowledge unless inferable from passage.

RC universe = passage.

## 52. EXTREME OPTION TRAP

Be suspicious of:

- all
- none
- always
- never
- completely
- only
- must

unless passage clearly supports them.

## 53. HALF-TRUE OPTION

An option may begin correctly but end incorrectly.

Example:

**The study found improved memory and therefore proved the treatment works for everyone.**

First part true.

Second part too strong.

Whole option:

\[ \boxed{wrong} \]

---

## 54. DISTORTED CAUSE OPTION

Passage:

**A happened before B.**

Option:

**A caused B.**

Not necessarily.

Temporal order "→" causality.

---

## 55. REVERSED RELATION OPTION

Passage:

**High costs reduce adoption.**

Option:

**Low adoption causes high costs.**

May be possible but not stated.

Reject.

---

## 56. SCOPE SHIFT

Passage:

**Indian urban commuters...**

Option:

**all people worldwide...**

Too broad.

---

## 57. CATEGORY SHIFT

Passage:

**smartphones**

---



Option:

**all electronic devices**

Too broad.

## 58. TIME SHIFT

Passage discusses:

**current findings.**

Option:

**this has always been true.**

Unsupported.

## 59. DEGREE SHIFT

Passage:

**modest improvement**

Option:

**dramatic transformation**

Too strong.

## 60. NEGATION TRAP

Passage:

**Evidence is insufficient to prove X.**

Option:

**X is false.**

Wrong.

“Not proven” does not equal “false.”

## 61. ABSENCE OF EVIDENCE

$\boxed{\text{No evidence for X}} \neq \text{Evidence against X}$

Very important in RC and reasoning.

---

## 62. AUTHOR VIEW VS SOMEONE ELSE'S VIEW

---

Passage may quote another researcher.

Question:

**What does the author believe?**

Do not confuse quoted opinion with author's stance.

Look for:

- however
- the author argues
- this view overlooks
- nevertheless

---

## 63. EXAMPLE

---

**Some economists argue that the policy will increase growth. However, recent evidence suggests the effect may be limited.**

Author's stance:

$\boxed{\text{skeptical/cautious about strong growth claims}}$

Not the economists' optimistic view.

---

## 64. FACT VS OPINION

---

Fact:

**The study involved 500 people.**

Opinion:

**The study was poorly designed.**

RC may ask which is author's judgment.

---

## 65. AUTHOR'S ASSUMPTION — BASIC

---

An assumption is an unstated idea necessary for an argument.

Example:

**The city should add buses because current buses are overcrowded.**

Possible assumption:

**Additional buses can reduce overcrowding.**

We will go deeper in Chapter 76.

## 66. READ QUESTION STEM CAREFULLY

Different stems mean different proof levels:

### **According to passage**

directly stated.

### **Can be inferred**

supported implication.

### **Most likely suggests**

weaker inference.

### **Author would agree**

match author's stance.

### **Except**

find one unsupported/opposite option.

## 67. "EXCEPT" QUESTIONS

If asked:

**All of the following are true EXCEPT...**

You need the **wrong/unsupported** option.

Circle mentally:

$\boxed{\text{EXCEPT} = \text{find exception}}$

Many careless errors happen here.

## 68. NEGATIVE-STEM QUESTIONS

Watch:

- NOT
- EXCEPT
- LEAST

- CANNOT

Treat them as high-risk words.

## 69. FAST RC METHOD — SHORT GATE PASSAGE

Use:

$\boxed{\text{RMAP}}$

**R — Read structure**

**M — Mark main claim**

**A — Answer from text**

**P — Prove with passage**

For each option, ask:

Where is the support?

## 70. LONG PLACEMENT RC METHOD

Use:

$\boxed{\text{PQMAP}}$

**P — Preview questions**

**Q — Quick passage map**

**M — Main idea/tone**

**A — Answer targeted questions**

**P — Prove/eliminate**

## 71. WHEN TO RE-READ

Do not reread whole passage for every question.

Re-read only:

- relevant paragraph
- one sentence above/below
- conclusion if inference/tone question

## 72. RC TIME MANAGEMENT

For a medium passage:

- first read: understand structure, not memorize

- factual questions: answer fast
  - inference/tone questions: spend more time
  - difficult vocabulary: use context, don't obsess
- 

## 73. GUESSING UNKNOWN WORDS

---

Use:

1. sentence tone
2. contrast word
3. examples
4. root/prefix
5. nearby explanation

This connects Chapter 72.

---

## 74. SHORT PASSAGE — LEVEL 0

---

### PASSAGE 1

Many organizations have adopted flexible working arrangements. Employees can often choose when or where they perform some of their work. This flexibility can reduce commuting time and help workers manage personal responsibilities. However, remote work may also reduce informal interaction among colleagues. For this reason, some organizations are experimenting with hybrid models that combine remote and office work.

#### Q1

What is the main idea?

A. Remote work should completely replace offices. B. Flexible work offers benefits but can reduce informal interaction, encouraging hybrid models. C. Commuting is the greatest problem in modern society. D. Offices are unnecessary.

#### Q2

The author's tone is best described as:

A. angry B. balanced C. sarcastic D. pessimistic

#### Q3

Why are some organizations trying hybrid models?

A. To combine flexibility with opportunities for in-person interaction B. To eliminate all technology C. Because remote work has no benefits D. To increase commuting time

---

## 75. ANSWERS — PASSAGE 1

---

### Q1

The passage covers:

- benefits
  - drawback
-

- hybrid response

$\boxed{B}$

---

## Q2

Both sides discussed.

$\boxed{B:\text{balanced}}$

---

## Q3

Hybrid model responds to remote-work interaction limitation while keeping flexibility.

$\boxed{A}$

---

## 76. SHORT PASSAGE — LEVEL 1

---

### PASSAGE 2

Researchers often use correlation to identify relationships between variables. If two variables change together, they are said to be correlated. Yet correlation alone does not establish that one variable causes the other. A third factor may influence both, or the direction of causation may be reversed. Establishing causality therefore generally requires stronger evidence.

### Q4

What is the passage primarily about?

- A. Correlation proves causation. B. Correlation can reveal relationships but is insufficient by itself to prove causation. C. Research should avoid studying variables. D. Third variables always cause correlations.

### Q5

Which can be inferred?

- A. Every correlation is meaningless. B. Causal claims require more evidence than mere correlation. C. Correlation is always caused by a third factor. D. Experimental research is useless.

### Q6

“Yet” primarily serves to:

- A. provide an example B. introduce a contrast/limitation C. state a cause D. define correlation
- 

## 77. ANSWERS — PASSAGE 2

---

### Q4

$\boxed{B}$

---

### Q5

---

Directly supported by final sentence.

\[ \boxed{B} \]

---

## Q6

“Yet” shifts from what correlation shows to what it cannot establish.

\[ \boxed{B} \]

---

## 78. LEVEL 2 PASSAGE

---

### PASSAGE 3

Artificial intelligence systems increasingly assist doctors by identifying patterns in medical images. In some tasks, these systems have achieved impressive levels of accuracy. However, performance can vary when the patient population or equipment differs from the data used during development. Errors may also be difficult to detect if clinicians rely too heavily on automated recommendations. AI tools should therefore be evaluated not only for average accuracy but also for reliability across different clinical settings.

### Q7

Best main idea?

A. AI should replace doctors immediately. B. Medical AI can perform well, but reliability across settings is essential. C. Medical imaging is inaccurate. D. AI is unsuitable for healthcare.

### Q8

Which is most strongly implied?

A. High average accuracy alone may not be enough to establish clinical reliability. B. Every AI model fails on new patients. C. Doctors should never use AI. D. Medical equipment causes all AI errors.

### Q9

Author's tone?

A. blindly enthusiastic B. cautiously analytical C. sarcastic D. hostile

### Q10

What role does the final sentence play?

A. Introduces an unrelated example B. Gives the author's main recommendation/conclusion C. Defines AI D. Provides historical background

---

## 79. ANSWERS — PASSAGE 3

---

### Q7

Benefits + reliability qualification.

\[ \boxed{B} \]

---

**Q8**

The paragraph explicitly says performance can vary across settings.

Thus:

$\boxed{A}$

**Q9**

Acknowledges accuracy but highlights limits.

$\boxed{B:\text{cautiously}\text{ analytical}}$

**Q10**

It synthesizes earlier concerns into evaluation criterion.

$\boxed{B}$

## 80. LEVEL 3 PASSAGE

### PASSAGE 4

Cities often respond to road congestion by expanding road capacity. Initially, additional lanes may reduce travel times. However, easier driving can encourage more people to use cars, and developers may build farther from city centers when travel becomes temporarily faster. Over time, this additional demand can consume much of the newly created capacity. This phenomenon does not imply that road expansion is never useful, but it suggests that increasing capacity alone may not provide a lasting solution to congestion.

**Q11**

The passage's central claim is:

A. Road expansion always worsens congestion. B. Road expansion can initially help, but induced demand may reduce its long-term effectiveness. C. Cities should prohibit cars. D. Developers are the sole cause of congestion.

**Q12**

Which statement would the author most likely agree with?

A. Road expansion should never be used. B. Congestion may require measures beyond simply adding road capacity. C. Additional roads always reduce traffic permanently. D. Travel times cannot be changed by infrastructure.

**Q13**

Which inference is NOT supported?

A. New road capacity can initially reduce travel time. B. Easier driving may increase car use. C. Added capacity necessarily makes congestion worse than before in every city. D. Road expansion alone may fail to provide a permanent solution.

**Q14**

Why does the author say “does not imply that road expansion is never useful”?

A. To qualify the argument and avoid an extreme conclusion B. To contradict the entire passage C. To claim road expansion is always effective D. To introduce an unrelated topic



---

## 81. ANSWERS — PASSAGE 4

---

### Q11

---

The passage is explicitly balanced:

initial benefit + long-run demand effect.

\[ \boxed{B} \]

---

### Q12

---

Final sentence supports:

\[ \boxed{B} \]

---

### Q13

---

Passage never claims universal worsening.

\[ \boxed{C} \]

---

### Q14

---

Author prevents overinterpretation.

\[ \boxed{A} \]

This is a classic **qualification** move.

---

## 82. QUALIFIERS — VERY IMPORTANT

---

Words like:

- often
- generally
- sometimes
- may
- can
- many
- some

limit a claim.

Do not silently convert:

\[ some \rightarrow all \]

or:

\[ may \rightarrow must \]

---

---

## 83. QUALIFICATION PHRASES

---

Examples:

**This does not mean...**

**Although...**

**In some cases...**

**On average...**

**Evidence remains limited...**

These often reveal author nuance.

---

## 84. MAIN IDEA ELIMINATION METHOD

---

Reject options that are:

**N — Narrow**

only one detail

**B — Broad**

beyond passage

**E — Extreme**

stronger than passage

**O — Opposite**

contradicts author

Use:

$\boxed{\text{NBEO}}$

---

## 85. INFERENCE ELIMINATION METHOD

---

Reject options that require:

- outside information
  - causal leap
  - universalization
  - reversed logic
  - speculation
-

Correct inference usually feels like:

$\boxed{\text{one small logical step}}$

---

## 86. TONE ELIMINATION METHOD

---

Check actual language.

If passage uses:

- “may”
- “however”
- “evidence suggests”
- “limitations”

likely:

- cautious
- analytical
- balanced

not:

- furious
  - sarcastic
  - celebratory
- 

## 87. RC MASTER CHECKLIST

---

Before answering:

### Main idea

Whole passage or one detail?

### Factual

Can I point to exact text?

### Inference

Small logical step or speculation?

### Tone

What language shows attitude?

### Title

Covers whole passage?

### Vocabulary

Meaning in this sentence?

### Reference

What exactly does pronoun point to?

---

---

## 88. COMMON RC TRAPS

---

### & pTRAP 1 — Outside knowledge

Use passage only.

### & pTRAP 2 — Extreme option

Watch all/always/never.

### & pTRAP 3 — Half-true option

Whole option must be true.

### & pTRAP 4 — Cause from correlation

Do not infer without support.

### & pTRAP 5 — Topic instead of main idea

Main idea needs author's point.

### & pTRAP 6 — Too broad title

Stay within scope.

### & pTRAP 7 — Quote mistaken for author opinion

Identify speaker.

### & pTRAP 8 — “Not proven” treated as false

Wrong.

### & pTRAP 9 — Tone overdramatized

Prefer precise moderate label.

### & pTRAP 10 — NOT/EXCEPT missed

Read stem carefully.

---

## CHAPTER 75 STATUS

---

Covered so far:

- passage-reading strategy '
- passage mapping '
- main idea '
- central theme '
- author's purpose '
- factual/direct questions '
- inference '
- correlation/causation trap '
- tone '
- title basics '
- vocabulary in context '

- reference questions '
- paragraph function '
- scope/certainty traps '
- author vs quoted view '
- qualifiers '
- NBEO elimination '
- GATE-style short RC '
- placement-style inference practice '

But **Chapter 75 is not complete yet.**

Still remaining:

- advanced inference
- assumptions inside RC
- strengthen/weaken within passages
- best title in difficult passages
- paragraph-summary questions
- author's agreement/disagreement
- fact vs inference vs opinion
- comparative passages
- longer placement RC
- time-pressure elimination
- final mixed RC set

## Progress

\[ \boxed{74/76} \text{ numbered chapters completed} \]

**Current:** 0=36 Chapter 75 — Reading Comprehension

## NEXT

0=64 **Chapter 75 Part 2 — Advanced Inference, Assumptions, Strengthen/Weaken, Comparative RC, Long Passage Strategy & Final RC Toolkit**

USER

next

ASSISTANT

## CHAPTER 75 — READING COMPREHENSION

### PART 2 — ADVANCED INFERENCE, ASSUMPTIONS, STRENGTHEN/WEAKEN, COMPARATIVE RC, LONG-PASSAGE STRATEGY & FINAL RC TOOLKIT

**Status:** 0=64 PROGRESS ! ' COMPLETE **GATE:** Very High **Placements:** Very High **Priority:** 0=74 1

This part handles the hardest RC questions.

At this stage, the passage may not ask:

**“What did the author say?”**

Instead, it may ask:

- What must follow?
- What is assumed?
- Which option strengthens the author's claim?
- Which weakens it?
- What would the author most likely agree with?
- How are two passages different?

The most important rule remains:

$\boxed{\text{Stay inside the evidence}}$

## 1. ADVANCED INFERENCE

An inference is not directly quoted from the passage.

It is something that follows logically.

Think:

$\boxed{\text{Passage facts} \rightarrow \text{Safe conclusion}}$

Not:

$\boxed{\text{Passage facts} \rightarrow \text{Interesting guess}}$

## 2. THE “ONE SMALL STEP” RULE

Suppose passage says:

**Several employees reported improved concentration when working from home.**

Safe inference:

**Working from home may improve concentration for some employees.**

Unsafe inference:

**Remote work improves productivity for every employee.**

The second jumps too far.

## 3. MUST BE TRUE VS COULD BE TRUE

### Must be true

Required by passage.

### Could be true

Merely possible.

RC inference questions generally want:

$\boxed{\text{strongly supported}}$

not just possible.

---

## 4. INFERENCE BY COMBINATION

Passage:

**The company raised prices. Sales volume fell during the next quarter. Competitors did not change their prices.**

Can we infer price increase caused the decline?

Not necessarily.

But we can safely say:

**Sales declined after the company increased prices.**

Temporal relation is supported.

Causality needs more evidence.

---

## 5. INFERENCE FROM QUANTIFIERS

Passage:

**Most participants completed the task.**

Safe:

$\boxed{\text{More than half completed}}$

Unsafe:

$\boxed{\text{All completed}}$

---

## 6. “ONLY SOME” / “SOME” LOGIC

If passage says:

**Some employees opposed the change.**

Can infer:

$\boxed{\text{At least one employee opposed}}$

Cannot infer:

- most opposed
- exactly some fixed percentage opposed
- everyone else supported it

---

## 7. NEGATIVE INFERENCE

Passage:

**None of the tested samples exceeded the safety threshold.**

Safe:

**All tested samples were at or below the threshold.**

Do not extend to untested samples.

---

## 8. BOUNDARY OF SAMPLE

Passage:

**All 50 tested devices passed.**

Cannot infer:

**Every device manufactured by the company will pass.**

Sample/population trap.

---

## 9. ADVANCED ASSUMPTION IN RC

An assumption is an unstated bridge required for an argument.

Structure:

$\boxed{\text{Premise} + \text{Assumption} \rightarrow \text{Conclusion}}$

---

## 10. EXAMPLE

Passage:

**City buses are overcrowded. Therefore, the city should add more buses.**

Possible assumption:



**Adding buses can increase passenger-carrying capacity and reduce overcrowding.**

Without that, recommendation weakens.

## 11. NEGATION TEST IN RC

To test assumption:

1. negate option
2. see whether argument still works

If argument collapses:

$\boxed{\text{Likely assumption}}$

## 12. EXAMPLE

Argument:

**The company should introduce remote work because commuting consumes significant employee time.**

Candidate assumption:

**Remote work can reduce commuting time.**

Negate:

**Remote work cannot reduce commuting time.**

Then argument collapses.

So assumption is necessary.

## 13. NOT EVERY SUPPORTING FACT IS AN ASSUMPTION

Argument:

**Sales declined, so the firm should change strategy.**

Option:

**The CEO dislikes the current strategy.**

May support action psychologically, but not logically required.

Not assumption.

## 14. STRENGTHEN QUESTIONS

---

A strengthening option makes the argument more believable.

It may:

- support causal link
  - remove an alternative explanation
  - increase sample reliability
  - show recommendation works
  - provide additional evidence
- 

## 15. STRENGTHEN — CAUSAL ARGUMENT

---

Claim:

**New training improved employee performance.**

Strongener:

**Performance improved substantially in trained employees but not in a similar untrained group.**

Why strong?

It controls an alternative explanation.

---

## 16. WEAKEN — CAUSAL ARGUMENT

---

Same claim.

Weakener:

**A new productivity bonus was introduced at the same time.**

This introduces another possible cause.

---

## 17. STRENGTHEN BY REPLICATION

---

Study claims method works.

Strongener:

**Independent teams reproduced the result.**

This increases confidence.

---

## 18. WEAKEN BY SAMPLE BIAS

---

Study claims:

---

**Students prefer online exams.**

Weakeners:

**Participants were recruited only from an online-learning club.**

Sample may be unrepresentative.

## 19. WEAKEN BY REVERSE CAUSATION

Claim:

**People who exercise more are happier; therefore exercise causes happiness.**

Weakeners:

**Happier people may be more likely to exercise.**

Reverse direction.

## 20. WEAKEN BY THIRD VARIABLE

Claim:

**People who own books perform better academically.**

Weakeners:

**Household income may influence both book ownership and academic opportunities.**

Confounding.

## 21. STRENGTHEN/WEAKEN MUST TARGET THE ARGUMENT

Irrelevant fact:

**Exercise has existed for centuries.**

Does not strengthen causal claim about exercise and mood.

Always ask:

$\boxed{\text{Does this option affect the conclusion?}}$

## 22. AUTHOR WOULD MOST LIKELY AGREE

This question tests author's stance.

Do:

1. identify central claim
2. identify tone/qualifiers
3. choose option consistent with both

---

## 23. EXAMPLE

Author says:

**AI can improve diagnosis but requires human oversight.**

Likely agreement:

**AI should be used as a support tool rather than assumed infallible.**

Not:

**AI should never be used.**

Not:

**AI should replace all doctors.**

---

## 24. AUTHOR WOULD DISAGREE

Look for option contradicting author's position.

Be careful:

An option can discuss same topic but go beyond author's certainty.

---

## 25. FACT VS INFERENCE VS OPINION

### Fact

directly verifiable statement.

### Inference

derived from facts.

### Opinion

judgment/value claim.

Example:

**The trial included 100 people. !' Fact**

The sample was probably too small for universal conclusions. !' Inference/evaluation

The study was excellent. !' Opinion

## 26. AUTHOR'S JUDGMENT WORDS

Watch:

- unfortunately
- importantly
- surprisingly
- inadequate
- promising
- excessive
- reasonable

These reveal author attitude.

## 27. PARAGRAPH SUMMARY

A good paragraph summary:

- captures main claim
- includes necessary qualification
- excludes detail/examples
- avoids new conclusions

## 28. SUMMARY TRAP — DETAIL

Paragraph contains:

- economic benefit
- environmental benefit
- implementation cost

Summary:

Implementation cost is <sup>1</sup> 5million.

Too narrow.

## 29. SUMMARY TRAP — OVERGENERALIZATION

Paragraph:

**This specific program improved attendance in two schools.**

Bad summary:

**Education reforms always improve attendance.**

Too broad.

## 30. SUMMARY METHOD

Ask:

**If I could preserve only one sentence from the paragraph, what should it say?**

That is close to the summary.

## 31. BEST TITLE — ADVANCED

A title should capture:

$\boxed{\text{Topic} + \text{angle}}$

Passage:

**benefits and risks of facial recognition.**

Better:

**Facial Recognition: Utility and Privacy Concerns**

Worse:

**Technology**

Worse:

**Facial Recognition Is Dangerous**

unless passage argues only that.

## 32. METAPHORICAL TITLES

Sometimes placement tests offer creative titles.

Use them only if they match the central theme precisely.

Do not choose a poetic title merely because it sounds interesting.

---

### 33. COMPARATIVE PASSAGES

You may get Passage A and Passage B.

Questions may ask:

- where they agree
  - where they disagree
  - difference in tone
  - how B responds to A
- 

### 34. COMPARATIVE RC METHOD

Create a mini-table:

	Passage A	Passage B
Main claim		
Tone		
Evidence		
Conclusion		

Then answer relationships.

---

### 35. AGREEMENT QUESTION

Passage A:

**Remote work increases flexibility.**

Passage B:

**Although remote work has limitations, flexibility is a major benefit.**

Both agree:

$\boxed{\text{Remote work can improve flexibility}}$

---

### 36. DISAGREEMENT QUESTION

A:

**Full remote work should become standard.**

B:

---

**Hybrid work is preferable because offices retain social benefits.**

Disagreement:

$\boxed{\text{whether fully remote work should replace office-based work}}$

## 37. SAME FACT, DIFFERENT INTERPRETATION

A and B can cite the same data but draw different conclusions.

Thus:

**Same evidence "same argument.**

## 38. LONG PLACEMENT RC — DON'T MEMORIZE DETAILS

For long passages, your first read should capture:

- subject
- paragraph role
- author's stance
- major contrast

You can return for details later.

## 39. LONG-PASSAGE MAP

Example:

### TEXT

P1 – Introduces problem  
P2 – Conventional explanation  
P3 – Criticism of conventional view  
P4 – New evidence  
P5 – Author conclusion

Now even a 700-word passage becomes manageable.

## 40. KEYWORD ANCHORS

For factual questions, remember locations:

- study names
- dates
- examples
- technical terms

Do not memorize numbers unless needed.



## 41. SKIMMING VS SCANNING

### Skimming

Read quickly for structure/main idea.

### Scanning

Search for specific fact/keyword.

RC needs both.

## 42. FIRST READ = SKIM FOR STRUCTURE

Second targeted read = scan for evidence.

This is usually faster than deep-reading every line first.

## 43. DIFFICULT SENTENCE METHOD

If a sentence is long:

1. find subject
2. find main verb
3. ignore examples/inserted clauses
4. identify connector
5. restate simply

## 44. EXAMPLE

**Although automation, particularly in industries characterized by repetitive tasks, has often been associated with fears of job displacement, its actual effect on employment depends heavily on how firms reorganize work.**

Simplify:

**Automation may displace jobs, but employment effect depends on reorganization.**

That's enough for RC.

## 45. AUTHOR'S POSITION MAY EMERGE LATE

Some passages begin by describing a common view and then reject it.

Example:

**It is often claimed that X. However, recent evidence suggests Y.**

Do not assume opening claim is author's view.

---

## 46. “SOME ARGUE...” WARNING

---

Statements attributed to others:

- some researchers claim
- critics argue
- proponents believe

may not be author's stance.

Wait for author's response.

---

## 47. CONCESSION STRUCTURE

---

Pattern:

**Although X, Y.**

Usually author grants X but emphasizes Y.

Example:

**Although the technology is promising, current evidence remains limited.**

Main emphasis:

$\boxed{\text{evidence remains limited}}$

---

## 48. “WHILE” COMPARISON

---

**While costs have fallen, reliability remains uncertain.**

Both true, but latter may be author's concern.

---

## 49. RC OPTION ELIMINATION — THE 6 BIG TRAPS

---

Use:

$\boxed{\text{BESORT}}$

**B — Broad**

too general

**E — Extreme**

too strong

**S — Scope shift**

wrong population/time/category

---

## **O — Opposite**

contradicts passage

## **R — Reversed**

cause/order/relationship reversed

## **T — True but irrelevant**

factually plausible but unsupported

---

## **50. “TRUE BUT IRRELEVANT” IS VERY DANGEROUS**

---

Option:

**Exercise improves cardiovascular health.**

Passage asks about:

**exercise and memory.**

Even if true, irrelevant.

Reject.

---

## **51. “PARAPHRASE” CORRECT OPTIONS**

---

Correct answers often use different words from passage.

Passage:

**adoption remained sluggish.**

Option:

**uptake was slow.**

Same meaning.

Do not reject because wording differs.

---

## **52. NEGATIVE QUESTION STEM**

---

Examples:

**Which is NOT supported?**

**Which CANNOT be inferred?**

All are true EXCEPT?

Write mentally:

$\boxed{\text{Find wrong option}}$

This prevents reversal mistakes.

## 53. TIME PRESSURE — QUESTION ORDER

For long RC:

1. factual/reference questions
2. vocabulary in context
3. main idea
4. inference
5. tone/author stance
6. difficult strengthen/weaken

This may vary, but factual questions usually provide quick marks.

## 54. DO NOT SPEND TOO LONG ON ONE QUESTION

If two options remain:

- locate passage evidence
- compare certainty/scope
- choose safer supported option

Do not reread entire passage repeatedly.

## 55. PASSAGE 1 — ADVANCED INFERENCE

### PASSAGE

A company introduced a four-day workweek without reducing employees' salaries. Six months later, average absenteeism had fallen and employee satisfaction scores had increased. Output per employee remained roughly unchanged. However, the company also reorganized teams and introduced new scheduling software during the same period.

### Q1

Which conclusion is best supported?

A. The four-day workweek definitely caused higher satisfaction. B. Employees became less productive. C. The company observed higher satisfaction and lower absenteeism after several workplace changes. D. Scheduling software caused all improvements.

### Q2

Why can the evidence not establish that the four-day week alone caused the changes?

A. Employees received salaries. B. Other changes occurred simultaneously. C. Output did not increase. D. Six months

is always too long.

### Q3

Which would most strengthen the claim that the four-day week itself improved satisfaction?

A. Similar improvements occurred in comparable teams that changed only to a four-day week. B. The company bought new computers. C. Employees worked in an office building. D. Competitors had five-day weeks.

---

## 56. ANSWERS — PASSAGE 1

---

### Q1

Safest description:

\[ \boxed{C} \]

No causal overclaim.

### Q2

Confounding variables:

\[ \boxed{B} \]

### Q3

Controls other explanations:

\[ \boxed{A} \]

---

## 57. PASSAGE 2 — ASSUMPTION / WEAKEN

---

### PASSAGE

A city council proposes charging private cars that enter the busiest downtown area during peak hours. Supporters argue that this will reduce congestion because some drivers will shift to public transport.

### Q4

Which assumption is most important?

A. At least some drivers can and will change travel mode in response to the charge. B. Every resident owns a car. C. Public transport is free. D. Traffic never occurs outside peak hours.

### Q5

Which most weakens the argument?

A. Most peak-hour drivers have no practical public-transport alternative and are unlikely to change travel behavior. B. The city has several parks. C. Some buses are blue. D. Congestion is unpopular.

---

## 58. ANSWERS — PASSAGE 2

---

### Q4

Argument depends on behavioral response:

---

$\boxed{A}$

### Q5

If drivers cannot shift mode, mechanism fails:

$\boxed{A}$

---

## 59. PASSAGE 3 — AUTHOR AGREEMENT

### PASSAGE

Online courses can expand access to education, particularly for learners who cannot relocate or attend fixed schedules. Yet access alone does not guarantee effective learning. Course design, feedback, learner motivation, and technical support all influence outcomes. The relevant question, therefore, is not whether online learning is inherently superior or inferior, but under what conditions it works well.

### Q6

Author would most likely agree:

A. Online learning is always inferior. B. Access is irrelevant. C. Effectiveness depends on implementation conditions. D. All learners benefit equally.

### Q7

Best tone:

A. dogmatic B. analytical and balanced C. sarcastic D. hostile

### Q8

Best title:

A. Why Online Learning Must Replace Universities B. Online Learning: Conditions for Effective Use C. The History of Computers D. Online Courses Never Work

---

## 60. ANSWERS — PASSAGE 3

### Q6

$\boxed{C}$

### Q7

$\boxed{B}$

### Q8

$\boxed{B}$

---

## 61. COMPARATIVE PASSAGE — LEVEL 3

### Passage A

Large language models can accelerate drafting, summarization, and information retrieval. Their productivity benefits are therefore likely to make them increasingly common in knowledge work. Organizations should focus on adopting

these tools quickly so they do not fall behind competitors.

### Passage B

Large language models can improve productivity, but their outputs may contain errors or unsupported claims. Rapid adoption without verification procedures can therefore create new risks. Organizations should adopt them selectively while building review and accountability mechanisms.

### Q9

Where do both passages agree?

A. Language models have no productivity value. B. Language models can improve productivity. C. Organizations should ban language models. D. Verification is unnecessary.

### Q10

Main disagreement?

A. Whether language models exist B. Whether they can produce text C. How rapidly and cautiously organizations should adopt them D. Whether knowledge work exists

### Q11

Passage B's tone relative to A is:

A. more cautious B. more enthusiastic C. more sarcastic D. completely unrelated

---

## 62. ANSWERS — COMPARATIVE PASSAGE

### Q9

\[ \boxed{B} \]

### Q10

\[ \boxed{C} \]

### Q11

\[ \boxed{A} \]

---

## 63. LONGER PLACEMENT-STYLE PASSAGE

### PASSAGE

For many years, economic growth was treated as closely linked to increasing material consumption. As incomes rose, households often bought more goods, firms expanded production, and resource use increased. This relationship led some observers to argue that environmental pressure could not be reduced without sacrificing economic prosperity.

More recently, however, researchers have examined whether economic activity can become less resource-intensive. Improvements in energy efficiency, shifts toward service industries, recycling, and cleaner technologies can reduce the amount of material or energy required for each unit of economic output. In some countries, certain environmental indicators have improved even while gross domestic product continued to rise.

These trends are sometimes described as evidence of “decoupling” economic growth from environmental harm. Yet the concept requires careful interpretation. A country may reduce emissions within its own borders while importing goods whose production generates emissions elsewhere. Likewise, improvements in efficiency can be partly offset if lower costs encourage greater consumption.

Decoupling is therefore not an all-or-nothing phenomenon. Evidence that some environmental pressures can decline while economies grow is important, but it does not show that all forms of environmental damage will automatically fall. Assessing progress requires attention to which indicator is being measured, where impacts occur, and whether reductions are absolute or merely relative to economic output.

---

## 64. QUESTIONS

---

### Q12

Main idea?

A. Economic growth always increases environmental damage. B. Decoupling is possible in some respects, but claims about it require careful measurement and qualification. C. Service industries cause no environmental harm. D. Economic growth should stop immediately.

### Q13

Why does the author mention imported goods?

A. To show that domestic environmental improvements may hide impacts occurring elsewhere. B. To argue imports should be banned. C. To explain currency exchange. D. To show manufacturing is impossible.

### Q14

What does “not an all-or-nothing phenomenon” imply?

A. Decoupling either happens perfectly or not at all. B. Different kinds/degrees of decoupling may occur. C. Economic growth is irrelevant. D. Environmental measurement is useless.

### Q15

Which would the author most likely reject?

A. Some environmental indicators can improve while GDP rises. B. Measuring only domestic emissions may give an incomplete picture. C. Efficiency improvements necessarily reduce total environmental pressure in every case. D. Different environmental indicators may show different trends.

### Q16

Tone?

A. analytical and cautious B. celebratory C. sarcastic D. dismissive

### Q17

Best summary?

A. Economic growth and environmental damage are completely unrelated. B. Some evidence supports decoupling, but its extent depends on definitions, indicators, locations, and behavioral responses. C. Imports are the only environmental problem. D. Efficiency has no value.

---

## 65. ANSWERS — LONG PASSAGE

---

### Q12

The entire final paragraph gives the qualification.

\boxed{B}



### Q13

Imported production can shift impacts geographically.

\[ \boxed{A} \]

### Q14

Decoupling may happen partially/by some indicators.

\[ \boxed{B} \]

### Q15

“Naturally/necessarily in every case” contradicts rebound-effect warning.

\[ \boxed{C} \]

### Q16

\[ \boxed{A} \]

### Q17

\[ \boxed{B} \]

---

## 66. RC FINAL SPEED SYSTEM

Use:

\[ \boxed{\text{MAP-E}} \]

### M — Map passage

What does each paragraph do?

### A — Author

What is author's stance?

### P — Proof

Where is textual support?

### E — Eliminate

Broad/extreme/scope/reversed/irrelevant options.

---

## 67. MAIN IDEA TOOLKIT

Use:

\[ \boxed{\text{Topic + Central claim + Qualification}} \]

Example:

**AI has benefits + risks + needs oversight.**

Do not reduce to only:

## 68. INFERENCE TOOLKIT

Correct inference should be:

- passage-based
- conservative
- one logical step
- same scope
- same certainty

## 69. ASSUMPTION TOOLKIT

Use:

$\neg \boxed{\text{Negation Test}}$

If negating option destroys argument:

likely assumption.

## 70. STRENGTHEN TOOLKIT

Look for:

- more evidence
- better control
- mechanism
- representative sample
- replication
- removal of alternative explanation

## 71. WEAKEN TOOLKIT

Look for:

- alternative cause
- reverse causation
- biased sample
- contradictory evidence
- flawed measurement
- implementation failure
- exception undermining universal claim

## 72. TONE TOOLKIT

---

Most common exam tones:

- objective
- analytical
- balanced
- cautious
- critical
- skeptical
- optimistic
- concerned

Extreme emotions are less common.

---

## 73. TITLE TOOLKIT

---

Reject title if:

- too broad
  - too narrow
  - extreme
  - focuses on example only
  - gives stance stronger than passage
- 

## 74. COMPARATIVE RC TOOLKIT

---

For each passage write mentally:

$\boxed{\text{Claim} + \text{Tone} + \text{Recommendation}}$

Then compare.

---

## 75. 10 MOST COMMON RC ERRORS

---

1. Using outside knowledge.
  2. Confusing topic with main idea.
  3. Choosing possible instead of supported inference.
  4. Correlation != causation.
  5. Some != all.
  6. May != must.
  7. Quoted view != author's view.
  8. Ignoring NOT/EXCEPT.
  9. Falling for half-true option.
  10. Spending too long rereading whole passage.
-

## 76. 15-SECOND OPTION CHECK

Before selecting an RC answer:

Ask:

- supported?
- same scope?
- same certainty?
- author's view?
- any extreme word?
- any causal leap?

If an option fails one badly, eliminate it.

## 77. FINAL MIXED PRACTICE

### Q18

Passage:

**A new tutoring program improved average test scores in one school. However, the school also hired additional teachers during the same term.**

Which is safest?

A. Tutoring definitely caused improvement. B. Additional teachers definitely caused improvement. C. Scores improved during a period in which both tutoring and staffing changed. D. Tutoring had no effect.

Answer:

\[ \boxed{C} \]

### 78. Q19

Passage:

**A medication reduced symptoms in 70% of participants, but the study had no control group.**

Best inference:

A. Medication definitely caused improvement. B. Treatment appears promising, but causality cannot be firmly established from this study alone. C. Medication is useless. D. All patients will improve.

\[ \boxed{B} \]

### 79. Q20

Passage:

**Some employees prefer flexible hours, while others value fixed schedules.**

What follows?

A. Everyone prefers flexibility. B. Preferences differ across employees. C. Fixed schedules are universally superior. D. No one likes flexible hours.

\[ \boxed{B} \]

---

## ' CHAPTER 75 COMPLETE

---

We have now covered:

- efficient passage reading '
- passage mapping '
- main idea '
- central theme '
- author's purpose '
- factual questions '
- advanced inference '
- quantifier/scope inference '
- tone '
- title '
- vocabulary in context '
- reference questions '
- paragraph function '
- author vs quoted view '
- assumptions in RC '
- strengthen/weaken '
- fact/inference/opinion '
- paragraph summary '
- author's agreement/disagreement '
- comparative RC '
- long placement passages '
- skimming/scanning '
- certainty/scope traps '
- BESORT elimination '
- MAP-E final toolkit '
- L0–L3 practice '

## Progress

---

\[ \boxed{75/76} \text{ numbered chapters completed} \]

## UNIT 4 — VERBAL APTITUDE

- Chapter 71 — Grammar for Aptitude '
- Chapter 72 — Vocabulary & Word Usage '
- Chapter 73 — Sentence Completion, Correction & Error Detection '

- Chapter 74 — Para Jumbles & Paragraph Completion '
- Chapter 75 — Reading Comprehension '
- Chapter 76 — Critical Reasoning & Verbal Deduction +

Only **one numbered chapter remains**.

## NEXT

### CHAPTER 76 — CRITICAL REASONING & VERBAL DEDUCTION

We will finish the numbered syllabus with:

- argument structure
- premise/conclusion
- assumptions
- strengthen/weaken
- inference
- evaluate-the-argument
- flaw questions
- paradox/resolution
- analogy-based reasoning
- necessary vs sufficient evidence
- causal reasoning
- author agreement
- verbal deduction
- GATE-style logical language
- placement-style CR sets
- final Unit 4 revision

After Chapter 76, we still continue with:

- **Unit 5 — GATE Special Training**
- **Unit 6 — Placement Special Training**
- **Unit 7 — Revision & Question Bank**
- **Unit 8 — Full Mocks**

USER

next

ASSISTANT

## CHAPTER 76 — CRITICAL REASONING & VERBAL DEDUCTION

### PART 1 — ARGUMENT STRUCTURE, PREMISE/CONCLUSION, ASSUMPTIONS, STRENGTHEN/WEAKEN, INFERENCE & CAUSAL REASONING

Status: 0% PROGRESS GATE: Very High Placements: Very High Priority: 0% Per 1

This is the **final numbered chapter** of the roadmap.

Critical Reasoning asks:

**Given some facts and an argument, what logically follows?**

It combines:

- assumptions
- conclusions
- cause-effect
- inference
- strengthen/weaken
- logical language
- evidence quality

The golden rule is:

$\boxed{\text{Separate what is GIVEN from what is CLAIMED}}$

## 1. WHAT IS AN ARGUMENT?

An argument usually contains:

### Premise

Evidence/reason.

### Conclusion

What the author wants you to accept.

Structure:

$\boxed{\text{Premise} \rightarrow \text{Conclusion}}$

Example:

**The road is flooded, so the school should remain closed.**

Premise:

$\boxed{\text{Road is flooded}}$

Conclusion:

$\boxed{\text{School should remain closed}}$

## 2. CONCLUSION SIGNAL WORDS

Watch for:

- therefore
- thus
- hence

- so
- consequently
- therefore we should
- this shows that
- it follows that

Example:

**Demand has fallen; therefore, production should be reduced.**

Conclusion follows “therefore.”

### 3. PREMISE SIGNAL WORDS

Watch:

- because
- since
- given that
- due to
- as
- for
- considering that

Example:

**The system should be upgraded because failures have increased.**

Premise:

$\boxed{\text{Failures increased}}$

Conclusion:

$\boxed{\text{System should be upgraded}}$

### 4. NOT EVERY SENTENCE IS PART OF THE ARGUMENT

Some sentences provide:

- background
- definitions
- examples

Example:

**The company operates in Hyderabad. Its servers failed six times last month. Therefore, the infrastructure should be reviewed.**

“Hyderabad” may just be background.

The real reasoning is:



## 5. ASSUMPTION

An assumption is an unstated belief required for the argument.

Example:

**The company should improve training because employee errors are increasing.**

Likely assumption:

**Better training can reduce at least some employee errors.**

## 6. NEGATION TEST

To identify necessary assumption:

1. take candidate assumption
2. negate it
3. see whether argument collapses

Candidate:

**Training can reduce employee errors.**

Negation:

**Training cannot reduce employee errors.**

Then recommendation loses its basis.

So:

[ \boxed{\text{Necessary assumption}} ]

## 7. ASSUMPTION "CONCLUSION

Example:

**Sales declined, so the company should advertise more.**

Conclusion:

**advertise more.**

Assumption:

**advertising can help sales.**

Do not confuse hidden bridge with final claim.

## 8. STRONG VS WEAK ASSUMPTION

If argument needs:

**Some customers will respond to lower prices.**

Do not choose stronger:

**Every customer will respond.**

Critical reasoning prefers the **minimum necessary assumption**.

## 9. STRENGTHEN QUESTION

A statement strengthens if it makes the conclusion more likely.

Example:

Argument:

**Flexible hours improve employee retention.**

Stronger:

**Companies adopting flexible hours showed lower resignation rates than similar companies without them.**

## 10. WEAKEN QUESTION

A statement weakens if it makes conclusion less convincing.

Same argument.

Weakener:

**Companies offering flexible hours also raised salaries substantially.**

Alternative cause.

## 11. STRENGTHEN DOES NOT MEAN “RELATED”

Argument:

**New training improved productivity.**

Option:

**The company has 500 employees.**

Related to company, but irrelevant.

A strengthening statement must affect:

$\boxed{\text{Premise}} \rightarrow \text{Conclusion link}$

## 12. WEAKEN DOES NOT NEED TO DISPROVE

To weaken:

**make the argument less convincing.**

You do not need to prove conclusion false.

Example:

**A new drug improved recovery.**

Weakeners:

**Patients also received another new treatment.**

This doesn't prove drug ineffective; it introduces uncertainty.

## 13. CAUSAL ARGUMENTS

Common form:

$A \text{ happened, then } B \text{ happened}$

Conclusion:

$A \text{ caused } B$

This is vulnerable.

## 14. CAUSAL WEAKENERS

Four major ones:

### 1. Alternative cause

C may cause B.

### 2. Reverse causation

B may cause A.

### 3. Coincidence

Association may be accidental.

### 4. Measurement/sample problem

Observed effect may not be real/generalizable.

---

## 15. ALTERNATIVE CAUSE

Claim:

**Productivity rose after a new software tool was introduced.**

Weakeners:

**Employee bonuses also increased that month.**

Maybe bonuses caused productivity change.

---

## 16. REVERSE CAUSATION

Claim:

**People who read more earn more; therefore reading causes higher income.**

Weakeners:

**Higher-income people may have more time/resources to read.**

Direction may be reversed.

---

## 17. COMMON CAUSE

Claim:

**Ice cream sales and drowning incidents both increase in summer.**

Not:

\[ Ice\ cream\ causes\ drowning \]

Common cause:

\[ \boxed{Hot\ weather} \]

## 18. POST HOC FALLACY

---

Pattern:

**A happened before B, therefore A caused B.**

Called:

$\boxed{\text{post hoc}}$

Temporal order is necessary for causation but not sufficient.

---

## 19. STRONG CAUSAL EVIDENCE

---

Causal argument becomes stronger if:

- control group exists
  - other factors held constant
  - mechanism known
  - effect repeats
  - dose-response exists
  - temporal order clear
  - random assignment used
- 

## 20. CONTROL GROUP

---

Suppose:

Group A receives treatment.

Group B doesn't.

All else similar.

If A improves more:

causal inference becomes stronger.

---

## 21. RANDOMIZATION

---

Random assignment helps balance hidden differences.

Thus strengthens causal studies.

---

## 22. REPLICATION

---

If independent studies find same result:

confidence increases.

---

## 23. DOSE-RESPONSE

If more exposure tends to produce larger effect:

supports causal mechanism.

Example:

**Increasing training hours consistently improved scores.**

## 24. INFERENCE QUESTIONS

Inference asks:

**Which statement follows from the given information?**

Correct inference must be:

- supported
- conservative
- within scope
- not stronger than evidence

## 25. EXAMPLE

Premises:

**All engineers in Team A know Python. Ravi is an engineer in Team A.**

Inference:

$\boxed{\text{Ravi knows Python}}$

## 26. INVALID INFERENCE

Premise:

**Some employees work remotely.**

Cannot infer:

**Most employees work remotely.**

## 27. QUANTIFIER DISCIPLINE

Remember:

## All

100%

## Most

more than half

## Some

at least one

## None

zero

Do not upgrade strength.

---

## 28. “NOT ALL”

---

$\boxed{\text{Not all A are B}}$

means:

$\boxed{\text{Some A are not B}}$

Important verbal deduction rule.

---

## 29. “NOT NONE”

---

$\neg(\text{No A are B})$

means:

$\boxed{\text{Some A are B}}$

---

## 30. NECESSARY VS SUFFICIENT

---

If:

$P \rightarrow Q$

Then:

- P sufficient for Q
- Q necessary for P

Example:

**If candidate gets rank 1, candidate qualifies.**

Rank 1 sufficient.

Qualification necessary consequence of rank 1.

---

## 31. “ONLY IF”

---

**A happens only if B happens.**

$\boxed{A \rightarrow B}$

Example:

**You can enter only if you have an ID.**

Entry !' ID.

## 32. "IF"

**A happens if B happens.**

$\boxed{B \rightarrow A}$

## 33. "UNLESS"

**A unless B**

usually:

$\boxed{\neg B \rightarrow A}$

Equivalent:

$\boxed{A \vee B}$

## 34. CONCLUSION-STRENGTH MATCH

Premise:

**Sales rose 5%.**

Conclusion:

**The strategy produced a modest improvement.**

Possible.

But:

**The strategy transformed the company completely.**

Too strong.



## 35. EVIDENCE STRENGTH

As conclusion becomes stronger, required evidence must become stronger.

$\boxed{\text{Stronger claim} \rightarrow \text{Stronger evidence}}$

## 36. EVALUATE-THE-ARGUMENT QUESTIONS

These ask:

**Which information would be most useful in judging the argument?**

Look for the key uncertainty.

Example:

**Company claims new ad campaign increased sales because sales increased after campaign.**

Best evaluation question:

**Did sales also rise in comparable markets where campaign was not used?**

This tests causal link.

## 37. EVALUATION= ASK WHAT WOULD CHANGE YOUR CONFIDENCE

Best evaluate question is one where:

- YES strongly supports
- NO strongly weakens

## 38. EXAMPLE

Argument:

**Opening a new bus route will reduce car traffic.**

Best evaluation:

**Are many current car commuters able and willing to use the new route?**

If yes !' stronger.

If no !' weaker.

## 39. FLAW QUESTIONS

You are asked:

## What is wrong with the reasoning?

Common flaws:

- correlation = causation
- hasty generalization
- false dilemma
- circular reasoning
- appeal to authority
- personal attack
- weak analogy
- sampling bias
- necessary/sufficient confusion

## 40. HASTY GENERALIZATION

Example:

**Two employees disliked remote work; therefore employees dislike remote work.**

Tiny sample !' huge conclusion.

## 41. SAMPLING BIAS

Survey asks:

**Should the gym expand?**

Only surveys regular gym users.

Cannot generalize to entire city.

Sample not representative.

## 42. FALSE DILEMMA

**Either we ban smartphones in class completely or students will never learn.**

Only two options presented when alternatives exist.

\[ \boxed{\text{False}} \text{ dilemma} \]

## 43. CIRCULAR REASONING

**This policy is good because it is an excellent policy.**

Conclusion restated as premise.

No independent support.

---

## 44. APPEAL TO AUTHORITY

**A famous actor says this medicine works, so it must work.**

Authority may not have relevant expertise/evidence.

---

## 45. AD HOMINEM

**Ignore Ravi's argument about taxation because Ravi is lazy.**

Attacks person, not reasoning.

---

## 46. APPEAL TO POPULARITY

**Everyone uses this app, therefore it is the best.**

Popularity does not prove quality.

---

## 47. WEAK ANALOGY

**A company is like a family, so managers should make all employee decisions exactly as parents decide for children.**

Analogy is too weak/incomplete.

---

## 48. PART-WHOLE FALLACY

**Every component is cheap, therefore entire system is cheap.**

Maybe integration and other costs matter.

---

## 49. WHOLE-PART FALLACY

**Company is profitable, so every division is profitable.**

Not necessarily.

---

---

## 50. PARADOX / RESOLVE THE DISCREPANCY

---

A paradox question gives two facts that seem inconsistent.

Example:

**Store reduced prices, yet revenue fell.**

Question:

**Which fact best explains this?**

Possible resolver:

**Sales volume fell sharply despite lower prices.**

No contradiction anymore.

---

## 51. PARADOX METHOD

---

Do not attack either fact.

Find an additional fact that allows both to be true.

$\boxed{\text{Fact1} + \text{NewFact} + \text{Fact2} \mid \text{all} \mid \text{coexist}}$

---

## 52. EXAMPLE

---

Fact 1:

**New road shortened travel distance.**

Fact 2:

**Average travel time increased.**

Possible explanation:

**Traffic volume increased dramatically.**

Both can be true.

---

## 53. STRENGTHEN VS RESOLVE

---

Strengthen asks:

**make conclusion stronger.**

Resolve asks:

**explain surprising coexistence.**

Different tasks.

## 54. COUNTEREXAMPLE

To disprove universal claim:

**All A are B.**

You need one case:

$\neg [A \wedge \neg B]$

That is enough.

## 55. EXAMPLE

Claim:

**All successful founders are extroverts.**

One successful introverted founder weakens/disproves universal claim.

## 56. ARGUMENT BY ANALOGY

Structure:

**A and B are similar in X,Y, therefore likely similar in Z.**

Strength depends on:

- relevance of similarities
- absence of important differences

## 57. STRONG ANALOGY

Two processors:

- same architecture
- same clock
- same cache
- same workload

Predict similar performance.

Reasonable.

---

## 58. WEAK ANALOGY

Two companies both have blue logos, therefore their profits will be similar.

Similarity irrelevant.

---

## 59. VERBAL DEDUCTION WITH CONDITIONALS

Given:

**If the server overheats, it shuts down.**

$\neg O \rightarrow \neg S$

If it did not shut down:

$\neg S \rightarrow \neg O$

Valid contraposition.

---

## 60. INVALID CONVERSE

Given:

$O \rightarrow S$

Server shut down.

Cannot conclude overheating.

There may be another cause.

---

## 61. DISJUNCTIVE REASONING

Given:

$A \vee B$

and:

$\neg A$

then:

$\boxed{B}$

Valid.

---

## 62. XOR REASONING

Exactly one of A,B.

If A true:

$\boxed{B=false}$

If A false:

$\boxed{B=true}$

---

## 63. CHAINED CONDITION

---

$A \rightarrow B$

$B \rightarrow C$

Then:

$\boxed{A \rightarrow C}$

---

## 64. CHAINED CONTRAPOSITION

---

From:

$A \rightarrow B \rightarrow C$

If:

$\neg C$

then:

$\boxed{\neg B, \neg A}$

---

## 65. “MUST BE TRUE” VERBAL DEDUCTION

---

Given:

- All managers are employees.
- No employees are minors.

Then:

$\text{Managers} \subset \text{Employees}$

and Employees disjoint Minors.

Therefore:

$\boxed{\text{No managers are minors}}$

Must be true.

---

## 66. “COULD BE TRUE”

---

Given:

- Some engineers are managers.
-

- No managers are interns.

Can some engineers be interns?

Yes, because engineers outside manager subset may be interns.

“Could” asks for one consistent model.

---

## 67. “CANNOT BE TRUE”

---

Given:

- All A are B.
- No B are C.

Then:

**Some A are C**

cannot be true, assuming such A exists in the proposed statement.

---

## 68. CONTRADICTION TEST

---

For “could be true”:

Assume option.

If no contradiction, it could be true.

For “cannot be true”:

Assume option.

If contradiction forced, answer.

---

## 69. STRENGTHEN — RECOMMENDATION ARGUMENT

---

Argument:

**Company should provide free shuttle buses because employees complain about commuting.**

Strongener:

**A large share of employees live along routes the shuttles would serve.**

This shows recommendation can address problem.

---

## 70. WEAKEN — RECOMMENDATION ARGUMENT

---

Weakener:

**Only 2% of employees commute along proposed routes.**



Recommendation less effective.

---

## 71. COST-BENEFIT ASSUMPTION

---

Recommendation often assumes:

$\boxed{\text{Benefits} \text{ outweigh costs}}$

or at least:

$\boxed{\text{Action} \text{ is feasible}}$

---

## 72. POLICY ARGUMENT

---

**City should build cycling lanes to reduce traffic.**

Possible assumptions:

- some drivers will switch
- lanes can be implemented
- cycling is feasible for some trips

Not:

**every person will cycle.**

Too strong.

---

## 73. PREDICTION ARGUMENT

---

**Demand will rise next year because incomes are expected to rise.**

Assumes:

**Other major determinants do not offset the income effect.**

Prediction often rests on “other things roughly equal.”

---

## 74. PLAN ARGUMENT

---

**We can finish project in one month because each module takes one week.**

May assume:

- modules can proceed in parallel
  - no major delays
  - resources available
-

Look for hidden scheduling/logistics assumptions.

---

## 75. SURVEY ARGUMENT

---

**80% of surveyed users liked product, so market will love it.**

Check:

- sample size
  - representativeness
  - question wording
  - response bias
- 

## 76. EXPERIMENT ARGUMENT

---

Check:

- control
  - randomization
  - sample
  - measurement
  - confounders
  - duration
- 

## 77. EXPERT-OPINION ARGUMENT

---

Ask:

- expert in relevant field?
  - unbiased?
  - claim within expertise?
  - evidence available?
- 

## 78. STATISTICS ARGUMENT

---

Watch:

- average vs individual
  - percentage vs absolute numbers
  - relative vs absolute change
  - denominator changes
- 

## 79. PERCENTAGE TRAP

---

**Complaints rose 100%.**

Sounds huge.

Could be:

\[ 1\rightarrow2 \]

Absolute increase only 1.

Always inspect base.

---

## 80. AVERAGE TRAP

**Average salary increased.**

Does not imply every salary increased.

---

## 81. TOTAL VS RATE

**Total accidents increased.**

Could be because number of drivers increased.

Accident rate may still decrease.

---

## 82. SELECTION BIAS

Data only includes successful cases.

Then conclusion about all attempts may be distorted.

Example:

**Studying famous startups only !' survivorship bias.**

---

## 83. SURVIVORSHIP BIAS

Looking only at survivors/successes and ignoring failures.

Classic reasoning flaw.

---

## 84. BASE-RATE NEGLECT

Ignoring background frequency.

Example:

A test is 99% accurate, but disease is extremely rare.

Positive result may still not mean near-99% chance of disease.

This connects probability.

---

## 85. CONFUSING NECESSARY WITH SUFFICIENT

---

Given:

**Degree required for job.**

Degree is necessary.

Does not mean:

**Anyone with degree gets job.**

Necessary "↯" sufficient.

---

## 86. CONCLUSION IDENTIFICATION PRACTICE

---

Argument:

**The new software reduced processing time in three independent trials. Therefore, the company should consider adopting it.**

Premise:

**reduced time in three trials.**

Conclusion:

$\boxed{\text{Company should consider adopting it}}$

---

## 87. ASSUMPTION PRACTICE

---

Same argument.

Possible assumption:

**Processing-time reduction is relevant to adoption decision.**

Negate:

**Processing time is irrelevant.**

Argument weakens sharply.

---

## 88. STRENGTHEN PRACTICE

---

Stronger:

---

**The software also integrates with the company's current systems at low cost.**

Supports feasibility.

## 89. WEAKEN PRACTICE

Weakeners:

**The trials used tiny files, while real production files are 100 times larger and show no speed benefit.**

Attacks generalizability.

## 90. EVALUATE PRACTICE

Question:

**Does the software maintain its speed advantage on actual production workloads?**

Very useful evaluation.

## 91. PARADOX PRACTICE

Fact:

**Restaurant had more customers but lower profit.**

Possible resolution:

**Average cost per meal rose sharply because ingredient prices increased.**

More customers + lower profit can coexist.

## 92. FLAW PRACTICE

Argument:

**Three students from College X got excellent jobs, so all students from College X receive excellent placements.**

Flaw:

$\boxed{\text{Hasty generalization}}$

## 93. GATE-STYLE FORMAL VERBAL DEDUCTION

Given:

1. If P then Q.
2. If Q then R.
3. R is false.

Then:

$\neg R \rightarrow \neg Q \rightarrow \neg P$

Therefore:

$\boxed{P, Q \text{ text}\{false\}}$

---

## 94. PLACEMENT-STYLE BUSINESS CR

Argument:

**Our competitor lowered prices and gained market share. We should lower prices too.**

Potential flaw:

**Assumes our customers, costs, products, and competitive context are sufficiently similar.**

This is analogy + causal assumption.

---

## 95. CR SOLVING FRAMEWORK

Use:

$\boxed{P-C-G}$

### **P — Premise**

What evidence is given?

### **C — Conclusion**

What is claimed?

### **G — Gap**

What unstated bridge connects them?

Most CR questions become easier once you find the gap.

---

## 96. FOR STRENGTHEN

Ask:

**Which option fills/supports the gap?**

---

## 97. FOR WEAKEN

Ask:

Which option attacks the gap?

## 98. FOR ASSUMPTION

Ask:

Which gap must be true?

Use Negation Test.

## 99. FOR EVALUATE

Ask:

Which question tests the gap?

## 100. FOR FLAW

Ask:

What is wrong with the gap?

## PRACTICE — LEVEL 0

### Q1

Argument:

The road is blocked; therefore, the bus should use another route.

What is conclusion?

A. Road is blocked B. Bus should use another route

### Q2

“All software engineers in this team know Python. Meera is a software engineer in this team.”

What follows?

A. Meera knows Python B. Meera may not know Python

### Q3

If  $(P \rightarrow Q)$  and  $Q$  is false, what follows?

A.  $P$  is false B.  $P$  is true

#### Q4

"Some students passed."

Can we infer all passed?

A. Yes B. No

#### Q5

If a claim is based only on two examples and generalized to everyone, what flaw?

A. Hasty generalization B. Circular reasoning

---

### LEVEL 1

#### Q6

Argument:

**Employee satisfaction increased after flexible hours were introduced.**

Which weakens causal claim most?

A. Salaries were increased at the same time. B. Office walls were repainted.

#### Q7

Which strengthens?

A. Similar teams without flexible hours did not show the same increase. B. Employees use computers.

#### Q8

Argument:

**The city should increase buses because buses are overcrowded.**

Which assumption?

A. More buses can help reduce overcrowding. B. Every citizen travels by bus.

#### Q9

Claim:

**All managers are employees. No employee is a minor.**

What follows?

A. No manager is a minor. B. Some managers are minors.

#### Q10

"Not all students passed."

Equivalent?

A. Some students did not pass. B. No student passed.



---

## LEVEL 2

---

### Q11

Argument:

**A survey of 50 premium-gym members found 90% support for more gyms, so 90% of the city supports more gyms.**

Main flaw?

A. Sampling bias B. Modus tollens

### Q12

Argument:

**Sales increased after a celebrity endorsement; therefore, endorsement caused the increase.**

Which most weakens?

A. The company also cut prices by 30%. B. The celebrity is famous.

### Q13

Paradox:

**A store lowered prices but total revenue fell.**

Which best resolves?

A. Sales quantity fell substantially. B. Customers like discounts.

### Q14

Argument:

**Since a degree is required for the job, everyone with a degree will be hired.**

Flaw?

A. Treats necessary condition as sufficient B. Hasty chronology

### Q15

If exactly one of A,B is true and A is false:

$\boxed{\phantom{?}}$

---

## LEVEL 3 — EXAM PRESSURE

---

### Q16

Argument:

**A school introduced tablet-based lessons. Average marks increased by 8%. Therefore, tablets improved learning.**

Which most weakens?

A. At the same time, class sizes were reduced from 40 to 20. B. Tablets have touchscreens. C. Students like technology. D. Teachers use lesson plans.

### Q17

Which most strengthens?

A. Similar classes that did not use tablets but had unchanged class sizes showed no score increase. B. The tablets were expensive. C. Students had different bag colors. D. The school has a library.

### Q18

Which question best evaluates the tablet argument?

A. Did other important teaching conditions change at the same time? B. What color were the tablets? C. Who manufactured classroom desks? D. How old is the school building?

### Q19

Argument:

**In one successful startup, employees worked 60-hour weeks. Therefore, startups that require 60-hour weeks will succeed.**

Best flaw?

A. Hasty generalization and causal oversimplification B. Modus ponens C. Valid deduction D. De Morgan's law

### Q20

Given:

$[A \rightarrow B, B \rightarrow C, \neg C]$

What must be true?

A.  $(\neg A, \neg B)$  B. A and B C. only  $(\neg A)$  D. C

## ANSWERS

### Q1

Conclusion is recommendation:

$\boxed{B}$

### Q2

All team software engineers know Python, and Meera belongs to that class.

$\boxed{A}$

### Q3

Contrapositive:

$\neg Q \rightarrow \neg P$

$\boxed{A}$

---

### Q4

Some does not imply all.

$\boxed{B}$

---

### Q5

Tiny sample != broad conclusion.

$\boxed{A}$

---

### Q6

Salary increase is alternative cause.

$\boxed{A}$

---

### Q7

Comparison group strengthens causal link.

$\boxed{A}$

---

### Q8

Recommendation assumes buses can address overcrowding.

$\boxed{A}$

---

### Q9

Managers != employees, employees exclude minors.

$\boxed{A}$

---

### Q10

$\neg(\text{All passed})$

means at least one did not.

$\boxed{A}$

---

---

**Q11**

Premium-gym users are not necessarily representative of entire city.

$\boxed{A: \text{Sampling bias}}$

---

**Q12**

Price cut provides alternative cause.

$\boxed{A}$

---

**Q13**

Revenue:

$\text{Price} \times \text{Quantity}$

If quantity falls enough, revenue can fall despite lower prices.

$\boxed{A}$

---

**Q14**

Degree is necessary, not sufficient.

$\boxed{A}$

---

**Q15**

Exactly one true.

A false !' B true.

$\boxed{B=\text{true}}$

---

**Q16**

Class-size reduction is a major confounder.

$\boxed{A}$

---

**Q17**

A similar no-tablet comparison group strengthens.

$\boxed{A}$

---

**Q18**

---

This directly tests alternative explanations.

$\boxed{A}$

---

## Q19

One startup cannot establish universal success mechanism.

$\boxed{A}$

---

## Q20

$A \rightarrow B \rightarrow C$

Given:

$\neg C$

Contrapositive chain:

$\neg C \rightarrow \neg B \rightarrow \neg A$

$\boxed{A}$

---

## CHAPTER 76 STATUS

Covered so far:

- argument structure '
- premise/conclusion '
- assumptions '
- negation test '
- strengthen '
- weaken '
- causal reasoning '
- alternative causes '
- reverse causation '
- confounders '
- inference '
- quantifiers '
- necessary/sufficient '
- evaluate argument '
- common flaws '
- paradox resolution '
- analogy reasoning '
- conditional deduction '
- must/could/cannot basics '
- sampling/statistical traps '
- P-C-G framework '

- L0–L3 practice '

But **Chapter 76 is not complete yet.**

Still remaining:

- advanced strengthen/weaken
- necessary vs sufficient assumptions
- evaluate-the-argument hard questions
- boldface/argument-role questions
- principle questions
- method-of-reasoning questions
- resolve-the-paradox advanced sets
- complete verbal deduction cases
- GATE-style formal verbal logic
- placement CR timed set
- **final Unit 4 revision**
- numbered-chapter completion audit

## Progress

\[ \boxed{75/76}\text{ numbered chapters completed} \]

**Current:** 0=0 Chapter 76 — Critical Reasoning & Verbal Deduction

## NEXT

0=0 Chapter 76 Part 2 — Advanced Critical Reasoning, Boldface Roles, Principle Questions, Method of Reasoning, Paradox, Final Verbal Deduction & Unit 4 Revision

USER

next

ASSISTANT

## CHAPTER 76 — CRITICAL REASONING & VERBAL DEDUCTION

### PART 2 — ADVANCED STRENGTHEN/WEAKEN, BOLDFACE ROLES, PRINCIPLE QUESTIONS, METHOD OF REASONING, PARADOX & FINAL UNIT 4 REVISION

**Status:** 0=0 **PROGRESS !** ' COMPLETE **GATE:** Very High **Placements:** Very High **Priority:** 0=0 **Yr 1**

This part completes:

- Chapter 76
- Unit 4 — Verbal Aptitude
- all **76 numbered chapters**

But remember:

**The full course is still NOT complete.**

After this, we continue to:

- Unit 5 — GATE Special Training
- Unit 6 — Placement Special Training
- Unit 7 — Revision & Question Bank
- Unit 8 — Mocks

## 1. ADVANCED STRENGTHEN — WHAT EXACTLY ARE YOU STRENGTHENING?

Every argument has a gap:

$\text{[ Premise } \rightarrow \boxed{\text{Gap}} \rightarrow \text{Conclusion ]}$

A strongener usually:

- fills the gap
- blocks an alternative explanation
- gives additional evidence
- confirms mechanism
- improves representativeness

## 2. EXAMPLE — RECOMMENDATION ARGUMENT

Argument:

**Employee turnover is high. Therefore, the company should introduce flexible working hours.**

Potential gap:

$\text{[ Flexible work } \rightarrow \text{ Lower turnover ]}$

Strongener:

**Exit surveys show that inflexible schedules are a major reason employees leave.**

This directly supports the proposed mechanism.

## 3. WEAKEN — ATTACK THE GAP

Same argument.

Weakeners:

**Most departing employees cite low salary, while very few mention work schedules.**

Now flexible hours may not solve the actual problem.

---

## 4. STRENGTHEN BY RULING OUT ALTERNATIVES

---

Claim:

**New fertilizer increased crop yield.**

Strongener:

**Fields receiving the new fertilizer and control fields were otherwise treated identically.**

This reduces alternative explanations.

---

## 5. WEAKEN BY ALTERNATIVE EXPLANATION

---

**Fields receiving the fertilizer also received more water.**

Now yield increase may be due to water.

---

## 6. STRENGTHEN BY MECHANISM

---

Claim:

**Exercise improves memory.**

Evidence:

**Exercise increases blood flow to brain regions involved in learning.**

This provides a plausible mechanism.

---

## 7. WEAKEN BY MEASUREMENT ERROR

---

Claim:

**Customer satisfaction increased dramatically.**

Weakener:

**The company changed the survey scale during the same period.**

Maybe apparent change comes from measurement, not satisfaction.

---



## 8. NECESSARY ASSUMPTION VS SUFFICIENT ASSUMPTION

This distinction is important.

### Necessary assumption

Argument requires it.

### Sufficient assumption

If true, may be enough to guarantee conclusion.

Critical-reasoning questions usually ask for:

$\boxed{\text{Necessary assumption}}$

## 9. EXAMPLE

Argument:

**We should open another cafeteria because the existing one is overcrowded.**

Necessary assumption:

**A second cafeteria could relieve at least some overcrowding.**

Not necessary:

**A second cafeteria would eliminate overcrowding completely.**

Too strong.

## 10. NEGATION TEST — ADVANCED

Candidate:

**Some users will use the second cafeteria.**

Negation:

**No users will use the second cafeteria.**

Then recommendation loses purpose.

Thus likely necessary.

## 11. WRONG ASSUMPTION — TOO STRONG

Argument:

**The policy may reduce traffic.**

Option:

**The policy will eliminate all traffic.**

The argument does not require such an extreme assumption.

Reject.

## 12. EVALUATE-THE-ARGUMENT — HARD VERSION

Argument:

**A company plans to move entirely to remote work because office rent is expensive.**

Best evaluation question:

**Will productivity, collaboration, and other operational costs remain acceptable under full remote work?**

Why?

Because recommendation depends on overall cost-benefit, not rent alone.

## 13. YES/NO TEST FOR EVALUATE QUESTIONS

A strong evaluate question has:

- one answer that strengthens
- opposite answer that weakens

Example:

**Are most employees able to perform their roles remotely?**

YES !' plan stronger.

NO !' plan weaker.

Excellent evaluation variable.

## 14. IRRELEVANT EVALUATION QUESTION

**What color is the office building?**

Doesn't affect recommendation.

Reject.

## 15. BOLDFACE / ARGUMENT-ROLE QUESTIONS

Some placement tests bold two statements and ask their roles.

Common roles:

- premise
- conclusion
- intermediate conclusion
- counterargument
- evidence
- background
- objection
- response to objection

## 16. INTERMEDIATE CONCLUSION

An intermediate conclusion is:

- supported by earlier premise
- then used as premise for final conclusion

Structure:

$[P_1 \rightarrow C_1 \rightarrow C_2]$

where  $(C_1)$  is intermediate.

## 17. EXAMPLE

**Customer complaints doubled last month. This suggests that service quality has declined. Therefore, management should investigate the support process.**

Roles:

- complaints doubled = premise
- service quality declined = intermediate conclusion
- investigate process = final conclusion

## 18. COUNTERARGUMENT

Example:

**Some managers argue that remote work reduces collaboration. However, recent team-level data show no decline in project completion.**

The first statement is:

$[\boxed{\text{opposing viewpoint / counterargument}}]$

Second responds to it.

---

## 19. CONCESSION

A concession admits a point but does not accept final opposing conclusion.

Example:

**Admittedly, the system is expensive. However, its long-term savings may justify the cost.**

“System is expensive” = concession.

Author's final stance still supports system.

---

## 20. BACKGROUND INFORMATION

Example:

**The company was founded in 1998. Last year, customer complaints doubled. Therefore, service procedures should be reviewed.**

“Founded in 1998” may simply be background.

Don't force every sentence into premise-conclusion role.

---

## 21. PRINCIPLE QUESTIONS

These ask:

**Which general rule best supports the argument?**

or:

**Which case best follows the stated principle?**

---

## 22. PRINCIPLE !' APPLICATION

Principle:

**If an action creates substantial risk to others without their consent, it should be restricted.**

Case:

**A factory releases toxic waste into a public river.**

Application:

$\boxed{\text{Restriction justified}}$

---

---

## 23. APPLICATION !' PRINCIPLE

Case:

**A company should disclose serious defects before selling a product.**

Underlying principle:

**People should be informed about risks that materially affect their decisions.**

---

## 24. PRINCIPLE QUESTION METHOD

Ask:

1. What general rule connects facts to conclusion?
2. Does the option cover this case?
3. Is it too broad or too narrow?

---

## 25. PRINCIPLE TRAP — OVERBROAD RULE

Argument:

**Drivers who repeatedly violate safety rules should face penalties.**

Bad principle:

**Anyone who ever makes any mistake should be punished severely.**

Much broader and more extreme.

---

## 26. METHOD-OF-REASONING QUESTIONS

These ask:

**How does the argument proceed?**

Common methods:

- gives example
- uses analogy
- eliminates alternatives
- attacks causal explanation
- generalizes from sample
- cites authority
- identifies contradiction

- derives consequence from rule

---

## 27. EXAMPLE — ANALOGY METHOD

**Just as preventive maintenance reduces machine breakdowns, regular software maintenance can reduce system failures.**

Method:

`\[ \boxed{\text{argument by analogy}} \]`

---

## 28. EXAMPLE — COUNTEREXAMPLE METHOD

Claim:

**All successful projects require large teams.**

Response:

**Project X succeeded with three people.**

Method:

`\[ \boxed{\text{counterexample to universal claim}} \]`

---

## 29. EXAMPLE — ELIMINATION

**The failure was not caused by hardware because all diagnostics passed, nor by network loss because connectivity was stable. The software update therefore remains the most likely cause.**

Method:

`\[ \boxed{\text{eliminates alternatives}} \]`

---

## 30. EXAMPLE — GENERALIZATION

**95% of a representative sample supports the proposal, so the proposal likely has broad public support.**

Method:

`\[ \boxed{\text{generalization from sample}} \]`

Quality depends on representativeness/sample size.

---

## 31. PARADOX — ADVANCED

A paradox question contains two facts:

\[ Fact\ A \]

and:

\[ Fact\ B \]

that appear inconsistent.

Correct answer adds:

\[ Fact\ C \]

so all three can coexist.

---

## 32. PARADOX EXAMPLE

**A restaurant increased prices, yet customer numbers also increased.**

Possible resolution:

**A famous review dramatically increased demand.**

Then higher price + more customers can both occur.

---

## 33. ANOTHER PARADOX

**Employees worked fewer hours, yet total output increased.**

Resolver:

**New automation substantially increased output per hour.**

---

## 34. PARADOX TRAP

Wrong answer:

**Employee output increased.**

This merely repeats one surprising fact.

It does not explain it.

---

## 35. PARADOX METHOD

Use:

\[ \boxed{BOTH} \]

---

### **B — Believe both facts**

Do not reject either.

### **O — Observe the tension**

Why do they seem inconsistent?

### **T — Third fact**

Find missing condition.

### **H — Harmonize**

Check if all coexist.

---

## **36. COMPLETE VERBAL DEDUCTION**

---

Example:

1. If A is approved, B is funded.
2. If B is funded, C begins.
3. C does not begin.

Then:

$\neg C \rightarrow \neg B \rightarrow \neg A$

So A cannot have been approved.

---

## **37. CONDITIONAL + OR**

---

Rules:

$A \vee B$

$A \rightarrow C$

$B \rightarrow D$

$\neg C$

From  $\neg C$ :

$\neg A$

Since A or B:

$B$

Then:

$D$

Therefore:

$\boxed{B, D \text{ must be true}}$

---

## **38. CONDITIONAL + XOR**

---



Exactly one of A,B.

If:

$A \rightarrow C$

and:

$\neg C$

then:

$\neg A$

Exactly one means:

$\boxed{B}$

---

## 39. CONDITIONAL + BICONDITIONAL

---

$A \leftrightarrow B$

$B \rightarrow C$

If A true:

$A \rightarrow B \rightarrow C$

If C false:

$\neg C \rightarrow \neg B \rightarrow \neg A$

---

## 40. NECESSARY/SUFFICIENT VERBAL LANGUAGE

---

**“requires”**

$A \text{ requires } B \rightarrow A \rightarrow B$

**“ensures”**

$A \text{ ensures } B \rightarrow A \rightarrow B$

Difference is semantic perspective:

- B required for A
- A enough to ensure B

---

## 41. “WITHOUT B, NO A”

---

Means:

$\boxed{A \rightarrow B}$

Example:

**Without registration, no candidate can appear.**

Appear !' registered.

---

## 42. “ONLY A CAN B”

Means:

$\boxed{B \rightarrow A}$

Example:

**Only registered users can vote.**

Vote if registered user.

---

## 43. “ALL BUT”

“All but one passed” means:

$\boxed{\text{Exactly one did not pass}}$

Placement tests may test such wording.

---

## 44. “NONE BUT”

“None but experts can solve it.”

Means:

$\boxed{\text{Only experts can solve it}}$

Solve if expert.

---

## 45. “NO FEWER THAN”

Means:

$\boxed{\text{at least}}$

Example:

**No fewer than 20 people attended.**

$\geq 20$

---

## 46. “NO MORE THAN”

Means:

$\boxed{\text{at most}}$

Example:

**No more than 5 errors.**

$\lceil \leq 5 \rceil$

---

## 47. “NOT LESS THAN”

---

Means:

$\lceil \boxed{\text{at least}} \rceil$

---

## 48. “NOT MORE THAN”

---

Means:

$\lceil \boxed{\text{at most}} \rceil$

---

## 49. “AT LEAST ONE”

---

$\lceil \geq 1 \rceil$

Negation:

$\lceil \boxed{\text{none}} \rceil$

---

## 50. “AT MOST ONE”

---

Possible:

- zero
- one

Not two or more.

---

## 51. “EXACTLY ONE”

---

$\lceil = 1 \rceil$

---

## 52. “AT LEAST TWO”

---

$\lceil \geq 2 \rceil$

Do not interpret as exactly 2.

---

## 53. ADVANCED QUANTIFIER DEDUCTION

---

Premise:

**All A are B.**

and:

---

**Some B are C.**

Can we infer some A are C?

No.

The B that are C may lie outside A.

## 54. SOME + ALL

Premise:

**Some A are B.**

**All B are C.**

Then:

$\boxed{\text{Some } A \text{ are } C}$

Because the known A's objects are also C.

## 55. NO + ALL

**All A are B.**

**No B are C.**

Then:

$\boxed{\text{No } A \text{ are } C}$

## 56. SOME-NOT DEDUCTION

**Some A are not B.**

Can we infer some B are not A?

No.

Do not reverse some-not.

## 57. UNIVERSAL EXISTENCE TRAP

**All dragons are blue.**

Does not prove dragons exist.

In formal GATE logic, universal statements do not assert existence.

---

## 58. BOLDFACE EXAMPLE

Argument:

**(1) The city's buses are frequently overcrowded.** Some officials argue that adding buses would be too expensive.  
**(2) Nevertheless, expanding bus capacity is likely to reduce delays and overcrowding.**

Role of (1):

$\boxed{\text{premise/evidence}}$

Role of (2):

$\boxed{\text{main conclusion}}$

---

## 59. BOLDFACE EXAMPLE 2

**(1) Some critics claim the policy will harm small businesses.** However, pilot data showed no decline in small-business revenue. Therefore, **(2) the criticism is not strongly supported by current evidence.**

(1):

$\boxed{\text{opposing claim}}$

(2):

$\boxed{\text{author's conclusion}}$

---

## 60. ADVANCED FLAW — SELF-SELECTION

Study:

**People who voluntarily joined a fitness app became healthier.**

Conclusion:

**App caused improvement.**

Possible flaw:

People who join voluntarily may already be more motivated.

$\boxed{\text{self-selection bias}}$

---

## 61. SURVIVORSHIP BIAS

Claim:

**We studied 100 successful companies and found all had aggressive growth plans. Therefore aggressive growth guarantees success.**

Missing failed companies with same strategy.

$\boxed{\text{Survivorship bias}}$

## 62. REGRESSION TO THE MEAN

Sometimes extreme performance naturally moves closer to average later.

Example:

**Worst-performing students received special intervention and improved.**

Could partly be regression to mean.

Advanced, but useful in data reasoning.

## 63. BASE-RATE TRAP

A rare event plus highly accurate test does not automatically imply high posterior probability.

Think Bayes.

This is more probability than verbal, but CR may test intuition.

## 64. ABSOLUTE VS RELATIVE RISK

“Risk increased by 100%” could mean:

$1\% \rightarrow 2\%$

Absolute rise:

$1 \text{ percentage point}$

Do not be misled by relative percentages.

## 65. MEAN VS MEDIAN

Argument:

**Average income increased, therefore typical worker is better off.**

Could be misleading if a few high earners drove mean.

Median may tell different story.

## 66. COMPOSITION EFFECT

Overall rate can change because group proportions changed, even if subgroup rates did not.

Advanced DI/CR connection.

---

## 67. TEMPORAL CONFOUNDING

If many things change simultaneously, attributing outcome to one is weak.

Always ask:

**What else changed?**

---

## 68. POLICY WEAKENER

Claim:

**A fine will reduce violations.**

Weakeners:

**Enforcement is so rare that violators are unlikely to face the fine.**

Mechanism fails.

---

## 69. POLICY STRENGTHENER

**Enforcement is reliable and violators are aware of the penalty.**

Makes deterrence more plausible.

---

## 70. FORECAST WEAKENER

Claim:

**Sales will increase because market grew last year.**

Weakeners:

**A major competitor is entering next month.**

New future factor undermines extrapolation.

---

## 71. TREND EXTRAPOLATION FLAW

**Sales increased for three months, so they will continue increasing forever.**

Past trend need not continue indefinitely.

## 72. FALSE PRECISION

Argument based on uncertain data gives exact forecast:

**Revenue will be  $1\ 1\ 2\ ,\ 3\ 4\ 7\ ,\ 8\ 9\ 2$  year.**

If inputs highly uncertain, excessive precision may be unjustified.

## 73. COMPARISON-GROUP FLAW

**School A uses tablets and scores higher than School B, therefore tablets caused better scores.**

Maybe schools differ in:

- teachers
- funding
- student background
- class size

Need comparable groups.

## 74. MISSING DENOMINATOR

**Hospital A had more complications than B.**

Need patient counts.

If A treated 10x more patients, complication rate may be lower.

## 75. RATE VS COUNT

Always distinguish:

$$\left[ \text{Rate} = \frac{\text{Count}}{\text{Exposure/Population}} \right]$$

Count alone may mislead.

## 76. ADVANCED CR SPEED METHOD

Use:

$$\boxed{\text{PCGA}}$$



## P — Premise

What do I know?

## C — Conclusion

What is claimed?

## G — Gap

Why doesn't premise automatically guarantee conclusion?

## A — Attack/Assist

Depending question:

- strengthen !' assist gap
- weaken !' attack gap
- assumption !' required gap
- evaluate !' test gap

## 77. QUESTION-STEM MAP

Stem	What to do
Assumption	Negation test
Strengthen	Support gap
Weaken	Attack gap
Inference	Safe conclusion
Evaluate	Test key uncertainty
Flaw	Name reasoning error
Paradox	Reconcile both facts
Principle	Find general rule
Method	Describe structure
Boldface	Identify roles

## 78. FINAL MIXED PRACTICE — LEVEL 0

### Q1

Argument:

**The server fails frequently, so it should be replaced.**

Premise?

A. Server fails frequently B. It should be replaced

## Q2

Conclusion?

A. Server fails frequently B. It should be replaced

## Q3

If all A are B and all B are C:

A. All A are C B. Some C are A necessarily

## Q4

"Only registered users may enter."

Means:

A. Enter !' registered B. Registered !' enter

## Q5

"No more than 4" means:

A.  $\leq 4$  B.  $\geq 4$

---

## LEVEL 1

### Q6

Claim:

**New advertising caused sales growth.**

Which weakens?

A. A major price cut occurred simultaneously. B. The ad used blue text.

### Q7

Which strengthens?

A. Sales rose only in regions exposed to the campaign, not similar unexposed regions. B. Company has an HR department.

### Q8

Argument:

**A city should build more bike lanes because many commuters say they would cycle if routes were safer.**

Necessary assumption?

A. At least some commuters will actually use safer bike routes. B. Every commuter owns a bicycle.

### Q9

Paradox:

**Product price increased, but demand also increased.**

Best resolver?

A. Product became fashionable after a celebrity endorsement. B. Price increased.

### Q10

"Not all machines failed" means:

A. At least one machine did not fail. B. No machine failed.

## LEVEL 2

### Q11

Argument:

**The university surveyed students attending an optional AI workshop. 95% wanted more AI electives. Therefore 95% of all university students want more AI electives.**

Flaw?

A. Sampling bias B. Contraposition

### Q12

Claim:

**New medication caused recovery.**

Weakeners:

A. Treatment group was also much younger than control group. B. Medication was in tablet form.

### Q13

Best evaluation:

**Will free shuttle buses reduce parking congestion?**

A. What proportion of drivers would switch to the shuttle? B. What color will buses be?

### Q14

Principle:

**People should bear costs when their actions impose avoidable harm on others.**

Which best fits?

A. Factory that pollutes a river should pay cleanup costs. B. Student should buy a red notebook.

### Q15

Method:

**Claim says all birds in region migrate. Researcher identifies one local non-migrating bird.**

Method?

A. Counterexample B. Circular reasoning

---

## LEVEL 3

---

### Q16

A company says:

**After we introduced a productivity app, output increased 15%; therefore, the app caused the increase.**

Which most weakens?

A. The company also hired 20% more staff during the same period. B. The app has a dark mode. C. Employees use laptops. D. Output is measured monthly.

### Q17

Which most strengthens?

A. Output per employee rose in app-using teams but remained unchanged in otherwise similar teams not using it. B. Company changed logo. C. Employees received new ID cards. D. Offices have air conditioning.

### Q18

Which is the most important assumption behind:

**“We should extend store hours because this will increase revenue.”**

A. Enough additional customers will shop during the extra hours to justify the added operating cost. B. Every customer prefers shopping at night. C. Competitors never change hours. D. Electricity is free.

### Q19

Argument:

**Every top-ranked student in one class used Strategy X. Therefore, Strategy X guarantees a top rank.**

Best flaw:

A. Confuses association with sufficient causation and ignores non-top students who may also use X. B. Valid modus ponens C. De Morgan's law D. No flaw

### Q20

Given:

$[ P \rightarrow Q, Q \rightarrow R, P \vee S, \neg R ]$

What must follow?

A.  $\neg P$  and therefore  $\neg S$  B. P and Q C. R D. Nothing

---

## ANSWERS

---

## Q1

$\boxed{A}$

## Q2

$\boxed{B}$

## Q3

Transitivity:

$A \subseteq B \subseteq C$

$\boxed{A}$

Note: does not necessarily prove any A exist.

## Q4

Only registered users may enter:

$\text{Enter} \rightarrow \text{Registered}$

$\boxed{A}$

## Q5

No more than 4:

$\boxed{x \leq 4}$

$\boxed{A}$

## Q6

Price cut is alternative cause.

$\boxed{A}$

## Q7

Comparison group strengthens causal claim.

$\boxed{A}$

## Q8

Argument requires at least some behavioral change.

$\boxed{A}$

---

**Q9**

Celebrity endorsement can raise willingness to buy despite higher price.

$\boxed{A}$

---

**Q10**

$\neg(\text{All failed})$

means:

$\boxed{\text{Some did not fail}}$

$\boxed{A}$

---

**Q11**

Workshop attendees are not necessarily representative.

$\boxed{A: \text{Sampling bias}}$

---

**Q12**

Age difference is a confounder.

$\boxed{A}$

---

**Q13**

This directly tests behavioral response.

$\boxed{A}$

---

**Q14**

Factory's action imposes external harm.

$\boxed{A}$

---

**Q15**

One counterexample defeats universal statement.

$\boxed{A}$

---

**Q16**

Hiring more staff provides obvious alternative cause for total output increase.

---

$\boxed{A}$

---

### Q17

Per-employee comparison + similar control group directly strengthens the app effect.

$\boxed{A}$

---

### Q18

Revenue benefit must exceed additional cost.

$\boxed{A}$

---

### Q19

Top students using X does not prove X is sufficient for top rank.

Also ignores base/comparison group.

$\boxed{A}$

---

### Q20

$P \rightarrow Q$

Given:

$\neg R$

Therefore:

$\neg Q, \neg P$

Given:

$P \vee S$

and:

$\neg P$

therefore:

$S$

So:

$\boxed{A}$

---

## ' CHAPTER 76 COMPLETE

We have now covered:

- premise/conclusion '
- assumptions '

- negation test '
- strengthen/weaken '
- causal reasoning '
- confounding/reverse causation '
- inference '
- quantifiers '
- necessary/sufficient conditions '
- evaluate-the-argument '
- flaw identification '
- paradox resolution '
- analogy reasoning '
- principle questions '
- method-of-reasoning questions '
- boldface/argument roles '
- conditional verbal deduction '
- XOR/OR/biconditional deduction '
- statistical reasoning traps '
- sample bias '
- survivorship bias '
- self-selection '
- rate/count traps '
- PCGA speed method '
- L0–L3 mixed practice '

---

## ' UNIT 4 — VERBAL APTITUDE COMPLETE

---

**Chapter 71 — Grammar for Aptitude '**

**Chapter 72 — Vocabulary & Word Usage '**

**Chapter 73 — Sentence Completion, Correction & Error Detection '**

**Chapter 74 — Para Jumbles & Paragraph Completion '**

**Chapter 75 — Reading Comprehension '**

**Chapter 76 — Critical Reasoning & Verbal Deduction '**

---

## UNIT 4 — 5-MINUTE REVISION

---

### Grammar

---

Check:

$\boxed{\text{Subject} \rightarrow \text{Verb} \rightarrow \text{Tense} \rightarrow \text{Pronoun}}$

Then:



- modifier
  - parallelism
  - comparison
  - preposition
  - article
  - idiom
- 

## Vocabulary

---

Unknown word?

Use:

\[ \boxed{\text{Grammar + Context + Tone + Root + Collocation}} \]

---

## Sentence Completion

---

Use:

\[ \boxed{\text{GCTC}} \]

- Grammar
  - Connector
  - Tone
  - Context/collocation
- 

## Error Detection

---

Scan:

\[ \boxed{\text{SVA\to Tense\to Pronoun\to Modifier\to Parallelism\to Usage}} \]

---

## Para Jumbles

---

Use:

\[ \boxed{\text{OPLCV}} \]

- Opener
  - Pairs
  - Logic
  - Chain
  - Verify
- 

## Paragraph Completion

---

Use:

\[ \boxed{\text{TPST}} \]

---

- Topic
  - Purpose
  - Scope
  - Tone
- 

## Missing Sentence

---

Use:

$\boxed{\text{Left} \text{ fit} + \text{Right} \text{ fit}}$

---

## Reading Comprehension

---

Use:

$\boxed{\text{MAP-E}}$

- Map
  - Author
  - Proof
  - Eliminate
- 

## RC Option Elimination

---

Use:

$\boxed{\text{BESORT}}$

- Broad
  - Extreme
  - Scope shift
  - Opposite
  - Reversed
  - True but irrelevant
- 

## Critical Reasoning

---

Use:

$\boxed{\text{PCGA}}$

- Premise
  - Conclusion
  - Gap
  - Attack/Assist
- 

## UNIT 4 — MUST-MEMORIZE VERBAL TRAPS

---

1. Each !' singular.
  2. A number of !' plural.
  3. The number of !' singular.
  4. Along with does not make subject plural.
  5. Either/or !' nearest subject agreement.
  6. a/an depends on sound.
  7. prefer X to Y.
  8. senior/superior/inferior to.
  9. discuss the issue, not "discuss about."
  10. look forward to + V-ing.
  11. No sooner...than.
  12. Hardly...when.
  13. Correlation "" causation.
  14. Some "" all.
  15. May "" must.
  16. Not proven "" false.
  17. Pronoun needs antecedent.
  18. Example usually follows general point.
  19. Main idea "" one detail.
  20. True option may still be irrelevant.
- 

## UNIT 4 — 15-MINUTE REVISION CHECKLIST

---

You should be able to answer:

- What controls subject–verb agreement?
- Since vs for?
- A vs an based on what?
- Who vs whom?
- Affect vs effect?
- Infer vs imply?
- What makes a modifier dangle?
- What is parallelism?
- What is a mandatory pair?
- Why can "however" reveal paragraph structure?
- Difference between fact and inference?
- Difference between main idea and topic?
- How do you identify tone?
- What is an assumption?
- How do you weaken causation?
- What is a confounder?
- What is sampling bias?
- Necessary vs sufficient?

- How to resolve a paradox?
- What does “only if” mean?

If these are comfortable, Unit 4 foundation is strong.

---

## ' NUMBERED ROADMAP COMPLETE

---

We have now completed:

$\boxed{76/76}$  text{ numbered chapters} ]

**Unit 0 — Calculation & Number Foundations '**

**Unit 1 — Quantitative Aptitude '**

**Unit 2 — Data Interpretation '**

**Unit 3 — Logical & Analytical Reasoning '**

**Unit 4 — Verbal Aptitude '**

But this means only the **core numbered teaching phase** is finished.

## & pTHE COMPLETE COURSE IS NOT YET FINISHED

---

The remaining compulsory units are:

### UNIT 5 — GATE SPECIAL TRAINING +

This will convert all knowledge into GATE marks:

- GATE GA paper pattern
- verbal + numerical reasoning integration
- MCQ/MSQ/NAT strategy
- graphical questions
- approximation
- option elimination
- back-solving
- sanity checking
- PYQ-style mixed questions
- time allocation
- negative-marking decisions

### UNIT 6 — PLACEMENT SPECIAL TRAINING +

- service-company aptitude
- product-company OA
- timed quant
- reasoning
- verbal
- DI
- section strategy
- speed vs accuracy

- company-style mixed tests

## UNIT 7 — REVISION & QUESTION BANK +

- formula book
- must-memorize tables
- 100 common traps
- mixed question bank
- error log
- weak-topic repair
- rapid revision plans

## UNIT 8 — MOCKS +

- GATE GA mocks
- placement aptitude mocks
- easy/standard/hard
- speed mocks
- post-mock analysis
- final weakness audit

## CURRENT COURSE STATUS

### Numbered conceptual chapters:

$\boxed{76/76 = 100\%}$

### Overall complete preparation system:

$\boxed{\text{Still in progress}}$

Concept learning is complete, but **exam conversion + revision + mocks remain**.

## NEXT

### UNIT 5 — GATE SPECIAL TRAINING

## PART 1 — GATE GENERAL APTITUDE EXAM STRATEGY, MCQ/MSQ/NAT, NEGATIVE MARKING, QUESTION SELECTION, APPROXIMATION & OPTION ELIMINATION

USER

next

### ASSISTANT

We're now shifting from learning concepts to **converting them into GATE marks**—especially question selection, negative-marking discipline, estimation, and option elimination.

## UNIT 5 — GATE SPECIAL TRAINING

### PART 1 — GATE GA PAPER STRATEGY, MCQ/MSQ/NAT, NEGATIVE MARKING, QUESTION SELECTION, APPROXIMATION & OPTION ELIMINATION

**Status:**  $\emptyset = \text{Not in PROGRESS}$  **Focus:** Converting aptitude knowledge into **GATE marks** **Priority:**  $\emptyset = \text{Not 1}$

For **GATE 2027**, the official pattern says the paper is a **3-hour CBT**, with **10 General Aptitude questions worth 15 marks** out of 100. GATE uses 1-mark and 2-mark questions, and the paper supports MCQ, MSQ, and NAT formats. Wrong MCQs carry negative marks:  $\setminus(-1/3\setminus)$  for a 1-mark MCQ and  $\setminus(-2/3\setminus)$  for a 2-mark MCQ. MSQ and NAT questions have no negative marking; MSQs have no partial credit.

That means:

$\setminus\boxed{\text{GA is only 10 questions but contributes 15\% of the entire paper}}\setminus$

So careless aptitude mistakes are expensive.

## 1. WHAT CHANGES NOW?

Until Chapter 76, our goal was:

**“Can I solve this concept?”**

From Unit 5 onward:

**“Can I recognize, solve, verify, and submit it quickly under GATE pressure?”**

There are now four skills:

$\setminus\boxed{\text{Knowledge + Selection + Speed + Accuracy}}\setminus$

A student can know every formula and still lose marks through poor selection.

## 2. THE GATE GA OBJECTIVE

You should not approach GA as:

**“I must solve all 10 immediately.”**

Instead:

**Collect the safest marks first.**

Think:

TEXT

Easy certain mark  
! ”

Moderate solvable mark  
!"  
Hard/time-consuming mark  
!"  
Risky guess only if justified

---

### 3. YOUR PRIMARY GA TARGET

---

Because the section is worth 15 marks, a strong preparation target should be:

$\boxed{12+\text{marks consistently in mocks}}$

and gradually push toward:

$\boxed{13-15}$

Do not make "15/15 every time" the minimum acceptable standard.

The first goal is **stable accuracy**.

---

### 4. NEVER START WITH THE HARDEST QUESTION

---

A difficult first question can consume:

- 4 minutes
- confidence
- attention
- working memory

while three easy questions may be waiting.

GATE is not a school exam where questions must be answered in order.

---

### 5. QUESTION-SELECTION MODEL

---

Classify every question quickly into:

#### **A — Attack Now**

You know method immediately.

#### **B — Return**

Solvable but needs calculation/thought.

#### **C — Leave Temporarily**

Unclear, lengthy, or high-risk.

Use:

$\boxed{A \rightarrow B \rightarrow C}$

---

### 6. FIRST-PASS RULE

---

On first seeing a GA question, ask:

**Do I know the approach within roughly 10–20 seconds?**

If yes:

$\boxed{\text{Attempt}}$

If no:

$\boxed{\text{Mark and move}}$

Do not begin experimenting blindly.

## 7. THE SUNK-COST TRAP

You spend 2 minutes.

You think:

**“I already spent 2 minutes, so I must finish.”**

Wrong.

If the remaining path is still unclear:

$\boxed{\text{Move on}}$

Those 2 minutes are already gone.

Do not lose another 3.

## 8. THREE-PASS GATE METHOD

### PASS 1 — Sure Shots

Attempt questions where:

- formula obvious
- reasoning direct
- RC answer clearly supported
- calculation short

Goal:

$\boxed{\text{Maximum marks with minimum mental friction}}$

### PASS 2 — Workable Questions

Return to:

- moderate calculations
- multi-step reasoning
- close verbal options



- DI requiring computation

## PASS 3 — Risk/Review

Finally handle:

- tough questions
- unresolved MCQs
- long calculations
- answers needing sanity check
- no-negative-mark questions you skipped

## 9. MCQ STRATEGY

MCQ has one correct choice.

Negative marking applies to wrong MCQ answers. For GATE 2027, a wrong 1-mark MCQ loses  $\frac{1}{3}$ , and a wrong 2-mark MCQ loses  $\frac{2}{3}$ .

Therefore:

**MCQ requires risk control.**

## 10. PURE BLIND GUESSING IN A 4-OPTION MCQ

Suppose:

- 4 choices
- 1 correct
- absolutely no knowledge

Probability correct:

$$\frac{1}{4}$$

Probability wrong:

$$\frac{3}{4}$$

For a 1-mark MCQ:

$$EV = \frac{1}{4}(1) + \frac{3}{4}\left(-\frac{1}{3}\right)$$

$$= \frac{1}{4} - \frac{1}{4} = 0$$

So:

$$\boxed{\text{Pure random guessing has zero expected mark}}$$

assuming four equally likely options.

## 11. WHY I STILL DON'T RECOMMEND BLIND GUESSING

Even with expected value 0:

- it adds variance
- it can convert a safe 0 into negative marks
- you may waste time
- you may psychologically start guessing too freely

So:

$\boxed{\text{No clue at all ! ' usually leave MCQ}}$

---

## 12. ELIMINATE ONE OPTION

---

Suppose four choices, and you confidently eliminate one.

Three remain.

If equally likely:

$$P(\text{correct}) = \frac{1}{3}$$

For 1-mark question:

$$EV = \frac{1}{3} - \frac{1}{3} \times \frac{1}{3}$$

$$= \frac{1}{3} - \frac{1}{9}$$

$$= \boxed{\frac{2}{9} > 0}$$

Now the gamble has positive expected value.

---

## 13. ELIMINATE TWO OPTIONS

---

Two choices remain:

$$P(\text{correct}) = \frac{1}{2}$$

$$EV = \frac{1}{2} - \frac{1}{2} \left( \frac{1}{3} \right)$$

$$= \frac{1}{2} - \frac{1}{6}$$

$$= \boxed{\frac{1}{3}}$$

Much better.

So **option elimination is not a side skill.**

It directly changes risk.

---

## 14. 2-MARK MCQ HAS SAME RISK STRUCTURE

---

Correct:

$$+2$$

Wrong:

$$-\frac{2}{3}$$

Blind 4-option guess:

$$\frac{1}{4}(2) + \frac{3}{4} \left( -\frac{2}{3} \right)$$


---

$\left[ =\frac{1}{2}-\frac{1}{2} \right]$

$\left[ =0 \right]$

Again.

## 15. MCQ DECISION TABLE

Situation	Recommendation
Know answer	Attempt
Know method, calculable	Attempt
Two options remain	Usually attempt
Three options remain after solid elimination	Reasonable attempt
Absolutely blind	Usually skip
Calculation inconsistent	Recheck before submit

## 16. “I FEEL B IS RIGHT” IS NOT ELIMINATION

Valid elimination requires reasoning:

**B violates sign.**

**C violates units.**

**D contradicts the passage.**

Not:

**“A looks weird.”**

Separate:

$\left[ \boxed{\text{Evidence-based elimination}} \right]$

from:

$\left[ \boxed{\text{intuition-only guessing}} \right]$

## 17. MSQ STRATEGY

In an MSQ, more than one option may be correct.

Official GATE 2027 rules say there is **no negative marking for MSQ**, but there is **no partial credit**.

Thus if correct answers are:

\[ A,C,D \]

and you mark only:

\[ A,C \]

you do not receive partial marks.

## 18. MSQ IS NOT “SELECT EVERYTHING THAT LOOKS POSSIBLE”

For each option, independently ask:

\[ \boxed{\text{Must this option be correct?}} \]

Evaluate A, B, C, D separately.

Do not think:

**“Probably two answers.”**

There may be a different number.

## 19. MSQ TRUTH-TABLE METHOD

Create mentally:

Option	True?
A	' / '
B	' / '
C	' / '
D	' / '

Only after testing all options should you submit the set.

## 20. MSQ HAS NO NEGATIVE MARKING — BUT TIME STILL MATTERS

Because there is no negative marking, an unanswered MSQ gives 0 and an incorrect attempted MSQ also gives 0.

So if time remains at the end:

\[ \boxed{\text{There is no marking penalty for making an attempt}} \]

But do not spend 5 minutes randomly toggling combinations when other solvable marks remain.

## 21. MSQ COMMON TRAP

Option A:

**Always true.**

Option B:

**Usually true.**

Option C:

**Could be true.**

If question asks correct statements:

Only actual truth matters.

“Could be” is not automatically correct unless question's condition supports it.

---

## 22. MSQ UNIVERSAL CLAIMS

Watch:

- always
- never
- for every
- only
- necessarily

A single counterexample can eliminate such an option.

---

## 23. NAT STRATEGY

Numerical Answer Type:

You enter a number rather than choose an option.

GATE 2027 has no negative marking for wrong NAT responses.

So NAT strategy is different:

$\boxed{\text{Solve ! ' Check units ! ' Check rounding ! ' Enter carefully}}$

---

## 24. NAT HAS NO OPTION SUPPORT

With MCQ, options may reveal:

- magnitude
- sign
- common mistakes

NAT gives you no such help.

Therefore sanity checking becomes more important.

---

## 25. NAT INPUT TRAPS

Be careful with:

- decimal place
- negative sign
- percentage vs decimal
- units
- rounding
- ratio order

Example:

If answer is:

$\backslash[ 25\% \backslash]$

Does system expect:

$\backslash[ 25 \backslash]$

or:

$\backslash[ 0.25? \backslash]$

Read question wording.

---

## 26. NAT RANGE

Official GATE answer keys may accept a numerical range for NAT responses rather than only one exact decimal representation.

But during the exam, do not rely on guessing the tolerance.

Calculate carefully.

---

## 27. NO NEGATIVE MARKING DOES NOT MEAN “DON'T CHECK”

A silly NAT error:

Correct:

$\backslash[ 2.5 \backslash]$

Youtype:

$\backslash[ 25 \backslash]$

No negative marking, but still:

$\backslash[ \boxed{0} \backslash]$

Accuracy matters.

---

## 28. THE FOUR GATE SOLVING MODES

For every quantitative question ask whether fastest route is:

### Mode 1 — Exact solution

---

## Mode 2 — Approximation

## Mode 3 — Option elimination

## Mode 4 — Back-solving

Do not automatically use full algebra.

---

## 29. EXACT SOLUTION

---

Use when:

- NAT
- options close together
- algebra short
- precision essential

Example:

$$\left[ \frac{23}{40} \times 100 = 57.5\% \right]$$

Exact is easy.

Do it.

---

## 30. APPROXIMATION

---

Use when:

- MCQ options far apart
- cumbersome numbers
- exact result unnecessary

Example:

$$\left[ \frac{498}{1003} \right]$$

Approximate:

$$\left[ \frac{500}{1000} = 0.5 \right]$$

If options:

- 0.12
- 0.31
- 0.50
- 0.88

Answer clear.

---

## 31. NEVER APPROXIMATE WHEN OPTIONS ARE CLOSE

---

Suppose options:

- 49.7
  - 49.9
-

- 50.1
- 50.3

Approximation to 50 is insufficient.

Now calculate accurately.

---

## 32. APPROXIMATION SAFETY RULE

---

Use approximation only if:

$\boxed{\text{Approximation error}} \ll \text{Option separation}$

---

## 33. ROUNDING DIRECTION

---

Know whether your estimate is above or below actual value.

Example:

$\frac{499}{1001}$

Replacing with:

$\frac{500}{1000} = 0.5$

Actual is slightly below 0.5.

This can resolve nearby options.

---

## 34. FRACTION BENCHMARKS

---

MUST MEMORIZE:

$\frac{1}{2} = 0.5$

$\frac{1}{3} \approx 0.333$

$\frac{2}{3} \approx 0.667$

$\frac{1}{4} = 0.25$

$\frac{3}{4} = 0.75$

$\frac{1}{5} = 0.2$

$\frac{1}{8} = 0.125$

$\frac{3}{8} = 0.375$

$\frac{5}{8} = 0.625$

$\frac{7}{8} = 0.875$

These save time constantly.

---

## 35. PERCENT BENCHMARKS

---

$10\% = 0.1$

---



$$\left[ 12.5\% = \frac{18}{100} \right]$$

$$\left[ 20\% = \frac{15}{100} \right]$$

$$\left[ 25\% = \frac{14}{100} \right]$$

$$\left[ 33.33\% = \frac{13}{100} \right]$$

$$\left[ 50\% = \frac{12}{100} \right]$$

$$\left[ 66.67\% = \frac{23}{100} \right]$$

$$\left[ 75\% = \frac{34}{100} \right]$$

---

### 36. OPTION ELIMINATION BY SIGN

Question asks value.

You derive expression:

$$\left[ \frac{\text{positive}}{\text{positive}} \right]$$

Then answer must be positive.

Immediately eliminate negative options.

---

### 37. OPTION ELIMINATION BY RANGE

Suppose:

$$\left[ 0 < x < 1 \right]$$

Options:

- -2
- 0.4
- 1.5
- 5

Only:

$$\left[ \boxed{0.4} \right]$$

No exact solving needed.

---

### 38. OPTION ELIMINATION BY PARITY

If answer must be even:

Options:

- 23
- 41
- 56
- 75

Answer:

$$\left[ \boxed{56} \right]$$

---

## 39. DIVISIBILITY ELIMINATION

---

If result must be divisible by 9:

Use digit sum.

Options:

- 214
- 315
- 422
- 601

315:

$$\backslash[ 3+1+5=9 \backslash]$$

So divisible by 9.

---

## 40. UNITS-DIGIT ELIMINATION

---

If:

$$\backslash[ 7^{\{103\}} \backslash]$$

only units digit needed.

Cycle:

$$\backslash[ 7,9,3,1 \backslash]$$

$$\backslash[ 103\bmod 4=3 \backslash]$$

Answer units digit:

$$\backslash[ \boxed{3} \backslash]$$

If only one option ends in 3, stop.

---

## 41. MAGNITUDE ELIMINATION

---

Suppose:

$$\backslash[ \sqrt{9999} \backslash]$$

Clearly close to:

$$\backslash[ 100 \backslash]$$

Options:

- 10
- 50
- 100
- 1000

No calculation required.

---

---

## 42. UNIT / DIMENSION ELIMINATION

---

Suppose asked speed.

Option units:

- km
- km/h
- hours
- km<sup>2</sup>

Only:

$\boxed{\text{km/h}}$

dimensionally valid.

Units can expose wrong options.

---

## 43. RATIO SCALE ELIMINATION

---

If:

$A:B=2:3$

and total = 100.

Then A:

40

B:

60

If options violate total/ratio, eliminate.

---

## 44. AVERAGE RANGE RULE

---

Average must lie between minimum and maximum values.

If numbers lie between 20 and 50:

$20 \leq \text{Average} \leq 50$

Option 72 impossible.

---

## 45. WEIGHTED AVERAGE RANGE

---

Weighted average with positive weights also lies between group averages.

If averages:

60, 80

combined average must be:

---

## 46. PROBABILITY RANGE

$$\boxed{0\leq P(E)\leq 1}$$

Options:

- -0.4
- 0.3
- 1.2
- 4

Only 0.3 valid.

## 47. PERCENTAGE RANGE — CONTEXT

If question asks:

**“percentage of total belonging to category A”**

usually:

$$\boxed{0\%-100\%}$$

But percentage increase can exceed 100%.

Do not misuse range rules.

## 48. TRIANGLE INEQUALITY ELIMINATION

For triangle sides:

$$\boxed{a+b>c}$$

If option gives:

$$\boxed{2,3,7}$$

Impossible.

## 49. LOGIC ELIMINATION

Question says:

**Some A are B.**

Option:

**No A are B.**

Direct contradiction.

Reject without drawing full diagram.

---

## 50. RC ELIMINATION

---

Passage uses:

**may improve**

Option says:

**always guarantees**

Reject through certainty mismatch.

No rereading needed.

---

## 51. BACK-SOLVING

---

Instead of solving equation, plug options.

Example:

**A number increased by 20% becomes 120. Original?**

Options:

80, 90, 100, 110

Test 100:

$\sqrt{100(1.2)=120}$

Done.

---

## 52. WHEN BACK-SOLVING IS GREAT

---

Use when:

- options numerical
- equation messy
- forward algebra slower
- answer values easy to test

---

## 53. START FROM MIDDLE OPTION

---

If options ordered:

$\sqrt{20,30,40,50}$

Test 40 first.

If result too high/low and relation monotonic, eliminate half.

This resembles binary search.

---

## 54. EXAMPLE

---

Find  $\backslash(x\backslash)$ :

$\backslash[3x+5=95\backslash]$

Options:

20, 25, 30, 35.

Test 30:

$\backslash[90+5=95\backslash]$

Done instantly.

---

## 55. MONOTONIC BACK-SOLVING

---

Suppose quantity increases with  $x$ .

Test middle option.

If resulting quantity too low:

eliminate all lower options.

Powerful.

---

## 56. OPTION DIFFERENCE METHOD

---

Options:

- 48
- 52
- 96
- 104

Often indicates common mistakes:

- half/double
- $\pm$  small adjustment

Understanding distractor design can reveal where to check.

But do not infer answer solely from option pattern.

---

## 57. SANITY CHECKING

---

Every numerical answer should survive at least one quick check.

Ask:

- sign?
  - size?
-

- units?
  - direction?
  - range?
- 

## 58. SIGN CHECK

---

If discount increases, selling price should generally decrease.

If your calculation gives higher selling price:

something is wrong.

---

## 59. DIRECTION CHECK

---

If speed increases for same distance:

time should decrease.

If your answer increases time:

recheck.

---

## 60. BOUND CHECK

---

Average of 10 and 20 cannot be 35.

Probability cannot be 1.4.

These checks take seconds.

---

## 61. REVERSE CHECK

---

You calculate original value = 80.

Question says after 25% increase result should be 100.

Check:

$$\backslash [ 80(1.25)=100 \backslash ]$$

Confirmed.

---

## 62. APPROXIMATE CHECK AFTER EXACT SOLUTION

---

You calculate:

$$\backslash [ \frac{497 \times 19}{101} = 93.47 \backslash ]$$

Quick estimate:

$$\backslash [ \frac{500 \times 20}{100} = 100 \backslash ]$$

93.47 is plausible.

If result were 9.347, likely decimal error.

---

---

## 63. COMMON GATE CARELESS ERRORS

---

1. Reading “increase by” as “increase to”.
  2. Percentage vs percentage points.
  3. km/h vs m/s.
  4. Minutes vs hours.
  5. Average speed formula misuse.
  6. Ratio order reversal.
  7. “At least” vs “at most”.
  8. NOT/EXCEPT missed.
  9. MCQ vs MSQ format overlooked.
  10. NAT decimal entered incorrectly.
- 

## 64. “INCREASE BY” VS “INCREASE TO”

---

Increase **by** 20%:

$$\text{New} = 1.2 \text{Old}$$

Increase **to** 120%:

same here.

But increase **to** 120% means final absolute value 120%.

Read carefully.

---

## 65. PERCENT VS PERCENTAGE POINT

---

Rate goes:

$$20\% \rightarrow 25\%$$

Increase in percentage points:

$$5 \text{ points}$$

Relative percentage increase:

$$\frac{5}{20} \times 100 = 25\%$$

Very different.

---

## 66. UNIT CONVERSION

---

$$1 \text{ m/s} = 3.6 \text{ km/h}$$

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

MUST MEMORIZE.

---



## 67. HOURS AND MINUTES

---

$\text{\texttt{\text{30}\text{min}=0.5h}}$

$\text{\texttt{\text{45}\text{min}=0.75h}}$

Not:

$\text{\texttt{0.45h}}$

Classic error.

---

## 68. AVERAGE SPEED TRAP

---

For equal distances at speeds  $(a,b)$ :

$\text{\texttt{\text{frac{2ab}{a+b}}}}$

not:

$\text{\texttt{\text{frac{a+b}{2}}}}$

unless equal times.

---

## 69. RATIO ORDER TRAP

---

If:

$\text{\texttt{B:A=3:2}}$

then:

$\text{\texttt{A:B=2:3}}$

Do not preserve the same order blindly.

---

## 70. “AT LEAST”

---

$\text{\texttt{\text{ge}}}$

“At least 3”:

$\text{\texttt{3,4,5,\dots}}$

---

## 71. “AT MOST”

---

$\text{\texttt{\text{le}}}$

“At most 3”:

$\text{\texttt{0,1,2,3}}$

depending context.

---

## 72. EXACTLY

---

$\boxed{=}$

Do not confuse:

- at least
- at most
- exactly

These appear heavily in reasoning.

---

## 73. NEGATIVE QUESTION STEM

If question asks:

**Which is NOT correct?**

Before solving, mentally write:

$\boxed{\text{Find false option}}$

This one habit saves marks.

---

## 74. MCQ VS MSQ IDENTIFICATION

Before answering option questions:

**Check whether it is MCQ or MSQ.**

Do not assume one answer automatically.

GATE supports both formats.

---

## 75. NAT BEFORE SUBMITTING

Check:

TEXT

Answer = 2.5

Did I type 2.5?

Not 25?

Not 0.25?

Not 2, 5?

Simple but crucial.

---

## 76. GATE CALCULATOR STRATEGY

Do not use the on-screen calculator for every arithmetic step.

Use mental arithmetic for:

- 25%

- simple ratios
- powers
- small multiplication
- cancellations

Calculator is valuable for cumbersome arithmetic, but unnecessary use slows you down.

---

## 77. CANCEL BEFORE MULTIPLYING

---

Instead of:

$$\left[ \frac{48 \times 75}{60} \right]$$

cancel:

$$\left[ \frac{48}{60} = \frac{4}{5} \right]$$

Then:

$$\left[ \frac{4}{5} (75) = 60 \right]$$

Faster and safer.

---

## 78. PERCENTAGE THROUGH FRACTIONS

---

Find 12.5% of 320.

Since:

$$\left[ 12.5\% = \frac{1}{8} \right]$$

$$\left[ 320 / 8 = 40 \right]$$

No calculator.

---

## 79. 16.67%

---

Approximately:

$$\left[ \frac{1}{6} \right]$$

So 16.67% of 600:

$$\left[ 100 \right]$$


---

## 80. 33.33%

---

$$\left[ \approx \frac{1}{3} \right]$$

33.33% of 90:

$$\left[ 30 \right]$$


---

## 81. GATE QUESTION VALUE VS TIME VALUE

---

A 2-mark question is not automatically worth spending twice the time.

If a 2-mark question takes 6 minutes and a 1-mark question takes 30 seconds:

do 1-marker first.

Think:

$\boxed{\text{Expected marks per minute}}$

---

## 82. MARKS-PER-MINUTE IDEA

---

Question A:

- 1 mark
- 90% confidence
- 30 sec

Very attractive.

Question B:

- 2 marks
- 40% confidence
- 5 min

Poor first-pass choice.

---

## 83. CONFIDENCE LABEL

---

After solving, mentally label:

### **H — High confidence**

No review unless time.

### **M — Medium**

Review later.

### **L — Low**

Recheck before final submission.

This prevents wasting equal review time on everything.

---

## 84. NEVER RE-SOLVE EVERYTHING

---

During review:

Do not recalculate every easy answer.

Prioritize:

- marked questions
  - suspicious answers
  - unit-sensitive NATs
-

- close-option MCQs
- MSQs where one option uncertain

## 85. CHANGE-ANSWER RULE

Do not change an answer because:

**“It suddenly feels wrong.”**

Change only if you identify:

$\boxed{\text{specific error or better reasoning}}$

## 86. COMMON REVIEW FAILURE

First solution:

correct.

During review:

student becomes anxious and changes it incorrectly.

Use evidence, not nervousness.

## 87. GATE GA FIRST-PASSEXAMPLE

Suppose 10 GA questions:

Q	Type	Difficulty
1	Grammar	Easy
2	Percentage	Easy
3	RC	Moderate
4	Work	Hard
5	Logic	Easy
6	DI	Moderate
7	Probability	Hard
8	Vocabulary	Easy
9	Ratio	Moderate
10	Reasoning	Moderate

First pass:

$\{1,2,5,8\}$

Then:

\[ 3,6,9,10 \]

Finally:

\[ 4,7 \]

This is much smarter than 1! 1 0 order.

---

## 88. WHEN TO ATTEMPT GA DURING FULL GATE PAPER

---

There is no universally correct order.

Possible strategies:

### Strategy A

GA first.

Good if it builds confidence.

### Strategy B

Subject first, GA later.

Good if technical mind is freshest initially.

### Strategy C

Easy global scan across paper.

Most flexible.

You must settle this through mocks.

---

## 89. DO NOT COPY SOMEONE ELSE'S ORDER

---

One topper may do GA first.

Another may do it last.

The correct strategy is:

\[ \boxed{\text{the one that repeatedly works in your mocks}} \]

---

## 90. MY DEFAULT RECOMMENDATION FOR TRAINING

---

During preparation:

**Practice GA as a compact timed section.**

Goal:

- solve quickly
- maintain accuracy
- learn when to move

Then full mocks will tell us where it should sit in your 3-hour paper.

---

## 91. GATE GA ACCURACY > RAW SPEED

---

Suppose:

Student A:

10 attempted, 6 correct.

Student B:

8 attempted, 8 correct.

Student B often scores better.

Do not confuse:

$\boxed{\text{Attempts}}$

with:

$\boxed{\text{Marks}}$

---

## 92. NEGATIVE-MARKING MINDSET

---

Your goal is not:

**“Avoid negative marking by attempting nothing uncertain.”**

That becomes too conservative.

Correct goal:

**Take calculated risks where evidence makes expected value favorable.**

---

## 93. CALCULATED RISK

---

Good:

**I eliminated 2 options using clear reasoning.**

Bad:

**Option C has appeared less often, so maybe C.**

Never use answer-pattern superstition.

---

## 94. OPTION-FREQUENCY FALLACY

---

Do not think:

**“Previous three answers were B, so this cannot be B.”**

Questions are not designed to balance answers in a predictable sequence.

Ignore patterns.

## 95. “LONG QUESTION = HARD” FALLACY

Some long verbal questions have easy conclusions.

Some one-line math questions are tricky.

Judge by solution path, not appearance.

## 96. “BIG NUMBERS = HARD” FALLACY

Numbers may cancel.

Example:

$$\left[ \frac{999 \times 35}{999} \right]$$

Answer:

$$[ 35 ]$$

Always simplify before calculating.

## 97. GATE-STYLE PRACTICE — Q1

A quantity increases from 80 to 100.

Percentage increase?

Normal:

$$\left[ \frac{20}{80} \times 100 = 25\% \right]$$

Fast:

$$[ 80 \rightarrow 100 ]$$

Increase is one-fourth of 80.

$$[ \boxed{25\%} ]$$

## 98. Q2 — OPTION ELIMINATION

Average of five positive numbers is 20.

Which could NOT be their sum?

A. 100 B. 20 C. 120 D. 80

Wait.

If average is exactly 20:



$$\backslash \text{ Sum}=5(20)=100 \backslash$$

So only 100 is possible.

If stem asks “could NOT”, then B,C,D all could not—bad single-answer design.

Lesson:

**Always verify the question itself gives unique MCQ structure.**

Let's use a proper question:

**What is their sum?**

A. 20 B. 40 C. 100 D. 120

$$\backslash \boxed{C} \backslash$$

## 99. Q3 — BACK-SOLVE

After a 25% discount, price is  $\text{\$}600$ .

Marked price?

Options:

$\text{\$}720, \text{\$}750, \text{\$}800, \text{\$}900$

After 25% discount:

$$\backslash \text{ SP}=75\% \backslash \text{ MP} \backslash$$

Test  $\text{\$}800$ :

$$\backslash 0.75(800)=600 \backslash$$

$$\backslash \boxed{\text{\$}800} \backslash$$

## 100. Q4 — SANITY CHECK

A car covers 120 km in 2 hours.

A student calculates speed as:

$$\backslash 240 \backslash \text{ km/h} \backslash$$

Sanity:

$$\backslash \text{ Speed}=\frac{\text{Distance}}{\text{Time}} \backslash$$

Dividing by 2 should make number smaller:

$$\backslash \boxed{60 \backslash \text{ km/h}} \backslash$$

## 101. Q5 — RC EXTREME ELIMINATION

Passage:

Exercise may improve concentration for some individuals.

Which inference safest?

A. Exercise always guarantees concentration. B. Exercise can potentially improve concentration for some people. C. Everyone must exercise before studying. D. Concentration depends only on exercise.

$\boxed{B}$

## 102. Q6 — LOGIC ELIMINATION

All A are B.

No B are C.

Which must be true?

A. Some A are C B. No A are C C. All C are A D. Some B are A necessarily

From:

$A \subseteq B$

and B disjoint C:

$\boxed{\text{No } A \text{ are } C}$

Answer:

$\boxed{B}$

## 103. Q7 — APPROXIMATION

$\frac{1998}{401}$

Closest?

A. 2 B. 5 C. 10 D. 20

Approximate:

$\frac{2000}{400} = 5$

$\boxed{B}$

## 104. Q8 — PROBABILITY SANITY

A student gets:

$P(E) = 1.25$

Immediately reject.

$0 \leq P(E) \leq 1$

## 105. Q9 — AVERAGE ELIMINATION

Numbers:

12, 18, 20, 30.

Average options:

A. 5 B. 15 C. 20 D. 40

Average must lie 12–30.

Eliminate A,D.

Exact:

$$\sqrt[4]{\frac{80}{4}}=20$$

$$\boxed{C}$$

---

## 106. Q10 — RATIO BACK-SOLVE

Boys:girls = 3:2.

Total = 50.

Each ratio block:

$$\frac{50}{5}=10$$

Boys:

$$30$$

Girls:

$$20$$

Do not form equations unless needed.

---

## 107. EXAM PRESSURE DRILL

For each question, train yourself to ask:

### TEXT

1. Exact or approximate?
2. Can I eliminate?
3. Can I back-solve?
4. Is there a domain/range check?
5. Is negative marking relevant?

This should become automatic.

---

## 108. MUST-MEMORIZE — CURRENT GATE 2027 PATTERN

For GATE 2027:

- total paper: **100 marks**
- duration: **3 hours**
- GA: **10 questions, 15 marks**

- question values: **1 or 2 marks**
- paper question formats: **MCQ, MSQ, NAT**
- wrong 1-mark MCQ:  $\frac{-1}{3}$
- wrong 2-mark MCQ:  $\frac{-2}{3}$
- MSQ: no negative marking
- NAT: no negative marking
- MSQ: no partial marking

The exact distribution of question formats within GA can vary, so do not assume a fixed yearly split.

---

## 109. 5-MINUTE GATE STRATEGY REVISION

---

### MCQ

Negative marking.

$\boxed{\text{Eliminate before risking}}$

### MSQ

No negative, no partial.

$\boxed{\text{Test each option independently}}$

### NAT

No negative.

$\boxed{\text{Calculate + check entry}}$

### Easy question

Do immediately.

### Hard question

Mark and return.

### Quant

Simplify before calculator.

### RC

Reject extreme/scope-shift answers.

### Numerical answer

Check sign, units, magnitude.

---

## 110. GATE EXAM-DAY MICRO-CHECKLIST

---

Before submitting any aptitude question:

### Quant

- units?
- sign?

- decimal?
- percentage?
- ratio order?

### Reasoning

- MUST vs COULD?
- left/right reference?
- all/some?
- if/only if?

### Verbal

- NOT/EXCEPT?
- author's view?
- extreme option?
- grammar subject?

### DI

- correct row/column?
- units?
- percent vs absolute?

---

## 111. PART 1 STATUS

We have completed:

- current GATE 2027 GA pattern '
- question-type strategy '
- MCQ negative-marking mathematics '
- MSQ strategy '
- NAT strategy '
- three-pass selection '
- calculated guessing '
- approximation '
- option elimination '
- back-solving '
- range/sign/unit checks '
- marks-per-minute thinking '
- confidence-based review '
- common GATE traps '
- first GATE-style pressure drill '

But **Unit 5 is not complete.**

---

## NEXT — UNIT 5 PART 2

### 0-60% GATE NUMERICAL & QUANTITATIVE SPEED TRAINING

We will now stop teaching chapters and start doing **exam conversion drills**, including:

- percentage/ratio questions without long algebra
- averages and weighted averages
- time-work shortcuts
- speed-distance shortcuts
- probability/counting under pressure
- number-system elimination
- approximation drills
- DI calculations
- 1-mark vs 2-mark strategy
- normal method vs GATE fastest method
- deliberate trap questions
- mixed timed sets.

USER

next

ASSISTANT

## UNIT 5 — GATE SPECIAL TRAINING

### PART 2 — GATE NUMERICAL & QUANTITATIVE SPEED TRAINING

**Status:**  $\emptyset=0\%$  PROGRESS **Focus:** Solve standard aptitude numericals with the **shortest valid GATE method**

**Priority:**  $\emptyset=0\%$  For 1

From now on, for every numerical question think:

$\boxed{\text{Can I avoid full algebra?}}$

Preferred order:

$\boxed{\text{Observation}} \rightarrow \text{Cancellation} \rightarrow \text{Fraction shortcut} \rightarrow \text{Approximation} \rightarrow \text{Algebra}$

## 1. PERCENTAGE SPEED METHOD

### Question

A salary increases from ₹ 8000 to ₹ 9200.

Percentage increase?

### Normal Method

Increase:

$\boxed{9200-8000=1200}$

$$\left[ \frac{1200}{8000} \times 100 \right]$$

$$\left[ = 15\% \right]$$

### Fast GATE Method

$$\left[ 8000 \times 15\% = 1200 \right]$$

Therefore:

$$\left[ \boxed{15\%} \right]$$

## 2. REVERSE PERCENTAGE

After a 20% increase, salary becomes ₹ 6000.

Original?

Do not subtract 20% of 6000.

Because:

$$\left[ \text{New} = 120\% \text{ of Original} \right]$$

$$\left[ \text{Original} = \frac{6000}{1.2} \right]$$

$$\left[ = \boxed{5000} \right]$$

## 3. REVERSE-PERCENTAGE TRAP

If ₹ 100 increases by 20%:

$$\left[ 100 \rightarrow 120 \right]$$

To return 120 to 100, decrease required:

$$\left[ \frac{20}{120} \times 100 = 16.67\% \right]$$

Not 20%.

## 4. SUCCESSIVE PERCENT CHANGE

Increase by  $(a\%)$ , then increase by  $(b\%)$ :

$$\left[ \boxed{a + b + \frac{ab}{100}} \right]$$

Example:

20% and 10%:

$$\left[ 20 + 10 + \frac{20(10)}{100} \right]$$

$$\left[ = 32\% \right]$$

## 5. INCREASE THEN DECREASE

Increase by  $(a\%)$ , decrease by  $(b\%)$ :

$$\left[ \boxed{a - b - \frac{ab}{100}} \right]$$

Example:

20% increase then 10% decrease:

$$20 - 10 - \frac{200}{100}$$

$$= \boxed{8\% \text{ increase}}$$

## 6. SAME % INCREASE AND DECREASE

If a quantity increases by  $(x\%)$ , then decreases by  $(x\%)$ :

Net loss:

$$\boxed{\frac{x^2}{100}\%}$$

Example:

20% up then 20% down:

$$\frac{20^2}{100} = 4\%$$

Net:

$$\boxed{4\% \text{ loss}}$$

## 7. PERCENTAGE-FRACTION CONVERSION

MUST MEMORIZE:

%	Fraction
10%	$\frac{1}{10}$
12.5%	$\frac{1}{8}$
16.67%	$\frac{1}{6}$
20%	$\frac{1}{5}$
25%	$\frac{1}{4}$
33.33%	$\frac{1}{3}$
37.5%	$\frac{3}{8}$
40%	$\frac{2}{5}$
50%	$\frac{1}{2}$
60%	$\frac{3}{5}$
62.5%	$\frac{5}{8}$
66.67%	$\frac{2}{3}$
75%	$\frac{3}{4}$

This table is one of the highest-value aptitude memory tables.



---

## 8. GATE SPEED EXAMPLE

---

37.5% of 320:

$$\left[ 37.5\% = \frac{3}{8} \right]$$

$$\left[ 320 \times \frac{3}{8} = 120 \right]$$

$$\left[ \boxed{120} \right]$$

No decimal multiplication.

---

## 9. RATIO — TOTAL PARTS METHOD

---

Ratio:

$$\left[ A:B=3:5 \right]$$

Total = 64.

Total parts:

$$\left[ 3+5=8 \right]$$

One part:

$$\left[ 64/8=8 \right]$$

Therefore:

$$\left[ A=24, \text{quad } B=40 \right]$$

---

## 10. RATIO — DIFFERENCE METHOD

---

Ratio:

$$\left[ A:B=7:4 \right]$$

Difference = 24.

Difference in parts:

$$\left[ 7-4=3 \right]$$

One part:

$$\left[ 24/3=8 \right]$$

Thus:

$$\left[ A=56, \text{quad } B=32 \right]$$

---

## 11. RATIO — DIRECT MULTIPLICATION

---

If:

$$\left[ A:B=2:3 \right]$$

and:

---

\[ B:C=4:5 \]

Make B equal.

LCM of 3 and 4 = 12.

$$\backslash [A:B=8:12 \backslash]$$

$$\backslash [ B:C=12:15 \backslash ]$$

Therefore:

$$\boxed{A:B:C=8:12:15}$$

## 12. PROPORTION CROSS-MULTIPLICATION

$\left[\frac{ab}{\phantom{x}}=\frac{cd}{\phantom{x}}\right]$

Then:

$$\boxed{ad=bc}$$

Use immediately.

Do not turn simple proportions into long equations.

### 13. AVERAGE — BASIC SPEED METHOD

Average of:

$$\setminus [18, 20, 22, 24, 26 \setminus]$$

Numbers are equally spaced.

Middle number:

$\boxed{22}$

No addition required.

## 14. AVERAGE OF CONSECUTIVE NUMBERS

For consecutive integers from  $(a)$  to  $(b)$ :

$$\boxed{\frac{a+b}{2}}$$

Example:

Average 31 to 51:

$$\lfloor \frac{31+51}{2} \rfloor = 41$$

## 15. DEVIATION METHOD

Find average:

\[ 48,51,52,49 \]

Take assumed average 50.

Deviations:

$$\{-2, +1, +2, -1\}$$

Sum = 0.

So average:

$$\boxed{50}$$

Much faster than full sum.

---

## 16. WEIGHTED AVERAGE

---

Group A:

40 students, average 70.

Group B:

60 students, average 80.

Combined:

$$\frac{40(70) + 60(80)}{100}$$

But use weighted movement.

From 70 to 80, second group is 60% of population.

Combined average:

$$70 + 0.6(10)$$

$$= \boxed{76}$$

---

## 17. COMBINED AVERAGE — ALLIGATION IDEA

---

If two group averages are:

$$60, 80$$

combined average:

$$68$$

Then group-size ratio:

$$(80 - 68) : (68 - 60)$$

$$12 : 8$$

$$\boxed{3 : 2}$$

Remember:

$$\boxed{\text{Lower} : \text{Higher} = (\text{Higher} - \text{Mean}) : (\text{Mean} - \text{Lower})}$$

---

## 18. PROFIT & LOSS BASE RULE

---

Profit/loss percentage is normally on:

---

$$\boxed{\text{Cost} \backslash \text{Price}}$$

Profit:

$$\boxed{\text{SP}-\text{CP}}$$

Profit %:

$$\boxed{\frac{\text{Profit}}{\text{CP}} \times 100}$$

---

## 19. PROFIT MULTIPLIER

20% profit:

$$\boxed{\text{SP}=1.2\text{CP}}$$

25% profit:

$$\boxed{\text{SP}=1.25\text{CP}}$$

20% loss:

$$\boxed{\text{SP}=0.8\text{CP}}$$

Use multipliers.

---

## 20. DISCOUNT MULTIPLIER

20% discount:

$$\boxed{\text{SP}=0.8\text{MP}}$$

Two successive discounts 20%, 10%:

$$\boxed{0.8 \times 0.9 = 0.72}$$

Overall discount:

$$\boxed{28\%}$$

---

## 21. DISCOUNT SHORTCUT

Successive discounts  $(a\%, b\%)$ :

$$\boxed{a+b-\frac{ab}{100}}$$

20% and 10%:

$$\boxed{20+10-2=28\%}$$

---

## 22. SIMPLE INTEREST SPEED

$$\boxed{\text{SI}=\frac{\text{PRT}}{100}}$$

If rate 10% for 3 years:

$$\boxed{\text{SI}=30\% \text{ of } P}$$

If  $(P=5000)$ :

$$\backslash SI=1500 \backslash$$

No lengthy substitution needed.

## 23. COMPOUND INTEREST — MULTIPLIERS

10% growth for 2 years:

$$\backslash 1.1^2=1.21 \backslash$$

21% total growth.

20% for 2 years:

$$\backslash 1.2^2=1.44 \backslash$$

44% growth.

## 24. TWO-YEAR CI SHORTCUT

For same annual rate  $\backslash(r\%)$ :

$$\backslash CI-SI=P\left(\frac{r}{100}\right)^2 \backslash$$

Example:

$$\backslash(P=10000, \backslash r=10\%)$$

$$\backslash CI-SI=10000(0.1)^2=100 \backslash$$

## 25. MIXTURE — ALLIGATION

Milk  $^{140}$  /  $^{160}$  mixed to obtain  $^{150}$  / L .

Ratio:

$$\backslash (60-50):(50-40) \backslash$$

$$\backslash 10:10 \backslash$$

$$\backslash \boxed{1:1} \backslash$$

## 26. MIXTURE SPEED RULE

Always write:

TEXT

Low ---- Mean ---- High

Ratio:

$$\backslash Low:High=(High-Mean):(Mean-Low) \backslash$$

Cross differences.

## 27. TIME & WORK — LCM METHOD

---

A finishes in 12 days.

B finishes in 18 days.

Instead of fractions, assume total work:

$$\boxed{\text{LCM}(12,18)=36}$$

A/day:

$$\boxed{36/12=3}$$

B/day:

$$\boxed{36/18=2}$$

Together:

$$\boxed{5/\text{day}}$$

Time:

$$\boxed{36/5=7.2\text{ days}}$$

---

## 28. TIME-WORK RECIPROCAL METHOD

If A takes  $(a)$ , B takes  $(b)$ :

Together:

$$\boxed{\frac{ab}{a+b}}$$

only when both work positively at constant rates.

For 12,18:

$$\boxed{\frac{12(18)}{30} = 7.2}$$

---

## 29. EFFICIENCY-TIME INVERSE

If A is twice as efficient as B:

$$\boxed{\text{Efficiency}_A:\text{Efficiency}_B=2:1}$$

Therefore time:

$$\boxed{\text{Time}_A:\text{Time}_B=1:2}$$

Memory:

$$\boxed{\text{Efficiency}\propto\frac{1}{\text{Time}}}$$

---

## 30. WORK-WAGE RULE

Wages are proportional to work done.

If equal time:

$$\boxed{\text{Wages}\propto\text{Efficiency}}$$

If different times:

$$\frac{Wages}{\text{propto Efficiency} \times \text{Time}}$$

---

## 31. PIPES

---

Treat inlet as positive.

Outlet as negative.

Example:

Pipe A fills in 6 h:

$$\frac{+1}{6}$$

Leak empties in 12 h:

$$\frac{-1}{12}$$

Net:

$$\frac{1}{6} - \frac{1}{12} = \frac{1}{12}$$

Tank fills in:

$$\boxed{12\text{h}}$$

---

## 32. SPEED-DISTANCE FUNDAMENTAL

---

$$D = ST$$

Always decide which variable is fixed.

This avoids most traps.

---

## 33. SAME DISTANCE

---

If distance fixed:

$$\text{Time} \propto \frac{1}{\text{Speed}}$$

If speed increases by 25%:

$$\text{Speed} = 1.25S = \frac{5}{4}S$$

New time:

$$\frac{4}{5}T$$

So time decreases by:

$$20\%$$

---

## 34. SPEED INCREASE !' TIME DECREASE SHORTCUT

---

If speed increases by  $x\%$ :

Time decrease:

---

$$\boxed{\frac{x}{100+x} \times 100\%}$$

For 25%:

$$\frac{25}{125} \times 100 = 20\%$$

---

### 35. SPEED DECREASE !' TIME INCREASE

---

If speed decreases by  $x\%$ :

Time increase:

$$\boxed{\frac{x}{100-x} \times 100\%}$$

Example 20% speed decrease:

$$\frac{20}{80} \times 100 = 25\%$$

---

### 36. AVERAGE SPEED — EQUAL DISTANCES

---

Speeds:

$$a, b$$

Average:

$$\boxed{\frac{2ab}{a+b}}$$

Example 60 and 40:

$$\frac{2(60)(40)}{100} = 48$$

Not 50.

---

### 37. AVERAGE SPEED — EQUAL TIMES

---

If equal time at  $a, b$ :

$$\boxed{\frac{a+b}{2}}$$

Example 60 and 40:

$$50$$

---

### 38. RELATIVE SPEED

---

Opposite directions:

$$\boxed{a+b}$$

Same direction:

$$\boxed{a-b}$$

---

### 39. TRAIN CROSSING POLE

---



$$\left[ \text{Time} = \frac{\text{Train length}}{\text{Speed}} \right]$$

Train crossing platform:

$$\left[ \text{Time} = \frac{\text{Train length} + \text{Platform length}}{\text{Speed}} \right]$$

---

## 40. BOATS

---

Downstream:

$$\left[ b + s \right]$$

Upstream:

$$\left[ b - s \right]$$

Hence:

$$\left[ \boxed{\text{Boat} = \frac{D+U}{2}} \right]$$

$$\left[ \boxed{\text{Stream} = \frac{D-U}{2}} \right]$$

---

## 41. NUMBER SYSTEM — DIVISIBILITY FIRST

---

Before long division, check divisibility.

For 2:

last digit even.

3:

digit sum divisible by 3.

4:

last two digits.

8:

last three digits.

9:

digit sum divisible by 9.

11:

alternating digit sum difference multiple of 11.

---

## 42. REMAINDER SPEED

---

For:

$$\left[ a^n \bmod m \right]$$

look for:

- small cycles
  - Fermat/Euler only when conditions fit
  - simplify base first
-

Example:

$$\backslash[ 23^{10}\bmod 5 \backslash]$$

$$\backslash[ 23\equiv 3\pmod 5 \backslash]$$

Cycle of 3 mod 5:

$$\backslash[ 3,4,2,1 \backslash]$$

$$\backslash[ 10\bmod 4=2 \backslash]$$

Answer:

$$\backslash[ \boxed{4} \backslash]$$

## 43. UNIT DIGIT

For powers, memorize cycles:

Base ending	Cycle
2	2,4,8,6
3	3,9,7,1
4	4,6
7	7,9,3,1
8	8,4,2,6
9	9,1

## 44. HCF-LCM CHECK

For two positive integers:

$$\backslash[ \boxed{HCF \times LCM = Product} \backslash]$$

Can solve missing quantity instantly.

## 45. COUNTING — FCP FIRST

Before permutation formulas, ask:

**How many independent choices at each step?**

If:

- 3 shirts
- 4 pants
- 2 shoes

Total:

$$[ 3 \times 4 \times 2 = 24 ]$$

---

## 46. PERMUTATION VS COMBINATION

---

Ask:

**Does order matter?**

If yes:

$$[ {}^nP_r ]$$

If no:

$$[ {}^nC_r ]$$

---

## 47. COMBINATION SYMMETRY

---

$$[ \boxed{{}^nC_r = {}^nC_{n-r}} ]$$

Example:

$$[ {}^{20}C_{18} = {}^{20}C_2 = 190 ]$$

Much faster.

---

## 48. PERMUTATION WITH REPEATED ITEMS

---

For  $(n)$  total items with repeats:

$$[ \boxed{\frac{n!}{p!q!\cdots}} ]$$

Example “LEVEL”:

5 letters.

L twice, E twice.

$$[ \frac{5!}{2!2!} = 30 ]$$

---

## 49. PROBABILITY — COMPLEMENT

---

“At least one” often easier using:

$$[ \boxed{1 - P(\text{None})} ]$$

Example:

Probability of at least one head in 3 fair tosses:

$$[ 1 - P(\text{TTT}) ]$$

$$[ 1 - \frac{1}{8} = \boxed{\frac{7}{8}} ]$$

## 50. EXACTLY VS AT LEAST

Exactly one:

count exactly one favorable case.

At least one:

often use complement.

Do not mix.

## 51. CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Speed idea:

Once B is known, shrink the universe to B.

## 52. SETS — TWO-SET FORMULA

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

For “neither”:

$$\text{Total} - n(A \cup B)$$

## 53. THREE-SET FORMULA

$$n(A \cup B \cup C)$$

$$= n(A) + n(B) + n(C)$$

$$- n(A \cap B) - n(B \cap C) - n(C \cap A)$$

$$+ n(A \cap B \cap C)$$

## 54. DI — DO NOT COMPUTE EVERYTHING

Suppose a table has 20 values.

Question asks only ratio of 2024 sales of A:B.

Read only:

- A 2024
- B 2024

Do not inspect unrelated cells.

## 55. DI — APPROXIMATE RATIOS

Values:

498 and 1004.

Ratio approximately:

$$\sqrt[5]{500:1000 = 1:2}$$

If options far apart, enough.

---

## 56. DI — PERCENT CHANGE

---

Old = 240

New = 300

Increase:

$$\sqrt[6]{60}$$

Instead of long calculation:

$$\sqrt[6]{60/240=1/4}$$

$$\sqrt[\boxed{25\%}]{} \sqrt[{}]{}$$

---

## 57. DI — PIE CHART

---

Angle:

$$\sqrt[\theta]{} \sqrt[{}]{}$$

Percentage:

$$\sqrt[\frac{\theta}{360}\times 100]{} \sqrt[{}]{}$$

Useful values:

$$\sqrt[36^\circ=10\%]{} \sqrt[{}]{}$$

$$\sqrt[72^\circ=20\%]{} \sqrt[{}]{}$$

$$\sqrt[90^\circ=25\%]{} \sqrt[{}]{}$$

$$\sqrt[120^\circ=33.33\%]{} \sqrt[{}]{}$$

$$\sqrt[180^\circ=50\%]{} \sqrt[{}]{}$$

---

## 58. DI — RATIOS BEFORE ABSOLUTE VALUES

---

If two sectors are:

$$\sqrt[72^\circ, 108^\circ]{} \sqrt[{}]{}$$

Ratio:

$$\sqrt[72:108 = 2:3]{} \sqrt[{}]{}$$

No need to convert both to percentages.

---

## 59. ALGEBRA — PLUGGING OPTIONS

---

Equation:

$$\backslash [ x^2-9x+20=0 \backslash ]$$

Options:

2,4,6,8.

Instead of quadratic formula:

Test 4:

$$\backslash [ 16-36+20=0 \backslash ]$$

So 4 is a root.

If question asks “which is a root?” done.

---

## 60. ALGEBRA — FACTOR RECOGNITION

---

$$\backslash [ x^2-9x+20 \backslash ]$$

Need two numbers:

product 20, sum 9.

$$\backslash [ 4,5 \backslash ]$$

Therefore:

$$\backslash [ (x-4)(x-5) \backslash ]$$

Roots:

$$\backslash [ 4,5 \backslash ]$$

---

## 61. IDENTITIES

---

MUST MEMORIZE:

$$\backslash [ (a+b)^2=a^2+2ab+b^2 \backslash ]$$

$$\backslash [ (a-b)^2=a^2-2ab+b^2 \backslash ]$$

$$\backslash [ a^2-b^2=(a-b)(a+b) \backslash ]$$

$$\backslash [ (a+b)^3=a^3+b^3+3ab(a+b) \backslash ]$$

$$\backslash [ a^3+b^3=(a+b)(a^2-ab+b^2) \backslash ]$$

$$\backslash [ a^3-b^3=(a-b)(a^2+ab+b^2) \backslash ]$$

---

## 62. QUICK SQUARES

---

Useful:

$$\backslash [ 35^2=1225 \backslash ]$$

$$\backslash [ 45^2=2025 \backslash ]$$

---

$$\backslash[ 55^2=3025 \backslash]$$

$$\backslash[ 65^2=4225 \backslash]$$

For numbers ending 5:

$$\backslash[ (10a+5)^2=a(a+1)\backslash,25 \backslash]$$

Example:

$$\backslash[ 85^2 \backslash]$$

$$\backslash[ 8\text{times}9=72 \backslash]$$

append 25:

$$\backslash[ \boxed{7225} \backslash]$$

---

## 63. MULTIPLYNEAR 100

---

$$\backslash[ 98\text{times}103 \backslash]$$

Use:

$$\backslash[ (100-2)(100+3) \backslash]$$

$$\backslash[ =10000+100-6 \backslash]$$

$$\backslash[ =\boxed{10094} \backslash]$$

---

## 64. DIFFERENCE OF SQUARES

---

$$\backslash[ 51\text{times}49 \backslash]$$

$$\backslash[ =(50+1)(50-1) \backslash]$$

$$\backslash[ =2500-1 \backslash]$$

$$\backslash[ =\boxed{2499} \backslash]$$

---

## 65. FRACTION COMPARISON

---

Compare:

$$\backslash[ \frac{5}{7}, \frac{7}{9} \backslash]$$

Cross multiply:

$$\backslash[ 5(9)=45 \backslash]$$

$$\backslash[ 7(7)=49 \backslash]$$

Thus:

$$\backslash[ \frac{7}{9}>\frac{5}{7} \backslash]$$

No decimal conversion.

---

## 66. GATE NUMERICAL TRIAGE

---

Before solving, classify:

### Type F

Formula direct.

### Type S

Shortcut obvious.

### Type O

Option testing.

### Type C

Calculation heavy.

Do:

$F \rightarrow S \rightarrow O \rightarrow C$

---

## 67. ONE-MARK MINDSET

---

For a typical 1-mark numerical:

**Avoid spending 4 minutes unless virtually certain and other easy questions are exhausted.**

A 1-mark question should usually feel compact.

---

## 68. TWO-MARK MINDSET

---

A 2-mark problem may justify more steps.

But still ask:

**Can a shortcut reduce this to 30–60 seconds?**

---

## 69. GATE TRAP DRILL 1

---

A product price increases by 25% and then decreases by 20%.

Net?

Multiplier:

$1.25 \times 0.8 = 1$

$\boxed{\text{No change}}$

Do not say 5% increase.

---

## 70. TRAP DRILL 2

---



Speed increases by 20%.

Time decreases by?

$$\left[ \frac{20}{120} \times 100 = 16.67\% \right]$$

$$\left[ \boxed{16.67\%} \right]$$

---

## 71. TRAP DRILL 3

---

Average of 10 numbers is 15.

One value 20 replaced by 40.

New average?

Total increases by:

$$\left[ 20 \right]$$

Average increases by:

$$\left[ 20/10=2 \right]$$

New average:

$$\left[ \boxed{17} \right]$$

No full total calculation.

---

## 72. TRAP DRILL 4

---

Average of 8 values = 25.

Add one value = 34.

New average:

Old total:

$$\left[ 8(25)=200 \right]$$

New total:

$$\left[ 234 \right]$$

New count:

$$\left[ 9 \right]$$

$$\left[ 234/9=26 \right]$$

$$\left[ \boxed{26} \right]$$

---

## 73. TRAP DRILL 5

---

A can do work in 10 days.

B is 25% more efficient.

B's time?

---

Efficiency multiplier:

$$\sqrt[1]{1.25}$$

Time multiplier:

$$\sqrt[1]{1/1.25=0.8}$$

$$\sqrt[1]{10(0.8)=8}$$

$$\sqrt[1]{\boxed{8 \text{ days}}}$$

---

## 74. TRAP DRILL 6

A train 150 m long moves at 54 km/h.

Time to cross pole?

Convert:

$$\sqrt[1]{54 \times \frac{5}{18} = 15 \text{ m/s}}$$

$$\sqrt[1]{T = 150/15}$$

$$\sqrt[1]{\boxed{10 \text{ s}}}$$

---

## 75. TRAP DRILL 7

Probability of at least one six in 2 dice throws?

Complement:

No six in one:

$$\sqrt[1]{5/6}$$

No six twice:

$$\sqrt[1]{25/36}$$

At least one:

$$\sqrt[1]{1 - \frac{25}{36} = \boxed{\frac{11}{36}}}$$

---

## 76. TRAP DRILL 8

How many ways choose 2 people from 10?

$$\sqrt[1]{{}^{10}C_2 = \frac{10(9)}{2} = \boxed{45}}$$

Do not use  $(10P2=90)$ .

---

## 77. TRAP DRILL 9

Number is increased from 50 to 75.

Increase:

$$\sqrt[1]{25}$$

---

Percentage:

$$\left[ \frac{25}{50} = 50\% \right]$$

Not:

$$\left[ \frac{25}{75} \right]$$

Base = original.

---

## 78. TRAP DRILL 10

---

1 5 0 Item discounted 20%, then 10%.

Price:

$$\left[ 500(0.8)(0.9) \right]$$

$$\left[ =360 \right]$$

Overall discount:

$$\left[ 140 \right]$$

$$\left[ 28\% \right]$$

---

## 79. MIXED SPEED SET — L1

---

### Q1

25% of 360 = ?

A. 72 B. 80 C. 90 D. 120

### Q2

A:B = 4:7, total = 55. A = ?

### Q3

Average of 11,13,15,17,19 = ?

### Q4

A takes 8 days, B 24 days. Together?

### Q5

A speed is 72 km/h. Convert to m/s.

---

## 80. ANSWERS

---

### Q1

$$\left[ \frac{1}{4}(360) = \boxed{90} \right]$$

### Q2

Total parts:

---

$\sqrt{11}$

One part:

$\sqrt{5}$

$A=4(5)=\boxed{20}$

### Q3

Symmetric AP.

Middle:

$\boxed{15}$

### Q4

$\frac{8(24)}{32}=6$

$\boxed{6 \text{ days}}$

### Q5

$72 \times \frac{5}{18}=20$

$\boxed{20 \text{ m/s}}$

---

## 81. MIXED SPEED SET — L2

### Q6

A price rises 10% and then 20%. Net rise?

### Q7

A number is decreased by 20%. By what percentage must the new number increase to restore original?

### Q8

Average of 6 numbers is 30. One value 42 is replaced by 24. New average?

### Q9

Equal distances are travelled at 30 km/h and 60 km/h. Average speed?

### Q10

Probability of at least one head in 4 fair coin tosses?

---

## 82. ANSWERS

### Q6

$10+20+\frac{10(20)}{100}=\boxed{32\%}$

### Q7

Take 100:

$100 \rightarrow 80$

Need +20 on 80:

$$\lfloor 20/80=25\% \rfloor$$

$$\lfloor \boxed{25\%} \rfloor$$

### Q8

Total decreases by:

$$\lfloor 42-24=18 \rfloor$$

Average decreases:

$$\lfloor 18/6=3 \rfloor$$

$$\lfloor \boxed{27} \rfloor$$

### Q9

$$\lfloor \frac{2(30)(60)}{90} = \boxed{40} \rfloor$$

### Q10

$$\lfloor 1-\left(\frac{1}{2}\right)^4 \rfloor$$

$$\lfloor =1-\frac{1}{16} \rfloor$$

$$\lfloor =\boxed{\frac{15}{16}} \rfloor$$

---

## 83. MIXED GATE PRESSURE SET — L3

### Q11

After a 20% discount, a seller still gains 20% profit. Marked price is what percentage of cost price?

### Q12

A completes work in 15 days. B is 50% more efficient than A. Together, how many days?

### Q13

A car travels first half of distance at 40 km/h and second half at 60 km/h. Average speed?

### Q14

If:

$$\lfloor {}^nC_2=45 \rfloor$$

find  $\lfloor (n) \rfloor$ .

### Q15

A fair die is thrown twice. Probability that sum is at least 11?

---

## 84. ANSWERS — L3

### Q11

Let:

$$\boxed{CP=100}$$

20% profit:

$$\boxed{SP=120}$$

20% discount means:

$$\boxed{SP=0.8MP}$$

$$\boxed{MP=120/0.8}$$

$$\boxed{=150}$$

So marked price:

$$\boxed{150\% \text{ of } CP}$$

---

## Q12

A efficiency:

$$\boxed{1}$$

B:

$$\boxed{1.5}$$

Together:

$$\boxed{2.5}$$

A alone time 15 means total-work rate scaling:

Combined time:

$$\boxed{15/2.5 = 6 \text{ days}}$$

---

## Q13

Equal distances:

$$\boxed{\frac{2(40)(60)}{100}}$$

$$\boxed{48 \text{ km/h}}$$

---

## Q14

$$\boxed{\frac{n(n-1)}{2}=45}$$

$$\boxed{n(n-1)=90}$$

$$\boxed{10(9)=90}$$

$$\boxed{n=10}$$

---

## Q15

Sum "e11:

- sum 11: (5,6),(6,5)
- sum 12: (6,6)

3 favorable outcomes.

Total:

$\sqrt[3]{36}$

$\boxed{\frac{3}{36} = \frac{1}{12}}$

## 85. 20-SECOND QUESTION RULE

For easy aptitude numericals, train toward:

- recognize method: 5 sec
- compute: 10 sec
- sanity check: 5 sec

Not every question will fit this, but many should.

## 86. 60-SECOND STOP RULE

If after ~60 seconds on a supposedly straightforward numerical you still have no clear path:

$\boxed{\text{Mark and move}}$

Likely either:

- you missed a shortcut
- question is harder than first-pass level

## 87. GATE SPEED DOES NOT MEAN RUSHING

Speed means:

$\boxed{\text{less unnecessary work}}$

not:

$\boxed{\text{faster careless arithmetic}}$

## 88. YOUR FINAL NUMERICAL SOLVING ORDER

For every quantitative aptitude problem:

**1. Identify topic.**

**2. Ask what is fixed:**

- total?
- distance?
- time?
- ratio?

- base?

**3. Convert percentages to fractions if useful.**

**4. Cancel before multiplying.**

**5. Ask if options can be used.**

**6. Approximate only if safe.**

**7. Perform sanity check.**

## 89. NUMERICAL MUST-MEMORIZE SHEET

### Percentage

$$\text{New} = \text{Old} \left( 1 \pm \frac{r}{100} \right)$$

### Successive increase

$$a + b + \frac{ab}{100}$$

### Successive discount

$$a + b - \frac{ab}{100}$$

### Average

$$\frac{\text{Sum}}{\text{Count}}$$

### Work

$$\text{Rate} = \frac{1}{\text{Time}}$$

### Distance

$$D = ST$$

### Equal-distance average speed

$$\frac{2ab}{a+b}$$

### Probability

$$P = \frac{\text{Favourable}}{\text{Total}}$$

### At least one

$$1 - P(\text{None})$$

### Combination

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

### Two-set union

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

## PART 2 STATUS

Completed:



- percentage speed methods '
- reverse percentage '
- successive changes '
- ratios/proportions '
- average shortcuts '
- weighted average/alligation '
- profit/loss/discount '
- SI/CI speed methods '
- mixtures '
- time-work/efficiency '
- pipes '
- speed-distance '
- trains/boats '
- number-system shortcuts '
- counting '
- probability '
- sets '
- DI speed methods '
- algebra shortcuts '
- mental arithmetic '
- numerical triage '
- GATE pressure drills '

## UNIT 5 STATUS

---

- Part 1 — GATE exam strategy '
- Part 2 — Numerical/Quant Speed Training '
- Part 3 — **GATE Verbal + Logical + Graphical Reasoning Under Time Pressure +**
- Part 4 — **Mixed GATE GA Sets + Final Unit 5 Strategy +**

## NEXT

---

### UNIT 5 — PART 3: GATE VERBAL, LOGICAL & GRAPHICAL REASONING UNDER TIME PRESSURE

Next:

- fastest verbal elimination
- grammar under 30 seconds
- RC under time pressure
- para-jumble shortcuts
- critical reasoning elimination
- syllogism/conditional reasoning speed
- graphical/visual reasoning
- data sufficiency in GATE style
- mixed verbal-logical traps
- 1-mark vs 2-mark decision drills.

USER

next

ASSISTANT

## UNIT 5 — GATE SPECIAL TRAINING

### PART 3 — GATE VERBAL, LOGICAL & GRAPHICAL REASONING UNDER TIME PRESSURE

**Status:**  $\emptyset = \text{IN PROGRESS}$  **Focus:** Solve verbal/reasoning questions with **fast elimination + minimum rereading**

**Priority:**  $\emptyset = \text{Tier 1}$

The rule for this part:

$\boxed{\text{Do not fully solve what you can eliminate}}$

For many GATE GA questions, the fastest path is:

$\boxed{\text{Read precisely}} \rightarrow \text{Find constraint} \rightarrow \text{Eliminate}$

#### 1. VERBAL QUESTIONS — FASTEST ORDER

For a sentence-based question, check:

1. grammar
2. connector
3. meaning
4. tone
5. collocation

This is faster than reading options repeatedly.

#### 2. GRAMMAR UNDER 30 SECONDS

Sentence:

Each of the applicants \_\_\_\_ submitted the form.

Options:

- have
- has
- are
- were

Key word:

$\boxed{\text{Each}}$

Singular.

Answer:

$\boxed{\text{has}}$

You do not need to analyze the whole sentence deeply.

---

### 3. SUBJECT-VERB SPEED TRICK

Strip the middle phrase.

Example:

**The quality of the products supplied by these vendors \_\_\_\_ improved.**

Ignore:

**of the products supplied by these vendors**

Core:

**The quality \_\_\_\_ improved.**

Thus:

$\boxed{\text{has}}$

---

### 4. “A NUMBER OF” / “THE NUMBER OF”

Fast memory:

$\boxed{\text{A number} \rightarrow \text{plural}}$

$\boxed{\text{The number} \rightarrow \text{singular}}$

Examples:

**A number of students are absent.**

**The number of students is increasing.**

---

### 5. CONNECTOR ELIMINATION

Sentence:

**The model performed well on training data; \_\_\_\_, its performance on new data was poor.**

Need contrast.

Answer candidates like:

- therefore 'L
- moreover 'L
- however '
- consequently 'L

No vocabulary depth needed.

## 6. CAUSE VS CONTRAST

Quick table:

Connector	Job
therefore	result
however	contrast
moreover	addition
because	cause
nevertheless	contrast
consequently	result

## 7. SENTENCE COMPLETION — POLARITY METHOD

Sentence:

**Despite repeated criticism, she remained \_\_\_\_.**

“Despite” implies contrast.

Criticism = negative.

Blank should likely be positive/strong:

$\boxed{\text{confident, calm, determined}}$

Eliminate negative options first.

## 8. DOUBLE-BLANK SPEED

Do not solve both independently.

Example:

**Although the proposal appeared \_\_, later evidence showed it was \_\_.**

Pattern:

$\boxed{\text{positive} \rightarrow \text{negative}}$

Therefore option pair like:

$\boxed{\text{promising, flawed}}$

is attractive immediately.

---

## 9. VOCABULARY UNDER PRESSURE

Unknown word?

Use:

$\boxed{\text{Context + Prefix/Root + Tone}}$

Example:

**His explanation was incoherent, and the audience struggled to follow it.**

Even if unfamiliar:

$\boxed{\text{in} = \text{not}}$

coherent = logically connected.

So:

$\boxed{\text{incoherent} = \text{not logically connected}}$

---

## 10. RC — DO NOT READ LIKE A NOVEL

For short GATE passages:

Read for:

- topic
- contrast
- conclusion
- author stance

Not every detail.

---

## 11. RC PASSAGE MAP IN 10–20 SECONDS

Example:

P1 — technology benefit P2 — limitation P3 — balanced conclusion

Mental summary:

$\boxed{\text{Benefit} \rightarrow \text{Risk} \rightarrow \text{Balanced}}$

Now main-idea questions become easier.

---

## 12. MAIN IDEA ELIMINATION

Reject options that are:

- too narrow
- too broad
- extreme
- opposite

Use:

$\boxed{\text{NBEO}}$

Narrow Broad Extreme Opposite

### 13. RC EXTREME-WORD FILTER

Be suspicious of:

- always
- never
- completely
- only
- all
- none
- guaranteed

Passage wording often uses:

- may
- can
- suggests
- some
- often

Preserve certainty.

### 14. RC “NOT PROVEN” TRAP

Passage:

**Evidence is insufficient to prove X.**

Do **not** infer:

$\boxed{X \text{ is false}}$

Only:

$\boxed{X \text{ is not established}}$

### 15. CORRELATION TRAP

Passage:

**X and Y are associated.**

Do not infer:

$\boxed{X \text{ causes } Y}$

unless evidence supports causation.

## 16. INFERENCE SPEED RULE

Correct inference is usually:

$\boxed{\text{one small logical step}}$

Wrong choices often require:

- outside knowledge
- causal leap
- universalization
- speculation

## 17. AUTHOR'S TONE — FAST METHOD

Look at qualifiers.

If author uses:

- may
- however
- evidence suggests
- limitation

Likely tone:

$\boxed{\text{cautious/analytical}}$

Not:

- furious
- celebratory
- sarcastic

## 18. PARA JUMBLE — FIRST FIND OPENER

Good opener:

- introduces topic independently
- no unexplained pronoun
- no “however/therefore/this” dependency

Bad likely opener:

**These results changed the field.**

What results?

Needs earlier sentence.

---

## 19. MANDATORY PAIR METHOD

Example:

A. A new battery design was introduced. B. This design reduced charging time.

Strong:

$\boxed{A \rightarrow B}$

Once pair found, treat AB as one block.

---

## 20. PARA JUMBLE SPEED ORDER

Use:

$\boxed{OPLCV}$

O — Opener P — Pair L — Logic C — Chain V — Verify

---

## 21. “FOR EXAMPLE” CLUE

Sentence beginning:

**For example...**

usually follows a general statement.

So:

$\boxed{\text{General} \rightarrow \text{Example}}$

This often eliminates half the options immediately.

---

## 22. “HOWEVER” CLUE

Needs a contrasting previous idea.

Thus:

$\boxed{\text{Claim} \rightarrow \text{However} \rightarrow \text{contrast}}$

---

## 23. “THEREFORE” CLUE

Needs cause/evidence before it.

Thus:

$\boxed{\text{Evidence} \rightarrow \text{Therefore} \rightarrow \text{conclusion}}$



---

## 24. CRITICAL REASONING — PCGA METHOD

---

Use:

$\boxed{\text{PCGA}}$

P — Premise C — Conclusion G — Gap A — Attack/Assist

For strengthen:

**assist gap.**

For weaken:

**attack gap.**

---

## 25. ASSUMPTION FAST METHOD

---

Use Negation Test.

Argument:

**The city should add buses because buses are overcrowded.**

Candidate:

**Extra buses can reduce overcrowding.**

Negate:

**Extra buses cannot reduce overcrowding.**

Argument collapses.

Therefore assumption.

---

## 26. STRENGTHEN — FASTEST PATTERNS

---

Look for:

- control group
- same result repeated
- mechanism
- representative sample
- alternative causes ruled out

---

## 27. WEAKEN — FASTEST PATTERNS

---

Look for:

- alternative cause
  - reverse causation
  - biased sample
  - measurement error
  - exception
  - implementation failure
- 

## 28. CAUSAL CLAIM CHECKLIST

---

Whenever you see:

$\backslash [ A \text{ happened, } \backslash \text{ then } \backslash B \backslash ]$

and conclusion:

$\backslash [ A \text{ caused } \backslash B \backslash ]$

ask immediately:

1. What else changed?
  2. Could B cause A?
  3. Is sample representative?
  4. Is there a control?
- 

## 29. SYLLOGISM SPEED

---

Translate to sets.

Example:

All A are B:

$\backslash [ A \text{ subseteq } B \backslash ]$

No B are C:

$\backslash [ B \cap C = \text{varnothing} \backslash ]$

Therefore:

$\backslash [ \boxed{A \cap C = \text{varnothing}} \backslash ]$

No diagram beyond this needed.

---

## 30. SOME + ALL

---

Some A are B.

All B are C.

Then:

$\backslash [ \boxed{\text{Some } A \text{ are } C} \backslash ]$

because those known A") objects also lie in C.

---

---

## 31. UNIVERSAL EXISTENCE TRAP

---

All A are B does not prove A exists.

So from:

**All unicorns are white**

you cannot infer:

**Some unicorns are white.**

Formal logic matters.

---

## 32. CONDITIONAL REASONING

---

Given:

$\lceil P \rightarrow Q \rceil$

Valid:

$\lceil \neg Q \rightarrow \neg P \rceil$

Invalid:

$\lceil Q \rightarrow P \rceil$

This single distinction solves many questions.

---

## 33. CONTRAPOSITION SPEED

---

If:

$\lceil A \rightarrow B \rightarrow C \rceil$

and:

$\lceil \neg C \rceil$

then immediately:

$\lceil \boxed{\neg B, \neg A} \rceil$

---

## 34. ONLY IF

---

“A only if B” means:

$\lceil \boxed{A \rightarrow B} \rceil$

Memory:

**B is necessary for A.**

---

---

## 35. IF

“A if B” means:

$\boxed{B \rightarrow A}$

---

## 36. UNLESS

“A unless B”:

$\boxed{\neg B \rightarrow A}$

Equivalent:

$\boxed{A \vee B}$

---

## 37. MUST / COULD / CANNOT

### MUST

true in every valid arrangement.

### COULD

true in at least one valid arrangement.

### CANNOT

true in none.

Do not solve all cases when question asks only one type.

---

## 38. “COULD BE TRUE” METHOD

Assume option.

If no contradiction appears:

$\boxed{\text{Could be true}}$

You only need one valid construction.

---

## 39. “MUST BE TRUE” METHOD

Try to break the option.

If you can build one valid counterexample:

$\boxed{\text{Not must}}$

Counterexample is often fastest.

## 40. DATA SUFFICIENCY — TARGET FIRST

Do not solve the entire system.

Ask:

**Can I determine the exact asked value?**

Example:

Question asks  $\sqrt{x+y}$ .

Statement gives:

$\sqrt{x+y=10}$

Sufficient immediately.

No need find  $x$  and  $y$  separately.

## 41. DS — NEVER LEAK INFORMATION

When testing Statement I:

Use only I.

When testing II:

Forget I completely.

This is one of the biggest DS mistakes.

## 42. YES/NO DATA SUFFICIENCY

If question asks:

**Is  $\sqrt{x>0}$ ?**

A statement is sufficient if it forces:

- always YES

or

- always NO.

You don't need a numeric value.

## 43. DS INSUFFICIENCY PROOF

Find two valid cases satisfying the statement but giving different target answers.

Then:

$\boxed{\text{Insufficient}}$

This is often faster than algebra.

---

## 44. GRAPHICAL REASONING — CORE IDEA

---

GATE may include graph/chart/visual interpretation.

First ask:

- what do axes represent?
- units?
- linear or categorical?
- absolute or percentage?
- cumulative or period value?

Misreading axes destroys the whole question.

---

## 45. GRAPH AXIS TRAP

---

Graph y-axis may start at 90, not 0.

A small difference may look visually huge.

Never judge by bar height alone.

Read numbers.

---

## 46. LINE GRAPH SLOPE

---

Positive slope:

quantity increasing.

Negative slope:

decreasing.

Zero slope:

constant.

Steeper slope:

greater rate of change, if scales comparable.

---

## 47. NONLINEAR AXIS WARNING

---

If log scale is used, equal vertical gaps do not represent equal absolute differences.

Always inspect scale labels.

---

## 48. BAR GRAPH SPEED

---

Do not calculate percentages unless asked.

If question asks:

---

### Which category is highest?

Just compare heights/values.

## 49. PIE CHART SPEED

Use angle fractions:

$$\left[ 90^\circ = \frac{1}{4} \right]$$

$$\left[ 72^\circ = \frac{1}{5} \right]$$

$$\left[ 120^\circ = \frac{1}{3} \right]$$

$$\left[ 180^\circ = \frac{1}{2} \right]$$

Avoid converting everything to percentages unnecessarily.

## 50. TABLE READING — ROW/COLUMN LOCK

Before computing, point mentally:

Year? Product? Unit?

A common error is reading correct number from wrong row.

## 51. MIXED GRAPH — FIRST IDENTIFY DIFFERENT UNITS

Bar may show revenue.

Line may show growth rate.

Do not compare:

$$\left[ 1 \text{ t e x t } \right]$$

directly with:

$$\left[ \% \right]$$

## 52. PERCENTAGE GROWTH FROM GRAPH

If value goes:

$$\left[ 80 \rightarrow 100 \right]$$

Growth:

$$\left[ \frac{20}{80} \times 100 = 25\% \right]$$

Always divide by old value.

## 53. VISUAL PATTERN REASONING

---

If a sequence of shapes is given, inspect separately:

1. rotation
2. reflection
3. count
4. position
5. shading
6. addition/removal
7. alternating pattern

Do not search for one giant pattern immediately.

---

## 54. ROTATION CHECK

---

If object rotates:

$[90^\circ, 180^\circ, 270^\circ, 360^\circ]$

track orientation rather than redrawing whole figure.

---

## 55. ALTERNATING PATTERN

---

Sequence may follow:

$[A, B, A, B, A, ?]$

So solve odd positions and even positions separately.

This applies to:

- numbers
  - letters
  - figures
- 

## 56. COUNTING ELEMENTS IN FIGURES

---

Track counts:

- dots
- lines
- shaded regions
- corners

Often pattern is numerical beneath visual presentation.

---

## 57. MIRROR VS ROTATION

---

Mirror reverses handedness.

Rotation preserves orientation relationship.

---



A letter or asymmetric shape can distinguish them quickly.

---

## 58. ODD-ONE-OUT VISUAL

---

Check:

- symmetry
- number of sides
- direction
- enclosure
- shading count
- connectivity

Do not rely only on appearance.

---

## 59. 1-MARK VS 2-MARK REASONING

---

A 1-mark logic question may still be tricky, but if it needs 10-case enumeration, likely not first-pass material.

For 2-mark:

more case work may be justified.

Still use constraints before enumeration.

---

## 60. REASONING CONSTRAINT-FIRST METHOD

---

Suppose 5 people seated.

Rules include:

- A left of B
- C not adjacent D
- E at an end

Start with strongest constraints:

- fixed end
- adjacency
- exact positions

Weak “left of” relation later.

---

## 61. STRONG CONSTRAINTS

---

Typically:

$\emptyset = \text{exactly}$   $\emptyset = \text{only}$   $\emptyset = \text{immediately}$   $\emptyset = \text{adjacent}$   $\emptyset = \text{opposite}$   $\emptyset = \text{fixed position}$   $\emptyset = \text{either/or}$

Weaker:

+Before/after +Somewhere left/right

---

## 62. OPTION TESTING IN SEATING

If options ask:

**Which arrangement could be valid?**

Do not build every arrangement.

Test each option against rules until one survives.

## 63. RANKING SPEED

If rank from top = L and from bottom = R:

$$\boxed{N=L+R-1}$$

Example:

7th from top, 12th from bottom:

$$\boxed{7+12-1=18}$$

## 64. PEOPLE BETWEEN

If positions are 5th and 12th:

people between:

$$\boxed{12-5-1=6}$$

## 65. DIRECTION SENSE

Use coordinates.

Take:

- East  $(+x)$
- West  $(-x)$
- North  $(+y)$
- South  $(-y)$

This prevents imagination errors.

## 66. DIRECTION EXAMPLE

Walk:

3 north 4 east

Distance from start:

$$\sqrt{3^2+4^2}=5$$

$$\boxed{5}$$

Classic 3–4–5.

---

## 67. BLOOD RELATION SPEED

---

Use generations:

TEXT

+1 parent  
0 same generation  
-1 child

Then track gender.

Do not narrate entire family story mentally.

---

## 68. SERIES — FIRST DIFFERENCE

---

Sequence:

2, 5, 10, 17, 26

Differences:

3,5,7,9

Next difference:

11

Next term:

\[ 37 \]

---

## 69. SERIES — SECOND DIFFERENCE

---

If first differences not obvious:

compute second differences.

Typical quadratic sequences have constant second difference.

---

## 70. LETTER SERIES

---

Convert letters:

\[ A=1,\ B=2,\dots,Z=26 \]

Use mod 26 when shifting beyond Z.

---

## 71. CODING-DECODING SPEED

---

Check likely operations:

- +1/-1 shifts
-

- reverse
- alphabet position
- alternate shifts
- pair swapping

Do not invent complex rules before testing simple ones.

## 72. GATE VERBAL PRESSURE DRILL 1

Sentence:

**Neither the manager nor the employees \_\_\_\_ available.**

Nearest subject:

employees.

Answer:

\[ \boxed{are} \]

## 73. DRILL 2

**Although the method was inexpensive, it was \_\_\_\_ unreliable.**

Need adverb modifying adjective “unreliable.”

Likely:

\[ \boxed{highly} \]

not “high”.

## 74. DRILL 3

Passage says:

**The treatment may reduce symptoms.**

Which option?

A. It always cures disease. B. It can potentially reduce symptoms. C. It is ineffective. D. Everyone should take it.

Answer:

\[ \boxed{B} \]

## 75. DRILL 4 — PARA JUMBLE

A. Therefore, adoption increased. B. The device became cheaper. C. Consumers could now afford it.

Order:

$\neg B \rightarrow C \rightarrow A$

---

## 76. DRILL 5 — CR

---

Argument:

**Test scores increased after coaching began.**

Weakeners:

A. School also reduced class size. B. Coaching center has air conditioning.

$\boxed{A}$

---

## 77. DRILL 6 — CONDITIONAL

---

If:

$P \rightarrow Q$

and:

$\neg Q$

Then:

$\boxed{\neg P}$

---

## 78. DRILL 7 — SYLLOGISM

---

All A are B.

Some A are C.

Then:

Those some A are both B and C.

Therefore:

$\boxed{\text{Some } B \text{ are } C}$

---

## 79. DRILL 8 — DS

---

Question:

What is  $(x+y)$ ?

Statement:

$2x+2y=20$

Divide 2:

$x+y=10$

Sufficient.

---

No need find x,y.

---

## 80. DRILL 9 — GRAPH

---

Sales:

2025 = 80 2026 = 100

Increase:

$\lfloor 25\% \rfloor$

not 20%.

---

## 81. DRILL 10 — RANKING

---

Ravi is 9th from top and 15th from bottom.

Total:

$\lfloor 9+15-1 = \boxed{23} \rfloor$

---

## 82. MIXED TIMED SET — L1

---

### Q1

Choose correct:

A. Each of the players are ready. B. Each of the players is ready.

### Q2

“All P are Q. No Q are R.” What must follow?

A. No P are R B. Some P are R

### Q3

If A only if B, which is correct?

A.  $\lfloor A \rightarrow B \rfloor$  B.  $\lfloor B \rightarrow A \rfloor$

### Q4

Passage says “some researchers disagree.” Which cannot be inferred?

A. At least one researcher disagrees B. All researchers disagree

### Q5

A is 8th from top, 10th from bottom. Total?

---

## 83. ANSWERS

---

### Q1

$\lfloor \boxed{B} \rfloor$

---

**Q2**

$\boxed{A}$

**Q3**

$\boxed{A}$

**Q4**

$\boxed{B}$

**Q5**

$8+10-1=17$

$\boxed{17}$

---

## 84. MIXED TIMED SET — L2

**Q6**

Argument:

Employees using software X finish work faster; therefore X causes higher productivity.

Best weakener:

A. More experienced employees are more likely to use X. B. X has a blue icon.

**Q7**

Arrange:

A. This result surprised researchers. B. Researchers repeated the experiment. C. The second trial produced the same outcome.

**Q8**

If:

$A \rightarrow B, B \rightarrow C, \neg C$

what follows?

**Q9**

Some A are B. All B are C. What must be true?

A. Some A are C B. All A are C

**Q10**

A graph shows value rises from 200 to 250. Percentage rise?

---

## 85. ANSWERS

**Q6**

Experience is an alternative explanation.

$\boxed{A}$

**Q7**

$B \rightarrow C \rightarrow A$

$\boxed{B-C-A}$

**Q8**

$\neg C \rightarrow \neg B \rightarrow \neg A$

$\boxed{\neg A, \neg B}$

**Q9**

$\boxed{A}$

**Q10**

$\frac{50}{200} \times 100 = \boxed{25\%}$

---

## 86. MIXED TIMED SET — L3

---

**Q11**

Passage:

**A study found that people who sleep longer report higher life satisfaction. The study was observational.**

Which is safest?

A. Longer sleep definitely causes greater satisfaction. B. Sleep and satisfaction were associated, but causation is not established. C. Satisfaction causes sleep. D. Everyone should sleep exactly 10 hours.

**Q12**

Argument:

**The city should ban cars downtown because this will reduce congestion.**

Best assumption:

A. Restricting cars will reduce at least some downtown car traffic. B. Every resident owns a car.

**Q13**

Data sufficiency:

What is  $\neg(x)$ ?

I.  $\neg(x^2=25)$  II.  $\neg(x>0)$

Which combination is sufficient?

**Q14**

All A are B. Some B are C. Which must be true?

A. Some A are C B. No definite A–C relationship follows



**Q15**

A line graph rises from 120 to 150 and then falls to 135. Net percentage change from start to end?

**87. ANSWERS****Q11**

Observational association "≠" causation.

$\boxed{B}$

**Q12**

$\boxed{A}$

**Q13**

I alone:

$x = \pm 5$

insufficient.

II alone insufficient.

Together:

$x = 5$

Therefore:

$\boxed{I+II \text{ together} \text{ sufficient}}$

**Q14**

Some B are C may be outside A.

$\boxed{B}$

**Q15**

Start:

120

End:

135

Increase:

15

$\frac{15}{120} \times 100 = 12.5\%$

$\boxed{12.5\% \text{ increase}}$

**88. 30-SECOND VERBAL CHECKLIST**

For grammar:

$\boxed{\text{Subject} \rightarrow \text{Verb} \rightarrow \text{Tense}}$

Then check:

- pronoun
  - preposition
  - modifier
  - parallelism
- 

## 89. 30-SECOND RC CHECKLIST

---

Ask:

1. What is author's central point?
  2. Is option too extreme?
  3. Does it stay in scope?
  4. Can I point to support?
- 

## 90. 30-SECOND REASONING CHECKLIST

---

Ask:

1. MUST/COULD/CANNOT?
  2. strongest constraint?
  3. can I eliminate using contradiction?
  4. can I test options directly?
- 

## 91. 30-SECOND CR CHECKLIST

---

$\boxed{\text{Premise} \rightarrow \text{Gap} \rightarrow \text{Conclusion}}$

Then:

- strengthen !' support gap
  - weaken !' attack gap
  - assumption !' required gap
  - evaluate !' test gap
- 

## 92. GRAPHICAL CHECKLIST

---

Before calculation:

- axes?
- units?
- scale?
- absolute or percentage?
- correct category/year?

Then calculate only what is needed.

---

## 93. WHEN TO SKIP A VERBAL QUESTION

---

Skip temporarily if:

- two options remain but distinction is subtle
- unfamiliar vocabulary dominates
- RC requires repeated rereading
- reasoning requires many cases

Return after easier marks.

---

## 94. WHEN NOT TO SKIP

---

Attempt now if:

- one grammar rule decides
  - one connector decides
  - direct passage fact
  - strong mandatory pair
  - obvious contradiction
  - simple conditional chain
- 

## 95. THE GATE REASONING MINDSET

---

Do not prove everything.

Instead:

$\boxed{\text{Prove only what the question asks}}$

If asked “which could be true,” build one valid case.

If asked “must be true,” try to break each option.

If asked “cannot be true,” find contradiction.

---

## 96. PART 3 STATUS

---

Completed:

- grammar speed elimination '
  - connector/polarity method '
  - vocabulary context speed '
  - RC under time pressure '
  - main idea/inference/tone elimination '
  - para-jumble shortcuts '
  - critical reasoning speed '
  - syllogism speed '
  - conditional deduction '
  - MUST/COULD/CANNOT strategy '
-

- data sufficiency speed '
- graphical reasoning '
- ranking/direction/blood relation shortcuts '
- series/coding quick methods '
- mixed timed L1–L3 drills '

## UNIT 5 STATUS

---

- Part 1 — GATE exam strategy '
- Part 2 — Numerical/Quant speed '
- Part 3 — Verbal/Logical/Graphical speed '
- Part 4 — **Mixed GATE GA Sets + Final Unit 5 Strategy +**

## NEXT

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### UNIT 5 — PART 4: MIXED GATE GENERAL APTITUDE SETS + FINAL UNIT 5 EXAM STRATEGY

Next we will do:

- realistic mixed 1-mark/2-mark style sets
- quant + verbal + reasoning + DI together
- deliberate GATE traps
- question-selection practice
- when to skip/return
- confidence marking
- final 15-mark GA attack plan
- post-set error classification
- Unit 5 completion audit.